All State Engineering Services & PSUs Examinations

MECHANICAL ENGINEERING

Previous Years

Chapterwise Objective Solved Papers

Volume-II

Useful for : **UPSC ESE, UPSC JWM, UPPSC AE, UPRVUNL AE, UP Jal Nigam AE, UKPSC AE, UJVNL AE, HPPSC AE, BPSC AE, MPPSC AE, JPSC AE, CGPSC AE, WBPSC AE, RPSC AE, MPSC AE, OPSC AEE, GPSC AE, TSPSC AEE, APPSC AEE, TNPSC AE, KPSC AE, Nagaland PSC CTSE, Assam PSC, ISRO Scientist/Engineer, TRB Polytechnic Lect., BHEL ET, CIL MT, SJVN ET, VIZAG Steel MT, DRDO Scientist, Haryana PSC, Mizoram PSC, Oil India Ltd., NPCIL ET, Sikkim PSC, TANGEDCO AE, TSGENCO AE, Punjab PSC, RRB SSE.**

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CONTENTS

- **Mechanical Engineering Exam Syllabus** **3-7**
- **Mechanical Engineering Exam Papers Analysis Chart** **8-13**
- **Mechanical Engineering Exam Papers Pie Chart and Bar Graph** **14**
- **Thermodynamics** **15-165**
  - Basic Concepts & Zeroth Law of Thermodynamics 15
  - First Law of Thermodynamics 60
  - Second Law of Thermodynamics 82
  - Entropy 114
  - Availability and Irreversibility 127
  - Properties of Pure Substances, Gases and Gas Mixture 136
  - Thermodynamic Relations 158
- **Power Plant Engineering** **166-297**
  - Steam Power Plant 166
  - Fuels and Combustion 209
  - Steam Turbine 215
  - Steam Nozzle 232
  - Gas Turbine 242
  - Jet Propulsion 261
  - Compressors 268
  - Nuclear Power Plant 290
- **Refrigeration and Air Conditioning** **298-412**
  - Introduction to Refrigerating machine 298
  - Vapour Compression System 325
  - Refrigerants and Components of Refrigeration System 346
  - Vapour Absorption System 366
  - Psychometric Chart and its Application 374
- **Internal Combustion Engine** **413-512**
  - Air Standard Cycles 413
  - SI and CI Engines 436
  - Fuels 461
  - Supercharging 471
  - IC Engine performance and Testing 475
  - Engine Cooling and Lubrication 491
  - Automobile Engineering 494
- **Fluid Mechanics** **513-766**
  - Properties of Fluid 513
  - Pressure and its Measurement 545
  - Hydrostatic Forces on Surfaces 561
  - Buoyancy and Floatation 576
  - Fluid Kinematics 595
  - Fluid Dynamics 621
  - Flow Through Pipes 663
  - Laminar Flow 693
  - Turbulent Flow 705
  - Boundary Layer Theory 712
  - Flow over Notches and Weirs 723
  - Dimensional and Model Analysis 735
  - Fluid Flow Around Submerged Bodies 749
  - Flow of Compressible Fluids 753
- **Fluid Machinery** **767-859**
  - Impact of Jets 767
  - Hydraulics Turbines 772
  - Reciprocating Pumps 820
  - Centrifugal Pumps 829
  - Miscellaneous Hydraulic Machines 854
- **Heat and Mass Transfer** **860-1005**
  - Steady and Unsteady Heat Conduction 860
  - Free and Forced Convection 926
  - Heat Transfer by Radiation 949
  - Boiling and Condensation 980
  - Heat Exchanger 983
  - Mass Transfer 1003
- **Industrial Engineering** **1006-1160**
  - Break Even Analysis 1006
  - Inventory Control 1015
  - Forecasting 1040
  - Queuing Theory 1047
  - Line balancing, Loading and Sequencing 1055
  - PERT and CPM 1061
  - Linear Programming, Transportation, Assignments 1079
  - Quality Control and Analysis 1096
  - Work Study, Work Measurement and Value Engineering 1111
  - Material Requirement Planning 1136
  - Production Planning and Control 1142
- **Miscellaneous Question** **1161-1200**

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सत्यमेव जयते

UPPSC

Uttar Pradesh Public Service Commission

Government of UP, Prayagraj

Combined State Engineering Services

(General Recruitment/Special Recruitment) Examination

Examination Pattern and Syllabus

The following two objective type papers will be for the Combined State Engineering Services Examination.

PAPER-I

Subject	No. of Questions	Marks	Total Marks	Time
General Hindi	25 (Each question of 3 marks)	75	375	2.30 Hours
Main Subject (Mech. Engg.-I)	100 (Each question of 3 marks)	300		

PAPER-II

Subject	No. of Questions	Marks	Total Marks	Time
General Studies	25 (Each question of 3 marks)	75	375	2.30 Hours
Main Subject (Mech. Engg.-II)	100 (Each question of 3 marks)	300		

Personal Examination (Interview) – 100 Marks

Total – 375 + 375 + 100 = 850 Marks

SYLLABUS

General Hindi – Hindi syllabus will be made in such a way that the candidates under standing of Hindi language and efficient use of words can be checked. Its level will be of high school.

General Studies – The question paper of General Studies will include information focusing on current events and their scientific aspects of such things which come in everyday experience and which can be expected from an educated person. The question paper will also included such questions in the history, politics and geography of India, to which candidates will be able to answer without special study.

MECHANICAL ENGINEERING

PAPER-I

1. **Engineering Mechanics:** Analysis of force systems, friction, cendtroid and centre of gravity, trusses and beams, principle of virtual work, kinematics and kinetics of particle, kinematics and kinetics of rigid bodies.
2. **Mechanisms and Machines:** Velocity and acceleration of links, cams and followers gears and gear trains clutches, belt drives, brakes and dynamometers, Flywheel and governors, balancing of rotating and reciprocating masses, balancing of multi cylinder engines, Free and forced vibration, damped vibration, whirling of shafts.

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3. **Mechanics of Solids:** Stresses and strains, compound stresses strains, Torsion of circular shafts, stresses and deflections in beams unsymmetrical bending, curved beams, Thin and thick cylinders and spheres, Buckling of columns, Energy methods, helical and leaf springs.

4. **Design of Machine Elements:** Design for Static and dynamic loading, Theories of failure, fatigue principles of design of rivetted, welded and bolted joints, shafts, springs, bearings, brakes, clutches and flywheels.

5. **Engineering Materials:** Crystal systems and crystallography, crystal imperfections, Alloys and phase diagrams,

   Heat treatment, ferrous and non ferrous metals and alloys, Mechanical properties and testing.

6. **Manufacturing:** Metal casting, metal forming, metal joining, Mechanics of metal cutting, machining and machine tool operations, unconventional machining methods limits, fits and tolerances, inspection: Surface roughness, comparators, computer integrated manufacturing, Flexible manufacturing systems, jigs and fixtures.

7. **Industrial Engineering:** Production, planning and control, inventory control and operation, research, CPM and PERT.

8. **Mechatronics and Robotics:** Microprocessors and microcontrollers, Architecture, Programming, Computer interfacing Programmable logic controller, sensors and actuators, Piezoelectric accelerometers, Hall effect sensors, optical encoder, resolver, Inductosyn, Pneumatic and Hydraulic Actuators, stepper motor, control system, mathematical modeling, control signals, controllability and observability, Robotics: Robot classification, robot specification. Notation: Direct and inverse kinematics homogeneous co-ordinates and arm equation of four axix SCARA Robot.

MECHANICAL ENGINEERING

PAPER-II

1. **Thermodynamics:** Thermodynamic systems and processes, properties of pure substances, concepts and applications of zeroth, first and second law of thermodynamics, entropy, availability and irreversibility, detailed analysis of thermodynamic cycles, ideal and real gases, fuels and combustion.

2. **Fluid Mechanics:** Basic concepts and properties of fluids, manometry, fluid statics, buoyancy, equations of motion, Bernoulli's equation and applications, viscous flow of incompressible fluids, laminar and turbulent flows, flow through pipes and head losses in pipes, dimensional analysis, Forces on immersed bodies and boundary layer over a flat plate, isentropic and adiabatic flows, normal shock waves.

3. **Heat Transfer:** Modes of heat transfer, steady and unsteady heat conduction, thermocouple time constant, critical thickness of insulation, heat transfer from fins, momentum and energy equations for boundary layer flow on a flat plate. Free and forced convection, radiation heat transfer, Stefan-Boltzmann law, shape factor, black and grey body radiation heat exchange, boiling and condensation, heat exchanger analysis, LMTD and NTU – effectiveness methods.

4. **Energy conversion:** SI and CI engines, performance characteristics and testing of IC engines, combustion phenomena in SI and CI engines, carburetion and fuel injection systems, emissions and emission control. Reciprocating and rotary pumps, pelton wheel, Francis and Kaplan turbines, velocity diagrams impulse and reaction principles steam and gas turbines; Rankine and Brayton cycles with regeneration and reheat, high pressure boilers, draft, condensers. Unconventional power systems, including nuclear, MHD, biomass, wind and tidal systems, utilization of solar energy; Reciprocating and rotary compressors; theory and applications, Theory of propulsions, pulsejet and ramjet engines.

5. **Environmental control:** Vapour compression, vapour absorption, steam jet and air refrigeration systems, properties of refrigerant and their nomenclature, psychometrics properties and processes, psychrometic relations, use of pschrometic chart, load estimation, supply air conditions, sensible heat factors, air conditioning system layout, comfort chart, comfort and industrial air conditioning.

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ESE MECHANICAL ENGINEERING

Contents for syllabi of both the Papers together for Preliminary Examination/ Stage-I (objective type Paper–II) and separately for Main/ Stage-II Examination (Conventional type Paper-I and Paper – II).

PAPER–I

1. Fluid Mechanics:

Basic Concepts and Properties of Fluids, Manometry, Fluid Statics, Buoyancy, Equations of Motion, Bernoulli's equation and applications, Viscous flow of incompressible fluids, Laminar and Turbulent flows, Flow through pipes and head losses in pipes.

2. Thermodynamics and Heat transfer:

Thermodynamic systems and processes; properties of pure substance; Zeroth, First and Second Laws of Thermodynamics; Entropy, Irreversibility and availability; analysis of thermodynamic cycles related to energy conversion: Rankine, Otto, Diesel and Dual Cycles; ideal and real gases; compressibility factor; Gas mixtures.

Modes of heat transfer, Steady and unsteady heat conduction, Thermal resistance, Fins, Free and forced convection, Correlationsfor convective heat transfer, Radiative heat transfer – Radiation heat transfer co - efficient; boiling and condensation, Heat exchanger performance analysis.

3. IC Engines, Refrigeration and Air conditioning:

SI and CI Engines, Engine Systems and Components, Performance characteristics and testing of IC Engines; Fuels; Emissions and Emission Control. Vapour compression refrigeration, Refrigerants and Working cycles, Compressors, Condensers, Evaporators and Expansion devices, Other types of refrigeration systems like Vapour Absorption, Vapour jet, thermo electric and Vortex tube refrigeration. Psychometric properties and processes, Comfort chart, Comfort and industrial air conditioning, Load calculations and Heat pumps.

4. Turbo Machinery:

Reciprocating and Rotary pumps, Pelton wheel, Kaplan and Francis Turbines, velocity diagrams, Impulse and Reaction principles, Steam and Gas Turbines, Theory of Jet Propulsion – Pulse jet and Ram Jet Engines, Reciprocating and Rotary Compressors – Theory and Applications

5. Power Plant Engineering:

Rankine and Brayton cycles with regeneration and reheat, Fuels and their properties, Flue gas analysis, Boilers, steam turbines and other power plant components like condensers, air ejectors, electrostatic precipitators and cooling towers – their theory and design, types and applications;

6. Renewable Sources of Energy:

Solar Radiation, Solar Thermal Energy collection - Flat Plate and focusing collectors their materials and performance. Solar Thermal Energy Storage, Applications – heating, cooling and Power Generation; Solar Photovoltaic Conversion; Harnessing of Wind Energy, Bio-mass and Tidal Energy – Methods and Applications, Working principles of Fuel Cells.

PAPER–II

7 . Engineering Mechanics:

Analysis of System of Forces, Friction, Centroid and Centre of Gravity, Dynamics; Stresses and Strains - Compound Stresses and Strains, Bending Moment and Shear Force Diagrams, Theory of Bending Stresses - Slope and deflection-Torsion, Thin and thick Cylinders, Spheres.

8. Engineering Materials:

Basic Crystallography, Alloys and Phase diagrams, Heat Treatment, Ferrous and Non Ferrous Metals, Non metallic materials, Basics of Nano-materials, Mechanical Properties and Testing, Corrosion prevention and control

9. Mechanisms and Machines:

Types of Kinematics Pair, Mobility, Inversions, Kinematic Analysis, Velocity and Acceleration Analysis of Planar Mechanisms, CAMs with uniform acceleration and retardation, cycloidal motion, oscillating followers; Vibrations –

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Free and forced vibration of undamped and damped SDOF systems, Transmissibility Ratio, Vibration Isolation, Critical Speed of Shafts. Gears – Geometry of tooth profiles, Law of gearing, Involute profile, Interference, Helical, Spiral and Worm Gears, Gear Trains- Simple, compound and Epicyclic; Dynamic Analysis – Slider – crank mechanisms, turning moment computations, balancing of Revolving & Reciprocating masses, Gyroscopes –Effect of Gyroscopic couple on automobiles, ships and aircrafts, Governors.

10. Design of Machine Elements:

Design for static and dynamic loading; failure theories; fatigue strength and the S-N diagram; principles of the design of machine elements such as riveted, welded and bolted joints. Shafts, Spur gears, rolling and sliding contact bearings, Brakes and clutches, flywheels.

11. Manufacturing ,Industrial and Maintenance Engineering:

Metal casting-Metal forming, Metal Joining, Machining and machine tool operations, Limits, fits and tolerances, Metrology and inspection, computer Integrated manufacturing, FMS, Production planning and Control, Inventory control and operations research – CPM-PERT. Failure concepts and characteristics-Reliability, Failure analysis, Machine Vibration, Data acquisition, Fault Detection, Vibration Monitoring, Field Balancing of Rotors, Noise Monitoring, Wear and Debris Analysis, Signature Analysis, NDT Techniques in Condition Monitoring.

12. Mechatronics and Robotics:

Microprocessors and Microcontrollers: Architecture, programming, I/O, Computer interfacing, Programmable logic controller. Sensors and actuators, Piezoelectric accelerometer, Hall effect sensor, Optical Encoder, Resolver, Inductosyn, Pneumatic and Hydraulic actuators, stepper motor, Control Systems - Mathematical modeling of Physical systems, control signals, controllability and observability. Robotics, Robot Classification, Robot Specification, notation; Direct and Inverse Kinematics; Homogeneous Coordinates and Arm Equation of four Axis SCARA Robot.

GATE MECHANICAL ENGINEERING

Section 1: Engineering Mathematics

Linear Algebra: Matrix algebra, systems of linear equations, eigen values and eigenvectors.

Calculus: Functions of single variable, limit, continuity and differentiability, mean value theorems, indeterminate forms; evaluation of definite and improper integrals; double and triple integrals; partial derivatives, total derivative, Taylor series (in one and two variables), maxima and minima, Fourier series; gradient, divergence and curl, vector identities, directional derivatives, line, surface and volume integrals, applications of Gauss, Stokes and Green's theorems.

Differential equations: First order equations (linear and nonlinear); higher order linear differential equations with constant coefficients; Euler-Cauchy equation; initial and boundary value problems; Laplace transforms; solutions of heat, wave and Laplace's equations.

Complex variables: Analytic functions; Cauchy-Riemann equations; Cauchy's integral theorem and integral formula; Taylor and Laurent series.

Probability and Statistics: Definitions of probability, sampling theorems, conditional probability; mean, median, mode and standard deviation; random variables, binomial, Poisson and normal distributions.

Numerical Methods: Numerical solutions of linear and non-linear algebraic equations; integration by trapezoidal and Simpson's rules; single and multi-step methods for differential equations.

Section 2: Applied Mechanics and Design

Engineering Mechanics: Free-body diagrams and equilibrium; friction and its applications including rolling friction, belt-pulley, brakes, clutches, screw jack, wedge, vehicles, etc.; trusses and frames; virtual work; kinematics and dynamics of rigid bodies in plane motion; impulse and momentum (linear and angular) and energy formulations; Lagrange's equation.

Mechanics of Materials: Stress and strain, elastic constants, Poisson's ratio; Mohr's circle for plane stress and plane strain; thin cylinders; shear force and bending moment diagrams; bending and shear stresses; concept of shear centre; deflection of beams; torsion of circular shafts; Euler's theory of columns; energy methods; thermal stresses; strain gauges and rosettes; testing of materials with universal testing machine; testing of hardness and impact strength.

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Theory of Machines: Displacement, velocity and acceleration analysis of plane mechanisms; dynamic analysis of linkages; cams; gears and gear trains; flywheels and governors; balancing of reciprocating and rotating masses; gyroscope.

Vibrations: Free and forced vibration of single degree of freedom systems, effect of damping; vibration isolation; resonance; critical speeds of shafts.

Machine Design: Design for static and dynamic loading; failure theories; fatigue strength and the SN diagram; principles of the design of machine elements such as bolted, riveted and welded joints; shafts, gears, rolling and sliding contact bearings, brakes and clutches, springs.

Section 3: Fluid Mechanics and Thermal Sciences

Fluid Mechanics: Fluid properties; fluid statics, forces on submerged bodies, stability of floating bodies; control-volume analysis of mass, momentum and energy; fluid acceleration; differential equations of continuity and momentum; Bernoulli's equation; dimensional analysis; viscous flow of incompressible fluids, boundary layer, elementary turbulent flow, flow through pipes, head losses in pipes, bends and fittings; basics of compressible fluid flow.

Heat-Transfer: Modes of heat transfer; one dimensional heat conduction, resistance concept and electrical analogy, heat transfer through fins; unsteady heat conduction, lumped parameter system, Heisler's charts; thermal boundary layer, dimensionless parameters in free and forced convective heat transfer, heat transfer correlations for flow over flat plates and through pipes, effect of turbulence; heat exchanger performance, LMTD and NTU methods; radiative heat transfer, Stefan- Boltzmann law, Wien's displacement law, black and grey surfaces, view factors, radiation network analysis

Thermodynamics: Thermodynamic systems and processes; properties of pure substances, behavior of ideal and real gases; zeroth and first laws of thermodynamics, calculation of work and heat in various processes; second law of thermodynamics; thermodynamic property charts and tables, availability and irreversibility; thermodynamic relations.

Applications: Power Engineering: Air and gas compressors; vapour and gas power cycles, concepts of regeneration and reheat. I.C. Engines: Air-standard Otto, Diesel and dual cycles. Refrigeration and air-conditioning: Vapour and gas refrigeration and heat pump cycles; properties of moist air, psychrometric chart, basic psychrometric processes. Turbomachinery: Impulse and reaction principles, velocity diagrams, Pelton-wheel, Francis and Kaplan turbines; steam and gas turbines.

Section 4: Materials, Manufacturing and Industrial Engineering

Engineering Materials: Structure and properties of engineering materials, phase diagrams, heat treatment, stress-strain diagrams for engineering materials.

Casting, Forming and Joining Processes: Different types of castings, design of patterns, moulds and cores; solidification and cooling; riser and gating design. Plastic deformation and yield criteria; fundamentals of hot and cold working processes; load estimation for bulk (forging, rolling, extrusion, drawing) and sheet (shearing, deep drawing, bending) metal forming processes; principles of powder metallurgy. Principles of welding, brazing, soldering and adhesive bonding.

Machining and Machine Tool Operations: Mechanics of machining; basic machine tools; single and multi-point cutting tools, tool geometry and materials, tool life and wear; economics of machining; principles of non-traditional machining processes; principles of work holding, jigs and fixtures; abrasive machining processes; NC/CNC machines and CNC programming.

Metrology and Inspection: Limits, fits and tolerances; linear and angular measurements; comparators; interferometry; form and finish measurement; alignment and testing methods; tolerance analysis in manufacturing and assembly; concepts of coordinate-measuring machine (CMM).

Computer Integrated Manufacturing: Basic concepts of CAD/CAM and their integration tools; additive manufacturing.

Production Planning and Control: Forecasting models, aggregate production planning, scheduling, materials requirement planning; lean manufacturing.

Inventory Control: Deterministic models; safety stock inventory control systems.

Operations Research: Linear programming, simplex method, transportation, assignment, network flow models, simple queuing models, PERT and CPM.

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Mechanical Engineering AE & PSC Previous Exam Papers Analysis Chart

Sl No.	Exam	Proposed Year	Question Paper	Total Question
Union Public Service Commission				
1.	UPSC ESE (PRE)	18.07.2021		150
2.	UPSC ESE (PRE)	05.01.2020		150
3.	UPSC ESE (PRE)	06.01.2019		150
4.	UPSC ESE (PRE)	07.01.2018		150
5.	UPSC ESE (PRE)	08.01.2017		150
6.	UPSC ESE	1992-2016		5400
7.	UPSC CSE (PRE)	1994-2010		1200
8.	UPSC JWM	03.09.2017		100
9.	UPSC JWM Advt. No.- 50/2010	2010		100
10.	UPSC JWM Advt. No.- 52/2010	2010		100
11.	UPSC JWM Advt. No.- 16/2009	2009		100
Uttar Pradesh Public Service Commission				
12.	UPPSC AE (Paper I)	13.12.2020		100
13.	UPPSC AE (Paper II)	13.12.2020		100
14.	UPPSC AE (Paper I)	12.04.2016		100
15.	UPPSC AE (Paper II)	12.04.2016		100
Uttar Pradesh JAL NIGAM				
16.	UP JAL NIGAM AE	2016		50
Uttar Pradesh Rajya Vidyut Utpadan Nigam Limited				
17.	UPRVUNL AE	05.07.2021		150
18.	UPRVUNL AE (UPJVUNL)	07.10.2016		150
19.	UPRVUNL AE	21.08.2016		150
20.	UPRVUNL AE	2014		150
Uttarakhand Public Service Commission				
21.	UKPSC AE (Paper I)	2013		180
22.	UKPSC AE (Paper II)	2013		180
23.	UKPSC AE (Paper I)	2012		200
24.	UKPSC AE (Paper II)	2012		200
25.	UKPSC AE (Paper I)	2007		200
26.	UKPSC AE (Paper II)	2007		200
UJVNL / PTCUL/UPCL Uttarakhand				
27.	UJVNL AE	12.09.2021		126
28.	UPCL AE	29.08.2021		50
29.	PTCUL AE	25.06.2017		126
30.	UJVNL AE	08.05.2016		126
Assam PSC				
31.	Assam PSC AE (IWT)	14.03.2021		100
32.	Assam PSC AE (PHED)	18.10.2020		100
33.	Assam PSC CCE Pre	2015		100
34.	APGCL Assistant Manager	31.01.2021		60
Andhra Pradesh PSC				
35.	APPSC Polytechnic Lect.	13.03.2020		150
36.	APPSC Polytechnic Lect. (Auto.)	12.03.2020		150
37.	APPSC AEE Screening	17.02.2019		100
38.	APPSC AEE (Mains) Mechanical	2016		150
39.	APPSC AEE Screening	2016		100
40.	APPSC AEE (Mains) Civil/Mechanical	2016		150
41.	APPSC AE Subordinate Services (Civil/Mechanical)	2016		150

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42.	APPSC Polytechnic Lect.	2012		150
43.	APPSC AEE (Civil/Mechanical)	2012		150
44.	APPSC AEE Paper - III (Mechanical)	2012		150
45.	APPSC AE(PHE)	04.12.2012		150
46.	APPSC Inspector of Factories	2009		150
47.	APPSC AE	2011		150
48.	APGENCO AE	2012		70
49.	APGENCO AE	2017		70
Airports Authority of India				
50.	AAI Junior Executive	29.11.2018		84
51.	AAI Junior Executive	26.03.2021		84
BHEL ET				
52.	BHEL ET	25.05.2019		120
Bihar Public Service Commission				
53.	BPSC AE (Paper V)	08.08.2019		50
54.	BPSC AE (Paper VI)	09.08.2019		50
55.	BPSC Poly. Tech. Lect.	24.07.2016		60
56.	BPSC Asstt. Prof.	29.11.2015		60
57.	BPSC AE (Paper V)	04.12.2012		50
58.	BPSC AE (Paper VI)	05.12.2012		50
Chhattishgarh PSC				
59.	CGPSC PSC AE	15.01.2021		150
60.	CGPSC PSC AE Shift-I	26.04.2015		150
61.	CGPSC PSC AE Shift-II	26.04.2015		150
62.	CGPSC AE Shift -II	25.02.2018		150
63.	CGPSC Poly. Lect.	13.10.2017		100
64.	CGPSC AE	16.10.2016		150
65.	CGPSC Poly. Lect.	22.05.2015		100
66.	CGPSC ARTO & TSI	07.01.2017		100
67.	CGPSC Asstt. Workshop Suptd.	17.07.2016		100
Coal India Ltd.				
68.	CIL MT	27.02.2020		100
69.	CIL MT	26.03.2017		100
Defence Research and Development				
70.	DRDO Scientist	2009		100
71.	DRDO Scientist	2008		100
Graduate Aptitude Test in Engineering (GATE)				
72.	GATE	1987-2021		1000
Gujarat Public Service Commission				
73.	GPSC DEE, Class-II (GWSSB)	04.07.2021		300
74.	GPSC Engineer, Class-II Pre	19.01.2020		80
75.	GPSC AE, Class-II	23.06.2019		200
76.	GPSC EE, Class-I, Pre	28.01.2017		100
77.	GPSC ARTO/Asstt. Director of Transport, Class-II Pre	30.12.2018		200
78.	GPSC AE	26.07.2015		100
79.	GPSC Asstt. Prof.	28.08.2016		100
80.	GPSC Poly. Lect	23.10.2016		100
81.	GPSC ARTO, Class-II	01.05.2016		200
82.	GPSC Asstt. Prof. (Production)	21.08.2016		100
83.	GMB AAE	25.06.2017		100

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84.	GPSC Asstt. Director of Transport	05.03.2017		200
85.	GPSC Poly. Lect. (Automobile)	16.10.2016		100
86.	GPSC Poly. Lect	07.12.2014		100
87.	GWSSB DEE	22.05.2016		100
88.	GWSSB AAE	17.06.2018		100
89.	GWSSB AAE	27.12.2015		100
90.	Vadodara Municipal Corporation DEE	13.05.2018		80
Haryana PSC				
91.	HPSC AE (PHED) Paper -I	04.09.2020		100
92.	HPSC AE (PHED) Paper -II	05.09.2020		100
93.	HPSC Civil Services Pre	2014		100
Hindustan Petroleum Corporation Limited				
94.	HPCL Engineer Shift-I	12.08.2021		85
95.	HPCL Engineer Shift-II	12.08.2021		85
96.	HPCL (HRRL) (E1 & E2)	07.08.2021		85
Himachal Pradesh PSC				
97.	HPPS Poly. Lect.	05.07.2021		80
98.	HPPSC Workshop Superintendent	08.07.2021		80
99.	HPPSC ADF	05.03.2019		80
100.	HPPSC AE	26.07.2018		80
101.	HPPSC Asstt. Prof.	18.02.2017		80
102.	HPPSC Asstt. Prof. (TE)	20.11.2017		80
103.	HPPSC AE (PWD)	10.08.2016		80
104.	HPPSC Asstt. Prof.	18.11.2016		80
105.	HPPSC Asstt. Prof. (TE)	29.10.2016		80
106.	HPPSC Poly. Lect.	10.07.2016		80
107.	HPPSC Workshop Superintendent	12.02.2015		80
108.	HPPSC AE	21.12.2014		80
109.	HPPSC Asstt. Prof. (TE)	11.02.2015		80
110.	HPPSC ITI Principal	2015		50
ISRO Scientist / Engineer				
111.	ISRO Scientist / Engineer	12.01.2020		70
112.	ISRO Scientist / Engineer	22.04.2018		70
113.	ISRO Scientist / Engineer	27.05.2017		70
114.	ISRO Scientist / Engineer	17.12.2017		70
115.	ISRO Scientist / Engineer	03.07.2016		70
116.	ISRO Scientist / Engineer	11.10.2015		70
117.	ISRO Scientist / Engineer	24.05.2014		70
118.	ISRO Scientist / Engineer	12.05.2013		70
119.	ISRO Scientist / Engineer	2012		70
120.	ISRO Scientist / Engineer	2011		70
121.	ISRO Scientist / Engineer	2010		70
122.	ISRO Scientist / Engineer	2009		70
123.	ISRO Scientist / Engineer	2008		70
124.	ISRO Scientist / Engineer	2007		70
125.	ISRO Scientist / Engineer	2006		70
126.	ISRO Scientist / Engineer (RAC)	12.01.2020		70
127.	ISRO Scientist / Engineer (RAC)	10.03.2019		70
128.	ISRO Scientist / Engineer (RAC)	22.04.2018		70
129.	ISRO Scientist / Engineer (RAC)	07.05.2017		70
130.	ISRO Scientist / Engineer (RAC)	29.11.2015		70
131.	ISRO Scientist / Engineer (RAC)	24.05.2014		70
132.	ISRO Scientist / Engineer (RAC)	2013		70

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Jharkhand Public Service Commission				
133.	JPSC AE Paper -I	10.04.2021		100
134.	JPSC AE Paper –II	10.04.2021		100
135.	JPSC AE Pre	19.01.2010		100
136.	JPSC AE (Paper V)	2013		50
137.	JPSC AE (Paper VI)	2013		50
138.	JUVNL AEE (GTO)	05.03.2017		75
Jammu & Kashmir PSC				
139.	JK PSC Civil Services Pre	2013		120
140.	JK PSC Civil Services Pre	2010		120
141.	JK PSC Civil Services Pre	2006		120
Karnataka Public Service Commission				
142.	Karnataka PSC AE (WRD)	31.07.2021		100
143.	Karnataka PSC AE	10.09.2017		100
144.	Karnataka PSC ADF	2015		100
145.	Karnataka PSC Poly. Lect.	27.05.2017		100
146.	KPCL AE	2016		100
Kerala PSC				
147.	Kerala PSC Asstt. Manager	09.08.2021		80
148.	Kerala PSC AE	05.04.2017		80
149.	Kerala PSC AE	31.12.2015		80
150.	Kerala PSC AE	07.07.2015		80
151.	Kerala PSC AE	06.08.2015		80
152.	Kerala PSC Poly. Lect.	14.08.2016		80
153.	Kerala PSC Asstt. Prof.	27.09.2016		80
154.	Kerala PSC IOF	19.04.2016		80
155.	Kerala PSC Poly. Lect.	09.12.2014		80
156.	Kerala PSC Poly. Lect. (Automobile)	05.05.2016		80
157.	Lal Bahadur Shastri Centre For Science & Technology (Kerala) Asstt. Prof.	2014		100
Maharashtra Public Service Commission				
158.	MPSC Poly. Lect. (Production)	2018		80
159.	MPSC Poly Lect.	2014		70
160.	MPSC (Deputy Engineer) (GWD)	2014		100
161.	MPSC (Deputy Engineer)	19.05.2013		80
162.	MPSC (HOD) Govt. Poly. Colleges	04.10.2014		80
Madhya Pradesh PSC				
163.	MPPSC Forest Service Exam	12.01.2016		150
164.	MPPSC AE	2016		100
165.	MPPSC AE	18.11.2015		100
Mizoram PSC				
166.	Mizoram PSC (Insp. of Factories) P-III	2019		50
167.	Mizoram PSC AE/SDO Paper-I	2014		50
168.	Mizoram PSC AE/SDO Paper-II	2014		50
169.	Mizoram PSC AE/SDO Paper-III	2014		50
Nagaland PSC				
170.	Nagaland PSC CTSE Paper-I	2018		200
171.	Nagaland PSC CTSE Paper-II	2018		200
172.	Nagaland PSC CTSE Paper-I	2017		200
173.	Nagaland PSC CTSE Paper-II	2017		200
174.	Nagaland PSC CTSE Paper-I	2016		200
175.	Nagaland PSC CTSE Paper-II	2016		200
176.	Nagaland PSC CTSE Paper-I	2015		200
177.	Nagaland PSC CTSE Paper-II	2015		200

Metallurgical & Engineering Consultant (India) Limited				
178.	MECON MT	17.03.2019		100
Nuclear Power Corporation of India Limited				
179.	NPCIL ET	2015		100
NLC India Limited				
180.	NLC GET Shift - I	17.11.2020		80
181.	NLC GET Shift - II	17.11.2020		80
Odisha Public Service Commission				
182.	OPSC AEE Paper-I	15.12.2019		180
183.	OPSC AEE Paper-II	15.12.2019		180
184.	OPSC AEE Paper-I	2015		90
185.	OPSC AEE Paper-II	2015		90
186.	OPSC Civil Services Pre	2011		120
187.	OPSC Civil Services Pre	2006		120
Oil India Limited				
188.	Senior Officer	23.12.2020		60
189.	Sr. Engineer (Drilling)	30.11.2019		60
190.	Sr. Engineer (Production)	30.11.2019		60
Punjab PSC				
191.	Punjab PSC Asstt. Municipal Engineer	15.06.2021		100
192.	Punjab PSC Poly. Lect.	06.08.2017		100
193.	Punjab PSC SDE	12.02.2017		100
194.	PSPCL AE	2012		60
Rajasthan Public Service Commission				
195.	RPSC Lecturer (Tech. Edu. Dept.) P-I	13.03.2021		150
196.	RPSC Lecturer (Tech. Edu. Dept.) P-II	13.03.2021		150
197.	RPSC ACF & FRO Gr.-I, P-III	26.02.2021		120
198.	RPSC Inspector Factory & Boilers	25.11.2020		200
199.	RPSC AE	17.12.2018		100
200.	RPSC VP ITI	06.11.2019		100
201.	RPSC Inspector Factory & Boilers	20.09.2017		100
202.	Rajasthan AE (Nagar Nigam) Shift –I	2016		84
203.	Rajasthan AE (Nagar Nigam) Shift –II	2016		84
204.	Rajasthan AE (Nagar Nigam) Shift –III	2016		84
205.	RPSC Lecturer(Tech. Edu. Dept.)	16.01.2016		100
206.	RPSC AE GWD	06.05.2016		100
207.	RPSC AE GWD	2011		100
208.	RPSC AE Drilling	05.05.2016		100
209.	RPSC AE	2013		100
210.	RPSC VP ITI	14.02.2016		100
211.	RPSC ACF	2011		200
212.	RPSC Lecturer(Tech. Edu. Dept.)	2011		100
RRB SSE				
213.	RRB SSE Shift-I	01.09.2015		16
214.	RRB SSE Shift-II	01.09.2015		15
215.	RRB SSE Shift-III	01.09.2015		15
216.	RRB SSE Shift-I	02.09.2015		15
217.	RRB SSE Shift-II	02.09.2015		15
218.	RRB SSE Shift-III	02.09.2015		15
219.	RRB SSE Shift-I	03.09.2015		16
220.	RRB SSE Shift-II	03.09.2015		15
221.	RRB SSE Shift-III	03.09.2015		15
222.	RRB SSE (Set Green)	21.12.2014		16

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223.	RRB SSE (Set Yellow)	21.12.2014		15
224.	RRB SSE (Set Red)	21.12.2014		15
SJVNL ET				
225.	SJVNL ET	28.05.2019		120
226.	SJVNL ET	06.11.2013		120
Sikkim PSC				
227.	Sikkim PSC AE	2017		75
Tamilnadu PSC				
228.	TNCSC AE	2020		25
229.	TNPSC AE	2019		200
230.	TNPSC AE	2018		200
231.	TNPSC AE	2017		200
232.	TNPSC AE	2014		200
233.	TNPSC AE	2013		200
234.	TNPSC AE (Industries)	2013		200
235.	TNPSC AE	2012		200
236.	TNPSC ACF	2012		200
237.	TNPSC AE	2008		200
238.	TANGEDCO AE	2015		60
239.	Tamilnadu TRB Poly. Tech. Lect.	2017		150
240.	Tamilnadu TRB Poly. Tech. Lect.	2012		150
241.	Tamilnadu TRB Asstt. Prof.	2012		150
Telangana PSC				
242.	TSPSC Manager (Engg.) in HMWSSB	12.11.2020		150
243.	TSPSC AEE (Civil/Mechanical)	28.08.2017		150
244.	TSPSC AEE	27.08.2017		150
245.	TSPSC AEE	18.10.2015		150
246.	TSPSC AE	2015		150
247.	TSPSC Manager (Engg.)	2015		150
TSGENCO				
248.	TSGENCO AE	14.11.2015		80
Vizag Steel MT				
249.	Vizag Steel MT Shift-I	24.01.2021		100
250.	Vizag Steel MT Shift-II	24.01.2021		100
251.	Vizag Steel MT	14.12.2020		100
252.	Vizag Steel MT	25.06.2017		100
253.	Vizag Steel MT	26.07.2015		100
254.	Vizag Steel MT (Re-Exam)	24.11.2013		100
255.	Vizag Steel MT	18.08.2013		100
256.	Vizag Steel MT	10.06.2012		100
257.	Vizag Steel MT	15.05.2011		100
West Bengal PSC				
258.	WBPSC AE	2017		200
259.	WBPSC AE	2012		100
260.	WBPSC AE	2008		100
261.	WBPSC AE	2007		100
262.	WBPSC AE	2003		100
			Total	**35301**

Note: After detailed analysis of above mentioned papers of UPSC and State Engineering Services Examination related to Mechanical Engineering **35301** (Volume-I + II) have been presented chapterwise. Questions of repeated and similar nature have been included so that the thechnique of asking question can benefit the competitors.

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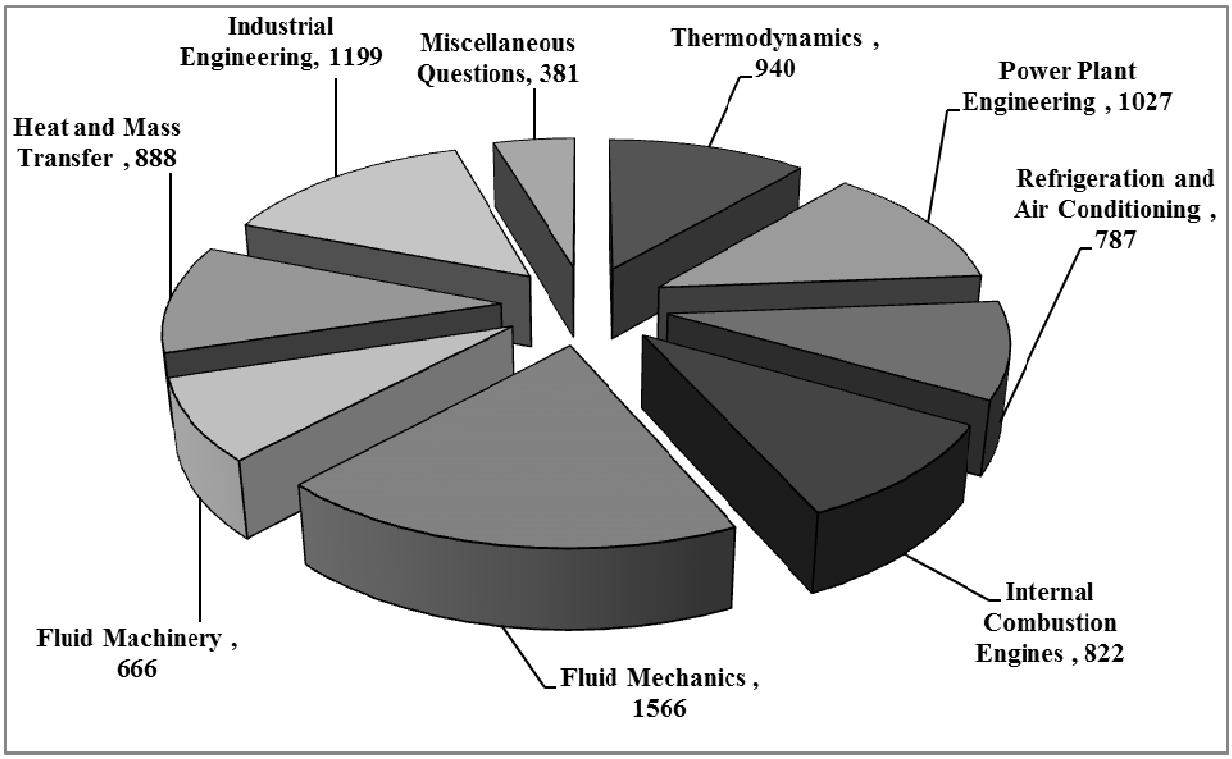
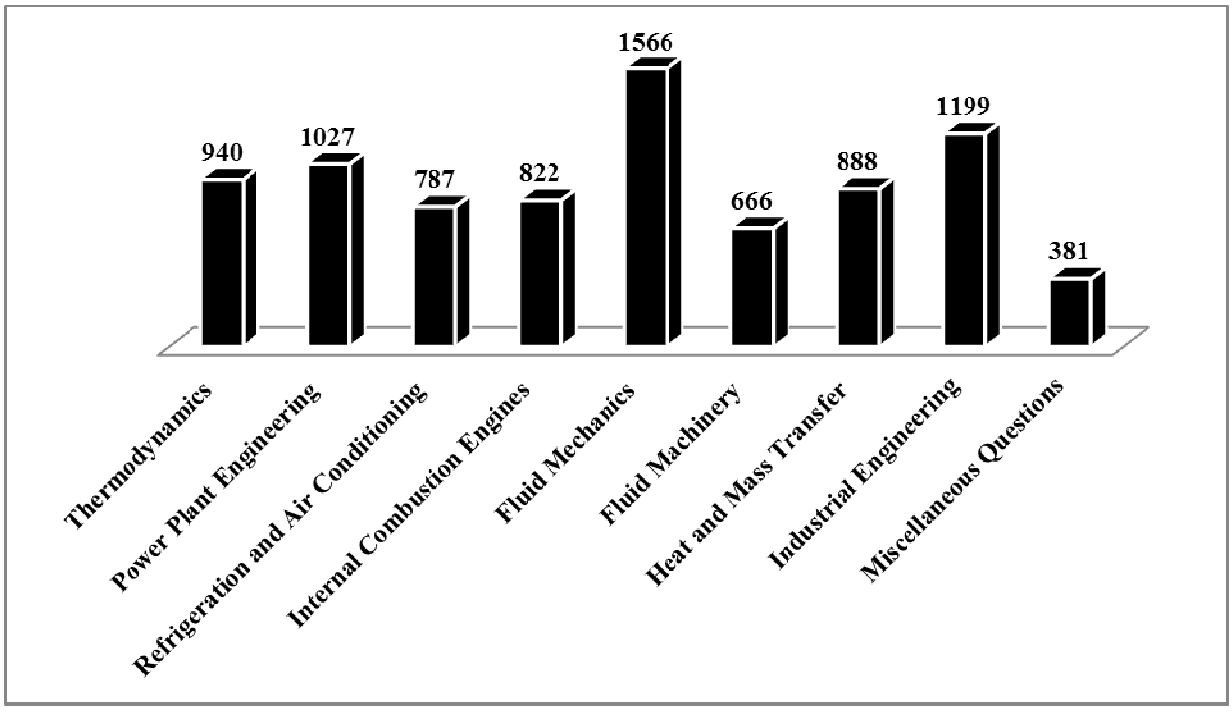
Trend Analysis of Mechanical Engineering Through Pie Chart and Bar Graph





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01.

Thermodynamics

1. Basic Concepts & Zeroth Law of Thermodynamics

1. Internal energy of a perfect gas depends on

(a) temperature, specific heats and pressure
(b) temperature, specific heats and enthalpy
(c) temperature, specific heats and entropy
(d) temperature only

Kerala PSC AM 09.08.2021
Assam Engg. College AP/Lect. 18.01.2021
Gujarat PSC AE 2019
TNPSC AE 2018, HPPSC AE 2018
WBPSC AE, 2017
GPSC Asstt. Director of Transport 05.03.2017
GATE 2016, GPSC Lect. 23.10.2016
TSPSC AEE 2015
OPSC AEE 2015 PAPER - II
GWSSB AAE, 27.12.2015
VIZAG STEEL MT 2015
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
Mizoram PSC 2014, Paper-I
WBPSC AE 2008
UKPSC AE 2007 Paper -II

Ans. (d) : For Perfect Gas internal energy,

$U = f(T)$

For Real Gas internal energy,

$U = f(T,V)$

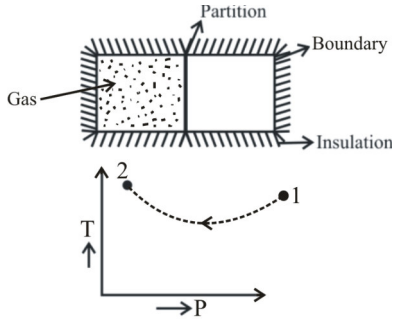
Where T = Temperature (K)
V = Volume

2. Work done in a free expansion process is

(a) Zero (b) minimum
(c) maximum (d) positive

Kerala PSC AM 09.08.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
JPSC AE, 10.04.2021, Paper-I
UPRVUNL AE 05.07.2021
APGCL AM, 2021
NLC GET 17.11.2020, Shift-II
JPSC AE PRE 2019
TNPSC AE 2018
ISRO Scientist/Engineer 22.04.2018
GPSC Asstt. Director of Transport 05.03.2017
Sikkim PSC (Under Secretary), 2017
GMB AAE 25.06.2017
CGPSC ARTO & TSI 07.01.2017
APPSC AEE Screening Test 2016
UKPSC AE 2007 Paper -II
OPSC Civil Services Pre 2006
CSE Pre-2002, 2001

Ans. (a) : Consider an insulated container divided in two parts, ideal gas is filled in one compartment other is vacuum. When partition is removed gas moved to vacuum there has no resistance.



For free expansion–

Heat (Q) = 0, workdone (W) = 0

3. During throttling process :

(a) internal energy does not change
(b) pressure does not change
(c) volume does not change
(d) enthalpy does not change

JPSC AE, 10.04.2021, Paper-I
Assam PSC AE (IWT) 14.03.2021
OPSC AEE 2019 PAPER - II
Gujarat PSC AE 2019
RPSC IOF & BOILER 20.09.2017
ISRO Scientist/Engineer 03.07.2016
Kerala PSC AE 06.08.2015
GWSSB AAE, 27.12.2015
Assam PSC CCE Pre 2015
(KPSC AE. 2015)
UPRVUNL AE 2014
UKPSC AE-2013, 2007 Paper-II
J&K PSC Civil Services Pre, 2010
TNPSC AE, 2008
J & K PSC Screening, 2006

Ans. : (d) During throttling process enthalpy does not change.

Throttling process:- Steam is said to be throttled when it passes through a restricted opening such as a narrow aperture or a slightly opened valve. The leakage of a fluid through a crack in the vessel is an example of throttling

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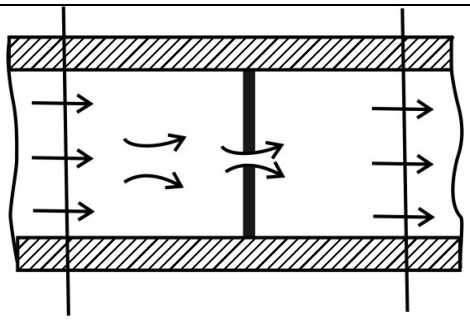

It may be noted that in throttling process:-

(i) No heat is supplied or rejected ($_1Q_2 = 0$)
(ii) No Work is done by the expanding fluid ($_1W_2=0$)
(iii) No Change in the internal energy (dU=0)
(iv) The enthalpy or total heat of the fluid remains constant ($h_1=h_2$)

4. Heat and work are :

(a) Intensive properties (b) Extensive properties
(c) Point functions (d) Path functions

UPCL AE 29.08.2021
Kerala PSC AM 09.08.2021
NLC GET 17.11.2020, Shift-II
OPSC AEE 2019 PAPER - II
GPSC ARTO Pre 30.12.2018
Nagaland PSC CTSE 2017 Paper-2
UP Jal Nigam AE 2016
Nagaland PSC (CTSE) 2016, Paper-II
UPRVUNL AE 2014
RPSC AEN Pre-2013, GATE 2011
UKPSC AE 2007 Paper -II

Ans. (d) : Heat and work both are path function.
Similarities between heat and work :
(i) Both are recognized at the boundary of the system, as they cross the boundary so they are considered as boundary phenomena.
(ii) System possesses energy, but neither heat and work.
(iii) Both are associated with process not state. Heat and work have no meaning at a state.
(iv) Both are path functions.

- **Path function-** Magnitude depends on the path followed during the process.
- **Point function-** Magnitude depends on state only and not a how the system approaches that state.

5. The measurement of temperature is based on

(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) None of these

Kerala PSC AM 09.08.2021
APPSC Poly Lect. Automobile Engg., 2020
TNPSC AE 2019, WBPSC AE, 2017
UPRVUNL AE 2014, 2016
GPSC Lect. (Auto) 16.10.2016
RRB SSE 01.09.2015 Shift-I
MPPSC State Forest Service Exam, 2014
Mizoram PSC 2014, Paper-I
J&K PSC Civil Services Pre, 2013
ESE 2009, CSE Pre-2009

Ans. (a)
1. Zeroth law of thermodynamics provides the basis of temperature measurement.
2. First law of thermodynamics provides conservation of law of energy.
3. Second law of thermodynamics provides the basic concept of entropy.

6. Which one of the following is the extensive property of the system?

(a) Density (b) Volume
(c) Pressure (d) Temperature

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II
GPSC Executive Engineer 23.12.2018
Assam PSC CCE Pre 2015
Kerala PSC Poly. Lect. 09.01.2014
RRB SSE 02.09.2015 Shift-I
UPRVUNL AE 2014
VIZAG STEEL MT 2011
TNPSC AE, 2008, ESE 1999

Ans. (b) : Density, pressure and temperature are Intensive properties because these are independent of mass. Volume is extensive property because it depends upon the mass.

7. An isolated system

(a) is a specified region where transfer of energy and/or mass take place
(b) is a region of constant mass and only energy is allowed to cross the boundaries
(c) cannot transfer either energy or mass to or from the surroundings
(d) none of the above

VIZAG Steel MT 24.01.2021
HPPSC Poly. Lect. 05.07.2021
APPSC Poly Lect. Automobile Engg., 2020
RPSC AE 2018
Karnataka PSC Lect., 27.05.2017
GWSSB AAE, 27.12.2015
TSPSC AEE 2015
WBPSC AE 2008, CSE Pre 2002

Ans. (c)			
Types of System	**Energy**	**Mass**	**Example**
1. Closed system	Yes	No	Piston cylinder arrangement without valves
2. Open system	Yes	Yes	Piston cylinder arrangement with valves, pump etc.
3. Isolated system	No	No	Universe, perfectly thermal flask

8. For a polytropic expansion (PV^n = constant), the temperature and pressure are related as.

(a) $\frac{T_1}{T_2}=\left(\frac{P_1}{P_2}\right)^{n-1}$ (b) $\frac{T_1}{T_2}=\left(\frac{P_1}{P_2}\right)^{\frac{n}{n-1}}$

(c) $\frac{T_1}{T_2}=\left(\frac{P_1}{P_2}\right)^{\frac{n-1}{n}}$ (d) $\frac{T_1}{T_2}=\left(\frac{P_1}{P_2}\right)^{\frac{n}{n+1}}$

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CGPSC AE 15.01.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC AE 2018
TNPSC AE (Industries) 09.06.2013
UKPSC AE-2013, Paper-II
UKPSC AE 2012 Paper–II
OPSC Civil Services Pre. 2011
ISRO Scientist/Engineer 2011, 2007

Ans. (c) : For Polytropic process, (PV^n = constant)

$$\frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{\frac{n-1}{n}} = \left(\frac{V_2}{V_1}\right)^{n-1}$$

9. Isentropic flow is

(a) Reversible adiabatic flow
(b) Irreversible adiabatic flow
(c) Frictionless fluid flow
(d) Reversible isothermal flow

NLCIL GET 17.11.2020, Shift-II
TSPSC Manager (Engg.) HMWSSB 12.11.2020
ISRO Scientist/Engineer (RAC) 22.04.2018
ISRO Scientist/Engineer 17.12.2017
UPPSC AE 12.04.2016 Paper-II
Mizoram PSC 2014 Paper-I
ESE 2011

Ans. (a) : Isentropic flow is reversible adiabatic flow.
We know that,
For isentropic flow $\Delta S = 0$

$$\Delta S = \frac{\delta Q}{T} + \delta \dot{S}_{gen}$$

$\delta Q = 0$ (for adiabatic flow)

$\delta \dot{S}_{gen} = 0$ (for reversible flow)

So $\Delta S = 0$

and we can say that flow will be isentropic.

10. An ideal gas at 27°C is heated at constant pressure till its volume becomes 3 times. What would be then the temperature of gas?

(a) 81°C (b) 627°C
(c) 543°C (d) 327°C

GPSC ARTO Pre 30.12.2018
GPSC EE Pre, 28.01.2017
RPSC IOF & BOILER 20.09.2017
RPSC VPITI 14.02.2016
KPSC ADF 2015
J&K PSC Civil Services Pre, 2013
ESE 2008

Ans. (b) : Given data
$T_1 = 27°C + 273 = 300$ K
Pressure (P) = constant i.e. $P_1 = P_2 = P$
Volume (V_1) = V and $V_2 = 3V$
temperature (T_2) = ?
Applying Ideal gas equation –

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\Rightarrow \quad \frac{P \times V}{300} = \frac{P \times 3V}{T_2}$$

$$\Rightarrow \quad T_2 = 900 \text{ K}$$

or $T_2 = 627°C$

11. Pressure reaches a value of absolute zero

(a) at a temperature of 273 K
(b) under vacuum condition
(c) at the earth's centre
(d) when molecular momentum of system becomes zero

Nagaland PSC (CTSE) 2016, Paper-II
APPSC AEE Mains 2016 Paper-III
Nagaland PSC CTSE 2015 Paper-II
GWSSB AAE, 27.12.2015
J&K PSC Civil Services Pre, 2013
ESE 2002

Ans. (d) : Pressure reaches a value of absolute zero of two possibilities.
Absolute zero temperature – The temperature at which motion of molecule completely stopped i.e., 0 K.
According to kinetic theory of gases, the P exerted by a gas of density ρ and rms velocity v is given by,

We know that, $P = \frac{1}{3}\rho v^2$

At 0 K molecule momentum is zero then v must be zero. So, pressure reaches a value of absolute zero.
Note : According to Third law of thermodynamics : Entropy of pure substance in thermodynamics equilibrium is zero at absolute zero temperature (0 K).

12. What is the standard fixed point of thermometry?

(a) The ice point
(b) The steam point
(c) The triple point of water
(d) None of the above

HPCL Engineer 12.08.2021 Shift-I
Assam PSC AE (PHED), 18.10.2020
AAI Jr. Executive 29.11.2018
Nagaland PSC (CTSE) 2016, Paper-II
Haryana PSC Civil Services Pre, 2014
UPSC JWM Advt. No.-52/2010

Ans. (c) : The triple point of water is the standard fixed point of thermometry triple point is the temperature at which all three phases of substance solid, liquid and gas (ice, water, steam) can co-exists in thermal equilibrium simultaneously.
Triple point of water = 0.01°C= 273.16K

13. Air is compressed adiabatically in a steady flow process with negligible change in potential and kinetic energy. The work done in the process is given by

(a) $-\int pdv$ (b) $+\int pdv$
(c) $-\int vdp$ (d) $+\int vdp$

UPPSC AE 12.04.2016 Paper-II
GPSC ARTU 01.05.2016

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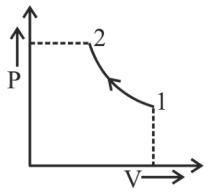
VIZAG Steel MT (Re Exam) 24.11.2013
TNPSC AE 2008
CSE Pre-2000, GATE 1996

Ans. : (c) Open system work:-

$$\int \delta W = -\int_1^2 Vdp$$



For reversible adiabatic compression work done

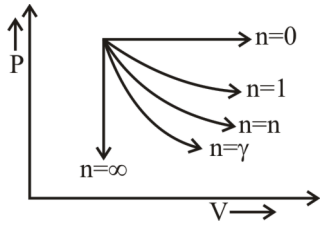
$$= \int dw = -\int vdp$$

14. If the value of n is infinitely large in a polytropic process $PV^n = C$, then the process is _____ .

(a) isothermal (b) isobaric
(c) isentropic (d) isochoric

UPPSC AE 13.12.2020, Paper-2
VIZAG MT, 14.12.2020
RPSC IOF, 2020
ISRO Scientist/Engineer 22.04.2018
RPSC AE 2018, TNPSC AE 2018
UKPSC AE 2007 Paper -II

Ans. (d) :



PV^n = Constant
$n = 0 \rightarrow$ Isobaric
$n = 1 \rightarrow$ Isothermal
$n = n \rightarrow$ Polytropic
$n = \gamma \rightarrow$ Isentropic
$n = \infty \rightarrow$ Isochoric (Constant volume)

15. Change in enthalpy of a system is the heat supplied at

(a) Constant entropy (b) Constant volume
(c) Constant temperature (d) Constant pressure

UPCL AE 29.08.2021
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
UPPSC AE 13.12.2020, Paper-2
WBPSC AE, 2017
Rajasthan AE (Nagar Nigam) 2016 Shift-3
MPPSC State Forest Service Exam, 2014
RPSC Vice Principal ITI 2008
ESE 2005

Ans. (d) According to first law of thermodynamics

$\partial Q = dU + \partial W$
$\partial Q = dU + PdV$

We know that -

$h = U + PV$
$dh = dU + PdV + VdP$

(VdP = 0 because of constant pressure)

$\boxed{dh = \partial Q}$

16. A process in which the working substance neither receives nor gives out heat to its surroundings during its expansion or compression, is called :

(a) isothermal process (b) hyperbolic process
(c) polytrophic process (d) adiabatic process

ISRO Scientist/Engineer (RAC) 12.01.2020
MECON MT 2019
AAI Jr. Executive 29.11.2018
Karnataka PSC Lect., 27.05.2017
RRB SSE 01.09.2015 Shift-I
VIZAG STEEL MT 2015
OPSC Civil Services Pre 2006

Ans. (d) : A process in which the working substance neither receives nor gives out heat to its surroundings during its expansion or compression, is called Adiabatic process.

17. A closed system is one in which :

(a) mass does not cross boundaries of the system, though energy may do so
(b) mass crosses the boundary but not the energy
(c) neither mass nor energy crosses the boundaries of the system
(d) both energy and mass cross the boundaries of the system

TSPSC AEE 2015, UPRVUNL AE 2014
J&K PSC Civil Services Pre, 2013
ESE 2011, 2010, 1999
OPSC Civil Service Pre-2006

Ans. (a) : Isolated system – Neither mass nor energy cross the system boundary.
Closed system – Only energy crosses the system boundary.
Open system – Mass and energy both crosses the boundary.

18. According to kinetic theory of gases, absolute zero temperature is obtained when :

(a) volume of the gas is zero
(b) pressure of the gas is zero
(c) kinetic energy of the gas is zero
(d) specific heat of the gas is zero

CGPSC AE 15.01.2021
GPSC Engineer, Class-II Pre-19.01.2020
CIL MT 27.02.2020
GPSC Executive Engineer 23.12.2018
Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : According to kinetic theory of gases Absolute zero temperature is obtained when the root mean square velocity of gas molecules reduces to zero.

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As we know that,

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

At T = 0 K (absolute zero temperature)

v_{rms} = zero

Thus, kinetic energy of the gas becomes zero.

19. The following are the examples of some intensive and extensive properties

1. pressure 2. Temperature
3. volume 4. Velocity
5. electric charge 6. Magnetization
7. viscosity 8. Potential energy

Which one of the following sets gives the correct combination of intensive and extensive

	Intensive	Extensive
(a)	1,2,3,4	5,6,7,8
(b)	1,3,5,7	2,4,6,8
(c)	1,2,4,7	3,5,6,8
(d)	2,3,6,8	1,4,5,7

GWSSB AAE, 27.12.2015
Haryana PSC Civil Services Pre, 2014
J&K PSC Civil Services Pre, 2010
UPSC JWM Advt. No.-52/2010
CSE Pre-1995

Ans. (c)
Intensive Properties :– Properties which are independent of mass
Ex.– Pressure, temperature, velocity, viscosity.
Extensive Properties :– Properties which depends upon mass.
Ex.– Volume, electric charge, magnetization, potential energy.

20. The definition of 1K as per the internationally accepted temperature scale is

(a) 1/100th the difference between normal boiling point and normal freezing point of water.
(b) 1/273.15th the normal freezing point of water
(c) 100 times the difference between the triple point of water and the normal freezing point of water.
(d) 1/273.16th of the triple point of water.

Nagaland PSC (CTSE) 2016, Paper II
Haryana PSC Civil Services Pre, 2014
ISRO Scientist/Engineer 12.05.2013
ESE 2004, GATE -1994

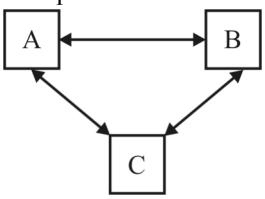
Ans. (d) : According to internationally accepted definition $1K = \left(\frac{1}{273.16}\right)^{th}$ of Triple point of water (H_2O)

21. When two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other. This statement is

(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) Not a law of thermodynamics

GPSC ARTO 01.05.2016
Nagaland PSC CTSE 2015 Paper-II
RRB SSE 02.09.2015 Shift-II
UPRVUNL AE 2014, APPSC Poly. Lect. 2013
WBPSC AE 2008, 2003

Ans. (a) : Zeroth law of thermodynamics- This law was given by RH follower
"According to this law if body A is in the thermal equilibrium with body B and body B is in thermal equilibrium with body C separately then body A and C will be in thermal equilibrium with each other".
Note- Zeroth law of the thermodynamics provide a concept of thermal equilibrium.



22. An open system

(a) is a specified region where transfers of energy and / or mass take place
(b) is a region of constant mass and only energy is allowed to cross the boundaries
(c) cannot transfer either energy or mass to or from the surroundings
(d) has an enthalpy transfer across its boundaries and the mass within the system is not necessarily constant

UJVNL AE 12.09.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
CIL MT 27.02.2020
BPSC Poly. Lect. 2016
HPPSC Lect. 2016
VIZAG STEEL MT 2015

Ans. : (a)

system	mass transfer	Energy transfer
open system	✓	✓
closed system	×	✓
Isolated system	×	×

23. The heat absorbed or rejected during a polytropic process is equal to :

(a) $\left(\frac{\gamma - n}{\gamma - 1}\right)^{1/2} \times$ workdone
(b) $\left(\frac{\gamma - n}{n - 1}\right) \times$ workdone
(c) $\left(\frac{\gamma - n}{\gamma - 1}\right)^{2} \times$ workdone
(d) $\left(\frac{\gamma - n}{\gamma - 1}\right) \times$ workdone

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Oil India Limited Sr. Engineer (Drilling) 30.11.2019
MPPSC AE 2016, ESE 2002

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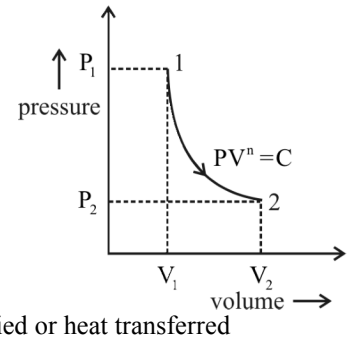
Ans. : (d)



Heat supplied or heat transferred

$$Q_{1\text{-}2}=\frac{\gamma\text{-}n}{\gamma\text{-}1}\times\frac{mR(T_1\text{-}T_2)}{n\text{-}1}$$

$$\text{work done during polytropic expansion}=\frac{mR(T_1\text{-}T_2)}{n\text{-}1}$$

$$Q_{1\text{-}2}=\frac{\gamma\text{-}n}{\gamma\text{-}1}\times\text{work done}$$

24. The net work done in a polytropic process is given by

(a) $\frac{P_1V_1-P_2V_2}{n-1}$ (b) $\frac{P_2V_2-P_1V_1}{n-1}$

(c) $\frac{P_1V_1-P_2V_2}{n+1}$ (d) $\frac{P_2V_2-P_1V_1}{n+1}$

WPSC AE 2017, APGENCO AE 2012
UKPSC AE 2007 Paper -II, WBPSC AE 2003

Ans. (a) : For polytropic process $[PV^n = C]$

$$[PV^n = C] = P_IV_I^n = P_FV_F^n \quad \leftarrow \begin{bmatrix} I = \text{initial} \\ F = \text{final} \end{bmatrix}$$

$$P=\frac{C}{V^n};\quad W=\int_{v_I}^{v_F}PdV=\int_{v_I}^{v_F}\frac{C}{V^n}dV$$

$$W=C\int V^{-n}dV=C\left(\frac{V^{-n+1}}{-n+1}\right)_{V_I}^{V_F}$$

$$W=C\left[\frac{V_F^{-n+1}-V_I^{1-n}}{1-n}\right]$$

$$=\frac{1}{1-n}\left[CV_F^{1-n}-CV_I^{1-n}\right] \quad ...(i)$$

$\because$ $PV^n = C = P_IV_I^n = P_FV_F^n$

keep the value of C in equation (i)

$$W=\frac{1}{1-n}\left[P_FV_F^nV_F^{1-n}-P_IV_I^nV_I^{1-n}\right]$$

$$=\frac{1}{1-n}\left[P_FV_F-P_IV_I\right]$$

$$\begin{vmatrix} P_F = P_2 \\ P_I = P_1 \end{vmatrix}$$

$$W=\frac{1}{1-n}\left[P_2V_2-P_1V_1\right]$$

$$\boxed{W=\frac{P_1V_1-P_2V_2}{n-1}}$$

25. For which of the following situations, zeroth law of thermodynamics will not be valid?

(a) 50 cc of water at 25°C is mixed with 150cc of water at 25°C

(b) 500 cc of milk at 15°C is mixed with 100cc of water at 15°C

(c) 5 kg of wet steam at 100°C is mixed with 50kg of dry and saturated steam at 100°C.

(d) 10 cc of water at 20°C is mixed with 10 cc of sulphuric acid at 20°C.

ISRO Scientist/Engineer (RAC) 12.01.2020
GPSC Asstt. Director of Transport 05.03.2017
MPSC Poly. Lect. 2014
ISRO Scientist/Engineer 2012

Ans. (d) : Two bodies in contact are at same temperature when in thermal equilibrium In case of 10cc of water at 20°C is mixed with 10 cc of sulphuric acid at 20°C there is no equilibrium between components because there will be an spontaneous ionic reaction.

26. A cylinder contains $5m^3$ of ideal gas at a pressure of 1 bar. This gas is compressed in a reversible isothermal process till its pressure increases to 5 bar.

(a) 804.7 (b) 953.2

(c) 981.7 (d) 1012.2

Vizag Steel (MT) 2017
GPSC Asstt. Director of Transport 05.03.2017
ISRO Scientist/Engineer (RAC) 07.05.2017
GATE 2013

Ans. (a) Volume $(V_1) = 5.0m^3$

Initial pressure (P_1) = 1 bar = 100 kPa

Final Pressure (P_2) = 5 bar = 500 kPa

We know that,

Work done (W) $=\int_{P_1}^{P_2}VdP$.....(i)

Since, process is isothermal

Therefore, PV = mRT = constant.....(ii)

Now equations (i) and (ii)

$$W=\int_{P_1}^{P_2}\frac{c}{P}dP=c\,ln\left(\frac{P_2}{P_1}\right)$$

$$W=P_1V_1ln\left(\frac{P_2}{P_1}\right)\quad(\because P_1V_1=P_2V_2)$$

$$W=(100\times5)\times ln\left(\frac{500}{100}\right)$$

W = 804.7 kJ

27. Which of the following sets have all intensive properties?

(a) Pressure, temperature, density, specific volume

(b) Pressure, volume, energy, specific volume

(c) Pressure, energy, volume, density

(d) Pressure, heat, density, specific volume

VIZAG Steel MT 24.01.2021 Shift-I

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(CGPSC Polytechnic Lecturer 2017)
TRB Polytechnic Lecturer 2017
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a) : Intensive Properties– It is the properties which is independent of mass is known as intensive properties
Ex.– Pressure, Temperature, Density, Thermal conductivity all form of specific energy etc.

28. Consider the following properties :

1. Energy	**2. Entropy**
3. Gibbs Energy	**4. Volume**
5. Pressure	**6. Temperature**
7. Viscosity	**8. Elasticity**

Which of the above are intensive properties?
(a) 1, 3, 5, 6 (b) 5, 6, 7, 8
(c) 1, 3, 5, 6, 7, 8 (d) 4, 5, 6, 8

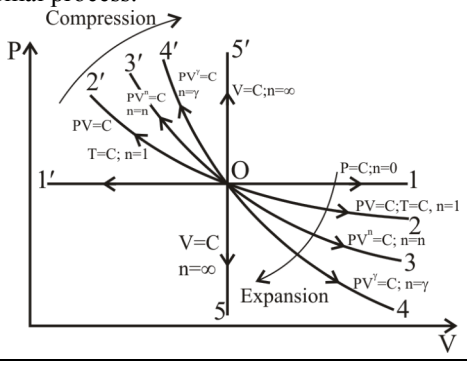
UPPSC AE 13.12.2020, Paper-2
GPSC ARTO Pre 30.12.2018
Nagaland PSC CTSE 2015 Paper-II
APPSC IOF, 2009

Ans. (b) : Intensive Properties– An intensive property is one that does not depend on the mass of substance or system.
Ex.: Temperature, pressure, density, specific volume, all specific quantity, viscosity, kinetic viscosity, specific heat (C_p & C_v) velocity, specific enthalpy, specific entropy, thermal diffusivity etc.
Extensive Property– An extensive property of a system depends on size or the amount of matter in the system.
Ex.: Volume, kinetic energy enthalpy, entropy, all type of energy electrical resistance.

29. The value of n=1 in the polytropic process indicates it to be
(a) Isothermal process (b) Reversible process
(c) Irreversible process (d) Adiabatic process

APGCL AM, 2021
Rajasthan AE (Nagar Nigam) 2016 Shift-3
ISRO Scientist/Engineer 2006
OPSC Civil Services Pre 2006

Ans. (a) : Polytropic process $PV^n = C$
if $n = 1$ then $PV^1 = C$
$PV = C$
In this equation temperature is constant so for $n = 1$ is isothermal process.



30. A mass m of a perfect gas at pressure p_1 and volume V_1 undergoes an isothermal process. The final pressure is p_2 and volume is V_2. The work done on the system is considered positive. If R is the gas constant and T is the temperature, then the work done in the process is _____

(a) $p_1V_1 \ell n\frac{V_2}{V_1}$ (b) $-p_1V_1 \ell n\frac{p_1}{p_2}$
(c) $RT \ell n\frac{V_2}{V_1}$ (d) $-mRT \ell n\frac{p_2}{p_1}$

Assam Engg. College AP/Lect. 18.01.2021
TRB Polytechnic Lecturer 2017
GATE 2017, TNPSC AE 2013

Ans. (b) : For isothermal process temperature is constant (T = C)
$PV = C$
or $P_1V_1 = P_2V_2 = \text{constant}$

Work done by the system $= \int_{v_1}^{v_2} Pdv$ $\quad p_1v_1 = p_2V_2 = C$

$= \int_{v_1}^{v_2} \frac{C}{V}dv$ $\quad \frac{V_2}{V_1} = \frac{p_1}{p_2}$

$= C \ell n\frac{V_2}{V_1}$

$= C \ell n\frac{p_1}{p_2}$

$= p_1V_1 \ell n\frac{p_1}{p_2}$ or $p_1V_1 \ell n\frac{V_2}{V_1}$

Work done in the process

$= -p_1V_1 \ell n\frac{p_1}{p_2}$ or $-p_1V_1 \ell n\frac{V_2}{V_1}$

31. Which thermometer is independent of the substance or material used in its construction?
(a) mercury thermometer
(b) alcohol thermometer
(c) ideal gas thermometer
(d) resistance thermometer

Nagaland PSC CTSE 2017 Paper-2
Nagaland PSC (CTSE) 2016, Paper-II
Haryana PSC Civil Services Pre, 2014
CSE Pre-2006

Ans. (c) Ideal gas thermometer is independent of substance or material used in it's construction. Ideal gas thermometers are based on Ideal gas equation $(PV = n\bar{R}T)$

32. For polytropic process (Pv^n = constant), the value of 'n' between two states (1 and 2) is given as:

(a) $n = \frac{ln(P_1 - P_2)}{ln(v_1 - v_2)}$ (b) $n = \frac{ln(P_1P_2)}{ln(v_1v_2)}$
(c) $n = \frac{ln\frac{(P_1)}{P_2}}{ln\frac{(v_1)}{v_2}}$ (d) $n = \frac{ln(P_1/P_2)}{ln(v_2/v_1)}$

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Oil India Ltd. Sr. Engg. (Production) 30.11.2019
CGPSC AE 25.02.2018
ISRO Scientist/Engineer 2010
WBPSC AE 2003

Ans. (d) : Polytropic process 1-2 follow $Pv^n = C$
Then the value of n
here n is polytropic process.

$$Pv^n = C$$

$$\Rightarrow \quad P_1v_1^n = P_2v_2^n$$

Take log both side

$$\ln P_1 + n \ln v_1 = \ln P_2 + n \ln v_2$$

$$\ln P_1 - \ln P_2 = n \ln v_2 - n \ln v_1$$

$$\ln\left[\frac{P_1}{P_2}\right] = n \ln\left[\frac{v_2}{v_1}\right]$$

$$n = \frac{\ln[P_1/P_2]}{\ln[v_2/v_1]}$$

33. A pump handling a liquid raises its pressure from 1 bar to 30 bar. Take the density of the liquid as 990 kg/m^3. The isentropic specific work done by the pump in kJ/kg is

(a) 0.10 (b) 0.30
(c) 2.50 (d) 2.93

GPSC ARTO Pre 30.12.2018
GPSC Executive Engineer 23.12.2018
GPSC Asstt. Director of Transport 05.03.2017
GATE 2011

Ans. (d) : Given, $P_2 = 30$ bar, $P_1 = 1$ bar, $\rho = 990$ kg/m^3

Specific workdone by pump $= \dfrac{(P_2 - P_1)}{\rho}$

$$= \frac{(30-1)\times 100 \text{ N/m}^2}{990 \text{ kg/m}^3}$$

$$= 2.93 \text{ kJ/kg}$$

34. The unit of temperature in S.I. unit is

(a) Centigrade (b) Celsius
(c) Fahrenheit (d) Kelvin

CGPSC AE 15.01.2021
Rajasthan Nagar Nigam AE 2016, Shift-II
MPPSC State Forest Service Exam, 2014

Ans. (d) : Kelvin is the unit of temperature in the SI system.

35. Heat flows from one body to other when they have

(a) Different specific heat
(b) Different heat contents
(c) Different atomic structure
(d) Different temperatures

GPSC Asstt. Prof. 28.08.2016
Rajasthan Nagar Nigam AE 2016, Shift-I
GPSC Lect. 23.10.2016

Ans. (d) : Heat flows from one body to other when they have different temperatures.

- Heat transfer has direction as well as magnitude. The rate of heat conduction in a direction is proportional to the temperature gradient.
- Heat $Q = mc\Delta T$
  $Q = mc(T_2 - T_1)$
  $Q \propto \Delta T$
  or $Q \propto (T_2 - T_1)$

36. The work done per unit mass of air in a compressor will be minimum, when n is equal to

(a) 1 (b) 1.2
(c) 1.3 (d) 1.4

JPSC AE, 10.04.2021, Paper-I
Assam PSC AE (PHED), 18.10.2020
WBPSC AE, 2017

Ans. (a) : Work done per unit mass of air in compressor will be minimum if value of n is equal to 1. The law followed by compressor is PV = C (Isothermal Process)
If value of n increases the work done on compressor increases.

37. If the internal energy and enthalpy of a gaseous substance are functions only of temperature and are independent of pressure, then the substance must be

(a) a liquid-vapour mixture
(b) an ideal gas
(c) a real gas
(d) saturated vapour

Haryana PSC AE (PHED) 04.09.2020, Paper-I
Nagaland PSC CTSE 2015 Paper-II
WBPSC AE, 2007

Ans. (b) : Enthalpy and internal energy of an ideal gas are function of temperature only and independent of pressure.
$\because$ For ideal gas, internal energy (U) = f(T)

38. A thermodynamic system refers to :

(a) Any defined region in space
(b) A specified mass in fluid flow
(c) A specified region of constant volume
(d) A prescribed and identifiable quantity of matter

JPSC AE, 10.04.2021, Paper-I
TSPSC Manager (Engg.) HMWSSB 12.11.2020
UPRVUNL AE 2014

Ans. :(d) A thermodynamic system refers to a prescribed and identifiable quantity of mater and in simple word we say that a specified region where we focus our study.

39. The thermodynamic process in which the heat energy supplied is used to increase the internal energy:

(a) Isochoric process (b) Isentropic process
(c) Isobaric process (d) Isothermal process

VIZAG Steel MT 24.01.2021
Nagaland PSC CTSE 2015 Paper-II
ESE 2008

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Ans. (a) : A constant volume (Isochoric) thermodynamic process in which the system is confined by mechanical rigid boundaries. No direct mechanical work can be done on the surroundings by a system with rigid boundaries, therefore the heat transferred into or out of the system equals to change in internal energy.

$Q = dU + PdV$

$V = Constant$

$dV = 0$

$$\boxed{Q = dU = U_2 - U_1}$$

40. The gas constant R is equal to the _____.

(a) Addition of specific heats at constant pressure and constant volume

(b) Ratio of specific heat at constant pressure and specific heat at constant volume

(c) Difference between the specific heat at constant pressure and specific heat at constant volume

(d) Product of specific heats at constant pressure and volume

UPCL AE 29.08.2021
HPCL Engineer 12.08.2021 Shift-I
GPSC DEE, Class-2 (GWSSB) 04.07.2021
Assam PSC AE (IWT) 14.03.2021
VIZAG MT, 14.12.2020
HPPSC Asstt. Prof. 18.09.2017
Assam PSC CCE Pre 2015
UKPSC AE 2012 Paper–II
ESE 2012, ISRO Scientist/Engineer 2010

Ans. (c) : According to Mayer's formula–

The gas constant (R) is equal to the difference of two specific heats.

$$\boxed{R = C_p - C_V} \text{ J/kgK}$$

$$\boxed{\gamma = \frac{C_P}{C_V}}$$

41. Air coming out of a punctured football

(a) becomes hotter

(b) becomes cooler

(c) remains at same temperature

(d) attains atmospheric temperature

AAI Jr. Executive 29.11.2018
WBPSC AE, 2017
Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : When air is coming out punctured tyre, puncture acts like a Nozzle when fluid is passed through nozzle, velocity increase and pressure decrease, resulting in decreasing of temperature. You can observe this phenomenon by keeping your hand near mouth and blowing.

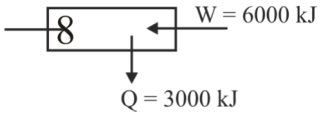
42. A tank containing air is stirred by a paddle wheel. The work input to the paddle wheel is 6000 kJ. The heat transferred to the surroundings from the tank is 3000 kJ. The external work done by the system is

(a) Zero (b) 3000 kJ

(c) 6000 kJ (d) 9000 kJ

UPPSC AE 13.12.2020, Paper-2
Rajasthan Nagar Nigam AE 2016 Shift-I
ESE 1999

Ans. (a)



This is a case of constant volume process (isochoric process) in irreversible constant volume process, the system does not perform work on the surrounding at the expense of internal energy.

43. In an adiabatic process, the work done for a given pair of end sates depends on:

(a) End states only (b) Mass of the system

(c) Value of index n (d) Heat transferred

CIL MT 27.02.2020, APGENCO AE, 2017
TNPSC ACF 2012

Ans. (a) : End state are determined by the internal energy (U) level at these states.

If heat energy ∂Q is given to a system by first law of thermodynamics

$$\partial Q = \partial W + dU$$

For adiabatic process $\partial Q = 0$

Hence $\partial W = dU$

Above equation shows that work done in an adiabatic process depends only upon end states (i.e. point function).

44. The specific heat of ideal gas depend on

(a) Temperature

(b) Volume

(c) Molecular weight and structure

(d) Pressure

ISRO Scientist/Engineer (RAC) 07.05.2017
TRB Poly. Lect., 2012, GATE 1996

Ans. (c) : The specific heat of ideal gas depend on molecular weight and structure.

45. Which one of the properties given is an intensive property of the system?

(a) Composition (b) Volume

(c) Kinetic energy (d) Entropy

HPPSC Poly. Lect. 05.07.2021
TSPSC AEE 2015, UPRVUNL AE 2014

Ans. : (a) Intensive Property – Composition
Extensive Property– Volume, Kinetic Energy, Entropy

46. In highly rarefied gases, the concept of this loses validity

(a) Thermodynamic equilibrium

(b) Continuum

(c) Stability

(d) Macroscopic view point

GPSC EE Pre, 28.01.2017
GPSC Asstt. Director of Transport 05.03.2017
ESE 2012

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Ans. (b) : Continuum- In this approach we consider system to be a continuous from of matter. It means that we do not consider the gaps/spaces between the atoms or molecules that are actually present. Instead, we consider a uniform distribution of matter with no spaces or gaps.
But in rarefied gas, gaps will be created among atoms and molecules invalid.

47. A control volume implies
(a) an isolated system
(b) a closed system
(c) a specified mass in fluid flow
(d) a fixed region in space

GPSC ARTO 01.05.2016
APPSC AEE Mains 2016 (Civil Mechanical)
ESE 2010

Ans. (d) : Control volume is a volume in space of special interest for particular Analysis. The surface of the control volume is referred as a control surface & is a closed surface. It implies a fixed region in space.

48. Which of the following is not a property of the system?
(a) Temperature (b) Pressure
(c) Volume (d) Heat

TNPSC AE 2017
Rajasthan Nagar Nigam AE 2016, Shift-II
UKPSC AE 2012, 2007 Paper–II

Ans. (d) : Heat is not a property of the system it is the form of energy whereas temperature, pressure and volume are the property of the system.

49. The value of the universal g as constant is
(a) 8.314 J/kg K (b) 83.14 kJ/kg K
(c) 848 kJ/kg K (d) 8.314 kJ/kg K

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Kerala PSC Poly. Lect. 09.01.2014
TNPSC AE (Industries) 09.06.2013

Ans. (d) :

50. A process in which no heat is supplied or rejected from the system and entropy is not constant is known as
(a) isothermal (b) isentropic
(c) polytropic (d) hyperbolic

ISRO Scientist/Engineer 11.10.2015
MPPSC State Forest Service Exam, 2014
Mizoram PSC 2014, Paper-I

Ans. (c) : Polytropic Process–The Polytropic process is also known as the general law for the expansion and compression of gases, and is given by the relation.

$$PV^n = \text{Constant}$$

- No heat is supplied or rejected from the system and entropy is constant then this process is called isentropic process.
- No heat is supplied or rejected from the system and entropy is not constant is is known as polytropic process.

51. The change in internal energy in a reversible process occurring in a closed system is equal to the heat transferred if the process occurs at constant
(a) temperature (b) pressure
(c) volume (d) enthalpy

CGPSC Poly. Lect. 22.05.2016
Assam PSC CCE Pre 2015
ESE 2005

Ans. (c) According to first law of thermodynamics for a process

$\delta Q = dU + \delta W$
$= dU + pdV$

For constant volume process

$\delta Q = dU + 0 \quad (dV = 0)$

$\delta Q = dU$

$dU = mc_v dT$ (For an ideal gas)

The change in internal energy in a closed system is equal to the heat transferred if the process occurs at constant volume.

52. If a gas is allowed to expand through a very minute aperture, then such a process is known as
(a) free expansion
(b) adiabatic expansion
(c) throttling
(d) parabolic expansion

Nagaland PSC (CTSE) 2018, Paper-II
TRB Asstt. Prof., 2012, TNPSC AE, 2008

Ans. (c) : When the air or gas flowing through a small or minute opening then enthalpy remains constant. It is known as isenthalpic or throttling process.

- It is highly irreversible process.
- There is no work and heat transfer takes place.
- During throttling process pressure always decreases.
- During throttling process temperature may or may not be increase or decrease or remains constant.
- For an ideal gas temperature remains constant.

53. In an isothermal process, the internal energy of molecules
(a) Increases
(b) Decreases
(c) Depends on temperature
(d) Remains constant

ISRO Scientist/Engineer (RAC) 07.05.2017
MPSC State Forest Service Exam 2014

Ans. (d) : We know that, internal energy U

$U = mc_v \Delta T$

For unit mass

$\Delta u = c_v \Delta T$

For a fluid, value of c_v will remain constant

$\Delta u \propto \Delta T$

For isothermal process $\Delta T = 0$

So,

$\Delta u = \text{constant}$

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54. The expression $\int_1^2 pdv$ gives the measure of work done during :

(a) steady flow reversible process
(b) non-flow reversible process
(c) open system and any process
(d) any system and any process

Nagaland PSC CTSE 2015, Paper-II
J & K PSC Screening, 2006

Ans. (b) : $\int_1^2 pdv$ is applicable in –

- Closed system or Non flow process
- Reversible process
- Quasistatic process

55. Cyclic integral of this is zero :

(a) Work transfer (b) Heat transfer
(c) Temperature (d) Latent heat

J&K PSC Civil Services Pre, 2010
UPSC JWM Advt. No.-52/2010

Ans. (c) : $\oint T = 0$

Temperature is a property i.e point function.

56. For steady flow through an insulated horizontal constant diameter pipe, which property remains constant?

(a) Enthalpy (b) Internal energy
(c) Entropy (d) Volume

GPSC ARTO Pre 30.12.2018
ESE 2012

Ans. (a) : For steady flow through an insulated horizontal constant diameters pipe, enthalpy remains constant.

Note-GPSC gives correct answer as option (d).

57. A frictionless piston-cylinder device contains a gas initially at 0.8 MPa and 0.015 m^3. It expands quasi-statically at constant temperature to final volume of 0.030 m^3. The work output (in kJ) during this process will be:

(a) 8.32 (b) 12.00
(c) 554.67 (d) 8320.00

GPSC ARTO Pre 30.12.2018, GATE 2009

Ans. (a) : Given, $P_1 = 0.8$ MPa
$V_1 = 0.015\ m^3$
$V_2 = 0.030\ m^3$

For isothermal process, $W = P_1V_1 \ln \frac{V_2}{V_1}$

$= 0.8 \times 0.015 \ln \left(\frac{0.030}{0.015}\right)$

$\boxed{W = 8.32\,kJ}$

58. A perfect gas at 27 ºC is heated at constant pressure till its volume is double. The final temperature is

(a) 54ºC (b) 108ºC
(c) 327ºC (d) 600ºC

UPCL AE 29.08.2021
ISRO Scientist/Engineer (RAC) 29.11.2015
ISRO Scientist/Engineer 2011

Ans. (c) : Gas equation,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Constant pressure $= P_1 = P_2 = P$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{V_2}{V_1} = \frac{T_2}{T_1}$$

$$\frac{2V_1}{V_1} = \frac{T_2}{300}$$

$T_2 = 600\ K = (600 - 273)°C$

$T_2 = 327°C$

59. Which of the following is not a path function

(a) Thermal conductivity (b) Kinetic Energy
(c) Heat (d) Work

APPSC AEE Screening Test 2016
BPSC AE 2012 Paper - V

Ans. (a) : Thermal conductivity – point function
Heat, work, – path function

60. A series of operations, which takes place in a certain order and restore the initial conditions at the end, is known as

(a) Reversible cycle
(b) Irreversible cycle
(c) Thermodynamic cycle
(d) None of these

Vizag Steel (MT) 2017
UPPSC AE 12.04.2016 Paper-II

Ans. (c) : Thermodynamic cycle– A series of operation which takes place in a certain order and restore the initial conditions at the end, is known as thermodynamic cycle.

61. Consider the following properties:

1. Temperature
2. Viscosity
3. Specific entropy
4. Thermal conductivity

Which of the above properties of a system is/are intensive?

(a) 1 only (b) 2 and 3 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4

Gujarat PSC AE 2019, ESE 2009

Ans. : (d) : Intensive Properties are as follows-

- Temperature
- Viscosity
- Specific entropy
- Thermal conductivity
- Specific volume
- Specific enthalpy

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62. The gauge pressure in a truck tire before and after the journey was recorded as 200 kPa and 220 kPa respectively at the location where atmospheric pressure and temperature was 100 kPa and 27ºC respectively. How much the rise in temperature of tire air after trip?

(a) 20ºC (b) 27ºC
(c) 47ºC (d) Insufficient data

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II

Ans. : (c) : Given

P_{atm} = 100 kPa, T_{atm} = 27 ºC = 300 K
P_{gauge1} = 200 kPa, P_{gauge2} = 220 kPa
$[P_1]_{Abs}$ = 300 kPa, $[P_2]_{Abs}$ = 320 kPa

Then,

$$V_1 = V_2$$

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$T_2 = \frac{T_1 \times P_2}{P_1} = \frac{300 \times 320}{300}$$

$$\boxed{T_2 = 320\,K = 47°C}$$

63. Air is compressed in a cylinder such that the volume changes from 0.2 to 0.02 m^3. The initial pressure is 200 kPa. If the pressure is constant, the approximate work is

(a) –36 kJ (b) –40 kJ
(c) –46 kJ (d) –52 kJ

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (a)

Work done, If the pressure is constant,

$$w = \int_1^2 PdV$$
$$= P(V_2 - V_1)$$
$$= 200 \times (0.02 - 0.2)$$
$$= -36 \text{ kJ}$$

64. Thermocouples are generally used for measuring temperature:

(a) 500 ºC (b) 1000 ºC
(c) 1500 ºC (d) 2000 ºC

TNPSC AE 2018, SJVN ET 2013

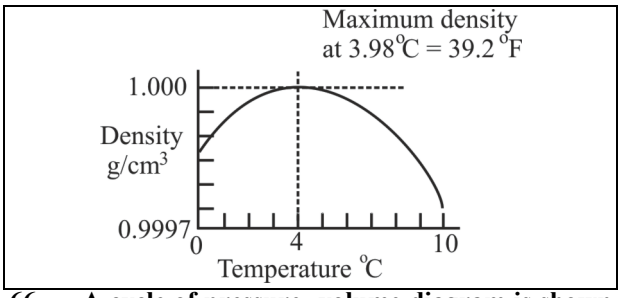
Ans. (c) : Thermocouples are generally used for measuring temperature up to 1500ºC

65. The density of water is maximum at

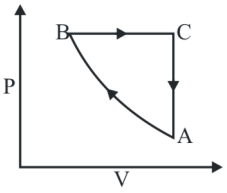
(a) 20 ^{0}C (b) 4 ^{0}C
(c) 0 ^{0}C (d) -4 ^{0}C

UPPSC AE 12.04.2016 Paper-II
GWSSB DEE 07.07.2016

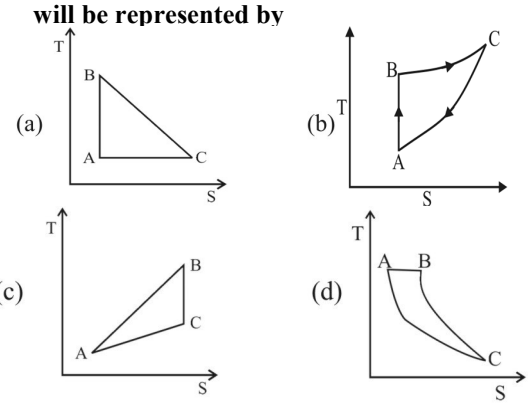
Ans. : (b) The density of water is maximum at 4 ^{0}C. In ice, the water molecules are in a crystal lattice that has a lot of empty space. When the ice melts to liquid water, the structure collapses and the density of the liquid increases.
Thus, the density of water is a maximum at 4 ^{0}C



66. A cycle of pressure- volume diagram is shown in the figure :

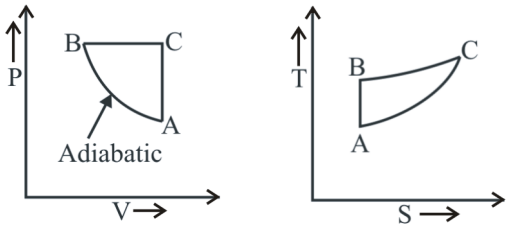


Same cycle on temperature-entropy diagram will be represented by



MPPSC AE 2016
ISRO Scientist/Engineer 2012, ESE 1995

Ans. (b)



Process A-B is adiabatic process so
In T-S diagram a vertical line is obtained (ΔS = Constant)
Process B-C is isobaric process (P = Constant)
So, $V \propto T$ $(V\uparrow\downarrow \rightarrow T\uparrow\downarrow)$
Process C-A is isochoric process (V = Constant)
$P \propto T$

67. A gas has

(a) two specific heat (b) three specific heat
(c) four specific heat (d) one specific heat

Rajasthan Nagar Nigam AE 2016, Shift-II
GWSSB AAE, 27.12.2015

Ans. (a) : A gas has two specific heat.
C_p = Specific heat at constant pressure.
C_v = Specific heat at constant volume.

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68. A system will be in thermodynamic equilibrium if it is in

(a) mechanical equilibrium only
(b) thermal equilibrium only
(c) chemical equilibrium only
(d) all of the above

TNPSC AE 2017, WBPSC AE, 2007

Ans. (d) : For a thermodynamic system to be in equilibrium it should be in :–
(1) Mechanical equilibrium i.e. no influence of external forces
(2) Thermal equilibrium i.e. equality of temperature
(3) Chemical equilibrium i.e. no chemical reaction potential should be present
(4) Phase equilibrium

69. 1 m^3 of air at a pressure of 10 kg/cm^2 is allowed to expand freely to a volume of 10 m^3. The work done will be

(a) Zero (b) –ve
(c) +ve (d) 10^5 kg m

GPSC EE Pre, 28.01.2017
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. (a) : In free expansion-
Work done (W) = 0
Q = 0

70. The value of $\frac{C_p}{C_v}$ for air is :

(a) 1 (b) 1.4
(c) 1.8 (d) 2.3

GMB AAE 25.06.2017, MPSC Poly. Lect. 2014

Ans. (b) : For air

$$\boxed{\frac{C_p}{C_v} = \gamma = 1.4}$$

$$C_p = \frac{\gamma R}{\gamma - 1} = 1.005 kJ/kgK$$

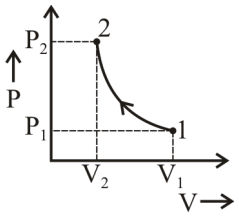
$$C_v = \frac{R}{\gamma - 1} = 0.718 kJ/kgK$$

71. In an adiabatic process, 5000 J of work is performed on a system. In the nonadiabatic process by which the system returns to its original state, 1000 J of heat is added to the system. The work done during the non-adiabatic process is

(a) +6000 J (b) –6000 J
(c) +4000 J (d) –4000 J

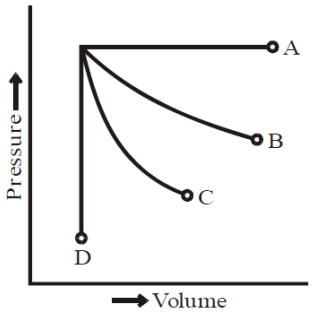
RPSC IOF & Boiler 20.09.2017
UPSC JWM Advt. No.-50/2010

Ans. (a) :



In adiabatic process,
$\delta Q = 0$
$\delta W = -5000$ J
We know that,
$\delta Q = U_2 - U_1 + \delta W$
$0 = (U_2 - U_1) - 5000$
$U_2 = U_1 + 5000$ J
$U_2 - U_1 = 5000$ J
From non-adiabatic 2 to 1,
$\delta Q = U_1 - U_2 + \delta W$
$1000 = -5000 + \delta W$
$\boxed{\delta W = 6000 J}$

72.

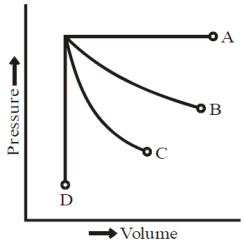


The figure shows thermodynamic expansion processes A, B, C and D. Which line is very close to the isentropic process?

(a) A (b) B
(c) C (d) D

TRB Poly. Lect., 2012
UPSC JWM Advt. No.-50/2010

Ans. (c) : For expansion process



Process	n	
Isobaric(A)	0	P = Constant
Isothermal(B)	1	T = C
Polytropic(C)	n	PV^n = Constant
Adiabatic(D)	γ	PV^γ = Constant
Isochoric(D)	∞	PV^∞ = Constant

73. An isochoric process is a

(a) Constant pressure process
(b) Constant volume process
(c) Constant temperature process
(d) Constant entropy process

GPSC Lect. 23.10.2016
WBPSC AE, 2007

Ans. (b) : Isochoric means constant volume.

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74. Intensive property of a system is one whose value :

(a) depends on the mass of the system, like volume

(b) does not depend on the mass of the system, like temperature, pressure, etc.

(c) is not dependent on the path followed but on the state

(d) is dependent on the path followed and not on the state

J&K PSC Civil Services Pre, 2013
OPSC Civil Services Pre 2006

Ans. (b) : Intensive property – Physical property that does not depend on the size or mass of sample like temperature, pressure, density, viscosity, velocity etc.

Extensive property- Property that depend on size or mass of sample like, volume, mass, entropy, energy etc.

75. Mole fraction of a component of gas equal to

(a) 1/f (b) f^2

(c) f (d) f/p

where f → volume fraction, and

p → pressure of the mixture

APPSC Poly. Lect. 2013, JPSC AE 2013 Paper-I

Ans. (c) : Mole fraction of a component of gas mixture is equal to volume fraction (f)

76. A reversible polytropic process is given by

(a) $\frac{T_1}{T_2}=\left(\frac{\rho_1}{\rho_2}\right)^{n-1}$ (b) $\frac{P_1}{P_2}=\left(\frac{\rho_1}{\rho_2}\right)^{n-1}$

(c) $\frac{T_1}{T_2}=\left(\frac{P_1}{P_2}\right)^{n-1}$ (d) $\frac{P_1}{P_2}=\left(\frac{\rho_1}{\rho_2}\right)^{\frac{n-1}{n}}$

BPSC Poly. Lect. 2016, VIZAG STEEL MT 2015

Ans. (a) : PV^n = Constant (for polytropic process)

PV/T = Constant

$$\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$$

$$\frac{P_1}{P_2}=\frac{V_2}{V_1}\times\frac{T_1}{T_2}$$

$$\left(\frac{P_1}{P_2}\right)=\left(\frac{V_2}{V_1}\right)^n$$

$$\frac{T_1}{T_2}\times\frac{V_2}{V_1}=\left(\frac{V_2}{V_1}\right)^n$$

$$\frac{T_1}{T_2}=\left(\frac{V_2}{V_1}\right)^{n-1}$$

$$V\propto\frac{1}{\rho}$$

$$\frac{T_1}{T_2}=\left(\frac{\rho_1}{\rho_2}\right)^{n-1}$$

77. The vapour pressure (mm Hg) of solid NH_3 is given by

$$\ln p = 23.05 - \frac{3754}{T}$$

and that of liquid ammonia is given by

$$\ln p = 19.49 - \frac{3063}{T}$$

Where, T is in Kelvin. What will be the triple point temperature of NH_3?

(a) 140.3 K (b) 195.2 K

(c) 225 K (d) 280 K

GATE 2021, APPSC Poly Lect. 13.03.2020

Ans. (b) : Given data :

$$\ln p = 23.03 - \frac{3754}{T} \quad \text{..... (1)}$$

Vapour pressure (mm Hg) of solid NH_3 is –

For liquid ammonia –

$$\ln p = 19.49 - \frac{3063}{T} \quad \text{.....(2)}$$

At triple point all three phases co-exist so all properties of substances should be identical

Subtracting equation (2) from equation (1)

$$= 3.54 - \frac{691}{T} = 0$$

$$\Rightarrow T = \frac{691}{3.54}$$

$$\Rightarrow T = 195.197\ K = 195.23\ K$$

78. A new temperature scale in degrees N is to be defined. The boiling and freezing points on this scale are 400°N and 100°N respectively. What will be the reading on new scale corresponding to 60°C?

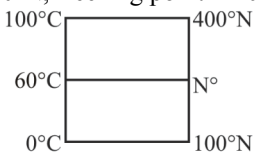
(a) 280°N (b) 180°N

(c) 220°N (d) 120°N

TSPSC AEE 2017, CSE Pre-2006, 1995

Ans. (a) : Given,

Boiling Point 400°N, freezing point = 100°N



$$\frac{60-0}{100-0}=\frac{N-100}{400-100}$$

$$\frac{60}{100}=\frac{N-100}{300}$$

$$N - 100 = 180$$

$$\boxed{N = 280°N}$$

79. Two gases having molecular weight 32 and 28 expand at constant pressure through the same temperature range. The ratio of work done by the two gases will be :

(a) 8 : 7 (b) 7 : 6

(c) 3 : 2 (d) 7 : 8

CGPSC AE 15.01.2021, CGPSC AE 16.10.2016

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Ans. (d) : Given,

Gas A, $(M_{wt})_A = 32$

Gas B, $(M_{wt})_B = 28$

Work done, $W_{1-2} = P(V_2 - V_1) = R(T_2 - T_1)$

$W \propto R$

$W \propto \frac{R_u}{M_{wt}}$ (R_u = Universal Gas Constant)

$\frac{W_1}{W_2} = \frac{(M_{wt})_2}{(M_{wt})_1} = \frac{28}{32} = \frac{7}{8}$

80. The ratio of specific heat of a gas at constant pressure and the specific heat of the gas at constant volume

(a) is always constant

(b) varies with temperature

(c) varies with pressure

(d) is not constant

GPSC Engineer Class-II Pre-19.01.2020

APPSC AEE Screening Test 2016

Ans. (a) : The ratio of specific heat of a gas at constant pressure and the specific heat of the gas at constant volume is always constant.

$$\gamma = \frac{C_p}{C_v}$$

81. Thermodynamic work is the product of

(a) Two intensive

(b) Two extensive properties

(c) An intensive property and change in an extensive property

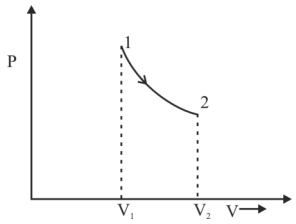
(d) An extensive property and change in an intensive property

UPPSC AE 12.04.2016 Paper-II

CSE Pre-1998

Ans. : (c) For quasistatic process work done is calculated by

$\int P.dV$



Pressure = Intensive property.

Volume (dV) = change in an extensive property.

Thermodynamic work is the product of an intensive property and change in an extensive property.

82. The maximum work by an expansion of a gas in a closed system is possible when process takes place at constant

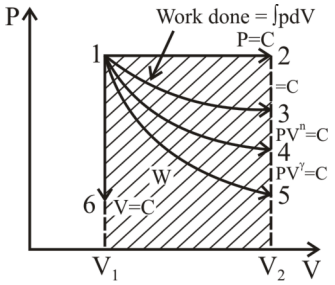
(a) Pressure (b) Enthalpy

(c) Temperature (d) Volume

Rajasthan AE (Nagar Nigam) 2016 Shift-3

UPPSC AE 12.04.2016 Paper-II

Ans. (a) : We know that,



Closed system work on P-V diagram, is the area under P-V should be projected on volume axis and mathematically expression for closed system work is

$\int_{V_1}^{V_2} PdV$

$\therefore \quad W_{closed} \int_{V_1}^{V_2} pdV$

According to graph,

$W_{1-2} > W_{1-3} > W_{1-4} > W_{1-5} > W_{1-6}$

So, max work obtain in expansion process is isobaric process & min work about to zero in isochoric process.

83. Consider the following statements about isolated system

1. No heat is transferred
2. No work is done
3. No mass flows across the boundary of the system
4. No chemical reaction takes place within the system

Of above, which are correct

(a) 1 and 2 (b) 1 and 3

(c) 1, 2 and 3 (d) 3 and 4

Nagaland PSC CTSE 2015 Paper-II

GATE 1999

Ans. (c) : Isolated System– It is a special case of closed system in which neither mass nor energy cross the boundary of the system.

Or

A system that does not interact with its surroundings, that is its total energy and mass stay constant.

Ex.: Universe, thermal flask.

84. The velocity of sound in an ideal gas does not depend on

(a) the specific heat ratio of the gas

(b) the molecular weight of the gas

(c) the temperature of the gas

(d) the density of the gas

Nagaland PSC CTSE 2015 Paper-II

CSE Pre-2002

Ans. (d) : Velocity of sound in an ideal gas

$C = \sqrt{\gamma RT}$; $C = f(\gamma, R, T)$

Where, γ = Specific heat ratio

R = Gas constant

T = Temperature

The velocity of sound in an ideal gas does not depend on the density of gases.

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85. The value of an extensive property is essentially dependent on :

(a) Mass or extent of the system
(b) Interaction of system with its surroundings
(c) Path followed by the system is going from one state to another
(d) Nature of boundaries, rigid or flexible

TSPSC Managers, 2015, ESE 2013

Ans. (a) : The value of an extensive property is essentially dependent on mass or extent of the system.

86. Match List I with List II and select the correct answer using the codes given below the lists:

A. Bomb calorimeter i. Pressure
B. Exhaust gas calorimeter ii. Enthalpy
C. Junker gas calorimeter iii. Volume
D. Throttling calorimeter iv. Specific heat

Code:

	A	B	C	D
(a)	iii	iv	i	ii
(b)	ii	iv	i	iii
(c)	iii	i	iv	ii
(d)	iv	iii	ii	i

APPSC Poly. Lect. 2013, ESE 1998

Ans. (a) :

List I		List II
A) Bomb calorimeter	-	Volume
B) Exhaust gas calorimeter	-	Specific heat
C) Junker gas calorimeter	-	Pressure
D) Throttling calorimeter	-	Enthalpy

87. The specific heat ratio k is given by the following expression :

(a) $1 + (C_P/R)$ (b) $\dfrac{1}{1-(R/C_v)}$

(c) $1 + (R/C_v)$ (d) $1 + (C_v/R)$

TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-1999

Ans. (c) :

$$C_p - C_v = R$$

$$\frac{C_p}{C_v} - 1 = \frac{R}{C_v}$$

$$k - 1 = \frac{R}{C_v}$$

$$k = 1 + \frac{R}{C_v}$$

88. A certain amount of an ideal gas is initially at a pressure P_1 and temperature T_1. First, it undergoes a constant pressure process 1-2 such that $T_2 = 3T_1/4$. Then, it undergoes a constant volume process 2-3 such that $T_3 = T_1/2$. The ratio of the final volume to the initial volume of the ideal gas is

(a) 0.25 (b) 0.75
(c) 1.0 (d) 1.5

GPSC Asstt. Director of Transport 05.03.2017
GATE 2014

Ans. (b) : For (1-2) process: Constant pressure process ($P_1 = P_2$)

$$\frac{T_1}{V_1} = \frac{T_2}{V_2} \Rightarrow V_2 = \frac{T_2}{T_1} \times V_1$$

For (2-3) : Constant volume process ($V_3 = V_2$)

Given, $T_2 = \dfrac{3T_1}{4}$

$$V_2 = \frac{3T_1}{4T_1} \times V_1 = \frac{3}{4} V_1$$

$$\frac{V_3}{V_1} = \frac{V_2}{V_1} = \frac{3}{4} = 0.75$$

89. The constant volume gas thermometer works on the principle that

(a) At low pressure, the temperature of the gas is independent of its pressure at constant volume
(b) At high pressure, the temperature of the gas is independent of its pressure at constant volume
(c) At low pressure, the temperature of the gas is proportional to its pressure at constant volume
(d) At high pressure, the temperature of the gas is proportional to its pressure at constant volume

GPSC Asstt. Director of Transport 05.03.2017
ESE 2012

Ans. (c) : The constant volume gas thermometer works on the principle that at low pressure, the temperature of gas is proportional to its pressure at constant volume.

90. The absolute zero pressure will be

(a) when molecular momentum of the system becomes zero
(b) at sea level
(c) at the temperature of –273K
(d) at the center of the earth

Nagaland PSC (CTSE) 2018, Paper-II
ESE 2002

Ans. (a) : Absolute zero pressure means there are no collisions between molecules to applied per unit area force occurs due to collision of molecules at absolute zero these is no momentum between molecules so no force developed. Atmospheric pressure is measured from absolute zero pressure.

91. What are the properties of a thermodynamic system whose value for the entire system is equal to the sum of their values for individual parts of the system?

(a) Thermodynamic properties
(b) Extensive properties
(c) Intensive properties
(d) Specific properties

TNPSC AE 2018, CSE Pre-2006

Ans. (b) : Extensive properties- Extensive properties of matter that changes as the amount of matter changes
Example- Volume, enthalpy, entropy etc.

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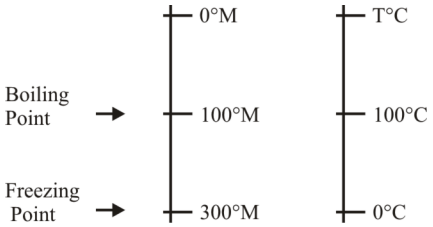
Intensive properties- An intensive properties is a bulk property, meaning that it is a local physical property of a system that does not depends on the system size or the amount of material in the system.
Example- Pressure, temperature, refractive index, density and hardness of object.

92. A new temperature scale say °M, the boiling and freezing point of water at 1 atm are 100°M and 300°M respectively. Correlate this scale with centigrade scale. The reading of 0°M on centrigrade scale is :

(a) 0°C (b) 50°C
(c) 100°C (d) 150°C

GPSC ARTO Pre 30.12.2018, ESE 2001

Ans. (d) :



$$\frac{300-0}{100-0}=\frac{0-T}{100-T}$$

$30000 - 300T = -100T$

$200T = 30000$

$\boxed{T = 150°C}$

93. Enthalpy in the fundamental unit is expressed as:

(a) MLT^{-2} (b) $ML^{-2}T^{-2}$
(c) ML^2T^{-2} (d) ML^2T^{-1}

AAI Jr. Executive 29.11.2018
CSE Pre-1994

Ans. (c) : Enthalpy is type of energy.
Energy and work is same unit

$W = F \times S$

$= [ML^2T^{-2}]$

94. A reversible process requires that

(a) there be no heat transfer.
(b) Newton's law of viscosity be satisfied.
(c) temperature of system and surrounding be equal.
(d) there will be no viscous or Coulomb friction in the system.

JPSC AE, 10.04.2021, Paper-I
UPRVUNL AE 05.07.2021

Ans. (d) : A reversible process requires that there will be no viscous or coulomb friction in the system.

95. Total heat of a substance is also known as :

(a) Internal energy (b) Entropy
(c) Thermal capacity (d) Enthalpy

CGPSC AE 15.01.2021
J & K PSC Screening, 2006

Ans. (d) : Total heat of a substance is known as Enthalpy
It is represented by H in kJ.

$\boxed{H = U + PV}$ → Flow work

Where, U = Internal Energy

96. For which of the following substances, the internal energy and enthalpy are functions of temperature only?

(a) Any gas (b) Saturated steam
(c) Water (d) Perfect gas

CGPSC AE 15.01.2021, WBPSC AE, 2017

Ans. (d) : For perfect gas or ideal gas the internal energy and enthalpy are functions of temperature only.
Then, Internal energy, $U = f(T)$
Enthalpy, $H = f(T)$

97. The sum of internal energy (U) and the product of pressure and volume (PV) is known as :

(a) Work done (b) Entropy
(c) Power (d) Enthalpy

CGPSC AE 15.01.2021, WBPSC AE 2003

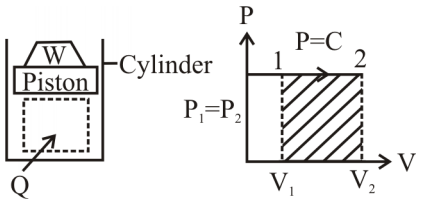
Ans. (d) : Enthalpy $\boxed{(H) = U + PV}$

98. Heat is supplied to air in a cylinder fitted with frictionless piston held by a constant weight. The process is

(a) Isothermal (b) Adiabatic
(c) Isobaric (d) Isochoric

Assam PSC AE (PHED), 18.10.2020
RPSC AEN Pre-2013

Ans. (c) : As weight is constant that means the pressure will be constant. The process is isobaric.



Constant pressure heat addition (at constant pressure) $W = P\int dv$

99. Work is a

(a) point function
(b) path function
(c) It depends on the state
(d) None of the above

Assam PSC AE (PHED), 18.10.2020
WBPSC AE, 2017

Ans. (b) : Heat and work both are path function it is not property it is a boundary phenomenon.

- Energy transfer due to temperature difference is called heat.

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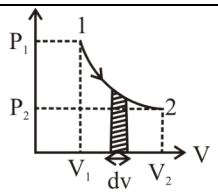


$dw = Pdv$

So, $\int_1^2 dw = \int_1^2 pdv$

100. The hyperbolic process is governed by -

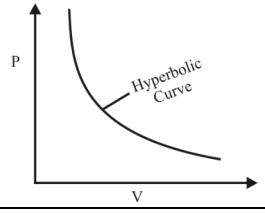
(a) Boyle's Law (b) Charles Law
(c) Gay-Lussac Law (d) Avogadro's Law

ISRO Scientist/Engineer 22.04.2018
GMB AAE 25.06.2017

Ans. (a) : According to the Boyles law at constant temperature pressure is inversely proportional to its given volume.

$$P \propto \frac{1}{V} \quad \text{(at T = const.)}$$

$$\boxed{PV = \text{constant}}$$



101. If a fluid expands suddenly into vacuum through an orifice of large dimension, then such a process is called

(a) free expansion
(b) hyperbolic expansion
(c) adiabatic expansion
(d) throttling

J&K PSC Civil Services Pre, 2013
APGCL AM, 2021

Ans. (a) :
Free expansion : Free expansion of gas occurs when it is subjected to expansion against the vacuum.
Hyperbolic expansion : It is Isothermal expansion (T = Constant).
Throttling : It is a process where a high pressure fluid is converted into low pressure by using throttle valve.

102. The gas with highest value of adiabatic index is:

(a) Helium (b) Nitrogen
(c) Oxygen (d) Methane

TSPSC Managers, 2015
UPSC JWM Advt. No.-52/2010

Ans. (a) : $\gamma \propto \frac{1}{\sqrt{M}}$

Element	Adiabatic Index (γ)
Helium (He)	1.67
Nitrogen (N_2)	1.4
Oxygen (O_2)	1.4
Methane	1.313

103. Identify open system and closed system from the following

(a) Blood circulation and respiration in human body
(b) Fuel system and radiator in cars
(c) Air compressor and boiler
(d) Shell and tube heat exchanger and blower

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (c) : Air compressor is an open system, while boiler can be a closed system if you include the pipes it is connected.

104. Flow of work may be expressed as

(a) Pdt (b) $(P_1V1 - P_2V_2)/n - 1$
(c) Td (d) Pdv

Nagaland PSC CTSE 2017, 2016 Paper-II

Ans. (d) : The work done in pushing the fluid is:
$W_{flow} = F.ds = P.A.ds. = P.dv$
Flow work is necessary for maintaining a continuous flow through a control volume.

105. If the time taken by a system to execute a process through a finite gradient is infinitely large, what is the nature of the process?

(a) becomes reversible
(b) remains irreversible
(c) becomes isothermal
(d) is adiabatic

UPSC JWM Adv. No-16/2009, ESE 2013

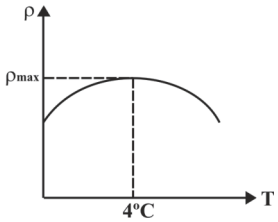
Ans. (a) : A reversible process is carried out infinitely slowly with an infinitesimal gradient, so that every state passed through by the system is an equilibrium state. So a reversible process coincides with a quasi-static process.

106. When water at 0°C is heated, its specific volume under standard atmosphere

(a) First increases then decreases
(b) First decreases then increases
(c) Increases steadily
(d) Decreases steadily

UPCL AE 29.08.2021
ISRO Scientist/Engineer 07.05.2017

Ans. (b) : Specific volume first decrease ↓ and then increase ↑.



Density of water first increases till 4ºC and then decreases.

$$\boxed{\uparrow \rho \propto \frac{1}{V\downarrow}}$$

Therefore, before 4ºC volume will decrease and increase on further heating.

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107. Which one of the following is the correct statement? Two adiabatic processes will _____

(a) Intersect at absolute zero temperature
(b) Never intersect
(c) Become orthogonal at absolute zero temperature
(d) Become parallel at absolute zero temperature

UPRVUNL AE 21.08.2016, CSE Pre-2007

Ans. : (b) Two adiabatic process will never intersect.

108. A battery is used to light a bulb, run a fan and heat an electric iron in case of a power failure. If each of the above, the systems has 100 W rating and is run for 15 minutes, what are the work done (W) and heat transferred (Q) by the battery?

(a) W = 90 kJ and Q = 90 kJ
(b) W = 180 kJ and Q = 0 kJ
(c) W = 270 kJ and Q = 0 kJ
(d) W = 90 kJ and Q = 180 kJ

Nagaland PSC CTSE 2015 Paper-II, CSE Pre-2005

Ans. (c) : Given, $P_{fan} = 100$ W

$P_{bulb} = 100$ W

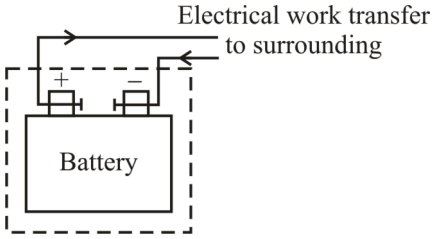
$P_{electric\ iron} = 100$ W

Total power $(P_{total}) = P_{fan} + P_{bulb} + P_{electric\ iron}$

$= 100 + 100 + 100$

$= 300$ W

Time = 15 minutes



Total workdone by battery in 15 minutes

$W = 300 \times 15 \times 60$

$= 270000$

$= 270$ kJ

There is flow of electrons to surrounding only from battery, so electrical workdone by battery and no heat transfer.

109. Which one of the following thermodynamic processes approximates the steaming of food in a pressure cooker?

(a) Isochoric (b) Isobaric
(c) Isothermal (d) Isenthalpic

GWSSB AAE, 27.12.2015, ESE 2007

Ans. (a) : Isochoric process is known as constant volume process or isometric process. In this process heat takes at given constant volume process for example pressure cooker.

110. An ideal gas as compared to a real gas at very high pressure occupies

(a) more volume (b) less volume
(c) same volume (d) no such correlation

Karnataka PSC Lect., 27.05.2017
TRB Asstt. Prof., 2012

Ans. (a) : For ideal gas,

$PV = nRT$

$P \propto \frac{1}{V}$ .....(i)

For real gas,

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$P \propto \frac{1}{V - b}$ .....(ii)

From (i) & (ii)

An ideal gas as compared to a real gas at very high pressure occupies more volume.

111. Absolute zero temperature is taken as _____.

(a) –373ºC (b) 273ºC
(c) –273.15ºC (d) 237ºC

UPRVUNL AE 05.07.2021

Ans. (c) : Absolute zero is the lowest limit of thermodynamic temperature scale, taken as zero Kelvin. At zero Kelvin (–273 ºC) the particle stop moving and all disorder disappears, thus nothing can be colder than absolute zero on the Kelvin scale.

112. Pyrometer works based on

(a) Laser technology (b) Photo conduction
(c) Thermal emission (d) Tyndall effect

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Pyrometer–A pyrometer is a type of thermometer used to measure high temperature. It is used for measuring temperature without any physical contact. It is used for measuring body temperature by measuring its electromagnetic radiation.

113. The temperature at which the Kelvin and Fahrenheit scales coincide is _____.

(a) 474 K (b) 350 K
(c) 474 F (d) 574 K

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : The temperature at which the Kelvin and Fahrenheit scales coincides is 574 Kelvin.

114. Energy that crosses the system boundaries is known as _____

(a) Energy in transit (b) global energy
(c) stored energy (d) local energy

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : Energy that crosses the system boundaries is known as energy in transit.

115. An absolute zero temperature, a reversible adiabatic line is also an :

(a) Isothermal Line (b) Isobaric Line
(c) Isochoric Line (d) Isenthalpic Line

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : At absolute zero temperature a reversible adiabatic line is also known as Isothermal line.

116. When the valve of an evacuated bottle is opened the atmospheric air rusher into it. If the atmospheric pressure is 100 kPa and 0.8 m^3 of air enters into the bottle, then the work done by the air will be

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(a) 80 kJ (b) 70 kJ
(c) 60 kJ (d) 50 kJ

APGCL AM, 2021

Ans. (a) : The displacement work done by air

$W_d = \int_{Bottle} Pdv + \int_{\substack{free\ air \\ boundary}} Pdv$

$= 0 + P\Delta V$

$= 0 + 100 \times 0.8$

$= 80$ kJ Ans.

117. Which of the following represents a closed system?

(a) Bomb calorimeter (b) Boiler
(c) Universe (d) Centrifugal pump

JPSC AE, 10.04.2021, Paper-I

Ans. (a) : Bomb calorimeter is closed system.

Open System–Type of system that allow mass and energy transfer.

Ex-Nozzle, turbine, piston cylinder with valve.

Closed System–Type of system that allow energy transfer.

Ex-Piston cylinder without valve, hot gases in rigid cylinder, bomb calorimeter.

Isolated System–Type of system that does not allow both mass and energy transfer.

Ex-Universe, thermoflask.

118. Which amongst the following is not a property of the system?

(a) Heat
(b) Composition
(c) Specific volume
(d) Thermal conductivity

JPSC AE, 10.04.2021, Paper-I

Ans. (a) : Heat is not a property of the system.

Heat and work are path functions.

Property is the characteristic of system which can be measured.

Example of property ⇒ Pressure, Volume, Temperature

119. In a thermoelectric thermometer for t°C temperature, the emf is given as: $E = 0.003t - 5 \times 10^{-7}t^2 + 0.5 \times 10^{-3}$ volts. Thermometer is having reference junction at ice point and is calibrated at ice point and steam point. What is the temperature shown by the thermometer for a substance at 30°C?

(a) 33.23°C (b) 36.28°C
(c) 41.23°C (d) 46.28°C

ESE 18.07.2021

Ans. (a) : $E = 0.003t - 5 \times 10^{-7}t^2 + 0.5 \times 10^{-3}$ volt

At ice point, when t = 0°C,

$E = 0.5 \times 10^{-3}$V

At steam point t = 100°C

$E = 0.003 \times 100 - 5 \times 10^{-7} \times (100)^2 + 0.5 \times 10^3$

$= 0.2955$V

At 30°C

$E = 0.003 \times 30 - 5 \times 10^{-7} \times (30)^2 + 0.5 \times 10^{-3}$

$= 0.09005$V

- When the gas thermometer reads 30°C the thermocouple will read

$= \frac{100}{0.2955} \times 0.09005$

$\approx 33.23\,^0$C

120. Consider the following statements for comparison of heat and work:

1. Both heat and work are transient phenomena.
2. Both heat and work are boundary phenomena.
3. Both heat and work are path functions and inexact differentials.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1, 2 and 3
(c) 2 and 3 only (d) 1 and 3 only

ESE 18.07.2021

Ans. (b) :

- Both heat and work are transient boundary phenomenon.
- Both heat and work are path function and inexact differentials.

121. What is the value of mass of the air in a room of size 4m × 5m × 6m at 100 kPa and 25°C? (Take R = 0.287 kPa.m^3.kg^{-1}.K^{-1}).

(a) 150 kg (b) 180 kg
(c) 140 kg (d) 130 kg

ESE 18.07.2021

Ans. (c) : Given, P = 100 kPa,

T = 25 + 273 = 298K

R = 0.287 kJ/kg

Volume (V) = 4 × 5 × 6 = 120 m^3

$P = \rho RT$

$PV = mRT$

$m = \frac{PV}{RT}$

$= \frac{100 \times 120}{0.287 \times 298}$

m = 140 kg

122. Statement (I): If two systems are in thermal Equilibrium with a third system, then they are not in thermal equilibrium with each other. Statement (II): Zeroth law of thermodynamics is the basis for temperature.

ESE 18.07.2021

Ans. (d) :

When a body A is in the thermal equilibrium with a body B, and separately with body C, then bodies B and C will be in thermal equilibrium with each other. This is known as zeroth law of thermodynamics. It is the basis of the temperature measurement.

Hence, statement (I) is incorrect.

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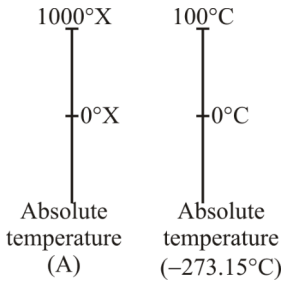
123. In an unknown temperature scale freezing point of water is 0°X and boiling point of water is 1000°X. The absolute zero in degree X is:

(a) −2731.5 (b) −273.15

(c) 0 (d) 2731.5

AAI Jr. Executive 26.03.2021

Ans. (a) :



$$\frac{100-0}{100+273.15}=\frac{1000-0}{1000-A}$$

$100000 - 100A = 373150$

$-100A = 273150$

$\boxed{A = -2731.5°X}$

124. A perfect gas is compressed to half of its initial volume. This is done using:

Process I– isothermal process and

Process II – polytropic process with index, n = 1.25

Larger work input is required in:

(a) Same in Process I and II

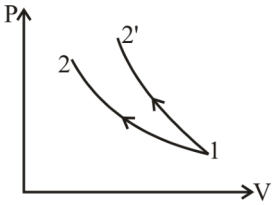
(b) Cannot be predicted

(c) Process I

(d) Process II

AAI Jr. Executive 26.03.2021

Ans. (d) :



1-2 – Isothermal process

1-2′ – Polytropic process

(n = 1.25)

From the PV diagram, it is clear that area under polytropic process is more than isothermal process.

125. Which of the following is a non quassi static process?

(a) Gradual compression of gas inside a piston cylinder arrangement

(b) Expansion of gas in a cylinder under constant pressure

(c) Free or unrestricted expansion of gas

(d) Rapid leakage of air from a bicycle tyre

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Free or unrestricted expansion of gas.

126. For an ideal gas as a working fluid for a given heat input Q, the process that gives the maximum work among the following four processes is

(a) Isothermal (b) Constant volume

(c) Constant pressure (d) Isentropic

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) : For an ideal gas as a working fluid for a given heat input Q. At constant pressure, the area under the P-V curve is maximum. Thus, maximum work is obtained.

127. An adiabatic system is a thermally insulated system with its surroundings and

(a) does not exchange work with its surroundings.

(b) exchanges work with surroundings.

(c) Both (a) and (b)

(d) None of the above

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : For thermally insulated system, neither mass nor energy exchanges with surroundings, but exchanges work with surroundings.

128. The term N.T.P. stands for_____

(a) Nominal Temperature and Pressure

(b) Natural Temperature and Pressure

(c) Normal Temperature and Pressure

(d) Normal Thermodynamic and Pressure

RPSC IOF, 2020

Ans. (c) : The term NTP stands for Normal Temperature and Pressure.

129. Which property is an intensive property of the system?

(a) Specific enthalpy (b) Volume

(c) Kinetic energy (d) Entropy

RPSC IOF, 2020

Ans. (a) : Specific enthalpy is the intensive property. Such as specific volume. Specific entropy etc. All specific property is intensive properties.

Extensive (Depend upon mass)	**Intensive (Do not depend upon mass)**
1. Mass	Temperature
2. Kinetic Energy	Pressure
3. Volume	Density
4. Entropy	Viscosity

130. Which of the following property is used as thermometric property in constant - volume gas thermometer?

(a) Volume (b) Pressure

(c) Electric resistance (d) Voltage

APPSC Poly Lect. 13.03.2020

Ans. (b) : In the constant volume gas thermometer pressure is used as thermometric property.

Thermometer	**Thermometric properties**
1. Constant volume gas thermometer	Pressure (P)
2. Constant pressure gas thermometer	Volume (V)

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3. Electric Resistance thermometer	Resistant (R)
4. Thermocouple	Thermal emf (e)
5. Mercury in glass thermometer	Length (L)

131. If specific heat ratio for a gas is γ, the change in internal energy of a mass of gas at constant pressure P, when volume changes from V to 2V is

(a) $\frac{PV}{\gamma-1}$ (b) $\frac{R}{\gamma-1}$

(c) PV (d) $\frac{\gamma PV}{\gamma-1}$

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : Given,

Specific heat ratio = γ

$V_i = V$

$V_f = 2V$

From ideal gas equation, $PV = mRT$

$V \propto T,$

Thus, $T_1 = T$

$T_2 = 2T$

Change in internal energy,

$\Delta U = m C_v dT$

$= m C_v (T_2 - T_1)$

$\Delta U = m C_v (2T - T)$

$\Delta U = m C_v T$

$$\Delta U = \frac{mRT}{\gamma-1} = \frac{PV}{\gamma-1}$$

132. Which one of the following is an example of a open system?

(a) a cylinder piston

(b) a mass sliding on an inclined plane

(c) a water tank with both inflow and outflow of water

(d) a rigid insulated vessel

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : An open system can exchange both energy and matter with its surrounding.

133. We would like to analyse an electric geyser (storage type water heater), for which a student chooses the system to be all the contents of the insulated container including the heater coil. In the first law analysis of this system, the electric power supplied to the geyser should be treated as

(a) heat input

(b) work input

(c) energy change

(d) internal sources of heat

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : Energy transfer due to temperature difference is taken as heat transfer while energy transfer other than temperature difference is taken as work transfer.

134. The density ρ and specific volume v are related by the expression

(a) ρv =1 (b) ρ/v = 1

(c) $\rho^2 = 1/v$ (d) $v^2 = 1/\rho$

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : $m = \rho V$

$$\frac{m}{V} = \rho \Rightarrow \frac{V}{m} = \frac{1}{\rho} \qquad \left[\because \text{sp.volume}(v) = \frac{V}{m}\right]$$

$$\text{Specific volume } (v) = \frac{1}{\rho}$$

135. Enthalpy of formation is defined as zero for

(a) Only oxygen in gas phase

(b) Only nitrogen in gas phase

(c) Only water in vapour phase

(d) All chemical elements in their naturally occurring phase

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : All element in their standard states (oxygen gas, solid carbon in the form of graphite etc) how a standard enthalpy of formation of zeros as there is no change invalid in their formation.

136. A perfect gas is heated at constant pressure from 30°C till its volume become double. Then the final temperature is:

(a) 333 °C (b) 120 °C

(c) 60 °C (d) 303 °C

CIL MT 27.02.2020

Ans. (a) : According to charles law at constant pressure the volume V of a gas is directly proportional to its absolute temperature T,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$V_1 = V$

$V_2 = 2V$

$$\frac{V}{303} = \frac{2V}{T_2} \qquad (T_1 = 273 + 30 = 303 \text{ K})$$

$T_2 = 606 - 273$

$T_2 = 333$ °C

137. When an ideal gas expands, its temperature

(a) remains constant

(b) increases

(c) decreases

(d) may increase or decrease

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : When an ideal gas expands its temperature may increase or decrease.

138. Which of the following statements is correct with respect to temperature?

(a) Temperature is an intensive property of the system

(b) Temperature is not a property of the system

(c) Temperature is an extensive property of the system

(d) Temperature does not change with volume

APPSC Poly Lect. Automobile Engg., 2020

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Ans. (a) : Intensive Property– Intensive property are independent of mass or size of the system.
Example– Pressure, Temperature, Density viscosity etc.

139. Quasi-static process
(a) is infinitely slow process
(b) experiences thermodynamic equilibrium at every small point on its path
(c) reversible process
(d) All of the above

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) : Quasi-static process is infinitely slow process. It experiences thermodynamic equilibrium at every small point on its path and it is reversible process.

140. An electrical resistance thermometer shows a resistance of 3 ohm at 0^0C and 6 ohm at 200^0C. What is the temperature when the resistance indicates 5 ohm?
(a) 233.3^0 (b) 133.3^0C
(c) 150^0C (d) 300^0C

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The resistance and temperature relation is given by

$R_t = R_o (1+\alpha t)$

R_0 = resistance at 0^0C

$R_0 = 3$ ohm

$R_{200} = 6$ ohm

So that,

$3 = R_0 (1+\alpha\times 0) \Rightarrow 3 = R_0$ --------- (I)

$6 = R_0 (1+\alpha\times 200)$ $6 = R_0 + 200\ R_0\alpha)$ ---------- (II)

On solving Equ. (I) and (II)

$6 = 3 + 200 \times 3 \times \alpha$

$6 = 3 + 600\alpha$

$3 = 600\alpha$

$$\alpha = \frac{3}{600} = 0.005$$

Now R = 5 ohm

$5 = R_0 (1+\alpha t)$

$5 = 3(1+0.005 \times t)$

$5 = 3 + 0.005 \times 3 \times t$

$2 = 0.005 \times 3 \times t$

$$t = \frac{2}{0.005\times 3}$$

$$= \frac{2}{15}\times 10^3$$

$t = 133.3^0$C

141. Efficiency of a heat engine is defined as the ratio of -
(a) Total heat input/Net work output
(b) Net work input/Total heat output
(c) Total heat output /Net work input
(d) Net work output/Total heat input

MECON MT 2019

Ans. (d) : Net work output/Total heat input

142. A non-flow device compresses air isothermally at a temperature of 500^oC from a pressure of 5.0 MPa to a final pressure of 15.0 MPa. If the device uses 200 kJ/kg of work input, then this process is
(a) impossible
(b) irreversible
(c) either reversible or irreversible
(d) reversible

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Ans. (a) : Given -

$T_1 = 500^0$C

$T_1 = 273 + 500$

$= 773$ K

$P_1 = 5.0$ MPa

$P_2 = 15.0$ MPa

work input = 200 kJ/kg

For non-flow isothermal process

$$\text{work done } W_{Isothermal} = P_1 V_1 ln \left(\frac{P_1}{P_2}\right)$$

$$= mRT_1 ln\left(\frac{P_1}{P_2}\right)$$

where, m = 1, R = 0.287 kJ/kg-K

$$W_{Isothermal} = 1 \times 0.287 \times 773\ ln\left(\frac{5}{15}\right)$$

$= -243.728$ kJ/kg

W = 243 kJ/kg (–ve means work done on the system)

Actual work input of the device

= 200 kJ/kg

Here, work done of the system 200 kJ/kg is less than actual work done of the system 243.7285 kJ/kg.

So, its impossible

143. Pick the correct statement regarding path function.
(a) The differentials of point functions are inexact differentials
(b) The differentials of point functions & path functions are exact differentials.
(c) The differentials of path function are inexact differentials.
(d) The differentials of path functions are exact differentials.

BHEL ET 2019

Ans. (c) : The differential of path functions are inexact differentials where as differential of point functions are exact differentials.

144. The pressure inside a balloon is proportional to the square of its diameter. It contains 2 kg of water at 150 kPa with 85% quality. The balloon and water are now heated so that a final pressure of 600 kPa is reached. the process undergone by the water is given by p-v equation as :
(a) $pv^{-2/3}$ = constant (b) $pv^{2/3}$ = constant
(c) $pv^{2/5}$ = constant (d) $pv^{-2/5}$ = constant

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Ans. (a) : Given as,

$P_1 = 150$ kPa

$P_2 = 600$ kPa

$m = 2$ kg

$x = 0.85$

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi\left(\frac{D}{2}\right)^3 = \frac{4}{3}\pi\frac{D^3}{8}$$

$$V = \frac{\pi}{6}D^3$$

$$D = \left(\frac{6V}{\pi}\right)^{\frac{1}{3}}$$

$$P \propto D^2$$

$$P = KD^2$$

$$P = K\left[\left(\frac{6V}{\pi}\right)^{\frac{1}{3}}\right]^2$$

$$\frac{P}{V^{2/3}} = \text{constant}$$

$$PV^{-2/3} = \text{constant}$$

145. A gas goes through a process given by $pv^2 = c$. It expands from state 1 of 350 kPa and 0.049 m^3 to a final pressure of 150 KPa. Work delivered during this process in kJ is :

(a) 4.765 (b) 8.5

(c) 5.915 (d) 3.897

BHEL ET 2019

Ans. (c) : Given - $PV^2 = C$

at state 1, –

$P_1 = 350$ kPa

$V_1 = 0.049\ m^3$

$P_2 = 150$ kPa

According to $PV^2 = C$, $n = 2$

$$P_1 V_1^2 = P_2 V_2^2$$

$$350 \times (0.049)^2 = 150 \times (V_2)^2$$

$$(V_2)^2 = \frac{350 \times (0.049)^2}{150}$$

$$V_2 = 0.07484\ m^3$$

$$\text{Work done } W = \frac{P_1V_1 - P_2V_2}{n-1}$$

$$= \frac{350 \times 0.049 - 150 \times 0.0744848}{2-1}$$

$$= 5.9222 \text{ kJ}$$

146. Pressure exerted by a gas in a closed container is:

(a) Weak function of Density and Temperature

(b) Weak function of Density and Volume

(c) Strong function of Density and Temperature

(d) Strong function of Density and Volume

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Ans. : (c) : Pressure exerted by a gas in a closed container is strong function of density and temperature. We know that,

$$PV = mRT$$

$$P = \frac{m}{V}RT \qquad \left[\because P = \frac{m}{V}\right]$$

$$P = \rho RT$$

$$\boxed{P = f(\rho, T)}$$

147. The condition for the reversibility of a cycle is

(a) the pressure and temperature of the working substance must not differ, appreciably, from those of the surroundings at any stage in the process

(b) all the processes, taking place in the cycle of operation, must be extremely slow

(c) the working parts of the engine must be friction free

(d) all of the above

Gujarat PSC AE 2019, ESE 2016

Ans. : (d) : The condition for the reversibility of a cycle is

- The pressure and temperature of the working substance must not differ, appreciably from those in the process.
- The working parts of the engine must be friction free.

148. Which of the following processes are thermodynamically reversible?

(a) Throttling

(b) Free expansion

(c) Constant volume and constant pressure

(d) Isothermal and adiabatic

TNPSC AE 2019

Ans. (d) : Throttling, Free expansion, Constant volume and constant pressure processes are thermodynamically irreversible whereas isothermal and adiabatic processes are thermodynamically reversible.

149. A piston cylinder arrangement has air at 600 kPa, 290 K and volume of 0.01 m^3. During a constant pressure process, if it gives 54 kJ of work, the final volume must be

(a) 0.10 m^3 (b) 0.05 m^3

(c) 0.01 m^3 (d) 0.15 m^3

JPSC AE PRE 2019

Ans. (a) : Given,

$P_1 = 600$ kPa

$T_1 = 290$ K

$W = 54$ kJ

$V_1 = 0.01\ m^3$

$V_2 = ?$

$W = PdV$

$= P(V_2 - V_1)$

$54 \times 10^3 = 600 \times 10^3\ (V_2 - 0.01)$

$V_2 = 0.1\ m^3$

150. A polytropic process with n = –1, initiates with P = V = 0 and ends with P = 600 kPa and V = 0.01 m^3. The work done is

(a) 2 kJ (b) 3 kJ

(c) 4 kJ (d) 6 kJ

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Ans. (b) : Given,

$n = -1$

$P_1 = 0$

$V_1 = 0$

$P_2 = 600$ kPa

$V_2 = 0.01\ m^3$

$$W = \frac{P_1V_1 - P_2V_2}{n-1} = \frac{0 - 600 \times 0.01}{-1-1}$$

$= 3$ kJ

151. For an ideal gas, enthalpy is represented by

(a) H = U – RT (b) H = U + RT

(c) H = RT – U (d) H = –(U + RT)

JPSC AE PRE 2019

Ans. (b) : H = U + PV

PV = mRT

for unity mass, m = 1

H = U + mRT

H = U + RT

152. Which one of the following represents the energy in storage?

(a) Work (b) Heat

(c) Energy (d) Internal energy

JPSC AE PRE 2019

Ans. (d) : Internal energy represents the energy in storage.

153. The pressure of air in an automobile tyre at temperature of 27°C is 1.75 bar (gauge). Due to running the temperature of air in the rises to 87°C. What will be the gauge pressure during this running? [P_{atm} = 1.01 bar, volume of tyre is assumed constant]

(a) 2.302 bar (b) 2.914 bar

(c) 1.677 bar (d) 3.180 bar

SJVN ET 2019

Ans. (a) : $P_{abs} = P_{atm} + P_{gauge}$

$= 1.01 + 1.75$

$= 2.76$ bar

$$\therefore \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$(P_2)_{abs} = \frac{(P_1)_{abs} \times T_2}{T_1} = \frac{2.76 \times 360}{300}$$

$$(P_2)_{gauge} = 3.312 - 1.01$$

$= 2.302$ bar

154. A non-flow reversible process takes place according to $V = \frac{15}{P}\ m^3$, where *P* is in bar. What will be the work done if pressure changes from 1 bar to 10 bar? [Given. ln (10) = 2.3025]

(a) 3.453 MN-m, expansion

(b) 3.453 N-m, compression

(c) 3.453 MN-m compression

(d) 3.453 N-m, expansion

SJVN ET 2019

Ans. (c) : Given,

$$V = \frac{15}{P}\ m^3$$

$P_1 = 1$ bar

$P_2 = 10$ bar

$P_1V_1 = P_2V_2$

$$\frac{P_1}{P_2} = \frac{V_2}{V_1},\ \frac{V_1}{V_2} = \frac{10}{1}$$

Non flow work $= \int PdV$

$$= \int_{10}^{1} \left(\frac{15}{V}\right) dV$$

$$= 15 \int_{10}^{1} \left(\frac{1}{V}\right) dV$$

$$= 15[\ell n 1 - \ell n 10]$$

$$= 15[0 - 2.3025]$$

$$= -34.538 \times 10^5\ Nm$$

$$= -3.4538\ MN\text{-}m$$

$$= 3.4538\ MN\text{-}m\ (Compression)$$

155. Certain quantities cannot be located on the graph by a point but are given by the area under the curve corresponding to the process. These quantities in concepts of thermodynamics are called as

(a) cyclic functions (b) point functions

(c) path functions (d) real functions

ESE 2019

Ans. (c) : Path functions like heat transfer and work transfer cannot be represented by points in the graph, but can be given by the area under the curve.

Heat transfer will be area under curve in T-s diagram while work transfer will be the area under curve in P-V diagram.

156. A sample of ideal gas has an internal energy U and is then compressed to one-half of its original volume while the temperature stays the same. What is the new internal energy of the ideal gas in terms of U ?

(a) U (b) 1/2U

(c) 1/4U (d) 2U

RPSC Vice Principal ITI 2018

Ans. (a) : Internal energy of a perfect gas is only function of temperature only.

For real gas internal energy is function of temperature and volume both U = f (T, V).

157. Specific heat at constant pressure can be given as (Where 'ϒ' is ratio of specific heats at constant pressure and constant volume, R is a gas constant, J is the Joule constant.)

(a) $\frac{\gamma R}{J(\gamma - 1)}$ (b) $\frac{JR}{\gamma(\gamma - 1)}$

(c) $\frac{\gamma R}{J(R - 1)}$ (d) $\frac{R}{J(\gamma - 1)}$

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Ans. (a) : We know that

$$C_p - C_v = \frac{R}{J} \quad \text{.....(i)}$$

and $\frac{C_p}{C_v} = \gamma$ .....(ii)

then $C_v = \frac{C_p}{\gamma}$

Putting the value of C_v in equation (i)

$$C_p - \frac{C_p}{\gamma} = \frac{R}{J}$$

$$C_p \frac{(\gamma - 1)}{\gamma} = \frac{R}{J}$$

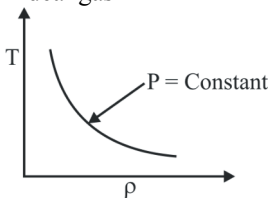
$$C_p = \frac{\gamma.R}{(\gamma - 1)} = \frac{\gamma.R}{J(\gamma - 1)}$$

158. The ordinate and abscissa of the diagram to depict the isobaric processes of an ideal gas as a hyperbola are, respectively

(a) temperature and entropy
(b) internal energy and volume
(c) temperature and density
(d) enthalpy and entropy

ESE 2018

Ans. (c) : For an ideal gas



$PV = mRT$

$P = \rho RT = \text{constant}$

$\rho T = \text{constant}$

i.e. equation of hyperbola.

159. Nitrogen at 25 atm and 37°C is contained in a cylinder of volume $10m^3$. The cylinder is placed in a large container of 150 times the cylinder volume. The container is perfectly evacuated and insulated. The nitrogen gas is allowed to discharge and fill the container. Calculate the final pressure after the system has reached equilibrium.

(a) 0.267 atm (b) 0.322 atm
(c) 0.259 atm (d) 0.167 atm

GPSC ARTO Pre 30.12.2018

Ans. (d) : Given, $T_1 = T_2 = 37°C = 310$ K

$P_1 = 25$ atm

$V_1 = 10\ m^3$

$V_2 = 150V_1 = 1500\ m^3$

As the system is insulated,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\Rightarrow \frac{25 \times 10}{T_1} = \frac{P_2 \times 1500}{T_2} \Rightarrow \boxed{P_2 = 0.167\,\text{atm}}$$

160. A container filled with 2 kg of O_2 is heated at constant pressure from 27 °C to 127 °C. The heat supplied in this process is:

(a) 225 kJ (b) 100 kJ
(c) 205 kJ (d) 182 kJ

AAI Jr. Executive 29.11.2018

Ans. (d) : Heat supplied at constant pressure

$Q = mc_pdT$

$= 2 \times 0.918 \times (127–27)$ [$\because$ C_p for oxygen = 0.918]

$= 183.6$ KJ

161. The unit of 'a' in van der Walls equation for one mole is:

(a) Nm^4 (b) Nm^5
(c) Nm^3 (d) Nm^2

AAI Jr. Executive 29.11.2018

Ans. (a) : Vander wall's equation one mole

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$$P = \frac{a}{V^2}$$

$a = (\text{Pressure}) \times (\text{Volume})^2$

$$= \frac{N}{m^2} \times (m^3)^2$$

$$= \frac{N}{m^2} \times m^6$$

$= Nm^4$

162. A 120-V electric resistance heater draws 10 A. It operates for 10 min in a rigid volume. Calculate the work done on the air in the volume.

(a) 720000 kJ (b) 720 kJ
(c) 12000 J (d) 12 kJ

RPSC Vice Principal ITI 2018

Ans. (b) :

$W = VIt = 120\ V \times 10\ A \times (10 \times 60)\ s$

$= 72 \times 10^4\ J = 720\ kJ.$

163. A steel tank placed in hot environment contains 5 kg of air at 4atm at 30°C. A portion of the air is released till the pressure becomes 2 atm. Later, the temperature of the air in the tank is found to be 150°C. The quantity of air allowed to escape is

(a) 4.72 kg (b) 4.12 kg
(c) 3.71 kg (d) 3.21 kg

ESE 2018

Ans. (d) : Given, $m_1 = 5$kg

$P_1 = 4$ atm

$= 4 \times 101.325$ kPa

$= 405.3$ kPa

$T_1 = 30 + 273 = 303$ K

$P_2 = 2$ atm

$= 2 \times 101.325$ kPa

$= 202.68$ kPa

$T_2 = 150 + 273 = 423$ K

From gas equation

$PV = mRT$

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$$V = \frac{m_1 R T_1}{P_1} = \frac{5 \times 0.287 \times 303}{405.3}$$

$$V = 1.072 \text{ m}^3$$

Again from gas equation for final state

$$m_2 = \frac{P_2 V}{R T_2}$$

$$= \frac{202.65 \times 1.072}{0.287 \times 423}$$

$$= 1.79 \text{ kg}$$

Amount of gas escaped = $m_1 - m_2$

$= 5 - 1.79$

$= 3.209$ kg

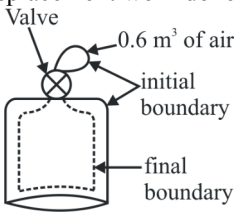
164. When the valve of an evacuated bottle is opened, the atmospheric air rushes into it. If the atmospheric pressure is 101.325 kPa and 0.6 m^3 of air enters into the bottle, then the work done by the air will be

(a) 80.8 kJ (b) 70.8 kJ

(c) 60.8 kJ (d) 50.8 kJ

ESE 2018

Ans. (c) : The displacement work done by air



$$W = \int_{\text{Bottle}} PdV + \int_{\text{Free air boundary}} PdV$$

$= 0 + P\Delta V$

$= 101.325 \times 0.6$

$= 60.8$ kJ

165. The heat energy stored in the gas and used for raising the temperature of the gas is known as

(a) external energy (b) internal energy

(c) kinetic energy (d) molecular energy

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : The heat energy stored in the gas and used for raising the temperature of the gas is known as internal energy.

166. When the gas is cooled at constant pressure

(a) its temperature increases but volume decreases

(b) its volume increases but temperature decreases

(c) both volume and temperature increases

(d) both volume and temperature decreases

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Charle's law states that for a gas at constant pressure. The volume V of a given quantity of gas is directly proportional to the absolute temperature of the gas. So, at constant pressure both temperature and volume decreases when gas is cooled.

167. Process is called reversible process if

(A) A system passes through a continuous series of equilibrium states during process.

(B) These equilibrium states cannot be located on property diagram, cannot be joined by line as a path.

(C) These equilibrium states can be located on property diagram, can be joined by line as a path.

Which of the statement given above is/are correct?

(a) (A) only (b) (A) and (C) only

(c) (B) and (C) only (d) (C) only

MPSC Lect. (Prod.) 2018

Ans. (b) : Process is called reversible process if,

(a) A system passes through a continuous series of equilibrium states during process.

(b) These equilibrium states can be located on property diagram, can be joined by line as a path.

168. Joules law states the specific internal energy of a gas depends only on

(a) the pressure of the gas

(b) the volume of the gas

(c) the temperature of the gas

(d) pressure and volume of the gas

TNPSC AE 2018

Ans. (c) : Joules law states the specific internal energy of a gas depends only on the temperature of the gas.

$u = f(T)$ only

169. The molecular mass expressed in gram (i.e. 1g-mole) of all gases, a NTP occupies a volume of

(a) 0.224 liters (b) 2.24 liters

(c) 22.4 liters (d) 224 liters

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The molecular mass expressed in gram of all gases at NTP occupies a volume of 22.4 liters.

170. The gauge pressure reading on an automobile tire is 240 kPa when the tire temperature is –30°C. The automobile is driven to a warmer climate and the tire temperature increases to 65°C. Assume the mass and the volume do not change. Then, estimate the gauge pressure in the tire.

(a) 480.5 kPa (b) 333.8 kPa

(c) 320.4 kPa (d) 280.6 kPa

HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : Given, $(P_g)_1 = 240$ kPa, $T_1 = -30°C = 243K$.

$T_2 = 65°C = 338$ K , $(P_g)_2 = ?$

From ideal gas equation

$PV = mRT$

V and m are constant,

$P \propto T$

$$\frac{(P_g)_2}{(P_g)_1} = \frac{T_2}{T_1}$$

$$(P_g)_2 = \frac{338}{243} \times 240 = 333.827 \text{ kPa} \approx 333.8 \text{ kPa}$$

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171. Which of the following does not transfer work to or from a system?

(a) A moving piston
(b) The expanding membrane of a balloon
(c) An electrical resistance heater
(d) A membrane that bursts

HPPSC Asstt. Prof. 20.11.2017

Ans. (d) : A membrane that bursts does not transfer work to or from a system because it is considered as free expansion process.
For free expansion process, W = 0

172. In a quasi equilibrium process, the pressure.

(a) remains constant
(b) varies with location
(c) is everywhere constant at an instant
(d) depends only on temperature

HPPSC Asstt. Prof. 20.11.2017

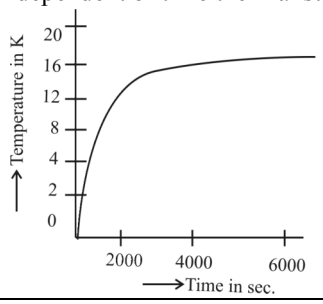
Ans. (c) : In a quasi equilibrium process, the pressure is everywhere constant at an instant.

173. The heat transfer problems in which the temperature at a particular point in the system periodically reaches the same value belong to

(a) Quasi-unsteady state (b) Quasi-steady state
(c) Steady process (d) Unsteady process

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (b) : Quasi steady state in which temperature at a particular point in the system periodically reaches the same value.
This state is independent of time thermal stress.



174. A bicycle pump forces air into the tyre against a pressure of 5 bar. If the piston is pushed slowly, the process can be approximated to

(a) Adiabatic (b) Isothermal
(c) Throttling (d) Free compression

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : When, air pushed in a tyre, the process is isentropic compression, pressure and temperature raised, but after some time temperature drop to ambient temperature, similarly. If we pushed slowly, the pressure increase but temperature is constant. So, the process is isothermal.

175. What will be the volume occupied by a given mass of air at a temperature of 400° C if the same mass of air occupied 2 m^3 at 15° C, while the pressure remain unchanged?

(a) 4.67 m^3 (b) 14.78 m^3
(c) 284.5 m^3 (d) 0.046 m^3

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Ans. (a) : Given, $T_1 = 15°C$ or $15 + 273 K = 288 K$
$T_2 = 400 + 273 = 673 K$
$V_1 = 2\ m^3$
$T_2 = 400°C = 673.15 K$
$V_2 = ?$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$V_2 = \frac{V_1}{T_1} \times T_2 = \frac{2 \times (273 + 400)}{288} = \frac{1346}{288}$$

$V_2 = 4.67\ m^3$

176. The molecular volume of a perfect gas at 600 kPa and 27° C will be

(a) 4.17 m^3/kg mol (b) 400 m^3/kg mol
(c) 0.15 m^3/kg mol (d) 41.7 m^3/kg mol

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Ans. (a) : We know,

$PV = n\bar{R}T$
$\bar{R} = 8314$ J/kg mol K
$T = 27 + 273 = 300$ K
$P = 600 \times 10^3\ N/m^2$
$PV = n\bar{R}T$

$$V = \frac{n\bar{R}T}{P}$$

$$\frac{V}{n} = \frac{\bar{R}T}{P} = \frac{8314 \times 300}{600 \times 10^3} = 4.17\ m^3/\text{kg-mol}$$

177. At Pressure P and absolute temperature T, the mass m of an ideal gas fills a closed container of volume V. An additional mass 2m of the same gas is introduced into the container and the volume is then reduced to V/3 and the temperature to T/3. The pressure of the gas will now be

(a) 3P (b) 2P
(c) P/3 (d) 9P

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Ans. (a) : If m_0 is molecular mass of the gas then for initial condition.

$$PV = \frac{m}{m_0} \cdot RT$$

After, 2m mass has been added

$$P' \cdot \frac{V}{3} = \frac{3m}{m_0} \cdot R \cdot \frac{T}{3}$$

So, comparing above two equations.
We get, $P' = 3P$

178. A reversible adiabatic path, on a P-V diagram, for an ideal gas passes through state A where P = 1 MPa and V = 0.0025 m^3. Given that the ratio of specific heats for the gas is 1.4, the slope of the path at A is

(a) 4×10^5 kPa/m^3 (b) -4×10^5 kPa/m^3
(c) -5.6×10^5 kPa/m^3 (d) 5.6×10^5 kPa/m^3

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Ans. (c) : Given, $P = 1\ MPa = 1 \times 10^6\ N/m^2$

$V = 0.0025\ m^3$

$$\gamma = \frac{C_p}{C_v} = 1.4$$

In reversible adiabatic process

$PV^\gamma = Constant, \quad \therefore V^\gamma dP + P_\gamma V^{\gamma-1} dV = 0$

$$\Rightarrow \quad \frac{dP}{dV} = \frac{-\gamma P}{V} = -\frac{1.4 \times 1 \times 10^6}{0.0025}$$

$$\frac{dP}{dV} = -5.6 \times 10^5\ kPa/m^3$$

179. An insulated rigid vessel contains some gas and an electrical resistor. In a certain interval of time the resistor connected to an external electric power source heats the gas. Considering the vessel and its contents as the system,

(a) Heat and work transfers are zero
(b) Heat and work transfers are positive
(c) Heat transfer is zero and work transfer is negative
(d) Heat transfer is positive and work transfer is zero.

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Ans. (c) : An insulated rigid vessel contains some gas and an electrical resistor in a certain interval of time the resistor connected to an external power source heats the gas. So, heat transfer is due to insulation and work done on the system, so, it is negative.

180. This process is one in which there is only work interaction between the system and the surroundings

(a) Diabetic process
(b) Adiabatic process
(c) Isothermal process
(d) Quasi-static process

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Ans. (b) : Adiabatic process is a type of thermodynamic process which occurs without transferring heat or mass between the system and its surroundings.

181. In which of the following processes, the heat is fully converted into work?

(a) Reversible adiabatic process
(b) Reversible isobaric process
(c) Reversible isometric process
(d) Reversible isothermal process

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Reversible means that in principle, the process is done infinitely slowly so that the microscopic reverse from the final state exactly regenerates the initial state. So, in reversible isothermal process the heat is fully converted into work.

182. Isochoric changes in ideal gas are characterized by

(a) $\frac{V}{T} = Const$ (b) $PV = Const$

(c) $\frac{P}{T} = Const$ (d) None of these

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : For isochoric process

Volume (V) = Constant

From ideal gas equation

$PV = nRT$

$$\frac{PV}{T} = Constant$$

as $V = Constant$

So $\frac{P}{T} = Constant$

183. Match List 1 with List 2 and choose the correct answer from the code-

List-I (laws of thermodynamics)	List-II (Defines)
A. First	(i) Absolute zero temperature
B. Second	(ii) Internal Energy
C. Zeroth	(iii) Temperature
D. Third	(iv) Entropy

	(A)	(B)	(C)	(D)
(a)	i	ii	iii	iv
(b)	iii	iv	ii	i
(c)	iv	ii	i	iii
(d)	ii	iv	iii	i

RPSC IOF & BOILER 20.09.2017

Ans. (d)

(A) First→(ii) Internal Energy
(B) Second→(iv) Entropy
(C) Zeroth→ (iii) Temperature
(D) Third → (i) Absolute zero Temperature

184. The heat capacity of the substance is given by expression:

(a) Heat capacity = (Mass) + (specific heat)
(b) Heat capacity = (Mass) – (specific heat)
(c) Heat capacity = (Mass) × (specific heat)
(d) Heat capacity = (Mass) / (specific heat)
(e) Heat capacity = $(Mass)^2 + (specific\ heat)^2$

(CGPSC Polytechnic Lecturer 2017)

Ans. (c) : Heat Capacity–Heat capacity or thermal capacity is a physical property of matter, defined as the amount of heat to be supplied to a given mass of a material to produce a unit change in its temperature. Heat capacity is an extensive property. It is denoted by C

$C = m \times c\ (J/k)$

Where

m → mass of the system
c → Specific heat

185. 1°C is equal to

(a) 273.15 K (b) 274.15 K
(c) 283.15 K (d) 263.15 K

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : 1°C is equal to Kelvin, 274.15 K.

186. Which of the following are intensive properties?

(1) Density **(2) Pressure**
(3) Energy **(4) Enthalpy**
(5) Temperature

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Options:

(a) 1, 2 and 4 (b) 1, 2 and 5
(c) 1, 3, 4 and 5 (d) 2, 3 and 4
(e) 1, 2, 3 and 4

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Intensive Properties :- It is the properties which is independent of mass is known as intensive properties.

Ex. Pressure, Temperature, Density, Thermal Conductivity and all specific energy etc.

Extensive properties :-It is the property which depend upon the mass of system is known as extensive properties.

Ex. Volume, Mass, All form of Energy [K.E., P.E, Enthalpy, Entropy and Internal Energy etc.

Note :-

1. The ratio of two extensive properties are known as intensive properties

$$\text{Density } (\rho) = \frac{\text{Mass(Extensive)}}{\text{Volume(Extensive)}}$$

↓

Intensive

2. All specific forms except specific density is always on intensive properties

Ex. Specific Enthalpy, Specific Entropy, Specific Internal Energy etc.

• Heat capacity is a extensive properties.

187. The measurement of elevated temperature is generally above 500°C. The instrument used for this purpose is known as:

(a) Monometer (b) Pyrometer
(c) Potentiometer (d) Tachometer
(e) Rotameter

CGPSC ARTO & TSI 07.01.2017

Ans. (b) :

- **Pyrometer :-** It is an instrument that measures temperature remotely, i.e by measuring radiation from the object without having to be in contact.
  The main difference between Thermometer and pyrometer is that the Thermometer is device to measure temperature and pyrometer is a remot-sensing thermometer used to measure high temperature.
- **Monometer-** It is an instrument used to measure and indicate pressure.
- **Potentiometer -**It is defined as a 3 terminal variable resistor in which the resistance is mainally varied to control the flow of electric current. A potentiometer acts as an adjustable voltage divider.
- **Tachometer -** It is an instrument designed to measure the rotation speed of a shaft or disk.
- **Rotameter -** It is a flow meter that measures volumetric flow of liquid and gases.

188. The speed of sound in air varies as

(a) $\sqrt{T}$ (b) $\sqrt{p}$
(c) $1/\sqrt{p}$ (d) p

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : $C = \sqrt{\gamma RT}$

$C \propto \sqrt{T}$

189. In a Steady flow process, the ratio of

(a) Heat transfer is constant
(b) Work transfer is constant
(c) Mass flow at inlet and outlet same
(d) All of these

Vizag Steel (MT) 2017

Ans. (d) : In a steady flow Process, the ratio of
■ Heat transfer is constant
■ Work transfer is constant
■ Mass flow at inlet and outlet is same.

190. The reading of temperature on the Celsius scale is 60° C. What is equivalent reading of temperature on the Fahrenheit scale?

(a) 130°F (b) 132°F
(c) 136°F (d) 140°F
(e) 146°F

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : We know that

$$\frac{C-0}{100-0} = \frac{F-32}{212-32}$$

$$\frac{C}{5} = \frac{F-32}{9}$$

C = 60°C then

$$\frac{60}{5} = \frac{F-32}{9} = 12$$

$$F - 32 = 108$$

$$\boxed{F = 140°F}$$

191. The root mean square velocity of the gas molecules is given by

where, k = Boltzmann's constant
T = absolute Temperature
m = mass of one molecule of a gas

(a) $\sqrt{\frac{kT}{m}}$ (b) $\sqrt{\frac{2kT}{m}}$
(c) $\sqrt{\frac{3kT}{m}}$ (d) $\sqrt{\frac{5kT}{m}}$

GMB AAE 25.06.2017

Ans. (c) : The root mean square velocity of the gas molecules is given by–

$$Q = V_{rms} = \sqrt{\frac{3kT}{m}}$$

192. The specific heat of air increases with increase in

(a) temperature
(b) pressure
(c) both pressure and temperature
(d) Variation of its constituents

WBPSC AE, 2017

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Ans. (a) : Air is diatomic gas (or non-monoatomic gas). As the temperature increases both C_p and C_v increases but their ratio γ decreases.

193. Work done is zero for the following process:

(a) Constant volume (b) Free expansion
(c) Throttling (d) All of the above

WBPSC AE, 2017

Ans. (d) :

- At constant volume,

$V = \text{Constant}$

$dV = 0$

$\therefore$ Work done, $W = PdV$

$= P \times 0 = 0$

$\boxed{W = 0}$

- Free expansion,

$\boxed{W = 0}$

$\boxed{Q = 0}$

In the free expansion of an ideal gas even though initial and final temperatures are same. The process is not isothermal because temperature reduces in between.

- Throttling process,

No change in enthalpy from one state to another state ($h_1 = h_2 = h = \text{Constant}$)

No workdone ($W = 0$)

Process is adiabatic ($Q = 0$)

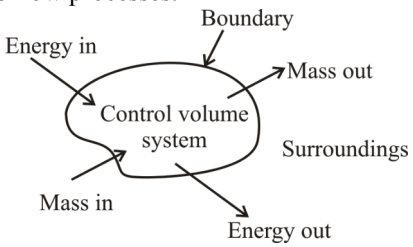
Process called isenthalpic

194. The processes occurring in open system which permit the transfer of mass to and from the system, are known as

(a) flow processes (b) non-flow processes
(c) adiabatic processes (d) none of these

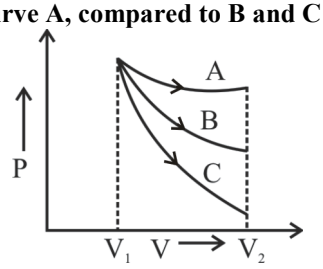
Sikkim PSC (Under Secretary), 2017

Ans. (a) : The process occurring in open system which permit the transfer of mass to and from the system are known as flow processes.



Example–Heat exchanger, pump, compressor, turbine and system with valve.

195. The curve A, compared to B and C represents



(a) Isobaric expansion
(b) Isochoric expansion
(c) Adiabatic expansion
(d) Isothermal expansion

Karnataka PSC Lect., 27.05.2017

Ans. (d) : Isothermal expansion

196. Following are the methods of measuring temperature:

A. By measuring the expansion of liquids like alcohol, pentane in capillary glass tubes
B. By measuring the expansion of mercury in capillary glass tube, radiation in pyrometers and production of emf in thermocouples
C. By measuring velocity of sound in gases and motion of electrons

Which of the statements given above is/are correct?

Select the code for the correct answer for the options given below:

(a) B alone (b) A and B only
(c) C alone (d) A, B and C

Karnataka PSC Lect., 27.05.2017

Ans. (d) : The methods of measuring temperature

(i) By measuring the expansion of liquids like alcohol, pentane in capillary glass tubes.
(ii) By measuring the expansion of mercury in capillary glass tube, radiation in pyrometers and production of 'emf' in thermocouples.
(iii) By measuring velocity of sound in gases and motion of electrons.

197. A diathermic wall is one which:

(a) permits thermal interactions
(b) prevents thermal interactions
(c) encourages thermal interactions
(d) does not exist

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (a) : A diathermic or dibatic wall is one that heat can pass through.

It is made from a thermal conductor it is essentially a thermal wall in other word diathermic wall is permit thermal interaction.

198. Statement (I) : The specific heat at constant pressure for an ideal gas is always greater than the specific heat at constant volume.

Statement (II) : Heat added at constant volume is not utilized for doing any external work.

ESE 2017

Ans. (a) : When gas is heated at constant pressure (C_P) it requires more heat energy as there is change is in internal energy as well as external work done.

But in case of constant volume (C_V) there is no external work done, so that the given heat has to increase only internal energy. Therefore C_P is always greater than C_V.

$$C_P = \left(\frac{dh}{dT}\right)_{P=\text{Const.}}$$

$$C_V = \left(\frac{du}{dT}\right)_{V=\text{Const.}}$$

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We know that enthalpy of fluid is more in value as compared to internal energy of the fluid.

$h > u$

So, $C_P > C_V$

199. All gases behave ideally under:

(a) high temperature conditions
(b) low pressure conditions
(c) high pressure conditions
(d) high pressure and high temperature conditions

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : Generally a gas behaves more like a ideal gas at higher temperature and lower pressure as the potential energy due to intermolecular forces becomes less significant compared with the particles kinetic energy and size of the molecule becomes less significant compared to the empty space between them.

200. During throttling process:

(a) there is no change internal energy
(b) no work is done
(c) heat exchange does not takes place
(d) all of the above

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : A throttling process is defined as the process in which there is no change in enthalpy from state one to state two ($h_1 = h_2$) no work is done, $W = 0$
Process is adiabatic, $Q = 0$

201. Pick up the correct relationship from the following

(a) $TdS = dH - Vdp$ (b) $TdS = dH + Vdp$
(c) $TdS = dU - Vdp$ (d) $TdS = dH - pdV$

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (a) : $TdS = du + PdV$
$TdS = dh - VdP$
above equations are applicable for reversible and irreversible process and for open and closed system both.
Where, T = Temperature of system
dS = Entropy of the system
du = Internal energy
dh = Enthalpy
V = Volume
P = Pressure

202. The specific heat of an ideal gas depend on its _____alone.

(a) Pressure (b) Volume
(c) Entropy (d) Temperature

APPSC AEE MAINS 2016 PAPER-III

Ans. (d) : Specific Heat– The specific heat is the amount of heat per unit mass required to rise the temperature by one degree Celsius.
$\Delta Q = mc\Delta T$

$$C = \frac{\Delta Q}{m\Delta T}$$

Where,
m = mass (kg)
C = Specific heat J/kg°C
ΔQ = Change in thermal energy (J)
ΔT = Change in temperature (°C)

203. A tyre with an inside volume of $1m^3$ contains air at 77^0C with a pressure of 5 bar. Air is an ideal gas under these conditions. The mass(kg) of air in the tyre is :

(a) 0.172 (b) 2.4
(c) 4.98 (d) 0.06

RPSC ADE 2016

Ans. (c) : $V_1 = 1\ m^3$
$T_1 = 273 + 77 = 350$ K
$P_1 = 5$ bar $= 5\times 10^5$ Pa

For air $R = \frac{8.314}{29} = 0.287$

For ideal gas equation –

$PV = mRT$

$5 \times 10^5 \times 1 = m \times 0.287 \times 350$

$$m = \frac{5\times 10^5}{0.287\times 350} = 4977.6 \text{ gram}$$

$m = 4.977 = 4.98$ kg

204. General gas equation is

(a) C_p-C_v= R (b) PV=nRT
(c) PV=zRT (d) PV=mRT

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (d) : General gas equation –
$PV = mRT$
where
P = pressure of gas
V = volume of gas
R = gas constant
T = temperature of gas
m = mass of 1 mole of gas

205. The amount of steam (at 100°C) required to raise the temperature of 200 g of water from 60°C to 100°C is

(a) 10 g (b) 16.8 g
(c) 20 g (d) 14.8 g

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Latent heat of steam = 2257 kJ/kg
Specific heat of water = 4.187 kJ/kg

$m \times 2257 = 0.2 \times 4.187 \times 40$

$m = 14.8$ g

206. Air is heated from 0°C to 100°C in a sealed metal container. Its density

(a) Increases slightly
(b) Decreases slightly
(c) Remains the same
(d) Change cannot be predicted

ISRO Scientist/Engineer 03.07.2016

Ans. (b) : Heated $0°C \xrightarrow{\Delta T} 100°C$
'Temperature Increased'
Ideal gas $PV = mRT$

$$P = \frac{m}{V}RT \qquad \left[\because P = \frac{m}{v}\right]$$

$P = \rho RT$
ρ = Density

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So, $\frac{P}{\rho} = RT$

$$\boxed{T \propto \frac{1}{\rho}}$$

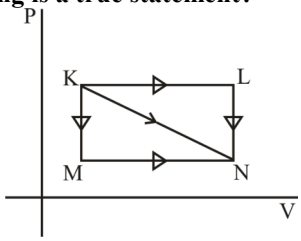
If temperature increase (↑) then
Density decreased (↓)

207. A perfect gas obeys
(a) Boyle's law only
(b) Charles' law only
(c) both Boyle's law and Charles' law
(d) None of the above

APPSC AEE Screening Test 2016

Ans. (c) : A perfect gas obeys both Boyle's law and Charles' law.

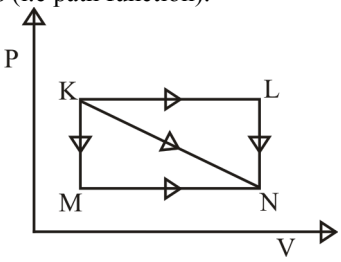
208. An ideal gas can be taken from point K to point N in three different paths: K→L→N, K→N, K→M→N. Which of the following is a true statement?



(a) The same work is done during each process.
(b) The same amount of heat is added to the gas during each process
(c) The same change in internal energy during each process
(d) The same entropy generated during each process

APPSC AEE MAINS 2016 PAPER-III

Ans. (c) : From this diagram we get that the same change in internal energy during each process as it is the property of the system (i.e point function) whereas work and heat for each process will be different because of area under P–V and T–S diagram will be different for each process (i.e path function).



209. Which of the following is correct statement for 'Energy (E)'?
(a) 'Energy (E)' is not the property of system
(b) 'Energy (E)' is path function
(c) Change in energy between two states of a system is different for different path followed
(d) $\oint dE = 0$

UPRVUNL AE 07.10.2016

Ans. (d) : • Energy E is the property of system and it is the point function.
• Change in energy between two states of a system is same for different path followed.
• For a cycle
$\oint dE = 0$

210. Which of the following equations is incorrect? (where V,P,T and Q are volume, pressure, temperature and heat transfer respectively)
(a) $\oint dV = 0$ (b) $\oint dP = 0$
(c) $\oint dT = 0$ (d) $\oint dQ = 0$

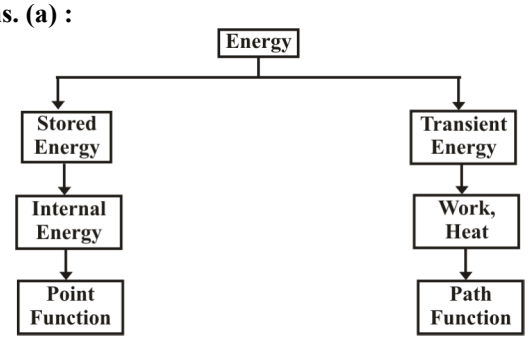
RPSC LECTURER 16.01.2016

Ans. (d) :
$\oint Q \neq 0$
$\oint V = 0$
$\oint P = 0$
$\oint T = 0$
Because V,P,T are properties of the system whereas Q and work (W) not a properties of the system.
Cyclic integration of the system properties will be equal to zero.

211. Which of the following statements is correct for "Energy"
(a) It is a point function
(b) It is a path function
(c) It is not a conserved quantity
(d) It can be measured by thermometer

RPSC LECTURER 16.01.2016

Ans. (a) :



212. Consider the following statements, which is the true statement?
(a) Dashpot is an open system
(b) Throttling is a reversible process
(c) Turbine is an isolated system
(d) Isothermal expansion is an irreversible process

GPSC ARTO 01.05.2016

Ans. (d) : Isothermal expansion means Temperature as the process are constant is an irreversible process.

213. Which is not a thermometric property
(a) Pressure (b) Wavelength
(c) Resistance (d) Flow rate

Nagaland PSC (CTSE) 2016, Paper-II

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Ans. (d) : Flow rate is not a thermometric property

Ex:-

Some Instrument	Thermometric property
(1)Constant Volume gas thermometer	Pressure
(2)Electrical resistance thermometer	Resistance
(3)Thermo Couple	Voltage (emf)
(4)Pyrometer	Wavelength

214. Which one is correct regarding the concept of continuum

(a) Mean free path of gas in a container is equal to or greater than the dimension of container

(b) Mean free path of gas in a container is less than or greater than the dimension of container

(c) Mean free path has no significance in continuum approach

(d) Mean free path is essential for calculating pressure in continuum approach

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : Mean free path of gas in a container is must be very small, compare to system dimension, Such that

$$(K_n) < 0.01$$

K_n – Knudson Number

215. The values of heat transfer and work transfer for the four processes of a thermodynamic cycle are given below :

Process	Heat transfer (kJ)	Work transfer (kJ)
1.	400	400
2.	zero	350
3.	–200	–200
4.	zero	–350

The thermal efficiency and work ratio for the cycle will be respectively.

(a) 50% and 0.27 (b) 33% and 0.27

(c) 33% and 0.37 (d) 50% and 0.37

UPRVUNL AE 21.08.2016

Ans. : (a) $\eta_{th} = \dfrac{\text{Net work done}}{\text{Heat supplied}}$

$$= \frac{400 + 350 - 200 - 350}{400}$$

$$\eta_{th} = \frac{200}{400} = 50\%$$

$$\text{Work ratio} = \frac{\text{Net work}}{\text{Positive work}} = \frac{W_{net}}{W_T} = \frac{200}{750}$$

Work ratio = 0.27

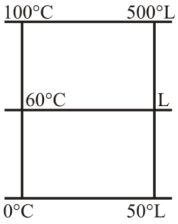
216. A new temperature scale in degree L is to be defined. The boiling and freezing points of water on this scale are 500°L and 50°L respectively. What will be the reading on new scale corresponding to 60^0C?

(a) 220^0L (b) 320^0L

(c) 180^0L (d) 280^0L

UPRVUNL AE 21.08.2016

Ans. : (b)



We know that,

$$\frac{60-0}{100-0} = \frac{L - 50°L}{500°L - 50°L}$$

$$\frac{60}{100} = \frac{L - 50°L}{450°L}$$

$$\frac{3}{5} = \frac{L - 50°L}{450°L}$$

$$3 = \frac{L - 50°L}{90°L}$$

$$270°L = L - 50°L$$

$$L = 270°L + 50°L$$

$$\boxed{L = 320°L}$$

217. Which of the following statement is correct?

(a) A sphere a cube and a thin circular plate of the same mass are made of the same material. If all of them are heated to the same high temperature the rate of cooling is maximum for the plate and minimum for the sphere.

(b) One mole of a monoatomic ideal gas is mixed with one mole of diatomic ideal gas. The molar specific heat of the mixture at constant volume is 2R. where R is the molar gas constant.

(c) Boilers are occasionally scrubbed by rapidly and artificially circulating water inside them to remove any thin water film which may have formed on their inside.

(d) The average kinetic energy of 1 kg of all ideal gases, at the same temperature is the same.

UPRVUNL AE 21.08.2016

Ans. : (b) One mole of a monoatomic ideal gas is mixed with one mole of diatomic ideal gas. The molar specific heat of the mixture at constant volume is 2R. where R is the molar gas constant.

218. Which one of the following is not a property of the system

(a) Pressure (b) Temperature

(c) Work transfer (d) Entropy

GPSC Lect. 23.10.2016

Ans. (c) : Properties of substances are things which depend only on point such as mass, temperature, volume and pressure.

219. The increase in pressure

(a) lowers the boiling point of a liquid

(b) raises the boiling point of a liquid

(c) does not affect the boiling point of a liquid

(d) none of these

GPSC Lect. 23.10.2016

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Ans. (b) : Increase in pressure raises the boiling point of a liquid.

220. A thermometer works on the principle of

(a) Law of stable equilibrium
(b) Zeroth law of thermodynamics
(c) First law of thermodynamics
(d) Second law of thermodynamics

ISRO Scientist/Engineer 03.07.2016

Ans. (b) : A thermometer works on the principle of Zeroth law of thermodynamic.

221. Properties of substances like pressure, temperature and density in thermodynamics coordinates are

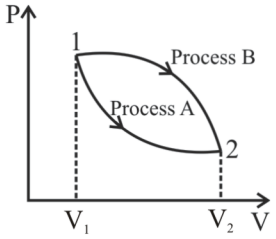
(a) Cyclic function (b) Point function
(c) Real Function (d) Path function

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (b) : **Point Functions**–They depends on state only and not on process (how a system reaches that state) and they are exact differentials.

Example–Pressure, temperature, enthalpy, volume, entropy etc.

i.e. $\int_1^2 dV = V_2 - V_1 = \Delta V$



Path Functions–The depends on how a system reaches that state (i.e. process). So path function boundary phenomenon and are inexact differentials. They are not properties of system.

Example–Heat work and entropy generation etc.

$$\int_1^2 \delta W = W_{1-2} \neq \Delta W = W_2 - W_1$$

or $$\int_1^2 \delta Q = Q_{1-2} \neq \Delta Q = Q_2 - Q_1$$

222. Which one of the following is the extensive property of Thermodynamic System?

(a) Heat (b) Density
(c) Pressure (d) Temperature

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : Extensive property – Depends of the mass of system.

Ex- Mass, volume, internal energy, heat capacity, enthalpy, entropy, Helmholtz energy, Gibbs energy

Intensive property – Independent of the mass of system Ex- pressure, temperature, density, concentration, melting point, boiling point, surface tension, viscosity.

223. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Throttling process	1. No work done
B. Isentropic process	2. No change in entropy
C. Free expansion	3. Constant internal energy
D. Isothermal process	4. Constant enthalpy

Code :

	A	B	C	D
(a)	4	2	1	3
(b)	1	2	4	3
(c)	4	3	1	2
(d)	1	3	4	2

(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (a)

Throttling process	Constant enthalpy ($h_1 = h_2 = h$)
Isentropic process	No change in entropy (S = Constant)
Free expansion	No work done [(W = 0, Q = 0)]
Isothermal process	Constant internal energy, It is U = f(T) for ideal gas

224. As differentials, heat and work would be described mathematically as

(a) exact (b) inexact
(c) point function (d) discontinuity

Assam PSC CCE Pre 2015

Ans. (b) : As differentials, heat and work would be described mathematically as inexact. Because they depends on path followed by processes.

225. The amount of heat required to raise the temperature of the unit mass of gas through one degree at constant volume is called-

(a) Specific heat at constant volume
(b) Specific heat at constant pressure
(c) Kilo Joule
(d) None of these

NPCIL ET, 2015

Ans. (a) : The amount of heat required to raise the temperature of the unit mass of gas through one degree at constant volume is called specific heat at constant volume.

$$\boxed{Q = mc_v\Delta T}$$

226. In free expansion of a ideal gas between two equilibrium states, the work transfer involved

(a) can be calculated by joining the two states on p-v coordinates by any path and estimating the area below

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(b) can be calculated by joining the two states by a quasi-static path and then finding the area below
(c) is zero
(d) is equal to heat generated by friction during expansion

Nagaland PSC CTSE 2015 Paper-II

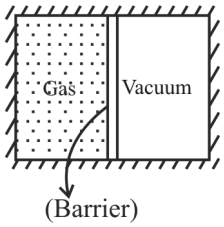
Ans. (c) : In free expansion of a gas between two equilibrium states, the work transfer involved is zero and heat transfer also zero.
Free expansion is an irreversible process so entropy is not constant for this process.

$(W)_{\text{free expansion}} = 0$ but $\left(\int pdV\right)_{\text{free expansion}} \neq 0$

227. The temperature in a process in which work is done by expanding a gas under adiabatic condition will
(a) Increase
(b) Decrease
(c) Remain unchanged
(d) Decrease/increase depending on properties of gas

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : For expansion of gas under adiabatic condition.



Let, process 1 - 2 is adiabatic expansion of ideal gas, $Q_{1\text{-}2} = 0$
We know, $Q_{1\text{-}2} = U_{1\text{-}2} + W_{1\text{-}2}$

$$-U_{1\text{-}2} = W_{1\text{-}2} \Rightarrow -mC_p dT = \int_1^2 pdV$$

$$dT = -\frac{p}{mC_p}\int_1^2 pdV \Rightarrow dT < 0$$

So, temperature will decrease during the adiabatic expansion.

228. Which of the following is not the intensive property?
(a) pressure (b) heat
(c) density (d) temperature

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (b) Intensive property which not depends on the mass.
Exp.- Pressure, Temperature, Density, viscosity, specific energy etc.
Extensive property-Which depends on the mass.
Exp.- Volume, Energy, entropy, enthalpy, heat etc.

229. The difference between two specific heats, $C_p - C_v = \frac{R}{J}$. This relation is valid for
(a) any gas (b) perfect gases
(c) real gases (d) pure gases

ISRO Scientist/Engineer 11.10.2015

Ans. (b) : $C_p - C_v = R \rightarrow$ "Ideal gas or Perfect gas"
All perfect gases are ideal gas and for perfect gas C_p & C_v not changes with temperature.

230. Work output from a system is at the expense of internal energy is a non flow process carried out
(a) at constant pressure (b) at constant volume
(c) adiabatically (d) polytropically

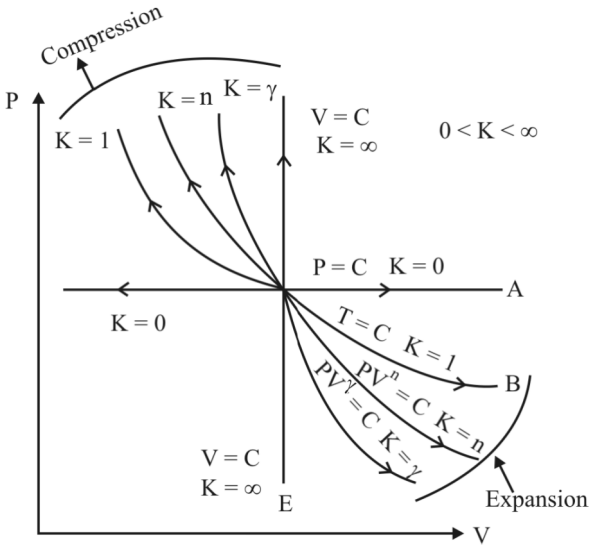
TSPSC AEE 2015

Ans. (c) : Work output from a system is at the expense of internal energy is a non flow process carried out adiabatically.

231. The general law of expansion or compression is $PV^n = C$. the process is said to be hyperbolic if n is equal to
(a) zero (b) 1
(c) γ (d) ∞

Assam PSC CCE Pre 2015

Ans. (b) : A general equation $PV^n = C$ is such that the following table gives different process for different value of K.
Let $PV^K = C$



n	Process
0	Isobaric
1	Isothermal
1<n<γ	Polytropic
γ	Adiabatic
∞	Isochoric

For P–V diagram

$$\left(\frac{dp}{dv}\right)_{\text{adiabatic}} = \gamma\left(\frac{dp}{dv}\right)_{\text{isothermal}}$$

∴ Slope of adiabatic curve in P–V curve > slope of isothermal in P–V curve.

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232. The gas at pressure P in the vessel is compressed suddenly to half of its original volume. Then pressure of the gas after a long time becomes :

(a) 1/2 P (b) 2P

(c) Less than 2P (d) Greater than 2P

TSPSC Managers, 2015

Ans. (b) : According gas equation.

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \quad \ldots\ldots (i)$$

Let $V_1 = V$, $P_1 = P$

after compressed $V_2 = \frac{V}{2}$, $P_2 = ?$

$\therefore$ From equation(i) $PV = \frac{V}{2} \times P_2$

$$\boxed{P_2 = 2P}$$

233. The efficiency of a Carnot engine using an ideal gas as the working substance is

(a) $\frac{(T_1 - T_2)}{T_1}$ (b) $\frac{T_1}{(T_1 - T_2)}$

(c) $\frac{(T_1T_2)}{(T_1 - T_2)}$ (d) $\frac{(T_1 - T_2)}{(T_1T_2)}$

VIZAG STEEL MT 2015

Ans. (a) : $\eta = \frac{T_1 - T_2}{T_1}$

234. 100% efficiency of a thermal cycle cannot be achieved because of

(a) it is not possible to achieve 0K temperature

(b) leakage

(c) non-availability of ideal substance

(d) frictional losses

GWSSB AAE, 27.12.2015

Ans. (a) : 100% efficiency of a thermal cycle cannot be achieved because it is not possible to achieve 0K temperature.

235. Thermocouples are used for measuring which of the following parameter?

(a) Humidity (b) Pressure

(c) Temperature (d) Density

UPRVUNL AE 2014

Ans : (c) Thermocouples are work like temperature sensor.

- From this, we can measure a temperature range from 200ºC to 1600ºC.
- It work on the principal of Seebeck effect.

236. Which among the following gives the correct relationship between Centigrade and Fahrenheit scales?

(Where C : degree Centigrade and F : degree Fahrenheit)

(a) C = 5/9 (F – 32) (b) F = (9/5)C–32

(c) C = (9/5) (F – 32) (d) C = 9/5 (F – 32)

UPRVUNL AE 2014

Ans. : (a) $\frac{C}{100} = \frac{F - 32}{180}$

$$C = \frac{100}{180}(F - 32) = \frac{5}{9}(F - 32)$$

$$\frac{C}{5} = \frac{F - 32}{9}$$

Where F = Fahrenheit

ºC= Centigrade

237. Mean square molecular speed is :

(a) directly proportional to density

(b) inversely proportional to density

(c) directly proportional to the square root of density

(d) inversely proportional to the square root of density

UPRVUNL AE 2014

Ans. : (d) Mean square molecular speed (V)–

$$V \propto \frac{1}{\sqrt{d}}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{d_2}{d_1}}$$

Where d = Density.

So, It is inversely proportional to the square root of density.

238. Bi-Metallic strips made of two different materials bend during a rise is temperature because of

(a) Differences in coefficient of linear expansion

(b) Differences in elastic properties

(c) Differences in thermal conductivities

(d) Difference in stress

TNPSC AE 2014

Ans. (a) : Bi-Metallic strips made of two different materials bend during a rise is temperature because of differences in coefficient of linear expansion.

239. The instrument which measures the temperature of the source without direct contact is

(a) Bi-metallic cut-out

(b) Vapour pressure thermometer

(c) Pyrometer

(d) Thin film thermometer

TNPSC AE 2014

Ans. (c) : Pyrometer–Pyrometer is a device use for measuring relatively high temperature, such as are encountered in furnace. Most pyrometer works by measuring radiation from the body whose temperature is to be measured. Radiation devices have the advantage of not touching the material being measured.

240. In a steady flow reversible adiabatic process work done is equal to

(a) Change in internal energy

(b) Change in entropy

(c) Change in enthalpy

(d) Heat transferred

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Ans. (c) : In a steady flow reversible adiabatic process work done is equal to change in enthalpy.

241. Which parameter remains constant during a reversible adiabatic process?

(a) Enthalpy (b) Temperature
(c) Entropy (d) Internal Energy

Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : Entropy remains constant for a reversible adiabatic process.

Change in entropy is given by- $\Delta S = \int_1^2 \frac{dQ}{T} + S_{generation}$ for adiabatic process heat transfer is zero (Q = 0), So, $\int_1^2 \frac{dQ}{T} = 0$

and for reversible process entropy generation is considered to be zero. So the $S_{gen} = 0$.

Therefore the overall change is entropy for the reversible adiabatic process becomes zero.

242. A system is capable of exchanging energy with its surroundings in the form of n-reversible work modes. The number of independent variables that completely specify the state of the system is

(a) n-2 (b) n-1
(c) n (d) n+1

Haryana PSC Civil Services Pre, 2014

Ans. (d) : Number of independent variables = number of work modes + 1 = n+1

243. When a gas is heated, change takes place in :

(a) pressure (b) volume
(c) temperature (d) All of these

MPSC Poly. Lect. 2014

Ans. (d) : When a gas is heated, changes takes place in pressure volume and temperature.

244. The atomic mass of oxygen is :

(a) 12 (b) 14
(c) 16 (d) 32

MPSC Poly. Lect. 2014

Ans. (c) : Atomic mass of Oxygen = 16
Atomic mass of Carbon = 12
Atomic mass of Nitrogen = 14
Atomic mass of Sulphur = 32.

245. The principle of working of the constant volume thermometer is based on

(a) Boyle's law (b) Charle's law
(c) Gay – Lussac's law (d) Equation of state

TNPSC AE 2014

Ans. (c) : Constant volume gas thermometer– This thermometer works on the principle of Law of Gay-Lussac. The law states that when the temperature of an ideal gas increases, there is a corresponding increase in pressure. Also, when the temperature decrease, the pressure too decrease correspondingly.

246. Which of the following is an example of irreversible process?

(a) Polytropic expansion of fluid
(b) Unrestricted expansion of gases
(c) Isothermal expansion
(d) Electrolysis

JPSC AE 2013 Paper-V

Ans. (b) : Irreversible process- In an irreversible process, there is a loss of heat due to friction, radiation and conduction.

In an actual practice, most of the processes are irreversible to some degree The main causes for the irreversibility may be.

(i) Mechanical and fluid friction
(ii) Unrestricted expansion
(iii) Heat transfer with a finite temperature difference.

247. Which of the following is considered as thermodynamic potential?

(a) Temperature (b) Internal energy
(c) Enthalpy (d) Entropy

JPSC AE 2013 Paper-V

Ans. (c) : Enthalpy is considered as thermodynamic potential.

248. 2 kg of substance receives 500 kJ and undergoes a temperature change from 100°C to 200°C. Then average specific heat of substance during the process will be:

(a) 5 kJ/kg °K (b) 2.5 kJ/kg °K
(c) 10 kJ/kg °K (d) 25 kJ/kg °K

SJVN ET 2013

Ans. (b) : m = 2kg
Q = 500 kJ
$T_1 = 100^oC$
$T_2 = 200^oC$
$Q = mc\Delta T$
$500 = 2 \times c\ (200 - 100)$
$\boxed{C = 2.5\ kJ/kg^oK}$

249. For non-flow closed system the value of net energy transferred as heat and work equals change in:-

(a) Enthalpy (b) Entropy
(c) Internal energy (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) : $\Delta Q - \Delta W = \Delta U$

250. The specific heats of gases increase with the rise in:

(a) Volume
(b) Temperature
(c) Volume and pressure
(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : The specific heat of gases increases with the rise in temperature.

251. An ideal gas is heated from temperature T_1 and T_2 by keeping its volume constant. The gas is expanded back to its initial temperature according to the law $PV^n = C$. If the entropy change in the two processes are equal, then the value of n in terms of the adiabatic index γ is

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(a) $n=\frac{\gamma+1}{2}$ (b) $n=\frac{\gamma-1}{2}$

(c) $n=\frac{\gamma+2}{4}$ (d) $n=\frac{\gamma+4}{2}$

JPSC AE 2013 Paper-V

Ans. (a) :

$$c_v \ell n\left(\frac{T_2}{T_1}\right)=c_v \ell n\left(\frac{T_1}{T_2}\right)+R\ell n\frac{V_1}{V_2}$$

$$\frac{V_1}{V_2}=\left(\frac{T_1}{T_2}\right)^{\frac{1}{n-1}}$$

$$c_v \ell n\left(\frac{T_2}{T_1}\right)=c_v \ell n\left(\frac{T_1}{T_2}\right)+\frac{R}{(n-1)}\ell n\frac{T_2}{T_1}$$

$$c_v \ell n\left(\frac{T_2}{T_1}\right)=-c_v \ell n\left(\frac{T_2}{T_1}\right)+\frac{R}{(n-1)}\ell n\left(\frac{T_2}{T_1}\right)$$

$$2c_v=\frac{R}{n-1}$$

$$c_p-c_v=R$$

$$2c_v=\frac{c_p-c_v}{n-1}$$

$$2(n-1)=\frac{c_p}{c_v}-1$$

$$2n-2=\gamma-1$$

$$2n=\gamma+1$$

$$\boxed{n=\frac{\gamma+1}{2}}$$

252. A vessel having a volume of 0.6 m^3 contains 3.0 kg of liquid water and water vapour mixture in equilibrium at a pressure of 0.5 MPa. The specific volume is

(a) 0.8 m^3/kg (b) 0.2 m^3/kg

(c) 0.6 m^3/kg (d) 0.2 kg/m^3

APPSC Poly. Lect. 2013

Ans. (b) : Given, V = 0.6 m^3, Pressure, P = 0.5 MPa = 5 bar, m = 3kg

Thus,

$$\text{Specific volume, } v=\frac{V}{m}=\frac{0.6}{3}=0.2 \text{ m}^3/\text{kg}$$

253. During a process involving one kg of working fluid in an insulated cylinder-piston arrangement the temperature of the fluid decreases from 350°C to 250°C. The specific heat of the fluid at constant volume is 0.8 kJ/kg°C. Considering the contents of the cylinder as the system, what will be the work done and its nature?

(a) 80 kJ (on the system)

(b) Zero

(c) 80 kJ (by the system)

(d) 125 kJ (by the system)

RPSC AEN Pre-2013

Ans. (c) : Given, m = 1kg

C_v = 0.8 kJ/kg°C

T_1 = 350°C , T_2 = 250°C

work done = Heat (Q) = $mC_v\Delta T$

= 1 × 0.8 × (350 – 250)

= 1 × 0.8 × 100

= 80 kJ (By the system)

254. The unit of work is

(a) kW (b) kWh

(c) kW/h (d) kJ/s

UKPSC AE 2012 Paper–II

Ans. (b) : kWh

255. The work in a closed system undergoing an isentropic process is given by

(a) $\frac{\gamma}{\gamma-1}mR(T_1-T_2)$ (b) $\frac{\gamma-1}{\gamma}mR(T_1-T_2)$

(c) $\frac{1}{\gamma-1}mR(T_1-T_2)$ (d) $\frac{\gamma}{\gamma-1}m(T_1-T_2)$

(Notations used have usual meaning)

UKPSC AE 2012 Paper–II

Ans. (c) : $\frac{1}{\gamma-1}mR(T_1-T_2)$

256. If 2 bar of air at 27°C is heated to 177°C at constant volume, the pressure will be

(a) 6.56 bar (b) 3 bar

(c) 9 bar (d) 13.11 bar

ISRO Scientist/Engineer 2012

Ans. (b) : P_1 = 2 bar

T_1 = 27°C

T_2 = 177°C

Gas equation, $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$

At constant volume $V_1 = V_2$

So, $\frac{P_1}{T_1}=\frac{P_2}{T_2}$

T_1 = 273 + 27 = 300 K

T_2 = 177 + 273 = 450 K

$\frac{2}{300}=\frac{P_2}{450}$

P_2 = 3 bar

257. The equation of state for a perfect gas is

(a) $\frac{P}{V}=RT$ (b) $\frac{P}{T}=\frac{V}{R}$

(c) PV = RT (d) $\frac{P}{\rho}=T$

APPSC AEE 2012

Ans. : (c) The equation of state for a perfect gas is

PV = mRT.

R = gas constant

Unit of R = $\frac{J}{kg-K}$

Value of R = 0.287 kJ/kg – K (for air)

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258. Extensive property of a system is one whose value

(a) depends on the mass of the system, like volume

(b) does not depend on the mass of the system like temperature

(c) is not dependent on the path followed but on the state

(d) is dependent on the path followed and not on the state

TRB Asstt. Prof., 2012

Ans. (a) : Extensive properties– It is the property which depends upon the mass of the system, is known as extensive properties.

Example– Volume, mass, all form of energy (K.E, P.E., Enthalpy, Entropy and Internal Energy).

Intensive Properties– It is the property which is independent of mass is known as intensive properties.

Example- Pressure, Temperature, Density, thermal conductivity, etc and all specific energy.

Note-

(1) The ratio of two extensive properties are known as intensive properties.

$$\text{Density}(\rho) = \frac{\text{Mass}}{\text{Volume}}$$

(Mass: Extensive; Volume: Extensive; Density: Intensive)

(2) All specific forms except specific density is always on intensive properties. Ex. Specific enthalpy, specific entropy etc.

(3) Heat capacity is a extensive properties.

259. Water is boiled at 1×10^5 Pa pressure in a coffee maker equipped with an immersion type electric heating element. The coffee maker initially contains 1 kg of water. Once boiling started, it is observed that half of the water in the coffee maker evaporated in 18 minutes. If the heat loss from the coffee maker is negligible, the power rating of the heating element is

(a) 0.90 kW (b) 1.52 kW

(c) 1.05 kW (d) 1.24 kW

ISRO Scientist/Engineer 2012

Ans. (c) : Pressure $P = 1 \times 10^5$ Pa

$m = 1$ kg

$t = 18$ minutes

Heat supplied for evaporating water $= Q = mH_v$

H_v = enthalpy of vaporization

$= 2257$ kJ/kg

$$\therefore \quad Q = \frac{1}{2} \times 2257$$

$H_v = 1128.5$ kJ

$$\text{Power (P)} = \frac{1128.5}{18 \times 60} = \frac{H_v}{t(\text{sec})}$$

$\simeq 1.05$ kW

260. Which pair of the following alternatives is correctly matched ?

List – I		List – II
(a) Heat	–	Point function
(b) Energy	–	Path function
(c) Entropy	–	Second law of thermodynamics
(d) Gibbs function	–	Path function

UKPSC AE 2012 Paper–II

Ans. (c) : Entropy – Second law of thermodynamics

261. For an adiabatic process,

(a) PV = a constant (b) PV^n = a constant

(c) $\frac{P}{\rho^\gamma}$ = a constant (d) $\frac{P}{K^\rho}$ = a constant

APPSC 2012

Ans. : (c) For an adiabatic process

$$\frac{P}{\rho^\gamma} = \text{constant}$$

γ = Ratio of the specific heat at constant pressure to the specific heat at Constant volume.

$C_p / C_v = 1.4$ for air.

262. When air expands from initial pressure P_1 and volume V_1 to final volume $5V_1$ following the law $PV^n = C$

(a) greater the value of n, greater the work obtained

(b) smaller the value of n, smaller the work obtained

(c) for n = 0, the work obtained is the greatest

(d) for n = 1.4, the work obtained is the greatest

ISRO Scientist/Engineer 2012

Ans. (c) : Given- Air expand

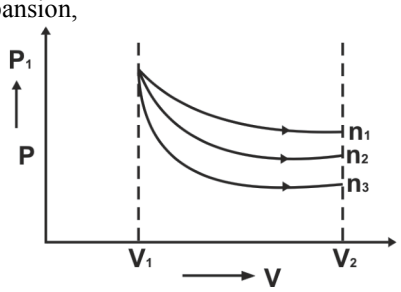
Initial pressure = P_1

Initial volume = V_1

Final volume = $5V_1$

$PV^n = C$

For Expansion,



Here, $n_1 < n_2 < n_3$

$\therefore$ Larger the value of n smaller is work obtained

$$W = \frac{P_1V_1 - P_2V_2}{n-1} \longrightarrow \text{Isotropic process}$$

$\therefore$ For n = 0 [P = Constant]

The work obtained is greatest.

263. Properties of substances like pressure temperature and density in thermodynamic co-ordinates are

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(a) path functions
(b) point functions
(c) cyclic functions
(d) thermodynamic functions

TNPSC ACF 2012

Ans. (b) : Point function
(a) Variable which depends on ends state only.
(b) All properties are point function.
(c) They should satisfy exact differential concleture.
Ex- Pressure, Temp, density.

264. An electric heater is left on for 30 minutes. During this time it uses 1200 W of electric energy. The total cost of the electricity will be if electricity costs 7 paisa per kWh
(a) 252 paisa (b) 25200 paisa
(c) 4200 paisa (d) 4.2 paisa

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : Given-

Time (t) = 30 min $= \frac{1}{2}$ hr Power (P) = 1200 W

Price of 1 kWh = 1 unit = 7 paisa

We know that,

$$\text{Total cost} = \frac{P \times t \times \text{unit cost}}{1000}$$

$$= \frac{1200 \times 0.5 \times 7}{1000} = 4.2 \text{ paisa}$$

265. Match the following :

1.	**Closed system**	**a.**	**Increase in static pressure**
2.	**Open system**	**b.**	**Increase in kinetic energy**
3.	**Pump**	**c.**	**Heat, mass and work interact**
4.	**Turbine**	**d.**	**Heat and work interact**
5.	**Nozzle**	**e.**	**Delivers work**

(a) 1–d, 2–c, 3–a, 4–e, 5–b
(b) 1–c, 2–d, 3–b, 4–e, 5–a
(c) 1–c, 2–a, 3–d, 4–e, 5–b
(d) 1–d, 2–c, 3–e, 4–a, 5–b

OPSC Civil Services Pre. 2011

Ans. (a) :

1. Closed system	Heat and work interact
2. Open system	Heat, mass and work interact
3. Pump	Increase in static pressure
4. Turbine	Delivers work
5. Nozzle	Increase in kinetic energy

266. Pressure of 1 (one) atmosphere is equivalent to:
(a) 0.101325 MPa (b) 750 mm of mercury
(c) 1 bar (d) 0.101325 kPa

OPSC Civil Services Pre. 2011

Ans. (a) :
1 atmosphere pressure = 101325 Pa
= 1.01325 bar
= 760 mm of mercury
= 760 Torr
= 0.101325 MPa
= 14.6959 Pound-force per square inch

267. What is the process in which the temperature of the working substance remains constant during its expansion or compression called as?
(a) Isothermal process (b) Hyperbolic process
(c) Adiabatic process (d) Polytrophic process

VIZAG STEEL MT 2011

Ans. (a) : In thermodynamics, an isothermal process is a type of thermodynamic process in which the temperature of system remains constant : $\Delta T = 0$
- $\Delta T = 0$, or T = constant
- $dU = 0 = mC_V\Delta T$ (for ideal gas only, internal energy)
- Q = 0 for adiabatic process.

268. Which law states that change of internal energy of a perfect gas is directly proportional to the change of temperature.
(a) Boyle's law (b) Charles law
(c) Gay-lussac law (d) Joule's law

VIZAG STEEL MT 2011

Ans. (d) : Boyle's law- The volume of a given mass of a gas varies as its absolute pressure, provided the moles and temperature remains constant.
Charle's law- The volume of a given gas varies directly as its absolute temperature provided the moles and pressure is kept constant.
Gay-lussac law- The pressure of a given mass of a gas varies directly as its absolute temperature provided the moles and volume is kept constant.
Joule's law- This law states that "The internal Energy of a given quantity of a gas depends on the temperature.

269. What happen when a gas is heated at constant pressure.
(a) Its temperature will increase
(b) Its volume will increase
(c) Both temperature and volume will increase
(d) Neither temperature nor volume will increase

VIZAG STEEL MT 2011

Ans. (c) : When a gas is heated at constant pressure then both temperature and volume will increase.

270. Which of the following gas has a minimum molecular mass?
(a) Oxygen (b) Nitrogen
(c) Hydrogen (d) Methane

VIZAG STEEL MT 2011

Ans. (c) : Atomic mass of hydrogen = 1.007
Atomic mass of Oxygen = 15.99
Atomic mass of Nitrogen = 14.006
Atomic mass of chlorine = 35.5
So molecular mass of hydrogen (H_2)= 2 × 1.007
= 2.014 u
molecular mass of Oxygen (O_2)= 2 × 15.99 = 31.99
molecular mass of Nitrogen (N_2) = 2 × 14.006 = 28.01u
molecular mass of Chlorine (Cl_2) = 2 × 35.5 = 71 u

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271. Which efficiency a machine can be called as an idea machine?

(a) Between 60 and 70%
(b) Between 70 and 80%
(c) Between 80 and 90%
(d) 100%

VIZAG STEEL MT 2011

Ans. (d) : An ideal machine, the work output is equal to work input, i.e. the efficiency of an ideal machine is 1 (or 100%).

272. What is the heat energy stored in the gas used for raising the temperature of the gas known as?

(a) External energy (b) Internal energy
(c) Kinetic energy (d) Molecular energy

VIZAG STEEL MT 2011

Ans. (b) : Internal energy is the heat energy stored in the gas used for raising the temperature of the gas.

273. ITPS defines which of the following as a reference for calibration of temperature scale?

(a) Ice Point
(b) Lamda Point
(c) Normal Boiling Point of Water
(d) Triple Point of Water

OPSC Civil Services Pre. 2011

Ans. (d) : Triple point of water is a reference for calibration of temperature scale.

274. The specific heat of an ideal gas C_p and C_v

(a) Vary with temperature
(b) Vary with pressure
(c) Vary with both pressure and temperature
(d) Are constant

UPSC JWM Advt. No.-52/2010

Ans. (d) : C_p, C_v and γ are constant for ideal gas.
C_p, C_v and γ are varies only for non -monoatomic real gases

275. Consider the following statements about thermodynamic properties of system :

1. Heat transferred at constant volume increase the internal energy of the system.
2. Heat transferred at constant pressure increases the enthalpy of a system.
3. The enthalpy of an ideal gas depends on the temperature only.

Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-52/2010

Ans. (a) : All options are correct.

276. Consider the following properties :

1. Specific heat 2. Thermal diffusivity
3. Dynamic viscosity 4. Density

Which of these properties of air increase with increase of temperature ?

(a) 1, 2, 3 and 4 (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3 only

UPSC JWM Advt. No.-50/2010

Ans. (d) : The properties of air increase with increase of temperature such as,

$$PV = mRT \Rightarrow P = \frac{m}{v}RT \Rightarrow P = \rho RT$$

$$\frac{P}{\rho} = RT \Rightarrow \boxed{T \propto \frac{1}{\rho}}$$

'T' is inversely proportional the density 'ρ'.

- T is directly proportional to the specific heat, thermal diffusivity & dynamic viscosity.

277. When 2 gm of a gas A is introduced into an evacuated flask kept at 25°C, the pressure is found to be 1 atmosphere. If 3 gm of another gas B is then added to the same flask, the total pressure becomes 1.5 atmospheres. Assuming ideal gas behaviour, the ratio of molecular weights, $M_A : M_B$ will be

(a) 1 : 3 (b) 3 : 1
(c) 2 : 3 (d) 3 : 2

UPSC JWM Advt. No.-50/2010

Ans. (a) : Given,

$W_A = 2$ gm, $W_B = 3$ gm
$P_A = 1$ atm
$P_{total} = 1.5$ atm

Ideal gas equation,

$PV = nRT$

$$\Rightarrow \quad n = \frac{W(\text{Weight of tubes})}{M(\text{Molecular weight})}$$

So, $$PV = \frac{W}{M}RT$$

$$M = \frac{WRT}{PV}$$

So, $$M_A = \frac{W_1RT}{P_1V} = \frac{2RT}{1\times V} \qquad \left[\because P_t = 1.5\,\text{atm and } \because P_t = P_A + P_B\right]$$

$$M_B = \frac{W_2RT}{P_2V} = \frac{3RT}{0.5V}$$

$$\frac{M_A}{M_B} = \frac{2RT}{V}\times\frac{0.5V}{3RT} = 1:3$$

278. Air with initial condition of P_1, V_1 expands to final condition of $\frac{P_1}{2}$, $3V_1$. The process is

(a) Hyperbolic
(b) Adiabatic
(c) Polytropic with $n > 1$
(d) Polytropic with $n < 1$

ISRO Scientist/Engineer 2010

Ans. (d) : Air,

Initial pressure = P_1

Final pressure = $\frac{P}{2}$

Expansion to this process,
Initial volume = V_1
Final volume = $3V_1$

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$$\frac{P_1}{P_2}=\left(\frac{V_2}{V_1}\right)^n$$

$$\frac{P_1}{P/2}=\left(\frac{3V_1}{V_1}\right)^n$$

$2 = 3^n$

$n = 0.63$

279. Consider the following statements about heat and work:

1. **Both are path functions**
2. **Both are exact differentials.**
3. **Work is low grade energy and heat is high grade energy.**
4. **Heat is low grade energy and work is high grade energy.**

Which of the following options are true?

(a) 1 and 2 (b) 1 and 3
(c) 1 and 4 (d) 2 and 3

DRDO Scientists 2009

Ans. (c) : Heat and work are-
1. Both are path function (inexact differential)
2. Heat is low grade energy and work is high grade energy.

280. The control volume of an open system may involve :

(a) Fixed and moving boundaries
(b) Real and imaginary boundaries
(c) Both (a) and (b)
(d) None of the above

APPSC IOF, 2009

Ans. (c) : Control Volume–The surface may be open or closed to mass flows and it may flows from energy in terms of heat transfer and work across it. The boundaries may be movable or stationary.

281. An adiabatic process is one in which :

(a) no heat enters or leaves the gas
(b) the change in internal energy is equal to work done
(c) the temperature of the gas changes
(d) all of the above

APPSC IOF, 2009

Ans. (d) : Adiabatic Process–
- No heat enters or leaves the gas
- The change in internal energy is equal to work done

$\delta Q = dU + \delta W$

$0 = dU + \delta W$

$\boxed{\delta W = -dU}$

- The temperature of the gas changes

282. The coefficient of linear expansion of a solid is L. The volume V of a cube of this solid on being heated by 1°C will change by

(a) VL (b) 3VL
(c) 3L (d) $\frac{VL}{3}$

ISRO Scientist/Engineer 2009

Ans. (b) Coefficient of linear expansion of a solid is = L.
Volume = V
$\Delta T = 1°C$.
We know that,
$V = a^3$
Take log both side
$\log V = \log a^3$
$= 3 \log a$
Partially differentiate both side

$$\frac{dV}{V}=3\frac{da}{a}$$

$$dV = 3\frac{da}{a}.V$$

$= 3 (L \times 1) \times V$
$= 3 VL$

283. The relation PV^γ = constant, where γ is the ratio of the specific heats of ideal gas, is applicable to

(a) Any adiabatic process
(b) Only reversible adiabatic process
(c) Only irreversible adiabatic process
(d) Only isothermal process

ISRO Scientist/Engineer 2009

Ans. (b) : PV^γ = constant
Its Adiabtaic process
γ = only reversible adiabatic process

284. The ratio of specific heats of a gas is 1.4 and the value of specific heat at constant pressure is 7.0 *cal/mole°C*. The difference between specific heat at constant pressure and specific heat at constant volume will be (in *cal/mole°C*)

(a) 1.4 (b) 1.7
(c) 5.0 (d) 2.0

ISRO Scientist/Engineer 2009

Ans. (d) : Ratio of specific heat of gas $\gamma=\frac{C_p}{C_v}=1.4$
Specific heat at constant pressure C_p = 7.0 cal/mole°C

$C_p - C_v = R$

or $\gamma=\frac{C_p}{C_v}$

$$1.4=\frac{7}{C_v}$$

$$C_v=\frac{70}{14}$$

$C_v = 5$

So, $C_p - C_v = 7 - 5 = 2$ cal/mole°C

285. By which one of the following the Kelvin temperature of a system can be measured?

(a) Mercury-in-glass thermometer
(b) Thermocouple
(c) Constant-volume gas thermometer
(d) Resistance thermometer

UPSC JWM Adv. No-16/2009

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Ans. (c) : The Kelvin temperature of a system can be measured by Constant Volume gas thermometer.

286. In the macroscopic approach, in the study of thermodynamics

i. the structure of matter is taken into account

ii. the structure of matter is not considered

iii. only a limited number of properties are used to describe the state of matter

iv. the values of the properties needed for the description of the system cannot be measured.

Of the statements :

(a) I alone is correct (b) I and II are correct

(c) II and III are correct (d) All are correct

TNPSC AE, 2008

Ans. (a) : In the macroscopic approach, the study of the thermodynamics the structure of matter is taken into account.

In this approach the individual molecular behaviour is neglected and average molecular behaviour is studied.

This approach is applicable for the continuum concept.

287. The generally used device for temperature measurement inside the furnace is

(a) gas thermometer

(b) optical pyrometer

(c) alcohol thermometer

(d) mercury thermometer

TNPSC AE, 2008

Ans. (b) : Optical pyrometer is a device generally used for temperature measurement inside the furnace.

288. Work done in reversible adiabatic process is given by

(a) $\frac{P_2V_2 - P_1V_1}{1-n}$ (b) $\frac{P_2V_2 - P_1V_1}{1-\gamma}$

(c) $\frac{P_2V_2 - P_1V_1}{\gamma-1}$ (d) $\frac{\gamma-1}{J}(P_2V_2 - P_1V_1)$

TNPSC AE, 2008

Ans. (b) : Workdone (W) in reversible adiabatic process is given by

$$W = \frac{P_1V_1 - P_2V_2}{\gamma-1} \quad \text{or} \quad W = \frac{P_2V_2 - P_1V_1}{1-\gamma}$$

289. The density of air at 10°C and 1MPa abs, in SI units is

(a) 1.231 (b) 12.31

(c) 118.4 (d) 65.0

ISRO Scientist/Engineer 2008

Ans. (b) : Density of air ρ = ?

T = 10°C

= 273 + 10

T = 283 K

P = 1MPa

PV = mRT

$$P = \frac{m}{V}RT$$

$$P = \rho RT$$

$$\rho = \frac{P}{RT}$$

$$= \frac{1\times10^6}{287\times283}$$

$$= 12.31 \text{ kg/m}^3$$

290. A well insulated vessel containing air is heated to double the pressure and temperature 150% in excess of initial. The amount of air that leaked is

(a) 20% (b) 22.5%

(c) 25% (d) 0%

ISRO Scientist/Engineer 2008

Ans. (c) : For ideal gas

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \text{ (air is an Ideal gas)}$$

$$\frac{P_1V_1}{T_1} = \frac{2P_1V_2}{1.5T_1}$$

$$V_1 = \frac{4}{3}V_2$$

The amount of air leaked

$$\frac{V_1 - V_2}{V_1}\%$$

$$\frac{V_1 - \frac{3}{4}V_1}{V_1}\%$$

= 25%

291. An unconstraint thin circular disc has a concentric hole in it. On heating the diameter of the hole will

(a) increase

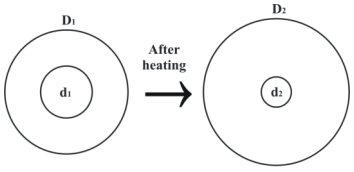
(b) decrease

(c) Remain constant

(d) first increase then decrease

ISRO Scientist/Engineer 2008

Ans. (b) : After heating the metal tends to elongate, it will also elongate towards the center of the concentric hole to fill some part of hole and make its diameter small.



($d_1 > d_2, D_2 > D_1$)

292. Kinetic energy of the molecules in terms of absolute temperature (T) is proportional to

(a) T (b) T^{-4}

(c) T^2 (d) $\sqrt{T}$

ISRO Scientist/Engineer 2008

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Ans. (a) : Kinetic energy of molecules given by

$$E = \frac{3}{2}kT$$

where k = Boltzman constant

T = Absolute temperature

So, $E \propto T$

293. The quantity of heat required to change the temperature of 1 gm of ice cream -6°C to -5°C is known as

(a) Latent heat of freezing
(b) Freezing heat
(c) Heat of vaporization
(d) Specific heat

ISRO Scientist/Engineer 2008

Ans. (d) :

294. A vessel of volume 1 m^3 contains oxygen (molecular weight = 32) at p = 1 bar and T = 47°C. The mass of oxygen in the vessel is (take universal gas constant as 8314 J/kmol-K)

(a) 40.0 kg (b) 3.0 kg
(c) 1.2 kg (d) 1.0 kg

DRDO Scientists 2008

Ans. (c) : Ideal gas equation

$$PV = mRT$$

$$R = \frac{R'}{M} \qquad (R' = \text{Universal gas constant})$$

$$1\times10^5\times1 = m\times\frac{8314}{32}(273+47)$$

$$m = 1.199 \text{ kg}$$

295. Temperature of a gas is due to

(a) its heating value
(b) kinetic energy of molecules
(c) repulsion of molecules
(d) attraction of molecules

TNPSC AE, 2008

Ans. (b) : The temperature of a gas is due to kinetic energy of molecules. As temperature increases, the molecules of the gas have the more kinetic energy.

296. The specific heat of diatomic gas at constant pressure is given by (R = universal gas constant and M = molecular weight of gas)

(a) $\frac{5}{2}\frac{R}{M}$ (b) $\frac{5}{2}MR$

(c) $\frac{7}{2}\frac{R}{M}$ (d) $\frac{7}{2}MR$

ISRO Scientist/Engineer 2008

Ans. (c) : We know that,

$$c_P = \frac{\gamma R_g}{(\gamma-1)}$$

For di-atomic

$$\gamma = \frac{7}{5} = 1.4$$

$$R_g = \frac{R}{M}$$

$$c_P = \frac{\gamma R_g}{(\gamma-1)} = \frac{1.4\times R_g}{(1.4-1)}$$

$$= \frac{1.4\times R_g}{0.4} = \frac{1.4\times R}{0.4M} \qquad (\because R_g = R/M)$$

$$c_P = \frac{7}{2}\frac{R}{M}$$

R = Universal Gas Constant

R_g = Characteristic Gas Constant

M = Molecular Weight

For, mono-atomic $\gamma = 5/3 = 1.66$, the minimum value of $\gamma = 1$ and maximum is 1.67.

The general formula for γ of any gas is given as

$$\gamma = 1 + \frac{2}{f}$$

Where,

f = Degree of freedom for mono-atomic f = 3 and for di-atomic f = 5.

297. The universal gas constant of a gas is the product of molecular weight of the gas and

(a) Gas constant
(b) Specific heat at constant pressure
(c) Specific heat at constant volume
(d) None of the above

ISRO Scientist/Engineer 2007

Ans. (a) : $PV = mRT$

We know that,

No. of mole (n) = $\frac{m}{M}$

so, m = nM

and $PV = n\bar{R}T$

$PV = nMRT$

where R = gas characteristic constant n = No. of mole, m = molecular mass, M = molecular weight, $\bar{R}$ = universal gas constant, T = Temperature

$$PV = nMRT$$

$$MR = \bar{R}$$

298. Bulk modulus K of a perfect gas in terms of pressure P in an isothermal process is related by the equation.

(a) K = 3P (b) K = P
(c) K = $\frac{P}{2}$ (d) K = 2P

WBPSC AE, 2007

Ans. (b) : K = P for isothermal process

K = γP for adiabatic process

299. A process in which all thermodynamic properties are same at the start and at the end of the process is known as

(a) reversible process (b) quasi-static process
(c) cyclic process (d) irreversible process

OPSC Civil Services Pre 2006

Ans. (c) : If properties at the start and at the end of process is same, the process is called cyclic process.

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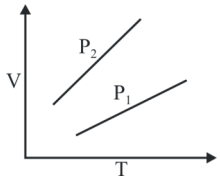
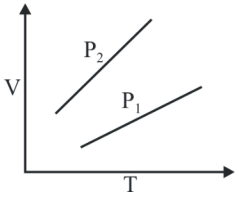
300. The volume versus temperature T graph for a certain amount of a perfect gas at two pressure P_1 and P_2 are as shown in the figure. It can be concluded that-



(a) The pressure P_1 is greater than the pressure P_2
(b) The adiabatic index for P_1 is higher than that for P_2
(c) P_1 represents monoatomic gas and P_2 represents diatomic gas
(d) None of the above

ISRO Scientist/Engineer 2006

Ans. (a) : Slope $= \left(\frac{V}{T}\right)$



Slope $(\text{Slope})_2 > (\text{Slope})_1$

$$\left(\frac{V}{T}\right)_2 > \left(\frac{V}{T}\right)_1$$

$$\frac{R}{P_2} > \frac{R}{P_1} \qquad (\because \text{Ideal gas equation})$$

$PV = mRT$

$\therefore \boxed{P_1 > P_2}$ $\qquad \frac{V}{T} = \frac{mR}{P} = \left(\frac{V}{T}\right)_1 = \frac{mR}{P_1}$

301. In case of ideal triatomic gas, the ratio of specific heat C_P/C_V would be-

(a) 1 (b) 1.33
(c) 1.40 (d) 1.41

ISRO Scientist/Engineer 2006

Ans. (b) : Specific heat

$$\gamma = \frac{C_P}{C_V}$$

$$\gamma = 1 + \frac{2}{f}$$

Where f is degree of freedom

f = 6 for triatomic gas

$$\gamma = 1 + \frac{2}{6} = 1.33$$

302. For polytropic process the specific heat is given by the relation :

$$C_n = C_p \frac{n-k}{(n-1)}$$

For $1 < n < k$, C_n will be :

(a) Zero (b) Positive
(c) Infinite (d) Negative

J & K PSC Screening, 2006

Ans. (d) : For polytropic process.

$$C_n = C_v\left(\frac{n-k}{n-1}\right) \qquad 1 < n < k$$

$$C_n = C_v\left(\frac{n-k}{n-1}\right) \qquad 1 < n < k$$

put $n = 1.2$ & $k = 1.4$

$$C_n = C_v\left[\frac{1.2-1.4}{1.2-1}\right] = -C_v$$

C_v is always positive Hence C_n will be negative.

2. First Law of Thermodynamics

303. A 100 W electric bulb was switched on in a 2.5m × 3m× 3m size thermally insulated from having temperature of 20°C. The room temperature at the end of 24 hrs will be

(a) 100°C (b) 470°C
(c) 370°C (d) 600°C

Consider density and specific heat of air as 1.2 kg/cm^3 and 0.718 kW/°C/kg

HPPSC ADF 05.03.2019
GPSC Executive Engineer 23.12.2018
ISRO Scientist/Engineer (RAC) 22.04.2018
AAI Jr. Executive 29.11.2018
VIZAG STEEL MT 10.06.2012
GATE 2006

Ans. (b) : Heat supply to thermally insulated room

$Q = 100 \times 24 \times 60 \times 60 = 8640$ kJ

Mass of air in room (m) $= \rho_{air} \times$ volume of room

$= 1.2 \times 2.5 \times 3 \times 3$

$m = 27$ kg

Heat supply to insulated room = rise internal energy of air inside the room

$8640 = 27 \times 0.718\,(T_2 - 20°)$

$T_2 = 465°C \simeq 470\,°C$

304. A steel ball of mass 1 kg and specific heat 0.4 kJ/kg °C is at a temperature of 60 °C. It is dropped into 1 kg water at 20 °C. The final steady state temperature of water is :

(a) 23.5 °C (b) 30 °C
(c) 35 °C (d) 40 °C

RPSC Vice Principal ITI 2018
GPSC ARTO Pre 30.12.2018
Vadodara Muncipal Corp. DEE, 2018
ISRO Scientist/Engineer (RAC) 07.05.2017
ISRO Scientist/Engineer 11.10.2015
GATE 1999

Ans. (a) $m_b = 1$ kg
$c_b = 0.4$ kJ/kgk
$T_{b_1} = 60°C$
$m_w = 1$ kg
$T_{w_1} = 20°C$

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Let the final steady state temperature of ball and water be T°C.

$$m_b c_b (T_{b_1} - T) = m_w c_w (T - T_{w_1})$$

$$1 \times 0.4\,(60° - T) = 1 \times 4.18\,(T - 20°)$$

$$\boxed{T = 23.5°C}$$

305. A mixture of gasses expands from 0.03 m^3 to 0.06 m^3 at a constant pressure of 1 MPa and absorbs 84 kJ of heat during the process. The change in internal energy of the mixture is

(a) 30 kJ (b) 54 kJ
(c) 84 kJ (d) 114 kJ

PPSC Asstt. Municipal Engineer 15.06.2021
ISRO Scientist/Engineer (RAC) 22.04.2018
TSPSC AEE 2017
Assam PSC CCE Pre 2015, TNPSC AE 2013
UKPSC AE 2012 Paper–II, CSE Pre-1994

Ans. (b) : Given,

$V_1 = 0.03m^3$ $V_2 = 0.06m^3$

$\delta W = PdV$

$P = 1000$ kPa

$W = 1000(0.06 - 0.03)$

$= 1000 \times 0.03$

$= 30$ kJ

$Q = 84$ kJ

So, $\delta Q = dU + \delta W$ (I^s Law of thermodynamics)

$84 = dU + 30$

$dU = 54$ kJ

306. The first law of thermodynamics is the law of

(a) Conservation of heat
(b) Conservation of momentum
(c) Conservation of energy
(d) Conservation of mass

HRRL (HPCL) Engineer 07.08.2021
Assam PSC AE (IWD) 14.03.2021
TSPSC Manager (Engg.) HMWSSB 12.11.2020
Rajasthan AE (Nagar Nigam) 2016 Shift-3
GPSC Lect. 23.10.2016
VIZAG Steel MT 2011
OPSC Civil Services Pre 2006

Ans. (c) : First law of thermodynamics is the law of conservation of energy which states that energy can neither can be created, nor destroyed however it can be converted from one form to another.

For a thermodynamic process:

$$\boxed{\partial Q = dE + \partial W}$$

For a thermodynamic cycle

$$\oint \delta Q = \oint \delta W$$

307. A 2 kW, 40 liters water heater is switched on for 20 minutes. The heat capacity C_P for water is 4.2 kJ/kgK. Assuming all the electrical energy has gone into heating the water, increase of water temperature in degree centigrade is

(a) 2.7 (b) 4.0
(c) 14.3 (d) 8.3

TSPSC AEE 2017, GPSC AE 26.07.2015
VIZAG STEEL MT 10.06.2012
PSPCL AE, 2012, GATE 2003

Ans. (c) : Energy given by heater = Energy taken by water

$P.t = mC_p\Delta T$

$2\times10^3\times20\times60 = 10^3\times 40\times10^{-3}\times 4.2\times\Delta T$

$\Delta T = 14.3°C$

308. The cyclic integral of (δQ – δW) for a process is equal to

(a) Positive
(b) Negative
(c) Zero
(d) Positive or negative depending upon the type of process

Nagaland PSC CTSE 2015 Paper-II
RRB SSE 03.09.2015 Shift-II
UKPSC AE-2013, Paper-II
ESE 2007

Ans. (c) : According to first law of thermodynamics

$\delta Q = dU + \delta W$

$\delta Q - \delta W = dU$

$$\therefore \quad \oint (\delta Q - \delta W) = \oint (dU)$$

Cyclic integration of a property of system is always zero. i.e. in a cyclic process no change in properties of system,

$$\oint (\delta Q - \delta W) = 0$$

$$\therefore \quad \boxed{\oint (\delta Q) = \oint (\delta W)}$$

or $(\Sigma Q)_{cyclic} = (\Sigma W)_{cyclic}$

309. For a closed system, difference between the heat added to the system and work done by the gas, is equal to the change in-

(a) Internal energy (b) Entropy
(c) Enthalpy (d) Temperature

ISRO Scientist/Engineer 2006
WBPSC AE 2003, ESE 2003

Ans. (a) : For a closed system, difference between the heat added to the system and work done by the gas, is equal to the change in Internal Energy.

Internal energy $\Delta E = Q - W$

310. 170 kJ of heat is supplied to a system at constant volume. Then the system rejects 180 kJ of heat at constant pressure and 40 kJ of work is done on it. The system is finally brought to its original state by an adiabatic process. If the initial value of internal energy is 100 kJ, then which one of the following statements is correct?

(a) the highest value of internal energy occurs at the end of the constant volume process
(b) the highest value of internal energy occurs at the end of the constant pressure process

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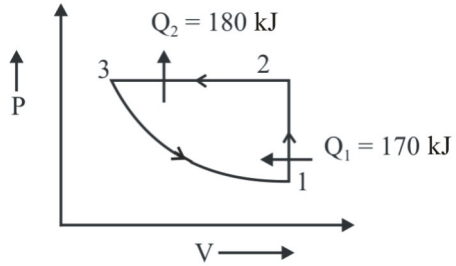
(c) the highest value of internal energy occurs after adiabatic expansion
(d) internal energy is equal at all the points.

Nagaland PSC (CTSE) 2016, Paper-II
Haryana PSC Civil Services Pre, 2014
ESE 2004

Ans. (a)



$1 \rightarrow 2$ Constant volume
$2 \rightarrow 3$ Constant pressure
$3 \rightarrow 1$ Adiabatic

$\boxed{U_1 = 100\,kJ}$

$Q_2 = 180$ kJ
$Q_1 = 170$ kJ

In process $1 \rightarrow 2$
$\delta Q = dU + \delta W$
$170 = dU + 0$
$170 = U_2 - U_1$
$U_2 = 170 + U_1$
$\boxed{U_2 = 170 + 100 = 270\,kJ}$

In process (2–3)
$\delta Q = dU + \delta W$
$-180 = U_3 - U_2 - 40$
$\boxed{U_3 = 130\,kJ}$

Means highest value of internal energy occurs at end of constant volume process which is $\boxed{U_2 = 270}$

$\boxed{U_2 > U_3 > U_1}$

311. A 10 kg solid at 100°C with a specific heat of 0.8 kJ/kg°C s immersed in 40 kg of 20°C liquid with a specific heat of 4.0 kJ/kg°C. The temperature after a long time if the container is insulated will be

(a) 30°C (b) 28°C
(c) 26°C (d) 24°C

ISRO Scientist/Engineer (RAC) 12.01.2020
GPSC ARTO 01.05.2016
ISRO Scientist/Engineer 11.10.2015

Ans. (d) : Equivalent Temperature $T_{equ.}$

$$= \frac{m_1 C_{p_1} T_1 + m_2 C_{p_2} T_2}{m_1 C_{p_1} + m_2 C_{p_2}}$$

$$T_e = \frac{10 \times 0.8 \times (100 + 273) + 40 \times 4 \times (20 + 273)}{10 \times 0.8 + 400 \times 4}$$

$$= \frac{8 \times 373 + 160 \times 293}{8 + 160}$$

$T_e \cong 296.81$ K

$T_e = 23.81^\circ C$

$\boxed{T_e \simeq 24^0 C}$

312. If δQ is the heat transferred to the system and δW is the work done by the system, then which of the following is an exact differential?

(a) δQ (b) δW
(c) δQ+ δW (d) δQ – δW

Nagaland PSC CTSE 2017, 2016 Paper-II
Haryana PSC Civil Services Pre, 2014

Ans. (d) : According to first law of thermodynamics.

$\boxed{dU = \delta Q - \delta W}$

dU = change in Internal Energy which is exact differential.
δQ = Inexact differential
δW = Inexact differential

313. A 1000 kg vehicle travelling at 80 m/s impacts a plunger attached to a piston-cylinder arrangement. If all of the energy of the vehicle is absorbed by the 20 kg of liquid contained in the cylinder, what is the maximum temperature rise of the liquid? (The specific heat of the liquid is 4.0 kJ/kg.°C)

(a) 55°C (b) 40°C
(c) 45°C (d) 50°C

Vododara muncipal Corp. DEE 2018
ISRO Scientist/Engineer 2009

Ans. (b) : Mass of vehicle (m) = 1000 kg
speed v = 80 m/s
Mass of liquid contained
m = 20 kg
c = 4.0 kJ/kg°C
ΔT = ?

$$\frac{1}{2}mv^2 = mc\Delta T$$

$$\frac{1}{2} \times 1000 \times 80^2 = 20 \times 4 \times (\Delta T)$$

$\Delta T = 40°C$

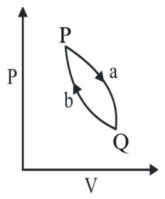
314. A system is taken from state "P" to state "Q" along two different paths "a" and "b". The heat absorbed and work done by the system along these paths are "Q_a", "Q_b" and "W_a", "W_b" respectively. Which among the following will be true?

(a) $W_a + Q_a = W_b + Q_b$
(b) $Q_a = Q_b$
(c) $W_a = W_b$
(d) $Q_a - W_a = Q_b - W_b$

UPRVUNL AE 2014
UKPSC AE 2012 Paper–II

Search on Telegram @apna_pdf

Ans. : (d) $Q_a - W_a = Q_b - W_b$



315. A system undergoes a change of rate during which 100 kJ of heat is transferred to it and does 50 kJ of work. The system is brought back to its original state through a process during which 120 kJ of heat is transferred to it. The work done by the system is

(a) 50 kJ (b) 79 kJ
(c) 170 kJ (d) none of these

ISRO Scientist/Engineer (RAC) 12.01.2020
TRB Poly. Lect., 2012

Ans. (c) $Q_1 = 100$ kJ, $W_1 = 50$kJ
$Q_2 = 120$kJ, $W_2 = ?$
As per Ist law of thermodynamics
Process 1-2 & Process (2-1)

$$\sum dQ = \sum dW$$

$$Q_{1-2} + Q_{2-1} = W_{1-2} + W_{2-1}$$
$$100 + 120 = 50 + W_{2-1}$$
$$220 = 50 + W_{2-1}$$
$$W_{2-1} = 220 - 50$$
$$\boxed{W_{2-1} = 170\text{kJ}}$$

316. A gas expands in a frictionless piston cylinder arrangement. The expansion process is very slow, and is resisted by an ambient pressure of 100 kPa. During the expansion process, the pressure of the system (gas) remains constant at 300 kPa. The change in volume of the gas is 0.01 m^3. The maximum amount of work that could be utilized from the above process is

(a) 0 kJ (b) 1 kJ
(c) 2 kJ (d) 3 kJ

TSPSC AEE 2017, GATE 2008

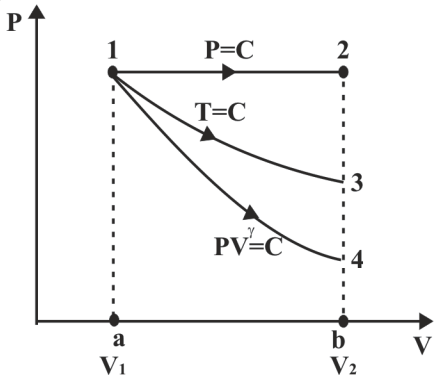
Ans. (c) : W = Resistance pressure × ΔV
W = P × ΔV
W = 100 × 0.01
W = 1 kJ This is required for atmosphere, means against atmosphere we have to do this but question is that how much we can use.
Now total work done = Total resistance × ΔV
= 300 × 0.01 = 3 kJ
1 kJ for atmosphere, 2 kJ for use.
So, answer is 2 kJ.

317. Starting from the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways, the work done by the gas is W_1 if the process is purely isothermal, W_2 if purely isobaric and W_3 is purely adiabatic, then

(a) $W_2 > W_1 > W_3$ (b) $W_2 > W_3 > W_1$
(c) $W_1 > W_2 > W_3$ (d) $W_1 > W_3 > W_2$

GPSC ARTO Pre 30.12.2018
ISRO Scientist/Engineer 03.07.2016

Ans. (a) :



We know that area under the P-V diagram will be equal to the work done for any process.
From above figure
Area under 1-2-b-a (P = constant) W_2
Area under 1-3-b-a-1 (T = constant) W_1
Area under 1-4-b-a-1 (PV^{γ} = constant) W_3
So, we can say that
$W_2 > W_1 > W_3$

318. According to first law of thermodynamics

(a) Mass and energy are mutually convertible
(b) Heat and work are mutually convertible
(c) Heat flows from hot substance to cold substance
(d) Carnot engine is most efficient

APPSC AEE 2012
ISRO Scientist/Engineer 2011

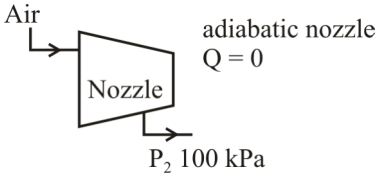
Ans. (b) : According to first law of thermodynamics–
- Total amount of energy the system is constant
- Energy can be transfer from one form to another
- Heat and work are mutually convertible

319. Air enters an adiabatic nozzle at 400 kPa and 900 K, with negligible velocity. If the flow is ideal and exit pressure is 100 kPa, the exit temperature in K and exit velocity in m/s are respectively

(a) 605.7, 768.7 (b) 225, 1164.8
(c) 516.9, 877.5 (d) 129.2,880.1

Nagaland PSC (CTSE) 2016, Paper-II
Haryana PSC Civil Services Pre, 2014

Ans. (a)
$P_1 = 400$ kPa
$T_1 = 900$ K
$V_1 = 0$



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$T_2 = ?$

$V_2 = ?$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{T_2}{900} = \left(\frac{100}{400}\right)^{\frac{1.4-1}{1.4}}$$

$$T_2 = 900 \times \left(\frac{100}{400}\right)^{\frac{0.4}{1.4}}$$

$T_2 = 605.7$ K

From SFEE

$$h_1 + \frac{c_1^2}{2000} + \frac{gz_1}{1000} + q_{cv} = h_2 + \frac{c_2^2}{2000} + \frac{gz_2}{1000} + W_{cv}$$

$q_{cv} = 0$, $W_{cv} = 0$ } adiabatic nozzle

$z_1 = z_2$

$c_1 = 0$

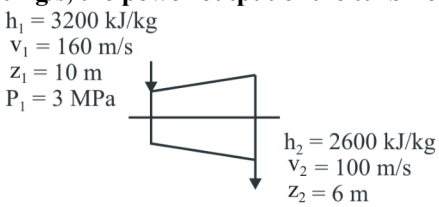
$$h_1 = h_2 + \frac{c_2^2}{2000}$$

$$c_2 = \sqrt{2000(h_1 - h_2)} = \sqrt{2000(C_p)[T_1 - T_2]}$$

$$= \sqrt{2000 \times 1.005 \times (900 - 605.66)}$$

$= 769.177$ m/s

320. The inlet and outlet conditions of steam for an adiabatic steam turbine are as indicated in the figure. The notations are as usually followed. If the mass rate of steam through the turbine is 20 kg/s, the power output of the turbine is



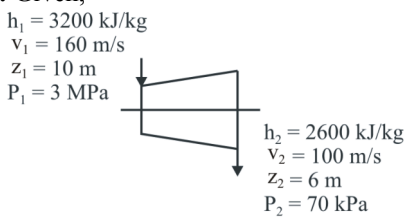
(a) 12.157 MW (b) 12.941 MW

(c) 168.001 MW (d) 168.785 MW

ISRO Scientist/Engineer 11.10.2015

GATE 2009

Ans. (a) : Given,



Applying the steady flow energy equation (SFEE)–

$$\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

Where $Q = 0$ For adiabatic flow

$$\dot{m}\left(h_1 + \frac{v_1^2}{2} + gz_1\right) = \dot{m}\left(h_2 + \frac{v_2^2}{2} + gz_2\right) + W$$

On unit balancing, we get

$$\dot{m}\left(h_1 + \frac{v_1^2}{2000} + \frac{gz_1}{1000}\right) = \dot{m}\left(h_2 + \frac{v_2^2}{2000} + \frac{gz_2}{1000}\right) + W$$

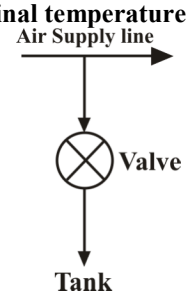
$$20\left(3200 + \frac{(160)^2}{2000} + \frac{9.81 \times 10}{1000}\right) = 20\left(2600 + \frac{(100)^2}{2000} + \frac{9.81 \times 6}{1000}\right) + W$$

$64257.96 = 52101.17 + W$

$W = 12156.79$ kW $= 12.1568 \times 10^6$ W

≈ 12.157 MW

321. A rigid, insulated tank is initially evacuated. The tank is connected with a supply line through which air (assumed to be ideal gas with constant specific heats) passes at 1 MPa, 350°C. A valve connected with the supply line is opened and the tank is charged with air until the final pressure inside the tank reaches 1 MPa. The final temperature inside the tank



(a) is greater than 350°C

(b) is less than 350°C

(c) is equal to 350°C

(d) may be greater than, less than or equal to 350°C, depending on the volume of the tank

Gujarat PSC AE 2019, GATE 2008

Ans. : (a) : The final temperature in the tank when initially evacuated tank is connected to pipe line.

$T_2 = \gamma \times$ temperature in pipe line

$= \gamma \times T_1$

$= 1.4 \times (350 + 273)$

$= 872.2$ K $= 599.2^\circ$C

$\boxed{T_2 = 600^\circ C}$

322. In a steady state flow process taking place in a device with a single inlet and single outlet, the work done per unit mass flow rate is given by $w = -\int_{inlet}^{Outlet} vdp$ where v is the specific volume and p is the pressure, the expression of w given above

(a) Is valid only if the process is both reversible and adiabatic

(b) Is valid only if the process is both reversible and isothermal

(c) Is valid for any reversible process

(d) Is incorrect; it must be $w = \int_{inlet}^{Outlet} pdv$

GPSC Asstt. Prof. 28.08.2016, GATE 2008

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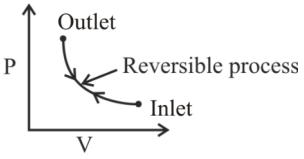
Ans. (c) : A reversible process is defined as a process in which the system and surroundings can be returned to the original conditions from the final state without producing any changes in the thermodynamic properties of the universe, if the process is reversed.

So, $w = -\int_{inlet}^{Outlet} vdp$, Represents a reversible process



323. The first law of thermodynamics takes the form W = –ΔH when applied to:

(a) A closed system undergoing a reversible adiabatic process

(b) An open system undergoing an adiabatic process with negligible change in kinetic and potential energies

(c) A closed system undergoing a reversible constant volume process

(d) A closed system undergoing a reversible constant pressure process

GPSC Asstt. Prof. 28.08.2016, GATE 1993

Ans. (b) : First law of thermodynamics defines, the internal energy (E) is equal to the difference of the heat transfer (Q) in a system and work (W) done by system.

W = –ΔH is an open system, undergoing an adiabatic process, ΔK.E and ΔP.E is zero.

Steady flow equation,

$H_1 + Q = H_2 + W$

In adiabatic process, Q = 0

So, $H_1 - H_2 = W \Rightarrow W = -\Delta H \ \therefore \ (H_2 > H_1)$

324. The internal energy of a certain system is a function of temperature alone and is given by the formula E = 25+0. 25t kJ. If this system executes a process for which the work done done by it per degree temperature increases is 0.75 KN-m, $\frac{dE}{dt} = Q - W$, the heat interaction per degree temperature increase, in kJ. is

(a) –1.00 (b) 1.00

(c) –0.50 (d) 0.50

MPPSC AE 2016, ESE 1995

Ans. : (b) E = 25 + 0.25t KJ

$$\frac{dE}{dt} = 0.25 kJ/°c$$

and

$$\frac{\delta W}{dt} = 0.75 kJ/°c$$

from the first law of thermodynamics

$\delta Q = \delta W + dE$

$$\frac{\delta Q}{dt} = \frac{\delta W}{dt} + \frac{dE}{dt}$$

$\delta Q = 0.25 + 0.75$

$\delta Q = 1.00$ kJ

325. A closed system receives 60 KJ heat but its internal energy decreases by 30 KJ. Then the work done by the system is

(a) 30 KJ (b) 90 KJ

(c) –30 KJ (d) –90 KJ

RPSC VPITI 14.02.2016, ESE 2010

Ans. (b) : Given Internal energy = – 30 KJ

Receive heat = + 60 KJ

Then work done in closed system

dQ = dU + dW

+ 60 = –30 + dW

dW = + 90 KJ

326. An ideal gas is filled in a balloon kept in an evacuated and insulated room When the balloon ruptures, the gas fills up the entire room. Now internal energy of gas..... and the enthalpy of gas..... at the end of this process.

(a) increases, increases

(b) constant, decreases

(c) constant, constant

(d) decreases, increases

(HPPSC LECT. 2016), GATE 2008

Ans. : (c) An ideal gas is filled in a balloon kept in an evacuated and insulated room. when the balloon ruptures, the gas fills up the entire room. Now internal energy of gas constant and the enthalpy of gas constant at the end of this process.

Internal energy of gas (dU) = $mc_v dt$

Enthalpy of gas (dH) = $mc_p dt$

Internal energy and Enthalpy of an ideal gas is a function of temperature

dU = f (T)

dH = f (T)

327. A steam turbine receives steam steadily at 10 bar with an enthalpy of 3000 kJ/kg and discharges at 1 bar with an enthalpy of 2700 kJ/kg. The work output is 250 kJ/kg. The change in KE and PE are negligible. The heat transfer from the turbine casing to the surroundings is equal to :

(a) 0 kJ (b) 50 kJ

(c) 150 kJ (d) 250 kJ

RPSC ADE 2016, GATE 2000

Ans. (b) : P_1 = 10 bar

h_1 = 3000 kJ/kg

P_2 = 1 bar

h_2 = 2700 kJ/kg

w = 250 kJ/kg

SFEE (Steady flow energy equation)

$$h_1 + \frac{c_1^2}{2} + gz_1 + q = h_2 + \frac{c_2^2}{2} + gz_2 + w$$

h_1 =Inlet enthalpy (kJ/kg)

c_1 = Inlet velocity m/s

z_1 = Potential energy at inlet

q = heat transfer

h_2 = outlet enthalpy (kJ/kg)

c_2 = outlet velocity (m/s)

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z_2= potential energy at outlet
w = work done
3000 + q = 2700 + 250
q = 2700 – 3000 + 250
= – 300 + 250
q = – 50
(-ve) sign denote heat transfer from the turbine
q = 50 kJ

328. A gas contained in a cylinder is compressed, the work required for compression being 5000 kJ. During the process, heat interaction of 2000 kJ causes the surroundings to be heated. The change in internal energy of the gas during the process is

(a) –7000 kJ (b) –3000 kJ
(c) 3000 kJ (d) 7000 kJ

TSPSC AEE 2017, GATE 2004

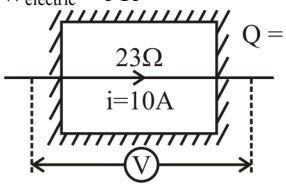
Ans. (c) : We know, From I[st] Law of thermodynamics
dQ = dU + dW
dW = – 5000 kJ (gas is compressing the cylinder)
dQ = –2000 kJ (Surrounding is being heated)
So,
–2000 = – 5000 + dU
dU = 3000 kJ

329. The contents of a well-insulated tank are heated by a resistor of 23W in which 10A current is flowing. Consider the tank along with its contents as a thermodynamic system. The work done by the system and the heat transfer to the system are positive. The rates of heat (Q), work (W) and change in internal energy (ΔU) during the process in kW are

(a) Q = 0, W = – 2.3, ΔU = + 2.3
(b) Q = + 2.3, W = 0, ΔU = + 2.3
(c) Q = – 2.3, W = 0, ΔU = – 2.3
(d) Q = 0, W = + 2.3, ΔU = – 2.3

GPSC Asstt. Director of Transport 05.03.2017
GATE 2011

Ans. (a) : Given,
$W_{electric} = i^2R$



Q = 0, Adiabatic

$W_{electric} = 10^2 \times 23 = -2.3$ kW (on system)
First law : Q – W = ΔU
$Q - (-W_{electric}) = \Delta U$
ΔU = +2.3 kW, $W_{electric}$ = –2.3 kW, Q = 0

330. An insulated rigid vessel contains a mixture of fuel and air. The mixture is ignited by a minute spark. The contents of vessel experience

(a) increase in temperature, pressure and energy
(b) decrease in temperature, pressure and energy
(c) increase in temperature and pressure but increase in energy
(d) increase in temperature and pressure but no change in energy

ISRO Scientist/Engineer (RAC) 07.05.2017
GATE 1993

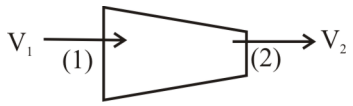
Ans. (d) : It is obvious that energy cannot be created its just transferred from one form to another.
So the mixture is ignited by a minute spark in an insulated rigid vessel contains a mixture of fuel and air, increase in temperature and pressure but not change in energy.

331. A small steam whistle (perfectly) insulated and doing no shaft work) causes a drop of 0.8 kJ/kg in enthalpy of steam from entry to exit. The kinetic energy of the steam at entry is negligible, the velocity of steam at exit is :

(a) 4 m/s (b) 40 m/s
(c) 80 m/s (d) 120 m/s

Vadodara Muncipal Corp. DEE, 2018
GATE 2001

Ans. (b) : From SFEE



$$h_1 + \frac{v_1^2}{2000} + \frac{gz_1}{1000} + q = h_2 + \frac{v_2^2}{2000} + \frac{gz_2}{1000} + w$$

$$h_1 = h_2 + \frac{v_2^2}{2000}$$

$$\frac{v_2^2}{2000} = h_1 - h_2 = \Delta h$$

$$v_2 = \sqrt{2000 \times \Delta h}$$

$$= \sqrt{2000 \times 0.8}$$

$$\boxed{v_2 = 40 m/s}$$

332. The first law of thermodynamics was developed by-

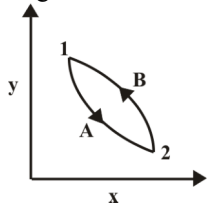
(a) Joule (b) Kelvin
(c) Charles (d) Carnot

RPSC AE 2018, UKPSC AE-2013, Paper-II

Ans. (a) : The first law of thermodynamics was developed by Joule.
First law of thermodynamics— When a closed system executes a complete cycle the sum of heat interactions is equal to the sum of work interaction.

$$\Sigma Q = \Sigma W$$

The summations being over the entire cycle.



$$Q_{A\,1-2} + Q_{B\,2-1} = W_{A\,1-2} + W_{B\,2-1}$$

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333. First law of thermodynamics for steady flow

(a) accounts for all energy entering and leaving a control volume

(b) is an energy balance for a specified mass of fluid

(c) is primarily concerned with heat transfer

(d) is an expression of the conservation of linear momentum

CGPSC AE 15.01.2021
TSGENCO AE 14.11.2015

Ans. (a) : First law of thermodynamic for steady flow energy equation, the total energy entering a system must be equal to total energy leaving the system. For a specified mass (i.e. unit mass)

$$u_1 + P_1V_1 + \frac{v_1^2}{2} + gz_1 + q_{1-2} =$$

$$u_2 + P_2V_2 + \frac{v_2^2}{2} + gz_2 + w_{1-2}$$

334. For an ideal gas with constant of specific heats, for calculation of the specific enthalpy,

(a) it is sufficient to know only the temperature.

(b) both temperature and pressure are required to be known.

(c) both temperature and volume are required to be known.

(d) both temperature and mass are required to be known.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : For ideal gas, specific enthalpy is a function of temperature only.
For real gases, specific enthalpy is function of temperature and pressure both.

335. In an internal combustion engine, if during compression, the heat rejected is 60 kJ/kg and work input is 120 kJ/kg, then what will be the change in internal energy of the working fluid?

(a) Loss of internal energy is 60kJ/kg

(b) Gain of internal energy is 60kJ/kg

(c) Loss of internal energy is 120kJ/kg

(d) Gain of internal energy is 120kJ/kg

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : Change in internal energy
$\Delta U = Q - W$
$= -60 - (-120)$
$= 60$ kJ/kg
it is gain of internal energy.

336. A constant volume chamber of 0.3 m^3 capacity contains 2 kg of gas at 278 K. Heat is transferred to the gas until the temperature is 373 K. the work done and the heat transferred during the process are _____, if the coefficient of specific heat at constant volume is 1.51 kJ/kg K.

(a) 287 kJ, 287 kJ (b) 143.5 kJ, 287 kJ

(c) 143.5 kJ, 430 kJ (d) 0,287 kJ

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : As we know that at constant volume, the work done of the system is zero.
V = constant $\Rightarrow dV = 0$
Work done $(W) = PdV$
$W = P \times 0 = 0$
Given- constant volume
$V = 0.3\ m^3$
$m = 2kg, \quad T_1 = 278$ k
$T_2 = 372$ k
$C_v = 1.51$ kJ/kgK
Heat transferred,
$Q = C_v \Delta T = mC_v(T_2 - T_1)$
$= 2 \times 1.51 \times (373 - 278)$
$= 2 \times 1.51 \times 95$
$= 287$ kJ

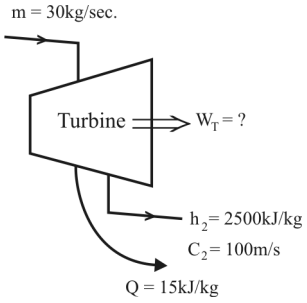
337. For an ideal gas turbine enthalpy of gas at entry and exit are 3000 kJ/kg and 2500 kJ/kg respectively. The velocity of gas are 200 m/s and 100 m/s respectively at the entry and exit. Heat loss of gas to the surrounding is 15 kJ/kg. The approximate power developed by turbine if mass flow rate of gas is 30 kg/s

(a) 16 MW (b) 0.55 MW

(c) 15 MW (d) 15.5 MW

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : $h_1 = 3000$ kJ/kg, $m = 30$ kg/sec.
$C_1 = 200$ m/s
using,



SFEE –

$$m\left[h_1 + \frac{C_1^2}{2000}\right] - Q = m\left[h_2 + \frac{C_1^2}{2000}\right] + W_T$$

$$m\left[h_1 + \frac{C_1^2}{2000}\right] = m\left[h_2 + \frac{C_2^2}{2000} + Q\right] + W_T$$

$$30 \times \left[3000 + \frac{(200)^2}{2000}\right] = 30 \times \left[2500 + \frac{100^2}{2000} + 15\right] + W_T$$

$$90600 = 75600 + W_T$$

$$W_T = 15000\text{kW}$$

$$\boxed{W_T = 15\text{MW}}$$

338. "The internal energy of a gas is a function of temperature."
Which of the following names is associated with the above law?

(a) Joule (b) Charles

(c) Boyle (d) Dalton

Assam PSC AE (IWT) 14.03.2021

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Ans. (a) : Joule's Law– Joule's law states that the internal energy of an ideal gas does not change if volume and pressure changes but does changes if temperature changes. The internal energy of ideal gas is a function of temperature only.

Charles Law– The volume of a given mass of a perfect gas varies directly as its absolute temperature when the absolute pressure remains constant.

$$V \propto T \text{ or } \frac{V}{T} = \text{Constant}$$

Boyle's Law– The absolute pressure of a given mass of a perfect gas varies inversely as its volume, when the temperature remains constant.

$$P \propto \frac{1}{V}$$

Dalton's Law–The total pressures exerted is equal to the sum of partial pressures of individual gases.

$$P = \sum_i P_i = P_1 + P_2 + P_3 + ...$$

339. A tank containing a fluid is stirred by a paddle wheel. The work input to the paddle wheel is 5090 kJ. The heat transfer from the tank is 1500 kJ. What is the change in internal energy of this control mass? (Consider the tank and the fluid inside a control surface)

(a) –3590 kJ (b) + 3590 kJ
(c) + 4590 kJ (d) –4590 kJ

ESE 18.07.2021

Ans. (b) : Given, work, W = 5090 kJ
Heat transfer, $\Delta Q = -1500$ kJ
$\Delta Q = \Delta U + \Delta W$
$\Delta U = \Delta Q - \Delta W = -1500 - (-5090)$
$= 3590$ kJ

340. During the charging of a storage battery, the current is 20 A and the voltage is 12.8V. The rate of heat transfer from the battery is 10W. At what rate is the internal energy increasing?

(a) –256 J/s (b) + 246 J/s
(c) + 256 J/s (d) –246 J/s

ESE 18.07.2021

Ans. (b) : Given,
I = 20 A
V = 12.8V
Q = 10W (heat transfer from battery)
Q = –10W
$\Delta Q = \Delta U + \Delta W$
$-10 = \Delta U - IV$
$-10 = \Delta U - (20)(12.8)$
$\Delta U = +246$ W
$\boxed{\Delta U = 246 \text{ J/s}}$

341. Assume air as a perfect gas. It flows into a diffuser at 300 m/s, 300 K and 100 kPa. At exit, the velocity is negligibly small, but the pressure is high. Assuming zero heat transfer, the exit temperature is:

(a) 452.6K (b) 344.8K
(c) 325K (d) 425K

AAI Jr. Executive 26.03.2021

Ans. (b) : Given, $v_1 = 300$ m/s, $T_1 = 300$ K
$T_2 = ?$, $v_2 \simeq 0$

$$C_pT_1 + \frac{v_1^2}{2000} = C_pT_2 + \frac{v_2^2}{2000}$$

$$C_pT_2 = C_pT_1 + \frac{v_1^2}{2000}$$

$$1.005T_2 = (1.005 \times 300) + \left(\frac{300^2}{2000}\right)$$

$$= 301.5 + 45$$

$$T_2 = \frac{346.5}{1.005}$$

$$T_2 = 344.77\,K$$

342. A closed system contains a simple compressible pure substance. The difference between work done by the system and heat transferred to the system is a _____ function.

(a) both point and path
(b) point
(c) path
(d) neither point nor path

AAI Jr. Executive 26.03.2021

Ans. (b) : From first law of thermodynamics,
dQ = dU + dW
dQ – dW = dU
Where dU is the internal energy and it is the point function of the system.

343. 1 kg of water falls from an altitude of 1000 m above ground level. What is the change in the temperature of water at the foot of the fall, if there are no losses during the fall? (Take specific heat of water as 1 kcal. kg^{-1}. K^{-1} and g = 9.81 m/s^2)

(a) 3.35°C (b) 2.35°C
(c) 3.32°C (d) 4.12°C

ESE 18.07.2021

Ans. (b) : Given, mass of water, m = 1 kg
Altitude, h = 1000 m
C_P = 1 kcal/kg
Since total energy is conserved
Loss of Potential energy = Gain in Thermal energy
$mgh = mc_p\Delta T$

$$1 \times 9.81 \times 1000 = \frac{1 \times 1000 \times 4.18\,J}{kg\,K} \times \Delta T$$

$\Delta T = 2.346°C$
$\boxed{\Delta T = 2.35^\circ C}$

344. A stationary mass of gas is compressed without friction from an initial state of 0.3 m^3 to 0.15 m^3 and 0.105 MPa, the pressure remaining constant during the process. There is a transfer of 40 kJ of heat from the gas during the process. How much does the internal energy of the gas change?

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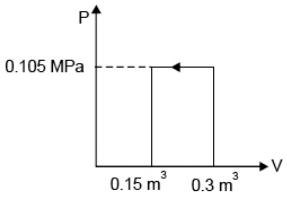
(a) –24.25 kJ (b) –19.62 kJ
(c) –15.91 kJ (d) –12.72 kJ

ESE 18.07.2021

Ans. (a) : Given, $V_1 = 0.3\ m^3$, $V_2 = 0.15\ m^3$
$P_2 = P_1 = 0.105\ MPa = 105\ kPa$
$\Delta Q = -40\ kJ$



Work done,
$W = PdV$
$= 0.105\ (0.15 - 0.3) \times 10^3$
$W = -15.75\ kJ$
From first law of thermodynamics,
$dQ = dU + dW$
$-40 = dU - 15.75$
$\boxed{dU = -24.25\ kJ}$

345. In a closed system, the heat and work for a process 1-2 is 10 kJ and 30 kJ respectively. What is the work during the return process 2-1, if the heat is 20 kJ?
(a) 40 kJ (b) 30 kJ
(c) 20 kJ (d) Zero

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Given
$Q_1 = 10\ kJ$ $Q_2 = 20\ kJ$ $W_1 = 30\ kJ$
By first law :
$$\oint Q = \oint W$$
$Q_1 + Q_2 = W_1 + W_2$
$10 + 20 = 30 + W_2$
$W_2 = 0\ kJ$

346. If a closed system receives 30 kJ of heat while undergoing 91 kJ of reduction in internal energy, then :
(a) 121 kJ of work is done by the system
(b) 61 kJ of work is done by the system
(c) 61 kJ of work is done on the system
(d) 121 kJ of work is done on the system

NLC GET 17.11.2020, Shift-II

Ans. (a) : In a closed system heat transfer is equal to the sum of internal energy and work transfer
So, $\delta Q = dU + \delta W$
Given, $\delta Q = 30\ kJ$
$dU = -91\ kJ$ (–ve sign indicates reduction in energy)
$\delta W = ?$
$\delta Q = dU + \delta W$
$30 = -91 + \delta W$
$\delta W = 121\ kJ$
δW = +ve so, 121 kJ work is done by the system
δW = – ve so, work done on the system.

347. If the work done on a closed system is 40 kJ/kg, and heat rejected from the system is 80 kJ/kg, then its internal energy is _____ .
(a) decreased by 120 kJ/kg
(b) increased by 120 kJ/kg
(c) increased by 40 kJ/kg
(d) decreased by 40 kJ/kg

VIZAG MT, 14.12.2020

Ans. (d) : Given,
$\delta W = -40\ kJ/kg$
$\delta Q = -80\ kJ/kg$
$du = ?$
From first law of thermodynamics,
$\delta Q = dU + \delta W$
$-80 = dU - 40$
$dU = 40 - 80$
$\boxed{dU = -40\,kJ/kg}$
Means internal energy decreased by 40 kJ/kg.

348. The statement of first law of thermodynamics for an isolated system is
(a) $\Delta E = 0$ (b) $\Delta E = \delta Q - \delta W$
(c) $\Delta E = \delta Q$ (d) $\Delta E = -\delta W$

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : The First law of thermodynamics states that the total energy of an isolated system is constant.
Isolated system, where no energy transfer is possible, (ΔE=0), meaning the energy of the isolated system always constant.

349. If in a constant volume expansion occurs at 10 m^3 and pressure varies from 1 bar to 2 bar, then the work transfer is:
(a) 0 kJ (b) 20 kJ
(c) 40 kJ (d) 10 kJ

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : Given, $V = 10\ m^3$
$P_1 = 1$ bar
$P_2 = 2$ bar
Work transfer in closed system = $\int P.dV$
Volume is constant. So, work transfer is zero.

350. The change in internal energy of the system, if the heat supplied to the system is 63 kJ and the work done by the system is 100 kJ, is:
(a) – 163 k (b) + 163 kJ
(c) – 37 kJ (d) + 37 kJ

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : Given, $dQ = 63\ kJ$
$dW = +100\ kJ$ (W.D by the system)
According to first law of thermodynamic
$dQ = dU + dW$
$dU = 63 - 100$
$\boxed{dU = -37\,kJ}$

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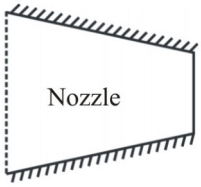
351. Air at 27°C and 100 kPa enters in a steady flow to a nozzle at a velocity of 100 m/s. If the inlet area of the nozzle is 0.5 m^2, what is the mass flow rate through the system

(a) 116 kg/s (b) 232 kg/s
(c) 58 kg/s (d) 143 kg/s

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Given, $P_1 = 100$ kPa, $T_1 = 300$ K
$v_1 = 100$ m/sec, $A_1 = 0.5\ m^2$



$$\rho_1 = \frac{P_1}{RT_1} = \frac{100}{0.287 \times 300}$$

$\rho_1 = 1.1614\ kg/m^3$

Mass flow rate, $\dot{m} = \rho_1 \times A_1 \times v_1$

$\dot{m} = 1.1614 \times 0.5 \times 100$

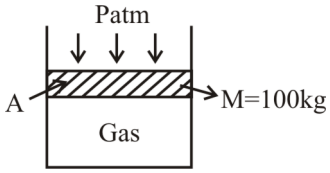
$\dot{m} = 58.07$ kg/sec.

352. A gas is contained in a cylinder with a moveable piston of 100 kg mass. When 2500 J of heat flow into the gas, the internal energy of the gas increases by 1500 J. What is the distance through which the piston rises?

(a) 2 m (b) 1 m
(c) 2.5 m (d) 0.5 m

ISRO Scientist/Engineer 12.01.2020

Ans. (b) :



Q = 2500 J

$(\Delta U) = 1500$ J

Applying first law of thermodynamics,

$Q = (\Delta U)_{gas} + W_{gas}$

$2500 = 1500 + W_{gas}$

$W_{gas} = 1000$ J

$W_{gas} = (F_{atm} + F_{piston}) \times \Delta x_{piston\ rise}$

As area of piston is not given, assuming W.D. on atmosphere to be zero.

$W_{gas} = F_{piston} \times \Delta x = mg.\ \Delta x$

$1000\ J = 100 \times 9.81 \times \Delta x$

$\Delta x = 1.01\ m \simeq 1\ m$

353. A piston-cylinder device with air at an initial temperature of 30°C undergoes an expansion process for which pressure and volume are related as given below:

p(kPa)	100	37.9	14.4
V (m^3)	0.1	0.2	0.4

The work done by the system for n = 1.4 will be

(a) 4.8 kJ (b) 6.8 kJ
(c) 8.4 kJ (d) 10.6 kJ

ESE 2020

Ans. (d) :

$$\text{Work done, } W = \frac{P_1V_1 - P_3V_3}{n-1}$$

$$= \frac{100 \times 0.1 - 14.4 \times 0.4}{1.4 - 1}$$

$$\boxed{W = 10.6\,kJ}$$

354. In a reversible non-flow process, the gas undergoes a process according to, pV = 2.8 Where, p is in bar and volume is in m^3. What will be the work done by the gas if pressure of gas increases from 0.7 bar to 7 bar? [Assuming $e^{2.3} = 10$]

(a) 6.4 kJ (b) –6.4 kJ
(c) –644 kJ (d) –64 kJ

APPSC Poly Lect. 13.03.2020

Ans. (c) : Given, PV = 2.8, $P_1 = 0.7$ bar, $P_2 = 7$ bar

Work done for non-flow process in hyperbolic process or isothermal process,

$$W = C \ln \frac{P_1}{P_2} \qquad \text{Where, } C = P_1V_1 = P_2V_2$$

$$W = P_1V_1 \ln \frac{P_1}{P_2}$$

$$= 2.8 \times 10^5 \ln \frac{0.7}{7}$$

$$= 2.8 \times 10^5 \ln(0.1)$$

$$= 2.8 \times 10^5 (-2.30) \qquad [\because e^{2.3} = \ln 10]$$

$$\boxed{W = -644.72\,kJ}$$

355. The rate of heat transfer between an electric motor and its surroundings is given by $Q = e^{-t} - 1$ Where, t is in seconds and Q is in kW. The shaft of motor rotates at a speed of 200 rad/s and applies a torque of 20 N–m on the load. The motor draws a constant electric power input of 5 kW. The time rate of change of energy (kW) of motor is given by :

(a) e^{-t} (b) $1 + e^{-t}$
(c) 1^{-et} (d) $2 + e^{-t}$

APPSC Poly Lect. 13.03.2020

Ans. (a) : Given : $Q = e^{-t} - 1$

Let, Initial time rate of change of energy = 0

Final time rate of change of energy = $\frac{dQ}{dt}$

$$\therefore \frac{dQ}{dt} = -e^{-t} - 0 = -e^{-t}$$

Rate of change of energy = Initial – Final

$= 0 - (-e^{-t})$

$= e^{-t}$

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356. A cylinder of 50 litre capacity contains oxygen at temperature of 27^0C and at absolute pressure 10 MPa. What will be the density of oxygen? [Gas constant for oxygen is 0.26 kJ/kg-K]

(a) 1.28 kg/m^3 (b) 12.8 kg/m^3
(c) 128.2 kg/m^3 (d) 1282 kg/m^3

APPSC Poly Lect. 13.03.2020

Ans. (c) : Given :
Pressure (P) = 10 MPa = 10×10^6 N/m^2
Temperature (T) = 27^0C + 273 = 300 K
Volume (V) = 50 litre = 0.05 m^3
Gas constant for oxygen (R) = 0.26 kJ/kg-K
$= 0.26 \times 10^3$ J/kg-K
PV = mRT (Let, m = mass of oxygen)

$$m = \frac{PV}{RT} = \frac{10\times10^6\times0.05}{0.26\times10^3\times300} = 6.41 \text{ kg}$$

$$\Rightarrow m = 6.41 \text{kg}$$

We know that–
mass (m) = volume (V) ×density (ρ)

$$\Rightarrow \rho = \frac{m}{V} = \frac{6.41}{0.05}$$

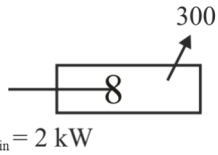
$$\Rightarrow \boxed{\rho = 128.2 \text{ kg/m}^3}$$

357. A 2 kW electric resistance heater submerged in 5 kg water is turned on and kept on for 10 minutes. During the process, 300 kJ of heat is lost from the water, the temperature rise of water is

(a) 0.40^0C (b) 43.1^0C
(c) 57.4^0C (d) 71.8^0C

UPPSC AE 13.12.2020, Paper-2

Ans. (b) : dW = 2 × 10 × 60 kJ
According to first law of thermodynamics,



$\delta Q = dU + \delta W$
$-300 = dU - 2 \times 10 \times 60$
As, heat is lost and work done on system; –ve sign takes.
$dU = 1200 - 300 = 900$
$mC_v dT = 900$

$$dT = \frac{900}{5\times4.2} = 42.85 \simeq 43.1^0\text{C}$$

358. If the fluid is an ideal gas, the temperature of gas in tank after it is charged ___ times initial temperature.

(a) 2 (b) 1.9
(c) 1.4 (d) 1.67

Oil India Senior Officer 23.12.2020

Ans. (c) : If the fluid is an ideal gas, the temperature of gas in tank after it is charged γ times initial temperature.
$T_2 = \gamma T_1$
$\gamma = 1.4$

359. Dry saturated steam enters a frictionless adiabatic nozzle with negligible velocity at a temperature of 300^oC [h_1 = 2751 kJ/kg]. It is expanded to a pressure of 5 MPa isentropically [h_2 = 2651 kJ/kg]. What will be the exit velocity of steam?

(a) 447.21 m/s (b) 572.33 m/s
(c) 14.14 m/s (d) 150.32 m/s

SJVN ET 2019

Ans. (a) : As per SFEE

$$h_1 + \frac{v_1^2}{2000} + q = w + \frac{v_2^2}{2000} + h_2 \quad \text{(if h in kJ/kg)}$$

$w = 0, q = 0, v_1 = 0$

$$v_2 = \sqrt{2000\times(2751-2651)} = 447.21 \text{ m/s}$$

360. Hydrogen gas is enclosed in piston cylinder arrangement at a pressure of 3 bar and 300K. The cylinder has a volume of 1.5 m^3. The process undergoes isothermal expansion to 3 m^3. Work done by gas is:

(a) 311 kJ (b) 251 kJ
(c) 350 kJ (d) 425 kJ

HPPSC ADF 05.03.2019

Ans. (a) : Given-
P_1 = 3 bar
T_1 = 300 k
V_1 = 1.5 m^3, V_2 = 3 m^3

$$W_{Isothermal} = P_1V_1 \ln\left(\frac{V_2}{V_1}\right)$$

$$= 3\times10^5\times1.5\ln\left(\frac{3}{1.5}\right)$$

$= 3 \times 1.5 \times 0.6931 \times 10^5$
= 311.85 kJ

361. The short coming of first law of thermodynamics is

(a) Direction of process
(b) Possibility of process
(c) Quality of energy
(d) Quantity of energy

JPSC AE PRE 2019

Ans. (a, b, c) : The short coming of first law of thermodynamics is–
- Direction of process
- Possibility of process
- Quality of energy

362. In a steady flow process, across the control volume mass and energy flow

(a) Varies continuously
(b) Remain constant
(c) Depends on control surface
(d) Depends on type of process

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Ans. (b)
Assumptions made in steady flow processes are
(i) Control volume moves relative to the coordinate frame.
(ii) The state of the mass at each discrete area of flow on the control surface does not vary with time. The rate at which heat and work cross the control surface remain constant.

363. What is the rise in temperature of 80 litres of water in 40 min by a heater of 2 kW. If whole of the heater energy used to raise the water temperature?
(a) 14.3ºC (b) 1.43ºC
(c) 2.52 ºC (d) 25.2 ºC

OPSC AEE 2019 PAPER - II

Ans. : (a) :
Heat taken by water = Heat given by heater
$\text{Heat (Q)} = mC_p\Delta T = \text{Power} \times \text{time}$
$80 \times 4.2 \times \Delta T = 2 \times 40 \times 60$
$\Delta T = \frac{2 \times 40 \times 60}{80 \times 4.2}$
$= 14.28^\circ C$
$\approx 14.3^\circ C$

364. A system executes a cycle during which there are four heat transfers Q_{12} = 220 kJ, Q_{23} = -25 kJ, Q_{34} = -180 kJ, Q_{41} = 50 kJ. The work during tube of the processes is W_{12} = 15 kJ, W_{23} = -10 kJ, W_{34} = 60 kJ. The work done during the process 4 – 1 is
(a) –230 kJ (b) 0 kJ
(c) 230 kJ (d) 130 kJ

TNPSC AE 2019

Ans. (b) : According to first law of thermodynamics
$[\Sigma Q] = [\Sigma W]$ [for cyclic process]
$Q_{1-2} + Q_{2-3} + Q_{3-4} + Q_{4-1} = W_{1-2} + W_{2-3} + W_{3-4} + W_{4-1}$
$220 + (-25) + (-180) + 50 = 15 + (-10) + 60 + W_{4-1}$
$\boxed{W_{4-1} = 0 \text{ kJ}}$

365. For a heat engine cycle, which of the following relation is always true [Q = heat transfer, W= work transfer]
(a) $\sum_{\text{cycle}} Q = 0 \sum_{\text{cycle}} W = 0$ (b) $\sum_{\text{cycle}} Q = \sum_{\text{cycle}} W$
(c) $\sum_{\text{cycle}} Q > \sum_{\text{cycle}} W$ (d) $\sum_{\text{cycle}} Q < \sum_{\text{cycle}} W$

SJVN ET 2019

Ans. (b) : $\sum_{\text{cycle}} Q = \sum_{\text{cycle}} W$

366. Flow work is analogous to
(a) Stirring work (b) Electrical work
(c) Displacement work (d) Shaft work

JPSC AE PRE 2019

Ans. (c) : Work in steady flow process = $\int VdP$
Work in non-flow process = $\int PdV$

367. An electric generator coupled to a windmill produces an average electric power of 6 kW. The power is used to charge a storage battery. Heat transfer from the battery to the surroundings is 0.3 kW. What will be the amount of energy stored in the battery in 2 hours?
(a) 41040 kW (b) 114 kW
(c) 11.4 kJ (d) 41040 kJ

SJVN ET 2019

Ans. (d) : Given,
$\delta Q = -0.3$ kW
$\delta W = -6$ kW
$dU = ?$
$dU = \delta Q - \delta W$
$dU = -0.3 + 6 = 5.7$ kW
$dU = 5.7 \times 2 \times 3600 = 41040$ kJ

368. For a non-flow constant pressure process the heat exchange is equal to:
(a) Zero
(b) The work done
(c) The change in internal energy
(d) The change in enthalpy

OPSC AEE 2019 PAPER - II

Ans. (d) Since enthalpy change of a system undergoing a process
$dH = Tds + VdP$
For reversible process
$\because Tds = \delta Q$
$\therefore dH = \delta Q + VdP$
If pressure is constant
$dP = 0$
$\boxed{dH = \delta Q}$

369. During steady flow compression process of a gas with mass flow rate of 2 kg/s, increase in specific enthalpy is 15 kJ/kg and decrease in kinetic energy is 2 kJ/kg. The rate of heat rejection to the environment is 3 kW. The power needed to drive the compressor is :
(a) 23 kW (b) 2 kW
(c) 29 kW (d) 37 kW

GPSC ARTO Pre 30.12.2018

Ans. (c) : From steady flow equation,
$$\dot{m}\left[h_1 + \frac{V_1^2}{2000}\right] + \dot{Q} = \dot{m}\left[h_2 + \frac{V_2^2}{2000}\right] + \dot{W}$$
$$\dot{W} = \dot{m}(h_1 - h_2) + \dot{m}\left(\frac{V_1^2}{2000} - \frac{V_2^2}{2000}\right) + \dot{Q}$$
$= 2(-15) + 2(2) - 3$
$= -30 + 4 - 3$
$\dot{W} = -29$ kW
The power needed to drive the compressor is 29 kW.

370. The specific heat of concentrated orange juice is 5.8 kJ/kg °C. How much heat must be removed to cool 2.0 kg of juice from 30.0°C to 0.0°C?
(a) 24kJ (b) 338kJ
(c) 348kJ (d) 600kJ

Vadodara Muncipal Corp. DEE, 2018

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Ans. (c) Heat removed = $mc\Delta T$
$= 2 \times 5.8 \times (30 - 0)$
$= 348$ kJ

371. In a cyclic process, heat transfers are 157 kJ, –262 kJ, –486 kJ and 315 kJ. Net work for this cyclic process is:
(a) 176 kJ (b) 151 kJ
(c) 276 kJ (d) 252 kJ

AAI Jr. Executive 29.11.2018

Ans. (c) : For a cyclic process
$\oint Q = \oint W$
$\oint Q = 157 - 262 - 486 + 315$
$= -276$ kJ
$\oint W = 276$ kJ

372. During the charging of a storage battery, the current *I* is 20A and the voltage E in 12.8V. The ratio of heat transfer from the battery is 16W. At what rate is the internal energy increasing?
(a) 256 Js^{-1} (b) 272 Js^{-1}
(c) 0 (d) 240 Js^{-1}

ISRO Scientist/Engineer 22.04.2018

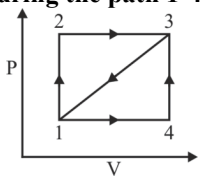
Ans. (d) : I = 20A
E = 12.8V
Q = 16W
U = ?
Energy store in battery during charging = $VI - Q$
$= 12.8 \times 20 - 16$
$= 256 - 16$
$= 240W\ (Js^{-1})$

373. Statement I: The energy of an isolated system is constant.
Statement II: The entropy of an isolated system can increase but cannot decrease.

ESE 2018

Ans. (b) : Energy of the isolated system is constant as there is no energy interaction. Entropy of an isolated system always increases and never decreases.

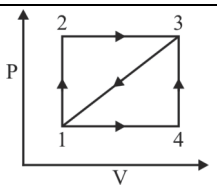
374. A system absorbs 100 kJ as heat and does 60 kJ work along the path 1-2-3. The same system does 20 kJ work along the path 1-4-3. The heat absorbed during the path 1-4- 3 is



(a) –140 kJ (b) –80 kJ
(c) 80 kJ (d) 60 kJ

ESE 2018

Ans. (d) : Given,
$Q_{123} = 100$ kJ
$W_{123} = 60$ kJ
$W_{143} = 20$ kJ



For path 1–2–3
$Q_{123} = U_3 - U_1 + W_{123}$
$U_3 - U_1 = 100 - 60 = 40$ kJ
For path 1–4–3
$Q_{143} = U_3 - U_1 + W_{143}$
$= 40 + 20 = 60$ kJ

375. Heat transferred to a closed stationary system constant volume is equal to
(a) work transfer
(b) increased in internal energy
(c) increase in enthalpy
(d) increase in gibbs function

RPSC IOF & BOILER 20.09.2017

Ans. : (b) $\delta Q = dU + \delta W$
for constant volume close system
$W = 0$
Hence,
$\delta Q = dU$

376. In a reversible isothermal expansion process, the fluid expands from 10 bar and 2 m^3 to 2 bar and 10 m^3. During this expansion process, 100 kW of heat is supplied. Then the work done during the process is
(a) 33.3 kW (b) 80 kW
(c) 100 kW (d) 20 kW

BPSC AE Mains 2017 Paper - V, ESE 2009

Ans. : (c) Assuming ideal gas and reversible isothermal process,
$\delta Q = dU + \delta W$
$\delta Q = \delta W$
for isothermal and ideal gas
$U = f(T)$
As heat transfer is 100 kW is given, work will also be equal to 100 kW.

377. Consider the following devices :
1. Internal combustion engine working on Otto cycle
2. Internal combustion engine working on Diesel cycle
3. Gas turbines
4. Steam turbines

In which of the above devices, the equation dQ = dU + Pdv is not applicable ?
(a) 2 and 3 (b) 1 and 4
(c) 3 and 4 (d) 1 and 2

UPSC JWM 2017

Ans. (c) : $\delta Q = dU + \delta W$ ...(1)
$\delta Q = dU + PdV$ ...(2)
The equation (2) is applicable for the closed system.

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378. A system is composed of a gas contained in a cylinder fitted with a piston. The gas expands from the state 1 for which $E_1 = 75$ kJ to a state 2 for which $E_2 = -25$ kJ. During the expansion, the gas does 60 kJ of work surroundings. The heat transferred to or from the systems during process is:

(a) – 30 kJ (b) – 40 kJ
(c) 30 kJ (d) 40 kJ

TRB Polytechnic Lecturer 2017

Ans. (b) : Given as,
$E_1 = 75$ kJ, $E_2 = -25$ kJ
$W = 60$ kJ
$Q = ?$
We know that
$Q = \Delta E + W$
$Q = (-25 - 75) + 60$
$Q = -40$ kJ

379. During a constant pressure expansion of a gas, 33.3% heat is converted into work while the temperature rises by 20K. The specific heat of the gas at constant pressure as a proportion of work, W is

(a) 8% (b) 10%
(c) 12% (d) 15%

ESE 2017

Ans. (d) : Work (W) = 33.3% Q
$W = 0.33\ Q$
Temperature rise (ΔT) = 20 K
We know that $Q = C_P\ \Delta T$

$$\frac{W}{0.33} = C_P \times 20$$

$$C_P = 0.15\ W$$

i.e. specific heat of the gas at constant pressure is 15% of the work.

380. A cylinder contains $10m^3$ of an ideal gas at a pressure of 2 bar. This gas is compressed in a reversible isothermal process till its pressure increases to 15 bar. What quantum of work will be required for this process?
(You can use the table given herewith.)

Number	2	2.5	3	5	7
Log_{10}	0.301	0.397	0.475	0.698	0.845

(a) 4500 kJ (b) 4030 kJ
(c) 450 kJ (d) 403 kJ

ESE 2017

Ans. (b) : Given,
$V_1 = 10\ m^3$
$P_1 = 2$ bar
$P_2 = 15$ bar
For reversible isothermal process work done is given by

$$W_{1\text{-}2} = P_1V_1\ ln\left(\frac{P_1}{P_2}\right)$$

$$2\times10^5\times10\ ln\left(\frac{2}{15}\right) = -4029.8 \simeq 4030\ \text{kJ (work done on the system)}$$

381. A gas is compressed in a cylinder by a movable piston to a volume one-half its original volume. During the process 300 kj heat left the gas and internal energy remained same. The work done on gas in Nm will be

(a) 300 Nm (b) 3000 Nm
(c) 30000 Nm (d) 300000 Nm

WBPSC AE, 2017

Ans. (d) : $\delta Q = dU + PdV$
$\delta Q = U_2 - U_1 + W$
$\delta Q = 0 + W$ $\quad [\because\ U = f(T)$
$W = -300\ kJ$ $\quad U = U_2 = U_1$
$= -300\times1000\ J$ $\quad$ or $T_2 = T_1 = T]$
$= -300000$ N-m
[–ve means work done on the gas]

382. Energy is added to 5 kg of air with a paddle wheel until $\Delta T = 100°C$. Find the paddle wheel work if the rigid container is insulated :

(a) –358 kJ (b) –382 kJ
(c) +358 kJ (d) +382 kJ

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : Given, m = 5kg, $\Delta T = 100$ °C
Because process is constant volume process.
$W = mc_v\Delta T$ $\quad (c_v)_{air} = 0.718$ kJ/kg-K
$= -5 \times 0.718 \times 100$
$W = -359$ kJ $\rightarrow$ (negative sign is because work is on the system)

383. The energy of the isolated system is always a constant, which is given by:

(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) Third law of thermodynamics
(e) Law of stable equilibrium

CGPSC Polytechnic Lecturer 2017

Ans. (b) : Isolated System–An isolated system is a thermodynamic system that cannot exchange either energy or mass outside the boundaries of the system.
So, $\Delta E = 0, \Delta m = 0$
So from the first law of thermodynamics energy of the isolated system will be remain constant.

384. A fan consumes 20 W of electric power and discharges air from a ventilated room at 0.25 kg/s. The maximum air outlet velocity is nearly

(a) 4.7 m/s (b) 8.7 m/s
(c) 10.2 m/s (d) 12.7 m/s

UPSC JWM 2017

Ans. (d) : Fan convert electrical energy into mechanical energy.

$$\text{Power} = \frac{1}{2}\times\text{Mass flow rate of air}\times\left(\text{discharge velocity}\right)^2$$

$$P = \frac{1}{2}\dot{m}v^2$$

$$v = \sqrt{\frac{2P}{\dot{m}}} = \sqrt{\frac{2\times20}{0.25}} = 12.65\text{m/s}$$

Maximum air outlet velocity $\simeq 12.7$m/s

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385. Which of the following is a low grade energy?

(a) Mechanical work (b) Electrical energy
(c) Wind power (d) Kinetic energy
(e) Heat energy

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : High Grade Energy–Energy that can be completely transformed into work without any loss i.e. fully utilizable.

Example–

1. Mechanical work
2. Electrical work
3. Wind energy
4. Tidal energy
5. Water energy
6. Kinetic energy

Low Grade Energy–Energy of which only a certain portion can be converted into mechanical work is called low grade energy.

Example–

1. Heat or thermal energy.
2. Heat drive from combustion of fossil fuels.
3. Heat drive from nuclear fission or fusion.

386. Oxygen expands in a reversible adiabatic manner through a nozzle from an initial pressure and initial temperature, and with an initial velocity of 50 m/s. There is a decrease of 38 K in temperature across the nozzle. (C_p for oxygen is 0.9185 kJ/kg.K). The exit velocity is :

(a) 268.9 m/s (b) 278.5 m/s
(c) 300 m/s (d) None of these

RPSC ADE 2016

Ans. (a) : For Nozzle –

$$h_1 + \frac{v_1^2}{2} = h_2 + \frac{v_2^2}{2}$$

$$\frac{v_2^2}{2} = c_p\left(T_1 - T_2\right) + \frac{v_1^2}{2}$$

$$= 0.9185 \times 10^3 \times (38) + \frac{50^2}{2}$$

$v_2 = 268.89$ m/s

387. Which one of the following statements holds good for the equation : dQ = dE + dW

(a) Any process undergone by a closed stationary system
(b) Any process reversible and irreversible and for any system
(c) A closed system when only pdV work is present
(d) Only reversible process

RPSC VPITI 14.02.2016

Ans. (b) : Equation dQ = dE + dW

For any process-Reversible or Irreversible system-Closed or opened system

dQ = dU + P.dV ⇒ Only for non flow/closed system stationary and reversible process.

dQ = dU + dW ⇒ For any process reversible or irreversible closed stationary system

388. Heat transfer in a cyclic process are +20 kJ, -5 kJ, -10 kJ and +15 kJ. Net work done for this cycle will be given by:

(a) +0 kJ (b) -20 kJ
(c) +20 kJ (d) -10 kJ

UPRVUNL AE 07.10.2016

Ans. (c) : From first law of thermodynamics.

$$\oint Q = \oint W$$

$Q_{Net} = W_{net}$

$W_{net} = 20 - 5 - 10 + 15$

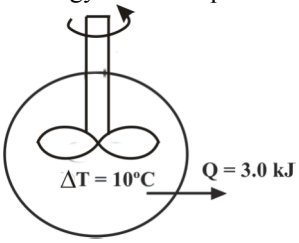
$W_{net} = +20$ kJ

389. 1 kg liquid (specific heat = 3.0 kJ/kg-K) is stirred in closed chamber and its temperature is raised by 10°C. Heat loss to the surrounding is 3.0 kJ. The work done on the water during the process will be:

(a) 36 kJ (b) 30 kJ
(c) 27 kJ (d) 33 kJ

UPRVUNL AE 07.10.2016

Ans. (d) : From energy balance equation.



Work done on the closed system (W) = Change in internal energy of the system + heat loss to the surrounding.

$W = mC_v\Delta T + Q_{Loss}$

$W = 1 \times 3 \times 10 + 3$

$W = 33$ kJ

390. A 500 W electric bulb was switched on in a 2 × 3 × 4 m size thermally insulated room having temperature of 25 °C. What will be the room temperature at the end of 16 hours?

(a) 120 °C (b) 420 °C
(c) 920 °C (d) 1120 °C
(e) 1420 °C

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) : Heat generated = P × 16 × 60 × 60 Joule

$$= \frac{500}{1000} \times 16 \times 60 \times 60\,\text{kJ}$$

Heat dissipated = $\rho \times V \times C_v \times (T - 25)$

$$= \frac{500}{1000} \times 16 \times 60 \times 60$$

$= 1.24 \times 2 \times 3 \times 4 \times 0.7\,(T - 25)$

$T = 1420°C$

391. A vertical cylinder with a freely floating piston contains 0.2 kg air at 1.1 bar and a small electric resistor. The resistor is wired to an external 12 volt battery. When a current of 2 amp is passed through the resistor for 100 sec, the piston sweeps a volume of 0.02 m^3.

Assume:

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(i) Piston and cylinder are insulated.
(ii) Air behaves as an ideal gas with C_v = 718J/kgK.

Find the rise in temperature of air, in °C.

(a) 1.4 (b) 1.6
(c) 1.7 (d) 1.9
(e) 2.1

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : Given,

m = 0.2 kg, P = 1.1 bar
E = 12 V, I = 2A
t = 100 sec, $V_s = 0.02\ m^3$,
C_v = 718 J/kgK

Insulation cond^n, $\delta Q = 0$

$$\delta Q = \delta U + \Delta W$$
$$0 = mC_v\Delta T + \delta W_{air} - \delta W$$
$$\Delta T = \frac{\delta W_{air} - \delta W}{mC_v}$$
$$= \frac{(1.1\times10^5\times0.02) - (12\times2\times100)}{0.2\times718}$$
$$\boxed{\Delta T = 1.4°C}$$

392. During charging of tank from a pipeline, the temperature of fluid is T in the pipeline. The final temperature in the tank will be T multiplied by

(a) C_p (b) C_v
(c) C_p/C_v (d) $C_p \times C_v$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : The final temperature in the tank will be $C_p/C_v.T$.

393. From the following, in which case uniform state uniform flow process model may be applied

(a) Flow through a partially opened valve
(b) Flow through a pipe line in an open tank
(c) Flow through a pipe line in an closed tank
(d) Flow through fully opened valve

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : In case uniform state uniform flow process model may be applied when flow through a pipe line in an open tank.

394. The.....corollary of the.....law of thermodynamics, that energy is conserved, is the application of the conservation of energy to closed systems.

(a) First, second (b) Second, first
(c) First , first (d) Second, second

HPPSC AE (PWD) 10.09.2016

Ans. (c) : First , first

395. A piston-cylinder containing air expands at a pressure of 150 kPa from a temperature of 285 K to a temperature of 550 K. The mass of air is 0.05 kg and C_V = 0.7176. Then the change in internal energy will be:

(a) 13.31 kJ (b) 3.8 kJ
(c) 9.51 kJ (d) 1.23 kJ

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Given than,

P = 150 kPa
$= 150 \times 10^3$ Pa
T_1 = 285 K
T_2 = 550 K
m = 0.05 kg
C_V = 0.7176

By the first law of thermodynamics–

$$dQ = du + dw$$
$$\because \quad du = mC_V dT$$
$$du = 0.05 \times 0.7176 \times (550–285)$$
$$du = 9.5008\ kJ$$

396. Specific heat at constant volume C_v =

(a) $\frac{R}{\gamma-1}$ (b) $\frac{\gamma-1}{\gamma-R}$
(c) $\frac{(\gamma-1)R}{\gamma}$ (d) $\frac{\gamma R}{\gamma-1}$
(e) None of these

Where R = Gas constant
γ = Adiabatic exponent

CGPSC AE 16.10.2016

Ans. (a) : $C_p - C_v = R$

$$\therefore \quad C_v = \frac{R}{\gamma-1}$$
$$C_p = \frac{\gamma R}{\gamma-1}$$

397. A closed system undergoes a process during which 200 kJ of heat is added to it. The system is then restored to its initial state. If the heat transfer and work transfer for the second process are -75 kJ and +100 kJ respectively what is the work transfer for the first process?

(a) 50 kJ (b) 25 kJ
(c) 75 kJ (d) 100 kJ

UPRVUNL AE 21.08.2016

Ans. : (b) Q_1 = 200 kJ W_1 = ?
Q_2 = –75 kJ W_2 = + 100 kJ
As per Ist law of thermodynamics
$\Sigma dQ = \Sigma dW$
$Q_1 + Q_2 = W_1 + W_2$
$200 - 75 = W_1 + 100$
$125 - 100 = W_1$
$\boxed{W_1 = 25\ kJ}$

398. Air at 100 kPa, 2°C occupies a 10 liter piston-cylinder device that is arranged to maintain constant air pressure. This device is now heated until its volume is 20 liters. The work produced by the air is

(a) 20 kJ (b) 10 kJ
(c) 2 kJ (d) 1 kJ

ISRO Scientist/Engineer 03.07.2016

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Ans. (d) : At, $P = 100$ kPa

$T = 2°C$

$V_1 = 10$ liter $= 10 \times 10^{-3}$ m^3

$V_2 = 20$ liter $= 20 \times 10^{-3}$ m^3

At Constant pressure–

Work done (W) $= P(V_2 - V_1)$

$= 100 \times 10^3 (20 \times 10^{-3} - 10 \times 10^{-3})$

$= 100 \times 10^3 \times 10 \times 10^{-3}$

$= 1000$ J

$W = 1$ kJ

399. A system is undergoing a cycle that consists of three processes. During the first process, the work is 5 kJ and the heat is 23 kJ. In the second process no work takes place and the heat interaction is –50 kJ. The third process is adiabatic. The work in the third process is

(a) –32 kJ (b) –23 kJ

(c) –22 kJ (d) zero

ISRO Scientist/Engineer 03.07.2016

Ans. (a) :

Process	Heat	Work
1	23 kJ	5 kJ
2	–50 kJ	0
3	0 (Adiabatic)	W

According to first low of thermodynamic

$$\Sigma Q = \Sigma W$$

$$23 - 50 + 0 = 5 + 0 + W$$

$W = -32$ kJ

400. Air at 100 kPa and 280K is compressed steady to 600 KPa and 400K. The mass flow rate of the air is 0.02 Kg/s, and a heat loss of 16 KJ/kg occurs during the process. Determine power input to the compressor.

(a) 3.2 KW (b) 2.7 KW

(c) 2.9 KW (d) 3.0 KW

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : $h_1 - \dot{q} = h_2 - w$

$$w = (h_2 - h_1) + \dot{q}$$

$$= \dot{m}\left[(C_pT_2 - C_pT_1) + \dot{q}\right]$$

$= 0.02[(1.005 \times 400 - 1.005 \times 280) + 16]$

$= 2.732$ KW

401. Which one of the following equations represents the first law of thermodynamics?

(a) $\partial Q = dU + PdV$ (b) $\partial W = dU + dV$

(c) $\partial Q = Tds$ (d) $Tds = dU + PdV$

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : First law for a process

$$\partial Q = dE + \partial W$$

Where, E = Total energy

$$E = U + KE + PE$$

If we consider change in KE and PE is zero(i.e. stationary system).

Then,

$$\partial Q = dU + \partial W$$

For closed system

$$\partial Q = dU + PdV$$

First law for a cycle

$$\oint \partial Q = \oint \partial W$$

Note-Official answer is given (d).

402. A closed system undergoes a process 1-2 for which the values of $Q_{1\text{-}2}$ and $W_{1\text{-}2}$ are +40 kJ and +60 kJ, respectively. If the system is returned to state 1, and $Q_{2\text{-}1}$ is –30 kJ, what is the value of the work $W_{2\text{-}1}$?

(a) +20 kJ (b) –50 kJ

(c) –80 kJ (d) +40 kJ

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : Initial and final condition is identical so it is a cyclic process.

First law of thermodynamics for cyclic process.

$$\oint \delta Q = \oint \delta W$$

or $$(\Sigma Q)_{cycle} = (\Sigma W)_{cycle}$$

Given, $Q_{1-2} = 40$ kJ

$W_{1-2} = 60$ kJ

$Q_{2-1} = -30$ kJ

Heat transfer equal to work transfer

$$Q_{1-2} + Q_{2-1} = W_{1-2} + W_{2-1}$$

$$40 - 30 = 60 + W_{2-1}$$

$$10 = 60 + W_{2-1}$$

$$W_{2-1} = -50 \text{ kJ}$$

The negative sign implies that the work is done on the system.

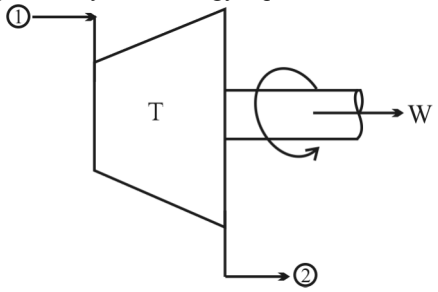
403. In a steady flow, adiabatic turbine, the changes in the internal energy, enthalpy, kinetic energy and potential energy of the working fluid, from inlet to exit, are –150 kJ/kg, –130 kJ/kg, –40 kJ/kg and 0 kJ/kg respectively. Which one of the following gives the amount of work developed by the turbine?

(a) 190 kJ/kg (b) 170 kJ/kg

(c) 150 kJ/kg (d) 90 kJ/kg

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : Steady flow energy equation of turbine –



$$h_1 + \frac{C_1^2}{2} + gz_1 + q = h_2 + \frac{C_2^2}{2} + gz_2 + w$$

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$$(q\text{-}w) = (h_2 - h_1) + \left(\frac{C_2^2}{2} - \frac{C_1^2}{2}\right) + g(z_2 - z_1)$$

For adiabatic turbine q = 0

$$-w = (h_2 - h_1) + \left(\frac{C_2^2 - C_1^2}{2}\right) + g(z_2 - z_1)$$

$-w = -130 - 40 - 0$

$w = 170$ kJ/kg

Change in internal energy i.e., 150 kJ / kg is superfluous data.

404. A mixture of gases expands from 0.03 m^3 to 0.08 m^3 at a constant pressure of 1 MPa and absorbs 76 kJ of heat during the process. The change in internal energy of the mixture is

(a) 116 kJ (b) 46 kJ
(c) 26 kJ (d) 126 kJ

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Given,

P = 1 MPa = 1000 kPa

dQ = 76 kJ

$dW = P(V_2 - V_1)$

$= 1000(0.08 - 0.03)$

$= 1000\ (0.05)$

dW = 50 kJ

dQ = dU + dW

dU = dQ – dW

dU = 76 – 50

dU = 26 kJ

405. Total enthalpy of steam at the inlet of a nozzle is 2800 kJ while static enthalpy at the exit is 2555 kJ. What is the steam velocity at the exit if expansion is isentropic?

(a) 70 m/s (b) 245 m/s
(c) 450 m/s (d) 700 m/s

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : Given, $h_1 = 2800$ kJ, $h_2 = 2555$ kJ

$$V_2 = \sqrt{2(h_1 - h_2)}$$

$$V_2 = \sqrt{2(2800 - 2555) \times 1000}$$

$$= \sqrt{2 \times 245 \times 1000}$$

$V_2 = 700$ m/sec

406. During a certain compression process, 1 kJ of mechanical work input is supplied to 2 kg of a gas enclosed in a cylinder piston assembly and 400 J of heat is rejected to the cooling water being circulated in the jacket encasing the cylinder. This brings about a change in the specific internal energy by:

(a) –700 J kg^{-1} (b) 600 J kg^{-1}
(c) 300 J kg^{-1} (d) –300 J kg^{-1}

OPSC AEE 2015 PAPER - II

Ans. : (c)

Work input = 1000J

Rejected heat = 400J

mass = 2 kg.

first law of thermodynamics

$\delta Q = dU + \delta W$.

$dU = \delta Q - \delta W$.

dU = -400+1000

dU = 600J

$$\text{Specific internal energy} = \frac{\text{Internal energy}}{\text{mass}} = \frac{600}{2}$$

u = 300 J/Kg

($\delta \rightarrow$ show that heat and work both are inexact differential)

407. A system undergoes a process in which the heat transfer to the system is 30 KJ and the work done by the system is 35000 Nm. The change in internal energy of the system is

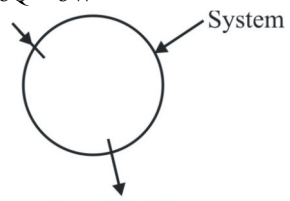
(a) +5 KJ (b) –5 KJ
(c) –10 KJ (d) + 10 KJ

TSPSC AEE 2015

Ans. : (b) According to first law of thermodynamic

$\delta Q = dU + \delta W$

$dU = \delta Q - \delta W$



dU = (30-35)

dU = –5 KJ.

408. First law of thermodynamics states that : (where $Q_{1\text{-}2}$ = Heat supplied, dU = Change in internal energy, and $W_{1\text{-}2}$ = Work done in heat units)

(a) $Q_{1\text{-}2} = dU + W_{1\text{-}2}$ (b) $Q_{1\text{-}2} = dU - W_{1\text{-}2}$
(c) $Q_{1\text{-}2} = dU/W_{1\text{-}2}$ (d) $Q_{1\text{-}2} = dU \times W_{1\text{-}2}$

NPCIL ET, 2015

Ans. (a) : $Q_{1\text{-}2} = dU + W_{1\text{-}2}$ – 1st Law of thermodynamics

409. In a given process of an ideal gas, $\delta W = 0$ and $\delta Q < 0$. Then for the gas

(a) the temperature will decrease
(b) the volume will increase
(c) the pressure will remain constant
(d) the temperature will increase

Haryana PSC Civil Services Pre, 2014

Ans. (a) : Given,

$\delta W = 0$ and $\delta Q < 0$

From first law of thermodynamics

$\delta Q = dU + \delta W$

$\delta Q = dU + 0$

$\because \delta Q < 0$ & $\delta Q = dU$

$dU < 0$

$m C_v (T_2 - T_1) < 0$

$\boxed{T_2 < T_1}$

Means temperature will decreases.

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410. An insulated 2 kg box falls from a balloon 3.5 km above the earth. The change in the internal energy of the box after it has hit the earth's surface will be :

(a) Zero (b) 70,000 kJ
(c) 68.6 kJ (d) 7 kJ

MPSC Poly. Lect. 2014

Ans. (c) : Change in internal energy = potential energy

$U = mgh$

$= 2 \times 9.81 \times 3.5 \times 10^3$

$= 68.67$ kJ

411. Which one of the following statements is not related to the first law of thermodynamics?

(a) Whenever energy is transformed form one to another, energy is always conserved
(b) Energy can neither be created nor destroyed
(c) The sum total of all energy remains constant
(d) Every process occurring in nature proceeds in the sense in which the sum of the entropies of all bodies taking part in the process is increased

UPRVUNL AE 2014

Ans. : (d) Entropy's law is not related to the first law of thermodynamic process. It is related to the second law of thermodynamic.

$$dQ = T.\, dS$$

• First law is related to energy conservation.

412. A sample of gas ($\gamma = 1.5$) is taken through an adiabatic process in which the volume is compressed from 1600 cm^3 to 400 cm^3. If initial pressure is 150 kPa, then how much will be the final pressure in the process?

(a) 1200 kPa (b) 1000 kPa
(c) 1100 kPa (d) 1400 kPa

ISRO Scientist/Engineer 24.05.2014

Ans. (a) : $\gamma = 1.5$

Adiabatic process = PV^γ = constant

$V_1 = 1600$ cm^3

$V_2 = 400$ cm^3

$P_1 = 150$ kPa

$P_2 = ?$

$$P_1V_1^\gamma = P_2V_2^\gamma$$

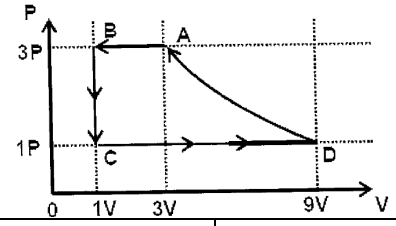
or $$P_2 = P_1\left(\frac{V_1}{V_2}\right)^\gamma$$

$$= 150\left(\frac{1600}{400}\right)^{1.5}$$

$$= 150 \times 4^{1.5}$$

$P_2 = 1200$ kPa

413. One mole of a monatomic gas is taken through a cycle ABCDA as shown in the P-V diagram. Column II give the characteristics involved in the cycle. Match A → B process with the processes given in Column II.



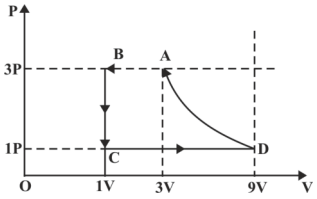
Column I	Column II
(A) Process A → B	(p) Internal energy decreases
(B) Process B → C	(q) Internal energy increases
(C) Process C → D	(r) Heat is lost
(D) Process D → A	(s) Heat is gained
	(t) Work is done on the gas

Process A → B signifies

(a) p,q (b) r,s
(c) p,r,t (d) q,t

ISRO Scientist/Engineer 24.05.2014

Ans. (c)



For process A → B

$$\frac{P_AV_A}{T_A} = \frac{P_BV_B}{T_B}$$

$P_A = P_B$ → At constant pressure

$$\frac{V_A}{T_A} = \frac{V_B}{T_B}$$

or $$\frac{T_B}{T_A} = \frac{V_B}{V_A} = \frac{1V}{3V} = \frac{1}{3}$$

or $$T_B = \frac{T_A}{3}$$

∵ $T_B < T_A$, so heat is 'lost'.

$$\Delta U = C_V \Delta T$$

$$= C_V(T_B - T_A) = -ve$$

So, here Internal energy decreases. As volume is compressed, work is done on the gas.

414. First law of thermodynamics is valid for

(a) all process
(b) reversible process only
(c) cyclic process only
(d) only cyclic process that are carried out reversibly

Haryana PSC Civil Services Pre, 2014

Ans. (a) First law of thermodynamics is based on energy conservation. It states that energy neither can be created nor destroyed. So, it is valid for all process.

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415. A company claims to have developed a revolutionary fan that consumes 25 W and delivers 0.8 kg/s of air at a velocity of 10 m/s. The claim has to be verified using appropriate law of thermodynamics. Which of the following combinations of conclusion and the law employed for the same is correct?

(a) Invalid – First law (b) Invalid – Second law
(c) Valid – First law (d) Valid – Second law

RPSC AEN Pre-2013

Ans. (a) : Invalid – First law

416. The steady flow energy equation of a perfect gas flowing through a nozzle with the initial and final velocities of V_1 and V_2 are given by

(a) $h_1 = h_2$ (b) $h_1 - h_2 = \frac{V_2^2 - V_1^2}{2}$

(c) $h_1 - h_2 = \frac{V_1^2 - V_2^2}{2}$ (d) $h_1 - h_2 = \frac{W_X}{m}$

TNPSC AE 2013

Ans. (b) : We know that SFEE,

$$m_1\left[h_1 + \frac{V_1^2}{2} + Z_1\right] + Q$$

$$= m_2\left[h_2 + \frac{V_2^2}{2} + Z_2\right] + W$$

For nozzle
We assume that
$Z_1 = Z_2$, $Q = 0$, $W = 0$
then we get

$$(h_1 - h_2) = \frac{V_2^2 - V_1^2}{2}$$

417. If H be the heat supplied to the system to do work W and change in internal energy as ΔU, then,

(a) $H = \Delta U + W$ (b) $\Delta U = H + W$
(c) $W = H + \Delta U$ (d) $H = W/\Delta U$

SJVN ET 2013

Ans. (a) : First Law of thermodynamics

$$\boxed{H = \Delta U + W}$$

where, H = Heat supplied
ΔU = Internal energy
W = work

418. A machine produces 100 kJ work by consuming 100 kJ heat. This machine will be called:-

(a) PMM - I (b) PMM - II
(c) PMM - III (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (b) : PMM-II (perpectual motion machine of second kind) is one which converts all the heat input into work while working in a cycle. A PMM-II has an η_{th} of 1.

419. Air is compressed isothermally by performing work equal to 16 kJ upon it. The change in internal energy is

(a) –16 kJ (b) Zero
(c) 16 kJ (d) 32 kJ

UKPSC AE 2012 Paper–II

Ans. (b) For isothermal process,
$T = C$; $\Delta T = 0$
$\Delta U = mC_v\Delta T$
$\Delta U = 0$

420. General energy equation for flow process is

(a) $dQ = du + pdv$
(b) $dQ = dh - vdp$
(c) $du + d(pv) + d\left(\frac{v^2}{2}\right) + d[gz] + dQ + dW = 0$
(d) $dQ = Tds$

TNPSC ACF 2012

Ans. (c) : Energy equation for flow process

$$du + d(pv) + d\left(\frac{v^2}{2}\right) + d(gz) + dQ + dW = 0$$

421. The heat transfer, the work done and the change in internal energy are all zero in case of

(a) a rigid vessel containing steam at 150ºC left in the atmosphere which is at 25ºC
(b) 1 kg gas contained in an insulated cylinder expanding as the piston moves slowly outwards
(c) 1 kg of air flowing adiabatically from atmosphere into a previously evacuated bottle
(d) a rigid vessel containing ammonia gas connected through a valve to an evacuated rigid vessel, the vessel, the valve and the connecting pipes being well insulated and the valve being opened slowly

APPSC AEE 2012

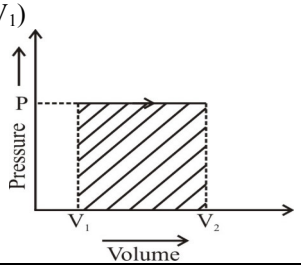
Ans. (d) :
(i) For a rigid vessel $\Delta V = 0$ then
Work done $dW = P\Delta V = 0$
(ii) Vessel are well insulated $(\Delta Q) = 0$
(iii) hence from first law of thermodynamics equation.
$dQ = dU + dW$
$0 = dU + o \Rightarrow dU = 0$

422. The work done in the expansion of a gas from volume V_1 to V_2 under constant pressure P is equal to

(a) Zero (b) $P(V_2 - V_1)$
(c) $P(V_2 + V_1)$ (d) $P/(V_2 - V_1)$

APPSC AEE 2012

Ans. (b) : Work done for non-flow process at constant pressure.
$W = PdV$
$W = P(V_2 - V_1)$



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423. Choose the correct alternative:

1. First law for a closed system undergoing a cycle $Q - W = \Delta E$
2. Two reversible adiabatic paths can not intersect each other.
3. If two fluids are mixed, the entropy of universe remains unchanged.
4. Clausius statement – Heat can flow from low to high temperature body without the aid of external work.
5. The efficiency of a reversible heat engine is independent of nature of working substance undergoing a cycle.

(a) All statements are true.
(b) Statements 2 and 5 are true.
(c) Statements 1, 3, 4 and 5 are true.
(d) Statements 1, 3 and 5 are true.

UKPSC AE 2012 Paper–II

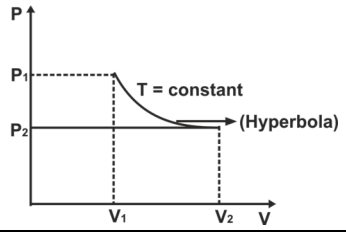
Ans. (b) : Statements 2 and 5 are true.

424. A gas is to expanded in a cylinder that its temperature remains constant. The resulting variation of pressure vs. volume is

(a) A parabola
(b) A hyperbola
(c) A straight line through origin
(d) None of these

ISRO Scientist/Engineer 2011

Ans. (b) : Temperature remain constant
PV = constant



425. Internal energy of an isolated system :

(a) Increases with addition on heat
(b) Increases with increase in mass of the system
(c) Remains same
(d) Increases with addition of work

OPSC Civil Services Pre. 2011

Ans. (c) : Internal energy of an isolated system remains same.
According to first law of thermodynamics
$\delta Q = dU + \delta W$
For isolated system
$\delta W = 0$
$\delta Q = 0$
$dU = 0$
Internal energy of perfect gas is only function of temperature only.

426. The identity $\delta Q = dU + PdV$ is valid for :

(a) Any process occurring in an open system
(b) Any process occurring in a closed system
(c) A quasi-static process without dissipation in a closed cycle
(d) Any process in any system

OPSC Civil Services Pre. 2011

Ans. (c) : $\delta Q = dU + PdV$
This equation is valid for closed system, quasi-static process without dissipation in a closed cycle and reversible process.

427. What is the following statement known as "Energy can neither be created nor destroyed but it can be transferred from one form to another?

(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) Kinetic theory of gases

VIZAG STEEL MT 2011

Ans. (b) : First Law of Thermodynamics-
⇒ Energy can neither be created nor destroyed but can be converted from one form to other.
⇒ It states that whenever a closed system executes a cycle process, the algebraic sum of work transfer is proportional to algebraic sum of heat transfer.

$$(\Sigma W)_{cycle} = (\Sigma Q)_{cycle}$$

428. 10 kg of air at 800 kPa are heated at a constant pressure from 450 K to 680 K. The work required is nearest to :

(a) 524 kJ (b) 960 kJ
(c) 715 kJ (d) 660 kJ

J&K PSC Civil Services Pre, 2010

Ans. (d) : Given,
For air,

$$m = 10 \text{ kg}, R = 0.287 \frac{\text{kPa.m}^3}{\text{kg.K}}$$

$P = 800$ kPa
$T_i = 450$ K and $T_f = 680$ K
For constant pressure,
Work done (W) = $P\Delta V$
$= P(V_f - V_i)$
From,

$$\therefore T_i = \frac{PV_i}{mR}, \Delta T = \frac{P}{mR}(V_f - V_i)$$

$$(680 - 450) = \frac{800}{10 \times 0.287}(V_f - V_i)$$

$(V_f - V_i) = 0.8251 \text{ m}^3$
$\therefore$ Work done (W) = $P\Delta V = P(V_f - V_i)$
$= 800 (0.8251)$
$= 660.10$ kJ

429. An air compressor compresses air with an enthalpy of 100 kJ/kg to a pressure and temperature that have an enthalpy of 200 kJ/kg. If 50 kJ/kg of heat is lost from the compressor as the air passes through it, the power required for an air mass flow of 0.5 kg/s is :

(a) 25 kW (b) 50 kW
(c) 75 kW (d) 100 kW

UPSC JWM Advt. No.-52/2010

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Ans. (c) : SFEE : applying

$h_1 + q = h_2 + W$ (for negligible KE and PE)

$\Rightarrow W = (h_1 - h_2) + q = (100 - 200) + (-50)$

Compressor work (W) = – 150 KJ/kg

$\therefore$ Compressor Power (P) = mW

= 0.5kg/sec × 150 kJ/kg = 75 kW

430. Steam at a pressure of 6 bars (enthalpy 2988 kJ/kg) passes through a frictionless nozzle to a pressure of 4 bars (enthalpy 2888 kJ/kg). Neglecting velocity of approach, the final velocity of steam at exit of nozzle will be

(a) 447.2 m/s (b) 44.72 m/s

(c) 4.472 m/s (d) 4472 m/s

UPSC JWM Advt. No.-50/2010

Ans. (a) : Given,

$P_1 = 6$ bar, $h_1 = 2988$ kJ/kg, $h_2 = 2888$ kJ/kg

Using S.F.E.E. for nozzle –

$$h_1 = h_2 + \frac{v_2^2}{2}$$

$$v_2 = \sqrt{2\times(2988-2888)\times10^3} = \sqrt{2\times100\times10^5}$$

$$\boxed{v_2 = 447.2\,m/s}$$

431. An inventor claims to have constructed a device that rejects 300 kJ of heat to a single reservoir while absorbing 300 kJ of work during a single cycle of the device. This device violates

(a) only the first law of thermodynamics and no other laws

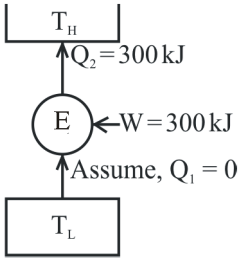
(b) only the second law of thermodynamics and no other laws

(c) both the first and the second law of thermodynamics

(d) neither the first nor the second law of thermodynamics

DRDO Scientists 2008

Ans. (a) : It is case of electric heater and it is possible without violating any law of thermodynamics.



432. One hundredth of a kilogram of air is compressed In a piston-cylinder device. At an instant of time when T= 400K, the rate at which work is being done on the air is 8.165 kW, and heat is being removed at a rate of 1.0 kW, the rate of temperature rise will be -

(a) 10 K/s (b) 100 K/s

(c) 1000 K/s (d) 10000 K/s

ISRO Scientist/Engineer 2006

Ans. (c) : Given, $m_{air} = \frac{1}{100} kg$

$\delta\dot{W} = -8.165$ kW

$\delta\dot{Q} = -1$ kW

According to first law of thermodynamics.

$$\delta\dot{Q} = d\dot{U} + \delta\dot{W}$$

$$-1 = \frac{mC_V dT}{t} - 8.165$$

$$\frac{mC_V dT}{t} = 7.165$$

So, $\frac{dT}{t} = \frac{7.165}{0.01\times0.718} \simeq 1000\,K/s$

For air $C_V = 0.718$ kJ/kg K

433. For an ideal gas, the change in enthalpy, dh is given by

(a) $C_v dT$ (b) $C_p dT$

(c) $\frac{C_p}{C_v} dT$ (d) $\frac{C_v}{C_p} dT$

WBPSC AE 2003

Ans. (b) : [P = C] for constant pressure.

$\because$ $\delta Q = dU + PdV$

$\delta Q = dU + d(PV)$

$= d(U + PV)$ $\because$ U+PV = h

$\delta Q = d(h)$ ...(i)

$\delta Q = mc_p dT$

$\boxed{dh = mc_p dT}$

or $[dh = C_p dT]$

Where, C_p = Heat capacity

3. Second Law of Thermodynamics

434. Carnot cycle consists of

(a) Two constant volume & two isentropic processes

(b) Two isothermal and two isentropic processes

(c) Two constant pressure and two isentropic processes

(d) One constant volume, one constant pressure and two isentropic processes

Assam PSC AE (IWT) 14.03.2021
ISRO Scientist/Engineer (RAC) 22.04.2018
TNPSC AE 2018
Vizag Steel (MT) 2017
Karnataka PSC Lect. 27.05.2017
Kerala PSC AE 05.04.2017
ISRO Scientist/Engineer (RAC) 29.11.2015
ISRO Scientist/Engineer 11.10.2015
TSGENCO AE 14.11.2015
KPSC AE 2015, TSPSC AEE 2015
Kerala PSC AE 06.08.2015
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
WBPSC AE 2008

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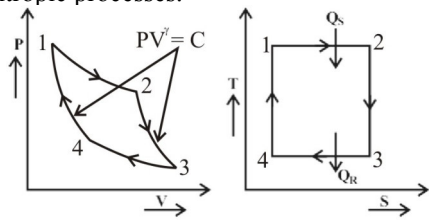
J & K PSC Screening, 2006
OPSC Civil Services Pre 2006

Ans. (b) : Carnot cycle consists of two isothermal and two isentropic processes.



Processes
1 - 2 → Isothermal expansion
2 - 3 → Isentropic expansion
3 - 4 → Isothermal compression
4 - 1 → Isentropic compression

435. The second law of thermodynamics defines
(a) enthalpy (b) internal energy
(c) entropy (d) specific heat

RPSC ACF & FRO, 26.02.2021
UPRVUNL AE 05.07.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC IOF, 2020
GPSC Engineer, Class-II Pre-19.01.2020
TSPSC Manager (Engg.) HMWSSB 12.11.2020
HPPSC AE 2018, HPPSC LECT. 2016
GPSC Lect. 23.10.2016
Rajasthan AE (Nagar Nigam) 2016 Shift-3
Assam PSC CCE Pre 2015
GWSSB AAE, 27.12.2015, APPSC Poly. Lect. 2013
UKPSC AE 2012 Paper-II, WBPSC AE 2008
J & K PSC Screening, 2006
WBPSC AE 2003

Ans. (c) : Entropy– It is an important thermodynamic property of a working substance, which increases with the addition of heat and decreases with its removal. As a matter of fact, it is tedious to define the term entropy. But it is comparatively easy to define change of entropy of a working substance. Over small range of temperature, the increase or decrease of entropy, when multiplied by the absolute temperature gives the heat absorbed or rejected by the working substance. Mathematically, heat absorbed or rejected by the working substance.

$\delta Q = Tds$

T = Absolute temperature
ds = Increase or decrease in entropy.

436. Kelvin-Planck law deals with
(a) conservation of energy
(b) conservation of heat
(c) conservation of mass
(d) conversion of heat into work

Assam PSC AE (IWT) 14.03.2021
GPSC DEE Class-2 (GWSSB) 04.07.2021
Oil India Limited Sr. Engineer (Drilling) 30.11.2019
GPSC Executive Engineer 23.12.2018
WBPSC AE, 2017
Rajasthan Nagar Nigam AE 2016, Shift-I
UPPSC AE, 12.04.2016 Paper-II
TSGENCO AE 14.11.2015
APPSC Poly. Lect. 2013
UKPSC AE 2012, 2007 Paper-II

Ans. (d) : The Kelvin-Planck statement of the second law states. It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.
So, Kelvin-Planck law deals with conversion of heat into work.

437. The transfer of heat takes place from the body of high temperature to the body of low temperature as per
(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) Stefan's law

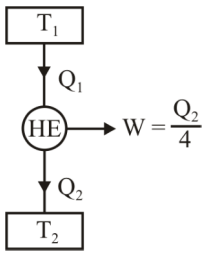
UPCL AE 29.08.2021, CGPSC AE 15.01.2021
JPSC AE, 10.04.2021, Paper - I
TPSC Manager (Engg.) HMWSSB 12.11.2020
Sikkim PSC (Under Secretary), 2017
CGPSC AE 16.10.2016
Rajasthan Nagar Nigam AE 2016, Shift-II
Rajasthan Nagar Nigam AE 2016, Shift-I
RRB SSE 02.09.2015 Shift-III
MPPSC AE 2016, TRB Asstt. Prof., 2012
APPSC AEE 2012

Ans. (c) : According to second law of thermodynamics, the transfer of heat takes place from the body of high temperature to the body of low temperature.

438. For a heat engine operating on a Carnot cycle, the work output is 1/4th of the heat rejected to sink. The thermal efficiency of the engine would be
(a) 10% (b) 20%
(c) 30% (d) 50%

JPSC AE, 10.04.2021, Paper-I
UPPSC AE 13.12.2020, Paper-2
ISRO Scientist/Engineer (RAC) 12.01.2020
GPSC ARTO Pre 30.12.2018
RPSC VPITI 14.02.2016
Rajasthan Nagar Nigam AE 2016, Shift-II
ESE 2003

Ans. (b)



$W = Q_1 - Q_2$

$Q_1 = W + Q_2 = \frac{Q_2}{4} + Q_2$

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$$Q_1 = \frac{5Q_2}{4}$$

$$\eta_{HE} = 1 - \frac{Q_2}{Q_1}$$

$$\Rightarrow 1 - \frac{Q_2}{\left(\frac{5Q_2}{4}\right)} = 1 - \frac{4}{5}$$

$$\Rightarrow \frac{1}{5} = 20\%$$

439. Which of the following is correct according to Clausius statement of second law of thermodynamics?

(a) Heat can be transferred from low temperature source to high temperature source without any work from external source

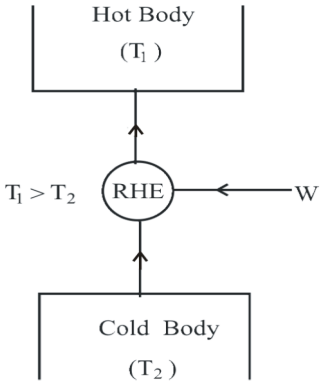
(b) It is impossible to transfer heat in a cyclic process from low temperature to high temperature without any work from external source

(c) Heat can be transferred from low temperature source to high temperature source if COP of process is more than unity

(d) Heat can be transferred from low temperature source to high temperature source with work being done from external source

Assam PSC AE (IWT) 14.03.2021
MECON MT 2019
Karnataka PSC AE, 10.09.2017
Rajasthan Nagar Nigam AE 2016, Shift-I
Rajasthan Nagar Nigam AE 2016, Shift-II
TNPSC AE, 2008

Ans. (b) : Clausius statement – It is impossible to transfer heat in a cyclic process from low temperature to high temperature without any work from external source.



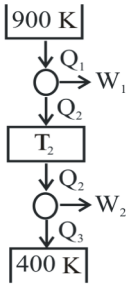
440. An engine operates between temperature limits of 900 K and T_2 and the other engine operates between T_2 and 400 K. For both engines to be equally efficient, T_2 should be equal to

(a) 600 K (b) 625 K
(c) 650 K (d) 700 K

BPSC AE Mains 2017 Paper - V

GPSC Lect. (Poly.) 07.12.2014
ISRO Scientist/Engineer 12.05.2013
UKPSC AE-2013, Paper-II, ESE 2010

Ans. : (a) : Given as,
$T_1 = 900$ K
$T_3 = 400$ K
If the efficiency are equal then we know that



$$T_2 = \sqrt{T_1 \times T_3}$$
$$T_2 = \sqrt{900 \times 400}$$
$$T_2 = 600\text{K}$$

441. Two Carnot heat engines are operating in series such that the heat sink of the first engine serves as the heat source of the second one. If the source temperature of the first engine is 1327°C, the sink temperature of the second engine is 127°C, and the thermal efficiency of both the engines are the same, the temperature of the intermediate reservoir is:

(a) 627°C (b) 410°C
(c) 577°C (d) 527°C

ISRO Scientist/Engineer (RAC) 22.04.2018
ISRO Scientist/Engineer 07.05.2017
Vizag Steel (MT) 2017, Punjab PSC (Lect.) 06.08.2017
OPSC AEE 2015 PAPER - II

Ans. (d) : Given,
$T_1 = 1327°C = 1600$ K
$T_3 = 127°C = 400$ K
If two engine connected in series and its thermal efficiency is same.
So, Intermediate temperature

$$T_2 = \sqrt{T_1 \times T_3}$$
$$= \sqrt{1600 \times 400} = 40 \times 20$$
$$= 800 \text{ K}$$
$$T_2 = 800 - 273$$
$$= 527°C$$

442. A solar energy based heat engine which receives 80 kJ of heat at 100°C and rejects 70 kJ of heat to the ambient at 30°C is to be designed. The thermal efficiency of the heat engine is :

(a) 70% (b) 1.88%
(c) 12.5% (d) Indeterminate

ISRO Scientist/Engineer (RAC) 22.04.2018
HPPSC Asstt. Prof. 18.11.2016
ISRO Scientist/Engineer 11.10.2015, GATE 1996

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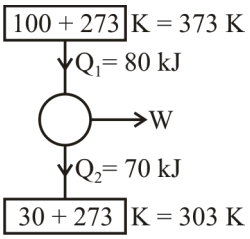
Ans. (c)



$$\eta_{actual} = \frac{W}{Q_s} \times 100$$

$$\eta_{actual} = \frac{80-70}{80} \times 100 = \frac{10}{80} \times 100$$

$$= 12.5\%$$

443. "It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature." This statement is known as:

(a) Clausius statement
(b) Kelvin-Plank statement
(c) Carnot statement
(d) None of the above

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Punjab PSC SDE 12.02.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
WBPSC AE 2003

Ans. (b) : Kelvin-Plank statement-
It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.

444. The efficiency of a Carnot engine depends on the

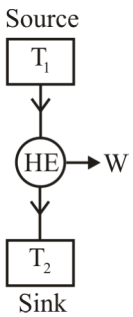
(a) Design substance
(b) Working substance
(c) Type of fuel fired
(d) Temperatures of source & sink

UPCL AE 29.08.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
Rajasthan Nagar Nigam AE 2016, Shift-I
UKPSC AE 2012 Paper-II, TRB Asstt. Prof., 2012

Ans. (d) : The efficiency of a carnot engine depends on the temperature of source and sink.

$$\eta = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}$$

Carnot engine is reversible engine so, $\eta = f(T_1, T_2)$



445. Carnot cycle efficiency is maximum when

(a) initial temperature is 0K
(b) final temperature is 0K
(c) difference between initial and final temperature is 0K
(d) final temperature is 0^oC

JPSC AE PRE 2019, TNPSC AE 2018
WBPSC AE, 2017, TNPSC ACF 2012

Ans. (b) : We know that Carnot cycle efficiency depends on source and sink temperature limits,

$$\eta_c = 1 - \frac{T_L}{T_H}$$

If $T_L \to 0$ K then

$$(\eta_c)_{max} = 100\%$$

446. What will be the change on efficiency of a Carnot engine, if the temperature of the sink is decreased?

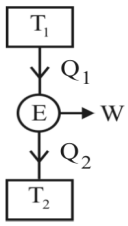
(a) It will decreases
(b) It will remain constant
(c) It will first increase and then decrease
(d) It will increases

Rajasthan Nagar Nigam AE 2016, Shift-I
APPSC AEE Mains 2016 Paper-III
MPPSC State Forest Service Exam, 2014

Ans. (d) : Carnot engine efficiency–

$$\eta = 1 - \frac{T_2}{T_1}$$

$T_2 \downarrow$, $\eta \uparrow$ (Assumed T_1 = constant)



447. A friction less heat engine can be 100 percent efficient only if its exhaust temperature is

(a) equal to its input temperature
(b) less than input temperature
(c) 0 K
(d) 0ºC

JPSC AE 2013 Paper-V
ISRO Scientist/Engineer 2012
UKPSC AE 2012 Paper–II

Ans. (c)

$$\eta = 1 - \frac{T_2}{T_1} \text{ (for frictionless engine)}$$

$\eta = 100\%$ (if, $T_2 = 0$ K)

448. If the thermal efficiency of Carnot heat engine is 50 percent, then coefficient of performance of a refrigerator working within the same temperature limit would be :

(a) 1 (b) 2
(c) 3 (d) 4

HPPSC Asstt. Prof. 18.09.2017
Kerala PSC AE 06.08.2015, *(KPSC AE. 2015)*

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Ans. (a) : Thermal efficiency of Carnot heat engine

$$\eta_c = \frac{Q_1 - Q_2}{Q_1} = \frac{T_H - T_L}{T_H} = 1 - \frac{T_L}{T_H} \quad ...(i)$$

Coefficient of performance (COP) of refrigerator

$$COP_R = \frac{T_L}{T_H - T_L} \quad ...(ii)$$

$$COP_R = \frac{T_L / T_H}{1 - T_L / - T_H} = \frac{T_L / T_H}{\eta_c} \quad \text{[from eq. (i)]}$$

from eqn. (i) $\left[1 - \eta_c = \frac{T_L}{T_H}\right] \leftarrow$ keep it above eqn

$$COP_R = \frac{T_L / T_H}{\eta_c} = \frac{1 - \eta_c}{\eta_c}$$

$$COP_R = \frac{1}{\eta_c} - 1$$

$$COP_R + 1 = \frac{1}{\eta_c}$$

$\because$ in question $\eta_c = 50\%$ $COP_R = ?$

$$0.5 = \frac{1}{COP_R + 1}$$

$$COP_R + 1 = 2$$

$$\boxed{COP_R = 1}$$

449. A reversible engine operates between temperatures 900 K & T_2 (T_2 < 900 K), & another reversible engine between T_2 & 400 K (T_2 > 400 K) in series. What is the value of T_2 if work output of both the engines are equal?

(a) 600 K (b) 625 K
(c) 650 K (d) 675 K

Nagaland PSC CTSE 2015 Paper-II
SJVN ET 2013, ESE 2005

Ans. (c) : Given,

$T_1 = 900$ K, $T_2 = 400$ K

Work output of both the engine are equal

$W_1 = W_2$

temperature difference should be equal

$900 - T_2 = T_2 - 400$

$2T_2 = 1300$

$T_2 = 650$ K

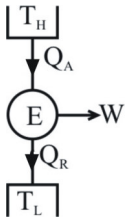
450. Three engines A, B and C operating on Carnot cycle respectively use air, steam and helium as the working fluid. If all the engines operate within the same high and low temperature limits. then which engine will have the highest efficiency?

(a) Engine A
(b) Engine B
(c) Engine C
(d) All engines will have the same efficiency

RPSC AE 2016, RPSC AEN Pre-2013
RPSC ACF-2011, ESE 2009

Ans. : (d)

Efficiency of Carnot cycle $(\eta) = \frac{T_H - T_L}{T_H}$



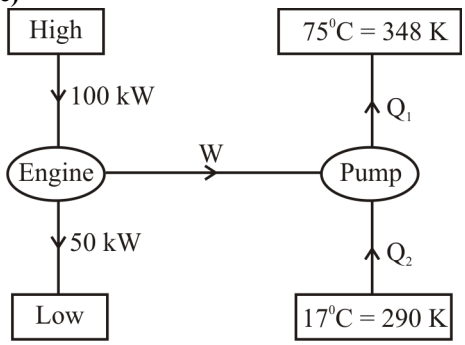
Efficiency of Carnot engines does not depend on the type of fuel use so engine A, B and C operating same high and low temperature limit then engine will have the same thermal efficiency.

451. An irreversible heat engine extracts heat from a high temperature source at a rate of 100 kW and rejects heat to a sink at a rate of 50 kW. the entire work output of the heat engine is used to drive a reversible heat pump operating between a set of independent isothermal heat reservoirs at 17°C and 75°C. The rate (in kW) at which the heat pump delivers heat to its high temperature sink is

(a) 50 (b) 250
(c) 300 (d) 360

GPSC ARTO Pre 30.12.2018
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012

Ans. (c)



$W = Q_1 - Q_2 = 100 - 50 = 50$ kW

$$(COP)_{pump} = \frac{348}{348 - 290} = \frac{Q_1}{W}$$

$$Q_1 = \frac{348 \times 50}{58}$$

$$\boxed{Q_1 = 300\,kW}$$

452. A reversible engine has ideal thermal efficiency of 30%. When it is used as refrigerating machine with all other conditions unchanged, the coefficient of performance will be

(a) 1.33 (b) 2.33
(c) 3.33 (d) 4.33

ISRO Scientist/Engineer (RAC) 07.05.2017
HPPSC Asstt. Prof. 18.09.2017
ESE 2013

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Ans. (b) : We know that for same temperature limt

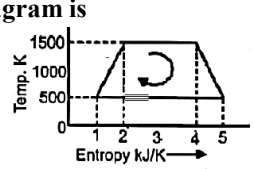
$$(COP)_{HP} = \frac{1}{\eta_C} = \frac{1}{0.3}$$

$$(COP)_{HP} = 3.33$$

$$(COP)_{ref} = (COP)_{HP} - 1$$

$$(COP)_{ref} = 3.33 - 1 = 2.33$$

453. The efficiency of a reversible cyclic process undergone by a substance as shown in the given diagram is



(a) 0.40 (b) 0.55
(c) 0.60 (d) 0.80

Nagaland PSC CTSE 2015 Paper-II, CSE Pre-1994

Ans. (c) : $\text{Efficiency} = \frac{\text{Area under 500 and 1500}}{\text{Area under 0 and 5}}$

$$= \frac{\frac{1}{2}\times\left[\{(5-1)+(4-2)\}\times(1500-500)\right]}{\frac{1}{2}\times\left[\{(5-1)+(4-2)\}\times(1500-500)\right]+(5-1)\times 500}$$

$$= \frac{3000}{5000} = 0.60$$

454. A carnot cycle refrigerator operates between 250 K and 300 K. Its coefficient of performance is

(a) 6.0 (b) 5.0
(c) 1.2 (d) 0.8

ISRO Scientist/Engineer 11.10.2015
VIZAG STEEL MT 10.06.2012

Ans. (b) : In carnot cycle,

$T_L = 250$ K, $T_H = 300$ K

$$(COP)_{Refrigerator} = \frac{T_L}{T_H - T_L}$$

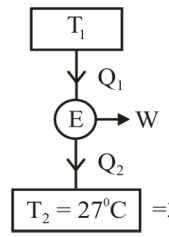
$$= \frac{250}{300-250} = \frac{250}{50} = 5$$

455. A Carnot engine operates between a source and sink. If 40% of heat is rejected at 27^0C, what will be the source temperature?

(a) 67^0C (b) 227^0C
(c) 477^0C (d) 757^0C

Assam PSC CCE Pre 2015, ESE 2009

Ans. (c)



The heat engine with source and sink is

$Q_2 = 0.4Q_1$

From carnot theorem for reversible engine -

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$\frac{Q_1}{T_1} = \frac{0.4Q_1}{300} \Rightarrow 0.4T_1 = 300$$

$$T_1 = \frac{300}{0.4}$$

$$= 750 \text{ K} = 477^0\text{C}$$

456. For a heat engine operating on Carnot cycle, the work output is 25% of heat rejected to the sink. The thermal efficiency of the engine would be :

(a) 10% (b) 20%
(c) 30% (d) 50%

RPSC AE 2016, RPSC ACF-2011

Ans. (b) : Given,

$$W_{out} = 25\% \text{ of } Q_R$$

$$W_{out} = \frac{25}{100}Q_R$$

$$W_{out} = \frac{1}{4}Q_R \quad \ldots(i)$$

we know, that

$$W = Q_S - Q_R$$

From (i) $\frac{1}{4}\times Q_R = Q_S - Q_R$

$$\frac{Q_R}{4} + Q_R = Q_S$$

$$Q_S = \frac{5}{4}Q_R \Rightarrow \frac{Q_R}{Q_S} = \frac{4}{5}$$

$$\eta_{Carnot} = 1 - \frac{Q_R}{Q_S}$$

$$= 1 - \frac{4}{5} \Rightarrow 1 - 0.8 = 0.20 = 20\%$$

457. The efficiency of a reversible heat engine depends on :

(a) the nature of the working fluid
(b) the quantity of the working fluid
(c) the temperature of the working fluid
(d) the phase of the working fluid

NLC GET 17.11.2020, Shift-II
RPSC IOF & Boiler 20.09.2017

Ans. (c) : The efficiency of reversible heat engine in which heat is received slowly at T_1 is found to be

$$\eta_{Rev} = \eta_{max} = 1 - \left(\frac{Q_2}{Q_1}\right)_{Rev} = 1 - \frac{T_2}{T_1}$$

$$\eta_{Rev} = \frac{T_1 - T_2}{T_1} \quad ; \eta_{rev} = f(T_1, T_2)$$

It is observed that as T_2 decreases and T_1 increases the efficiency of the reversible cycle increases.

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Since η is always less than unity.

T_2 is always greater than zero and positive since efficiency depends on operating temperature efficiency of all reversible heat engines operating between the same heat reservoirs are the same, the efficiency of the reversible heat engine, is independent of the nature or amount of the working substance undergoing the cycle.

458. One reversible heat engine operates between 1000 K and T K and another reversible heat engine operates between T K and 400 K. If both heat engines have same heat input and output, the value of T is

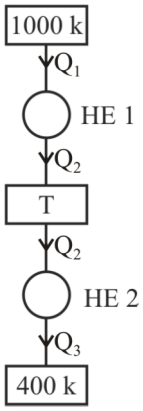
(a) 582.7 K (b) 632.5 K

(c) 682.8 K (d) 732.5 K

UPPSC AE 13.12.2020, Paper-2

ESE 2016

Ans. (b)



For both heat engines, have same heat input and output.

So, $T = \sqrt{T_1 T_2}$

$= \sqrt{(1000)(400)}$

$T = 632.5\ K$

459. A Carnot engine receiving heat at 400 K has an efficiency of 25%. The COP of a Carnot refrigerator working between the same temperature limits is :

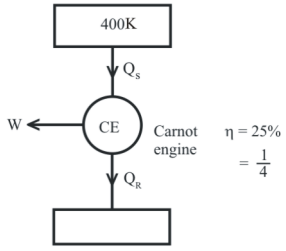
(a) 1 (b) 2

(c) 3 (d) 4

Vadodara Muncipal Corp. DEE, 2018

TSPSC AEE 2017

Ans. (c)



$$\eta_{HE} = \frac{1}{(COP)_{Heat\ pump}}$$

$$(COP)_{Heat\ Pump} = \frac{1}{\eta_{engine}}$$

$$= \frac{1}{1/4} = 4$$

$$(COP)_{Heat\ pump} = (COP)_{RE} + 1$$

$$4 = (COP)_{RE} + 1$$

$$\boxed{(COP)_{RE} = 4 - 1 = 3}$$

460. If the thermal efficiency of a Carnot engine is 1/5, the COP of a Carnot refrigerator is :

(a) 4 (b) 5

(c) 6 (d) 3

J&K PSC Civil Services Pre, 2010

UPSC JWM Advt. NO. - 52/2010

Ans. (a) : Given,

$$(COP)_{H.P} = \frac{1}{\eta} = 5$$

$$(COP)_R = (COP)_{H.P} - 1$$

$$(COP)_R = 5 - 1 = 4$$

461. An inventor claims that his heat engine has the following specifications :
Power developed : 50 kW
Fuel burned per hour : 3 kg
Heating value of fuel : 75000 kJ/kg
Temperature limits : 627^0C and 27^0C
This heat engine is

(a) reality (b) impossible

(c) costly (d) imaginary

RPSC IOF, 2020

TRB Asstt. Prof., 2012

CSE Pre-2002

Ans. (b) : Given,

Power (P) = 50kW

Fuel Burned /hr = 3 kg/hr

or Fuel Burned /sec = $\frac{3}{3600}$ kg/sec

Heating value of fuel = 75000 kJ/kg

Temperature limit = 627°C and 27°C

$$\eta_{Theorectical} = \frac{P}{Q \times m} = \frac{50}{\frac{3}{3600} \times 75000} = 0.8$$

$$\boxed{\eta_{Theorectical} = 80\%}$$

$$\eta_{ideal} = \left(1 - \frac{T_L}{T_H}\right) \times 100 = \left[1 - \frac{(27+273)}{(627+273)}\right] \times 100$$

$$= \boxed{\eta_{ideal} = 66.67\%}$$

$$\boxed{\eta_{Theorectical} > \eta_{ideal}}$$

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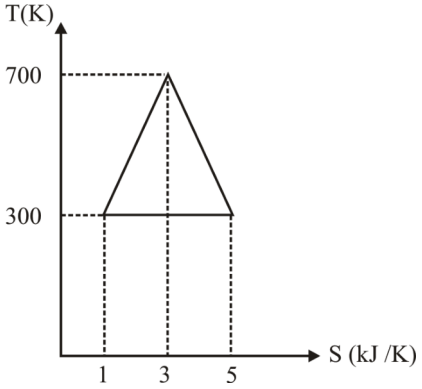
462. The thermal efficiency of the hypothetical cycle shown is



(a) 0.6 (b) 0.5
(c) 0.4 (d) 0.3

GPSC ARTO Pre 30.12.2018
ESE 2017

Ans. (c) : $Q_{rejected} = 300\,(5-1) = 300 \times 4 = 1200$ kJ

$$Q_{supplied} = \frac{1}{2}(700-300)(5-1) + 300(5-1)$$

$$= \frac{1}{2} \times 400 \times 4 + 1200$$

$$Q_{supp} = 2000\text{kJ}$$

$$\eta_{cycle} = 1 - \frac{Q_{rej}}{Q_{supp}} = 1 - \frac{1200}{2000} = 0.4$$

$$\boxed{\eta_{cycle} = 40\%}$$

463. According to the Clausius statement of second law of thermodynamics, the COP of a refrigerator is never

(a) ∞ (b) 1
(c) < 1 (d) > 1

Nagaland PSC (CTSE) 2016, Paper-II
Haryana PSC Civil Services Pre, 2014

Ans. (a) : According to the Clausius statement of second law of thermodynamics, the COP of a refrigerator is never be infinite (∞).

464. While deriving an expression for efficiency of a carnot cycle. Which one of the following is not a correct assumption?

(a) The cold body has infinite capacity
(b) The cylinder head is a perfect insulator
(c) The hot body has infinite capacity
(d) Both piston and cylinder are perfect insulators

UPRVUNL AE 21.08.2016
CSE Pre-2008

Ans. : (b) Assumption carnot cycle
(i) The gas in the cylinder is perfect gas.
(ii) The physical constants of gas in the engine cylinder are same as those at air at moderate temperature.
(iii) All the compression and expansion process are adiabaitc and they take place without any friction.
(iv) Cylinder head is a perfect diathermic.

465. A Carnot engine rejects 30% of absorbed heat to a sink at 30°C. The temperature of the heat source is

(a) 100°C (b) 433°C
(c) 737°C (d) 1010°C

UJVNL AE 2016
ISRO Scientist/Engineer 12.05.2013, ESE 2010

Ans. (c) : Given, Q_{rej} = 30% of $Q_{supp.}$

$$Q_{rej} = 0.3\ Q_{supp.}$$

$$\eta = 1 - \frac{Q_{rej}}{Q_{supp.}}$$

$$\eta = 1 - 0.3$$

$$\eta = 0.7$$

$$\eta = \left(1 - \frac{T_2}{T_1}\right)$$

So, $$0.7 = 1 - \frac{303}{T}$$

or $$\frac{303}{T} = 0.3$$

or $$T = 1010\ K$$

$$T = 1010 - 273$$

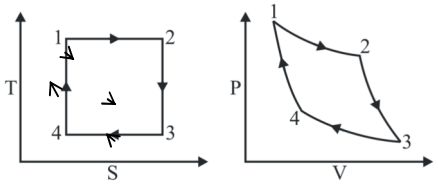
$$T = 737°C$$

466. The area of a p-v diagram for a Carnot cycle represents

(a) Heat supplied (b) Heat rejected
(c) Work done (d) Temperature drop

UPPSC AE 12.04.2016 Paper-II, TPSC AE 2015

Ans. : (c) The area on a P-V diagram for a Carnot cycle represents work done.



(i) Isothermal expansion:-

$$Q_{1-2} = P_1 V_1 \ln(V_2 / V_1)\ \text{[Heat supplied]}$$

(ii) Isentropic expansion:-

Decrease in internal energy = $\frac{mR(T_2 - T_3)}{\gamma - 1}$

(iii) Isothermal compression:-

$$Q_{3-4} = P_3 V_3 \ln(V_3 / V_4)$$

(iv) Isentropic compression:-

Increase in internal energy = $\frac{mR(T_1 - T_3)}{\gamma - 1}$

467. A heat engine is supplied with 250 kJ/s of heat at constant fixed temperature of 227°C. The heat is rejected at 27°C. The cycle is reversible, if the amount of heat rejected is

(a) 273 kJ/s (b) 200 kJ/s
(c) 180 kJ/s (d) 150 kJ/s

ISRO Scientist/Engineer (RAC) 22.04.2018
UPPSC AE 12.04.2016 Paper-II
ESE 2009, CSE Pre-1995

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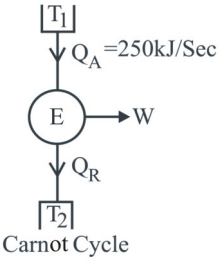
Ans. : (d)



$T_1 = 227+273=500K$
$T_2 = 27+273=300K$
$Q_A = 250$ kW
Efficiency of heat engine

$$\eta_\varepsilon = \frac{T_1 - T_2}{T_1} = \frac{Q_A - Q_R}{Q_A}$$ (Because Cycle is Reversible)

$$0.4 = \frac{250 - Q_R}{250}$$

$$\boxed{Q_R = 150 kJ/Sec}$$

468. The efficiency of Carnot cycle may be highest :

(a) By increasing the source temperature
(b) By increasing sink temperature
(c) By increasing source temperature and decreasing sink temperature simultaneously
(d) By decreasing source temperature and increasing sink temperature simultaneously

GPSC Engineer, Class-II Pre-19.01.2020
APPSC IOF, 2009

Ans. (c) : $\eta_{carnot} = \frac{T_H - T_L}{T_H} \times 100\% = 1 - \frac{T_L}{T_H} \times 100\%$

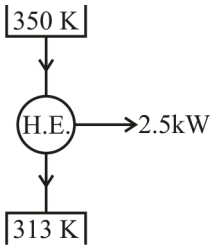
As T_H increases and T_L decreases, the $\frac{T_L}{T_H}$ decreases, and efficiency increases.

469. A solar collector receiving solar radiation at the rate of 0.6 kW/m^2 transforms it to the internal energy of a fluid at an overall efficiency of 50%. The fluid heated to 350 K is used to run a heat engine which rejects heat at 313 K. If the heat engine is to deliver 2.5 kW power, the minimum area of the solar collector required would be

(a) 8.33 m^2 (b) 16.66 m^2
(c) 39.68 m^2 (d) 79.36 m^2

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2004, ESE 2016

Ans. (d) :



Internal energy of fluid after observing solar radiation

$= 0.6 \times \eta = 0.6 \times 0.5 = 0.3$ kW/m^2

$$\eta = 1 - \frac{T_2}{T_1} = \frac{2.5}{Q_1} = 1 - \frac{313}{350}$$

$Q_1 = 23.648$ kJ
$23.809 = 0.3 \times A$
$A = 78.82 \simeq 79$ m^2

470. The efficiency of carnot cycle when it operates between the temperature of 373 and 273 K is:

(a) 0.268 (b) 0.366
(c) 0.134 (d) 0.183

APPSC Poly Lect. Automobile Engg., 2020
CIL MT 27.02.2020

Ans. (a) : The efficiency of carnot cycle

$$\eta_{carnot\ cycle} = \frac{T_H - T_L}{T_H}$$

Given, $T_H = 373$ K
$T_L = 273$ K

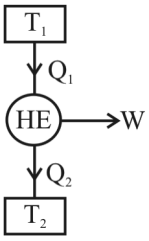
$$\eta_{carnot\ cycle} = \frac{373 - 273}{373}$$
$$= 0.268$$

471. An inventor claims a thermal engine operates between ocean layers at 27°C and 10°C. It produces 10 kW and discharges 9900 kJ/min. Such an engine is

(a) Impossible b) Reversible
(c) Possible (d) Probable

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (a) :



$T_1 = 27°C = 300$ K
$T_2 = 10°C = 283$ K
Maximum or **Carnot** efficiency

$$\eta_{max} = 1 - \frac{T_2}{T_1} = 1 - \frac{283}{300}$$
$$= 0.056 = 5.66\%$$

Actual efficiency,
$Q_s = W + Q_{rej.}$

$$Q_s = 10 + \frac{9900}{60} = 175 kW$$

$$\eta_a = \frac{W_{net}}{Q_s} = \frac{10}{175} = 5.71\%$$

Actual efficiency is more than the Carnot efficiency which is impossible.

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472. A heat engine is supplied with 2512kJ/min of heat at 650°C. Heat rejection with 900kJ/min takes place at 100°C. This type of heat engine is

(a) Ideal (b) Irreversible
(c) Impossible (d) Practical

GPSC Asstt. Director of Transport 05.03.2017
ESE 2012

Ans. (c) : Given, $T_1 = 373$ K, $T_2 = 923$ K

$Q_s = 2512$ kJ/min

$$\eta = 1 - \frac{Q_1}{Q_2} = 1 - \frac{T_1}{T_2}$$

$$\eta_a = 1 - \frac{900}{2512} = 64.17\%$$

$$\eta_{ideal} = 1 - \frac{373}{923} = 59.58\%$$

$\eta_a > \eta_{ideal}$

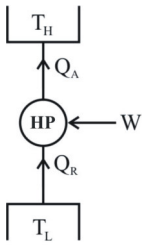
So, it is impossible.

473. A heat pump operating on Carnot cycle pump heat from a reservoir at 300 K to a reservoir at 600 K. The coefficient of performance is

(a) 1.5 (b) 0.5
(c) 2 (d) 1.0

TSPSC AEE 2017, MPPSC AE 2016

Ans. : (c)



$T_H = 600$K

$T_L = 300$K

$$(COP)_{HP} = \frac{T_H}{T_H - T_L}$$

$$(COP)_{HP} = \frac{600}{600 - 300} = 2$$

474. The efficiency of Carnot cycle is maximum for

(a) Reversible engine
(b) Gas engine
(c) Petrol engine
(d) Well lubricated engine

Vizag Steel (MT) 2017
Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : In reversible engine efficiency of carnot cycle is maximum.

$\because \eta_{reversible} > \eta_{irreversible}$

Each process is reversible $\oint\left(\frac{dQ}{T}\right) = 0$

475. According to Kelvin-Planck's statement of second law of thermodynamics.

(a) It is impossible to construct an engine working on a cyclic process, whose sole purpose is to convert heat energy into work.
(b) It is possible to construct an engine working on a cyclic process, whose on a cyclic process, whose sole purpose is to convert heat energy into work.
(c) It is impossible to construct a device which operates in a cyclic process and produces no effect other than the transfer of heat from a cold body to a hot body.
(d) None of the above

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : Second Law of Thermodynamics –

- **Kelvin Planck Statement–** It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.
- **Clausius Statement –** It is impossible to construct a device which operating in a cycle will produce no effect other than the transfer of heat from cooler body to a hotter body.

476. A heat engine is provided with a heat input of 1800 kJ/min and the output from the engine is 9 kW. Thermal efficiency of the engine is _____.

(a) 66.66% (b) 30%
(c) 70% (d) 33.33%

VIZAG Steel MT 24.01.2021

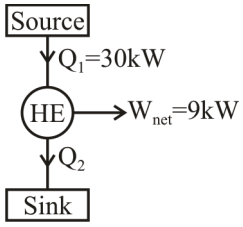
Ans. (b) : Given,

Heat input $(Q_1) = 1800$ kJ/min

$$= \frac{1800}{60} \text{kJ/sec}$$

$(Q_1) = 30$ kW

$W_{net} = 9$ kW



$\therefore$ Efficiency of heat engine $\eta_{HE} = \frac{W}{Q_1}$

$$= \frac{9}{30} \times 100 = 30\%$$

477. An engine operates between temperature limits of 900 K and T_2 and another between T_2 and 200 K. For both engine to have same efficiency, T_2 should be.

(a) 552 K (b) 320 K
(c) 424 K (d) 350 K

RPSC ACF & FRO, 26.02.2021

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Ans. (c) : If efficiency is same ($\eta_I = \eta_{II}$)

Intermediate temperature

$$T_2 = \sqrt{T_H T_L}$$

$$= \sqrt{900 \times 200}$$

$$\boxed{T_2 = 424.26K}$$

478. A Carnot Cycle operates between two temperature T_1 and T_2 ($T_1 > T_2$). If T_1 is increased by ΔT and T_2 is decreased by ΔT. The efficiency η_2 in the second case and the efficiency η_1 in the first case are related by :

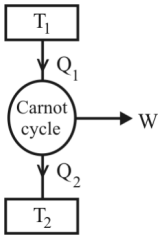
(a) $\eta_1 > \eta_2$ (b) $\eta_2 > \eta_1$

(c) $\eta_1 = \eta_2$ (d) Unpredictable

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : Efficiency of Carnot cycle

$$\eta_1 = 1 - \frac{T_2}{T_1}$$



$$\eta_2 = 1 - \frac{(T_2 - \Delta T)}{(T_1 + \Delta T)}$$

$$\boxed{\eta \uparrow \;\; T_2 \downarrow \& T_1 \uparrow}$$

$$\boxed{\eta_2 > \eta_1}$$

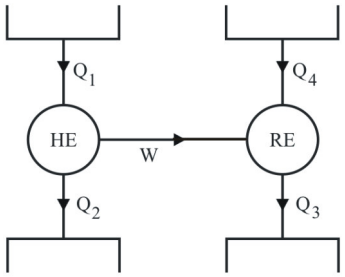
479. If an engine of 40% thermal efficiency drives a refrigerator having a coefficient of performance of 5, then the heat input to engine for each kJ of heat removed from the cold body of the refrigerator is :

(a) 0.50 kJ (b) 0.75 kJ

(c) 1 kJ (d) 1.25 kJ

RPSC ACF & FRO, 26.02.2021

Ans. (a) : Given,



$\eta_{HE} = 0.4$

$Q_4 = 1kJ$

$(COP)_{RE} = 5$

$$(COP)_{RE} = \frac{Q_4}{W} \qquad \eta = \frac{W}{Q_1}$$

$$5 = \frac{1}{0.4Q_1} \qquad W = 0.4Q_1$$

$$Q_1 = \frac{1}{0.4 \times 5}$$

$$\boxed{Q_1 = 0.5kJ}$$

480. Statement (I): A thermal energy reservoir is a system that always remains at constant temperature even though the heat is added to or removed from it.

Statement (II): A thermal reservoir that supplies heat energy is called sink and one that absorbs the heat energy is called source.

ESE 18.07.2021

Ans. (c) :

A thermal reservoir that supplies energy is termed as source and one that absorbs the heat energy is called sink.

481. An engine works on Carnot cycle. Initially it had efficiency of 40 percent. Keeping lower temperature fixed, higher temperature is raised by 10 percent. New efficiency will be:

(a) 35.35 percent (b) 30 percent

(c) 40 percent (d) 45.45 percent

AAI Jr. Executive 26.03.2021

Ans. (d) : $\eta_{carnot} = 1 - \frac{T_L}{T_H}$

$$0.4 = 1 - \frac{T_L}{T_H} \Rightarrow \frac{T_L}{T_H} = 1 - 0.4$$

$$= 0.6$$

$$T_L = 0.6\, T_H$$

$$(\eta_{carnot})_{New} = 1 - \frac{T_L}{(1.1T_H)}$$

$$(\eta_{carnot})_{New} = 1 - \left(\frac{0.6}{1.1}\right)$$

$= 45.45\%$

482. All reversible engines have _____ efficiencies.

(a) negative (b) positive

(c) zero (d) same

UPRVUNL AE 05.07.2021

Ans. (d) : All reversible engines have same efficiencies.

483. The coefficient of performance (COP) of a Carnot refrigerator is 5. If the refrigerator, operating between the same two thermal reservoirs, is used as a Carnot heat pump, the COP of the heat pump will be

(a) 4 (b) 5

(c) 6 (d) 8

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Ans. (c) : COP of heat pump

$COP_{HE} = 1 + COP_R$

$= 1 + 5$

$= 6$

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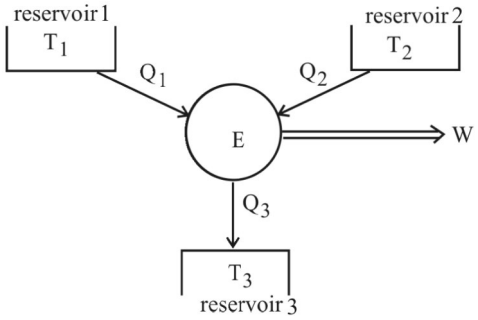
484. Which law of thermodynamics can be termed as law of degradation of energy?

(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) Third law of thermodynamics

VIZAG Steel MT 24.01.2021
Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (c) : The second law of thermodynamic is termed as law of degradation of energy. This law states if energy is transferred to the system then whole energy cannot be converted to the work, some energy will move into the sink.

Study the given figure and the following problem statement in order to answer the question no. 485-487.



Consider the heat engine E shown in the figure to be a reversible one. If you are given that T_1 = 600 K, T_2 = 540 K and T_3 = 300 K and Q_1 and Q_2 are 100 kJ and 90 kJ respectively, answer the following question.

485. The heat rejected Q_3 is

(a) 190 kJ (b) 10 kJ
(c) 100 kJ (d) 90 kJ

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : $T_1 = 600$ K
$T_2 = 540$ K
$T_3 = 300$ K
$Q_1 = 100$ kJ
$Q_2 = 90$ kJ
Energy balance
$Q_1 + Q_2 = W + Q_3$
second law of thermodynamics (Clausius Inequality)

$$\frac{Q_1}{T_1}+\frac{Q_2}{T_2}-\frac{Q_3}{T_3}=0$$

$$\frac{100}{600}+\frac{90}{540}=\frac{Q_3}{300}$$

$Q_3 = 100$ kJ

486. The work output W is

(a) Zero (b) 90 kJ
(c) 100 kJ (d) 180 kJ

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : Energy balance $Q_1 + Q_2 = W + Q_3$
100 + 90 = W + 100
W = 90 kJ

487. The thermal efficiency of the cycle E is

(a) 18/19 (b) 10/19
(c) zero (d) 9/19

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) :

$$\text{Thermal efficiency} = \frac{W}{Q_1+Q_2} = \frac{90}{100+90} = \frac{9}{19}$$

488. What is the highest possible theoretical efficiency of a heat engine operating with a hot reservoir of furnace gives at 1500 K when the cooling water available is at 300 K?

(a) 60% (b) 80%
(c) 50% (d) 70%

VIZAG MT, 14.12.2020

Ans. (b) : Highest possible theoretical efficiency

$$\eta_{carnot} = 1-\frac{T_{sink}}{T_{source}}$$

$$= 1-\frac{300}{1500}$$

$$= 0.8$$

$$\boxed{\eta_{carnot} = 80\%}$$

489. The efficiency of Carnot engine is 25%. The efficiency is increased to 30% when the sink temperature is reduced by 20^0C . What will be the source temperature?

(a) 300 K (b) 400 K
(c) 500 K (d) 600 K

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given data : Efficiency (η_1) = 0.25
(η_2) = 30 % = 0.30
Let, T_1 = Source temperature
T_2 = Sink temperature

$$\eta_{max} = 1-\frac{T_2}{T_1}$$

$$0.25 = 1-\frac{T_2}{T_1} \quad \text{..... (1)}$$

when sink temperature reduce by 20^0C then –

$$0.30 = 1-\frac{(T_2-20)}{T_1} \quad \text{.....(2)}$$

Subtract equation (1) from equation (2)

$$\Rightarrow \frac{-(T_2-20)}{T_1}+\frac{T_2}{T_1} = 0.05$$

$$\Rightarrow \frac{-T_2+20+T_2}{T_1} = 0.05$$

$$\Rightarrow \frac{20}{T_1} = 0.05$$

$$\Rightarrow T_1 = \frac{20}{0.05} = \frac{2000}{5}$$

$\Rightarrow \boxed{T_1 = 400\,K}$ source temperature

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490. A heat engine is supplied with 1200 kW of heat at a constant temperature of 600 K and it rejects 700 kW of heat at 300 K. Which of the following is true for this heat engine?

(a) It is reversible cycle
(b) It is an irreversible cycle
(c) It is an impossible cycle
(d) The efficiency of this cycle is greater than 50%

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given data

Heat supplied (Q_s) = 1200 kW; $T_H = 600$ K

Heat rejected (Q_R) = 700 kW; $T_L = 300$ K

$$\eta_1 = \left(\frac{Q_s - Q_R}{Q_s}\right)\times 100$$

$$= \frac{500}{1200}\times 100$$

$$\eta_1 = 41.66\%$$

$$\eta_2 = \left(1 - \frac{T_L}{T_H}\right)\times 100$$

$$= 1 - \frac{300}{600} \times 100$$

$$= \frac{1}{2}\times 100 = 50\%$$

$\because \eta_1 \neq \eta_2$

Therefore, It is an irreversible cycle.

491. A reversible heat engine is operating between 800°C and 30°C. The least rate of heat rejection per kW of net output of the engine is:

(a) 0.6 kW
(b) 0.8 kW
(c) 0.2 kW
(d) 0.4 kW

NLC GET 17.11.2020, Shift-II

Ans. (d) : Given, $T_1 = 800°C$
$= 1073K$
$T_2 = 30°C$
$= 303K$
$W_{net} = 1$ kW

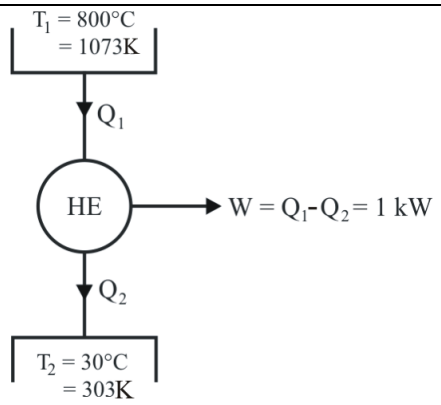
For reversible engine, the rate of heat rejection will be minimum

$$\eta_{max} = \eta_{Rev} = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{303}{1073}$$

$$= 0.718$$

Now, $\frac{W_{net}}{Q_1} = \eta_{max} = 0.718$



$$Q_1 = \frac{1}{0.718}$$

$$= 1.392kW$$

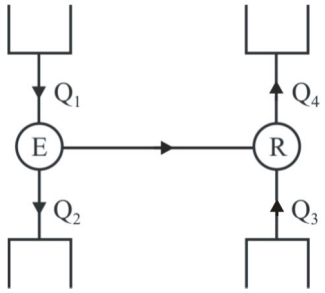
$$Q_2 = Q_1 - W_{net}$$

$$= 1.392 - 1$$

$$Q_2 = 0.392kW \approx 0.4kW$$

This is the least rate of heat rejection.

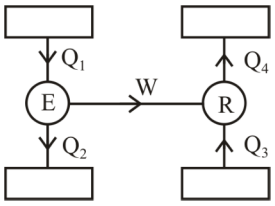
492. In the figure shown below, 'E' is the heat engine with efficiency of 0.4 and 'R' is the refrigerator, if $Q_2 + Q_4 = 3Q_1$, the COP of the refrigerator will be



(a) 3.0 (b) 4.5
(c) 5.0 (d) 5.5

UPPSC AE 13.12.2020, Paper-2, ESE 2014

Ans. (c) : Given,



$\eta_{HE} = 0.4$

$$\eta_{HE} = \frac{W}{Q_1} = 0.4$$

$$\boxed{W = 0.4Q_1}$$

$Q_1 - Q_2 = 0.4Q_1$

So, $Q_2 = 0.6\ Q_1$

$Q_2 + Q_4 = 3Q_1$

$0.6Q_1 + Q_4 = 3Q_1$

$$\boxed{Q_4 = 2.4Q_1}$$

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$$COP = \frac{Q_3}{W} = \frac{Q_4 - W}{W} = \frac{Q_4}{W} - 1$$
$$= \frac{2.4Q_1}{0.4Q_1} - 1$$
$$\boxed{COP = 5}$$

493. If a heat engine is receiving 900 kJ and operating at 60% efficiency, the heat rejected is:
(a) 160 kJ (b) 260 kJ
(c) 360 kJ (d) 540 kJ

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : Heat received $Q = 900$ kJ
$\eta = 60\% = 0.60$
$$\eta = \frac{W}{Q_1} \Rightarrow 0.60 = \frac{W}{900}$$
$W = 540$ kJ
$W = Q_1 - Q_2$
$Q_2 = 900 - 540$
$\Rightarrow$ $\boxed{Q_2 = 360\,kJ}$ heat rejection

494. A heat pump
(a) can be used as an air conditioner as well
(b) cannot be used as an air conditioner
(c) is a refrigerator operating in a reversed Carnot cycle
(d) a refrigerator requiring no energy

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : A heat pump can be used as air conditioner as well.
For same temperature limits,
$$(COP)_{HP} = 1 + (COP)_R = \frac{1}{\eta_{HE}}$$

495. 0.5 kg of air (ideal gas) executes a Carnot power cycle having a thermal efficiency of 50 percent. The heat transfer to the air during the isothermal expansion is 40 kJ. At the beginning of the isothermal expansion the pressure is 7 bar and the volume is 0.12 m^3. Determine the maximum and minimum temperature for the cycle in K.
(a) 292.7 K and 585.4 K
(b) 585.4 K and 292.7 K
(c) 192.3 K and 787.4 K
(d) 787.4 K and 192.3 K

RPSC IOF, 2020

Ans. (b) : The maximum and minimum temperature T_1 and T_2
$P_1V_1 = mRT_1$
$7\times10^5 \times 0.12 = 0.5 \times 287 \times T_1$
Maximum temperature
$T_1 = 585.4$ K
$$\eta_{cycle} = \frac{T_1 - T_2}{T_1}$$
$$0.5 = \frac{585.4 - T_2}{585.4}$$
Minimum temperature $T_2 = 292.7$K

496. Among the following, the low grade energy is :
(a) Water energy (b) Tidal energy
(c) Heat energy (d) Wind energy

NLC GET 17.11.2020, Shift-II

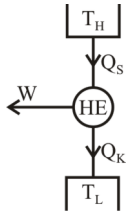
Ans. (c) : Examples of high grade energy are:
1. Mechanical work
2. Electrical work
3. Water power
4. Wind power
5. Tidal power

Examples of low grade energy are:
1. Heat or thermal energy
2. Heat derived from combustion of fossil fuels
3. Heat derived from nuclear fission or fusion

497. A heat engine is supplied heat at the rate of 1700 kJ/min and gives an output of 9 kW. The thermal efficiency will be around
(a) 31.76% (b) 35.76%
(c) 42.85% (d) 45.85%

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) :



$Q_s = 1700$ kJ/min
$$= \frac{1700}{60} kW = 28.33 kW$$
$W = 9$ kW
$$\eta = \frac{9}{\frac{1700}{60}} = 0.3176$$
$\eta = 31.76\%$

498. A Carnot cycle which operates between the temperatures of 400 K and 300 K. If 20000 kcal/hr of heat is supplied, the work output will be
(a) 3000 kcal/hr (b) 4000 kcal/hr
(c) 5000 kcal/hr (d) 6000 kcal/hr

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (c) : $$\eta = 1 - \frac{300}{400} = \frac{W_{output}}{20000}$$
$W_{output} = 5000$ kcal/hr.

499. The efficiency of Carnot Cycle may be increased by
(a) Increasing the highest temperature
(b) Decreasing the highest temperature
(c) Increasing the lowest temperature
(d) Decreasing the lowest temperature

RPSC IOF, 2020

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Ans. (a & d) : Efficiency of Carnot cycle between two (T_1 and T_2) temperature limit

$$\eta_c = 1 - \frac{T_2}{T_1}$$

Where, T_2 - lowest temperature

T_1- highest temperature

Efficiency may increased by lowering the lowest temperature and increase the highest temperature.

The efficiency of Carnot cycle is mainly depends upon lower temperature. because highest temperature is depending upon metallurgical property. so we cannot increase it beyond a limit.

500. If the COP of a Carnot refrigerator is 6, then the ratio of higher temperature to lower is :

(a) 6 : 1 (b) 3 : 2

(c) 4 : 3 (d) 7 : 6

BHEL ET 2019

Ans. (d) : Given - $(COP)_{Ref} = 6$

$$(COP)_{Ref} = \frac{T_L}{T_H - T_L}$$

$$6 = \frac{1}{\frac{T_H}{T_L} - 1}$$

$$\frac{T_H}{T_L} - 1 = \frac{1}{6}$$

$$\frac{T_H}{T_L} = \frac{1}{6} + 1$$

$$\boxed{\frac{T_H}{T_L} = \frac{7}{6}}$$

501. A thermal reservoir is a body of

(a) Small heat capacity

(b) Large heat capacity

(c) Infinite heat capacity

(d) Large work capacity

JPSC AE PRE 2019

Ans. (c) : Thermal Reservoir—It is a large body with infinite heat capacity which can supply or absorbs unlimited quantity of heat without appreciable change in its temperature.

Example—Atmosphere, ocean, river, industrial furnace, two phase mixture of liquid and vapour.

• It supplies heat then it is called a source.

• It absorbs heat then it is called a sink.

502. The statement "The efficiency of all reversible heat engines operating between the same temperature levels is the same" is known as:

(a) Corollary of Carnot theorem

(b) Zeroth law of thermodynamics

(c) First law of thermodynamics

(d) Third law of thermodynamics

SJVN ET 2019

Ans. (a) : The efficiency of all reversible heat engines operating between the same temperature level is the same is known as corollary of carnot theorem.

503. A heat engine using lake water at 12 °C as source and the surrounding atmosphere at 2°C as sink executes 1080 cycles per min. If the amount of heat drawn a per cycle is 57 J, then the output of the engine will be :

(a) 66 W (b) 46 W

(c) 56 W (d) 36 W

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (d) : Given,

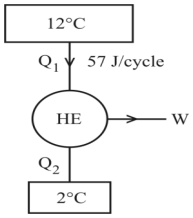
1080 cycles /min.

$Q_1 = 57 \times 1080$ J/min.

$= 61560$ J/min.

$Q_1 = 1026$ W

$$\eta = 1 - \frac{T_2}{T_1}$$



$$= 1 - \left(\frac{2+273}{12+273}\right)$$

$$\eta = 0.035$$

$$\eta = \frac{W}{Q_1}$$

$$W = \eta \times Q_1$$

$$= 0.035 \times 1026$$

$$\boxed{W = 36W}$$

504. In a Carnot cycle operating between 200°C and –20°C. If the system receives 90 kJ from the source then the net work done is :

(a) 38.65 kJ (b) 36.5 kJ

(c) 42.84 kJ (d) 62.54 kJ

HPPSC ADF 05.03.2019

Ans. (c) : $T_H = 200°C = 273 + 200 = 473$ K

$T_L = -20°C = 273 - 20 = 253$ K

$Q_s = 90$ kJ

$$\eta_{carnot} = \frac{W_{net}}{Q_s} = \frac{T_H - T_L}{T_H}$$

$$\frac{W_{net}}{90} = \frac{473 - 253}{473}$$

$$W_{net} = \frac{220}{473} \times 90$$

$$W_{net} = 41.8 \text{ kJ}$$

505. A heat engine receives 1 kW of heat transfer at 1200 K and gives out 600 W as work, with the rest as heat transfer to the ambient at 300 K. The second law efficiency of the engine is :

(a) 70% (b) 90%

(c) 80% (d) 60%

BHEL ET 2019

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Ans. (c) : Heat supplied (Q_S) = 1 kW

$W_{net} = 600$ W

$T_H = 1200$ K, $T_L = 300$ K

$$\eta_{Actual} = \frac{W_{net}}{Q_S} = \frac{600}{1000} = 0.60$$

$$\eta_{maximum} = \frac{T_H - T_L}{T_H} = 1 - \frac{T_L}{T_H}$$

$$= 1 - \frac{300}{1200} = 1 - 0.25$$

$$\eta_{maximum} = 0.75$$

Second law of efficiency of engine

$$\eta_{II} = \frac{\eta_{actual}}{\eta_{maximum}} = \frac{0.60}{0.75} \times 100 = 80\%$$

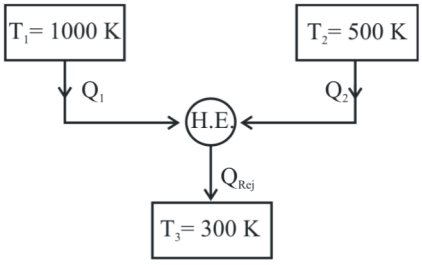
506. A heat engine receives half of its heat supply at 1000 K and remaining half at 500 K. Heat is rejected to the sink at 300 K. The maximum thermal efficiency of the heat engine will be:

(a) 55% (b) 10%

(c) 45% (d) 65%

SJVN ET 2019

Ans. (a) :



$Q_1 = Q_2 = Q/2$

From clausius inequality for reversible heat engine,

$\Delta S_{rev} = 0$

$$\frac{Q}{2000} + \frac{Q}{1000} - \frac{Q_R}{300} = 0$$

$$\frac{3Q}{2000} = \frac{Q_R}{300}$$

$$\frac{Q_R}{Q} = \frac{9}{20}$$

We know that,

$$\eta_{HE} = 1 - \frac{Q_R}{\left(\frac{Q}{2} + \frac{Q}{2}\right)}$$

$$= 1 - \frac{Q_R}{Q}$$

$$= 1 - \frac{9}{20} = 55\%$$

507. Consider the following statements:

1. The entropy of a pure crystalline substance at absolute zero temperature is zero.
2. The efficiency of a reversible heat engine is independent of the nature of the working substance and depends only on the temperature of the reservoirs between which it operates.
3. Carnot's theorem states that of all heat engines operating between a given constant temperature source and a given constant temperature sink, none has a higher efficiency than a reversible engine.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only

(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : (i) The third law of thermodynamics defines the absolute zero of entropy. The entropy of a pure crystalline substance at absolute zero temperature is zero.

(ii) Efficiency of Carnot $= 1 - \frac{T_L}{T_H}$

Here, it is cleared that efficiency of Carnot engine depends upon the temperature of reservoirs. Working substance does not play any role in the efficiency of Carnot engine. Every heat engine works between two temperature limits and give some work out.

Carnot is a ideal case of engines that is not possible. No heat engine can be efficient as Carnot.

508. A source at a temperature of 500 K provides 1000 kJ of heat. The temperature of environment is 27°C. The maximum useful work (in kJ) that can be obtained from the heat source is

(a) 600 kJ (b) 500 kJ

(c) 400 kJ (d) 270 kJ

GPSC Executive Engineer 23.12.2018

Ans. (c) : $\eta = \frac{W_{max}}{Q_s}$

$$\eta = 1 - \frac{T_L}{T_H}$$

$$= 1 - \frac{300}{500}$$

$$= \frac{2}{5}$$

$$\frac{2}{5} = \frac{W_{max}}{Q_s}$$

$W_{max} = 400$ kJ

509. A reversible heat engine receives 2 kJ of heat from a reservoir at 1000 K and a certain amount of heat from a reservoir at 800 K. It rejects 1 kJ of heat to a reservoir at 400 K. The net work output (in kJ) of the cycle is

(a) 0.8 (b) 1.0

(c) 1.4 (d) 2.0

UJVNL AE 12.09.2021

GPSC Executive Engineer 23.12.2018

Ans. (c) : We know that for reversible heat engine

$\Delta S = 0$

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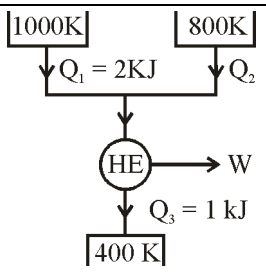

$$\frac{Q_3}{T_3}-\left(\frac{Q_1}{T_1}+\frac{Q_2}{T_2}\right)=0$$

$$\frac{1}{400}-\left(\frac{2}{1000}+\frac{Q_2}{800}\right)=0$$

$Q_2 = 0.4$

then $W = (Q_1 + Q_2) - Q_3$

$W = (2 + 0.4) - 1$

$= 1.4$ kJ

510. Compute the power required to drive a reversible refrigerator if 100 kJ/min is absorbed from the cold region and the isothermal process occurs at 40°C and -5°C.

(a) 17 kJ/min (b) 35 kJ/min

(c) 51 kJ/min (d) 25 kJ/min

GPSC ARTO Pre 30.12.2018

Ans. (a) : Given,

$Q_{absorbed} = 100$ kJ/min

$T_L = -5°C = 268$ K

$T_H = 40°C = 313$ K

$(COP)_{th} = (COP)_{act}$

$$\frac{T_L}{T_H - T_L} = \frac{Q_{absorbed}}{Power}$$

$$\frac{268}{313-268} = \frac{100}{P}$$

$$P = \frac{4500}{268} = 16.79$$

$$\boxed{P \approx 17\,kJ/min.}$$

511. An ideal Carnot engine operates between reservoirs at temperature 20 °C and 200 °C. If 10 kW of power is produced, the rate at which heat is rejected is approximately:

(a) 14.3 kJ/s (b) 12.3 kJ/s

(c) 15.3 kJ/s (d) 16.3 kJ/s

AAI Jr. Executive 29.11.2018

Ans. (d) : $T_1 = 20°C$

$T_1 = 20 + 273 = 293$ K

$T_2 = 200°C$

$T_2 = 200 + 273 = 473$ K

$W = 10$ kW

$Q_2 = ?$

$$W = Q_1\left(1-\frac{T_2}{T_1}\right)$$

$$10kW = Q_1\left(\frac{473-293}{473}\right)$$

$Q_1 = 26.27$ kW

$Q_2 = Q_1 - W = 16.27$ kW

512. A heat Engine gives an output of 3 kW when the input is 10000 joules/sec. The thermal efficiency of the engine is

(a) 30% (b) 20%

(c) 40% (d) 76.7%

ISRO Scientist/Engineer (RAC) 22.04.2018

CSE Pre-1995

Ans. (a) : $Q_{in} = 10$ kJ/s

$W_{output} = 3$ kW

$$\eta_{Therm} = \frac{W_{output}}{Q_{in}} = \frac{3}{10} = 0.3$$

$\eta_{Therm} = 30\%$

513. Which of the following devices complies with the Clausius statement of the second law of thermodynamics?

(a) Closed-cycle gas turbine

(b) Internal combustion engine

(c) Steam power plant

(d) Domestic refrigerator

ESE 2018

Ans. (d) : Clausius statement is related to refrigerator and heat pump not with heat engine.

514. A thermodynamic cycle is composed of four processes. The heat added and the work done in each process are as follows:

Process	Heat transfer (J)	Work done (J)
1- 2	0	50 (by the gas)
2 - 3	50 (from the gas)	0
3 - 4	0	20 (on the gas)
4 - 1	80 (to the gas)	0

The thermal efficiency of the cycle is

(a) 20.3% (b) 37.5%

(c) 40.3% (d) 62.5%

ESE 2018

Ans. (b) : Heat rejected $(Q_2) = 50$

Heat added $(Q_1) = 80$

Thermal efficiency of the cycle

$$\eta = 1 - \frac{Q_2}{Q_1}$$

$$= 1 - \frac{50}{80} = 37.5\%$$

515. Consider the following statements:

1. Heat pumps and air conditioners have the same mechanical components.
2. The same system can be used as heat pump in winter and as air conditioner in summer.
3. The capacity and efficiency of a heat pump fall significantly at high temperatures.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only

(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

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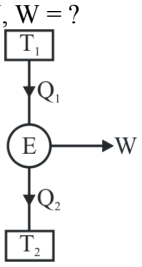
Ans. (d) : Heat pumps and air conditioners have the same mechanical components.
Therefore it is not economical to have two separate systems to meet the heating and cooling requirements of a building. One system can be used as a heat pump in winter and an air conditioner in summer.
Heat pump is a device used to maintain the temperature of system higher than that of surroundings by supplying heat to system To evaluate the performance of a heat pump, COP is calculated. It is defined as the ratio of desired effect to external work supplied.
Note : The answer given by UPSC is option (b).

516. An engine works on the basis of Carnot cycle operating between temperatures of 800 K and 400 K. If the heat supplied is 100 kW, the output is
(a) 50 kW (b) 60 kW
(c) 70 kW (d) 80 kW

ESE 2018

Ans. (a) : Given,
$T_1 = 800$ K, $T_2 = 400$ K
$Q_1 = 100$ kW, $W = ?$



We know that

$$\eta = \frac{\text{Work output}}{\text{Heat supplied}}$$

$$\eta = \frac{W}{Q_1} = 1 - \frac{T_2}{T_1}$$

$$\frac{W}{100} = 1 - \frac{400}{800} = 0.5$$

$W = 50$ kW

517. If the temperature of the source is increased, the efficiency of the Carnot engine-
(a) decreases
(b) increases
(c) will be equal to the efficiency of a practical engine
(d) does not change

RPSC AE 2018

Ans. (b) : We know that the efficiency of Carnot engine.

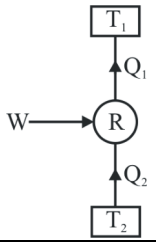
$$\eta_c = 1 - \frac{T_{sink}}{T_{source}}$$

If temperature of source T_{source} increase than $\eta_c \uparrow$

518. The coefficient of performance of a heat pump working on reversed Carnot cycle is 6. If this machine works as a refrigerator with work input of 10 kW, the refrigerating effect will be
(a) 35 kW (b) 40 kW
(c) 45 kW (d) 50 kW

ESE 2018

Ans. (d) : $(COP)_{HP} = 1 + (COP)_R$
$(COP)_R = 6 - 1 = 5$



$$(COP)_R = \frac{\text{Refrigerating effect } (R_e)}{\text{Work input}}$$

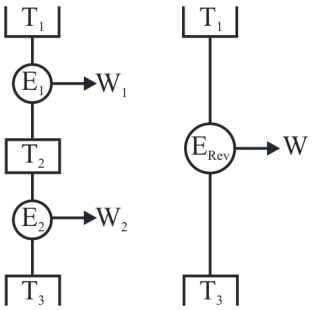
$$5 = \frac{R_e}{10}$$

$R_e = 50$ kW

519. Two reversible engines are connected in series between a heat source and a sink. The efficiencies of these engines are 60% and 50%, respectively. If these two engines are replaced by a single reversible engine, the efficiency of this engine will be
(a) 60% (b) 70%
(c) 80% (d) 90%

ESE 2018

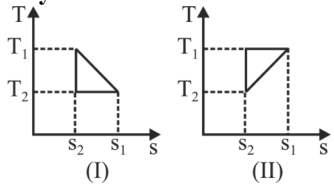
Ans. (c) : Given,
$\eta_1 = 0.6$
$\eta_2 = 0.5$



Overall efficiency of the combined engine

$\eta_0 = \eta_1 + \eta_2 - \eta_1 \cdot \eta_2$
$= (0.6 + 0.5) - (0.6 \times 0.5)$
$= 0.8$
$\Rightarrow 80\%$

520. What is the ratio of the efficiencies $\left(\frac{\eta_I}{\eta_{II}}\right)$ for the two cycles as shown in the T-s diagrams?



(a) $\frac{T_1 + T_2}{2T_1}$ (b) $\frac{T_1 + T_2}{2T_2}$
(c) $\frac{2T_2}{T_1 + T_2}$ (d) $\frac{2T_1}{T_1 + T_2}$

ESE 2018

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Ans. (d) : Work done is same in both the cycle but heat rejected is more in cycle-II when compared to cycle-I

$$\eta_I = \frac{\frac{1}{2}(T_1 - T_2)(S_1 - S_2)}{\frac{1}{2}(T_1 - T_2)(S_1 - S_2) + T_2(S_1 - S_2)}$$

$$= \frac{\frac{1}{2}(T_1 - T_2)}{\frac{1}{2}(T_1 - T_2) + T_2} = \frac{T_1 - T_2}{T_1 + T_2}$$

$$\eta_{II} = \frac{\frac{1}{2}(T_1 - T_2)(S_1 - S_2)}{T_1(S_1 - S_2)} = \frac{T_1 - T_2}{2T_1}$$

$$\frac{\eta_I}{\eta_{II}} = \frac{2T_1}{T_1 + T_2}$$

521. One reversible heat engine operates between 1600 K and T_2K and another reversible heat engine operates between T_2K and 400 K. If both the engines have the same heat input and output, then temperature T_2 is equal to

(a) 800 K (b) 1000 K
(c) 1200 K (d) 1400 K

Nagaland PSC (CTSE) 2018, Paper-II
ESE 1993

Ans. (a) : For same heat input and output

$$\eta_I = 1 - \frac{Q_2}{Q_1} = \eta_{II}$$

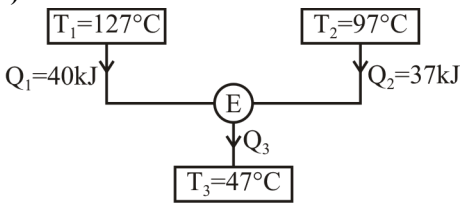
$$1 - \frac{T_2}{1600} = 1 - \frac{400}{T_2} \Rightarrow T_2\sqrt{1600 \times 400} = 800\,K$$

522. A reversible cycle receive 40 kJ of heat from one heat source at a temperature of 127°C and 37 kJ from another heat source at 97°C. The heat rejected (in kJ) to the heat sink at 47°C is

(a) 64 kJ (b) 320 kJ
(c) 640 kJ (d) 960 kJ

GPSC Executive Engineer 23.12.2018
GATE 2016

Ans. (a) :



$$\frac{Q_1}{T_1} + \frac{Q_2}{T_2} - \frac{Q_3}{T_3} = 0$$

$$\frac{40}{127 + 273} + \frac{37}{97 + 273} - \frac{Q_3}{47 + 273} = 0$$

$$\boxed{Q_3 = 64\,kJ}$$

523. A Scientist says that the efficiency of his heat engine, which works at source temperature 127°C and sink temperature 27°C is 26%, then

(a) It is impossible
(b) It is quite possible
(c) It is possible but less probable
(d) Data incomplete

GPSC EE Pre, 28.01.2017

Ans. (a) : Given,

$T_H = 127°C = 273 + 127$
$= 400K$
$T_L = 27ºC = 273 + 27$
$= 300K$
$\eta = 26\%$

We know that-

$$\eta_{carnot} = 1 - \frac{T_L}{T_H}$$

$$= 1 - \frac{300}{400} = 1 - \frac{3}{4}$$

$= 1 - 0.75$
$= 0.25$
$= 25\%$

Hence, given heat engine efficiency (η = 26%) is greater than (η_{carnot} = 25%). So, its impossible.

524. Choose the incorrect statement. The Second Law of Thermodynamics indicates that

(a) complete conversion of low grade energy into high grade energy is impossible
(b) heat and work are not completely interchangeable
(c) reversible and irreversible are two distinct processes
(d) it is possible to transfer heat from low temperature to high temperature if heat extracted is exactly equal to work delivered

Punjab PSC (Lect.) 06.08.2017

Ans. (d) : Second Law of Thermodynamics –

- **Kelvin Planck Statement–** It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.
- **Clausius Statement –** It is impossible to construct a device which operating in a cycle will produce no effect other than the transfer of heat from cooler body to a hotter body.

525. An inventor claims that heat engine will develop 9 kW receiving 17500 J/s of heat while working between temperatures of 130°C and –40°C his claim

(a) Is not possible
(b) Is possible
(c) Cannot be judged
(d) Is possible with further improvements

GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : Given,

$T_1 = 130 + 273 = 403$ K
$T_2 = 273 - 40 = 233$ K
P = 9 kW,

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$Q_s = 17.5$ kW

$$\eta = \frac{P}{Q_s} = \frac{9}{17.5} = 0.514$$

$$\eta_c = \frac{T_1 - T_2}{T_1} = \frac{403 - 233}{403} = 0.421$$

$\eta > \eta_c$, Impossible violates second law.

526. An inventor states that his new conceptual engine, while operating between temperature limits of 377°C and 27°C, will reject 50% of heat absorbed from the source. What type of cycle will this engine have?

(a) Carnot cycle (b) Stirling cycle
(c) Impossible cycle (d) Possible cycle

GPSC Asstt. Director of Transport 05.03.2017
ESE 2012

Ans. (d) : Given,

$\eta_a = 50\% \rightarrow$ because 50% absorbed

$T_2 = 300$ K, $T_1 = 650$ K

$$\eta_c = \frac{T_1 - T_2}{T_1} = \frac{650 - 300}{650} \times 100$$

$\eta_c = 53.84\%$

So, $\eta_{actual} < \eta_{carnot}$ so, it is possible cycle.

527. For a given temperature T_1, as the difference between T_1 and T_2 increase, the COP of a carnot heat pump

(a) Increases
(b) Decreases
(c) Does not change
(d) First decrease, then increases

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : $COP_{HP} = \frac{T_H}{T_H - T_C} = \frac{T_1}{T_1 - T_2}$

$(T_1 - T_2)$ increases COP decreases for constant T_1.

528. What is the maximum rate that a heat pump which uses 1 kW of electric power can supply heat to a house at 27^0C when the outside temperature is 12^0C

(a) 50 J/s (b) 25 kJ/s
(c) 20 kJ/s (d) 30 J/s

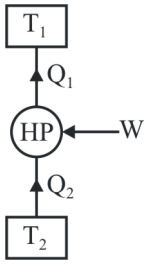
ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Given,

W = 1 kW = 1 kJ/s

$T_1 = 27 + 273 = 300$ K

$T_2 = 12 + 273 = 285$ K



$$(COP)_{HP} = \frac{\text{Heating Effect}}{\text{Work Input}} = \frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2} = \frac{T_1}{T_1 - T_2} = \frac{300}{300 - 285}$$

$(COP)_{HP} = 20$

$$\frac{Q_1}{W} = 20$$

$Q_1 = 20 \times W = 20 \times 1 = 20$ kJ/s

529. During January, at a location in Rohtang, winds at –6°C can be observed. Several meters below the ground, the temperature remains at 27°C. A scientist claims to have devised a power cycle exploiting this situation that has a thermal efficiency of 10%. Whether the claim is true?

(a) True (b) False
(c) Can't say (d) Insufficient data

ISRO Scientist/Engineer 07.05.2017

Ans. (a) : $\eta_{Ideal} = \frac{T_{max} - T_{min}}{T_{max}} = \frac{27 - (-6)}{(27 + 273)} = \frac{33}{300} \times 100 = 11\%$

$\eta_{scientist} = 10\% < \eta_{Ideal}$

Scientist claim is true since power cycle having 10% thermal efficiency is possible because Carnot efficiency is 11%.

530. Which one of the following statements is correct?

(a) A machine which violates Clausius statement will violate the first law of thermodynamics
(b) A machine which violates Kelvin-Plank statement will violate the first law of thermodynamics
(c) A machine which violates the second law of thermodynamics will violate the first law of thermodynamics
(d) A machine which violates Kelvin-Plank statement will violate the Clausius statement

UPSC JWM 2017

Ans. (d) : Kelvin Plank Statement- It is impossible for a heat engine to produced net work in a complete cycle if it exchange heat only with bodies at a single fixed temperature.

Clausius statement- It is impossible to construct an engine which operates in a cycle, transfer heat from cooler body to hotter body without any work input.

A machine which violated Kelvin plank statement will violate the clausius statement.

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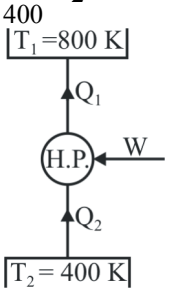
531. A heat pump operates between two heat reservoirs, one at 800 K and other at 400 K. What will be the coefficient of performance (COP) of the heat pump?

(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0
(e) 2.5

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : $[COP]_{H.P.} = \frac{T_1}{T_1 - T_2} = \frac{800}{800 - 400}$

$[COP]_{H.P.} = \frac{800}{400} = 2$



532. The efficiency (η) of a reversible heat engine receiving heat solely at T_1 and rejecting heat solely at T_2 is given by:

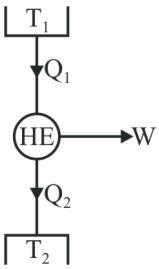
(a) $\eta = \frac{T_1 - T_2}{T_2}$ (b) $\eta = \frac{T_2 - T_1}{T_2}$
(c) $\eta = \frac{T_1 - T_2}{T_1}$ (d) $\eta = \frac{T_2 - T_1}{T_1}$
(e) $\eta = \frac{T_2 - T_1}{T_1^2}$

(CGPSC Polytechnic Lecturer 2017)

Ans. (c) : $\eta_{rev.} = \frac{W}{Q_1}$

$= \frac{Q_1 - Q_2}{Q_1}$

$\eta_{rev.} = 1 - \frac{Q_2}{Q_1}$



For reversible engine

$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$, So

$\boxed{\eta_{rev.} = 1 - \frac{T_2}{T_1}}$

$\boxed{\eta_{rev.} = \frac{T_1 - T_2}{T_1}}$

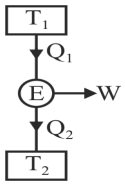
533. A heat engine working on the Carnot cycle receives heat at the rate of 50 kW from a source at 1300K and rejects it to a sink at 400K. The heat rejected is

(a) 20.3 kW (b) 15.4 kW
(c) 12.4 kW (d) 10.8 kW

ESE 2017

Ans. (b) : Given,

$T_1 = 1300$ K
$T_2 = 400$ K
$Q_1 = 50$ kW



Efficiency of engine

$\eta = \frac{\text{Work done}}{\text{Heat Supplied}}$

$\eta = \frac{W}{Q_1} = \frac{T_1 - T_2}{T_1}$ For carnot engine

$\frac{Q_1 - Q_2}{Q_1} = \frac{1300 - 400}{1300}$

$1 - \frac{Q_2}{Q_1} = \frac{9}{13} \Rightarrow \frac{Q_2}{Q_1} = 0.3076$

$Q_2 = 50 \times 0.3076$

$Q_2 = 15.38$ kW

534. A reversible heat engine rejects 80% of the heat supplied during a cycle of operation. If the engine is reversed and operates as a refrigerator, then its coefficient of performance shall be

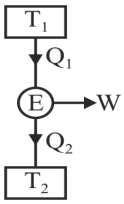
(a) 6 (b) 5
(c) 4 (d) 3

ESE 2017

Ans. (c) : Given,

$Q_2 = 0.8\ Q_1$

For heat engine



$Q_1 = W + Q_2$

$W = Q_1 - Q_2$

$W = 0.2\ Q_1$

$\eta_E = \frac{W}{Q_1}$, $\eta_E = \frac{0.2Q_1}{Q_1}$

$\eta_E = 0.2$

$1 + (COP)_R = \frac{1}{\eta_E}$

$(COP)_R = \frac{1}{0.2} - 1$, $(COP)_R = 4$

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535. A Carnot refrigeration cycle is used to estimate the energy requirement in an attempt to reduce the temperature of a specimen to absolute zero. Suppose that we want to remove 0.01 J of energy from the specimen when it is at 2×10^{-6} K. How much work is necessary if the high-temperature reservoir is at 20°C?

(a) 622 kJ (b) 864 kJ
(c) 1170 kJ (d) 1465 kJ

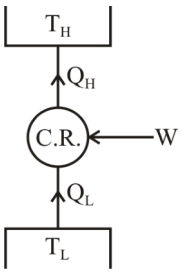
HPPSC Asstt. Prof. 20.11.2017

Ans. (d) : Given, $T_L = 2 \times 10^{-6}$ K

$T_H = 20°C = 293$ K

$Q_L = 0.01$ J

$W = ?$



$$COP = \frac{T_L}{T_H - T_L} = \frac{Q_L}{W}$$

$\therefore \quad T_H >>> T_L$

$$\frac{T_L}{T_H} = \frac{Q_L}{W}$$

$$W = \frac{Q_L}{T_L} \times T_H = \frac{0.01 \times 293}{2 \times 10^{-6}}$$

$W = 1465000$ J $= 1465$ kJ

536. A carnot engine receives 1200kJ and 3600kJ heat from reservoirs which are at 600 K and x K respectively and rejects heat of 3200 kJ to the sink which is at 400 K. Calculate x in Celsius.

(a) 800 (b) 600
(c) 427 (d) 327
(e) 227

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : From Clausius Inequality

For Reversible Process

$$\int \frac{dQ}{T} = 0$$

$$\frac{1200}{600} + \frac{3600}{x} - \frac{3200}{400} = 0$$

$x = 600$ K

$x = 600 - 273$

$x = 327°C$

537. A perfect heat engine works on the Carnot cycle between 1000^0 C and 200^0 C. The efficiency of the engine will be

(a) 80% (b) 60%
(c) 42.8% (d) 37.2%

WBPSC AE, 2017

Ans. (b) : $T_H = 1000$ °C

$= 1000 + 273 = 1273$ K

$T_L = 200°C = 200 + 273 = 473$ K

We know that,

$$\eta_{carnot} = 1 - \frac{T_L}{T_H}$$

$$= 1 - \frac{473}{1273}$$

$= 1 - 0.371$

$= 0.6284$

$\eta_{carnot} = 62.84\%$

538. A reversible engine receives heat from a reservoir at 727^0C and rejects heat at a temperature T_2. A second reversible engine operating on Carnot cycle receives heat rejected by the first engine and rejects heat to a sink at 27^0C. If work output from each of the engines is equal what is the value of temperature T_2?

(a) 775 K (b) 650 K
(c) 750 K (d) 625 K

UPRVUNL AE 21.08.2016

Ans. : (b) For Engine first–

$$\eta_1 = \frac{T_H - T_L}{T_H}$$

$$\eta_1 = \frac{1000 - T_2}{1000} = \frac{\text{work output}}{\text{Heat supplied}}$$

For Engine IInd –

$$\eta_2 = \frac{T_2 - 300}{T_2} = \frac{\text{work output}}{\text{Heat supplied}}$$

Both engine work output same

$1000 - T_2 = T_2 - 300$

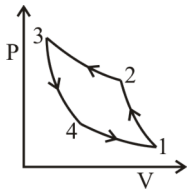
$T_2 = 650$ K

539. If the ratio of higher to lower absolute temperature is 1.5. Then COP of a heat pump working on reversed Carnot cycle will be

(a) 6 (b) 4
(c) 3 (d) 2

GPSC ARTO 01.05.2016

Ans. (c) :



For Carnot cycle 4 – 1 & 2 – 3 at constant temperature & 1 – 2 & 3 – 4 are isentropic process.

Highest temperature T_H, Lowest temperature = T_L

$T_H / T_L = 1.5$

$$\therefore \quad (COP)_{Heat\ pump} = \frac{T_H}{T_H - T_L} = \frac{T_H}{T_L(T_H/T_L - 1)}$$

$$\boxed{(COP)_{H.P} = \frac{1.5}{(1.5-1)} = 3}$$

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540. A heat engine is supplied heat at the rate of 1700 kJ/min and gives an output of 9 kW. The rate of heat rejection will be

(a) 17.33 kJ/sec (b) 18.33 kJ/sec
(c) 19.33 kJ/sec (d) 10.33 kJ/sec

GPSC ARTO 01.05.2016

Ans. (c) : Heat supplied $Q_s = 1700$ kJ/min

$$= \frac{1700}{60} = 28.33 \text{ kW}$$

Work output W = 9 kW

Work output $W = Q_S - Q_R$

$$\Rightarrow Q_R = Q_S - W = 28.33 - 9$$

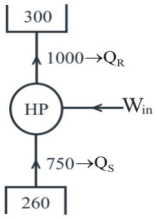
Heat rejection $\boxed{Q_R = 19.33 \text{ kW}}$

541. An industrial heat pump operates between the temperatures of 27°C and –13°C. The rates of heat addition and heat rejection are 750 W and 1000W respectively. The COP for the heat pump is :

(a) 7.5 (b) 6.5
(c) 4.0 (d) 3.0

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) : $W_{in} = Q_R - Q_S = 1000 - 750 = 250$ W



$$(COP)_{HP} = \frac{Q_R}{W_{in}}$$

$$= \frac{1000}{250}$$

$$(COP)_{HP} = 4$$

542. A heat engine works on a Carnot cycle with a heat sink at a temperature of 27°C. What would be the temperature of the heat at source if efficiency of the engine is 10%?

(a) 100°C (b) 90°C
(c) 60°C (d) 30°C

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : Given,

Sink temperature $(T_2) = 27 + 273 = 300$ K

We know that,

$$\eta_{\text{heat engine}} = 1 - \frac{T_2}{T_1} \qquad \text{(Reversible heat engine)}$$

$$10\% = \frac{10}{100} = 1 - \frac{300}{T_1}$$

$$T_1 = 333 \text{ K}$$

$$\boxed{T_1 = 60^0 \text{C}}$$

$\therefore$ Temperature of the heat at source 60°C.

543. A engine works on Carnot cycle between 727°C and 227°C. The efficiency of the engine is

(a) 50% (b) 75.4%
(c) 31.2% (d) 68.8%

ISRO Scientist/Engineer 03.07.2016

Ans. (a) $\eta = 1 - \frac{T_C}{T_H}$

$T_C = 227 + 273 = 500$ K

$T_H = 727 + 273 = 1000$ K

$$\eta = 1 - \frac{500}{1000}$$

$\eta = 0.5 = 50\%$

544. A heat engine working with a thermal efficiency of 35% receives 2 kW of heat from a furnace. The waste heat rejected from the engine is

(a) 0.7 kW (b) 0.35 kW
(c) 1.3 kW (d) 1.65 kW

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Thermal efficiency = 35%

= 0.35

$Q_S = 2$ kW

$Q_R = ?$

$$\eta_{th} = 1 - \frac{Q_R}{Q_S}$$

$$0.35 = 1 - \frac{Q_R}{2 \times 10^3}$$

$$\frac{Q_R}{2} = 1 - 0.35$$

$Q_R = 2 \times 0.65$

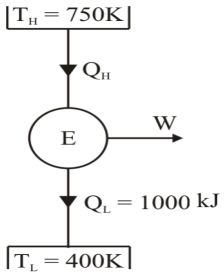
$Q_R = 1.3$ kW

545. If carnot engine rejects heat at temperature of 400 K and accepts at 750K. What shall be heat absorbed, if heat rejected is 1000 kJ.

(a) 946 kJ (b) 800 kJ
(c) 1875 kJ (d) 750 kJ

APPSC AEE Screening Test 2016

Ans. (c) :



For Carnot engine,

$\eta_{ideal} = \eta_{actual}$

$$1 - \frac{T_L}{T_H} = 1 - \frac{Q_L}{Q_H}$$

$$Q_H = \frac{1000 \times 750}{400}$$

$\boxed{Q_H = 1875 \text{ kJ}}$

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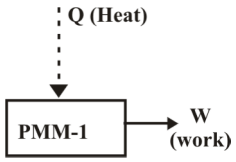
546. The device that produces net work in a complete cycle by exchanging heat only with single thermal energy reservoir is known as:

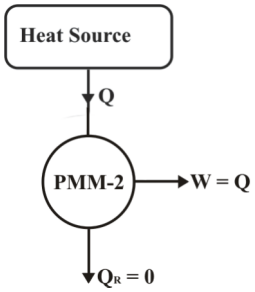
(a) PMM3 (b) Heat pump
(c) PMM2 (d) PMM1

UPRVUNL AE 07.10.2016

Ans. (c) : PMM-1—Perpetual motion machine of first kind does not exist because such machines violate the first law of thermodynamics such machines will produce the energy by itself and as we know that according to the law of energy conservation, energy could not be created or destroyed but could be converted from one form of energy to other form of energy.



PMM-2—Perpetual motion machines of second kind are those machines that violate the second law of thermodynamics because such machines will absorb continuously heat energy from a single thermal reservoir and will convert the absorbed heat energy completely into work energy.
Such machine will have 100% efficiency.



PMM-3—Perpetual motion machine of third kind of imaginary machine that has zero friction.

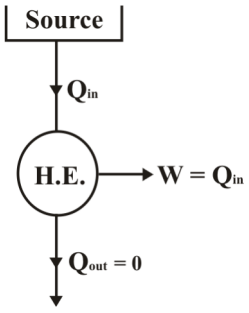
547. Perpetual motion machine of second kind (PMM-II) violates the

(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) Third law of thermodynamics

RPSC LECTURER 16.01.2016

Ans. (c) : Perpetual motion machine of second kind (PMM-II) violates the second law of thermodynamics.

$\eta_{HE} = 100\%$



548. What will be the thermal efficiency of heat engine if heat engine is supplied heat at the rate 30,000 J/s and gives out output of 9kW?

(a) 0.6 (b) 0.5
(c) 0.3 (d) 0.4

Rajasthan AE (Nagar Nigam) 2016 Shift-3
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. (c) $\eta_{thermal} = \dfrac{\text{work output}}{\text{supplied heat}}$

Given, $W = 9\ kW = 9 \times 10^3$ J/sec
$Q_A = 30000$ J/sec

$$\eta_{thermal} = \frac{9\times10^3}{30000}$$

$$\boxed{\eta_{th} = 0.3}$$

549. Three carnot engines A, B and C having source temperatures of 750 K, 700 K and 650 K have sink temperatures respectively 400 K, 350 K and 300 K . Which engine is least efficient?

(a) Engine A (b) Engine B
(c) Engine C (d) none of these

RPSC ADE 2016

Ans. (a) :

$T_{(max)_A} = 750K$ $T_{(min)_A} = 400K$
$T_{(max)_B} = 700K$ $T_{(min)_B} = 350K$
$T_{(max)_C} = 650K$ $T_{(min)_C} = 300K$

$$\eta_A = \frac{T_{max} - T_{min}}{T_{max}} = \frac{350}{750} = 0.46 = 46\%$$

$$\eta_B = \frac{700-350}{700} = \frac{350}{700} = \frac{1}{2} = 50\%$$

$$\eta_C = \frac{650-300}{650} = \frac{350}{650} = \frac{7}{13} = 53\%$$

$\eta_C > \eta_B > \eta_A$
the least efficient engine is A.

550. Which of the following second-law statement is incorrect

(a) The entropy of an isolated system must remain constant or increase
(b) The entropy of a hot copper block decreases as it cools down
(c) If ice is melted in water in an insulated container, the net entropy decreases
(d) Work must be input if energy is transferred from a cold body to a hot body

HPPSC Asstt. Prof. 20.11.2017
RPSC VPITI 14.02.2016

Ans. (c) : We have $ds = \dfrac{dQ}{T} + s_{gen}$ (fixed mass entropy)

(i) For isolated/insulated system $\dfrac{dQ}{T} = 0$

$ds = s_{gen} > 0$ (always) Hence entropy always increases

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(ii) For closed system

$$\frac{dQ}{T} \Rightarrow \text{can be positive, negative or zero}$$

Hence $ds = \frac{dQ}{T} + s_{gen}$

Entropy can be positive, negative or zero.

551. An inventor claims to have developed a heat engine that receives 700 kJ of heat from a source at 500 K and produces 300 kJ of net work while rejecting the waste heat to a sink at 290 K. Which of the following is true?

(a) The claim made by inventor is false because such a heat engine violates first law of thermodynamics.

(b) The claim made by inventor is false because such a heat engine violates second law of thermodynamics.

(c) The heat engine can be possible only if all the processes are reversible.

(d) The invented engine does not violate any law so there is nothing new in the claim of the inventor.

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : For a heat engine,

$$\eta_{ideal} = 1 - \frac{T_L}{T_H} = 1 - \frac{290}{500} = 1 - 0.58 = 0.42 \simeq 42\%$$

$$\eta_{actual} = 1 - \frac{Q_R}{Q_A} = 1 - \frac{300}{700} = 1 - 0.42 = 0.57 \simeq 57\%$$

$\eta_{reversible} < \eta_{cycle}$, which is not possible because it violates second law of thermodynamics.

552. Two cycles coupled in series have efficiencies 35% and 25% respectively. What is the overall efficiency?

(a) 42.5% (b) 51.25%

(c) 55% (d) 57.25%

(e) 60%

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : As we know,

Overall efficiency $(\eta_0) = \eta_1 + \eta_2 - \eta_1 \times \eta_2$

$= 0.35 + 0.25 - (0.35 \times 0.25)$

$= 0.5125$

$\eta_0 = 51.25\%$

553. " All reversible heat engines operating between the same temperature have the same efficiency". This is :

(a) Rankine's theorem 2

(b) Carnot's theorem 2

(c) Brayton's theorem

(d) Sterling's theorem

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) Ist Theorem– The efficiency of an irreversible engine can not be more than a reversible engine operating b/w same temperature limits.

2nd Theorem – Efficiency of all reversible engines operating b/w same temperature limits is same.

554. Carnot cycle operates between the temperature of 1000 K and 500 K. The efficiency of the cycle is-

(a) 0.5 (b) 0.66

(c) 0.75 (d) 0.33

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Ans. (a) : Given $T_L = 500$ K, $T_H = 1000$ K

$$\eta_{carnot\ cycle} = 1 - \frac{T_L}{T_H} = 1 - \frac{500}{1000}$$

$$\boxed{\eta_{carnot\ cycle} = 0.5}$$

555. Efficiency of an engine is defined as

(a) output/ input

(b) output/ (output + losses)

(c) both A and B

(d) None of the above

VIZAG STEEL MT 2015

Ans. (c) $\eta_{engine} = \frac{output}{input}$

$\because$ input = output + losses

or $\eta_{engine} = \frac{output}{output + losses}$

or $\eta_{engine} = \frac{input - losses}{input}$

556. A manufacturer claims to have a heat engine capable of developing 15 kW by receiving heat input of 1700 kJ/min. and working between the temperature limits of 227°C and 27°C. His claim is:

(a) Justified

(b) Not possible

(c) May be possible with lot of sophistications

(d) Theoretically possible

NPCIL ET, 2015

Ans. (b) : Given- Output = 15 kW

Input = 1700 kJ/min

$= \frac{1700}{60}$ kJ/sec

$= 28.333$ kW

Higher Temperature (T_H) = 227°C = 273 + 227 = 500K

Lower Temperature (T_L) = 27°C = 273 + 27 = 300K

Ideal efficiency of heat engine $(\eta)_{ideal}$

$$= 1 - \frac{T_L}{T_H} = 1 - \frac{300}{400}$$

$= 0.4$

$$(\eta)_{Actual} = \frac{Output}{Input}$$

$$= \frac{15}{28.333}$$

$= 0.529$

Here, the actual efficiency is greater than ideal efficiency which is not possible.

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557. The work done in a Carnot heat engine cycle is given by

(a) $R \log r(T_1 - T_2)$ (b) $R \log_e r(T_1 - T_2)$

(c) $R.\log(T_1 - T_2)$ (d) $R \log_e(T_1 - T_2)$

TANGEDCO AE 2015

Ans. (b) : Carnot heat engine cycle consists of four process – Two isothermal and two isentropic process.

Work done during isothermal

$$W = P_1V_1 \ln\frac{V_2}{V_1} = RT_1 \ln\frac{V_2}{V_1}$$

Work done during isentropic

$$W = \frac{P_2V_2 - P_1V_1}{\gamma - 1} = \frac{R(T_2 - T_1)}{\gamma - 1}$$

Total work by the gas (including both isentropic & both isothermal process)

$$W = RT_1 \ln\frac{V_2}{V_1} + \frac{R(T_2 - T_1)}{\gamma - 1} + RT_2 \ln\frac{V_4}{v_3} + \frac{R(T_1 - T_2)}{\gamma - 1}$$

$$W = RT_1 \ln\frac{V_2}{V_1} + RT_2 \ln\frac{V_4}{V_3} \quad \text{——(I)}$$

Now– $TV^{\gamma-1} = C$

$$\frac{T_2}{T_1} = \left(\frac{V_2}{V_3}\right)^{\gamma-1} \quad \text{——(II)}$$

and

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_4}\right)^{\gamma-1} \quad \text{——(III)}$$

from equation (II) & (III)

$$\frac{V_2}{V_3} = \frac{V_1}{V_4}$$

$$\frac{V_2}{V_1} = \frac{V_3}{V_4} = \text{constant} = r \quad \text{——(IV)}$$

From equation (I) & (IV)

$$W = RT_1 \ln r + RT_2 \ln\frac{1}{r} = R \ln r(T_1 - T_2)$$

558. A reversible heat engine operating between hot and cold reservoirs delivers a work output of 74 kJ, while it rejects a heat of 86 kJ. The efficiency of this engine is

(a) 0.46 (b) 0.51

(c) 0.75 (d) 0.82

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Given, $W_{net} = 74$ kJ, $Q_2 = 86$ kJ

$W_{net} = Q_1 - Q_2$

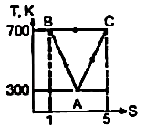
$74 = Q_1 - 86$

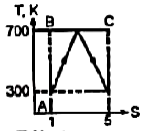
$Q_1 = 160$ kJ

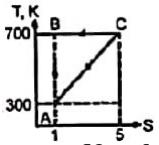
$$\eta = \frac{W_{net}}{Q_1}$$

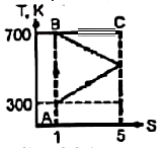
$$\eta = \frac{74}{160} = 0.46$$

559. Which one of the above hypothetical heat engine cycle represents maximum efficiency.

(a) 

(b) 

(c) 

(d) 

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : Work done by all cases is same

$$= \frac{1}{2} \times (5-1) \times (700 - 300)$$

$= 800$ units

Heat input = Area of B-C - 5-1= $700 \times (5 - 1) = 2800$

Heat input

$$\left\{300 \times \left(\frac{5-1}{2}\right) + \frac{1}{2} \times \left(\frac{5-1}{2}\right)(700 - 300)\right\}$$

$= 1000$ units

Heat input for d is

$$\left\{300 \times (5-1) + \frac{1}{2} \times (5-1) \times (700 - 300)\right\}$$

$= 2000$

$\therefore$ Hypothetical heat engine cycle represents maximum efficiency when minimum heat input is case (b).

560. The heat to be supplied to a Carnot engine operating between 800 and 400 K and producing 100 kJ of work, is-

(a) 100 kJ (b) 200 kJ

(c) 300 kJ (d) 400 kJ

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (b) $\eta_c = 1 - \frac{T_L}{T_H} = \frac{W}{Q_S}$

$$= 1 - \frac{400}{800} = \frac{100}{Q_S} = \frac{1}{2}$$

$Q_S = 200$ kJ

561. A house requires 60 M cal/hr in winter for heating. Heat pump absorbs heat from cold air outside and requires 8M cal/hr of work. the COP will be-

(a) 0.75 (b) 7.5

(c) 6.5 (d) 10

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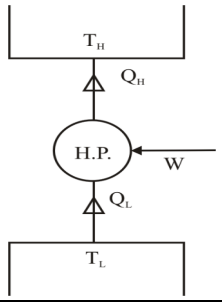
Ans. : (b)



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$Q_H = 60$ M cal/hr
$W = 8$ M cal/hr

$$(COP)_{H.P.} = \frac{60}{8} = 7.5$$

562. 300 kJ/s of heat is supplied at constant temperature of 500K to a heat engine. The heat rejection takes place at 300K. The following results are obtained (a) 210kJ (b) 180 kJ (c) 150kJ. Identify whether result is a reversible cycle, irreversible cycle or impossible cycle respectively-

(a) reversible/irreversible/impossible respectively
(b) irreversible/reversible/impossible respectively
(c) irreversible/impossible/reversible respectively.
(d) none of the above

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Ans. : (b) According to Classious inequality

$$\oint \frac{\delta Q}{T} \le 0$$

(a) $+\frac{300}{500} - \frac{210}{300} < 0$ (Irreversible)

(b) $\frac{300}{500} - \frac{180}{300} = 0$ (Reversible)

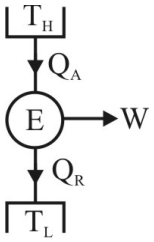
(c) $\frac{300}{500} - \frac{150}{300} > 0$ (Impossible)

563. A Carnot engine Working between 600K and 300K produces 200 KJ of work. The heat supplied is

(a) 200 KJ (b) 400KJ
(c) 2000J (d) 400 J

TSPSC AEE 2015

Ans. : (b)



$T_H = 600$ K
$T_L = 300$K
$W = 200$KJ

$$\eta_c = \frac{T_H - T_L}{T_H}$$

$$\eta_c = 0.5$$

$$\eta_c = \frac{\text{Work output}}{\text{heat supplied}}$$

$$0.5 = \frac{200}{Q_A}$$

$Q_A = 400$KJ

564. Two heat engines with efficiencies $\eta_1 = 0.8$ and $\eta_2 = 0.5$ are coupled together, then overall efficiency (η) of the system is :

(a) 0.65 (b) 0.95
(c) 0.70 (d) 0.90

KPSC ADF 2015

Ans. (d) : Given;
Efficiency of two heat engine :

$\eta_1 = 0.8$ and $\eta_2 = 0.5$

$\therefore$ overall efficiency of coupled engine (η_0)

$$= \eta_1 + \eta_2 - \eta_1 \times \eta_2$$

$$\Rightarrow \quad \eta_0 = 0.8 + 0.5 - (0.8 \times 0.5)$$

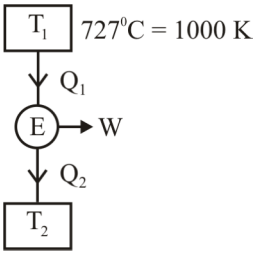
$$\boxed{\eta_0 = 0.90}$$

565. A Carnot cycle is having an efficiency of 0.75. If the temperature of high-temperature reservoir is 727^0C, what is the temperature of low-temperature reservoir?

(a) 23^0C (b) -23^0C
(c) 0^0C (d) 250^0C

Assam PSC CCE Pre 2015

Ans. (b) We know that,



$$\eta = \frac{T_1 - T_2}{T_1}$$

$$\frac{1000 - T_2}{1000} = 0.75$$

$750 = 1000 - T_2$

$T_2 = 250\text{K} = -23^0$C

566. The most efficient cycle is :

(a) Carnot Cycle
(b) Rankine Cycle
(c) Brayton Cycle
(d) Vapour Compression Cycle

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : The most efficient cycle is Carnot cycle.

567. A Carnot cycle engine receives and rejects heat with a 20^0C temperature differential between itself and the thermal energy reservoirs. The expansion and compression processes have a pressure ratio of 50. For 1 kg of air as the working substance, cycle temperature limits of 1000 K and 300 K and T_0= 280 K, determine the second law efficiency.

(a) 0.935 (b) 0.945
(c) 0.955 (d) 0.965
(e) 0.975

CGPSC AE 26.04.2015 SHIFT-II

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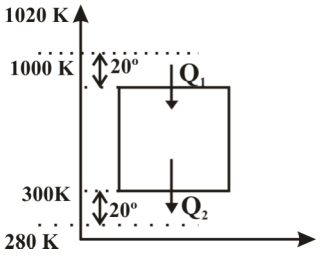
Ans. (d)



$$\frac{W}{Q_1} = \frac{T_1 - T_2}{T_1} = \frac{(1000-300)}{1000} = 0.7$$

$$\boxed{W = Q_1 \times 0.7}$$

$$\frac{W_{max}}{Q_1} = \frac{(1020-280)}{1020}$$

$$W_{max} = 0.725\ Q_1$$

$$\eta_{II} = \frac{W}{W_{max}} = \frac{0.70 Q_1}{0.725\ Q_1}$$

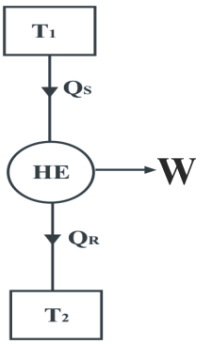
$$\boxed{\eta_{II} = 0.965}$$

568. A heat engine operates between a cold reservoir at temperature T_2 = 300K and a hot reservoir at Temperature T_1. It takes 200J of heat from the hot reservoir and deliver 120J of heat to the cold reservoir in a cycle. What will be the efficiency of the engine?

(a) 30% (b) 40%
(c) 60% (d) 20%

ISRO Scientist/Engineer 24.05.2014

Ans. (b) :



$Q_S = 200$ J
$Q_R = 120$ J
$T_2 = 300$ K

We know that,

$$\eta = \frac{W}{Q_S} = \frac{Q_S - Q_R}{Q_S}$$

$$= 1 - \frac{Q_R}{Q_S} = 1 - \frac{120}{200} = \frac{2}{5}$$

= 0.4 or 40%

569. The efficiency of carnot engine working between 800K and 400K is :

(a) 100% (b) 50%
(c) 0.5% (d) None of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : $\eta_{carnot} = 1 - \frac{T_2}{T_1}$

$$= 1 - \frac{400}{800}$$

$$= 1 - \frac{1}{2}$$

$$= \frac{1}{2}$$

$$\boxed{\eta_{carnot} = 50\%}$$

570. In a Carnot engine, when heat is supplied by a source :

(a) The temperature of the source decreases
(b) The temperature of the source remains constant
(c) The temperature of sink increases
(d) The temperature of both source and sink decreases

TNPSC AE (Industries) 09.06.2013

Ans. (b) : In a Carnot engine, when heat is supplied by a source the temperature of the source remains constant.

571. T_1 is high and T_2 is low temperature of a Carnot heat engine. Which is the most efficient way to increase efficiency?

(a) To increase T_1
(b) To decrease T_2
(c) To increase T_1 and T_2 both
(d) To decrease T_1 and T_2 both

UKPSC AE-2013, Paper-II

Ans. (b) : T_1 is high and T_2 is low temperature of a Carnot heat engine. The most efficient way to increase efficiency to decrease T_2.

572. A heat engine receives 1120 kJ of heat and rejects 840 kJ of heat while operating between two temperature limits of 560 K and 280 K. It indicates that the engine operates on the following cycle:-

(a) Reversible cycle
(b) Irreversible cycle
(c) Impossible cycle
(d) Unpredictable cycle

UKPSC AE-2013, Paper-II

Ans. (b) : Given as

$Q_s = 1120$ kJ, $Q_R = 840$ kJ
$T_1 = 560$ K, $T_2 = 280$ K

$$\eta_{ideal} = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{280}{560}$$

$\eta_{ideal} = 0.50$

$$\eta_{actual} = \frac{W_{net}}{Q_s} = \frac{1120 - 840}{1120} = 0.25$$

$\eta_{actual} < \eta_{ideal}$

It indicates that the engine operates on the irreversible cycle.

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573. A heat engine developes 60 kW work having an efficiency of 60%, Amount of heat rejected will be:-

(a) 400 kW (b) 10 kW
(c) 40 kW (d) 20 kW

UKPSC AE-2013, Paper-II

Ans. (c) : Given as

$W = 60\ kW$

$\eta = 60\%$

$$\eta = \frac{W}{Q_{in}}$$

$$Q_{in} = \frac{60 \times 100}{60}$$

$$\boxed{Q_{in} = 100\,kW}$$

Then, $Q_R = Q_{in} - W$

$Q_R = 100 - 60$

$$\boxed{Q_R = 40\,kW}$$

574. A heat engine operates at 75% of the maximum possible efficiency. The ratio of the heat source temperature to the heat sink temperature (in Kelvin) is 5/3. The fraction of the heat supplied, that is converted to work is :

(a) 0.6 (b) 0.4
(c) 0.3 (d) 0.7

APGENCO AE 2012

Ans. (c) : Given, $\eta = 0.75$

$$\frac{T_1}{T_2} = \frac{5}{3}$$

$$\Rightarrow \quad \eta = 1 - \frac{T_2}{T_1} = 1 - \frac{3}{5}$$

$\eta = 0.4$

$$\eta = \frac{\text{Work done}}{\text{Heat supplied}}$$

Since the engines uses only 75% of maximum possible efficiency,

$$\eta = \frac{W}{Q_s}$$

$$0.75 \times 0.4 = \frac{W}{Q_s}$$

$W = 0.3\ Q_s$

575. Efficiency of Carnot engine with $T_1 = 200^0C$, $T_2 = 30^0C$ is–

(a) 85% (b) 36%
(c) 80% (d) 12%

TRB Asstt. Prof., 2012

Ans. (b) : $\eta = \left(1 - \frac{T_2}{T_1}\right) \times 100$

Given, $T_1 = 200°C \Rightarrow 200 + 273 = 473\ K$

$T_2 = 30°C \Rightarrow 30 + 273 = 303\ K$

$$\eta = \left(1 - \frac{303}{473}\right) \times 100$$

$\eta = 35.94 \approx 36\%$

576. In a Carnot cycle, heat is transferred at

(a) constant pressure
(b) constant volume
(c) constant enthalpy
(d) constant temperature

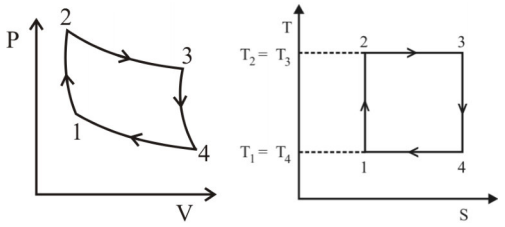
TRB Asstt. Prof., 2012

Ans. (d) : Carnot Cycle– It is an ideal cycle of work producing device.

It is not a practical working cycle because reversible adiabatic process is very fast and reversible isothermal process is very slow.

So it cannot takes place in one cycle.

Carnot consist of four process.



Process (1–2) = Reversible adiabatic compression
Process (2–3) = Reversible Isothermal heat addition
Process (3–4) = Reversible adiabatic Expansion.
Process (4–1) = Reversible Isothermal Heat Rejection.

577. An ideal engine absorbs heat at 127ºC and rejects at 77ºC. The efficiency is

(a) 13% (b) 39%
(c) 50% (d) 40%

ISRO Scientist/Engineer 2012

Ans. (a) : $T_1 = 127ºC$

$T_1 = 127 + 273 = 400K$

$T_2 = 77ºC$

$= 273 + 77 = 350K$

Efficiency $\eta = 1 - \frac{T_2}{T_1}$

$$= 1 - \frac{350}{400}$$

$\eta = 12.5\% = 13\%$

578. If maximum surface temperature of sea is 30 °C and temperature in depth is 4 °C, how much can be the maximum thermal efficiency of Ocean Thermal Conversion (OTEC) system?

(a) 8.58% (b) 13.3%
(c) 86.7% (d) none of the above

UKPSC AE 2012 Paper–II

Ans. (a) : $\eta = \frac{T_1 - T_2}{T_1} = \frac{30 - 4}{303} = \frac{26}{303} = 8.58\%$

579. Carnot cycle is not considered as a practical cycle because

(a) its p – V diagram is narrow
(b) its thermal efficiency is low
(c) heat addition takes place at high pressure
(d) heat rejection takes place at high temperature

UKPSC AE 2012 Paper–II

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Ans. (a) : Carnot cycle is not considered as a practical cycle because its p – V diagram is narrow.

580. An engine receives 15152 J/s of heat and produces 5 kW of power. The efficiency of the engine is

(a) 25% (b) 27.5%
(c) 30% (d) 33%

UKPSC AE 2012 Paper–II

Ans. (d) : Efficiency of engine

$$\eta = \frac{\text{Power}}{\text{heat supply}} = \frac{5}{15.152}$$

$$= 0.329 \simeq 33\%$$

581. An inventor claims to have developed an engine that takes in 105 MJ at a temperature of 400 K, rejects 42 MJ at a temperature of 200 K and delivers 15 kWh of mechanical work. Is this engine feasible ?

(a) Yes
(b) No
(c) The information given is not sufficient to find answer.
(d) May be or may not be depends upon several factors.

UKPSC AE 2012 Paper–II

Ans. (b) : $\eta_{ideal} = 1 - \frac{T_L}{T_H} = 1 - \frac{200}{400} = 0.5$

$$\eta_{actual} = 1 - \frac{Q_L}{Q_H} = 1 - \frac{42}{105} = 0.6$$

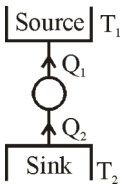
$\eta_{ideal} < \eta_{actual}$ not possible

582. If Q_1 is the heat transfer between hot temperature source & machine and Q_2 between cold temperature source & machine, then for heat pump, COP will be equal to

(a) $\frac{Q_1}{Q_1 - Q_2}$ (b) $\frac{Q_2}{Q_1 - Q_2}$
(c) $\frac{Q_1}{Q_2 - Q_1}$ (d) $\frac{Q_2}{Q_2 - Q_1}$

TNPSC ACF 2012

Ans. (a) :



$$(COP)_{HP} = \frac{Q_1}{Q_1 - Q_2}$$

$$(COP)_R = \frac{Q_2}{Q_1 - Q_2}$$

583. The Thermal efficiency of an engine which is supplied heat at the rate of 15,000 N.m/sec and gives an output of 4500 W, is

(a) 45% (b) 50%
(c) 30% (d) 33%

APPSC AEE Screening Test 2016
TNPSC ACF 2012

Ans. (c) : Q_S = 15000 Nm/sec = 1500 J/sec = 1500 W
W = 4500 W

$$\eta = \frac{W}{Q_s} = \frac{4500}{15000} = 30\%$$

584. A Carnot cycle having efficiency of 0.75. if the temperature of the high temperature reservoir is 727°C, what is the temperature of low temperature reservoir?

(a) 23°C (b) –23°C
(c) 0°C (d) 250°C

VIZAG STEEL MT 10.06.2012

Ans. (b) : Efficiency of carnet cycle (η) = 0.75

$T_H = T_{High} = 727°C$
$= 727 + 273$
$T_H = 1000$ K
$T_{Low} = T_L = ?$

$$\text{Efficiency } (\eta) = \frac{T_H - T_L}{T_H}$$

$$\Rightarrow 0.75 = \frac{1000 - T_L}{1000}$$

$\Rightarrow 750 = 1000 - T_2$
$\Rightarrow T_L = 1000 - 750 = 250$ K
$\Rightarrow T_L = 250 - 273$

$$\boxed{T_L = -23^0 C}$$

585. A Carnot engine produces 70 kW while operating between temperature limits of 927°C and 27°C. The heat rejected by the engine per second is :

(a) 10 kW (b) 20 kW
(c) 30 kW (d) 40 kW

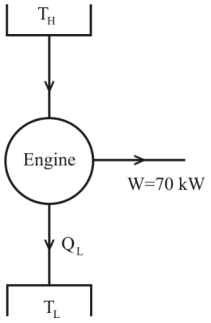
UPSC JWM Advt. No.-52/2010

Ans. (b) : Solution :

$T_H = 927°C + 273 = 1200$ K
$T_L = 27°C + 273 = 300$ K

$$\eta_{HE} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{1200} = 1 - \frac{1}{4}$$

$$\eta_{HE} = 3/4$$



also, $\eta_{HE} = \frac{Q_H - Q_L}{Q_H} = \frac{3}{4}$

$$\Rightarrow \frac{W}{Q_H} = \frac{3}{4}$$

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$$Q_H = W \times \frac{4}{3} = \frac{70 \times 4}{3} = \frac{280}{3}$$

$\therefore \quad Q_H - Q_L = 70$

$\Rightarrow \quad Q_L = Q_H - 70 = \frac{280}{3} - 70$

$$Q_L = \frac{70}{3}$$

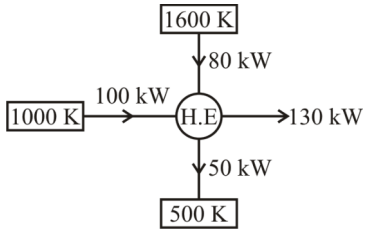
$\Rightarrow \quad \boxed{Q_L = 23.33\text{kW}} \approx 20\text{kW}$

586. An engineer claims to have developed an heat engine, which absorbs 100 kW of heat from a thermal reservoir at 1000 K, 80 kW of heat from another thermal reservoir at 1600 K, produces 130 kW of work output, and rejects heat to the thermal reservoir at 500 K. Which of the following statement(s) is (are) correct?

(a) This heat engine violates second law of thermodynamics
(b) The amount of heat rejected to the sink is 50 kW
(c) Cyclic integral of dQ/T is 0.05
(d) All of the above

APPSC IOF, 2009

Ans. (d) :



Heat rejected to sink = 100 + 80 – 130
= 50 kW

$$\oint \frac{\delta Q}{T} = \frac{100}{1000} + \frac{80}{1600} - \frac{50}{500} = 0.05$$

From Clausius inequality $\oint \frac{\delta Q}{T} \le 0$

But, here $\oint \frac{\delta Q}{T} > 0$ so violets the second law so all options are correct.

587. A heat engine is supplied with heat from a source at 1000 K and rejects heat to a sink at 300 K and this engine is operating with a thermal efficiency of 30%. The second law efficiency of this heat engine is :

(a) 60% (b) 43%
(c) 30% (d) 70%

APPSC IOF, 2009

Ans. (b) : Given, $T_1 = 1000\text{K}$, $T_2 = 300\text{K}$

$\eta_{actual} = 30\% = 0.30$

$$\eta_{carnot} = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{1000} = 1 - 0.30$$

$\eta_{carnot} = 0.70$

We know that,

$$\eta_{II} = \frac{\eta_{actual}}{\eta_{carnot}}$$

$$= \frac{0.3}{0.7}$$

$= 43\%$

588. The efficiency of a thermodynamic cycle cannot be unity due to which of the following reasons?

(a) Violates the first law of thermodynamics
(b) Violates the second law of thermodynamics
(c) Violates the third law of thermodynamics
(d) Violates the zeroth law of thermodynamics

UPSC JWM Adv. No-16/2009

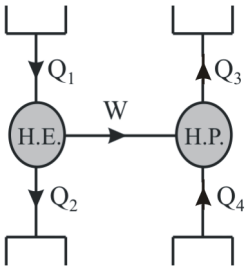
Ans. (b) : The efficiency of a thermodynamic cycle can not be unity because it violates the second law of thermodynamics.

589. A heat engine is used to drive a heat pump. The heat transfers from the heat engine and from the heat pump are used to heat the circulating water in a plant. The efficiency of the heat engine is 30% whereas the COP of the heat pump is 5. The heat transfer to the circulating water and the heat transfer to the engine bears a ratio of

(a) 0.7 (b) 0.8
(c) 1.5 (d) 2.2

DRDO Scientists 2009

Ans. (c) :



$$\frac{Q_3}{Q_1} = ?$$

$$\eta_{HE} = 0.3 = \frac{W}{Q_1}$$

$$(COP)_{HP} = 5 = \frac{Q_3}{W}$$

$$\frac{Q_3}{Q_1} = 5 \times 0.3 = 1.5$$

590. The COP of a Carnot heat pump operating between –3°C and 27°C is

(a) 10.0 (b) 9.0
(c) 0.111 (d) 0.10

DRDO Scientists 2008

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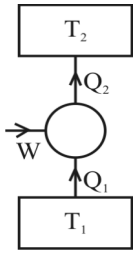
Ans. (a)



$T_1 = -3°C = 270\ K$

$T_2 = 27°C = 300\ K$

$$(COP)_{HP} = \frac{Q_2}{Q_2 - Q_1} = \frac{T_2}{T_2 - T_1} = \frac{300}{30} = 10$$

591. An inventor claims to have devised an engine that produces 2000 kJ of work while receiving 1900 kJ of heat from a single reservoir during a complete cycle of the engine. Such an engine would violate

(a) only the first law of thermodynamics and no other laws

(b) only the second law of thermodynamics and no other laws

(c) both the first and the second law of thermodynamics

(d) neither the first nor the second law of thermodynamics

DRDO Scientists 2008

Ans. (a) : An inventor claims to have developed an engine that produces 2000 kJ of work while receiving 1900 kJ, of heat. The proposed engine claims to produce more work in a cyclic process than that the amount of heat that is supplied. So it is in violation of the first law of thermodynamics.

$$\Delta U = Q - W$$

592. The efficiency η of any heat engine and efficiency η_R of a reversible heat engine operating between common heat source and heat sink are related as

(a) $\eta > \eta_R$ (b) $\eta < \eta_R$

(c) $\eta \geq \eta_R$ (d) $\eta \leq \eta_R$

ISRO Scientist/Engineer 2007

Ans. (d) : η = efficiency of any heat engine

η_R = Reversible heat engine

So, $\eta \leq \eta_R$

Any heat engine must also include reversible heat engine or Carnot engine

593. 300 kJ/sec of heat is applied at a constant fixed temperature of 290°C to a heat engine. The heat rejection takes place at 8.5°C. Then match the following :

Results obtained	Cycle
(a) 215kJ/sec are rejected	(1) reversible
(b) 150kJ/sec are rejected	(2) irreversible
(c) 75kJ/sec are rejected	(3) impossible

	A	B	C
(a)	1	2	3
(b)	2	1	3
(c)	3	2	1
(d)	1	3	2

ISRO Scientist/Engineer 2007

Ans. (b) : $Q_1 = 300$ kJ/sec

$T_1 = 290 + 273 = 563\ K$

$T_2 = 8.5 + 273 = 281.5\ K$

We know that,

$\int \frac{dQ}{T} = 0$ For reversible

$\int \frac{dQ}{T} < 0$ For irreversible

$\int \frac{dQ}{T} > 0$ Impossible

If $Q_2 = 215, Q_1 = 300$ kJ/sec

$$\frac{Q_1}{T_1} - \frac{Q_2}{T_2} = +\frac{300}{563} - \frac{215}{281.5} = -0.23$$ irreversible

When, $Q_2 = 150, Q_1 = 300$

$$\frac{Q_1}{T_1} - \frac{Q_2}{T_2} = +\frac{300}{563} - \frac{150}{281.5} = 0$$ reversible

When, $Q_2 = 75, Q_1 = 300$

$$\frac{Q_1}{T_1} - \frac{Q_2}{T_2} = \frac{300}{563} - \frac{75}{281.5} = 0.27 > 0$$ impossible

594. Which of the following is an irreversible cycle

(a) Carnot (b) Stirling

(c) All of the above (d) None of the above

J & K PSC Screening, 2006

Ans. (d) : Carnot and Stirling are reversible cycle.

595. If other factors are constant, with increase in the temperature of sink, the efficiency of Carnot engine

(a) increases

(b) remains same

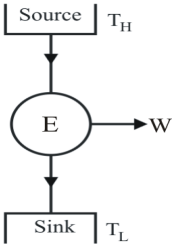
(c) decreases

(d) may increase or decrease

WBPSC AE 2003

Ans. (c) : $\downarrow \eta_{carnot} = \frac{T_H - T_L}{T_H} = 1 - \frac{T_L}{T_H} \uparrow$

If (T_L) increases then (η_c) of carnot engine decreases.



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4. Entropy

596. The condition for reversibility of a cycle is-

(a) Cyclic $\oint \frac{dQ}{T} < 0$ (b) Cyclic $\oint \frac{dQ}{T} > 0$

(c) Cyclic $\oint \frac{dQ}{T} = 0$ (d) None of these

TNPSC AE 2017
Nagaland PSC (CTSE) 2016, Paper-II
MPPSC AE 08.11.2015
UKPSC AE 2012 Paper–II
ISRO Scientist/Engineer 2006
WBPSC AE 2003

Ans. (c) : According to Clausius inequality–

$\oint \frac{dQ}{T} = 0$ Reversible cycle

$\oint \frac{dQ}{T} < 0$ Irreversible cycle

$\oint \frac{dQ}{T} > 0$ Impossible cycle

597. The value of $\oint \frac{dQ}{T}$ for an irreversible process is

(a) 0 (b) > 0
(c) < 0 (d) 1

RPSC IOF, 2020, Gujarat PSC AE 2019
JPSC AE PRE 2019, APPSC Poly. Lect. 2013
WBPSC AE, 2007, ESE 2011, 1994
GATE 1993

Ans. : (c) : $\oint \frac{dQ}{T} = 0$, the cycle is reversible

$\oint \frac{dQ}{T} < 0$, the cycle is irreversible and possible

$\oint \frac{dQ}{T} > 0$, the cycle is impossible.

598. An ideal gas of mass m and temperature T_1 undergoes a reversible isothermal process from an initial pressure P_1 to final pressure P_2. The heat loss during the process is Q. The entropy change ΔS of the gas is

(a) $mR\ell n\left(\frac{P_2}{P_1}\right)$ (b) $mR\ell n\left(\frac{P_1}{P_2}\right)$

(c) $mR\ell n\left(\frac{P_2}{P_1}\right) - \frac{Q}{T_r}$ (d) Zero

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
UPPSC AE 13.12.2020, Paper-2
OPSC AEE 2019 PAPER - II
BPSC AE Mains 2017 Paper - V
GATE 2012

Ans. (b) : Change in Entropy

$$S_2 - S_1 = C_p \ln\left(\frac{T_2}{T_1}\right) - mR \ln\left(\frac{P_2}{P_1}\right)$$

For Isothermal process

$$T_1 = T_2 = \text{constant}$$

$$S_2 - S_1 = -mR \ln\left(\frac{P_2}{P_1}\right)$$

$$\boxed{S_2 - S_1 = mR \ln\left(\frac{P_1}{P_2}\right)}$$

599. The increase in entropy of a system represents

(a) increase in availability of energy
(b) increase in temperature
(c) decrease in pressure
(d) degradation of energy

APGENCO AE, 2017
APPSC AEE MAINS 2016 PAPER-III
Assam PSC CCE Pre 2015
VIZAG STEEL MT 2011
CSE Pre-1994

Ans. (d) : Entropy is the measure of degree of disorderness. Increase in entropy of a system represents degradation of energy.

Entropy principle, $dS \geq \frac{dQ}{T}$

600. Area under T - S diagram represents

(a) heat transfer for reversible process
(b) heat transfer for irreversible process
(c) heat transfer for all process
(d) heat transfer for adiabatic process

Oil India Senior Officer 23.12.2020
HPPSC Asstt. Prof. 29.10.2016
UKPSC AE-2013, 2007 Paper-II
TNPSC AE 2013

Ans. (a) : For reversible process only,

$Q = \int Tds$

Hence, heat transfer for reversible process is represented by area under T-S curve.

601. The Heat transfer takes place due to

(a) Change of temperature
(b) Change in Entropy
(c) Change of pressure
(d) Mass transfer

CIL MT 27.02.2020, RPSC AE 2018
WBPSC AE, 2017
APPSC AEE Screening Test 2016

Ans. (b) : Heat transfer–

$\delta Q = Tds$

T – Absolute temperature
ds – Change in entropy

602. If the value of $\oint \frac{dQ}{T}$ is greater than zero, the nature of the thermodynamic cycle is

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(a) Reversible
(b) Irreversible
(c) Both reversible and irreversible
(d) Impossible

HPPSC AE 2018
TNPSC AE (Industries) 09.06.2013
UPSC JWM Adv. No-16/2009
ESE 1997

Ans. (d) : We know that Clausius inequality–

$$\oint \frac{dQ}{T} = 0 \quad [\text{Reversible cycle}]$$

$$\oint \frac{dQ}{T} < 0 \quad [\text{Irreversible cycle}]$$

$$\oint \frac{dQ}{T} > 0 \quad [\text{Impossible cycle}]$$

603. When two gases suddenly mix up with each other then resultant entropy of the system will:
(a) decrease
(b) remain same
(c) increase
(d) may increases of decrease depending upon the initial value

GPSC Engineer, Class-II Pre-19.01.2020
Kerala PSC AE (Irrigation) 05.04.2017
Kerala PSC Asstt. Prof. 27.09.2016
J&K PSC Civil Services Pre, 2013

Ans. (c) : Entropy increases when two gas mix with each other, but entropy doesn't increase when the two gases mixing are at same temperature. It is an irreversible process; can not separate them again after mixing up without external aid and changing the surroundings.
So the system prefers to be in the mixed up state than in the initial state. From second law we know that processes which increase entropy are favored over the opposite.

604. Which of the following processes is irreversible process
(a) Isothermal (b) Adiabatic
(c) Throttling (d) All of the above

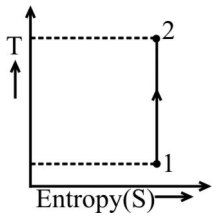
CGPSC AE 15.01.2021, WBPSC AE, 2017
Vizag Steel (MT) 2017

Ans. (c) :
Throttling process in which
- No change in enthalpy from state one to state two ($h_1 = h_2$)
- No work is done (W = 0)
- Process called isenthalpic
- Process is adiabatic (Q = 0)

605. The change in entropy is zero during ..... process.
(a) Polytropic (b) Adiabatic
(c) constant pressure (d) Hyperbolic

(HPPSC LECT. 2016), TNPSC AE 2014
UKPSC AE 2012 Paper–II

Ans. : (b)



In a reversible adiabatic process, no heat enters or leaves the gas
$\delta Q = 0$
$\delta S = 0$
Change of entropy during a reversible adiabatic process is zero. The reversible adiabatic on T-S graph is shown by a vertical straight line 1-2.

606. A system undergoes a state change from 1 to 2. According the second law of thermodynamics for the process to be feasible, the entropy change, S_2-S_1 of the system
(a) is positive or zero
(b) is negative or zero
(c) is zero
(d) can be positive, negative or zero

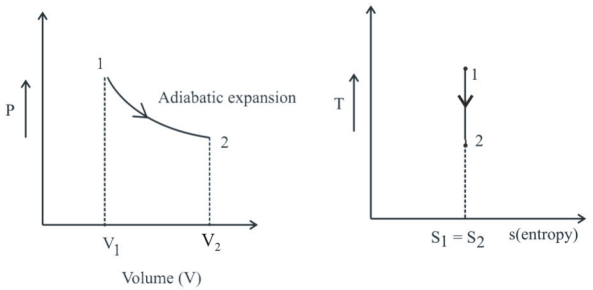
CGPSC Asstt. Workshop Supt., 17.07.2016
ISRO Scientist/Engineer 12.05.2013
GATE 1997

Ans. (d) : Entropy change $\Delta s \geq 0$ (for universe) but for system it can be (+ve), (-ve) or zero.
(Δs) universe ≥ 0
By second law of Thermodynamics.

607. During adiabatic expansion
(a) internal energy remains constant
(b) temperature remains constant
(c) entropy remains constant
(d) enthalpy remains constant

BPSC AE Mains 2017 Paper - V
TNPSC AE, 2008

Ans. (c) : During adiabatic expansion, entropy constant



608. The entropy may be expressed as a function of
(a) pressure and temperature
(b) temperature and volume
(c) heat and work
(d) All of the above

TNPSC AE 2014
APPSC Poly. Lect. 2013

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Ans. (a) : The entropy may be expressed as a function of pressure and temperature & also for volume.

$$\Delta s = C_p \ln\frac{T_2}{T_1} - R \ln\frac{P_2}{P_1}$$

$$\Delta s = C_v \ln\frac{T_2}{T_1} + R \ln\frac{V_2}{V_1}$$

Note-But the official answer key is given by commission TNPSC AE option is (a).

609. A process 'A' is irreversible and adiabatic process 'B' is reversible and adiabatic. The entropy change in process 'A' and process 'B', respectively are

(a) Zero and positive (b) Zero and negative
(c) Negative and zero (d) Positive and zero

Nagaland PSC (CTSE) 2016, Paper-II
Haryana PSC Civil Services Pre, 2014

Ans. (d) : If a process 'A' is irreversible and adiabatic process 'B' is reversible and adiabatic. The entropy change in process 'A' and process 'B', is positive and zero, respectively.

Process A $\rightarrow$ Irreversible and adiabatic $\rightarrow \Delta S > 0$ (positive)

Process B $\rightarrow$ Reversible and adiabatic $\rightarrow \Delta S = 0$ (zero)

610. The entropy of universe tends to

(a) become zero
(b) remain constant
(c) be maximum
(d) attain a certain finite minimum value

MPPSC AE 08.11.2015
TSPSC Managers, 2015

Ans. (c) : The entropy of universe increases and tends towards the maximum value.

According to the second law of thermodynamics, the total entropy of the system and its surroundings (universe) increases in a spontaneous process.

$$\Delta S_{uni} = \Delta S_{total} = \Delta S_{sys} + \Delta S_{surr}$$

$$\Delta S_{uni} > 0$$

Note-Option (a) is given by the commission.

611. For reversible adiabatic process, change in entropy is

(a) positive (b) negative
(c) zero (d) depends on pressure

UJVNL AE 12.09.2021
ISRO Scientist/Engineer (RAC) 07.05.2017
APPSC IOF, 2009

Ans. (c) : For reversible adiabatic process, change in entropy is zero.

We know that

$$\Delta S = \frac{\delta Q}{T} + \delta \dot{S}_{gen}$$

For adiabatic process, $\delta Q = 0$

and for reversible process, $\delta \dot{S}_{gen} = 0$

So, for reversible adiabatic process

$$\Delta S = 0$$

612. When a system undergoes a process such that $\int \frac{dQ}{T} = 0$ and $\Delta S > 0$, the process is :

(a) isothermal
(b) reversible adiabatic
(c) irreversible adiabatic
(d) isobaric

MPPSC AE 2016, CSE Pre-1999

Ans. : (c) $\int \frac{dQ}{T} = 0$ and $\Delta S > 0$

the process is irreversible adiabatic

i. $\Delta S > 0$ (Irreversible process)

ii. $\Delta S = 0$ (Reversible process)

iii. $\Delta S < 0$ (Not possible)

We know that,

$$\Delta S = \int \frac{\delta Q}{T} + \delta \dot{S}_{gen}$$

$\int \frac{\delta Q}{T}$ = change in entropy due to external heat exchange by the system

$\delta \dot{S}_{gen}$ = change in entropy due to irreversibility like friction, viscosity, etc

$\delta \dot{S}_{gen}$ = entropy generation is a path function & inexact differential.

613. In an experimental set-up, air flows between two stations P and Q adiabatically. The direction of flow depends on the pressure and temperature conditions maintained at P and Q. The conditions at station P are 150 kPa and 350 K. The temperature at station Q is 300 K. The following are the properties and relations pertaining to air:

C_p = 1.005 kJ/kgK
C_v = 0.718 kJ/kgK
R = 0.287 kJ/kgK

If the pressure at station Q is 50kPa, the change in entropy (SQ – SP) in kJ/kgK is

(a) – 0.155 (b) 0
(c) 0.160 (d) 0.355

GPSC Asstt. Director of Transport 05.03.2017
GATE 2011

Ans. (c) : Given, $P_1 = 150$ kPa, $T_1 = 350$ K

$P_2 = 50$ kPa, $T_2 = 300$ K

$$S_Q - S_P = C_v \ln\frac{T_2}{T_1} + R \ln\left[\frac{V_2}{V_1}\right]$$

$$\frac{V_2}{V_1} = \frac{P_1 T_2}{P_2 T_1} = 2.57$$

$S_Q - S_P = -0.1107 + 0.287 \ln(2.57)$

$= 0.16$ kJ/kgK

614. In an experimental set-up, air flows between two stations P and Q adiabatically. The direction of flow depends on the pressure and temperature conditions maintained at P and Q. The conditions at station P are 150 kPa and 350 K. The temperature at station Q is 300 K.

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The following are the properties and relations pertaining to air:
C_p = 1.005 kJ/kgK
C_v = 0.718 kJ/kgK
R = 0.287 kJ/kgK
If the air has to flow from station P to station Q, the maximum possible value of pressure in kPa at station Q is close to
(a) 50 (b) 87
(c) 128 (d) 150

GPSC Asstt. Director of Transport 05.03.2017
GATE 2011

Ans. (b) : Given, $\left[\gamma = \frac{C_P}{C_V} = 1.4\right]$

$$\frac{T_Q}{T_P} = \left(\frac{P_Q}{P_P}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow \frac{300}{350} = \left(\frac{P_Q}{150}\right)^{\frac{1.4-1}{1.4}}$$

P_Q = 87.45 kPa

615. In which one of the following processes is there an increase in entropy with no degradation of energy?
(a) Polytropic expansion
(b) Isothermal expansion
(c) Isochoric heat addition
(d) Isobaric heat addition

HPPSC Asstt. Prof. 18.09.2017
CSE Pre-1996

Ans. (b) : Isothermal expansion

616. A system of 100 kg mass undergoes a process in which its specific entropy increases from 0.3 kJ/kg K to 0.4k J/kg K. At the same time, the entropy of the surrounding decreases from 80 kJ/K to 75 kJ/K. The process is :
(a) Reversible and isothermal
(b) Irreversible
(c) Reversible only
(d) Isothermal only

GPSC ARTO Pre 30.12.2018
ESE 2017, 1997

Ans. (b) : $(dS)_{univ} = (dS)_{system} + (dS)_{surrounding}$
$= 100\,(0.4 - 0.3) + (75 - 80)$
$= 10 - 5$
$(dS)_{univ} = 5$
$(dS)_{univ} > 0 \Rightarrow$ the process is irreversible

617. In which of the following situations, the entropy change is negative?
(a) Air expands isothermally from 6 bar to 3 bar
(b) Air is compressed to half of its volume at constant pressure
(c) Air is supplied with heat at constant volume till its pressure is doubled
(d) Air expands adiabatically from 6 bar to 3 bar

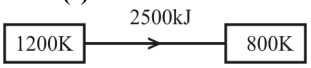
UPPSC AE 13.12.2020, Paper-2
ESE 2000

Ans. (b) : During compression process, as air is compressed to half its volume at constant pressure entropy change is negative.

618. Consider the following two processes.
(I) A heat source at 1200 K loses 2500 KJ of heat to sink at 800 K.
(II) A heat source at 800 K loses 2000 kJ of heat to sink at 500 K.
Which of the following statement is TRUE?
(a) Process (I) is more irreversible than Process (II)
(b) Process (II) is more irreversible than Process (I)
(c) Irreversibility associated in both the process is equal.
(d) Both the processes are reversible

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
GATE 2010

Ans. (b): Process-(I)

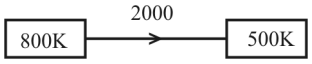


$I = T_0 \Delta S$
$I \propto \Delta S$

$$(\Delta S)_I = \left[-\frac{2500}{1200} + \frac{2500}{800}\right] = 1.04$$

$\boxed{(\Delta S)_I = 1.04}$

Process-(II)



$$(\Delta S)_{II} = \left[-\frac{2000}{800} + \frac{2000}{500}\right]$$

$\boxed{(\Delta S)_{II} = 1.5}$

$(\Delta S)_{II} > (\Delta S)_I$

Hence process (II) is more irreversible then

619. Ninety kilograms of ice at 0ºC are completely melted. Find the entropy change, in kJ/K, if T_2 = 0ºC. (Latent heat of fusion is 318.5 kJ/kg.K)
(a) 0 (b) 45
(c) 105 (d) 85

ISRO Scientist/Engineer (RAC) 12.01.2020
ISRO Scientist/Engineer 2011

Ans. (c) : mass, m = 90 kg
T = 0ºC = (0 + 273) K
= 273 K
Latent heat of fusion
l = 318.5 kJ/kgK

Entropy change $ds = \frac{ml}{T} = \frac{dQ}{T}$

$$= \frac{90 \times 318.5}{273}$$

= 105 kJ/K

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620. The change of entropy, when heat is absorbed by the gas, is

(a) positive (b) negative

(c) positive or negative (d) zero

TNPSC AE 2017, WBPSC AE 2008

Ans. (a) : The change of entropy, when heat is absorbed by the gas, is positive

We know that

$$\Delta S = \frac{\delta Q}{T} + \delta S_{gen}$$

then $\delta Q \uparrow \rightarrow \Delta S \uparrow$

621. Entropy generated (S_{gen}) can be taken as a criterion to indicate feasibility of process. Which of the following conditions are correct?

1. If $S_{gen} = 0$, then the process is a reversible process.

2. If $S_{gen} > 0$, then the process is an irreversible process.

3. If $S_{gen} < 0$, then the process is impossible.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (d) : All statements are correct-

- If $S_{gen} = 0$, then the process is a reversible process.
- If $S_{gen} > 0$, then the process is an irreversible process.
- If $S_{gen} < 0$, then the process is impossible.

622. Total energy of the universe

(a) is always increasing

(b) is always decreasing

(c) either increases or decreases

(d) is always constant

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : The total amount of energy and matter in the universe remains constant, merely changing from one form to another. The first law of thermodynamics states that energy is always conserved, it can not be created or destroyed. In essence energy can be converted from one form into another.

623. When the potential gradient in a process is infinitesimal or zero, what will be the change in entropy of the universe?

(a) Entropy increases

(b) Entropy decreases

(c) No change in entropy

(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : When the potential gradient in a process is infinitesimal or zero, then there is no change in the entropy of universe.

Note-Official answer is given as option (a).

624. One kilomole of an ideal gas is throttled from an initial pressure of 0.5 MPa to 0.1 MPa. The initial temperature is 300K. The entropy change of the universe is

(a) 13.38 kJ/K (b) 4014.3 kJ/K

(c) 0.0446 kJ/K (d) 0.0446 kJ/K

GATE, 1995

Ans. (a) Given,

M = 1 kilomole

As, $T_2 = T_1$ since ideal gas as for throttling (isenthelpic) process temperature remains constant,

$$\therefore \quad \Delta s = C_p l\,n\frac{T_2}{T_1} - R l\,n\frac{P_2}{P_1}$$

$$\Delta s = \frac{R}{M} l\,n\frac{0.1}{0.5}$$

$$= -\overline{R}\; l\,n\frac{1}{5}$$

$$\Delta s = 13.38\,kJ/K$$

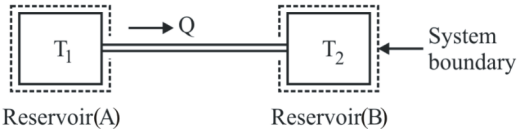
625. If heat of Q is transferred from high temperature reservoir at T_1 to the low temperature reservoir at T_2, then the entropy change of universe is :

(a) $\frac{T_1 - T_2}{T_1 T_2} Q$ (b) $\frac{T_1 - T_2}{T_1 / T_2} Q$

(c) $\frac{T_1 T_2}{T_1 - T_2} Q$ (d) $\frac{T_1 - T_2}{T_2 / T_1}$

NLC GET 17.11.2020, Shift-II

Ans. (a) : Heat Q be the rate of heat transfer reservoir at T_1 to the Flow temperature reservoir T_2



For reservoir A $\quad \Delta S_A = -\frac{Q}{T_1}$

It is negative because heat flows out of the reservoir

For reservoir B $\quad \Delta S_B = +\frac{Q}{T_2}$

It is positive because heat flows into the reservoir.

For isolated system at steady state,

$$S = S_A + S_B$$

$$(\Delta S)_{univ} = (\Delta S)_A + (\Delta S)_B$$

$$(\Delta S)_{universe} = -\frac{Q}{T_1} + \frac{Q}{T_2}$$

$$(\Delta S)_{universe} = Q\left[\frac{T_1 - T_2}{T_1 . T_2}\right]$$

$T_1 > T_2$ so $(\Delta S)_{univ}$ = +ve

So the process is irreversible and possible.

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626. Which of the following states, "It is impossible, by any means, to reduce any system to the absolute zero of temperature in a finite number of operations"?

(a) Kelvin - Planck's statement of II law
(b) Carnot Theorem
(c) Clausius statement of II law
(d) Fowler - Guggenheim Statement of III law

VIZAG MT, 14.12.2020

Ans. (d) : Fowler-Guggenheim Statement of III[rd] Law– It states that it is impossible, by any means, to reduce any system to the absolute zero temperature in a finite number of operations.

627. A mass of 5 kg of water at 293 K is turned completely to ice at 273 K. The entropy change in the process is

(a) 6.13479 kJ/K (b) –1.4847 kJ/K
(c) –7.6195 kJ/K (d) 8.3195 kJ/K

UPPSC AE 13.12.2020, Paper-2

Ans. (b) : Given,

Mass of water, m = 5 kg

$T_1 = 293$ K

$T_2 = 273$ K

$C_P = 4.2$ kJ/kg

Entropy change, $\Delta S = m C_p \ln\left(\frac{T_2}{T_1}\right)$

$= 5 \times 4.2 \ln\left(\frac{273}{293}\right)$

$= -1.484$ kJ/K

628. A perfect gas having $P_1 = 0.1$ N/mm^2, $V_1 = 0.18$m^3, $T_1 = 20^0$C is compressed to $\frac{1}{10}$ of its volume in an isothermal process. The change in entropy is

(a) – 141.45 J/K (b) 141.45 J/K
(c) – 4144.23 J/K (d) Zero

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : Given,

$P_1 = 0.1$ N/mm^2

$V_1 = 0.18$ m^3

$T_1 = 20^0$C = 293 K

$V_2 = \frac{V_1}{10}$

Change in entropy

$\Delta S = mR \ln\left(\frac{V_2}{V_1}\right)$

$= \frac{P_1 V_1}{T_1} \ln\left(\frac{V_1}{10V_1}\right)$

$= \frac{0.1 \times 0.18}{213} \times \ln\left(\frac{1}{10}\right)$

$\boxed{\Delta S = -141.45 \text{ J/K}}$

629. If the entropy decreases in a process, the following statement is true

(a) the process is impossible
(b) the process is likely to be exothermic
(c) the process is likely to be endothermic
(d) the process cannot be irreversible

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : In exothermic process heat is removed from the system.

So, $\delta Q = -ve$

$dS = \frac{\delta Q}{T} = -ve$

$dS < 0 \Rightarrow S_2 - S_1 < 0 \Rightarrow S_2 < S_1$

630. When 7200 kJ of heat is transferred from 1000 K source to 500 K sink, the net change in entropy is about:

(a) +3 kK (b) +7 kJ
(c) –7 kJ (d) –3 kJ

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Net change in entropy = $(\Delta S)_{source} + (\Delta S)_{sink}$

$= -\frac{7200}{1000} + \frac{7200}{500}$

$= -7.2 + 14.4$

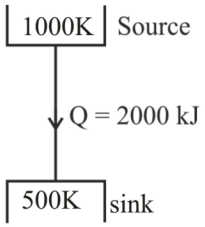
$(\Delta S)_{net} = +7.2$

631. A hot body at 1000 K transfers 2000 kJ of heat to a body at 500 K. Determine the net entropy change?

(a) +4 kJ/kg (b) –2 kJ/kg
(c) +2 kJ/kg (d) –4 kJ/kg

ISRO Scientist/Engineer 12.01.2020

Ans. (c) :



$\Delta S_{net} = \Delta S_{source} + \Delta S_{sink}$

$= \frac{-2000}{1000} + \frac{2000}{500} = -2 + 4$

$\Delta S_{net} = +2$ kJ/kg

632. Entropy of the system ..... with the addition of heat.

(a) increases
(b) decreases
(c) remains the same
(d) first increases and then decreases

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : Entropy of the system increase with the addition of heat.

Heat addition to the system is positive i.e δQ is positive

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$$ds = \left(\frac{dQ}{T}\right)$$

$ds = +ve$

$ds > 0$

$S_2 - S_1 > 0$

$\boxed{S_2 > S_1}$

633. The entropy of 1 kg water (s_f) at any temperature T K in terms of specific heat of water (c_{pw}) can be expressed as :

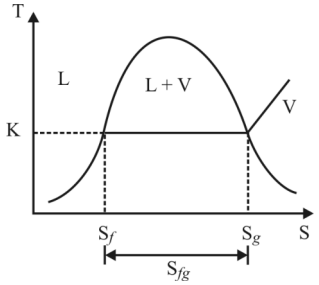
(a) $S_f = C_{pw} \log_e \frac{T}{273}$ (b) $S_f = C_{pw} \log_{10} \frac{273}{T}$

(c) $S_f = C_{pw} \log_{10} \frac{T}{273}$ (d) $S_f = C_{pw} \log_e \frac{273}{T}$

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (a) : We know that,

$ds = \int_{T_1}^{T_2} \frac{dQ}{T}$ (For Reversible system)



$T_1 = 0°C = 273\ K;\ \ T_2 = T\ K$

$(S_f) = \int_{T_1}^{T_2} \frac{mC_{pw}dT}{T}$

$\boxed{S_f = C_{pw} \log_e \frac{T}{273}}$

634. Which of the following represents the slope of constant pressure line on T-s diagram of an ideal gas? [where, C_p and C_v are specific heat of gas]

(a) $\frac{T}{C_P}$ (b) $\frac{T}{C_P^2}$

(c) $\frac{T}{C_v}$ (d) $\frac{T^2}{C_p}$

SJVN ET 2019

Ans. (a) : For unit mass, heat addition on during constant pressure process -

$\delta q = C_p dT$

$\frac{\delta q}{T} = C_p \frac{dT}{T}$

$ds = C_p \frac{dT}{T}$ or $\frac{dT}{ds} = \frac{T}{C_p}$

635. The entropy will usually increase when

1. **A molecule is broken into two or more smaller molecules.**
2. **A reaction occurs that results in an increase in the number of moles of gas.**
3. **Process is adiabatic**
4. **Process is reversible adiabatic**
5. **A liquid changes to a gas**

(a) 1, 2, 4 and 5 (b) 2, 3 and 5

(c) 1, 2, 3 and 5 (d) 1, 2 and 4

Gujarat PSC AE 2019

Ans. : (c) : The entropy will usually increase when-

- A molecule is broken into two or more smaller molecules.
- A reaction occurs that results in an increase in the number of moles of gas.
- Process is adiabatic.
- A liquid changes to a gas.

636. If 'T' refers to temperature and S refers to heat, then for a process from state 1 to state 2 heat transfer in a reversible process is given by

(a) $T_0 \times (S_2 - S_1)$ (b) $\frac{T_0}{(S_2 - S_1)}$

(c) $\frac{S_2 - S_1}{T_0}$ (d) $S_2(T_2 - T_1)$

TNPSC AE 2019

Ans. (a) : We know that

$\frac{\delta Q}{T} = \Delta S$ (for reversible process)

So heat maximum heat transfer

$\delta Q = T_0 \times (S_2 - S_1)$

637. Anything that generates entropy always.

(a) increases enthalpy

(b) decreases pressure

(c) decreases energy

(d) lowers chemical reaction

TNPSC AE 2018

Ans. (b) : Anything that generates entropy always decreases pressure.

638. The entropy of water at 0 K is assumed to be

(a) 1 JK^{-1} (b) -1 JK^{-1}

(c) 10 JK^{-1} (d) 0

ISRO Scientist/Engineer 22.04.2018

Ans. (d) : The entropy of water at 0 K is assumed to be zero.

639. A simple compressible substance inside a cylinder undergoes a change of state quasi-statistically, isothermally and at constant pressure of 1.5 MPa. If the enthalpy change for the system is 4000 kJ at 150°C, what is the corresponding change in its entropy?

(a) 9.45 kJ K^{-1} (b) 26.67 kJ K^{-1}

(c) 14.17 kJ K^{-1} (d) 0

ISRO Scientist/Engineer 22.04.2018

Ans. (a) : Firstly the substance is not ideal gas, so don't try to use formula for change in entropy for ideal gas. The situation is basically phase change as temperature is constant.

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We know that **dH = TdS + VdP**
At constant pressure dH = TdS (dP = 0)

$$dS = \frac{dH}{T}$$

The entropy change is given as –

$$dS = \frac{4000}{150+273}$$

dS = 9.45 kJ/K

640. The entropy __ in an irreversible cyclic process.
(a) remains constant (b) decreases
(c) increases (d) None of the above

Sikkim PSC (Under Secretary), 2017

Ans. (c) : The entropy increases in an irreversible process. $\boxed{\begin{matrix} dS > 0 \\ \frac{dQ}{T} < 0 \end{matrix}}$

641. 273 kg of ice at 0ºC is completely melts into water at 0ºC. Find the entropy change, in kJ/K. (Take Latent heat of fusion of ice = 335 kJ/kg)
(a) 0 (b) 335
(c) 91.5 (d) 0.335

GPSC EE Pre, 28.01.2017

Ans. (b) : Given,
m = 273 kg
T = 0ºC = 273 K
L = Q =Latent heat = 335 kJ/kg

$$\text{Change entropy (ds)} = \frac{mL}{T}$$

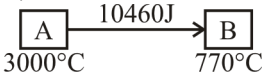
$$= 273 \times \frac{335}{273}$$

= 335 kJ/K

642. 10460 J of heat from a body of 3000°C is transferred to another body at 770°C. The change of entropy in this case is
(a) 3486 J/K (b) 35 J/K
(c) 11.6 J/K (d) 6.1 J/K

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Given, T_A = 3000°C, T_B = 770°C
Heat transfer (ΔQ) = 10460 J



Entropy change for A,

$$\Delta s_1 = \frac{\Delta Q}{T_A} = \frac{-10460}{3000+273} = \frac{-10460}{3273}$$

$$= -3.19 \text{ J/k}$$

Entropy change for B, $\Delta s_2 = \frac{\Delta Q}{T_B} = \frac{10460}{770+273} = \frac{10460}{1043}$

= 10.02 J/K

Entropy change for system = $\Delta s_1 + \Delta s_2$
= –3.18 + 10.02
= 6.83 J/K

643. The entropy of a hot baked potato decreases as it cools.
(a) The above statement is correct
(b) The above statement is incorrect as it violates the increase in entropy principle
(c) Too less information to comment
(d) The above statement is incorrect as it violates second law of thermodynamics

ISRO Scientist/Engineer 07.05.2017

Ans. (a) : The entropy of a baked potato decreases as it cools. This statement is correct because with decrease in temperature, potato entropy decrease, but heat released into surroundings will increase the entropy of surrounding. So, as per entropy principle

$$(\Delta S)_{universe} \geq 0$$

where $(\Delta S)_{universe} = (\Delta S)_{system} + (\Delta S)_{surrounding}$

644. Entropy per unit mass is
(a) An extensive property
(b) An intensive property
(c) It may be intensive or extensive
(d) All of the above

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : Entropy per unit mass, is specific entropy which is a intensive property. Intensive property are those which are independent of mass.

645. Entropy generation in closed system is:
(a) equal to zero for irreversible process
(b) less than zero for irreversible process
(c) greater than zero for reversible process
(d) greater than zero for irreversible process
(e) Less than zero for reversible process

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : For closed system
(a) For Reversible Process

S.No.	Conditioned	Change in Entropy
1.	Heat is Supplied to the system	Increases
2.	Heat Rejected by the system	Decreases
3.	Adiabatic Process	Constant

(b) For Irreversible Process.

S.No.	Conditioned	Change in Entropy
1.	Heat is supplied to the system	Increases
2.	Heat is rejected by the system	Increases, Decreases Remains Constant
3.	Adiabatic Process	Increases

- $(ds)_{rev} = \frac{dQ}{T}$
- $(ds)_{irrev} = \frac{dQ}{T} + (\delta s)_{generation}$.

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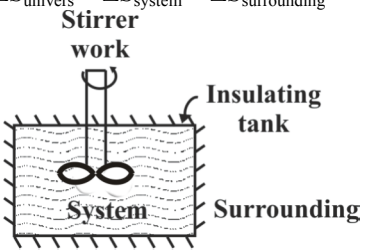
646. Stirrer work supplied to liquid in insulating chamber increases its temperature from T_1 to T_2. The change in entropy of universe will be: (where C_p = specific heat of liquid)

(a) $C_p \ln \frac{T_1}{T_2}$ (b) Zero

(c) Negative (d) $C_p \ln \frac{T_2}{T_1}$

UPRVUNL AE 07.10.2016

Ans. (d) : $\Delta S_{univers} = \Delta S_{system} + \Delta S_{surrounding}$



$$\Delta S_{system} = C_P \ln \frac{T_2}{T_1}$$

$\Delta S_{surrounding}$ = zero

$$\Delta S_{universe} = C_P \ln \frac{T_2}{T_1}$$

647. The entropy increase of the solid substance as it melts into liquid at 27°C (latent heat of fusion of substance = 400 kJ/kg)

(a) 14.8 kJ/kg-K (b) 120 MJ/kg-K

(c) 10.8 MJ/kg-K (d) 1.33 kJ/kg-K

RPSC LECTURER 16.01.2016

Ans. (d) :

$$\Delta S = \frac{\delta Q}{T} = \frac{L.H.F.}{T} = \frac{400}{300}$$

ΔS = 1.33 kJ/kg-K

648. During fusion, the entropy of the system:

(a) decreases (b) increases

(c) always remains constant (d) none of these

(HPPSC LECT. 2016)

Ans. : (b) During fusion, the entropy of the system increases.

649. When the air is throttled, there is an entropy increase. For 2 kg/s of air the entropy increase by 0.06 kW/K . The pressure ratio (final to initial state) is [for air, characteristic gas constant= 0.287 kJ/kgK]

(a) 0.9 (b) 1.0

(c) 0.8 (d) None of these

RPSC ADE 2016

Ans. (a) : Throttling process dH = 0 = m c_p (T_2-T_1)

or $T_1 = T_2$

$$S_2 - S_1 = m c_p \ln \frac{T_2}{T_1} - mR \ln \frac{P_1}{P_2} \text{ (for any process)}$$

0.06 = 0 + 2 × 0.287 ln (1/pressure ratio)

Pressure ratio = 0.9007

650. Entropy at absolute zero temperature is zero for:

(a) Glass fibre

(b) Natural perfect gas

(c) Imperfect and impure substance

(d) Perfect crystal

HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : Entropy at absolute zero temperature is zero for perfect crystal.

Third law of thermodynamics states that the entropy of a system approach a constant value as the temperature approaches zero. At zero temperature the system must be in a state with minimum thermal energy.

651. A rigid insulated cylinder has two compartments separated by a thin membrane. While one compartment contains one kmol of nitrogen at certain temperature and pressure, the other contains one kmol of carbon dioxide at the same pressure and temperature. The membrane is ruptured and the two gases are allowed to mix. Assume that the gases behave as ideal gases. Calculate the increase in entropy of the contents of the cylinder. Universal gas constant is 8314.35/kmol-K.

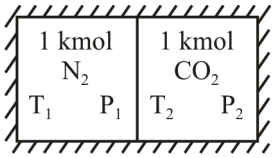
(a) 9.525 kJ/K (b) 10.525 kJ/K

(c) 11.525 kJ/K (d) 12.525 kJ/K

(e) 13.525 kJ/K

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) :



Irreversibility = $T_0 S_{gen}$

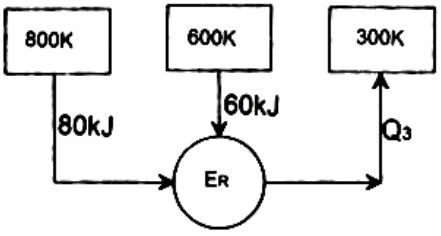
$$S_{gen} = \Delta S_{system}$$

$$= -R\left(n_1 \ln \frac{x_1}{x_2 + x_1} + n_2 \ln \frac{x_2}{x_1 + x_2}\right)$$

$$= -8.314\left[1 \ln\left(\frac{1}{2}\right) + 1 \ln\left(\frac{1}{2}\right)\right]$$

= –8.314 [2 × (–0.693)] = 11.525 kJ/K

652. Calculate the thermal efficiency of reversible heat engine shown in the figure below.



(a) 52% (b) 54%

(c) 57% (d) 61%

(e) 65%

CGPSC Asstt. Workshop Supt., 17.07.2016

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Ans. (c) : As we know,

$$Q_1 + Q_2 - Q_3 = W$$

So, $$\frac{Q_1}{800} + \frac{Q_2}{600} - \frac{Q_3}{300} = 0$$

$$\Rightarrow \quad \frac{80}{800} + \frac{60}{600} = \frac{Q_3}{300} \Rightarrow Q_3 = 60\,\text{kJ}$$

$$W = 80 + 60 - 60 = 80\ \text{kJ}$$

$$\eta = \frac{W}{Q_1 + Q_2} = \frac{80}{80 + 60} \times 100 = 57\%$$

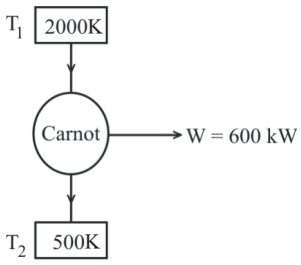
653. A carnot cycle operates between the temperature limits of 500 K and 2000 K and produces 600 kW of net power. The rate of entropy change of the working fluid during heat addition process is :

(a) 1.2 kJ/kg-K (b) 0.3 kJ/kg-K

(c) 2.5 kJ/kg-K (d) Data insufficient

GPSC Lect. (Auto) 16.10.2016

Ans. (*)



$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{500}{2000} = 1 - 0.25$$

$$\eta = 0.75$$

$$\eta = \frac{W}{Q_s}$$

$$Q_s = \frac{W}{\eta} = \frac{600}{0.75} = 800\text{kW}$$

Entropy during heat addition $= \frac{Q_s}{T_1} = \frac{800}{2000} = \frac{0.4\text{kW}}{\text{K}}$

$0.4\frac{\text{kW}}{\text{K}}$ will be correct answer.

Note-Official answer given by commission is (c).

654. Which of the following is a thermodynamic process where the entropy of the system remains constant throughout?

(a) Isentropic process (b) Isosceles process

(c) Isoclinic process (d) None of these

KPCL AE 2016

Ans. (a) : Entropy of the system remains constant (s = constant) is isentropic process.

655. Which of the following statements is FALSE?

(a) A mixture of an ideal gas also behaves as an ideal gas

(b) The enthalpy of an ideal gas is a function of temperature only

(c) The entropy of an ideal gas is a function of temperature only

(d) The temperature of an ideal gas always decreases during isentropic expansion

Nagaland PSC (CTSE) 2016, Paper-II

Haryana PSC Civil Services Pre, 2014

Ans. (c) : Entropy of an ideal gas is not a function of temperature. Enthalpy & internal energy is a function of temperature only.

656. 1 kg of water at room temperature is brought into contact with a high temperature thermal reservoir. The entropy change of the universe is :

(a) Equal to entropy change of the reservoir

(b) Equal to entropy change of water

(c) Equal to zero

(d) Always positive

HPPSC Asstt. Prof. 18.11.2016

ISRO Scientist/Engineer (RAC) 2013

GATE 2010

Ans. (d) : Entropy change of universe is always positive

$$(\Delta S)_{\text{universe}} = (\Delta S)_{\text{system}} + (\Delta S)_{\text{surrounding}} > 0$$

657. Which of the following is incorrect?

(a) Entropy is an extensive property

(b) Entropy depends on mass of the system

(c) Entropy of a system increases with absorption of heat

(d) All of the above

(e) None of these

CGPSC AE 16.10.2016

Ans. (e) : As we know, entropy is an extensive property it depends on the mass & entropy of a system increase with the absorption of heat. So (a), (b) & (c) are correct statements.

658. Which of the followings is correct statement?

(a) Entropy of isolated system always decreases

(b) Energy always degrades during the real process

(c) Energy always destroyed during the real process

(d) Heat transfer through a finite temperature difference is reversible process

UPRVUNL AE 07.10.2016

Ans. (b) : • Entropy of isolated system always increase or remain constant.

- Energy always degrades during the real process.
- Energy losses occur during the real process not destroyed.
- Heat transfer through a finite temperature difference is irreversible process.

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659. Which one of the following is incorrect with reference to Clausius inequality statement?

(a) For reversible process : $\Delta S = 0$
(b) For irreversible process : $\Delta S > 0$
(c) For impossible process : $\Delta S < 0$
(d) None of the above

TANGEDCO AE 2015

Ans. (d) :

660. An electric current of 1 amp flows through a resistor of 300 ohm, which is in contact with a reservoir at 300 K. At steady state, the rate of entropy generation of the universe is

(a) 1 W/K (b) 2 W/K
(c) 2 J/K (d) 1 J/K

Assam PSC CCE Pre 2015

Ans. (a) Given,
I = 1 amp, R = 300 Ohm, T = 300 K
We know that,

$$Q = I^2Rt$$
$$= 1 \times 300 \times 1$$
$$= 300 \text{ J}$$
$$(\Delta S)_{universe} = (\Delta S)_{system} + (\Delta S)_{surrounding}$$
$$= 0 + \frac{300}{300}$$
$$= 1 \text{ J/K}$$

So rate of entropy generation of universe is
$(\Delta s)_{universe} = 1$ J/sec K = 1 W/K

661. An electric motor of 15 kW is subjected to a braking test for 1/2 hour. The heat generated by the frictional forces in the process is transferred to the surroundings at 27°C. The resulting entropy change will be

(a) 90 kJ/K (b) 180 kJ/K
(c) 55.5 kJ/K (d) 1000 kJ/K

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Given, Electric power (P) = 15 kW

$$t = \frac{1}{2}\text{hour} = 1800 \text{ second}$$
$$T = 27 + 273$$
$$T = 300 \text{ K}$$

Total energy supplied to electric motor

$$Q = Pt = 15 \times 1800 = 27000 \text{ kJ}$$
$$dS = \frac{dQ}{T} = \frac{15 \times 1800}{300} = 90 \text{ kJ/K}$$

Resulting entropy change will be,
dS = 90 kJ/K

662. The correct statement regarding entropy is that:

(a) Entropy is a path function
(b) Entropy can be obtained from a direct measurement of Q and T
(c) During a change of a state of a system, the entropy change is the same whether that change has occurred via a reversible process or an irreversible one
(d) Entropy of an isolated system is zero

OPSC AEE 2015 PAPER - II

Ans. : (c) Entropy:– During a change of a state of a system, the entropy change is the same whether that change has occurred via a reversible process or an irreversible one.
Clausies theorem:- The cyclic intergal of the quantity δQ/T for a reversible process in zero. This suggest that the quantity δQ/T is a point function and hence a property of the system.

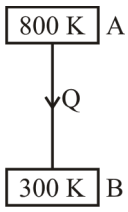
$$\int_1^2 \left(\frac{\delta Q}{T}\right)_{rev} = \int_1^2 dS = S_2 - S_1$$

663. Heat flows between two reservoirs having temperatures 800 K and 300 K, respectively. If the entropy change of the cold reservoir is 20 kJ/K, then what is the entropy change for the hot reservoir?

(a) −7.5 kJ/K (b) −5 kJ/K
(c) 5 kJ/K (d) 7.5 kJ/K

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) :



$$(ds)_B = \frac{Q}{T} = 20$$
$$= \frac{Q}{300} = 20$$
$$Q = 6000 \text{ kJ/K}$$
$$(ds)_A = -\frac{Q}{800} = -\frac{6000}{800} = -7.5$$
$$\boxed{(ds)_A = -7.5 \text{ kJ/K}}$$

664. Identify the effect on the entropy, when a closed system is undergoing an irreversible process.

(a) Entropy will rise
(b) Entropy remains constant
(c) Value of entropy will fall
(d) Can increase, decrease or remain constant

UPRVUNL AE 2014

Ans. : (d) Close system, irreversible process

$$\frac{dQ}{T} \le 0 \qquad (dQ = T.\, dS)$$

It means, Entropy can increase, decrease or remains constant.

665. As shown in Fig. below, an ideal gas is confined to a cylinder by a piston. The piston is pushed down slowly so that the gas temperature remains at 27°C. During the compression, 900 J of work is done on the gas. Find the entropy change of the gas.

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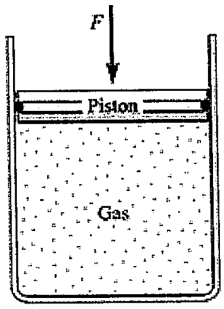
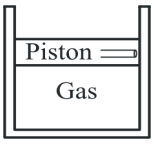


(a) 3 J/K (b) –3 J/K
(c) –6 J/K (d) 6 J/K

ISRO Scientist/Engineer 24.05.2014

Ans. (b) :



Apply first law of thermodynamic process

$$\partial Q = dU + \partial W$$

$dU = 0$ ($\because$ T = Constant)

$$\partial Q = \partial W$$

$\partial Q = -900$ ($\because$ work is done on the gas)

$$dS = \frac{\partial Q}{T}$$

$$= -\frac{900}{300} = -3\ J/K$$

666. The change in entropy of the universe, when any irreversible process takes place is

(a) Positive (b) Zero
(c) Negative (d) Unity

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a) : The change in entropy of the universe, when any irreversible process takes place is positive.

667. Third law of thermodynamics is:-

(a) An extension of second law
(b) An extension of first law
(c) An independent law of nature
(d) An extension of zeroth law

UKPSC AE-2013, Paper-II

Ans. (c) : Third law of thermodynamics is an independent law of nature.

668. Consider the relationship TdS = dU + pdV, between entropy (S), internal energy (U), pressure (p), temperature (T) and volume (V), which of the following statements is correct :

(a) it is a applicable only for a reversible process
(b) it is also valid for any ideal gas
(c) it holds good for any process reversible or irreversible
(d) none of these

PSPCL AE, 2012

Ans. (c) : Entropy equation Tds = dU + PdV.
holds for any reversible or irreversible process.

669. Air is compressed to half volume at constant pressure, then the change in entropy

(a) increase (b) decrease
(c) does not change (d) not predictable

ISRO Scientist/Engineer 2012

Ans. (b) : At constant pressure,
change entropy

$$s_2 - s_1 = c_p \ln\frac{T_2}{T_1} - R\ln\left(\frac{P_2}{P_1}\right)$$

$\therefore$ $P_2 = P_1$ = Constant pressure

And Let initial temperature = T_1

Then final temperature $T_2 = \frac{T_1}{2}$

$$\Delta s = c_p \ln\left(\frac{T_2}{T_1}\right) - R\ln\left(\frac{P_2}{P_1}\right)$$

$$= c_p \ln\left(\frac{T_2}{T_1}\right) - R\ln 1$$

$$= c_p \ln\left(\frac{1}{2}\right) - R\ln 1$$

$$= c_p \ln(0.5) - 0$$

$$= c_p \ln(0.5) = -0.693\, c_p.$$

So, Δs = - (Negative)

670. A heat engine transfers 15 kJ of heat to a thermal reservoir at 300 K. The change of entropy of the reservoir in the process is :

(a) $\Delta S_{reservoir} = -50\ JK^{-1}$
(b) $\Delta S_{reservoir} = +50\ JK^{-1}$
(c) $\Delta S_{reservoir} = +200\ KJ^{-1}$
(d) $\Delta S_{reservoir} = +4500\ kJ.K$

OPSC Civil Services Pre. 2011

Ans. (b) : Given as,
Q = 15 kJ
T = 300 K

$$\Delta S = \frac{Q}{T} = \frac{15000}{300}$$

ΔS reservoir = + 50 JK^{-1}

671. What is the property of a working substance which increases or decreases as the heat is supplied or removed in a reversible manner known as ?

(a) Enthalpy
(b) Internal energy
(c) Entropy
(d) External energy

VIZAG STEEL MT 2011

Ans. (c) : Entropy is the property of a working substance which increases or decreases as the heat is supplied or removed in a reversible manner.

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672. The entropy of a closed system can increased by

1. Heat transfer

2. Work transfer

3. Internal irreversibilities

Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only

(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-52/2010

Ans. (a) : All statements are correct.

673. When a system is in equilibrium, any conceivable change in entropy would be:

(a) Maximum (b) Zero

(c) Positive (d) Negative

UPSC JWM Adv. No-16/2009

Ans. (b) : When a system is in equilibrium state then change in entropy would be zero.

674. A certain amount of fluid at temperature T_1 is mixed with a equal amount of the same fluid at temperature T_2 is an insulated container. With total fluid as the system, consider the following statements:

(1) Energy of the system is conserved

(2) Entropy of the system is conserved

(3) Entropy of the system increases

(4) Entropy of the system decreases

Which of the following options is true?

(a) 1 and 2 (b) 1 and 3

(c) 1 and 4 (d) 1 only

DRDO Scientists 2009

Ans. (b) :

675. Carnot engine is irreversible due to

(a) friction between moving parts

(b) losses from working fluid in transit

(c) high speed

(d) both (A) and (B)

TNPSC AE, 2008

Ans. (d) : Carnot engine is irreversible due to –

- Friction between moving parts
- Losses from working fluid in transit

676. In a reversible cycle the entropy of the system

(a) increases

(b) decreases

(c) does not change

(d) first increases and then decreases

WBPSC AE 2008

Ans. (c) : For reversible process,

$$\frac{dQ}{T} = 0$$

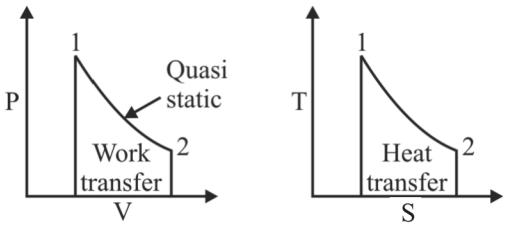
677. In case of reversible heat transfer, the integrating factor is given by:

(a) 1/P (b) 1/V

(c) 1/(PV) (d) 1/T

AAI Jr. Executive 29.11.2018

Ans. (d) :



$$W_{1-2} = \int_1^2 dW = \int_1^2 pdV$$

It is valid for a quasi-static or reversible process. The work transfer is equal to the integral of the product of the intensive property P and the differential change in the extensive property, dV just like displacement work, the heat transfer can also written as.

$$Q_{1-2} = \int_1^2 TdS = \int_1^2 dQ$$

$$dQ = TdS$$

$$dS = \frac{1}{T}dQ$$

$$\int_1^2 dS = \int_1^2 \frac{1}{T}dQ$$

Where $\frac{1}{T}$ is integrating factor.

678. A reversible Carnot engine operates between 27°C and 1527°C, and produces 400 kW of net power. The change of entropy of the working fluid during the heat addition process is

(a) 0.222 kW/K (b) 0.266 kW/K

(c) 0.288 kW/K (d) 0.299 kW/K

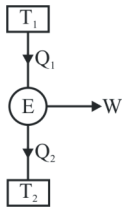
ESE 2018

Ans. (b) : Given,

$T_1 = 1527 + 273 = 1800$ K

$T_2 = 27 + 273 = 300$ K

W = 400 kW



Efficiency $\eta = \frac{W}{Q_1} = 1 - \frac{T_2}{T_1}$ (For Carnot engine)

$$\frac{W}{Q_1} = 1 - \frac{T_2}{T_1}$$

$$\frac{400}{Q_1} = 1 - \frac{300}{1800}$$

Heat input (Q_1) = 480 kW

Entropy change (ΔS) $= \frac{Q_1}{T_1} = \frac{480}{1800} = 0.266$ kW/K

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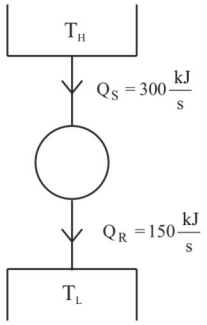
679. 300 kJ/s of heat is supplied at a constant fixed temperature of 290°C to a heat engine. 150 kJ/s of heat are rejected at 8.5 °C. Then the cycle is reported as

(a) Reversible (b) Irreversible
(c) Impossible (d) Random

TNPSC AE 2018

Ans. (a) : Given, $T_H = 290^oC = 563\ K$
$T_L = 8.5\ ^oC = 281.5\ K$

$$\Delta S = \frac{\delta Q_S}{T_H} + \frac{\delta Q_R}{T_L}$$



$$= \frac{300}{563} + \left(\frac{-150}{281.5}\right)$$

$\boxed{\Delta S = 0}$

It means the cycle is reversible

680. In Carnot cycle, the algebraic sum of the entropy changes for the cycle is

(a) Positive
(b) Negative
(c) Zero
(d) Depends on temperature of sink

MPPSC AE 08.11.2015
MPPSC State Forest Service Exam, 2014

Ans. (c) : Thermal efficiency $= 1 - \frac{Q_c}{Q_h} = 1 - \frac{T_c}{T_h}$

$$-\frac{Q_c}{Q_h} = -\frac{T_c}{T_h} \Rightarrow \frac{Q_c}{T_c} = \frac{Q_h}{T_h}$$

$$\frac{Q_c}{T_c} - \frac{Q_h}{T_h} = 0$$

So, $\oint \frac{dQ}{dT} = 0$

The algebraic sum of the entropy changes for the cycle is zero.

As entropy is a point function does not depend on the path.

681. Gas A at 1 MPa, 100°C and Gas B at 5 MPa, 100°C are mixed such that final temperature after mixing remains 100°C. The process is adiabatic. The entropy of the gases after mixing:-

(a) Will increase
(b) Will remain same
(c) Will decrease
(d) Cannot be calculated

UKPSC AE-2013, Paper-II

Ans. (b) : Entropy change while mixing of fluid,

$$\Delta S = 2mC\ln\left(\frac{T_1 + T_2}{2\sqrt{T_1 T_2}}\right) = 2mC\ln\left(\frac{2T}{2T}\right)$$

$\Delta S = 0$

682. A carnot engine operates between 327°C and 27°C. If the entropy change during heat addition is 1 kJ/K, the work produced by the engine is :

(a) 100 kJ (b) 200 kJ
(c) 300 kJ (d) 400 kJ

UPSC JWM Advt. No.-52/2010

Ans. (c) : Given, About Carnot engine-

$T_H = 327 + 273 = 600K, T_L = 27 + 273 = 300K$

Entropy change (ds) = 1 kJ/K

Let; W = Work produced by the engine.

$Q = Tds$

$$\eta_{carnot} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{600} = \frac{1}{2}$$

$$\therefore \quad \eta_{carnot} = \frac{W}{Q_H}$$

$\therefore$ $Tds = Q \Rightarrow Q_H = T_H.\ ds = 600 \times 1$

$\Rightarrow$ $Q_H = 600$ kJ

$$\therefore \quad \eta_{carnot} = \frac{W}{Q_H} \Rightarrow W = \eta_{carnot} \times Q_H$$

$W = \frac{1}{2} \times 600 \quad \Rightarrow \boxed{W = 300kJ}$

5. Availability and Irreversibility

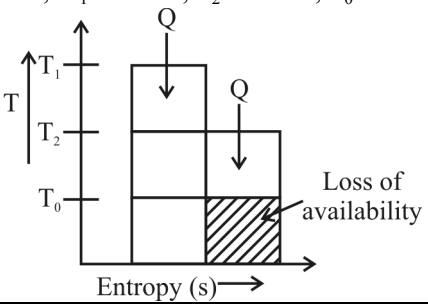
683. The loss of available energy associated with the transfer of 1000 kJ of heat from a constant temperature system at 600 K to another at 400 K, when the environmental temperature is 300 K is

(a) 166.67 kJ (b) 250.00 kJ
(c) 500.00 kJ (d) 750.00 kJ

UPPSC AE 13.12.2020, Paper II
MPPSC AE 2016
MPPSC AE 08.11.2015
MPSC Poly. Lect. 2014, ESE 2008, 2004

Ans. (b) : Given data,

$Q = 1000$ kJ, $T_1 = 600$ K, $T_2 = 400$ K, $T_0 = 300$ K



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Loss of available energy,

$I = T_0 (\Delta S_{uni})$;

$= T_0 [\Delta S_{sys} + \Delta S_{sur}]$;

Where, ΔS_{uni} = Change of entropy of universe.

ΔS_{sys} = Change of entropy of system

ΔS_{sur} = Change of entropy of surrounding

$$= T_0\left[\frac{-Q}{T_1}+\frac{Q}{T_2}\right]$$

$$= 300\left[\frac{-1000}{600}+\frac{1000}{400}\right]$$

$$= 300[-1.667+2.5] = 249.9\,\text{kJ}$$

$I \approx 250$ kJ

684. Availability of a system at any given stage is

(a) a property of the system

(b) the total energy of the system

(c) the maximum work obtainable as the system goes to dead state

(d) the maximum useful work obtainable as the system goes to dead state

GPSC Executive Engineer 23.12.2018
TSPSC AEE 2015
MPPSC AE 08.11.2015, GATE 2000

Ans. (d) : Availability of a system at any given stage is the maximum useful work obtainable as the system goes to dead state.

685. Availability function is expressed as

(a) $a = u + P_0v - T_0s$

(b) $a = u + P_0du - T_0ds$

(c) $a = du + P_0du - T_0ds$

(d) $a = u + P_0u + T_0s$

UJVNL AE 2016, RPSC AE 2016
RRB SSE 03.09.2015 Shift-I
WBPSC AE 2008, ESE 2002

Ans. (a) : Availability function,

For closed system, $\phi_{c.s} = u + P_0v - T_0s$

For open system, $\phi_{o.s} = h - T_0s$

686. In an irreversible process there is

(a) Heat loss (b) Heat gain

(c) No heat loss (d) No transfer of heat

MPPSC State Forest Service Exam, 2014
WBPSC AE 2008
UKPSC AE 2012, 2007 Paper–II

Ans. (a) : An irreversible process can be defined as a process in which the system and the surrounding do not return to their original condition once the process is initiated. In an irreversible process there is heat loss.

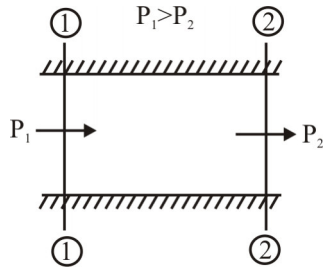
687. Pressure of steady and adiabatic flow of an ideal gas through pipe is reduced from P_1 to P_2. If T_0 = environment temperature, its irreversibility per unit mass flow rate is given by:

(a) $T_0 R \ln\left(\frac{P_1 - P_2}{P_2}\right)$ (b) $T_0 R \ln\left(\frac{P_1}{P_1 - P_2}\right)$

(c) $T_0 R \ln\frac{P_1}{P_2}$ (d) $T_0 R \ln\frac{P_2}{P_1}$

Oil India Limited Sr. Engineer (Production) 30.11.2019
UPRVUNL AE 07.10.2016

Ans. (c) :



Given,

Adiabatic flow of an ideal gas from P_1 to P_2

If T_0 = environment temperature.

$\therefore$ We know that

Irreversibility $(I) = T_0(\Delta S)_{univ.} = T_0[(\Delta S)_{system} + (\Delta S)_{surr}]$

For steady & adiabatic flow

$(\Delta S)_{surr} = 0$

$\therefore I = T_0 [(\Delta S)_{syst.}]$

For process. 1-2

$$(\Delta S)_{system} = R \ln\frac{P_1}{P_2}$$

$I = T_o (\Delta S)_{system}$

$$I = T_o R \ln\frac{P_1}{P_2}$$

688. Consider the following statements:

i. Slow heating of water from an electric heater.

ii. Isentropic expansion of air.

iii. Evaporation of a liquid from heat source at the evaporation temperature.

iv. Constant pressure heating of a gas by a constant temperature source.

Which of these processes is/are reversible?

(a) ii and iv (b) iv only

(c) iii and iv (d) ii and iii

Oil India Limited Sr. Engineer (Production) 30.11.2019
ESE 2010, 2005

Ans. (d) : A process is said to be reversible when system reversed to its initial condition without leaving any traces on surrounding i.e. both system and surrounding should return to their initial state at the end of process.

Slow and friction less processes considered as reversible process.

(i) Isentropic expansion of air is a reversible adiabatic process

(ii) Evaporation of liquid from a heat source at the evaporation temperature is also a reversible process because it very slow & initial condition can be restored.

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689. The pressure, temperature and velocity of air flowing in a pipe are 5 bar, 500 K and 50 m/s, respectively. The specific heats of air at constant pressure and at constant volume are 1.005 kJ/kgK and 0.718 kJ/kgK, respectively. Neglect potential energy. If the pressure and temperature of the surroundings are 1 bar and 300 K, respectively, the available energy in kJ/kg of the air stream is

(a) 170 (b) 187
(c) 191 (d) 213

GPSC Asstt. Director of Transport 05.03.2017
GATE 2013

Ans. (b) : Given,
For flow stream,
Available energy = $(h_2 - h_1) + K.E. - T_o(s_2 - s_1)$

$$= C_p(T_2 - T_1) + \frac{1}{2} \times V^2 + T_o\left[C_p \ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1}\right]$$

$C_p = 1.005$, $C_v = 0.718$, $V = 50$ m/s
$P_1 = 1$ bar, $P_2 = 5$ bar, $T_2 = 500$ K, $T_1 = 300$ K
A.E. = 187 kJ/kg

690. A system is said to be in the dead state when it is in

(a) thermal equilibrium with the environment
(b) chemical equilibrium with the environment
(c) mechanical equilibrium with the environment
(d) thermodynamic equilibrium with the environment

TSPSC Manager (Engg.) HMWSSB 12.11.2020
NLCIL GET 17.11.2020, Shift-II

Ans. (d) : A system is said to be in the dead state when it is in thermodynamic equilibrium with the environment.

691. The maximum possible work a system delivers as it undergoes a reversible process from the specified initial state to the state of its environment, (dead state) is

(a) Entropy (b) Enthalpy
(c) Exergy (d) Energy

RPSC ACF & FRO, 26.02.2021
ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) : Exergy- Exergy or availability of a given system is defined as the maximum useful work that is obtainable in a process in which the system comes to equilibrium with its surroundings (dead state).

692. For a steady process, the conditions at stage 1 and stage 2 are, respectively, $h_1 = 300$ kJ/kg, $h_2 = 150$ kJ/kg, $S_1 = 1.25$ kJ/kg-K and $S_2 = 0.8$ kJ/kg-K. The availability at the ambient temperature 300 K will be

(a) 15 kJ/kg (b) 20 kJ/kg
(c) 25 kJ/kg (d) 35 kJ/kg

RPSC IOF, 2020, ESE 2017

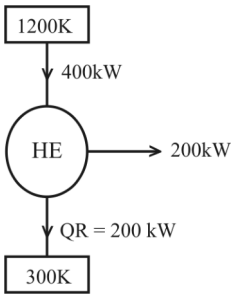
Ans. (a) :
Availability $\Psi = \Psi_1 - \Psi_2 = (h_1 - h_2) - T(S_1 - S_2)$
Given,
$h_1 = 300$ kJ/kg
$h_2 = 150$ kJ/kg
$S_1 = 1.25$ kJ/kg-K
$S_2 = 0.8$ kJ/kg-K
$T = 300$ K
$\Psi = \Psi_1 - \Psi_2$
$= (h_1 - h_2) - T(S_1 - S_2)$
$= (300 - 150) - 300(1.25 - 0.8)$
$\Psi = 15$ kJ/kg

693. A heat engine receives heat at the rate of 400 kW from a source at 1200 K and reject waste heat to a sink at 300 K. If power output from the engine is 200 kW. What is the reversible power and the irreversibility rate for the process?

(a) 200 kW; 100 kW (b) 300 kW; 200 kW
(c) 300 kW; 100 kW (d) 400 kW; 200 kW

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) :



$$\eta_{rev} = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{1200}$$

$$\eta_{rev} = 0.75$$

$$\eta_{rev} = \frac{W_{rev}}{Q_S}$$

$$W_{rev} = \eta_{rev} \times Q_S$$

$$= 0.75 \times 400$$

$$\boxed{W_{rev} = 300\text{kW}}$$

$$I_{rr} = W_{max} - W_{actual}$$

$$= 300 - 200$$

$$\boxed{I_{rr} = 100\text{ kW}}$$

694. Mechanical and fluid friction are some of the causes of _____

(a) irreversibility (b) decrease in efficiency
(c) Reversibility (d) thermal degradation

UPRVUNL AE 05.07.2021

Ans. (a) : Main cause of irreversibility-
→ Unrestricted expansion
→ Mechanical and fluid friction
→ Heat transfer with a finite temperature difference.

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695. Irreversibility caused by effects like turbulence and friction is known as

(a) Internal irreversibility
(b) Dissipative irreversibility
(c) External irreversibility
(d) None of these

Oil India Senior Officer 23.12.2020

Ans. (a) : Irreversibility caused by effects like turbulence and friction is known as Internal irreversibility.

696. Among the following which is NOT the reason for the irreversibility.

(a) Isothermal heat transfer
(b) Free expansion
(c) Paddle wheel work transfer
(d) Friction

NLC GET 17.11.2020, Shift-II

Ans. (a) : An irreversible process is a process that cannot return both the system and the surroundings to their original conditions. That is, the system and the surroundings could not return to their original conditions if the process was reversed. There are many factors that make a process irreversible. Five of the most common causes of irreversibility are–

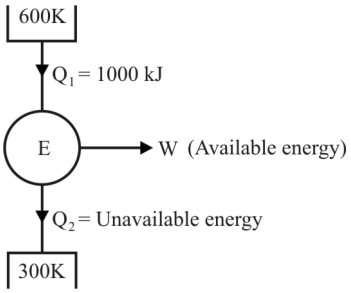
- Friction
- Unrestrained expansion of fluid
- Heat transfer through a finite temperature difference
- Mixing of two different substances
- Paddle wheel work transfer

697. The available energy of the source that can supply 1000 kJ at 600 K in the surrounding of 27°C

(a) 300 kJ (b) 1000 kJ
(c) 600 kJ (d) 500 kJ

NLC GET 17.11.2020, Shift-II

Ans. (d) : The maximum work output obtainable from a certain heat input in a cyclic heat engine is called available energy(AE).



$$W_{max} = \left(1 - \frac{T_2}{T_1}\right).Q_1$$

$$= \left[1 - \left(\frac{300}{600}\right)\right] \times 1000$$

$$W_{max} = 500kJ$$

698. A vapour while condensing at 420^{0}C, transfers heat to water evaporating at 250^{0}C. If ambient is at 35^{0}C, what fraction of available energy is lost in the process?

(a) 0.18 (b) 0.22
(c) 0.26 (d) 0.30

UPPSC AE 13.12.2020, Paper-2

Ans. (c) : Given,

$T_C = 420^0C$, $T_E = 250^0C$, $T_O = 35^0C$

$T_C = 693$ K, $T_E = 523$ K, $T_O = 308$ K

Fraction of available energy lost in process is given by

$$\Delta E = \frac{T_O(T_C - T_E)}{T_E(T_C - T_O)} = \frac{308(693-523)}{523(693-308)}$$

$$\boxed{\Delta E = 0.26}$$

699. Irreversibility in a system:

(a) reduces friction
(b) reduces temperature
(c) increases entropy
(d) increases pressure

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : An irreversible process increases the entropy of universe due to increase temperature and randomness of the molecules.

Because the entropy is a point function the change in entropy of the system is the same, whether the process is reversible or irreversible.

700. If a system has a available energy 949.2 kJ and change in availability energy 1319.2 kJ and Effectiveness ∈ is

(a) 71.9% (b) 50%
(c) 35.95% (d) 0

TNPSC AE 2019

Ans. (a) : Effectiveness (ε)- The effectiveness of system is the ratio of the usefull or actual work done to the maximum or reversible work.

$$\%\varepsilon = \frac{\text{Available energy}}{\text{change in available energy}}$$

$$\%\varepsilon = \frac{949.2}{1319.2} \times 100$$

$$\boxed{\%\varepsilon = 71.95\%}$$

701. The internal irreversibility of Rankine cycle is caused by

1. fluid friction, 2. throttling, 3. mixing

Select the correct answer using the code given below.

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2019

Ans. (d) : Internal irreversibility of Rankine cycle is caused by fluid friction, throttling and mixing.

702. A heat engine receives heat from a source at 1200 K at the rate of 500 kW and rejects the waste heat to a medium at 300 K. The power output of the heat engine is 180 kW. What will be the irreversibility of the system?

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(a) 195 kW (b) 340 kW
(c) 125 kW (d) 320 kW

SJVN ET 2019

Ans. (a) : Efficiency of Carnot Engine

$$\frac{\text{Carnot output}}{\text{Input}} = 1 - \frac{T_2}{T_1}$$

$$\text{Carnot output} = 500 \times \left(1 - \frac{300}{1200}\right) = 375 \text{ kW}$$

Irreversibility of system = Carnot output - actual output
= 375 – 180 = 195 kW

703. When 25 kg of water at 95°C is mixed with 35 kg of water at 35°C, the pressure being taken as constant at surrounding temperature of 15°C and C_p of water is 4.2 kJ/kg K, the decrease in available energy due to mixing will be nearly

(a) 270.5 kJ (b) 277.6 kJ
(c) 281.8 kJ (d) 288.7 kJ

ESE 2019

Ans. (c) : Given,
$m_1 = 25$ kg
$m_2 = 35$ kg
$T_0 = 15 + 273 = 288$ K
$T_1 = 95 + 273 = 368$ K
$T_2 = 35 + 273 = 308$ K

After mixing let T is the final temperature
Heat lost = Heat gained
$m_1 c_p (T_1 - T) = m_2 c_p (T - T_2)$
$25 (368 - T) = 35 (T - 308)$
$T = 333$ K

Entropy generation

$$\Delta S_{gen} = \Delta S_1 + \Delta S_2 = m_1 c_p \ln\left(\frac{T}{T_1}\right) + m_2 c_p \ln\left(\frac{T}{T_2}\right)$$

$$\Delta S_{gen} = 25 \times 4.2 \ln\left(\frac{333}{368}\right) + 35 \times 4.2 \ln\left(\frac{333}{308}\right)$$

$\Delta S_{gen} = 0.978$ kJ/k

Decrease in available energy due to mixing
Irreversibility = $T_0 \Delta S_{gen}$
= 288 × 0.978 = 281.8 kJ

704. Irreversibility of the process is equal to

(a) W_{max} - W (b) W - W_{max}
(c) W_{max} (d) W

TNPSC AE 2018

Ans. (a) : The actual work which a system does is always less than the idealized reversible work, and the difference between the two is called the irreversibility of the process.

$I = W_{max} - W$

This is also, referred as degradation, or dissipation.

705. Unavailable energy is the portion of energy that

(a) cannot be converted into work by turbine
(b) cannot be converted into work even by reversible heat engine
(c) cannot be converted into work by Rankine engine
(d) cannot be converted into work by a pump

TNPSC AE 2018

Ans. (b) : Unavailable energy is the portion of energy that cannot be converted into work even by reversible heat engine.

706. A heat reservoir is maintained at 927°C. If the ambient temperature is 27°C, the availability of heat from the reservoir is limited to

(a) 57% (b) 66%
(c) 75% (d) 88%

ESE 2018

Ans. (c) : Availability of heat $= 1 - \frac{T_2}{T_1}$

$$= 1 - \frac{300}{1200}$$

$= 0.75 = 75\%$

707. A hot reservoir at 900 K is brought into contact with the ambient at 300 K for a short time. During this period 9000 kJ of heat is lost by the hot reservoir. The total loss in availability due to this process is

(a) 2400 kJ (b) 1800 kJ
(c) 6000 kJ (d) 9000 kJ

GPSC Executive Engineer 23.12.2018
GATE 1995

Ans. (c) : Entropy change for hot reservoir (ΔS_h)

$$\Delta S_h = \frac{Q}{T} = \frac{-9000}{900} = -10 \text{ kJ/K}$$

Entropy change in cold reservoir $\Delta S_c = \frac{Q}{T} = \frac{+9000}{300}$

$$\Delta S_c = +30 \text{ kJ/k}$$

Loss in availability = $T_0 [\Delta S_c + \Delta S_h]$
$\Rightarrow$ Loss in availability = 300 (30–10)
= 6000 kJ.

708. One side of a wall is maintained at 400 K and the other at 300 K. The rate of heat transfer through the wall is 1000 W and the surrounding temperature is 25°C. Assuming no generation of heat within the wall, the irreversibility (in W) due to heat transfer through the wall is

(a) 0.833 (b) 248.23 W
(c) 20.825 W (d) 25.833 W

GPSC Executive Engineer 23.12.2018
GATE 2015 Set-3

Ans. (b) : Given,
Q = 1000W, $T_0 = 25 + 273 = 298$ K
Irreversibility I = $T_0 \Delta S_{universe}$
$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{gen}$

$$\Delta S_{uni} = \Delta S_{sys} = \sum \frac{Q}{T} = Q\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\Delta S_{uni} = 1000 \left(\frac{1}{300} - \frac{1}{400}\right) = 0.833 \text{ W/K}$$

$I = T_0 (\Delta S_{uni})$
$= 298 \times 0.833$
$= 248$ W

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709. Which one of the following represents unavailability?

(a) $T_0(\Delta S_0)$ (b) $T(\Delta S)$

(c) $T_0(\Delta S)$ (d) $T(\Delta S_0)$

TRB Polytechnic Lecturer 2017

Ans. (c) : Unavailability = $T_0(\Delta S)$

Where T_0 = environment temperature

$$(\Delta S)_{universe} = \left[(\Delta S)_{system} + (\Delta S)_{surrounding}\right]$$

710. Consider the following statements:
On heating an elastomer under tensile load, its shrinkage

1. maximizes the enthalpy
2. maximizes the entropy
3. minimizes the free energy
4. avoids breaking

Which of the above statements are correct?

(a) 1 and 2 (b) 2 and 3

(c) 3 and 4 (d) 1 and 4

ESE 2017

Ans. (b) : In heating elastomer under tensile load entropy get decrease because of only one possible microstate but by shrinkage entropy get increased because of many possible microstate.

The internal energy of an ideal gas is a function of temperature only.

By Helmholtz free energy equation

Available energy is given by

$A = U - Ts$

Where U = Internal energy

T = Temperature

s = Entropy

So, by increase in entropy free energy get reduced.

On heating an elastomer under tensile load, its shrinkage maximize the entropy and minimize the free energy.

711. The maximum work output from two finite bodies-one at temperature T_1 and the other at temperature T_2 is

(a) $W_{max} = C_p\left(\sqrt{T_1} - \sqrt{T_2}\right)^2$

(b) $W_{max} = C_p\left(\sqrt{T_1} + \sqrt{T_2}\right)^2$

(c) $W_{max} = C_p\left(\sqrt{T_1}\sqrt{T_2}\right)^{\frac{1}{2}}$

(d) $W_{max} = \frac{1}{2}C_p\left(\sqrt{T_1} - \sqrt{T_2}\right)^2$

BPSC AE Mains 2017 Paper - V

Ans : (a) : Maximum work output will be obtained when there will be no irreversibility.

We can extract work till both bodies reaches at same temperature let it be T_f.

Entropy change of body 1

$$= c_p \ell n \frac{T_f}{T_1}$$

Entropy change of body 2

$$= c_p \ell n \frac{T_f}{T_2}$$

For reversible process entropy change is zero, so

$$c_p \ell n \frac{T_f}{T_1} + c_p \ell n \frac{T_f}{T_2} = 0$$

$$\Rightarrow T_f = \sqrt{T_1 T_2}$$

Work done = (Energy given by body 1) – (Energy absorbed body 2)

$$= c_p(T_1 - T_f) - c_p(T_f - T_2)$$

$$= c_p(T_1 + T_2 - 2T_f)$$

$$= c_p(T_1 + T_2 - 2\sqrt{T_1 T_2})$$

$$= c_p(\sqrt{T_1} - \sqrt{T_2})^2$$

Note we have assumed $T_1 > T_2$.

We will get same answer for $T_2 > T_1$

712. An ideal gas is flowing through an insulated pipe at the rate of 3.3 kg/s. There is a pressure drop of 15% from the inlet to the outlet of the pipe. What is the rate of energy loss because of this pressure drop due to friction, given that R_{gas}=0.287 kJ/kg K and the reference temperature T_0 is 300K?

(a) 42.62 kW (b) 40.26 kW

(c) 38.14 kW (d) 35.13 kW

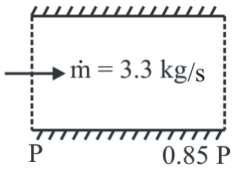
ESE 2017

Ans. (a) : Given,

R_{gas} = 0.287 kJ/kg K

$\dot{m}$ = 3.3 kg/s

T_0 = 300 K



$$\text{Change in entropy } (S_2 - S_1) = R\,\ell n\left(\frac{P_1}{P_2}\right)$$

$$= -R\,\ell n\left(\frac{P_2}{P_1}\right)$$

$$= -R\,\ell n\left(\frac{P_1 - \Delta P}{P_1}\right)$$

$$= -R\,\ell n\left(1 - \frac{\Delta P}{P_1}\right)$$

$$= \frac{R\Delta P}{P_1}$$

Change in entropy of surrounding $(\Delta S)_{surr} = 0$

Entropy generation $\Delta S_{gen} = (\Delta S)_{system} + (\Delta S)_{surr}$

$$= \dot{m}(S_2 - S_1) = \dot{m}R\frac{\Delta P}{P_1}$$

$$= 3.3 \times 0.287 \times \frac{0.15P_1}{P_1}$$

$$= 0.142065$$

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Rate of energy loss i.e. loss of availability
$= T_0 (\Delta S)_{gen}$
$= 300 \times 0.142065$
$= 42.619$ kW

713. Determine the available energy of furnace gas, C_p = 1.046 kJ/kg.K, when is cooled from 1260 K to 480 K at constant pressure. The surroundings are at 295 K.

(a) – 518.1 KJ/kg (b) 428.3 KJ/kg
(c) –1113.7 KJ/kg (d) None of these

RPSC ADE 2016

Ans. (a) : Given, $T_1 = 1260$ K, $T_2 = 480$ K, $T_0 = 295$ K
$C_p = 1.046$ kJ/kg-K

$Q = C_p \Delta T$
$= 1.046 (T_2 - T_1)$
$= 1.046 (480 - 1260)$
$= -815.88$ kJ/kg

Irreversibility,

$I = T_0 \Delta s$

$$= 295\left[C_p \ln\frac{T_2}{T_1}\right]$$

$$= 295\left[1.046 \ln\left(\frac{480}{1260}\right)\right]$$

$I = -297.77$ kJ/kg

Available energy,

$AE = Q - T_0 \Delta s$
$= -815.88 - (-297.77)$
$AE = -518.1$ kJ/kg

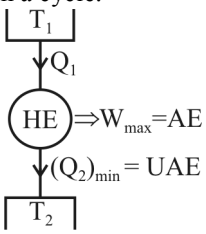
Note-The official answer given by commission is (d).

714. The minimum energy that has to be rejected to the sink (as per 2nd law of thermodynamics) is called?

(a) Minimum work
(b) Unavailable energy (UAE)
(c) Available energy (AE)
(d) Constant energy

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (b) : Available Energy (AE) or Exergy–It is the maximum amount of work that can be extracted from a certain heat input in a cycle.



Unavailable Energy (UAE)–It is the minimum amount of heat that has been rejected to the sink

$$\boxed{UAE = T_0 \Delta s}$$

$T_0 \rightarrow$ Ambient temperature

$\Delta s \rightarrow$ Change in entropy

The minimum energy that has to be rejected to the sink as per second law of thermodynamics is called unavailable energy.

715. Neglecting changes in kinetic energy and potential energy, for unit mass the availability in a non-flow process becomes $a=\varphi - \varphi_0$, where φ is the availability function of the

(a) Steady flow process system
(b) Closed system
(c) Isolated system
(d) Open system

Rajasthan Nagar Nigam AE 2016, Shift-II
ESE 1998

Ans. (b)

$$A = \left(H_1 + \frac{mu_1^2}{2} + mgz_1 - T_0 S_1\right) - \left[H_2 + \frac{mu_2^2}{2} + mgz_2 - T_0 S_L\right]$$

$$\boxed{A = \psi_1 - \psi_2} ----(I)$$

Where ψ is the energy (available it) function of a steady flow system.

Neglecting the change in kinetic energy and potential energy.

$$A = (U_1 + P_o V_1 - T_o S_1) - (U_2 + P_o V_2 - T_o S_2)$$

$$\boxed{A = \varphi_1 - \varphi_2} ----(II)$$

φ_1, φ_2 are energy availability/function

(II) is valid for only closed system.

716. Law of degradation of energy says that unavailable energy is gradually decreasing due to

(a) Increase in reversible processes
(b) Increase in irreversible processes
(c) Increase in unavailable energy
(d) None of these

UPPSC AE 12.04.2016 Paper-II

Ans : (a) Law of degradation of energy says that unavailable energy is gradually decreasing due to Increase in reversible processes.

Unavailable energy $= T_0 ds$

Availability $\left(W_{max}\right) = Q - T_o ds$

717. The irreversibility associated with the process is defined as

(a) Loss of qualitative energy
(b) Loss of energy
(c) Loss of enthalpy
(d) Loss of entropy

APPSC AEE Mains 2016 Paper-III

Ans. (a) : The irreversibility associated with the process is defined as loss of qualitative energy.

718. The exergy of the system will depend on

(a) Entropy generation
(b) Surrounding temperature
(c) Surrounding pressure
(d) All the given answers

APPSC AEE Mains 2016 Paper-III

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Ans. (d) : Exergy– Exergy is defined as the amount of work a system can perform when it is brought into thermodynamic equilibrium with its environment. It depends on entropy generation, pressure and temperature of surrounding.

719. Which of the following statement is incorrect ?

(a) The greater pressure difference in throttling the lesser the irreversibility

(b) The primary source of internal irreversibility in power is fluid friction in rotary machine

(c) The greater the irreversibility the greater the increase in adiabatic process

(d) The entropy of the universe in continuously on the increase

KPSC ADF 2015, ESE 1992

Ans. (a) : The greater pressure difference in throttling the lesser the irreversibility. This is incorrect statement. When pressure difference in throttling increases, irreversibility increases.

720. A system is working between 800 K and 400 K. If the heat supplied to the system is 200 kJ, then the unavailable energy is

(a) 150 kJ (b) 125 kJ

(c) 75 kJ (d) 100 kJ

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (d)

$$(W_{net})_{A.E} = Q_1\left(1-\frac{T_2}{T_1}\right)$$

Given

$Q = 200$ kJ

$T_1 = 800$ K

$T_2 = 400$ K

$$(W_{net})_{A.E.} = 200\left(1-\frac{400}{800}\right)$$

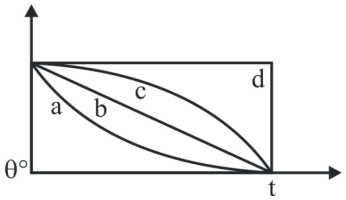
$= 100$ kJ

UAE = Q – A.E.

UAE = 200 – 100

UAE = 100 kJ.

721. A hot liquid is kept in a big room. Its temperature (θ) is plotted as a function of time (t), which of the following curve may represent the plot?



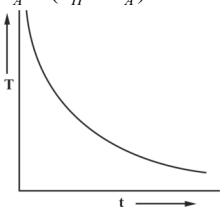
(a) Curve d (b) Curve b

(c) Curve c (d) Curve a

ISRO Scientist/Engineer 24.05.2014

Ans. (d) The graph drawn between the temperature of the body and time is known as cooling curve. The slope of the tangent to the curve at any point gives the rate of fall of temperature.

In general $T(t) = T_A + (T_H - T_A)e^{-kt}$



Where,

T(t) = temperature at time t

T_A = Ambient temperature

T_H = Temperature of hot object at time 0

k = position constant

t = time

722. Consider the following statements:

I. Availability is generally conserved.

II. Availability can either be negative or positive.

III. Availability is the maximum theoretical work obtainable.

IV. Availability can be destroyed in irreversibilities.

(a) III and IV are correct

(b) I and III are correct

(c) I and IV are correct

(d) II and IV are correct

APPSC Poly. Lect. 2013, ESE 1996

Ans. (a) : (i) Availability is defined as the maximum useful work that is obtainable in a process in which the system comes to equilibrium with its Surroundings.
(ii) Availability can be destroyed by irreversibilities.

723. Steam flows through an adiabatic steady flow turbine from state 1 to state 2. with respect to a base temperature T_0, the unavailable energy is

(a) $T_0 (I_1 - I_2)$

(b) $T_0 (S_1 - S_2)$

(c) $(I_1 - I_2) - T_0 (S_1 - S_2)$

(d) $I_2 + I_0 (S_1 - S_2)$

UKPSC AE 2012 Paper–II

Ans. (b) : $T_0 (S_1 - S_2)$

724. The loss in available energy associated with a given process is :

(a) $T_0\Delta S_{uni}$ (b) $Q - T_0\Delta S_{uni}$

(c) $Q + T_0\Delta S_{uni}$ (d) $T_0\Delta S_{system}$

APGENCO AE 2012

Ans. (a) : The loss of available energy

$= T_0 (\Delta S)_{universe}$

$= T_0 (S_{system} + S_{surrounding})$

725. The exergy of an isolated system in a process :

(a) can never increase

(b) can never decrease

(c) always remain constant

(d) always positive

J&K PSC Civil Services Pre, 2010

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Ans. (a) : It is the counter part of the entropy principle which states that the exergy of an isolated system can never increases.

726. Read the given information followed by a question and select the most appropriate option.

A heat source at 1000K loses 2500 kJ heat to a sink at:

(I) 500K

(II) 800K

In which case is the heat transfer more irreversible?

(a) Same in both (I) and (II)

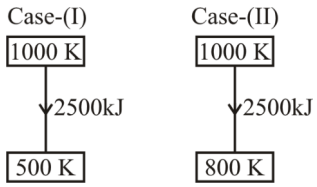
(b) Only (I)

(c) Cannot be predicted

(d) Only (II)

AAI Jr. Executive 26.03.2021

Ans. (b) :



$$(ds)_I = -\frac{2500}{1000} + \frac{2500}{500} = -2.5 + 5 = 2.5$$

$$(ds)_{II} = -\frac{2500}{1000} + \frac{2500}{800} = -2.5 + 3.125 = 0.625$$

$(ds)_I > (ds)_{II}$

In case(I) heat transfer is more irreversible.

727. A reversible engine is operating with temperature limits of 800 K and 300 K. If it takes 560 kJ of heat, then unavailable work would be:

(a) 175 kJ (b) 200 kJ

(c) 210 kJ (d) 150 kJ

AAI Jr. Executive 29.11.2018

Ans. (c) : For reversible engine

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$\frac{560}{800} = \frac{Q_2}{300}$$

$$Q_2 = 210 \text{ kJ}$$

Unavailable work = 210 kJ

728. In a reversible cycle, the source temperature is 227ºC and the sink temperature is 27ºC. The maximum available work for a heat input of 100 kJ will be

(a) 100 kJ (b) 60 kJ

(c) 40 kJ (d) 88 kJ

TSPSC AEE 2017

Ans. (c) : Maximum efficiency for 227ºC and 27ºC sources

$$\eta_{engine} = \frac{(227+273)-(27+273)}{(227+273)}$$

$$= \frac{200}{500} = \frac{2}{5}$$

$$= 0.4 = 40\%$$

So, maximum work available for a heat input

$$\eta = \frac{W}{Q}$$

$$W = \eta \times Q = 0.4 \times 100 \text{ kJ}$$

$$W = 40 \text{ kJ}$$

729. An ideal gas is throttled through an insulated pipe at the rate of 3 kg/s. There is a 10% pressure drop from an inlet to exit of the pipe. The values of R = 0.287 kJ/kgK and T_o = 300 K. The rate of energy loss for the pressure drop due to friction, will be nearly

(a) 34 kW (b) 30 kW

(c) 26 kW (d) 22 kW

ESE 2020

Ans. (c) : Due to throttling of an ideal gas, the process is isothermal so,

Change in entropy of gas,

$$\Delta S = -mR \ln\left(\frac{P_2}{P_1}\right)$$

$$\Delta S = -3 \times 0.287 \times \ln\left(\frac{0.9P_1}{P_1}\right)$$

$$\Delta S = 0.091 \text{ kW/K}$$

Rate of energy loss = $T_0 \Delta S = 300 \times 0.091$

$$= 27.21$$

$$\approx 26 \text{ kW}$$

730. 150 kJ of heat is transferred from a heat source of 527°C to a heat sink at 127°C. If the ambient temperature is 47°C, the loss of available energy during the process will be

(a) 40 kJ (b) 50 kJ

(c) 60 kJ (d) 70 kJ

UPSC JWM 2017

Ans. (c) : Loss of available energy = $T_0 \Delta S$

Ambient temperature T_0 = 47 + 273 = 320 K

and $\Delta S = \frac{Q_1}{T_1} - \frac{Q_1}{T_2}$

T_1 = 127 + 273 = 400 K

T_2 = 527 + 273 = 800 K

Q = 150 KJ

Loss of available energy = $T_0\left(\frac{Q_1}{T_1} - \frac{Q_2}{T_2}\right)$

Where $Q_1 = Q_2$,

$$= 320 \times 150\left(\frac{1}{400} - \frac{1}{800}\right)$$

$$= 60 \text{ kJ}$$

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731. The main cause of the irreversibility is

(a) Mechanical and Fluid Friction
(b) Unrestricted expansion
(c) Heat transfer with a finite temperature difference
(d) All of the above

UKPSC AE 2012 Paper–II

Ans. (d) : All of the above

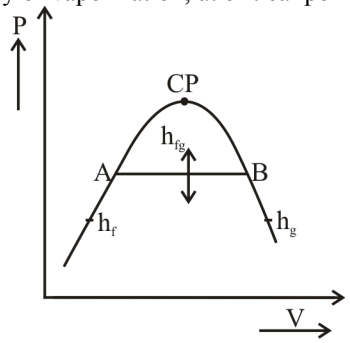
6. Properties of Pure Substances, Gases and Gas Mixture

732. The enthalpy of vaporization, at critical point is

(a) maximum (b) minimum
(c) zero (d) none of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC AE 2019, 2017
RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 18.09.2017
Rajasthan Nagar Nigam AE 2016, Shift-II
ESE 2008

Ans. (c) :
The enthalpy of vaporization, at critical point is zero



The line AB denotes enthalpy of vaporization. At critical point (CP), the length become zero so enthalpy of vaporization is zero.
Enthalpy of vaporization

$$h_{fg} = h_g - h_f$$

At critical point, the liquid and vapour have same properties, so

$$h_f = h_g$$

$$\boxed{h_{fg} = 0}$$

733. Constant pressure lines in the super-heated region of the Mollier diagram will have

(a) A positive slope
(b) A negative slope
(c) Zero slope
(d) Both positive and negative slope

RPSC Vice Principal ITI 2018
GPSC Executive Engineer 23.12.2018
Nagaland PSC CTSE 2015 Paper-II
HPPSC Asstt. Prof. 2014

Ans. (a) : Mollier diagram for a pure substance drawn on h-s coordinate.
According to Tds equation–

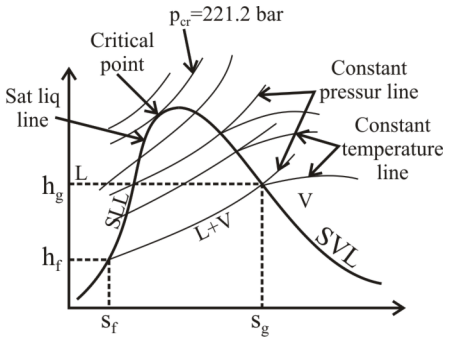
$$Tds = dh - vdp$$

$$T = \left(\frac{dh}{ds}\right) - v\left(\frac{dp}{ds}\right)$$

For constant pressure line $\left(\frac{dp}{ds}\right)_p = 0$

$$\boxed{T = \left(\frac{\partial h}{\partial s}\right)_p}$$

It is a positive slope which is equal to T(absolute temperature).
T is always positive, so slope will also be always positive.



734. Critical temperature is the temperature above which

(a) A gas will never liquify
(b) A gas will immediately liquify
(c) Water will evaporate
(d) Water will never evaporate

GPSC Executive Engineer 23.12.2018
Kerala PSC AE (Irrigation) 05.04.2017
HPPSC Asstt. Prof. 18.09.2017
RPSC VPITI 14.02.2016

Ans. (a) : The critical temperature of substance is the temperature above which vapour/gas of substance can not be liquified no matter what pressure is applied. Every substance has a critical temperature.

735. The value of compressibility factor (z) for an ideal gas may be :

1. less or more than one
2. equal to one
3. zero
4. less than zero

The correct value(s) is/are given by

(a) 1 and 2 (b) 1 and 4
(c) 2 only (d) 1 only

OPSC AEE 2019 PAPER - II
Nagaland PSC CTSE 2015 Paper-II
Kerala PSC Poly. Lect. 09.01.2014
ESE 2002

Ans. (c) : Compressibility Factor (z)–It is defined as the ratio of product PV and RT mathematically,

$$\boxed{z = \frac{PV}{nRT}}$$

For ideal gas, z = 1
For real gas, z > 1 or z < 1

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736. According to Dalton's law, the total pressure of the mixture of gases is equal to

(a) greater of the partial pressure of gases
(b) average of the partial pressure of gases
(c) Sum of the partial pressure of all
(d) Sum of the partial pressure of all divided by average molecular weight

TNPSC AE 2019
Kerala PSC AE (Irrigation) 05.04.2017
Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : According to Dalton's law, the total pressure of the mixture of gases is equal to sum of the partial pressure of all.

737. A standard vapour is compressed to half its volume without changing its temperature. The result is that

(a) All the vapour condenses to liquid
(b) Some of the liquid evaporates and the pressure does not change
(c) The pressure is double its initial value
(d) Some of the vapour condenses and the pressure does not change

TSPSC AEE 2017, APPSC Poly. Lect. 2013
ESE 1997

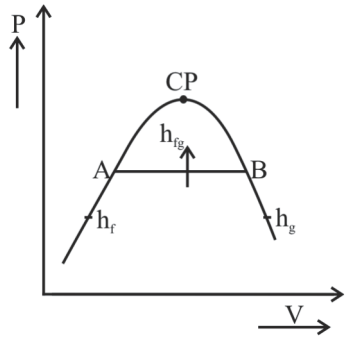
Ans. (d) : By compressing a standard vapour some of the vapour condenses and the pressure does not change.

738. The latent heat of steam with increase of pressure ..... .

(a) remains same
(b) increases
(c) decreases
(d) behaves unpredictably

GWSSB DEE 07.07.2016
UKPSC AE 2007 Paper -II
ESE 2002

Ans. (c) : The latent heat of steam with increase of pressure decreases.



739. Mixture of Ice and Water form a

(a) Closed System
(b) Heterogeneous system
(c) Isolated System
(d) Open system

WBPSC AE, 2017
Rajasthan AE (Nagar Nigam) 2016 Shift-3
Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (b) : Homogeneous System–When a system consisting of a single phase is called a homogeneous system.
Example–Dry air.
Heterogeneous System–While a system consisting of more than one phase is known as a heterogeneous system.
Example–Mixture of ice and water, humid air steam-water mixture.
Phase–A quantity of matter homogeneous throughout in chemical composition and physical structure is called a phase.
Every substance can exist in any one of the three phase, i.e. solid, liquid and gas.

740. The equation PV = mRT is essentially valid for

(a) Perfect gas (b) Real gas
(c) Monoatomic gas (d) Mixture of gases

UJVNL AE 2016, (KPSC AE 2015)
TNPSC ACF 2012

Ans. : (a) The equation PV = mRT is essentially valid for perfect gas.
The equation PV = mRT may also be expressed in another form

$$P = \frac{m}{V}RT$$

$$\boxed{P = \rho RT}$$

741. In throttling of fluid the following property remains unchanged before and after throttling:

(a) temperature
(b) entropy
(c) enthalpy
(d) both (b) and (c) above

(CGPSC Polytechnic Lecturer 2017)
WBPSC AE, 2007

Ans. (c) : During throttling enthalpy is constant it is also known as isenthalpic process. It is an irreversible adiabatic process.

742. The ideal gas laws can be applied, with least error, to

(a) Boiling water
(b) Wet steam
(c) Dry saturated steam
(d) Superheated steam

WBPSC AE, 2017, GPSC AE 26.07.2015

Ans. (d) : The dry superheated steam behaves as ideal gas and follows ideal gas laws.

743. Wet clothes are hung on a clothesline outdoors in sub-zero weather. After a day, the clothes are brought into the house and observed to be dry. The process of drying is best explained as

(a) vaporization (b) sublimation
(c) melting (d) condensation

Rajasthan Nagar Nigam AE 2016, Shift-II
ISRO Scientist/Engineer 11.10.2015

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Ans. (b) : On cold days, the wet clothes dry due to the water in the clothes that freezes, then sublimates, to create water vapour that escapes the clothes, therefore drying it.

744. Reduced pressure is

(a) always less than atmospheric pressure
(b) always unity
(c) an index of molecular position of a gas
(d) dimensionless

TRB Poly. Lect., 2012, ESE 2011

Ans. (d) Reduced pressure (P_r) is defined as the ratio of the actual pressure of the system to the critical pressure of the system. It is a dimensionless quantity.

745. For a pure substance at its triple point, the number of degrees of freedom is

(a) 0 (b) 1
(c) 2 (d) None of these

GPSC Executive Engineer 23.12.2018
Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Gibb's phase rule is given by,

$$P + F = C + 2$$

For pure substance (C) = 1
At triple point (P) = 3

$$3 + F = 1 + 2$$

$$\boxed{F = 0}$$

746. Super heated vapours behave:

(a) Exactly as a gas
(b) As steam
(c) As ordinary vapour
(d) Approximately as a gas

SJVN ET 2013, TRB Asstt. Prof., 2012

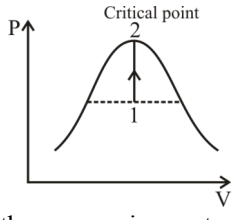
Ans. (d) : Superheated vapours behave approximately as a gas.

747. If a pure substance contained in a rigid vessel passes through the critical state on heating, its initial state should be

(a) sub-cooled water (b) saturated water
(c) wet steam (d) saturated steam

Nagaland PSC CTSE 2015 Paper-II
CSE Pre-1998

Ans. (c)



In a rigid vessel the process is constant volume process initial state is wet state (wet steam).

748. Dryness fraction of steam expressed in terms of mass of dry steam (m_s) in steam and mass of water particle (m_w) in steam is define as :

(a) $m_s/(m_s + m_w)$ (b) $m_s/(m_s - m_w)$
(c) $m_w/(m_s + m_w)$ (d) none of the above

GPSC Lect. (Auto) 16.10.2016
TNPSC AE 2019, 2014

Ans. (a) : $$\text{Dryness fraction} = \frac{m_s}{m_s + m_w}$$

m_s = mass of dry steam
m_w = mass of water particle

749. Saturated liquid at high-pressure P_1 having enthalpy of saturated liquid 1000 kJ/kg is throttled to a lower pressure P_2. At pressure P_2, enthalpy of saturated liquid and that of the saturated vapour are 800 and 2800 kJ/kg respectively. The dryness fraction of vapour, after throttling process is :

(a) 0.1 (b) 0.5
(c) 18/28 (d) 0.8

RPSC ADE 2016, ESE 2003

Ans. (a) : In process of throttling, enthalpy remains constant

$$h_1 = h_2 = h_{f_2} + x\left(h_{g_2} - h_{f_2}\right)$$

If dryness fraction is x

$$1000 = 800 + x(2800 - 800)$$

$$x = 0.1$$

750. The total volume of a mixture of gases is equal to the sum of the volumes that would be occupied by each component at the mixture temperature T and pressure P applicable to ideal gases is:

(a) Dalton's law (b) Avogadro's law
(c) Amagat's law (d) Kelvin's law

HPPSC Asstt. Prof. 29.10.2016
HPPSC AE (PWD) 10.09.2016

Ans. (c) : Amagat's law– Amagat's law assumes that the volumes of the component gases (again at the same temperature and pressure) are additive, the interactions of the different gases are the same as the average interactions of the component.

$$V_{min} = \sum_{i}^{k} V_i = V_1 + V_2 + V_3 + \ldots\ldots\ldots\ldots + V_k$$

751. In thermodynamics, pure substance is the one which :

(a) Behaves like a perfect gas
(b) Consists of only a single chemical species
(c) Has no dissolved impurities
(d) Has a number of chemical species but has constant composition

Kerala PSC Asstt. Prof. 27.09.2016
APPSC AEE MAINS 2016 PAPER-III

Ans. (d) : Thermodynamic, pure substance is the one which has a number of chemical species but has constant composition.

Pure Substance–

- A substance that has a fixed chemical composition throughout is called a pure substance i.e. water, nitrogen, helium and CO_2.

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- Substance which are mixture of various elements or compounds also qualifies as pure substances as long as mixture is homogeneous.

Properties of Pure Substance–

- A pure substance is a material with homogenous and invariable composition.
- Pure substances can have multiple phases, an ice water mixture is still a pure substance but an air-steam is not a pure substance.

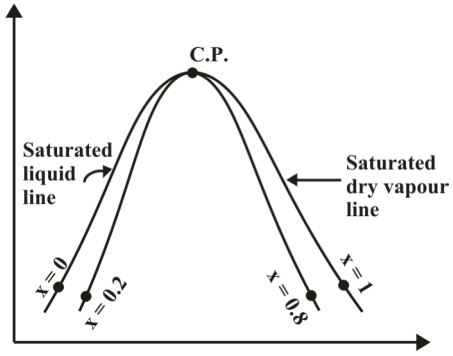
752. For dry saturated vapour, the value of dryness fraction will be

(a) 1.0 (b) 0.75
(c) 0.5 (d) 0

RPSC Lecturer 16.01.2016
TNPSC AE (Industries) 09.06.2013

Ans. (a) : Dryness fraction (x)

$$= \frac{\text{Mass of vapour}}{\text{Mass of vapour + Mass of liquid}}$$



$$x = \frac{m_v}{m_L + m_v}$$

at saturated vapour line

$m_L = 0$

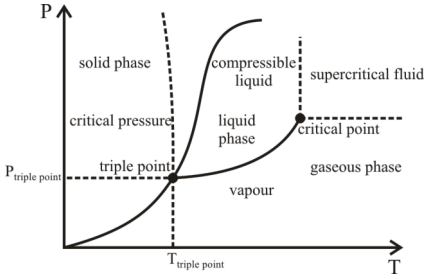
$$x = \frac{m_v}{m_v + 0} = 1$$

753. No substance can exist in the liquid phase in stable equilibrium

(a) at temperature below the triple point temperature
(b) at pressure below the triple point pressure
(c) at pressure above the triple point pressure
(d) at temperature above the triple point temperature

GPSC EE Pre, 28.01.2017, ESE 2012

Ans. (b) : About to three phase equilibrium



- No substance can exist in the liquid phase in stable equilibrium at pressure below the triple point pressure.
- The same can be said for temperature for substance that contract on freezing.
- Substances at high pressure can exist in the liquid phase at temperature below the triple point temperature.

754. Critical point of water refers to the state in which-

(a) Liquid and solid coexist
(b) Solid and liquid coexist
(c) Liquid and vapour coexist
(d) All the three coexist

NPCIL ET, 2015
UKPSC AE 2012 Paper–II

Ans. (c) : Critical point of water refers to state in which-liquid and vapour co exist.
Critical point is point where liquid directly converts into vapour and triple point of water is the point where three phase exist. i.e. solid, liquid and gas.

755. No liquid can exist as a liquid at

(a) 0°C temperature (b) 200°C temperature
(c) Zero pressure (d) Zero viscosity

Nagaland PSC CTSE 2017 Paper-2
APPSC AEE Mains 2016 Paper-III

Ans. (c) : No liquid can exist as a liquid at zero pressure:-
Generally absolute zero pressure is the point where there exist a minimum temperature i.e. zero. That can be possible only when molecular momentum of system become zero. There should not be any motion of particles so there is no collision of particles, kinetic energy nullifies and the temperature becomes zero.

756. If 'h' refers total heat, T refers to temperature, then in the throttling process,

(a) $h_1^2 = h_2$ (b) $h_1 = h_2$
(c) $h_1 = h_2 + \frac{h_{fg}}{T_s}$ (d) $h_2 = h_1 + \frac{h_{fg}}{T_s}$

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC AE 2019

Ans. (b) : The throttling process is a constant enthalpy process.

$h_1 = h_2$

757. With the increase in pressure -

(a) The boiling point of water decreases and enthalpy of evaporation increases.
(b) The boiling point of water increases and enthalpy of evaporation decreases
(c) Both the boiling point of water and enthalpy of evaporation decreases.
(d) Both the boiling point of water and enthalpy of evaporation increases

RPSC ACF & FRO, 26.02.2021
TNPSC AE 2014

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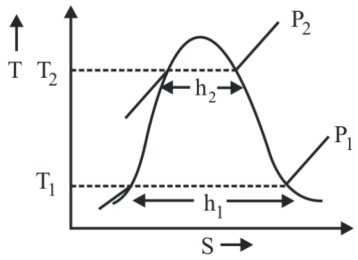
Ans. (b) : With the increase in pressure the boiling point of water increases and enthalpy of evaporation decreases



$P_2 > P_1$
$T_2 > T_1$
$h_2 < h_1$

758. Consider the following statements :
When dry saturated steam is throttled from a higher pressure to a lower pressure, the :
(1) Pressure decreases and the volume increases.
(2) Temperature decrease and the steam becomes super heated.
(3) Temperature and the dryness fraction increases.
(4) Entropy increases without any change in Enthalpy.
Which of these statements are correct?
(a) (1) & (4) (b) (1), (2) & (4)
(c) (1) & (3) (d) (2) & (4)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
ESE 2000

Ans. (b) : When dry saturated steam is throttled from a higher pressure to lower pressure, the
(1) Pressure decreased and volume increases
(2) Without any change in enthalpy.
(3) Dryness fraction increases.

759. Which of the following laws is applicable for the behavior of a perfect gas?
(a) Boyle's law (b) Charles' law
(c) Gay-Lussac law (d) all of the above

APGCL AM, 2021
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) :
Boyle's Law : It states that the pressure exerted by a gas (of a given mass, kept at a constant temperature) is inversely proportional to the volume occupied by it.

$$P_1V_1 = P_2V_2$$

Charles Law : It states that the volume of an ideal gas is directly proportional to the absolute temperature at constant pressure.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Gay-Lussac's Law : The pressure of given mass of gas varies directly with the absolute temperature of the gas, when the volume is kept constant.

760. If a pure substance is below the triple point temperature, the solid on being heated will only
(a) temperature remain constant
(b) liquify
(c) vapourize or sublimate
(d) have its temperature increase

UPPSC AE 13.12.2020, Paper-2
ESE 2013

Ans. (c) : If a pure substance is below the triple point temperature, the solid on being heated will only vapourize or sublimate.

761. Highest ratio of specific heat is possible for?
(a) Argon (b) Helium
(c) Hydrogen (d) Air

ISRO Scientist/Engineer 12.01.2020
J&K PSC Civil Services Pre, 2010

Ans. (a,b) :

Gas	Ratio of specific heat	Examples
Mono atomic	$\gamma = 1.667$	He, Ar, Neon
Diatomic	$\gamma = 1.4$	O_2, N_2, H_2, Air
Tri-atomic	$\gamma = 1.33$	O_3

762. The specific heat of a gas remains constant at all temperatures and pressures is discussed in:
(a) Joule's Law (b) Boyle's Law
(c) Charles' Law (d) Regnault's Law

CIL MT 27.02.2020, TSPSC AEE 2015

Ans. (d) : Regnault's law states that the specific heat capacity of a gas at constant pressure (C_P) and specific heat capacity of a gas at constant volume (C_V) do not change with change in pressure and temperature.
According to this law, C_P and C_V values of gas remain constant.

763. The slope drawn on sublimation curve on P-T diagram for all substances gives _____ value.
(a) infinite (b) zero
(c) negative (d) positive

RPSC ACF & FRO, 26.02.2021
AAI Jr. Executive 29.11.2018

Ans. (d)
1. Slope of sublimation curve for all substances is positive.
2. Slope of fusion curve for most of substances is positive but for water it is negative.
3. Those substances whose triple point pressure lies above atmospheric pressure will directly sublimate.

764. In Vander Waals equation of state the two constant are determined from the behaviour of substance at :
(a) Saturated point (b) Triple point
(c) Critical point (d) Never determined.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II

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Ans. (c) : Vander Waals equation-

$$\left(P+\frac{a}{V^2}\right)(V-b)=RT$$

Force of Cohesion $=\frac{a}{V^2}$

Volume of molecules = b
(Co- volume)

where $a=3P_cV_c^2$ and $b=\frac{V_c}{3}$

Critical pressure $(P_c)=\frac{a}{27b^2}$

Critical Volume $(V_c)=3b$

Critical temperature $(T_c)=\frac{8a}{27bR}$

765. The triple point temperature for water is

(a) 0°K (b) 0°C
(c) 0.01°C (d) 0.1°C

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) : 0.01 °C (273.16 K, 32.01 °F)

766. The specific heat at constant pressure is _____ that of specific heat at constant volume.

(a) equal to (b) less than
(c) more than (d) None of these

RPSC ACF & FRO, 26.02.2021

Ans. (c) : The specific heat at constant pressure is more than that of specific heat at constant volume.

$C_p > C_v$

$C_p - C_v = R$

$C_p = R + C_v$

$\frac{C_p}{C_v}=\gamma$

$C_p = \gamma C_v$

C_p is greater than C_v because when a gas is heated at constant volume, no external work is done. So, heat supplied is consumed only in increasing the internal energy of gas.
But if gas is heated at constant pressure, external work is done as well as increasing the internal energy of gas.

767. The enthalpy of dry saturated steam ___ with the increase in pressure.

(a) decreases (b) increases
(c) remains constant (d) All of these

RPSC ACF & FRO, 26.02.2021

Ans. (a) : The enthalpy of dry saturated steam decrease with the increase in pressure.

768. Which combination of the following statements is correct?

P. A gas cools upon expansion only when its Joule-Thomson coefficient is positive in the temperature range of expansion.

Q. For a system undergoing a process, its entropy remains constant only when the process is reversible.

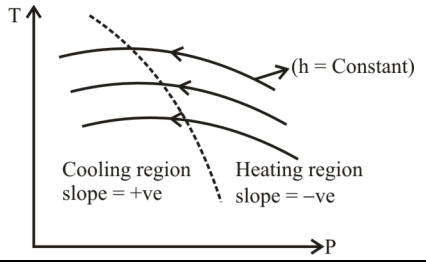
R. the work done by a closed system in an adiabatic process is a point function.

S. A liquid expands upon freezing when the slope of its fusion curve on Pressure Temperature diagram is negative.

(a) R and S (b) P and Q
(c) Q, R and S (d) P, Q and R

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : For a system undergoing a process, its entropy remains constant in two possible conditions.
(i) Reversible and adiabatic.
(ii) Irreversible and non-adiabatic.
But in irreversible and non-adiabatic process, the entropy generation and entropy generates by surrounding is equal in magnitude and opposite in direction.
Q statement is incorrect.

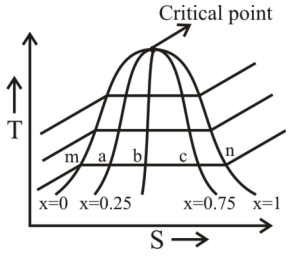


769. The dryness (x) fraction of superheated steam is taken as

(a) x = 0 (b) x = 0.9
(c) x = 0.999 (d) x = 1

Assam PSC AE (IWT) 14.03.2021

Ans. (d) :



The value of x varies between zero and one. For saturated water, when water just starts boiling, x = 0, and for saturated vapour when vaporisation is complete, x = 1. For which the vapour is said to be dry saturated. Point m indicate the saturated liquid state with x = 0 and point n indicate the saturated vapour state with x = 1 the line mn indicating the transition from liquid to vapour.

770. What is the state at which saturated liquid line with respect to vaporization and saturated vapor line on p-v diagram of pure substance made called?

(a) Saturation state
(b) Critical state
(c) Vaporization state
(d) Superheated vapor state

Assam PSC AE (IWT) 14.03.2021

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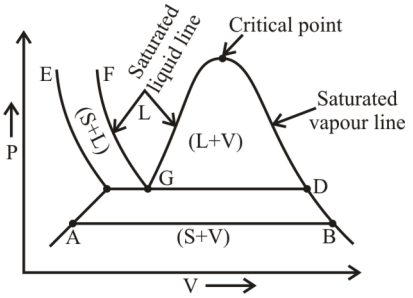
Ans. (b)



The point at which the saturated liquid line with respect to vapourisation and saturated vapour on a P-V diagram of a pure substance is called critical point.

The temperature, pressure and specific volume corresponding to the critical point are termed as critical properties.

771. In throttling calorimeter, the thermodynamic process used to throttle the wet steam is _____.

(a) constant volume process
(b) process with constant specific heat
(c) constant entropy process
(d) constant enthalpy process

VIZAG Steel MT 24.01.2021

Ans. (d) : In throttling process,
Enthalpy (h) = Constant

772. The state of a simple compressible pure substance can be fixed by specifying

(a) one independent property.
(b) two independent properties.
(c) three independent properties.
(d) four independent properties.

ESE 18.07.2021

Ans. (b) : The state of pure substance of a given mass can be fixed by specifying two independent intensive properties provided the system is in equilibrium. This is known as 2-properties rule.

Temperature and specific volume, for example, are always independent properties, and together they can fix the state of a simple compressible system.

773. A vessel has two compartments. First compartment contains 2 kg of dry saturated steam at 7 bar [$v_{g@7bar}$= 0.27 m^3/kg] and second compartment contains 1 kg of steam of dryness fraction of 0.8 at 3.5 bar [$v_{g@3.5bar}$= 0.55 m^3/kg]. After the partition is removed, the pressure of the mixture is found to be 5 bar [$v_{g@5bar}$ = 0.37 m^3/kg]. Neglecting the volume of water, what will be specific volume of mixture?

(a) 0.281 m^3/kg (b) 0.326 m^3/kg
(c) 0.462 m^3/kg (d) 0.521 m^3/kg

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given,
$m_1 = 2$ kg
$x_1 = 1$
$P_1 = 7 \text{ bar} = 7 \times 10^5 \text{ N/m}^2$
v_{g_1} at 7 bar = 0.27 m^3/kg
$m_2 = 1$ kg
$x = 0.8$
$P_2 = 3.5 \text{ bar} = 3.5 \times 10^5 \text{ N/m}^2$
v_{g_2} at 3.5 bar = 0.55 m^3/kg
After mixing
$P_m = 5$ bar
V_g at 5 bar = 0.37 m^3/kg
$v_m = ?$
We know total mass (m) = $m_1 + m_2$ = 1+2 = 3kg
Neglecting volume of water

$$V_m = \frac{m_1\left(x_1 v_{g_1}\right) + m_2\left(x_2 v_{g_2}\right)}{m}$$

$$= \frac{2\times(1\times 0.27)+1\times(0.8\times 0.55)}{3}$$

$$= 0.3267 \text{ m}^3/\text{kg}$$

774. The pressure and temperature of mixture of 4 kg O_2 and 6 kg N_2 are 4 bar and 27^0C. What will be the value of molecular weight of mixture?

(a) 40.67 (b) 39.87
(c) 29.47 (d) None of the above

UPPSC AE 13.12.2020, Paper-2

Ans. (c) : Given,
mass of O_2 = 4 kg
mass of N_2 = 6 kg
Molecular weight of mixture,

$$\Rightarrow \frac{m_{o_2}\times\left(M_{wt}\right)_{o_2} + m_{N_2}\times\left(M_{wt}\right)_{N_2}}{m_{o_2} + m_{N_2}}$$

$$\Rightarrow \frac{(4\times 32)+(6\times 28)}{4+6}$$

$$\Rightarrow 29.6$$

775. According to the state postulate, the number of independent properties required to determine the state of a system containing a pure substance, with two of its phases in equilibrium, is

(a) 1 (b) 2
(c) 3 (d) zero

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : The number of properties required to fixed the state of a system is given by the state postulate. The slope of a simple compressible system is completely specified by two independent, intensive properties.

776. Ice, when heated, directly sublimates into water vapour without going through liquid phase when

(a) temperature is lower than critical temperature
(b) pressure is lower than critical pressure
(c) temperature is lower than triple point temperature
(d) pressure is lower than triple point pressure

Haryana PSC AE (PHED) 04.09.2020, Paper-I

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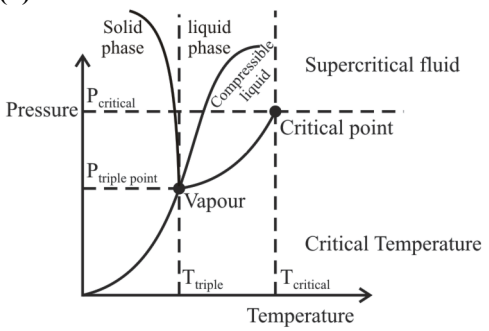
Ans. (d)



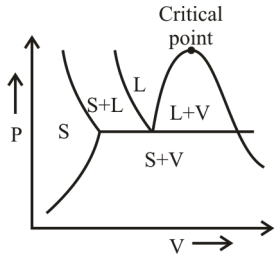
On decreasing temperature below triple point it solidifies while decreasing pressure below triple point it changes directly into vapour state i.e. sublimation.

777. In the P-V diagram for water, the horizontal portion of an isothermal line indication

(a) solid phase
(b) vapour phase
(c) phase change from solid to liquid or liquid to vapour
(d) liquid phase

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) :



778. When a material solidify due to latent heat of fusion is known as:

(a) Thermometry (b) Super cooling
(c) Under cooling (d) Thermal arrest

CIL MT 27.02.2020

Ans. (d) : • The cooling curve formed when a material is solidified due to latent heat of fusion is called as thermal arrest.

• Thermometry is the process of measuring temperature.

• Subcooling or undercooling is the process of lowering the boiling point.

779. If the value of a gas constant is 0.1 kJ/kg.K and specific heat at constant volume is 0.2 kJ/kg.K, the specific heat ratio will be:

(a) 1.5 (b) 1.1
(c) 3.3 (d) 2.5

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : Given, $C_v = 0.2$ kJ/kgK

$R = 0.1$ kJ/kgK

We know that,

$C_p - C_v = R$

$C_p = 0.1 + 0.2$

$C_p = 0.3$

Ratio of specific heat

$$\frac{C_p}{C_v} = \gamma = \frac{0.3}{0.2}$$

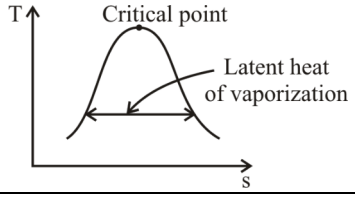
$$\boxed{\frac{C_p}{C_v} = 1.5}$$

780. At critical pressure

(a) the entropy of a substance is zero
(b) the latent heat of vaporization is zero
(c) saturated vapour becomes a gas
(d) a real gas obeys ideal gas laws

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : At critical pressure the latent heat of vaporization is zero.



781. Gas laws are least valid in the cases of

(a) mono atomic gases (b) polyatomic gases
(c) vapours (d) gas mixtures

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : Gas laws are least valid in the case of vapours.

782. Critical point of water is

(a) 22.06 kPa (b) 22.06 MPa
(c) 22.06 atm (d) 22.06 bar

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : $\Rightarrow$ Critical pressure of water = 220.6 bar
= 22.06 MPa
$\Rightarrow$ Critical temperature of water = 374ºC

783. In a shower mixer, cold water at 27 deg C, flowing at 5 kg/min is mixed with hot water at 77 deg C flowing at 15 kg/min. The exit temperature of the mixture is

(a) 45.4 deg C (b) 64.5 deg C
(c) 34.5 deg C (d) 68.4 deg C

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : For steady state,
The rate of enthalpy entering the mixture
= The rate of enthalpy leaving mixture
$(5\times C_{pw}\times 27) + (15\times C_{pw}\times 77) = (5+15)\times C_{pw}\times (T_{exit})_{mixture}$
$(T_{exit})_{mixture} = 64.5$ºC

784. When a liquid and its vapour are in equilibrium at a certain pressure and temperature, then which of the following is required to identify the saturation state?

(a) Pressure
(b) Temperature
(c) Pressure or temperature
(d) Both (a) and (b)

MECON MT 2019

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Ans. (c) : When a liquid and its vapour are in equilibrium at a certain pressure and temperature, only the pressure or the temperature is sufficient to identify the saturation state.

785. In a single component system, the maximum number of phases that can coexist together in equilibrium is _____.

(a) 5 (b) 2
(c) 3 (d) 4

VIZAG MT, 14.12.2020

Ans. (c) : At triple point single component exist in three phases. (DOF) at triple point = 0.

786. Triple point of a substance

(a) Occurs in a mixture of two or more gases
(b) Is the point where three phases exists together
(c) Occurs in sublimation
(d) None of the above

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (b) : Triple point of a substance is the point where three phases (solid, liquid & gas) exist together.
Triple point of water = 0.01°C = 273.16 K

787. For sublimation process which of the following are equilibrium?

(a) Solid and liquid phase
(b) Solid and vapour phases
(c) Solid, liquid and vapour phases
(d) None of the above

HPPSC ADF 05.03.2019

Ans. (b) : For sublimation process- Solid directly converted into vapour phase.

788. Nitrogen at an initial state of 10 bar, 1m^3 and 300 K is expanded isothermally to final value of 2m^3. The p-v-T relation is $(p + a/v^2)\, v = RT$, where a > 0. The final pressure;

(a) Will be slightly less than 5 bar
(b) Will be slightly more than 5 bar
(c) Will be exactly 5 bar
(d) Cannot be ascertained in absence of value of a.

HPPSC ADF 05.03.2019
GATE 2005

Ans. (b) : Given,

$P_1 = 10$ bar, $V_2 = 2\ m^3$
$V_1 = 1\ m^3$, $T = 300$ k

$$\left(P+\frac{a}{V^2}\right)(V) = RT$$

[According to Vanderwall's equation]

$$\left(P+\frac{a}{V^2}\right)V = \text{Constant} \Rightarrow$$

$$\left(P_1+\frac{a}{V_1^2}\right)V_1 = \left(P_2+\frac{a}{V_2^2}\right)V_2 \Rightarrow \left(10+\frac{a}{1^2}\right)1 = \left(P_2+\frac{a}{2^2}\right)2$$

$$\Rightarrow \frac{10+a}{2} = P_2+\frac{a}{4} \Rightarrow 5+\frac{a}{2} = P_2+\frac{a}{4}$$

$$\Rightarrow 5+\frac{a}{4} = P_2 \text{ and } a > 0 \text{ is given}$$

$\therefore$ The value of P_2 should be slightly greater than 5 bar

789. On Mollier chart a flow though a turbine is represented by:

(a) Horizontal line
(b) Vertical line
(c) Curved line convex up
(d) Curved line concave up

HPPSC ADF 05.03.2019

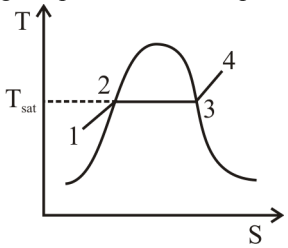
Ans. (b) : On Mollier chart a flow through, a turbine is represented by vertical straight line because isentropic expansion in turbine.

790. Saturated state of pure substance (water) is a:

(a) State from which a change of phase may occur without a change in pressure or temperature
(b) State from which a change of phase occur with a change in pressure of temperature.
(c) State from which a change of phase may occur without a change in pressure and temperature.
(d) State from which a change of phase may occur with a change in pressure and temperature

Oil India Limited Sr. Engineer (Production) 30.11.2019

Ans. (c) : Saturated state of pure substance (water) is the state from which change of phase may occur without a change in pressure and temperature.



On T-S diagram horizontal line 2-3 represent saturation state of pure substance through pressure and temperature not varies.

791. Which of the following options can always be approximated to be an ideal gas?

(a) Highly superheated vapour
(b) Dry saturated vapour
(c) Supercritical fluid
(d) Saturated vapour

BHEL ET 2019

Ans. (a) : Assumption of Ideal gas-

- Intermolecular force should be negligible
- Volume occupied by the molecular should be negligible compared to container volume.
- Water vapour mixed with air is considered as ideal gas.
- Steam should never be considered as ideal gas unless mentioned.
- Steam is the special term given to the gases form of water only when present at high temperature.

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792. Pick the correct statement about pure substances.

(a) A mixture of liquid air and gaseous air is a pure substances.
(b) A mixture of ice and liquid water is not a pure substance.
(c) A mixture of two or more phases of pure substances is not a pure substance even though the chemical composition of all the phases is the same throughout.
(d) A mixture of two or more phases of pure substance is still a pure substance as long as the chemical composition of all the phases is the same throughout.

BHEL ET 2019

Ans. (d) : Pure substance– A mixture of two or more phases of pure substance is still a pure substance as long as the chemical composition of all the phases is the same throughout.

793. Water vapour can be considered as Ideal Gas.

(a) Never (b) Always
(c) At high pressure (d) At low pressure

OPSC AEE 2019 PAPER-II

Ans. (d) : At very low pressure and high temperature all gases and vapour approaches to ideal gas behaviour.

794. Specific heat of mono atomic gases:

(a) Increase with temperature rise
(b) Decrease with temperature rise
(c) Does not depends on change in temperature
(d) None of these

OPSC AEE 2019 PAPER - II

Ans. (c) : Specific heat for mono atomic ideal gas is $\frac{5R}{2}$ and for diatomic ideal gas is $\frac{7R}{2}$

795. Calculate the dryness fraction of steam which has 1.5 kg of water in suspension with 50 kg of steam

(a) 0.971 (b) 1
(c) 0 (d) 0.485

TNPSC AE 2019

Ans. (a) : Data Given

$m_w = 1.5$ kg, $m_v = 50$ kg

$$\text{Dryness fraction (x)} = \frac{m_v}{m_w + m_v}$$

$$x = \frac{50}{1.5 + 50} = 0.971$$

796. On Mollier chart, slope of an isobar on h-s diagram is equal to: [where T = absolute temperature]

(a) T^4 (b) T^2
(c) T (d) T^3

SJVN ET 2019

Ans. (c) : dh = Tds + vdP, For constant pressure/ isobaric process

dP = 0

So, dh = Tds

$$\text{or } \left.\frac{dh}{ds}\right|_p = T$$

797. An ideal gas having the weight of 20 N/m^3 at the temperature of 27°C and pressure of 0.206 N/mm^2 (abs). The gas constant will be [Consider g = 9.81 m/s^2]

(a) 912 kJ/kg-K (b) 0.912 kJ/kg-K
(c) 0.4251 kJ/kg-K (d) 425.1 kJ/kg-K

SJVN ET 2019

Ans. (c) : PV = mRT

$$\frac{0.206}{10^{-6}} \times 1 = \frac{20}{9.81} \times R \times 300 \text{ or}$$

R = 425.1 J/kgK

= 0.4251 kJ/kgK

798. A gas mixture consists of 3 kg of O_2, 5kg of N_2 and 12 kg of CH_4. The mass fraction and mole fraction of O_2 are

(a) 0.25 and 0.125 (b) 0.15 and 0.092
(c) 0.25 and 0.092 (d) 0.15 and 0.125

ESE 2019

Ans. (b) : Mass fraction of $O_2 = \frac{m_{O_2}}{m_{O_2} + m_{N_2} + m_{CH_4}}$

$$= \frac{3}{3+5+12} = 0.15$$

Mole fraction of $O_2 = \frac{n_{O_2}}{n_{O_2} + n_{N_2} + n_{CH_4}}$

$$= \frac{\left(\frac{3}{32}\right)}{\left(\frac{3}{32}\right) + \left(\frac{5}{28}\right) + \left(\frac{12}{6}\right)}$$

= 0.092

799. What is the lowest pressure at which water can exist in liquid phase in stable equilibrium?

(a) 101.325 kPa (b) 0.311 kPa
(c) 22.06 kpa (d) 0.611 kPa

BHEL ET 2019

Ans. (d) : The lowest pressure at which water can exist in liquid phase in stable equilibrium of 0.611 kPa.

800. The statement that molecular weight of all gases occupy the same volume is known as-

(a) Avogardo's hypothesis
(b) Gas law
(c) Dalton's law
(d) Thermodynamics law

RPSC AE 2018

Ans. (a) : The statement that molecular weight of all gases occupy the same volume is known as Avogardo's hypothesis.

801. The point that connects the saturated-liquid line to the saturated-vapour line is called the

(a) triple point
(b) critical point
(c) superheated point
(d) compressed liquid point

RPSC Vice Principal ITI 2018

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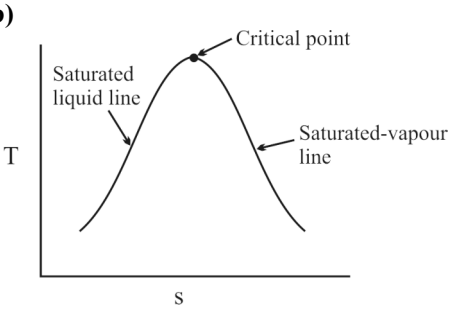
Ans. (b)



The properties of liquid and vapour are identical at critical point.

802. The value of characteristics constant of oxygen would be

(a) 0.412 kJ/kg-K (b) 0.262 kJ/kg-K
(c) 1.004 kJ/kg-K (d) 0.624 kJ/kg-K

TNPSC AE 2018

Ans. (b) : The value of characteristics constant of oxygen would be 0.262 kJ/kg-K.

803. It is desired to store 28 kg of nitrogen at 14 MPa pressure and 27°C in a cylinder. Assuming that nitrogen behaves like an ideal gas, determine the size of the cylinder.

(a) 0.01782 m^3 (b) 0.1782 m^3
(c) 1.782 m^3 (d) 17.82 m^3

RPSC AE 2018

Ans. (b) : Data given:

m = 28 kg, P = 14 MPa, T = 27°C = 300 K

Ideal gas equation

$PV = mRT$

where

$$R = \frac{\bar{R}}{M}$$

molecular weight (M) for

$N_2 = 28$

$\bar{R}$ (Universal gas constant)

= 8.314 kJ/k-m K

$$R = \frac{8.314}{28} = 0.2969\ kJ/K\,kg$$

then,

$$V = \frac{28 \times 0.2969 \times 300}{14 \times 1000}$$

$V = 0.178157\ m^3$

804. Which of the following law governs the isothermal process

(a) Boyle's Law (b) Charle's law
(c) Joule's law (d) Gay Lussac's law

HPPSC AE 2018

Ans. (a) : Boyle's law governs the isothermal process.

T = Constant

$P_1V_1 = P_2V_2$ = Constant

Charle's Law – P = Constant

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} = \text{Constant}$$

Gay Lussac's Law – Gay Lussac's Law states that the pressure of a given mass of a gas varies directly with absolute temperature of the gas when the volume is kept constant.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} = \text{Constant}$$

805. Which one of the following substances has constant specific heat at all pressures and temperature?

(a) Mono-atomic gas (b) Di-atomic gas
(c) Tri-atomic gas (d) Poly-atomic gas

ESE 2018

Ans. (a) : The specific heats, c_p and c_v vary with the temperature, the variation being different for each gas. For monoatomic gases, such as He, Ne, Ar and most metallic vapours, specific heats are constant.

806. According to kinetic theory of gases, the kinetic energy of translation of a molecule is proportional to

(a) T (b) $\sqrt{T}$
(c) 1/T (d) $1/\sqrt{T}$

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : According to kinetic theory of gases, the kinetic energy of translation of a molecule is proportional to temperature (T).

Kinetic energy, $K.E. = \frac{3}{2}\frac{R}{N_A} \times T \Rightarrow K.E. \propto T$

Where, R = Gas constant

N_A = Avogardro's number

807. The ratio of molar specific heats for monoatomic gas is

(a) 1 (b) 1.4
(c) 1.67 (d) 1.87

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The ratio of the specific heats $\gamma = C_P/C_V$ is a factor in adiabatic engine processes. This ratio $\gamma = 1.66$ air, which is predominantly a diatomic gas.

808. If a_1 and a_2 are the dryness fractions obtained in separating calorimeter and throttling calorimeter respectively, then the actual dryness fraction of steam will be:

(a) $(a_1 + a_2)/2$ (b) $(a_1a_2)/2$
(c) $a_1 + a_2$ (d) a_1a_2

AAI Jr. Executive 29.11.2018

Ans. (d) : The separating calorimeter when used to alone does not give a very accurate result because we have to assume that all the water is separated out from the steam while passing through the calorimeter which is not the case also the throttling calorimeter is only able to measure the dryness fraction of steam in the steam gets superheated after throttling the actual dryness fraction of steam.

Actual dryness fraction (x_{actual}) = $a_1 \times a_2$

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809. The ..... states that change of internal energy of a perfect gas is directly proportional to the change of temperature

(a) Boyle's law (b) Charles's law
(c) Gay-Lussac law (d) Joule's law

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Joule's First law – A physical law expressing the relationship between the heat generated and current flowing through a conductor.

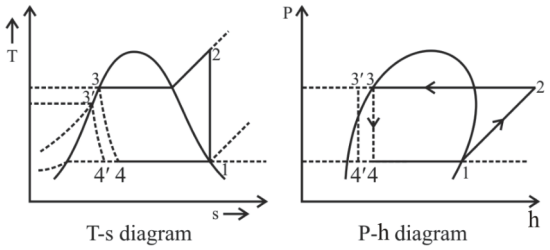
Joule's second law – This law states that change of internal energy of a perfect gas is directly proportional to the change of temperature and independent of its volume and pressure.

810. On cooling some amount of liquid water, its saturation temperature was observed to be lower for a given pressure of the liquid water. This liquid water is said to be

(a) sub cooled liquid
(b) saturated liquid
(c) compressed liquid
(d) superheated vapour

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : The term subcooling refers to a liquid existing at a temperature below its normal boiling part.



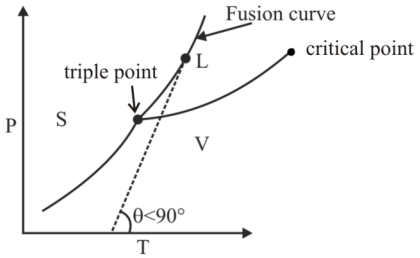
Subcooling 1-2-3-3'-4'-4-1

811. During melting the volume of pure substance other than water :

(a) Decreases
(b) Increases
(c) Remains constant
(d) First increases and then decreases

TRB Polytechnic Lecturer 2017

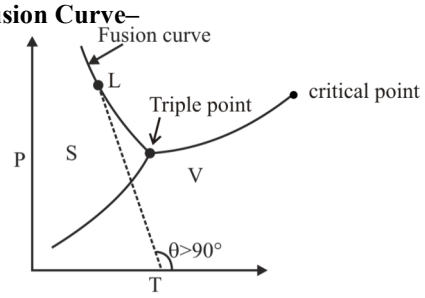
Ans. (b) : General Substance Fusion Curve–



$$\left(\frac{\partial P}{\partial T}\right)_{\text{General substance}} = +ve$$

Liquids → Solids [Contract]
Solid → Liquid [Expands]

H_2O Fusion Curve–



$$\left(\frac{\partial P}{\partial T}\right)_{H_2O} = -ve$$

Solid → Liquid [Contracts]
Liquid → Solid [Expands]

812. The temperature of an ideal gas always decreases during

(a) Isobaric expansion
(b) Isothermal expansion
(c) Adiabatic expansion
(d) Isentropic expansion

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : The temperature of an ideal gas always decreasing during isentropic expansion.

813. Statement (I): A homogeneous mixture of gases that do not react within themselves can be treated as a pure substance.
Statement (II): Flue gases can be treated as a homogeneous mixture of gases.

ESE 2017

Ans. (b) : The composition of pure substance is invariable and same through out the sample i.e. constituents of pure substance do not react themselves. Hence statement-I is definition of pure substance and statement-II is example, so both are true.

814. What is the degree of freedom at critical point of water (pure substance)?

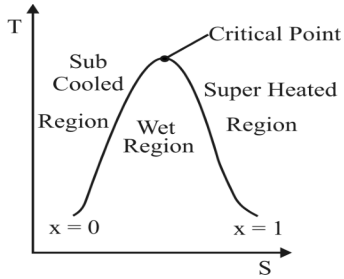
(a) 1 (b) 2
(c) 0 (d) 4

CGPSC ARTO & TSI 07.01.2017

Ans. (c) : Critical point :-It is the intersection point of saturated liquid line and saturated vapour line. or It is a point above which the liquid directly flash off into the vapour.

- At the critical point the degree of freedom is zero.

At the Critical point
$P + F = C + 1$ (Not = C + 2)
$2 + F = 1 + 1$
$\boxed{F = 0}$



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- At the critical point

Water Temperature (T_c) = 374ºC

And Pressure (P_c) = 221.2 bar

- At the critical point enthalpy all vaporisation (latent heat) will be zero
- On increasing the pressures of liquid the boiling point increases and latent heat decreases.

Degree of freedom (F)	Conditioned
0	At critical Point
2	Sub-Cooled Region
1	Wet Region
2	Super Heated Region

815. According to Avogadro's law

(a) the product of the gas constant and the molecular mass of an ideal gas is constant.

(b) the sum of partial pressure of the mixture of two gases is the sum of the two.

(c) equal volume of all gases, at the same temperature and pressure, contain equal number of molecules.

(d) All of the above

Karnataka PSC AE, 10.09.2017

Ans. (c) : Avogadro's law state that equal volumes of all gases at the same temperature and pressure, contain equal, number of molecules.

816. In the particular case of water, the sublimation curve is called the

(a) frost line (b) streak line

(c) ice line (d) steam line

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : In the particular case of water the sublimation curve is called frost line. The vaporization curve is called steam line.

The fusion curve is called the ice line.

817. It is not possible to liquify hydrogen by application of pressure, at normal room temperatures, because

(a) It has high thermal conductivity

(b) It is a diatomic molecule

(c) Its critical temperature is lower than the room temperature

(d) It forms water under pressure

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Critical temperature of a substance is the temperature above which vapour of the substance cannot be liquified, No matter how much pressure is applied, critical temperature of Hydrogen is –239.95°C, So, it is impossible to liquify hydrogen at room temperature.

818. In steam tables, the entropy is shown as zero for

(a) Saturated vapour at atmospheric pressure

(b) Saturated liquid at atmospheric pressure

(c) Saturated vapour at 0°C

(d) Saturated liquid at 0°C

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : In steam tables, the entropy is shown as zero for, a saturated liquid at 0°C ≈ 0.01°C.

819. All gases have compressibility factor of nearly unity

(a) At very high temperatures and pressures

(b) At very low pressures approaching zero

(c) Near the critical point

(d) Above the critical point

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Compressibility factor [z] also known as compression factor, is a correction factor which describes the deviation of a real gas from ideal gas,

For ideal gas 'z' = 1.

$$z = \frac{PV}{RT}$$

From above for, z is nearly unity at very low pressure approaching zero.

820. Most of the gases exhibit drop in temperature upon expansion. However, this may not be true in case of

(a) carbon dioxide (b) oxygen

(c) hydrogen (d) helium

BPSC AE Mains 2017 Paper - V

Ans. (c) : In thermodynamics, the- Joule-Thomson (also known as the Joule-Kelvin effect or Kelvin-Joule effect) describes the temperature change of real gas or liquid when it is forced through a value or porous plug while keeping it insulated so that no heat is exchanged with the environment. This procedure is called a throttling process or Joule-Thomson process. At room temperature, all gases except hydrogen, helium, and neon gives heating effect upon expansion by the Joule-Thomson process when being throttled through an orifice, there are three gases experience the same effect but only at lower temperature.

Note-Option (c&d) both are correct. But official answer is (c).

821. If the dryness fraction of a sample by throttling calorimeter is 0.8 and that by separating calorimeter is also 0.8, then the actual dryness fraction of sample will be taken as

(a) 0.8 (b) $\sqrt{0.8}$

(c) 0.64 (d) 0.5

BPSC AE Mains 2017 Paper - V

UPPSC AE 12.04.2016 Paper-II

Ans. (c) : x_1 = Dryness fraction of steam considering separating calorimeter.

x_2 = Dryness fraction of steam entering the throttling calorimeter.

$x = x_1x_2$

Actual dryness fraction of steam in the sample

$x = 0.8 \times 0.8$

$= 0.64$

822. The value of dryness fraction at critical point for water-steam phase transformation may be

(a) 0 (b) 1

(c) either 0 or 1 (d) all of these

TNPSC AE 2017

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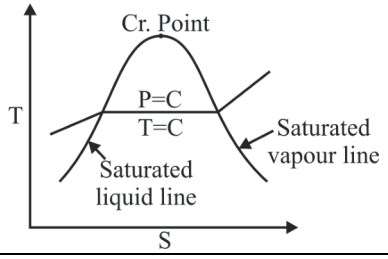
Ans. (d) : The value of dryness fraction at critical point for water-steam phase transformation is undefined.

823. A Saturation state of pure substance (water) is a:

(a) State from which a change of phase may occur with a change in pressure or temperature.
(b) State from which a change of phase may occur with a change in pressure and temperature.
(c) State from which a change of phase may occur with a change in pressure and volume.
(d) State from which a change of phase may occur without a change in pressure and temperature.
(e) State from which a change of phase may occur without a change in pressure or temperature.

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : Saturation State–A saturation state of pure substance (water) is a state from which a change of phase may occur without a change in pressure and temperature.



824. Universal gas constant is defined as equal to product of the molecular weight of the gas and

(a) specific heat at constant pressure
(b) specific heat at constant volume
(c) ratio of two specific heat
(d) gas constant

WBPSC AE, 2017

Ans. (d) : The ideal gas constant is given by
$PV = nRT$
$\rightarrow$ The universal gas constant or molar constant (R_u) of gas is the product of the specific gas constant (R) and the molecular mass of the gas (M)
$\because R_u = MR$ $[R_u = 8.314$ kJ/kg- mol.K$]$

825. Consider the following statements for a throttling process.

1. It is an adiabatic process
2. There is no work transfer in the process
3. Entropy increases in throttling process

Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : Throttling process, is an adiabatic process and have no work transfer in the process and entropy increases in this due to heat transfer.

826. Volume of 1 kg of dry steam is known as :

(a) Total volume (b) Saturated volume
(c) Specific volume (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (c)

$$\text{Specific volume}(v) = \frac{\text{Volume of steam}}{\text{Mass of steam}} (m^3/kg)$$

827. The process of superheating follows :

(a) Reynold's law
(b) Stefan-Boltzmann's law
(c) Charles's law
(d) Ernst law

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : The process of superheating follow Charle's law.
According to Charles's law (P = Constant)

$$\frac{V}{T} = \text{constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

828. Sublimation is the process of

(a) Changing from gas state to solid state
(b) Changing from solid state to gas state
(c) Changing from liquid to vapour state
(d) Existence of solids, liquids and gases simultaneously

APPSC AEE Mains 2016 Paper-III

Ans. (b) : Sublimation– when a solid turns into a gas without first becoming liquid, that's sublimation.

829. Ideal gas equation may be written as [where, P = absolute pressure, v = specific volume, R = characteristic gas constant, T = absolute temperature, m = mass of gas, n = number of moles of gas, Z = compressibility factor]:

(a) Pv = nRT (b) Pv = ZRT
(c) Pv = mRT (d) Pv = RT

UPRVUNL AE 07.10.2016

Ans. (d) : Compressibility factor $Z = \frac{Pv}{RT}$
For ideal gas Z = 1
Then, $\boxed{Pv = RT}$

830. Generalized compressibility chart is drawn between:

(a) Reduced pressure (P_r) on y-axis and reduced temperature (T_r) on x-axis
(b) Compressibility factor (Z) on y-axis and reduced temperature (T_r) on x-axis
(c) Compressibility factor (Z) on y-axis and reduced pressure (P_r) on x-axis
(d) Compressibility factor (Z) on x-axis and reduced pressure (P_r) on y-axis

UPRVUNL AE 07.10.2016

Ans. (c) : Generalized compressibility chart is drawn between compressibility factor (Z) on y-axis and reduced pressure (P_r) on x-axis.

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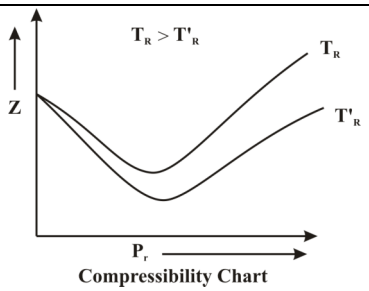


Compressibility Chart

$$Z = \frac{PV_{actual}}{RT}$$

$$Z = \frac{V_{actual}}{V_{ideal}}$$

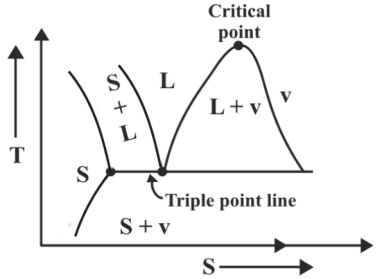
Z = 1 - for ideal gas
Z > 1 - less compressible
Z < 1 - more compressible

831. Which of the following is true statement for phase diagram of pure substance?

(a) Sublimation curve, fusion curve and vaporization curve meets at critical point
(b) Extreme points of vaporization curve are triple point and critical point
(c) Extreme points of fusion curve are triple point and critical point
(d) Fusion curve for water has positive slope

UPRVUNL AE 07.10.2016

Ans. (b) : In phase diagram of pure substance, extreme points of vaporization curve are triple point and critical points.



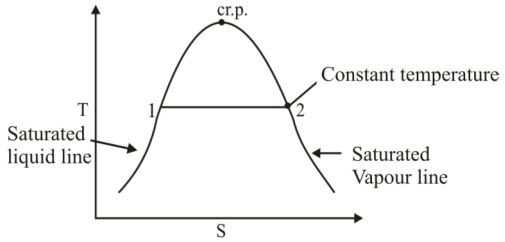
- Fusion curve for water has negative slope.

832. If the degree of saturation of air is zero, the air is said to be

(a) superheated air (b) unsaturated air
(c) saturated air (d) atmospheric air

APPSC AEE Mains 2016 Paper-III

Ans. (c) : If the degree of saturation of air is zero, the air is said to be saturated air.



We know that degree of saturation (DOS) = $(T_1 - T_2)$
When, $T_1 = T_2$
then, DOS = 0
It means air is said to be saturated air.

833. What do you mean by antifreeze chemicals?

(a) Substance that will not affect the freezing point of water
(b) Substance that evaporates the water
(c) Substance that lowers the freezing point of water
(d) Substance that increases freezing point of water

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : An antifreeze is an additive which lowers the freezing point of water based liquid. An antifreeze mixture is used to achieve freezing point depression for cold environments common antifreeze increase the boiling point of the liquid allowing higher coolant temperature.

834. Consider the liquid vapour saturation curve of a pure substance on the P-T diagram. The magnitude to the slope of the tangent line to this curve at temperature T (in Kelvin) is

(a) Proportion of the enthalpy of vaporization at that temperature
(b) Proportional to the square of temperature T
(c) Proportional to the temperature T
(d) Proportional to the volume change at the temperature

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Liquid on vapour saturation of a pure substance on the P-T diagram. The magnitude of the slope of the tangent line to this curve at temperature T (in Kelvin) is proportional of the enthalpy of vaporization at the temperature.

835. The mass of vapour in the system divided by the total system mass (the mass of vapour plus the mass of liquid) is defined as:

(a) Quantity (b) Quality
(c) Mass mix ratio (d) Vapour mix ratio

HPPSC AE (PWD) 10.09.2016

Ans. (b) : The mass of vapour in the system divided by the system mass is known as quality. It is also known as dryness fraction of steam.
Quality (x) can be calculated by

$$x = \frac{m_{vapour}}{m_{total}} = \frac{m_{vapour}}{m_{vapour} + m_{liquid}}$$

836. The correct sequence of the decreasing order of the value of characteristic gas constants given is

(a) hydrogen, nitrogen, air, carbon dioxide
(b) carbon dioxide, hydrogen, nitrogen, air
(c) air, nitrogen, carbon dioxide, hydrogen
(d) nitrogen, air, hydrogen, carbon dioxide
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (a) : We know that –
Gas characteristic constant

$$(R) = \frac{\bar{R}}{M}$$

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M = molecular weight

For Hydrogen, M = 1

$(R)_{H_2} = \bar{R}$

For Nitrogen, m = 14

$$(R)_{N_2} = \frac{\bar{R}}{14}$$

For air, M = 28

$$(R)_{air} = \frac{\bar{R}}{28}$$

For carbon dioxide , M = 44

$$(R)_{CO_2} = \frac{\bar{R}}{44}$$

837. In a coupled separating and throttling calorimeter. 0.89 is the dryness fraction of steam in the separating unit and 0.93 is the dryness fraction in the throtting unit. What is the approximate value of the dryness fraction of the steam sample?

(a) 0.93 (b) 0.89

(c) 0.91 (d) 0.83

UPRVUNL AE 21.08.2016

Ans. : (d) Separating calorimeter dryness fraction $(x_1) = 0.89$

throttling calorimeter dryness fraction $(x_2) = 0.93$

Approximate value of dryness fraction $(x) = x_1 \times x_2$

$x = 0.89 \times 0.93$

$x = 0.8277$

$x = 0.83$

838. The temperature above which vapour of the substance CANNOT be liquified, no matter how much pressure is applied is called

(a) Absolute Temperature

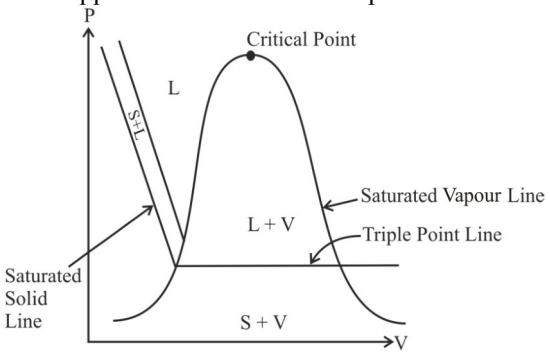
(b) Critical Temperature

(c) Equilibrium Temperature

(d) Osmotic Temperature

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : The temperature above which vapour of substance can not be liquified no matter how much pressure applied is called critical temperature.



$P_{critical\ point} = 221.1$ bar

$T_{critical\ point} = 374^0C$

839. If one kg of steam sample contains 0.8kg of dry steam, then what will be it's dryness fraction?

(a) 0.6 (b) 1

(c) 0.8 (d) 0.2

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : Dryness fraction $x = \frac{m_g}{m_g + m_\ell}$

Where m_g = mass of actual dry stream

m_ℓ = mass of water in steam

Given,

$m_g = 0.8$ kg

$m_\ell = 1 - 0.8 = 0.2$

then $x = \frac{m_g}{m_g + m_\ell} = \frac{0.8}{0.8 + 0.2}$

$x = 0.8$

840. The amount of heat absorbed when 1 kg of saturated liquid water boils at pressure P is Q_1. The amount of heat released when 1 kg of saturated water vapour condenses at same pressure P is Q_2, then

(a) $Q_1 = Q_2$ for all values of P

(b) $Q_1 > Q_2$ for all values of P

(c) $Q_1 < Q_2$ for all values of P

(d) $Q_1 > Q_2$ or $Q_1 < Q_2$ depends upon value of Pressure P.

GPSC Asstt. Prof. 28.08.2016

Ans. (a) : For a same saturated pressure, the heat released and absorbed is same, from one state/phase to another.

$Q_1 \rightarrow$ heat absorbed

$Q_2 \rightarrow$ heat released

$Q_1 = Q_2$ at same pressure

841. In the Unary Phase diagram of water, the point at which ice, water and water vapour coexist in equilibrium, the degrees of freedom are :

(a) 0 (b) 1

(c) 2 (d) 3

KPSC ADF 2015

Ans. (a) : $P + F = C + 2$

$3 + F = 1 + 2$

$F = 0$

The point (Triple point) at which ice, water and water vapour coexist in equilibrium, the degree of freedom is zero.

842. Select the correct statement as per Charles's law.

(a) PV = Constant, if T is kept constant.

(b) $\frac{V}{T}$ = constant, if P is kept constant.

(c) $\frac{P}{V}$ = constant, if T is kept constant.

(d) $\frac{T}{P}$ = constant, if T is kept constant.

TSPSC AEE 2015

Ans. (b) : Charles's law:- The volume of a given mass of a perfect Gas varies directly as its absolute temperature, when the absolute pressure remains constant

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$V \propto T$

$\frac{V}{T} = \text{costant}$

$\boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2} = \frac{V_3}{T_3} = \text{constant}}$

843. Which of the following gasses will have the maximum value of gas constant R

(a) nitrogen (b) carbon dioxide
(c) sulpher dioxide (d) oxygen

TSPSC AEE 2015

Ans. (a) :

Element	Molecular weight (M)
Nitrogen (N_2)	28
Carbon Dioxide (CO_2)	44
Sulpher Dioxide (SO_2)	64
Oxygen (O_2)	32

$$\text{gas constant } R = \frac{\bar{R}}{M}$$

$$\boxed{R \propto \frac{1}{M}}$$

So, Nitrogen gas will have the maximum value of gas constant because its molecular weight is 28.

844. The process of converting ice directly to steam is called?

(a) Equivalent evaporation
(b) Sublimation
(c) Condensation
(d) Critical evaporation

TSPSC Managers, 2015

Ans. (b) : The process of converting ice directly to steam is called sublimation process.

845. To convert volumetric analysis to gravimetric analysis, the relative volume of each constituent of the flue gases is-

(a) multiplied by its molecular weight
(b) divided by its molecular weight
(c) multiplied by its density
(d) multiplied by its specific weight

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a) To convert volumetric analysis to gravimetric analysis, the relative volume of each constituent of the flue gases is multiplied by its molecular weight

846. The dryness fraction of steam within a turbine is not allowed to fall below

(a) 0.95 (b) 0.50
(c) 0.99 (d) 0.90

GPSC AE 26.07.2015

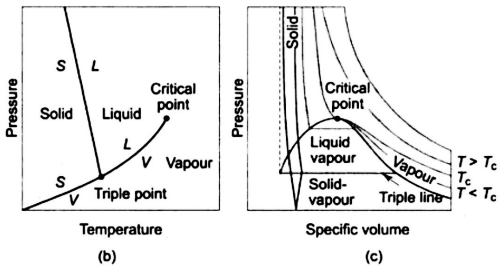
Ans. (d) :

847. A pure substance cannot remain in liquid phase if it is below the

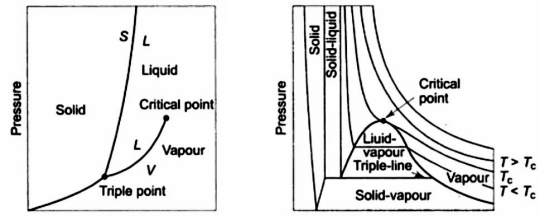
(a) critical state
(b) triple point
(c) saturated liquid line
(d) saturated vapour line

Assam PSC CCE Pre 2015

Ans. (b) : According to given diagram graph for a substance that expands on freezing and substance that contracts on freezing. A pure substance cannot remain in liquid phase if it is below the triple point.



p–v–T surface and projections for a substance that expands on freezing
(a) Three-dimensional view (b) Phase diagram (c) p-v diagram



p–v–T surface and projections for a substance that contracts on freezing
(a) Three-dimensional view (b) Phase diagram (c) p-v diagram

848. During addition of latent heat, for a wet steam, there is change in :

(a) temperature only
(b) temperature & pressure
(c) phase only
(d) temperature & phase

KPSC ADF 2015

Ans. (c) : During addition of latent heat for a wet steam, there is change in phase only and at constant temperature and pressure.

849. Which of the following expands in volume upon freezing?

(a) Mercury (b) Alcohol
(c) Water (d) Chloroform

Assam PSC CCE Pre 2015

Ans. (c) : Water expands in volume upon freezing.

850. Sensible heat is the heat required to -

(a) change liquid into vapour
(b) increase the temperature of a liquid or vapour
(c) none of three mentioned here.
(d) change vapour into liquid

GWSSB AAE, 27.12.2015

Ans. (b) : Sensible heat refers to the amount of energy needed to increase or decrease the temperature of some substances, independent of phase change.

851. For steam

(a) The critical temperature is 221.2°C and critical pressure is 374.15 bar
(b) The critical temperature is 374.15°C and critical pressure is 221.2 bar

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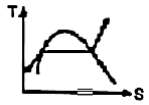
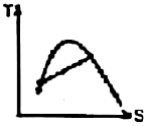
(c) The critical temperature is 221.2°C and critical pressure is 221.2 bar
(d) The critical temperature is 374.15°C and critical pressure is 374.15 bar

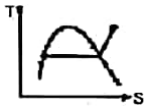
Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : At critical point, Critical temperature (T_{cr}) = 374.15°C, Specific volume (v_{cr}) = 0.003155 m^3/kg
Critical pressure (P_{cr}) = 221.2 bar.

852. The conversion of water from 40°C to steam at 200°C at a pressure of bar is best represented as

(a) 

(b) 

(c) 

(d)

Nagaland PSC CTSE 2015 Paper-II
ESE 1994

Ans. (a) :

853. Dryness fraction is the ratio of :
(a) mass of dry steam to the mass of water vapour in suspension
(b) mass of water vapour in suspension to the mass of water vapour and mass of dry steam
(c) mass of dry steam to the mass of dry steam and mass of water vapour in suspension
(d) mass of water vapour in suspension to the mass of dry steam

Kerala PSC AE 06.08.2015

Ans. (c) : Dryness fraction- The term dryness fraction is related with the cut steam. It is defined as the ratio of 'mass of dry steam to the sum of mass of dry steam and mass of water vapour is suspension.

$$x = \frac{m_s}{m_s + m_w}$$

Where, x = dryness fraction
m_s = mass of steam
m_w = mass of water

854. Which one of the following statements is correct?
(a) Pressure and temperature are independent during phase change
(b) An isothermal line is also a constant pressure line in the wet vapour region
(c) Entropy decreases during evaporation
(d) The term dryness fraction is used to specify the fraction by mass of liquid in a mixture of liquid and vapour

Nagaland PSC CTSE 2015 Paper-II
ESE 1994

Ans. (b) :

855. A system comprising of a single phase, is known as
(a) Open system
(b) Closed system
(c) Homogeneous system
(d) Heterogeneous system

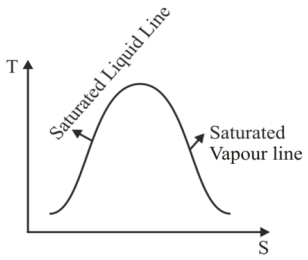
MPPSC State Forest Service Exam, 2014

Ans. (c) : A system comprising a single phase is called a homogeneous system.

856. A state of a pure substance at which a phase change ends or begins is called
(a) Critical state (b) Saturated state
(c) Triple point (d) Gibb's state

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : The saturated state is a state where phase change begins or ends.



- A saturated liquid line where any further addition of energy will cause a small portion of the liquid to convert to vapour.
- Similarly a saturated vapour line, further heating will result superheated vapour state.

857. The heating of a gas at constant pressure is governed by
(a) Boyle's law (b) Charles's law
(c) Gay-Lussac law (d) Joule's law

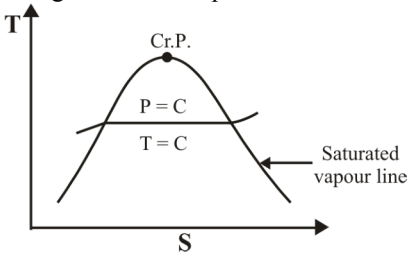
TNPSC AE 2014

Ans. (b) : Charles's Law– The heating of a gas at constant pressure.
Boyle's Law– The heating of a gas at constant temperature.
Gay-Lussac Law– The heating of a gas at constant volume.

858. The heating of wet steam at constant temperature till it becomes dry saturated is similar to that of heating at a
(a) constant volume (b) constant pressure
(c) constant entropy (d) constant enthalpy

TNPSC AE 2014

Ans. (b) : The heating of wet steam at constant temperature till it becomes dry saturated is similar to that of heating at a constant pressure.



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859. The kinetic energy lost in friction is transformed into heat which tends to
(a) cool or condense the steam
(b) dry or superheat the steam
(c) increase the pressure of the steam
(d) reduce the dryness fraction
(e) decrease the specific volume of steam

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : The kinetic energy lost in friction is transformed into heat which tends to dry or superheat the steam.

860. Which of the following can be regarded as gas so that gas law could be applicable within the commonly encountered temperature limits ?
(a) O_2, N_2, steam, CO_2
(b) O_2, N_2, water vapour
(c) SO_2, NH_3, CO_2, moisture
(d) O_2, N_2, H_2, air

J&K PSC Civil Services Pre, 2013

Ans. (d) :

861. Water boils when its vapour pressure :
(a) equals that of the surroundings
(b) equals 760 mm of mercury
(c) equals to atmospheric pressure
(d) equals the pressure of water in the container

J&K PSC Civil Services Pre, 2013

Ans. (c) : Any liquid is in an open container and exposed to normal atmospheric pressure, the liquid boils when its saturated vapour pressure becomes equal to 1 atmospheric pressure.

862. When the fuel is burned and water is release in the liquid phase, the heating value of fuel is called
(a) higher heating value
(b) lower heating value
(c) enthalpy of formation
(d) None the above

JPSC AE 2013 Paper-V

Ans. (a) : higher heating value

863. The ratio of $\frac{PV}{RT}$ is
(a) Equations of state
(b) Compressibility factor
(c) Reduced properties
(d) Critical compressibility factor

TNPSC AE 2013

Ans. (b) : Compressibility factor (Z)– Compressibility factor is also known as the compression factor or the gas deviation factor, is a correction factor which describe the deviation of a real gas from ideal gas behaviour. It is simply defined as the ratio of the molar volume of a gas to the molar volume of an ideal gas at the same temperature and pressure.

$$Z = \frac{PV}{RT}$$

For ideal gas $\rightarrow$ $Z = 1$

864. A certain gas has C_p value of 1968 J/kgK and C_v value of 1507 J/kgK. The value of R is
(a) 0.461 KJ/kgK (b) 1307 J/kgK
(c) 1 (d) 461 KJ/kgK

TNPSC AE 2013

Ans. (a) : We know that-
Mayer's formula is given as-
$C_p - C_V = R$
$1968 - 1507 = R$
$R = 461$ J/kgK
$R = 0.461$ kJ/kgK

865. RMS velocity of hydrogen gas at NTP is:
(a) 526 m/s (b) 932 m/s
(c) 1356 m/s (d) 1839 m/s

SJVN ET 2013

Ans. (d) : RMS velocity of hydrogen gas at NTP is 1839 m/s.

866. Triple point and critical point pressure of carbon dioxide are:-
(a) 4.58mm Hg and 221.2 bar respectively
(b) 5.18 bar and 221.2 bar respectively
(c) 1 bar and 50 bar respectively
(d) 5.18 bar and 73.8 bar respectively

UKPSC AE-2013, Paper-II

Ans. (d) : Triple point and critical point pressure of carbon dioxide are 5.18 bar and 73.8 bar respectively.

867. Which one of the following is weaker than hydrogen bonds?
(a) Ionic bond (b) Vander Waals bond
(c) Covalent bond (d) Metallic bond

UKPSC AE-2013, Paper-I

Ans. (b) : Vander waals bond is weaker than hydrogen bonds.

868. Which water becomes ice very fast when it is kept inside the freezer ?
(a) Warm water (b) Cold water
(c) Water with salt (d) Water with ethanol

TNPSC AE (Industries) 09.06.2013

Ans. (a) : Warm water becomes ice very fast when it is kept inside the freezer and it is called Mpemba effect.

869. Mollier chart is drawn between
(a) temperature and pressure
(b) temperature and entropy
(c) enthalpy and entropy
(d) pressure and enthalpy

TRB Poly. Lect., 2012

Ans. (c) : Mollier chart is drawn between enthalpy (h) & entropy (s)

870. Under thermal equilibrium, flow of steam is
(a) isentropic (b) adiabatic
(c) hyperbolic (d) polytropic

BPSC AE 2012 Paper - V

Ans. : (a) : Under thermal equilibrium, flow of steam is isentropic.

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871. Critical pressure of a liquid is the pressure
(a) above which liquid will remain liquid
(b) above which liquid will become gas
(c) above which liquid becomes vapour
(d) above which liquid becomes solid

BPSC AE 2012 Paper - V

Ans : (c) : Critical pressure of a liquid is the pressure above which liquid becomes vapour.

872. A perfect gas obeys
(a) Boyle's law only
(b) Charles law only
(c) Boyle's Charles' laws both
(d) none of these

TNPSC ACF 2012

Ans. (c) : A perfect gas obeys Boyle's and Charles's law Boyle's Law- Robert Boyle states that at constant temperature, the pressure P and gas varies inverse with its volume or $\boxed{PV = K}$.

Charles Law- Charles statics that at constant pressure, the volume of gas is directly proportional to its absolute temperature T.

873. Gases have
(a) only one value of specific heat
(b) two values of specific heat
(c) three values of specific heat
(d) no values of specific heat

TNPSC ACF 2012

Ans. (b) : Gases have two value of specific heat C_P and C_V. C_P is always greater than C_V.

$\boxed{C_P > C_V}$

874. Steam has a specific saturation temperature at
(a) each volume (b) each pressure
(c) each enthalpy (d) each entropy

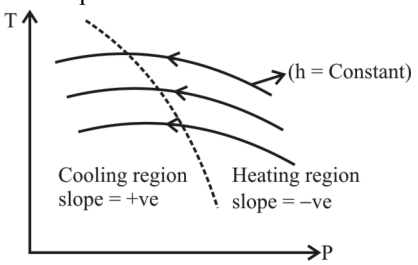
TRB Poly. Lect., 2012

Ans. (b) : Steam has a specific saturation temperature at each atmospheric pressure. Saturation state exist when the air is holding the maximum amount of water vapour possible at the existing temperature and pressure. When DPT and air temperature are equal, the air is said to be saturated.

875. When wet steam is throttled to a low pressure, its temperature
(a) Increases (b) Does not change
(c) Gets halved (d) Decreases

APGENCO AE 2012

Ans. (d) : When wet steam is throttled valve to a low pressure, its temperature decreases.



876. Which is a single phase system ?
(a) Mixture of water and alcohol
(b) Mixture of oil and water
(c) Liquid water, ice and water vapour
(d) Water and ice

OPSC Civil Services Pre. 2011

Ans. (a) : Single phase system is mixture of water and alcohol.

877. A fluid behave as an ideal gas provided it is at:
(a) High Temperature and Pressure
(b) High Temperature and Moderate Pressure
(c) Low Temperature and Pressure
(d) High Temperature and Low Pressure

OPSC Civil Services Pre. 2011

Ans. (d) : A real gases obeys perfect gas law at high temperature and low pressure.
- Inter molecular attraction between molecules are negligible.
- Volume occupied by molecules as compared to total volume is negligible.

878. Match the following :

1. Empirical temperature	**a. Path function**
2. Polarization	**b. Permeable to heat transfer**
3. Heat transfer	**c. Mechanical equivalent of heat**
4. Joule's experiment	**d. Isotherm**
5. Diathermic	**e. Dipole moment**

(a) 1–d, 2–a, 3–e, 4–c, 5–b
(b) 1–d, 2–e, 3–a, 4–c, 5–b
(c) 1–d, 2–b, 3–a, 4–c, 5–e
(d) 1–d, 2–c, 3–b, 4–a, 5–e

OPSC Civil Services Pre. 2011

Ans. (b) :

1. Empirical temperature	Isotherm
2. Polarization	Dipole moment
3. Heat transfer	Path function
4. Joule's experiment	Mechanical equivalent of heat
5. Diathermic	Permeable to heat transfer

879. For real gases $C_P = C_V$ at :
(a) Critical temperature
(b) Triple point
(c) Absolute zero temperature
(d) All temperature

J&K PSC Civil Services Pre, 2010

Ans. (c) : The heat capacity depends on the nature, size and the composition or the substance in the system.
The equation for heat capacity is as follows–

$$Q = nC\Delta T$$

Where Q is the heat supplied
n is the no. of moles
C is the molar heat capacity
ΔT is the change in temperature
We know that C_P is the heat capacity at constant pressure. C_P is the amount of heat absorbed or released by unit mass of a substance with change in temperature at constant pressure. The change in temperature causes a change in the enthalpy of the system.

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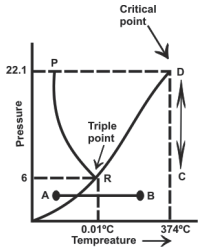
We know that C_V is the heat capacity at constant volume. C_V is the amount of heat absorbed or released by unit mass of a substance at constant volume.
The value of C_P is always greater than C_V.
But at absolute zero temperature
$C_P = C_V$

880. The critical point and triple point data for water are, T_c = 374ºC, P_c = 22.1 MPa, T_T = 0.01ºC, P_T = 0.6 kPa indicate the phase change that will occur in following cases,

(i) Ice at 0.5 kPa is heated isobarically

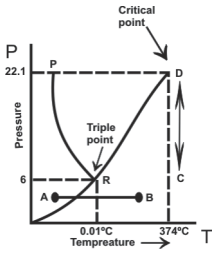
(ii) Water vapour at 400ºC is compressed isothermally



(a) (i) along AB, (ii) along CD
(b) (i) along CD, (ii) along AB
(c) (i) along OD, (ii) along PR
(d) (i) along RD, (ii) along OR

ISRO Scientist/Engineer 2009

Ans. (a) : T_c = 374ºC, P_c = 22.1 kPa, T_T = 0.01ºC, P_T = 0.6 kPa



(I) Ice at 0.5 kPa is heated Isobarically i.e. at constant pressure with 0.5 kPa.
Heat Isobarically represented from P-T diagram along (AB)
(II) Water vapour at 400ºC i.e. water vapour region got after critical point temperature alone 370ºC.
Its represented from P-T diagram (along-CD).

881. The magnitude of which one of the following quantity does not indicate the departure of a real gas from the ideal gas behavior:

(a) Compressibility factor
(b) Joule-Kelvin coefficient
(c) Fugacity
(d) Entropy

UPSC JWM Adv. No-16/2009

Ans. (c) : The magnitude of the Fugacity does not indicate the departure of a real gas from the ideal gas behavior.

882. In order to determine the quality of wet steam by a separating and throttling calorimeter, the steam should be throttled such that the final state is :

(a) Saturated vapour only
(b) Superheated vapour only
(c) At a pressure higher than the original pressure
(d) A mixture of saturated liquid and saturated vapour

UPSC JWM Adv. No-16/2009, ESE 2015

Ans. (b) In order to determine the quality of wet steam by a separating and throttling calorimeter, the steam should be throttled such that the final state is Superheated vapour only.

883. A pressure-cooker contains water at 100ºC with liquid volume being one-tenth of vapour volume. At 100ºC, the specific volume for saturated liquid and saturated vapour are 0.001 m^3/kg and 1.6 m^3/kg, respectively. The dryness fraction for the mixture is

(a) $\frac{1}{160}$ (b) $\frac{1}{161}$
(c) $\frac{160}{161}$ (d) 1

DRDO Scientists 2009

Ans. (b) : $v_f + v_g = v, \quad v_f = \frac{v_g}{10}$

$\therefore \quad v_g = \frac{10v}{11}, \; v_f = \frac{v}{11}$

$x = \frac{m_g}{m_l + m_g}, \; m_g = \frac{10}{11} \times \frac{v}{1.6\text{kg}}$

$m_l = \frac{1}{11} \times \frac{v}{0.001}\text{kg}$

$x = \frac{\frac{10}{1.6}}{\frac{10}{1.6} + 1000} = \frac{100}{16100} = \frac{1}{161}$

884. Which one of the following remains constant for a system undergoing phase change like melting or vaporization?

(a) Enthalpy (b) Entropy
(c) Volume (d) Gibbs function

UPSC JWM Adv. No-16/2009

Ans. (d) : Since the Gibbs function depends naturally on the Thermodynamic variables p and T and since common phase transitions occurs at a constant p and T, thus, the molar/specific Gibbs function tends to be a constant during the phase change.

885. In the Vander Waal's gas equation $\left(p + \frac{a}{V^2}\right)(v - b) = RT$ the constant 'a' is introduced to compensate for

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(a) Reduction in specific volume
(b) Inter-molecular forces
(c) Reduction in specific heat
(d) All of the above

ISRO Scientist/Engineer 2009

Ans. (b) : Vender Waal's gas equation

$$\left(P+\frac{a}{V^2}\right)(V-b)=RT$$

where a = Account for intermolecular force
b = Volume occupied by the gas

886. For water boiling at constant pressure of 1 bar, the specific heat is
(a) zero
(b) very small but positive
(c) negative
(d) infinitely large

DRDO Scientists 2008

Ans. (d) : For water boiling at constant pressure of 1 bar, the specific heat is infinitely large during boiling, pressure and temperature will not change.

$$C_p=\frac{dQ}{dT}$$

$$dT=0 \Rightarrow C_p=\infty$$

887. For pure water the triple point temperature is
(a) 273 K (b) 273.16 K
(c) 273.1 K (d) 273.2 K

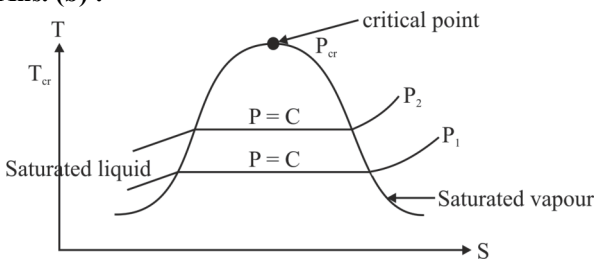
WBPSC AE, 2007

Ans. (b) :

888. Which of the given statements is true?
(a) At the critical point, all the three phases of water coexist in equilibrium.
(b) At the critical point, saturated liquid and saturated vapour phases are identical
(c) At the triple point, all the phases of water coexist in non-equilibrium
(d) At the triple point, saturated liquid and saturated vapour phases are identical.

BHEL ET 2019

Ans. (b) :



- Critical point is a point where saturated liquid and saturated vapour curve meet.
- At critical point a liquid directly flash into vapour.
- At critical point latent heat of vapourization is zero.

889. The equation $\left\{P+\frac{a}{V^2}\right\}(V-b)=RT$ is known as
(a) perfect gas equation
(b) Maxwell's equation
(c) kinetic theory of gases equation
(d) Vander walls equation

BPSC Poly. Lect. 2016
VIZAG STEEL MT 2015
J & K PSC Civil Services Pre, 2013
UKPSC AE 2012 Paper–II
TNPSC ACF 2012
ISRO Scientist/Engineer 2007

Ans : (d) Vander walls equation

$$=\left\{P+\frac{a}{V^2}\right\}(V-b)=RT$$

The ideal gas law treats that the molecules of a gas as point particles with perfectly elastic collisions. This works well for dilute gases in many experimental circumstances. But gas molecules are not point masses, and there are circumstances where the properties of the molecules have an experimentally measurable effect. A modification of the ideal gas law as proposed by Johannes D. van der Waals in 1873 to take into account molecular size and molecular interaction forces. It is usually referred to as the Vander Walls equation of state.

890. The volume of a perfect gas varies directly as its absolute temperature when the absolute pressure remains constant. This law is called
(a) Charles Law (b) Joule's Law
(c) Gay Lussac Law (d) Boyle's Law

VIZAG Steel MT 24.01.2021, Shift-I
TNPSC AE (Industries) 09.06.2013

Ans. (a) : Charles's law states that the volume of a gas is directly proportional to its absolute temperature assuming the quantity of gas and pressure remains constant.

$$\boxed{\frac{V_1}{T_1}=\frac{V_2}{T_2}}$$

891. Isothermal compressibility of an ideal gas in
(a) $\frac{nR}{VP}$ (b) $\frac{nRT}{VP^2}$
(c) $\frac{nR}{VP^2}$ (d) $\frac{nRT}{VP}$

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Isothermal compressibility

$$=-\frac{1}{V}\frac{\partial V}{\partial P}\Big|_{T=C} \qquad =-\frac{1}{V}\frac{\partial}{\partial P}\left(\frac{nRT}{P}\right)$$

$$=-\frac{1}{V}\left(-\frac{nRT}{P^2}\right) \qquad =\frac{nRT}{VP^2}$$

892. For a perfect gas, according to Boyle's law
(a) PV = constant, if T is kept constant
(b) V/T= constant, if P is kept constant
(c) P/T = constant, if V is kept constant
(d) T/P = constant, if V is kept constant

GMB AAE 25.06.2017

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Ans. (a) : Boyles law – At constant temperature pressure is inversely proportional to its volume.

Mathematically– $P \propto \frac{1}{V}$

$P = \frac{k}{V}$

$PV = k$ where k is a constant.

$\boxed{PV = \text{constant}}$

7. Thermodynamic Relations

893. Joule-Thomson coefficient is defined as :

(a) $\left(\frac{\partial T}{\partial P}\right)_h$ (b) $\left(\frac{\partial H}{\partial P}\right)_T$

(c) $\left(\frac{\partial H}{\partial T}\right)_P$ (d) $\left(\frac{\partial P}{\partial T}\right)_h$

Oil India Limited Sr. Engineer (Drilling) 30.11.2019
SJVN ET 2019
CGPSC Poly. Lect. 22.05.2016
BPSC AE Mains 2017 Paper - V
MPPSC AE 2016
Vizag Steel MT (Re-Exam) 24.11.2013
ESE 2012, 1995

Ans : (a) Joule Thomson co-efficient:-
The change in temperature with drop in pressure at constant enthalpy is termed as Joule-Thomson coefficient (μ).

$$\mu = \left(\frac{\partial T}{\partial P}\right)_h$$

It varies with both the temperature and pressure of the gas.
* Joule Thomson coefficient for an ideal gas is zero.

894. Which relationship defines Gibbs free energy G:

(a) G = H + TS (b) G = H - TS
(c) G = U + TS (d) F = U + TS

Kerala PSC AM 09.08.2021
Kerala PSC AE (Irrigation) 05.04.2017
APPSC AEE Mains 2016 Paper-III
SJVN ET 2013
OPSC Civil Services Pre 2006
ESE 1999

Ans. (b) : Gibbs free energy–In thermodynamics, the Gibb's free energy is a thermodynamic potential that can be used to calculate the maximum reversible work that may be performed by a thermodynamic system of a constant temperature and pressure Gibbs free energy.
G = H – TS.

895. Joule-Thomson coefficient for an ideal gas is

(a) higher than zero (b) less than zero
(c) zero (d) 1

GPSC Asstt. Director of Transport 05.03.2017
ISRO Scientist/Engineer 03.07.2016
CGPSC AE 16.10.2016, APPSC AEE 2012
DRDO Scientists 2009, WBPSC AE, 2007

Ans. (c) : Joule-Thomson coefficient $(\mu) = \left(\frac{\partial T}{\partial P}\right)_h$

h = Constant

For ideal gas $\boxed{\mu = 0}$

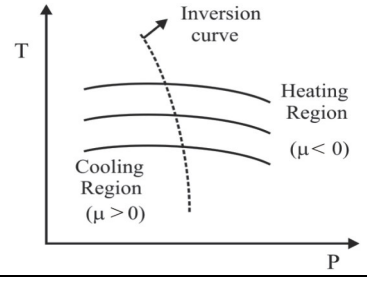
Note : Cooling region, μ = (+ve)
Heating region, μ = (–ve)

896. A gas having a negative Joule-Thomson effect ($\mu < 0$), when throttled will :

(a) Become cooler
(b) Become warmer
(c) Remain in the same temperature
(d) Become Cooler or warmer depending upon the type of gas

Gujarat PSC AE 2019
GPSC ARTO Pre 30.12.2018
TANGEDCO AE 2015
ISRO Scientist/Engineer 11.10.2015
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012

Ans. (b) : A gas having negative Joule-Thomson effect ($\mu < 0$)
When throttled will become warmer.



897. The Helmholtz function F is defined as

(a) F = H + TS (b) F = H – TS
(c) F = U + TS (d) F = U – TS

HPCL Engineer 12.08.2021 Shift-II
OPSC AEE 2019 Paper - II
Nagaland PSC CTSE 2015 Paper-II
Kerala PSC Poly. Lect. 09.01.2014
BPSC AE 2012 Paper - V, ESE 2007

Ans. (d) : Helmholtz function = F
F = U – TS
The difference in Helmholtz function gives maximum work in closed system.
Gibbs function (G)
G = H – TS
The difference in Gibbs function gives maximum work in open system.

898. A substance whose Joule- Thomson coefficient is negative, is throttled to a lower pressure. During this process

(a) the entropy of the substance will decrease
(b) the entropy of the substance will remain constant
(c) the temperature of the substance will decrease
(d) the temperature of the substance will increase

UPPSC AE 13.12.2020, Paper-2

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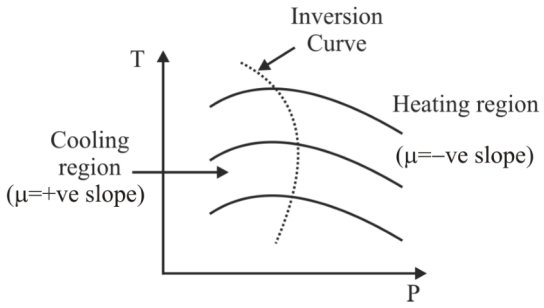
GPSC ARTO Pre-30.12.2018
VIZAG Steel MT 18.08.2013
VIZAG Steel MT 10.06.2012
GATE 2001

Ans. (d) Joule- Thomson coefficient, $= \left(\frac{\partial T}{\partial P}\right)_h = \mu$



During heating region, Joule-Thomson coefficient is negative, the temperature of the substance will increase.

899. A positive value of Joule-Thomson coefficient of a fluid means

(a) temperature drops during throttling
(b) temperature remains constant during throtting
(c) temperature rises during throttling
(d) none of these

OPSC AEE 2019 Paper - II
HPPSC Asstt. Prof. 18.11.2016
ISRO Scientist/Engineer 12.05.2013
GATE 2002

Ans. (a) : Joule - Thomson coefficient

$$\mu_J = \left(\frac{\partial T}{\partial P}\right)_h$$

$\mu_J = +ve$ or $\mu > 0$

It means temperature drop during throttling.

$\mu_J < 0$ for heating region

900. The equation which perfectly represents Gibbs Phase Rule for a process where the pressure is also a variable is expressed as :

(a) F = P – C + 2 (b) F = C + P – 1
(c) F = P – C – 1 (d) F = C – P + 2

BHEL ET 2019
OPSC AEE 2019 Paper-I
GPSC Lect. (Auto) 16.10.2016
ESE 2001, 1999

Ans. (d) : Gibbs phase rule-

$$P + F = C + 2$$

or $\boxed{F = C - P + 2}$

P = Number of phase
C = Number of chemical components
n = number of non-compositional variable (i.e. n=2)
h = 2 i.e., pressure and temperature
F = Degree of freedom

901. The specific heat relation is

(a) $C_P - C_V = \frac{vT\beta^2}{k}$ (b) $C_P - C_V = \frac{vTk}{\beta^2}$

(c) $C_P - C_V = \frac{pTk}{\beta^2}$ (d) $C_P - C_V = \frac{v^2T\beta}{k}$

JPSC AE 2013 Paper-V
ESE 2003, CSE PRE 1998

Ans. (a) :

$$C_P - C_V = -T\left(\frac{\partial V}{\partial T}\right)_P^2 \left(\frac{\partial V}{\partial T}\right)_T$$

$$\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$$

$$K_T = -\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_T$$

We get,

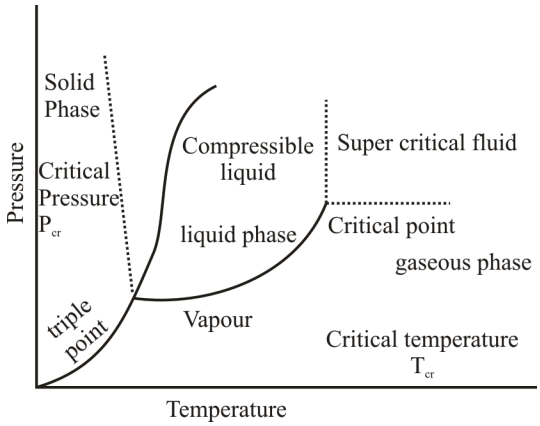
$$C_P - C_V = \frac{Tv\beta^2}{k_T}$$

902. The slope of curve is given by the Clausius-Clapeyron on:

(a) P-T plot (b) ρ-T plot
(c) V-T plot (d) s-T plot

AAI Jr. Executive 29.11.2018
BPSC AE Mains 2017 Paper - V
RPSC Lect. (Tech. Edu. Dept.) 2011
ESE 1999

Ans. (a) :



Clausius-Clapeyron equation

$$\frac{dP}{dT} = \frac{s_g - s_f}{v_g - v_f} \quad , \; v_g >>> v_f$$

$$\frac{s_g - s_f}{v_g} = \frac{L.H.}{T_s.v_g} = \frac{h_{fg}}{T_s.v_g}$$

or $\frac{dP}{dT} = \frac{(L.H.)P_s}{R.T_s^2}$

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903. The pressure 'p' of an ideal gas and its mean kinetic energy E per unit volume are related by the relation

(a) $p = \frac{4}{3}E$ (b) $p = \frac{3E}{2}$

(c) $p = \frac{E}{3}$ (d) $p = \frac{2E}{3}$

ISRO Scientist/Engineer 22.04.2018
Nagaland PSC CTSE 2015 Paper II
ISRO Scientist/Engineer 11.10.2015

Ans. (d) : We know that,

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

rms = 'root mean square'

M = Molar mass

$E_{avg.}$ is the "Average kinetic energy" of each 'molecule'.

$$E_{avg.} = \frac{1}{2}mV_{rms}^2$$

Where, $m = \frac{M}{\text{Avogadro no.}}$

$$= \frac{M}{N_A}$$

$$E_{avg.} = \frac{1}{2}\frac{M}{N_A} \times \frac{3RT}{M} = \frac{3}{2}\frac{RT}{N_A}$$

$PV = nRT$

$\therefore \quad RT = \frac{PV}{n}$

$$E_{avg.} = \frac{3}{2N_A} \cdot \frac{PV}{n}$$

$$\frac{E_{avg.}}{V} = \frac{3}{2}\frac{P}{nN_A}$$

nN_A = Total number of molecules

$$\frac{E_{avg.} \times nN_A}{V} = \frac{3}{2}P$$

$\therefore \quad E = \frac{3}{2}P$

or $\boxed{P = \frac{2}{3}E}$

904. Which one of the following parameter is significant to ascertain chemical equilibrium of a system?

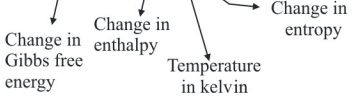
(a) Clapeyron of equation
(b) Maxwell relation
(c) Gibb's function
(d) Helmholts function

BPSC AE Mains 2017 Paper - V
Kerala PSC AE 06.08.2015
Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : Gibb's function–Gibb's energy, or free enthalpy is a quantity that is used to measure the maximum amount of work done in a thermodynamic system when the temperature and pressure are kept constant. Gibbs free energy is denoted symbol G :

$$\boxed{G = H - Ts}$$

$$\boxed{\Delta G = \Delta H - T\Delta S}$$



905. The thermodynamic parameters are

I. Temperature II. Specific volume
III. Pressure IV. Enthalpy
V. Entropy

The Clapeyron equation of state relationship between.

(a) I and II (b) II, III and V
(c) III, IV and V (d) I, II, III and IV

APPSC Poly. Lect. 2013, ESE 1997

Ans. (d) : Clapeyron's eq^n. of state relationship b/w

$$\frac{dP}{dT} = \frac{\left(h_g - h_f\right)}{T_{sat}\left(v_g - v_f\right)}$$

$v_g >>> v_f$

i) Pressure
ii) Temperature
iii) Enthalpy
iv) Specific volume

906. The slope of constant pressure curve on enthalpy-entropy diagram gives:

(a) Density (b) Specific volume
(c) Absolute temperature (d) Entropy

UPRVUNL AE 07.10.2016
UKPSC AE 2012 Paper–II

Ans. (c) : We know that

$Tds = dh - VdP$

If P = constant

$Tds = dh$

$$\boxed{\left(\frac{dh}{ds}\right)_P = T}$$

An enthalpy-entropy chart also known as h-s or Mollier diagram.

907. Consider the following statements

(1) Availability is the maximum theoretical work obtainable
(2) Clapeyron's equation for dry saturated steam is given by $\left(v_s - v_f\right) = \frac{dT_s}{dP}\left(\frac{h_s - h_f}{T_s}\right)$
(3) A gas can have any temperature at a given pressure unlike a vapour, which has a fixed temperature at a given pressure.
(4) Joule Thomson coefficient is expressed as $\mu = \left(\frac{\partial s}{\partial P}\right)_h$ of these statements :

(a) 1, 2, 3 are correct
(b) 1, 3 and 4 are correct
(c) 2 and 3 are correct
(d) 1, 2, and 4 are correct

MPPSC AE 2016, ESE 2001

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Ans. : (a)
i. Clapeyron's equation for dry saturated steam is

$$v_s - v_f = \frac{dT_s}{dP}\left(\frac{h_s - h_f}{T_s}\right)$$

ii. Joule Thomson coefficient (μ)

$$\mu = \left(\frac{\partial T}{\partial P}\right)_h$$

iii. The availability of a given system is defined as the max. Useful work that is obtainable in a process in which the system comes to equilibrium with its surrounding.

908. Which of the following Maxwell relation is INCORRECT?

(a) $\left(\frac{\partial T}{\partial V}\right)_s = -\left(\frac{\partial P}{\partial s}\right)_V$ (b) $\left(\frac{\partial T}{\partial P}\right)_s = \left(\frac{\partial V}{\partial s}\right)_P$

(c) $\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial s}{\partial V}\right)_T$ (d) $\left(\frac{\partial V}{\partial T}\right)_P = \left(\frac{\partial P}{\partial s}\right)_T$

APPSC Poly Lect. 13.03.2020
SJVN ET 2019

Ans. (d) : Maxwell Equations–There are four Maxwell equations which are applicable for any substances.

(1) $\left(\frac{\partial T}{\partial V}\right)_s = -\left(\frac{\partial P}{\partial s}\right)_V$

(2) $\left(\frac{\partial T}{\partial P}\right)_s = \left(\frac{\partial V}{\partial s}\right)_P$

(3) $\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial s}{\partial V}\right)_T$

(4) $\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial s}{\partial P}\right)_T$

$\left(\frac{\partial V}{\partial T}\right)_P = \left(\frac{\partial P}{\partial s}\right)_T$ is incorrect regarding to Maxwell equation.

909. The value of compressibility factor, Z at the critical state of a Vander Waal's gas is

(a) 3.735 (b) 0.735
(c) 3.375 (d) 0.375

UPPSC AE 13.12.2020, Paper-2
UKPSC AE 2012 Paper II

Ans. (d) : The value of compressibility factor, Z at critical state of Vander waal's equation is

$$Z = \frac{P_c V_c}{RT_c} = \frac{3}{8} = 0.375$$

910. The 'specific heat at constant pressure' is defined as :

(a) $T\,(\partial s/\partial T)_P$ (b) $(\partial h/\partial T)_P$
(c) Both (a) and (b) (d) None of the above

OPSC Civil Services Pre, 2011
APPSC IOF, 2009

Ans. (c) : The specific heat at constant pressure is defined as :

$$c_P = \left.\frac{\partial h}{\partial T}\right|_P,\ c_P = T\left(\frac{\partial s}{\partial T}\right)_P, c_v = T\left(\frac{\partial s}{\partial T}\right)_v$$

911. Gibb's function is expressed as

(a) (u +pv + Ts) (b) (u + pv – Tds)
(c) (u + pdv –Tds) (d) (u + pv – sdT)

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b)

912. The value of the compressibility of an ideal fluid is

(a) zero
(b) unity
(c) infinity
(d) more than that of a real fluid

Assam PSC AE (PHED), 18.10.2020

Ans. (a) : In thermodynamic and fluid mechanics compressibility (also known as the coefficient of compressibility or isothermal compressibility) is a measure of relative volume change of fluid or solid as response to a pressure change. The value of compressibility of an ideal fluid is zero.

913. Latent heat of vaporization of water can be calculated from the p-T data of water using the following equation.

(a) Third law of thermodynamics
(b) Clausius - Clapeyron equation
(c) Kelvin-Planck statement of second law of thermodynamics
(d) Clausis statement of second law of thermodynamics

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : Clausius Clapeyron equation gives the latent heat of vaporization from pressure and temperature data.

$$\frac{dp}{dT} = \frac{h_{fg}}{T(V_g - V_f)}$$

914. Which of the following can be used to determine the variation of saturation pressure with temperature along phase boundaries?

(a) Joule-Thomson relation
(b) Carnot equation
(c) Rankine- Hougoniot's relation
(d) Clausius- Clapeyron relation

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : Clausius Clapeyron Equation-

$$\frac{dP}{dT} = \frac{h_g - h_f}{T_{sat}(v_g - v_f)} = \frac{(\text{Latent heat}) \times P_{sat}}{RT_{sat}^2} = \frac{(h_g - h_f)P_{sat}}{RT_{sat}^2}$$

$\Rightarrow$ Clausius Clapeyron equation is used to determine enthalpy change associated with phase change (Latent heat) using pressure temperature and volume. It may also be used to predict the boiling temperature.

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915. Which one of the following properties remains unchanged for a real gas during Joule-Thomson process

(a) Temperature (b) Enthalpy
(c) Entropy (d) Pressure

TNPSC AE 2019

Ans. (b) : Enthalpy properties remains unchanged for a real gas during Joule-Thomson process.

$$\left(\frac{\partial T}{\partial P}\right)_{h=const.} = \mu$$

916. In a single-component condensed system, if degree of freedom is zero, maximum number of phases that can co-exist_____

(a) 0 (b) 1
(c) 2 (d) 3

OPSC AEE 2019 Paper-I

Ans : (d) : By using Gibb's phase rule

$P + F = C + 2$

$P + 0 = 1 + 2$

$\boxed{P = 3}$

917. The difference between two specific heats $C_P - C_V = R/J$. This relation is valid for:

(a) Any gas (b) Perfect gas
(c) Ideal gas (d) Pure gas

GPSC ARTO Pre 30.12.2018

Ans. (*) : When the gas is heated at constant pressure, these is increase in internal energy and piston moves outward and work is performed by gas.

$$dQ' = dQ + \frac{dW}{J}$$

Where $\frac{dW}{J}$ is the heat equivalent of the work

$$C_p dT = C_v dT + \frac{PdV}{J}$$

$$C_p dT = C_v dT + \frac{RdT}{J}$$

$$C_p = C_v + \frac{R}{J}$$

$$\boxed{C_p - C_v = \frac{R}{J}}$$

This relation is valid for ideal gas and perfect gas both.

918. The equation of state

$$pv = RT\left(B_0 + \frac{B_1}{v} + \frac{B_2}{v^2} + \frac{B_3}{v^3}\right)$$ **is known as**

(a) Vander wall's equation
(b) Benedict-Webb-Rubin equation
(c) Gibbs equation
(d) Virial equation

TNPSC AE 2018
CSE Pre-2005

Ans. (d) : Virial equation,

$$pv = RT\left(B_0 + \frac{B_1}{v} + \frac{B_2}{v^2} + \frac{B_3}{v^3}\right)$$

The virial equation of state is valid for gases at low density only.

919. The relationship $(dT/dP)_h = 0$ holds good for

(a) An ideal gas at any state
(b) A real gas at any state
(c) Any gas at its critical state
(d) Any gas at its inversion point

GPSC Executive Engineer 23.12.2018
GATE 1993

Ans. (a & d) :

920. Specific heat at constant pressure is given by

(a) $\left(\frac{\partial s}{\partial t}\right)_p$ (b) $\left(\frac{\partial v}{\partial t}\right)_p$
(c) $\left(\frac{\partial h}{\partial t}\right)_p$ (d) $\left(\frac{\partial v}{\partial t}\right)_p$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Specific heat at constant pressure is given by $\left(\frac{\partial h}{\partial t}\right)_p$, Specific heat at constant pressure represents the heat supplied to a unit mass of the system to raise its temperature through 1K, keeping the pressure constant.

921. Which one is not a Maxwell equation

(a) $\left(\frac{\partial V}{\partial S}\right)_p = \left(\frac{\partial T}{\partial p}\right)_S$ (b) $\left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial S}{\partial p}\right)_T$
(c) $\left(\frac{\partial T}{\partial V}\right)_s = -\left(\frac{\partial p}{\partial S}\right)_V$ (d) $\left(\frac{\partial P}{\partial T}\right)_p = \left(\frac{\partial S}{\partial V}\right)_T$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : $\left(\frac{\partial P}{\partial T}\right)_p = \left(\frac{\partial S}{\partial V}\right)_T$ is not a Maxwell equation among the given option.

922. ..... equation of state for a gas is an improvement of the ideal gas law.

(a) Beattie-Bridgeman (b) Dalton
(c) Clausius (d) Vander Waals

HPPSC AE (PWD) 10.09.2016

Ans. (d) : Vander Waals Equation- The Vander Waals equation is an equation of state that corrects for two properties of real gases. The excluded volume of gas particle and attractive force between gas molecules.

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = n\bar{R}T$$

Where
P = pressure (atm)
n = number of moles (mol)
$\bar{R}$ = 8.3145 J/mole-K (universal gas constant)
V = volume
T = temperature (K)
a and b = constant

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923. A car tube contains air at 200 kPa pressure in the cold night of temperature (–10°C). During the day time, the volume of the tube is found to be increased by 9.5%. Find out day temperature and change in pressure.

(a) 15°C, 219 kPa (b) 15°C, 210 kPa
(c) 19°C, 219 kPa (d) 19°C, 210 kPa
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : $P_1 = 200$ kPa, $T_1 = 263$ K, $T_2 = ?$

$V_2 = 1.095\ V_1$

$PV = nRT$

$$\Rightarrow \quad \frac{PV}{T} = \text{Constant}$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}\left(\frac{V_2}{V_1}\right)$$

(By Hit & trial method, check option 'a').

For $T_2 \Rightarrow P_2V_2 = nRT_2$

$T_2 = 298$ K $= 15°C$

$\Rightarrow \quad P_2 = 200 + 19 = 219$ kPa

924. Clapeyron equation represents the dependence of

(a) pressure on temperature
(b) saturation pressure on saturation temperature
(c) vapor pressure on temperature
(d) entropy generation on temperature

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : According to Maxwells equation (3rd relationship)

$$\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$$

When saturated liquid convert to saturated vapour of constant temperature and pressure, during the evaporation the pressure and temperature is independent of volume.

$$\left(\frac{dP}{dT}\right)_{sat.} = \frac{S_g - S_f}{V_g - V_f} = \frac{S_{fg}}{V_{fg}}$$

but $S_g - S_f = \frac{h_{fg}}{T}$

$$\boxed{\left(\frac{dP}{dT}\right)_{sat.} = \frac{h_{fg}}{T\left(V_g - V_f\right)}}$$

This is Clausis-Clapeyron equation. So this equation represents the dependence of saturation pressure on saturation temperature. Clapeyron equation provides relationship between temperature, specific, volume, pressure and enthalpy.

925. Match List-I with List-II and select the correct answer :

List-I		List-II
A.	Bottle filling of gas	1. Absolute zero temperature
B.	Nernst-Simon statement	2. Variable flow
C.	Joule-Thomson effect	3. Quasi-static path
D.	$\int pdV$	4. Isenthalpic process
		5. Dissipative effect
		6. Low grade energy
		7. Process and temperature during phase change

	A	B	C	D
(a)	6	5	4	3
(c)	2	5	7	4

	A	B	C	D
(b)	2	1	4	3
(d)	6	1	7	4

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) :

	List-I		List-II
A.	Bottle filling of gas	2.	Variable flow
B.	Nernst-Simon statement	1.	Absolute zero temperature
C.	Joule-Thomson effect	4.	Isenthalpic process
D.	$\int pdV$	3.	Quasi-static path

926. For a perfect gas match list I with list II:

List I	List II
(A) Isobaric thermal expansion coefficient	**(1) 0**
(B) Isothermal compressibility	**(2) ∞**
(C) Isentropic compressibility	**(3) 1/v**
(D) Joule - Thomson coefficient	**(4) 1/T**
	(5) 1/p
	(6) $1/\gamma p$

(a) A-4, B-3, C-2, D-1 (b) A-1, B-2, C-4, D-6
(c) A-4, B-5, C-6, D-1 (d) A-3, B-4, C-6, D-5

ISRO Scientist/Engineer 12.05.2013

Ans. (c) :

(A) Isobaric thermal expansion coefficient – 1/T
(B) Isothermal compressibility – 1/p
(C) Isentropic compressibility – $1/\gamma p$
(D) Joule - Thomson coefficient – 0

927. The maximum work, that can be obtained from a system during a given process in which the initial and final temperatures of the system are equal to the surrounding temperature, is equal to the decrease in :

(a) Internal energy
(b) Helmholtz Free energy
(c) Gibbs Free energy
(d) Enthalpy

APGENCO AE 2012

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Ans. (b) The Helmholtz free energy is a thermodynamic potential that measures the useful work obtainable from a closed thermodynamic system at a constant temperature (isothermal).
The Gibbs free energy or free enthalpy is most commonly used as a measure of thermodynamic potential (especially in chemistry) when it is convenient for applications that occur at constant pressure.

928. The expression indicating Y as a thermodynamic property is :

(a) $Y = XZdZ + \frac{1}{2}Z^2dX$

(b) $Y = XZ^2dZ + Z^2dX$

(c) $Y = pdv - vdp$

(d) $Y = p^2dv + vdp$

OPSC Civil Services Pre. 2011

Ans. (a) : $Y = XZdZ + \frac{1}{2}Z^2dX$

929. $\left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial T}{\partial P}\right)_V$ **is equal to**

(a) zero (b) 1

(c) –1 (d) Infinity

ISRO Scientist/Engineer 2008

Ans. (c) : We know that

$PV = mRT$

$PV = RT$

$P = \left(\frac{RT}{V}\right); \quad \left(\frac{\partial P}{\partial V}\right)_T = -\frac{RT}{V^2}$

$V = \left(\frac{RT}{P}\right); \quad \left(\frac{\partial V}{\partial T}\right)_P = \frac{R}{P}$

$P = \frac{PV}{R}; \quad \left(\frac{\partial T}{\partial P}\right)_V = \frac{V}{R}$

$\left(\frac{\partial P}{\partial v}\right)_T \times \left(\frac{\partial V}{\partial T}\right)_P \times \left(\frac{\partial T}{\partial P}\right)_V = -\frac{RT}{V^2} \times \frac{R}{P} \times \frac{V}{R}$

$= -\frac{RT}{PV}$

$\because \quad PV = RT$

$= -1$

930. The state of a real gas if changed from pressure P_1, temperature T_1 to pressure P_2 temperature T_2. The change in enthalpy, h_2-h_1, is given by

(a) $\int_{T_2}^{T_1} C_p dT$

(b) $\int_{T_1}^{T_2} C_P\, dT + \int_{T_1}^{T_2} \left(\frac{\partial V}{\partial p}\right)_T dp$

(c) $\int_{T_1}^{T_2} C_v\, dT + \int_{P_1}^{P_2} \left[V - T\left(\frac{\partial V}{\partial p}\right)_p\right] dp$

(d) $\int_{T_1}^{T_2} C_p\, dT + \int_{P_1}^{P_2} \left[V - T\left(\frac{dV}{dT}\right)_p\right] dp$

ISRO Scientist/Engineer 2006

Ans. (c) : The change in enthalpy,

$h_2\text{-}h_1 = \int_{T_1}^{T_2} C_v\, dT + \int_{P_1}^{P_2} \left[V - T\left(\frac{\partial V}{\partial p}\right)_p\right] dp$

931. The Joule-Thomson effect is not exhibited by

(a) ideal gases (b) real gases

(c) carbon dioxide (d) refrigerant vapours

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Joule-Thomson Coefficient– The temperature behaviour of the fluid during the throttling (h = constant) process is described by the Joule-Thomson coefficient,

$\mu = \left(\frac{\partial T}{\partial P}\right)_h = \frac{1}{C_P}\left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right]$

For ideal gas, $\mu_{ideal\ gas} = 0$

- This coefficient is positive in cooling region and negative in heating region. It is zero on the inversion curve.

932. The Gibbs free-energy function is a property comprising:

(a) Pressure, volume and temperature

(b) Enthalpy, temperature and entropy

(c) Temperature, pressure and enthalpy

(d) Volume, enthalpy and entropy

HPPSC Asstt. Prof. 18.09.2017

Ans. (b) :

933. For a refrigerant being used in a vapour compression refrigeration system, the Joule-Thomson coefficient should be

(a) positive (b) negative

(c) zero (d) infinity

Haryana PSC Civil Services Pre, 2014

Ans. (b) : In refrigeration the desired effect is cooling but refrigerant gets heat from the refrigerated room. So, for heating of refrigerant–

$\boxed{\mu_j < 0}$

$\boxed{\mu_j = \left(\frac{\partial T}{\partial P}\right)_h}$

μ_j – Joule Thomson coefficient

$\mu_j > 0 \rightarrow$ Cooling

$\mu_j < 0 \rightarrow$ Heating

$\mu_j = 0 \rightarrow$ for Ideal gas

934. When a real gas follows Joule Thomson expansion process, the temperature

(a) always increases

(b) always decreases

(c) remains same

(d) may increase or decrease

(e) becomes zero

CGPSC AE 26.04.2015 SHIFT-II

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Ans. (d) : When a real gas follows Joule Thomson expansion process, the temperature may increase or decrease.

Joule- Thomson coefficient (μ)- Joule- Thomson Coefficient (μ) is defined as the ratio of the temperature change to the pressure drop, and is expressed in terms of the thermal expansion coefficient. The Joule - Thomson Coefficient will be zero at a point called Inversion point for all real gases, Expansion of most real gases causes cooling when the joule-Thomson coefficient is positive and the gas temperature is below the Inversion temperature. However, at atmosphere pressure, as the inversion temperature for hydrogen is low and hence hydrogen will warm during a Joule- Thomson expansion at room temperature. Since there is no change of temperature when an ideal gas expands through throttling device, a non-zero Joule-Thomson coefficient refers to a real gas.

935. For achieving the cooling effect by Joule-Kelvin, expansion the initial temperature of gas must be below the

(a) Boiling point temperature
(b) Freezing point temperature
(c) Maximum inversion temperature
(d) Saturation temperature

TNPSC AE 2013

Ans. (c) : For achieving the cooling effect by Joule-Kelvin, expansion the initial temperature of gas must be below the maximum inversion temperature.

936. Joule-Thomson process is

(a) Throttling process (b) Heating process
(c) Compression process (d) Expansion process

Gujarat PSC AE 2019

Ans : (a) : $\boxed{\mu = \left(\frac{\partial T}{\partial P}\right)_h}$ constant enthalpy 'h' process.

937. The change in specific volume during vaporization of a refrigerant is 0.035 m^3/kg. The slope of the saturation curve on a 'pressure-temperature' diagram at 20°C is 17.69 kPa/K. The enthalpy of vaporization of the refrigerant is:

(a) 9.99 kJ/kg (b) 181.5 kJ/kg
(c) 45.37 kJ/kg (d) 99.44 kJ/kg

APPSC IOF, 2009

Ans. (b) : $\left(\frac{dP}{dT}\right) = \frac{h_{fg}}{T \times V_{fg}}$

$(17.69) \times (273 + 20) \times 0.035 = h_{fg}$

$\boxed{h_{fg} = 181.5\,kJ/kg}$

938. When an ideal gas undergoes Joule-Kelvin expansion its :

(a) Temperature decreases
(b) Temperature increases
(c) Temperature does not change
(d) Pressure increases

UPSC JWM Adv. No-16/2009

Ans. (c) : When an ideal gas undergoes Joule-Kelvin expansion i.e., throttling, there is no change in temperature. For an ideal gas, the value of Joule-Kelvin coefficient (μ) comes out to be zero.

Hence, $\boxed{\mu = \left(\frac{dT}{dP}\right)_h = 0}$

939. For an ideal gas, the slope of the constant-volume line in the T-s diagram is

(a) higher than the slope of the constant pressure line
(b) lower than the slope of the constant pressure line
(c) equal to the slope of the constant pressure line
(d) equal to the slope of the constant temperature line

DRDO Scientists 2009

Ans. (a) : $\left(\frac{dT}{dS}\right)_v = \frac{T}{C_V}$

$\left(\frac{dT}{dS}\right)_p = \frac{T}{C_p}$

Since, $C_p > C_v$

$\left(\frac{dT}{dS}\right)_v > \left(\frac{dT}{dS}\right)_p$

Hence, For an ideal gas, the slope of the constant-volume line in the T-s diagram is higher than the slope of the constant pressure line.

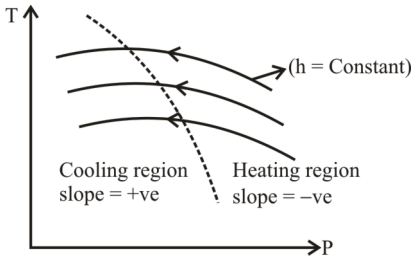
940. Region inside the inversion curve is represented by : (where μ is Joule - Kelvin coefficient)

(a) Cooling region, $\mu < 0$
(b) Heating region, $\mu > 0$
(c) Cooling region, $\mu > 0$
(d) Heating region, $\mu < 0$

UPRVUNL AE 07.10.2016

Ans. (c) : Joule - Kelvin coefficient (μ)

$\mu = \left(\frac{\partial T}{\partial P}\right)_h$



For ideal gas

$\mu = 0$

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02.

Power Plant Engineering

1. Steam Power Plant

1. In the thermal power plant, the deaerator is used mainly to

(a) remove air from condenser
(b) increase firewater temperature
(c) reduce steam pressure
(d) remove dissolved gases from the feed water

Assam PSC AE (PHED), 18.10.2020
RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017
MPPSC AE 2016, GPSC AE 26.07.2015
OPSC Civil Services Pre 2006

Ans. (d) : Presence of dissolved gases like O_2 and CO_2 in water makes water corrosive the solubility of these gases in water decreases with increase in temperature and becomes zero at saturation temperature.
The deaerator is used mainly to remove dissolved gases from the feed water.

2. Which of the following is not a boiler mounting?

(a) Fusible plug (b) Blowdown valve
(c) Water level indicator (d) Preheater

VIZAG STEEL MT 24.01.2021, Shift-II
RPSC IOF, 2020, SJVN ET 2019
RPSC AE 2018, 2016, HPPSC Lect. 2016
UJVNL AE 2016

Ans. (d) : Boiler Mounting-
(i) Water level indicator
(ii) Pressure gauge
(iii) Safety valve
(iv) Steam stop valve
(v) Blow off -cock
(vi) Feed check valve
(vii) Fusible plug
(viii) A man hole
(ix) Mud holes or slight holes

Boiler accessories
(i) Feed pump
(ii) Superheater
(iii) Economiser (Feed Water Preheater)
(iv) Air Preheater
(v) Injector
(vi) Steam separator
(vii) Steam strap

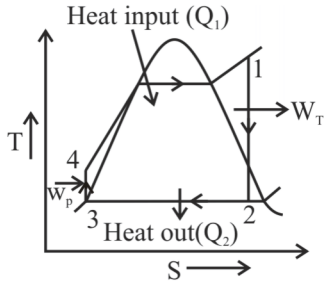
So, Preheater is not boiler mountings it is boiler accessories.

3. In a steam power plant the ratio of isentropic heat drop in the prime mover to the amount of heat supplied per unit mass of steam is known as

(a) Stage efficiency (b) Degree of reaction
(c) Rankine efficiency (d) Relative efficiency

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017
RPSC Lect. (Tech. Edu. Dept.) 2011
RPSC AE GWD, 2011
UKPSC AE 2007 Paper -II, ESE 2000

Ans. (c) :



So, Rankine efficiency, is the ratio, of net work done to net heat transfer.

$$\eta_{rankine} = \frac{W_T - W_P}{Q_1}$$

$$= \frac{(h_1 - h_2) - (h_4 - h_3)}{(h_1 - h_4)}$$

If pump work neglected then,

$$\eta_{rankine} = \frac{h_1 - h_2}{h_1 - h_4}$$

4. An economizer in a steam generator performs the function of :

(a) Preheating the combustion air
(b) Preheating the feed water
(c) Preheating the input fuel
(d) Raising the temperature of steam

ESE 2020
Jharkhand Urja Vikas Nigam Ltd. AE 2017
HPPSC Asstt. Prof. 18.11.2016
MPPSC State Forest Service Exam, 2014
RPSC ACF-2011, GATE 1989

Ans. (b) : Economiser–It is placed in the path of the flue gases in between the exit from the boiler and entry of the chimney. It's function is preheating the feed water.

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5. In high pressure natural circulation boilers, the flue gas flow through the following boiler accessories.

1. Superheater
2. Air heater
3. Economiser
4. ID fan

The correct sequence of the flow of flue gases through these boiler accessories is :

(a) 1, 3, 4, 2 (b) 3, 1, 4, 2
(c) 3, 1, 2, 4 (d) 1, 3, 2, 4

RPSC ACF & FRO, 26.02.2021
RPSC IOF & Boiler 20.09.2017
Nagaland PSC CTSE 2015 Paper-II
APPSC IOF, 2009, ESE 1997

Ans. (d) : Superheater → Economiser → Air preheater → ID Fan

6. Which of the following are fire tube boilers

(a) Cochran (b) Lancashire
(c) Locomotive (d) All of these

Nagaland PSC CTSE 2017 Paper-2
J&K PSC Civil Services Pre, 2013
UKPSC AE 2007 Paper-II, ESE 2005

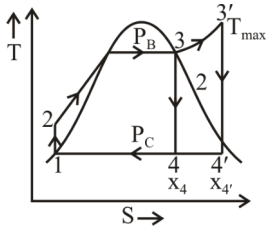
Ans. (d) : Cochran, Lancashire, Locomotive all are fire tube boilers. Fire tube boiler are internally fire tube boilers.

7. The efficiency of superheat Rankine cycle is higher than that of simple Rankine cycle because.

(a) the enthalpy of main steam is higher for superheat cycle
(b) the mean temperature of heat addition is higher for superheat cycle
(c) the temperature of steam in the condenser is high
(d) the quality of steam in the condenser is low.

RPSC ACF & FRO, 26.02.2021, ESE 2019
GPSC Executive Engineer 23.12.2018
RPSC Vice Principal ITI 2018
TSPSC Managers, 2015, GATE 2002

Ans. (b) :



- W_P-Constant
- W_T-↑, W_{net}-↑, η-↑, x-↑, Moisture-↓

The efficiency of superheat Rankine cycle is higher than that of simple Rankine cycle because the mean temperature of heat addition is higher for superheat cycle.

8. Which of the following is a water-tube boiler?

(a) Cochran boiler
(b) Babcock and Wilcox boiler
(c) Lancashire boiler
(d) Locomotive boiler

APPSC Poly Lect. 13.03.2020
GMB AAE 25.06.2017
Nagaland PSC (CTSE) 2016, Paper-II
MPPSC State Forest Service Exam, 2014

Ans. (b) : Water tube boiler– In water tube steam boiler the water is contained inside the tube which are surrounded by flames and hot gases from outside.

	Water Tube Boiler		Fire Tube Boiler
(1)	Babcock and Wilcox boiler	(1)	Cochran boiler
(2)	Stirling boiler	(2)	Lancashire boiler
(3)	Lamont boiler	(3)	Locomotive boiler
(4)	Benson boiler		

9. Which one of the following sequences indicates the correct order for flue gas flow in the steam power plant layout?

(a) Superheater, Economiser, Air preheater.
(b) Economiser, Air preheater, Superheater.
(c) Air preheater, Economiser, Superheater.
(d) Economiser, Superheater, Air preheater.

RPSC ACF & FRO, 26.02.2021
RPSC IOF, 2020
OPSC AEE 2015 Paper - II, ESE 2001

Ans. (a) : Sequence of flue gases flow in steam power plant →
Super heater → Economiser → Air preheater.

10. Which of the following is a fire tube boiler?

(a) Lancashire boiler
(b) Babcock and Wilcox boiler
(c) Both of the above
(d) none of the above

JPSC AE, 10.04.2021, Paper-I
Punjab PSC SDE 12.02.2017
Nagaland PSC (CTSE) 2016, Paper-II
UKPSC AE 2007 Paper -II

Ans. (a) : Fire tube Boiler –
- Cochran Boiler
- Lancashire Boiler
- Cornish Boiler
- Scotch marine Boiler
- Locomotive Boiler

Water tube steam Boilers–
- Babcock and Wilcox
- Stirling Boiler
- La-mont Boiler
- Benson Boiler
- Yarrow Boiler
- Loeffler Boiler

11. In steam power cycle, reheat factor is usually in the range:

(a) 1.02 to 1.05 (b) 1.12 to 1.15
(c) 1.5 to 1.8 (d) 1.9 to 2.1

JPSC AE, 10.04.2021, Paper-I
HPPSC Asstt. Prof. 2014, ESE 2004

Ans. (a) : In steam power cycle, reheat factor is usually in the range of 1.02 to 1.05.

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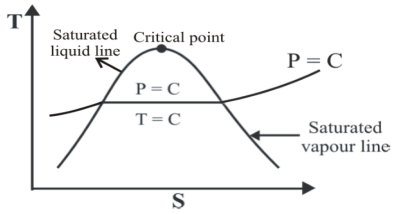
12. Steam super heating is done at constant

(a) Pressure (b) Mass flow rate
(c) Entropy (d) Temperature

HPPSC AE 2018, TNPSC AE 2014
J & K PSC Screening, 2006

Ans. (a) : Steam super heating is done at constant pressure.



Degree of super heat = $(T_{superheat} - T_{saturated\ vapour})$

13. In steam power cycle, the process of removing non-condensable gases is called

(a) scavenging process
(b) deaeration process
(c) exhaust process
(d) condensation process

HPPSC Asstt. Prof. 18.09.2017
GPSC ARTO 01.05.2016
ESE 2019, 1996, GATE 1992

Ans. (b) : The removal of dissolved gases from boiler feedwater is an essential process in a steam system. The presence of dissolved oxygen in feedwater causes rapid localized corrosion in boiler tubes. This mechanical process is known as deaeration and will increase the life of a steam system dramatically.

14. In a boiler, feed water supplied per hour is 205 kg while coal fired per hour is 23 kg. The net enthalpy rise per kg of water is 145 kJ. If the calorific value of coal is 2050 kJ/kg, then boiler efficiency will be

(a) 56% (b) 75%
(c) 48% (d) 63%

ISRO Scientist/Engineer (RAC) 22.04.2018
ISRO Scientist/Engineer (RAC) 07.05.2017
GPSC Lect. (Poly.) 07.12.2014
VIZAG STEEL MT 2011

Ans. (d) : Heat generated by coal per hour is given by

$$Q = 2050 \times 23 \text{ kJ/hr}$$

If total heat generated will be used completely to increase the enthalpy of water.

Then per kg increase in enthalpy of water is

$$\Delta h = \frac{2050 \times 23}{205} = 230 \text{ kJ/kg}$$

Actual increase in enthalpy is given as 145 kJ/kg

So efficiency of boiler is

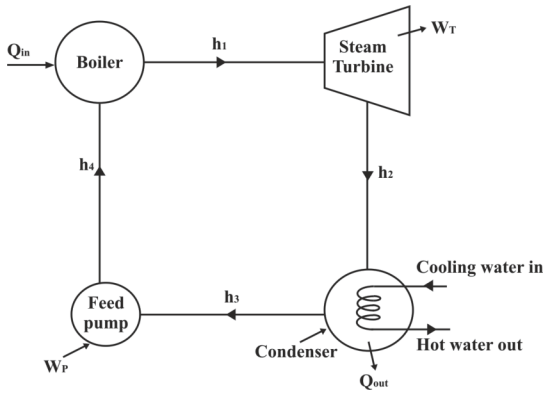
$$\eta = \frac{145}{230} = 0.63 = 63\%$$

15. Thermal power plant works on

(a) Carnot cycle (b) Joule cycle
(c) Rankine cycle (d) Brayton cycle

ISRO Scientist/Engineer (RAC) 07.05.2017
Rajasthan Nagar Nigam AE 2016, Shift-II
ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Thermal power plant works on Rankine cycle.



Heat supply to Boiler — $Q_{in} = h_1 - h_4$
Heat rejected from condenser — $Q_{out} = h_2 - h_3$
Turbine Work — $W_T = h_1 - h_2$
Pump Work — $W_P = h_4 - h_3$
Network — $W_{net} = W_T - W_P$

16. A device which is used to drain off water from steam pipes without escape of steam is called

(a) Steam separator
(b) Steam trap
(c) Pressure reducing valve
(d) Injectors

RPSC IOF, 2020, GPSC AE 26.07.2015
ESE 2002

Ans. (b) : Steam trap are used in such applications to ensure that steam is not wasted. Steam trap is a device which is used to drain off water from steam pipes without escape of steam.

17. Principle purpose of installing Electrostatic Precipitators is

(a) To reduce operating load on induced draft fan
(b) To reduce flue gas flow through stack
(c) To reduce stack emission of suspended particulate matters
(d) To improve balance draught control in fluctuating boiler load

Haryana PSC AE (PHED) 04.09.2020, Paper-I
Rajasthan AE (Nagar Nigam) 2016 Shift-3
OPSC Civil Services Pre 2006

Ans. (c) : Electrostatic precipitator collect dust in flue gas produced by boiler.
To reduce stack emission of suspended particulate matters.

18. An air preheater is installed

(a) before the economiser
(b) before the superheater
(c) between the economiser and chimney
(d) none of these

Sikkim PSC (Under Secretary), 2017
Assam PSC CCE Pre 2015
GPSC Lect. (Poly.) 07.12.2014
ESE 1992

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Ans. (c) : Boiler → Preheater → economizer → air preheater → chimney
So that exhaust air from the economizer will transfer it's heat from to the air preheater. The heated air in the air preheater is fed to the furnace. This will improve burning of fuel.

19. Which is not a boiler accessory?
(a) Economiser (b) Fusible plug
(c) Superheater (d) Air Preheater

VIZAG MT, 14.12.2020
Mizoram PSC AE/SDO 2014, Paper-I
SJVN ET 2013

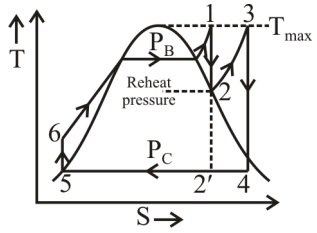
Ans. (b) : Boiler mountings-
(i) Two safety valves.
(ii) Two water level indicators
(iii) A pressure gauge
(iv) A steam stop valves
(v) A feed check valves
(vi) A blow off cock
(vii) An attachment for inspectors test gauge
(viii) A main hole
(ix) Mud holes or sight holes

Boiler accessories-
(i) Feed pumps
(ii) Injector
(iii) Economiser
(iv) Air pre heater
(v) Super heater
(vi) Steam separator
(vii) Steam trap

20. Reheating of Rankine Cycle will
(a) not alter turbine efficiency
(b) improve the steam quality
(c) decrease the nozzle and blade efficiency
(d) decrease the turbine efficiency

TNPSC AE 2018, TRB Poly. Lect. 2017
Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : Reheating of Rankine cycle will improve the steam quality.



- W_T-↑, W_P-Constant, W_{net}-↑
- Q_s-↑, Q_R-↑, T_{MA}-↓↑, η-↑↓ (Can't say)
- $\downarrow \text{Specific Steam Consumption} = \frac{3600}{W_{net}\uparrow}$
- x-↑, ↓errosion of turbine blade

21. The draught in locomotive boiler is produced by
(a) Chimney (b) Centrifugal fan
(c) Steam jet (d) Locomotion
(e) None of the above

RPSC ACF & FRO, 26.02.2021
CGPSC Poly. Lect. 22.05.2016, ESE 2003

Ans. (c) : In a locomotive boiler, the draught is produced by a steam jet.
Steam jet draught is classified as– (a) induced steam jet, (b) forced steam jet
(a) Induced steam jet– The draught produced by a steam jet issuing from a nozzle placed in the chimney.
(b) Forced steam jet– The draught produced by a steam jet issuing from a nozzle placed in the ashpit under the fire grate of the furnace.

22. The values of enthalpy of steam at inlet and outlet of a steam turbine in a Rankine cycle are 2800 kJ/kg and 1800 kJ/kg respectively. Neglecting pump work, the specific steam consumption in kg/kW-hour is :
(a) 3.60 (b) 0.36
(c) 0.06 (d) 0.01

ISRO Scientist/Engineer (RAC), 10.03.2019
TSPSC AEE 2017
GPSC Asstt. Director of Transport 05.03.2017
VIZAG Steel MT 10.06.2012, GATE 2011

Ans. (a) : Given,
$h_1 = 2800$ kJ/kg & $h_2 = 1800$ kJ/kg
$W_{net} = h_1 - h_2$

$$\text{specific steam consumption} = \frac{3600}{W_{net}} kg/kW\text{-}hr$$

$$= \frac{3600}{2800-1800}$$
$= 3.6$ kg/kWh

23. Benson boiler is the one of the high pressure boiler having :
(a) no drum
(b) three drums
(c) one drum
(d) one water drum and one steam drum

RPSC IOF & Boiler 20.09.2017
APPSC IOF, 2009, ESE 1999

Ans. (a) : Benson boiler is a high pressure drum-less, super critical, water tube boiler with forced circulation.

24. Steam coming out of whistle of pressure cooker.
(a) wet vapour (b) superheated vapour
(c) ideal gas (d) Dry saturated vapour

GWSSB AAE, 27.12.2015
VIZAG Steel MT (Re-Exam) 24.11.2013
UKPSC AE 2012 Paper–II

Ans. (a) : In pressure cooker there is always mixture of steam and water so quality of steam is wet vapour.

25. In a Rankine cycle, regeneration results in higher efficiency because
(a) Pressure inside the boiler increases
(b) Heat is added before steam enters the low pressure turbine
(c) Average temperature of heat addition in the boiler increases
(d) Total work delivered by the turbine increases

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
UPPSC AE 13.12.2020, Paper-II
GPSC ARTO Pre 30.12.2018

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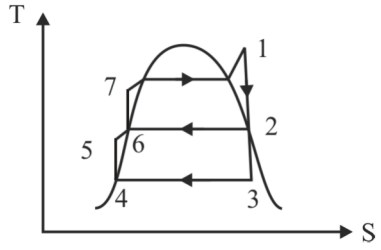
Ans. (c)



In Rankine cycle, regeneration results in higher efficiency because mean temperature of heat addition in boiler increases

26. Reheat factor is defined as the :

(a) ratio of cumulative enthalpy drop to isentropic enthalpy drop
(b) product of cumulative entropy drop to isentropic enthalpy drop
(c) ratio of isentropic enthalpy drop to cumulative enthalpy drop
(d) sum of isentropic and cumulative enthalpy drops

CGPSC AE 15.01.2021, APGENCO AE 2017
CGPSC AE 16.10.2016

Ans. (a) : $\text{Reheat factor} = \dfrac{\text{Cumulative Enthalpy drop}}{\text{Isentropic Enthalpy drop}}$

27. The device used to heat feed water by utilizing the heat of the exhaust flue gases before leaving through the chimney is called:

(a) Super heater (b) Economizer
(c) Air preheater (d) Evaporator

ISRO Scientist/Engineer (RAC), 10.03.2019
ESE 1999

Ans. (b) : Economizer– Economiser is a boiler accessories which is used to heat feed water by utilizing the heat of flue gases.

28. Overall efficiency of a thermal power plant is equal to :

(a) Rankine cycle efficiency
(b) Carnot cycle efficiency
(c) Regenerative cycle efficiency
(d) Boiler efficiency × turbine efficiency × generator efficiency

RPSC AE 2016, RPSC ACF-2011

Ans. (d) : $\eta_{overall} = \eta_{boiler} \times \eta_{turbine} \times \eta_{generator}$

29. Load factor of a power station is usually :

(a) equal to unity (b) less than unity
(c) more than unity (d) zero

RPSC AE 2016, RPSC ACF-2011

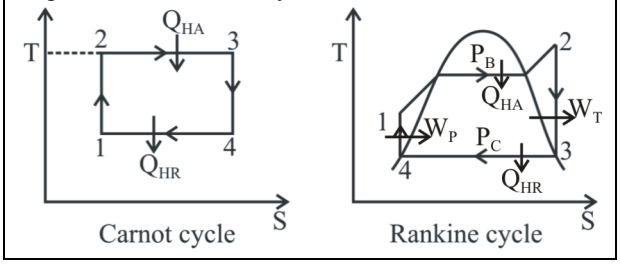
Ans. (b) : Load factor of a power station is defined as average load to maximum demand. Load factor of a power station is generally less than unity.

30. The thermal efficiency of an ideal Rankine cycle is less than that of Carnot cycle operating between the same maximum and minimum temperature limits because :

(a) Heat addition does not takes place at constant temperature
(b) The expansion process is not reversible and adiabatic
(c) Heat rejections does not take place at constant temperature
(d) The compression process is not reversible and adiabatic.

TSPSC Managers, 2015, APGENCO AE 2012

Ans. (a) : The thermal efficiency of an ideal Rankine cycle is less than that of carnot cycle operating between the same maximum and minimum temperature limit because heat addition does not take place at constant temperature in Rankine cycle.



31. Rankine cycle efficiency of a steam power plant may be in the range of

(a) 15 to 20% (b) 35 to 45%
(c) 70 to 80% (d) 90 to 95%

APPSC Poly. Lect. 2013
UKPSC AE 2007 Paper-II

Ans. (b) : Rankine cycle efficiency of a steam power plant may be in the range of 35 to 45%.

32. Which of the following boiler accessories utilizes the waste heat of flue gases to heat the feed water?

(a) Water preheater (b) Economiser
(c) Feed heater (d) Superheater

Punjab PSC (Lect.) 06.08.2017
RPSC AEN Pre-2013

Ans. (b) : In boilers, economiser are heat exchanger devices that heat fluids, usually water, but not normally beyond the boiling point of that fluid. They are a device fitted to boiler which saves energy utilizes the waste heat of flue gases to heat the feed water.

33. Cooling tower is used to :

(a) cool the space or room
(b) cool the hot water coming from the condenser
(c) freeze the water
(d) increase the pressure of water

APPSC Poly Lect. 13.03.2020
GWSSB AAE, 27.12.2015

Ans. (b) : Cooling tower is used to cool the hot water which coming from the condenser.

34. The draught which a chimney produces is called-

(a) Induced draught (b) Natural draught
(c) Forced draught (d) Balanced draught

NPCIL ET, 2015, TNPSC AE 2014

Ans. (b) : Draught produced by chimney is called natural draught.

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35. In fire tube boilers, pressure is limited to

(a) 16 bar (b) 32 bar
(c) 48 bar (d) 64 bar

Nagaland PSC CTSE 2017 Paper-2
Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : In fire-tube boiler, pressure is limited to 16 bar.

36. Attempering is

(a) reduction of steam pressure
(b) reduction of steam temperature
(c) discharge of flue gases at certain height
(d) conditioning of disposable fluids for minimum pollution

Nagaland PSC CTSE 2015 Paper-II
OPSC Civil Services Pre. 2011

Ans. (b) : Attempering is reduction of steam temperature.

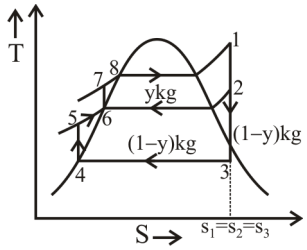
37. Consider an actual regenerative Rankine cycle with one open feed water heater. For each kg steam entering the turbine. If m kg steam with a specific enthalpy of h_1 is bleed from the turbine and the specific enthalpy of liquid water entering the heater is h_2, then h_3 specific enthalpy of saturated liquid leaving the heater is equal to -

(a) $mh_1 - (h_2 - h_1)$ (b) $h_1 - m\ (h_2 - h_1)$
(c) $h_2 - m\ (h_2 - h_1)$ (d) $mh_2 - (h_2 - h_1)$

CGPSC Asstt. Workshop Supt., 17.07.2016
TSGENCO AE 14.11.2015
GATE 1997

Ans. (c) : Open feed water heater (OFWH)

$$h_3 \times 1 = mh_1 + (1-m)h_2$$
$$= mh_1 + h_2 - mh_2$$
$$h_3 = h_2 - m(h_2 - h_1)$$



38. In a steam power plant, feed water heater is a heat exchanger to preheat feed water by:

(a) live steam from steam generator
(b) hot air from air preheater
(c) hot flue gases coming out of the boiler
(d) extracting steam from turbine

Nagaland PSC CTSE 2015 Paper-II
HPPSC Asstt. Prof. 2014

Ans. (d) : Feed water heaters are heat exchangers designed to preheat feed water by means of condensing the extract steam from turbine.
A feed water, heater, is a power plant component used to pre-heat water delivered to a steam generating boiler.

39. Equivalence evaporation is defined as

(a) the ratio of heat actually used in producing steam to the heat liberated in the furnace
(b) the amount of water evaporated in kg per kg of coal burnt
(c) evaporation of water from and at 100ºC into dry saturated steam
(d) evaporation of 15.653 kg of water per hour from and at 100ºC

OPSC AEE 2019 PAPER - II
BPSC AE Mains 2017 Paper - V

Ans. : (c) : Equivalent Evaporation– It is the amount of water evaporated from feed water at 100°C at normal atmospheric pressure (1.01 bar). It is usually written as "from at 100°C".
Mathematically equivalent evaporation from and at 100°C.

$$E = \frac{m_c\left(h - h_{f_1}\right)}{2257}$$

Where,
m_C = Mass of water actually evaporated or steam produced in kg/hr or kg/kg of fuel burnt.
h = Total heat of steam in kJ/kg of steam and
h_{f_1} = Sensible heat of feed water in kJ/kg of steam.

40. The ratio of energy produced by a power plant to the installed capacity of plant is called as:

(a) Load factor (b) Use factor
(c) Average load factor (d) Demand factor

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II

Ans. : (b) : Plant use factor- It is the ratio of energy produced in a given time to maximum possible energy that could be produced during actual number of hours of operation.

$$\text{Plant use factors} = \frac{\text{Annual Production of energy}}{\text{Operational hours in year} \times \text{Capacity of plant}}$$

41. In a power plant, the efficiency of the electric generator, turbine, boiler, cycle and the overall plant are 0.97, 0.95, 0.92, 0.42 and 0.33 respectively. In the generated electricity, the auxiliaries will consume nearly

(a) 7.3% (b) 6.5%
(c) 5.7% (d) 4.9%

ESE 2020, 2019

Ans. (a) :

$$\eta_{overall} = \eta_{boiler} \times \eta_{turbine} \times \eta_{cycle} \times \eta_{generator} \times \eta_{auxiliaries}$$
$$0.33 = 0.92 \times 0.95 \times 0.42 \times 0.97 \times \eta_a$$
$$\eta_{aux} = \frac{0.33}{0.92 \times 0.95 \times 0.42 \times 0.97} = 0.9267$$

Power consumed (%) = $1 - \eta_a = 1 - 0.9267$
$= 0.0733$
$= 7.33\%$

42. The efficiency of chimney is approximately

(a) 80% (b) 40%
(c) 20% (d) 0.25%

(KPSC AE 2015), TNPSC AE 2014

Ans. (d) : The efficiency of chimney is approximately 0.25%

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43. Induced draught fans of a large steam power plant has

(a) forward curved blade
(b) backward curved blade
(c) radial blade
(d) double curved blade
(e) straight blade

CGPSC AE 26.04.2015 Shift-II, ESE 1996

Ans. (a) : Induced draught fans of a large steam power plant has forward curved blade.

44. Fusible plug is fitted in a boiler to

(a) extinguish fire in case of low water level
(b) prevent leakage of steam from boiler
(c) allow passage of only super heated steam from the boiler
(d) keep the boiler pressure within the prescribed limits

(KPSC AE 2015), TSPSC AEE 2015

Ans. : (a) Fusible plug is fitted in a boiler to extinguish fire in case of low water level. Fusible plug:- It is fitted to the crown plate of the furnace of the fire box. Its object is to put off the fire in the furnace of the boiler when the level of water in the boiler falls to an unsafe limit and thus avoids the explosion which may take place due to overheating of the furnace plate.

45. For the safety of a steam boiler the number of safety valves fitted are

(a) One (b) Two
(c) three (d) Four

RPSC AE 2016, RPSC ACF-2011

Ans. : (b) For the safety of a steam boiler the number of safety valves and water level indicator fitted are two.
These are the devices attached to the steam chest for preventing explosions due to excessive internal pressure of steam. The following are the four types of safety valves.
(i) Lever safety valve (ii) Dead weight safety valve (iii) High steam and low water safety valve (iv) spring loaded safety valve.

46. Match list-I with list-II select the correct answer using the codes given below the list:

List -I	**List -II**
(a) Lancashire	1. High pressure water tube
(b) Cornish	2. Horizontal double fire tube
(c) La-mont	3. Vertical multiple fire tube
(d) Cochran	4. Low pressure inclined water tube
	5. Horizontal single fire tube

Code :

	A	**B**	**C**	**D**
(a)	2	5	1	3
(b)	2	4	3	1
(c)	1	5	2	3
(d)	5	4	1	3

UJVNL AE 2016, ESE 1999

Ans. (a) :

List -I	**List -II**
Lancashire	Horizontal double fire tube
Cornish	Horizontal single fire tube
La-mont	High pressure water tube
Cochran	Vertical multiple fire tube

47. Size of boiler tubes is specified by

(a) Mean diameter and thickness
(b) Outside diameter and thickness
(c) Insider diameter and thickness
(d) Outside diameter and inside diameter

Nagaland PSC CTSE 2017 Paper-2
GWSSB AAE, 27.12.2015

Ans. (b) : Size of boiler tubes is specified by outside diameter and thickness.

48. Thermal efficiency of steam turbine can be increased by :

(a) Reheating
(b) Regeneration
(c) Increasing the inlet pressure
(d) All of the above

MPPSC AE 08.11.2015
APGENCO AE 2012

Ans. (d) : Thermal efficiency of steam turbine can be increase by–
– Reheating
– Regeneration
– Inlet pressure (P_1)↑ increasing

49. In a glass tube type water indicator for a boiler, one end of the tube is connected to water space and the other end is connected to

(a) water space also (b) chimney
(c) steam space (d) superheater

Sikkim PSC (Under Secretary), 2017
GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : The water of the boiler comes into the glass tube through the lower tube and the steam through the upper tube. The water then stands in the glass tube at the same level as in the boiler. Two cocks are used to control the passage of between the boiler and the glass tube while the third cock is used to discharge some of the water from inside the boiler to see whether the gauge is in proper order or not. The glass tube is protected by means of a cover, made of specially toughened glass, which will prevent any accident that may happen due to breaking of glass tube. It is used for ordinary boiler.

50. A device used to increase the temperature of saturated steam without raising its pressure, is called

(a) blow of cock (b) fusible plug
(c) superheater (d) stop value

OPSC AEE 2019 PAPER - II
Sikkim PSC (Under Secretary), 2017
TNPSC ACF 2012

Ans. (c) : • A device used to increase the temperature of saturated steam with out raising it's pressure is called superheater.

- Superheater is a boiler accessory used to increase the steam temperature without effecting it's pressure. This is done by double heating of steam.
- In superheater gases is passes over the super heater tube and remainder will pass directly to the bottom tube.

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51. The equivalent evaporation rate (kg/hr) of a boiler producing 2000 kg/hr of steam with enthalpy content of 2426 kJ/kg from feed water at temperature 40 deg. C (liquid enthalpy = 168 kJ/kg and enthalpy of vaporization of water at 100 deg. C = 2258 kJ/kg) is :

(a) 2000 (b) 2249
(c) 1658 (d) 1649

ISRO Scientist/Engineer (RAC), 10.03.2019
GPSC ARTO 01.05.2016, GATE 1993

Ans. (a) : Given,
mass of steam (m) = 2000 kg/hr.
$h_1 = 2426$ kJ/kg
$t_1 = 40°C$
$h_{feed} = 168$ kJ/kg

$$\left(h_{fg}\right)_{water}|_{100°C} = 2258\,kJ/kg$$

Equivalent evaporation $\dfrac{\left(h_1 - h_{feed}\right)}{h_{fg}} \times m$

$$= \frac{2000(2426-168)}{2258}$$

= 2000 kg/hr.

52. A pump handling a liquid raises its pressure from 1 bar to 30 bar. Take the density of the liquid as 990 kg/m^3. The isentropic specific work done by the pump in kJ/kg is

(a) 0.10 (b) 0.30
(c) 2.50 (d) 2.93

VIZAG STEEL MT 18.08.2013, GATE 2011

Ans. (d) : $P_1 = 1$ bar, $P_2 = 30$ bar, $\rho = 990$ kg/m^3

Isentropic specific work (w) $= \dfrac{\Delta P}{\rho} = \dfrac{(30-1)}{990} \times 100$

done by pump $\boxed{w = 2.93\ kJ/kg}$

53. In a pulverized-fuel-fired large power boiler, the heat transfer from the burning fuel to the walls of the furnace is

(a) by conduction only
(b) by convection only
(c) by conduction and convection
(d) predominantly by radiation

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 1999

Ans. (d) : Predominantly by radiation

54. Economizer in the boiler heats the

(a) steam (b) air
(c) feed water (d) coal

Assam PSC AE (IWT) 14.03.2021
TNPSC AE (Industries) 09.06.2013

Ans. (c) : A common application of economizers in steam power plants is to capture the waste heat from boiler flue gases and transfer it to the boiler feed water. This raises the temperature of the boiler feed water, lowering the needed energy input, in turn reducing the firing rates needed for the rated boiler output.

55. De-aeration of feed water is carried out because it reduces :

(a) cavitation of boiler feed pumps
(b) corrosion caused by oxygen
(c) heat transfer coefficient
(d) pH value of water

HPPSC W.S. Poly. 2016
J&K PSC Civil Services Pre, 2013

Ans. (b) : De-aeration is the process to remove dissolved gases (particularly oxygen) in feed water to reduce corrosion caused by oxygen.

56. Water (C_p = 4 kJ/kgK) is fed a boiler at 30°C, the enthalpy of vaporization at atmospheric pressure in the boiler is 2400 kJ/kg; the steam coming from the boiler is 90% dry. What is the net heat supplied in the boiler?

(a) 2160 kJ/kg (b) 24 kJ/kg
(c) 2440 kJ/kg (d) 2280 kJ/kg

HPPSC Asstt. Prof. 2014
ISRO Scientist/Engineer (RAC) 12.01.2020
ESE 2009

Ans. (c) : Heat supplied for a substance undergoing phase change of $Q = m[C_p dT + x \cdot h_{fg}]$
$= 1[4 \times (100 - 30) + 0.9 \times 2400]$
$= 280 + 2160$
$= 2440$ kJ/kg

57. Consider the following statements:

1. **Fusible plug is a device used to put off fire in the furnace of the boiler when the level of water in the boiler fails to an unsafe limit.**
2. **Super heater is a device used to increase the temperature of saturated steam without raising its pressure.**
3. **An economizer decreases the steam raising capacity of boiler.**

Which of the above statements is/are correct?

(a) 1 and 2 (b) 2 and 3
(c) 1 only (d) 1, 2 and 3

UPRVUNL AE 21.08.2016, CSE Pre-2005

Ans. : (a)
- Super heater is a device used to increase the temperature of saturated steam without raising its pressure.
- Fusible plug is a device used to put off fire in the furnace of the boiler when the level of water in the boiler fails to an unsafe limit.

58. The efficiency of regenerative rankine cycle is higher than simple rankine cycle because

(a) total work delivered by the turbine increases
(b) heat is added before steam enters the low pressure turbine
(c) average temperature of heat addition in the boiler increases
(d) total work delivered by the turbine decreases
(e) pressure inside the boiler increases

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
CGPSC AE 26.04.2015 Shift-II, GATE 2003

Ans. (c) : The efficiency of regenerative Rankine cycle is higher than simple Rankine because average temperature of heat addition in the boiler increase.

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59. Which of the following is the boiler accessory?

(a) Blow - off cock
(b) Pressure gauge
(c) Water level indicator
(d) Economiser

APPSC Poly Lect. 13.03.2020
TSPSC AEE 2015

Ans. : (d) Boiler Accessory– Boiler Accessory are those device which increase the efficiency of the boiler.

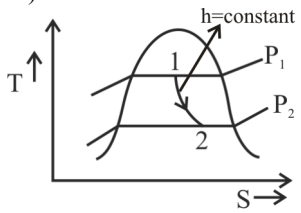
(i) Superheater (ii) Economiser
(iii) Air Preheater (iv) Feed pump.
(v) Injector (vi) Steam Separator
(vii) Steam Trap

60. When wet steam flows through a throttle valve and remains wet at exit:

(a) its temperature and quality increases
(b) its temperature decreases but quality increases
(c) its temperature increases but quality decreases
(d) its temperature and quality decreases

HPPSC Asstt. Prof. 2014
GATE 1996

Ans. (b) : Throttling is a perfect example of isenthalpic process. So, when wet steam flows through a throttle valve and remains wet at exit then its temperature decreases, entropy increases and quality of steam (dryness fraction) increases.



Note-Answer (b) is correct. but in Himachal PSC, (d) is given as correct answer.

61. Which one of the following boiler accessories does not need 'flue-gas' for its operation?

(a) Economizer (b) Preheater
(c) Injector (d) Super heater

Vizag Steel MT (Re-Exam) 24.11.2013
ESE 2007

Ans. (c) : Economiser, preheater and superheater are boiler accessories. They need 'flue gas' to their operation so that they can increase the efficiency.

62. Which device is used in thermal power plants to reduce level of pollution ?

(a) induced draft fan (b) precipitator
(c) chimney (d) pulveriser

GWSSB DEE 07.07.2016
J&K PSC Civil Services Pre, 2013

Ans. (b) : Induced fan – Provide negative pressure in the furnace.
Precipitator -device for removing dust from gas/smoke.
Chimney – Ventilation structure that isolates hot toxic exhaust gases or smoke from human living areas.
Pulveriser – Device used to crush the coal.

63. For a given set a operating pressure limits of a Rankine cycle, the highest efficiency occurs for

(a) saturated cycle (b) super heated cycle
(c) reheat cycle (d) regenerative cycle

TRB Poly. Lect., 2012, GATE 1994

Ans. (d) : For a given set of operating pressure limits of a Rankine cycle the highest efficiency occurs for regenerative cycle. In regenerative rankine cycle thermal efficiency is increased since the mean temperature of heat addition to the cycle is increased.

64. In steam power station, the choice of high temperature steam is for

(a) increasing the efficiency of boiler
(b) increasing the efficiency of turbine
(c) increasing the efficiency of condenser
(d) increasing the overall efficiency

RPSC AE 2016, RPSC ACF-2011

Ans. (d) : The binary vapor cycle is the combination of two cycle. One operates in the high temperature region, another operates in the low temperature. Thus the plant efficiency improvement in thermal power. Use of steam for heating of boiler feed water.

65. Presence of moisture in fuel oil would

(a) keep the burner tips cool
(b) aid in proper combustion
(c) cause sputtering, possibly extinguishing flame
(d) clean the nozzles

RPSC IOF, 2020, BPSC AE 2012 Paper - V

Ans. (c) : Presence of moisture in fuel oil would because sputtering, possibly extinguishing flame.

66. The boiler mounting used to empty the boiler drum when necessary for cleaning, inspection and repair is _____.

(a) Blow off cock (b) safety valve
(c) steam stop valve (d) fusible plug

VIZAG STEEL MT 24.01.2021, Shift-II, Shift-I

Ans. (a) : The boiler mounting used to empty the boiler drum when necessary for cleaning, inspection and repair is blow off cork.
Boiler mountings are a safety equipment installed for safe boiler operations and efficient working.

67. A high pressure drumless water tube steam boiler using forced circulation is _____.

(a) Benson Boiler
(b) La Mont Boiler
(c) Babcock and Wilcox Boiler
(d) Loeffler Boiler

VIZAG STEEL MT 24.01.2021, Shift-II, Shift-I

Ans. (a) : Benson Boiler–

- Benson boiler is high pressure drumless supercritical, water tube steam boiler with forced circulation.
- The bubbles do not form because at supercritical pressure the density of water and steam becomes the same.

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68. _____ **can be measured only by throttling calorimeter.**

(a) Dryness fraction of high- pressure stream
(b) Very high dryness fraction of up to 0.98
(c) Dryness fraction of low pressure stream
(d) Very low dryness fraction of up to 0.4

UPRVUNL AE 05.07.2021

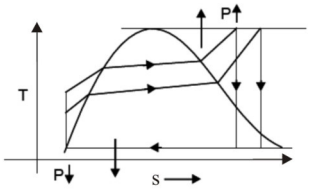
Ans. (b) :

69. **The efficiency of any cycle increases with**

(a) the decrease of maximum pressure and the constant of exhaust pressure.
(b) the decrease of maximum pressure and the decrease of exhaust pressure.
(c) the increase of maximum pressure and the decrease of exhaust pressure
(d) the increase of maximum pressure and the constant of exhaust pressure

ESE 18.07.2021

Ans. (c) :



When the maximum pressure increases and decreases the exhaust pressure then mean temperature of heat addition increases and mean temperature of heat rejection decreases. Thus, efficiency of cycle increase.

70. **In a power plant, the efficiencies of the electric generator, turbine (mechanical), boiler, cycle and the overall plant are 0.97, 0.95, 0.92, 0.42 and 0.33, respectively. What is the efficiency of auxiliaries?**

(a) 98.14% (b) 92.68%
(c) 83.41% (d) 75.14%

ESE 18.07.2021

Ans. (b) : Given,

$\eta_g = 0.97$, $\eta_m = 0.95$, $\eta_{boiler} = 0.92$

$\eta_{isen} = 0.42$, $\eta_{overall} = 0.33$,

$$\eta_{overall} = \eta_g \times \eta_m \times \eta_{boiler} \times \eta_{aux}$$

$$\eta_{aux} = \frac{0.33}{0.97 \times 0.95 \times 0.92 \times 0.42}$$

$$\approx 92.68\%$$

71. **Consider the following statements for analysis of steam cycles:**

(1) A steam power plant continuously converts the energy stored in fossil fuels or fissile fuels into shaft work.
(2) Steam power plants work on Brayton cycle
(3) In supercritical steam cycle, steam is generated in a 'once-through' boiler at a pressure above the critical point of 27.5 bar.
(4) Deaerator is used for the purpose of deaerating the feedwater

Which of the above statements are correct?

(a) 2 and 3 only (b) 1 and 3 only
(c) 1 and 4 only (d) 2 and 4 only

ESE 18.07.2021

Ans. (c) :

- Steam power plant works on Rankine cycle.
- A steam power plant continuously converts the energy stored in fossil fuels or fissile fuels into shaft work.
- Deaeratar is used for the purpose of removal of air (deaerating) the feed water entering to the boiler.
- Critical point of water is 22.12 MPa or 220 bar and 373.14°C.

72. **Consider the following statements for fluidized bed boilers:**

(a) Fluidized bed boilers produce steam from fossil and waste fuels by using a technique called fluidized bed combustion
(b) Cyclone separators are gas cleaning devices that utilize the centrifugal force created by a spinning gas stream to separate particles from a gas.
(c) In a pressurized fluidized bed boiler, the combustion process takes place in a pressurized environment resulting in a compact furnace and improved combustion efficiency.

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (d) : All statements are correct as per their definition.

73. **In a closed feed water heater used in a regenerative cycle, the temperature of the feed water leaving a heater must be:**

(a) lesser than the saturation temperature at the steam extraction pressure.
(b) higher than the saturation temperature at the steam extraction pressure.
(c) equal to the saturation temperature at the steam extraction pressure.
(d) equal to or higher than the saturation temperature at the steam extraction pressure.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : In a closed feed water heater used in a regenerative cycle, the temperature of the feed water leaving a heater must be lesser than the saturation temperature at the steam extraction pressure.

Their difference is known as terminal temperature difference of heater.

74. **A good steam power plant having a Rankine cycle may have efficiency in the range of _____.**

(a) 35-45 percent (b) 70–80 percent
(c) 5–20 percent (d) 90-95 percent

UPRVUNL AE 05.07.2021
UKPSC AE 2007 Paper -II

Ans. (a) : A good steam power plant having a Rankine cycle may have efficiency in the range of 35-45 percent.

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75. The thermal efficiency of Rankine cycle is increased by

(a) Decreasing the pressure and volume at which mass is added to the cycle
(b) Decreasing the temperature at which mass is added to the cycle
(c) Increasing the average temperature at which heat is added to the cycle
(d) Increasing the pressure and volume at which is added to the cycle

HPPSC Workshop Suptd. 08.07.02021

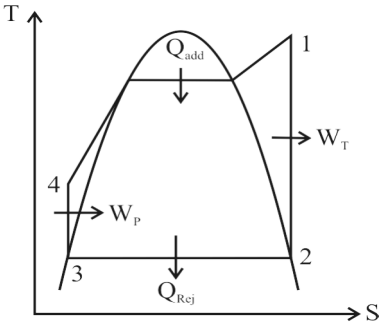
Ans. (c) : The thermal efficiency of Rankine cycle is increased by increasing the average temperature at which heat is added to the cycle.

76. Rankine cycle comprises

(a) Two isentropic processes and two constant volume processes
(b) Two isentropic processes and two constant pressure processes
(c) Two isothermal processes and two constant pressure processes
(d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b)



Process 1-2 Isentropic expansion.
Process 2-3 Constant pressure heat rejection.
Process 3-4 Pump work.
Process 4-1 Constant pressure heat addition.

77. The capacity of steam plant is expressed in terms of steam rate. Its units will be

(a) kg/h (b) kg/kWh
(c) kWh/kJ (d) kW/kg

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Steam Rate $= \frac{3600}{W_{net}}\left(\frac{kg}{KWh}\right)$

78. With increase of mean temperature of heat addition, the efficiency of Rankine vapour power cycle will

(a) decrease
(b) independent of mean temperature of heat addition
(c) increase
(d) none of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

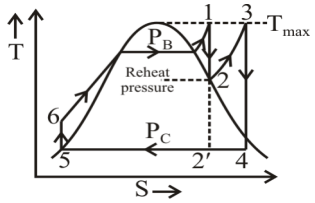
Ans. (c) : $\uparrow \eta = 1 - \frac{T_L}{T_{MA}\uparrow}\downarrow$

79. With reheating the rankine cycle efficiency will

(a) increase depending on maximum pressure
(b) decrease depending on minimum pressure
(c) no change
(d) may increase or decrease depending on mean temperature of heat addition

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) :



- W_T-↑, W_P-Constant, W_{net}-↑
- Q_s-↑, Q_R-↑, T_{MA}-↓↑, η-↑↓ (Can't say)
- $\downarrow \text{Specific Steam Consumption} = \frac{3600}{W_{net}\uparrow}$
- x-↑, ↓ errosion of turbine blade

80. The ratio of thermal power discharged to environment to electrical power output is called

(a) collector efficiency
(b) particulate efficiency
(c) pollution efficiency
(d) Thermal discharge index

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : Thermal discharge index

81. The tariff method which charges the consumer according to his maximum demand and energy consumption is

(a) Hopkinson demand rate method
(b) Doherty rate method
(c) Wright demand rate method
(d) Block meter rate

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Hopkinson rate is most common method of pricing electricity for industrial use. It consist of an energy charge for total kilowatt hour consumption plus an additional demand charge based on maximum usage by the plant during any quarterly hour period during the month.

82. The maximum demand of a power station is 96000 kW and the load is seven intervals of 6 hrs, 2 hrs, 4 hrs, 2 hrs, 4hrs, 4 hrs, 2 hrs each measured in MW was 48,60,72,60, 84, 96 and 48. The load factor will be

(a) 0.71 (b) 0.6
(c) 0.8 (d) 0.54

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Given, Load and time–

Time(hr)	6	2	4	2	4	4	2
Load(MW)	48	60	72	60	84	96	48

Energy generated = 48 × 6 + 60 × 2 + 72 × 4 + 60 × 2 + 84 × 4 + 96 × 4 + 48 × 2

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$= 1632$ MWh

$$\text{Load factor} = \frac{\text{Average load}}{\text{Peak load}}$$

$$\text{Average load} = \frac{1632}{24} = 68\,\text{MW}$$

Maximum demand = 96000 kW = 96 MW

$$\text{Load factor} = \frac{68}{96} = 0.71$$

83. Which of the following statements are correct regarding the Rankine cycle?
For a given set of operating pressure limits of a Rankine cycle, superheating the steam beyond the saturation temperature at the boiler pressure will result in:
(1) an increase in the moisture content of steam leaving the turbine
(2) an increase in thermal efficiency of the Rankine cycle
(3) an increase in the mean temperature of heat addition
(4) a decrease in the amount of heat rejected in the condenser
(a) (1) and (4) (b) (3) and (4)
(c) (1) and (2) (d) (2) and (3)

AAI Jr. Executive 26.03.2021

Ans. (d) : For a given set of operating pressure limits of a Rankine cycle, superheating the steam beyond the saturation temperature at the boiler pressure result in–
(a) Decrease in the moisture content and increases quality of steam leaving the turbine.
(b) Increase in thermal efficiency of rankine cycle.
(c) An increase in mean temperature of heat addition.

84. Which one of the boiler is not a high pressure boiler?
(a) Velox boiler (b) Lamont boiler
(c) Cochran boiler (d) Loeffler boiler

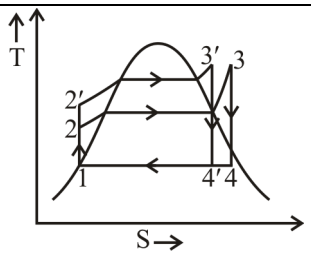
VIZAG STEEL MT 24.01.2021, Shift-II

Ans. (c) : A boiler is called a high pressure boiler when it operates with a steam pressure above 80 bar. e.g. Velox boiler, Lamont boiler, Loeffler boiler.

85. As the boiler pressure increases :
(a) the heat absorbed in the evaporator increases.
(b) the friction of the total heat absorbed in the superheater increases
(c) the latent heat of vaporization increases
(d) for the given superheated steam temperature, the quality of steam at turbine exhaust decreases.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : As the boiler pressure increases for the given superheated steam temperature the quality of steam at turbine exhaust decreases.
As the boiler pressure increases mean temperature of heat addition increases due to which efficiency of rankine cycle increases but at turbine exhaust quality of steam decreases to avoid this reheat cycle is adopted. Quality of steam should not be below 85% to avoid corrosion of turbine blade.



- W_P-↑
- W_T-↑↑, W_{net}-↑, η-↑, x-↓, Moisture-↑
- ↓Errosion of turbine blade

$$x(\text{dryness fraction}) = \frac{m_v}{m_v + m_a}$$

86. Which type of Power Plant is preferred as peak load type ?
(a) Nuclear (b) Gas
(c) Hydraulic (d) Thermal

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : A list of the most popular peak load power plants–
- Gas
- Solar
- Wind turbines
- Diesel generators

Other, base load power plant are–
1. Coal, 2. Nuclear, 3. Hydroelectric, 4. Geothermal

87. Power plant has load factor as 0.6, capacity factor as 0.4, maximum demand as 20 MW. The average load and reserve capacity over and above peak load is :
(a) 30 MW, 12 MW (b) 30 MW, 10 MW
(c) 12 MW, 20 MW (d) 12 MW, 10 MW

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : Given,
Load factor = 0.6
Capacity factor = 0.4
Maximum demand = 20 MW

$$\text{Load factor} = \frac{\text{Average load}}{\text{Maximum demand}}$$

Average load = Maximum demand × Load factor
$= 20 \times 0.6$
$= 12$ MW

$$\text{Plant capacity} = \frac{\text{Average load}}{\text{Capacity factor}} = \frac{12}{0.4} = 30\,\text{MW}$$

Reserve capacity = Plant capacity – Maximum demand
$= 30 - 20 = 10$ MW

88. Compared to conventional sources of power such as thermal plants, the cost of wind power is :
(a) 100 times (b) 10 times
(c) comparable (d) 1/10 times

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : Compared to conventional sources of power such as thermal plants, the cost of wind power is 1/10 times.

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89. A thermal power plant of 100 MW capacity uses coal of 25 MJ/kg colorific value. If thermal efficiency of power plant is 40% and generator efficiency is 90%. What will be approximate coal consumption per hr when plant run at full load?

(a) 11.11 tons/hr (b) 11.11 kg/hr
(c) 40.0 tons/hr (d) 40.0 kg/hr

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : Given,
Plant capacity = 100 MW
Coal calorific value = 25 MJ/kg

$$\eta_{th} = 40\%$$
$$\eta_g = 90\%$$
$$\eta_0 = 0.40 \times 0.90$$
$$\eta_0 = 0.36$$
$$\eta_0 = \frac{100\text{MW}}{25 \times \dot{m}}$$
$$\dot{m} = \frac{100}{25 \times 0.36}$$
$$\dot{m} = 11.11\,\text{kg/sec}$$
$$\dot{m} = \frac{11.11}{1000} \times 3600\,\text{tons/hr}$$
$$\boxed{\dot{m} = 40\,\text{tons/hr}}$$

90. Compared to a conventional steam plant, the temperature and pressure in a geothermal plant are:

(a) comparable (b) much higher
(c) higher (d) lower

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : In geothermal power plant pressure and temperature is low as compare to conventional steam power plant.

91. Which is not a correct option for thermo-electric solar power system?

(a) Tower type solar plant
(b) Concentrator collector solar plant
(c) Flat plate collector solar plant
(d) Photo –voltaic solar plant

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : Correct statements for thermo solar power systems,
- Concentrator collector solar plant
- Flat plate collector solar plant
- Photovoltaic solar plant

92. Compared to a conventional steam plant, the efficiency of geothermal plant is :

(a) very high (b) higher
(c) comparable (d) lower

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : The thermal efficiency of geothermal electric station is low, around 7-10%, because geothermal fluids are at low temperature compared with steam from boilers.

93. The expression $\Delta h = v\Delta p$ is utilised by which component of the steam power plant cycle?

(a) Boiler (b) Turbine
(c) Condenser (d) Pump

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

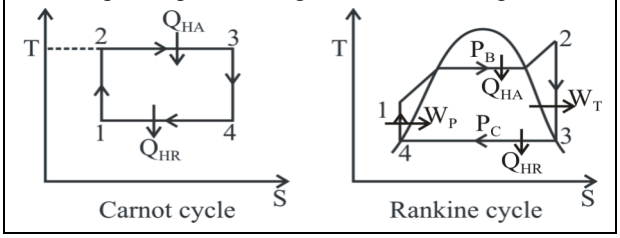
Ans. (d) : The expression $\Delta h = v\Delta p$ is utilised in pump of steam power plant.

94. Carnot cycle is different from Rankine Cycle in steam power plant during the following process:

(a) Heat Addition (b) Expansion Work
(c) Heat Rejection (d) Pump Work

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : Carnot cycle is different from Rankine Cycle in steam power plant during the heat addition process.

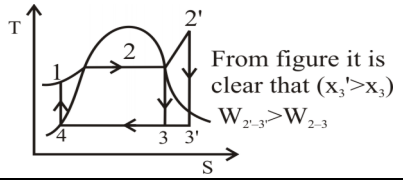


95. Which one of the following modifications of the simple ideal Rankine cycle increases the thermal efficiency and reduces the moisture content of the steam at the turbine outlet?

(a) Increasing the turbine inlet temperature
(b) Increasing the boiler pressure
(c) Decreasing the boiler pressure
(d) Decreasing the condenser pressure

Assam Engg. College AP/Lect. 18.01.2021

Ans. (a) : As temperature at inlet of turbine increases then thermal efficiency increases and moisture content reduces.



96. Cornish boiler is

(a) Fire tube boiler (b) High pressure boiler
(c) Water tube boiler (d) Locomotive boiler

JPSC AE, 10.04.2021, Paper-I

Ans. (a) : Cornish boiler is type of fire tube boiler.

Fire Tube Boiler	**Water Tube Boiler**
Locomotive	Babcock & Wilcox
Cornish	LaMont
Cochran	Velox
Lancashire	Benson

97. In locomotive boiler, the maximum pressure is of the order of

(a) 5 kg/cm^2 (b) 10 kgf/cm^2
(c) 20 kgf/cm^2 (d) 100 kgf/cm^2

JPSC AE, 10.04.2021, Paper-I

Ans. (c) : In locomotive boiler, the maximum pressure is of the order of 20 kgf/cm^2.

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98. Which of the following as referred to steam boiler are defined as mountings?

(a) Superheater (b) Economiser
(c) Preheater (d) Fusible plug

JPSC AE, 10.04.2021, Paper-I

Ans. (d) : Fusible plug

Boiler mountings – are safety device

- Water level indicator
- Pressure gauge
- Safety valve
- Fusible plug
- Blow off cock
- Feed check valve

Boiler accessories – for enhancing the performance

- Injector
- Economiser
- Air preheater
- Superheater
- Steam separator
- Steam trap

99. The work output from the turbine in a Rankine cycle is given by the change of ____ between inlet and outlet.

(a) Temperature (b) Entropy
(c) internal energy (d) enthalpy

UPRVUNL AE 05.07.2021

Ans. (d) : The work output from the turbine in a Rankine cycle is given by the change of enthalpy between inlet and outlet.

100. If the mass of a total steam-water mixture is 55 kg and the steam vapour is 5 kg, then the dryness fraction is:

(a) 0.09 (b) 0.10
(c) 0.08 (d) 0.06

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : Dryness fraction $x = \frac{m_v}{m_\ell + m_v}$

Mass of vapour $m_v = 5$ kg

Mass of mixture $(m_\ell + m_v) = 55$ kg

So $\boxed{x = \frac{5}{55} = 0.09}$

101. In a solar tower power system, each mirror is mounted on a system called

(a) Regenerator (b) Linear Fresnel
(c) Dish (d) Heliostat

ESE 2020

Ans. (d) : Heliostat

102. The partial vacuum created by the fan in the furnace and flues, draws the products of the combustion from the main flue and allows them to pass up to the chimney. Such a draught is called

(a) Balanced draught (b) Forced draught
(c) Induced draught (d) Artificial draught

ESE 2020

Ans. (c) Induced draught

103. The higher power requirements for compression in a steam power plant working on Carnot vapour cycle

(a) Increases the plant efficiency as well as work ratio
(b) Reduces the plant efficiency as well as work ratio
(c) Does not affect the plant efficiency as well as work ratio
(d) Increases the plant efficiency and reduces work ratio

ESE 2020

Ans. (b) : Reduces the plant efficiency as well as work ratio.

104. Reheating should be carried out

(a) before the HP steam is first introduced to the turbine
(b) before the LP steam is drained to the condenser
(c) at an optimum pressure
(d) at any stage, pressure does not matter

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : Reheating should be carried out at an optimum pressure.

105. A steam ejector which removes air and other non-condensable gases from the condenser is known as

(a) Wet air pump (b) Dry air pump
(c) Centrifugal pump (d) Circulating pump

ESE 2020

Ans. (b) : Dry air pumps are used to remove air and other non-condensable gases.

106. The process of abstracting steam at a certain section of the turbine and subsequently using it for heating feed water supplied to the boiler is called

(a) Reheating (b) Regeneration
(c) Bleeding (d) Binary vapour cycle

ESE 2020

Ans. (b) : Regeneration
Note : As per UPSC (c) is correct answer.

107. The actual thermal efficiency of a cycle is ..... the air standard efficiency

(a) equal to (b) less than
(c) more than (d) not comparable to

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : It happens due to various irreversibilities in real engine that hinder then to achieve efficiency because before draw a p-v curve of air standard cycle we assume that all processes are internally reversible and working fluid is an ideal gas. But in actual practice in every operation in real engine have finite losses like heat losses friction losses pumping losses, losses at inlet and outlet valves. Time losses because fuel take little bit time to ignite so actual efficiency differs from an ideal one.

108. The India Boilers Act _____ was enacted with the objective to provide mainly for the safety of life and property of persons from the danger of explosions of steam boilers and for achieving uniformity in registration and inspection during operation and maintenance of boilers in India.

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(a) 1903 (b) 1914
(c) 1918 (d) 1923

RPSC IOF, 2020

Ans. (d) : The Indian boilers act 1923.

109. Out of the following impurities generally found in water, which one requires special consideration in case of very high pressure boiler?

(a) Hydrogen (b) Ammonia
(c) Silica (d) Dissolved salt

RPSC IOF, 2020

Ans. (c) : Silica requires special consideration in case of very high pressure boilers.

110. Locomotive boiler is a

(a) Single tube, horizontal, internally fired and stationary boiler
(b) Single tube, vertical, externally fired and stationary boiler
(c) Multi-tubular, horizontal, internally fired and mobile boiler
(d) Multi-tubular, horizontal, externally fired and stationary boiler

RPSC IOF, 2020

Ans. (c) : Locomotive boilers is multi tubular, horizontal internally fired and stationary boiler. Locomotive boiler is a type of boiler that is used to heat up the water and create steam from this water. This boiler sits horizontally just like Lancashire boiler and Cornish boiler.

111. A turbine generator unit has output of 150 MW and efficiency of 0.80. Calculate energy supplied per hour by steam generator.

(a) 675×10^9 J (b) 576×10^9 J
(c) 605×10^9 J (d) 705×10^9 J

RPSC IOF, 2020

Ans. (a) : Given,

Output = 150 MW

$= 150 \times 10^6 \times 3600$ J/hour

$\eta_T = 0.80$

$$\eta_T = \frac{\text{output}}{\text{energy supplied/h}}$$

$$0.80 = \frac{150\times10^6\times3600\,\text{J/h}}{\text{energy supplied per hour}}$$

Energy supplied per hour $= 6.75 \times 10^{11}$ J

$= 675 \times 10^9$ J

112. In most steam power plants, closed feed water heaters are favoured, but one open heater is used

(a) To reduce cost
(b) To achieve better heat transfer
(c) To provide high feed water temperature
(d) To deareate the feed water by removing the dissolved gas

RPSC IOF, 2020

Ans. (d) : Open feed water heater works as a dearator. To deareate the feed water by removing the dissolved gas.

113. These tubes in a natural circulation water tube boiler are smaller in number and bigger in diameter

(a) Downcomers
(b) Risers
(c) Both downcomers and risers
(d) Superheaters

RPSC IOF, 2020

Ans. (a) : Downcomers- A pipe to conduct something downwards. These tubes in a natural circulation water tube boiler are smaller in number and bigger in diameter.

114. A boiler evaporates 4 kg of water per kg of coal into dry steam at 10 bar (h_g = 2776.2 kJ/kg). The feed water temperature is 32°C (h_{fg} = 134 kJ/kg). The factor of evaporation is _____.

(a) 1.17 (b) 2.34
(c) 1.76 (d) 0.59

VIZAG MT, 14.12.2020

Ans. (a) : m = 4 kg

$h_g = 2776.2$ kJ/kg

$h_{fg} = 134$ kJ/kg

$$\text{Factor of evaporation } (F_e) = \frac{h_g - h_{fg}}{2257}$$

$$= \frac{2776.2 - 134}{2257}$$

$$\boxed{F_e = 1.17}$$

115. Which one is NOT a reason behind the fact that the ideal regenerative cycle is practically not possible?

(a) The reversible heat transfer takes infinite time
(b) It is mechanically impossible to exchange heat in the turbine
(c) The moisture content in the turbine will be high
(d) The steam specific volume will be too high

UPPSC AE 13.12.2020, Paper-2

Ans. (d) : The ideal regenerative cycle is practically not possible because–

(i) The moisture content in the turbine will be high. So, erosion of turbine blade takes place.
(ii) It is mechanically impossible to exchange heat in the turbine
(iii) The reversible heat transfer takes infinite time.

116. If η_1 and η_2 are the thermal efficiencies of two individual power plants. When they are coupled in series, the overall thermal efficiency η_O of the combined plant is given by

(a) $\eta_O = \eta_1\eta_2$ (b) $\eta_O = \eta_1 + \eta_2$
(c) $\eta_O = \eta_1 + \eta_2 - \eta_1\eta_2$ (d) $\eta_O = \dfrac{\eta_1 + \eta_2}{\eta_1\eta_2}$

UPPSC AE 13.12.2020, Paper-2

Ans. (c) : η_1 and η_2 are the thermal efficiencies of two individual power plants,

$$\boxed{(\eta_O)_{\text{series}} = \eta_1 + \eta_2 - \eta_1\eta_2}$$

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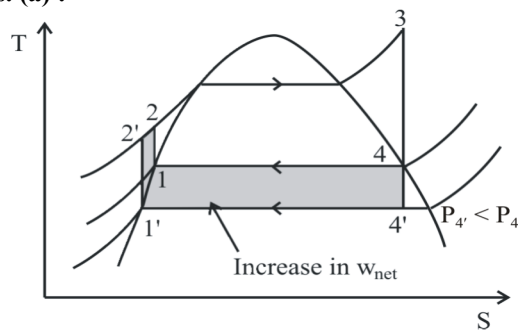
117. Keeping all other parameters constant, if the temperature of the condenser of a Rankine cycle is decreased, the cycle efficiency :

(a) increase
(b) remains constant
(c) decreases
(d) cannot be determined from above inputs

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) :



(1) $W_T\uparrow$, $W_P\uparrow$, $W_{net}\uparrow$
(2) $W_{net}\uparrow = \uparrow W_T - W_P\uparrow$, $Q_S\uparrow$, Q_R (Cannot say)
(3) $\eta\uparrow$

$$\eta_{th} = \frac{W_{net}}{Q_{in}}$$

Lowering the operating pressure of the condenser lowers the temperature, at which heat is rejected. The overall effect of lowering the condenser pressure is an increase in thermal efficiency of the cycle and efficiency portion shows the amount of increase in work output.

118. The refrigeration cycle differs from reversed Rankine cycle in the following process

(a) condensation (b) expansion
(c) evaporation (d) compression

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : In refrigeration cycle, throttling is done at constant enthalpy where as in reversed Rankine cycle expansion is done in turbine of constant entropy.

119. Furnaces of most pulverized-coal-fired thermal power plant boiler are

(a) induced draught furnaces
(b) Natural draught furnaces
(c) Forced draught furnaces
(d) Balanced draught furnaces

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Force draft fan are used to regulate proper combustion and maximize efficiency of the fuel.

120. Induced draft fans typically are centrifugal fans with

(a) Radial vanes
(b) Backward vanes
(c) Forward curved vanes
(d) None of the above

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Induced draft fans typically are centrifugal fans forward curved vanes.

121. In forced circulation boilers, about 90% of water is recirculated without evaporation. The circulation ratio is

(a) 0.1 (b) 0.9
(c) 9 (d) 10

RPSC IOF, 2020

Ans. (d) :

$$\text{Circulation ratio} = \frac{\text{Flow rate of saturated steam in downcomers}}{\text{Flow rate of steam released from drum}}$$

$$= \frac{m_g + m_p}{m_g} \Rightarrow \frac{1}{\frac{m_g}{m_g + m_p}} \quad .....(i)$$

$$= \frac{1}{TDF}$$

Given,

$m_g = 0.1 \rightarrow$ Mass of stream released from drum

$m_\ell = 0.9 \rightarrow$ Mass of liquid at riser exit

TDF – Top dryness fraction

From equation (i) –

$$CR = \frac{0.1 + 0.9}{0.1} = 10$$

122. Which of the following is true for steam condenser?

(a) Exhaust steam and cooling water do not come in direct contact in jet condenser
(b) Exhaust steam and cooling water come in direct contact and mixed together in surface condenser
(c) Evaporative condenser is a type of jet condenser
(d) Evaporative condenser is a type of surface condenser

APPSC Poly Lect. 13.03.2020

Ans. (d) : Evaporative condenser is a type of surface condenser.

123. Net-work from a steam turbine is 1000 kW boiler supplies 2800 kJ/kg of steam, and cooling water in condenser takes 2100 kJ/kg of steam. Neglecting the pump work, the steam flow rate should be

(a) 0.357 kg/s (b) 0.476 kg/s
(c) 1.428 kg/s (d) 1.678 kg/s

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Turbine work = Heat supplied in the boiler – heat rejected to the condenser (If pump work is neglected)

$\dot{m}$ = mass flow rate of steam

$1000 = 2800 \times \dot{m} - 2100 \times \dot{m} = 700\,\dot{m}$

$\dot{m} = 1.428$ kg/sec

124. Steam rate is defined as [where Q = Heat input to cycle (kW), W_{net} = shaft output (kW)]

(a) $\frac{1}{W_{net}}(kg/kWh)$ (b) $\frac{3600(Q)}{W_{net}}(kJ/kWh)$

(c) $\frac{Q}{W_{net}}(kJ/kWh)$ (d) $\frac{3600}{W_{net}}(kg/kWh)$

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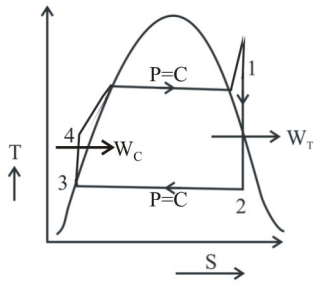
Ans. (d) : $\frac{3600}{W_{net}}(kg/kWh)$

125. If enthalpy of steam at the entry and exit of steam turbine is respectively 3000 kJ/kg and 2000 kJ/kg, if pump work is ignored, what will be the specific steam consumption?

(a) 3.60 kg/kW.hr (b) 3.60 kg/kW.s
(c) 0.360 kg/kW.s (d) Insufficient data

OPSC AEE 2019 PAPER - II

Ans : (a) : Given,
$h_1 = 3000$ kJ/kg
$h_2 = 2000$ kJ/kg



Net output from turbine

$W_T = h_1 - h_2$
$= 3000 - 2000$
$= 1000$ kJ/kg

So,
Specific Steam Consumption (SSC) in kg/kW.hr

$$SSC = \frac{3600}{W_T} = \frac{3600}{1000} = 3.6 \text{ kg/kW.hr}$$

126. For an ideal turbine enthalpy of gas at entry and exit is 3,000 kJ/kg and 2,500 kJ/kg respectively and having velocity 200 m/s and 100 m/s respectively at entry and exit. If mass flow rate of gas is 30 kg/s, the approximate power output of turbine is:

(a) 16 MW (b) 15 MW
(c) 15.5 MW (d) Insufficient data

OPSC AEE 2019 PAPER- II

Ans : (c) : Given,
$h_1 = 3000$ kJ/kg
$h_2 = 2500$ kJ/kg
$v_1 = 200$ m/s
$v_2 = 100$ m/s
$m = 30$ kg/s
Applying SFEE between inlet and outlet, power output of turbine,

$$h_1 + \frac{v_1^2}{2000} + \frac{gz_1}{1000} + q = h_2 + \frac{v_2^2}{2000} + \frac{gz_2}{1000} + w$$

$[\because z_1 = z_2, q = 0]$

$$3000 + \frac{200^2}{2000} = 2500 + \frac{100^2}{2000} + w$$

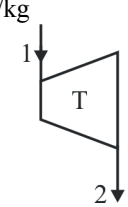
$w = 515$ kJ/kg
$P = mw$
$= 30 \times 515$
$= 15.45$ MW
≈ 15.5 MW

127. In a steam turbine with steam flow rate of 1 kg/s, inlet velocity of steam of 100 m/s, exit velocity of steam of 150 m/s, enthalpy at inlet of 2900 kJ/kg, enthalpy at outlet of 1600 kJ/kg, the power available from the turbine will be nearly

(a) 1575.5 kW (b) 1481.6 kW
(c) 1387.7 kW (d) 1293.8 kW

ESE 2019

Ans. (d) : Given,
$\dot{m} = 1$ kg/s
$c_1 = 100$ m/s
$c_2 = 150$ m/s
$h_1 = 2900$ kJ/kg
$h_2 = 1600$ kJ/kg



From steady flow energy equation

$$\dot{m}\left[h_1 + \frac{c_1^2}{2} + gz_1\right] + \dot{q} = \dot{m}\left[h_2 + \frac{c_2^2}{2} + gz_2\right] + \dot{W}_T$$

We assume that
$z_1 = z_2$, No heat transfer take place $\dot{q} = 0$

$$\dot{m}\left[h_1 + \frac{c_1^2}{2}\right] = \dot{m}\left[h_2 + \frac{c_2^2}{2}\right] + \dot{W}_T$$

$$\dot{m}\left(h_1 + \frac{c_1^2}{2000}\right) = \dot{m}\left(h_2 + \frac{c_2^2}{2000}\right) + \dot{W}_T$$

$$1\left(2900 + \frac{(100)^2}{2000}\right) = 1\left(1600 + \frac{(150)^2}{2000}\right) + \dot{W}_T$$

$\dot{W}_T = 1293.75$ kW

128. The efficiency of the vapour power Rankine cycle can be increased by

1. **increasing the temperature of the working fluid at which heat is added**
2. **increasing the pressure of the working fluid at which heat is added**
3. **decreasing the temperature of the working fluid at which heat is rejected**

Which of these statements is/are correct?

(a) 2 and 3 (b) 1 alone
(c) 1 and 2 (d) 1, 2 and 3

Gujarat PSC AE 2019, ESE 1999

Ans. : (d) : The efficiency of the vapour power Rankine cycle can be increased by-

(i) Increasing the temperature of working fluid, the mean temperature of heat addition increases and heat transfer irreversibility reduces. This results in improved efficiency.

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(ii) Since in Rankine cycle, there is phase change during heat addition or rejection. So by increasing pressure of fluid during heat addition, the temperature also increases and efficiency increases.
(iii) Reducing temperature of heat rejection will also increases the efficiency of cycle. These points are clear by

$$\eta = 1 - \frac{T_{\text{rejection (mean)}}}{T_{\text{addition (mean)}}}$$

129. Reheat cycle is essential in a steam power plant to
(a) utilize the heat of flue gases
(b) heat the feed water on its way to boiler
(c) increase the plant efficiency
(d) to reduce the moisture content at the low pressure stages of turbines so that erosion and corrosion problems can be eliminated

Gujarat PSC AE 2019

Ans. : (d) : To reduce the moisture content at the low pressure stages of turbines so that erosion and corrosion problems can be eliminated

130. Draught in a boiler is important for:
(a) Removing combustible gases from boiler
(b) For proper combustion of fuel
(c) Both (A) and (B)
(d) None of these

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Ans. : (c) : Draught in a boiler is important for
- Removing combustible gases from boiler
- For proper combustion of fuel.

To discharge these gases to the atmosphere through the chimney.

131. On a boiler one end of the water indicator (Glass Tube Type) is connected to water space in boiler and other end is connected to:
(a) Open to atmosphere
(b) Steam space of boiler
(c) Superheater
(d) Another part of water space

OPSC AEE 2019 PAPER - II

Ans. : (b) : On a boiler one end of the water indicator (glass tube type) is connected to water space in boiler and other end is connected to steam space of boiler.

132. A steam jet issuing from a nozzle placed under the fire grate, in the ash pit of the furnace is used to produce which type of draught?
(a) Natural (b) Induced
(c) Forced (d) None of the options

OPSC AEE 2019 PAPER-II

Ans. : (c) : Forced steam jet draught- When the steam jet issuing from a nozzle is placed in the ash pit under the fire grate of the furnace, it is then known as forced steam jet draught.

133. In a locomotive and marine boilers the safety valve used are mainly:
(a) Leaver operated safety valve
(b) High steam and low water safety valve
(c) Spring loaded safety valve
(d) Dead weight safety valve

OPSC AEE 2019 PAPER - II

Ans. : (c) : In a locomotive and marine boilers the safety valve used are mainly spring loaded safety valve.

134. Low grade waste heat in bottoming cycle of cogeneration is used for:
(a) Feed water heating (b) Processing
(c) Power generation (d) Not for any use

OPSC AEE 2019 PAPER - II

Ans. : (a) : Low grade waste heat in bottoming cycle of cogeneration is used for feed water heating.

135. The following is (are) ash handling system(s).
(a) Hydraulic system (b) Pneumatic system
(c) Steam jet system (d) All of these

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Ans. : (d) : In thermal power plants, an ash handling system is used to collect and dispose off discharged ash. Once it has been cooled down to a manageable temperature. It is used in various industries like construction, cement plants and other allied industries.
Ash handling system in power plant –
- Hydraulic system
- Pneumatic system
- Steam jet system

136. In a thermal power plant feed water heater is used to pre heat the feed water by:
(a) Hot gases from boiler furnace
(b) Steam from boiler
(c) Hot air from air pre heater
(d) Turbine exhaust steam

OPSC AEE 2019 PAPER - II

Ans. : (d) : In a thermal power plant feed water heater is used to pre heat the feed water by turbine exhaust steam.

137. Complete the sentence : "The function of the _____ is to increase the temperature for air before it enters the furnace".
(a) Air preheater (b) Super heater
(c) Injector (d) Economizer

SJVN ET 2019

Ans. (a) : Air preheater is to increase the temperature for air before it enters the furnace.

138. Steam enters a condenser at 35° C [$P_{\text{sat}@35^{o}C}$ = 42.21 mm of Hg]. Barometer reading is 760 mm of Hg and vacuum of 690 mm Hg is recorded in the condenser. The vacuum efficiency will be given by:
(a) 86.01% (b) 82.30%
(c) 96.10% (d) 80.23%

SJVN ET 2019

Ans. (c)

$$\text{Vacuum efficiency} = \frac{\text{Actual vacuum}}{\text{Barometer} - \text{Absolute pressure of steam}}$$

P_{Actual} = 690 mm Hg
$P_{\text{Barometer}}$ = 760 mm Hg
P_{Steam} = 42.21 mm Hg

$$\eta_{\text{vacuum}} = \frac{690}{760 - 42.21} = \frac{690}{717.79} = 96.10\%$$

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139. Rankine cycle efficiency for a power plant is

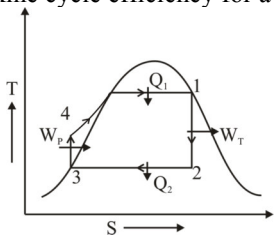
Q_1 - heat transferred to the working fluid
Q_2 - heat rejected from the working fluid
W_T - work transferred from the working fluid
W_P - work transferred into the working fluid

(a) $\eta = \dfrac{W_T - W_P}{Q_1}$ (b) $\eta = \dfrac{Q_2}{W_T - W_P}$

(c) $\eta = W_T - W_P$ (d) $\eta = \dfrac{Q_2 - Q_1}{W_T}$

TNPSC AE 2019

Ans. (a) : Rankine cycle efficiency for a power plant.



$$\eta_R = \frac{W_{net}}{Q_{supplied}}$$

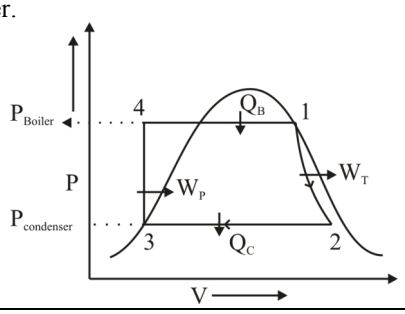
$$\eta_R = \frac{W_T - W_P}{Q_1}$$

140. At ideal condition of vapour power cycle heat rejection at reversible constant pressure occurs at

(a) turbine (b) pump
(c) condenser (d) boiler

TNPSC AE 2019

Ans. (c) : At ideal condition of vapour power cycle heat rejection at reversible constant pressure occurs at condenser.



141. Which of the following statements is/are correct regarding superheater in boilers?

1. It is heat exchanger in which heat is transformed to the saturated steam to increase its temperature.
2. It raises the overall efficiency.
3. It reduces turbine internal efficiency.

Select the correct answer using the code given below.

(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) 1 only

ESE 2019

Ans. (a) : The superheater is a heat exchanger in which heat is transferred to the saturated steam to increase its temperature. It raises the overall efficiency. It reduces the moisture content in last stages of the turbine and thus increases the turbine internal efficiency.

142. A 2 kg of steam occupying 0.3 m^3 at 15 bar is expanded according to the law $pv^{1.3}$ = constant to a pressure of 1.5 bar. The work done during the expansion will be

(a) 602.9 kJ (b) 606.7 kJ
(c) 612.5 kJ (d) 618.3 kJ

ESE 2019

Ans. (d) : Given,

$m = 2$ kg
$V_1 = 0.3\ m^3$
$P_1 = 15$ bar $= 1500$ kPa
$P_2 = 1.5$ bar $= 150$ kPa

$$P_1V_1^{1.3} = P_2V_2^{1.3}$$

$$V_2 = V_1\left(\frac{P_1}{P_2}\right)^{\frac{1}{1.3}}$$

$$= 0.3\left(\frac{15}{1.5}\right)^{\frac{1}{1.3}}$$

$$= 1.7634\ m^3$$

$$W = \frac{P_1V_1 - P_2V_2}{n-1} = \frac{(1500\times 0.3) - (150\times 1.763)}{1.3-1}$$

$$= 618.5\ kJ$$

143. A boiler is having a chimney of 35 m height. The draught produced in terms of water column is 20 mm. The temperature of flue gas inside the chimney is 365°C and that of air outside the chimney is 32°C. The mass of air used will be nearly

(a) 10.3 kg/kg of fuel (b) 12.5 kg/kg of fuel
(c) 14.7 kg/kg of fuel (d) 16.9 kg/kg of fuel

ESE 2019

Ans. (d) : Given, $h_w = 20$ mm $= 0.02$ m

$T_a = 273 + 32 = 305$ K
$T_g = 365 + 273 = 638$ K

$$h_w = 353\,H\left[\frac{1}{T_a} - \frac{1}{T_g}\frac{M_a+1}{M_a}\right]$$

$$0.02 = 353\times 35\left[\frac{1}{305} - \frac{1}{638}\left(\frac{M_a+1}{M_a}\right)\right]$$

$$\frac{0.02}{353\times 35} = \frac{1}{305} - \frac{1}{638}\left(\frac{M_a+1}{M_a}\right)$$

$$\frac{M_a+1}{M_a} = 1.059$$

$M_a = 16.9$ kg/kg of fuel

144. A frictionless piston cylinder device contains 5 kg of steam at 400 kPa and 200°C. The heat is now transferred to the steam until the temperature reaches 250°C. If the piston is not attached to a shaft, its mass is constant and by taking the values of specific volume v_1 as 0.53434 m^3/kg and v_2 as 0.59520 m^3/kg the work done by the steam during this process is

(a) 121.7 kJ (b) 137.5 kJ
(c) 153.3 kJ (d) 189.1 kJ

ESE 2019

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Ans. (a) : Given,

$m = 5$ kg $\quad V_1 = 0.53434\ m^3/kg$

$P = 400$ kPa $\quad V_2 = 0.59520\ m^3/kg$

$T_1 = 200°C \quad T_2 = 250°C$

Since the weight of the piston and the atmosphere pressure are constant,

So assuming it is a constant pressure process

Work done $W = P(V_2 - V_1)$

$= mP(V_2 - V_1)$

$= 400 \times 10^3 \times 5\ (0.59520 - 0.53434)$

$= 121.72$ kJ

145. 5 kg steam expands in a piston cylinder device from a pressure of 5 bar and 700^0 C (h = 3925.97 kJ/kg, s = 8.5892 kJ/kg-K, u = 3477.52 kJ/kg) to a pressure and temperature of 3 bar and 600^0 C (h = 3703.20 kJ/kg, s = 8.5892 kJ/kg-K, u = 3300.79 kJ/kg). If the process is reversible and adiabatic, the total work (kJ) for the process will be:

(a) 883.65 (b) 88.365

(c) 558.8 (d) 55.88

BHEL ET 2019

Ans. (a) : Mass of the steam (m) = 5 kg

$u_1 = 3477.52$ kJ/kg

$u_2 = 3300.79$ kJ/kg

For reversible and adiabatic process–

$Q = 0$

From non flow equation–

$Q = (u_2 - u_1) + W$

$U = m \times u$

$W = m(u_1 - u_2)$

$W = 5 \times (3477.52 - 3300.79)$

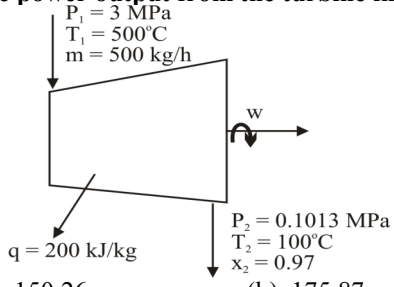
$W = 883.65$ kJ

146. Steam flows through a turbine as shown in the figure. The walls of the turbine are not insulated, so that there is heat transfer through the walls as shown. The properties of steam are as given below :

P = 3 MPa. T = 500 ^{0}C (h = 3456.48 kJ/kg, s = 7.2337 kJ/kg-K)

P = 0.1013 MPa, T = 100 ^{0}C (h_f = 419.1 kJ/kg. h_g = 2676 kJ/kg)

The power output from the turbine in kW is :



(a) 150.26 (b) 175.87

(c) 90.03 (d) 125.76

BHEL ET 2019

Ans. (c) : $h_f = 491.1$ kJ/kg, $h_g = 2676$ kJ/kg

$\dot{m} = 500\ kg/hr$

$= \frac{500}{3600}$

$\dot{m} = 0.139\ kg/sec$

$q = -200$ kJ/kg

Enthalpy at point –

$h_2 = h_f + x_2(h_g - h_f) = 491.1 + 0.9\ (2676 - 491.1)$

$= 2610.45$

From steady flow energy equation –

$(h_1 + q) = (h_2 + w)$

$(3456.48 - 200) = 2610.45 + w$

$w = 646.03$ kJ/kg

Work done in kW –

$\dot{w} = w \times \dot{m} = 646.03 \times 0.139$

$\boxed{\dot{w} = 89.79\ kW}$

147. A pump handling a liquid raises its pressure from 1 bar to 30 bar. Take the density of liquid as 990 kg/m^3. The isentropic specific work done by the pump in kJ/kg is :

(a) 3.93 (b) 1.93

(c) 4.93 (d) 2.93

ISRO Scientist/Engineer (RAC), 10.03.2019

GATE 2011

Ans. (d) : work $= \int v dP$

$= \int \frac{dP}{\rho}$

$= \frac{P_2 - P_1}{\rho} = \frac{(30-1) \times 100}{990} = 2.929\ kJ/kg$

148. A steam turbine receives steam at 15 bar, 350^{o}C [h = 3147.5 kJ/kg, s = 7.100 kJ/kg-K] and exhaust to the condenser at 0.06 bar. [h_f = 151.5 kJ/kg, h_{fg} = 2416.0 kJ/kg, s_f = 0.520 kJ/kg-K, s_{fg} = 7.800 kJ/kg-K] What will be the approximate thermal efficiency of ideal Rankine cycle operating between these two limits? [Neglect the pump work]

(a) 49.79% (b) 45.76%

(c) 22.76% (d) 31.97%

SJVN ET 2019

Ans. (d) : Given,

$h_1 = 3147.5$ kJ/kg, $\quad s = 7.100$ kJ/kg-K

$h_f = 151.5$ kJ/kg, $\quad s_f = 0.520$ kJ/kg-K

$h_{fg} = 2416.0$ kJ/kg, $\quad s_{fg} = 7.800$ kJ/kg-K

$\because$ In turbine, entropy is constant,

So, $s_1 = s_2 = s$

$\because \quad s = s_f + x s_{fg}$

$7.100 = 0.520 + x(7.800)$

$x = 0.8435$

Enthalpy of turbine at the exhaust,

$h_2 = h_f + x h_{fg}$

$= 151.5 + 0.8435\ (2416)$

$h_2 = 2189.396$ kJ/kg

Efficiency, $\eta = \frac{W_T - W_P}{Q_S}$

If pump work is neglected than–

$\eta = \frac{W_T}{Q_S}$ or $\eta = \frac{h_1 - h_2}{h_1}$

$\eta = \frac{3147.5 - 2189.396}{3147.5}$

$\boxed{\eta = 30.45\%}$

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149. **Statement I:** Rankine efficiency of a steam power plant increases in winter compared to summer.
Statement II: The increase in Rankine efficiency is due to lower condenser temperature.

ESE 2018

Ans. (a) : In winter condenser temperature decreases as cooling water temperature decreases, hence thermal efficiency increases.

150. **Statement I:** Direct condensers are more efficient than surface condensers.
Statement II: In condenser, the momentum pressure drop opposes the frictional pressure drop.

ESE 2018

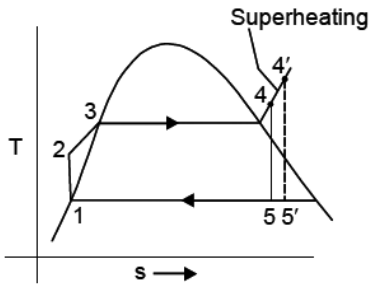
Ans. (b) : Direct condensers are more efficient than contact type condenser as velocity of water plus vapour at inlet is more than velocity of water at exit from condenser.
Hence momentum pressure drop is negative and friction drop is positive.

151. **All power plants use superheated steam due to which of the following advantages?**
1. **Superheating is mostly done from waste heat of boiler without additional cost of fuel.**
2. **The plant efficiency increases due to higher temperature of steam.**
3. **There is less corrosion and erosion of equipment due to absence of moisture in the steam.**

Select the correct answer using the code given below:
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (d) :



- 1 – 2 – 3 – 4 – 5 is normal Rankine cycle
- 1– 2 – 3 – 4'– 5' is superheated Rankine cycle because of superheating mean temperature (T_m) of heat addition ↑.

As $\eta = f(T_m)$
So, h ↑ for superheated cycle

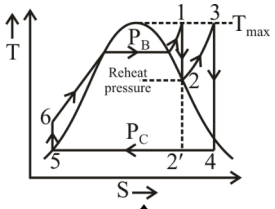
- at point 5', dryness fraction is more than at point 5, so moisture content is less after the turbine exit, so less erosion and corrosion is there in superheated Rankine cycle than Rankine cycle.
- Superheaters (CSH) are placed after the reheater and before the economizer which uses waste heat releasing to the stack.

152. **Consider the following statements regarding Reheat Rankine Steam Cycle:**
1. The main purpose of reheat in Rankine cycle is to increase the efficiency of the cycle
2. In practice, the reheat is generally limited to one point of expansion
3. Due to reheat, the steam rate (specific steam consumption) is reduced.

Which of the above statements are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (b) : The main purpose of reheating is to avoid excess moisture in steam at the end of expansion to protect the turbine. But it need not improve the cycle efficiency.



- W_T-↑, W_P-Constant, W_{net}-↑
- Q_s-↑, Q_R-↑, T_{MA}-↓↑, η-↑↓ (Can't say)
- $\downarrow \text{Specific Steam Consumption} = \frac{3600}{W_{net} \uparrow}$
- x-↑, ↓errosion of turbine blade

153. **Statement I:** The condenser in a steam power plant is always filled with a mixture of water, steam and air.
Statement II: Slightly wet steam enters the condenser where in the pressure is below the atmospheric conditions, causing some leakage of air through the glands and also the release of some air dissolved in the boiler feed water.

ESE 2018

Ans. (a) : Steam enters the condenser in wet state, hence we have water, vapour and air.
At lower pressures air leaks through glands and also releases some air dissolved in boiler feed water.
Hence both the statements are correct and II is correct explanation of I.

154. **Which one of the following methods is more effective to improve the efficiency of the Rankine cycle used in thermal power plant?**
(a) Increasing the condenser temperature
(b) Decreasing the condenser temperature
(c) Decreasing the boiler temperature
(d) Increasing the boiler temperature

ESE 2018

Ans. (b) : Method to increase the efficiency of the Rankine cycle
(i) Lowering the condenser pressure
(ii) Superheating the steam to high temperature
(iii) Increasing the Boiler pressure

155. **Statement I:** Reheating between the high-pressure and low-pressure turbines increases the turbine work output.
Statement II: The constant pressure lines on T-s diagram diverge away from the origin.

ESE 2018

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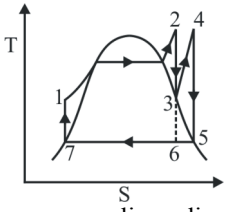
Ans. (a)

■ Work output of the reheat cycle is increased as

$$(h_3 - h_6) < (h_4 - h_5')$$



■ The constant pressure line diverge on the T-s diagram from the origin.

156. Which of the following process is represented by a horizontal line parallel to the abscissa of a Mollier-diagram?

(a) constant pressure
(b) constant temperature
(c) free expansion on throttling
(d) adiabatic expansion

GPSC Executive Engineer 23.12.2018

Ans. (c) : Free expansion on throttling.

157. The thermal efficiency of a steam plant is defined as ?

(a) The ratio of heat equivalent of electrical output to the heat of combustion of coal
(b) The ratio of heat of combustion of coal to the heat equivalent of electrical output.
(c) The ratio of heat equivalent of mechanical energy transmitted to the turbine shaft to the heat of combustion of coal.
(d) The ratio of heat of combustion of coal to the heat equivalent of mechanical energy transmitted to the turbine shaft.

Vadodara Muncipal Corp. DEE, 2018

Ans. (c) : Thermal efficiency of steam power plant is the ratio of heat equivalent of mechanical energy transmitted to the turbine shaft and heat of the combustion of coal.

158. In Rankine cycle, adiabatic enthalpy drop across the prime mover is 840 kJ/kg. The enthalpy of steam supplied is 2940 kJ/kg. If the back pressure is 0.1 bar, then the specific steam consumption is (assume h_f at 0.1bar = 191.8 kJ/kg):

(a) 2.4 kg/kWh (b) 3.4 kg/kWh
(c) 12.4 kg/kWh (d) 4.2 kg/kWh

AAI Jr. Executive 29.11.2018

Ans. (d) : Adiabatic enthalpy drop = $(h_1 - h_2)$= 840 kJ/kg
Enthalpy steam supplied (h_1) = 2940 kJ/kg
Back pressure P_2 = 0.1 bar
Corresponding 0.1 bar and h_f = 191.8 kJ/kg
Useful work done per kg steam = 840 kJ/kg
∴ Specific steam consumption or steam rate,

$$= \frac{3600}{W_{net}} \text{kg/kW-hr}$$

$$= \frac{3600}{840}$$

$$= 4.285 \text{kg/kWh}$$

159. Which of the following is incorrect for Rankine vapor power cycle?

(a) It has two reversible adiabatic and two reversible isobaric process.
(b) Its efficiency is dependent on mean temperature of addition and temperature of heat rejection.
(c) Constant pressure heat rejection takes place in the turbine.
(d) The efficiency of Rankine cycle can be increased by increasing the degree of superheat at constant pressure.
(e) Heat addition takes place in the boiler at constant pressure.

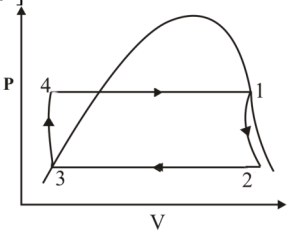
CGPSC AE 25.02.2018

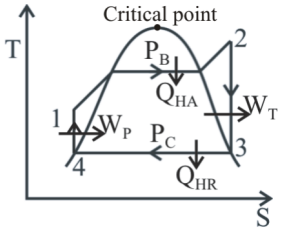
Ans. (c) : Process 1 - 2 – Reversible adiabatic expansion [Turbine work]

Process 2-3– Reversible isobaric heat rejection [isothermal heat rejection]

Process 3-4– Reversible adiabatic compression [Pump work]

Process 4-1– Reversible isobaric heat addition [In Boiler]





160. A curve showing the variation of load on a power station with respect to time is known as-

(a) Load curve (b) Load duration curve
(c) Diversity factor (d) Performance curve

RPSC AE 2018
UPPSC AE 12.04.2016 Paper-II

Ans. (a) : A curve showing the variation of load on a power station with respect to time is known as Load curve.

161. A power generation station is to supply four regions of loads with peak demand of 10 MW, 15 MW, 20 MW and 30 MW. If the diversity factor is 1.5, the maximum demand on the station is-

(a) 70 MW (b) 60 MW
(c) 50 MW (d) 40 MW

RPSC AE 2018

Ans. (c) : Diversity factor—Diversity factor is the ratio of the sum of the individual non-coincident maximum loads of various sub-divisions of the system to the maximum demand of complete system.

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The diversity factor is almost always larger than 1, since all components would have to be on simultaneously at full load for it to be one.

Data given:-

Peak demands—10 MW, 15 MW, 20 MW, 30 MW

Diversity factor = 1.5

$$\text{Diversity factor} = \frac{\text{Sum of all peak demands}}{\text{The maximum demand on the station}}$$

The maximum demand on the station

$$= \frac{10+15+20+30}{1.5} = \frac{75}{1.5}$$

$$= 50 \text{ MW}$$

162. In a steam generator, feed water supply rate is 205 kg/h and rate of coal fired is 23 kg/h. The net enthalpy rise of water for conversion to steam is 145 kJ/kg. If the calorific value of coal is 2050 kJ/kg, the steam generator efficiency will approximately be:

(a) 52% (b) 42%

(c) 72% (d) 62%

AAI Jr. Executive 29.11.2018

Ans. (d) : $\text{Efficiency of boiler} = \frac{\text{Enthalpy rise}}{\text{Heat extracted}}$

$$= \frac{205 \times 145}{23 \times 2050}$$

$$= 0.63$$

$$= 63\%$$

163. Which one of the following modifications to a Rankine cycle would upgrade/enhance its efficiency so as to approach that of Carnot cycle?

(a) Incomplete expansion of steam

(b) Reheating of steam

(c) Regenerative feed heating by steam

(d) Partial condensation of steam

HPPSC Asstt. Prof. 18.09.2017

Ans. (c) : Regenerative feed heating by steam.

164. If reheating is done in Rankine cycle, then:

(a) work output decreases

(b) pump work increases

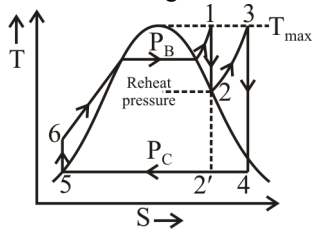
(c) heat rejection remains constant

(d) dryness fraction increases

(e) heat rejection decreases

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : Reheating :- It is process by which we can increase the dryness fraction of steam at the turbine exhaust is known as reheating.



- W_T-↑, W_P-Constant, W_{net}-↑
- Q_s-↑, Q_R-↑, T_{MA}-↓↑, η-↑↓ (Can't say)
- $\downarrow \text{Specific Steam Consumption} = \frac{3600}{W_{net} \uparrow}$
- x-↑, ↓errosion of turbine blade

165. If the boiler pressure increases at fixed maximum temperature, what will be the impact on work done for Rankine cycle?

(a) Increases linearly

(b) Decreases linearly

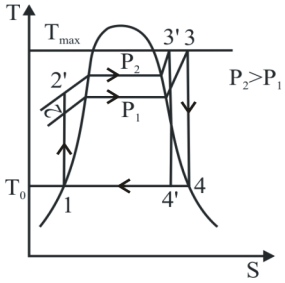
(c) Increases exponentially

(d) Decreases exponentially

(e) Almost remains constant

CGPSC ARTO & TSI 07.01.2017

Ans. (e) :



Here,

$$(W_T)_{P_1} = h_3 - h_4$$

$$(W_T)_{P_2} = h_{3'} - h_4$$

$$\boxed{h = f(T)}$$

(T_{max} - T_o) are same for both

i.e (1, 2, 3, 4) & (1 2' 3' 4')

166. In a steam engine the change in internal energy equal to:

(a) The work done during whole of the Rankine cycle

(b) The work done during adiabatic expansion only

(c) The work done during compression

(d) The change of enthalpy

PTCUL AE 25.06.2017

Ans. (b) :

167. One kilowatt is equal to -

(a) 1 N-m/s (b) 100 N-m

(c) 1000 N-m/s (d) 1×10^6 N-m/s

GMB AAE 25.06.2017

Ans. (c) : 1 kilowatt = 1000 Nm/sec.

One kilowatt is defined as energy consumption of 1000 Jule for 1 second.

1 kW = 1000 J/sec.

1 kW = 1000 W

168. On a Mollier chart the flow through turbine is represented by

(a) horizontal line (b) vertical line

(c) curved line (d) Any one of above

WBPSC AE, 2017

Ans. (b) : On a Mollier chart the flow through turbine is represented by vertical line.

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169. Boiler rating is usually defined in terms of :

(a) Maximum temperature of steam in Kelvin
(b) Heat transfer rate in kJ/hr
(c) Heat transfer area in meter^2
(d) Steam output in kg/hr

HPPSC Asstt. Prof. 20.11.2017, GATE 1992

Ans. (d) : Boiler rating is usually defined in terms of steam output in kg/hr.

170. In a steam generation process, in case of a natural draught system, the heat carried away by flue gases is

(a) Less due to lower flue gas temperatures
(b) Less due to higher flue gas temperatures
(c) More due to lower flue gas temperature
(d) More due to higher flue gas temperatures

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : Draught is the pressure difference which cause a flow of air or gases from one point to another point in the boiler system.
Draught is required in a boilers system mainly to remove flue gases from the system after combustion and the heat exchange.

171. Safety valves release excess steam when the pressure of steam inside the boiler exceeds rate pressure and as per boiler regulations every boiler must be fitted with two safety valves. Which of the following is NOT an example of safety valve?

(a) Dead weight
(b) Lever
(c) Low steam and high water
(d) Spring loaded

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) : Dead weight, lever, spring loaded is the type of safety valve.

- Low steam and high water is example of water level indicator.

172. Blowing down of boiler water is the process to

(a) Reduce the boiler pressure
(b) Increase the steam temperature
(c) Control the solids concentration in the boiler water
(d) Control the drum level

GPSC Asstt. Director of Transport 05.03.2017

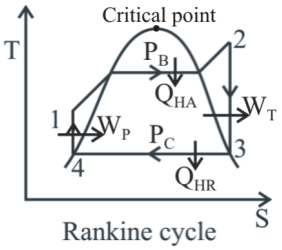
Ans. (c) : Blowing down is a process which involves draining a small amount higher hardness and solids water from the boiler and replacing it with lower hardness and solids makeup water to maintain a predetermined concentration of hardness and solids, so we can say it controls the solids concentration in the boiler water.

173. Rankine cycle efficiency for fixed steam temperature of any value up to critical temperature will be maximum for steam pressure of

(a) 9.8 MPa
(b) 19.61 MPa
(c) Half the critical pressure
(d) Critical pressure

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Rankine cycle efficiency for fixed steam temperature of any value upto critical temperature will be maximum for steam pressure of critical pressure.



174. Which of the following is most appropriate for vapour power cycles?

(a) In Carnot cycle, expansion of vapour occur without change in entropy
(b) In Rankine cycle, transformation of liquid water in steam generator occurs at constant temperature
(c) In Carnot cycle heating of vapour occurs at constant volume
(d) In Rankine cycle, entropy of vapour increases during its expansion in steam turbine

ISRO Scientist/Engineer 17.12.2017

Ans. (a)

- In Carnot cycle, compression and expansion takes place isentropically while Rankine cycle, transformation of liquid water in steam generator occurs at constant pressure.
- In Carnot cycle heating of vapour occurs at constant temperature while Rankine cycle, entropy of vapour remain constant during its expansion in steam turbine

175. A steam power plant is a classic example of _____.

(a) Heat reservoir (b) Heat pump
(c) Heat engine (d) Refrigerator
(e) Compressor

(CGPSC Polytechnic Lecturer 2017)

Ans. (c) : A steam power plant is a classic example of a heat engine. External fired heat engine are generally steam engines, and they differ from internal combustion engine in that the heat source is separate from the fluid that does work.
For Example–An external combustion engine would use a flame to heat water into steam, then using the steam to run a turbine.

176. The fusible plug of a boiler is an example of

(a) Safe life approach
(b) Fail safe approach
(c) Reliability approach
(d) Maintainability approach

UPSC JWM 2017

Ans. (b) : Fusible plug in boiler is a fitting used in steam boiler to enhance the safety during operation. Fusible plug is installed in small horizontal fire tube boilers between furnance and boiler water drum for protection of boiler while lower water level in drum. Fusible plug is an example of fail safe approach.

177. Consider the following statements:

1. For a given power, a fire tube boiler occupies less space than a water tube boiler,

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2. Steam at a high pressure and in large quantities can be produced with a simple vertical boiler
3. A simple vertical boiler has a fire tube

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

UPSC JWM 2017

Ans. (c) : For a given power, a fire tube boiler occupies more space than a water tube boiler.
Fire tube boiler- In fire tube boiler, the flue gases flow in the tubes and water flows from the shell. Fire tube boiler is a vertical boiler.
Water tube boiler- in water tube boiler, water flows from the tubes and the flue gases from the shell or passes over the tubes.

178. Consider the following statements with regard to reheat Rankine cycle:

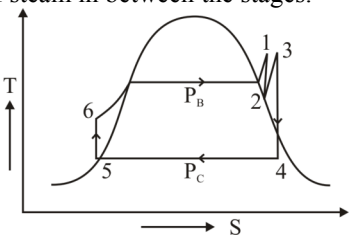
1. All the steam is taken out of the turbine after partial expansion for reheating at constant volume
2. The cycle permits the use of very high pressure without excessive moisture at condenser inlet
3. The efficiency of the reheat cycle may or may not exceed that of a simple cycle operating between the same maximum temperature and pressure
4. If the reheat pressure is more than 80% of the maximum system pressure, the thermal efficiency will be more than that for a simple cycle

Which of the above statements are correct?

(a) 2, 3 and 4 only (b) 1 and 2 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

UPSC JWM 2017

Ans. (a) : Reheat Rankine cycle- In reheat cycle, the total expansion of steam from boiler to condenser pressure take place in more than 1st stage with reheating of steam in between the stages.



As steam quality x is increases, the moisture content at the inlet of condenser decreases.
For the same maximum pressure and temperature limit, the efficiency of Rankine cycle increase or decrease.

- All the steam is taken out of the turbine after partial expansion for reheating at constant pressure.

179. Statement (I): Water entering into a condenser from the cooling tower has much dissolved impurities.
Statement (II): In a closed cooling system, the water is continuously aerated, therefore, there is abundant dissolved oxygen in this water.

ESE 2017

Ans. (c) : A cooling tower is an open cooling system and water can dissolved impurities from air because of direct contact.
A condenser a closed cooling system and there exists a vacuum in the condenser during operation. Hence there exist a possibilities of air leakage which is continuously deaerated.

180. Statement (I): Heat carried away by hot gases in chimney draught is much greater than the work required for lifting the same gases through the height of the chimney. Yet artificial draught is not preferred.
Statement (II): Artificial draught involves large initial cost as well as large maintenance cost.

ESE 2017

Ans. (d)

181. Statement (I): The overall combustion efficiency of a fuel oil based plant is less as compared to that of a coal burning plant.
Statement (II): Fuel oils contain comparatively larger percentage of hydrogen, which produce more moisture per kg of fuel burnt.

ESE 2017

Ans. (d)

182. Statement (I): Providing reheat in a Rankine cycle would increase the efficiency of the cycle.
Statement (II): Reheat in Rankine cycle reduces specific steam consumption.

ESE 2017

Ans. (d) : Advantage of Reheating–
(i) By reheating there is gain of 4 to 7% of thermal efficiency take place as the output of turbine is increased.
(ii) Reheating may also shortout blade erosion and corrosion problem in turbine.

$\because$ Specific steam consumption (S.S.C) $= \frac{3600}{W_{net}}$ kg/kW-hr

Due to reheating W_{net} increases and specific steam consumption decreases.

183. Consider the following statements in respect of regenerative Rankine cycle:

1. Regeneration increases the efficiency of the cycle.
2. The boiler capacity is increased for a given output.
3. The capacity of the condenser is reduced.

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

ESE 2017

Ans. (a) :

- Regeneration increases the mean temperature of heat addition of cycle, hence the efficiency of cycle increases.
- For same power output steam flow rate requires bigger boiler.
- Because of steam flow rate to condenser decreases so small condenser is required.

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184. Consider the following statements:
The presence of air inside condensers
1. remains as a non-condensable gas
2. reduces the condensing coefficient
3. tends to cling to the surface
4. introduces large thermal resistance

Which of the above statements are correct?
(a) 1, 2, 3 and 4 (b) 1, 2 and 3 only
(c) 3 and 4 only (d) 1, 2 and 4 only

ESE 2017

Ans. (a) : Effect of air inside the condenser :
(i) The performance of condenser due to presence of air gets affected badly.
(ii) The heat transfer surfaces such as outside the surface of condenser tubes which reduces the condensing heat transfer co-efficient.
(iii) As air tend to cling the surface which reduces the condensing co-efficient.
(iv) Because of low thermal conductivity of air, air imparts large thermal resistance.

185. The boiler flue gas is source of
(a) HCl (b) NO
(c) HF (d) Volatile organic compounds

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (b) : Nitrogen oxide (NO) compound is the main source of boiler flue gas.

186. The boiler consists of horizontal fire tubes is
(a) Cochran (b) Cornish
(c) Babcock and Wilcox (d) Stirling

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (a) : Cochran boiler is multi-tubular vertical fire tube boiler having a number of horizontal fire tubes.

187. The Grate in the boilers
(a) used as a base on which the burning of the solid fuel takes place
(b) used for superheating the steam
(c) used as a water storing device
(d) used for collecting the mud in present in the water

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : The grate in the boilers used as base on which the burning of the Solid fuel take place.

188. A steam engine is a
(a) External combustion engine
(b) Internal combustion engine
(c) It may be internal or external type
(d) All of the above

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : Steam engine is a heat engine that performs mechanical work using steam as a working fluid.

189. A simple steam power cycle receives 1,00,000 kJ/min as heat transfer from hot combustion gases and rejects 66,000 kJ/min as heat transfer to the environment. If the pump power required is 1400 kJ/min, the thermal efficiency of the cycle and turbine power output is
(a) 51.5% and 590 kW
(b) 34% and 590 kW
(c) 51.5% and 566.6 kW
(d) 34% and 566.6 kW

ISRO Scientist/Engineer 03.07.2016

Ans. (b) : $Q_S = 100{,}000$ kJ/min
Pump work required = 1400 kJ/min
$Q_R = 66000$ kJ/min
$W_{net} = Q_S - Q_R$
$= 34000$ kJ/min
Turbine work = W_{net} + Pump work required
$= 34000 + 1400$
$= 35400$ kJ/min

$$\text{Turbine work} = \frac{35400}{60} = 590 \text{ kW}$$

$$\text{Thermal efficiency} = 1 - \frac{Q_R}{Q_S}$$

$$= 1 - \frac{66000}{100{,}000} = 34\%$$

190. Regarding the energy scenario of India, the power installed capacity as on 31st December 2015 approximately is
(a) 280×10^3 MW (b) 320×10^3 MW
(c) 400×10^3 MW (d) 470×10^3 MW

GPSC ARTO 01.05.2016

Ans. (a) : 280×10^3 MW

191. A boiler in India should confirm to safety regulations of ..... .
(a) DIN (b) BS
(c) ASTM (d) IBR

GWSSB DEE 07.07.2016

Ans. (d) : Indian Boiler Regulation (IBR) Governs design, manufacture, installation and operation of all steam producing vessels.

192. Thermal efficiency of a thermal power plant is of the order of ..... .
(a) 15% (b) 20%
(c) 30% (d) 45%

GWSSB DEE 07.07.2016

Ans. (c) : Thermal efficiency of a thermal power plant is 30%.

193. The height of chimney in a power plant s governed by ..... .
(a) control of pollution
(b) the draught to be created
(c) limitation o construction facilities
(d) quantity of flue gases to be handled

GWSSB DEE 07.07.2016

Ans. (b) : For producing a natural draught, height of chimney should be very high.
$\uparrow \Delta P = (\rho_0 - \rho_1) g H \uparrow$

194. The boiler commonly used in sugar mills is
(a) Cochran (b) Cornish
(c) Lancashire (d) Stirling

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Lancashire, a fire tube boiler is used in sugar mills & industries. It is reliable and simplicity is design easy in operation and less operating and maintenance cost. This Lancashire boiler is used, where a lot steam is required.

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195. Efficiency of cycle will increase due to Reheating if

(a) Mean temperature of heat addition in boiler is higher than in reheat process
(b) Mean temperature of heat addition in boiler is lower than in reheat process
(c) it will increase without any conditions
(d) it will not increase

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (a) : Efficiency of cycle will increase due to reheating if temperature of heat addition in boiler is higher than in reheat process

$$\text{Efficiency} = \frac{\text{Work output}}{\text{Heat supplied is less in this condition only}}$$

196. Why don't use more than two reheats?

(a) Not economical
(b) Increased and mechanical stresses
(c) Net efficiency is decreased
(d) All of the mentioned

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : We never use two reheat of same time, because the mechanical stresses increases in much higher proportion than the pressure because of the prevailing high temperature, not economical also, net efficiency is decreased.

197. The cycle efficiency in a single reheat plant is influenced by

(a) Pressure at turbine inlet
(b) Pressure at which steam is reheated
(c) temperature at which steam is reheated
(d) temperature at turbine inlet

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : Reheating below a certain pressure reduces the mean temperature of heat addition below the earlier value causing decrease in efficiency.

198. The optimum reheat pressure for the most of the modern power plants is

(a) 0.1-0.5 (b) 0.2-0.3
(c) 0.2-0.25 (d) 0.25-0.3

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : The optimum reheat pressure for the most of the modern power plants is 0.2 – 0.25 of the initial steam pressure. In this range of reheat pressure efficiency is maximum.

199. What should be the dryness fraction at the time of exhaust?

(a) 0.88 (b) 0.80
(c) 0.90 (d) 0.75

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : For modern turbines, the admissible dryness fraction of exhaust steam (at the turbine exit) should be not less than, $x = 0.86$ to 0.88

200. Reheat is preferred for the plants having efficiency greater than

(a) 150 MW (b) 200 MW
(c) 100 MW (d) 250 MW

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Reheat is preferred for the plants having efficiency greater than 100 MW, for producing power less than this reheat will not be economical.

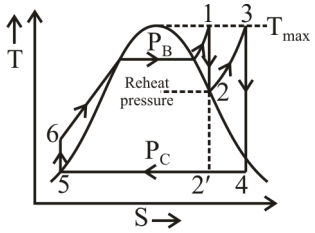
201. Which one of the following is an advantage of reheat cycle

(a) The size of the Low pressure (LP) turbine blades is reduced because specific volume is reduced by 7 to 8%
(b) The reheating reduce 4 to 5% fuel consumption with a corresponding reduction in the fuel handling
(c) Lower steam pressures and temperatures and less costly materials can be used to obtain the required thermal performance
(d) All of the mentioned

Nagaland PSC (CTSE) 2016, Paper-II

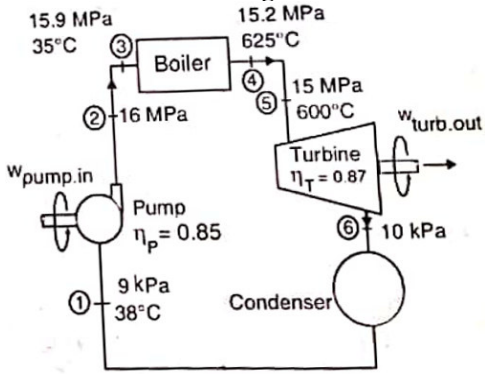
Ans. (d) : Advantage of Reheat Cycle are:-

(i) Size of the Low Pressure (LP) turbine blades is reduced because specific volume is reduced by 7 to 8%.
(ii) The reheating reduce 4 to 5% fuel consumption with a corresponding reduction in the fuel handling.
(iii) Lower steam pressure and temperature and less costly materials can be used to obtain the required thermal performance.



- W_T-↑, W_P-Constant, W_{net}-↑
- Q_s-↑, Q_R-↑, T_{MA}-↓↑, η-↑↓ (Can't say)
- $\downarrow \text{Specific Steam Consumption} = \frac{3600}{W_{net}\uparrow}$
- x-↑, ↓errosion of turbine blade

202. A steam power plant operates on the cycle shown in given below. If the isentropic efficiency of the turbine is 87 percent and the isentropic efficiency of the pump is 85 percent, determine the thermal efficiency of the cycle and the net power output of the plant for a mass flow rate of 15 kg/s.



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(a) 0.361 and 18.9 MW
(b) 0.301 and 18.9 MW
(c) 0.361 and 16.9 MW
(d) 0.3 and 18.9 MW
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : $W_{pump\,in} = \frac{W_{s_{pump\,in}}}{\eta_p} = \frac{V_1(P_2 - P_1)}{\eta_p}$

$= \frac{0.001009 \times (16000 - 9)}{0.85} = 19.0\,kJ/kg$

$W_{turb\,out} = \eta_t \; W_{s_{turb\,out}} = \eta_T(h_5 - h_6)$ (from steam table)

$= 0.87(3583.1 - 2115.3)$

$= 1277.0$ kJ/kg

$q_{in} = h_4 - h_3 = (3647.6 - 160.1)$

$= 3487.5$ kJ/kg

$W_{net} = W_{turbine} - W_{pump}$

$= 1277 - 19 = 1258$ kJ/kg

Thermal efficiency,

$\eta_{th} = \frac{W_{net}}{q_{in}} = \frac{1258}{3487.5} = 0.361 = 36.1\%$

$\dot{W}_{net} = \dot{m}(W_{net})$

$= 15 \times 1258.0 = 18.9$ MW

Note-This question has been deleted by CGPSC.

203. What is the efficiency of an ideal regenerative Rankine cycle power plant using saturated steam at 377^0C and pressure 135 bars at the inlet to the turbine and condensing temperature of 25^0C (corresponding saturation pressure of 3.6kPa)?
(a) 92.3% (b) 45.8%
(c) 85.6% (d) 63.8%

UPRVUNL AE 21.08.2016

Ans. : (b) The efficiency of an ideal regenerative Rankine cycle power plant = 45.8%

204. The number of fire tubes in a Cochran boiler are
(a) 75 (b) 115
(c) 165 (d) 225

APPSC AEE Screening Test 2016

Ans. (c) : The number of the fire tubes in a Cochran boiler are 165.

205. Thermal efficiency of Rankine cycle can be enhanced by
(a) Decreasing the average temperature of heat addition
(b) Increasing the superheat of constant pressure
(c) Increasing the average temperature of heat rejection
(d) The efficiency of Rankine cycle does not depend on average temperature of heat addition

RPSC LECTURER 16.01.2016

Ans. (b) : Thermal efficiency of Rankine cycle can be enhanced by—
1. Decrease in condenser pressure
2. Increase the boiler pressure
3. Increasing the superheat at constant pressure

206. The ratio of average load to the maximum load is known as
(a) Utilization factor (b) Diversity factor
(c) Plant capacity factor (d) Load factor

RPSC LECTURER 16.01.2016

Ans. (d) : Load factor— The ratio of average load to the maximum load is known as load factor.
Utilization factor— It is defined as the ratio of the time that a equipment is in use to the total time that it could be in use.
Diversity factor— Diversity factor is defined as the ratio of sum of individual maximum demand to maximum demand on power station.

$$\text{Diversity factor} = \frac{\text{Installed load}}{\text{Running load}}$$

207. Use of modified Rankine cycle causes the
(a) Reduction of the bore of cylinder
(b) Reduction of the stroke of cylinder
(c) Increase the work output of an engine
(d) Increase the efficiency of an engine

RPSC LECTURER 16.01.2016

Ans. (b) : Use of modified Rankine cycle causes the reduction of the stroke of cylinder.

208. Regeneration of simple Rankine cycle leads to
(a) The heating process in the boiler be less irreversible
(b) The heating process in the boiler be more irreversible
(c) The reduction in average temperature of heat addition
(d) The reduction in efficiency of cycle

RPSC LECTURER 16.01.2016

Ans. (a) : Regeneration—It is a process of steam extracting from intermediate of turbine which is used for increase the temperature of feed water. It tends to heating process in the boiler be less irreversible.

209. "Sinking fund method" is used to calculate the
(a) Initial cost of the power plant
(b) Installation cost of the power plant
(c) Depreciation cost of the power plant
(d) Interest on the loan borrowed to install the power plant

RPSC LECTURER 16.01.2016

Ans. (c) : "Sinking fund method" is used to calculate the depreciation cost of the power plant.

210. Fire tube boilers are those in which-
(a) flue gases pass through tubes and water around it
(b) water passes through the tubes and flue gasses around it
(c) forced circulation takes place
(d) tubes are laid vertically

RPSC IOF & Boiler 20.09.2017

Ans. : (a) Fire tube boilers are those in which flue gases pass through tubes and water around it.
Fire tube boilers are-
1. Cornish Boiler
2. Cochran Boiler
3. Locomotive Boiler
4. Lancashire Boiler
5. Scotch marine Boiler

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211. When a liquid boils at constant pressure, the following parameter increases

(a) temperature
(b) latent heat of vapourization
(c) kinetic energy
(d) entropy

RPSC IOF & Boiler 20.09.2017

Ans. : (d) In Rankine vapour cycle liquid boils at constant pressure in boiler and entropy increases.

212. Volume of 1 kg of dry steam is known as:

(a) total volume (b) saturated volume
(c) specific volume (d) none of these

(HPPSC LECT. 2016)

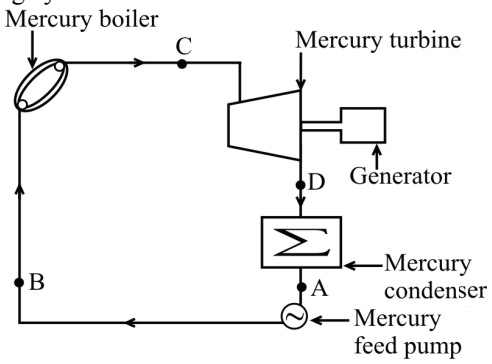
Ans. : (c) Volume of 1 Kg of dry steam is known as specific volume.

213. In binary vapour cycle

(a) Mercury is used in the bottoming cycle
(b) Steam is used in topping cycle
(c) Mercury is used in topping cycle
(d) Either mercury or steam is used in topping cycle

HPPSC W.S. Poly. 2016

Ans. : (c) In binary vapour cycle Mercury is used in topping cycle.



Binary vapour plant
In binary vapour cycle mercury is used in topping cycle.

214. The process of maintaining the speed of a steam turbine constant for various load conditions is known as :

(a) Reheating (b) Bleeding
(c) Governing (d) Cooling

HPPSC W.S. Poly. 2016

Ans. : (c) The process of maintaining the speed of a turbine for various load conditions is known as governing.
The throttle governing of a steam turbine is a method of controlling the turbine output by varying the quantity of steam entering into the turbine. This method is also known as servomotor method.

215. Economizer is used in a steam power plant to heat

(a) air (b) feed water
(c) flue gases (d) steam

RPSC AE 2016

Ans. : (b) Economizer is used in steam power plant to heat feed water.
An economiser is a device used to heat feed water by utilising the heat in the exhaust flue gases before leaving through the chimney. As the name indicates, the economiser improves the economy of the steam boiler.

216. Depreciation charges are high in case of

(a) thermal plant (b) diesel plant
(c) hydroelectric plant (d) gas turbine plant

RPSC AE 2016

Ans. : (a) Depreciation charges are high in case of thermal plant.

217. Which of the following is the commercial unit used to measure electricity consumption?

(a) Kilowatt-hour (b) Kilowatt
(c) Joule per second (d) Mega watts

UJVNL AE 2016

Ans. : (a) Kilowatt- hour is the Commercial unit used to measure electricity consumption.
1 kW.h=1kW.3600s=3600kWs=3600kJ=3.6MJ

218. For a given set of operating pressure limits of a Rankine cycle the highest efficiency occurs

(a) Saturated cycle (b) Superheated cycle
(c) Reheat cycle (d) Regenerative cycle

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
UPPSC AE 12.04.2016 Paper-II

Ans. : (d) Operating pressure limits of a Rankine cycle the highest efficiency occurs in regenerative cycle. In practical regenerative cycle the feed water enters the boiler at high temperature as compared to the corresponding temperature in case of simple Rankine cycle and it is heated by steam extracted or bleed from intermediate stages of the turbine.

219. A pipe carrying steam (h=3000kJ/kg) at flow rate of 1 kg/s and another pipe carrying steam (h=2500kJ/kg) at flow rate of 2 kg/s mixes adiabatically to flow in single pipe. Find the enthalpy of resulting steam:

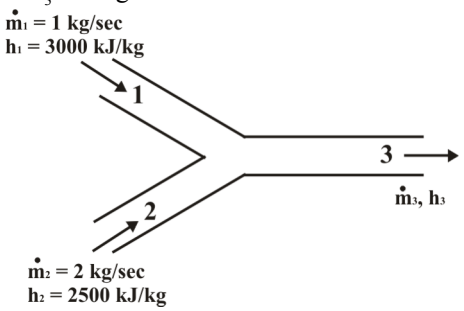
(a) 2767 kJ/kg (b) 2667 kJ/kg
(c) 2867 kJ/kg (d) 8000 kJ/kg

UPRVUNL AE 2016

Ans. (b) : Mass conservation law

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3 = 1 + 2$$

$$\dot{m}_3 = 3 \text{ kg/sec}$$



Energy conservation law,

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3$$

$$1 \times 3000 + 2 \times 2500 = 3 \times h_3$$

$$3 \times h_3 = 8000$$

$$h_3 = 2667 \text{ kJ/kg}$$

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220. The behaviour of superheated vapour is similar to that of

(a) perfect gas (b) steam
(c) air (d) ordinary gas

APPSC AEE MAIN 2016 PAPER-III

Ans. (a) : The behaviour of superheated vapour is similar to that of perfect gas (ideal gas).

221. Rankine efficiency of steam power plant

(a) improves in summer as compared to that in winter
(b) improves in winter as compared to that in summer
(c) in unaffected by climatic conditions
(d) none of these

RPSC IOF & Boiler 20.09.2017

Ans. : (b) Improves in winter as compared to that in summer

222. Steam enters adiabatic turbine operating steady state with an enthalpy of 3251.0 kJ/Kg leaves as a saturated mixture of 15 kPa with dryness faction (quality) 0.9. The enthalpy of the saturated liquid and vapour at 15 kPa are h_f = 225.94 kJ/Kg and h_g= 2598.3 kJ/Kg respectively. The mass flow rate of steam is 10 Kg/sec. Kinetic & Potential energy changes are negligible. The power output of Turbine in Mega Watt is–

(a) 6.5 (b) 9.1
(c) 8.9 (d) 27.0

RPSC IOF & Boiler 20.09.2017
GATE 2012

Ans. : (c) Given,

$h_1 = 3251.0$ kJ/kg
$h_f = 225.94$ kJ/kg
$h_g = 2598.3$ kJ/kg
$x = 0.9$
$h_2 = h_f + x\,(h_g - h_f)$
$= 225.94 + 0.9\,(2598.3 - 225.94)$
$= 2361.064$
$h_2 = 2361.064$ kJ/kg
$\therefore$ Power output from turbine,
P = Mass flow rate $\times (h_1 - h_2)$
$= 10 \times (3251 - 2361.064)$
$= 8899.36$ kJ/Sec.
$= 8899.36$ kJ/Sec.

$P \approx 8.9\,MW$

223. A small steam whistle (perfectly insulated) and doing no shaft work causes a drop of 0.8 kJ/kg in the enthalpy of steam from entry to exit. If the kinetic energy of steam at entry is negligible, the velocity of steam at exit is–

(a) 4 m/s (b) 20 m/s
(c) 80 m/s (d) 120 m/s

RPSC IOF & Boiler 20.09.2017

Ans. : (*) Velocity $= 44.72\sqrt{\Delta h}$
$= 44.72\sqrt{0.8}$
$= 39.99\ m/s$
$\approx 40 m/s$

Where, Δh is in kJ/kg and velocity in m/s.

224. Ratio of indicated work to hypothetical Indicated work in a steam engine is the:

(a) Indicated thermal efficiency
(b) Friction factor
(c) Mechanical efficiency
(d) Diagram factor

UJVNL AE 2016
ESE 1996

Ans. : (d) Diagram factor $= \dfrac{\text{Actual indicated work}}{\text{Hypothetical indicated work}}$

225. A solar thermal collector

(a) Collects the solar energy and reflects it back
(b) Absorbs the solar radiation and dissipates it to the ambient
(c) Collects and converts the solar energy into electrical energy
(d) Collects and converts the solar energy into thermal energy and delivers it to the next stage of the system

UPPSC AE 12.04.2016 Paper-II

Ans. : (d) A solar thermal collector collects and converts the solar energy into thermal energy and delivers it to the next stage of the system.

226. A solar cell is basically

(a) A voltage source, controlled by flux of radiation
(b) A current source, controlled by flux of radiation
(c) An uncontrolled current source
(d) An uncontrolled voltage source

UPPSC AE 12.04.2016 Paper-II

Ans. : (b) A solar cell is basically current source, controlled by flux of radiation.

227. The capacity of generators being installed in super thermal power plant is

(a) 100 MW (b) 200 MW
(c) 400 MW (d) 500 MW

UPPSC AE 12.04.2016 Paper-II

Ans : (d) The capacity of generators being installed in super thermal power plant is 500 MW.

228. Rankine cycle uses steam. The "high" temperature and pressure are 500 K and 35 bar; the "low" pressure is 1 bar. The pump work (kJ/kg) for the cycle is (specific volume of water = 1.043×10^{-3} m^2/kg)-

(a) 3.55 (b) 3.4
(c) 3.6 (d) None of these

RPSC ADE 2016

Ans. (a) : Suction side pressure in pump (P_1) = 1 bar
Delivery side pressure in pump (P_2) = 35 bar
Pump work = $-vdp = -1.043 \times 10^{-3} \times (35-1) \times 10^5$
$= -3545.2$ Joule/kg
$= -3.546$ kJ/kg
Note – Negative sign is due to work done on the pump.

229. Consider the following statements: 1. Boiler mountings are mainly protective devices 2. Steam stop Valve is an accessory 3. Feed water pump is an accessory 4. Economiser is a mounting which of these statements are correct?

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(a) 1, 2, 3 and 4 (b) 1, 2 and 4
(c) 2 and 3 (d) 3 and 1

RPSC VPITI 14.02.2016

Ans. (d) : Boiler mountings are protective/safety devices whereas accessories increases boiler efficiency.

Boiler Mountings–

Safety valve, water level indicator, pressure gauge, fusible plug, blow off cock, feed check valve, steam stop valve.

Boiler Accessories–

Superheater, economiser, injector, airpreheater, feed pump.

230. Steam lap is

(a) similar to missing quantity
(b) there is nothing like steam lamp
(c) the amount of opening in the steam port when the piston is exactly at the end of the stroke
(d) the distance the valve must move from its mid-position before steam is admitted to the cylinder

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (d) : Steam lap is the distance that valve must move from its mid position before steam is admitted to the cylinder.

231. The size of coal fed in a pulverized coal fired boiler for efficient combustion should be

(a) 75 -85% of coal passing through 100 mesh BIS sieve
(b) 85 - 95% of coal passing through 150 mesh BIS sieve
(c) 70-75% of coal passing through 200 mesh BIS sieve
(d) 70 - 80% coal passing through 50 mesh BIS sieve

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : The size of coal feed in pulverized coal fired boiler for efficient combustion is 70–75 % at coal passing through 200 mesh BIS sieve.

232. A vessel of volume 1.5 m^3 contains a mixture of liquid water and steam in equilibrium at 1 bar given that 80% of the volume is occupied by the steam. Find the dryness fraction of the mixture. Assume at 1 bar; $v_f = 0.002\ m^3/kg$ and $v_g = 2\ m^3/kg$

(a) 0.8 (b) 0.08
(c) 0.04 (d) 0.004
(e) 0.008

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (d) : Given, $v_f = 0.002\ m^3/kg$, $v_g = 2\ m^3/kg$

$V_g = 0.80\ m^3$, $V_f = 0.20\ m^3$

$$\text{So, mass of mixture} = \frac{V_g}{v_g} + \frac{V_f}{v_f}$$

$$= \frac{0.80}{2} + \frac{0.20}{0.002}$$

$$= 0.4 + 100$$

$$= 100.4 \text{ kg}$$

$$\text{Specific volume (v)} = \frac{1}{m} = \frac{1}{100.4}$$

$$= 0.00996\ m^3/kg$$

We know that,

$$v = v_f + x(v_f - v_g)$$

$$0.00996 = 0.002 + x(2 - 0.002)$$

$$0.00996 - 0.002 = x(2 - 0.002)$$

$$x \times 1.998 = 0.00796$$

$$x = \frac{0.00796}{1.998}$$

$$x = 0.00398 \approx 0.004$$

233. A steam power plant has the boiler efficiency of 90%, turbine efficiency of 92%, generator efficiency of 87% and cycle efficiency of 47%. If 8% of the generated power is used to run the auxiliaries, what is the overall plant efficiency?

(a) 31% (b) 35%
(c) 38% (d) 41%
(e) 45%

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : As we know,

$$\eta_{overall} = \eta_{boiler} \times \eta_{turbine} \times \eta_{generator} \times \eta_{aux}$$

$$\eta_{aux} = \frac{\text{Net power transmitted by generator}}{\text{Gross power produced by plant}}$$

$$= 0.92$$

$$\eta_{overall} = 0.90 \times 0.92 \times 0.87 \times 0.47 \times 0.92$$

$$= 31\%$$

234. Natural Circulation through water wall in boiler operating in subcritical regime is caused by

(a) Temperature difference between economizer inlet water and down comer tube water
(b) Density difference between furnace wall water and downcomer tube water
(c) Water and steam flow difference between economiser and final superheater
(d) Pressure difference between boiler drum pressure and superheater inlet steam pressure

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : Natural circulation of water in boiler due to density difference furnace wall water and downcomer tube water.

235. Water circulation in a thermo-syphon cooling system caused by :

(a) Conduction currents
(b) A belt driven water pump
(c) A belt driven impeller
(d) Change in density of water

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) Water circulation in a thermo-syphon cooling system caused by change in density of water.

236. Furnace wall of water tube boilers receives mainly

(a) Sensible heat (b) Radiant heat
(c) Convection heat (d) Conduction heat

Rajasthan Nagar Nigam AE 2016, Shift-I

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Ans. (b) : As mentioned earlier, water tube boilers are used for high output.

- These are invariably used in all the power plants.
- These are either forced circulation type or natural circulation type.
- 'Steam drum' is the heart of these boilers and stores wet and dry steam.

237. Steam flow in a steam generator is controlled by using :

(a) Drum level sensor
(b) Feed water flow sensor
(c) Steam flow sensor
(d) All of the above

OPSC AEE 2015 Paper - II

Ans : (c) Steam flow in a steam generator is controlled by using steam flow sensor.

238. In the design of tall chimney for a factory, in addition to strength and rigidity considerations, care should be taken to check its safety against wind, because.

(a) the wind creates a large bending moment on the chimney
(b) the wind can have any direction
(c) additional forces are exerted on the chimney due to mixing of the effluent gases with the surrounding air
(d) of the effect of vortex shedding on the chimney

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : In the design of tall chimney for a factory, in addition to strength and rigidity considerations, care should be taken to check its safety against wind, because of the effect of vortex shedding on the chimney.

239. What are the effects of regenerative feed water heating for the same turbine output?

1. It significantly increases the cycle efficiency and reduces the heat rate.
2. It increases the steam flow rate.
3. It increases the steam flow to the condenser.
4. If there is no change of boiler output, the turbine output drops.

Select the correct answer using the code given below:

(a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 1, 3 and 4 only (d) 2, 3 and 4 only

ESE 18.07.2021

Ans. (b) Effect of regeneration:

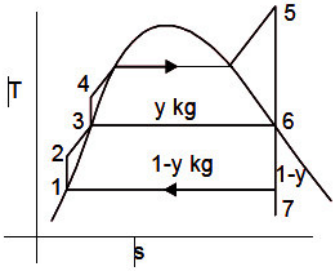
(i) $\eta \uparrow$ & Heat rate $\downarrow$ Due to $\uparrow$ in mean temperature of heat addition.

$$\eta \propto \frac{1}{\text{Heat rate}}$$

(ii) For the same turbine output steam flow $\uparrow$ (bigger boiler is required) due to bleeding of steam to heat the feed water.

(iii) Steam flow rate in Condenser is reduces as each stage of regeneration, steam is bleeding off because of this less amount of steam condense in condenser.

(iv) If boiler size is fixed and same amount of steam is produced then because of bleeding, less amount of steam available at turbine at each stage leads to $\uparrow$ in turbine output.



- Due same boiler output

$$\eta = \frac{W_T \rightarrow \text{reduced}}{Q \rightarrow \text{more reduced}}$$

- Since the amount of heat input in the boiler reduced significantly compared to work lost,

the efficiency$\uparrow$.

240. The diameter of fire tubes in Cochran boiler is of the order of

(a) 2 cm (b) 6 cm
(c) 8 cm (d) 12 cm

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : The diameter of fire tubes in Cochran boiler is of the order of 6 cm.

241. Which is not correct statement about the function of furnace

(a) To provide proper conditions for continuous complete combustion
(b) mix fuel with air and ignite
(c) Separate ash from coal
(d) Maintain heat supply to prepare and ignite the incoming fuel

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Not correct statement option (c) separate ash from coal.

242. In steam power plant, what is the outcome of regenerative feed heating :

(1) Increase in specific output
(2) Increase in cycle efficiency
(3) Improved quality of exhaust steam
(4) Reduced condenser load

Select the correct answer

(a) (1) and (3) only (b) (2) only
(c) (2) and (3) (d) (1), (2) and (3)

TSPSC Managers, 2015

Ans. (c) : In steam power plant, regenerating feed heating-

- Increase in cycle thermal efficiency $\eta\uparrow$
- Reduced condenser load,
- decreases in specific output.

243. The average load factor of thermal power plants in India is

(a) 100% (b) 80 to 95%
(c) 50 to 60% (d) 35 to 45%

VIZAG STEEL MT 2015

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Ans. (c) : The average load factor of thermal power plants in India is 50 to 60%.

244. The basic job of feed water treatment in boilers is to overcome the problem of :

(a) carryover
(b) embrittlement
(c) All of three mentioned here
(d) corrosion

GWSSB AAE, 27.12.2015

Ans. (c) : The basic of feed water treatment in boiler to reduce carryover, embrittlement and corrosion.

245. Fire tube boilers are those in which:

(a) water passes through the tubes and flue gases around it :
(b) forced circulation takes place
(c) None of three mentioned here.
(d) flue gases pass through tubes and water around it.

GWSSB AAE, 27.12.2015

Ans. (d) : Flue gases passes through tubes and water around it.

246. A regenerative steam cycle results in

(a) increased work output per unit mass of steam
(b) decreased work output per unit mass of steam
(c) increased thermal efficiency
(d) decreased work output per unit mass of steam as well as increased thermal efficiency

Nagaland PSC CTSE 2015 Paper-II

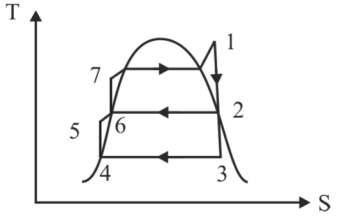
Ans. (d) : Effect of Regeneration–

(1) It increases the cycle efficiency because supplying heat rate decreases.

$$\eta \uparrow = \frac{W_{net}}{Q_s \downarrow}$$

(2) There is no change in boiler output but turbine output decreases.

(3) It decreases the steam flow rate to the condenser, thus requiring smaller condenser.



247. Consider the following statements:

1. In a steam power plant, the volume flow at the low pressure end is considerably reduced by bleeding.
2. Theoretically, the efficiency of a steam power plant increases with the increasing number of feed heaters but the gain is in diminishing return.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) :

248. A thermal electric power plant produces 1500 MW of power. If the coal releases 900 × 10^7 KJ/h of energy, then what is the rate at which heat is rejected from the power plant?

(a) 500 MW (b) 1000 MW
(c) 1500 MW (d) 2000 MW

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : Energy released by coal = 900 × 10^7 kJ/hr
= 2500 MW

Heat rejected from power plant
2500 – 1500
= 1000 MW

249. Consider the following statements:

Expansion joints in steam pipelines are installed to

1. Allow for future expansion of plant.
2. Take stresses away from flanges and fittings.
3. Permit expansion of pipes due to temperature rise.

Of these statements, which are correct

(a) 1, 2 and 3 (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Expansion joints in steam pipelines are installed to,
1. Allow for future expansion of plant.
2. Take stresses away from flanges and fittings.
3. Permit expansion of pipes due to temperature rise.

250. Compound steam engine is one in which

(a) Steam expands twice
(b) Two engines are combined together
(c) Major parts are repeated twice
(d) Two units are put together

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Compound steam engine is one in which steam expands twice.

251. What is the efficiency of an ideal regenerative Rankine cycle power plant using saturated steam at 327^0C and pressure 135 bar at the inlet to the turbine and condensing temperature of 27^0C (corresponding saturation pressure of 3.6 kPa) ?

(a) 92% (b) 33%
(c) 50% (d) 42%

TSGENCO AE 14.11.2015

Ans. (c) : Ideal regenerative Rankine cycle efficiency is same as carnot cycle

$$\eta = 1 - \frac{T_2}{T_1}$$

Given, –

T_1 = 327°C = 327 + 273
= 600 K

T_2 = 27^0C = 273 + 27 = 300 K

$$\eta = \left(1 - \frac{300}{600}\right)$$

$\eta = 50\%$

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252. Degree of superheating obtained using a liquid-to suction heat exchanger is:

(a) Always greater than the degree of sub cooling
(b) Always less than degree of sub cooling
(c) Always equal to degree of sub cooling
(d) Depends on the effectiveness of heat exchanger
(e) Depends on the type of the refrigerant

CGPSC AE 26.04.2015 Shift-II

Ans. (a) : Degree of Superheating obtained using a liquid-to-suction heat exchanger is always greater than the degree of sub cooling.

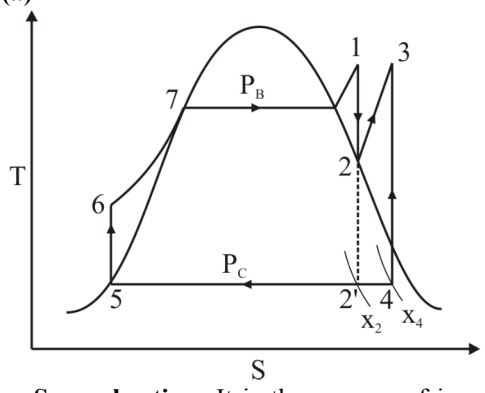
253. Consider the following statement regarding superheating in Rankine cycle

(1) Reduces specific steam consumption
(2) Increase dryness fraction at exhaust for the same condenser pressure
(3) It reduces cycle efficiency, of these statement

(a) 1 and 2 are correct
(b) 2 and 3 are correct
(c) 1 and 3 are correct
(d) 1, 2 and 3 are correct

TSPSC AEE 2015, CSE Pre-1995

Ans. (a)



Super heating- It is the process of increasing the temperature at constant pressure above saturated vapour.

Effect of superheating in Rankine cycle-

$W_T \uparrow$, W_P = constant
$W_{net} \uparrow$, $Q_r \uparrow$, $Q_S \uparrow$, $T_{MA} \uparrow$, $T_{MR} \uparrow$
$x \uparrow \rightarrow$ erosion of blade$\downarrow$

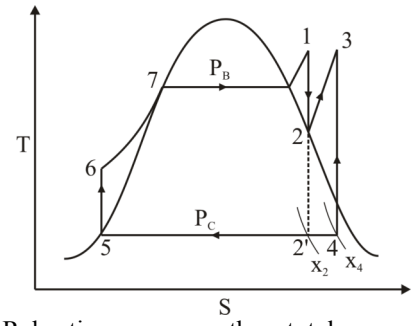
254. Which combination of the following statements are correct. The incorporation of reheater in a steam power plant

P. Always increase thermal efficiency
Q. Always increase dryness fraction at condensed inlet
R. Increase the mean temperature of heat addition
S. Always increases specific work output

(a) P and S (b) Q and S
(c) P only (d) Q only

TSPSC AEE 2015, GATE 2007

Ans. (b) : Reheating- The primary aim of the reheating is to improve the quality (x) of the working fluid [steam] at the exist of turbine. So that less erosion of blade.



Reheating means the total expansion of working fluid takes place in more than one stage there for with the use of reheating the specific steam consumption [S.S.C.] and make the plant compact.

Effect of reheating in Rankine cycle –

$W_T \uparrow$, W_P = constant, $Q_R \uparrow$, $\eta \uparrow\downarrow$, $W_{net} \uparrow$, $T_{MA} \uparrow\downarrow$, $x \uparrow \rightarrow$ vapour $\uparrow \rightarrow$ liquid $\downarrow \rightarrow$ erosion of turbine blade$\downarrow$.

255. Dryness fraction is the ratio of :

(a) mass of dry steam to the mass of water vapour in suspension
(b) mass of water vapour in suspension to the mass of water vapour and mass of dry steam
(c) mass of dry steam to the mass of dry steam and mass of water vapour in suspension
(d) mass of water vapour in suspension to the mass of dry steam

(KPSC AE. 2015)

Ans. : (c) Dryness fraction is the ratio of mass of dry steam to the mass of dry steam and mass of water vapour in suspension.

$$x = \frac{m_g}{m_g + m_f} = \frac{m_g}{m}$$

m_g = Mass of actual dry steam
m_f = Mass of water in suspension
$m = m_g + m_f$

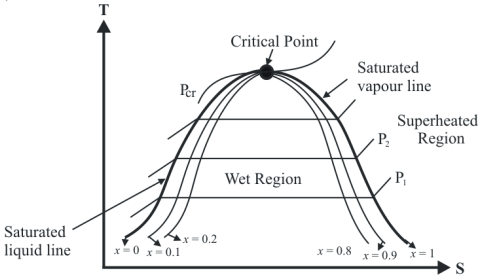
The value of dryness fraction. In case of dry steam is unity. At this stage, the mass of water in suspension (m_f) is zero.

256. If the dryness fraction of steam (x) is less than '1' then the steam is called as

(a) Dry steam (b) super heated steam
(c) Wet steam (d) Both (a) & (b)

TSPSC AEE 2015

Ans. : (c) If the dryness fraction of steam (x) is less then 1, then steam is called as wet steam.



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257. A Carnot Cycle is having an efficiency of 0.75. If the temperature of the high temperature reservoir is 727°C, what is the temperature of low temperature reservoir?

(a) 23°C (b) 0°C
(c) –23°C (d) 250°C

GPSC AE 26.07.2015

Ans. (c) : $\eta = 1 - \frac{T_L}{T_H}$

$0.75 = 1 - \frac{T_L}{(727+273)}$

$T_L = 250K = -23^0C$

258. According to IBR the thickness of boiler shell is not to be less than

(a) 5 mm (b) 7 mm
(c) 10 mm (d) 12 mm

GPSC AE 26.07.2015

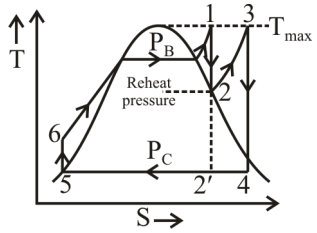
Ans. (b) : According to IBR the thickness of boiler shell is not to be less than 7 mm.

259. In a Rankine cycle, with maximum steam temperature being fixed from metallurgical considerations, as the boiler pressure increases:

(a) the condenser load will increase
(b) the quality of turbine exhaust will decrease
(c) the quality of turbine exhaust will increase
(d) the quality of turbine exhaust will remain unchanged

UPPSC AE 13.12.2020 Paper-II
HPPSC Asstt. Prof. 2014
ESE 1997

Ans. (b)



- Dryness fraction (quality of steam) x↑
- Moisture content ↓

As pressure increases quality of steam at exhaust will decreases to increase quality of steam at outlet of turbine reheat is adopted.

260. Rankine cycle efficiency for a power plant is 29%. The carnot cycle efficiency will be

(a) 15% (b) 25%
(c) 29% (d) None of these

MPPSC State Forest Service Exam, 2014

Ans. (d) : $\eta_{rankine} = \frac{Q_1 - Q_2}{Q} = 1 - \frac{Q_2}{Q_1} = 29\%$

$\eta_{carnot} = 1 - \frac{T_2}{T_1}$

But here we can't find the efficiency because both process have different.

261. An example of forced circulation boiler is

(a) Cochran boiler
(b) Lancashire boiler
(c) Babcock and Wilcox boiler
(d) Lamont boiler

MPPSC State Forest Service Exam, 2014

Ans. (d) : Lamont boiler is a high pressure forced circulation, water tube boiler with internally fired furnace, an external pump is used to circulate the pressurized water.

262. Function of air pump in condenser is to

(a) Remove water
(b) Maintain vacuum
(c) Remove condensate
(d) None of these

MPPSC State Forest Service Exam, 2014

Ans. (b) : The main function which an air pump performs is that it maintains vacuum in the condenser nearly as possible equal to that corresponding to the exhaust steam temperature by removing air from the condenser.

263. For a pure substance, what are the numbers of the thermodynamic degree of freedom for saturated vapour and superheated vapour, respectively?

(a) 1 and 1 (b) 1 and 2
(c) 2 and 1 (d) 2 and 2

HPPSC Asstt. Prof. 2014, ESE 2007

Ans. (b) :

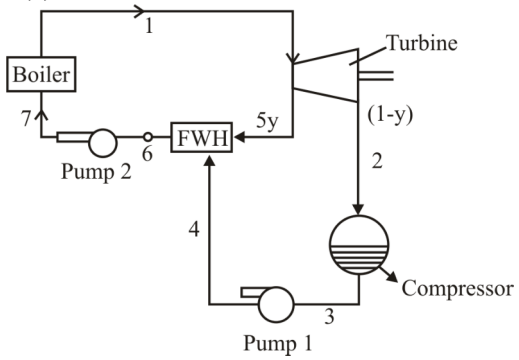
Degree of freedom	
Saturated vapour-	1
Superheated vapour -	2

264. In regenerative Rankine's cycle

(a) Feed water is heated with the help of steam
(b) The steam is extracted at a suitable point in a turbine and is reheated
(c) The steam is heated after expansion in the turbine
(d) The steam is heated before entering the turbine

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a)



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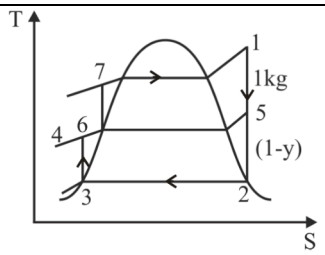

FWH = Feed water Heater

- In feed water heater, water is heated with the help of steam.

265. Demand factor is defined as
(a) Average load/ maximum demand
(b) Maximum demand/ connected load
(c) Connected load/ maximum demand
(d) Maximum demand × connected load

TNPSC AE 2014

Ans. (b) : Demand factory is defined as the ratio of maximum demand to connected load.

266. In the operation of steam engines the vapour cycle adopted is
(a) Carnot cycle
(b) Rankine cycle
(c) Modified Rankine cycle
(d) Regenerative cycle

TNPSC AE 2014

Ans. (c) : In the operation of steam engines the vapour cycle adopted is modified Ranking cycle.

267. Without the use of superheater a boiler produces steam of about
(a) 80% dryness fraction
(b) 90% dryness fraction
(c) 98% dryness fraction
(d) 88% dryness fraction

TNPSC AE 2014

Ans. (c) : Without the use of superheater a boiler produces steam of about 98% dryness fraction.

268. The density of super saturated steam in comparison to ordinary saturated steam at the corresponding pressure is about.
(a) 0.5 times (b) 1.0 time
(c) 8.0 times (d) 4.0 times

GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : The density of super saturated steam in comparison to ordinary steam at the corresponding pressure is about 8 time as sp. volume of supersaturated steam is greater than saturated steam at same pressure.

269. An ideal fluid for vapour turbine cycle should have
(a) High critical temperature with low pressure
(b) High critical temperature with high pressure
(c) Low critical temperature with high pressure
(d) Low critical temperature with low pressure

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : An ideal fluid for vapour turbine cycle should have – high critical temperature and low pressure as pressure increase its latent heat of evaporation decreases and at critical points it becomes zero.

270. A power plant giving least running cost of production of electric power is
(a) Nuclear power plant
(b) Steam power plant
(c) Hydro electric power plant
(d) Gas turbine power plant

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : A power plant giving least running cost of production of electric power is nuclear power plant but installation cost is very high

271. The reheat cycle helps to reduce
I. fuel consumption.
II. steam flow.
III. the condenser size.
Of these statements, which is correct?
(a) I and II (b) I and III
(c) II and III (d) I, II and III

APPSC Poly. Lect. 2013, ESE 2000

Ans. (a) : The reheat cycle helps to reduce, fuel consumption and steam flow. In reheat cycle condenser size increases while in regeneration cycle condenser size reduces.

272. Hydro electric power plants classified on the basis of head of operation, are..
(a) ultra low, low, medium and light only
(b) low, medium and light only
(c) ultra low, low and medium only
(d) ultra low and medium

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Hydro electric power plants are classified on the basis of head of operation, are –
(1) ultra low, (2) low, (3) medium and light.

273. The locomotive boiler has
(a) 137 fire tubes and 44 super-heated tubes
(b) 147 fire tubes and 34 super-heated tubes
(c) 157 fire tubes and 24 super-heated tubes
(d) 167 fire tubes and 14 super-heated tubes

JPSC AE 2013 Paper-V

Ans. (c) : 157 fire tubes and 24 super-heated tubes

274. In a boiler, various heat losses take place. The biggest loss is due to
(a) moisture in fuel (b) dry flue gases
(c) steam formation (d) unburnt carbon

JPSC AE 2013 Paper-V

Ans. (b) : The biggest loss is due to dry flue gases.

275. The maximum percentage gain in regenerative feed heating cycle, the thermal efficiency
(a) increases with number of feed heaters increasing
(b) decreases with number of feed heaters increasing
(c) remains same unaffected by number of feed heaters
(d) None of the above

JPSC AE 2013 Paper-V

Ans. (a) : In regenerative cycles, the increase in efficiency due to regeneration is proportional to increases in temperature of feed water in a feed water heater.

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276. The reheat factor depends upon
(a) initial pressure and super-heat
(b) exit pressure
(c) turbine stage efficiency
(d) All of the above

JPSC AE 2013 Paper-V

Ans. (d) : The reheat factor (R.F.)

$$R.F. = \frac{\text{Cumulative heat drop}}{\text{Isentropic heat drop}}$$

It depends on:
(i) Initial pressure and superheat.
(ii) Exit pressure.
(iii) Turbine stage efficiency.
It increase with the increase in number of stages. The value of reheat factor varies from 1.02 to 1.06.

277. Which power plant has maximum life expectancy?
(a) Hydro-Power Plant
(b) Thermal Power Plant
(c) Nuclear Power Plant
(d) Same for all

SJVN ET 2013

Ans. (a) : Hydro-power plant has maximum life expectancy.

278. The degree of collection for a given ash collector in which 75 kg/s of ash is entering and 25 kg/s of ash leaving is
(a) 0.67 (b) 0.75
(c) 0.52 (d) 0.93

TNPSC AE 2013

Ans. (a) : $m_{in} = 75$ kg/s
$m_0 = 25$ kg/s
Collection of ash in given ash collector is given as

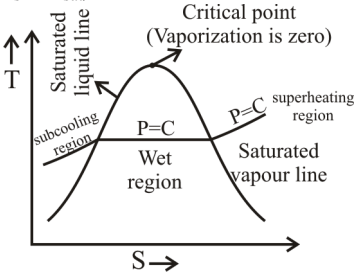
$$\text{Degree of collection (DOC)} = \frac{\dot{m}_{in} - \dot{m}_o}{\dot{m}_{in}}$$

$$= \frac{75 - 25}{75}$$
$$= 0.6667$$

279. The excess temperature ΔT_e
T_s - Surface temperature
T_{sat} - Saturation temperature
T_{sup} - Super heated
(a) T_{sup} (b) $(T_s + T_{sat})$
(c) $(T_{sup} - T_{sat})$ (d) $(T_s - T_{sat})$

TNPSC AE 2013

Ans. (d) : Excess temperature ΔT_e or degree of superheating
$= (T_s - T_{sat})$



280. The heat carried away by coolant is called as
(a) Unaccounted Loss (b) Dead Loss
(c) Minor Loss (d) Major Loss

TNPSC AE 2013

Ans. (b) : The heat carried away by coolant is called as dead loss.

281. The following boiler operates in supercritical range of pressure:-
(a) Benson (b) Babcock and Wilcox
(c) Loeffler (d) Cornish

UKPSC AE-2013, Paper-II

Ans. (a) : Benson boiler operates in supercritical range of pressure.

282. The capacity of boiler is defined as:-
(a) The volume of feed water inside the shell
(b) The volume of the steam space inside the shell
(c) The maximum pressure at which steam can be generated
(d) Amount of water converted into steam from 100°C to 110°C in one hour

UKPSC AE-2013, Paper-II

Ans. (d) : The capacity of boiler is defined as amount of water converted into steam from 100°C to 110°C in one hour.

283. The type of safety valve recommended for high pressure boiler is:-
(a) Dead weight safety valve
(b) Lever safety valve
(c) Spring loaded safety valve
(d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) :

284. Babcock-Wilcox boiler is
(a) water tube boiler (b) once through boiler
(c) supercritical boiler (d) fire tube boiler

VIZAG STEEL MT 18.08.2013

Ans. (a) : Water tube bailer– It is the type as boiler in which water circulates in tubes heated externally by the fire fuel is burned inside the furnace, creating hot gas which boils water in the steam-generating tubes.

285. Water requires for attemperation is taken from
(a) reheater (b) preheater
(c) condenser (d) economizer

VIZAG STEEL MT 18.08.2013

Ans. (b)
- Boiler feed water is used attemperation and water required for attemperation is taken from preheater.
- A direct contact attemperator injects a stream of high purity water into the superheated steam.
- Attemperation is the primary means for controlling the degree of superheat in a superheated boiler.

286. Super critical boilers are
(a) fire tube boilers
(b) water tube boilers
(c) a combination of (A) and (B)
(d) once through boilers

VIZAG STEEL MT 18.08.2013

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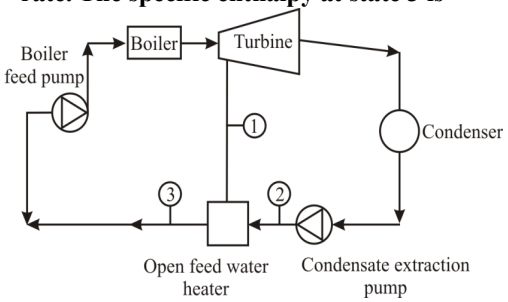
Ans. (d) : Supercritical Boiler– A super critical steam generator is a type of boiler that operates at supercritical pressure, frequently used in the protection of electric power. In contrast to a subcritical boiler in which bubbles can form, a super critical steam generator operator at pressure above the critical pressure 22 MPa.

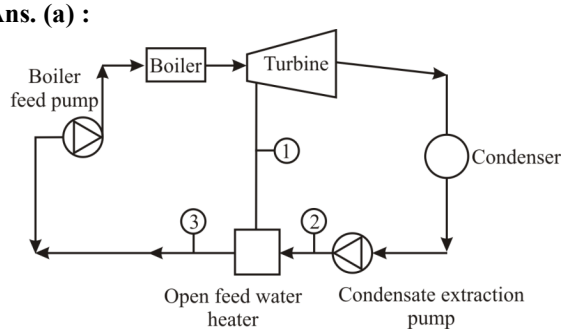
287. A thermal power plant operates on a regenerative cycle with a single open feed water heater as shown in the figure. From the state points shown, the specific enthalpies are: h_1 = 2800 kJ/kg, and h_2 = 200 kJ/kg. The bleed to the feed water s 20% of the boiler steam generation rate. The specific enthalpy at state 3 is



(a) 720 kJ/kg (b) 2280 kJ/kg
(c) 1500 kJ/kg (d) 3000 kJ/kg

VIZAG STEEL MT 18.08.2013
GATE 2008

Ans. (a) :



$h_1 = 2800$ kJ/kg $h_2 = 200$ kJ/kg $h_3 = 0.2h_1 + h_2 \times 0.8$

$= 0.2 \times 2800 + 200 \times 0.8 = 560 + 160$

$\boxed{h_3 = 720 \text{ kJ/kg}}$

288. Within the boiler, the temperature of steam is highest in

(a) water drum (b) water tubes
(c) water walls (d) super heater

VIZAG STEEL MT 18.08.2013

Ans. (d) : Super Heater – Super heater tubes have steam on one side and hot combustion gases on the other. surface temperature are higher than the boiler tubes. The passage of heat from the flue gas through tube wall increases the temperature to about 565°C.

289. Reheat cycle in steam power plant is used to :

(a) prevent excess of 10-12% moisture content in last stages of turbine
(b) utilize heat of the flue gases
(c) increase plant efficiency
(d) improve condenser performance

J&K PSC Civil Services Pre, 2013

Ans. (a) : Reheat cycle in steam power plant is used to prevent excess of 10-12% moisture content in last stages of turbine.

290. The maximum percentage gain in regenerative feed heating cycle thermal efficiency

(a) increases with number of feed heaters increasing
(b) decreases with number of feed heaters increasing
(c) remains same unaffected by feed heaters
(d) none of the above

APPSC Poly. Lect. 2013

Ans. (a) : The maximum percentage gain in regenerative feed heating cycles thermal efficiency increase with number of feed heaters increasing.

291. In steam tables the entropy is shown a zero for

(a) Saturated vapour at atmospheric pressure
(b) Saturated liquid at atmospheric pressure
(c) Saturated vapour at 0°C
(d) Saturated liquid at 0°C

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (d) : In steam tables the entropy is shown a zero for saturated liquid at 0°C.
The unit of specific entropy in the steam tables is kJ/kg-K and the values are given relative to an arbitrary reference state (i.e. triple point).
In the steam tables, the entropy of saturated liquid at 0.01°C ≈ 0°C is given by value of zero.

292. Choose the wrong statement about the regenerative vapour power cycle employing bleed steam for feed water heating:

(a) It increases the thermodynamic efficiency of the cycle.
(b) The thermal stresses in the boiler are reduced
(c) Work done per kg of steam is increased.
(d) It requires boiler of large capacity for same level of power output as compared to a simple Rankine cycle.

RPSC AEN Pre-2013

Ans. (c) : About the regenerating vapour power cycle employing bleed steam for feed water heating.

- The regeneration process in steam power plants is accomplished by extracting (or bleeding) steam from turbine at various stages and feed that steam in heat exchangers where the feed water is heated. These heat exchangers are called regenerator or Feed Water Heater (FWH).
- It increases the thermodynamic efficiency of the cycle.
- The thermal stresses in the boiler are reduced.
- It requires boiler of large capacity for same level of power output as compared to a simple Rankine cycle.

293. In a steam condenser, the partial pressure of steam and air are 0.06 bar and 0.007 bar respectively. The condenser pressure is :

(a) 0.067 bar (b) 0.06 bar
(c) 0.053 bar (d) 0.007 bar

TNPSC AE (Industries) 09.06.2013

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Ans. (a) : $P_C = P_a + P_s$
$= 0.06 + 0.007$
$= 0.067$ bar

294. Given below an extract from steam tables.

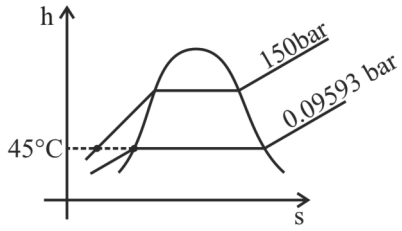
		Specific Volume (m^3/kg)		Enthalpy (KJ/kg)	
Temperature (°C)	P saturated (bar)	Saturated liquid	Saturated vapour	Saturated liquid	Saturated vapour
45	0.09593	0.001010	15.26	188.45	2394.8
342.24	150	0.001658	0.10337	1610.5	2610.5

The specific enthalpy of water in KJ/kg at 150 bar and 45°C will be

(a) 203.60 (b) 200.53
(c) 196.38 (d) 188.45

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2006

Ans. (a) :



$\Rightarrow (h)_{\text{at 45°C and 150bar}}$
$= (h)_{\text{at 45° and 0.0959bar}} + (v_f)_{\text{at 45°C and 0.0959 bar}} \times (dP)$
$= 188.45 + 0.00101\,(150 \times 100 - 9.59)$
$= 188.45 + 15.14 = 203.59$ kJ/kg $\simeq 203.6$ kJ/kg

295. The diameter of Cornish boiler is of the order of :

(a) 1-2 m (b) 1.5-2.5 m
(c) 2-3 m (d) 2.5-3.5 m

J&K PSC Civil Services Pre, 2013

Ans. (a) : Cornish boiler consist of an iron tube of approximate 1.5 m dia and 8.5 m long.

296. Which of the following is a fire tube type high pressure boiler?

(a) Velox boiler (b) Benson boiler
(c) Loeffler boiler (d) Lancahsire boiler

RPSC AEN Pre-2013

Ans. (a) : Velox Boiler– Velox boiler is a high pressure, forced circulation water tube boiler.
Benson boiler- Its a high pressure water tube boiler with forced circulation.
Lancahsire– Lancashire boiler is a horizontal and stationary type fire tube boiler.
Loeffler boiler- Loeffler is a water tube boiler used for generating steam by the burning of fuel which is coal and further that used for generation of power.

297. The peak loaded on a power plant is 40 MW and a group of four loads having maximum demand of 15 MW, 12 MW, 18 MW and 5 MW respectively are connected to the plant. The capacity of the plant is 60 MW. What is diversity factor of the plant?

(a) 1.5 (b) 1.25
(c) 0.8 (d) 0.75

RPSC AEN Pre-2013

Ans. (b) : Diversity factor, (DF)

$$= \frac{\sum(\text{Individual maximum demands})}{\text{maximum demand of the system}}$$

$$= \frac{12+15+18+5}{40}$$

$$= \frac{50}{40} = 1.25$$

298. The function of a condenser in thermal power plant is

(a) to act as reservoir to receive steam for turbine
(b) to condense steam into condensate to be reused again
(c) to create vacuum
(d) all of the above

BPSC AE 2012 Paper - V

Ans. : (d) : The function of a condenser in thermal power plants is -
■ to act as reservoir to receive steam for turbine.
■ to condense steam into condensate to be reused again.
■ to create vacuum.

299. After turbine trip, the cooling water to condenser

(a) is stopped immediately
(b) is stopped after 15-30 minutes
(c) is never stopped
(d) may be stopped anytime

BPSC AE 2012 Paper - V

Ans. : (b) : After turbine trip, the cooling water to condenser is stopped after 15-30 minutes.

300. Which of the following are the advantages of pulverized coal firing?
1. Higher boiler efficiency.
2. Fast response for no load changes.
3. Ability to use low preheated air reducing internal losses.
4. Ability to release large amounts of heat enabling it to generate about 2000 t/h of steam in one boiler.
Select the correct answer using the code given below:

(a) 1 and 2 only (b) 1 and 3 only
(c) 1 and 4 only (d) 2, 3 and 4 only

ESE 18.07.2021

Ans. (c) : Pulverized coal firing system has following advantages:
- It has better and far response over firing with load variation.
- Ability to use high preheated air reducing exhaust losses.
- Ability to release large amount of heat enabling it to generate about 2000 t/h of steam or more in one boiler.

301. One kg steam sample contains 0.4 kg water vapor. Its dryness fraction is

(a) 0.4 (b) 0.6
(c) $\frac{0.4}{1.4}$ (d) 0.4×0.6

BPSC AE 2012 Paper - V

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Ans. : (a)

$$\text{Dryness fraction} = \frac{\text{mass of vapour}}{\text{total mass}}$$

$$= \frac{m_v}{m_v + m_a} = \frac{0.4}{1} = 0.4$$

302. For high boiler efficiency the feed water is heated by

(a) recuperator (b) convective heater
(c) super heater (d) economiser

UKPSC AE 2012 Paper–II

Ans. (d) : economiser

303. Which of the following is not a high pressure boiler ?

(a) Lancashire boiler (b) La-mont boiler
(c) Benson boiler (d) Loeffler boiler

UKPSC AE 2012 Paper–II

Ans. (a) : Lancashire boiler

304. The most practical vapour power cycle is

(a) Carnot (b) Joule
(c) Rankine (d) Binary

UKPSC AE 2012 Paper–II

Ans. (c) : Rankine

305. The shell diameter and length of locomotive boiler are

(a) 1.5 m, 4 m (b) 1.5 m, 6 m
(c) 1 m, 4 m (d) 2 m, 4 m

BPSC AE 2012 Paper - V

Ans. : (a) : Shell diameter of locomotive boiler is 1.5 m and length of locomotive boiler is 4 m.

306. Consider the following statements for cooling towers:

(1) Cooling tower is an artificial device used to cool the hot cooling water coming out of condenser more effectively.
(2) The amount of water usually lost with induced draft cooling tower ranges from 5% to 6% by evaporation.
(3) The amount of water usually lost with induced draft cooling tower ranges from 7% to 8% by drift losses.
(4) The rate of evaporation of water and its cooling effect on the remaining water depends upon the relative humidity of air passing through the tower.

Which of the above statements are correct?

(a) 1 and 4 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

ESE 18.07.2021

Ans. (a) : The amount of water usually lost with induced draft cooling towers ranges from 1 to 2% by evaporation and 0.5 to 2% by drift losses.

- Cooling tower is an artificial device used to cool the hot cooling water coming out of condenser more effectively.
- The rate of evaporation of water and its cooling effect on the remaining water depends upon the relative humidity of air passing through the tower.

307. Super heating of steam for steam engine results in

(a) reducing misting quantity
(b) more balanced engine
(c) less work output
(d) increase in efficiency of Rankine cycle

APPSC AEE 2012

Ans. (d) : A common modification to the Rankine cycle is superheating the steam in the boiler increase the efficiency of cycle.

308. Water tube boilers are

(a) internally fired
(b) externally fired
(c) internally as well as externally fired
(d) none of these

TNPSC ACF 2012

Ans. (b) : Water tube boilers are externally fired where as fire tube boilers are internally fired.

309. The draught may be produced by

(a) mechanical fan (b) chimney
(c) a steam jet (d) all of these

TNPSC ACF 2012

Ans. (d) : The draught may be produced by
(1) Mechanical fan (2) Chimney (3) Steam jet

310. Balanced draught system uses

(a) steam jet draught
(b) induced draught
(c) forced draught
(d) both induced and force draught

TRB Poly. Lect., 2012

Ans. (d) : Balanced draught uses both forced & induced draught to overcome the resistance in air pre-heater and grate, while the induced draught fan overcome draught losses through boiler, economiser & connecting flue passage etc.

311. Economizer is generally placed between :

(a) Last superheater / reheater and air preheater
(b) Air preheater and chimney
(c) Electrostatic precipitators
(d) Induced draft fan and forced draft fan

PSPCL AE, 2012

Ans. (a) : Last superheater / Reheater and air preheater
Boiler feed water → Radiant super heater → Reheater Constructive superheater → Economizer → Air preheater → Electrostatic precipitator → Fan → Chimney

312. Steam flow is controlled by using :

(a) Drum level sensor
(b) Feedwater flow sensor
(c) Steam flow sensor
(d) All of these

OPSC Civil Services Pre. 2011

Ans. (d) : steam flow is control by using :
■ Drum level sensor
■ Feedwater flow sensor
■ Steam flow sensor

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313. Plants A, B and C require 3 kg/s, 8 kg/s and 28 kg/s of saturated water flow in the downcomer to produce 1 kg/s steam respectively. The most feasible plant will be :

(a) Plant B (b) Plant A
(c) Plant C (d) Plant A and B

OPSC Civil Services Pre. 2011

Ans. (b) : The most feasible plant will be plant A.

314. Load centre in a power station is :

(a) centre of coal fields
(b) centre of maximum load of equipments
(c) centre of gravity of electrical load
(d) centre of power station

RPSC ACF-2011

Ans. (c) : Load centre in a power station is centre of gravity of electrical load.

315. Diversity factor is always :

(a) equal to unity (b) less than unity
(c) more than unity (d) zero

RPSC AE 2016
RPSC ACF-2011

Ans. (c) : Diversity is usually more than one. Diversity factor is the ratio of sum of the individual maximum demands of the various subdivisions of a system (or part of a system) to the maximum demand of the whole system (or part of the system).

316. High load factor indicates that :

(a) cost of generation per unit power is increased
(b) total plant capacity is utilized for most of the time
(c) total plant capacity is not properly utilized for most of the time
(d) load on the plant is high

RPSC ACF-2011

Ans. (b) : High load factor indicates total plant capacity is utilized for most of the time.

317. Which one of the following does not cause thermal pollution?

(a) Nuclear power plant
(b) Hydro power plant
(c) Coal-fired power plant
(d) Solar-thermal power plant

RPSC AE GWD, 2011

Ans. (b) : Hydro power plant does not cause thermal pollution.

318. Centralised collection device for generating power from the Sun is called

(a) Fuel cell (b) Power tower
(c) Solar collector (d) Solar photovoltaic

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Centralised collection device for generating power from sun is called power tower, it uses a concave mirror or lenses to concentrate a large area of sunlight onto a receiver, which is converted into heat & then into electricity using electric generator.

319. The future growth of large scale hydro power plants in the developed world probably will be limited because

(a) the release of impounded water alters stream flow patterns
(b) most economical sites are already used
(c) reservoirs eventually fill with sediments
(d) all of the above

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : The future growth of large scale hydro power plants is limited due to many reason,

- Release of impounded water alters stream flow patterns.
- Most economical sites are already used.
- Reservoirs eventually fill with sediments and sources are limited etc.

320. Consider the following statements for reheating in steam power plants :

1. Reheating is a must for modern high pressure power plants using pressure above 100 bar.
2. Not more than two reheats are used in a steam power plant.
3. Reheating reduces the steam rate.

Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-52/2010

Ans. (a) : Explanation- Incase of the reheating in steam power plants-

(1) Reheating is a must for modern high pressure power plants using pressure above 100 bar. (also above 250 bar)
(2) Not more than two reheats are used in a steam power plant because its thermal efficiency reduced.
(3) Reheating reduces the steam rate also.

321. Consider the following statements for regeneration in steam power plant cycle :

1. The steam rate increases by regeneration
2. The heat rate decreases by regeneration.
3. The condenser size is reduced due to regeneration.

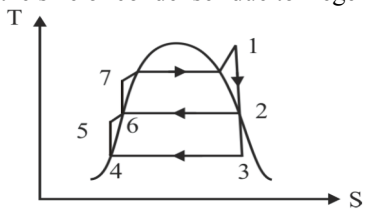
Which of these statements are correct ?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-52/2010

Ans. (a) : Explanation- Due to the Regeneration in steam power plant.

- Increase the cycle efficiency due to regeneration.
- Reduces the Heat rate.
- It increase the steam flow rate
- Reduced the size of condenser due to Regeneration.



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322. Consider the following statements for Brayton cycle and Rankine cycle :

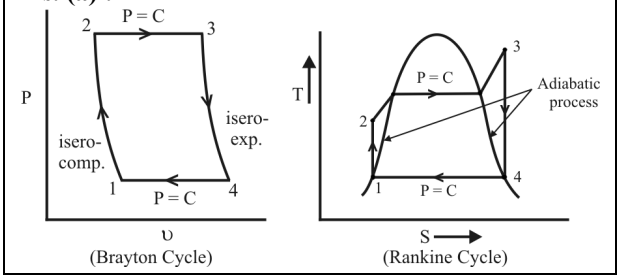
1. Both Rankine cycle and Brayton cycle consist of two reversible isobars and two reversible adiabatic.
2. The work ratio of Rankine cycle is higher than that of Brayton cycle
3. The efficiency of Rankine cycle steam plants using steam turbines is higher than that of Brayton cycle plants using gas turbines.

Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-52/2010

Ans. (a) :

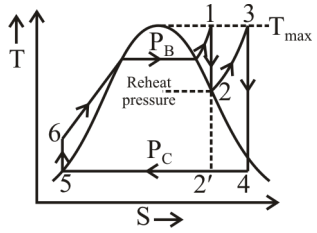


323. How do the pump work change when a simple Rankine cycle is modified with reheating ?

(a) increase (b) decrease
(c) remains same (d) none of these

J&K PSC Civil Services Pre, 2010

Ans. (c) : Pump work remains same when a simple rankine cycle is modified with reheating.



- W_T-↑, W_P-Constant, W_{net}-↑
- Q_s-↑, Q_R-↑, T_{MA}-↓↑, η-↑↓ (Can't say)
- $\downarrow \text{Specific Steam Consumption} = \dfrac{3600}{W_{net}\uparrow}$
- x-↑, ↓errosion of turbine blade

324. For the same power output operating charges are minimum for a :

(a) thermal plant (b) nuclear plant
(c) hydel plant (d) gas turbine plant

APPSC IOF, 2009

Ans. (c) : Hydroelectric power plant has minimum charges for same power output operating charges.

325. The draught produced by chimney of given height at given outside temperature :

(a) decreases if the chimney gas temperature increases
(b) remains same irrespective of chimney gas temperature
(c) increases if the chimney gas temperature increases
(d) may increase or decrease

APPSC IOF, 2009

Ans. (c) : The draught produced by chimney of given height at given outside temperature increases if the chimney gas temperature increases.

326. The power station supplies loads having maximum demands of 45 MW, 30 MW, 25 MW and 10 MW. The annual peak load is 70 MW. The demand factor is :

(a) 1.571 (b) 0.636
(c) 0.786 (d) 1.273

APPSC IOF, 2009

Ans. (a) : $\text{Demand factor} = \dfrac{\text{Actual max. consumption}}{\text{Total connected load}} = \dfrac{45+30+25+10}{70} = 1.571$

327. In a simple Rankine cycle, steam enters the condenser at saturated vapour and leaves as saturated liquid. The condenser and boiler pressure are maintained at p_1 and p_2, respectively. If v_f and v_g are the specific volumes of saturated liquid and saturated vapour at condenser pressure, then pump work can be approximated as :

(a) $v_g(p_2-p_1)$ (b) $v_f(p_2-p_1)$
(c) $p_1(v_g-v_f)$ (d) $p_2(v_g-v_f)$

DRDO Scientists 2009

Ans. (b) : Pump work $W_p = \int v dp$

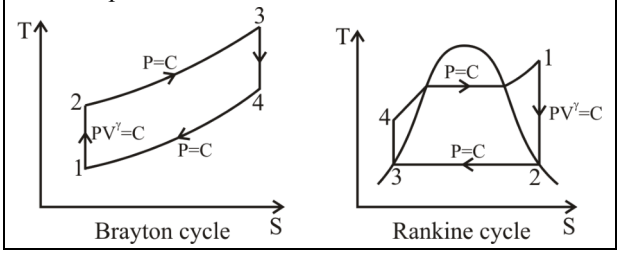
$$\boxed{W_p = V_f(P_2 - P_1)}$$

328. Two reversible isobars and two reversible adiabatics constitute the Brayton cycle. The same processes also make up the:

(a) Stirling cycle (b) Ericssion cycle
(c) Otto cycle (d) Rankine cycle

UPSC JWM Adv. No-16/2009

Ans. (d) : Rankine cycle and Brayton cycle constitute the same processes.



329. The efficiency of reheat cycle is given by

(a) $\dfrac{\text{Work done}}{\text{Heat supplied}}$

(b) $\dfrac{\text{Total useful heat drop}}{\text{Heat supplied}}$

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(c) $\frac{\text{Aditabatic heat drop}}{\text{Heat supplied}}$

(d) $\frac{\text{Total useful heat drop}}{\text{Total aditabatic heat drop}}$

TNPSC AE, 2008

Ans. (a) : $\eta_{\text{reheat cycle}} = \frac{\text{Work done}}{\text{Heat supplied}}$

330. The ideal cycle for a steam power plant is the Rankine cycle instead of the Carnot cycle because
(a) the Rankine cycle has higher efficiency
(b) the Rankine cycle efficiency equals the Carnot cycle efficiency
(c) the Rankine cycle has higher work ratio and is easier to implement
(d) the Carnot cycle gives lower turbine work

DRDO Scientists 2008

Ans. (c) : The ideal cycle for a steam power plant is the Rankine cycle instead of the Carnot cycle because the Rankine cycle has higher work ratio and is easier to implement.

331. The modern steam power plant runs on
(a) Carnot cycle
(b) Ericson cycle
(c) Modified Rankine cycle
(d) Otto cycle

WBPSC AE, 2007

Ans. (c) : The modern steam power plant runs on modified Rankine cycle.

332. Corrosion in boilers is minimised by maintaining boiler water as
(a) Acidic (b) Alkaline
(c) At high pressure (d) None of the above

UKPSC AE 2007 Paper -II

Ans. (b) : Alkaline

333. An air preheater
(a) increases evaporative capacity of a boiler
(b) increases the efficiency of boiler
(c) enables low grade fuel to be burnt
(d) all of the above

UKPSC AE 2007 Paper -II

Ans. (d) : An air preheater –
- Increases evaporative capacity of a boiler.
- Increases the efficiency of boiler.
- Enables low grade fuel to be burnt.

334. An economizer in a boiler
(a) increases steam pressure
(b) increase steam flow
(c) decreases fuel consumption
(d) decreases boiler efficiency

J & K PSC Screening, 2006

Ans. (c) : Economiser is a device which utilize heat of exhaust flue gases in order to increase temperature of feed water entering into the boiler so less heat will be required to convert water into steam. So it decreases fuel consumption in boiler.

335. The term heating surface in a boiler means.
(a) area of grate
(b) volume of furnace
(c) outer surface area of boiler shell
(d) surface area which is in contact with flue gases.

J & K PSC Screening, 2006

Ans. (d) : The term heating surface in boiler means surface area which is in contact with flue gases.

336. The efficiency of Rankine cycle
(a) increases with increase in exhaust pressure
(b) increases with decrease temperature of heat rejection
(c) decreases with increase in temperature of heat rejection
(d) is independent of temperature of heat rejection and exhaust pressure

J & K PSC Screening, 2006

Ans. (b & c) : The efficiency of rankine cycle is increased by increasing the mean temperature of heat addition or decreasing the mean temperature of heat rejection.

337. The burner tips in corner fired boiler are tilted to control
(a) steam temperature
(b) steam quantity
(c) air fuel ratio
(d) atmospheric pollution

OPSC Civil Services Pre 2006

Ans. (a) : The burner in the corners are tilted up or down to increase the radiant going to the reheater surface. This also affects the super heat absorption and temperature of super heater.

338. At a pressure of 221.2 bar and temperature of 374.15°C, the change of volume accompanying evaporation of water is
(a) very high (b) very low
(c) negligible (d) zero

OPSC Civil Services Pre 2006

Ans. (d) : At critical point latent heat of vaporisation is zero and at this point density of liquid and vapour is same for water critical point occurs at 374.14°C temperature and 221.2 bar pressure.

339. Hydraulic testing of boilers is done at pressures-
(a) Below and above atmosphere
(b) Slightly above atmospheric pressure
(c) At half the working pressures of boiler
(d) At 1.5 to 2 times the working pressure

ISRO Scientist/Engineer 2006

Ans. (d) : Hydraulic test shall be carried as–
- The test pressure shall be raised gradually.
- It shall not exceed by more than 6% of the required pressure.
- The test pressure shall be maintained for 30 minutes.
- The pressure shall be reduced to maximum allowable working pressure and maintained for close visual inspection of the pressure parts.

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■ Hydraulic testing before assembly into the boiler if the completed boiler is tested hydraulically to 1.5 times the maximum permissible working pressure at site.

340. Which one of the following is true in an isentropic process of expansion from state (1) to state (2) ?

(a) $\frac{P_2}{P_1}=\left(\frac{T_1}{T_2}\right)^{\frac{K-1}{K}}$ (b) $\frac{P_2}{P_1}=\left(\frac{T_2}{T_1}\right)^{\frac{K-1}{K}}$

(c) $\frac{P_2}{P_1}=\left(\frac{T_1}{T_2}\right)^{\frac{K}{K-1}}$ (d) $\frac{P_2}{P_1}=\left(\frac{T_2}{T_1}\right)^{\frac{K}{K-1}}$

TNPSC AE, 2008

Ans. (d) $\frac{P_2}{P_1}=\left(\frac{T_2}{T_1}\right)^{\frac{K}{K-1}}$

Where, K = Adiabatic index

2. Fuels and Combustion

341. Coal fired power plant boilers manufactured in India generally use :

(a) Pulverized fuel combustion
(b) Fluidize bed combustion
(c) Circulating fluidized bed combustion
(d) Moving stoker fired system

RPSC IOF & Boiler 20.09.2017
RPSC AE GWD, 2011, ESE 1997

Ans. (a) : Coal fired power plant boilers manufactured in India generally use Pulverized fuel combustion get high burning rate.

342. Which of the following coal has the highest calorific value ?

(a) Anthracite (b) Bituminous
(c) Lignite (d) Peat

HPPSC W.S. Poly. 2016
TNPSC AE (Industries) 09.06.2013

Ans. (a) :

Coal	Calorific Value (kJ/kg)
Peat	23000
Lignite	25000
Bituminous	33500
Anthracite	36000

Anthracite Coal has the highest Calorific Value.

343. The excess air required for combustion of pulverised coal is of the order of

(a) 100 to 150% (b) 30 to 60%
(c) 15 to 40% (d) 5 to 10%

RPSC IOF, 2020, ESE 1996

Ans. (c) : The amount of air more than the theoretical requirement is referred to as excess air. Power plant boilers normally run about 10 to 20 percent excess air. Natural gas-fired boilers may run as low as 5 percent of excess air.

Pulverized coal fired boilers may run with 15 to 40% of excess air.

344. Calorific value of fuel obtained by using Bomb Calorimeter is:

(a) Constant Volume Lower Calorific Value
(b) Constant Volume Higher Calorific Value
(c) Constant Pressure Lower Calorific Value
(d) Constant Pressure Higher Calorific Value

OPSC AEE 2019 PAPER - II
ESE 1992

Ans. : (b) : The calorific value obtained in a bomb calorimeter test represents the gross heat of combustion, per unit mass of fuel sample. This is the heat produced when the sample burns plus the heat given up when the newly formed water vapour condenses and cools to the temperature of bomb.

345. Volumetric analysis of sample of dry products of combustion gave the following results:
CO_2 = 10%, CO = 1%, O_2 = 8%, N_2 = 81%
Their proportion by weight will be

(a) 440 : 28 : 256 : 2268
(b) 22 : 14 : 256 : 1134
(c) 440 : 14 : 28 : 2268
(d) 22 : 28 : 14 : 1134

Nagaland PSC CTSE 2015 Paper-II
ESE 1993

Ans. (a) : Given, Volumetric analysis
CO_2 = 10%, CO = 1%, O_2 = 8%, N_2 = 81%
Proportion ratio of weight
= [12 + 32] × 10 : [12+16] ×1 : [32] × 8 : [28] × 81
= 440 : 28 : 256 : 2268

346. The amount of CO produced by 1 kg of carbon on incomplete combustion is

(a) 3/11kg (b) 3/8 kg
(c) 8/3 kg (d) 7/3 kg
(e) 11/3 kg

CGPSC AE 26.04.2015 Shift-II
ESE 1999

Ans. (d) : The Stoichiometric equation of carbon,

$\underset{12}{C}+\underset{16}{O}\rightarrow\underset{28}{CO}$

CO Produced by one kg Carbon-

$=\frac{28}{12}=\frac{7}{3}$kg CO/kg of Carbon

347. Consider the following statements:
The maximum temperature produced by combustion of a unit mass of fuel depends on
I. LCV
II. ash content
III. mass of air supplied.
IV. pressure in the furnace.
Of these statements

(a) I alone is correct
(b) I and III are correct
(c) II and IV are correct
(d) III and IV are correct

APPSC Poly. Lect. 2013
ESE 1998

Ans. (b) : The maximum temperature produced by the combustion of an unit mass of fuel depend upon lower calorific value (LCV) and mass of air supplied.

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348. Lignite, bituminous and anthracites are different type of

(a) nuclear fuels (b) coal
(c) natural gas (d) biogas

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : Lignite, bituminous and anthracites are different types of coal.
The rank of coal (from most to least carbon content) are as follows-anthracite, bituminous coal, sub-bituminous coal and lignite.

349. Depletion of the ozone layer is damaging to human health. Negative effects include

(a) skin cancer (b) osteoporosis
(c) dyspepsia (d) AIDS

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : The ozone layer in the upper atmosphere or stratosphere absorbs harmful ultraviolet rays from the sun and protect us from skin cancer. CFCs are linked to the depletion of this ozone layer.
Depletion of the ozone layer is damaging to human health negative effects include skin cancer.

350. A fuel consists of 92% carbon, 7% hydrogen and remaining residual matter by mass. Working from first principles, the higher calorific value of the fuel is

(a) 40176 kJ/kg (b) 41176 kJ/kg
(c) 40876 kJ/kg (d) 41678 kJ/kg

ESE 18.07.2021

Ans. (a) : 1 mole of carbon = 393.5 kJ/mol
1 mole of hydrogen = 286 kJ/mol

$$\text{Moles of carbon} = \frac{920}{12}$$

$$\text{Moles of } H_2 = \frac{70}{2}$$

$$\text{Energy released} = \frac{920}{12} \times 393.5 + \frac{70}{2} \times 286$$
$$= 40178.33 \text{ kJ/kg.}$$

351. In order to burn a fuel completely, which of the following basic conditions must be fulfilled?

1. **Supply enough air for complete combustion of fuel.**
2. **Secure low turbulence for through mixing of fuel and air.**
3. **Maintain a furnace temperature high enough to ignite the incoming fuel air mixture.**
4. **Provide a furnace volume large enough to allow time for combustion to be completed.**

Select the correct answer using the code given below:

(a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 1, 3 and 4 only (d) 2, 3 and 4 only

ESE 18.07.2021

Ans. (c) : Basic conditions must be fulfilled-
(i) Supply enough air for combustions of fuel.
(ii) Maintain a furnace temperature high enough to ignite the incoming fuel air mixture.
(iii) Provide a furnace volume large enough to allow time for combustion to be completed.

352. Consider the following statements for solid fuels:

(1) Peat is the first stage in the formation of coal from wood.
(2) The average calorific value of bituminous coal is 1524 kJ/kg.
(3) Anthracite is very hard coal and has a shining black lustre.
(4) Wood charcoal is obtained by destructive distillation of wood.

Which of the above statements are correct?

(a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 2 and 3 only (d) 1, 2 and 4 only

ESE 18.07.2021

Ans. (b) :
- Anthracite is the highest grade of coal, it is a shiny black dense, hard, brittle coal that borders on graphite.
- The calorific value of coals range from 25600 to 32600 kJ/kg.
- Peat is considered as the first geological step in coal's formation.
- Wood charcoal is obtained by destructive distillation of wood.

353. Which of the following are the essential functions of fuel cells?

1. **The charging (or electrolyser) function in which the chemical AB is decomposed to A and B**
2. **The storage function in which A and B are held apart.**
3. **The charge function in which A and B are charged with the simultaneous generation of electricity.**

(a) 1 and 3 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1, 2 and 3

ESE 2020

Ans. (c) : 1 and 2 only
The discharge function in which A and B are charged with the simultaneous generation of electricity.

354. Which of the following are related to the Proton Exchange Membrane Fuel Cell (PEMFC)?

1. Polymer electrolyte
2. Hydrogen fuel and oxygen
3. Pure water and small amount of electricity
4. Nitrogen gas

(a) 1 and 3 only (b) 2 and 4 only
(c) 1 and 2 only (d) 3 and 4 only

ESE 2020

Ans. (c) : 1 and 2 only

355. A fluidized bed combustion system having an output of 35 MW at 80% efficiency when using a coal of heating value 26 MJ/kg with a sulphur content of 3.6% requires a particular limestone to be fed to it at a calcium-sulphur molar ratio of 3.0 so as to limit emissions of SO_2 adequately. The limestone used contains 85% $CaCO_3$. The required flow rate of limestone will be

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(a) 2405 kg/h (b) 2805 kg/h
(c) 3205 kg/h (d) 3605 kg/h

ESE 2020

Ans. (a) : Coal burning rate = $\frac{35}{0.80\times 26} = 1.683 \text{ kg/s}$
$= 6057.7 \text{ kg/h}$
$\therefore$ Flow rate of sulphur = $\frac{6057.7\times 0.036}{32}$
$= 6.81 \text{ kmol/h}$
for a Ca:S molar ratio of 3.0, the flow rate of calcium is $6.81\times 3.0=20.44$ kmol/h
$\therefore$ Mass flow rate of $CaCO_3$ required = 20.44×100
= 2044 kg/h (Since molecular weight of $CaCO_3$ is 100)
$\therefore$ Mass flow of limestone = 2044/0.85 = 2404.7 kg/h which is about 40% of the coal burning rate.

356. A thermal power plant generates power of 1000 MW. Coal releases 900 × 10^7 kJ/h of energy. The rate at which heat is rejected from the this power plant will be:
(a) 500 MW (b) 1000 MW
(c) 1500 MW (d) 250 MW

AAI Jr. Executive 29.11.2018
ESE 2009

Ans. (c) :
Energy released by coal $= \frac{900\times 10^7\times 1000}{3600} \text{ J/sec}$
$= 2500 \text{ MW}$
Heat rejected from the power plant = 2500 – 1000
= 1500 MW

357. A base load power station of capacity 35 MW has annual output of 150 ×10^6 kWh. The plant capacity factor of the power plant is
(a) 68% (b) 35%
(c) 49% (d) 25%

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (c) : Capacity Factor—The capacity factor is defined as the ratio of the total actual energy produced or supply over a definite period, to the energy that would have been produced if the plant has operated continuously at the maximum rating. The capacity factor mainly depends on the type of the fuel used in the circuit.
So, actual annual output is given as
$$Q_O = 150\times 10^6\times 3600 kJ$$
Ideal annual output is given as
$$Q_i = 35\times 10^3\times 365\times 24\times 3600 \ kJ$$
So capacity factor
$$CF = \frac{Q_o}{Q_i} = 0.489 \approx 0.49$$

358. Liquefied Petroleum Gas is a mixture of
(a) Propane and butane
(b) Propane and ethane
(c) Methane and isopropane
(d) Butane and ethane

HPPSC AE 2018

Ans. (a) : Liquefied Petroleum gas is a mixture of propane and butane are flammable mixture of hydrocarbon gases used as fuel in heating appliances, cooking equipment and vehicles.

359. The natural solid fuel is
(a) wood (b) coke
(c) anthracite coal (d) pulverized coal

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : Solid fuel refers to various types of solid materials that one used as fuel to produce energy and provide heating. Usually released through combustion solid fuels include wood, charcoal, peat coal, hexamine fuel tablets and pellets made from wood, corn, wheat rye and other grains. So, Natural solid fuel is wood.

360. The fuel mostly used in cement industry and in metallurgical process is
(a) wood charcoal (b) bituminous coke
(c) pulverized coal (d) coke

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The fuel mostly used in cement industry and in metallurgical process is pulverized coal. A pulverized coal fired in the form of powdered coal or coal dust it is fine as face powder in cosmetic makeup. Pulverized coal firing ensures complete combustion of coal, thus ensuring higher efficiency of stream generators. It is predominantly adopted in large coal fired utility boilers. The finer the grinding of coal, the more efficient its combustion.

361. Coke is produced
(a) when coal is first dried and then crushed to a fine powder by pulverising machine
(b) from the finely ground coal by moulding under pressure with or without a binding material
(c) when coal is strongly heated continuously for 42 to 48 hours in the absence of air in a closed vessel
(d) by heating wood with a limited supply of air to a temperature not less than 280ºC.

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Coke is produced by destructive distillation of coal in coke ovens for 42 to 48 hours in the absence of air in a closed vessel until all volatile components in the coal evaporate. The material remaining is called coke.

362. Steam coal is a
(a) pulverized coal
(b) brown coal
(c) coking bituminous coal
(d) non-coking bituminous coal

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Steam coal is a non-coking bituminous coal. Steam coal is a grade of steam to create electricity. Most of the coals used in power plant.

363. Producer gas is obtained by
(a) partial combustion of coal, coke, anthracite coal or charcoal in a mixed air steam blast
(b) carbonization of bituminous coal
(c) passing steam over incandescent coke
(d) passing air and a large amount of steam over waste coal at about 650ºC

Nagaland PSC (CTSE) 2018, Paper-II

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Ans. (a) : Producer gas is obtained by partial combustion of coal, coke anthracite coal or charcoal in a mixed air steam blast. Producer gas is a gas mixture containing carbon monoxide, hydrogen, carbon dioxide and nitrogen.

364. In Pulverized fuel burners, the rate of combustion is high in

(a) Long flame burner (b) Cyclone burner
(c) Tangential burner (d) Turbulent burner

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : In pulverized fuel burner, the rate of combustion is high in turbulent burner because the fuel and air mixed in this burner.

365. A diesel power station has fuel consumption of 0.2 kg per kWh. If the calorific value of diesel is 11000 kcal per kg, overall efficiency of power station is

(a) 43.3% (b) 65.5%
(c) 39.2% (d) 25.6%

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (c) : Power plant output is given as

$$P_o = \frac{1}{0.2}\frac{kWh}{kg}$$

$$= \frac{3600}{0.2}\frac{kJ}{kg}$$

Power plant input is given as

$$P_i = 11000 \times 4.18 \frac{kJ}{kg}$$

1 Cal = 4.18 J

Then efficiency as

$$\eta = \frac{P_o}{P_i} \approx 0.395$$

366. Which fuel cell is suitable for spacecraft applications?

(a) Direct methanol fuel cell
(b) Proton exchange membrane fuel cell
(c) Alkaline fuel cell
(d) Phosphoric acid fuel cell

ESE 2017

Ans. (c) : Fuel Cell– A fuel cell converts hydrogen and oxygen into water, producing electricity and heat in the process.

Alkaline Fuel Cell– Long used by NASA on space missions, alkaline fuel cells can achieve power generating efficiencies of up to 70%. They were used on the Apollo spacecraft to provide both electricity and drinking water.

Proton Exchange Membrane Fuel Cell (PEM)–These fuel cells operate at relatively low temperatures, have high power density, can vary their output quickly to meets shifts in power demand, and are suited for applications, such as automobiles, where quick startup is required.

Direct Methanol Fuel Cell–These cell are similar to the PEM cells in that they both use a polymer membrane as the electrolyte.

Phosphoric Acid Fuel Cell–This type of fuel cell is used in stationary power generators with output in the 100 kW to 400 kW. range to power many commercial premises around the world.

367. Volumetric analysis of a certain flue gas gave CO_2 15%, O_2 5% and rest as N_2. The gas was at a temperature of 200°C and a pressure of 5 bar. The partial pressure of N_2 in the flue gas is

(a) 250 kN/m^2 (b) 300 kN/m^2
(c) 350 kN/m^2 (d) 400 kN/m^2

ESE 2017

Ans. (d) : Given,

$X_{CO_2} = 0.15$

$X_{O_2} = 0.05$

$X_{N_2} = 0.8$

$P_{total} = 5$ bar

Partial pressure of N_2

$$= \frac{\text{No. of moles of } N_2}{\text{Total no. of moles}} \times P_{Total}$$

$$= \frac{0.8}{0.15+0.05+0.8} \times 500 = 400 \text{ kPa}$$

368. Statement (I): Proximate analysis of coal is done to determine its calorific value.

Statement (II): In proximate analysis of coal, the percentages of moisture, volatile matter, fixed carbon and ash are determined.

ESE 2017

Ans. (d) : Proximate analysis is done to examine four factors i.e. Moisture, volatile compound ash content and fixed carbon.
Calorimeters are used for calorific value estimation.
So, statement-I is wrong.

369. Continuous single effect evaporator is to be fed with 5000 Kg/hour of the solution containing 1 wt% solute and is concentrated with 5 wt% solute. The vapour production rate equals to

(a) 4000 kg/hour (b) 2000 kg/hour
(c) 3000 kg/hour (d) 1000 Kg/hour

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : Feed (F) - 5000 kg/hr

x_F = 1 wt % = 0.01 kg solute / kg feed

x_L = 5 wt % = 0.05 kg solute / kg liquid product

Solute balance

$F\,x_F = L\,X_L$ (No solute in the vapour)

$5000 \times 0.01 = L \times 0.05$

L = 1000 kg/ hr

Overall material balance

$F = L + V \Rightarrow 5000 = 1000 + V$

V = 4000 kg/hr

370. According to EPA of USA, the following is not one of the six major pollutants?

(a) Ozone (b) Carbon monoxide
(c) Nitrogen oxides (d) Carbon dioxide

Nagaland PSC (CTSE) 2016, Paper-II

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Ans. (d) : According to EPA of USA, the six pollutants majorly
(i) Carbon monoxide
(ii) Nitrogen oxide
(iii) Sulphur dioxide
(iv) Lead
(v) Ground level ozone
(iv) Particulate matter

371. The Pollution Standard Index (PSI) scale has span from
(a) 0-200 (b) 0-300
(c) 0-400 (d) 0-500

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : The Pollution Standard Index (PSI) scale has span from 0 to 500. The index figures enables the public to determine whether the air pollution levels in a particular location are good, unhealthy, hazardous or worse.

372. Which of the following is/are inorganic gases?
(a) Carbon monoxide (b) Hydrogen sulphide
(c) Chlorine (d) All of the above

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : Some of the inorganic gases are:-
(i) Carbon monoxide
(ii) Hydrogen chloride
(iii) Chlorine
(iv) Hydrogen sulphide
(v) Fluorine

373. The major contributor of Carbon monoxide is
(a) Motor vehicle
(b) Industrial processes
(c) Stationary fuel combustion
(d) None of the above

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Carbon monoxide (CO), a colorless, odorless tasteless and toxic air pollutant is produced in the incomplete combustion of carbon containing fuels. The major contributor of carbon monoxides is motor vehicle.

374. Fugitive emissions consist of
(a) Street dust
(b) Dust from construction activities
(c) Dust from farm cultivation
(d) All of the above

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : Fugitive emissions are emissions of gases or vapors from pressurized equipment due to leaks and other unintended or irregular releases of gases, mostly from industrial activities, dust, fine particles.

375. The principal source of volatile organics (Hydrocarbons) is
(a) Transportation
(b) Industrial processes
(c) Stationary fuel combustion
(d) Volcanoes

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : The principal source of volatile organics (hydrocarbons) is crude oil & through industrial process, where there is a large of decomposed organic matters contains hydrogen and carbon in abundance.
When they are made to bond, hydrocarbons are formed others key sources of natural hydrocarbons are coal petroleum and natural gas.

376. The list of industrial sources of air pollution and their emissions are given. Match the following.

A. Ammonia	**1.Carbon monoxide**
B. Plating	**2. Particulates**
C. Fertilizers	**3. Metal fumes**

The correct order is

	A	B	C
(a)	1	2	3
(b)	3	2	1
(c)	1	3	2
(d)	3	1	2

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) :

List of Industrial Sources	Emissions of air pollutions
A. Ammonia	1.Carbon monoxide
B. Plating	2. Metal fumes
C. Fertilizers	3. Particulates

377. The threshold concentration of sulphur dioxide in any industrial activity should not be permitted beyond
(a) 2 ppm (b) 3 ppm
(c) 4 ppm (d) 5 ppm

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : The threshold concentration of sulphur dioxide in any industrial activity should not be permitted beyond the 5 ppm. (range).

378. The following method is used for analysis of combustion products (flue gases)?
(a) Non Dispersive Infra-Red Analyzer
(b) Flame Ionization Detector
(c) Flue Gas Analyzer for Boiler Flue Gases
(d) All of the above

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : Following methods used for analysis of combustion products (flue gases) are: -
(i) Non dispersive infra- red analyzer
(ii) Flame ionization detector
(iii) Flue gas analyzer for boiler flue gases

379. Consider the following statement:
The difference between higher and lower heating values of the fuels is due to _____
1. heat carried by steam from the moisture content of fuel
2. Sensible heat carried away by the flue gases
3. heat carried away by steam from the combustion of hydrogen in the fuel
4. heat lost by radiation
(a) 1, 2, 3 and 4 are correct
(b) 1 and 2 are correct
(c) 3 alone is correct
(d) 2, 3 and 4 are correct

UPRVUNL AE 21.08.2016

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Ans. : (c) The difference between higher and lower heating values of the fuels is due to heat carried away by steam from the combustion of hydrogen in the fuel

380. The ratio of "additional input required" to "increase an additional output" in power plant is known as

(a) Heat rate (b) Incremental heat rate
(c) Steam rate (d) Efficiency

RPSC Lecturer 16.01.2016

Ans. (b) : Incremental heat rate—Incremental heat rate is the ratio of the heat input required for a step change in net power output for power plant.
Incremental heat rates are normally given in units of Btu/kWh, or kJ/kWh.

381. Junker's gas calorimeter is used to determine the calorific value of:

(a) gaseous fuels (b) petrol
(c) coke (d) all fuels

(HPPSC LECT. 2016)

Ans. (a) : (i) Junker's gas calorimeter is used to determine the calorific value of gaseous fuels.
(ii) Bomb calorimeter is used for finding the higher calorific value of solid and liquid fuels.
(iii) The modification of Boy's gas calorimeter is known as Junker's colorimeter.

382. Which of the following is considered to be superior quality coal for power plants?

(a) Bituminous coal (b) Peat
(c) Coke (d) Lignite

UPPSC AE 12.04.2016 Paper-II

Ans. : (a) Non- coking bituminous coal is the fuel mostly used in boilers. The colorific value of the common coal varieties are
Peat:- 21000 kJ/kg, Lignite:- 25000 kJ/kg
Bituminous:- 33500 kJ/kg

383. Low grade fuels have

(a) Low moisture content
(b) Low ash content
(c) Low calorific value
(d) High carbon content

UPPSC AE 12.04.2016 Paper-II

Ans. : (c) Low grade fuels have low calorific value.

384. The presence of Carbon monoxide (CO) in the flue gas show inefficient combustion. This is because,

(a) The CO is slow to exchange heat through the heat exchange surfaces than CO_2
(b) The combustion of coal to CO releases less heat as compared to that released during combustion of CO to carbon-dioxide (CO_2)
(c) The combustion of coal to CO releases more heat as compared to that released during combustion of coal to carbon-dioxide (CO_2)
(d) The CO carries away more heat as compared to equal amount of carbon dioxide (CO_2).

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (b) : CO is form of unburnt fuel so combustion is in efficient.
The combustion of coal to CO releases less heat as compared to that released during combustion of CO to CO_2 (carbon dioxide).

385. Orsat gas analyzer determines volumetric composition of the flue gases in respect of

(a) O_2 and N_2
(b) O_2 and CO_2
(c) CO, CO_2 and N_2
(d) N_2, O_2, CO_2 and CO

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : The volumetric composition of flue gases N_2, CO_2, CO and O_2.

386. Solar cells are made of

(a) silica (b) steel
(c) sulphur (d) antimony

VIZAG STEEL MT 2015

Ans. (a) : Solar cells are made of silica.

387. Gasohol is a mixture of:

(a) 90% ethanol+ 10% gasoline
(b) 10% ethanol + 90% gasoline
(c) 40% ethanol + 60% gasoline
(d) 50% ethanol + 50 % gasoline

OPSC AEE 2015 PAPER-II

Ans. (b) : Gasohol is a mixture of 10% ethanol +90% gasoline.

388. Major disadvantage of LPG as a fuel in automobile is:

(a) Reduction in life of the engine
(b) Less power compared to gasoline
(c) Both (A) and (B)
(d) Knocking tendency

OPSC AEE 2015 PAPER - II

Ans. (c) : Major disadvantage of LPG as a fuel in automobile is less power compared to gasoline.

389. Dry flue gases with a composition of CO_2 = 10.4%, O_2 = 9.6% and N_2 = 80%, indicate that

(a) excess air is used
(b) air is insufficient
(c) hydrogen is not present in the coal
(d) air is just sufficient

Nagaland PSC CTSE 2015 Paper-II, ESE 1995

Ans. (a) : Presence of O_2 in dry flue gases shows that the excess air is used.

390. Which of the following gases has the highest calorific value ?

(a) producer gas (b) coal gas
(c) coke oven gas (d) blast furnace gas

J&K PSC Civil Services Pre, 2013

Ans. (b) : Gas/fuel	Calorific value
Producer gas	5800 kJ/m^3
Coal gas (N_2 + CH_4 +CO)	10 to 20 MJ / m^3
Blast furnace gas	3900 KJ/m^3

391. For each mole of oxygen, number of moles of nitrogen required for complete combustion of carbon are

(a) 20/21 (b) 2/21
(c) 77/21 (d) 79/21

APPSC Poly. Lect. 2013

Ans. (d) : For each mole of Oxygen number of moles of nitrogen required for combustion of Carbon are 79/21 = 3.76 moles of nitrogen.

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392. Anthracite coal is:
(a) Clean, dense and hard
(b) Non-caking and hard to ignite
(c) Burns smokelessly
(d) All of these

SJVN ET 2013

Ans. (d) : Anthracite, often referred to as hard coal, compact variety of coal that has a submetallic luster, the fewest impurities and the highest energy density of all types of coal and is the highest ranking of coals.

393. Examples of slurry type fuels for thermal plants are:
(a) Coal and oil (b) Coal and water
(c) Coal and methanol (d) All of these

SJVN ET 2013

Ans. (d) : Examples of slurry type fuels for thermal plants are-
(i) coal and oil
(ii) coal and water
(iii) coal and methanol

394. The permissible maximum concentration of SO_2 at ground level are
(a) 0.05 to 0.08 ppm for 24 Hrs
(b) 0.07 to 0.09 ppm for 1 Hr
(c) 0.15 to 0.2 ppm for 24 Hrs
(d) 0.08 to 1.15 ppm for 1 Hr

TNPSC AE 2013

Ans. (a) : The permissible maximum concentration of SO_2 at ground level are 0.05 to 0.08 ppm for 24 Hrs.

395. Chloro-fluorocarbons are banned due to
(a) High cost
(b) Toxicity
(c) High Boiling Point
(d) Preventing Infrared rays to escape and allowing UV rays to Earth

TNPSC AE 2013

Ans. (d) : Chloro-fluorocarbons are banned due to preventing Infrared rays to escape and allowing UV rays to Earth.

396. The specific heat of the products of combustion increases with increase in
(a) pressure (b) temperature
(c) fuel-air ratio (d) both (b) and (c)

BPSC AE 2012 Paper - V

Ans. : (d) : The specific heat of the products of combustion increases with increase in temperature and fuel-air ratio.

397. What two gases are typically combined in a fuel cell?
(a) Hydrogen and nitrogen
(b) Hydrogen and oxygen
(c) Oxygen and nitrogen
(d) Helium and nitrogen

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Fuel cell is an electrochemical cell that converts the chemical energy of fuel into electrical energy. Here fuel is hydrogen & the oxidising agent is oxygen.

398. Correct statement is :
(a) MOX is multi - oxide fuel
(b) MOX is molten oxide fuel
(c) MOX can utilize surplus weapon grade plutonium
(d) MOX is not commercially prepared till to date

OPSC Civil Services Pre. 2011

Ans. (c) : Mixed oxide fuel (MOX), common referred to as mox fuel, is nuclear fuel that contains more than one oxide of fissile material, usually consisting of plutonium blended with natural uranium, reprocessed uranium or depleted uranium.

399. Which is not a minor actinide ?
(a) Uranium (b) Americium
(c) Curium (d) Neptunium

OPSC Civil Services Pre. 2011

Ans. (a) : The minor actinides are the actinide elements in used nuclear fuel other than uranium and plutonium, which are termed the major actinides. The most important isotope in spent nuclear fuel are neptunium, americium, curium, Through– 248, Through –252, Californium.

400. Sulphur in coal results in
(a) causing clinkering and slagging
(b) corroding air heaters
(c) spontaneous combustion during coal storage
(d) all of the above

TNPSC AE, 2008

Ans. (d) : Sulphur in coal results in –
- Causing clinkering and slagging
- Corroding air heaters
- Spontaneous combustion during coal storage

3. Steam Turbine

401. Bleeding in turbine means
(a) leakage of steam
(b) steam doing no useful work
(c) extracted steam for pre-heating feed water
(d) removal of condensed steam

CGPSC AE 15.01.2021, HPPSC ADF 05.03.2019
RPSC IOF & Boiler 20.09.2017
Kerala PSC Asstt. Prof. 27.09.2016
UKPSC AE 2012 Paper–II
UKPSC AE 2007 Paper -II

Ans. : (c) Bleeding in turbine means extracted steam for pre-heating feed water.
The effects of Bleeding are –
(i) It increases thermodynamic efficiency of the turbine.
(ii) The boiler is supplied with hot water.
(iii) A small amount of work is lost by the turbine, which decreases the power developed.

402. The compounding of steam turbine is usually done for:
(a) Reduction of steam consumption
(b) Reduction in turbine size
(c) Reduction of motor speed
(d) Efficiency improvement

OPSC AEE 2019 PAPER - II

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HPPSC Asstt. Prof. 20.11.2017
RPSC AE 2016, PSPCL AE 2012
RPSC ACF 2011
UKPSC AE 2007 Paper-II, ESE 1999

Ans. : (c) : Since the speed of single stage impulse turbine is very high. To bring this speed in practical application range, compounding (Pressure and velocity) is done. Here either the pressure of steam is reduced in steps (Pressure compounding) or high velocity is used in steps (Velocity compounding). Gear box is out of option because for such high reduction it is not an economical idea.

403. Which of the following is correct about Rateau turbine?

(a) Rateau turbine is a simple turbine.
(b) Rateau turbine is velocity compounded.
(c) Rateau turbine is pressure compounded.
(d) Rateau turbine is both, pressure and velocity compounded.
(e) Rateau turbine is 50% reaction turbine.

GPSC ARTO Pre-30.12.2018
CGPSC ARTO & TSI 07.01.2017
Nagaland PSC CTSE 2015 Paper-II
CGPSC AE 2014 Paper-II
VIZAG Steel MT Re-Exam 24.11.2013
RPSC IOF 2010, GATE 2000

Ans. (c) :
- Simple Impulse Turbine - De- laval turbine
- Velocity compounded impulse turbine -Curtis
- Pressure compounding impulse turbine- Rateau Turbine
- 50% Reaction Turbine- Parson's Turbine
- 100% Reaction turbine - Hero's Turbine

404. De Laval turbine is

(a) simple single wheel impulse turbine
(b) simple single wheel reaction turbine
(c) velocity compounded impulse turbine
(d) pressure compounded impulse turbine

JPSC AE, 10.04.2021, Paper-I
GPSC Lect. 23.10.2016
TSPSC AEE 2015, (KPSC AE 2015)
UKPSC AE 2012 Paper–II

Ans. (a) : De-Laval turbine is simple single wheel impulse turbine. Expansion of steam takes place in one set of nozzle and moving blades. At nozzle exit steam pressure drop to condenser pressure & velocity is very high.

405. The maximum efficiency for Person's reaction turbine is given by

(a) $\eta_{max} = \frac{\cos\alpha}{1+\cos\alpha}$ (b) $\eta_{max} = \frac{2\cos\alpha}{1+\cos\alpha}$

(c) $\eta_{max} = \frac{2\cos^2\alpha}{1+\cos^2\alpha}$ (d) $\eta_{max} = \frac{1+\cos^2\alpha}{2\cos^2\alpha}$

BPSC Poly. Lect. 2016
CGPSC AE 26.04.2015 Shift-II
TNPSC AE 2014, JPSC AE 2013 Paper-V
ESE 2006, 2004, 1999, CSE Pre-2001, 1996

Ans. (c) : The maximum efficiency for Person's reaction turbine is given by $\eta_{max} = \frac{2\cos^2\alpha}{1+\cos^2\alpha}$

406. The condition for maximum blade efficiency for a single-stage impulse turbine is that the speed ratio should be equal to :

(a) $\cos 2\alpha$ (b) $\frac{\cos\alpha}{2}$

(c) $\frac{1}{2}\cos 2\alpha$ (d) $2\cos\alpha$

where α = angle made by absolute velocity at inlet

CGPSC AE 15.01.2021
NLC GET 17.11.2020, Shift-II
APPSC AEE Screening Test 2016

Ans. (b) : The condition for maximum blade efficiency for a single stage turbine is that the speed ratio should be equal to :

$$\rho_{optimum} = \frac{U}{V_1} = \frac{\cos\alpha}{2}$$

407. The net result of pressure-velocity compounding of steam turbine is

(a) less number of stages
(b) large turbine for a given pressure drop
(c) shorter turbine for a given pressure drop
(d) lower friction loss

RPSC IOF, 2020
J&K PSC Civil Services Pre, 2010
ESE 1997

Ans. (b) : Principle of compounding is decreasing rotor speed by increasing number of stages of expansion hence larger turbines.

408. Curtis turbine is

(a) reaction steam turbine
(b) pressure velocity compounded steam turbine
(c) pressure compounded impulse steam turbine
(d) velocity compounded impulse steam turbine

Haryana PSC AE (PHED) 04.09.2020, Paper-I
OPSC AEE 2015 Paper-II
APPSC Poly. Lect. 2013

Ans. (d) : De-laval - simple impulse turbine
Curtis- Velocity compounded impulse turbine
Rateau- Pressure compounded impulse trubine

409. In case of reaction steam turbine

(a) There is enthalpy drop in fixed and moving blades
(b) There is enthalpy drop only in fixed blades
(c) There is enthalpy drop only in moving blades
(d) There is no enthalpy drop

GPSC ARTO Pre-30.12.2018
HPPSC Asstt. Prof. 18.09.2017
RPSC AE 2016, RPSC ACF-2011
MPPSC State Forest Service Exam, 2014

Ans. (a) : For a reaction steam turbine, there is enthalpy drop in fixed & moving blades.

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410. Parson's turbine is a :

(a) impulse turbine of 50 percent degree of impulse
(b) reaction turbine of varying degree of reaction.
(c) reaction turbine of 50 percent degree of reaction
(d) impulse turbine of 50 percent degree of reaction.

NLC GET 17.11.2020, Shift-II
MPPSC State Forest Service Exam, 2014
JPSC AE 2013 Paper-V

Ans. (c) : Parson's turbine is a reaction turbine of 50 percent degree of reaction thus a blade with 50% degree of reaction or a 50% reaction stage is one in which half the enthalpy drop of the stage occurs in the fixed blade and half in the moving blades. A very widely used design has half degree of reaction or 50% reaction and this is known as parson's turbine.

411. In a single-stage impulse turbine, which of the following is correct? [where η_s = Stage efficiency, η_b = Blade efficiency, η_N = Nozzle efficiency]

(a) $\eta_s = \eta_b \times \eta_N$ (b) $\eta_s = \eta_b/\eta_N$
(c) $\eta_s = \eta_N/\eta_b$ (d) $\eta_s = \eta_N = \eta_b$

SJVN ET 2019, ESE 2008
UKPSC AE 2007 Paper-II

Ans. (a) : Stage consists is combination of blade and nozzle and both arranged in series.
So stage efficiency = blade efficiency × nozzle efficiency.

412. The diagram efficiency is highest for simple impulse turbine stage having smooth and symmetrical blade when blade steam speed ratio can be given as

(a) $\cos\alpha_1$ (b) $\frac{\cos\alpha_1}{4}$
(c) $\frac{\cos\alpha_1}{2}$ (d) None of these

Where α_1 is the angle of absolute velocity at inlet.

UPPSC AE 12.04.2016 Paper-II
OPSC AEE 2015 PAPER - II
ESE 2001

Ans. : (c) The diagram efficiency is highest for simple impulse turbine stage having smooth and symmetrical blade when blade steam speed ratio can be given as

$$\frac{\cos\alpha_1}{2}$$

Impulse Turbine:-

* Blade velocity $(V_b) = \frac{\pi DN}{60}$

* The area of flow $(A_b) = \frac{\pi}{4}(D_2^2 - D_1^2)$

* Optimum velocity ratio $(\rho_{opt}) = \frac{\cos\alpha_1}{2}$

* Maximum efficiency $(\eta_{max}) = \frac{1+k_b}{2}.\cos^2\alpha$

413. In case of impulse steam turbine there is :

(a) enthalpy drop in fixed and moving blades
(b) enthalpy drop only in moving blades
(c) enthalpy drop in nozzles
(d) no enthalpy drop

RPSC AE 2016, RPSC ACF-2011, ESE 2012

Ans. (c) : In case of impulse steam turbine enthalpy drop in nozzles. In a impulse turbine there is no change in the pressure of the steam as it passes through the moving blades. There is change only in the velocity.

414. Degree of reaction is defined as the ratio of :

(a) Enthalpy drop in moving blades to enthalpy drop in fixed blades
(b) Enthalpy drop in fixed blades to total enthalpy drop in moving and fixed blades
(c) Enthalpy drop in fixed blades to enthalpy drop in moving blades
(d) Enthalpy drop in moving blades to total enthalpy drop in fixed and moving blades

GPSC ARTO Pre 30.12.2018
RPSC LECTURER 16.01.2016
CSE Pre-1996, 1994

Ans. (d) :

$$\text{Degree of Reaction} = \frac{\Delta h_{\text{moving blades}}}{\Delta h_{\text{moving blades}} + \Delta h_{\text{fixed blades}}}$$

415. De laval turbine is used for applications requiring

(a) Low power, low speed
(b) Low power, high speed
(c) High power, high speed
(d) High power, low speed

APPSC AEE Screening Test 2016
JPSC AE 2013 Paper-I, ESE 2018

Ans. (b) : De Laval turbine is an impulse turbine in which steam impinges upon revolving blades from a flared nozzle. Causes expansion of the steam, and hence changes its pressure energy into kinetic energy.
De laval turbine is used for low power and high speed.

416. Degree of reaction of the turbine if the isentropic enthalpy drop in fixed blade is 1.5 times of the isentropic enthalpy drop in the moving blade.

(a) 0.6 (b) 0.5
(c) 0.4 (d) 0.66

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper-II, ESE 1993

Ans. (c) : Degree of reaction (R)

$$= \frac{(\Delta h)_{\text{isen MB}}}{(\Delta h_{\text{isen}})_{MB} + (\Delta h_{\text{isen}})_{FB}}$$

$$= \frac{(\Delta h)_{MB}}{(\Delta h)_{MB} + 1.5(\Delta h)_{MB}}$$

$$= \frac{1}{1+1.5}$$

$$\boxed{R = 0.4}$$

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417. If the enthalpy drop in the moving blades and fixed blades of a steam turbine is 20 kJ/kg and 30 kJ/kg respectively then what is the degree of reaction?

(a) 33% (b) 40%
(c) 60% (d) 67%
(e) 77%

CGPSC AE 26.04.2015 Shift-II
TNPSC AE (Industries) 09.06.2013

Ans. (b) : We know that

$$\text{DOR} = \frac{\text{Heat drop in moving blades}}{\text{Head drop in a stage}}$$

$$\text{DOR} = \frac{20}{(20+30)} = 0.40$$

$$\boxed{\text{DOR} = 40\%}$$

418. In steam turbines the reheat factor:

(a) Increases with the increase in number of stages
(b) Decreases with the increase in number of stages
(c) Remains same irrespective of number of stages
(d) None of the above

OPSC AEE 2015 PAPER - II
UKPSC AE 2007 Paper-II

Ans. : (a) In steam turbine the reheat factor increase with the increase in number of stages.
Reheat factor:- It is an important term used for the multi-stage turbines, which may be broadly defined as the ratio of cumulative heat drop to the isentropic heat drop.

419. The isentropic heat drop in the nozzle of an impulse steam turbine with a nozzle efficiency 0.9, blade velocity ratio = 0.5, the mean blade velocity 150 m/s in kJ/kg is

(a) 50 (b) 60
(c) 40 (d) 75

ISRO Scientist/Engineer (RAC) 12.01.2020
GPSC ARTO 01.05.2016

Ans. (a) : Blade velocity ratio $= \frac{U}{V_1} = 0.5$

Where, U = mean blade velocity and

V_1 = Absolute velocity of steam at inlet $= \frac{150}{0.5} = 300$ m/sec.

By SFEE actual heat drop –

$$\frac{V_1^2}{2} = \frac{(300)^2}{2} \text{ J}$$

Nozzle efficiency, $= \frac{\text{Actual heat drop}}{\text{Isentropic heat drop}}$

$\therefore$ Isentropic heat drop in nozzle $= \frac{(300)^2}{2\times 0.9} = \frac{90000}{2\times 0.9}$

$$\boxed{\text{Isentropic heat drop} = 50 \text{ kJ/kg}}$$

420. Blade erosion in steam turbines occurs due to:

(a) high temperature steam
(b) droplets in steam
(c) high flow rate
(d) high rotational speed
(e) high kinetic energy

JPSC AE 10.04.2021 Paper-I
CGPSC ARTO & TSI 07.01.2017, ESE 2007

Ans. (b) : Steam turbine blade erosion is of two types. Erosion by particulates is due to oxides scales and is found at the front end of turbines. Erosion by water droplets is essentially a back-end problem.
Stream turbine blades are highly subjected to Water Droplet Erosion (WDE) caused by high energy impingement of liquid water droplets.

421. Pressure on two sides of impulse wheel of a steam turbine

(a) is same
(b) is different
(c) increase from one side to the other side
(d) decrease from one side on the other side

RPSC AE 2016, RPSC ACF-2011

Ans. (a) : Pressure on two side of impulse wheel of a steam turbine is same.

422. Commonly used method of governing in steam turbines is by :

(a) throttle governing
(b) nozzle control governing
(c) bypass governing
(d) hydraulic governing

Kerala PSC Poly. Lect. (Auto) 05.05.2016
J&K PSC Civil Services Pre, 2013

Ans. (a) : Throttle governing is commonly used method of governing in steam turbines.

423. Reheat factor in steam turbine depends upon :

(a) exit pressure only
(b) stage efficiency only
(c) initial temperature and pressure only
(d) all of the above

JPSC AE, 10.04.2021, Paper-I
J&K PSC Civil Services Pre, 2010

Ans. (d) $\text{Reheat factor} = \frac{\text{Commulative enthalpy drop}}{\text{isentropic enthalpy drop}}$

Reheat factor depends upon-
(i) Stage efficiency
(ii) Initial pressure and condition of steam
(iii) final pressure
It increases with increase in number of stages, the value of reheat factor varies from 1.02 to 1.06.

424. If the pressure of steam at the exhaust from the turbine is saturated corresponding to the temperature of process heat requirement, then this turbine is known as:

(a) Reaction turbine
(b) Impulse turbine
(c) Pass out turbine
(d) Back pressure turbine

ESE 2020, UPRVUNL AE 2016

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Ans. (d) : If the pressure of steam at the exhaust from the turbine is saturated corresponding to the temperature of process heat requirement, then this turbine is known as back pressure turbine.

425. When two rotors of the outward radial flow turbine is rotating in opposite direction then it is known as:

(a) Pass out turbine
(b) Ljungstrom turbine
(c) 50% reaction turbine
(d) Condensing turbine

OPSC AEE 2019 PAPER - II, ESE 1994

Ans. : (b) : When two rotors of the outward radial flow turbine is rotating in opposite direction then it is known as Ljungstrom turbine.

426. Consider the following statements : In an impulse turbine :

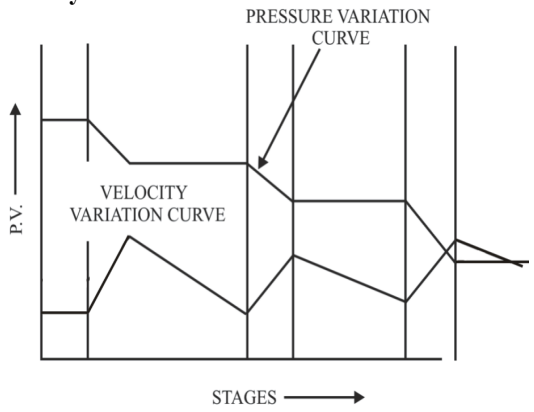
1. The relative velocity of steam at inlet and outlet of moving blades are equal
2. The moving blades are symmetrical
3. The outlet area of the moving blades is smaller than the inlet area.

(a) 1, 2 and 3 are correct
(b) 2 and 3 are correct
(c) 1 and 2 are correct
(d) 1 and 3 are correct

GPSC ARTO Pre 30.12.2018, CSE Pre-1997

Ans. (c) : In Impulse turbine,
(i) The relative velocity at inlet and outlet of moving blades are equal.
(ii) The moving blades are symmetrical.
(iii) The outlet area of moving blades is equal to the inlet area.

427. The given figure represents pressure and velocity variation for a :



(a) Reaction type turbine
(b) Velocity compounded impulse turbine
(c) Pressure-velocity compounded impulse turbine
(d) Pressure compounded impulse turbine

GPSC ARTO Pre 30.12.2018, CSE Pre-1995

Ans. (d) : Pressure Compounding Impulse Turbine.
Rateau Impulse Turbine (Pressure Compounding)– In pressure compounded turbine, moving blades and fixed blades are shown in fig. (given in question)

The steam of boiler pressure enters the first set of nozzle but does not drop the total pressure in one set of nozzles. The total steam pressure is dropped by number of stages and each stage consists of a set of nozzles and a ring of moving blades.

428. What is the cause of reheat factor in a steam turbine?

(a) Supersaturation (b) Reheating
(c) Superheating (d) Blade friction

UPRVUNL AE 21.08.2016
CSE Pre-2005

Ans. : (d) Reheat factor – It is an important term used for the multi stage turbine, which may be broadly defined as the ratio of cumulative heat drop to the isentropic heat drop.

429. In a Parson's turbine stage, the blade velocity is 320 m/s at the mean radius and the rotor blade exit angle is 30°. For minimum kinetic energy of the steam leaving the stage, the steam velocity at the exit of the rotor will be

(a) $160/\sqrt{3}$ m/s (b) $320/\sqrt{3}$ m/s
(c) $640/\sqrt{3}$ m/s (d) $640/\sqrt{3}$ m/s

APPSC AEE MAIN 2016 PAPER-III
ESE 1994

Ans. (c) : Data given
u = 320 m/s
$\alpha = 30°$
then steam velocity at the exit of the ratio will be V then
V cos 30° = u = 320

$$V = \frac{320}{\cos 30°} = \frac{320 \times 2}{\sqrt{3}}$$

$$V = \frac{640}{\sqrt{3}} \text{ m/s}$$

430. The following data refer to an axial flow turbine stage:
Relative velocity of steam at inlet to the rotor = 79.0 m/s., Relative velocity at the rotor exit = 152 m/s rotor mean peripheral velocity = 68.4 m/s, work output per kg of steam = 14100 J. What is the approximate degree of reaction?

(a) 0.9 (b) 0.8
(c) 0.7 (d) 0.6

Nagaland PSC CTSE 2015 Paper-II
ESE 1995

Ans. (d) : Enthalpy drop in moving blades

$$= \frac{V_{r_2}^2 - V_{r_1}^2}{2 \times 1000} = \frac{(152)^2 - (79)^2}{2000} = 8.43 \text{kJ/kg}$$

$$\text{Degree of reaction} = \frac{8.43}{14.1} = 0.597 \cong 0.6$$

431. Which one of the following sketches represents an impulse blade?

(a) (b)
(c) (d)

Nagaland PSC CTSE 2015 Paper-II
ESE 1995

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Ans. (a) : Figure (a) impulse turbine blade.
As no pressure change from inlet to outlet no area change is allowed.

432. Governing of steam turbines can be done by the following :

1. Nozzle control
2. Throttle control
3. Providing additional valve of passage

The correct answer will be –

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

PSPCL AE, 2012, ESE 1992

Ans. (a) : Nozzle control : In nozzle governing the flow rate of stream is regulated by opening shutting of sets of nozzles rather than regulating its pressure. In this method groups of two, three or more nozzles form a set and each set is controlled by a separate value.
Throttle control : In throttle governing the pressure of steam is reduced at the turbine entry thereby decreasing the availability of energy the flow rate is controlled using a partially opened steam control value.

433. Given that
α_1 = nozzle angle
n = number of rows of moving blades, in a velocity compounded impulse turbine, the optimum blade speed ratio is

(a) $2\cos\alpha_1.n$ (b) $\frac{n\cos\alpha_1}{2}$
(c) $\frac{\cos\alpha_1}{2(n+1)}$ (d) $\frac{\cos\alpha_1}{2}$

ISRO Scientist/Engineer 2007
ESE 1994

Ans. (d) : α_1 = Nozzle Angle
n = No. of moving blades
In velocity of compound impulse turbine—
So, optimum blade speed ratio $= \frac{\cos\alpha_1}{2}$

434. If the enthalpy drop across the fixed blades in a turbine is zero, the degree of reaction is :

(a) zero (b) 0.5
(c) Unity (d) Infinity

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
RPSC IOF, 2020

Ans. (c) : Degree of reaction $(R) = \frac{(\Delta h)_{MB}}{(\Delta h)_{MB} + (\Delta h)_{FB}}$

$(\Delta h)_{FB} = 0$
MB = Moving blade
FB = Fixed blade

Degree of reaction $(R) = \frac{(\Delta h)_{MB}}{(\Delta h)_{MB} + 0}$

$\boxed{R = 1}$

435. Consider the following statements for steam turbines:

(1) The ratio of actual enthalpy drop to isentropic enthalpy drop is known as mechanical efficiency.
(2) The ratio of enthalpy drop in moving blades to enthalpy drop in the stage is known a degree of reaction.
(3) Rateau turbine is the example of reaction turbine.
(4) Curtis turbine is the example of impulse turbine.

Which of the above statements are correct?

(a) 2 and 4 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

ESE 18.07.2021

Ans. (a) :
- Rateau turbine is a pressure compounding multistage impulse turbine.
- Curtis turbine is velocity compounding multistage impulse turbine.

Degree of reaction-

$$R = \frac{(\Delta h)_{\text{moving blade}}}{(\Delta h)_{\text{stage}}}$$

$$R = \frac{(\Delta h)_{\text{moving blade}}}{(\Delta h)_{\text{moving blade}} + (\Delta h)_{\text{fixed blade}}}$$

436. The pressure on the two sides of the moving blades of a reaction steam turbine is

(a) same
(b) higher at inlet
(c) lower at inlet
(d) may be higher or lower depending on quality of entry steam

JPSC AE, 10.04.2021, Paper-I

Ans. (b) : The pressure on the two sides of the moving blades of a reaction steam turbine is higher at inlet.

437. For Parson's reaction steam turbine, degree of reaction is

(a) 60% (b) 50%
(c) 100% (d) 75%

JPSC AE, 10.04.2021, Paper-I

Ans. (b) : Degree of reaction
$= \frac{\text{Isentropic heat drop in moving blade}}{\text{Sum of drop in fixed and moving blade}}$
For Parson's turbine degree of reaction is 50%.
For Hero's reaction turbine, degree of reaction is 100%.

438. 'Bleeding' or 'Extracting' steam from the turbine at various points is done for the purpose of :

(a) Reheating (b) Regeneration
(c) Moisture removal (d) Pressure reduction

RPSC ACF & FRO, 26.02.2021

Ans. (b) : Bleeding or extracting steam from turbine at various points is done for the purpose of Regeneration. In this heat of extracting steam is utilized in feed water heating.

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439. The velocity triangles at the inlet and exit of the rotor of a turbo machine are shown. V denotes the absolute velocity of the fluid, W denotes the relative velocity of the fluid, and U denotes the blade velocity. Subscripts 1 and 2 refer to inlet and outlet respectively. If $V_2 = W_1$ and $V_1 = W_2$, then the degree of reaction is _____.

(a) 0 (b) 1
(c) 0.5 (d) 0.25

Assam Engg. College AP/Lect. 18.01.2021

Ans. (c) : If $V_2 = W_1$ and $V_1 = W_2$
Also $\alpha_1 = \beta_2$ and $\beta_1 = \alpha_2$
Then, it represents parson's reaction turbine which has degree of reaction as 0.5.

440. Which of the following are applied (used) ways of compounding steam turbines?
1. Pressure compounding
2. Temperature compounding
3. Velocity compounding

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

ESE 2020

Ans. (d) : There are three types of compounding used in steam turbines which are :
- Pressure compounding
- Velocity compounding
- Pressure-Velocity compounding

441. Whirl velocity in a steam turbine refers to :

(a) the component of the velocity of steam in the direction of blade velocity
(b) the velocity of whirling turbine wheel.
(c) the component of the velocity of turbine wheel in the direction of blade velocity.
(d) the component of the velocity of steam in the direction normal to the blade velocity.

NLC GET 17.11.2020, Shift-II

Ans. (a) : Whirl velocity in a steam turbine refers to the component of the velocity of steam in the direction of blade velocity. The whirl velocity is the tangential component of absolute velocity at the blade inlet and outlet. This component of velocity is responsible for the whirling or rotating of the turbine rotor.

442. Symmetric velocity diagram in a frictionless reaction turbine implies that

(a) the blade angles at inlet and outlet are equal
(b) the flow angles at inlet and outlet are equal
(c) blade angle at inlet is equal to flow angle at outlet
(d) the flow does not accelerate through the rotor vanes

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : This can achieve in inflow impulse turbine absolute inlet velocity = outlet relative velocity inlet relative velocity = outlet absolute velocity blade angle at inlet is equal to flow angle of outlet. It has 50% degree of reaction.

$\alpha_1 = \beta_2$ and $\beta_1 = \alpha_2$

443. Blade speed ratio is the ratio of

(a) blade velocity to whirl component of inflow velocity
(b) whirl component of inflow velocity to blade velocity
(c) inflow absolute velocity to blade velocity
(d) blade velocity to in flow absolute velocity

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Blade ratio is the ratio of blade velocity to inflow absolute velocity $\left(\frac{u_1}{v_1}\right)$.

444. Diagram efficiency of a steam turbine stage is defined as the ratio of

(a) Diagram power to inflow rate of kinetic energy
(b) Diagram power to inflow steams enthalpy
(c) Diagram power to inflow steam stagnation enthalpy
(d) Diagram power to total power input from inflow steam

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Diagram efficiency is also known as blade efficiency or rotor efficiency. It is defined as the ratio of work done per kg of steam by the rotor to the energy available at the inlet, per kg of steam.

445. If blade velocity coefficient of a blade is less than 1, it means that

(a) the blade is smooth and frictionless
(b) the blade is symmetric
(c) the blade is asymmetric
(d) the blade is not smooth

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Blade velocity co-efficient is the ratio of relative velocity of steam at outlet tip of the blade to the relative velocity of steam at inlet tip of the blade if blades velocity co-efficient is less than one it means blade is not smooth.

446. Degree of reaction of smooth, symmetric reaction turbine is

(a) infinite (b) 0.5
(c) zero (d) 0.25

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : For smooth, symmetric reaction turbine
$\alpha = \theta$, $v_1 = v_{r2}$, $u_1 = u_2$
$\beta = \phi$, $v_2 = v_{r1}$
Degree of reaction = 50%

447. Leakage losses are a matter of concern in

(a) velocity compound impulse turbines
(b) simple impulse turbine
(c) reaction turbines
(d) pressure - compound impulse turbines

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : Leakage losses are a matter of concern in impulse turbine.
Note – minor leakage losses also occur in reaction turbine.

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448. Which of the following is not achieved through velocity compounding of impulse steam turbine.

(a) Reducing the rotor speed
(b) Reducing blade losses
(c) Reducing the back pressure
(d) Reducing the blade size

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : In impulse turbine, pressure remain same at all place equal to atmospheric pressure so type of compounding in impulse turbine does not reduce the back pressure.

449. In an axial flow impulse turbine stage, if ΔC_w is the change in whirl or circumferential velocity from inlet to outlet and C_b is the blade velocity, the power produced with mass flow rate ($\dot{m}$) of working fluid is

(a) $\dot{m}\Delta C_w \times \Delta C_w$ (b) $\dot{m}\Delta C_w$
(c) $\dot{m}\Delta C_w \times C_b$ (d) $\dot{m}C_b \times C_b$

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : For axial flow circumferential velocity at inlet and outlet is remain same and constant.

$$\text{work} = \dot{m}\left(V_{w_2}u_2 - V_{w_1}u_1\right)$$

$$\dot{m}u\left(V_{w_2} - V_{w_1}\right) \qquad [\because u_1 = u_2]$$

$$= \dot{m} \times \Delta C_w \times C_b$$

450. For a single stage of axial flow impulse turbine with symmetrical blade profile, having angle of attack α and axial flow velocity C_a, the blade velocity C_b should be _____ for maximum power.

(a) $C_a \cos\alpha$ (b) $\frac{C_a \cos\alpha}{2}$
(c) $C_a \cos(\alpha/2)$ (d) $(C_a/\cos\alpha)$

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) :

451. In an isentropic flow of ideal gas though turbine blade without any heat addition and work, if the temperature is constant

(a) internal energy can change
(b) Pressure cannot change
(c) Enthalpy cannot change
(d) Pressure and velocity can change

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : This is similar to nozzle when no heat added / rejected and no work is done. But across section changes pressure and velocity changes.

452. In an impulse reaction turbine stage, the heat drop in fixed and moving blades are 20 kJ/kg and 30 kJ/kg respectively. The degree of reaction for this stage will be :

(a) 3/5 (b) 5/3
(c) 2/5 (d) 5/2

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) :

$$\text{Degree of reaction} = \frac{(\Delta h)_{\text{moving blade}}}{(\Delta h)_{\text{moving blade}} + (\Delta h)_{\text{fixed blade}}}$$

$$= \frac{30}{20+30} = \frac{30}{50} = \frac{3}{5}$$

453. Critical speed of a turbine is

(a) same as run away speed
(b) speed that will cause mechanical failure of the shaft
(c) speed at which natural frequency of vibrations equals the number of revolutions at the same time
(d) speed equal to synchronous speed of the generator

Gujarat PSC AE 2019

Ans. : (c) : Critical speed of a turbine is speed at which natural frequency of vibrations equals the number of revolutions at the same time.

454. Which one of the following statements is correct with respect to axial flow 50% reaction turbine?

(a) The combined velocity diagram is symmetrical.
(b) The outlet absolute velocity should not be axial for maximum utilization
(c) Angles of both stator and rotor are not identical.
(d) For maximum utilization, the speed ratio $\frac{U}{v_1} = \sin^2\alpha$

ESE 2019

Ans. (a) : Efficiency of 50% reaction turbine is –

$$\eta = \frac{2\left(\frac{2V_1}{U}\sin\alpha_1 - 1\right)}{\sigma^2}$$

For maximum efficiency –

$$\Rightarrow \quad \frac{V_1}{U} = \frac{1}{\sin\alpha_1}$$

α_1 is large and efficiency is maximum when –

$$\frac{V_1}{U} \approx 1$$

So maximum efficiency –

$$\alpha_1 = 90°$$

Hence, combined velocity diagram is symmetrical.

455. The degree of reaction is given by:

(a) $\frac{\text{Actual enthalpy change in rotor}}{\text{Actual enthalpy change in stator}}$
(b) $\frac{\text{Actual enthalpy change in rotor}}{\text{Actual enthalpy change in state}}$
(c) $\frac{\text{Actual enthalpy change in state}}{\text{Actual enthalpy change in rotor}}$
(d) $\frac{\text{Actual enthalpy change in stator}}{\text{Actual enthalpy change in rotor}}$

SJVN ET 2019

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Ans. (b)
The degree of reaction is given by
$$= \frac{\text{Actual enthalpy change in rotor}}{\text{Actual enthalpy change in state}}$$

456. An impulse turbine of single stage having 1.2 m diameter runs at 1000 rpm. What will be the inlet steam velocity if the blade speed ratio is 0.314?
(a) 200 m/s (b) 100 m/s
(c) 150 m/s (d) 250 m/s

OPSC AEE 2019 PAPER - II

Ans. : (a) : Given,
D = 1.2 m
N = 1000 rpm
$$\frac{u}{v_1} = \text{Speed ratio} = 0.314$$
$$u = \frac{\pi DN}{60}$$
$$= \frac{\pi \times 1.2 \times 1000}{60} = 62.83 \text{ m/s}$$
$$\frac{u}{v_1} = 0.314$$
$$\frac{62.83}{v_1} = 0.314$$
$$v_1 = 200 \text{ m/s}$$

457. A throttle governed steam turbine develops 30 IHP by consuming 300 kg/hr. steam and 60 IHP with 570 kg/hr. How much steam is required to develop 20 IHP?
(a) 210 kg/hr (b) 230 kg/hr
(c) 300 kg/hr (d) 270 kg/hr

OPSC AEE 2019 PAPER - II

Ans. : (a) : Relation between the steam consumption per hour (m) and the Indicated Power (I.P.) is (Willian's law) –
$m = a + b \times I.P.$ Where a and b are constant
Find the value a and b –
$570 = 60 \times b + a$ .....(i)
$300 = 30 \times b + a$ .....(ii)
From equation (i) and (ii) –
$a = 30, b = 9$
$\because$ Steam consumption when developing 20 IHP
$m = 30 + 20 \times 9$
$\boxed{m = 210 \text{ kg/hr}}$

458. The coefficient of velocity for steam turbine is the ratio of:
(a) Relative velocity leaving the blade to the velocity of whirl at inlet of moving blade
(b) Velocity of whirl at inlet of moving blade to the relative velocity leaving the blade
(c) Relative velocity at inlet of moving blade to the relative velocity leaving the blade
(d) Relative velocity leaving the blade to the relative velocity at inlet of moving blade

OPSC AEE 2019 PAPER - II

Ans. : (d) : The coefficient of velocity for steam turbine is the ratio of relative velocity leaving the blade to the relative velocity at inlet of moving blade.

459. In a steam turbine, the nozzle angle at the inlet is 18°. The relative velocity is reduced to the extent of 6% when steam flows over the moving blades. The output of the turbine is 120 kJ/kg flow of steam. If the blades are equiangular, the speed ratio and the absolute velocity of steam ratio and the absolute velocity of steam at inlet for maximum utilization are nearly
(a) 0.42 and 230.2 m/s (b) 0.48 and 230.2 m/s
(c) 0.42 and 515.1 m/s (d) 0.48 and 515.1 m/s

ESE 2019

Ans. (d) : Given,
Nozzle angle at inlet (α) = 18°
Loss of relative velocity = 6%
Output power = 120 kJ/kg
Optimum blade speed, $\left(\frac{V_b}{V_1}\right) = \frac{\cos\alpha}{2}$
$$\left(\frac{V_b}{V_1}\right) = \frac{\cos 18°}{2} = 0.475$$
$$\frac{V_b}{V_1} = 0.48$$
Power developed –
$$\left(V_1 \cos\alpha - V_b\right)\left(2 - \frac{\text{Loss}}{100}\right) \times V_b = 120 \times 10^3$$
Putting, $V_b = 0.48\ V_1$
$\boxed{V_1 = 515.1 \text{ m/s}}$

460. A turbine in which steam expands both in nozzle as well as in blades is called as
(a) impulse reaction turbine
(b) reciprocating steam turbine
(c) gas turbine
(d) Curtis turbine

ESE 2019

Ans. (a) : Impulse reaction turbine
- In impulse turbine steam expands in nozzles only.
- In impulse reaction turbine steam expands in nozzles as well as it passes through blades.

461. A steam turbine is supplied with steam at a pressure of 20 bar gauge. After expansion in the steam turbine, the steam passes to condenser which is maintained at a vacuum of 250 mm of mercury by means of pumps. The inlet and exhaust steam pressures will be nearly
(a) 2101 kPa and 68 kPa
(b) 2430 kPa and 78 kPa
(c) 2101 kPa and 78 kPa
(d) 2430 kPa and 68 kPa

ESE 2019

Ans. (a) : Inlet pressure (P_1) = 20 bar gauge
Absolute inlet pressure
$(P_1)_{abs} = P_1 + P_{atm}$
= 20 + 1.01325
= 21.01325 bar
= 2101 kPa

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Exit pressure (P_2) = 250 mm of Hg (vacuum)
Absolute exit pressure $(P_2)_{abs} = P_{atm} - P_{vac}$
$= (101.325 - 13.6 \times 9.81 \times 0.25)$ kPa
$= 67.996$
$= 68$ kPa

462. Consider the following statements regarding compounding in steam turbines:
1. In impulse turbine, steam pressure remains constant between ends of the moving blades.
2. In reaction turbine, steam pressure drops from inlet to outlet of the blade.
3. In velocity compounding, partial expansion of steam takes place in the nozzle and further expansion takes place in the rotor blades.

Which of the above statements are correct?
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2019

Ans. (a) : (i) As the steam flows over the moving blades nearly all its velocity is absorbed. However, the pressure remains constant during the process. It is three stage pressure compounded impulse turbine.
(ii) In reaction turbine steam pressure drops from inlet to outlet of the blade.
(iii) In velocity compounding whole expansion takes place in nozzle and velocity is reduced subsequent stages.

463. The maximum blade efficiency of single stage impulse turbine in terms of nozzle angle α under ideal conditions is proportional to :
(a) $\cos\alpha$ (b) $\sin\alpha$
(c) $\sin^2\alpha$ (d) $\cos^2\alpha$

GPSC ARTO Pre 30.12.2018

Ans. (d) : Max. blade efficiency of single stage impulse turbine.

$$\eta_{max} = \cos^2\alpha$$

For Parson's turbine $\eta_{max} = \dfrac{2\cos^2\alpha}{1+\cos^2\alpha}$

464. In an ideal impulse turbine, the
(a) Absolute velocity at the inlet of moving blade is equal to that at the outlet
(b) Relative velocity at the inlet of the moving blade is equal to that at the outlet
(c) Axial velocity at the inlet is equal to that at the outlet
(d) Whirl velocity at the inlet is equal to that at the outlet

GPSC ARTO Pre 30.12.2018

Ans. (b) : In an ideal impulse turbine, the relative velocity at the inlet of moving blade is equal to that at the outlet.

$$v_{r_1} = v_{r_2}$$

465. In an axial flow impulse turbine, energy transfer takes place due to
(a) Change in relative kinetic energy
(b) Change in absolute kinetic energy
(c) Change in pressure energy
(d) Change in energy because of centrifugal force

GPSC ARTO Pre 30.12.2018

Ans. (a) :

466. A steam turbine rotor of mass 6000 kg is running at 8000 RPM out to the extent of 20 m. The centrifugal force tending to lift the turbine from its foundation will be approximately;
(a) 4.4 MN (b) 8.4 MN
(c) 6.4 MN (d) 10.4 MN

AAI Jr. Executive 29.11.2018

Ans. (b) : $F = mr\omega^2$

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 8000}{60}$$
$$= 837.33$$
$$= 6000 \times 20 \times (837.33)^2$$
$$= 8.4 \text{ MN}$$

467. To maximize the efficiency, the speed ratio in Parson's reaction turbine is
(a) $\dfrac{\cos\alpha}{2}$ (b) $\dfrac{\cos^2\alpha}{2}$
(c) $\cos\alpha$ (d) $\cos^2\alpha$

HPPSC AE 2018

Ans. (c) : To maximize the efficiency, the speed ratio in Parson's reaction turbine is cos α.

468. A steam turbine, in which a part of the steam after partial expansion, is used for process heating and the remaining steam is further expanded for power generation, is known as .....
(a) Back pressure turbine
(b) Impulse turbine
(c) Low pressure turbine
(d) Pass out turbine

RPSC AE 2018

Ans. (d) : A steam turbine, in which a part of the steam after partial expansion, is used for process heating and the remaining steam is further expanded for power generation, is known as pass out turbine.

469. Which of the following is true for steam turbines?
(a) In reaction turbine, gradual pressure drop takes place over fixed blades only
(b) In reaction turbine, gradual pressure drop takes place over moving blades only
(c) In impulse turbine, pressure drop takes place in nozzle as well as in the blades
(d) In impulse turbine, pressure drop takes place in nozzle, and increases in the blades
(e) In reaction turbine, gradual pressure drop takes place in moving as well as in fixed blades

CGPSC AE 25.02.2018

Ans. (e) : In reaction turbine, gradual pressure drop takes place in moving as well as in fixed blade whereas in impulse turbine pressure drop takes place in nozzle only.

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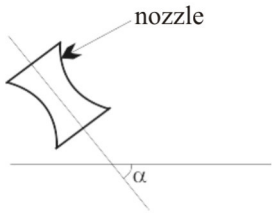
470. Angle made by the absolute velocity of steam to the horizontal is called as

(a) blade entrance angle
(b) nozzle angle
(c) outlet angle
(d) none of the mentioned

RPSC Vice Principal ITI 2018

Ans. (b) :



Angle made by the absolute velocity of steam to the horizontal is called as nozzle angle.

471. In a simple impulse turbine, the nozzle angle at the entrance is 30°. For the maximum diagram efficiency, what is the blade speed ratio?

(a) 0.259 (b) 0.75
(c) 0.3 (d) 0.433

BPSC AE Mains 2017 Paper - V

Ans. : (d) : Given,
Nozzle angle $\alpha = 30°$
for maximum efficiency

Blade speed ratio $= \frac{u}{v} = \frac{\cos\alpha}{2}$

So, $\frac{u}{v} = \frac{\cos 30°}{2}$

$= 0.433$

472. In a steam turbine when steam expand, the entropy

(a) increases linearly
(b) decreases linearly
(c) increases exponentially
(d) does not change

WBPSC AE, 2017

Ans. (d) : In a steam turbine when steam expand, the entropy does not change.

473. The impulse turbine blade efficiency is $\cos^2\alpha$, where α is the nozzle exit angle, if

(a) blade solidity is 0.65
(b) blades are equiangular
(c) blades are equiangular and frictionless
(d) blade velocity coefficient is unity

APGENCO AE, 2017

Ans. (c) : Blades are equiangular and frictionless

Impulse turbine blade efficiency $\boxed{\eta = \cos^2\alpha}$

Where, α = Nozzle angle

474. The isentropic enthalpy drop in fixed blades is one fourth of the isentropic enthalpy drop in moving blades of a turbine. The degree of reaction is

(a) 0.25 (b) 0.50
(c) 0.75 (d) 0.85

APGENCO AE, 2017

Ans. (*)

Degree of reaction $= \frac{\Delta h_{mb}}{\Delta h_{mb} + \Delta h_{fb}}$

Given that, $\Delta h_{fb} = \frac{1}{4}\Delta h_{mb}$

$= \frac{4\Delta h_{mb}}{4\Delta h_{mb} + 1\Delta h_{mb}}$

$= \frac{4\Delta h_{mb}}{5\Delta h_{mb}} = \frac{4}{5} = 0.8$

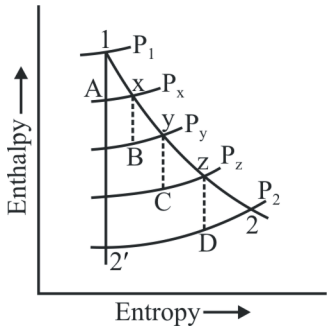
Note-This questions has been excluded for evaluation.

475. The reheat factor R_f for a steam turbine is usually in the range of

(a) 0 to 1 (b) 1 to 1.065
(c) 2 to 2.065 (d) 5 to 10

ISRO Scientist/Engineer 17.12.2017

Ans. (b) :



Reheat factor (RF)

$= \frac{\text{Cumulative enthalpy drop (Isentropic)}}{\text{Isentropic enthalpy drop (overall)}}$

$RF = \frac{\Delta h_{1A} + \Delta h_{xB} + \Delta h_{yC} + \Delta h_{zD}}{\Delta h_{12}}$

It lies between 1 to 1.04

476. Statement (I) : The efficiency of steam turbine considerably reduces if throttle governing is carried out at lower loads.

Statement (II) : Nozzle control can only be applied to the first stage of turbine.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I).
(c) Statement (I) is true but Statement (II) is false.
(d) Statement (I) is false but Statement (II) is true.

UPSC JWM 2017

Ans. (b) : In throttle governing, the flow rate of steam is controlled using a partially opened steam control valve. If throttle governing carried out at lower load, the efficiency of steam turbine reduced.
Nozzle governing is applied only to the first stage whereas the subsequent stages remain unaffected.

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477. From the consideration of the erosion of blades in the later stages of a steam turbine, the maximum moisture content at the turbine exhaust is limited to:

(a) 20% (b) 12%
(c) 88% (d) 5%

TRB Polytechnic Lecturer 2017

Ans. (b) : From the consideration of the erosion of blades in the later stages of a steam turbine, the maximum moisture content at the turbine exhaust is limited to 12%.

478. The turbine having identical fixed and moving blade is

(a) De-laval turbine
(b) Parson's reaction turbine
(c) Rateau's turbine
(d) Zoelly turbine

GPSC ARTO 01.05.2016

Ans. (b) : Turbine having identical fixed and moving blade is parson's reaction turbine.

479. The shaft of a steam turbine is usually rotated at

(a) Natural frequency of vibration
(b) Much below the natural frequency of vibration
(c) Much above the natural frequency of vibration
(d) There is no such eriterion
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (c) : In a shaft of turbine, the critical speed of vibration is higher. So it is operated at much higher frequency.

480. In reaction turbine, a stage is represented by

(a) each row of blades
(b) number of entries of steam
(c) number of easings
(d) number of exits of steam

GPSC Lect. 23.10.2016

Ans. (a) : In reaction turbine, whenever fluid passes through a row of blades, it represents a stage.

481. In reaction turbines-

(a) the steam is expanded in nozzles and there is no fall pressure as the steam passes over the rotor blades
(b) steam is directed over bucket-like-blades which propels the rotor
(c) expansion of steam takes place as it passes through the moving blades on the rotor as well as through the guide blades fixed to the casing
(d) steam pressure remains constant

RPSC 2016

Ans. : (c) Expansion of steam takes place it passes through the moving blades on the rotor as well as through the guide blades fixed to the casing.

- In a reaction turbine there is enthalpy drop both in fixed and moving blades.
- In a reaction turbine the expansion of steam takes place partly in the fixed blades and Partly in the moving blades.
- In case of impulse steam turbine there is enthalpy drop in nozzle.

482. In Rankine cycle, the work output from the turbine is given by

(a) Change in internal energy between inlet and outlet
(b) Change in enthalpy between inlet and outlet
(c) Change in entropy between inlet and outlet
(d) Change of temperature between inlet and outlet

UPPSC AE 12.04.2016 Paper-II
TSGENCO AE 14.11.2015

Ans. : (b) In Rankine cycle, the work output from the turbine is given by change in enthalpy between inlet and outlet

Turbine work $(W_T) = (h_1 - h_2)\,kJ/kg.$

483. The function of fixed blades on reaction turbine is to:

(a) Alter the direction of steam and decrease its velocity
(b) Alter the direction of steam and increase its pressure
(c) Alter the direction of steam and expand it to higher velocity
(d) Only to alter the direction of steam

UPRVUNL AE 2016

Ans. (c) : The function of fixed blades on reaction turbine is to alter the direction of steam and expand it to higher velocity.

484. In Curtis turbines

(a) The velocity of steam drops gradually after passing over the rows of moving blades
(b) The pressure of steam drops gradually after passing over the rows of moving blades
(c) The mass of steam drops gradually after passing over the rows of moving blades
(d) Both pressure and velocity of steam drops gradually after passing over the rows of moving blades

RPSC LECTURER 16.01.2016

Ans. (a) : In Curtis turbines the velocity of steam drops gradually after passing over the rows of moving blades.

- Compounding is done to reduce the rotational speed of the impulse turbine to practical limits.
- Three main types of compounded impulse turbine are:

(a) Pressure compounded steam turbine : The Rateau turbine
(b) Velocity compounded steam turbine: The Curtis turbine
(c) Pressure velocity compounded impulse turbine: The Rateau Curtis turbine.

485. Power developed per unit mass flow rate of steam by an stage of impulse turbine is 60 kW. The velocity of steam from nozzle is 400 m/s. Diagram efficiency will be:

(a) 75% (b) 95%
(c) 65% (d) 85%

UPRVUNL AE 2016

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Ans. (a) : Diagram efficiency—It is also known as blade efficiency or rotor efficiency. It is defined as the ratio of work done per kg of steam by the rotor to the energy available at the inlet per kg of steam

$$\eta_D = \frac{\text{Work done/kg of steam}}{\text{Energy available at the inlet per kg of steam (V^2/2)}}$$

$$\eta_D = \frac{2\times 60\times 10^3}{(400)^2}$$

$$\eta_D = \frac{120\times 10^3}{160000} = \frac{3}{4}$$

$$\eta_D = 0.75$$

$$\boxed{\eta_D = 75\%}$$

486. The component of velocity which is responsible for producing the work in steam turbine is called

(a) Axial velocity (b) Whirl velocity
(c) Relative velocity (d) Absolute velocity

RPSC LECTURER 16.01.2016

Ans. (b) : Whirl velocity component which is responsible for producing the work is steam turbine. Whirl velocity is also known as tangential velocity.

487. In a reaction stage turbine, the velocity

(a) increases in stator and remains constant in rotor
(b) increases in stator and decreases in rotor
(c) decreases in stator and remains constant in rotor
(d) remains constant throughout

RPSC IOF & Boiler 20.09.2017

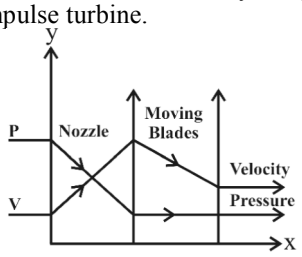
Ans. : (b) Increases in stator and decreases in rotor.

488. In an impulse turbine stage, pressure-

(a) increases in stator and decreases in rotor
(b) decreases in stator and increases in rotor
(c) decreases in stator and remains constant in rotor
(d) remains constant in both stator and rotor

RPSC IOF & Boiler 20.09.2017

Ans. : (c) Pressure and velocity diagram along the length of impulse turbine.



The figure clearly indicates that pressure are constant and velocity reduces in moving blades of impulse turbine.

489. The effect of blade friction in a steam turbine is to

(a) reheat the gas
(b) increase the specific output
(c) reduce the exhaust pressure
(d) reduce work done

BPSC Poly. Lect. 2016

Ans. (d) :

i) Some resistance is always offered by the blade surface to the idling steam jet, whose effect is to reduce the velocity of the Jet.
ii) The effect of blade friction in a steam turbine is to reduce work done.
iii) The blade friction in the impulse turbine reduces the velocity of steam by 10 to 15% passes over the blade.

490. High load factor indicates that

(a) cost of generation per unit power is increased
(b) total plant capacity is utilized for most of the time
(c) total plant capacity is not properly utilized for most of the time
(d) load on the plant is high

RPSC AE 2016

Ans. : (b) High load factor indicates that total plant capacity is utilized for most of the time.

491. The ratio of work done to the energy supplied to rotor in a turbine stage is called

(a) Blade efficiency (b) Stage efficiency
(c) Nozzle efficiency (d) None of these

UPPSC AE 12.04.2016 Paper-II

Ans. : (a)

- The ratio of work done to the energy supplied to rotor in a turbine stage is called blade efficiency.
- The ratio of rate of work done and energy supplied to the stage in a turbine is called stage efficiency.

492. Surging is the phenomenon of ..... .

(a) air stream blocking the passage
(b) motion of air at sonic velocity
(c) unsteady, periodic and reversed flow
(d) production of no air pressure

GWSSB DEE 07.07.2016

Ans. (c) : Surging is the phenomenon of unsteady, Periodic & Reversed flow. It is a momentary back flow through compressor from the discharge to the suction, forward discharge gas will flow backwards through the compressor towards the suction side.

493. Thermodynamic efficiency of a steam turbine as compared to steam engine is :

(a) equal
(b) less
(c) None of three mentioned here.
(d) more.

GWSSB AAE, 27.12.2015

Ans. (d) : Thermodynamic efficiency of a steam turbine as compared to steam engine is more.

494. The value of reheat factor in steam turbines normally varies from

(a) 0.42 to 6.4 (b) 0.61 to 0.82
(c) 1.02 to 1.06 (d) 1.2 to 1.6

TSGENCO AE 14.11.2015

Ans. (c) : Reheat factor > 1
R.F. = 1.04 to 1.08

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495. Which one of the following is the correct statement?
The degree of reaction of an impulse turbine.
(a) is less than zero
(b) is greater than zero
(c) is equal to zero
(d) increases with steam velocity at the inlet

TSGENCO AE 14.11.2015

Ans. (c) : The degree of reaction of an impulse turbine is equal to zero.

496. In a two-row Curtis stage with symmetrical blading
(a) Work done by both rows of moving blades are equal
(b) Work done by the fist row of moving blades is double of the work done by second row of moving blades
(c) Work done by the first row of moving blades is three times the work done by second row of moving blades
(d) Work done by the first row of moving blades is four times the work done by the second row of moving blades

TSGENCO AE 14.11.2015

Ans. (c) : Curtis turbine is a velocity compounding turbine. In a Curtis turbine, the work produced in each stage is not the same. The ratio of work produce in a 2 stage curtis turbine with symmetrical blading, the work done by the first row of moving blades is three time as work done by second row of moving blades. So, the work ratio of 2-stage turbine is 3 : 1. The ratio is 5 : 3 : 1 in three-stage turbine and the ratio is 7 : 5 : 3 : 1 in a four stage turbine.

497. Steam turbines are classified according to
(a) direction of flow
(b) principle of action
(c) number of cylinder
(d) arrangement of pressure drop

MPPSC AE 08.11.2015

Ans. (d) : Steam turbines are classified according to arrangement of pressure drop like HP, IP and LP turbine.

498. Steam enter a single stage impulse steam turbine with an inlet velocity of 50 m/s and leaves with and outlet velocity of 20 m/s. The work done by 100kg of steam is
(a) 85kJ (b) 95kJ
(c) 105kJ (d) 115kJ
(e) 125kJ

CGPSC AE 26.04.2015 Shift-II

Ans. (c) :

$$W = m\frac{\left[v_2^2 - v_1^2\right]}{2}$$

$$W = 100\frac{\left[50^2 - 20^2\right]}{2}$$

$$\boxed{W = 105 \text{ kJ}}$$

499. The process of maintaining the speed of the turbine constant for various load conditions is
(a) bleeding (b) reheating
(c) governing (d) compounding

TSPSC AEE 2015

Ans. : (c) The process of maintaining the speed of the turbine constant for various load conditions is governing.
Throttle Governing of steam turbine:- The throttle governing of a steam turbine is a method of controlling the turbine output by varying the quantity of steam entering into the turbine. This method is also know as servo motor method.

500. Due to the presence of friction over the blade of steam turbine, blade velocity coefficient exists and is defined as :
(a) $\frac{\text{Relative velocity of the steam at the exit of blade}}{\text{Relative velocity of the steam at the inlet of blade}}$
(b) $\frac{\text{Absolute velocity of the steam at the exit of blade}}{\text{Absolute velocity of the steam at the inlet of blade}}$
(c) $\frac{\text{Relative velocity of the steam at the inlet of blade}}{\text{Relative velocity of the steam at the exit of blade}}$
(d) $\frac{\text{Absolute velocity of the steam at the inlet of blade}}{\text{Absolute velocity of the steam at the exit of blade}}$
(e) $\frac{\text{Relative velocity of the steam at the exit of blade}}{\text{Absolute velocity of the steam at the inlet of blade}}$

CGPSC AE 26.04.2015 Shift-II

Ans. (a) : Due to presence of friction over the blade of steam turbine blade velocity coefficient exists and is defined as the ratio of relative velocity of the steam at the exit of blade (V_{re}) to relative velocity of the steam at the inlet of blades (V_{ri}).

$$\text{Blade velocity coefficient} = \frac{V_{re}}{V_{ri}}$$

501. If V_N and α are the nozzle exit velocity and nozzle angle in an impulse turbine, the optimum blade velocity is given by
(a) $V_N \cos 2\alpha$ (b) $V_N \sin 2\alpha$
(c) $\frac{V_N \cos\alpha}{2}$ (d) $\frac{V_N^2}{2}$

MPPSC AE 08.11.2015

Ans. (c) : Given, V_N = Nozzle exit velocity
α = Nozzle angle
$\therefore$ In case of Impulse turbine-

$$\rho = \frac{u}{V_N} = \frac{\cos\alpha}{2}$$

$$u = \frac{V_N \cos\alpha}{2}$$

Note- Option (d) is given by the commission.

502. The expansion of steam, as it flows over the blades in reaction turbine, represents
(a) isothermal process
(b) isentropic process
(c) throttling process
(d) free-expansion process

Assam PSC CCE Pre 2015

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Ans. (b) : The expansion of steam as it flows over the blades in reaction turbine represents isentropic process.

503. The thermal property that remains same on the two sides of the impulse wheel of a steam turbine is

(a) heat (b) temperature
(c) velocity (d) pressure

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : When the steam passes through the moving blades the velocity drops but the pressure remains same. The fact that the pressure does not drop across the moving blades is the distinguishing feature of the impulse turbine. The pressure at the inlet of the moving blades is the same as the pressure at the outlet of moving blades.

504. For a given pressure and temperature of steam, the thermal efficiency of the steam engine with condenser compared to a steam engine without condenser is always

(a) lower (b) higher
(c) same (d) none of these

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (b : The thermal efficiency of the steam engine with condenser compared to a steam engine without condenser is always higher.

505. As a result of blade friction the relative velocity at outlet of impulse turbine compared to inlet relative velocity is :

(a) nearly same (b) 2% less
(c) 10-15% less (d) 30% less

J&K PSC Civil Services Pre, 2013

Ans. (c) :

506. The super saturation of steam results in slight :

(a) increase in entropy
(b) increase of final dryness fraction
(c) increase of discharge
(d) all of the above

J&K PSC Civil Services Pre, 2013

Ans. (d) : Supersaturated or metastable flow has following effects –

(i) density increases hence increase in mass of steam discharged.
(ii) Increase entropy and specific volume of steam.
(iii) Reduce heat drop tends exit velocity of steam reduced.
(iv) Increase dryness fraction.

507. Absolute velocity Vector =

(a) Peripheral Velocity Vector + Relative Velocity Vector
(b) Relative Velocity Vector
(c) Axial Velocity Vector
(d) Peripheral Velocity Vector

TNPSC AE 2013

Ans. (a) : Absolute velocity Vector = Peripheral Velocity Vector + Relative Velocity Vector.

508. Multi-stage reaction turbines employ

(a) with large pressure drop
(b) with lower pressure drop
(c) constant pressure drop
(d) with high rotational speed

TNPSC AE 2013

Ans. (a) : Multi-stage reaction turbines employ with large pressure drop.

509. The entire change in the static properties occur in the rotor implies

(a) zero degree reaction
(b) hundred percent degree of reaction
(c) degree of reaction
(d) negative degree of reaction

TNPSC AE 2013

Ans. (b) : The entire change in the static properties occur in the rotor implies hundred percent degree of reaction.
This type turbine which have 100% DOR also known as Hero's Turbine.

510. In a reaction turbine, when the degree of reaction is zero, then there is

(a) no heat drop in the moving blades
(b) no heat drop in the fixed blades
(c) maximum heat drop in the moving blades
(d) maximum heat drop in the fixed blades

JPSC AE 2013 Paper-V

Ans. (a) : No heat drop in the moving blades

$$\text{Degree of Reaction} = \frac{\Delta h_{mb}}{\Delta h_{fb} + \Delta h_{mb}}$$

$$\Delta h_{mb} = 0,$$

$$\boxed{DOR = 0}$$

511. The purpose of governing in steam turbine is to

(a) reduce the effective heat drop
(b) reheat the steam and improve its quality
(c) completely balance against end thrust
(d) maintain the speed of the turbine

JPSC AE 2013 Paper-V

Ans. (d) : The purpose of governing in steam turbine is to maintain the speed of the turbine.

512. Pressure compounding can be done in the following type of turbines:-

(a) Impulse turbines
(b) Reaction turbines
(c) Both impulse and reaction turbines
(d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) : Pressure compounding can be done in impulse as well as the reaction turbine.

513. Blade efficiency of steam turbine is equal to:-

(a) $\frac{V(V_{w1} - V_{w2})}{2g}$ (b) $\frac{2V(V_{w1} + V_{w2})}{V_1^2}$

(c) $\frac{V(V_{w1} - V_{w2})}{V_1^2}$ (d) $\frac{V^2(V_{w1} - V_{w2})}{2V_1}$

UKPSC AE-2013, Paper-II

Ans. (b) : Blade or diagram efficiency,

$$(\eta_b) = \frac{\text{Power developed by runner}}{\text{Kinetics energy}}$$

$$= \frac{2(V_{w_1} + V_{w_2})V}{V_1^2}$$

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514. The ratio of static or pressure head change occuring in the rotor to the total change across the stage :

(a) Pressure Ratio (b) Degree of reaction
(c) Reheat factor (d) Aspect ratio

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

$$\text{Degree of reaction} = \frac{\text{Static pressure drop in the rotor}}{\text{Static pressure drop in the stage}}$$

515. In an impulse steam turbine, the steam expands in:

(a) Nozzles
(b) Moving blades
(c) Nozzle and moving blades
(d) Moving and fixed blades

APGENCO AE 2012

Ans. (a) : In an impulse steam turbine the steam expands wholly in nozzle and there is a pressure drop and heat drop.

516. Steam turbines are used for

(a) Electric power generation
(b) Direct drive for fans, compressors
(c) Marine propulsion
(d) All of the above

APGENCO AE 2012

Ans. (d) : Steam turbine (Rankine cycle) are used for electric power generation, Direct drive for fans, compressors and marine propulsion.

517. Steam expands in a turbine from 40 bar, 500°C to 0.10 bar isentropically. Assuming ideal conditions, given enthalpy drop 1198.8 kJ/kg, nozzle angle 16° and N as 3000 rpm. Mean diameter of the wheel, if the turbine were of simple impulse stage, will be

(a) 3.73 m (b) 4.73 m
(c) 5.63 m (d) 4.96 m

ISRO Scientist/Engineer 2012

Ans. (b) : V_1 = rotor velocity
V_b = blade velocity
α = nozzle angle

$$V_1 = 44.72(\Delta h)^{1/2} = 44.72(1198.8)^{1/2} = 1548.37 \text{m/s}$$

$$V_b = \frac{V_1 \cdot \cos\alpha}{2} = \frac{1548.37 \times \cos 16^\circ}{2} = 744.2 \text{ m/s}$$

$$V_b = \frac{\pi DN}{60} = \frac{\pi D \times 3000}{60} = 744.2$$

$$\Rightarrow \quad D = 4.73 \text{ m}$$

518. The ratio of the increase in pressure in rotor blades to total increase in pressure in the stage is called

(a) Pressure ratio
(b) pressure coefficient
(c) degree of reaction
(d) slip factor

BPSC AE 2012 Paper - V

Ans. : (c) : The ratio of the increase in pressure in rotor blades to total increase pressure in the stage is called degree of reaction.

519. In pressure-compounded turbine

(a) pressure drop in each stage is equal
(b) pressure increases as steam flows over blades
(c) most of the kinetic energy of steam is absorbed as it passes over moving blades
(d) None of these

BPSC AE 2012 Paper - V

Ans. : (c) : Pressure compounded turbine- In this type of turbine, the compounding is done for pressure of steam only i.e., to reduce the high rotational speed of turbine the whole expansion of steam is arranged in a number of steps by employing a number of simple turbine in a series keyed on the same shaft. Each of these simple impulse turbine consisting of one set of nozzle and one row of moving blades is known as a stage of the turbine and thus this turbine consists of several stages.

520. The degree of reaction for a turbomachinary in which heat drop in moving blades is 8 kJ/kg and in fixed blades 12 kJ/kg, would be

(a) 66.6% (b) 150%
(c) 40% (d) 166.6%

UKPSC AE 2012 Paper–II

Ans. (c) : Degree of reaction

$$= \frac{\text{Heat drop in moving blade}}{\text{Heat drop in moving blade} + \text{Heat drop in fixed blade}}$$

$$= \frac{8}{12+8} = \frac{8}{20} = \frac{2}{5} = 0.4 = 40\%$$

521. Parson's reaction turbine has

(a) fixed blades only
(b) moving blades only
(c) identical fixed and moving blades
(d) fixed and moving blades of different shapes.

UKPSC AE 2012 Paper–II

Ans. (c) : identical fixed and moving blades

522. Both pressure and velocity of steam drops in the moving blades of

(a) Impulse turbine
(b) Velocity compounded impulse turbine
(c) Reaction turbine
(d) Pressure compounded impulse turbine

ISRO Scientist/Engineer 2012

Ans. (c) : Both pressure and velocity of steam drops in the moving blades of reaction turbine.

523. In parson's steam turbine, steam expands in

(a) nozzles only
(b) blades only
(c) partly in nozzles and partly in blades
(d) none of the above

UKPSC AE 2012 Paper–II

Ans. (c) : In parson's steam turbine, steam expands in partly in nozzles and partly in blades.

524. Wrong statement for the Curtis turbine is :

(a) Curtis turbine consists of a set of nozzles followed by a set of moving blades
(b) In conventional design only two rows of moving blades are used
(c) There is no restriction on the number of rows of moving blades
(d) Curtis stage is also known as velocity compounding

OPSC Civil Services Pre. 2011

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Ans. (c) : Curtis turbine –
- Curtis turbine consists of a set of nozzle followed by a set of moving blades.
- In conventional design only two rows of moving blades.
- Curtis stages is also known as velocity compounding.

525. What is the action of steam in a steam turbine?
(a) Static
(b) Dynamic
(c) Static and dynamic
(d) Neither static nor dynamic

VIZAG STEEL MT 2011

Ans. (b) : A steam turbine works on the dynamic action of steam. Steam is caused to fall in pressure in a passage or nozzle. Due to this fall in pressure, certain amount of heat energy is converted into kinetic energy to give it high velocity.

526. For a given power an impulse turbine as compared to a reaction turbine has ___ row of blades :
(a) less (b) more
(c) equal (d) none of above

J&K PSC Civil Services Pre, 2010

Ans. (b) : An impulse turbine as compared to a reaction turbine, for a given power has more row of blades.

527. For reaction turbines which of the following governing methods is not used ?
(a) Throttle governing
(b) By-pass governing
(c) Nozzle control governing
(d) Throttle and by pass governing

J&K PSC Civil Services Pre, 2010

Ans. (c) : Governing method for reaction turbine –
(i) Throttle governing
(ii) By-pass governing
(iii) Throttle and by pass governing
Note - A nozzle used only impulse turbine, and not in reaction turbine.

528. Pressure drop in nozzle occurs in
(a) reaction turbine (b) impulse turbine
(c) steam turbine (d) all of those

TNPSC AE, 2008

Ans. (b) : Pressure losses occurs when fluid flows from convergent to divergent.

529. In a 50% reaction stage of a gas turbine
(a) the static enthalpy drop in the stator is greater than that in the rotor
(b) the static enthalpy rise in the stator is greater than that in the rotor
(c) the static enthalpy rise in the stator is greater than that in the rotor
(d) the stage static enthalpy drop is equally shared by the rotor and stator

DRDO Scientists 2008

Ans. (d) : Degree of reaction

$$R = \frac{\text{Static enthalpy change in rotor}}{\text{Static enthalpy change in stage}}$$

$$= \frac{\Delta h_R}{\Delta h_R + \Delta h_S}$$

$$0.5 = \frac{\Delta h_R}{\Delta h_R + \Delta h_S}, \quad \Delta h_R = 0.5\,\Delta h_R + 0.5\,\Delta h_S$$

$$\Delta h_R = \Delta h_S$$

530. What is the effect of bleeding?
(a) It decreases the power developed by the turbine
(b) The boiler is supplied with hot water
(c) It increases thermodynamic efficiency of turbine
(d) All of above

UKPSC AE 2007 Paper -II

Ans. (d) : All of above

531. Steam turbines are governed by which of following methods?
(a) Throttle governing
(b) Nozzle control governing
(c) Bypass governing
(d) All of above

UKPSC AE 2007 Paper -II

Ans. (d) : All of above

532. For a single stage impulse turbine, having nozzle angle a, maximum blade efficiency under ideal conditions is given by
(a) $\frac{\cos\alpha}{2}$ (b) $\frac{\cos^2\alpha}{2}$
(c) $\frac{\cos 2\alpha}{2}$ (d) $\cos^2\alpha$

UKPSC AE 2012, 2007 Paper -II

Ans. (d) : $\cos^2\alpha$

533. In the impulse turbine the steam expands in
(a) blade
(b) nozzle
(c) partly in nozzle and partly in blade
(d) none of the above

UKPSC AE 2007 Paper -II

Ans. (b)
- In impulse turbine the steam expands in nozzle only.
- In reaction turbine the steam expands in fixed as well as moving blade.

534. Compared to reciprocating steam engines, steam turbines
(a) have higher efficiency
(b) have low steam consumption
(c) have less frictional losses due to sliding parts
(d) all of above

UKPSC AE 2007 Paper -II

Ans. (d) : All of above

535. The condition for maximum efficiency of reaction turbine is given by
(a) $V_b = V_1 \cos\alpha$ (b) $V_b = V_1^2 \cos\alpha$
(c) $V_b = (V_1 \cos\alpha)/2$ (d) $V_b = \frac{V_1^2 \cos 2\alpha}{2}$
[V_1 being the absolute speed of the steam entering the blades and α being nozzle angle]

UKPSC AE 2007 Paper -II

Ans. (a) : $V_b = V_1 \cos\alpha$

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536. In steam turbines the ratio of blade length divided by the steam passage width is

(a) aspect ratio (b) diagram factor
(c) unity (d) none of these

OPSC Civil Services Pre 2006

Ans. (a) : Ratio of blade length to the steam passage width is known as aspect ratio.

537. Reheating of steam is used

(a) to increase efficiency
(b) to increase work output
(c) to increase turbine inlet temperature
(d) to reduce amount of fuel used

Nagaland PSC (CTSE) 2017, 2016, Paper-II

Ans. (b) : With reheating of steam turbine is used to increase the work output.
Work output
= Work done by turbine – Work done by compressor

538. Reheat of steam under ideal conditions takes place at constant

(a) Temperature (b) Enthalpy
(c) Pressure (d) Entropy

Nagaland PSC CTSE 2017 Paper-2

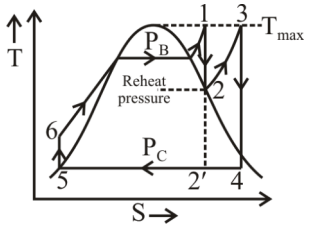
Ans. (c) : Reheating of steam under ideal conditions take place at constant pressure.

539. Reheating of steam

(a) Decreases steam rate
(b) Increases steam rate
(c) No effect on steam rate
(d) None of the mentioned

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Main purpose of reheating is to avoid excess moisture in steam at the end of expansion to protect the turbine. Since, by reheating the work output increase the steam rate decreases.



- W_T-↑, W_P-Constant, W_{net}-↑
- $\downarrow \text{Specific Steam Consumption} = \frac{3600}{W_{net}\uparrow}$
- η-↑↓ (Can't say)

4. Steam Nozzle

540. The critical pressure ratio for maximum discharge in steam nozzle is

(a) $\frac{2}{\gamma+1}$ (b) $\left(\frac{2}{\gamma+1}\right)^{\gamma}$

(c) $\left(\frac{2}{\gamma+1}\right)^{\gamma-1}$ (d) $\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$

where 'γ' is the isentropic index

ISRO Scientist/Engineer 17.12.2017
ISRO Scientist/Engineer 03.07.2016
RPSC LECTURER 16.01.2016
BPSC Poly. Lect. 2016, ESE 2002, 2000

Ans. (d) : Critical pressure ratio for maximum discharge

$$\frac{P_2}{P_1} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

541. Shock effect in a nozzle is felt in

(a) divergent portion (b) straight portion
(c) convergent portion (d) throat

BPSC AE Mains 2017 Paper - V
TANGEDCO AE 2015, TNPSC AE 2014
BPSC AE 2012 Paper - V

Ans. : (a) : Shock effect in a nozzle is felt in divergent portion.
For a nozzle to have a supersonic outlet, the speed at the throat should be definitely sonic and the nozzle should be divergent from the throat section.

542. The maximum velocity attainable at the throat of a steam nozzle is

(a) Much less than sonic velocity
(b) Slightly less than sonic velocity
(c) Supersonic velocity
(d) Sonic velocity
(e) None of the above

UPSC JWM 2017
CGPSC Poly. Lect. 22.05.2016
J&K PSC Civil Services Pre, 2013
BPSC AE 2012 Paper-V

Ans. (d) : The maximum velocity attainable at the throat of a steam nozzle is sonic velocity. Sonic is stated that sonic velocity of sound in that fluid (a gas/vapour). Its a maximum that can be achieved by the fluid.

543. The function of steam nozzle is to convert :

(a) heat energy steam into kinetic energy
(b) heat energy steam into rotational energy
(c) kinetic energy into heat energy of steam
(d) heat energy steam into pressure energy

APPSC-AE-2019, Kerala PSC AE 06.08.2015
KPSC AE. 2015, TSPSC AEE 2015

Ans. (a) : Steam nozzle- Steam nozzle is a passage of varying cross-section by means of which heat energy of steam is converted into kinetic energy as the steam expands from higher pressure to a lower pressure.

Therefore nozzle is designed to increase the velocity of steam.

544. Convergent-divergent nozzle is said to be choked when .....

(a) critical pressure is attained at the exit and Mach number at this section is sonic
(b) velocity at the throat becomes supersonic
(c) exit velocity becomes supersonic
(d) mass flow rate through the nozzle reaches a maximum value

RPSC AE 2018
Punjab PSC (Lect.) 06.08.2017, ESE 2016

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Ans. (d) : In the choked condition, a constant and maximum mass flow rate of air passes through nozzle. In this condition,

- critical pressure at throat and after that pressure reduces.
- mach number at throat is one i.e. velocity of flow is sonic.
- After the throat, the flow velocity may be subsonic or supersonic depending upon exit pressure of nozzle.

545. Pressure loss occurs in nozzle due to .....

(a) Due to viscous force
(b) When fluid flows from convergent to divergent
(c) Frictional forces
(d) None of the mentioned

Nagaland PSC (CTSE) 2017, 2016, Paper-II

Ans. (b) : Pressure loss generally occurs when fluid flow from convergent section to divergent section in nozzle.

546. The effect of friction in nozzle is to :

(a) keep dryness fraction constant
(b) increase dryness fraction
(c) decrease dryness fraction
(d) first increase dryness fraction up to a certain limit and then decrease

J&K PSC Civil Services Pre, 2013
ESE 1995

Ans. (b) : Effect of friction on nozzles–

- Entropy is increased
- The energy available decreases
- Reduction in enthalpy
- Steam is reheated i.e. quality of vapour at the exit is improved means dryness fraction increases.
- Decreases in mass flow rate.

547. In a nozzle super heated steam expanding from 10 bar to 2 bar, the pressure at the throat will be

(a) 1.66 bar (b) 3.26 bar
(c) 5.46 bar (d) 8.26 bar

RPSC IOF, 2020, APGENCO AE, 2017

Ans. (c) Saturated value n = 1.135
Super heated value n = 1.3

$$\text{Critical pressure} = \frac{P_2}{P_1} = \left(\frac{2}{n+1}\right)^{\frac{n}{n-1}}$$

$$\frac{P_2}{P_1} = (0.869)^{4.33}$$

$$\frac{P_2}{10} = 0.544$$

$P_2 = 5.44$ bar ≈ 5.46 bar

548. In a steam nozzle, inlet pressure of superheated steam is 10 bar. The exit pressure is decreased from 3 bar to 1 bar. The discharge rate will :

(a) remain constant
(b) decrease
(c) increase slightly
(d) increase or decrease depending on whether the nozzle is convergent or convergent-divergent

GPSC ARTO Pre 30.12.2018, ESE 1999

Ans. (a) : In steam nozzle, as the steam pressure is decreased at the exit but according to conservation of mass discharge rate will remain constant.

549. In a steam nozzle, to increase the velocity of steam above sonic velocity by expanding steam below critical pressure.

(a) a vacuum pump is added
(b) ring diffusers are used
(c) divergent portion of the nozzle is necessary
(d) abrupt change in cross-section is needed

GPSC ARTO Pre 30.12.2018, ESE 2006

Ans. (c) : In a steam nozzle, to increase the velocity of steam above sonic velocity by expanding steam below critical pressure, divergent portion of the nozzle is necessary.

550. The critical pressure ratio for initially dry saturated steam is

(a) 0.528 (b) 0.546
(c) 0.577 (d) 0.582

JPSC AE 2013 Paper-V, ESE 1994

Ans. (c) : For dry saturated steam
n = 1.135

$$\frac{P_2}{P_1} = \left(\frac{2}{n+1}\right)^{\frac{n}{n-1}}$$

$$\frac{P_2}{P_1} = \left(\frac{2}{2.135}\right)^{\frac{1.135}{0.135}}$$

$$\frac{P_2}{P_1} = (0.9367)^{8.407}$$

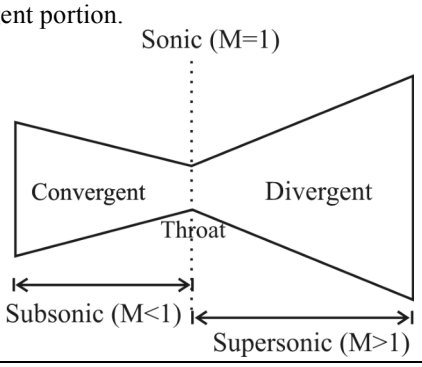
$= 0.577$

551. Mach number is more than unity in which of the following portions of a convergent-divergent nozzle?

(a) Convergent portion (b) Straight portion
(c) Throat (d) Divergent portion

TNPSC AE (Industries) 09.06.2013
BPSC AE 2012 Paper - V

Ans. : (d) : Mach number is more than unity, the nozzle is divergent portion.



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552. In a nozzle designed for the maximum discharge conditions, the flow velocity in the convergent section of the nozzle

(a) is sonic
(b) is subsonic
(c) is supersonic
(d) depends upon the initial pressure and condition of steam

BPSC AE Mains 2017 Paper - V
ESE 2013

Ans. : (b) : In a nozzle designed for the maximum discharge conditions, the flow velocity in the convergent section of the nozzle is subsonic.

553. A correctly designed convergent-divergent nozzle working at a designed load is

(a) always isentropic (b) always chocked
(c) never chocked (d) never isentropic

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012

Ans. (b) : Always chocked– Divergent portion will be act as nozzle only if flow is supersonic for that must be chocked.

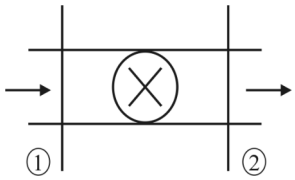
554. If steam is throttled, its.

(a) Pressure and enthalpy remain unchanged.
(b) Temperature and entropy remain unchanged
(c) Enthalpy remains unchanged but other property changes.
(d) Enthalpy remains unchanged but pressure may or may not change.

RPSC ACF & FRO, 26.02.2021

Ans. (c) : If steam is throttled its enthalpy remains unchanged but other property changes.

Throttling Process-



$P_2 < P_1$ (Pressure drop)
$V_2 < V_1$ (Expansion process)
$S_2 > S_1$ (Irreversible process)
$h_1 = h_2$ (Isenthalpic process)

555. If the turbine nozzle operates at a higher pressure ratio than the design value, then it is called _____.

(a) sub critical condition
(b) critical condition
(c) super critical condition
(d) abnormal condition

VIZAG STEEL MT 24.01.2021, Shift-II, Shift-I

Ans. (c) : If the turbine operates at a higher pressure ratio than the design value, then its called super critical condition

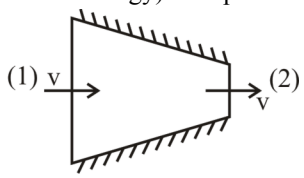
$$\frac{P_c}{P_1} = \left(\frac{2}{n+1}\right)^{\frac{\gamma}{\gamma-1}}$$

556. Which condition is not satisfied for the nozzle?

(a) Velocity decreases
(b) Velocity increases
(c) Heat transfer is zero
(d) Work transfer is zero

VIZAG STEEL MT 24.01.2021, Shift-II

Ans. (a) : Nozzle is a device which increases velocity (i.e. increases kinetic energy) in expanse of pressure.



= Heat transfer zero
= Work transfer zero

557. Air flows isentropically through a duct. At a section, the area is 0.05 m^2, velocity v = 180 m/s and temperature T = 470 K. Compute the stagnation temperature T_0

(a) 486 K (b) 286 K
(c) 556 K (d) 508 K

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : $V_1 = 180$ m/sec, $C_p = 1.005$ kJ/kgK
$T_1 = 470$ K

Stagnation enthalpy $h_0 = h_1 + \frac{V_1^2}{2000}$

$$C_pT_0 = C_pT_1 + \frac{V_1^2}{2000}$$

Stagnation temperature, $T_0 = T_1 + \frac{V_1^2}{2000C_p}$

$$T_0 = 470 + \frac{180^2}{2000 \times 1.005}$$

$$T_0 = 486.119 \text{ K}$$

558. The quality of vapour at the exit of nozzle ------ due to nozzle friction.

(a) Increases (b) Decreases
(c) Does not change (d) Unpredictable

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : The quality of vapour at the exit of nozzle increases due to nozzle friction.
Due to friction, reheating of steam takes place leads to effect of super saturation in nozzle.

559. Which of the following are effects of nozzle friction?

1. Enthalpy drop decreases
2. Exit velocity reduces
3. Decrease in specific volume
4. Decrease in mass flow rate

Select correct code

(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) 1, 2 and 4

UPPSC AE 13.12.2020, Paper-2

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Ans. (d) : Effect of Nozzle Friction–
(1) Actual enthalpy drop decreases.
(2) Reduction in the exit velocity of fluid.
(3) Increase in specific volume (density decrease).
(4) Decrease in mass flow rate.
(5) Increase in dryness fraction.
(6) Increase in entropy.

560. When a steam nozzle is supersaturated, the mass flow rate through it, when compared to equilibrium conditions
(a) Does not change
(b) is lower
(c) is higher
(d) may be higher or lower depending on inlet temperature

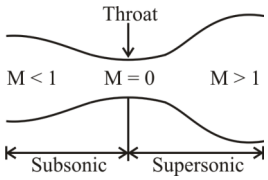
Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : During super saturation or condition of metastable the density is increased (about eight times), which gives the increase in mass of steam discharged.

561. To achieves highest exit flow velocity in application using liquids, the nozzles shape requires is _____
(a) A convergent - divergent passage
(b) A divergent - convergent passage
(c) A divergent passage only
(d) A convergent passage only

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a)



To achieve highest velocity the nozzle should be convergent divergent type. In which velocity in convergent portion subsonic at throat sonic and in divergent portion supersonic.

562. Critical pressure ratio in a nozzle is ratio of pressure at _____ to the inlet pressure
(a) Exit of the nozzle
(b) Minimum area
(c) Location of sonic velocity
(d) Atmospheric

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Critical pressure ratio in a nozzle is ratio of pressure at location of sonic velocity (throat) to the inlet pressure.

563. An aircraft is in its take off roll at sea level with ambient temperature of 18°C. What is the approximate speed of the aircraft if the temperature measured by a probe at the exit of the engine diffuser is 36°C? (Assuming air stagnation at diffuser outlet and C_P of air as 1.0 kJ/kg.K)
(a) 12 m/s (b) 3 m/s
(c) 6 m/s (d) 18 m/s

ISRO Scientist/Engineer 12.01.2020

Ans. (*) : $T_1 = 18°C = 291K$ $C_P = 1.0$ kJ/kgK
Temperature measured by a probe at the exit of the engine diffuser = stagnation temperature = 36°C = 309K

$$T_0 = T_1 + \frac{v^2}{2C_P}$$

$$36 = 18 + \frac{v^2}{2\times1\times10^3}$$

$$v = \sqrt{2\times10^3\times18} = 60\sqrt{10} \text{ m/sec.}$$

v = 189.7 m/sec

Note-This question is deleted by ISRO.

564. If m_f is the mass of fuel supplied per kg of air in one second, then the mass of gases leaving the nozzle of turbojet will be
(a) $(1-m_f)$kg/s (b) $\frac{1}{(1+m_f)}$kg/s
(c) $(1+m_f)$kg/s (d) $\frac{1}{(1-m_f)}$kg/s

ESE 2020

Ans. (c) : $(1+m_f)$kg/s

565. The diameter of a nozzle d for maximum transmission of power through it, is
(a) $\left[\frac{D^5}{8fL}\right]^{\frac{1}{4}}$ (b) $\left[\frac{D^5}{8fL}\right]^{\frac{1}{2}}$
(c) $\left[\frac{D^5}{fL}\right]^{\frac{1}{4}}$ (d) $\left[\frac{8D^5}{fL}\right]^{\frac{1}{2}}$

where :
D = Diameter of pipe
f = Coefficient of friction
L = Length of pipe

ESE 2020

Ans. (a) : $\left[\frac{D^5}{8fL}\right]^{\frac{1}{4}}$

566. A 120 mm diameter jet of water is discharging from a nozzle into the air at a velocity of 40 m/s. The power in the jet with respect to a datum at the jet will be
(a) 380 kW (b) 360 kW
(c) 340 kW (d) 320 kW

ESE 2020

Ans. (b) : Given:
d = 120 mm = 0.12 m
V = 40 m/s

$$P = \frac{1}{2}\dot{m}V^2 \qquad \because \dot{m} = \rho aV$$

$$= \frac{1}{2}(\rho aV)V^2$$

$$= \frac{1}{2}\rho aV^3$$

$$= \frac{1}{2}\times\frac{(1000)}{1000}\times\frac{\pi(0.12)^2}{4}\times(40)^3$$

$$P = 361.91 \text{ kW}$$

$$\approx 360 \text{ kW}$$

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567. Subsonic and supersonic diffusers have the following geometry

(a) Divergent and convergent respectively
(b) Both divergent
(c) Both convergent
(d) Convergent and divergent respectively

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a)

- For subsonic compressible flow, converging passage act as nozzle and diverging passage as a diffuser.
- For supersonic compressible flow, converging passage act as diffuser and diverging passage act as nozzle.

568. For a super sonic flow if the area of flow increases then.

(a) Velocity remains constant
(b) Velocity increases
(c) Velocity decreases
(d) Velocity initially increases and after attaining peak decreases.

OPSC AEE 2019 PAPER - II

Ans. : (b) : For a supersonic flow if the area of flow increases pressure decreases and velocity increases.

569. The region inside the Mach cone is called:

(a) Zone of action (b) Zone of silence
(c) Control volume (d) None of these

OPSC AEE 2019 PAPER - II

Ans. : (a) : The region inside the mach cone is called zone of action.

570. At critical pressure ratio, the velocity at the throat of a nozzle is:

(a) Equal to the sonic speed
(b) Less than the sonic speed
(c) More than the sonic speed
(d) None of these

OPSC AEE 2019 PAPER - II

Ans. : (a) : Mach number at throat is one i.e. velocity of flow is sonic.

571. In a supersonic flow, a diffuser is a conduit having:

(a) Constant area throughout its length
(b) Converging-diverging passage
(c) Gradually decreasing area
(d) Diverging-converging passage

OPSC AEE 2019 PAPER - II

Ans. : (c) : In a supersonic flow, a diffuser is a conduit having gradually decreasing area.

572. The sonic velocity in a fluid medium is directly proportional to:

(a) Mach number
(b) Pressure
(c) Temperature
(d) Square root of temperature

OPSC AEE 2019 PAPER - II

Ans. : (d) : $\text{Sonic Velocity} = \sqrt{\gamma RT}$

573. Speed of sound is maximum in

(a) Liquid (b) Gas
(c) Solid (d) Semisolid

Gujarat PSC AE 2019

Ans. (c) : Speed of the sound depends on the density of medium through which it is travelling more the density more is the speed. As density of solid is more than liquid and gases, sound travels with more speed in solids.

574. For nozzle governed turbine the efficiency is mainly affected due to the losses during:

(a) Throttling
(b) Inter stage pressure drop
(c) Stage condensation
(d) Partial admission

OPSC AEE 2019 PAPER - II

Ans. : (d) : For nozzle governed turbine the efficiency is mainly affected due to losses during partial admission.

575. The nozzle efficiency of a turbine is:

(a) Ratio of work done on moving blades to actual enthalpy drop
(b) Ratio of work delivered by blades to Isentropic enthalpy drop
(c) Ratio of actual to Isentropic enthalpy drop
(d) None of these

OPSC AEE 2019 PAPER - II

Ans. : (c) :

$$\text{Nozzle efficiency} = \frac{\text{Useful heat drop}}{\text{Isentropic heat drop}}$$

576. What will be the approximate velocity (in m/s) of bullet fired in standard air (Temperature of air = 300 K) if the mach angle is 30° [R = 0.28 kJ/kg-K, γ = 1.4]

(a) $560\sqrt{6}$ (b) $140\sqrt{6}$
(c) $70\sqrt{6}$ (d) $280\sqrt{6}$

SJVN ET 2019

Ans. (d) : Given, T = 300 K
Mach angle (θ) = 30°
R = 0.28 kJ/kgK = 280 J/kgK
γ = 1.4

$$\sin\theta = \frac{1}{M}$$

$$\sin 30^\circ = \frac{1}{M}$$

$$\frac{1}{2} = \frac{1}{M}$$

$$M = 2$$

$$C = \sqrt{\gamma RT}$$

$$= \sqrt{1.4 \times 280 \times 300}$$

$$= 14 \times 10\sqrt{6}$$

$$= 140\sqrt{6}$$

$$V = MC$$

$$= 2 \times 140\sqrt{6}$$

$$= 280\sqrt{6}$$

577. In an isentropic flow through a nozzle, air flows at the rate of 600 kg/hr. At inlet to nozzle, the pressure is 2 MPa and the temperature is 127°C. The exit pressure is of 0.5 MPa. If the initial velocity of air is 300 m/s, the exit velocity will be

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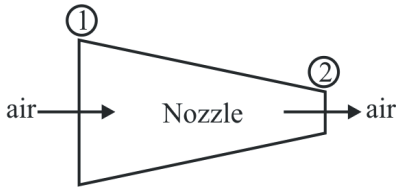
(a) 867 m/s (b) 776 m/s
(c) 685 m/s (d) 594 m/s

ESE 2019

Ans. (d) :



Given,

$\dot{m} = 600$ kg/hr
$= 0.167$ kg/s
$p_1 = 2$ MPa
$p_2 = 0.5$ MPa
$T_1 = 127 + 273 = 400$ k
$C_1 = 300$ m/s

We know that

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{T_2}{400} = \left(\frac{0.5}{2}\right)^{\frac{1.4-1}{1.4}}$$

$T_2 = 269.18$ K

From steady flow energy equation

$$\dot{m}\left[h_1 + \frac{C_1^2}{2000}\right] = \dot{m}\left[h_2 + \frac{C_2^2}{2000}\right]$$

$$h_1 + \frac{C_1^2}{2000} = h_2 + \frac{C_2^2}{2000}$$

$$C_p T_1 + \frac{(300)^2}{2000} = C_p T_2 + \frac{C_2^2}{2000}$$

$$(1.005 \times 400) + 45 = 1.005 \times 269.19 + \frac{C_2^2}{2000}$$

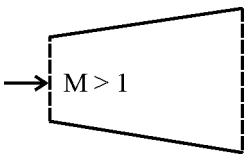
$C_2 = 594.09$ m/s

578. In flow through a nozzle, the Mach number is more than unity

(a) in the converging section
(b) in the diverging section
(c) at the throat
(d) can be in any section of the nozzle depending upon the nozzle profile and geometry

GPSC Executive Engineer 23.12.2018

Ans. (b) :



When mach number (M) > 1

Pressure	Decreases
Velocity	Increases
Mach No.	Increases
Temperature	Decreases
Density	Decreases

579. The velocity of sound in a gaseous medium at a temperature of 400K in 400ms^{-1}. If the temperature of the medium is increased to 600 K, the velocity of sound will be nearly

(a) 400 ms^{-1} (b) 490 ms^{-1}
(c) 600 ms^{-1} (d) None of there

ISRO Scientist/Engineer 22.04.2018

Ans. (b) : We know that velocity of sound in a gaseous medium.

$$C = \sqrt{\gamma RT}$$

$$C \propto \sqrt{T} \qquad \therefore \ \gamma R = Constant$$

$$\frac{C_2}{C_1} = \sqrt{\frac{T_2}{T_1}}$$

$$C_2 = 400 \times \sqrt{\frac{600}{400}}$$

$$C_2 = 400 \times 1.2247$$

$$C_2 = 489.89 \approx 490 \ m/s$$

580. For air (k = 1.4) the critical pressure ratio P_1/P_0 for isentropic flow is

(a) 0.833 (b) 0.728
(c) 0.628 (d) 0.528

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : Critical pressure in nozzle flow

$$\frac{P_1}{P_o} = \left(\frac{2}{k+1}\right)^{k/k-1}$$

Given,

$k = 1.4$

$$\frac{P_1}{P_o} = \left(\frac{2}{1.4+1}\right)^{\frac{1.4}{1.4-1}} = 0.528$$

581. The expansion of steam in a Nozzle follows-

(a) Carnot cycle (b) Joule cycle
(c) Rankine cycle (d) Stirling cycle

GMB AAE 25.06.2017

Ans. (c) : The expansion of steam in a nozzle follows Rankine cycle. The friction in the nozzle increases the dryness fraction of steam. The pressure of steam at throat is called critical pressure.

582. A converging diverging nozzle of a supersonic solid propellant rocket is designed for an exit pressure of 35 KPa. The rocket flying at an altitude having an ambient pressure of 28 KPa would cause

(a) A normal shock in the diverging portion of the nozzle
(b) A normal shock at the exit of the nozzle
(c) A normal shock at the throat of the nozzle
(d) No Shock

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Shock wave will be produced, when the designed pressure is lower than the ambient pressure.

So, $P_{designed} > P_{ambient}$

Here, No, shock will produce.

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583. The expansion in a nozzle is a:

(a) isentropic process (b) adiabatic process
(c) isochoric process (d) isobaric process

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (a) :

584. In an atmosphere the speed of sound is 300 m/s. If a plane travels at 1620 km/h in this atmosphere, the Mach angle is

(a) 30.5º (b) 56.3º
(c) 10.7º (d) 41.8º

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) :

$$\sin\alpha = \frac{1}{m} = \frac{1}{\frac{v}{c}} = \frac{c}{v}$$

$$\alpha = \sin^{-1}\left(\frac{c}{v}\right)$$

$$= \sin^{-1}\left(\frac{300}{1620\times\frac{5}{18}}\right) = \sin^{-1}\left(\frac{300}{450}\right)$$

$$= \sin^{-1}(0.667) = 41.8^0$$

585. In a standard atmosphere the temperature is 15 ºC at sea level and –56ºC at an altitude of 20 km. A supersonic plane has the same speed at the sea level as well as at an altitude of 20 km. If its Mach number is 1.5 at sea level, its Mach number at an altitude of 20 km in standard atmosphere is

(a) 2.80 (b) 1.30
(c) 1.99 (d) 1.73

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : $T_1 = 15°C = 15 + 273 = 288k$

$T_2 = -56°C = -56 + 273 = 217k$

$M_1 = 1.5$

$M_2 = ?$

$$M_1 = \frac{V}{C_1} = \frac{V}{\sqrt{\gamma RT_1}}$$

$$M_2 = \frac{V}{C_2} = \frac{V}{\sqrt{\gamma RT_2}}$$

$$\frac{M_2}{M_1} = \sqrt{\frac{T_1}{T_2}}$$

$$M_2 = 1.5\sqrt{\frac{288}{217}} = 1.73$$

586. The difference of level of tip of main nozzle and fuel level in float chamber of a simple carburetor is called :

(a) Nozzle lip (b) Throttle lip
(c) Throttle dip (d) Nozzle dip

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (a) The difference of level of tip of main nozzle and fuel level in float chamber of a simple carburetor is called Nozzle lip.

587. Fire hose nozzle is generally made of ..... .

(a) divergent shape
(b) convergent shape
(c) convergent-divergent shape
(d) cylindrical shape

GWSSB DEE 07.07.2016

Ans. (b) : Fire hose nozzle is generally made of convergent shape.

588. Supersonic diffusers have ….. geometry.

(a) Only convergent
(b) Only divergent
(c) Divergent and convergent respectively
(d) Uniform

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) :

589. Head loss in nozzles occurs due to .....

(a) Pipes head loss
(b) Pressure loss
(c) Friction loss
(d) None of the mentioned

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Head loss in nozzles occurs due to pipelines.

590. The purpose of 'diffuser' is opposite to that of:

(a) Compressor (b) Nozzle
(c) Valve (d) Condenser

HPPSC AE (PWD) 10.09.2016

Ans. (b) : Nozzle- A steam nozzle converts heat energy of steam into kinetic energy. Where as diffuser convert kinetic energy into heat energy. A nozzle increase the velocity and decrease the pressure of a fluid while a diffuser decreases the velocity and increase the pressure of a fluid.

591. Air is accelerated isentropically from 100 m/s to 300 m/s in a nozzle. If the temperature at the inlet is 127°C, the inlet Mach number is (take R = 287 J/(kg K) and specific heat ratio = 1.4)

(a) 0.249 (b) 0.442
(c) 0.747 (d) 0.333

ISRO Scientist/Engineer 03.07.2016

Ans. (a) : Actual velocity $V_1 = 100$ m/s

Final velocity $V_2 = 300$ m/s

For an ideal gas, in an isentropic process–

$C = \sqrt{\gamma RT}$

R = 287 J/kg-K, T = 127°C = 127 + 273 = 400 K

$\gamma = 1.4$

$C = \sqrt{1.4\times 287\times 400}$

C = 400.898

Mach number $(M) = \left(\frac{V}{C}\right)$

V = Actual velocity

$M = \frac{100}{400.898}$

M = 0.249

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592. Gaseous Nitrogen at 30°C is expanded through a converging nozzle from a total pressure of 0.5 MPa to a back pressure chamber. If the back pressure is increased from 0.1 MPa to 0.4 MPa in steps of 0.02 MPa, the trend in mass flow rate through the nozzle

(a) will gradually decrease
(b) will gradually increase
(c) will remain constant till a point and then gradually decrease
(d) will remain constant till a point and then gradually increase

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Back pressure chamber

$P_b = 0.5$, $\gamma = 1.4$ "for air"

$$\frac{P_c}{P_b} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

$$P_c = P_b\left(\frac{2}{2.4}\right)^{\frac{1.4}{0.4}}$$

$$P_c = 0.5\left(\frac{2}{2.4}\right)^{\frac{1.4}{0.4}}$$

Critical pressure

$P_c = 0.2641$ MPa

When we start increasing pressure from 0.1 MPa to 0.2641 MPa i.e. upto critical pressure mass flow rate is constant, beyond 0.2641 MPa to 0.4 MPa mass flow rate decreases.

593. The value of index n for isentropic expansion of superheated steam through the nozzle is

(a) 1.4 (b) 1.3
(c) 1.135 (d) 1.113
(e) 1.125

CGPSC AE 26.04.2015 Shift-II

Ans. (b) : The value of index n for isentropic expansion of superheated steam and saturated steam through the nozzle are 1.3 and 1.135 respectively.

594. Semi-cone angle of the divergent part of the convergent divergent steam nozzle is of the order of

(a) 3 to 10 (b) 13 to 20
(c) 23 to 30 (d) 33 to 40
(e) 43 to 50

CGPSC AE 26.04.2015 Shift-II

Ans. (a) : Semi-Cone angle of the divergent part of the convergent divergent steam nozzle is of the order of 3 to 10.

595. Nozzle is designed for

(a) maximum pressure at outlet
(b) maximum discharge
(c) minimum discharge
(d) none of the above

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : A nozzle is a duct of smoothly varying cross sectional area, in which a steadily flowing fluid can be accelerated to a high velocity by a drop in pressure of the fluid along the duct.

Nozzles are used in steam and gas turbines, jet engines, rockets, in flow measurement, in injectors for pumping water, in ejectors for removing air from condensers etc.

Types of Nozzles–

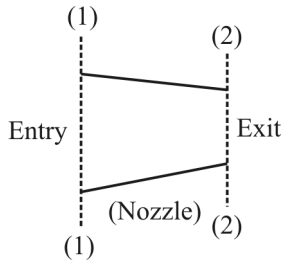
1. Convergent nozzle
2. Convergent-divergent nozzle
3. Divergent nozzle

596. A perfect gas flows through a nozzle where it expands in reversible adiabatic manner. The inlet conditions are 22 bar, 500°C, 38 m/s. At exit the pressure is 2 bar. Given R = 190J/kg K and γ = 1.35. The exit velocity will be

(a) 700 m/s (b) 726 m/s
(c) 801 m/s (d) 701 m/s

MPPSC AE 08.11.2015

Ans. (b) :



$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{T_2}{(500+273)} = \left(\frac{2}{22}\right)^{\frac{1.35-1}{1.35}}$$

$T_2 = 415.134$ K

$$\gamma = \frac{C_p}{C_v} = 1.35$$

$C_p - C_v = R = 0.190$ kJ/kg-K

$$C_p - \frac{C_p}{1.35} = 0.19$$

$C_P = 0.733$

$$C_p(T_2 - T_1) + \frac{1}{2000}(38)^2 = \frac{C^2}{2000}$$

$$0.733(773 - 415.134) + \frac{1}{2000}(38)^2 = \frac{1}{2000}C^2$$

$$262.32 + \frac{1}{2000}(38)^2 = \frac{1}{2000}C^2$$

$C = 725.31$ m/s.

597. At normal atmospheric conditions, the ratio of the speed of sound in water to that in air is about

(a) 1.5 (b) 2.0
(c) 4.0 (d) 7.0

TNPSC AE 2014

Ans. (c) : At normal atmospheric conditions, the ratio of the speed of sound in water to that in air is about 4.0.

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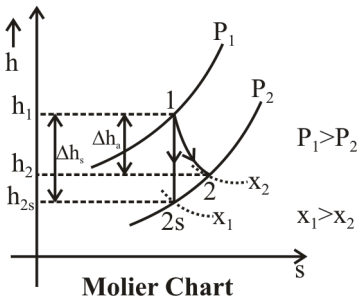
598. In steam nozzle for the same pressure ratio the dryness fraction of exit steam increases due to considering the effect of

(a) friction loss (b) contraction loss
(c) velocity loss (d) pressure loss

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (a) :



Molier Chart

Nozzle Efficiency–In actual working there is always friction while flow takes places through nozzle. Friction is mainly due to surface of nozzle and fluid and also due to internal fluid friction. Nozzle efficiency is defined as a ratio of actual enthalpy drop to the isentropic enthalpy drop possible in a nozzle.

$$\eta_N = \frac{\text{Actual enthalpy drop}}{\text{Isentropic enthalpy drop}}$$

$$\eta_N = \frac{\Delta h_a}{\Delta h_s} = \frac{h_1 - h_2}{h_1 - h_{2s}}$$

The effect of considering friction in steam nozzles for the same pressure ratio leads to increase in dryness fraction of exit steam.

599. The ratio of exit pressure to inlet pressure for maximum mass flow rate per unit area of steam through a nozzle when steam is initially dry saturated is

(a) 0.6 (b) 0.578
(c) 0.555 (d) 0.5457

MPPSC State Forest Service Exam, 2014

Ans. (b) : As we know,
The mass flow rate is maximum where,

$$\frac{d}{dy}\left[y^{2/n} - y^{\frac{n+1}{n}}\right] = 0, \qquad y = \frac{p_2}{p_1}$$

$$y = \left[\frac{2}{n+1}\right]^{\frac{n}{n-1}}$$

For maximum,

$$\frac{p_{cr}}{p_1} = \left[\frac{2}{n+1}\right]^{\frac{n}{n-1}} \Rightarrow n = 1.135$$

$$p_{cr} = 0.578\, p_1$$

600. The isentropic expansion of steam through nozzle for the steam initially dry saturated at inlet is approximated by equation

(a) $pv = C$ (b) $pv^{1.4} = C$
(c) $pv^{1.3} = C$ (d) $pv^{1.135} = C$

MPPSC State Forest Service Exam, 2014

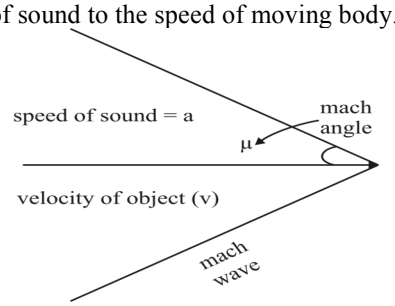
Ans. (d) : $\boxed{PV^n = C}$
$n = 1.3 \rightarrow$ Superheated steam
$n = 1.135 \rightarrow$ Dry saturated steam
$n = 1.035 + 0.1x \rightarrow$ Wet steam

601. Mach angle (α) is defined as the

(a) Quarter angle of the mach cone
(b) Zero angle of the mach cone
(c) Half of the angle of the mach cone
(d) Full angle of the mach cone

TNPSC AE 2014

Ans. (c) : Definition of mach angle– Half of the vertex angle of a mach cone whose sine is the ratio of the speed of sound to the speed of moving body.



$$\sin\mu = \frac{a}{v}$$

$$\sin\mu = \frac{1}{M}$$

602. Steam flow through a nozzle is considered as :

(a) constant volume flow
(b) constant pressure flow
(c) adiabatic flow
(d) isothermal flow

J&K PSC Civil Services Pre, 2013

Ans. (c) : Adiabatic flow is considered for flow through nozzle where heat transfer is taken negligible.

603. The ratio of exit pressure to inlet pressure for maximum mass flow rate per unit area of steam through nozzle when steam is initially superheated is

(a) 0.555 (b) 0.578
(c) 0.5457 (d) 0.6

APPSC Poly. Lect. 2013

Ans. (c) : The critical pressure ratio is defined as the ratio of pressure at the throat to inlet pressure for choked flow (max. mass flow condition) i.e.

$$\text{critical pressure ratio } (P_{cr}) = \left(\frac{2}{n+1}\right)^{\frac{n}{n-1}}$$

for superheated steam, n=1.3$\Rightarrow$ (P_{cr}) = 0.545
for wet steam, n= 1.127$\Rightarrow$ (P_{cr}) = 0.58
for dry saturated steam, n=1.135$\Rightarrow$ (P_{cr}) = 0.577

604. The factors on which nozzle efficiency depends are

I. Reynolds number of flow.
II. Nature of fluid flow and its state.
III. Turbulence in fluid and its state.
IV. Workmanship of the manufacturing of nozzle.

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Of these statements, which are correct?

(a) I, II, III and IV (b) I, II and III
(c) II, III and IV (d) none of the above

APPSC Poly. Lect. 2013

Ans. (a) : The nozzle efficiency depends upon the following factors-

i) Material of the nozzle
ii) Size & Shape of the nozzle
iii) Finish of the nozzle
iv) Angle of divergence
v) Nature of the fluid and its state
vi) Friction
vii) Fluid velocity
viii) Turbulence in the flow passages
ix) Reynold number of the flow

605. At some section of convergent-divergent nozzle, the temperature is 200ºC. Then sonic velocity is

(a) 871.8 m/s (b) 435.9 m/s
(c) 217.95 m/s (d) 1600 m/s

APPSC Poly. Lect. 2013

Ans. (b) : Given, $R = 0.287 \times 10^3$ J/kg-K

$T = 200°C$

$= 200 + 273 = 473$

$C = \sqrt{\gamma RT} = \sqrt{1.4 \times 0.287 \times 10^3 \times 473} = 435.9\,m/s$

606. The device which converts kinetic energy of the fluid into pressure energy is :

(a) Nozzle (b) Diffuser
(c) Compressor (d) Turbine

TNPSC AE (Industries) 09.06.2013

Ans. (b) : The device which converts kinetic energy of the fluid into pressure energy is Diffuser.

607. The region outside the Mach cone is called.

(a) zone of action (b) zone of silence
(c) control volume (d) None of the above

JPSC AE 2013 Paper-V

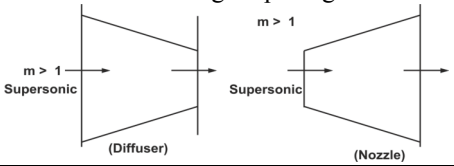
Ans. (b) : The region outside the Mach cone is called zone of silence.

608. When a supersonic flow is admitted into a convergent, the flow will

(a) Accelerate (b) Decelerate
(c) No change (d) None of the above

ISRO Scientist/Engineer 2012

Ans. (b) : For supersonic flow the convergent passage acts as diffuser. And a divergent passage as a nozzle.



609. The sound of explosion on a ship located 10km away from the beach was heard 28.6 seconds after the explosion was seen. Estimate the temperature of air at that time.

(a) 31ºC (b) 31K
(c) 26.2 ºC (d) 288.5K

ISRO Scientist/Engineer 2012

Ans. (a) : Speed of sound at the location

$C = \sqrt{\gamma RT}$

$\gamma = 1.4$ (air)

$R = 287$ J/kgK

$T = ?$

located on ship i.e. Distance

$d = 10$ km

$d = 10 \times 10^3$m

time $t = 28.6$ sec.

So, velocity i.e. $C = \frac{10 \times 10^3}{28.6} = \frac{Distance}{time}$

$C = \frac{10^4}{28.6}$

$\therefore \quad C = \sqrt{\gamma RT}$

$\therefore \quad \sqrt{1.4 \times 287 \times T} = \frac{10^4}{28.6}$

$\therefore \quad T = 304.27k$

$T = 31.1ºC$

610. The flow of steam in nozzle is subsonic at

(a) throat
(b) entrance
(c) convergent portion
(d) divergent portion

BPSC AE 2012 Paper - V

Ans. : (c) : The flow of steam in a nozzle is subsonic at convergent portion.
The flow of steam in a nozzle is supersonic at divergent portion.

611. For a convergent - divergent nozzle, critical pressure ratio occurs when

(a) velocity at exit becomes sonic
(b) shock at exit becomes
(c) nozzle efficiency is maximum
(d) increase in exit and inlet pressure ratio does not increases steam flow rate

BPSC AE 2012 Paper - V

Ans. : (d) : For a convergent divergent nozzle, critical pressure ratio occurs when increase in exit and inlet pressure ratio does not increase steam flow rate.

612. Ideal isentropic velocity of air jet from a nozzle for a drop in enthalpy of 400 kJ/kg in meter per second is

(a) 894.4 (b) 632.5
(c) 2801.7 (d) 400.8

TNPSC ACF 2012

Ans. (a) : $h_1 - h_2 = 400$ kj/kg

$\frac{v^2}{2} = h_1 - h_2$

$v = \sqrt{2(h_1 - h_2)}$

$v = \sqrt{2(400) \times 10^3}$

$v = 894.4$ m/s

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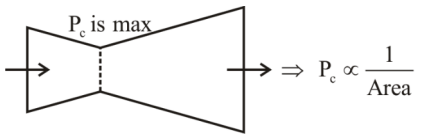
613. In the frictionless flow of the steam, the critical pressure ratio will occur

(a) At exit of the nozzle
(b) Where area is minimum
(c) Where discharge is minimum
(d) Where condition of steam is dry saturated

UPSC JWM Advt. No.-50/2010

Ans. (b) : In the frictionless flow of the steam, the critical pressure ratio will occur where area is minimum in a nozzle.



614. Assertion (A) : While steam flows through a nozzle, its velocity increases and specific volume decreases.

Reason (R) : In a nozzle when steam expands, there is drop in its pressure and enthalpy.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (d) : As we know, while streams flow through a nozzle its velocity increases and due to change in mass flow rate its specific volume increases. A nozzle is an element whose primary function is to convert enthalpy into KE. When steam flows through the nozzle from high pressure to

$$h_1 = h_2 + \frac{v_2^2}{2g}$$

For nozzle one at low pressure, its velocity specific volume both increases.

615. In a nozzle, if back pressure is equal to inlet pressure.

(a) no flow occurs
(b) maximum flow occurs
(c) flow is subsonic in diverging section
(d) flow is supersonic in converging as well as diverging sections

TNPSC AE, 2008

Ans. (a) : In a nozzle if back pressure is equal to inlet pressure ($P_b = P_1$). There will be no flow and pressure distribution will be uniform along with the nozzle.

5. Gas Turbine

616. In a two stage gas turbine plant with inter-cooling and reheating

(a) both work ratio and thermal efficiency improve
(b) work ratio improves but thermal efficiency decreases
(c) thermal efficiency improves but work ratio decreases
(d) both work ratio and thermal efficiency decrease

TNPSC AE 2019, GPSC ARTO 30.12.2018
GPSC Executive Engineer 23.12.2018
RPSC AE 2016, TSGENCO AE 14.11.2015
ESE 2014, JPSC AE 2013 Paper-V
RPSC ACF-2011, RPSC AE GWD 2011
CSE Pre-2006

Ans. (b) : In a two stage gas turbine plant with intercooling and reheating, work ratio improves but thermal efficiency decreases.

• Effect of intercooling –
$W_C\downarrow$, W_T-same, $W_{net}\uparrow$, $Q_S\uparrow$, $Q_R\uparrow$, $\eta\downarrow$
• Effect of reheating –
W_C-same, $W_T\uparrow$, $W_{net}\uparrow$, $Q_S\uparrow$, $Q_R\uparrow$, $\eta\downarrow$

$\because$ $\text{Work ratio } (r_w) = \dfrac{\text{Net work}}{\text{Positive work(Turbine)}} = \dfrac{W_{net}}{W_T}$

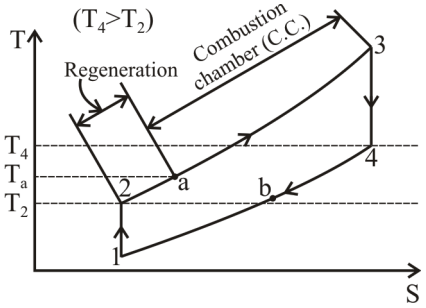
$\eta\downarrow$

617. In a gas turbine plant, a regenerator increases

(a) work output (b) pressure ratio
(c) thermal efficiency (d) all of the above

CGPSC AE 15.01.2021, APGENCO AE, 2017
CGPSC ARTO & TSI 07.01.2017
CGPSC AE 16.10.2016, RPSC AE 2016
TSPSC Managers, 2015, KPSC ADF 2015
TNPSC AE (Industries) 09.06.2013
RPSC ACF-2011, ESE 2008

Ans. (c) :



In a gas turbine power plant, a regenerator increases ($\uparrow$)
W_c = constant
W_T = constant, $W_{net} = W_T - W_C$ = constant
$Q_S\downarrow$ $\eta\uparrow$

618. Reheating in gas turbine results in

(a) Increase in work ratio but decrease in thermal efficiency
(b) Increase of thermal efficiency
(c) Decrees of work
(d) Decrease of thermal efficiency

Nagaland PSC CTSE 2017 Paper-2
Rajasthan Nagar Nigam AE 2016, Shift-I
APPSC AEE 2016
OPSC AEE 2015 PAPER - II
JPSC AE 2013 Paper-V, SJVN ET 2013
BPSC AE 2012 Paper - V

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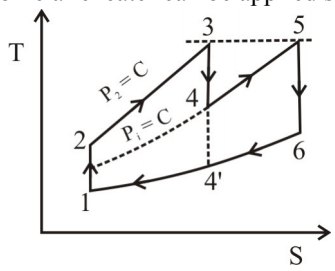
Ans. (a) : Effect of Reheating in a gas turbine– By staging the heat supply process with a combustion and a reheater, the cycle efficiency decreases, but it permits more heat recovery from the turbine exhaust gases.

Reheating is applied in a gas turbine in such a way that it increases the turbine work without increasing the compressor work or melting the turbine material. When a gas turbine plant has a high pressure and low pressure turbine a reheater can be applied successfully.



Effect of Reheating-

W_c - same	Q_s - ↑	T_{mR} - ↑↑
W_T - ↑	Q_R - ↑	P_{mean} - ↓
W_{net} - ↑	T_{mA} - ↑	η- ↓s

619. Thermal efficiency of a gas turbine plant as compared to diesel engine plant is :

(a) higher (b) lower

(c) same (d) un-predictable

GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC AE 2016
APPSC Poly Lect. 2013
RPSC ACF-2011, WBPSC AE 2008
UKPSC AE 2007 Paper-II

Ans. (b) : • The thermal efficiency at 20%, 26% is however, generally lower than comparable steam turbine plant and much lower than equivalent diesel plant.

• Gas turbine plant is significantly lower in capital cost than similarly rated steam turbine and above 5MW has considerable cost advantages over diesel engine.

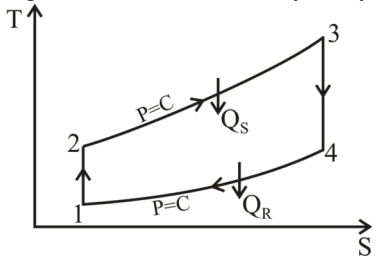
620. A gas turbine works on which one of the following cycles :

(a) Brayton (b) Rankine

(c) Stirling (d) Otto

JPSC AE, 10.04.2021, Paper-I
VIZAG STEEL MT 24.01.2021, Shift-II
Haryana PSC AE (PHED) 04.09.2020, Paper-I
Rajasthan Nagar Nigam AE 2016 Shift II
UJVNL AE 2016
ESE 18.07.2021, 2020, 1998

Ans. (a) : A gas turbine works on Brayton cycle.



1-2 → Isentropic compression
2-3 → Constant pressure heat addition
3-4 → Isentropic expansion
4-1 → Constant pressure heat rejection

621. Intercooling in gas turbine system:

(a) decreases net output but increases thermal efficiency

(b) increases net output but decreases thermal efficiency

(c) decreases net output and thermal efficiency both

(d) increases net output and thermal efficiency both

OPSC AEE 2019 PAPER - II
Punjab PSC SDE 12.02.2017
Kerala PSC AE (Irrigation) 05.04.2017
BPSC AE 2012 Paper - V, ESE 1993

Ans. (b) : Effects of intercooling in Gas Turbine-

- The compressor work (W_C) decreases.
- The Turbine work (W_T) remains same.
- Heat supplied and heat rejected both increases.
- The temperature of heat addition and heat rejection both decreases.
- The scope of regeneration increases as the outlet temperature of compressor decreases from 2 to 4°C.

622. Efficiency of Brayton cycle is given by [where 'T' is the pressure ratio, γ = specific heat ratio]

(a) $\eta = 1 - \frac{1}{r^{\gamma-1}}$ (b) $\eta = 1 - \frac{1}{r}$

(c) $\eta = 1 - \frac{1}{r^{\gamma}}$ (d) $\eta = 1 - \frac{1}{r^{(\gamma-1)/\gamma}}$

CGPSC AE 15.01.2021
ISRO Scientist/Engineer (RAC) 22.04.2018
RPSC LECTURER 16.01.2016
UKPSC AE 2007 Paper -II, GATE 2004

Ans. (d) :

$$\eta_{Brayton} = \frac{q_{in} - q_{out}}{q_{in}}$$

$$= 1 - \frac{q_{out}}{q_{in}}$$

$$= 1 - \frac{C_p(T_3 - T_4)}{C_p(T_2 - T_1)}$$

$$= 1 - \frac{T_3 - T_4}{T_2 - T_1}$$

For Otto and Brayton cycle

$$T_1 \times T_3 = T_2 \times T_4$$

$$= 1 - \frac{T_4\left(\frac{T_3}{T_4} - 1\right)}{T_1\left(\frac{T_2}{T_1} - 1\right)}$$

$$\frac{T_3}{T_4} = \frac{T_2}{T_1}$$

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then

$$\eta_{Brayton} = 1 - \frac{1}{\left(\frac{T_1}{T_4}\right)}$$

We know that

$$\left(\frac{T_1}{T_4}\right) = \left(\frac{P_1}{P_4}\right)^{\left(\frac{\gamma-1}{\gamma}\right)}$$

$$\eta_{Brayton} = 1 - \frac{1}{r^{(\gamma-1)/\gamma}}$$

623. In a gas turbine power plant, reheating of gases between high pressure and low pressure turbine stages will

(a) Improve turbine output
(b) Decrease turbine output
(c) Increase compressor work
(d) Decrease compressor work

JPSC AE, 10.04.2021, Paper-I
Haryana PSC AE (PHED) 04.09.2020, Paper-I
ISRO Scientist/Engineer 07.05.2017
TSPSC AEE 2015

Ans. (a) : For gas Turbine–

	Work output	ηth
Regeneration	No effect	↑
Intercooling	↑	↓
Reheating	↑	↓

Following are the effect of reheating in a gas turbine power plant.

- No change in compressor work
- Turbine work output increases
- Network output increases
- Both mean temperature of heat addition and heat rejection increases
- Both heat supplied and heat rejected increases
- Efficiency decreases
- Scope of regeneration increases

624. In a two stage gas turbine plant, reheating after first stage

(a) decrease thermal efficiency
(b) increases thermal efficiency
(c) does not affect thermal efficiency
(d) none of the above

RPSC AE 2016, BPSC Poly. Lect. 2016
TNPSC ACF 2012, RPSC ACF-2011

Ans. (a) : In a two stage gas turbine plant, reheating after first stage decrease thermal efficiency.

• Effect of reheating –

W_C-same, $W_T\uparrow$, $W_{net}\uparrow$, $Q_S\uparrow$, $Q_R\uparrow$, $\eta\downarrow$

625. A gas turbine working on Brayton cycle produces 4000 kW of net power. If its work ratio is 40%, what is the power consumed by the compressor?

(a) 2000 kW (b) 4000 kW
(c) 6000 kW (d) 8000 kW

RPSC Vice Principal ITI 2018
GPSC Executive Engineer 23.12.2018
HPPSC Asstt. Prof. 18.09.2017
TSPSC Managers, 2015

Ans. (c) : Given, Net Power (W_{net}) = 4000 kW

$$r_w = 40\% = \frac{40}{100} = 0.4$$

$$\text{Work ratio } (r_w) = \frac{W_{net}}{W_T}$$

$$\frac{4000}{W_T} = 0.4$$

$$W_T = \frac{4000}{0.4} = 10000\text{kW}$$

We know that, $W_{net} = W_T - W_C$

$$W_C = W_T - W_{net}$$

$$= 10000 - 4000$$

$$W_C = 6000\text{kW}$$

626. In a gas turbine cycle, the turbine output is 600 kJ/kg, the compressor work is 400 kJ/kg and the heat supplied is 1000 kJ/kg. The thermal efficiency of the cycle is :

(a) 80% (b) 60%
(c) 40% (d) 20%

BPSC Mains 2017 Paper-V
KPSC AE 2015
J&K PSC Civil Services Pre, 2010
ESE 2016, 1997

Ans. (d) : Given,

Turbine work (W_T) = 600 kJ/kg

Compressor work (W_C) = 400 kJ/kg

Heat supplied (Q_S) = 1000 kJ/kg

$$\text{Thermal efficiency } (\eta_{thermal}) = \frac{W_{net}}{\text{Heat supplied}\,(Q_s)}$$

$$= \frac{W_T - W_C}{Q_S}$$

$$= \frac{600 - 400}{1000} = \frac{200}{1000}$$

$$= \frac{2}{10} \times 100$$

$$\boxed{\eta_{thermal} = 20\%}$$

627. A gas turbine cycle with infinitely large number of stage during compression and expansion leads to :

(a) Stirling cycle (b) Atkinson cycle
(c) Ericsson cycle (d) Brayton cycle

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
CSE Pre-2008, 2003, GATE 1994

Ans. (c) : Ericsson Cycle– The Ericsson cycle consists of two isothermal and two constant pressure processes. The processes are:

Process 1-2: Reversible isothermal compression.

Process 2-3: Constant pressure heat addition.

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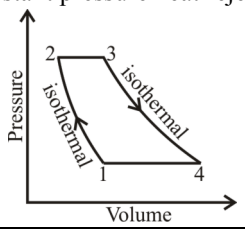
Process 3-4: Reversible isothermal expansion.
Process 4-1: Constant pressure heat rejection.



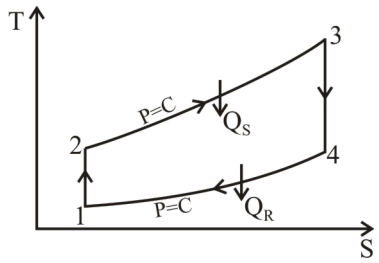
628. The work ratio in a gas turbine plant is equal to
where
r_p = pressure ratio
T_1 = Compressor inlet temperature
T_3 = Turbine inlet temperature

(a) $1-r_p^{\frac{\gamma-1}{\gamma}}$ (b) $1-\frac{T_1}{T_3}r_p^{\frac{\gamma-1}{\gamma}}$

(c) $1+\frac{T_1}{T_2}r_p^{\frac{\gamma-1}{\gamma}}$ (d) $1+r_p^{\frac{\gamma-1}{\gamma}}$

UPPSC AE 13.12.2020, Paper-2
BPSC AE Mains 2017 Paper-V
ESE 2009

Ans. (b)



For gas turbine,

$$\text{Work Ratio } (r_w) = \frac{\text{Net work}}{\text{Turbine work}} = \frac{W_{net}}{W_T}$$

$$r_w = \frac{W_T - W_C}{W_T} = 1 - \frac{W_C}{W_T}$$

$$r_w = 1 - \frac{mC_p(T_2 - T_1)}{mC_p(T_3 - T_4)}$$

$$r_w \Rightarrow 1 - \frac{(T_2 - T_1)}{(T_3 - T_4)} = 1 - \frac{T_1}{T_4}\left(\frac{\frac{T_2}{T_1} - 1}{\frac{T_3}{T_4} - 1}\right)$$

As, $\frac{T_2}{T_1} = \frac{T_3}{T_4}$

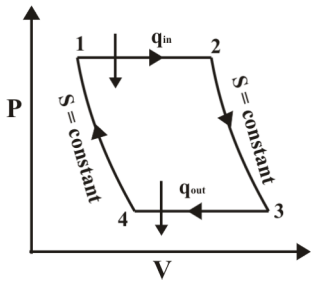
$$r_w \Rightarrow 1 - \frac{T_1}{T_4} = 1 - \frac{T_1}{T_3}\left(r_p\right)^{\frac{\gamma-1}{\gamma}}$$

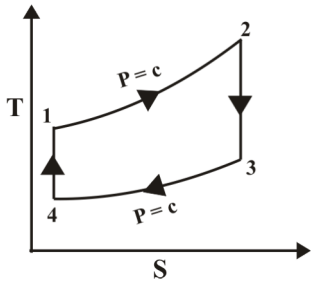
Work ratio, $\boxed{r_w = 1 - \frac{T_1}{T_3}\left(r_p\right)^{\frac{\gamma-1}{\gamma}}}$

629. Brayton cycle consists of
(a) Two reversible isobars and two reversible adiabatic process
(b) Two reversible isotherm and two reversible adiabatic process
(c) Two reversible isotherm and two reversible isobars process
(d) Two reversible adiabatics and one reversible isotherm, one reversible isobaric process

Oil India Senior Officer 23.12.2020
RPSC Lecturer 16.01.2016
TRB Asstt. Prof., 2012

Ans. (a) : Brayton Cycle–Brayton cycle consists of two reversible isobars and two reversible adiabatic process.





630. For air standard Brayton cycle, increase in the maximum temperature of the cycle, while keeping the pressure ratio the same would result in
(a) Increase in air standard efficiency
(b) Decrease in air standard efficiency
(c) No change in air standard efficiency
(d) Increase in the efficiency but reduction in net work
(e) Increase in the net work but reduction in efficiency

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 18.09.2017
CGPSC AE 26.04.2015 Shift-II
CSE Pre-1998

Ans. (c) : Efficiency of Brayton cycle,

$$\eta_{Brayton} = 1 - \frac{1}{\left(r_p\right)^{\frac{\gamma-1}{\gamma}}}$$

For air standard Brayton cycle, air standard efficiency is the function of pressure ratio.

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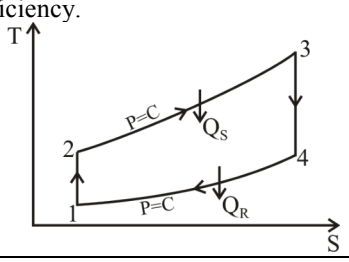
So increase in the maximum temperature of the cycle, while keeping the same pressure ratio no change in air standard efficiency.

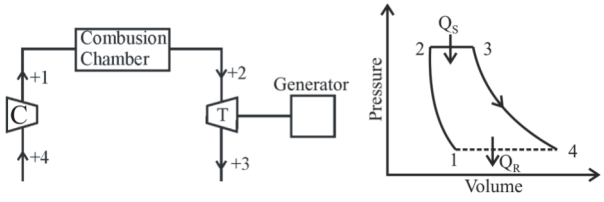


631. An open cycle gas turbine works on

(a) Carnot cycle (b) Otto cycle
(c) Joule's cycle (d) Stirling cycle

COPSC AEE 2019 Paper-II
Nagaland PSC (CTSE) 2018, Paper-II
VIZAG Steel (MT) 2017

Ans. (c) : Open cycle gas turbine works on joule's or Brayton cycle.



Joule cycle has two isentropic and two isobaric process.

632. Open cycle Gas turbine works on :

(a) Carnot cycle (b) Rankine cycle
(c) Stirling cycle (d) Brayton cycle

CGPSC AE 15.01.2021
Kerala PSC AE (Irrigation) 05.04.2017
GWSSB AAE, 27.12.2015

Ans. (d) : Open cycle Gas turbine works on brayton cycle.

633. High air-fuel ratio in gas turbine is used to :

(a) Reduce exhaust temperature after combustion
(b) Improve thermal efficiency
(c) Reduce power input of compressor
(d) Improve net power output of the cycle.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper-II
Nagaland PSC (CTSE) 2018, 2016, Paper-II

Ans. (a) : High air –fuel ratio in gas turbine reduces exhaust temperature after combustion.

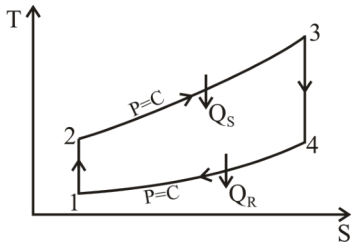
634. The work ratio of closed cycle gas turbine plant depends upon

(a) pressure ratio of the cycle and specific heat ratio
(b) temperature ratio of the cycle and specific heat ratio
(c) pressure ratio temperature ratio and specific heat ratio
(d) only on pressure ratio

TSPSC Manager (Engg.) HMWSSB 12.11.2020
RRB SSE 03.09.2015 Shift-II
TANGEDCO AE 2015

Ans. (c) : 'Work ratio' defined as the ratio of net work to the total turbine work.

- Higher the value of work ratio lesser the effect on the variation of compressor, efficiency and the performance.



Work ratio $(r_w) = 1 - \frac{W_C}{W_T}$

$$r_w = 1 - \frac{T_2 - T_1}{T_3 - T_4}$$

$$r_w = 1 - \frac{T_1}{T_4}\left(\frac{\frac{T_2}{T_1} - 1}{\frac{T_3}{T_4} - 1}\right)$$

$$\frac{T_2}{T_1} = \frac{T_3}{T_4} = (r_p)^{\frac{\gamma-1}{\gamma}}$$

Work ratio $(r_w) = 1 - \frac{T_1}{T_4}$

$$\boxed{r_w = 1 - \frac{T_1}{T_3}(r_p)^{\frac{\gamma-1}{\gamma}}}$$

The work ratio of closed cycle gas turbine plant depends upon pressure ratio, temperature ratio and specific heat ratio.

635. The overall pressure ratio through a three-stage gas turbine is 10 and the efficiency is 0.86. The inlet temperature is 1400 K. Assuming the temperature rise in each stage to be the same, the pressure ratio and the stage efficiency at first stage will be respectively.

(a) 2.487 and 0.8341 (b) 1.895 and 0.8285
(c) 1.895 and 0.8341 (d) 2.12 and 0.8308

CGPSC AE 15.01.2021, CGPSC AE 16.10.2016

Ans. (b) : Given, $\eta = 0.86$, $T_1 = 1400$ K

$$\frac{T_{4'}}{T_1} = \left(\frac{P_{4'}}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{T_{4'}}{T_1} = \left(\frac{1}{10}\right)^{0.286} \Rightarrow T_{4'} = 724.64\,\text{K}$$

$$\eta_T = \frac{T_1 - T_4}{T_1 - T_{4'}} \Rightarrow 0.86 = \frac{1400 - T_4}{1400 - 724.64}$$

$T_4 = 819.19$ K

As temperature rise same in each stage,

$3(T_1 - T_2) = 1400 - 819.19$

$(T_1 - T_2) = 193.60$ K

$T_2 = 1206.39$ K

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Pressure ratio $\boxed{\frac{P_1}{P_2}=\left(\frac{1400}{1206.39}\right)=1.896}$

Stage efficiency for each stage,

$$\eta_{poly}=\frac{\gamma}{\gamma-1}\times\frac{\ln\left(\frac{1400}{819.19}\right)}{\ln 10}$$

$$\eta_{poly}=0.813$$

$$\frac{P_1}{P_2}=\left(\frac{T_1}{T_2}\right)^{\frac{1}{0.286\times0.813}}$$

Stage efficiency,

$$\eta_s=\frac{1-\left(\frac{P_2}{P_1}\right)^{\frac{0.4\times0.813}{1.4}}}{1-\left(\frac{P_2}{P_1}\right)^{\frac{0.4}{1.4}}}=\frac{1-\left(\frac{1}{1.896}\right)^{\frac{0.4\times0.813}{1.4}}}{1-\left(\frac{1}{1.896}\right)^{\frac{0.4}{1.4}}}$$

$$\boxed{\eta_s=0.828}$$

636. An ideal closed cycle gas turbine plant working between temperatures 927^0C and 27^0C using air as working fluid. The pressure ratio for maximum work output is

(a) 11.3 (b) 13.3
(c) 15.3 (d) 17.3

UPPSC AE 13.12.2020, Paper-2
ESE 2018

Ans. (a) : Given,

$T_{max} = 927^0C = 1200$ K

$T_{min} = 27^0C = 300$ K

Optimum pressure ratio, for maximum work output is

$$(r_p)_{opt}=\left(\frac{T_{max}}{T_{min}}\right)^{\frac{\gamma}{2(\gamma-1)}}$$

$$(r_p)_{opt}=\left(\frac{1200}{300}\right)^{\frac{1.4}{2(1.4-1)}}=(4)^{1.75}=11.3$$

637. Gas turbines for power generation are normally used :

(a) to supply peak load requirements
(b) to enable start thermal power plant
(c) when other sources of power fail
(d) to supply base load requirements

GWSSB DEE 07.07.2016
GWSSB AAE, 27.12.2015

Ans. (a) : To fulfill peak or excess load requirement gas turbine power generation are used.

638. The maximum temperature in a gas turbine is

(a) 200ºC (b) 500ºC
(c) 700ºC (d) 1000ºC

Sikkim PSC (Under Secretary), 2017
Nagaland PSC CTSE 2017, 2016 Paper -II

Ans. (d) : The maximum temperature of gas turbine is 1000ºC.

639. The Maximum net specific work obtained in an ideal Brayton cycle for max temperature 900 K and minimum temperature 400 K is given by :

(a) $100C_p$ (b) $500C_p$
(c) $800C_p$ (d) $700C_p$

GPSC ARTO Pre 30.12.2018
ESE 2015

Ans. (a) : Given, $T_{max} = 900$ K

$T_{min} = 400$ K

Maximum specific work $= C_p[\sqrt{T_{max}}-\sqrt{T_{min}}]^2$

$= C_p[\sqrt{900}-\sqrt{400}]^2$

$= C_p[10]^2$

$= 100\ C_p$

640. Increasing the number of reheating stages in a gas turbine to infinity, makes the expansion tending:

(a) Reversible adiabatic (b) Isothermal
(c) Isobaric (d) Isochoric

GPSC ARTO Pre 30.12.2018
TSPSC AE 2015

Ans. (b) : • Increasing the number of reheating stages in a gas turbine to infinity makes the expansion tending to isothermal.

• Increasing the intercooling in compressor to infinity also makes the compression tending to isothermal.

641. Work ratio is the ratio of

(a) Net work/turbine work
(b) Turbine work/compressor work
(c) Net work/compressor
(d) Compressor work/net work

Gujarat PSC AE 2019, TNPSC AE 2018

Ans. (a) : Work ratio is the ratio of network to turbine work.

$$\text{Work ratio, } r_w=\frac{W_{turbine}-W_{compressor}}{W_{turbine}}$$

$$W_r=1-\frac{W_{compressor}}{W_{turbine}}$$

$W_r = 1 - W_{br}$

where $W_{br} \rightarrow$ back work ratio

642. For the same compression ratio, the efficiency of the brayton cycle is–

(a) Equal to diesel cycle
(b) Equal to Otto cycle
(c) Equal to dual cycle
(d) Greater than diesel cycle

RPSC IOF & Boiler 20.09.2017
ESE 2020

Ans. : (b) For the same compression ratio the efficiency of Brayton cycle is equal to that of Otto cycle.

$$\text{Otto cycle efficiency } (\eta_{otto})=1-\frac{1}{r_C^{\gamma-1}}$$

$$\text{Brayton cycle efficiency } (\eta_{Brayton})=1-\frac{1}{r_p^{\frac{\gamma-1}{\gamma}}}$$

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r_C = Compression ratio
r_p = Pressure ratio
γ = Adiabatic index

643. Thermal efficiency of a gas turbine cycle with ideal regenerative heat exchanger is
(a) Equal to work ratio
(b) Less than work ratio
(c) More than work ratio
(d) Unpredictable

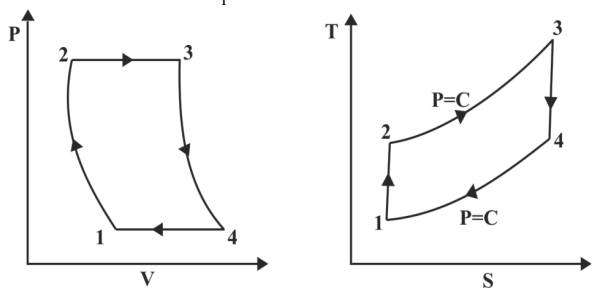
UPPSC AE 12.04.2016 Paper-II
GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Work ratio is defined as the ratio of the network ($W_T - W_C$) to the turbine work developed (W). It is denoted by WR.

$$\text{Work ratio, } r_w = \frac{\text{Net work}}{\text{Turbine work}}$$

$$= \frac{W_T - W_C}{W_T}$$

$$= 1 - \frac{W_C}{W_T}$$



Gas turbine work on Brayton cycle

$$\text{W.R.} = 1 - \frac{C_P(T_2 - T_1)}{C_P(T_3 - T_4)} = 1 - \frac{T_1\left[\frac{T_2}{T_1} - 1\right]}{T_3\left[1 - \frac{T_4}{T_3}\right]}$$

For Otto and Brayton cycle

$$T_1 \times T_3 = T_2 \times T_4$$

then

$$r_P = \frac{P_2}{P_1} = \frac{P_3}{P_4}$$

$$W.R. = 1 - \frac{T_1}{T_3} \times \frac{T_2}{T_1}$$

$$W.R. = 1 - \frac{T_1}{T_3} \times r_P^{\left(\frac{\gamma-1}{\gamma}\right)}$$

So work ratio of gas turbine depends on both pressure ratio and temperature.
Thermal efficiency of a gas turbine cycle with ideal regenerative heat exchanger is equal to work ratio.
Efficiency of ideal regenerative cycle –

$$\boxed{\eta_{\text{ideal Reg.}} = 1 - \frac{T_{\min}}{T_{\max}}\left(r_p\right)^{\frac{\gamma-1}{\gamma}}}$$

$$\eta_{\text{ideal Reg.}} \to \max \quad \text{for} \quad \boxed{r_p = 1}$$

644. The efficiency of a simple gas turbine can be improved by using a regenerator, because the:
(a) Work of compression is reduced
(b) Heat required to be supplied is reduced
(c) Work out put of the turbine is increase
(d) Heat rejected is increased

GPSC Executive Engineer 23.12.2018
MPPSC AE 2016

Ans. : (b) The efficiency of simple gas turbine can be improved by using a regenerator, because the heat required to be supplied is reduced.

645. A gas turbine cycle with heat exchanger and reheating improves
(a) only the thermal efficiency
(b) only the specific power output
(c) both thermal efficiency and specific power output
(d) neither thermal efficiency nor specific power output

GPSC Executive Engineer 23.12.2018
GATE 1993

Ans. (c) : A gas turbine cycle with heat exchanger and reheating improves both thermal efficiency and specific power output.

646. Which of the following cycle is used in aircraft?
(a) Brayton cycle (b) Rankine cycle
(c) Atkinsion cycle (d) Stirling cycle
(e) Carnot cycle

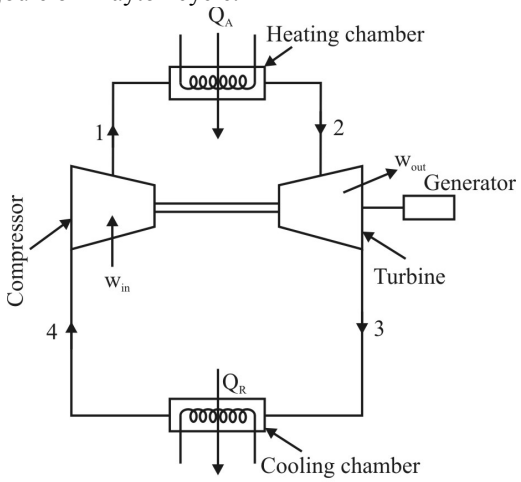
CGPSC ARTO & TSI 07.01.2017
APGENCO AE 2012

Ans. (a) : Brayton cycle -Air craft
Rankine cycle - Power plant

647. A closed cycle gas turbine works on
(a) Carnot cycle (b) Rankine cycle
(c) Ericsson cycle (d) Joule cycle

TSPSC AEE 2015
UKPSC AE 2007 Paper -II

Ans. : (d) Joule cycle:- A close cycle gas turbine works on joule or Brayton cycle.



Close cycle gas turbine

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648. Pressure ratio in gas turbines is the ratio of

(a) Compressor pressure ratio
(b) Highest pressure to exhaust pressure
(c) Inlet pressure to exhaust pressure
(d) Pressures across the turbine

Nagaland PSC (CTSE) 2017, 2016, Paper-II

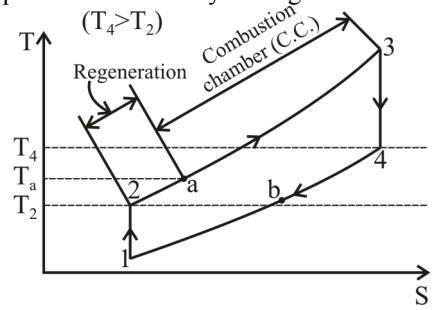
Ans. (a) : Pressure ratio in gas turbines is the ratio of compressor pressure ratio. The pressure ratio of gas turbines ranges from 11 to 16.

649. Thermal efficiency of closed cycle gas turbine plant is increased by the following:-

(a) Reheating (b) Intercooling
(c) Regenerator (d) None of the above

UKPSC AE-2013, 2007 Paper-II

Ans. (c) : Thermal efficiency of closed cycle gas turbine plant is increased by the regenerator.



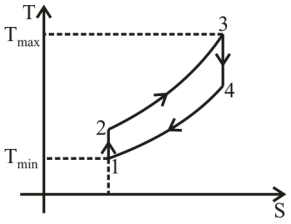
• Effect of regeneration –
W_T-same, W_{net}-same, $Q_S\downarrow$, $Q_R\downarrow$, Specific fuel consumption (sfc)$\downarrow$, $\eta\uparrow$

650. Consider a simple gas turbine (Brayton cycle) and a gas turbine with perfect regeneration. In both the cycles, the pressure ratio is 8 and the ratio of specific heats of working medium is 1.5. The ratio of minimum to maximum temperature is 0.2 (with temperature in K) in the regenerative cycle. The ratio of the thermal efficiency of the simple cycle to that of regenerative cycle is

(a) 0.63 (b) 0.73
(c) 0.83 (d) 0.93

ISRO Scientist/Engineer (RAC) 12.01.2020
GATE 2016

Ans. (c) :



$r_p = 8, \gamma = 1.5$

$$\frac{T_{min}}{T_{max}} = 0.2$$

$$\eta_{cycle} = 1 - \frac{1}{(r_p)^{\frac{\gamma-1}{\gamma}}} = 1 - \frac{1}{(8)^{1/3}} = 0.5$$

$$\eta_{ideal\ regeneration} = 1 - \frac{T_{min}}{T_{max}}(r_p)^{\frac{\gamma-1}{\gamma}} = 0.6$$

$$\frac{\eta_{cycle}}{\eta_{ideal\ regeneration}} = \frac{0.5}{0.6} = 0.8333$$

651. The compression ratio of Brayton cycle corresponding to maximum work output for the given temperature limits of T_{min} and T_{max} will be:

(a) $\left(\frac{T_{max}}{T_{min}}\right)^{\frac{\gamma}{2(\gamma-1)}}$ (b) $\left(\frac{T_{min}}{T_{max}}\right)^{\frac{\gamma}{2(\gamma-1)}}$

(c) $\left(\frac{T_{max}}{T_{min}}\right)^{\frac{\gamma}{(\gamma-1)}}$ (d) $\left(\frac{T_{min}}{T_{max}}\right)^{\frac{\gamma}{(\gamma-1)}}$

GPSC Asstt. Prof. 28.08.2016
GATE 2004

Ans. (a) : To obtain maximum work output for the given temperature limits of T_{min} and T_{max}, the compression ratio should be optimum pressure ratio

$$r_{opt} = \sqrt{r_{max}} = (r_{max})^{1/2}$$

$$r_{opt} = \left(\frac{T_{max}}{T_{min}}\right)^{\frac{\gamma}{2(\gamma-1)}}$$

652. Consider the following statements with reference to Gas turbine cycle

1. **Regeneration increases thermal efficiency**
2. **Reheating decreases thermal efficiency**
3. **Cycle efficiency increases when maximum temperature of the cycle is increased.**

Of these statements

(a) 1, 2 and 3 are correct
(b) 2 and 3 are correct
(c) 1 and 2 are correct
(d) 1 and 3 are correct
(e) None of the above

UPSC JWM 2017
CGPSC Poly. Lect. 22.05.2016

Ans. (a) : In gas turbine cycle –
For effect of Regeneration, – Compressor work

- W_C = same
- W_T = same
- W_{net} = same
- $Q_s \rightarrow\downarrow$
- $Q_R -\downarrow$
- $T_{MA}-\uparrow$
- $T_{MR}-\downarrow$
- cycle (η) $-\uparrow$

For effect of Reheating –

- W_c – same
- $W_T -\uparrow$
- $W_{net} -\uparrow$
- $Q_s -\uparrow$

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- $Q_R - \uparrow$
- $T_{MA} - \uparrow$
- $T_{MR} - \uparrow\uparrow$
- $P_{mean} - \downarrow$
- $\eta - \downarrow$

Note- But official answer is option (d).

653. A gas turbine cycle with regeneration and reheating improves

(a) only thermal efficiency
(b) only specific power output
(c) both efficiency and power output
(d) Neither efficiency nor power output

TSPSC AEE 2015, GATE 1993

Ans. (c) : A gas turbine cycle with regeneration and reheating improves both efficiency and power output.

654. When a heat exchanger is added to an ideal simple cycle,

(a) power output increases and the efficiency increases
(b) power output decreases, but the efficiency increases
(c) power output increases, but the efficiency decreases
(d) power output remains constant, but the efficiency increases

VIZAG STEEL MT 24.01.2021, Shift-II, Shift-I

Ans. (d) : When a heat exchanger is added to an ideal simple cycle power output remains constant but the efficiency increases.
e.g. Brayton cycle with Regeneration.

655. A gas turbine plant operates on Brayton cycle between T_{min} = 300 K and T_{max} = 1073 K. Find the maximum work done per kg of air (C_p = 1.005 kJ/kg-K)

(a) 27.9 kJ/kg (b) 239.5 kJ/kg
(c) 537.3 kJ/kg (d) 776.8 kJ/kg

RPSC ACF & FRO, 26.02.2021

Ans. (b) : Given,
$T_{max} = 1073$ K
$T_{min} = 300$ K
For air, $C_p = 1.005$ kJ/kg-K

$$W_{max} = C_p\left[\sqrt{T_{max}} - \sqrt{T_{min}}\right]^2$$

$$= 1.005\left[\sqrt{1073} - \sqrt{300}\right]^2$$

$$\boxed{W_{max} = 239.4667} \text{ kJ/kg}$$

656. Gas turbines are suitable for aircraft propulsion because

(a) gas turbines are lightweight
(b) gas turbines are compact in size
(c) gas turbines have a high power to weight ratio
(d) All of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : Gas turbines are particularly suited for aircraft propulsion because they are light weight in size compact and have a high power-to-weight ratio.

657. In a gas turbine plant, heat supplied is 667.2 kJ/kg, and heat rejected is 391.43 kJ/kg. What is the thermal efficiency of the plant?

(a) 57.29% (b) 72.51%
(c) 41.33% (d) 32.83%

ESE 18.07.2021

Ans. (c) : Given,
$Q_{supp.} = 667.2$ kJ/kg
$Q_{rej} = 391.43$ kJ/kg

$$\eta_{th} = \frac{Q_{supp.} - Q_{rej.}}{Q_{supp.}}$$

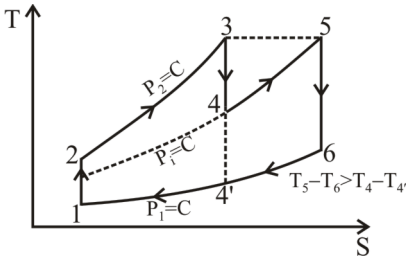
$$\eta = \frac{667.2 - 391.43}{667.2} = 41.33\%$$

658. Which one following statements is incorrect regarding reheating process in gas turbine?

(a) It uses multistage expansion
(b) It reduces moisture which cause erosion of turbine blades
(c) It increases thermal efficiency
(d) It increases specific work output

VIZAG STEEL MT 24.01.2021, Shift-II

Ans. (c) : Reheating in Gas Turbine–Reheating mean expansion of working fluid takes place in more than one stage.



Effect of Reheating–
(1) W_c – Same (2) $W_T - \uparrow$ (3) $W_{net} - \uparrow$
(4) $Q_s - \uparrow$ (5) $Q_R - \uparrow$ (6) $T_{MA} - \uparrow$
(7) $\eta \downarrow$

659. The efficiency of gas power cycle working on Brayton is a function of

(a) pressure ratio only
(b) maximum temperature only
(c) maximum to minimum temperature ratio
(d) pressure ratio and maximum to minimum temperature ratio

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : $\boxed{\eta = 1 - \frac{1}{(r_p)^{\frac{\gamma-1}{\gamma}}}}$

$r_p \rightarrow$ Pressure ratio
$\gamma \rightarrow$ Adiabatic index

660. A simple open cycle type gas turbine power plant consists of _____.

(a) turbine, combustion chamber and heat exchanger
(b) compressor, combustion chamber and turbine

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(c) turbine, compressor and heat exchanger
(d) turbine, combustion chamber and charge cooler

VIZAG MT, 14.12.2020

Ans. (b) : A simple open cycle gas turbine power plant consist of–
(1) Compressor, (2) Combustion chamber, (3) Turbine

661. For a pressure ratio, the cycle efficiency of a gas turbine can be increased by
(a) increasing the cycle peak temperature
(b) increasing the compressor inlet temperature
(c) decreasing the cycle peak temperature
(d) increasing the compressor output temperature

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : Efficiency of gas turbine can increase by –
(i) Increasing turbine work
(ii) Decreasing heat supplied
(iii) Decreasing compressor work.
The best way to increase turbine work which can achieve by increasing the cycle temperature.

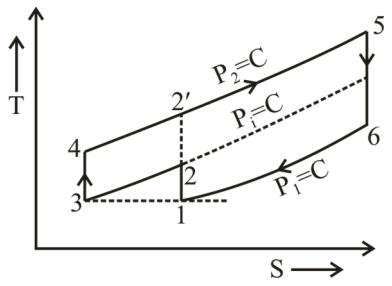
$$\eta_{Brayton} = 1 - \frac{1}{\left(r_p\right)^{\frac{\gamma-1}{\gamma}}}$$

Where, r_p = Pressure ratio
γ = Specific heat ratio

662. The use of inter cooling in gas turbine is mainly for
(a) increasing the pressure ratio
(b) reducing the turbine inlet temperature
(c) increasing overall work output
(d) bringing down combustion chamber temperature

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) :



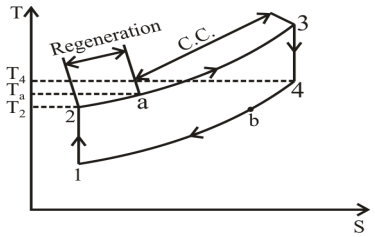
Effect of intercooling,
- W_T = Constant
- $W_C \downarrow$
- $W_{net}\uparrow$; $Q_S\uparrow$; $Q_R\uparrow$; $\eta\downarrow$

Intercooler is used to reduce compressor work, hence increases overall work output and efficiency.

663. Regeneration of exhaust gas energy to incoming gases in gas turbine is done
(a) Before the compressor
(b) After the compressor
(c) After the combustor
(d) Outside the gas turbine

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) Heating the compressed air after compressor using available heat energy from turbine exhaust is called regeneration.



- $W_C = h_2 - h_1$ (Constant)
- $W_T = h_3 - h_4$ (Constant)
- $W_{net} = W_T - W_C$ (Constant)
- $Q_S = (h_3 - h_a)\downarrow$
- $\boxed{\uparrow \eta = \frac{W_{net}}{Q_S \downarrow}}$

664. Air enters the compressor of a gas turbine plant operating on Brayton cycle at 1 bar and 27°C. The pressure ratio in the cycle is 6. If the relation between the turbine work W_T and compressor work W_C is $W_T = 3W_C$ and $\gamma = 1.4$, the cycle efficiency will be nearly
(a) 40% (b) 50%
(c) 60% (d) 70%

ESE 2020

Ans. (a) : Given, P_1= 1 bar
$T_1 = 27^0C = 300$ K

$$\eta = 1 - \left(\frac{1}{r_p}\right)^{\frac{\gamma-1}{\gamma}}$$

$$= 1 - \left(\frac{1}{6}\right)^{\frac{0.4}{1.4}} = 0.4 = 40\%$$

665. Which of the following changes in maximum T_1 and minimum T_2 temperature leads to maximum improvement in the efficiency of an ideal turbine?
(a) $T_1 + \Delta T$
(b) $T_2 + \Delta T$
(c) $T_2 + \Delta T$ and $T_1 - \Delta T$
(d) $T_1 + \Delta T$ and $T_2 - \Delta T$

OPSC AEE 2019 PAPER - II

Ans. : (d) :

666. A gas turbine plant operates on Brayton cycle between temperature limit of 1100 K and 310 K. The maximum work done per kg of air (kJ/kg) and corresponding cycle efficiency (in %) will be:
(a) 301.3 and 51.2 (b) 243.3 and 46.9
(c) 297.6 and 41.2 (d) 26.7 and 57.6

HPPSC ADF 05.03.2019

Ans. (b) :

$$\text{Maximum work}\left(W_{max}\right) = Cp\left[\sqrt{T_{max}} - \sqrt{T_{min}}\right]^2$$

$$= 1.005x\left[\sqrt{1100} - \sqrt{310}\right]^2$$

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$\boxed{W_{max} = 243.3 kJ/kg}$

At optimum condition,

$$(r_p)_{opt.} = \left(\frac{T_{max}}{T_{min}}\right)^{\frac{\gamma}{2(\gamma-1)}}$$

$$= \left(\frac{1100}{310}\right)^{\frac{1.4}{2\times0.4}}$$

Pressure ratio $\boxed{(r_p)_{opt} = 9.17}$

$$\eta = 1 - \frac{1}{(r_p)^{\frac{\gamma-1}{\gamma}}}$$

$$= 1 - \frac{1}{(9.17)^{\frac{0.4}{1.4}}}$$

$$\eta = 0.469$$

$\boxed{\eta = 46.9\%}$

667. Gas turbines are preferred in aircraft propulsion, due to

(a) They are heavy
(b) They have low power to weight ratio
(c) They have high power to weight ratio
(d) They are efficient

JPSC AE PRE 2019

Ans. (c) : Gas turbines are preferred in aircraft propulsion, due to they have high power to weight ratio.

668. An isentropic air-turbine is used to supply 0.1 kg/s of air at 0.1 MN/m^2 and 285 K to a cabin. The pressure at inlet to the turbine is 0.4 MN/m^2. What will be the power developed by the turbine? [Consider C_P = 1 kJ/kg-K. γ = 1.41, $4^{\left(\frac{0.41}{1.41}\right)} = 1.5$]

(a) 14.25 kW (b) 10.25 kW
(c) 6.75 kW (d) 8.25 kW

SJVN ET 2019

Ans. (a) : Given,
$P_2 = 0.4\times10^6$ N/m^2
$P_1 = 0.1\times10^6$ N/m^2
$T_1 = 285$ K
m = 0.1 kg/s

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{T_2}{T_1} = \left(\frac{0.4\times10^6}{0.1\times10^6}\right)^{\frac{0.41}{1.41}}$$

$\therefore T_2 = T_1 \times 1.5$
$= 285 \times 1.5$
$= 427.5$ K

Work done = $mc_P (T_2 - T_1)$
$= 0.1\times1\times10^3\times(427.5 - 285)$
$= 14250$ W
$= 14.25$ kW

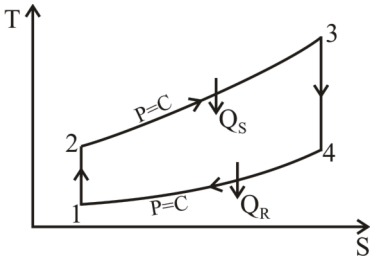
669. The pressure ratio and maximum temperature of Brayton cycle are 5 : 1 and 928 K respectively. Air enters the compressor at 1 bar and 300 K. What will be the net work output of the cycle per kg of air flow?

[consider, C_p = 1.0 kJ/kg-K γ = 1.41, $5^{\left(\frac{0.41}{1.41}\right)} = 1.6$]

(a) 256 kJ/kg (b) 205 kJ/kg
(c) 120 kJ/kg (d) 168 kJ/kg

SJVN ET 2019

Ans. (d) :
P_1= 1 bar
T_1= 300 K
T_3= 928 K
C_p = 1.0 kJ/kg-K
γ = 1.41



$$\eta = 1 - \frac{1}{(r_p)^{\frac{\gamma-1}{\gamma}}}$$

$$= \frac{\text{Net work output}}{\text{Heat supplied}}$$

$$\frac{T_2}{T_1} = (r_p)^{\frac{\gamma-1}{\gamma}}$$

$$T_2 = 300\times1.6 = 480 K$$

$$\eta = 1 - \frac{1}{(r_p)^{\frac{\gamma-1}{\gamma}}} = \frac{\text{Net work output}}{c_p(T_3 - T_2)}$$

$$1 - \frac{1}{1.6} = \frac{\text{Net work output}}{1\times(928 - 480)}$$

Net work output = 168 kJ/kg

670. Mechanical Efficiency of a gas turbine as compared to internal combustion engine is

(a) higher (b) lower
(c) same (d) unpredictable

TNPSC AE 2019

Ans. (a) : The mechanical efficiency of a gas turbine (95%) is quite high as compared with I.C. engine (85%) since I.C. engine has large number of sliding parts.
The thermal efficiency of a simple gas turbine is low (15 to 20%) as compared with I.C. engine (25 to 30%).

671. The following is (are) the limitation(s) of gas turbines:

(a) They are not self starting
(b) Higher rotor speeds
(c) Low efficiencies at part loads
(d) All of these

OPSC AEE 2019 PAPER - II

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Ans. : (d) : Limitations of gas turbines are as follows-
(i) They are not self starting
(ii) Higher rotor speeds
(iii) Low efficiencies at part loads
(iv) The efficiencies depends on the ambient condition (P_a and T_a).
(v) Compressor work required is quite large, which tells upon the efficiency of the plant.

672. Which of the following method (s) can be used to improve the thermal efficiency of open cycle gas turbine plant?
(a) Inter cooling (b) Reheating
(c) Regeneration (d) All of these

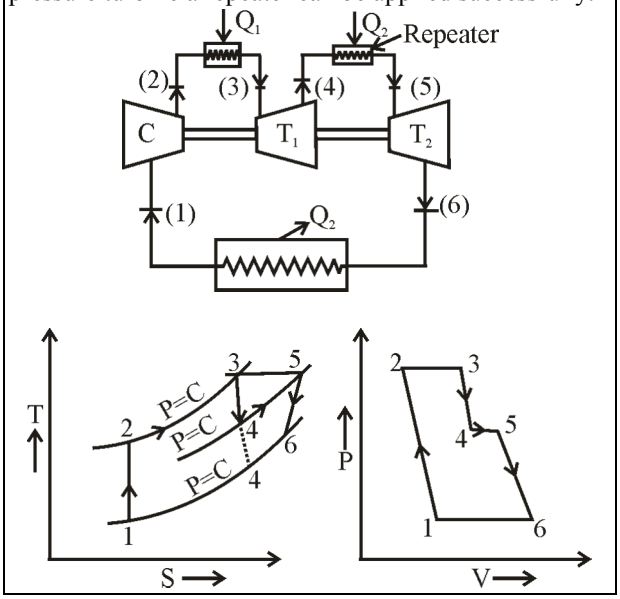
MECON MT 2019

Ans. (d) : The following method (s) can be used to improve the thermal efficiency of open cycle gas turbine plant : Inter cooling, Reheating, Regeneration.

673. Reheating in a gas turbine
(a) Increases the thermal efficiency
(b) Increases the compressor work
(c) Increases the turbine work
(d) Decreases the thermal efficiency

Nagaland PSC (CTSE) 2018, Paper-II
ESE 1998

Ans. (c) : Reheating is applied in a gas turbine in such a way that it increases the turbine work without increasing the compressor work.
When a gas turbine plant has a high pressure and low pressure turbine a repeater can be applied successfully.



674. An open cycle pressure gas turbine uses a fuel of calorific value 40000 kJ/kg with air fuel ratio of 80 : 1 and develops a net output of 80 kJ/kg of air. The thermal efficiency of cycle is :
(a) 12% (b) 16%
(c) 20% (d) 18%

GPSC ARTO Pre 30.12.2018

Ans. (b) : Given, Calorific value = 40000 kJ/kg
AFR = 80 : 1
Net output = 80 kJ/kg of air

$$\eta_{thermal} = \frac{\text{Net Output} \times \text{AFR}}{\text{Calorific value}} = \frac{80 \times 80}{40000} = 0.16 = 16\%$$

675. In a gas turbine plant, compressor power, turbine power and heat input are respectively given as 300 kJ/kg, 500 kJ/kg and 700 kJ/kg. The thermal efficiency and back work ratio of the plant will be:
(a) 71.4%; 0.4 (b) 42.8%; 0.6
(c) 71.4%; 0.6 (d) 28.6%; 0.4
(e) 28.6%; 0.6

CGPSC AE 25.02.2018

Ans. (e) : Data given-
$W_C = 300$ kJ/kg, $W_T = 500$ kJ/kg, $Q_s = 700$ kJ/kg

$$\text{Then, } \eta = \frac{W_T - W_{comp}}{Q_S} = \frac{500 - 300}{700} = \frac{200}{700}$$

$$\eta = 0.286 = 28.6\%$$

$$\text{Back work ratio} = \frac{W_{Comp}}{W_T} = \frac{300}{500} = 0.6$$

676. Gas turbine cycle with regeneration refers to:
(a) Preheating of the air leaving the compressor with the help of turbine exhaust gases
(b) Precooling of the air leaving the compressor with the help of cold water
(c) Heating the exhaust gases between two turbines
(d) Intercooling the air between two compressors
(e) Cooling of the exhaust gases between two turbines

CGPSC AE 25.02.2018

Ans. (a) : Gas turbine cycle with regeneration refers to preheating of the air leaving the compressor with the help of turbine exhaust gases.

677. Statement I: The air-standard efficiency of Brayton cycle depends only on the pressure ratio.
Statement II: For the same compression ratio, the air-standard efficiency of Brayton cycle is equal to that of Otto cycle.

ESE 2018

Ans. (b) : Air standard efficiency of Brayton cycle depends only on pressure ratio.

$$\eta = 1 - \left(\frac{1}{r_c}\right)^{\frac{\gamma-1}{\gamma}}$$

In theoretical point of view both will have same efficiency only when the compression ratio is equal to pressure ratio.

678. Statement I: The air-fuel ratio employed in a gas turbine is around 60:1.
Statement II: A lean mixture of 60:1 in a gas turbine is mainly used for complete combustion.

ESE 2018

Ans. (c) : Air fuel ratio used in gas turbines vary from 50 : 1 to 120 : 1.
Very lean mixtures are used in gas turbine to reduce the peak temperature of the cycle.
Hence statement-II is wrong.

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679. For a gas turbine the pressure ratio may be in the range

(a) 18 to 22 (b) 16 to 18
(c) 3 to 5 (d) 2 to 3

GPSC Executive Engineer 23.12.2018

Ans. (a) :

- For a gas turbine the pressure ratio may be in the range 18 to 22.
- Normal air to fuel ratio of gas turbine is 60 : 1 to 100 : 1.
- Temperature at the inlet to turbine is very high (1300K – 1500 K).

680. With the increase in pressure ratio thermal efficiency of a simple gas turbine plant with fixed turbine inlet temperature

(a) Increases
(b) Decreases
(c) First increases and then decreases
(d) First decreases and then increases

GPSC Executive Engineer 23.12.2018

Ans. (a) : With the increase in pressure ratio thermal efficiency of a simple gas turbine plant with fixed turbine inlet temperature increases.

681. The efficiency of Joule cycle is

(a) Greater than Carnot cycle
(b) Less than Carnot cycle
(c) Equal to Carnot cycle
(d) None of these

Sikkim PSC (Under Secretary), 2017

Ans. (b) : The efficiency of Joule cycle is less than Carnot cycle.
In joule cycle air is used as working fluid which have specific volume which may required huge back work from turbine which may lead to decrease in network output of the cycle.

682. In a gas turbine plant, regeneration is done to

(a) increase compression work
(b) decrease turbine work
(c) limit the maximum temperature
(d) improve plant efficiency

GPSC EE Pre, 28.01.2017

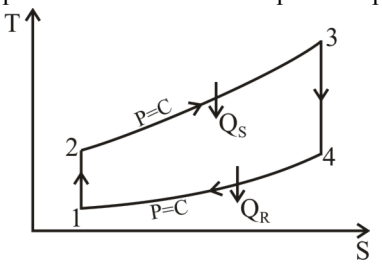
Ans. (d) : In regeneration gas turbine, as an exhaust gas heat exchanger a regenerator recovers heat from the exhaust and uses it do preheat the compressed air combustion thus regenerators can used to increase the thermal efficiencies and power output of gas turbine.

683. Assertion (A) : In constant pressure type gas turbines, large quantity of air is used, in excess of its combustion requirements.
Reason (R) : Excess air is used to compensate for inevitable air-loss due to leakage in the system.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of a
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

Punjab PSC (Lect.) 06.08.2017

Ans. (c) : In constant pressure type gas turbines, large quantity of air is used, in excess of its combustion requirements. Gas turbine cycle consists of two isentropic process and two constant pressure process.



Process $1 \rightarrow 2$: – Isentropic compression
Process $4 \rightarrow 1$: – Constant pressure heat rejection
Process $3 \rightarrow 4$: – Isentropic expansion
Process $2 \rightarrow 3$: – Constant pressure heat addition

684. The output of a gas turbine is 300 kW and its efficiency is 20 percent, the heat supplied is

(a) 600 kW (b) 6000 kW
(c) 150 kW (d) 15 kW

GPSC Asstt. Director of Transport 05.03.2017

Ans. (*) :

$$\text{Efficiency } (\eta) = \frac{\text{Output}}{\text{Heat supplied}} \Rightarrow 20\% = \frac{300}{\text{Input}}$$

Input = 1500 kW

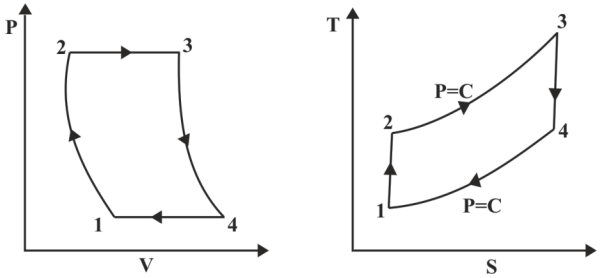
685. The work ratio of a gas turbine is a function of

(a) temperature ratio and pressure ratio
(b) pressure ratio
(c) temperature ratio
(d) none of these

ISRO Scientist/Engineer 17.12.2017

Ans. (a) : Work ratio is defined as the ratio of the network ($W_T - W_C$) to the turbine work developed (W). It is denoted by WR.

$$WR = \frac{\text{Net work}}{\text{Turbine work}} = \frac{W_T - W_C}{W_T} = 1 - \frac{W_C}{W_T}$$



Gas turbine work on Brayton cycle

$$\text{Work ratio } = 1 - \frac{C_P(T_2 - T_1)}{C_P(T_3 - T_4)} = 1 - \frac{T_1\left[\frac{T_2}{T_1} - 1\right]}{T_3\left[1 - \frac{T_4}{T_3}\right]}$$

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For Otto and Brayton cycle

$T_1 \times T_3 = T_2 \times T_4$

Then,

$$r_P = \frac{P_2}{P_1} = \frac{P_3}{P_4}$$

$$\text{Work ratio} = 1 - \frac{T_1}{T_3} \times \frac{T_2}{T_1}$$

$$\text{Work ratio} = 1 - \frac{T_1}{T_3} \times r_P^{\left(\frac{\gamma-1}{\gamma}\right)}$$

So, work ratio of gas turbine depends on both pressure ratio and temperature.

686. Efficiency of gas turbines lies between

(a) 85 to 95% (b) 50 to 60%
(c) 30 to 50% (d) 15 to 18%

ISRO Scientist/Engineer 17.12.2017
Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Overall efficiency of gas turbine plant can be about 35%.
Note-Official answer given by ISRO is (d).

687. In an ideal gas turbine cycle with inter-cooling, reheating and regeneration as the number of compression and expansion states are increased, which of the following statements are correct?

I. The Brayton cycle reduces to Ericsson cycle.
II. The Brayton cycle reduces to Stirling cycle.
III. The cycle efficiency becomes equal to Carnot cycle efficiency.

(a) Statement I alone (b) Statement III alone
(c) Statement I and II (d) Statement II and III

ISRO Scientist/Engineer 07.05.2017

Ans. (*) : Note : ISRO's option excluded for evaluation.

688. The amount of energy added by heat transfer to the cycle to produce unit of network output is called

(a) Heat rate (b) Work ratio
(c) Back work ratio (d) Thermal efficiency

TNPSC AE 2017

Ans. (a) : $\text{Heat rate} = \frac{\text{Heat added}}{\text{work done}}$

If work done=1
Then, Heat rate= Heat added

689. In Brayton cycle, heat addition is a _____.

(a) Constant enthalpy process
(b) Constant entropy process
(c) Constant volume process
(d) Constant pressure process

CIL MT 26.03.2017

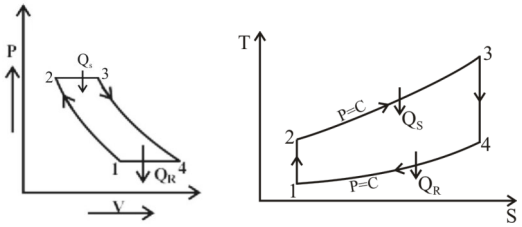
Ans. (d) : Brayton cycle is a theoretical cycle for gas turbines. This cycle consists of two reversible adiabatic or isentropic processes and two constant pressure processes. The heat addition and rejection takes place at constant pressure.

690. The gas in cooling chamber of a closed cycle gas turbine is cooled at

(a) Constant volume
(b) Constant temperature
(c) Constant pressure
(d) None of these

Vizag Steel (MT) 2017

Ans. (c) : The gas in cooling chamber of a closed cycle gas turbine is cooled at constant pressure.



This cycle is known as Joule's cycle or Brayton cycle

691. Consider the following statements in respect of ideal and practical gas turbine cycles:

1. In the ideal cycle case, the cycle efficiency depends on the pressure ratio only.
2. In the practical cycle case (with irreversibilities in the compression and expansion processes), the cycle efficiency depends on the maximum temperature as well as on the pressure ratio.
3. In the practical cycle case, at a given maximum temperature, the maximum efficiency and the maximum work done occur at a same pressure ratio.

Which of the above statements are correct?

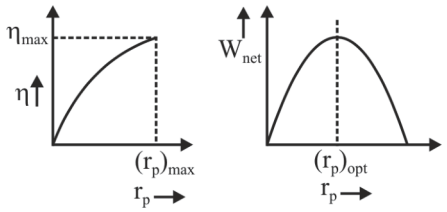
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (a) : (i) The efficiency of ideal gas turbine cycle depends on the pressure ratio.

$$\eta = 1 - \frac{1}{r_p^{\gamma-1/\gamma}}$$

(ii) The efficiency of turbine and compression came in picture due to irreversibility so final efficiency depend not only pressure ratio but also maximum temperature. Therefore statement (1) and (2) are correct while statement (3) is wrong because maximum efficiency occurs at $(r_p)_{max}$ and maximum work done occurs at $(r_p)_{opt}$



692. For the same efficiency of the Brayton cycle and the Carnot cycle working between temperature limits of T_{max} and T_{min}, the power contribution of the Brayton cycle will be

(a) zero (b) maximum
(c) minimum (d) 50% of the Carnot cycle

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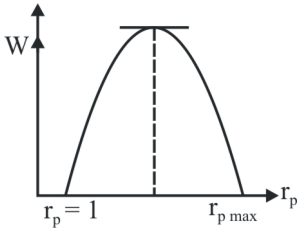
Ans. (a) : If efficiency is same then power contribution of Brayton cycle will be zero.



at $(r_p)_{max}$

$\eta_{brayton} = \eta_{carnot}$

Network is zero as turbine work = compressor work

693. The air-fuel ratio for a gas turbine is generally kept closer to

(a) 20 : 1 (b) 30 : 1

(c) 40 : 1 (d) 60 : 1

BPSC AE Mains 2017 Paper - V

Ans. : (d) : The air-fuel ratio for a gas turbine is generally kept closer to 60 : 1.

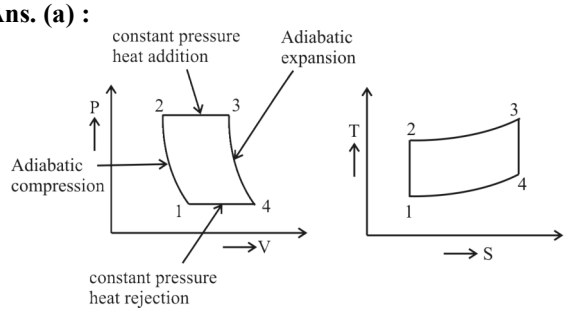
694. An air standard Brayton cycle has air entering the compressor at 27 °C and 100 kPa. The pressure ratio is 10 and maximum allowable temperature in the cycle is 1350 K. The temperature after expansion in the turbine will be :

(a) 699 K (b) 650 K

(c) 630 K (d) 600 K

RPSC ADE 2016

Ans. (a) :



$T_1 = 27 + 273 = 300$ K

$P_1 = 100$ KPa

$r_p = 10 = \dfrac{P_2}{P_1} = \dfrac{P_3}{P_4}$

$T_3 = T_{max} = 1350$ K

$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow (10)^{\frac{1.4-1}{1.4}}$$

$$\frac{T_3}{T_4} = (10)^{0.2857} = 1.93$$

$$T_4 = \frac{1350}{1.93} = 699\,\text{K}$$

695. Condenser is not used in

(a) Steam power plant

(b) Vapour compression refrigeration system

(c) Gas turbine power plant

(d) Vapour absorption refrigeration system

RPSC LECTURER 16.01.2016

Ans. (c) : In gas turbine power plant condenser is not used.

Condenser—A condenser is a device or unit used to condense a gaseous substance into liquid state through cooling. In so doing, the latent heat is released by the substance and transferred to the surrounding environment.

696. The maximum net specific work obtainable in an idle Brayton cycle T_{max} = 1600K and T_{min} = 400K is given by:

(a) 400 C_p (b) 500 C_p

(c) 600 C_p (d) 800 C_p

RPSC VPITI 14.02.2016

Ans. (a) : Given $T_{max} = 1600$K

$T_{min} = 400$K

Specific work done $W = C_p\left[\sqrt{T_{max}} - \sqrt{T_{min}}\right]^2$

$= C_p\left[\sqrt{1600} - \sqrt{400}\right]^2$

$= C_p[40 - 20]^2$

$W = 400\ C_p$

697. An open cycle pressure gas turbine uses a fuel of calorific value of 36000 kJ/kg with air-fuel ratio of 60:1 and develops a net output of 60 kJ/kg of air. The thermal efficiency of the cycle is:

(a) 6% (b) 10%

(c) 12% (d) 16%

RPSC VPITI 14.02.2016

Ans. (b) : In an open cycle gas turbine

Given, Calorific value = 36000 kJ/kg

Air-fuel ratio $\left(\dfrac{A}{F}\right) = 60 : 1$

Net output W = 60 kJ/kg

Mass of air = 60 kg

Mass of fuel = 1 kg

$$\eta = \frac{W_{net} \times m}{\frac{F}{A} \times 36000} \times 100$$

$$\eta = \frac{60 \times 60}{1 \times 36000} \times 100$$

$= 10\%$

698. The operating pressure in gas turbine as compared to IC engine is-

(a) Double (b) Same

(c) More (d) Less

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (d) : The operating pressure in gas turbine as compared to IC engine is less.

699. The effective power of gas turbines is increased by adding the following in compressor

(a) Ammonia and water vapour

(b) Carbon dioxide

(c) Nitrogen

(d) Hydrogen

Nagaland PSC (CTSE) 2016, Paper-II

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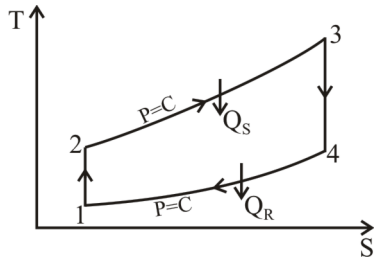
Ans. (a) The effective power of turbine = Power produce by turbine – Power consume by compressor

Power consume by compressor = $h_2 - h_1$

$= \dot{m}C_p(T_2 - T_1)$



The effective power of gas turbine is increased when power consumed by compressor is minimum i.e. C_p value should be minimum for that.

$$(C_p)_{H_2} > (C_p)_{N_2} > (C_p)_{CO_2} > (C_p)_{NH_3 \text{ and } H_2O \text{ vapour}}$$

700. The thermal efficiency of a gas turbine with regenerator is maximum when pressure ratio is :

(a) less than 1.3 (b) more than 1.0

(c) equal to 1.0 (d) zero

HPPSC W.S. Poly. 2016

Ans. : (c) The thermal efficiency of a gas turbine with regenerator no effect on the work output but thermal efficiency increase.

Efficiency of ideal regenerative cycle –

$$\eta_{\text{ideal Reg.}} = 1 - \frac{T_{min}}{T_{max}}\left(r_p\right)^{\frac{\gamma-1}{\gamma}}$$

$\eta_{\text{ideal Reg.}} \rightarrow$ maximum for $\boxed{r_p = 1}$

The thermal efficiency of a gas turbine with regenerator is maximum when pressure ratio is equal to 1.0

701. The ideal efficiency of simple gas turbine cycle depends upon

(a) Pressure ratio (b) Cut-off ratio

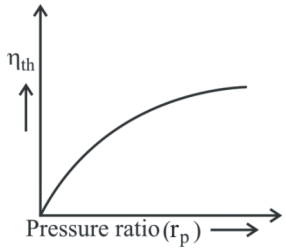
(c) Both (a) and (b) (d) None of the above

UPPSC AE 12.04.2016 Paper-II

Ans. : (a) Thermal efficiency at gas turbine:-

$$\eta_{th} = 1 - \frac{1}{r_p^{(\gamma-1)/\gamma}}$$

Pressure ratio $\left(r_p\right) = P_2 / P_1$



* Optimum pressure ratio for maximum specific output in simple gas turbine.

$$(r_p)_{optimum} = \left(\frac{T_{max}}{T_{min}}\right)^{\frac{\gamma}{2(\gamma-1)}}$$

702. If the temperature at the turbine inlet is kept constant, the net output of a simple gas turbine plant would

(a) Increase with increasing pressure ratio

(b) Decrease with increasing pressure ratio

(c) First increase and then decrease with increasing pressure ratio

(d) remains unaffected with changes in pressure ratio

UPPSC AE 12.04.2016 Paper-II

Ans. : (c) If the temperature at the turbine inlet is kept constant, the net output of a simple gas turbine plant would first increase and then decrease with increasing pressure ratio.

703. Which of the following cycles is the basis for the working of a closed cycle gas turbine?

(a) Otto Cycle (b) Joule Cycle

(c) Brayton Cycle (d) Both B and C

KPCL AE 2016

Ans. (d) : Joule cycle and Brayton cycle is basis for the working for a closed cycle gas turbine.

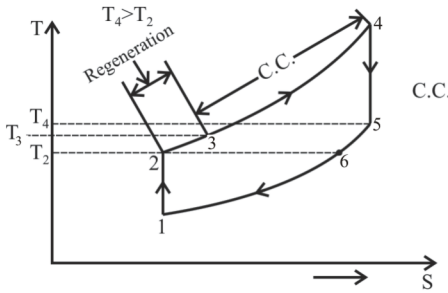
704. The air exit from the compressor and turbine in a gas turbine plant is at T_2 and T_5 respectively, while air enters from regenerator into combustion chamber at T_3. The effectiveness of regenerator is:

(a) $\frac{T_3 - T_2}{T_5 - T_2}$ (b) $\frac{T_5 - T_2}{T_3 - T_2}$

(c) $\frac{T_5 - T_3}{T_5 - T_2}$ (d) $\frac{T_5 - T_3}{T_5 - T_2}$

NPCIL ET, 2015

Ans. (a) :



Effectiveness of regenerator $(\varepsilon) = \frac{T_3 - T_2}{T_5 - T_2}$

705. The efficiency of a Brayton cycle increases with-

(a) Decrease in compression ratio

(b) Decrease in pressure ratio

(c) Increase in pressure ratio

(d) Both increase and decrease of pressure ratio

NPCIL ET, 2015

Ans. (c) : $\uparrow \eta_{\text{Brayton}} = 1 - \frac{1}{\left(\uparrow r_p\right)^{\frac{\gamma-1}{\gamma}}}$

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706. Gas turbine used in an aircraft is :

(a) close-cycle turbine
(b) open-cycle turbine
(c) constant-volume turbine
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : Gas turbine used in an aircraft is open-cycle turbine. Gas turbine cycle consists of two isentropic process and two constant pressure process.

707. Thermal efficiency of closed cycle gas turbine increase by

(a) reheating (b) intercooling
(c) regenerator (d) all of the above

TSGENCO AE 14.11.2015

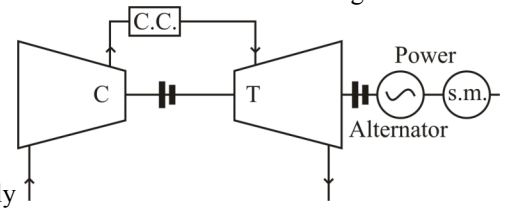
Ans. (c) : Thermal efficiency of closed cycle gas turbine increases by regenerator while decreases thermal efficiency in case of reheating and intercooling.

708. Which of the following is the part of gas turbine plant?

(a) Compressor
(b) Condenser
(c) Pelton turbine
(d) Thermostatic expansion valve
(e) Kaplan turbine

CGPSC AE 26.04.2015 Shift-II

Ans. (a) : Gas Turbine Plant– In the gas turbine power plant air is compressed in a compressor(C). This compressed air then passes through a combustion chamber (C.C.) where the temperature of the compressed air rises. That high pressure and high temperature air is passed through gas turbine (T). In gas turbine (T) compressed air suddenly expended; hence it gains kinetic energy, and because of this kinetic energy the air can do mechanical work for rotating the turbine.



709. The output in a gas turbine cycle is 800kJ/kg. The compressor work is 500kJ/kg whereas heat supplied is 1500kJ/kg. The thermal efficiency of the cycle is

(a) 10% (b) 20%
(c) 40% (d) 60%
(e) 80%

CGPSC AE 26.04.2015 Shift-II

Ans. (b) : Data given as,
$W_T = 800$ kJ/kg , $W_c = 500$ kJ/kg
$Q_s = 1500$ kJ/kg

$$\eta = \frac{W_{net}}{Q_{supplied}} = \frac{800-500}{1500}\times 100 = \frac{300}{1500}\times 100$$

$$\boxed{\eta = 20\%}$$

710. Consider the following statements:

1. **Gas turbine plants are heavier and larger in size than steam turbine plants.**
2. **The speed of rotation of the moving elements of gas turbines is much higher than those of steam turbines.**
3. **Gas turbines require 'cooling water' for its operations.**
4. **Almost any kind of fuel can be used with gas turbine.**
5. **Gas turbine plants are generally used for meeting peak power load.**

Of these, which are correct

(a) 1 and 2 (b) 1 and 3
(c) 1 and 4 (d) 3 and 4
(e) 2, 4 and 5

CGPSC AE 26.04.2015 Shift-II

Ans. (e)
1. Steam turbine plants are heavier and larger in size than gas turbine plants.
2. The speed of rotation of the moving elements of gas turbines is much higher than those of steam turbines.
3. Gas turbine not required 'Cooling water' for its operations.
4. Almost any kind of fuel can be used with gas turbine.
5. Gas turbine plants are generally used for meeting peak power load.

711. Back work ratio of gas turbine plant is:

(a) $\frac{\text{Turbine work}}{\text{Compressor work}}$ (b) $\frac{\text{Turbine work}}{\text{Heat input}}$
(c) $\frac{\text{Compressor work}}{\text{Heat input}}$ (d) $\frac{\text{Compressor work}}{\text{Turbine work}}$
(e) $\frac{\text{Turbine work - Compressor work}}{\text{Turbine work}}$

CGPSC AE 26.04.2015 Shift-II

Ans. (d) : Back work ratio- It is defined as the ratio of negative work to positive work.

$$r_{\text{back ratio}} = \frac{\text{Compressor work}}{\text{Turbine work}}$$

712. Gas turbine as compared to an internal combustion engine

(a) torque produced is uniform
(b) can be driven at high speed
(c) has more efficiency
(d) All the above

TSPSC AEE 2015

Ans. : (d) Gas turbine as compared to an internal combustion engine
(i) Torque produced is uniform
(ii) Can be driven at high speed
(iii) has more efficiency

713. The use of regeneration in a gas turbine cycle increases :

(a) Efficiency but has no effect on output
(b) Output but has no effect on efficiency
(c) Both efficiency and output
(d) None

Kerala PSC Poly. Lect. 09.01.2014

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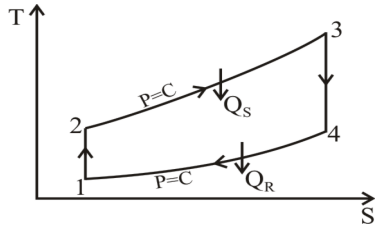
Ans. (a) : The use of regeneration in a gas turbine cycle increases efficiency but has no effect on output.

714. In an ideal gas turbine plant, it is assume that the compression and expansion process are.

(a) isothermal (b) isentropic
(c) isobaric (d) None of these

MPSC Poly. Lect. 2014

Ans. (b) : Ideal gas turbine power plant works on Brayton cycle.



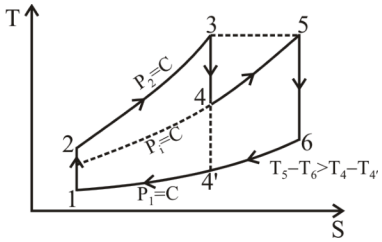
1 – 2– Isentropic compression.
2 – 3– Isobaric heat addition.
3 – 4– Isentropic expansion.
4 – 1– Isobaric heat rejection.

715. In a two-stage gas turbine plant, reheating after first stage

(a) increases work ratio
(b) is less than work ratio
(c) does not affect
(d) in unpredictable

APPSC Poly. Lect. 2013

Ans. (a) : In two-stage gas turbine plant, reheating after first stage increase work ratio.



• Effect of reheating –
W_C-same, $W_T\uparrow$, $W_{net}\uparrow$, $Q_S\uparrow$, $Q_R\uparrow$, $\eta\downarrow$

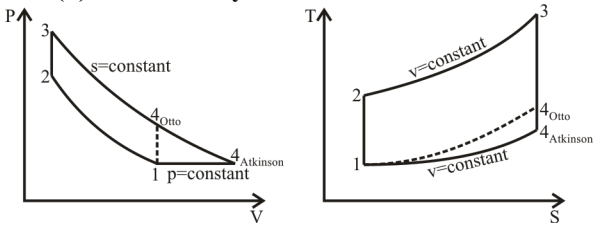
$\because$ Work ratio $(r_w) = \dfrac{W_{net}}{W_T}$

716. Which of the following gas power cycles has unequal expansion and compression strokes?

(a) Stirling cycle (b) Atkinson cycle
(c) Joule cycle (d) Ericsson cycle

RPSC AEN Pre-2013

Ans. (b) : Atkinson cycle-



Atkinson cycle has unequal expansion an compression strokes.

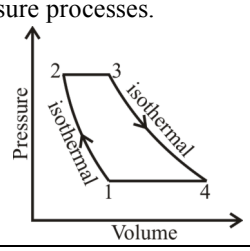
717. Which of the following gas power cycles, in ideal conditions has, thermodynamic efficiency equal to a Carnot cycle operating between the same temperature limits?

(a) Ericsson cycle (b) Joule cycle
(c) Atkinson cycle (d) Dual cycle

RPSC AEN Pre-2013

Ans. (a) : Ericsson cycle in ideal conditions has thermodynamic efficiency equal to a Carnot cycle operating between the same temperature.

• Ericsson cycle consists of two isothermal and two constant pressure processes.



718. Which of the following is not an advantage of gas turbines over IC engines?

(a) The weight of turbines per unit power is much less.
(b) There is no problem of knocking or other abnormal combustion.
(c) No or greatly reduced cooling requirements.
(d) Better part load efficiency.

RPSC AEN Pre-2013

Ans. (d) : Advantage of gas turbine over IC engine-

- The weight of turbines per unit power is much less.
- There is no problem of knocking or other abnormal combustion
- No or greatly reduced cooling requirement.

719. Gas turbine blade material must possess:

(a) High creep rate
(b) Low creep rate
(c) Plasticity at high temperature
(d) Electrical conductivity

SJVN ET 2013

Ans. (b) : Modern turbine blades often use nickel based super alloys that incorporate chromium, cobalt and rhenium.
Gas turbine blade material must possess following properties :

(i) Material must withstand high temperature and high stress.
(ii) It must have low creep rate.
(iii) It must have high resistance to oxidation corrosion and erosion.
(iv) It must neither be brittle at ordinary temperature nor plastic when hot.
(v) It must have good cast ability or forge ability characteristics.
(vi) It must have good machine ability to achieve precise dimension.
(vii) It must have high resistance to fatigue failure.
(viii) It must maintain structural stability when exposed to varying temperature.

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720. In a gas turbine plant, fuel can be:
(a) High octane gasoline
(b) Heavy diesel oil
(c) Pulverized Coal
(d) All of these

SJVN ET 2013

Ans. (d) : In a gas turbine, fuel can be :
■ High octane gasoline
■ Heavy diesel oil
■ Pulverized coal

721. An engine working on stirling cycle between the same temperature limits as that of Carnot cycle, has air standard efficiency:-
(a) More than Carnot cycle
(b) Equal to Carnot cycle
(c) Less than Carnot cycle
(d) Depends upon other factor also

UKPSC AE-2013, Paper-II

Ans. (b) : For Carnot cycle–
• The efficiency of all reversible heat engine operating between same temperature limit is same.

722. A gas turbine plant operates between 27°C, minimum temperature and 927°C maximum temperature. Calculate the cycle efficiency:-
(a) 25% (b) 50%
(c) 75% (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) : Given, $T_{min} = 27°C = 27 + 273 = 300\ K$
$T_{max} = 927°C = 927 + 273 = 1200\ K$

$$\eta = 1 - \frac{T_{min}}{T_{max}}$$
$$= 1 - \frac{300}{1200}$$
$$= 1 - \frac{1}{4} = \frac{3}{4}$$
$$\eta = 75\%$$

723. The effect of regeneration on efficiency and work output of Brayton cycle is that its:-
(a) Efficiency decreases but work output increases
(b) Efficiency as well as work output increases
(c) Efficiency increases but work output remains same
(d) Efficiency remains same but work output increases

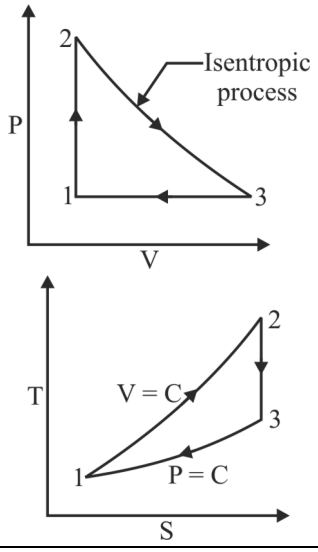
UKPSC AE-2013, Paper-II

Ans. (c) : The effect of regeneration on efficiency and work output of Brayton cycle is that its efficiency increases but work output remains same.

724. Which cycle consists of three processes?
(a) Ericsson cycle (b) Stirling cycle
(c) Atkinson cycle (d) Lenoir cycle

UKPSC AE-2013, Paper-II

Ans. (d) : The Lenoir cycle is an idealized thermodynamic cycle often used to model a pulse jet engine.



725. Constant volume gas turbine works on:-
(a) Joule cycle (b) Carnot cycle
(c) Otto cycle (d) Atkinson cycle

UKPSC AE-2013, Paper-II

Ans. (d) : Constant volume gas turbine works on atkinson cycle.

726. The thermal efficiency of an open cycle gas turbine increases with the:-
(a) Increase in inlet temperature of atmospheric air
(b) Decrease in inlet temperature of atmospheric air
(c) Remain same for all temperatures of inlet air
(d) None of the above

UKPSC AE-2013, Paper-II

Ans. (b) : The thermal efficiency of an open cycle gas turbine increases with the decrease in inlet temperature of atmospheric air.

727. Gas turbine blades are given a rake
(a) equal to zero
(b) in the direction of motion of blades
(c) opposite to the direction of motion of blades
(d) depending on the velocity

BPSC AE 2012 Paper - V

Ans. : (b) : Gas turbine blades are given a rake in the direction of motion of blades.

728. The blades of gas turbine are made of
(a) mild steel
(b) stainless steel
(c) carbon steel
(d) high-nickle alloy (neimonic)

BPSC AE 2012 Paper - V

Ans. : (d) : Turbine blades are made of nickel or rhenium alloys capable of withstanding high heat without distortion.

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729. Which of the following statements is not true for gas turbines ?
(a) Low full load thermal efficiency
(b) Costly machines
(c) Self starting unit
(d) Slow in its response to acceleration

UKPSC AE 2012 Paper–II

Ans. (c) : Self starting unit

730. Work ratio is a guide in the determination of
(a) the size of the gas turbine
(b) overall efficiency of the turbine
(c) mechanical efficiency of the turbine
(d) compressor efficiency

UKPSC AE 2012 Paper–II

Ans. (a) : the size of the gas turbine

731. Which phenomenon has the most adverse effect on volumetric efficiency when engine works at high speeds?
(a) Flow friction and choking
(b) Ram effect and choking
(c) Flow friction and charge heating
(d) Charge heating and back flow

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Volumetric efficiency reduces with increase in temperature. Flow friction and choking causes temperature to rise and hence the volumetric efficiency is adversely affected due to flow friction and choking.
Ram Effect–Ram effect is used or works in a way by using the dynamic air pressure created by the vehicle motion and increasing the static air pressure in combustion chamber it allows a greater mass flow through the engine.
Thus, ram effect pushes more air into the cylinder to increase volumetric efficiency.

732. In a two-stage turbine plant operating on a Brayton cycle with regeneration :
(a) work output increases, but cycle efficiency decreases
(b) work output increases, but cycle efficiency remains unchanged
(c) cycle efficiency increase, but work output remains unchanged
(d) both cycle efficiency and work output increase

DRDO Scientists 2009

Ans. (c) : $\eta = 1 - \frac{1}{(r_p)^{\frac{\gamma-1}{\gamma}}}$
Where, γ = adiabatic index
r_p = pressure ratio

733. The air standard cycle used in gas turbine is called
(a) Otto cycle (b) Lenoir cycle
(c) Stirling cycle (d) None of the above

WBPSC AE, 2007

Ans. (d) : The air standard cycle used in gas turbine is called Brayton cycle.

734. Reheating in gas turbines is done
(a) between compressor stages
(b) between turbine stages
(c) between compressor and turbine
(d) within the combustion chamber

OPSC Civil Services Pre 2006

Ans. (b) : Reheating - between turbine stage
Inter cooling - between compressor stage
Regeneration - between compressor and combustion chamber

6. Jet Propulsion

735. Efficiency of a jet engine is higher at :
(a) lower altitude (b) higher altitude
(c) low speed (d) high speed

RPSC AE 2016, TNPSC ACF 2012
RPSC ACF-2011, ESE 2003

Ans. (b) : The combustion efficiency of most aircraft gas turbine engine at sea level take off conditions is almost 100%. It decreases nonlinearly to 98% at altitude cruise conditions. Air –fuel ranges from 50:1 to 130 : 1

736. Propulsion efficiency of a jet engine is given by (where u is flight velocity and v is jet velocity relative to aircraft)
(a) $\frac{2u}{v-u}$ (b) $\frac{v+u}{2u}$
(c) $\frac{2u}{v+u}$ (d) $\frac{v-u}{2u}$

RPSC AE 2016, UJVNL AE 2016, ESE 1995

Ans. : (c) Propulsive efficiency of jet engine = $\frac{2u}{v+u}$
u = flight velocity
V = Jet velocity.

737. For speed above 3000 km/hour, it is more advantageous to use
(a) Turbojet engine (b) Ramjet engine
(c) Propellers (d) Rockets

Rajasthan AE (Nagar Nigam) 2016 Shift-3
TNPSC ACF 2012, TNPSC AE, 2008

Ans. (b) : For speed above 3000 km/hr, it is more advantageous to use Ramjet engine. Ramjet engine is not self operating to zero flight speed.

738. A rocket engine for the combustion of its fuel
(a) Carries its own oxygen
(b) Uses surrounding air
(c) Uses compressed atmospheric air
(d) Does not require oxygen

Nagaland PSC (CTSE) 2016, Paper-II
TNPSC AE (Industries) 09.06.2013
TNPSC ACF 2012

Ans. (a) : The petroleum used as rocket-fuel is a type a highly refined kerosene, called RP-1 in the United State petroleum fuel are usually used in combination with liquid oxygen as the oxidizer.

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739. For a jet propulsion unit ideally the compressor work and turbine work are

(a) unequal
(b) equal
(c) not related to each other
(d) None of the above

UPRVUNL AE 05.07.2021
BPSC Poly. Lect. 2016
J&K PSC Civil Services Per-2010

Ans. : (b) For a jet propulsion unit ideally the compressor work and turbine work are equally.

Jet Propulsion : -

The propulsive work needed for the propulsion of aircraft, missiles and spaceships is generally achieved by

(i) Using an engine or a gas turbine to drive the air screw or the propeller.
(ii) Expending high temperature and high pressure gases that are discharged in the rearward direction as high velocity jet (jet propulsion)

740. The motion of rocket is based on the principal of conservation of

(a) Linear momentum
(b) Angular momentum
(c) Kinetic energy
(d) Mass

TNPSC AE 2019
GPSC EE Pre, 28.01.2017

Ans. (a) : Rocket work on the principal of conservation of linear momentum. Rocket ejaculates gases in backward direction which creates momentum of the gases backward and thus by conservation in the forward direction making it moves forward.

741. The velocity of air entering in a rocket is

(a) less compared to an aircraft
(b) zero compared to an aircraft
(c) more compared to an aircraft
(d) same compared to an aircraft

TNPSC AE 2013, TNPSC ACF 2012

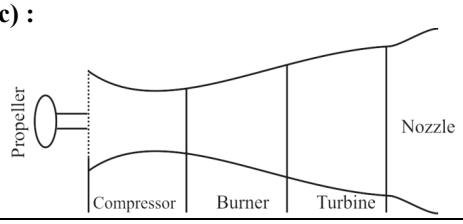
Ans. (b) : The velocity of air entering in a rocket is zero compared to an aircraft.

742. Which one of the following is the correct sequence of the position of the given components in a turboprop?

(a) Propeller, Compressor, Turbine, Burner
(b) Compressor, Propeller, Burner, Turbine
(c) Propeller, Compressor, Burner, Turbine
(d) Compressor, Propeller, Turbine, Burner

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (c) :



743. Propulsive efficiency is defined as the ratio of

(a) Thrust power and fuel energy
(b) Engine output propulsive power
(c) Thrust power and propulsive power
(d) Propulsive power and fuel input

APPSC AEE 2012, TNPSC ACF 2012

Ans. (c) : Propulsive efficiency $= \frac{\text{Thrust Power}}{\text{Propulsive Power}}$

$$\eta_p = \frac{2}{1+\frac{v_e}{v_0}}$$

Where, v_e = exhaust expulsion velocity
v_0= Air speed at the inlet

744. The essential difference between the turbofan engine and turboprop engine is :

(a) the bypass ratio is more for turboprop engine than the turbofan engine
(b) the bypass ratio is low for turboprop engine than the turbofan engine
(c) the turbofan engine uses big propellers
(d) both of them are ramjet engines

CGPSC AE 15.01.2021

Ans. (a) :

745. A radial IC engine arrangement is used in

(a) Turbojet Engines
(b) Aircraft propeller engines
(c) Large marine ships
(d) Formula -1 racing

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : The radial engine is a reciprocating type IC engine configuration in which cylinders radiate outward from a control crank case like the spokes of a wheel,. The radial configuration was commonly used for aircraft engines before gas turbine engines become predominant no unbalance couple generated in radial engine.

746. A nozzle used in Rocket motor is typically

(a) A convergent passage
(b) A divergent passage
(c) A convergent divergent passage
(d) a divergent convergent passage

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Only convergent divergent nozzle gives super sonic speed so rocket used convergent divergent nozzle.

747. In a turbo-prop aircraft, the aircraft speed is 720 km/hr and air leaves from propeller at 1400 km/hr relative to the propeller, Assuming air density 1 kg/m^3 and ignoring pressure effects, calculate the thrust produced by propeller of 2-meter diameter.

(a) 720 π N (b) 1440 π N
(c) 40000 π N (d) 518400 π N

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Velocity of air flow at the propeller disc = C = 720 km/h = 200 m/s

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Theoretical value of mass flow rate

$\dot{m} = \rho AC$

$\dot{m} = 1 \times \frac{\pi}{4} \times 2^2 \times 200 = 200\pi$

Thrust = $\dot{m}\left(2C_j - u\right)$

$= 200\ \pi\ (1440 - 720) \times \frac{5}{18}$

$= 200\pi\left(720 \times \frac{5}{18}\right)$

$= 40000\ \pi$ N

748. Specific impulse of a rocket is

(a) Rocket thrust per unit mass of propellant
(b) Rocket thrust integrated over flight time per unit mass of propellant
(c) Maximum rocket thrust divided by mass of vehicle
(d) Average rocket thrust divided by mass of vehicle

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : Specific impulse is a measure of how effectively a rocket uses propellant or a jet engine uses fuel.

749. Difference between turboprop and a turbofan propulsion device is

(a) Use of high pressure for reaction force
(b) Difference in the nature of the fan blades
(c) Location on the aircraft
(d) Presence of a surrounding duct over the inlet fan

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Turboprops consist of a huge propeller in front of the engine core and turbofans have a small fan in front of the core propeller is much move gigantic than the fan. All the gas sucked in a turboprop go through the core and produce power when incident on turbine blades. The turbine power is used to run a propeller. But in turbofan, a certain volume of air bypasses the core.

750. The pressure at the exit of rocket nozzle should be____ for maximum efficiency.

(a) Above atmospheric pressure
(b) Below atmospheric pressure
(c) Equal to atmospheric pressure
(d) Equal to critical pointing pressure

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : The pressure at the exit of rocket nozzle should be equal to atmospheric pressure for maximum efficiency.

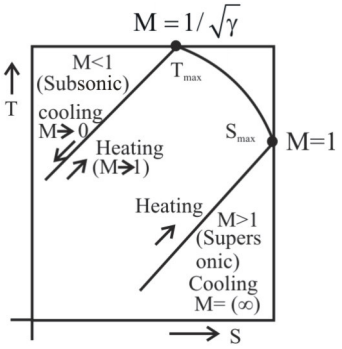
751. Which of the following statement is true for a Rayleigh flow at M = 1?

(a) Enthalpy is maximum
(b) Entropy is maximum
(c) Enthalpy is minimum
(d) Both (b) and (c)

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : Rayleigh flow-

M → Mach No.



Entropy increases heat gain and thus we proceed to the right on the Rayleigh line as heat is transferred to the liquid. The mach no. equal to 1, which is point of maximum entropy.

752. A fluid passes through a well insulated tube of 4.7 cm^2 cross section area and 5 m length. If the pipe friction coefficient is 0.07 and the flow velocity is 3 m/s, then the flow can be represented using?

(a) Rayleigh flow model
(b) Isentropic flow model
(c) Gino flow model
(d) Fanno flow model

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : (i) Fanno flow-

(a) Process is adiabatic
(b) There is friction exist ($\mu \neq 0$)
(c) $h_{01} = h_{02} \Rightarrow T_{01} = T_{02}$
(d) $P_{01} \neq P_{02}$

Fanno flow is the adiabatic flow through a constant area duct where the effect of friction is considered.

(ii) Rayleigh flow:

(a) Heat transfer takes place
(b) Frictionless ($\mu = 0$)
(c) $h_{01} \neq h_{02}$
(d) $P_{01} = P_{02}$

Flow in a constant area duct with heat transfer and without friction is called Rayleigh flow.

753. Which one of the following may be considered as a single cylinder two stroke reciprocating engine running at 2400 rpm to 2700 rpm for rapid chain of impulses?

(a) Turbo jet (b) Pulse jet
(c) Ram jet (d) Athodyd jet

ESE 2020

Ans. (b) : Pulse jet is intermittent flow engine similar to a reciprocating engine and pulse jet operates similar to Otto cycle. Hence, it can be called as a reciprocating engine.

754. In jet propulsion of ships, when the inlet orifices are at right angles to the direction of motion of the ships, the efficiency of propulsion η is

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(a) $\frac{2u^2}{V+u}$ (b) $\frac{2Vu}{(V+u)^2}$

(c) $\frac{2u}{(V+u)^2}$ (d) $\frac{2Vu}{V+2u}$

where :
V = Absolute velocity of the issuing jet
u = Velocity of the moving ship

ESE 2020

Ans. (b) : $\frac{2Vu}{(V+u)^2}$

755. Which of the following are essential for a good combustion chamber of turbojet engine?
1. **It should allow complete combustion of fuel.**
2. **It should maintain sufficiently high temperature in the zone of combustion in addition to proper atomization of fuel thus leading to continuous combustion.**
3. **It should not have high rate of combustion.**
4. **The pressure drop should be as small as possible**

(a) 1, 2 and 4 only (b) 1, 2 and 3 only
(c) 1, 3 and 4 only (d) 2, 3 and 4 only

ESE 2020

Ans. (a) : 1, 2 and 4 only

756. Speed of sound in water is equal to (*K*-bulk modulus, ρ-density)

(a) $\sqrt{\frac{K}{\rho}}$ (b) $\sqrt{K\rho}$

(c) $\sqrt{\frac{\rho}{K}}$ (d) $\frac{K}{\rho}$

ISRO Scientist/Engineer 22.04.2018

Ans. (a) : Speed of sound in water $C = \sqrt{\frac{K}{\rho}}$

where, ρ = Density
K = Bulk modulus

757. Consider the following statements regarding a Ram Jet:
1. The engine has neither a compressor nor a turbine
2. It operates at much higher temperature than a gas turbine
3. It cannot operate statically. It needs to be put in flight by some means at sufficiently high speed before it produces any thrust and propels itself.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d)
(i) In Ramjet engine there is no. compressor or turbine. The compression process is achieved by Ram effect and expansion occurs in nozzle.
(ii) In gas turbines the maximum temperature is limited by the metallurgical limits of turbine blades.
(iii) Ramjet engine cannot generate thrust in static conditions. It needs to be put in flight by some means at sufficiently high speed before it produces any thrust.

758. Air enters a turbojet engine at the rate of 40 kg/s with a velocity of 250 m/s relative to an aircraft which is moving at 300 km/hr. Exhaust of the engine has a velocity of 700 m/s relative to the moving aircraft. The thrust developed by the engine is

(a) 24 kN (b) 18 kN
(c) 12 kN (d) 9 kN

ESE 2018

Ans. (b) : Given,
$\dot{m} = 40$ kg/s
Velocity of air entering relative to flight
$V_{inlet} = 250$ m/s
Velocity of exhaust gases relative to flight
$V_{exit} = 700$ m/s
Thrust (F) = $\dot{m}(V_{exit} - V_{inlet})$
$= 40 (700 - 250) = 18000$ N = 18 kN

759. Greater the difference between jet velocity and aeroplane velocity

(a) Less the propulsive efficiency
(b) Greater the propulsive efficiency
(c) Unaffected is the propulsive efficiency
(d) None of the above

GPSC Executive Engineer 23.12.2018

Ans. (a) : Less the propulsive efficiency.

760. Air-fuel ratio in a jet engine will be of the order of

(a) 10 : 1 (b) 15 : 1
(c) 20 : 1 (d) 60 : 1

Nagaland PSC CTSE 2017 Paper-2

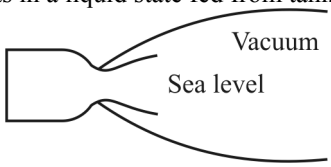
Ans. (d) : Air-fuel ratio in a jet engine, in the order of 60:1
Range (50:1 to 130:1)

761. Which of the following is true with respect to the rocket engines?

(a) They can operate in airless vacuum
(b) They have their own captive oxygen supply
(c) Both A and B
(d) None of these

KPCL AE 2016

Ans. (c) : With respect to Rocket engine
- They can operate in airless vacuum
- They have their own captive oxygen supply
- Solid-rocket fuels are chemical rockets which are used as propellant in a solid phase.
- Liquid propellant rockets uses one or more propellants in a liquid state fed from tanks.



difference between- a sea- level and vacuum optimised nozzle.

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762. Which of the following does not use ambient air for propulsion?
(a) Turbo jet (b) Pulse jet
(c) Turbo-prop (d) Rocket

UPPSC AE 12.04.2016 Paper-II

Ans. : (d) Rocket does not use ambient air for propulsion

Rocket engine :- In a rocket propulsion system, the entire mass of the fuel and oxidizing agent is carried by the rocket body itself. As such, the rocket does not depend on any surrounding air for burning the fuel.

763. Which of the following is recommended for refrigeration of supersonic aircraft?
(a) Reduced ambient system
(b) Simple system
(c) Regenerative system
(d) Bootstrap system

UPRVUNL AE 2016

Ans. (a) : Reduced ambient system is recommended for refrigeration of supersonic aircraft.

764. Modern large turbojet and turbofan engines of aeroplane uses which type of compressor?
(a) Centrifugal compressor only
(b) Combination of radial and centrifugal
(c) Axial flow compressor
(d) Radial flow compressor only

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : Modern turbojet and turbofan engine uses axial flow compressor.
⇒ Used in high speed ship engines.
⇒ Used in small scale powers stations.

765. In open cycle turbo-jet engines used in military aircraft, reheating the exhaust gas from the turbine by Burning more fuel is used to increase.
1. Thrust.
2. The efficiency of engine.
3. The range of aircraft.

Of above statement, which are correct
(a) 1 and 3 (b) 1 and 2
(c) 2 are 3 (d) 1, 2 and 3

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : In open cycle turbojet engines used in military aircraft, reheating the exhaust gas from the turbine by burning, more fuel is used to increase the thrust and range of aircraft.

766. Consider the following statements relating to rocket engines:
1. The combustion chamber in a rocket engine is directly analogous to the reservoir of a supersonic wind tunnel.
2. The stagnation conditions exist at the combustion chamber.
3. The exit velocities of exhaust gases are much higher than those in jet engines.
4. Efficiency of rocket engines is higher than that of jet engines.

Of these statements, which are correct?
(a) 1, 3 and 4 (b) 2, 3 and 4
(c) 1, 2 and 3 (d) 1, 2 and 4

Nagaland PSC CTSE 2015 Paper-II

Ans. (c)

767. In a turbo-prop system, the expansion of gases takes place in the following manner
(a) 80% in turbine and 20% in nozzle
(b) 70% in turbine and 30% in nozzle
(c) 60% in turbine and 40% in nozzle
(d) 50% in turbine and 50% in nozzle

TSGENCO AE 14.11.2015

Ans. (a) : In a turbo-propulsion system, the expansion of gas takes place in the 80% in turbine and 20% in nozzle.

768. The effective jet velocity from a rocket is 4000m/s. The forward velocity is 1500m/s and propellant consumption is 80kg/s. The propulsive efficiency is.
(a) 66% (b) 62%
(c) 63% (d) 61%
(e) 68%

CGPSC AE 26.04.2015 Shift-II

Ans. (a) : Propulsive efficiency for Rocket engine -

$$\eta_p = \frac{2\times\frac{V_e}{V_f}}{1+\left(\frac{V_e}{V_f}\right)^2} = \frac{2\times\left(\frac{4000}{1500}\right)}{1+\left(\frac{4000}{1500}\right)^2}\times 100$$

$$\eta_p = 65.75\% \approx 66\%$$

Propulsive efficiency for Jet Engine

$$\eta_p = \frac{2}{1+\left(\frac{V_e}{V_f}\right)} = \frac{2}{1+\left(\frac{4000}{1500}\right)}\times 100$$

$$\eta_p = 54.55\%$$

769. The following type of aircraft requires longest runway for take off
(a) the aircraft that uses turboprop
(b) the aircraft that uses turbojet
(c) the aircraft that uses piston engine
(d) the aircraft that uses turbofan

TANGEDCO AE 2015

Ans. (b) : The aircraft that uses turbojet requires longest runway for takeoff. Ex.: Boeing 747 uses turbofan engine.

770. A hypersonic airbreathing propulsion requires the following type of combustion chamber
(a) can type gas turbine combustor
(b) annular type gas turbine combustor
(c) ramjet combustor
(d) supersonic combustor

TANGEDCO AE 2015

Ans. (c) : A hypersonic air breathing propulsion requires Ramjet combustor.

771. The thrust coefficient in rocket propulsion is computed by using the following values
(a) Combustion pressure and throat area of nozzle
(b) Combustion pressure and exhaust gas velocity

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(c) Combustion pressure, thrust and nozzle throat area
(d) Thrust, effective exhaust velocity and exit area of nozzle

TNPSC AE 2014

Ans. (c) : By using the values of combustion pressure, thrust and nozzle throat area, the thrust coefficient in rocket propulsion is computed.

772. Fuel oxidizer combination for hybrid propellant rockets is
(a) Liquid hydrogen – Liquid oxygen
(b) Lithium hydride (LiH) – chlorine trifluoride (ClF_3)
(c) Hydrazine – Liquid Fluorine
(d) Alcohol - WFNA

TNPSC AE 2014

Ans. (b) : Fuel oxidizer combination for hybrid propellant rockets is Lithium hydride (LiH) – chlorine trifluoride (ClF_3).

773. An aircraft cannot be designed without the part of
(a) Turbine (b) Compressor
(c) Combustion chamber (d) Propeller

TNPSC AE 2014

Ans. (c) : An aircraft cannot be designed without the part of combustion chamber.

774. The inverse of Specific Propellant Consumption (SPC) of rocket engine is called
(a) Impulse to Weight Ratio (IWR)
(b) Specific impulse
(c) Thrust coefficient
(d) Weight flow coefficient

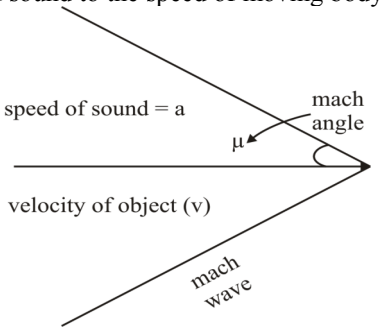
TNPSC AE 2014

Ans. (b) : The inverse of Specific Propellant Consumption (SPC) of rocket engine is called specific impulse. Specific impulse is a measure of how effectively a rocket uses propellant or a jet engine uses fuel.

775. An aircraft moves through the atmosphere with a velocity of 450 m/s. If the speed of sound in this medium is 300 m/s, the mach angle would be
(a) $\cos^{-1}(2/3)$ (b) $\sin^{-1}(2/3)$
(c) $\tan^{-1}(2/3)$ (d) $\cos^{-1}(3/2)$

TNPSC AE 2014

Ans. (b) : Definition of mach angle– Half of the vertex angle of a mach cone whose sine is the ratio of the speed of sound to the speed of moving body.



$$\sin\mu = \frac{a}{v}$$

$$\sin\mu = \frac{1}{M}$$

$$\mu = \sin^{-1}\frac{1}{\left(\frac{450}{300}\right)}$$

$$\boxed{\mu = \sin^{-1}\left(\frac{2}{3}\right)}$$

776. The propulsive efficiency is :
(a) $\frac{m\left(c_j^2 - u^2\right)}{2m_f Q_f}$ (b) $\frac{2\sigma}{1+\sigma}$
(c) $\frac{Q_j - Q_r}{Q_j}$ (d) $\frac{m\left(c_i^2 - u^2\right)}{2m_f Q_f}$

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

777. A jet engine works on the principle of conservation of :
(a) Mass (b) Discharge
(c) Energy (d) Momentum

TNPSC AE (Industries) 09.06.2013

Ans. (d) : A jet engine works on the principle of conservation of momentum.

778. A turbo prop has an additional feature compared to turbo jet in having :
(a) Grid passages (b) Inlet diffuser
(c) Inter cooler (d) Reduction Gear

TNPSC AE (Industries) 09.06.2013

Ans. (d) : Reduction Gear

779. In Rayleigh flow heating process, the heat added is accounted from static or stagnation temperature as
(a) $C_P (T_{02}–T_{01})$ (b) $C_P (T_2–T_1)$
(c) $C_P (T_2–T0_1)$ (d) $C_P (T_{01}–T_{01})$

TNPSC ACF 2012

Ans. (a) : Rayleigh flow- Constant Area duet flow with heat transfer let without friction heat added from static $= C_p(T_{02}–T_{01})$

780. Ramjet engine propels at a Mach number 3 in a atmosphere at high altitude of –11ºC and 0616 bar. The acoustic velocity and light speed measured in metres per second are
(a) 324,5, 1080.2 (b) 324.5, 973.4
(c) 337.2, 1011.6 (d) 300, 900

TNPSC ACF 2012

Ans. (b) : $C = \sqrt{\gamma RT}$

$$C = \sqrt{1.4 \times 287 \times (-11 + 273)}$$

$$\boxed{C = 324.45 \text{ m/s}}$$

$$m = \frac{V}{C}$$

$$3 = \frac{V}{324.45}$$

$$\boxed{V = 973.367 \text{ m/s}}$$

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781. Rocket carrying propellants with payload has applications in

(a) carrying passengers and payload
(b) cargo and missiles only
(c) space explorations and communication satellites
(d) Warfare only

TNPSC ACF 2012

Ans. (c) : Rocket carrying propellants with payload has application in space explorations and communication satellites.

782. The turbojet engine takes power for propulsion from

(a) Propeller (b) Turbine
(c) Nozzle thrust (d) Compressor

TNPSC ACF 2012

Ans. (c) : The turbojet engine takes power for propulsion from nozzle thrust.

783. The maximum propulsive efficiency of a turbojet is attained at around speed.

(a) 550 km/hr (b) 1050 km/hr
(c) 1700 km/hr (d) 2400 km/hr

TNPSC ACF 2012

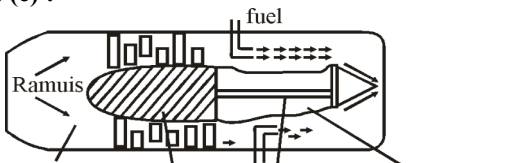
Ans. (d) : The maximum propulsion efficiency of a turbojet is attained at around following speed 2400 km/hr.

784. In jet engines the products of combustion after passing through the gas turbine are discharged into

(a) atmosphere
(b) back to the compressor
(c) discharge nozzle
(d) vacuum

TNPSC ACF 2012

Ans. (c) :



Get exit nozzle

785. In jet engines the compression ratio

(a) varies as square root of the speed
(b) varies linearly to speed
(c) varies as square of the speed
(d) varies as cube of the speed

TNPSC ACF 2012

Ans. (c) : In jet engines the compression ratio varies as square of the speed.

786. Pick up the correct statement :

(a) Closed cycle gas turbine is an I.C. engine
(b) Air-fuel ratio in gas turbine is 100, 1
(c) Ideal efficiency of closed cycle gas turbine plant is more than Carnot cycle efficiency
(d) Thrust in turbojet is produced by nozzle exit gases

TNPSC ACF 2012

Ans. (d) : The turbojet engine consist of a diffuser at the entrance which slows down the entrance air and thereby compresses it, called the ramming effect; a simple open cycle gas turbine and an exit nozzle which expands the gas and converts the thermal energy of the exit. Gas into kinetic energy when emmited by out in the from of jet the increased velocity of air there by produces thrust.

787. The overall efficiency of rocket is maximum when aircraft velocity compared to jet velocity is

(a) same (b) half
(c) double (d) one fourth

TNPSC ACF 2012

Ans. (b) : $\eta_{overall} = \eta_{th} \times \eta_p$

η_{th} = thermal efficiency

η_p = propulsive efficiency

$\eta_{overall}$ efficiency is maximum when

$$\boxed{C_{aircraft} = \frac{C_{jet}}{2}}$$

788. In gas turbine jet propulsion units, when the velocity of discharged jet equals the flight speed, then the propulsion efficiency and thrust become

(a) 100%, zero (b) 100%, maximum
(c) 50%, maximum (d) 0%, maximum

APPSC AEE 2012

Ans. (a) :

(i) Propulsive efficiency $\eta_p = \dfrac{2}{1+\dfrac{V_e}{V_0}} \times 100\%$

If $V_e = V_0$ then $\eta_p = \dfrac{2}{1+1} \times 100$

$$\boxed{\eta_p = 100\%}$$

(ii) Thrust – If the $\dot{m}_e = \dot{m}_0$ and $V_e = V_0$ then thrust $F = \dot{m}_e V_e = \dot{m}_0 V_0$

$F = 0$

789. The critical angle of attack of an aerofoil is the where

(a) the lift becomes zero
(b) the drag becomes zero
(c) the drag begins to rise
(d) the lift begins to drop

APPSC AEE 2012

Ans. (a) : The angle of attack of an aerofoil at which the flow of air about the aerofoil changes abruptly so that lift is sharply reduced and drag is sharply increased also known as stalling angle of attack.

790. Ramjet engine

(a) is self - operating at zero flight speed
(b) is not self-operating at zero flight speed
(c) requires no air for its operation
(d) produces a jet consisting of plasma

TNPSC AE, 2008

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Ans. (b) : Ramjet engine is not self operating at zero flight speed. A Ramjet uses this high pressure in front of the engine to force air through the tube, where it is heated by combusting some of it with fuel. It is then passed through a nozzle to accelerate it to supersonic speeds. Ramjet work more efficiently at supersonic speed at M = 3 nearly.

791. A rocket exists exhaust gases with ρ = 0.5 kg/m^3 out a 50 cm diameter nozzle at a velocity of 1200 m/s. The thrust, is

(a) 420 kN (b) 280 kN
(c) 140 kN (d) 90 kN

ISRO Scientist/Engineer 2008

Ans. (c) : ρ = 0.5 kg/m^3
diameter d = 50 cm = 0.5 m
velocity v = 1200 m/s
Thrust = ρAV^2
$= 0.5 \times 0.785 \times 0.25 \times (1200)^2$
= 141300 N
= 141.3 kN

7. Compressors

792. The most efficient method of compressing air is to compress it

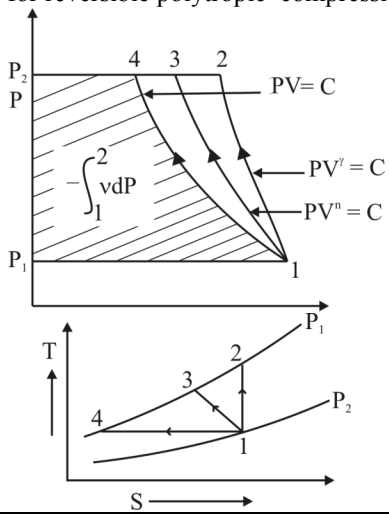
(a) adiabatically (b) isentopically
(c) isothermally (d) isochorically

TNPSC AE 2019
HPPSC Lect. (Auto) 23.04.2016
CGPSC Poly. Lect. 22.05.2016
GPSC Lect. 23.10.2016
J&K PSC Civil Services Pre-2013

Ans. (c) : The work of compression or the steady flow work input to the gas.

$$W_c = \frac{\gamma}{\gamma-1} P_1V_1\left[\left[\frac{P_2}{P_1}\right]^{\frac{\gamma-1}{\gamma}} - 1\right]$$

For reversible adiabatic compression.
Similarly, for reversible polytropic compression



$$W_n = \frac{n}{n-1} P_1V_1\left[\left(\frac{P_2}{P_1}\right)^{\left(\frac{n-1}{n}\right)} - 1\right]$$

For reversible isothermal compression of an ideal gas $W_t = P_1V_1 \ell n (P_2/P_1)$.
So, for same pressure ratio P_2/P_1 the isothermal compression needs the minimum work, whereas adiabatic compression needs the maximum work, while the polytropic compression work lies between isothermal and adiabatic.

793. Which one of the following is the effect of blade shape on the performance of a centrifugal compressor?

(a) Backward curved blades have poor efficiency
(b) Backward curved blades lead to stable performance
(c) Forward curved blades have higher efficiency
(d) Forward curved blades produce lower pressure ratio

RPSC Lect. Tech. Edu. Dept. 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II
Karnataka PSC AE, 10.09.2017
APPSC IOF, 2009, ESE 1996

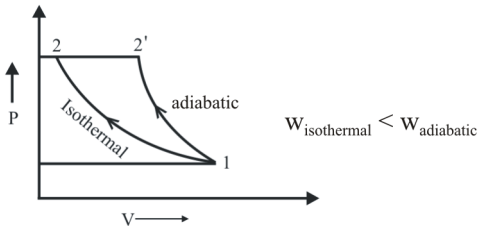
Ans. (d) : Forward curved blades produces lower pressure ratio in centrifugal compressor.

794. The power consumed by a reciprocating air compressor will be minimum, if the compression follows the following thermodynamic process :

(a) isothermal (b) isentropic
(c) adiabatic (d) polytropic

ISRO Scientist/Engineer (RAC), 10.03.2019
GPSC EE PRE 28.01.2017
CGPSC ARTO & TSI 07.01.2017
GMB AAE 25.06.2017
UKPSC AE 2012 Paper–II , ESE 2012

Ans. (a) : isothermal– Compressor work will be least when it is followed by isothermal process.



795. A centrifugal compressor is suitable for

(a) high pressure ratio, low mass flow
(b) low pressure ratio, low mass flow
(c) high pressure ratio, high mass flow
(d) low pressure ratio, high mass flow

JPSC AE, 10.04.2021, Paper-I
Assam PSC AE (PHED), 18.10.2020
BPSC AE Mains 2017 Paper - V,
ISRO Scientist/Engineer 07.05.2017, ESE 2008

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Ans. (d) : Centrifugal compressor increases the kinetic energy of the gas with a high speed impeller and then convert this energy into increased pressure in a divergent outlet passage called the diffuser centrifugal compressor are particularly suited for compressing low pressure ratio and high mass flow.

796. The volumetric efficiency of a compressor with increase in pressure ratio-

(a) Increases
(b) Decreases
(c) Remains the same
(d) May increase or decrease depending on the clearance volume

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Punjab PSC (Lect.) 06.08.2017
NPCIL ET, 2015
J&K PSC Civil Services Pre, 2010

Ans. (b) : $\downarrow \eta_{volumetric} = 1 + C - C\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}}_{\uparrow}$

$\frac{P_2}{P_1} = \text{Pressure ratio}$

797. The degree of reaction in an axial flow compressor is defined as the ratio of static enthalpy rise in the

(a) rotor to static enthalpy rise in the stator
(b) stator to static enthalpy rise in the rotor
(c) rotor to static enthalpy rise in the stage
(d) stator to static enthalpy rise in the stage

ISRO Scientist/Engineer 22.04.2018
Karnataka PSC AE, 10.09.2017
TSPSC AEE 2015, APPSC IOF, 2009

Ans. (c) :
Degree of reaction for Axial flow compressor

$$= \frac{\text{Static Enthalpy rise in the rotor}}{\text{Static enthalpy rise in the stage}}$$

798. The pressure of the intercooler in multistage reciprocating air compressor at which work required to compress the air is minimum is given by

(a) $P_2 = P_3P_1$ (b) $P_2 = \frac{P_3}{P_1}$

(c) $P_2 = \frac{P_1}{P_3}$ (d) $P_2 = \sqrt{P_1P_3}$

Where
P_2 = pressure of air in the intercooler
P_1 = initial pressure of air
P_a = delivery pressure of air

ISRO Scientist/Engineer 22.04.2018
Karnataka PSC Lect., 27.05.2017
MPPSC AE 2016
MPSC Dy. Engg. (GWD) 19.05.2013

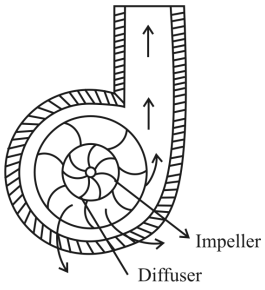
Ans. (d) : $P_2 = \sqrt{P_1P_3}$

799. The stagnation pressure rise in a centrifugal compressor takes place

(a) in the diffuser only
(b) in the impeller only
(c) in the diffuser and impeller
(d) in the inlet guide vanes only

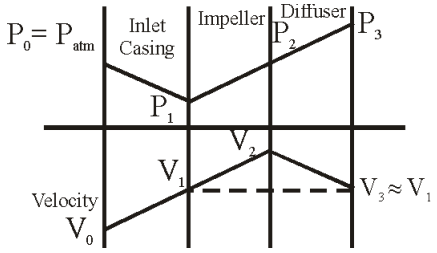
Sikkim PSC (Under Secretary), 2017
HPPSC Asstt. Prof. 18.09.2017
TNPSC AE 2013, ESE 1994

Ans. (c) :



Centrifugal Compressor

The mechanical energy provided by impeller from some external source, as the rotor rotates, it sucks air through it's eye increases it's pressure due to centrifugal force and forces the air to flow over the diffuser. The pressure of air is further increased during its flow over the diffuser.



In the diffusion of centrifugal compressor, the air is decelerated with a rise in static pressure.

800. In a four stage compressor, if the pressure at the first and third stage are 1 bar and 16 bar, then the delivery pressure at the fourth stage will be

(a) 1 bar (b) 16 bar
(c) 64 bar (d) 256 bar

OPSC AEE 2019 PAPER - II
APGENCO AE, 2017
Nagaland PSC (CTSE) 2016, Paper-II
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : We know that in multistage compressor

$$\frac{P_2}{P_1} = \frac{P_3}{P_2} = \frac{P_4}{P_3}$$

Given, $P_1 = 1$ bar
$P_3 = 16$ bar

$$\frac{P_2}{P_1} = \frac{P_3}{P_2}$$

$$P_2 = \sqrt{P_1P_3}$$

$$= \sqrt{16 \times 1}$$

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$P_2 = 4$ bar

$$\frac{P_3}{P_2} = \frac{P_4}{P_3}$$

$$P_4 = \frac{P_3^2}{P_2} = \frac{16^2}{4} = 64 \text{ bar}$$

$\boxed{P_4 = 64 \text{ bar}}$

801. A large clearance volume in reciprocating compressor results in

(a) Reduced volume flow rate
(b) Increased volume flow rate
(c) Lower suction pressure
(d) Lower delivery pressure
(e) None of the above

ISRO Scientist/Engg. (RAC) 10.03.2019
CGPSC Poly. Lect. 22.05.2016
KPSC AE 2015, ESE 1995

Ans. (a) : A large clearance volume in low volumetric efficiency and hence large cylinder dimensions increase the contact area between the piston and cylinder and so, increase friction and work.
Volumetric efficiency is given by–

$$\eta_{volumetric} = 1 + C - C\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}}$$

where,
C = clearance ratio

$\left(\frac{P_2}{P_1}\right)$ = pressure ratio

As the clearance ratio increases, the volumetric efficiency of the reciprocating compressor decreases as the pressure ratio increases. The clearance volume reduces the capacity of the compressor and volume flow rate.

802. Losses in centrifugal compressor are due to

(a) Impeller channel losses
(b) diffuser losses
(c) all types of losses
(d) inlet losses

Rajasthan Nagar Nigam AE 2016, Shift-I
BPSC AE 2012 Paper - V, ESE 2012

Ans. (c) : The losses in a centrifugal compressor are almost of the same type as those in a centrifugal pump. However, the following features are to be noted. Frictional losses : A major portion of the losses is due to fluid friction in stationary and rotating blade passages.

803. The usual assumption in elementary compressor cascade theory is that :

(a) axial velocity through the cascade changes
(b) with no change in axial velocity between inlet and outlet, the velocity diagram is formed
(c) axial velocity through the cascade theory does not change
(d) for elementary compressor theory, the pressure rise across the cascade is given by equation of state

APPSC AEE 2012, APPSC IOF, 2009
ESE 1996

Ans. (c) : The usual assumption in elementary compressor cascade theory is that axial velocity through the cascade theory does not change.

804. Stalling of blades in an axial flow compressor is the phenomenon of

(a) airstream blocking the passage
(b) motion of air at sonic velocity
(c) unsteady, periodic and reversible flow
(d) airstream not able to follow the blade contour

Gujarat PSC AE 2019
BPSC AE Mains 2017 Paper - V, ESE 2012

Ans. : (d) : Stalling is flow separation from blade contour profile. This happens when the flow velocity is low or any other reason due to which inlet blade angle deviates from designed value. Under this condition air stream is not able to follow blade contour. Unsteady periodic and flow reversal is surging not stalling.

805. The work done in compressing a gas isothermally is given by:

(a) $\frac{\gamma}{\gamma-1} P_1 V_1 \left[\left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]$ (b) $mRT_1 \log_e \frac{P_2}{P_1}$

(c) $mC_p(T_2 - T_1)$ (d) $mRT_1\left(1 - \frac{T_2}{T_1}\right)$

MPPSC AE 2016, (UJVNL AE 2016)
Nagaland PSC CTSE 2015 Paper-II

Ans. : (b) (i) Work done during isothermal Compression

$$W = P_1 V_1 \ln\left(\frac{P_2}{P_1}\right) = mRT_1 \ln\left(\frac{P_2}{P_1}\right)$$

$$= mRT_1 \ln\left(\frac{P_2}{P_1}\right)$$

(ii) Work done during polytropic compression (PV^n=C)

$$W = \frac{n}{n-1} \times mRT_2\left(1 - \frac{T_1}{T_2}\right) = \frac{n}{n-1} mR(T_2 - T_1)$$

(iii) Work done during isentropic compression

$$W = \frac{\gamma}{\gamma-1} \times mR(T_2 - T_1) = mC_p(T_2 - T_1)$$

806. Roots blower is an example of :

(a) Reciprocating (positive displacement) compressor
(b) Rotary (positive displacement) compressor.
(c) Centrifugal compressor
(d) Axial compressor.

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017, ESE 2003

Ans. (b) : Roots blower is an example of Rotary (Positive displacement) compressor.
Positive displacement compressor- Rise in pressure is obtained by positively displacing fluid to a lower volume.

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807. The choking in compressors implies

(a) no flow of air
(b) fixed mass flow rate regardless of pressure ratio
(c) reducing mass flow rate regardless of pressure ratio
(d) increased inclination of chord with air stream

Punjab PSC (Lect.) 06.08.2017
APPSC AEE Screening Test 2016
GPSC AE 26.07.2015, ESE 1996

Ans. (b) : Choking or Stonewall–The choking occurs in the compressor which operates at a low discharge pressure and maximum flow rate. It is the limit of the performance curve, at this point velocity of the impeller reaches a velocity of sound of the gas at that condition within the compressor (Mach number reaches unity). This point is also known as stonewalling of compressor.
In choking flow rate is maximum and pressure ratio is minimum.

808. With increase in clearance volume, the ideal work of compressing 1 kg of air

(a) increases
(b) decreases
(c) remains same
(d) first increases and then decreases

OPSC AEE 2019 PAPER - II
APPSC Poly. Lect. 2013

Ans. (c) : In a reciprocating air compressor the compression work per kg of air is independent of clearance volume. So, it will be remain the same.

$$W_{comp} = \frac{n}{n-1} P_1 V_{act} \left[\left(\frac{P_H}{P_L} \right)^{\frac{n-1}{n}} - 1 \right]$$

W_{comp} is independent of clearance volume.

809. Surging basically implies

(a) Unsteady periodic and reversed flow
(b) Forward motion of axial speed above sonic velocity
(c) The surging action due to the blast of air produced in compressor
(d) Forward movement of aircraft

RPSC Lect. (Tech. Edu. Dept.) 2011
ESE 1996

Ans. (a) : Surging is the phenomenon of unsteady, Periodic & Reversed flow. It is a momentary back flow through compressor from the discharge to the suction, forward discharge gas will flow backwards through the compressor towards the suction side.

810. In multistage compressor, the isothermal compression is achieved by

(a) Employing intercooler and insulating the cylinder
(b) by only insulating the cylinder
(c) by constant cooling the cylinder
(d) by running the compressor at very low speed

Rajasthan Nagar Nigam AE 2016, Shift-II
APPSC AEE 2012

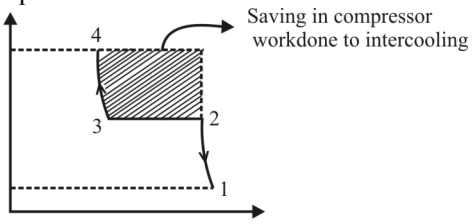
Ans. (d) : Multistaging is simply the compression of air or gas in two or more cylinder in place of a single cylinder compressor.
Isothermal compression is achieved by running the compressor at very low speed.

811. In an air compressor, intercooling is done for the following effect.

(a) to reduce the work input
(b) to reduce coolant temperature
(c) to increase cooling tower efficiency
(d) to increase the work input

J & K PSC Screening, 2006
OPSC Civil Services Pre 2006

Ans. (a) : With the use of inter cooler the work input to the compressor reduces.



812. Reciprocating air compressor is best suited for:

(a) small quantity of air at high pressure
(b) small quantity of air at low pressure
(c) large quantity of air at low pressure
(d) large quantity of air at high pressure.

VIZAG STEEL MT 24.01.2021, Shift-II
GWSSB AAE, 27.12.2015

Ans. (a) :

Compressor	useful for
centrifugal	moderate discharge and head
Axial	low head high discharge
reciprocating	low discharge and high head.

813. If an axial flow compressor is designed for a constant velocity through all stages, then the area of annulus of the succeeding stages will :

(a) progressively increase
(b) progressively decrease
(c) depend upon the number of stages
(d) remain the same

APPSC IOF, 2009, ESE 1998

Ans. (b) : If an axial flow compressor is designed for a constant velocity through all stages, then the area of annulus the succeeding stages will be progressively decrease .

814. For a multistage compressor, the polytropic efficiency is

(a) the efficiency of all stages combined together.
(b) the efficiency of one stage.
(c) constant throughout for all the stages.
(d) a direct consequence of the pressure ratio.

JPSC AE, 10.04.2021, Paper-I
ESE 1996

Ans. (c) : For multistage compressor, the polytropic efficiency is constant throughout for all the stages.

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815. The clearance volume of a reciprocating compressor directly affects

(a) Piston speed
(b) Noise level
(c) Volumetric efficiency
(d) Temperature of air after compression

JPSC AE, 10.04.2021, Paper-I
CSE Pre-1998

Ans. (c) : The clearance volume of a reciprocating compressor directly affects volumetric efficiency.

$$\eta_v = 1 + C - C\left(\frac{P_H}{P_L}\right)^{1/n} \Rightarrow \eta_v = 1 - C\left[\left(\frac{P_H}{P_L}\right)^{1/n} - 1\right]$$

816. Reciprocating compressors are provided with

(a) Simple disc plate valve
(b) Poppet valve
(c) Spring-loaded disc valve
(d) Solenoid valve

JPSC AE, 10.04.2021, Paper-I
ESE 2000

Ans. (a) : Reciprocating compressors are provided with simple disc plate valve.

817. In reciprocating air compressor the method of controlling the quantity of air delivered is done by

(a) Throttle control (b) Blow-off control
(c) Clearance control (d) All of the above

JPSC AE, 10.04.2021, Paper-I
ESE 2007

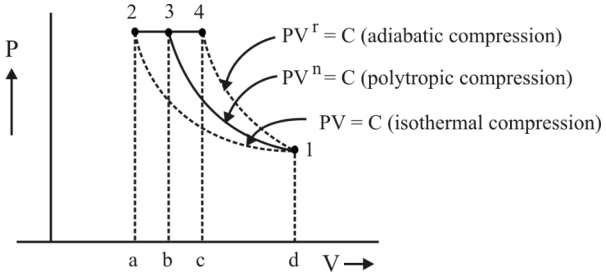
Ans. (d) : In reciprocating air compressor the method of controlling the quantity of air delivered is done by–
(i) Throttle control
(ii) Blow-off control
(iii) Clearance control

818. Work input to the air compressor with n as index of compression:

(a) increases with increase in value of *n*
(b) decreases with increases in value of *n*
(c) remains same whatever the value of *n*
(d) first increase and then decreases with value of *n*

Kerala PSC AE 06.08.2015, (KPSC AE. 2015)

Ans. (a) : Work input to the air compressor with n index of compression increases with increase in value of n.



Work done in increasing order.
(1-4-c-d) < (1-3-b-d) < (1-2-a-d)

819. The main reason for adopting the axial flow compressors instead of centrifugal compressors in aircraft turbine is that :

(a) starting torque for axial flow compressor is high
(b) the frontal area of axial flow compressor is considerably less
(c) the efficiency of middle speed range is higher
(d) pressure ratio per state is high.

Kerala PSC AE 06.08.2015, KPSC AE 2015

Ans. (b) : The frontal area of axial flow compressor is considerably less, this is the main reason of using axial flow compressor instead of centrifugal compressors in aircrafts turbine.
While a centrifugal compressor can achieve greater compression than a single compressor stage in a axial flow engine, the axial flow design allows for multiple compressor stages achieving higher overall compressor of the air and consequently greater efficiency.

820. The suction pressure is 1 bar and delivery pressure is 125 bar. What is the ideal intermediate pressure at the end of first stage for a 3-stage air compressor?

(a) 25 bar (b) 5 bar
(c) 10 bar (d) 20 bar
(e) None of the above

CGPSC Poly. Lect. 22.05.2016, ESE 2008

Ans. (b) : Given, $P_1 = 1$ bar, $P_4 = 125$ bar

$$\frac{P_4}{P_3} = \frac{P_3}{P_2} = \frac{P_2}{P_1} = k$$

$$\frac{P_4}{P_1} = \frac{P_4}{P_3} \times \frac{P_3}{P_2} \times \frac{P_2}{P_1} = k \times k \times k = k^3 = \left(\frac{P_2}{P_1}\right)^3$$

$$P_2 = P_1 \times \left(\frac{P_4}{P_1}\right)^{1/3}$$

$$= 1 \times \left(\frac{125}{1}\right)^{1/3} = 5\,\text{bar}$$

821. Generally, which type of compressor is used in a gas turbine

(a) Reciprocating (b) Centrifugal
(c) Axial flow (d) Lob type

HPPSC AE 2018, ESE 2012

Ans. (c) : Axial Flow Compressor– An axial compressor that can continuously pressurize gases. It is a rotating, aerofoil based compressor in which the gas or working fluid principally flows parallel to the axis of rotation, or axially. The energy level of the fluid increases as it flows through the compressor due to the action of the rotor blades which exert a torque on the fluid. The stationary blades slow the fluid, converting the circumferential component of flow into pressure.
Axial flow compressor produce a continuous flow of compressed gas, and the benefits of high efficiency and large mass flow rate, particularly in relation to their size and cross-section. Axial compressor are integral to the design of large gas turbine such as Jet engines.

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822. The isothermal efficiency of a reciprocating compressor is defined as

(a) $\frac{\text{Actual workdone during compression}}{\text{Isothermal workdone during compression}}$

(b) $\frac{\text{Adiabatic workdone during compression}}{\text{Isothermal workdone during compresseion}}$

(c) $\frac{\text{Isothermal workdone during compression}}{\text{Actual workdone during compression}}$

(d) $\frac{\text{Isothermal workdone during compression}}{\text{Actual workdone during adiabatic compression}}$

TNPSC AE 2014, CSE Pre-1994

Ans. (c) : The isothermal efficiency of a reciprocating compressor is defined as

$$\eta_{\text{isothermal}} = \frac{\text{Isothermal workdone during compression}}{\text{Actual workdone during compression}}$$

823. The clearance volume in reciprocating air compressor is provided

(a) to reduce the work done per kg of air delivered
(b) to increase the volumetric efficiency of the compressor
(c) to accommodate the valves in the head of the compressor
(d) to create turbulence in the air to be delivered

ESE 2018, TSGENCO AE 14.11.2015

Ans. (c) : The clearance volume in reciprocating air compressor is provided to accommodate the valves in the head of the compressor.

824. The performance of a reciprocating compressor is expressed by

(a) $\frac{\text{isothermal work}}{\text{indicated work}}$ (b) $\frac{\text{adiabatic work}}{\text{indicated work}}$

(c) $\frac{\text{isothermal work}}{\text{adiabatic work}}$ (d) $\frac{\text{adiabatic work}}{\text{frictionless work}}$

BPSC Poly. Lect. 2016, CSE Pre-2003

Ans. : (a) Isothermal efficiency. OR Compressor efficiency $= \frac{\text{Isothermal work done}}{\text{Indicated work done}}$

825. When the outlet angle from the rotor of a centrifugal compressor is more than 90º, then the blades are said to be?

(a) Forward curved (b) Backward curved
(c) Radial (d) None of these

OPSC AEE 2019 PAPER - II, ESE 1992

Ans. : (a) : When the outlet angle from the rotor of a centrifugal compressor is more than 90º then the blades are said to be forward curved.

826. Why intercooling in multistage compressors is done ?

(a) To minimize the work of compression
(b) To cool the air delivery
(c) To cool the air during compression
(d) None of these

HPPSC Asstt. Prof. 18.09.2017
UKPSC AE 2012, 2007 Paper–II

Ans. (a) : To minimize the work of compression

827. The volumetric efficiency of a reciprocating compressor is equal to:-

(a) $1+K-K(p_2/p_1)$ (b) $1+K-K(p_2/p_1)^n$
(c) $1+K+K(p_2/p_1)^{1/n}$ (d) $1+K-K(p_2/p_1)^{1/n}$

APPSC AEE 2016
UKPSC AE-2013, Paper-II

Ans. (d) : The volumetric efficiency of a reciprocating compressor,

$$\eta_v = 1 + K - K\left[\frac{p_2}{p_1}\right]^{\frac{1}{n}}$$

Where K is clearance ratio

828. The rotary compressors are used for delivering.

(a) Small quantities of air at high pressure
(b) Large quantities of air to high pressure
(c) Small quantities of air at low pressure
(d) Large quantities of air at low pressure

Rajasthan AE (Nagar Nigam) 2016 Shift-3
MPSC Dy. Engg. (GWD) 19.05.2013

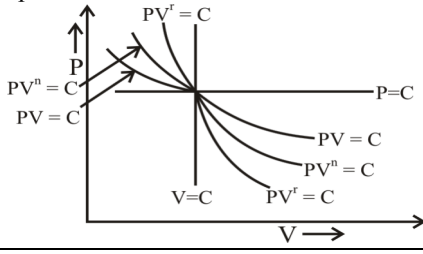
Ans. (d) : The rotary compressors are used for delivering large quantities of air at low pressure.

829. When a gas is to be stored, the type of compression that would be ideal is

(a) Isothermal (b) Adiabatic
(c) Polytropic (d) Constant volume

JPSC AE, 10.04.2021, Paper-I
J & K PSC Screening, 2006

Ans. (a) : When a gas is to be stored, isothermal compression requires minimum work and maximum for isochoric process.



830. In a reciprocating compressor the effect of increase in delivery pressure with fixed clearance volume and suction pressure will be:

(a) increase in volumetric efficiency
(b) decrease in volumetric efficiency
(c) initially volumetric efficiency increases then decreases
(d) There will be no change in volumetric efficiency.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : In a reciprocating compressor the effect of increase in delivery pressure with fixed clearance volume and suction pressure will decrease in volumetric efficiency.

$$\eta_V = 1 + C - C\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}} \qquad C = \frac{V_c}{V_s} = \frac{\text{Clearance volume}}{\text{Swept volume}}$$

$\eta_V \downarrow$ $P_2 \uparrow$

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831. For axial flow compressor, reaction ratio in terms of rotor blade angle at inlet (β_1), at outlet (β_2) and flow coefficient (ϕ) is :

(a) $2\phi (\tan\beta_1 - \tan\beta_2)$
(b) $2\phi (\tan\beta_1 + \tan\beta_2)$
(c) $\phi/2 (\tan\beta_1 - \tan\beta_2)$
(d) $\phi/2 (\tan\beta_1 + \tan\beta_2)$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) :

$$\text{Degree of reaction} = \frac{\text{Enthalpy increases in rotor}}{\text{Enthalpy increases in stage}}$$

$$R = \frac{C_a}{2U}(\tan\beta_1 + \tan\beta_2)$$

$$\boxed{R = \frac{\phi}{2}(\tan\beta_1 + \tan\beta_2)}$$

$$\phi = \frac{C_a}{U} \text{ (flow coefficient)}$$

C_a = Flow velocity
U = Velocity of rotor at outlet
β_1= Blade angle at inlet
β_2= Blade angle at outlet

832. A fan consume 200 W of electric power and discharge air from a ventilated room at the rate of 1 kg/s. What will be the possible air velocity from the fan?

(a) 200 m/s (b) 20 m/s
(c) 42 m/s (d) 4.2 m/s

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : $\text{Power} = \frac{1}{2}mV^2$

$$200 = \frac{1}{2}\times 1 \times V^2$$

$$V = \sqrt{200\times 2}$$

$$V = \sqrt{400}$$

$$\boxed{V = 20 m/s}$$

833. Car A.C. compressor shaft gets power from _____.

(a) solenoid (b) multi plate clutch
(c) engine shaft (d) steering gear

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) : engine shaft

834. Work done in compression is maximum if the process is

(a) Isothermal (b) $PV^{1.2} = C$
(c) Adiabatic (d) $PV^{1.5} = C$

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) :

835. An air compressor compresses air from 0.98 bar to 4 bar and has a 4% of stroke clearance volume. If polytropic index is 1.25, volumetric efficiency will be

(a) 91.68% (b) 61.98%
(c) 96% (d) 82.83%

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Given

$$\frac{P_2}{P_1} = \left(\frac{4}{0.98}\right)$$

$$V_C = 0.04V_s$$

$$C = \frac{V_C}{V_s} = \frac{0.04V_s}{V_s} = 0.04$$

$$\eta_v = 1 + C - C\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}}$$

$$= 1 + 0.04 - 0.04\left(\frac{4}{0.98}\right)^{\frac{1}{1.25}}$$

$$\boxed{\eta_v = 0.916 = 91.6\%}$$

836. In a reciprocating compressor, air is compressed from 1 bar to 5 bar. The expansion and compression are isentropic. The clearance volume is 5% of the stroke volume. The volumetric efficiency is _____. (assume 5^(1/1.14) as 3.1)

(a) 84.5% (b) 72.65%
(c) 89.5% (d) 80.05%

VIZAG STEEL MT 24.01.2021, Shift-II, Shift-I

Ans. (c) : Given,
P_1 = 1 bar
P_2 = 5 bar
clearance volume (v_c) 5% of stroke volume

$$\therefore \text{ clearance ratio} = \frac{v_c}{v_s}$$

$$\text{clearance ratio (c)} = \frac{0.05 \cdot v_s}{v_s} = 0.05$$

$$\text{volumetric efficiency } (\eta)_{\text{volumetric}} = 1 + c - c\left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}}$$

$$= 1 + 0.05 - 0.05\left(\frac{5}{1}\right)^{\frac{1}{1.14}}$$

$$= 1.05 - 0.05 \times 3.1 \qquad \left[\because 5^{\frac{1}{1.14}} = 3.1\right]$$

$$= 1.05 - 0.155$$
$$= 0.895$$
$$= 89.5\%$$

837. Volumetric efficiency of an ideal air compressor depends on :

(a) clearance ratio and speed
(b) speed and pressure ratio
(c) clearance ratio and pressure ratio
(d) cylinder aspect ratio and pressure ratio

NLC GET 17.11.2020, Shift-II

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Ans. (c) : Volumetric efficiency depends on ambient pressure and temperature, suction pressure and temperature. Ratio of clearance to swept volume and pressure limit. Volumetric efficiency increases with decrease in pressure ratio in compressor.

$$\eta_{vol} = 1 + C - C\left[\frac{P_2}{P_1}\right]^{1/n}$$

$$\eta_{vol} = f\left(C, r_p, n\right)$$

838. If the air enters the reciprocating compressor at 2 bar and leaves at 8 bar, the optimal intermediate pressure for two stage compression is :

(a) 3.5 bar (b) 4.0 bar
(c) 6 bar (d) 3.0 bar

NLC GET 17.11.2020, Shift-II

Ans. (b) : Optimum intermediate pressure $= \sqrt{P_1 \times P_2}$

$= \sqrt{2 \times 8}$

$$\boxed{P_{opt} = 4.0\,bar}$$

839. For a large mass flow rate requirement of an engine, which of the following compressors is preferred?

(a) reciprocating compressor
(b) multi-stage centrifugal compressor
(c) axial compressor
(d) centrifugal compressor

VIZAG MT, 14.12.2020

Ans. (c) :

Compressor	useful for
centrifugal	moderate discharge and head
Axial	low head high discharge
reciprocating	low discharge and high head.

For a large mass flow rate requirement axial flow compressors are used.

840. Air is drawn in a compressor at the rate of 0.8 kg/s at a pressure of 1 bar and temperature of 20^0C. Delivering temperature is 90^0C and pressure is 10 bar. The air is delivered through an area of 2×10^{-3} m^2. If R = 287 J/kgK, the air exit velocity is

(a) 41.7 m/s (b) 35.8 m/s
(c) 29.7 m/s (d) 27.3 m/s

UPPSC AE 13.12.2020, Paper-2

Ans. (a)

$T_1 = 293$ K $\quad T_2 = 90^0 = 363$ K

$P_1 = 1$ bar $\quad P_2 = 10$ bar

$m_a = 0.8$ kg/sec

At exit,

$P_2V_2 = m\,RT_2$

$$\frac{10 \times \left(10^5\right)}{287 \times 363} = \rho_2 \qquad \left[\because \rho = \frac{m}{v}\right]$$

Density, $\rho_2 = 9.59$ kg/m^3

Mass flow, $m = \rho_2 A v_2$

$0.8 = 9.59 \times 2 \times 10^{-3} \times v_2$

$v_2 = 41.7$ m/s

841. If maximum of 6 m/s flow velocity is allowed, calculate approximately the inlet pipes are for an air compressor delivering 0.4 kg/s of air coming at 1 bar, 300 K.

(a) 258 cm^2 (b) 566 cm^2
(c) 874 cm^2 (d) 1074 cm^2

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : $\dot{m} = \rho AV \Rightarrow$

$$0.4 = \frac{P}{RT} \times A \times V \qquad \left[\begin{array}{l}\because PV = mRT \\ P = \rho RT \\ \rho = P/RT\end{array}\right]$$

$$0.4 = \frac{1 \times 10^5}{287 \times 300} \times A \times 6$$

$A = 0.0574$ m^2 $= 574$ cm^2

842. The specific internal energy of air leaving an air-compressor is 80 kJ/kg higher than the inlet. Calculate the power to drive the compressor of air flow rate is 0.4 kg/s cooling water absorbs 55 kW

(a) 25 kW (b) 32 kW
(c) 62 kW (d) 87 kW

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Work done = change in internal energy + heat absorb by cooling water

$= 80 \times 10^3 \times 0.4 + 55 \times 10^3$

$= 87000$ J/s

$= 87$ kW

843. The ideal process for the compression of air in an ideal reciprocating compressor is:

(a) adiabatic process (b) isentropic process
(c) polytropic process (d) isothermal process

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : Ideally, there is no heat loss to the surrounding from the air. Both intake and exhaust valve closed. The air is compressed according to isothermal process. Its pressure is increased from P_1 to P_2 and temperature T is constant.

844. A _____ compressor is suitable for very high volume flow rate with low pressure rise.

(a) reciprocating (b) centrifugal
(c) vane (d) axial flow

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : Axial flow compressor is suitable for very high volume flow rate with low pressure.
In the axial-flow compressor the blades are arranged in much the same way as in a reaction turbine and the flow of air is along the axis of compressor. The velocity of air is changed during its passage through the set of blades and pressure lift occurs both in the moving blade passages and in fixed blade passage which act as diffuser.

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845. The whirl component of the velocity in the velocity triangle is the component of:

(a) the velocity of the flow in the direction of blade speed
(b) the absolute velocity of the fluid in the direction of blade speed
(c) the absolute velocity of the fluid in the radial direction
(d) the absolute velocity of the flow in the axial direction

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : The whirl velocity is the tangential component of absolute velocity at the blade inlet and outlet. This component of the velocity is responsible for the whirling or rotating of the turbine rotor.

846. In reciprocating air compressors following is NOT a method of controlling the quantity of air delivered

(a) throttle valve (b) blow-off control
(c) clearance control (d) flow control valve

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Reciprocating Air Compressor– Energy flow in rotary air compressor is continuous but in reciprocating air compressor, it is intermittent.

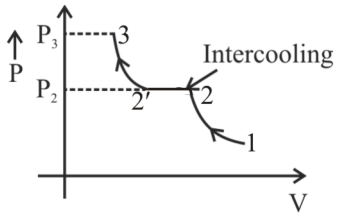
- It is a positive displacement compressor that compresses the air in a fixed volume.
- In reciprocating air compressor flow control valve not a method of controlling the quantity of air delivered.

847. In a P-V diagram of a steady flow compressor, the intercooling can be represented as

(a) Vertical line (b) Horizontal line
(c) Parabolic line (d) None of the above

ISRO Scientist/Engineer 12.01.2020

Ans. (b) :



In a PV diagram of a steady flow compressor the intercooling can be represented by horizontal line.

848. Flow coefficient in axial flow compressor is defined as _____ .[Axial velocity of flow = v_f, Blade velocity at periphery = u]

(a) $\frac{v_f}{u}$ (b) $\frac{u}{v_f}$
(c) $\frac{v_f + u}{u}$ (d) $\frac{u^2}{v_f}$

APPSC Poly Lect. 13.03.2020

Ans. (a) : Flow coefficient (Ψ) in axial flow compressor is defined as the ratio of axial velocity of flow (v_f) to the blade velocity at peripheral (u)

$$\psi = \frac{V_f}{u}$$

849. A single acting two-stage air compressor deals with 4 m^3/min of air at 1.013 bar and 15°C with a speed of 250 rpm. The delivery pressure is 80 bar. If the inter cooling is complete, the intermittent pressure after first stage will be.

(a) 9 bar (b) 8 bar
(c) 7 bar (d) 6 bar

ESE 2020

Ans. (a) : For two stage air compressor

$P_i = \sqrt{P_1 P_2} = \sqrt{1.013 \times 80} = 9\,\text{bar}$

850. Which of the following statement regarding centrifugal compressor is true?

(a) Conversion of dynamic pressure into static pressure takes place in the volute casing due to its convergent shape
(b) In multi stage centrifugal compressor, the width of the blades reduces progressively in the direction of flow
(c) In multi stage centrifugal compressor, the width of the baldes increases progressively in the direction of flow
(d) Multi staging in centrifugal compressors is commonly used for high refrigerant capacity applications

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) : Multistage centrifugal compressor is used for high compression ratio, the impeller diameter of all stages remains same, but the width of the impeller or blades becomes progressively in the direction of flow so as to get constant mass flow rate.

851. Which one of the following statements is true?

(a) Compared to reciprocating compressor, the performance of centrifugal compressor is less sensitive to speed
(b) Surging in centrifugal compressor takes place as evaporator pressure increases and condenser pressure decreases
(c) Surging in centrifugal compressor takes place as evaporator presssure decreases and condenser pressure increases
(d) For a given flow rate, frictional losses decreases as the number of blades increases

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (c) : (a) Compared to reciprocating compressor, the performance of centrifugal compressor is more sensitive to speed.

(b) **Surging in Centrifugal Compressor–**Surging is a complete breakdown of steady flow through the compressor due to periodic flow reversal. This reversal of flow occurs during closing of valve in the operating range having mass flow rate less than corresponding to maximum pressure ratio. This reversal of flow causes decrease in efficiency, abnormal sound vibration and the intensity is more, it may lead to mechanical damage. Surging is more likely to occure when the refrigeration load low (i.e. evaporator pressure is low) and the condensing pressure is high.

(d) For a given flow rate, frictional losses decreases as the number of blade decreases.

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852. The blade height of axial compressor decreases in axial direction to maintain

(a) Maximum mass flow rate
(b) Lower mass flow rate
(c) Constant speed
(d) Constant mass flow rate

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (d) : Height of blade is decreases from inlet to outlet as pressure increases to get a constant axial flow velocity in order to neutralise axial thrust on the rotor and also maintain constant mass flow rate.

$$\dot{m} = \rho A V_f$$

$$A = \pi D_m H$$

$$D_m = \frac{D_t + D_h}{2}$$

$D_t \rightarrow$ Tip diameter

$D_h \rightarrow$ Hub diameter

So the blade height of axial compressor decreases in axial direction to maintain constant mass flow rate.

853. A two stage centrifugal compressor operating at 3000 rpm is to compress refrigerant R134a from an evaporator temperature of 0°C (h_i = 398.6 kJ/kg) to a condensing temperature of 32°C (h_c = 419.8 kJ/kg). If the impeller diameter of both the stages have to be same, what is the diameter of the impeller?

Assume suction condition to be dry saturated, compression process isentropic, impeller blades radial and refrigerant enters impeller radially

(a) 0.65 m (b) 0.8 m
(c) 1.0 m (d) 1.5 m

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : $W_{total} = (419.8 - 398.6) \times 10^3$ J/kg

$= 21200$ J/kg

Work per stage $= \frac{W_{total}}{2} = u_2^2$

$W_{total} = \phi_s \phi_p \dot{m} U_2^2$

Where, $\phi_s = 1$, $\phi_p = 1$ and $\dot{m} = 1$

(Because two stage centrifugal compressor)

$W_{per\ stage} = 10600$ J/kg

$u_2 = 102.95$ m/sec $= \frac{\pi D_2 N}{60}$

$D_2 = 0.655$ m

854. The centrifugal compressors are generally used for refrigerants that require.

(a) Small displacement and low condensing pressures
(b) Large displacements and high condensing pressures
(c) Small displacements and high condensing pressures
(d) Large displacements and low condensing pressures

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (d) : Large displacements and low condensing pressures

855. In a two stage reciprocating air compressor with a suction pressure of 4 bar and delivery pressure of 16 bar, the ideal intercooler pressure will be:

(a) 10 bar (b) 8 bar
(c) 32 bar (d) 20 bar

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (b) : $P_H = 16$ bar, $P_L = 4$bar

Intercooler Pressure $(P_2) = \sqrt{P_H \times P_L}$

$= \sqrt{16 \times 4}$

$= \sqrt{64}$

$= 8$ bar.

856. Loading coefficient of an axial flow compressor for a given stage work is [where, u is the peripheral, velocity of rotor]

(a) inversely proportional to $\sqrt{u}$
(b) inversely proportional to u^2
(c) directly proportional to u^2
(d) directly proportional to u

SJVN ET 2019

Ans. (b) : The blade Loading coefficient (ψ) for an axial flow compressor can be defined as the ratio of work done to the square of peripheral velocity of rotor therefore,

$$\psi = \frac{\text{Work done}}{(\text{Peripheral velocity of rotor})^2}$$

$$\psi = \frac{W}{u^2}$$

857. The work input in reciprocating air compressor for same inlet state and same exit pressure is minimum when:

(a) Compression follows $Pv^{1.4}$ = Constant
(b) Compression follows Pv = Constant
(c) Compression follow $Pv^{1.3}$ = Constant
(d) Compression follow $Pv^{1.2}$ = Constant

SJVN ET 2019

Ans. (b) : The work input in reciprocating air compressor for same inlet state and same exit pressure is minimum when compression follow Pv = constant

858. A centrifugal compressor develops a pressure ratio of 5 and air consumption of 30 kg/s. The inlet temperature and pressure are 15°C and 1 bar respectively. For an isentropic efficiency of 0.85, the power required by the compressor will be nearly

(a) 5964 kW (b) 5778 kW
(c) 5586 kW (d) 5397 kW

ESE 2019

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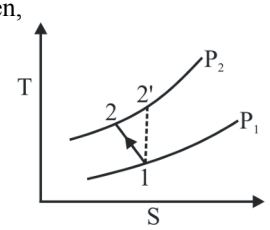
Ans. (a) : Given,



$\frac{P_2}{P_1} = 5$

$\dot{m} = 30$ kg/s

$T_1 = 15 + 273 = 288$ k

$P_1 = 1$ bar

$\eta = 0.85$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{T_2}{288} = (5)^{\frac{0.4}{1.4}}$$

$T_2 = 456.14$ k

Ideal work $\left(\dot{W}_c\right) = C_p(T_2 - T_1)$

$= 1.005\ (456.14 - 288)$

$= 168.98$ kJ/kg

$$\eta_c = \frac{\dot{W}_c}{\dot{W}_a} = \frac{\text{Ideal work}}{\text{Actual work}}$$

$$\dot{W}_a = \frac{W_c}{\eta_c} = \frac{168.98}{0.85} = 198.8 \text{ kJ/kg}$$

Actual power for compressor

$W_a = \dot{m}_a \dot{W}_a = 30 \times 198.8 = 5964.02$ kW

859. In centrifugal compressors, there exists a loss of energy due to the mismatch of direction of relative velocity of fluid at inlet with inlet blade angle. This loss is known as

(a) frictional loss (b) incidence loss

(c) clearance loss (d) leakage loss

ESE 2019

Ans. (b) : Additional losses that occur in a row of blades in a centrifugal compressor stage on account of incidence (mismatch of direction of relative velocity of fluid of inlet with inlet blade angle) termed as incidence losses. It is conventionally known as shock losses.

860. A compressor delivers 4 m^3 of air having a mass of 5 kg. The specific weight and specific volume of air being delivered will be nearly

(a) 12.3 N/m^3 and 0.8 m^3/kg

(b) 14.6 N/m^3 and 0.4 m^3/kg

(c) 12.3 N/m^3 and 0.4 m^3/kg

(d) 14.6 N/m^3 and 0.8 m^3/kg

ESE 2019

Ans. (a) : Given,

$V = 4\ m^3$, $m = 5$ kg

(i) Specific volume $(v) = \frac{V}{m} = \frac{4}{5} = 0.8\ m^3/kg$

(ii) Specific weight $(w) = \rho g$

$$= \frac{1}{v} g = \frac{9.81}{0.8} = 12.26 \text{ N/m}^3$$

861. An air compressor compresses atmospheric air at 0.1 MPa and 27°C by 10 times of air inlet pressure. During compression, the heat lost to the surrounding is estimated to be 5% of compression work. Air enters the compressor with a velocity of 40 m/s and leaves with 100 m/s. The inlet and exit cross sectional areas are 100 cm^2 and 20cm^2 respectively. The temperature of air at the exit from the compressor will be

(a) 1498 K (b) 1574 K

(c) 1654 K (d) 1726 K

ESE 2019

Ans. (a) : Given,

$P_1 = 0.1$ MPa $P_2 = 10P_1 = 1$ MPa

$T_1 = 300$ k $Q_2 = 5\%\ W_c$

$V_1 = 40$ m/s $A_1 = 100\ cm^2$

$V_2 = 100$ m/s $A_2 = 20\ cm^2$

By continuity equation

$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$

$$\frac{P_1}{RT_1} \cdot A_1 \cdot V_1 = \frac{P_2}{RT_2} \cdot A_2 \cdot V_2$$

$$\frac{0.1}{300} \times 100 \times 40 = \frac{1}{T_2} \times 20 \times 100$$

$T_2 = 1500$ K

862. Which one of the following compressors will be used in vapour compression refrigerator for plants up to 100 tonnes capacity?

(a) Reciprocating compressor

(b) Rotary compressor

(c) Centrifugal compressor

(d) Double-acting compressor

ESE 2019

Ans. (a) : Reciprocating compressors are employed in plants with capacity up to 100 tonnes.
For plants with higher capacities, centrifugal compressors are used.

863. For a centrifugal compressor with radial vanes, slip factor is the ratio of

(a) isentropic work to Euler work

(b) whirl velocity to the blade velocity at the impeller exit

(c) stagnation pressure to static pressure

(d) isentropic temperature rise to actual temperature rise

Gujarat PSC AE 2019, ESE 2013

Ans. : (b) : Slip factor 'ϕ_s' is defined as the ratio of work done (when the number of blades are finite) to the work done under ideal conditions (when the number of bladed are infinite)

$$\phi_s = \frac{\dot{m}U_2 V'_{w2}}{\dot{m}U_2 V_{w2}} = \frac{V'_{w2}}{V_{w2}} = \frac{V'_{w2}}{U_2}$$

864. The incorrect option related to Prewhirl of centrifugal compressor:

(a) To maintain the Mach number less than unity at impeller eye tip

(b) To increase the relative velocity at inlet guide vane

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(c) To avoid chocking at compressor inlet
(d) None of these

OPSC AEE 2019 PAPER - II

Ans. : (b) : Prewhirl of centrifugal compressor-
- To maintain the mach number less than unity at impeller eye tip.
- To avoid chocking at compressor inlet.
- To decrease the relative velocity at inlet guide vane.
- The prewhirl reduces the work input of the compressor. The inlet velocity of the impeller eye is inclined by an angle, known as prewhirl.

865. In a centrifugal compressor, an increase in speed at a given pressure ratio causes:
(a) Increase in efficiency
(b) Decrease in flow
(c) Increase in flow
(d) Increase in flow and decrease in efficiency

OPSC AEE 2019 PAPER - II
ESE 2013

Ans. : (d) : Increased in speed gives increases flow. As work of compression increases then it reduces compressor efficiency. (Eddies are formed)

866. A fan supplies 250 CFM air flow rate. It is equal to _____ m^3/s.
(a) 0.095 (b) 0.118
(c) 0.127 (d) 0.175

BHEL ET 2019

Ans. (b) : Given - 250 CFM
Note :– The volumetric flow rate of a liquid or gas in cubic feet per minute
so, 1 CFM = 0.47 litre/sec.
250 CFM = 0.47 × 250 litre/sec.
= 117.5 litre/sec.
= 117.5×10^{-3} m^3/s
= 0.118 m^3/sec.

867. The optimum intermediate pressure P_2 in a two stage air compressor having P_1 = 4 bar and P_3 = 9 bar, as suction and delivery pressure respectively, is equal to :
(a) 13 bar (b) 9 bar
(c) 4 bar (d) 6 bar

BHEL ET 2019

Ans. (d) : Given - Two stage air compressor
$P_1 = 4$ bar
$P_3 = 9$ bar
optimum intermediate pressure
$$P_2 = P_4 = \sqrt{P_1 P_3}$$
$$= \sqrt{4 \times 5}$$
$$= \sqrt{36} = 6 \text{ bar.}$$

868. Statement I: In air-blast injection, a separate compressor is used to create an air blast at a pressure of 6 MPa.
Statement II: The solid injection system is heavier as it needs increasing the fuel pressure to 30 MPa.

ESE 2018

Ans. (c) : In air blast injection, a separate air compressor issued to inject the fuel into the cylinder to increase the pressure of air to 6 MPa.
Solid injection system does not requires a separate compressor hence its weight is less.

869. Air is drawn in a compressor at the rate of 0.8 kg/s at a pressure of 1 bar and temperature of 20 °C, and is delivered at a pressure of 10 bar and temperature of 90 °C. This air delivery is through an exit valve of area 2×10^{-3} m^2. If R is 287kJ/kg-K, the exit velocity of the air is
(a) 41.7 m/s (b) 35.8 m/s
(c) 29.7 m/s (d) 27.3 m/s

ESE 2018

Ans. (a) : Given, $\dot{m}_1 = 0.8$ kg/s
$p_1 = 100$ kPa
$$\dot{v}_1 = \frac{\dot{m}_1 R T_1}{p_1} = \frac{0.8 \times 0.287 \times 293}{100}$$
$$= 0.6727 \text{ m}^3/\text{s}$$
$$\dot{v}_2 = \frac{\dot{m}_1 R T_2}{p_2} =$$
$$\frac{0.8 \times 0.287 \times 363}{1000} = 0.0833 \text{ m}^3/\text{s}$$
$$\dot{v}_2 = A_2 v_2$$
$$v_2 = \frac{\dot{v}_2}{A_2} = \frac{0.0833}{2 \times 10^{-3}} = 41.67 \text{ m/s}$$

870. Which of the following is a "Dynamic compressor"?
(a) Centrifugal compressor
(b) Screw compressor
(c) Root blower
(d) Reciprocating compressor
(e) Vane type compressor

CGPSC AE 25.02.2018

Ans. (a) : Screw compressor, Root blower compressor, reciprocating compressor and vane type compressor are positive displacement compressor whereas centrifugal compressor, and axial flow compressor are "Dynamic compressor"

871. In the axial flow compressor, the efficiency can be maximized by
(a) Increasing the speed
(b) Decreasing the speed
(c) Maintaining the speed constant and moderate
(d) None of these

HPPSC AE 2018

Ans. (a) : In the axial flow compressor, the efficiency can be maximized by increasing the speed of rotor of axial flow compressor.

872. If ε is the clearance ratio for a reciprocating compressor, the volumetric efficiency will be equal to
(a) $1-\varepsilon\left(1+r^{\frac{1}{n}}\right)$ (b) $1+\varepsilon\left(1-r^{\frac{1}{n}}\right)$
(c) $1+\varepsilon\left(1-\frac{1}{r^{\frac{1}{n}}}\right)$ (d) $1-\varepsilon\left(1-r^{\frac{1}{n}}\right)$

HPPSC AE 2018

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Ans. (b) : We know that, volumetric efficiency of reciprocating compressor

$$\eta_V = 1 + \varepsilon - \varepsilon\left[\frac{P_H}{P_L}\right]^{\frac{1}{n}}$$

where

$$\frac{P_H}{P_L} = r = \text{(Pressure ratio)}$$

$$\eta_V = 1 + \varepsilon\left[1 - r^{\frac{1}{n}}\right]$$

873. The application of centrifugal compressor in aircraft leads to

(a) Large frontal area of aircraft
(b) Higher flow rate through the engine
(c) Higher aircraft speed
(d) Lower frontal area of aircraft

RPSC Vice Principal ITI 2018

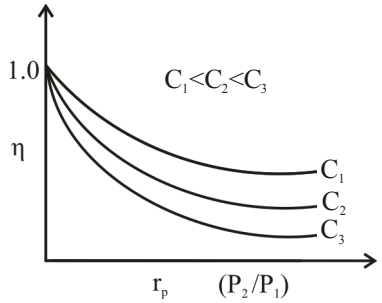
Ans. (a) : Because of higher pressure ratio per stage of centrifugal compressor, the size of impeller is large which leads to increase in frontal area of turbine.

874. Which of the following statement does not apply to the volumetric efficiency of a reciprocating air compressor?

(a) It decrease with increase of inlet temperature
(b) It increase with decrease of pressure ratio
(c) It increase of decrease in clearance ratio
(d) It decrease with increase in clearance to stroke ratio

RPSC Vice Principal ITI 2018

Ans. (a) : The variation of volumetric efficiency with pressure ratio and clearance ratio is show in figure.



$$\eta_v = 1 + C - C\left(\frac{P_2}{P_1}\right)^{1/n}$$

$$= 1 + C\left(1 - \left(\frac{P_2}{P_1}\right)^{1/n}\right)$$

- At constant clearance 'C' and decreases in pressure ratio the volumetric efficiency increases.
- At constant pressure ratio as clearance 'C' reduces. The volumetric efficiency increases.
- Increases in delivery pressure means increases in pressure ratio i.e. decreases in volumetric efficiency.
- Inlet temperature does not affect volumetric efficiency.

875. For actual single stage centrifugal compressor, the maximum power ratio is of the order of

(a) 1 : 1.2 (b) 1 : 2
(c) 1 : 4 (d) 1 : 10

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : The maximum compression ratio in an actual single stage axial flow compressor is of the order of 1:1.2

876. For the same pressure ratio during compression, compressing in two-stage compressor with perfect inter-cooling instead of a single-stage compressor is characterized by:

1. Increase in volumetric efficiency
2. Decrease in work input
3. Decrease in discharge temperature

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

UPSC JWM 2017

Ans. (d) : On comparing two compressor with perfect inter cooling instead of single stage, compressor with perfect inter cooling at same pressure ratio found following characteristics-

- Increase in volumetric efficiency
- Decrease in work input
- Decrease in discharge temperature.

877. Consider the following statements in respect of maximum efficiency of a two-stage reciprocating compressor:

1. The pressure ratios are same for each stage.
2. The work done is same in each stage.
3. The inter cooling is perfect.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (d) : For ideal two stages reciprocating air compressor the pressure ratio is same in all stages. The work done is same in all stages and for ideal situation inter cooling is perfect.

878. Consider the following statements for single-stage reciprocating compressors:

1. Isothermal process is the most desirable process for compression.
2. The size of clearance volume provided in the compressor has no effect on work done per kg of air delivered.
3. The Volumetric efficiency of the compressor decreases with increasing pressure.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

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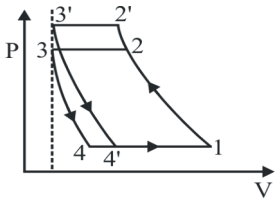
Ans. (d)



In this above fig. show that volumetric efficiency decreasing with increase in pressure.

Since the area onto y-axis i.e. $\int Vdp$ being minimum most for isothermal compression $\frac{\text{W.D.}}{\text{Cycle}}$ is least.

$$\frac{\text{Work done}}{\text{Cycle}} = \frac{n}{n-1} P_1 (V_1 - V_4)\left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$$

So work done is independent of clearance volume.

Volumetric efficiency of a reciprocating compressor is

$$\eta_V = 1 + C - C\left(\frac{P_2}{P_1}\right)^{1/n}$$

$$C = \frac{V_C}{V_S} = \text{Clearance ratio}$$

P_2 = Delivery pressure

P_1 = Inlet pressure

As P_2 increases, η_{vol} decreases.

879. Consider the following statements:
The volumetric efficiency of a reciprocating compressor can be improved by
1. decreasing the clearance volume
2. cooling the intake air
3. heating the intake air

Which of the above statements is / are correct?
(a) 2 only (b) 3 only
(c) 1 and 2 (d) 1 and 3

ESE 2017

Ans. (c) : The volumetric efficiency of reciprocating compressor,

$$\eta_v = 1 + C - C\left(\frac{P_2}{P_1}\right)^{1/n}$$

Since pressure ratio $\frac{P_2}{P_1} > 1$ so, by decreasing the clearance volume, volumetric efficiency get increased. Since volumetric efficiency is a geometrical parameter involving expansion exponent 'n' of gas. The value of n is higher at low temperature, so by cooling the intake air volumetric efficiency may increase slightly.

880. In a two-stage reciprocating air compressor, the suction and delivery pressure are 1 bar and 4 bar respectively. For the maximum efficiency, the intercooler pressure is
(a) 1.5 bar (b) 2.5 bar
(c) 2.0 bar (d) 3.0 bar

BPSC AE Mains 2017 Paper - V

Ans. : (c) : For minimum work consumption and maximum efficiency, the intermediate pressure

$P_i = \sqrt{P_1.P_2} = \sqrt{1 \times 4} = 2$ bar

881. High positive incidence in an axial compressor blade leads to
(a) suppression of separation of flow on the blade
(b) choking of the flow
(c) separation of flow on the suction side of the blade
(d) separation of flow on the pressure side of the blade

BPSC AE Mains 2017 Paper-V

Ans. : (c) : Stalling is the separation of flow from the blade surface. At high incidence, flow separation occurs on the suction side of the blades which is referred as positive stalling.

882. The volumetric efficiency of a reciprocating compressor
(a) increases with increasing clearance ratio
(b) increases with increase in compression index
(c) does not change with change in clearance ratio and pressure ratio
(d) decreases both with increasing clearance ratio and pressure ratio

BPSC AE Mains 2017 Paper - V

Ans. : (d) : Volumetric efficiency

$$\eta = 1 - \frac{v_c}{v_s}\left[\left(\frac{P_2}{P_1}\right)^{1/n} - 1\right]$$

$$= 1 - c\left[\left(\frac{P_2}{P_1}\right)^{1/n} - 1\right]$$

$$= 1 + c - c\left(\frac{P_2}{P_1}\right)^{1/n}$$

Hence volumetric efficiency decreases with increasing clearance ratio and pressure ratio both.

883. The mean effective pressure of the compressor is the :
(a) actual volume of air delivered by the compressor when reduced to normal temperature and pressure conditions.
(b) volume of air delivered by the compressor
(c) volume of air sucked by the compressor during its suction stroke
(d) none of the above

Punjab PSC SDE 12.02.2017

Ans. (d) : The mean effective pressure is a quantity relating to the operation of a reciprocating engine and is a valuable measure of an engine's capacity to do work that is independent of engine displacement.

884. Separators are generally used in air compressor installations :
(a) before the intercooler
(b) after the intercooler
(c) between the after cooler and receiver
(d) before first stage suction

Punjab PSC SDE 12.02.2017

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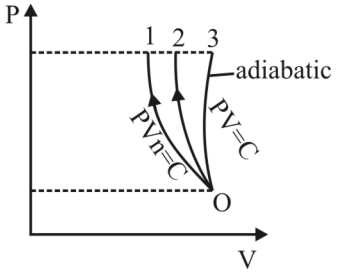
Ans. (c) : Separators are generally used in air compressor installations between the after cooler and receiver.

885. In single stage reciprocating compressor, work done on air will be minimum when compression is:

(a) isothermal (b) isochoric
(c) polytropic (d) adiabatic
(e) isentropic

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Process (0-1)- Isothermal
Process (0-2)- Polytropic
Process (0-3) - adiabatic



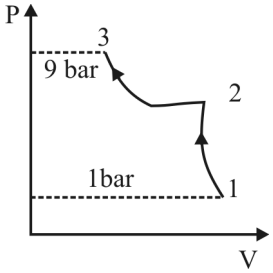
It is clear from the figure.
Isothermal process required minimum work during compression.

886. In two stage reciprocating compressor, suction and delivery pressure are 1 bar and 9 bar respectively. What will be the inter cooling pressure for maximum efficiency?

(a) 2 bar (b) 3 bar
(c) 4.5 bar (d) 5 bar
(e) 3.5 bar

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : $P_{Suction} = 1$ bar
$P_{Delivery} = 9$ bar



$$P_{intermediate} = \sqrt{P_{suc} \times P_{dev}}$$

$$P_{int} = \sqrt{1 \times 9}$$

$$= \mathbf{3\ bar}$$

887. The blade passages in a compressor are_____

(a) converging (b) diverging
(c) constant (d) unpredictable

WBPSC AE, 2017

Ans. (b) : The blade passages in a compressor are diverging.

888. An axial flow compressor has _____

(a) a druth type rotor
(b) a disc type rotor
(c) a drum type or disc type rotor
(d) None of the above

WBPSC AE, 2017

Ans. (c) : An axial flow compressor has a drum type or disc type rotor.

889. Which of the following statements is *wrong*?

(a) In a two-stage reciprocating air compressor with complete intercooling, maximum work is saved.
(b) The minimum work required for a two-stage reciprocating air compressor is double the work required for each stage.
(c) The ratio of the volume of free air delivery per stroke to the swept volume of the piston is called volumetric efficiency.
(d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (d) : None of the above

890. The compressor capacity is defined as the :

(a) actual volume of the air delivered by the compressor when reduced to normal temperature and pressure conditions.
(b) volume of air delivered by the compressor
(c) volume of air sucked by the compressor during its suction stroke
(d) None of the above

Punjab PSC SDE 12.02.2017

Ans. (b) : The compressor capacity is defined as the volume of air delivered by the compressor.

891. The overall isothermal efficiency of the compressor is defined as the ratio of :

(a) work required to compress the air isothermally to the actual work required to compress the air for the same pressure ratio.
(b) isothermal horse power to the shaft horsepower of B.H.P. of the motor or engine required to drive the compressor
(c) volume of free air delivery per stroke to the swept volume of the piston.
(d) adiabatic horsepower to the horsepower required to drive the compressor.

Punjab PSC SDE 12.02.2017

Ans. (b) : The overall isothermal efficiency of the compressor is defined as the ratio of isothermal horse power to the shaft horsepower or B.H.P. of the motor or engine required to drive the compressor.

892. The volumetric efficiency of a compressor :

(a) increases with decrease in compression ratio
(b) decreases with decrease in compression ratio
(c) increases with increase in compression ratio
(d) decreases with increases in compression ratio

Punjab PSC SDE 12.02.2017

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Ans. (a & d) : Volumetric Efficiency (η_V)– It is defined as the ratio of actual volume at the entry of compressor to that of the theoretical swept volume.

$$\eta_V = \frac{\text{Actual volume at entry}}{\text{Theoretical swept volume}} = \frac{V_a}{V_s}$$

$$\eta_V = \frac{V_a}{\frac{\pi}{4} D^2 L \times k \times \frac{N}{60 \times n}}$$

Volumetric efficiency of a reciprocating compressor is,

$$\eta_V = 1 + C - C\left(\frac{P_H}{P_L}\right)^{1/n}$$

- $P_2 \uparrow \rightarrow \downarrow \eta_V = \left[1 - \left[C \uparrow \left(\frac{P_2 \uparrow}{P_1}\right)^{1/n} - 1\right] \uparrow\right] \downarrow$
- $n \downarrow \rightarrow \eta_V \uparrow = \left[1 - \left[C \downarrow \left(\frac{P_2}{P_1}\right)^{\frac{1}{n\downarrow}} - 1\right] \downarrow\right] \uparrow$
- $C \uparrow \rightarrow \eta_V \downarrow = \left[1 - \uparrow C \left[\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}} - 1\right] \uparrow\right] \downarrow$

893. If the flow of air through the compressor is perpendicular to its axis, then it is a :

(a) reciprocating compressor
(b) centrifugal compressor
(c) axial flow compressor
(d) turbo compressor

Punjab PSC SDE 12.02.2017

Ans. (b) : If the flow of air through the compressor is perpendicular to its axis, then it is a centrifugal compressor.

894. Compression efficiency is compared against

(a) Ideal compression
(b) Adiabatic compression
(c) Isentropic compression
(d) Isothermal compression

Nagaland PSC CTSE 2017 Paper-2

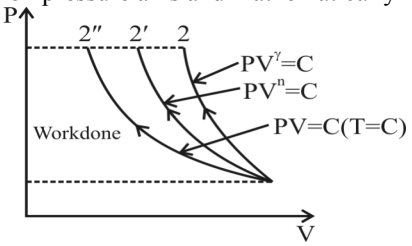
Ans. (d) : Compression efficiency is compared against isothermal compression.

895. Maximum work is done in compressing air when the compression is

(a) Adiabatic (b) Polytropic
(c) Isentropic (d) Isothermal

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (a) : **Open System Work or Steady (Flow) Work–**To obtain this work area under P-V should be projected on pressure axis and mathematically



$W_{open} = -\int v dp$

∴ According to graph maximum work is done in adiabatic compression of air.

So, Maximum work done in compressing air when the compression is adiabatic. In adiabatic system compression will generate heat due to internal friction but heat will not escape since there is no heat transfer and provide more work to resist the compression due to heat generated.

896. Clearance volume of a reciprocating compressor is 120 ml and the volume of cylinder at bottom dead centre is 960 ml. What is the clearance ratio of the compressor?

(a) $\frac{1}{9}$ (b) $\frac{1}{8}$
(c) $\frac{1}{7}$ (d) $\frac{1}{6}$
(e) $\frac{1}{5}$

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : As we know,

$$\text{Clearance ratio} = \frac{\text{Clearance volume}}{\text{Swept volume}}$$

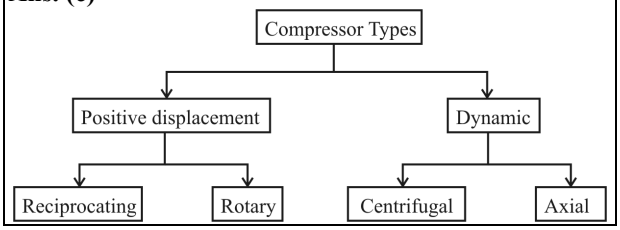
$$= \frac{120}{960 - 120} = \frac{1}{7}$$

897. Which of the following belong to the Roto-dynamic compressor family?

(a) Centrifugal compressors
(b) Axial flow compressors
(c) Both A and B
(d) None of these

KPCL AE 2016

Ans. (c)



898. What is the double acting compressor?

(a) It is a compressor where air is compressed on both sides of the piston
(b) It is a compressor where air is compressed by two pistons working inside a single cylinder
(c) Both A and B
(d) None of these

KPCL AE 2016

Ans. (a) : A compressor in which suction, compression and delivery of air take place on both sides of the piston is known as double acting compressor. It has two delivery strokes for such revolution of the crank shaft.

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899. In which of the following compressors, the flow of air through the compressor is parallel to its axis?

(a) Axial flow compressor
(b) Centrifugal compressor
(c) Both A and B
(d) None of these

KPCL AE 2016

Ans. (a) : If the flow of air through the compressor is parallel to its axis, then the compressor is axial flow compressor

900. The degrees of reaction is usually kept ..... for all types of axial flow compressors.

(a) 0.2 (b) 0.3
(c) 0.4 (d) 0.5

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : The degrees of reaction is usually kept 0.5 for all types of axial flow compressors.

901. The volumetric efficiency for reciprocating air compressors is about

(a) 10 to 40% (b) 40 to 60%
(c) 60 to 70% (d) 70 to 90%

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : The volumetric efficiency for reciprocating air compressors is about 70 to 90%.

902. Induced draught fans of a large steam generator have

(a) Backward curved blades
(b) Forward curved blades
(c) Straight of radial blades
(d) Double curved blades
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (a) : Induced draught fan is normally located at the outlet between dust collector and chimney. The fan takes hot flue gases from the boiler through dust collector and delivers it to the chimney into the open atmosphere ID fan creates negative pressure or suction to discharge the gases out after combustion from the furnace. Since ID fans can handle hot flue gases, they have more corrosion and errosion problems even when used with electrostatic precipitaters. The most common type of fan used as ID fans are the backward inclined blade.

903. A multistage compressor is to be designed for a given flow rate pressure ratio. If the compressor consists of axial flow stages followed by centrifugal instead of only axial flow stages then the _____

(a) Axial length of the compersor would be increased
(b) Overall diameter would be increased
(c) Overall diameter would be decreased
(d) Axial length of the compressor would be decreased

UPRVUNL AE 21.08.2016

Ans. : (b) A multistage compressor is to be designed for a given flow rate pressure ratio. If the compressor consists of axial flow stages followed by centrifugal instead of only axial flow stages then the overall diameter would be increased.

904. Cylinder clearance in a compressor should be

(a) as large as possible
(b) as small as possible
(c) about 50% of swept volume
(d) about 40% of swept volume

GPSC Lect. 23.10.2016

Ans. (b) : Cylinder clearance in a compressor should be as small as possible, because the high pressure, is produced and increase its efficiency.

905. Match list 1 with list 2 and select the answer from the code given below:

List-I (Equipment in a refrigeration system)	List-II (Purpose)
A. Compressor	i. Enthalpy remains constant
B. Evaporator	ii. Enthalpy increases
C. Throttle valve	iii. Enthalpy increases but pressure remains constant
D. Condenser	iv. Enthalpy decreases but pressure remains constant

	A	B	C	D
(a)	iii	ii	i	iv
(b)	ii	iii	iv	i
(c)	ii	iii	i	iv
(d)	iv	ii	iii	i

RPSC IOF & Boiler 20.09.2017

Ans. : (c)

A	B	C	D
ii	iii	i	iv

906. The work done in compressing 1 kg of air in compression will be least when the value of index is

(a) 1 (b) 1.1
(c) 1.2 (d) 1.41

BPSC Poly. Lect. 2016

Ans : (a) The work done on the air is minimum when the compression is Isothermal (when n=1) and it is maximum when the Compression is isentropic (when $n=\gamma$) because isothermal line has less slope then isentropic line

$$\left(\frac{dP}{dV}\right)_{T=C} = -\left(\frac{P}{V}\right)$$

$$\left(\frac{dP}{dV}\right)_{Adiabatic} = -\gamma\left(\frac{P}{V}\right)$$

We seen that area under adiabatic curve in P-V diagram is more as compared with isothermal curve in P-V diagram.

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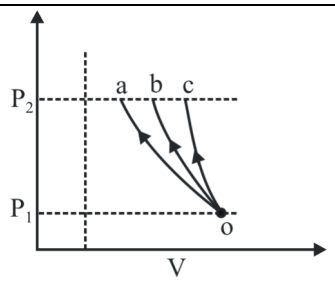


process (o-a) → Isothermal process → n = 1
process (o-b) → Polytropic process → n = 1.3
process (o-c) → Adiabatic process → n = 1.4

907. Multistage compression leads to:

(a) decreased vol. efficiency for a given pressure ratio
(b) increased vol. efficiency for a given pressure ratio
(c) more cost
(d) more noise

(HPPSC LECT. 2016)

Ans. : (b) Advantages of multistage compression:-
(i) It improves the volumetric efficiency for the given pressure ratio.
(ii) It reduces the leakage loss considerably
(iii) It gives more uniform torque and hence a smaller size flywheel is required
(iv) It reduces the cost of compressor

908. What will happen to the volumetric efficiency with increasing pressure ratio in case of single stage compression in compressor?

(a) Decreases (b) Increases
(c) Remains unaffected (d) None of these

UPPSC AE 12.04.2016 Paper-II

Ans. : (a) Volumetric efficiency:- It is the ratio of volume of free air delivery per stroke to the swept volume of the piston. The volumetric efficiency of a reciprocating air compressor with clearance volume is given by

$$\eta_{vol} = 1 + C - C\left(P_2/P_1\right)^{1/n}$$

$$C = \text{clearance ratio} = \frac{\text{Clearance volume}}{\text{Swept volume of the piston}}$$

P_1= Initial pressure of air
P_2 = Final pressure of air
n = Polytropic index
if (P_2/P_1) increases then volumetric efficiency decreases.

$$\eta_{vol} \uparrow \text{ when } \left(P_2/P_1\right) \downarrow$$

909. The compression work requirement is minimum in case of compression process being

(a) Adiabatic (b) Isochoric
(c) Isothermal (d) Hyperbolic

UPPSC AE 12.04.2016 Paper-II

Ans. : (c) The compression work requirement is minimum is case of compression process being isothermal.

(i) Work done during adiabatic compression

$$= \frac{\gamma}{\gamma - 1}(p_2 v_2 - p_1 v_1)$$

(ii) Work done during polytrophic compression

$$= \frac{n}{n - 1}(p_2 v_2 - p_1 v_1)$$

(iii) Work done during isothermal compression

$$= p_1 v_1 \ln(p_2/p_1)$$

910. The mass flow rate of air compressed in axial flow compressor is _____ centrifugal compressor.

(a) Less than (b) Equal to
(c) Greater than (d) Unpredictable

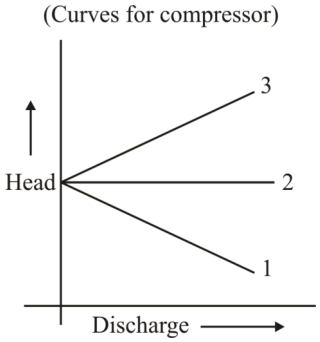
APPSC AEE 2016

Ans. (c) : The mass flow rate of air compressed in axial flow compressor is greater than centrifugal compressor.

911. List - I

(a) $\beta_2 < 90^o$
(b) $\beta_2 = 90^o$
(c) $\beta_2 > 90^o$

List - II



(a) A-1, B-2, C-3 (b) A-1, B-3, C-2
(c) A-2, B-1, C-3 (d) A-3, B-2, C-1

TSPSC AEE 2015

Ans. (a)

Curve	Outlet vane angle (β_2)
1	$\beta_2 < 90^o$
2	$\beta_2 = 90^o$
3	$\beta_2 > 90^o$

912. A three stage compressor with perfect intercooling between stages, compresses air from 1 bar to 27 bar. The optimum pressure in the last inter cooler will be

(a) 1 bar (b) 3 bar
(c) 9 bar (d) 27 bar
(e) $\sqrt{27}$ bar

CGPSC AE 26.04.2015 Shift-II

Ans. (c) : given as,
$\eta = 3$, P_1= 1 bar , P_6= 27 bar
For perfect intercooling

$$\left[\frac{P_2}{P_1}\right]^3 = \left[\frac{P_6}{P_1}\right] = \left[\frac{27}{1}\right]$$

$$\boxed{P_2 = 3\text{bar}}$$

$$\frac{P_4}{P_3} = \frac{P_2}{P_1}$$

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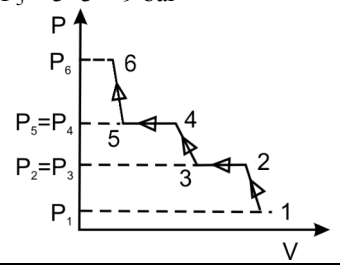
$$P_4 = \frac{P_2 \times P_3}{P_1} = \frac{P_2 \times P_2}{P_1} \qquad (P_2 = P_3)$$

$P_4 = P_5 = 3 \times 3 = 9$ bar



913. Consider the following statements. The volumetric efficiency of a reciprocating compressor depends on
(1) Clearance factor
(2) Pressure ratio
(3) Index of expansion
(4) Index of compression
of these statements
(a) (1) and (2) are correct
(b) (1) and (3) are correct
(c) (1), (2) and (3) are correct
(d) (1), (2) and (4) are correct

TSPSC AEE 2015

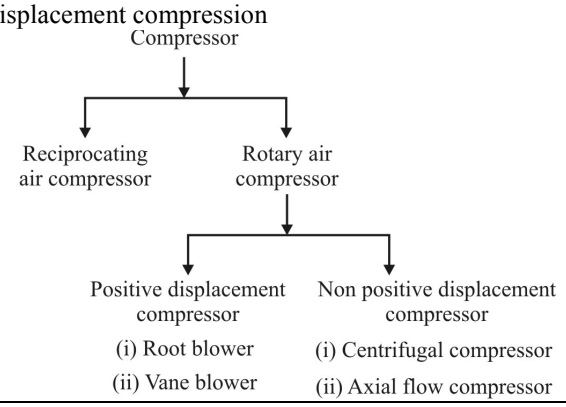
Ans. (c) : The volumetric efficiency of a reciprocating compressor depends on clearance factor, pressure ratio and index of expansion.

$$\eta_v = 1 + C - C\left[\frac{P_H}{P_L}\right]^{\frac{1}{n}}$$

914. The positive displacement compressor is
(a) Roots blower compressor
(b) Axial flow compressor
(c) Centrifugal compressor
(d) Condenser

TSPSC AEE 2015

Ans. : (a) Roots blower compressor is positive displacement compression



915. The volumetric efficiency of a reciprocation compressor depends upon
(a) Clearance volume
(b) Pressure ratio
(c) Index of expansion
(d) All of the above

GPSC AE 26.07.2015

Ans. (d) : $\eta_v = 1 + C - C\left(\frac{P_H}{P_L}\right)^{\frac{1}{n}}$

$\frac{P_H}{P_L} \rightarrow$ Pressure ratio

$C \rightarrow$ Clearance ratio

$\eta \rightarrow$ Polytropic index

916. The capacity of a reciprocating air compressor expressed in m^3/min refers to volume of
(a) Receiver tank
(b) Compressor displacement
(c) Free air delivered
(d) Compressed air discharge

GPSC AE 26.07.2015

Ans. (c) : The capacity of a reciprocating air compressor expressed in m^3/min refers to volume of free air delivered.

917. In axial flow fans and turbines fluid enters and leaves as follows
(a) radially, axially
(b) axially, axially
(c) axially, radially
(d) combination of axial and radial

TNPSC AE 2014

Ans. (b) : An axial flow fans and turbines fluid enters and leaves as axially and axially respectively.

918. Aeroplanes employ following type of compressor :
(a) radial flow (b) axial flow
(c) centrifugal (d) combination of above

J&K PSC Civil Services Pre, 2013

Ans. (b) : Aeroplanes employ following type of compressor axial flow because high discharge at low pressure.

919. A compressor at high altitude will draw :
(a) more power (b) less power
(c) same power (d) none of the above

J&K PSC Civil Services Pre, 2013

Ans. (b) : A compressor at high altitude will draw less power because density of air is decreases.

920. Select the wrong statement about the clearance volume and volumetric efficiency of a reciprocating, air compressor.
(a) The volumetric efficiency decreases with increased clearance volume.
(b) The volumetric efficiency decreases with increased pressure ratio.
(c) The volumetric efficiency increases with increase pressure ratio.
(d) The clearance volume has no effect on work done per kg of air delivered.

RPSC AEN Pre-2013

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Ans. (c) : $\eta_{Volumetric} = 1 + c - c\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}}$

$$\eta_{Volumetric} = 1 - c\left[\left(\frac{P_2}{P_1}\right)^{1/n} - 1\right]$$

$r_p = \frac{P_2}{P_1}$ = Pressure ratio

c = clearance volume

- $\eta \downarrow$ with $(r_p) \uparrow$
- $\eta \downarrow$ with $c \uparrow$

921. Screw compressors are used for.

(a) air compression
(b) refrigerant compression
(c) Slurry motion
(d) all of the above

MPSC Dy. Engg. (GWD) 19.05.2013

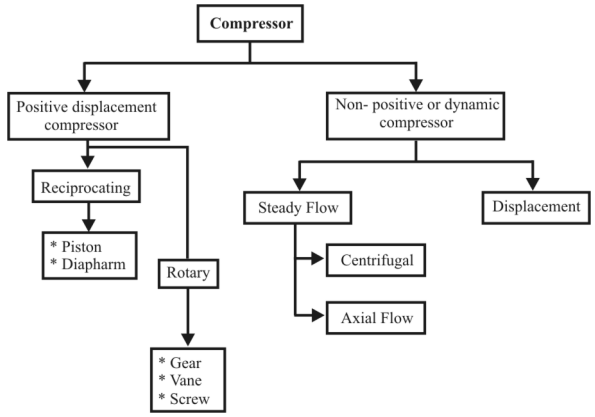
Ans. (b) : Screw Compressor–Screw compressors are used for refrigerant compression. Rotary screw compressors contain two meshed male and female rotors that rotate together in opposite direction. As the air travels through the rotors, the volume of space between the rotors decreases, compressing the refrigerant.

922. Which of the following is a non-positive (Dynamic) type compressor?

(a) Axial flow (b) Vane type
(c) Lobe type (d) Screw type

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Axial flow is a (non-positive) or rotor dynamic type compressor.



923. Method adopted for increasing isothermal efficiency of compressor.

(a) Water jacketing (b) Inter-cooling
(c) External fins. (d) All of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Method adopted for increasing isothermal efficiency of compressor.
1. Water jacketing
2. Inter-cooling
3. External fins.

924. A $3m^3$/ min compressor means that it.

(a) Compresses $3m^3$/min of standard air.
(b) Compresses $3m^3$/ min of free air.
(c) delivers $3m^3$/min of compressed air.
(d) delivers $3m^3$/ min of compressed air at delivery pressure.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : A $3m^3$/min compressor means that it compresses $3m^3$/min of free air.

925. Centrifugal compressor does not have advantage over reciprocating compressor for/in

(a) Considerably high efficiency
(b) Large size compressor needs less floor area
(c) Higher pressure increase per stage
(d) quiet in operation.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : Advantage of centrifugal compressor over reciprocating compressor.
(1) Centrifugal compressor have better isothermal and isentropic efficiency.
(2) Centrifugal compressor requires less floor area.
(3) Centrifugal compressors are quiet in operating comparatively.

926. The static pressure rise in a centrifugal compressor occurs in

(a) Impeller and diffuser
(b) Diffuser and volute
(c) Impeller and volute
(d) Impeller, diffuser and volute

TNPSC AE 2013

Ans. (a) : The static pressure rise in a centrifugal compressor occurs in impeller, diffuser.

927. The work done factor for an axial compressor varies from

(a) 0.5 to 0.75 (b) 0.6 to 0.8
(c) 0.82 to 0.73 (d) 0.98 to 0.85

TNPSC AE 2013

Ans. (d) : The work done factor for an axial compressor varies from 0.98 to 0.85.

928. The separation of flow from the blade surface

(a) Surging (b) Priming
(c) Stalling (d) Governing

TNPSC AE 2013

Ans. (c) : Stalling– It is a situation of separation of air flow at the aero-foil blades of the compressor. This phenomenon depending upon the blade-profile leads to reduced compression and drop in engine power. Flow separation occur on the suction side of the blade.

929. A blower handles 1 kg/s of air at 20°C and consumes 10 kW of power. If the inlet and exit velocities are 100 m/s and 50 m/s respectively and the specific heat is 1 kJ/kg K, the exit temperature is

(a) 299.25 K (b) 306.75 K
(c) 299.75 K (d) 279.25 K

TNPSC AE 2013

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Ans. (a) : Data given as—

$\dot{m}$ = 1 kg/s $T_1 = 20^oC$

$P_i = 10$ kW $T_2 = ?$

$V_1 = 100$ m/s

$V_2 = 50$ m/s

c = 1 kJ/kgK

Power consumes by blower is used to change in kinetic energy and change in internal energy of the air.

$$P_i = \dot{m}c[T_2 - T_1] + \frac{1}{2}\dot{m}[V_2^2 - V_1^2]$$

$$10\times10^3 = 1\times1000\times[T_2 - 20] + \frac{1}{2}\times1\times[50^2 - 100^2]$$

$$10000 = 1000\times[T_2 - 20] - \left(\frac{1}{2}\times150\times50\right)$$

$T_2 = 26.25^oC$

$\boxed{T_2 = 299.25\ K}$

930. A single-stage reciprocating air compressor is required to compress 1 kg of air from 1 bar to 4 bar. The initial temperature is 27 ºC. The work developed during isentropic compression is

(a) 146.96 kJ (b) 146.04 kJ

(c) 146.63 kJ (d) 146.89 kJ

JPSC AE 2013 Paper-V

Ans. (c) :

$$w = \frac{n}{n-1}mRT_1\left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$$

$$= \frac{1.4}{0.4}\times1\times0.287\times300\left[\left(\frac{4}{1}\right)^{\frac{0.4}{1.4}} - 1\right]$$

$$= 146.45\,kJ$$

$$\approx 146.63\,kJ$$

931. What is the minimum work required to compress 1 kg of air from 1 bar, 27 ºC to 16 bar in two stages, if the law of compression is $pv^{1.25}$ =constant and the inter-cooling is perfect?

[Take : R = 287 J/kg-K]

(a) 275.09 kN-m (b) 175.09 kN-m

(c) 375.09 kN-m (d) 475.09 kN-m

JPSC AE 2013 Paper-V

Ans. (a) : In case of perfect intercooling in multi stage compressor

$W_{Total} = n \times W_{each\ stage}$

n = No. of stage

Work done in each stage is same.

Given,

$P_1 = 1$ bar

$P_3 = 16$ bar

m = 1 kg

n = 1.25

$P_2 = \sqrt{P_1.P_3}$

$$P_2 = \sqrt{16\times1}$$

$$= 4\,bar$$

$$W = \frac{n}{n-1}P_1V_1\left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$$

$$= \frac{n}{n-1}mRT_1\left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$$

$$= \frac{1.25}{1.25-1}\times1\times0.287\times300\left[\left(\frac{4}{1}\right)^{\frac{1.25-1}{1.25}} - 1\right]$$

$$= 137.54\,kN-m$$

$$W_{Total} = 2\times W$$

$$= 137.54\times2$$

$$= 275.09\,kN-m$$

932. In centrifugal compressors, pressure ratio may be increased:-

(a) By decreasing slip factor

(b) By increasing slip factor

(c) By making rough blades

(d) By none of the above

UKPSC AE-2013, Paper-II

Ans. (a) : In centrifugal compressors, pressure ratio may be increased by decreasing slip factor.

933. Which of the following compressors has highest capacity?

(a) Reciprocating (b) Centrifugal

(c) Axial flow (d) Fan

UKPSC AE-2013, Paper-II

Ans. (c) : Axial flow compressors has highest capacity.

934. Surging can be reduced by :

(a) Increasing the speed of the fan

(b) Reducing the speed of the fan

(c) Employing fixed guide vanes

(d) Reducing the flow rate

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

935. Centrifugal compressor is suitable for :

(a) Low specific speed, high pressure ratio and low mass flow rate

(b) High specific speed

(c) Low pressure ratio

(d) High specific speed

TNPSC AE (Industries) 09.06.2013

Ans. (c) : Centrifugal compressor is suitable for low pressure ratio.

936. In an aircraft gas turbine, the axial flow compressor is preferred because of :

(a) High pressure rise (b) Low frontal area

(c) High thrust (d) High propulsion

JPSC AE, 10.04.2021 Paper-I

APGENCO AE 2012

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Ans. (b) : In an aircraft gas turbine, the axial flow compressor is preferred because of low frontal area. In axial compressors the gas flows parallel to the axis of rotation of the rotor, which many rows of aerodynamically shaped blades extending radially outward.

937. The pressure ratio of an ideal vaned compressor with increase in mass flow rate
(a) increases
(b) decreases
(c) remains constant
(d) first decreases and then increases

BPSC AE 2012 Paper - V

Ans. : (c) : The pressure ratio of an ideal vaned compressor with increase in mass flow rate remains constant.

938. In centrifugal blowers/compressors, the ratio of outlet whirl velocity to the blade velocity is known as
(a) work factor (b) slip factor
(c) degree of reaction (d) pressure coefficient

UKPSC AE 2012 Paper–II

Ans. (b) : In centrifugal blowers/compressors, the ratio of outlet whirl velocity to the blade velocity is known as slip factor.

939. When an impeller has backward curved vanes in a centrifugal blower, then with an increase in flow rate, Euler head H
(a) increases (b) decreases
(c) remains constant (d) none of the above

UKPSC AE 2012 Paper–II

Ans. (b) : decreases

940. Performance of an air compressor at high altitudes as compared to that at sea level is
(a) better
(b) inferior
(c) same
(d) depends on type of compressor

UKPSC AE 2012 Paper–II

Ans. (b) : Performance of an air compressor at high altitudes as compared to that at sea level is inferior.

941. Unstable operation of the compressor manifested by a cycle reversal of pressure gradient and flow in the delivery pipe is known as
(a) Knocking (b) Chocking
(c) Surging (d) none of these

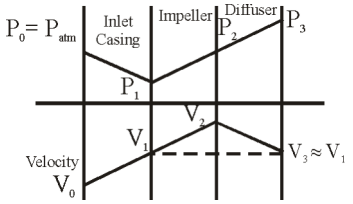
TNPSC ACF 2012

Ans. (c) : Surging- It is phenomenon which is caused due to unsteady periodic and reversal of flow through the compressor when compressor has to operated at less mass flow rate than a predetermined value.

942. In the diffuser of centrifugal compressor, the air is
(a) accelerated with a rise in static pressure
(b) accelerated with a decrease in static pressure
(c) decelerated with a decrease in static pressure
(d) decelerated with a rise in static pressure

TNPSC ACF 2012

Ans. (d)



In the diffusion of centrifugal compressor, the air is decelerated with a rise in static pressure.

943. The capacity of a compressor is specified as 50 cubic metre per minute. It implies that
(a) It is capable of supplying 50 cu. m of compressed air per min
(b) Compressor is capable of compressing 50 cu. m of free air per min
(c) Compressor has storage tank capable of storing 50 cu. m of air
(d) One kg of compressed air as delivered by the compressor would expand to 50 cu.m under NTP conditions

APPSC AEE 2012

Ans. (b) : The capacity of a compressor is specified as 50 Cubic meter/minute, it implies that compressor is capable of compressing 50 Cu. m of free air per min.

944. Degree of reaction of an axial flow compressor is defined as the ratio of
(a) static enthalpy rise in the rotor to that in the whole stage
(b) static enthalpy rise in the whole stage to that in the rotor
(c) static enthalpy decrease in the rotor to that in the whole stage
(d) static enthalpy decrease in the whole stage to that in the rotor.

TNPSC ACF 2012

Ans. (a) : Degree of reaction of an axial flow compressor defined as the ratio of static enthalpy rise in the rotor to that in the whole stage.

$$R_D = \frac{(\Delta h)_{MB}}{(\Delta h)_{stage}} = \frac{(\Delta h)_{MB}}{(\Delta h)_{MB} + (\Delta h)_{FB}}$$

[MB = Moving Blade, FB = Fixed Blade]

945. Which is not true in case of Multi-stage compression ?
(a) Improves overall volumetric efficiency
(b) Reduce total driving power
(c) Requires a big flywheel
(d) Reduces leakage losses

OPSC Civil Services Pre. 2011

Ans. (c) :
- In multistage compression due to low pressure ratio, the volumetric efficiency improves.
- Inter cooling between stages reduces work consumption. since compression is in multi stages, the torque fluctuations will be small i.e. uniform torque. So, small flywheel required.

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946. How is the air compressed in a double acting compressor?
(a) In one cylinder
(b) In two cylinders
(c) In a single cylinder on both sides of the piston
(d) In two cylinders on both sides of the piston

VIZAG STEEL MT 2011

Ans. (c) : Air compressed in a single cylinder on both side of piston is called double acting compressor.

947. In a centrifugal compressor the stagnation temperature across the impeller
(a) increases
(b) decreases
(c) does not show a monotonic trend
(d) remains constant

DRDO Scientists 2008

Ans. (a) : In a centrifugal compressor the stagnation temperature across the impeller increases.

948. The performance of reciprocating compressor is compared on the basis of efficiency
(a) volumetric (b) mechanical
(c) overall (d) Isothermal

UKPSC AE 2007 Paper -II

Ans. (d) : Isothermal

949. In a centrifugal compressor the pressure developed depends on
(a) Impeller tip velocity (b) Inlet temperature
(c) Compression index (d) All of the above

UKPSC AE 2007 Paper -II

Ans. (d) : All of the above

950. While measuring the volume flow rate of air in a compressor a drum is added to the system. The objective is :
(a) to increase pressure
(b) to increase mass flow
(c) to increase swept volume
(d) to reduce fluctuations

J & K PSC Screening, 2006

Ans. (d) :

951. The suction pipe diameter of refrigerating unit compressor in comparison to delivery pipe diameter is
(a) larger
(b) smaller
(c) equal
(d) dependant on capacity

J & K PSC Screening, 2006

Ans. (a) : The suction pipe diameter of refrigerating unit compressor in comparison to delivery pipe diameter is larger because during suction refrigerant in form of vapour but during delivery it is in liquid phase.

952. The capacity of a compressor is $5m^3$/ min refers to:
(a) Standard air
(b) Free air delivery
(c) Compressed air
(d) Compressed air at delivery pressure

J & K PSC Screening, 2006

Ans. (b) : The capacity of a compressor is 5m/min refers to free air delivery.

953. In refrigeration, reciprocating compressors are very suitable for
(a) Small displacement and low condensing pressure
(b) Small displacement and high condensing pressure
(c) Large displacement and low condensing pressure
(d) Large displacement and high condensing pressure

OPSC Civil Services Pre 2006

Ans. (b) : Reciprocating compressor are used for high pressure ratio and low discharge (or displacement)

954. If n is the index of compression, then the work done in compressing air is proportional to
(a) n (b) n^2
(c) $\left(\frac{n}{n-1}\right)$ (d) $\left(\frac{n-1}{n}\right)$

OPSC Civil Services Pre 2006

Ans. (c) : Work done during polytropic compression

$$W = \frac{n}{n-1}\left(P_1V_1 - P_2V_2\right)$$

$$W \propto \frac{n}{n-1}$$

8. Nuclear Power Plant

955. The function of a moderator in a nuclear reactor is to
(a) slow down the fast moving electrons
(b) speed up the slow moving electrons
(c) start the chain reaction
(d) transfer heat produced inside the reactor to a heat exchanger

RPSC IOF, 2020, RPSC AE 2016
TNPSC AE 2014
TNPSC AE (Industries) 09.06.2013
TNPSC ACF 2012, RPSC ACF-2011
CSE Pre-2003, 2001

Ans. : (a) The function of a moderator in a nuclear reactor is to slow down the fast moving electrons.
Moderator:- A moderator is a material introduced into the fuel mass in order to slow down the speed of fast moving electrons. the slowing down of fast neutrons is desirable because slow moving electrons are more effective than fast neutrons in triggering fission. A moderator, apart from its high neutron slowing power and low non-productive neutron, should be
(i) Stable under nuclear radiation
(ii) Corrosion resistant
(iii) Good thermal conductor
(iv) Available in adequate

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956. In a pressurized water reactor

(a) coolant water is pressurized to work as moderator
(b) coolant water boils in the core of the reactor
(c) coolant water is pressurized to prevent boiling of water in the core
(d) no moderator is used

Nagaland PSC (CTSE) 2016, Paper-II
RPSC AE 2016
TNPSC AE 2014
RPSC ACF-2011
APPSC IOF, 2009

Ans. (a) : The pressurized water reactor (PWR) is a type of nuclear reactor used to the generate electricity and propel nuclear submarines and novel vessels. They make used of light water (Ordinary as opposed to heavy water) as their coolant and neutron moderator. Nuclear fission produces heat inside the reactor.

957. The most commonly used moderator in nuclear power plants is

(a) heavy water (b) graphite
(c) steel (d) concrete and bricks

GPSC DEE Class-2 (GWSSB) 04.07.2021
Assam PSC AE (PHED), 18.10.2020
GPSC Executive Engineer 23.12.2018
ESE 2000

Ans. (a) : The most commonly used moderator in nuclear power plants is heavy water.

958. The combination of fuel and moderator for CANDU reactor is:

(a) Enriched Uranium, Hard water
(b) Natural Uranium, Hard water
(c) Enriched Uranium, Water
(d) Natural Uranium, Water

OPSC AEE 2019 PAPER - II
RPSC AE 2016, RPSC ACF-2011
CSE Pre-2004

Ans. : (b) : The combination of fuel and moderator for CANDU reactor is natural uranium and hard water.

959. In a nuclear reactor the function of a reflector is to:

(a) reduce the speed of the neutrons
(b) stop the chain reaction
(c) reflect the escaping neutrons back into the core
(d) all of the above

RPSC AE 2016
UKPSC AE-2013, Paper-II
RPSC ACF-2011

Ans. (c) : A reflector is a region of unfudded material surrounding the core. It function is to scatter neutrons that leak from the core, thereby returning some of them back into the core. This design feature allows for a smaller core size. In addition, reflectors "smooth out" the power density by utilizing neutrons that would otherwise leak out through fissioning within fuel material located near the core's outer region.

960. Which material is the most commonly used moderator?

(a) Graphite (b) Sodium
(c) Deuterium (d) any of the above

RPSC AE 2016, RPSC ACF-2011
TNPSC AE, 2008

Ans. (a) : Moderator is used to slow down the fast moving neutrons. Most commonly used moderator, are graphite and heavy water (D_2O). Water cannot be used as moderator in a nuclear reactor.

961. A boiling water reactor uses which of the following as fuel?

(a) Enriched uranium (b) Plutonium
(c) U^{234} (d) U^{235}

BPSC AE Mains 2017 Paper - V
TNPSC AE 2013
BPSC AE 2012 Paper - V

Ans. : (a) : Boiling water reactors must use enriched uranium as their nuclear fuel, due to their use of light water. This is because light water absorbs too many neutrons to be used with natural uranium, so the fuel content of fissile uranium 235 must be increased.

962. The first nuclear power plant in India is located at

(a) Kota (b) Kalapakkam
(c) Tarapur (d) Bareilly

TNCSC AE 2020
GWSSB AAE, 27.12.2015
BPSC AE 2012 Paper - V

Ans. : (c) : The first nuclear power plant in India is located at Tarapur (Maharashtra)

963. If k is the ratio of the rate of production of neutrons to the rate of loss of neutrons, the reactor is called a critical reactor when -

(a) $k = 1$ (b) $0 < k < 1$
(c) $k = 0$ (d) $k > 1$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
BPSC AE Mains 2017 Paper - V
CSE Pre-2003

Ans. (a) : The multiplication factor is

$$k = \frac{\text{Rate of production of neutrons}}{\text{Rate of loss of neutrons}}$$

If $k = 1 \rightarrow$ Critical reactor
$k > 1 \rightarrow$ Explosion occurs
$k < 1 \rightarrow$ The chain reaction comes to half

964. The function of control rods in nuclear power plant is to:

(a) Control absorption of neutrons
(b) Control temperature
(c) Control the fuel consumption
(d) None of the above.

ISRO Scientist/Engineer (RAC) 22.04.2018
RPSC AE GWD, 2011

Ans. (a) : Control rods are used in nuclear reactors to control the fission rate of uranium and plutonium. They are composed of chemical elements such as boron, silver, indium and cadmium that are capable of absorbing neutrons without themselves fissioning.

Search on Telegram @apna_pdf

965. The function of a moderator is to

(a) Absorb the part of the Kinetic energy of the neutrons
(b) Extract the heat
(c) Reflect back some of the neutrons
(d) Start the reactor

Nagaland PSC (CTSE) 2017, 2016 Paper-II
RPSC AE GWD, 2011

Ans. (a) : Function of moderator is used to slow down the neutrons by absorbing some of the kinetic energy of neutrons by direct collision, which result the increasing the chances of fission.

966. Enriched uranium is required as a fuel in a nuclear reactor, if light water is used as a coolant and a moderator, because light water has :

(a) high neutron absorption cross-section.
(b) low moderating efficiency
(c) low neutron absorption cross -section
(d) high neutron scatter cross-section.

RPSC ACF & FRO, 26.02.2021
ESE 1994

Ans. (b) : Enriched uranium is required as a fuel in a nuclear reactor, if light water is used as moderator and coolant, because light water has low moderating efficiency.

967. The energy released during the fission of one atom of uranium-235 in million electron volt is about

(a) 100 (b) 200
(c) 300 (d) 400

Assam PSC AE (PHED), 18.10.2020
ESE 1999, 1995, 1993

Ans. (b) : The total energy released in fission of an atomic nucleus varies with the precise break up, but average about 200 million electron volt.
For U-235 or 3.2×10^{-11} joule. This is about 82 TJ/kg that from U-233 is about the same, and that from P_u -239 is about 210 million electron volt per fission.

968. Fast breeder reactions use

(a) water as moderator
(b) carbon dioxide as moderator
(c) graphite as moderator
(d) no moderator

TNPSC AE, 2008, CSE Pre-2002

Ans. (d) : No moderator is used in the breeder reactor since fast neutrons are more efficient in transmitting U-238 to PU-239. At this concentration of U-235, the cross-section for fission with fast neutrons is sufficient to sustain the chain reaction. Using water as coolant would slow down the neutrons, but the use of liquid sodium avoids that moderation and provides a very efficient heat transfer medium.

969. Assertion (A) : Pressurised water reactor (PWR) nuclear power plant use superheated steam.
Reason (R) : An increase in superheat at constant pressure increases the cycle efficiency.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

Punjab PSC (Lect.) 06.08.2017
CSE Pre-2001

Ans. (a) : Both (A) and (R) are true and (R) is the correct explanation of (A).

970. Which one of the following pairs of materials is used as moderators in nuclear reactors?

(a) Cadmium and heavy water
(b) Heavy water and zirconium
(c) Zirconium and beryllium
(d) Beryllium and heavy water
(e) Cadmium and beryllium

CGPSC AE 26.04.2015 Shift-II
CSE Pre-1995

Ans. (d) : Beryllium and heavy water, pairs of materials is used as moderators in nuclear reactors.

971. Shielding in a nuclear reactor is generally done to protect against

(a) excess electrons
(b) X-rays
(c) α and β rays
(d) neutron and gamma rays

BPSC AE Mains 2017 Paper - V
CSE Pre-2000

Ans. : (d) : The shielding of charged particle radiation is therefore much less severe problem and shielding is done to protect against highly energetic neutron and gamma rays.

972. In nuclear reactors, control rod is made of

(a) lead and tin
(b) boron and cadmium
(c) graphic
(d) zinc

TNPSC AE (Industries) 09.06.2013
TNPSC AE, 2008

Ans. (b) : Control rods are used in nuclear reactors to control the fission rate of Uranium and Plutonium. They are composed of chemical elements such as Boron, Cadmium, silver and Indium that are capable of absorbing many neutrons without themselves fissioning.

973. Fast breeder reactors are best suited for India because of :

(a) large thorium deposits
(b) large uranium deposits
(c) large plutonium deposits
(d) all of the above

RPSC AE 2016, RPSC ACF-2011

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Ans. (a) : Fast breeder reactors are most suitable in India, because of largest thorium deposits.

974. Thermal shielding is provided to :
(a) protect the walls of the reactor from radiation damage
(b) prevent melt down of the core of the reactor
(c) protect the operating personnel from exposure to radiation
(d) all of the above

RPSC AE 2016, RPSC ACF-2011

Ans. (d) : Thermal shielding is provided to –
(i) Protect the walls of the reaction from radiation damage.
(ii) Prevent melt down of the case of the reactor.
(iii) Protect the operating personnel from exposure to radiations.

975. Fission chain reaction is possible when :
(a) fission produces the same number of neutrons which are absorbed
(b) fission produces more electrons than are absorbed
(c) fission produces less electrons than are absorbed
(d) none of the above

RPSC AE 2016, RPSC ACF-2011

Ans. (b) : In nuclear fission, an unstable atoms splits into two or more smaller pieces that are more stable and releases energy in the process, the fission process also releases extra neutrons, which can then split additional atoms resulting in a chain reaction that releases a lot of energy.

976. The conversion ratio of a breeder reactor is :
(a) equal to unity (b) more than unity
(c) less than unity (d) none of the above

RPSC AE 2016, RPSC ACF-2011

Ans. (b) : In a breeder reactor the conversion ratio is higher than 1. Break-even is achieved when the conversion ratio reaches 1.0 and the reactor produces as much fissile material as it uses.

977. Which is not a preferable property of coolant used in nuclear reactor?
(a) Must have low boiling point
(b) Must have low melting point
(c) Must not absorb neutrons
(d) Must be non-oxidising

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II

Ans. (a) : Nuclear reactor have certain specific requirements of coolant–
(1) High specific heat thermal conductivity and low viscosity.
(2) Low corrosive and errosive effects on all the reactor materials.
(3) High boiling and low melting point.
(4) High thermal and radiation resistance.
(5) Low neutron absorption & low neutron activation.

978. When the control rods are inserted into the reactor, K (Multiplication factor) becomes
(a) 0 (b) >1
(c) 1 (d) <1

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (d) : When the control rods are inserted into the reactor i.e. neutron production is less than losses due to obsorbtion and leakage, then the reactor is subcritical and the multiplication factor becomes less than 1.

979. The function of coolant is to
(a) Extract heat from reactor
(b) Slow down neutrons
(c) Control the reaction
(d) Reflect the neutrons

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (a) : The function of coolant is to extract heat from reactor and provide system at stable temperature.

980. Reactors designed for propulsion applications uses.
(a) Natural uranium (b) Any uranium
(c) Pure uranium (d) Enriched uranium

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Reactors designed for propulsion applications are designed for enriched uranium.

981. The correct sequence for radioactive isotopes in decreasing half life time :
(a) U^{239} Pu^{239} U^{233} (b) U^{233} Pu^{239} U^{233}
(c) U^{239} Pu^{233} U^{239} (d) U^{233} Pu^{239} U^{239}

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : $U^{233} > Pu^{239} > U^{239}$

982. The correct combination for moderator and reactor are :

(1) Graphite		(I) Pressurised water
(2) Water		(II) Liquid metal cooled
(3) No moderator		(III) Breeder

	(1)	(2)	(3)
(a)	III	II	I
(b)	I	II	III
(c)	II	III	I
(d)	II	I	III

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : Graphite → Liquid metal cooled
Water → Pressurized water
No moderator → Breeder

983. Boiling water reactor and pressurized water reactor are
(a) nuclear reactors (b) solar reactors
(c) OTEC (d) biogas reactors

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : Boiling water reactor and pressurized water reactor are nuclear reactors. A boiling water reactor is a type of light water nuclear reactor used for the generation of electric power. The main difference between a BWR and PWR is that in a BWR, the reactor core heats water. Which turns to steam and then drives a steam turbine. In a PWR the reactor core heats water, which does not boil.

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984. With reference to nuclear fuels, U-238 has a _____ half life than U-235.

(a) longer
(b) same
(c) shorter
(d) incorrect equation, one is non-radioactive

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) :
Half life of U- 235 = 700 million years = 7×10^8 years
Half life of U- 238 = 4.46×10^9 years

985. The rate of nuclear fission reaction is varied by changing

(a) Coolant temperature
(b) Control rod position
(c) Amount of fuel
(d) Pressure relief valve setting

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : The rate of nuclear fission reaction is varied by changing control rod position.

986. A moderator, in nuclear power plants, is a medium introduced into the fuel mass in order to-

(a) Slow down the speed of fast moving neutrons
(b) Control the reaction
(c) Reduce the temperature
(d) Extract heat from nuclear reaction

MECON MT 2020

Ans. (a) : A moderator, in nuclear power plants, is a medium introduced into the fuel mass in order to slow down the speed of fast moving neutrons.

987. The nuclear radiators shielded are

(a) Electrons
(b) α, β and γ rays
(c) Neutrons and gamma rays
(d) Protons decay

TNCSC AE 2020

Ans. (c) : The nuclear radiators produced in a reactor which must be shielded are neutron and gamma rays. Depleted uranium is used for shielding of a reactor care can be provided by materials of reactor pressure vessel, reactor internals (neutron reflector).

988. U_{233} is produced

(a) artificially
(b) as basic raw material
(c) when thorium is irradiated by neutrons
(d) by fission of U_{238}

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Uranium-233 is produced by the neutron irradiation of thorium-232. When thorium-232 absorbs a neutron. It becomes thorium-233, which as a half-life of only 22 minutes. Thorium-233 decays into protactinium-233 through beta decay.

989. A nuclear reaction is said to be critical when the neutron population in the reactor core is

(a) rapidly increasing leading to the point of explosion
(b) decreasing from the specified value
(c) reduced to zero
(d) constant

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Criticality means that a reactor is controlling a sufficient number of neutrons to maintain an ongoing series of reactions this is the normal state of nuclear power generation.

990. Reactors for propulsion applications are designed for

(a) any form of uranium (b) natural uranium
(c) enriched uranium (d) plutonium

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Reactors for propulsion applications are designed for enriched uranium.

991. Enriched uranium may contain fissioinable contents from

(a) 1 to 40% (b) 10 to 50%
(c) 1 to 80% (d) 1 to 99%

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Natural uranium is 99.284% U^{238} isotope with U^{235} only constituting about 0.711% of its mass. U^{235} is the only nuclide existing in nature that is fissile with thermal neutrons.

992. The main function of shielding in nuclear reactor is provide protection against:

(a) α-rays (b) β-rays
(c) γ-rays (d) None of these

PTCUL AE 25.06.2017

Ans. (c) : Shielding in nuclear reactor is provided to protect against all the harmful radiation but the highly penetrating neutrons and gamma rays are majorly focused.

993. A nuclear fission produces energy

(a) 20 Mev (b) 200 MeV
(c) 2000 MeV (d) 20,000 MeV

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : One fission event results in the release of about 200 MeV of energy.

994. The efficiency of a nuclear power plant in comparison to a conventional thermal power plant is

(a) Same (b) More
(c) Less
(d) May be less or more depending on size

Nagaland PSC CTSE 2017 Paper-2

Ans. (c) : The efficiency of nuclear power plant in comparison to conventional thermal power plant is less.
Typically :- Nuclear power plant efficiency– 33-37%
conventional thermal power plant (fossil fueled power plant) efficiency– 45%

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995. In nuclear plants, MOX fuels are

(a) Blend of plutonium and natural or depleted uranium
(b) Blend of carbon and depleted uranium
(c) Blend of metal fuels and natural uranium
(d) Blend of natural uranium and uranium carbide

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : In nuclear Plant, MOX (mixed oxide fuel), commonly that contain more than one oxide of material consisting of plutonium blended with natural uranium, reprocessed uranium, or depleted uranium.

996. Gas cooled reactors use the following materials as moderator and coolant respectively :

(a) Heavy water and CO_2
(b) Graphite and air
(c) Graphite and hydrogen
(d) Graphite and CO_2

Karnataka PSC AE, 10.09.2017

Ans. (d) : Gas cooled reactors use the following materials as moderator and coolant respectively graphite and CO_2.

997. Thorium breeder reactors are more promising for India essentially because

(a) They develop more power
(b) The technology there of is simple
(c) Thorium deposit are abundantly available in the country
(d) These can be easily designed

UPSC JWM 2017

Ans. (c) : Thorium deposit are abundantly available in India that's why thorium breeder reactors are more promising for India.

998. The presence of reflector in nuclear power plant results in

(a) Increase production of neutrons
(b) Controlled production of neutrons
(c) Decreases leakages of neutrons
(d) Decreases the speed of neutrons

RPSC VPITI 14.02.2016

Ans. (c) : Reflector, which is medium of flow neutron absorption and high neutrons scattering cross actions.
Some of the neutrons leaking out of the core are scattered back into it by reflector nuclei. This reduces the neutron leakage.

999. How many stable isotopes does uranium has?

(a) Two (b) Eight
(c) Twenty six (d) None of these

KPCL AE 2016

Ans. (d) : Uranium ($_{92}U$) is naturally occurring radioactive element that has no stable isotope. It has three primordial isotopes, uranium- 238, uranium- 235 and uranium-234 that have long half-life and are found in appreciable quantity in the earth curst. The decay product uranium- 234 is also found.

1000. Which of the following elements are used in control rods that are used in nuclear reactors to control the fission rate?

(a) Cadmium (b) Boron
(c) Indium (d) All the above

KPCL AE 2016

Ans. (d) : Control rods are used in nuclear reactors to control the fission rate of uranium or plutonium. Their compositions includes chemical elements such as boron, cadmium, silver or indium, that are capable of absorbing many neutrons without themselves fissioning.

1001. Which of the following is not used as moderator

(a) Water (b) Heavy water
(c) Graphite (d) Boron

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : Moderator used in nuclear reactors–
(1) Light water (H_2O)
(2) Heavy water (D_2O)
(3) Graphite (C)

- Boron is a primary neutron absorber used in construction of control rods in the core of a nuclear reactor. The control rods includes chemical elements such as boron, cadmium, silver, hafnium or indium.

1002. The reactor performs the following function as that of ..... in a steam power plant.

(a) Furnace (b) Turbine
(c) Electric generator (d) Boiler

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : The reactor produces and controls the release of energy from splitting the atoms of certain elements, the energy released is used as heat to make steam to generate electricity, like the furnace in a steam power plant.

1003. Which of the followings is not the desirable feature of moderator in nuclear reactor?

(a) It has low thermal conductivity
(b) It slows down the neutron
(c) It is non corrosive
(d) It has good chemical stability

RPSC LECTURER 16.01.2016

Ans. (a) : Modertor—The moderator, which is of importance in thermal reactors, is used to moderate, that is, to slow down, neutron from fission to thermal energies.

Key properties of moderator—

- High melting and boiling point
- High specific heat capacity
- Low viscosity
- Low corrosive
- Cheap
- Good chemical stability

1004. Which of the following is not the component of nuclear power plant

(a) Steam generator (b) Steam turbine
(c) Nuclear reactor (d) Penstock

RPSC LECTURER 16.01.2016

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Ans. (d) : Steam generator, steam turbine, nuclear reactor are the components of nuclear power plant whereas penstock is the main part of hydro-electric-power plant.

1005. Which process is responsible for production of energy in the Sun

(a) Nuclear fission reaction
(b) Nuclear fusion reaction
(c) Exothermal chemical reaction
(d) All of the above

UPPSC AE 12.04.2016 Paper-II

Ans. : (b) Nuclear fusion reaction is responsible for production of energy in the sun.
Nuclear fusion : In Nuclear physics, Nuclear fusion is a reaction in which two or more atomic nuclei come close enough to form one or more different atomic nuclei and subatomic particles (neutrons and/or protons). The difference in mass between the products and reactions is manifested as the release of large amounts of energy. This difference in mass arises due to the difference in atomic "binding energy' between the atomic nuclei before and after the reaction. Fusion in the process that powers active or "main sequence" starts, or other high magnitude starts.

1006. The working fluid used in an MHD system coupled to a fast breeder reactor is

(a) Hot flue gases (b) Seeded inert gas
(c) Liquid metal inert gas (d) Liquid metal only

UPPSC AE 12.04.2016 Paper-II

Ans. : (d) The working fluid used in an MHD system coupled to a fast breeder reactor is liquid metal only.

1007. Which impurity in water requires critical attention on very high pressure boilers

(a) Hydrogen (b) Ammonia
(c) Silica (d) Dissolved salts

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Silica in high-pressure water voltilizers and pressure over with steam to the turbine as the steam expands and drops in pressure and temperature the silica condenses and deposits on the turbine blade, cutting the machine efficiency.

1008. If B is magnetic field strength, the theoretical power output of MHD generator is proportional to:-

(a) B (b) B^2 (c) B^3 (d) $\sqrt{B}$

UKPSC AE-2013, Paper-II

Ans. (b) :

1009. The gas which is used as a coolant in a nuclear power plant is:-

(a) Freon (b) Ammonia
(c) Helium (d) Chlorine

UKPSC AE-2013, Paper-II

Ans. (c) : Helium is used as a coolant in a nuclear power plant.

1010. From the view point of moderating ratio only, which of the following is best moderator?

(a) Ordinary water (b) Carbon
(c) Heavy water (d) Beryllium

RPSC AEN Pre-2013

Ans. (c) : Heavy water (D_2O) is best moderator in nuclear power plant.

1011. Which of the following statements is not correct about MHD power generation ?

(a) Lesser thermal pollution
(b) No moving parts
(c) High operation efficiency
(d) No direct conversion of heat into electrical energy

UKPSC AE 2012 Paper–II

Ans. (d) : No direct conversion of heat into electrical energy

1012. The controls rods in the control system of nuclear reactors are used to

(a) absorb excess neutrons
(b) control fuel consumption
(c) control temperature
(d) none of these

TNPSC ACF 2012

Ans. (a) : Cadmium rod is used as an absorbing material to control the reaction rate of neutron.

1013. Spent fuel from the nuclear thermal reactor contains

(a) plutonium (b) unused fuel
(c) fission products (d) All of the above

APPSC AEE 2012

Ans. (d) : Spent fuel from the nuclear reactor contains-
- Plutonium
- Unused fuel
- Fission products

1014. Breeder reactor employs liquid-metal coolant, because it

(a) acts as good moderator
(b) produces maximum steam
(c) transfer heat from core at a fast rate
(d) increases rate of reaction in core

BPSC AE 2012 Paper - V

Ans. : (c) : Breeder reactors employ liquid metal coolant because it transfer heat from core at a fast rate. It has a large specific heat so that it is an efficient heat transfer fluid. In practice those reactors which have used liquid metal coolants have been fast-neutron reactors.

1015. Solid fuel for nuclear reactions may be fabricated into various small shapes, such as

(a) plates (b) pallets
(c) pins (d) any one of the above

BPSC AE 2012 Paper - V

Ans. : (d) : Solid fuel for nuclear reactions may be fabricated into various small shapes, such as plates, pallets and pins.

1016. The coolant used in boiling water reactor is

(a) CO_2
(b) Pressurised water
(c) mixture of water and steam
(d) liquid metal

BPSC AE 2012 Paper - V

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Ans. : (c) : A boiling water reactor (BWR) used demineralized water as a coolant and neutron moderator. Heat is produced by nuclear fission in the reactor core and this causes the cooling water to boil, producing steam.

1017. What type of energy is derive from Ocean ground water?

(a) Solar energy (b) Geo-thermal energy

(c) Nuclear energy (d) OTEC

RPSC AE GWD, 2011

Ans. (d) : OTEC (Ocean thermal energy conversion) is derived from ocean ground water.

1018. Isotope that can be fissioned by neutrons of all energies is :

(a) Pu – 239 (b) Pu – 240

(c) U – 238 (d) Th –238

OPSC Civil Services Pre. 2011

Ans. (a) : Pu - 239, isotope can be fissioned by neutrons of all energies.

1019. The false statement is :

(a) Alpha particles are helium nuclei

(b) Beta decay is accompanied by emission of neutrino and gamma radiation

(c) Neutron is converted into a proton in a positive beta decay

(d) Gamma ray is emitted from the nucleus while X–ray is emitted from the atom

OPSC Civil Services Pre. 2011

Ans. (c) : Neutron is converted into a proton in a positive beta decay.

1020. Graphite is used as a moderator in :

(a) Boiling water reactor

(b) Pressurized water reactor

(c) Light water reactor

(d) Gas cooled reactor

OPSC Civil Services Pre. 2011

Ans. (d) : Graphite is used as a moderator in gas cooled reactor.

1021. Operating charges for same power output are minimum for :

(a) gas turbine plant (b) hydroelectric plant

(c) thermal plant (d) nuclear plant

RPSC ACF-2011

Ans. (b) : Hydroelectric power plant has lowest running cost because it does not needs any fuel and can be operated by few number of persons. Nuclear plant and thermal power plant requires fuels also the handling cost of fuels is added with total cost.

1022. Heavy water is used in atomic power plants as

(a) Fuel (b) Source of energy

(c) Lubricant (d) Moderator

ISRO Scientist/Engineer 2008

Ans. (d) : Heavy water is used in atomic power plants as moderator.

1023. The size of the reactor is said to be critical when

(a) chain reaction can be initiated

(b) it becomes uncontrollable

(c) it explodes

(d) it produces no power

TNPSC AE, 2008

Ans. (a) : When the atom-splitting reactor of a nuclear power plant is operating normally, it is said to be critical or in a state of critically. Critically means that a reactor is controlling a sustained fission chain reaction, where each fission event release a sufficient number of neutrons to maintain on ongoing series reactions.

1024. The most practical fuel for a thermonuclear reactor, both from economical and nuclear considerations, is

(a) Plutonium (b) Uranium

(c) Deuterium (d) Thorium

TNPSC AE, 2008

Ans. (c) : Most commercial nuclear reactors use normal water (also called light water) as a neutron moderator.

Some reactor designs, such as the CANDU reactor, use heavy water.

A heavy water molecule contains deuterium (Hydrogen-$_1D^2$), which is an isotope of hydrogen that has a neutron in its nucleus in addition to the proton.

1025. Which of the following is the primary fuel in nuclear power?

(a) U^{233} (b) U^{238}

(c) U^{235} (d) Pu^{239}

UKPSC AE 2007 Paper -II

Ans. (c) : The primary fuel used in nuclear power plants Uranium (U_{235}).

* Isotopes of uranium are U_{234}, U_{235} and U_{238} etc.

* The secondary fuel used in nuclear power plants is U_{235} and plutonium (Pu_{239}.).

1026. In MHD power generation system, the electrical conductivity of air is increased by

(a) Heating it to very high temperatures

(b) Seeding

(c) Both (a) and (b)

(d) None of these measures

UKPSC AE 2007 Paper -II

Ans. (c) : Both (a) and (b)

1027. A reactor that produces more fissionable material than it consumes is called

(a) breeder reactor (b) fast breeder reactor

(c) water cooled reactor (d) nuclear reactor

OPSC Civil Services Pre 2006

Ans. (b) : If the reactor is configurated to produce more fissionable material (Plutonium) than it consumes is called Fast Breeder Reactor (FBR).

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03.

Refrigeration and Air-Conditioning

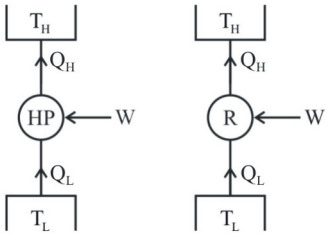
1. Introduction to Refrigerating machine

1. Consider a refrigerator and a heat pump working on the reversed carnot cycle between the same temperature limits. Then:

(a) COP of Heat Pump=COP of Refrigerator
(b) COP of Heat Pump=COP of Refrigerator–1
(c) COP of Heat Pump=COP of Refrigerator+1
(d) COP of Heat Pump= 1/(COP of Refrigerator)

TNCSC AE 2020
Oil India Senior Officer 23.12.2020
Haryana PSC AE (PHED) 04.09.2020 Paper-I
VIZAG MT, 14.12.2020
GPSC Executive Engg. 23.12.2018
Vadodara Muncipal Corp. DEE, 2018
Punjab PSC SDE 12.02.2017
(HPPSC LECT. 2016)
GPSC Asstt. Prof. (Prod.) 21.08.2016
UJVNL AE 2016
OPSC AEE 2015 Paper-II
GPSC AE 26.07.2015
Kerala PSC Poly. Lect. 09.01.2014
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
WBPSC AE 2007, GATE 1995
ESE 1992, 1994, 2000

Ans. : (c)



$$(COP)_R = \frac{T_L}{T_H - T_L}$$

$$(COP)_{H.P} = \frac{T_H}{T_H - T_L}]$$

$$(COP)_{H.P} - (COP)_R = 1]$$

$$\boxed{(COP)_{H.P} = 1 + (COP)_R}$$

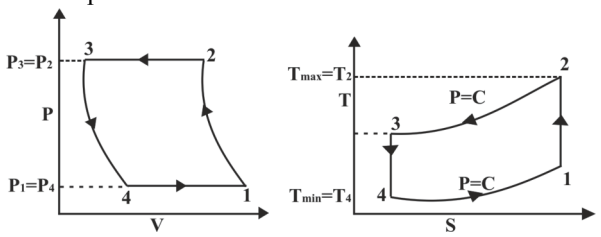
2. Bell Coleman cycle used for air refrigeration system is a

(a) Reversed Carnot cycle
(b) Reversed Otto cycle
(c) Reversed Joule cycle
(d) Reversed Rankine cycle

Nagaland PSC (CTSE) 2018, Paper-II
ISRO Scientist/Engineer (RAC) 22.04.2018
Sikkim PSC Under Secretary 2017
ISRO Scientist/Engineer (RAC) 07.05.2017
CGPSC Poly. Lect. 22.05.2016
Rajasthan AE (Nagar Nigam) 2016 Shift-3
APPSC AEE MAINS 2016 PAPER-III
GPSC Lect. 23.10.2016
UKPSC AE-2013, Paper-II
OPSC Civil Services Pre. 2011, TNPSC AE, 2008

Ans. (c) : Bell-Coleman Cycle—The Bell-Coleman cycle (also called as the Joule or "reverse" Brayton cycle) is a refrigeration cycle where the working fluid is a gas that is compressed and expanded, but does not change phase. Air is most often this working fluid. As there is no condensation and evaporation intended in this gas cycle, components corresponding to the condenser and evaporator in a vapour compression cycle are the hot and cold gas-to-gas heat exchangers.

This cycle may be thought of as a modification of reversed Carnot cycle, as the two isothermal processes of Carnot cycle are replaced by two isobaric heat transfer processes.



Process 1-2: Isentropic Compression

The air drawn from the refrigerator to air compressor cylinder where it compressed isentropically (constant entropy). No heat transfer by the air. During compression, the volume decreases while the pressure and temperature of air increases.

Process 2-3: Constant pressure heat rejection

The warm compressed air is then passed through cooler, where it cooled down at constant pressure. The heat rejected per kg of air during this process is equal to: $Q_{(2-3)} = C_p (T_2 - T_3)$

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Process 3-4: Isentropic expansion
No heat transfer takes place. The air expands isentropically in expander cylinder. During expansion, the volume increases, Pressure P_3 reduces to P_4. Temperature also falls during expansion from T_3 to T_4.

Process 4-1: Constant pressure heat addition
Heat transfer from the refrigerator to air. The temperature increases from T_4 to T_1. Volume increases to V_1 due to heat transfer. Heat absorbed by air per kg during this process is equal to: $Q_{(4\text{-}1)} = C_p (T_1\text{-}T_4)$

3. One ton of refrigeration corresponds to :

(a) 210 kJ/min (b) 210 kJ/h
(c) 335 kJ/min (d) 335 kJ/h

NLCIL GET 17.11.2020 Shift-II
Assam PSC AE (PHED) 18.10.2020
Nagaland PSC (CTSE) 2018, Paper-II
TNPSC AE 2017, Kerala PSC IOF 19.04.2016
Kerala PSC AE 06.08.2015, KPSC AE 2015
J&K PSC Civil Services Pre, 2013
TNPSC AE (Industries) 09.06.2013
TNPSC ACF 2012, ESE 1999

Ans. (a) : One ton of refrigeration corresponds to 210 kJ/min.

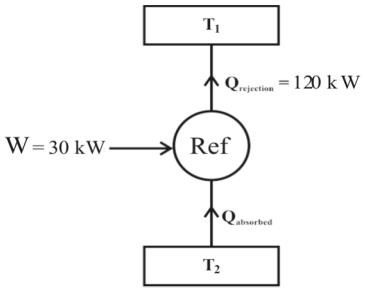
1TR = 3.5 kJ/sec
= 3.5 × 60 kJ/min
= 210 kJ/min
= 50 kcal/min = 12000 BTU/hr

4. A condenser of a refrigeration system rejects heat at a rate of 120 kW while compressor consumes power of 30 kW. the coefficient of performance of the system would be.

(a) 1/4 (b) 4
(c) 1/3 (d) 3

Assam PSC (PHED) 18.10.2020
GPSC Executive Engineer 23.12.2018
Nagaland PSC (CTSE) 2018, Paper-II
RPSC IOF & Boiler 20.09.2017
HPPSC Asstt. Prof. 10.11.2016
Assam PSC CCE Pre 2015
HPSC Asstt. Prof. 2014, ESE 1995, GATE 1992

Ans. : (d)



Heat rejection rate (Q_{rej})= 120 kW
Work Supplied (W) = 30 kW
$Q_{absorbed} = 120 - 30$
$= 90$ kW

$$(COP)_{Ref.} = \frac{Q_{absorbed}}{W} = \frac{90}{30} = 3$$

$$(COP)_{Ref.} = 3$$

5. One tonne of refrigeration is equivalent to

(a) 1.5 kW (b) 2.5 kW
(c) 3.5 kW (d) 5 kW

CIL MT 27.02.2020
UPPSC AE 12.04.2016 Paper-II
GPSC ARTO 01.05.2016
MPPSC State Forest Service Exam, 2014
MPSC HOD (Govt. Poly Colleges) 04.10.2014
Mizoram PSC AE/SDO 2014, Paper-I
UKPSC AE 2012 Paper–II
OPSC Civil Services Pre-2011
ESE 1999

Ans. (c) : TR is the equivalent to the rate of heat transfer needed to produce 1 ton (2000 Lb) of ice at 0°C water to 0°C ice in one day (24 hr).

$$1TR = \frac{1000 \times 335}{24 \times 3600} = 3.5\,kJ/s$$

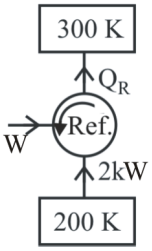
1TR = 3.5 kW = 210 kJ/min = 50 kcal/min

6. A refrigerating machine working on a reversed carnot cycle takes out 2kW of heat from the cold body while working between the temperature limits of 300 K and 200 K. The COP and power consumed by the cycle will be respectively.

(a) 1 and 1 kW (b) 1 and 2 kW
(c) 2 and 1 kW (d) 2 and 2 kW

Gujarat PSC AE 2019
TRB Poly Lect. 2012, 2017
Nagaland PSC (CTSE) 2016 Paper-II
RPSC IOF & Boiler 20.09.2017
Haryana PSC Civil Services Pre, 2014
UPSC JWM Advt. No. 50/2010
CSE Pre-2004, ESE 1997

Ans. (c)



$$(COP)_{Carnot} = \frac{T_L}{T_H - T_L}$$

$$= \frac{200}{300 - 200} = 2$$

$$(COP) = 2 = \frac{\text{Refrigeration effect}}{\text{Power Input}}$$

$$2 = \frac{2\,kW}{\text{Power Input}}$$

$$\text{Power Input} = \frac{2}{2} = 1\,kW$$

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7. The efficiency of Carnot cycle is 0.75. If the cycle is reversed, its COP as heat refrigerator is

(a) 0.25 (b) 0.33

(c) 1.33 (d) 4

Assam PSC AE (PHED), 18.10.2020
Gujarat PSC AE 2019
APPSC AEE MAINS 2016 PAPER-III
UPPSC AE 12.04.2016 Paper-II
HPPSC W.S. Poly. 2016
ESE 2010, CSE Pre-2002

Ans. (b) : Efficiency of the Carnot cycle $= 1 - \frac{T_2}{T_1}$

$$(COP)_{\text{Heat pump}} = \frac{1}{\eta_c} = \frac{T_1}{T_1 - T_2} = \frac{1}{0.75}$$

$(COP)_{HP} = 1.33$

$(COP)_{Ref} = (COP)_{HP} - 1$

$= 1.33 - 1$

$(COP)_{Ref} = 0.33$

8. A house refrigerator with its door open is switched on in a closed room. The air in the room is :

(a) Cooled

(b) Remains at same temperature

(c) Heated

(d) Heated or cooled depending on atmospheric pressure

GPSC ARTO Pre 30.12.2018, WBPSC AE, 2017
GWSSB DEE 07.07.2016
ISRO Scientist/Engineer 03.07.2016
Kerala LBS Centre for Science and Tech. Asst. Prof 2014
TRB Poly. Lect., 2012

Ans. (c) : A house refrigerator with its door open is switched on in a closed room, the temperature of the room increases.

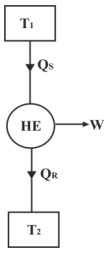
9. Efficiency of Carnot engine is given as 80%. If the cycle direction is reversed, what will be the value of COP of reversed Carnot cycle

(a) 1.25 (b) 0.5

(c) 0.25 (d) none of the above

RPSC IOF, 2020
ISRO Scientist/Engineer (RAC) 22.04.2018
ISRO Scientist/Engineer (RAC) 29.11.2015
ISRO Scientist/Engineer 12.05.2013
SJVN ET 2013

Ans. (c) :



$$\eta_{\text{carnot}} = 0.8 = 1 - \frac{T_2}{T_1}$$

or $\frac{T_2}{T_1} = 0.2, \frac{T_1}{T_2} = 5$

$$(COP)_{\text{carnot}} = \frac{T_2}{T_1 - T_2}$$

$$\therefore \quad (COP)_{\text{carnot}} = \frac{1}{\frac{T_1}{T_2} - 1} = \frac{1}{4} = 0.25$$

10. If a Carnot cycle has coefficient of performance 4, the ratio of maximum temperature to minimum temperature will be

(a) 0.25 (b) 1.25

(c) 5 (d) 3

HPPSC AE 2018
UKPSC AE-2013, Paper-II
CSE Pre-2003, APPSC AEE 2012
ESE 1999

Ans. (b) : We know that

$$[COP]_{\text{carnot cycle}} = \frac{T_L}{T_H - T_L}$$

$$\frac{T_L}{T_H - T_L} = 4$$

$$\frac{1}{\left[\frac{T_H}{T_L}\right] - 1} = 4$$

$$\frac{T_H}{T_L} - 1 = \frac{1}{4} = 0.25$$

$$\left[\frac{T_H}{T_L}\right] = 1.25$$

11. In an ideal refrigeration cycle, the condenser and evaporator temperatures are 27^0C and -13^0C respectively. The COP of cycle is :

(a) 6.5 (b) 7.5

(c) 10.5 (d) 15

OPSC AEE 2015 Paper-II
APPSC Poly Lect. 2013
OPSC Civil Services Pre. 2011
UKPSC AE 2007 Paper-II, ESE 1997

Ans. (a) : Given

$T_1 = 27^0\text{C} = 300$ K

$T_2 = -13^0\text{C} = 260$ K

$$\text{COP} = \frac{T_2}{T_1 - T_2} = \frac{260}{300 - 260} = \frac{260}{40}$$

$\boxed{\text{COP} = 6.5}$

12. While designing the refrigeration system of an aircraft, the prime consideration is that the

(a) system has high C.O.P

(b) H.P./ton is low

(c) weight of refrigerant circulated in the system is low

(d) weight of the refrigeration equipment is low

ISRO Scientist/ Engineer (RAC) 10.03.2019
APGENCO AE, 2017

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HPPSC Asstt. Prof. 20.11.2017
Nagaland PSC CTSE 2015 Paper-II, ESE 1993

Ans. (d) : While designing the refrigeration system of an aircraft, the prime consideration is that the weight of the refrigeration equipment is low.

13. A heat pump working on a reversed Carnot cycle has C.O.P. of 5. If it works as a refrigerator taking 1 kW of work input, the refrigerating effect will be

(a) 4 kW (b) 3 kW
(c) 1 kW (d) 2 kW

OPSC AEE 2019 Paper-II, TSPSC AEE 2017
Kerala PSC Poly. Lect. 09.01.2014
UKPSC AE 2007 Paper-II, ESE 1993

Ans. (a) : We know, $COP_{HP} = 1 + COP_R$

$5 = 1 + COP_R$

$COP_R = 4$

$$COP_R = \frac{R.E}{W_{in}}$$

$$4 = \frac{R.E}{1}$$

$$\boxed{R.E = 4\,kW}$$

14. When the lower temperature of refrigerating machine is fixed then the coefficient of performance can be improved by:

(a) Operating the machine at high speed
(b) Operating the machine at low speed
(c) Raising the higher temperature
(d) Lowering the higher temperature

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper-II
Rajasthan Nagar Nigam AE 2016 Shift-I
ESE 1992

Ans. (d) : $\boxed{(COP)_{RE} = \frac{T_L}{T_H - T_L}}$

If T_L is fixed

$T_H \uparrow$ then $(COP)_{RE} \downarrow$

$T_H \downarrow$ then $(COP)_{RE} \uparrow$

15. Air refrigeration cycle is generally employed in

(a) domestic refrigerator
(b) commercial refrigerator
(c) air-conditioning
(d) gas liquefaction

Assam PSC AE (PHED), 18.10.2020
Nagaland PSC (CTSE) 2018, Paper-II
APPSC Poly Lect. 2013, ESE 2006, 1998

Ans. (d) : The processes are used for industrial and commercial purposes.

Many gases can be put into a liquid state at normal atmospheric pressure by simple cooling.

Gas cycle refrigeration system find applications in air craft cabin cooling and also in the liquefaction of various gases.

16. A refrigerator working on a reversed Carnot cycle has a COP of 4. If it works as a heat pump and consumes 1 kW, the heating effect will be

(a) 1 kW (b) 4 kW
(c) 5 kW (d) 6 kW

UPPSC AE 13.12.2020, Paper-2
Assam PSC CCE Pre 2015
UKPSC AE 2012 Paper-II, ESE 2003

Ans. (c) : Given,

$(COP)_R = 4$

$(COP)_{HP} = 1 + (COP)_R$

$\Rightarrow 1 + 4 = 5$

If $P_{consumed} = 1$ kW

$$(COP)_{HP} = \frac{\text{Heating Effect}}{P_{consumed}} \Rightarrow 5 = \frac{\text{Heating Effect}}{1}$$

$$\boxed{\text{Heating Effect} = 5\,kW}$$

17. Air-refrigerator works on

(a) Carnot cycle (b) Rankine cycle
(c) Brayton cycle cycle (d) Bell-Coleman

CGPSC AE 15.01.2021
Nagaland PSC CTSE 2015 Paper-II
J&K PSC Civil Services Pre, 2013
OPSC Civil Services Pre 2006

Ans. (d) : Air refrigeration works on Bell-Coleman cycle which is also known as reversed Brayton or Joule cycle.

18. In air-conditioning of aeroplanes, using air as a refrigerant, the cycle used is

(a) reversed Carnot cycle
(b) reversed Joule cycle
(c) reversed Brayton cycle
(d) reversed Otto cycle

Nagaland PSC (CTSE) 2018, Paper-II
Nagaland PSC (CTSE) 2017, 2016 Paper-II
TNPSC ACF 2012

Ans. (c) : In air conditioning of aeroplanes using air as a refrigerant, the cycle is reversed Brayton cycle.

19. Which of the following theories is related to the theory of thermocouple?

(a) Piezoelectric effect (b) Skin effect
(c) Seebeck effect (d) Faraday's law

Assam PSC AE (PHED), 18.10.2020
Vadodara Muncipal Corp. DEE, 2018
GPSC Lect. (Auto) 16.10.2016

Ans. (c) : The seebeck effect is when electricity is created between a thermocouple when the ends are subjected to a temperature difference between them.

Piezoelectric effect- Piezoelectricity is the electric charge that accumulates in certain solid materials in response to applied mechanical stress. The word piezoelectricity means electricity resulting from pressure and latent heat.

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Skin effect- Skin effect is the tendency of an alternating electric current (AC) to become distributed within a conductor such that the current density is largest near the surface of the conductor and decreases exponentially with greater depths in the conductor.

Faraday's Law- Faraday's law of induction is a basic law of electromagnetism predicting how a magnetic field will interact with an electric circuit to produce an electromotive force.

20. In a one ton capacity water cooler, water enters at 30 °C at the rate of 200 litre/hour. Taking specific heat of water as 4.16 kJ/kg K, the outlet temperature of water will be

(a) 3.5 °C (b) 6.3 °C
(c) 23.7 °C (d) 15 °C

Assam PSC AE (PHED) 18.10.2020
UKPSC AE 2012 Paper–II, ESE 2003

Ans. (d) : Given,

$Q = 1\ \text{Ton} = 3.5\ \text{kW}$

$T_1 = 30°C$

$C_{pw} = 4.16\ \text{kJ/kg K}$

Volume flow rate $\dot{Q} = 200\ \text{litre/hour}$

$$= \frac{200 \times 10^{-3}}{3600}\ m^3/\text{sec}$$

$$\dot{Q} = \frac{1}{18000}\ m^3/\text{sec}$$

$$\therefore \quad \dot{m}_w = \rho A V$$

$$= \rho \dot{Q} \qquad [\because \ \dot{Q} = AV\ ]$$

$$= 1000 \times \frac{1}{18000}\ \text{kg/sec}$$

$$\dot{m}_w = \frac{1}{18}\ \text{kg/sec}$$

$$Q = \dot{m}_w C_{pw} \Delta T$$

$$3.5 = \frac{1}{18} \times 4.16 \times (30 - T_2)$$

$$63 = 124.8 - 4.16T_2$$

$$4.16T_2 = 124.8 - 63$$

$$4.16T_2 = 61.8$$

$$T_2 = \frac{61.8}{4.16} = 14.85$$

$$T_2 \approx 15°C$$

21. The COP of a refrigerator which operates on reversed Carnot cycle whose highest temperature is 27°C and lower temperature is –23°C.

(a) 0.2 (b) 5
(c) 5.75 (d) 6

RPSC IOF & Boiler 20.09.2017
TNPSC AE 2013, TRB Asst. Prof. 2012

Ans. (b)

$$[COP]_{Ref.} = \frac{T_L}{T_H - T_L}$$

$$= \frac{(-23 + 273)}{[(273 + 27) - (-23 + 273)]}$$

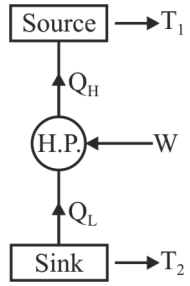
$$= \frac{250}{50} = 5$$

22. If T_1 and T_2 be the highest and lowest absolute temperatures encountered in a heat pump operating on a reversed Carnot cycle, then its COP is equal to

(a) $\frac{T_1 - T_2}{T_1}$ (b) $\frac{T_1 - T_2}{T_2}$
(c) $\frac{T_2}{T_1 - T_2}$ (d) $\frac{T_1}{T_1 - T_2}$

CGPSC AE 15.01.2020
Nagaland PSC CTSE 2015 Paper-II
RPSC AE 2008

Ans. (d) : Given,



$$(COP)_{HP} = \frac{\text{High temperature}}{\text{High temperature} - \text{low temperature}}$$

$$(COP)_{HP} = \frac{T_1}{T_1 - T_2}$$

23. A Carnot refrigerator works between the temperatures of 200 K and 300 K. If the refrigerator receives 1 kW of heat, the work requirement will be

(a) 0.5 kW (b) 0.67 kW
(c) 1.5 kW (d) 3 kW

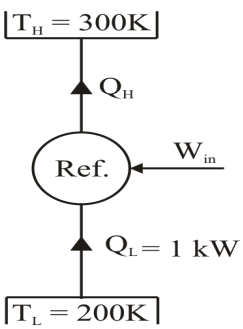
TNPSC 2019
RPSC Vice Principle ITI 2018
CSE Pre-2000

Ans. (a) : For Carnot refrigerator

$$[COP]_{ideal} = [COP]_{actual}$$

$$\frac{T_L}{T_H - T_L} = \frac{Q_L}{W_{in}}$$

$$W_{in} = \frac{1}{\left[\frac{200}{300 - 200}\right]}$$



$$W_{in} = 0.5\text{kW}$$

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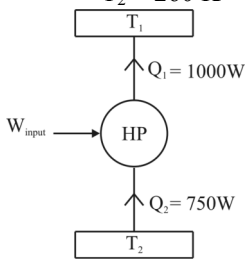
24. An industrial heat pump operates between the temperatures of 27°C and -13°C. The rates of heat addition and heat rejection are 750 W and 1000 W respectively. The COP for the heat pump is

(a) 7.5 (b) 6.5
(c) 4.0 (d) 3.0

Gujarat PSC AE 2019
PSPCL AE, 2012, GATE 2003

Ans. : (c) : Given $Q_1 = 1000$ W
$Q_2 = 750$ W
$T_1 = 300$ K
$T_2 = 260$ K



$$\therefore (COP)_{HP} = \frac{Q_1}{\text{work input}}$$

$$= \frac{1000}{Q_1 - Q_2}$$

$$= \frac{1000}{1000 - 750}$$

$$= \frac{1000}{250}$$

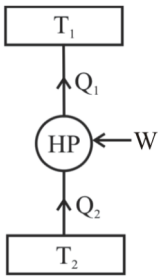
$$\boxed{(COP)_{HP} = 4}$$

25. An ideal refrigerator is operating between a condenser temperature of 37°C and an evaporator temperature of –3°C. If the machine is functioning as a heat pump, its coefficient of performance (COP) will be

(a) 6.0 (b) 6.75
(c) 7.0 (d) 7.75

BPSC AE Mains 2017 Paper - V
UKPSC AE 2012 Paper-I, ESE 2010

Ans. : (d) : Given,
$T_1 = 37°C = 310K$
$T_2 = -3°C = 270K$



$$(COP)_{HP} = \frac{T_1}{T_1 - T_2} = \frac{310}{310 - 270}$$

$$= 7.75$$

26. The relative Coefficient of Performance (COP) is

(a) Actual COP - Theoretical COP
(b) Actual COP /Theoretical COP
(c) Actual COP x Theoretical COP
(d) Theoretical COP/ Actual COP

Nagaland PSC (CTSE) 2018, Paper-II
Rajasthan AE (Nagar Nigam) 2016 Shift-3
NPCIL ET, 2015

Ans. (b) : The ratio of the actual COP to the theoretical COP is known as relative coefficient of performance, mathematically,

$$(COP)_{Relative} = \frac{\text{Actual COP}}{\text{Theoretical COP}}$$

27. One ton of refrigeration is equal to the refrigeration effect corresponding to melting of 1000 kg of ice in

(a) 10 minute (b) 10 hour
(c) 12 hours (d) 24 hours

GPSC Executive Engineer 23.12.2018
Rajasthan Nagar Nigam AE 2016, Shift-I
TRB Asstt. Prof., 2012

Ans. (d) : Unit of Refrigeration (TR)- 1 Ton of refrigeration means the amount of heat that is to be removed from 1 American tonne (2000/bs = 907 kg) of water at 0^0C is order to convert it into ice at 0^0 in 1 day (24 hours).

Therefore ton of refrigeration represents heat transfer rate but not mass.

$$1 \text{ TR} = \frac{907 \times 334}{24 \times 3600} = 3.5 \text{ kW}$$

1 TR = 3.5 kJ/sec
1 TR = 210 kJ/min.
1 TR = 50 k-cal/min. (1 k-cal = 4.18 cal)

Note-1 British tonne = 2220 lbs = 1000 kg)

28. A carnot-refrigerator has a COP of 6. What is the ratio of the lower to the higher absolute temperature?

(a) 1/6 (b) 7/8
(c) 6/7 (d) 1/7

APPSC Poly Lect. 13.03.2020
AAI Jr. Executive 29.11.2018
ESE 2006

Ans. (c) : Given; $(COP)_{Ref} = 6$

$$(COP)_{Ref} = \frac{T_L}{T_H - T_L}$$

$$\frac{T_H - T_L}{T_L} = \frac{1}{(COP)_{Ref}} = \frac{1}{6}$$

$$\frac{T_H}{T_L} - 1 = \frac{1}{6} \Rightarrow \frac{T_H}{T_L} = \frac{7}{6} \Rightarrow \boxed{\frac{T_L}{T_H} = \frac{6}{7}}$$

29. A 1 ton capacity water cooler cools water steadily from 35^0C to 20^0C. The specific heat of water is 4.18 kJ/kg-K. The water flow rate will be nearly

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(a) 13.33 liter/hr (b) 33.3 liter/hr
(c) 200 liter/hr (d) 250 liter/hr

Nagaland PSC CTSE 2016 Paper-II
Haryana PSC Civil Services Pre, 2014, ESE 2001

Ans. (c) Water cooler capacity = 1 ton

RE = 1 ton = 1 × 3.5 kW

$3.5 = \dot{m}\, Cp\,(T_2 - T_1)$

$3.5 = \dot{m} \times 4.18 \times (308 - 293)$

$\dot{m} = 0.05582$ kg/sec.

$\dot{m} = \rho V$

$0.05582 = 1000 \times V$

$V = 5.582 \times 10^{-5}\ m^3/sec.$

$$= \frac{5.582 \times 10^{-5} \times 10^3}{\frac{1}{3600}} \text{ liter}$$

$$\boxed{V = 200 \text{ liter/hr}}$$

30. The ratio of heat extracted in the refrigerator to the work done on refrigerant is called :

(a) Coefficient of performance of heat pump
(b) Coefficient of performance of refrigeration
(c) Refrigerating efficiency
(d) Compressor efficiency

CGPSC AE 15.01.2021, Punjab PSC SDE 12.02.2017
CGPSC AE 16.10.2016

Ans. (b) : The ratio of heat extracted in the refrigerator to the work done on refrigerant is called coefficient of performance of refrigeration.

$$\boxed{COP = \frac{\text{Refrigeration Effect}}{\text{Work done}}}$$

31. A Bell Coleman cycle is a reversed

(a) Brayton cycle (b) Akinson cycle
(c) Ericsson cycle (d) Carnot cycle

J & K PSC Screening, 2006, ESE 2005

Ans. (a) : A Bell Coleman cycle is a reversed Brayton Cycle.

32. The COP of a domestic refrigerator

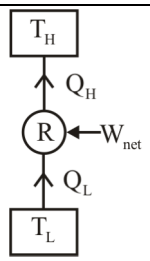
(a) Is more than 1
(b) Is less than 1
(c) Is equal to 1
(d) Depends upon the weather conditions

Rajasthan Nagar Nigam AE 2016, Shift-I
UKPSC AE 2007 Paper-II

Ans. (a) Coefficient of Performance–The performance of refrigerators and heat pumps is expressed in terms of the coefficient of performance (COP), defined as.

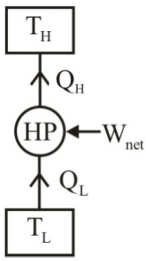
$$(COP)_{Ref.} = \frac{\text{Desired Output}}{\text{Required input}} = \frac{\text{Cooling Effect}}{\text{Work Input}}$$

$$= \frac{Q_L}{W_{net}}$$



$$(COP)_{HP} = \frac{\text{Desired Output}}{\text{Required Input}} = \frac{\text{Heating Effect}}{\text{Work Input}}$$

$$= \frac{Q_H}{W_{net}}$$



Both $(COP)_R$ and $(COP)_{HP}$ can be greater than 1.
For fixed values of Q_L and Q_H

$$\boxed{(COP)_{HP} = (COP)_R + 1}$$

33. A refrigerating machine having coefficient of performance equal to 2 is used to remove heat at the rate of 1200 kJ/min. What is the power required for this machine?

(a) 80 kW (b) 60 kW
(c) 20 kW (d) 10 kW

RPSC IOF, 2020, ESE 2007

Ans. (d) : Given,

$Q_2 \Rightarrow$ heat removed = 1200 kJ/min

$$= \frac{1200}{60} \text{kJ/sec}$$

$Q_2 = 20$ kW

$(COP)_R = 2$

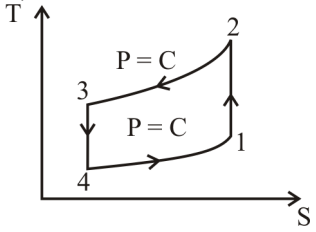
$$(COP)_R = \frac{Q_2}{\text{Power input (P)}}$$

$$2 = \frac{20}{P}$$

$$P = \frac{20}{2}$$

Power required P = 10 kW

34. The thermodynamic cycle shown below (T-S diagram) indicates



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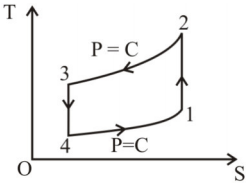
(a) Reversed Carnot cycle
(b) Vapor compression cycle
(c) Reversed Brayton cycle
(d) Vapor absorption cycle

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (c) : Reversed Brayton cycle



Reverse Brayton cycle
1-2 Isentropic compression
2-3 Isobaric heat rejection
3-4 Isentropic expansion
4-1 Isobaric heat absorption

35. A boot strap air cooling system has-
(a) One heat exchanger
(b) Two heat exchangers
(c) Three heat exchangers
(d) Four heat exchangers

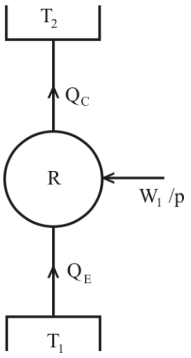
GMB AAE 25.06.2017, JPSC AE 2013 Paper-V

Ans. (b) : Boot strap air cooling system has two heat exchanges one is air cooler and another is after cooler. The air bled from the compressor and gets cooled by the first heat exchanger. Then the air is compressed at high temperature and pressure in turbine. A substantial amount of heat is removed in the second heat exchanger and air is further cooled through the turbine.

36. The Heat Rejection Factor (HRF) is given by -
(a) $1+COP$ (b) $1-COP$
(c) $1+\frac{1}{COP}$ (d) $1-\frac{1}{COP}$

Nagaland PSC (CTSE) 2018, Paper-II
GMB AAE 25.06.2017

Ans. (c) : Heat Rejection factor or ratio-
Q_C = Heat of condenser
RE = Refrigeration effect



It is defined as the ratio of heat rejected in condenser to the refrigeration effect.

$$H.R.F. = \frac{Q_{condenser}}{RE}$$

$$= \frac{RE + W}{RE}$$
$$= 1 + \frac{W}{RE}$$
$$= 1 + \frac{1}{(COP)_R} \quad \left[COP = \frac{RE}{W}\right]$$

37. If a Carnot refrigeration cycle is to have a COP of 5 the ratio of maximum temperature to minimum temperature in the cycle should be :
(a) 1.2 (b) 1.5
(c) 2.0 (d) 2.5

GPSC EE Pre 28.01.2017
J&K PSC Civil Services Pre, 2010

Ans. (a) : Given -
$(COP)_{ref} = 5$

$$(COP)_{ref} = \frac{T_L}{T_H - T_L}$$

$$= \frac{T_L}{T_L\left[\frac{T_H}{T_L} - 1\right]} = 5$$

$$\frac{1}{\left(\frac{T_H}{T_L} - 1\right)} = 5$$

$$\left(\frac{T_H}{T_L} - 1\right) = \frac{1}{5}$$

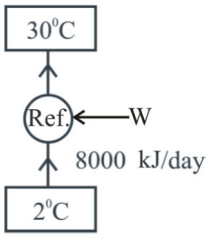
$$\frac{T_H}{T_L} = \frac{1}{5} + 1 = \frac{6}{5}$$

$$\frac{T_H}{T_L} = 1.2$$

38. A domestic refrigerator, set at 2^0C, handles on an average a thermal load of 8000 kJ per day. The ambient temperature is 30^0C and the COP of the refrigerator is 0.15 times that of a Carnot refrigerator. The daily electricity consumption in kWh is approximately.
(a) 1.47 (b) 1.51
(c) 3.28 (d) 2.86

Nagaland PSC CTSE 2016 Paper-II
Haryana PSC Civil Services Pre, 2014

Ans. (b) : $T_L = 2 + 273 = 275$ K
$T_H = 30 + 273 = 303$ K
(COP) of refrigerator = 0.15 × COP of Carnot refrigerator



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$$= 0.15 \times \frac{T_L}{T_H - T_L}$$

$$= 0.15 \times \frac{275}{303 - 275}$$

$$(COP) = 1.4732 = \frac{RE}{\text{work input}}$$

$$\text{Work input } W = \frac{8000}{1.4732}$$

$$W = 5430.355 \text{ kJ/day}$$

$$W = \frac{5430.355}{24 \times 60 \times 60}$$

$$= 0.0628 \text{ kW}$$

Daily consumption in kWhr

$$W = 0.0628 \times 24$$

$$= 1.51 \text{ kWhr}$$

39. COP of a reversed Carnot cycle refrigerator working between higher temperature T_2 and lower temperature T_1

(a) will increase with increase in T_1 keeping T_2 fixed

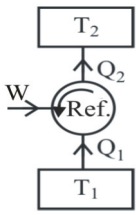
(b) will decrease with increase in T_1 keeping T_2 fixed

(c) will first increased with increase in T_1 and then decrease with increase in T_1 keeping T_2 fixed

(d) none of the above

Nagaland PSC (CTSE) 2016 Paper-II
Haryana PSC Civil Services Pre, 2014

Ans. (a)



$$COP = \frac{T_1}{T_2 - T_1}$$

$T_1 \uparrow \rightarrow$ COP $\uparrow$ and T_2 is fixed

40. The heat removal rate from a refrigerated space is 7.2 kW and the power input to the compressor is 1.8 kW. The coefficient of performance (COP) of the refrigerator is

(a) 4 (b) 5

(c) 1.25 (d) 0.25

ISRO Scientist/Engineer 22.04.2018
GPSC Executive Engineer 23.12.2018

Ans. (a) :

$$[COP]_{ref} = \frac{Heat\ removed\ from\ space}{Work\ done}$$

$$= \frac{7.2}{1.8}$$

$$[COP]_{ref} = 4$$

41. The COP of gas refrigeration cycle [r = pressure ratio]

(a) increases with increase in pressure ratio

(b) decreases with increase in pressure ratio

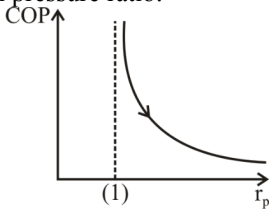
(c) remains constant with increase in pressure ratio

(d) is given by, COP $= \frac{1}{\left(r^{\gamma}\right) - 1}$

(e) is given by, COP = $= \frac{1}{\left(r^{1/\gamma}\right) - 1}$

CGPSC 25 Apr. (Shift II) 2018
CGPSC AE 25.02.2018

Ans. (b) : The COP of gas refrigeration cycle decreases with increase in pressure ratio.



42. Thermal efficiency of a Carnot engine is 30%. If the engine is reversed to operate as heat pump with same operating conditions, what will be the COP of the heat pump?

(a) 0.3 (b) 1.33

(c) 2.33 (d) 3.33

UKPSC AE-2013, Paper-II, CSE Pre-1995, 1994

Ans. (d) : Given as, $\eta_{carnot} = 30\%$

For same temperature limit, $[COP]_{HP} = \frac{1}{\eta_{carnot}} = \frac{100}{30}$

$$[COP]_{HP} = 3.33$$

43. If the capacity of a refrigerating machine is one ton, then the rate of heat abstraction is equal to:-

(a) 50 kcal/min (b) 100 kcal/min

(c) 150 kcal/min (d) 500 kcal/min

Rajasthan AE (Nagar Nigam) 2016 Shift-3
UKPSC AE-2013, Paper-II

Ans. (a) : One ton = 210 kJ/min

1 calories = 4.18 J

Then the rate of heal abstraction $= \frac{210}{4.18}$ kcal/min

$= 50.239$ kcal/min $\simeq 50$ kcal/min

44. A refrigerator storage is supplied with 3600 kg of fish at a temperature of 27°C. The fish has to be cooled to – 23°C for preserving it for a long period without deterioration. The cooling takes place in 10 hours. The specific heat of fish is 2.0 kJ/kg K above freezing point of fish and 0.5 kJ/kg K below freezing point of fish, which is –3°C. The latent heat of freezing is 230 kJ/kg. What is the power to drive the plant if the actual COP is half that of the ideal COP?

(a) 30 kW (b) 15 kW

(c) 12 kW (d) 6 kW

Nagaland PSC CTSE 2015 Paper-II, CSE Pre-2002

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Ans. (c) : Net Refrigeration Effect (NRE)

$$= mC_p\Delta T + mL + mC'_p\Delta T'$$

$$= \frac{m}{3600\times10}\left[C_P\left(T_1 - T_f\right) + \left(L + C_P\left(T_f - T_2\right)\right)\right]$$

$$= \frac{1}{10}\left[2\times30 + 230 + 10\right]$$

$$= 30 \text{ kW}$$

$$(COP)_{carnot} = \frac{T_2}{T_1 - T_2} = \frac{273-23}{(273+27)-(273-23)}$$

$$= 5$$

$$(COP)_{actual} = \frac{1}{2}(COP)_{ideal} = \frac{1}{2}\times\frac{T_2}{T_1 - T_2} = \frac{1}{2}\times\frac{250}{300-250}$$

$$= 2.5$$

$$(COP)_{actual} = \frac{NRE}{W}$$

$$\therefore W = \frac{NRE}{(COP)_{actual}} = \frac{30}{2.5} = 12\,kW$$

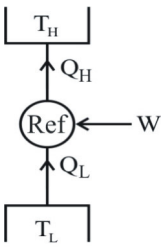
45. The C.O.P of a refrigerator working on the reversed carnot cycle is

(a) $\frac{T_L}{T_H - T_L}$ (b) $\frac{T_H - T_L}{T_L}$

(c) $\frac{T_H}{T_H - T_L}$ (d) $\frac{T_L}{T_L - T_H}$

GPSC Executive Engineer 23.12.2018
TSPSC AEE 2015

Ans. : (a)



$$(C.O.P.)_{ref} = \frac{T_L}{T_H - T_L}$$

$$(C.O.P.)_{ref} = \frac{\text{Refrigeration effect}}{\text{work input}}$$

46. The power (kW) required per ton of refrigeration is N/COP. Where COP is the coefficient of performance then N is equal to

(a) 2.75 (b) 4.75

(c) 5.25 (d) 3.50

UPRVUNL AE 21.08.2016, CSE Pre-2001

Ans. : (d)

$$COP = \frac{\text{Refrigeration effect}}{\text{Work input}} = \frac{3.5}{W_{in}}$$

$$COP = \frac{3.5\times COP}{N}$$

$$\boxed{N = 3.5}$$

47. Heat rejection ratio of the condenser is 1.2. The COP of this refrigeration plant will be:

(a) 3 (b) 4

(c) 5 (d) 6

(e) 2

CGPSC AE 25.02.2018

Ans. (c) : Heat Rejection Ratio- It is defined as the ratio of heat rejected by condenser to refrigeration effect produce.

$$HRR = \frac{Q_{cond.}}{Q_{R.E.}} > 1$$

$$HRR = \frac{Q_{R.E.} + W_{in}}{Q_{R.E.}} = 1 + \frac{W_{in}}{Q_{R.E.}} = 1 + \frac{1}{[COP]_{Ref}}$$

$$1.2 = 1 + \frac{1}{[COP]_{Ref}} \Rightarrow \boxed{(COP)_{Ref} = 5}$$

48. In aircraft, air refrigeration is used because of

(a) low weight per ton of refrigeration

(b) high heat transfer rate

(c) low temperature at high altitudes

(d) higher coefficient of performance

ISRO Scientist/Engineer (RAC) 07.05.2017
ESE 1995

Ans. (a) : Due to low weight per ton of refrigeration air is used as a refrigerant in aircraft.

49. The COP of a domestic refrigerator as compared to that of an air conditioner is in general :

(a) High

(b) Low

(c) Same

(d) COP depends on dry bulb temperature

WBPSC AE 2017, TSPSC Managers, 2015

Ans. (b) : The COP of domestic refrigerator is less than the COP of air conditioner because the lower temperature maintained in the refrigerator is less than the temperature maintained in the air conditioner.

$$\boxed{(COP)_{Ref.} = \frac{T_L}{T_H - T_L}}$$

$T_L \downarrow \quad (COP)_{Ref.} \downarrow$

50. The ratio of high temperature to low temperature for reversed Carnot refrigerator is 1.25. The COP will be

(a) 2 (b) 3

(c) 4 (d) 5

OPSC Civil Services Pre 2011
TNPSC AE, 2008

Ans. (c) :

Let, T_H = High temperature of reversed Carnot refrigerator.

T_L = Low temperature of reversed Carnot refrigerator.

Given ; $\frac{T_H}{T_L} = 1.25$

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$$\therefore (COP)_R = \frac{T_L}{T_H - T_L} = \frac{1}{\frac{T_H}{T_L} - 1} = \frac{1}{1.25 - 1}$$

$$= \frac{1}{0.25} = \frac{100}{25}$$

$$(COP)_R = 4$$

51. A cold storage is to be maintained at –5°C while the surroundings are at 35°C. The heat leakage from the surroundings into the cold storage is estimated to be 29 kW. The actual COP of the refrigeration plant used is $\frac{1}{3}$rd that of an ideal plant working between the same temperatures. The power required to drive the plant will be

(a) 13 kW (b) 14 kW
(c) 15 kW (d) 16 kW

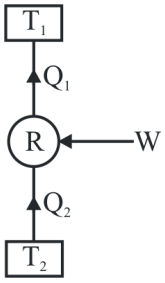
ESE 2020, 2019

Ans. (a) : Given,

$T_1 = 35 + 273 = 308$ K

$T_2 = -5 + 273 = 268$ K

$Q_2 = 29$ kW



$$(COP)_{ideal} = \frac{T_2}{T_1 - T_2}$$

$$= \frac{268}{308 - 268} = 6.7$$

$$(COP)_{actual} = \frac{1}{3}(COP)_{ideal}$$

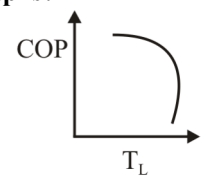
$$= 2.23$$

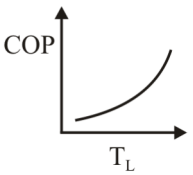
$$(COP)_{actual} = \frac{RE}{W_{input}} = \frac{Q_2}{W}$$

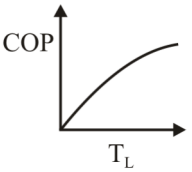
$$W = \frac{Q_2}{(COP)_{actual}} = \frac{29}{2.23}$$

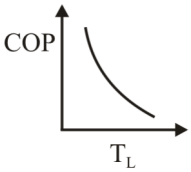
$W = 12.987 \Rightarrow 13$ kW

52. For a given value of T_H (source temperature) for a reversed Carnot cycle, the variation of T_L (sink temperature) for different values of COP is represented by which one of the following graphs?

(a)


(b)


(c)


(d)


UPPSC AE 13.12.2020, Paper-2, ESE 2009

Ans. (b)

$$COP = \frac{T_L}{T_H - T_L}$$

As, T_H is constant

$$COP = \frac{1}{\left(\frac{T_H}{T_L} - 1\right)}$$

$COP \propto T_L$

So, As T_L, increases COP increases

53. A cold coffee shop uses a refrigerator with COP 2.5 for cooling drinking water at 0.42 litre/min from 18°C to 10°C. The specific heat capacity of water is 4.18 kJ/kg-K. Power consumption by refrigerator will be:

(a) 85W (b) 75W
(c) 92.9W (d) 120W

AAI Jr. Executive 26.03.2021

Ans. (c) : Given, COP = 2.5, V = 0.42 litre/min

$(C_p)_{water} = 4.18$ kJ/kg-K

$$Q = \dot{m} C_p dT$$

$$Q = \frac{0.42 \times 10^{-3} \times 10^3 \times 4.18 \times (18 - 10)}{60}$$

$Q = 0.23408$ kW

$$COP = \frac{Q}{\text{Power consumption}}$$

$$2.5 = \frac{0.23408}{P}$$

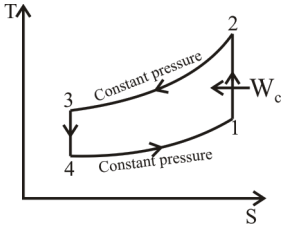
$$\boxed{P = 92.9\,W}$$

54. Air enters the compressor of an ideal gas refrigeration cycle at 10 °C and 80 kPa. The maximum and minimum temperature in cycle are 250 °C and –50°C. Assuming the value of specific heat at constant pressure as equal 1.0 kJ/kgK, the compressor work is (nearest) :

(a) 170 kJ/kg (b) 190 kJ/kg
(c) 240 kJ/kg (d) 220 kJ/kg

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) :



$T_1 = 10°C + 273 = 283$ K

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$T_2 = 250°C + 273 = 523\ K$
$T_4 = -50°C + 273 = 223\ K$
$C_p = 1\ kJ/kg\text{-}K$
$W_c = mC_p(T_2 - T_1)$
$= 1 \times 1 \times [523 - 283]$
$= 240\ kJ/kg$

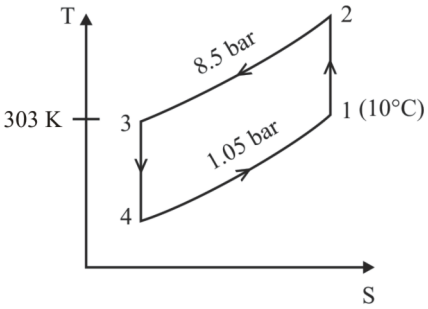
55. A refrigerator working on Bell-Coleman cycle operates between pressure limits of 1.05 bar and 8.5 bar. Air is drawn from the cold chamber at 10°C, compressed and then it is cooled to 30°C before entering the expansion cylinder. The expansion and compression follow the $pV^{1.3}$ = constant. Assuming $\gamma = 1.4$, the COP of the system will be :

(a) 1.87 (b) 1.3
(c) 1.03 (d) 1.62

CGPSC AE 15.01.2021

Ans. (b) : Given,

$T_3 = 30°C = 303\ K$
$T_1 = 10°C = 283\ K$



$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} \qquad [\because\ n = 1.3]$$

$$\frac{T_2}{283} = \left(\frac{8.5}{1.05}\right)^{\frac{1.3-1}{1.3}} \Rightarrow T_2 = 458.54\,K$$

$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4}\right)^{\frac{n-1}{n}}$$

$$\frac{303}{T_4} = \left(\frac{8.5}{1.05}\right)^{\frac{1.3-1}{1.3}} \Rightarrow T_4 = 187\,K$$

$$COP = \frac{(T_1 - T_4)}{\frac{n}{n-1} \times \left(\frac{\gamma - 1}{\gamma}\right)\left[(T_2 - T_1) - (T_3 - T_4)\right]}$$

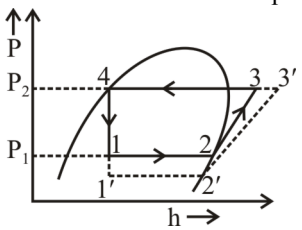
$$COP = \frac{(283 - 187)}{\frac{1.3}{1.3-1} \times \left(\frac{1.4-1}{1.4}\right)\left[(458.54 - 283) - (303 - 187)\right]} = 1.3$$

56. On reduction in suction pressure, the refrigerating effect and COP___.

(a) increase (b) decrease
(c) remain constant (d) become zero

UPRVUNL AE 05.07.2021

Ans. (b) : Effect of decrease in suction pressure–



RE_1 = 1-2 $(W_{in})_1$ = 2-3
RE_2 = 1′-2′ $(W_{in})_2$ = 2′-3
$RE_1 > RE'_2$ $(W_{in})_2 > (W_{in})_1$

Hence, from above we conclude that on decreasing suction pressure both RE and COP decreases.

57. When compared to vapour compression refrigeration, the gas cycle refrigeration system has the following feature.

(a) Higher COP
(b) Low power requirement
(c) Pressurization at high altitude difficult
(d) Light weight and compact.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : • Gas cycle refrigeration system are light weight than vapour compression refrigeration system.
• $(COP)_{VCRS} > (COP)_{gas/air\ refrigeration.}$
• $(Power\ requirement)_{gas} > (Power)_{VCRS}$

58. A refrigerator system operating on reversed cannot cycle has a refrigeration temperature of 258 K and the cooling water temperature is 308 K. The capacity of the system is 10 tonnes. The power required to un the unit is _____.

(a) 6.78 kW (b) 5.68 kW
(c) 6.43 kW (d) 9.46 kW

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Given-
$T_L = 258\ K$
$T_H = 308\ K$
COP of Reverse Carnot cycle.

$$(COP)_{ref} = \frac{T_L}{T_H - T_L}$$

$$(COP)_{ref.} = \frac{258}{308 - 258} = \frac{258}{50}$$

$(COP)_{ref.} = 5.16$
And given refrigeration effect
(RE) = 10 tonne
$= 10 \times 3.5\ kW$
$= 35\ kW$

$$\therefore (COP)_{ref.} = \frac{\text{Refrigeration effect (RE)}}{\text{power input}}$$

$$5.16 = \frac{35}{P_{input}}$$

$$\text{or } P_{input} = \frac{35}{5.16} = 6.78\ kW$$

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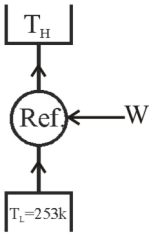
59. A refrigerating machine works on a reversed Carnot cycle. It consumes 6 kW and the produces a refrigerating effect of 100 kJ/min. The sink temperature is 233 K. The COP-coefficient of performance of the engine is ____.

(a) 2.78 (b) 3.26
(c) 2.48 (d) 2.38

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Given- work input to the refrigerator, Win = 6 kW

Refrigeration effect, (RE) = 1000 kJ /min
= 16.666 kJ/sec
(RE) = 16.666 kW



$$(COP)_{Ref.} = \frac{RE}{W_{input}} = \frac{16.66}{6}$$

$$= 2.777$$

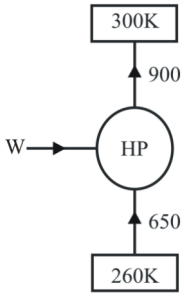
$$(COP)_{ref.} = 2.78$$

60. A heat pump operates between the temperature of 27°C and – 13°C. The rates of heat addition and heat rejection are 650 W and 900 W, respectively. The COP for the heat pump is :

(a) 7.5 (b) 6.5
(c) 3.6 (d) 2.6

RPSC ACF & FRO, 26.02.2021

Ans. (c) :



Work input, W = 900 – 650
= 250 W

$$(COP)_{actual} = \frac{\text{Desired effect}}{\text{work input}}$$

$$= \frac{900}{250}$$

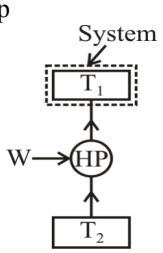
$$\boxed{(COP)_{actual} = 3.6}$$

61. Which is the case where low grade (heat) be more than the high grade (work) is the possibility?

(a) Heat Pump (b) Refrigerator
(c) Heat Engine (d) Heat Exchanger.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : Heat Pump



Heat supply in the system is more than workdone.

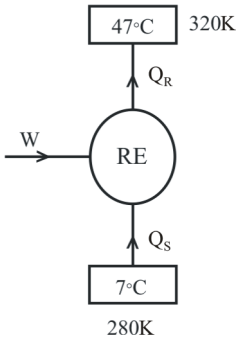
62. A refrigerating system of 1TR capacity operates between temperature range of 7 °C and 47 °C. The minimum amount of heat rejected to the surrounding will be

(a) 3 kW
(b) 3.5 kW
(c) 4 kW
(d) Not possible to determine with given data

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : Given,

$Q_S = 1$ TR
$= 1 \times 3.5$
$Q_S = 3.5$ kW



$$\frac{Q_S}{Q_R} = \frac{T_S}{T_R}$$

$$Q_R = \frac{3.5 \times 320}{280} = 4\,kW$$

63. If the efficiency of engine is 40%, then $(COP)_{HP}$ is :

(a) 0.5 (b) 1.5
(c) 2.5 (d) 3.5

CGPSC AE 15.01.2021

Ans. (c) : Given,

$\eta_E = 40\% = 0.4$

$$(COP)_{HP} = (COP)_R + 1 = \frac{1}{\eta_E}$$

$$(COP)_{HP} = \frac{1}{0.4} = 2.5$$

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64. The refrigerating capacity of 165 litres domestic refrigerator is approximately equal to:

(a) 0.1 tonne (b) 1.15 tonnes
(c) 5 tonnes (d) 8 tonnes

CGPSC AE 15.01.2021

Ans. (a) : The refrigerating capacity of 165 litres domestic refrigerator is approximately equal to 0.1 tonne.

65. In Bell-Colemann refrigerator for the same temperature range

(a) COP of dense air system is higher than COP of open air system.
(b) COP of dense air system is lower than COP of open air system.
(c) COP of dense air system is equal to COP of open air system.
(d) COP of dense air system may be higher or lower than COP of open air system depending upon pressure ratio.

JPSC AE, 10.04.2021, Paper-I

Ans. (a) : In Bell-Colemann refrigerator for the same temperature range COP of dense air system is higher than COP of open air system.

66. An air cooled condenser has 6 m^2 of surface with a removal of 50 kJ hr^{-1}. m^{-2}.°C^{-1}. What is the refrigerant temperature to dissipate 5235 kJ/hr, if the room temperature is 25°C?

(a) 24.31°C (b) 35.82°C
(c) 42.45°C (d) 56.94°C

ESE 18.07.2021

Ans. (c) : Given, Area = 6 m^2

The removal rate of condenser = $50\frac{\text{kJ}}{\text{hr.m}^2.^\circ\text{C}}$

$\therefore$ Total removal rate = $(50\times 6) = 300 \text{ kJ/hr.°C}$

Heat to be removed = 5235 kJ/hr

$$\therefore \quad \Delta T = \frac{5235}{300} = 17.45^\circ\text{C}$$

$$(T - T_{room}) = 17.45^\circ\text{C}$$

$$T = (17.45 + 25) = 42.45^\circ\text{C}$$

67. The actual and theoretical COP of rolling piston compressor are 3.6 and 4.7 respectively. What is the relative COP?

(a) 8.3 (b) 16.92
(c) 1.3 (d) 0.76

ESE 18.07.2021

Ans. (d)

$$(COP)_{rel} = \frac{\text{Actual COP}}{\text{Theoretical COP}}$$

$$= \frac{3.6}{4.7} = 0.76$$

68. A refrigerator operates on Reversed Carnot cycle. What is the power required to drive the refrigerator between temperatures of 42°C and 4°C, if heat at the rate of 2 kJ/s is extracted from the low temperature region?

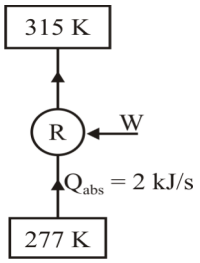
(a) 0.174 kW (b) 0.374 kW
(c) 0.274 kW (d) 0.474 kW

ESE 18.07.2021

Ans. (c) : Given,

$T_H = 42°C = 315 \text{ K}$

$T_L = 4°C = 277 \text{ K}$



$$COP = \frac{Q_{abs}}{W}$$

$$\frac{T_L}{T_H - T_L} = \frac{Q_{abs}}{W}$$

$$\frac{277}{315 - 277} = \frac{2}{W}$$

$$W = 0.274 \text{ kW}$$

69. Ice is formed at 0°C from water at 20°C temperature of the brine is –8°C. The refrigeration cycle used is perfect reversed Carnot cycle. Latent heat of ice = 335 kJ/kg, and C_{pw} = 4.18 kJ/kgK The ice formed per kWh will be nearly

(a) 81.4 kg (b) 76.4 kg
(c) 71.8 kg (d) 68.8 kg

ESE 2020

Ans. (a) : $Q_{Ref} = 335\times m + 4.18(20-0)m$

So,

$$(COP)_{ideal} = \frac{(-8+273)}{[(273+20)-(-8+273)]} = \frac{265}{28}$$

$$\because (COP)_{ideal} = \frac{Q_{Ref}}{\text{Power Input}}$$

$$\frac{265}{28} = \frac{335\times m + 4.18\times 20\times m}{3600}$$

$$m = \frac{265\times 3600}{28\times(335 + 4.18\times 20)} = 81.39$$

$$\approx 81.4 \text{ kg}$$

70. A domestic food freezer maintains a temperature of –15°C. The ambient air temperature is 30°C. If heat leaks into the freezer at the continuous rate of 1.75 kJ/s, the least power necessary to pump this heat out continuously will be nearly

(a) 0.1 kW (b) 0.2 kW
(c) 0.3 kW (d) 0.4 kW

ESE 2020

Ans. (c) :

$$(COP)_{max.} = \frac{T_2}{T_1 - T_2} = \frac{Q_2}{W_{min}} = \frac{258}{303 - 258} = \frac{1.75}{W_{min}}$$

$$W_{min} = 0.305 \text{ kW}$$

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71. The power per tonne of refrigeration (in kW) is

(a) 3.5/COP (b) COP/3.5

(c) 3.5 × COP (d) None of these

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : $COP = \frac{\text{Desired effect}}{\text{Work input}}$

$$COP|_{\text{refrigeration}} = \frac{\text{Refrigeration effect}}{\text{Work input}} = \frac{RE}{W_{in}}$$

R_e = 1 tonne = 3.5 kW

$$W_{in} = \frac{RE}{(COP)_R} = \frac{3.5}{(COP)_R}$$

72. Gas refrigeration cycles

(a) operate with simple and light components

(b) Cannot incorporate regeneration

(c) have higher COPs than vapour compression cycles operating between the same temperature limits

(d) are not suitable for cryogenic applications

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Gas refrigeration cycles operate with simple and light components.

73. For high C.O.P.

(a) Critical temperature in general should be low

(b) Critical temperature in general should be high

(c) Critical temperature is insignificant

(d) Critical pressure is important

RPSC IOF, 2020

Ans. (b) : A high critical temperature yields higher COP due to smaller compressor superheat and smaller flash gas losses on the other hand since the vapour pressure will be low when critical temperature is high, the volumetric capacity will be lower for refrigerants with high critical temperature.

74. The refrigeration cycle suitable for aircraft applications is _____.

(a) vapour absorption refrigeration

(b) vapour compression refrigeration

(c) air compression refrigeration

(d) vapour absorption refrigeration

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : The refrigeration cycle suitable for aircraft applications is air compression refrigeration.

75. A house requires 60 M cal/hr in winter for heating. Heat pump absorbs heat from cold air outside and requires 8 Mcal/hr of work. The COP will be

(a) 0.75 (b) 6.5

(c) 7.5 (d) 10

RPSC IOF, 2020

Ans. (c) : Given, Q_1 = 60 Mcal/hr.

W = 8 Mcal/hr.

$$COP = \frac{Q_1}{W}$$

$$COP = \frac{60}{8} = 7.5$$

76. A Carnot refrigerator operating between –10°C and 25°C will have a COP of :

(a) 2.5 (b) 5.3

(c) 7.5 (d) 8.1

NLC GET 17.11.2020, Shift-II

Ans. (c) : Given,

$T_1 = 25°C = 298K$

$T_2 = -10°C = 263K$

We know when Carnot refrigeration operating between T_2 and T_1

$$(COP)_{Ref} = \frac{T_2}{T_1 - T_2}$$

$$= \frac{263}{298 - 263}$$

$$(COP)_{Ref} = 7.5$$

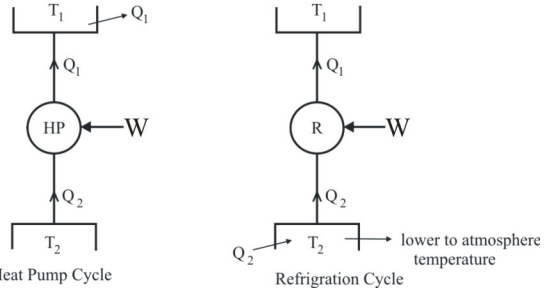
77. If a refrigerator and heat pump operate between the same temperature limits, then the difference between the COP of heat pump and refrigerator is:

(a) 2 (b) 1

(c) 4 (d) 3

NLC GET 17.11.2020, Shift-II

Ans. (b) : According to question,



We know that

$$COP = \frac{\text{desired effect}}{\text{work input}} \quad \text{[For reversible]}$$

$$(COP)_{Heat\ Pump} = \frac{T_1}{T_1 - T_2} \quad \text{[For reversible]}$$

$$(COP)_{\text{Refrigation}} = \frac{T_2}{T_1 - T_2}$$

$$(COP)_{\text{H.P.}} - (COP)_R = \frac{T_1}{T_1 - T_2} - \frac{T_2}{T_1 - T_2}$$

$$= \frac{T_1 - T_2}{T_1 - T_2}$$

$$\boxed{(COP)_{HP} - (COP)_{\text{Ref}} = 1}$$

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78. A refrigerator operates between the temperature of –23°C and 27°C. If one TR = 3.5 kW, the minimum power required (in kW) per TR to operate the refrigerator is

(a) 0.5 (b) 0.7
(c) 0.9 (d) 1.0

Assam PSC AE (PHED), 18.10.2020

Ans. (b) : Given-

$Q_2 = 3.5$ kW

$T_L = -23°C = 250$ K

$T_H = 27°C = 300$ K

$$(COP)_R = \frac{T_L}{T_H - T_L}$$

$$= \frac{250}{300 - 250}$$

$(COP)_R = 5$

$$(COP)_R = \frac{Q_2}{W}$$

$$W = \frac{Q_2}{(COP)_R}$$

$$W = \frac{3.5}{5} = 0.7$$

79. For a refrigerator working on ideal Carnot cycle, having evaporator coil at 240 K and the condenser coil at 300 K, what is approximately the theoretical COP?

(a) 0.5 (b) 2.0
(c) 4.0 (d) 5.0

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : $$COP = \frac{T_2}{T_1 - T_2} = \frac{\text{Heat absorbed}}{\text{Work done}}$$

$$COP = \frac{240}{300 - 240} = 4$$

80. One tonne refrigeration machine means:

(a) One tonne of water is converted into ice
(b) One tonne of ice melts from 0°C in 24 hours
(c) Mass of the machine
(d) One tonne of refrigerant is used

CIL MT 27.02.2020

Ans. (b) : A tonne is the cooling capacity of an air conditioning system. One tonne is equal to the amount of heat required to melt one tonne of ice in a 24-hour period.

81. If the cooling rate in the refrigeration cycle is 420 kJ/min, then the ton of refrigeration is:

(a) 8 (b) 6
(c) 4 (d) 2

NLCIL GET 17.11.2020, Shift-II

Ans. (d)

1 ton of refrigeration = 3.5 kJ/sec = 210 kJ/min

$$\text{Cooling rate} = \frac{420}{210} = 2 \text{ TR}$$

82. The ideal gas-refrigeration cycle is the same as the

(a) Brayton cycle
(b) Reversed Brayton cycle
(c) Vapour compression refrigeration cycle
(d) Vapour absorption refrigeration cycle

ESE 2020

Ans. (b) : Reversed Brayton cycle

83. For refrigeration system work done per kg of air is 30 kcal and heat extracted per kg of air is 45 kcal. Amount of refrigerant used is 10 kg. The coefficient of performance of system is:

(a) 6.52 (b) 1.50
(c) 10.56 (d) 0.67

HPPSC ADF 05.03.2019

Ans. (b) : Given-

Work done (W_{in}) = 30 k cal.

Heat extracted (Q) = 45 k cal

$$COP = \frac{\text{Heat extracted}}{W_{in}} = \frac{45}{30} = 1.50$$

84. A reversed Carnot engine is used for heating a building. It supplies 210×10^3 kJ/hr of heat to the building at 20°C. The outside air is at –5°C. The heat taken from the outside will be nearly

(a) 192×10^3 kJ/hr
(b) 188×10^3 kJ/hr
(c) 184×10^3 kJ/hr
(d) 180×10^3 kJ/hr

ESE 2019

Ans. (a) : Given,

$T_1 = 20 + 273 = 293$ K

$T_2 = -5 + 273 = 268$ K

$Q_1 = 210 \times 10^3$ kJ/hr

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \Rightarrow \frac{210 \times 10^3}{Q_2} = \frac{293}{268}$$

$Q_2 = 192 \times 10^3$ kJ/hr

85. If a Carnot refrigerator and a Carnot heat pump are operating between the same two thermal reservoirs, and the COP of the refrigerator is 2.5, then COP of the heat pump will be :

(a) 4.5 (b) 1.5
(c) 2.5 (d) 3.5

BHEL ET 2019

Ans. (d) : Given $(COP)_{Ref} = 2.5$

$(COP)_{HP} = 1 + (COP)_{Ref} = 1 + 2.5$

$(COP)_{HP} = 3.5$

86. A refrigerator with a COP of 2 removes the heat from the refrigerated space at the rate of 100 kJ/min. The amount of heat rejected to surrounding will be:

(a) 2.0 kW (b) 3.0 kW
(c) 1.5 kW (d) 2.5 kW

SJVN ET 2019

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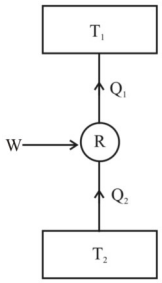
Ans. (d) : Given,



COP = 2

Q_2 = 100 kJ/min

$= \frac{100}{60} = 1.67$ kJ/sec

$COP = \frac{Q_2}{W}$

$W = \frac{1.67}{2} = 0.834$ kJ/sec

$Q_1 = W + Q_2$

$W = Q_1 - Q_2$

$Q_1 = 0.834 + 1.67 = 2.50$ kJ/sec

87. If the operating temperatures for a Carnot pump is –3 °C and 27 °C, then its COP is:

(a) 10 (b) 8

(c) 4 (d) 6

AAI Jr. Executive 29.11.2018

Ans. (a) : $COP = \frac{\text{Desired effect}}{W_{input}}$

$(COP)_{HP} = \frac{T_H}{T_H - T_L} = \frac{300}{300-270} = \frac{300}{30}$

$(COP)_{HP} = 10$

88. A computer lab having a load of 450 MJ/day is cooled round the clock. It requires an air-conditioning plant of capacity about:

(a) 2 ton (b) 1.5 ton

(c) 1.75 ton (d) 2.75 ton

AAI Jr. Executive 29.11.2018

Ans. (b) : In a air conditioning, load of refrigeration = 450 MJ / day

$= \frac{450 \times 1000}{24 \times 3600} = 5.2083 \text{ kJ/sec} = 5.2083 \text{ kW}$

[$\because$ 1 Tonne of refrigeration = 3.5 kW]

So,

Load required capacity of air conditioning =

$\frac{5.2083}{3.5} = 1.488 \approx 1.5$ Tonne

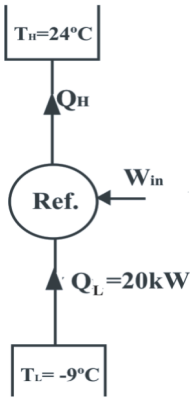
89. The operating temperature of a cold storage is -9ºC and load for the refrigeration plant is 20 kW for ambient temperature of 24ºC. The actual COP of the plant used is one forth that of ideal plant working between the same temperature. The power required to drive the plant is:

(a) 2.5 kW (b) 5 kW

(c) 10 kW (d) 3 kW

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (c) : COP of ideal refrigeration



$(COP)_{\text{ideal Ref.}} = \frac{T_L}{T_H - T_L}$

$(COP)_{\text{Ideal Ref.}} = \frac{(-9+273)}{(24+273)-(-9+273)}$

$(COP)_{\text{Ideal Ref.}} = 8$

$(COP)_{\text{actual Ref.}} = \frac{(COP)_{\text{ideal Ref.}}}{4} = \frac{8}{4}$

$2 = \frac{20}{W_{in}}$

$\boxed{W_{in} = 10\text{kW}}$

90. A Carnot refrigerator working between 300 K and 250 K has the cooling capacity of 10 TR. Rate of heat rejected to the high temperature reservoir will be:

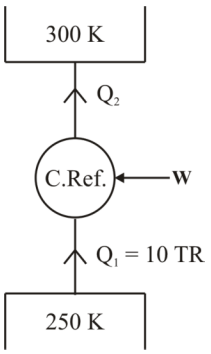
(a) 35.1 kW (b) 7.02 kW

(c) 42.12 kW (d) 28 kW

(e) 30 kW

CGPSC AE 25.02.2018

Ans. (c) :



$[COP]_{Ref} = [COP]_{actual}$

[1 TR = 3.51685 kW]

$\frac{250}{300-250} = \frac{10 \times 3.51685}{Q_2 - 10 \times 3.51685} = 5$

$\boxed{Q_2 = 42.12 \text{ kW}}$

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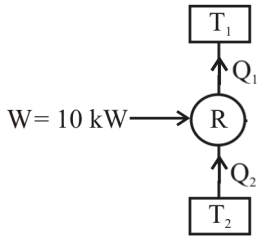
91. A refrigerator working on a reversed Carnot cycle absorb 10 kW power and has a COP of 4. What will be the quantity of heat delivered to a building if it works as a heat pump?

(a) 30 kW (b) 40 kW
(c) 50 kW (d) 60 kW

RPSC Vice Principal ITI 2018

Ans. (c)



$(COP)_{Ref} = 4$

$$4 = \frac{RE}{W}$$

$RE = 4 \times 10 = 40$ kW

Hence heat supplied to building

$= 40 + 10 = 50$ kW

92. A cold storage has capacity for food preservation at a temperature of –3 °C when the outside temperature is 27°C. The minimum power required to operate with a cooling load of 90 kW is

(a) 20 kW (b) 15 kW
(c) 10 kW (d) 5 kW

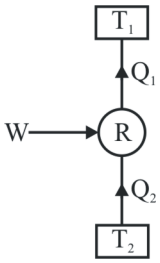
ESE 2018

Ans. (c) : Given,

$Q_2 = 90$ kW

$T_2 = -3 + 273 = 270$ K

$T_1 = 27 + 273 = 300$ K



minimum power is when COP is maximum and maximum COP is Carnot COP

$$(COP)_R = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}$$

$$(COP)_{carnot} = \frac{T_2}{T_1 - T_2} = \frac{270}{300 - 270} = 9$$

$$(COP)_{carnot} = \frac{Q_2}{W}$$

$$W = \frac{90}{9} = 10 \text{ kW}$$

93. An ideal refrigerator working on a reversed Rankine cycle has a capacity of 3 tons. The COP of the unit is found to be 6. The capacity of the motor required to run the unit is (take 1 tonn = 210 kJ/min)

(a) 1.85 kW (b) 1.75 kW
(c) 1.65 kW (d) 1.50 kW

ESE 2018

Ans. (b) : Given,

Refrigerating effect (RE) = $3 \times 210 = 630$ kJ/min

$(COP)_R = 6$

$$(COP)_R = \frac{RE}{W}$$

$$W = \frac{RE}{(COP)_R} = \frac{630}{6} = 105 \text{ kJ/min} = 1.75 \text{ kW}$$

94. The Bell Coleman cycle consists of:

(a) Two adiabatic processes and two constant pressure processes
(b) Two adiabatic processes and two constant volume processes
(c) Two isothermal processes and two adiabatic processes
(d) Two constant volume processes and two constant pressure processes

PTCUL AE 25.06.2017

Ans. (a) : The Bell Coleman cycle consists of two adiabatic processes and two constant pressure processes.

95. In a boot-strap air evaporative cooling system, the evaporator is provided

(a) between the combustion chamber and the first heat exchanger
(b) between the first heat exchanger and the secondary compressor
(c) between the secondary compressor and the second heat exchanger
(d) between the second heat exchanger and the cooling turbine

Karnataka PSC AE, 10.09.2017

Ans. (d) : Evaporator is placed between the second heat exchanger and the cooling turbine.

96. Which of the following conditions clearly indicates the difference between a refrigerator and a heat pump?

(a) COP of refrigerator > 1 and COP of heat pump < 1
(b) COP of refrigerator < 1 and COP of heat pump > 1
(c) COP of refrigerator ≥ 1 and COP of heat pump ≤ 1
(d) COP of refrigerator < or > 1 and COP of heat pump > 1

Karnataka PSC AE, 10.09.2017

Ans. (d) (COP) Refrigerator is > or < 1

$(COP)_{VCRS} > 1$

$(COP)_{VARS} < 1$

$(COP)_{HP} > 1$

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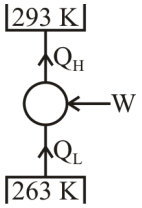
97. A perfect reversed heat engine is used for making ice at –5°C from water available at 20° C. The temperature of brine is –10°C. If the specific heat for ice is 2 kJ/kg and latent heat of ice is 335 kJ/kg, then the quantity of ice formed per kW hour will be (Specific heat of water = 4.2 kJ/kg)

(a) 94 kg (b) 74 kg
(c) 88 kg (d) 86.4 kg

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Given, $T_H = 273 + 20 = 293$ K

$T_L = 273 - 10 = 263$ K



$$COP = \frac{263}{293-263} = 8.76$$

Heat extracted $= mC_p\Delta T + mL$

$= m[4.2 \times 20 + 335 + 2 \times 5]$

$Q_L = m[84 + 335 + 10]$

$Q_L = m[429]$

Work input, W = 1 kWhr = 3600 kJ

$$COP = \frac{Q_L}{W} = \frac{m \times 429}{3600}$$

$$8.76 = \frac{m \times 429}{3600}$$

$m = 73.51 \simeq 74$ kg

98. A room contains 25 kg of air at 100 kPa and 14°C. The room has a 250 W refrigerator and 1 kW electric resistance heater. During a cold winter day, it is observed that both the refrigerator and the heater are running continuously, but the air temperature in the room remains constant. The rate of heat loss from the room during that day is

(a) 3600 kJ/hr. (b) 1250 kJ/hr.
(c) 4500 kJ/hr. (d) 2700 kJ/hr.

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : To keep the temperature of the room as constant, the heat loss must be equal to the total heat dissipated in room.

Total heat generated in the room $= Q_R + Q_{heater}$

$= 250 + 1000 = 1250$W

$$= \frac{1250 \times 3600}{1000} = 4500 \text{kJ/hr}$$

99. An ideal refrigerator has a capacity of 6 tons, with a COP of 6. The capacity of the prime mover to run the machine should be at least

(a) 2.5 kW (b) 3.5 kW
(c) 4.5 kW (d) 5.5 kW

JWM 2017

Ans. (b) : Coefficient of Performance, COP = 6

Refrigeration capacity = 6 tons.

$= 6 \times 3.5$ kW

$$COP = \frac{\text{Refrigeration capacity}}{\text{Power input}}$$

$$6 = \frac{6 \times 3.5}{P_{input}}$$

$P_{input} = 3.5$ kW

100. 'COP' of a reversible heat pump is 1.2. If it is reversed to run as reversible heat engine then its efficiency shall be

(a) 0.833 (b) 0.2
(c) 1.2 (d) 0.5

TNPSC AE 2017

Ans. (a) : $[COP]_{H.P.} = 1.2$

We know that for same temperature limits,

$$\eta_{H.E.} = \frac{1}{[COP]_{H.P.}}$$

$$\eta_{H.P.} = \frac{1}{1.2} = 0.833$$

$\boxed{\eta_{H.E.} = 83.3\%}$

101. A heat pump working on a reversed Carnot cycle has a COP of 5. If it works as a refrigerator and consumes 2 kW power, the heat absorbed by the refrigerator will be

(a) 6 kW (b) 7 kW
(c) 8 kW (d) 9 kW

JWM 2017

Ans. (c) : COP of heat pump, $COP_{HP} = 5$

COP of Refrigerator, $COP_{Ref.} = COP_{HP} - 1$

$= 5 - 1 = 4$

$$COP = \frac{\text{Refrigerating effect}}{\text{Work done}}$$

Refrigerating effect $= COP_{Ref.} \times$ Work done

Heat absorbed by refrigerator $= 4 \times 2 = 8$ kW

102. The refrigeration system of an ice plant working between temperatures of -5°C and 25°C produces 20kg of ice per minute from water at 20°C. The specific heat of water is 4.2 kJ/kg and latent heat of ice is 335 kJ/kg. The refrigeration capacity of the refrigeration plant is

(a) 9040 kJ/min (b) 8750 kJ/min
(c) 8380 kJ/min (d) 8010 kJ/min

ESE 2017

Ans. (c) : Given,

$T_H = 25°C$ $(\dot{m})_{ice} = 20$ kg/min

$T_L = -5°C$ $T_{initial} = 20°C$

$(C_P)_{water} = 4.2$ kJ/kg

$(L)_{ice} = 335$ kJ/kg

Refrigeration capacity $= \dot{m} C_P \Delta T + \dot{m}L$

$= 20 \times 4.2 \times (20 - 0) + 20 \times 335 = 8380$ kJ/min

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103. Consider the following statements for refrigeration and air-conditioning :

1. In a refrigerating machine, the heat exchanger that absorbs heat is connected to a conditioned space.
2. A refrigerating cycle operating reversibly between two heat reservoirs has the highest coefficient of performance.
3. The lower the refrigeration required and the higher the temperature of heat rejection to the surroundings, the larger the power consumption.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (d) : All three statements are correct.
For lower refrigeration requirement and higher heat rejection, work input required will get increase because of high compression ratio.

104. In a refrigerator, the evaporator and condenser coil temperatures are -33°C and 27°C, respectively. Assuming that the maximum COP is realized, the required power input for a refrigerating effect of 4 kW is

(a) 8 kW (b) 4 kW
(c) 2 kW (d) 1 kW

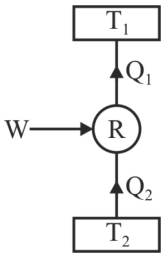
ESE 2017

Ans. (d) : Given,

$Q_2 = 4$ kW

$T_1 = 27 + 273 = 300$ K

$T_2 = -33 + 273 = 240$ K



$$(COP)_R = \frac{\text{Refrigerating effect}}{\text{Work input}}$$

$$(COP)_R = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$$

$$= \frac{240}{300 - 240} = \frac{240}{60} = 4$$

$$(COP)_R = \frac{Q_2}{W} = \frac{4}{W}$$

$$W = \frac{4}{4} = 1 \text{ kW}$$

105. Which of the following statements is true for throttling process?

(a) Enthalpy remains constant during throttling process.
(b) Entropy remains constant during throttling process.
(c) Pressure remains constant during throttling process.
(d) It is a reversible process.
(e) It is an isentropic process.

CGPSC ARTO & TSI 07.01.2017

Ans. (a) :Throttling Process :- When the air or gas flowing through an small opening then enthalpy remains constant. It is known as isenthalpic or throttling Process.

Note :- In throttling process

- There is no work and heat transfer.
- Pressure always decreases.
- During throttling process temperature may or may not be increase or decrease or remains constant but for an ideal gas temperature remain constant.

S.No.	Process	Parameter Remains Constant
1.	Isobaric Process	P (Pressure)
2.	Isochoric Process	V (Volume)
3.	Isothermal Process	T (Temperature)
4.	Isentropic Process	S (Entropy)
5.	Throttling Process	h (Enthalpy)

106. A Carnot cycle refrigerator operates between 250 K and 300 K. Its co-efficient of performance is

(a) 6.0 (b) 1.2
(c) 5.0 (d) 0.8

GPSC ARTO 01.05.2016

Ans. (c) : Lower temperature, $T_L = 250$K
Higher temperature, $T_H = 300$K

$$(Cop)_{refrigerato} = \frac{T_L}{T_H - T_L} = \frac{250}{300 - 250} = 5$$

107. A refrigeration cycle operates between condenser temperature of 27 deg C and evaporator temperature of -23 deg C. The carnot efficiency of performance of cycle will be

(a) 1.2 (b) 0.16
(c) 0.6 (d) 6

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (b) Given-

$T_H = 27 + 273 = 300$ K

$T_L = -23 + 273 = 250$ K

$$(COP)_R = \frac{T_L}{T_H - T_L}$$

$$= \frac{250}{300 - 250} = \frac{250}{50} = 5$$

$$\eta_{carnot} = \frac{1}{1 + (COP)_R} = \frac{1}{1+5} = \frac{1}{6}$$

$$\eta_{carnot} = 0.16$$

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108. Air – conditioning load for a vehicle is lower when it is moving and higher when it is at stationary condition. Choose appropriate option.

(a) False
(b) True
(c) It depends upon ambient conditions and speed of the vehicle
(d) Incomplete information

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : For moving vehicle load on vehicle AC is less as compared stationary condition.

109. What is one ton of refrigeration equal to?

(a) approximately 12,000 BTU/hr
(b) approximately 120 BTU/hr
(c) approximately 12 BTU/hr
(d) none of these

KPCL AE 2016

Ans. (a) : 1TR = 12000 BTU/hr
= 200 BTU/min
(BTU- British Thermal unit)

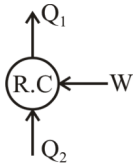
110. A ton of refrigeration is defined as_____

(a) the heat of fusion absorbed by melting 1 ton of pure ice at 32ºC in 24 hours
(b) the heat of fusion absorbed by melting 1 ton of pure ice at 0ºC in 1 hour
(c) the heat of fusion absorbed by melting 1 ton of pure ice at 0ºC in 24 hours
(d) none of these

KPCL AE 2016

Ans. (c) : A ton of refrigeration is defined as the heat of fusion observed by melting 1 ton of pure ice at 0ºC in 24 hours.
1 Ton = 3.5 kW = 210 kJ/min

111. Observe the figure below and choose the correct option



(a) $COP_{Refrigerator} = \frac{Q_1}{W}$

(b) $COP_{Refrigerator} = \frac{Q_1 - Q_2}{W}$

(c) $COP_{Heat\ pump} = \frac{Q_2}{W}$

(d) $COP_{Heat\ pump} = \frac{Q_1}{W}$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : $COP_{Heat\ pump} = \frac{Q_1(High)}{Work\ Input} = \frac{Desired\ effect}{Work\ Input}$

112. A domestic food freezer maintains a temperature of –23ºC. The ambient air temperature is 27ºC. If heat leaks into the freezer at continuous rate of 2kJ/sec. What is the least power is necessary to pump this heat out continuously

(a) 40 W
(b) 400 W
(c) 2100 W
(d) 210 W

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : $T_L = 250K$, $T_H = 300K$, $Q = 2kJ/sec$,

$$COP = \frac{250}{300-250} = \frac{250}{50} = 5$$

$$COP = \frac{Q}{W_{in}} \Rightarrow W_{in} = \frac{2\times1000}{5} = 400\,W$$

113. 1.5 kW per tonne of refrigeration is required to maintain the temperature of –40ºC in the refrigerator. If the refrigerator cycle works on Carnot cycle, determine the temperature of the sink.

(a) 233 K
(b) 333 K
(c) 240 K
(d) 340 K

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : $W_{in} = 1.5kW$, $Q = 3.5kW$, $T_L = 233K$

$$COP = \frac{3.5}{1.5} = 2.33$$

$$COP = \frac{T_L}{T_H - T_L} \Rightarrow T_H - 233 = \frac{233}{2.33}$$

$$T_H = 100 + 233 = 333K$$

114. The capacity of a refrigerator is 200 TR when working between –6°C and 25°C. Determine the mass of ice produced per day from water at 25°C. Assume that the cycle operates on reversed Carnot cycle and latent heat of ice is 335 kJ/kg.

(a) 137.55 tonnes
(b) 127.83 tonnes
(c) 187.34 tonnes
(d) 157.78 tonnes
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : Q = 200 TR, $T_1 = –6°C = 267$ K, $T_2 = 25°C$
$h_{fg} = 335$ kJ/kg
Heat extraction capacity = 200 × 210
= 42000 kJ/min
Heat removed from 1 kg,
= mCdT + mh_{fg}
= 1 × 4.186 (25 – 0) + 1 × 335 = 439.7 kJ/kg
Mass of ice produced per min

$$= \frac{42000}{439.7} = 95.52\,kg/min$$

Mass per day = 95.52 × 60 × 24
= 137.55 tonnes

115. Carnot refrigeration cycle absorbs heat at 270 K and rejects heat at 300 K. If the cycle is absorbing 1130 kJ/min at 270 K, how many kJ of work is required per second?

(a) 1.8 kJ/s
(b) 2.1 kJ/s
(c) 3.4 kJ/s
(d) 4.8 kJ/s
(e) None of these

CGPSC AE 16.10.2016

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Ans. (b) : $COP_R = \frac{T_1}{T_2 - T_1} = \frac{270}{300 - 270} = 9$

W_R = Work per second

$Q_1 = 1130$ kJ/min = 18.83 kJ/s

$$COP_R = \frac{Q_1}{W_R}$$

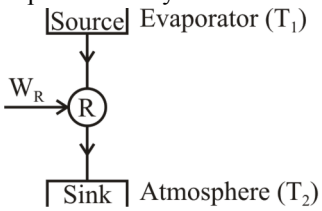
$$W_R = \frac{18.83}{9} = 2.1 kJ/s$$

116. With regard to a refrigeration system, which of the following statements is correct?

(a) It rejects energy to a low temperature body
(b) It removes heat from a high temperature body and delivers it to a low temperature body
(c) It removes heat from a low temperature body and delivers it to a high temperature body
(d) All of the above
(e) None of the above

CGPSC AE 16.10.2016

Ans. (c) : With regards to a refrigeration system. It removes heat from a low temperature body and delivers it to a high temperature body.



117. Which one is incorrect statement?

(a) $COP_{HP} = COP_R + 1$
(b) Efficiency of carnot cycle is given by $\eta_0 = 1 - \frac{T_2}{T_1}(T_1 > T_2)$
(c) Efficiency of Brayton cycle is given by $\eta_{brayton} = 1 - \frac{1}{(r_p)^{\frac{\gamma}{\gamma-1}}}$
(d) Efficiency of Otto cycle is given by $\eta_{Otto} = 1 - \frac{1}{r^{\gamma-1}}$
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : We know that,

$COP_{HP} = COP_R + 1$

Efficiency of carnot cycle is,

$$\eta_c = 1 - \frac{T_2}{T_1}$$

Efficiency of otto cycle

$$\eta_{otto} = 1 - \frac{1}{r_p^{\gamma-1}}$$

Efficiency of Brayton cycle

$$\eta_{Brayton} = 1 - \frac{1}{r_p^{\left(\frac{\gamma-1}{\gamma}\right)}}$$

118. The capacity of a refrigerator is one tone. This means that the heat removing capacity of the refrigerator is

(a) 1000 kCal/hr (b) 2000 kCal/hr
(c) 3000 kCal/hr (d) 4000 kCal/hr

APPSC AEE Screening Test 2016

Ans. (c) : The capacity of a refrigeration is one tonne

1 ton = 3.5 kW = 210 kJ/min = 50 kcal/min

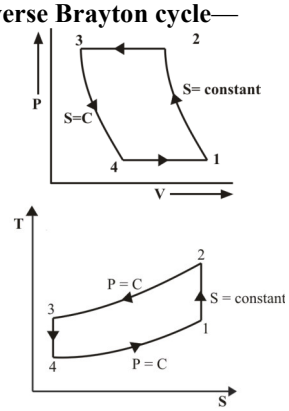
$$\Rightarrow \quad \frac{50 kcal}{\frac{1}{60} hr} = 3000 kcal/hr$$

119. Which of the following is not true for reverse Brayton cycle?

(a) It has very low COP relative to vapour compression cycle
(b) It consists of two constant pressure process and two isentropic process
(c) It's working fluid is generally ecofriendly
(d) Heat transfer in evaporator is at constant temperature

UPRVUNL AE 2016

Ans. (d) : Reverse Brayton cycle—



Reverse Bryton cycle consists of two constant pressure process and two isentropic process.
Heat transfer in evaporator is at constant pressure.

120. Which of the following is not true for heat pump?

(a) Low temperature reservoir is environment
(b) Low temperature reservoir is conditioned space
(c) It is a cyclic device
(d) Work done is negative

UPRVUNL AE 2016

Ans. (b) : Heat pump is a cyclic device which consume work to maintain a space above the surrounding temperature. In which low temperature reservoir is environment.

121. HP/TR of refrigeration system is:

(a) Inversely proportional to the COP of refrigeration system
(b) Directly proportional to the COP of refrigeration system
(c) Independent of COP of refrigeration system
(d) 1+ { COP of refrigeration system}

UPRVUNL AE 2016

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Ans. (a) : We know that

$$(COP)_{ref.} = \frac{\text{Desired effect}}{\text{Work done}} = \frac{TR}{HP}$$

So, HP/TR of refrigeration system inversely proportional to the COP of refrigeration.

122. 500 kg of fruits are supplied to a cold storage at 20°C. The cold storage is maintained at 5°C and the fruits get cooled to the storage temperature in 10 hr. The latent heat of freezing is 105 kJ/kg and specific heat of fruit is 1.256 kJ/kgk. Find the refrigeration capacity of the plant.

(a) 0.82 TR (b) 0.91 TR
(c) 0.13 TR (d) 0.54 TR
(e) None of these

CGPSC AE 16.10.2016

Ans. (e) m = 500 kg, $T_2 = 20°C = 293$ K, $T_1 = 278$ K

$h_{fg} = 105$ kJ/kg, $C_f = 1.256$ kJ/kg-K

$Q_1 = mC_f(T_2 - T_1)$

$= 500 \times 1.256 \times (20 - 5) = 9420$ kJ

$Q_2 = m.h_{fg} = 500 \times 105 = 52500$ kJ

Total heat removed in 10 hr.

$Q = Q_1 + Q_2 = 9420 + 52500$

$= 61920$ kJ

Total heat removed in one min.

$$= \frac{61920}{10 \times 60} = 103.2\,\text{kJ/min}$$

Refrigeration capacity,

$$= \frac{103.2}{210} = 0.4914\,\text{TR}$$

123. The working temperatures in evaporator and condenser coils of a refrigerator are –25°C and 30°C, respectively. The COP of the refrigerator is 0.85 of the maximum COP for a power input of 2 kW. The refrigeration effect produced will be

(a) 7.6 kW (b) 9 kW
(c) 10.2 kW (d) 12 kW

ISRO Scientist/Engineer 03.07.2016

Ans. (a) : $T_c = -25 + 273$

$= 248$ K

$T_E = 30 + 273 = 303$ K

$$COP = \frac{T_L}{T_H - T_L}$$

$$= \frac{248}{303 - 248} = \frac{248}{55} = 4.5$$

$(COP)_{Actual} = 0.85 \times 4.5$

$$(COP)_A = \frac{Q(\text{Ref. effect})}{W}$$

$Q = 3.825 \times 2$

$\boxed{Q = 7.6\,\text{kW}}$

124. Assertion (A) : Heat pump used for heating is a definite advancement over the simple electric heater.

Reasons (R) : The heat pump is far more economical in operation than electric heater.

(a) Both A & R are correct and R is correct explanation of A.
(b) Both A & R are correct ad R is not correct explanation of A.
(c) A is true R is false
(d) A is false R is true

TSPSC Managers, 2015

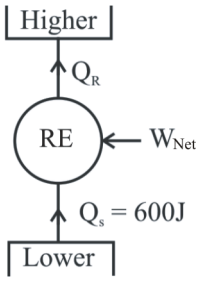
Ans. (a) : Heat pump used for heating is a definite advancement over the simple electric heater because the heat pump is more economical in operation than electric heater.

125. A refrigerator does 150 J of work to transfer 600 J of heat from its cold compartment. Calculate the refrigerator's COP and heat exhausted in J.

(a) 4.00 & 700 (b) 4.65 & 435
(c) 5.62 & 650 (d) 2.34 & 800

NPCIL ET, 2015

Ans. (*) : Given-



Net Work (W_{Net}) = 150 J

$Q_R = Q_s + W_{Net}$

$Q_R = 600 + 150$

$Q_R = 750$ J

$$(COP)_{Ref} = \frac{600}{150} = 4$$

126. Relation between coefficient of performance [COP] of refrigeration and heat engine efficiency (η) working in between same environment is :

(a) $\eta = 1 - COP$ (b) $COP = 1 + \eta$
(c) $1 + COP = \frac{1}{\eta}$ (d) $COP = \frac{1}{\eta}$

KPSC ADF 2015

Ans. (c) : Relation between coefficient of performance $(COP)_{ref}$ of refrigeration and heat engine efficiency (η_{HE}) is–

$$\boxed{1 + (COP)_{Ref} = \frac{1}{\eta_{HE}}}$$

127. Horse power per ton of refrigeration is expressed as

(a) $\frac{4.75}{COP}$ (b) $\frac{COP}{4.75}$
(c) $4.75 \times COP$ (d) $\frac{47.5}{COP}$

Nagaland PSC CTSE 2015 Paper-II

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Ans. (a) : We know that,

$$COP = \frac{RE}{W_{in}} \Rightarrow \frac{W_{in}}{RE} = \frac{1}{COP}$$

$\therefore$ Horse power per ton of refrigeration

$$= \frac{1}{COP} \times \left(\frac{3500}{746}\right) = \frac{4.7}{COP}$$

(1 ton = 210 kJ/min = 3.5 kW, 1 HP = 746 Watt)

128. A water chiller with a capacity of 30 tons of refrigeration cools 20 m^3/hr of water entering at 12°C. What is the temperature of water leaving the chiller?

(a) 5°C (b) 6°C
(c) 7°C (d) 7.5°C

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : Given,

$Q = 30 \times 3.5167$
$= 105.5$ kW

$Q = M_w \times C_p (T_i - T_{outlet})$

$$105.5 = \frac{20 \times 1000}{3600} \times 4.18(12 - T_{outlet})$$

$T_{outlet} = 7.456 \approx 7.5°C$

129. A cold storage plant of 20 tonne of refrigerator capacity operates between 200 and 300 K. The power required to run the plant if the plant has half COP of a Carnot cycle (Tonne of refrigeration= 3.5 kW) is

(a) 50kW (b) 60kW
(c) 70kW (d) 80kW

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (c) COP of Carnot refrigerator is given as

$$(COP)_c = \frac{T_L}{T_H - T_L} = \frac{200}{(300 - 200)} = 2$$

$$(COP)_{Actual\ ref.} = \frac{(COP)_C}{2} = \frac{2}{2}$$

$(COP)_{actual\ ref.} = 1$

$$(COP)_{act.\ ref.} = 1 = \frac{\text{Ref. capacity}}{\text{work done}(W)/\text{sec.}}$$

Power = 20 × 3.5
Power = 70 kW

130. A refrigerator operates on a reverse Carnot cycle between 900K and 300K. If heat at the rate of 3 kJ/s is extracted from the low temperature space, then the power required to drive the refrigerator-

(a) 3kW (b) 6kW
(c) 9kW (d) 4kW

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (b) $[COP]_{Ref} = \frac{T_L}{T_H - T_L} = \frac{\text{Refrigeration effect}}{\text{work done}}$

$$= \frac{300}{900 - 300} = \frac{3}{W}$$

$\boxed{W = 6\ kW}$

131. A refrigerator with a COP of 3 uses 2.4 kg/min refrigerant extracting 150 kJ/kg heat in the evaporator. Assuming compressor efficiency of 80% the minimum size of the motor is

(a) 1.5 kW (b) 2.5 kW
(c) 6 kW (d) 4.5 kW

TSPSC AEE 2015

Ans. (b) : Data given-

COP = 3 m = 2.4 kg/min
Extracting heat = 150 kJ/kg
$\eta_c = 80\%$

$$COP = \frac{\text{Refrigeration capacity}}{W_{in}} = \frac{2.4 \times 150}{W_{in} \times 60} = 3$$

$\boxed{W_{in} = 2kW}$

We know that

$$\eta_c = \frac{W_{in}}{[W_{in}]_{act.}}$$

$$[W_{in}]_{act.} = \frac{2 \times 100}{80} = 2.5\ kW$$

132. A Carnot refrigerator rejects 3000 kJ of heat at 400 K while using 1000 kJ of work. The lowest operating temperature in the cycle should be about :

(a) 288 K (b) 300 K
(c) 267 K (d) 273 K

Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : Given, $Q_R = 3000$ kJ, W = 1000 kJ
$Q_a = 3000 - 1000 = 2000$ kJ
$T_H = 400$ K, $T_L = ?$

$$(COP)_{Ref.} = \frac{T_L}{T_H - T_L} = \frac{Q_a}{W}$$

$$\frac{T_L}{400 - T_L} = \frac{2000}{1000}$$

$T_L = 2 \times (400 - T_L)$
$T_L = 800 - 2T_L$
$3T_L = 800$
$T_L = 266.67$ K

133. A domestic refrigerator of 0.75 tonne capacity is found to have a Co-efficient of performance of 7.5. It's power consumption is

(a) 750 W (b) 550 W
(c) 350 W (d) 150 W

MPPSC State Forest Service Exam, 2014

Ans. (c) : $COP = \frac{Q_H}{W}$

So, 0.75 tonne capacity means, 0.75 × 3.51 = 2.63 kW
COP = 7.5

So, $W = \frac{2.63 \times 10^3}{7.5} = 350\ W$

134. A reversed Carnot cycle removes 40,000 W from a heat sink. The temperature of the heat sink is 260 K and the temperature of the heat reservoir is 320 K. The COP of the engine is :

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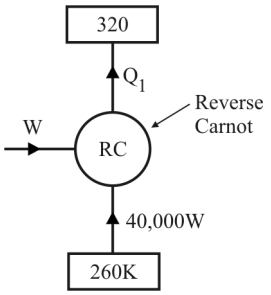
(a) 2.33 (b) 3.33
(c) 4.33 (d) 5.33

MPSC Poly. Lect. 2014

Ans. (c) : $COP = \frac{T_L}{T_H - T_L} = \frac{260}{320 - 260} = 4.33$



135. A condenser of a refrigeration system rejects heat at rate of 150 kW, while its compressor consumes 60 kW power. What would be the coefficient of performance of the system?

(a) 1 (b) 1.5
(c) 2 (d) 2.5

UPRVUNL AE 2014

Ans. : (b) Coefficient of performance (COP)

$$= \frac{\text{Refrigeration Effect}}{\text{Work done}}$$

$$COP = \frac{150 - 60}{60} = \frac{90}{60} = 1.5$$

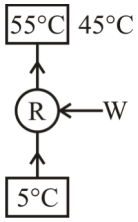
COP= 1.5

136. The surrounding air temperature in summer is 45°C in a locality. The refrigeration temperature is 5°C. Assume 10°C temperature difference in the heat exchanger that exchanges heat with surroundings. The COP of the Carnot refrigerator under these conditions, will be:-

(a) 6.95 (b) 7.95
(c) 6.56 (d) 5.56

UKPSC AE-2013, Paper-II

Ans. (d) :



$$(COP)_R = \frac{(273+5)}{(273+55)-(273+5)} = 5.56$$

137. A heat pump operating on Carnot cycle pumps heat from a reservoir at 300 K to a reservoir at 600 K. The coefficient of performance is

(a) 1.5 (b) 0.5
(c) 2 (d) 1

APPSC Poly. Lect. 2013

Ans. (c) : Heat pump

$$(COP)_{HP} = \frac{T_H}{T_H - T_L} = \frac{600}{600 - 300}$$

$$(COP)_{HP} = \frac{600}{300} = 2$$

138. A loaded domestic refrigerator consumes 1.0 kWh of energy during a certain period. The internal energy of the system drops by 5000 kJ. The quantity of heat transferred from the system to the surroundings will be

(a) 86 MJ (b) 860 MJ
(c) 8.6 MJ (d) 0.86 MJ

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : W_{input} = 1 kW-hr = 3600 kJ

$\Delta U = -5000$ kJ

$\delta Q = \Delta U + \delta W$

$\delta Q = -5000 - 3600 = -8600 = -8.6$ MJ

The heat transferred from the system to the surrounding is 8.6 MJ.

139. Thermo electric refrigeration system is based on :

(a) Joule effect
(b) Peltier effect
(c) Joule – Thomson effect
(d) Adiabatic demagnetisation

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Peltier Effect–The peltier effect is the reverse phenomenon of the see beck effect. The peltier effect occurs when a temperature difference is created between the junctions by applying a voltage difference across the terminals.

Thermoelectric refrigeration is the process of pumping heat energy out of insulated chamber in order to reduce the temperature of the chamber below that of the surrounding air.

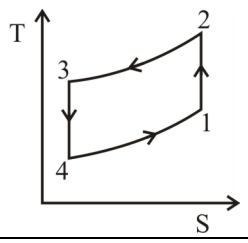
Thermoelectric refrigeration uses a principal called the peltier effect to pump heat electronically.

140. The ratio of temperatures at the beginning and end of isentropic expansion process in Bell-Coleman air refrigeration cycle is 1.25. What is the COP

(a) 0.25
(b) 4.0
(c) 5.0
(d) The data are insufficient.

RPSC AEN Pre-2013

Ans. (b) :



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Given,

$$\frac{T_3}{T_4}=\frac{T_2}{T_1}=\left(r_p\right)^{\frac{\gamma-1}{\gamma}}=1.25$$

$$(COP)_{Bell-Coleman}=\frac{1}{\left(r_p\right)^{\frac{\gamma-1}{\gamma}}-1}$$

$$=\frac{1}{1.25-1}=4$$

141. The main characteristic feature of Air Refrigeration System is
(a) High COP
(b) throughout the cycle the refrigerant remains in gaseous state
(c) Weight of air circulate is less
(d) High Refrigerating cost

TNPSC AE 2013

Ans. (b) : The main characteristic feature of Air Refrigeration System is throughout the cycle the refrigerant remains in gaseous state.

142. A refrigerator (COP=2.5) removes energy by heat transfer from freezer at 0°C at a rate of 9000 kJ/hr and rejects energy to the surroundings. Determine power input to the refrigerator:-
(a) 614.3 kJ/hr (b) 3600 kW
(c) 659.3 kJ/hr (d) 1 kW

UKPSC AE-2013, Paper-II

Ans. (d) : Given,
COP = 2.5
Refrigeration effect = 9000 kJ/hr

$$=\frac{9000}{3600}=2.5\,kW$$

$$COP=\frac{\text{Refrigeration effect}}{W_{net}}$$

$$W_{net}=\frac{2.5}{2.5}=1\,kW$$

143. A Carnot refrigerator operates between 300.3K and 273K. The fraction of cooling effect required as work input is:-
(a) 20% (b) 10%
(c) 50% (d) Cannot be calculated

UKPSC AE-2013, Paper-II

Ans. (b) : $T_1 =$ 300.3 K
$T_2 = 273$ K

$$(COP)_{Ref}=\frac{T_2}{T_1-T_2}=\frac{273}{300.3-273}=\frac{273}{27.3}=10$$

We know that

$$(COP)_{Ref}=\frac{Q_2}{W_{input}}$$

$$W_{input}=\frac{Q_2}{(COP)_{Ref}}=\frac{Q_2}{10}$$

$$W_{input}=\frac{Q_2}{10}$$

$$W_{input}=0.1\,Q_2$$

$$\boxed{W_{input}=10\%Q_2}$$

144. The refrigeration system works on:-
(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) : The refrigeration system works on second law of thermodynamics.

145. Two reversible refrigerators are arranged in series and their COP are 4 and 5 respectively. The COP of the composite refrigeration system would be
(a) 1.5 (b) 2
(c) 3 (d) 4.5

UKPSC AE 2012 Paper–II

Ans. (b) : The COP of the composite refrigeration system $=\frac{[COP]_1\times[COP]_2}{1+[COP]_1+[COP]_2}=\frac{4\times5}{1+4+5}=\frac{20}{10}=2$

146. The Bell-Coleman refrigeration cycle uses _____ as the working fluid.
(a) Hydrogen (b) Carbon dioxide
(c) Air (d) Any inert gas

UKPSC AE 2012 Paper–II

Ans. (c) : The Bell-Coleman refrigeration cycle uses air as the working fluid.

147. Effects of heat pump and refrigeration respectively are obtained at
(a) compressor and condenser
(b) evaporator and condenser
(c) condenser and evaporator
(d) compressor and evaporator

UKPSC AE 2012 Paper–II

Ans. (c) : Effects of heat pump and refrigeration respectively are obtained at condenser and evaporator.

148. A carnot engine with an efficiency of 0.6 drives a carnot refrigerator, having a COP of 5. The energy absorbed from the cold body by the refrigerator for each kJ energy absorbed from the source by the engine is :
(a) 2 kJ (b) 8 kJ
(c) 0.12 kJ (d) 3 kJ

APGENCO AE 2012

Ans. (d) : Given, $\eta = 0.6$, $Q_s = 1$ kJ
$(COP)_{ref.} = 5$

$$(COP)_{carnot}=\frac{Q_{cold\ side}}{W_{net}}$$

$$5=\frac{Q_c}{W_{net}}$$

$$\because\quad \eta_{carnot}=\frac{W_{net}}{Q_s}=0.6=\frac{W_{net}}{1}$$

$W_{net} = 0.6$ kJ
$Q_{cold} = W_{net} \times 5 = 0.6 \times 5$
$= 3$ kJ

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149. A refrigerator removes heat from a refrigerated space at -5°C at a rate of 0.35 kJ/s and rejects it to an environment at 20°C. The minimum required power input is

(a) 30W (b) 33W
(c) 56W (d) 124W

ISRO Scientist/Engineer 2012

Ans. (b) : In Refrigerator,

$T_L = -5°C, Q = 0.35$ kJ/sec

$T_L = 273 - 5$

$T_L = 268K$

$T_H = 20°C$

$= 273 + 20 = 293K$

$$(COP)_{Ref} = \frac{Desired\ effect}{Work\ input}$$

$$= \frac{T_L}{T_H - T_L}$$

$$= \frac{268}{293-268} = \frac{268}{25} = 10.72$$

$$\therefore \quad (COP)_{ref.} = \frac{Q}{P}$$

$$P = \frac{0.35}{10.72} \times 1000W$$

$= 32.65$ W = 33W

150. A domestic refrigerator maintains a temperature of -20^0C. The ambient air temperature is 25^0C. If heat leaks into the refrigerator at the continuous rate of 1 kJ/s, the least power required to pump this heat out from the refrigerator continuously is :

(a) 18 watt (b) 180 watt
(c) 1.8 kilowatt (d) 18 kilowatt

OPSC Civil Services Pre. 2011

Ans. (b) : Given as,

$Q_2 = 1$ kJ/s = 1000 J/s

$T_1 = 25°C = 298K$

$T_2 = -20°C = 253K$

$$\Rightarrow \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$$

$$\Rightarrow \frac{1\times1000}{W} = \frac{253}{298-253}$$

$$\Rightarrow W = \frac{45\times1000}{253} = 177.7 \text{ watt}$$

$\boxed{\approx 180 \text{ watt}}$

151. 1 ton of refrigerant is

(a) the SI unit used in refrigeration problem
(b) the cooling effect produced by melting 1 ton of ice
(c) the refrigeration effect to freeze 1 ton of water at 0°C into ice at 0°C in 24 hrs.
(d) the refrigeration effect to produce 1 ton of ice at NTP condition

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : One ton of refrigeration in the amount of heat that is needed to extracted from 1 ton of water at 0°C to convert to ice at 0°C in 24 hrs. Latent heat from 0°C water to 0°C ice only.

$\therefore$ 1 TR = 210 kJ/min. = 3.5 kW = 50 kcal/min

152. A Carnot heat pump is used to heat a house in winter months. The average outside temperature is –3°C and indoor temperature is maintained at 27°C. If the power required by the heat pump is 10 kW, the heat supplied to the house is :

(a) 25 kW (b) 50 kW
(c) 75 kW (d) 100 kW

UPSC JWM Advt. No.-52/2010

Ans. (d) : Given, $T_L = -3°C + 273 = 270$ K

$T_H = 27°C + 273 = 300K$

Power Required by heat pump (P) = 10 kW = W_{input}

$$\therefore \quad (COP)_{HP} = \frac{T_H}{T_H - T_L} = \frac{300}{300-270} = \frac{300}{30}$$

$$(COP)_{HP} = 10 = \frac{Q_1}{W_{input}}$$

$\Rightarrow \quad Q_1 = 10 \times 10$

$\boxed{Q_1 = 100 \text{ kW}}$ $\rightarrow$ Heat supplied to the house.

153. A 2-ton capacity water cooler has water entering at 30° C and leaving at 15° C. What is the water flow rate in Lph?

(a) 40 (b) 66.6
(c) 402 (d) 502

UPSC JWM Advt. No.-50/2010

Ans. (c) : As we know,

Capacity = 2 Ton

$t_{in} = 30°C$

$t_{out} = 15°C$

$\dot{m} = ?$

$Q = \dot{m}C_p\Delta t$

$$2\times3.5\times10^3\times3600 = \dot{m}\times4186\times15$$

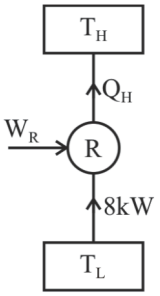
$\dot{m} = 402$ Lph

154. A refrigerating unit has a COP of 5. It extracts 480 kJ of heat from the cold reservoir in one minute. As a heat pump, how much heat it would deliver to the environment ?

(a) 1.6 kW (b) 6.67 kW
(c) 9.6 kW (d) 576 kW

UPSC JWM Advt. No.-50/2010

Ans. (c) : Desired effect $= \frac{480}{60} = 8$kW



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$$(COP)_R = \frac{\text{Desire Effect}}{W}$$

$$(COP)_R = \frac{8}{W} = 5$$

$W_R = 1.6\ kW \Rightarrow W_R = Q_H - Q_R$

$Q_H = Q_R + W$

$Q_H = 8 + 1.6 = 9.6\ KW$

155. A Carnot refrigerator requires 1.5 kW/ton of refrigeration to maintain a region at a temperature of –30°C. The COP of the Carnot refrigerator is :

(a) 1.42 (b) 2.33
(c) 2.87 (d) 3.27

APPSC IOF, 2009

Ans. (b) : 1 ton = 3.51 kW

So, COP will be $= \frac{\text{Desired effect}}{\text{Work input}} = \frac{3.51}{1.5} = 2.34$

156. It is proposed to heat a house in the winter with a heat pump. The house is to be maintained at 20°C at all times. When the ambient temperature out side drops to –10°C the rate at which heat is lost from the house is estimated to be 25 kW. The minimum electrical power required to drive the heat pump is :

(a) 2.56 kW (b) 4 kW
(c) 5 kW (d) 3.56 kW

APPSC IOF, 2009

Ans. (a) : Minimum power if we assume a Carnot cycle,

$$\dot{Q}_H = Q_{leak} = 25\,kW$$

$$COP = \frac{\dot{Q}_H}{W_{in}} = \frac{T_H}{T_H - T_L} = \frac{293}{30} = 9.767$$

$$W_{in} = \frac{25}{9.767} = 2.56\,kW$$

157. The coefficient of performances of Carnot refrigerator operating between the reservoirs at -10^0C and 40^0C is

(a) 0.526 (b) 5.26
(c) 52.6 (d) 526.0

TNPSC AE, 2008

Ans. (b) : Given, $T_L = -10^0C + 273 = 263$ K

$T_H = 40^0C + 273 = 313$ K

$$(COP)_R = \frac{T_L}{T_H - T_L} = \frac{263}{313 - 263} = \frac{263}{50} = \frac{26.3}{5}$$

$$(COP)_R = 5.26$$

158. Air refrigeration works on which cycle?

(a) Rankine
(b) Bell-Coleman cycle
(c) Reversed Carnot cycle
(d) Both (b) and (c)

UKPSC AE 2007 Paper -II

Ans. (d) : Both (b) and (c)

159. A simple refrigeration system operates between a condenser temperature of 40ºC and evaporator temperature of 0ºC respectively. The maximum C.O.P. is

(a) 7.8 (b) ∞ (Infinity)
(c) 0 (Zero) (d) 6.8

OPSC Civil Services Pre 2006

Ans. (d) : Maximum COP of refrigerator

$$COP = \frac{T_2}{T_1 - T_2} = \frac{273}{40} = 6.825$$

160. 'TR', Ton of Refrigeration, is the unit of refrigeration. (1 TR = 3.5 kW). Ten (10) kg of water is cooled from 20ºC to ice at 0ºC (Twenty ºC to Zero ºC) in one hour by an ice-making plant. The capacity of refrigeration of the ice-making plant in 'TR' is

(a) 0.33 TR (b) 1.15 TR
(c) 10 TR (d) 0.1 TR

OPSC Civil Services Pre 2006

Ans. (a) : Given, m = 10 kg, $\Delta T = 20°C$, 1 TR = 3.5 kW

Latent heat for water (L_f) = 334.4 kJ

$Q = mc\Delta T + mL_f$

$= m(c\Delta T + L_f)$

$= 10(4.18 \times 20 + 334.4)$

$Q = 4180$ kJ

$$Q = \frac{4180}{3600} = 1.1611\ kW$$

$$\text{Capacity of refrigeration} = \frac{1.1611}{3.5}$$

$= 0.331$ TR

2. Vapour Compression Refrigeration System

161. During a refrigeration cycle, heat is rejected by the refrigerant in a :

(a) condenser (b) compressor
(c) evaporator (d) expansion valve

CGPSC AE 15.01.2021
JPSC AE 10.04.2021 Paper-I
GPSC DEE, Class-2 (GWSSB) 04.07.2021
Sikkim PSC (Under Secretary), 2017
GPSC Lect. 23.10.2016
GPSC Lect. (Poly.) 07.12.2014
UKPSC AE 2007 Paper-II
OPSC Civil Services Pre 2006

Ans. (a) : During a refrigeration cycle, heat is rejected by the refrigerant in condenser and heat is absorbed by refrigerant in evaporator.

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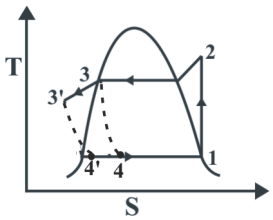
162. The effect of Sub cooling in a refrigeration cycle will result in

(a) More COP
(b) Less COP
(c) No change in COP
(d) More power consumption

PPSC Asstt. Municipal Engineer 15.06.2021
ISRO Scientist/Engineer (RAC) 22.04.2018
Sikkim PSC (Under Secretary), 2017
OPSC AEE 2015 PAPER-II
TANGEDCO AE 2015

Ans. (a) : The effect of Sub cooling in a refrigeration cycle will result in more COP.



Cycle 1-2-3-4-1 without sub-cooling.
So refrigeration effect
$= (h_1 - h_4)$
Cycle 1-2-3-3'-4'-4-1 with sub-cooling
Refrigeration effect = $h_1 - h_{4'}$

$$COP = \frac{\text{Ref. effect}}{\text{Work input}}$$

$\uparrow COP \propto$ Refrigeration effect $\uparrow$

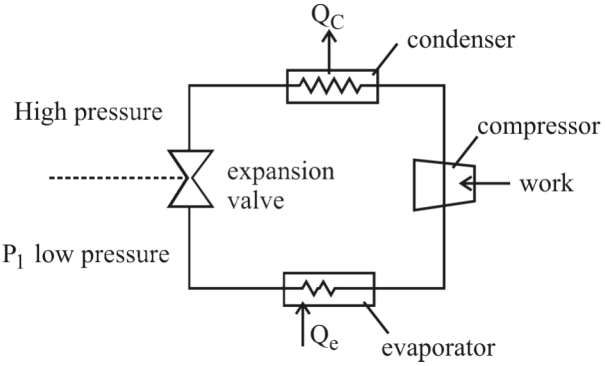
where compression work will be same in both cases
So, COP will be more in sub-cooling cycle as compared without sub-cooling cycle.

163. Where does the lowest temperature occur in a vapour compression cycle?

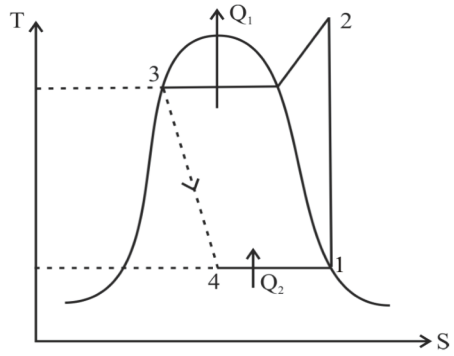
(a) Evaporator (b) Condenser
(c) Expansion valve (d) Compressor

CIL MT 27.02.2020, WBPSC AE, 2017
Kerala PSC Asstt. Prof. 27.09.2016
Rajasthan Nagar Nigam AE 2016, Shift-II
Kerala PSC Asstt. Prof. 27.09.2016
RPSC Lect. (Tech. Edu. Dept.) 2011
UKPSC AE 2007 Paper-II

Ans. (a) :



In vapour compression cycle lowest temperature occur in evaporator.

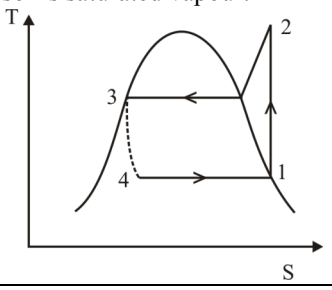


164. In a vapour compression refrigeration system the condition of refrigerant before entering the compressor is :

(a) wet vapour
(b) dry saturated vapour
(c) superheated vapour
(d) saturated liquid

CGPSC AE 15.01.2021
TSPSC Manager Engg. HMWSSB 12.11.2020
TRB Polytechnic Lecturer 2017
Nagaland PSC CTSE 2017, 2016, 2015 Paper-II
Kerala PSC AE (Irrigation) 05.04.2017
Punjab PSC SDE 12.02.2017
UKPSC AE 2007 Paper-II

Ans. (b) : In vapour compression refrigeration system, the condition of refrigerant before entering the compressor is saturated vapour.



165. What is the idea behind replacing an expander with a throttle valve in vapour compression refrigeration systems?

(a) Because it considerably reduces the system weight
(b) Because it improves the COP as the condenser is small
(c) Because the positive work in Isentropic expansion of liquid is very small
(d) Because it leads to significant cost reduction

HPPSC Asstt. Prof. 18.09.2017
UPRVUNL AE 2014, APPSC AEE 2012
UKPSC AE 2007 Paper-II, ESE 1996

Ans : (c) The idea behind is replacing an expander with a throttle valve in vapour compression refrigeration systems because the positive work in Isentropic expansion of liquid is very small and the work output does not justify the cost of expander.

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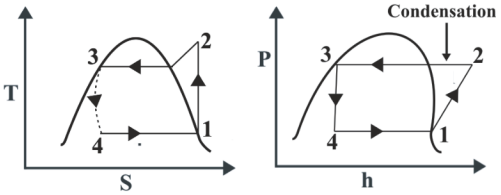
166. In vapour compression system, the condition of refrigerant before passing through the condenser is

(a) Saturated liquid
(b) wet vapour
(c) Dry saturated vapour
(d) superheated vapour

Nagaland PSC (CTSE) 2018, Paper-II
ISRO Scientist/Engineer (RAC) 07.05.2017
VIZAG STEEL MT 2017
RPSC IOF & Boiler 20.09.2017

Ans. (d) : In vapour compression system, the condition of refrigerant before passing through the condenser is super heated vapour.



Process (1-2) - Adiabatic compression
Process (2-3) - Condensation
Process (3-4) - Throttling
Process (4-1) - Evaporation

167. Coefficient of performance (COP) of vapour compression refrigeration system with T_e as evaporator temperature and T_c as condenser temperature.

(a) Increases with increase in T_e at constant T_c
(b) Does not change with variation of T_e at constant T_c
(c) Increases with increase in T_c at constant T_e
(d) Increases with decrease in T_e at constant T_c

Oil India Ltd. Sr. Engg. (Production) 30.11.2019
CGPSC AE 25.02.2018
RPSC LECTURER 16.01.2016

Ans. (a) : We know that,
Coefficient of performance of vapour Compression refrigeration system.

$$(COP)_{VCRS} = \frac{T_e}{T_c - T_e}$$

$\therefore \quad (COP)_{VCRS} = f(T_c, t_e)$

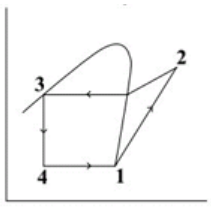
If T_c is constant.

$$(COP)_{VCRS} \uparrow = \frac{T_e \uparrow}{(T_c - T_e \uparrow)\downarrow}$$

Then

$$T_e \uparrow \Rightarrow (COP)_{VCRS} \uparrow$$

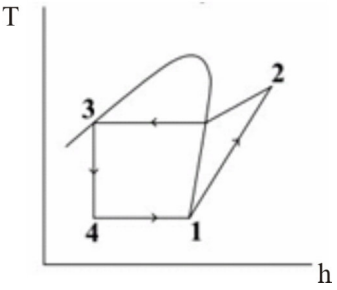
168. The vapour compression refrigeration cycle is represented as shown in the figure below, with state 1 being the exit of the evaporator. The coordinate system used in this figure is



(a) p-h (b) T-s
(c) p-s (d) T-h

UPPSC AE 13.12.2020, Paper-2
Gujarat PSC AE 2019
BEL PE 2015, GATE 2005

Ans : (d)



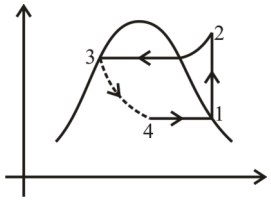
Reasons-
(i) The throttling process 3-4 is vertical i.e., at constant enthalpy. So abscissa must be enthalpy.
(ii) Condensation process (P = constant) 2-3 is not horizontal so ordinate does not represented by pressure but temperature.

169. In an ideal vapour compression refrigeration cycle, the specific enthalpy of refrigerant in kJ/kg at the following states is given as
Inlet of condenser : 283
Exit of condenser : 116
Exit of evaporator : 232
The COP of the cycle is

(a) 2.27 (b) 2.75
(c) 3.27 (d) 3.75

ISRO Scientist/Engineer (RAC) 12.01.2020
HPPSC Asstt. Prof. 18.11.2016
UPPSC AE 12.04.2016 Paper-II
GATE 2009

Ans. (a)



Enthalpy at inlet of condenser (h_2) = 283 kJ/kg enthalpy at exit of evaporator (h_1) = 232kJ/kg enthalpy as exit of condenser (h_3) = enthalpy as inlet of evaporator (h_4) = 116 kJ/kg

$h_3 = h_4 = 116$ kJ/kg

$$COP = \frac{h_1 - h_4}{h_2 - h_1} = \frac{232 - 116}{283 - 232} = 2.274$$

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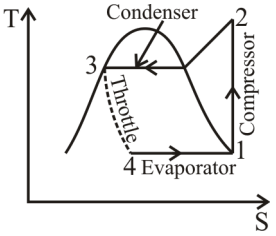
170. In a refrigeration cycle, the heat is absorbed by a refrigerant in a

(a) Evaporator (b) Condenser
(c) Compressor (d) Expansion valve

RPSC VPITI 14.02.2016
MPPSC State Forest Service Exam, 2014
UKPSC AE 2012 Paper-II
J&K PSC Civil Services Pre, 2010

Ans. (a) :



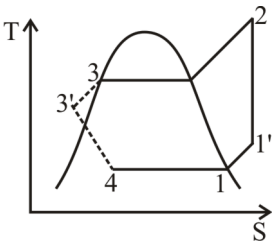
1-2 is the process of heat addition to the refrigerant; the heat is absorbed before compression in the evaporator.

171. Sub-cooling heat exchanger is used in a refrigeration cycle. The enthalpies at condenser outlet and evaporator outlet are 78 kJ/kg and 182 kJ/kg respectively. The enthalpy at outlet of isentropic compressor is 230 kJ/kg and enthalpy of sub-cooled liquid is 68 kJ/kg. The COP of the cycle is

(a) 3.25 (b) 2.16
(c) 3.0 (d) 3.5

Nagaland PSC CTSE 2016 Paper-II
Haryana PSC Civil Services Pre, 2014
TRB Asst. Prof. 2012, ESE 2002

Ans. (c)



Given,
$h_3 = 78$ kJ/kg
$h_1 = 182$ kJ/kg
$h_2 = 230$ kJ/kg
$h_4 = h_{3'} = 68$ kJ/kg

Degree of subcooling = Degree of superheating

$$h_3 - h_{3'} = h_{1'} - h_1$$

$$78 - 68 = h_{1'} - 182$$

$$\boxed{h_{1'} = 192\,kJ/kg}$$

$$COP = \frac{h_1 - h_4}{h_2 - h_{1'}}$$

$$COP = \frac{182 - 68}{230 - 192}$$

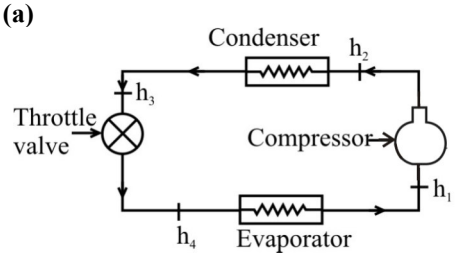
$$\boxed{COP = 3}$$

172. A simple saturated refrigeration cycle has the following state points, enthalpy after compression = 425 kJ/kg, enthalpy after throttling = 125 kJ/kg, enthalpy before compression = 375 kJ/kg. The COP of cycle will be :

(a) 5 (b) 3.5
(c) 6 (d) 10

GPSC DEE, Class-2 (GWSSB) 04.07.2021
HPPSC W.S. Poly. 2016
UKPSC AE 2012 Paper-II

Ans : (a)



$h_2 = 425$kJ/kg
$h_4 = 125$ kJ/kg
$h_1 = 375$ kJ/kg

$$COP = \frac{\text{Refrigeration effect}}{\text{work input}} = \frac{h_1 - h_4}{h_2 - h_1}$$

$$COP = \frac{375 - 125}{425 - 375}$$

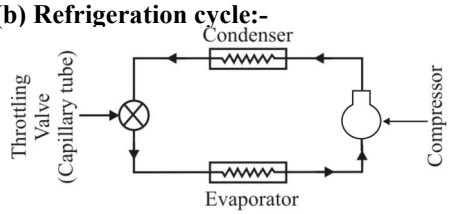
$$COP = 5$$

173. The throttling operation in a refrigeration cycle is carried out in

(a) evaporator (b) capillary tube
(c) discharge valve (d) expansion valve

(KPSC AE 2015), Assam PSC CCE PRE. 2015
UKPSC AE 2012 Paper-II

Ans : (b) Refrigeration cycle:-



The throttling operation in a refrigeration cycle is carried out in capillary tube.

174. In a vapour compression system, the condition of refrigerant before entering the expansion or throttle valve is

(a) high pressure saturated liquid
(b) wet vapour
(c) very wet vapour
(d) dry vapour

OPSC AEE 2019 Paper-II
Nagaland PSC (CTSE) 2018, Paper-II
Punjab PSC SDE 12.02.2017

Ans. (a) : In a vapour compression system the condition of refrigerant before entering the expansion or throttle valve is high pressure saturated liquid.

⇒ After leaving the expansion valve the condition of refrigerant is low pressure saturated liquid and vapour mixture.

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175. The pressure at the inlet of a refrigerant compressor is called

(a) Suction pressure (b) Discharge pressure
(c) Critical pressure (d) Back pressure

CGPSC AE 15.01.2021
Nagaland PSC (CTSE) 2018, Paper-II
JPSC AE 2013 Paper-V

Ans. (a) : The pressure at the inlet of a refrigerant compressor is called suction pressure in case of a compressor, the pressure in the line connected to the inlet (suction) of the compressor determines the pressure at the suction of the compressor.

176. When the discharge pressure is too high in a refrigeration system, high pressure control is installed to

(a) stop the cooling fan
(b) stop the water pump
(c) regulate the flow of cooling water
(d) stop the compressor

UPRVUNL AE 2014
APPSC Poly. Lect. 2013, ESE 1996

Ans. (d) : When the discharge pressure is too high in a refrigeration system, high pressure control is installed to stop the compressor.

177. The flash chamber in a single stage simple vapour compression system

(a) increases the refrigerating effect
(b) decreases the refrigerating effect
(c) increases the work of compression
(d) has no effect on refrigerating effects

GPSC ARTO Pre 30.12.2018
TSPSC AEE 2015, ESE 1998

Ans. (d) : Flash chamber separates the liquid from vapour refrigerant and sends only liquid refrigerant to evaporator & vapour to compressor by passing evaporator coil.

178. In a vapour compression cycle the condition of refrigerant is saturated liquid :

(a) before entering the compressor
(b) before passing through the condenser
(c) after passing through the condenser
(d) after passing through the expansion valve.

Kerala PSC AE 06.08.2015
KPSC AE 2015, TNPSC ACF 2012

Ans. (c) : Vapour Compression cycle- In vapour compression cycle, after passing through the condenser the refrigerant is saturated liquid.

179. The vapour compression refrigerator employs the following cycle-

(a) Carnot (b) Reversed Carnot
(c) Rankine (d) Reversed Rankine

GPSC Executive Engg. 23.12.2018
Kerala PSC AE (Irrigation) 05.04.2017
ISRO Scientist/Engineer (RAC) 29.11.2015

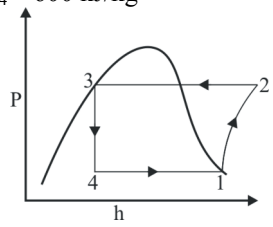
Ans : (d) The vapour compression refrigerator employs reversed Rankine cycle

180. For simple vapour compression cycle, enthalpy at suction = 1600kJ/kg, enthalpy at discharge from the compressor = 1800kJ/kg, enthalpy at exit from condenser = 600 kJ/kg. What is the COP for his refrigeration cycle?

(a) 3.3 (b) 5.0
(c) 4.0 (d) 4.5

MPPSC AE 2016, ESE 2008

Ans : (b) $h_1 = 1600$ kJ/kg
$h_2 = 1800$ kJ/kg
$h_3 = h_4 = 600$ kJ/kg



$$(COP)_{ref} = \frac{\text{Refrigeration effect}}{\text{Work input}}$$

$$(CoP)_{ref} = \frac{1600-600}{1800-1600} = \frac{1000}{200} = 5$$

$$\boxed{(COP)_{ref} = 5}$$

181. In refrigeration, if the vapour is not superheater after compression it is

(a) Dry Compression
(b) Wet Compression
(c) Superheated Compression
(d) Saturated Compression

UPRVUNL AE 05.07.2021
TNPSC AE 2013

Ans. (b) : It is an isentropic compression process of the refrigerant where work is supplied to the compressor to increase the pressure. If the refrigerant is found in both liquid and vapour from at this stage, it is called wet compression. If the refrigerant is purely in a vapour from, it is called dry compression.

182. The performance of which of the following units in a vapour compression system is adversely affected by the presence of moisture?

(a) Condenser (b) Expansion valve
(c) Evaporator (d) Compressor

OPSC AEE 2015 Paper-II
OPSC Civil Services Pre. 2011

Ans. (b) : Presence of moisture in a refrigerant affects the working of expansion valve.

183. Co-efficient of performance is always

(a) more than unity
(b) less than unity
(c) equal to unity
(d) none of these

GPSC ARTO 01.05.2016
TNPSC AE (Industries) 09.06.2013

Ans. (a) : Coefficient of performance (COP) > 1

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184. The formation of frost on cooling coils in a refrigerator.

(a) Increases the heat transfer
(b) Improves the C.O.P. of the system
(c) Increases the power consumption
(d) Reduces the power consumption

ISRO Scientist/Engineer (RAC) 22.04.2018
GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : The formation of frost on cooling coils in a refrigerator reduces the heat transfer, hence for same heat transfer rate work required will increased hence power consumption increased.

185. The C.O.P. of Carnot refrigerator in winter will be _____ as compared to C.O.P. if summer.

(a) Lower (b) Higher
(c) Same (d) Unpredictable

OPSC AEE 2019 Paper - II
GPSC Lect. (Poly.) 07.12.2014

Ans. (b) : $(COP)_R = \dfrac{T_L}{T_H - T_L}$

$= \dfrac{1}{\dfrac{T_H}{T_L} - 1}$

(T_H) summer > (T_H) winter

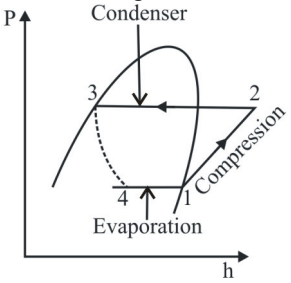
i.e. surrounding temperature in summer is more than the winter hence $[(COP)_R]_{winter} > [(COR)_R]_{summer}$

186. The enthalpies at the beginning of compression, at the end of compression and at the end of condensation are respectively 185kJ/kg, 210kJ/kg and 85kJ/kg. What is COP of VCRS?

(a) 0.25 (b) 5.4
(c) 4 (d) 1.35
(e) 5

CGPSC ARTO & TSI 07.01.2017
ESE 2005

Ans. (c) : $h_1 = 185$ kJ/kg
$h_2 = 210$ kJ/kg
$h_4 = h_3 = 85$ kJ/kg



$COP = \dfrac{h_1 - h_4}{h_2 - h_1}$

$COP = \dfrac{185 - 85}{210 - 185}$

$COP = \dfrac{100}{25}$

$COP = 4$

187. The sub-cooling is a process of cooling the refrigerant in vapor compression refrigeration system

(a) Before compression (b) After compression
(c) Before throttling (d) After throttling

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC AE (Industries) 09.06.2013

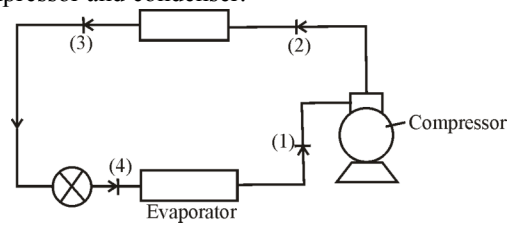
Ans. (c) : The sub-cooling is a process of cooling the refrigerant in vapor compression refrigeration system before throttling.

188. In mechanical refrigeration system, the refrigerant has the maximum temperature

(a) in evaporator
(b) before expansion valve
(c) between compressor and condenser
(d) between condenser and evaporator

Nagaland PSC (CTSE) 2018, Paper-II
UKPSC AE 2012 Paper–II

Ans. (c) : In mechanical refrigeration system the refrigerant has the maximum temperature between compressor and condenser.

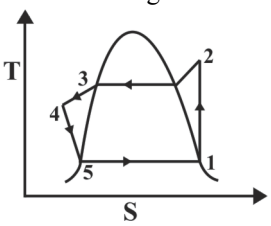


189. In refrigeration cycle, sub cooling occurs when

(a) Latent heat from refrigerant is removed
(b) Sensible heat from refrigerant is removed
(c) Refrigerant has low latent heat removed
(d) Refrigerant has high thermal conductivity

ISRO Scientist/Engineer (RAC) 07.05.2017
BPSC AE 2012 Paper-V

Ans. (b) : In refrigeration cycle, sub cooling occurs when sensible heat from refrigerant is removed.



In above figure process 3-4, show sub-cooling which remove sensible heat from refrigerant.

Process 1-2 → compression
Process 2-3 → condensation
Process 3-4 → sub-cooling
Process 4-5 → throttling
Process 5-1 → evaporation

190. In a vapor compression refrigeration plant, the refrigerant leaves the evaporator at 195 kJ/kg and the condenser at 65 kJ/kg. For every kg of refrigerant the plant can supply per second a cooling load of :

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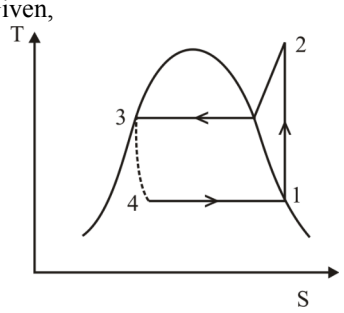
(a) 70 kW (b) 100 kW
(c) 130 kW (d) 160 kW

GPSC ARTO Pre 30.12.2018, ESE 2005

Ans. (c) : Given,



$h_1 = 195$ kJ/kg

$h_3 = h_4 = 65$ kJ/kg

$\dot{m} = 1$ kg/s

Cooling load, $\Rightarrow RE = \dot{m}(h_1 - h_3)$

$= 1(195 - 65)$

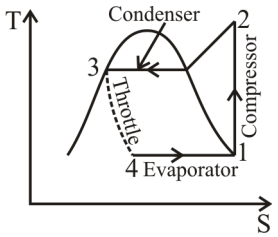
$= 130$ kW

191. In a VCR plant, the refrigerant leaves the compressor and evaporator with enthalpy of 205 kJ/kg and 177 kJ/kg respectively. Enthalpy of the refrigerant leaving the condenser is 105 kJ/kg. If the mass flow rate of the refrigerant is 0.2 kg/s, the refrigeration effect will be

(a) 12.2 kW (b) 14.4 kW
(c) 16.4 kW (d) 20.2 kW

GPSC ARTO Pre 30.12.2018
GPSC EE Pre, 28.01.2017

Ans. (b) :



Throttling device so,

$h_3 = h_4 = 105$ kJ/kg

$h_1 = 177$ kJ/kg, $h_2 = 205$ kJ/kg

$R.E = \dot{m}(h_1 - h_4)$

$= 0.2(177 - 105)$

$= 0.2 \times 72$

$= 14.4$ kW

192. In a vapour compression refrigeration system, liquid to suction heat exchanger is used to

(a) Keep the COP constant.
(b) Prevent the liquid refrigerant from entering the compressor.
(c) Sub-cool the liquid refrigerant leaving the condenser.
(d) Sub- cool the vapour refrigerant from the evaporator.

Karnataka PSC Lect., 27.05.2017
HPPSC Asst. Prof. 18.11.2016

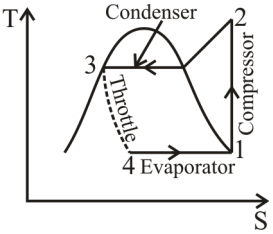
Ans. (c) : In a vapour compression refrigeration system liquid to suction heat exchanger is used to sub-cool the liquid refrigerant leaving the condenser.

193. The condition of a refrigerant after passing through the expansion or throttle valve in a vapour compression refrigeration system is

(a) High pressure saturated liquid
(b) Very wet vapour
(c) Dry vapour
(d) None of the above

Karnataka PSC Lect., 27.05.2017
CSE Pre-2002

Ans (b) :



The refrigerant in VCRS immediately after expansion is wet vapour as shown below:
Point '4' shows the wet vapour after expansion valve.

194. The lowest temperature during the cycle in a vapour compression system occurs after.

(a) Compression (b) Expansion
(c) Condensation (d) Evaporation

ISRO Scientist/Engineer (RAC), 10.03.2019
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : Expansion– Lowest temperature during the cycle in vapour compression system occurs after (expansion).

195. In simple saturated vapour compression refrigeration cycle the following results were obtained. Heat rejected in condenser = 160 kJ/kg, compressor work = 32 kJ/kg. The COP of refrigeration is :

(a) 5
(b) 4
(c) 6
(d) not possible to find with this data

UKPSC AE 2013 Paper-II, APPSC IOF, 2009

Ans. (b) : $COP = \dfrac{\text{Ref. effect}}{\text{Compressor work}} = \dfrac{160 - 32}{32}$

$= \dfrac{128}{32} = 4$

196. The COP of a vapour compression plant in comparison to vapour absorption plant is :

(a) more
(b) less
(c) same
(d) more/less depending on size of plant

OPSC AEE 2019 Paper-II
J & K PSC Screening, 2006

Ans. (a) : COP of vapour compression plant – 4 to 6
COP of vapour absorption plant – 0.8 to 1.2

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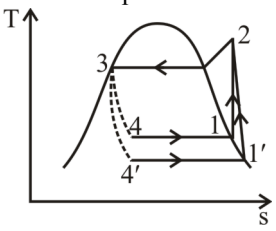
197. If the evaporator temperature of a plant is lowered keeping the condenser temp. constant, the power input of compressor required will be

(a) same (b) more
(c) less (d) unpredictable

ISRO Scientist/Engineer (RAC) 22.04.2018
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : If the evaporator temperature of a plant is lowered, by keeping condenser temperature constant, it means you are increasing effort by compressor. So, compressor required more power for desired output.



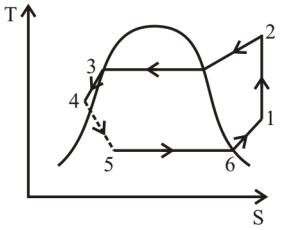
$$(W_{comp.})_{1-2} < (W_{comp.})_{1'-2}$$

198. The effects of superheating of vapour in the evaporator and sub-cooling of condensate in the condenser :

(a) Decrease the COP
(b) Increase the COP
(c) Superheating increases COP, but sub-cooling decrease COP
(d) Superheating decrease COP, but sub-cooling decreases COP

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
ESE 2010

Ans. (b) :



When superheating of vapour in the evaporator and sub-cooling of condensate in the condenser is done simultaneously then it is known as Grindley cycle and it's sole effect to increase the COP.

199. Which of the following statement is wrong?

(a) The performance of the vapour compression refrigerator varies considerably with both vaporising and condensing temperatures.
(b) In vapour compression cycle, the useful part of the heat transfer is at the condenser.
(c) In ammonia hydrogen (Electrolux) refrigerator, no compressor is required.
(d) The effect of under cooling the refrigerant is to decrease the coefficient of performance.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper-II

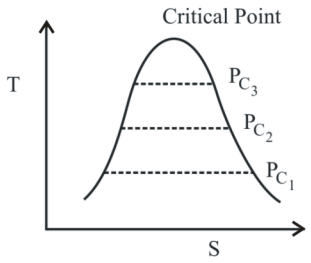
Ans. (d) : The effect of under cooling the refrigerant is to increase the coefficient of performance.

200. In a vapour compression refrigeration cycle for making ice, the condensing temperature for higher COP.

(a) should be near the critical temperature of the refrigerant
(b) should be above the critical temperature of the refrigerant
(c) should be much below the critical temperature of the refrigerant.
(d) could be of any value as it does not affect the COP.

RPSC ACF & FRO, 26.02.2021, ESE 2006

Ans. (c) :



$P_C \rightarrow$ Critical pressure

As critical pressure or critical temperature increases, the refrigeration effect decreases and work input increases. So the condensing temperature should be much below the critical temperature.

201. The highest temperature during the cycle, in a vapor compression refrigeration system, occurs after

(a) Compression (b) Condensation
(c) Expansion (d) Evaporation

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : The highest temperature during the cycle, in a vapor compression refrigeration system, occurs after compression.

202. Wet compression vapor compression refrigeration cycle means

(a) Vapor compression takes place in wet region
(b) Vapor compression in dry region, but evaporation in wet region
(c) Vapor compression in wet region, but leaves in super heated region
(d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Wet compression vapor compression refrigeration cycle means vapor compression takes place in wet region.

203. The order in which main components of vapor compression of refrigeration system are used is

(a) Compressor-Evaporator-Condenser-Throttle valve
(b) Compressor-Condenser-Evaporator-Throttle valve
(c) Compressor-Condenser-Throttle valve-Evaporator
(d) Compressor-Throttlevalve-Evaporator-Condenser

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) :

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204. The flow rate of refrigerant R-12 in a refrigerator is 0.03 kg/s. The refrigerant enters the compressor as saturated at 150.9 kPa. After adiabatic compression, it enters the condenser as superheated vapour at 500 kPa and 100°C and leaves the condenser as saturated liquid at the same pressure.

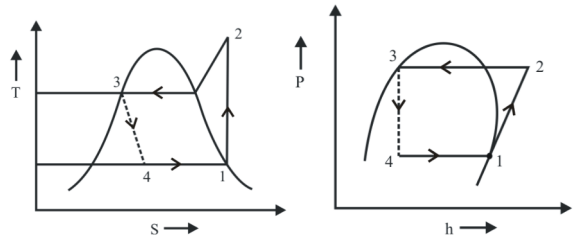
Pressure (kPa)	Temperature (°C)	Specific enthalpy	
		h_ℓ (kJ/kg)	h_g (kJ/kg)
150.9	–20	17.82	178.74
500	15.6	50.64	195.01
For the saturated vapour at 500 kPa and 100 °C, h = 252.05 kJ/kg.			

Using the data given above, the refrigeration effect in kW will be :

(a) 3.70 (b) 3.84
(c) 4.00 (d) 3.25

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) :



Refrigeration effect (RE) = m [$h_1 - h_4$]
= m [$h_1 - h_3$]
$h_3 = h_4$ (Isenthalpic process)
= 0.03 [178.74 – 50.64]
RE = 3.84 kW

205. The flash chamber intercooler allows the refrigerant to enter the 2nd state of compressor of vapour compression refrigeration system at the state of :

(a) Dry saturated vapour
(b) Saturated liquid
(c) Wet vapour
(d) Superheated vapour

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

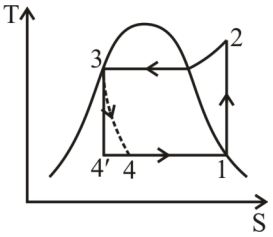
Ans. (b) : Flash chamber is a device which is used to separate the liquid refrigerant from a mixture of refrigerant at the entry of evaporator, It allows only the liquid refrigerant to enter into evaporator which results in absorption of heat.

206. In vapour compression refrigeration system, if expansion cylinder is used in place of throttle valve, the COP will

(a) increase
(b) decrease
(c) will remain same
(d) not be predictable unless refrigerant used is known

JPSC AE, 10.04.2021, Paper-I

Ans. (a)



- When we use throttle valve,

$$(COP)_1 = \frac{\text{Desired effect}}{\text{Power input}} = \frac{h_1 - h_4}{h_2 - h_1}$$

- When we use expansion cylinder (isentropic expander)

$$(COP)_2 = \frac{\text{Desired effect}}{\text{Power input}} = \frac{h_1 - h_{4'}}{h_2 - h_1}$$

$(COP)_2 > (COP)_1$ because desired effect while using throttle valve is more than desired effect when we use expansion cylinder.

In VCRS if expansion cylinder is used in place of throttle valve the COP will increase.

207. Freon-12 is used in a simple saturation cycle, with suction saturation temperature of – 10°C and condensing saturation temperature of 30°C. If the clearance volume is 6% of the stroke volume, what is the volumetric efficiency? (Consider specific volume at suction and discharge to be 0.07815 m^3/kg) and 0.025 m^3/kg respectively)

(a) 87.24% (b) 71.31%
(c) 64.85% (d) 55.43%

ESE 18.07.2021

Ans. (a)

Volumetric efficiency (η_v)

$$= 1 + C - C\frac{V_{suction}}{V_{discharge}}$$

$$C = \text{Clearance factor} = \frac{\text{Clearance volume}}{\text{Swept volume}} = 0.06$$

Stroke volume = swept volume

$$\therefore \eta_v = 1 + C - C\left(\frac{V_{suction}}{V_{discharge}}\right)$$

$$= 1 + 0.06 - \left(0.06 \times \frac{0.07815}{0.025}\right)$$

= 0.8724 = 87.24%

208. The heat rejection factor in vapour compression refrigeration system will be

(a) equal to 1
(b) more than 1
(c) less than 1
(d) depends on cooling medium of the condenser

ISRO Scientist/Engineer (RAC) 12.01.2020

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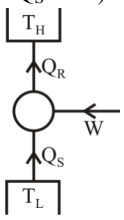
Ans. (b) : Heat rejection ratio $(HRR) = \frac{Q_R}{Q_S}$

$Q_R > Q_S$ $(Q_R = Q_S + W)$



So HRR in vapour compression refrigeration system will be more than one.

Alternate solution,

$$\boxed{HRR = 1 + \frac{1}{COP}}$$

COP of VCRS is always greater than one

So, $\boxed{HRR > 1}$

209. The power consumption of Vapour Compression Refrigeration System (VCRS) with air cooled condenser compared to VCRS with water cooled system will be higher due to

(a) High discharge temperature
(b) High condensing temperature
(c) Low suction temperature
(d) High refrigeration load

ISRO Scientist/Engineer (RAC) 12.01.2020

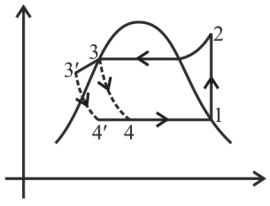
Ans. (a) : The power consumption of Vapour Compression Refrigeration System (VCRS) with air cooled condenser compared to VCRS with water cooled system will be higher due to 'High discharge temperature', VCRS with water is high efficient than VCRS with air.

210. Liquid sub cooler in Vapour Compression Refrigeration System will

(a) Prevent the entry of liquid into compressor
(b) Increases Net Refrigeration Effect
(c) Reduce work of compression
(d) Increase work of compression

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) :



VCRS cycle with subcooling ⇒ 1–2–3–4

VCRS cycle with subcooling ⇒ 1–2–3′–4′

It is clear from the graph, liquid subcooler increases net refrigeration effect and it does not affect the work of compression.

211. A refrigerator is operating on ideal vapour compression cycle with R-134a as working fluid at flow rate of 0.1 kg/s. Data given are :

Enthalpy at the inlet of the compressor = 240 kJ/kg

Enthalpy at the outlet of the compressor = 280 kJ/kg

Enthalpy at the outlet of the condenser = 100 kJ/kg

What will be the refrigeration capacity and COP of the refrigerator, respectively?

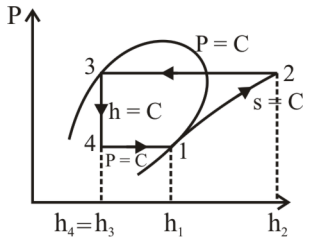
(a) 14 kW, 3.5 (b) 4 kW, 4
(c) 10 kW, 2 (d) 5 kW, 3.5

APPSC Poly Lect. 13.03.2020

Ans. (a) : Given, $h_1 = 240 kJ/kg$, $h_2 = 280$ kJ/kg,

Mass of refrigerant $(m_{ref}) = 0.1$ kg/sec,

$h_3 = h_4 = 100$ kJ/kg



$$COP = \frac{\text{Refrigerating Effect (R.E.)}}{\text{Work input } (W_{in})}$$

$$= \frac{h_1 - h_4}{h_2 - h_1}$$

$$= \frac{240 - 100}{280 - 240}$$

COP = 140 / 40 = 3.5

Now,

Refrigeration capacity (RC) = R.E. × $\dot{m}_{ref}$

$$RC = (h_1 - h_4) \times \dot{m}_{ref}$$

$\because$ RC = (240–100) × 0.1

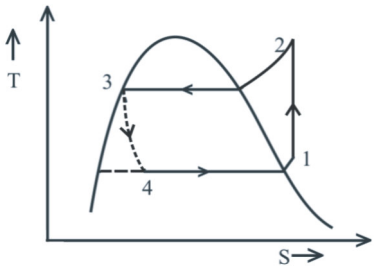
RC = 14 kW

212. Which one of the following is the effect of suction vapour superheat?

(a) Decreases the refrigeration effect
(b) Decreases the specific volume
(c) Decreases the energy for compression
(d) Increases the refrigeration effect

UPPSC AE 13.12.2020, Paper-2

Ans. (d)



The effect of suction vapour superheat increases the refrigeration effect.

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213. For a vapor compression refrigeration cycle, compressor work for the same cooling capacity could be reduced by

(a) Superheating in evaporator
(b) Subcooling in condenser
(c) Reducing condenser pressure
(d) Increasing evaporator pressure

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) :

214. A Freon 12 simple saturation cycle operates at temperature of 35°C and –15°C for the condenser and evaporator. If the refrigeration effect produced by the cycle is 111.5 kJ/kg and the work required by the compressor is 27.2 kJ/kg, the value of COP will be nearly

(a) 4.1 (b) 3.6
(c) 3.1 (d) 2.6

ESE 2020

Ans. (a) :

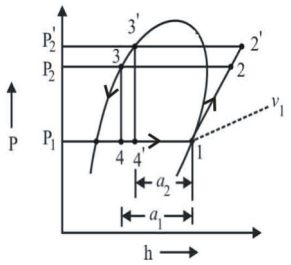
$$COP = \frac{RE}{W} = \frac{111.5}{27.2} = 4.099$$

215. In vapour compression refrigeration system, increase in compressor delivery pressure :

(a) decreases the heat rejection.
(b) increases the refrigeration effect.
(c) decreases the refrigeration effect
(d) increases the heat rejection.

NLC GET 17.11.2020, Shift-II

Ans. (c) :



An increase in condensing pressure result in a decrease of the refrigerating capacity (h_1–h_4) and increase in power consumption as seen From the cycle 1-2-3-4 changed to cycle 1-2'-3'-4' shows decrease in refrigerating capacity effect and volumetric efficiency. The increase in power consumption is due to the increased mass Flow due to decreased refrigerating effect and an increase in specific work due to increased pressure ratio although the isentropic line remains the same.

216. If the pressure range of compressor is low, then the COP will be

(a) low
(b) high
(c) remains unchanged
(d) Cannot be determined

JPSC AE PRE 2019

Ans. (b) : If the pressure range of compressor is low, then the COP will be high.

217. The principle of vapour compression system is that fluids absorb heat while changing from a :

(a) Vapour phase to liquid phase and give up heat in changing from liquid phase to vapour phase
(b) Liquid phase to vapour phase and give up heat in changing from a vapour phase to liquid phase
(c) Vapour phase to liquid phase only
(d) Liquid phase to solid state phase and give up heat in changing from a solid state to liquid phase

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (b) : VCRS system absorb heat during change of liquid phase to vapour phase (In evaporator) and release heat in changing from vapour phase to liquid phase in condenser.

218. In vapour compression refrigeration system, heat rejected as compared to heat absorbed is

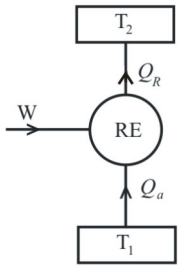
(a) More
(b) Less
(c) Equal
(d) Depends on the refrigerant used

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) : In VCRS system Heat rejected is more as compared to heat absorbed.

$$Q_R = Q_a + w$$

$$\boxed{Q_R > Q_a}$$



219. The coefficient of performance of vapour compression refrigeration system:

(a) decreases with the decrease in evaporator temperature at fixed condenser temperature
(b) Increases with the increase in condenser temperature at fixed evaporator temperature
(c) Remains constant with the decrease in evaporator temperature at fixed condenser temperature
(d) Increases with the decrease in evaporator temperature at fixed condenser temperature

SJVN ET 2019

Ans. (a) : Effect of decrease in evaporator pressure-

- Refrigerating effect will decrease.
- COP will decrease.
- Volumetric efficiency will decrease.
- Work done will increase.
- Effect of increase in condense pressure
- Refrigeration effect will decrease.
- COP will decrease.
- Volumetric efficiency will decrease.
- Work done will increase.

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220. The compressor of an ammonia refrigerating machine has a volumetric efficiency of 85% and swept volume of 0.28m^3/min. Ammonia having a dry specific volume of 0.25m^3/kg enters the compressor with a dryness fraction of 0.7. The mass flow rate of ammonia through the machine is

(a) 1.28 kg/min (b) 1.36 kg/min
(c) 1.42 kg/min (d) 1.57 kg/min

ESE 2018

Ans. (b) : $\eta_{vol.} = \dfrac{\text{Actual volume}}{\text{Theoretical swept volume}}$

$= \dfrac{m_r . v_1}{v_s}$

$\dot{m}_r \times v_1 = \eta_{vol.} \times V_s$

$\dot{m}_r \times x \times v_1 = \eta_{vol.} \times V_s$

$\dot{m}_r \times 0.7 \times 0.25 = 0.85 \times 0.28$

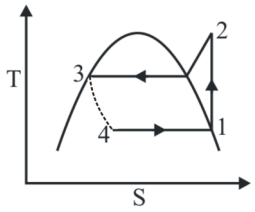
$\boxed{\dot{m}_r = 1.36 \text{ kg/min}}$

221. The following data refer to a vapour compression refrigerator:
Enthalpy at compressor Inlet = 1200 kJ/kg
Enthalpy at compressor Outlet = 1400 kJ/kg
Enthalpy at condenser Outlet = 200 kJ/kg
The COP of the refrigerator is

(a) 7 (b) 6
(c) 5 (d) 4

ESE 2018

Ans. (c) : Given,



$h_1 = 1200$ kJ/kg
$h_2 = 1400$ kJ/kg
$h_3 = h_4 = 200$ kJ/kg

Because of constant enthalpy expansion take place

$$COP = \frac{h_1 - h_4}{h_2 - h_1} = \frac{1200 - 200}{1400 - 1200} = 5$$

222. Consider the following statements:

1. The operation of a refrigerator unit at more than one temperature can be accomplished by using different throttling valves and a separate compressor for each 'temperature range'.
2. The refrigerated space must be maintained above the ice point to prevent freezing.
3. In domestic refrigerators, the refrigerant is throttled to a higher pressure in the freezer followed by full expansion in the refrigerated space.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

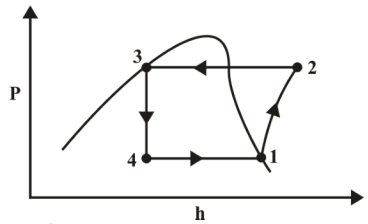
Ans. (a) : In domestic refrigerators refrigerant is throttled to lower pressure in freezer when compared to refrigerator, statement-3 is wrong. Hence by elimination method only option (a) is true.

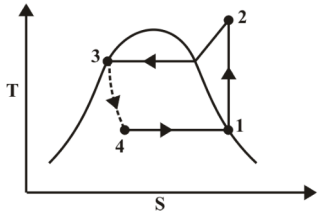
223. Which one of the following expansion process takes place in a vapour compression cycle?

(a) Polytropic process with change in temperature
(b) Adiabatic process with work transfer
(c) Isentropic process with change in enthalpy
(d) Adiabatic process with constant enthalpy

RPSC AE 2018

Ans. (d) : Adiabatic process with constant enthalpy expansion process takes place in a vapour compression cycle.





Process (1 - 2) → Isentropic compression process
Process (2 -3) → Constant pressure heat rejection process
Process (3 - 4) → Adiabatic process with constant enthalpy
Process (4 - 1) → Constant pressure heat addition process

224. In Vapour Compression Refrigeration System (VCRS), throttle valve is used in place of isentropic expander because:

(a) throttling process in throttle valve is reversible.
(b) throttling valve improves the refrigeration effect.
(c) positive work in throttle valve is very large.
(d) it reduces the compressor work.
(e) it leads to cost reduction.

CGPSC ARTO & TSI 07.01.2017

Ans. (e) : Isenthalpic expansion is used in place of isentropic expansion of liquid is very less and the work output does not justify the cost expander.
Hence isenthalpic expansion is used in place of isentropic expansion and it leads to cost reduction.

225. Refrigerant entering the compressor of Vapour Compression Refrigeration System is:

(a) superheated to reduce the work input of compressor.
(b) superheated to reduce the refrigerating effect.

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(c) superheated to ensure dry compression of refrigerant.
(d) subcooled to reduce the work input of compressor.
(e) subcooled to reduce the refrigerating effect.

CGPSC ARTO & TSI 07.01.2017

Ans. (c) : Refrigerant entering the compressor of Vapour Compression Refrigeration System is superheated to ensure dry compression of refrigerant.

226. Which of the following statements is wrong?
(a) The value of C.O.P. is always greater than one.
(b) In a vapour compression system, the condition of refrigerant before entering the compressor is dry saturated vapour
(c) The space between the saturated liquid line and saturated vapour line in a pressure enthalpy chart is the wet vapour region.
(d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (a) :
(COP) Refrigerator is > or < 1
$(COP)_{VCRS} > 1$
$(COP)_{VARS} < 1$
$(COP)_{HP} > 1$

227. An ideal vapour compression refrigerator operates between a condenser pressure p_1 and an evaporator pressure p_2. Which of the following changes would increase its COP?
(a) Increasing p_1 by Δp and keeping p_2 constant
(b) Decreasing p_2 by Δp and keeping p_1 constant
(c) Adopting wet compression
(d) Sub cooling the refrigerant

GPSC EE Pre, 28.01.2017, ESE 2012

Ans. (d) : COP decreases with the decreases pressure of evaporator and increasing condenser pressure. The (COP) increases by installed a sub-cooling process between the condenser and expansion value. The sub cooling reduce flashing of the liquid during expansion and increases refrigeration effect.

228. In system A vapour are superheated by 10°C in the evaporator while in system B vapour are superheated by 10°C in a liquid vapour regenerative heat exchanger, other conditions being the same. Then :
(a) COP of A = COP of B
(b) COP of both A and B > COP of Reversed Carnot Cycle.
(c) COP of A > COP of B
(d) COP of A < COP of B

HPPSC Asstt. Prof. 20.11.2017

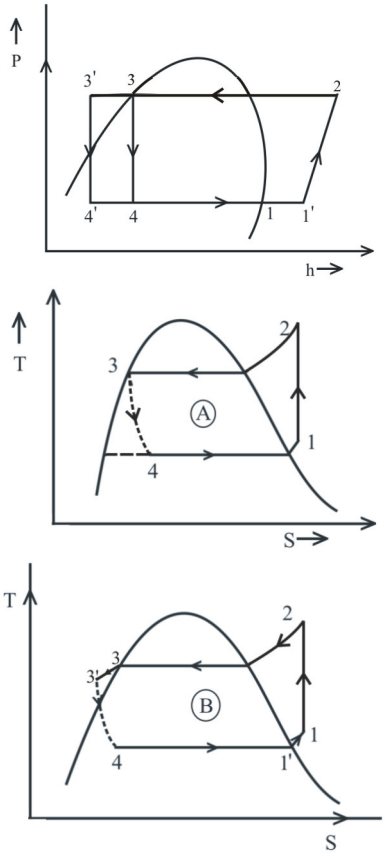
Ans. (a) : Method : 1–

$$h_1' - h_1 = h_3 - h_3'$$

For Regeneration as

$$h_1' - h_4 = h_1 - h_4'$$

$\therefore$ COP is same.



Method : 2–

$RE = h_1 - h_4 \qquad W_{net} = h_2 - h_1$

The heat gain in liquid vapour heat exchanger causes equivalent sub-cooling in refrigerator.

$$h_1 - h_1' = h_3 - h_3'$$

$$RE = h_1' - h_4 = h_1 - h_4$$

$$W = h_2 - h_1$$

229. The C.O.P. of a refrigerator cycle with increase in evaporator temperature, keeping the condenser temperature constant will
(a) Increase
(b) Decrease
(c) Remain unaffected
(d) May increase or decrease depending upon the type of refrigerant used

Nagaland PSC CTSE 2017 2nd Paper

Ans. (a) : The C.O.P. of a refrigerator cycle with increase in evaporator temperature, keeping the condenser temperature constant will increase.

230. In a vapour compression system, superheating of the refrigerant decreases COP, the reason being:
1. Increase of work input to the compression
2. It does not depend on refrigeration effect

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

JWM 2017

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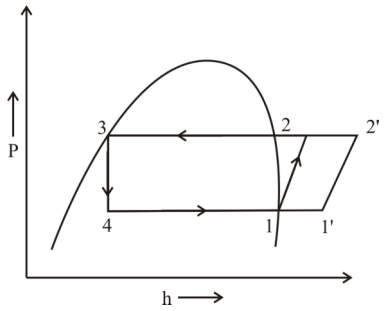
Ans. (a) : COP of vapour compression system

$$= \frac{h_1 - h_4}{h_2 - h_1}$$



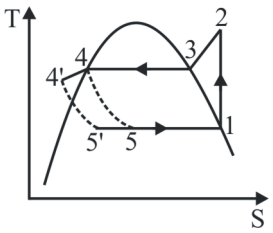
Super heating effect-

- Refrigerating effect will increase
- Work done will be directly proportional to inlet temperature and COP is inversely proportional to the work done. So, as a result of increase of work input to the compression, it will decrease the COP.

231. **Statement (I):** Sub-cooling of a refrigerant liquid increases the coefficient of performance of a refrigeration cycle.
Statement (II): Sub-cooling reduces the work requirement of the refrigeration cycle.

ESE 2017

Ans. (c) :



Process 4–4' → Subcooling
Process 1–2 → Work requirement
The subcooling increases refrigerant effect from ($h_1 - h_5$) to ($h_1 - h_{5'}$) and no work requirement.

232. **Statement (I):** Use of non-azeotropic mixtures used as the refrigerant in a vapour-compression system improves the coefficient of performance.
Statement (II): The increase in this coefficient is attributable to reduction in volume.

ESE 2017

Ans. (c) : The evaporation takes place over a temperature range increasing direction. Hence mean temperature of evaporation increases and condensation takes place over a temperature range in decreasing direction. Mean temperature of heat rejection decreases which results in increased COP not reduction in volume.
So, statement-II is wrong.
The non-azeotropic mixtures are non isothermal refrigerant i.e. temperature of refrigerant varies during heat rejection and addition.

233. **In the working of a vapour-compression refrigeration plant, the following enthalpies are recorded at salient points in the cycle:**

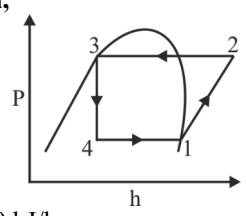
1. Enthalpy at inlet to compressor (saturated vapour), h_1=300kJ/kg.
2. Enthalpy at outlet of compressor (after isentropic compression), h_2=330kJ/kg.
3. Enthalpy at exit of condenser (saturated liquid), h_3=150 kJ/kg.

What is the COP of the plant?

(a) 3 (b) 4
(c) 5 (d) 6

ESE 2017

Ans. (c) : Given,

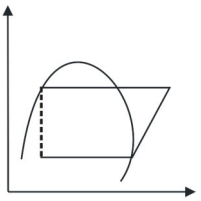


$h_1 = 300$ kJ/kg
$h_2 = 330$ kJ/kg
$h_3 = h_4 = 150$ kJ/kg

$$(COP)_{VCRS} = \frac{\text{Refrigerating effect (RE)}}{\text{Work Input}(W_{in})} = \frac{h_1 - h_4}{h_2 - h_1}$$

$$= \frac{300 - 150}{330 - 300} = \frac{150}{30} = 5$$

234. **The ordinates and abscissa of the diagram given for the vapour-compression refrigeration cycle represent**



(a) pressure and volume
(b) temperature and entropy
(c) enthalpy and entropy
(d) pressure and enthalpy

ESE 2017

Ans. (d) : The given cycle is vapour compression refrigeration cycle. The left side vertical process shown by dotted line is throttling.
So ordinate is pressure and abscissa is enthalpy.

235. **In compound compression system with intercooling in a refrigeration cycle the suction vapour to the second stage of compression is :**

(a) Superheated (b) Wet
(c) Dry saturated (d) Subcooled

HPPSC W.S. Poly. 2016

Ans : (c) In compound compression system with intercooling in a refrigeration cycle the suction vapour to the seconds stage of compression is dry saturated.

236. **At least pressure and temperature, the latent heat of vaporisation of a refrigerant**

(a) Depends on other factor
(b) Increases
(c) Decreases
(d) Remains same

Rajasthan AE (Nagar Nigam) 2016 Shift-3

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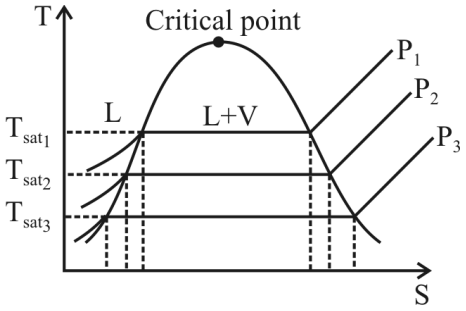
Ans. (b) : According to graph as temperature and pressure decreases, the latent heat of vaporisation of refrigerant increases.



$P_1 > P_2 > P_3$, $T_{sat_1} > T_{sat_2} > T_{sat_3}$

So, at least pressure and temperature latent heat of vaporisation maximum (increases).

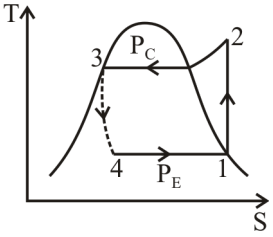
Note-At critical point latent heat vaporisation of substance is zero.

237. Refrigerant vapor enters into the compressor of a standard vapor compression cycle at -10^0C (h = 402 kJ/kg) and leaves the compressor at 50^0C (h = 432 kJ/kg). It leaves the condenser at 30^0C(h = 237 kJ/kg). The COP of the cycle is ---

(a) 4 (b) 3.5
(c) 6 (d) 5
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (e) : Given,



$h_1 = 402$ kJ/kg
$h_2 = 432$ kJ/kg
$h_3 = 237$ kJ/kg

$$(COP)_{cycle} = \frac{h_1 - h_3}{h_2 - h_1}$$

$$= \frac{402 - 237}{432 - 402}$$

$$= \frac{165}{30} = 5.5$$

238. Replacing a water-cooled condenser with an air-cooled one in a vapor compression refrigeration system with constant evaporator pressure result in

(a) Decrease in pressure ratio
(b) Increase in condensation pressure
(c) Increase in pressure ratio
(d) Increase in condensation temperature

UPRVUNL AE 21.08.2016

Ans. : (d) Replacing a water-cooled condenser with an air-cooled one in a vapour compression refrigeration system with constant evaporator pressure result increase in condensation temperature.

239. In a vapour compression cycle, a good refrigerator should have a

(a) Large latent heat of vaporization at condenser pressure.
(b) Large latent heat at evaporator pressure
(c) Condenser pressure close to critical
(d) Low critical pressure

UPPSC AE 12.04.2016 Paper-II
ESE 2013

Ans. : (b) In a vapour compression cycle, a good refrigerator should have a large latent heat at evaporator pressure.

240. COP of air refrigerator is related with COP of vapour compression refrigerator as

(a) $(COP)_{air} > (COP)_{vap.c}$
(b) $(COP)_{air} < (COP)_{vap.c}$
(c) $(COP)_{air} = (COP)_{vap.c}$
(d) None of the above

UPPSC AE 12.04.2016 Paper-II

Ans. : (b) COP of air refrigerator is related with COP of vapour compression refrigerator as

$$\boxed{(COP)_{air} < (COP)_{vap.c}}$$

241. Specific isentropic work of compression in vapour compression refrigeration system decreases with

(a) Decrease in evaporator temperature at constant condenser temperature
(b) Increase in evaporator temperature at constant condenser temperature
(c) Increase in condenser temperature at constant evaporator temperature
(d) Decrease in evaporator temperature and increase in condenser temperature

RPSC Lecturer 16.01.2016

Ans. (b) : Specific isentropic work of compression in vapour compression refrigeration system decreases with increase in evaporator temperature at constant condenser temperature.

242. Refrigerating effect in vapour compression refrigeration system increases with

(a) Increase in evaporator temperature at constant condenser pressure
(b) Decrease in evaporator pressure at constant condenser pressure
(c) Increase in condenser temperature at constant evaporator temperature
(d) Increase in mass flow rate of refrigerant in the system

RPSC Lecturer 16.01.2016

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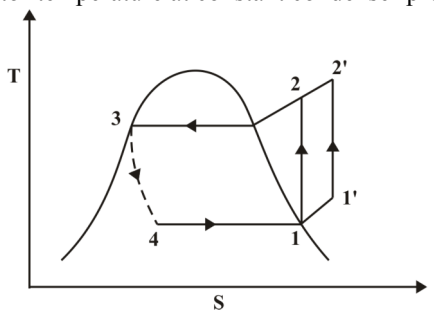
Ans. (a) : Refrigerating effect in vapour compression refrigeration system increases with increase in evaporator temperature at constant condenser pressure.

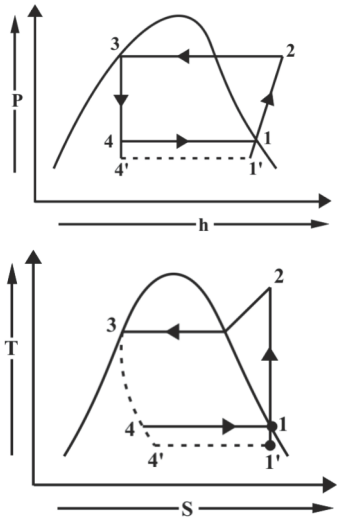


Refrigeration effect
4 - 1' > 4 - 1

243. At constant condenser pressure, reduction in evaporator pressure of vapour compression refrigeration cycle does not lead to:
(a) Decrease in evaporation temperature
(b) Decrease in refrigeration effect
(c) Increase in specific volume of suction vapour
(d) Decrease in specific compressor work

UPRVUNL AE 2016

Ans. (d) : Specific compressor work increase, if condenser pressure remain constant, reduction in evaporator. Pressure of vapour compression refrigeration cycle.

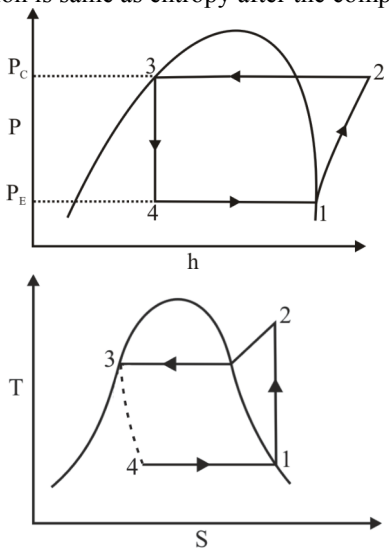


COP ↓, Refrigeration effect ↓

244. In the compression process of vapour compression refrigeration system
(a) Enthalpy before compression is same as enthalpy after the compression
(b) Internal energy before compression is same after the compression
(c) Temperature before compression is same as temperature after the compression
(d) Entropy before compression is same as entropy after the compression

APPSC AEE MAINS 2016 PAPER-III

Ans. (d) : In the compression process of vapour compression refrigeration system entropy before compression is same as entropy after the compression.



From T–S diagram
$S_2 = S_1$ = constant
compression process (1 – 2)

245. In a refrigeration system, expansion valve is incorporated between-
(a) evaporator and compressor
(b) condenser and evaporator
(c) compressor and condenser
(d) superheater and subcooler

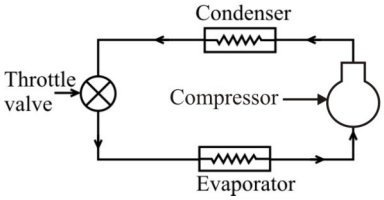
RPSC IOF & Boiler 20.09.2017

Ans : (b) In a refrigeration system, expansion valve is incorporated between condenser and evaporator.

246. A refrigerant moving in a refrigerator follows:
(a) open system
(b) closed system
(c) both open and closed system exists
(d) none of the above

(HPPSC LECT. 2016)

Ans : (b) A refrigerant moving in a refrigerator follows close system.



247. In case of a refrigerator higher COP can be achieved with
(a) lower evaporator temperature and higher condenser temperature
(b) higher evaporator temperature and lower condenser temperature
(c) higher evaporator temperature and higher condenser temperature
(d) lower evaporator temperature and lower condenser temperature

GPSC ARTO 01.05.2016

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Ans. (b) : COP of refrigerator

$$\frac{T_E}{T_C - T_E}$$

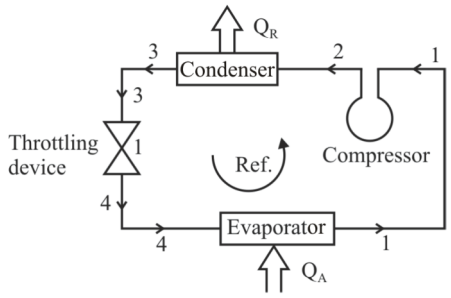
for higher COP of Refrigerator $T_E \uparrow$ & $T_C \downarrow$es

248. In a refrigeration cycle heat is ---------- by a refrigerant in evaporator and heat is ---------- by refrigerant in a condenser.

(a) Absorbed, Absorbed (b) Absorbed, Rejected
(c) Rejected, Absorbed (d) Rejected, Rejected

APPSC AEE Screening Test 2016

Ans. (b) : In a refrigeration cycle heat is absorbed by a refrigerant in evaporator and heat is rejected by refrigerant in a condenser.



249. Dry compression is preferred due to:

(a) High volumetric efficiency
(b) High mechanical efficiency
(c) Less chance of damage of the compressor
(d) All of the above

OPSC AEE 2015 PAPER - II

Ans : (c) Dry compression is preferred due to less chance of damage of the compressor.

250. Most of the domestic refrigerators work on the following refrigeration system

(a) vapour absorption
(b) Electrolux refrigerator
(c) dual cycle
(d) vapour compression

GWSSB AAE, 27.12.2015

Ans. (d) : Electrolux – No need of electricity
Vapour absorption – little electricity for pump
Vapour compression – Now a days

251. Highest temperature encountered in refrigeration cycle should be

(a) Near critical temperature of refrigerant
(b) Above critical temperature
(c) At critical temperature
(d) Much below critical temperature

Nagaland PSC CTSE 2015 Paper-II

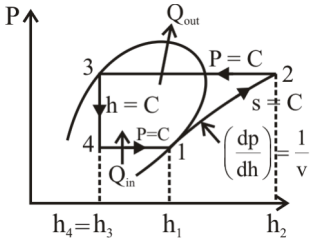
Ans. (d) : Highest temperature encountered in refrigeration cycle should be much below critical temperature.

252. The lower horizontal line of the refrigeration cycle plotted on pressure-enthalpy diagram represents

(a) Condensation of the refrigerant vapour
(b) Evaporation of the refrigerant liquid
(c) Compression of the refrigerant vapour
(d) Metering of the refrigerant liquid

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : In given P-h diagram there are four process
Process 1-2 : Reversible adiabatic compression
Process 2-3 : Constant pressure heat rejection
Process 3-4 : Throttling (Isenthalpic expansions in throttle valves
Process 4-1 : Constant pressure heat absorption or evaporation of refrigerant in evaporator.



The lower horizontal line of the refrigeration cycle plotted on pressure enthalpy diagram represents evaporation of refrigerant liquid.

253. Consider the following statements pertaining to a vapour compression type refrigerator.

1. The condenser rejects heat to the surroundings from the refrigerant.
2. The evaporator absorbs heat from the surroundings to be cooled.
3. Both the condenser and evaporator are heat exchangers with refrigerant as a common medium.
4. The amount of heat exchanged in condenser and evaporator are equal under steady conditions.

Which of the above statements are correct?

(a) 1 and 2 (b) 1, 2 and 3
(c) 1, 2 and 4 (d) 2, 3 and 4

Nagaland PSC CTSE 2015 Paper-II

Ans. (b)
(1) The condenser rejects heat to the surroundings from the refrigerant.
(2) The evaporator absorbs heat from the surroundings to be cooled.
(3) Both the condenser and evaporator are heat exchangers with refrigerant as a common medium.

254. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Bell Coleman refrigeration	1. Expansion cylinder
B. Reversed carnot cycle	2. Heat Pump
C. Milk chilling plant	3. Decreases COP
D. Superheating in refrigeration cycle	4. Compressor
	5. Brine
	6. Vortex refrigerator
	7. Increases COP

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	A	B	C	D		A	B	C	D
(a)	1	6	5	3	(b)	4	2	3	7
(c)	1	2	5	3	(d)	4	6	3	7

Nagaland PSC CTSE 2015 Paper-II
CSE Pre-2006

Ans. (c) :

Bell Coleman refrigeration	–	Expansion cylinder
Reversed carnot cycle	–	Heat Pump
Milk chilling plant	–	Brine
Superheating in refrigeration cycle	–	Decreases COP

255. In a vapour compression refrigeration system, the compressor capacity is 420 kJ/min and refrigerating effect is 2100 kJ/minute and heat rejection factor is 1.2. What will be the heat rejected from the condenser and C.O.P respectively?

(a) 5040 kJ/minute and 5
(b) 2520 kJ/minute and 5
(c) 2520 kJ/minute and 4
(d) 2540 kJ/minute and 4

Nagaland PSC CTSE 2015 Paper-II

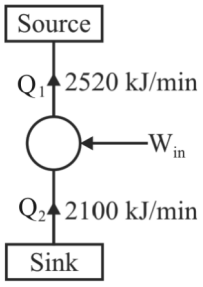
Ans. (b) : Given, W_{in} = 2100 kJ/min, G = 1.2

$$\text{Heat rejection factor (G)} = \frac{\text{Heat rejected in condenser}}{\text{Refrigeration effect}}$$

$$1.2 = \frac{Q_1}{2100}$$

$$Q_1 = 1.2 \times 2100$$

$$= 2520 \text{ kJ/min}$$



$$W_{in} = Q_1 - Q_2$$

$$= 2520 - 2100$$

$$= 420 \text{ kJ/min}$$

$$COP = \frac{2100}{420}$$

$$COP = 5$$

256. In ammonia vapour compression system, the temperature of ammonia after compression will be

(a) 0º to 40ºC (b) 40º to 50ºC
(c) 50º to 70ºC (d) 70º to 110ºC

MPPSC AE 08.11.2015

Ans. (d) : In ammonia vapour compression system, the temperature of ammonia after compression will be 70ºC to 110ºC.

257. In two-stage compression system with flash gas removal

(a) Refrigerant mass flow rates in both low and high stage compressors are equal
(b) Refrigerant mass flow rates in high stage compressors is smaller than that in low stage compressor
(c) Mass flow rates in low and high stage compressors are equal if the pressure ratios are equal
(d) Refrigerant mass flow rates in high stage compressors is greater than that in low stage compressor
(e) Mass flow rates in low and high stage compressors equal if the entropy are equal

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : In two-stage compression system with flash gas removal refrigerant mass flow rates in high stage compressor is smaller than that in low stage compressor.

258. Multi-evaporator systems with a single compressor and a pressure reducing valve:

(a) Yield very high COPs compared to multi-evaporator, single stage systems
(b) Yield lower compressor discharge temperature compared to single stage systems
(c) Yield slightly higher refrigeration effect in the low temperature evaporator
(d) Yield slightly higher refrigeration effect in the high temperature evaporator compared to single stage systems
(e) Yield slightly lower refrigeration effect in the high temperature evaporator compared to single stage systems

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d): Multi-evaporator system with a single compressor and a pressure reducing valve yield slightly higher refrigeration effect in the high temperature evaporator compared to single stage systems.

259. Consider the following in cascade systems:

1. Different refrigerants are used in individual cascade cycles
2. There is no mixing of refrigerants and no migration of lubricating oil
3. Higher COPs compared to multi-stage systems can be obtained
4. Operating pressures need not be too high or too low

(a) 1 and 2 (b) 1 and 3
(c) 1 and 4 (d) 1, 2 and 3
(e) 1, 2 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (e) : The cascade refrigeration system is a freezing system that use two kinds of refrigerants having different boiling points, which run through their own independent freezing cycle and are joined by a heat exchanger. This system is employed to obtain temperature of -40 to -80^{0}C or ultra – low temperature lower than them.

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260. For a two-stage cascade system working on Carnot cycle and between low and high temperatures of -90^{0}C and 50^0 C, the optimum cascade temperature at which the COP will be maximum is given by:

(a) -20^{0}C (b) -30^{0}C
(c) -67^{0}C (d) -78^{0}C
(e) 0^{0}C

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : $T_H = 50^0C = 323K$
$T_L = -90^0C = 183K$
For Cascade, optimum Cascade Temperature for maximum COP

$$T_{c.c.\ opt.} = \sqrt{T_H \times T_L}$$

$$T_{c.c.\ opt.} = \sqrt{323 \times 183} = 243.123\ K$$

$$T_{c.c.\ opt} = -29.87 \approx -30^0C$$

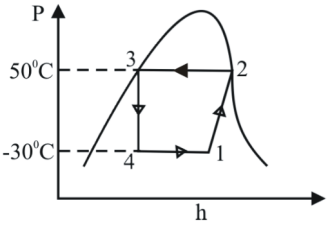
261. A refrigerator based on ideal vapour compression cycle operates between the temperature limits of –30°C and 50°C. The refrigerant enters the condenser as saturated vapour and leaves as saturated liquid. The enthalpy and entropy values at these temperatures are given in the table below. The refrigeration effect per kg of refrigerant (kJ/kg) is around

T(°C)	h_f (kJ/kg)	h_{fg} (kJ/kg)	S_f (kJ/kgK)	S_{fg} (kJ/kg)
-30	**22**	**161**	**0.07**	**0.66**
50	**82**	**123**	**0.31**	**0.38**

(a) 174 (b) 67
(c) 87 (d) 91
(e) 69

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : We know that,



$h_3 = h_4 = 82$ kJ/kg
$S_1 = S_2 = [S_f + x\ S_{fg}]_2$
$S_2 = 0.31 + 1 \times 0.38$
$S_2 = 0.69$ kJ/kgK
$S_1 = [S_f + xS_{fg}]$
$0.69 = 0.07 + x \times 0.66$
$\boxed{x = 0.9393}$

$$h_1 = h_{f_1} + xhf_{g_2}$$

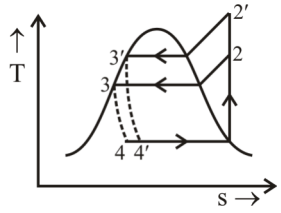
$h_1 = 22 + 0.9393 \times 161 = 173.22$ kJ/kg
Then,
Ref. Effect = $h_1 - h_4 = 173.22 - 82 = 91.22$ kJ/kg

262. In a vapour compression refrigeration plant, if the condenser temperature increases, the power input to the compressor will

(a) Decrease (b) Increase
(c) Remain the same (d) Be unpredictable

MPPSC State Forest Service Exam, 2014

Ans. (b)



1-2 : Compressor
2-3 : Condenser
3-4 : Expansion
4-1 : Evaporator
As condenser temperature increases, the power input to the compressor will increases. ($W_{1-2'} > W_{1-2}$)
The refrigeration effect decreases ($RE_{1-4} > RE_{1-4'}$)

263. Which of the following is used in vapour compression refrigeration system?

(a) Throttle valve
(b) Expansion cylinder
(c) Both throttle valve and expansion cylinder
(d) Pump

UPRVUNL AE 2014

Ans: (a) Throttle valve is used in vapour compression refrigeration system. In throttle valve enthalpy is constant (i.e. $h_1 = h_2$).

264. Which of the following statements are correct?

I. Practically all common refrigerants have approximately the same COP and power requirement.
II. Ammonia mixes freely with lubricating oil and this helps lubrication of compressors.
III. Dielectric strength of refrigerants is an important property in hermetically sealed compressor units.
IV. Leakage of ammonia can be detected by halide torch method.

Choose the correct answer from the options given below.

(a) I and II (b) I and III
(c) III and IV (d) I, II, III and IV

UPRVUNL AE 2014

Ans : (b) (I) Practically all common refrigerants have approximately the same COP and power requirement. (III) Dielectric strength of refrigerants is an important property in hermetically sealed compressor units.

265. Consider the following for the evaporator of a vapour compression refrigeration system:

1. A low temperature is maintained so that heat can flow from the external fluid
2. Refrigeration effect is produced as the refrigerant liquid vapourizes
3. A low pressure is maintained so that the compressor can run

Of the above, which is correct

(a) 1 (b) 2
(c) 3 (d) 1 and 2
(e) 1 and 3

CGPSC AE 26.04.2015 SHIFT-II

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Ans. (d) : The evaporator of a vapor compression refrigeration system maintained a low temperature so that heat can flow from the external fluid and produce refrigeration effect as the refrigerant liquid vapourizes.

266. Match List I with List II and select the correct answer using the codes given below the lists:

List I	List II
A. Bell Coleman refrigeration	i. Compressor
B. Vapour compression	ii. Generator
C. Absorption refrigeration	iii. Flash chamber
D. Jet refrigeration	iv. Expansion cylinder

Code :

	A	B	C	D
(a)	i	iv	iii	ii
(b)	iv	i	iii	ii
(c)	i	iv	ii	iii
(d)	iv	i	ii	iii

APPSC Poly. Lect. 2013

Ans. (d) : List-I **List-II**
i) Bell Coleman refrigeration- Expansion cylinder
ii) Vapour compression refrigeration- Compressor
iii) Absorption refrigeration- Generator
iv) Jet refrigeration- Flash chamber

267. Choose the correct statement about the effect of changing various operating conditions of a vapour compression refrigeration cycle.
(a) Lowering the evaporator pressure will always increase the COP.
(b) Increasing the condenser pressure will always result in increase the COP.
(c) Superheating the suction vapour always increases the COP.
(d) Lowering the evaporator always decreases the COP.

RPSC AEN Pre-2013

Ans. (d) : Note- Deleted by commission.

268. The process of cooling the liquid refrigerant below the condensing temperature
(a) Super cooling (b) Sub-cooling
(c) Condensation (d) Saturation

TNPSC AE 2013

Ans. (b) : The term subcooling also called undercooling refers to a liquid existing at a temperature below its normal boiling point.

269. In case of multi-stage vapour compression system, flash intercooling is beneficial in case of those refrigerants only for which:-
(a) Latent heat of vapourization is more
(b) Latent heat of vapourization is less
(c) Thermal conductivity is high
(d) Thermal conductivity is low

UKPSC AE-2013, Paper-II

Ans. (a) : In case of multi-stage vapour compression system, flash intercooling is beneficial in case of those refrigerants only for which latent heat of vapourization is more.

270. In an ammonia vapour compression system, wet compression process is used instead of dry compression. The C.O.P.:-
(a) Will increase
(b) Will decrease
(c) Can increase or decrease
(d) Remains unaffected

UKPSC AE-2013, Paper-II

Ans. (a) : In an ammonia vapour compression system, wet compression process is used instead of dry compression. The C.O.P. will increase because compressor work in wet compression is less.

271. Vapour compression cycle using R-12 gives maximum COP when:-
(a) Suction state to compressor is in wet region
(b) Suction state to compressor is in superheat region
(c) Suction state to compressor is dry saturated
(d) None of the above

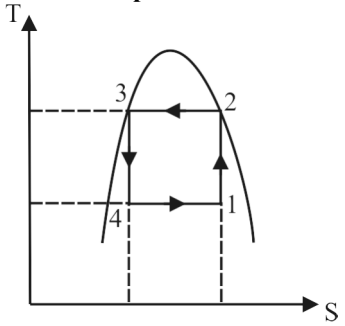
UKPSC AE-2013, Paper-II

Ans. (b) : Vapour compression cycle using R-12 gives maximum COP when suction state to compressor is in superheat region.

272. During which of the following process does heat rejection take place in Carnot Vapour Cycle?
(a) Constant volume (b) Constant pressure
(c) Constant temperature (d) Constant entropy

PSPCL AE, 2012

Ans. (c) : Constant temperature –



1 – 2 Process = Reversible adiabatic compression.
2 – 3 Process = Constant temperature heat rejection.
3 – 4 Process = Reversible adiabatic expansion.
4 – 1 Process = Constant temperature heat addition.

273. In case of ejector–compression system the power input is in the form of
(a) electric power (b) heat
(c) mechanical work (d) steam power

UKPSC AE 2012 Paper–II

Ans. (d) : steam power

274. Pressure drop in capillary tube (used in refrigeration systems) takes place due to
(a) friction
(b) change in momentum
(c) both (a) & (b) above
(d) none of the above

UKPSC AE 2012 Paper–II

Ans. (c) : both (a) & (b) above

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275. In a refrigeration cycle the effect of superheating on COP is :

(a) Decrease (b) Remains change
(c) Increase (d) None of these

J&K PSC Civil Services Pre, 2010

Ans. (d) : Effect of superheating–

(1) Refrigeration effect of superheating increases if superheating occurs in evaporator.
(2) Work input to the compressor increases because, it is function of inlet temperature to the compressor.
(3) COP may increase or decrease depending upon the refrigerant. For example, in case of R-12 refrigerant supercharging would result increase in COP whereas in case of ammonia superheating would result a decrease in COP.

276. A vapour compression refrigerator having a COP of 5 rejects 30 kW of heat in the condenser. The refrigerating effect is :

(a) 20 kW (b) 25 kW
(c) 30 kW (d) 35 kW

UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation- Given; $(COP)_R = 5$

(For VCR system)

Rejected heat $(Q_R) = 30$ kW

Let; RE = Refrigerating effect

$$(COP)_R = \frac{R.E}{W_{input}} = \frac{Q_2}{Q_1 - Q_2}$$

Q_1 = Heat absorbed by Evaporator = RE

Q_1 = Heat rejected to condenser

$$\therefore \quad 5 = \frac{Q_2}{30 - Q_2} \Rightarrow 150 - 5Q_2 = Q_2$$

$$\Rightarrow 6Q_2 = 150 \quad \Rightarrow Q_2 = 25kW$$

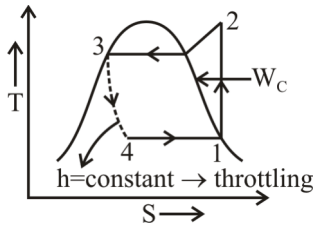
$$\boxed{\because \quad R.E. = Q_2 = 25\ kW}$$

277. In a vapour-compression refrigerator, the design pressure at the condenser will depend on

(a) Type of compressor
(b) Refrigerant used
(c) Its capacity
(d) Its coefficient of performance

UPSC JWM Advt. No.-50/2010

Ans. (a)



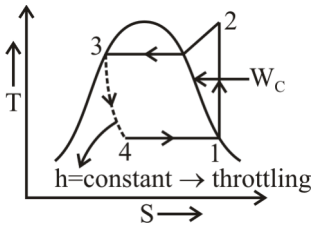
Pressure at a point '2', P_2 is the designed pressure in a VCRS system, which is depend upon the type of compressor used for the compression of refrigerant.

278. In a vapour-compression refrigeration system, the thermodynamic process which the refrigerant undergoes while passing through the expansion valve is

(a) Isentripic (b) Adiabatic
(c) Throttling (d) Free expansion

UPSC JWM Advt. No.-50/2010

Ans. (c) :



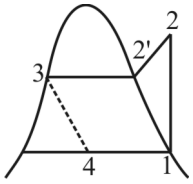
As we know, in a VCRS, the thermodynamic process, which the refrigerant undergoes while passing through the expansion value is throttling, which is highly irreversible.

279. In a vapour-compression refrigeration cycle, if h_1 and h_2 denote the enthalpies at inlet and exit of the compressor, respectively, h_3 is the enthalpy at the exit of the condenser and h_4 is the enthalpy at the inlet of the evaporator, then COP for the cycle is :

(a) $\frac{h_1 - h_3}{h_2 - h_3}$ (b) $\frac{h_1 - h_4}{h_2 - h_3}$
(c) $\frac{h_1 - h_3}{h_2 - h_1}$ (d) $\frac{h_2 - h_1}{h_1 - h_4}$

DRDO Scientists 2009

Ans. (c) :



$$\text{COP} = \frac{\text{Refrigerant effect}}{\text{Work done on compressor}}$$

$$\frac{h_1 - h_4}{h_2 - h_1}$$

Since process 3-4 is an isenthalpic throttling process

$\therefore \quad h_3 = h_4$

Hence

$$\text{COP} = \frac{h_1 - h_3}{h_2 - h_1}$$

280. In vapour compression cycle, heat rejected is 65 kW and work done in compression is 10 kW, the COP of refrigerator will be

(a) 6.5 (b) 5.5
(c) 5.0 (d) None of the above

UKPSC AE 2007 Paper -II

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Ans. (b)

$$COP = \frac{\text{desired effect}}{\text{work input}}$$

$$= \frac{65-10}{10}$$

=5.5

281. Super heating in a refrigeration cycle

(a) increases COP
(b) decreases COP
(c) COP remains unaltered
(d) other factors decide COP

J & K PSC Screening, 2006

Ans. (d) : COP may increase or decrease depending on the retrigerant.

For R12, R134 → COP↑

And For Freon, R22, NH_3 → COP↓

282. In vapour compression refrigeration system, the refrigerant is superheated mainly to

(a) Increase the volumetric efficiency of the compressor
(b) Avoid wet compression
(c) Decrease compressor work input
(d) Decrease load on the condenser

OPSC Civil Services Pre 2006

Ans. (b) : By superheating specific volume at inlet to compressor and work of compression increases and COP increases or not depends upon the relative increase in refrigerating effect and work of compression which is depends upon the nature of the refrigerant used.

A minimum amount of superheat is desirable as it prevents the entry of liquid droplet into the compressor. Compressor are designed to handle the vapour not liquid.

3. Refrigerants and Components of Refrigeration System

283. Which of the following refrigerant has the maximum ozone depletion potential in the startophere?

(a) Ammonia (b) Carbon dioxide
(c) Sulphur dioxide (d) Flourine

UJVNL AE 12.09.2021
OPSC AEE 2019 Paper-II
Nagaland PSC (CTSE) 2018, Paper-II
RPSC Vice Principal ITI 2018
HPPSC Asst. Prof. 20.11.2017
GPSC Asst. Prof. 28.08.2016
PSPCL AE, 2012, ESE 1992

Ans. (d) : Fluorine has the maximum ozone depletion potential in the stratosphere. R-11 or CFC 11 has the maximum potential amongst chlorocarbons because of the presence of three chlorine atoms in the molicule.

284. In the window air conditioner, the expansion device used is :

(a) Capillary tube
(b) Thermostatic expansion valve
(c) Automatic expansion valve
(d) Float valve

Nagaland PSC CTSE 2017 paper-II
TSPSC AEE 2017
HPPSC Asstt. Prof. 18.11.2016
UPPSC AE 12.04.2016, APGENCO AE 2012
VIZAG STEEL MT 10.06.2012, GATE 2004

Ans. (a) : Capillary tube is used as expansion device in the window air conditioner.

- **Thermostatic expansion valve** is a throttling device used in refrigerator and air conditioning which controls the amount of refrigerant liquid injected into a system's evaporator based on the evaporator outlet temperature and pressure called superheat.
  Automatic Expansion Valve–A valve designed to maintain a constant pressure in the evaporator and to meter the refrigerant by using a pressure sensing device. Also called a constant pressure expansion valve.
- **Float valve** are used in domestic water system. A float valve maintains the level in the cold water header tank which regulates the pressure of the hot water system.

285. The color of the flame of halide torch, in case of leakage of Freon refrigerant, will change to:

(a) bright green (b) yellow
(c) red (d) orange

Nagaland PSC (CTSE) 2018, Paper-II
GPSC ARTO Pre 30.12.2018
RPSC Vice Principal ITI 2018
OPSC Civil Services Pre.2011
J&K PSC Civil Services Pre, 2010
UKPSC AE 2007 Paper-II, CSE Pre-1996

Ans. (a) : The colour of the flame of halide torch, in case of leakage of Freon refrigerant, will change to bright green. The flame of the torch will change to a green color when it is exposed to a refrigerant with chlorine. The intensity of the change in the flame will vary depending on the size of the leaks.

286. For Ammonia refrigeration system, the pipe for carrying refrigerant shall be made of following material:

(a) brass (b) copper
(c) steel or wrought iron (d) aluminium

ISRO Scientist/Engineer (RAC) 22.04.2018
Kerala PSC Poly. Lect., 2017
Sikkim PSC (Under Secretary), 2017
GPSC EE Pre, 28.01.2017
GPSC ARTO 01.05.2016, ESE 2012, CSE Pre-1998

Ans. (c) : Ammonia react with copper pipes because of corrosive in nature. Aluminium is generally used in air cooling coils and not for liquids. So steel is most commonly used for ammonia system.

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287. The chemical formula of Freon-12 is -

(a) CCl_2F (b) CCl_2F_2

(c) CCl_2F_3 (d) $CClF$

RPSC IOF 2020, GMB AAE 25.06.2017
OPSC AEE 2015 Paper-II
APPSC IOF 2009
UKPSC AE 2012, 2007 Paper-II
CSE Pre-1994

Ans. (b) : The chemical formula of F – 12 $C_mH_nF_p\ Cl_q$

for Hydro carbon R_{012} $n + p + q = 2m + 2$

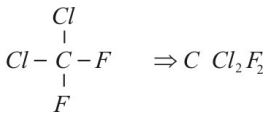
$C_{m-1}\ H_{n+1}\ F_n$ …(i) R (m–1) (n+1)p

$m-1=0 \quad \Rightarrow m=1$

$n+1=1 \quad \Rightarrow n=0 \qquad 0 + 2 + q = 2 \times 1+2$

$p=2 \quad \Rightarrow p=2 \qquad q = 2$

$C_1H_0F_2Cl_2$



$\Rightarrow C\ Cl_2F_2$

288. Environment friendly refrigerant R 134a is used in the new generation domestic refrigerators. Its chemical formula is

(a) $CHCIF_2$ (b) $C_2CI_3F_3$

(c) $C_2CI_2F_4$ (d) $C_2H_2F_4$

UPPSC AE 13.12.2020, Paper-2
APPSC Poly Lect. 13.03.2020
Nagaland PSC CTSE 2017 2nd Paper
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2004

Ans. (d)

R-134a

For saturated hydrocarbon refrigerators

$\therefore$ $R – (m – 1) (n + 1)p$

$m – 1 = 1 \Rightarrow m = 2$

$n + 1 = 3 \Rightarrow n = 2$

$p = 4$

And $n + p + q = 2m + 2$

$2 + 4 + q = 2 \times 2 + 2$

$q = 0$

$\therefore$ $C_mH_nF_p\ Cl_q$

Put the values R-134a $\Rightarrow C_2H_2F_4$

289. In milk chilling plants, the usual secondary refrigerant is

(a) ammonia solution (b) sodium silicate

(c) glycol (d) brine

TNPSC AE 2018, GPSC ARTO Pre. 30.12.2018
OPSC AEE 2015 Paper-II, APPSC Poly. Lect. 2013
APPSC AEE 2012, ESE 1998

Ans. (d) : In milk chilling plants, the usual secondary refrigerant is brine solution & primary refrigerant is ammonia.

290. The leaks in a refrigeration system using Freon are detected by:

(a) A halide torch which on detection produces greenish flame light

(b) Sulphur sticks which on detection gives white smoke

(c) Using reagents

(d) Sensing reduction in pressure

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II
RPSC VPITI 14.02.2016
MPPSC AE 2016, ESE 2006

Ans. (a)

1. Freon compound leaks are detected by Halide torch test (in the presence of Freon leaks the colour of light will change from blue to bluish green).
2. NH_3 leaks are detected by sulphur stick method. In the presence of NH_3 leaks white fumes of ammonium sulphite is formed.
3. SO_2 leaks are detected by ammonia swab test.
4. Halocarbon compound leak are detected by soap bubble method.

291. Domestic refrigerator working on vapour compression cycle uses which one of the following type of expansion device?

(a) Capillary tube

(b) Thermostatic valve

(c) Electrically operated throttling valve

(d) Manually operated valve

Rajasthan AE (Nagar Nigam) 2016 Shift-3
GWSSB DEE 07.07.2016, TANGEDCO AE 2015
GWSSB AAE, 27.12.2015, CSE Pre-2002

Ans. (a) : Capillary tube is a tiny narrow hollow copper tube with an internal diameter range from 0.5 to 2.0 mm. The refrigeration system work on vapour compression cycle comprising of evaporator compressor, condenser and expansion device

292. The refrigerant commonly used in commercial ice plants

(a) CO_2 (b) Air

(c) Ammonia (d) Freon-12

KPSC AE 2015
MPPSC State Forest Service Exam, 2014
J&K PSC Civil Services Pre.2013
UKPSC AE 2012 Paper-II

Ans. (c) : Ammonia is a refrigerant so called halogen free chemicals, it perhaps the most common refrigerant in industrial. Cooling plants and among the oldest ones.

293. Dry ice is

(a) Ice free from water

(b) Ice free from dissolved air or gases

(c) Ice prepared from filtered water

(d) Solid carbon dioxide

Gujarat PSC AE 2019, WBPSC AE, 2017
TNPSC AE 2014, UKPSC AE 2012 Paper–II

Ans. (d) : Solid carbon dioxide (CO_2) is also known as ice free from water.

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294. A refrigerant should have:

(a) Low specific heat of liquid
(b) High boiling point
(c) High latent heat of vaporisation
(d) Higher critical temperature

OPSC AEE 2019 PAPER - II
UKPSC AE 2013 Paper-II
J&K PSC Civil Services Pre, 2013
J &K PSC Screening 2006

Ans. : (c) : <u>Properties of Refrigerant</u>

- Low boiling point.
- High latent heat.
- Low freezing point.
- High critical temperature.
- Positive evaporator and condenser pressures, high density.
- Low specific volume.
- High thermal conductivity.
- Low viscosity.
- Low specific heat.

295. The capacity of a domestic refrigerator is in the range of

(a) 0.1 to 0.3 ton (b) 1 to 3 tons
(c) 3 to 5 tons (d) 5 to 7 tons

JPSC AE, 10.04.2021, Paper-I
Assam PSC AE (PHED), 18.10.2020
RPSC IOF 2020
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : The capacity of domestic refrigerator is in the range of 0.1 to 0.3 ton the capacity of domestic refrigerator is generally rated in litre. Thus litre capacity indicates the volume or space to be cooled by the domestic refrigerator. On the other hand the AC capacity is given by the term Ton of refrigeration.

296. The deposition of frost on evaporator tubes of an air conditioner will result in :

(a) Decreases in heat transfer
(b) Increases in heat transfer
(c) No change in heat transfer
(d) Increase in capacity of evaporator

RPSC Vice Principal ITI 2018, TNPSC AE 2017
PSPCL AE, 2012, ESE 1992

Ans. (a) : A decrease in heat transfer happens due to the addition of the frost layer around the evaporator.
Evaporator is a industrial apparatus for converting liquid into vapour.
Condenser is a device for converting gaseous into liquid.
Evaporator and condenser are opposite in nature.

297. The flow of refrigerant in a refrigeration cycle is controlled by

(a) Compressor (b) Condenser
(c) Evaporator (d) Expansion valve

Gujarat PSC AE 2019, HPPSC ADF 05.03.2019
Rajasthan AE Nagar Nigam 2016 Shift-I
OPSC AEE 2015 Paper - II

Ans. (d) : Refrigerant flow is controlled by expansion valve. The refrigerant vaporize from the heat it absorbs heat in the evaporator flow control device (expansion valve). This controls the flow of the liquid refrigerant into the evaporation. Control device usually are thermostatic, meaning that they are responsive to the temperature of the refrigerant.

298. The boiling point of ammonia is :

(a) 0 ºC (b) –50 ºC
(c) –33.3 ºC (d) 33.3 ºC

JPSC AE Pre. 2019
Nagaland PSC (CTSE) 2018, Paper-II
Rajasthan AE Nagar Nigam 2016 Shift-III
J&K PSC Civil Services Pre, 2013

Ans. (c) : The boiling point of ammonia is –33.3 ºC.

299. Which of the following refrigerant has the highest critical point pressure?

(a) Freon- 11 (b) Freon- 12
(c) Freon- 22 (d) Ammonia

CGPSC AE 15.01.2021
RPSC Lect. (Tech. Edu. Dept.) 2011
J&K PSC Civil Services Pre.2010

Ans. (d) :

	Refrigerant	Critical Pressure (bar)
(i)	R-11	43.8
(ii)	R-12	41.2
(iii)	R-22	49.38
(iv)	R-717 (Ammonia)	113.86

300. Which of the refrigerant has the least freezing point

(a) Freon-12 (b) Freon-22
(c) CO_2 (d) NH_3

Nagaland PSC (CTSE) 2018, Paper-II
Rajasthan Nagar Nigam AE 2016, Shift-I
UKPSCE AE 2007 Paper-II

Ans. (b) : Freon–22

- Refrigerants in increasing order of freezing point
  $CO_2 > NH_3 > R–11 > R–12 > R–22$
- Refrigerants in decreasing order of critical temperature
  $R–11 > NH_3 > R–12 > R–22 > CO_2$
- Refrigerants in decreasing of critical pressure
  (113–86) $NH_3 > CO_2$ (73.8) $> R–22 > R–11 > R–12$ (41.2)
- Refrigerants in decreasing of COP
  $R–11 > NH_3 > R–12 > R–22 > CO_2$

Refrigerant	Freezing point	Boiling point
R-12 (CF_2Cl_2)	157.8 ºC	29.8 ºC
NH_3 (R-717)	77.8 ºC	33.3 ºC
CO_2 (R744)	56.7 ºC	78.0 ºC
R22 ($CHClF_2$)	160.0 ºC	41.3 ºC

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301. Which one of the following refrigerants has the highest critical temperature?

(a) H_2O (b) CO_2
(c) R-12 (d) NH_3

Nagaland PSC (CTSE) 2016 Paper-II
Haryana PSC Civil Services Pre, 2014
CSE Pre-1996

Ans. (a) : H_2O has highest critical temperature

Refrigerant		Critical Temperature (^{0}C)
H_2O	–	374.4
NH_3	–	132.4
R-12	–	112
CO_2	–	31.0

302. In a domestic vapour compression refrigerator the refrigerant used is

(a) CO_2 (b) Freon-12
(c) Ammonia (d) All of the above

JPSC AE, 10.04.2021, Paper-I
MECON MT 2019, MPPSC 08.11.2015

Ans. (b) : In a domestic vapour compression refrigerator the refrigerant used is freon-12.

Refrigerants	Application
NH_3	Cold storage, ice plants
H_2O	Li-Br absorption system
CO_2	Used as dry ice in transport
R-11	Central air conditioning
R-12	Domestic refrigerator
R-22	Window type air conditioner
R-113	Air conditioning

303. A good refrigerant should have :

(a) High latent heat of vaporization and low freezing point
(b) High operating pressure and low freezing point
(c) High specific volume and high latent heat of vaporization
(d) Low COP and low freezing point

ISRO Scientist/ Engineer (RAC) 10.03.2019
Nagaland PSC (CTSE) 2018, Paper-II, PSPCL AE, 2012

Ans. (a) : High latent heat of vaporization and low freezing point.

– The refrigerant should have low boiling point and low freezing point .
– It must have low specific heat and high latent heat because high specific heat decreases the refrigerating effect per kg of refrigerant and high latent heat at low temperature increase refrigeration effect per kg of refrigerant.
– It should have low specific volume to reduce the size of compressor.
– It should have high critical pressure and temperature to avoid large power requirements.

304. The significant advantage of using ammonia as a refrigerant is its :

(a) characteristic odor (b) high latent heat
(c) solubility (d) inflammability

GPSC ARTO Pre 30.12.2018
UKPSC AE-2013, Paper-II
CSE Pre-1996

Ans. (b) : The significant advantage of using ammonia as a refrigerant is its high latent heat.

Properties of good refrigerants–
(i) Low boiling point
(ii) High critical temperature
(iii) High latent heat of vaporization
(iv) Low specific heat of liquid
(v) Non-corrosive to metal
(vi) Non-flammable
(vii) Non-explosive

305. Which of the following properties of a refrigerant is undesirable?

(a) high critical temperature
(b) low specific heat of liquid
(c) low specific volume vapour
(d) high boiling point

Kerala PSC AE 06.08.2015
Nagaland PSC CTSE 2015 Paper-II
KPSC AE 2015

Ans. (d) : Desirable Propenties of Refrigerents are –

→ high critical temperature
→ low specific volume vapour
→ low specific heat of liquid
→ low freezing point
→ High latent heat of vapourisation
→ large conductivity to reduce size of condenser and evaporator.
→ Non-corrosive
→ Non-toxic
→ Low boiling point
→ Non-flammable and on-explosive
→ low cost and easily available
→ To liquefy at moderate pressure and temperature low viscosity.

306. Ammonia refrigerant is

(a) Non-toxic
(b) Non-inflammable
(c) Toxic and non-inflammable
(d) Highly toxic and inflammable

VIZAG STEEL MT 2011
Nagaland PSC (CTSE) 2018, Paper-II
UKPSC AE 2007 Paper -II

Ans. (d) : Highly toxic and inflammable

307. A thermostatic expansion valve in refrigeration system :

(a) Ensures the evaporator completely filled with refrigerant of the load.
(b) Is suitable only for constant load system
(c) Maintains different temperatures in evaporator in proportion to load
(d) None of the above

Nagalanad PSC CTSE 2015 Paper-II
PSPCL AE, 2012, ESE 1992

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Ans. (a) : A thermostatic expansion valve is a component in refrigeration and air conditioning systems that controls the amount of refrigerant released in to the evaporator there by keeping superheat that is the difference between the current refrigerant.

308. The domestic refrigerator uses the following type of compressor:

(a) Centrifugal
(b) Axial
(c) Piston type reciprocating moved no box
(d) Screw

OPSC AEE 2019 PAPER - II
Nagaland PSC CTSE 2017 Paper-II
J&K PSC Screening 2006

Ans : (c) : The domestic refrigerator uses reciprocating type compressor. The compressor constructs the refrigerant vapour raising its pressure and temperature and pushes it into the coils of the condenser on the outside of the refrigerator. The refrigerant absorbs the heat inside the fridge when it flows through the evaporator coils, cooling down the air inside the fridge.

309. The thermostatic expansion valve operates based on

(a) temperature of the evaporator
(b) pressure in the evaporator
(c) degree of superheat at exit of evaporator
(d) degree of super heat at exit of the compressor

ISRO Scientist/Engineer (RAC) 12.01.2020
OPSC AEE 2019 PAPER - II
ESE 2005

Ans. (c) : Thermostatic Expansion Valve–It is used to maintain constant degree of superheat in the evaporator irrespective of load.

310. Global warming is caused by

(a) ozone (b) carbon dioxide
(c) nitrogen (d) carbon monoxide

TRB Poly. Lect., 2012, GATE 2000

Ans. (b) : Global warming is caused by CO_2 & other air pollutant & green house gas.

311. A good refrigerant should have:

(a) High Latent heat
(b) High Thermal conductivity
(c) Low Specific heat and Low Viscosity
(d) All of these

GMB AAE 25.06.2017
RPSC VPITI 14.02.2016

Ans. (d) : A good refrigerant should have-
1. Latent heat should be high
2. Its thermal conductivity should be high
3. Its specific heat should be low in liquid phase but high in vapour phase
4. Its viscosity should be low.

312. Chemical formula of R-22 is given by :

(a) CCl_4F (b) CCl_2F_2
(c) $CHCl_2F$ (d) $CHClF_2$

APPSC Poly Lect. 13.03.2020
APGENCO AE 2017

Ans. (d) : Chemical formula of R–22 is $\rightarrow CHClF_2$
Saturated hydro carbon
$C_m H_n F_p Cl_q$ .....(i) & $n + p + q = 2m + 2$
Given,
$R_{22} \Rightarrow R_{022}$ compared with $R(m-1)(n+1)p$
$m - 1 = 0$
$\boxed{m = 1}$
$n + 1 = 2$
$\boxed{n = 1}$
$p = 2$
Put the value of m and n in given equation
$n + P + q = 2m + 2$
$1 + 2 + q = 2 \times 1 \times 2$
$3 + q = 4$
$\boxed{q = 1}$
Putting the value of m, n, p & q in equation (i)
Formula become CH Cl F_2 .

313. The refrigerant used in steam jet refrigeration is

(a) Brine (b) Ammonia
(c) Freon-12 (d) Water

GPSC Executive Engineer 23.12.2018
TSPSC AEE 2017

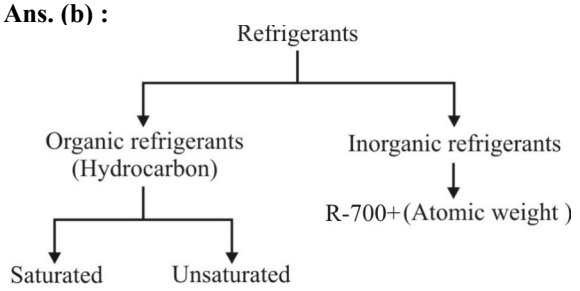
Ans. (d) : Steam jet or Ejector refrigeration system works on low grade energy, uses water as refrigerant. It uses the basic principle of boiling of liquid at lower temperature by reducing pressure on its surface.

314. Refrigerant R-717 is

(a) Air (b) Ammonia
(c) Carbon dioxide (d) Freon-12

UPPSC AE 13.12.2020, Paper-2
OPSC AEE 2015 Paper-II

Ans. (b) :



Refrigerants	Chemical formula
Carbon dioxide	R - 744
Carbon mono-oxide	R-728
Ammonia	R-717

315. When two refrigerants are mixed in proper proportions, the mixture forms a third refrigerant called

(a) Auxiliary refrigerant (b) An azeotrope
(c) Synthetic refrigerant (d) None of these

MPPSC State Forest Service Exam, 2014
J&K PSC Civil Services Pre, 2013

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Ans. (b) : Azeotrope–Mixture of refrigerants which behaves like pure substance. Their designation started with R-500.

316. Ozone depletion by CFC occurs by breakdown of:

(a) Chlorine atoms from refrigerant by UV radiation and reaction with Ozone in troposphere
(b) Fluorine atoms from refrigerant by UV radiation and reaction with Ozone in troposphere
(c) Chlorine atoms from refrigerant by UV radiation and reaction with Ozone in stratosphere
(d) Fluorine atoms from refrigerant by UV radiation and reaction with Ozone in stratosphere

GPSC ARTO Pre 30.12.2018, ESE 2002

Ans. (c) : Ozone depletion by CFC occurs by breakdown of chlorine atoms from refrigerant by UV radiation and reaction with ozone in stratosphere.

317. The use of refrigerant R-22 for temperature below –30°C is not recommended due to its

(a) good miscibility with lubricating oil
(b) poor miscibility with lubricating oil
(c) low evaporating pressure
(d) high compressor discharge temperature

MPPSC AE 08.11.2015, GATE 1993

Ans. (d) :

318. Presence of moisture in Freon refrigeration system causes

(a) Ineffective refrigeration
(b) Damage to compressor
(c) Freezing at automatic regulating valve
(d) None of the above

GPSC Asstt. Director of Transport 05.03.2017
APPSC AEE Screening Test 2016
Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : Moisture that enters the system can cause many and serious problems, such as moisture freezing in the refrigerant control orifice, acid and oil sludging. In Freon refrigeration system causes freezing at automatic regulating valve.

319. A good refrigerant should have-

(a) Large latent heat of vaporization and low operating pressures
(b) Small latent heat of vaporization and high operating pressures
(c) Large latent heat of vaporization and high operating pressures
(d) Small latent heat of vaporization and low operating pressures

RPSC AE 2018, ESE 1992

Ans. (a) : Desirable properties of refrigerants—
Thermodynamics Property—

1. **Critical Temperature**—The critical temperature of the refrigerant should be as high as possible.
2. **Specific heat**—The specific heat of the vapour should be high.
3. **Enthalpy of vaporization**—It should be as high as possible.
4. **Thermal conductivity**—It should be high as possible because it decrease size of evaporator.

$$Q = kA\frac{dT}{dx}$$

$$k \propto \frac{1}{A}$$

5. **Condensing pressure and high evaporator pressure**—Both should be positive if the evaporator pressure is less than atmospheric pressure then possibility of leakage of air their for evaporator pressure should be kept almost equal to atmospheric pressure where as condenser pressure should have some moderate value.
6. **Freezing point**—Low freezing point is desirable.

Chemical properties

1. **Toxicity**—The refrigerants should be non-poisonous to humans and food stuff.
2. **Flammability**—It should be non-flammable in nature.

Action with oil

Refrigerants + Fully miscible with oil $\rightarrow$ R-11, R-12
Refrigerants + Fully Immiscible with oil $\rightarrow$ NH_3, CO_2
Refrigerants + Partially miscible $\rightarrow$ R-22

There are some refrigerants with fully immiscible with oil and some refrigerant which are fully miscible does not create problem but the refrigerant like R-22 which are partially miscible with oil create problems and most chocking problems are occurs with R-22.

Physical properties

1. **Cost**—It should be low cost.
2. **Viscosity**—It should be low for easy flow of the refrigerant.
3. **Leak detection**—First of all they should not be leakage of refrigerant at any cost but it leaks out then its detection should be fast as possible by the simple method.

320. Azeotropes are:

(a) mixture of refrigerant and lubricating oil
(b) mixture of primary and secondary refrigerant
(c) mixture of refrigerants with unique boiling point
(d) mixture of refrigerant with glide in temperature
(e) mixture of refrigerant of low latent heat of vaporization

CGPSC AE 25.02.2018

Ans. (c) : Mixture of refrigerants with unique boiling point called Azeotropes.

321. Refrigerant CF_3 Br is designated as:

(a) R131 (b) R13B1
(c) R23B1 (d) R11B1
(e) R717

CGPSC AE 25.02.2018

Ans. (b) : Refrigerant CF_3Br is designated as R13B1

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322. Pull-down period in refrigeration system is:
(a) the time for which the compressor is not working
(b) the time required to reach the specified temperature inside the cabinet after switching on the unit
(c) the time required for charging the refrigerant in the system
(d) time required to leak the refrigerant from system
(e) time required by the system to consume 1 unit of electricity

CGPSC AE 25.02.2018

Ans. (b) : Pull-down period in refrigeration system is the time required to reach the specified temperature inside the cabinet after switching on the unit.

323. Which of the following is 'HCFC' refrigerant?
(a) R11 (b) R290
(c) R12 (d) R717
(e) R22

CGPSC AE 25.02.2018

Ans. (e) : Hydro chlorofluorocarbons or HCFC and chemicals that are mainly used as refrigerant unfortunately, releases of HCFC (R-22) deplete the earth's protective ozone layer and contribute to climate change. R-22 is an HCFC refrigerant that is often used in air conditioning equipments.

324. Environmental protection agencies advise against the use of chlorofluoro carbon refrigerants since these react with :
(a) Water vapour and cause acid rain
(b) Plants and cause green house effect
(c) Oxygen and cause its depletion
(d) Ozone layer

OPSC Civil Services Pre. 2011, ESE 2008

Ans. (d) : In chloro fluoro-carbon, chlorine atom reacts with ozone in upper atmosphere and causes its depletion.

325. Which of the following refrigerant characteristics change constantly during the cooling cycle?
(a) Pressure and phase
(b) Phase and flow
(c) Flow and temperature
(d) Temperature and pressure

Nagaland PSC CTSE 2017 2nd Paper
BPSC AE 2012 Paper - V

Ans : (d) : Temperature and pressure refrigerant characteristics change constantly during the cooling cycle.

326. Which of the following is the desirable property of refrigerant in vapor compression refrigeration system?
(a) Low vapor density
(b) Low freezing temperature
(c) Low latent heat of vaporization
(d) Condenser and evaporator pressure should be below atmospheric pressure
(e) GWP and ODP should be high

CGPSC AE 25.02.2018

Ans. (b) : Low freezing temperature is the desirable property of refrigerant in vapor compression refrigeration system.

327. The refrigerant used for absorption refrigerator working on solar heat is a mixture or water and
(a) Carbon dioxide (b) Freon 12
(c) Lithium bromide (d) Sulphur dioxide

APPSC AEE 2012, ESE 1996

Ans. (c) : Refrigerant used for absorption refrigerator working an solar heat – Water – Refrigerant
Lithium bromide – absorbent

328. Which of the following element in the refrigerant is the main contributing factor for ozone depletion of the stratosphere?
(a) Carbon (b) Chlorine
(c) Fluorine (d) Hydrogen

ISRO Scientist/Engineer (RAC) 12.01.2020
ESE 2009

Ans. (b) : The chlorine element which is present in commonly used refrigerant attacks the ozone layer which is located in stratosphere, if filters out the harmful ultraviolet (UV) radiations which were emitted by the sun and the chlorine element reduces the thickness of ozone layer. Therefore we have to use those refrigerant which are having lower value of ozone depletion (ODP). The other alternatives are fluorocarbon, hydrocarbon etc.

329. In a flooded type evaporator of a refrigerator, an accumulator at suction of the compressor is used to :
(a) collect liquid refrigerant and prevent it from going to compressor
(b) detect liquid in vapour
(c) superheat the vapour
(d) collect vapours

ESE 2018
J&K PSC Civil Services Pre, 2013

Ans. (a) : In a flooded type evaporator of a refrigerator, an accumulator at suction of the compressor is used to collect liquid refrigerant and prevent it from going to compressor.

330. For large tonnage (more than 200 tons) air conditioning applications, which one of the compressors is recommended?
(a) Reciprocating (b) Rotating
(c) Centrifugal (d) Screw

TSPSC AEE 2017, APPSC Poly. Lect. 2013

Ans. (c) : For large tonnage (more than 200 tons) air-conditioning applications, the compressor recommended is centrifugal compressor.

331. The capacity of refrigerating machine is expressed as
(a) inside volume of cabinet
(b) lowest temperature attained
(c) gross weight of machine in tons
(d) rate of abstraction of heat from the space being cooled

Assam PSC AE (PHED), 18.10.2020
Assam PSC CCE Pre-2015

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Ans. (d) : The capacity of refrigerating machine is expressed as the rate of abstraction of heat from the space being cooled.
The commonly used unit for expressing the capacity of a refrigerating machine is ton of refrigeration to freeze one metric ton of water at 0ºC into ice in 24 hours.

332. What is hunting of thermostatic expansion valve?
(a) A variation of evaporative load with degree of superheat
(b) A variation of pressure of the evaporator with variation of load
(c) Alternate overfeeding and starving of refrigerant flow in the evaporator
(d) This is used in connecting with expansion valve.

GPSC ARTO Pre 30.12.2018
ESE 2008

Ans. (c) : Hunting of thermostatic expansion valve is alternate overfeeding and starving of refrigerant flow in the evaporator.

333. Thermostatic expansion valve is called as
(a) Constant superheat valve
(b) Constant pressure valve
(c) Constant temperature valve
(d) Constant flow rate valve

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (a) : Thermostatic expansion valve is called as constant superheat valve.

334. The capillary tube used as expansion device in vapour compression system works on the principle of
(a) Isothermal expansion causing pressure drop
(b) Adiabatic expansion causing pressure drop
(c) Flow through pipe with friction causing pressure drop
(d) Throttle expansion causing pressure drop

Nagaland PSC (CTSE) 2017, 2016, Paper-II

Ans. (c) : The capillary tube used as expansion device in vapour compression system works on the principle of flow through pipe with friction causing pressure drop.

335. Flooded evaporator is used in A.C. Units of capacity
(a) < 3 tons (b) < 10 tons
(c) > 10 tons (d) None

Nagaland PSC (CTSE) 2017, 2016, Paper-II

Ans. (c) : Flooded evaporator is used in A.C. Units of capacity is greater than 10 tons.

336. Consider the following statements:
1. In Freon 22 system, moisture choking generally does not occur.
2. Freon 11 is mainly used in large capacity air-conditioning plants with centrifugal compressors.
3. Presence of lubricating oil in evaporator will increase the heat transfer coefficient.
4. Refrigerants that are completely miscible with oil, do not cause oil choking.

Of these statements
(a) 1, 2 and 3 are correct
(b) 1, 2 and 4 are correct
(c) 2, 3 and 4 are correct
(d) 1, 3 and 4 are correct

Nagaland PSC CTSE 2015 Paper-II
CSE Pre-1999

Ans. (b) : True statements are–
1. In Freon 22 system, moisture choking generally does not occur.
2. Freon 11 is mainly used in large capacity air-conditioning plants with centrifugal compressors.
3. Refrigerants that are completely miscible with oil, do not cause oil choking.

337. In on-off control refrigeration system, which one of the following expansion devices is used?
(a) Capillary tube
(b) Thermostat
(c) Automatic expansion valve
(d) Float valve

Nagaland PSC CTSE 2015 Paper-II
CSE Pre-2004

Ans. (a) : In on-off control refrigeration system, which one of the following expansion devices is used capillary tube.

338. A valve which maintains a constant degree of superheat at the end of the evaporator coil, is called
(a) automatic expansion valve
(b) high side float valve
(c) thermostatic expansion valve
(d) low side float valve

Nagaland PSC CTSE 2015 Paper-II
ESE 1993

Ans. (c) : A valve which maintains a constant degree of superheat at the end of the evaporator coil, is called thermostatic expansion valve.

339. The formation of frost on cooling coils in a refrigerator
(a) Increases heat transfer
(b) Improves C.O.P. of the system
(c) Reduces power consumption
(d) Increases power consumption

RPSC AE 2018
APPSC AEE Screening Test 2016

Ans. (d) :
The formation of frost on cooling coils in a refrigerator increases power consumption.

340. The capillary tube as an expansion device, is used in
(a) Domestic refrigeration (b) Water cooler
(c) Room cooler (d) All of these

Vizag Steel (MT) 2017
UKPSC AE 2007 Paper -II

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Ans. (d) : In the air conditioner the expansion device used for constant enthalpy expansion process. Here the capillary tube is used as an expansion device in the following :-
i. Domestic refrigeration,
ii. Water cooler,
iii. Room air conditioners,
iv. Freezer.

341. Superheating in a refrigeration cycle
(a) Decreases COP
(b) Increases COP
(c) COP remains unchanged
(d) Unpredictable

Nagaland PSC CTSE 2017 2nd Paper
(KPSC AE 2015)

Ans : (d) Superheating in a refrigeration cycle unpredictable COP.
Effect of super heating:-
1. Refrigeration effect will increase
2. Work done will be directly proportional to inlet temperature than work done will increase.
3. COP will depend on refrigerating substance. for R-12 it will increase and for NH_3 it will decrease
4. Volumetric efficiency will remain almost constant.

342. A hermetically sealed unit implies
(a) Compressor motor is sealed
(b) Compressor is sealed
(c) Refrigerant cycle is sealed
(d) Compressor and motor are sealed

OPSC AEE 2015 Paper-II
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : A hermetically sealed compressor is one in which both compressor and motor are confined in a single outer welded steel shell.
It is used for electrical components (compressor and motor) that must be secure against water vapour & foreign bodies to maintain proper function and reliability.

343. The moisture in a refrigerant is removed by
(a) Safety relief valve (b) Driers
(c) Expansion valves (d) Evaporator

Rajasthan AE (Nagar Nigam) 2016 Shift-3
UKPSC AE 2007 Paper -II

Ans. (b) : The drier in a refrigeration or air condition system has two essential function
(1) Absorbs systems contaminant such as water.
(2) Provide physical filtration.

344. In a refrigeration cycle, the moisture is to be removed before it enters the
(a) Cold side of the system
(b) Evaporator
(c) Compressor
(d) Condenser

GPSC ARTO 01.05.2016
MPPSC State Forest Service Exam, 2014

Ans. (c) : As we know, in a refrigeration the moisture is to be removed before it enters the compressor because it is difficult to compress, the refrigerant having moisture, & compressor get damaged.

345. In vapour compression refrigeration system, oil-separator is placed between
(a) Compressor and condenser
(b) Condenser and throttling valve
(c) Throttling valve and evaporator
(d) Evaporator and compressor

APSC AEE 2012
OPSC Civil Services Pre 2006

Ans. (a) : The function of an oil separator is to separate oil from the hot gas in the discharge line and return it to the compressor crankcase or to the oil reservoir in system with multiple compressors. Oil separator is installed immediate after the compressor.

346. Which of the following is correct for capillary tube used in vapor compression refrigeration system?
(a) It is very expensive
(b) It requires high maintenance
(c) The starting torque requirement of the motor of the compressor is low
(d) Its bore is large
(e) It is not used with hermetically sealed compressor

CGPSC AE 25.02.2018

Ans. (c) : Capillary tube - The capillary tube is a copper tube of very small internal diameter. It is of very long length and it is coiled to several turns so that it would occupy less space. The internal diameter of the capillary tube used for the refrigeration and air conditioning application varies from 0.5 to 23.28 mm. In any refrigerator the capillary is connected in between a condenser and evaporator to expand the refrigerant inside.

347. In the expansion valve of a refrigerator, the following property remains constant.
(a) Entropy (b) Enthalpy
(c) Internal energy (d) Product PV

RPSC ACF & FRO, 26.02.2021

Ans. (b) : In the expansion valve of a refrigerator enthalpy remains constant.

348. Which is/are the wrong statements for required refrigerant properties?
(1) High latent heat of vaporisation
(2) Higher critical temperature
(3) High boiling point
(4) High specific heat of liquid
(a) (1), (2) (b) (1), (3)
(c) (2), (4) (d) (3), (4)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : Required refrigerant properties
(1) High latent heat of vaporization.
(2) High critical temperature
(3) Low boiling point.
(4) Specific heat of refrigerant in liquid phase must be low and in vapour phase must be high.

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349. What is the correct combination of refrigerants?

(a)	R-11	(I)	Centrifugal compressor & large capacity system
(b)	R-12	(II)	Reciprocating compressor & small capacity system
(c)	R-502	(III)	Frozen food
(d)	R-718	(IV)	Steam ejector

(a) (a-I), (b-II), (c-IV), (d-III)
(b) (a-II), (b-I), (c-IV), (d-III)
(c) (a-I), (b-II), (c-III), (d-IV)
(d) (a-II), (b-I), (c-III), (d-IV)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a):

350. Low boiling point refrigerants are preferably suited for :
(a) large centrifugal compressors
(b) large axial flow rotary compressors
(c) reciprocating compressors
(d) All of the above

CGPSC AE 15.01.2021

Ans. (c) : Low boiling point refrigerants are preferably suited for reciprocating compressors.

351. Which property is suitable for refrigerants?
(a) Low specific volume
(b) Low critical temperature
(c) Low thermal conductivity
(d) Low latent heat

VIZAG Steel MT 24.01.2021

Ans. (a) : Property is suitable for refrigerants low boiling point, high critical temperature, high latent heat of vaporisation and low specific volume.

352. Relative ozone destruction efficiency of R-12 is
(a) 0.29 (b) 0.86
(c) 0.05 (d) 0.57

ESE 18.07.2021

Ans. (b) : The relative ozone destruction efficiency of R-12 ranges in 0.82 – 0.86.

353. The refrigerant Freon-12 belongs to the family of
(a) Aldehydes (b) ketones
(c) methanes (d) ethanes

HPPSC Poly. Lect. 05.07.2021

Ans. (c) : Freon 12(CF_2Cl_2) also known as dichlorodifluoromethane.

354. The Ozone Depletion Potential (ODP) among the refrigerants are measured relatively to the following refrigerant due to its maximum potential?
(a) R 717 (b) R 123
(c) R 11 (d) R 22

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : CFC11 (Trichlorofluoro methane) or R-11 has the maximum potential amongst chlrofluorocarbons because of the presence of three chlorine atoms in the molecule. So, the ozone depletion potential (ODP) among the refrigerants are measured relatively to the R-11 refrigerant due to the its maximum potential.

355. An air-air heat pump system adopted for year-round-conditioning can be achieved, if the following devices reverses its duties during summers and winters :
(a) Evaporator, Condenser
(b) Compressor, Expander
(c) Compressor, Condenser
(d) Evaporator, Expander

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) :

356. In turbo machine used to circulate refrigerant in large refrigeration plant is
(a) a centrifugal compressor
(b) a radial turbine
(c) an axial compressor
(d) an axial turbine

JPSC AE, 10.04.2021, Paper-I

Ans. (c) : In turbo machine, an axial compressor is used to circulate refrigerant in large refrigeration plant.

357. The purpose of lowering the pressure at refrigeration in an expansion valve is to
(a) Improve the boiling point
(b) Lower the freezing point
(c) to improve rate of condensation
(d) lower the rate of evaporation

TNCSC AE 2020

Ans. (c) : The purpose of lowering the pressure at refrigerant in an expansion valve is to improve rate of condensation.
Expansion valve reduce condensing pressure and temperature to evaporation temperature and pressure to get cooling.

358. For the same pressure the saturation temperature of ammonia is
(a) Higher than saturation temperature of water
(b) Lower than saturation temperature of water
(c) Same as the saturation temperature of water
(d) Depends on concentration of ammonia in water

UPPSC AE 13.12.2020, Paper-2

Ans. (b) : For same pressure, the saturation temperature of ammonia is lower than saturation temperature of water.

359. Choose undesirable properties of a secondary refrigerants.
(a) low freezing point
(b) high viscosity
(c) good stability
(d) low vapour pressure

UPPSC AE 13.12.2020, Paper-2

Ans. (b) : Desirable properties of a secondary refrigerant–
(i) low freezing point
(ii) good stability
(iii) low vapour pressure
High viscosity is the undesirable property of secondary refrigerant.

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360. The boiling point of Refrigerant R-32 is:

(a) –43 °C (b) –52 °C
(c) –10 °C (d) –76 °C

CIL MT 27.02.2020

Ans. (b) : The boiling point of refrigerant R – 32 is –52 °C

361. The compressor in the Carnot vapour cycle has to be large as it has to handle a

(a) dry saturated steam
(b) dry saturated liquid
(c) liquid + vapour system
(d) high temperature condenser exhaust

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : The compressor in the carnot vapour cycle has to be large as it has to handle a liquid and vapour system.

362. In food refrigeration facilities, most widely used fluid is,

(a) Ammonia (b) Water
(c) R-134a (d) Ethane

Oil India Senior Officer 23.12.2020

Ans. (a) : In food refrigeration facilities, most widely used fluid is ammonia.

363. A single-stage, single acting reciprocating refrigerant compressor has the following data:
Clearance volume = 0.0005 m^3
Stroke volume = 0.01 m^3
Suction volume = 0.0084 m^3
The volumetric efficiency of compressor is given by

(a) 84% (b) 94%
(c) 78% (d) 90%

SJVN ET 2019

Ans. (a) : Volumetric efficiency of compressor

$$= \frac{\text{Amount of gas leaving the compressor}}{\text{Amount of gas entering the compressor}}$$

$$\eta_v = \frac{0.0084}{0.01} \times 100 = 84\%$$

364. Air is designated as

(a) R729 (b) R764
(c) R732 (d) R744

SJVN ET 2019

Ans. (a) : Air is an inorganic compound
So
R-(700 + Molecular weight of air)
R-[700 + 29]
R-729

365. Which of the following is not the arrangement of supply ducts used in air conditioning?

(a) Extended plenum duct system
(b) Axial perimeter duct system
(c) Loop perimeter duct system
(d) Radial perimeter duct system

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (b) : Types of Supply Duct System– These are basic type of supply and return duct system. Any one of the system types, or a combination of different types, can be utilized to fit the need of a particular structure. The general types of supply duct system include.
(i) Radial perimeter duct system
(ii) Extended plenum duct system
(iii) Reducting plenum duct system
(iv) Reducting trunk duct system
(v) Loop perimeter duct system,

366. Consider the following statements:
An expansion device in a refrigeration system

(1) reduces the pressure from the condenser to the evaporator
(2) regulates the flow of the refrigerant to the evaporator depending on the load
(3) is essentially a restriction offering resistance to flow

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2019

Ans. (d) : An expansion device in a refrigeration system expands the liquid refrigerant from the condenser pressure to the evaporator pressure.
The expansion device also controls the supply of the liquid to the evaporator at the rate at which it is evaporated. The expansion device is essentially a restriction.

367. The centrifugal compressors are generally used for refrigerants that require:

(a) Small displacements and low condensing pressures
(b) Large displacements and high condensing pressures
(c) Small displacements and high condensing pressures
(d) Large displacements and low condensing pressures

OPSC AEE 2019 PAPER - II

Ans : (d) : The centrifugal compressors are generally used for refrigerants that require large displacements and low condensing pressures.

368. Most air cooled condensers are designed to operate with a temperature difference of

(a) 5ºC (b) 8ºC
(c) 14ºC (d) 22ºC

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Most air cooled condensers are designed to operate with a temperature difference of 14ºC. In air cooled condensers the temperature difference between the ambient and the condensing temperature is referred to as the condenser split.

369. In case of leakage of ammonia refrigerant, a sulphur stick produces:

(a) blue smoke (b) yellow smoke
(c) white smoke (d) red smoke

AAI Jr. Executive 29.11.2018

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Ans. (c) : Sulphur stick are used industrial ammonia refrigeration system to detect minor ammonia leaks. A sulphur stick is made from a wick which contain particle of sulphur.
The sulphur stick is lit and smolders some thing like the taper that is used to light fire works. The colour of sulphur smoke is used to find the leak when there is no ammonia. The smoke is colourless but when ammonia is present the combined sulphur and ammonia vapour produce a white smoke.

370. If a gas is to be liquefied, its temperature must be .....
(a) increased to two times of critical temperature
(b) increased to five times of critical temperature
(c) raised to ten times of critical temperature
(d) decreased below the critical temperature

RPSC AE 2018

Ans. (d) : If a gas is to be liquefied, its temperature must be decreased below the critical temperature.

371. The freezing point of R-12 is
(a) –86.6ºC (b) –95.2ºC
(c) –107.7ºC (d) –135.8ºC

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : The freezing point of R–12 is –135.8ºC.

372. The refrigerant used in small-tonnage commercial machines (hermetically sealed units) is
(a) Ammonia (b) Carbon dioxide
(c) Sulphur dioxide (d) R-12

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The refrigerant used in small tonnage commercial machines is sulphur dioxide.
Being easily condensed and possessing a high heat of evaporation, sulfur dioxide is a candidate material for refrigerants. Prior to the development of chlorofluorocarbons sulfur dioxide was used as a refrigerant in home refrigerators.

373. In order to collect liquid refrigerant and to prevent it from going to a ..... a device known as accumulator is used at the suction of compressor
(a) compressor (b) condenser
(c) expansion valve (d) evaporator

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : In order to collect liquid refrigerant and to prevent it from going to a compressor, a device known as accumulator is used at the suction of compressor. The accumulator is installed in the suction line, close to the compressor. It usually is a vertical container with top connection. This protects the compressor against liquid slugging by providing additional superheat to the suction gas.

374. An ordinary passenger aircraft requires a cooling system of capacity
(a) 2 TR (b) 4 TR
(c) 8 TR (d) 10 TR

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : An ordinary passenger aircraft requires a cooling system capable of 8 TR capacity and a super constellation requires a cooling system of more than 8 TR capacity.

375. Which is the type of condenser used for ≥3 ≤50 Tons cooling capacity of a refrigeration unit?
(a) Water cooled (b) Air cooled
(c) Water and air cooled (d) None

Nagaland PSC CTSE 2017 2nd Paper

Ans. (a) : Ton's of Refrigeration Capacity :-
i) Water cooled condenser – ≥ 3 - ≤ 50 tons.
ii) Air cooled condenser – ≤ 3 tons.
iii) Water and air cooled condenser – > 50 tons.
iv) Shell & tube type condenser – 10 – 50 tons.

376. Which is the type of condenser used for > 50 Tons cooling capacity of a refrigeration unit?
(a) Water cooled (b) Air cooled
(c) Water and air cooled (d) None

Nagaland PSC CTSE 2017 2nd Paper

Ans. (c) : Water cooled type of condensers having refrigeration capacity is around > 50 tons.

377. The material of pipelines for a system using Freon as refrigerant should be
(a) Aluminium (b) Copper
(c) Steel (d) Brass

TSPSC AEE 2017

Ans. (b) : The material of pipelines for a system using Freon as a refrigerant should be copper, because copper conducts heat better than other materials.

378. The refrigerant R-764 stands for-
(a) Ammonia (b) Carbon dioxide
(c) Sulphur dioxide (d) Methyl chloride

GMB AAE 25.06.2017

Ans. (c) : R – 764
For inorganic substance
R – 700 + Molecular weight
Molecular weight of SO_2 = 64
R -764 = SO_2
$NH_3 \Rightarrow 17 \Rightarrow$ R717
$CO_2 \Rightarrow 44 \Rightarrow$ R744

379. Which of the following is not used as a refrigerant?
(a) Ammonia (b) Carbon Monoxide
(c) Carbon Dioxide (d) Sulphur Dioxide

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Carbon monoxide is not used as a refrigerant.

380. Consider the following statements in respect of (l) the temperature of the medium, (m) the refrigerant and (n) the condenser and absorption system – in a refrigeration unit:
1. Temperature of the medium being cooled must be below that of the evaporator.
2. Refrigerant leaves the condenser as liquid.
3. Any solar thermally operated absorption system is capable only of intermittent operation.

Which of the above statements are correct?
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

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Ans. (c) : In absorption system the temperature of medium being cooled is more than the evaporator temperature, so statement-1 is wrong.

381. CFC's were banned due to

(a) Global warning (b) Ozone depleting
(c) Both (1) and (2) (d) None of these

Nagaland PSC CTSE 2017 2nd Paper

Ans. (c) : CFC banned due to reason because they damage the earth's ozone layer, CFC are also greenhouse gas that affects the environment by contributing to global warming.

382. Which of the following is eco-friendly refrigerant?

(a) R-11 (b) R-718
(c) R-12 (d) R-22
(e) R-110

CGPSC ARTO & TSI 07.01.2017

Ans. b : Eco- friendly refrigerators those refrigerant which are not lead to deplete ozone layer.
Some Eco friendly Refrigerants.
1. R- 290 – Propane
2. R-600 A – Isobutane
3. R-134 A – Tetraflouroethane
4. R-718 – Water

383. In larger industrial vapour compression refrigeration systems, the refrigerant widely used is

(a) R-12 (b) Carbon dioxide
(c) Ammonia (d) Sulphur dioxide

APGENCO AE, 2017

Ans. (c) : In larger industrial vapour compression refrigeration systems, the refrigerant widely used is ammonia.

384. Thermostatic expansion valves are set mostly for a super heat of

(a) 5°C (b) 10°C
(c) 15°C (d) 20°C

Kerala PSC Poly. Lect., 2017

Ans. (a) : Mostly thermostatic expansion valves are set for a super heat of 5°C.

385. In a Refrigeration system with multiple evaporators at different temperature with compound compression and flash intercooler, the suction vapour to the high pressure stage is:

(a) Wet state (b) Dry saturated
(c) Super heated (d) Saturated liquid

Kerala PSC Poly. Lect., 2017

Ans. (b) : In a Refrigeration system with multiple evaporators at different temperature with compound compression and flash intercooler, the suction vapour to the high pressure stage is dry saturated.

386. Finned tube evaporates are used in a

(a) Fridge (b) Window air conditioner
(c) Water cooler (d) None

Nagaland PSC CTSE 2017 2nd Paper

Ans. (b) : Finned tube evaporators are used in a windows air conditioner.

387. Bare tube evaporator is used in a

(a) Cold storage plant (b) Ice plant
(c) Milk Plant (d) None

Nagaland PSC CTSE 2017 2nd Paper

Ans. (b) : Bare tube evaporator is used in a ice plant. In this type of evaporator, the entire surface of the coil is in contact with the refrigerant inside. This type of evaporator is also called prime surface evaporator. They are easier to clean, therefore its use in household purposes.

388. In a dry evaporator refrigerant at the outlet is

(a) in the wet state (b) saturated state
(c) super heated state (d) none

Nagaland PSC CTSE 2017 2nd Paper

Ans. (c) : In dry evaporator, refrigerant at the outlet is super heated state.

389. Forced air cooled condenser in refrigeration unit has a

(a) Counter flow (b) Parallel flow
(c) Cross flow (d) None

Nagaland PSC CTSE 2017 2nd Paper

Ans. (c) : Forced air cooled condenser in refrigeration unit has a cross flow.

390. Water cooled condenser in refrigeration unit

(a) Counter flow (b) Parallel flow
(c) Cross flow (d) None

Nagaland PSC CTSE 2017 2nd Paper

Ans. (a) : Water cooled condenser in refrigeration unit has a counter flow.

391. Flooded evaporator needs at the outlet a

(a) Flash chamber (b) Accumulator
(c) Inter cooler (d) None

Nagaland PSC CTSE 2017 2nd Paper

Ans. (b) : Flooded evaporator needs at outlet of accumulator.

392. A condenser manufacturer gives a 'heat rejection capacity factor' at 26°C wet bulb temperature of 1.22. What is the condenser duty if the cooling capacity is 350 kW?

(a) 387 kW (b) 296 kW
(c) 427 kW (d) 572 kW
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : As we know,
Condenser duty = Cooling capacity × Factor
= 350 × 1.22
= 427 kW

393. Freon group of refrigerants have a property of

(a) Non-Toxic and Non-Inflammable
(b) Inflammable
(c) Toxic
(d) Toxic and Non- Inflammable

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : The Freon are colourless, odourless non-flammable, non-corrosive gases or liquid of low toxicity that are introduced as refrigerant.
Non toxic (due to presence of fluorine)
Non flammable (due to absence of hydrogen)

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394. Which of the following are blamed for ozone layer depletion?

(a) Chloro-fluoro-carbon refrigerants
(b) Hydrofluorocarbons
(c) Both A and B
(d) None of these

KPCL AE 2016

Ans. (c) : The main cause of ozone layer depletion and the ozone hole is manufactured chemicals, especially manufactured halocarbon refrigerants, solvents propellants and foam blowing agents (Chlorofluorocarbons) (CFCs) (HCFCs) referred to as ozone depleting substance (ODS).

395. Which one of the following is a CFC refrigerant?

(a) R 744 (b) R 290
(c) R 502 (d) R 718
(e) None of the above

Kerala PSC AM 09.08.2021
CGPSC Poly. Lect. 22.05.2016

Ans. (c) : R502 – Azeotropes as refrigerants
R-502 → R-(500 + random number)
$CHClF_2 + CClF_2CF_3 \rightarrow CFC \rightarrow C_3HCl_2F_7$

396. Ozone depletion potential (ODP) is the measure of ozone depletion capability of a refrigerant compared to that of

(a) R11 (b) R718
(c) R717 (d) R22

RPSC LECTURER 16.01.2016

Ans. (a) : Ozone depletion potential (ODP) is the measure of ozone depletion capability of a refrigerant compared to that of R-11 [$CFCl_3$]
R-717 → NH_3
R-718 → H_2O
R-22 → CHF_2Cl

397. Which of the following is not the desirable property of refrigerant

(a) Low latent heat
(b) High vapour density
(c) Low freezing temperature
(d) Low condenser pressure

RPSC Lecturer 16.01.2016

Ans. (a) : Refrigerant should be high latent heat of vaporisation, high vapour density, low freezing temperature these are the desirable property of refrigerant.

398. The leakage in a refrigeration system using ammonia is detected by

(a) hilide torch (b) sulphur sticks
(c) soap solution (d) hot water

APPSC AEE MAINS 2016 PAPER-III
ESE 2000

Ans. (b) : The leakage in a refrigeration system using ammonia is detected by sulphur sticks.

399. The capillary tube is not used in large capacity refrigeration systems because

(a) Cost is too high
(b) Capacity control is not possible
(c) It is made of copper
(d) Required pressure drop cannot be achieved

APPSC AEE MAINS 2016 PAPER-III

Ans. (b) : The capillary tube is not used in large capacity refrigeration systems because capacity control is not possible.

400. R-12 is preferred over R-22 in deep freezer, because

(a) It has lower operating pressure
(b) It gives higher COP
(c) It is miscible with oil over a large range of temperature
(d) All of the above

UPPSC AE 12.04.2016 Paper-II

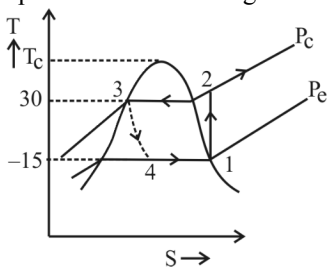
Ans : (c) R-12 is preferred over R-22 in deep freeze, because it is miscible with oil over a large range of temperature.
R-12:- Dichloro - difluore - methane (CCl_2F_2)
* Non-toxic, non corrosive and non-flammable
* Operates at low pressure
* Relatively low latent heat
* Equally amenable with all types of compressor and condensing tubes
* Good miscibility with oil
* Leakage easy detected with halide torch or by an electronic detector

401. Which one of the following is WRONG statement?

(a) A refrigerant should have higher critical temperature
(b) A refrigerant should have high boiling point
(c) A refrigerant should have high latent heat
(d) A refrigerant should have low specific heat

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : Properties of refrigerant–
1. Critical temperature should be high



2. Latent heat should be high.
3. Refrigerant should have low specific heat.
4. Refrigerant should have low boiling point.

402. Dry evaporator is used in A.C. Units of capacity

(a) < 3 tons (b) < 10 tons
(c) > 10 tons (d) None

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Dry evaporator is used is A.C. Units of capacity is less than 3 tons.

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403. Heat lost in the condenser is

(a) < (N + W) (b) = (N + W)
(c) > (N + W) (d) None

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) :

$$Heat\ Loss\left(Q_L\right)=\left(N+W\right)$$

404. Heat lost in the condenser is due to

(a) Decrease in the degree of super heat
(b) Decrease in the degree of sub cooling
(c) Decrease in the degree of super heat + Latent heat + Increase in degree of sub cooling
(d) None of these

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Heat loss in condenser is due to decrease in degree of super heat + Latent heat + Increase of degree of sub-cooling.

405. The purpose of installing a flash chamber in the refrigeration circuit is to

(a) Improve overall heat transfer coefficient
(b) Reduce pressure losses through evaporator
(c) Reduce the size of evaporator by avoiding vapours going to evaporator
(d) All the above

APPSC AEE Screening Test 2016

Ans. (d) : At entry evaporator (liquid + vapour) particle is present. So with the use of flash chamber which is installed at the entry evaporator allow only liquid particle to enter into the evaporator which results in absorption of heat

- With the use of flash chamber their is no impact on (COP)
- It helps to reducing the size of evaporator.

406. In an air craft refrigeration system the pressure at the cooling turbine outlet is equal to

(a) Ambient pressure
(b) Cabin pressure
(c) Pressure at inlet to compressor
(d) None of the above

UPPSC AE 12.04.2016 Paper-II

Ans : (b) In an air craft refrigeration system the pressure at the cooling turbine outlet is equal to the cabin pressure.

407. The presence of moisture in refrigeration effect is felt at:

(a) Evaporator (b) Condenser pipes
(c) Expansion valve (d) Compressor unit

UPPSC AE 12.04.2016 Paper-II

Ans : (c) The presence of moisture in refrigeration effect is felt at Expansion valve.

408. The refrigeration effect in a dry evaporator compared to flooded evaporator in a similar plant is

(a) Same
(b) More
(c) Less
(d) More or less depending on ambient conditions

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : The advantage of flooded evaporator is that the whole surface of the evaporator coil is in contact with liquid so refrigeration effect per unit input is high.
It is expensive but COP of this refrigerator is high.
The refrigeration effect in a dry evaporator compared to flooded evaporator in a similar plant is less.

409. Match List-I with List-II and select the correct answer :

	List-I		List-II
A.	Dichlorodifluoro-methane	1.	R 718
B.	Water	2.	R 22
C.	Methyl chloride	3.	R 40
D.	Monochlorodifluoro-methane	4.	R 12

	A	B	C	D		A	B	C	D
(a)	4	1	2	3	(b)	1	4	3	2
(c)	1	4	2	3	(d)	4	1	3	2

Nagaland PSC CTSE 2015 Paper-II
CSE Pre-1999

Ans. (d) :

Dichlorodifluoro-methane	–	R 12
Water	–	R 718
Methyl chloride	–	R 40
Monochlorodifluoro-methane	–	R 22

410. The refrigerant R - 744 is

(a) ammonia (b) Freon-12
(c) carbon dioxide (d) sulfur dioxide

(KPSC AE 2015)

Ans : (c) for Inorganic refrigerants
R- 700 + Atomic weight)
R - 700 + (CO_2 Atomic weight)
R - 744

411. Pipe material for the refrigerator with ammonia solution as working fluid should not be:

(a) Mild steel (b) Copper
(c) Aluminum (d) Brass

OPSC AEE 2015 PAPER - II

Ans : (b) Pipe material for the refrigerator with ammonia solution as working fluid should not be copper. An Ammonia refrigerating system the tubes of a shell and tube condenser are made of steel. Every vapour-compression refrigeration system or unit ever built will have at least one each of these four components:
* Compressor * Condenser
* Expansion * Evaporator

412. An evaporator pressure regulator performs the following function

(a) Monitors evaporator temperature
(b) Maintains compressor inlet pressure
(c) Maintains expansion valve
(d) Keeps pressure up for system needs

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : An evaporator pressure regulator Keeps pressure up for system needs

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413. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Sulphur candle test	1. Propane
B. Halide torch test	2. Ammonia
C. Soap and water test	3. Halocarbon refrigerants
D. Ammonia swab test	4. Sulphurdioxide

	A	B	C	D		A	B	C	D
(a)	2	3	1	4	(b)	4	1	3	2
(c)	2	1	3	4	(d)	4	3	1	2

Nagaland PSC CTSE 2015 Paper-II, CSE Pre-2004

Ans. (a) :

Sulphur candle test	–	Ammonia
Halide torch test	–	Halocarbon refrigerants
Soap and water test	–	Propane
Ammonia swab test	–	Sulphurdioxide

414. In split system of air - conditioner, the condensing unit is located:

(a) Inside the area to be conditioned
(b) Outside the area to be conditioned
(c) Near the refrigerator
(d) None of the above

OPSC AEE 2015 PAPER - II

Ans : (b) In split system of air- conditioner, the condensing unit is located outside the area to be conditioned.

415. The ideal location for a filter-drier in a refrigeration cycle is the point where it will remove moisture before the moisture enters the

(a) Condenser
(b) Compressor
(c) Cold side of system
(d) Warm side of system

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : The ideal location for a filter-drier in a refrigeration cycle is the point where it will remove moisture before the moisture enters the cold side of the system.

416. Use of flash chamber in Vapour Compression Refrigeration (VCR) is best justified with :

(a) Increased refrigeration effect
(b) Reduced compression work
(c) Reduced size of evaporator
(d) Increased (COP)

KPSC ADF 2015

Ans. (c) : Use of flash chamber in vapour compression refrigeration (VCR) is best justified with reduced size of evaporator.

A flash chamber is a device which separates liquid from vapours. Only liquid is passed to evaporator and the vapours will be passed to the compressor directly. This flash chamber between the condenser and evaporator is known as flash inter cooling.

417. Consider the following for use of refrigeration system with liquid sub cooler

1. Prevent the entry of liquid into compressor
2. Prevent flashing of refrigerant liquid ahead of low stage expansion device
3. Promotes flashing of refrigerant liquid ahead of low stage expansion device
4. Reduce work of compression

Of the above, which is correct

(a) 1 (b) 2
(c) 3 (d) 4
(e) 1 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Use of refrigeration system with liquid sub cooler prevent flashing of refrigerant liquid ahead of low stage expansion device.

418. Consider the following for flash tank:

1. Flash gas formed during expansion can be removed at an intermediate pressure
2. Quality of refrigerant at the evaporator inlet can be increased
3. Temperature of refrigerant vapour at the inlet to higher stage compressor can be reduced
4. Pressure drop in evaporator be reduced

Of the above, which is correct

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3
(e) 1, 3 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (e) : 1. **Function of flash tank**- Flash gas formed during expansion can be removed at an intermediate pressure.
2. Temperature of refrigerant vapour at the inlet to higher stage Compressor can be reduced.
3. Pressure drop in evaporator can be used.

419. Which of the following refrigerant has the lowest boiling point?

(a) R-134a (b) R-407c
(c) R-410a (d) R-417a

MPPSC State Forest Service Exam, 2014

Ans. (c) : R-410a is azeotropic mixture of R-32 & R-125, used as refrigerant in air conditioning having low boiling point as compared to all among.

420. Which of the following refrigerant has minimum ozone depletion potential?

(a) R-11 (b) R-12
(c) R-22 (d) R-134a

MPPSC State Forest Service Exam, 2014
ESE 1992

Ans. (d) : R-134a is known as tetrafluoro ethane from the family of HFC refrigerant which is a replacement to CFCs & HCFCs because it have very less or minimum ozone depletion, potential.

421. The Refrigerant R – 718 is

(a) Ammonia (b) Water
(c) Air (d) Carbon dioxide

TNPSC AE 2014

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Ans. (b) : R–718—R-718 is a organic compound.
R – [700 + moleculer weight]
R – [700 + moleculer weight of water (H_2O)]
R – [700 + 1 × 2 + 16]
R – 718 → water

422. Match List I (Containing designation of refrigerants) with List II (containing chemical formula and name of refrigerant) and select the correct answer:

List I		List II	
A	**R-22**	**1**	**CCl_2F_2**
B	**R-11**	**2**	**SO_2**
C	**R-12**	**3**	**CCl_3F**
D	**R-764**	**4**	**CH_2FCl**
		5	**$CHClF_2$**

(a) A-4, B-2, C-3, D-5 (b) A-1, B-3, C-2, D-4
(c) A-5, B-3, C-1, D-2 (d) A-1, B-4, C-5, D-3
(e) A-4, B-5, C-1, D-3

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) :
R – 22 $CHClF_2$
R – 11 CCl_3F
R – 12 CCl_2F_2
R – 764 SO_2

423. Consider the following for Ammonia as a refrigerant
1. It offers excellent performance
2. It is a natural refrigerant
3. It is inexpensive
4. It deplete ozone layer

Of the above, which is correct
(a) 1 and 2 (b) 1 and 3
(c) 1 and 4 (d) 1, 2 and 3
(e) 3 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : Properties and Advantages of Ammonia Refrigerant—
1. **Small piston displacement**- Ammonia has the highest refrigerating effect per pound compared to all the refrigerants being used including the halocarbon. Even though the specific volume of ammonia is high, the compressor displacement required per ton of refrigeration is quite small, due to this small compressor is required per ton of the refrigeration capacity. This saves lots of Power in the long run.
2. Since specific volume of Ammonia is high it is used mostly with rotary and the centrifugal compressor.
3. **Environment friendly**- It is safe to the environment and does not cause and depletion of the ozone layer.
4. **Corrosive nature of Ammonia**-An hydro ammonia is non-corrosive in nature, however, in presence of moisture it tends to become corrosive to copper, brass and other non-ferrous materials.
5. **Miscibility with oil**-Ammonia refrigerant is non-miscible with oil.
6. **Ammonia is Cheap and available readily**-Ammonia is available almost every where and is the cheapest of all the commonly used refrigerants.

424. Which of the following …… has the highest freezing point?
(a) Sulphur dioxide (b) Ammonia
(c) Carbon dioxide (d) R.12

GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : Sulphur dioxide – 75.2 C
Ammonia – 28 C
Carbon dioxide – 109.4C
R– 21.8 C

425. The critical temperature for Freon-12 is
(a) Around - 20º C (b) Above 100º C
(c) 0ºC (d) Around 50º

GPSC Lect. (Poly.) 07.12.2014

Ans. (b) : The critical temperature for Freon-12 is
R – 12 – 111.97ºC

426. Consider the following statements :
Dry compression in reciprocating compressor is preferred because it
1. **prevents valve damage**
2. **enables use of thermostatic expansion valve**
3. **minimize irreversibility in the compressor**
4. **prevents washing out of the lubricating oil from cylinder walls**

Which of these statements are correct?
(a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 3 and 4

Haryana PSC Civil Services Pre, 2014
CSE Pre-1999

Ans. (c)

427. Consider the following statements concerning fixed vane and rotary compressors as used in vapour compression refrigeration system.
1. These compressors are used in small capacity systems (less than 2kW)
2. They require suction value, but do not require discharge value
3. Refrigerant leakage is minimized by hydrodynamic lubrication
4. Compared to reciprocating compressors, the re-expansion losses are high in rotary vane compressor

Of the above, which is correct
(a) 1 and 2 (b) 1 and 3
(c) 1 and 4 (d) 2 and 3
(e) 2 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Fixed vane and rotary compressor as used in vapour Compression refrigeration system because these compressors are used in small capacity system (less than 2 kW) and refrigerant leakage is minimized by hydrodynamic lubrication.

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428. Hermetic compressors are used mainly in smaller systems as they:
(a) Yield higher COP
(b) Do not require frequent servicing
(c) Offer the flexibility of using any refrigerant
(d) Can be used under different load conditions efficiently
(e) result in better refrigeration capacity

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Hermetic compressors are used mainly in smaller system as they do not require frequent servicing.

429. Actual expansion process in a throttling device is:-
(a) Reversible adiabatic expansion
(b) Isenthalpic expansion
(c) Fanno - line expansion
(d) Isothermal expansion

UKPSC AE-2013, Paper-II

Ans. (c) : Actual expansion process in a throttling device is fanno - line expansion,

430. The system used for cooling of supersonic aircraft is:-
(a) Simple system
(b) Bootstrap system
(c) Reduced ambient system
(d) Regenerative system

UKPSC AE-2013, Paper-II

Ans. (c) : The system used for cooling of supersonic aircraft is reduced ambient system.

431. External equalizer in thermostatic expansion valve (TEV) is used so that the TEV senses the pressure of evaporator:-
(a) At the outlet
(b) At the inlet
(c) In the middle
(d) At outlet as well as inlet

UKPSC AE-2013, Paper-II

Ans. (a) : External equalizer in thermostatic expansion valve (TEV) is used so that the TEV senses the pressure of evaporator at the outlet.

432. Refrigeration in aeroplanes usually employs the following refrigerant :
(a) CO_2 (b) Freon-11
(c) Freon-22 (d) Air

J&K PSC Civil Services Pre, 2013

Ans. (d) : Refrigeration in aeroplanes usually employs the following refrigerant air.

433. Moisture should be removed from refrigerants to avoid
I. Compressor seal failure.
II. freezing at the expansion valve.
III. restriction to refrigerant flow.
IV. corrosion of steel parts.
Of these statements
(a) I, II, III and IV are correct
(b) I and II are correct
(c) II, III and IV are correct
(d) I, III and IV are correct

APPSC Poly. Lect. 2013

Ans. (a) : Moisture should be removed from refrigerants to avoid
(i) Compressor seal failure.
(ii) freezing at the expansion valve
(iii) restriction to refrigerant flow
(iv) Corrosion of steel parts

434. In the following refrigeration x represents any digit. Which of these is a hydrocarbon refrigerant?
(a) R-xx 1 (b) R-xx0
(c) R-7xx (d) R-x2

RPSC AEN Pre-2013

Ans. (b)
- Hydrocarbon refrigerant – (Saturated refrigerant)
  R – xx0
- Unsaturated (Hydrocarbon) refrigerant
  R – xxx0
- If the refrigerant is inorganic (Compound) refrigerant
  R – 700 + Mol. weight

435. The working fluid in refrigeration cycle is
(a) Refrigerator (b) Refrigerant
(c) Absorbent (d) Lubricant

TNPSC AE 2013

Ans. (b) : The working fluid in refrigeration cycle is known as refrigerant.

436. The boiling point of carbon dioxide is
(a) –20.5 ºC (b) – 50 ºC
(c) – 73.6 ºC (d) – 78.3 ºC

JPSC AE 2013 Paper-V

Ans. (c) : – 73.6 ºC

437. Normal boiling point of different refrigerants are given below. Which pairs are correct?
(1) NH3 – 40.8°C (2) R-12 – 29.8°C
(3) R-22 – 33.3°C (4) R-134a – 26.2°C
(a) (1) and (2) are correct
(b) (3) and (4) are correct
(c) (2) and (4) are correct
(d) (1) and (4) are correct

UKPSC AE-2013, Paper-II

Ans. (c) :

438. A regenerative liquid – vapour heat exchanger is used in a vapour compression system operating on ammonia refrigerant, the COP:-
(a) Will increase
(b) Will decrease
(c) Can increase or decrease
(d) Remains unaffected

UKPSC AE-2013, Paper-II

Ans. (b) :

439. The refrigerant C_2H_4 is designated as:-
(a) R 24 (b) R 15
(c) R 1150 (d) R 150

UKPSC AE-2013, Paper-II

Ans. (c) : $C_mH_nF_pCl_q$ m = 2, n = 4, p = 0, q = 0
n + p + q = 2m
R1(m – 1) (n + 1)p
R1150

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440. Which of the refrigerant is more toxic?
(a) Carbon dioxide (b) Ammonia
(c) Freon-12 (d) Freon-22
UKPSC AE-2013, Paper-II

Ans. (b) : Ammonia as a refrigerant is more toxic.

441. Which of the following refrigerant is more costly?
(a) Ammonia (b) Carbon dioxide
(c) Methyl chloride (d) Freon-12
UKPSC AE-2013, Paper-II

Ans. (d) :

442. Pick up the correct statement:-
(a) The refrigerant should have high thermal conductivity and low freezing temperature
(b) The refrigerant should have low heat transfer coefficient and high latent heat
(c) The refrigerant should have high specific volume and high latent heat
(d) The refrigerant should have high specific volume and low latent heat
UKPSC AE-2013, Paper-II

Ans. (a) :

443. The most commonly used refrigerant in vapour compression system is :
(a) Ammonia (b) R-12
(c) CO_2 (d) Ethylene
TNPSC AE (Industries) 09.06.2013

Ans. (b) :

444. Finned evaporators are used on air conditioning application to
(a) equalize air flow over the cooling coil surface
(b) prevent moisture carry over
(c) extend the effective area of the cooling surface
(d) increase the dehumidifying capacity
UKPSC AE 2012 Paper–II

Ans. (c) : Finned evaporators are used on air conditioning application to extend the effective area of the cooling surface.

445. What is the storage temperature for milk ?
(a) 4 °C (b) 7 °C
(c) 2 °C (d) 0.5 °C
UKPSC AE 2012 Paper–II

Ans. (d) : 0.5 °C

446. Due to suction vapour superheating in vapour compression cycle the COP increases in case of the following refrigerant :
(a) R 22 (b) NH_3
(c) R – 12 (d) None of the above
UKPSC AE 2012 Paper–II

Ans. (c) : For R 12 & R 134 COP↑
and for Freon, R22 and NH_3 COP↓

447. Which one of the following is not a desirable property of a good refrigerant ?
(a) low specific heat
(b) high specific volume of vapour
(c) large latent heat at evaporator pressure
(d) high critical temperature
UKPSC AE 2012 Paper–II

Ans. (b) : high specific volume of vapour

448. Which refrigerant would you choose for 800 TR air conditioning plant using centrifugal compressor ?
(a) NH_3 (b) CO_2
(c) CFC 11 (d) CFC 114
UKPSC AE 2012 Paper–II

Ans. (c) : CFC 11

449. The brine is an aqueous solution of ___ in water.
(a) Calcium chloride (b) Sodium chloride
(c) Calcium carbonate (d) Sodium carbonate
UKPSC AE 2012 Paper–II

Ans. (b) : Sodium chloride

450. Go through the following statements and choose the correct alternative :
1. Wet compression increases COP of ammonia vapour compression system.
2. Use of liquid-vapour heat exchanger in vapour compression system decreases COP in case of ammonia refrigerant.
3. For good performance, a refrigerant must have high critical pressure and low critical temperature.
4. Refrigerants that are not miscible with oils, presents many problems.
5. In flooded evaporators, the liquid refrigerant covers the entire heat transfer surface.

(a) Statements 1, 2 and 5 are true
(b) Statements 1, 2 and 3 are true
(c) Statements 3, 4 and 5 are true
(d) Statements 2, 4 and 5 are true
UKPSC AE 2012 Paper–II

Ans. (a) : Statements 1, 2 and 5 are true

451. The function of solenoid valve in a refrigeration system is to
(a) control the flow of refrigerant in suction line
(b) control the flow of refrigerant through expansion valve
(c) stop the flow of refrigerant when there is no load on the evaporator
(d) stop the flow of refrigerant in liquid line when compressor stops
UKPSC AE 2012 Paper–II

Ans. (d) : stop the flow of refrigerant in liquid line when compressor stops

452. An evaporator pressure regulator performs the following function
(a) Monitors evaporator temp
(b) Maintains compressor inlet pressure
(c) Keeps pressure up for system needs
(d) All of these
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : An evaporator pressure regulator, monitors evaporator temperature by changing throttling pressure, maintains compressor inlet pressure, due to change in throttling pressure & keeps pressure up for system needs and constant surface temperature at evaporator, by throttling in suction line.

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453. When a R–22 refrigeration system is charged with R–12, the system will :
(a) Not Function
(b) Operate at low temperature
(c) Have considerable fluctuations in the operating pressures
(d) Have its refrigeration capacity reduce to about one-third

OPSC Civil Services Pre. 2011

Ans. (d) : When a R–22 refrigeration system is charged with R–12, the system will have its refrigeration capacity reduce to about one-third.

454. The refrigeration system of a passenger aircraft works on reversed :
(a) Brayton cycle (b) Atkinson cycle
(c) Carnot cycle (d) Otto cycle

J&K PSC Civil Services Pre, 2010

Ans. (a) : The refrigeration system of a passenger aircraft works on reversed brayton cycle.

$$(COP) = \frac{1}{\left[\left(r_p\right)^{\frac{\gamma-1}{\gamma}} - 1\right]}$$

Where, r_p = pressure ratio

$$r_p = \frac{P_2}{P_1}$$

455. Assertion (A) : Sometimes particular combination of refrigerant and piping will result in a resonant frequency which may amplify the sound and vibration to an undesirable level.
Reason (R) : Muffler is used to solve the above problem.
Codes :
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (b) : We know that sometimes particular combination of refrigerant & piping will result in a resonant frequency which may amplify the sound & vibration to an undesirable level for so for solving these problem we use muffler which decrease the resonating frequency.

456. In a domestic refrigerator, periodical defrosting is necessary to
(a) Protect the coil from damage
(b) Avoid wastage of power
(c) Reduce evaporator pressure
(d) Reduce condenser pressure

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know, in a domestic refrigerator, the periodical defrosting is necessary to reduce the wastage of power. In this, we remove the ice that is deposited on the evaporator tube. Since ice is bad conductor of heat.

457. Which is usually costliest item in a refrigeration system ?
(a) condenser (b) capillary tube
(c) compressor (d) evaporator

J&K PSC Civil Services Pre, 2010

Ans. (c) : Compressor is the costliest item in a refrigerating system.

458. At the back of domestic refrigerator, the bank of tubes are :
(a) Evaporator tubes
(b) Condenser tubes
(c) Refrigeration cooling tubes
(d) Capillary tubes

J&K PSC Civil Services Pre, 2010

Ans. (b) : The bank of tubes at the back of domestic refrigerator is condenser tubes

459. The number of hydrogen atoms in the refrigerant R-12 is :
(a) 4 (b) 2
(c) 1 (d) 0

DRDO Scientists 2009

Ans. (d) : Chemical formula of F-12 is CCl_2F_2
So, there is no hydrogen atom.

460. With an increase in suction pressure, the volumetric efficiency of a reciprocating refrigeration compressor
(a) increases
(b) decreases
(c) remains constant
(d) may increase or decrease depending upon the type of refrigerant

UKPSC AE 2007 Paper -II

Ans. (a) : $\eta = 1 + C - C\left(\frac{P_{condenser}}{P_{evaporator}}\right)^{1/n}$

$P_{evaporator}$ = Suction pressure of compressor
$P_{evaporator}\uparrow$, $\eta_v\uparrow$

461. Dry ice is produced by expanding
(a) Liquid ammonia
(b) Liquid Freon-22
(c) Liquid carbon dioxide
(d) Liquid Freon-12

UKPSC AE 2007 Paper -II

Ans. (c) : Liquid carbon dioxide

462. Which of the refrigerant is least used these days?
(a) Freon-12 (b) Sulphur dioxide
(c) Carbon dioxide (d) Ammonia

UKPSC AE 2007 Paper -II

Ans. (c) : Carbon dioxide

463. Among the refrigerants namely R - 11, R - 12, R - 22 and R - 717, the ratio of specific heats is highest for
(a) R - 12 (b) R - 22
(c) R -11 (d) R - 717

UKPSC AE 2007 Paper -II

Ans. (d) : R - 717

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464. The following refrigerant is not miscible with oil.
(a) R - 717 (b) R - 11
(c) R - 22 (d) R - 12
UKPSC AE 2007 Paper -II

Ans. (a) : R - 717

465. It is desirable for a good refrigerant to have
(a) High critical temperature
(b) Low critical temperature
(c) High freezing point
(d) High melting point
OPSC Civil Services Pre 2006

Ans. (a) : Properties of a good refrigerants
(i) Low boiling point.
(ii) High critical temperature.
(iii) High latent heat of vaporisation.
(iv) Low specific heat of liquid.
(v) Low specific volume of vapour.
(vi) Easy to liquefy at moderate pressure and temperature.

466. Which of the following is NOT an ozone friendly refrigerant?
(a) Freon-12 (b) Ammonia
(c) Air (d) Carbon dioxide
OPSC Civil Services Pre 2006

Ans. (a) : CFC-R12 (Freon-12) was widely used in refrigerators, inhalers and aerosol spray cans.
CFCs were not widely used in air-conditioning system, but are worth talking about here to understand their effect on the environment.
When CFCs, which are made chlorine, fluorine and carbon, break down over time due to exposure to UV radiation, they release chlorine atoms which react with the ozone molecule to destroy it.

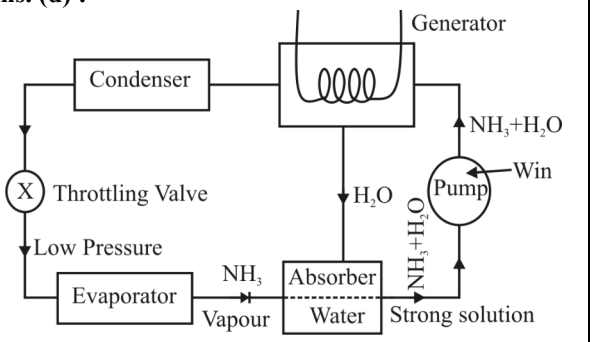
4. Vapour Absorption Refrigeration System

467. In an Electrolux system:
(a) ammonia is absorbent and water is refrigerant
(b) lithium bromide is refrigerant and water is absorbent
(c) water is refrigerant and lithium bromide is absorbent
(d) ammonia is refrigerant and water is absorbent
NLCIL GET 17.11.2020, Shift-II, Vizag Steel MT 2017
HPPSC W.S. Poly. 2016, KPSC AE 2015
ISRO Scientist/Engineer (RAC) 29.11.2015
APPSC IOF 2009, ESE 2005

Ans. (d) : The electrolux refrigeration system is also called three fluid absorption system. The three fluids used are ammonia, hydrogen and water. The main purpose of this system is to make the machine noiseless by eliminating moving parts (pump).
Water is used as absorbent because it has ability to absorbs ammonia.

468. In vapour absorption refrigeration, heat is rejected in:
(a) condenser only
(b) generator only
(c) absorber only
(d) condenser and absorber
(e) generator and absorber
CGPSC AE 25.02.2018
CGPSC ARTO & TSI 07.01.2017
BPSC AE Mains 2017 Paper-V
Nagaland PSC CTSE 2017, 2016 Paper-II
Haryana PSC Civil Services Pre, 2014

Ans. (d) :



→ In vapour absorption system compressor is replaced by absorber, pump and generator.
→ In VARS heat is rejected in condenser and absorber.

469. Waste heat can be effectively used in which one of the following refrigeration systems?
(a) vapour compression refrigeration cycle
(b) air refrigeration cycle
(c) vapour absorption refrigeration cycle
(d) vortex refrigeration cycle
VIZAG STEEL MT 2017
Nagaland PSC CTSE 2017, 2016 Paper-II
UJVNL AE 2016
CGPSC Poly. Lect. 22.05.2016
Haryana PSC Civil Services Pre, 2014
UPRUVNL AE 2014

Ans. (c) : VARS system is based on Low grade energy (heat) or solar based energy.
VARS = Vapour Absorption Refrigeration System.

470. The refrigerant used for absorption refrigerators working on heat from solar collectors is a mixture of water and which of the following substance?
(a) Carbon dioxide (b) Sulphur dioxide
(c) Lithium bromide (d) Freon 12
Nagaland PSC (CTSE) 2018, Paper-II
UJVNL AE 2016, UPRVUNL AE 2014
UKPSC AE 2007 Paper -II, ESE 1996

Ans : (c) Lithium Bromide use as absorber in absorption refrigerators system and H_2O is use as refrigerant.
An absorption refrigerator is a refrigerator that uses a heat source to provide the energy needed to drive the cooling process.

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471. Lithium bromide in vapour absorption refrigeration system is used as:

(a) refrigerant (b) cooling substance
(c) auxiliary refrigerant (d) absorbent

ISRO Scientist/ Engineer (RAC) 07.05.2017
UKPSC AE 2012 Paper-II
J&K PSC Civil Services Pre. 2010
J & K PSC Screening, 2006

Ans. (d) : LiBr (Lithium Bromide) in vapour Absorption System is used as absorbent and water (H_2O) is used as refrigerant.

472. In aqua-ammonia and lithium bromide water absorption refrigeration systems, the refrigerants are respectively:

(a) Ammonia and water
(b) Water and water
(c) Water and lithium bromide
(d) Ammonia and lithium bromide

OPSC AEE 2019 PAPER - II
Nagaland PSC (CTSE) 2018, Paper-II
APPSC AEE 2016
UKPSC AE 2013 Paper-II

Ans. : (a) : In aqua ammonia and lithium bromide water absorption refrigeration system the refrigerants are respectively Ammonia and water.

473. In a lithium bromide absorption refrigeration system

(a) lithium bromide is used as a refrigerant and water as an absorbent
(b) water is used as a refrigerant and lithium bromide as an absorbent
(c) ammonia is used as a refrigerant and lithium bromide as an absorbent
(d) none of the above

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 18.09.2017
GPSC ARTO 01.05.2016

Ans. (b) : Solar refrigeration systems are based on vapour absorption refrigeration system. Two most commonly refrigerant used in vapour absorption systems are the mixture of
(i) Ammonia and water
(ii) Water and Lithium

- In a Lithium bromide absorption refrigeration system water is used as a refrigerant and lithium bromide as on absorbent.

474. The most suitable pair of refrigerant and absorbent combination for solar refrigeration is

(a) Ammonia and water
(b) Ammonia and sodium thiocyanide
(c) Water and lithium bromide
(d) R22 and dimethyl formamide

Gujarat PSC AE 2019
GPSC EE PRE.28.01.2017, ESE 2012

Ans : (c) : Solar heat input is used in LiBr - H_2O (Lithium bromide and water) refrigeration system.

475. In a vapour absorption refrigerator, the temperature of evaporator and ambient air are 10 °C and 30 °C, respectively. For obtaining COP of 2 for this system, the temperature of the generator is to be nearly

(a) 90 °C (b) 85 °C
(c) 80 °C (d) 75 °C

UPPSC AE 13.12.2020, Paper-2
ESE 2018

Ans. (c) : Given,

$T_e = 10 + 273 = 283\ K$

$T_o = 30 + 273 = 303\ K$

$COP = 2$

$(COP)_{VARS} = \eta_E\ (COP)_R$

$$= \frac{T_G - T_o}{T_G} \times \frac{T_e}{T_o - T_e} = \frac{T_e}{T_G}\left(\frac{T_G - T_o}{T_o - T_e}\right)$$

$$2 = \frac{283(T_G - 303)}{T_G(303 - 283)}$$

$T_G = 352.87\ K$

$T_G = 79.87°C \simeq 80°C$

476. In the absorption refrigeration cycle, the compressor of vapour compression refrigeration cycle is replaced by

(a) Liquid pump
(b) Generator
(c) Absorber and generator
(d) Absorber, liquid pump and generator

UKPSC AE 2012, 2007 Paper–II

Ans. (d) : Absorber, liquid pump and generator

477. A vapour absorption refrigeration system

(a) gives noisy operation
(b) gives quiet operation
(c) requires little power consumption
(d) cools below 0ºC

Nagaland PSC (CTSE) 2018, Paper-II
UKPSC AE 2012 Paper–II

Ans. (b) : A vapour absorption refrigeration system gives quiet operation. The vapour compression refrigeration system requires the expenditure of high-grade energy in the form of shaft work to drive the compressor with the concommitant disadvantage of vibration and noise, the absorption refrigeration system requires only low grade energy in the form of heat to drive it and it is relatively silent in operation and subject little wear.

478. A vapour absorption refrigerator uses ..... as a refrigerant

(a) water (b) ammonia
(c) freon (d) aqua-ammonia

Nagaland PSC (CTSE) 2018, Paper-II
UKPSC AE 2007 Paper -II

Ans. (d) : A vapour absorption refrigerator uses aqua-ammonia as a refrigerant. In the vapour absorption system the refrigerant used is ammonia, water or lithium bromide. The refrigerant gets condensed in the condenser and it gets evaporated in the evaporator. The refrigerant produces cooling effect in the evaporator and releases the heat to the atmosphere via the condenser.

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Vapour absorption refrigeration system may have two types–
(1) NH_3-Water
(2) LiBr-Water
In first type NH_3 (Ammonia) is used as refrigerant & in second type water is used as refrigerant. Hence in VAS water and ammonia (aqua-ammonia) is used as refrigerant.

479. In Electrolux refrigerator
(a) ammonia is absorbed in hydrogen
(b) ammonia is absorbed in water
(c) ammonia is evaporates in hydrogen
(d) hydrogen evaporates in ammonia

Nagaland PSC (CTSE) 2018, Paper-II
ESE 2005

Ans. (c) : An electrolux refrigerator works on three fluid system. There is no circulation pump. The total pressure is the same throughout the system. The third fluid remains mainly in the evaporator. Which reduces the partial pressure of refrigerant to enable it to evaporate at low pressure and hence low temperature.
Liquid NH_3 evaporates in the evaporator in the presence of H_2. Hydrogen is chosen because it is non-corrosive and insoluble in water. Thus a complete leak proof silent system is achieved.

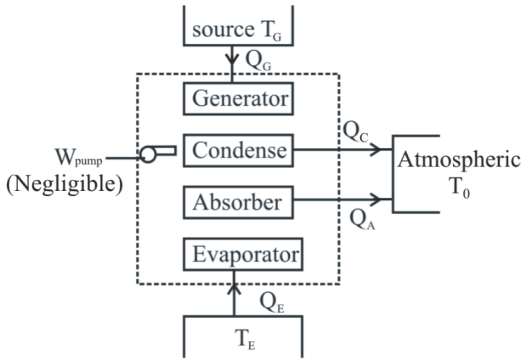
480. The C.O.P. of an absorption type refrigerator is given by (where T_1 = Temperature at which the working substance receives heat, T_2 = Temperature of cooling water, and T_3 = Evaporator temperature)

(a) $\dfrac{T_1(T_2-T_3)}{T_3(T_1-T_2)}$ (b) $\dfrac{T_3(T_1-T_2)}{T_1(T_2-T_3)}$

(c) $\dfrac{T_1(T_1-T_2)}{T_3(T_2-T_3)}$ (d) $\dfrac{T_3(T_2-T_3)}{T_1(T_1-T_2)}$

Sikkim PSC (Under Secretary), 2017
ESE 2012, 2003

Ans. (b) :



I[st] law of thermodynamic–

$Q_E + Q_G = Q_C + Q_A$ .....(i)

$\dfrac{Q_E}{T_E}+\dfrac{Q_G}{T_G}=\dfrac{Q_C}{T_C}+\dfrac{Q_A}{T_0}$ .....(ii) [Clausius inequality]

$\dfrac{Q_E}{T_E}+\dfrac{Q_G}{T_G}=\dfrac{Q_C+Q_A}{T_0}\Rightarrow\dfrac{Q_E+Q_G}{T_0}=\dfrac{Q_E}{T_0}+\dfrac{Q_G}{T_0}$

$\dfrac{Q_E}{T_E}-\dfrac{Q_E}{T_0}=\dfrac{Q_G}{T_0}-\dfrac{Q_G}{T_G}$

$Q_E\left[\dfrac{1}{T_E}-\dfrac{1}{T_0}\right]=Q_G\left[\dfrac{1}{T_0}-\dfrac{1}{T_G}\right]$

$Q_E\left[\dfrac{T_0-T_E}{T_E\cdot T_0}\right]=Q_G\left[\dfrac{T_G-T_0}{T_G\cdot T_0}\right]$

$(COP)=\dfrac{Q_E}{Q_G}=\dfrac{T_G-T_0}{T_G}\times\dfrac{T_E}{T_0-T_E}$

$\boxed{(COP)_{VARS}=\dfrac{Q_E}{Q_G}=\dfrac{T_E}{T_G}\left(\dfrac{T_G-T_0}{T_0-T_E}\right)}$

$(COP)_{VARS}=\dfrac{T_3(T_1-T_2)}{T_1(T_2-T_3)}$

481. In ammonia-water and lithium bromide-water absorption refrigeration systems the absorbents are, respectively.
(a) ammonia and water
(b) water and water
(c) ammonia and lithium bromide
(d) water and lithium bromide

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Ammonia –water absorption system-
Refrigerant → Ammonia (NH_3)
Absorbent → Water (H_2O)
Lithium bromide water absorption system-
Refrigerant → Water
Absorbent → Lithium bromide.

482. In vapour absorption refrigeration system, the refrigeration temperature is 7 °C, generator temperature 127 °C temperature of heat sink 47 °C. The maximum possible COP of the system
(a) 0.714 (b) 0.86
(c) 1.4 (d) 0.11

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : Given, $T_E = 7°C = 7°C + 273 = 280K$
$T_G = 127°C = 400K$
$T_O = 47°C = 320K$

$(COP)_{max}=\dfrac{T_G-T_O}{T_G}\times\dfrac{T_E}{T_O-T_E}$

$=\dfrac{400-320}{400}\times\dfrac{280}{320-280}$

$(COP)_{max}=1.4$

483. In an absorption type refrigeration system, heating in generator, refrigeration in evaporator and cooling by cooling water in condenser, take place at 95°C, –5°C and 30°C respectively. What is the maximum COP of the system?
(a) 1.17 (b) 1.35
(c) 1.52 (d) 1.78

ESE 18.07.2021

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Ans. (b)

Temperature of evaporator (T_E) = –5°C = 268 K

Temperature of generator (T_G) = 95°C = 368 K

Temperature of surrounding (T_o) = 30°C = 303 K

$$\text{Maximum COP} = \frac{T_E(T_G - T_O)}{T_G(T_O - T_E)}$$

$$= \frac{268\times(368-303)}{368\times(303-268)} = 1.352$$

484. The operating temperatures of a single stage Vapour Absorption Refrigeration System (VARS) are : Generator 90 °C, condenser and absorber 40°C, evaporator 0°C. If the pump work is negligible, find the ideal COP of this VARS

(a) 0.94 (b) 1.4

(c) 0.84 (d) 2.4

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) :

$$COP|_{VARS} = \frac{T_E}{T_g}\left(\frac{T_g - T_o}{T_o - T_E}\right)$$

$T_g = 90°C = 363$ K

$T_o = 40°C = 313$ K

$T_E = 0°C = 273$ K

$$COP|_{VARS} = \frac{363-313}{363}\times\frac{273}{313-273}$$

$$COP|_{VARS} = 0.94$$

485. A solar thermal operated vapour absorption system is capable of

(a) Continuous operation

(b) Both continuous operation and intermittent operation

(c) No operation

(d) Intermittent operation

JPSC AE PRE 2019

Ans. (d) : A solar thermal operated vapour absorption system is capable of intermittent operation. Solar refrigeration systems are based on vapour absorption refrigeration system.

486. In an Electrolux refrigerator, a thermo-siphon bubble pump is used to lift the

(a) weak aqua solution from the generator to the separator

(b) weak aqua solution from the separator to the absorber

(c) strong aqua solution from the generator to the separator

(d) strong aqua solution from the generator to the evaporator

ESE 2019

Ans. (a) : A thermo-siphon bubble pump is used to lift weak aqua solution from the generator to the separator and then to the absorber.

487. In vapor absorption refrigeration system, heat input, rejection and refrigeration occurs at 400 K, 300 K and 270 K respectively. Ideal COP of the system will be:

(a) 1.25 (b) 2.25

(c) 3.25 (d) 4.25

(e) 4.75

CGPSC AE 25.02.2018

Ans. (b) : For vapor absorption refrigeration system-

$T_G \rightarrow \eta_{H.E.}\left[1-\frac{T_o}{T_G}\right]$

T_o

$T_{ref} \rightarrow [COP]_{ref.} = \frac{T_{ref.}}{(T_o - T_{Ref.})}$

Then Ideal COP of VARS.

$$[COP]_{VARS} = \left[1-\frac{T_o}{T_G}\right]\times\left[\frac{T_{Ref}}{T_o - T_{Ref.}}\right]$$

$$= \frac{T_{Ref.}}{T_G}\left[\frac{T_G - T_o}{T_o - T_{Ref.}}\right]$$

Where, $T_{Ref.} = 270$ K

$T_o = 300$ K

$T_G = 400$ K

$$[COP]_{VARS} = \frac{270}{400}\left[\frac{400-300}{300-270}\right] = \frac{270}{400}\times\frac{100}{30}$$

$$\boxed{[COP]_{VARS} = 2.25}$$

488. A vapor absorption refrigeration system with COP of 0.85 and refrigeration capacity of 1 kW. The heat input at generator (Q_g) and summation of heat rejected at absorber & condenser (Q_a + C) is given as:

(a) $Q_g = 1.25$ kW; $Q_a + c = 1.25$ kW

(b) $Q_g = 1.25$ kW; $Q_a + c = 1$ kW

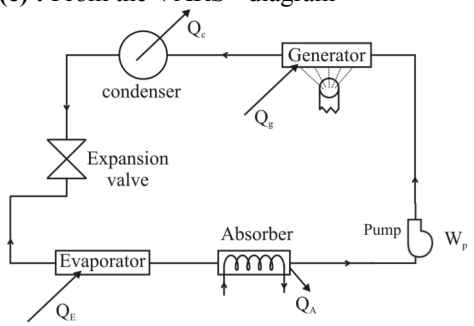
(c) $Q_g = 0.8$ kW; $Q_a + c = 1.8$ kW

(d) $Q_g = 0.8$ kW; $Q_a + c = 1$ kW

(e) $Q_g = 1.25$ kW; $Q_a + c = 2.25$ kW

CGPSC AE 25.02.2018

Ans. (e) : From the VARS - diagram



Data given-

R.E. = 1 kW

$[COP]_{VARS} = 0.8$

If, we assume W_P [pump work] equal to zero then

$$[COP]_{VARS} = \frac{R.C.}{Q_g + W_p} = \frac{R.C.}{Q_g} \qquad W_p = 0$$

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$$0.8 = \frac{1}{Q_g} \Rightarrow Q_g = 1.25\text{kW}$$

We know that,

$$Q_E + Q_g = Q_A + Q_C$$

$$Q_A + Q_C = 1.25 + 1 = 2.25\text{kW}$$

489. Consider the following statements regarding vapour absorption systems in the field of refrigeration:

1. In ammonia-water absorption system, ammonia is the refrigerant.
2. In water-lithium bromide system, water is the refrigerant.
3. Ammonia-water absorption reaction is endothermic.
4. The amount of ammonia absorbed by water is inversely proportional to the temperature of ammonia.

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1, 3 and 4
(c) 1, 2 and 4 (d) 2, 3 and 4

ESE 2018

Ans. (c) : Ammonia water absorption is exothermic hence it's wrong statement.

490. COP of simple vapour absorption cycle is around

(a) 0.6 (b) 4
(c) 2 (d) 0.1

Vadodara Muncipal Corp. DEE, 2018

Ans. (a) : COP of vapour absorption cycle is around 0.6.

491. Maximum possible COP of a solar absorption refrigeration system with generator temperature of 360 K, absorber temperature of 300 K. condenser temperature of 300 K and evaporator temperature of 270 K is :

(a) 9 (b) 6
(c) 3 (d) 1.5

GPSC ARTO Pre 30.12.2018

Ans. (d) : Given,

$$T_G = 360\text{K}, \quad T_E = 270\text{K}, \quad T_0 = 300\text{K}$$

COP of Vapour Absorption Refrigeration system,

$$\text{COP} = \frac{T_G - T_O}{T_G} \times \frac{T_E}{T_O - T_E}$$

$$\text{COP} = \frac{360 - 300}{360} \times \frac{270}{300 - 270}$$

$$\text{COP} = \frac{270 \times 60}{360 \times 30}$$

COP = 1.5

492. Aqua Ammonia is used as refrigerant in the following type of refrigerant system

(a) Absorption (b) Compression
(c) Direct (d) Indirect

TSPSC AEE 2017

Ans. (a) : Aqua Ammonia is used as refrigerant in the absorption type of refrigeration system.

493. Which of the following statement is/are correct?

(a) In a vapour absorption refrigerator, the compression of refrigerant is avoided.
(b) Sub-cooling can be achieved by circulating more quantity of cooing water through the condenser.
(c) In a vapour compression refrigerator, the vapour is drawn in the compressor cylinder during its suction stroke and is compressed adiabatically during the compression stroke.
(d) All of the above

Karnataka PSC AE, 10.09.2017

Ans. (d) : In VARS absorption of refrigerator, the compression of refrigerant is avoided.

- Sub-cooling can be achieved by circulating more quantity of cooling water through the condenser.
- Vapour is compressed adiabatically during the compression stroke.

494. Consider the following statements in respect of a vapour – absorption refrigeration cycle:

1. The absorption refrigeration cycle is generally used when waste heat is available from an existing source or when free energy like solar energy is to be used.
2. There are no moving parts in the absorption refrigeration plant except a small liquid pump.
3. The value of the coefficient of performance is nearly the same in both vapour – absorption and vapour – compression refrigeration plants.

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

ESE 2017

Ans. (b) :

- An absorption refrigerator is a refrigerator that uses a heat source (e.g. solar energy, a fossil-fueled flame, waste heat from factories) to provide the energy needed to drive the cooling process.
- The absorption cycle use a liquid pump, not a compressor to create the pressure rise between evaporator and condenser. Pumping a liquid is much easier and cheaper than compressing a gas, so the system takes less work input.
- The COP of absorption system is about 20 to 30% of the vapour compression system.

495. What is the critical temperature and critical pressure of ammonia respectively?

(a) 132.4°C and 111.3 atm
(b) –132.4°C and 11.13 atm
(c) 132.4°C and 11.3 atm
(d) none of these

KPCL AE 2016

Ans. (a) : For ammonia, the critical temperature and critical pressure is 132.4°C and 111.3 atm.

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496. During the liquefaction of gases, if the gas is at a sufficiently low temperature and high pressure, the throttling process would bring the gas into the saturated-mixture region, where the vapour and liquid could be separated using:

(a) Linde liquefier (b) Hamstead liquefier
(c) Thomson liquefier (d) Slinger liquefier

HPPSC Asstt. Prof. 29.10.2016

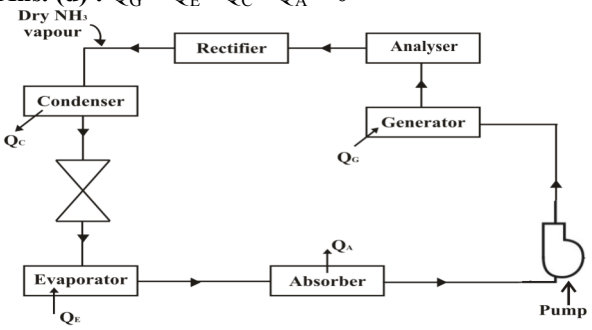
Ans. (a) : Linde liquefier.

497. Neglecting the pump work, Energy equation for vapour absorption refrigeration system may be written as (where Q = heat transfer rate into the system)

(a) $Q_{generator} + Q_{evaporator} + Q_{condenser} + Q_{absorber}$
(b) $Q_{generator} + Q_{evaporator} + Q_{condenser} - Q_{absorber}$
(c) $Q_{generator} + Q_{evaporator} - Q_{condenser} + Q_{absorber}$
(d) $Q_{generator} + Q_{evaporator} - Q_{condenser} - Q_{absorber}$

RPSC Lecturer 16.01.2016

Ans. (d) : $Q_G + Q_E - Q_C - Q_A = 0$

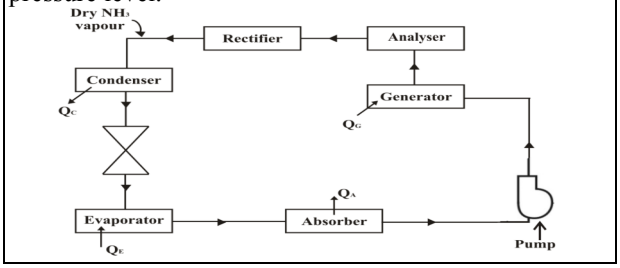


498. Which of the following components of vapour absorption refrigeration system has same pressure level (neglecting the pipe loss)

(a) Generator and absorber
(b) Evaporator and absorber
(c) Generator and evaporator
(d) Condenser and evaporator

RPSC Lecturer 16.01.2016

Ans. (b) : Evaporator and absorber components of vapour absorption refrigeration system has same pressure level.



499. Ammonia absorption refrigeration cycle requires

(a) Very little work input
(b) Same work input as for vapour compression cycle
(c) Zero work input
(d) Maximum work input

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (a) : Ammonia absorption refrigeration cycle requires very little work input.

500. An Electrolux refrigerator is called as

(a) Two-fluid absorption system
(b) No fluid absorption system
(c) Three-fluid absorption system
(d) Single-fluid absorption system

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : **Electrolux Refrigerator–**It is based on vapour absorption system. It is a three fluid VARS system. i.e. water (H_2O), ammonia (NH_3) and hydrogen (H_2).
Water (H_2O) acts as absorbent, ammonia acts as refrigerant and hydrogen is used to create low partial pressure for NH_3 refrigerant.
Note-Flow takes place under the action of gravity so no pump is used.

501. C.O.P. of a vapour absorption system can be increased by using

(a) Vapour compression
(b) Heat exchanger
(c) Both vapour compression and Heat exchanger
(d) None of these is correct

MPPSC AE 08.11.2015

Ans. (b) : C.O.P. of a vapour absorption system can be increased by using heat exchanger.

502. Consider the following for an ideal refrigerant-absorbent mixture:

1. There is neither expansion nor contraction upon mixing
2. The mixing process is exothermic
3. The mixing process is endothermic
4. Obeys Raoult's law in liquid phase and Dalton's law in vapour phase of the above,

Which is correct

(a) 1 (b) 2
(c) 3 (d) 1 and 3
(e) 1 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (e) : For an ideal refrigerant- absorbent mixture there is neither expansion nor contraction upon mixing and obeys Raoult's law in liquid phase and Dalton's law in vapour phase.

503. In an ammonia-water system, a rectification column is used mainly to:

(a) To improve the COP of the system
(b) To reduce the operating pressures
(c) To minimize the concentration of water in refrigeration circuit
(d) To maximize the concentration of water in refrigeration circuit
(e) To minimize the space of the system

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : In an ammonia- water system, a rectification column is used mainly to minimize the concentration water in refrigeration circuit.

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504. An Electrolux refrigerator uses

(a) one pump (b) two pump
(c) no pump (d) three pump

Haryana PSC Civil Services Pre, 2014

Ans. (c) : Electrolux refrigerator system has three fluids (H_2, H_2O & NH_3), in which pump is not used.

505. Which of the following has the lowest COP?

(a) Vapour absorption cycle
(b) Vapour compression cycle dry compression
(c) Vapour compression cycle with superheated vapour at the end of compression
(d) Vapour compression cycle with sub cooling

TNPSC AE 2014

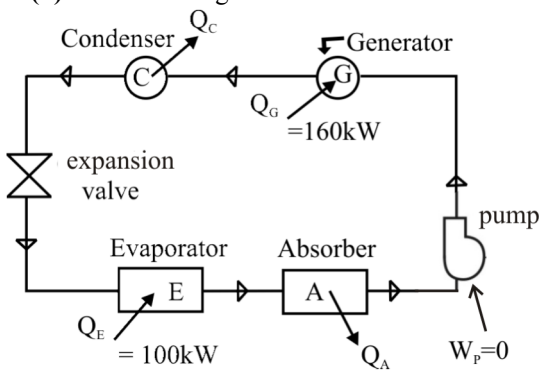
Ans. (a) : Vapour absorption cycle has the lowest COP.

506. The operating temperatures of a single stage vapour absorption refrigeration system are: generator: 90^0C; condenser and absorber: 40^0C; evaporator: 0^0C. The system has a refrigeration capacity of 100 kW and the heat input to the system is 160 kW. The solution pump work is negligible. Find the COP of the system and the total heat rejection rate from the system.

(a) 0.525, 160kW (b) 0.625, 260kW
(c) 1.6, 100kW (d) 1.6, 160kW
(e) 0.625, 160kW

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : VARS – diagram



From this fig –

$$[COP]_{VARS} = \frac{\text{Ref.Effect}}{Q_G + W_P} = \frac{100}{160 + 0} = 0.625$$

from energy Conservation

$$Q_C + Q_A = Q_G + Q_E$$
$$= 160 + 100$$
$$\boxed{Q_C + Q_A = 260\,kW}$$

507. Match List-I (Basic components of Aqua-ammonia refrigeration system) with List-II (Function of the components in the system) and select the correct answer using the codes given the lists:

List I		List II	
A	Generator	1	Dehydration
B	Analyser	2	Removal of vapour from strong aqua-ammonia solution
C	Rectifier	3	Producing dry ammonia vapour by removing traces of water particles completely
D	Receiver	4	Storage of high pressure liquid ammonia
		5	Formation of liquid ammonia from high pressure vapours

(a) A-3, B-1, C-2, D-5 (b) A-5, B-3, C-4, D-5
(c) A-1, B-3, C-2, D-5 (d) A-2, B-1, C-3, D-4
(e) A-2, B-4, C-1, D-3

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) :

Components	Function
1. **Generator**	Removal of vapour from strong aqua- ammonia solution
2. **Analyser**	Dehydration
3. **Rectifier**	Producing dry ammonia vapour by removing traces of water particles completely.
4. **Receiver**	Storage of high pressure liquid ammonia.

508. Consider the following for absorption refrigeration system as compared to compression system

1. Higher COPs
2. Lower refrigeration temperatures
3. Possibility of using low-grade energy sources
4. Possibility of using in vehicles

(a) 1 (b) 2
(c) 3 (d) 4
(e) 1 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : Vapour Absorption Refrigeration system (VARS) belong to class of vapour cycles similar to vapour Compression refrigeration systems. Since these systems run on low-grade thermal energy, they are preferred when low grade energy such as waste heat or solar energy is available.

509. Consider the following for water-lithium bromide systems:

1. Crystallization of solution is likely to occur in absorber
2. Crystallization of solution is likely to occur in solution heat exchanger
3. Crystallization is likely to occur when generator temperature falls
4. Crystallization is likely to occur when condenser pressure falls

Of the above, which is correct

(a) 1 (b) 2
(c) 4 (d) 1 and 2
(e) 1 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (e) : In water – lithium bromide systems crystallization of solution is likely to occur in absorber and crystallization is likely to occur when condenser pressure falls.

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510. Absorption of energy in solar collectors can be increased to a maximum value by coating the absorber surface by:-

(a) Black paint (b) Lamp black
(c) Rough black coating (d) Selective coating

UKPSC AE-2013, Paper-II

Ans. (d) : Absorption of energy in solar collectors can be increased to a maximum value by coating the absorber surface by selective coating.

511. On increasing the temperature of absorber plate of solar collector the efficiency of collector:-

(a) Increases
(b) Decreases
(c) May increase or decrease depending upon atmospheric temperature
(d) Does not depend on absorber plate temperature

UKPSC AE-2013, Paper-II

Ans. (b) : On increasing the temperature of absorber plate of solar collector the efficiency of collector decreases.

512. Solar photovoltaic cell is made up of semiconductor materials. The efficiency of conversion of solar energy into electrical energy is of the order of:-

(a) 7% (b) 21%
(c) 70% (d) 90%

UKPSC AE-2013, Paper-II

Ans. (b) :

513. In vapour absorption system, the compressor is the vapour compressor cycle is replaced by :

(a) Condensor
(b) Absorber – Generator Assambly
(c) Superheater
(d) Evaporator

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

514. What is the function of rectifier in a practical vapour absorption refrigeration system?

(a) To convert AC supply to DC supply.
(b) To condense the weak ammonia solution.
(c) To remove the water vapours from the ammonia coming out from the analyser.
(d) to control the pressure of vapours entering the condenser.

RPSC AEN Pre-2013

Ans. (c) Analyser– Dehydration
Rectifier– To remove water vapour from the ammonia coming out from analyser.
Generator– Removal of vapour from strong aqua ammonium solution.
In vapour absorption refrigeration system to remove the water vapour from the ammonia coming out from analyser. The compressor is replaced by absorber, pump and generator. Absorbent Refrigerant (ammonia – water),
Ammonia – refrigerant
Water – absorber

515. Ammonia is used in Aqua - ammonia system because it is

(a) Vigorously absorbed in water (b) Low cost
(c) Toxic (d) Low ODP

TNPSC AE 2013

Ans. (a) : Ammonia is used in Aqua - ammonia system because it is vigorously absorbed in water.
Ammonia - water vapor
Absorption Refrigeration system—In this system ammonia is used as the refrigerant and water is used as the absorbent. The ammonia - water absorption system is used in the domestic as well as the commercial application where the requirement of the temperature is above 320F.

516. In a vapour absorption refrigeration system, the refrigeration temperature is – 15 °C and the generator temperature is 110 °C. If sink temperature is 55 °C, the maximum COP of the system will be

(a) 1.00 (b) 3.69
(c) 0.34 (d) 0.90

UKPSC AE 2012 Paper–II

Ans. (*) : $T_G = 110°C = 110 + 273 = 383\ K$
$T_O = 55°C = 55 + 273 = 328\ K$
$T_R = -15°C = -15 + 273 = 258\ K$

$$[COP]_{max.} = \left[1 - \frac{T_o}{T_G}\right]\left[\frac{T_R}{T_o - T_R}\right]$$

$$= \left[1 - \frac{328}{383}\right]\left[\frac{258}{328 - 258}\right] = 0.52$$

517. In a lithium bromide absorption refrigeration system, the refrigerant is :

(a) Lithium bromide (b) Water
(c) Ammonia (d) None of these

OPSC Civil Services Pre. 2011

Ans. (b) : In a Lithium bromide absorption refrigeration system the refrigeration is water and lithium bromide is absorbent.

518. In an ammonia-hydrogen refrigeration system the hydrogen

(a) helps evaporation of NH_3
(b) acts as refrigerant
(c) helps NH_3 to flow through the circuit
(d) burns to supply heat

TNPSC AE, 2008

Ans. (a) : In an ammonia-hydrogen refrigeration system, the hydrogen helps evaporation of NH_3.

519. In aqua-ammonia absorption refrigeration system, the refrigerant is

(a) Water (b) aqua-ammonia
(c) Ammonia (d) air

OPSC Civil Services Pre 2006

Ans. (c) : In aqua- ammonia (NH_3-water) absorption system, Ammonia act as refrigerant and water as absorbent. Ammonia has highest affinity to absorb in water.

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5. Psychometric Chart and its Applications

520. The wet bulb depression is zero, when relative humidity is equal to

(a) Zero (b) 50%
(c) 75% (d) 100%

RPSC IOF, 2020, OPSC AEE 2019 Paper-II
ISRO Scientist / Engineer (RAC) 07.05.2017
TSPC AE 2017, GPSC ARTO 01.05.2016
UKPSC AE 2012 Paper-II
OPSC Civil Service Pre-2011
UKPSC AE 2007 Paper-II, ESE 2006

Ans. (d) : Wet bulb depression is the difference between the dry bulb temperature and wet bulb temperature. Hence, when DBT = WBT implies that relative humidity = 100%
Therefore when the wet bulb depression is zero the relative humidity is equal to 100%.

521. Dew point temperature is the temperature at which condensation begins when air is cooled at constant

(a) volume (b) entropy
(c) pressure (d) enthalpy

UJVNL AE 12.09.2021, Gujarat PSC AE 2019
RPSC AE 2018, TSPSC AEE 2017
Nagalanad PSC CTSE 2017, 2016 Paper-II
UPPSC AE 12.04.2016 Paper-II
Haryana PSC Civil Services Pre, 2014
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
PSPCL AE 2012, UKPSC AE 2012 Paper-II
GATE 2006

Ans. (c) Dew point temperature is the temperature at which condensation begins when air is cooled at constant pressure.

522. The sensible heat factor is the ratio of :

(a) Sensible heat and latent heat
(b) latent heat and sensible heat
(c) sensible heat and total heat
(d) latent heat and total heat

APPSC Poly Lect. 13.03.2020
Haryana PSC AE (PHED) 04.09.2020 Paper-I
VIZAG MT 14.12.2020
ISRO Scientist/Engineer (RAC), 10.03.2019
TANGEDCO AE 2015
UKPSC AE 2013 Paper-II
TNPSC AE, 2008

Ans. (c) : The sensible heat factor (SHF) is the ratio of the sensible heat and total heat.

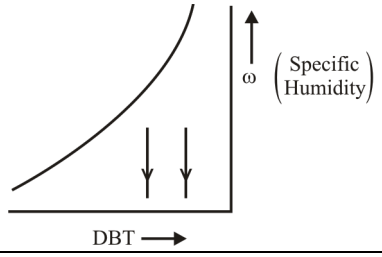
$$SHF = \frac{\text{sensible heat}}{\text{total heat}\left(\text{sensible heat + latent heat}\right)}$$

523. The vertical and uniformly spaced lines on a psychrometric chart indicate :

(a) wet-bulb temperature
(b) dew-point temperature
(c) specific humidity
(d) dry –bulb temperature

GPSC DEE, Class-2 (GWSSB) 04.07.2021
CGPSC AE 15.01.2021
Nagaland PSC (CTSE) 2018, Paper-II
CGPSC AE 16.10.2016
MPPSC State Forest Service Exam 2014

Ans. (d) : The vertical and uniformly spaced lines on a psychrometric chart indicate DBT
(Dry Bulb Temperature)

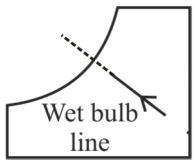


524. During adiabatic cooling of moist air

(a) DBT remains constant
(b) Specific humidity remains constant
(c) Relative humidity remains constant
(d) WBT remains constant
(e) None of the above

UPPSC AE 13.12.2020
GPSC Executive Engineer 23.12.2018
CGPSC Poly. Lect. 22.05.2016
UPRUVNL AE 2014, HPPSC AE 2014
APPSC AEE 2012

Ans. (d) : During adiabatic cooling enthalpy remains constant and in the psychrometric chart the WBT lines and the enthalpy lines have the same slope therefore the WBT will also remain constant during the adiabatic cooling process.



525. In Psychrometric chart, relative humidity lines are:

(a) curved
(b) horizontal
(c) straight inclined from left to right
(d) vertical

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TSPSC Manager (Engg.) HMWSSB 12.11.2020
ISRO Scientist/Engineer (RAC) 22.04.2018
KPSC AE 2015
TRB Poly. Lect. 2012
UKPSC AE 2012, 2007 Paper–II

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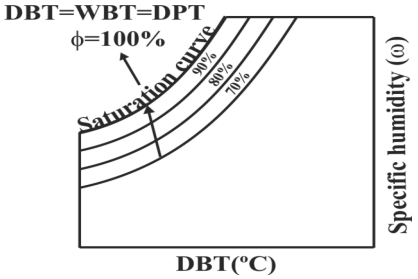
Ans. (a) : In Psychrometric chart, relative humidity lines are curve.



526. During an adiabatic saturation process of an unsaturated air, the parameters which remains constant is

(a) Dry bulb temperature
(b) Dew point temperature
(c) Thermodynamic wet bulb temperature
(d) Relative humidity

PPSC Asstt. Municipal Engineer 15.06.2021
UPPSC AE 13.12.2020, Paper-2
RPSC IOF 2020, RPSC AE 2018
KPSC AE 2015

Ans. (c) : During an adiabatic saturation process of an unsaturated air, the thermodynamic wet bulb temperature remains constant.

- Dry bulb temperature, t↓
- Specific humidity ratio, ω↑
- Dew point temperature (DPT↑)
- Relative humidity, (ϕ)↑
- h (constant)
- Specific volume, v↓
- WBT (Constant)

527. The difference between dry bulb temperature and wet bulb temperature, is called :

(a) dry bulb depression
(b) wet bulb depression
(c) dew point depression
(d) degree of saturation

OPSC AE 2019 Paper-II
Punjab PSC SDE 12.02.2017
GPSC LECT 23.10.2016
TNPSC AE 2013, UKPSC AE 2007 Paper-II

Ans. (b) : Wet bulb depression = DBT – WBT

- WBT is an indirect measure of the dryness of the moist air.

528. The minimum temperature to which air can be cooled under ideal conditions in evaporative cooling is

(a) Dew point temperature of the inlet air
(b) Wet bulb temperature of the inlet air
(c) Dry bulb temperature of the inlet air
(d) All the above

OPSC AEE 2019 Paper-II
Nagaland PSC CTSE 2015 Paper-II
KPSC ADF 2015
OPSC Civil Services Pre 2006
ESE 1995

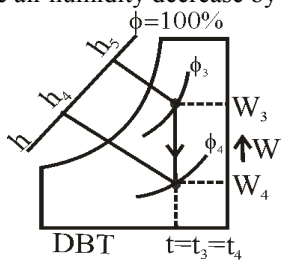
Ans. (b) : Evaporative cooling is used in air washer. In which air can cooled theoretically upto wet bulb temperature of entering air. Similarly in cooling tower water can be cooled upto wet bulb temperature of entering air.

529. During dehumidification process ..... remains constant

(a) Wet bulb temperature
(b) Relative humidity
(c) Dry bulb temperature
(d) Specific humidity

Nagaland PSC (CTSE) 2018, Paper-II
Vizag Steel (MT) 2017
TSPSC AEE 2017, KPSC AE 2015
Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : When the state of air is altered along DBT = Constant line such as 3-4 in Fig. moisture in the form of water vapour is transferred to change the specific humidity of the air humidity decrease by w_3 to w_4.



530. When the outside air is introduced for ventilation purpose there is :

(a) Sensible heat gain
(b) Latent heat gain
(c) Sensible and latent heat gain
(d) No heat gain

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper-II, CGPSC AE 16.10.2016
UKPSC AE 2007 Paper-II

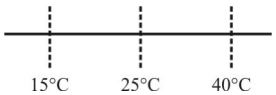
Ans. (c) : Sensible heat and latent heat both gain when outside air is introduced for ventilation purpose. Sensible heat gain due to temperature difference between fresh air and air in space. Latent heat gain due to humidity.

531. The atmospheric air at dry bulb temperature of 15 °C enters a heating coil maintained at 40°C. Air leaves the heating coil at 25 °C. The bypass factor of heating coil is

(a) 0.376 (b) 0.60
(c) 0.40 (d) 0.67

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper-II, HPPSC AE 2014
ISRO Scientist / Engineer (RAC) 22.4.2018

Ans. (b) :



$$\text{By pass factor (BPF)} = \frac{40°C - 25°C}{40°C - 15°C}$$

$$(BPF) = 0.60$$

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532. The expression $\frac{0.622p_v}{p_1 - p_v}$ is used to determine

(a) relative humidity (b) specific humidity
(c) degree of saturation (d) partial pressure

CGPSC AE 2014 Paper-II, APPSC Poly. Lect. 2013
CSE Pre-2001, ESE 1998

Ans. (b) : The expression $\frac{0.622p_v}{p_1 - p_v}$ is used to determine specific humidity (ω).

533. The most commonly used method for the design of duct size is the

(a) velocity reduction method
(b) equal friction method
(c) static regain method
(d) dual or double method

UJVNL AE 2016, UPRUVNL AE 2014
APPSC Poly. Lect. 2013, ESE 1996

Ans. (b) : Equal friction method- where a constant head loss is taken and duct sizes are taken considering this constant loss throughout the ducting. Most commonly used method because of easier calculation.

534. For air with a relative humidity of 80%

(a) The dry bulb temperature is less than wet bulb temperature
(b) The dew point temperature is less than wet bulb temperature
(c) The dew point temperature and wet bulb temperature are equal
(d) The dry bulb and dew point temperature are equal

ISRO Scientist/Engineer (RAC) 07.05.2017
UPPSC AE 12.04.2016 Paper-II
TSPSC Manager 2015, GATE 2003

Ans. (b) : RH (ϕ) = 80%
So, air is unsaturated
DBT > WBT > DPT

535. The pressure, dry bulb temperature and relative humidity of air in a room are 1 bar, 30°C and 70% respectively. If the saturated steam pressure at 30°C is 4.25 kPa, the specific humidity of the room air in kg water vapour/kg dry air is:

(a) 0.0083 (b) 0.0101
(c) 0.0191 (d) 0.0242

GPSC Asst. Director Of Transport 05.03.2017
VIZAG Steel MT 2017
UP Jal Nigam AE 2016, GATE 2013

Ans. (c) : Relative humidity (ϕ) = 0.7
Saturated steam pressure (P_{v_s}) = 4.25 kPa
Total pressure P_t = 1 bar = 100 kPa
∴ Specific humidity,

$$\omega = 0.622 \times \frac{P_v}{P_t - P_v} \quad ...(i)$$

Again from relative humidity,

$$\phi = \frac{P_v}{P_{v_S}}$$

$P_v = 0.74 \ P_{v_s} = 2.975$ kPa
From equation (i),

$$\omega = 0.622 \times \frac{2.975}{100 - 2.975} = 0.091$$

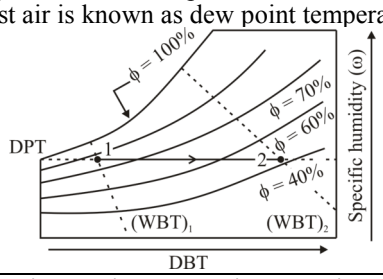
ω = 0.0191 kgw.v/kg.d.a.

536. Saturation temperature of the water vapour in the moist air is known as :

(a) Dry bulb temperature
(b) Dew point temperature
(c) Wet bulb temperature
(d) Critical temperature
(e) Adiabatic saturation temperature

CGPSC AE 25.02.2018
APPSC AEE Screening Test 2016
RPSC Insp. of Factories And Boiler 2016

Ans. (b) : Saturation temperature of the water vapour in the moist air is known as dew point temperature.

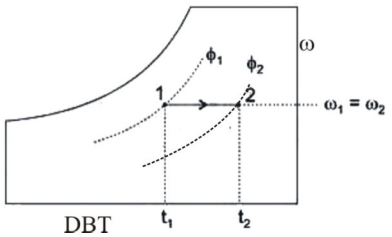


537. During a simple sensible heating process, the relative humidity _____ .

(a) Increases (b) Decreases
(c) Is zero (d) Remains constant

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Nagaland PSC (CTSE) 2018, Paper-II
CIL MT 26.03.2017, UPPSC AE 12.04.2016 Paper-II

Ans. (b) : Sensible heating- It is the process of increasing the Dry bulb temperature at constant specific humidity.



Effects of sensible heating-

i) Dry bulb temperature increases
ii) Sp. Humidity or humidity ratio remains constant
iii) Dew point temperature is constant.
iv) Relative humidity decreases
v) Enthalpy increases
vi) Specific volume increases.
vii) Wet bulb temperature increases.

538. If we have 100% RH, then DPT, WBT and DBT can be related to each other as:

(a) DBT < DPT < WBT
(b) DBT = DPT = WBT
(c) DPT > WBT > DBT
(d) None of the above

OPSC AEE 2019 PAPER - II
Nagaland PSC CTSE 2017 Paper-II
OPSC AEE 2015 PAPER - II
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

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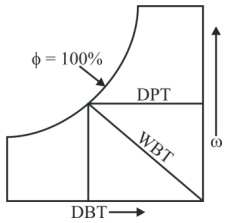
Ans : (b) When 100% Relative humidity then saturation condition



(i) At saturation conditions
DBT = WBT = DPT
(ii) At unsaturated conditions
DBT > WBT > DPT.
DBT = Dry bulb temperature
WBT = Wet bulb temperature
DPT = Dew point temperature.

539. In a saturated air–water mixture the :
(a) dry bulb temperature is higher than the wet bulb temperature
(b) dew point temperature is lower than the wet bulb temperature
(c) dry bulb temperature, wet bulb temperature and dew point temperature are the same
(d) dry bulb temperature is higher than the dew point temperature

TANGEDCO AE 2015, (HPPSC AE 2014)
OPSC Civil Service Pre-2011, ESE 1993

Ans. : (c) (i) At saturated condition
DBT = WBT = DPT
WBD = 0
(ii) At unsaturated condition
DBT > WBT > DPT
DBT = Dry bulb temperature
WBT = wet bulb temperature
DPT = Dew point temperature
WBD = wet bulb Depression
(Saturated ϕ = 100%)

540. Which of the following processes is generally used in winter air conditioning?
(a) dehumidification
(b) humidification
(c) cooling and dehumidification
(d) heating and humidification

TSPSC Manager Engg. HMWSSB 12.112020
VIZAG Steel MT 2017
Kerala PSC AE 06.08.2015, KPSC AE 2015

Ans. (d) : Heating and humidification process is used in winter air conditioning

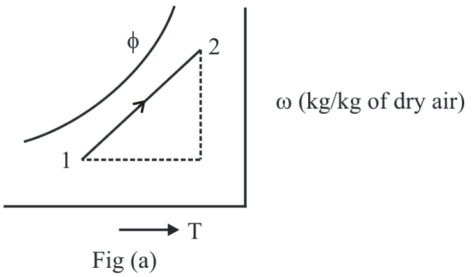


Fig (a)

Fig (a) shows the heating & humidification process.

541. The comfort conditions in air conditioning are at
(a) 0ºC DBT and 0% RH
(b) 20ºC DBT and 60% RH
(c) 30ºC DBT and 80% RH
(d) 40ºC DBT and 90% RH

MPPSC AE 08.11.2015, KPSC AE 2015
J&K PSC Civil Services Pre, 2013
ESE 2012

Ans. (b) : The comfort conditions in air conditioning are at 20ºC DBT and 60% RH.
DBT → Dry bulb temperature
RH → Relative humidity

542. The latent heat load in a auditorium is 25% of sensible heat load. The value of sensible heat factor is :
(a) 0.3 (b) 0.5
(c) 0.8 (d) 1.0

GPSC ARTO Pre 30.12.2018
BPSC AE MAINS 2017 Paper-V
ESE 2016, CGPSC Poly Lect 22.05.2016

Ans. (c) : Given, LH = 25% of SH

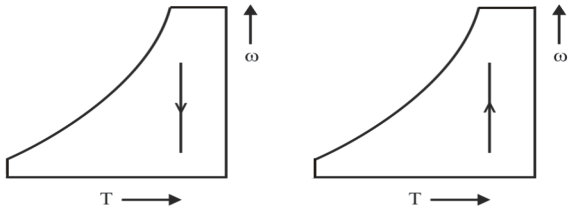
$$SHF = \frac{SH}{LH + SH} = \frac{SH}{0.25SH + SH}$$

$$SHF = \frac{1}{1.25} = 0.8$$

543. In psychrometric charts, what does a vertical downward line represent ?
(a) Sensible cooling process
(b) Adiabatic saturation process
(c) Humidification process
(d) Dehumidification process

GPSC ARTO Pre 30.12.2018
BPSC AE MAINS 2017 Paper-V
UKPSC AE 2012 Paper-II, ESE 2009

Ans. (d) : Dehumidification process



Vertical downward line represents dehumidification process and vertical upward line represents humidification process.

544. The relative humidity is defined as :
(a) the mass of water present in $1m^3$ of dry air
(b) the mass of water present in 1 kg of dry air.
(c) the ratio of the actual mass of water vapour in a unit mass of dry air to the mass of water vapour in the same mass of dry air when it is saturated at the same temperature and pressure.

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(d) the ratio of the actual mass of water vapour in a given volume of moist air to the mass of water vapour in the same volume of saturated air at the same temperature and pressure.

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Nagaland PSC (CTSE) 2018, Paper-II
Punjab PSC SDE 12.02.2017
APGENCO AE 2012

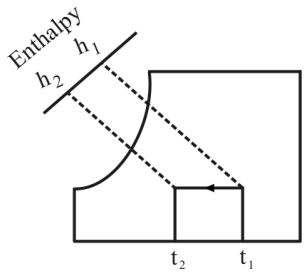
Ans. (d) : The relative humidity is defined as the ratio of the actual mass of water vapour in a given volume of moist air to the mass of water vapour in the same volume of saturated air at the same temperature and pressure.

545. During sensible cooling, wet bulb temperature-

(a) Decreases
(b) Increases
(c) Remains constant
(d) Can increase or decrease

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GMB AAE 25.06.2017
J&K PSC Civil Services Pre. 2010
ESE 1998

Ans. (a)



Dew point temperature remains contant during sensible cooling and wet bulb temperature decreases.

546. Dew point is

(a) the temperature at which condensation of steam in saturated air will start
(b) the lowest temperature attainable for a mixture of air and steam
(c) dependent on pressure of air and volume of air
(d) highest temperature in an air conditioning cycle

GPSC Lect. 23.10.2016
TRB Asstt. Prof., 2012, TNPSC AE 2008
UKPSC AE 2007 Paper-II

Ans. (a) : Dew Point- The dew point is the temperature at which air is saturated with water vapour, which is the gaseous state of water.

When air has reached the dew point temperature at particular pressure, the water vapour in the air is in equilibrium with liquid water meaning water vapour is condensing at the same rate at which liquid water is evaporating.

547. The effective temperature is a measure of the combined effects of

(a) Dry bulb temperature and relative humidity
(b) Dry bulb temperature and air motion
(c) Wet bulb temperature and air motion
(d) Dry bulb temperature, relative humidity and air motion

GPSC ARTO 30.12.2018, TSPSC AEE 2017
CGPSC ARTO & TSI 07.01.2017
ESE 1998

Ans. (d) : A measure of Comfort that involves the combined effects of the dry bulb temperature, relative humidity and air movements.

548. In a psychrometric process, the sensible heat added is 30 kJ/sec and latent heat added is 20 kJ/sec. The sensible heat factor for the process is

(a) 0.3 (b) 0.6
(c) 0.67 (d) 1.5

RPSC IOF, 2020, JWM 2017
ISRO Scientist/Engg. (RAC) 07.05.2017

Ans. (b) : Sensible heat factor is the ratio of sensible heat to the total heat.

So, $$SHF = \frac{30}{30+20}$$

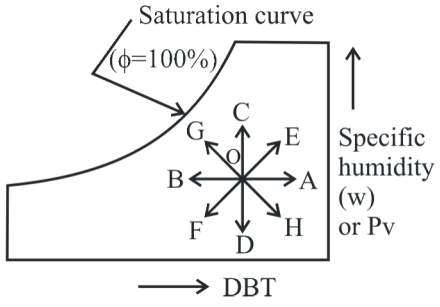
$$SHF = 0.6$$

549. Which of the following parameters-remains constant during sensible cooling or heating process?

(a) Dry-bulb temperature
(b) Wet-bulb temperature
(c) Humidity ratio
(d) Relative humidity

BPSC AE MAINS 2017 Paper-V
Assam PSC CCE Pre 2015
RPSC AEN Pre-2013

Ans. (c)



Process in diagram		Type of Air Conditioning
OA	–	Sensible Heating
OB	–	Sensible Cooling
OC	–	Humidification
OD	–	Dehumidification
OE	–	Heating & Humidification
OF	–	Cooling & Dehumidification

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OG – Cooling & Humidification
OH – Heating & Dehumidification

Specific Humidity or Humidity Ratio (ω)

$$\omega = \frac{\text{Mass Vapour}}{\text{Mass of dry air}} = \frac{m_v}{m_a}$$

$$\omega = 0.622\frac{P_v}{P_a} = 0.622\frac{P_v}{P_t - P_v}$$

$P_t = P_v + P_a$

P_v = Partial Pressure of Vapour

P_a = Partial Pressure of Air

P_t = Total Pressure (Atmospheric)

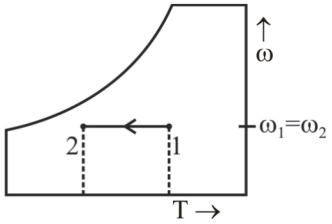
550. In sensible cooling process, specific humidity of the system

(a) Increases (b) Decreases

(c) Remains constant (d) None of these

MPPSC State Forest Service Exam, 2014
ESE 2012
UKPSC AE 2007 Paper-II

Ans. (c) :



1-2 → Sensible cooling $T_1 > T_2$

1-2 → Specific humidity is remain constant ($\omega_1 = \omega_2$)

551. The bypass factor of cooling coil

(a) Decrease with increase in velocity of air passing through it

(b) Increase with increase in velocity of air passing through it.

(c) Remains unchanged with increase in velocity of air passing through it

(d) May increase or decrease with increase in velocity of air passing through it depending upon the condition of air entering

ISRO Scientist/Engineer (RAC), 10.03.2019
MPPSC AE 2014, PSPCL AE 2012

Ans. (b) : Bypass factor of cooling coil increases with increase in velocity of air passing through it.

552. The ratio of partial pressure of water vapour (p_v) to the saturation pressure of water vapour (p_s) at same temperature is

(a) relative humidity

(b) degree of saturation

(c) specific humidity

(d) absolute humidity

UPPSC AE 13.12.2020, Paper-2
Oil India Senior Officer 23.12.2020
TNPSC 2019, 2013

Ans. (a) : Relative Humidity– The ratio of partial pressure of water vapour (P_v) to the saturation pressure of water vapour (P_{vs}) at same temperature

$$\boxed{\phi = \frac{P_v}{P_{vs}}}$$

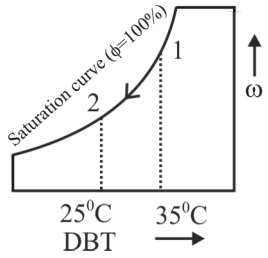
553. Moist air at 35^0C and 100% relative humidity is entering a psychrometric device and leaving at 25^0C and 100% relative humidity.
The name of the device is

(a) Humidifier (b) Dehumidifier

(c) Sensible heater (d) Sensible cooler

UPPSC AE 13.12.2020, Paper-2
GUJARAT PSC AE 2019
GATE 2014

Ans. (b) : From point (1) 35^0C to point (2) 25^0C at 100% relative humidity, dehumidification takes place. So, most appropriate option should be option (b)



554. On psychrometric chart wet bulb temperature lines are :

(a) horizontal

(b) vertical

(c) straight inclined sloping downward to the right

(d) curved

VIZAG MT 14.12.2020, ESE 2007
J & K PSC Screening, 2006

Ans. (c) : Wet Bulb Temperature– Straight Inclined and sloping downwards towards right

Dry Bulb Temperature – Vertical

Specific humidity – Horizontal

Relative humidity – Curved

555. The air conditioning process during summer is _____.

(a) cooling and humidification

(b) heating and dehumidification

(c) cooling and dehumidification

(d) heating and humidification

VIZAG Steel MT 24.01.2021
Kerala PSC Asstt. Prof. 27.09.2016
TNPSC AE (Industries) 09.06.2013

Ans. (c) : The air conditioning process during summer is cooling and dehumidification and in winter is heating and humidification.

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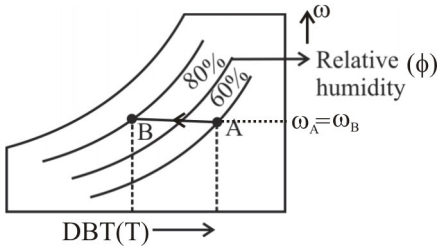
556. The relative humidity during sensible cooling process

(a) increases (b) decreases

(c) remains same (d) unpredictable

Kerala PSC IOF 19.04.2016
RPSC Lect. (Tech. Edu. Dept.) 2011
UKPSC AE 2007 Paper-II

Ans. (a) : From, Psychrometric chart,
The relative humidity during sensible cooling process increases.
A-B → sensible cooling.



557. The ratio of the actual mass of water vapour in a unit mass of dry air to the mass of water vapour in the same mass of dry air when it is saturated at the same temperature and pressure, is called:

(a) humidity ratio
(b) relative humidity
(c) absolute humidity
(d) degree of saturation

Nagaland PSC (CTSE) 2018, Paper-II
Punjab PSC SDE 12.02.2017
Kerala PSC IOF 19.04.2016
ESE 2004

Ans. (d) : Degree of Saturation – It is the mass of water vapour in a sample of air to the mass of water vapour in the same air when it is saturated at the same temperature

• The capacity of air to absorb moisture is called degree of saturation

$$\mu = \frac{w}{w_S}\bigg|_T = \frac{P_V}{P_S}\left[\frac{1-P_S/P}{1-P_V/P}\right]$$

w = Specific humidity at vapour pressure P_V ' w_S = Specific humidity at saturation pressure P_S.

Relative Humidity (R_H or ϕ)- It is the ratio of mass of water vapour in a given volume of air at any temperature and pressure to the maximum amount of mass of water vapour which is the same volume of air can hold at the same temperature conditions. The air contains maximum amount of water vapour at the saturations conditions.

$$R_H/\phi = \frac{\text{mass of water vapour in a fixed volume of moist air at temp. T}}{\text{mass of water vapour in a saturated sample of moist air at same temp. and vol.}}$$

$$= \frac{m_v}{m_s}\bigg|_T = \frac{P_v V/RT}{P_s V/RT} = \frac{P_v}{P_s} = \frac{V_s}{V_v}$$

$$\mu = \phi\left[\frac{1-P_s/P}{1-P_v/P}\right]$$

Humidity ratio (Specific Humidity)– It is the ratio of mass of water vapour to the mass of dry air contained in the sample air. It is denoted '*w*'. It is normally expressed in g/kg of dry air.

$$w = \frac{\text{mass of water vapour in air}}{\text{mass of dry air in air}}$$

Specific humidity / humidity ratio

$$w = \frac{m_w}{m_a} = \frac{V_a}{V_w} = 0.622\frac{P_v}{P-P_v}$$

P_v = Partial pressure of vapour
P = Total pressure of moist air.

Absolute humidity– It is the mass of water vapour present in one cubic metre of dry air. It is expressed in terms of gram per cubic metre of dry air (g/m^3 of dry air) many a time it is expressed in terms of gram per m^3 of dry air one kg of water vapour is equal to 15,430 grams.

$$P_V V = m_V R_V T$$

where, P_v = Vapour pressure in air saturation pressure at dew point.
V = Volume of air, which is also of water vapour.
T = Dry bulb temp.
R_V = Gas constant of water vapour (462 kJ/kg-K)
m_V = mass of water vapour in kg.

$$\text{Vapour Density} = \frac{m_V}{V} = \frac{P_v}{R_v T}$$

558. The equivalent diameter (D) of a circular duct corresponding to a rectangular duct having longer side 'a' and shorter side 'b' for the same velocity and pressure drop is given by

(a) $D = \frac{a+b}{ab}$ (b) $D = \frac{ab}{a+b}$

(c) $D = \frac{2ab}{a+b}$ (d) $D = \frac{a+b}{2ab}$

Nagaland PSC (CTSE) 2016, Paper-II
Rajasthan AE (Nagar Nigam) 2016 Shift-3
ESE 2004

Ans. (c) : Equivalent diameter (D) of a circular duct corresponding to a rectangular duct having longer side 'a' and shorter side 'b' for the same velocity and pressure drop is given by

$$D = \frac{4A}{P} = \frac{4ab}{2(a+b)} = \frac{2ab}{a+b}$$

559. It is desired to condition the outside air form 70% RH and 45°C DBT to 50% RH and 25° DBT at room condition. The practical arrangement would be

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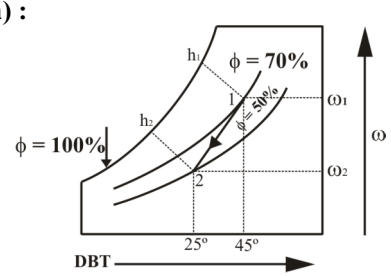
(a) Cooling and dehumidification
(b) Cooling and humidification
(c) Heating and dehumidification
(d) heating and humidification

TNPSC AE 2014, APPSC IOF 2009
CSE Pre-1994

Ans. (a) :

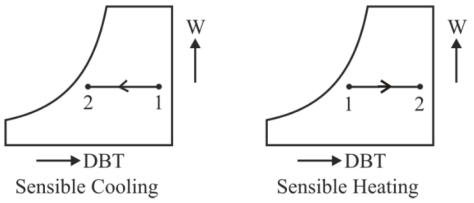


$\omega \downarrow \rightarrow$ Dehumidification
DBT $\downarrow \rightarrow$ Cooling
$h_1 > h_2 \rightarrow h \downarrow$
WBT $\downarrow$

560. In a psychrometric chart sensible heating or cooling is represented by
(a) Vertical line (b) Horizontal line
(c) Inclined line (d) Curved lines

VIZAG Steel MT 24.01.2021, Shift-I
GUJARAT PSC AE 2019
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

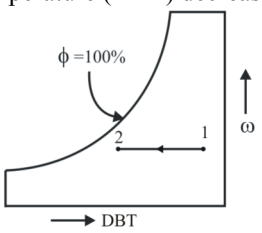
Ans. (b) In a psychrometric chart sensible heating or cooling is represented by horizontal line.



561. During the process of sensible cooling :
(a) The specific humidity increase and dry bulb temperature decreases
(b) The specific humidity remains constant and dry bulb temperature decreases
(c) The specific humidity decreases and dry bulb temperature increase
(d) The specific humidity remains constant and dry bulb temperature increases.

Oil India Limited Sr. Engineer (Drilling) 30.11.2019
PSPCL AE 2012, ESE 2012

Ans. (b) : According to psychrometric chart during sensible cooling 1-2 specific humidity remains constant and dry bulb temperature (DBT) decreases.



562. During chemical dehumidification process of air
(a) dry-bulb temperature and specific humidity decrease
(b) dry-bulb temperature increases and specific humidity decreases
(c) dry-bulb temperature decreases and specific humidity increases
(d) dry-bulb temperature and specific humidity increase

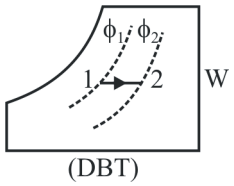
BPSC AE Mains 2017 Paper - V
UPPSC AE 12.04.2016 Paper-II, GATE 2004

Ans : (b) : Air dehumidified while enthalpy remains constant. Certain chemicals like silica gel and alumina absorb moisture from air and hence its 'ω' value decreases but at the same time latent heat of condensation liberates and added to air (self-heating) and hence its dry bulb temperature increases.

563. If a certain mass of moist air in an air tight vessel is heated to a higher temperature, then
(a) Specific humidity of the air increases
(b) Specific humidity of the air decreases
(c) Relative humidity of the air increases
(d) Relative humidity of the air decreases

RPSC AE 2018
UPPSC AE 12.04.2016 Paper-II
ISRO Scientist/Engineer 03.07.2016

Ans. (d) :



$\phi_2 < \phi_1$

■ Moist air is a composition of dry air and water vapour. Dry air is a pure substance but moist air is not a pure substance because the percentage of water vapour content varies from place to place. As we have seen that at some places their is a high humidity and at some places their is a low humidity.
"Specific humidity indicates the actual amount of water vapour present in the air where as, relative humidity indicates indirectly the moisture absorption capacity of the present air."
Note : Generally the state of moist air is superheated.

564. Water at 42°C is sprayed into a stream of air at atmospheric pressure, dry bulb temperature of 40°C and a wet bulb temperature of 20°C. The air leaving the spray humidifier is not saturated. Which of the following statements is true?
(a) Air gets cooled and humidified
(b) Air gets heated and humidified
(c) Air gets heated and dehumidified
(d) Air gets cooled and dehumidified

Gujarat PSC AE 2019
UPPSC AE 12.04.2016 Paper-II, GATE 2005

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Ans : (b) : Spray water temperature = 42°C
Air stream condition
DBT = 40°C
WBT = 42°C
Since, temperature of spraying water is greater than that of air stream temperature. Hence, air gets heated and humidified.

565. Humidity ratio can be given in terms of partial pressures of dry air (P_a) and water vapour (P_v) as

(a) $0.622\left(\frac{P_a}{P_v}\right)$ (b) $0.622\left(\frac{P_v}{P_a}\right)$

(c) $0.622\left(\frac{P_v}{P_v - P_a}\right)$ (d) None of the above

UPPSC AE 12.04.2016 Paper-II
Nagaland PSC (CTSE) 2016 Paper-II
ESE 2001

Ans : (b) Humidity ratio = $\frac{m_v}{m_a}$

$PV = mRT$

For vapour $P_vV = m_v R_vT$

$$m_v = \frac{P_v \cdot V_v}{R_v T_v}$$

For air $m_a = \frac{P_a \cdot V_a}{R_a T_a}$

$V_v = V_a$
$T_v = T_a$

We know that
$R_a = 0.287$ kJ/kgK
$R_v = 0.416$ kJ/kgK

$$\omega = \frac{m_v}{m_a} = \frac{P_v \cdot R_a}{R_v \cdot P_a}$$

Humidity ratio = $\frac{0.622 P_v}{(P_t - P_v)}$

$P_t = P_v + P_a$

m_a = mass of dry air
P_v = Partial pressure of vapour
P_a= Partial Pressure of air
P_t= Total pressure

566. In a spray washing system, if the temperature of water is higher than the dry bulb temperature of entering air, then the air is

(a) Heated and dehumidified
(b) Heated and humidified
(c) Cooled and humidified
(d) Cooled and dehumidified

Nagaland PSC (CTSE) 2018, Paper-II
Vizag Steel (MT) 2017, ESE 1993

Ans. (b) : In spray washing system, if the temperature of water is higher than the dry bulb temperature of entering air then the air is heated and humidified.

567. As relative humidity decreases, the dew point will be :

(a) lower than WBT (b) equal to WBT
(c) higher than WBT (d) unpredictable

RPSC ACF & FRO, 26.02.2021
(KPSC AE. 2015)
Kerala PSC AE 06.08.2015

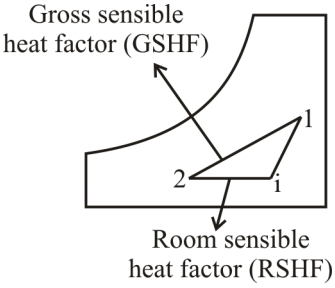
Ans. (a) : As relative humidity decreases, the dew point temperature will be lower than wet. bulb temperature.

568. The supply air state of cooling coil with a bypass factor (BPF) lies at

(a) Intersection of RSHF line with saturation curve
(b) Intersection of GSHF line with saturation curve
(c) Point dividing RSHF line in proportion of BPF and (1-BPF)
(d) Intersection of RSHF line and GSHF line

RPSC Lect (Tech. Edu Dept) 13.03.2021
OPSC AEE 2019 PAPER - II
Kerala PSC Poly. Lect., 2017

Ans. (d) : The supply air state of cooling coil with a bypass factor (BPF) lies of intersection of RSHF line and GSHF line.



569. The moisture content of air is indicated by

(a) dry bulb temperature
(b) wet bulb temperature
(c) dew point temperature
(d) both wet bulb and dew point temperature

TNPSC AE, 2008
J & K PSC Screening, 2006

Ans. (c) : The moisture content of air is indicated by Dew Point Temperature (DPT). The higher the dew point, the higher the moisture content of the air at a given temperature.

570. If the specific humidity of moist air remains same but its dry bulb temperature increases.

(a) its dew point temperature increases
(b) its dew point temperature decreases
(c) its dew point temperature remains same
(d) its dew point temperature may increase or decrease depending upon increase or decrease of relative humidity.

UPPSC AE 13.12.2020 Paper-II
TNPSC AE, 2008

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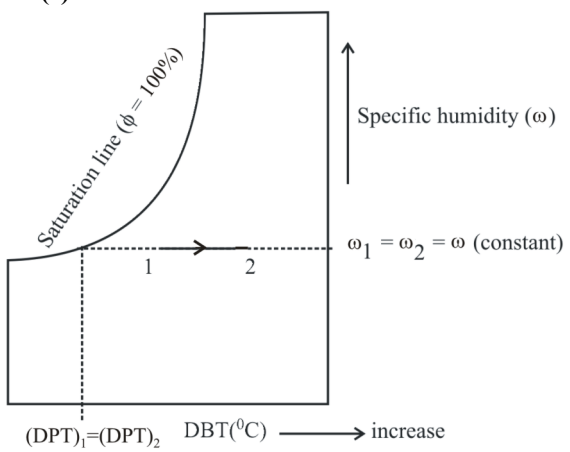
Ans. (c)



If the specific humidity (ω) of moist air remains same but its dry bulb temperature (DBT) increases, then its dew point temperature (DPT) remains same.

i.e., Process is sensible heating.

i.e., $\omega_1 = \omega_2 = \omega$ (same)

DBT increase

$(DPT)_1 = (DPT)_2 = DPT$ (same)

571. If the specific heats of dry air and water vapour are 1kJ/kg-K and 1.88 kJ/kg-K and the humidity ratio is 0.011, the specific heat of moist air will be:

(a) 1.0207 kJ/kg-K (b) 1.869 kJ/kg-K
(c) 1.891 kJ/kg-K (d) 0.9793 kJ/kg-K

TSPSC Managers, 2015
ESE 1994

Ans. (a) : C_p of Moist air = C_p of dry air + humidity ratio × C_p of water vapour.

$= 1.00 + 0.011 \times 1.88$
$= 1.00 + 0.02068$
$= 1.0207$ kJ/kg-K

572. If 1 kg of air having specific humidity as 0.03 kg/kg of dry air mixes with 2 kg of air having specific humidity as 0.015 kg/kg of dry air. The specific humidity of mixture in kg/kg of dry air will be approximately

(a) 0.025 (b) 0.015
(c) 0.02 (d) 0.03

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper-II

Ans. (c) : Specific humidity for mixture

$$\omega_{mix} = \frac{m_1\omega_1 + m_2\omega_2}{m_1 + m_2}$$

$$= \frac{1\times 0.03 + 2\times 0.015}{1+2}$$

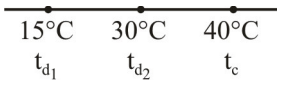
$$\omega_{mix} = 0.02$$

573. The atmospheric air at dry bulb temperature 15° C enters a heating coil maintained at 40° C. If the air leaves the heating coil at 30° C, the coil efficiency is _____.

(a) 0.6 (b) 0.4
(c) 0.2 (d) 0.75

VIZAG MT, 14.12.2020, ESE 2013

Ans. (a) :



Coil efficiency

$$\eta = \frac{30-15}{40-15} = \frac{15}{25}$$

$$\boxed{\eta = 0.6}$$

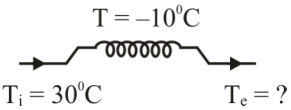
574. In an air condition unit air enters the cooling coil a temperature 30^0C. The coil surface temperature is -10^0C. If the cooling coil bypass factor is 0.45, then the temperature at the exit will be

(a) 6^0C (b) 8^0C
(c) 10^0C (d) 12^0C

UPPSC AE 13.12.2020, Paper-2
ESE 2018

Ans. (b) : Given,

$T_i = 30^0$C
$T_e = ?$
$t = -10^0$C
BPF = 0.45



$$BPF = \frac{T_e - t}{T_i - t}$$

$$0.45 = \frac{T_e + 10}{30 + 10} \Rightarrow T_e = 8^0\text{C}$$

575. If the volume of moist air with 50% RH is isothermally reduced to half its original volume, then relative humidity of moist air becomes

(a) 25% (b) 60%
(c) 70% (d) 100%

UPPSC AE 13.12.2020, Paper-2
ESE 2003

Ans. (d) : Given,

$\varphi_1 = 50\%$

If volume is reduced to half its original isothermally

$\left(\text{i.e.}\, V_2 = \frac{V_1}{2}\right)$

$$P_1V_1 = P_2V_2$$

$$P_1V_1 = P_2 \times \frac{V}{2}$$

$$P_2 = 2P_1$$

$$\varphi_2 = \frac{P_{v_2}}{P_{v_s}} = \frac{2P_{v_1}}{P_{v_s}} = 2\times 50 = 100\%$$

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576. The room sensible heat loss is 30,000 kJ/hr and the latent heat loss is 20,000 kJ/hr. Then the sensible heat factor is

(a) 0.667 (b) 0.60
(c) 0.30 (d) 3.00

UPPSC AE 13.12.2020, Paper-2, ESE 2013

Ans. (b) : Given,
Sensible heat loss, (SHL) = 30,000 kJ/hr
Latent heat loss, (LHL) = 20,000 kJ/hr

$$SHF = \frac{SHL}{SHL + LHL}$$

$$SHF = \frac{30,000}{30,000 + 20,000} = 0.6$$

577. For an air conditioned space, RTH = 100 kW, RSHF = 0.75, volume flow rate of air is 100 m^3/minute and room specific humidity is 0.01 kg/kg of dry air. The specific humidity of supply air in kg/kg of dry air is

(a) 0.0100 (b) 0.0075
(c) 0.0050 (d) 0.0025

UPPSC AE 13.12.2020, Paper-2, ESE 2005

Ans. (c) Given,
RTH = 100 kW, RSHF = 0.75
C_{mm} = Volume flow rate of air = 100 m^3/min
ω_2 = 0.01 kg/kg of dry air
Supply Air, $\Delta\omega = \omega_1 - \omega_2$

$$RSHF = \frac{RSH}{RTH}$$

RSH = 0.75 × 100 = 75 kW
RLH = RTH – RSH
= 100 – 75 = 25 kW
We know that,
RLH = 50 $C_{mm}\Delta\omega$ kW
25 = 50 × 100 × $\Delta\omega$
$\Delta\omega$ = 0.005 kg/kg of dry air

578. The sensible heat factor for auditorium or cinema hall is generally kept as

(a) 0.6 (b) 0.7
(c) 0.8 (d) 0.9

Assam PSC AE (PHED), 18.10.2020
OPSC AEE 2019 Paper-II

Ans. (b) : The sensible heat factor for auditorium or cinema hall is generally kept as 0.70.

579. The atmospheric air at 35ºC and 60% RH can be brought to 20ºC and 60% RH by

(a) cooling and humidification process
(b) cooling and dehumidification process
(c) sensible cooling process
(d) adiabatic saturation process

Assam PSC AE (PHED), 18.10.2020, ESE 2006

Ans. (b) : The atmospheric air at 35ºC and 60% RH can be brought to 20ºC and 60% RH by cooling and dehumidification process.

580. Humidity ratio is defined as the ratio of

(a) Mass of water vapour to the mass of air water vapour mixture
(b) Mass of water vapour to the mass of dry air in a given mixture of air-water vapour mixture
(c) Mass of dry air to the mass of air water vapour mixture
(d) Mass of dry air to mass of water vapour in an air-water vapour mixture

APGCL AM, 2021, OPSC AEE 2019 Paper-II

Ans. (b) Humidity Ratio– It is the ratio of mass of vapour to the mass of dry air in a given mixture of air water vapour . It is represented by (ω)

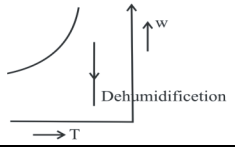
$$\omega = \frac{m_v}{m_a} \text{kgvapour/kg dry air}$$

581. The process of removing moisture from air at constant dry bulb temperature is known as

(a) Sensible heating (b) Sensible cooling
(c) Humidification (d) dehumidification

APGCL AM, 2021, HPPSC W.S Poly. 2016

Ans. (d) :
⇒ Process of removing moisture from air at constant dry bulb temperature is known as dehumidification.
⇒ Pure Humidification & dehumidification processes are not possible practically.



582. Dew point temperature of air at one atmospheric pressure (1.013 bar) is 18°C. The air dry bulb temperature is 30°C. The saturation pressure of water at 18°C and 30°C are 0.02062 bar and 0.04241 bar respectively. The specific heat of air and water vapour respectively are 1.005 and 1.88 kJ/kgK and latent heat of vaporization of water at 0°C is 2500 kJ/kg. The specific humidity (kJ/kg of dry air) and enthalpy (kJ/kg of dry air) of the moist air respectively are :

(a) 0.01051, 52.64 (b) 0.01291, 63.15
(c) 0.01481, 78.66 (d) 0.01532, 81.40

APPSC IOF, 2009, GATE 2004

Ans. (b) : P = 1.013 bar, $(P_s)_{DPT} = (P_v)_{DBT}$ = 0.02062 bar

$$\text{Specific humidity } (\omega) = 0.622 \times \frac{P_v}{P_t - P_v}$$

$$\omega = \frac{0.622 \times 0.02062}{1.013 - 0.02062} = 0.01291 \text{kg/kg of dry air}$$

h = 1.005 DBT + w(2500 + 1.88 DBT)
h = 1.005 × 30 + 0.01291 × (2500 + 1.88 × 30)
h = 3015 + 33.029
h = 63.179 kJ/kg d.a.

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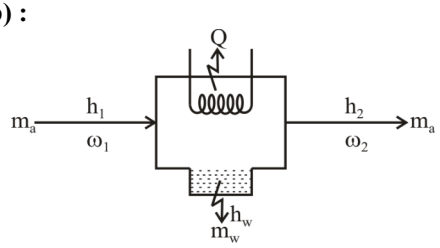
583. Atmospheric air at a flow rate of 3 kg/s (on dry basis) enters a cooling and dehumidifying coil with an enthalpy of 85 kJ/kg of dry air and humidity ratio of 19 gram/kg of dry air. The air leaves the coil with an enthalpy of 43 kJ/kg of dry air and a humidity ratio of 8 gram/kg of dry air. If the condensate water leaves the coil with an enthalpy of 67 kJ/kg, the required cooling capacity of the coil in kW is :

(a) 75 (b) 123.8
(c) 128.2 (d) 159

APPSC IOF, 2009, GATE 2007

Ans. (b) :



Mass flow rate of water, $m_w = m_a(\omega_1 - \omega_2)$
$= 3 \times (0.019 - 0.008)$
$= 0.033$ kg/s

Energy balance equation,
$m_a h_1 = m_a h_2 + m_w h_w + Q$
$3 \times 85 = 3 \times 43 + 0.033 \times 67 + Q$
$Q = 123.8$ kW

584. Dew point temperature lines on the psychrometric chart are :

(a) Horizontal
(b) Vertical
(c) Curved
(d) Straight inclined slopping downward to the right.

ISRO Scientist/Engineer (RAC), 10.03.2019
Kerala PSC IOF 19.04.2016

Ans. (a) : Horizontal– Dew point temperature lines on psychrometric charts are horizontal lines.

585. Taking B as bypass factor in sensible cooling or heating of air, the coil efficiency is equal to

(a) B-1 (b) 1-B
(c) B+1 (d) 1/B

MPPSC State Forest Service Exam, 2014
HPPSC AE 2014

Ans. (b) : If B as by pass factor in sensible cooling or heating of air, the coil efficiency is equal to,

$\Rightarrow \quad \eta_{coil} = (1 - B) \times 100\%$

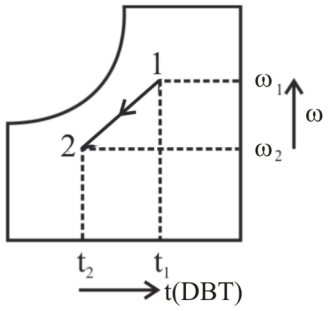
586. A stream of moist air at dry-bulb temperature of 40°C and dew-point temperature of 25°C passes through a water shower whose temperature is maintained at 20°C. The stream of air will undergo a process of

(a) Sensible cooling
(b) Evaporative cooling
(c) Cooling and humidification
(d) Cooling and dehumidification

RPSC IOF 2020, GPSC EE Pre, 28.01.2017

Ans. (d) : Cooling and Dehumidification of moist air-

- Air is to be cooled and during cooling water vapour is to be separated from the air.
- Since moisture separation will occur only when moist air is cooled to a temperature below its DPT. Therefore, the effective surface temperature of the cooling coil kept below the initial DPT of the air and the surface temperature is called apparatus dew point temperature (t_{ADP}).



587. Two air streams with mass flow rates of 36 kg/min and 14 kg/min with respective enthalpies of 36 kJ/kg and 50 kJ/kg are mixed. The enthalpy of mixture is nearly

(a) 64 kJ/kg (b) 55 kJ/kg
(c) 46 kJ/kg (d) 40 kJ/kg

GPSC ARTO Pre 30.12.2018, ESE 2017

Ans. (d) : Given, $m_1 = 36$ kg/min $h_1 = 36$ kJ/kg
$m_2 = 14$ kg/min $h_2 = 50$ kJ/kg

$$m_1h_1 + m_2h_2 = (mh)_{mix}$$
$$36 \times 36 + 14 \times 50 = 50\, h_{mix}$$
$$h_{mix} = 39.92 \simeq 40\, kJ/kg$$

588. The optimum effective temperature for human comfort is

(a) Higher in winter than the summer
(b) Lower in winter than the summer
(c) Same in winter and summer
(d) Not dependent on season

Nagaland PSC (CTSE) 2018, Paper-II
ESE 2005

Ans. (b) : Optimum effective temperature is dependent on seasonal changes. So, For human comfort the optimum effective temperature is lower Human generally fell comfortable between temperatures of 22°C to 27°C and a relative humidity of 40% to 60%.

589. Sensible heat factor is given by

(a) S.H/(S.H + L.H) (b) (S.H+L.H)/S.H
(c) (L.H–S.H)/S.H (d) S.H/(L.H–S.H)

Nagaland PSC (CTSE) 2018, Paper-II
ESE 1999

Ans. (a) : Sensible heat ratio is defined as the sensible heat or cooling load divided by the total heat or cooling load.

$$SHF = \frac{SH}{S.H. + L.H.}$$

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590. During sensible cooling

(a) Relative humidity remains constant
(b) Wet bulb temperature increases
(c) Specific humidity increases
(d) Partial pressure of vapor remains constant

Nagaland PSC (CTSE) 2018, Paper-II
HPPSC Asstt. Prof. 18.09.2017

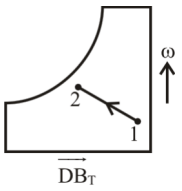
Ans. (d) : During sensible cooling process the dry bulb temperature and wet bulb temperature of the air reduces, while the latent heat of the air and the dew point temperature of the air remains constant. In general the sensible cooling process is carried out by passing the air over the coil.

591. When the air is passed through an insulated chamber having sprays of water maintained at a temperature higher than the dew point temperature of entering air but lower than the dry bulb temperature, then the air is said to be

(a) cooled and dehumidified
(b) cooled and humidified
(c) heated and humidified
(d) heated and dehumidified

ISRO Scientist/Engineer (RAC) 12.01.2020
ESE 1994

Ans. (b) : Because, insulated chamber is having sprays of water, the process will humidification and the dry bulb temperature of spray water is less than the air. Then the process will also be cooling process.
So air is said to be cooled and humidified.



592. Specific humidity is defined as

(a) Mass of water vapor contained in air-vapor mixture per kg of dry air
(b) Mass of water vapor contained per kg of air-vapor mixture
(c) Mass of dry air contained per kg of air-vapor mixture
(d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : Specific humidity is defined as mass of water vapor contained in air-vapor mixture per kg of dry air.

593. For a typical sample of ambient air (at 35°C, 75% relative humidity and standard atmospheric pressure), the amount of moisture in kg per kg of dry air will be approximately.

(a) 0.002 (b) 0.027
(c) 0.25 (d) 0.75

VIZAG STEEL MT 10.06.2012
GATE 2005

Ans. (b) : Given data-Dry bulb temperature $T_{db} = 35°C$
Relative humidity $\phi = 75\% = 0.75$
at 35°C, $P_s = 0.05628$ bar (from psychrometric chart)

$$\phi = \frac{P_v}{P_s} \Rightarrow 0.75 = \frac{P_v}{0.05628}$$

$P_V = 0.75 \times 0.05628$
$P_V = 0.04221$ bar

$$W = \frac{0.622\,P_V}{P - P_V} = \frac{0.622 \times 0.04221}{1.01325 - 0.04221}$$

$$\boxed{W = 0.0270 \text{ kg / kg of dry air}}$$

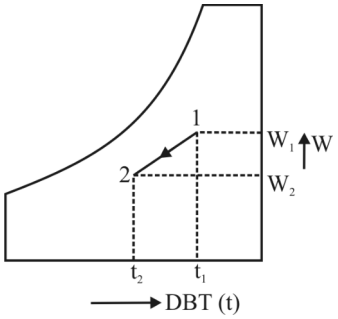
594. Air at dry bulb temperature of 35ºC and dew point temperature of 25ºC passes through an air washer whose temperature is maintained at 20ºC. What is the nature of the process?

(a) Cooling humidification
(b) Sensible cooling
(c) Heating and dehumidification
(d) Heating and humidification
(e) Cooling and dehumidification

CGPSC ARTO & TSI 07.01.2017
ESE 2005

Ans. (e) : Cooling and Dehumidification of moist air :-

- Air is to be cooled and during cooling water vapour is to be separated From air.



- Since moisture separation will occur only when moist air is cooled to a temperature below its DPT.

595. Humidity ratio W is expressed as
(P_v = Partial pressure of vapour
Pa = partial pressure or air
Pb = barometric pressure of air)

(a) $W = 0.622\ P_v / P_b$
(b) $W = 0.622\ P_v / P_a$
(c) $W = 0.622 \dfrac{P_v}{P_b + P_v}$
(d) $W = 0.622 \dfrac{P_v}{(P_b - P_v)}$

GPSC ARTO 01.05.2016
OPSC AEE 2015 PAPER - II

Ans. (d) : Humidity ratio/specific Humidity (W) =

$$\frac{M_{vapour}}{M_{dry\ air}} = \frac{M_V}{M_a}$$

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P_V = Pressure of water vapour
P_a = Pressure of dry air
P = Total atmosphere pressure
$P = P_V + P_a$

$$W = \frac{M_V}{M_a} = \left(\frac{P_v.V/R_{vT}}{P_a V/R_{aT}}\right) \quad \{\because PV = MRT\}$$

$$= \frac{P_V}{P_a}\frac{R_a}{R_V} = \frac{18}{29}.\frac{P_V}{P_a} \quad \{\because P_a = P - P_v\}$$

$$\boxed{W = 0.622.\frac{P_V}{P - P_V}}$$

596. If the specific humidity of the moist air remains same but its dry bulb temperature increases
(a) its dew point temperature increases
(b) its dew point temperature decreases
(c) its dew point temperature remains constant
(d) none of the above

ESE 2011, UKPSC AE 2007 Paper -II

Ans. (c) : Its dew point temperature remains constant

597. The enthalpy of moist air with normal notation as is given by
(a) $h = (1.005 + 1.88\omega)\, t + 2500\omega$
(b) $h = 1.88\omega t + 2500\omega$
(c) $h = 1.005\omega t$
(d) $h = (1.88 + 1.005\omega)\, t + 2500\omega$

ESE 2019, RPSC Lect. 16.01.2016

Ans. (a) : Enthalpy of moist air
$h = h_a + \omega h_v = 1.005\, t + \omega\,[2500 + 1.88t]$
$= [1.005 + 1.88\omega]t + 2500\,\omega$

598. If moist air is passed over chemicals like silica gel, the process which takes place is
(a) humidification
(b) dehumidification
(c) cooling and dehumidification
(d) heating and dehumidification

UKPSC AE 2012 Paper–II
ESE 2010

Ans. (d) : heating and dehumidification

599. The amount of moisture in air can be measured by
(a) Sling psychrometer (b) Mass spectrometer
(c) Photometer (d) Thermistor

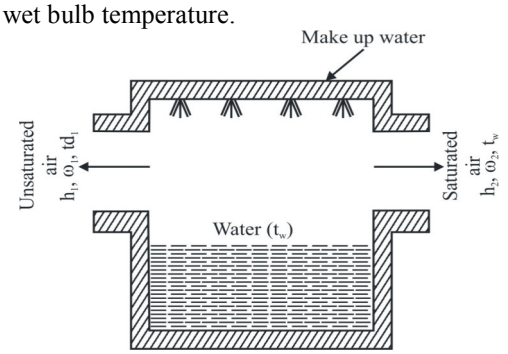
GPSC Executive Engineer 23.12.2018
KPSC AE 2015

Ans : (a) The amount of moisture in air can be measured by sling psychrometer.

600. In cooling tower, "approach" is the temperature difference between the:
(a) Hot inlet water and cold outlet water.
(b) Hot inlet water and wet bulb temperature
(c) Cold outlet water and wet bulb temperature
(d) Dry bulb and wet bulb temperature

OPSC AEE 2015 PAPER - II
APPSC AEE 2012

Ans : (c) In cooling tower "approach" is the temperature difference between the cold outlet water and wet bulb temperature.



The Thermodynamic wet bulb temperature or adiabatic saturation temperature is the temperature at which the air can be brought to saturation state, adiabatically, by the evaporator of water into the flowing air.

601. For completely dry air total heat is
(a) Total latent heat
(b) Sum of latent heat and sensible heat
(c) Total sensible heat
(d) Difference of sensible heat and latent heat

RPSC Vice Principal ITI 2018, NPCIL ET, 2015

Ans. (c) : For completely dry air, since latent part will not present hence total heat will be equal to total sensible heat.

602. When air is adiabatically saturated, the temperature attained is
(a) dew point temperature
(b) dry bulb temperature
(c) wet bulb temperature
(d) triple point temperature

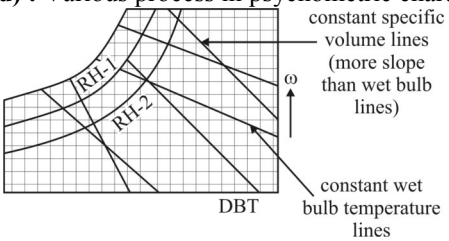
Gujarat PSC AE 2019, ESE 2011

Ans : (c) : When air is adiabatically saturated, the temperature attained is wet bulb temperature.

603. The statements concern psychrometric chart
1. Constant relative humidity lines are uphill straight lines to the right.
2. Constant wet bulb temperature lines are downhill straight lines to the right.
3. Constant specific volume lines are downhill straight lines to the right.
4. Constant enthalpy lines are coincident with constant wet bulb temperature lines.
Which of the statements are correct?
(a) 2 and 3 (b) 1 and 2
(c) 1 and 3 (d) 2 and 4

Gujarat PSC AE 2019, GATE 2006

Ans : (d) : Various process in psychometric charts



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(i) Constant RH lines are curved and uphill to right.
(ii) Constant WBT line and specific volume lines both are straight lines but different slopes.
(iii) Constant enthalpy and constant WBT line are coincident.

604. Evaporative regulation of body temperature fails when the body temperature is :
(a) more than wet bulb temperature but less then dry bulb temperature
(b) more than dew point temperature but less than wet bulb temperature
(c) more than dew point temperature but less than dry bulb temperature
(d) less than dew point temperature.

GPSC ARTO Pre 30.12.2018
CSE Pre-1999

Ans. (d) : Evaporative regulation of body temperature fails when the body temperature is less than dew point temperature.

605. On Psychrometric chart, the constant wet bulb temperature lines coincides with (which one of the following is correct)
(a) Constant relative humidity liners
(b) Constant enthalpy lines
(c) Constant dew point temperature line
(d) Constant volume lines

Nagaland PSC (CTSE) 2017, 2016 Paper-II
ESE 2008

Ans. (b) : On Psychrometric chart, the constant wet bulb temperature lines coincides with constant enthalpy lines.

606. For cooling and dehumidifying of unsaturated moist air, it must be passed over a coil at a temperature
(a) Of adiabatic saturation of incoming stream
(b) Which is lower than the dew point of incoming steam
(c) Which lies between dry bulb and wet bulb temperature
(d) Constant which lies between wet bulb and dew point temperature of incoming stream

ISRO Scientist Engineer 22.04.2018
Nagaland PSC (CTSE) 2017, 2016, 2015 Paper-II

Ans.(b) : For cooling and dehumidifying of unsaturated moist air, it must be passed over a coil at a temperature which is lower than the dew point of incoming steam.

607. Effective temperature depends on dry bulb temperature and
(a) Wet bulb temperature only
(b) Relative humidity and specific volume
(c) Specific humidity and enthalpy
(d) Wet bulb temperature and air motion

Nagaland PSC (CTSE) 2016, Paper-II, ESE 2006

Ans. (d) : Effective temperature depends on dry bulb temperature and Wet bulb temperature and air motion

608. Equal friction method of designing ducts is preferred
(a) When system is balanced
(b) When system is not balanced
(c) Only for return ducts
(d) For any system

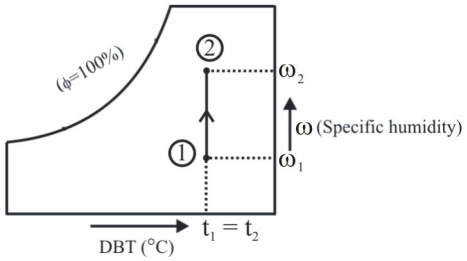
Nagaland PSC (CTSE) 2016, Paper-II
ESE 2008

Ans. (a) : Equal friction method of designing ducts is preferred it is quite easy to use, total pressure drop is low and flow is balanced.

609. A humidification process means :
(a) An increase in specific humidity
(b) A decrease in relative humidity
(c) A decrease in temperature
(d) An increase in temperature

RPSC IOF & Boiler 20.09.2017
UPSC JWM Advt. No.-52/2010

Ans. (a) : Explanation : Humidification means adding the water vapour or moisture to the humid air. During the humidification process, specific humidity should be increases



(humidification process express in Psychometric chart)

610. A cooling coil with a bypass factor of 0.1 and apparatus dew point (adp) of 12°C comes in contact with air having a dry-bulb temperature of 38°C and dew point of 9°C. Over the cooling coil, the air would undergo
(a) sensible cooling
(b) cooling and humidification
(c) cooling and dehumidification
(d) adiabatic saturation

Nagaland PSC CTSE 2015 Paper-II, CSE Pre-2001

Ans. (a) : Given,
BPF = 0.1
ADP = 12 °C
DBT = 38 °C
DPT = 9°C

$$\text{BPF for cooling coil} = \frac{t_{d_2} - t_{d_3}}{t_{d_1} - t_{d_3}}$$

$$0.1 = \frac{t_{d_2} - 12}{38 - 12}$$

$$t_{d_2} = 2.6 + 12$$

$$t_{d_2} = 14.6\ °C > DPT$$

Sensible cooling.

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611. Moist air exists at a pressure of 1.01 bar. The partial pressure and saturation pressure of water vapour are 0.01 bar and 0.02 bar, respectively. What are the relative humidity and humidity ratio of the moist air, respectively?

(a) 50% and 0.00622 (b) 100% and 0.0126
(c) 50% and 0.0126 (d) 100% and 0.00622

Nagaland PSC CTSE 2015 Paper-II
ESE 2004

Ans. (a) : Given,

$P_{atm} = 1.01$ bar

$P_v = 0.01$ bar

$P_{sat} = 0.02$ bar

$$RH = \frac{P_v}{P_{sat}} = \frac{0.01}{0.02}$$

$$= 50\%$$

$$\text{Humidity ratio } (\omega) = \frac{0.622 P_v}{P_{atm} - P_v}$$

$$= \frac{0.622 \times 0.01}{1.01 - 0.01}$$

$$= 0.00622$$

612. When a GSHF line is extended, it may strike the saturation curve at a point, which is called

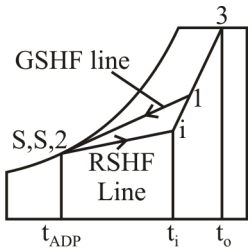
1. Effective surface temperature.
2. Air saturation temperature.
3. Water boiling temperature.
4. Apparatus dew point.

Which of the above is correct?

(a) 1 and 2 (b) 1 and 4
(c) 2 and 3 (d) 3 and 4

Nagaland PSC CTSE 2015 Paper-II
CSE Pre-2007

Ans. (b) :



GSHF → Grand sensible heat factor
RSHF → Room sensible heat factor
ADP → Apparatus dew point

613. Consider the following statements

1. The human body can lose heat even if temperature is less than the atmospheric temperature.
2. Relative humidity can be increased by cooling dehumidification.
3. Warm air increases the rate of radiation of heat from the human body.
4. Increase in air movement increases the evaporation from the human body.

Which of the above are correct

(a) 1 and 4 (b) 2 and 4
(c) 1 and 3 (d) 2 and 3

Nagaland PSC CTSE 2015 Paper-II
ESE 1994

Ans. (a) : ■ The human body can lose heat even if temperature is less than the atmospheric temperature.
■ Increase in air movement increases the evaporation from the human body.
■ It has to do with the temperature sensing mechanism of human body.

614. Which of the following statements is correct

(a) Dew point temperature can be measured with the help of thermometer
(b) Dew point temperature is the saturation temperature corresponding to the partial pressure of water vapors in moist air
(c) Dew point temperature is the same as the thermodynamic wet bulb temperature
(d) For saturated air, dew point temperature is less than the wet bulb temperature

RPSC VPITI 14.02.2016
TSGENCO AE 14.11.2015

Ans. (b) : Dew point temperature is the saturation temperature corresponding to the partial pressure of water vapour in moist air. It is the saturation temperature corresponding to the initiation of the condensation or water particle just start to condense.

615. For low bypass factor a cooling coil, the fin spacing and the number of tube rows will be respectively

(a) High and high (b) High and low
(c) Low and high (d) Low and low
(e) None of the above

BPSC AE Mains 2017 Paper-V
CGPSC Poly. Lect. 22.05.2016

Ans. (c) : The bypass factor is a function of the airflow, number of rows, surface temperature, number of fins per inch, height of fins and many other construction attributes of coils.
The bypass factor of a cooling coil decreases with decrease in fin spacing and increase in number of rows.

616. What are the general comfort conditions in an air-conditioning system?

(a) 20 DBT, 80% RH (b) 24 DBT, 60% RH
(c) 25 DBT, 40% RH (d) 25 DBT, 100% RH
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
ESE 2006

Ans. (b) : The comfort conditions in air conditioning are at 24 DBT and 60% RH.

617. As an index of comfort, the temperature of saturated air at which a person would experience the same feeling of comfort as experienced in the actual unsaturated environment is called

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(a) Comfort temperature
(b) Effective temperature
(c) Wet bulb temperature
(d) Soothing temperature

UPPSC AE 12.04.2016 Paper-II
ESE 2010

Ans : (b) Effective temperature is the temperature of saturated air at which a person would experience same feeling of comfort as in unsaturated environment.

618. The phenomenon that enables cooling towers to cool water to a temperature below the dry bulb temperature of air is termed as
(a) Chemical dehumidification
(b) Adiabatic evaporative cooling
(c) Cooling and dehumidification
(d) Sensible cooling

RPSC Lect (Tech. Edu. Dept.) 13.03.2021
ESE 2020

Ans. (b) : Water cools due to evaporation and air gets heated and humidified and the phenomenon is called adiabatic evaporative water cooling.

619. In air-conditioning systems, air may be cooled and dehumidified by
1. Spraying chilled water to air in the form of fine mist
2. Circulating chilled water or brine in a tube placed across the air flow.
3. Placing the evaporator coil across the air flow.
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

RPSC Lect (Tech. Edu. Dept.) 13.03.2021
ESE 2020

Ans. (d) : 1, 2 and 3

620. A measure of feeling warmth or coolness by the human body in response to the air temperature, moisture content and air motion is called
(a) Dry bulb temperature
(b) Effective temperature
(c) Wet bulb temperature
(d) Dew point temperature

ESE 2020, CGPSC AE 16.10.2016

Ans. (b) : A measure of feeling warmth or coolness by the human body in response to the air temperature, moisture content and air motion is called effective temperature.

621. Sensible Heat Factor (SHF) is defined as:
(a) $SHF = \frac{LH}{SH + LH}$ (b) $SHF = \frac{LH}{SH - LH}$
(c) $SHF = \frac{SH}{SH + LH}$ (d) $SHF = \frac{SH}{SH - LH}$

Where $SH \rightarrow$ Sensible Heat
$LH \rightarrow$ Latent Heat

TRB Polytechnic Lecturer 2017
ESE 1999

Ans. (c) : $SHF = \frac{\text{Sensible heat}}{\text{Total heat}}$

$= \frac{SH}{LH + SH}$

622. In a desert cooler, the main process is
(a) sensible cooling
(b) dehumidification
(c) adiabatic saturation
(d) cooling and dehumidification

Nagaland PSC CTSE 2017 2nd Paper
HARYANA PSC Civil Services Pre. 2014

Ans. (c): In dessert cooler, cooling and humidification process used (which responsible in increasing humidity) while, coolers used in normal regions which apply cooling + dehumidification (decrease humidity)

623. In a sample of moist air at standard atmospheric pressure of 101.5 kPa and 28^0C, the partial pressure of water vapour is 1.445 kPa. If the saturation pressure of water vapour is 3.55 kPa at 28^0C, then what are the humidity ratio and relative humidity of moist air sample?
(a) 0.00836 and 1.32% (b) 0.00836 and 40%
(c) 0.00898 and 41% (d) 0.01344 and 1.32%
(e) 0.01344 and 40%

CGPSC AE 26.04.2015 SHIFT-II
ESE 2009

Ans. (c) : Data given as,
P = 101.325 kPa at 28^0C
P_v = 1.445 kPa
$P_{v\,sat.}$ = 3.55 kPa at 28^0C
We know that

Humidity ratio (ω) = $\frac{0.622P_v}{(P - P_v)} = \frac{0.622 \times 1.445}{(101.325 - 1.445)}$

$\omega = 0.00898$

Relative humidity (ϕ)

$\phi = \frac{P_v}{P_{vsat.}} = \frac{1.445}{3.55} \times 100$

$\phi = 40.7 \approx 41\%$

624. Index of human comfort is given by:
(a) Relative humidity
(b) Dry bulb temperature
(c) Air velocity
(d) Effective temperature
(e) Dew point temperature

CGPSC AE 25.02.2018

Ans. (d) : Index of human comfort is given by effective temperature.

625. Which of the following is true for sensible heating process?
(a) DBT increases, WBT increases, RH increases, Dew point temperature remains same
(b) DBT increases, WBT decreases, RH increases, Dew point temperature increases
(c) DBT increases, WBT increases, RH decreases, Dew point temperature remain same

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(d) DBT increases, WBT decreases, RH decreases, Dew point temperature increases
(e) DBT decreases, WBT increases, RH increases, Dew point temperature remains same

CGPSC AE 25.02.2018

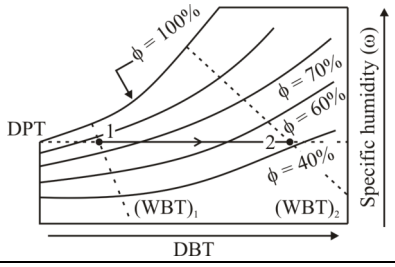
Ans. (c) : For sensible heating process-

DBT ↑ , WBT ↑

DPT → Constant

ω → constant

ϕ ↓, h ↑, v ↑



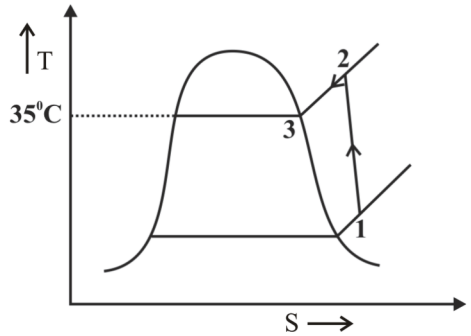
626. Moist air at a pressure of 100 kPa is compressed to 500 kPa and then cooled to 35°C in an aftercooler. The air at the entry to the aftercooler is unsaturated and becomes just saturated at the exit of the aftercooler. The saturation pressure of water at 35°C is 5.628 kPa. The partial pressure of water vapour (in kPa) in the moist air entering the compressor is closest to

(a) 0.57 (b) 1.13
(c) 2.26 (d) 4.52

VIZAG STEEL MT 10.06.2012
GATE 2008

Ans. (b) : Assuming that compression is isentropic in air compressors, the process can be described on the T–S diagram.

The process in the intercooler is constant pressure.



$P_2 = P_3$

But, $P_2 = 5P_1$

$$P_1 = \frac{P_2}{5} = \frac{P_3}{5}$$

$$P_1 = \frac{5.628}{5}$$

$$\boxed{P_1 = 1.13 \text{ kPa}}$$

627. The by- pass factor for a cooling coil

(a) decreases with increase in velocity of air passing through it.
(b) increases with increase in velocity of air passing through it.
(c) remains unchanged with increase in velocity of air passing through it.
(d) May increase of decrease with increase in velocity of air passing through it depending upon the condition of air entering.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : The by- pass factor is defined as amount of uncontained air which pass the coil increase in velocity of air by pass factor will increase.

628. During sensible heating of moist air, enthalpy

(a) increases. (b) decreases.
(c) remains constant. (d) None of the above

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : During sensible heating of moist air, enthalpy increases.

629. In a psychrometric process, the sensible heat added is 40 kJ/sec and the latent heat added is 20 kJ/sec. The sensible heat factor for the process will be :

(a) 0.3 (b) 0.6
(c) 0.67 (d) 1.5

RPSC ACF & FRO, 26.02.2021

Ans. (c) : Sensible heat factor (SHF)

$$= \frac{\text{Sensible heat}}{\text{sensible heat} + \text{latent heat}}$$

$$= \frac{40}{40 + 20}$$

$$= 0.666$$

$$\boxed{\text{SHF} = 0.67}$$

630. Which one of the following conditions is not satisfied for saturated air?

(a) 100 percent relative humidity
(b) Specific humidity is constant
(c) Zero wet bulb depression
(d) DBT is equal to DPT

VIZAG Steel MT 24.01.2021

Ans. (b) : The maximum limit of water vapour that a given quantity of air hold at a particular pressure or temperature is termed as saturated vapour. In such a case, relative humidity will be 100%.

- For saturated vapour all the temperatures are equal.

$$\boxed{\text{DBT} = \text{WBT} = \text{DPT}}$$

- In case of saturated air the value of WBD

WBD = DBT – WBT

$$\boxed{\text{WBD} = 0}$$

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631. Which is the correct combinations for psychometric processes?

(a)	Activated alumina	(I)	Heating & humidification	
(b)	Water injection	(II)	Heating & dehumidification	
(c)	Steam injection	(III)	Cooling & dehumidification	
(d)	Hygroscopic	(IV)	Cooling & humidification	

(a) (a-I), (b-II), (c-III), (d-IV)
(b) (a-III), (b-IV), (c-I), (d-II)
(c) (a-IV), (b-III), (c-II), (d-I)
(d) (a-II), (b-IV), (c-I), (d-III)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) :

632. Consider the following statements for sensible heat factor:

1. **Sensible heat factor will be negative if sensible heat and latent heat are both negative.**
2. **Sensible heat factor will be negative if sensible heat is negative and latent heat is positive.**
3. **Sensible heat factor will be negative if sensible heat is positive and latent heat is negative.**
4. **Sensible heat factor will be negative if sensible heat and latent heat are both positive.**

Which of the above statements are correct?

(a) 2 and 3 only (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 4 only

ESE 18.07.2021

Ans. (a) : Sensible heat factor (SHF)

$$= \frac{\text{Sensible heat (SH)}}{\text{Sensible Heat (SH)} + \text{Latent Heat (LH)}}$$

(1) when SH is -ve & LH is -ve SHF $\rightarrow$ +ve
(4) when SH $\rightarrow$ +ve & LH is +ve SHF $\rightarrow$ +ve
Hence, statement 1 & 4 are incorrect and statement 2 & 3 are correct.

633. If the air initially at dry bulb temperature 35°C and wet bulb temperature 26.1° as it enters an air washer which has a humidifying efficiency of 85%, then what is the final dry bulb temperature of air washed with recalculated spray water?

(a) 26.81°C (b) 27.43°C
(c) 32.83°C (d) 30.49°C

ESE 18.07.2021

Ans. (b) :

$$\text{Humidifying efficiency} = \frac{DBT - T_{out}}{DBT - WBT}$$

$$0.85 = \frac{35^0 - T_{out}}{(35^0 - 26.1^0)}$$

$$\Rightarrow T_{out} = 35° - 0.85\,(35° - 26.1°)$$
$$= 27.435°C$$

634. When air passes over a dry surface which is at a temperature greater than its (air) dry bulb temperature, it undergoes _____

(a) sensible heating (b) sensible cooling
(c) condensation (d) none of the above

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : When air passes over a dry surface which is at a temperature greater than its (air) dry bulb temperature, it undergoes sensible heating.

635. For saturated air

(a) Wet bulb depression is zero
(b) Wet bulb depression is positive
(c) Wet bulb depression is positive
(d) Wet bulb depression can be either positive or negative

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : For saturated air, the wet bulb depression is zero.

636. The temperature of air recorded by a thermometer, when it is not affected by the moisture present in it is called

(a) Wet bulb temperature
(b) Dry bulb temperature
(c) Dew Point Temperature
(d) None of these

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :

637. Room sensible heat factor is defined as : (RSH=Room sensible heat, RL.H=Room Latent Heat)

(a) RSH/(RSH + RLH)
(b) RLH/(RSH + RLH)
(c) RSH/RLH
(d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Room sensible heat factor

$$RSHF = \frac{RSH}{RSH + RLH}$$

638. The Apparatus Dew Point (ADP) for the cooling coil of an air conditioning system with high latent load compared to a system with high sensible load will be

(a) More
(b) Less
(c) Remains same
(d) Depends on air flow through the coil

ISRO Scientist/Engineer (RAC) 12.01.2020

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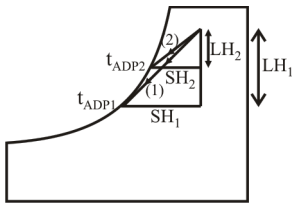
Ans. (b) :



Process (1) → ADP with high latent load.
Process (2) → ADP with high sensible load.
ADP of process (1) is lower than ADP of process (2).

639. Psychrometer is useful to measure.
(a) Dry bulb temperature
(b) Dry and wet bulb temperature
(c) Wet bulb temperature
(d) Dew point temperature

Oil India Senior Officer 23.12.2020

Ans. (b) : Dry and wet bulb temperature

640. The primary factors responsible for human comfort are
(a) dry bulb temperature, relative humidity and air motion
(b) dry bulb temperature, dew point temperature and air motion
(c) dry bulb temperature, relative humidity and latitude of the place
(d) dry bulb temperature, relative humidity, air motion and elevation of the place

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : The primary factor responsible for human comfort are–
(i) Dry Bulb Temperature (DBT)
(ii) Relative Humidity (ϕ)
(iii) Air motion

641. Partial pressure of water vapour at dew point temperature of moist air is 1.5×10^{-3} MPa. The barometric pressure is 0.1 MPa. The specific humidity of air is
(a) 15.225 gm/kg da (b) 9.47 gm/kg da
(c) 15.00 gm/kg da (d) 9.33 gm/kg da

UPPSC AE 13.12.2020, Paper-2

Ans. (b) : Given,

$P_v = 1.5 \times 10^{-3}$ MPa

$P_{atm} = 0.1$ MPa

Specific humidity, $\omega = \dfrac{0.622\, P_v}{P_t - P_v}$

$$\omega = \frac{0.622 \times 1.5 \times 10^{-3}}{\left(0.1 - 1.5 \times 10^{-3}\right)}$$

$$\boxed{\omega = 9.47 \text{gm/kg dry air}}$$

642. If air at dry bulb temperature of 35^0C and dew point temperature of 20^0C passes through air washer in which water is sprayed at 25^0C, then the process would be
(a) sensible cooling
(b) cooling and dehumidification
(c) cooling and humidification
(d) cooling at constant dew point temperature

UPPSC AE 13.12.2020, Paper-2

Ans. (c) : As water sprayed is at 25^0C < (DBT = 35^0C) and dew point temperature is 20^0C
So, cooling and humidification process takes place.

643. For summer air-conditioning, the relative humidity should not be less than
(a) 40% (b) 60%
(c) 75% (d) 90%

Assam PSC AE (PHED), 18.10.2020

Ans. (b) : Relative humidity should not be less than 60% according to the comfort conditions for summer air conditioning.
⇒ For winter air conditioning, the relative humidity should not be more than 40%.

644. Dew point temperature of a moist air mixture is achieved by
(a) Providing cold water of varying temperature in contact with it
(b) Cooling moist air without varying its humidity
(c) Adiabatic saturation of air
(d) Cooling as well as saturation of air

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : For an air mixture with constant mixture content when the temperature decrease to (T_d) the moist air is saturated (ϕ = 100%) and water will leave the air and condense into "dew" if the temperature continues to fall. Therefore dew point is the critical point of water condensation.

645. A sling psychometer measures
(a) Specific humidity
(b) Relative humidity
(c) Wet bulb temperature
(d) Dew point

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : Sling psychometer is used to measure relative humidity of moist air which is represented in percentage (ϕ in %).

646. In a system with outdoor air for ventilation, apparatus dew point (ADP) of the evaporator coil is _____ the room ADP
(a) equal to
(b) higher than
(c) lower than
(d) invariant of (subject to more that)

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : In outdoor air ventilation, cooling and dehumidification is done which can achieve only if ADP of the evaporator is less than the room ADP.

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647. In a system without any outdoor air for ventilation, the cooling load of the building is _____ the coil load

(a) lower than
(b) equal to
(c) higher than
(d) invariant of (subject to more data)

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : For without outdoor air for ventilation, the load on cooling coils is only sensible cooling and no latent cooling, so the cooling load of the building is equal to the coil load.

648. Air-conditioning systems with reheat are used when the_____ loads are high

(a) Latent (b) Sensible
(c) Total (d) Bypass air

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Reheating is done when incoming air is too much cold. In this, air is initially heated and then adiabatically humidified and final position is achieved after reheating.

649. If the atmospheric conditions are 20°C, 1.013 bar and specific humidity of 0.0095 kg/kg of dry air, the partial pressure of vapour will be nearly

(a) 0.076 bar (b) 0.056 bar
(c) 0.036 bar (d) 0.016 bar

ESE 2020

Ans. (d) : Given, DBT = 20°C = 293 K

$P_t = 1.013$ bar

$$\omega = 0.622 \times \frac{P_v}{P_t - P_v}$$

$$0.0095 = 0.622 \times \frac{P_v}{1.013 - P_v}$$

$$0.0152 = \frac{P_v}{1.013 - P_v}$$

$$0.0153 - 0.0152P_v = P_v$$

$$0.0153 = 1.0152P_v$$

$P_v = 0.016$ bar

650. A duct of rectangular cross-section 600 mm × 400 mm carries 90 m^3/min of air having density of 1.2 kg/m^3. When the quantity of air in both cases is same, the equivalent diameter of a circular duct will be nearly

(a) 0.86 m (b) 0.76 m
(c) 0.64 m (d) 0.54 m

ESE 2020

Ans. (d) :

$$D = \frac{1.3(ab)^{0.625}}{(a+b)^{0.25}}$$

$$= \frac{1.3(0.6 \times 0.4)^{0.625}}{(0.6+0.4)^{0.25}} = 0.53 \text{ m}$$

651. A room having dimensions of 5 m × 5 m × 3 m contains air at 25°C and 100 kPa at a relative humidity of 75%. The corresponding value of P_S is 3.169 kPa. The partial pressure of dry air will be nearly

(a) 106 kPa (b) 98 kPa
(c) 86 kPa (d) 78 kPa

ESE 2020

Ans. (b) :

$$\phi = \frac{P_v}{P_s}$$

$$0.75 = \frac{P_v}{3.169}$$

$P_v = 2.376$ kPa

$P_t = P_a + P_v$

$100 = P_a + 2.376$

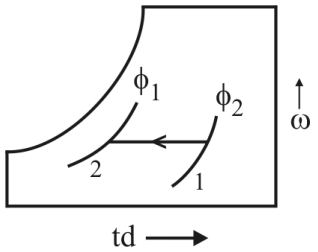
$P_a = 97.624 \approx 98$ kPa

652. As warm air cools, it's relative humidity

(a) Increases
(b) Decrease
(c) Remains same
(d) Depends on wet bulb temperature

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) :



$\phi_2 < \phi_1 \rightarrow$ Relative humidity increases

653. Atmospheric air flowing at rate of 4 kg/s enters the cooling and dehumidifying coil with enthalpy value of 90 kJ/kg of dry air and has humidity rate of 20 gram/kg dry air. When it leaves the coil it has humidity ratio 10 gram/kg of dry air and enthalpy of 50 kJ/kg or dry air. Condensate water leaving coil has enthalpy of 70 kJ/kg. The cooling capacity of coil in kW should be:

(a) 110.1 (b) 132.4
(c) 162.8 (d) 115.4

HPPSC ADF 05.03.2019

Ans. (*) : $m_a = 4$ kg/s

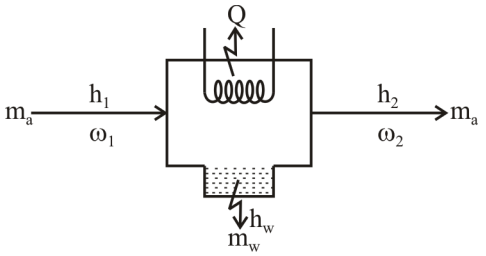
$\omega_1 = 20$ gram/kg of dry air

$h_1 = 90$ kJ/kg of dry air

$\omega_2 = 10$ gram/kg of dry air

$h_2 = 50$ kJ/kg of dry air

$h_w = 70$ kJ/kg



$m_w = (\omega_1 + \omega_2)m_a = (20 - 10) \times 4 = 40$ gram/s

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Energy balance,

$m_a h_1 = m_a h_2 + Q + m_w h_w$

$4 \times 90 = 4 \times 50 + Q + 0.04 \times 70$

$\boxed{Q = 157.2\,kW}$

654. A room contains 40 kg of dry air and 0.5 kg of water vapour. The total pressure and temperature of air in the room are 100 kPa and 27° C, respectively. Given that the saturation pressure for water at 27°C is 3.2 kPa, the relative humidity of the room is:

(a) 64.32% (b) 52.34%

(c) 61.56% (d) 67.37%

HPPSC ADF 05.03.2019

Ans. (c) : Relative humidity $\phi = \frac{P_v}{P_{vs}}$

$P_{vs} = 3.2$ Kpa Pv=?

Specific humidity $\omega = \frac{0.622 P_v}{P - P_v}$

$$\frac{0.5}{40} = \frac{0.622 \times P_v}{100 - P_v}$$

$50 - 0.5\,P_v = 24.88\,P_v$

$P_v = 1.97$ KPa

$$\phi = \frac{1.97}{3.2} = 0.615$$

$\boxed{\phi = 61.5\%}$

655. Match the list- I and list- II.

List-I	List-II
P : Cooling and dehumidification	1: DB increases and DP decreases
Q : Chemical dehumidification	2: DP increases and DB is constant
	3: DB and WB both decreases
	4: DB decreases and DP increases

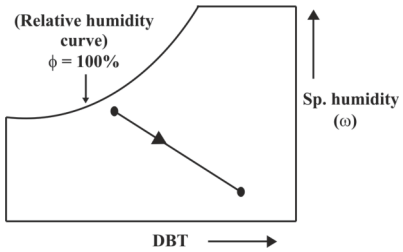
Here, DB = Dry bulb temperature : DP = Dew point temperature : WB = Wet bulb temperature

(a) P- 3, Q- 1 (b) P- 1, Q- 3

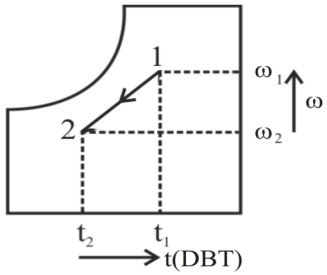
(c) P- 4, Q- 2 (d) P- 2, Q- 4

HPPSC ADF 05.03.2019

Ans. (a) :



Chemical dehumidification



Cooling and dehumidification

In cooling and dehumidification DB and WB both decreases.

In chemical dehumidification DB increases and DP decreases.

656. The humidity ratio of atmospheric air at 28°C [$P_{sat@28^oC,\ water} = 28.34$mm Hg] dry bulb temperature and 760 mm Hg pressure is 0.018 kg/kg of dry air. What will be the relative humidity of air?

(a) 96.13% (b) 75.42%

(c) 55.25% (d) 88.35%

SJVN ET 2019

Ans. (b) : Given, $P_t = 760$ mm Hg

$\omega = 0.018$ kg/kg of dry air

$P_s = 28.34$ mm Hg

$$\omega = \frac{0.622 P_v}{P_t - P_v}$$

$$0.018 = \frac{0.622\,P_v}{P_t - P_v}$$

$\Rightarrow (760 - P_v)0.018 = 0.6122\,P_v$

$\Rightarrow 760 \times 0.018 - 0.018 P_v = 0.622\,P_v$

$P_v = 21.375$ mm Hg

$$RH(\phi) = \frac{P_v}{P_s} = \frac{21.375}{28.34} = 75.42\%$$

657. Amount of moisture that air can hold increases by:

(a) Decreasing both Saturation Pressure and Temperature

(b) Decreasing Saturation Pressure and Increasing Temperature

(c) Increasing Saturation Pressure and Decreasing Temperature

(d) Increasing both Saturation Pressure and Temperature

OPSC AEE 2019 PAPER - II

Ans : (d) : Increasing both saturation pressure and temperature

658. In air conditioning design hunt load for an average adult male during idle condition can be considered as:

(a) 75 W (b) 115 W

(c) 175 W (d) 215 W

OPSC AEE 2019 PAPER - II

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Ans : (d) : In air conditioning design hunt load for an average adult male during idle condition can be considered as 215 W

659. For comfort condition the effective temperature for child as compared to adult is:
(a) Same (b) Higher
(c) Lower (d) Insufficient data

OPSC AEE 2019 PAPER - II

Ans : (b) : The children require higher effective temperature compared with adults.

660. During an air-conditioning of a plant, the room sensible heat load is 40 kW and room latent heat load is 10 kW, ventilation air is 25% of supply air. At full load, the room sensible heat factor will be
(a) 0.9 (b) 0.8
(c) 0.7 (d) 0.6

ESE 2019

Ans. (b) : $RSHF = \frac{RSH}{RSH + RLH} = \frac{RSH}{RTH}$

Where RSHF = Room sensible heat factor
RSH = Room sensible heat
RLH = Room latent heat
RTH = Room total heat

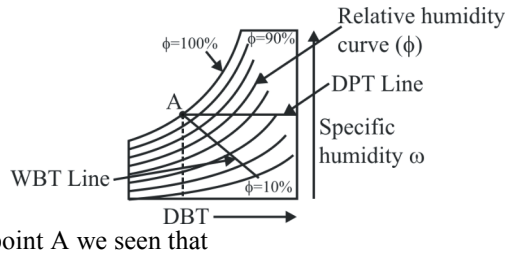
Given,
RSH = 40 kW
RTH = RSH + RLH = 50 kW

$RSHF = \frac{40}{50} = 0.8$

661. If the relative humidity of atmospheric air is 100%, then the wet-bulb temperature will be
(a) more than dry-bulb temperature
(b) equal to dew-point temperature
(c) equal to dry-bulb temperature
(d) less than dry-bulb temperature

ESE 2019

Ans. (b,c) : If relative humidity of atmospheric air is 100% then air is saturated and for saturated air DBT = WBT = DPT



At point A we seen that
DBT = WBT = DPT

662. Statement I: Referring to vapour compression refrigeration system, the coefficient of performance (COP) of a domestic refrigerator is less than that of a comfort air-conditioning plant.
Statement II: In domestic refrigerator, the work required for pumping the same amount of heat is more than that in an air-conditioning plant because of greater difference between condenser and evaporator temperatures.

ESE 2018

Ans. (a) : Evaporator temperature of refrigeration system is less than that of air conditioning system.
Hence work input in refrigerator is more than that of air conditioning system.
Therefore, COP of air conditioner is more than that of refrigerator. Compressor and evaporator temperature difference for domestic refrigerator is more than air-conditioning.

663. In a steam condenser, the partial pressure of steam and air are 0.06 bar and 0.007 bar, respectively. The condenser pressure is :
(a) 0.067 bar (b) 0.06 bar
(c) 0.053 bar (d) 0.007 bar

GPSC ARTO Pre 30.12.2018

Ans. (a) : Given,
Partial pressure of steam, $p_s = 0.06$ bar
Partial pressure of air, $p_a = 0.007$ bar
Condense pressure, $P_c = p_s + p_a$
$= 0.06 + 0.007$
$P_c = 0.067$ bar

664. A conference hall of the size 7m × 4m × 4m is filled with an air-water vapour mixture of 38 °C, atmospheric pressure of 1bar and RH 70%. The mass of air will be (assume Ps at 38°C = 0.06624 bar):
(a) 149.2 kg (b) 83.2 kg
(c) 119.7 kg (d) 169.7 kg

AAI Jr. Executive 29.11.2018

Ans. (c) : Sol. $0.7 = \frac{P_v}{P_{vs}} \Rightarrow P_v = 0.7 \times 0.06624 \times 10^5$

$P_V = 4636.8$
$PV = mRT$
$V = 7 \times 4 \times 4 = 112$
R = 0.286 kJ/kgK

$m_v = \frac{4636.8 \times 112}{0.286 \times 311}$

$m_v = 3.614 kg$

Specific humidity $(\omega) = \frac{m_v}{m_a} = \frac{0.622 \times P_v}{P_t - P_v}$

$P_t = 1 bar = 10^5 N/m^2$

$\frac{3.614}{m_a} = \frac{0.622 \times 4636.8}{95363.2}$

$m_a = 119.51 kg$

665. In a chemical dehumidification process, the dry bulb temperature of air
(a) Decreases
(b) Remains constant
(c) Increases
(d) Depends on wet bulb temperature

ISRO Scientist/Engineer (RAC) 22.04.2018

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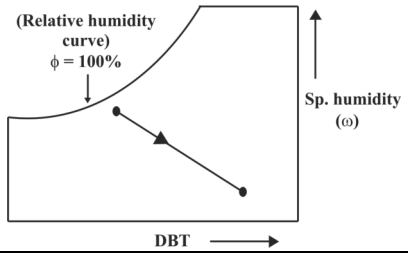
Ans. (c) : In a chemical dehumidification process, the dry bulb temperature of air increases.



666. During sensible heating of moist air, enthalpy-

(a) increases (b) decreases
(c) remains constant (d) none of the above

RPSC AE 2018

Ans. (a) : During sensible heating, the enthalpy of the moist air will increase.

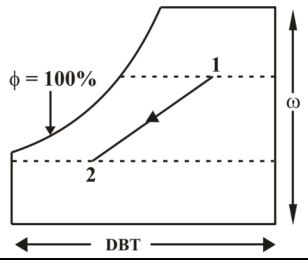
667. The relative humidity, during cooling and dehumidification of moist air-

(a) increases
(b) decreases
(c) can increase or decrease
(d) remains constant

RPSC AE 2018

Ans. (c) : The relative humidity, during cooling and dehumidification of moist air can increase or decrease

$\omega \downarrow, DBT \downarrow, WBT \downarrow$



668. In an unsaturated air the state of vapour is-

(a) Wet (b) Superheated
(c) Saturated (d) Unsaturated

RPSC AE 2018

Ans. (b) : In an unsaturated air the state of vapour is superheated.

669. Statement I: In air-conditioning, the atmospheric air (mixture of dry air and water vapour) can be considered as mixture of two ideal gases.

Statement II: In the temperature range used in air conditioning, the partial pressure of the water vapour is very low and it follows the ideal gas relation with negligible error.

ESE 2018

Ans. (a) : The air to be processed in air conditioning system is a mixture of dry air and water vapour.

When air and water vapour exist alone at moderate or low pressure, always the case for our conditions, they closely approximate ideal gases. In that case the relationship between pressure, temperature and mass can be expressed as,

$PV = mRT$

At low pressure, if their and water vapour are mixed together the interaction between the molecules of the two gases is modest so that the mixture can be considered an ideal gas.

670. Consider the following statements:

1. The relative humidity of air does not change with temperature as long as specific humidity remains constant.
2. Dew-point temperature is the temperature at which air is cooled at constant volume
3. Saturated air passing over a water surface does not cause change of air temperature
4. For saturated air, dry bulb, wet-bulb and dew point temperatures are identical

Which of the above statements are correct?

(a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 3 and 4

ESE 2018

Ans. (d) : (i) Relative humidity changes with DBT hence statement (1) is wrong.

(ii) Dew point temperature is the temperature at which when condensation happens at constant pressure. Hence statement (2) is also wrong therefore option (d) is correct.

671. The difference between dry bulb temperature and dew point temperature, is called

(a) Dry bulb depression
(b) Wet bulb depression
(c) Dew point depression
(d) Degree of saturation

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The difference between dry bulb temperature and dew point temperature is called Dew Point depression. For a constant temperature, the smaller the difference the more moisture is there and the higher the relative humidity.

672. The wet bulb depression is zero when relative humidity is

(a) 0 (b) 0.5
(c) 0.75 (d) 1.0

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Wet bulb depression is the difference between the dry bulb and wet bulb temperature. Hence, when DBT = WBT implies that relative humidity is 100% therefore when the wet bulb depression is zero the relative humidity is equal to one.

673. The alignment circle is marked on the psychometric chart at

(a) 20ºC DBT and 50% RH
(b) 26ºC DBT and 50% RH
(c) 20ºC DBT and 60% RH
(d) 26ºC DBT and 60% RH

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : Alignment circle is a reference point on the psychometric chart it generally lies within 24ºC to 26ºC DBT and 50% Relative humidity.

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674. In a psycrometric chart, specific humidity lines are:

(a) vertical and uniformly space
(b) horizontal and uniformly space
(c) horizontal and non-uniformly space
(d) curved lines

HPPSC Asstt. Prof. 18.09.2017

Ans. (b) :

675. The degree of warmth or cold felt by a human body depends mainly on :

(a) DBT (b) Relative humidity
(c) Air velocity (d) All of the above

Kerala PSC Poly. Lect., 2017

Ans. (d) : The degree of warmth or cold felt by a human body depends mainly on 'DBT', 'Relative humidity' and 'Air-velocity'.

Effective Temperature–It is the temperature of saturated air at which a person or human being would feel same level of comfort as in actual environment. It includes comfort temperature velocity, humidity and air motion.

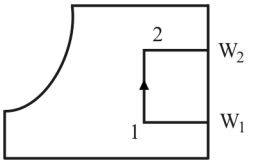
It is found that at an effective temperature of 20°C, 99% of people are comfortable in winter and in summer this temperature is found to be 21.6°C. The comfortable relative humidity conditions are 50% to 60%.

676. The humidification process on the psychrometric chart is shown by-

(a) Horizontal line (b) Veritical line
(c) Inclined line (d) Curved line

GMB AAE 25.06.2017

Ans. (b) : The humidification process on the psychometric chart shown by the vertical line



677. A low wet bulb temperature indicates

(a) very cool air
(b) very low humidity
(c) very low atmospheric temperature
(d) Both (a) and (b)

WBPSC AE, 2017

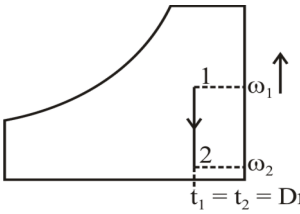
Ans. (d) : A low wet bulb temperature indicates very cool air and very low humidity.

678. A dehumidifier is required in

(a) A winter air-conditioning system
(b) A summer air-conditioning system
(c) Filters for removing dust particles
(d) A duct where air is circulated into required space

Karnataka PSC AE, 10.09.2017

Ans. (b) : The purpose of dehumidifier is to reduce humidity in the environment and used mainly in A.C. in summers.

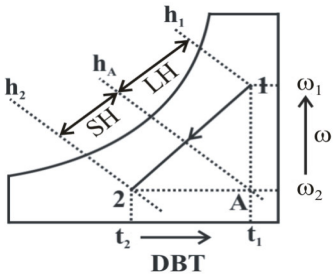


679. The sensible heat factor during cooling and dehumidification process is given by (where h_1 = Enthalpy of air entering the cooling coil, h_2 = Enthalpy of air leaving the cooling coil, and h_A = Enthalpy of air at the end of dehumidification process)

(a) $\frac{h_A - h_2}{h_1 - h_2}$ (b) $\frac{h_2 - h_A}{h_1 - h_2}$
(c) $\frac{h_1 - h_2}{h_A - h_2}$ (d) $\frac{h_A - h_1}{h_2 - h_1}$

Karnataka PSC AE, 10.09.2017

Ans. (a) :



$$SHF = \frac{\text{Sensible heat}}{\text{Total heat}} = \frac{SH}{SH + LH}$$

$$= \frac{h_A - h_2}{h_1 - h_2}$$

680. Which of the following is NOT a method for the design of Air-conditioning system?

(a) Velocity reduction method
(b) Equal friction method
(c) Equal pressure method
(d) Static regain method

Punjab PSC (Lect.) 06.08.2017

Ans. (c) : Method for the design of Air-conditioning system is

(i) Velocity Reduction Method
(ii) Equal Friction Method
(iii) Static Regain Method

681. The Dew point temperature of humid air is

(a) Always higher than the wet bulb temperature
(b) Same as the wet bulb temperature
(c) The temperature of air obtained on cooling it till saturation at constant pressure
(d) The temperature of air obtained on cooling it till saturation at constant volume

GPSC Asstt. Director of Transport 05.03.2017

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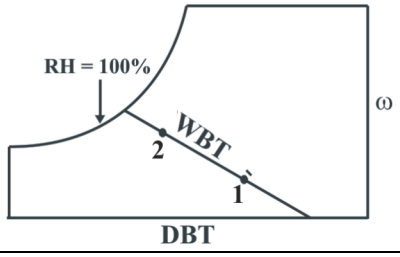
Ans. (c) : From psychrometric chart Dew Point Temperature (DPT) lines on psychometric charts are horizontal line upto the saturation curve.
So, for humid Air, the dew point temperature obtained on cooling it till saturation at constant pressure.

682. During evaporative cooling process, wet bulb temperature
(a) Increases (b) decreases
(c) remains constant (d) unpredictable
ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (C) : During evaporative cooling process, wet bulb temperature remains constant.

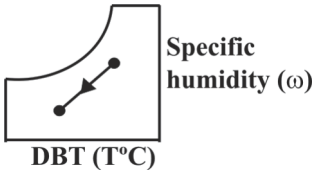


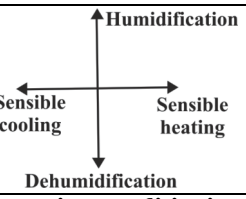
683. The bypass factor of cooling coil decreases with
(a) Decrease in fin spacing and increase in number of rows
(b) Increase in fin spacing and increase in number of rows
(c) Increase in fin spacing and decrease in number of rows
(d) Decrease in fin spacing and decrease in number of rows
ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (a) : The bypass factor of cooling coil decrease with decrease in fin spacing and increase in number of rows.

684. The air inside a space capsule is maintained at the required condition by passing it through a set of heat exchangers. The heat exchangers are designed such that the humid air passing through them is cooled and partial condensation of the water vapour occurs. On a psychometric chart, this process appears as a line which is
(a) Horizontal to the left
(b) Vertical downwards
(c) Diagonal downwards to the left
(d) Diagonal upwards to the right
ISRO Scientist/Engineer 07.05.2017

Ans. (c) : On a psychrometric chart, this process appears as a line which is diagonal to the left and this process is called cooling and dehumidification.





685. For winter air conditioning, the relative humidity should not be more than :
(a) 60% (b) 75%
(c) 40% (d) 90%
TRB Polytechnic Lecturer 2017

Ans. (c) : For winter air conditioning, the relative humidity should not be more than 40%.

686. The object for air conditioning a car is to control these in the
(a) Temperature and Pressure
(b) Pressure and Humidity
(c) Humidity and Temperature
(d) Humidity and Pressure
TNPSC AE 2017

Ans. (c) : The object for air conditioning a car is to control these in the Humidity and Temperature.

687. Air at 30°C and 1 bar has a specific humidity of 0.016 kg/kg of dry air. By considering the saturation pressure of water vapour at 30°C as 4.246kPa, the relative humidity of the air will be
(a) 66.1% (b) 60.2%
(c) 58.8% (d) 56.8%
ESE 2017

Ans. (c) : Given,
$T_{sat} = 30°C$, $P = 1$ bar
$P_{sat} = 4.246$ kPa
Specific humidity (ω) = 0.016 kg/kg of dry air
We know that

$$\omega = 0.622\left(\frac{P_V}{P - P_V}\right)$$

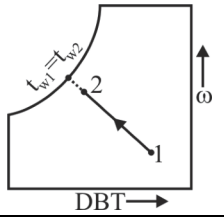
$$0.016 = 0.622\left(\frac{P_V}{1 - P_V}\right)$$

$$P_V = 0.025 \text{ bar} = 2.5 \text{ kPa}$$

$$\text{Relative humidity } (\phi) = \frac{P_V}{P_{sat}} = \frac{2.5}{4.246} = 0.5887 = 58.87\%$$

688. Consider the following statements in respect of an evaporative cooling process:
1. The wet-bulb temperature remains constant.
2. The dew-point temperature remains constant.
3. The enthalpy remains constant.
Which of the above statements are correct?
(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only
ESE 2017

Ans. (d) :



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The evaporative cooling process wet bulb temperature and enthalpy remains constant for process 1 – 2
Cooling and humidification

$T_{w1} = T_{w2}$

$h_1 = h_2$

Note- In this process (1–2)
- Relative humidity (RH) increases
- Specific humidity (ω) increases
- DBT decreases.

689. The atmospheric air at dry bulb temperature of 15ºC enters a heating coil maintained at 40ºC. The air leaves the heating coil at 22.5ºC. What is the by-pass-factor of the heating coil?

(a) 0.35 (b) 0.4
(c) 0.6 (d) 0.7
(e) 0.75

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : By pass Factor $= \frac{t_2 - t_3}{t_1 - t_3}$

Where,
t_1= Dry bulb temperature of air entering the heating Coil.
t_2= Dry bulb temperature of air leaving the heating coil
t_3= Dry bulb temperature of the heating coil.
t_1= 15ºC, t_2= 22.5ºC, t_3= 40ºC

By Pass factor $= \frac{22.5 - 40}{15 - 40}$
$= 0.7$

690. The by pass factor (B.P.E) is expressed as:

(a) $\frac{dt_1 - t_{coil}}{dt_2 - t_{coil}}$ (b) $\frac{dt_1 - t_{coil}}{dt_2 + t_{coil}}$
(c) $\frac{dt_1 + t_{coil}}{dt_2 + t_{coil}}$ (d) None of these

PTCUL AE 25.06.2017

Ans. (a) :

691. For air conditioning the operation theatre in a hospital, the percentage of outside air in the air supplied is:

(a) Zero (b) 20
(c) 50 (d) 100

HPPSC Asstt. Prof. 18.09.2017

Ans. (d) : For air conditioning the operation theatre in a hospital, the percentage of outside air in the air supplied is 100.

692. A human body feels comfortable when the heat produced by the metabolism of human body is equal to :

(a) Heat dissipated to surroundings
(b) Heat stored in the human body
(c) Sum of Heat dissipated to surroundings and Heat stored in the human body
(d) Difference of Heat dissipated to surrounding and Heat stored in the human body

MPPSC AE 2016

Ans. : (c) A human body feels comfortable when the heat produced by the metabolism of human body is equal to sum of heat dissipated to surrounding and heat in the human body.

693. Where, H_1 = Total heat of air entering the heating coil
H_2 = Total heat of air leaving the heating coil
H_3 = Total heat of air at the end of the humidification
What is the sensible heat factor during the heating and humidification process equal to:

(a) $(H_3 - H_1)/(H_3 + H_1)$ (b) $(H_2 - H_1)/(H_3 - H_1)$
(c) $(H_3 + H_1)/(H_3 + H_2)$ (d) $(H_1 + H_2)/(H_1 + H_3)$

RPSC VPITI 14.02.2016

Ans. (b) : Sensible heat factor is the ratio of sensible heat to total heat.
Sensible heat = $H_2 - H_1$
Total heat = $H_3 - H_1$
Sensible heat factor during heating and humidification

$= \frac{H_2 - H_1}{H_3 - H_1}$

694. For the unsaturated air, wet bulb temperature is

(a) More than dry bulb temperature
(b) Twice the dew point temperature
(c) Less than dry bulb temperature
(d) Less than dew point

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : For unsaturated air wet bulb temperature is less than dry bulb temperature because of evaporative cooling.
In case of unsaturated air, DBT > WBT > DPT
In case of saturated air DBT = WBT = DPT

WBD = 0

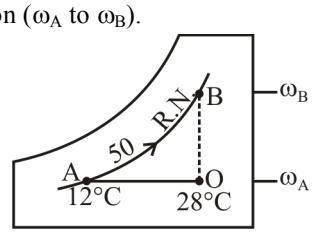
WBD = Wet Bulb depression

695. Atmospheric air from 12 °C and 50% relative humidity can be brought to 28 °C and 50% relative humidity by:

(a) Heating and humidification process
(b) Heating and dehumidification process
(c) Sensible heating process
(d) Adiabatic saturation process
(e) Chemical heating

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : Path A to B is a process of heating as well as humidification (ω_A to ω_B).



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696. When atmospheric air is heated at constant pressure, then which one is not correct

(a) Humidity ratio does not change
(b) Relative humidity increases
(c) Dew point temperature does not change
(d) Dry bulb temperature remain constant

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b,d) : When atmospheric air is heated at constant pressure, then relative humidity decreases & dry bulb temperature also increases.

697. To determine the relative humidity of an air-water vapour mixture, a psychrometer uses..... thermometer(s).

(a) one (b) two
(c) three (d) zero

HPPSC AE (PWD) 10.09.2016

Ans. (b) : To determine the relative humidity of an air water vapour mixture a psychrometer uses two thermometers.

698. Which of the following parameter increases during heating and humidification process?

(a) Humidity ratio
(b) Dry bulb temperature
(c) Saturation temperature
(d) All of the above
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (d) : During heating and humidification process,
- Humidity ratio
- Dry bulb temperature
- Saturation temperature

699. A cooling coil with a bypass factor of 0.15 and apparatus dew point (adp) of 10^0C comes in contact with air having a dry-bulb temperature of 35^0C and dew point of 15^0C. Over the cooling coil the air would undergo

(a) Sensible cooling
(b) Cooling and dehumidification
(c) Cooling and humidification
(d) Adiabatic saturation

UPRVUNL AE 21.08.2016

Ans. : (b) Given, BPF = 0.15, ADP = 10°C, DBT = 35°C, DPT = 15°C

$$\text{BPF} = \frac{t_{d_2} - t_{d_3}}{t_{d_1} - t_{d_3}}$$

$$0.15 = \frac{t_{d_2} - 10}{35 - 10} \Rightarrow t_{d_2} = 13.75$$

$$t_{d_2} < \text{DPT}$$

Hence, cooling and dehumidification takes place.

700. If wet bulb depression is equal to the sum of range and approach of a cooling tower then the water

(a) Outlet temperature is equal to dry bulb temperature of ambient air
(b) Inlet temperature is equal to the wet bulb temperature of ambient air
(c) Outlet temperature is equal to the wet bulb temperature of ambient air
(d) Inlet temperature is equal to dry bulb temperature of ambient air

UPRVUNL AE 21.08.2016

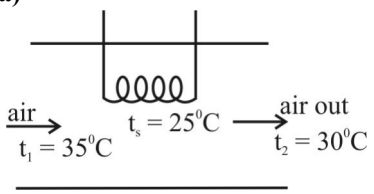
Ans. : (d) If wet bulb depression is equal to the sum of range and approach of a cooling tower, then the water inlet temperature is equal to dry bulb temperature of ambient air.

701. When 35^0C and 30^0C are the dry bulb temperatures of air entering and leaving a cooling coil, then the cooling coil temperature is 25^0C. What is the bypass factor for the case of sensible cooling?

(a) 0.50 (b) 0.20
(c) 0.25 (d) 0.33

UPRVUNL AE 21.08.2016

Ans. : (a)



$$\text{By pass factor (BPF)} = \frac{t_2 - t_s}{t_1 - t_s}$$

$$= \frac{30 - 25}{35 - 25} = 0.5$$

By pass factor = 0.5

702. Ratio of latent heat transfer to total heat transfer is given by

(a) Sensible heat factor (SHF)
(b) 1 - SHF
(c) $(SHF)^2$
(d) $1 - (SHF)^2$

RPSC LECTURER 16.01.2016

Ans. (b) : We know that
Sensible heat factor (SHF)

$$= \frac{\text{Sensible heat}}{\text{Sensible heat + Latent heat}} = \frac{\text{SH}}{\text{SH+LH}}$$

$$(1 - SHF) = 1 - \frac{SH}{SH + LH}$$

$$(1 - SHF) = \frac{SH + LH - SH}{SH + LH}$$

$$\frac{LH}{\text{Total heat}} = (1 - SHF)$$

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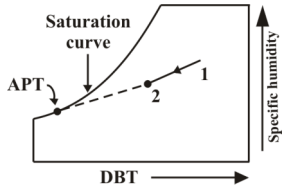
703. On pyschormetric chart, when condition line is extended to meet the saturation curve at a point, the temperature of this point is known as

(a) Dry bulb temperature
(b) Wet bulb temperature
(c) Apparatus dew point
(d) Atmospheric temperature

RPSC Lecturer 16.01.2016

Ans. (c) : On pyschormetric chart, when condition line is extended to meet the saturation curve at a point, the temperature of this point is known as Apparatus dew point temperature.



704. If sensible and latent heating transfer in an air conditioning process is 2.5 kW and 1.5 kW respectively. Then sensible heat factor (SHF) will be

(a) 1 (b) Zero
(c) 0.5 (d) 0.625

UPRVUNL AE 2016

Ans. (d) : We know that

$$\text{Sensible heat factor} = \frac{\text{Sensible heat}}{\text{Total heat}}$$

$$= \frac{\text{Sensible heat}}{\text{Sensible heat + Latent heat}}$$

$$= \frac{2.5}{2.5+1.5} = \frac{2.5}{4} = 0.625$$

705. Sensible heat factor (SHF) line is inclined at an angle of 45° with temperature axis on psychrometric chart. It's SHF will be:

(a) 0.59 (b) 0.49
(c) 0.29 (d) 0.39

UPRVUNL AE 2016

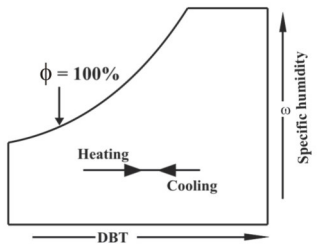
Ans. (b) : Sensible heat factor (SHF) line is inclined at an angle of 45° with temperature axis on psychrometric chart. It's SHF will be 0.50.

706. Sensible heating/cooling process on psychometric chart is:

(a) Along constant relative humidity line
(b) Along constant specific humidity line
(c) Along constant specific volume line
(d) Along constant enthalpy line

UPRVUNL AE 2016

Ans. (b) :



Sensible heating/cooling process on psychometric chart is along constant specific humidity line.

707. The constant partial pressure lines on the psychrometric chart are

(a) horizontal lines (b) vertical lines
(c) inclined lines (d) curved lines

APPSC AEE 2016

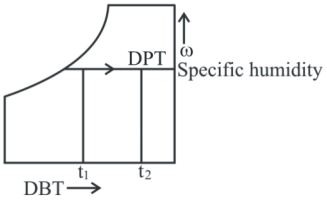
Ans. (a) : The constant partial pressure lines on the psychrometric chart are horizontal lines.

708. If the specific humidity of moist air remains the same but its DBT increases, its DPT

(a) Remains the same
(b) Increases
(c) Decreases
(d) May increase or decrease depending on its relative humidity

UPPSC AE 12.04.2016 Paper-II

Ans : (a) For the psychometric chart, we have seen that dew point temperature remains same.



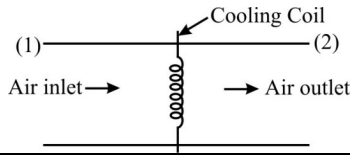
Dew point temperature line is horizontal and parallel to specific humidity line

709. If the air is passed over the cooling coils then this process is termed as

(a) Sensible heating
(b) Cooling with humidification
(c) Cooling with dehumidification
(d) None of the above

UPPSC AE 12.04.2016 Paper-II

Ans : (c) If the air is passed over the cooling coils then this process is termed as cooling with dehumidification.



Dehumidification is process of remaining mixture from the air at constant dry bulb temperature. there for, dehumidification, of air will take place if the coil surface temperature is below the dew point temperature of entering air.

710. What is air conditioning means?

(a) Maintaining comfort temperature and humidity
(b) maintaining odorless and clean air at comfortable velocity
(c) Both, option A and B
(d) None of the above

GPSC Lect. (Auto) 16.10.2016

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Ans. (c) : Air conditioning is the control of temperature, humidity purity and motion of air in an enclosed space.

711. In a moist air mixture, the dew point temperature is ..... the wet bulb temperature.

(a) more than (b) less than
(c) always equal to (d) not comparable to

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : For unsaturated moist air mixture-

$$DBT > WBT > DPT$$

For saturated moist air mixture-

$$DBT = WBT = DPT$$

712. Sensible heat is the heat required to ..... .

(a) change vapour into liquid
(b) change liquid into vapour
(c) increase the temperature of a liquid or vapour
(d) None of three mentioned here

GWSSB DEE 07.07.2016

Ans. (c) : Sensible heat is the heat required to increase the temperature of a liquid or vapour.

713. In a split type air conditioner, the unit kept outside the room is

(a) compressor, condenser and fan
(b) evaporator, fan
(c) condenser and fan
(d) compressor and fan

GPSC AE 26.07.2015

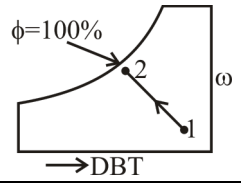
Ans. (a) : In a split type air conditioner, the unit kept outside the room is compressor, condenser and fan.

714. Adiabatic saturation process for moist air involves:

(a) Cooling and humidification
(b) Cooling and dehumidification
(c) Change in WBT
(d) Constant relative humidity

TSPSC Managers, 2015

Ans. (a) : Cooling and humidification process is one of the most commonly used air conditioning application for the cooling purpose. In this process the moisture is added to the air by passing it over the stream or spray of water at which temperature lower than the dry bulb temperature of the air.



715. On a particular time of a day, if the dry bulb temperature is equal to the wet bulb temperature, then it can safely concluded that-

(a) Specific humidity of air was 0.010 kg per kg of dry air
(b) Dew point was less than 20°C
(c) Relative humidity of air was 100 %
(d) Atmospheric air was free from moisture

NPCIL ET, 2015

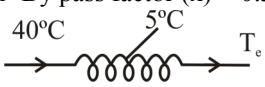
Ans. (c) : On a particular day (time), if the dry bulb temperature is equal to the wet bulb temperature, then safely concluded that relative humidity of air was 100%
At DBT = WBT = Relative humidity 100%

716. The bypass factor of a cooling coil is 0.3, when its coil surface temperature is 5°C and air enters at 40°C. The exit air temperature is:

(a) 10°C (b) 12.5°C
(c) 15.5°C (d) 20°C

NPCIL ET, 2015

Ans. (c) : Given- By pass factor (x) = 0.3



$$\text{By pass factor (x)} = \frac{T_e - 5}{40 - 5}$$

$$0.3 = \frac{T_e - 5}{35}$$

$$T_e - 5 = 10.4$$

$$T_e = 10.5 + 5$$

$$T_e = 15.5^{\circ}C$$

717. Free air is the air at :

(a) 20°C and 1 kg /cm^2 and relative humidity of 36%
(b) 25°C 1 kg/cm^2 and relative humidity of 50%
(c) 0°C and standard atmospheric conditions
(d) atmospheric conditions at any specific location.

GWSSB AAE, 27.12.2015

Ans. (d) : free air refer to air at atmospheric condition at any specific location.

718. The weight of the water vapour in kg contained in 1 m^3 of air-vapour mixture at its total pressure is known as

(a) Degree of saturation (b) Percent humidity
(c) Relative humidity (d) Vapour density

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : The weight of the water vapour in kg contained in 1m^3 of air-vapour mixture at its total pressure is known as vapour density.

719. Consider the following statements pertaining to Room Sensible Heat Ratio (RSHR) line:

1. It is the line draw on psychrometric chart through the room conditions.
2. Its slope is the ratio of sensible cooling load to room latent heat load.
3. The supply air condition must lie on this line for satisfactory performance.

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Which of the statements given above are correct?

(a) 1, 2 and 3 (b) 1 and 2
(c) 1 and 3 (d) 2 and 3

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : ■ It is the line drawn on psychometric chart through the room condition.
■ The supply air condition must lie on this line for satisfactory performance.

720. Which of the parameter remains constant during evaporative cooling process through an air washer with the same water recirculated again and again

(a) Wet bulb temperature
(b) Dry bulb temperature
(c) Dew point
(d) Absolute humidity

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Evaporative Cooling–This system is based on the principle that when moist but unsaturated air comes in contact with a wetted surface whose temperature is higher than the dew point temperature of air. These systems are attractive for hot and dry climates.

721. A cooling tower is to be installed in a place where dry bulb temperature and wet bulb temperature are almost constant throughout? Such a proposition is

(a) Excellent
(b) Not desirable
(c) Can be considered
(d) Other data are required to determine the same

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : Wet bulb temperature is an important factor in performance of evaporative water cooling equipment. It is a controlling factor from the aspect of minimum cold water temperature to which water can be cooled by the evaporative method. Thus wet bulb temperature of the air entering the cooling tower determines operating temperature levels throughout the plant. Theoretically a cooling tower will cool water to the entering wet bulb temperature, when operating without a heat load.

722. Delignification (eating up of wood) in cooling tower is controlled by

(a) Controlling blow down to keep the concentration of soluble and scale forming solids below a limit
(b) Softening the makeup water with lime several phosphates
(c) Adjusting pH between 7 to 7.5
(d) Adding chlorine, copper sulphate, potassium permanganate etc. to the circulating water

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Delignification (eating up of wood) in cooling tower is controlled by adjusting pH between 7 to 7.5.

723. If air is heated without changing its moisture content, the dew point will-

(a) increases (b) remain same
(c) decrease (d) unpredictable

ISRO Scientist/Engineer (RAC) 29.11.2015

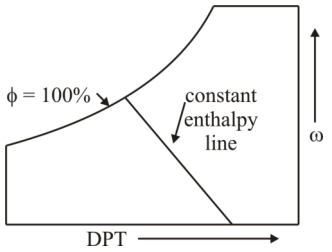
Ans : (b) If air is heated without changing its moisture content means specific humidity remains constant and therefore its dew point temperature will be remain same after heating at constant moisture content.

724. On a Psychrometric chart, adiabatic process follows

(a) constant DBT lines
(b) Constant relative humidity lines
(c) Constant DPT lines
(d) Constant enthalpy lines

TSPSC AEE 2015

Ans. (d) :



In psychrometric charge adiabatic process follows constant enthalpy lines.

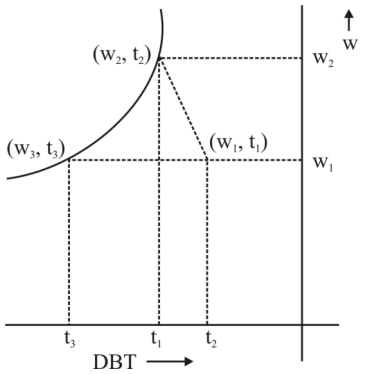
725. An air water vapour mixture has a DBT of 60° c and a DPT of 40°c. The WBT (tw) for the above mixture would be

(a) Less than 40°C
(b) 40°C
(c) 60°C
(d) $40^{o}C < T_w < 60^{o}C$

TSPSC AEE 2015

Ans. (d) : We know that
DBT > WBT > DPT
So, $40^{o}C < T_W < 60^{o}C$
For saturated vapour-
DBT = WBT = DPT

726. Psychrometric chart



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(a) $t_1 \rightarrow DBT, t_2 \rightarrow DBT, t_3 \rightarrow WBT$

(b) $t_1 \rightarrow DBT, t_2 \rightarrow WBT, t_3 \rightarrow DBT$

(c) $t_1 \rightarrow WBT, t_2 \rightarrow DBT, t_3 \rightarrow DPT$

(d) $t_1 \rightarrow DBT, t_2 \rightarrow WBT, t_3 \rightarrow DPT$

TSPSC AEE 2015

Ans. (d) : $t_1 \rightarrow DBT, t_2 \rightarrow WBT, t_3 \rightarrow DPT$

727. Heating of air without changing its moisture content takes place on a psychometric chart along :

(a) A horizontal line of constant dew point

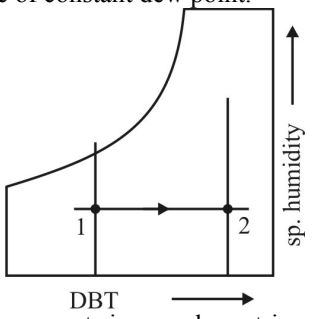
(b) Rising line

(c) Falling line

(d) Curved line

OPSC AEE 2015 PAPER - II

Ans : (a) Heating of air without changing its moisture content takes place on a psychometric chart along a horizontal line of constant dew point.



line – 1-2 represent in psychometric chart heating without changing moisture content.

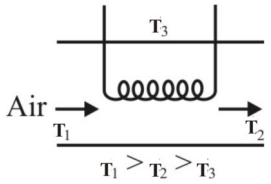
728. If T_1 and T_2 are dry bulb temperature of air entering and leaving the coil and T_3 is ADP temperature (temperature of cooling coil), then T_1 bypass factor for sensible cooling of air is:

(a) $\frac{T_3 - T_1}{T_2 - T_1}$ (b) $\frac{T_1 - T_2}{T_2 - T_3}$

(c) $\frac{T_2 - T_3}{T_1 - T_3}$ (d) $\frac{T_2 - T_3}{T_1 - T_2}$

OPSC AEE 2015 PAPER - II

Ans : (c)



$$\text{By pass factor}(BPF) = \frac{T_2 - T_3}{T_1 - T_3}$$

729. During mixing of steams in case of air conditioning, the process is associated with:

(a) Throttling (b) Adiabatic

(c) Isobaric (d) Isochoric

OPSC AEE 2015 PAPER - II

Ans : (b) During mixing of steams in case of air conditioning, the process is associated with adiabatic.

730. The desuperheating coil in a condenser performs the function of:

(a) improving the heat of superheat of the refrigerant before it goes into condenser

(b) Removing heat from condenser

(c) Improving condenser performance

(d) To humidity the refrigerant before it goes into condenser

OPSC AEE 2015 PAPER - II

Ans : (c) The desuperheating coil in a condenser performance the function of improving condenser performance.

731. Consider the following statements.

(1) As the by-pass factor (BPF) of the cooling coil increases, temperature difference between air at the outlet of the coil and coil ADP decreases

(2) The BPF of the coil increases as the velocity of air through the coil increases

(3) The BPF of the coil increases as the fin pitch increases

(4) The BPF of the coil decreases as the number of rows in the flow direction increase

Of the above, which is correct?

(a) 1, 2 and 3 (b) 1 and 3

(c) 2 and 4 (d) 2, 3 and 4

(e) 1, 2, 3 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : By-pass Factor (BPF) is part of the total air through the coil which fails to come into contact with the surface of the cooling coil.

1. The BPF of the coil increase as the velocity of air through the coil increase.
2. The BPF of the coil increase as the fin pitch increase.
3. The BPF of the coil decrease as the number of rows in the flow direction increase.

732. Consider the following statements.

1. The air conditioning load on a building increases, if 0.4% design value is used for outside conditions instead of 1.0% value for summer
2. The air conditioning load on a building decreases, if 0.4% design value is used for outside conditions instead of 1.0% value for summer
3. For winter air conditioning, a conservative approach is to select 99.6% value for the outside design conditions instead of 99% value
4. For winter air conditioning, a conservative approach is to select 99% value for the outside design conditions instead of 99.6% value

Of the above, which is correct

(a) 1 and 2 (b) 2 and 3

(c) 2 and 4 (d) 1 and 3

(e) 1, 3 and 4

CGPSC AE 26.04.2015 SHIFT-II

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Ans. (d) : 1. The air conditioning load on a building increases, if 0.4% design value is used for outside conditions instead of 1.0% value for summer
2. For winter air conditioning, a conservative approach is to select 99.6% value for the outside design conditions instead of 99% value

733. Consider the following statements

1. Thermodynamic WBT is a property of moist air, while WBT as measured by wet bulb thermometer is not a property
2. Both the thermodynamic WBT and WBT as measured by wet bulb thermometer are properties of moist air
3. Under no circumstances, dry bulb and wet bulb temperatures are equal
4. Wet bulb temperature is always lower than dry bulb temperature, but higher than dew point temperature

Of the above, which is correct?

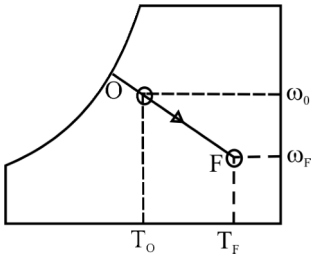
(a) 1 (b) 2
(c) 3 (d) 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : Wet bulb temperature is always lower than dry bulb temperature, but higher than dew point temperature

$t_b > t_w > t_d$

734. The process shown in the psychometric process is



(a) Sensible heating
(b) Sensible cooling
(c) Cooling and humidification
(d) heating with humidification
(e) Chemical dehumidification

CGPSC AE 26.04.2015 SHIFT-II

Ans. (e) : The process shown in the psychometric process is chemical dehumidification.

$T_d \uparrow$, T_w = Const, $\omega\downarrow$, $\phi\downarrow$
h = Constant

735. Water in an insulated evaporative cooler evaporates at the rate of 0.005 kg/s. Air flow rate is 1.2 kg/s. What is the air temperature decrease if the specific heat of humid air is 1kJ/kg K and latent heat of water is 2700 kJ/kg?

(a) 2.7^0C (b) 5.25^0C
(c) 11.25^0C (d) 13.25^0C
(e) 14.75^0C

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : Data given,

$\dot{m}_w = 0.005\,kg/s$

$\dot{m}_A = 1.2\,kg/s$

$(LH)_w$ = 2700 kJ/kg
C_p = 1 kJ/kg K
ΔT = ?
We know that,
Heat gained by water will be equal to heat lost by air

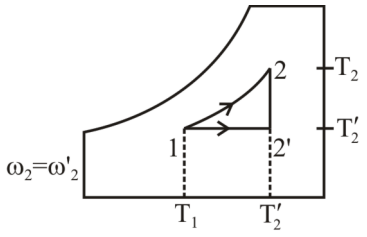
$$\dot{m}_w \times (LH)_w = \dot{m}_A\, C_p \Delta T$$
$$0.005 \times 2700 = 1.2 \times 1 \times \Delta T$$
$$\boxed{\Delta T = 11.25^0 C}$$

736. In spray humidification process, total heat

(a) Remains same (b) Decreases
(c) Increases (d) Unpredictable

MPPSC State Forest Service Exam, 2014

Ans. (a) :



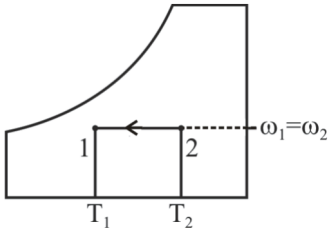
1-2' is heating, $T_{2'} > T_1$
2'-2 is humidification
So, in spray humidification process total heat remains same.

737. Cooling of air without any change in its specific humidity, is called

(a) Sensible heating (b) Humidification
(c) Dehumidification (d) Sensible cooling

MPPSC State Forest Service Exam, 2014

Ans. (d) : From psychrometric chart,



$\Delta\omega = 0$ or constant ω
1 to 2 is a sensible cooling process in which the specific humidity remains constant.

738. During air conditioning of a space the sensible heat added is 100 Kcal /sec and the latent heat added is 50 cal/sec, then the sensible heat factor is :

(a) 0.25 (b) 0.33
(c) 0.50 (d) 0.67

MPSC Poly. Lect. 2014

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Ans. (d) : Given,
Sensible Heat (SH) = 100 Kcal/sec.
Latent Heat (LH) = 50 Kcal/sec.
Sensible Heat Factor (SHF)

$$= \frac{SH}{SH + LH} = \frac{100}{100 + 50} = 0.6667$$

739. Grand total heat is :
(a) RLH+TSH (b) TLH+RLH
(c) TSH-TLH (d) TSH+TLH

TNPSC AE (Industries) 09.06.2013

Ans. (d) : Grand total heat = TSH+TLH

740. Dew point temperature is constant as long as there is :
(a) no change in moisture content of the air
(b) no change in volume of air
(c) no change in dry bulb and wet bulb temperature
(d) no change in relative and specific humidity of air

J&K PSC Civil Services Pre, 2013

Ans. (a) : Dew point temperature is constant as long as there is no change in moisture content of the air.

741. Air is normally dehumidified by :
(a) injecting water (b) passing steam
(c) heating (d) cooling

J&K PSC Civil Services Pre, 2013

Ans. (d) : Air is normally dehumidified by cooling.
When temperature of air becomes less than Dew point temperature then air will be dehumidified.

742. Consider the following psychometric terms:
A. Degree of saturation B. Specific humidity
C. Humidity ratio D. Relative humidity
Which of the above terms have the same meaning?
(a) (A) and (B) (b) (B) and (C)
(c) (A), (B) and (C) (d) (C) and (D)

RPSC AEN Pre-2013

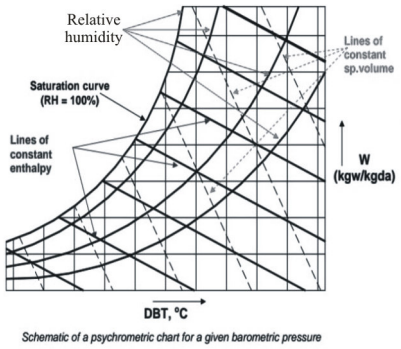
Ans. (b) : Specific humidity or humidity ratio-
It is defined as the ratio of mass of water vapour per kg of dry air in the given volume and given temperature.
Specific humidity / humidity ratio (ω).

$$(\omega) = \frac{m_v}{m_a} = \frac{0.622 P_v}{P - P_v}$$

743. Which of the following psychrometric properties is not shown as straight inclined lines on the psychometric chart?
(a) Wet bulb temperature
(b) Enthalpy
(c) Specific volume
(d) Relative humidity

RPSC AEN Pre-2013

Ans. (d) :



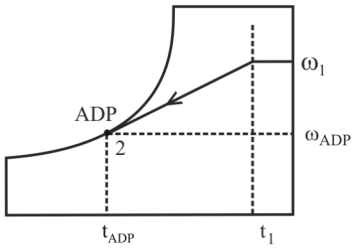
Schematic of a psychrometric chart for a given barometric pressure

(h) Enthalpy = Inclined uniformly spaced
WBT = Inclined but non uniformly space
(Sp. volume) (v) = Highest degree of inclination
(h) Enthalpy, WBT = same degree of inclination.

744. What is the full form of the term ADP employed in air conditioning?
(a) Apparatus dew point
(b) Adiabatic dew point
(c) Ambient design point
(d) Air conditioning design point.

RPSC AEN Pre-2013

Ans. (a) : ADP employed for air conditioning-
ADP = Apparatus dew point.

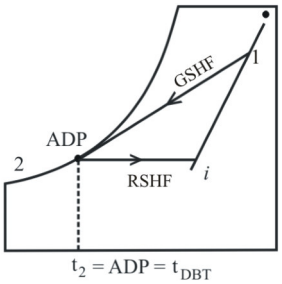


Apparatus dew point is obtained when cooling & dehumidification line cut the saturation curve.

745. What is the line on the psychometric chart joining the state of mixture of fresh air and re-circulated air entering the cooling coil to the state of ADP of cooling coil called ?
(a) Mean SHF line
(b) Room SHF line
(c) Effective SHF line
(d) Grand SHF line

RPSC AEN Pre-2013

Ans. (d) : Grand SHF line –



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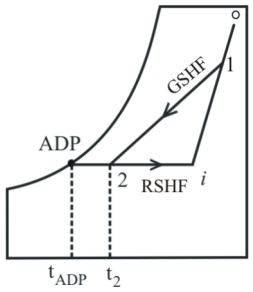
RSHF – For non- zero bypass factor.



746. In winter air conditioning, the inside design conditions are given by the following:-
(a) 25°C DBT, 50% RH
(b) 25°C DBT, 60% RH
(c) 21°C DBT, 50% RH
(d) 21°C DBT, 60% RH

UKPSC AE-2013, Paper-II

Ans. (d) :

747. Effective temperature in respect of air conditioning is the temperature of:-
(a) Unsaturated air
(b) Saturated air
(c) Dry air
(d) Either (b) or (c) above

UKPSC AE-2013, Paper-II

Ans. (b) : Effective temperature in respect of air conditioning is the temperature of saturated air.

748. During adiabatic saturation process of air in air washer, the water is sprinkled at a surface temperature equal to —— of incoming air:-
(a) DBT (b) WBT
(c) DPT (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (b) :

749. When steam is injected into the incoming air, the following changes occur in the condition of air:-
(a) Temperature increases significantly but specific humidity increases negligibly.
(b) Temperature increases negligibly but specific humidity increases appreciably.
(c) Both temperature and sp. humidity increase appreciably.
(d) Both temperature and sp. humidity increase negligibly.

UKPSC AE-2013, Paper-II

Ans. (b) : When steam is injected into the incoming air then temperature increase negligibly but specific humidity increases appreciably of the air.

750. In the concept of effective temperature, the relative humidity of air is considered to be:-
(a) 0% (b) 50%
(c) 75% (d) 100%

UKPSC AE-2013, Paper-II

Ans. (b) :

751. For atmospheric air, following conditions are given:-
DBT = 35°C, DPT = 23°C and barometer reads 750 mm Hg. Saturation pressure at 23°C and 35°C is 21.06 mm Hg and 42.4 mm Hg respectively. The relative humidity and partial pressure of water vapour will be:-
(a) 49.7% and 21.06 mm Hg respectively
(b) 35.5% and 42.4 mm Hg respectively
(c) 35.5% and 15.06 mm Hg respectively
(d) Impossible to calculate

UKPSC AE-2013, Paper-II

Ans. (a) : Partial Pressure corresponds to DPT
P_v= 21.06 mm Hg
P_s= 42.4 mm Hg

$$\phi = \frac{P_v}{P_S} = \frac{21.06}{42.4} = 0.4967$$

$$\boxed{\phi \simeq 49.67\%}$$

752. A sample of moist air is at a dry bulb temperature of 21°C and dew point temperature of 15°C. Specific humidity of air is 0.011 kg w.v./kg d.a. What will be the enthalpy of moist air?
(a) 59.4 kJ/kg d.a. (b) 323.4 kJ/kg d.a.
(c) 52.5 kJ/kg d.a. (d) 49.0 kJ/kg d.a.

UKPSC AE-2013, Paper-II

Ans. (d) : Given as, t_{dbt} = 21°C, t_{dpt} = 15°C
ω = 0.011 kg w.v./kg d.a.
We know that enthalpy of moist air

$$h = 1.005\,t_{dbt} + \omega\left[2500 + 1.88\,t_{dbt}\right]$$

$$h = 1.005 \times 21 + 0.011 \times [2500 + 1.88 \times 21]$$

h = 49 kJ/kg d.a.

753. The necessary data required for the design of the air conditioning apparatus is/are:-
(a) Grand total heat
(b) Dehumidified air quantity
(c) Apparatus dew point
(d) All of the above

UKPSC AE-2013, Paper-II

Ans. (d) :

754. The ratio of the actual partial pressure of the water vapour of the air-vapour mixture, to the pressure of the saturated water vapour at the same dry bulb temperature of the air is known as
(a) relative humidity
(b) absolute humidity
(c) specific humidity
(d) degree of saturation

TNPSC ACF 2012

Ans. (a) : Relate humidity $= \frac{P_V}{P_{VS}}$

P_V = actual pressure of the water vapour
P_{VS} = the pressure of the saturated water vapour.

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755. The dry bulb, wet bulb and dew point temperature will all the same when the air is

(a) completely dry
(b) fully saturated
(c) partially wet
(d) at atmopsheric pressure

TRB Poly. Lect., 2012

Ans. (b) : The dry bulb, wet bulb & dew point temperature will be all are same, when the air is completely saturated.

756. When air is at saturated state, which pair, out of the given alternatives is not correctly matched ?

List-I		List-II
(a) Relative humidity	-	100%
(b) DBT	-	WBT
(c) Degree of saturation	-	1
(d) Specific humidity	-	0.01 kg w.v/kg d.a

UKPSC AE 2012 Paper–II

Ans. (d) : Specific humidity - 0.01 kg w.v/kg d.a

757. In case of air conditioning of auditoriums, the cooling load which is predominant is

(a) lighting load
(b) occupancy load
(c) load due to fans
(d) load due to electronic equipments

UKPSC AE 2012 Paper–II

Ans. (b) : occupancy load

758. The design condition of temperature for winter air conditioning is

(a) 25 ± 1 °C (b) 27 °C
(c) 21 °C (d) none of the above

UKPSC AE 2012 Paper–II

Ans. (c) : 21 °C

759. In summer air conditioning system fresh air is introduced into the recirculated air to

(a) reduce load on equipment
(b) exercise easy control over equipment
(c) improve air quality by diluting odour and contaminants
(d) reduce quantity of supply air

UKPSC AE 2012 Paper–II

Ans. (c) : improve air quality by diluting odour and contaminants

760. Temperature recorded by a thermometer which is not affected by moisture is

(a) dry bulb temperature
(b) wet bulb temperature
(c) dew point temperature
(d) adiabatic saturation temperature

UKPSC AE 2012 Paper–II

Ans. (a) : The dry bulb temperature refers basically to the ambient air temperature. It is called "Dry Bulb" because the air temperature is indicated by a thermometer not affected by the moisture of the air.

761. In cooling and dehumidifying apparatus, the effect of the by pass factor is to

(a) lower the ADP of the cooling coil
(b) decrease the COP of the system
(c) both (a) & (b) above
(d) increase the ADP of the cooling coil and to improve the COP of the system

UKPSC AE 2012 Paper–II

Ans. (c) : both (a) & (b) above

762. Ambient air dry bulb temperature is 45°C and wet bulb temperature is 27°C. Select the lowest possible condensing temperature from the following for an evaporative cooled condenser.

(a) 25°C (b) 30°C
(c) 42°C (d) 48°C

VIZAG STEEL MT 10.06.2012

Ans. (d) : Condensing temperature should be greater than ambient dry bulb temperature.

In the option only (D) is satisfied, which have temperature $T_c > 45°$

So, $\boxed{t_{condensing} = 48^0 C}$

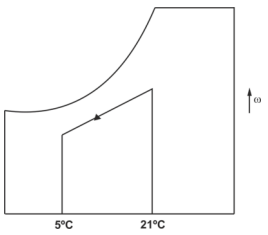
763. 1kg of moist air of RH 70% at 21°C is cooled at constant pressure of 1 bar to 5°C. The vapour pressure at 21°C and 5°C are 0.025 bar and 0.0087 bar. The percentage of water vapour that condenses into water, at 5°C is

(a) 66% (b) 85%
(c) 51% (d) 17%

ISRO Scientist/Engineer 2011

Ans. (a) : Moist air, mass (m) = 1 kg

RH=70%



Specific humidity is given by

$$\omega = \frac{0.622 P_v}{P_t - P_v}$$

Percentage of water vapour condenses into water.

$$\frac{\omega_1 - \omega_2}{\omega_1} \times 100$$

$$= \frac{\frac{0.025}{1-0.025} - \frac{0.087}{1-0.087}}{\frac{0.025}{1-0.025}} \times 100$$

= 66%

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764. The conditioned air supplied to the room must have the capacity to take up

(a) Room sensible heat load only
(b) Room latent heat only
(c) Both room sensible heat and latent heat loads
(d) None of the above

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : Conditioned air supplied to the room must have the capacity to take up, both room sensible heat & latent heat loads, due to the presence of water vapour in the air.

765. Consider the following statements for moist air, when it has attained the dew point temperature.

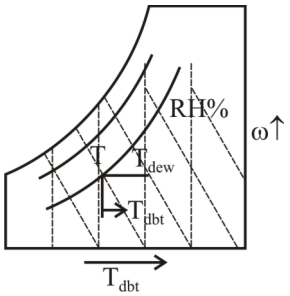
1. Dry-and wet-bulb temperatures are indentical.
2. Relative humidity is 100%.
3. Air is fully saturated with water vapour.
4. Humidity ratio is 100%.

Which of the above statements is/are correct ?

(a) 1 only (b) 2 and 3 only
(c) 1, 2 and 3 (d) 3 and 4

UPSC JWM Advt. No.-50/2010

Ans. (c) :



- When a moist air attains the Dew Point Temperature (DPT), the DBT is equal to Wet Bulb Temperature (WBT) & equal to dew point temperature.
- RH (Relative Humidity) is 100%.
- And is fully saturated.

766. Consider the following properties :

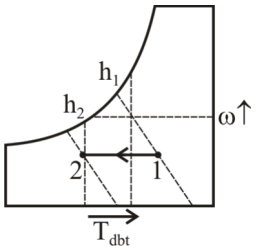
1. Wet-bulb temperature
2. Specific enthalpy of air-vapour mixture

Which of these properties decrease(s) during sensible cooling of air-water vapour mixture?

(a) Neither 1 nor 2 (b) Both 1 and 2
(c) 1 only (d) 2 only

UPSC JWM Advt. No.-50/2010

Ans. (b) :



From phychrometric chart, we know that, these properties decrease during sensible cooling of air water vapour is wet bulb temperature & specific enthalpy of air vapour mixture.

767. Assertion (A) : Comfort is a function of temperature, humidity and velocity in the conditioned space.

Reason (R) : By controlling temperature, humidity and velocity in the conditioned space, one can select a combination which consumes less energy without compromising the comfort.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (a) : As we know, human comfort is a function of temperature, humidity and velocity of the air this is decided by the factor called is effective temperature. By controlling these, in a combination, which consume less energy without compromising the comfort. Both sentences related to each other.

768. In an extra humid atmosphere which one of the following motor would you select?

(a) Open type
(b) Semi-protected
(c) Drip proof
(d) Totally enclosed fan cooled

ISRO Scientist/Engineer 2010

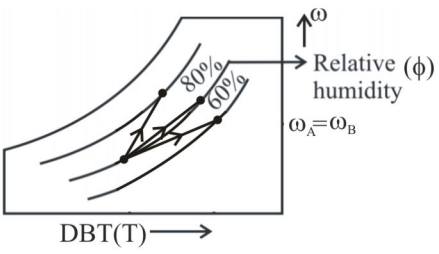
Ans. (d) : Totally enclosed fan cooled. This type of motor will not permit extra moisture from the humid atmosphere to come inside the motor and hence protects it from corrosion and other factors which help in reducing its life.

769. The relative humidity during heating and humidification :

(a) Increases
(b) Decreases
(c) May increase or decrease
(d) Remains constant

J&K PSC Civil Services Pre, 2010

Ans. (c) :



The specific humidity during heating and humidification process may increase or decrease.

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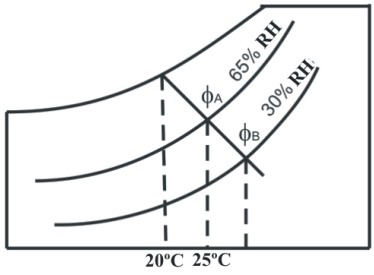
770. Two systems have the same wet bulb temperature at 20°C. System 'A' has Dry bulb temperature of 25°C since it has a relative humidity of 65%. System 'B' has a relative humidity of 30%. What will be its dry bulb temperature

(a) 33°C
(b) 25°C
(c) 27°C
(d) 22°C

ISRO Scientist/Engineer 2009

Ans. (a) : Two system A and B
for system 'A' WBT = 20°C
DBT = 25°C
Relative humidity
$\phi_A = 65\%$
for system 'B'
WBT = 20°C
DBT = ?
Relative humidity
$\phi_B = 30\%$



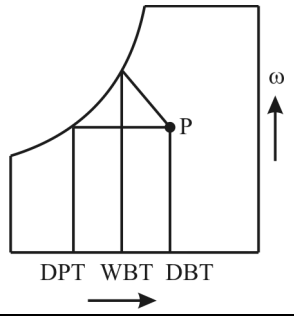
So here we need to take a guess between (a) & (c), as we are going from 65% RH to 30% RH the DBT for system B should be higher i.e. not 27°C, as it is very close. Hence the correct option is (a).

771. Which of the following options is true for the dry bulb temperature (T_{DBT}), the wet bulb temperature (T_{DPT}) for a mixture of air and water vapour ?

(a) $T_{DPT} < T_{DBT} < T_{WBT}$
(b) $T_{DPT} < T_{WBT} < T_{DBT}$
(c) $T_{WBT} < T_{DPT} < T_{DBT}$
(d) $T_{WBT} < T_{DBT} < T_{DPT}$

DRDO Scientists 2009

Ans. (b) : $T_{DPT} < T_{WBT} < T_{DBT}$



772. In a counter-flow cooling tower, warm water enters at temperature of T_w. The dry bulb and the wet bulb temperatures of air entering the tower are T_{DBT} and T_{WTB}, respectively. The minimum temperature of water at the exit of the cooling tower is :

(a) T_{WBT} (b) T_{DBT}
(c) $T_W - T_{DBT}$ (d) $T_W - T_{WBT}$

DRDO Scientists 2009

Ans. (c) :

773. For a given dry bulb temperature as the relative humidity increases, the wet bulb depression will be

(a) more
(b) less
(c) same
(d) more / less depending on other factors

TNPSC AE, 2008

Ans. (b) : Wet bulb Depression = Dry bulb temperature – Wet bulb temperature
For a given dry bulb temperature ($DBT_1 = DBT_2$), as the relative humidity ($\phi_2 > \phi_1$) increases, the wet bulb depression (WBD) will be decreases or less.

774. If pure humidification is desired, the spray water through which air passes, to be maintained as

(a) WBT (b) DBT
(c) DPT (d) None of these

TNPSC AE, 2008

Ans. (b) : If pure humidification is desired, the spray water through which air passes, has to be maintained as DBT. i.e, constant Dry Bulb Temperature (DBT) line.

775. In adiabatic saturation

(a) the process consists of increasing the humidity ratio without change in its heat content
(b) the process includes change in DBT keeping the humidity ratio of air as constant
(c) the process consists of heating the air without changing the humidity ratio
(d) the process can be accomplished either by the addition of steam or heated water in air

TNPSC AE, 2008

Ans. (a) : Adiabatic saturation is the process consists of increasing the humidity ratio (ω) without change in its heat content.

776. Determining moisture in steam by measuring the temperature in throttling calorimeter is an example of

(a) direct measurement
(b) indirect measurement
(c) measurement by comparison
(d) measurement by calibration

TNPSC AE, 2008

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Ans. (b) : The percentage by weight of steam in a mixture of steam and water is called the quality of the steam. The apparatus used to determine the moisture content of steam is called a calorimeter through since it may not measure the heat in the steam, the name is not descriptive of the function of the apparatus.

777. Clothes dry relatively easily in an environment where the

(a) relative humidity is high
(b) specific humidity is high
(c) relative humidity is low
(d) relative humidity fluctuates

DRDO Scientists 2008

Ans. (c) :

778. If P_v is partial pressure of water vapour in air and P_s is the saturation pressure of water vapour at the same temperature, the relative humidity is equal to

(a) $1-\frac{P_v}{P_s}$ (b) $1-\frac{P_s}{P_v}$ (c) $P_s - P_v$ (d) P_v / P_s

UKPSC AE 2007 Paper -II

Ans. (d) : P_v / P_s

779. Foggy condition in atmosphere results when

(a) Hot and humid air mixes with cold dry air
(b) Cold and dry air mixes with hot and dry air
(c) Hot and humid air mixes with hot and dry air
(d) None of the above

UKPSC AE 2007 Paper -II

Ans. (a) : Hot and humid air mixes with cold dry air

780. The wet bulb temperature is a measure of which humidity?

(a) Relative (b) Absolute
(c) Specific (d) None of above

UKPSC AE 2007 Paper -II

Ans. (b) : Absolute

781. The dry bulb temperature during heating and dehumidification

(a) decreases
(b) increases
(c) remains constant
(d) unpredictable

UKPSC AE 2007 Paper -II

Ans. (b) : During the heating and dehumidification process dry bulb temperature of the air increases while its dew point and wet bulb temperature reduces.

782. The specific humidity is the mass of water vapour present in (per)

(a) kg of dry air
(b) m^3 of dry air
(c) m^3 of wet air
(d) kg of wet air

UKPSC AE 2007 Paper -II

Ans. (a) : Kg of dry air

783. Air conditioning is concerned with maintaining the following:

(a) Temperature (b) Humidity
(c) Cleanliness (d) All the above

UKPSC AE 2007 Paper -II

Ans. (d) : All the above.

784. Air in domestic Window Air Conditioner is dehumidified by :

(a) heating (b) cooling
(c) Injecting water (d) Injecting steam

J & K PSC Screening, 2006

Ans. (b) : Air in domestic window air conditioner is dehumidified by cooling.
When temperature of air becomes less than Dew Point Temperature then air will be dehumidified.

785. Heating and dehumidification can be achieved simultaneously if air is passed through :

(a) sprays of water maintained at a temperature higher then the dew point temperature of the entering air
(b) a solid absorbent surface
(c) a liquid absorbent spray
(d) any one of the (b) and (c)

J & K PSC Screening, 2006

Ans. (d) : Heating and dehumidification can be achieved simultaneously by passing air through.
(1) A solid absorbent surface
(2) A liquid absorbent surface
This process can be achieved by using hygroscopic material This material can be solid or liquid.

786. For summer air conditioning in coastal regions, the psychrometric process employed is

(a) cooling with humidification
(b) cooling with dehumidification
(c) sensible cooling
(d) sensible heating

OPSC Civil Services Pre 2006

Ans. (b) : In coastal areas relative humidity and moisture contents in the air is high so in summer air conditioning cooling and dehumidification process is employed.

787. In psychrometry, during a sensible cooling process, the dew point temperature

(a) Decreases
(b) Increases
(c) May increase or decrease
(d) Remains constant

OPSC Civil Services Pre 2006

Ans. (d) : During sensible cooling or heating, dew point temperature and specific humidity (moisture content) remain same. In sensible cooling relative humidity, wet bulb temperature and enthalpy decreases and relative humidity increases.

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04.

Internal Combustion Engine

1. Air Standard Cycles

1. For same compression ratio thermal efficiency of:

(a) Otto cycle is less than Dual cycle
(b) Otto cycle is same as that for Diesel cycle
(c) Otto cycle is less than that of Diesel cycle
(d) Otto cycle is more than that of Diesel cycle

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Gujarat PSC AE 2019, RPSC AE 2018
Nagaland PSC CTSE 2017 Paper-2
TNPSC AE 2017
CGPSC Poly. Lect. 22.05.2016
RPSC Lecturer 16.01.2016
Rajasthan Nagar Nigam AE 2016, Shift-I
Kerala PSC AE 06.08.2015
OPSC AEE 2015 PAPER - II
MPPSC State Forest Service Exam, 2014
TNPSC AE 2013, WBPSC AE 2008
ISRO Scientist/Engineer 2012
UKPSC AE 2007 Paper -II
OPSC Civil Services Pre 2006

Ans. (d) : For same compression ratio, the efficiency of otto cycle is more efficient than diesel cycle. Due the large amount of heat transfer.

$\eta_{otto} > \eta_{diesel}$ (same compression ratio)

(i) For same compression ratio and heat addition

$\eta_{otto} > \eta_{dual} > \eta_{diesel}$

(ii) for same maximum temperature and heat rejection

$\eta_{diesel} > \eta_{dual} > \eta_{otto}$

2. For the same maximum pressure and the heat input, the efficiencies of the power cycles are

(a) $\eta_{diesel} > \eta_{dual} > \eta_{otto}$
(b) $\eta_{dual} > \eta_{diesel} > n_{otto}$
(c) $\eta_{otto} > \eta_{diesel} > \eta_{dual}$
(d) $\eta_{otto} > \eta_{dual} > \eta_{diesel}$

Assam PSC AE (PHED), 18.10.2020
TNPSC AE 2019
BPSC AE Mains 2017 Paper - V
WBPSC AE, 2017, UPPSC AE. 2016
RPSC AE 2016, (HPPSC AE 2014)
RPSC ACF-2011

Ans. (a) : For the same maximum pressure and heat input

$\eta_{diesel} > \eta_{dual} > \eta_{Otto}$

For same compression ratio and heat output

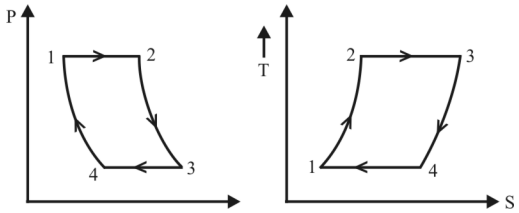
$\eta_{Otto} > \eta_{dual} > \eta_{diesel}$

3. A cycle consisting of two reversible isothermal processes and two reversible isobaric processes is known as :

(a) Atkinson cycle (b) Stirling cycle
(c) Brayton cycle (d) Ericsson cycle

ISRO Scientist/Engineer (RAC), 10.03.2019
Vadodara Muncipal Corp. DEE, 2018
Kerala PSC Asstt. Prof. 27.09.2016
APGENCO AE 2012, ESE 2009
WBPSC AE, 2007, GATE 1996

Ans. (d) : Ericsson cycle consist of two reversible isothermal and two reversible isobaric.



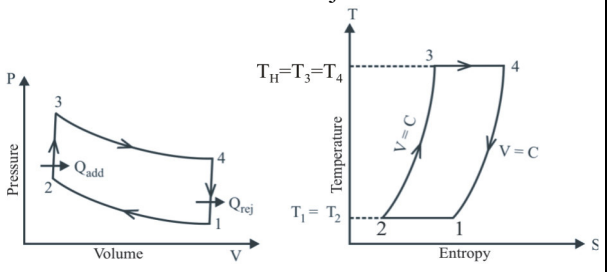
1–2 → Constant Pressure Heat addition
2–3 → Isothermal Expansion
3–4 → Constant Pressure Heat rejection
4–1 → Isothermal compression.

4. Stirling cycle consists of :

(a) Two constant volume and two adiabatic processes
(b) Two constant volume and two isothermal processes
(c) Two constant pressure and two isothermal processes
(d) One constant volume, one constant pressure and two adiabatic processes

VIZAG Steel MT, 14.12.2020
Nagaland PSC (CTSE) 2018, Paper-II
Punjab PSC SDE 12.02.2017
GMB AAE 25.06.2017, GPSC Lect. 23.10.2016
TSPSC Managers, 2015

Ans. (b) : Stirling cycle– It consists of four processes.
1–2→ Isothermal compression.
2–3→ Constant volume heat addition
3–4→ Isothermal expansion.
4–1→ Constant volume heat rejection.



At the same temperature limits carnot cycle efficiency is equal to stirling cycle efficiency.

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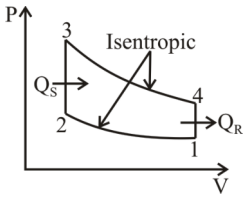
5. The constant volume cycle is also called

(a) Carnot cycle (b) Joule cycle
(c) Diesel cycle (d) Otto cycle

CGPSC AE 15.01.2021
Haryana PSC AE (PHED) 04.09.2020, Paper-I
APPSC AEE Screening Test 2016
MPPSC AE 08.11.2015, TSPSC AEE 2015
UKPSC AE-2013, Paper-II

Ans. (d) :



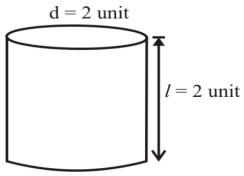
It has two constant volume process one for heat addition and other for heat rejection and two reversible adiabatic process.

6. An IC engine has a bore and stroke each equal to 2 units. The total area to calculate heat loss from the engine can be taken as

(a) 4π (b) 5π
(c) 6π (d) 8π

GPSC EE Pre, 28.01.2017, (HPPSC AE 2014)
ESE 2012, APGENCO AE 2012
GATE 1998

Ans. (c) :



Stroke length (l) = 2 unit
Base diameter (d) = 2 unit
r = 1 unit
Total area heat loss from engine
$= 2\pi r\,[r + l]$
$= 2\pi \times 1\,[1 + 2]$
$= 2\pi \times [3]$
$= 6\pi$

7. The crank radius of a single cylinder IC engine is 60 mm and the diameter of the cylinder is 80 mm. The swept volume of the cylinder in cm^3 is :

(a) 48 (b) 96
(c) 301 (d) 603

GPSC Asstt. Director of Transport 05.03.2017
UP Jal Nigam AE 2016
VIZAG STEEL MT 10.06.2012
VIZAG STEEL MT 18.08.2013
GATE 2011

Ans. (d) : Given,

Diameter of cylinder (D)	= 80 mm	= 8 cm
Radius of crank (r)	= 60 mm	= 6 cm
Length of stroke (L)	= 2 r	= 12 cm

Sewpt volume of the cylinder,

$$V_S = \frac{\pi}{4}D^2L = \frac{\pi}{4}\times(8)^2\times 12$$

$$V_S = 603.18\ cm^3$$

8. If the compression ratio of an engine working on otto cycle is increased 5 to 7 then the percentage increase in efficiency will be–

(a) 2% (b) 4%
(c) 8% (d) 14%

RPSC ACF & FRO, 26.02.2021
TNPSC AE 2017, HPPSC AE 2014
RPSC IOF & Boiler 20.09.2017

Ans : (d)

C-I, $\eta_1 = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{1}{5^{1.4-1}} = 47.4\%$

C-II, $\eta_2 = 1 - \frac{1}{7^{1.4-1}} = 54.08\%$

$$\%\text{increase} = \frac{\eta_2 - \eta_1}{\eta_1} = \frac{\eta_2}{\eta_1} - 1$$

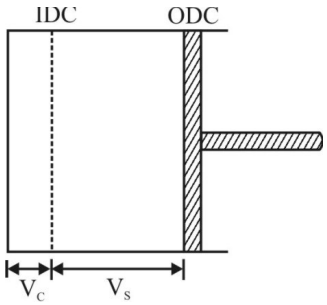
$$= \frac{54.08}{47.4} - 1 = 14.09 \approx 14\%$$

9. A 100 cc engine has the following parameter as 100 cc

(a) cylinder volume (b) clearance volume
(c) swept volume (d) full tank capacity

Rajasthan Nagar Nigam AE 2016, Shift-I
(KPSC AE 2015), GWSSB AAE, 27.12.2015
TNPSC AE 2014

Ans. : (c)



$$V_s = \frac{\pi}{4}D^2 \times L$$

$$V_s = \text{Swept volume}\left(cm^3\right)$$

D = bore dia (cm)
L = Stroke length (cm)

10. The Otto cycle is more efficient than Diesel cycle for the condition.

(a) Same maximum pressure and heat input
(b) Same maximum pressure and maximum temperature
(c) Same heat input and compression ratio
(d) Same heat output and maximum pressure

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
GPSC ARTO Pre 30.12.2018
HPPSC Asstt. Prof. 29.10.2016
GPSC Lect. (Auto) 16.10.2016
MPPSC AE 08.11.2015

Ans. (c) : For same compression ratio and Heat addition.

$$\eta_{otto} > \eta_{dual} > \eta_{diesel}$$

For same compression ratio and Heat rejection.

$$\eta_{otto} > \eta_{dual} > \eta_{diesel}$$

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For same maximum temp. and heat rejection.

$$\eta_{diesel} > \eta_{dual} > \eta_{otto}$$

For same maximum temp. and heat addition.

$$\eta_{diesel} > \eta_{dual} > \eta_{otto}$$

11. The air standard efficiency of a standard otto cycle for compression ratio of 5.5 is

(a) 25% (b) 50%

(c) 70% (d) 90%

ISRO Scientist/Engineer (RAC) 07.05.2017
RPSC AE 2016
MPPSC State Forest Service Exam, 2014

Ans. (b) : We know that

$$\eta_{otto} = 1 - \frac{1}{r^{(\gamma-1)}}$$

$$= 1 - \frac{1}{(5.5)^{1.4-1}}$$

$$\eta_{otto} \approx 0.5 = 50\%$$

12. In an air standard Otto cycle, r is the volume compression ratio and γ is an adiabatic index (c_p/c_v), the air standard efficiency is given by :

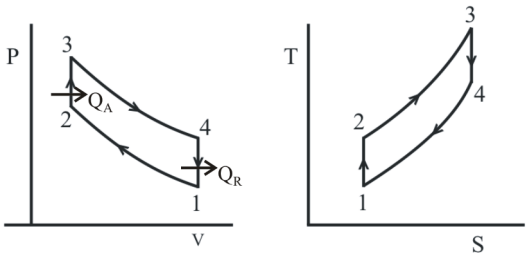
(a) $\eta = 1 - 1/(r^{\gamma} - 1)$ (b) $\eta = 1 - 1/r^{\gamma}$

(c) $\eta = 1 - 1/r^{(\gamma-1)}$ (d) $\eta = 1 - 1/r^{2\gamma-1}$

GPSC ARTO Pre 30.12.2018
TNPSC AE 2014, WBPSC AE 2003

Ans. (c) : Air standard efficiency of I.C. engine.

$$\eta_{sta} = \left[1 - \frac{1}{r^{\gamma-1}}\right]$$



1 - 2 → Compression process

2 - 3 → heat addition (combustion)

3 - 4 → expansion process

4 - 1 → exhaust process

13. In an air standard Diesel cycle at fixed compression ratio and fixed value of adiabatic index (γ)

(a) Thermal efficiency increases with increases in heat addition cut-off ratio

(b) Thermal efficiency decreases with increase in heat addition cut-off ratio

(c) Thermal efficiency remains same with increase in heat addition cut-off ratio

(d) None of the above

Nagaland PSC CTSE 2017 Paper-2
TNPSC AE 2017
UPPSC AE 12.04.2016 Paper-II

Ans.: (b) In an air standard diesel cycle at fixed compression ratio and fixed value of adiabatic index then thermal efficiency decreases with increase in heat addition cut–off ratio.

• $\eta\uparrow$ with decrease cut-off ratio (ρ)

14. In an air-standard Otto cycle, the compression ratio is 10. The condition at the beginning of the compression process is 100 kPa and 27°C. Heat added at constant volume is 1500 kJ/kg, while 700 kJ/kg of heat is rejected during the other constant volume process in the cycle. (Specific gas constant for air = 0.287 kJ/kg K)

The mean effective pressure (in kPa) of the cycle is

(a) 103 (b) 310

(c) 515 (d) 1032

VIZAG STEEL MT 10.06.2012
VIZAG STEEL MT 18.08.2013
GATE 2009

Ans. (d) : $V_1 = \frac{RT_1}{P_1} = \frac{0.287 \times 300}{100} = 0.861 \text{ m}^2/\text{kg}$

$$\frac{V_1}{V_2} = r = 10$$

$$V_2 = \frac{V_1}{r} = \frac{0.861}{10} = 0.0861 \text{ m}^3/\text{kg}$$

Mean effective pressure

$$P_m = \frac{W_{net}}{V_1 - V_2}$$

$$= \frac{Q_S - Q_R}{0.816 - 0.0861}$$

$$= \frac{1500 - 700}{0.7749}$$

$$= \frac{800}{0.7749}$$

$$P_m \cong 1032 \text{ kPa}$$

15. The efficiency of Diesel cycle approaches to Otto cycle efficiency when

(a) Cut-off is increased

(b) Cut-off is decreased

(c) Cut-off is zero

(d) Cut-off is constant

RPSC IOF, 2020
Nagaland PSC (CTSE) 2018, Paper-II
Nagaland PSC CTSE 2015 Paper-II

Ans. (c) :

$$\eta_{diesel} = 1 - \frac{1}{r^{\gamma-1}} \frac{(\rho^{\gamma} - 1)}{\gamma(\rho - 1)}$$

The diesel cycle efficiency depends upon the cut-off ratio. So when the cut-off ratio is equal to one then it approaches to Otto cycle efficiency as then

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Cut-off ratio = $\frac{V_3}{V_2} = 1$.

So, $V_3 = V_2$

Which ultimately ensure that the heat addition is done at constant volume and approaches to the Otto cycle at the same compression ratio.

16. The order of values of thermal efficiency of Otto, Diesel and Dual cycle, when they have equal compression ratio and heat rejection, is given by:

(a) $\eta_{Otto} > \eta_{Diesel} > \eta_{Dual}$
(b) $\eta_{Diesel} > \eta_{Dual} > \eta_{Otto}$
(c) $\eta_{Dual} > \eta_{Diesel} > \eta_{Ottal}$
(d) $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$

Nagaland PSC (CTSE) 2018, Paper-II
MPPSC AE 2016
UKPSC AE-2013, 2012 Paper-II

Ans : (d)
i. Equal compression ratio and heat rejection order of values of thermal efficiency:-

$\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$

ii. For same maximum temperature and heat addition

$\eta_{Diesel} > \eta_{dual} > \eta_{Otto}$

iii. Same maximum pressure and heat input

$\eta_{diesel} > \eta_{dual} > \eta_{Otto}$

17. The accumulation of carbon in a cylinder results in increase of

(a) clearance volume
(b) ignition delay
(c) effective compression ratio
(d) volumetric efficiency

TNPSC AE (Industries) 09.06.2013
UKPSC AE-2013, Paper-II
APPSC AEE 2012

Ans. (c) : The accumulation of carbon in a cylinder results in-

- Increase of effective compression ratio
- Decreases fuel efficiency
- Decreased compression
- Increased exhaust pollutants
- Decreased oil life
- Decreased ring life
- Decreased spark plug life

18. The thermal efficiency of a 2 stroke cycle engine as compared to a 4 stroke cycle engine is

(a) More (b) Less
(c) Equal (d) Much less

JPSC AE, 10.04.2021, Paper-I
Nagaland PSC CTSE 2017 Paper-2
TNPSC ACF 2012

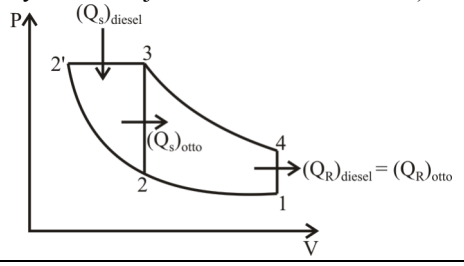
Ans. (b) : The thermal efficiency of a 2 stroke cycle engine as compared to 4 stroke cycle is always less. Thermal efficiency of four stroke is higher part load efficiency is better than two stroke cycle engine.

19. For the same maximum pressure and temperature of Otto, Diesel and dual combustion air standard cycles :

(a) The compression ratio will be same
(b) Heat supplied to the cycles will be same
(c) Air standard efficiency will have same value
(d) Heat rejected by the engine will be same

Kerala PSC Poly. Lect., 2017
JPSC AE 2013 Paper-V

Ans. (d) : For the same maximum pressure and temperature of Otto, Diesel and dual combustion air standard cycles heat rejected by the engine will be same (i.e. all cycle heat rejected at constant volume).



20. An engine is having clearance volume as 20 cm^3 and total volume 320 cm^3, mechanical efficiency 90% and volumetric efficiency as 80%. The volume of charge that can be available in the engine after suction.

(a) 320 cc (b) 300 cc
(c) 270 cc (d) 240 cc

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II

Ans. (d) : Given,

$V_C = 20cm^3$

$V_T = V_S + V_C = 320cm^3$

Where, V_C = Clearance volume

V_S = Swept volume

V_T = Total volume

$\eta_{mech} = 0.90$

$\eta_{vol.} = 0.80$

$V_S + V_C = 320$

$V_S = 320 - V_C$

$= 320 - 20$

$V_S = 300\ cm^3$

$$\eta_V = \frac{\text{Actual suction volume}}{\text{Theoretical swept volume}}$$

Actual suction volume $= \eta_V \times V_S$

$= 0.80 \times 300$

$= 240$ cc

21. An engine operating on Otto cycle has clearance volume as 10% of the swept volume. If $\gamma = 1.4$ the air standard cycle efficiency is

(a) 38.3% (b) 39.8%
(c) 60.2% (d) 61.7%

UPPSC AE 13.12.2020, Paper-2
GATE 2003

Ans. (d) : Given

$\gamma = 1.4$, $V_c = 10\%\ V_s$

$V_c = 0.1\ V_s$

$\frac{V_c}{V_s} = 0.1 = c$ (clearance ratio)

Compression ratio $(r) = 1 + \frac{1}{c}$

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$$r = 1 + \frac{1}{0.1}$$

$$\boxed{r = 11}$$

$$\eta_{otto} = 1 - \left(\frac{1}{r}\right)^{\gamma-1} = 1 - \left(\frac{1}{11}\right)^{1.4-1}$$

$$= 61.67\%$$

22. The efficiency of Diesel cycle with decrease in cut off

(a) Increases
(b) Decreases
(c) Remain unaffected
(d) First increases and then decreases

WBPSC AE, 2017
MPPSC State Forest Service Exam, 2014

Ans. (a) : $\boxed{\eta_{diesel} = 1 - \frac{(\rho^{\gamma} - 1)}{\gamma \cdot r^{(\gamma-1)} \cdot (\rho - 1)}}$

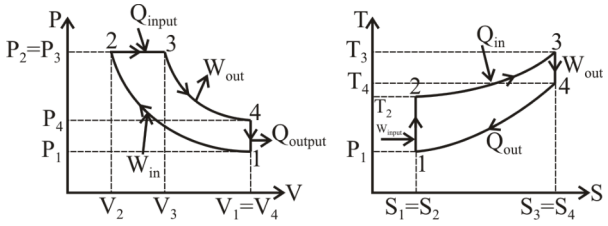
So, if cutoff ratio(ρ)↓, η_{diesel}(↑)

23. A cycle consisting of one constant pressure, one constant volume and two isentropic processes is known as

(a) Carnot cycle (b) Stirling cycle
(c) Otto cycle (d) Diesel cycle

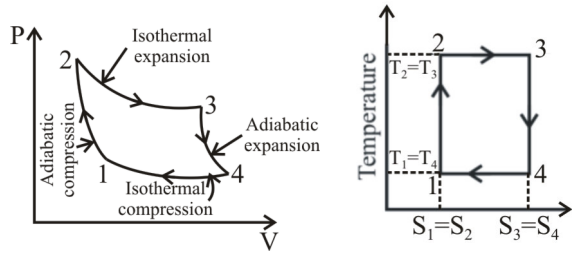
ISRO Scientist/Engineer (RAC), 10.03.2019
Sikkim PSC (Under Secretary), 2017

Ans. (d) : Diesel cycle- Diesel cycle consist of two reversible adiabatic or isentropic and one constant pressure, and one constant volume.

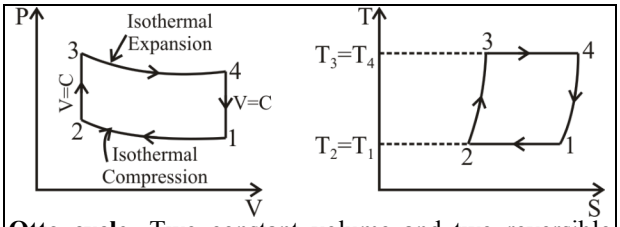


1-2 : Isentropic compression
2-3 : Constant pressure heat addition
3-4 : Isentropic expansion
4-1 : Constant volume heat rejection

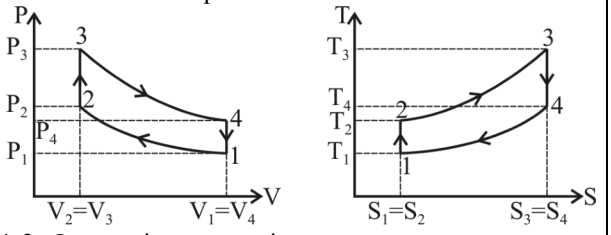
Carnot cycle- Two adiabatic process and two isothermal process.



Stirling cycle- Two isothermal process and two constant volume process.



Otto cycle- Two constant volume and two reversible adiabatic or isentropic.



1-2 : Isentropic compression
2-3 : Constant volume heat addition
3-4 : Isentropic expansion
4-1 : Constant volume heat rejection

24. For the same values of peak pressure, peak temperature and heat rejection, the correct order of efficiencies for Otto, Dual and Diesel cycles.

(a) Otto > Dual > Diesel
(b) Dual > Diesel > Otto
(c) Diesel > Dual > Otto
(d) Diesel > Otto > Dual

ISRO Scientist/Engineer (RAC), 10.03.2019
GATE 2015

Ans. (c) : For same peak pressure, peak temperature and heat rejected

$$\eta_{Diesel} > \eta_{Dual} > \eta_{Otto}$$

for same compression ratio

$$\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$$

25. Which of the following cycle has maximum efficiency

(a) Rankine (b) Brayton
(c) Carnot (d) Joule

Rajasthan Nagar Nigam AE 2016, Shift-I
MPPSC AE 08.11.2015

Ans. (c) : Carnot cycle has maximum efficiency. This cycle is practically not possible so it is called hypothetical cycle.

26. In air-standard Otto cycle the terminal pressure at the end of compression, heat release and expansion are respectively P_2, P_3 and P_4. If the corresponding values are P'_2, P'_3 and P'_4 taking into account the effect of variable specific heat and dissociation of the working fluid, then

(a) $P_2 < P'_2$ and $P_3 < P'_3$
(b) $P_3 < P'_3$ and $P_4 < P'_4$
(c) $P_2 > P'_2$, $P_3 > P'_3$ and $P_4 < P'_4$
(d) $P_2 > P'_2$ and $P_3 > P'_3$

GPSC Asstt. Director of Transport 05.03.2017
GATE 1989

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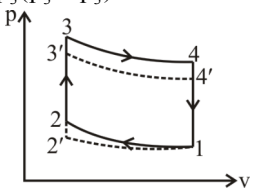
Ans. (d) : All gases except monoatomic gases shows an increase in specific heat at higher temperature. At the sometime the difference ($c_p - c_v = R$) remains constant. But their ratio (c_p/c_v) decreases with increase in temperature.

The Otto cycle with the constant specific heat is represented by solid lines and the dotted lines represents the Otto cycle with the variable specific heats. So at the end of the compression, p_2 will be greater than $p_2'(p_2 > p_2')$ and at the end of the heat addition, p_2 will be greater than $p_3'(p_3 > p_3')$.



27. Air standard analysis of high speed diesel engines can be done using-

(a) Otto cycle
(b) Diesel cycle
(c) Dual combustion cycle
(d) None of the above

CIL MT 27.02.2020
GPSC Lect. (Auto) 16.10.2016

Ans. (c) : Air standard analysis of high speed diesel engines can be done by using dual combustion cycle.

28. Efficiency of Diesel cycle depends on

(a) Compression ratio
(b) Compression ratio and cutoff ratio
(c) Cut off ratio
(d) Cut off and pressure ratio

GMB AAE 25.06.2017, TNPSC AE 2013

Ans. (b) : Efficiency of diesel cycle depends on compression ratio and cut off ratio

$$\eta_{diesel} = 1 - \frac{1}{r^{(\gamma-1)}} \frac{\left(\rho^{\gamma} - 1\right)}{\gamma(\rho - 1)}$$

$$\eta_{diesel} = f(r, \rho)$$

Where r = Compression ratio
ρ = Cutoff ratio

29. Thermal efficiency of Otto cycle _____that of Brayton cycle for the given same compression process

(a) Same as (b) less than
(c) greater than (d) not comparable to

APPSC AEE MAINS 2016 PAPER-III
UPPSC AE 12.04.2016 Paper-II

Ans. (a) : For same compression ratio (r), thermal efficiency of Otto cycle same as Brayton cycle.

$$\eta_{Otto} = 1 - \frac{1}{r^{(\gamma-1)}}$$

$$\eta_{Brayton} = 1 - \frac{1}{(r_p)^{(\gamma-1)/\gamma}}$$

Where r = Compression ratio
r_P = Pressure ratio

We know that

$$r_p = r^{\gamma}$$

$$\left[\frac{P_2}{P_1}\right] = \left[\frac{V_1}{V_2}\right]^{\gamma}$$

$$\eta_{Brayton} = 1 - \frac{1}{(r)^{\gamma\times(\gamma-1)/\gamma}}$$

$$= 1 - \frac{1}{r^{\gamma-1}}$$

Then,

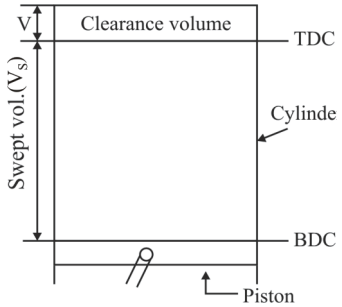
$$\eta_{Brayton} = \eta_{Otto}$$ (For same compression ratio)

30. The stroke of an engine is the

(a) volume of the cylinder
(b) length of the connecting rod
(c) internal diameter of the cylinder
(d) distance between TDC and BDC

Kerala PSC Poly. Lect. (Auto) 05.05.2016
TSPSC AEE 2015

Ans. : (d) The stroke of an engine is the distance between TDC and BDC



Distance between TDC to BDC is called stroke length

$$V_s = \frac{\pi}{4} D^2 \times L_s$$

V_s = Swept volume
L_s = stroke length
D = Bore diameter

31. The bore and stroke of the cylinder of a six cylinder engine working on an Otto cycle are 17 cm and 30 cm respectively, total clearance volume is 9225 cm^3, and then what is the compression ratio?

(a) 7.8 (b) 6.2
(c) 15.8 (d) 5.4

RPSC Lect. (Tech. Edu. Dept.) 2011
ESE 2009

Ans. (d) : (C.R.) Compression Ratio

$$= \frac{\text{Volume of the cylinder (total)}}{\text{Clearance volume}}$$

Clearance volume (V_C) = 9225 cm^3
Volume of the Cylinder (Total) $V_T = V_S + V_C$

V_S = Swept volume = $\frac{\pi}{4} D^2 \times L \times n$

D = Diameter of bore, L = Length of the bore
n = No. of cylinder.

$$C.R. = \frac{V_T}{V_C} = \frac{V_s + V_c}{V_c} = \frac{9225 + \frac{\pi}{4}(17)^2 \times 30 \times 6}{9225}$$

C.R. = 5.4

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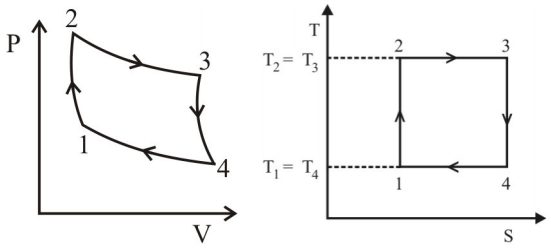
32. If both Stirling and Carnot cycles operate within the same temperature limilt, then the efficiency of Stirling cycle as compared to Carnot cycle is

(a) more (b) less
(c) equal (d) anything, more or less

WBPSC AE, 2017
APPSC AEE MAINS 2016 PAPER-III

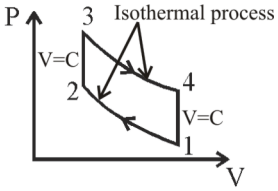
Ans. (c) : • A Carnot cycle consists of four process :- Two isothermal process and two isentropic process

$$\eta_{Carnot} = \frac{T_3 - T_1}{T_3}$$



• Stirling cycle is the modified form of Carnot cycle. Stirling cycle consists of the reversible isothermal and two reversible constant volume (isochoric) process.

$$\eta_{Stirling} = \frac{T_3 - T_1}{T_3}$$



Which is same, as Carnot efficiency.

33. An ideal air standard Otto cycle has a compression ratio of 8.5. If the ratio of the specific heats (γ) is 1.4, then what is the thermal efficiency (in percentage) of the Otto cycle?

(a) 57.5 (b) 45.7
(c) 52.5 (d) 95

VIZAG STEEL MT 10.06.2012
GATE 2002

Ans. (a) : Given r = 8.5, $\gamma = 1.4$

$$\eta_{Otto} = \left(1 - \frac{1}{r^{\gamma-1}}\right) \times 100$$

$$= \left(1 - \frac{1}{8.5^{1.4-1}}\right) \times 100$$

$$\boxed{\eta_{Otto} = 57.5\%}$$

34. Thermal efficiency of a diesel cycle having fixed compression ratio, with increase in cut-off ratio will :

(a) Increase
(b) Decrease
(c) Independent
(d) May increase or decrease depending on the other factors

Kerala PSC AE (Irrigation) 05.04.2017
KPSC ADF 2015

Ans. (b) : In case of a diesel cycle having fixed compression ratio, with increase in cut-off ratio (r_c) thermal efficiency decreases.

$$\text{Efficiency } (\eta) = 1 - \frac{1}{(r)^{\gamma-1}}\left(\frac{r_c^{\gamma} - 1}{\gamma(r_c - 1)}\right)$$

where, r_c = cut off ratio

$$\boxed{r_c \uparrow \eta \downarrow}$$

35. In a two-stroke cycle engine the operations namely suction compression, expansion, exhaust are completed in the number of revolutions of crankshaft equal to

(a) Four (b) Three
(c) Two (d) One

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : In a two-stroke cycle engine the operations namely suction compression, expansion, exhaust are completed in the number of revolutions of crankshaft equal to one revolution.

36. A gas engine has a swept volume of 300 cm^3 and its volumetric efficiency is 0.88 and mechanical efficiency is 0.90. The volume of mixture taken in per stroke is

(a) 248 cm^3 (b) 252 cm^3
(c) 264 cm^3 (d) 286 cm^3

ISRO Scientist/Engineer (RAC) 22.04.2018
ESE 1995

Ans. (c) : We know that volumetric efficiency is given by

$$\eta_{vol} = \frac{\text{Actual Volume of mixture taken per stroke (V)}}{\text{Swept Volume } (V_S)}$$

$$0.88 = \frac{V}{V_S} = \frac{V}{300}$$

$$\boxed{V = 264\,cm^3}$$

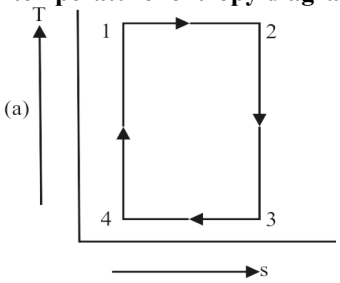
37. In case of diesel cycle, increasing the cutoff ratio will increase

(a) Efficiency
(b) Mean effective pressure
(c) The maximum pressure
(d) The engine weight

TSPSC Managers, 2015, TSPSC AEE 2015

Ans. (b) : Mean effective pressure will increase in case of diesel cycle if increasing the cut off ratio.

38. Which of the following represents Otto cycle on temperature- entropy diagram?



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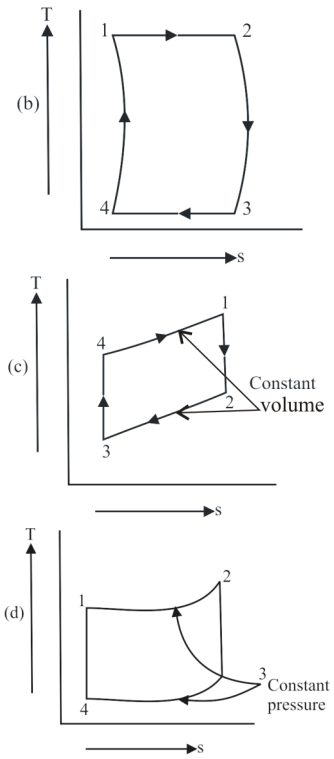


Ans. (c) :

39. The air standard otto cycle comprises

(a) Two constant pressure processes and two constant volume processes.

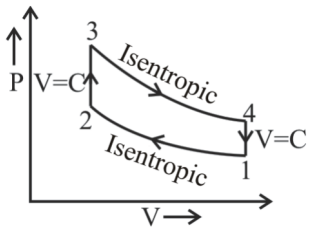
(b) Two constant pressure processes and two constant entropy processes.

(c) Two constant volume processes and two constant entropy processes

(d) None of the above.

Nagaland PSC CTSE 2017 Paper-2
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : Otto Cycle –



1-2– Isentropic compression

2-3– Constant volume Heat addition.

3-4 – Isentropic expansion

4-1 –Constant volume Heat rejection.

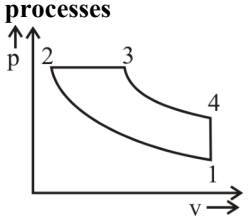
$$\eta_{otto} = 1 - \frac{1}{r^{\gamma-1}}$$

Compression ratio (r) = $\frac{V_1}{V_2} = \frac{V_s + V_c}{V_c}$

Where, V_s = Swept volume

V_c = Clearance volume

40. An air standard Diesel cycle consists of the following processes



1-2: Air is compressed isentropically

2-3: Heat is added at constant pressure

3-4: Air expands isentropically to the original volume

4-1: Heat is rejected at constant volume

If γ and T denotes the specific heat ratio and temperature respectively the efficiency of the cycle is

(a) $1-\frac{T_4-T_1}{\gamma(T_3-T_2)}$ (b) $1-\frac{T_4-T_1}{T_3-T_2}$

(c) $1-\frac{\gamma(T_4-T_1)}{T_3-T_2}$ (d) $1-\frac{T_4-T_1}{(\gamma-1)(T_3-T_2)}$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (a) : $\eta = 1 - \frac{Q_{rejection}}{Q_{supplied}}$

$= 1 - \frac{mc_v(T_4-T_1)}{mc_P(T_3-T_2)}$

$= 1 - \frac{T_4-T_1}{\frac{c_p}{c_v}(T_3-T_2)} \quad \left[\because \gamma = \frac{C_p}{C_v}\right]$

$\eta = 1 - \frac{T_4-T_1}{\gamma(T_3-T_2)}$

41. Match List-I (details of the processes of the cycle) with List-II (name of the cycle) and select the correct answer using the codes given below the List:

List I		List II	
A	**Two isothermals and two constant volumes**	**1**	**Carnot**
B	**Two isentropic and two constant volumes**	**2**	**Joule**
C	**Two isentropic and two constant pressures**	**3**	**Otto**
D	**Two isothermals and isentropic**	**4**	**Diesel**
		5	**Striling**

(a) A-5, B-3, C-1, D-2 (b) A-5, B-3, C-2, D-1

(c) A-3, B-4, C-2, D-1 (d) A-5, B-4, C-2, D-1

Nagaland PSC CTSE 2015 Paper-II
CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) :

Cycle	**Processes**
Stirling cycle	Two isothermal and two constant volumes.
Otto cycle	Two isentropic and two Constant Volume

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Diesel cycle	Two isentropic and one Constant Pressure and one constant volume
Carnot cycle	Two isothermals and two isentropic
Joule cycle	Two constant Pressure and two isentropic

42. Which factor leads to lower thermal efficiency on an actual Otto cycle in comparison of ideal Otto cycle?

(a) Increase in specific heat ratio of working fluid with rise in temperature.
(b) Decrease in specific heat ratio of working fluid with rise in temperature.
(c) Decrease in specific heat of working fluid with rise in temperature.
(d) No change in specific heat with change in temperature.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

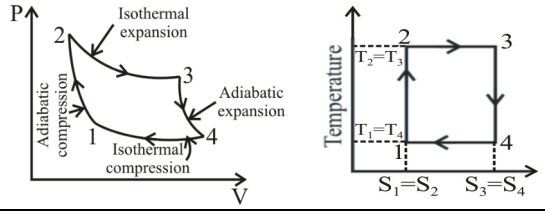
Ans. (b): Actual Otto cycle is less efficient than ideal Otto cycle because in actual cycle specific heat ratio decreases as temperature increases while in ideal cycle remains constant.

43. The air standard cycles which consist of two isothermal process and two isentropic process:

(a) Carnot cycle (b) Diesel cycle
(c) Otto cycle (d) Ericsson cycle

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (a) : The air standard cycle of carnot, consists of two isothermal process and two isentropic process.



44. The stroke volume is 500 cc. The clearance is 50 cc. The compression ratio is

(a) 10 (b) 11
(c) 9 (d) 15

JPSC AE, 10.04.2021, Paper-I

Ans. (b) : Given, $V_S = 500$ cc
$V_C = 50$ cc

$$\text{Compression ratio, } r = \frac{V_S + V_C}{V_C} = \frac{V_1}{V_2}$$

$$r_C = \frac{550}{50}$$

$$\boxed{r_C = 11}$$

45. Lenoir cycle is used for

(a) Gas turbines (b) Pulse jet engines
(c) S.I engines (d) C.I engines

ESE 18.07.2021

Ans. (b) : Gas turbine → Brayton cycle
S.I. engine → Otto cycle
C.I engine → Diesel cycle
Pulse jet engine → Lenoir cycle.

46. A diesel engines has a compression ratio of 20 and cut-off takes place at 5% of the stroke. What is the cut-off ratio

(a) 1.21 (b) 1.47
(c) 1.73 (d) 1.95

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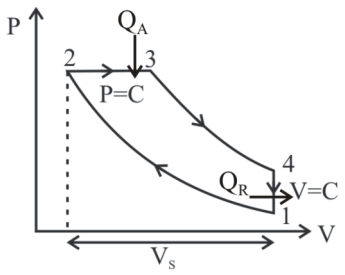
Ans. (d) :



$$\frac{V_1}{V_2} = 20 = r \qquad (r = \text{compression ratio})$$

$$V_3 - V_2 = \frac{5}{100}(V_1 - V_2)$$

$$\frac{V_3}{V_2} - 1 = \frac{5}{100}\left(\frac{V_1}{V_2} - 1\right)$$

$$\text{or } \frac{V_3}{V_2} = \frac{1}{20}(20-1) + 1 = \frac{19}{20} + 1$$

$$= 1.95$$

47. The cubic capacity of a four-stroke oversquare spark-ignition engine is 275 cc. The clearance volume is 25 cc. What is the compression ratio of the engine?

(a) 8 (b) 10
(c) 12 (d) 14

ESE 18.07.2021

Ans. (c) : Given, swept volume, $V_s = 275$ cc
Clearance volume, $V_c = 25$ cc

$$\text{Clearance ratio (C)} = \frac{V_c}{V_s}$$

$$= \frac{25}{275} = \frac{1}{11}$$

$$\text{Compression ratio (r)} = 1 + \frac{1}{C}$$

$$= 1 + 11$$

$$r = 12$$

48. An ideal air-standard diesel engine has a compression ratio of 14:1. The state of air at the beginning of the compression stroke is 400 kPa and 400 K. The highest temperature during the cycle is 1500 K. Assume air to be an ideal gas with constant specific heats. The gas constant of air is R = 0.287 kJ/kg-K and specific heat at constant volume, c_v = 0.717 kJ/kg-K. Air temperature at the end of the compression stroke is:

(a) 1050K (b) 1500K
(c) 1149.51K (d) 560K

AAI Jr. Executive 26.03.2021

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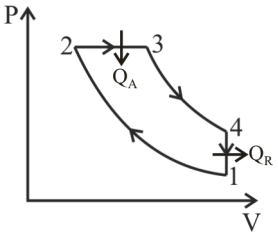
Ans. (c)



Given, $r_c = 14$
$P_1 = 400$ kPa
$T_1 = 400$ k
$T_3 = 1500$ k

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = (r_c)^{\gamma-1}$$

$$\frac{T_2}{400} = (14)^{1.4-1} \quad [\gamma = 1.4]$$

$T_2 = 400 \times (14)^{0.4}$
$T_2 = 1149.51$ K
Air temperature at end of compression stroke.

49. The following conclusion on efficiency (η) of gas power cycles working between same temperature limits can be drawn
(a) $\eta_{carnot} < \eta_{stirling} < \eta_{ericsson}$
(b) $\eta_{carnot} > \eta_{stirling} > \eta_{ericsson}$
(c) $\eta_{ericsson} < \eta_{stirling} < \eta_{carnot}$
(d) $\eta_{carnot} = \eta_{stirling} = \eta_{ericsson}$

GPSC DEE, Class-2 (GWSSB) 04.07.2021

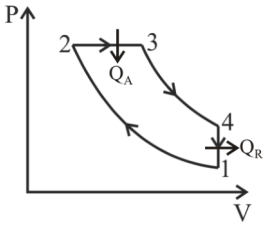
Ans. (d) : gas power cycles working between same temperature

$$\eta_{carnot} = \eta_{stirling} = \eta_{ericsson}$$

50. A compression ignition engine working on an air standard Diesel cycle has the following particulars
Cylinder bore = 15 cm
Stroke = 25 cm
Clearance volume=400 cm³
The fuel injection takes place at a constant pressure of 5% of the stroke. The cut off ratio will be about
(a) 0.82 (b) 1.55
(c) 2.1 (d) 12.0

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :



$$V_s = \frac{\pi}{4} d^2 \ell$$

$$= \frac{\pi}{4} \times (15)^2 \times 25 \text{cm}^3$$

$= 4417.86$ cm³
$V_c = 400$ cm³

$$r = 1 + \frac{V_s}{V_c}$$

$$= 1 + \frac{4417.86}{400}$$

$r = 12.04$
ρ = Cut off at 5% of stroke
$\therefore \quad V_3 - V_2 = 0.05 \times V_s$
$\therefore \quad V_3 = V_c + 0.05\ V_s$
$= 620.90$ cm³

$$\rho = \frac{V_3}{V_2} = \frac{620.90}{400}$$

$\rho = 1.55$

51. In a dual cycle of limited pressure air standard cycle at fixed compression ratio, the heat added at constant volume is x and at constant pressure is y. Also x + y = constant. Choose the correct statement.
(a) If y is increased, thermal efficiency is increased
(b) If x = y, the thermal efficiency is maximum
(c) If x is increased, the thermal efficiency is increased
(d) All the above statements are true

CGPSC AE 15.01.2021

Ans. (c) :

52. The thermal efficiency of the dual combustion cycle depends on _____.
(a) Compression ratio, cutoff ratio and pressure ratio together
(b) Only Compression ratio and cutoff ratio
(c) Only pressure ratio and cutoff ratio
(d) Only Compression ratio

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (a) : The thermal efficiency of dual combustion cycle depends on compression ratio, cutoff ratio and pressure ratio together.

$$\eta_{dual} = 1 - \frac{1}{r^{\gamma-1}} \left[\frac{r_p (r_c^{\gamma} - 1)}{(r_p - 1) + \gamma r_p (r_c - 1)} \right]$$

r = Compression ratio
r_c = Cutoff ratio
r_p = Pressure ratio or explosion ratio

53. The cylinder of an IC engine has 40 cm bore and 50 cm stroke. The clearance is 2832 cm³. The swept volume of the engine is
(a) 600000 cm³ (b) 30000 cm³
(c) 6000 cm³ (d) 60000 cm³

Assam PSC AE (PHED), 18.10.2020

Ans. (d) : Swept volume = Total volume – Clearance volume

Total Volume = $\pi r^2 h$
$= \pi \times 20 \times 20 \times 50$
$= 62831.85$ cm³
Swept volume = 62831.85 – 2832
= 59999.85
Swept volume ≈ 60000 cm³

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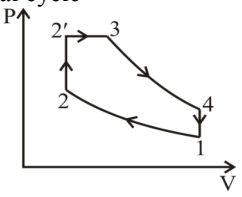
54. Which of the following is true for dual cycle? [x = compression ratio, y=expansion ratio, z = cut-off ratio]

(a) $x = yz$ (b) $x = y + z$

(c) $x = y/z$ (d) $x = y - z$

APPSC Poly Lect. 13.03.2020

Ans. (a) : For dual cycle –



Compression ratio (x) $= \frac{V_1}{V_2}$

Expansion ratio(y)= $\frac{V_4}{V_3}$

Cutt off ratio (z) $= \frac{V_3}{V_{2'}} = \frac{V_3}{V_2}$

Expansion Ratio (y) $= \frac{V_4}{V_3} = \frac{V_1}{V_3} = \frac{V_1}{V_2} \times \frac{V_2}{V_3}$

$$y = x \times \frac{1}{z} = \frac{\text{Compression Ratio}}{\text{Cut off Ratio}}$$

$\therefore$ Compression ratio (x)

= Expansion ratio (y) × Cut-off ratio(z)

$\Rightarrow \boxed{x = yz}$

55. Compression ratio of an internal combustion engine is defined as _____ . [V_S = Swept volume, V_C = Clearance volume]

(a) $\frac{V_c + V_s}{V_s}$ (b) $\frac{V_c}{V_s + V_c}$

(c) $\frac{V_c + V_s}{V_c}$ (d) $\frac{V_s}{V_c}$

APPSC Poly Lect. 13.03.2020

Ans. (c) : Compression ratio (r) $= \frac{V_c + V_s}{V_c}$

Where ;

V_c = clearance volume

V_s = swept volume

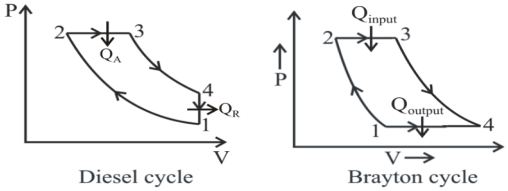
56. The difference between Diesel cycle and Brayton cycle is in the following process :

(a) Heat addition (b) Expansion

(c) Heat rejection (d) Compression

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : In Brayton cycle heat is rejected at constant pressure. In diesel heat is rejected of constant volume.



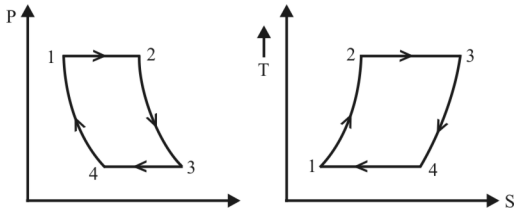
57. Which does not constitute the theoretical cycle for the working of a practical I.C. engine?

(a) Otto Cycle (b) Diesel Cycle

(c) Dual Cycle (d) Ericsson Cycle

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Ericsson Cycle– It has two constant pressure processes and two reversible isotherms.



- Basic Ericsson cycle is externally irreversible.
- By incorporating a perfect heat exchanger, it is made reversible and efficiency equal to carnot cycle.

58. Which of the following uses a regenerator?

(a) Brayton cycle (b) Ericsson cycle

(c) Stirling cycle (d) Both (b) and (c)

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : Stirling and Ericsson cycle sometime called the regenerative cycle, a heat exchanger which utilizes waste heat.

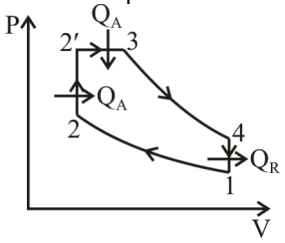
59. An ideal cycle based on the concept of combination of two heat transfer processes, one at constant volume and the other at constant pressure, is called

(a) Otto cycle (b) Dual cycle

(c) Diesel cycle (d) Carnot cycle

ESE 2020

Ans. (b) : In a dual cycle a part of the heat is first supplied to the system at constant volume and then the remaining part at constant pressure.



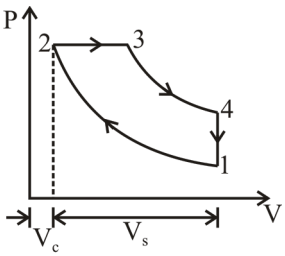
60. A diesel engine has a compression ratio of 16 and cut-off takes place at 6% of the stroke. What will be the cut-off ratio?

(a) 1.6 (b) 1.9

(c) 2.1 (d) 2.4

APPSC Poly Lect. 13.03.2020

Ans. (b) :



Diesel cycle

$(V_3 - V_2) = x\%$ of $(V_1 - V_2)$

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$$\left(\frac{V_3}{V_2}-1\right)=0.06\times\left(\frac{V_1}{V_2}-1\right)$$

$(\rho - 1) = 0.06 \times (r - 1)$ $(\rho = V_3/V_2,\ r = V_1/V_2)$

$\rho - 1 = 0.06 \times (16 - 1)$ (Given, r = 16)

$\rho - 1 = 0.06 \times 15$

$\rho = 1 + 0.9$

$\boxed{\rho = 1.9}$

61. A mass of 1 kg of air at 27ºC and 0.98 atm is taken through a diesel cycle. If the compression ratio of the engine is 16, calculate the temperature of the air after compression? (for calculation, take the ratio of specific heats of air as 1.5)

(a) 1200 deg C (b) 1473 deg C

(c) 927 deg C (d) 768 deg C

ISRO Scientist/Engineer 12.01.2020

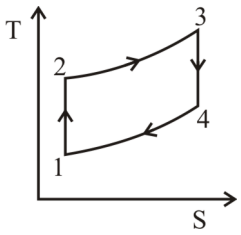
Ans. (c) : m = 1kg of air

$T_1 = 27ºC = 300$ K

$P = 0.98$ atm $= 0.98 \times 101.325$ kPa

$r_c = 16$ $T_2 = ?$

$n = 1.5$



Compression is isentropic process,

$$\frac{T_2}{T_1}=\left(\frac{V_1}{V_2}\right)^{(n-1)}=(r_c)^{n-1}$$

$$\frac{T_2}{T_1}=16^{0.5}=4$$

$T_2 = 4 \times 300 = 1200$ K

$T_2 = 1200 - 273 = 927ºC$

62. In an engine working on an ideal Otto cycle, the temperature at the beginning and at the end of compression are 300 K and 600 K respectively. What will be the air standard efficiency of the engine? [Consider $\gamma = 1.4$]

(a) 50% (b) 45%

(c) 55% (d) 40%

SJVN ET 2019

Ans. (a) : Given,

$T_1 = 300$ K

$T_2 = 600$ K

$$\frac{T_1}{T_2}=\left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$\frac{600}{300}=\left(\frac{V_1}{V_2}\right)^{1.4-1}$$

$$\frac{V_1}{V_2}=(2)^{\frac{1}{0.4}}$$

$$\frac{V_1}{V_2}=r_c=5.656$$

$$\eta_{otto}=1-\frac{1}{(r_c)^{r-1}}$$

$$=1-\frac{1}{(5.65)^{1.4-1}}$$

$= 49.99\%$

$= 50\%$

63. In an Otto cycle the compression ratio is 9 : 1 and the pressure and temperature at the beginning of the compression process are 100 KPa and 10^0 C respectively. The heat addition by combustion gives highest temperature as 2500 K. The highest cycle pressure is

(a) 7.95 MPa (b) 2.17Mpa

(c) 1.39 MPa (d) 8.50 MPa

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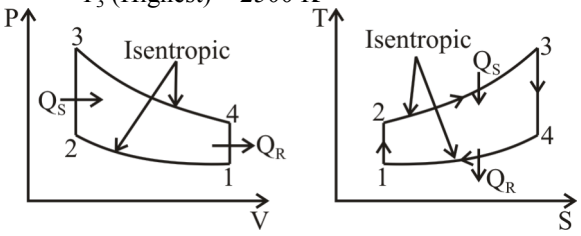
Ans. (a) : Given, Compression ratio (r) $=\frac{9}{1}=9$

$P_1 = 100$ kPa, $T_1 = 10°C = 10 + 273 = 283$ K

T_3 (Highest) $= 2500$ K



For isentropic process, $PV^{\gamma} = \text{Constant}$

or $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$$\left(\frac{T_2}{T_1}\right)=\left(\frac{V_1}{V_2}\right)^{\gamma-1}=r^{\gamma-1}=9^{1.4-1}=9^{0.4}$$

$\therefore$ $T_1 \times 9^{0.4} = 283 \times 9^{0.4} = 681.527$ K

Also, $\frac{P_2}{P_1}=\left(\frac{V_1}{V_2}\right)^{\gamma}=9^{1.4}$

$\therefore$ $P_2 = P_1 \times 9^{1.4} = 100 \times 9^{1.4}$

$P_2 = 2167.4$ kPa

Now we know that,

From process, 2-3 i.e. V = Constant

$P \propto T$

i.e. $\frac{P_2}{P_3}=\frac{T_2}{T_3}$

$$\frac{2167.4}{P_3}=\frac{681.527}{2500}$$

$\boxed{P_3 = 7.95\,MPa}$

64. In an Otto cycle the compression ratio is 9 : 1 and the pressure and temperature at the beginning of the compression process are 100 kPa and 10^o C. The heat addition by combustion gives the highest temperature as 2500 K. Thermal efficiency of the cycle is :

(a) 68.47% (b) 48.57%

(c) 38.4% (d) 58.47%

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Ans. (d) : Given - Compression ratio $(r) = \frac{v_1}{v_2} = \frac{9}{1}$

air (γ) = 1.4

So, Otto cycle thermal efficiency

$$\eta_{th} = 1 - \frac{1}{(r)^{\gamma-1}}$$

$$= 1 - \frac{1}{(9)^{1.4-1}} = 1 - \frac{1}{9^{0.4}}$$

$$\eta_{th} = 0.5847 = 58.47\%$$

65. A 4-stroke reciprocating engine has cylinder diameter of 4 cm, stroke length of 7 cm and clearance volume 2 cm^3. The engine capacity in cc is:

(a) 110 (b) 252

(c) 400 (d) 352

BHEL ET 2019

Ans. (d) : Given - Diameter (d) = 4 cm

Stroke length (L) = 7 cm

Clearance volume (V_C) = 2 cm^3

Engine capacity in cc = Swept volume × No. of cylinder

$$= \frac{\pi}{4} \times d^2 \times L \times 4 = \frac{\pi}{4} \times 4 \times 4 \times 7 \times 4$$

$$= 351.85$$

$$\approx 352 \text{ cc}$$

66. In an Otto cycle the compression ratio is 9 : 1 and the pressure and temperature at the beginning of the compression process are 100 kPa and 10^o C. The heat addition by combustion gives the highest temperature as 2500 K. Specific heat added by combustion is :

(a) 0.762 MJ/kg (b) 1.303 MJ/kg

(c) 0.286 MJ/kg (d) 1.048 MJ/kg

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Ans. (b) : Given -

In Otto cycle -

Compression ratio $(r) = \frac{9}{1}$

$$\frac{v_1}{v_2} = \frac{9}{1}$$

P_1 = 100 kPa, $T_1 = 10^0$ C = 273 + 10 = 283 K

T_3 = 2500 K

C_P= 1.005 kJ/kgK , C_V = 0.718 kJ/kgK. γ = 1.4

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = (9)^{1.4-1} = 9^{0.4}$$

$$\frac{T_2}{T_1} = 2.4082$$

$T_2 = 2.40822 \times 283$

$T_2 = 681.5275$ K

Specific heat addition by combustion -

$Q_{ha} = mC_v \times (T_3 - T_2)$

$= 1 \times 0.718 [2500 - 681.5275]$

$= 0.718 \times 1818.4724 = 1.305$ MJ/kg.

67. A diesel engine has a compression ratio of 14 and cutoff takes place at 6% of the stroke. The air standard efficiency will be

(a) 74.5% (b) 60.5%

(c) 52.5% (d) 44.5%

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Ans. (b) : Given,

r = 14

$(V_3 - V_2) = 0.06 (V_1 - V_2)$

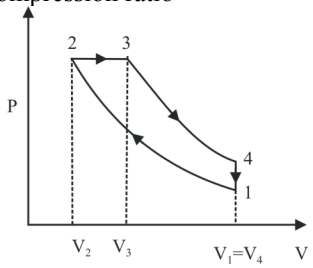
$$\left(\frac{V_3}{V_2} - 1\right) = 0.06\left(\frac{V_1}{V_2} - 1\right)$$

$(\rho - 1) = 0.06 (r - 1)$

$(\rho - 1) = 0.06 (14 - 1)$

$\rho = 1.78$

r = compression ratio



ρ = cut off ratio

$$\eta_{diesel} = 1 - \frac{1}{r^{\gamma-1}}\left[\frac{\rho^{\gamma} - 1}{\gamma(\rho - 1)}\right]$$

$$= 1 - \frac{1}{14^{(1.4-1)}}\left[\frac{1.78^{1.4} - 1}{1.4(1.78 - 1)}\right]$$

= 60.4%

68. An IC engine works with a compression ratio of 16. If cut off happens at 8% of the stroke, then the cut-off ratio of this engine is :

(a) 1.2 (b) 2.2

(c) 4.2 (d) 3.2

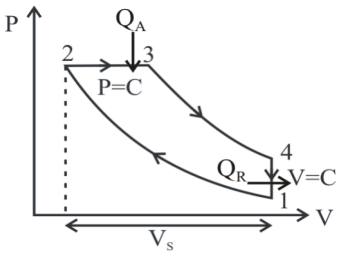
BHEL ET 2019

Ans. (b) : Given -

Compression ratio (r) = 16

cut-off ratio (ρ) = 8% of stroke

$$(V_3 - V_2) = \frac{8}{100} \times V_S$$



$V_3 - V_2 = 0.08 V_s$

$V_3 = 0.08 V_s + V_2$ ...(1)

compression ratio $r = 16 = \frac{V_c + V_s}{V_c}$

$16 V_c = V_c + V_s$

$15 V_c = V_s$

$$V_c = \frac{V_s}{15}$$

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$[\because V_c = V_2]$

$$V_2 = \frac{V_S}{15}$$

From equation (1)

$$V_3 = 0.08V_S + \frac{V_S}{15}$$

$$= \frac{1.2V_S + V_S}{15} = \frac{2.2V_S}{15}$$

$$\text{Cut-off ratio } (\rho) = \frac{V_3}{V_2} = \frac{\frac{2.2V_S}{15}}{\frac{V_S}{15}}$$

$$= \frac{2.2V_S}{15} \times \frac{15}{V_S}$$

$$\boxed{\text{cut} - \text{off ratio}(\rho) = 2.2}$$

69. Air enters diesel engine with a density of 1 kg/m^3. The compression ratio is 21. At steady state, the air intake is 21 × 10^{-3} kg/s and the network output is 20 kW. The Mean effective pressure (kPa) is :

(a) 500 (b) 1000
(c) 1500 (d) 2000

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (b) : Given, Compression ratio (r) = 21, ρ = 1kg/m^3,

m = 21 × 10^{-3}kg/sec.

$$r = \frac{V_s + V_c}{V_c} = 21$$

$$V_s = 20V_c \quad \ldots(i)$$

$$\text{Volume intake} = \frac{m}{\rho} = \frac{21\times10^{-3}}{1} = 21\times10^{-3}$$

$$V_s + V_c = 21\times10^{-3}$$

$$V_s + \frac{V_s}{20} = 21\times10^{-3}$$

$$V_s\left[1 + \frac{1}{20}\right] = 21\times10^{-3}$$

$$\boxed{V_s = 0.02}$$

Mean effective Pressure

$$P_{mean} = \frac{\text{work done}}{\text{swept volume}(V_s)}$$

$$= \frac{20\times10^3}{0.02}$$

$$= 1000000$$

$$\boxed{P_{mean} = 1000 \text{ kPa}}$$

70. The compression ratio of diesel engine as compared to expansion ratio is:

(a) Low
(b) Same
(c) High
(d) Depend on engine size

OPSC AEE 2019 PAPER - II

Ans. (c) : The compression ratio of diesel engine as compared to expansion ratio is high.

71. An IC engine having 6 cylinders, works on Otto cycle. It has a bore of 20 cm and a stroke 40 cm. If the clearance volume is 9000 cm^3, the compression ratio is :

(a) 10.03 (b) 8.53
(c) 9.38 (d) 7.33

BHEL ET 2019

Ans. (c) : Given - No. of cylinder = 6
Bore diameter (d) = 20 cm
Stroke (L) = 40 cm
Clearance volume (v_c) = 9000 cm^3

$$\text{Swept volume } (v_s) = \frac{\pi}{4}\times\left(20^2\right)\times40\times\text{No. of cylinder}$$

$$= \frac{\pi}{4}\times20\times20\times40\times6$$

$$v_s = 75398.2236 \text{ cm}^3$$

$$\text{Compression ratio (r)} = \frac{\text{Volume before compression}}{\text{Volume after compression}}$$

$$\text{Compression ratio (r)} = \frac{v_s + v_c}{v_c}$$

$$= \frac{75398.2236 + 9000}{9000}$$

$$= \frac{84398.223}{9000}$$

$$= 9.3775$$

$$\boxed{r = 9.38}$$

72. The crank radius of a single cylinder IC engine is (60/π) mm and diameter of the cylinder is 40 mm. The swept volume of the cylinder in cm^3 is:

(a) 48 (b) 58
(c) 68 (d) 78

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) : Given, Crank radius (r) $= \frac{60}{\pi} mm$

Bore dia (D) = 40 mm

$$\text{Swept volume } (V_s) = \frac{\pi}{4}D^2L$$

$$= \frac{\pi}{4}D^2\times(2r)$$

$$= \frac{\pi}{4}\times40^2\times2\times\frac{60}{\pi}$$

$$= 48000 \text{ mm}^3$$

$$\boxed{V_s = 48 \text{ cm}^3}$$

73. An IC engine has a bore and stroke length of 4 cm each. Total surface area through which heat transfer takes place in cm^2 is

(a) 18π (b) 24π
(c) 28π (d) 8π

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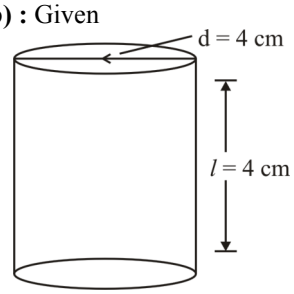
Ans. (b) : Given



Stroke length (ℓ) = 4 cm
bore diameter (d) = 4 cm
Heat transfer take place
at total surface area of cylinder

$= 2\pi r[r+\ell]$

$= 2\times\pi\times 2[2+4]$

$= 2\times\pi\times 2\times 6 \;\; = 24\pi \;\; cm^2$

74. The engine of a Lexus car has three cylinders with total displacement of 770 cc. If the compression ratio is 8.7, then each cylinder has a clearance volume of:

(a) 36.3 cc (b) 33.33 cc
(c) 44.4 cc (d) 3.33 cc

AAI Jr. Executive 29.11.2018

Ans. (b) : Compression Ratio = 8.7

$$CR = \frac{V_c + V_s}{V_c}$$

$$8.7 = \frac{V_c + V_s}{V_c}$$

$$8.7V_c = V_c + V_s$$

$$7.7V_c = 770$$

$$V_c = 100$$

$$\text{Each cylinder} = \frac{100}{3}$$

$$V_c = 33.33 \text{ cc}$$

75. Which one of the following cycles has the highest thermal efficiency for given maximum and minimum cycle temperature?

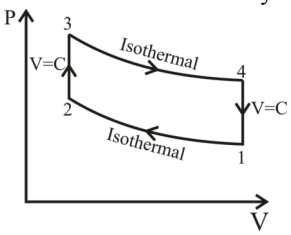
(a) Brayton cycle (b) Otto cycle
(c) Diesel cycle (d) Stirling cycle

GPSC ARTO Pre 30.12.2018

Ans. (d) : Stirling cycle has the highest thermal efficiency for given maximum and minimum cycle temperature.

Stirling cycle that features isothermal heat addition and heat rejection processes,

It is an alternate version of the carnot cycle.



76. Compression ratio of I.C. engine is .....

(a) the ratio of the volume of its combustion chamber from its largest capacity to its smallest capacity
(b) the ratio of clearance volumes available in the combustion cylinder
(c) the ratio of pressure after compression to pressure before compression
(d) the ratio of pressure before compression to pressure after compression

RPSC AE 2018

Ans. (a) : Compression ratio (r)—It is defined as the ratio of the volume of its combustion chamber from its largest capacity to its smallest capacity.

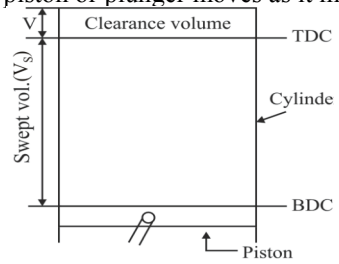
r = 7 - 12 (For SI Engine)
r = 12 - 17 (For Dual cycle Engine)
r = 16 - 21 (For CI Engine)

77. If V = volume at TDC and r = compression ratio. Then swept volume will be given by:

(a) $\frac{rV}{(r-1)}$ (b) $\frac{(r-1)v}{r}$
(c) Vr (d) (r – 1)V
(e) V ($r^2 - 1$)

CGPSC AE 25.02.2018

Ans. (d) : Swept Volume (V_S)– The volume through which a piston or plunger moves as it makes a stroke.



We know that compression ratio (r)

$$= \frac{V + V_S}{V}$$

$$r = 1 + \frac{V_S}{V}$$

$$\boxed{V_S = (r-1).V}$$

78. For the same maximum pressure and temperature

(a) Otto cycle is more efficient than Diesel cycle
(b) Diesel cycle is more efficient than Otto cycle
(c) Dual cycle is more efficient than Otto and Diesel cycle
(d) Dual cycle is less efficient than Otto and Diesel cycle

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : For the same maximum pressure and temperature Diesel cycle is more efficient than otto cycle.

$\eta_{diesel} > \eta_{dual} > \eta_{otto}$

79. The two stroke cycle engine has

(a) one suction valve and one exhaust valve operated by one cam
(b) one suction valve and one exhaust valve operated by two cams

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(c) only ports covered and uncovered by piston to effect charging and exhausting
(d) none of the above

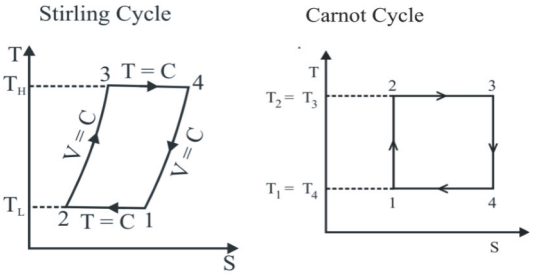
Nagaland PSC CTSE 2017 Paper-2

Ans. (c) : Two-stroke cycle has only ports covered and fin covered by piston to effect charging and exhausting.

80. Which of the following ideal cycles has the same efficiency as Stirling cycle?
(a) Diesel cycle (b) Otto cycle
(c) Joule cycle (d) Carnot cycle
(e) Dual cycle

CGPSC ARTO & TSI 07.01.2017

Ans. (d) :



$$\eta = 1 - \frac{T_L}{T_H}$$

81. Otto cycle efficiency is higher than diesel cycle efficiency for the same compression ratio and heat input because in Otto cycle.
(a) Combustion is at constant volume
(b) Expansion and compression are isentropic
(c) Maximum temperature is higher
(d) Heat rejection is lower

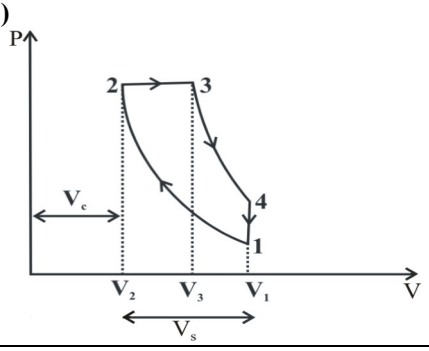
TSPSC AEE 2017

Ans. (d) : Although for a given compression ratio the Otto cycle has higher efficiency, because diesel engine, can be operated to higher compression ratio.
But for the same compression ratio and heat input the otto cycle efficiency is higher than diesel cycle due to heat rejection is lower so its increases Otto cycle efficiency.

82. In a diesel engine, the compression ratio is 13 : 1 and the fuel is cut-off at 8% of the stroke. γ for air = 1.4 and assuming the clearance volume = 1 m^3, the cut-off ratio is
(a) 1.51 (b) 1.96
(c) 0.196 (d) 0.51

Karnataka PSC AE, 10.09.2017

Ans. (b)



Compression ratio (r) = 13

$$r = \frac{V_1}{V_2} = 13$$

$$\text{Cutt-off}\,(\rho) = \frac{V_3}{V_2}$$

Given, $V_3 - V_2 = 0.08(V_1 - V_2)$

$V_3 = 0.08V_1 + 0.92V_2$

$V_3 = 0.08 \times 13V_2 + 0.92V_2$

$$\rho = \frac{V_3}{V_2} = 1.96$$

83. Increasing the compression ratio of Otto engine from 5 to 6 will change the air-standard efficiency by
(a) 20% (b) 8%
(c) 25% (d) –20%

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Given, $C.R_1 = r_1 = 5$ Let $\gamma = 1.4$

$C.R_2 = r_2 = 6$

Case-I $\eta_{otto} = 1 - \frac{1}{r^{\gamma-1}}$

$$\eta_I = 1 - \frac{1}{5^{1.4-1}} = 47.5\%$$

Case-II $\eta_{II} = 1 - \frac{1}{6^{1.4-1}} = 51.2\%$

$\eta_{II} - \eta_I = 51.2 - 47.5$

$= 3.7\%$

$$\% \text{ change in efficiency} = \frac{51.2 - 47.5}{47.5}$$

$= 7.8\% \approx 8\%$

84. An Engine operating on Otto cycle is having a thermal efficiency of 0.3 and relative efficiency of 0.5. The air-standard efficiency of the cycle is
(a) 0.15 (b) 0.2
(c) 0.6 (d) 0.8

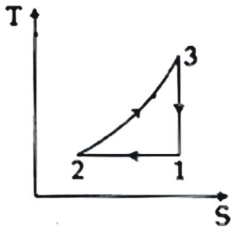
GPSC Asstt. Director of Transport 05.03.2017

Ans. (c)

$$\text{Relative efficiency} = \frac{\text{Indicated thermal efficiency}}{\text{Air standard efficiency}}$$

$$0.5 = \frac{0.3}{\eta_{air}} \Rightarrow \eta_{air} = 0.6$$

85. An ideal air standard cycle is shown in figure.



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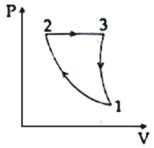
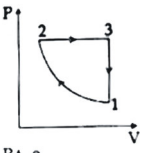
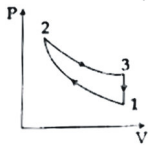
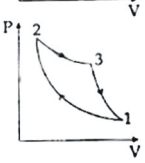
(a) 

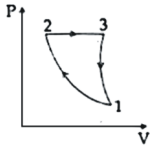
(b) 

(c) 

(d) 

TRB Polytechnic Lecturer 2017

Ans. (a)



86. The work output of theoretical Otto cycle
- (a) Increases with increase in compression ratio
- (b) Increases with increase in pressure ratio
- (c) Increases with increase in adiabatic index
- (d) Follows all the above

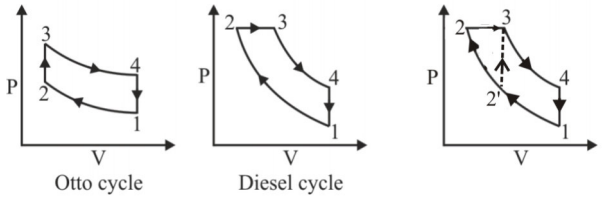
Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : Work output of theoretical otto cycle increase with increasing compression ratio, as well increases with increase in pressure ratio and increases with increase in adiabatic index.

87. If the maximum pressure in both air standard Otto and Diesel cycles is the same, then the relations for compression ratio r and the efficiency η between the two cycles are
- (a) $r_{Diesel} > r_{Otto}$ and $\eta_{Diesel} > \eta_{Otto}$
- (b) $r_{Otto} > r_{Diesel}$ and $\eta_{Diesel} > \eta_{Otto}$
- (c) $r_{Diesel} > r_{Otto}$ and $\eta_{Otto} > \eta_{Diesel}$
- (d) $r_{Otto} > r_{Diesel}$ and $\eta_{Otto} > \eta_{Diesel}$

ESE 2017

Ans. (a) :



Process 1 – 2 – 3 – 4 – 1 Diesel cycle
Process 1 – 2' – 3 – 4 – 1 Otto cycle
Compression ratio, $r_{diesel} > r_{otto}$
In this case by considering same maximum pressure and heat input or same maximum pressure and work output compression ratio and efficiency of diesel cycle is more than otto cycle.

88. The air standard efficiency of Ericsson cycle for same temperature limits is equal to the efficiency of:
- (a) Otto cycle
- (b) Rankine cycle
- (c) Carnot cycle
- (d) Dual cycle

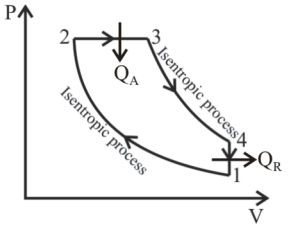
PTCUL AE 25.06.2017

Ans. (c) :

89. Cut-off ratio which is used to describe the diesel engine performance is defined as:
- (a) Volume at the start of heat addition/volume at the end of heat addition
- (b) Pressure at the start of heat addition/Pressure at the end of heat addition
- (c) Volume at the end of heat addition/volume at the start of heat addition
- (d) Pressure at the end of heat addition/pressure at the start of heat addition

HPPSC AE (PWD) 10.09.2016

Ans. (c) : The cut-off ratio is the ratio of the volume after combustion to the volume before combustion.
Air standard diesel cycle



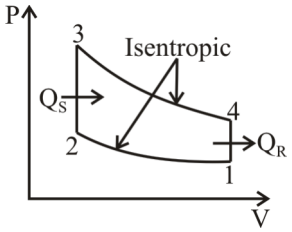
Compression ratio (r) $= \frac{V_1}{V_2}$

cut off ratio $(r_c) = \frac{V_3}{V_2}$

90. In an otto cycle, twice as much energy is generated per cycle as is rejected through the exhaust. What is the compression ratio? (for air , k = 1.4
- (a) 5.66
- (b) 7.66
- (c) 11.2
- (d) None of these

RPSC ADE 2016

Ans. (a) :



Efficiency of Otto cycle $= 1 - \frac{1}{(r)^{\gamma-1}}$

Efficiency $= \frac{\text{work output}}{\text{Heat supplied}} = \frac{\text{Heat generated - Heat rejected}}{\text{Heat supplied}} = \frac{2-1}{2}$

$\frac{1}{2} = 1 - \frac{1}{(r)^{1.4-1}}$ or r = compression ratio = 5.66

91. An otto cycle with a specific compression ratio is executed using air, argon, ethane and biogas. For which case will the thermal efficiency be highest?
- (a) Air
- (b) Argon
- (c) Ethane
- (d) Biogas

GPSC Lect. (Auto) 16.10.2016

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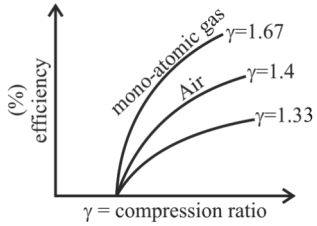
Ans. (b) : $\eta_{otto} = 1 - \left(\frac{1}{r}\right)^{\gamma-1}$

For Argon $\gamma = 1.66$

Air $\gamma = 1.4$

As γ increases, efficiency also increases.



92. Air standard diesel cycle consist of :

(a) Constant volume heat addition
(b) Isothermal heat addition
(c) Adiabatic heat addition
(d) Constant pressure heat addition

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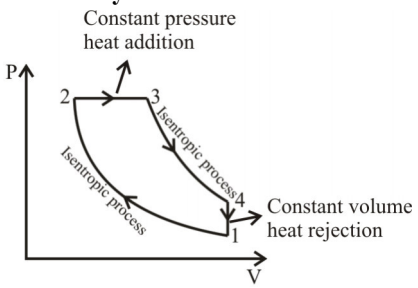
Ans. (d) : Diesel Cycle –



1-2 Isentropic compression.
2-3 Constant pressure heat addition.
3-4 Isentropic expansion.
4-1 Constant volume heat rejection.

$$\eta_{diesel} = 1 - \frac{(\rho^{\gamma} - 1)}{\gamma \cdot r^{(\gamma-1)} \cdot (\rho - 1)}$$

r = compression ratio = V_1/V_2

ρ = cut-off ratio $= \frac{V_3}{V_2}$

93. What is the Otto Cycle efficiency if the compression ratio is 8 and the ratio of the molar specific heats is 1.8?

(a) 19% (b) 81%
(c) 18% (d) None of these

KPCL AE 2016

Ans. (b) : Given, Compression ratio (r) = 8

$\gamma = 1.8$

$$\eta_{otto} = 1 - \frac{1}{r^{\gamma-1}}$$

$$= 1 - \frac{1}{(8)^{1.8-1}}$$

$$\eta_{otto} = 81.05\%$$

94. The dual combustion cycle is also known as the_____

(a) Trinkler cycle
(b) Seiliger cycle
(c) Limited pressure or mixed cycle
(d) All the above

KPCL AE 2016

Ans. (d) : Dual combustion cycle (also known as the mixed cycle, Trinkler cycle, Seiliger cycle or sabathe cycle) is a thermal cycle which is combination of the Otto cycle and the diesel cycle.

95. Thermal efficiency of Otto cycle can be represented by $\eta_{otto} = 1 - \frac{T_a}{T_b}$. Which of the following element is correct for state point a and b?

(a) Constant volume heat addition process will start from 'b'
(b) Constant volume heat rejection process will end at point 'b'
(c) Isentropic compression process will end at point 'a'
(d) Isentropic compression process will start from point 'b'

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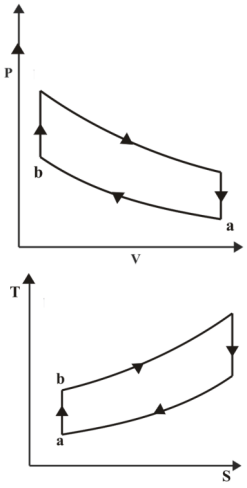
Ans. (a) : The thermal efficiency of Otto cycle.

$$\eta_{Otto} = 1 - \frac{T_a}{T_b}$$



$$\eta_{otto} = 1 - \frac{1}{\left(\frac{T_b}{T_a}\right)}$$

$$\eta_{otto} = 1 - \frac{1}{(r_c)^{\gamma-1}}$$

It means that constant volume heat addition process will start from 'b' and adiabatic compression will start from point 'a' to point 'b'.

96. Swept volume of engine may be written as:

(a) $V_{BDC} - V_{TDC}$
(b) Volume of cylinder at BDC (V_{BDC})
(c) $V_{BDC} + V_{TDC}$
(d) Volume of cylinder at TDC (V_{TDC})

UPRVUNL AE 2016

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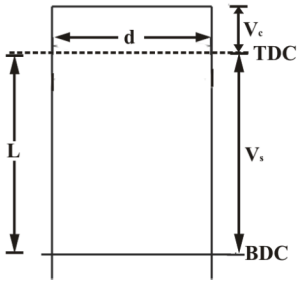
Ans. (a) : Swept volume of engine may be written as $V_{BDC} - V_{TDC}$



V_c = Clearance volume
V_s = Swept volume
d - Bore of cylinder
L - Stroke length

97. Thermal efficiency of diesel cycle is

(a) Reduced at high compression ratio for same cut off ratio and heat capacity ratio
(b) Independent of variation of cut off ratio and heat capacity ratio
(c) Reduced at higher cut off ratio for same compression ratio and heat capacity ratio
(d) Reduced at high heat capacity ratio for same compression ratio and cut off ratio

RPSC LECTURER 16.01.2016

Ans. (c) $\eta_{Diesel} = 1 - \frac{1}{r^{(\gamma-1)}} \times \frac{(\rho^{\gamma}-1)}{\gamma(\rho-1)}$

Thermal efficiency of diesel cycle is reduced at higher cut off ratio for same compression ratio and heat capacity ratio.

98. If stroke volume of engine = 300 cc. Clearance volume of engine = 20 cc. The compression ratio of engine will be

(a) 16 (b) 14
(c) 15 (d) 17

UPRVUNL AE 2016

Ans. (a) : Data given,
$V_s = 300$ cc
$V_c = 20$ cc
Then compression ratio (r_c)

$$r_c = \frac{V_s + V_c}{V_c}$$

$$r_c = 1 + \frac{V_s}{V_c} = 1 + \frac{300}{20}$$

$\boxed{r_c = 16}$

99. Air is compressed in a car engine from 22°C and 95 kPa in a reversible and adiabatic manner. If the compression ratio V_1/V_2 of this engine is 8, determine the final temperature of the air

(a) 632.7 K (b) 562.7 K
(c) 662.7 K (d) 762.7 K
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : $T_1 = 295$ K, $T_2 = ?$

$$\frac{V_1}{V_2} = 8$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$T_2 = (8)^{1.4-1} \times 295$$

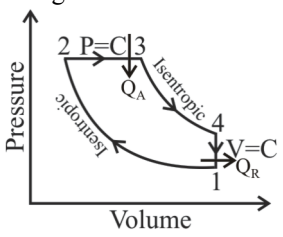
$T_2 = 662.7$ K

100. Diesel engine works on :

(a) Constant volume heat addition and constant volume heat rejection.
(b) Constant pressure heat addition and constant volume heat rejection
(c) Constant pressure heat addition and constant pressure heat rejection.
(d) None of the above

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : Diesel engines works on-



Process 2-3- const pressure heat addition.
Process 4-1- const volume heat rejection.

Efficiency of diesel engine, $\eta_{diesel} = 1 - \frac{1}{r^{\gamma-1}}\left[\frac{\rho^{\gamma}-1}{\gamma(\rho-1)}\right]$

Where, r = Compression ratio
ρ = Cut-off ratio

101. If compression ratio is increased :

(a) Thermal efficiency will be increased
(b) Polytropic efficiency will be increased
(c) Petrol efficiency will be decreased
(d) Air standard efficiency will be increased

HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : Air standard efficiency

$$\boxed{\eta = 1 - \left(\frac{1}{r}\right)^{\gamma-1}}$$

with increase in compression ratio ($r\uparrow$), air standard efficiency ($\eta\uparrow$) will increased.
r = compression ratio
γ = specific heat ratio = C_p/C_v

102. A cross sectional area of a single cylinder engine by its stroke length is known as :

(a) Clearance volume
(b) Combustion volume
(c) Swept volume
(d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : Swept Volume– Swept volume can be defined as the volume swept by the engine piston during one stroke. Swept volume is also the product of piston area and stroke

$$V_s = \frac{\pi}{4} D^2 \times L$$

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Where, D – Bore dia
L – Length of stroke

Clearance volume- Clearance volume can be defined as the volume that remains in the cylinder when the engine piston is at the top dead centre position.

103. If clearance volume of IC engines is increased, the compression ratio will

(a) Increase (b) Decrease
(c) Remain same (d) Be doubled

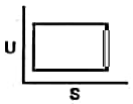
Assam PSC CCE Pre 2015

Ans. (b) :

Compression ratio, $r_c \downarrow = \frac{V_c \uparrow + V_s}{V_c \uparrow} = 1 + \frac{V_s}{V_c}$

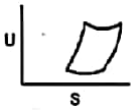
If clearance volume of I.C. engines is increased, the compression ratio will decreases.

104. An Otto cycle on internal energy (U) and entropy (S) diagram is shown in

(a) 

(b)

(c)

(d) 

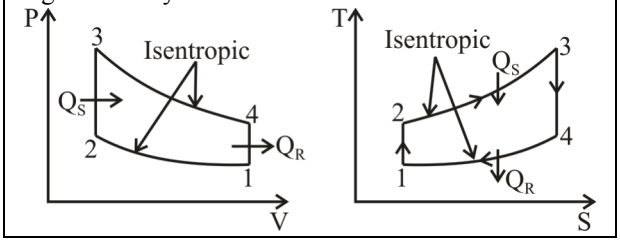
Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : In air standard otto cycle air is used as working fluid which is considered as ideal gas.

We know,

For ideal gas (U) = f(T) only (Joule's law)

U-S diagram of otto cycle should be similar to T-S diagram otto cycle.



105. The temperature of interior surface of cylinder wall in normal operation is not allowed to exceed

(a) 80°C (b) 120°C
(c) 180°C (d) 240°C

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : The temperature of interior surface of cylinder wall in normal operation is not allowed to exceed 180°C.

106. A diesel and Otto cycle have the same compression ratio. The cutoff ratio of the cycle is 'S'. The air standard efficiency of the cycle will be equal when

(a) $S^r - r(S - 1) - 1 = 0$
(b) $S^r - r(S - 1) + 1 = 0$
(c) $S^r - r(S + 1) + 1 = 0$
(d) $S^r - r(S - 1) + r = 0$

TSPSC AEE 2015

Ans. (a) : If

$$\eta_{diesel} = \eta_{otto}$$

$$1 - \frac{1}{x^{(r-1)}} \frac{\left(S^r - 1\right)}{r\left(S-1\right)} = 1 - \frac{1}{x^{(r-1)}}$$

$$S^r - 1 = r\left(S-1\right)$$

$$\boxed{S^r - r\left(S-1\right) - 1 = 0}$$

107. The thermal efficiency of a diesel cycle can be increased by

(a) increasing both compression ratio and cutoff ratio
(b) decreasing both compression ratio and cutoff ratio
(c) decreasing compression ratio and increasing cutoff ratio
(d) increasing compression ratio and decreasing cutoff ratio

TSPSC AEE 2015

Ans. (d) : The thermal efficiency of a diesel cycle can be increase by increasing compression ratio and decreasing cutoff ratio (ρ):

$$\eta_{diesel} = 1 - \frac{1}{r^{(\gamma-1)}} \frac{\left(\rho^\gamma - 1\right)}{\gamma\left(\rho - 1\right)}$$

- $\uparrow\eta$ with $\uparrow$ r(compression ratio)
- $\uparrow\eta$ with $\downarrow$ ρ(cut-off)

108. Which gas power cycle consists of 4 processes during which work alone is transferred during two processes and heat alone is transferred during the other two processes?

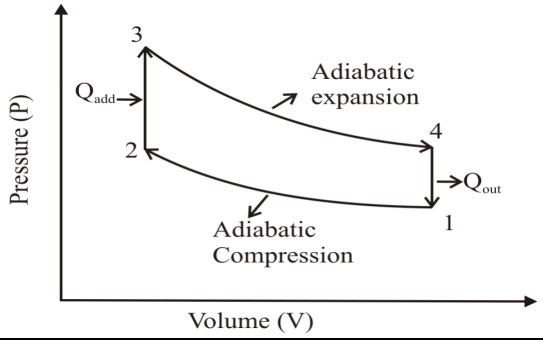
(a) Atkinson cycle (b) Carrot cycle
(c) Diesel cycle (d) Otto cycle

TSPSC Managers, 2015

Ans. (d) : Otto cycle-

- two constant volume process.
- two reversible adiabatic process.

Efficiency of otto cycle $\eta_o = 1 - \frac{1}{r^{\gamma-1}}$



109. The bore and stroke of the cylinder of a 6 cylinder engine working on an otto cycle are 20 cm and 30 cm respectively The total clearance volume is 3000π cm^3. What is the compression ratio?

(a) 5 (b) 6
(c) 7 (d) 8

TSPSC Managers, 2015

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Ans. (c) : Given- Bore diameter (d) = 20 cm
Stroke length (l) = 30 cm
No. of cylinder (n) = 6
total clearance volume (V_c) = 3000 π cm^3

$\therefore$ Swept volume = $n \times \frac{\pi}{4} \times d^2 \times l$

$= 6 \times \frac{\pi}{4} \times (20)^2 \times (30)$

$V_s = 18000\pi \text{ cm}^3$

We know that,

Compression ratio (r) $= \frac{V_s + V_c}{V_c}$

$(r) = \frac{18000\pi + 3000\pi}{3000\pi}$

$r = \frac{21000\pi}{3000\pi} = 7$

110. The thermal efficiency of Otto cycle increases with increase in
(a) Compression ratio
(b) Pressure ratio
(c) Adiabatic index
(d) All options are correct

MPPSC State Forest Service Exam, 2014

Ans. (d) : As we know, the thermal efficiency of otto cycle increase with increase in compression ratio, pressure ratio, adiabatic index.

$\Rightarrow \quad \eta_{otto} = 1 - \frac{1}{r^{\gamma-1}}$

- $\uparrow \eta_{otto}$ with $\uparrow$ r(compression ratio)
- $\uparrow \eta_{otto}$ with $\uparrow \gamma$
- $\uparrow \eta_{otto}$ with pressure ratio increases

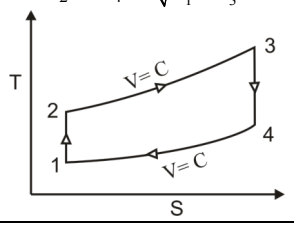
111. For maximum specific output an air standard Otto cycle.
(a) the working fluid should be air
(b) the speed be high
(c) suction temperature should be high
(d) temperature of the working fluid at the end of compression be the geometric mean of minimum and maximum temperature of cycle
(e) temperature of the working fluid at the end of compression be the arithmetic mean of minimum and maximum temperature of cycle.

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) For maximum specific output of an air standard Otto cycle temperature of the working fluid at the end of compression be the geometric mean of minimum and maximum temperature of cycle

For otto cycle $\rightarrow T_1 \times T_3 = T_2 \times T_4$

For max. work $\rightarrow T_2 = T_4 = \sqrt{T_1 \times T_3}$



112. Air standard efficiency of an I.C. Engine depends on
(a) Compression ratio (b) Speed of operation
(c) Fuel used (d) Torque

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Air standard efficiency of an I.C. Engine depends on $\boxed{\eta = 1 - \left(\frac{1}{r}\right)^{\gamma-1}}$ hence depends upon compression ratio (r) and specific heat ratio (γ).

113. The swept volume and clearance volume of an Otto cycle are 0.0184 m^3 and 0.00262 m^3 respectively. The value of compression ratio is:
(a) 7 (b) 8
(c) 9 (d) 5.99

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Compression ratio $= \frac{V_c + V_s}{V_c}$

$= \frac{0.00262 + 0.0184}{0.00262}$

$= \frac{0.02102}{0.00262}$

$= 8.022$

114. If air is compressed adiabatically from atmospheric condition in a cylinder having compression ratio of 6, then pressure at the end of compression shall be :
(a) 6 atm
(b) Less than 6 atm
(c) More than 6 atm
(d) Less or more than 6 at depending on temperature at the end of compression

J&K PSC Civil Services Pre, 2013

Ans. (c) : $PV^{\gamma} = \text{constant}$

$P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$\left(\frac{V_1}{V_2}\right)^{\gamma} = \left(\frac{P_2}{P_1}\right)$

Compression ratio (r) $= \frac{V_1}{V_2}$

$[\because \gamma = 1.4 \text{ for air}]$

$(6)^{1.4} = \left(\frac{P_2}{P_1}\right)$

$\frac{P_2}{P_1} = 12.28 \Rightarrow P_2 = 13P_1$

115. In an Otto cycle, the temperature at the beginning and end of the isentropic compression are 316 K and 596 K respectively. The compression ratio will be
(a) 4.588 (b) 4.858
(c) 4.885 (d) 4.558

JPSC AE 2013 Paper-V

Search on Telegram @apna_pdf

Ans. (c) : Given,

$T_1 = 316$ K

$T_2 = 596$ K

Compression ratio $= \frac{V_1}{V_2}$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$\left(\frac{596}{316}\right) = \left(\frac{V_1}{V_2}\right)^{0.4}$$

$$\frac{V_1}{V_2} = \left(\frac{596}{316}\right)^{\frac{1}{0.4}}$$

$$\boxed{\frac{V_1}{V_2} = 4.885}$$

116. Increase in compression ratio in CI engines

(a) decreases turbulence
(b) increases mechanical efficiency
(c) decreases mechanical efficiency
(d) decreases thermal efficiency

TNPSC AE 2013

Ans. (c) : Decrease mechanical efficiency when increase in compression ratio of CI engines.

117. Air standard efficiency of an Otto cycle having compression ratio 4 and adiabatic exponent for air $\gamma = 1.5$ will be:-

(a) 50% (b) 75%
(c) 25% (d) 5.0%

UKPSC AE-2013, Paper-II

Ans. (a) : We know that efficiency of an otto cycle

$$\eta_{otto} = 1 - \frac{1}{[C.R.]^{\gamma-1}} = 1 - \frac{1}{[4]^{1.5-1}} = 0.50$$

$\eta_{otto} = 50\%$

118. The thermal efficiency of an ideal Otto cycle:-

(a) Increases with compression ratio
(b) Increases with specific heat ratio
(c) Increases with compression ratio and specific heat ratio
(d) Increases with compression ratio but decreases with specific heat ratio

UKPSC AE-2013, Paper-II

Ans. (c) : The thermal efficiency of an ideal Otto cycle increases with compression ratio and specific heat ratio.

119. For the same heat input and compression ratio the most efficient cycle is

(a) Otto cycle (b) Diesel cycle
(c) Dual cycle (d) None of these

TRB Poly. Lect., 2012

Ans. (a) : For same heat input & compression ratio, efficiency of Otto cycle is higher than that of diesel cycle. Dual cycle efficiency always lies between the Otto cycle and diesel cycle.

$$\boxed{\eta_{otto} > \eta_{dual} > \eta_{diesel}}$$

120. If T_{max} and T_{min} be the maximum and minimum temperature in an Otto cycle, then for the ideal conditions, the temperature after compression should be

(a) $\frac{T_{max} + T_{min}}{2}$
(b) $\sqrt{\frac{T_{max}}{T_{min}}}$
(c) $\sqrt{T_{max} \times T_{min}}$
(d) $T_{min} + \frac{T_{max} - T_{min}}{2}$

BPSC AE 2012 Paper - V

Ans : (c) : If the T_{max} and T_{min} be the maximum and minimum temperature in an Otto cycle. For the ideal conditions the temperature after compression should be

$= \sqrt{T_{max} \times T_{min}}$

121. In an Otto cycle, the heat addition and heat rejection take place at

(a) constant volume and at constant pressure respectively
(b) constant volume and at constant volume respectively
(c) constant pressure and at constant volume respectively
(d) constant pressure and at constant pressure respectively

UKPSC AE 2012 Paper–II

Ans. (b) : constant volume and at constant volume respectively

122. The air standard efficiency of an Otto cycle for a compression ratio of 5 and index $\gamma = 1.4$ is

(a) 60% (b) 50%
(c) 47.47% (d) 40%

UKPSC AE 2012 Paper–II

Ans. (c) : Given, r = 5

Air standard efficiency of an Otto cycle

$$\eta_{otto} = 1 - \frac{1}{r^{\gamma-1}}$$

$$= 1 - \frac{1}{(5)^{1.4-1}}$$

$= 47.47\%$

123. Consider the following statements :

1. An I.C. engine transforms chemical energy into mechanical energy.
2. A compressed spring possess potential energy.
3. A football rolling on the ground performs plane motion.
4. Strain gauges are used to measure torque.

Following are the correct statements :

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 and 4 only (d) 1, 2 and 4 only

UKPSC AE 2012 Paper-I

Ans. (d)

124. The ratio of actual cycle efficiency to that of the ideal cycle efficiency is called

(a) effectiveness (b) work ratio
(c) efficiency ratio (d) net efficiency

TNPSC ACF 2012

Ans. (c) : Efficiency ratio $= \frac{\text{Actual cycle efficiency}}{\text{Ideal cycle efficiency}}$

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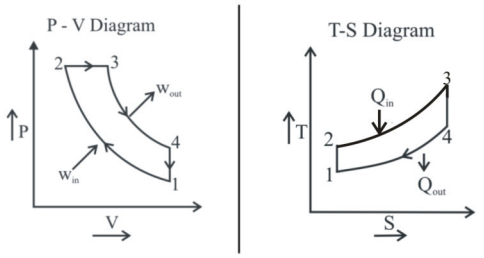
125. The air standard cycle with the following processes in sequences :
(1) Reversible adiabatic compression
(2) Reversible constant pressure heat addition
(3) Reversible adiabatic expansion
(4) Reversible constant volume heat rejection is:
(a) Otto Cycle (b) Diesel Cycle
(c) Dual Cycle (d) Brayton Cycle

OPSC Civil Services Pre. 2011

Ans. (b) : (1) Reversible adiabatic compression
(2) Reversible constant pressure heat addition
(3) Reversible adiabatic expansion
(4) Reversible constant volume heat rejection



126. For specific limits for the maximum and minimum temperature, the ideal cycle with the lowest thermal efficiency is :
(a) Carnot (b) Otto
(c) Diesel (d) Dual

J&K PSC Civil Services Pre, 2010

Ans. (b) : Otto cycle has the lowest thermal efficiency for fixed maximum and minimum temperature.

$$\eta_{otto} = 1 - \frac{1}{\left(r^{\gamma-1}\right)}$$

$$\eta_{otto} = 1 - \left(\frac{1}{r}\right)^{\gamma-1}$$

$$\left[\because \left(\frac{1}{r}\right)^{\gamma-1} = \frac{T_1}{T_2} = \frac{\text{minimum temperature}}{\text{maximum temperature}}\right]$$

127. An oil engine working on ideal diesel cycle has 20 cm bore and 30 cm stroke and the clearance volume is 420 cm^3. The fuel supply is 50% of the stroke. If the fuel supply is delayed from 5% to 8%. Find the percentage change in thermal efficiency :
(a) 64.7% (b) 7.75%
(c) 52.4% (d) 58.0%

APPSC IOF, 2009

Ans. (b) : Given, d = 20 cm = 0.2 m, L = 30 cm = 0.3 m

$$v_2 = 420\ cm^3 = 4.2 \times 10^{-4}\ m^3$$

$$r = \frac{v_1}{v_2} = \frac{v_c + v_s}{v_c}$$

v_s = area × length

$$v_s = \frac{\pi}{4} d^2 \times L$$

$$v_s = \frac{\pi}{4}(0.2)^2 \times 0.3$$

$$v_s = 9.4 \times 10^{-3}\ m^3$$

$$r = \frac{4.2\times10^{-4} + 9.42\times10^{-3}}{4.2\times10^{-4}} = 23.42$$

Cuttoff ratio, $\rho = \frac{v_3}{v_2}$ for 5%

$$= \frac{\left(v_c + 5\% v_s\right)}{v_c}$$

$$\rho_{5\%} = 2.12$$

For 8%

$$\rho_{8\%} = 1 + \frac{(0.08 \times 9.42 \times 10^{-3})}{4.2 \times 10^{-4}} = 2.79$$

$$\eta_{\text{diesel } 5\%} = -\frac{1}{r^{(\gamma-1)}} \times \frac{(\rho^{\gamma} - 1)}{\gamma(\rho - 1)} = 66.34\%$$

Similarly $\eta_{\text{diesel } 8\%} = 58.58\% \rightarrow \Delta\eta = 7.75\%$

128. In an air-standard Diesel cycle, the compression ratio is 16, the cylinder bore is 200 mm and the stroke is 300 mm. Compression begins at 1 bar and 27°C. The cut-off takes place at 8 per cent of the stroke. The work done per cycle is :
(a) 51.56 kJ (b) 8.21 kJ
(c) 7.73 kJ (d) 30.16 kJ

APPSC IOF, 2009

Ans. (a) : d = 200 mm = 0.2 m, L = 300 mm = 0.3 m

$$r = \frac{v_1}{v_2} = \frac{v_c + v_s}{v_c} = 16$$

$$\rho = \frac{v_3}{v_2} \quad \text{for 8\%}$$

$$= \frac{v_c + 8\% v_s}{v_c} \qquad \therefore v_s = 9.4 \times 10^{-3}\ m^3$$

$$\rho = \frac{0.66 v_s + 0.08 v_s}{0.66 v_s} = 1.12 \qquad v_c = \frac{9.4}{15} \times 10^{-3}$$

$$v_c = \frac{v_s}{15}$$

$$\eta_{diesel} = 1 - \frac{1}{\gamma r^{(\gamma-1)}} \times \frac{(\rho^{\gamma} - 1)}{\rho - 1}$$

$$= 1 \frac{1}{1.4 \times 16^{1.4-1}} \times \frac{1.12^{1.4} - 1}{(1.12 - 1)}$$

$$= 1 - 1.43 \times \frac{1}{4.24} = 66.2\%$$

$$\eta = \frac{W}{Q}$$

$$\Rightarrow \quad W = \frac{p_1 v_1 r^{\gamma-1}\left[\gamma(\rho-1) - r^{1-\gamma}(\rho^{\gamma-1})\right]}{\gamma - 1}$$

$$= 51.56\ kJ$$

129. An oil engine working on Diesel cycle has compression ratio of 13.5. It uses A : F = 28 : 1. The CV of fuel used = 42 MJ/kg. The temperature at the end of compression is 800°C. Take $C_p = (1 + 18.8 \times 10^{-5}T)$ kJ/kg K, where T is in K. The percentage of stroke at which cut-off occurs is :

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(a) 8% (b) 16.7%
(c) 2.8% (d) 8.25%

APPSC IOF, 2009

Ans. (d) : $C_p = (1 + 18.8 \times 10^{-5}T)$ kJ/kg-K

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{r-1} \Rightarrow T_2 = 800°C = 1073\,K$$

$$T_1 = \frac{1073}{(13.5)^{1.4-1}} = 379.15\,K$$

$dQ = mC_p\,dT,$

$Q = \int_2^3 C_p dT$ for 1 kg of mixture

$$Q = \frac{C.V}{A+F} = \frac{42000}{28+1} = 1448.29\,kJ/kg$$

$$1448.29 = \int_2^3 (1+18.8\times10^{-5}T)\,dT$$

T_3 = We get

$$V_3 = V_2 \times \frac{T_3}{T_2}$$

Now, $(V_1 - V_2) = V_2\left(\frac{V_1}{V_2} - 1\right) = 12.502$

$$\rho\% = \frac{V_3 - V_2}{V_1 - V_2} = 8.25\%$$

130. For the same maximum pressure and same output :

(a) Otto cycle is less efficient than diesel cycle
(b) Dual cycle is more efficient than diesel cycle
(c) Otto cycle is more efficient than diesel cycle
(d) Dual cycle is less efficient than Otto cycle

APPSC IOF, 2009

Ans. (a) : $\Rightarrow$ For the same maximum pressure and heat input

$\eta_{diesel} > \eta_{dual} > \eta_{otto}$

$\Rightarrow$ For the same maximum pressure and heat output

$\eta_{diesel} > \eta_{dual} > \eta_{otto}$

$\Rightarrow$ For same maximum temperature and heat rejection

$\eta_{diesel} > \eta_{dual} > \eta_{otto}$

$\Rightarrow$ For same compression ratio and heat addition

$\eta_{otto} > \eta_{dual} > \eta_{diesel}$

$\Rightarrow$ For same compression ratio and heat rejection.

$\eta_{otto} > \eta_{dual} > \eta_{diesel}$

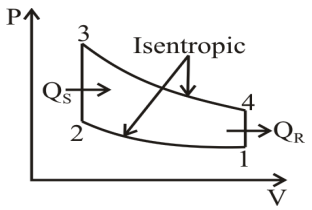
Diesel engine has higher thermal efficiency than otto cycle because of higher compression ratio.

131. Which process is not associated with diesel cycle?

(a) Constant pressure (b) Constant volume
(c) Adiabatic (d) Isothermal

UKPSC AE 2007 Paper -II

Ans. (d) :



Process (1-2) = Reversible adiabatic compression
Process (2-3) = Constant pressure heat addition
Process (3-4) = Reversible adiabatic expansion
Process (4-1) = Constant volume heat rejection

132. Diesel cycle efficiency is maximum when cut off is

(a) minimum (b) maximum
(c) zero (d) none of the above

UKPSC AE 2007 Paper -II

Ans. (a) : Diesel cycle efficiency is maximum when cut off is minimum.

2. SI and CI Engines

133. At the time of starting, idling and low-speed operation, the carburetor supplies a mixture which can be termed as

(a) Lean
(b) Slightly leaner than stoichiometric
(c) Stoichiometric
(d) Rich

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
Haryana PSC AE (PHED) 04.09.2020, Paper-I
Gujarat PSC AE 2019
OPSC AEE 2019 PAPER - II
ISRO Scientist/Engineer (RAC), 10.03.2019
GPSC ARTO Pre 30.12.2018
CGPSC AE 2014- II
RPSC AE 2016, (KPSC AE 2015)

Ans. (d) : In a SI engine carburetor, the idling line is used for providing rich mixture at low load. The no load running mode of the engine is called idling.

- At normal load (cruising) lean mixture required.
- For max. power rich mixture required.

The problem of dilution of charge by residual gases becomes more pronounced at low loads and idling hence richer mixture (Air-fuel ratio around 12.5:1) is required

134. In spark ignition engines knocking can be reduced by:

(a) Increasing the compression ration
(b) Increasing the cooling water temperature
(c) Retarding the spark advance
(d) Increasing the inlet air temperature

MPPSC AE 2016, RPSC AE 2016
KPSC ADF 2015, KPSC AE. 2015
Kerala PSC AE 06.08.2015
RPSC ACF-2011, TNPSC AE, 2008

Ans : (c) In spark ignition engines knocking can be reduced by Retarding the spark advance.

Factor tending to reduce knocking in SI engine

* Self ignition temperature of fuel	-	High
* Compression ratio	-	Low
* Inlet temperature and pressure	-	Low
* Super charging	-	No
* Spark advance	-	No
* Flame travel distance	-	Small
* Engine size	-	Small
* Engine Speed	-	High
* Octane rating	-	High

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135. Compression ratio in SI engine varies from

(a) 6 to 10 (b) 10 to 14
(c) 14 to 22 (d) 22 to 32

JPSC AE, 10.04.2021, Paper-I
HPPSC AE 2018
UKPSC AE 2007 Paper -II
Kerala PSC Poly. Lect., 2017
WBPSC AE 2008

Ans. (a) : Compression ratio for SI Engine (r) is 6 to 10
C.R. for CI engine is 16 to 21
C.R. for Dual cycle engine is 12 to 17

136. If petrol is used in a diesel engine, then :

(a) Low power will be produced
(b) Efficiency will be low
(c) Higher knocking will occur
(d) Black smoke will be produced

RPSC ACF & FRO, 26.02.2021
RPSC IOF, 2020
Nagaland PSC (CTSE) 2018, Paper-II
HPPSC Asstt. Prof. 2014
TNPSC ACF 2012

Ans. (c) : If petrol is used in diesel engine then higher knocking will occur because in diesel engine compression ratio is high.

137. The compression ratio for diesel engines is close to

(a) 3 to 6 (b) 5 to 8
(c) 15 to 20 (d) 20 to 30

ISRO Scientist/Engineer 22.04.2018
RPSC AE 2016, KPSC ADF 2015
UKPSC AE 2012 Paper–II , RPSC ACF-2011

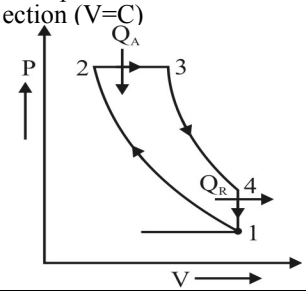
Ans. (c) : For diesel engine, $r_c = 15 – 22$
For Otto engine, $r_c = 7 – 12$
For Dual cycle engine, $r_c = 13 – 17$

138. In a four stroke cycle diesel engine during suction stroke

(a) Only air is sucked in
(b) Only fuel is sucked in
(c) Mixture of fuel and air is sucked in
(d) First air is sucked in, later fuel is sucked in

RPSC ACF & FRO, 26.02.2021
Nagaland PSC CTSE 2017 Paper-2
Rajasthan Nagar Nigam AE 2016, Shift-II
GPSC Lect. 23.10.2016, TSPSC AEE 2015

Ans. : (a) In a four stroke cycle diesel engine during suction stroke only air is sucked.
Diesel cycle:-
1-2 :- Isentropic compression
2-3:- heat addition (p = C)
3-4:- Isentropic Expansion
4-5:- Heat rejection (V=C)



139. With increase in the intake air temperature of I.C. engine, its efficiency will –

(a) Remain same (b) Decreases
(c) Unpredictable (d) Increases

Jharkhand Urja Vikas Nigam Ltd. AE 2017
Rajasthan Nagar Nigam AE 2016, Shift-I
(HPPSC AE 2014), ESE 2018, GATE 1998

Ans. (b) : With increase intake air temperature, density of air decreases and hence volumetric efficiency and power output decreases.

140. Knocking tendency in an SI engine reduces with increasing

(a) Supercharging (b) Wall temperature
(c) Compression ratio (d) Engine speed

UPPSC AE 13.12.2020, Paper-2
Karnataka PSC Lect., 27.05.2017
HPPSC Lect. (Auto) 23.04.2016
(KPSC AE 2015), WBPSC AE 2008

Ans. : (d) Knocking tendency in an SI engine reduces with increasing engine speed.

Factor tending to reduce knocking in SI engine		
* Self ignition temperature of fuel	-	High
* Compression ratio	-	Low
* Inlet temperature and pressure	-	Low
* Super charging	-	No
* Spark advance	-	No
* Flame travel distance	-	Small
* Engine size	-	Small
* Engine Speed	-	High
* Octane rating	-	High

141. Scavenging is usually done to increase

(a) Thermal efficiency (b) Speed
(c) Power output (d) Fuel consumption

RPSC ACF & FRO, 26.02.2021
Nagaland PSC CTSE 2017 Paper-2
Rajasthan AE (Nagar Nigam) 2016 Shift-3
WBPSC AE 2008

Ans. (c) : Scavenging is a process of pushing exhaust gas out of the cylinder and drawing in fresh draught of air or air fuel mixture for the next. Scavenging is usually done to increase power output.

142. For same power output and same compression ratio four stroke S.I. engine as compared to two stroke engine has :

(a) Higher fuel consumption
(b) Lower thermal efficiency
(c) Higher thermal efficiency
(d) Higher exhaust temperature

TNPSC AE 2017, GPSC ARTO 01.05.2016
RPSC AE 2016, RPSC ACF-2011

Ans : (c) For same power output and same compression ratio four stroke SI engine as compared to two stroke engine has high thermal efficiency.

(i) A two stroke cycle engine gives twice the number of power stroke than the four stroke cycle engine at the same engine speed. Theoretically, a two-stroke cycle engine should develop twice the power as that of a four-stroke cycle.

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143. The knocking tendency in CI engine increases with

(a) Decrease of compression ratio
(b) Increase in coolant water temperature
(c) Increase of compression ratio
(d) Increase of temperature of inlet air

BPSC AE Mains 2017 Paper - V
Punjab PSC SDE 12.02.2017
TNPSC AE 2014
UKPSC AE 2012 Paper–II

Ans. (a) : Following parameters will minimize knocking in CI engines :

S.No.	Parameters	Knocking
(i) Compression ratio	High	↓
(ii) Self-ignition temperature	Low	↓
(iii) Delay Period	Less	↓
(iv) Inlet temp and pressure	High	↓
(v) Combustion chamber temperature	High	↓
(vi) RPM	Low	↓
(vii) Engine size	Small	↓
(viii) Cetane Rating	High	↓
(ix) Supercharging	Yes	↓

144. By advancing the spark timing in an S.I. engine, the possibility of knock will

(a) increase (b) decrease
(c) not change (d) be eliminated

APPSC AEE MAINS 2016 PAPER-III
APPSC AEE 2012, WBPSC AE 2008
UKPSC AE 2013, 2007 Paper -II

Ans. (a) : Advancing the timing means the plug fires earlier in the compression stroke (further from TDC). Advance is required because the air/fuel mixture does not burn instantly. It takes time for the flame to ignite the all the mixture. However, if the timing is advanced too far, it will cause an engine knock.

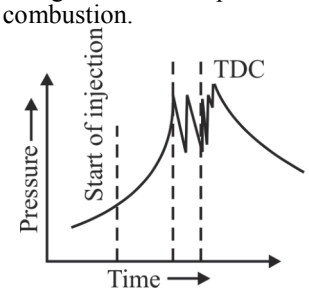
145. The knock in Diesel engine occurs due to

(a) Instantaneous and rapid burning of the first part of the charge
(b) Instantaneous auto ignition of last part of charge
(c) Delayed burning of the first part of the charge
(d) Reduction of delay period

CIL MT 27.02.2020
GWSSB DEE 07.07.2016
GPSC Asstt. Prof. 28.08.2016
MPPSC AE 08.11.2015

Ans. (a) : If the ignition delay quite long, so much fuel can accumulate that the rate of pressure rise is almost instantaneous. Such situation produce the extreme pressure differential and violent gas vibration known as knocking.

The knock in diesel engine occurs due to instantaneous and rapid burning of the first part of the charge i.e uncontrolled combustion.



146. During idling of a petrol engine requires :

(a) Chemically correct mixture
(b) Variable mixture
(c) Lean mixture
(d) Rich mixture

PPSC Asstt. Municipal Engineer 15.06.2021
RPSC IOF, 2020
Kerala PSC AE 06.08.2015, RPSC ACF-2011

Ans. (d) : Petrol Engine- During idling of a petrol engine requires rich mixture, to compensate for the charge dilution due to exhaust gases from the combustion chamber. exhaust gas dilution prevents efficient combustion by reducing the contact between the fuel and air particles.

147. In spark ignition engine the possibility of knocking can be reduced by :

(a) Increasing compression ratio
(b) Decreasing compression ratio
(c) Increasing the coolant temperature
(d) Advancing the spark timing

RPSC ACF-2011
J & K PSC Screening, 2006, ESE 2002

Ans. (b) : In spark ignition engine the possibility of knocking can be reduced by-

- Self ignition temperature of fuel – **Low**
- Compression ratio – **Low**
- Spark advancement – **NO**
- Engine speed – **High**
- Inlet pressure and temp – **Low**

148. Flame speed in SI engine increases:

(a) with increase in compression ratio
(b) with decrease in intake temperature and pressure of the mixture
(c) with decrease in engine speed
(d) with decrease in turbulence in mixture
(e) at very lean mixture

RPSC IOF, 2020, CGPSC AE 25.02.2018
OPSC AEE 2015 PAPER - II

Ans. (a) : Factor influencing the flame speed.

1. Turbulence– Turbulence ↑→ speed ↑→ heat flow to cylinder wall ↑→ flame speed ↑

2. Compression ratio-

(C.R.) ↑→ P ↑ ,T ↑→ flame speed ↑

3. Engine output- Engine output ↑→ turbulence ↑→ flame speed ↑

4. Engine speed- ↑→ V_{flame} ↑

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149. Carburetor is used for

(a) SI engines (b) Gas engines
(c) C.I. engines (d) None of the above

JPSC AE, 10.04.2021, Paper-I
Oil India Limited Sr. Engineer (Drilling) 30.11.2019
WBPSC AE 2008

Ans. (a) : Carburetor is used in SI engines.
The function of carburetor is to supply air and petrol mixture.

150. Voltage developed to strike spark is in the spark plug is in the range :

(a) 6 to 12 volts (b) 100 to 200 volts
(c) 1000 to 2000 volts (d) 20000-25000 volts

RPSC AE 2018, 2016
APPSC Poly. Lect. 2013, RPSC ACF-2011

Ans. (d) : Spark plugs usually require voltage of 12000-25000 volts.

151. What is the theoretical cycle on which the SI engines work ?

(a) Otto cycle (b) Diesel cycle
(c) Ericson cycle (d) Carnot cycle

GMB AAE 25.06.2017
RPSC AE (GWD) 06.05.2016
GWSSB AAE, 27.12.2015

Ans. (a) : Theoretical cycle on which the S.I. (spark Ignition) engine work is Otto cycle.

152. Combustion in compression ignition engine is :

(a) Homogeneous (b) Heterogeneous
(c) Laminar (d) Turbulent

Gujarat PSC AE 2019, MPSC Dy. Engineer 2014
UKPSC AE 2007 Paper -II

Ans. (b) : Combustion in CI engine is heterogeneous because of low physical delay period causes not proper mixing of air and fuel and results in uncontrolled heterogeneous combustion.

153. In carburetors the meter rod and economizer is used to supply air fuel mixture during:

(a) Cold starting (b) Acceleration
(c) Idling (d) Power enrichment

OPSC AEE 2019 PAPER - II
RPSC AE 2016, UJVNL AE 2016

Ans : (d) : Economiser is basically a valve that remains closed at normal cruise operation and gets opened to supply rich mixture at full throttle operation. It regulates the additional fuel supply during full throttle operation. So it can be called a power enrichment system.

154. The actual volume of fresh charge admitted in 4-stroke petrol engine is:

(a) Equal to stroke volume
(b) Equal to stroke volume and clearance volume
(c) Less than stroke volume
(d) More than stoke volume

CIL MT 27.02.2020
GPSC Asstt. Prof. 28.08.2016
Nagaland PSC CTSE 2015 Paper-II

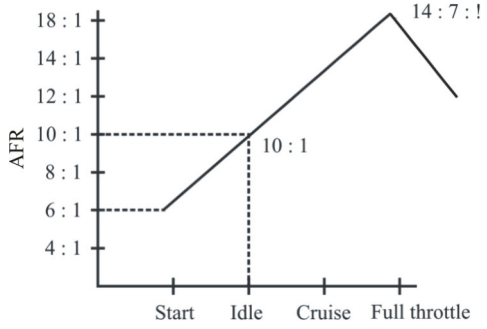
Ans. (c) : The actual volume of fresh charge admitted in 4- stroke petrol engine is less than stroke volume due to presence of clearance volume, and presence of some burned gas due to valve timing.

155. The air-fuel ratio for idling speed of an automobile petrol engine is closer to :

(a) 10 : 1 (b) 15 : 1
(c) 17 : 1 (d) 21 : 1

GPSC ARTO Pre 30.12.2018
SJVN ET 2013, APGENCO AE 2012

Ans. (a) : The Air Fuel Ratio (AFR) for idling speed of an automobile petrol engine is closer to 10 : 1



156. Detonation in spark ignition engine is due to early auto-ignition of the charge near

(a) The spark plug
(b) The suction valve
(c) The exhaust valve
(d) The combustion chamber cylinder walls
(e) None of these

CGPSC AE 15.01.2021, CGPSC AE 16.10.2016
UKPSC AE 2007 Paper -II

Ans. (d) : Due to early auto-ignition of the charge near the combustion chamber cylinder walls. Auto ignition may overheat the spark plug and exhaust valve and it remains so hot that its temperature is sufficient to ignite the charge in next cycle during the compression stroke before spark occurs and this causes the pre-ignition of the charges. Detonation is produced (knocking).

157. Which part is not common between the petrol and Diesel engine?

(a) Air cleaner (b) Exhaust silencer
(c) Battery (d) Fuel injector

Vadodara Muncipal Corp. DEE, 2018
MPPSC AE 08.11.2015
J & K PSC Screening, 2006

Ans. (d) : Fuel injector is not common part between the petrol and diesel engine. Because fuel injector used to injecting the fuel in the diesel engine. But in the petrol engine carburetor used to atomized the fuel.

158. In case of petrol engines, at starting

(a) Rich fuel-air ratio is needed
(b) Weak fuel-air ratio is needed
(c) Chemically correct fuel-air ratio is needed
(d) Any fuel-air ratio will do

RPSC AE 2016, TNPSC AE 2014, RPSC ACF-2011

Ans. : (a) Petrol engine at starting rich fuel air ratio is needed.

Operation	**A/F ratio**
Idling	12 - 12.5
Cruising	16 - 16.5
Max power	12 - 13
Starting	3 - 5

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159. A diesel engine in a power plant is more efficient than a spark engine because :

(a) air standard efficiency for a diesel cycle is higher than an Otto cycle at a fixed compression ratio

(b) diesel gives more heat per kg of fuel than gasoline

(c) higher temperature in diesel engine compared to a Spark engine

(d) compression ratio of a CI engine is higher than that of an SI engine

APPSC Poly Lect. Automobile Engg., 2020
UPPSC AE 13.12.2020, Paper-2
GPSC ARTO Pre 30.12.2018

Ans. (d) : A diesel engine is usually more efficient than a spark ignition engine because a diesel being a heavier hydrocarbon release more heat per kg than gasoline. The compression ratio of a diesel engine is higher than that of an spark ignition engine.

160. In a diesel engine, the fuel is ignited by

(a) Heater

(b) Spark plug

(c) Injected fuel

(d) Heat resulting from compressing air that is supplied for combustion.

GMB AAE 25.06.2017
Rajasthan Nagar Nigam AE 2016, Shift-II
GWSSB AAE, 27.12.2015

Ans. (d) : In a diesel engine, the mixture (fuel vapour + air), which is mostly air, is compressed by piston, due to this high pressure by piston, heat is produced and temperature reaches to fuel's ignition temperature and combustion starts. Hence in a diesel engine the fuel is ignited by heat resulting from compressing the air that is supplied for combustion.

161. In S.I. engine, to develop high voltage for spark plug

(a) Battery is installed

(b) Distributer is installed

(c) Carburetor is installed

(d) Ignition coil is installed

GPSC DEE, Class-2 (GWSSB) 04.07.2021
OPSC AEE 2019 PAPER - II
WBPSC AE 2008

Ans. (d) : In S.I. engine, to develop high voltage for spark plug Ignition coil is installed.

162. A Spark plug gap should be :

(a) 0.2 to 0.8mm (b) 0.3 to 0.7mm

(c) 0.4 to 0.9mm (d) 0.6 to 1.0mm

Kerala PSC Poly. Lect., 2017
UKPSC AE 2007 Paper -II

Ans. (b) : Spark plugs gap in automobiles generally have a gap between 0.3 and 0.7 mm.

Note-As per given option, suitable answer is option (b). But Kerala PSC cancelled this equation in official answer key.

163. In petrol engines, the knocking tendency will increase when

(a) Speed is increased

(b) Speed is decreased

(c) Fuel air ratio is made rich

(d) Fuel air ratio is made lean

(e) None of these

CGPSC AE 16.10.2016
RPSC AE (GWD) 06.05.2016

Ans. (b&c) : The tendency of knocking in petrol engines will depend on following factors –

(i) Compression ratio (r_p)↑ → Knocking↑

(ii) Speed↑ → Knocking↓

(ii) Load↑ → Knocking↑

(iv) Spark advancement↑ → Knocking↑

(v) FAR is made rich↑ → Knocking↑

In this question, both (b) & (c) options are correct.

Note-This question has been deleted by CGPSC.

164. An air-fuel ratio 8 will have

(a) Chemically correct air

(b) Excess fuel

(c) Excess air

(d) Any of the above

RPSC AE 2016, RPSC ACF-2011

Ans : (b)

(i) A mixture which contains less air than the stoichiometric requirement is called a rich mixture (Excess fuel)
Example: A/F ratio of 12:1, 8:1, 10:1 etc)

(ii) A mixture which contains more air than the stoichiometric requirement is called a lean mixture (Excess air)
Example: A/F ratio of 17:1, 20:1 etc)

165. In naturally aspirated engine, pressure inside cylinder at the end of suction stroke is

(a) less than atmospheric

(b) same as atmospheric

(c) more than atmospheric

(d) depends on atmospheric conditions

RPSC AE 2016, RPSC ACF-2011

Ans : (a) In naturally aspirated engine, pressure inside cylinder at the end of suction stroke is less than atmospheric.

166. In a 4–cylinder, 4–stroke Diesel engine operating at 1200 r.p.m., the duration of fuel injection is 20^o. The time in seconds during which fuel is injected would be

(a) $\frac{1}{360}$ sec (b) $\frac{1}{180}$ sec

(c) $\frac{1}{720}$ sec (d) $\frac{1}{1440}$ sec

Nagaland PSC CTSE 2015 Paper-II
TNPSC AE 2014

Ans. (a) : First Method–

Given, N = 1200 rpm

Time taken by one revolution $= \frac{60}{1200} = \frac{1}{20}$ sec

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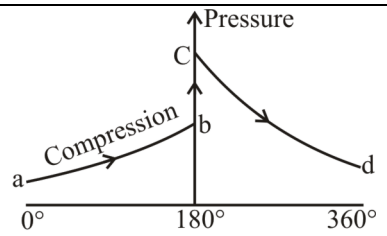

Crank angle (deg)

In four stroke engine fuel injection that is part of power stroke that lasts a stroke that is half a revolution of 180°.

Duration of power stroke $= \frac{1}{2} \times \frac{1}{20}$

Fuel injection of $20° = \frac{20}{180} \times \frac{1}{40} = \frac{1}{360}$ sec

Second Method– Angle turned by crank during fuel injection

$$\theta = 20^0 = 20° \times \frac{\pi}{180°} \text{ rad.}$$

N = 1200 rpm

We know that

$$\frac{\theta}{t} = \omega$$

$$t = \frac{\theta}{\omega} = \frac{\left[20° \times \frac{\pi}{180°}\right]}{\left[\frac{2\pi \times 1200}{60}\right]} = \frac{1}{360} \text{ sec}$$

Where t is time in seconds during fuel is injected.

167. Consider the following statements: Knock in the S.I. engine can be reduced by

(1)Supercharging
(2)Retarding the spark
(3)Using a fuel of long straight chain structure
(4)Increasing the engine speed
Other these correct statements are

(a)1 and 2 (b) 2 and 3
(c)1, 3 and 4 (d) 2 and 4

ESE 2017
UPPSC AE 12.04.2016 Paper-II

Ans.: (d)
Factor Tending to reduce knocking in SI engine

* Self ignition temperature of fuel	- High
* Compression ratio	- Low
* Inlet temperature and pressure	- Low
* Super charging	- No
* Spark advance	- No
* Flame travel distance	- Small
* Engine size	- Small
* Turbulence	- High
* Engine speed	- High
* Octane rating	- High

168. A diesel engine has

(a) one valve (b) two valves
(c) three valves (d) four valves

Nagaland PSC (CTSE) 2018, Paper-II
KPCL AE 2016

Ans. (c) : A diesel engine has three valves inlet valve, exhaust valve and fuel injector valve.
There are different valves used in a diesel engines Inlet valve-inlet valve generally allow the intake air to enter into combustion chamber.
Exhaust valve – likewise they allow the exhaust gas to allow burnt out gas from chamber to exhaust manifold.

169. The air-fuel ratio of the petrol engine is controlled by

(a) Fuel pump (b) Governor
(c) Injector (d) Carburetor

Gujarat PSC AE 2019
J&K PSC Civil Services Pre, 2013

Ans : (d) : The air-fuel ratio of the petrol engine is controlled by carburetor.

170. The function of a distributor in a coil ignition system of I.C. engines is

(a) to distribute spark
(b) to distribute power
(c) to distribute current
(d) to time the spark

Nagaland PSC (CTSE) 2018, Paper-II
TNPSC AE, 2008

Ans. (d) : The function of a distributor in a coil ignition system of I.C. engines is to time the spark.
A distributor is an enclosed rotating shaft used in spark-ignition internal combustion engines that have mechanically timed ignition. The distributor's main functions is to route secondary or high voltage, current from the ignition coil to the spark plugs in the correct firing order and For the correct amount of time.

171. In a petrol engine, the tendency for detonation increases with

(a) Retarded spark timing
(b) Running the engine at high speed
(c) Supercharging
(d) Increasing the cooling rate

Nagaland PSC CTSE 2017 Paper-2
GPSC EE Pre, 28.01.2017

Ans. (c) : In a petrol engine, the tendency for detonation increases with supercharging. Supercharging is operation of forcing additional air under pressure in the engine cylinder is known as supercharging.

172. Which of the following is not a function of nozzle in fuel injection

(a) Atomization
(b) Distribution of fuel to the required area
(c) Impingement on the walls
(d) No dribbling

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (c) : The function of nozzle in fuel injection system are to filter the fuel, to time the fuel injection, to break up the fuel into small particles. Non-impingement on the walls function of nozzle.

173. For S.I. engine carburetor the float is usually made of

(a) Brass sheet (b) Nylon
(c) Cast iron (d) Either (a) or (b)

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (d) : S.I. engine carburetor the float usually made of brass sheet, nylon.

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174. The intensity of detonation will depend mainly upon the amount of

(a) Energy contained in the end mixture
(b) Rate of chemical reaction
(c) Both energy contained in the end mixture and rate of chemical reaction
(d) None of the above

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (c) : The intensity of detonation will depend mainly upon the amount of energy contained in the end-mixture and the rate of chemical reaction which release it in the form of heat and a high intensity pressure-wave.

175. The exhaust valve usually opens ..... in a four stroke S.I. engine

(a) At TDC
(b) At BDC
(c) At 35º to 60º after BDC
(d) At 35º to 60º before BDC

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (d) : The exhaust valve in a four-stroke cycle petrol engine open at 50º before BDC (Bottom Dead Centre) and closes at 15º after TDC (Top Dead Centre)

176. Stoichiometric air-fuel ratio of petrol is roughly

(a) 50:1 (b) 25:1
(c) 15:1 (d) 1:1

GPSC Lect. 23.10.2016
GPSC Asstt. Prof. 28.08.2016

Ans. (c) : The stoichiometric mixture for a gasoline engine is the ideal ratio of air to fuel that burns all fuel with no excess air. Ratio is 14.7:1 (A/F).

177. Lean mixture in an automobile is required during :

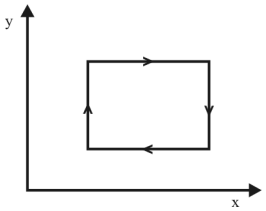
(a) Idling (b) Starting
(c) Accelerating (d) Cruising

HPPSC W.S. Poly. 2016, (KPSC AE 2015)

Ans : (d) Lean mixture in an automobile is required during cruising

Operation	**A/F ratio**
Idling	12-12.5
Cruising/normal	16- 16.5
Maximum power	12-13
Starting	3-5

178.

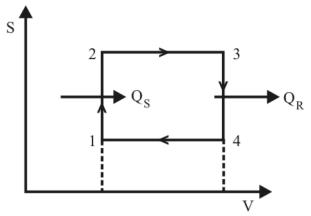


An air standard Otto cycle has the following shape on a thermodynamic plane. The X and Y coordinates respectively are:

(a) V and P (b) V and S
(c) V and T (d) S and P

PPSC Asstt. Municipal Engineer 15.06.2021
TSPSC Managers, 2015

Ans. (b)



(V-S diagram in otto cycle)

179. Duration between the time of injection and time of ignition in a diesel engine is called:

(a) Lead period
(b) Delay period
(c) Combustion period
(d) Period of compression

Sikkim PSC (Under Secretary), 2017
OPSC AEE 2015 PAPER - II

Ans : (b) Duration between the time of injection and time of ignition in a diesel engine is called delay period. Delay period in compression ignition engines depends upon

(i) Temperature and pressure in the cylinder at the time of injection.
(ii) Nature of the fuel mixture strength
(iii) Relative velocity between the fuel injection and air turbulence pressure of residual gases.

180. For same power and same speed, the flywheel of a four-stroke engine as compared to two-stroke I.C. engine will be :

(a) Smaller
(b) Bigger
(c) Same size
(d) Dependent on other engine parameters

OPSC AEE 2019 PAPER - II
(HPPSC AE 2014)

Ans : (b)

Four-stroke Engine	**Two-stroke Engine**
(i) Turning moment is not so uniform and hence a heavier flywheel is needed	(i) Turning moment is more uniform and hence a lighter flywheel can be used.
(ii) Higher volumetric efficiency due to more time for ixture intake	(ii) Lower volumetric efficiency. due to lesser time for mixture intake
(iii) Part load efficiency is better	(iii) Part load efficiency is poor.
(iv) Four- stroke engine have valves and valve mechanism	(iv) Two- Stroke engines have no valve but only ports.
(v) One power stroke in two revolutions lesser cooling and lubrication requirement lower rate of wear and tear.	(v) One power stroke in one revolution of crank shaft. Greater cooling and lubrication requirement. Higher rate of wear and tear.

181. The injection pressure in diesel engine is of the order of:-

(a) 30 – 40 bar (b) 100 – 150 bar
(c) 200 – 300 bar (d) 400 – 600 bar

UKPSC AE-2013, 2007 Paper-II

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Ans. (b) : The injection pressure in diesel engine is of the order of 100 – 150 bar.

182. Solid injection in C.I. engine refers to injection of

(a) Liquid fuel only (b) Liquid fuel and air
(c) Solid fuel (d) Solid fuel and air

MPSC Dy. Engg. (GWD) 19.05.2013
UKPSC AE 2007 Paper-II

Ans. (a) : Solid Injection System–It is the injection of atomized fuel oil into the combustion chamber of a diesel engine under the pressure of the liquid fuel itself. In this system, the liquid fuel is injected directly into the combustion chamber without the aid of compressed air.

183. Highest useful compression ratio is the compression ratio at which

(a) engine can be safely operated
(b) engine gives maximum thermal efficiency
(c) engine operates smoothly
(d) detonation first becomes audible

RPSC AE 2016, RPSC ACF-2011

Ans : (d) Highest useful compression ratio is the compression ratio at which detonation first becomes audible.

184. Which of the following are the assumptions involved in the auto-ignition theory put forth for the onset of knock in SI engines?

I. Flame velocity is normal before the onset of auto-ignition.
II. A number of end-gas elements auto ignite simultaneously.
III. Preflame reactions are responsible for preparing the end gas to ignite.

Of these statements, which are correct?

(a) I and II (b) I and III
(c) II and III (d) I, II and III

APPSC Poly. Lect. 2013, ESE 1998

Ans. (d) : Assumption which are involved in Auto-ignition theory put forth for the on set of knock is S.I. engines.

(1) Flame velocity in normal before the onset of auto Ignition.
(2) A number of end-gas elements auto ignite simultaneously.
(3) Pre-flame reactions are responsible for preparing the end-gas to ignite.

185. In petrol engines, the knocking tendency will increase when :

(a) Speed is decreased
(b) Fuel air ratio is made rich
(c) Fuel –air ratio is made lean
(d) Speed is increased

CGPSC AE 15.01.2021
J&K PSC Civil Services Pre, 2010

Ans. (a) : In SI engines, the knocking tendency will increase when speed is decreased.
As speed decreased turbulence decreases and temperature of cylinder increases, knocking tendency increase.

186. When a petrol engine is supplied with diesel fuel :

(a) The engine will not run
(b) The engine will consume more fuel
(c) The engine will operative with reduced output
(d) The exhaust will have dense black smoke

OPSC AEE 2015 PAPER - II
J & K PSC Screening, 2006

Ans. (a) : When a petrol engine is supplied with diesel fuel then combustion not takes place which results in high knocking and engine will not run.

187. Fuel is injected in the CI engine at the end of :

(a) suction stroke (b) compression stroke
(c) expansion stroke (d) exhaust stroke

APPSC Poly Lect. Automobile Engg., 2020
OPSC AEE 2015 PAPER - II

Ans. (b) : Fuel is injected in the CI engine at the end of the compression stroke. Fuel is injected directly into the combustion chamber by the injector.

188. Consider the following:

1. Increased cetane number
2. Increased compression ratio
3. Increased injection advance
4. Increased air turbulence

Which of the factors will reduce the physical delay in diesel engine?

(a) 1, 2 and 3 only (b) 2 and 4 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4

RPSC Lect. (Tech. Edu. Dept.) 2011, ESE 2010

Ans. (b) : Physical delay means physical preparation of mixture e.g. mixing, heating, fuel vapourizing, atomising and how much fast this occure. From given factors, compression ratio and turbulence (or swirl) and the only factors which enhances this physical preparation fast i.e. reduced physical delay.
Cetane number is fuel characteristic which influences chemical delay.
Increased injection advance means longer physical delay because it starts from start of injection and increased knock.

189. The essential function of the carburetor in a S.I. engine is to :

(a) Mix the fuel into air stream and amount dictated by the load and speed
(b) Vapourise fuel
(c) Distribute fuel uniformly into all cylinders
(d) Both (B) and (C)

HPPSC Asstt. Prof. 20.11.2017
APGENCO AE 2012

Ans. (a) : The main functions of a carburetor are–

(i) The main function of carburetor is to mix air and gasoline and provides a high combustion mixture.
(ii) It controls the engine speed.
(iii) It also regulates the air-fuel ratio
(iv) Increase or decrease the amount of mixture according to the engine speed and load changing.
(v) To supply the correct amount of air-fuel mixture at the correct strength under all conditions of load and speed of the engine.

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190. Match List-I with List-II and select the correct answer :

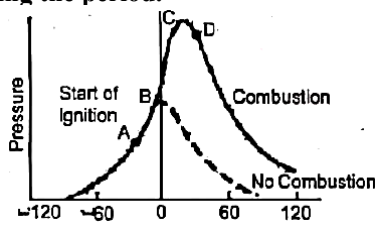
List-I	List-II
A. Idling	1. 12.5
B. Cold starting	2. 9.0
C. Cruising	3. 16.0
D. Maximum power	4. 22.0
	5. 3.0

	A	B	C	D
(a)	2	4	5	1
(b)	1	3	4	2
(c)	5	2	1	3
(d)	2	5	3	1

Nagaland PSC CTSE 2015 Paper-II
ESE 1995

Ans. (d) : Idling — 9.0
Cold starting — 3.0
Cruising — 16.0
Maximum power — 12.5

191. Hypothetical pressure diagram for a compression ignition engine is shown in the given figure. The diesel knock is generated during the period.



(a) AB (b) BC
(c) CD (d) after D

Nagaland PSC CTSE 2015 Paper-II
ESE 1998

Ans. (b) : Combustion is responsible for detonation in CI engine.

192. Ignition lag is
(a) Time before actual fuel injection and the pump plunger starts to pump fuel
(b) Time corresponding to actual ignition
(c) Time corresponding to actual injection and TDC
(d) Time taken by fuel after injection (before TDC) to reach up to auto ignition temperature

APPSC AEE MAINS 2016 PAPER-III
Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : Ignition lag is time taken by fuel after injection (before TDC) to reach up to auto ignition temperature.

193. For a given weight and displacement, a well-designed two –stroke engine can provide significantly ____ power than its four-stroke counterpart.
(a) Less (b) Equal
(c) More (d) Unpredictable

RPSC ACF & FRO, 26.02.2021

Ans. (c) : For a given weight and displacement, a well designed two stroke engine can provide significant more power than it's four stroke counterpart.

194. In two wheelers, which scavenging method is used?
(a) Uniform scavenging
(b) Reverse flow scavenging
(c) Loop scavenging
(d) Cross scavenging

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Scavenging is the process of replacing the exhaust gas in a cylinder of an internal combustion engine with the fresh air/fuel mixture (or fresh air in the case of direct injection engine) for the next cycle. If scavenging is incomplete the remaining exhaust gases can cause improper combustion for the next cycle leading to reduce power output. Scavenging is equally important for both two stroke and four stroke engines. Most modern four stroke engine use cross flow cylinder heads and valve timing overlap to scavenge the cylinders. Modern two-stroke engine use either schnuerl scavenging (also known as loop scavenging)

195. Decrease of air-fuel ratio in spark ignition engines results in
(a) Increase of NO_x
(b) A decrease of CO and unburnt hydrocarbon
(c) An increase of CO and unburnt hydrocarbon
(d) None of the above

UPPSC AE 13.12.2020, Paper-2

Ans. (c) : Decrease of air fuel ratio in spark ignition engine results in an increase of CO and unburnt hydrocarbon.

196. Achieving high compression ratio is feasible in compression ignited engines due to
(a) Non-premixed combustion
(b) Premixed combustion
(c) Detonation
(d) Deflagration

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : C.I. engines can run at higher compression ratio since they are based on the auto ignition principle and are not limited by the premixed fuel air combustion knock as in S.I. engines.

197. Equivalence ratio is the ratio of
(a) Actual to stoichiometric air-fuel ratio
(b) Actual mass of fuel to actual mass of air
(c) Actual to stoichiometric fuel-air ratio
(d) Actual mass of air to actual mass of fuel

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Equivalence ratio is defined as the ratio of the actual fuel/air ratio to stoichiometric fuel/ air ratio.
For equivalence ratio is less than one, mixture is lean with excess air.

198. The effect of dissociation in combustion process is for:
(a) Separating the product of combustion
(b) Reducing the flame temperature
(c) Reducing the use of excess air
(d) Reducing CO_2 percentage in the gas

OPSC AEE 2019 PAPER - II

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Ans. (b) : Dissociation reaction is endothermic CO_2 absorbs heat and becomes ($CO + O_2$). H_2O absorbs heat and becomes ($H_2 + O_2$). These reactions reduce peak temperatures produced.

199. Which option is not correct for I head combustion chamber for SI engine?
(a) Higher volumetric efficiency
(b) Lower pumping losses
(c) Higher surface to volume ratio
(d) Less knocking

OPSC AEE 2019 PAPER - II

Ans : (c)
- Lower pumping losses and higher volumetric efficiency from better breathing of the engine from larger valves or valve lifts and more direct passageways.
- Less distance for the flame to travel and therefore greater freedom from knock, or in other words, lower octane requirements.
- Lower surface-volume ratio and, therefore, less heat loss and less air pollution.
- Easier to cast and hence lower casting cost.

200. Assertion (A): After burning increases the thrust of a jet engine.
Reason (R) : The air fuel ratio of jet engine is high.
(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true and R is not the correct explanation of A.
(c) A is true but R is false
(d) A is false but R is true

Gujarat PSC AE 2019

Ans : (b) : After burning in jet engine is equivalent to reheat in gas turbine. Since air fuel ratio of jet engine is high i.e., very lean mixture having excess air or oxygen, some extra fuel can be burnt in exhaust gases. So in after burning, secondary fuels is injected in exhaust from turbine, which increases jet velocity from nozzle and so the thrust.

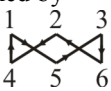
201. In a 6 cylinder petrol engine, the standard firing order is :
(a) 1-6-2-5-4-3 (b) 1-2-3-4-5-6
(c) 1-5-3-6-2-4 (d) 1-3-6-5-4-2

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (c) : Firing order is the sequence of power delivery of each cylinder in multi cylinder IC engine.

Firing order of V-shape 6-cylinder engine → 1-5-3-6-2-4 (crank-pins having piston mounted are arranged in every 60° of the crank shaft).

Also standard firing order for six cylinder engine can be determined by –



= 1-5-3-6-2-4

For four cylinder engine

= 1-3-4-2

202. Consider the following statements:
Combustion chamber is
(1) the volume between TDC and BDC during the combustion process.
(2) the space enclosed between the upper part of the cylinder and the top of the piston during the combustion process
(3) the space enclosed between TDC and the top of the piston during the combustion process
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

ESE 2019

Ans. (b) : A combustion chamber is that part of an internal combustion engine or a reaction engine in which the fuel/air mixture is burned.
It is the space enclosed between the upper part of the cylinder and the top of the piston during the combustion process.

203. Which design of the combustion chamber has less problem in cold starting?
(a) M Combustion Chamber
(b) Open Combustion Chamber
(c) Air-Cell Combustion Chamber
(d) Any design of Combustion Chamber

OPSC AEE 2019 PAPER - II

Ans : (a) : M - Combustion chamber has less problem in cold starting.

204. Which type of nozzle is preferred to avoid weak injection and dribbling of fuel in combustion chamber?
(a) Single hole (b) Multi hole
(c) Pintle (d) Pintaux

OPSC AEE 2019 PAPER - II

Ans : (c) : Pintle type of nozzle is preferred to avoid weak injection and dribbling of fuel in combustion chamber.

205. Which device can be used to measure aromatic hydrocarbon in exhaust gas?
(a) Dispersion analyzer
(b) Non-dispersive infrared analyzer
(c) Flame ionization detector
(d) All of these

OPSC AEE 2019 PAPER - II

Ans : (d) : All of these

206. The spark advance is specified in:
(a) Time in seconds
(b) Degree of crank rotation
(c) Percentage of engine speed in rpm
(d) None of these

OPSC AEE 2019 PAPER - II

Ans : (b) : The spark advance is specified in degree of crank rotation.

207. Which is the best option to avoid knock in SI engine?
(a) Supercharging of Engine
(b) Delay period should low
(c) Delay period should high
(d) Spark should be advanced

OPSC AEE 2019 PAPER - II

Ans : (c) : Delay period should be high is the best option to avoid knock in SI engine.

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208. Assertion (A) : A simple or elementary carburetor provides progressively rich mixture with increasing air flow.
Reason (R) : The density of the air tends to increase as the rate of air flow increases.
(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true and R is not the correct explanation of A.
(c) A is true but R is false
(d) A is false but R is true

Gujarat PSC AE 2019

Ans : (c) : At higher speed a simple carburetor provide rich mixture because of reduced mass flow rate of air. This reduced mass flow rate is due to decreases in density at higher speed. The density of air decrease at higher speed due to compressibility effect.

209. In a four stroke cycle diesel engine, the compression
(a) starts at 40º after bottom dead center and ends at 10º before top dead center
(b) starts at 40º before bottom dead center and ends at 10º after top dead center
(c) starts at top dead center and ends at 40º before bottom dead center
(d) may start and end anywhere

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : In four stroke cycle diesel engine the compression start at 40º after bottom dead center and ends at 10º before top dead center.

210. In IC engine
1. the ideal air capacity of a two-stroke engine is the mass of air required to concurrently fill the total cylinder volume at inlet temperature and exhaust pressure
2. with increase in air-fuel ratio beyond the value for maximum power, there is a fall in power developed and this fall is more with higher values of air-fuel ratio.
3. the volumetric efficiency of the engine depends on the design of intake and exhaust manifold.

Which of the above are correct?
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : Volumetric efficiency indicates the breathing ability of the engine.
Intake system and exhaust manifold must be designed in such a way that the engine must be able to take in as much air as possible to the rate at which the volume is displaced by the system.

211. The essential function of the carburetor in a spark ignition engine is to :
(a) Meter the fuel into air stream and amount dictated by the load and speed
(b) Bring about mixing of air and fuel to get a homogeneous mixture
(c) Vaporise the fuel
(d) Distribute fuel uniformly to all cylinders in a multi cylinder engine and also vaporise it.

RPSC Vice Principal ITI 2018

Ans. (a) : The essential function of the carburetor in a spark ignition engine is to meter the fuel into air stream and amount dictated by the load and speed.

212. Consider the following statements:
1. Recycling exhaust gases by partial mixing with the intake gases increases the emission of oxides of nitrogen from the engine.
2. The effect of increase in altitude of operation on the carburetor is to enrich the entire port-throttle operation.
3. When the carburetor throttle is suddenly opened, the air-fuel mixture may lean out temporarily resulting in engine stall.
4. Use of multi-venturi system makes it possible to obtain a high velocity air stream when the fuel is introduced at the main venturi throat.

Which of the above statements are correct?
(a) 1 and 3 (b) 1 and 4
(c) 2 and 3 (d) 2 and 4

ESE 2018

Ans. (c) : (i) Exhaust Gas Recirculation (EGR) is an effective pretreatment technique, which is being used widely to reduce and control the oxides of nitrogen (NO_x) emission from diesel engine. Exhaust gas recycle is done by ducting some of the exhaust flow back into the intake system, usually immediately after the throttle. Hence statement-1 is wrong.
(ii) An altitude compensating system which can vary the operating characteristic of a carburetor for obtaining a desirable enriched air fuel ratio in accordance with changes in altitude.
(iii) In conventional carburetors, the air fuel ratio is enriched for obtaining increased engine power when the throttle valve of the carburetor is opened.
(iv) Multiple venturi systems having three venturies if fuel is injected in third venturi air flow rate will increases if fuel injection in main venturi air flow rate will not change.
Hence statement-4 is also wrong.

213. Which of the following actions will help to reduce the black smoke emission of a diesel engine?
1. Run at lower load, i.e., derating
2. Have regular maintenance of the diesel engine, particularly of injection system
3. Use diesel oil of higher cetane number

Select the correct answer using the code given below.
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (c) : The visible black smoke is generally due to injecting too much fuel for the amount of air available in the cylinder or due to poor mixing between fuel and air. This black smoke emission can be avoided by the proper design and maintenance of the injection system.
Derating reduces the power output, therefore, there will be a lean mixture of fuel and air, the combustion of fuel will be proper and there will be lesser black smoke emissions.

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214. If one cylinder of a diesel engine receives more fuel than the others, it is a serious condition for that cylinder and can be checked by

1. judging the seizure of the piston
2. checking incomplete combustion in that cylinder
3. checking cylinder exhaust temperature with a pyrometer

Which of the above is/are correct?

(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

ESE 2018

Ans. (c) : • If the exhaust temperature is high it can cause catastrophic engine failure.
• Therefore the exhaust gas temperature is checked whenever there is more fuel injected into the cylinder.

215. Consider the following statements with reference to combustion and performance in a four-stroke petrol engine:

1. The auto-ignition temperature of petrol as a fuel is higher than that of diesel oil as a fuel.
2. The highest compression ratio of petrol engines is constrained by the possibility of detonation.
3. A petrol engine is basically less suitable for supercharging than a diesel engine.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : Petrol have higher self ignition temperature, high volatile to burn when you compress that much it may burst out. Whereas diesel is low volatile and fire up at the compression without any external source. Supercharging is preferred in CI engine because it leads to decrease delay period, therefore decrease detonation where as in SI engine it increases compression ratio, increases detonation.
Hence all statements are correct.

216. In a four stroke cycle petrol engine, the compression

(a) starts at 40º after bottom dead center and ends at 30º before top dead center
(b) starts at 40º before bottom dead center and ends at 30º after top dead center
(c) starts at bottom dead center and ends at top dead center
(d) may start and end anywhere

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : In four stroke cycle petrol engine, the compression starts at 40º after top dead center while the expansion stats at 30º before top dead center and ends at 50º before bottom dead center.

217. In a four stroke cycle, the minimum temperature inside the engine cylinder occurs at the

(a) beginning of suction stroke
(b) end of suction stroke
(c) beginning of exhaust stroke
(d) end of exhaust stroke

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : In a four stroke cycle, the minimum temperature inside the engine cylinder occurs at the beginning of suction stroke while end of suction stroke air absorbs heat from cylinder and get warmed.

218. Which of the following statement is wrong?

(a) In compression ignition engines, detonation occurs near the beginning of combustion.
(b) Since the fuel in compression ignition engines is injected at the end of compression stroke, therefore there will be no pre-ignition.
(c) To eliminate knock in compression ignition engines, we want to achieve auto-ignition not early and desire a long delay period.
(d) In compression ignition engines, because of heterogeneous mixture, the rate of pressure rise is comparatively lower.

Karnataka PSC AE, 10.09.2017

Ans. (c) : To eliminate knock in compression ignition engines we want to have smaller delay period and nearly auto-ignition.

219. To mix air and petrol in the required proportion and to supply it to the engine during suction stroke, ... is employed.

(a) Injector (b) Carburetor
(c) Fuel pump (d) Spark plug

APGENCO AE, 2017

Ans. (b) : In petrol engine mixing of fuel and air in carburetor.

220. In the operation of four-stroke diesel engines, the term 'squish' refers to the :

(a) Injection of fuel in the pre-combustion chamber
(b) Discharge of gases from the pre-combustion chamber
(c) Entry of air into the combustion chamber
(d) Stripping of fuel from the core

HPPSC Asstt. Prof. 20.11.2017

Ans. (c) : In the operation of four stroke diesel engines, the term 'squish' refers to the entry of air into the combustion chamber.
Squish is an effect in internal combustion engines which creates sudden turbulence of the air-fuel mixture as the piston approaches top dead centre (TDC).

221. The most perfect method of scavenging is

(a) Cross scavenging
(b) Uniflow scavenging
(c) Loop scavenging
(d) Reverse flow scavenging

TNPSC AE 2017

Ans. (b) : The most perfect method of scavenging is uniflow scavenging.
Uniflow scavenging is a design used by some two-stroke engines where the fresh charge enters near the bottom of the cylinder and the exhaust gases exit through a valve situated at the top of the cylinder.

222. In a typical medium speed 4-stroke cycle diesel engine, the inlet valve [TDC-Top Dead Centre, BDC-Bottom Dead Centre]

(a) Opens at 20º before TDC and closes at 35º after BDC
(b) Opens at TDC and closes at BDC

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(c) Opens at 10º after TDC and closes at 20º before BDC
(d) Remain open for 200º

TNPSC AE 2017

Ans. (a) : In a typical medium speed 4-stroke cycle diesel engine, the inlet valve [TDC-Top Dead Centre, BDC-Bottom Dead Centre] Opens at 20º before TDC and closes at 35º after BDC.

223. Petrol engines are not suitable for part-load operation, because
(a) Mechanical efficiency is poor due to increasing internal losses at increased throttling
(b) Of fear of pre-ignition
(c) Of huge knocking
(d) Of increased detonation tendency

TNPSC AE 2017

Ans. (a) : Petrol engines are not suitable for part-load operation, because mechanical efficiency is poor due to increasing internal losses at increased throttling.

224. A distributor in spark ignition engines performs the function of
(a) Distributing the right quantity of fuel oil to the desired cylinder
(b) Distributing the air requirement appropriately
(c) Adding additives to fuel oil
(d) Providing the correct firing order in the engine

TNPSC AE 2017

Ans. (d) : A distributor in spark ignition engines performs the function of providing the correct firing order in the engine.

225. Engine knock can be measured by using
(a) Combustion pressure sensor
(b) Mechanical vibration sensor
(c) Ion current measurement
(d) All the above

TNPSC AE 2017

Ans. (d) : Engine knock can be measured by using
Combustion pressure sensor
Mechanical vibration
Ion current measurement

226. Statement (I): Air-blast injection in diesel engines could reduce engine efficiency.
Statement (II): Air-blast injection in diesel engines is not instantaneous but happens when the piston moves outward with the injection valve remaining open for whatever reason.

ESE 2017

Ans. (*) :

227. The knocking sensitivity of engines could be reduced by
(a) Compact combustion chamber geometry
(b) Central position of the spark plug
(c) Increased turbulence
(d) All the above

TNPSC AE 2017

Ans. (d) : The knocking sensitivity of engines could be reduced by compact combustion chamber geometry, central position of the spark plug, and increased turbulence.

228. Knocking takes place in C.I. Engines
(a) at the start of combustion
(b) at the end of combustion
(c) during combustion
(d) during the delay period

TNPSC AE 2017

Ans. (a) : Knocking takes place in C.I. Engines at the start of combustion.

229. Which of the following is the most effective air cleaner in case of diesel engine
(a) Dry type (b) Wet type
(c) Whirl type (d) Oil bath type

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : Oil bath type most effective air-cleaner in diesel engine, because when air-fuel ratio is about 10% rich mixture, then a high flame speed is obtained in diesel engine.

230. Carburetion is process of
(a) Forming a combustible air-fuel mixture by mixing the proper amount of fuel and air before admitting it into the engine cylinder
(b) Igniting the air-fuel mixture into the engine cylinder during power stroke
(c) Combusting the carbon in the atmosphere
(d) None of these

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : Carburetion is process of forming a combustible air-fuel mixture by mixing the proper amount of fuel and air before entering to engine cylinder.

231. Modern CRDI engines uses injection pressure of the order of
(a) 400 bar (b) 800 bar
(c) 1000 bar (d) 1600 bar

TNPSC AE 2017

Ans. (d) : CRDI Engine- CRDI stands for common rail direct injection which means direct injection of fuel into the cylinder of a diesel engine through a single common line called rail which is connected to fuel injectors.

232. Reducing the combustion-chamber surface are
(a) reduces the amount of unburned HC in the exhaust gas
(b) increases the amount of unburned HC in the exhaust gas
(c) reduces the amount of NO_x in the exhaust gas
(d) increases the amount of unburned HC and NO_x in the exhaust gas

TNPSC AE 2017

Ans. (a) : Reducing the combustion-chamber surface are reduces the amount of unburned HC in the exhaust gas.

233. In the Passenger cars, the following type of carburetor is preferred
(a) Horizontal type
(b) Upward draught type
(c) Downward draught type
(d) Inclined draught type

TNPSC AE 2017

Ans. (c) : Downward draught type of carburetor is preferred in the passenger cars.

234. Statement (I) : Knocking and pre-ignition are the two words explaining the same phenomena.
Statement (II) : Pre-ignition occurs only in petrol engines.

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(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I)
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I)
(c) Statement (I) is true but Statement (II) is false
(d) Statement (I) is false but Statement (II) is true

JWM 2017

Ans. (b) : Knocking and pre-ignition are the two words explaining the same phenomena. Knocking occur only in the diesel engine while Pre-ignition occur only in petrol engines. Pre-ignition is the ignition of the air-fuel charge prior to the spark plug firing.

235. C.I. engine as compared to S.I. engine require
(a) No flywheel (b) Same size flywheel
(c) Smaller flywheel (d) Bigger flywheel

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : C.I. engine compare S.I. engine require bigger flywheel.

236. The volume of air required to consume 1 litre of fuel by a four stroke engine is
(a) 5–6 m^3 (b) 9–10 m^3
(c) 2–3 m^3 (d) 15–18 m^3

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : Volume of 9-10 m^3 air is required for consuming 1 litre of fuel by a four-stroke engine.

237. The power to weight ratio in a two stroke engine as compared to four stroke engine is
(a) More
(b) Less
(c) Equal
(d) Depends on the power rating

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : The power to weight ratio is double in two stroke engine as compared to four stroke engine.

238. Which one of the following form a part of the fuel supply system for a diesel engine?
(a) Supply pump (b) Filter
(c) Spray Nozzle (d) All of the above

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : Following are the part use to form the fuel supply system for a diesel engine.
• Supply pump • Filter • Injector Nozzle

239. The valve timing is controlled by
(a) camshaft (b) crank shaft
(c) governor (d) all of the above

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : In 4-stroke cycle engines and some two-stroke engines, the valve timing controlled by the camshaft. This can varied by modifying the camshaft, or it can be varied during engine by variable valve timing.

240. If delivery ratio is 1.7, what will be the excess air factor in case of two stroke engine?
(a) 0.3 (b) 0.7
(c) 3.4 (d) 1.4
(e) 0.6

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Delivery ratio = 1.7
Excess air factor (λ) = ?
$\because$ $\lambda = R_{del} - 1$
$= 1.7 - 1$
$= 0.7$

241. Diesel engine can work on very lean air fuel ratio of the order of 30 : 1. A petrol engine can also work on such a lean ratio provided ..... .
(a) it is properly designed
(b) best quality fuel is used
(c) can not work as it is impossible
(d) flywheel size is proper

GWSSB DEE 07.07.2016

Ans. (c) :

242. Which of the following statements is true?
(a) In diesel engines, thermal losses are lower in the direct injection as compared to indirect injection
(b) In diesel engines, thermal losses are higher in the direct injection as compared to indirect injection
(c) In diesel engines, thermal losses in the direct injection are same as compared to indirect injection
(d) In diesel engines, thermal losses in the direct injection are either lower or higher as compared to indirect injection depending on the external temperature

KPCL AE 2016

Ans. (a) : In diesel engines, thermal losses are lower in the direct injection as compared to indirect injection.

243. Which of the following is the correct formula for Engine Volumetric Efficiency, given that V_a is the volume of air taken in the cylinder and V_c is the maximum possible volume in the cylinder?
(a) V_c/V_a (b) V_a/V_c
(c) $V_c . V_a$ (d) None of the above

KPCL AE 2016

Ans. (b) :

$$\eta_{volumetric} = \frac{\text{volume of air taken in the cylinder } (V_{air})}{\text{Maximum possible volume } (V_c) \text{ in the cylinder}}$$

$$\eta_{volumetric} = \frac{V_a}{V_c}$$

$$= \frac{V_a}{\frac{\pi}{4} D^2 \times L} \qquad \left[\because V_c = V_s = \frac{\pi}{4} D^2 L\right]$$

244. Which of the following denotes the power developed by combustion of fuel in the combustion chamber?
(a) Brake power (b) Indicated power
(c) Operational power (d) None of these

KPCL AE 2016

Ans. (b) : Indicated power is defined as the total power developed by combustion of fuel in the combustion chamber.

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245. To which of the following categories does a petrol, petrol engine belong?
(a) Compression engines
(b) Spark ignition engine
(c) Both A and B
(d) None of these

KPCL AE 2016

Ans. (b) : Petrol engine belongs to spark ignition engine (S.I.) while diesel engine is a compression ignition engine (C.I.).

246. Air fuel ratio of about ..... means too rich mixture for a S.I. engine
(a) 1:10 (b) 1:12
(c) 1:14 (d) 1:13

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : The reaction of mixture of fuel with Oxygen in the air is cause from burning of fuel. This is called air-fuel ratio (AFR).
A rich mixture would be one with a (AFR) ratio less than 10 : 1, which means 10 part of air to one part of fuel

247. ..... is the ignition of homogeneous mixture in the cylinder, before the timed ignition spark occurs, caused by the local overheating of the combustible mixture
(a) Detonation (b) Post detonation
(c) Pre-ignition (d) Before ignition

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Pre-ignition is the ignition of the homogeneous mixture in the cylinder, before the timed ignition spark occurs, caused by the local overheating of the combustible mixture.

248. Which of the following is not an affect of pre-ignition
(a) It increases tendency of detonation in engines
(b) Pre-ignition is a serious type of abnormal combustion
(c) Pre-ignition in single cylinder engine will result in a steady reduction in speed and power output
(d) The real desirable effects of pre-ignition are when it occurs only in one or more cylinders in a multi cylinder engine

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : The real undesirable effects of pre-ignition are which it occurs only in one or more cylinder in a multi cylinder engine.

249. At present the amount of power that can be developed in the cylinder of a petrol engine is fixed by the liability of a fuel to
(a) Detonate (b) Pre-ignition
(c) Unburnt charge (d) Slow ignition

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (a) : present the amount of power that can be developed in the cylinder of a petrol engine is fixed by the liability of a fuel to detonate

250. Which one of the following methods cannot control Detonation
(a) Increase engine speed
(b) Retarding spark
(c) Reducing pressure in the inlet manifold by throttling
(d) Making air fuel ratio neither too lean nor too rich

Nagaland PSC CTSE 2017, 2016 Paper-II

Ans. (d) : Making air-fuel ratio neither too lean nor too rich is wrong statement here. Right statement, says that air-fuel ratio either too lean or too rich, preferably latter.

251. Most engines operates with at least ..... excess air due to difficulty of introducing sufficient exposed oxygen to the fuel vapour in the given time available so that the combustion process can be completed before the exhaust valve opens.
(a) 10% (b) 20%
(c) 30% (d) 40%

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Excess air ensure that there is enough air for complete combustion. Excess air is expressed as a percentage of theoretical air required. Thus, 10% excess air indicates that 110% total air is being supplied. In a combustion process the complete burning cannot be accomplished with the exact theoretical amount of air

252. In C.I. engine, the fuel which is automised form is considerably colder than the hot compressed air in the cylinder. Although the actual ignition is almost instantaneous, an appreciable time elapses before the combustion is in full progress. This time occupied is called the
(a) Delay period
(b) Ignition lag
(c) Delay period or Ignition Lag
(d) Elapsed Time

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : In C.I. engine, the fuel which is automised form is considerably colder than the hot compressed air in the cylinder. Although the actual ignition is almost instantaneous, an appreciable time elapses before the combustion is in full progress. This time occupied is called the delay period or Ignition lag.

253. The power output from a spark ignition engine is varied by :
(a) Changing the ignition temperature
(b) Regulating the amount of air fuel inducted
(c) Regulating the amount of air fuel mixture
(d) Regulating the amount of fuel

HPPSC Asstt. Prof. 18.11.2016

Ans. (c) Increasing the power output can be achieved by-
1. Increasing the compression ratio.
2. Increasing the breathing capacity of engine (super charging).
3. Increasing the bore of engine.
4. Adding more fuel.

254. Which of the following factor (s) increase(s) the tendency for knocking in the C.I. engine?
(a) Increasing the speed injection advance and coolant temperature
(b) Increasing both the compression ratio and the coolant temperature

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(c) Increasing both the speed and the injection advance
(d) Increasing the compression ratio

UPRVUNL AE 21.08.2016

Ans. : (c) Factors tending to reduce knocking in CI engine
(i) Self ignition temperature of fuel – Low
(ii) Cylinder Size – Large
(iii) Speed – Low
(iv) Inlet temperature and pressure – High
(v) Compression ratio – High

255. Consider the following statements relevant to the ignition system of SI engine
1. Too large a dwell angle will result in misfiring
2 Too large a dwell angle will result in burning of condenser and contact points
3. Too small a dwell angle will lead to the burning of condenser and contact points.
4. Too small a dwell angle will result in misfiring

Which of the above statements are correct?
(a) 2 and 3 (b) 1 and 2
(c) 3 and 4 (d) 4 and 1

UPRVUNL AE 21.08.2016

Ans. : (b) • Too large a dwell angle will result in burning of condenser and contact points
• Too small a dwell angle will result in misfiring

256. Fuel injection pressure in solid injection system is approximately in the range of
(a) <10.5 bar (b) 10.5-21 bar
(c) 30 - 50 bar (d) 200 - 246 bar

UPPSC AE 12.04.2016 Paper-II

Ans.: (d) Fuel injection pressure in solid injection system is approximately in the range of 200-246 bar.

257. Problem of scavenging appears in case of
(a) Petrol engine (b) Diesel engine
(c) Four stroke engine (d) Two stroke engine

UJVNL AE 2016

Ans : (d) Problem of scavenging appears in case of two stroke engine
Scavenging:-The process of removing burnt gases, from the combustion chamber of the engine cylinder, is known as scavenging.
Scavenging are two types
1) Cross flow scavenging
2) Back or loop scavenging

258. In C.I. Engine, the function air swirl is
(a) Haphazard motion of air in the cylinder
(b) rotary motion of the air for proper distribution of fuel in the cylinder
(c) radial motion of air in the cylinder
(d) Mixing of air and fuel

APPSC AEE MAINS 2016 PAPER-III

Ans. (b) : In C.I. Engine, the function air swirl is rotary motion of the air for proper distribution of fuel in the cylinder.

259. Precise petrol injection system is :
(a) Direct injection
(b) Sequential injection
(c) Throttle body injection
(d) Port injection

HPPSC W.S. Poly. 2016

Ans : (c) Precise petrol injection system is throttle body injection.
The injector or atomizer is also an important part of the diesel engine which breaks up the fuel and sprays into the cylinder into a very fine divided particles.

260. In a three cylinder petrol engine the standard firing order is :
(a) 3-2-1 (b) 1-2-3
(c) 1-3-2 (d) None of the above

GPSC Lect. (Auto) 16.10.2016

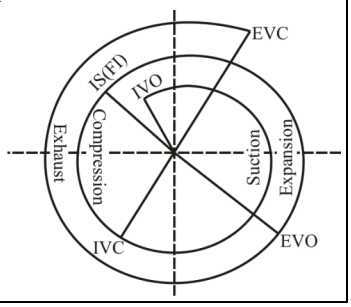
Ans. (c) : In a three cylinder petrol engine the standard firing order is 1-3-2.

261. The inlet valve of a four stroke cycle I.C. engine remains open approximately
(a) 205° (b) 190°
(c) 235° (d) 285°

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : The inlet valve at a four stroke cycle I.C. engine remains open approximately 235°.
Valve Timing Diagram–

IVO = Inlet valve open
IVC = Inlet valve close
EVO = Exhaust valve open
EVC = Exhaust valve close



262. Which of the following is associated with the valve operating mechanism?
(a) Gudgeon pin (b) Crown
(c) Tappet (d) Distributor

RPSC AE (GWD) 06.05.2016

Ans. (c) : A tappet is most commonly component in an internal combustion engine which converts the rotating motion of the camshaft into linear motion of valves, either directly or indirectly.

263. The maximum pressure of air fuel mixture at the end of compression in petrol engine varies from :
(a) 10-30 kg/cm^2 (b) 30-100 kg/cm^2
(c) 6-10 kg/cm^2 (d) 10-20 kg/cm^2

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) The maximum pressure of air fuel mixture at the end of compression in petrol engine varies from 10-20 kg/cm^2.

264. Which of the following is not true for internal combustion engine as compare to external combustion engine?
(a) It has a lower ratio of weight to output
(b) Higher overall efficiency
(c) They are all self starting
(d) Mechanical simplicity

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c)

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265. Which of the following is NOT a disadvantage of two stroke engines as compared to four stroke engines ?

(a) More cooling requirement
(b) Heavier fly wheel required
(c) More wear and tear
(d) Lower thermal efficiency

RPSC AE (GWD) 06.05.2016

Ans. (b) : Advantages

- A two stroke cycle engine has twice the number of power stroke than the four stroke cycle engine at the same on engine speed.
- For same power developed a two stroke cycle engine is lighter, less bulky and occupies less floor area.
- A two stroke engine has less maintenance cost since it requires only few parts.
- The weight of the two stroke engine is less than four stroke cycle engine because of the lighter flywheel due to more uniform torque on the crankshaft.

266. Pick the wrong statement about CI engines :

(a) They operate at higher compression ratio.
(b) The charge sucked in suction stroke consists of air only.
(c) They have lower thermal efficiency than SI engines of comparable size.
(d) They are more suitable for two-stroke operation.

RPSC AE (GWD) 06.05.2016

Ans. (c) : C.I. have higher thermal efficiency than S.I. engine of comparable size.

267. Consider the following statements about SI engines : (a) The air is sucked in the suction stroke and fuel is injected into the cylinder to obtain a homogenous mixture of air and fuel. (b) A homogeneous mixture of air and fuel is prepared externally and sucked in the cylinder during suction stroke. (c) Ignition is by a spark plug exactly at the end of compression stroke. (d) Ignition is by a spark plug little before the end of the compression stroke. Which of the above statement(s) are correct?

(a) Both (a) and (c) (b) Both (b) and (c)
(c) (a), (b) and (c) (d) Both (b) and (d)

RPSC AE (GWD) 06.05.2016

Ans. (d) : To about S.I. engine-

- A homogeneous mixture of air and fuel is prepared externally and sucked in the cylinder suction stroke.
- Ignition is by a spark plug little before the end of the compression stroke.

268. For a diesel engine, the maximum efficiency occurs at

(a) Very low loads (b) Around 50% load
(c) Around 80% load (d) Full load

RPSC AE (GWD) 06.05.2016

Ans. (c) : around 80% load

269. The air requirement of a petrol engine during starting compared to theoretical air required for complete combustion is:

(a) More
(b) Less
(c) Same
(d) May be more or less depending on engine capacity

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : The ratio of mass of air sucked by engine to mass of fuel injected is air-fuel ratio. The air requirement of a petrol engine during starting compared to theoretical air (15:1- stoichiometric air fuel ratio) required for complete combustion is less.

270. The pressure inside the engine cylinder is …… atmospheric pressure during suction stroke.

(a) Above
(b) Below
(c) Equal
(d) Either below or above

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : The pressure inside the engine cylinder is below the atmospheric pressure during suction stroke. The gas is continuously sucked inside cylinder till its pressure not becomes more than atmospheric pressure.

271. Which of the following is not difference between four stroke cycle CI engine and SI engine?

(a) Four stroke CI works on diesel cycle and SI works on otto cycle.
(b) Four stroke CI are heavier due to higher peak pressure and SI are lighter due to lower peak pressure.
(c) Four stroke CI are self –ignition due to high compression ratio, on the other side SI requires ignition system with spark plug.
(d) Four stroke CI is internal combustion while SI engine is external combustion engine.

GPSC Lect. (Auto) 16.10.2016

Ans. (d) :

272. Single cylinder two stroke SI engine has :

(a) Transfer and exhaust ports
(b) Inlet and exhaust valve
(c) Spark distributor
(d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Two stroke SI engine have port mechanism. It consist of two ports. Transfer and exhaust ports. Four stroke SI engine have valve mechanism. It consist of inlet and exhaust valve.

273. A two stroke engine may be easily identified by:

(a) Absence of piston rings
(b) Size of flywheel
(c) Piston
(d) Absence of valves

KPSC ADF 2015

Ans. (d) : A two stroke engine may be easily identified by absence of valve. In 2-strokes engine inlet and exhaust ports are present.

274. Which of the following are the assumptions involved in the auto-ignition theory put forth for the onset of knock in SI engines?

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1. Flame velocity is normal before the onset of auto ignition.
2. A number of end-gas elements auto ignite simultaneously.
3. Pre flame reactions are responsible for preparing the end-gas to ignite.

Select the correct answer using the codes given below.

(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) 1, 2 and 3

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : Assumptions–
(1) Flame velocity is normal before the onset of auto ignition.
(2) A number of end-gas elements auto ignite simultaneously.
(3) Pre flame reactions are responsible for preparing the end-gas to ignite.

275. In the opposed piston diesel engine, the combustion chamber is located

(a) Above the piston (b) Below the piston
(c) Between the pistons (d) Anywhere

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : In opposed piston engine the combustion chamber is the space left between the two pistons when both are at inner dead centre position. It is this place between the pistons where in the fuel injection volume, air starting valve, air starting valve, pressure relief valve and indicating cocks are fixed.

276. Match List-I with List-II and select the correct answer :

	List-I		List-II
A.	Highest useful compression ratio	1.	Ignitable mixture
B.	Dopes	2.	Knock rating of fuels
C.	Limiting mixture strength	3.	Detonation
D.	Delay period	4.	Chain of chemical reactions in combustion chamber

	A	B	C	D
(a)	2	3	1	4
(c)	2	3	4	1

	A	B	C	D
(b)	3	2	1	4
(d)	3	4	2	1

Nagaland PSC CTSE 2015 Paper-II

Ans. (a)
- **Highest Useful Compression Ratio–** SI engine fuel is limited to knocking tendency of fuel as it is used as one of the methods for knocking rating of fuel.
- Dopes are chemical additive which are used to improve combustion performance of fuel.
- Limiting mixture strength is the range for ignitable mixture. The ignition limits are wider at increased temperature because of higher rates of reaction and high thermal diffusivity coefficient of mixture.
- Delay period is related with detonation in CI engine. Higher the delay period more is the tendency of knocking in CI engine.

277. Compensating devices are provided in carburetors

(a) To charge the quantity of mixture depending upon load
(b) To provide always an economy mixture
(c) To modify the mixture strength depending upon requirements under various operating conditions
(d) To supply extra fuel during acceleration only

MPPSC AE 08.11.2015

Ans. (c) : Compensating devices are provided in carburetors to modify the mixture strength depending upon requirements under various operating condition.

278. Which is not a part of magneto-ignition system?

(a) Condenser (b) Battery
(c) Circuit breaker (d) Induction coil

(KPSC AE 2015)

Ans : (b) Battery is not a part of magneto- ignition system.
Magneto ignition system:- The magneto ignition system has the same principle of working as that of coil ignition system. except that no battery is required, as the magneto acts its own generator.
This type of ignition system is generally employed in small spark ignition engines such as scooters, motor cycles and small motor boat engines.

279. A temperature indictor is usually provided for automobiles, which indicates temperature of

(a) Lubrication oil
(b) Engine cylinder wall
(c) Jacket cooing water
(d) Engine piston

GPSC AE 26.07.2015

Ans. (d) : A temperature indictor is usually provided for automobiles, which indicates temperature of engine piston.

280. Consider the following statements:
Detonation in the SI engine can be suppressed by
1. Retarding the spark timing
2. increasing the engine speed
3. using 10% rich mixture
4. using a fuel of high octane rating of these statements

(a) 1 and 3 are correct
(b) 2 and 3 are correct
(c) 1, 2 and 3 are correct
(d) 1 and 2 are correct
(e) 1, 2, 3 and 4 are correct

CGPSC AE 26.04.2015 SHIFT-II

Ans. (e) :
Detonation in the SI engine can be suppressed by
1. Retarding the spark timing
2. Increasing the engine speed
3. Using 10% rich mixture
4. Using a fuel of high octane rating.
5. Reduction in compression ratio, temperature of cylinder wall and over load on the engine.
6. Increasing turbulence in engine cylinder.

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281. Self-ignition temperature of petrol is of the order of :
(a) 150°C (b) 370°C
(c) 450°C (d) more than 500°C
(HPPSC AE 2014)

Ans : (d) The Self Ignition auto ignition temperature is lowest temperature at which a fuel such as petrol or diesel spontaneously ignites in normal atmosphere without an external source of ignition, such as a flame or spark.

282. The function of carburettor is to:-
(a) Refining the fuel
(b) Increase the pressure of fuel vapours
(c) Inject petrol in cylinder
(d) Automise and vapourise the fuel and to mix it with air in proper ratio
UKPSC AE-2013, Paper-II

Ans. (d) : The function of carburettor is to automise and vapourise the fuel and to mix it with air in proper ratio.

283. For compression ignition Engines minimum compression ratio is :
(a) 7 (b) 9
(c) 12 (d) 10
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : Compression ratio

$$(r) = \frac{V_T}{V_C} = \frac{V_S + V_C}{V_C} = 1 + \frac{V_S}{V_C}$$

V_S = Swept volume
V_C = Clearance volume
V_T = Total volume
Compression ratio of SI engine = 6 to 10
Compression ratio for CI engine = 16 to 22
Compression ratio for dual engine = 12 to 16
So, minimum compression ratio of CI engine is 12 as per question.

284. Modern cars, using petrol as a fuel, has compression ratio of about
(a) 6-7 : 1.0 (b) 9-10:1.0
(c) 18.0:1.0 (d) 12.0:1.0
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Modern cars, using petrol as a fuel, has compression ratio of about 9-10:1.0.

285. If A_p is the area of piston & A_r is the area of piston rod, then for extension stroke effective area 'A' is.
(a) $A = A_p$ (b) $A = A_p - A_r$
(c) $A = A_r$ (d) $A = A_p + A_r$
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : A_p = Area of piston
A_r = Area of piston rod.
Then for extension stroke effective area A is
$A = A_p$
Stroke Volume $(V_s) = A_p \times L$

$$V_s = \frac{\pi}{4} D^2 \times L$$

286. Scavenging air in diesel engine means:
(a) Air used for combustion sent under pressure
(b) Forced air for cooling cylinder
(c) Burnt air containing product of combustion
(d) Air used for forcing burnt gases out of engine cylinder during the exhaust period.
SJVN ET 2013

Ans.(d) : In diesel engine, scavenging is the process of removing the burnt gases from the combustion chamber of the engine cylinder during exhaust period and fresh air coming inside the cylinder.

287. Which is correct for diesel power plant?
(a) No ash handling, but high operating cost
(b) Difficult to design, but low maintenance cost
(c) Low efficiency at part loads, but noise problem
(d) All of these
SJVN ET 2013

Ans. (a) : For diesel power plant no ash handling, but high operating cost.

288. In a low speed S.I. engine, the inlet valve closes approximately:-
(a) 40° after BDC (b) 30° before BDC
(c) 10° after BDC (d) 10° before BDC
UKPSC AE-2013, Paper-II

Ans. (c)

289. The function of fuel injector is to:-
(a) Pump the fuel at high pressure
(b) Mix diesel with air
(c) Automoise the fuel
(d) Ignite the fuel
UKPSC AE-2013, Paper-II

Ans. (c) : The function of fuel injector is to automoise the fuel.

290. The process of replacement of combustion products with fresh air charge
(a) Turbocharging (b) Idling
(c) Supercharging (d) Scavenging
TNPSC AE 2013

Ans. (d) : The process of replacement of combustion products with fresh air charge is known as scavenging.

291. In a cycle, the spark lasts roughly for
(a) 1 sec (b) 0.1 sec
(c) 0.01 sec (d) 0.001 sec
APPSC AEE 2012

Ans. (d) : In a cycle, the spark lasts roughly for 0.001 sec.

292. For low load operation, most economical engine is
(a) an S.I. engine
(b) a C.I. engine
(c) a two-stroke engine
(d) both S.I. and C.I. engines are equally good
APPSC AEE 2012

Ans. (b) : For low load operation most economical engine is a CI engine.

293. Ignition accelerators are substance which
(a) increase the rate of pre-flame reaction and reduce the ignition lag
(b) increase knock
(c) reduce detonation
(d) increase thermal efficiency
BPSC AE 2012 Paper - V

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Ans : (a) : Ignition accelerators are substance which increase the rate of pre-flame reaction and reduce the ignition lag.

294. Diesel, as compared to petrol is
(a) highly ignitable
(b) more difficult to ignite
(c) less difficult to ignite
(d) none of these

TNPSC ACF 2012

Ans. (b) : Diesel as compared to petrol is more difficult to ignite.
Method of ignition- In petrol engine spark plug is used to produce spark for the ignition. In CI engine heat of compressed air is used for the ignition.

295. In order to prevent knocking in spark ignition engines, the charge away from the spark plug should have
(a) low density (b) low temperature
(c) long ignition delay (d) all of these

TNPSC ACF 2012

Ans. (d) : Following parameter will minimize knocking in SI engine.
Compression ratio- Low
Self ignition temperature- High
Mass of charge- Low
Inlet temperature of charge- Low-

296. How many cylinders does a rotary internal combustion engine has ?
(a) 4 (b) 5
(c) 6 (d) 7

VIZAG STEEL MT 2011

Ans. (c) :

297. Consider the following factors:
Diesel engine knock can be reduced by increasing
1. Engine speed
2. Compression ratio
3. Degree of supercharge
4. Injection advance
Which of the statements is/are true?
(a) 1, 2 and 3 only (b) 2 and 3 only
(c) 1, 3 and 4 only (d) 2 only

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Diesel knocking can be reduced by increasing compression ratio (which reduce the delay period) & Degree of supercharging increases the charge and encourage the atomization of fuel & prevent knocks.
Decreasing engine speed reduce knock & similar in injection advance.

298. When does ignition initiates in a Spark Ignition Engine?
(a) Immediately after spark
(b) Immediately after completion of physical and chemical delay
(c) Immediately after the completion of Ignition lag
(d) Just after completion of flame propagation

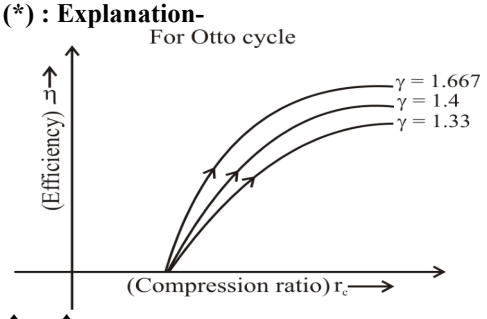
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : In Spark Ignition engine, ignition initiates immediately after the completion of ignition lag. Ignition lag in terms of crank angle is 10º to 20º & in terms of second 0.0015 sec. before reaching to top dead centre of cylinder.

299. Consider the following statements for the thermal efficiency of the Otto cycle :
1. The efficiency increases with increase in compression ratio.
2. The efficiency decreases with increase in specific heat ratio of working fluid.
3. The efficiency is independent of the temperature levels at which the cycle operates :
Which of these statements are correct?
(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-52/2010

Ans. (*) : Explanation-



(1) $\eta \uparrow, r_c \uparrow$
(2) $\eta \uparrow, \gamma \uparrow$
(3) $\eta = 1-\frac{T_L}{T_H} = 1-\frac{1}{(r_c)^{\gamma-1}}$
$\eta = f(T_L, T_H)$
So only statement (1) is correct. But no suitable option is given.

300. The flame travel in SI engines is very slow initially due to :
(a) Low transposition rate and low turbulence
(b) Spark plug being located very close to the cylinder wall
(c) Low turbulence and low heat liberation
(d) Fraction of unburned mass which is a function of crank angle

APPSC IOF, 2009

Ans. (d) : The burned gas fraction in the unburned mixture, due to the residual gas fraction and any recycled exhaust gases recirculation (EGR), slow down both flame development and propagation. Residual gas fraction increases at part loads in SI-Engines, reducing flame propagation.

301. A simple carburettor has a venturi throat diameter of 22 mm and the coefficient of flow is 0.82. The fuel orifice has a diameter of 1.2 mm and the coefficient of fuel flow is 0.70. The petrol surface is 4 mm below the throat. Taking nozzle tip into account, the air/fuel ratio for a pressure drop of 0.075 bar is :
(a) 12.75 (b) 15.78
(c) 7 (d) 1.6155

APPSC IOF, 2009

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Ans. (b) : Using $\frac{A}{F}=\frac{C_{dv}A_v}{C_{dn}A_n}\sqrt{\frac{\rho_a}{\rho_f}},\ A_v=\frac{\pi}{4}d_v^2$

$$=\frac{0.82\times0.785\times(22\times10^{-3})^2}{0.70\times0.785\times(1.2\times10^{-3})^2}\sqrt{\frac{1.2}{750}}$$

$$\frac{A}{F}=15.74:1$$

For taking into account, x = 4 mm,

Using $\frac{A}{F}=\frac{C_{dv}}{C_{dn}}\frac{A_v}{A_n}\times\sqrt{\frac{\rho a(p_1-p_2)}{\rho_f(p_1-p_2-\rho_f.g.x)}}$

$$\frac{A}{F}=15.78:1$$

302. The temperature of the cooling water leaving the diesel engine should not be more than :
(a) 30°C (b) 60°C
(c) 74°C (d) 45°C

APPSC IOF, 2009

Ans. (b) : The temperature of cooling water leaving the diesel engine should not exceed 60°C or 140º F.

303. The fuel air ratio for maximum power of S.I. engines, should be
(a) Lean (b) Rich
(c) May be lean or rich (d) Chemically correct

WBPSC AE 2008

Ans. (b) : The fuel air ratio for maximum power of S.I. engines, should be rich.

304. Due to which of the following reasons a diesel engine gives a smoke exhaust
(a) Water in the fuel
(b) Fuel injection is late
(c) Fuel is not distributed equally to all the cylinders
(d) Exhaust valve receives too much lube oil

WBPSC AE 2008

Ans. (b) : Due to lack of appropriate time to burn the fuel a black or gray smoke may comes out in exhaust.

305. The thermal efficiency of petrol engines is about
(a) 15% (b) 30%
(c) 50% (d) 70%

TNPSC AE, 2008

Ans. (b) : The thermal efficiency of petrol engines is about 30%.

306. Thermal efficiency of a diesel engine used for power generation is
(a) Less than air standard value
(b) More than air standard value
(c) Equal to that of air standard value
(d) Twice that of air standard value

TNPSC AE, 2008

Ans. (a) : Thermal efficiency of a diesel engine used for power generation is less than air standard value.

307. Delay period of CI engine fuel is affected by
(a) Properties of fuel
(b) Atomization of fuel
(c) Compression ratio of engine
(d) All the above

UKPSC AE 2007 Paper -II

Ans. (d) : All the above

308. The amount of air which is supplied for complete combustion is called
(a) Primary air (b) Secondary air
(c) Tertiary air (d) None of these

UKPSC AE 2007 Paper -II

Ans. (b) : Secondary air

309. Source of electrical energy in SI engine is
(a) battery (b) magneto
(c) generator (d) any of the above

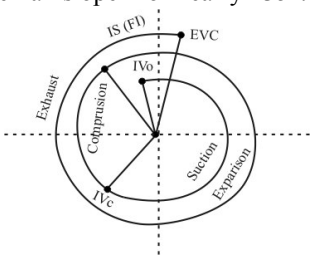
UKPSC AE 2007 Paper -II

Ans. (d) : Any of the above

310. The inlet valve of a four stroke cycle I.C. engine remains open for nearly :
(a) 180° (b) 125°
(c) 235° (d) 200°

J & K PSC Screening, 2006

Ans. (c) : The inlet valve for a four stroke cycle IC engine remains open for nearly 235°.



311. A four stroke diesel engine does not have
(a) Cooling (b) Ignition coil
(c) Cam shaft (d) Piston rings

WBPSC AE 2003

Ans. (b) : 4- stroke diesel engine have
→ Piston, Crankshaft, Camshaft, cylinder, valves, fly wheel connecting rod, fuel injectors thus 4-stroke diesel engine does not have Ignition coil.

312. In a naturally aspirated diesel engine, the air is supplied by :
(a) A supercharger
(b) A centrifugal blower
(c) A vacuum chamber
(d) An injection tube

J&K PSC Civil Services Pre, 2013

Ans. (c) : A vacuum is created in the cylinder during suction stroke in naturally aspirated diesel engine. The speed of diesel engine increases, this vacuum pulls in even more air and diesel fuel. So, the engine gas faster developing higher horsepower levels. The volumetric efficiency is less than 100%.
Note – Centrifugal blower are also used in super charger and air is supplied to piston as more pressure than atmospheric pressure. So, volumetric efficiency is more than 100% .

Search on Telegram @apna_pdf

313. Four cylinder four stroke petrol engine running at 1000 rpm will have in all how many sparks in one minute?

(a) 250 (b) 500
(c) 1000 (d) 2000

RPSC ADE 2016

Ans. (d) : Sparking take place one in each cycle in each cylinder and four stroke complete one cycle in two revolution of crankshaft.

So, number of sparks = $\frac{1000 \times 4}{2} = 2000$

314. Which of the following statement about two-stroke engines is wrong ?

(a) Valves are absent.
(b) The cycle is completed in two strokes or one revolution of crankshaft.
(c) Lighter than a four-stroke engine of same power.
(d) Due to absence of valves, a complex mechanism for admission of charge and discharge of burnt gases is required.

RPSC AE (GWD) 06.05.2016

Ans. (d) : About Two-stroke engine-

- Valves are absent
- In two stroke engine, the air-fuel mixture enters through the inlet port.
- The cycle is completed in two stroke or one revolution of crank shaft.
- Lighter than a four stroke engine of same power.

315. For a four stroke diesel engine exhaust gas temperature

(a) Increases with load
(b) Decreases with load
(c) Is constant throughout the variation of load
(d) No prediction is possible

UKPSC AE 2007 Paper -II

Ans. (a) :

316. Ignition in a SI engine takes place when the piston is :

(a) Exactly at TDC position on its compression stroke
(b) Leaving the TDC position on its compression stroke
(c) Approaching TDC position on its exhaust stroke
(d) Approaching TDC position on its Compression stroke

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) Ignition in a SI engine takes place when the piston is approaching TDC position on its compression stroke.

317. In a four-stroke cycle petrol engine, during suction stroke

(a) Only air is sucked in
(b) Only petrol is sucked in
(c) Mixture of petrol and air is sucked in
(d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : In a four-stroke cycle petrol engine, during suction stroke Mixture of petrol and air is sucked in.

318. Ignition timing of a multi cylinder petrol engine can be adjusted by :

(a) Rotating the crank
(b) Adjusting the spark plug gap
(c) Adjusting ignition coil position
(d) Rotating the distributor

J&K PSC Civil Services Pre, 2013

Ans. (d) : Ignition timing of a multi cylinder petrol engine can be adjusted by rotating the distributor.

319. Knocking is compression ignition engine will increase:

(a) To a certain limits and then no change
(b) By increasing compression ratio
(c) Both fuel and air is sucked
(d) None of the above

HPPSC ADF 05.03.2019

Ans. (d) : The knocking tendency in compression ignition engines for a given fuel will be enhanced by decreasing compression ratio. If compression ratio is low in C.I. engines, there will not be enough temperature generation for ignition of fuel air mixture.

320. Which of the following factors increase detonation in the SI engine?

1. Increased spark advance
2. Increased engine rpm
3. Increased compression ratio.

Select the correct answer using the codes given below:

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

RPSC ACF & FRO, 26.02.2021

Ans. (c) : Factors increasing detonation in SI engine →
(1) Increased spark advance.
(2) Increased compression ratio.
(3) Increase inlet measure and temperature
(4) Low engine speed
(5) Large flame travel distance.

321. In CI engines with increase in compression ratio the delay period -

(a) First increases and then decreases
(b) Increases
(c) Decreases
(d) Not affected

RPSC ACF & FRO, 26.02.2021

Ans. (c) : In CI engines with increase in compression ratio the delay period decreases.

322. Which is the best option to avoid knock is S.I. engine?

(a) Supercharging of engine
(b) Low delay period
(c) Higher spark advance
(d) Increase in engine speed

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : In SI engine increase in engine speed reduces knocking.

Search on Telegram @apna_pdf

323. Which of the following actions will not reduce the tendency of knocking in an SI engine?
(a) Reducing compression ratio.
(b) Reducing load
(c) Retarding spark timing.
(d) Reducing engine speed.

RPSC AEN Pre-2013

Ans. (d) : To reduce the tendency of knocking is an S.I. engine.
- Reducing compression ratio.
- Reduced load.
- Retarding spark ignition timing.
- Increase engine speed.

324. Which one of the following set of materials is most commonly used in catalytic converters for CI engines?
(a) Platinum, palladium and rhodium
(b) Palladium, rhodium and ruthenium
(c) Rhodium, ruthenium and platinum
(d) Ruthenium, platinum and palladium

GPSC ARTO Pre 30.12.2018

Ans. (a) : The materials most commonly used in catalytic convertors for CI engine are Platinum, Palladium and Rhodium.

325. In S.I. engine, the throttle valve of carburetor controls the quantity of:-
(a) Fuel (b) Air
(c) Fuel and air mixture (d) Lubricating oil

UKPSC AE-2013, Paper-II

Ans. (c) : In S.I. engine, the throttle valve of carburetor controls the quantity of fuel and air mixture.

326. In petrol engine, governor manipulates
(a) fuel pump (b) automiser
(c) filter (d) throttle valve

UKPSC AE 2007 Paper -I

Ans. (d) : Throttle valve

327. In C.I. engines, ignition delay can be reduced by:-
(a) Decreasing compression ratio
(b) Increase in air inlet temperature
(c) Decrease in coolant temperature
(d) Decreasing the engine speed

UKPSC AE-2013, Paper-II

Ans. (b) : In C.I. engines, ignition delay can be reduced by increase in air inlet temperature.

328. Primary cause of an excessively high pressure rise in diesel knock is due to
(a) Prolonged Delay period
(b) Pre ignition
(c) Pre ignition delay
(d) None of the above

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Diesel knock is the sound produced by the very rapid rate of pressure rise during the early part of the uncontrolled second phase of combustion. The primary cause of an excessively high pressure rise is due to a prolonged delay period.

329. Detonation is harmful as
(a) Increase in the rate of heat transfer causes a reduction in the power output and efficiency of the engine.
(b) Excessive turbulence removes most of the insulating gas boundary layer from the cylinder walls.
(c) High intensity of knock causes crankshaft vibration and the engine runs rough.
(d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (d) : None of the above option are correct.

330. The knocking in S.I. engines increases with
(a) Increase in inlet air temperature
(b) Increase in compression ratio
(c) Increase in cooling water temperature
(d) All of the above

JPSC AE, 10.04.2021, Paper-I

Ans. (d) : Knocking in S.I engine increases with–
- Increase in inlet air temperature.
- Increase in compression ratio.
- Increase in cooling water temperature.
- By supercharging.

331. In an S.I. engine, when most economical mixture is supplied, the colour of exhaust gases can be expected to be
(a) Bluish
(b) Dark red
(c) Dark blue and reddish yellow
(d) White

Karnataka PSC AE, 10.09.2017

Ans. (c) : Most economical mixture for S.I. engine is 15 : 1 to 16 : 1 where the colour of exhaust gases are dark blue and reddish yellow.

332. To have best efficiency in S.I. engine air-fuel mixture should be :
(a) Rich in fuel (b) Lean in fuel
(c) Both (a) and (b) (d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : For best efficiency air fuel mixture should be lean in fuel. For max power air fuel mixture should be rich in fuel.

333. For maximum power generation, the air fuel ratio for petrol engine vehicles is of the order of:
(a) 9 : 1 (b) 12 : 1
(c) 15 : 1 (d) 18 : 1

J&K PSC Civil Services Pre, 2013

Ans. (b) : Air fuel ratio to petrol engine
For max power – 12.6 : 1
For best fuel economy – 15.4 : 1

334. Which one of the following is not the chief effect of detonation?
(a) A loud pulsating noise, which may be accompanied by a vibration of the engine
(b) A loud pulsating noise from the wheel assembly
(c) An increase in the heat lost to the surface of combustion chamber
(d) An increase in carbon deposits

JPSC AE 2013 Paper-V

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Ans. (b) : A loud pulsating noise from the wheel assembly is not the chief effect of detonation.

335. In a diesel engine, during idling, the air fuel ratio may be of the order of

(a) 30 : 1 (b) 15 : 1
(c) 10 : 1 (d) 8 : 1

GPSC ARTO 01.05.2016

Ans. (c) : The rich air-fuel mixture is required during idling which is order of 10:1.

336. Increase in compression ratio in a diesel engine:

(a) Increases ignition delay
(b) Reduces ignition delay
(c) Keeps ignition delay constant
(d) Has no effect on ignition delay

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (b) : Increase in compression ratio (CR = 14 to 18) in a diesel engine reduce ignition delay, improve efficiency of engine and increase the brake thermal efficiency.

337. The tendency of knocking in CI engine is reduced by:

(a) high self-ignition temperature of fuel
(b) decrease in cooling water temperature
(c) short ignition lag of the fuel
(d) decrease in engine cylinder diameter
(e) low compression ratio

CGPSC ARTO & TSI 07.01.2017

Ans. (c) : The tendency of knocking in CI Engine is reduced by short ignition lag of the fuel.

- Knocking in an internal combustion engine, sharp caused by premature combustion of part of the compressed air fuel mixture in the Cylinder.

These pressure waves force parts of the engine to vibrate, which produce on audible knock.

338. Due to detonation the following parameters attains very high value:

(a) rate of temperature rise
(b) peak temperature
(c) rate of rise of horse power
(d) rate of pressure rise

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : Detonation or engine knock, occurs simply when the fuel preignates before the piston reaches schedule spark ignition detonation is generally caused by excessive heat excessive cylinder pressure, improper, ignition timing inadequate fuel octane or a combination of these. So, As a result of detonation in an I.C. engine following parameter attains very high value rate of rise of pressure.

339. In order to eliminate knocking in compression ignition engines, there should be

(a) short delay period
(b) late auto-ignition
(c) low compression ratio
(d) high self ignition temperature of fuel

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : In order to eliminate knocking in compression ignition engines there should be short delay period Add 1% of ethyl nitrate or any nitrate to accelerate the combustion of fuel this process is known as doping and reduce delay period considerably and hence prevent diesel knock. For spontaneous ignition of the fuel, we require much higher temperature this can be achieved by compression ratio.

340. A higher compression ratio causes

(a) An acceleration in the rate of combustion
(b) Tendency of an engine to increase detonation
(c) Pre-ignition
(d) All of these

TNPSC AE, 2008

Ans. (d) : A higher compression ratio causes –

- An acceleration in the rate of combustion
- Tendency of an engine to increase detonation
- Pre-ignition

341. In diesel engine the duration between the start of injection and ignition is known as

(a) Physical delay (b) Chemical delay
(c) Ignition delay (d) Period of ignition

RPSC IOF, 2020

Ans. (c) : The ignition delay in diesel engine the duration between the start at injection and ignition.

This delay period consists of physical delay whereas automisation vaporization and mixing of air fuel occur and chemical delay attributed to precombustion reactions.

342. Which of the following will reduce knocking in diesel engines ?

(a) High compression ratio and short delay period of fuel.
(b) High compression ratio and long delay period of fuel.
(c) Low compression ratio and short delay period of fuel.
(d) Low compression ratio and long delay period of fuel.

RPSC AE (GWD) 06.05.2016

Ans. (a) : To reduce knocking in diesel engines, high compression ratio and short delay period of fuel.

343. The angle of rotation from the instant of spark to TDC position of the piston is known as :

(a) Spark delay (b) Spark advance
(c) Spark inlay (d) Spark timer

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : The angle of rotation from the instant of spark to TDC position of the piston is known as spark advance. The angle will be described as a certain angle advanced before top dead centre. Advancing the spark BTDC means that the spark is energized prior to the point where the combustion chamber reaches its minimum site.

344. With …… speed of rotation the spark will be advanced in I.C. engine.

(a) Increased (b) Decreased
(c) Zero (d) None of these

HPPSC Lect. (Auto) 23.04.2016

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Ans. (a) : Effect of spark timing on performance spark advance is the time before top dead centre (TDC) when the spark is initiated. It is usually expressed in number of degree of crank shaft rotation relative to TDC. Hence with increased speed of rotation the spark will be advanced in I.C. engine.

345. If a large amount of mixture in an engine cylinder auto ignites rapid rise in pressure occurs causing a direct blow on the engine structure is known as :

(a) Knock (b) Block
(c) Choke (d) Pinkerling

HPPSC Lect. (Auto) 23.04.2016

Ans. (a) : Knocking in spark ignition I.C. engine occurs when combustion of some of the air/fuel mixture in the cylinder does not result from propagation of the flame front ignited by the spark plug, but one or more pockets of air/fuel mixture explode outside the envelope of the normal combustion front.

346. Though thermal efficiency of the constant volume combustion is higher than the constant pressure combustion, it is not preferred due to which of the following reasons?

1. Mechanical difficulties requiring valves to isolate the combustion chamber from the compressor and turbine
2. Combustion is intermittent impairing the smooth running of the machine

(a) neither 1 nor 2 (b) 1 only
(c) 2 only (d) 1 and 2

VIZAG MT, 14.12.2020

Ans. (d) : Constant volume combustion is not preferred because

(1) Mechanical difficulties requiring valves to isolate the combustion chamber from the compressor and turbine.
(2) Combustion is intermittent impairing the smooth running of the machine.

347. In multi cylinder SI engine required firing order is achieved with the help of :

(a) Carburetor (b) Battery
(c) Ignition coil (d) Distributor

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : In multicylinder SI engine required firing order is achieved with the help of distributor.
Firing order for a 4-cylinder engine – 1–3–4–2
Firing order for a 6-cylinder engine – 1–5–3–6–2–4

348. Which category of combustion is desired to avoiding knocking or detonation in engines?

(a) Premixed combustion
(b) Partially premixed combustion
(c) Non-premixed combustion
(d) HCCI combustion

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Homogeneous change compression ignition (HCCI) combustion is required to avoid knocking in engine.

349. Knocking in a spark ignition engine is

(a) Promoted by a short flame travel length
(b) Normally at the beginning of the combustion process
(c) Desirable for improving thermal efficiency
(d) Promoted by reduced turbulence of the fuel-air mixture during combustion.

WBPSC AE 2003

Ans. (d) : Knocking in a SI engine is promoted by reduced turbulence of the fuel air mixture during combustion.
Factors affecting knocking in SI engine–
(1) Compression (6-10)↑ → Knocking↑
(2) Spark timing advancing↑ → Knocking↑
(3) Load↑ → Knocking↑
(4) Turbulence (speed)↑ → Knocking↓
(5) Flame travel distance↑ → Knocking↑
(6) Octane rating of fuel↑ → Knocking↓

350. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Association	1. Pseudo shock
B. Dissociation	2. Knock
C. Flame front	3. Endothermic
D. Abnormal combustion	4. Exothermic

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	1	2
(c)	3	4	2	1
(d)	4	3	2	1

Nagaland PSC CTSE 2015 Paper-II

Ans. (b)

Association	—	Exothermic
Dissociation	—	Endothermic
Flame front	—	Pseudo shock
Abnormal combustion	—	Knock

351. In an internal combustion engine, firing order depends upon

(a) Crank shaft design
(b) Arrangement of cylinder
(c) Number of cylinders
(d) All of the above

UKPSC AE 2012 Paper–II

Ans. (d) : All of the above

352. Very high compression ratios are not used with petrol engines because :

(a) Self ignition would take place before the spark occurs
(b) Power required for compression ratio very high
(c) Lower thermal efficiency
(d) Cylinder walls are to be made thicker

TNPSC AE (Industries) 09.06.2013

Ans. (a) : Very high compression ratios are not used with petrol engines because self ignition would take place before the spark occurs.

353. The crankcase of many diesel engines are kept under a slight vacuum to

(a) Improve fuel economy
(b) Improve air charge velocity
(c) Reduce the risk of crankcase explosion
(d) All of the above

RPSC Lect. (Tech. Edu. Dept.) 2011

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Ans. (d) : The crankcase of many diesel engines are kept under a slight vacuum to improve fuel economy, air charge velocity and reduce the risk of crank case explosion.

354. In S.I. engine if the spark plug deposit indicates black coating of soot, it therefore indicates that the engine has been generally operating on

(a) Too rich mixture
(b) Too lean mixture
(c) Stoichiometric mixture
(d) Most economical mixture

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : If the Spark plug deposit indicates black coating of soot, in S.I. engine, it indicates that the engines has been generally operating on too rich mixture.

3. Fuels

355. In a SI Engine, which one of the following is the correct order of the fuels with increasing detonation tendency?

(a) Paraffins, Olefins, Naphthalenes, Aromatics
(b) Aromatics, Naphthalenes, Paraffins, Olefins
(c) Naphthalenes, Olefins, Aromatics, Paraffins
(d) Aromatics, Naphthalenes, Olefins, Paraffins.

GPSC ARTO Pre 30.12.2018, (KPSC AE. 2015)
Kerala PSC AE 06.08.2015, ESE 1997

Ans. (d) : The correct order of fuels with increasing detonation tendency.

Aromatics, Naphthalenes, Olefins, Paraffins.

356. Ignition quality of fuels for petrol engines in determined by

(a) Cetane number rating
(b) Octane number rating
(c) Calorific value rating
(d) Volatility of fuel

RPSC AE 2018, 2016
Mizoram PSC AE/SDO 2014, Paper-I
J&K PSC Civil Services Pre, 2013
UKPSC AE 2012 Paper–II
RPSC ACF-2011

Ans : (b) Ignition quality of fuels of petrol engine is determined by octane rating.
Octane number:- The Knocking tendency of a fuel in SI engines is generally expressed by its octane number The percentage by volume of iso- octane in a mixture of iso-octane and normal heptane, iso-octane is a very good anti-knock fuel, therefore it is assigned a rating of 100 octane number. on other hand normal heptane has very poor anti-knock qualities there for it is given a rating of 0 (Zero) octane number.

357. By higher octane number of SI fuel, it is meant that the fuel has

(a) Longer ignition delay
(b) Higher flash point
(c) Lower viscosity
(d) Higher heating value

Assam PSC AE (PHED), 18.10.2020
Punjab PSC (Lect.) 06.08.2017
GPSC ARTO 01.05.2016
UJVNL AE 2016, ESE 1995

Ans. (a) : By higher octane number of SI fuel, it is meant that the fuel has longer ignition delay.

358. The ignition quality of diesel fuel is expressed by

(a) Octane number (b) Cetane number
(c) Carbon percentage (d) Calorific value

RPSC IOF, 2020, BPSC Poly. Lect. 2016
GPSC Lect. 23.10.2016
OPSC AEE 2015 PAPER - II

Ans : (b) (i) The ignition quality of diesel fuel is expressed by cetane number. It is defined as the percentage, by volume of cetane in a mixture of cetane and alpha-methyl-napthalene.
The cetane which is a straight chain paraffin which good ignition quality is assigned a cetane number of 100 and alpha-methyl-napthalene which is hydrocarbon with is poor ignition quality is assigned 0 cetane number.
(ii) The knocking tendency of a fuel in SI engine is generally expressed by its octane number. The percentage, by volume of iso-octane in a mixture of iso-octane and normal heptane.

359. Velocity of flame propagation in the SI engine is maximum for a fuel-air mixture which is

(a) 10% richer than stoichiometric
(b) Equal to stoichiometric
(c) More than 10% richer than stoichiometric
(d) 10% leaner than stoichiometric

Gujarat PSC AE 2019
RPSC Vice Principal ITI 2018, ESE 1999

Ans. (a) : Flame velocity in SI engine combustion is maximum when mixture strength is 10% rich. Because in this condition, the temperature is maximum and combustion is very good resulting maximum combustion products. But at the same time, this is also the situation of detonation.

360. Stoichiometric ratio is :

(a) chemically correct air-fuel ratio by volume
(b) chemically correct air-fuel ratio by weight
(c) theoretical mixture of air for complete combustion
(d) actual ratio of air to fuel for maximum efficiency

RPSC AE 2016
OPSC AEE 2015 PAPER - II, RPSC ACF-2011

Ans. (a) : Stoichiometric ratio is chemically correct air-fuel ratio by volume.

361. Iso-octane content in a fuel for S.I. engines :

(a) Retards auto-ignition
(b) Accelerates auto-ignition
(c) Does not affect auto-ignition
(d) None of the above

OPSC AEE 2019 PAPER - II
HPPSC W.S. Poly. 2016, UKPSC AE 2012 Paper–II

Ans : (a) Iso-octane content in a fuel for S.I. engines retards autoignition.
Iso-octane is a very good anti-knock fuel, therefore it is assigned a rating of 100 octane number. on the other hand, normal heptane has very poor anti-knock qualities, therefore it is given a rating of zero octane number. These two fuels i.e. iso-octane and normal heptane are known as primary reference fuels.

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362. The two reference fuels used for cetane rating are

(a) Cetane and iso-octane
(b) Catane and tetraethyl lead
(c) Cetane and n-heptane
(d) Cetane and α-methyl naphthalene
(e) None of the above

OPSC AEE 2019 PAPER - II
CGPSC Poly. Lect. 22.05.2016
ESE 1995

Ans. (d) : Cetane and α methyl naphthalene are the reference fuel taken for cetane rating, fuel of cetane number 40 has the same ignition quality as a mixture of 40% cetance and 60% alpha-methyl naphthalene as cetance number is counted on cetance percentage.

363. The octane number 100 is assigned to

(a) Iso - octane (b) Normal heptane
(c) Cetane (d) Tetraethyl lead

TNPSC AE 2019
AAI Jr. Executive 29.11.2018
TNPSC AE 2013

Ans. (a) : 2, 2, 4- Trimethylpentane, also known as isooctane or iso-octane, is an organic compound with the formula C_8H_{18}. It is one of several isomers of octane have standard 100 point on the octane rating scale. Higher the octane number, lesser is the tendency to knock during combustion.

364. At flash point temperature:

(a) Fuel ignites without any spark
(b) Fuel emits vapours which produce an inflammable mixture with air
(c) Fuel vaporize sat very high rate
(d) the fuel spontaneously ignites for longer time

GPSC ARTO Pre 30.12.2018
OPSC AEE 2015 PAPER - II

Ans : (b) At flash point temperature fuels emits vapours which produce an inflammable mixture with air.

365. What does the composition of dry gas $CO_2 = 10.4\%$, $O_2 = 9.6\%$, $N_2 = 80\%$ indicates?

(a) Air is just sufficient
(b) Excess air is used
(c) Air is insufficient
(d) Hydrogen is not present in the fuel

OPSC AEE 2019 PAPER - II
ESE 1995

Ans : (b) : In dry flue gases analysis, water vapours are not present. The presence of oxygen and absence of carbon monoxide is present in the gases, it indicates complete combustion i.e. sufficient oxygen.

366. The theoretically correct mixture of air and petrol is

(a) 10 : 1 (b) 15 : 1
(c) 20 : 1 (d) 25 : 1

Nagaland PSC (CTSE) 2018, 2017, 2015 Paper-II
TNPSC AE 2014

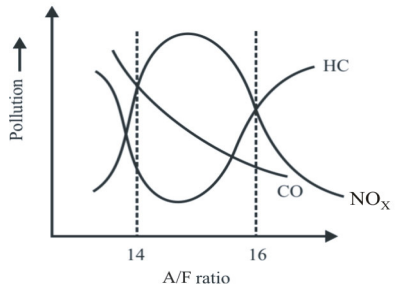
Ans. (b) : The theoretically correct mixture of air and petrol is 15:1.

367. With increase in A/F ratio the level of HCs in engine exhaust gas

(a) Increases
(b) Decreases
(c) Initially increase till chemically corrected ratio than reduces
(d) Initially reduces till chemically corrected ratio than increase

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II

Ans. (d)



368. Scavenging means:

(a) Using fresh air for compressor
(b) To reduce detonation
(c) Using air for throwing burnt gases out of cylinder during exhaust stroke
(d) Using correct air fuel mixture

OPSC AEE 2015 PAPER - II
WBPSC AE 2003

Ans : (c) Scavenging of I.C. Engine:- The process of removing burnt gases form the combustion chamber of the engine cylinder, is known as scavenging.
Types of Scavenging:-
(i) Cross flow scavenging
(ii) Backflow or loop Scavenging
(iii) Uniflow scavenging.

369. Stochiometric air-fuel ratio by volume for combustion of methane in air is :

(a) 15:1 (b) 17.16:1
(c) 9.52:1 (d) 10.58:1
(e) None of the above

CGPSC Poly. Lect. 22.05.2016, ESE 2002

Ans. (c) : For combustion of methane –

$$CH_4 + O_2 \rightarrow CO_2 + H_2O$$
$$16 \quad 32 \quad 44 \quad 18$$
$$1 \quad 2 \quad \frac{44}{16} \quad \frac{18}{16}$$

Stochiometric air-fuel ratio by volume for combustion of methane –

$$\frac{2}{0.21} \Rightarrow 9.52:1$$

370. Which of the following symptoms shows that the combustion is necessarily complete?

(a) Presence of free carbon in exhaust
(b) Presence of CO in exhaust
(c) Presence of Oxygen in exhaust
(d) Presence of Nitrogen in exhaust

RPSC Lect. (Tech. Edu. Dept.) 2011
ESE 2009

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Ans. (c) : The combustion takes place in the presence of air which contain oxygen. If the oxygen is left after complete combustion, then remaining oxygen passes through exhaust. This condition/symptoms shows that the combustion is necessarily complete.

371. Alcohols are unsuitable as diesel engine fuels because

(a) The cetane number of alcohol fuels is very low which prevents their ignition by compression

(b) The cetane number of alcohol fuels is very high which prevents their ignition by compression

(c) The octane number of alcohol fuels is very low which prevents their ignition by compression

(d) None of the above

GPSC ARTO Pre 30.12.2018
GATE 1992

Ans. (a) : In diesel engines, self ignition of the fuel happens due to compression. This is due to lower ignition delay i.e., the time between rise in temperature and commencement of combustion.
This ability for self ignition under compression is better expressed as cetane number. This is where alcohol lacks, having a very low cetane number and hence poor ignition upon compression.

372. The cetane number of automotive diesel fuel used in India lies in which of the following ranges ?

(a) 30-40 (b) 41-50
(c) 51-60 (d) 61-70

UKPSC AE 2012 Paper–II
ESE 2007

Ans. (b) : Cetane number 41-50 octane number 91-93.

373. Which among the following automobile exhaust gas pollutants is a major cause of photochemical smog?

(a) CO (b) CO_2
(c) NO_2 (d) SO_2

ISRO Scientist/Engineer (RAC), 10.03.2019
ESE 2007

Ans. (c) : NO_x are major pollutant which cause photo chemical smog.

374. In order to achieve maximum possible fuel economy the design features which will be affected are

(a) Volumetric efficiency
(b) Compression ratio
(c) Method of charging
(d) Atomization

RPSC AE 2016, RPSC ACF-2011

Ans : (d) In order to achieve maximum possible fuel economy the design features will be affected are atomization.

375. Specific fuel consumption is defined as

(a) Fuel consumption per hour
(b) Fuel consumption per BHP
(c) Fuel consumption per hour per BHP
(d) Fuel consumption per hour per HP

BPSC AE 2012 Paper - V

Ans : (c) : Specific fuel consumption is defined as fuel consumption per hour per BHP

376. A fuel of cetane number 40 has same ignition quality as a mixture of

(a) 40% cetane and 60% alpha methyl naphthalene

(b) 40% alpha methyl napthalene and 60% cetane

(c) 40% petrol and 60% diesel

(d) 40% diesel and 60% petrol

GPSC EE Pre, 28.01.2017

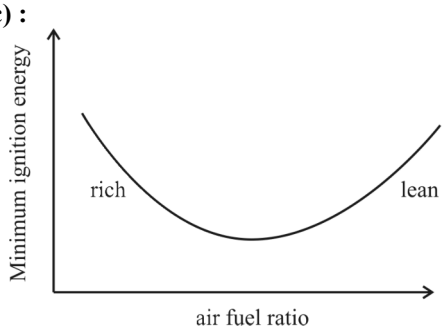
Ans. (a) : A fuel of cetane number 40 has the same ignition quality as a mixture of 40% cetane and 60% alpha methyl naphthalene.

377. The spark energy required for igniting the fuel-air mixture is minimum at :

(a) Equivalence ratio << 1
(b) Equivalence ratio >> 1
(c) Equivalence ratio nearly = 1
(d) Independent of the equivalence ratio

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) :



The equivalence ratio is defined as the ratio of the actual fuel/air ratio to the stoichiometric fuel air ratio (ϕ < 1) it is less than one the combustion is lean with excess air.
From figure minimum spark energy required for igniting the fuel is at equivalence ratio one.

378. Barium based fuel additives in C.I. engines contributes to :

(a) Control of emission
(b) More pollution
(c) Insufficient combustion
(d) Knocking

HPPSC Lect. (Auto) 23.04.2016

Ans. (a) : Barium based fuel additives in CI engine contribute to control of emission.

379. Which of the following is a desired quality of SI engine fuel for reduced knocking?

(a) Higher octane number
(b) Lower octane number
(c) Higher cetane number
(d) Higher Analine point.

RPSC AEN Pre-2013

Ans. (a) : Higher octane number is a desired quality of S.I. engine fuel for reduced knocking. Octane number is also known as octane rating. Octane numbers are based on a scale on iso-octane is 100 (minimal knock) and heptane is 0 (bad knock).

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380. Which is correct statement about reaction time for auto ignition of fuel and the fuel air ratio?
(a) Lean mixture has low reaction time
(b) Rich mixture has high reaction time
(c) Chemically correct mixture has minimum reaction time
(d) All of the above

(HPPSC AE 2014)

Ans : (d) Correct statement about reaction time for auto ignition of fuel and the fuel air ratio.
(i) lean mixture has low reaction time
(ii) Rich mixture has high reaction time
(iii) Chemically correct mixture has minimum reaction time.

381. Tetra ethyl lead in fuel is used to
(a) Increase the lubrications
(b) Decrease octane number
(c) Increase octane number
(d) None of the above

UKPSC AE 2007 Paper -II

Ans. (c) : Increase octane number

382. Fuel injector is used in
(a) steam engines
(b) gas engines
(c) spark ignition engines
(d) compression ignition engines

UKPSC AE 2012 Paper–II

Ans. (d) : compression ignition engines

383. Some qualities of C.I. engine fuels are given below. Select the quality which is not desirable:-
(a) Good ignition quality (b) High volatility
(c) High viscosity (d) Low flash point

UKPSC AE-2013, Paper-II

Ans. (c) : High viscosity of C.I. engine fuels are not desirable.

384. Major constituent of natural gas is:
(a) Ethane (b) Methane
(c) Propane (d) Butane

OPSC AEE 2015 PAPER - II

Ans : (b) Major constituent of natural gas is methane. The natural gas is usually. Found in or near the petroleum fields under the earth's surface. It essentially, consists of mars gas or methane (CH_4). Together with small amounts of other gases such as ethane (C_2H_6), Carbon dioxide (CO_2) and Carbon mono-oxide (CO).

385. Flash point of a volatile material is the lowest temperature at which material
(a) ignites in presence of ignition source
(b) evaporates
(c) ignites without the presence of ignition source
(d) none of the above

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : The flash point of a volatile material is the lowest temperature at which vapour of materials will ignite, when given an ignition source.

386. Auto ignition point of the gasoline lies between the temperature range
(a) 147-200 ºC (b) 247-280 ºC
(c) 347-380 ºC (d) 347-447 ºC

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : Auto ignition point of the gasoline lies between the temperature range 247-280ºC.

387. The correct statement is
(a) Fire point is higher than the flash point
(b) Fire point is lower that the flash point
(c) There is no relation between Fire point and the flash point
(d) Sometimes Fire point is higher and some time is Flash point

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : The fire point is higher than the flash point because at the flash point more vapor may not be produced rapidly enough to sustain combustion.

388. Flammable fuels have
(a) Flash point more the than 37.8ºC
(b) Flash point lesser than 37.8ºC
(c) No temperature limits
(d) only pressure dependency

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : Flammable fuels have a flash point below 37.8ºC (100ºF).

389. The mass of excess air supplied is equal to
(a) $\frac{23}{100} \times$ mass of excess carbon
(b) $\frac{23}{100} \times$ mass of excess oxygen
(c) $\frac{100}{23} \times$ mass of excess carbon
(d) $\frac{100}{23} \times$ mass of excess oxygen

JPSC AE 2013 Paper-V

Ans. (d) : In order to ensure complete and rapid combustion of fuel. Some quantity of air, in excess of the theoretical minimum air, is supplied. We know that in order to supply one kg of oxygen, we need 100/23 of air, similarly, the mass of excess air supplied.
$= \frac{100}{23} \times$ mass of excess oxygen

390. For each mole of oxygen, number of moles of nitrogen required for complete combustion of carbon are
(a) 20/21 (b) 2/21
(c) 77/21 (d) 79/21

TNPSC AE 2017

Ans. (d) : Most commonly, the fuel is a hydrocarbon a compound of hydrogen and carbon. The ideal oxidizer is pure oxygen, but most commonly, it is air. For our work, air will be regarded as a mixture of nitrogen and oxygen, with respective mole fraction of 79% and 21%.
For each mole of oxygen in air, there was 79/21 = 3.76 moles of nitrogen.

391. Which of the following is the indicator of the combustion speed of diesel fuel?
(a) Heptane number (b) Cetane number
(c) Octane number (d) All the above

KPCL AE 2016

Ans. (b) : Centane number (CN) is an indicator of the ignitibility of diesel fuels. It provides information about the ignition delay i.e. the speed of self-ignition of diesel fuel when injected into hot air through the fuel injector.

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392. The probability of knocking in spark ignition engines, the charge away from the spark plug should have :

(a) High self ignition temperature
(b) Low volatility
(c) Low ignition delay
(d) All of the above

Punjab PSC SDE 12.02.2017

Ans. (d) : The probability of knocking in spark ignition engines, the charge away from the spark plug should have–

- high self ignition temperature
- low volatility
- low ignition delay

393. In case of hydrocarbon fuel the stoichiometric fuel air ratio is about :

(a) 1 : 15 (b) 1.12
(c) 12 : 1 (d) 1 : 18

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : In case of hydrocarbon fuel the stoichiometric fuel-air ratio is about 1 : 15

394. Pre-ignition is caused by the spontaneous combustion of the mixture before the end of the compression stroke, and is due to

(a) Cylinder walls being too hot
(b) Overheated spark plug points
(c) Red hot carbon deposits on cylinder walls
(d) Any one of these

Karnataka PSC AE, 10.09.2017

Ans. (d) : Pre-ignition caused by spontaneous combustions of the mixture before the end of compression stroke because of.

- Cylinder walls being too hot.
- Overheated spark plug points.
- Red hot carbon deposits on cylinder walls.

395. Fuel having maximum resistance to detonation is

(a) benzene (b) toluene
(c) iso-octance (d) n-heptane

TNPSC AE, 2008

Ans. (c) : For SI fuel having maximum resistance to detonation is iso-octane. Iso-octane has little tendency towards detonation gasoline with the same knocking characteristics as a mixture of 90% iso-octane and 10% heptane would have an octane rating of 90. Because some fuels are more knock-resistant than iso-octane, the definition has been extended to allow for octane numbers higher than 100.

396. The cetane values for high speed engines vary between

(a) 5 to 11 (b) 15 to 21
(c) 25 to 34 (d) 45 to 54

TNPSC AE, 2008

Ans. (d) : The cetane values for high speed engines vary between 45 to 54.

397. The mass of carbon per kg of flue gas is given by:

(a) $\frac{3}{7}CO_2+\frac{11}{3}CO$ (b) $\frac{7}{3}CO_2+\frac{7}{3}CO$
(c) $\frac{3}{11}CO_2+\frac{7}{3}CO$ (d) $\frac{11}{3}CO_2+\frac{3}{7}CO$

Kerala PSC Poly. Lect., 2017

Ans. (*) : $C + O_2 \rightarrow CO_2$

$12 \quad 32 \quad 44$

$1 \quad \frac{32}{12} \quad \frac{11}{3}$

Hence 1 kg of carbon produces $\frac{11}{3}$ kg of CO_2

$C+\frac{1}{2}O_2 \rightarrow CO$

$12 \quad 8 \quad 28$

$1 \quad \frac{8}{12} \quad \frac{28}{12}$

$1 \quad \frac{4}{3} \quad \frac{7}{3}$

Hence 1 kg of carbon produces $\frac{7}{3}$ kg of CO

Note- Official answer given by commission is (d).

398. Which of the following coal variety has least calorific value?

(a) Steam coal (b) Lignite
(c) Bituminous coal (d) Anthracite

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (b) :

Coal type	**- Calorific value**
1. Lignite	– 4000 kcal/ kg
2. Bituminous	– 5800 kcal/kg – 8000 kcal/kg
3. Anthracite	– 7800 Kcal/kg – 8000 kcal/kg
4. Coal	– 7800 kcal/kg – 8000 kcal /kg

399. What is the octane number of a mixture containing 85% of iso-octane and 15% of normal heptanes?

(a) 15 (b) 42.5
(c) 85 (d) None of these

KPCL AE 2016

Ans. (c) :Octane no.- It is the percentage of iso-octane in the fuel containing iso-octane and normal heptane.

400. What is the octane number of compressed natural gas?

(a) 80 (b) 120
(c) 180 (d) None of these

KPCL AE 2016

Ans. (d) : The octane number of compressed natural gas (CNG) is 130. Natural gas is a smart, low emission and alternative to gasoline and diesel. Pneumatic gasoline has 91 octane number.

401. In a petrol engine, the gas which does not burn and passes out without transformation is

(a) Oxygen (b) Carbon dioxide
(c) Carbon monoxide (d) Nitrogen

Nagaland PSC (CTSE) 2016, Paper-II

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Ans. (d) : In a petrol engine, the gas which does not burn and passes out without transformation is Nitrogen

402. The threshold limit of benzene is

(a) 15 ppm (b) 20 ppm
(c) 25 ppm (d) 30 ppm

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : The threshold limit of benzene is 25 ppm

403. Which of the following is used as antiknock compound in gasoline?

(a) Tetramethyl lead (b) Tetraethyl lead
(c) Trimethyl lead (d) Triethyl lead

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (b) : Organic lead (tetraethyl lead) is used as antiknock in gasoline and jet fuels.

404. Iso octane is

(a) C_8H_{18} (b) $C_{12}H_{26}$
(c) $C_{10}H_{22}$ (d) $C_{15}H_{32}$

WBPSC AE 2003

Ans. (a) : Iso-Octane – C_8H_{18}
The Octane rating of a spark ignition engine fuel is the detonation resistance (anti-knock rating) compared to a mixture of iso-octane.

405. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Semi-bituminous coal	1. Methane and carbon dioxide
B. High speed diesel oil	2. Propane and butane
C. Biogas	3. Calorific value of 45,000 kJ/kg
D. LPG	4. Power plants

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	2	1
(c)	3	4	2	1
(d)	4	3	1	2

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) :

List-I	List-II
A. Semi-bituminous coal	4. Power plants
B. High speed diesel oil	3. Calorific value of 45,000 kJ/kg
C. Biogas	1. Methane and carbon dioxide
D. LPG	2. Propane and butane

406. Octane number is determined by comparing the performance of the petrol with the following hydrocarbons

(a) Iso-octane
(b) Mixture of normal heptanes and iso-octane
(c) Alpha methyl napthalene
(d) Mixture of methane and ethane

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : The octane number is determined by comparing under standard conditions the knock intensity of the fuel with that of blends of two reference fuels.
Iso Octane– Which resists knocking tendency and heptane, which knocks readily.

407. Detonation is caused by the following unstable compounds

(a) Peroxides, aldehydes and ketones
(b) Peroxides, aldehydes, oxides and sulphides
(c) Aldehydes, oxides and ketones
(d) Ketones and sulphur compounds

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) :

408. Combustion of 1 kg of hydrogen needs _____ of oxygen and generates _____ water.

(a) 6 kg, 7 kg (b) 8 kg, 9 kg
(c) 9 kg, 10 kg (d) 4 kg, 5 kg

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : For combustion of hydrogen –

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$$
$$2 \quad 16 \rightarrow 18$$
$$1 \quad 8 \quad 9$$

From above, we can say that for combustion of 1 kg of hydrogen needs 8 kg of oxygen and generates 9 kg of water.

409. When the fuel is burned and water is released in the liquid phase, the heating value of fuel is called –

(a) Higher heating value
(b) Lower heating value
(c) Enthalpy of formation
(d) Latent heat value

TSGENCO AE 14.11.2015

Ans. (a) : The heat of combustion is the amount of energy released when a substance is burned in the presence of oxygen. The heat of combustion also known as heating value, is reported in two forms. The higher or gross, heating values assumed that all of the water produced by combustion is in the liquid phase. The lower or net heating value assumes the water is vapour.

410. A 5 BHP engine running at full load would consume diesel of the order of:

(a) 0.3 kg/hr (b) 1 kg/hr
(c) 3 kg/hr (d) 5 kg/hr

HPPSC Asstt. Prof. 2014

Ans. (b) : 1 kg/hr

411. When the mixture is lean:

(a) Efficiency is less
(b) Power output is less
(c) Maximum temperature and pressure are higher
(d) All of the above

HPPSC ADF 05.03.2019

Ans. (b) When the mixture is lean power output is less.

412. How does cetane number varies with octane number in fuels in IC engines?

(a) No relation
(b) Directly proportional
(c) Inversely proportional
(d) Both are equal

ISRO Scientist/Engineer 07.05.2017

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Ans. (c) : $\text{Cetane Number} \propto \frac{1}{\text{Octane Number}}$

413. 1 kg of carbon produces following percentage of CO_2 in flue gases:
(a) 11% (b) 18%
(c) 23% (d) 29%

SJVN ET 2013

Ans. (d) : Complete combustion equation for carbon –

$$C + O_2 + \frac{79}{21}N_2 \rightarrow CO_2 + \frac{79}{21}N_2$$

$$\% \text{ of } CO_2 = \frac{44}{44 + \frac{79}{21} \times 28} \times 100$$

$$= 29.47\% \simeq 29\%$$

414. The chemically correct A/F ratio by weight for methanol is
(a) 10:1 (b) 8:1
(c) 6.5:1 (d) 9.0:1
(e) 12.0:1

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : The chemically correct A/F ratio by weight for methanol is 6.5 : 1.

415. The range of A/F ratio for SI engine is:
(a) 9-18 (b) 1-9
(c) 18-50 (d) 50-100
(e) 20-30

CGPSC AE 25.02.2018

Ans. (a) • The range of A/F ratio for SI engine is 9-18
• The range of A/F ratio for CI engine is 18:1 to 70:1.

416. The specific gravity of commonly used diesel is
(a) 1 (b) 0.7
(c) 0.85 (d) 0.5

UKPSC AE 2007 Paper -II

Ans. (c) : 0.85

417. Anti knock property of C.I. engine fuel can be improved by adding:-
(a) Tetra – ethyl lead (b) Amylnitrate
(c) Hexadecane (d) Trimethyl pentane

UKPSC AE-2013, Paper-II

Ans. (b) :

418. Disadvantage of hydrogen as a fuel in IC engine:
(a) Storage is easy (b) Low NO_x emission
(c) Detonating tendency (d) Easy handling

Ans : (c) Detonation tendency increase if hydrogen as a fuel in IC Engine.

419. For complete burning of 1 kg of carbon, the air required will be about
(a) 2.67 kg (b) 11.6 kg
(c) 12.7 kg (d) 14.5 kg

UKPSC AE 2012 Paper–II

Ans. (b) : Combustion of 1 kg of carbon –

$$C + O_2 \rightarrow CO_2$$
$$12 \quad 32 \quad 44$$
$$1 \quad \frac{32}{12} \quad \frac{44}{12}$$

The air require for complete combustion –

$$\frac{32}{12 \times 0.23} = 11.6\,\text{kg}$$

420. Vapour lock in a gasoline - fueled internal combustion engine is basically .....
(a) Due to partial stoppage of fuel supply due to vaporization of fuel in the supply
(b) Due to mechanical seizure in the exhaust system
(c) Due to accelerated supply of fuel to the engine
(d) Due to freezing of liquid fuel

RPSC AE 2018

Ans. (a) : Vapour lock in a gasoline - fueled internal combustion engine is basically due to partial stoppage of fuel supply due to vaporization of fuel in the supply.

421. The specification for diesel fuel for India as per BIS is given by
(a) IS:1464 (b) IS:1460
(c) IS:1468 (d) IS:1462
(e) IS:1461

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : The specification for diesel fuel for India as per BIS is given by IS : 1460.

422. The normal operating range of air/fuel ratio for a CI engine with diesel fuel is
(a) 8 : 1 to 12 : 1 (b) 12 : 1 to 22 : 1
(c) 20 : 1 to 30 : 1 (d) 18 : 1 to 70 : 1

BPSC AE Mains 2017 Paper - V

Ans : (d) : 18 : 1 to 70 : 1

423. Consider the following statements:
1. Octane rating of ethanol is more than 100.
2. Cetane number of fuel used in diesel engines in India is in the range of 80 to 90.
3. Biodiesel in mono alkyl ester of vegetable oil.
4. Compressed natural gas is mainly composed of methane.
5. Motor gasoline is a mixture of various hydrocarbons with a major proportion being aromatic hydrocarbons.

Which of these statements are correct?
(a) 1 and 2 (b) 1, 2 and 3
(c) 1, 2, 3 and 4 (d) 1, 3 and 4
(e) 2, 3, 4 and 5

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) :
- Octane rating, ethanol has a rating of 113
- Biodiesel is mono alkyl ester of vegetable Oil.
- CNG is mainly composed of methane. [CH_4 - 84.5%, Ethane [C_2H_6] - 7.7%]
- Propane (C_3H_8) - 2.4%, Butane (C_4H_{10}) - 0.58%, pentane (C_5H_{12}) - 0.37%]

424. CFR engine is used to measure the following quality of diesel
(a) Cold Filter Plugging Point
(b) Cloud Point
(c) Cetane Number
(d) CHNS Ratio
(e) Lubricity

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : CFR Engine - Cooperative Fuel Research [CFR] engine, the source for fuel rating equipment.
CFR engine is used to measure cetane number, quality of diesel and octane number, quality of Petrol and gasoline knock.

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425. Calorific values of gaseous fuels is determined by
(a) Bomb Calorimeter
(b) Junker's Calorimeter
(c) Separating Calorimeter
(d) Throttling Calorimeter

TSPSC AEE 2015

Ans : (b) Calorific values of gaseous fuels is determined by Junker's calorimeter.
Bomb Calorimeter–Bomb Calorimeter used to measure calorific value of solid and liquid fuels i.e.– Coal, gasoline, kerosene etc.
Junker Calorimeter–Junker calorimeter used to find out calorific value of gases fuel.
Throttling and separating calorimeter– Throttling and separating calorimeter are used to measure the dryness fraction of steam.

426. In the Orsat apparatus KOH solution is used to absorb :
(a) Carbon monoxide (b) Carbon dioxide
(c) Oxygen (d) None of the above

(KPSC AE. 2015)

Ans : (b) In the orsat apparatus KOH solution is used to absorb Carbon dioxide (CO_2) and Sulfur dioxide (SO_2)
Orsat Gas Analysis:- Orsat gas analyzer measures the volume or more fraction of CO_2,CO and O_2 in the dry flue gas. An orsat analyzer contains three pipes Containing chemical solutions.
- KOH absorbs CO_2, Pyragallol absorbs O_2 and $CuCl_2$ absorbs CO.

427. Which of the following statement is not correct with respect to alcohol as alternate fuels in I.C. Engine?
(a) Alcohols are corrosive in nature
(b) Alcohol contains about half the heat energy of gasoline
(c) Auto-knock characteristics of alcohol is poor
(d) Alcohol does not vaporize as easily as gasoline

TNPSC AE 2017

Ans. (c) : Alcohol– Alcohols are alternate fuels because they can be obtained from both natural and manufactured source. Methanol and ethanol are two kinds of alcohols that seem most promising.
Advantages-
1. It is high octane fuel, so it can be used efficiently for higher compression ratio.
2. It produce less overall emissions compared to gasoline.
3. It gives higher pressure and more power in the expansion stroke.
4. It has high latent heat of vaporization which results in a cooler intake process. This rise the volumetric efficiency of the engine and reduce the required work in the compression stroke.

Disadvantages-
1. The calorific value (29700) of the fuel is less.
2. With equal thermal efficiency and similar engine output usage, twice as much fuel would have to be purchased.
3. Alcohol is much more corrosive then gasoline on copper, brass, aluminium, rubber and many plastics.
4. It has poor cold weather starting characteristics due to low vapour pressure and evaporation.

428. Lead compound were added in gasoline to
(a) Reduce HC emissions
(b) Reduce knocking
(c) Reduce exhaust temperature
(d) Increase power output

TNPSC AE 2017

Ans. (b) : To reduce knocking, lead compounds were added in gasoline.

429. Hydro carbon emission in CI Engine is mainly due to
(a) Over mixing of fuel and air
(b) Under mixing of fuel and air
(c) Constant mixing of fuel and air
(d) Both (A) and (B)

TNPSC AE 2017

Ans. (d) : Hydro carbon emission in CI Engine is mainly due to over mixing of fuel and air, under mixing of fuel and air.

430. When air is used as oxidiser of a fuel, the nitrogen?
(a) Decrease to the inert level of the fuel
(b) Gets oxidized to NO
(c) Increases the inert content of the fuel
(d) May react with the fuel elements forming nitride

TNPSC AE 2019

Ans. (c) : When air is used as oxidiser of a fuel, the nitrogen increases the inert content of the fuel.

431. A 1 g sample of fuel is burned in a bomb calorimeter containing 1.2 kg of water at an initial temperature of 25°C. After the reaction, the final temperature of the water is 33.2°C. The heat capacity of the calorimeter is 837 J/°C. The specific heat of water is 4.18 J/ °C. The heat released by the fuel will be nearly
(a) 36 kJ/g (b) 42 kJ/g
(c) 48 kJ/g (d) 54 kJ/g

ESE 2019

Ans. (c) : Given,
$m_f = 1$ gm
$m_w = 1.2$ kg
$T_1 = 25°C$
$T_2 = 33.2°C$
Heat capacity of calorimeter (mc) = 837 J/°C
= 0.837 kJ/°C
Specific heat of water is taken as 4.18 kJ/°C
$m_f (c_v)_f = m_w c_p \Delta T + mc \Delta T$
$1 \times (c_v)_f = [1.2 \times 4.18 \times (33.2 - 25)] + [0.837 \times (33.2 - 25)]$
$= 47.995$
$(c_v)_f \simeq 48$ kJ/g

432. Flash point for diesel fuel should be
(a) Minimum 49ºC (b) Maximum 49ºC
(c) Minimum 99ºC (d) Maximum 99ºC

UKPSC AE 2007 Paper -II

Ans. (a) : Minimum 49ºC

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433. Sulphur content in diesel fuel should not be more than
(a) 10% (b) 5%
(c) 1% (d) 0.01%
UKPSC AE 2007 Paper -II

Ans. (c) : 1%

434. In Orsat apparatus, when the percentage of carbon dioxide, oxygen and carbon monoxide are known, the remaining gas is assumed to be
(a) Hydrogen (b) Sulphur dioxide
(c) Nitrogen (d) Air
ESE 2020

Ans. (c) : Orsat apparatus gives dry analysis of flue gases and in the flasks CO_2, O_2 and CO are absorbed.

435. Which property is related to Alcohol?
(a) Lower vapour pressure
(b) Higher cetane number
(c) Smaller octane number
(d) Good lubricating quality
OPSC AEE 2019 PAPER - II

Ans : (a) : Lower vapour pressure property is related to Alcohol.

436. Which property is not related with LPG as fuel for automotive applications?
(a) Lower knock resistance
(b) Lower crank case dilution
(c) Lower volumetric efficiency
(d) Lesser residue and oil contamination
OPSC AEE 2019 PAPER - II

Ans : (a) :

437. High API gravity of fuel means:
(a) Lower Diesel Index
(b) Lower Brake specific fuel consumption
(c) Lower heat of combustion
(d) Lower ignition quality
OPSC AEE 2019 PAPER - II

Ans : (b) : API Gravity– The American Petroleum Institute gravity, or API gravity, is measure of how heavy or light a petroleum liquid is compared to water. High API gravity of fuel means lower brake specific fuel consumption.

438. For same output the specific fuel consumption of petrol engine as compared to diesel engine is:
(a) Higher (b) Lower
(c) Same (d) None of these
OPSC AEE 2019 PAPER - II

Ans : (a) : The specific fuel consumption per B.P. hour for a petrol engine is about 0.25 kg. The specific fuel consumption per B.P. hour for a diesel engine is about 0.2 kg.

439. The problem of engine that can be reduces with the rise in front end volatility of fuel:
(a) Cold starting (b) Vapour lock
(c) Hot starting (d) Spark plug fouling
OPSC AEE 2019 PAPER - II

Ans. : (a) : The problem of engine that can be reduces with the rise in front end volatility is cold starting.

440. Which is not the correct option for fuel injection system of petrol engine?
(a) Increase in power
(b) Improve distribution of the mixture
(c) Reduce volumetric efficiency
(d) Reduce mixture temperature in the cylinder
OPSC AEE 2019 PAPER - II

Ans. : (c) : Reduce volumetric efficiency

441. During distillation process which fuel vaporise in the end?
(a) Kerosene (b) Aviation fuel
(c) Diesel (d) Petrol
OPSC AEE 2019 PAPER - II

Ans : (c) : During distillation process diesel fuel vaporise in the end.

442. Which one of the following fuels can be obtained by fermentation of vegetable matter?
(a) Benzene (b) Diesel
(c) Gasoline (d) Alcohol
Gujarat PSC AE 2019

Ans : (d) : Alcohol.

443. The preference of diesel engines for road transport is due to
(a) Easy starting
(b) Low specific fuel consumption over a large range of load
(c) Complete combustion of charge
(d) Harmless exhaust
Gujarat PSC AE 2019

Ans : (b) : The preference of diesel engine for road transport is due to low specific fuel consumption over a large range of load.

444. The specific fuel consumption per kW-hr for diesel engine is very close to
(a) 0.4 kg/kW-hr (b) 0.33 kg/kW-hr
(c) 0.27 kg/kW-hr (d) 0.47 kg/kW-hr
BPSC AE Mains 2017 Paper - V

Ans : (c) : 0.27 kg/kW-hr

445. Which of the following is the lightest and most volatile liquid fuel?
(a) Diesel (b) Kerosene
(c) Fuel oil (d) Gasoline
BPSC AE 2012 Paper - V

Ans : (d) : Gasoline is highly volatile due to its low flash point and easily vaporize when exposed to air.

446. Pour point of fuel oil is the
(a) Minimum temperature to which oil is heated in order to give off inflammable vapours in sufficient quantity to ignite momentarily when brought in contact with a flame
(b) Temperature at which it solidifies or congeals
(c) It catches fire without external aid
(d) Temperature at which it flows easily
BPSC AE 2012 Paper - V

Ans : (b) : Pour point of fuel oil is the temperature at which it solidifies or congeals. Pour point is the temperature at which a liquid becomes semisolid and loses its flowing characteristics. As the paraffin content is high in crude oil, thus it has a high pour point.

447. Which of the following fuels detonates readily?
(a) Benzene (b) Iso-octane
(c) n-heptane (d) Alcohol
BPSC AE 2012 Paper - V

Ans : (c) : Normal heptane fuel detonates readily. Octane number equates to the anti-knock qualities of a compressible mixture of heptane and iso-octane which is expressed as the percentage of isooctane in heptane and is listed on pumps for gasoline (petrol) dispensed globally.

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448. The commonly used ignition accelerator is
(a) Acetone peroxide
(b) Ethyl nitrate
(c) Isoamyl nitrate
(d) Any one of the above

BPSC AE 2012 Paper - V

Ans : (d) : The commonly used ignition accelerator is-
- Acetone peroxide
- Ethyl nitrate
- Isoamyl nitrate

449. The missing equality per stroke is equal to
(a) Cylinder feed – indicated mass of steam
(b) Cylinder feed + indicated mass of steam
(c) Mass of cushion steam + indicated mass of steam
(d) Mass of cushion steam – cylinder feed

JPSC AE 2013 Paper-V

Ans. (a) : Cylinder feed – indicated mass of steam

450. The temperature of the products of combustion when the maximum amount of chemical energy is converted to thermal energy is
(a) Higher than adiabatic flame temperature
(b) Lower than adiabatic flame temperature
(c) Equal to adiabatic flame temperature
(d) Independent of adiabatic flame temperature

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Maximum amount of chemical energy is converted to thermal energy when the air fuel ratio is stoichiometric. At that ratio, the adiabatic flame temperature is equal to the temperature of the product of combustion. When the air fuel ratio is less or more the temperature of products of combustion is always less than the adiabatic flame temperature.

451. LPG mainly consists of:
(a) Methane and ethane
(b) Propane and butane
(c) Petrol and diesel
(d) Hexane and butadiene

RPSC ADE 2016

Ans. (b) : Propane and butane, LPG mainly consists of propane and butane
CH_4 = methane
C_3H_8 = propane
C_4H_{10} = Butane
C_2H_6 = ethane

452. The fuel injection pressure in a diesel engine is in the range of :
(a) 20 to 25 atm (b) 30 to 40 atm
(c) 60 to 70 atm (d) 100 to 170 atm

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Injection pressure during the whole process should be above 1000–1200 bar for a good spray formation and air-fuel mixture, a tendency in practice to 1600–1800 bar and higher is noted. Total nozzle area should be as small as possible in relation to cylinder diameter for good combustion particularly at part load.

453. The freezing temperature of petrol is nearly
(a) 1°C (b) 10°C
(c) –10°C (d) below 30°C

GPSC ARTO 01.05.2016

Ans. (d) : Without knowing the composition of the particular fuels it's not possible to give a very precise answer. However petrol has much power freezing point that diesel fuel. The range for petrol freezes b/w –40°C to –60°C.

454. The type of pump used for fuel injection in C.I. engine is
(a) Plunger pump (b) Gear pump
(c) Vane pump (d) Centrifugal pump

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : The pump is used for fuel injection in CI engine is plunger pump
An injection pump is a device that pumps fuel into the cylinders of a diesel engine.

455. Which of the following symptoms shows that the combustion in air is necessarily complete?
(a) Absence of oxygen in exhaust
(b) Absence of nitrogen in exhaust
(c) Absence of free carbon in exhaust
(d) Absence of carbon monoxide in exhaust

HPPSC Asstt. Prof. 18.09.2017

Ans. (d) : Absence of carbon monoxide in exhaust

456. The percentage of energy in the petrol burnt in the engine which is actually utilized in propelling the vehicle is :
(a) 40% (b) 60%
(c) 25% (d) 20%

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c)

457. Which of the following gases is used as a fuel in internal combustion engine?
(a) Liquefied petroleum gas
(b) Blast furnace gas
(c) Coke even gas
(d) All of the above

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) :
- Liquefied petroleum gas
- Blast furnace gas
- Coke even gas

458. A mixture of theoretical air and fuel for complete combustion of fuel is called :
(a) Rich mixture
(b) Lean mixture
(c) Air-Fuel mixture
(d) Stoichiometric mixture

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) : A mixture of theoretical air and fuel for complete combustion of fuel is called stoichiometric mixture.

459. The process of breaking up of liquid fuel into droplets of fine spray is known as :
(a) Vaporization (b) Atomization
(c) Carburetion (d) Injection

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (b) : The process of breaking up of liquid fuel into droplets of fine spray is known as atomization.

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460. The heat energy released by a fuel is measured by :

(a) Energy meter (b) Thermometer
(c) Calorimeter (d) Anemometer

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c) :

	Devices	Used to Measure
1.	Energy meter	Amount of electric energy consumed by building etc.
2.	Thermometer	Temperature
3.	Calorimeter	Calorific valve (heat energy)
4.	Anemometer	Wind speed

The heat energy released by a fuel is measured by calorimeter.

461. The anti knock property of the fuel depends on its:

(a) Molecular structure
(b) Self ignition temperature
(c) Chemical composition
(d) All of the above

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) : • Molecular structure
• Self ignition temperature
• Chemical composition

462. A coal containing high amount of volatile matter will have

(a) high ignition temperature
(b) low ignition temperature
(c) high ash fusion temperature
(d) very low ash content

APPSC AEE 2012

Ans. (b) : A coal containing high amount of volatile matter will have low ignition temperature.

463. HUCR is the highest compression ratio at which the

(a) Engine can be run
(b) Engine gives maximum power output
(c) Engine is most efficient
(d) Fuel can be used in a test engine without knocking

APPSC AEE 2012

Ans. (d) : HUCR (Highest Useful Compression Ratio) is the highest compression ratio at which the fuel can be used in a test engine without knocking.

464. Which of the following air-fuel ratio indicate that the mixture is rich?

(a) 10:1 (b) 14.7: 1
(c) 15:1 (d) 20:1

GPSC AE 26.07.2015

Ans. (a) : When the air-fuel ratio in mixture is less than stiochiometric ratio, then fuel is rich mixture.
We know, 14:7 is stiochiometric ratio, ratio lesser than this i.e., 10:1 is showing richer air-fuel ratio.

465. Carbon monoxide formation in internal combustion engine is due to

(a) insufficiency of air
(b) insufficiency of oxygen
(c) insufficiency of air and temperature
(d) insufficiency of oxygen and temperature

TANGEDCO AE 2015

Ans. (a) : Incomplete combustion of fuel is due to insufficient air and carbon monoxide which takes place in internal combustion engine.

466. Petrol is distilled at

(a) 65º to 220ºC (b) 220º to 345ºC
(c) 345º to 470ºC (d) 470º to 550ºC

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : Petrol is distilled at a temperature in range of 65–220ºC While kerosene is distilled at 220–350ºC.

467. The normal heptane (C_7H_{16}) is given a rating of ..... octane number

(a) 0 (b) 50
(c) 100 (d) 120

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : The normal heptanes (C_7H_{16}) is given a rating of 0 (zero) octane number, n-heptanes is defined as the zero point of has an octane rating scale 2,2,4 – Trimethylpentane has an octane rating. 100, whereas n-heptane has an octane rating of zero. The octane number is based on an arbitrary scale which is a measure of its ability to resist knocking as it burns on combustion.

468. One kg of carbon mono oxide requires how much quantity of oxygen to produce 11/7 kg of carbon dioxide gas?

(a) 4/7 kg (b) 11/4 kg
(c) 9/7 kg (d) None of the above

ISRO Scientist/Engineer 22.04.2018

Ans. (a) $CO + \frac{1}{2}O_2 \rightarrow CO_2$

So,

$\frac{1}{2}$ mole of O_2 would make 1 mole CO_2

So, 16 kg O_2 would make 44 kg CO_2

Then $\frac{16}{44}$ kg O_2 would make 1 kg CO_2

Therefore,

$$\frac{11}{7} \text{ kg } CO_2 \text{ make by } \frac{11}{7} \times \frac{16}{44} \text{ kg } O_2 = \frac{4}{7} \text{ kg } O_2$$

469. The octane number of petrol generally available is

(a) 20 to 40 (b) 40 to 60
(c) 60 to 80 (d) 80 to 100

RPSC IOF, 2020

Ans. (d) : The octane number of petrol generally available is 80 to 100.

4. Supercharging

470. In I.C. engine, Supercharging is used for:

(a) supplying compressed air to remove combustion products
(b) providing forced cooling to avoid excessive heating
(c) supplying air to the engine at higher density to increase its mass
(d) Injecting excess fuel for more load

ISRO Scientist/Engineer (RAC), 10.03.2019

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GPSC Executive Engineer 23.12.2018
APPSC AEE MAINS 2016 PAPER-III
GPSC Asstt. Prof. 28.08.2016
CGPSC AE 2014 II

Ans. (c) : Supplying air to the engine at higher density to increases its mass.

471. Which type of compressor mostly used in supercharger of internal combustion engines?
(a) Roots blower
(b) Axial flow compressor
(c) Radial flow compressor
(d) Reciprocating compressor

Kerala PSC Poly. Lect., 2017
Punjab PSC SDE 12.02.2017
J&K PSC Civil Services Pre, 2010
ESE 1996

Ans. (c) : A compressor mostly used for supercharging of I.C. engines is radial flow compressor (centrifugal compressor).

472. Purpose of supercharging an engine is to
(a) Increase power output of engine
(b) Reduce specific fuel consumption
(c) Reduce noise of engine
(d) Improve cooling of cylinders

RPSC AE 2016, RPSC ACF-2011
RPSC Lect. (Tech. Edu. Dept.) 2011
ESE 2009

Ans. : (a) Following are the objects of supercharging the engines
(i) To reduce mass of the engine per brake power.
(ii) To maintain power of aircraft engines at high Altitudes where less oxygen is available for combustion.
(iii) To reduce space occupied by the engine.
(iv) To reduce the consumption of lubricating oil
(v) To increase the power output of an engine when greater power is required.

473. The objective of supercharging the engine is
(a) To reduce mass of the engine per brake power
(b) To reduce space occupied by the engine
(c) To increase the power output of an engine when greater power is required
(d) All of the above

Sikkim PSC (Under Secretary), 2017
ESE 2007

Ans. (d) : The objective of supercharging is to increase the charge density of the working medium (air or air fuel mixture) by any means and with the help of a suitable system, before it enters the work cylinder i.e. to precompress the charge. In doing so the temperature of the working medium should not be marked raised, since this would influence the temperature profile of high pressure work cycle.
Supercharging process increases power output by increase the inlet pressure of the air in the C.I. engine during suction stroke. It does not response to the size and shape of engine.

474. The operation of forcing additional air under pressure, into the engine cylinder is known as
(a) Carburation (b) Supercharging
(c) Dissociation (d) Turbulence

MECON MT 2019, TNPSC AE, 2008

Ans. (b) : Supercharging– The operation of forcing additional air under pressure, into the engine cylinder is known as supercharging.

475. How can the output of a diesel engine be improved without increasing the engine revolution?
(a) Supercharging
(b) Increase flywheel size
(c) Scavenging
(d) Feeding more fuel

Rajasthan Nagar Nigam AE 2016, Shift-I
HPPSC Asstt. Prof. 2014

Ans. (a) : Supercharging – To increase the power output of the engine, following can be done–
- Increase the air inducted per unit time to burn more fuel.
- Improve the degree of utilization of air.
- Improve the thermal efficiency.
- Increasing the density of air to increase the power output of the engine is called supercharging.
- In CI engine, supercharging reduces detonation tendency. So it is always useful for CI engine i.e. more power, more economy and less detonation. The limit of supercharging is governed by thermal loading due to very high temperature.

476. Supercharging is essential in
(a) Diesel engines (b) Gas turbines
(c) Petrol engines (d) Aircraft engines

TNPSC AE, 2008

Ans. (d) : Supercharging is essential in aircraft engines. The operation of forcing additional air under pressure, into the engine cylinder is known as supercharging.

477. Supercharging increases the power output of the engine by:
(a) Increasing the charge temperature
(b) Increasing the charge pressure
(c) Increasing the speed of engine.
(d) Quality of fuel admitted.

RPSC ACF & FRO, 26.02.2021

Ans. (b) : A super charger is an air compressor that increases the pressure or density of air supplied to an internal combustion engine. This gives each intake cycle of the engine more oxygen, letting it burn more fuel and do more work, thus increasing power.

478. In a turbocharger, the exhaust gases
(a) Run the centrifugal pump rotor
(b) Heat the incoming air
(c) Run the turbine rotor
(d) Both (a) and (b) above

WBPSC AE, 2007

Ans. (d) :

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479. …. turbo charging is based upon the idea of increasing the expansion ratio relative to compression ratio by means of early closure of inlet valve as the boost pressure is increased.

(a) Pulse turbocharger
(b) Hyper bar turbocharger
(c) Two-stage turbocharger
(d) Miller turbocharger

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : In miller turbocharger, turbo charging is based upon the idea of increasing the expansion ratio relative to compression ratio by means of early closure of inlet valve as the boost pressure is increased.

480. Which of the following is used to given more oxygen in each intake cycle of the engine, letting it burn more fuel and do more work

(a) Additional Oxygen cylinder
(b) Supercharger
(c) Extra cam on the intake valve
(d) None of these

KPCL AE 2016

Ans. (b) : Supercharger is used to given more oxygen in each intake cycle of the engine, letting it burn more fuel and do more work.

481. Which of the following is true about compressors used as supercharger

(a) Positive displacement type
(b) Axial flow type
(c) Centrifugal type
(d) All of the above

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (d) : The apparatus used to increase the air density is known as a supercharger. It is merely a compressor, which provides a denser charge to the engine, there by enabling the consumption of a greater mass of charge with the same total piston displacement.

482. By supercharging

(a) Power stroke becomes stronger
(b) Loss in exhaust is reduced
(c) Engine can be made to run smoother
(d) Thermal efficiency of the engine can be improved

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : It is the process of increasing the mass (density) of the air fuel mixture (in the S.I. engines) or air (in C.I engines) induced into the engine cylinder. This is usually close with the help of a compressor or blower known as supercharger

(i) Supercharging increases the power developed by the engine
(ii) To maintain the power of aircraft engine at high a altitudes where less oxygen is available for combustion
(iii) To reduce mains of the engine per brake power.
(iv) To reduce space occupied by the engines.
(v) To reduce consumption of lubricating oil.
(vi) To increase the power output of an engine when greater power is required.

483. For the same compression ratio and the same engine speed, supercharging I.C. engine leads to

(a) Increase in mechanical efficiency
(b) Better gas exhaust process
(c) Higher thermal loading
(d) All of the above
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : Supercharging is the general term for an air compressor to increase the pressure or density of air entering an engine, caused increase in mechanical efficiency, better exhaust & higher thermal loading.

484. What is the main objective of supercharging of the engine?

(a) To reduce the mass of the engine per brake power
(b) To reduce space occupied by engine
(c) To increase the power output of engine
(d) All of the above
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (d) : The objective of supercharging the engine is–

– To reduce mass of the engine per brake power.
– To reduce space occupied by the engine.
– To increase the power output of an engine when greater power is required.

485. The process of supercharging is meant for

(a) Raising exhaust pressure
(b) Increasing density of intake air
(c) Increasing quantity of fuel going into cylinder
(d) Providing air for cooling

UKPSC AE 2012 Paper–II

Ans. (b) : Increasing density of intake air

486. Supercharging is usually not suggested for a petrol engine due to:

(a) Decrease in volumetric efficiency
(b) Increase in specific fuel consumption
(c) Increase in power loss
(d) Increase in knocking

OPSC AEE 2019 PAPER - II

Ans. : (d) : The degree of supercharging in S.I. engines is limited by the knock. The increase in pressure and temperature reduces ignition delay and consequently increases the knocking tendency.

487. The compressor of turbocharger is driven

(a) By mechanical power of engine
(b) Manually
(c) By gas turbine mounted in the exhaust flow of engine
(d) By separate electrical motor driven by battery

SJVN ET 2019

Ans. (c) : Turbocharger increases the amount of air entering the engines to create more power. A turbocharger has the compressor powered by a turbine which is driven by exhaust gas from engines.

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488. The purpose of supercharging is to
(a) Increase detonation
(b) Increase the volumetric efficiency
(c) Decrease the volumetric efficiency
(d) Decrease mechanical efficiency

TNPSC AE 2013

Ans. (b) : Advantage of supercharging–
1. High power output.
2. Increase the volumetric efficiency.
3. Better mixing of air and fuel.
4. Better atomization of fuel.
5. Quicker acceleration of vehicle.

489. The process of increasing the mass of charge introduced into the same volume of an engine is called
(a) Supercharging (b) Scavenging
(c) Idling (d) Tumbling

TNPSC AE 2014

Ans. (a) : The process of increasing the mass of charge introduced into the same volume of an engine is called supercharging.

490. Which of the following is not the supercharge?
(a) Root type supercharge
(b) Vane type supercharge
(c) Centrifugal type supercharge
(d) Reciprocating supercharge
(e) Screw type supercharge

CGPSC AE 2014- II

Ans. (d) : Reciprocating supercharge is not the supercharge because of high pressure and low discharge of delivery air.

491. Turbo-charger is used
(a) In gas turbines for compression of air
(b) For supercharging diesel engine
(c) In jet propulsion units
(d) In rockets for producing air fuel mixture

RPSC AE 2016

Ans : (b) Turbo-charger is used for supercharging diesel engine.

492. Supercharging for CI engines is preferred because
(a) It decreases the knocking tendency
(b) It leads to reduction in smoke
(c) It improves combustion
(d) All of these

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : Supercharging for C.I. engine is preferred because
(I) It decrease the knocking tendency
(II) It lead to reduction in smoke
(III) It improves combustion
Explanation :
Super charging is the process of increasing the mass (or in other words density) of the air fuel mixture, which responsible to improvement in combustion and reduction of smoke and achieve higher power output.

493. What are the types of supercharger?
(a) Gear and Vane type
(b) Root and Reciprocating type
(c) Reciprocating and Centrifugal type
(d) Gear, Vane and Reciprocating type
(e) Centrifugal, Root and Vane type

CGPSC ARTO & TSI 07.01.2017

Ans. (e) : There are three main types of supercharger,
1. Root- type
2. Centrifugal
3. Twin Screw

494. Which of the following is not an objective of super charging?
(a) To increase the compression ratio
(b) To overcome power losses at high altitudes
(c) To increase the power output
(d) To reduce the weight to power ratio

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (a) : To increase the compression ratio is not an objective of super charging.

495. For the same power developed in I.C. engines, the cheaper system is ..... .
(a) naturally aspirated (b) supercharger
(c) centrifugal pump (d) turbo charger

GWSSB DEE 07.07.2016

Ans. (b) : For the same power developed in I.C. engines, the cheaper system is supercharger.

496. The operation of forcing additional air under pressure into the engine cylinder is known as
(a) supercharging (b) carburetion
(c) turbulence (d) delay period

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : The operation of forcing additional air under pressure into the engine cylinder is known as supercharging.

497. By supercharging the diesel engine, the possibility of knocking
(a) Decreases (b) Increases
(c) Remains constant (d) Does not change

Assam PSC AE (PHED), 18.10.2020

Ans. (a) : If the delay period is long, a large amount of fuel will be injected and accumulated in the chamber.
The auto ignition of this large amount of fuel may cause high rate of pressure rise and high maximum pressure which may cause knocking.
Super charging is the process of increasing the density of incoming air/charge in the cylinder by increasing the pressure to increase the power output of the engine.
⇒ Super charging is not preferred in petrol engine (SIE), because by increasing pressure of incoming charge the chances of knocking increases in SIE.
⇒ Supercharging is generally preferred in CIE (diesel engine) In fact by increasing pressure of incoming air, chances of knocking is decreases.

498. In SI engines, supercharging leads to :
(a) Increase in BHP and decrease in thermal efficiency
(b) Increase in thermal efficiency and decrease in BHP
(c) Increase in both thermal efficiency and BHP
(d) None of the above

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (a) : Basically supercharging means supplying high pressure or high density air to the engine. When the high pressure air is supplied to the engines, the mass flow rate of fuel also increases results increase in BHP and decrease in thermal efficiency.

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499. The volumetric efficiency of super charged engine is

(a) Between 100–110% (b) Between 90–100%
(c) Between 80–90% (d) Between 70–80%

Assam PSC AE (PHED), 18.10.2020

Ans. (a) : The volumetric efficiency of super charged engine is between 100-110%

500. In supercharged IC engines, the volumetric efficiency is

(a) 80% (b) 90%
(c) 100% (d) More than 100%

APGENCO AE, 2017

Ans. (d) : Superchargers are used to improve volumetric efficiency more than 100%

501. A turbo charger in a diesel engine is for :

(a) Fuel economy
(b) Avoiding knock
(c) Increased efficiency
(d) Increased power output for a given size of the engine

TNPSC AE (Industries) 09.06.2013

Ans. (d) : A turbo charger in a diesel engine is for increased power output for a given size of the engine.

502. Which are fuel cells? 1. Methane 2. Phosphoric acid fuel cell 3. Solid Oxide fuel cell

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) All 1, 2 and 3

RPSC VPITI 14.02.2016

Ans. (d) : Methane
Phosphoric acid fuel cell
Solid oxide fuel cell
All are fuel cell so option (d) is correct.

5. IC Engine performance and Testing

503. A two stroke engine has a speed of 750 rpm. A four stroke engine having an identical cylinder size running at 1500 rpm. The theoretical output of the two stroke engine will be :

(a) Twice that of the four stroke engine.
(b) Half that of the four stroke engine.
(c) The same as that of the four stroke engine
(d) Depend upon whether it is a CI or SI engine.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
(KPSC AE. 2015)
Nagaland PSC CTSE 2015 Paper-II
Kerala PSC AE 06.08.2015, ESE 1997

Ans. (c) : given,
Two stroke engine-
Speed 750 rpm
Four stroke engine –
Speed 1500 rpm
$\because$ indicated power of two stroke engine

$$\left[IP = \frac{PLANK}{60}\right] \text{ (kW)}$$

P = mean effective pressure, N/m^2
L = length of stroke, m
A = area of cross-section of the cylinder, sqm (m^2)
N = rpm, k = No. of cylinder

$N = \frac{N}{2}$ for four stroke engine, N for two stroke engine.

Now, $IP = \frac{PLANK}{60}$ (2Stroke)

$= \frac{PLAK}{60} \times (750)$ ....(i)

$IP = \frac{PLAK}{60} \times \frac{N}{2}$ (4Stroke)

$= \frac{PLAK}{60} \times \frac{1500}{2}$

$= \frac{PLAK}{60} \times 750$ ...(ii)

$\because$ equation (i) & (ii) are equal, if all the parameters are constant and speed is 750 & 1500 for two stroke & 4-stroke engine respectively.

504. The ratio of brake power to indicated power of an IC engine is called :

(a) Mechanical efficiency
(b) Thermal efficiency
(c) Volumetric efficiency
(d) Relative efficiency

Nagaland PSC CTSE 2017 Paper-2
Rajasthan AE (Nagar Nigam) 2016 Shift-3
RPSC AE 2016, TSPSC AEE 2015
UKPSC AE 2012, 2007 Paper -II
RPSC ACF-2011

Ans. (a) : The ratio of brake power to indicated power of an I.C. engine is called mechanical efficiency.

$$\eta_{mechanical} = \frac{B.P.}{I.P.}$$

505. Morse test is applicable only to -

(a) Single cylinder SI engines
(b) Single cylinder CI engines
(c) Multi-cylinder CI engines
(d) Single and multi-cylinder SI and CI engines

RPSC ACF & FRO, 26.02.2021
Harayana PSC AE (PHED) 04.09.2020, Paper-I
KPCL AE 2016, GPSC Lect. (Auto) 16.10.2016
WBPSC AE 2003

Ans. (c) : Morse test can be conducted for multi cylinder engines for SI and CI engine both. The purpose of this test is to obtain the approximate indicated power of a multi cylinder engine.

506. In a four stroke cycle engine, the four operations suction, compression, expansion and exhaust are completed in the number of revolutions of crankshaft equal to :

(a) 1 (b) 2
(c) 3 (d) 4

Nagaland PSC CTSE 2017, 2016 Paper-2
GWSSB AAE, 27.12.2015
TNPSC ACF 2012, RPSC ACF-2011

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Ans. (b) : In a four stroke engine, the engine requires four stroke of the piston (intake, compression, power, exhaust) to complete one cycle. Hence, it requires two rotations of crankshaft to complete one cycle.

507. In a four stroke cycle. S.I. engine the camshaft runs at

(a) same speed as crankshaft
(b) half the speed of crankshaft
(c) twice the speed of crankshaft
(d) any speed irrespective of crankshaft speed

Nagaland PSC (CTSE) 2018, Paper-II
TNPSC AE 2017, RPSC AE 2016
RPSC ACF-2011

Ans : (b) In a four stroke cycle SI engine the camshaft runs at half the speed of crankshaft.

508. In a variable speed SI engine, the maximum torque occurs at the maximum

(a) Speed
(b) Brake power
(c) Indicated power
(d) Volumetric efficiency
(e) None of the above

GPSC ARTO Pre 30.12.2018
CGPSC Poly. Lect. 22.05.2016
UKPSC AE 2012 Paper–II
ESE 1999

Ans. (b) : In a variable speed SI engine, the maximum torque occurs at the maximum brake power.

$$\boxed{BP = T \times \omega}$$

509. Brake Horse Power (BHP) of a diesel engine can be increased by :

(a) Increasing the pressure of intake air
(b) Increasing the temperature of intake air
(c) Decreasing the density of intake air
(d) Increasing temperature and decreasing density of intake air.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
APGENCO AE, 2017
Karnataka PSC AE, 10.09.2017
CGPSC Asstt. Workshop Supt., 17.07.2016
WBPSC AE 2003, GATE 1991

Ans. (a) : The BHP of diesel can be increased mostly by burning more fuel and that can be achieved by increasing the density of intake air and increasing pressure of intake air.

510. In I.C. Engines power developed inside the cylinder is called:

(or)

The power actually developed by the engine cylinder of an I.C. engine is known as

(a) Brake horse power
(b) Pumping power
(c) Indicated horse power
(d) None of these

JPSC AE, 10.04.2021, Paper-I
ISRO Scientist/Engineer (RAC) 07.05.2017
GPSC Asstt. Prof. 28.08.2016
GPSC Lect. 23.10.2016

Ans. (c) : Indicated power developed in the cylinder of an engine as calculated from the average pressure of the working fluid, piston area, the stroke, and no. of working strokes per minute. It is the theoretical power of reciprocating engine, converting gas (fuel) into energy.

511. The method of determination of indicated power of multi cylinder SI engine is by the use of :

(a) Morse test (b) Prony brake test
(c) Motoring test (d) Heat balance test

CGPSC Poly. Lect. 22.05.2016
UJVNL AE 2016, MPPSC AE 2016
ESE 1995

Ans : (a) The method of determination of indicated power of multicylinder SI engine is by the use of morse test.
The mores test is adopted to find the indicated power of each cylinder of a high speed I.C. engine without using Indicator diagram.

512. A 40 kW engine has a mechanical efficiency of 80%. If the frictional power is assumed to be constant with load what is the approximate value of the mechanical efficiency at 50% of the rated load?

(a) 55% (b) 45%
(c) 65% (d) 75%

GPSC ARTO Pre 30.12.2018
UPRVUNL AE 21.08.2016, ESE 2005

Ans. : (c) Brake Power = 40 kW

$$\eta_{mech} = 0.8 = \frac{\text{Brake power}}{\text{Indicated power}}$$

$$0.8 = \frac{40}{IP}$$

IP = 50 kW

Frictional power = Indicated power – Brake power

FP = 10 kW

At 50% of the rated load

BP = 20 kW

IP = BP + FP = 30 kW

$$\eta_{mech} = \frac{20}{30} \times 100 = 66.6\% = 65\%$$

513. The ratio of the indicated thermal efficiency to the air standard efficiency is known as

(a) Mechanical efficiency (b) Overall efficiency
(c) Volumetric efficiency (d) Relative efficiency

RPSC IOF, 2020, CIL MT 27.02.2020
PTCUL AE 25.06.2017

Ans. (d) : Relative efficiency- It is also known as efficiency ratio. The relative efficiency of an I.C. engine is the ratio of the indicated thermal efficiency to the air standard efficiency.

Mechanical efficiency- Ratio of brake power to the indicated power.

Overall efficiency- It is the ratio of the work obtained at the crank shaft in a given time to the energy supplied by the fuel during the same time.

Volumetric efficiency- It is the ratio of the actual volume of charge admitted during the suction stroke at N.T.P. to the swept volume of the piston.

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514. In a Morse test, for a 2 cylinder, 2 stroke spark ignition engine, the brake power was 9 kW whereas the brake power of individual cylinders with spark cut off were 4.25 kW and 3.75 kW respectively. The mechanical efficiency of the engine is

(a) 90% (b) 80%
(c) 45.5% (d) 52.5%

RPSC IOF, 2020
GPSC ARTO Pre 30.12.2018
ESE 2017, 1999

Ans. (a) : Indicated power of second cylinder is
9 – 4.25 = 4.75 kW
First engine IP = 9 – 3.75 = 5.25 kW
Total indicated power of engine
Total IP = 4.75 + 5.25 = 10 kW
Mechanical efficiency of engine

$$\eta_{mec} = \frac{9}{10} \times 100$$

$$\eta_{mech} = 90\%$$

515. A turbo-charged four-stroke direct injection diesel engine has a displacement volume of 0.0259 m^3 (25.9 litres). The engine has an output of 950 kW at 2200 rpm. The mean effective pressure in MPa is closest to

(a) 2 (b) 1
(c) 0.2 (d) 0.1

HPPSC ADF 05.03.2019
VIZAG STEEL MT 10.06.2012
GATE 2010

Ans. (a) : $BP = \frac{n.P_m.VN.k}{60}$
Where, P_m = mean effective pressure
n = number as cylinder = 1
V = Displace volume = 0.2510 cm^3
N = r.p.m
K = 1/2 for 4-stroke
= 1 for 2-stroke engine

$$\therefore P_m = \frac{B.P \times 60}{n.V.N.K.}$$

$$= \frac{950 \times 60}{0.0259 \times 2200 \times \left(\frac{1}{2}\right)}$$

= 2000 KPa

$\boxed{P_m = 2MPa}$

516. Specific fuel consumption of a diesel engine as compared to that for petrol engine is

(a) Lower
(b) Higher
(c) Same for same power output
(d) May be lower or higher

RPSC AE 2016
UKPSC AE 2012 Paper–II
RPSC ACF-2011

Ans : (a) Specific fuel consumption is amount of fuel consumed to produce unit power in one hour.

$$SFC = \frac{kg/h \text{ fuel consumed}}{\text{power (kW)}}$$

Specific fuel consumption of a diesel engine as compared to that for petrol engine is lower.

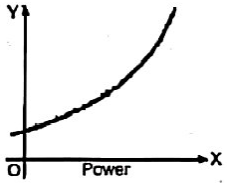
517. Morse test is used to determine the I.P. of a

(a) single cylindrical petrol engine
(b) four stroke engine
(c) single cylindrical diesel engine
(d) multi-cylindrical engine

Nagaland PSC (CTSE) 2018, Paper-II
HPPSC W.S. Poly. 2016, ESE 1995

Ans. (d) : Morse test can be conducted for multi cylinder engines the purpose approximate indicated power of a Multicylinder engine. It consist of running the engine against the dynamo-meter at a particular speed, cutting out the firing of each cylinder in turn and noting the fall in BP each time while maintaining the speed constant.

518. The curve show in the given figure is characteristic of diesel engines

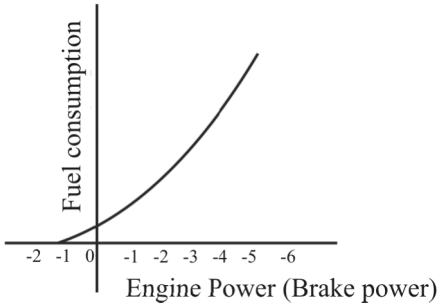


What does the Y-axis represent?

(a) Efficiency
(b) Air-fuel ratio
(c) Specific fuel consumption
(d) Total fuel consumption

Nagaland PSC CTSE 2015 Paper-II
ESE 1995

Ans. (d) :



This graph is extrapolated to get friction power.

519. The ratio of the work obtained at the crankshaft in a given time to the energy supplied during the same time is called

(a) mechanical efficiency
(b) overall efficiency
(c) indicated thermal efficiency
(d) volumetric efficiency

Nagaland PSC (CTSE) 2018, Paper-II
TNPSC ACF 2012

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Ans. (b) : The ratio of the work obtained at the crankshaft in a given time to the energy supplied during the same time is called overall efficiency.

$$\eta_0 = \frac{B.P. \times 3600}{m_f \times C}$$

Where B.P. = Brake power
m_f = mass of fuel consumed in kg
C = calorific value of fuel.

520. An engine indicator is used to determine the following
(a) Speed
(b) Temperature
(c) Mean effective pressure and indicated horse power
(d) Brake horse power

Nagaland PSC CTSE 2017 Paper-2
J&K PSC Civil Services Pre, 2010

Ans. (c) : Mean effective pressure and indicated horse power.

521. Which one of the following methods of determining the engine friction is applicable for diesel engines only?
(a) Morse test
(b) Motoring test
(c) Deceleration Method
(d) Willian's line method

OPSC AEE 2019 PAPER - II
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Willian's line method is only applicable in CI engine for calculation of engine friction power.
Morse test – Determines Indicated power for multi cylinder engine.
Motoring test- used to find friction power of engine for SI & CI both.

522. In a Morse test for a 2-cylinder, 4-stroke, compression ignition engine, the brake power was 12 kW whereas the brake powers of individual cylinders with injection cut off were 4.25 kW and 4.75 kW respectively. The mechanical efficiency of the engine is
(a) 75.9% (b) 77.9%
(c) 76.9% (d) 80.0%

CGPSC AE 2014-II
Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : Given,
Brake power = 12 kW
Power supplied by cylinder No. 1 = 12–4.25 = 7.75 kW
Power supplied by cylinder No. 2 = 12–4.75 = 7.25 kW
Total indicated power = 7.75 + 7.25 = 15 kW

$$\eta_{mechanical} = \frac{BP}{IP} = \frac{12}{15}$$
$$= .8 = 80\%$$

523. A four stroke engine having a brake power of 105 kW is supplied with a fuel at a rate of 4.4 kg per 10 minutes. The brake specific fuel consumption of the engine is
(a) 0.18 kg/kW-hr (b) 0.25 kg/kW-hr
(c) 0.36 kg/kW-hr (d) 0.42 kg/kW-hr

UPPSC AE 13.12.2020, Paper-2
ESE 2018

Ans. (b) : Given,
BP = 105 kW
$\dot{m}_f$ = 4.4 kg per 10 minutes

$$\dot{m}_f = \frac{4.4}{10} \times 60 \text{ per hour}$$

$\therefore$ (BSFC) brake specific fuel consumption

$$= \frac{m_f(kg/hr)}{BP(kW)} = \frac{4.4 \times 60}{10 \times 105}$$
$$= 0.25 \text{ kg/kW-hr}$$

524. If the speed of the engine is increased, the indicated power will :
(a) Increase (b) Decrease
(c) Remain same (d) All of these

RPSC ACF & FRO, 26.02.2021
RPSC IOF, 2020

Ans. (a) : If speed of an engine is increased, the indicated power will increase

$$\text{Indicated power (IP)} = \frac{P_m LAnk}{60}$$

n = N (for 2-stroke engine) (N = RPM)

$$n = \frac{N}{2} \text{ (for 4-stroke engine)}$$

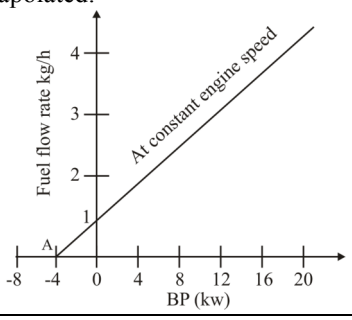
k= no of cylinder
IP $\propto$ N
If speed increases then IP increases

525. The measurement of frictional power by William's Line is applicable only to
(a) SI engines at a particular speed
(b) CI engines at a particular speed
(c) Any engine at a particular speed only
(d) None of the above

GPSC Lect. (Auto) 16.10.2016, TNPSC AE 2014

Ans. (b) : A graph between fuel consumption rate (kg/h) taken on y-axis and BP (kW) on x-axis is drawn, while energy is made to run at same constant speed, say 1500 rpm.

- A graph is extrapolated back to zero fuel consumption which cut on -ve x-axis at a point 'A'.
- The -ve intercept on x-axis represents FP at that speed of the engine.
- Although when BP = 0, some fuel consumption is there. This fuel is consumed to overcome engine friction.
- Only for CI engine to be run at a constant speed as fuel consumption rate v/s BP plot is almost straight line in case of diesel engine, hence can be extrapolated.



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526. If performance of S.I. engines of different manufacturers having different capacities, sizes and systems are to be compared, the common parameter would be.

(a) Engine cylinder diameter
(b) Brake power
(c) Mean effective pressure
(d) Weight of engine

RPSC AE 2016, RPSC ACF-2011

Ans. : (c) If performance of SI engine of different manufacturers having different capacities, size and systems are to be compared, the common parameter is the mean effective pressure.

$$\text{mean effective pressure} = \frac{\text{Work done per cycle}}{\text{Stroke Volume } (V_s)}$$

$$P_m = \frac{\text{WD / Cycle}}{V_s}$$

527. Morse test measures the indicated power of :

(a) SI engine (b) CI engine
(c) Steam engines (d) Steam turbine

Kerala PSC AE 06.08.2015 (KPSC AE. 2015)

Ans : (a, b) Morse test measure the indicated power of SI and CI engine
The morse test is adopted to find the indicated power of each cylinder of a high speed I.C. engine, without using indicator diagram.
■ But KPSC given option (a) as answer.

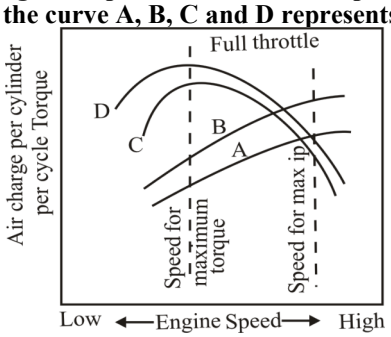
528. Torque developed by the engine is maximum at

(a) Minimum speed of engine
(b) Maximum speed of engine
(c) Maximum volumetric efficiency speed of engine
(d) Maximum power speed of engine

RPSC AE 2016, RPSC ACF-2011

Ans : (a) Torque developed by the engine is maximum at minimum speed of engine.

529. The figure shows typical performance plot against speed for a variable speed SI Engine. the curve A, B, C and D represents



(a) Air Consumption, Indicated Power, Indicated Torque, Air Charge
(b) Air Charge, Indicated Power, Indicated Torque, Air Consumption
(c) Indicated Torque, Indicated Power, Air Charge, Air Consumption
(d) Indicated Power, Air Consumption, Indicated Torque and Air Charge
(e) Indicated Power, Air Charge, Air Consumption, Indicated Torque

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) :
Curve A → Curve A represents Indicated power
Curve B - Curve B represents Air Consumption
Curve C - Indicated Torque
Curve D - Air Charge

530. During a morse test on a 4-cylinder engine, the following measurements of brake power were taken at constant speed.

All cylinders firing	3000 kW
Number 1 cylinder not firing	2062.5 kW
Number 2 cylinder not firing	2062.5 kW
Number 3 cylinder not firing	2061.5 kW
Number 4 cylinder not firing	2063.5 kW

The mechanical efficiency of the engine is :

(a) 91.53% (b) 85.00%
(c) 80.00% (d) 78.00%

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : Given B.P. = 300 kW

$(I.P.)_1 = 3000 - 2062.5 = 937.5$ kW
$(I.P.)_2 = 3000 - 2062.5 = 937.5$ kW
$(I.P.)_3 = 3000 - 2061.5 = 938.5$ kW
$(I.P.)_4 = 3000 - 2063.5 = 936.5$ kW

$$\eta_{mech} = \frac{BP}{IP} = \frac{3000}{937.5 + 937.5 + 938.5 + 936.5}$$
$$= 0.80$$
$$\boxed{\eta_{mech} = 80\%}$$

531. The following data is given, Brake power is 80 kW, Frictional power is 20 kW, then the mechanical efficiency is _____.

(a) 40 percent (b) 60 percent
(c) 80 percent (d) 100 percent

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (c) : Given,
Brake power (BP) = 80 kW
Friction power (FP) = 20 kW
Indicated power (IP) = BP + FP
= 80 + 20 = 100 kW

$$\text{Mechanical efficiency } (\eta_{mechanical}) = \frac{BP}{IP} = \frac{80}{100} \times 100$$
$$= 80\%$$

532. Ratio of brake power to swept volume is known as :

(a) Input (b) Specific input
(c) Specific output (d) Relative efficiency

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : $\text{Specific output} = \frac{\text{Brake power (BP)}}{\text{Swept volume } (V_{Swept})}$

533. A single-cylinder engine running at 1800 r.p.m develops a torque of 8 N-m. The indicated power of the engine is 1.8 kW. The loss of power due to friction is

(a) 0.4 kW (b) 0.35 kW
(c) 0.292 kW (d) 0.15 kW

Assam PSC AE (PHED), 18.10.2020

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Ans. (c) : Given- Single cylinder engine
Speed engine N = 1800 rpm
Torque T = 8 N-m
Indicated power IP = 1.8 kW

$$\text{Brake power B.P.} = \frac{2\pi NT}{60} = \frac{2\pi \times 1800 \times 8}{60}$$

B.P. = 1.50796 kW
Friction power F.P. = I.P. – B.P.
= 1.8 – 1.50796
F.P. = 0.29204 kW

534. The b.p of a four-cylinder engine is 30 with all cylinders firing and 20 with one cylinder cut. The mechanical efficiency is
(a) 75% (b) 65%
(c) 60% (d) 80%

Assam PSC AE (PHED), 18.10.2020

Ans. (a) : Indicated power for one cylinder = 30 – 20
= 10 kW
Total indicated power = 4 × 10 = 40 kW

$$\eta_{mech} = \frac{\text{Brake power}}{\text{Indicated power}} \times 100$$

$$= \frac{30}{40} \times 100 = 75\%$$

535. An IC engine gives an output of 3 kW when the input is 10000 J/s. The thermal efficiency of the engine is
(a) 30% (b) 33.3%
(c) 60% (d) 66.6%

Assam PSC AE (PHED), 18.10.2020

Ans. (a) : Thermal efficiency of engine

$$\eta_{th} = \frac{\text{Output}}{\text{Input}} \times 100$$

$$= \frac{3 \times 10^3}{10 \times 10^3} \times 100$$

$\eta_{th} = 30\%$

536. Find the piston speed of a diesel engine running at 1440 rpm. The engine has a 100 mm bore and stroke length to bore ratio is 1.5.
(a) 7.54 m/s (b) 7.2 m/s
(c) 6.54 m/s (d) 7.5 m/s

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : Given, D = 100 mm

$$\frac{L}{D} = 1.5, \; L = 150 \text{ mm}$$

$$\text{Average speed} = \frac{2LN}{60}$$

$$= \frac{2 \times 150 \times 10^{-3} \times 1440}{60}$$

= 7.2 m/s

537. In four-stroke diesel engine.
1. At high load. NO_x emission increases due to increases in combustion temperature.
2. At high speed, hydrocarbon emission increases due to low combustion temperature.
Choose most appropriate option.
(a) Statement 1 is right but 2 is wrong
(b) Statement 1 is wrong but 2 is right
(c) Both statements 1 and 2 are right
(d) Can't say

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : In four stroke diesel engine.
(1) At high load, NO_x emission increases due to increases in combustion temperature.
(2) At high speed hydrocarbon emission increases due to low combustion temperature.
Both statements are correct.

538. The frictional power (F.P.) is given by-
(a) F.P = B.P. – I. P. (b) F.P. = I.P. – B. P.
(c) F.P. = B.P./I.P. (d) F.P. = I.P./B.P.

GMB AAE 25.06.2017

Ans. (b) : The frictional power (F.P) is given by –
F. P. = I.P. – B.P.
Frictional power is defined as the how much power will be used for running the engine. Engine having some friction in gears chain drives and shaft, mating parts. It consumes some energy from the power.

539. What is the brake thermal efficiency of an engine which consumes 7 kg of fuel in 20 minutes and develops BP of 65 kW ? CV of fuel used = 42 MJ/kg :
(a) 0.265 (b) 0.83
(c) 73% (d) 0.33

APPSC IOF, 2009

Ans. (a) : Fuel consumption $= \frac{7}{\left(\frac{20}{60}\right)} = 21 \text{kg/hr}$

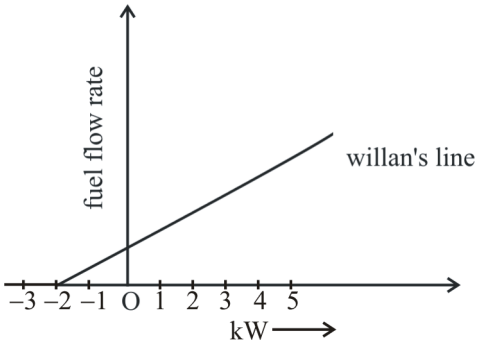
Brake thermal efficiency

$$\eta = \frac{\text{B.P}}{m_f \times \text{C.V}} = \frac{65 \times 3600}{21 \times 42000} = 0.265$$

540. If the fuel consumption of an engine is 0.2 g/s at no load, and 0.4 g/s while delivering 30 kW, the frictional power of the engine using William's line method is
(a) 30 kW (b) 45 kW
(c) 60 kW (d) 90 kW

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) :



The line can assume straight line.
Power = friction + constant × fuel flow
At no load,

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0 = friction + constant × 0.2
At P = 30 kW,
30 = friction + constant × 0.4
Solving above equation –
friction = –30
Constant = +15

541. The actual power supplied by the engine crankshaft is called

(a) Indicated power (b) Brake power
(c) Frictional power (d) None of these

Sikkim PSC (Under Secretary), 2017

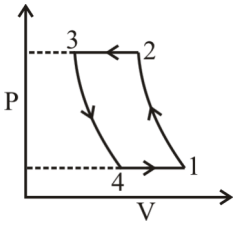
Ans. (b) : The actual power supplied by the engine crankshaft is called brake power.
It is considered as the backbone of an I.C. engine whose function is to convert the reciprocating motion of the position in to the rotary motion with the help of connecting rod. This shaft contains one or more eccentric portion is called crank.

542. With suction pressure being atmospheric, increase in delivery pressure with fixed clearance volume

(a) Increases volumetric efficiency
(b) Decreases volumetric efficiency
(c) Does not change
(d) None of these

MPPSC State Forest Service Exam, 2014

Ans. (b) : $\eta_v = \dfrac{v_1 - v_4}{v_1 - v_3} = \dfrac{\text{Effective swept vol.}}{\text{Swept volume}}$



$$\eta_v = 1 - c\left[\left(\frac{P_2}{P_1}\right)^{1/n} - 1\right]$$

- $P_2\uparrow$, $\eta_v\downarrow$
- $P_1\uparrow$, $\eta_v\uparrow$
- $n\uparrow$, $\eta_v\uparrow$
- $c\uparrow$, $\eta_v\downarrow$

543. What is brake power?

(a) It is the power required to bring the engine to a stop
(b) It is torque developed when the brake is applied, measured at the brake shoe
(c) It is the power developed by an engine and measured at the output shaft
(d) None of these

KPCL AE 2016

Ans. (c) : Brake power- The power developed by an engine at the output shaft is called the brake power.

$$\text{Brake power (B.P.)} = \frac{2\pi NT}{60000}\text{kW}$$

T = torque in 'N-m'
The difference between I.P. and B.P. is called friction power (F.P.)
F. P. = I.P. – B.P.
The ratio of B.P. to I.P. (Indicated power) is called mechanical efficiency.

$$\eta_{\text{mechanical}} = \frac{\text{B.P.}}{\text{I.P.}} \times 100\%$$

544. If the approximate mean pressure during induction, compression, power and exhaust strokes of an I.C. engine are respectively 15 kN/m^2 below atmosphere, 200 kN/m^2 above atmosphere, 1000 kN/m^2 above atmosphere and 20 kN/m^2 above atmosphere, then the mep of the engine in kN/m^2 is

(a) 765 (b) 795
(c) 800 (d) 805

WBPSC AE 2003

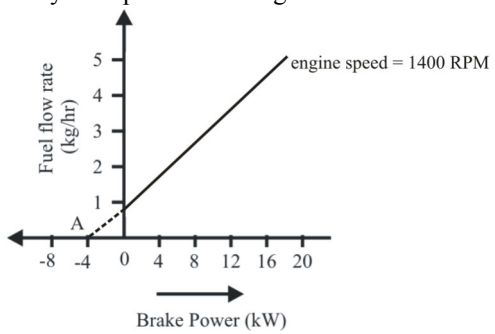
Ans. (a) :
Induction Pressure = 15 kN/m^2 below atmosphere
during Compression = 200 kN/m^2 above the atmosphere
during power = 1000 kN/m^2 above the atm
during exhaust = 20 kN/m^2 above the atm
(mep) mean effective pressure of the engine
mep = 1000 – 200 – (15 + 20) =1000–200–35
mep= 765 kN/m^2

545. Which one of the following quantities is assumed constant for an I.C. engine while estimating its friction power by extrapolation through William's line?

(a) Brake thermal efficiency
(b) Indicated thermal efficiency
(c) Mechanical efficiency
(d) Volumetric efficiency

WBPSC AE 2003

Ans. (b) : Indicated thermal efficiency is assumed constant for an I.C. engine while estimating its friction power by extrapolation through William's line.



$$\dot{m}_f = \lambda BP + \dot{m}_{0f}$$

Where, λ is the slope
→ Negative line on point A shows the friction power at the same speed of engine.
→ Willan's line method is only used in compression ignition (CI) engine. It is not applicable for SI engine.

546. If the mechanical efficiency of an IC engine is 80 percent and the indicated power is100 kW, the frictional power is:

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(a) 40 kW (b) 8 kW
(c) 10 kW (d) 20 kW

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : Brake power = Mechanical efficiency × indicated power

$= 0.80 \times 100$

$= 80$ kW

Frictional power = Indicated power – Brake power

$= 100 - 80$

$\boxed{\text{Frictional power} = 20\,\text{kW}}$

547. An engine produces 12 kW brake power while working with a brake thermal efficiency of 36%. If the calorific value of the fuel used is 48 MJ/kg then what is the fuel consumption?

(a) 2.5 kg/hour (b) 1.5 kg/hour
(c) 2.0 kg/hour (d) 3.0 kg/hour

UPRVUNL AE 21.08.2016

Ans. : (a) Break power = 12 kW

η_{bth} (Brake thermal efficiency) = 36%

$C_V = 48$ MJ/kg

$$\eta_{bth} = \frac{BP}{m_f \times C_V}$$

C_v = Calorific value

m_f = mass of fuel

$$0.36 = \frac{12}{m_f \times 48000}$$

$$m_f = \frac{12}{0.36 \times 48000} \times 3600$$

$\boxed{m_f = 2.5\,\text{kg/hour}}$

548. A jet engine consumes 1 kg of fuel for each 40 kg of air intake. Fuel consumption is 1 kg/sec. When aircraft travels in still air at 200 m/sec, velocity of discharge gases with respect to engine is 700 m/sec. The power developed by engine is

(a) 7200 kW (b) 5600 kW
(c) 2070 kW (d) 4140 kW

ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Given,

$\dot{m}_f = 1$ kg

$\dot{m}_a = 40$ kg

Velocity of aircraft $V_a = 200$ m/s

Velocity of discharge $V_d = 700$ m/s

$$F_{net} = (\dot{m}_a + \dot{m}_f) \times V_{exhaust(d)} - \dot{m}_a \times V_{jet(a)}$$

$$\text{Power} = \left[(\dot{m}_a + \dot{m}_f)V_d - \dot{m}_a V_a\right] V_a$$

$= [(40 + 1)\,700 - 40 \times 200] \times 200$

$= 4140000$ W

$= 4140$ kW

549. A large diesel engine runs on a stroke cycle at 2000 rpm. The engine has a displacement of 25 litre and a brake mean effective pressure of 0.6 MN/m^2. It consumes 0.018 kg/s of fuel (calorific value = 42000 kJ/kg). Determine the brake power.

(a) 250 kW (b) 225 kW
(c) 275 kW (d) none of the above

ISRO Scientist/Engineer 11.10.2015

Ans. (a) : Given, Speed (N) = 2000 rpm

Volume (V_S) = 25 litre

$$= \frac{25}{1000}\,m^3$$

$P_{mef.} = 0.6$ MPa $= 0.6$ MN/m^2

Consumes, $(\dot{m}_g) = 0.018$ kg/s

Calorific value (C_f) = 42000 kJ/kg

$\boxed{\text{B.P.} = P_{mef.} \times V_S \times \text{Number of cycle/sec}}$

$$\text{Brake power (BP)} = 0.6 \times 10^3 \times \frac{25}{1000} \times \frac{2000}{2 \times 60}$$

$= 250$ kW

550. Work done by an engine in 20 sec is 1000 Joules. What is the power generated by the engine in Watt?

(a) 50 Watt (b) 70 Watt
(c) 100 Watt (d) 120 Watt

MECON MT 2019

Ans. (a) : Power generated, $P = \frac{W}{t}$

$$= \frac{1000}{20} = 50\,\text{Watt}$$

551. The indicated mean effective pressure for four stroke single cylinder engine that produces 24 kW of power at engine crank shaft at 2000 rpm. having mechanical efficiency as 80%. Consider engine swept volume as 900 cc and stroke length as 9 cm:

(a) 0.33 bar (b) 10 bar
(c) 20 bar (d) 0.167 bar

OPSC AEE 2019 PAPER - II

Ans : (c) : Given,

$N = 2000$ rpm

B.P. $= 24$ kW

$V_s = 900$ cc

$L = 9$ cm

$$\eta_m = \frac{\text{B.P.}}{\text{I.P.}}$$

$$\text{I.P.} = \frac{24}{0.8}$$

$= 30$ kW

$V_s = 900$ cc

$$= \frac{0.0009 \times 1 \times 2000}{60 \times 2}$$

$= 0.015$ m^3/sec

I.P. $= P_{imep} \times V_s$

$$P_{imep} = \frac{30}{0.015}$$

$P_{imep} = 2000$ kPa $= 20$ bar

552. The volumetric efficiency of a well-designed SI engine lies in the range of

(a) 40%-50% (b) 51%-60%
(c) 61-%-70% (d) 71%-90%

BPSC AE Mains 2017 Paper - V

Ans : (d) : The normal range of volumetric efficiency at fuel throttle for SI engines is between 80 to 85% where as for CI engines it is between 85 to 90%.

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553. An increase in the mean effective pressure of a diesel engine with fixed compression ratio can be obtained with increase in

(a) Speed of the engine
(b) Charge density
(c) Cut-off ratio
(d) Back pressure

BPSC AE Mains 2017 Paper - V

Ans : (c) : By increasing the cut-off ratio, the work done increases and hence mean effective pressure increases.

554. The power to weight ratio of diesel engine compared to petrol engine is

(a) High (b) Low
(c) Same (d) Not comparable

BPSC AE 2012 Paper - V

Ans : (b) : The power to weight ratio of diesel engine compared to petrol engine is low.

555. A 2-stroke cycle internal combustion engine has a mean effective pressure of 6 bar. The speed of the engine is 1000 r.p.m. If the diameter of the piston and stroke are 110 mm and 140 mm respectively, then the Indicated power developed by the engine is

(a) 6.7 kW (b) 13.3 kW
(c) 26.6 kW (d) 39.9 kW

TNPSC AE 2014

Ans. (b) : Data given -

$P_{mep} = 6$ bar

N = 1000 r.p.m.

d = 110 mm = 0.11 m

L = 140 mm = 0.14 m

I.P. of the engine = ?

We know that

$$P_{mep} = \frac{\text{I.P.}}{\text{Swept volume}}$$

$$600 \times 10^3 = \frac{\text{I.P.}}{\frac{\pi}{4} d^2 \times L \times \frac{N}{60}}$$

$$600 \times 10^3 = \frac{\text{I.P.}}{\frac{\pi}{4} \times (0.11)^2 \times (0.14) \times \frac{1000}{60}}$$

I. P. = 13.2979 kW

556. Rate of work done on the piston by burning of the fuel inside the cylinder of IC engine is called

(a) Friction power (b) Indicated power
(c) Brake power (d) Mechanical power

RPSC LECTURER 16.01.2016

Ans. (b) : Indicated power—Indicated power is the total power available from the expanding of the gases in cylinders. Rate of work done on the piston by burning of the fuel inside the cylinder of IC engine is called Indicated power.

Brake Power (BP)—Brake Power can be defined as the power available at the crank shaft. Since it is measured with the help of Brake dynamometer so it is referred as brake power.

Friction Power (FP)—The power required or power lost to overcome the friction between the mating members of engine is called Friction Power.

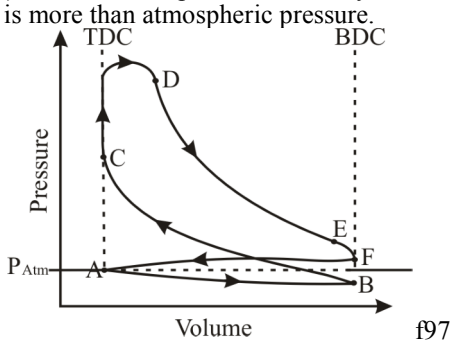
FP = IP - BP

557. The exhaust pressure in the cylinder of an IC engine is

(a) More than atmospheric pressure
(b) Less than atmospheric pressure
(c) Equal to atmospheric pressure
(d) Zero (absolute)

RPSC LECTURER 16.01.2016

Ans. (a) : The exhaust pressure in the cylinder of an IC engine is more than atmospheric pressure.



f97

Process BC - Compression

CD → Combustion

DE → Expansion

FA → Exhaust

From this actual combustion cycle

$P_F > P_{atm}$

558. If ip → indicated power, bp → brake power, fp → frictional power. Which of the following statement is true for IC engine:

(a) ip < bp (b) fp = ip + bp
(c) ip < fp (d) ip > bp

UPRVUNL AE 2016

Ans. (d) : We know that,

Indicated power (ip) = Brake power (bp) + Frictional power (fp)

So, ip > bp

559. Mechanical efficiency of a diesel engine is defined as

(a) $\frac{\text{Indicative Horse Power}}{\text{Brake Horse Power}}$

(b) $\frac{\text{Brake Horse Power}}{\text{Indicative Horse Power}}$

(c) $\frac{\text{Brake Horse Power}}{\text{Fuel Input Power}}$

(d) Brake Horse Power × Indicative Horse Power

TNPSC AE 2013

Ans. (b) : Mechanical efficiency of a diesel engine is defined as the ratio of brake horse power to indicative horse power.

$$\eta_m = \frac{BHP}{IHP}$$

560. The brake power of an engine whose mechanical efficiency is 80% and indicated power is 125000 W is

(a) 100 W (b) 10,000 kW
(c) 100 kW (d) 125 kW

TNPSC AE 2013

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Ans. (c) : Data given

$\eta_{m=}$ 80%

IP = 125000 W

BP = ?

We know that

$$\eta_m = \frac{BP}{IP}$$

$$BP = 125000 \times \frac{80}{100}$$

BP = 100000 W

$$\boxed{BP = 100\ kW}$$

561. An engine consumes 5 kg/hr of fuel and its brake power is 20 kW. The value of brake specific fuel consumption is

(a) 0.25 kg/kWh (b) 4 kg/kWh

(c) 2 kg/kWh (d) 15 kg/kWh

TNPSC AE 2013

Ans. (a) : Data given as-

$\dot{m}_f$ = 5 kg/hr

BP = 20 kW

bpsfc = ?

We know that

$$bpsfc = \frac{\dot{m}_f}{BP} = \frac{5}{20}$$

$$bpsfc = 0.25 \frac{kg}{kWhr}$$

562. Morse test in Multi cylinder engines is used to determine

(a) Volumetric efficiency

(b) Brake thermal efficiency

(c) Mechanical efficiency

(d) Brake power

TNPSC AE 2014

Ans. (c) : Morse test aim–

1. To study indicated power of individual cylinder.
2. Indicated power of engine.
3. Frictional power of engine.
4. Mechanical efficiency.

563. The output of an I.C. Engine is measured by a rope brake dynamometer. The diameter of the brake pulley is 750 mm and rope diameter is 50mm. The dead load on the tight side is 400 N and the spring balance reading is 50 N. The engine consumes 4.2 kg/h of fuel at rated speed of 1000 rpm. Then the brake specific fuel consumption of the engine is

(a) 0.143 kg/kWh (b) 0.286 kg/kWh

(c) 0.268 kg/kWh (d) 0.134 kg/kWh

TNPSC AE 2014

Ans. (b) : Data given-

D = 750 mm, d = 50 mm

P = 400 N, N = 1000 rpm

S = 50 N

$$\text{Brake power (BP)} = \frac{(P-S)\pi D_m N}{1000 \times 60} kW$$

where $D_m = (D + d) = 750 + 50 = 800$ mm

$$BP = \frac{(400-50) \times 3.14 \times 0.8 \times 1000}{1000 \times 60}$$

$$\boxed{BP = 14.6533\ kW}$$

then $$bsfc = \frac{m_c}{BP} = \frac{4.2}{14.653}$$

bsfc = 0.2866 kg/kWh

564. A 4-stroke diesel engine, when running at 2500 rpm has injection duration of 2.0 ms. What is the corresponding duration of the crank angle in degrees?

(a) 30 (b) 31

(c) 29 (d) 28

(e) 32

CGPSC AE 26.04.2015 SHIFT-II

Ans. (a : Data given

N = 2500 rpm

t [injection time] = 2×10^{-3} s

Rotation during Injection time

$$\theta = \omega \times t = \frac{2 \times \pi \times 2500}{60} \times 2 \times 10^{-3} \times \frac{180}{\pi}$$

$$\boxed{\theta = 30^0}$$

565. What is the compression ratio of a 4-stroke engine whose capacity is 245cc clearance volume is 27.2cc and diameter of bore is 10% higher than stroke?

(a) 9 (b) 10

(c) 11 (d) 12

JWM 2017

Ans. (b) : Compression ratio of engine(r)

$$= \frac{\text{Total volume}}{\text{clearance volume}}$$

$$r = \frac{\text{Swept volume + clearance volume}}{\text{clearance volume}}$$

$$r = \frac{245}{27.2} + 1 = 10$$

Note- Engine capacity is to be considered as swept volume.

566. What is the number of cycles completed per second for a four stroke diesel engine running at 6000 rpm?

(a) 50 (b) 500

(c) 6000 (d) 3000

CIL MT 26.03.2017

Ans. (a) : Engine speed = 6000 rpm

One cycle in four stroke requires two revolutions. So Number of cycles per minute = 6000/2 = 3000 cycles.

So number of cycles per second = 3000/60 = 50 cycles.

567. Automatic transmission as compared to manual transmission are usually

(a) More fuel efficient

(b) Less fuel efficient

(c) Equally efficient

(d) None of the above

TNPSC AE 2017

Ans. (b) : Automatic transmission as compared to manual transmission are usually Less fuel efficient.

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568. A 4-stroke diesel engine has length of 20 cm and diameter of 16 cm. The engine is producing power of 25 kW when it is running at 2500 r.p.m. The mean effective pressure of the engine will be nearly

(a) 5.32 bar (b) 4.54 bar
(c) 3.76 bar (d) 2.98 bar

ESE 2019

Ans. (d) : Given,

$L = 20\ cm = 0.20\ m$ $N = 2500\ rpm$

$D = 16\ cm = 0.16\ m$

$BP = 25\ kW$

Swept volume $(V_s) = \frac{\pi}{4}D^2L$

n = No. of Cylinder

n = 1

$$V_s = \frac{\pi}{4}(0.16)^2 \times 0.20$$

$$V_s = 0.004\ m^3$$

Mean effective pressure

$$p_m = \frac{BP}{V_s \times \frac{N}{120} \times n}$$

$\frac{N}{120}$ = For Four stroke engine (rps)

$$= \frac{25}{0.004 \times \left(\frac{2500}{120}\right)}$$

$= 298.41\ kPa$

$= 2.98\ bar$

569. A 4-stroke, 6-cylinder gas engine with a stroke volume of 1.75 litres develops 26.25 kW at 506 r.p.m. and the MEP is 600 kN/m^2. The number of misfires per minute per cylinder will be

(a) 3 (b) 4
(c) 5 (d) 6

ESE 2019

Ans. (a) : Given,

n = 6

$V_s = 1.75\ lit = 1.75 \times 10^{-3}\ m^3$

$BP = 26.25\ kW$

$p_m = 600\ kN/m^2$

$N = 506\ rpm$

$$p_m = \frac{BP}{V_s \times n \times \frac{N_1}{120}}$$

$$600 = \frac{26.25}{1.75 \times 10^{-3} \times 6 \times \frac{N_1}{120}}$$

$N_1 = 500\ rpm$

but given N = 506 rpm

For developing the given mean effective pressure only 500 rpm is required, where as actual is 506 rpm

Missed firing cycles $= \frac{N - N_1}{2}$

$$= \frac{506 - 500}{2} = 3$$

570. A 2-stroke oil engine has bore of 20 cm, stroke 30cm, speed 350 r.p.m., i.m.e.p. 275 kN/m^2, net brake load 610 N, diameter of brake drum 1m, oil consumption 4.25 kg/hr, calorific value of fuel 44 × 10^3 kJ/kg. The indicated thermal efficiency will be

(a) 29.1% (b) 31.3%
(c) 33.5% (d) 35.7%

ESE 2019

Ans. (a) : 2-stroke oil engine

$d = 0.2\ m$ $N = 350\ rpm$ $k = 1$

$\ell = 0.3\ m$ $p_m = 275\frac{kN}{m^2}$

$\dot{m} = 4.25\ kg/hr$

$CV = 44 \times 10^3\ kJ/kg$

$$\dot{m} = \frac{4.25}{3600} = 0.00118\ kg/s$$

Indicated power (IP) = $p_m \cdot v_s$

$$= p_m \times \left(\frac{\pi}{4}d^2 \times \ell \times k \times \frac{N}{60}\right)$$

$$= 275 \times \left[\frac{\pi}{4} \times (0.2)^2 \times 0.3 \times 1 \times \frac{350}{60}\right]$$

$= 15.118\ kW$

Indicated thermal efficiency

$$\eta_{ith} = \frac{IP}{\dot{m}\,CV} = \frac{15.118}{0.00118 \times 44 \times 10^3} = 0.291 = 29.1\%$$

571. The following are the results of a Morse test conducted on a four-cylinder, four-stroke petrol engine at a common constant speed in all cases: The brake power of the engine when all the cylinders are firing is 80 kW. The brake power of the engine when each cylinder is cut off in turn is 55 kW, 55.5 kW, 54.5 kW and 55 kW, respectively The mechanical efficiency of the engine when all the cylinders are firing will be

(a) 90% (b) 85%
(c) 80% (d) 75%

ESE 2018

Ans. (c) : BP = 80 kW

$IP_1 = 25\ kW = 80 - 55$

$IP_2 = 24.5\ kW = 80 - 55.5$

$IP_3 = 25.5\ kW = 80 - 54.5$

$IP_4 = 25\ kW = 80 - 55$

$IP = IP_1 + IP_2 + IP_3 + IP_4$

$= 25 + 24.5 + 25.5 + 25 = 100\ kW$

Mechanical efficiency $\eta_m = \frac{BP}{IP} = \frac{80}{100} = 80\%$

572. A 4-cylinder diesel engine running at 1200 r.p.m. developed 18.6 kW. The average torque when one cylinder was cut out was 105 Nm. If the calorific value of the fuel was 42000 kJ/kg and the engine used 0.34 kg of diesel/kW hr, the indicated thermal efficiency was nearly.

(a) 29% (b) 26%
(c) 31% (d) 23%

ESE 2017

Ans. (a) : Given,

$N = 1200\ rpm$ $T_{avg} = 105\ Nm$

$B.P. = 18.6\ kW$ $CV = 42000\ kJ/kg$

brake-specific fuel consumption (bsfc)

$= 0.34\ kg/kW\ hr$

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BP when one cylinder is cut off

$$= \frac{2\pi NT}{60\times1000} = \frac{2\pi\times1200\times105}{60\times1000}$$

$$= 13.19 \text{ kW}$$

Indicated power of one-cylinder

$$(IP)_{1\text{-cylinder}} = (BP)_{4\text{-cylinder}} - (BP)_{3\text{-cylinder}}$$

$$= 18.6 - 13.19 = 5.41 \text{ kW}$$

Indicated power for 4-cylinder

$$(IP)_{4\text{-cylinder}} = 5.41 \times 4 = 21.64 \text{ kW}$$

Brake specific fuel consumption

$$\text{bsfc} = \frac{\dot{m}_f}{BP}$$

$$0.34 = \frac{\dot{m}_f}{18.6}$$

$$\dot{m}_f = 6.324 \text{ kg/hr}$$

$$= \frac{6.324}{3600}\text{kg/s} = 0.00175 \text{ kg/s}$$

Indicated thermal efficiency

$$\eta = \frac{IP}{\dot{m}_f CV} = \frac{21.64}{0.00175\times42000}$$

$$= 0.2933$$

$$\eta = 29.33\%$$

573. The following data was observed for a CI engine :
Fuel-air ratio : 0.04, Volumetric efficiency : 85%, Indicated mean effective pressure; 8 bar Fuel calorific value : 44000 kJ/kg. Density of air : 1 kg/m^3. What is the indicated thermal efficiency of this engine?

(a) 43% (b) 64%
(c) 53% (d) 69%

BHEL ET 2019

Ans. (c) : Fuel-air ratio $= \frac{m_f}{m_a} = 0.04$

$$\eta_{ith} = \frac{P_{mean}\times m_a}{m_f.C.V.\rho_a\times\eta_{vol}}$$

$$= \frac{8\times10^5}{0.04\times44000\times10^3\times1\times0.85}$$

$$\eta_{ith} = 53.47\%$$

574. An IC engine produces an indicated power of 12 kW. If the mechanical efficiency of the engine is 90%, then loss of power due to friction is given as :

(a) 1.2 kW (b) 2.2 kW
(c) 3.2 kW (d) 0.2 kW

BHEL ET 2019

Ans. (a) : Given - Indicated power (IP) = 12 kW

$$\eta_m = 90\%$$

mechanical efficiency $\eta_m = \frac{BP}{IP}$

$$0.9 = \frac{BP}{12}$$

BP = 12 × .90 = 10.8 kW

Then power due to friction, FP = IP – bP

FP = 12 – 10.8

$$\boxed{FP = 1.2\text{kW}}$$

575. A single-cylinder, two-stroke petrol engine develops 4 kW indicated power. If the mean effective pressure is 6.5 bar and piston diameter is 100 mm, the average speed of the piston is

(a) 23.5 m/s (b) 47 m/s
(c) 94 m/s (d) 186 m/s

JPSC AE 2013 Paper-V

Ans. (c) : V = 2LN

$$I.P. = \frac{P_m LAN}{60}$$

$$LN = \frac{I.P.\times60}{P_m.A}$$

$$V = \frac{2\times I.P.\times60}{P_m.A} = \frac{2\times4\times10^3\times60}{6.5\times10^5\times\frac{\pi}{4}\times(0.1)^2}$$

$$= \frac{2\times4\times60\times4}{6.5\times\pi} = 94.0238 \text{ m/sec}$$

576. A four-stroke single-cylinder SI engine of 6 cm diameter and 10 cm stroke running at 4000 rpm develops power at a mean effective pressure of 10bar. The power developed by the engine is

(a) 9.42 kW (b) 5.54 kW
(c) 4.92 kW (d) 2.94 kW

ESE 2018

Ans. (a) : Given,
D = 0.06 m
L = 0.1 m
N = 4000 rpm
P_m = 1000 kPa

Brake power for 4-stroke engine $= \frac{P_m LAN}{120}$

$$= \frac{1000\times0.1\times\frac{\pi}{4}\times(0.06)^2\times4000}{120} = 9.42 \text{ kW}$$

577. The mechanical efficiency of a single-cylinder four-stroke engine is 60%. The frictional power is estimated to be 30 kW. What is the indicated power?

(a) 120 kW (b) 75 kW
(c) 150 kW (d) 130 kW

ESE 18.07.2021

Ans. (b) : $\eta_m = \frac{BP}{IP} = 0.6$

BP = 0.6 IP

IP – FP = 0.6 IP

$$IP = \frac{FP}{0.4} = \frac{30}{0.4} = 75\text{kW}$$

578. A four-stroke petrol engine at full load delivers 100 kW. It requires 10 kW to rotate it without load at same speed. What is the mechanical efficiency at half load?

(a) 67.82% (b) 70.24%
(c) 77.32% (d) 83.33%

ESE 18.07.2021

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Ans. (d) : BP = 100 kW at full load

$BP = \frac{100}{2}$ = 50kW at half load

at half load-

FP = constant = 10 kW

Mechanical efficiency at half load-

$$\eta_m = \frac{BP}{IP} = \frac{BP}{BP + FP} = \frac{50}{50 + 10} = 83.33\%$$

579. If crank shaft of four stroke S.I. engine rotates at 1000 rpm then cam shaft of the same engine rotates at :

(a) 1000 rpm (b) 100 rpm

(c) 2000 rpm (d) 500 rpm

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Cam shaft rotates at $= \frac{1000}{2} = 500$rpm

580. The ratio of work done per cycle to the stroke volume of a compressor is known as.

(a) Compressor capacity

(b) Compression ratio.

(c) Compressor efficiency

(d) Mean effective pressure

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Mean Effective Pressure-The mean effective pressure (mep) is useful parameter in the study of the reciprocating devices that are used for power production.

WD per cycle = $p_m \times$ piston area $\times$ stroke

$$WD_{cycle} = p_m \times A \times L$$

$$p_m = \frac{WD_{cycle}}{AL} = \frac{WD}{V_s} = \frac{\text{Workdone per cycle}}{\text{Stroke volume}}$$

581. A four stroke single cylinder engine produces 24kW power at engine crank shaft at 2000 rpm. The mechanical efficiency of engine at 80%, swept volume 800 cc and stroke length as 9 cm. The indicated mean effective pressure of engine will be :

(a) 6.4 bar (b) 10 bar

(c) 12.8 bar (d) 20 bar

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : Given,

BP = 24 kW

N = 2000 rpm

$\eta_{mech} = 0.80$ $V_S = \frac{\pi}{4} D^2 \times 2$

$V_S = 800$ cc $V_S = A \times L$

$L = 9 \text{ cm} = 9 \times 10^{-2}$ $A = \frac{V_S}{L} = \frac{800}{9} = 88.88 \text{cm}^2$

$= 88.88 \times 10^{-4} \text{ m}^2$

Indicated mean effective pressure (P_m) = ?

$$\eta_{mech} = \frac{BP}{IP}$$

$$IP = \frac{BP}{\eta_{mech}} = \frac{24}{0.80} = 30\text{kW}$$

$$IP = \frac{P_m LA(N/2)}{60}$$

$$30 \times 10^3 = \frac{P_m \times 9 \times 10^{-2} \times 88.88 \times 10^{-4} \times \left(\frac{2000}{2}\right)}{60}$$

$P_m = 22.5$ bar ≈ 20 bar

582. The bore and stroke of a single cylinder four-stroke engine are 100 mm and 160 mm respectively. If the brake torque is 50 N-m the bmep is

(a) 15 bar (b) 10 bar

(c) 5 bar (d) 7.6 bar

RPSC IOF, 2020

Ans. (c) : Given,

Bore, D = 100 mm = 0.1 m

Stroke, L = 160 mm = 0.16 m

Brake torque = 50 N-m

Brake power, BP = Torque $\times \omega$

For four stroke, speed $= \frac{N}{2}$

$$BP = \frac{50 \times 2\pi N}{60}$$

$$BP = \frac{B_{MEP} \times LAN}{60}$$

$$\frac{50 \times 2\pi N}{60} = \frac{B_{MEP} \times 0.16 \times \frac{\pi}{4} \times (0.1)^2 \times \frac{N}{2}}{60}$$

Brake mean effective pressure,

$B_{MEP} = 5 \times 10^5$ N/m^2

$= 5$ bar

583. A 4-stroke, 4- cylinder diesel engine running at 1200 rpm develops 70 kW. Given parameter are:

Brake thermal efficiency = 30%;

Calorific value of fuel = 42,000 kJ/kg;

Engine bore = 150 mm;

Engine stroke = 120 mm;

Air-fuel ration = 15:1;

Density of air = 1.15 kg/m^3:

Air consumption (in m^3/S) is:

(a) 0.07246 (b) 0.06124

(c) 0.1294 (d) 0.007432

HPPSC ADF 05.03.2019

Ans. (a) :

584. An engine working on Otto cycle consumes 8 litres of fuel per hour and develops 25 kW. The specific gravity of fuel is 0.75 and its calorific value is 44000 kJ/kg. The indicated thermal efficiency of the engine is:

(a) 34.1% (b) 24.2%

(c) 28.4% (d) 54.5%

AAI Jr. Executive 29.11.2018

Ans. (a) : Fuel consumption per hour (m_f) = Volume flow rate $\times$ density

$= 8 \times 10^{-3}$ (m^3/hr) $\times 0.75 \times 10^3$ (kg/m^3)

= 6 kg/hr

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$$\text{Indicated thermal efficiency} = \frac{IP}{m_f(kg/sec)\times C.V.}$$
$$= \frac{25}{\frac{6}{3600}\times 44000}$$
$$= 0.3409$$
$$= 34.09\% \approx 34.1\%$$

585. A 16 kW engine with 80% mechanical efficiency has the frictional power constant for all loads. If the engine is loaded with 75% of its rated power, then mechanical efficiency of this engine is :

(a) 65% (b) 80%
(c) 75% (d) 60%

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Given that
I. P = 16 kW
B.P = 75% of its rated power
= 0.75 × 16 = 12 kW B.P. = Break power

$$\eta_{mech} = \frac{B.P}{I.P} = \frac{12}{16}$$ I.P. = Indicated power

$\eta_{mech} = 75\%$

586. An engine running at 3000 rpm has an injection duration 1 mS. What is the corresponding injection duration in crank angle in degrees?

(a) 16 (b) 18
(c) 30 (d) 4

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b)
1 min = 3000 revolution

$$1 \text{ sec} = \frac{3000}{60} = 50 \frac{\text{revolution}}{\text{sec}}$$

Then angle turned by crank in 1 mili second
$= 360 \times 50 \times 10^{-3}$
$= 18^0$

587. The cubic capacity of a four stroke over-square S.I. engine is 245 cc. The clearance volume is 27.2cc. The compression ratio of the engine is about:

(a) 7.8 (b) 6.5
(c) 9 (d) 10
(e) 12.9

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : Given
Total volume = 245cc
Clearance Volume = 27.2cc
Swept Volume = 245-27.2
= 217.8cc

$$\text{Compression Ratio} = \frac{\text{Total Volume}}{\text{Clearance Volume}}$$
$$= \frac{2450}{27.2}$$
$$= 9.0073$$

Compression Ratio = 9

588. A 4-stroke diesel engine, when running at 1500 rpm has a duration of 2 mili-seconds. What is the corresponding duration of the crank angle, in degrees?

(a) 18º (b) 9º
(c) 38º (d) 15º
(e) 20º

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Given.
N = 1500 rpm ; t = 2 mili second $= 2\times10^{-3}$ second

$$\omega = \frac{2\pi N}{60} = \frac{2\times\pi\times1500}{60}$$
$$t = \frac{\theta}{\omega}$$
$$2\times10^{-3} = \frac{\theta}{\frac{2\pi\times1500}{60}}$$
$$\theta = 0.314159$$
$$\theta = \frac{03.4159\times180}{\pi}$$
$$\theta = \mathbf{18^o}$$

589. Engine misfiring is likely to result from

(a) spark plug gas too small
(b) spark plug gas too wide
(c) vapour lock in the fuel line
(d) stoichiometric fuel air mixture

APPSC AEE 2012

Ans. (c) : Engine misfiring is likely to result from vapour lock in the fuel line.

590. The Volumetric efficiency of an engine whose swept volume is 0.5 m^3/min and volume of intake air 0.4 m^3/min :

(a) 80% (b) 20%
(c) 10% (d) 50%

TNPSC AE (Industries) 09.06.2013

Ans. (a) :

591. The mechanical efficiency of a single-cylinder, four -stroke engine is 90%. The frictional power is 30 kW. What will be the brake power developed by the engine?

(a) 240 kW (b) 270 kW
(c) 300 kW (d) 330 kW

APPSC Poly Lect. 13.03.2020

Ans. (b) : Let;
BP = Brake power
IP = Indicated power
FP = Frictional Power = 30 kW
Given;
Mechanical efficiency (η_m) = 90% = 0.9
We know that,

$$\eta_m = \frac{BP}{IP}$$
$$\frac{BP}{IP} = 0.9 \Rightarrow IP = \frac{10\ BP}{9} \quad \text{..... (i)}$$

Now, IP = BP + FP

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$$\Rightarrow \frac{10\,BP}{9} = BP + 30kW \quad \text{[from equation (i)]}$$

$$\Rightarrow \frac{BP}{9} = 30\ kW$$

$$\Rightarrow \boxed{B.P. = 270kW}$$

592. The straight line plot of experimental values of total fuel consumption and the brake power is called

(a) Atkinson line (b) Otto Line
(c) Willans' line (d) Diesel line

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (c) : The straight line plot of experimental values of total fuel consumption and the brake power is called Willan's line

Figure

593. A rope brake dynamometer is used to measure the brake power of a single cylinder four stroke petrol engine. It is found that the torque due to brake load is about 200 Nm and the engine makes and angular velocity of 52.36 rad/s The brake power developed by the engine is _____.

(a) 20.94 kW (b) 10.47 kW
(c) 6.98 kW (d) 5.24 kW

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (b) : Given
torque, T = 200 N-m
angular velocity, $\omega = 52.36$ rad/sec
$\therefore$ Brake power, $P = T\omega$
$= 200 \times 52.36$
$= 10472$ W
$= 10.472$ kW

594. The ratio of the volume of charge admitted at N.T.P. to the swept volume of the piston is called

(a) mechanical efficiency
(b) overall efficiency
(c) volumetric efficiency
(d) relative efficiency

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The ratio of the volume of charge admitted at N.T.P. to the swept volume of the piston is called volumetric efficiency. Volumetric efficiency is the ratio of the volume of fluid actually displaced by a piston or plunger to its swept volume.

595. The thermal efficiency of diesel engines is about

(a) 15% (b) 30%
(c) 50% (d) 70%

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : The thermal efficiency of petrol and gas engines is about 30% and the thermal efficiency of diesel engines is about 70%.

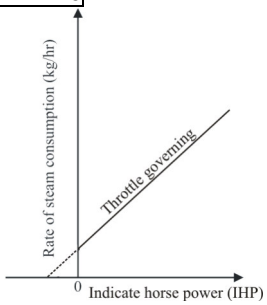
596. William's law gives a straight line graph between the rate of steam consumption and

(a) pressure of steam
(b) temperature of steam
(c) volume of steam
(d) indicated horse power

UKPSC AE 2012 Paper–II

Ans. (d) : William's law is hold's good for throttle governing.

$\boxed{W = a \times IP + b}$



597. The stroke of an engine is increased when the:

(a) Piston is shortened
(b) Connecting rod is lengthened
(c) Crank shaft throw is lengthened
(d) Gudgeon pin is moved nearer to the crank shaft

RPSC VPITI 14.02.2016

Ans. (c) : If crank shaft throw is lengthened the stroke of an engine is increased as L = 2r.

598. Which of the following is true for four-stroke engines?

(a) One power stroke is obtained in each revolution of crankshaft
(b) One power stroke is obtained in every two revolutions of crankshaft
(c) One power stroke is obtained in every four revolutions of crankshaft
(d) One power stroke is obtained in every six revolutions of crankshaft

APPSC Poly Lect. 13.03.2020

Ans. (b) : • In four-stroke engines, one power stroke is obtained in every two revolutions of crankshaft.
• For two-stroke engine one power stroke is obtained in each revolution of crank shaft.

599. A fictitious pressure that, if it acted on the piston during the entire power stroke, would produce the same amount of net work as that produced during the actual cycle is called

(a) Quasi equivalent pressure
(b) Mean equivalent pressure
(c) Mean effective pressure
(d) Quasi static pressure

ESE 2020

Ans. (c) : Mean effective pressure,

$$P_m = \frac{\text{Work output}}{\text{Swept volume}} = \frac{W_{net}}{V_s}$$

600. In a four stroke IC engine turning 1500 rev/min, how many revolutions the crankshaft will have to make for every power stroke?

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(a) 1500 (b) 375
(c) 3000 (d) 750

HPPSC Poly. Lect. 05.07.2021

Ans. (d) : In 4 stroke 2 revolution of crank shaft takes place.

$\therefore$ For every power stroke revolution is $\frac{1500}{2} = 750.$

601. Diagram factor is defined as the ratio of

(a) theoretical m.e.p. and actual m.e.p.
(b) actual m.e.p. and theoretical m.e.p.
(c) theoretical m.e.p. and swept volume
(d) actual m.e.p. and swept volume

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (b) Diagram Factor- Diagram Factor is defined as the ratio of actual m.e.p. and theoretical m.e.p., where m.e.p. is mean effective pressure.

602. The number of power strokes per minute in a two stroke engine is

(a) equal to the rpm (b) half of the rpm
(c) twice of the rpm (d) four times of the rpm

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (a) : Number of working strokes per min for a two stroke cycle engine are equal to the speed of the engine in r.p.m while the number of working strokes per min for a four stroke cycle engine are one half of the speed of the engine in r.p.m.

603. The cam shaft of a four stroke I.C. engine running at 2000 rpm will run at:

(a) 1500 rpm
(b) 1000 rpm
(c) 2000 rpm
(d) Any value independent of engine speed

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : The Camshaft is either chain or gear driven from the crank shaft. Because the engine is a four stroke, the camshaft will rotate at half the speed of the camshaft. So engine speed

$N_{engine} = 2 \times N_{camshaft}$

$N_{camshaft} = \frac{N_{engine}}{2} = \frac{2000}{2}$

$= 1000$ rpm.

604. The Brake Mean effective Pressure (BMEP) of 1200 CC four stroke engine delivering 20 kW at 2000 rpm is :

(a) 5 bar (b) 8 bar
(c) 10 bar (d) 12 bar

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : BMEP (brake mean effective pressure)

Brake power $= \frac{P_m LAnK}{60}$

n = N/2 for four stroke; n = N for two stroke

K = No. of cylinder

$\because P_m$ = BMEP

$20 \times 10^3 = P_m \times 1200 \times 10^{-6} \times \frac{2000}{2} \times 1 \times \frac{1}{60}$

$P_m = 1 \times 10^6$ N/m^2

$= 10$ bar

605. A diesel engine develops a Brake power of 4.5 kW. Its indicated thermal efficiency is 30% and the mechanical efficiency is 85%. Considering calorific value of the fuel as 40000 kJ/kg, the fuel consumption will be :

(a) 2.1 kg/hr (b) 3.9 kg/hr
(c) 1.6 kg/hr (d) 4.6 kg/hr

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) : Given, B.P. = 4.5 kW

Indicated thermal efficiency η_{ith}= 30%

Mechanical efficiency η_{mech}= 85%

Calorific value = 40000 kJ/kg

$$\eta_{mech} = \frac{BP}{IP}$$

$$0.85 = \frac{4.5}{IP}$$

$$IP = \frac{4.5}{0.85} = 5.294 kW$$

$$= 5.294 \frac{kJ}{\text{sec}} = 5.294 \times 3600 \frac{kJ}{hr}$$

$$= 19058.82 \frac{kJ}{hr}$$

$$\eta_{ith} = \frac{IP}{m_f . c_v}$$

$$m_f = \frac{IP}{\eta_{ith} \times c_v} = \frac{19058.82}{0.30 \times 40000} = 1.58 \simeq 1.6 kg/hr$$

606. A tank contains a mixture of CO_2 and N_2 in the mole proportions of 0.2 and 0.8 at 1 bar and 290 K. It is connected by a duct of sectional area 0.1 m^2 to another tank containing a mixture of CO_2 and N_2 in the mole proportion of 0.8 and 0.2. The duct is 0.5 m long. The diffusion of CO_2 is. (Take Diffusion coefficient = 0.16×10^{-4} m^2/s)

(a) 3.43×10^{-6} kg/s (b) 2.23×10^{-6} kg/s
(c) 3.5×10^{-6} kg/s (d) 3.8×10^{-6} kg/s
(e) 2.5×10^{-6} kg/s

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) Given, L = 0.5 m p = 1 bar

$A_c = 0.1$ m^2 T = 290 K

$D_{AB} = 0.17 \times 10^{-4}$ m^2/s

The partial pressure of two gases in two tanks are :

$p = p_1 + p_2$, $p_i = x_i p$

$p_{A,1} = 0.2$ bar, $p_{A,2} = 0.8$ bar

$p_{B,1} = 0.8$ bar, $p_{B,2} = 0.2$ bar

For the rate of molar diffusion of CO_2 using equation,

$$\dot{N}_A = \frac{D_{AB} \cdot A}{R_u T}\left[\frac{p_{A_2} - p_{A_1}}{L}\right]$$

$$= \frac{0.16 \times 10^{-4} \times 0.1(0.8 - 0.2) \times 10^5}{8314 \times 290 \times 0.5}$$

$= 7.96 \times 10^{-8}$ kg-mol/s

And, $\dot{m}_A = \dot{N}_A M_A = 7.96 \times 10^{-8} \times 44$

$= 3.503 \times 10^{-6}$ kg/s

$\simeq 3.5 \times 10^{-6}$ kg/s

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607. The brake of an IC Engine having speed 1200 rpm with torque 15 Nm is:

(a) 300 π watts (b) 450 π watts
(c) 150 π watts (d) 600 π watts

Kerala PSC IOF 19.04.2016

Ans. (d) $BP = \frac{2\pi N}{60} \times T$

$= 2\pi \times \frac{1200}{60} \times 15$

$= 600\pi$ watts

608. Diesel engine having brake thermal efficiency of 30% and the calorific value of fuel used is 10000 kcal/kg the brake specific fuel consumption will be

(a) 0.21 kg/HP.hr (b) 0.29 kg/HP.hr
(c) 0.39 kg/HP.hr (d) 0.49 kg/HP.hr

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (a) : We know that

Brake thermal efficiency is given as

$$\eta_b = \frac{BP}{m_f \times CV}$$

and Brake sp. fuel consumption is given as

$$bsfc = \frac{m_f}{BP}$$

So using above formulas we have,

$$bsfc = \frac{1}{\eta_b \times CV}$$

then

$CV = 10000 \times 10^3 \times 4.184$ J/kg

$$\left[\frac{J}{kg} = \frac{J}{\sec} \times \frac{\sec}{kg} = W\frac{\sec}{kg}\right]$$

$$CV = 10000 \times 10^3 \times 4.184 \times \frac{1}{746}\ HP \times \frac{hr}{3600 kg}$$

$= 15.579$ HP-hr/kg

Now we have

$$bsfc = \frac{1}{\eta_b \times CV} = \frac{1}{0.3 \times 15.579} = 0.214\ kg/HP.hr$$

609. A four-cylinder petrol engine is tested for its mechanical efficiency. During the test, spark plugs of each cylinder were inoperative and the corresponding brake loads noted are 100, 120, 130, 120 N. If the net brake load is 180 N, then the mechanical efficiency of the engine is :

(a) 75% (b) 66.6%
(c) 60% (d) 72%

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d)

$I_1 = 180 - 100 = 80$
$I_2 = 180 - 120 = 60$
$I_3 = 180 - 130 = 50$
$I_4 = 180 - 120 = 60$

Total IP = 80 + 60 + 50 + 60 = 250

$$\eta_{mech} = \frac{B.P}{I.P} = \frac{180}{250} = 0.72$$

$\eta_{mech} = 72\%$

610. Engine operating conditions may be indicated by the colour of exhaust smoke. Black smoke could indicate

(a) Insufficient speed droop setting
(b) Overloaded engine
(c) Clogged drain holes in the oil control rings
(d) Complete combustion

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Black exhaust smoke can appear when the vehicle is burning to much fuel/gasoline in order to make the engine run. This condition is occurred during the overloaded engine & this could be caused by a clogged air filter, malfunctioning fuel injection system and a blocked manifold.

611. A Mercedes-Benz car engine has swept volume of 300 cc and clearance of 25 cc. Its volumetric efficiency is 0.88 and mechanical efficiency is 0.90. The volume of mixture taken in per stroke is:

(a) 198 cc (b) 164 cc
(c) 264 cc (d) 284 cc

AAI Jr. Executive 29.11.2018

Ans. (c) : $V_s = 300$ cc

$V_c = 25$ cc

Volumetric efficiency = 0.88

Mechanical efficiency = 0.90

The volume of mixture taken per stroke = Actual volume

$$\eta_{vol} = \frac{\text{Actual volume}}{\text{Swept volume}}$$

$$0.88 = \frac{\text{Actual volume}}{300}$$

Actual volume = 264 cc

6. Engine Cooling and Lubrication

612. Which is more viscous lubricant oil given below?

(a) SAE 30 (b) SAE 40
(c) SAE 70 (d) SAE 80

TNPSC AE 2017, GWSSB AAE, 27.12.2015
GPSC AE 26.07.2015

Ans. (d) : According to The society of Automotive engineers (SAE), SAE 80 is more viscous lubricant oil.

613. The following type of Lubrication system is used in two stroke engines.

(a) Petroil (mist) system
(b) Wet sump system
(c) Dry sump system
(d) All of these

RPSC ACF & FRO, 26.02.2021
HPPSC W.S. Poly. 2016
GPSC Lect. 23.10.2016

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Ans. (a) : Petroil (mist) lubrication system is used in two stroke engine.

Wet sump lubrication system & dry sump lubrication system is used in four stroke engine.

Wet sump system- Small engines such as automobile engines.

Dry sump system- Large stationary, marine and air craft engines.

Mist/Petroil lubrication – Two stroke

Wet sump, Dry sump } → Four stroke.

614. In automobile engines a thermostat is provided for :

(a) Regulating the temperature of suction air
(b) Regulating the temperature of lubrication oil
(c) Controlling the temperature of the cooling system
(d) Regulating the temperature of exhaust gases

RPSC ADE 2016, RPSC AE 2016
RPSC ACF-2011

Ans. (c) : When the engine is cold, the thermostat is closed and prevents coolant from flowing into the engine and allows the engine to warm up

Once the engine has warmed, the thermostat will open and allow coolant to flow and cool the engine when it starts to get too hot. This function is vital for optimal engine performance.

615. The purpose of thermostat in an engine cooling system is to

(a) Indicate coolant temperature
(b) Prevent coolant from boiling
(c) Allow the engine to warm up quickly
(d) Pressurize coolant for effective cooling

RPSC IOF, 2020
ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (c) : Any liquid-cooled car engine has a small device called thermostat that sits between the engine and radiator. The thermostat is most in cars about 2 inches in diameter. It used to block the flow of coolant to the radiator until the engine has warmed up. When the engine is cold, no coolant flows through the engine. Once the engine reaches its operating temperature (generally about 95ºC), the thermostat opens. By lifting the engine warm up of quickly as possible, the thermostat reduces engine wear, deposits and emissions.

616. Detergents are oil additives used to

(a) Reduce viscosity
(b) Increase fire point
(c) Prevent sludge formation
(d) Prevent foaming

RPSC IOF, 2020

Ans. (c) : Detergent additives are used to clean and neutralize oil impurities which would normally cause deposits (oil study) on vital engine part typical detergents are magnesium sulfonates. Corrosion or rust inhibiting additives retard the oxidation of metals inside an engine.

617. In a normally operating diesel engine the main source of lubricating oil contamination in the crank case is a result of the

(a) Metal particles loosened by wear
(b) Condensation of water vapour
(c) Break down of the lubricating oil itself
(d) Fuel dilution

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Note- No answer is given by commission.

618. The lower cylindrical portion of the piston which improves piston cooling performance is called:

(a) Piston crown (b) Connecting rod
(c) Piston pin boss (d) Piston skirt

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : Piston skirt is the cylindrical walls of a piston. The surface of the piston skirt is left slightly rough which helps retain lubrication. Due to lubrication, it improves piston cooling performance.

619. The function of oil ring provided on piston is :

(a) To provide gas seal
(b) To meter and distribute oil directed on cylinder wall
(c) To conduct heat from piston to cylinder wall
(d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : The function of oil ring provided on the piston is to meter and distribute oil directed on cylinder wall.

620. Additives are added in an aqueous ethylene glycol solution, which is used as engine coolant.

(a) To prevent metal corrosion
(b) To prevent a buffering action against acids
(c) To dissolve certain additives to improve antiform property and ready identification
(d) All options are correct.

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Additives added in engine coolant for following purposes-

(1) To prevent metal corrosion
(2) To prevent a buffering action against acids
(3) To dissolve certain additives to improve antiform property and ready identifications.
(4) Antifreeze material.

621. The volumetric efficiency of an air cooled engine is lower due to..

(a) Simple engine design
(b) Use of fins on cylinder
(c) Higher cylinder head temperatures
(d) Higher specific output from engine.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : The volumetric efficiency of an air cooled engine is lower due to higher cylinder head temperatures.

Volumetric efficiency $(\eta_v) = \frac{V_a}{V_s}$

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Where, V_a = Actual volume
V_s = Swept volume
Volumetric efficiency- It is the ratio of the actual volume of charge admitted during the suction stroke at NTP to the swept volume of piston.

622. Accumulation of rust and scale in the engine cooling system will cause
(a) Slow worm up
(b) Reduced output
(c) Low exhaust temperature
(d) Overheating

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : In engine cooling system, accumulation of rust and scale in the engine block the flow of cooling causing the engine to overheat and leak, which lead to very expensive repairs.

623. Mark the incorrect statement :
In air cooling of IC engines, the heat dissipated depends on
(a) The area of the cooling surface in contact with air
(b) Mass rate of flow of air
(c) Temperature difference between the cylinder and air
(d) Conductivity of the metal
(e) None of these

CGPSC AE 16.10.2016

Ans. (e) : In an air cooled system, heat generated due to combustion in the engine, will be conducted to the fins when air flow over fins and heat dissipated depends on the area of cooling surface in contact with air, mass rate of flow of air, temperature difference between both air & engine cylinder & conductivity of metal.

624. Force feed lubrication system means that oil is delivered to the engine by
(a) Gravity
(b) Pressure created by oil pump
(c) Splashing of the crank shaft
(d) None of the above

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (b) : Pressure created by oil pump for force feed lubrication system in engine.

625. Lubrication system for high capacity engines
(a) Wet Sump Lubrication
(b) Mist Lubrication
(c) Splash system
(d) Dry Sump Lubrication

TNPSC AE 2013

Ans. (d) : Dry sump lubrication– Dry sump lubrication system is an engine lubrication system in which the lubrication oil is carried in an external tank and not internally in a sump. The sump is kept relatively free from oil by scavengine pumps, which return the oil to the tank after cooling. This is the opposite of a wet pump system.

626. In petrol engine, increases of cooling water temperature will
(a) Increase the knocking tendency
(b) Decrease the knocking tendency
(c) Not affect the knocking tendency
(d) Increase or decrease knocking tendency depending on strength and time of spark

TNPSC AE 2017

Ans. (a) : In petrol engine, increases of cooling water temperature will increase the knocking tendency.

627. The thermostat in I.C. engines permitting hot water to go to radiator is set around
(a) 70 - 80º C (b) 80 - 85º C
(c) 85 - 95º C (d) Above 100º C

TNPSC AE 2017

Ans. (b) : The function of thermostat in I.C. engine is block the flow of coolant to the radiator until the engine has warmed up once the engine reaches its operating temperature (generally 80 - 85°C), the thermostat opens.

628. Most commonly used lubrication system in heavy vehicles is the
(a) Splash Lubrication system
(b) Pressure Lubrication system
(c) Gravity Lubrication system
(d) Petrol Lubrication system

TNPSC AE 2017

Ans. (b) : Most commonly used lubrication system in heavy vehicles is the Pressure Lubrication system.

629. A system in which the lubricating oil is carried in separate tanks from where it is fed to the engine is called the :
(a) dry sump system
(b) wet sump system
(c) splash system
(d) mist lubrication system

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : In a dry-sump system the oil still falls to the base of the engine, but into a much shallower sump. Where one or more scavenge pump draw it away and transfer it to a (usually external) reservoir where it is both cooled and de-aerated before being re-circulated through the engine by a pressure pump.

630. The function of oil scrapper ring is to :
(a) Retain compression
(b) Lubricate cylinder walls
(c) Maintain vaccum
(d) Reduce piston wear

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (b) The function of oil scrapper ring is to lubricate cylinder walls.

631. Excessive pressure build up in force feed lubrication system prevented by :
(a) Relief valve
(b) Stop valve
(c) Balancer
(d) Pressure reducing valve

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (a) Excessive pressure build up in force feed lubrication system prevented by relief valve.

632. Which type of pump is preferred for pumping lubricating oil in automobile:
(a) Open impeller type
(b) Centrifugal pump with impeller shrouded on one side only

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(c) Centrifugal pump with impeller shrouded on both side only
(d) Gear pump

RPSC ADE 2016

Ans. (d) : Gear pump is preferred for pumping lubricating oil in automobile.

7. Automobile Engineering

633. For a four cylinder vertical engine, the commonly used firing order is

(a) 1 – 2 – 3 – 4 (b) 3 – 1 – 2 – 4
(c) 4 – 2 – 3 – 1 (d) 1 – 3 – 4 – 2

Assam PSC AE (PHED), 18.10.2020
RPSC IOF, 2020, RPSC AE 2016
TNPSC AE 2014, APGENCO AE 2012
RPSC ACF- 2011
J&K PSC Civil Services Pre, 2010

Ans. (d) : 1 – 3 – 4 – 2 firing order is commonly used for a four cylinder engine. For the four-cylinder engine, the firing order is decided such that a power stroke occurs every 180 degrees.

634. Piston compression rings are manufactured by:

(a) Aluminium –WC (b) Cast iron
(c) HSS (d) Any material

HPPSC Lect. (Auto) 23.04.2016
UKPSC AE 2012 Paper–II
UKPSC AE 2007 Paper -II, ESE 2007

Ans. (b) : Piston compression rings are manufactured by cast iron has high damping capacity. Ductile material causes blow by loss.

635. The NOx level in engine exhaust gas can be reduces by following method :

(a) Thermal reactor
(b) Exhaust gas recirculation
(c) Advance spark timing or fuel injection
(d) Use of after burner

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
Assam PSC AE (PHED), 18.10.2020
OPSC AEE 2019 PAPER - II

Ans. (b) : The NO_X level in engine exhaust gas can be reduces by Exhaust gas recirculation (EGR).

636. The three way catalytic converter cannot control which one of the following?

(a) HC emission (b) CO emission
(c) NO_x emission (d) PM emission

GPSC ARTO Pre 30.12.2018
MPPSC AE 2016, ESE 2008

Ans : (d) PM emission

637. The method of governing for a petrol engine is:

(a) Hit and misgoverning
(b) Quality governing
(c) Quantity governing
(d) Misgoverning

APGENCO AE, 2017
OPSC AEE 2015 PAPER - II
TSPSC AEE 2015

Ans : (c) Governing of I.C. Engine:- In order to have a high efficiency of an I.C. engine at different load conditions, its speed must be kept constant as far as possible. The process of providing any arrangement, which will keep the speed constant is known as governing of I.C Engines.

(i) The method of governing for a petrol engine is Quantity governing.
(ii) The method of governing for a diesel engine is Quality governing.

638. Consider the following emissions of an IC engine:

1. CO 2. Smoke Emissions
3. No_x 4. Particulate Matter
5. HC

Which of these emissions cause photochemical smog?

(a) 1 and 2 (b) 1, 2 and 3
(c) 2, 3 and 4 (d) 2, 3 and 5
(e) 3 and 5

UPPSC AE 12.04.2016 Paper-II
CGPSC AE 26.04.2015 SHIFT-II
ESE 1999

Ans. (e) : Nitrogen oxide (NO_x) , hydrocarbons, carbon monoxide, particulate matter, hydro fluorocarbon 134a, CH_4, and nitrous oxide emit by vehicles. but NO_x and HC emission cause photochemical smog.

639. Longitudinal distance between the centers of front and rear axles of an automotive vehicle is called as.....

(a) Wheel track (b) Wheel base
(c) Wheel over hang (d) None of these

GPSC Lect. 23.10.2016
RPSC AE 2016, RPSC ACF-2011

Ans. (b) : Wheel base is the horizontal distance between the centre of the front and rear wheels or distance between the centers of front and rear axles.

640. Engine pistons are usually made of aluminum alloy because it :

(a) Is lighter (b) Wears less
(c) Absorbs shocks (d) Is stronger

GWSSB DEE 07.07.2016
GWSSB AAE, 27.12.2015
J&K PSC Civil Services Pre, 2013

Ans. (a) : Aluminum alloy piston has high thermal conducting so less variation in temperature from the crown to piston ring. It is also less density therefore light weight construction and less inertia force.

641. The design of piston head is based on,

(a) strength and rigidity considerations
(b) bending and torsional moments
(c) buckling consideration
(d) strength and heat transfer considerations

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (d) : The piston head of an internal combustion engine was designed to withstand the cylinder pressure and dissipate the heat as quickly as possible. Therefore the thickness of piston head was determined based on two criteria, strength to withstand the cylinder pressure and heat dissipation.

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642. Which one of the following in-line engine working on a four stroke cycle is completely balanced inherently?

(a) 2 cylinder engine (b) 3 cylinder engine
(c) 4 cylinder engine (d) 6 cylinder engine

Assam PSC CCE Pre 2015
UKPSC AE-2013, Paper-I

Ans. (d) : In-line engine working on a four stroke cycle is completely balanced inherently 6 cylinder engine.

643. Which of the following set of material in most commonly used in catalytic converter is for CI engines?

(a) Platinum, Palladium and Rhodium
(b) Palladium, Rhodium and Ruthenium
(c) Rhodium, Ruthenium and Platinum
(d) Ruthenium, platinum and Palladium

MPPSC AE 2016, ESE 2008

Ans. : (a) Platinum, Palladium, and Rhodium set of materials in most commonly used in catalytic converters for CI engine.
Platinum, Rhodium for-NO_x elimination.
Platinum, Palladium for-CO, HC elimination.

644. In loop scavenging top of the piston is :

(a) Flat (b) Contoured
(c) Slanted (d) Depressed

Kerala PSC Poly. Lect. (Auto) 05.05.2016
TANGEDCO AE 2015

Ans. (b) : In loop scavenging top of the piston is contoured.

645. The most commonly used supplementary restraint system is

(a) Seat belt
(b) Disc brakes
(c) Air bags
(d) Telescopic steering column

TNPSC AE 2017, KPCL AE 2016

Ans. (c) : The most commonly used supplementary restraint system is Air bags.

646. The top ring nearest to the piston crown is known as:

(a) Compression ring (b) Oil ring
(c) Scrapper ring (d) Grove ring

RPSC ACF & FRO, 26.02.2021
TNPSC AE (Industries) 09.06.2013

Ans. (a) : The top ring nearest to the piston crown is known as compression rings.

647. Toe-in is adjusted by :

(a) Adding slims
(b) Changing the effective length of the tie rods
(c) Changing gear trains with crank
(d) None of the above

HPPSC Lect. (Auto) 23.04.2016
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (b) : Toe-in and Toe-out adjustment are made by screwing the track rod ball joint end up or down the track rods to alter their length. It is not possible to make totally accurate adjustments at home without proper wheel aligning equipment. Hence Toe-in is adjusted by changing the effectives length of the tie rod.

648. Purpose of using differential gear in automobile is to.

(a) Help in turning
(b) Control speed
(c) Avoid jerks
(d) Obtain rear movement

UPRVUNL AE 05.07.2021
ISRO Scientist/Engineer 2006

Ans. (a) : The purpose of differential gear on rear axle is to allow the back wheels on at different speed during turn and at same speed in straight motion. Because at turn the outer wheel moves faster than inner to cover more distance and inner wheel moves slow to cover less distance.

649. In automobiles Hooke's joint is used between which of the following?

(a) Clutch and gear box
(b) Differential and wheels
(c) Flywheel and clutch
(d) Gear box and differential

UPRVUNL AE 21.08.2016
TSPSC AEE 2015

Ans. : (d) In Automobiles, Hooke's Joint is used between gear box and differential.
Hooke's Joint – A hooke's joint is used to connect two shaft which are interesecting at a small angle
Application – The main application of the universal or Hooke's joint is found in the transmission from the gear box to the differential or back axle of the Automobiles.

650. A Gasoline engine running in a closed room is dangerous because the exhaust gas contains mainly

(a) Blue smoke (b) Water vapour
(c) Carbon monoxide (d) Air

TNPSC AE 2017, GPSC AE 26.07.2015

Ans. (c) : A Gasoline engine running in a closed room is dangerous because the exhaust gas contains mainly carbon monoxide.

651. The effective torque at the contact between road and driving wheel is known as :

(a) engine torque (b) friction power
(c) tractive effort (d) brake power

APPSC Poly Lect. Automobile Engg., 2020
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c) : Tractive Effort– Tractive Effort is defined as the force at the rims or the outer edges of the driving wheels of the moving trains. In the other words it is the sum of the tractive force and rolling effort on the road surface.

652. The dynamo in an automobile :

(a) Converts mechanical energy into electrical energy
(b) Continually recharge the battery
(c) Acts as a reservoir of electrical energy
(d) Supplies electric power

RPSC AE 2016, RPSC ACF-2011

Ans. (a) : The dynamo in an automobile converts mechanical energy into electrical energy.

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653. An automobile moving at a velocity of 40 km/hr is experiencing a wind resistance of 2 kN. If the automobile is moving at a velocity of 50 km/hr, the power required to overcome the wind resistance is

(a) 43.4 kW (b) 3.125 kW
(c) 2.5 kW (d) 27.776 kW

UPPSC AE 13.12.2020, Paper-2
ISRO Scientist/Engineer 2006

Ans. (a) : Given,

$F_1 = 2$ kN

$v_1 = 40$ km/hr

$v_2 = 50$ km/hr

$$\text{Wind Resistance } (F) = \frac{1}{2} C_D \rho A v^2$$

$$F \propto v^2$$

$$\frac{F_1}{F_2} = \frac{v_1^2}{v_2^2}$$

$$\frac{2}{F_2} = \left(\frac{40}{50}\right)^2 \Rightarrow F_2 = 3.125\,\text{kN}$$

Power required, $P = F \times v$

$$= 3.125 \times 50 \times \frac{5}{18}$$

$P = 43.4$ kW

654. Precise petrol injection system is :

(a) Direct injection
(b) Sequential injection
(c) Throttle body injection
(d) Port injection

HPPSC W.S. Poly. 2016

Ans : (c) Precise petrol injection system is throttle body injection.
The injector or atomizer is also an important part of the diesel engine which breaks up the fuel and sprays into the cylinder into a very fine divided particles.

655. For premature ignition of any local hot-spot to occur in advance of the time spark on the combustion stroke it must attain a minimum temperature of something like

(a) 700-800ºC (b) 300-400ºC
(c) 400-500ºC (d) 900-1000ºC

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Pre-ignition is the ignition of the homogenous mixture in the cylinder, before the time ignition spark occurs, caused by the local overheating of the combustible mixture.
For pre-mature ignition of any local hot-spot to occur in advance of the timed spark on the combustion stroke, it must attain a minimum temperature of something like 700-800ºC

656. The float of a carburetor controls :

(a) Fuel flow rate
(b) Fuel and air mixture flow rate
(c) Air flow rate
(d) Level of fuel

BPSC Asstt. Prof. 29.11.2015

Ans. (d) : The float of a carburetor controls level of fuel in float (fuel) chamber.
The main function of carburetor to mix air and gasoline and provides a high combustion mixture. There are also addition function of carburetor.

- It controls the engine speed.
- It regulates the air-fuel ratio.
- It increases/decreases the amount of mixture according to the engine speed and load changing.

657. Dynamo in automobile is a :

(a) Series generator
(b) Shunt generator
(c) Either of series generator or shunt generator
(d) none of these

RPSC ADE 2016

Ans. (b) : Dynamo in automobile is a shunt generator dynamo convert mechanical energy into electrical energy.

658. The fins at the top of a motor cycle engine cylinder are longer than those at the bottom because:

(a) Hot air rises
(b) The top is the hottest part
(c) They are in an unexposed position
(d) Extra strength is required at the top

RPSC VPITI 14.02.2016

Ans. (b) : Longer fins provided on the top of motor cycle engine cylinder because the top of the engine is hottest part so that larger heat convection area can be provided.

659. The word "castor" in automobile is associated with which of the following:

(a) Transmission system
(b) Oils for automobile
(c) Front axle alignment
(d) Propeller shaft for power transmission

RPSC VPITI 14.02.2016

Ans. (c) : Castor angle is the angular displacement of the steering axis from the vertical axis of a steered wheel in a car, motor cycle, bicycle as seen from the side of the vehicle.
Caster causes a wheel to align with the direction of travel and can be accomplished either by caster displacement or caster angle.
Caster displacement moves the steering axis ahead of the axis of wheel rotation, caster angle moves the steering axis from vertical.

660. For cars, which is the most popular manual steering gear?

(a) Worm and nut type
(b) Worm and wheel type
(c) Cam and roller type
(d) Rack & pinion type

RPSC VPITI 14.02.2016

Ans. (d) : Many modern cars uses rack and pinion steering mechanism where the steering wheels turn on pinion gear. The pinion moves the rack. Which is linear gear that meshes with the pinion, converting circular motion into linear motion along the transverse axis of cars.

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661. The angle between the King-pin centre line and the vertical in the plane of the wheel is called :

(a) King –pin inclination (b) Caster angle
(c) Camber angle (d) Included angle

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : King Pin Inclination–It is the angle between king pin centre line and vertical line when seen from the front of the vehicle. It is also called steering axle inclinator.

Caster Angle–Caster is the angle between the vertical line and king pin centre line in the plane of wheel when viewed from the side is called the caster angle.

Camber Angle–Camber is the angle at which wheel and the stored relative to the road-assuming it is perfectly flat.

Included Angle–It is sum of camber and S.A.I. (Steering axis inclination).

662. Which is the combination for following statements for suspension system?

(a)	Antiroll bar	(I)	High wheel wobbling
(b)	Independent suspension	(II)	High damping
(c)	Rigid axle suspension	(III)	Reduce rolling effect
(d)	Leaf spring	(IV)	High riding comfort

(a) (a-III), (b-II), (c-IV), (d-I)
(b) (a-III), (b-IV), (c-I), (d-II)
(c) (a-IV), (b-III), (c-II), (d-I)
(d) (a-IV), (b-III), (c-I), (d-II)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) :

663. Maintenance that involves a system of periodic inspection and maintenance designed to keep a machine in operation is called.

(a) Preventive maintenance
(b) Total productive maintenance
(c) Predictive maintenance
(d) Breakdown maintenance

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Preventive maintenance is maintenance that is regularly performed on a piece of equipment to lessen the likelihood of it failing. It is performed while the equipment is still working so that it does not break down unexpectedly, to keep a machine in operation.

664. Condition for correct steering for a four-wheeler is:

(a) $\sin\phi - \sin\theta = \frac{W}{H}$

(b) $\tan\phi + \cot\theta = \frac{W}{H}$

(c) $\cot\phi - \cot\theta = \frac{W}{H}$

(d) $\cos\phi - \cos\theta = \frac{W}{H}$

Where W = wheel base : H = distance between the pivots of front axle, and ϕ, θ are angles through which axes of outer and inner wheels turn, respectively.

CGPSC AE 15.01.2021

Ans. (c) : Condition for correct steering for a four wheeler is

$$\boxed{\cot\phi - \cot\theta = \frac{W}{H}}$$

where, W = wheel base
H = distance b/w pivots of front axle
ϕ & θ are angles through which axes of outer and inner wheels turn respectively.

665. ……. type nozzle can be used in automobiles for converting the high pressure of fuel into velocity.

(a) Prandtl (b) Bezier
(c) Pintle (d) Ericsson

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : Types of Nozzle in IC engine
- Pintle nozzle
- Single hole nozzle
- Multi hole nozzle
- Pintaux nozzle.

Nozzle is that part of an injector through which the liquid fuel is sprayed into the combustion chamber. It is used in diesel engine in which fuel is drawn separately through injector of end of compression stroke and air is drawn into cylinder in suction stroke.

Hence pintle type nozzle can be used in automobiles for converting the high pressure of fuel into velocity.

666. The point at which the centre line of the wheel and the centre line of the ball joints cross is called the point of :

(a) Intersection (b) Aperture
(c) Cross-section (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (a) : The point at which the centre line of the wheel and the centre line of the ball joints cross is called the point of intersection.

667. In order to facilitate starting of locomotive in any position, the cranks of a locomotive with two cylinders are placed at-

(a) 45° to each other (b) 90° to each other
(c) 120° to each other (d) 180° to each other

NPCIL ET, 2015

Ans. (b) : In order to facilitate starting of locomotive in any position, the cranks of a locomotive with two cylinders are placed at 90º to each other.

668. The choke in an automobile is meant for supplying

(a) Lean mixture
(b) Weak mixture
(c) Stoichiometric mixture
(d) Rich mixture

Assam PSC AE (PHED), 18.10.2020

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Ans. (d) : In an automobile engine such as in an internal combustion engine a choke valve modifies the air pressure in the intake manifold, thereby altering the ratio of fuel and air quantity entering the engine. Choke valves are generally used in naturally aspirated engines with carburetors to supply a richer fuel mixture at starting the engine.

669. The device used to capture fuel vapour from carburetor is

(a) Lambda sensor (b) Catalytic converter
(c) Canister (d) Scrubber

Assam PSC AE (PHED), 18.10.2020

Ans. (c) Canister- An activated carbon canister is used to capture hydrocarbon vapor emissions from the carburetor or fuel tank as part of an evaporative emission control system.

Lambda Sensor- An oxygen sensor is an electronic device that measures the proportion of oxygen in the gas or liquid being analysed.

Catalytic convertor- A catalytic converter is an exhaust emission control device that reduces toxic gases and pollutant in exhaust gas from an internal combustion engine into less toxic pollutant by catalyzing a Redox reaction.

Scrubber- Scrubbers comprise those devices in which a power driven rotor produces the fine spray and the contacting of gas and liquid. As in other types of scrubber, it is the droplets that are the principal collecting bodies for the dust particles.

670. Catalytic converters are used to convert____ through____ reaction at the exhaust of the engine.

(a) CO and HC; reduction
(b) CO_2 and H_2O; oxidation
(c) CO and HC; oxidation
(d) CO_2 and H_2O; reduction

Assam PSC AE (PHED), 18.10.2020

Ans. (c) : Catalytic converters are used to convert CO and HC through oxidation reaction at the exhaust of the engine.

671. The gearbox in automobile is placed between.

(a) Clutch and differential
(b) Steering and engine
(c) Engine and clutch
(d) Differential and Hooke's joint

Assam PSC AE (PHED), 18.10.2020

Ans. (c) : The main function of the clutch is to transmit the power from engine to the transmission system. The clutch is engaged when the vehicle is running.

The purpose of the gear box is to give means so that the torque can be varied as required between engine and the road wheels.

The gear box is placed between engine and clutch.

672. Gudgeon pin forms the link between

(a) Piston and big end of the connecting rod
(b) Piston and small end of the connecting rod
(c) Connecting rod and crank
(d) Big end and small end

Assam PSC AE (PHED), 18.10.2020

Ans. (b) : In internal combustion engine the gudgeon pin, wrist pin or piston pin connects the piston to the small end of the connecting rod to pivot upon as the piston moves.

A gudgeon is a pivot or journal.

673. Advantages of radial ply-tyres over cross ply-tyres is/are :

(a) Less rolling resistance
(b) Longer life
(c) Higher braking efficiency
(d) All of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : A radial ply tyre is a particular design of vehicular tyre. In this design the cord plies are arranged at 90° to the direction of travel or radially from the centre of tyre.

Advantage of radial ply tyres over cross ply tyres are :

1. Less rolling resistance.
2. Longer life
3. Higher breaking efficiency.
4. Reduced fuel consumption due to less rolling resistance.
5. Radial ply tyres provides good steering control because of the flexibility that the tyres often which results in better control of vehicle.

674. Aerodynamic drag caused by the wind resistance depends upon the :

(a) Shape of the vehicle
(b) Frontal area of the vehicle
(c) Air density and vehicle speed
(d) All of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Aerodynamic drag caused by the wind resistance depends upon the-

1. Shape of the vehicle
2. Frontal area of the vehicle
3. Air density and vehicle speed.

Factors that affect Drag of –

The object of – Shape and size

The motion of – Velocity and indination to flow.

The Air of – mass, viscosity, compressibility.

675. Full floating axle

(a) Transmit torque only
(b) Take side thrust and transmit torque
(c) Take weight of vehicle side thrust and transmit torque
(d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Full Floating Axle – A Full Floating axle consist of a wheel hub assemble that is separate from the axle shaft. A spindle bolted to the axle tube supports the wheel hub by means of a pair of wheel bearings Therefore, the weight of the vehicle and its cargo is transfered to the axle tube, rather than the axle shaft itself. As a result, a full floating axle shaft is not subjected to the bending moment or shear force. It's only task is to transmit power to the wheel hub. As a result, the shaft is only subjected to torsional loads.

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676. If the brake wheels gets locked before vehicle stops the vehicle is said to be :

(a) Skidding (b) Rolling
(c) Pitching (d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : If the brake wheels gets locked before vehicle stops the vehicle is said to be skidding.

677. What is a cause of brake disc run out; which happens for brakes employed in automotives? Choose most appropriate option :

(a) Ineffectiveness of the brakes
(b) Judder while braking
(c) Wearing of brake pads at restricted areas
(d) Quick wearing of brake pads.

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : Judder during braking is caused as a result of brake disc run out.

678. What is purpose of differential gear box?

(a) To control the speed and torque of the vehicle as per requirement
(b) To transfer the power from engine to front wheel.
(c) To allow each wheel to rotate independently each at its own speed so there will not "wind-up"
(d) It is equally applicable for straight roads as well as at the time turn for smooth power transmission.

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : Differential gear, in automotive mechanics, gear arrangement that permits power from the engine to be transmitted to a pair of driving wheels, dividing the force equally between them but permitting them to follow paths of different lengths as when turning a corner or traversing an uneven road. On a straight road the wheels rotate at the same speed, when turning a corner the outside wheel has farther to go and will run faster than inner wheel.

679. For wheel alignment, which of the following parameters are most important ?

1. Camber
2. Steering axis inclination (SAI)
3. Toe
4. Caster

(a) Statement 1, 2, 3 and 4
(b) Statement 1 and 3
(c) Statement 1, 2 and 4
(d) Statement 1

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : For wheel alignments important parameters are –
1. Camber
2. Steering axis inclination (SAI)
3. Toe
4. Caster

680. Suspension systems have following primary functions.

(a) Isolate passengers and cargo from vibration and shock.
(b) Improve mobility and provides lateral and longitudinal stability and resists chassis roll.
(c) Provide the vehicle control
(d) All above options are correct.

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Primary Function of suspension systems-
1. Isolate passengers and cargo from vibration and shock.
2. Improve mobility and provides lateral and longitudinal stability and resist chassis roll.
3. Provide the vehicle control.

681. What are the factors affecting the rolling resisting of vehicle?

(a) Tyre material
(b) Flex side wall and contact area of the tyre
(c) Pressure in tyre may change inflation
(d) All above options are correct

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Factors affecting rolling resistances of vehicle-
1. Tyre material
2. Tyre diameter, tyre width.
3. Flex side wall and contact area of the tyre.
4. Tyre construction, tyre thread
5. Tyre inflation pressure.

682. An ideal antilock braking system measures skid by:

(a) Measuring the difference between wheel speed and vehicle speed
(b) Differentiating vehicle speed with respect and vehicle speed
(c) Measuring crank shaft angular speed
(d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : An ideal antilock braking system (ABS) measures skid by measuring the difference between wheel speed and vehicle speed.
The function of anti-lock brake system (ABS) is that it maintains directional control during braking by preventing the wheels from locking.

683. Springing and damping on a vehicle are not responsible for

(a) Ride comfort and dynamic wheel load.
(b) Kinematic changes and elasto-kinematics of the wheels as they hung and jounce
(c) Direction change of the vehicle
(d) The tendency of the body to roll and pitch.

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : Springing and damping suspension on a vehicle are responsible for :
1. Comfort ride and dynamic wheel
2. Kinematic changes and elasto kinematics of the wheels as they hung and jounce.
3. The tendency of the body to roll and pitch.
It is not responsible for direction change of vehicle.

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684. The distance between front of the front tyres and rear of the front tyre is called :

(a) Toe in or toe-out
(b) King pin inclination
(c) Camber angle
(d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : The distance between front of the front tyres and rear of the front tyre is called toe-in or toe-out.

685. Which is the odd option with respect to parameters affecting the engine heat transfer?

(a) Fuel air ratio, compression ratio
(b) Cam shaft and material of crank shaft
(c) Spark advance, pre-ignition and knocking
(d) Engine output, cylinder wall temperature

GPSC Lect. (Auto) 16.10.2016

Ans. (b) :

686. The wheel at a moving car posses?

(a) Potential energy
(b) Kinetic energy of translation and Rotation both
(c) Kinetic energy of translation
(d) Kinetic energy of rotation

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : The wheel of a moving car posses kinetic energy of translation and Rotation both.

687. With which of the following engines term scavenging is associated ?

(a) Aero Engine
(b) Diesel Engine
(c) High efficiency engines
(d) None of these

J&K PSC Civil Services Pre, 2010

Ans. (c) : The term scavenging is generally associated with high efficiency engines. In automobiles, scavenging is a process of pushing exhaust gas out of the cylinder and drawing in fresh draught of air or air fuel mixture for the next cycle. The process is essential in having a smooth running of I.C. engine. If scavenging is incomplete of the following stroke will begin with mix of exhaust fumes rather than clean air. This may be insufficient for poor combustion which leading to poor running condition on an engine. Scavenging is equally important for both 2 strokes and 4-stroke engine, However, its more difficult to achieve in 2 stroke engine.

688. What is the purpose of a particulate trap?

(a) Filter the impurities, oxidize them and cleans the filter
(b) It is a three way catalyst which removes all the three pollutants simultaneously
(c) It is a single converter used to oxidize carbon monoxide and hydrocarbons
(d) It is a thin steel linear in a thermal reactor to minimize heat losses by thermally isolating the core

APPSC IOF, 2009

Ans. (a) : A DPF typically uses, a substrate made up of a ceramic material formed into a honey comb structure in order to reduce emissions from diesel vehicles and store exhaust soot, which must be periodically burned off to regenerate the filter.

689. Exhaust Gas Recirculation is integrated in automobiles in order to :

(a) Increase the NO_x emision
(b) Reduce exhaust temperature of flue gases
(c) Increase engine peak operating temperature
(d) Reduce engine peak operating temperature

KPSC ADF 2015

Ans. (d) : Exhaust gas Recirculation is integrated in automobiles in order to reduce engine peak operating temperature. The main advantage of exhaust gas recirculation is reduce the emission of NO_x.

690. What is the function of a shackle with a leaf spring?

(a) Control sideways.
(b) Allow spring to expand in length.
(c) Allow pivoting of spring end
(d) Articulated vehicle's

MPSC Dy. Engg. (GWD) 19.05.2013

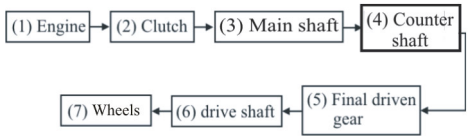
Ans. (b) : The-Shackle used in leaf spring to takes up the tendency to elongate or expand the leaf spring when compressed and support vehicles weight.

691. The correct flow of power through the driven train in automobile is.

(a) Engine-Driveshaft-Clutch-Mainshaft-Countershaft-Final driven gear-Wheels
(b) Engine-Clutch-Main Shaft-Countershaft-Final driven gear-Driveshaft-Wheels.
(c) Engine-Clutch-Countershaft-Main Shaft-Final driven gear-Driveshaft-Wheels.
(d) Engine-Main shaft-Countershaft-Clutch-Final driven gear-Driveshaft-Wheels.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Correct flow of power in Automobile –



692. Which of the following is not a defect of a chassis frame?

(a) Broken weld (b) Dislocated parts
(c) Cracks (d) Knock

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Defects of a chassis frame are –
(1) Broken weld
(2) dislocated parts
(3) cracks

693. Due to driving or braking efforts, a load is acting at the centre of the wheel called as.

(a) Fore and aft load (b) Vertical load.
(c) Side thrust (d) Shock load

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Due to driving or braking efforts, a load is acting at centre of the wheel called as fore and aft load.

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694. Incorrect caster angle causes

a) Hard steering and develop a tendency to wander

b) Vehicle would pull to one side when the brakes are applied

c) Clutch fails to disengage

d) Excessive tension loss at engine operating temperature

Which of the above statements is /are correct.

(a) a only (b) a & b only

(c) a, b and c (d) All of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Incorrect caster angle causes-

(a) Hard steering and develop a tendency to wander

(b) Vehicle would pull to one side when the brakes are applied.

695. ….are used to connect leaf springs with the chassis frame.

(a) Spring shackles

(b) Torsion bars

(c) Helper springs.

(d) Rubber torsion units

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Spring shackles are used to correct leaf spring with the chassis frame. It takes up the tendency to elongate the leaf spring when compressed and thus makes for soften springness.

696. As per Indian Automobiles engine specifications, small cars are considered…

(a) up to 1000 cc (b) 1000-2000 cc

(c) More than 2000 cc (d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : As per Indian Automobiles engine specifications, small cars are considered up to 1000 cc.

697. The steering and braking systems of automobile are parts of

(a) Power plant of engine

(b) Basic structure of vehicle

(c) Control system of vehicle

(d) Accessories of vehicle

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : The steering and braking systems of automobile are parts of control system of vehicle.

698. The commonly used material for frame construction is …

(a) Cold rolled open hearth steel

(b) Brass

(c) Copper

(d) Hot rolled steel

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : The commonly used material for frame construction is cold rolled open hearth steel.

699. In cars, generally…. chassis is used.

(a) Full forward (b) Semi forward

(c) Central (d) Overhang

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : In cars, generally full forward chassis is used.

700. The Vahan and Sarathi are conceptualized to capture the functionality as mandated by which Act :

(a) Central Motor Vehicle Act, 1988

(b) Central Motor Vehicle Act, 1999

(c) Central Motor Vehicle Act, 1989

(d) None of the above

GPSC ARTO Pre 30.12.2018

Ans. (a) : The Vahan and Sarathi are conceptualized to capture the functionality as mandated under central Motor Vehicle Act, 1988.

701. Which rule has been amended to standardise the form and the manner in which registration numbers are displayed in the country :

(a) Central Motor Vehicle Act, 1988

(b) Central Motor Vehicle Act, 1999

(c) Central Motor Vehicle Act, 1989

(d) None of the above

GPSC ARTO Pre 30.12.2018

Ans. (c) : Central Motor Vehicle Rule, 1989 has been amended to standardise the form and the manner in which registration number are displayed in the country.

702. The basic requirement of a good combustion chamber is :

(a) Minimum turbulence

(b) Low compression ratio

(c) High thermal efficiency and power output.

(d) Low volumetric efficiency

Punjab PSC SDE 12.02.2017

Ans. (c) : The basic requirements of a good combustion chamber are to provide–

(i) High Power output

(ii) High thermal efficiency

(iii) Low specific fuel consumption

(iv) Smooth engine operation

(v) Reduced exhaust pollutants

703. What are Swirl flaps?

(a) They are small butterfly valves fitted to the intake manifold just before the cylinder head intake ports of modern automotive diesel engines

(b) They are small butterfly valves fitted to the intake manifold just before the cylinder head intake ports of modern automotive petrol engines

(c) They are small fuel injectors fitted to the intake manifold just before the cylinder head intake ports of modern steam engines

(d) None of these

KPCL AE 2016

Ans. (a) : Swirl flaps are small butterfly valves fitted to intake manifold just before the cylinder head, intake ports of modern automotive (Mazda, Audi, BMW, Vauxhall and Alfa Romeo) diesel engines.

704. Which of the following connects the steering rack to the knuckles in an automobile?

(a) Spindle (b) Tie rod

(c) Differential (d) None of these

KPCL AE 2016

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Ans. (b) : Tie rod ends are used to connect one steering rod to another steering rod or the steering knuckle. Basically the tie rod end is the ball socket with a threaded sleeve or rod attached. It is generally caused metal.

705. Which of the following type of joints the ball joints are used on the tie-rod ends, because they can deal with movement of suspension not only vertically but also in other direction?

(a) T-joints (b) Ball joints
(c) Straight joints (d) None of these

KPCL AE 2016

Ans. (b) : The ball joints are used on the tie rod ends because they can deal with movement of the suspension both vertically and in other directions.

706. The braking system that maintains directional control during braking is called_____

(a) Self locking braking system
(b) Skid locking braking system
(c) Anti locking braking system
(d) None of these

KPCL AE 2016

Ans. (c) : The function of anti-lock brake system (ABS) maintains directional control during braking by preventing the wheels from locking. An anti-lock braking system is safety anti-skid braking system used on aircraft and on land vehicles.

707. Which of the following is true with a V6 engine?

(a) It is more compact than the inline 4 and more compact than the V8 engine
(b) It is widely used as a transverse engine with front-wheel drive layout
(c) It has two banks of three cylinders each, set at either a 60 or 90 degree angle to each other
(d) All the above

KPCL AE 2016

Ans. (d) : V6 engine is more compact than the inline 4 and more compact than the V8 engine.

- Its widely used as a transverse engine with front-wheel drive layout.
- It has two banks of three cylinders each set at either a 60 or 90 degree angle to each other.

708. In an automobile "wheel base" is the distance between centre of the .....wheel to the centre of the .....wheel.

(a) Left front, left rear
(b) Left front, right rear
(c) Right front, left rear
(d) Left rear, right rear

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : A vehicle's wheel base is the distance between the centers of the front and rear wheels.

709. Which one of the following automobile exhaust gas pollutants is a major cause of photochemical smog?

(a) CO (b) HC
(c) NO_X (d) SO_X

HPPSC Asstt. Prof. 20.11.2017

Ans. (c) : NO_X pollutants is a major cause of photochemical smog.

710. A motor car moving at a certain speed takes a left turn in a curved path. If the engine rotates in the same direction as that of wheels, then due to the centrifugal forces :

(a) The reaction on the inner wheels increases and on the outer wheels decreases
(b) The reaction on the outer wheels increases and on the inner wheels decreases.
(c) The reaction on the front wheels increases and on the rear wheels decreases
(d) The reaction on the rear wheels increases and on the front wheels decreases.

HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : Because centrifugal forces acts outward direction from the centre of the curved path (radially outward). So vehicle has the tendency to slightly tilt on outer side. So that the reaction on the outer wheels increases and on the inner wheels decreases.

711. Which one of the following does not form a part of the fuel supply system for a diesel engine?

(a) Supply pump (b) Filter
(c) Air Cleaner (d) Spray Nozzle

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Fuel Supply System in diesel engine is made of the following part are:-
(i) Diesel tank or reservoir.
(ii) Low pressure pump.
(iii) Filter.
(iv) Fuel injection pump.
(v) Fuel injection nozzle.

712. In the fuel injection pumps for diesel engines the effective pressure strokes of the plungers in the fuel injection pump are regulated by means of a

(a) Fuel pump (b) Control Rack
(c) Needle valve (d) Cam shaft

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b): In fuel injection pumps for diesel engines the effective pressure strokes of the plungers in the fuel injections pump are regulated. By means of a control rack.

713. Worn out carburetor jets are likely to cause

(a) Excess engine output
(b) Excess fuel consumption
(c) More noise
(d) Knocking

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : Worn out carburetor jets are likely to cause excess fuel consumption, responsible & effect on engine performance. Black smoke from exhaust are comes out, back firing or overheating in the carburetor arises.

714. Automobile radiator is filled with

(a) Alkaline water (b) Acidic water
(c) Hard water (d) Soft water

Nagaland PSC CTSE 2017, 2016 Paper-II

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Ans. (d) : Automobile radiator is filled with soft water. (mixture of 60% coolant and 40% water)

715. The function of automobile catalytic converter is to control emissions of

(a) Carbon dioxide and hydrogen
(b) Carbon monoxide and hydrogen
(c) Carbon monoxide and carbon dioxide
(d) Carbon monoxide and nitrogen dioxide

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : A catalytic converter is an exhaust emission control device that reduces toxic gases and pollutants in exhaust gas from an internal combustion engine into less-toxic pollutant by catalyzing a redo reaction (an oxidation and control and a reduction reaction). Example to convert carbon monoxide, hydrocarbons and NO_x (NO_2/NO).

716. An automobile heats up while lying in a parking lot on a sunny day. The process can be assumed to be

(a) Isothermal (b) Isobaric
(c) Isometric (d) Isentropic

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : An automobile heats up while lying in a parking lot on a sunny day. The process can be assumed to be isobaric.
As we know that heat from sun in the mode of radiative nature and adds up to the car, due to which temperature as well as pressure both.

717. Rubber-tired mountings are not useful where.

(a) Onsite and job to job mobility is important
(b) Rocky and hard surface
(c) Speed of movement is important
(d) High moisture /wet surface

MPSC Dy. Engineer 2014

Ans. (d) : Rubber tired mountings are not useful where high moistured surfaces are present.
Extremely humid conditions should be avoided. Humidity in store room air must not be so high that condensation occurs on tyres must not be stored where they are exposed to rain, splashes etc.

718. Where is the Hooke's joint used in an automobile?

(a) Between gearbox and propeller shaft
(b) Between flywheel and clutch
(c) Between differential gear and wheel
(d) Between clutch and gear box

Kerala PSC AE 06.08.2015

Ans. (a) : Hooke's joint – It is also known as universal gear box and propeller shaft.
Hooke's joints whose axes are inclined to each other, and is commonly used in shaft that transmit rotary motion.

719. A differential gear in an automobile is :

(a) Simple gear train
(b) Epicyclic gear train
(c) Compound gear train
(d) None of these

Kerala PSC AE 06.08.2015

Ans. (b) : A differential gear in automobile is a epycyclic gear train.
The epycyclic gear train are effect for transmitting high velocity ratio with gears of moderate size in a comparatively lesser space. The epicyclic gear train used in the back gear of lathe, differential gear of the automobiles, hoists, pulley blocks, wrist watch etc.

720. The ratio between the width of the front axle and that of the wheel base of a steering mechanism is 0.3. At the instance when the front inner wheel is turned by 15°, the angle by which the outer front wheel will turn for perfect steering is:

(a) 13.9° (b) 15.9°
(c) 14.9° (d) 12.9°

AAI Jr. Executive 29.11.2018

Ans. (a) : Given,

$\text{Ratio} = \frac{w}{l} = 0.3$, $\theta_{inner} = 15^0$

According to Ackerman condition

$$\cot\theta_{outer} - \cot\theta_{inner} = \frac{w}{l}$$

$$\cot\theta_{outer} - \cot 15^0 = 0.3$$

$$\cot\theta_{outer} = 0.3 + \cos 15^0 = 0.3 + 3.732$$

$$\cot\theta_{outer} = 4.032$$

$$\theta_{outer} = 13.92^0$$

721. Wankel engine operates by transferring..... motion.

(a) Rotary (b) Translation
(c) Eccentric (d) Linear

HPPSC AE (PWD) 10.09.2016

Ans. (a) : The wankel engine is a type of internal combustion engine using an eccentric rotary design to convert pressure into rotating motion compared to the reciprocating piston engine the wankel engine has more uniform torque and less vibration and for a given power is more compact and weight less.

722. Power impulses from an I.C. engine are smoothed out by

(a) Governor (b) Crank shaft
(c) Gear Box (d) Flywheel

GPSC AE 26.07.2015

Ans. (d) : Power impulses from an I.C. engine are smoothed out by flywheel.

723. The ratio of air rotational speed to crankshaft rotational speed is :

(a) Swirl Ratio (b) Speed ratio
(c) Turbulence Ratio (d) Compression ration

TNPSC AE (Industries) 09.06.2013

Ans. (a) : The ratio of air rotational speed to crankshaft rotational speed is swirl ratio.

724. What is the main objectives for engine control?

(a) Users demand for precise controls
(b) The automotive industry's need to do creative research for sustainable development.

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(c) Government regulation concerning emissions and fuel economy
(d) None of the above.

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : The main objectives for engine control, is government regulations concerning emissions and fuel economy .
From the emission perspective, the goal of the engine control system is to provide the demanded quantity of fuel air, and EGR (if only) at the required time and in the required temperature and pressure state.
EGR – Exhaust gas recirculation.

725. In the present day racing cars ….. arrangement is preferred
(a) Central engine
(b) Rear engine rear wheel/drive
(c) Front engine from wheel drive
(d) Articulated vehicle's

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : In the present day racing cars central engine arrangement is preferred.

726. The angle of inclination of the front wheel tyre with respect to the vertical plane is
(a) Caster (b) Camber
(c) Wheel track (d) Toe-out

TNPSC AE 2017

Ans. (b) : The angle of inclination of the front wheel tyre with respect to the vertical plane is Camber.

727. In constant speed test, the vehicle is driven with
(a) Constant speed at various steer angle
(b) Constant speed at constant steer angle
(c) Constant speed at various turing radii
(d) Constant speed at constant steer angle with constant radius

TNPSC AE 2017

Ans. (c) : In constant speed test, the vehicle is driven with Constant speed at various turing radii.

728. Choose one feature that improves the forward visibility of a vehicle.
(a) Brake light (b) Hazard lights
(c) Turn indicators (d) Cornering head light

TNPSC AE 2017

Ans. (d) : Cornering head light that improves the forward visibility of a vehicle.

729. Which type of bus is more suited for the following features?
Engine is front of passenger compartment
Low ratio of useful length to overall length
Poor aerodynamic shape and high tyre weight
(a) Classic type bus (b) Double ducker bus
(c) Split level bus (d) Articulated bus

TNPSC AE 2017

Ans. (a) : Classic type bus

730. In work cell control checking the continuation of work cycle is known as
(a) Interlock (b) Bottleneck
(c) Automatic lock (d) Check point

TNPSC AE 2018

Ans. (a) : In work cell control checking the continuation of work cycle is known as interlock.

731. Which of the following chasis layout is fitted with transfer case?
(a) Front engine - Front wheel drive
(b) Rear engine - Rear wheel drive
(c) Front engine - All wheel drive
(d) Front engine - Rear wheel drive

TNPSC AE 2017

Ans. (c) : Front engine - all wheel drive chasis layout is fitted with transfer case.

732. The slots or openings in a disc wheel enhances
(a) Vehicle body cooling
(b) Passenger compartment cooling
(c) Engine - Radiator cooling
(d) Brake system cooling

TNPSC AE 2017

Ans. (d) : The slots or openings in a disc wheel enhances brake system cooling.

733. Air brakes are mostly used in case of
(a) Cars (b) Jeeps
(c) Trucks (d) Three-wheelers

TNPSC AE 2017

Ans. (c) : Air brakes are mostly used in case of trucks.

734. The purpose of preventive maintenance is to
(a) Help schedule breakdowns
(b) Eliminate routine service work
(c) Force the driver to use his own service station
(d) Help prevent failure

TNPSC AE 2017

Ans. (d) : The purpose of preventive maintenance is to help prevent failure.

735. Service specifications are set by the
(a) Vehicle manufacturer
(b) Technician
(c) Service manager
(d) Society of Automotive Engineers

TNPSC AE 2017

Ans. (a) : Service specifications are set by the Vehicle manufacturer.

736. Which of the following is not a part of the vehicle chassis?
(a) Wheels (b) Front axle
(c) Steering system (d) Seats

TSPSC AEE 2015

Ans : (d) Seats is not a part of the vehicles chassis.

737. The basic function of the suspension in an automobile is to
(a) Absorb vibration and impact forces from the road surface
(b) Automatically correct the effect of over steering
(c) Ensure that the wheel alignment is not disturbed during driving
(d) Ensure that the steering wheel can deliver a suitable amount of steering force

TSPSC AEE 2015

Ans : (a) The basic function of the suspension is an automobile is to absorb vibration and impact forces from the road surface.

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738. Gearbox is produced by–
(a) Design by drawing
(b) Design by craft evolution
(c) Design synthesis
(d) Simultaneous design

Nagaland CTSE 2017 Ist Paper

Ans. (a) : Gearbox is produced by design by drawing.

739. Modern automobile car is produced by–
(a) Sequential design
(b) Design by craft evolution
(c) Design synthesis
(d) Simultaneous design

Nagaland CTSE 2017 Ist Paper

Ans. (d) : Modern automobile car is produced by "simultaneous design".

740. In a test to determine braking efficiency of a vehicle weighing 1200 kg is placed on a brake testing machine. The brake tester shows the following reading. Front right : 2120 N; Front left 2080 N; Rear Right. : 1490 N; Rear left : 1510 N; Then the braking efficiency is
(a) 50% (b) 60%
(c) 70% (d) 80%

TNPSC AE 2017

Ans. (b) : 60%

741. Crankshaft journals should be reground if they are tapered or out of round by more than
(a) 0.0076mm (b) 0.025mm
(c) 0.25mm (d) 0.76mm
(e) 0.076mm

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Crankshft journals should be reground if they are tapered or out of round by more than 0.025 mm.

742. The tyre is designated as "175/65 R14 82s", then the load index for the tyre is
(a) 175 (b) 65
(c) 14 (d) 82

TNPSC AE 2017

Ans. (d) : "175/65 R14 82s" tyre 175 is the width of the contact path or the area of the tyre that touches the ground, 65 is the profile of the tyre or the distance between the inner and outer circle of the tyre, both of these are in millimeters, while, R 14 is the diameter of the wheel or the inner circle of the tyre, it means, in this case, the tyre has an inner diameter of 14 inches and the load index for the tyre is 82.

743. Metallic noise in torque converter is due to-
(a) Seal leakage in torque converter
(b) Spring of outlet relief valve being too weak
(c) Loose stator blades.
(d) Screen in scavenge line not open.

MPSC Dy. Engineer 2014

Ans. (c) : Metallic noise in torque converter is due to loose stator blades.

744. A leaf spring in automobiles is used
(a) to apply forces
(b) to measure forces
(c) to absorb shocks
(d) to store strain energy

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) :

745. The magneto in an automobile is basically :
(a) Transformer (b) D.C. generator
(c) Magnetic circuit (d) A.C. generator

(HPPSC AE 2014)

Ans : (b) The magneto in an automobile is basically D.C. generator

Magneto Ignition system:- The magneto ignition system has the same principle of working as that of coil ignition system, except that no battery is required as the magneto acts as its own generator.

It consists of either rotating magnets in fixed coils, or rotating coils in fixed magnets. The current produced by the magneto is made to flow to the induction coil which works in the same way as that of coil ignition system. The high voltage current is then made to flow to the distributor, which connects the sparking plugs in rotation depending upon the firing order of the engine.

This type of ignition system is generally employed in small spark ignition engines such as scooters, motor cycle and small motor boat engine.

746. In a metal-shell type ignition coil. the core is wound with
(a) Primary winding
(b) Secondary winding
(c) Primary and secondary winding together
(d) None of the above

WBPSC AE, 2007

Ans. (c) :

747. Nipping of leaf springs in a vehicle is to :
(a) vary the interleaf sliding
(b) increase the load carrying capacity
(c) keep stresses uniform in the springs
(d) vary the effective length of spring

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : One of the methods of equalizing the stresses in different leaves is to pre-stress the spring. The initial gap between the extra full length leaf and the graduate-length before the assembly is called a nip. Such pre-stressing achieved by a difference in raddi of curvature is known as nipping.

748. Which of the following is a part of chassis?
(a) Doors (b) Differential
(c) Brakes (d) Wind shields

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Part of chassis are like, wheels, front axle steering system, brakes.

749. Which of the following is NOT an arrangement pattern of an IC engine cylinder?
(a) Opposed cylinder engine
(b) Radial
(c) V-type engine
(d) Circular

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Cylinder arrangement– Vertical, Horizontal, Inline, V-type radial, opposed cylinder or piston engines.

Volve or port design and location overhead in two stroke engine, cross scavenging, loop scavenging, uniflow scavenging.

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750. The speed of a vehicle is 'S' in km/hr and the wheel radius is 'R'. Then the rpm of the wheel is :

(a) $S/(120\pi R)$ (b) $1000S/(120\pi R)$
(c) $1000S/(2\pi R)$ (d) $1000S/(60\pi R)$

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The speed of a vehicle is 'S' in km/hr and wheel radius is R then the RPM

$$RPM = \frac{1000S}{120\pi R}$$

751. In an automobile pneumatic governor, the amount of vacuum applied to the diaphragm is controlled by the :

(a) spring (b) lever
(c) accelerator pedal (d) butterfly valve

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : The accelerator is also known as gas pedal. It is the pedal located on the floor or the far-right. This pedal controls the amount of gas being fed into the engine and thereby controls the speed of the vehicle. In an automobile pneumatic governor, the amount of vacuum applied to the diaphragm is controlled by the accelerator pedal.

752. A condition that causes vapour locking in a braking system is due to :

(a) overcooling of the brakes during high speed driving
(b) keeping the vehicle without use for an extended period
(c) an excessively high engine speed on a downhill road
(d) overheating of the fluid due to frequent brake application

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Vapour locking in a brake system is overheating of the fluid due to frequent brake application.

753. The material used for gas-filled cartridge in a car bumper is :

(a) aluminium (b) HDPE
(c) LDPE (d) steel

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : HDPE stands for high density poly ethylene is a thermoplastic polymer made from petroleum. As one of the most versatile plastic material around HDPE plastic is used in a wide variety of applications including plastic, bottles, milk jugs shampoo bottles, bleach bottles, cutting boards and piping.

754. In front wheel drive vehicles, which type of axle is used to transmit power from the drive shaft to wheels?

(a) Real drive axle
(b) Dead axle
(c) Front drive axle
(d) One piece axle housing

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Front wheel drive axles, also called axle shafts or front drive shaft, transfer power from the transaxle differential to the Hub and Wheel of a vehicle. Front wheel drive axles turn much slower than a drive shaft for a rear-wheel drive vehicle, universal joints that connect the drive axle are called CV joint.

755. Which component is located between the axles which permits wheel fitted on one axle to turn at a different speed from that of the other wheel fitted on the another axle during power transmission through shaft in heavy vehicles?

(a) Universal joint (b) Journal bearing
(c) Differential gear (d) Rack and pinion

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Differential gear component is located between the axles which permits wheel fitted on one axle to turn at a differential speed from that of the outer wheel fitted on the another axle during power transmission through the shaft in heavy vehicles.
Differential gear in automotive mechanics gear arrangement that permit power from the engine to be transmitted to a pair of driving wheels dividing the force equally between them but permitting them to follow paths of different lengths as when turning a corner or traversing an uneven load.

756. Why are hydraulic and electric brakes usually used only in heavy vehicles such as locomotives and heavy trucks?

(a) They are complex in design
(b) They are very costly to operate
(c) These brakes are so strong that vehicles stop suddenly and the chance of an accident in light vehicles is high if used
(d) These brakes are only used when large amount of energy is required to retard the speed of the vehicles and it takes time to stop the vehicle

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Hydraulic and electric brakes usually used only in heavy vehicles such as locomotives heavy trucks these brakes are only used when the large amount of energy is required to retard the speed of the vehicles and it takes time to stop the vehicle.

757. In an automobile, choke is applied for

(a) increasing speed
(b) fuel economy
(c) starting in cold weather
(d) starting in hot weather

GPSC ARTO 01.05.2016

Ans. (c) : Fact is that fuel in the engine is cold when the engine is first started and in order to heat it up, a fuel and air mixture is required, and that's what a choke is designed to do.

758. Which of the following is a type of chassis of an automobile?

(a) Semiforward (b) Centrifugal
(c) Steering (d) Wind
(e) Automatic

CGPSC ARTO & TSI 07.01.2017

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Ans. (a) : Chassis is the main support structure of the vehicle which is also known as frame. It bears all the stresses on the vehicle in both static and dynamic condition.
Semi Forward Control chassis :-
In which engine is mounted that half of it is in the driver's cabin where as the other half is in front, outside the driver's cabin.
Eg- Tata SE series of vehicles.
Full forward control chassis :-
- In which engine is mounted completely inside the driver's cabin
- Maximum utilization of space is achieved.

Eg - Tata E series of vehicle.

759. Cylinder block construction by mono block is preferred to individual engine construction because of :
(a) Compact engine size
(b) Stiff and combined structure
(c) Easy manufacturing operations
(d) All of the above
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) • Compact engine size
• Stiff and combined structure
• Easy manufacturing operations

760. In the transmission system provision of slip joints allows a change in the :
(a) Angle of shaft
(b) Length of shaft
(c) Speed of rotation
(d) Amount of power transmission
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (b) In the transmission system provision of slip joints allows a change in the length of shaft.

761. The most effective section to build chassis frame against bending is :
(a) Rectangular bar
(b) Round Hollow tube
(c) Square hollow section
(d) Round bar
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c) The most effective section to build chassis frame against bending is square hollow section.

762. An under inflated tyre will wear the tread most:
(a) Near the centre
(b) In the cross section direction
(c) In the lateral direction
(d) Near the edges
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) An under inflated tyre will wear the tread most near the edges.

763. Trafficators are light singal used for :
(a) Indicating the direction in which turning
(b) Light traffic
(c) Reversing the car in traffic
(d) Heavy Traffic
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (a) Trafficators are light singal used for indicating the direction in which turning.

764. The term ply rating with reference to a tyre refers to the :
(a) Actual number of plies
(b) Recommended inflation pressure
(c) Rated strength
(d) Aspect ratio
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c) Rated strength

765. In the recirculating ball type steering gear, the balls travel between the ball nut and the :
(a) Gear rack (b) Worm wheel
(c) Steering wheel shaft (d) Worm shaft
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) Worm shaft

766. The included angle in steering geometry is the sum of the :
(a) Camber and caster
(b) Caster and Steering axis inclination
(c) Camber and Steering axis inclination
(d) Camber and toe-in
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c) : The included angle in steering geometry is the sum of the camber and Steering axis inclination.

767. Valve overlap is the number of degrees of camshaft rotation during which
(a) Both valves are closed
(b) Both valves are open
(c) Both (A) and (B)
(d) All of the above
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (b) : Both valves are open

768. In vehicles with till steering, the steering column is pivoted in :
(a) Upper bracket (b) Lower bracket
(c) Tile bracket (d) Steering yoke joint
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d) : In vehicles with till steering, the steering column is pivoted in steering yoke joint.

769. The caster is called as positive when the top of the kingpin is inclined to the :
(a) Rear of the wheel (b) Front of the wheel
(c) Left of the wheel (d) Right of the wheel
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (a) : The caster is called as positive when the top of the kingpin is inclined to the rear of the wheel.

770. The vehicle ride will be comfortable if :
(a) Unsprung weight is kept minimum
(b) Sprung weight is kept minimum
(c) Vehicle weight is kept minimum
(d) All of the above
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (a) : The vehicle ride will be comfortable if unsprung weight is kept minimum.

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771. The most commonly used cranking motor drive is:

(a) Barrel type
(b) Bendix drive
(c) Friction clutch drive
(d) Over running clutch

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (b) The most commonly used cranking motor drive is bendix drive.

772. Tractive effort in an automobile is maximum in

(a) first gear (b) second gear
(c) third gear (d) top gear

APPSC AEE 2012

Ans. (a) : Tractive effort is defined as the force at the rims or the outer edges of the driving wheels of moving trains. The maximum allowable tractive effort that can be applied without wheel slip is $f_m = 9.81\mu M_t N$ f_{max} at the first gear,
Where,
μ = co-efficient of adhesion

773. 'Ladder Frame' is an example for _____.

(a) supports of machine tool
(b) chassis of automobile
(c) suspension
(d) roof

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : 'Ladder Frame' is an example for chassis of automobile.

774. Which aspect is not correct in the context of manual transmission; conventionally called gear boxes ?

(a) lot of fatigue to driver in operation
(b) simplicity of design
(c) low initial cost
(d) high friction losses

HPPSC Poly. Lect. 05.07.2021

Ans. (d) : high friction losses

775. The automobile gears are generally made of

(a) mild steel (b) alloy steel
(c) cast steel (d) cast iron

HPPSC Poly. Lect. 05.07.2021

Ans. (b) : alloy steel

776. A brake commonly used in motor cars is

(a) shoe brake
(b) band brake
(c) band and block brake
(d) internal expanding brake

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : The internal expanding type of brake is commonly used in motor car and light trucks. When the break are applied to all the four wheels, the braking distance in which the vehicle is brought to rest after applying the break will be the least.

777. Identify the correct order of classification of Internal Combustion Engines :

(a) Reciprocating type, internal Combustion Engine, non-reciprocating type
(b) Reciprocating type, external combustion engine, Non-reciprocating Type
(c) SI Engine, CI Engine, internal combustion engine
(d) Internal combustion engine, reciprocating type, non-reciprocating type

TNPSC AE, 2008

Ans. (d) : Internal combustion engine, reciprocating type, non-reciprocating type.

778. The specific gravity of the completely discharged battery is of order of

(a) 1.11 to 1.14 (b) 1.26 to 1.299
(c) 1.17 to 1.205 (d) 1.205 to 1.235

GPSC AE 26.07.2015

Ans. (a) : The specific gravity of the completely discharged battery is of order of 1.11 to 1.14

779. In a fire hazard area, prime mover used is :

(a) Petrol engine (b) Diesel Engine
(c) Air motor (d) All of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : In a fire hazard area, prime mover used is Air motor.

780. The purpose of large spring loaded valve in a radiator cap is to:

(a) Lower temperature at which the coolant boils
(b) Prevent the coolant escaping when it boils
(c) Reduce the risk of the rubber hoses collapsing when the pressure is low
(d) Pressurize the system which raises the boiling point of the coolant

RPSC VPITI 14.02.2016

Ans. (*) : No answer is given by commission.

781. Flat head engines are also known as ….. engine.

(a) G-Head (b) O-Head
(c) L-Head (d) Z-Head

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : A flat head engine also known as a side valve engine or valve in block engine is an internal combustion engine with its poppet valve contained with the engine valve, instead of in the cylinder head as in an overhead valve engine.
The advantages of a side valve engine include, simplicity, reliability, low part count, low cost, low weight, compactness, responsive low-speed power, low mechanical engine noise and insensitivity to low octane fuel. As the cylinder cross-section has the shake of an inverted 'L' other name such as L-block or L-head engine.

782. The ignition coil in an auto mobile acts as a

(a) Rectifier
(b) Switch
(c) Step up transformer
(d) DC to AC converter

WBPSC AE 2003

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Ans. (c) : The ignition coil in an auto mobile acts as a step up transformer.
Ignition coil is an induction coil in an automobiles ignition system that transforms the battery's voltage to the thousands of volts which are needed to create an electric spark in spark plugs to ignite the fuel.

783. The silencer of internal combustion engine :
(a) Reduces noise
(b) Decrease break specific fuel consumption
(c) Increase break specific fuel consumption
(d) Has no effect on its efficiency

HPPSC Asstt. Prof. 18.11.2016

Ans. (a) : Muffler, also called silencer, device through which the exhaust gases from an internal combustion engine are passed to attenuate (reduce) the airborne noise of the engine.

784. Aneroid is a device
(a) For cold starting
(b) For controlling emission
(c) For decompression
(d) None of the above

WBPSC AE, 2007

Ans. (b) : For controlling emission

785. Piston in IC Engine is usually made of
(a) Copper alloy (b) Tungsten alloy
(c) Nickel alloy (d) Aluminium alloy

Gujarat PSC AE 2019

Ans. : (d) : Piston in IC engine is usually made of aluminium alloy because it is lighter aluminium alloy. Commonly used materials for IC engine piston are cast iron, cast steel, forged steel, cast aluminium alloys and forged aluminium alloy. Advantages of aluminium over cast iron high thermal conductivity.

786. Piston pins are usually either a press fit in the rod or:
(a) locked to the rod with a bolt
(b) free floating in the rod and piston
(c) locked to the piston with a bolt
(d) a press fit in the piston

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : The small end attaches to the piston pin or wrist pin, which is currently most often press fit into the con rod but can swivel in piston, a floating wrist pin design.

787. A Jerk pump used for fuel injection system is
(a) Vane type (b) Rotary type
(c) Reciprocating type (d) None of the above

UKPSC AE 2007 Paper -II

Ans. (c) : Reciprocating type

788. The elements of most concern in regard to pollution caused by engines are :
(a) CO and CO_2
(b) CO and hydrocarbons
(c) CO_2 and hydrocarbons
(d) hot products of combustion

J&K PSC Civil Services Pre, 2013

Ans. (b) : Major pollutant from motor vehicle.
(a) Volatile organic compounds
(b) Carbon monoxide (CO)
(c) Nitrogen oxides (NO_X)
(d) sulphur dioxide (SO_2)
(e) Green house gases
(f) lead
(g) hydro carbons (HC)

789. Consider the following:
1. Catalytic convertor
2. Use of high compression ratio
3. Oxidation catalyst in the exhaust manifold
4. Use of high degree of supercharger
By which of these techniques, can the oxides of nitrogen in the exhaust be controlled?
(a) 1 only (b) 2 and 4 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Catalytic converters change harmful substances in a car's exhaust gases such as carbon monoxide, Nitric oxide, Nitrogen dioxide and hydrocarbons, into less harmful substances like CO_2 and water.

790. In emission measuring equipments, the HC is measured using
(a) Gas Chromatograph
(b) Flame Ionization Detector
(c) Chemiluminescence Detector
(d) NDIR Analyzer
(e) HFFR Test Rig

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : With the help of Flame Ionization Detector, the HC is measured in emission.

791. Rhodium in the catalytic converter promotes the reduction of
(a) HC (b) CO
(c) NO_x (d) Smoke

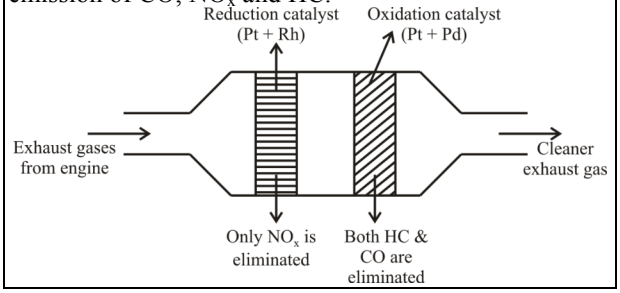
TNPSC AE 2017

Ans. (c) : Rhodium is the catalytic convertor promotes the reduction of NO_x.

792. Three way catalytic converters reduce the emission of
(a) CO, CO_2 and soot (b) CO, NO_x and HC
(c) CO_2, NO_x and HC (d) CO, HC and soot

TNPSC AE 2017

Ans. (b) : Three way catalytic converters reduce the emission of CO, NO_x and HC.



793. NO_x emission is maximum in S.I. engines when the air fuel ratio is
(a) Exactly stoichiometric
(b) Lean mixture
(c) Rich mixture
(d) Nearby stoichiometric

TNPSC AE 2017

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Ans. (d) : NO_x emission is maximum in S.I. engines when the air fuel ratio is nearby stoichiometric.

794. Reducing the compression ratio of an engine reduces the combustion temperature, and this
(a) Reduces the amount of NO_x formed
(b) Increases the amount of NO_x formed
(c) Reduces the amount of HC formed
(d) Increases the amount of H_2O formed

TNPSC AE 2017

Ans. (a) : Reducing the compression ratio of an engine reduces the combustion temperature, and this reduces the amount of NO_x formed.

795. The function of the air aspirator system is
(a) Furnishes the addition air for reduce HC and NO_x emission
(b) Furnishes the addition air for reduce HC and CO emission
(c) Furnishes the addition air for reduce HC and CO but increase of NO_x
(d) Furnishes the addition air for increase of HC and reduce CO and NO_x

TNPSC AE 2017

Ans. (b) : The function of the air aspirator system is Furnishes the addition air for reduce HC and CO emission.

796. NO_x emission in SI engines will be lowest during
(a) Acceleration (b) Deceleration
(c) Cruising (d) Idling

TNPSC AE 2017

Ans. (d) : During idling NO_x emission in SI engines will be lowest.

797. The important requirement of a catalytic convertor is
(a) High surface area and low volume heat capacity
(b) Low surface area and low volume heat capacity
(c) High surface area and high volume heat capacity
(d) Low surface area and high volume heat capacity

TNPSC AE 2017

Ans. (a) : The important requirement of a catalytic convertor is High surface area and low volume heat capacity

798. The capacity of battery is expressed in terms of:
(a) Current rating (b) Voltage rating
(c) Ampere-hour-rating (d) None of the above

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c) Battery capacity is measured in milliampere × hours (MAH)

799. MPFI used in automobile engines stands for
(a) Multipoint Petrol Flow Injector
(b) Micro Point Fuel Injector
(c) Multi Point Fuel Injector
(d) Micro Point Flow Injector

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) : Multi Point Fuel Injector

800. Carbon monoxide and Cabron-di-oxide emissions can be measured using.
(a) Infra red method
(b) Chemiluminescent method
(c) Flame ionization method
(d) Ultra violet method

TANGEDCO AE 2015

Ans. (a) : Infrared method are used for measuring carbon monoxide and carbon-di-oxide emission from internal combustion engine.

801. Consider the following probable causes of sticking injection plunger in a shovel.
(a) Improper injection adjustment
(b) Plugged injection spray hole
(c) Incorrect valve and injection timing
(d) Water in the fuel
Which of the following is correct?
(a) a and b (b) a, b and c
(c) c and d (d) d and a

MPSC Dy. Engineer 2014

Ans. (d) : The cause of sticking injection plunger in a shovel–
(1) Improper injection adjustment.
(2) Water in the fuel.

802. The valve of a CI engine nozzle has a cylindrical or conical projection protruding through the mouth of the nozzle and the nozzle body has an auxiliary hole. What is this type of nozzle called?
(a) Multihole nozzle (b) Pintle nozzle
(c) Pintaux nozzle (d) Open nozzle

RPSC AEN Pre-2013

Ans. (c) : In pintaux nozzle, the valve of a C.I. engine-nozzle has a cylindrical or conical projection protruding through the mouth of nozzle and nozzle body has an auxiliary hole.

803. Common rail injection system uses injection pressures of the order
(a) 100 – 200 bar (b) 200 – 400 bar
(c) 400 – 600 bar (d) 1500 bar

RPSC IOF, 2020

Ans. (d) : Generally, the common rail injection system uses injection pressure of the order of 1500 bar.

804. The maximum temperature of I.C engine cylinder is
(a) 1500-1800 deg C (b) 1800-2000 deg C
(c) 500-1500 deg C (d) 2000-2500 deg C

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (d) :

805. Ribs are provided inside below the piston head of an I.C. Engine:
(a) To increase strength
(b) To increase stiffness
(c) To increase toughness
(d) To increase surface area for dissipation of heat.

GPSC Asstt. Prof. 28.08.2016

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Ans. (d) : Ribs strengthen the piston head against the gas pressure they increase the rigidity and prevent the distortion of the piston head and also transmit a large portion of combustion heat from piston head.

806. In two-stroke engines the valve ports are provided at :

(a) Crankshaft (b) Piston
(c) Cylinder walls (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : In two stroke engine the valve ports are provided at cylinders wall.
There are two ports
(i) Inlet port, (ii) Exhaust port

807. The injected fuel evaporates and it is injected at various points along the stream in :

(a) Carburetor (b) Economiser
(c) MPFI (d) MDIF

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : Multi point fuel injection (MPFI) system of injecting fuel into internal combustion engine through multi ports situated on intake valve of each cylinder, it delivers an exact quantity of fuel in each cylinder at the right time.
There are three types of MPFI system.
(i) Batched
(ii) Simultaneous
(iii) Sequential.
Batched MPFI- In it fuel is injected to the groups of the cylinder without bringing their intake stroke together.
Simultaneous MPFI– In it, fuel is inserted to all cylinder at same time.
Sequential MPFI– In it injection is timed to overlap with intake stroke of each cylinder.
Hence, the injected fuel evaporates and it is injected of various points along the stream in MPFI.

808. Ante-chambers are sometimes provided in an I.C. Engine as :

(a) Pre-combustion chamber
(b) Post-combustion chamber
(c) Combustion chamber
(d) Exhaust chamber

HPPSC Lect. (Auto) 23.04.2016

Ans. (a) : Pre-combustion chamber is an auxiliary space in which combustible gases are ignited and combustion started ahead of the main combustion chamber of a jet or gas engine. Hence Ante-chamber are sometimes provided in an IC-engine as Pre-combustion chamber.

809. Poppet valve is also known as :

(a) Star valve (b) Mushroom valve
(c) Direct valve (d) Booster valve

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : A poppet valve is a valve typically used to control the timing and quantity of gas or vapour flow into an engine. It consists of a hole or open ended chamber usually round or oval in cross-section and a plug usually a disk shape on the end of a shaft known as a valve system.
A poppet valve is also known as mushroom valve.

810. In S.I. engine ….. ignition system can be used.

(a) Thermo-fluid (b) Magneto
(c) Colman (d) Condensate

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : An ignition system generates a spark heats an electrode to a high temperature to ignite a fuel air mixture in spark ignition I.C. engine.
Magneto ignition system is used in SI engine. The basic component of a magneto ignition system consists of a magneto, breaker points, capacitor, ignition switch, distributer, spark plug leads and spark give.

811. If the spark plug on a petrol engine runs hotter, the engine may-

(a) Stop (b) Run smoothly
(c) Misfire (d) Detonate

NPCIL ET, 2015

Ans. (d) : The operating temperature of a spark plug is the actual physical temperature at the tip of the spark plug within the running engine. If the tip of the spark plug is too hot it can cause pre-ignition leading to detonation/knocking and damage may occur.

812. The voltage required to produce a spark across the electrode gap of the spark plug is:

(a) 2000 to 4000 V (b) 4000 to 6000 V
(c) 6000 to 10000 V (d) 10000 to 12000 V

NPCIL ET, 2015

Ans. (c) : The voltage required to produce a spark across the gap, between the sparking points is 6000 to 10,000 volts.

813. The objective of providing masked inlet valve in the air passage of compression ignition engine is to:

(a) Enhance flow rate
(b) Control air flow
(c) Induce primary swirl
(d) Induce secondary turbulence

NPCIL ET, 2015

Ans. (c) : The object of providing masked inlet valve in the pair passage of compression ignition engines is to primary swirl.

814. List the required operation of a catalytic convertors on HC emissions.

(a) Oxidation to H_2O and CO_2
(b) Reduction to H and C
(c) Reaction with NO_x
(d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Catalytic Convertors- Catalytic Convertor is an exhaust emission control device that reduced toxic gases and pollutants in exhaust gas from an internal Combustion engine into less toxic pollutants by catalyzing a redox reaction (an oxidation and a reduction reaction)

$2CO + O_2 \rightarrow CO_2$ (Oxidation)

$C_xH_y + O_2 \rightarrow CO_2 + H_2O$ (Reduction)

The catalytic convertor combine oxygen (O_2) with carbon monoxide (CO) and unburned Hydrocarbons (C_nH_n) to produce Carbon dioxide (CO_2) and water (H_2O).

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815. To improve fuel economy …..... choose most appropriate option.

(a) While driving avoid quick starts and sudden stop
(b) Keep tires inflated to the recommended maximum pressure and don't carry unnecessary weight in or on the vehicle
(c) Option (B) is right and A is wrong
(d) Both option A and B are right.

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : To improve fuel economy
1. Avoid quick starts and sudden stop.
2. Keeps tyre inflation proper as per recommendation
3. Avoid over loadings
4. proper maintenance of vehicle

816. The skirt of piston.....

(a) Is used to withstand the pressure of gas in the cylinder
(b) Acts as a bearing for the side thrust of the connecting rod
(c) Is used to seal the cylinder in order to prevent leakage of the gas past the piston
(d) None of the above

GPSC EE Pre, 28.01.2017

Ans. (b) : Piston skirt is fitted in both two stroke and four stroke engines. It has different function for different engines. In large cross head two stroke engine with uniflow scavenging these skirts are short in length and are fitted to act as a guide and stablise the position of the piston inside the linear. It is generally made of cast iron. The diameter of the skirt is usually kept slightly larger than that of the piston. This is done to prevent damage to the linear surface due to the piston movement.

817. As per the technical specifications of automotive vehicles, what is the Engine type used in TATA INDICA V2'02?

(a) 4 cylinder in-line
(b) 5-cylinder, 4 stroke, in-line
(c) 6-cylinder in-line
(d) One of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : 4 cylinder in-line engine is used in TATA INDICA V2'02.

818. Identify the correct set of approximations made in the thermodynamic analysis of internal combustion engines:

P. The combustion process is replaced by an equivalent energy addition process
Q. The working fluid is a mixture of carbon dioxide and water vapor.
R. The exhaust process is replaced by an equivalent energy rejection process.
S. The working fluids have constant heat capacities.

(a) P, R, S (b) P, R
(c) R, S (d) P, Q, R

APGENCO AE 2012

Ans. (a) : Thermodynamics Analysis of Internal Combustion Engine–

- The combustion process is replaced by an equivalent energy addition process.
- The exhaust process is replaced by an equivalent energy rejection process.
- The working fluids have constant heat capacities

$$(mC_p)_A = (mC_p)_R$$

819. Which one of the following is incorrect with respect to painting of vehicles?

(a) Paints creates a thermal boundary layer on the surface
(b) Paints prevents rapid corrosion of parts
(c) Paints colour increases the ability to be seen
(d) Paint colour increases the aesthetic look

TNPSC AE 2017

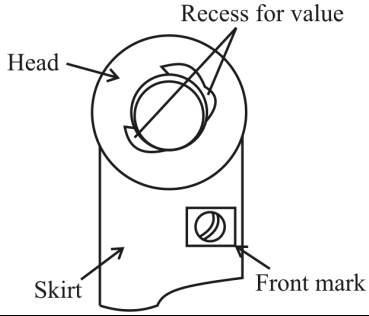
Ans. (a) : Paints creates a thermal boundary layer on the surface is incorrect with respect to painting of vehicles.

820. What is the name given to the crescent-shaped cavity on the top surface of the piston head?

(a) Valve recess (b) Crescent hole
(c) Valve cup (d) None of these

KPCL AE 2016

Ans. (a) : The crescent shaped cavity on the piston head top surface is called as valve recess



821. Which one of the following valves is provided for starting the engine manually, during cold weather conditions?

(a) Starting jet valve
(b) Compensating jet valve
(c) Choke valve
(d) Auxiliary air valve

ESE 2020

Ans. (c) : Choke valve

822. Engine overheating may be due to

(a) Struck radiator pressure cap
(b) Open thermostat
(c) Excess coolant
(d) Broken fan belt

TNPSC AE 2017

Ans. (d) : Engine overheating may be due to broken fan belt.

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05.

Fluid Mechanics

1. Properties of Fluid

1. Viscosity has the following dimensions:

(a) MLT^{-2} (b) $ML^{-1}T^{-2}$

(c) $ML^{-2}T^{-2}$ (d) $ML^{-1}T^{-1}$

Assam PSC AE (PHED), 18.10.2020
VIZAG STEEL MT 2015
TNPSC AE 2014
ISRO Scientist/Engineer 24.05.2014
APPSC Poly. Lect. 2013
APPSC AEE 2012
J&K PSC Civil Services Pre, 2010
UKPSC AE 2007 Paper -II

Ans. (d) : Unit of viscosity(dynamic viscosity) $\Rightarrow \frac{N-s}{m^2}$

$$1\frac{N-s}{m^2} = 10 \text{ poise}$$

At 20^0C viscosity of water is 1 C.P. (Centi Poise).

$$\text{Dynamic viscosity} = \frac{\tau}{\frac{du}{dy}}$$

$$\mu = \frac{\text{Force}}{\text{Area} \times \text{Velocity gradient}}$$

Force = $[MLT^{-2}]$

Area = $[L^2]$

$$\text{Velocity gradient } \frac{du}{dx} = \frac{[LT^{-1}]}{[L]} = [T^{-1}]$$

Therefore

$$\mu = \frac{[MLT^{-2}]}{[L^2][T^{-1}]}$$

$$\mu = [ML^{-1}T^{-1}]$$

* Dimension of viscosity $\Rightarrow ML^{-1}T^{-1}$.

2. Newton's law of viscosity depend upon the :

(a) Stress and strain in a fluid

(b) Shear stress pressure and velocity.

(c) Shear stress and rate of strain

(d) Viscosity and shear stress

JPSC AE, 10.04.2021, Paper-I
RPSC ACF & FRO, 26.02.2021
TSPSC Manager (Engg.) HMWSSB 12.11.2020
Gujarat PSC AE 2019
Sikkim PSC (Under Secretary), 2017
UJVNL AE 2016
BPSC Asstt. Prof. 29.11.2015
OPSC Civil Services Pre 2006, ESE 1998

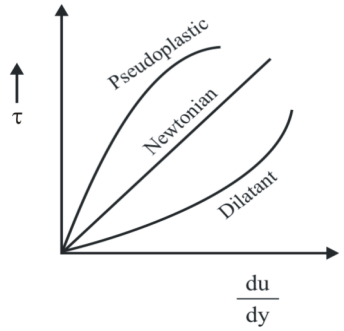
Ans. (c) : Newton's law of viscosity- A fluid is said to be Newtonian fluid if,

Shear stress $\propto$ Rate of shear strain $\left(\frac{d\phi}{dt}\right)$

Shear stress $\propto$ velocity gradient $\left(\frac{du}{dy}\right)$

$$\boxed{\tau = \mu\frac{du}{dy} = \mu\frac{d\phi}{dt}}$$

Where, τ = Shear stress



3. Unit of Kinematic viscosity is

(a) m/sec^2 (b) m^2/sec

(c) m/sec (d) m^2/sec^3

Vadodara Muncipal Corp. DEE, 2018
TSPSC AEE 2017
BPSC Asstt. Prof. 29.11.2015
CGPSC AE 26.04.2015 SHIFT-II
UPRVUNL AE 2014
UKPSC AE-2013, Paper-II
RPSC Lect. (Tech. Edu. Dept.) 2011
GATE 2001

Ans. (b) : Kinematic Viscosity (ν)

$$= \frac{\text{Dynamic Viscosity}(\mu)}{\text{Density of fluid}(\rho)}$$

$$\nu = \frac{\mu}{\rho} \qquad (\mu = \text{kg/m-sec})$$

$$(\rho = \text{kg/m}^3)$$

$$\nu = \frac{\text{kg/m-sec}}{\text{kg/m}^3} = \text{m}^2/\text{sec}$$

Unit of kinematic viscosity is "m^2/sec".

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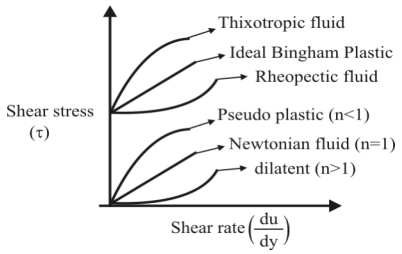
4. The stress-strain relation of Newtonian fluid is

(a) linear (b) parabolic

(c) hyperbolic (d) cubic

HPPSC ADF 05.03.2019, HPPSC AE 2018
ISRO Scientist/Engineer 22.04.2018
Nagaland PSC (CTSE) 2018, Paper-II
Vizag Steel (MT) 2017
TRB Asstt. Prof., 2012, WBPSC AE 2003

Ans. (a) : A Newtonian fluid is one where there is a linear relationship between shear stress and strain-rate. The ratio of shear stress to strain-rate is the viscosity of the fluid. A fluid where there is no linear relationship between shear stress & strain-rate is called a Non-Newtonian fluid.



5. Which one of the following sets of condition clearly apply to an ideal fluid?

(a) Viscous and compressible
(b) Non Viscous and incompressible
(c) Non Viscous and compressible
(d) Viscous and incompressible

CIL MT 27.02.2020, RPSC AE 2018
WBPSC AE, 2017
Rajasthan AE (Nagar Nigam) 2016 Shift-3
UKPSC AE 2012 Paper–II
OPSC Civil Services Pre 2006, CSE Pre-1995

Ans. (b) : Condition for an ideal fluid

- Fluid must be non-viscous (inviscous)
  $\mu = 0, \tau = 0$
- Fluid must be incompressible.
- $\Delta\rho = 0, \rho =$ constant
- Fluid should be irrotational.
  $\omega = 0$

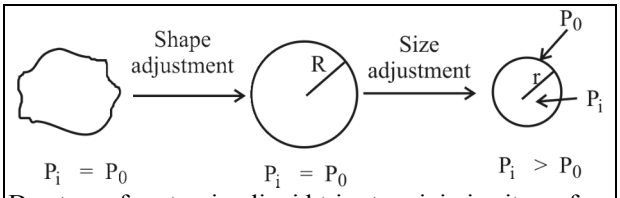
6. Due to which one of the following properties, falling drops of water become spherical in shape?

(a) Viscosity (b) Surface tension

(c) Cohesion (d) Adhesion

Nagaland PSC (CTSE) 2018, Paper-II
Kerala PSC IOF 19.04.2016,
GWSSB DEE 07.07.2016
Rajasthan Nagar Nigam AE 2016, Shift-II
GWSSB AAE, 27.12.2015, TSPSC AEE 2015

Ans. (b) : Any system tries to attain the condition of stable equilibrium, keeping its potential energy as minimum as possible. Thus the quantity of liquid will adjust its shape until its surface area and its surface energy becomes minimum. This minimization of surface area is possible only due to surface tension.



Due to surface tension liquid tries to minimize its surface area. This is reason why drops are spherical in shape.

7. Newton's Law of viscosity states that

(a) Shear stress is directly proportional to velocity
(b) Shear stress is directly proportional to velocity gradient
(c) Shear stress is directly proportional to shear strain
(d) Shear stress is directly proportional to viscosity

GPSC Executive Engineer 23.12.2018
APGENCO AE, 2017
HPPSC Lect. (Auto) 23.04.2016
TNPSC AE 2014, UPRVUNL AE 2014
GATE 1995

Ans. (b) : Newton's Law of viscosity states that shear stress is directly proportional to rate of shear strain of velocity gradient.

$$\tau \propto \frac{du}{dy}$$

$$\tau = \mu\frac{du}{dy} = \mu\frac{d\phi}{dt}$$

8. Dimension of surface tension is:

(a) ML^2T^{-2} (b) MT^{-2}

(c) $ML^{-1}T^{-2}$ (d) MT^{-1}

CIL MT 27.02.2020, SJVN ET 2019
GPSC Executive Engineer 23.12.2018
GMB AAE 25.06.2017
Assam PSC CCE Pre 2015, GATE 1996

Ans. (b) :

Surface tension $\sigma = \frac{\text{Force}}{\text{Length}} = \frac{MLT^{-2}}{L} = MT^{-2}$

9. For a Newtonian fluid

(a) shear stress is proportional to acceleration.
(b) rate of shear stress is proportional to shear strain
(c) shear stress is proportional to density.
(d) shear stress is proportional to rate of shear strain.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
APPSC AEE SCREENING 17.02.2019
ISRO Scientist/Engineer (RAC) 07.05.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
UPRVUNL AE 2014, GATE 2006

Ans. (d) : For a Newtonian fluid
Shear stress ∝ rate of shear strain
Shear stress ∝ velocity gradient

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$\tau \propto \frac{d\phi}{dt}$ or $\tau \propto \frac{du}{dy}$

$\boxed{\tau = \mu \frac{d\phi}{dt}}$ $\boxed{\tau = \mu \frac{du}{dy}}$

10. S.I. Unit of absolute (dynamic) viscosity is..

(a) N/mm^2 (b) N.mm/s

(c) $N.s/m^2$ (d) N.mm

Haryana PSC AE (PHED) 04.09.2020, Paper-I
Nagaland PSC (CTSE) 2018, 2015 Paper-II
WBPSC AE, 2017
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : Unit of dynamic Viscosity $= \frac{N.s}{m^2} = 10$ poise

Unit of kinematic viscosity = $m^2/s = 10^4$ stokes

11. Kinematic viscosity is equal to

(a) Density/dynamic viscosity

(b) Dynamic viscosity/density

(c) Dynamic viscosity × density

(d) None of these

Punjab PSC SDE 12.02.2017, (HPPSC LECT. 2016)
KPCL AE 2016, (KPSC AE 2015)
Assam PSC CCE Pre 2015

Ans : (b) Kinematic viscosity,

$$\nu = \frac{\text{Dynamic Viscosity or absolute viscosity}}{\text{Density}} = \frac{\mu}{\rho}$$

Unit of kinematic viscosity = m^2/sec. (SI unit)

Unit of kinematic viscosity = cm^2/sec or stoke (CGS unit)

$1 m^2/sec = 10^4 cm^2/sec = 10^4$ stoke

12. Kinematic viscosity of air at 20°C is given to be 1.6×10^{-5} m^2/s. Its kinematic viscosity at 70°C will be approximately-

(a) 2.2×10^{-5} m^2/s (b) 1.6×10^{-5} m^2/s

(c) 1.2×10^{-5} m^2/s (d) 3.2×10^{-5} m^2/s

APPSC AE Subordinate Service Civil/Mech. 2016
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 1999

Ans. (a) : We know,

Kinematic viscosity $(\nu) = \frac{\mu}{\rho}\ m^2/s$

Relation between dynamic viscosity & temperature,

i.e. $\mu \propto \sqrt{T}$

Relation between density & temperature,

$PV = mRT$

$P = \rho RT$

$\rho \propto \frac{1}{T}$

$\nu = \frac{\sqrt{T}}{1/T} = T^{1/2} \times T$

$\nu = T^{3/2}$

Given, $\nu_1 = \nu_{at\ 20°C} = 1.6 \times 10^{-5}$ m^2/s, $\nu_2 = \nu_{at\ 70°C} = ?$

$$\frac{\nu_1}{T_1^{3/2}} = \frac{\nu_2}{T_2^{3/2}}, \quad \begin{bmatrix} T_1 = 20+273 = 293\,K \\ T_2 = 70+273 = 343\,K \end{bmatrix}$$

$$\frac{1.6\times10^{-5}}{293^{3/2}} = \frac{\nu_2}{343^{3/2}}$$

$\nu_2 = 2.02 \times 10^{-5}$ m^2/s

13. If temperature increases, viscosity of the gases _____ and the viscosity of liquids _____.

(a) decreases, decreases

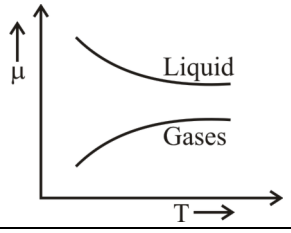
(b) decreases, increases

(c) increases, increases

(d) increases, decreases

VIZAG MT, 14.12.2020, WBPSC AE, 2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
RPSC VPITI 14.02.2016

Ans. (d) : If temperature increases, viscosity of the gases increases and the viscosity of liquid decreases.



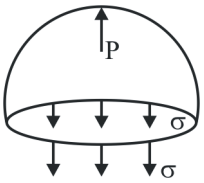
14. The pressure difference between the inside and outside of a liquid droplet of diameter 'd' due to surface tension (σ) is given by the relation

(a) $\Delta P = \frac{4\sigma}{d}$ (b) $\Delta P = \frac{8\sigma}{d}$

(c) $\Delta P = \frac{2\sigma}{d}$ (d) $\Delta P = \frac{6\sigma}{d}$

Oil India Senior Officer 23.12.2020
ISRO Scientist/Engineer 03.07.2016
TSPSC AEE 2015, ESE 2008

Ans. (a) : Pressure force = Surface tension force



$P \times \frac{\pi}{4} d^2 = \sigma \pi d$

$\boxed{P = \frac{4\sigma}{d}}$

15. Surface tension is due to

(a) Cohesion

(b) Viscous force

(c) Adhesion

(d) The difference between adhesive and cohesive force

JPSC AE, 10.04.2021, Paper-I
MPPSC AE 2016
UKPSC AE 2012 Paper–II
ESE 1998

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Ans : (a) Surface tension is due to cohesion force.
Surface tension:- Surface tension occur at the interface of liquid and gas or at the interface of two liquid.
* It is a surface phenomenon
* It is force per unit length (N/m)
* For water-air interface at 20°C, its value is 0.0736 N/m
* It is due to cohesion only.

16. Which of the following instrument is used to measure the specific gravity of liquid?
(a) Hydrometer
(b) Densitometer
(c) Differential manometer
(d) Hygrometer

APPSC AE Subordinate Service Civil/Mech. 2016
Rajasthan Nagar Nigam AE 2016, Shift-I
GWSSB DEE 07.07.2016

Ans. (a) :

Instrument	Used for Measure
Hydrometer	Specific gravity
Densitometer	Degree of darkness
Differential Manometer	Pressure difference
Hygrometer	Humidity

17. The bulk modulus of a fluid
(a) is independent of pressure and density
(b) is higher is the fluid is more compressible
(c) increases with pressure of fluid
(d) has the dimensions of volumetric chain

OPSC AEE 2019 PAPER - II
TNPSC ACF 2012, ESE 2011

Ans. (c) : Bulk modulus –

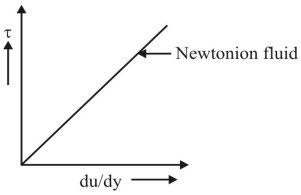
$$K = \frac{dP}{\left(\frac{dV}{V}\right)}$$

$$\boxed{K \propto dP}$$

18. Newton's law of viscosity is given by the relation
(a) $\tau = \mu^2 \frac{du}{dy}$ (b) $\tau = \sqrt{\mu} \frac{du}{dy}$
(c) $\tau = \mu \frac{du}{dy}$ (d) $\tau = \mu^3 \frac{du}{dy}$

TNPSC AE 2018, TPSC AE. 2015, CSE Pre-1995

Ans : (c)



$\tau \propto \frac{du}{dy}$ (According to Newton's law of viscosity)

$$\tau = \mu . \frac{du}{dy}$$

τ = shear stress

$\frac{du}{dy}$ = Velocity gradient.

19. Study of fluid at rest is known as
(a) Kinematics (b) Dynamics
(c) Statics (d) None of the above

NPCIL ET, 2015, APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (c) : Fluid statics or hydrostatics is the branch of fluid mechanics that studies fluid at rest.

20. Select the correct statement
(a) Viscosity of gas increases with temperature .
(b) Density of gas increase with temperature.
(c) Surface tension of liquid increases with temperature.
(d) Bulk modulus is independent of temperature.

RPSC IOF & Boiler 20.09.2017
APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (a) : Viscosity of gases increases with increase in temperature.
In case of gases the cohesive forces are small and molecular momentum transfer predominates.
With the increase in temperature, molecular momentum transfer increases and hence viscosity increases.
For a gas; $\mu = \mu_0 + \alpha t - \beta t^2$
Where; α and β are constant for gas
μ_0 = Viscosity of gas at 0ºC in poise
μ = Viscosity of gas at tºC in poise

21. The mercury does not wet the glass. This is due to the property of the liquid known as
(a) Cohesion (b) Adhesion
(c) Viscosity (d) Surface tension

Nagaland PSC (CTSE) 2018, Paper-II
Rajasthan Nagar Nigam AE 2016, Shift-II
GWSSB AAE, 27.12.2015

Ans. (d) : The mercury does not wet the glass. This is due to the property of the liquid known as surface tension. The liquid molecules on the surface experience a net inward pull due to which there will be tension in the liquid surface. This tension is referred as surface Tension.

22. A fluid is a substance that
(a) always expands until it fills any container.
(b) has the same shear stress at a point regardless of its motion.
(c) cannot remain at rest under the action of any shear force.
(d) cannot be subjected to shear forces.

HPPSC Poly. Lect. 05.07.2021
Karnataka PSC Lect., 27.05.2017
TSPSC Managers, 2015

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Ans. (c) : A fluid is a substance that cannot remain at rest under the action of only shear force.
A substance in liquid or gaseous phase is referred as fluid. They are capable of deforming continuously under the action of tangential force or shear force.

23. If the surface tension of water-air interface is 0.073 N/m, the gauge pressure inside a rain drop of 1mm diameter will be

(a) 0.146 N/m^2 (b) 73 N/m^2
(c) 146 N/m^2 (d) 292 N/m^2

UPPSC AE 13.12.2020, Paper-2
ISRO Scientist/Engineer 2011
ESE 1999

Ans. (d) : Surface tension (σ) = 0.073 N/m
d = 1 mm
Pressure of rain droplet

$$P = \frac{4\sigma}{d}$$

$$P = \frac{4 \times 0.073}{1 \times 10^{-3}}$$

P = 292 N/m^2

24. The dimension of kinematic viscosity in terms of the dimension of mass. M, length L and time T is

(a) $ML^{-1}T^{-1}$ (b) $ML^{-1}T^{-2}$
(c) L^2T^{-1} (d) LT^{-2}

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
GATE 2001, ESE 2007

Ans. (c)

$$\upsilon = \frac{\mu}{\rho}$$

$$\upsilon = \frac{N-s}{m^2} \times \frac{m^3}{kg}$$

$$= \frac{kg-m-sec}{sec^2 . m^2} \times \frac{m^3}{kg}$$

$$= \frac{m^2}{sec}$$

$$= \frac{L^2}{T} = \left[L^2T^{-1}\right]$$

25. A static fluid can have

(a) non-zero normal and shear stress.
(b) negative normal stress and zero shear stress.
(c) positive normal stress and zero shear stress.
(d) zero normal stress and zero shear stress.

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2001

Ans. (b) : negative normal stress and zero shear stress.

26. One poise is equal to

(a) 0.1 N-s/m^2 (b) 1 N-s/m^2
(c) 10 N-s/m^2 (d) 100 N-s/m^2

Assam PSC CCE Pre 2015
(HPPSC LECT. 2016)
J & K PSC Screening, 2006

Ans. (a)

$$1 \text{ Poise} = \frac{\text{Dyne} - \text{sec}}{\text{cm}^2}$$

$$\left\{\text{Dyne} = \frac{\text{gm} - \text{cm}}{\text{sec}^2}\right\}$$

$$1 \text{ Poise} = \frac{\text{gm}}{\text{cm} - \text{sec}}$$

Conversion

$$1 \text{ Poise} = \frac{1}{10} \text{ Pa} - \text{sec}$$

Dimension of dynamic viscosity

$$\left[ML^{-1}\ T^{-1}\right]$$

27. Viscosity of water in comparison to mercury is:

(a) higher (b) same
(c) unpredictable (d) lower

Gujarat PSC AE 2019
GWSSB AAE, 27.12.2015
ISRO Scientist/Engineer 12.05.2013

Ans. (d) :

Dynamic Viscosity	Kinematic Viscosity
$\mu_{water} = 1.0 \times 10^{-3} \frac{N-s}{m^2}$	$v_{water} = 1 \times 10^{-6} \frac{m^2}{sec}$
$\mu_{Hg} = 1.56 \times 10^{-3} \frac{N-s}{m^2}$	$v_{Hg} = 1.15 \times 10^{-7} \frac{m^2}{sec}$

28. A liquid compressed in cylinder has a volume of 0.04m^3 at 50 kg/cm^2 and a volume of 0.039m^3 at 150 kg/cm^2. The bulk modulus of elasticity of liquid is

(a) 400 kg/cm^2 (b) 40 × 10^6 kg/cm^2
(c) 40 × 10^5 kg/cm^2 (d) 4000 kg/cm^2

SJVN ET 2013
ISRO Scientist/Engineer 2011, 2007

Ans. (d) : V_1 = 0.04 m^3, P_1 = 50kg/cm^2, V_2 = 0.039 m^3, P_2 =150 kg/cm^2
Bulk Modulus

$$K = \frac{dP}{-\frac{dV}{V}}$$

$$= \frac{P_2 - P_1}{-\frac{(V_2 - V_1)}{V_1}} = \frac{150 - 50}{-\left(\frac{0.039 - 0.04}{0.04}\right)}$$

$$= \frac{100}{\left(\frac{0.001}{0.04}\right)}$$

= 4000 kg/cm^2

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29. Inside a soap bubble, the pressure is

(a) $\frac{4\sigma}{d}$ (b) $\frac{2\sigma}{d}$

(c) $\frac{8\sigma}{d}$ (d) $\frac{6\sigma}{d}$

Nagaland PSC CTSE 2017, 2016 Paper-II
APPSC AEE Mains 2016 (Civil Mechanical)
TNPSC AE 2013

Ans. (c) :
- Pressure inside drop (solid like sphere)

$P = \frac{4\sigma}{d}$

- Pressure inside soap bubble

$P = \frac{8\sigma}{d}$

- Pressure inside jet

$P = \frac{2\sigma}{d}$

30. A liquid would wet the solid,

(a) If adhesion force is less than cohesion force
(b) If adhesion force is zero
(c) If adhesion force is more than cohesion force
(d) If adhesion force is equal to cohesion force

Vadodara Muncipal Corp. DEE, 2018
Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : A liquid would wet the solid, when
– Adhesive force greater than cohesive force
– Contact angle $\theta < 90^0$ (Acute angle)
– Water is a wetting fluid

31. Which of the following is unitless?

(a) Kinematic viscosity (b) Bulk modulus
(c) Surface tension (d) Strain

Rajasthan Nagar Nigam AE 2016, Shift-II
(KPSC AE 2015)

Ans. (d) : Strain is the unitless quantity–

$\text{Strain} = \frac{\text{Change in dimension}}{\text{original dimension}} = \frac{\Delta L}{L}$

- Bulk modulus – Unit N/m^2 or Pa

$K = \frac{\Delta P}{\left(-\frac{\Delta V}{V}\right)}$

- Unit of kinematic viscosity –

$\upsilon = \frac{\mu}{\rho}$

MKS – m^2/sec
S.I. – m^2/sec
CGS – cm^2/sec = stokes

- Unit of surface tension = N/m or J/m^2

$T = \frac{F}{\ell}$

$\text{Also tension} = \frac{\text{Energy}}{\text{Area}} = \frac{E}{A}(J/m^2)$

32. The density of water is maximum at

(a) 100^0C (b) 0^0C
(c) 0^0K (d) 4^0C

GMB AAE 25.06.2017
Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (d) : The density of water is maximum at 4^0C. At 4^0C, the density of water starts reducing due to hydrogen bond formation.

33. A fluid having no viscosity is known as

(a) real fluid
(b) ideal fluid
(c) Newtonian fluid
(d) non-Newtonian fluid

CGPSC AE 15.01.2021
Assam PSC CCE Pre 2015

Ans. (b) : Properties of an ideal fluid
- Non viscous, frictionless and incompressible.
- Does not offer shear resistance against flow.
- Bulk modulus is infinite.
- No such fluid exist in practical situation.

34. If the fluid is at rest, then

(a) The shear stress changes according to the coefficient of viscosity
(b) Value of shear stress is zero
(c) Shear stress along the horizontal plane is zero
(d) All of the above options

Nagaland PSC CTSE 2017 Paper-2
GPSC Lect. (Auto) 16.10.2016

Ans. (b) : If the fluid is at rest then value of shear stress is zero.

35. The property of fluid which determines its resistance to shearing stress is called.

(a) Surface tension (b) Viscosity
(c) Compressibility (d) None of the above

GPSC Lect. (Auto) 16.10.2016
VIZAG STEEL MT 2011

Ans. (b) : Viscosity- Viscosity is a measure of a fluids resistance to flow. It describes the internal friction of a moving fluid. A fluid with large viscosity resist motion because its molecules gives it a lot of internal friction. It offers resistance to shearing stress.

36. If the diameter of a capillary tube is doubled, the capillary rise will become

(a) $\sqrt{2}$ times less (b) double
(c) half (d) $\sqrt{2}$ times more

JPSC AE, 10.04.2021, Paper-I
CSE Pre-2001

Ans. (c) : Capillary rise $= \frac{4\sigma\cos\theta}{\rho Dg} \Rightarrow h_1 \propto \frac{1}{D}$

If tube diameter is double

$h_2 \propto \frac{1}{2D}$

$\boxed{h_2 = \frac{h_1}{2}}$

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37. In CGS unit, dynamic viscosity is expressed as:

(a) poise (b) Pa-s
(c) stoke (d) None of the above

BPSC Poly. Lect. 2016
BPSC Asstt. Prof. 29.11.2015

Ans. (a) : Dynamic viscosity (μ)

SI system : Pa – sec, or $\frac{N-sec}{m^2}$ or $\frac{kg}{m-sec}$

CGS system : 1 Poise $= \frac{Dyne-sec}{cm^2}\left\{Dyne = \frac{gm-cm}{sec^2}\right\}$

$1\ Poise = \frac{gm}{cm-sec}$

1 poise = 0.1 Pa-sec

Dimension of dynamic viscosity

$[ML^{-1}T^{-1}]$

38. Pick the correct dimension of surface tension.

(a) N/m^2 (b) J/m
(c) W/m^2 (d) J/m^2

AAI Jr. Executive 29.11.2018
GATE 1997

Ans. (d) :

N/m^2	–	Pressure
J/m	–	Force
W/m^2	–	Heat flux
J/m^2	–	Surface tension

39. The shear developed in lubricating oil, of viscosity 9.81 poise, filled between two parallel plates 1 cm apart and moving with relative velocity of 2 m/s is :

(a) 196.2 N/m^2 (b) 20 N/m^2
(c) 40 N/m^2 (d) 29.62 N/m^2

APPSC AE Subordinate Service Civil/Mech. 2016
ESE 2001

Ans. (a) : Given : Viscosity (μ) = 0.981 Ns/m^2

Distance between plates = dy = 1 cm = 1×10^{-2} m

Relative velocity of plates = du = 2 m/s

According to Newton's law of viscosity

$$\tau = \mu\frac{du}{dy} = 0.981 \times \frac{2}{1\times10^{-2}} = 196.2\ N/m^2$$

40. Fluid is a substance which offers no resistance to change of

(a) Pressure (b) Flow
(c) Volume (d) Shape

TNPSC AE 2019
Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (d) : Fluid is a substance which offers no resistance to change of shape.

41. A fluid which obeys Newton's law of viscosity is termed as

(a) Real fluid
(b) Newtonian fluid
(c) Non-Newtonian fluid
(d) Ideal fluid

JPSC AE Pre 2019
ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (b) : A fluid which obeys Newton's law of viscosity is termed as Newtonian fluid.

42. Viscosity of the gas

(a) decreases with increases in temperature
(b) increases with increase in temperature
(c) remains same with increase in temperature
(d) may increase or decrease with increase in temperature depending upon the atomic structure of the gas

Assam PSC AE (IWT) 14.03.2021
WBPSC AE, 2017
CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Viscosity (dynamic viscosity) of gas increase with increase in temperature because viscosity of gas depends on molecular velocity or molecular momentum transfer

Temperature	Liquid (μ)	Gas (μ)
T ↑	↓	↑

43. S.I. unit of viscosity is:

(a) 1 poise (b) 10 poise
(c) Centipoise (d) None of the above

OPSC AEE 2015 PAPER - II
J & K PSC Screening, 2006

Ans : (b) Unit of viscosity:-

(i) MKS unit of viscosity $= \frac{Kgf-sec}{m^2}$

(ii) CGS unit of viscosity $= \frac{dyne-sec}{cm^2}$

(iii) SI unit of viscosity $= \frac{N-sec}{m^2}$

$$\frac{1kgf-sec}{m^2} = \frac{9.81N-sec}{m^2} = 9.81\ poise$$

$$\boxed{10poise = 1\frac{N-sec}{m^2}}$$

44. The capillary rise at 20°C in a clean glass tube of 1 mm bore containing water is approximately :

(a) 5 mm (b) 10 mm
(c) 20 mm (d) 30 mm

HPPSC W.S. Poly. 2016, ESE 2001

Ans : (d) Given, t = 20°c

glass tube dia = 1 mm

$\sigma = 0.0736$ N/m

$$h = \frac{4\sigma\cos\theta}{\rho dg}$$

$\cos\theta = 1$ $(\theta = 0^0)$

$$h = \frac{4\sigma}{\rho gd}$$

$$h = \frac{4\times0.0736}{9810\times1\times10^{-3}}\ m$$

h = 30 mm

45. Units for surface tension is expressed in

(a) N/m (b) N/m^2
(c) m/N (d) n^2/m

APPSC AEE Screening Test 2016
TSPSC AEE 2015

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Ans : (a) Surface tension:- Surface tension occur at the interface of liquid and gas or at the interface of two liquid.
* It is a surface phenomenon
* It is force per unit length (N/m)
* For water-air interface at 20°C its value is 0.0736 N/m
* It is due to cohesion only.

46. If P is the gauge pressure within a spherical droplet, then gauge pressure within a bubble of the same fluid and of same size will be :
(a) P/4 (b) P/2
(c) P (d) 2P

HPPSC Asstt. Prof. 18.11.2016, GATE 1999

Ans. (d) : Gauge pressure for spherical droplet–

$$\Delta P = \frac{4\sigma}{D}$$

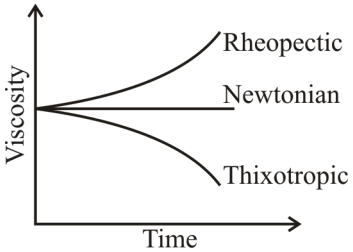
On soap bubble $\Delta P = \frac{8\sigma}{D}$

So 2P

47. The printer's ink is an example of :
(a) Newtonian fluid
(b) ideal fluid
(c) Thixotropic fluid
(d) Plastic fluid

Kerala PSC Asstt. Prof. 27.09.2016
WBPSC AE 2008

Ans. (c) : The printer's ink is an example of thixotropic fluid. Thixotropic is time dependent Non-newtonian fluids. Its viscosity decreases with time.
Thixotropic–Viscosity decrease with stress overtime.
e.g. Honey – Keep stirring and solid becomes liquid.
Rheopectic– Viscosity increases with stress overtime.
e.g. Cream– The longer you whip, thicker it gets.



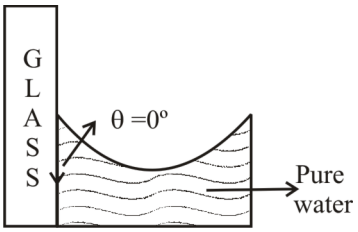
48. The angle of contact between pure water and clean glass is

(OR)

In the equation for capillary rise or fall of a liquid h = 4 σ cosθ/wd, the value of θ for water is :
(a) 0° (b) 45°
(c) 90° (d) 130°

APPSC AEE Mains 2016 (Civil Mechanical)
TSPSC Managers, 2015

Ans. (a) : In case of pure water & clean glass, the angle of contact will be '0°' since if is pure & does not have any type of impurities dissolve in it.



49. Cavitation will take place if the pressure of the flowing fluid at any point is :
(a) more than vapour pressure of the fluid
(b) equal to vapour pressure of the fluid
(c) less than vapour pressure of the fluid
(d) None of the above

GPSC Executive Engineer 23.12.2018
APPSC AE 04.12.2012

Ans. (c) : If the pressure of the flowing fluid at any point is less than the vapour pressure of the fluid then there cavitation will take place.
Due to the cavitation erosion occurs in the runner blade of turbine etc.

50. Compressibility is equal to :
(a) $\frac{dV/V}{dp}$ (b) $\frac{dp}{-(dV/V)}$
(c) $\frac{dp}{dp}$ (d) $\sqrt{\frac{dp}{dp}}$

APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (a) : Compressibility (β) : It is inverse of bulk modulus of elasticity (K).

$$\therefore \boxed{\beta = \frac{1}{K} = \frac{dV/V}{dp}}$$

Where, dV/V = Change in volume per unit volume.
dp = Change in the pressure.

51. One stoke is equal to
(a) 10^{-2} m^2/s (b) 10^{-3} m^2/s
(c) 10^{-4} m^2/s (d) 10^{-6} m^2/s

CGPSC AE 15.01.2021
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : One stoke is equal to 10^{-4} m^2/sec.

52. An ideal fluid is one which
(a) is incompressible
(b) has negligible surface tension
(c) is compressible
(d) is non viscous and incompressible

GPSC Asstt. Prof. 28.08.2016
UKPSC AE 2007 Paper -II

Ans. (d) : An ideal fluid is one which is non viscous and incompressible and also has negligible surface tension.

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53. The capillary rise or depression in a small diameter tube is :

(a) Directly proportional to the specific weight of the fluid
(b) Inversely proportional to the surface tension
(c) Inversely proportional to the diameter
(d) Directly proportional to the surface area

RPSC ACF & FRO, 26.02.2021, ESE 2008

Ans. (c) : Capillarity rise or depression.

$$h = \frac{4\sigma\cos\theta}{\rho g d}$$

$$\boxed{h = \frac{4\sigma\cos\theta}{wd}}$$

$$h \propto \frac{1}{w(\text{specific weight})}$$

$$h \propto \sigma(\text{surface tension})$$

$$h \propto \frac{1}{d(\text{Diameter})}$$

54. For a fluid, the shear stress was found to be directly proportional to the rate of angular deformation. The fluid is classified as

(a) Newtonian (b) non-Newtonian
(c) dilatant (d) thixotropic

Assam PSC AE (PHED), 18.10.2020
Oil India Ltd. Sr. Engg. (Production) 30.11.2019

Ans. (a) : For a fluid, the shear stress was found to be directly proportional to the rate of angular deformation. The fluid is classified as Newtonian fluid.

55. The product of Kinematic Viscosity and mass density is:

(a) Specific viscosity
(b) viscosity index
(c) Coefficient of viscosity
(d) Absolute viscosity

CIL MT 27.02.2020, TNPSC AE 2019

Ans. (d) : Kinematic viscosity (v) is the ratio of absolute viscosity or dynamic viscosity (μ) to the mass density (ρ).

Then, absolute viscosity = Kinematic viscosity × Mass density

$$\mu = v \times \rho$$

56. A soap bubble of radius 25 mm has inside pressure 25 N/m^2 above atmospheric pressure. The surface tension in the soap film will be:

(a) 2.156 N/m (b) 0.156 N/m
(c) 0 N/m (d) 1.156 N/m

AAI Jr. Executive 29.11.2018, ESE 2014

Ans. (b) : r = 25 mm, d = 2r, d = 50 mm

$d = 50 \times 10^{-3}$ m = 0.05 m

$P = 25$ N/m^2

$\sigma = ?$

Soap bubble,

$$P = \frac{8\sigma}{d}$$

$$25 = \frac{8\sigma}{0.05}$$

$$8\sigma = 1.25$$

$$\sigma = \frac{1.25}{8}$$

$$\sigma = 0.156 \text{ N/m}$$

57. The ratio of specific weight of a liquid to the specific weight of pure water at a standard temperature is called

(a) Density of liquid
(b) Specific gravity of liquid
(c) Compressibility of liquid
(d) Surface tension of liquid

Nagaland PSC (CTSE) 2018, Paper-II
VIZAG STEEL MT 2011

Ans. (b) : The ratio of specific weight of a liquid to the specific weight of pure water at a standard temperature is called specific gravity of liquid.

The usual standard of comparison For solids and liquids is water at 4ºC which has a density of 1.0 kg per litre.

58. Oil in a hydraulic cylinder is compressed from an initial volume 2 m^3 to 1.96 m^3 and the pressure increase is from 40 MPa to 80 MPa. The bulk modulus of elasticity of oil is

(a) 1000 MPa (b) 2000 MPa
(c) 4000 MPa (d) 8000 MPa

ISRO Scientist/Engineer (RAC) 07.05.2017
GATE 2007

Ans. (b) : Given,

$V_1 = 2\text{m}^3$, $P_1 = 40$ MPa

$V_2 = 1.96$ m^3, $P_2 = 80$ MPa

Bulk modulus of elasticity (K)

$$K = -\frac{Change\ in\ pressure}{Volumetric\ strain}$$

$$= -\frac{(P_2 - P_1)}{\frac{(V_2 - V_1)}{V_1}} = -\frac{(80-40)}{\left(\frac{1.96-2}{2}\right)}$$

$$= \frac{40 \times 2}{0.04}$$

K = 2000 MPa

59. An ideal fluid is

(a) One which obeys Newton's law of viscosity
(b) Frictionless and incompressible
(c) Very viscous
(d) Frictionless and compressible

Nagaland PSC CTSE 2017 Paper-2
Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : An ideal fluid is an incompressible fluid & frictionless which has a zero viscosity (μ = 0). In ideal flow, there is no existence of shear force because of absence of viscosity.

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60. A fluid in which resistance of deformation is independent of the shear stress, is known as

(a) Bingham plastic fluid (b) Pseudo plastic fluid
(c) Dilatent fluid (d) Newtonian fluid

TNPSC AE 2014
Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : For Newtonian fluid there is constant viscosity for different fluid of particular temp which resist the deformation without any influence of shear stress. In other words dynamic viscosity is constant for particular fluid at particular temperature.

61. A fluid is a substance which deforms continuously under the action of :

(a) Shearing Force (b) Tensile Force
(c) Compressive Force (d) None of these

HPPSC Poly. Lect. 05.07.2021
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : Fluid- A fluid does not have any preferred shape. It can deform continuously when subjected to infinitesimally small shear force. The space between fluids molecules is relatively large i.e. intermolecular force or cohesive force is low.

62. Pascal-Second (Pa-s) is the unit of

(a) Pressure
(b) Kinematic Viscosity
(c) Dynamic Viscosity
(d) Surface tension

RPSC IOF, 2020, UPRVUNL AE 2014

Ans. (c) : Pascal second is the unit of (viscosity) dynamic viscosity.

- Pascal is the unit of pressure
- m^2/sec is the unit of kinematic viscosity
- N/m is unit of surface tension

63. The viscosity of a liquid is

(a) Decreases with increase in temperature
(b) Remains practically constant with temperature rise of fall
(c) Fairly large as compared to viscosity for gases
(d) Consider body influenced by molecular momentum transfer

TNPSC AE 2019, 2018

Ans. (a) : The viscosity of a liquid is decreases with increase in temperature because of molecular cohesive force of liquid decrease with increase in temperature of liquid.

For liquid $T \uparrow \mu \downarrow$

For gas $T \uparrow \mu \uparrow$ [Molecular momentum transfer rate $\uparrow$]

64. For which of the following fluids, the apparent viscosity can be considered to be independent of the rate of shear strain and equal to the fluid's viscosity?

(a) Ketchup (b) Water
(c) Corn Starch Solution (d) Blood

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (b) : Apparent viscosity is the shear stress applied to fluid divided by the rate of shear strain $\left(\mu = \frac{\tau}{du/dy}\right)$. For a Newtonian fluid, the apparent viscosity is constant and equal to Newtonian viscosity of the fluid.
Exp.- Water, air, alcohol, oil etc.
Ketchup, corn starch & blood are example of non Newtonian fluid for that apparent viscosity depends on shear rate.

65. Which of the following is a property of a fluid?

(a) Eddy viscosity
(b) Eddy diffusivity
(c) Kinematic viscosity
(d) Turbulent Prandtl number

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : Kinematic viscosity is the property of fluid.

66. A Newtonian fluid is one for which :

(a) $\tau = \mu\left(\frac{\partial u}{\partial y}\right)^m . m \neq 1$

(b) $\tau = \tau_c + \mu\frac{\partial u}{\partial y}$

(c) $\tau = \mu\frac{\partial u}{\partial y}$

(d) $\tau = \tau_c + \mu\left(\frac{\partial u}{\partial y}\right)^m . m \neq 1$

CGPSC AE 15.01.2021

Ans. (c) : A Newtonian fluid is one which obeys Newton's law of viscosity.
Acc. to Newton's law of viscosity

$$\tau \propto \frac{du}{dy}$$

$$\boxed{\tau = \mu\frac{du}{dy}}$$

where, μ is the dynamic viscosity of fluid.

67. Dimension of viscosity is :

(a) L^2/T (b) $N\text{-}s^2/m$
(c) kg/m-s (d) kg-s/m

CGPSC AE 15.01.2021

Ans. (a) : Dimension of kinematic viscosity = L^2/T
Unit of kinematic viscosity = m^2/sec
Unit of viscosity (Dynamic viscosity) = $N\text{-}s/m^2$
= kg/m-s
Dimension of viscosity = $[ML^{-1}T^{-1}]$
And,
Dimension of kinematic viscosity = $[L^2T^{-1}]$

68. The property of the liquid surface film to exert a tension so that even small loads can be supported over it is known as

(a) Viscosity (b) Adhesion
(c) Surface tension (d) Capillary

APGCL AM, 2021

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Ans. (c) : The property of the liquid surface film to exert a tension so that even small loads can be supported over it is known as Surface tension.
It is due to unalarmed cohesive forces at the liquid gas interface or liquid-liquid interface, in liquid side.

69. A liquid has a specific gravity of 1.9 and a kinematic viscosity of 6 stokes. What is the dynamic viscosity?

(a) 1.14 Ns/m^2 (b) 2.44 Ns/m^2
(c) 3.40 Ns/m^2 (d) 11.40 Ns/m^2

ESE 18.07.2021

Ans. (a) : Given, Specific gravity = 1.9
Kinematic viscosity, $\nu = 6$ stokes

$$\nu = 6 \times 10^{-4}\ m^2/s$$

$$\nu = \frac{\mu}{\rho} \ (\rho = \text{density})$$

Dynamic viscosity, $\mu = \nu\rho$

$$= 6 \times 10^{-4} \times 1.9 \times 10^3 = 1.14\ N\text{-}s/m^2$$

70. Statement (I): The viscosity of liquids decreases with the increase of temperature while the viscosity of gases increases with the increase of temperature.
Statement (II): The viscous forces in a fluid are due to cohesive forces and molecular Momentum transfer.

ESE 18.07.2021

Ans. (a) :

71. Fluids for which the flow behaviour index 'n' is greater than unity are called _____

(a) dilutant (b) coherent
(c) dilatant (d) none of the above

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) :

72. A 2 liter liquid weights 15N. Its specific gravity will be

(a) 0.764 (b) 7500N m^{-3}
(c) 764.5kgm^{-3} (d) 764

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) :

73. The surface tension at the soap-air interface is 0.1 N/m. The internal pressure in the soap bubbles of 32 mm diameter is _____ N/m^2.

(a) 12.5 (b) 25
(c) 37.5 (d) 50

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (b) : Given,
Surface tension, $\sigma = 0.1$ N/m
Diameter, d = 32 mm

$$d = 32 \times 10^{-3}\ m$$

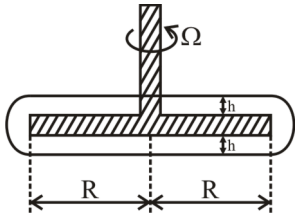
Pressure in the soap bubble,

$$P = \frac{8\sigma}{d}$$

$$= \frac{8 \times 0.1}{32 \times 10^{-3}} = \frac{800}{32}$$

$$P = 25\ N/m^2$$

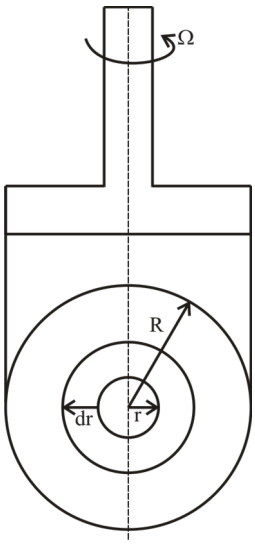
74. A disc of radius R rotates at an angular velocity Ω inside a disc shaped container filled with oil of viscosity μ, as shown in figure. Assuming a linear velocity profile and neglecting shear stress on the outer disc edges, the formula for viscous torque on the disc is



(a) $\frac{\pi\mu\Omega R^2}{h}$ (b) $\pi\mu\Omega h R^3$
(c) $\frac{\pi\mu\Omega R^5}{h^2}$ (d) $\frac{\pi\mu\Omega R^4}{h}$

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (d) :



From Newton law of viscosity,

$$\gamma = \mu \cdot \frac{du}{dy} = \mu \cdot \frac{r\Omega}{h} \qquad du = v - 0 = r\omega = r\Omega$$

$$dy = h$$

Consider a strip of thickness dr at radius r,
Friction force on strip, $dF_f = \tau.2\pi rdr$
Frictional torque on strip $dT_f = dF_f \times r$

$$dT_f = \tau.2\pi rdv.r$$

$$dT_f = \mu \cdot \frac{r\Omega}{h} \cdot 2\pi r^2 dr$$

$$\int dT_f = \frac{2\pi\mu\Omega}{h} \int_0^R r^3 dr$$

One surface torque $T_f = \frac{\pi\mu\Omega R^4}{2h}$

So viscous torque on the both surface of the disc,

$$T_f = \frac{\pi\mu\Omega R^4}{h}$$

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75. Find the change in height (h) in a circular tube (of radius R = 1 mm) filled with liquid (water) with surface tension T = 0.073 N/m. Contact angle of water-air-glass interface is 0°

(a) 1.5 cm (b) 2.5 cm
(c) 0.5 cm (d) 0.15 cm

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : Capillary rise (h) $= \frac{4\sigma\cos\theta}{\rho g D}$

R = 1 m, D = 2 mm, $\theta = 0°$C, g = 9.81 m/sec^2

σ = 0.073 N/m, ρ = 1000 kg/m^3

$$h = \frac{4\times0.073\times\cos\theta}{1000\times9.81\times2\times10^{-3}} = 0.0148\,m$$

h = 1.5 cm

76. Which of the following is a non-Newtonian fluid?

(a) Air (b) Water
(c) Gasoline (d) None of the above

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : Examples of various types of fluids.
(1) Newtonian fluid- Water, Air, kerosene, gasoline, diesel, petrol, methyl alcohol, ethyl alcohol, etc.
(2) Non Newtonian fluid
(a) Time dependent fluid
(i) Thixotropic : Printer ink
(ii) Rheopectic : Gypsum paste
(b) Time independent fluid:
(i) Pseudo plastic : Blood, tomato, ketchup, paint etc.
(ii) Dilatant : Sugar solution, Butter solution, starch solution, quick sand etc.
(iii) Ideal plastic or Bingham plastic : Tooth paste.

77. A flat plate 0.1 m^2 area is pulled at 30 cm/s relative to another plate located at a distance of 0.01 cm from it, the fluid separating them being water with viscosity of 0.001 Ns/m^2. The power required to maintain velocity will be

(a) 0.05 W (b) 0.07 W
(c) 0.09 W (d) 0.11 W

ESE 2020

Ans. (c) :

$\because$ Shear stress, $\tau = \mu\frac{du}{dy} = \mu\frac{u}{y}$

Power, $P = (\tau\times A)\times u$

$$= \left(\mu\frac{u}{y}\times A\right)\times u \quad = \mu A\frac{u^2}{y}$$

Given : μ = 0.001 N-s/m^2
y = 0.01 cm
u = 30 cm/s
A = 0.1 m^2
So, putting the respective values,

$$P = 0.001\times0.1\times\frac{(30)^2\times10^{-4}}{0.01\times10^{-2}}$$

$\boxed{P = 0.09\,W}$

78. When the pressure of liquid is increased from 3 MN/m^2 to 6 MN/m^2, its volume is decreased by 0.1%. The bulk modulus of elasticity of the liquid will be

(a) 3×10^{12} N/m^2
(b) 3×10^{9} N/m^2
(c) 3×10^{8} N/m^2
(d) 3×10^{4} N/m^2

ESE 2020

Ans. (b) : $\because$ Bulk modulus of elasticity,

$$K = -\frac{dp}{\left(\frac{dv}{v}\right)} = -\frac{(6-3)\times10^6}{\left(\frac{0.1}{100}\right)} = 3\times10^9\ N/m^2$$

79. The dynamic viscosity of a liquid is 1.6 × 10^{-4} Ns/m^2, whereas the density is 400 kg/m^3. The kinematic viscosity in m^2/s is :

(a) 64×10^{-8} (b) 40×10^{8}
(c) 64×10^{-6} (d) 40×10^{-8}

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Given that dynamic viscosity (μ) = 1.6×10^{-4} N sec/m^2, density (ρ) = 400 kg/m^3

Kinematic viscosity (ν) = $\frac{\mu}{\rho}$

$$\nu = \frac{1.6\times10^{-4}}{400}$$

$$\nu = \frac{1.6\times10^{-4}\times10^{-2}}{4}$$

$\nu = 4\times10^{-7}$ m^2/sec. $= 40\times10^{-8}$

80. A substance whose specific volume is constant is called

(a) Incompressible substance
(b) Ideal substance
(c) Pure substance
(d) Actual substance

TNCSC AE 2020

Ans. (a) : A substance whose specific volume (or density) is constant is called an incompressible substance, the specific volumes of solids and liquids essential remain constant during a process.

81. Two large horizontal surfaces are 2 cm apart. The space between the surface is filled with the fluid of dynamic viscosity of 8 poise. What will be the force required to large a very thin plate of surface area 1 m^2 between the two large plane surface with the speed of 0.5 m/s. The thin plate is equidistance of 1 cm from each large plane surface. Buoyancy and gravitational effect are considered to the negligible.

(a) 0.8 N (b) 8 N
(c) 80 N (d) 800 N

APPSC Poly Lect. 13.03.2020

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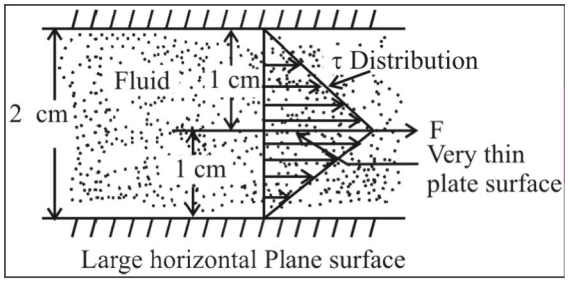
Ans. (c) : Given data :

Dynamic viscosity (μ) of fluid = 0.8 N-s/m^2 = 8 poise



Very thin plate surface area (A) = 1 m^2

speed (v)= 0.5 m/s

distance (y) = 1 cm = 0.01 m

let,

F = force required to drag a very thin plate

$\because \quad \tau = \frac{F}{A} = \mu\frac{du}{dy}$ or $\mu\left(\frac{v}{y}\right)$

Drag force, $F_D = \mu A\left(\frac{v}{y}\right)$

Drag force will act both side of the plate –

Total drag force (F) = $2\mu A\left(\frac{v}{y}\right)$ (i.e. both side of plate)

$$= 2\times 0.8\times 1\times\left(\frac{0.5}{0.01}\right) = 2\times 0.8\times 50$$

F = 80 N

82. The pressure inside a droplet of water is 0.2 kPa greater than outside atmospheric pressure. What will be the diameter of droplet of water? [Surface tension of water in contact with air is 0.07 N/m]

(a) 0.7 mm (b) 1.4 mm

(c) 14 mm (d) 0.14 mm

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given data :

Pressure (ΔP) = 0.2 kPa = 0.2×10^3 Pa

Surface tension (σ) = 0.07 N/m

Let,

d= diameter of the droplet of water

$\because$ Pressure inside droplet

$$(\Delta P) = \frac{4\sigma}{d} \qquad \text{(In case of water droplet)}$$

$$\therefore \quad d = \frac{4\sigma}{\Delta P} = \frac{4\times 0.07}{0.2\times10^3} = \frac{0.28}{0.2}\times10^{-3}\,m$$

$d = 1.4\times10^{-3}$ m

d = 1.4 mm

83. The pH for the potable water should be in the range of

(a) 5-6 (b) 7-8.5

(c) 9-10 (d) 12-13

Assam PSC AE (PHED), 18.10.2020

Ans. (b) : The pH for the potable water (drinking water) lies within the range 7 to 8.5.

Natural waters can be of lower pH, as a result of for example acid rain or higher pH in limestone areas.

pH < 7 $\rightarrow$ Acidic

pH > 7 $\rightarrow$ Basic

84. A small shear force is applied on an element and then removed. If the element regains its original position, what kind of an element can it be?

(a) Solid (b) Liquid

(c) Fluid (d) Gaseous

Assam PSC AE (PHED), 18.10.2020

Ans. (a) : Fluid such as liquid fluid and gaseous cannot resist even a small shear force and get permanently deformed. Hence, the element must be a solid element.

85. A beaker is filled with a liquid up to the mark of one litre and weighed. The weight of the liquid is found to be 6.5 N. The specific gravity of the liquid will be

(a) 0.65 (b) 0.66

(c) 0.67 (d) 0.68

Assam PSC AE (PHED), 18.10.2020

Ans. (b) : Specific gravity of a liquid is defined as the ratio of the density of the liquid to that of water density

of the liquid (ρ_l) = $\frac{6.5/9.81}{10^3}$ kg/m^3

density of water (ρ_w) = 1000 kg/m^3

Thus specific gravity S = 0.66

86. For an incompressible fluid, the density

(a) Varies with temperature only

(b) Varies with pressure only

(c) Varies with both pressure and temperature

(d) Remain constant

JPSC AE PRE 2019

Ans. (d) : Incompressible flow—A flow in which the volume of a fluid and its density does not change during the flow is called an incompressible flow. All the liquids are considered to have incompressible flow.

87. Pick the correct statement about viscosity.

(a) In general, it decreases with temperature for liquids because the internal force of attraction between two fluid layers gets decreased.

(b) In general, it decreases with temperature for gases because the internal force of attraction between two fluid layers gets decreased.

(c) In general, it increases with temperature for liquids because the internal force of attraction between two fluid layers gets increased.

(d) In general, it increases with temperature for gases because the internal force of attraction between two fluid layers gets increased.

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Ans. (a) : In general, it decreases with temperature for liquid because the internal force of attraction between two fluid layers gets decreased.
In case of gases as the temperature increases the molecular momentum transfer increases so viscosity of gases increases.

88. Calculate the specific weight of one litre of petrol of specific gravity 0.7
(a) 700 N/m^3 (b) 6867 N/m^3
(c) 9810 N/m^3 (d) 6.8 N/m^3
TNPSC AE 2019

Ans. (b) : sp. gravity of petrol

$$= \frac{\text{Density of petrol}\left(\rho_p\right)}{\text{Density of water}\ \left(\rho_w\right)}$$

$$\rho_p = \rho_w \times \text{ sp. gravity of petrol}$$

$$\rho_p = 1000 \times 0.7 = 700 \frac{kg}{m^3}$$

Specific weight of petrol = $\rho_p \times g$
$= 700 \times 9.81$
$= 6867\ N/m^3$

89. In an incompressible fluid flow, the density of the medium is
(a) Zero (b) Infinity
(c) Constant (d) Unity
APPSC AEE SCREENING 17.02.2019

Ans. (c) : For incompressible flow
density (ρ) = constant

90. If σ is the surface tension and R is the cylinder radius, the pressure increase (ΔP) in the interior of a liquid cylinder is given by
(a) $\frac{\sigma}{R}$ (b) $\frac{2\sigma}{R}$
(c) $\frac{3\sigma}{R}$ (d) $\frac{4\sigma}{R}$
APPSC AEE SCREENING 17.02.2019

Ans. (a) : The pressure rise across cylindrical liquid jet is given by

$$\Delta P = \frac{2\sigma}{D} = \frac{\sigma}{R}$$

91. For pseudoplastic non-Newtonian fluids, the apparent viscosity
(a) increases with increasing deformation rate
(b) decreases with increasing deformation rate
(c) is independent of the deformation rate
(d) decreases with time
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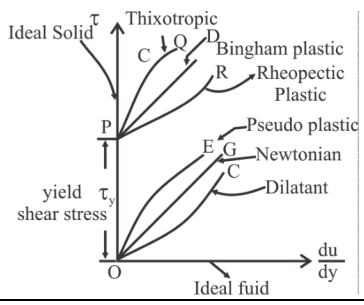
Ans. (b) :



Pseudo plastic fluid : (represented by line OE).
- $\tau_y = 0$ and $n < 1$
- The apparent viscosity decreases with the rate of deformation or rate of shear strain. Hence, Pseudo plastic fluids are also called shear thinning fluids.

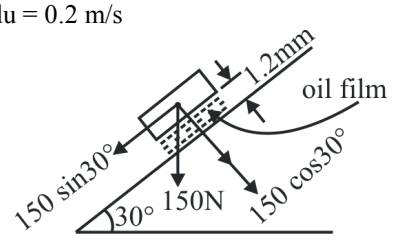
e.g. Blood, milk, Paper Pulp, Polymeric solutions such as rubber, suspension paints.

92. Which fluid does not experience stress during flow?
(a) Dilatant (b) Bingham
(c) Viscoplastic (d) Inviscid
JPSC AE Pre 2019

Ans. (d) : $\tau = \mu \frac{du}{dy}$
for Inviscid, $\mu = 0$, Then $\tau = 0$

93. A plate weighing 150 N and measuring 0.8 ×0.8 m^2 just slides down an inclined plane over an oil film of 1.2 mm thickness for an inclination of 30° and velocity of 0.2 m/s. Then the viscosity of the oil used is
(a) 0.3 Ns/m^2 (b) 0.4 Ns/m^2
(c) 0.5 Ns/m^2 (d) 0.7 Ns/m^2
ESE 2019

Ans. (d) : Given, W = 150 N
$A = 0.8 \times 0.8\ m^2$
dy = 1.2 mm
$\theta = 30°$
du = 0.2 m/s



For Newtonian fluid

$$\tau = \mu \frac{du}{dy}$$

where τ = shear stress $= \frac{F}{A} = \frac{150 \sin 30}{0.8 \times 0.8}$

$$\tau = \frac{15000}{128} \frac{N}{m^2}$$

$$\frac{15000}{128} = \mu \frac{du}{dy} = \mu \frac{0.2}{1.2 \times 10^{-3}}$$

$$\mu = 0.70 \frac{N-s}{m^2}$$

94. If the bulk modulus of elasticity of the water is 2.2 × 10^6 kN/m^2, the speed of pressure wave is given by
(a) 22000 m/s (b) 1483.2 m/s
(c) 3561.2 m/s (d) 500.3 m/s
APPSC AEE SCREENING 17.02.2019

Ans. (b) : The speed of pressure wave (or sonic speed) is given by

$$v = \sqrt{\frac{K}{\rho}} = \sqrt{\frac{2.2 \times 10^9}{1000}} = 1483.2\ m/s$$

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95. Pick the correct statement about the bulk modulus of elasticity :

(a) it is a dimensionless number
(b) is independent of pressure and viscosity
(c) it is larger if fluid is more compressible
(d) it is higher if the fluid is less compressible

BHEL ET 2019

Ans. (d) : Bulk modulus of elasticity (K) is defined as the ratio of compressive stress to voloumetric strain.

$$\text{Bulk modulus} = K \uparrow = \frac{dp}{-\frac{dV}{V}} = -\frac{Vdp}{dV \downarrow}$$

96. In the surface tension of a soap bubble is 0.035 N/m, then work done in blowing the soap bubble of radius 5 cm in air is :

(a) 220 mJ (b) 2.2 mJ
(c) 22 mJ (d) 0.22 mJ

BHEL ET 2019

Ans. (b) : Given -

Surface tension of soap bubbles

$\sigma = 0.035$ N/m

Radius of soap bubble R = 5 cm

Work done $= \sigma[\text{change of surface area}]$

$= \sigma\left[2 \times 4\pi R^2\right]$

$= 0.035 \times (2 \times 4 \times \pi \times 0.05^2)$

$= 0.002199$

$= 2.199 \times 10^{-3}$ J

$= 2.2$ mJ

97. Pick the correct statement about mechanical pressure of fluid element.

(a) It is a first order tensor and depends upon the orientation of the surface upon which it acts.
(b) It is a zero order tensor and does not depend upon the orientation of the surface upon which it acts.
(c) It is zero order tensor and depends upon the orientation of the surface upon which it acts.
(d) It is a first order and does not depend upon the orientation of the surface upon which it acts.

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Ans. (b) :

- Net pressure acting upon element doesn't depend on its inclination.
- Pressure is a scalar quantity and it have only magnitude not direction.
- Pressure is a zero order tensor.

98. 'A fluid is at rest' means that :

(a) it has zero normal stress and non-zero shear stress.
(b) it has non-zero normal stress and zero shear stress.
(c) it has non-zero normal stress and shear stress.
(d) it has zero-normal stress and zero shear stress.

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Ans. (b) : A fluid is at rest means, it has non-zero normal stress and zero shear stress.

99. Select the quantity in the following that is not a dimensionless parameter

(a) Froude number (b) Weber number
(c) Kinematic viscosity (d) Pressure coefficient

Gujarat PSC AE 2019

Ans : (c) : Kinematic viscosity 'v'

$$\nu = \frac{\text{Dynamic viscosity}}{\text{Density}}$$

Kinematic viscosity has the dimensions $L^2\ T^{-1}$ and the units $m^2\ s^{-1}$.

100. A cubical wooden block of edge 100 mm and weight 1 kN sliding down on an inclined plane of inclination 30^0 with the horizontal. A Newtonian fluid with a viscosity of 0.2 Ns/m^2 is layered on the incline plane. If the thickness of the fluid layer is 0.02 mm, then terminal velocity of the block in m/s is :

(a) 0.25 (b) 2.5
(c) 5 (d) 0.5

BHEL ET 2019

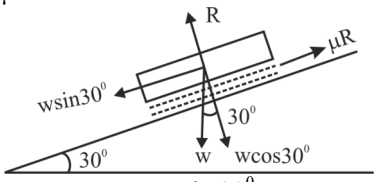
Ans. (c) : Given -

Cubic block, a = 100 mm = 0.1 m

Weight (W) = 1 kN, y = 0.02 mm = 0.02×10^{-3}m

$\theta = 30^0$

$\mu = 0.2$ N-s/m^2



$F_x = W \sin 30^0$

$= 1 \times 10^3 \times \frac{1}{2}$

$F_x = 1000 \times \frac{1}{2} = 500\text{N}$

Area $= a \times a$

$= 0.1 \times 0.1$

$= 0.01\ m^2$

According to Newtonian fluid

$$\tau = \mu \frac{du}{dy}$$

$$\tau = \mu \frac{v}{y}$$

$$\frac{F_x}{A} = \mu \frac{v}{y}$$

$$v = \frac{F_x y}{A\mu}$$

$$= \frac{500 \times 0.02 \times 10^{-3}}{0.01 \times 0.2} = \frac{500 \times 0.02}{0.002}$$

$$= \frac{10}{2} = 5\ \text{m/sec}$$

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101. One litre of a certain fluid weighs 12 N. What is its specific volume?

(a) $8.175 \times 10^{-4} m^3/kg$ (b) $8.175 \times 10^{-3} m^3/kg$

(c) $69.24 \times 10^{-4} m^3/kg$ (d) $69.24 \times 10^{-3} m^3/kg$

MECON MT 2019

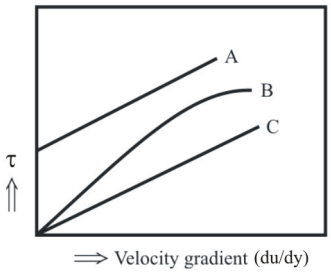
Ans. (a) : Specific volume $= \frac{V}{m}$

Where, $V = 1 \times 10^{-3}\ m^3$

$$m = \frac{12}{9.81}\ kg$$

$$\text{Specific volume} = \frac{1 \times 10^{-3} \times 9.81}{12} = 8.175 \times 10^{-4}\ m^3/kg$$

102. Figure shows a plot of shear stress (τ) versus velocity gradient (du/dy). Name the type of fluids corresponding to A, B and C



(a) A = ideal plastic fluid, B = Newtonian fluid and C = Non–Newtonian fluid

(b) A = ideal plastic fluid, B = Non- Newtonian fluid and C = Newtonian fluid

(c) A = ideal fluid, B = Non- Newtonian fluid and C = Newtonian fluid

(d) A = ideal fluid, B = Newtonian fluid and C= Non-Newtonian fluid

Oil India Ltd. Sr. Engg. (Drilling) 30.11.2019

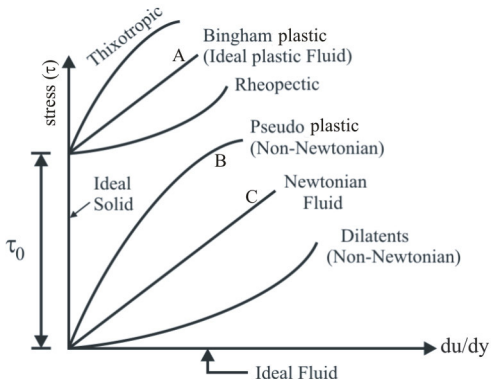
Ans. (b) : Shear stress and velocity gradient diagram for various types of fluid.

For Newtonian Fluid.

$$\tau = \mu\left(\frac{du}{dy}\right)$$

For Non-Newtonian Fluid-

$$\tau = B + k\left(\frac{du}{dy}\right)^n$$



- Dilatent (shear thicking fluid) → B = 0; n > 1
  Example– Honey, Sugar solution, quick sand etc.
- Pseudoplastic (shear thining fluid) → B = 0; n < 1
  Example– Blood, milk, polymer solution (colloidal solution) etc.
- Bingham plastics → B ≠ 0; B = τ_0, n = 1
  Example- Toothpaste, gel, cream, drilling mud etc.

103. The specific gravity of mercury is

(a) 1360 (b) 13600

(c) 1.36 (d) 13.6

APPSC AEE SCREENING 17.02.2019

Ans. (d) : $S_m = \frac{\rho_m}{\rho_w} = \frac{13600}{1000} = 13.6$

104. If σ refers to the surface, tension of the water, ρ refers to mass density, g refers to gravitational accelerations, d refers to the diameter of the glass tube, then the capillary rise of water in the glass tube h is given by

(a) $h = \frac{2\sigma}{\rho g d}$ (b) $h = \frac{3\sigma}{\rho g d}$

(c) $h = \frac{4\sigma}{\rho g d}$ (d) $h = \frac{6\sigma}{\rho g d}$

TNPSC AE 2019

Ans. (c) : The capillary rise of water in the glass tube h is given as

$$h = \frac{4\sigma \cos\theta}{\rho g.d}$$

$$\text{So, } h = \frac{4\sigma}{\rho g d}$$

$\theta = 0^o$ [For water in glass tube]

105. What will be the density of fluid having dynamic viscosity of 0.05 poise and kinematic viscosity of 0.04 stokes?

(a) 125 kg/m^3 (b) 12.5 kg/m^3

(c) 1250 kg/m^3 (d) 1.25 kg/m^3

(e) 12500 kg/m^3

CGPSC AE 25.02.2018

Ans. (c) : We know that

$$\text{kinematic vicsocity}(\nu) = \frac{\text{Dynamic viscosity}(\mu)}{\text{Density}(\rho)}$$

$$\rho = \frac{0.05 \times 10^{-1}}{0.04 \times 10^{-4}} = 1250\ kg/m^3$$

$$1\ \text{Poise} = 10^{-1}\ \frac{N\text{-}s}{m^2}$$

$$1\ \text{Stokes} = 10^{-4}\ \frac{m^2}{s}$$

106. A glass tube of 8 mm diameter is immersed in water. Surface tension of water is 0.0075 kg m^{-1} and rise of water in the tube due to capillary effect is

(a) 7.5 mm (b) 3.75 mm

(c) 11.25 mm (d) None of these

ISRO Scientist/Engineer 22.04.2018

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Ans. (b) : Rise of water in the tube due to capillary effect

$$h = \frac{4\sigma\cos\theta}{\rho g d}$$ For water in glass tube θ = 0°,

Cos 0° = 1

$$h = \frac{4\times 0.0075\times 1}{1000\times 0.008} \qquad (\sigma = 0.0075\times g \;\; N/m)$$

h = 0.00375 m

$$\boxed{h = 3.75\,mm}$$

107. Property of a fluid by which molecules of different kinds of fluids are attracted to each other is called

(a) adhesion (b) cohesion
(c) viscosity (d) surface tension

TNPSC AE 2018

Ans. (a) : Adhesion, is the property of fluid by which molecules of different kinds of fluids are attracted to each other.

108. Compressibility is the reciprocal of

(a) bulk modulus of elasticity
(b) shear modulus of elasticity
(c) young's modulus of elasticity
(d) viscosity

TNPSC AE 2018

Ans. (a) : Compressibility is the reciprocal of bulk modulus of elasticity.

$$\beta = \frac{1}{K} = \frac{\left[\frac{\Delta V}{V}\right]}{\Delta P}$$

109. The normal stresses within an isotropic Newtonian fluid are related to

1. Pressure
2. Viscosity of fluid
3. Velocity gradient

Which of the above are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : The normal stresses of an isotropic Newtonian fluid are related to pressure, viscocity and velocity gradient. The relationships for the normal stresses is

$$\sigma_x = -p + 2\mu\frac{\partial u}{\partial x}$$

110. The mass of 2.5 m^3 of a certain liquid is 2 tonnes. Its mass density is

(a) 200 kg/m^3 (b) 400 kg/m^3
(c) 600 kg/m^3 (d) 800 kg/m^3

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Mass of liquid = 2 tonn = 2000 kg

volume of the solution = 2.5 m^3

mass density is defined as mass per unit volume.

So mass density = $\frac{M}{V} = \frac{2000}{2.6} = 800\ kg/m^3$

111. A vessel of 4 m^3 contains an oil which weights 30 kN. The specific weight of the oil is

(a) 4.5 k N/m^3 (b) 6 k N/m^3
(c) 7.5 k N/m^3 (d) 10 k N/m^3

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Specific weight is the amount of weight per unit volume of a substance

$$\gamma = \frac{w}{v}$$

$$\gamma = \frac{30}{4} = 7.5 \;\; kN/m^3$$

112. The variation in the volume of a liquid with the variation of pressure is called its

(a) Surface tension (b) Compressibility
(c) Capillary (d) Viscosity

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : The variation in the volume of a liquid with the variation of pressure is called its compressibility.

113. One litre of water occupies a volume of

(a) 100 cm^3 (b) 250 cm^3
(c) 500 cm^3 (d) 1000 cm^3

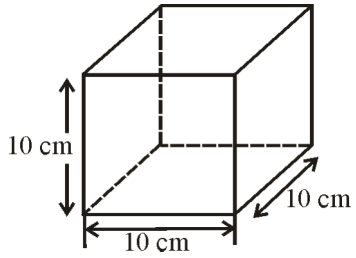
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : One liter is the volume of a cube that is 10 cm on each side.

Volume of cube = length × width × height

1 liter = 1000 cm^3

1 liter = 1000 × 10^{-6} m^3



114. Bulk modulus of a fluid is the ratio of

(a) Shear stress to shear strain
(b) Increase in volume to the viscosity of fluid
(c) Increase in pressure to the volumetric strain
(d) Critical velocity to the viscosity of fluid

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Bulk modulus is a measure of the resistance of a fluid to compression. It is defined as the ratio of pressure stress to volumetric strain.

115. How does the viscosity of fluids vary with increase in temperature?

(a) Increases for both liquids and gases
(b) Increases for liquids, decreases for gases
(c) Increases for gases, decreases for liquids
(d) Decreases for both liquids and gases

ISRO Scientist/Engineer 07.05.2017

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Ans. (c) :
Temperature↑= μ_{gas} ↑ (viscosity of gas increases)
and Temperature↑= μ_{liquid} ↓ (Viscosity of liquid decreases)

116. Viscosity is the property of fluid evolved due to
(a) Resistance offered by the layers of fluid
(b) Resistance offered by sidewalls of the channel
(c) Spontaneous reactions in the fluid
(d) None of the above

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : Viscosity is defined as the property of fluid evolved due to internal resistance offered by one layer of the fluid to the adjacent layer.

117. The viscosity of liquids decreases with increase in temperature due to:
(a) Decreased cohesive forces
(b) Increased cohesive forces
(c) Decreased molecular momentum transfer
(d) Increased molecular momentum transfer

CIL MT 26.03.2017

Ans. (a) : In liquids main cause of viscosity is **cohesion** between the molecules. With increase in temperature the cohesive force decreases as the energy of particles become more, hence movement of particles become easy. Hence viscosity of water decreases with increase in temperature.
In gases the important cause of viscosity is **randomness/molecular** collision due to C_{rms}(root mean square) velocity. Due to rise in temperature, kinetic energy of molecules increases which makes C_{rms} increase, hence randomness and collision of molecules increased. This makes the flow difficult and hence viscosity in gases increases with temperature.

118. One kgf/cm^2 when converted to SI units is
(a) 0.0981 MPa (b) 0.98 MPa
(c) 10^4 Pa (d) 1 Pa

TNPSC AE 2017

Ans. (a) : $\text{kgf/cm}^2 = \dfrac{9.81}{10^{-4}}\dfrac{\text{N}}{\text{m}^2} = 9.81\times10^4\text{Pa}$
$= 0.0981 \times 10^6$ Pa = 0.0981 MPa

119. The excess pressure in a droplet of 0.002m diameter a fluid with surface tension of 0.01 N/m is
(a) 10 (b) 20
(c) 4 π (d) 0.0004 π

TNPSC AE 2017

Ans. (b) :
$$P_{gauge} = \frac{4\sigma}{d}$$
$$= \frac{4\times0.01}{0.002}$$
$$= 20 \text{ Pa}$$

120. Statement (I): Non viscous flow between two plates is an example for irrotational flow.
Statement (II): Forced vortex is irrotational in nature.
(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I)
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I)
(c) Statement (I) is true but Statement (II) is false
(d) Statement (I) is false but Statement (II) is true

UPSC JWM 2017

Ans. (c) : Non-viscous flow is a flow has zero viscosity
$$\mu = 0$$
$$\tau = \mu\frac{du}{dy};\quad \tau = 0;\quad \frac{F}{A} = 0$$
Shear force, F = 0
In non-viscous flow– fluid particle do not revolve its own centre of mass as absence of shear force which makes it irrotational flow.
Forced vortex flow– When a fluid is rotated about a vertical axis with constant speed and every particle of this fluid has some angular velocity.
Forced vortex is rotational in nature.

121. A plate 0.02 mm distance from a fixed plate moves at 20 cm/s requires a shear stress of 4 N/m^2 to maintain this speed. Viscosity of fluid available inside these plates is :
(a) 4×10^{-4} Poise (b) 4×10^{-6} Poise
(c) $4\times10^{-5}\ \dfrac{\text{Ns}}{\text{m}^2}$ (d) $4\times10^{-4}\ \dfrac{\text{Ns}}{\text{m}^2}$

TRB Polytechnic Lecturer 2017

Ans. (d) : Data given,
$y = 0.02 \times 10^{-3}$ m, u = 0.2 m/s, $\tau = 4\dfrac{\text{N}}{\text{m}^2}$, $\mu = ?$
We know that,
$$\mu = \frac{\tau}{u/y} = \frac{4}{0.2/\left(0.02\times10^{-3}\right)} = 4\times10^{-4}\frac{\text{Ns}}{\text{m}^2}$$

122. If one litre of a fluid has a mass of 7.5 kg then its specific gravity is:
(a) 0.75 (b) 7.5
(c) 75 (d) 750

TRB Polytechnic Lecturer 2017

Ans. (b) : Data given,
m = 7.5 kg, V = 1 litre = 10^{-3}m^3, $\rho = \dfrac{m}{V} = 7.5\times10^3\ \dfrac{\text{kg}}{\text{m}^3}$
Then, Specific gravity of fluid
$$= \frac{\rho_{fluid}}{\rho_{water}} = \frac{7.5\times10^{+3}}{1000} = 7.5$$

123. For the conversion of one poise into MKS unit of dynamic viscosity dividing factor will be
(a) 9.81 (b) 0.981
(c) 98.1 (d) 981

Nagaland PSC CTSE 2017 Paper-2

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Ans. (c) : The dividing factor of converting one poise into MKS unit of dynamic viscosity is 98.1.

$1 \text{ poise} = \frac{1}{10}\frac{\text{Ns}}{\text{m}^2} = \frac{1}{98.1}\frac{\text{kgf}-\text{s}}{\text{m}^2} \quad (\because 1\text{kgf} = 9.81\text{N})$

124. A Spherical water drop of 1 mm in diameter splits up in air in to 64 smaller drops of equal size. The surface tension coefficient of water in air is 0.073 N/m. The work required in splitting up the drop is

(a) 0.96×10^{-6} J (b) 0.69×10^{-6} J
(c) 0.32×10^{-6} J (d) 0.23×10^{-6} J

ESE 2017

Ans. (b) : Given–

D = 1 mm
R = 0.5 mm
n = 64
σ = 0.073 N/m

Volume before splitting = Volume after splitting

$$\frac{4}{3}\pi R^3 = n\cdot\frac{4}{3}\pi r^3$$

$$R^3 = nr^3$$

$$r = \frac{R}{n^{1/3}} = \frac{0.5}{(64)^{1/3}} = \frac{1}{8}$$

$$r = 0.125 \text{ mm}$$

Surface energy before splitting $E_1 = \sigma.4\pi R^2$
Surface energy after splitting $E_2 = n.(\sigma 4\pi r^2)$
So, work require in splitting = $E_2 - E_1$
$= 4\pi\sigma\,[nr^2 - R^2]$
$= 4 \times 3.14 \times 0.073\,[64\,(0.125)^2 - (0.5)^2] \times (10^{-3})^2$
$E_2 - E_1 = 0.687 \times 10^{-6}$ J

125. If no resistance is encountered by displacement, such a substance is known as:

(a) perfect solid (b) perfect liquid
(c) gas (d) ideal fluid

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : If no resistance is encountered by displacement, such a substance is known as ideal fluid.
Fluid is said to be ideal if it is assumed to be both incompressible and non-viscous.
Its bulk modulus is infinite. Ideal fluid has no surface tension.

126. Liquids are generally used as insulating materials upto stress of about:

(a) 100 MV/cm (b) 50 MV/cm
(c) 50 KV/cm (d) 500 MV/cm

PTCUL AE 25.06.2017

Ans. (c) :

127. Unit of mass density is -

(a) kg /km (b) kg/m^3
(c) $kg.sec^2/m^3$ (d) $kg.sec^2/m^4$

GMB AAE 25.06.2017

Ans. (b) : kg/m^3

128. The kinematic viscosity is dependent upon

(a) pressure (b) distance
(c) flow (d) density

WBPSC AE, 2017

Ans. (d) : Kinematic viscosity = $\frac{\text{Dynamic viscosity}}{\text{Density}}$

$$\nu = \frac{\mu}{\rho}$$

$$\nu \propto \frac{1}{\rho}$$

129. Which of the following is the unit of kinematic viscosity?

(a) Pascal (b) Poise
(c) Stoke (d) Faraday

WBPSC AE, 2017

Ans. (c) : Kinematic Viscosity–It is the ratio of dynamic viscosity to density of substance.

- Formula $\nu = \frac{\mu}{\rho}$
- Unit of kinematic viscosity $= \frac{\text{Unit of dynamic viscosity}}{\text{Unit of mass density}}$

$$\text{SI/MKS unit} = \frac{\text{kg/m.s}}{\text{kg/m}^3}$$

$$\text{SI/MKS unit} = \frac{\text{m}^2}{\text{s}}$$

$$\text{CGS unit} = \frac{\text{cm}^2}{\text{s}} = \text{stokes}$$

- Dimensional formula $\Rightarrow L^2T^{-1}$ or $M^0L^2T^{-1}$

$1\ m^2/s = 10^4$ stoke
1 stoke = $10^{-4}\ m^2/s$

130. A fluid whose viscosity does not change with the rate of deformation or shear strain is known as

(a) real fluid
(b) ideal fluid
(c) Newtonian fluid
(d) Non-Newtonian fluid

Sikkim PSC (Under Secretary), 2017

Ans. (c) : A fluid whose viscosity does not change with the rate of deformation or shear strain is known as Newtonian fluid.
A real fluid, in which shear stress is directly proportional to velocity gradient (Rate of shear stress is known as Newtonian fluid)

$$\tau = \mu\frac{du}{dy}$$

e.g. Newtonian fluid– Oil, water, petrol, diesel, Hg etc.

131. In fluid statics

(a) There is no relative motion between adjacent layers of fluid where shear stress is maximum
(b) Velocity gradient and shear stress is zero
(c) Liquid particles are at rest with no velocity gradient with maximum shear stress

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(d) Forces acting on fluid particles are due to gravity which cause spinning of molecules leading to steady increase in velocity

Karnataka PSC Lect., 27.05.2017

Ans (b) : For a static fluid, i.e. a body of fluid that is at rest or has zero velocity the shear stress is always zero. Also for static fluids, the normal stress is always positive.

132. Which of the following forces does NOT act on a fluid in rest?

(a) Gravity force
(b) Hydrostatic force
(c) Surface tension force
(d) Viscous force

Punjab PSC (Lect.) 06.08.2017

Ans. (d) : Viscous force does not act on a fluid in rest.

133. Which of the following liquids is classified as Newtonian fluid?

(a) Honey (b) Petrol
(c) Glycerine (d) Paint

HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : Examples of Newtonian Fluid- Water, Air, Kerosene, Diesel, Petrol, Methyl Alcohol, ethyl alcohol etc.

134. The density of a liquid is 1000 kg/m^3. At location where g = 5 m/s^2, the specific weight of the liquid will be :

(a) 200 N/m^3
(b) 5000 × 9.81/5 N/m^3
(c) 5000 N/m^3
(d) 5000 × 5/9.81 N/m^3

HPPSC Asstt. Prof. 20.11.2017

Ans. (c) : Given, $\rho = 1000$ kg/m^3, $g = 5$ m/sec^2

Specific weight (w) = ρg

= 1000 × 5 = 5,000 N/m^3

135. The density of one litre of liquid which weighs 7.5 N is

(a) 764.53 Kg/m^3 (b) 754.55 Kg/m^3
(c) 760.53 Kg/m^3 (d) 750.55 Kg/m^3

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : Given –

Weights = 7.5 N

$$\text{Volume} = \frac{1}{1000}\text{m}^3 = 10^{-3}\ \text{m}^3$$

$$\text{Specific weight } w = \frac{\text{weight}}{\text{volume}}$$

$$= \frac{7.5}{10^{-3}} = 7500\ \text{N/m}^3$$

$$\text{Now density } \rho = \frac{w}{g} = \frac{7500}{9.81} = 764.53\ \text{kg/m}^3$$

136. What happens to the liquid level, when a small diameter tube is inserted into a liquid whose contact angle is 125°?

(a) Liquid level in the tube will fall first and then rise.
(b) Liquid level in the tube falls.
(c) Liquid level in the tube remains constant.
(d) Liquid level in the tube rises.

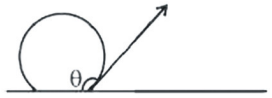
CIL MT 26.03.2017

Ans. (b) : Case – 1 contact angle is acute

Wetting fluid

In case of wetting fluid, level in capillary tube will rise.

Case – 2 Contact angle is obtuse



Non - wetting fluid

In case of non – wetting fluid, level in capillary tube will fall.

137. The pressure in metres of oil (specific gravity 0.85) equivalent to 85 meters of water is :

(a) 100m (b) 85m
(c) 8.5 m (d) none of the above

Kerala PSC IOF 19.04.2016

Ans. (a) : $(\rho gh)_w = (\rho gh)_{oil}$

$1000 \times g \times 85 = 850 \times g \times h$

$h_{oil} = 100$ m

138. The temperature influence of dynamic viscosity on a fluid is as follows :

(a) For gases viscosity increases with increase in temperature and for liquids it decreases with increase in temperature
(b) For gases viscosity increases with increase in temperature and for liquids it increases with increase in temperature
(c) For gases viscosity decreases with increase in temperature and for liquids it increases with increase in temperature
(d) For gases viscosity decreases with increases in temperature and for liquids it decreases with increases in temperature.

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Effect of temp on viscosity of fluids –

In case of liquid, cohesive forces between molecules cause viscosity. Hence at higher temp when molecule possesses more energy, they can oppose the large intermolecular cohesive force more strongly. As a result, energized liquid molecule can move more freely, thus viscosity of liquid decreases with increase in temperature.

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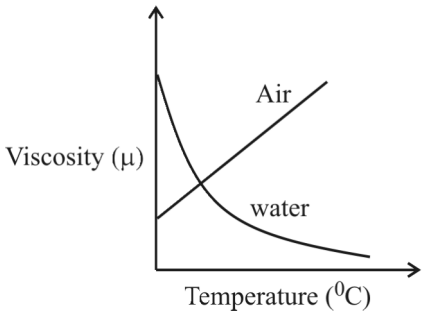
In case of gases, viscosity is caused by molecular collisions. At high temp, more molecular collision per unit time occurs. This results in greater resistance to flow.



139. Which one of the following with reference to fluid property is true?

(a) Capillarity is due to cohesion and adhesion and surface tension is due to cohesion

(b) Capillarity is due to adhesion and surface tension is due to cohesion

(c) Capillarity is due to adhesion and surface tension is due to cohesion and adhesion

(d) Capillarity is due to cohesion and surface tension is due to adhesion

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : The rise or fall of liquid in a small dia. tube when it is immersed in liquid is known as capillarity.

Capillary rise is due to adhesion and capillary fall is due to cohesion.

Hence capillarity is due to cohesion and adhesion whereas surface tension is due to cohesion only.

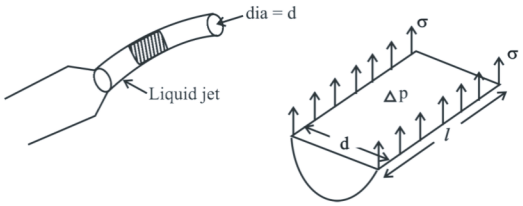
140. What is the internal pressure intensity p in excess of the outside pressure intensity for a jet a liquid of radius R and length l?

(a) $\frac{\sigma}{R}$ (b) $\frac{\sigma}{Rl}$

(c) $\frac{\sigma}{2R}$ (d) $\frac{4\sigma}{R}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Pressure intensity inside the liquid above outside pressure –



Force due to pressure

$= \Delta p \times l \times d$

Force due to surface tension = σ × 2l

$= \Delta p \times l \times d = \sigma \times 2l$

$\Delta p = \frac{2\sigma}{d} = \frac{\sigma}{R}$

141. The capillary rise of water h, in a glass tube can be determined using

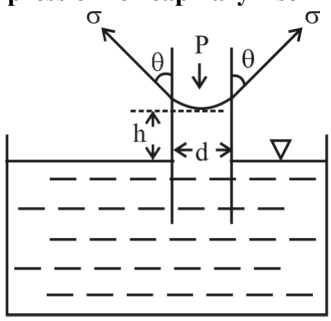
(Where σ is the surface tension; θ is contact or wetting angle; w is specific weight of liquid and R is radius of tube.)

(a) $h = \frac{2\sigma\cos\theta}{Rw}$ (b) $h = \frac{\sigma\cos\theta}{Rw}$

(c) $h = \frac{4\sigma\cos\theta}{Rw}$ (d) $h = \frac{\sigma\cos\theta}{2Rw}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Expression for capillary rise –



d = dia of glass tube

h = height of liquid in the tube

σ = surface tension of liquid

θ = angle of contact between liquid and glass tube.

The weight of liquid of height h in the tube

= (Area of tube × h) × ρ × g

$= \frac{\pi}{4}d^2 \times h \times \rho g$ ..... (i)

Vertical component of surface tensile force

$= (\sigma \times \text{circumference}) \times \cos\theta$

$= \sigma \times \pi d \times \cos\theta$ ..... (ii)

For equilibrium equating,

Equation (i) and (ii)

$\Rightarrow \quad \frac{\pi}{4}d^2 \times h \times \rho g = \sigma \times \pi d \times \cos\theta$

$h = \frac{4\sigma\cos\theta}{\rho g d}$

$h = \frac{2\sigma\cos\theta}{Rw}$

142. If mercury is filled in a small circular glass tube, the value of contact angle for mercury and the glass is equal to

(a) 130^0 (b) 0^0

(c) 75^0 (d) 90^0

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Value of contact angle –

Mercury and glass = 130^0

Water and glass = 0^0

Kerosene and glass = 260^0

Water and silver = 90^0

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143. How much pressure is required to cause 1% change in the density of water?

(a) 21 MPa (b) 2.1 MPa
(c) 210 MPa (d) 2100 MPa

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : $m = d \times v = \text{Constant}$

$v\Delta d + d\Delta v = 0$

$$\frac{\Delta v}{v} = -\frac{\Delta d}{d} = -\frac{1}{100} = -10^{-2}$$

$$k = -\frac{\Delta p}{\left(\frac{\Delta v}{v}\right)}$$

$$\Delta p = k\left(-\frac{\Delta v}{v}\right)$$

For water $k = 2.1$ GPa

$\Delta p = 2.1 \times 10^9 \times 10^{-2}$

$= 2.1 \times 10^7$

$\Delta p = 21$ MPa

A pressure of about 21 MPa is required to cause 1% change in the density of water.

144. What is poise?

(a) Unit of viscosity (b) Dyne-sec/cm^2
(c) 0.1 times Ns/m^2 (d) All the above

APPSC AEE Screening Test 2016

Ans. (d) : Unit of viscosity "poise".

$$1 \text{ poise} = 0.1\frac{N-s}{m^2}$$

$$1 \text{ poise} = 1\frac{\text{Dyne}-\text{sec}}{cm^2}$$

145. Rate of deformation of fluid element is equal to

(a) Shear stress
(b) Coefficient of dynamic viscosity
(c) Coefficient of kinematic viscosity
(d) Velocity gradient

RPSC LECTURER 16.01.2016

Ans. (d) : Rate of angular deformation
or
Rate of shear deformation
or
Rate of shear strain

$$\dot{\gamma} = \frac{d\phi}{dt} = \left(\frac{du}{dy}\right)$$

So, rate of angular deformation will be equal to velocity gradient in transverse direction.

As per Newton's law of viscosity

$$\tau \propto \frac{d\phi}{dt} \Rightarrow \tau = \mu\frac{d\phi}{dt}$$

This equation valid for 1-D, 2-D and 3-D flow

$$\tau = \mu\frac{du}{dt}$$ valid only for 1-D flow.

146. Which of the following is not the unit of dynamic viscosity?

(a) N-s/m^2 (b) Poise
(c) Pa.s (d) Nm^2/s^2

UPRVUNL AE 2016

Ans. (d) : Unit of dynamic viscosity is given as

We know that

$$\tau = \mu\left(\frac{du}{dy}\right)$$

$$\frac{N}{m^2} = (\mu) \times \frac{m}{\sec \times m} \qquad \boxed{\mu = \frac{N}{m^2} \times \sec}$$

$$1Pa = 1\frac{N}{m^2} \qquad \boxed{\mu = Pa - \sec}$$

In CGS unit of dynamic viscosity is Poise $\left(\frac{\text{dyne}-\text{sec}}{cm^2}\right)$

147. A cube of 1 m side has weight 1000 N in water. What will be its weight in air (g = 10 m/s^2)?

(a) 9,000 N (b) 1,000 N
(c) 11,000 N (d) 10,000 N

UPRVUNL AE 2016

Ans. (c) : In equilibrium

weight in air = Weight in water + weight of displaced water

$= W + \rho_W \times V_W \times g$

$= 1000 + 1000 \times (1)^3 \times 10$

$= 1000 + 10000 = 11000$ N

148. For liquids, the values of dynamic viscosity (μ) and kinematic viscosity (v) are

(a) Highly dependent of variation of pressure
(b) Practically independent of variation of pressure
(c) Practically independent of variation of temperature
(d) Increases with increase in temperature

RPSC LECTURER 16.01.2016

Ans. (b) : For liquids—

(a) Dynamic viscosity (μ)

• Temperature ↑, molecular cohesive force ↓, μ↓

• Pressure ↑, μ = constant

(b) Kinematic viscosity— $\upsilon = \frac{\mu\downarrow}{\rho\downarrow}$

Temperature ↑, μ ↓ ρ ↓

So

Temperature ↑, v ↓

• Effect of pressure.

$$\nu = \frac{\mu}{\rho}$$

P ↑ μ = constant, P ↑ ρ = constant

So variation of pressure does not effect μ and v for liquids.

149. Match list-I with list-II select the correct Answer

List -I	List -II
(a) Dynamic viscosity	1. Pa
(b) Kinematic viscosity	2. m^2/Sec
(c) Torsional stiffness	3. Ns/m^2
(d) Modulus of rigidity	4. N/m
	5. Nm

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Code

	A	B	C	D
(a)	3	2	4	1
(b)	5	2	4	3
(c)	3	4	2	3
(d)	5	4	2	1

UJVNL AE 2016

Ans : (a)

List -I	List-II
i) Dynamic viscosity	Ns/m^2
ii) Kinematic viscosity	m^2/Sec
iii) Torsional stiffness	N/m
iv) Modulus of rigidity	Pa

150. Mercury does not wet the glass due to

(a) Its cohesion is zero
(b) Its surface tension is zero
(c) Its adhesion is zero
(d) It is a solid metal at room temperature.

UJVNL AE 2016

Ans : (c) Mercury does not wet the glass due to its adhesion is zero. Adhesion is the tendency of dissimilar particles or surfaces to cling to one another (cohesion refers to the tendency of similar or identical particles/surfaces to cling to one another).

151. A block of volume V mm^3 is subjected to hydrostatic pressure P MPa modulus of elasticity is E GPa and Poisson's ratio ν = 0.5. Which statement is true about the block?

(a) Bulk modulus K=∞, perfectly incompressible and change in volume is zero.
(b) Bulk modulus K=1, perfectly incompressible and change in volume is zero.
(c) Bulk modulus K=0, perfectly incompressible and change in volume is ∞.
(d) Bulk modulus K= ∞, perfectly compressible and change in volume is zero.

BPSC Poly. Lect. 2016

Ans : (a) Block volume = V mm^3
hydrostatics pressure = P MPa
Modulus of Elasticity (E) = E GPa
Poisson's ratio (ν) = 0.5

$$\text{Bulk modules (K)} = \frac{\textit{normal stress}}{\textit{Volumetric strain}}$$

If Volumetric strain Should be zero then Bulk Modulus is ∞.
we know that for incompressible

$$\frac{\Delta V}{V} = 0$$

$$K = \frac{\sigma_n}{0}$$

$$\boxed{K = \infty}$$

152. Compressibility of a liquid is expressed by its:

(a) Density
(b) Pressure
(c) Volume
(d) Bulk modulus of elasticity

(HPPSC LECT. 2016)

Ans : (d) $\text{Compressibility} = \frac{1}{\text{Bulk modulus}}$

$$\text{Bulk modulus} = \frac{\text{Increase of pressure}}{\text{Volumetric strain}}$$

Compressibility is the reciprocal of the bulk modulus of elasticity.

153. Surface tension is expressed as:

(a) Force per unit area
(b) Force per unit volume
(c) Force per unit length
(d) Force only

(HPPSC LECT. 2016)

Ans : (c) Surface tension:- Surface tension is defined as the tensile force acting on the surface of a liquid in contact with a gas or on the surface between two immiscible liquid such that the contact surface behaves like a member under tension.

$$\sigma = \frac{\text{Tensile force}}{\text{unit length}} \text{ oR } \frac{\text{Surface energy}}{\text{Unit area.}}$$

154. The density of a fluid is sensitive to changes its pressure. The fluid will be known as :

(a) Newtonian fluid (b) Perfect fluid
(c) Compressible fluid (d) Real fluid

MPPSC AE 2016

Ans : (c) The density of a fluid is sensitive to change its pressure. The fluid will be known as Compressible fluid.

155. A plate 1 mm distant from a fixed plate moves at 0.25 m/s and requires a force/unit area of one Pascal to maintain this speed. Viscosity of the fluid between the plate will be

(a) 4 Ns/m^2 (b) 0.4 Ns/m^2
(c) 0.04 Ns/m^2 (d) 0.004 Ns/m^2

GPSC ARTO 01.05.2016

Ans. (d) : $\tau = \frac{\mu du}{dy}$

du = 0.25 m/sec., dy = 1 mm = 0.001m, μ = ?
τ = 1 Pascal

$$\tau = \frac{\mu du}{dy} \Rightarrow \mu = \frac{\tau dy}{du} = \frac{1 \times 0.001}{0.25}$$

$$\boxed{\mu = 0.004 \text{ Ns/m}^2}$$

156. If a solid steel sphere of 10,000 cm^3 subjected to a uniform hydrostatic pressure of 700 × 10^4 kg(f)/sq.m, then decrease in volume of the sphere for 1/K=6x10^{-11} (kg(f)/sq.m)$^{-1}$

(a) 2.4 cm^3 (b) 8.4 cm^3
(c) 4.2 cm^3 (d) 1.2 cm^3

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : We know,

$$K = \frac{P}{\left(\frac{\Delta V}{V}\right)} \qquad [1/K = \beta]$$

$$\Delta V = P \times V \times \beta$$

$$= 700 \times 10^4 \times 10000 \times 10^{-6} \times 6 \times 10^{-11}$$

$$\Delta V = 4.2 \times 10^{-6} m^3 = 4.2 cm^3$$

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157. In Newtonian fluids, for a given shear stress, the rate at which the fluid deforms is

(a) Inversely proportional to the dynamic viscosity
(b) Directly proportional to the dynamic viscosity
(c) Independent of kinematic viscosity
(d) None of the given answers

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : A Newtonian fluid is defined as one with constant viscosity with zero shear Rate at zero shear stress, that is the shear rate is directly proportional to the shear stress. The Rate at which the fluid deforms is inversely proportional to the dynamic viscosity.

$$\boxed{\tau = -\mu\frac{du}{dy}}$$

158. Two horizontal plates are placed 14 mm apart and the space between them is filled with an oil of viscosity of 14 poise. If the upper plate moves with 2.5 m/s velocity, the shear stress in the oil is

(a) 2500 N/sq.m
(b) 250 N/sq.m
(c) 25 N/sq.m
(d) 2.5 N/sq.m

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : Dynamic viscosity = 14 poise = 1.4 N-s/m^2

V = 2.5 m/s, h = 1.4 cm = 1.4 × 10^{-2} m

Newton's law of viscosity

$$\tau = \mu \times \frac{du}{dy}$$

$$\frac{du}{dy} = v/h = \text{Constant}$$

$$\frac{du}{dy} = \frac{2.5}{1.4\times10^{-2}} \Rightarrow \tau = \frac{1.4\times2.5}{1.4\times10^{-2}}$$

$$\boxed{\tau = 250\ \text{N/m}^2}$$

159. Fluid flow analysis is valid as long as the smallest length dimension of the problem is much larger than the distance between molecules. This concept is known as

(a) Control volume
(b) Ideal fluid
(c) Fluid continuum
(d) Homogeneous fluid

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : The continuum assumption is an idealization of continuum mechanism, under which fluids can be treated as continuous, even though, on a macroscopic scale, fluid properties can vary continuously from one volume element to Another & are average values of the molecular properties.

160. Match List I and List II and select the correct answer using the codes given below the lists -

List I description	List II property of fluid
P. Property which explains the spherical shape of the drop of a liquid	1. Viscosity
Q. Property which explains cavitation in fluid flow	2. Surface Tension
R. Property which explains rise of sap in free	3. Compressibility
S. Property which explains flow of oil jet	4. Vapour pressure
	5. Capillarity

(a) P Q R S
1 2 4 5
(b) P Q R S
2 4 5 1
(c) P Q R S
3 4 5 1
(d) P Q R S
1 2 3 4

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : Surface Tension :- is a property which explain the spherical shape of the drop of a liquid.
Vapour pressure :- Property which explains cavitation in fluid flow.
Capillarity :- Property which explains the rises or fall of liquid in a narrow space.
Viscosity :- Property which explains flow of oil jet.

161. When a pressure of 2×10^4 kN/m^2 is applied to 50liters of a liquid, its volume is decreases by 0.5 liter. The bulk modulus of liquid in N/m^2 is

(a) 20×10^9 (b) 2×10^9
(c) 4×10^9 (d) 40×10^9

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : Given, P = 2 × 10^4 kN/m^2
V = 50 L
dV = 0.5 L

$$K = \frac{P}{-(dV/V)} = \frac{2\times10^4\times10^3}{0.5/50} = 2\times10^9\ \text{N/m}^2$$

162. What is the SI unit of measurement of dynamic viscosity?

(a) Kilogram per meter (b) Newton-second
(c) Newton (d) Pascal-second

KPCL AE 2016

Ans. (d) : S.I. unit of dynamic viscosity (μ)
= (Pa. sec)

$$= \frac{\text{N}-\text{sec}}{\text{m}^2} \text{ or } \frac{\text{kg}}{\text{m}-\text{sec.}}$$

163. What is the SI unit of measurement of specific gravity?

(a) Kilograms
(b) Meters per second^2
(c) Kilograms per cubic meter
(d) None of these

KPCL AE 2016

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Ans. (d) : S.I. unit of specific gravity "unit less"

$$\text{Specific gravity (s)} = \frac{\text{density of fluid}}{\text{density of standard fluid}}$$

164. A container of 5 m^3 contains a particular vegetable oil which weights 32 kN. What is the specific weight of the oil?

(a) 4.5 kN/m^3 (b) 6.4 kN/m^3
(c) 0.15 kN/m^3 (d) None of these

KPCL AE 2016

Ans. (b) : Specific weight $(w) = \frac{\text{Weight}}{\text{Volume}}$

$$= \frac{32}{5} = 6.4\,kN/m^3$$

165. The relation for shear stress (τ) for a non-Newtonian fluid is given by the relation

(a) $\tau = \left(\frac{dv}{dy}\right)^{\frac{1}{n}}$ (b) $\tau = \left(\frac{dv}{dy}\right)^{n}$

(c) $\tau = \mu\frac{dv}{dy}$ (d) $\tau = \mu\left(\frac{dv}{dy}\right)^{\frac{1}{n}}$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : shear stress (τ) for a non-Newtonian fluid is given as:-

$$\tau = A\left(\frac{dv}{dy}\right)^{n} + B$$

166. What is the density of dry air at 101 kPa and 25°C?

(a) 1.18 kg/m^3 (b) 1.14 kg/m^3
(c) 1.08 kg/m^3 (d) 1.24 kg/m^3
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : As we know,

$$P = \rho RT \qquad \therefore \left[P = \frac{m}{v}RT\right]$$

$$\text{Density of air } (\rho) = \frac{101\times1000}{298\times287} = 1.18\,kg/m^3$$

167. What is the mass of air contained in a room of dimensions 4 × 6 × 3 m if the specific volume of the air is 0.83 m^3/kg?

(a) 85.3 (b) 92.4
(c) 86.7 (d) 74.8
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : Volume of room = 4 × 6 × 3 = 72 m^3
Density of air at NTP = 1.3 kg/m^3
Specific volume = 0.83 m^3/kg
Mass of air = Volume × density

$$\text{Mass of air} = \frac{\text{Volume}}{\text{Specific volume}} = \frac{72}{0.83}$$

$$= 86.7\ kg$$

168. A hydraulic pressure has a ram of 15 cm diameter and plunger of 1.5 cm. It is required to lift a weight of 1 tonne. The force required on plunger is equal to

(a) 10 Kg (b) 100 Kg
(c) 1000 Kg (d) 1 Kg

RPSC VPITI 14.02.2016

Ans. (a) : Weight on the ram F = mg

$$F = 1000 \times 10 = 10000$$

Using Pascal's law $\frac{F}{A} = \text{constant}$

where = $A = \pi D^2$

$$\frac{F}{A} = \frac{F_p}{A_p}$$

Where F_p = Plunger force
A_p = Plunger Area

$$\frac{10000}{(15)^2} = \frac{F_p}{(1.5)^2}$$

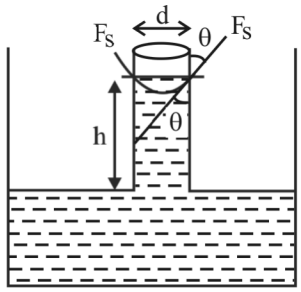
$F_p = 100N$
$F_p = 10$ kg

169. The rise or fall of head 'h' in a capillary tube of diameter 'd' and liquid surface tension 'σ' and specific weight 'w' is equal to

(a) $\frac{4\sigma}{wd}$ (b) $\frac{4d\sigma}{w}$
(c) $\frac{4wd}{\sigma}$ (d) $\frac{4w\sigma}{b}$

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Capillarity is defined as rise (or fall) of a liquid surface in a small tube when tube is hold vertical in liquid.



$$h = \frac{4\sigma\cos\theta}{\rho g d}$$

$$h = \frac{4\sigma}{\omega d}\ (\text{Let } \theta = 0) \quad \because \omega = \rho g$$

h = Capillary rise
σ = Surface tension
d = Diameter
θ = Angle of contact water and glass.

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170. The atmospheric pressure with rise in altitude decreases

(a) Linearly
(b) First slowly and then steeply
(c) First steeply and then gradually
(d) Unpredictable

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : Pressure decreases with increasing altitude. The pressure at any level in the atmosphere may be interpreted as the total weight of the air above a unit area at any elevation. At higher elevations, there are fewer air molecules above a given surface than a similar surface at lower levels. So, as we go up at higher altitude the pressure decreases.

171. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Ideal fluid	1. Viscosity does not changes with rate of deformation.
B. Newtonian fluid	2. Fluid of zero viscosity.
C. $\frac{\mu}{\rho}$	3. Dynamic viscosity
D. Mercury in glass	4. Capillary depression.
	5. Kinematic viscosity.
	6. Capillary rise.

	A	B	C	D
(a)	1	2	4	6
(c)	2	1	3	6

	A	B	C	D
(b)	1	2	3	4
(d)	2	1	5	4

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) :

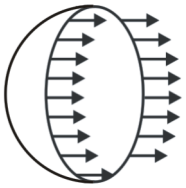
Ideal fluid	–	Fluid of zero viscosity.
Newtonian fluid	–	Viscosity does not changes with rate of deformation.
$\frac{\mu}{\rho}$	–	Kinematic viscosity.
Mercury in glass	–	Capillary depression.

172. The expression for relation between the gauge pressure p inside a liquid droplet (i.e. difference of pressure between the inside and outside of a liquid drop) of diameter d and the surface tension σ is

(a) $\sigma = pd$ (b) $\sigma = \pi pd$
(c) $\sigma = 4pd$ (d) $\sigma = \frac{pd}{4}$

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : Surface tension on liquid droplet



d = dia of droplet
σ = Surface tension of liquid
Pressure inside drop

$$F = \sigma \times \pi d = \rho \times \frac{\pi}{4} d^2$$

$$d = \frac{4\sigma}{P}$$

$$\sigma = \frac{Pd}{4}$$

173. In equilibrium condition, fluids are not able to sustain

(a) shear force
(b) resistance to viscosity
(c) surface tension
(d) geometric similitude

MPPSC AE 08.11.2015

Ans. (c) : In equilibrium condition (dynamic case), fluids are not able to sustain surface tension because in equilibrium condition cohesive forces resist the surface tension.

174. A liquid compressed in a cylinder has a volume of 0.04 m^3 at 100 kg/cm^2 and a volume of 0.039 m^3 at 200 kg/cm^2. The bulk modulus of elasticity of liquid is-

(a) 4000 kg/cm^2 (b) 400 kg/cm^3
(c) 400×10^5 kg/cm^2 (d) 40×106 kg/cm^2

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (a) $P_1 = 200$ kg/cm^2, $P_2 = 100$ kg/cm^2
$V_1 = 0.039$ m^3 $= 39000$ cm^3
$V_2 = 0.04$ m^3 $= 40000$ cm^3

$$K = \frac{-dP}{\left(\frac{dV}{V}\right)} = \frac{-(100-200)}{\left(\frac{0.04-0.039}{0.039}\right)}$$

$$\boxed{K = 4000\,kg/cm^2}$$

175. A pressure of 500 kPa applied to a $2m^3$ if the liquid results in a volume change of 0.004 m^3, the bulk modulus in MPa is :

(a) 2.5 (b) 25
(c) 250 (d) 2500

MPSC Poly. Lect. 2014

Ans. (c) : Given,
P = 500 kPa
$V = 2m^3$
$\Delta V = 0.004$ m^3

$$\text{Bulk modulus } k = \frac{P}{-\left(\frac{\Delta V}{V}\right)}$$

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'–' sign represents decrease in volume

$$= \frac{500 \times 10^3}{\left(\frac{0.004}{2}\right)}$$

$$= 250 \times 10^6 \text{ Pa}$$

$$\boxed{k = 250 \text{ MPa}}$$

176. The specific gravity of water is taken as :

(a) 0.001 (b) 0.01

(c) 0.1 (d) 1

MPSC Poly. Lect. 2014

Ans. (d) : Specific gravity (s) $= \frac{\rho_{fluid}}{\rho_{standard\ fluid}}$

for water (S = 1)

177. A liquid compressed in a cylinder has initially a volume of 20 m^3 at a pressure of 100 Pa. If the new volume is 40 m^3 at a pressure of 50 Pa, the bulk modulus of elasticity would be :

(a) 20 Pa (b) – 20 Pa

(c) 50 Pa (d) – 50 Pa

UPRVUNL AE 2014

Ans : (c)

Bulk modulus of elasticity $(K) = \frac{\text{change in pressure}}{\text{Volumetric strain}}$

$$(K) = \frac{\Delta P}{-\frac{\Delta V}{V}}$$

$\Delta P = P_2 - P_1 = 50 - 100 = -50 Pa$

$\Delta V = V_2 - V_1 = 40 - 20 = 20 m^3$

$\left[\begin{matrix}\Delta P = \text{change in pressure} \\ \Delta V = \text{change in volume}\end{matrix}\right]$

$$K = \frac{\Delta P}{\frac{\Delta V}{V}} = \frac{-50}{-\frac{20}{20}} = 50 \text{ Pa}$$

178. Bulk modulus 'K' is :

(a) Ratio of normal stress to Volumetric strain

(b) Ratio of change in volume to normal stress

(c) Ratio of shear to stress

(d) Ratio of stress to shear

UPRVUNL AE 2014

Ans : (a) Bulk Modulus (K) = It is the ratio of Normal stress to volumetric strain

$$K = \frac{\text{Normal stress}}{\text{Volumetric strain}}$$

179. The velocity distribution flow over a flat plate is given by U = (3/4)y - y^2 in which u is the velocity in m/s at a distance y meter above the plate. If the dynamic viscosity of the fluid is 10 poise, then the shear stress at 0.2m is

(a) 0.45 N/m^2 (b) 0.35 N/m^2

(c) 0.55 N/m^2 (d) 0.25 N/m^2

ISRO Scientist/Engineer 24.05.2014

Ans. (b) : Velocity Distribution

$$U = \left(\frac{3}{4}\right) y - y^2$$

Dynamic viscosity $\mu = 10$ Poise

$$= 10 \times \frac{1}{10} \text{ Pa.sec}$$

$$\mu = 1 \text{ Pa.sec}$$

$$U = \frac{3}{4} y - y^2$$

$$\frac{dU}{dy} = \frac{3}{4} - 2y$$

At, y = 0.2m

$$\left.\frac{dU}{dy}\right|_{y=0.2} = \frac{3}{4} - 2 \times 0.2$$

$$= 0.3551$$

$$\boxed{\tau = \mu \frac{dU}{dy}}$$

Shear stress $\tau = 1 \times 0.35$

$$= 0.35 \text{ N/m}^2$$

180. Which of the following fluid is heaviest?

(a) Air (b) Castor oil

(c) Glycerine (d) Carbon tetra chloride

TNPSC AE 2014

Ans. (d) : Carbon tetra chloride

181. Determine the bulk modulus of elasticity of liquid, if the pressure of liquid is increased from 60N/cm^2 to 120N/cm^2. The volume of liquid was found to decrease by 0.20%.

(a) 1×10^4 N/cm^2 (b) 2×10^4 N/cm^2

(c) 3×10^4 N/cm^2 (d) 4×10^4 N/cm^2

(e) 5×10^4 N/cm^2

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : Data given,

$P_1 = 60$ N/cm^2, $P_2 = 120$ N/cm^2

$$\frac{\Delta V}{V} = \frac{0.2}{100}$$

$$K = \frac{\Delta P}{\frac{\Delta V}{V}} = \frac{(120 - 60)}{\frac{0.2}{100}} = +\frac{60}{0.002} = +\frac{60 \times 10^3}{2}$$

$$\boxed{K = +3 \times 10^4 \text{ N/cm}^2}$$

182. A clean glass tube of 2 mm diameter contains water at 40^0C. The capillary rise is approximately

(a) 5mm (b) 10mm

(c) 15mm (d) 20mm

(e) 25mm

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : Given,

Dia of glass tube (d) = 2 mm

for water $\theta = 0^0$

$\sigma = 0.073$ N/m

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So,

$$h = \frac{4\sigma\cos\theta}{\rho g d} = \frac{4\times 0.073\times\cos 0^0}{1000\times 9.81\times 2\times 10^{-3}}$$

$h = 14.88 \text{ mm} \approx 15 \text{ mm}$

183. Match List I with List II and elect the correct answer:

List I		List II	
A	Ideal fluid	1	Shear stress does not vary linearly with the rate of strain
B	Newtonian fluid	2	Tensile stress varies linearly with the rate of strain
C	Non-Newtonian fluid	3	Shear stress is zero
D	Bingham plastic	4	Viscosity decreases with increase in temperature
		5	Shear stress varies linearly with the rate of strain
		6	Fluid behave like a solid until a minimum yield stress beyond which it exhibits a linear relationship between shear stress and the rate of strain

(a) A-3, B-5, C-6, D-1 (b) A-3, B-5, C-1, D-6
(c) A-5, B-3, C-4, D-2 (d) A-5, B-4, C-3, D-2
(e) A-2, B-6, C-5, D-3

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Ideal fluid - Shear stress is zero
Newtonian fluid - Shear stress varies linearly with the rate of strain $\left[\tau \propto \frac{du}{dy}\right]$
Non- Newtonian fluid - Shear stress does not vary linearly with the rate of strain.
Bingham plastic - Fluid behaves like a solid until a minimum yield stress beyond which it exhibits a linear relationship between shear stress and the rate of shear strain.

184. A fluid in equilibrium cannot sustain :
(a) tensile stress (b) compressive stress
(c) shear stress (d) bending stress

J&K PSC Civil Services Pre, 2013

Ans. (c) : A fluid in equilibrium cannot sustain shear stress.

185. The capillary rise in the glass tube is not to exceed 0.2 mm of water. Determine minimum size given that surface tension for water in contact with air = 0.0725 N/m.
(a) 12.8 cm (b) 13.8 cm
(c) 14.8 cm (d) 15.8 cm

APPSC Poly. Lect. 2013

Ans. (c) : Capillary rise, $h = 0.2 \text{ mm} = 0.2 \times 10^{-3}\text{m}$
Surface tension (σ) = 0.0725 N/m
$\rho_{water} = 1000 \text{ kg/m}^3$

$$h = \frac{4\sigma}{\rho g d}$$

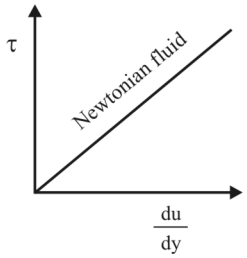
$$0.2\times 10^{-3} = \frac{4\times 0.0725}{1000\times 9.81\times d}$$

$$d = \frac{4\times 0.0725}{1000\times 9.81\times 0.2\times 10^{-3}} = 0.148\text{m} = 14.8\text{cm}$$

186. Which of the following is true for a Newtonian fluid?
(a) Viscous shear stress is independent of velocity gradient.
(b) Viscous shear stress depends linearly on velocity gradient.
(c) Viscous shear stress is zero at all velocity gradients.
(d) Viscous shear stress decreases with velocity gradient.

RPSC AEN Pre-2013

Ans. (b) : According to Newtonian fluid-
Viscous shear stress (τ) depends linearly on velocity gradient $\left(\frac{du}{dy}\right)$



187. The tank of size 2m × 2m × 2m hold, how much litre of water?
(a) 8000 litres (b) 6000 litres
(c) 4000 litres (d) 2000 litres

TNPSC AE 2013

Ans. (a) : Given, $V = 2 \times 2 \times 2 \text{ m}^3$
$V = 8 \text{ m}^3$
we know that
$1 \text{ litter} = 10^{-3} \text{ m}^3$
$V = 8 \times 10^3 \text{ litre}$
$V = 8000 \text{ litre}$

188. The density of water is 1000 kg/m^3 at
(a) 0^oC (b) 0^oK
(c) 4^oC (d) 20^oC

SJVN ET 2013

Ans. (c) : The density of water is 1000 kg/m^3 at 4^oC or 277^oK.

189. Which of the following expression correctly defines the relationship between internal pressure intensity (p) and the radius of spherical droplet (r)?

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(a) $p \propto \frac{1}{r}$ (b) $p \propto r$

(c) $p \propto \frac{1}{r^2}$ (d) $p \propto r^2$

UKPSC AE-2013, Paper-II

Ans. (a) : For a bubble–

Internal pressure $\Delta P = \frac{4\sigma}{r}$

$\left[\Delta P \propto \frac{1}{r}\right]$

For a droplet which has only one surface, $\Delta P = \frac{2\sigma}{r}$

190. The velocity gradient in the transverse direction for a fluid flow equals

(a) the pressure gradient in the flow
(b) the rate of shear strain
(c) the stress at that point
(d) the strain at that point

JPSC AE 2013 Paper-V

Ans. (b) : Newton's law of viscosity, $\tau \propto \gamma$

Where, τ = Shear stress

Rate of shear strain $\dot{\gamma} = \frac{d\phi}{dt} = \frac{du}{dy}$

$\tau = \mu \frac{d\phi}{dt} = \mu \frac{du}{dy}$

It is velocity gradient but physically it is rate of shear strain of fluid.

191. Consider two flat parallel plates placed in horizontal condition 1.2 cm apart and the space between them is filled with oil of viscosity 15.0 poise. The upper plate is moved with a velocity of 3.25 m/s, then the shear stress in the oil is

(a) Between 100 – 150 N/m^2
(b) Between 151 – 200 N/m^2
(c) Between 201 – 350 N/m^2
(d) Between 351 – 450 N/m^2

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (d) : Given, Thickness(t) = 1.2 cm = 0.012 m

μ = 15 poise = 1.5 N-s/m^2

ΔV = 3.25 m/sec

$\tau = \mu \times \frac{\partial v}{\partial y} = \mu \times \frac{\Delta v}{t}$

$\tau = 1.5 \times \frac{3.25}{0.012} = 406.25 N/m^2$

192. Surface tension of a fluid

(a) depends on the forces of molecular attraction
(b) is inversely proportional to fluid density
(c) is zero when the fluid at rest
(d) is the result of interaction between viscosity and atmospheric pressure

TNPSC ACF 2012

Ans. (c) : Surface Tension : It is the elastic tendency of a fluid surface which makes it acquire the least surface area possible. Surface tension allow insects to float and stride on the water surface.

193. In a static fluid

(a) resistance to shear stress is small
(b) fluid pressure is zero
(c) linear deformation is small
(d) only normal stresses can exist

TRB Poly. Lect., 2012

Ans. (d) : A fluid at rest, can have only normal stresses exist this Condition is known as hydrostatics.

194. Property of a fluid by which its own molecules are attracted is called

(a) adhesion (b) cohesion
(c) viscosity (d) compressibility

TRB Asstt. Prof., 2012

Ans. (b) : • **Cohesion**– It is the action or property of like molecules sticking together. eg.- water molecules

- **Adhesion–** It is the action or property of alike molecules sticking together.
  Example- Chalk and blackboard.
- **Viscosity –** Viscosity is defined as the measure of the resistance of a fluid to gradual deformation by shear or tensile stress.
- **Compressibility (capability of compression)–** The ability of something (such as a fluid) to be reduced in volume or size under pressure.

195. In the phenomenon of cavitation, the characteristics fluid property involved is

(a) surface tension
(b) viscosity
(c) bulk modulus of elasticity
(d) vapour pressure

TRB Asstt. Prof., 2012

Ans. (d) : In the phenomenon of cavitation, the characteristics fluid property involved is vapour pressure.

196. An oil of specific gravity 0.9 is contained in a vessel. At a point the height of the oil is 40m. The corresponding height of water at that point is:

(a) 40 m (b) 4 m
(c) 36 m (d) 3.6 m

APGENCO AE 2012

Ans. (c) : Given, S_{oil} = 0.9

h_1 = 40 m

S_w = 1, h_2 = ?

According to Hydrostatic law –

$S_{oil} \times h_1 = S_w \times h_2$

$0.9 \times 40 = 1 \times h_2$

$h_2 = 36m$

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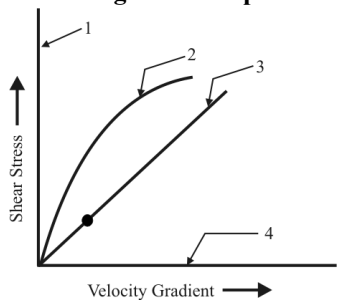
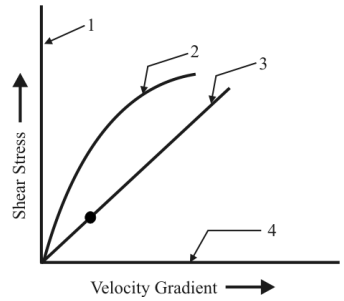
197. Curve "2" in figure corresponds to :



(a) Ideal fluid
(b) Newtonian fluid
(c) Non-Newtonian fluid
(d) Ideal solid

APGENCO AE 2012

Ans. (c) :



Curve – 1 – Ideal solid
Curve – 2 – Pseudoplastic (Non-Newtonian fluid)
Curve – 3 – Newtonian fluid
Curve – 4 – Ideal fluid

198. An increase in pressure of a liquid from 7.5 MPa to 15 MPa results into 0.2 percent decrease in its volume. The coefficient of compressibility of the liquid in m^2/N is

(a) 0.267×10^{-9} (b) 2.67×10^{-9}
(c) 1×10^{-9} (d) None of the above

ISRO Scientist/Engineer 2012

Ans. (a) : $P_1 = 7.5$ MPa
$P_2 = 15$MPa
$dP = P_2 - P_1$
Pressure difference $dP = (15 - 7.5) \times 10^6$
$dP = 7.5 \times 10^6$ N/m^2

Volumetric $\frac{dV}{V} \times 100 = -0.2$

Compressibility $\beta = \frac{-\frac{dV}{V}}{dP} = \frac{0.2/100}{(15-7.5)\times 10^6}$

$\beta = 0.267 \times 10^{-9} m^2/N$

199. A reservoir of capacity 0.01 m^3 is completely filled with a fluid of coefficient of compressibility $0.75 \times 10^{-9} m^2/N$. The amount of fluid that will spill over (in m^3), if pressure in the reservoir is reduce by $2 \times 10^7 N/m^2$ is

(a) 1×10^{-4} (b) 1.5×10^{-4}
(c) 0.15×10^{-4} (d) None of the above

ISRO Scientist/Engineer 2012

Ans. (b) : Given
$V = 0.01$ m^3
compressibility $\beta = 0.75 \times 10^{-9}$ m^2/N
Let, volume spill dV(m^3)
$dP = 2 \times 10^7$ N/m^2

Compressibility $\beta = \frac{-\frac{dV}{V}}{dP}$

$0.75 \times 10^{-9} = \frac{\frac{dV}{0.01}}{2 \times 10^7}$

$dV = 0.75 \times 10^{-9} \times 0.01 \times 2 \times 10^7$
$= 1.5 \times 10^{-4} m^3$

200. An oil of specific gravity 0.7 and pressure 0.14 kgf/cm^2 will have the weight of oil as

(a) 70 cm of oil (b) 2 m oil
(c) 20 cm of oil (d) 80 cm of oil

APPSC AEE 2012

Ans. (b) : $P = 0.14$ kgf/cm^2 $= 0.14 \times 9.81 \times 10^4$ Pascal
Specific gravity of oil = 0.7
$P = \rho gh$

$h = P/\rho g = \frac{0.14 \times 9.81 \times 10^4}{700 \times 9.81} = 2$

h = 2 m of oil

201. Schlieren flow visualization technique operates by employing

(a) density variation in flow
(b) density gradients in flow
(c) second derivatives of density
(d) higher derivatives of density

APPSC AE 04.12.2012

Ans. (b) : Schlieren is a device which indicates the density gradient $\frac{d\rho}{dy}$ in flow.

The illumination or contrast on a screen is dependent upon the focal length of a lens, adjustment of a knife edge and the density gradient in fluid flow.

202. The height through which water rises by capillary action in a glass tube of 2 mm bore if the surface tension at the prevailing temperature is 0.075 g/cm, is

(a) 1.5 cm (b) 3 cm
(c) 0.75 cm (d) 10.3 cm

APPSC AE 04.12.2012

Ans. (a) : Given,
Glass tube bare (d) = 2 mm = 2×10^{-3} m
Surface tension (σ) = 0.075 g/cm
= 0.07354 Newton/meter
= 0.07354 kg-m/sec^2/m
Let,
h = height of rise water due to capillary action

$\Rightarrow \quad h = \frac{4\sigma \cos\theta}{\rho g d}$

For water glass tube; $\theta = 0^\circ$ and $\rho = 1000$ kg/m^3

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$$\therefore\ h = \frac{4\sigma}{\rho g d} = \frac{4\times 0.07354\ \text{kg}-\text{m}/\text{s}^2/\text{m}}{1000\text{kg}/\text{m}^3\times 9.81\text{m}/\text{sec}^2\times 2\times 10^{-3}\text{m}}$$

$h = 0.01499$ m

$= 1.499$ cm

≈ 1.5 cm

203. The pressure at a point in a fluid is not, equal in all direction if :

(a) the fluid is at rest
(b) there are shear stresses
(c) the fluid is accelerated
(d) the fluid is rotated at a constant speed

APPSC AE 04.12.2012

Ans. (b) : The pressure at a point in a fluid is not equal in all direction if there are shear stresses because when we applying the shear on the fluid, then in the fluid continuous deformation occurs.

204. The diameter of a soap bubble with has an inside pressure of 2.5 N/m^2 over the atmospheric pressure and a surface tension of 0.0125 N/m is

(a) 40 mm (b) 4 mm
(c) 16 mm (d) 60 mm

ISRO Scientist/Engineer 2012

Ans. (a) : $P = 2.5$ N/m^2

Surface tension $\sigma = 0.0125$ N/m

For Soap Bubble,

$$P = \frac{8\sigma}{d}\ \text{(P = Excess pressure)}$$

$$\therefore\quad d = \frac{8\sigma}{P} = \frac{8\times 0.0125}{2.5}$$

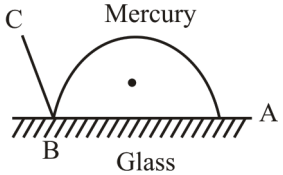
$d = 0.04$m $= 40$ mm

205. There is a pin hole of diameter 0.1 mm at the bottom of a glass vessel containing mercury. What depth of mercury can be put in the vessel without any leakage occurring? Take $\sigma = 0.55$ N/m.

(a) $h = 0.165\cos\theta$ m (b) $h = 0.23\cos\theta$ m
(c) $h = 0$ m (d) $h = 10.3$ m

APPSC AEE 2012

Ans. (a) :



F_s = Surface tension force

F_h = Cohesion force

$F_s = \pi d\ .\ \sigma\cos\theta$

$= \pi \times 0.0001 \times 0.55\cos\theta$

$$\Rightarrow\quad F_h = P\times A = \rho g h \times \frac{\pi}{4}d^2$$

$$= 13.6\times 10^3 \times 9.81 \times h \times \frac{\pi}{4}\times 0.0001^2$$

$$\Rightarrow\quad F_s = F_h$$

$$\Rightarrow\quad \pi \times 0.0001 \times 0.55\cos\theta = 13.6\times 10^3 \times 9.81\times \frac{\pi}{4}\times 0.0001^2 \times h$$

$h = 0.165\cos\theta$ m

206. What would happen to the rise or depression of liquid the tube due to surface tension if the size of the tube is increased.

(a) Decreased
(b) Increased
(c) Remains unchanged
(d) Depends upon the characteristics of liquid

VIZAG STEEL MT 2011

Ans. (a) :

207. Which of the following property of air does not increase with rise in temperature?

(a) Specific gravity
(b) Kinematic viscosity
(c) Thermal conductivity
(d) Thermal diffusivity

ISRO Scientist/Engineer 2010

Ans. (a) : $\upsilon = \frac{\mu}{\rho}$

μ for air increases with rise in temperature and ρ decreases with rise in temperature so, value of kinematic viscosity increases.

Thermal diffusivity

$$\alpha = \frac{k}{\rho C_p}$$

For air k increases with temperature and ρ decreases with rise in temperature.

So value of thermal diffusivity increases.

208. Assertion (A) : The pressure of flowing liquid in any part of the hydraulic system should not be allowed to fall below its vapour pressure.

Reason (R) : The absolute pressure head for water should not be below 2.5 m of water.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know the pressure of flowing liquid in any part of the hydraulic system should not be allowed to fall below its vapour pressure because if the pressure of flowing fluids falls below, its up than cavitation will occur.

The absolute head is equal to $10.3 - 7.6 = 2.7$ m of water. If the pressure at summit become less than 2.1 m of water absolute gases & air obstructed in the flow.

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209. The increase in pressure required to decrease unit volume of mercury (bulk modulus = 28.5 MPa) by 0.1% is

(a) 285 GPa (b) 285 MPa
(c) 2.85 GPa (d) 28.5 kPa

DRDO Scientists 2009

Ans. (d) : $K = \frac{dPV}{dV} = \frac{dP}{0.001} = 28.5 \times 10^6$

Or $dP = 28.5$ kPa

210. Consider the following assumptions :

1. Fluid is Newtonian.
2. Fluid is incompressible
3. Fluid experiences turbulent flow:
4 : No slip at the boundary surfaces.

Reynolds equation for one dimensional flow in hydrodynamic lubrication is based on which of these assumptions?

(a) 1, 2 and 3 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) 1, 2 and 4

UPSC JWM Adv. No-16/2009

Ans. (d) : Reynolds equation for one-dimensional flow in hydrodynamic lubrication is based on Assumptions –
- Fluid is Newtonian
- Fluid is incompressible
- No slip at the boundary surface

211. A real practical fluid possesses which of the following?

(a) Viscosity (b) Surface tension
(c) Compressibility (d) All of the above

WBPSC AE 2008

Ans. (d) : A real practical fluid possesses following properties–
1. Viscosity
2. Surface tension
3. Compressibility

212. If a liquid is compressed isothermally to a very high value, the dynamic viscosity of the liquid will

(a) increase (b) decrease
(c) remain unchanged (d) none of the above

WBPSC AE, 2007

Ans. (c) : remain unchanged

213. The viscosity of water with respect to air is about

(a) 50 times (b) 55 times
(c) 60 times (d) 65 times

UKPSC AE 2007 Paper -II

Ans. (b) : Viscosity is the propriety of the fluid by virtue of which the fluid offers resistance against shear force.
$\Rightarrow$ The viscosity of water is almost 55 times more than the viscosity of air.
i.e. $\mu_{water} > \mu_{air}$

214. One stoke is equal to

(a) 1 cm^2/s
(b) 1 m^2/s
(c) 1 ft^2/s
(d) 1 $m.m^2/s$

UKPSC AE 2007 Paper -II

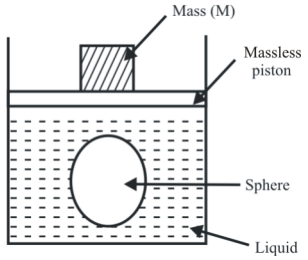
Ans. (a) : 'Stoke' is a unit if kinematic viscosity.
1 stoke = 10^{-4} m^2/s = 1 cm^2/s
1 C.S. = 10^{-2} stokes
* At 20^0C Kinematic viscosity of water is 1 C.S.
* Its dimension $\Rightarrow M^0L^2T^{-1}$.

215. A solid sphere of radius R and made of a material of bulk modulus K is completely immersed in a liquid in a cylindrical container. A mass less piston of area A floats on the surface of the liquid. When a mass M is placed on the piston to compress the liquid, the fractional change in the radius of the sphere, $\frac{dR}{R}$ is given by-

(a) $\frac{Mg}{KA}$ (b) $\frac{Mg}{2KA}$
(c) $\frac{Mg}{4KA}$ (d) $\frac{Mg}{3KA}$

WBPSC AE 2003

Ans. (d) :



$\because$ $\text{Bulk modulus}(K) = \frac{dP}{-dV/V}$

$V = \frac{4}{3}\pi R^3$ $P = \frac{W}{A} = \frac{Mg}{A}$ ...(ii)

$\frac{dV}{V} = \frac{3dR}{R}$ ...(i)

keep the both equation in $\left[\frac{dP}{-dV/V} = K\right]$

$K\frac{dV}{V} = dP$

$K\left(3\frac{dR}{R}\right) = \frac{Mg}{A}$

$3\frac{dR}{R} = \frac{Mg}{AK}$

$\boxed{\frac{dR}{R} = \frac{Mg}{3AK}}$

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2. Pressure and its Measurement

216. Atmospheric pressure held in terms of water column is

(a) 7.50 m (b) 9.30 m
(c) 10.30 m (d) 11.30 m

CIL MT 27.02.2020
APPSC Poly Lect. Automobile Engg., 2020
APPSC Poly. Lect. 2013
APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (c) : We know that–

Pressure head (h) = $\frac{P}{\rho g}$ in meter.

Let, $P = P_{atm}$ = Atmospheric pressure = 1.013 bar
$= 1.013\times10^5$ N/m^2

For water; Density (ρ) = 100 kg/m^3
and $g = 9.81$ m/s^2

$\therefore \quad h_w = \frac{1.013\times10^5}{1000\times9.81} = \frac{101.3}{9.81} = 10.32$ m

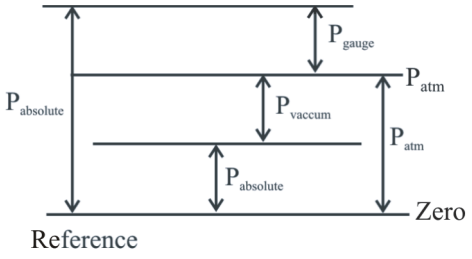
$\boxed{h_w \simeq 10.30 \text{ m}}$

217. Absolute pressure is measured as

(a) Gauge pressure- Atmospheric pressure
(b) Gauge pressure + Atmospheric pressure
(c) Gauge pressure/Atmospheric pressure
(d) None of the above

UPRVUNL AE 05.07.2021
Nagaland PSC (CTSE) 2018, Paper-II
Karnataka PSC Lect., 27.05.2017
CGPSC Polytechnic Lecturer 2017
TSPSC AEE 2015

Ans (b) : Absolute Pressure-



- Pressure measured with respect to complete vacuum or zero absolute pressure is called absolute pressure.
- Complete vacuum or zero absolute pressure means when there is no molecular momentum.
- Absolute pressure never be negative

$P_{absolute} = P_{atm} + P_{gauge}$

- **Gauge Pressure-** When pressure in a fluid is measured with respect to the atmospheric pressure, then it is called gauge pressure. Gauge pressure can be positive, negative or zero.
- **Vacuum Pressure-** It is defined as the pressure below the atmospheric pressure.

$P_{vacuum} = P_{atm} - P_{absolute}$

218. Barometer is used to measure.

(a) atmospheric pressure.
(b) very low pressure
(c) difference of pressure between two points
(d) pressure in pipes, channels etc.

GPSC Lect. 23.10.2016, GWSSB AAE, 27.12.2015
MPSC Poly. Lect. 2014, SJVN ET 2013

Ans. (a) : Barometer– A barometer is a scientific instrument that used to measure atmospheric pressure, also called barometric pressure.

1 atmospheric pressure = 1.01325 bar
= 101.325kPa
= 101325 Pa

$\boxed{1 \text{ bar} = 10^5 \text{ Pascal}}$

219. Pascal law is applicable when :

(a) fluid is at rest
(b) fluid is in motion
(c) fluid is compressible
(d) fluid is incompressible

HPPSC Workshop Suptd. 08.07.02021
HPPSC Asstt. Prof. 20.11.2017
TSPSC Managers 2015, ESE 2011

Ans. (a) : Pascal law–It states that pressure at a point in a static fluid has the same magnitude in all the direction.

$\boxed{P = P_x = P_y = P_z}$

In other words, if a certain pressure is applied at any point in a static fluid then it gets equally transmitted in all the direction.

Pascal law applied for both compressible and incompressible fluid.

220. The reading of the pressure gauge fitted on a vessel is 25 bar. The atmospheric pressure is 1.03 bar and the value of 'g' is 9.81 m/s^2. The absolute pressure in the vessel in bar

(a) 23.97 (b) 25.0
(c) 34.84 (d) 26.03

RPSC Vice Principal ITI 2018
TSPSC AEE 2017, CSE Pre-1994

Ans. (d) : We know,

$P_{abs} = P_{atm} + P_{gauge}$
$= 1.03 + 25$
$= 26.03$ bar

221. Piezometer is used to measure

(a) Pressure in pipe
(b) very low pressure
(c) atmospheric pressure
(d) difference of pressure between two points

WBPSC AE, 2017
Nagaland PSC (CTSE) 2016 Paper-II
SJVN ET 2013

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Ans. (b) : A Piezometer is simple glass tube is open at both the ends, whose one end of the glass tube is connected to the pipe line, while other end is kept open to the atmosphere. Piezometer is used to measure moderate pressure, if in the question, 'moderate' not given then it would be used to measure low pressure also.
Piezometer is used to measure very low pressure or gauge pressure.

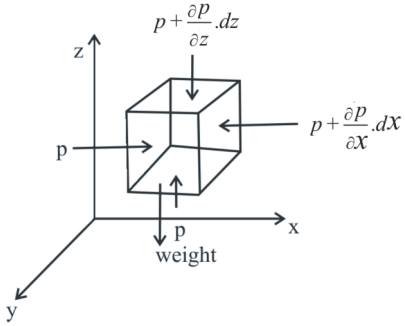
222. If 'Z' is vertically upwards, 'ρ' is the density and 'g' is gravitational acceleration then pressure $\frac{\delta P}{\delta Z}$ in fluid at rest due to gravity is given by :

(a) $\frac{\rho g Z^2}{2}$ (b) $-\rho g$

(c) $-\rho g Z$ (d) $\frac{\rho g}{Z}$

MPPSC AE 2016, KPSC ADF 2015
ESE 1995

Ans : (b)



Hydrostatics law $\frac{\partial p}{\partial z} = -\rho g$

223. As per Pascal's law :

(a) intensity of pressure at a point in a moving fluid is equal in the direction of applied force.
(b) intensity of pressure at a point in a static fluid is equal in the direction of applied force.
(c) intensity of pressure at a point in a static fluid is equal in all directions.
(d) intensity of pressure at a point in a moving fluid is equal in all directions.

NLC GET 17.11.2020, Shift-II
Karnataka PSC Lect., 27.05.2017
GPSC Lect. 23.10.2016

Ans. (c) : According to the Pascal's law, the pressure at a point in a static fluid is equal in all directions.

224. Hydrostatic law of pressure is given by :

(a) $\frac{\partial P}{\partial z} = \rho g$ (b) $\frac{\partial P}{\partial z} = 0$

(c) $\frac{\partial P}{\partial z} = z$ (d) $\frac{\partial P}{\partial z} = c$

APPSC AE 04.12.2012
APPSC AEE 2012, ESE 2002

Ans. (a) : Hydrostatic law of pressure– Pressure in a static mass of fluid increasing in the vertically downward direction at a rate is equal to the specific weight of fluids. It is the Hydrostatic law.

$\therefore \quad \frac{\partial P}{\partial Z} = w$

Where, Specific weight (w) = ρg

$\therefore \quad \boxed{\frac{\partial P}{\partial Z} = \rho g}$

225. An inclined manometer, inclined at 30° to the horizontal, measure the pressure differential between two locations of pipe carrying water. If the manometric liquid is mercury (specific gravity 13.6) and manometer showed a level difference of 20 cm, then the pressure head difference of water between the two tapping will be

(a) 1.26 m
(b) 1.36 m
(c) 2.52 m
(d) 2.72 m

TSPSC AEE 2017, CSE Pre-1994

Ans. (a) : Given, $\ell_1 - \ell_2 = 20$ cm

$S_{Hg} = 13.6$

We know, $S_w = 1$

Pressure head difference $= (S_{Hg} - S_w) \times (\ell_1 - \ell_2)\sin 30$

$= (13.6 - 1) \times (0.20)\sin 30$

$= 12.6 \times 0.2 \times \frac{1}{2}$

$= 1.26$ m

226. The intensity of pressure at any point in a liquid at rest is the same in all directions. The above statement is known as

(a) Newton's law
(b) Reynold's law
(c) Pascal's law
(d) None of the above

TNPSC AE 2018
GPSC Lect. (Auto) 16.10.2016

Ans. (c) : Pascal's Law- It state that pressure at any point in a static fluid is the same in all directions

$P = \rho g h = P_x = P_y = P_z$

227. If a hydraulic press has a ram of 12.5 cm diameter and plunger of 1.25 cm diameter, what force would be required on the plunger to raise a mass of 1 tonne on the ram?

(a) 981 N
(b) 98.1 N
(c) 9.81 N
(d) 0.98 N

JPSC AE, 10.04.2021, Paper-I
CSE Pre-1998

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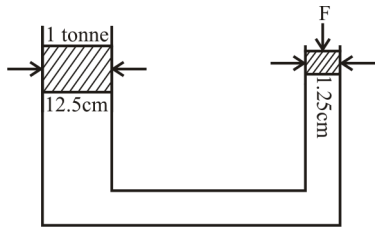
Ans. (b) :



The pressure is same in both ram and plunger–

$$\left(\frac{F}{A}\right)_1 = \left(\frac{F}{A}\right)_2$$

$$\frac{F}{\frac{\pi}{4}(1.25)^2} = \frac{1000\times 9.81}{\frac{\pi}{4}\times(12.5)^2}$$

$$\boxed{F = 98.1N}$$

228. Differential manometer is used to measure

(a) pressure in pipes, channels etc
(b) atmospheric pressure
(c) very low pressure
(d) difference of pressure between two points

Nagaland PSC (CTSE) 2018, Paper-II
TRB Asstt. Prof., 2012

Ans. (d) : Differential manometer- A differential manometer is a device that measures the difference of pressure between two places (point).

229. At what depth below the free surface of oil having a density of 784 kg/m^3 will the pressure be very nearly equal to 1 bar?

(a) 10 metres (b) 14 metres
(c) 13 metres (d) 7.84 metres

GPSC EE Pre, 28.01.2017, ESE 2012

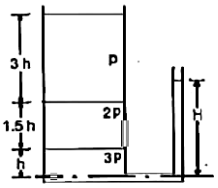
Ans. (c) : Given- Density (ρ) = 784 kg/m^3
Pressure (p) = 1 bar
$P = 1\times10^5$ N/m^2
We know that,
Pressure, $P = \rho gh$

$$1\times10^5 = 784 \times 9.81 \times h$$

$$h = \frac{1\times10^5}{784\times9.81}$$

$h = 13.002$ m

230. Three immiscible liquids of specific densities p, 2p and 3p are kept in a jar. The height of the liquids in the jar and at the piezometer fitted to the bottom of the jar is as shown in the given figure. The ratio H/h is



(a) 4 (b) 3.5
(c) 3 (d) 2.5

Nagaland PSC CTSE 2015 Paper-II
ESE 2001

Ans. (c) : Pressure = ρgh

$p \times 3h + 1.5h \times 2p + h \times 3p - H \times 3p = 0$

$9ph = 3pH$

$9h = 3H$

$$\frac{H}{h} = \frac{9}{3}$$

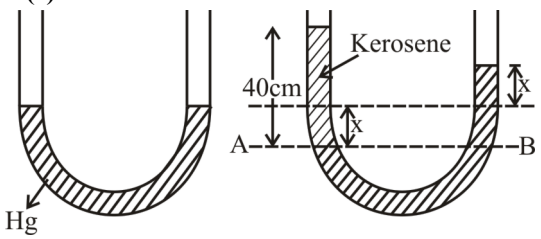
$$\frac{H}{h} = 3$$

231. An open U tube is containing mercury, Kerosene of specific gravity 0.8 is poured into one of its limbs so that the length of column of kerosene is about 40 cm. The level of mercury column in that limb is lowered approximately by how much?

(a) 2.35 cm (b) 3.6 cm
(c) 1.2 cm (d) 0.6 cm

APPSC AE Subordinate Service Civil/Mech. 2016
ESE 2008

Ans. (c) :



$0.4 \times 0.8 = 13.6 \times 2x$

$x = 0.0117$ m

$x = 1.17 \approx 1.2$ cm

232. For manometer, a better liquid combination is one having

(a) higher surface tension
(b) lower surface tension
(c) high density and high dynamic viscosity
(d) low density and low dynamic viscosity

WBPSC AE, 2017
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (b) : For manometer, a better liquid combination is one having lower surface tension.

233. A manometer measures the pressure differential between two locations of a pipe carrying water. If the manometric liquid is mercury (specific gravity 13.6) and the manometer showed a level difference of 20cm, then the pressure head difference of water between the two tapings will be

(a) 1.26m (b) 2.72m
(c) 1.36m (d) 2.52m

ISRO Scientist/Engineer 2011, ESE 2009

Ans. (d) : $S_{Hg} = 13.6$ (specific gravity of mercury)
$h_{Hg} = 20$ cm = 0.20 m
$S_w = 1$ (specific gravity of water)
$h_w = ?$

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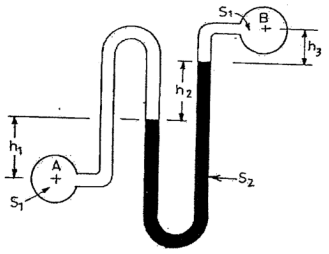
$$h_w = x\left(\frac{S_{Hg}}{S_w} - 1\right)$$
$$= 0.2\left(\frac{13.6}{1} - 1\right)$$
$$= 0.2(12.6) = 2.52 \text{ m}$$

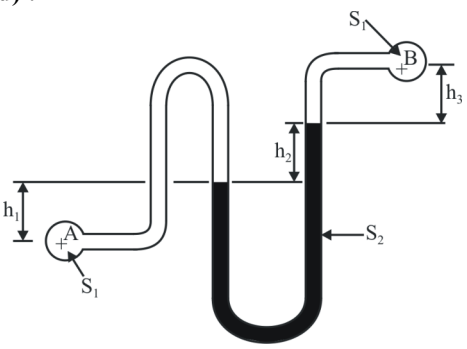
234. Two pipe lines at a different pressures, p_A and p_B, each carrying the same liquid of specific gravity of S_1 are connected to a U tube with a liquid of specific gravity of S_2 resulting in the level differences h_1, h_2 and h_3 as shown in the figure. The difference is pressure head between points A and B in terms of head of water is.



(a) $h_1S_2+h_2S_1+h_3S_1$ (b) $h_1S_1+h_2S_2-h_3S_1$
(c) $h_1S_1-h_2S_2-h_3S_1$ (d) $h_1S_1+h_2S_2+h_3S_1$

ISRO Scientist/Engineer 2006
ESE 1993

Ans. (d) :



$P_A - S_1h_1 - S_2h_2 - S_1h_3 = P_B$

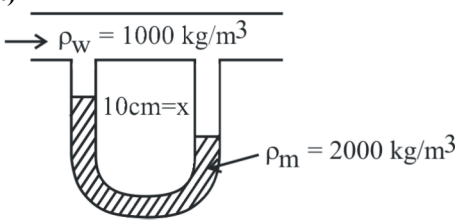
$$\boxed{P_A - P_B = h_1S_1 + S_2h_2 + h_3S_1}$$

235. Pressure drop of water, having a density of 1000 kg/m^3, flowing through a pipe between two points is measured by using a vertical U-tube manometer. Manometer uses a liquid with density 2000 kg/m^3. The difference in height of Manometric liquid in the two limbs of the manometer is observed to be 10 cm. The pressure drop between the two points is :

(a) 981 N/m^2
(b) 98.1 N/m^2
(c) 1962 N/m^2
(d) 19620 N/m^2

APPSC AE Subordinate Service Civil/Mech. 2016
ESE 2002

Ans. (a)



Density of water (ρ_w) = 1000 kg/m^3
Density of manometric liquid (ρ_m) = 2000 kg/m^3
Difference in height of manometric liquid (x) = 10 cm
pressure drop between two points = $P_1 - P_2$

Difference in pressure head = $h = \frac{P_1 - P_2}{\rho g}$

When manometer contains heavier liquid, the difference in pressure head is given by

$$h = x\left[\frac{\rho_m}{\rho_w} - 1\right]$$

$$h = 0.1\left[\frac{2000}{1000} - 1\right] = 0.1$$

Now, $\frac{P_1 - P_2}{\rho g} = h \Rightarrow P_1 - P_2 = 0.1 \times 1000 \times 9.81$

$$P_1 - P_2 = 981 \text{ N/m}^2$$

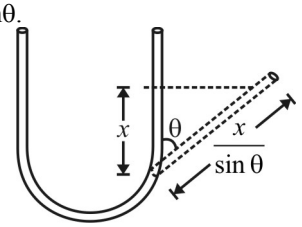
Pressure drop between two points = 981 N/m^2

236. In order to increase sensitivity of U-tube manometer, one leg is usually inclined by angle θ. Sensitivity of inclined tube to sensitivity of U-tube is equal to:

(a) sin θ (b) cos θ
(c) tan θ (d) 1/sin θ

OPSC AEE 2015 PAPER - II
ESE 2009

Ans : (d) In order to increase sensitivity of U - tube manometer, one leg is usually inclined by angle θ. Sensitivity of inclined tube to sensitivity of U-tube is equal to 1/Sinθ.



(sensitivity)$_{\text{U-tube}} \propto x$

$(\text{sensitivity})_{\text{inclined tube}} \propto \frac{x}{\sin\theta}$

237. Which property of mercury is the main reason for use in barometers?

(a) High density
(b) Negligible capillary effect
(c) Very low vapour pressure
(d) Low compressibility

UPPSC AE 12.04.2016 Paper-II
ESE 2007

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Ans : (c) Very low vapour pressure of mercury is the main reason for use in barometers.
Vapour Pressure : Molecules of liquid have tendency to escape from the free surface and get deposited on the top of the container if it is closed. These molecules exert partial pressure, which is known as vapour pressure.
The saturation vapour of Hg. at 20^0C is 1.2×10^{-6} m of Hg.

238. If a mercury-oil differential manometer shows a 20 cm difference of mercury level, the difference in the pressure head is (consider the specific gravity of oil = 0.8)
(a) 2.0 m of oil (b) 2.5 m of oil
(c) 3.2 m of oil (d) 4.2 m of oil
APPSC AEE SCREENING 17.02.2019
ESE 2009

Ans. (c) : By using formula
$$h = x\left(\frac{\rho_m}{\rho_o}-1\right) = 0.2\left(\frac{13600}{800}-1\right)$$
= 3.2 m of oil.

239. The pressure at a point is equal in all directions
(a) only when the fluid is inviscid
(b) when the fluid is incompressible
(c) when the fluid is at rest
(d) in a laminar flow
UPPSC AE 13.12.2020, Paper-2, ESE 2011

Ans. (c) : According to Pascal's law
The pressure at a point is equal in all directions when the fluid is at rest.

240. The difference in pressure head, measures a mercury water differential manometer for 20 cm difference of mercury will be :
(a) 2.72 m (b) 2.52 m
(c) 2.0 m (d) 0.2 m
APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (b) : Let, x = difference of mercury level
= 20 cm = 0.2 m
$\therefore$ The difference in pressure head = $\frac{P_1}{\rho g}-\frac{P_2}{\rho g}$ = head (h)
$$\Rightarrow \frac{P_1-P_2}{\rho g} = h = x\left(\frac{S_m}{S}-1\right)$$
Where,
S_m = Specific gravity of mercury = 13.6
S = Sq. gravity of water = 1
$$\therefore \frac{P_1}{\rho g}-\frac{P_2}{\rho g} = 0.2\times\left(\frac{13.6}{1}-1\right)$$
= 2.52m

241. Pressure in Pascal at a depth of 1 m below the free surface of water will be:
(a) 1 pa (b) 9810 Pa
(c) 98.1 Pa (d) 981 Pa
Kerala PSC IOF 19.04.2016
WBPSC AE 2003

Ans. (b) P = ρgh
= 1000 × 9.81 × 1
$\boxed{P = 9810\,Pa}$

242. Assertion (A): pressure is equal in all direction at a point in an ideal fluid flow.
Reason (R) : Pascal's law is valid for all cases where shear stresses are zero.
(a) A is true; R is true, and is and it explains A
(b) A is true; R is true, but it does not explain A
(c) A is true ; R is false
(d) A is false; R is true
APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (d) : When an ideal fluid flows through a horizontal tube of variable cross-section, the pressure becomes different in different points. So option (d) is correct. Whereas Pascal's law is valid for all cases where shear stresses are zero.

243. What is the approximate height of mountain if barometer pressure at sea level and at mountain is respectively 760 mm and 724 mm of Mercury considering air density as 1.2 kg/m^3?
(a) 36 m (b) 41 m
(c) 360 m (d) 410 m
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (d) : $P = \rho g\,(h_1 - h_2)$ $\rho_{Hg} = 13600$ kg/m^3
$= 13600 \times 9.81 \times (0.760 - 0.724)$
P = 4802.976 Pa
$P = \rho_{air} g \times x$ $\rho_{air} = 1.2$ kg/m^3
$4802.976 = 1.2 \times 9.81 \times x$
$\boxed{x = 408\ mtr}$ ≈ 410 m

244. The unit of pressure in S.I. unit is :
(a) Pascal
(b) mm of water column
(c) kg/cm^2
(d) Dyne per square centimeter
CGPSC AE 15.01.2021

Ans. (a) : SI unit of pressure is N/m^2 or Pascal (Pa).

245. What is the force needed to apply to a piston of 2 cm radius in order to result a force of 6000 N at the working piston of radius 6 cm?
(a) 1334 N (b) 333 N
(c) 1050 N (d) 667 N
ESE 18.07.2021

Ans. (d) : Pressure will be same
$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$
$$\frac{F_1}{\frac{\pi}{4}(2)^2} = \frac{6000}{\frac{\pi}{4}(6)^2}$$
F_1=666.67 N

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246. _____ **fluids are used in a manometer to keep their size manageable, in case large pressure differences are anticipated**

(a) Moderate (b) Light
(c) Heavy (d) Extremely light

UPRVUNL AE 05.07.2021

Ans. (c) : Heavy

247. Which pressure measurement device has its construction where in, a hollow metal tube is bent like a hook and the end is closed, connected to dial indicator?

(a) Bourdon tube (b) Manometer
(c) Barometer (d) Pirani gauge

UPRVUNL AE 05.07.2021

Ans. (a) :

248. A unit of pressure, 1 mm of Hg equals 1 __

(a) Roentgen (b) PSI
(c) Torr (d) Pascal

UPRVUNL AE 05.07.2021

Ans. (c) : A unit of pressure, 1 mm of Hg equals 1 torr.

249. Which of the following can withstand large pressure differences?

(a) Trapezoidal duct (b) triangular duct
(c) circular pipes (d) Rectangular duct

UPRVUNL AE 05.07.2021

Ans. (c) :

250. The value of one bar (in SI units) is equal to

(a) $100 N/m^2$ (b) $1000 N/m^2$
(c) $1\times10^4 N/m$ (d) $1\times10^5 N/m^2$

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : The value of one bar (in SI units) is equal to $1\times10^5 N/m^2$.

251. Bourdon gauge measures

(a) absolute pressure
(b) gauge pressure
(c) local atmospheric pressure
(d) standard atmospheric pressure

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : The pressure measured with the help of a pressure gauge is known as gauge pressure, in which atmospheric pressure is taken as datum. Bourdon gauge measures gauge pressure.
Barometer is used to measure atmospheric pressure.

252. Vacuum pressure is defined as the pressure

(a) below the atmospheric pressure
(b) above the atmospheric pressure
(c) between the atmospheric pressure and gauge pressure
(d) above the absolute pressure

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : Vacuum pressure is any pressure less than the local atmospheric pressure. It is defined as the difference between the local atmospheric pressure and the point of measurement.
Absolute pressure = Atmospheric pressure – Vacuum pressure

253. A hydraulic press has a ram of 30 cm diameter and plunger of 3 cm diameter. Find the weight lifted by the hydraulic press when the force applied at the plunger is 500N.

(a) 30 kN (b) 50 kN
(c) 20 kN (d) 40 kN

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (b) : According to Pascal's law–

$$\frac{F_1}{A_1}=\frac{F_2}{A_2}$$

Given, $d_1 = 3$ cm
$d_2 = 30$ cm
$F_1 = 500$ N
$F_2 = ?$

$$\frac{F_1}{\frac{\pi}{4}d_1^2}=\frac{F_2}{\frac{\pi}{4}d_2^2}$$

$$\frac{500}{\frac{\pi}{4}(3)^2}=\frac{F_2}{\frac{\pi}{4}(30)^2}$$

$$F_2=\frac{500\times900}{9}=50000N$$

$$\boxed{F_2=50kN}$$

254. A barometer reads 765 mm of Hg. The atmospheric pressure is:

(a) 1.02 bar (b) 3.82 bar
(c) 7.65 bar (d) 0.51bar

CIL MT 27.02.2020

Ans. (a) : $P = \rho gh = 13.6 \times 9.81 \times 765 = 1.02$ bar

255. A vacuum gauge fixed on a steam condenser reads 80 kPa. The barometer indicates 1.013 bar. The absolute pressure in terms of mercury head

(a) 160 mm of Hg (b) 190 mm of Hg
(c) 380 mm of Hg (d) 760 mm of Hg

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : Given,
$P_v = 80$ kPa (vacuum)
$P_{atm} = 1.013$ bar $= 101.3$ kPa
$P_v = P_{atm} - P_{abs}$
$P_{abs} = P_{atm} - P_v$
$= 101.3 - 80$
$P_{abs} = 21.3$ kPa
$\rho_m g h_m = 21.3 \times 10^3$

$$h_m=\frac{21.3\times10^3}{13.6\times10^3\times9.81}$$

$h_m = 0.1596$m
Absolute pressure of mercury head, $h_m = 159.6$ mm

$$\boxed{h_m=160mm}$$

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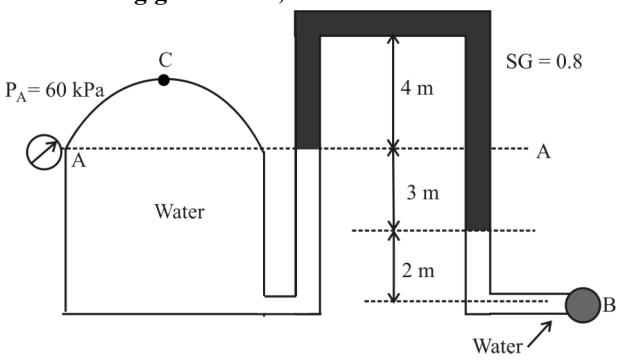
256. In the figure below, the pressure in pipe B, taking g = 10 m/s^2, is



(a) 16 kPa (b) 44 kPa
(c) 104 kPa (d) 76 kPa

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Taking reference as A
Pressure in right limb = pressure on left limb
$P_A = P_B - 800 \times 10 \times 3 - 2 \times 1000 \times 10$
$60 \times 10^3 = P_B - 24000 - 20000$
$P_B = 104 \times 10^3$ Pa = 104 kPa
Note - Distance measured downward from reference is taken negative.

257. The following observations are recorded for steam condenser :
Condenser vacuum = 680 mm (Hg)
Barometer reading = 760 mm (Hg)
Condenser temperature = 35°C
$P_{sat\ @35°C}$ = 42 mm Hg
What will be the vacuum efficiency?
(a) 82.4% (b) 88.4%
(c) 94.7% (d) 98.4%

APPSC Poly Lect. 13.03.2020

Ans. (c) : We know that,
Vacuum efficiency

$$= \frac{\text{Actual vacuum pressure}}{\left(\begin{matrix}\text{Barometer}\\ \text{pressure}\end{matrix}\right) - \left(\begin{matrix}\text{Absolute}\\ \text{pressure of steam}\end{matrix}\right)}$$

$$= \frac{680}{760-42} \times 100$$
$$= 94.7\%$$

258. A U-tube differential manometer
(a) it used upright if the pressure difference is small
(b) is used inverted if the pressure difference is small
(c) cannot be inclined at any angle
(d) must be fitted with a well for the sake of accuracy

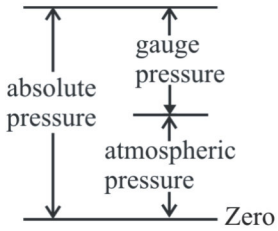
TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : A U-tube differential manometer is used inverted if the pressure difference is small.

259. Gauge pressure at a point is equal to
(a) Absolute pressure plus atmospheric pressure
(b) Absolute pressure minus atmospheric pressure
(c) Vacuum pressure plus absolute pressure
(d) None of these

RPSC IOF, 2020

Ans. (b) : Absolute pressure = atmospheric pressure + gauge pressure
Gauge pressure = absolute pressure – atmospheric pressure



260. Bourdon tube is used to measure :
(a) Volume flow rate (b) Temperature of gas
(c) Velocity of flow (d) Pressure

SJVN ET 2019

Ans. (d) : Bourdon tube is used to measure pressure.

261. A water lake has a maximum depth of 100 m. If the atmospheric pressure is 101 kPa, the absolute pressure at this depth is
(a) 1082 kPa (b) 881 kPa
(c) 900 kPa (d) 678 kPa

APPSC AEE SCREENING 17.02.2019

Ans. (a) : $P = P_{atm} + \rho gh$
$= 101 \times 10^3 + 1000 \times 9.81 \times 100$
$= 1082$ kPa

262. If the atmospheric pressure at sea level is 10.144 N/cm^2, what will be the absolute pressure at the height of 3 km from sea level? [Pressure variation may be assumed to follow hydrostatic law, density of air is assumed constant as 1.2 kg/m^3]
(a) 7.612 N/m^2 (b) 13.675 N/cm^2
(c) 8.612 N/cm^2 (d) 6.612 N/cm^2

SJVN ET 2019

Ans. (d) : $P_{abs} = P_{atm} - \rho gh$

$$= 10.144 - \frac{1.2 \times 9.81 \times 3000}{10000}$$
$$= 10.144 - 3.5316$$
$$= 6.612 \text{ N/cm}^2$$

263. The below figure shows multi-tube manometer using water and mercury that is used to measure the pressure of air in a vessel. It is given that h_1 = 0.4 m, h_2 = 0.5 m, h_3 = 0.3 m, h_4 = 0.7 m, h_5 = 0.1 m and h_6 = 0.5 m. Assuming the density of water to be 1000 kg/m^3 & the density of mercury to be 13600 kg/m^3, for the given values of heights, the gauge pressure in kN/m^2 will be :

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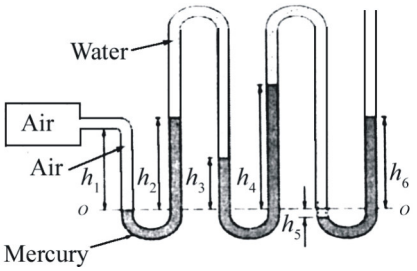


(a) 90 (b) 1900
(c) 19 (d) 190

BHEL ET 2019

Ans. (d) : Applying hydrostatic law,
$P_{air}+1\times9.81\times0.4-13600\times9.81\times0.5+1000\times9.81\times(0.5-0.3) -13600\times9.81\times(0.7-0.3)+1000\times9.81(0.7+0.1) -13600\times9.81\times(0.1+0.6) = P_0$
$P_g = P_{air} - P_0 = 190.31$ kN/m^2

264. Pirani gauge is used for the measurement of:
(a) Very high pressure
(b) High vacuum
(c) Liquid level under pressure
(d) Liquid level at atmospheric pressure

OPSC AEE 2019 PAPER - II

Ans : (b) : The pirani gauge is a robust thermal conductivity gauge used for the measurement of the pressure in vacuum systems.

265. A force of 400 N is required to open a process control valve. What is the area of diaphragm needed for a diaphragm actuator to open the valve with a control gauge pressure of 70 kPa?
(a) 0.0095 m^2 (b) 0.0086 m^2
(c) 0.0057 m^2 (d) 0.0048 m^2

ESE 2018

Ans. (c) : Given,
F = 400 N

$$P = 70 \text{ kPa} = 70\times10^3 \frac{N}{m^2}$$

$$\text{Area of diaphragm } A = \frac{F}{P}$$

$$= \frac{400}{70\times10^3}$$

$$= 0.0057 \text{ m}^2$$

266. A force of 10kN is required to move a work piece. What is the needed working pressure, if the piston diameter is 100 mm?
(a) 1.55 MPa (b) 1.46 MPa
(c) 1.27 MPa (d) 1.12 MPa

ESE 2018

Ans. (c) : Given,
F = 10 kN = 10 × 10^3 N
d = 100 mm = 0.1 m

$$\text{Area (A)} = \frac{\pi d^2}{4} = 7.85\times10^{-3} \text{ m}^2$$

Let the needed working pressure is P

$$\text{So} \quad P = \frac{F}{A}$$

$$P = \frac{10\times10^3}{7.85\times10^{-3}} = 1.27\times10^6 \frac{N}{m^2}$$

P = 1.27 MPa

267. Barometric reading shows 740 mm of Hg at a gravitational acceleration of 9.81 m/s^2. If mercury temperature is 10^{0}C and density of mercury at this temperature is 13570 kg/m^3 then the atmospheric pressure will be
(a) 98.5 kPa (b) 9.85 kPa
(c) 0.985 kPa (d) 0.0985 kPa

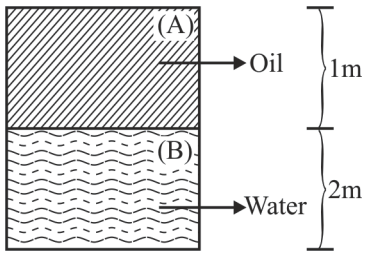
MPSC Lect. (Prod.) 2018

Ans. (a)
h = 740 mm g = 9.81 m/sec^2, ρ = 13570 kg/m^3
P = ρgh = 13570 × 9.81 × 0.74 = 98.5 kPa.

268. A container contains water and oil of specific gravity 0.9. Oil is at the top with a depth of 1m and water at the bottom with a height 2 m. Assuming g = 9.81 m/s^2, the pressure intensity at the bottom of the container in kPa is will be:
(a) 28.45 (b) 13.2
(c) 21.54 (d) 15.6

AAI Jr. Executive 29.11.2018

Ans. (a) :



$\rho_1 = 1000$ kg/m^3
$\rho_2 = (Sg)_{oil} \times \rho_\omega$
$= 0.9 \times 1000$
$\rho_2 = 900$ kg/m^3
$h_1 = 2$m, $h_2 = 1$m

Point (A)
$P_A = \rho_2 gh$
$= 900 \times 9.81 \times 1$
$= 8829$ N/m^2

Point (B)
$P_B = \rho_1 gh$
$= 1000 \times 9.81 \times 2$
$= 19620$

$P_{bottom} = P_A + P_B$
$= 8829 + 19620$
$= 28449$

$P_{bottom} = 28.449$ kPa.

269. The gauge pressure at the surface of a liquid of density 900 kg/m^3 is 0.4 bar. If the atmospheric pressure is 1×10^5 Pa, the absolute pressure at a depth of 50 m is
(a) 5.81 bar (b) 15.8 bar
(c) 25.8 bar (d) 0.5 bar

ISRO Scientist/Engineer (RAC) 22.04.2018

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Ans. (a) : We know that absolute pressure is

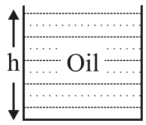
$$P_{abs} = P_{atm} + P_{gauge} + P_h$$
$$= 1\times10^5 + 0.4\times10^5 + 900\times9.81\times50$$
$$= 5.81 \times 10^5 \text{ Pa}$$
$$= 5.81 \text{ bar}$$

270. An oil of specific gravity 0.9 is contained in a vessel. At a point the height of oil is 40m. Find the corresponding height of water at the point.

(a) 900 m (b) 36 m
(c) 10.33 m (d) 18 m

TNPSC AE 2018

Ans. (b) : $\rho_{oil} = 900\frac{kg}{m^3}$



$h = 40$ m

Then pressure at bottom

$P_{gauge} = \rho gh$

$= 900 \times 9.81 \times 40$

For height of water at that point, pressure will be same.

$$\rho_w \times g \times h_w = \rho_{oil} \times g \times h_{oil}$$
$$h_w = \frac{900\times9.81\times40}{1000\times9.81}$$
$$\boxed{h_w = 36m}$$

271. If a pipe contains an oil of sp. gravity 0.9 and a differential manometer connected at the two points A and B shows a difference in mercury level as 15 cm, the difference of pressure at the two points will be

(a) 1.32435 N/m^2 (b) 1905 N/m^2
(c) 18688 N/m^2 (d) 124587 N/m^2

TNPSC AE 2018

Ans. (c) : $(P_A - P_B) = (\rho_m - \rho_{oil}) \times g \times h$

$= (13600 - 900) \times 9.81 \times 0.15$

$= 18688.05$ N/m^2

272. Which of the following is correct for the unit of pressure?

(a) 1 bar = 10^5 N/m^2 = 10^5 Pa
(b) 1 bar = 10^5 N/mm^2 = 10^5 Pa
(c) 1 MPa = 100 bar = 1000 kN/m^2
(d) 1 MPa = 10 bar = 100 kN/m^2
(e) 1 bar = 10^5 kN/m^2 = 10^5 Pa

CGPSC AE 25.02.2018

Ans. (a) : Unit of pressure-

1. $1 \text{ Pa} = 1\frac{N}{m^2}$

2. $1 \text{ bar} = 10^5\frac{N}{m^2}$

3. 1 atm. = 101325 Pa

4. $1\frac{kgf}{cm^2} = \frac{9.81}{10^{-4}}\frac{N}{m^2} = 9.81\times10^4 \text{ Pa}$

5. $1\frac{lbf}{(inch)^2} = \frac{pound}{(inch)^2} = \frac{0.453}{(2.54)^2}\frac{kgf}{cm^2}$

$$= \frac{0.453}{(2.54)^2}\times9.81\times10^4 \text{ Pa}$$

PSI → Pound per square inch

1 atm = 14.7 PSI

273. Statement I: A differential inverted U-tube manometer determines the difference in pressures between two points in a flow section to which it is connected.

Statement II: The sensitivity of an inclined gauge depends on the angle of inclination.

ESE 2018

Ans. (b) : (i) The differential inverted U tube manometer measure the pressure difference between the two points where the manometer is connected. The pressure difference between the two points is given by

$$\Delta P = \rho g x\left(1 - \frac{\rho_m}{\rho}\right)$$

(ii) The sensitivity of the inclined manometer is given by

$$\text{Sensitivity} = \frac{l}{\sin\theta}$$

Hence both the statements are individually true and second is not the correct explanation or first.

274. When the pressure intensity at a point is more than the local atmospheric pressure, then the difference of these two pressures is called

(a) Gauge pressure
(b) Absolute pressure
(c) Positive gauge pressure
(d) Vacuum pressure

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Positive gauge pressure- Pressure more than atmospheric is known as positive gauge pressure.

If $P > P_{atm}$

Positive gauge pressure $p_{gauge} = P - P_{atm}$

Gauge Pressure– Pressure measured with respect to atmospheric pressure is known as gauge pressure.

Absolute pressure– The pressure measured with respect to zero pressure is known as absolute pressure.

Vacuum pressure – When the pressure less than atmospheric pressure is known as vacuum pressure $P < P_{atm} \Rightarrow$ Vacuum Pressure $P_{vacuum} = P_{atm} - P$

275. The value of one bar (in S.I. units) is equal to-

(a) 1×10^2 N/m^2 (b) 1×10^3 N/m^2
(c) 1×10^4 N/m^2 (d) 1×10^5 N/m^2

GMB AAE 25.06.2017

Ans. (d) : The bar is a unit of pressure defined as 100 kilopascal (kPa). It is about to the equal of atmospheric pressure on earth at sea level.

1 bar = 10,0000 Pa = 1×10^5 N/m^2

1 Pa = 1×10^5 N/m^2

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276. The mass of a piston that can be supported by a gas entrapped under the piston in a 0.1 sq.m area vertical cylinder, when a manometer indicates a difference of 100 mm of Hg column for the gas pressure is

(a) 0.136 kg (b) 136 kg
(c) 13.6 kg (d) 1.36 kg

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (b) : Vertical cylinder engine
Piston area = 0.1 m^2
Manometer indicate pressure in Hg column
h = 100 mm of Hg column
Then piston weight is balanced by negative pressure force in cylinder.
Pressure force in cylinders = weight of piston
Area × $\rho_m gh = mg$
$m = \rho_m \times h \times 0.1$
$m = 13.6 \times 10^3 \times 100 \times 10^{-3} \times 0.1$
$m = 13.6 \times 100 \times 0.1$
m = 136 kg

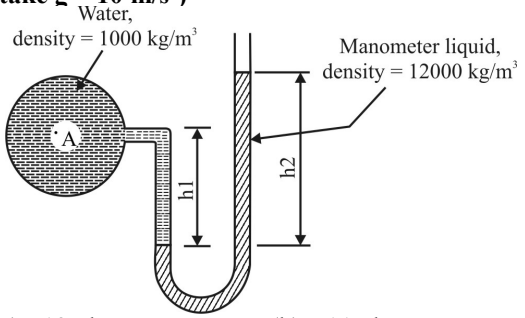
277. Assuming constant temperature condition and air to be an ideal gas, the variation in atmospheric pressure with height calculated from fluid statics is

(a) Linear (b) Exponential
(c) Quadratic (d) Cubic

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : $\Delta p = \rho gh$
$p = p_o e^{-mgh/kT}$,
It is more accurately represented as an exponential function of height.

278. A U tube manometer shown in figure is used to measure the gauge pressure of water density $\rho_1 = 1000$ kg/m^3. If the density of manometer liquid ρ_2 is 12000 kg/m^3, $h_1 = 0.5$ & $h_2 = 1.0$ m, gauge pressure at 'A', the centre of tube is (take g = 10 m/s^2)



(a) 125 kPa (b) 115 kPa
(c) 60 kPa (d) 5 kPa

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : $P_x = P_{x'}$
$P_A + \rho_1 gh_1 = P_0 + \rho_2 gh_2$
Neglecting atmospheric pressure
$P_A + 1000 \times 10 \times 0.5 = 0 + 12000 \times 10 \times 1$
$P_A = 120000 - 5000$
= 115000 Pa
= 115 kPa

279. Statement (I): For measurement of pressure at a section of a pipe running full, several openings round the periphery of pipe are often connected to a manometer through a common annulus.
Statement (II): Manometer connections to a pipe should be perpendicular to the axis of flow, edges smooth and free from burrs.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I)
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I)
(c) Statement (I) is true but Statement (II) is false
(d) Statement (I) is false but Statement (II) is true

UPSC JWM 2017

Ans. (d) : Manometer connected to a pipe should be perpendicular to the axis of flow, edges smooth and free from burs.

280. It is required to find pressure difference between two horizontal pipes through which water is flowing using inverted manometer. Two manometric fluids with specific gravities 13.6 and 0.8 and are available.
Which manometric fluid is to be used now?

(a) Fluid with specific gravity 13.6
(b) Fluid with specific gravity 0.8
(c) Both fluids can be used
(d) None can be used

TRB Polytechnic Lecturer 2017

Ans. (b) : Fluid with specific gravity 0.8 is used as a manometric fluid in inverted manometer because this fluid have low specific density than flowing fluid [specific gravity of water S = 1] in pipes.

281. The barometric pressure at the base of a mountain is 750 mm Hg and at the top 600 mm. If the average air density is 1 kg/m^3, the height of the mountain is approximately:

(a) 2000 m (b) 3000 m
(c) 4000 m (d) 5000 m

HPPSC Asstt. Prof. 18.09.2017

Ans. (a) : Given,
At bottom = 750 mm of Hg
At top = 600 mm of Hg
Average air density = 1 kg/m^3
$\rho_{Hg} \times g (H_B - H_T) = \rho_{air} \times g \times h$
Where h = height of mountain
$13.6(750 - 600) = 1 \times h$
$h = 150 \times 10^{-3} \times 13.6$
$h = 2040$ m ≈ 2000 m

282. Local atmospheric pressure is measured by

(a) Hydrometer (b) Barometer
(c) Hygrometer (d) Altmeter

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : local atmospheric pressure is measured by 'Barometer'.

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283. A barometer at a given location

(a) shows the local atmospheric pressure which is invariant with time

(b) always shows the local atmospheric pressure which may change with time

(c) shows the standard atmospheric pressure, if it is of aneroid type

(d) shows the local temperature if it is of mercury column type

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Barometer stress local Atmospheric pressure Barometer always shows the local atmospheric pressure which may change with time.

284. The standard atmospheric pressure is 760 mm of mercury. At a certain location the barometer reads 710 mm of mercury. At this place an absolute pressure of 360 mm of mercury corresponds to a gauge pressure, in mm of mercury

(a) 400 mm of vacuum (b) 350 mm of vacuum

(c) 360 mm of vacuum (d) 710 mm

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : $P_{abs} = P_{atm\ local} + P_{gauge}$

$360 = 710 + P_{gauge}$

Gauge pressure (P_g) = 350 mm of vacuum

285. An inclined manometer contains a liquid of relative density 0.8 and has an inclination of 30° to the horizontal. For a certain pressure the column length was 10 cm. If there is an uncertainty of 1° in the measurement of the angle of inclination, the calculated pressure would have an uncertainty of

(a) 1% (b) 0.28%

(c) 1.75% (d) 3.33%

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : $P = \rho g L \sin\theta$

$P \propto \rho g L \sin\theta$ ————(1)

$dP = k.\cos\theta\, d\theta$ ————(2)

(2) ÷ (1) for small Angles

$$\frac{dP}{P} = \frac{d\theta}{\theta} = \frac{1^\circ}{30^\circ} \times 100\% = 3.33\%$$

286. The pressure of a liquid measured with the help of a piezometer tube is:

(a) Vacuum pressure

(b) Gauge pressure

(c) Absolute pressure

(d) Atmospheric pressure

HPPSC Asstt. Prof. 18.09.2017

Ans. (b) :

287. Barometric pressure is equal to

(a) 735.6 mm hg (b) 100 mm hg

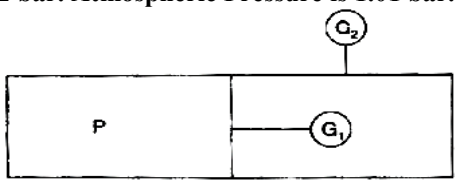
(c) 250 mm hg (d) 760mm hg

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (d) : 1 Barometric pressure = 760 mm of Hg atmospheric pressure known as barometric pressure.

1 atm = 1.01 bar = 101325 Pa = 14.696 psi

288. The pressure gauges G_1 and G_2 installed on the system show pressure of P_{G1} = 6 bar and P_{G2} = 2 bar. Atmospheric Pressure is 1.01 bar.



What is the value of the unknown pressure P?

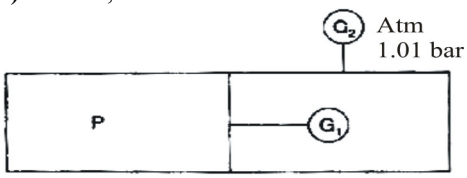
(a) 4.99 bar (b) 6 bar

(c) 7.01 bar (d) 8.01 bar

(e) 9.01 bar

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) : Given,



P_{G1} = 6 bar

P_{G2} = 2 bar

For partial pressure,

$P = P_{G1} + P_{G2} + P_{atm}$

= 6 + 2 + 1.01 = 9.01 bar

289. One atmospheric pressure at STP condition is not equal to :

(a) 1.01325 kg/cm^2

(b) 10^4 N/m^2

(c) 760 mm of mercury column at mean sea level

(d) 10.333 m of water column

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : 1 atm pressure = 10^5 pascal = 10^5 N/m^2

= 1.01325 kg/cm^2

= 760 mm of Hg

= 10.333m of water column

290. Positive gauge pressure is

(a) Pressure less than atmospheric pressure

(b) Pressure equal to atmospheric pressure

(c) Absolute zero pressure

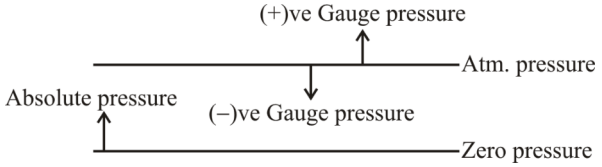
(d) Pressure greater than atmospheric pressure

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : All those pressure that are measured with respect to Atmospheric pressure line are known as gauge pressure. Gauge pressure can be positive & negative.

(i) Positive gauge pressure are measured above or greater than atmospheric pressure line.

(ii) Negative pressure are lie below or less than the atmospheric pressure line.



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291. A Piezometer is used for measuring ..... pressures.

(a) Low (b) Moderate
(c) High (d) High as well as low

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : A Piezometer is simple glass tube is open at both the ends, whose one end of the glass tube is connected to the pipe line, which other end is kept open to the atmosphere. Piezometer is used to measure moderate pressure, if in the question, 'moderate' not given then it would be used to measure low pressure also.

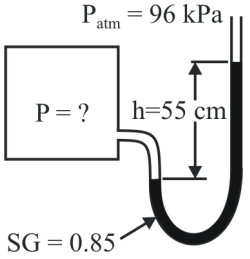
292. The Pascal's law states that the

(a) Pressure at any point in a fluid at rest has the same magnitude in all directions
(b) Pressure at any point in a fluid at rest has the dissimilar magnitude in all directions
(c) Pressure at any point in a fluid at rest has the same magnitude in transverse and longitudinal directions only
(d) Pressure at any point in a fluid at rest has the same magnitude in perpendicular direction only

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Pascal's Law – It states that the pressure or intensity of pressure at a point in a static fluid is equal in all directions.

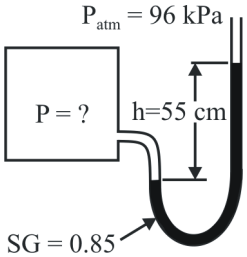
293. A manometer is used to measure the pressure of a gas in a tank. The fluid used has a specific gravity of 0.85 and the manometer column height is 55 cm, as shown in figure. If the local atmospheric pressure is 96 kPa, what is the absolute pressure in the tank?



(a) 4.6 kPa (b) 98.6 kPa
(c) 100.6 kPa (d) 200 kPa

ISRO Scientist/Engineer 03.07.2016

Ans. (c) :



SG = 0.85
$\rho = 0.85 \times 1000$ kg/m^3
$\rho = 850$ kg/m^3
$P = P_{atm} + \rho gh$
$= 96 \times 10^3 + 850 \times 9.81 \times 0.55$
$= 100.6$ kPa

294. What is the minimum angle of inclination of an inclined manometer to avoid the force of surface tension?

(a) 4^0 (b) 3^0
(c) 5^0 (d) 6^0

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : To avoid the force of surface tension in an inclined manometer the minimum angle of inclination is 4^0.

295. McLeod gauges are used for measuring:

(a) RPM of shaft (b) Surface roughness
(c) Vacuum (d) Velocity

(HPPSC LECT. 2016)

Ans : (c) McLeod gauges used for measuring vacuum. A McLeod gauge is a scientific instrument used to measure very low pressures. It was invented in 1874 by Herbert McLeod (1841-1923). These gauges have largely been replaced by electronic vacuum gauges. The design of a McLeod gauge is somewhat similar to a that of a mercury-column manometer. Typically it is filled with mercury. If used incorrectly, this mercury can escape and contaminate the vacuum system attached to the gauge.

296. Micro-manometer is used to

(a) determine low pressure difference
(b) determine higher pressure difference
(c) pressure measurement for gases only
(d) measure pressure in congested areas

APPSC AEE Screening Test 2016

Ans. (a) : Micro Manometer– When an additional gauge liquid is in a U-tube manometer, a large difference in meniscus levels may be obtained for a very small pressure difference.

297. A machine requires an effort of 10 kg to lift a load of 250 kg and an effort of 13 kg for a load of 400 kg. The effort required to lift a load of 500 kg will be

(a) 15 kg (b) 25 kg
(c) 35 kg (d) 45 kg

GPSC ARTO 01.05.2016

Ans. (a) : E = Effort, W = Load
Given, $E_1 = 10$ kg, $W_1 = 250$ kg, $E_2 = 13$ kg,
$W_2 = 400$ kg, $W_3 = 500$ kg, $E_3 = ?$
Total difference of load $W_2 - W_1 = 400 - 250 = 150$ kg
Total Difference of Effort $E_2 - E_1 = 13 - 10 = 3$ kg
$\because$ 3 kg of Effort = 150 kg of Load
So, $W_3 - W_1 = 250$ kg of Load $= \frac{250}{150} \times 3 = 5$ kg of Effort.
So, total effort of lift the load "500 kg" = 10 + 5

$$\boxed{E_3 = 15\,\text{kg}}$$

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298. A simple Bourdon tube pressure gauge is a

(a) closed loop control system
(b) open loop control system
(c) manually operated system
(d) none of the above

GPSC ARTO 01.05.2016

Ans. (b) : Bourdon tube pressure gauge– The measurement of static pressure by boundary tube pressure gauge is a typical simple and inexpensive design of the system that is open loop system.

299. With reference to the containers of different shapes having the same base area and filled with the same liquid for equal depths, the apparent contraction in the hydrostatic force on the base of a liquid container and the weight of liquid in the container is known as

(a) D' Alembert's paradox
(b) Hydrodynamic paradox
(c) Elevator Paradox
(d) Hydrostatic paradox

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : The hydrostatic paradox states the fact that in different shaped containers, with the same base area, which are filled with a liquid of the same height, the applied force by the liquid on the base of each container is exactly the same.

300. Select the correct statement

(a) Absolute pressure = Gauge pressure - Atmospheric pressure
(b) Gauge pressure = Absolute pressure - Atmospheric pressure
(c) Absolute pressure = Atmospheric pressure + vacuum pressure
(d) Gauge pressure = Atmospheric pressure + vacuum pressure

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : As we know,
Gauge Pressure = Absolute Pressure – Atmospheric Pressure

301. Which of the following pressure gauge is generally used for the calibration of other pressure measuring gauges

(a) Bourdon's tube pressure gauge
(b) Diaphragm pressure gauge
(c) Dead weight pressure gauge
(d) Manometer gauge

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Generally dead weight pressure gauge used for calibration.
Dead weight testers are system that physically generate to know pressure. They may also be used to accurate measure system pressure. These device do not require a display as the combination of the masses is used to determine the output pressure.

302. The atmospheric pressure with rise in altitude decreases ..... .

(a) linearly
(b) first slowly and then steeply
(c) first steeply and then gradually
(d) unpredictable

GWSSB DEE 07.07.2016

Ans. (b) : The atmospheric pressure with rise in altitude decreases first slowly and then steeply.

303. The atmospheric pressure at sea level is

(a) 101.3 kN/m^2 (b) 10.3m of water
(c) 760 mm of Hg (d) All of these

TSPSC AEE 2015

Ans : (d) Atmospheric pressure:- The atmospheric air exerts a normal pressure upon all surface with which it is in contact and it is known as atmospheric pressure. it is also known as barometric pressure.
Standard atmospheric pressure = 101.3 kN/m^2
= 10.3 m of water
= 760 mm of Hg

304. Working principle of dead weight pressure gauge tester is based on :

(a) Dalton's law of partial pressure
(b) Newton's law of viscosity
(c) None of three mentioned here.
(d) Pascal's law

GWSSB AAE, 27.12.2015

Ans. (d) : Dead weight pressure gauge tester works on Pascal's law & use to calibrate all kind of pressure gauge.

305. Choose the wrong statement. Alcohol is used in manometer, because

(a) Its vapour pressure is low
(b) It provides suitable meniscus for the inclined tube
(c) Its density is less
(d) It provides longer length for a given pressure difference

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Some of the important and desirable properties of the manometric liquids are:
(i) High chemical stability
(ii) Low viscosity
(iii) Low capillary constant
(iv) Low coefficient of thermal expansion
(v) Low volatility
(vi) Low vapour pressure

306. The pressure in the air space above an oil (sp. Gravity 0.7) surface in a tank is 0.1 kg/cm^2. The pressure at 5 m below the oil surface will be-

(a) 5 meters of water column
(b) 4 meters of water column
(c) 3.5 meters of water
(d) 4.5 meters of water column

ISRO Scientist/Engineer (RAC) 29.11.2015

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Ans : (d) Total pressure can be written as

$P_{abs} = P_{atm} + P$

$= 0.1\times10^4\times g + 5\times0.7\times1000\times g$

$= 4500\times g$

$1000\times g\times h = 4500\times g$

$\boxed{h = 4.5 \text{ m of water column}}$

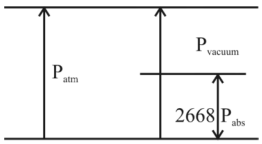
307. The absolute pressure in the condenser of a steam power plant is 2668 Pa and line barometer reading is 760 mm of mercury, then the vacuum recorded in the condenser is

(a) 74 cm of mercury (b) 72 cm of mercury

(c) 75 cm of mercury (d) none of these

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (a) :



$h_{atm} = 760$ mm

$P_{abs} = 2668 \text{ Pa} = 13600 \times 9.81 \times h_{abs}$

$h_{abs} = 20$ mm

$h_{vacuum} = h_{atm} - h_{abs}$

$= 760 - 20 = 740 \text{ mm} = 74 \text{ cm}$

308. Approximate working pressure in the domestic pressure cooker is

(a) 1.1 to 1.3 bar (b) 2.0 to 3.0 bar

(c) 4.0 to 6.0 bar (d) 6.0 to 10.0 bar

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Approximate working pressure in the domestic pressure cooker is 1.1 to 1.3 bar

309. The specific weight W of ocean water changes according to the equation $W = W_0 + C\sqrt{h}$. W_0 = Specific weight of ocean water at free surface and C is a constant. The pressure as a function of depth h is given by

(a) $W_0 h$ (b) $W_0 h + (5/2)Ch$

(c) $W_0 h + (2/3)Ch^{3/2}$ (d) $W_0 h - (2/3)Ch^{2/3}$

ISRO Scientist/Engineer 24.05.2014

Ans. (c) : Function of specific weight

$W = W_o + C\sqrt{h}$

W = sp. weight of ocean water changes

W_0 = sp. wt. of ocean water of free surface. 'c' is constant.

$P = \rho gh$

Pressure is a function of 'h'.

So, $P = \int \rho gh$

sp. wt. $\rightarrow$ $W = \rho g$

$P = \int W dh$

$= \int \left(W_o + C\sqrt{h}\right) dh$

$= W_o h + C\dfrac{h^{3/2}}{(3/2)}$

$\therefore P = W_o h + \dfrac{2}{3}Ch^{3/2}$

310. An inverted U tube manometer is used

(a) To measure large pressure difference

(b) When the specific weight of manometric fluid is less than the specific weight of flowing fluid

(c) To measure pressure at a point in a flowing fluid

(d) To measure the dynamic pressure of a moving stream of fluid

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Inverted U-Tube Manometer.

- It is used for measuring low pressure difference.
- Manometric fluid is lighter than the other fluids in the bulb of container.

311. In a differential manometer, a head of 1.8m of fluid A in limb 1 is found to balance a head of 0.9 m of fluid B in limb 2. The ratio of specific gravities of A to B is

(a) 2 (b) 0.5

(c) 3 (d) 0.18

(e) 0.33

CGPSC AE 26.04.2015 SHIFT-II

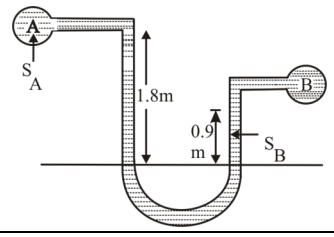
Ans. (b) : According to question

Head of 1.8m of fluid A in limb - 1 = Head of 0.9 m of fluid B in limb - 2

$1.8 \times S_A = S_B \times 0.9$

$\dfrac{S_A}{S_B} = \dfrac{0.9}{1.8} = \dfrac{1}{2}$

$\boxed{\dfrac{S_A}{S_B} = 0.5}$



312. A gauge indicates 15kPa as the fluid pressure while, the ambient pressure is 160 kPa. The absolute pressure in the fluid in kgf/cm^2.

(a) 1.68 (b) 1.78

(c) 1.73 (d) 1.83

(e) 1.64

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Given, $P_g = 15$ kPa , $P_{atm} = 160$ kPa

$P_{abs.} = P_{atm} + P_g = 160 + 15 = 175$ kPa

$P_{abs} = \dfrac{175\times10^3}{9.81\times10^4}\ \dfrac{\text{kgf}}{\text{cm}^2} = 1.78 \text{ kgf/cm}^2$

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313. What will be the depth of a point below water surface in sea, where pressure intensity is 1.006 MN/m^2? (Specific gravity of sea water = 1.025)

(a) 10 m (b) 100 m
(c) 1000 m (d) 1 m

UKPSC AE-2013, Paper-II

Ans. (b) : Given as,

$P = 1.006 \times 10^6$ Pa

$\rho_{sea\ water} = 1025 \frac{kg}{m^3}$

Then, $h = \frac{P}{\rho_{sea\ water} \times g} = \frac{1.006 \times 10^6}{1025 \times 9.81}$

$h = 100.047 \simeq 100$ m

314. The height of water column corresponding to a pressure equivalent of 60 cm of mercury column will be:-

(a) 816 cm (b) 8160 cm
(c) 81.6 cm (d) 7996.0 cm

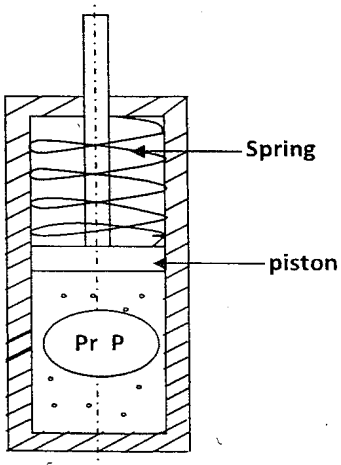
UKPSC AE-2013, Paper-II

Ans. (a) : $[\rho gh]_{water} = [\rho gh]_{mercury}$

$1000 \times g \times h_w = 13600 \times g \times 0.6$

$h_w = 816$ cm

315. A hydraulic jack is used to compress a spring as shown fig. Stiffness of spring is 10^5 N/m, by applying a pressure 'p' in the hydraulic, the spring gets compressed by 10mm. The cross sectional area of the piston is 25cm^2. The applied pressure 'p' is



(a) 4×10^5 Pascals (b) 40 Pascals
(c) 250 Pascals (d) 25 Pascals

ISRO Scientist/Engineer 12.05.2013

Ans. (a) : Force due to pressure

$F = 10^5 \times 10 \times 10^{-3} = 1000$N

$A = 25 \times 10^{-4}$ m^2

Applied pressure $P = \frac{F}{A}$

$= \frac{1000}{25 \times 10^{-4}}$

$= 4 \times 10^5$ Pa.

316. High vacuum pressure is most commonly expressed in:

(a) Pascal (b) cm of water
(c) Torr (d) micron

SJVN ET 2013

Ans. (c) : High vacuum pressure is most commonly expressed in Torr.

$1\ Torr = 133.32\ Pa$

317. Which of the following is a Mechanical Gauge?

(a) Diaphragm gauge
(b) Dead weight pressure gauge
(c) Bourdon tube pressure gauge
(d) All the above

UKPSC AE 2012 Paper–II

Ans. (d) : All the above

318. The pressure at a point in a fluid is not equal in all directions if

(a) the fluid is at rest
(b) there are shear stresses
(c) the fluid is accelerated
(d) the fluid is rotated at a constant speed

APPSC AEE 2012

Ans. (c) : The Pascal law states that the liquid at rest applies pressure at a point is same in all directions but pressure is not equal in all direction when the fluid is accelerated.

319. What liquid used in manometers

(a) Low density
(b) High density
(c) Low surface tension
(d) High surface tension

VIZAG STEEL MT 2011

Ans. (c) :

320. The pressure in meters of oil (specific gravity 0.85) equivalent to 42.5 m of water is :

(a) 42.5 m
(b) 50 m
(c) 52.5 m
(d) 85 m

RPSC AE GWD, 2011

Ans. (b) : $S_1h_1 = S_2h_2$

$S_{oil}\ h_{oil} = S_{water}\ h_{water}$

$0.85\ h_{oil} = 1 \times 42.5$

$h_{oil} = \frac{42.5}{0.85} = 50$m

321. An open tank contains water to a depth of 2m and oil over it to a depth of 1m. If the specific gravity of oil is 0.8, then the pressure intensity at the interface of the two fluid layers will be

(a) 9750 N/m^2
(b) 8720 N/m^2
(c) 9347 N/m^2
(d) 7848 N/m^2

ISRO Scientist/Engineer 2011

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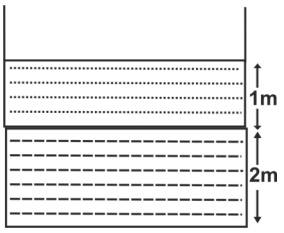
Ans. (d)



Given Sp. gravity of oil = 0.8

Sp. gravity of water = $S_w = 1$

Pressure intensity of two fluid layer

$= 0 + \rho gh$

$= 0.8 \times 1000 \times 9.8 \times 1$

$= 7848\ N/m^2$

322. The height of water column corresponding to a pressure equivalent of 750 mm of mercury :

(a) 10.2 m (b) 10 m

(c) 10.4 m (d) 10.6 m

J&K PSC Civil Services Pre, 2010

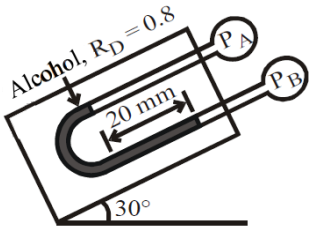
Ans. (a) : $\rho_w \times g \times h_w = \rho_{Hg} \times g \times h_{Hg}$

$1000 \times h_w = 13.6 \times 10^3 \times 750 \times 10^{-3}$

$h_w = \dfrac{10200}{1000}$

$h_w = 10.2$ m

323.



For an inclined manometer as shown in the figure, the pressure difference $P_A - P_B$ will be

(a) $50\ N/m^2$ (b) $78\ N/m^2$

(c) $128\ N/m^2$ (d) $156\ N/m^2$

UPSC JWM Advt. No.-50/2010

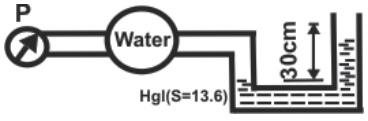
Ans. (b) : From the manometer,

$h = 0.02 \sin 30° = 0.2 \times \dfrac{1}{2} = 0.01$

$P = \rho gh = 800 \times 9.81 \times 0.01$

$P = 78.48\ N/m^2$

324. In the given figure pressure p, in kPa, is



(a) 37 (b) 48.0

(c) 45.2 (d) 51.3

ISRO Scientist/Engineer 2010

Ans. (a) : $P_{gauge} + \rho_w gh - \rho_{Hg} gH = 0$

$P_{gauge} = 13.6 \times 10^3 \times 9.8 \times 0.3 - 1000 \times 9.8 \times 0.30$

$P_{gauge} = 37081.8\ Pa \approx 37\ kPa$

325. Given that: Specific gravity of mercury = 13.6; Intensity of pressure = 40 kPa Express the intensity of pressure (gauge) in various units (S.I.)

(a) 0.3 bar, 3.077 m of water, 0.15 m of mercury

(b) 0.4 bar, 4.077 m of water, 0.299 m of mercury

(c) 0.5 bar, 5.077 m of water, 0.339 m of mercury

(d) None of the above

ISRO Scientist/Engineer 2010

Ans. (b) : Specific gravity of mercury = 13.6

Intensity of pressure = 40kPa

$P_m = \rho_m gh$

$40 \times 10^3 = 13.6 \times 9.81 \times h \times 1000$

$h = \dfrac{40 \times 10^3}{13.6 \times 9.81 \times 10^3} = \dfrac{40}{13.6 \times 9.81}$

h = 299.81 mm

h = 0.299m

For water,

P= 40 kPa

$\rho_W = 1000\ kg/m^3$

$P_W = \rho_W gh_w$

$40 \times 10^3 = 1000 \times 9.81 \times h_w$

$h_w = \dfrac{40 \times 10^3}{9810}$

$h_w = 4.077$m

Intensity of pressure (gauge) in (S.I.)

P= 40 kPa

P = 0.4 bar

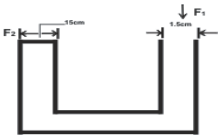
326. A hydraulic press has a ram of 15 cm diameter and plunger of 1.5 cm. It is required to lift a mass of 1000 kg. The force required on plunger is nearly equal to

(a) 100N (b) 1000N

(c) 10000N (d) 10N

ISRO Scientist/Engineer 2009

Ans. (d) :



By Archimedes principle

$\dfrac{F_1}{A_1} = \dfrac{F_2}{A_2}$

$\dfrac{F_1}{\frac{\pi(1.5)^2}{4}} = \dfrac{F_2}{\frac{\pi(15)^2}{4}} \Rightarrow \dfrac{F_1}{1.5^2} = \dfrac{1000}{15^2}$

or $F_1 = 10$ N

327. The pressure difference measured by a mercury oil differential manometer on the two sides of the orifice meter gives a reading of 50 cm of mercury. Determine the differential pressure head in orifice. Take the specific gravity of oil as 0.9.

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(a) 7.05m of oil (b) 5.05m of oil
(c) 4.75m of oil (d) 3.35m of oil

ISRO Scientist/Engineer 2008

Ans. (a) : $S_m = 13.6$, x = 50cm

$S_{oil} = 6.9$

$$h = \left(\frac{S_m}{S_{oil}} - 1\right)x$$

$$= \left(\frac{13.6}{0.9} - 1\right)50$$

h = 7.05 m of oil

328. An open rectangular box of base 2m × 2m contains a liquid of specific gravity 0.80 up to a height of 2.5 m. If the box is imparted vertically upward acceleration of 4.9m/s^2, what will be the pressure on the base of the tank?

(a) 9.81 kPa (b) 19.62 kPa
(c) 36.80 kPa (d) 29.40 kPa

ISRO Scientist/Engineer 2007

Ans. (d) : h = 2.5m, a = 4.9 m/s^2

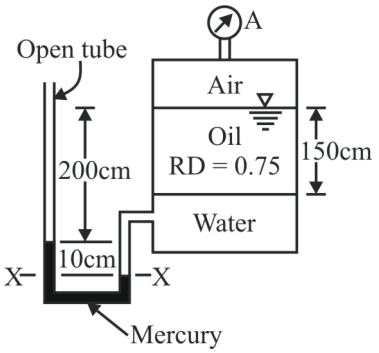
Pressure exerted on the base

$$= \rho gh\left(1 + \frac{a}{g}\right)$$

$$= 1000 \times 0.8 \times 9.8 \times 2.5\left(1 + \frac{4.9}{9.8}\right)$$

= 29.4 kPa

329. The tank shown in the figure below is closed at top and contains air at a pressure of P_A. The value of P_A for the manometer readings shown will be



(a) -3.573 kPa (b) -4.573 kPa
(c) -6.573 kPa (d) -7.573 kPa

ISRO Scientist/Engineer 2007

Ans. (a) : Using jumping of fluid technique

$P_A + 0.75 \times 1000 \times 9.8 \times 1.5 + 1000 \times 9.8 \times 0.60 - 13.6 \times 1000 \times 9.8 \times 0.10 = 0$

$P_A = -3.577$ kPa

330. A pressure of 25m of head of water is equal to:

(a) 25 kN/m^2 (b) 245 kN/m^2
(c) 2500 kN/m^2 (d) 2.5 kN/m^2

J & K PSC Screening, 2006

Ans. (b) : Given,

h = 25m

$\rho_{water} = 1000$ kg/m^3

$P = \rho gh$

$= 1000 \times 9.81 \times 25$

$$= 245250 \frac{N}{m^2}$$

$$= 245.25 \frac{kN}{m^2}$$

331. A mercury water manometer indicates a gauge difference of 400 mm. The difference in pressure measured in meters of waters is-

(a) 0.4 (b) 0.8
(c) 10.80 (d) 5.04

J & K PSC Screening, 2006

Ans. (d) : $P = \rho gh$

$$P = \rho g \times x\left[\frac{\rho_m}{\rho} - 1\right]$$

$$P = 1000 \times 9.81 \times 0.4\left[\frac{13600}{1000} - 1\right]$$

$$P = 49442.4 \frac{N}{m^2}$$

$$\text{in head of water} = \frac{P}{\rho g} = \frac{49442.4}{1000 \times 9.81} = 5.04\text{m of water}$$

332. 100 m of water column is equal to

(a) 1000 kN/m^2 (b) 100kN/m^2
(c) 10kN/m^2 (d) 1kN/m^2

ISRO Scientist/Engineer 2006

Ans. (a) : h = 100 m

Pressure (P) = ρgh = 1000 × 9.81 × 100

$P = 981$ kN/m^2 $\simeq 1000\ kN/m^2$

3. Hydrostatic Forces on Surfaces

333. The resultant hydrostatic force acts through a point known as…..

(a) Center of gravity (b) Center of buoyancy
(c) Center of pressure (d) None of above

APPSC Poly Lect. Automobile Engg., 2020
Vadodara Muncipal Corp. DEE, 2018
GPSC Asstt. Prof. 28.08.2016
Nagaland PSC CTSE 2017, 2016, 2015 Paper-II
MPPSC State Forest Service Exam, 2014
Mizoram PSC AE/SDO 2014, Paper-I
APPSC AE 04.12.2012, APPSC AEE 2012
J&K PSC Civil Services Pre, 2010
ESE 2007

Ans. (c) : Hydrostatic force are the resultant force caused by the pressure of a liquid acting on submerged surface.

$F_R \rightarrow$ It act on the centre of pressure.

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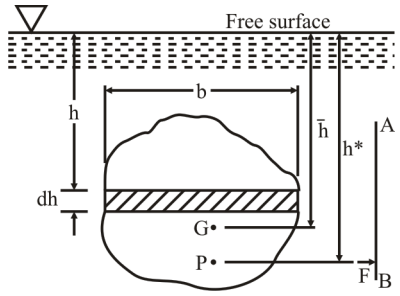
Center of pressure is given by,

$$h^* = \bar{h} + \frac{I_G}{A\bar{h}}\sin^2\theta$$



334. In an inclined plane submerged in water, the centre of pressure is located

(a) at the centroid
(b) below the centroid
(c) above the cetnroid
(d) anywhere in the plane

OPSC AEE 2019 PAPER - II
TSPSC AEE 28.08.2017 (Civil/Mechanical)
WBPSC AE, 2017, APGENCO AE 2017
APPSC AEE Screening Test 2016
MPPSC State Forest Service Exam, 2014
APPSC AEE 2012, ESE 2011

Ans : (b) In an inclined planes submerged in water, the centre of pressure is located below the centroid.

$$\bar{h} = \frac{I_G \sin^2\theta + A\bar{x}^2}{A\bar{x}} = \frac{I_G \sin^2\theta}{A\bar{x}} + \bar{x}$$

In equation $\bar{x}$ is the distance of C.G. of the area of the vertical surface from free surface of the liquid. Hence from equation it is clear that

(i) Centre of pressure $(\bar{h})$ lies below the centre of gravity of the vertical surface.
(ii) The distance of centre of pressure from free surface of liquid is independent of the density of the liquid.

335. The location of the center of pressure over a surface immersed in a liquid is

(a) always above the centroid
(b) will be at the centroid
(c) will be below the centroid
(d) depends upon the density of fluid

APGENCO AE, 2017, (HPPSC LECT. 2016)
Mizoram PSC AE/SDO 2014, Paper-I
ESE 2013

Ans. (c) : Location of center of pressure from free surface of the liquid,

$$y_{cp} = \frac{I_G \cdot \sin^2\theta}{A\bar{h}} + \bar{h}$$

Where, $\bar{h}$ = Distance between center of body to free surface of the liquid.

$\because \quad \frac{I_G \sin^2\theta}{A\bar{h}} \geq 0$

$\therefore \quad y_{cp} > \bar{h}$

So location of the centre of pressure over a surface immersed in a liquid is will be below the centroid.

336. The vertical component of the hydrostatic force on a submerged curved surface is

(a) The force on a vertical projection of the surface
(b) The product of pressure at centroid and the surface area
(c) The mass of the liquid vertically above it
(d) The weight of the liquid vertically above it

OPSC Civil Services Pre 2006
CSE Pre-1995, ESE 1993

Ans. (d) : Vertical component of the hydrostatic force on the submerged curved surface is the weight of the liquid vertically above it and horizontal component is force on the submerged body in horizontal direction on projected area.

337. Which of the following forces act on a fluid at rest?

1. Gravity force
2. Hydrostatic force
3. Surface tension
4. Viscous force

Select the correct answer using the codes given below.

(a) 1, 2, 3 and 4 (b) 1, 3 and 4
(c) 1, 2 and 3 (d) 2 and 4

UPRVUNL AE 21.08.2016
CGPSC Poly. Lect. 22.05.2016
ESE 1998

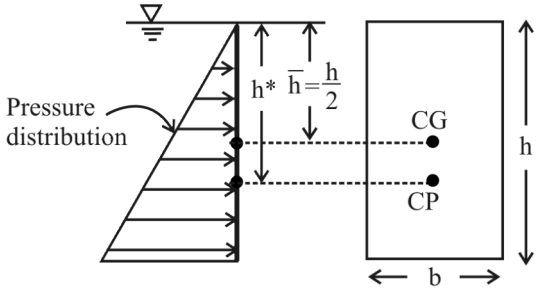
Ans. : (c) The following forces act on a fluid at rest
(i) Gravity force
(ii) Hydrostatics force
(iii) Surface tension

338. The depth of centre of pressure in a rectangular lamina immersed vertically in water up to height h is given by

(a) h/4 (b) 2h/3
(c) 3h/4 (d) h/2

Assam PSC CCE Pre 2015
ESE 2012, CSE Pre-2008

Ans. (b) We know that,
For vertical immersed rectangular lamina.



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Centre of pressure (C.P.),

$$h^* = \frac{I_{GG}}{A\bar{h}} + \bar{h}$$

$$h^* = \frac{bh^3}{12 \times bh \times \frac{h}{2}} + \frac{h}{2} = \frac{h}{6} + \frac{h}{2} = \frac{4h}{6} = \frac{2h}{3}$$

339. Consider a hollow hemispherical object of a diameter d immersed in a liquid (specific weight w) with its plane surface coinciding with the free surface. The vertical component of force on the curved surface would be

(a) $\frac{1}{12} w\pi d^3$

(b) $\frac{3}{8} w\pi d^3$

(c) $\frac{4}{3} w\pi d^3$

(d) Zero

(e) None of the above

TSPSC AEE 28.08.2017 (Civil/Mechanical)
CGPSC Poly. Lect. 22.05.2016

Ans. (a) Given :

Diameter = d

Specific weight = w

Vertical component of force on curved surface

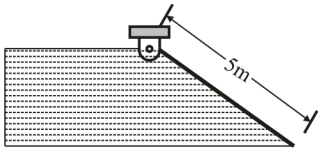
$= \rho g V$

$= w \times \frac{2}{3}\pi R^3 = w \times \frac{2}{3} \times \pi \times \left(\frac{d}{2}\right)^3$

$= w \times \frac{2}{3} \times \pi \times \frac{d^3}{8}$

$= \frac{1}{12} w\pi d^3$

340. A hinged gate of length 5 m, inclined at 30° with the horizontal and with water mass on its left, is shown in figure below. Density of water is 1000 kg/m^3. The minimum mass of the gate in kg per unit width (perpendicular to the plane of paper), required to keep it closed is_____



(a) 9623

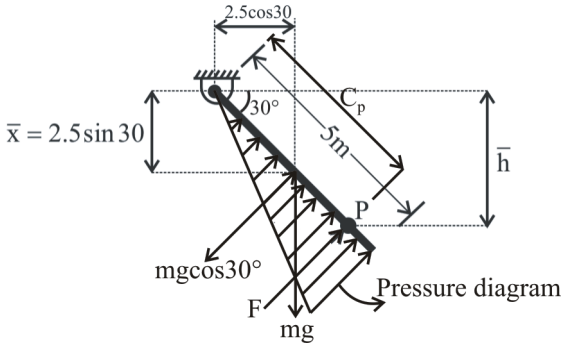
(b) 6600

(c) 7546

(d) 5000

Assam Engg. College AP/Lect. 18.01.2021
GATE 2013

Ans. (a) :



Pressure force acting at center of pressure,

$$F = \rho g A\bar{x} = 1000 \times g \times (5 \times 1) \times \frac{5}{2}\sin 30° = 6250g$$

From pressure diagram center of pressure,

$$C_p = \frac{2h}{3} = \frac{2 \times 5}{3}$$

Taking moment about hinge,

$$F \times C_p = mg\cos 30 \times \frac{h}{2}$$

$$6250g \times \frac{2}{3} \times 5 = m \times g \times \cos 30 \times \frac{5}{2}$$

$m = 9622.5 \approx 9623$ kg

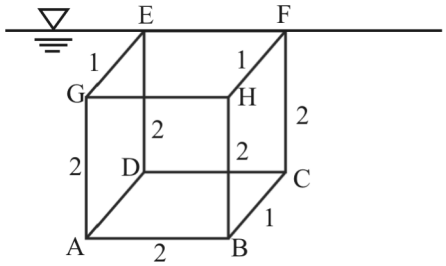
341. A rectangular water tank, full to the top, has its length, breadth and height in the ratio of 2:1:2. The ratio of hydrostatic forces at the bottom to that at any larger vertical surface is

(a) 1/2 (b) 1

(c) 4 (d) 2

Nagaland PSC CTSE 2015 Paper-II
ESE 2015, 1996

Ans. (b) : $\bar{x}$ = vertical distance of centre of gravity from the free surface.



Force on the bottom (ABCD)

$(F_1) = \rho g A\bar{x}$

$= \rho g\ 2 \times 2 \times 1$

$F_1 = 4\rho g$

Force on the larger vertical surface (ABGH)

$F_2 = \rho g A\bar{x}$

$= \rho g\ 2 \times 2 \times 1$

$F_2 = 4\rho g$

$$\frac{F_1}{F_2} = \frac{4\rho g}{4\rho g} = 1$$

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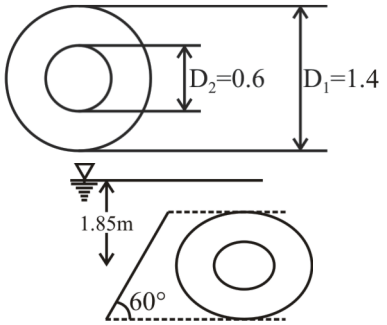
342. A circular annular plate having outer and inner diameters of 1.4 m and 0.6 m respectively is immersed in water with its plane making an angle of 60° with the horizontal. The centre of the circular annular plate is 1.85 m below the free surface. The hydrostatic thrust on one side of the plate is

(a) 1975 N (b) 19.75 N
(c) 11.4 N (d) 22.8 N

Nagaland PSC CTSE 2015 Paper-II
ESE 1998

Ans. (d) : Given,



$$F_H = \rho g A \bar{x}$$

$$= 9.81 \times 1000 \times \left[\frac{\pi}{4}\left((1.4)^2 - (0.6)^2\right)\right] \times 1.85$$

$$F_H = 22794.516$$

$$\approx 22.8 \text{ kN}$$

343. A vertical sluice gate, 2.5 m wide and weighing 500 kg is held in position due to horizontal force of water on one side and associated friction force, when the water level drops down to 2 m above the bottom of the gate, the gate just starts sliding down. The coefficient of friction between the gate and the supporting structure is

(a) 0.20 (b) 0.10
(c) 0.05 (d) 0.02

Nagaland PSC CTSE 2015 Paper-II
ESE 1999

Ans. (b) : Given, w = 500 kg, b = 2.5 m, μ = ?
Hydrostatic force due to water head,

$$F = \bar{h}\rho g A$$

$$= \frac{2}{2} \times 1000 \times G \times 2 \times 2.5$$

$$= 5000 \text{ gN}$$

For limiting condition,

$$\mu F = mg$$

$$\mu \times 5000g = 500 \text{ g}$$

$$\mu = 0.1$$

344. A bucket of water is hanging from a spring balance. An iron piece is suspended into water without touching sides of bucket from another support. The spring balance reading will

(a) Remain same
(b) Decrease
(c) Increase
(d) Increase/decrease depending on depth of immersion

WBPSC AE, 2017
ISRO Scientist/Engineer 2009

Ans. (a) : It will Increase—
When the piece of iron is lowered into the water, the water level in the bucket increase and so that the pressure at the bottom of bucket also increase. Thus total at the bottom also increase. Therefore the spring balance reading increase.

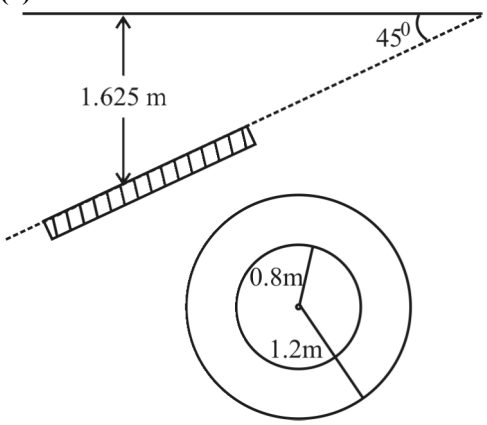
Remain same—
It will remain unchanged, assuming that the water completely fills the bucket. In this case the amount equal to extra weight of iron piece, an equivalent weight of water will spill out maintaining the overall weight constant.

345. A circular annular plate bounded by two concentric circles of diameter 1.2 m and 0.8 m is immersed in water with its plane making an angle of 45^0 with the horizontal. The centre of the circle is 1.625 m below the free surface. What will be the total pressure force on the face of the plate?

(a) 10.00 kN (b) 7.07 kN
(c) 14.141 kN (d) 18.00 kN

APPSC AE Subordinate Service Civil/Mech. 2016
ESE 2004

Ans. (a) :



Total pressure force (F) = $\rho g A \bar{h}$

projected area (A) = $\frac{\pi}{4}(1.2^2 - 0.8^2)$

$\bar{h}$ = 1.625 m

$$F = 1000 \times 9.81 \times \frac{\pi}{4}\left(1.2^2 - 0.8^2\right) \times 1.625$$

F = 10016.18 N = 10.01 kN ≃ 10 kN

So, total pressure force = $\boxed{F = 10 \text{ kN}}$

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346. A rectangular plate 0.75 m × 2.4 m is immersed in a liquid of relative density 0.85 with its 0.75 m side horizontal and just at the water surface. If the plane of the plate makes an angle of 60° with the horizontal, the pressure force on one side of the plate, in kN, is

(a) 15.6 (b) 7.8
(c) 24.0 (d) 18.0

TSPSC AEE 28.08.2017 (Civil/Mechanical)
ESE 2008

Ans. (a) : $F = \rho.g.\bar{h}.A$

$$= (0.85\times1000)(9.81)\left(\frac{2.4\times\sin 60}{2}\right)(0.75\times2.4)$$

$= 15598.1\ N = 15.6\ kN$

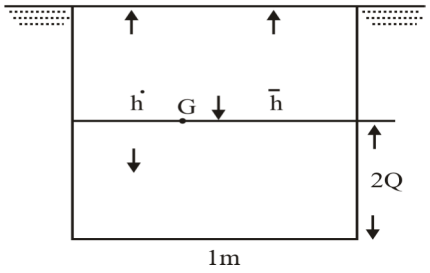
347. A rectangular plate, 1 m wide and 2m deep, is held just below the surface of the water. The total pressure on this lamina is :

(a) 19.62 kN (b) 2 kN
(c) 9.81 kN (d) 98.1 kN

APPSC Poly Lect. Automobile Engg., 2020
CIL MT 27.02.2020

Ans. (a)

Free water surface



Total pressure is given by equation as

$$F = \rho A g\bar{h}$$

$\rho = 1000\ kg/m^3$

$g = 9.8\ m/sec^2$

$A = 2 \times 1 = 2m^2 \qquad \bar{h} = \frac{1}{2}\times 2 = 1m$

$F = 1000 \times 2 \times 9.8 \times 1$

$= 19.6\times10^3 = 19.6\ kN$

348. If a water tank contains 1.3 m deep water. The pressure exerted by the water per metre length of the tank would be?

(a) 2.89 kN (b) 8.29 kN
(c) 9.28 kN (d) 28.9 kN

ISRO RAC 2018
VIZAG STEEL MT 2011

Ans. (b) : $h = 1.3$

$\ell = 1$ (per meter length)

$F = \rho g A\bar{h}$

$$F = 1000\times9.81\times(1.3\times1)\times\frac{1.3}{2}$$

$F = 8.29\ kN$

349. The total force on a rectangular plane surface 4 m wide and 6 m deep which lies in vertical plane of water when its upper edge is horizontal and coinciding with water surface.

(a) 1412640 N (b) 470880 N
(c) 941760 N (d) 706320 N

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : $F = \rho g A\bar{h}$

$$= 1000\times9.81\times4\times6\times\frac{6}{2}$$

$= 706320\ N$

350. Total pressure on a lamina of area A is WAy, where W = specific weight and y = depth of the centroid of the lamina below the free surface.

(a) only when the lamina is horizontal
(b) only when the lamina is at 45° with the horizontal
(c) whether the lamina is horizontal, vertical or inclined.
(d) only when the lamina is vertical

CGPSC AE 15.01.2021

Ans. (c) : Total pressure on a lamina of area A is WAy, where
W= Specific weight
y = depth of centroid
acts whether the lamina is horizontal, vertical or inclined.

351. Which of the following statements is False, in case of centre of pressure of a submerged vertical circular plate from free surface of the liquid?

(a) Position of the centre of pressure is directly proportional to the density of the liquid
(b) Position of the centre of pressure is calculated by using the principle of moments
(c) Position of the centre of pressure is directly proportional to h from free surface of the liquid
(d) Position of the centre of pressure lies below the centre of gravity of vertical surface

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (a) : For centre of pressure,

$$\bar{h} = \bar{x} + \frac{I_g \sin^2\theta}{A\bar{x}}$$

Where, $\bar{h}$ = Centre of pressure
$\bar{x}$ = Centre of gravity

From here, it is clear that $\bar{h}$ will always be greater than $\bar{x}$.

352. A tank with four equal vertical faces of width t and depth h is filled up with a liquid. If the force on any vertical side is equal to the force at the bottom, then the value of h/t will be _____.

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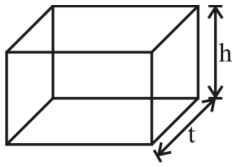
(a) 0.5 (b) 1.5
(c) 1 (d) 2

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (d) : $P_{bottom} = P_{vertical\ side}$

$(\rho g A\bar{x})_{bottom} = (\rho g A\bar{x})_{side}$



$$\rho g \times h \times t \times t = \rho g \times h \times t \times \frac{h}{2}$$

$$h \times t \times t = h \times t \times \frac{h}{2}$$

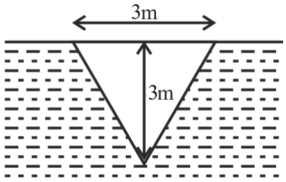
$$\boxed{\frac{h}{t} = 2}$$

353. An isosceles triangular plate of base 3 m and altitude of 3 m is immersed vertically in an oil of specific gravity 0.8. The base of the plate coincide with the free surface of the oil. The total pressure on the plate is _____. (Assume g = 10 m/s^2)

(a) 18 kN (b) 48 kN
(c) 36 kN (d) 72 kN

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (c) : Given,



Specific gravity of oil, $S_{oil} = 0.8$

$P_{oil} = 0.8 \times 1000$

$= 800\ kg/m^3$

Area of isosceles triangle $= \frac{1}{2} \times 3 \times 3$

$$A = \frac{9}{2} m^2$$

$\therefore\ \bar{x} = h/3 = 3/3 = 1$ m

total pressure, $P = \rho g A\bar{x}$

$$= 800 \times 10 \times \frac{9}{2} \times 1$$

$$= 400 \times 10 \times 9$$

$$= 36000\ N$$

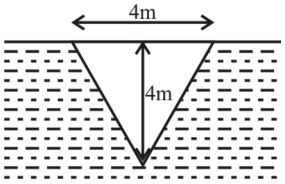
$$= 36\ kN$$

354. Find centre of pressure of an isosceles triangular plate of base 4 m and altitude 4 m. When it is immersed vertically in an oil of specific gravity 0.9. The base of plate coincide with the free surface of oil.

(a) 1.99 m (b) 1.75 m
(c) 2.81 m (d) 1.50 m

Oil India Senior Officer 23.12.2020

Ans. (a)



$$h^* = \frac{I_G}{A\bar{h}} + \bar{h}$$

Where, h^* = Centre of pressure

$\bar{h}$ = Centre of gravity

A = Area

Given, altitude (h) = 4m

Base (b) = 4m

$$\bar{h} = \frac{h}{3} = \frac{4}{3}$$

$$I_G = \frac{bh^3}{36} = \frac{4 \times 4^3}{36} = \frac{64}{9}$$

$$A = \frac{1}{2} b \times h = \frac{1}{2} 4 \times 4 = 8$$

$$h^* = \frac{64/9}{8 \times \frac{4}{3}} + \frac{4}{3}$$

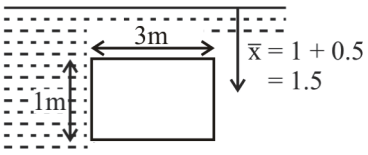
$$= \frac{2}{3} + \frac{4}{3} = \frac{6}{3} = 2m$$

355. A rectangular plate 3 m long and 1 m wide is immersed vertically in water in such a way that its 3 m side is parallel to the water surface and the plate is 1 m below it. The total pressure on the plate is _____.

(a) 88.28 kN (b) 44.14 kN
(c) 66.21 kN (d) 22.07 kN

VIZAG MT, 14.12.2020

Ans. (b) :



$F_h = \rho g A\bar{x}$

$= 1000 \times 9.81 \times 3 \times 1 \times [1 + 0.5]$

$= 44145\ N$

$\boxed{F_h = 44.145 kN}$

356. When the depth of immersion of a plane surface is increased the centre of pressure will

(a) come closer to centroid
(b) move farther away from the centroid
(c) will remain unchanged
(d) depends on the specific weight of the liquid

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : When the depth of immersion of a plane surface is increased the centre of pressure will come closer to centroid.

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357. The centre of pressure of a vertical rectangular plate with height of h m from its base is at

(a) h/2 from base
(b) h/3 from base
(c) 2h/3 from base
(d) 3h/4 from base

UPPSC AE 13.12.2020, Paper-2

Ans. (b) Vertical rectangular plate ($\theta = 90°$) –

$$\text{Centre of pressure, } h_{CP} = h_{CG} + \frac{I\sin^2\theta}{A \times h_{CG}}$$

$$= \frac{h}{2} + \frac{\frac{bh^3}{12}}{(bh)\times\frac{h}{2}}$$

$$h_{CP} = \frac{h}{2} + \frac{h}{6} = \frac{2h}{3}$$

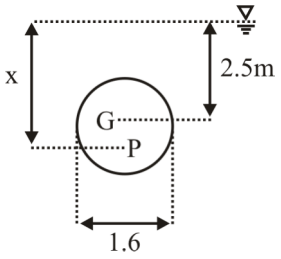
Centre of pressure from its base $\Rightarrow$ h – 2h/3 = h/3

358. A circular plate of diameter 1.6 m is placed vertically in water in such a way that the centre of the plate is 2.5m below the free surface of the water. The location of the centre of pressure is

(a) 2.564 m
(b) 2.5 m
(c) 2.864 m
(d) 2.654 m

UPPSC AE 13.12.2020, Paper-2

Ans. (a)



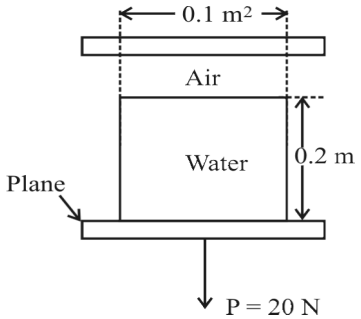
dia = 1.6 m

centre of pressure

$$h_{CP} = h_{CG} + \frac{I}{A\,h_{CG}}$$

$$2.5 + \frac{\frac{\pi d^4}{64}}{\frac{\pi}{4}(1.6)^2 \times 2.5} = 2.5 + \frac{0.3216}{2.0106 \times 2.5} = 2.564 \text{ m}$$

359. In the figure below, a cylindrical vessel of area 0.1 m^2 is inverted on a plate, so that some air is trapped at the closed top and of the beaker. Take g = 10 m/s^2. If the force needed to pull out the plate against atmosphere pressure is 20 N, the gauge pressure of trapped air is

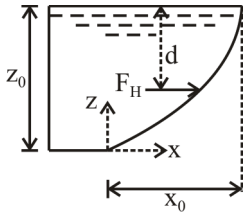


(a) -1.8 kPa (b) -2.2 kPa
(c) 2.2 kPa (d) 1.8 kPa

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : Force balance
20 + weight of water = force due to air in container
20 + 1000 × 0.2 × 0.1 × 10 = P × 0.1
P = 2.2 kPa (negative gauge pressure)
P = –2.2 kPa gauge

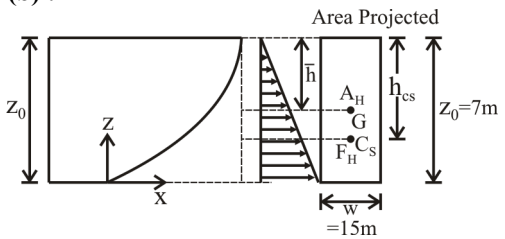
360. A dam has a parabolic shape $\frac{z}{z_0} = \left(\frac{x}{x_0}\right)^2$ as in shown in figure, with x_0 = 3 m and z_0 = 7 m. The fluid is water (specific weight = 9810 N/m^3) and atmospheric pressure may be omitted. Compute the horizontal force F_H and its line of action from surface of water 'd'. Assume width of dam is 15 m.



(a) F_H = 3605 kN, d = 1.16 m
(b) F_H = 3605 kN, d = 4.66 m
(c) F_H = 4300 kN, d = 4.66 m
(d) F_H = 4300 kN, d = 1.16 m

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) :



Hydrostatic force on horizontal plane surface is,

$$\bar{h} = 3.5, A_H = 15 \times 7 = 105 \text{ m}^2$$

$$F_H = \rho g \bar{h} A_H = \gamma \bar{h} A_H$$

$$= 9810 \times 3.5 \times 15 \times 7$$

$$F_H = 3605.17 \text{ kN}$$

Location of centre of pressure for

$$h_{cp} = \frac{2}{3} \times 7 = 4.66 \text{ m}$$

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361. For an inclined plane for which position, maximum total pressure acts on it?

(a) Horizontal (b) Vertical
(c) Skewed (d) Inclined

JPSC AE PRE 2019

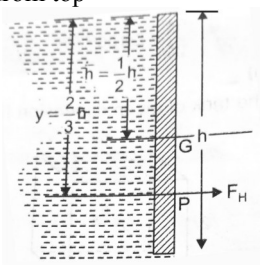
Ans. (b) : Vertical

362. A beaker contains water upto h height. The location of centre of pressure is

(a) h/3 from top (b) h/2 from top
(c) 2h/3 from top (d) 3h/4 from top

JPSC AE PRE 2019

Ans. (c) : 2h/3 from top



Assuming width of Lamina is 'b'.

∴ Centre of pressure for vertical plane surface (θ = 90°)

$$y = \bar{h} + \frac{I_{NA}}{\bar{h}A} = \frac{h}{2} + \frac{\frac{1}{12}bh^3}{\frac{h}{2}bh} = \frac{h}{2} + \frac{1}{6}h = \frac{2}{3}h$$

363. A rectangular plane surface of width 2m and height 3 m is placed vertical in water. What will be the location of centre of pressure of the surface when its upper edge is horizontal and lies 2.5 m below the free surface of water?

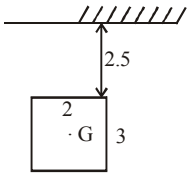
(a) 4.0755 m (b) 4.0125 m
(c) 4.2525 m (d) 4.1875 m

SJVN ET 2019

Ans. (d) Given A = 2 ×3 m^2

$$I_G = \frac{2\times 3^3}{12}$$

$$\bar{h} = 2.5 + \frac{3}{2} = 4$$



$$h^* = \bar{h} + \frac{I_G}{A\bar{h}} = 4 + \frac{\frac{2\times 3^3}{12}}{6\times 4}$$

$$h^* = 4.1875$$

364. The pressure at a point in a liquid, if density of water is 1000 kg/m^3 and vertical height from free surface of fluid is 0.5m.

(a) 6000 N/m^2 (b) 500 N/m^2
(c) 4025 N/m^2 (d) 4905 N/m^2

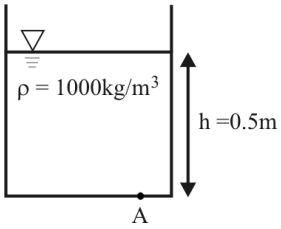
Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (d) : Given, ρ = 1000 kg/m^3

h = 0.5 m

P_A = ?

we know that,



$P_A = \rho gh = 1000 \times 9.81 \times 0.5$

$$\boxed{P_A = 4905 N/m^2}$$

365. Which explanation is true for the mechanical pressure (p_m) and thermodynamic pressure (P_{th})?

(a) A pressure sensor records p_{th}. However for most of the practical engineering cases, the system at the intermediate states of a process gets enough opportunity to attain equilibrium. Thus both P_{th} are equivalent in these cases.

(b) P_{th} is the average mechanical pressure exerted at a point in the fluid but P_m is defined as the force exerted on container walls as fluid molecules coincide with it during their random movement.

(c) P_m is a variable quantity and it depends upon the coordinate system used.

(d) A pressure sensor records P_{th}. However for most of the practical engineering cases, the system at the intermediate states of a process does not get enough opportunity to attain equilibrium. Thus both P_{th} and P_m are equivalent in these cases.

BHEL ET 2019

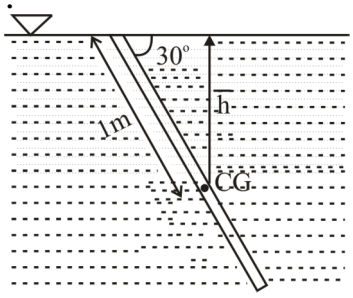
Ans. (a) :

366. A square plate (2m × 2m) lies in water such that its plane makes an angle of 30° with free surface of water and one edge of the plate coincide with the free surface. The total pressure force and centre of pressure is:

(a) 9.81 kN; 0.666 m (b) 19.62 kN; 0.166 m
(c) 9.81 kN; 0.166 m (d) 19.62 kN; 0.566 m
(e) 19.62 kN; 0.666 m

CGPSC AE 25.02.2018

Ans. (e) :



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$\bar{h} = 1 \times \sin 30^0$

$\boxed{\bar{h} = 0.5\text{m}}$

Then total pressure force

$F = \rho \times g \times \bar{h} \times A$

$= 1000 \times 9.81 \times 0.5 \times 2 \times 2 = 19.62$ kN

We know that center of pressure

$$\overset{*}{h} = \frac{I_{CG.} \sin^2\theta}{A\bar{h}} + \bar{h}$$

$$\overset{*}{h} = \frac{\frac{2^4}{12} \times (\sin 30^\circ)^2}{2 \times 2 \times 0.5} + 0.5$$

$$= \frac{\frac{2^4}{12} \times \frac{1}{4}}{2} + 0.5$$

$$\overset{*}{h} = 0.166 + 0.5$$

$$\overset{*}{h} = 0.666 \text{ m}$$

367. A square plate (side =a) lies in the vertical position in the water with its upper horizontal edge coincides with water surface. Total pressure force (F) and position of centre of pressure (h*) on the plane surface will be respectively given by :

[Where, ρ is the density of water]

(a) $F = \frac{\rho g a^3}{3}, h^* = \frac{2}{3}a$ (b) $F = \frac{\rho g a^3}{4}, h^* = \frac{2}{5}a$

(c) $F = \frac{\rho g a^3}{2}, h^* = \frac{2}{5}a$ (d) $F = \frac{\rho g a^3}{2}, h^* = \frac{2}{3}a$

(e) $F = \frac{\rho g a^3}{5}, h^* = \frac{2}{7}a$

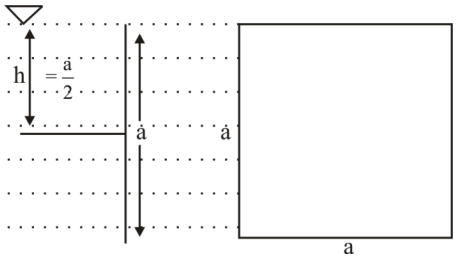
CGPSC AE 25.02.2018

Ans. (d) : Total pressure force–

$F = \rho \times g \times \bar{h} \times A$

$= \rho \times g \times \frac{a}{2} \times a^2$

$F = \frac{1}{2}\rho g a^3$



Position of centre of pressure $(\overset{*}{h})$

$$\overset{*}{h} = \frac{I_{C.G.} \times \sin^2\theta}{\bar{h} \times A} + \bar{h}$$

$$(\overset{*}{h}) = \frac{\frac{a^4}{12} \times 1}{\frac{a}{2} \times a^2} + \frac{a}{2}$$

$$(\overset{*}{h}) = \frac{a}{6} + \frac{a}{2} = \frac{2}{3}a \qquad \left(\theta = 90^\circ, I = \frac{a^4}{12}\right)$$

368. A rectangular plate 30 cm × 50 cm is immersed vertically, in water with its longer side vertical. The total force on one side of the plate is estimated as 17.6 kN. If the plate is turned in the vertical plane at its centre of gravity by 90º and if all other factors remain the same, the total force on one side of the plane would now be

(a) 8.8 kN (b) 15.6 kN

(c) 17.6 kN (d) 19.6 kN

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : The total hydrostatic force on a plane surfaces doesn't depend on the orientation of surfaces so force 17.6 kN acts on the rectangular plate at every orientation.

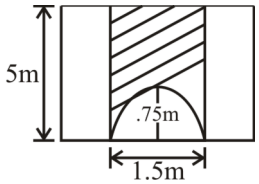
369. A hemispherical shape of diameter 1.5m is provided in the bottom of a tank. If the depth of water above the horizontal floor of tank is 5m. What is the magnitude of resultant force of hemisphere?

(a) 78 kN (b) 74 kN

(c) 76 kN (d) 75 kN

Kerala PSC Poly. Lect., 2017

Ans. (a) :



Horizontal force is balance

Then vertical force (F_v)

$F_v = \rho g V$

$$= \rho g \left[\frac{\pi}{4} D^2 \times 5 - \frac{1}{2}\left(\frac{4}{3}\pi(.75)^3\right)\right]$$

$$= 1000 \times g \left[\frac{\pi}{4}(1.5)^2 \times 5 - \frac{1}{2}\left(\frac{4}{3}\pi(.75)^3\right)\right]$$

$= 78$ kN

370. Hydrostatic force on plane submerged surface

(a) remains the same, no matter how the surface is turned as long as the centroid is unaltered

(b) acts vertically down no matter low the body is inclined

(c) changes with the shape of the area even if the centroid is unaltered

(d) depends upon the weight of the fluid displaced

Karnataka PSC Lect., 27.05.2017

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Ans (a) : Hydrostatic force on a plane submerged surface remains the same, no matter how the surface is turned as long as the centroid is unaltered.

371. In which of the following cases is the total pressure force acting on a submerged plane surface = $\rho.g.\bar{h}.A$?

Here,

ρ = density of fluid,

A = area of surface,

g = acceleration due to gravity,

$\bar{h}$ = distance of CG of area from free surface of fluid.

(a) Vertical plane surface submerged in liquid
(b) Horizontal plane surface submerged in liquid
(c) Both (a) and (b)
(d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (c) : Hydrostatic force (F) = $\rho g \bar{h} A$
It is valid for both vertical or horizontal surface submerged in liquid and doesn't depends on the orientation of plane surfaces.

372. Statement (I): Depth of centre of pressure of any immersed surface is independent of the density of the liquid.
Statement (II): Centre of area of the immersed body lies below the centre of pressure.

ESE 2017

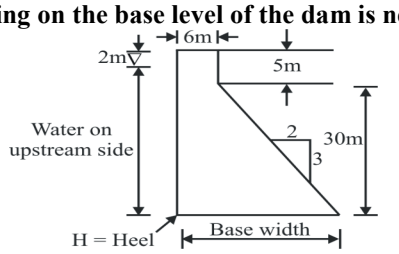
Ans. (c) : (i) The depth of centre of pressure of immersed surface in liquid

$$= \text{depth of centre of area} + \frac{I_{CG}\sin^2\theta}{A\bar{h}}$$

$$h = \bar{h} + \frac{I_{CG}\sin^2\theta}{A\bar{h}}$$

So, centre of pressure is independent of density of liquid.
(ii) Centre of area lies above the centre of pressure.

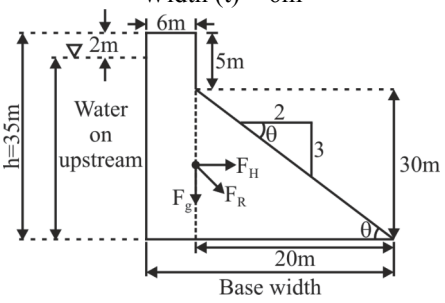
373. A section of a dam made of concrete, ρ = 2.6, total height = 35m, with top walkway width of 6m, is shown. The upstream bottom most point is called the Heel of the dam. The sloped part on downstream side is 3 vertical on 2 horizontal. Water stands till 2 m short of the top of the dam section. The net resultant force acting on the base level of the dam is nearly.



(a) 1370 k kgf (b) 1385 k kgf
(c) 1400 k kgf (d) 1433 k kgf

ESE 2017

Ans. (d) : Given-
Relative density (RD) = 2.6
Total height (h) = 35 m
Width (t) = 6m



Water on upstream = 35 – 2 = 33 m

$$\text{Base width of dam} = 6 + \frac{30}{\tan\theta}$$

$$= 6 + \frac{30}{(3/2)} = 26 \text{ m}$$

$$\text{Submerged volume } (\forall) = (6\times 35\times 1) + \left(\frac{1}{2}\times 20\times 30\right)\times 1$$

$$= 510 \text{ m}^3$$

Horizontal force act on vertical surface

$$F_H = \rho g \bar{h} A$$

$$= 1000\times 9.81\times\left(\frac{33}{2}\right)\times(33\times 1)$$

$$= 5341545 \text{ N}$$

$$= 544500 \text{ kgf}$$

$$= 544.5 \text{ k kgf}$$

Weight of dam (F_g) = mg

$$= (RD)\,\forall g$$

$$= 2.6 \times 1000 \times 510 \times 9.81$$

$$= 1.3008 \times 10^7 \text{ N}$$

$$= 1326000 \text{ kgf}$$

$$= 1326 \text{ k kgf}$$

$$\text{Net resultant force } (F_R) = \sqrt{F_g^2 + F_H^2}, \quad F_g = F_v$$

$$= \sqrt{(1326)^2 + (544.5)^2}$$

$$= 1433 \text{ k kgf}$$

374. The hydrostatic pressure on a plane surface is

(a) wAh (b) $wAh\,Sin^2\theta$
(c) wAh/2 (d) $wAh\,Sin\,\theta$

Where A = area of plane surface and h depth of centroid of the area below the liquid-free surface

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : Hydrostatic pressure on a plane surface is,

$\boxed{F = wAh}$

A = area of plan surface
h = depth of centroid of the plane area below the liquid-free surface.
w = specific weight

375. For a submerged plane in a liquid, the resultant hydrostatic force F on one side of the plane is related to area A, centroidal depth h, depth of the centre of pressure h_{cp} and depth of bottom edge h_b as
(where γ is specific weight)

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(a) γAh_{cp} (b) γAh

(c) γA_b (d) Ah/γ

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Hydrostatic force F on one side of plane area submerged = $\rho ghA = \gamma Ah$

376. When a force acting on the curved surfaces, ... force equals the force on the projected area of the curved surface and acts at the centre of pressure of the projected area.

(a) Vertical

(b) Horizontal

(c) Only 30 degree inclined

(d) None of the above

HPPSC AE (PWD) 10.09.2016

Ans. (b) : When a force acting on the curved surfaces horizontal force equals the force on the projected area of the curved surface and acts at centre of pressure of the projected area.

The vertical component of hydrostatic force on a submerged curved surface is equal to the weight of the liquid volume vertically above the submerged surface to the level of the free surface of liquid and acts through the centre of gravity of the liquid in that volume.

377. Centre of pressure for an inclined immersed surface expressed in terms of depth is given by

(a) $\bar{h} = \frac{I_G \sin\theta}{A\bar{X}} + \bar{X}$ (b) $\bar{h} = \frac{I_G \sin^2\theta}{A\bar{X}} + \bar{X}$

(c) $\bar{h} = \frac{I_G \cos\theta}{A\bar{X}} + \bar{X}$ (d) $\bar{h} = \frac{I_G \cos^2\theta}{A\bar{X}} + \bar{X}$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : Centre of pressure for an inclined immersed surface expressed in terms of depth is given by,

$$\bar{h} = \frac{I_G \sin^2\theta}{A\bar{X}} + \bar{X}$$

$\bar{h}$ = Vertical distance of centre of pressure

$\bar{X}$ = Vertical distance of centre of Gravity

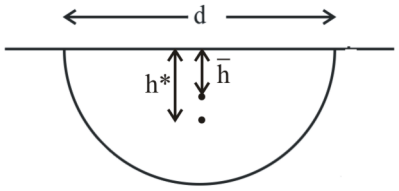
378. For a vertical semi-circular plate, submerged in a homogenous liquid with its diameter 'd' at the free surface, the depth of centre of pressure from the free surface is

(a) $3\pi d/32$ (b) $3\pi d/12$

(c) $3\pi d/16$ (d) $\pi d/32$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Depth of centre of pressure for a vertical semi-circular plate–



$\bar{h}$ = height of centre of gravity

h* = height of centre of pressure

$$h^* = \frac{I_G}{A\bar{h}} + \bar{h} = \frac{I_{base} - A\bar{h}^2}{A\bar{h}} + \bar{h} = \frac{I_{base}}{A\bar{h}}$$

[From parallel axes theorem $I_G + A\bar{h}^2 = I_{base}$]

$$h^* = \frac{I_{base}}{A\bar{h}}$$

$$I_{base} = \frac{1}{2} \times I_{circular} = \frac{\pi d^4}{128}$$

$$A = \frac{\pi}{8} d^2, \ \bar{h} = \frac{2d}{3\pi}$$

$$h^* = \frac{\frac{\pi d^4}{128}}{\frac{\pi}{8} d^2 \times \frac{2d}{3\pi}} = \frac{3\pi d}{32}$$

379. Magnitude of hydrostatic resultant force (F) acting on completely submerged plane surface is:

(a) (Equal to the pressure at centroid of surface) × (Area of surface)

(b) (Lower than the pressure at centroid of surface) × (Area of surface)

(c) (Higher than the pressure at centroid of surface) × (Area of surface)

(d) (Atmospheric pressure) × (Area of surface)

UPRVUNL AE 2016

Ans. (a) : Magnitude of hydrostatic resultant force (F) acting on completely submerged plane surface is equal to the pressure at centroid of surface × Area of surface.

380. A circular annular plate having outer and inner diameters of 1.5 m and 0.5m respectively is immersed in water with its plane making an angle of 60^0 with the horizonal. The centre of the circular annular plate is 2m below the free surface. The hydrostatic thrust on one side of the plate is _____

(a) 31.4 kN

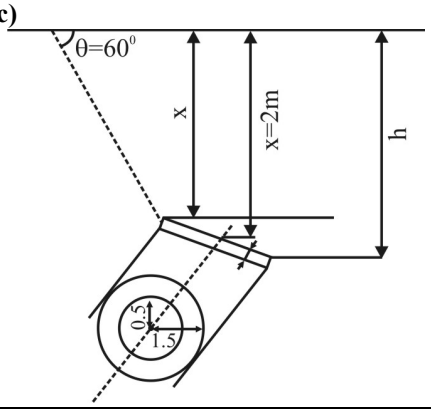
(b) 30.8 N

(c) 30.8 kN

(d) 31.4 N

UPRVUNL AE 21.08.2016

Ans. : (c)



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Area of plate = $\frac{\pi}{4}\left(d_0^2 - d_i^2\right) = \frac{\pi}{4}\left(1.5^2 - 0.5^2\right)$

$= \frac{\pi}{2}\ m^2$

Total hydrostatics force on plate in one side = $\rho g A \bar{x}$

$P = 1000 \times 9.81 \times \frac{\pi}{2} \times 2$

$P = 30.8$ kN

381. For a closed single pair of lock gates having 120° between them when symmetrically placed, the resultant pressure force (P) acting on a gate and the reaction force (F) between the butting edges of the two gates are related as

(a) F = P/2
(b) F = 2P
(c) $F = \sqrt{3}P/2$
(d) F = P

APPSC AEE Mains 2016 (Civil Mechanical)

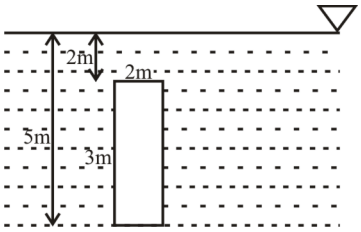
Ans. (d) :

382. A rectangular plate of 2m wide is submerged in water vertically such that its top and bottom surfaces are 2m and 5m respectively below the free surface. The total pressure on plate is

(a) 20601 N
(b) 20601 kN
(c) 206.01 kN
(d) 206.01 N

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) :



Given,

b = 2 m, d = 3 m

$\bar{h} = 2 + \frac{3}{2} = 3.5$m

$A = 2 \times 3 = 6\ m^2$

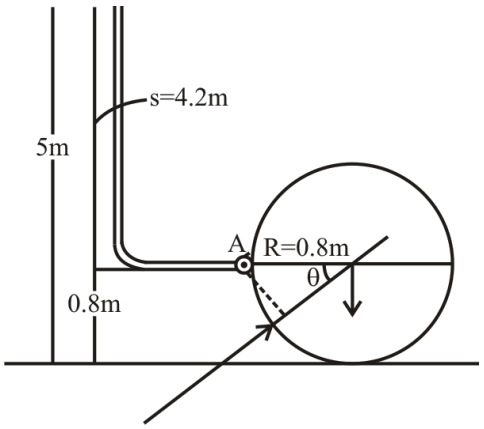
$P = \rho g \bar{h} A$

$= 1000 \times 9.81 \times 3.5 \times 6$

$= 206010$ N

$= 206.01$ kN

383. A long solid cylinder of radius 0.8 m hinged at point A is used as an automatic gate, as shown in Fig. When the water level reaches 5 m, the gate opens by turning about the hinge at point A. Determine the magnitude and direction of the hydrostatic force acting on the cylindrical surface.



(a) $F_R = 52.3$ kN, $\theta = 46.4°$
(b) $F_R = 55.3$ kN, $\theta = 46.4°$
(c) $F_R = 52.3$ kN, $\theta = 36.4°$
(d) $F_R = 55.3$ kN, $\theta = 36.4°$
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : Given,

Density of water = 10^3 kg/m^3

Radius of solid cylinder = 0.8 m

Horizontal component force on the surface,

$F_H = \rho g h_{CG} (A)_{projected}$

$= 10^3 \times 9.81 \times \left(4.2 + \frac{0.8}{2}\right) \times (0.8 \times 1)$

$F_H = 36.1008$ kN

Vertical component on the surface,

$F_V = \rho g \forall$

$= 10^3 \times 9.81 \times \left[(4.2 \times 0.8 \times 1) + \frac{1}{4}(\pi \times 0.8^2 \times 1)\right]$

$F_V = 37.892$ kN

Resultant hydrostatic force acting on the cylinder,

$F_R = \sqrt{F_H^2 + F_V^2}$

$= \sqrt{(36.1008)^2 + (37.892)^2} = 52.33$kN

Direction, $\theta = \tan^{-1}\left(\frac{F_V}{F_H}\right) = \tan^{-1}\left(\frac{37.892}{36.1008}\right)$

$\boxed{\theta = 46.38°}$

384. A circular plate 1.5 m diameter is submerged in water with its greatest and least depths below the surface being 2 m & 0.75 m respectively. What is the approximate total pressure on one face of the plane?

(a) 24 kN
(b) 12 kN
(c) 16 kN
(d) 32 kN

APPSC AE Subordinate Service Civil/Mech. 2016

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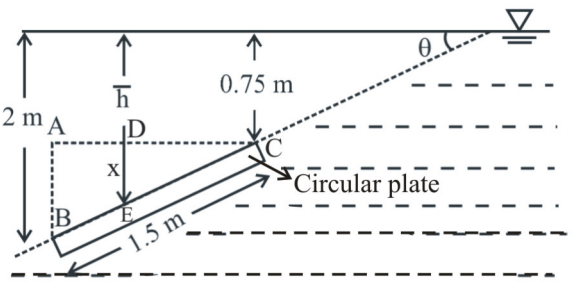
Ans. (a) :



Total pressure on the face of plate,

$$F = \rho g A \bar{h}$$

$\rho = 1000\ kg/m^3$, $g = 9.81\ m^2/s$, $A = \frac{\pi}{4}\times 1.5^2\ m^2$

ΔABC & CDE are similar

$$\frac{AB}{BC} = \frac{DE}{EC}$$

$$\frac{(2-0.75)}{1.5} = \frac{x}{(1.5/2)}$$

$$\frac{1.25}{1.5} = \frac{x}{0.75}$$

$$x = 0.625\ m$$

$$\bar{h} = 0.75 + x = 0.75 + 0.625 = 1.375\ m$$

$$F = 1000 \times 9.81 \times \frac{\pi}{4} \times (1.5)^2 \times 1.375$$

$$F = 23836.59,\ F = 23.84\ kN \simeq 24\ kN$$

385. The position of centre of pressure on a plane surface immersed vertically in a static mass of liquid is given by (where A is the total area of the surface, I_G is the moment of inertia of the area about an axis passing through the centroid of the area and parallel to the free liquid surface, and $\bar{x}$ is the vertical depth of centroid of surface below the free surface of liquid)

(a) $\bar{X} + \frac{I_G}{A\bar{x}}$ (b) $\bar{X} - \frac{I_G}{A\bar{x}}$

(c) $\bar{X} + \frac{A\bar{x}}{I_G}$ (d) $\bar{X} - \frac{A\bar{x}}{I_G}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Total pressure and centre of pressure –

Inclined surface	Vertical surface	Horizontal surface
Total pressure (F) $= \rho g A\bar{h}$	$F = \rho g A\bar{h}$	$F = \rho g A\bar{h}$
Centre of pressure (h*) =	When $\theta = 90^0$	When $\theta = 0^0$
$h^* = \frac{I_G \sin^2\theta}{A\bar{h}} + \bar{h}$	$\sin 90^0 = 1$	$\sin 0 = 0$
	$h^* = \frac{I_G}{A\bar{h}} + \bar{h}$	$h^* = \bar{h}$

386. The depth of centre of pressure in rectangular lamina of 3m deep fully immersed in a liquid having one side in line with free surface is :

(a) 1m (b) 1.5m

(c) 2m (d) 2.5 m

Kerala PSC IOF 19.04.2016

Ans. (c) $h^* = \frac{2h}{3} = \frac{2\times 3}{3} = 2m$

387. The forces being dealt with in fluid statics are:

(a) Gravity forces and surface tension

(b) Gravity forces and viscous forces

(c) Gravity forces and pressure forces

(d) None of these

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) : The forces being dealt with in fluid statics are gravity forces and pressure forces.

388. The centre of pressure of a surface subjected to fluid pressure is the point

(a) On the surface at which resultant pressure acts

(b) On the surface at which gravitational force acts

(c) At which all hydraulic forces meet

(d) Where pressure equivalent to hydraulic thrust will act

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : The center of pressure of surface subjected to fluid pressure is the point on the surface at which resultant pressure act.

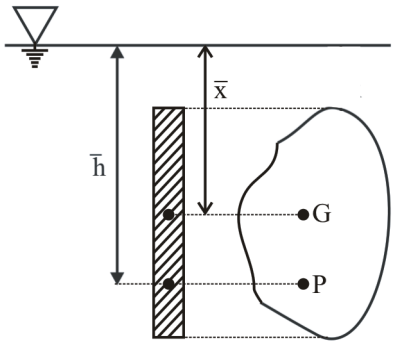
389. The centre of pressure for a vertically immersed surface lies at following distance from c.g.

(a) $\frac{I_G}{Ah_G}$ below (b) $\frac{I_G}{Ah_G}$ above

(c) 0 (d) $\frac{Ah_G}{I_G}$ below

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Vertical Plane Submerged in Liquid–



$$F = \rho g A\bar{x}$$

centre of pressure, $\bar{h} = \bar{x} + \frac{I_G}{A\bar{x}}$

I_G = Area moment of inertia about centroidal axis and parallel to free axis

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A = Area of surface of immersed body

$\bar{x}$ = Vertical distance of centre of gravity (CG) of body from free surface.

$\bar{h}$ = Vertical distance of centre of pressure (CP) from free surface.

390. A rectangular plate surface 2 m wide and 4 m deep lies in vertical plane in water. What will be the pressure and centre of pressure when the upper edge is 2 m below the surface? (Assume g = 10 m/s^2)

(a) 80 kN (b) 120 kN
(c) 160 kN (d) 320 kN

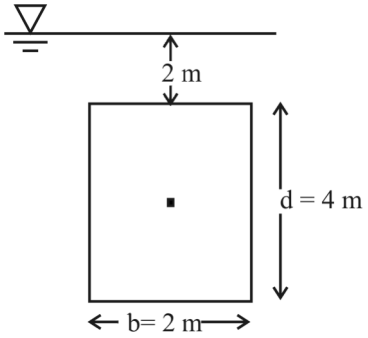
TSGENCO AE 14.11.2015

Ans. (d) : Given,

A rectangular plate,

Wide (b) = 2m, depth (d) = 4 m

Upper edge is 2 m below the surface –



$\bar{h}$ = height from the free surface.

$\bar{h}$ = 2 m + 2 m = 4 m

$F = \rho g A \bar{h}$

$= 1000 \times 10 \times 2 \times 4 \times 4$

$= 320000$ N

$= 320$ kN

391. Match list I with List II and select the correct answer:

	List I		List II
A	Centroid	1	always positive
B	Centre of pressure	2	area moment zero
C	Free surface	3	Constant area
D	Second moment of area	4	constant pressure
		5	resultant force

(a) A-2, B-4, C-5, D-3 (b) A-4, B-2, C-4, D-1
(c) A-2, B-5, C-4, D-1 (d) A-5, B-3, C-2, D-4
(e) A-3, B-4, C-5, D-2

CGPSC AE 26.04.2015 SHIFT-II

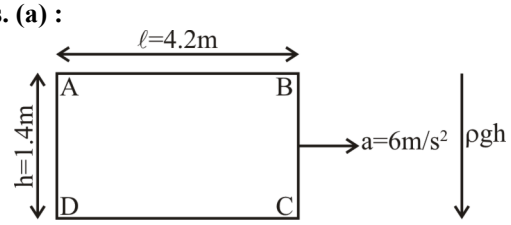
Ans. (c) : A - 2, B - 5, C - 4, D - 1

392. A 1.4 m high, 4.2 m long enclosed tank is filled with water and accelerated horizontally at 6m/s^2. If the top of the tank has a small slit across the front, the maximum pressure in the tank is nearly

(a) 38.9 kPa (b) 45.8 kPa
(c) 59.7 kPa (d) 66.7 kPa
(e) 71.4 kPa

CGPSC AE 26.04.2015 SHIFT-II

Ans. (a) :



$P_B = P_{atm}$ (because of small opening)

$P_A = P_B + \rho a \ell = \rho a \ell$

$P_C = P_B + \rho g h = \rho g h$

$P_D = P_A + \rho g h = \rho a \ell + \rho g h$

Maximum pressure in tank is at point D,

$P_D = \rho a \ell + \rho g h$

$P_D = 1000 \times 6 \times 4.2 + 1000 \times 9.81 \times 1.4$

$P_D = 38.9$ kPa

393. A vertical rectangular plane surface is submerged in water such that its top and bottom surfaces are 1.5 m and 6.0 m respectively below the free surface. The position of centre of pressure below the free surface will be at a distance of

(a) 3.8 (b) 4.0m
(c) 4.2m (d) 4.5m
(e) 4.8m

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) :

$$\bar{h} = 1.5 + \left[\frac{6-1.5}{2}\right] = 3.75 m$$

Then position of center of pressure.

$$\overset{*}{h} = \frac{I_{CG} \times \sin^2\theta}{A \times \bar{h}} + \bar{h}$$

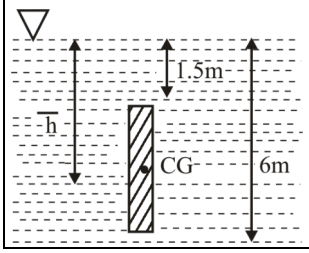
width of plate W = 1 m

$$I_{C.G.} = \frac{1 \times 4.5^3}{12} m^4, \theta = 90^0$$

$A = 4.5 \times 1$ m^2

$$\overset{*}{h} = \frac{\left[\frac{1 \times 4.5^3}{12}\right][\sin 90^\circ]^2}{4.5 \times 1 \times 3.75} + 3.75 = 0.45 + 3.75$$

$$\boxed{\overset{*}{h} = 4.2 \text{m}}$$



394. Which of the following is a measure of the pressure exerted by a state of stress?

(a) Deviatoric stress (b) Hydrostatic stress
(c) Von Mises stress (d) Cauchy stress

HPPSC Asstt. Prof. 2014

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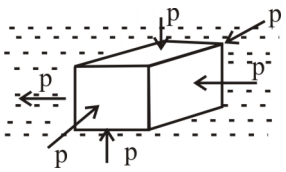
Ans. (b)



Pressure is a negative normal stress. According to Pascal's law, pressure at any point in a static fluid have equal magnitude in all direction.

Condition of Hydrostatic- $(\sigma_x = \sigma_y = \sigma_z)$

$$\Rightarrow -p_x = -p_y = -p_z$$

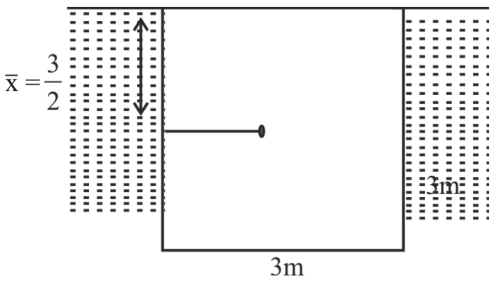
$$\Rightarrow -p = -p = -p$$

395. A square surface 3m × 3 m lies in a vertical line in water with its upper edge at water surface. The hydrostatic force on square surface is :

(a) 28,000 kg
(b) 13,500 kg
(c) 17,000 kg
(d) 21,350 kg

UPRVUNL AE 2014

Ans : (b)



Given

Pressure force $F = \rho g A \bar{x}$

$$= 1000 \times 10 \times 3 \times 3 \times \frac{3}{2}$$

F = 135000 Newton

or

$$F = \frac{135000}{10}$$

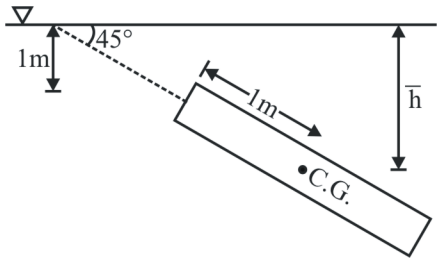
$$\boxed{F = 13500\,kg}$$

396. An inclined plate 2 m long and 1 m wide lies with its length inclined at 45° to the surface of water and its nearest edge is 1m below it. If specific weight of water is 1000kg/m³, then total pressure force on the plate (in kg) is approximately

(a) 2150
(b) 2510
(c) 3220
(d) 3420

ISRO Scientist/Engineer 24.05.2014

Ans. (d) :



Given,

$\rho = 1000\ kg/m^3$

$A = 2 \times 1 = 2m^2$

$\bar{h} = (1 + 1 \times \sin\theta)$

$= (1 + 1 \times \sin 45°)$

$F = \rho A \bar{h}$

$= 1000 \times 2 \times (1 + 1 \sin 45°)$

$F = 3414\,kg$

397. A container carrying water is moved in a horizontal direction with an acceleration of 2.45 m/s². The angle of inclination of free water surface to the horizontal is

(a) 14.03° (b) 67.8°
(c) 45° (d) 0°

UKPSC AE 2012 Paper–II

Ans. (a) : $\tan\theta = \frac{a}{g} = \frac{2.45}{9.81}$

$\theta = 14.03°$

398. A box of rectangular base 2m × 3m contains gasoline (Sp. Gravity 0.7) upto a height of 5m. The force on the base and two vertical surfaces, 2m × 5m and 3m × 5m respectively.

(a) 206 kN, 258 kN and 172 kN
(b) 21 kN, 17.5 kN and 26.3 kN
(c) 258 kN, 172 kN and 206 kN
(d) 206 kN, 172 kN and 258 kN

UKPSC AE 2012 Paper–II

Ans. (d) : Force on base (2h × 3m),

$F_b = 0.7 \times 9.81 \times (2 \times 3) \times 5 = 206$ kN

Force on vertical faces,

(2m × 5m),

$$F = 0.7 \times 9.81 \times (2 \times 5) \times \frac{5}{2} = 171.6 \simeq 172\ kN$$

(3m × 5m),

$$F = 0.7 \times 9.81 \times (3 \times 5) \times \frac{5}{2} = 257.51 \simeq 258\ kN$$

399. The ratio of pressures between two points X and Y located respectively at depths of 0.5 m and 8 m below water level in a tank is

(a) $1:\sqrt{2}$ (b) 1 : 2
(c) 1 : 8 (d) 1 : 16

APPSC AEE 2012

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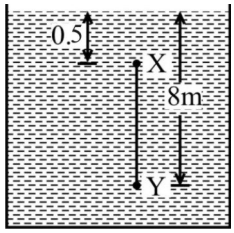
Ans : (d)



Pressure at a point X

$P_x = \rho g h_1$

$P_x = \rho g \times 0.5$

$P_x = 0.5\rho g$ ..... (i)

Pressure at a point Y

$P_y = \rho g h_2$

$P_y = 8\rho g$ ..... (ii)

Ratio of pressure between x and y.

$\frac{P_x}{P_y} = 1:16$

400. The hydrostatic pressure in kgf exerted on one side of an annular area enclosed by concentric circles of radii 2 m and 1 m, and having its centroid 4 m below water surface is

(a) 10000π (b) 11000π

(c) 12000π (d) 24000π

APPSC AEE 2012

Ans : (c) Area = $\frac{\pi}{4}\left(4^2 - 2^2\right)$

Area = $\frac{12\pi}{4} m^2$

$\bar{x} = 4m$

hydrostatics pressure (P) = $\rho g A \bar{x}$

$= 1000 \times \frac{12\pi}{4} \times 4 \text{ kgf}$

Hydrostatic pressure = $12000\ \pi$ kgf

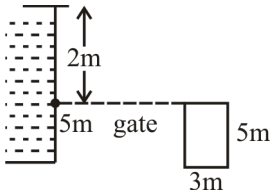
401. A water tank has a gate in its vertical wall 5 m high and 3 m wide. The top edge of the gate is 2 m below the water surface. The hydrostatic force on the gate surface is

(a) 147 kN (b) 367 kN

(c) 490 kN (d) 662 kN

UPSC JWM Advt. No.-50/2010

Ans. (d) :



Given, $\bar{x} = 2 + 2.5 = 4.5\,m$

$F = wA\bar{x} = 9810 \times 3 \times 5 \times 4.5$ [$\because w = \rho g$]

F = 662 kN

402. Hydrostatic pressure is

(a) a scalar quantity

(b) a vector quantity

(c) a second order tensor

(d) both (a) and (c) above

WBPSC AE, 2007

Ans. (a) : Pressure is a zero order tensor or scalar quantity.

403. The horizontal component of force on a curved surface is equal to the-

(a) Product of pressure at its centroid and area

(b) Weight of liquid retained by the curved area

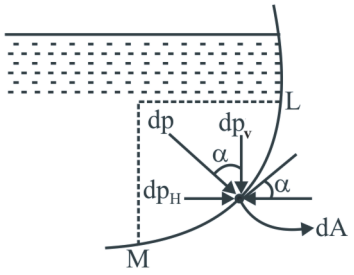
(c) Force on a vertical projection of the curved surface

(d) Weight of liquid vertically above the curved surface

ISRO Scientist/Engineer 2006

Ans. (c)

- The horizontal component of force on a curved surface is equal to the force on a vertical projection of the curved surface.
- Horizontal component P_H = Total pressure force On the projected area of the curved surface on vertical plane.



- Vertical component P_V = Weight of the liquid supported by the curved surface upto free surface of liquid.

4. Buoyancy and Floatation

404. A floating body is in stable equilibrium when

(a) the metacentric height is zero.

(b) the metacentre is below its centre of gravity

(c) centre of gravity is below the centre of buoyancy

(d) the metacentre is above its centre of gravity

Assam Engg. College AP/Lect. 18.01.2021
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021
VIZAG Steel MT 24.01.2021 Shift-II, Shift-I
TSPSC Manager (Engg.) HMWSSB 12.11.2020
HPPSC ADF 05.03.2019, GATE 2017, 2010
GPSC Lect. (Auto) 16.10.2016
Nagaland PSC (CTSE) 2016, 2015 Paper-II
KPSC AE. 2015, Kerala PSC AE 06.08.2015
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
ISRO Scientist/Engineer 12.05.2013
TRB Asstt. Prof., 2012, ESE 2010, 2005, 1994
WBPSC AE 2003

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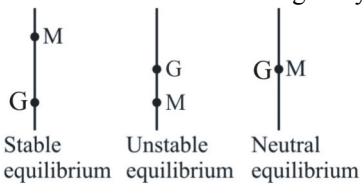
Ans. (d) : Equilibrium of floating body-

1. Stable – Metacentre is above centre of gravity.
2. Unstable – Metacentre is below the centre of gravity.
3. Neutral – Metacentre coincides with centre of gravity.

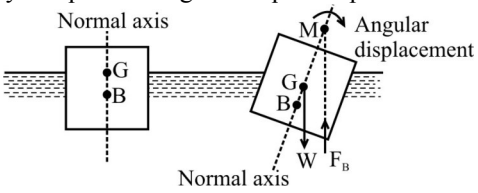
A floating body will remain in stable equilibrium the metacentre M lies above the Centre of gravity .



M = Metacentre

G = Centre of gravity

The stability of floating body is determined from position of metacentre (M). In floating body the weight of body is equal to weight of liquid displaced.



■ Metacentre is above centre of gravity (CG).

405. For the stability of a floating body, under the influence of gravity alone, which of the following is true?

(a) Metacentre should be below the centre of gravity

(b) Metacentre should be above the centre of gravity

(c) Metacentre and centre of gravity must lie on the same horizontal line

(d) Metacentre and centre of gravity must lie on the same vertical line

PPSC Asstt. Municipal Engineer 15.06.2021
Gujarat PSC AE 2019
OPSC AEE 2019 PAPER - II
Nagaland PSC CTSE 2017 Paper-2
UPRVUNL AE 2016
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2010

Ans : (b) : For the stability of a floating body metacentre should be above the centre of gravity.

■ Stable equilibrium of a floating body metacentre should be above centre of gravity.

■ Unstable equilibrium of a floating body metacentre should be below centre of gravity.

■ Neutral equilibrium of a floating body metacentre and centre of gravity must coincides.

406. Unstable equilibrium of a floating body implies

(a) Metacentre M is above centre of gravity G

(b) Metacentre M is below centre of gravity G

(c) Metacentre M coincides with centre of gravity G

(d) None of these

APPSC AEE SCREENING 17.02.2019
Nagaland PSC CTSE 2017 Paper-2
Karnataka PSC AE, 10.09.2017
APPSC AEE Screening Test 2016
ISRO Scientist/Engineer 24.05.2014
TNPSC AE 2013

Ans. (b) : <u>For "Floating body"-</u>

Metacentre M is above centre of gravity G
(M > G) "Stable Equilibrium"
Metacentre M is below centre of gravity G
(M<G) "Unstable Equilibrium"
Metacentre M coincides with centre of gravity
(M = G) "Neutral"

407. The metacentric height of a floating body is

(a) the distance between metacentre and centre of buoyancy

(b) the distance between the centre of buoyancy and centre of gravity

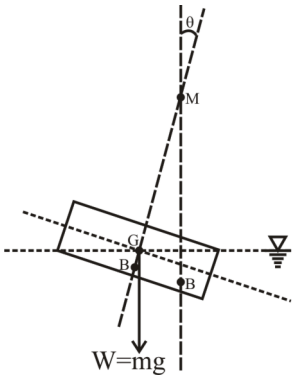
(c) the distance between metacentre and centre of gravity

(d) none of these

Nagaland PSC (CTSE) 2018, 2017, 2016 Paper-II
GPSC Lect. 23.10.2016
TNPSC AE, 2008, ESE 2009

Ans. (c) : Metacentre and Metacentric Height–

Metacentric Height (GM)–The distance between the center of gravity (G) of floating body and the metacenter (M) is called metacentric height (i.e distance GM).



Metacentric height (GM) = BM – BG

408. The buoyancy depends on :

(a) viscosity of the liquid

(b) pressure of the liquid displaced

(c) depth of immersion

(d) mass of liquid displaced

OPSC AEE 2019 PAPER - II
GWSSB AAE, 27.12.2015
SJVN ET 2013
ISRO Scientist/Engineer 2007

Ans. (d) : Mass of liquid displaced– According to Archimedes principle, the buoyant force of a partially immersed body is equal to the weight of the displaced liquid.

Where, $F_b = \rho_f V g$

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F_b = Buoyancy force (N)
ρ_f = Density of liquid (kg/m^3)
V = Displaced volume of liquid (m^3)
$\rho_f \times V$ = mass of liquid displaced
g = Standard gravity = 9.81 m/s^2

409. For a floating body, buoyant force acts at the
(a) Centroid of the floating body
(b) Center of gravity of the body
(c) Centroid of the fluid vertically below the body
(d) Centroid of the displaced fluid

GPSC Asstt. Director of Transport 05.03.2017
GATE 2016, JPSC AE 2013 Paper-V
CSE Pre-1995

Ans. (d) : For a floating body, buoyant force acts at the centroid of the displaced fluid.

410. The Centre of buoyancy is:
(a) Centroid of displaced volume of fluid
(b) A point slightly above meta-centre
(c) Centre of gravity of the body
(d) All of the above

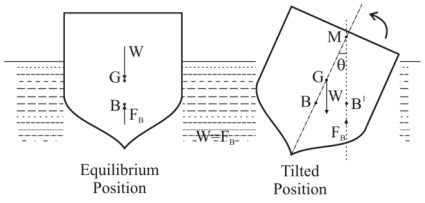
OPSC AEE 2015 PAPER - II
CGPSC AE 26.04.2015 SHIFT-II
ESE 2010, CSE Pre-2009

Ans : (a) The centre of buoyancy is centroid of displaced volume of fluid.
Centre of buoyancy:- It is defined as the point, through which the force of buoyancy is supposed to act. As the force of buoyancy is a vertical force and is equal to the weight of the fluid displaced by the body. The Centre of buoyancy will be the centre of gravity of the fluid displaced.

411. If a body is in stable equilibrium the metacentric height should be
(a) zero
(b) positive
(c) negative
(d) depends on the fluid

Assam Engg. College AP/Lect. 18.01.2021
TNPSC AE 2017
Mizoram PSC AE/SDO 2014, Paper-I
ESE 2011

Ans. (b) :



If,
GM > 0 (M is above G) → Stable equilibrium
GM = 0 (M coinciding with G) →Neutral equilibrium
GM < 0 (M is below G) → Unstable equilibrium

412. A piece of wood of volume V and specific gravity 0.87 floats on the surface of a liquid of specific gravity 1.31. The portion of the wood which is submerged in the liquid will be
(a) 0.335 V (b) 0.665 V
(c) 0.87 V (d) 0.13 V

UPPSC AE 13.12.2020, Paper-2
GPSC EE Pre, 28.01.2017
ESE 2012

Ans. (b) : Given,
Specific gravity of wood = 0.87
Specific gravity of liquid = 1.31
wt. of body = wt. of vol. displaced
$\rho_b g V_b = \rho_f g V_{displaced}$

$$V_{displaced} = \frac{0.87}{1.31} V_b$$

$$\boxed{V_{displaced} = 0.6641 V_b}$$

The portion of wood which is submerged in the liquid will be 0.6641 V

413. Resultant pressure of the liquid in case of an immersed body acts through which one of the following?
(a) Centre of gravity (b) Centre of pressure
(c) Metacentre (d) Centre of buoyancy

JPSC AE, 10.04.2021, Paper-I
APPSC AEE SCREENING 17.02.2019
ESE 2007

Ans. (b) : Centre of Pressure– It is defined as the point of application of the total hydrostatic pressure on the surface. Resultant pressure of the liquid in case of an immersed body acts through centre of pressure.

414. The following terms relate to floating bodies : Centre of gravity-G. Metacentre-M, weight of the floating body – W. Buoyancy force-F_n. Match List-I and List-II and select the correct answer :

List-I (Condition)	List-II (Result)
A. G is above M	1. Stable equilibrium
B. G and M coincide	2. Unstable equilibrium
C. G is below M	3. Floating body
D. $F_B > W$	4. Neutral equilibrium

Code :

	A	B	C	D
(a)	1	3	2	4
(b)	3	1	4	2
(c)	2	3	4	1
(d)	2	4	1	3

(e) None of the above

CGPSC Poly. Lect. 22.05.2016
ISRO Scientist/Engineer 2007
ESE 1995

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Ans. (d)

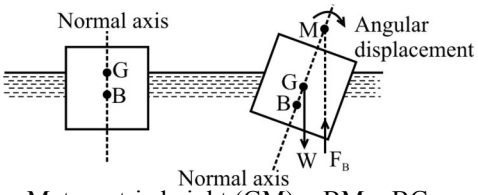
G is above M	–	Unstable equilibrium
G and M coincide	–	Neutral equilibrium
G is below M	–	Stable equilibrium
$F_B > W$	–	Floating body

415. When a body floating in a liquid is given a small angular displacement, it stands oscillating about a point known as

(a) centre of pressure (b) centre of gravity
(c) centre of buoyancy (d) metacentre

ISRO Scientist/Engineer 11.10.2015
APPSC AEE 2012
J&K PSC Screening, 2006

Ans : (d) When a body floating in a liquid is given a small angular displacement, it stands oscillating about a point known as metacentre.



Metacentric height (GM) = BM – BG

* Metacentric height for rolling condition will be less than metacentric height for pitching condition.

416. For a body completely submerged in a fluid, the centre of gravity (G) and centre of buoyancy (B) are known. The body is said to be in stable equilibrium, if

(a) B does not coincides with centre of mass of the displaced fluid
(b) G coincides with centre of mass of the displaced fluid
(c) B lies above G
(d) B lies below G

APPSC AEE Mains 2016 (Civil Mechanical)
GATE 2014, ESE 2011

Ans. (c) : A body in a fluid is said to be stable when given small displacement, it returns to its original position.
Stability of completely submerged Bodies.
The center of gravity 'G' is below the center of Buoyancy B.

417. The buoyancy depends upon the

(a) Weight of the liquid displaced
(b) Pressure with which the liquid is displaced
(c) Viscosity of the liquid
(d) Compressibility of the liquid

Nagaland PSC (CTSE) 2018, Paper-II
KPCL AE 2016, VIZAG STEEL MT 2011

Ans. (a) : The buoyancy depends on the weight of the liquid displaced. According to Archimedes principle, the bouyant force of a partially immersed body is equal to the weight of the displaced liquid.
Therefore the buoyant force depends upon the density of the fluid and the submerged volume of the body.

418. A body floating in a liquid is said to be in neutral equilibrium, if its metacenter

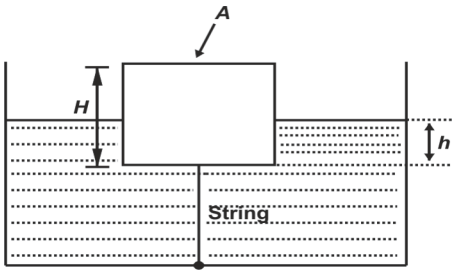
(a) Coincides with its center of gravity
(b) Lies above its center of gravity
(c) Lies below its center of gravity
(d) Lies between the center of buoyancy and center of gravity

Assam PSC AE (IWT) 14.03.2021
Nagaland PSC (CTSE) 2018, Paper-II
APPSC AEE 2012

Ans. (a) : A body floating in a liquid is said to be in neutral equilibrium, if its metacentre coincides with its centre of gravity.

Equilibrium	Submerged body	Floating body
Stable	Centre of gravity below center of buoyancy	Centre of gravity below metacentre
Unstable	Centre of gravity above centre of buoyancy	Centre of gravity above metacentre.
Neutral	Centre of gravity coincides with centre of buoyancy	Centre of gravity lies at metacentre

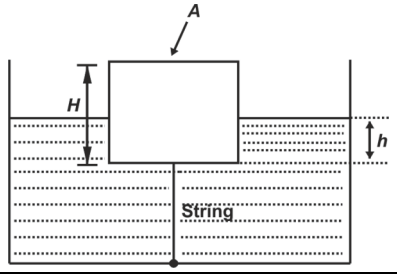
419. A cylindrical body of cross-sectional area A height H and density ρ_s, is immersed to a depth h in a liquid of density ρ, and tied to the bottom with a string. The tension in the string is



(a) ρghA (b) $(\rho h - \rho_s H)gA$
(c) $(\rho - \rho_s) ghA$ (d) $(\rho_s - \rho) ghA$

UPRVUNL AE 2014
ISRO Scientist/Engineer 2011, GATE 2003

Ans. (b) : Cross-section area = A
Height = H
Immersed depth = h
Density of liquid = ρ
Density of solid body = ρ_s
Tension of string = T



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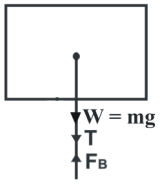
FBD →"Body Immersed"



$T + W = F_B$

$T = F_B - W$

$= (\rho gAh - \rho_s gHA)$

$T = gA(\rho h - \rho_s H)$

420. A piece weighing 3 kg in air was found to weigh 2.5 kg when submerged in water. Its specific gravity is-

(a) 1 (b) 5

(c) 6 (d) 7

RPSC IOF, 2020

ISRO Scientist/Engineer 2006, WBPSC AE 2003

Ans. (c) : weight of piece in air

$\rho_P \times V \times g = 3 \times g$ …(i)

Loss in weight of piece in water

= weight of water displaced.

$\rho_W \times V \times g = (3 - 2.5) \times g$

$\rho_W \times V \times g = 0.5 \times g$ …(ii)

Equation (i) divide by (ii)

$$SG = \frac{\rho_P}{\rho_W} = 6$$

421. A body weight is 392.4 N in air and 196.2 N in water, the volume of body in (m^3) is:

(a) 0.05 (b) 0.10

(c) 0.02 (d) 0.20

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

OPSC AEE 2019 PAPER - II

Ans. (c) : weight of body in air (W) = 392.4N

weight of body in water (N) = 196.2 N

loss of weight = buoyancy force

$392.4 - 196.2 = 1000 \times V \times g$

$V = 0.02 m^3$

422. Stability of freely falling object is assured if its centre of

(a) Buoyancy lies below its centre of gravity

(b) Gravity coincides with its centre of gravity

(c) Gravity lies below its metacentre

(d) Buoyancy lies below its metacentre

RPSC Lect. (Tech. Edu. Dept.) 2011

ESE 1999

Ans. (c) : Stability of freely falling object is assured if its centre of gravity lies below its metacentre.

423. A floating body has the centre of buoyancy at B, centre of gravity at G and metacentre at M. For stable equilibrium of the body

(a) MG = 0 (b) M is below G

(c) BG = 0 (d) M is above G

Assam PSC CCE Pre 2015, ESE 2007

Ans. (d)

(i) Stable equilibrium
- M above G
- BM > BG
- GM (+ve)

(ii) Unstable equilibrium
- M below G
- BM < BG
- GM (–ve)

(iii) Neutral equilibrium
- M at G
- GM = 0

424. The metacentric height (GM) is expressed as

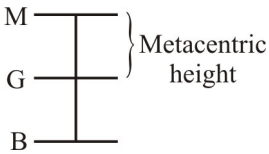
(a) GM = BG – (I/V) (b) GM = (V/I) – BG

(c) GM = (I/V) – BG (d) GM = BG – (V/I)

Assam PSC AE (PHED), 18.10.2020

ESE 2008

Ans. (c) :



Metacentric height

$GM = BM - BG$

$BM = \frac{I}{V}$

Where I = area moment of inertia at top view about longitudinal axis

V = volume of the fluid displaced by body

Hence,

$GM = \frac{I}{V} - BG$

425. The centre of gravity of the volume of the liquid displaced is called

(a) centre of pressure (b) centre of buoyancy

(c) metacentre (d) none of these

Sikkim PSC (Under Secretary), 2017

J & K PSC Screening, 2006

Ans. (b) : Centre of Buoyancy- Centre of buoyancy is the point through which buoyant force act. It is centroid of liquid displaced by body.

Metacentre- Metacentre is defined as the point about which a body starts oscillating when the body is tilted by small angle.

Centre of pressure is defined as point of application of total pressure on the surface.

426. A metallic piece weighs 80 N in air and 60 N in water. The relative density of the metallic piece is about

(a) 8 (b) 6

(c) 4 (d) 2

JPSC AE, 10.04.2021, Paper-I

CSE Pre-2002

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Ans. (c) : Weight in air (W_a) = 80 N
Weight in water (W_w) = 60 N
Relative density of metallic piece,

$$= \frac{W_a}{W_a - W_w} = \frac{80}{80-60} = 4$$

427. A piece of metal of specific gravity 7 floats in mercury of specific gravity 13.6. What fraction of its volume is under mercury?
(a) 0.5 (b) 0.4
(c) 0.515 (d) 0.6

TRB Asstt. Prof., 2012, ESE 2013

Ans. (c) : Principal of Flotation– A body will float in a liquid when weight of the body is equal to this force of buoyancy.

$\rho_m \times g \times V = \rho_{Hg} \times g \times V_{Hg}$, where V=Total volume
V_{Hg} = volume under mercury

$$\frac{V_{Hg}}{V} = \frac{\rho_m}{\rho_{Hg}} = \frac{7}{13.6} = 0.515$$

428. A piece of wood having a weight of 5 kg floats in water with 60% of its volume under the liquid. The specific gravity of wood is
(a) 0.83 (b) 0.6
(c) 0.4 (d) 0.3

SJVN ET 2013, TRB Asstt. Prof., 2012

Ans. (b) : $S_b \times g \times V = S_w \times g \times V_w$
V_w = Volume of wood under water
$V_w = 0.6\ V$
V = Total volume of wood.
$S_b \times g \times V = 1000 \times g \times 0.6V$
$S_b = 600$
Specific Gravity of wood = 0.6

429. As a ship enters into a river from sea, one can expect that :
(a) It rises a little
(b) It sinks a little
(c) It remains at the same level
(d) It's level depends on the material used for construction

KPSC ADF 2015, ESE 2011

Ans. (b) : As a ship enters into a river from sea, it will sink a little because density of sea water is high due to uniform solution of salt than river. So the displaced volume of liquid or sink volume of ship will be more in river than sea to balance weight of ship by buoyancy force.

430. The least radius of gyration of a ship is 9 m and the metacentric height is 750 mm. The time period of oscillation of the ship is
(a) 42.41 s (b) 75.4 s
(c) 20.85 s (d) 85 s

Nagaland PSC CTSE 2015 Paper-II
ESE 1999

Ans. (c) : Given,
Radius of gyration (K) = 9 m
Metacentric height (h) = 750 mm = 0.750 m

$$T = 2\pi\sqrt{\frac{K^2}{gh}}$$

$$T = 2\pi\sqrt{\frac{9^2}{0.750 \times 9.81}}$$

T = 20.85 sec

431. When a body is immersed wholly or partially in a liquid, it is lifted up by a force equal to the weight of liquid displaced by the body. This statement is called
(a) Pascal's law
(b) Archimedes's principle
(c) Principle of flotation
(d) Bernoulli's theorem

GPSC Lect. 23.10.2016
VIZAG STEEL MT 2011

Ans. (b) : Archimedes's principle states that when a body is immersed fully or partially in a liquid it is lifted up by a force equal to the weight of liquid displaced by the body.

432. A small plastic boat loaded with pieces of steel rods is floating in a bath tub. If the cargo is dumped into the water allowing the boat to float empty, the water level in the tub will
(a) Rise
(b) Remains same
(c) Fall
(d) Cannot be estimated from the information

ISRO Scientist/Engineer 12.05.2013, 2011, 2010

Ans. (c) : By using Archimedes Principal, the volume of liquid displaced by boat and rods is greater than the volume of liquid displaced by empty boat and cargo dumped in water. Therefore, water level of tub will fall.

433. A buoyant force is :
(a) Equal to volume of liquid displaced
(b) Force necessary to maintain equilibrium of a submerged body
(c) The resultant force acting on floating body
(d) The resultant force on a body due to the fluid surrounding it

OPSC Civil Services Pre. 2011, ESE 2011

Ans. (d) : The resultant force on a body due to the fluid surrounding it.
Buoyant force (F_B) = $\rho_f . g \times V_{fd}$

434. The necessary and sufficient condition for bodies in flotation to be in stable equilibrium is that the centre of gravity is located below the
(a) metacentre (b) centre of buoyancy
(c) epicenter (d) centroid

Oil India Ltd. Sr. Engg. (Production) 30.11.2019
ESE 2018

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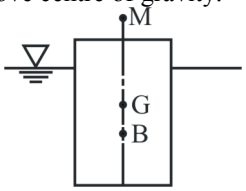
Ans. (a) : For stable equilibrium of floating bodies, metacentre is above centre of gravity.



Note : The distance between the centre of gravity G and metacentre M of a floating body (ie GM) is known as metacentric height.
Higher the metacentric height, higher will be the stability.

435. If a submerged body is in unstable equilibrium, then
(a) the centre of buoyancy is below the centre of gravity.
(b) the centre of buoyancy is above the centre of gravity.
(c) meta-centre is below the centre of buoyancy.
(d) meta-centre is above the centre of buoyancy.

ESE 18.07.2021
Oil India Senior Officer 23.12.2020

Ans. (a) : For unstable equilibrium of submerged body centre of gravity should be above centre of Buoyancy.
When the centre of buoyancy is below the centre of gravity then, the submerged body is in unstable equilibrium.

436. For a submerged body, if the centre, of buoyancy coincides with the centre of gravity, the equilibrium is called :
(a) stable (b) unstable
(c) neutral (d) None of the above

APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (c) : For a submerged body, if the centre of buoyancy coincides with the centre of gravity, then the body is in the neutral equilibrium.

437. A bucket of water hangs with a spring balance. If an iron piece is suspended into water from another support without touching the sides of the bucket the spring balance will show
(a) A decreased reading
(b) An increased reading
(c) Increased or decreased reading depending on the depth of immersion
(d) No change in reading

UPRVUNL AE 21.08.2016, ESE 2012, 1999

Ans. : (d) A bucket of water hangs with a spring balance. If an iron piece is suspended into water from another support without touching the sides of the bucket, the spring balance will show no change in reading.

438. Forces acting on a solid submerged body in a fluid and on a fluid body of the same shape at the same depth are always_____.
(a) Negative (b) identical
(c) zero (d) different

UPRVUNL AE 05.07.2021

Ans. (b) :

439. A uniform body 3m long, 2m wide and 1m deep floats in water. If the depth of immersion is 0.6 m then the weight of the body is :
(a) 3.53 kN (b) 33.53 kN
(c) 35.3 kN (d) 3.53 kN

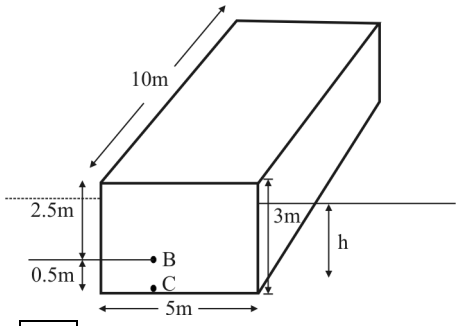
RPSC ACF & FRO, 26.02.2021

Ans. (c) : Archimedes Floating Principle
weight of body (W) = weight of fluid displaced by the body
$= \rho_{water} \times V_{submerged} \times g$
$= 1000 \times [3 \times 2 \times 0.6] \times 9.81$
$= 35316$ N
$\boxed{W = 35.316 \text{ kN}}$

440. A rectangular block of width = 5m, height = 3m and length = 10m (into the plane of the paper) is floating in a liquid of density 1500 kg/m^3. If the centre of buoyancy is located at a vertical distance of 2.5 metres from the top edge of the block, then the density of the material making up the block is :
(a) 1500 kg/m^3 (b) 500 kg/m^3
(c) 750 kg/m^3 (d) 1700 kg/m^3

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (b) : h = depth of immersion.
CB = h/2
h = 2 × CB
= 2 × 0.5



$\boxed{h = 1}$

Weight of body = Buoyancy force
$\rho_b \times V_b \times g = \rho_f \times \forall \times g$
($\forall$ =submerged volume)
$\rho_b \times (5 \times 3 \times 10) \times 9.81 = 1500 \times (1 \times 5 \times 10) \times 9.81$
$\boxed{\rho_b = 500 \text{ kg/m}^3}$

441. How much of concrete with $\gamma = 25$ kN/m^3 must be attached to a beam having a volume of 0.1 m^3 and specific gravity 0.6 to cause both to sink in water ? (Take g = 9.81 m/s^2)
(a) 0.825 kN (b) 0.745 kN
(c) 0.525 kN (d) 0.645 kN

ESE 18.07.2021

Ans. (d) : For Sinking of both beam & the attached concrete, the weight should be greater than or equal to the Buoyant force.
Weight = $[V_{conc} \times 25 + 0.1 \times (0.6 \times 9.81)]$ kN

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Buoyant force = $(V_{conc} + 0.1) \times 9.81$ kN

Hence,

$(V_{conc} \times 25) + (0.1 \times 0.6 \times 9.81) \geq V_{conc} \times 9.81 + 0.981$

$\Rightarrow V_{conc} \geq 0.02583\ m^3$

$\Rightarrow$ Wt of concrete $\geq 0.02583 \times 25 = 0.6458$ kN

$\Rightarrow (\text{Wt of concrete to be attached})_{min} = 0.6458$kN

442. What is the depth of immersion in water of the wooden block (density 650 kg/m^3) of 2.5 m breadth and 1.5m height and 6m long?

(a) 0.557 m (b) 0.256m

(c) 0.975 m (d) 0.776 m

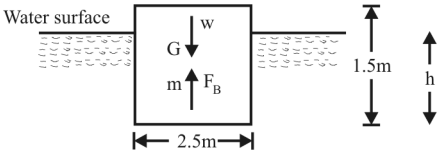
NLC GET 17.11.2020, Shift-II

Ans. (c) : Given,

Breadth = 2.5 m

Height = 1.5 m

Length = 6m



Volume of block(V) = $2.5 \times 1.5 \times 6 = 22.50 m^3$

Density of wood(ρ) = 650 kg/m^3

Weight of the block(W) = $\rho g V$

$= 650 \times 9.81 \times 22.50$

$= 143471$N

For equilibrium

Weight of water displaced(W_{fd}) = Weight of wooden block(W)

$= 14347$ N

Volume of water displaced(V_{fd}) = Weight of water displace/weight density of water

$$= \frac{143471}{1000 \times 9.81} = 14.625 m^3$$

Position of centre of Buoyance $\Rightarrow$

Volume of wooden block in water = Volume of water displaced

$2.5 \times h \times 6 = 14.625 m^3$

$$h = \frac{14.625}{2.5 \times 6}$$

$$\boxed{h = 0.975 m}$$

443. An object weighing 100 N in air was found to weight 75 N when fully submerged in water. The relative density of the object is

(a) 4.0 (b) 4.5

(c) 2.5 (d) 1.125

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : Given,

Wt. of object in air = 100 N

Wt. of object in water = 75 N

Bouyant force of water = 100 – 75

$\rho g V_{body} = 25$

$$V_{body} = \frac{25}{1000 \times 9.81} = 2.548 \times 10^{-3} m^3$$

$$\text{density of body} = \frac{(\text{mass})_{air}}{V_b} = \frac{W_{air}}{g \times V_b}$$

$$\Rightarrow \frac{100}{9.81 \times 2.548 \times 10^{-3}}$$

$\rho_{body} \Rightarrow 4 \times 10^3$ kg/m^3

So, specific gravity of body = $\frac{4 \times 10^3}{10^3} = 4$

$\left[\because \rho_w = 1000 kg/m^3\right]$

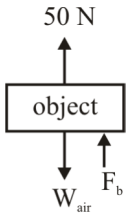
444. An object weights 50 N in water. Its volume is 15.3 liter. Its weight when fully immersed in an oil by specific gravity 0.8 will be

(a) 40 N (b) 62.5 N

(c) 80 N (d) 65 N

UPPSC AE 13.12.2020, Paper-2

Ans. (c)



wt. of object in water = 50 N

vol. of object = 15.3 litres

W_{air} = wt. of object in air

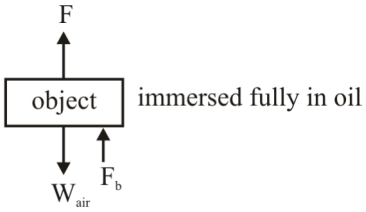
$W_{air} = F + 50$

$50 = W_{air} - \rho_f g V_{displaced}$

$50 = W_{air} - 10^3 \times 9.81 \times 15.3 \times 10^{-3}$

$\boxed{W_{air} = 200.093\,N}$

Now,



Weight when fully immersed in oil

$W_{air} = F_b + F_{oil}$

$F_{oil} = W_{air} - F_b$

$= 200.093 - 800 \times 9.81 \times 15.3 \times 10^{-3}$

$= 79.925$ N

$= 80$ N

445. When a body is completely or partially immersed in a fluid, how much its weight be distributed for it to be in stable equilibrium?

(a) Around the lower part

(b) Around the upper part

(c) It is independent of weight distribution

(d) None of the above

Assam PSC AE (PHED), 18.10.2020

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Ans. (a) : When the weight distribution is around the lower part the centre of gravity is at lower portion and hence below the centre of buoyancy which is condition for stable equilibrium.

446. Ice floats on water because

(a) It is colder than water
(b) Water shrinks on freezing
(c) Water has a high surface tension
(d) water expands on freezing

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Density of ice is less than that of water. A substance with lower density than water can float on water. Therefore, ice floats on water.

447. Oscillations of a floating body are small when the metacentric height is:

(a) Negative (b) Very small
(c) Large (d) Zero

CIL MT 27.02.2020

Ans. (c) : The metacentric height is the direct measure of stability of floating body. Therefore, the larger the metacentric height. The body will be more stable oscillations of floating body are small.

448. A body of specific weight w_1 floats on a liquid of specific weight w_2. Then the ratio of the volume of the object above the liquid to its volume is:

(a) $(w_2 - w_1)/w_1$ (b) $w_2/(w_2 - w_1)$
(c) $w_1/(w_2 - w_1)$ (d) $(w_2 - w_1)/w_2$

CIL MT 27.02.2020

Ans. (d) : Specific weight of liquid $\rho_1 g = w_1$

Specific weight of liquid $\rho_2 g = w_2$

$(\rho_1 g)v_1 = \rho_2 g \times$ Volume of water displaced

$w_1 v_1 = w_2 \times v'$

$$v' = \frac{w_1}{w_2} v_1$$

and the ratio of volume of body above to liquid to total volume of body $= \dfrac{v_1 - v'}{v_1} = \dfrac{v_1 - \frac{w_1}{w_2} \times v_1}{v_1}$

$$\boxed{\text{Ratio} = \frac{w_2 - w_1}{w_2}}$$

449. A cube of ice floating in a glass of water melts. After melting, the water level

(a) rises
(b) remains constant
(c) falls
(d) initially falls and then rises

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : A cube of ice floating in a glass of water melts. After melting, the water level remains constant.

According to the Archimedes principle, the floating substance displaces some liquid, so when the ice melts, there will be no change in the water level as the melted ice will occupy the same volume as it was occupying earlier.

450. A stone weighs 400 N in air and when immersed in water it weighs 225 N. If the specific weight of water is 9810 N/m^3, the relative density of the stone will be nearly

(a) 5.9 (b) 4.7
(c) 3.5 (d) 2.3

ESE 2020

Ans. (d) : Given $W_{real} = 400\ N = mg = \rho_{body} \times g \times V_{body}$

$W_{apparent} = 225\ N$

$W_{apparent} = W_{real} -$ Buoyant force

So,

Buoyant force $= 400 - 225$

$= \rho_{fluid} \times g \times V_{fd}$ —(ii)

By (i) and (ii)

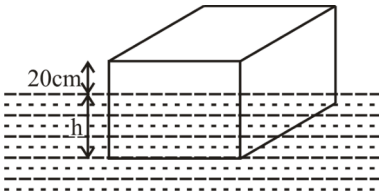
$$\text{R.D.} = \frac{\rho_{body} \times g \times V_{body}}{\rho_{fluid} \times g \times V_{fluid}} = \frac{\rho_{body}}{\rho_{fluid}} = \frac{400}{175} = 2.38 \approx 2.3$$

451. A large iceberg floating on sea water is of cubical shape and its specific gravity is 0.9. If 20 cm proportion of the iceberg is above the sea surface, the volume of iceberg is ____ (Assume specific gravity of sea water is 1).

(a) 9 m^3 (b) 6 m^3
(c) 7 m^3 (d) 8 m^3

VIZAG MT, 14.12.2020

Ans. (d) :



Specific gravity of sea water = 1

Specific gravity of iceberg = 0.9

$\rho_{sw} = 1000\ kg/m^3$

$\rho_{ice} = 900\ kg/m^3$

Archimedes principle

Weight of body = Buoyancy force

$$\rho_{ice} \times A \times \left(h + \frac{20}{100}\right) = \rho_{sw} \times A \times h$$

$$900 \times \left[h + \frac{20}{100}\right] = 1000\,h$$

$$900\,h + \frac{900 \times 20}{100} = 1000\,h$$

$$1000\,h - 900\,h = \frac{900 \times 20}{100}$$

$100\ h = 180$

$$\boxed{h = 1.8}$$

$$H = h + \frac{20}{100} = 1.8 + 0.20$$

$$\boxed{H = 2\,m}$$

$$\boxed{\text{Volume of cube} = 2^3 = 8\,m^3}$$

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452. A pontoon of 15696 kN displacement is floating in water. A weight of 245.25 kN is moved through a distance of 8 m across the deck of pontoon, which tilts the pontoon through an angle 4°. The meta-centric height of the pontoon will be

(a) 1.788 m (b) 2.788 m
(c) 3.788 m (d) 4.788 m

RPSC IOF, 2020

Ans. (a) : Given, weight of pontoon = displacement
W = 15696 kN
Movable weight = 245.25 kN
Distance moved by weight = 8m
Angle of heel = 4°
Metacentric height

$$GM = \frac{\text{Movable weight} \times \text{Distance moved by weight}}{\text{Weight of pontoon} \times \tan\theta}$$

$$GM = \frac{245.25 \times 8}{15696 \times \tan 4^\circ}$$

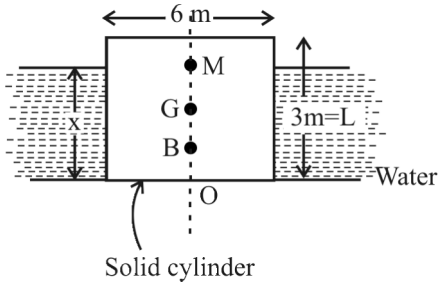
GM = 1.788 m

453. A solid cylinder has following specifications :
Diameter = 6 m
Length = 3 m,
Specific gravity of cylinder material = 0.6
The cylinder is floating in water with its axis vertical. Which of the following is true for the cylinder?
[M = Metacentre of cylinder, G= Centre of gravity of cylinder]

(a) Metacentre (M) is 0.65 m below the centre of gravity (G) and it is unstable equilibrium
(b) Medicine (M) is 0.65 m below the centre of gravity (G) and it is in stable equilibrium
(c) Metacentre (M) is 0.65 m above the centre of gravity (G) and it is in stable equilibrium
(d) Metacentre (M) is 0.65 m above the centre of gravity (G) and it is in unstable equilibrium

APPSC Poly Lect. 13.03.2020

Ans. (c) :



Let,
G = Centre of gravity of cylinder
V = Volume of displaced water
For stable equilibrium, Metacentre 'M' is above the centre of gravity (G). When the cylinder is floating in water.

Let,
x = length of submerged cylinder which is inside the water.
Given, Length or height of cylinder (L) = 3 m
Diameter of cylinder (d) = 6 m
Specific gravity of cylinder = 0.6
$\therefore$ Density of cylinder (ρ_c) = 600 kg/m^3
& Density of water (ρ_w) = 1000 kg/m^3
$\therefore$ In equilibrium condition –
Buoyant force = weight of the cylinder

$$F_B = (mg)_{cylinder}$$

$$\Rightarrow w\forall = (\rho_c V_c g)$$

$$\rho_w g \times \frac{\pi}{4} d^2 \times x = \rho_c g \times \frac{\pi}{4} d^2 L$$

$$\rho_w x = \rho_c L$$

$$x = \frac{\rho_c L}{\rho_w} = \frac{600 \times 3}{1000} = 1.8$$

Therefore, $OB = \frac{x}{2} = \frac{1.8}{2} = 0.9m$

$$\therefore BG = OG - OB$$

$$= \frac{3}{2} - 0.9 = 1.5 - 0.9$$

$$\Rightarrow \boxed{BG = 0.6m}$$

$$\text{Metacentric height (GM)} = \left(\frac{I}{\forall} - BG\right)$$

I = Moment of Inertia of cylinder about longitudinal axis

$$= \frac{\pi}{64} d^4$$

$$GM = BM - BG$$

$$\therefore \quad G.M. = \frac{\frac{\pi}{64} d^4}{\left(\frac{\pi}{4} d^2 \cdot x\right)} - BG \qquad = \frac{d^2}{16x} - BG$$

$$= \frac{(6)^2}{16 \times 1.8} - 0.6 = \frac{36}{28.8} - 0.6$$

$$= 1.25 - 0.6$$

$\boxed{GM = 0.65m}$; GM > 0

Hence, Metacentre (M) is 0.65 m above the centre of gravity (G) and it is in stable equilibrium.

454. The upward force exerted by a liquid on a body when the body is immersed in the liquid is known as :

(a) Buoyancy (b) Water force
(c) Pressure (d) Hydrostatic force

GWSSB DEE 07.07.2016
TNPSC AE (Industries) 09.06.2013
TRB Asstt. Prof., 2012

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Ans. (a) : Buoyancy is the force that causes objects to float. It is the force exerted on an object in upward direction that is partly or wholly immersed in a fluid.

455. A body weights 20 N and 10 N when submerged in the liquids of specific gravity 0.8 and 1.2 respectively. The volume of the body will be given by

(a) $2.548 \times 10^{-3}\ m^3$ (b) $2.935 \times 10^{-3} m^3$
(c) $3.215 \times 10^{-3} m^3$ (d) $2.875 \times 10^{-3} m^3$

SJVN ET 2019

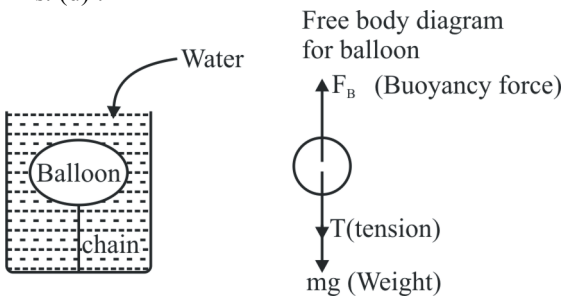
Ans. (a) : Let V_{body} = Volume of body at equilibrium
Wt. of submerged body = Wt. of body – Buoyancy force
$(Wt.\ of\ body)_1$ = Wt. in submerged + Buoyancy force....(i)
$(Wt.\ of\ body)_2$ = Wt. in submerged + Buoyancy force...(ii)
Wt. of body = Wt. in submerged + density of fluid $\times V_{body} \times g$
$(Wt.\ of\ body)_1 = (Wt.\ of\ body)_2$
$20 + 800\ V_{body} \times g = 10 + 1200 \times V_{body} \times g$ = Wt. of body
$V_{body} = 2.548 \times 10^{-3}\ m^3$

456. A spherical balloon of 1.5 m diameter is completely immersed in water and chained to the bottom. If the chain has a tension of 10 kN, the weight of the balloon will be nearly

(a) 9.11 kN (b) 8.22 kN
(c) 6.44 kN (d) 7.33 kN

ESE 2019

Ans. (d) :



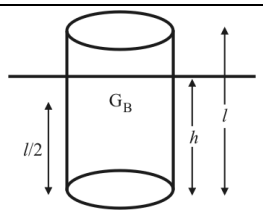
$F_B = mg + T$
Weight (mg) $= F_B - T$
$= \rho gV - T$
$= 1000 \times 9.81 \times \left[\frac{4}{3}\pi(0.75)^3\right] - 10 \times 10^3$
$= 7.33$ kN

457. A uniform wooden cylinder has a specific gravity of 0.5. Find the ratio of diameter to length of the cylinder so that it will just float upright in a state of neutral equilibrium in water :

(a) $\sqrt{2}$ (b) $\sqrt{3}$
(c) 2 (d) 3

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) : Given, specific gravity of cylinder = 0.5
Density of cylinder ($\rho_{cyl.}$) = 500
For floating body weight = Buyancy force



$mg = \rho_w Vg$
$\rho_{cyl} \times \cancel{V} \times g = \rho_w \cancel{V} g$
$500 \times \frac{\pi}{4} d^2 \times l \times g = 1000 \times \frac{\pi}{4} d^2 \times h \times g$
$\boxed{l = 2h}$
$BG = l/2 - h/2$
$= \frac{2h}{2} - h/2$
$BG = h/2$
For Neutral equilibrium GM = 0
$\frac{I_{mini}}{\cancel{V}} - BG = 0$
$\frac{\pi d^4}{64 \times \frac{\pi}{4} d^2 \times h} - \frac{h}{2} = 0$
Put h = l/2
$\frac{\pi d^4}{64 \times \frac{\pi}{4} d^2 \times \frac{l}{2}} - \frac{l/2}{2} = 0$
$\frac{d^2}{8l} - \frac{l}{4} = 0$
$\frac{d^2}{8l} = \frac{l}{4}$
$4d^2 = 8l^2$
$d^2 = 2l^2$
$\boxed{d = \sqrt{2}\, l}$
$\boxed{\frac{d}{l} = \sqrt{2}}$

458. A body of volume 3.0 m^3 weighs 2 kN in water. The body's weight in air is

(a) 6.0 kN (b) 20.4 kN
(c) 5.0 kN (d) 31.4 kN

APPSC AEE SCREENING 17.02.2019

Ans. (d) : We know that,
$W_{water} = W_{air} - F_B$
$W_{air} = W_{water} + F_B$
$= 2 \times 10^3 + 1000 \times 9.81 \times 3$ $(\because F_B = \rho gV)$
$W_{air} = 31.4$ kN

459. The point, through which the whole weight of the body acts, irrespective of its position, is known as

(a) Moment of inertia (b) Centre of gravity
(c) Centre of percussion (d) Centre of mass

GPSC Executive Engineer 23.12.2018

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Ans. (b) : The point through which the whole weight of the body acts irrespective of its position is known as centre of gravity.

460. Wooden block of width 2.5 m and of depth 1.5 m; floats horizontally on water. The density of wooden block is 650 kg/m^3 and its length is 6.0 m, then volume of water displaced is

(a) 14.625 m^3 (b) 15.723 m^3
(c) 13.625 m^3 (d) 12.323 m^3

MPSC Lect. (Prod.) 2018

Ans. (a) : As per the Archimedes principle,
buoyant force = weight of fluid displace by the body.

$$\rho_{body} \cdot V_{body} \times g = \rho_{water} \times V_{displaced} \times g$$

$$V_{displaced} = \frac{\rho_{body} \times V_{body}}{\rho_{water}} = \frac{650 \times 2.5 \times 1.5 \times 6}{1000}$$

$$V_{displaced} = 14.625 \text{ m}^3$$

461. An irregular shape body weighing 7.5 kg occupies a volume of 0.01 m^3. It will be completely submerged in a fluid having specific gravity of:

(a) 0.75 (b) 0.5
(c) 1.3 (d) 1.05

AAI Jr. Executive 29.11.2018

Ans. (a) : Density of liquid $(\rho) = \frac{m}{v}$

$$= \frac{7.5}{0.01}$$

$$= 750 \text{ kg/m}^3$$

$$\text{Specific gravity} = \frac{750}{1000}$$

$$= 0.75$$

462. A ship will sink if it does not displace water equal to its own

(a) volume (b) density
(c) surface area (d) weight

TNPSC AE 2018

Ans. (d) : A ship will sink if it does not displace water equal to its own weight.
For floating of bodies any body displace water equal to its own weight.

463. A stone weights 392.4 N in air and 196.2 N in water. Find the weight of water displaced.

(a) 392.4 N (b) 196.2 N
(c) 3.924 N (d) 1.962 N

TNPSC AE 2018

Ans. (b) : Weight of displaced water
= weight of stone in air – weight of stone in water
= 392.4 – 196. 2 = 196.2 N

464. Statement I: If a boat, built with sheet metal on wooden frame, has an average density which is greater than that of water, then the boat can float in water with its hollow face upward but will sink once it overturns.
Statement II: Buoyant force always acts in the upward direction.

ESE 2018

Ans. (b) : When boat is floating in upright position it displaces more water as compared to overturned position. The buoyancy force is directly proportional to the displaced volume. Hence upright position has more buoyancy force and the boat is able to float.
Thus both statements are individually true.

465. A piece of ice is floating in a beaker containing thick sugar solution of water. As the ice melts, the total level of the liquid

(a) Increases (b) Decreases
(c) Remains unchanged (d) Insufficient data

GPSC EE Pre, 28.01.2017

Ans. (a) : Let ρ = density of sugar solution > ρ_{water}
m = mass of ice
In floating condition weight = upthrust

$\therefore$ $mg = V_1 \rho g$

or $V_1 = \frac{m}{\rho}$ .....(i)

When ice melts, m mass of ice becomes m mass of water. Volume of this water formed.

$$V_W = \frac{m}{\rho_W} \text{ .....(ii)}$$

Since, $\rho_W < \rho$, $V_W > V_I$
Hence, level will increases.

466. A boat carrying a number of large stones is floating in a water tank. What will happen to the water level, if the stones are unloaded into water?

(a) It will rise
(b) It will remain unchanged
(c) It will rise till half the number of stones are unloaded and then begin to fall
(d) It will fall

GPSC EE Pre, 28.01.2017

Ans. (d) : Volume of stone = V_s
Density of stone = ρ_s
Volume of liquid = V_l
density of liquid = ρ_l
$\because$ When the stones were in the boat, the weight of stones were balanced by the buoyant force.

$\because$ $V_s \rho_s = V_l \rho_l$

If $\rho_s > \rho_l$
$V_s < V_l$

Hence, when the stones are unloaded into water, the water level falls because the volume of the water displaced by stones in water will be less than the volume of water displaced when stones are in boat.

467. A carton carrying vessel of mass 10 tons (width = length = 5 m, height = 6 m), floats on water. The mass of contents of the vessel are symmetrically placed. If the vessel tilts by 0.01 radians by moving a 50 kg carton from the centre by a distance of 2 m towards side in horizontal direction, the metacentric height of the vessel is (g = 10 m/s^2)

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(a) 1 m (b) 3 m
(c) 1.5 m (d) 2 m

ISRO Scientist/Engineer 17.12.2017

Ans. (a) : Mass of vessel (M) = 10 ton
$= 10 \times 10^3$ kg
Mass of carton (m) = 50 kg
$b = \ell = 5$ m
$h = 6$ m
$\theta = 0.01$ rad $\simeq \tan\theta$
$w \times x \times \cos\theta = W \times GM \times \sin\theta$

$$GM = \frac{w \times x}{W \times \tan\theta}$$

$$= \frac{50 \times g \times 2}{10 \times 10^3 \times g \times 0.01}$$

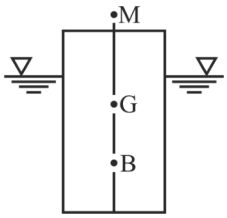
$= 1$

468. Consider the following statements pertaining to stability of floating bodies:
1. A floating body will be stable when the centre of gravity is above the centre of buoyancy.
2. The positions of metacentre corresponding to different axes of rotation are generally different for the same floating object.
3. For cargo ships, the metacentric height varies with loading.

Which of the above statements are correct?
(a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

ESE 2017

Ans. (d) :



Statement I is incorrect because stability of floating body is measured by metacenter with respect to center of gravity.

469. A piece of ore weighing 1.5 N in air and 1.1 N in water. Its volume is :
(a) 40.8 m^3 (b) 40.8 cm^3
(c) 40.8 mm^3 (d) 4.08 mm^3

TRB Polytechnic Lecturer 2017

Ans. (b) : Given,
Weight of a piece in air $W_a = 1.5$ N
Weight of piece in water $W_w = 1.1$ N
Then,
Weight of water displaced = 1.5 – 1.1 = 0.4 N
$\rho_w \times g \times V = 0.4$

$$V = \frac{0.4}{1000 \times 9.81}$$

$\boxed{V = 40.8\,cm^3}$

470. An odd shaped container weight 2N when empty. If it is full of water it weights 4907 N. Container's volume is _____. [take density of water as 1000 kg/m^3 and gravitational acceleration at 9.81 m/s^2] :
(a) 250 litres (b) 500 litres
(c) 750 litres (d) 1000 litres

TRB Polytechnic Lecturer 2017

Ans. (b) : Weight of empty container w = 2N
[Weight of water + Weight of container] = 4907
$w + \rho_{water} \times g \times V = 4907$
$1000 \times 9.81 \times V = 4907 - 2$
$V = 0.5\ m^3$
$\boxed{V = 500\,litres}$

471. A symmetrical right circular cone of wood floats in fresh water with axis vertical and apex downwards. The axial height of the centre is 'L'. The submerged portion has a height 'X', measured upwards from the apex. What would be the height of the centre of buoyancy from the apex?
(a) $\frac{1}{2}X$ (b) $\frac{3}{4}X$
(c) $\frac{2}{3}X$ (d) $\frac{1}{3}X$
(e) $\frac{3}{5}X$

CGPSC ARTO & TSI 07.01.2017

Ans. (b) :

472. A merchant ship has meta centric height of 2.7m. After unloading it increased to 3.6 m. The ratio of periodic times before and after unloading the cargo ship if radious of gyration is not altered is
(a) 1 (b) 0.75
(c) $\sqrt{(3/4)}$ (d) $\sqrt{(4/3)}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : $T = 2\pi\sqrt{\frac{K^2}{g(GM)}} \Rightarrow T \propto \frac{1}{\sqrt{GM}}$
K = Radius of gyration
GM = Metacentric height

$$\frac{T_1}{T_2} = \sqrt{\frac{(GM)_2}{(GM)_1}} = \sqrt{\frac{3.6}{2.7}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{4}{3}}$$

473. The solid P floats with half its volume immersed in water and solid Q floats with one third of its volume above the water surface. The densities of solid P and Q are in the ratio
(a) 1:3 (b) 2:3
(c) 3:4 (d) 1:2
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

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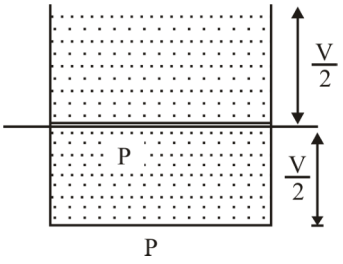
Ans. (c)



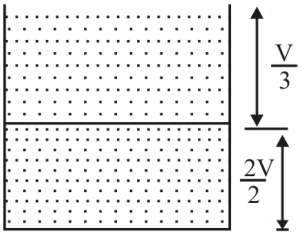
P

Immersed volume

of 'P' $= \frac{V_P}{2}$

$m_P = \frac{V_P}{2} \times \rho$

$\therefore V_p \rho_p = \frac{V_P}{2} \times 1 \Rightarrow \rho_p = \frac{1}{2}$



Q

Immersed volume of

$$Q = \frac{2V_Q}{3}$$

$$m_Q = \frac{2V_Q}{3} \times \rho_Q$$

$$V_Q.P_Q = \frac{2V_Q}{3} \times 1 \Rightarrow P_Q = \frac{2}{3}$$

Their density ratio, $\frac{\rho_P}{\rho_Q}$

$$\frac{\rho_P}{\rho_Q} = \frac{1}{2} \times \frac{3}{2}$$

$$\boxed{\frac{\rho_P}{\rho_Q} = \frac{3}{4}}$$

474. A rectangular pontoon 10 m by 4 m in plan weighs 280 kN. A steel tube weighing 34 kN is placed longitudinally on the deck. When the tube is in a central position, the centre of gravity for the combined weights lies on the vertical axis of symmetry 250 mm above the water surface. Find the metacentric height.

(a) 1.67 m (b) 1.47 m
(c) 2.67 m (d) 0.67 m
(e) None of these

CGPSC AE 16.10.2016

Ans. (e) :
We know, weight of pontoon + load = 280 + 34
= 314 kN

Weight of seawater = 1025 × 9.8 × 10 × 4 × draught

$$\text{Draught} = \frac{314 \times 1000}{1025 \times 9.8 \times 10 \times 4} = 0.781\,\text{m}$$

$$BM = \frac{I}{V} = \frac{\frac{1}{12} \times 10 \times 4^3}{4 \times 10 \times 0.781} = 1.707\,\text{m}$$

$$BG = 0.25 + \frac{0.781}{2} = 0.640\,\text{m}$$

GM = BM – BG
= 1.707 – 0.640
= 1.067 m

475. A body weighing 40 N is losing 5 N weight when submerged in water. Then its specific gravity of the body is equal to for a closed single pair of lock gates having 120° between them

(a) 8 (b) 6
(c) 4 (d) 2

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : We know, Weight = 40 N
Buoyant force (F_b) = 5N

$F_b = \rho_w \times g \times V_b$

$$V_b = \frac{5}{9810}$$

In Air

$F = mg = 40$ N

$$\Rightarrow m = \frac{40}{9.81}$$

$$\text{Density of body} = \frac{m}{V_b}$$

$$= \left(\frac{40}{9.81}\right) \div \left(\frac{5}{9810}\right)$$

$$= 8000\ \text{kg/cm}^3$$

$$\text{Specific gravity} = \frac{\text{density of body}}{\text{density of water}}$$

$$\text{Specific gravity} = \frac{8000}{1000} = 8$$

476. A 10 cm side cube weighing 5N is immersed in a liquid of relative density 0.8 contained in a rectangular tank of cross-sectional area 15cm × 15 cm. If the tank contained liquid to a height of 8 cm before the immersion. The buoyant force on the cube is

(a) 10 N (b) 5000 N
(c) 5 N (d) zero

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : As we know the Cube's weight = 5N is immersed in liquid.
Weight of cube= Buoyant force = 5N

477. An iceberg floats in sea water. If the specific gravity of iceberg and sea water are 0.9 and 1.03 respectively, The percentage of total volume of the iceberg below the sea water surface is

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(a) 87.38% (b) 8.738%
(c) 7.38% (d) 78.38%

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : Acc. to Archimedes Principle weight of iceberg = weight of heater displaced

$$V \times 0.9 \times g = x \times 1.03 \times g$$

$$\frac{x}{v} = \frac{90}{103}$$

% of volume of iceberg submerged in sea water = $\frac{90}{103} \times 100 = 87.381$

478. A solid is suspended by a thread and the body is submerged in a liquid. If V and γ_b all the volume and specific weight of the body are 1 m^3 and 25 kN and the specific weight of the liquid is 10 kN, the tension in the thread is

(a) 15 kN (b) 10 kN
(c) 5 kN (d) 25 kN

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : The tension in the thread is given by
= Specific weight of the body – specific weight of liquid
= 25 kN – 10 kN
= 15 kN

479. A body weighing 40 N is losing 5 N weight when submerged in water. Then its specific gravity of the body is equal to for a closed single pair of lock gates having 120° between them

(a) 8 (b) 6
(c) 4 (d) 2

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : We know, Weight = 40 N
Buoyant force (F_b) = 5N

$$F_b = \rho_w \times g \times V_b$$

$$V_b = \frac{5}{9810}$$

In Air

$$F = mg = 40 \text{ N}$$

$$\Rightarrow m = \frac{40}{9.81}$$

$$\text{Density of body} = \frac{m}{V_b}$$

$$= \left(\frac{40}{9.81}\right) \div \left(\frac{5}{9810}\right)$$

$$= 8000 \text{ kg/cm}^3$$

$$\text{Specific gravity} = \frac{\text{density of body}}{\text{density of water}}$$

$$\text{Specific gravity} = \frac{8000}{1000} = 8$$

480. A 10 cm side cube weighing 5N is immersed in a liquid of relative density 0.8 contained in a rectangular tank of cross-sectional area 15cm × 15 cm. If the tank contained liquid to a height of 8 cm before the immersion. The buoyant force on the cube is

(a) 10 N (b) 5000 N
(c) 5 N (d) zero

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : As we know the Cube's weight = 5N is immersed in liquid.
Weight of cube= Buoyant force = 5N

481. An iceberg floats in sea water. If the specific gravity of iceberg and sea water are 0.9 and 1.03 respectively, The percentage of total volume of the iceberg below the sea water surface is

(a) 87.38%
(b) 8.738%
(c) 7.38%
(d) 78.38%

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : Acc. to Archimedes Principle weight of iceberg = weight of heater displaced

$$V \times 0.9 \times g = x \times 1.03 \times g$$

$$\frac{x}{v} = \frac{90}{103}$$

% of volume of iceberg submerged in sea water = $\frac{90}{103} \times 100 = 87.381$

482. If the centre of gravity is beneath the center of buoyancy, then the buoyant object will be

(a) Stable
(b) Unstable
(c) Whether stable or not depends on the quantity of fluid displaced
(d) None of these

KPCL AE 2016

Ans. (a) : The weight on the object acts through its gravity. A buoyant object will be stable if the center of gravity is beneath the center of buoyancy (B > G) because angular displacement will pull back to equilibrium.

483. If small disturbances do not create any additional forces and so the body remains in the disturbed position, is a case of:

(a) stable equilibrium
(b) unstable equilibrium
(c) neutral equilibrium
(d) none of these

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : If small disturbances do not create any additional forces and so the body remains in the disturbed position, is a case of neutral equilibrium for fully submerge body.

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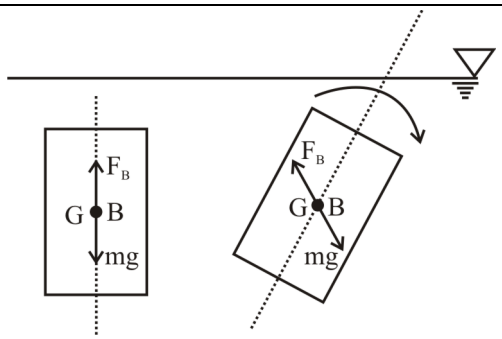
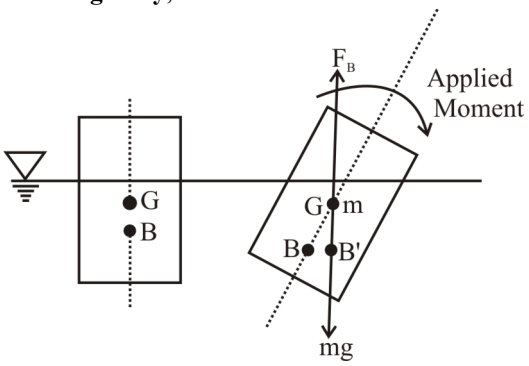


There is no resisting moment due to mg.

For floating body,



GM = 0

There is no moment developed due to mg of body about it.

Body will gain new equilibrium position.

484. A wooden block of width 1.25 m, depth 0.75 m and length is 3.0 m is floating in water. Specific weight of the wood is 6.4 KN/m^3. What is the volume of water displaced?

(a) 2.65 m^3 (b) 3.92 m^3
(c) 1.83 m^3 (d) 4.63 m^3

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Use, Archimedes principle of floating Bodies,

F_B = wt. of Body (fluid displaced)

$\rho_{water} \times g \times (V)_{displaced\ water} = \omega.(V)_{wooden\ Black}$

$1000 \times 9.81 \times (V)_{water\ displaced} = 6.4 \times 1000 \times (1.25 \times 0.75 \times 3)$

Volume of water displaced = 1.83m^3

485. The point at which the line of action of buoyant force meets the centroidal axis of the body, when disturbed, is defined as.....

(a) COG
(b) Instantaneous centre
(c) Metacentre
(d) Point of inflexion

HPPSC AE (PWD) 10.09.2016

Ans. (c) : The point at which the line of action of buoyant force meets the centroidal axis of the body when disturbed is defined as metacentre.

Meta-centric height (GM) = BM – BG

In ship, metacentric height for rolling condition will be less than metacentric height for pitching condition.

486. A wooden block floats in water with 5 cm height projecting above the water surface. The block is next launched in glycerine (relative density = 1.35) and projects 7.5 cm above the surface of glycerine. The height of the block and its relative density will be respectively

(a) 1.46 cm and 0.657
(b) 14.6 cm and 0.657
(c) 14.6 cm and 0.0657
(d) 1.46 cm and 0.0657
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) : As we know,

$W = BF_1$ (Buoyancy force)

$(S_{P.}gr)_{water} \times (W_{water})\ (V_1) = (W_{water})V_D$

$(S_{P.}gr)_{water} \times Ah = A(h - 5)$

$$(S_{P.}gr)_{water} = \frac{h-5}{h} \quad \text{.....(i)}$$

$W = BF_2$ (Buoyancy force)

$(S_{P.}gr)_{water}\ (W_{water}) \times V_t = (S_{P.}gr)_{glycerine}\ (W_{water}) \times V_0$

$$(S_{P.}gr)_{water} = \frac{1.35(h-7.5)}{h} \quad \text{.....(ii)}$$

From (i) & (ii)

$$\frac{h-5}{h} = \frac{1.35(h-7.5)}{h}$$

$\Rightarrow$ h = 14.64 cm

$$(S_{P.}gr)_{water} = \frac{14.64-5}{14.64} = 0.658$$

487. The point through which the resultant gravitational force acts for any orientation of the body is termed as

(a) Centre of gravity
(b) Centroid
(c) Origin
(d) Centre pair of plane area

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Centre of gravity / Centre of mass – It is the point where the whole mass of the body is concentrate. This is where the gravitational force (Weight) of the body acts for any orientation of the body.

Centroid – It is the geometrical centre of the object. It is the centre of gravity for objects of uniform density.

488. A crane is used to lower weights into the sea (sea water density = 1025 kg/m^3) for an underwater construction project. What is the percentage reduction in the tension in the rope of the crane due to a rectangular 0.4m × 0.4m × 3m concrete block (density = 2300kg/m^3) when it is completely immersed in water compared to the tension in the rope when it was suspended in air

(a) 45% (b) 55%
(c) 65% (d) 75%

ISRO Scientist/Engineer 03.07.2016

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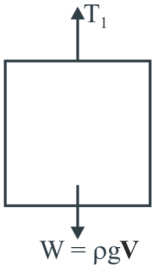
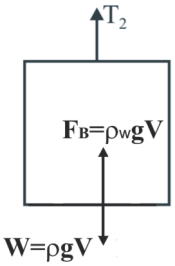
Ans. (a) : When in air–



When immersed in water–



$\rho = 1025 \text{ kg/m}^3$

$T_1 = \rho gV$

ρ_B = Buoyancy force

$\rho_B = 2300 \text{ kg/m}^3$

$T_2 = W - F_B$

$$\therefore \quad \frac{T_2}{T_1} = \frac{W - F_B}{\rho gV} = \frac{(\rho_B - \rho)gV}{\rho gV}$$

$$= \left(\frac{2300 - 1025}{2300}\right)$$

$$\frac{T_2}{T_1} = 0.554$$

$$\text{Percentage reduction} = \left(1 - \frac{T_2}{T_1}\right) \times 100$$

$$= (1 - 0.554)$$

$$= 44.5\ \% \approx 45\%$$

489. When a body is immersed in a fluid, the buoyant force experienced by it, is proportional to

(a) Volume of the body
(b) Volume of the fluid displaced
(c) Weight of the body
(d) Velocity of immersion

UJVNL AE 2016

Ans : (b) Buoyant force OR Buoyancy:- When a fluid is immersed in a fluid, an upward force is exerted by the fluid on the body. This upward force is equal to the **Weight of fluid displaced** by the body is called the force of buoyancy or simply buoyancy

$F_B = \rho_l V_b g$

ρ_l = density of liquid

V_b = Volume of fluid displaced.

490. A wooden plank (sp. gravity 0.5) 1m × 1m × 0.5, float in water with 1.5 kN load on it with 1m × 1m surface horizontal. The depth of plank lying below water surface shall be

(a) 0.178 m (b) 0.25 m
(c) 0.4 m (d) 0.5 m

GPSC AE 26.07.2015

Ans. (c) : Given,

Specific gravity of wooden plank = 0.5 = 0.5 × 100

$= 500 \text{ kg/m}^3$

Dimension of wooden plank = 1 × 1 × 0.5

Weight on the wooden plank = 1.5 kN

Weight at the wooden plank = Weight of water displaced by wooden plank

$1.5 \times 1000 + 500 \times g \times 1 \times 1 \times 0.5 = 1 \times 1 \times h \times g \times 1000$

$1500 + 250g = 1000g \times h$

$6 + g = 4g \times h$

$$h = \frac{6+g}{4g}$$

$$= \frac{6}{4g} + \frac{1}{4}$$

$h = 0.403$ m

$= 0.4$ m

491. A piece of wood having a mass of 4 kg floats in a liquid of specific gravity 0.8 gm/cm^3 what will be the specific gravity of the wood piece if 75% of its volume is inside the liquid (Assume g = 10 m/s^2)

(a) 0.4 gm/cm^3 (b) 0.6 gm/cm^3
(c) 0.8 gm/cm^3 (d) 1.0 gm/cm^3

TSGENCO AE 14.11.2015

Ans. (b) : Given,

Mass (m) = 4 kg

$S_{liquid} = 0.8 \text{ gm/cm}^3$

$S_{wood} = ?$

$g = 10 \text{ m/s}^2$

$g = 1000 \text{ cm/s}^2$

According to be buoyancy force due to floating.

Buoyancy force = weight of fluid displaced

If wood 75% volume inside the liquid.

$\rho_{wood} \times g \times V = \rho_{liquid} \times g \times 0.75\ V$

$S_{wood} \times V = S_{liquid} \times 0.75\ V$

$S_{wood} = 0.8 \times 0.75 = 0.6 \text{ gm/cm}^3$

492. What will be volume of water displaced for buoyancy for a wooden block of width 2 m and depth 1 m. Density of wooden block is 700 kg/m^3 and its length is 4 m (Assume g = 10 m/s^2)

(a) 2.8 m^3 (b) 5.6 m^3
(c) 11.2 m^3 (d) 22.4 m^3

TSGENCO AE 14.11.2015

Ans. (b) : Given,

Width (w) = 2 m

depth (d) = 1m

density of wooden block = 700 kg/m^3

length (ℓ) = 4 m

$g = 10 \text{ m/s}^2$

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Buoyancy force (F) = Weight of fluid displaced

$\rho_{wood} \times g \ V_{block} = \rho_{wood} \times g \ V_{water}$

$700 \times 10 \times 2 \times 1 \times 4 = 1000 \times 10 \times V_{water}$

$V_{water} = \frac{700 \times 10 \times 2 \times 1 \times 4}{1000 \times 10}$

$= \frac{700 \times 8}{1000}$

$= 5.6 \ m^3$

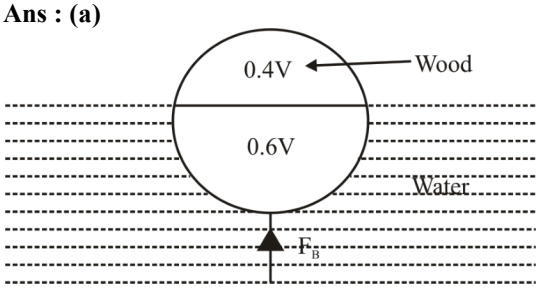
493. A piece of wood having weight of 10 kg floats in water with 60% of its volume under the liquid. Determine the specific Gravity of wood.

(a) 0.6 (b) 0.83

(c) 0.4 (d) 0.3

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a)



$m = 10 \ kg$

$F_B = mg$

$\rho_{wood} \times V \times g = \rho_{water} \times 0.6V \times g$

$\frac{\rho_{wood}}{\rho_{water}} = 0.6$

494. Match List-I with List-II regarding a body party submerged in a liquid and select the correct answer using the codes given below the lists:

List-I		List-II	
A.	Centre of pressure	1.	Point of application of the weight of displaced liquid
B.	Centre of gravity	2.	Point about which the body starts oscillating when tilted by a small angle
C.	Centre of buoyancy	3.	Point of application of hydrostatic pressure
D.	Metacentre	4.	Point of application of the weight of the body

	A	B	C	D
(a)	4	3	1	2
(c)	3	4	1	2

	A	B	C	D
(b)	4	3	2	1
(d)	3	4	2	1

Nagaland PSC CTSE 2015 Paper-II

Ans. (c)

- **Centre of Pressure–** A point on a surface through which the resultant force due to pressure passes.
- **Centre of Gravity–** A point from which the weight of body or system may be considered to act in uniform gravity it is the same as the centre of mass.
- **Centre of Buoyancy–** The centroid of the immersed part of a ship or other floating body.
- **Metacentre–** The point of intersection between an imaginary line drawn vertically through the centre of buoyancy of a floating vessel and a corresponding line through the new centre of buoyancy when the vessel is tilted.

495. A stone weighs 392.4 N in air and 196.2 N in water. The specific gravity of the stone will be

(a) 2.5 (b) 3.0

(c) 3.5 (d) 2.0

GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : Wt in air $(W_a) = 392.4$ N

Wt in water $= (W_a - F_B) = 196.2$ N

$F_B = 196.2$

$\rho_f \ g.V = 196.2$

$\rho_b \ g.V = 392.4$

$Sp.gr = \frac{\rho_b}{\rho_f} = \frac{392.4}{196.2}$

$Sp.gr = 2$

496. Match list I with List II and select the correct answer:

List I		List II	
A	Metacentric height	1	weight of displaced volume
B	G below B	2	CG of displaced volume
C	Centre of buoyancy	3	Stability
D	M and G coincides	4	Unstable
		5	Always stable
		6	Always Neutral

(a) A-3, B-2, C-6, D-5

(b) A-3, B-5, C-2, D-6

(c) A-4, B-3, C-5, D-2

(d) A-1, B-4, C-6, D-5

(e) A-6, B-2, C-4, D-5

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Metacentric height– Metacentric height tell us about stability of submerged body.

- When G below B, then body always get stable condition
- Center of buoyancy always CG of displaced volume of fluid.
- M and G coincides then body always in neutral condition.

497. A wooden plank (sp. graivty = 0.5) of 1m × 1m × 0.5 m floats in water with 1.5 kN load on it with 1 m × 1m surface horizontal. The depth of plank lying below water surface shall be

(a) 0.178 m (b) 0.250 m

(c) 0.403 m (d) 0.5 m

ISRO Scientist/Engineer 24.05.2014

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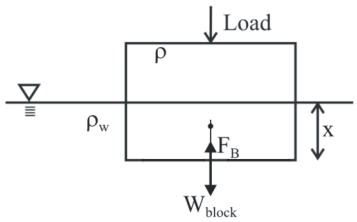
Ans. (c)



Bouyant weight = weight of wooden plank + load

$$= \rho_w V + \frac{1.5\times10^3}{9.81}$$

$$\rho_w \times 1\times 1\times x = \rho\times 1\times 1\times 0.5 + \frac{1.5\times10^3}{9.81}$$

$$x = 0.5\times 0.5 + \frac{1.5\times10^3}{9.81\times10^3}$$

$$x = 0.4029\text{m}$$

498. A body of dimensions 2.0 m × 1.0 m × 3.0 m weight 3924 N in water. Find its weight in air.

(a) 62784 N (b) 52874 N
(c) 6278 N (d) 627.8 N

APPSC Poly. Lect. 2013

Ans. (a) : Weight of stone in air = weight of water displaced + weight of stone in water.

V = 2 × 1 × 3 = 6

(w_a) weight in air = 1000 × 9.81 × 6 + 3924
= 62784 N

499. For a floating body to in equilibrium.

(a) Metacentre should be above CG.
(b) Centre of Buoyancy and CG must lie on same vertical plane
(c) A rotating couple should be formed
(d) All of above

SJVN ET 2013

Ans. (d) : For a floating body to be in equilibrium
- Metacentre should be above C.G.
- Centre of buoyancy and C.G. must lie on same vertical plane.
- A rotating couple should be formed.

500. A spherical balloon is filled with helium at sea level. Helium and balloon material together weigh 500 N. If the net upward lift force on the balloon is also 500 N, what is the diameter of the balloon?

(a) 5.42m (b) 6.78m
(c) 3.43m (d) 2.34m

ISRO Scientist/Engineer 2012

Ans. (a) : Net upward lift = Force of buoyancy - weight of balloon

$$500 = \left(\rho_{air} \times \frac{4}{3}\pi R^3\right) \times 9.81 - 500$$

$$\frac{1000\times 3}{1.22\times 9.81\times 4\times \pi} = R^3$$

[$\because \rho_{air} = 1.22$ kg/m^3]

R = 2.71 m

D = 5.45 m

501. A metal piece having density exactly equal to the density of a fluid is placed in the liquid. The metal piece will

(a) sink to the bottom
(b) float on the surface
(c) will be partly immersed
(d) will be wholly immersed

UKPSC AE 2012 Paper–II

Ans. (d) : will be wholly immersed

502. A solid body sinks in a fluid when

(a) the specific gravity of its material is greater than unity
(b) the buoyance force does not pass through the metacentre
(c) the weight of the fluid displaced is less than the weight of the body
(d) the metacentre lies below the C.G.

APPSC AEE 2012

Ans. (c) : A solid body sinks in a fluid when the weight of the fluid displaced is less than the weight of the body.

503. The resultant force on a floating body will act

(a) vertically upwards through centre of buoyancy
(b) vertically downwards through centre of buoyancy
(c) vertically upwards through meta centre
(d) vertically downwards through meta centre

UKPSC AE 2012 Paper–II

Ans. (a) vertically upwards through centre of buoyancy

504. The metacentric height of a ship is 0.6 m and the radius of gyration is 4 m. What would be the time of rolling of the ship?

(a) 4.1 s (b) 5.2 s
(c) 10.4 s (d) 14.1 s

VIZAG STEEL MT 2011

Ans. (c) : Given,

Metacentric height, GM = 0.6 m

Radius of gyration, k = 4 m

Time of rolling of the ship,

$$T = 2\pi\sqrt{\frac{k^2}{g(GM)}}$$

$$= 2\pi\sqrt{\frac{(4)^2}{9.81\times 0.6}}$$

T = 10.35 sec

505. If two objects are weighed in water and both of them lose the same weight, then the two objects must have identical

(a) Specific gravity (b) Weight in air
(c) Density (d) Volume

ISRO Scientist/Engineer 2011

Ans. (d) : Because two object lose same weight

So, $F_{B_1} = F_{B_2}$

$\rho g V_1 = \rho g V_2$

$V_1 = V_2$

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506. Four cars, with a mass of 1500 kg each, are loaded on a 6 m wide, 12 m long small car ferry. How far, in cm will it sink in the water?

(a) 15.2 (b) 11.5

(c) 10.2 (d) 8.3

ISRO Scientist/Engineer 2010

Ans. (d) : Four cars,

Each mass = 1500 kg

width = 6m

length = 12m

From Archimedes principle of boundary

$F_B = W$

Let height at depth ferry car is 'h' m

$F_B = W$

$1000 \times g \times (6 \times 12 \times x) = 4 \times 1500 \times 9.8$

$x = 8.3$cm

507. A metallic cube of side 10 cm, density 6.98 gm/cc is floating in liquid mercury (density 13.6 gm/cc), with 5 cm height of cube exposed above the mercury level. Water (density 1 gm/cc) is filled over this, to submerge the cube fully. The new height of cube exposed above mercury level is

(a) 4.6 cm (b) 5.4 cm

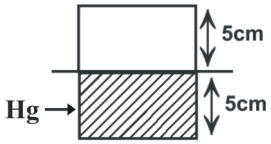
(c) 5.0 cm (d) 5.8 cm

ISRO Scientist/Engineer 2009

Ans. (b) : Cube of side = 10 cm, density of liquid = 6.8 gm/cc is floating.

Mercury (density 13.6 gm/cc) with height = $h_{mercury}$ = 5 cm

Water (density 1 gm/cc) is filled fully submerge the cube fully h_{water} = ?

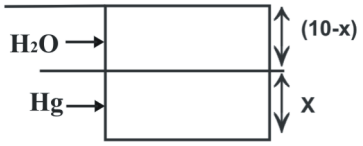


The weight of the cube W_c = The weight of mercury $W_{mercury}$ displaced.

$W_c = W_{mercury} = \rho g \times$ vol.

$= 13.6 \times 10^3 \times 9.81 \times (10 \times 10 \times 5)$

$W_{mercury} = 66708000$ gm



In second case the weight of cube is balanced by the buoyancy force by water and mercury as given below—

$W_c = 13.6 \times x \times 100 \times 9810 + 1 \times 9810 \times (10 - x) \times 100$

$66708000 = 13.6 \times x \times 100 \times 9810 + 1 \times 9810 \times (10 - x) \times 100$

$66708000 = 13.6 \times x \times 100 \times 9810 + 9810 \times 10 \times 100 - 9810 \times x \times 100$

$68 = 13.6\ x - 1 + 10 - x$

Or $x = 4.6$ cm

Then new height of mercury,

$10 - x = 10 - 4.6 = 5.4$ cm

508. For a cylindrical wooden block completely submerged under water, the only correct statement among the following is

(a) Buoyancy is the largest when the axis is horizontal

(b) Buoyancy is the largest when the axis is vertical

(c) Buoyancy increases with the depth at which the body is kept

(d) Buoyancy is independent of the vertical position and orientation

DRDO Scientists 2008

Ans. (d) : Buoyancy is independent of the vertical position and orientation

509. A solid body sinks in a fluid when

(a) The specific gravity of its material greater than unity

(b) The buoyancy force does not pass through the metacentre

(c) The weight of the fluid displaced is less than the weight of the body

(d) The metacentre lies below the centre of gravity

OPSC Civil Services Pre 2006

Ans. (c) : A body will sink in the fluid only if body has more sp. gravity than the fluid or weight of fluid displaced is less than the weight of the body.

5. Fluid Kinematics

510. The imaginary line drawn in the fluid in such a way that the tangent to any point gives the direction of motion at the point, is known as

(a) Path line (b) Stream line

(c) Steak line (d) Potential line

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I
ESE 2020
OPSC AEE 2019 PAPER - II
Nagaland PSC (CTSE) 2018, Paper-II
APGENCO AE, 2017
TRB Polytechnic Lecturer 2017
GPSC Asstt. Prof. 28.08.2016
WBPSC AE 2003

Ans. (b) : Stream line – The imaginary line drawn in the fluid in such a way that the tangent to any point gives the direction of motion at that point is known as stream line.

Path line – The path traced by a single fluid particle at different instant of time is known as path line.

Streak line – It is the locus of different fluid particles passing through a fixed point is called streak line.

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511. Existence of velocity potential implies that :

(a) Fluid is continuum
(b) Fluid is irrotational
(c) Fluid is ideal
(d) Fluid is compressible

APGCL AM, 2021
UPPSC AE 13.12.2020, Paper-2
APPSC AEE Mains 2016 (Civil Mechanical)
UPRVUNL AE 2016
HPPSC Asstt. Prof. 18.11.2016
J&K PSC Civil Services Pre, 2010
WBPSC AE, 2017, GATE 1994

Ans. (b) : Velocity potential function (ϕ)

$$\phi = f(x, y, z, t)$$

$$\phi = f(x, y, z)$$

$$\frac{d\phi}{dx} = -u, \frac{d\phi}{dy} = -v, \frac{d\phi}{dz} = -w$$

If an incompressible fluid flow is possible that means div.V = 0

$$\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} = 0$$

$$\frac{d^2\phi}{dx^2} + \frac{d^2\phi}{dy^2} + \frac{d^2\phi}{dz^2} = 0 \qquad \{\text{Laplace equation}\}$$

- If ϕ exist for an incompressible fluid flow then it satisfies the Laplace equation.
- If ϕ exist for a fluid means that fluid flow is irrotational fluid flow therefore irrotational fluid flow is called potential flow.

512. The continuity equation is the result of application of the following law to the flow field

(a) law of conservation of momentum
(b) law of conservation of energy
(c) law of conservation of mass
(d) first law of thermodynamics

CIL MT 27.02.2020, TNPSC AE 2019
GWSSB DEE 07.07.2016
TANGEDCO AE 2015
Mizoram PSC AE/SDO 2014, Paper-I
TNPSC AE (Industries) 09.06.2013
UKPSC AE 2012 Paper-II
RPSC AE GWD 2011
OPSC Civil Services Pre 2011

Ans. (c) : Continuity equation–

- It is based on the principle of conservation of mass.
- For a fluid flowing through pipe at all the cross-section the quantity per second is constant.
- The continuity equation is given by

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

513. Steady flow occurs when

(a) velocity does not change
(b) pressure does not change along the flow
(c) conditions change gradually with time
(d) conditions do not change with time

Assam PSC AE (PHED), 18.10.2020
TNPSC AE 2019, WBPSC AE, 2017
Kerala PSC IOF 19.04.2016
TSPSC AEE 2015, TRB Asstt. Prof., 2012

Ans. (d) : Steady flow refers to the condition where the fluid properties at a point in the system do not change at any time.
In steady flow velocity and acceleration of fluid does not change with respect to time.

514. If the stream function is given by $\psi = 3xy$, then the velocity at point (2, 3) will be

(a) 7.21 unit (b) 10.82 unit
(c) 18 unit (d) 54 unit

UPPSC AE 13.12.2020, Paper-2
APGCL AM, 2021, UJVNL AE 2016
MPPSC AE 2016
ISRO Scientist/Engineer 2009, ESE 2015, 1998

Ans. (b) Given, Stream function –

$$\psi = 3xy$$

$$u = -\frac{\partial\psi}{\partial y} = -3x; \qquad v = \frac{\partial\psi}{\partial x} = 3y$$

$$v = \sqrt{u^2 + v^2}$$

At point (2, 3),

$$v = \sqrt{(-3\times 2)^2 + (3\times 3)^2}$$

$$v = \sqrt{36+81} \Rightarrow v = 10.82\,\text{unit}$$

515. Streamlines, path lines and streak lines are virtually identical for

(a) Steady flow (b) Uniform flow
(c) Flow of ideal fluids (d) Non uniform flow

BHEL ET 2019
APPSC AE Subordinate Service Civil/Mech. 2016
RPSC AE GWD 2011
J&K PSC Civil Services Pre, 2010
J&K PSC Screening, 2006
GATE 1994

Ans. (a) :
Stream line – It is an imaginary line such that the tangent at any point on it will give the direction of velocity of flow.
Path line – Actual path travelled by any individual fluid particle over some time period is called path line.
Streak line – It is the locus of particles that have passed sequentially through a prescribed point in the flow.
Note : For steady flow stream line path line and streak line are same or virtually identical.

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516. A streamline and an equipotential line in a flow field-

(a) are parallel to each other
(b) are perpendicular to each other
(c) intersect at an acute angle
(d) are identical

JPSC AE PRE 2019
Nagaland PSC CTSE 2017 Paper-2
GPSC Asstt. Director of Transport 05.03.2017
RPSC IOF & Boiler 20.09.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
GATE 2011

Ans : (b)

slope of velocity potential

$$u = -\frac{d\phi}{dx},\ v = -\frac{d\phi}{dy}$$

$$\left(\frac{dy}{dx}\right)_{\phi=c} = -\frac{u}{v}$$

slope of stream function

$$\frac{d\psi}{dx} = -v,\ \frac{d\psi}{dy} = u$$

$$\left(\frac{dy}{dx}\right)_{\psi=c} = \frac{v}{u}$$

Hence,

$$\left(\frac{dy}{dx}\right)_{\phi=c} \times \left(\frac{dy}{dx}\right)_{\psi=c} = \left(\frac{u}{v}\right) \times \left(\frac{-v}{u}\right) = -1$$

These are perpendicular to each other.

517. Irrotational flow is characterised as the one in which

(a) the fluid flows along a straight line
(b) the fluid does not rotate as it moves along
(c) the net rotation of fluid particles about their mass centres remains zero
(d) the streamlines of flow are curved and closely spaced

Punjab PSC (Lect.) 06.08.2017
UPRVUNL AE 21.08.2016
Nagaland PSC CTSE 2015 Paper-II
APPSC AEE 2012, ESE 2004, 1998

Ans. (c) : Irrotational flow is a flow in which each element of the moving fluid undergoes no net rotation with respect to a chosen co-ordinate axes from one instant to other.

518. A flow whose stream line is represented by a curve is called

(a) one dimensional flow
(b) two dimensional flow
(c) three dimensional flow
(d) four dimensional flow

Nagaland PSC (CTSE) 2018, Paper-II
APPSC AEE 2012
UKPSC AE 2012, 2007 Paper-II
VIZAG STEEL MT 2011
WBPSC AE 2008

Ans : (b) A flow whose stream line is represented by a curve is called two dimensional flow.

(i) For a steady one dimensional flow, the fluid parameter are function of single space co-ordinate x only.

$$u = f(x), v = 0 \text{ and } w = 0$$

(ii) for a steady two dimensional flow, the fluid parameter are function of two space co-ordinates (x and y only).

$$u = f_1(x,y),\ v = f_2(x,y) \text{ and } w = 0$$

(iii) For a steady three dimensional flow, the fluid parameter are function of three space co-ordinates (x, y and z only)

$$u = f_1(x,y,z), v = f_2(x,y,z) \text{ and } w = f_3(x,y,z)$$

519. In a flow field, the streamlines and equipotential lines

(a) Cut orthogonally except at the stagnation points
(b) Are parallel
(c) Are orthogonal everywhere in the flow field
(d) Cut at any angle

APPSC AE Subordinate Service Civil/Mech. 2016
TRB Poly. Lect., 2012, ESE 1997
GATE 1994

Ans. (a) : In a flow field, the stream lines and equipotential lines cut orthogonally at all points except at the stagnation point because at stagnation point the velocity of flow is zero, that means there will be no flow in that direction and hence no flow lines or stream lines.

520. If a two-dimensional flow field has velocities along the x and y directions given by $u = x^2t$ and $v = -2xyt$ respectively, where t is the time, then the equation of streamline is :

(a) x^2y = constant
(b) xy^2 = constant
(c) xy = constant
(d) Not possible to determine

BHEL ET 2019
APPSC AE Subordinate Service Civil/Mech. 2016
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021
GATE 2006

Ans. (a) : Given,

$$u = x^2t \ \&\ v = -2xyt$$

Equation of stream line–

$$\boxed{\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}}$$

$$\frac{dx}{x^2t} = \frac{dy}{-2xyt}$$

$$\frac{dx}{x} = \frac{dy}{-2y}$$

Integrating to the side

$$\int \frac{dx}{x} = \frac{-1}{2}\int \frac{dy}{y}$$

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$$\ln x = \frac{-1}{2}\ln y + \ln C_1$$

$$\ln x = \ln \frac{1}{\sqrt{y}} + \ln C_1$$

$$\ln x = \ln \frac{C_1}{\sqrt{y}}$$

$$x = \frac{C_1}{\sqrt{y}}$$

$$x^2 = \frac{C_1^2}{y}$$

$$\boxed{x^2 y = C}$$

$$\boxed{x^2 y = \text{Constant}}$$

521. The 2D flow with velocity

$$\vec{V} = (x + 2y + 2)\hat{i} + (4 - y)\hat{j}$$

(a) Compressible and irrotational
(b) Compressible and rotational
(c) Incompressible and irrotational
(d) Incompressible and rotational

RPSC ADE 2016
Vizag Steel MT (Re-Exam) 24.11.2013
ISRO Scientist/Engineer 12.05.2013, GATE 2001

Ans. (d) : Given,
Velocity component

$$\vec{V} = (x + 2y + 2)\hat{i} + (4 - y)\hat{j}$$

Here, $u = x + 2y + 2$
$v = 4 - y$

(i) Thus $\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1$

Therefore

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Since the velocity field satisfy the continuity equation for incompressible flow.
Thus the flow is incompressible.

(ii) $\frac{\partial v}{\partial x} = 0, \quad \frac{\partial u}{\partial y} = 2$

The rotation is given by

$$w_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

$$= \frac{1}{2}(0 - 2) = -1$$

Since the rotation is non zero so the flow is rotational.

522. A flow in which each liquid particle has a definite path and the paths of individual particles do not cross each other, is called

(a) Steady flow (b) Uniform flow
(c) Streamline flow (d) Turbulent flow

Nagaland PSC (CTSE) 2018, Paper-II
Punjab PSC SDE 12.02.2017
GPSC Lect. 23.10.2016
Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Stream Line Flow– It is an imaginary line/curve drawn in space such that a tangent drawn to it at any point gives direction of velocity.
Two stream lines can never intersect each other or a single stream line can never intersect with itself because at any given instant at any point the velocity is unique.
Steady flow– A flow is said to be a steady flow if fluid properties do not vary with respect to time.
Uniform Flow – A flow is said to be uniform flow if the velocity remains constant at different section at any given instant at time otherwise the flow is non-uniform.
Turbulent Flow – When fluid particles move in highly disorganized manner then that flow is known as turbulent flow. Turbulent flow generally occur at high velocities.

523. For an irrotational flow the equation $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$ is known as

(a) Cauchy Riemann's equation
(b) Euler's equation
(c) Laplace equation
(d) Poisson's equation

ISRO Scientist/Engineer 22.04.2018
TNPSC AE 2014, CSE Pre 2009
ESE 2007

Ans. (c) : Laplace equation is

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

524. The basic continuity equation for compressible fluid is. (Symbols used have usual meaning)

(a) $\frac{dV}{V} + \frac{dA}{A} + \frac{d\rho}{\rho} = 1$

(b) $\frac{dV}{2V} + \frac{dA}{2A} + \frac{d\rho}{\rho} = 0$

(c) $VdV + AdA + \rho d\rho = 0$

(d) $\frac{dV}{V} + \frac{dA}{A} + \frac{d\rho}{\rho} = 0$

TSGENCO AE 14.11.2015
CSE Pre 2003, 2004, 2008, DRDO Scientists 2008

Ans. (d) : Continuity equation for compressible fluid –

$$\frac{dv}{v} + \frac{dA}{A} + \frac{d\rho}{\rho} = 0$$

$$\boxed{\rho_1 A_1 V_1 = \rho_2 A_2 V_2}$$

525. ..... lines provide an instantaneous picture of the particles, which have passed through a given point like the injection point of a dye in a flow.

(a) Stream (b) Path
(c) Time (d) Streak

HPPSC Workshop Suptd. 08.07.02021

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APPSC AEE SCREENING 17.02.2019
HPPSC AE (PWD) 10.09.2016

Ans. (d) : A streak line is the locus of the temporary location of all particles. That have passed through a fixed point in the flow field at any instant of time.
While path line refer to the identity of a fluid particle.

526. The velocity potential function in a two dimensional flow fluid field is given by $\phi = x^2 - y^2$. The magnitude of velocity at point P(1, 1) is:

(a) zero (b) 2

(c) $2\sqrt{2}$ (d) 8

KPSC ADF 2015, ESE 2013, CSE Pre-2002

Ans. (c) : Given,
Velocity potential function–

$\phi = x^2 - y^2$; Point P(1, 1)

Let, V = Magnitude of the velocity

$\therefore \quad \frac{\partial \phi}{\partial x} = -u$

$\frac{\partial}{\partial x}\left(x^2 - y^2\right) = -u$

$\Rightarrow \quad u = -2x$

$\therefore \quad u_{(1,1)} = -2$

$\frac{\partial \phi}{\partial y} = -v$

$\Rightarrow \quad \frac{\partial}{\partial y}\left(x^2 - y^2\right) = -v$

$-v = -2y$

$\Rightarrow \quad v = 2y$

$\Rightarrow \quad v(1, 1) = 2$

$\therefore \quad V = \sqrt{u^2 + v^2} = \sqrt{(-2)^2 + (2)^2}$

$= \sqrt{4+4}$

$\Rightarrow \quad \boxed{V = 2\sqrt{2}}$ unit

527. Shear strain rate in a fluid is given by

(a) $\frac{1}{2}\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$

(b) $\frac{1}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)$

(c) $\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$

(d) $\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)$

UPPSC AE 13.12.2020, Paper-2
APPSC AEE 2012, APPSC AE 04.12.2012

Ans. (b)

Shear strain rate, $\boxed{\varepsilon_{xy} = \frac{1}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)}$

528. Velocity field can be related to stream function as

(a) $u = \frac{\partial \psi}{\partial y}; v = -\frac{\partial \psi}{\partial x}$

(b) $u = -\frac{\partial \psi}{\partial y}; v = -\frac{\partial \psi}{\partial x}$

(c) $u = -\frac{\partial \psi}{\partial y}; v = \frac{\partial \psi}{\partial x}$

(d) $u = \frac{\partial \psi}{\partial y}; v = \frac{\partial \psi}{\partial x}$

APPSC AE 2019, TNPSC AE 2018, ESE 1995

Ans. (a & c) : Stream function is defined as

$$u = \frac{-\partial \psi}{\partial y}, \; v = \frac{\partial \psi}{\partial x}$$

Note: Some authors define it as $u = \frac{\partial \psi}{\partial y}, \; v = \frac{-\partial \psi}{\partial x}$

Both the definitions are correct as velocity components defined in above manner satisfy two dimensional continuity equation for incompressible flow.

For example $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial}{\partial x}\left(\frac{-\partial \psi}{\partial y}\right) + \frac{\partial}{\partial y}\left(\frac{\partial \psi}{\partial x}\right) = 0$

529. The necessary condition for the flow of a liquid to be uniform is that

(a) The velocity is constant at a point with respect to time
(b) The velocity is constant at a point in the flow field with respect to space
(c) The velocity changes at point with respect to time
(d) The velocity changes at point with respect to space

TNPSC AE 2019
Nagaland PSC CTSE 2015 Paper-II
UKPSC AE 2012 Paper-II

Ans. (b) : The necessary condition for the flow of a liquid to be uniform is that the velocity is constant at a point in the flow field with respect to space.

$$\left(\frac{\partial V}{\partial S}\right)_t = 0$$

530. In a two-dimensional velocity field with velocities u and v along the x and y directions respectively, the convective acceleration along the x-direction is given by-

(a) $u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}$ (b) $u\frac{\partial u}{\partial x} + u\frac{\partial v}{\partial y}$

(c) $u\frac{\partial v}{\partial x} + u\frac{\partial u}{\partial y}$ (d) $u\frac{\partial u}{\partial x} + u\frac{\partial u}{\partial y}$

RPSC IOF & Boiler 20.09.2017
OPSC AEE 2015 PAPER - II
GATE 2006

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Ans : (a) Convective acceleration is defined as the rate of change of velocity due to change of position of fluid particle in a fluid flow.
Convective acceleration along the x- direction is given by (for 3D)

$$a_x = \underbrace{u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}}_{\text{convective acceleration}} + \underbrace{\frac{\partial u}{\partial t}}_{\text{local acceleration}}$$

(for 2D convective acceleration along the x-direction)

$$a_x = u\frac{du}{dx} + v\frac{du}{dy}$$

531. If $\Psi = (x^2 - y^2)$ is the stream function in a two-dimensional flow field, then the magnitude of velocity vector at point (1,1) would be

(a) zero (b) $2\sqrt{2}$
(c) 4 (d) 8

BPSC AE Mains 2017 Paper - V
ESE 2010, GATE 1989

Ans : (b) : Given,

$$\psi = x^2 - y^2$$

$$u = \frac{-d\psi}{dy} = 2y = 2 \text{ at } (1,1)$$

$$v = \frac{d\psi}{dx} = 2x = 2 \text{ at } (1,1)$$

$$|v| = \sqrt{u^2 + v^2}$$

$$= 2\sqrt{2}$$

532. A one dimensional flow is one which-

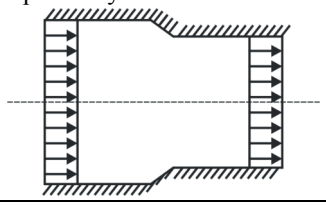
(a) Is uniform flow
(b) is steady uniform flow
(c) Takes place in straight lines
(d) Involves zero transverse component of flow

RPSC VPITI 14.02.2016
Nagaland PSC CTSE 2015 Paper-II
ISRO Scientist/Engineer 2006

Ans. (d) : One Dimensional Flow–It is that type of flow in which the flow parameter such as velocity is a function of time and one space co-ordinate only.
Mathematically,

$u = f(x)$, $v = 0$ and $w = 0$

Where u, v and w are velocity components in x, y and z directions respectively.



533. A streamlined body is defined as a body about which

(a) The vortex drag is zero
(b) The flow is along the streamlines
(c) The flow separation is suppressed
(d) The flow is laminar

GPSC Executive Engineer 23.12.2018
TNPSC AE 2018, GATE 1987

Ans. (c) : A stream lined body is defined as that body whose surface coincides with the streamlines, when the body is placed in a flow. In that case the separation of flow will take place only at the trailing edge, i.e., the flow separation is suppressed.

534. An ideal flow of any fluid must fulfil the following

(a) Newton's law of viscosity
(b) Pascal's Law
(c) Continuity equation
(d) Boundary layer theory

WBPSC AE, 2017, RPSC VPITI 14.02.2016
SJVN ET 2013

Ans. (c) : An ideal flow of any fluid must fulfill the continuity equation AV = constant
Generalized continuity equation–

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

$$\frac{D\rho}{Dt} + \rho(\nabla \cdot \vec{v}) = 0$$

Divergence $\vec{v} = \nabla \cdot \vec{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$

For steady flow,

$$\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

For incompressible flow–

$$\rho = c$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (3D)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (2D)$$

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad (1D)$$

535. Vorticity is

(a) two times the rotation
(b) 1.5 times the rotation
(c) three times the rotation
(d) equal to rotation

Assam PSC AE (IWT) 14.03.2021
Oil India Senior Officer 23.12.2020
Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : Vorticity is given by two times of the rotation.
Vorticity $\xi = 2\omega = \text{Curl } V$
It measures local rotation of fluid tendency of a fluid particle to rotate or circulate at a particular point.

536. A steady, incompressible flow, is given by $u = 2x^2 + y^2$ and $v = -4xy$. What is the convective acceleration along x- direction at point (1, 2)?

(a) $a_x = -8$ unit (b) $a_x = 6$ unit
(c) $a_x = 24$ unit (d) $a_x = -24$ unit

APPSC AE Subordinate Service Civil/Mech. 2016
ESE 2015, 2008

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Ans. (a) : Given,

$u = 2x^2 + y^2$

$v = -4xy$

Convective acceleration along x - direction

$$a_x = u\frac{du}{dx} + v\frac{du}{dy} \quad \text{..... (i)}$$

$\frac{du}{dx} = 4x$, $\frac{du}{dy} = 2y$

at (1, 2) , at (1, 2)

$\left(\frac{du}{dx}\right)_{(1,2)} = 4$ $\left(\frac{du}{dy}\right)_{(1,2)} = 4$

$u_{(1,2)} = 2 \times 1^2 + 2^2 = 6$

$v_{(1,2)} = -4 \times 1 \times 2 = -8$

Put the values in equation (i)

$a_x = 6 \times 4 + (-8) \times 4 = 24 - 32 = -8$ unit

537. Local acceleration in fluid is due to

(a) unsteady nature of flow

(b) non-uniformity of flow

(c) turbulence in flow

(d) irrotational nature of flow

TSPSC Manager (Engg.) HMWSSB 12.11.2020

UKPSC AE 2012 Paper-II

Ans. (a) : Local acceleration in fluid $\left(a = \frac{du}{dt}\right)$ is due to unsteady nature of flow.

538. A stream function is given by ($x^2 - y^2$). The potential function of the flow will be

(a) $2xy + f(x)$ (b) $2(x^2 - y^2)$

(c) 2xy + constant (d) $2xy + f(y)$

ISRO Scientist/Engineer 2011

ESE 2000

Ans. (c) : Stream function, $\psi = x^2 - y^2$

$\because$ $u = \frac{\partial\psi}{\partial y} = \frac{\partial\phi}{\partial x}$

$\frac{\partial\phi}{\partial x} = -(-2y)$

$\frac{\partial\phi}{\partial x} = 2y$

Integrate both sides,

$\phi = 2xy + f(y)$ .....(i)

$\because$ $v = \frac{\partial\psi}{\partial x} = \frac{\partial\phi}{\partial y}$

Integration,

$\phi = 2xy + f(x)$ .....(ii)

$\therefore$ From equation (i) & (ii)

$f(x) = f(y) = \text{Constant}$

$\therefore$ $\phi = 2xy + \text{Constant}$

539. The general equation of continuity for three-dimensional flow of a compressible fluid for steady flow is (where u, v, w are components of velocity in x, y and z directions respectively)

(a) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

(b) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial w}{\partial z} = 0$

(c) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 1$

(d) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = u.v.w.$

PTCUL AE 25.06.2017

ISRO Scientist/Engineer 2008, 2007

Ans. (a) : (1) Generalized continuity equation

$$\frac{\partial\rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

(2) Steady flow continuity equation (compressible/incompressible)

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

540. If the governing equation for a flow field is given by $\nabla^2\phi = 0$ and the velocity is given by $\vec{V} = \nabla\phi$, then

(a) $\nabla \times \vec{V} = 0$ (b) $\nabla \times \vec{V} = 1$

(c) $\nabla^2 \times \vec{V} = 1$ (d) $(\vec{V}.\nabla)\vec{V} = \frac{\partial\vec{V}}{\partial t}$

ISRO Scientist/Engineer 2007, ESE 1992

Ans. (a) : Governing equation is $\nabla \times \vec{V} = 0$.

541. If the stream function is ϕ = 2xy then the velocity at a point (1, 2) is equal to :

(a) 2 (b) 4

(c) $\sqrt{20}$ (d) 16

TSPSC AEE 2015

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : Stream function $\phi = 2xy$

$u = -\frac{\partial\phi}{\partial y}$ $v = \frac{\partial\phi}{\partial x}$

$u = -\frac{\partial}{\partial y}(2xy)$ $v = \frac{\partial}{\partial x}(2xy)$

$u = -2x$ $v = 2y$

$u_{(1,2)} = -2 \times 1$ $v_{(1,2)} = 2 \times 2$

$= -2$ $= 4$

$V = \sqrt{u^2 + v^2} = \sqrt{(-2)^2 + 4^2} = \sqrt{20}$

542. Impingement of a jet on a flat plate may be expressed by:-

(a) $\psi = xy$ (b) $\psi = x^2 - y^2$

(c) $\psi = x^3 - y^3$ (d) $\psi = x / y$

UKPSC AE-2013, Paper-II

APPSC AE 04.12.2012

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Ans. (a) : Impingement of a jet on a flat plate may be expressed by $\psi = xy$.

543. A flow field satisfying $\nabla \cdot \vec{V} = 0$ as the continuity equation represents always

(a) A steady compressible flow
(b) An incompressible flow
(c) an unsteady and incompressible flow
(d) an unsteady and compressible flow

ESE 2019, GATE 2008

Ans. (b) : The general form of the continuity equation is

$$\frac{\partial \rho}{\partial t} + \vec{\nabla}(\rho \vec{v}) = 0$$

For incompressible flow ρ = constant and the above equation reduces to

$$\vec{\nabla} \cdot \vec{v} = 0$$

Hence, the above equation always represents an incompressible flow whether the flow is steady or unsteady.

544. A two dimensional flow field is described by the velocity components u = 2x and v = –2y. The corresponding velocity potential function will be

(a) $\phi = y^2 - x^2$ (b) $\phi = 2(x + y)$
(c) $\phi = x^2 - y^2$ (d) $\phi = x^2 + y^2$

ESE 2010, UKPSC AE 2007 Paper -II

Ans. (a) : $\phi = f(x, y)$ for steady flow

$$d\phi = \frac{\partial \phi}{\partial x}dx + \frac{\partial \phi}{\partial y}dy \qquad \left\{\frac{\partial \phi}{\partial x} = -u, \ \frac{\partial \phi}{\partial y} = -v\right\}$$

see, $d\phi = -u\,dx + (-v)\,dy$

$$\int d\phi = -\int u dx - \int v dy \Rightarrow \int d\phi = -\int 2x dx + \int 2y dy$$

$$\phi = -x^2 + y^2$$

$$\therefore \boxed{\phi = y^2 - x^2}$$

545. At given instant the flow parameters remains same at every point for:

(a) Laminar flow (b) Uniform flow
(c) Steady state flow (d) Quasi state flow

OPSC AEE 2019 PAPER - II
WBPSC AE, 2017

Ans. : (b) : At given instant the flow parameters remain same at every point for uniform flow.

546. Flow occurring in a pipeline when a valve is being opened is

(a) Steady (b) Unsteady
(c) Laminar (d) Vortex

TNPSC AE 2019, Vizag Steel (MT) 2017

Ans. (b) : Since valve of pipe is being opened and closed gradually so the discharge is not constant. Hence it is a non-steady flow.

$\phi = f(t)$,

$$\therefore \frac{d\phi}{dt} = 0$$

547. If u and v represent velocity components in x and y directions of a two-dimensional potential flow, then $\frac{\partial u}{\partial x}$ is equal to:

(a) $\frac{\partial v}{\partial x}$ (b) $\frac{\partial v}{\partial y}$
(c) $\frac{\partial u}{\partial y}$ (d) $-\frac{\partial v}{\partial y}$

CIL MT 26.03.2017, GATE 2005

Ans. (d) : According to continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{\partial u}{\partial x} = \frac{-\partial v}{\partial y}$$

548. A fluid flow is represented by the velocity field $V = ax\hat{i} + ay\hat{j}$, where 'a' is a constant. The equation of streamline passing through a point (1, 2) is

(a) x – 2y = 0 (b) 2x + y = 0
(c) 2x – y = 0 (d) x + 2y = 0

TSPSC AEE 2017, GATE 2004

Ans. (c) : We know, $u_x = ax$, $u_y = ay$

Equation of stream line is

$$\frac{dx}{u_x} = \frac{dy}{u_y}, \qquad \frac{dx}{ax} = \frac{dy}{ay} \quad \text{.....(1)}$$

Integrating both sides of eq(1)

$$\ln x = \ln y + \ln c$$

$$\ln\left(\frac{x}{y}\right) = \ln c$$

$$x = cy$$

for (1, 2), $1 = 2c$

$$c = \frac{1}{2}$$

Now, $x = \frac{y}{2}$ so,

2x – y = 0 is equation

549. For continuity equation given by $\nabla.V = 0$ to be valid, where V is the velocity vector, which one of the following is a necessary condition?

(a) Steady flow
(b) Irrotational flow
(c) Inviscid flow
(d) Steady Incompressible flow

TSPSC AEE 2017, CSE Pre-2007

Ans. (d) : Continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial_x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

For incompressible flow ρ = constant

$$\rho\left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right] = 0$$

$$\nabla.V = 0$$

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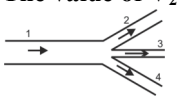
550. Pipe 1, branches to three pipes as shown in the given figure. The areas and corresponding velocities are as given in the following table. The value of V_2 in cm per second will be

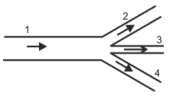


Pipe	Velocity (cm per second)	Area (sq cm)
1.	50	20
2.	V_2	10
3.	30	15
4.	20	10

(a) 15 (b) 20
(c) 30 (d) 35

ISRO Scientist/Engineer 2008, CSE Pre-1995

Ans. (d) :



$Q_1 = Q_2 + Q_3 + Q_4$
$\rho A_1V_1 = \rho (A_2V_2 + A_3V_3 + A_4V_4)$
$50 \times 20 = V_2 \times 10 + 30 \times 15 + 20 \times 10$
$V_2 = 35$ cm/sec

551. Linear momentum equation for steady flow with fixed control volume is given by [Where, V : Velocity vector, $\hat{n}$: outward normal unit vector of area dA. F = forces on control volume]

(a) $\sum F = \int_{cs} \rho(V \times V).\hat{n}\, dA$

(b) $\sum F = \int_{cs} \rho V(V.\hat{n})\, dA$

(c) $\sum F = \int_{cs} \rho(V.\hat{n})\, dA$

(d) $\sum F = \int_{cs} \rho V(V \times \hat{n})\, dA$

UPRVUNL AE 2016
UPRVUNL AE II 07.10.2016

Ans. (b) : Linear momentum equation for steady flow with fixed control volume is given by

$$\sum F = \int_{cs} \rho V(V.\hat{n})\, dA$$

552. In a stream line steady flow two points A and B on a stream line are 1m apart and the flow velocity varies from 2m/s to 5m/s. What is the acceleration of fluid at B in meter per second square?

(a) 3 (b) 6
(c) 9 (d) 15

RPSC ACF & FRO, 26.02.2021
ESE 2015

Ans. (d) : Given,
$V_A = 2$ m/s, $V_B = 5$ m/s, ds = 1 m
If the flow is along stream line,

$$\vec{a}_B = V_B \frac{d\vec{V}}{ds}$$
$$= 5 \times \frac{(5-2)}{1}$$
$$= 5 \times 3$$
$$a_B = 15 \text{ m/s}^2$$

553. The volumetric flow rate (per depth) between two streamlines having stream function and is Ψ_1 and Ψ_2 is

(a) $|\psi_1 + \psi_2|$ (b) $|\psi_1 - \psi_2|$
(c) $\frac{\psi_1}{\psi_2}$ (d) $\Psi_1 \Psi_2$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2016

Ans. (b) : Volumetric flow rate or discharge = $|\psi_1 - \psi_2|$

554. For an incompressible flow field, $\vec{v}$, which one of the following conditions must be satisfied?

(a) $\frac{\partial v}{\partial t} + (\vec{v}.\nabla)\vec{v} = 0$ (b) $\nabla \times \vec{v} = 0$
(c) $(\vec{v}.\nabla) \times \vec{v} = 0$ (d) $\nabla.\vec{v} = 0$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2014

Ans. (d) : $\nabla.\vec{v} = 0$
For incompressible flow
If $\vec{v} = u\hat{i} + v\hat{j} + w\hat{k}$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$
$$\left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right).\left(u\hat{i} + v\hat{j} + w\hat{k}\right) = 0$$
$$\nabla.\vec{v} = 0 \text{ or div } \vec{v} = 0$$

This is known as continuity equation for incompressible flow.

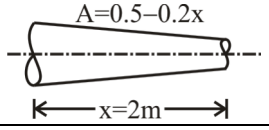
555. The area of a 2 m long tapered duct decreases as A = (0.5 – 0.2x) where 'x' is the distance in meters. At a given instant, a discharge of 0.5 m^3/s is flowing in the duct and is found to increase at a rate of 0.2 m^3/s. The local acceleration (in m^2/s) at x = 0 will be

(a) 1.4 (b) 1.0
(c) 0.4 (d) 0.66

Nagaland PSC CTSE 2015 Paper-II, ESE 1998

Ans. (c) : Given, Area (A) = 0.5 – 0.2x

$$A|_{x=0} = 0.5\text{m}, A|_{x=2} = 0.5 - 0.2 \times 2 = 0.1$$



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u = local acceleration

$$u = \frac{Q}{(0.5 - 0.2x)}$$

$$\frac{\partial u}{\partial t} = \frac{1}{(0.5 - 0.2x)} \times \frac{\partial Q}{\partial t}$$

$$\frac{\partial u}{\partial t} = \frac{1}{0.5} \times 0.2$$

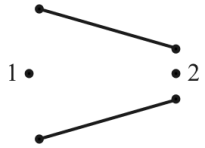
$$= 0.4$$

556. Flow of a liquid at a constant rate through a conical pipe is classified as :

(a) Unsteady, non uniform flow
(b) Unsteady, uniform flow
(c) Steady, non uniform flow
(d) Steady, uniform flow

ISRO Scientist/Engineer (RAC), 10.03.2019
RPSC ADE 2016

Ans. (c) : Steady, non uniform flow– Flow will be steady because liquid flows at a constant rate.
non uniform because of conical cross-section of pipe.



$\because \quad A_2 < A_1$

$\therefore \quad V_2 < V_1$ (From continuity equation)

$\boxed{V_2 \neq V_1}$ So, flow is non uniform

557. In one dimensional flow, the flow

(a) is steady and uniform
(b) takes place in straight line
(c) takes place in curve
(d) takes place in one direction

Nagaland PSC (CTSE) 2018, Paper-II
Sikkim PSC (Under Secretary), 2017

Ans. (d) : • The flow in a pipe is considered one-dimensional when variations of pressure and velocity occur along the length of the pipe, but any variation over the cross-section is assumed negligible.

• The flow in which the velocity is the function of time and one space co-ordinate (x) is called as one-dimensional flow.

Example–Flow through the pipe is consider as a one dimensional flow.

u = f(x), v = 0, w = 0

558. The stream function in a two-dimensional flow field is given by $\Psi = x^2 + y^2$. The magnitude of the velocity at point (1, 1) will be

(a) 2 (b) $2\sqrt{2}$
(c) 4 (d) $4\sqrt{2}$

APPSC Poly Lect. 13.03.2020
UPSC JWM Advt. No.-50/2010

Ans. (b) : Given, $\psi = x^2 + y^2$, v = ? point (1, 1)
We know that, for stream function (2D) –

$$u = -\frac{\partial \psi}{\partial y} \;\&\; v = \frac{\partial \psi}{\partial x}$$

We get,

$$u = -2y \qquad \Rightarrow \qquad v = 2x$$

$$\vec{v} = u\hat{i} + v\hat{j}$$

$$\vec{v}_{(1,1)} = -2(1)i + 2(1)j \qquad \left[\because \text{Given, at point } (1,1)\right]$$

$$= -2i + 2j$$

$$|\vec{v}| = \sqrt{(-2)^2 + 2^2} = 2\sqrt{2}\ \text{m/s}$$

559. Of the possible irrotational flow functions given below, the incorrect relation is (where ψ = stream function and ϕ = velocity potential)

(a) $\psi = xy$
(b) $\psi = A(x^2 - y^2)$
(c) $\phi = ur\cos\theta + \frac{u}{r}\cos\theta$
(d) $\phi = \left(r - \frac{2}{r}\right)\sin\theta$
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
ESE 1995

Ans. (d) : If the stream function satisfy the Laplace equation the flow is irrotational, otherwise rotational.

If $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$ (Laplace equation)

Then, $\omega_z = 0$

a. $\psi = xy$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \frac{\partial^2 (xy)}{\partial x^2} + \frac{\partial^2 (xy)}{\partial y^2}$$

$$= 0 + 0 = 0$$

Satisfy the Laplace equation therefore flow is irrotational.

B. $\psi = A(x^2 - y^2)$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \frac{\partial^2 \left[A\left(x^2 - y^2\right)\right]}{\partial y^2} + \frac{\partial^2 \left[A\left(x^2 - y^2\right)\right]}{\partial y^2}$$

$= 2A - 2A = 0$

Satisfy Laplace equation, therefore flow irrotational.

C. $\nabla^2 \phi = 0$

$$\left.\begin{aligned} V_r &= -\frac{\partial \phi}{\partial r} \\ V_\theta &= -\frac{1}{r}\frac{\partial \phi}{\partial \theta} \end{aligned}\right\} \text{Cylindrical coordinate}$$

$$2\omega_z = \frac{1}{r}\frac{\partial}{\partial_r}\left[-\frac{1}{r}\frac{\partial \phi}{\partial \theta}\right] - \frac{1}{r^2}\frac{\partial}{\partial \theta}\left[-\frac{\partial \phi}{\partial r}\right]$$

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$$\phi = ur\cos\theta + \frac{u}{r}\cos\theta$$

$$\frac{\partial\phi}{\partial r} = u\cos\theta - \frac{u}{r^2}\cos\theta$$

$$\frac{\partial^2\phi}{\partial r^2} = \frac{2u}{r^3}\cos\theta$$

$$\frac{\partial\phi}{\partial\theta} = -ur\sin\theta - \frac{u}{r}\sin\theta$$

$$\frac{\partial^2\phi}{\partial\theta^2} = -ur\cos\theta - \frac{u}{r}\cos\theta$$

The Laplace equation, in radial coordinates (r, θ) is given by,

$$\frac{1}{r}\frac{\partial\phi}{\partial r} + \frac{\partial^2\phi}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} = 0$$

Substituting for L.H.S. terms, we get :

$$= \frac{1}{r}\left(u\cos\theta - \frac{u}{r^2}\cos\theta\right) + \frac{2u}{r^3}\cos\theta + \frac{1}{r^2}\left(-ur\cos\theta - \frac{u}{r}\cos\theta\right)$$

$$u\cos\theta\left(\frac{1}{r} - \frac{1}{r^3} + \frac{2}{r^3} - \frac{1}{r} - \frac{1}{r^3}\right) = 0$$

The Laplace equation is satisfied and hence the given function ϕ represents a possible irrotational flow.

D. $\phi = \left(r - \frac{2}{r}\right)\sin\theta$

Laplace equation in radial coordinates (r, θ) is given as:

$$\frac{1}{r}\frac{\partial\phi}{\partial r} + \frac{\partial^2\phi}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} = 0$$

$$\frac{\partial\phi}{\partial r} = \left(1 + \frac{2}{r^2}\right)\sin\theta$$

$$\frac{\partial^2\phi}{\partial r^2} = -\frac{2}{r^3}\sin\theta$$

$$\frac{\partial\phi}{\partial\theta} = \left(r - \frac{2}{r}\right)\cos\theta$$

$$\frac{\partial^2\phi}{\partial\theta^2} = -\left(r - \frac{2}{r}\right)\sin\theta = \left(\frac{2}{r} - r\right)\sin\theta$$

L.H.S of Laplace equation is :

$$\frac{1}{r}\times\left(1 + \frac{2}{r^2}\right)\sin\theta - \frac{2}{r^3}\sin\theta + \frac{1}{r^2}\left(\frac{2}{r} - r\right)\sin\theta$$

$$= \sin\theta\left(\frac{1}{r} + \frac{2}{r^3} - \frac{2}{r^3} + \frac{2}{r^3} - \frac{1}{r}\right)$$

$$= \sin\theta\left(\frac{2}{r^3}\right) \neq 0$$

Hence, the given function does not represent any possible irrotational flow.

560. A flow field which has only convective acceleration is

(a) a steady uniform flow
(b) an unsteady uniform flow
(c) a steady non-uniform flow
(d) an unsteady non-uniform flow
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
GATE 2014

Ans. (c)

Type of flow	Acceleration	
	Temporal	Convective
steady uniform flow	0	0
steady non-uniform flow	0	exists
unsteady uniform flow	exists	0
unsteady non-uniform flow	exists	exists

561. The equation of streamline passing through (1, 1) for the velocity function $V = -y^2\hat{i} - 6x\hat{j}$ is:

(a) $10x^2 - 2y^3 = 8$ (b) $9x^3 - y^3 = 8$
(c) $9x^2 - y^3 = 8$ (d) $9x^2 - y^2 = 8$

SJVN ET 2019
AAI Jr. Executive 29.11.2018

Ans. (c) : $V = -y^2\hat{i} - 6x\hat{j}$

Stream function,

$$\frac{dx}{u} = \frac{dy}{v}$$

$$\frac{dx}{-y^2} = \frac{dy}{-6x}$$

$$\int dx(-6x) = \int -y^2 dy$$

$$-6\frac{x^2}{2} = \frac{-y^3}{3} + C$$

At point (1, 1)

$$-3x^2 = -\frac{y^3}{3} + C \qquad \text{.....(1)}$$

$$-3 = \frac{-1}{3} + C$$

$$C = -\frac{8}{3}$$

Equation (1)

$$-3x^2 = -\frac{y^3}{3} - \frac{8}{3}$$

$$-9x^2 = [-y^3 - 8]$$

$$9x^2 - y^3 = 8$$

562. If the total acceleration of fluid flow is always zero, then it is:

(a) unsteady and uniform flow
(b) steady and uniform flow
(c) steady but non-uniform flow
(d) unsteady and non-uniform flow

AAI Jr. Executive 26.03.2021
BHEL ET 2019

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Ans. (b) :

	Convective Acceleration	Local Acceleration or temporal	Total Acceleration
Steady and uniform	0	0	0
Unsteady and uniform	0	exist	exist
steady and non-uniform	exist	0	exist
unsteady and non-uniform	exist	exist	exist

563. The tangent drawn to the instantaneous velocity in a flow field is called _____ line.

(a) path (b) streak
(c) time (d) stream

AAI Jr. Executive 26.03.2021

Ans. (d) : The tangent drawn to the instantaneous velocity in a flow field is called stream line.

564. A flow is said to be rotational when

(a) the streamlines are curved
(b) a velocity gradient in the normal direction to flow exists
(c) every fluid element has finite angular velocity about its mass centre
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : Rotational Flow–Rotational flow is that type of flow in which the fluid particles has finite angular velocity about its mass centre. Flow along steamlines, also rotate about their own axis.

565. Which of the following characteristics regarding fluid kinematics is/are correct?

1. **Streamline represents an imaginary curve in the flow field so that the tangent to the curve at any point represents the direction of instantaneous velocity at that point.**
2. **Path lines, streamlines and streak lines are identical in steady flow.**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

ESE 2020

Ans. (c) : Both 1 and 2

566. A velocity field is given as

$\vec{V} = Axy\hat{i} - Byzt\hat{j}$

Where, A and B are constants, x, y, z are in meter and t is in seconds. Which of the following is true of this flow field?

(a) Steady and 3-dimensional
(b) Unsteady and 2-dimensional
(c) Steady and 2-dimensional
(d) Unsteady and 3-dimensional

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given, velocity field equation –

$\vec{V} = Axy\hat{i} - Byzt\hat{j}$ ..... (1)

Let In x, y and z direction velocities u, v and w are respectively.

Compare with, $\vec{V} == u\hat{i} + v\hat{j} + w\hat{k}$

From equation 1st

$u = Axy$

$v = -Byzt$

$w = 0$

For steady, incompressible density will be constant thus applying continuity equation for 3-D –

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\therefore \frac{\partial}{\partial x}(Axy) + \frac{\partial}{\partial y}(-Byzt) + \frac{\partial(0)}{\partial z} = 0$$

$= Ay - Bzt + 0$

$= Ay - Bzt$

$\neq 0$

Hence flow is unsteady because given velocity field is not satisfied the continuity equation.

Result : Hence the flow is unsteady and 2-dimensional

567. The velocity potential function of a 2D flow is given by $\phi = 4xy$. The velocity at a point (5,0) is _____.

(a) 20 units (b) 10 units
(c) 30 units (d) 40 units

VIZAG MT, 14.12.2020

Ans. (a) : $\phi = 4xy$, $v(5, 0) = ?$

$$u = -\frac{d\phi}{dx} = -\frac{d}{dx}4xy = -4y$$

$u_{(5, 0)} = 0$

$$v = -\frac{d\phi}{dy} = -\frac{d}{dy}(4xy) = -4x$$

$v_{(5, 0)} = -4 \times 5 = -20$

Velocity, $V = \sqrt{u^2 + v^2}$

$= \sqrt{(-20)^2 + 0^2}$

$\boxed{V = 20 \text{ units}}$

568. If the x and y components of velocity in a two-dimensional incompressible flow are given as u = 3x + 3y and v = 2x – 3y the value for vorticity ω_z is

(a) 5 (b) –5
(c) 1 (d) –1

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : $u = 3x + 3y$

$$\frac{\partial u}{\partial y} = 3$$

$v = 2x - 3y$

$$\frac{\partial v}{\partial x} = 2$$

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Vorticity = 2 × rotational component $= 2\times\frac{1}{2}\left[\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right]$

$= \frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}$

$= 2 - (3) = -1$

569. For a two-dimensional irrotational velocity flow with potential function given by $\phi = 2(x^2+2y-y^2)$, the velocity vector is given as

(a) $-4xi - 4(1-y)j$

(b) $-4xi - 4(1-y)j$

(c) $4xi + 4(1-y)j$

(d) $4xi - 4(1-y)j$

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : $\phi = 2(x^2 + 2y - y^2)$

$u = -\frac{\partial\phi}{\partial x}, \qquad v = \frac{-\partial\phi}{\partial y}$

$u = -2(2x) = -4x$

$v = -2(2-2y) = -4 + 4y = -4(1 - y)$

Net velocity $V = ui+vj = -4xi-4(1-y)j$

570. Which method is most commonly used in fluid mechanics for analysis?

(a) Lagrangian method

(b) Eulerian method

(c) Control volume analysis

(d) None of the above

Assam PSC AE (PHED), 18.10.2020

Ans. (b) : In a fluid mechanics the matter of concern is the general state of motion at various points in the fluid system (as in Eulerian approach) rather than the motion of each particle (as in lagrangian approach). Hence, the Eulerian method is most commonly used in fluid mechanics for variance analysis.

In this method we describes velocity, acceleration pressure etc. at a point in flow field.

571. An incompressible flow is represented by the velocity potential function $\phi = 4x^2 + 4y^2 + 17t$. For the flow, which one of the combinations of the following statement holds true?

i. Flow is physically possible;

ii. Flow is physically not possible;

iii. Flow satisfies the continuity equation;

iv. Flow does not satisfy the continuity eq.

(a) (i) and (iv) (b) (i) and (iii)

(c) (ii) and (iii) (d) (ii) and (iv)

HPPSC ADF 05.03.2019

Ans. (d) : $\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2} = 8+8 = 16$

$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2} \neq 0$

So, flow is physically not possible and flow does not satisfy the continuity eq.

572. What will be the approximate resultant velocity at point (1, 2), if the stream function for a two dimension flow is given by $\Psi = x^3 + y^2$?

(a) 3.0 (b) 5.0

(c) 7.0 (d) None of these

OPSC AEE 2019 PAPER - II

Ans : (b) : Given, $\psi = x^3 + y^2$

$u = \frac{\partial\psi}{\partial y} = 2y$

$v = \frac{-\partial\psi}{\partial x} = -3x^2$

At point (1,2) u = 4, v = –3,

Resultant velocity,

$V = \sqrt{u^2+v^2} = \sqrt{(4)^2+(-3)^2} = \sqrt{16+9} = \sqrt{25}$

$\boxed{V = 5}$

573. A Streamline is defined as:

(a) Line parallel to outer surface of pipe

(b) Line parallel to central axis of flow

(c) Line along which pressure drop is uniform

(d) Line of equal velocity in a flow

OPSC AEE 2019 PAPER - II

Ans : (d) : Streamline– streamline, in fluid mechanics, the path of imaginary particles suspended in the fluid and carried along with it. In steady flow, the fluid is in motion but the streamlines are fixed.

574. In which type of flow the stream function satisfies the Laplace equation?

(a) Rotational Flow Crest

(b) Irrotational Flow

(c) Circular Flow

(d) Never Possible

OPSC AEE 2019 PAPER - II

Ans. : (b) : The stream function satisfies the Laplace equation in irrotational flow.

575. The equation, $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y} = 0$ represents

(a) two-dimensional, compressible, continuity equation

(b) two-dimensional, incompressible, continuity equation

(c) two-dimensional, compressible, momentum equation

(d) two-dimensional, incompressible, momentum equation

APPSC AEE SCREENING 17.02.2019

Ans. (b) : $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y} = 0$ represents 2D continuity equation for incompressible flow.

576. If dr is a directed fluid element, the equation, dr × V = 0 refers to a

(a) Pathline (b) Timeline

(c) Streakline (d) Streamline

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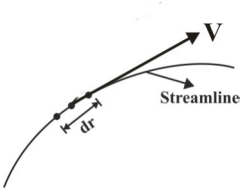
Ans. (d) : As per definition of streamline velocity is tangential to it. Hence $\vec{V}$ and $d\vec{r}$ are parallel to each other. If two vectors are parallel then their cross product is zero.



577. If the flow field is steady, the fluid particle will undergo only a
(a) Convective acceleration
(b) Local acceleration
(c) Total acceleration
(d) All of the above

APPSC AEE SCREENING 17.02.2019

Ans. (a) : Total acceleration is sum of local and convective acceleration. If flow is steady then local acceleration is zero. Hence only convective acceleration can be present.

578. Which of the following function is valid potential function (ϕ)?
(a) $x^2 - 3x^2y$ (b) $y^4 - 6x^2y^2$
(c) $y^3 - 3x^2y$ (d) $x^3 - y^3$

SJVN ET 2019

Ans. (c) : For potential function validation

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0$$

Given,

$$\phi = y^3 - 3x^2y$$

$$\frac{\partial\phi}{\partial x} = 0 - 6xy$$

$$\frac{\partial^2\phi}{\partial x^2} = -6y \text{ ————(i)}$$

$$\frac{\partial\phi}{\partial y} = 3y^2 - 3x^3$$

$$\frac{\partial^2\phi}{\partial y^2} = 6y - 0$$

$$\frac{\partial^2\phi}{\partial y^2} = 6y \text{ ————(ii)}$$

From (i) + (ii)

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2} = -6y + 6y$$

$$\boxed{\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0}$$

579. Which of the following sets of equations represents the possible 2-D, incompressible flow?
(a) $u = 4x + y, v = x - y^2$
(b) $u = x + 2y, v = x^2 - y^2$
(c) $u = x + y, v = x - y$
(d) $u = xt + 2y, v = x^2 - yt^2$

SJVN ET 2019

Ans. (c) : Possible case of 2-D fluid flow satisfy continuity Eq.

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

580. The stream function for a flow field is $\psi = 3x^2y + (2+t)y^2$. The velocity at a point P for position vector $r = i + 2j - 3k$ time $t = 2$ will be
(a) $19i - 12j$ (b) $21i - 12j$
(c) $19i + 22j$ (d) $21i + 22j$

ESE 2019

Ans. (a) : Given,

$$\psi = 3x^2y + (2+t)y^2$$
$$r = i + 2j - 3k$$
$$t = 2$$

On comparing position vector

$$r = xi + yj + zk$$
$$x = 1 \quad y = 2 \quad z = -3$$

$$u = \frac{\partial\psi}{\partial y} = 3x^2 + 2(2+t)y$$

at $x = 1$ and $t = 2$

$$u = 3(1)^2 + 2(2+2)2 = 19$$

$$v = -\frac{\partial\psi}{\partial x} = -6xy$$

at $x = 1, y = 2$

$$v = -6(1)(2) = -12$$

Velocity vector $\vec{v} = u\hat{i} + v\hat{j}$

$$= 19\hat{i} - 12\hat{j}$$

581. A flow through an expanding tube at constant rate is called
(a) Steady uniform flow
(b) Steady non-uniform flow
(c) Unsteady uniform flow
(d) Unsteady non-uniform flow

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : A flow through an expanding tube at constant rate is called steady flow.
Steady non-uniform flow condition change from point to point in the stream but do not change with time.

582. In which method of describing fluid motion, the observer remains stationary and observe changes in the fluid parameters at a particular point only?
(a) Lagrangian method (b) Eulerian method
(c) Stationary method (d) All of the above

Vadodara Muncipal Corp. DEE, 2018

Ans. (a) : In Lagrangian method of describing fluid motion observer remains stationary and observe changes in the fluid parameters at a point only.

583. If the stream function of two dimensional flow is $\psi = 5xy$, then the velocity at a point (3, 4) is
(a) 25 ms^{-1} (b) 20 ms^{-1}
(c) 7 ms^{-1} (d) 5 ms^{-1}

ISRO Scientist/Engineer 22.04.2018

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Ans. (a) : Given, $\psi = 5xy$, at point (3, 4) velocity in x-direction.

$$U = \frac{\partial \psi}{\partial y} = 5x$$

$$U_{(3,4)} = 15 m/s$$

Velocity in y-direction

$$V = -\frac{\partial \psi}{\partial x} = -5y$$

$$V_{(3,4)} = -20 m/s$$

$$V'_{(3,4)} = \sqrt{U^2 + V^2} = \sqrt{(15)^2 + (-20)^2}$$

$V'_{(3,4)} = 25$ m/s.

584. The continuity equation is satisfied by

(a) $u = A \sin xy, v = -A \sin xy$

(b) $u = x + y, v = x - y$

(c) $u = 2x^2 + cy, v = 3y^2$

(d) $u = x + 2y, v = 2x + y$

RPSC Vice Principal ITI 2018

Ans. (b) : For incompressible possible flow, continuity equation must be satisfied.

$$\boxed{\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0}$$

1. $u = A \sin xy, \quad v = -A \sin xy$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = yA \cos xy - xA \cos xy,$$ Not satisfied

2. $u = x + y, v = x - y$

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 1 - 1 = 0,$$ Satisfied

3. $u = 2x^2 + cy, v = 3y^2$

$$\frac{\partial u}{\partial x} = 4x + 0 = 4x$$

$$\frac{\partial v}{\partial y} = 2 \times 3y = 6y$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 4x + 6y,$$ Not satisfied

4. $u = x + 2y, v = 2x + y$

$$\frac{\partial u}{\partial x} = 1 + 0 = 1$$

$$\frac{\partial v}{\partial y} = 0 + 1 = 1$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 1 + 1 = 2,$$ Not satisfied

585. Three dimensional flow field is described by $V = \left(y^2 + z^2\right)\hat{i} + \left(x^2 + z^2\right)\hat{j} + \left(x^2 + y^2\right)\hat{k}$. **The component of rotation at (1, 2, 3) is given as:**

(a) $\omega_x = -1, \omega_y = 2, \omega_z = -1$

(b) $\omega_x = -1, \omega_y = -2, \omega_z = 1$

(c) $\omega_x = -1, \omega_y = -2, \omega_z = -1$

(d) $\omega_x = 1, \omega_y = 2, \omega_z = -1$

(e) $\omega_x = 1, \omega_y = -2, \omega_z = -1$

CGPSC AE 25.02.2018

Ans. (a) : Given

$$u = \left(y^2 + z^2\right), v = \left(x^2 + z^2\right), w = x^2 + y^2$$

$$\omega_x = \frac{1}{2}\left[\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right] = \frac{1}{2}\left[2y - 2z\right]_{(1,2,3)}$$

$$\omega_x = \frac{1}{2} \times \left[2 \times 2 - 2 \times 3\right] = -1$$

$$\boxed{\omega_x = -1}$$

$$\omega_y = \frac{1}{2}\left[\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right] = \frac{1}{2}\left[2z - 2x\right]_{(1,2,3)}$$

$$\omega_y = \frac{1}{2} \times \left[2 \times 3 - 2 \times 1\right]$$

$$= \frac{1}{2} \times 4 = 2$$

$$\boxed{\omega_y = 2}$$

$$\omega_z = \frac{1}{2}\left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right] = \frac{1}{2}\left[2x - 2y\right]_{(1,2,3)}$$

$$\omega_z = \frac{1}{2}\left[2 \times 1 - 2 \times 2\right] = \frac{1}{2} \times (-2)$$

$$\boxed{\omega_z = -1}$$

$$\boxed{\omega_x = -1, \omega_y = 2, \omega_z = -1}$$

586. If the velocity vector in a two dimensional flow field is given by $\vec{v} = 2xy\hat{i} + (2y^2 - x^2)\hat{j}$; then the vorticity vector curl $\vec{v}$ will be-

(a) $2y^2\,\hat{j}$ (b) $6y\,\hat{k}$

(c) Zero (d) $-4x\,\hat{k}$

RPSC AE 2018

Ans. (d) : Given, $\vec{v} = 2xy\hat{i} + (2y^2 - x^2)\hat{j}$

$u = 2xy$

$v = 2y^2 - x^2$

then

Vorticity vector

Vorticity $(\xi) = 2\vec{\omega}$

$$\xi = 2 \times \frac{1}{2} curl\ \vec{V}$$

$$= curl\ \vec{V}$$

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$$= \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & [2y^2 - x^2] & 0 \end{bmatrix}$$

$$= \hat{i}[0-0] - \hat{j}[0-0] + k[-2x-2x]$$

$$\xi = -4x\hat{k}$$

587. Statement I: When flow is unsteady, both normal and tangential components of acceleration will occur.
Statement II: During unsteady flow, in addition to the change of velocity along the path, the velocity will also change with time.

ESE 2018

Ans. (d) : Unsteady flow does not necessarily mean that normal acceleration is non-zero.
For example if streamlines are parallel then even in unsteady flow the normal acceleration is zero.
Normal component will only occur in case of a curved streamline.

588. A velocity field is described as

$$\vec{V} = 3x^2y\hat{i} - 6xyz\hat{k}$$

Where x, y, z are in m and V in m/s. The above equation represents

(a) A rotational flow
(b) An irrotational flow
(c) A steady flow
(d) An incompressible flow

GPSC Executive Engineer 23.12.2018

Ans. (d) : Given:

velocity field $(\vec{v}) = (3x^2y)\hat{i} - (6xyz)\hat{k}$

(i) check for incompressible flow:- It is satisfies continuity equation for steady and incompressible flow.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\frac{\partial}{\partial x}(3x^2y) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(-6xyz)$$

$$6xy + 0 + (-6xy) \Rightarrow 6xy - 6xy = 0$$

The flow velocity components satisfied law of conservation of mass i.e., continuity equation hence the given flow field represents and "incompressible flow".

(ii) Check for irrotational flow:-

$$w_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)$$

$$w_y = \frac{1}{2}\left(\frac{\partial}{\partial z}(3x^2y) = \frac{\partial}{\partial x}(-6xyz)\right)$$

$$w_y = \frac{1}{2}(0 + 6yz)$$

$$w_y \neq 0$$

Hence, flow is "rotational".
(iii) Check for steady flow:- The given flow is steady since the velocity components of each does not contain time 't' (i.e., "t" appears explicitly in case of un-steady flow). Hence, given flow field is "steady flow".

589. A point moves according to the relation $x = 8t^2 \cos(\omega t)$, $y = -4t^3 \sin(\omega t)$, where x and y are in meters, and t in seconds. If angular velocity $\omega = 8\pi$ rad/sec, velocity vector of the point at t = 0.5 sec is given by

(a) $8 + j4\pi$ (b) $6 + j20\pi$
(c) $8 - j4\pi$ (d) $-20\pi - j6$

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : $x = 8t^2 \cos(\omega t)$
$y = -4t^3 \sin(\omega t)$
$\omega = 8\pi$
$t = 0.5$

$$u = \frac{dx}{dt} = 16t\cos(\omega t) - 8t^2 \cdot \omega \cdot \sin(\omega t)$$

$$v = \frac{dy}{dt} = -12t^2 \sin(\omega t) - 4t^3 \cdot \omega \cdot \cos(\omega t)$$

Velocity vector at t = 0.5 sec

$V = ui + vj$
$= [16 \times 0.5 \cos(8\pi \times 0.5) - 8(0.5)^2 \times 8\pi \sin(8\pi \times 0.5)]i + [-12(0.5)^2 \sin(8\pi \times 0.5) - 4(0.5)^3 \times 8\pi \times \cos(8\pi \times 0.5)]j$
$= [8\cos(4\pi) - 16\pi \sin(4\pi)]i + [-3\sin(4\pi) - 4\pi \cos(4\pi)]j$

as $\sin(n\pi) = 0$
$= 8\cos(4\pi) - 4\pi j \cos(4\pi)$
$\cos(n\pi) = 1$ if n is even
$= 8 - 4\pi j$

590. In a doublet equipotential lines are :

(a) Circles tangent to y-axis
(b) Circles tangent to x-axis
(c) Concentric circle with centre on y-axis
(d) Concentric circle with centre on x-axis

Kerala PSC Poly. Lect., 2017

Ans. (a) : It is seen that the streamlines are circles which are tangential to the x-axis while the equipotential lines are also circles but tangential to y-axis.

591. A flow through a long pipe at constant rate is called

(a) steady uniform flow
(b) steady non-uniform flow
(c) unsteady uniform flow
(d) unsteady non-uniform flow

Sikkim PSC (Under Secretary), 2017

Ans. (a) : A flow through a long pipe at constant rate is called steady uniform flow.

592. A path line describes :

(a) The velocity direction at all points on the line.
(b) The path followed by particles in a flow

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(c) the path over a period of time of a single particle that has passed out at a point
(d) The instantaneous position of all particles that have passed a point.

HPPSC Asstt. Prof. 20.11.2017

Ans. (c) : A path line describes the path over a period of times of a single particle that has passed out at a point.

593. In a steady flow of incompressible fluid, as the diameter is doubled, the velocity will
(a) be halved (b) be doubled
(c) increase four fold (d) decrease four fold

TNPSC AE 2017

Ans. (d) : We know that for incompressible fluid,

$Q = A_1V_1 = A_2V_2 = \text{constant}$

$$\frac{\pi}{4}d_1^2 \times V_1 = \frac{\pi}{4}d_2^2 \times V_2$$

$$\left[\frac{V_2}{V_1}\right] = \left[\frac{d_1}{d_2}\right]^2 = \left[\frac{d}{2d}\right]^2$$

$$\boxed{V_2 = \frac{V_1}{4}}$$

594. Which one of the following is a valid potential function?
(a) $\phi = c\ln x$ (b) $\phi = c\cos x$
(c) $\phi = 3xy$ (d) $\phi = c(x^2+y^2)$

TNPSC AE 2017

Ans. (c) : For possible velocity field ϕ satisfied Laplace equation

$$\frac{\partial \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

then, $\phi = 3xy$
satisfied Laplace equation.

595. Consider the following statements:
1. Stream lines and path lines are instantaneous lines
2. Path lines and streak line are generated by passage of time
3. Path line can be found by time exposure of a marked particle
4. Stream lines, path lines and streak lines are identical in steady flow

Which of the above statements are correct?
(a) 1, 2 and 3 only (b) 2, 3 and 4 only
(c) 1 and 4 only (d) 1, 2, 3 and 4

UPSC JWM 2017

Ans. (b) : Stream line is one that drawn tangential to the velocity vector at every point in the flow at a given instant.
- Path line is the line traced by a given particle.
- Streak line concentrated on fluid particles that have gone through a fixed point.
- Path lines and streak line are generated by passage of time.
- Path lines can be found by time exposure of a marked particle.
- Stream lines, path lines and streak line are identical in steady flow.

596. A steady, incompressible flow is given by $u = x^2 + y^2$ and $v = 2xy$. What is the convective acceleration along x direction at the point (1, 1)?
(a) 12 unit (b) 6 unit
(c) 8 unit (d) 24 unit

CIL MT 26.03.2017

Ans. (c) : Convective acceleration along x direction

$$a_x = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}$$

$$u = x^2 + y^2 \Rightarrow \frac{du}{dx} = 2x, \quad \frac{du}{dy} = 2y$$

$a_x = (x^2+y^2)2x + 2xy \times 2y$
$= 2x^3 + 2xy^2 + 4xy^2$
$(a_x)_{1,1} = 2 + 2 + 4 = 8$ unit.

597. A two dimensional flow field is defined as $\vec{V} = \vec{i}x - \vec{j}y$. The equation of the stream line passing through the point (1, 2) is
(a) $xy + 2 = 0$ (b) $x^2y + 2 = 0$
(c) $xy - 2 = 0$ (d) $x^2y - 2 = 0$

ESE 2017

Ans. (c) : Given, $V = x\hat{i} - y\hat{j}$ point = (1, 2)

$u = x$

$v = -y$

Streamline equation $\frac{dx}{u} = \frac{dy}{v}$

$$\frac{dx}{x} = \frac{dy}{-y}$$

Integrate on both the sides $\int\frac{dx}{x} = -\int\frac{dy}{y}$

$$\ell nx = -\ell ny + \ell nc$$

$$\ell nx + \ell ny = \ell nc$$

$$xy = c \quad \text{.....(1)}$$

at point (1, 2) from equation (1)

$$(1)\ (2) = c$$

$$c = 2$$

$$xy = 2$$

$$xy - 2 = 0$$

598. A stream line is defined as the line:
(a) parallel to central axis of flow
(b) parallel to outer surface of pipe
(c) along which the pressure drop is uniform
(d) none of the above

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : A stream line is an imaginary line drawn in a flow field such that a tangent drawn at any point on this line represents the direction of velocity vector at that point.
There is no velocity component normal to stream lines.
There is no discharge across a streamline.

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599. If flow conditions satisfy 'Laplace equation', then :

(a) flow is rotational
(b) flow does not satisfy continuity equation
(c) flow is irrotational and satisfy continuity equation
(d) flow is irrotational and does not satisfy continuity equation

(HPPSC LECT. 2016)

Ans : (c) If flow conditions satisfy Laplace equation then flow is irrotational and satisfy continuity equation. Laplace equation is given by

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0$$

if velocity potential function satisfies Laplace equation then it also satisfies continuity equation and hence the flow is possible.

600. If Velocity potential satisfies Laplace equation the flow will be

(a) Irrotational flow (b) Free vortex flow
(c) Continuous flow (d) Forced vortex flow

APPSC AEE Screening Test 2016

Ans. (a) : Velocity Potential (ϕ)–

- Flow always occurs in the direction of decreasing potential
- If velocity potential (ϕ) exists and satisfy continuity equation, then it represents the possible irrotational flow.
- If (ϕ) satisfy Laplace equation then it is the case of steady incompressible and irrotational flow.
- Laplace equation

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0$$

601. The velocity potential for a two dimensional flow is $\phi = x(2y-1)$ determine the velocity at a point P(4,5).

(a) 10.25 unit (b) 12.04 unit
(c) 15.35 unit (d) 18.74 unit

RPSC IOF & Boiler 20.09.2017

Ans : (b) Velocity potential of flow

$$\phi = x(2y-1)$$
$$= 2xy - x$$
$$u = \frac{-\partial\phi}{\partial x} = -(2y-1)$$

At point P (4, 5)

$$u = -(2\times 5 - 1) = -9$$
$$v = -\frac{\partial\phi}{\partial y} = -2x = -2\times 4 = -8$$

Resultant velocity at point 'P'

$$v = \sqrt{u^2+v^2} = \sqrt{(-9)^2+(-8)^2}$$
$$\boxed{v = 12.04\,\text{unit}}$$

602. Fluid velocity along a streamline is given by $V = (V_0 xt)\hat{i}$. Its acceleration will be given by:

(a) $V_0 x + V_0^2 xt^2$ (b) $V_0^2 t^2$
(c) $V_0 x$ (d) $V_0 x + V_0 xt^2$

UPRVUNL AE 2016

Ans. (a) : $V = (V_o xt)\hat{i} + o.\hat{j} + o\hat{k}$

$u = V_o xt$, v = 0, w = 0

then acceleration in x-direction

$$\frac{dU}{dx} = u\frac{\partial u}{\partial x} + v.\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}$$
$$\frac{dU}{dx} = (V_o xt)\times\left[\frac{\partial}{\partial x}(V_o xt)\right] + \frac{\partial}{\partial t}(V_o xt)$$
$$= (V_o xt)\times V_o t + V_o x$$
$$\frac{dU}{dx} = V_o x + V_0^2 xt^2$$

603. A potential function exists for

(a) Steady flow only
(b) Two dimensional irrotational flow only
(c) Irrotational flow of fluid whether compressible or incompressible
(d) Irrotational flow of incompressible fluids only

UPPSC AE 12.04.2016 Paper-II

Ans : (d) A potential function exists for irrotational flow of incompressible fluids only.

Velocity potential function (ϕ) : -

$$u = \frac{-\partial\phi}{\partial x}, v = \frac{-\partial\phi}{\partial y}, w = \frac{-\partial\phi}{\partial z}.$$

* Flow always occurs in the direction of decreasing potential function.
* Velocity potential function exists only for irrotational flow.
* Laplace equation is given by ;

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0$$

If velocity potential function satisfies Laplace equation then it also satisfies continuity equation and hence the flow is possible.

604. A stream function

(a) Is a mathematical function which has no physical equivalence
(b) Is defined only for steady and incompressible flow
(c) Satisfies Laplace equation for rotational motion
(d) May not remain constant for a streamline

UPPSC AE 12.04.2016 Paper-II

Ans : (b) A stream function is defined only for steady and incompressible flow.

stream function (ψ) :-

$$v = \frac{\partial\psi}{\partial x}, u = -\frac{\partial\psi}{\partial y}$$

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* If ψ exists, then it satisfies continuity equation and flow may be rotational or irrotational.
* If stream function satisfies Laplace equation then it is case of irrotational flow

605. A body is called streamline body when

(a) It is symmetrical about the axis along the free stream
(b) Surface of the body coincides with the streamlines
(c) Flow is laminar around it
(d) It produces no drag for flow around it

UPPSC AE 12.04.2016 Paper-II

Ans : (b) A body is called streamline body when Surface of the body coincides with the streamlines.

606. The equation of continuity is (a = cross sectional area, V = velocity, W = weight per unit area)

(a) $W_1a_1 = W_2a_2$ (b) $W_1V_1 = W_2V_2$
(c) $a_1/V_1 = a_2/V_2$ (d) $a_1V_1 = a_2V_2$

GPSC ARTO 01.05.2016

Ans. (d) : Continuity equation $\rightarrow$ Rate of flow is constant.

So, $Q_1 = Q_2$

$$\boxed{A_1V_1 = A_2V_2}$$

Where,

Q_1 = Rate of flow in (m^3/sec)
A = Area of cross-section (m^2)
V = Velocity of flow (m/sec.)

607. If the velocity distribution over parallel planes is identical in a flow region, then the flow is treated as

(a) one dimensional (b) two dimensional
(c) three dimensional (d) Axis-Symmetric

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : 2D- flow fluid motion can be said to be a two-Dimensional flow when the flow velocity at every point is parallel to a fixed plane.

608. The velocity field for a flow is given by v = (5x + 6y + 7z) i+(6x + 5y + 9z)j+(3x + 2y + λz)k and density varies as $\rho = \rho_0 e^{-2t}$. In order that the mass is conserved, the value of λ should be

(a) –12 (b) –10
(c) –8 (d) 10

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : $v = (5x + 6y + 7z)\,i + (6x + 5y + 9z)\,j + (3x + 2y + \lambda z)k$

$$\rho = \rho_0 e^{-2t}$$

So, using continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} + \frac{\partial \rho v}{\partial y} + \frac{\partial \rho w}{\partial z} = 0$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}\rho(5x + 6y + 7z) + \frac{\partial \rho}{\partial y}(6x + 5y + 9z) + \frac{\partial y}{\partial z}(3x + 2y + \lambda z) = 0$$

$$\rho_0 e^{-2t}(-2 + 5 + 5 + \lambda) = 0 \Rightarrow \lambda = -8$$

609. Uniform flow occurs

(a) when the spatial rate of the change of velocity is zero
(b) when the temporal rate of the change of velocity is zero
(c) when the velocity changes steadily along the direction of the flow
(d) only when velocity vector at any point remains constant.

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : Uniform flow is defined as the type of flow in which the velocity of any given time does not change with respect to space. It will occur when the spatial rate of the change of velocity is zero.

610. The x-component of velocity in a 2D incompressible flow is given by u = 1.5x. At the point (x,y) = (1,0), the y-component of velocity v = 0. The equation for the y-component of the velocity is

(a) v = 0 (b) v = 1.5y
(c) v = –1.5y (d) v = –1.5y

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : As we know,

$$\frac{du}{dx} + \frac{dv}{dy} = 0 \text{ for incompressible flow.}$$

$\Rightarrow u = 1.5x$

$$\frac{d}{dx}(1.5x) + \frac{dv}{dy} = 0$$

$$1.5 + \frac{dv}{dy} = 0$$

$$\boxed{v = -1.5y}$$

611. For a 2D irrotational flow, the velocity potential is defined as $\varphi = \log_e (x^2 + y^2)$. Which of the following is possible stream function ψ, for this flow?

(a) $\frac{1}{2}\tan^{-1}\left(\frac{y}{x}\right)$ (b) $\tan^{-1}\left(\frac{y}{x}\right)$
(c) $2\tan^{-1}\left(\frac{y}{x}\right)$ (d) $2\tan^{-1}\left(\frac{x}{y}\right)$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : We know,

$$\frac{\partial \psi}{\partial y} = \frac{\partial \phi}{\partial x}$$

$$\frac{\partial \psi}{\partial y} = \frac{\partial}{\partial x}\left[\log e\left(x^2 + y^2\right)\right]$$

$$\frac{1}{x^2 + y^2} \times \frac{d}{dx}\left(x^2 + y^2\right)$$

$$= \frac{1}{x^2 + y^2} \times 2x$$

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Integrating Both side.

$$\int \partial\psi = \int \frac{2x}{x^2+y^2}\partial y$$

$$\psi = 2x.\frac{1}{x}\tan^{-1}\left(\frac{y}{x}\right)+C$$

$$\psi = 2\tan^{-1}\left(\frac{y}{x}\right)+C$$

612. A stream function is given by $\psi = 4x - 3y$. The resultant velocity at any point is

(a) 7.81 units/s (b) 7 units/s
(c) 3.5 units/s (d) 5 units/s

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Stream function = $\psi = 4x - 3y$

$$u = -\frac{\partial\psi}{\partial y} = 3$$

$$v = \frac{\partial\psi}{\partial x} = 4$$

Resultant velocity

$$v = \sqrt{u^2+v^2}$$

$$= \sqrt{3^2+4^2}$$

$$= 5 \text{ units}$$

613. The continuity equation

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0$$

is valid for

(a) Ideal fluid flow only
(b) Incompressible fluids, whether the flow is steady or unsteady
(c) Steady flow, whether compressible or incompressible
(d) Incompressible fluids and steady flow only
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (d) : The continuity equation

$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0$ is valid for steady and incompressible fluid.

614. Vorticity is equal to

(a) Twice the angular velocity of fluid particle
(b) Half of the angular velocity of a fluid particle
(c) One third of the angular velocity of a fluid particle
(d) Thrice the angular velocity of a fluid particle
(e) None of these

CGPSC AE 16.10.2016

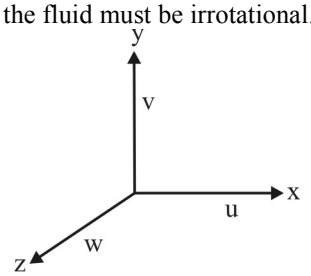
Ans. (a) : Vorticity is a measure of rotation of a fluid particle. Particles rotate as they revolve around the centre of the flow.
Vorticity is twice the average angular velocity of the angular deformation.

615. The realisation velocity potential on fluid flow indicates that the _____

(a) Flow is rotational and satisfies the continuity equation
(b) Vorticity must be non-zero
(c) Circulation around any closed curve must have a finite value
(d) Flow must be irrotational

UPRVUNL AE 21.08.2016

Ans. : (d) The realisation velocity potential on fluid indicates that the fluid must be irrotational.



$$\frac{d\phi}{dx} = -u, \frac{d\phi}{dy} = -v, \frac{d\phi}{dz} = -w$$

(i) Velocity potential function exists, the flow should be irrotational.
(ii) If velocity potential (ϕ) satifies the Laplace equation, it represent the possible steady incompressible irrotational flow.

616. Identify the following statements as true : (i) Axisymmetric are two dimensional (ii) One dimensional flow can take place in a curved passage (iii) Local acceleration is zero in steady flow (iv) Uniform flow can take place in a conical passage

(a) (i) and (ii) are true
(b) (i), (iii) and (iv) are true
(c) (ii) and (iv) are true
(d) (i), (ii), (iii) and (iv) are true

RPSC VPITI 14.02.2016

Ans. (*)

(1) Axisymmetric flows are two dimensional
$u = f_1(x, y)$
$v = f_2(x, y)$ and $w = 0$
(2) Local acceleration is zero in steady flow

$$\frac{\partial v}{\partial t} = 0 \Rightarrow \frac{\partial^2 v}{\partial t^2} = 0$$

(3) Uniform flow can take place in a uniform cross-section passage.
(4) Two dimensional flow can takes place in curved passage.

Note-The options are not appropriate. Only (i) & (iii) points are correct.

617. The velocity components in a two-dimensional steady incompressible flow are :
$u = 2x^2 - axy$, $v = y^2 - 4xy$. What should be the value of 'a' so that the equation of continuity is satisfied?

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(a) 3
(b) 2
(c) 1
(d) Any positive integer

RPSC AE (GWD) 06.05.2016

Ans. (b) : Given-

$u = 2x^2 - axy,\ v = y^2 - 4xy$

$\frac{\partial u}{\partial x} = 4x - ay,\ \frac{\partial v}{\partial y} = 2y - 4x$

According to continuity equation–

$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

$4x - ay + 2y - 4x = 0$

$-ay + 2y = 0$

$a = 2$

618. A Newtonian fluid has the following velocity field:

$$V = X^2Z\,i + 2XY^2Z\,j - XZ^3\,k$$

The rate of shear deformation ε_{YZ} at the point X = 2, Y = –2, Z = 3 is:

(Take $\mu = 0.5$ for the given flow)

(a) –11.5 (b) –23
(c) –46 (d) 11.5
(e) 46

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (*) : Velocity field,

$\vec{V} = x^2z\hat{i} + 2xy^2z\hat{j} - xz^3\hat{k}$

Velocity component in x-direction, $u = x^2z$

Velocity component in y-direction, $v = 2xy^2z$

Velocity component in z-direction, $w = -xz^3$

Rate of shear deformation,

$$\varepsilon_{yz} = \frac{1}{2}\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right)$$

$$\varepsilon_{yz} = \frac{1}{2}\left(2xy^2 + 0\right) = xy^2$$

At (2, –2, 3)

$\varepsilon_{yz} = 2(-2)^2 = 8$

619. The velocity equation for a pipe whose valve is being opened or closed gradually is given by

(a) $V = ax^4 + bx^2t$ (b) $V = ax^2 + bxt$
(c) $V = ax^3 + bxt$ (d) $V = ax + bxt$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : The velocity equation for a pipe whose valve is being opened or closed gradually is given by

$$V = ax + bxt$$

620. The equation to the streamlines for the velocity field is given as $V = 2x^3\hat{i} - 6x^2\hat{j}$

(a) $YX^2 = C$ (b) $YX^3 = C$
(c) $YX = C$ (d) $Y^2X^2 = C$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (*) : streamlines equation, for the velocity field,

$V = 2x^3\hat{i} - 6x^2\hat{j}$

$\frac{dy}{dx} = \frac{v}{u} = \frac{-6x^2}{2x^3} = \frac{-3}{x}$ $(u = 2x^3,\ v = -6x^2)$

$\int dy = -3\int \frac{1}{x}dx$

$y = -3 \log x + \log c$

$y = \log \frac{1}{x^3} + \log c$

$y = \log \frac{c}{x^3} \Rightarrow e^y = \frac{c}{x^3} \Rightarrow e^y x^3 = c$

Note:- Official Answer is (b).

621. While dealing with kinetics of particles, the forces in which the work is dependent upon the path followed by the particles is known as:

(a) Conservative forces
(b) Non-conservative forces
(c) Coplanar forces
(d) Non-collinear forces

HPPSC AE (PWD) 10.09.2016

Ans. (b) : Non-conservative force are dissipative force such as friction or air resistance. These forces take energy from the system as the system progresses (energy that you can't get back). These forces are path dependent therefore it matters where the object starts and stops.

Example- Friction force, air resistance rocket propulsion push or pull.

622. Continuity equation for a one dimensional flow through uniform cross sectional area is (Where ρ is the mass density of fluid; and u is the velocity component in x direction; and t is time variable)

(a) $\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} = 0$ (b) $\frac{\partial \rho}{\partial x} + \frac{\partial \rho u}{\partial x} = 0$
(c) $\frac{\partial \rho}{\partial t} - \frac{\partial \rho u}{\partial x} = 0$ (d) $\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial y} = 0$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Most general form of continuity equation for three dimensional flow.

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

For one dimensional flow, flow in y and z direction is zero v = 0, w = 0

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) = 0$$

623. Flow of liquid flowing through a long pipe of constant diameter at either increasing or decreasing rate is

(a) unsteady uniform flow
(b) steady uniform flow

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(c) steady non-uniform flow
(d) unsteady non-uniform flow

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Steady and unsteady flow – If flow and fluid properties at any given location does not change with time, flow is called steady otherwise, unsteady.

$$\left.\begin{aligned}\frac{dp}{dt}&=0\\ \frac{dv}{dt}&=0\\ \frac{d\rho}{dt}&=0\end{aligned}\right\}\text{flow is steady otherwise unsteady.}$$

624. A steady, incompressible, two-dimensional velocity field is given by,

$$\vec{V}(u,v)=(0.5+0.8x)\vec{i}+(1.5-0.8y)\vec{j}$$

The number of stagnation points there in the flow field is

(a) zero (b) many
(c) 1 (d) 2

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Given Field–

$$\vec{V}(u,v)=(0.5+0.8x)\hat{i}+(1.5-0.8y)\hat{j}$$

For stagnation point local velocity should be zero.

$\therefore \quad u = 0.5 + 0.8x = 0$

$v = 1.5 - 0.8y = 0$

This is linear equation we get only one value of x and y. Hence,

"Only one stagnation point is possible".

625. The equation of continuity is applicable when the flow:

(a) Velocity is uniform at all cross sections
(b) Is steady
(c) Is one dimensional
(d) All of the above

OPSC AEE 2015 PAPER - II

Ans. : (d)

The equation of continuity is applicable when:-

(i) Flow is steady
(ii) Flow is one dimensional.

Continuity equation :-

$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ (for compressible fluid)

$A_1 V_1 = A_2 V_2$ (for incompressible liquid)

The equation based on the principle of conservation of mass is called continuity equation. Thus for a fluid flowing through the pipe at all the cross- section, the quantity of fluid per second is constant.

626. Specify which of the following must be fulfilled by the flow of any fluid, real or ideal, laminar or turbulent.

(a) Newton's law of Viscosity
(b) Velocity at boundary must be zero relatively to boundary.
(c) The continuity equation
(d) Velocity normal to a solid boundary is zero.

TSPSC Managers, 2015

Ans. (c) : The continuity equation (conservation of mass) specify the flow of any fluid, real or ideal, laminar or turbulent.

627. In case of a two dimensional flow, the components of velocity are given by u = ax; v = by, the streamlines will consist of a series of

(a) Circular arcs (b) Parabolic arcs
(c) Hyperbolic arcs (d) Elliptical arcs

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Given, u = ax, v = by

Stream line equation -

$$\frac{dx}{u}=\frac{dy}{v}$$

$$\frac{dx}{ax}=\frac{dy}{by}$$

$$\Rightarrow b\frac{dx}{x}=a\frac{dy}{y}$$

$$\Rightarrow \int b\frac{dx}{x}=\int a\frac{dy}{y}$$

$b \ln x = a \ln y + \ln c$

$x^b = y^a c$

$x = y^{a/b} c^{1/b}$

The streamlines will consist of a series of hyperbolic arcs.

628. A practical example of steady non-uniform flow is given as the

(a) Motion of the river around the bridge piers
(b) Steadily increasing flow through the pipe
(c) Steadily increasing flow through a reducing section
(d) Constant discharge through a long, straight tapering pipe

TSPSC AEE 2015

Ans. (d) : Constant discharge through a long, straight tapering pipe is the example of steady non-uniform flow.

629. The curl of velocity vector is also known as

(a) linear deformation rate
(b) angular velocity
(c) shear rate
(d) vorticity

TANGEDCO AE 2015

Ans. (d) : $\vec{w} = \vec{\nabla} \times \vec{v}$

Vorticity is the curl of velocity vector signifies the tendency of something to rotate about a point.

630. A velocity field in a plane flow is given by V=2yti+xj m/s. The magnitude of the acceleration at the point (4, 2) t =3 s is

(a) $52.5 m/s^2$ (b) $48.5 m/s^2$
(c) $30.5 m/s^2$ (d) $24.5 m/s^2$
(e) $36.5 m/s^2$

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : Given,

V = 2yti + xj Compare to V = ui + vj

then u = 2yt , v= x

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Then total acceleration in x- direction

$$a_x = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{\partial u}{\partial t} = 2yt \times 0 + x \times 2t + 2y$$

$$a_x = x \times 2t + 2y$$

Putting $x = 4$, $y = 2$, $t = 3$

$$a_x = 28\ m/s^2$$

Total acceleration in y - direction

$$a_y = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + \frac{\partial v}{\partial t}$$

$$a_y = 2yt \times 1 + 0 + 0 = 12\ m/s^2$$

$$a_x = 28 m/s^2$$

$$a_y = 12\ m/s^2$$

$$a = a_x \hat{i} + a_y \hat{j},$$

$$a = \sqrt{a_x^2 + a_y^2}$$

$$a = \sqrt{(28)^2 + (12)^2}$$

$$a = 30.46 m/s^2$$

631. The stream function is

(a) constant along an equipotential line
(b) along a stream line
(c) defined only in irrotational flow
(d) defined only for incompressible flow

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : The stream function is defined only for incompressible flow.

632. In a steady flow in a varying section pipe if the diameter is doubled the kinetic energy will

(a) be doubled
(b) increase 4 times
(c) increase 8 times
(d) decrease to one sixteenth

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : Continuity equation–

$$V_1A_1 \times \frac{3}{4}d_1^2 = V_2 \times \frac{3}{4}d_2^2$$

$$V_2 = V_1\left(\frac{d_1}{d_2}\right)^2$$

$$V_2 = V_1\left[\frac{d_1}{2d_1}\right]$$

$$V_2 = \frac{V_1}{4}$$

$$K_2 = \frac{1}{2}mv_2^2 \qquad K_2 = \frac{1}{2}m\frac{V_1^2}{16}$$

$$\boxed{K_2 = \frac{K_1}{16}}$$

633. In steady flow of a fluid, the acceleration of any fluid particle is

(a) Constant (b) Variable
(c) Zero (d) Negative

MPPSC State Forest Service Exam, 2014

Ans. (c) : Steady flow in which the velocity of the fluid at a particular fixed point does not changes with time,

$$a = \frac{dv}{dt} = 0 \qquad \therefore\ v = \text{Constant}$$

634. The velocity potential function ϕ satisfy the Laplace equation: $\delta^2\phi/\delta x^2 + \delta^2\phi/\delta y^2 + \delta^2\phi/\delta z^2 = 0$. Then the flow is

(a) Unsteady, incompressible, rotational
(b) Steady, compressible, rotational
(c) Steady, incompressible, irrotational
(d) Unsteady, compressible, irrotational

ISRO Scientist/Engineer 24.05.2014

Ans. (c) : 3-D continuity equation which satisfies Laplace equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

or

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0$$

635. For a two dimensional potential flow, the velocity potential is given by: $\phi = 4x(3y-4)$. The numerical value of stream function at the point (2,3) is

(a) 10 Units (b) 20 Units
(c) 18 Units (d) 16 Units

ISRO Scientist/Engineer 24.05.2014

Ans. (c) : $\phi = 4x(3y-4)$

$$u = -\frac{\partial\phi}{\partial x} = -\frac{\partial\psi}{\partial y}$$

$$v = -\frac{\partial\phi}{\partial y} = \frac{\partial\psi}{\partial x}$$

$$u = 4(3y-4) = \frac{\partial\psi}{\partial y}$$

$$v = -12x = \frac{\partial\psi}{\partial x}$$

$$\frac{\partial\psi}{\partial x} = -12x \Rightarrow \psi = -6x^2 + g(y)$$

$$\frac{\partial\psi}{\partial y} = 0 + g'(y) = -16 + 12y$$

or $g(y) = -16y + 6y^2$

$$\psi = 6(-x^2 + y^2) - 16y$$

$$\psi|_{(2,3)} = 6(-4+9) - 16 \times 3$$

$$= 6 \times 5 - 48$$

$$= -18 \quad \text{i.e. 18 units}$$

636. Circulation is defined as the line integral of tangential component of velocity about a

(a) Centre
(b) close contour in a fluid flow
(c) velocity profile
(d) pressure profile

ISRO Scientist/Engineer 24.05.2014

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Ans. (b) : Circulation is defined as the line integral of the tangential component of the velocity taken around a closed contour.

$\Gamma = \oint_c \vec{V}.\, d\vec{S}$

637. Which of the following flow is rotational (where u and v are velocity components in x and y directions respectively)

(a) $u = y;\ v = \left(\frac{3}{2}x\right)$ (b) $u = xy^2; v = x^2y$

(c) Both (a) and (b) (d) None of these

ISRO Scientist/Engineer 24.05.2014

Ans. (a) :

for option (a)

$$\omega_z = \frac{1}{2}\left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right]$$

$$\omega_z = \frac{1}{2}\left[\frac{3}{2} - 1\right] = \frac{1}{4} \neq 0$$

For option (b)

$$\omega_z = \frac{1}{2}\left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right]$$

$$= \frac{1}{2}[2xy - 2xy] = 0$$

$\boxed{\omega = 0} \rightarrow$ "Irrotational motion"

638. For irrotational fluid flow curl of velocity vector is

(a) 0 (b) > 0

(c) < 0 (d) None of these

ISRO Scientist/Engineer 24.05.2014

Ans. (a) : Rotational flow

Here $\nabla \times \vec{V} \neq 0$ at every point in the flow. The fluid elements move and deform, and also rotate.

Irrotational flow

Here $\nabla \times \vec{V} = 0$ at every point in the flow. The fluid elements move and deform, but do not rotate.

639. If $\psi = 3x^2y - y^3$, the values of u and v are

(a) $6xy, 3x^2-3y^2$ (b) $3x^2-3y^2, 6xy$

(c) $3x^2-3y^2, -6xy$ (d) $3y^2-3x^2, 6xy$

(e) $6xy, 3y^2-3x^2$

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : Given, $\Psi = 3x^2y - y^3$

We know that

$$u = \frac{\partial \psi}{\partial y}, \qquad v = \frac{-\partial \psi}{\partial x}$$

So

$$u = 3x^2 - 3y^2, \qquad v = -6xy$$

640. Type of flow in which the fluid particles while flowing along stream lines, do not rotate about their own axis

(a) Rotational flow (b) Irrotational flow

(c) Non-uniform flow (d) Turbulent flow

TNPSC AE 2013

Ans. (b) : An irrotational flow the fluid particles while flowing along stream lines, do not rotate about their own axis.

641. The stream function for a 2 dimensional flow is given by $\Psi = 4xy$. Choose the wrong conclusion about the nature of the flow?

(a) The flow satisfies continuity equation

(b) The flow is a rotational flow

(c) The flow is a potential flow

(d) The flow is irrotational flow.

RPSC AEN Pre-2013

Ans. (*) : Given, stream function for a 2-D flow,

$$\Psi = 4xy$$

$$\frac{\partial \Psi}{\partial x} = +v \quad \Big| \quad \frac{\partial \Psi}{\partial y} = -u$$

$$4y = +v \quad \Big| \quad 4x = -u$$

$$v = +4y \quad \Big| \quad u = -4x$$

To satisfy continuity equation,

$$\frac{du}{dx} + \frac{dv}{\partial y} = 0$$

$$-4 + 4 = 0$$

So, the flow satisfies continuity equation.

We know,

$$\omega_{xy} = \frac{1}{2}\left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right]$$

$$= \frac{1}{2}(0 - 0) = 0$$

The flow is irrotational also.

Both option (a) & (d) are correct.

Note- Official answer given by commission is (b)

642. Fluid flow is said to be irrotational when

(a) the flow is both steady and uniform

(b) the shear stress varies imearly across the section

(c) the fluid obeys Newton's law of viscosity

(d) the fluid has zero vorticity

TNPSC ACF 2012

Ans. (d) :

Steady flow	Unsteady flow
fluid properties do not change with time	Fluid properties change with time.
Uniform flow	Non uniform flow
Velocity at any given time does not change with respect to space.	Velocity at any given time change with respect to space.

Irrotational flow- Fluid particles do not rotate about its mass centre in both circular as well as straight line motion.

$\omega = 0$

Vorticity (Ω) = $2\bar{\omega} = 2 \times 0 = 0$

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643. In an ideal fluid flow

(a) velocity is zero at the boundary
(b) velocity potential exists
(c) motion is uniform
(d) particles move is straight lines

TNPSC ACF 2012

Ans. (b) : In an ideal fluid flow velocity potential exist.

$$\frac{d\theta}{dx} = U\,,\ \frac{d\theta}{dy} = V\,,\ \frac{-d\theta}{dz} = U$$

644. Steady flow occurs when

(a) the direction and magnitude of the velocity at all points are identical
(b) the velocity of successive fluid particles, at any point, is the same at successive periods of time
(c) the magnitude and direction of the velocity do not change from point to point in the fluid
(d) the fluid particles move in plane and the streamline patterns are identical

TRB Asstt. Prof., 2012

Ans. (b) : • For steady flow,

Fluid properties do not change with time

$$\frac{\partial V}{\partial t} = 0, \frac{\partial P}{\partial t} = 0, \frac{\partial \rho}{\partial t} = 0$$

• For unsteady flow,

Fluid properties change with time

$$\frac{\partial V}{\partial t} \neq 0, \frac{\partial P}{\partial t} \neq 0, \frac{\partial \rho}{\partial t} \neq 0$$

A flow in which the velocity of the fluid at a particular fixed point does not change with time called steady flow.

645. For irrotational and incompressible flow, the velocity potential and stream function are given by ϕ and ψ respectively. Which one of the following sets is correct ?

(a) $\nabla^2 \phi = 0\,,\ \nabla^2 \psi = 0$
(b) $\nabla^2 \phi \neq 0\,,\ \nabla^2 \psi = 0$
(c) $\nabla^2 \phi = 0\,,\ \nabla^2 \psi \neq 0$
(d) $\nabla^2 \phi \neq 0\,,\ \nabla^2 \psi \neq 0$

UKPSC AE 2012 Paper–II

Ans. (a) : $\nabla^2 \phi = 0\,,\ \nabla^2 \psi = 0$

646. Which of the following statements is true for two-dimensional flow of ideal fluids?

(a) Potential function exists if stream function exists
(b) Stream function may or may not exist
(c) Both potential function and stream function must exist
(d) Stream function will exist but potential function may not exist

APPSC AEE 2012

Ans. (c) : Both potential function and stream function must exist.

647. The dividing streamline for a uniform flow superimposed over a two-dimensional droplet is

(a) a straight line (b) a circle
(c) a sphere (d) an ellipse

APPSC AEE 2012

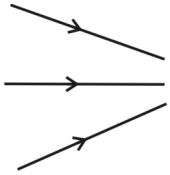
Ans. (b) : Streamline for a uniform flow superimposed over a two dimensional droplet is a circle.

648. In a converging steady flow, there is

(a) no acceleration
(b) no temporal acceleration
(c) only convective acceleration
(d) convective and temporal acceleration

APPSC AE 04.12.2012

Ans. (c) :



[Straight converging flow]

In a converging steady flow, there is change in magnitude w.r.t space. Therefore, only convective acceleration is exist and temporal acceleration is zero because there is no any normal component.

649. In uniform flow, the velocities of fluid particles are :

(a) Equal at all sections
(b) Dependent on time
(c) Random at all sections
(d) None of these

OPSC Civil Services Pre. 2011

Ans. (a) : The flow in which velocity at a given time does not change with respect to space.

650. Which of the following statements is FALSE about stream function (ψ) and potential function (ϕ) for a steady, incompressible and inviscid fluid?

(a) Lines of constant ϕ and ψ are mutually orthogonal
(b) ψ is defined for two-dimensional flows whereas ϕ is defined for three-dimensional flows
(c) Both ψ and ϕ exists for three-dimensional flows
(d) Both ψ and ϕ exists for two-dimensional flows

DRDO Scientists 2009

Ans. (c) : (i) Line of constant ϕ and ψ are mutually orthogonal.

(ii) 'ψ' stream function is defined for 2-D flows whereas 'ϕ' potential function is defined for 3-D flows.

(iii) Both 'ψ' and 'ϕ' exists for 2-D flows.

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651. The continuity equation :

(a) applies only for irrotational flow

(b) expresses the relation between mass and energy

(c) is applicable for steady, one dimensional flow with uniform flow over the cross-section

(d) relates the mass flow rate along a streamline

APPSC IOF, 2009

Ans. (c) : The continuity equation is the statement of conservation of mass. It is applicable for both compressible as well as incompressible.

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Where ρ = Density

A = Area

V = Velocity

652. Which is the continuity equation?

(a) $h_0 = h + \frac{V^2}{2g_c J}$ (b) $G = \frac{m}{A} = \rho V$

(c) $M = \frac{V}{a}$ (d) $S = S_0$

TNPSC AE, 2008

Ans. (b) : Continuity equation : $G = \frac{m}{A} = \rho V$

653. The velocity components in x and y directions in terms of velocity potential (ϕ) are

(a) $u = -\frac{\partial \phi}{\partial x}, \; v = \frac{\partial \phi}{\partial y}$

(b) $u = \frac{\partial \phi}{\partial y}, \; v = \frac{\partial \phi}{\partial x}$

(c) $u = -\frac{\partial \phi}{\partial y}, \; v = -\frac{\partial \phi}{\partial x}$

(d) $u = -\frac{\partial \phi}{\partial x}, \; v = -\frac{\partial \phi}{\partial y}$

TNPSC AE, 2008

Ans. (d) : The velocity components in x and y directions in terms of velocity potential (ϕ) for 2-D –

$\boxed{u = -\frac{\partial \phi}{\partial x}}, \boxed{v = -\frac{\partial \phi}{\partial y}}$

654. For a 2D rotational flow the Laplacian of the stream-function at any point in the flow-field.

(a) is always very large

(b) is always very small

(c) equals zero

(d) equals vorticity

DRDO Scientists 2008

Ans. (c) : For a 2D rotational flow the Laplacian of the stream-function at any point in the flow-field equals zero.

655. An equipotential line

(a) has no velocity component normal to it

(b) has no velocity component tangential to it

(c) is a line of constant velocity

(d) is a line of zero normal acceleration

UKPSC AE 2007 Paper -II

Ans. (b) : Has no velocity component tangential to it

656. Along a stream line

(a) Velocity is constant (b) ϕ is zero

(c) ψ is zero (d) ψ is constant

UKPSC AE 2007 Paper -II

Ans. (d) : Along a stream line, flow of fluid the velocity of all fluid particles at a given instant is constant. Along streamline potential function (ϕ) is constant.

657. The two stagnation points for ideal flow over a circular cylinder coincide when

(a) the free stream is uniform

(b) the circulation about the cylinder is very small

(c) the circulation about the cylinder exceeds a critical value

(d) the circulation about the cylinder is a critical value

UKPSC AE 2007 Paper -II

Ans. (d) : The circulation about the cylinder is a critical value

658. The pressure gradient for developed flow in a conduit is linear since it satisfies

(a) the equation of continuity

(b) the equation of momentum

(c) the equation of energy

(d) none of the above

WBPSC AE, 2007

Ans. (a) : The pressure gradient for developed flow in a conduit is linear since it satisfies the equation of continuity.

659. In the case of steady flow of a fluid the acceleration of any fluid particle is :

(a) Constant

(b) Variable

(c) Zero

(d) Zero under limiting conditions

J & K PSC Screening, 2006

Ans. (d) : $a_x = \underbrace{u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{w\partial u}{\partial z}}_{\text{Convective acceleration}} + \underbrace{\frac{\partial u}{\partial t}}_{\text{Local acceleration}}$

In steady flow, local acceleration is zero.

In uniform flow, convective acceleration is zero.

for steady & uniform flow local and convective both acceleration will be zero.

660. Which one of the following velocity fields does not represent a possible motion of an incompressible fluid

(a) $\overline{U} = 5x^3\hat{i} - 15x^2y\hat{j} + t\hat{k}$

(b) $\overline{U} = 5x^3\hat{i} - 15x^2y\hat{j}$

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(c) $\overline{U} = 2x\hat{i} - 2y\hat{j}$

(d) $\overline{U} = 2x\hat{i} + 2y\hat{j}$

Where $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors in x, y and z directions and 't' is the time variable.

OPSC Civil Services Pre 2006

Ans. (d) : If fluid flow satisfy continuity equation, it will be possible case of fluid flow.
For incompressible, steady flow

$$\text{continuity equation} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\vec{U} = 2x\hat{i} + 2y\hat{j}$$

$$u = 2x \;\&\; v = 2y$$

$$\frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(2y)$$

$= 2 + 2 = 4$ which is not equal to zero.

661. The vector field $\vec{F} = x\hat{i} - y\hat{j}$ (where $\hat{i}$ and $\hat{j}$ are unit vectors) is-

(a) Divergence free, but not irrotational
(b) irrotational, but not divergence free
(c) Divergence free and irrotational
(d) Neither divergence free nor irrotational

ISRO Scientist/Engineer 2006

Ans. (c) : Vector field

$\vec{F} = x\hat{i} - y\hat{j}$ (Where $\hat{i}$ and $\hat{j}$ are unit vector)

Divergence $\oint \text{div}(\vec{F}) = \nabla \cdot \vec{F}$

$$= \frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(0)$$

$= 1 - 1 = 0$

$$\text{Curl } (\nabla \times F) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & -y & 0 \end{vmatrix}$$

$$= \vec{i}\left[-\frac{\partial(y)}{\partial z}\right] - \vec{j}\left[-\frac{\partial(x)}{\partial z}\right] + \vec{k}\left[\frac{\partial(-y)}{\partial x} - \frac{\partial(x)}{\partial y}\right]$$

$= 0$

So, vector field is divergence free and irrotational.

6. Fluid Dynamics

662. Bernoulli's theorem deals with the principle of conservation of

(a) energy (b) Momentum
(c) mass (d) force

Assam PSC AE (IWT) 14.03.2021
JPSC AE, 10.04.2021, Paper-I
UPRVUNL AE 05.07.2021
CIL MT 27.02.2020, TNPSC AE 2019
BPSC Poly. Lect. 2016
Rajasthan AE (Nagar Nigam) 2016 Shift-3

APPSC AEE Screening Test 2016
VIZAG STEEL MT 2015, SJVN ET 2013
APPSC AEE 2012, CSE Pre-2003

Ans. : (a) Bernoulli's theorem deals with the principal of conservation of energy.
Bernoulli's equation is obtained by integrating the Euler's equation of motion.

$$\int \frac{dp}{\rho} + \int g.dz + \int v.dv = constant$$

$$\frac{P}{\rho g} + \frac{V^2}{2g} + Z = constant$$

$$\frac{P}{\rho g} = pressure\,head.$$

$$V^2/2g = kinetic\,head.$$

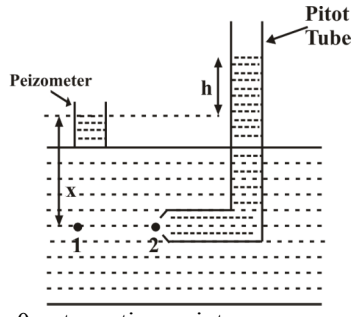
$$Z = Potential\,head.$$

663. Pitot tube is used to measure the

(a) Flow rate of fluid
(b) Velocity of the fluid
(c) Density of fluid
(d) Pressure inside the pipe

APGCL AM, 2021, SJVN ET 2019
GPSC Executive Engineer 23.12.2018
Karnataka PSC Lect., 27.05.2017
ISRO Scientist/Engineer (RAC) 07.05.2017
RPSC LECTURER 16.01.2016
Kerala PSC AE 06.08.2015
TPSC AE. 2015, OPSC AE 2015
GPSC Lect. (Poly.) 07.12.2014
UKPSC AE 2012 Paper–II
J&K PSC Civil Services Pre, 2010

Ans. (b) : Pitot tube—Pitot tube can be used to measure fluid flow velocity and dynamic pressure difference.
A pitot tube is measuring velocity of flow.
* Pitot tube is based on principle of conservation of kinetic head into pressure head. The point at which velocity reduces zero is called stagnation point
* Velocity head is indicated by the difference in liquid level between the pitot tube and the piezometer. The pitot tube measures the total head and therefore known as total head tube.



$V_2 = 0$ - stagnation point

So,

$$V_1 = \sqrt{2gh}$$

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664. A rotameter is used to measure

(a) rpm of engine
(b) rotation of shafts
(c) twist due to torque applied on shafts
(d) flow of liquids and gases

GPSC Executive Engineer 23.12.2018
Nagaland PSC CTSE 2017 Paper-2
GMB AAE 25.06.2017, RPSC AE. 2016
GWSSB DEE 07.07.2016
GWSSB AAE, 27.12.2015
VIZAG STEEL MT 2015
SJVN ET 2013, RPSC ACF-2011
OPSC Civil Services Pre. 2011

Ans. : (d) A rotameter (variable area meter) is used to measure flow of liquids and gases. There is no difference between rotameter and flow meter.

Device	Measurement
Venturimeter	Rate of flow
Flow nozzle	Rate of flow
Orifice meter	Rate of flow
Bend meter	Rate of flow
Rotameter	Rate of flow
Pitot tube	Velocity
Hot wire Anemometer	air and gas velocity

665. Hot wire anemometer is used to :

(a) measure pressure of liquid
(b) measure velocity of air stream
(c) measure temperature of moving fluid
(d) measure thermal conductivity of solid

GMB AAE 25.06.2017, WBPSC AE, 2017
RPSC AE 2016, RPSC ADE 2016
GWSSB DEE 07.07.2016, APPSC Poly. Lect. 2013
RPSC ACF-2011, RPSC AE GWD, 2011

Ans. (b) : A hot-wire anemometer is a thermal transducer which has been widely used to measure the gas or air velocity.

666. In Navier Strokes equation the considered fluid forces are:

(a) Gravity, Pressure, Turbulent
(b) Viscous, Pressure, Elastic
(c) Gravity, Pressure, Elastic
(d) Gravity, Pressure, Viscous

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
RPSC IOF, 2020
OPSC AEE 2019 PAPER - II
MPPSC AE 08.11.2015
Mizoram PSC AE/SDO 2014 Paper-I
RPSC AE GWD, 2011
UKPSC AE 2007 Paper -II

Ans. (d) : Navier Stoke's equation – Gravity, pressure, viscous
Reynold's equation – Gravity, pressure, viscous, turbulent
Euler's equation – Gravity, pressure
Navier-Stokes equation describes the motion of viscous fluid substances.

These balance equation arise from applying Newton's second law to fluid motion, together with assumption that the stress in the fluid is the sum of a diffusing viscous term (proportional to gradient of velocity) and a pressure term, hence describing the viscous flow

$$\rho\frac{du}{dt} = \tau B_x - \frac{\partial P}{\partial x} + \mu\Delta^2 u$$

Where, B_x is body force (e.g. gravity)

$F_i = F_p + F_g + F_v$ valid for viscous force only.

667. In what conditions Bernoulli's equation is applicable.

(a) Rotational flow with incompressible fluid
(b) Irrotational flow with compressible or incompressible fluid
(c) Steady state rotational flow with incompressible fluid
(d) Steady state, Irrotational flow with incompressible fluid

ESE 2018, GPSC Lect. (Auto) 16.10.2016
Nagaland PSC CTSE 2017, 2016 Paper-2
RPSC AE (GWD) 06.05.2016
Mizoram PSC 2014, Paper-I
UKPSC AE 2012 Paper–II
ESE 2014, 2012, 2009

Ans. (d) : Bernoulli's equation is applicable for-
1. Steady flow
2. Irrotational flow
3. Incompressible flow
4. Inviscid flow
5. Laminar flow

668. In general Bernoulli's equation each of the terms represents

(a) Energy in kg.m/kg mass of fluid
(b) Energy in N.m/kg mass of fluid
(c) Energy in N.m/N weight of fluid
(d) Power in kW/kg mass of fluid

CIL MT 26.03.2017
Nagaland PSC CTSE 2017 Paper-2
Nagaland PSC (CTSE) 2016, Paper-II
APPSC AEE Mains 2016 (Civil Mechanical)
APPSC AEE 2012, ESE 1995

Ans. (c) : In general Bernoulli's equation each of the terms represents (N.m)/weight (N)

$$\frac{P}{\rho g} + \frac{V^2}{2g} + z = \text{constant}$$

669. Bernoulli's equation is derived making assumptions that

(a) the flow is uniform, steady and incompressible
(b) the flow is non-viscous, uniform and steady
(c) the flow is steady, non-viscous, incompressible and irrotational
(d) none of these

UPRVUNL AE 05.07.2021
HPPSC Asstt. Prof. 20.11.2017

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ESE 2009, TNPSC AE, 2008
J & K PSC Screening, 2006, CSE Pre-2006

Ans. (c) : Bernoulli's equation is derived making assumptions that the flow is steady, non-viscous, incompressible and irrotational.

670. Rotameter is used for measuring
(a) Density of fluids
(b) Velocity of fluid in pipes
(c) Viscosity of fluids
(d) Discharge of fluids

BPSC Asstt. Prof. 29.11.2015
GPSC AE 26.07.2015
MPPSC State Forest Service Exam, 2014
TNPSC AE (Industries) 09.06.2013
ESE 2006

Ans. (d) : Rotameters are simple industrial flow meters that measures the flow rate of liquid or gas in a closed tube. They give direct reading & shows straight line variation.

671. In fluid mechanics, the continuity equation is a mathematical statement embodying the principle of
(a) Conservation of momentum
(b) Conservation of weight
(c) Conservation of energy
(d) Conservation of mass

TNPSC AE 2019
TNPSC AE (Industries) 09.06.2013
UKPSC AE 2012 Paper–II
OPSC Civil Services Pre. 2011

Ans. (d) : In fluid mechanics, the continuity equation is a mathematical statement embodying the principle of conservation of mass.
$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ [For compressible fluid]
$A_1 V_1 = A_2 V_2$ [For incompressible fluid]

672. Euler's equation in the differential from can be expressed as (Where dP is the pressure, ρ is the density of the fluid, v is the velocity, dz is the differential head, g is the gravitational acceleration.)
(a) $\frac{P}{\rho}+\frac{v^2}{2}+gz=0$
(b) $\frac{dP}{\rho}+vdv+gdz=0$
(c) $\frac{dp}{\rho}+vdv+gdz=0$
(d) $\frac{d\rho}{P}+vdv+gdz=0$

HPPSC AE 2018
OPSC AEE 2015 PAPER - II
VIZAG STEEL MT 2015
WBPSC AE 2003

Ans. (b) : Euler's equation of motion:- This is equation of motion in which the forces due to gravity and pressure are taken into consideration. This is derived by considering the motion of a fluid element along a stream - line as.
$\frac{dP}{\rho}+g.dz+v.dv=o$
Bernoulli's equation is obtained by integrating the Euler's equation of motion and assume fluid is incompressible.
$\int\frac{dp}{\rho}+\int g.dz+\int v.dv=\text{constant}$

673. Navier Stoke's equation represents the conservation of
(a) Mass (b) Momentum
(c) Pressure (d) Energy

VIZAG Steel MT 24.01.2021 Shift-II
GPSC Executive Engineer 23.12.2018
Nagaland PSC CTSE 2015 Paper-II
GATE 2000

Ans. (b) :
- Navier Stoke's equation represents law of conservation of momentum.
- Continuity equation represents law of conservation of mass.
- Bernoulli's equation represents law of conservation of energy.

674. The pitot static tube measures
(a) Dynamic pressure
(b) Static pressure
(c) Total pressure
(d) Difference in static & dynamic pressure

RPSC IOF, 2020
TSPSC AEE 28.08.2017 (Civil/Mechanical)
CGPSC AE 26.04.2015 SHIFT-II
CSE Pre-2006

Ans. (a) : The pitot tube is a differential pressure measuring device.
The pitot tube installed in the flow stream measures the direct pressure at the contact pitot tube hole and a second measurement is required being of static pressure. The difference between two measurements gives a value for dynamic pressure.

675. For measuring flow by a venturimeter. It should be installed in
(a) vertical line
(b) horizontal line
(c) inclined line with flow downward
(d) in any direction and in any location

Vizag Steel (MT) 2017
HPPSC Asstt. Prof. 18.09.2017
TRB Poly. Lect., 2012
J & K PSC Screening, 2006

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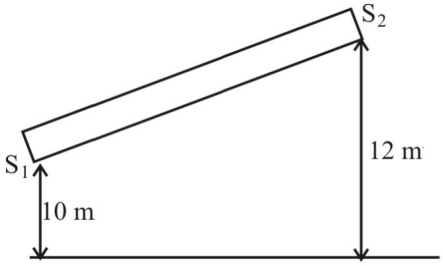
Ans. (d) : Venturimeter should be installed in any direction and in any location while for measuring flow.

676. A smooth pipe of diameter 200 mm carries water. The pressure in the pipe at section S_1 (elevation=10m) is 50 kPa. At section S_2 (elevation=12 m) the pressure is 20 kPa and velocity 2m/s. Density of water is 1000 kg/m³ and acceleration due to gravity is 9.81 m/s². Which of the following is true?

(a) Flow is from S1 to S2 and head loss is 0.53 m
(b) Flow is from S2 to S1 and head loss is 0.53 m
(c) flow is from S1 to S2 and head loss is 1.06 m
(d) flow is from S2 to S1 and head loss is 1.06 m

RPSC ADE 2016
ISRO Scientist/Engineer 12.05.2013
GATE 2010

Ans. (c) : $P_1 = 50$ kPa
$P_2 = 20$ kPa
$v_1 = v_2$ = uniform section
$h_1 = 10$ m
$h_2 = 12$ m



The flow will be from high energy to low energy.

$$E_1 = \frac{50\times10^3}{\rho g} + 10 + \frac{v^2}{2g}$$

$$= \frac{50}{9.81} + 10 + \frac{v^2}{2g}$$

$$= 15.096 + \frac{(2)^2}{2\times9.81}$$

$E_1 = 15.3007$ ..... (i)

$$E_2 = \frac{20\times10^3}{9.81\times1000} + 12 + \frac{v^2}{2g}$$

$$= 14.038 + \frac{(2)^2}{2\times9.81}$$

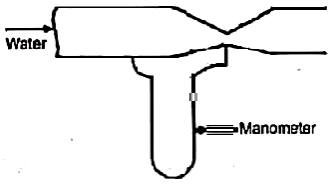
$E_2 = 14.2426$ .....(ii)

As $E_1 > E_2$

Head loss, $E_1 - E_2 = 15.3007 - 14.2426 = 1.058$

So, flow will take from (i) to (ii) with head loss of 1.058 meter.

677. A horizontal pipe of cross-sectional area 5 cm² is connected to a venturi meter of throat area 3 cm² as shown in the given Fig. The manometer reading is equivalent to 5 cm of water. The discharge in cm³/s nearly.



(a) 0.45 (b) 5.5
(c) 21.0 (d) 370

APPSC AE Subordinate Service Civil/Mech. 2016
Nagaland PSC CTSE 2015 Paper-II
ESE 1998

Ans. (d) : Given,
$A = 5$ cm²
$A_T = 3$ cm²
$h = 5$ cm
$Q = ?$

$$Q_{actual} = C_d \times \frac{A_1 \times A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}$$

Given, $C_d = 1$
$g = 981$ cm/sec²

$$Q_{actual} = \frac{3\times5}{\sqrt{5^2 - 3^2}} \times \sqrt{2\times981\times5}$$

$Q_{actual} = 371.42$ cm³/sec

678. When venturimeter is inclined, then for a given flow it will show

(a) Less reading
(b) More reading
(c) Same reading
(d) Inaccurate reading

Vizag Steel (MT) 2017
Nagaland PSC CTSE 2015 Paper-II
APPSC AEE 2012

Ans. (c) : When the venturimeter is inclined, then for a given flow it will show same reading.
For venturimeter,

$$Q_{act} = C_d \frac{A_1A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2gH} \quad (\text{Where } A_1 < A_2)$$

$C_d = 0.96$ to 0.98; $A_1 > A_2$
Venturimeter is not suitable for very low velocities due to variation C_d.
For any venturimeter (Vertical, inclined etc), with differential manometer we get 'h' directly $h = y\left(\frac{S_m}{S_\ell} - 1\right)$ no correction need for its orientation.

679. For venturimeter, the range for coefficient of discharge, C_d is:

(a) 0.95 – 0.99 (b) 0.85 – 0.94
(c) 0.55 – 0.81 (d) 0.73 – 0.88

AAI Jr. Executive 29.11.2018
APPSC AEE 2012
J&K PSC Civil Services Pre, 2010

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Ans. (a) :

Coefficient of discharge (C_d)	Value
Venturimeter	0.94 – 0.99
Orificemeter	0.64 – 0.76

680. Speed of an aeroplane is measured by

(a) pitot tube
(b) hot wire anemometer
(c) venturimeter
(d) rotameter

ISRO Scientist/Engineer 22.04.2018
J&K PSC Civil Services Pre, 2013
TNPSC AE (Industries) 09.06.2013

Ans. (a) : Pitot tube—It is used to measure velocity of aeroplanes and boats.
Hot Wire Anemometer—It is used for measuring the velocity of the fluid.
Venturi meter—It is used for measuring the rate of flow of fluid flowing through pipe.
Rotameter—It is used to measures the volumetric flow rate of fluid in a closed tube.

681. A venturimeter of 20 throat diameter is used to measure the velocity of water in a horizontal pipe of 40 mm diameter. If the pressure difference between the pipe and throat sections is found to be 30 kPa, then, neglecting losses, the flow velocity is

(a) 0.2 m/sec (b) 1.0 m/sec
(c) 1.4 m/sec (d) 2.0 m/sec

ISRO Scientist/Engineer 12.05.2013
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012

Ans. (d) : Given, $d_1 = 40$ mm, $d_2 = 20$ mm
Apply Bernoulli's equation –

$$\frac{p_1}{\rho g}+\frac{V_1^2}{2g}+z_1=\frac{p_2}{\rho g}+\frac{V_2^2}{2g}+z_2+h_f$$

$\because$ $z_1 = z_2$ and neglecting frictional losses

$$\frac{p_1-p_2}{\rho g}=\frac{V_2^2-V_1^2}{2g} \quad \text{.....(i)}$$

Continuity equation –

$$A_1V_1 = A_2V_2$$

$$\frac{\pi}{4}(40)^2\times V_1=\frac{\pi}{4}(20)^2\times V_2$$

$$4V_1 = V_2 \quad \text{.....(ii)}$$

From equation (i) & (ii) –

$$\frac{30\times10^3}{10^3}=\frac{V_2^2-V_1^2}{2}$$

$$V_2^2 - V_1^2 = 60$$

Putting $V_2 = 4V_1$

Therefore, $16V_1^2 - V_1^2 = 60$

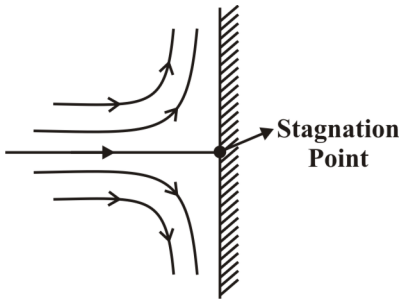
$$V_1 = 2 \text{ m/s}$$

682. A stagnation point is a point where

(a) pressure is zero
(b) total energy is zero
(c) total energy is maximum
(d) velocity of flow reduces to zero

APPSC AEE Mains 2016 (Civil Mechanical)
OPSC AEE 2015 PAPER - II
APPSC AEE 2012

Ans. (d) :



A stagnation point is a point where velocity of flow reduces to zero.
Pitot tube is based on principal of conservation of kinetic head into pressure head. The point at which velocity reduces zero is called stagnation point.

683. The kinetic energy correction factor α is a measure of effect of non-uniform distribution of velocity which is caused on account of viscous and other resistance. It is expressed by

(a) $\frac{1}{A}\int_A\left(\frac{v}{V}\right)^2 dA$ (b) $\frac{1}{A}\int_A\left(\frac{v}{V}\right)dA$

(c) $\frac{1}{A}\int_A\left(\frac{v}{V}\right)^3 dA$ (d) $\frac{1}{A}\int_A\left(\frac{V}{v}\right)^3 dA$

TSPSC AE Civil/Mechanical 28.08.2017
Nagaland PSC (CTSE) 2016, Paper-II
APPSC AEE 2012

Ans. : (c) Kinetic energy correction factor (α) :-

$$\alpha=\frac{\text{K.E/sec based on actual velocity}}{\text{K.E/sec based on average velocity}}$$

$$\alpha=\frac{1}{A}\int_A\left(\frac{v}{V}\right)^3.dA$$

Kinetic energy correction factor for laminar flow through a circular pipe is 2.

684. An assumption not made in the derivation of Bernoulli's equation is:

(a) Steady flow
(b) Uniform flow
(c) Inviscid flow
(d) Two-dimensional flow

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Kerala PSC Asstt. Prof. 27.09.2016
APPSC AEE 2012

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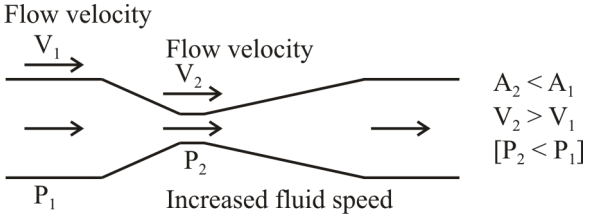
Ans. (b) : Assumptions in Bernoulli's equation.

- Fluid is ideal
- Flow is steady
- Flow is continuous
- Fluid is incompressible (density remain constant)
- Flow is non-viscous
- Flow is irrotational
- Applicable along a stream line.



Energy per unit volume before = Energy per unit volume after

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g h_2$$

Pressure energy | Kinetic energy per unit volume | Potential energy per unit volume

685. Venturimeter is used to

(a) Measure the velocity of a flowing liquid
(b) Measure the pressure of the flowing liquid
(c) Measure the discharge of liquid flowing in a pipe
(d) Measure the pressure difference of liquid flowing between two points in a pipe line

Nagaland PSC (CTSE) 2018, 2016 Paper-II
Mizoram PSC AE/SDO 2014 Paper-I
ESE 2012

Ans. (c) : Venturimeter measures the flow speed of the incompressible fluid. There are 'two, and discharge of liquid flowing in a pipe. It is gradually converging and gradually diverging device used for the measurement of flow in a pipeline.
It is based on the principle of Bernoulli's theorem.

686. Hot wire anemometer is used to measure

(a) pressure in gases (b) liquid discharge
(c) pressure in liquids (d) gas velocities

APPSC AEE Mains 2016 (Civil Mechanical)
WBPSC AE 2003

Ans. (d) : The Hot wire Anemometer is a device used for measuring the velocity and direction of the fluid. This can be done by measuring the heat loss of the wire which is placed in the fluid stream.

687. The speed of a submarine can be measured by:

(a) Pitot tube
(b) Hot wire anemometer
(c) Pirani gauge
(d) Any of the above

WBPSC AE, 2017
J & K PSC Screening, 2006

Ans. (a) : The speed of a submarine can be measured by pitot tube.
Hot wire anemometer– air and gas velocity measurement.
Pirani gauge- Measures pressure It is mostly used in the high vacuum range. (1 torr – 10^{-3} torr)

688. Which mouthpiece is having maximum co-efficient of discharge?

(a) External mouthpiece
(b) Convergent-divergent mouthpiece
(c) Internal mouthpiece
(d) None of these

RPSC IOF, 2020, APPSC Poly. Lect. 2013

Ans. (b) : In convergent divergent mouthpiece as the area of outlet increased, the discharge is increased and the velocity V_c is increased and hence the pressure head (H_c) is decreased.
Hence for the condition of maximum discharge pressure head remains minimum.
So, in convergent divergent mouthpiece co-efficient of discharge is maximum.

689. The theoretical velocity of jet at vena contracta in terms of Head of water is:

(a) $\sqrt{(2gh)}$ (b) $\sqrt{(2g)}h$
(c) 2gh (d) $\sqrt{2}(gh)$

CIL MT 27.02.2020
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : The theoretical velocity of jet at vena contracta is given by the relation $v = \sqrt{2gh}$ where h is the head of water at vena contracta. The value of coefficient of velocity varies slightly with different shapes of the edges of the orifice.

690. An instrument which offers no obstruction to the flow, offers no additional loss and is suitable for flow rate measurement is:

(a) Venturimeter
(b) Rotameter
(c) Magnetic flow meter
(d) Bend meter

HPPSC Asstt. Prof. 18.09.2017
CSE Pre-1997

Ans. (c) : An instrument which offers no obstruction to the flow, offers no additional loss and is suitable for flow rate measurement is magnetic flow meter.

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691. Venturimeter (V), flow nozzle (N) and orificemeter (O) arranged in increasing order of co-efficient of discharge are

(a) V, N, O
(b) N, O, V
(c) O, N, V
(d) O, V, N

RPSC ADE 2016, APPSC AEE 2012

Ans. (c) : Coefficient of discharge, C_d measuring equipment

$(C_d)_{venturimeter} = 0.986$
$(C_d)_{flow\ nozzle} = 0.90$
$(C_d)_{orificemeter} = 0.61$

Increasing order of C_g $O < N < V$

692. Rotameter is a

(a) drag force flow meter
(b) variable area flow meter
(c) variable head flow meter
(d) rotating propeller type flow meter

TNPSC AE 2018
GPSC Lect. (Auto) 16.10.2016

Ans. (b) : A rotameter is a device that measure the volumetric flow rate of fluid in a closed tube. It belongs to a class of meters called variable area meters, which measure flow rate by allowing the cross-sectional area the fluid travels through to vary, causing a measurable effect.

693. All the terms of energy in Bernoulli's equation have dimension of:

(a) Force
(b) Work
(c) Length
(d) Energy

Rajasthan Nagar Nigam AE 2016 Shift-II
OPSC AEE 2015 PAPER - II

Ans. : (c) All the terms of energy in Bernoulli's equation have dimension of length.

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \text{constant}$$

$\frac{p}{\rho g}$ = pressure energy per unit weight $\left[\frac{N-m}{N} \text{OR } m\right]$

$v^2/2g$ = Kinetic energy per unit weight $\left[\frac{N-m}{N} \text{OR } m\right]$

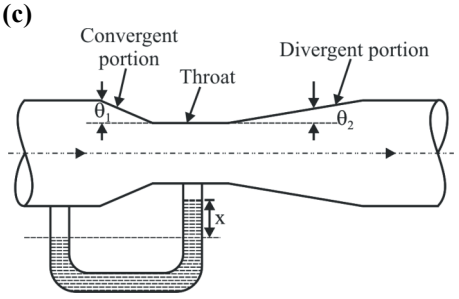
Z = Potential energy per unit weight $\left[\frac{N-m}{N} \text{OR } m\right]$

694. The divergent portion of a venturimeter is made longer than convergent portion in order to

(a) avoid the tendency of breaking away the stream of liquid
(b) to minimise frictional losses
(c) both (1) & (2)
(d) to maximize frictional losses

Punjab PSC SDE 12.02.2017
TSPSC AEE 2015

Ans. : (c)



$$d_2 = \left(\frac{1}{3} \text{to} \frac{1}{2}\right) d_1$$

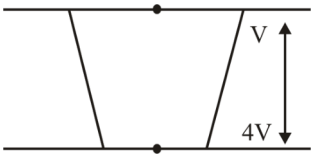
Angle of convergence $(\theta_1) = 18° - 22°$

Angle of divergence $(\theta_2) = 5° - 7\frac{1°}{2}$

The divergent portion of venturimeter is made longer than convergent portion in order to

(i) Avoid the tendency of breaking away the stream of liquid
(ii) To minimize frictional losses.

695. A liquid flows downward through at tapered vertical portion of a pipe. At the entrance and exit of the pipe, the static pressures are equal. If for a vertical height 'h' the velocity becomes four times, then the ratio of 'h' to the velocity head at entrance will be

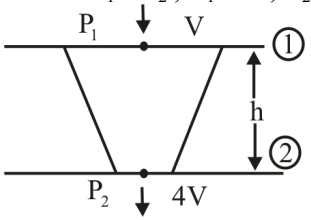


(a) 3
(b) 8
(c) 15
(d) 24
(e) 27

CGPSC AE 26.04.2015 SHIFT-II
ESE 1998

Ans. (c) : Given as

$P_1 = P_2$, $V_1 = V$, $V_2 = 4V$



then

Appling Bernoulli's equation

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\frac{V^2}{2g} + h = \frac{(4V)^2}{2g} + 0 = 16\frac{V^2}{2g}$$

$$15\frac{V^2}{2g} = h$$

then

$$\frac{h}{(v^2/2g)} = 15$$

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696. Euler's equation of motion is a statement of :

(a) linear momentum conservation for the flow of an inviscid fluid
(b) mass conservation
(c) energy conservation
(d) linear momentum conservation for a real fluid

BHEL ET 2019, ESE 2005

Ans. (a) : Assumption of Euler's equation of motion
(1) Flow is non-viscous fluid
- Derive the Euler's equation by linear momentum conservation.
- Flow is steady $\left(\frac{\partial v}{\partial t}=0\right)$ and incompressible (ρ = C)

697. Velocity head is given by

(a) $\frac{v}{g}$ (b) $\frac{v^2}{2g}$
(c) $\frac{v^3}{2g}$ (d) $\frac{v^2}{2g^2}$

APPSC AEE 2012
UKPSC AE 2007 Paper -II

Ans. (b) : Velocity head or kinetic head is in the form at kinetic energy per unit weight.

* Velocity head $\frac{v^2}{2g}$ (in meter)

* Pressure head $=\frac{P}{\rho g}$ (in meter)

* Potential head = z (in meter)

698. Match List I (measuring Devices) with List II (Measuring parameter) and select the correct answer using the codes given below the lists

List I (Device)	List II (Parameter)
A. Pitot tube	1. Flow static pressure
B. Micro-manometer	2. Rate of flow (indirect)
C. Pipe bend meter	3. Differential pressure
D. Wall pressure tap	4. Flow stagnation pressure

	A	B	C	D
(a)	1	3	2	4
(b)	4	3	2	1
(c)	1	2	3	4
(d)	4	2	3	1

ISRO Scientist/Engineer 2008
ESE 2004

Ans. (b) :

699. Bernoulli's equation is applied to

(a) venturimeter (b) orifice meter
(c) pitot tube meter (d) all of the above

TRB Polytechnic Lecturer 2017
ISRO Scientist/Engineer 11.10.2015

Ans. (d) : Bernoulli Theorem–For a steady non-viscous, incompressible fluid flow along a stream line the summation of kinetic energy, potential energy, pressure energy is constant and this is known as conservation of energy.

$$\boxed{\frac{P}{\rho}+gz+\frac{V^2}{2}=\text{Constant}}\ \frac{\text{Joule}}{\text{kg}}$$

Divide by 'g'

$$\boxed{\frac{P}{w}+z+\frac{V^2}{2g}=C}\ \text{J/N}$$

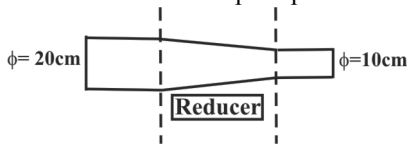
This is classical Bernoulli equation.
It is applicable to venturimeter, orificemeter, pitot tube.

700. Consider steady, incompressible and irrotational flow through a reducer in a horizontal pipe, where the diameter is reduced from 20 cm to 10 cm. The pressure in the 20cm pipe just upstream of the reducer is 150 kPa. The fluid has a vapour pressure of 50 kPa and a specific weight of 5kN/m^3. Neglecting frictional effects, the maximum discharge (in m^3/sec) that can pass through the reducer without causing cavitations is

(a) 0.05 (b) 0.16
(c) 0.27 (d) 0.38

ISRO Scientist/Engineer 12.05.2013
GATE 2009

Ans. (b) : For the max. discharge can passes through the reducer without causing cavitations means pressure at reducer exit will be fluid vapour pressure of 50 kPa.



P_1 = 150 kPa P_2 = 50kPa
sp. wt. $w = \rho g = 5$ kN/m^3
Applying Bernoulli's equation.

$$\frac{P_1}{\rho g}+\frac{V_1^2}{2g}+z_1=\frac{P_2}{\rho g}+\frac{V_2^2}{2g}+z_2$$

For Horizontal pipe $z_1 = z_2$

or, $$\frac{P_1-P_2}{\rho g}=\frac{V_2^2-V_1^2}{2g}$$

or, $$V_2^2-V_1^2=\frac{2\times g(P_1-P_2)}{\rho\times g}$$

$$=\frac{2\times 9.81\times(150-50)\times 10^3}{5\times 10^3}$$

$$V_2^2-V_1^2=392.4 \quad \text{.....(i)}$$

So, apply continuity equation.

$$A_1V_1=A_2V_2$$

$$\frac{\pi}{4}(0.2)^2\times V_1=\frac{\pi}{4}(0.1)^2V_2$$

$$V_2=4V_1$$

or $15V_1^2=392.4$ or $V_1^2=26.16$

or $V_1 = 5.1147$ m/s

$$Q=A_1V_1$$

$$=\frac{\pi}{4}(0.2)^2\times 5.1147$$

$Q = 0.16$ m^3/sec

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701. Bernoulli's equation represent the

(a) Energies at any point in the flow field and is obtained by integrating the Euler equations

(b) Forces at any point in the flow field and is obtained by integrating the momentum equation for viscous flows

(c) Momentum at any point in the flow field and is obtained by integrating the equation of continuity

(d) Moment of momentum and is obtained by integrating the energy equation

ESE 18.07.2021

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Euler's equation of motion–

$$\frac{dP}{\rho} + gdz + vdv = 0$$

Bernoulli's equation is obtained by integrating the Euler's equation

$$\int \frac{dP}{\rho} + \int gdz + \int v\,dv = \text{constant}$$

If flow is incompressible ρ is constant

$$\frac{P}{\rho} + gz + \frac{v^2}{2} = \text{constant}$$

$$\boxed{\frac{P}{\rho g} + z + \frac{v^2}{2g} = \text{constant}}$$

Each term in Bernoulli's equation represents energy per unit weight

702. Which of the following equation gives actual discharge through a venturimeter? (Where a_1 and a_2 are the cross sectional areas at the inlet section and throat of the venturimeter, g is the acceleration due to gravity. H is the venturihead; and C_d is the coefficient of discharge.)

(a) $C_d \dfrac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 - a_2^2}}$ (b) $\dfrac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 - a_2^2}}$

(c) $C_d \dfrac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 + a_2^2}}$ (d) $\dfrac{a_1 a_2 \sqrt{2gH}}{2\sqrt{a_1^2 - a_2^2}}$

TNPSC AE 2019

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : $C_d \dfrac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 - a_2^2}}$

703. Change in angular momentum of fluid flowing in curved path results in a

(a) change in pressure

(b) change in total energy

(c) dynamic force passing through its center of curvature

(d) torque

TSPSC Manager (Engg.) HMWSSB 12.11.2020

APPSC AEE 2012

Ans. (d) : Change in angular momentum of fluid flowing in a curved path results in a torque.

704. Water is coming out from a tap and falls vertically downwards. At the tap opening, the stream diameter is 20mm with uniform velocity of 2 m/s Acceleration due to gravity is 9.81 m/s^2. Assuming steady, inviscid flow, constant atmospheric pressure everywhere and neglecting curvature and surface tension effects, the diameter in mm of the stream 0.5m below the tap is approximately

(a) 10 (b) 15

(c) 20 (d) 25

GPSC Asstt. Director of Transport 05.03.2017

GATE 2013

Ans. (b) : Given, $d_1 = 20$ mm, $d_2 = ?$

$$v_1 = 2 \text{ m/s}, \; v_2 = \sqrt{2gh}$$

$$Q = A_1V_1 = A_2V_2$$

$$\frac{\pi}{4} \times (20)^2 \times 2 = \frac{\pi}{4} \times d_2^2 \times \sqrt{2 \times 9.8 \times 0.5}$$

$$d_2 = 15 \text{ mm}$$

705. For measurement of fluid flow, venturimeter is preferred over an orifice plate because.

(a) it is cheaper and easy to install

(b) Co-efficient of discharge is constant

(c) Energy or head losses are lower

(d) Accuracy is very high

GPSC Lect. (Auto) 16.10.2016

SJVN ET 2013

Ans. (c) : Energy or head losses are lower in venturimeters as compared to orificemeter hence it is preferred over orificemeter.

Venturimeter and orificemeter both are used to measure rate of flow or discharge.

706. In a pipe of 90 mm diameter water is flowing with a mean velocity of 2 m/s and a gauge pressure of 350 kN/m^2, The total head, if pipe is 8 metres above the datum line.

(a) 23.88 m (b) 43.88 m

(c) 33.88 m (d) 53.88 m

TSPSC AEE 2017

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : We know,

$$H = Z + \frac{v^2}{2g} + \frac{P}{\rho g} = 8 + \frac{2^2}{2 \times 9.81} + \frac{350 \times 10^3}{9.81 \times 10^3}$$

$$H = 43.88\text{m}$$

707. The equations of motion for a viscous fluid are known as

(a) Euler's equation

(b) Reynolds equation

(c) Navier-Stokes equation

(d) Hagen-Poiseuille equation

Assam PSC CCE Pre 2015

OPSC Civil Services Pre. 2011

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Ans. (c) : The Navier-Stoke's equation of motion of fluid considered only gravity, pressure and viscous forces only. Hence these equations are applicable in flows either laminar by viscous flows. There is no term in these equations to count compressibility turbulence and surface tension forces. But the viscous or linear flow has very low R_e. Hence the Navier stokes equations are solved analytically for $R_e << 1$.

708. Coefficient of velocity (C_v) is defined as the ratio between _____.

(a) Theoretical velocity of jet at vena contracta to the actual velocity
(b) Actual velocity of jet at vena contracta & Ideal velocity of the jet
(c) Area of jet at vena contracta to the actual velocity
(d) Theoretical velocity of jet at vena contracta to Area of jet at vena contracta

VIZAG Steel MT 24.01.2021 Shift-II
MECON MT 2019

Ans. (b) : Coefficient of velocity (C_v) is defined as the ratio between actual velocity of jet at vena-contracta and ideal velocity of jet.

$$C_v = \frac{V_{actual}}{V_{ideal}}$$

709. If a fluid jet discharging from a 50 mm diameter orifice has a 40 mm diameter at its vena contracta, then its coefficient of contraction will be _____.

(a) 0.52 (b) 0.32
(c) 1.56 (d) 0.64

VIZAG Steel MT 24.01.2021 Shift-II
CSE Pre-1996

Ans. (d) : Given,
Dia at vena contracta (d) = 40 mm
Dia of orifice (d_1) = 50 mm
Coefficient of contraction (C_c)

$$= \frac{\text{Area of jet at vena contracta}}{\text{Area of jet at orifice}}$$

$$= \frac{\frac{\pi}{4}\times(40)^2}{\frac{\pi}{4}\times(50)^2} = \frac{1600}{2500}$$

$$C_c = \frac{16}{25} = 0.64$$

710. Flow of water in a pipe about 3m in diameter can be measured by:

(a) orifice plate (b) venturimeter
(c) rotameter (d) pitot tube

MPPSC State Forest Service Exam, 2014
J & K PSC Screening, 2006

Ans. (d) : Pitot tube is used to measure velocity but indirectly it can measure discharge as well using continuity equation. Orifice, venturimeter and rotameter are used for direct measurement of discharge.

711. Navier Stokes equations are associated with:

(a) Buoyancy (b) Turbulance
(c) Viscosity (d) Compressiblity

UKPSC AE-2013, Paper-II
J & K PSC Screening, 2006

Ans. (c) : Navier stokes equations are associated with viscosity.

712. Velocity at a point in a pipe flow may be measured by installing

(a) a Pitot probe at that point
(b) a wall trap
(c) a stagnation pressure probe at that point
(d) a Prandtl probe at that point

UPPSC AE 13.12.2020, Paper-2
OPSC Civil Service Pre-2006

Ans. (a) : Velocity at a point in a pipe flow may be measured by installing a pitot probe at that point.

713. A pipe has 10 mtr velocity head at outlet. What height will be reached by fluid stream, if pipe is kept vertical?

(a) 100 mtr
(b) 50 mtr
(c) 10 mtr
(d) 14.10 mtr

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
JPSC AE, 10.04.2021, Paper-I

Ans. (c) : Velocity head (h) $= \frac{v^2}{2g} = 10\,\text{mtr}$

Height reached by fluid stream = Velocity head = 10 mtr.

714. A fluid jet is discharging from a 100 mm nozzle and the vena-contracta formed has a diameter of 90 mm. If the coefficient of velocity is 0.95, then the coefficient of discharge for the nozzle is

(a) 0.7695 (b) 0.81
(c) 0.9025 (d) 0.855

ISRO Scientist/Engineer 2011
CSE Pre-1994

Ans. (a) : Diameter of Jet = 100 mm
Diameter of vena-contracta = 90 mm
Coefficient of velocity $C_v = 0.95$
$C_d = C_c \times C_v$
C_c = Coefficient of contraction

$$C_c = \frac{\text{Area of vena contracta}}{\text{Area of Jet}}$$

$$= \frac{\pi(90)^2}{4} / \frac{\pi(100)^2}{4}$$

$= 0.81$
$C_d = C_c \,.\, C_v$
$= 0.81 \times 0.95$
$C_d = 0.7695$

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715. Which is the false statement about a torque converter?

(a) the blades have a curved shape
(b) the stator unit redirects the flow of oil to the impeller
(c) maximum torque multiplication occurs at low speed
(d) the oil is driven by the impeller unit

HPPSC Poly. Lect. 05.07.2021

Ans. (c) :

716. The pressure that would be reached if the local flow is imagined to slow down to zero velocity frictionlessly is defined as :

(a) Static pressure (b) Stagnation pressure
(c) Dynamic pressure (d) Kinematic pressure

RPSC ACF & FRO, 26.02.2021

Ans. (b) : The pressure that would be reached if the local flow is imagined to slow down to zero velocity frictionless is defined as stagnation pressure.

717. A Prandtl tube (Pitot-static tube with C_v =1) is used to measure the velocity of water. The differential manometer reading is 10mm of liquid column with a relative density of 11. Assuming g = 9.8 m/s^2, the velocity of water (in m/s) is :

(a) 1.32 (b) 1.4
(c) 1.6 (d) 1.5

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : Velocity of water,

$$v = C_v\sqrt{2gh} \qquad \because h = x\left(\frac{S_m}{S} - 1\right)$$

$$= C_v\sqrt{2gx\left(\frac{S_m}{S} - 1\right)}$$

$$= 1\times\sqrt{2\times 9.81\times 0.010\times(11-1)}$$

$$\boxed{v = 1.40\,m/s}$$

718. Of the various method of measuring discharge through a pipeline, the one with least loss of energy and direct reading is by means of :

(a) orifice plate
(b) notch
(c) traversing a Pitot-static probe
(d) venturimeter

CGPSC AE 15.01.2021

Ans. (d) : On various methods of measuring discharge through a pipe line, loss of energy and direct reading is in order.

$$\underset{\text{Loss of energy increases}}{\text{Venturimeter} < \text{Nozzle} < \text{Orificemeter}} \longrightarrow$$

719. Which fluid mechanics principle is involved in the game of cricket, tennis and table tennis?

(a) Rotational flow and circulatory flow on the ball
(b) Flow with circulation only
(c) Irrotational flow
(d) None of these

CGPSC AE 15.01.2021

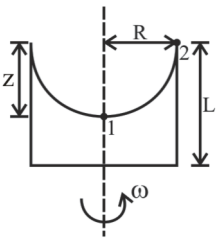
Ans. (d) : The Magnus effect explains commonly observed deviations from the typical trajactories or paths of spinning balls in sport, football, table tennis, volleyball, golf, baseball and cricket.

720. An open circular tank of radius R mm and long L mm contains water up to a height of x mm (where, x<L). The tank is rotated about its vertical axis at ω r.p.s, the depth of parabola (Z) varies with _____.

(a) the square of R & the square root of ω
(b) the square root of ω only
(c) the square of R only
(d) the square of R & ω

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (d) :



Depth of parabola (z) $= \dfrac{R^2\omega^2}{2g}$

721. A liquid mass readjusts itself and undergoes a rigid body type of motion when it is subjected to a

(a) constant angular velocity
(b) constant angular acceleration
(c) linearly varying velocity
(d) linearly varying acceleration

JPSC AE, 10.04.2021, Paper-I

Ans. (a) : A liquid mass readjusts itself and undergoes a rigid body type of motion when it is subjected to a constant angular velocity.

722. In a rotameter as the flow rate increases, the float

(a) rotates at higher speed
(b) rotates at lower speed
(c) rises in the tube
(d) drops in the tube

JPSC AE, 10.04.2021, Paper-I

Ans. (c) : In rotameter as the flow rate increases, the float rises in the tube.

723. The differential manometer connected to pitot static tube used for measuring fluid velocity gives

(a) static pressure
(b) total pressure
(c) dynamic pressure
(d) difference between total pressure & dynamic pressure

JPSC AE, 10.04.2021, Paper-I

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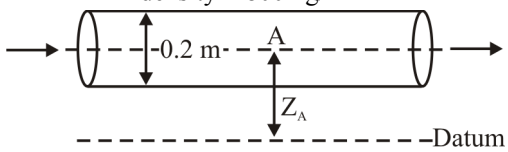
Ans. (c) : The differential manometer connected to pitot static tube used for measuring dynamic pressure.

724. Oil of specific gravity 0.8 flows through a 0.2 m diameter pipe under a pressure of 100 kN/ m^2. If the datum is 5 m below the centerline of the pipe and the total energy with respect to the datum is 35 Nm/N, the discharge is (Take g = 9.81 m/s^2)

(a) 0.58 m^3/sec (b) 0.47 m^3/sec
(c) 0.31 m^3/sec (d) 0.22 m^3/sec

ESE 18.07.2021

Ans. (a) : Given, Specific gravity = 0.8
density = 800 kg/m^3



Applying Bernoulli's equation-

$$\frac{P_A}{\rho g}+\frac{V_A^2}{2g}+Z_A=35$$

$$\frac{100\times10^3}{800\times9.81}+\frac{V_A^2}{2\times9.81}+5=35$$

$$12.74+\frac{V_A^2}{19.62}+5=35$$

$$\frac{V_A^2}{19.62}=17.26$$

$$V_A = 18.402 \text{ m/s}$$

Discharge, $Q = V_A \times \frac{\pi}{4}d^2$

$$= 18.402 \times \frac{\pi}{4}\times(0.2)^2$$

$$Q = 0.578 \text{ m}^3/\text{sec}.$$

725. The velocity of water is found by using a pitot-static tube. The stagnation pressure head and static pressure head was found to be 5 m and 3 m respectively. If the coefficient of tube is 0.97, the velocity water is

(a) 4.34 ms^{-1} (b) 6.07 ms^{-1}
(c) 5.67 ms^{-1} (d) 1.38 ms^{-1}

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Given, $C_v = 0.97$, $P_{stag} = 5$ m, $P_{static} = 3$ m
Velocity of water,

$$v = C_v\sqrt{2g(h_{stag}-h_{static})}$$

$$= 0.97\sqrt{2g(5-3)}$$

$$\boxed{v = 6.07 \text{ m/s}}$$

726. Find the velocity of the flow of an oil through a pipe, when the difference of mercury level in a differential U-tube manometer connected to the two, tapings of the pitot tube is 100 mm. Take coefficient of pitot tube 0.98 and specific gravity of oil = 0.8.

(a) 4.39 m/s
(b) 6.8 m/s
(c) 5.49 m/s
(d) 9 m/s

Oil India Senior Officer 23.12.2020

Ans. (c) : $V = C_v\sqrt{2gh}$

$$V = C_v\sqrt{2gy\left(\frac{\rho_m}{\rho_\ell}-1\right)}$$

$$V = 0.98\sqrt{2\times9.81\times0.1\left(\frac{13600}{800}-1\right)}$$

$$V = 5.49 \text{ m/s}$$

727. Find the velocity of the flow of an oil through a pipe, when the difference of mercury level in a differential U-tube manometer connected to the two tappings of pitot-tube is 100 mm. Take co-efficient of pitot tube 0.98 and specific gravity of oil = 0.8

(a) 5.49 m/sec
(b) 5.94 m/sec
(c) 4.49 m/sec
(d) 4.94 m/sec

RPSC IOF, 2020

Ans. (a) : Manometer reading = 100 mm = 0.1 m
Specific gravity of oil = 0.8
Specific gravity of mercury = 13.6
Co-efficient of pitot tube (C) = 0.98
Density of water = 1000 kg/m^3
Density of oil = 0.8×1000 = 800 kg/m^3
Density of mercury = 13.6 × 1000 = 13600 kg/m^3

Velocity of flow of an oil $V = C\sqrt{\frac{2g(\rho_m-\rho_{oil})\times h}{\rho_{oil}}}$

$$= 0.98\sqrt{\frac{2\times9.81(13600-800)\times0.1}{800}}$$

$$V = 5.49 \text{ m/sec}$$

728. Water is flowing in a smooth pipe of uniform diameter 10 cm, has following data :

At section 1 : Pressure = 50 kPa, Elevation =10m

At section 2 : Pressure = 20 kPa, Elevation =12m

The velocity of flow is 2 m/s. Which of the following is correct for this pipe?

(a) Flow is taking place from section 1 to section 2 with a head loss of 2 m
(b) Flow is taking place from section 1 to section 2 with a head loss of 1 m
(c) Flow is taking place from section 2 to section 1 with a head loss of 1 m
(d) Flow is taking place from section 2 to section 1 with a head loss of 2 m

APPSC Poly Lect. 13.03.2020

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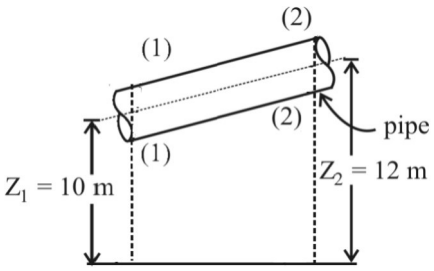
Ans. (b)



$P_1 = 50$ kPa $= 50 \times 10^3$ N/m^2
$P_2 = 20$ kPa $= 20 \times 10^3$ N/m^2
$v_1 = v_2 = 1$ m/s
Let,
h_2 = head loss
for water,
Density (ρ) = 1000 kg/m^2
let g = 10 m/s^2
Applying Bernoulli's equation between

$$\frac{P_1}{\rho g}+\frac{v_1^2}{2g}+z_1=\frac{P_2}{\rho g}+\frac{v_2^2}{2g}+z_2+h_2 \quad (\because v_1=v_2)$$

$$\Rightarrow h_2=\frac{P_1-P_2}{\rho g}+(z_1-z_2)$$

$$=\frac{(50-20)\times10^3}{1000\times10}+(10-12)$$

$$\Rightarrow h_2=\frac{30}{10}-2=3-2=1$$

$h_2 = 1$ m

Now, Flow is taking place from section (1) to section (2) with a head loss (h_2) of 1 m.

729. Assumptions for validity of Bernoulii equation along a streamline include :

(a) Incompressible, viscous, isothermal
(b) Compressible, viscous, adiabatic
(c) incompressible, inviscid, isothermal
(d) compressible, inviscid, isothermal

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Assumptions of Bernoulli's equation are steady (no change with time), incompressible, inviscid & isothermal.

730. What will be the velocity of an oil (specific gravity = 0.8) through a pipe, if the differences of mercury level in a differential U-tube manometer connected to the two tapping of the pitot-static tube is 50 mm. [Coefficient of pitot tube = 1, g = 10 m/s^2]

(a) 1 m/s (b) 2 m/s
(c) 3 m/s (d) 4 m/s

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Ans. (d) : Given,
Specific gravity oil (S_o) = 0.8
Coefficient of pitot (C_v) = 1
Acceleration due to gravity (g) = 10 m/s^2
Let mercury level difference (x) = 50 mm = 50×10^{-3}m
specific gravity of Manometric fluid
(mercury - S_m) = 13.6
Now,
We know that,

$$h = x\left(\frac{S_m}{S_0}-1\right) \quad (\because S_m > S_o)$$

$$v_{actual} = C_v v_{th}$$

$$= C_v\sqrt{2gh}$$

$$= C_v\sqrt{2gx\left(\frac{S_m}{S_o}-1\right)}$$

Put the value –

$$v = 1\sqrt{2\times10\times0.05\left(\frac{13.6}{0.8}-1\right)}$$

$v = 4$ m/s

731. An open cylindrical vessel of 20 cm diameter and 100 cm long contains water up to a height of 80 cm. What will be the rotational speed of the vessel about its vertical axis such that no water spills? [g = 10 m/s^2]

(a) $10\sqrt{2}$ rad/s (b) $20\sqrt{2}$ rad/s
(c) $30\sqrt{2}$ rad/s (d) $40\sqrt{2}$ rad/s

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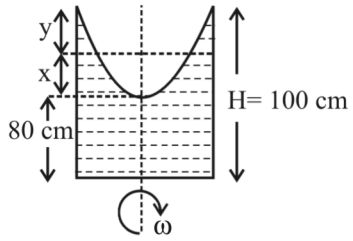
Ans. (b) :



Cylinder fill with water

Given data : Dia of cylinder (d) = 20 cm
Lets ω = Rotational speed of vessel.
For the condition of water is
no spills –
x = y = 20 cm
$\therefore$ Height of parabola (z) = x + y
= 20 + 20 = 40 cm = 0.4 m

$$\therefore z=\frac{r^2\omega^2}{2g} \qquad g = 10 m/s^2$$

r = d/2 = 20/2 = 10 cm = 0.1 m

$$\Rightarrow \omega^2=\frac{2gz}{r^2}=\frac{2\times10\times0.4}{(0.1)^2}$$

$\omega^2 = 2 \times 10 \times 40$

$\boxed{20\sqrt{2} \text{ rad/sec}}$

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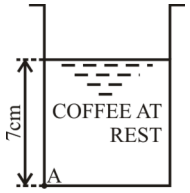
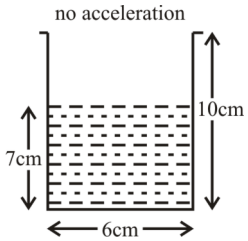
732. A person rests her coffee mug on a horizontal tray while she accelerates at $\frac{10}{\sqrt{3}}$ m/s^2. The mug is 10 cm deep and 6 cm in diameter and contains coffee 7 cm deep at rest. Assuming rigid body acceleration, calculate gauge pressure in the bottom corner point A as shown in figure if density of coffee is 1010 kg/m^3. Assume acceleration due to gravity g = 10 m/s^2.



(a) 965 Pa (b) 1005 Pa
(c) 865 Pa (d) 765 Pa

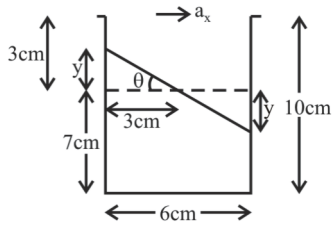
ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (c) :



having a acceleration

$\rho_{coffee} = 1010\,kg/m^3, a_x = \frac{10}{\sqrt{3}} m/sec^2$



$\tan\theta = \frac{a}{g}$

$\tan\theta = \frac{10}{\sqrt{3}\times 10} = \frac{y}{x}$

$\theta = 30°$ $\quad \frac{y}{x} = \tan 30°$

$y = 3 \times \tan 30°$

$y = 1.732$ cm

$y < 3$ cm, So No spillage,

For No spillage, Rise = Fall

Height (h) at A = 7 + 1.732 = 8.732 cm

p_{gauge} at A = $\rho gh = 1010 \times 10 \times 0.08732 = 881.932$ Pa

733. Which of the following methods is used for the measurement of flow rate of a liquid?

(a) Coriolis method
(b) Conveyor-based method
(c) Thermal mass flow method
(d) Bourdon tube method

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a, c) : Coriolis method is used for the measuring liquid flow rate. Orifice plate, venturi tube and flow nozzle all these device are used for flow abstraction.

734. An open tank is filled with water to a height of 20 m. What is the velocity of the water flow at the outlet, if the outlet is at the base of the tank?

(a) 40 m/s (approx.) (b) 20 m/s (approx.)
(c) 10 m/s (approx.) (d) 5 m/s (approx.)

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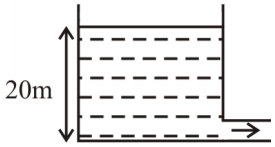
Ans. (b) :



$V = \sqrt{2gh} = \sqrt{2\times 10\times 20}$

V = 20 m/sec.

735. Which of the following equation states, "In an ideal incompressible fluid, when the flow is steady and continuous, the sum of pressure energy, kinetic energy and potential energy is constant along a stream line"?

(a) Bernoulli's equation
(b) Euler's equation
(c) Cartesian Equation
(d) Continuity equation

VIZAG MT, 14.12.2020

Ans. (a) : Bernoulli's equation

736. In a pipe of 100 mm diameter, water is flowing with a mean velocity of 2 m/s and at a gauge pressure of 350 kN/m^2. The total energy of water if the pipe is 8 m above the datum line is ____ (Assume 'g' as 10 m/s^2).

(a) 43.88 m (b) 4.388 m
(c) 0.4388 m (d) 438 m

VIZAG MT, 14.12.2020

Ans. (a) : Total energy head $= \frac{P}{\rho g} + \frac{v^2}{2g} + z$

$= \frac{350\times 10^3}{1000\times 10} + \frac{2^2}{2\times 10} + 8$

$= 35 + 0.2 + 8$

$= 43.2$ m.

Note-Question removed by RINL.

737. Which one of the following is an irrotational flow?

(a) Free vortex flow (b) Forced vortex flow
(c) Coutte flow (d) Wake flow

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : Free vortex flow is an irrotational flow.

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738. Which one of the following is not a case of ideal fluid flow?

(a) Inviscid (b) Incompressible
(c) Forced vortex flow (d) Super critical flow

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Ans. (c) : Ideal fluid—A fluid is said to be ideal if it is assumed to be both incompressible and non viscous.
- Its bulk modulus is infinite.
- Ideal fluid has no surface tension.
- Ideal fluid are imaginary and do not exist in nature.

739. The angle of the diffuser portion of a venturimeter is less than the angle of the converging portion to :

(a) increase to velocity of flow in the direction of flow at the diverging part
(b) avoid the situation where flow may become compressible
(c) minimize the loss of energy due to flow separation
(d) decrease the pressure in the direction of flow at diverging part

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Ans. (c) : The angle of the diffuser portion (6^o to 7^o) of a venturimeter is less than the angle of the convergent portion (20^o to 22^o) to minimize the loss of energy due to flow separation.

740. The flow through a Venturimeter is proportional to:

(a) $H^{1/5}$ (b) H
(c) $H^{3/2}$ (d) $H^{1/2}$

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Ans. : (d) : We know that discharge through venturimeter,

$$Q = \frac{c_d . A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2gH}$$

$Q \propto H^{1/2}$

741. A nozzle has 100 m velocity head at outlet. What height will be reached by fluid stream if the nozzle is kept vertical?

(a) 100 m (b) 10 m
(c) 141 m (d) Data insufficient

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Ans. : (a) : Given as,

$$\frac{u^2}{2g} = 100\text{m}$$

$u^2 = 2 \times 10 \times 100$

$u = 44.72$ m/s

then,

$v^2 = u^2 - 2g \times h$ (v = 0)

$u^2 = 2 \times 10 \times h$

$(44.72)^2 = 2 \times 10 \times h$

$h = 99.99 \simeq 100\,\text{m}$

742. In a flowing fluid, a particle may possess

(a) Inertial energy (b) Pressure energy
(c) Kinetic energy (d) All of the above

Gujarat PSC AE 2019

Ans. : (d) : In a flowing fluid, a particle may possess-
- Inertial energy
- Pressure energy
- Kinetic energy
- Elevation or gravitational potential energy.

743. If P_1 and P_2 refers to the pressure V_1, V_2 refers to the velocities ρ_1 and ρ_2 refers to specific gravity and Z_1 and Z_2 refers to elevation of two points 1 and 2 in a pipe, then the Bernoulli's equation can take the form

(a) $\frac{P_1}{\rho_1} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho_2} + \frac{V_2^2}{2g} + Z_2$

(b) $\frac{P_1}{\rho_1 g} + \frac{V_1^2}{2} + Z_1 = \frac{P_2}{\rho_2 g} + \frac{V_2^2}{2} + Z_2$

(c) $\frac{P_1}{\rho_1 g} + \frac{V_1^2}{2g} + gZ_1 = \frac{P_2}{\rho_2 g} + \frac{V_2^2}{2g} + Z_2$

(d) $\frac{P_1}{\rho_1 g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho_2 g} + \frac{V_2^2}{2g} + Z_2$

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Ans. (d) : Bernoulli's theorem [When no loss]

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

When loss of energy take place during flow of liquid from (1) to (2)

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h$$

Where $h \rightarrow$ head loss

744. Consider a tank attached with an orifice of diameter d. If H is the head of the liquid above the centre of the orifice, the theoretical velocity of the discharged flow through the orifice is given by

(a) $\sqrt{gH}$ (b) $\sqrt{2gH}$
(c) $\sqrt{gD}$ (d) $\sqrt{2gd}$

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Ans. (b) : As per Torricelli's law, velocity through orifice at bottom of the tank is given by

$$V = \sqrt{2gH}$$

where,

H = Head above orifice

745. In the category of flow meters, head loss is at the least for

(a) orifice meter (b) nozzle flow meter
(c) venturi meter (d) manometer

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Ans. (c) : Venturi meter does not have flow separation due to gradual expansion hence head loss is minimum.

746. Navier-Stokes equation in fluid mechanics is derived from the

(a) Newton's first law of motion
(b) Newton's second law of motion
(c) Newton's third law of motion
(d) Newton's law of viscosity

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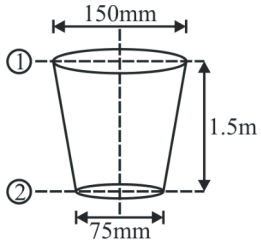
Ans. (b) : Navier - stokes equation is application of Newton's second law to the motion of fluid element.

747. A vertical water pipe 1.5 m long, taper from 75 mm diameter at the bottom to 150 mm diameter at the top and the rate of flow is 50 L/s upwards. If the pressure at the bottom end is 150 kN/m^2, the pressure at the top will be nearly

(a) 195.2 kN/m^2 (b) 191.4 kN/m^2
(c) 187.6 kN/m^2 (d) 183.8 kN/m^2

ESE 2019

Ans. (a) :



Given,

$Z_1 = 1.5$ m

$Z_2 = 0$

$Q = 50 \ \ell/s = 0.05 \ m^3/s$

$d_2 = 75$ mm $= 0.075$ m

$d_1 = 150$ mm $= 0.15$ m

$p_2 = 150$ kN/m^2

$\because \quad Q = A_1V_1 = A_2V_2$

$$V_1 = \frac{Q}{A_1} = \frac{4Q}{\pi d_1^2} = 2.83 \text{ m/s}$$

$$V_2 = \frac{Q}{A_2} = \frac{4Q}{\pi d_2^2} = 11.32 \text{ m/s}$$

Applying Bernoulli's equation between (1) and (2)

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$\frac{p_1}{\rho g} + \frac{(2.83)^2}{2\times 9.81} + 1.5$$

$$= \frac{150\times 10^3}{1000\times 9.81} + \frac{(11.32)^2}{2\times 9.81} + 0$$

$$\frac{p_1}{\rho g} + 0.408 + 1.5$$

$= 15.29 + 6.531$

$$\frac{p_1}{\rho g} = 19.913$$

$p_1 = 195346$ N/m^2

$p_1 = 195.34$ kN/m^2

748. The coefficient of discharge for orifice meter as compared to coefficient of discharge for Venturimeter is:

(a) Equal (b) Very High
(c) Very Small (d) High

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Ans. : (c) : For Venturimeter $C_{dv} = 0.94 - 0.98$
For orificemeter $C_{dv} = 0.64 - 0.76$

749. Pick the correct statement about the piezometric head in a static liquid.

(a) It remains constant only in a horizontal plane.
(b) It remains constant at all points in the liquid.
(c) It decreases linearly with depth below a free surface.
(d) It increases linearly with depth below a free surface.

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Ans. (b) : It remains constant at all points in the liquid.

750. In a "free-vortex", velocity potential line (ϕ) is a function of

(a) angle (b) radius
(c) angle and radius (d) velocity

APPSC-AE-2019

Ans. (a) : The velocity potential for free vortex is given by

$$\phi = \frac{\Gamma}{2\pi}\theta$$

751. The radial component of velocity everywhere in a free vortex Motion is

(a) None-zero and finite (b) Zero
(c) Maximum (d) Minimum

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Ans. : (b) : The radial component of velocity everywhere in a free vortex motion is zero.

752. What is the approximate value of coefficient of velocity for sharp edge orifice, if a jet from under constant head of 25 cm have horizontal and vertical coordinate of vena contract at 18 cm and 4 cm respectively?

(a) 0.95 (b) 1.00
(c) 0.90 (d) Insufficient data

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Ans : (c) :

$$C_v = \frac{x}{\sqrt{4hy}} = \frac{18}{\sqrt{4\times 25\times 4}} = 0.90$$

753. For an ideal and steady flow, "the total energy of a fluid at a point is constant" is the statement of

(a) Euler's equation
(b) Pascal's law
(c) Navier-Stokes equation
(d) Bernoulli's theorem

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Ans. (d) : As per Bernoulli's equation sum of pressure energy, kinetic energy and potential energy is constant along streamline for steady incompressible flow.

754. The fluid property responsible for the problem of water hammer is

(a) viscosity (b) density
(c) elasticity (d) surface tension

ISRO Scientist/Engineer (RAC), 10.03.2019

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Ans. (b) : Water Hammer–When a liquid flow in a long pipeline is reduced suddenly, due to the compressibility of the liquid, the sudden change in momentum would cause a pressure surge to develop.

This pressure moves through the pipe at high speed and undergoes reflection at the ends and the phenomenon is known as water hammer.

The pressure rise in water hammer depend upon–

- The velocity of flow water in pipe
- Time taken to close the valve
- The length of pipe
- Elastic properties of the material of the pipe
- The density of fluid

Thus the fluid property which affects the water hammer is density.

755. An orifice is said to be large, if

(a) The size of orifice is large
(b) The velocity of flow is large
(c) The available head of liquid is more than 5 times the height of orifice
(d) The available head of liquid is less than 5 times the height of orifice

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : An orifice is a small aperture through which the fluid passes. The thickness of an orifice in the direction of flow is very small in comparison to its other dimensions. An orifice is said to be large if the available head of liquid is less than 5 times the height of orifice.

756. The discharge through a convergent mouthpiece is ..... the discharge through an internal mouthpiece of the same diameter and head of water

(a) Equal to (b) One half
(c) Three fourth (d) Double

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : The discharge through a convergent mouthpiece is double the discharge through an internal mouthpiece of the same diameter and head of water.

757. Which of the following is correct for venturimeter?

(a) It is used to measure the velocity of the flow
(b) The difference of pressure head at section 1 and 2 is given by, $x\left[\frac{s_m}{s}-1\right]$ Where, x =difference in level of manometric head between two limbs of U-tube manometer, S_m = specific gravity of manometric fluid, and S = specific gravity of flowing fluid ($S_m > S$)
(c) Its convergent cone has included angle less than that of divergent cone
(d) The length of convergent cone is larger than that of divergent cone
(e) Actual discharge through venturimeter is, $Q_a = \frac{c_d a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}}$

CGPSC AE 25.02.2018

Ans. (b,e) : Venturimeter– A venturimeter is a device used to measure the fluid flow through pipes. This flow measurement device is based on the principle of Bernoulli's theorem.

- The difference of pressure head at section 1 and 2 is given by, $x\left[\frac{s_m}{s}-1\right]$ Where, x =difference in level of manometric head between two limbs of U-tube manometer, S_m = specific gravity of manometric fluid, and S = specific gravity of flowing fluid ($S_m > S$)
- Its convergent cone has included angle (20° ± 1°) more than that of divergent cone (5° to 7°)
- The length of convergent cone is smaller than that of divergent cone
- Actual discharge through venturimeter is, $Q_a = \frac{c_d a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}}$

Note-The option (b), (e) both are correct.

758. If coefficient of discharge for an orifice is 0.6, coefficient of velocity is 0.95, then the coefficient of contraction will be :

(a) 0.57 (b) 0.631
(c) 1.58 (d) 1
(e) 0.921

CGPSC AE 25.02.2018

Ans. (b) : We know that -

Coefficient of discharge (C_d) = coefficient of velocity (C_V) × coefficient of contraction (C_C)

$C_d = C_c \times C_v$

$0.6 = 0.95 \times C_c$

$C_c = \frac{0.6}{0.95} \Rightarrow \boxed{C_c = 0.6315}$

759. Statement I: There exists a positive pressure difference between the inlet and throat of a venturi meter.
Statement II: The coefficient of discharge of a venturi meter accounts for the non-uniformity of flow at both inlet and throat.

ESE 2018

Ans. (b) : From inlet to throat area decreases, velocity increase and pressure decreases. Hence statement-I is correct.

The co-efficient of discharge considers all losses occurring due to non ideal behaviour of fluid. It includes surfaces friction as well as internal friction occurring due to non uniform velocity at each section. Hence both the statements are correct but statement-II is not related to statement-I.

760. Statement I: Non-viscous flow between two plates held parallel with a very small spacing between them is an example of irrotational flow.
Statement II: Forced vortex implies irrotational flow.

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Ans. (c) : Non viscous flow between two parallel plates has uniform velocity at each section. In uniform flow all velocity gradients are zero hence it is irrotational.
Forced vortex is not irrotational flow. It is always rotational flow.
Hence statement-II is wrong.

761. The total energy of each particle at various places in the case of a perfect incompressible fluid flowing in a continuous stream
(a) keeps on increasing
(b) keeps on decreasing
(c) remains constant
(d) may increase or decrease

ESE 2018

Ans. (c) : For a perfect incompressible liquid, flowing in a continuous stream, the total energy of a particle remains the same, while the particle moves from one point to another. This statement is called Bernoulli's equation.

762. The pressure of the liquid flowing through the divergent portion of a venturimeter
(a) Remains constant
(b) Increases
(c) Decreases
(d) depends upon mass of liquid

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : The function of converging cone is to accelerate the flow and create pressure difference between the inlet converging cone and the throat.
The function of throat is to stabilize the flow and facilitate the provision of pressure tapping. The function of diverging cone is to reduce the velocity and increase the pressure to its original value to the extent it is possible practically.

763. In a steady flow along a stream line at a location in the flow, the velocity head is 6 m, the pressure head is 3 m, and the potential head is 4 m. The height of hydraulic gradient line at this location will be
(a) 13 m (b) 9 m
(c) 10 m (d) 7 m

RPSC Vice Principal ITI 2018

Ans. (d) : The height of hydraulic gradient line at this location will be,
= Pressure head + Potential head
= 3 + 4 = 7 m

764. The discharge through an orifice fitted in a tank can be increase by
(a) Fitting a short length of pipe to the outside
(b) Sharpening the edges of orifice
(c) Fitting a long length of pipe to the outside
(d) Fitting a long length of pipe to the inside

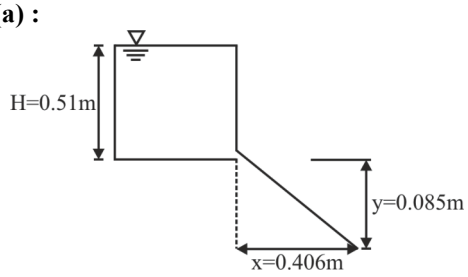
Vizag Steel (MT) 2017

Ans. (a) : The discharge through an orifice fitted in a tank can be increased by fitting a short length of pipe to the outside.

765. A jet of water issues from a sharp-edged vertical orifice under a constant head of 0.51m. At a certain point of the issuing jet, the horizontal and vertical coordinates measured from vena contracta are 0.406 m and 0.085m, respectively. What is the value of the coefficient of velocity?
(a) 0.975 (b) 0.925
(c) 0.875 (d) 0.825

ESE 2017

Ans. (a) :



Theoretical velocity $(V_{th}) = \sqrt{2gH}$
$= \sqrt{2 \times 9.81 \times 0.51}$
$= 3.16$ m/s
Let V_{act} is the actual velocity of water at orifice
By using projectile motion

$$y = \frac{1}{2}gt^2, \qquad t = \sqrt{\frac{2y}{g}}$$

Actual velocity $(V_{act}) = \frac{x}{t} = \frac{x}{\sqrt{\frac{2y}{g}}}$

$$= \frac{0.406}{\sqrt{\frac{2 \times 0.085}{9.81}}} = 3.084 \text{ m/s}$$

Coefficient of velocity $C_V = \frac{V_{act}}{V_{th}}$

$$= \frac{3.084}{3.16} = 0.9749 = 0.975$$

766. A fluid jet discharging from 20 mm diameter orifice has a diameter 17.5 mm at its Vena-Contracta. The coefficient of contraction will be nearly:
(a) 0.98 (b) 0.88
(c) 0.77 (d) 0.67

UPSC JWM 2017

Ans. (c) : Coefficient of contraction

$$= \frac{\text{Area of jet at vena-contrata}}{\text{Area of orifice}}$$

$$= \frac{A_C}{A_O} = \frac{\pi/4\, d_c^2}{\pi/4\, d_o^2} = \left(\frac{d_c}{d_o}\right)^2$$

$$= \left(\frac{17.5}{20}\right)^2 = 0.765$$

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767. The generation of lift by spinning cylinder in a fluid stream is called as

(a) Magnus effect (b) Raman's effect
(c) Reynolds effect (d) Blasius effect

TSPSC AEE 2017

Ans. (a) : Magnus effect is an observable phenomenon that is commonly associated with a spinning object moving through air or another fluid, it generates lift.

768. For a perfect incompressible liquid flowing in a continuous stream, the total energy of a particle remains the same while the particle moves from one point to another. This statement is called

(a) Continuity equation
(b) Bernoulli's equation
(c) Pascal's law
(d) Archimedes' principle

Karnataka PSC AE, 10.09.2017

Ans. (b) : Total energy of a particle remains same while moving in an incompressible fluid from one point to another point.

$$\frac{P}{\rho g}+\frac{V^2}{2g}+z = \text{constant}$$

769. The tangential velocity of the water element having a free vortex is :

(a) directly proportional to its distance from the centre
(b) inversely proportional to its distance from the centre
(c) directly proportional to its (distance)^2 from the centre
(d) inversely proportional to its (distance)^2 from the centre

Punjab PSC SDE 12.02.2017

Ans. (b) : Free vortex flow – In this flow fluid mass rotates due to conservation of angular momentum. The velocity profile is inversely proportional with the radius.

$vr = \text{constant}$

The point at the centre of Rotation is called singular point, where velocity approaches to infinite medium.

770. The flow of liquid around a circular bend in a pipe is an example of flow where no external torque is required to rotate the fluid mass. This flow is known as

(a) Free Vortex flow
(b) Uniform Potential flow
(c) Forced Vortex flow
(d) Superimposed Flow

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : When no external torque required to rotate the fluid mass that type of flow is called free vortex flow–

Example of free vortex flow -

1. Flow at liquid through a hole provide at the bottom of contains .
2. Flow of liquid around a circular bend in pipe.
3. A whirlpool in a river.
4. Flow of fluid in a centrifugal casing.

771. A water tank is located 3 m above ground and depth of water in tank is 2 m. Exit pipe is 50 mm in dia and 50 m long. Velocity of water at the exit at the ground level is

(a) 1.5 m/s (b) 2 m/s
(c) 0.7 m/s (d) ¼ m/s

ISRO Scientist/Engineer 17.12.2017

Ans. (*) : Note : This question is excluded for evaluation by the ISRO.

772. The energy loss in flow through nozzle as compared to venturimeter is

(a) same (b) more
(c) Laminar (d) Vortex

Vizag Steel (MT) 2017

Ans. (b) : The energy loss in flow through nozzle as compared to venturimeter is more.

773. Which of the following device is used to measure the airflow velocity in wind tunnel testing?

(a) Anemometer (b) Altimeter
(c) Barometer (d) Steam generator

TNPSC AE 2017

Ans. (a) : Anemometer is a device used to measure the airflow velocity in wind tunnel testing.

774. Statement (I) : When a liquid is poured in different shaped vessels with base area same and equal height, the apparent contradiction of equal force at base, but different weights of liquids is hydrostatic paradox.
Statement (II) : Bernoulli's equation is valid for inviscid liquids.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I).
(c) Statement (I) is true but Statement (II) is false.
(d) Statement (I) is false but Statement (II) is true.

UPSC JWM 2017

Ans. (b) : The Bernoulli's equation is an approximate equation that is valid only in inviscid regions of flow where net viscous forces are negligibly small compared to inertial, gravitational, or pressure forces.

775. Coefficient of contraction of nozzle is ____

(a) $C_c = C_v/C_d$ (b) $C_c = C_d/C_v$
(c) $C_c = C_d \times C_v$ (d) None of these

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : Coefficient of contraction of nozzle is (C_c)

$$= \frac{\text{Coefficient of discharge}}{\text{Coefficient of Velocity}}$$

$$C_c = \frac{C_d}{C_v}$$

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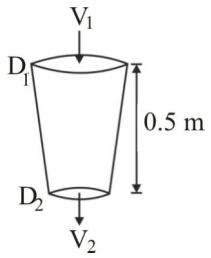
776. Water is coming out from a tap and falls vertically downwards. At the tap opening, the stream diameter is 20 mm with uniform velocity of 2 m/s. Assuming steady inviscid flow, constant pressure atmosphere everywhere, and neglecting curvature and surface tension effects, the diameter of the stream 0.5m below the tap opening is nearly.

(a) 11.7 mm (b) 14.6 mm
(c) 17.5 mm (d) 20.4 mm

ESE 2017

Ans. (b) :



Given, $D_1 = 20$ mm $V_1 = 2$ m/s
$Z_1 - Z_2 = 0.5$ m $P_1 = P_2 = P_{atm}$

Apply Bernoulli's equation between (1) & (2)

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$\frac{V_1^2}{2g} + Z_1 = \frac{V_2^2}{2g} + Z_2$$

$$\frac{V_2^2}{2g} = \frac{V_1^2}{2g} + (Z_1 - Z_2)$$

$$\frac{V_2^2}{2\times 10} = \frac{(2)^2}{2\times 10} + 0.5$$

$V_2 = 3.741$ m/s

Apply continuity equation between (1) & (2)

$$A_1V_1 = A_2V_2$$

$$D_1^2V_1 = D_2^2V_2$$

$$D_2 = D_1\sqrt{\frac{V_1}{V_2}} = 20\sqrt{\frac{2}{3.741}}$$

$D_2 = 14.62$ mm

777. Consider the following statements regarding Bernoulli's equation:

1. **It is assumed that no energy has been supplied.**
2. **The velocity of a steady stream of fluid flow will depend on the cross-sectional area of the stream.**
3. **Consider two sections 1 and 2 along a flow stream. In this reach, if q is work done by a pump, w is work absorbed by turbine, ρ is density of water and g is acceleration of gravity, with p, v and z carrying standard meanings, Bernoulli's equation will read**

$$\frac{p_1}{\rho} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho} + \frac{v_2^2}{2g} + z_2 + w + g$$

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (a) : In third statement Bernoulli equation is wrong. Since the Bernoulli's equation is derived from a force balance on fluid particle moving along a stream line. Hence it is not applicable in flow sections involving pump, turbine etc. If flow section involves such devices, the steady flow energy equation is used as

$$\frac{P_1}{\rho} + \frac{1}{2}V_1^2 + gZ_1 + \text{pump work}$$

$$= \frac{P_2}{\rho} + \frac{1}{2}V_2^2 + gZ_2 + \text{turbine work}$$

778. A liquid in a circular container is given a rigid body rotation about the axis of the cylinder. The piezometric line in a cross section is

(a) a horizontal line (b) a circular arc
(c) a parabola (d) a vertical line

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Constant pressure surface in forced vortex is a paraboloid of revolution. A circle drawn on the surface such that its centre is an the axis of paraboloid has a constant plevation. As the pressure is already same on the parabolic surface the piezometric head is constant along the circle.

779. An open cylindrical tank with its axis vertical is 0.89 m high and is 0.8 m in diameter. It is filled with an oil of density 800 kg/m^3 and is rotated at 120 rpm about the axis of the cylinder. The gauge pressure at the centre of the bottom of the tank is

(a) 0.49 m of oil
(b) zero
(c) 1.29 m of oil
(d) 6.27 mPa

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : $Z_{max} = \frac{\omega^2R^2}{2g} = \left(\frac{2\pi\times 120}{60}\right)^2 \times \frac{0.4^2}{2\times 9.81}$

i.e. $Z_{max} = 1.29$ m > 0.89 m

$\therefore$ Surface at the centre of base exposed to atmosphere.

$\therefore P_{gauge} = 0$

780. An open circular cylinder 1.2 m high is filled with a liquid to its top. The liquid is given a rigid body rotation about the axis of the cylinder and the pressure at the centre of the bottom is found to be 0.3 m of liquid. The ratio of the volume of liquid spilled out of the cylinder to the origin volume is

(a) 3/8
(b) 3/4
(c) 1/2
(d) 1/4

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : $V_{spilled} = V_{paraboloid}$

$\frac{1}{2}\pi R^2 \times 0.9$

$V_{original} = \pi R^2 \times 1.2$

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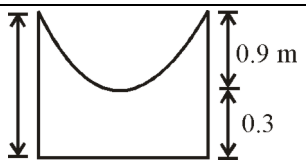

$$\frac{V_{spilled}}{V_{original}} = \frac{\frac{1}{2}\times\pi\times R^2\times 0.9}{\pi R^2\times 1.2} = \frac{9}{24} = \frac{3}{8}$$

781. A 20 cm diameter open cylindrical container contains kerosene (relative density = 0.80) to a height of 20 cm. It is rotation about a vertical axis of the cylinder. If the bottom of the cylinder at the axis is just exposed, the speed of rotation is

(a) 94.6 rpm (b) 267.5 rpm
(c) 133.8 rpm (d) 535.0 rpm

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : $Z_{max} = \frac{\omega^2 R^2}{2g}$

Given, $\frac{y}{2} = 0.2m$

$\therefore y = 0.4$ m

$0.4 = \frac{\omega^2\times 0.1^2}{2\times 9.81}$

$\omega = 28.01$ rad/sec

$N = \frac{\omega\times 60^o}{2\pi}$

$= 267.5$ rpm

782. An orifice is discharging under a head of 1.25 m of water. A pitot tube kept at its centre line at the vena contracta indicates a head of 1.20 m of water. The coefficient of velocity of the orifice is

(a) 0.990 (b) 0.980
(c) 0.965 (d) 0.960

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : $V = C_v\sqrt{2gH}$

Where, H = head at orifice

$V = \sqrt{2gh}$

Where h = head measured by pitot tube

$C_v = \sqrt{\frac{h}{H}} = \sqrt{\frac{1.2}{1.25}} = 0.98$

783. A venturimeter has a C_d = 0.95. For a differential head of 2.8 m across the inlet and the throat, the loss of head between the inlet and throat is

(a) 0.273 m (b) 0.140 m
(c) 0.302 m (d) 0.95 m

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : $c_d = \sqrt{\frac{H-h_f}{H}}$

$0.95 = \sqrt{\frac{2.8-h_f}{2.8}}$

$h_f = 0.273$ m

784. If any venturimeter carrying a liquid at sufficiently high discharges, cavitation conditions can occur at the throat. This occurs due to the pressure at the throat reaching a critical value controlled by

(a) the viscosity of the fluid
(b) eddy viscosity of the flow
(c) volumetric bulk modulus of the fluid
(d) surface tension of the fluid

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (*) : Cavitations happens when pressure inside flow falls below vapour pressure. The vapour pressure for particular fluid depends only on temperature of the fluid. None of the option provided gives correct explanation.

785. The stagnation pressure in front of an object in a fluid flow is equal to

(a) static pressure
(b) dynamic pressure
(c) sum of the static and dynamic
(d) Piezometric head

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) :

$P_o = P + \frac{PV^2}{2}$

i.e. stagnation pressure = static pressure + dynamic pressure.

786. A pitot-static tube indicates a differential head of 0.75 m of water between its two openings when inserted in a stream of water. If the coefficient of the tube is 0.99, the velocity in m/s, at the location of the tube is

(a) 4.43 (b) 0.78
(c) 3.84 (d) 3.80

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : $v = \sqrt{2gh}$

$= 0.99\sqrt{2\times 9.81\times 0.75}$

$= 3.8$ m/s

787. A horizontal cylinder half filled with fuel is having an acceleration of 10 m/s^2. The gravitational forces are negligible. The free surface of the liquid will be

(a) horizontal
(b) slopes in the direction of acceleration
(c) vertical
(d) slopes in the direction opposite of acceleration

TNPSC AE 2017

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Ans. (c) : $\tan\theta = \frac{a}{g}$

$g \approx 0$

$\theta = 90°$

A horizontal cylinder half filled with fuel is having an acceleration of 10 m/s^2. The gravitational forces are negligible. The free surface of the liquid will be vertical.

788. Which of the following statements is wrong?

(a) A flow whose streamline is represented by a curve, is called two-dimensional flow.

(b) The total energy of a liquid particle is the sum of potential energy, kinetic energy and pressure energy.

(c) The length of divergent portion in a venturimeter is equal to the convergent portion.

(d) A pitot tube is used to measure the velocity of flow at the required point in a pipe.

Karnataka PSC AE, 10.09.2017

Ans. (c) : Angle of convergence is 15°-20° and angle of divergence is 6°-7° and length of both convergent pipe and divergent pipe is different. Length of divergent portion is more than convergent portion to avoid eddies formation.

789. The velocity in a cross section is non-uniform. In one—dimensional method of analysis, the kinetic energy of fluid per unit weight is

(a) $V^2/2g$ (b) $\alpha V^2/2g$

(c) $\beta V^2/2g$ (d) p/γ

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : If velocity at a cross-section not uniform the actual kinetic energy is different from kinetic energy calculated based on average velocity.

Hence kinetic correction factor (usually denoted by α) must be used to get actual kinetic energy.

790. Indicate the incorrect statement:

For Bernoulli equation as $p/\gamma + V^2/2g + Z$ = constant to be applicable (where γ is specific weight)

(a) the flow must be steady

(b) the fluid must be an ideal gas

(c) the flow must be irrotational

(d) the fluid must be incompressible

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : The fluid must be ideal (frictionless) but not necessarily ideal gas. Hence second statement is incorrect.

Irrotational flow is not necessary condition for the Bernoulli's equation. Here third statement is also wrong. (However if flow is irrotational then the Bernoulli's equation can be applied between any two points even through these points are not on same stream line).

791. In a flow of a real fluid with no addition of energy

(a) the energy line will be horizontal or sloping upward in the direction of the flow

(b) the energy line can never be horizontal or sloping upward in the direction of the flow

(c) the piezometric line can never be horizontal or sloping downward in the direction of the flow

(d) the centre line of the pipe can never be above the energy line

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Energy line can never be horizontal or sloping upward because energy always decreases in the direction of flow due to friction. The piezometric line can increase in the direction of flow if velocity in diffuser velocity decreases, pressure increases hence piezometric line can rise in the direction of flow.

Pipe centerline can be above energy line for example in siphon. In such a case pressure for the pipe portion above energy line will be below atmospheric.

792. The difference between the total head line and the hydraulic grade line represents

(a) the velocity head

(b) the piezometric head

(c) the pressure head

(d) the elevation head

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : The total energy line represents variation of total energy $\left(\frac{P}{\rho g} + \frac{V^2}{2g} + z\right)$ and hydraulic grade line represents variation of piezometric head $\left(\frac{P}{\rho g} + z\right)$ in the direction of flow. Hence the difference between the two line is kinetic energy head $\left(\frac{V^2}{2g}\right)$.

793. At a distance of 10 cm from the axis of a whirlpool in a ideal liquid, the velocity is 5 m/s. At a radius of 30 cm the depression of the free surface below the surface of the liquid at a very large distance is

(a) 7.98 cm (b) 3.33 cm

(c) 14.16 cm (d) 21.37 cm

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : Assuming the flow as free vortex

i.e. $V = \frac{k}{r}$

$\frac{V_2}{V_1} = \frac{r_1}{r_2}$

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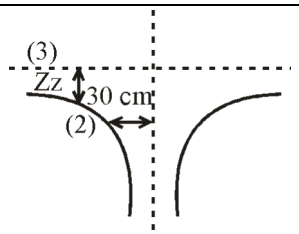

$$\frac{V_2}{5}=\frac{10}{30}$$

$$V_2=\frac{5}{3}\text{ m/s}$$

As the free vortex is an irrotational vortex flow the Bernoulli's equation can be applied between any two points. Considering undisturbed free surface as a reference level application of Bernoulli's equation between the given point and far away point from the centre gives.

$$\frac{P_2}{\rho g}+\frac{v_2^2}{2g}+z_2=\frac{P_3}{\rho g}+\frac{v_3^2}{2g}+2_3$$

$$\frac{P_{atm}}{\rho g}+\frac{v_2^2}{2g}+z_2=\frac{P_{atm}}{\rho g}+0+z_3$$

$$\because\ Z_3–Z_2=\frac{v_2^2}{2g}=\frac{\left(\frac{5}{3}\right)^2}{2\times 9.81}$$

$$=0.1416\text{ m}=14.16\text{ cm}$$

794. A 20 cm diameter orifice discharging from a tank issues out a jet of 15.75 cm diameter at vena contracta. The coefficient of contraction is

(a) 0.520 (b) 0.620
(c) 0.790 (d) 0.887

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Coefficient contraction (C_C)

$$=\frac{\text{Area Vena Contracta}}{\text{Area of orifice}}$$

$$=\frac{\frac{\pi}{4}\times 15.75^2}{\frac{\pi}{4}\times 20^2}$$

$$=0.62$$

795. Fire hose nozzle is generally of-

(a) Divergent shape
(b) Convergent shape
(c) Convergent – Divergent shape
(d) Parabolic shape

GMB AAE 25.06.2017

Ans. (b) : A fire hose nozzle is a device which generate and direct a stream of water foam, powder or any other type of fire extinguishing substance.
The hose nozzle is cylindrical in shape with the continue decrease in its size i.e. the convergent shape.

796. The coefficient of discharge for an orifice meter is in the range

(a) 0.97 to 0.99 (b) 0.93 to 0.98
(c) 0.62 to 0.65 (d) 0.51 to 0.55

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : The coefficient of discharge for an orifice meter is in the range of 0.62 to 0.65 which depends on Reynolds number and area ratio.

797. If a body moving in a horizontal line with a certain velocity starts ejecting mass downwards at a constant rate the horizontal velocity of the body will _____

(a) Remain unchanged
(b) Start increasing at an increasing rate
(c) Start increasing at a constant rate
(d) Start decreasing at a constant rate

UPRVUNL AE 21.08.2016

Ans. : (a) If a body moving in a horizontal line with a certain velocity starts ejecting mass downwards at a constant rate the horizontal velocity of the body will remain unchanged.

798. The point in a fluid stream where the diameter of the stream is the least, and fluid velocity is at its maximum is known as_____

(a) Squeeze point
(b) Vena Contracta
(c) Coefficient of contraction
(d) None of these

KPCL AE 2016

Ans. (b) : Vena Contracta is the point in a fluid stream where the diameter of the stream is the least and fluid velocity is at its maximum, such as in case of a stream issuing out of a nozzle (orifice). It is place where the cross-section area is minimum.

799. A 25 mm diameter nozzle discharges 0.76 m^3 of water per minute when the head is 60 m. The diameter of the jet is 22.5mm. Determine the values of co-effieient C_c, C_v and Cd

(a) $C_c = 0.81$ $C_v = 0.96$ and $C_d = 0.72$
(b) $C_c = 0.81$ $C_v = 0.75$ and $C_d = 0.92$
(c) $C_c = 0.85$ $C_v = 0.95$ and $C_d = 0.75$
(d) $C_c = 0.81$ $C_v = 0.92$ and $C_d = 0.75$

UPRVUNL AE 21.08.2016

Ans. : (d) dia nozzle (D) = 0.025 m

Actual discharge (Q_{act}) = $0.76 m^3/min = \frac{0.76}{60}$

$= 0.01267\ m^3/sec$

Head (H) = 60 m
Dia of Jet (d) = 0.0225 m

$$\text{Coefficient of contraction } (C_c)=\frac{\text{Area of Jet}}{\text{Area of nozzle}}$$

$$C_c=\frac{\frac{\pi}{4}d^2}{\frac{\pi}{4}D^2}$$

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$$C_c = d^2/D^2 = \frac{0.0225^2}{0.025^2} = 0.81$$

$$\text{Coefficient of discharge } (C_d) = \frac{\text{Actual discharge}}{\text{theoretical discharge}}$$

$$C_d = \frac{0.01267}{\frac{\pi}{4}D^2 \times \sqrt{2gH}}$$

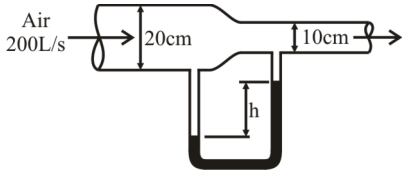
$$C_d = \frac{0.01267}{\frac{\pi}{4}(0.025)^2 \times \sqrt{2g \times 60}}$$

$$C_d = 0.752$$

$$\text{Coefficient of velocity } (C_v) = \frac{C_d}{C_c} = \frac{0.752}{0.81}$$

$$C_v = 0.928$$

800. Air flows through a pipe at a rate of 200 L/s. The pipe consists of two sections of diameters 20 cm and 10 cm with a smooth reducing section that connects them. The pressure difference between the two pipe sections is measured by a water manometer. Neglecting frictional effects, determine the differential height of water between the two pipe sections. Take the air density to be 1.20 kg/m^3



(a) 4.7 cm (b) 3.2 cm
(c) 3.7 cm (d) 3.9 cm
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) $Q = 200$ L/s $= 0.2$ m^3/s, $D_1 = 0.2$ m, $D_2 = 0.1$ m & air $= 1.2$ kg/m^3

$$Q = A_1V_1 = A_2V_2 \Rightarrow V_1 = \frac{Q}{A_1} = \frac{0.2}{\frac{\pi}{4} \times (0.2)^2} = 6.366\,\text{m/s}$$

$$V_2 = \frac{Q}{A_2} = 25.465\,\text{m/s}$$

Now, $$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\frac{P_1 - P_2}{\rho g} = \frac{V_2^2 - V_1^2}{2g}$$

$$\Rightarrow \quad P_1 - P_2 = \frac{\rho_{air}}{2}\left(V_2^2 - V_1^2\right) \qquad \therefore [z_1 = z_2 = 0]$$

$$P_1 - P_2 = \frac{1.2}{2}(25.465^2 - 6.366^2)$$

$$P_1 - P_2 = 364.764 \text{ Pa}$$

$$h = \frac{P_1 - P_2}{\rho_w g - \rho_{air} g} = \frac{364.764}{1000 \times 9.81 - 9.81 \times 1.2}$$

$$h = 0.0372 \text{ m} = 3.72 \text{ cm}$$

801. Rotameter is a :
(a) variable head flow meter
(b) variable area flow meter
(c) drag force flow meter
(d) rotating propeller type flow meter

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : Rotameter are simple industrial flow meter that measure the flow rate of liquid or gas in a closed tube.

802. Three litres of oil flows through an orifice of 20 mm diameter/second. The density of the oil is 800 kg/m^3. If coefficient of discharge is 0.62, then the head of oil maintained will be:
(a) 3.02 meter of oil (b) 6.04 meter of oil
(c) 9.08 meter of oil (d) 12.09 meter of oil

HPPSC Asstt. Prof. 29.10.2016

Ans. (d) :

$$C_d = \frac{Q_{actual}}{Q_{theoretical}}$$

$$0.62 = \frac{3 \times 10^{-3}}{\frac{\pi}{4} \times 0.02^2 \times \sqrt{2 \times 9.8 \times h}}$$

h = 12.09 meter of oil

803. The coefficient of discharge through an orifice is given by
(a) $C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}}$
(b) $C_d = \frac{\text{Theoretical discharge}}{\text{Actual discharge}}$
(c) C_d = Theoretical discharge × Actual discharge
(d) C_d = Theoretical discharge - Actual discharge

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : The Coefficient of discharge through an orifice is,

$$C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}}$$

804. Find the discharge through a totally drowned orifice 2.0 in wide and 1m deep if the difference of water levels on both the sides of the orifice be 3 m. Take $C_d = 0.62$ and $g = 10$ m/s^2
(a) 0.92 m^3/sec (b) 0.98 m^3/sec
(c) 0.96 m^3/sec (d) 1.02 m^3/sec

UPRVUNL AE 21.08.2016

Ans. : (*) Given, H = 3 m, b = 2 m, $H_2 - H_1 = 1$ m

$$Q = C_d \times b \times (H_2 - H_1) \times \sqrt{2gH}$$

$$Q = 0.62 \times 2 \times 1 \times \sqrt{2 \times 10 \times 3}$$

$= 9.60$ m^3/sec

805. The coefficient of velocity for an orifice is given by (where x and y are the coordinates of a point in the jet measured from the vena - contracta and h is the orifice head)

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(a) $\sqrt{\frac{x^2}{4hy}}$ (b) $\sqrt{\frac{y^2}{4hx}}$

(c) $\sqrt{\frac{hx^2}{4y}}$ (d) $\sqrt{\frac{3x^2}{4hy}}$

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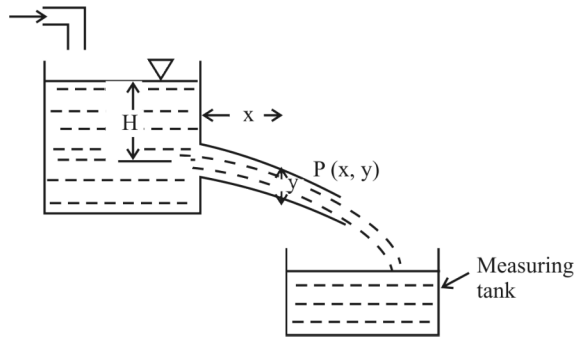
Ans. (a) : Determination of coefficient of velocity –



Consider a liquid particle which is at vena contracta at any time and takes the position at P along the jet in time 't'.

Let, x = horizontal distance travelled by the particle in time 't'

y = vertical distance between P and orifice.

v = actual velocity of jet at vena contracta.

$x = v \times t$

$y = \frac{1}{2}gt^2$ .....(i)

$t = \frac{x}{v}$ ..... (ii)

Put the value of 't' in equation ..... (i)

$y = \frac{1}{2}g \times \left(\frac{x}{v}\right)^2$

$v^2 = \frac{gx^2}{2y}$

$v = \sqrt{\frac{gx^2}{2y}}$

But theoretical velocity , $V_{th} = \sqrt{2gH}$

Coefficient of velocity $(C_v) = \frac{v}{v_{th}}$

$C_v = \sqrt{\frac{gx^2}{2y}} \times \frac{1}{\sqrt{2gH}} = \sqrt{\frac{x^2}{4yH}}$

Coefficient of velocity $(C_v) = \frac{v}{v_{th}}$

$C_v = \sqrt{\frac{gx^2}{2y}} \times \frac{1}{\sqrt{2gH}} = \sqrt{\frac{x^2}{4yH}}$

$C_v = \frac{x}{\sqrt{4yH}}$

806. For external mouthpiece absolute pressure head at vena-contracta is given as (where H_a is the atmospheric pressure head and H is the height of liquid surface above the centre of the mouthpiece)

(a) $H_a - 0.89\ H$ (b) $H_a + 0.33\ H$

(c) $H_a + 0.95\ H$ (d) $H_a - H$

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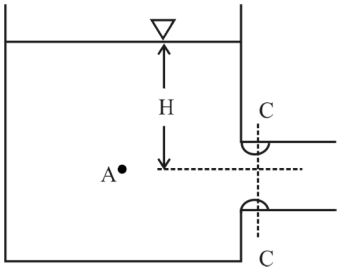
Ans. (a) : A mouthpiece is a short length of a pipe which is two or three times its diameter in length. If this pipe is fitted externally to the orifice, the mouthpiece is called external cylindrical mouthpiece and if it is fitted internally then mouthpiece is called internal mouthpiece or Borda's or Re-entrant mouthpiece.

Pressure head at Vena - contracta –



Applying Bernoulli's equation at A and C-C, we get

$\frac{P_A}{\rho g} + \frac{V_A^2}{2g} + Z_A = \frac{P_C}{\rho g} + \frac{V_c^2}{2g} + Z_c$

$\frac{P_A}{\rho g} = H_a + H \quad \begin{bmatrix} V_A = 0 \\ Z_A = Z_C \end{bmatrix}$

$H_a + H + 0 = \frac{P_C}{\rho g} + \frac{V_C^2}{2g} = H_c + \frac{V_C^2}{2g}$

$H_c = H_a + H - \frac{V_C^2}{2g}$

C_C For external mouthpiece = 0.62

Hence, $v_c = \frac{V_1}{0.62}$

$H_c = H_a + H - \left(\frac{V_1}{0.62}\right)^2 \times \frac{1}{2g}$

$H_a = H + H - \left(\frac{V_1}{2g}\right)^2 \times \frac{1}{(0.62)^2}$

$H = 1.375\ \frac{V_1^2}{2g}$

$\frac{V_1^2}{2g} = \frac{H}{1.375} = 0.7272\ H$

$H_c = H_a + H - 0.7272H \times \frac{1}{(0.62)^2}$

$= H_a + H - 1.89H$

$H_c = H_a - 0.89\ H$

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807. In an orifice, the coefficient of resistance is defined as the ratio of

(a) Loss of kinetic energy as the liquid flow through an orifice and the actual kinetic energy possessed by the flowing fluid.

(b) Theoretical kinetic

(c) Actual kinetic energy possessed by the flowing fluid and the loss of kinetic energy as the liquid flow through an orifice

(d) Theoretical kinetic energy as the liquid flow through an orifice and the loss of kinetic energy as the liquid flow through an orifice

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Coefficient of resistance (C_r) – It is ratio of loss of head (loss of K.E.) in the orifice to the head of water (actual K.E.) available at the exit of the orifice.

$$C_r = \frac{\text{Loss of K.E.}}{\text{Actual K.E.}} = \frac{C_c^2 - C_d^2}{C_d^2}$$

C_r is generally neglected, while solving problems.

808. The theoretical value of coefficient of contraction of sharp edged orifice is equal to

(a) $\pi/(\pi+2)$ (b) $(\pi+2)/\pi$

(c) $(\pi-2)/\pi$ (d) $\pi/(\pi-2)$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : The theoretical value of coefficient of contraction for a sharp edge orifice is equal to 0.611 or $\frac{\pi}{\pi+2}$

809. The energy loss in an

(a) orifice meter is more than that of venturimeter

(b) venturimeter is more than that of orifice meter

(c) orifice meter is same as that of venturimeter

(d) orifice meter and that of venturimeter is zero

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : The energy losses or head loss in orifice meter is more than venturimeter due to flow separation.

$C_d = 0.98 \rightarrow$ For venturimeter

$C_d = 0.60 \rightarrow$ For orifice meter

810. If a tank discharge water from an orifice under variable head h, the water will be lowered at a constant velocity, if the surface area of the tank varies as

(a) $\sqrt{h}$

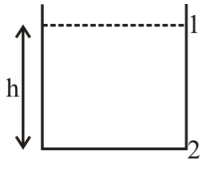
(b) $\frac{1}{\sqrt{h}}$

(c) $\frac{1}{h}$

(d) h

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) :



Applying Bernoulli's equation at (1) and (2)

$$\frac{P_0}{\rho g} + \frac{V_1^2}{2g} + h = \frac{P_0}{\rho g} + \frac{V_2^2}{2g} + 0$$

$$V_2^2 - V_1^2 = 2gh \qquad \text{.....(3)}$$

According to continuity equation,

$$A_1V_1 = A_2V_2$$

V_1 = Constant

$$V_2 = \left(\frac{A_1}{A_2}\right)V_1$$

By using equation (3)

$$\left(\frac{A_1}{A_2}\right)^2 V_1^2 - V_1^2 = 2gh$$

$$V_1^2\left(A_1^2 - A_2^2\right) = 2ghA_2^2$$

As area of orifice is negligible, $A_2 = 0$

$$V_1^2\left(A_1^2\right) \simeq 2ghA_2^2$$

$$\boxed{A_1 \propto \sqrt{h}}$$

Surface area of tank is directly proportional to square root of height.

811. An orifice discharge under a head of 1.25m of water. A pitot tube kept at its centre line at the vena contracta indicates a head of 1.20m of water. The coefficient of velocity of the orifice is

(a) 0.990 (b) 0.980

(c) 0.956 (d) 0.965

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : 'h_0' be the head of orifice & h_p = head of pitot tube

We know, $V = \sqrt{2gh_p}$

or $V = \sqrt{2 \times g \times 1.2}$

Also, $V = C_V\sqrt{2gh_0}$

So, $C_V\sqrt{2gh_0} = \sqrt{2gh_p}$

$$C_v = \sqrt{\frac{h_p}{h_0}} = \sqrt{\frac{1.2}{1.25}} = 0.9797$$

$C_v = 0.98$

812. Mouthpiece has increased net head when compared to small orifice due to

(a) turbulence in the tube

(b) vena contracta occured within the tube

(c) smaller length of the tube

(d) smooth boundary of the tube

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Ans. (b) : Mouthpiece has increased net head when compared to small orifice due to vena contracta occurred within the tube. A mouthpiece is a short length of pipe which is two or three times of its diameter length, fitted in a tank or vessel containing the fluid.
It is used for measuring the flow rate of liquid.

813. Coefficient of velocity of nozzle is .....

(a) $\frac{\text{Theoretical velocity}}{\text{Actual velocity}}$

(b) None of the mentioned

(c) It is an experimental value

(d) $\frac{\text{Actual velocity}}{\text{Theoretical velocity}}$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : Coefficient of velocity of nozzle (C_v) is the ratio of :-

$$C_v = \frac{\text{Actual velocity}}{\text{Theoretical velocity}}$$

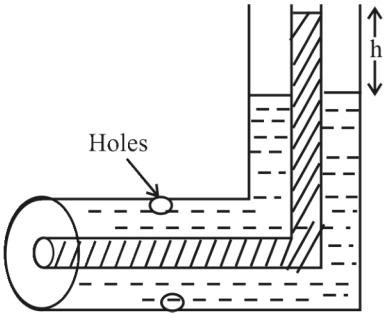
814. The coefficient of Prandtl Pitot tube is equal to

(a) 1.0 (b) 0.80

(c) 0.87 (d) 0.95

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Prandtl tube or pitot static tube is used to measure velocity at a point in fluid flow. The front portion of the tube is rounded to avoid separation of flow and on its shaft, holes are provided at certain distance where stream lines become parallel. The two tube are enclosed in the same tube.



$$h = \frac{v_A^2}{2g}$$

$$v_A = \sqrt{2gh}$$

Actual velocity, $V_{actual} = C_v\sqrt{2gh}$

$$\boxed{C_v = 0.99 \text{ to } 1.0}$$

815. The length of the convergent cone parallel to the axis of the venturimeter is approximately equal to
(Where D is the diameter of the inlet section and d is the diameter of the throat)

(a) 2.7 (D – d) (b) 2.2 (D – d)

(c) 2.7 (D + d) (d) 2.2 (D – d)

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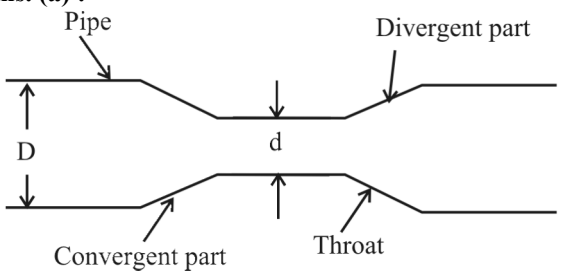
Ans. (a) :



Length of convergent cone = 2.7 (D-d)

dia of throat = $\left(\frac{1}{3}\text{to}\frac{3}{4}\right)$ of D

816. The ratio of diameter of throat to the diameter of the pipe in a Venturi meter may vary from

(a) 1/3 to 3/4 (b) 1/2 to 1

(c) 1/3 to 1/2 (d) 1/2 t 2/3

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Diameter of throat of venturimeter is 1/3 to 3/4 times dia of pipe.

$$\frac{\text{Dia. of throat}}{\text{Dia. of pipe}} = \frac{1}{3}\text{to}\frac{3}{4}$$

817. Piezometric head in fluid motion is equal to the sum of

(a) Static head and potential head

(b) Kinetic head and static head

(c) Velocity head and potential head

(d) Kinetic head, static head and potential head

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Ans. (a) :

$$\underbrace{\frac{P}{\rho g}}_{\text{static head}} + \underbrace{z}_{\text{Potential head}} = \text{Piezometric head}$$

Hydraulic head or piezometric head is specific measurement of liquid pressure above a vertical datum.

818. The pressure at any point in a static mass of fluid varies in vertical direction as indicated by which of the following equation?
(where γ is the specific weight ρ is the mass density; and g is the acceleration due to gravity)

(a) $\frac{dp}{dz} = -\rho g$ (b) $\frac{dp}{dz} = \rho g$

(c) $\frac{dp}{dz} = \gamma g$ (d) $\frac{dp}{dz} = -\gamma g$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : The pressure variation in vertical direction is related as $\frac{dp}{dz} = -\rho g$.

819. A piezometer and a Pitot tube are tapped into a horizontal water pipe, as shown in figure. The velocity of water at the center of the pipe is

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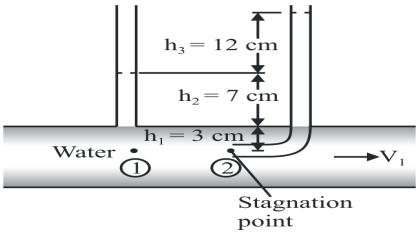
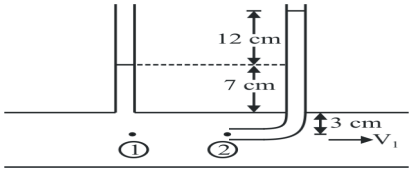


(a) 2.4 m/s (b) 1.53 m/s
(c) 2.07 m/s (d) 1.93 m/s

ISRO Scientist/Engineer 03.07.2016

Ans. (b) :



Dynamic head $h_{dynamic} = 12$ cm
$= 0.12$ m

Velocity of water $= \sqrt{2g\,h_{dynamic}}$
$= \sqrt{2 \times 9.81 \times 0.12}$
$= 1.53$ m/s

820. Prandlt's Pitot tube measures
(a) Stagnation head only
(b) Pressure head only
(c) Velocity head and Stagnation head
(d) Pressure head and Stagnation head

APPSC AEE Screening Test 2016

Ans. (c) : Pitot tube is a device used for measuring the velocity of flow at any point in a pipe or a channel. For an open stream of liquid with a free surface, this single tube is sufficient to determine the velocity, but for a fluid flowing though a closed duct, the Pitot tube measures the velocity head and stagnation point and so the static pressure must be measured separately.

821. Bernoulli's equation is not applicable for
(a) Steady flow
(b) Incompressible flow
(c) Flow with work transfer
(d) Frictionless flow

RPSC LECTURER 16.01.2016

Ans. (c) : Bernoulli's equation assumption:-
• Steady flow
• Incompressible
• Inviscid flow
• Irrotational flow
• Frictionless flow
So, Bernoulli's equation is not applicable for flow with work transfer.

822. The most appropriate governing equation of ideal fluid flow are:
(a) Euler's equation
(b) Navier Stokes' equation
(c) Reynolds' equation
(d) Hage Poisullie equation

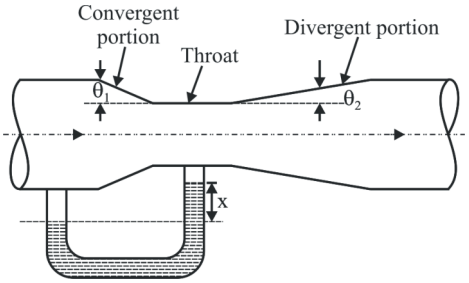
(HPPSC LECT. 2016)

Ans. : (a) The most appropriate governing equation of ideal fluid flow are Euler's equation.
Euler's equation:- This is equation of motion in which the force due to gravity and pressure are taken into consideration. This is derived by considering the motion of a fluid element along a stream line

823. The size of a venturimeter is specified by
(a) Fluid pressure
(b) Discharge
(c) Piper diameter and throat diameter
(d) Length of venturimeter

UPPSC AE 12.04.2016 Paper-II

Ans. : (c)
The size of a venturimeter is specified by pipe diameter and throat diameter.



$d_2 = \left(\frac{1}{3}\text{ to }\frac{1}{2}\right) d_1$

Angle of convergence $= 20^0$–22^0
Angle of divergence $= 6^0$–7^0 and it should be not greater than 7^0 to avoid flow separation.

824. In a flow field at the stagnation point
(a) Pressure is zero
(b) Total energy is zero
(c) Pressure head is equal to velocity head
(d) All the velocity head is converted into pressure head

UPPSC AE 12.04.2016 Paper-II

Ans. : (d) In a flow field at the stagnation point all the velocity head is converted into pressure head.
Pitot-tube is based on principle of conversion of kinetic head into pressure head. The point at which velocity reduces to zero is called stagnation point

825. For the flow to occur between two points in a pipeline, the differential pressure between these points should be more than
(a) Surface friction (b) Viscosity force
(c) Frictional force (d) All of the above

UPPSC AE 12.04.2016 Paper-II

Ans. : (d) For the flow to occur between two points in a pipeline, the differential pressure between these points should be more than– (i) Surface friction, (ii) Viscosity force, (iii) Frictional force

826. An unsymmetrical sprinkler has frictionless and equal flow through each of the two nozzles. If the absolute velocities are 6.0 and 9.0 m/s, the ratio of the distances of the nozzles from the rotating shaft
(a) $9 : 6$ (b) $9^{1/2} : 6^{1/2}$
(c) $9^2 : 6^2$ (d) $9^3 : 6^3$

APPSC AEE Mains 2016 (Civil Mechanical)

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Ans. (a) : Given,

Absolute velocities 6 m/s & 9 m/s

We know,

$m_1v_1r_1 = m_2v_2r_2$

For equal flow, $m_1 = m_2$

$$\frac{r_1}{r_2} = \frac{v_2}{v_1} = \frac{9}{6}$$

827. The pressure variation along the radial direction for vortex flow along a horizontal plane is given by

(a) $\partial p/\partial r = \rho v^2/r$ (b) $\partial p/\partial r = \rho v/r$

(c) $\partial p/\partial r = \rho v^2/r^2$ (d) $\partial p/\partial r = \rho v^3/r^2$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : We know,

Equation for two radial motion

$$\left(P + \frac{\partial P}{\partial r}\Delta r\right)\Delta A = P\Delta A - \rho\Delta r\Delta A\frac{v^2}{r}$$

$$\left(\frac{\partial P}{\partial r}\Delta r\right)\Delta A = \rho\Delta r\Delta A\frac{v^2}{r}$$

$$\frac{\partial P}{\partial r} = \rho\frac{v^2}{r}$$

828. The loss of energy in Orifice meter is more than in the case of Venturimeter because

(a) Sudden obstruction (b) Turbulence

(c) More friction (d) pressure varation

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : The loss of energy in orifice meter is more than in the case of venturimeter because of sudden Obstruction, cause positive pressure Gradient.

829. At two points 1 and 2 in a pipeline, the velocities are V and 2V respectively. Both the points are at the same elevation. The fluid density as ρ. The difference between pressure at 1 and 2 is

(a) $\frac{1}{2}\rho V^2$ (b) $\frac{3}{2}\rho V^2$

(c) $2\rho V^2$ (d) $3\rho V^2$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : Using Bernoulli's equation

$$\frac{P_1}{\rho g} + \frac{V^2}{2g} = \frac{P_2}{\rho g} + \frac{(2V)^2}{2g} = \frac{P_2}{\rho g} + \frac{4V^2}{2g}$$

$$\frac{P_1 - P_2}{\rho g} = \frac{3V^2}{2g}$$

$$P_1 - P_2 = \rho \times 1.5V^2 = \frac{3}{2}\rho V^2$$

830. A pitot static tube having coefficient of velocity 0.98, measures velocities of water in a pipe. The stagnation pressure recorded is 3 m and static pressure 2m. The actual velocity in pipe is

(a) 4.429 m/s (b) 4.341 m/s

(c) 4.431 m/s (d) zero

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : We know,

$$V = C_v\sqrt{2g\left(h_{stag} - h_{static}\right)}$$

$$= 0.98\sqrt{2 \times 9.81 \times (3-2)}$$

$$\boxed{V = 4.429 m/s}$$

831. The reading of differential manometer of a venturimeter, placed at 45° to the horizontal is 11cm. If the venturimeter is turned to horizontal position, the manometer reading will be

(a) zero (b) $\frac{1}{\sqrt{2}}$cm

(c) 11 cm (d) $11\sqrt{2}$cm

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : The inclination of pipe does not affect the reading of the manometer, since it measure pressure difference between two points & will not change with orientation.

832. Water is flowing with a velocity of 9.81 m/s. under a pressure of 9.81 kPa. If the height above the datum is 30 m, calculate the total energy per unit weight of water.

(a) 35.905 Nm/N (b) 40.00 Nm/N

(c) 30.905 Nm/N (d) 40.905 Nm/N

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (a) : Total Energy per unit weight of water is obtained from Bernoulli's equation.

$$\frac{P}{\rho g} + \frac{V^2}{2g} + Z_1 = \text{const}$$

$$\frac{9.81 kPa}{1000 \times 9.81} + \frac{(9.81)^2}{2 \times 9.81} + 30m$$

$\Rightarrow$ 1m + 4.905 m + 30 m

$\Rightarrow$ 35.905 Nm/N

833. For a control volume with an inlet area of 0.1 m^2, exit area of 0.05 m^2 and the inlet velocity 30 m/sec, the force on the control volume will be (take density = 3.89 kg/m^3 and constant):

(a) 175 N (b) 350 N

(c) 700 N (d) 525 N

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : Continuity eq.

$A_1V_1 = A_2V_2$

$0.1 \times 30 = 0.05 \times V_2$

$V_2 = 60$ m/sec

$F = \rho A_2V_2^2 - \rho A_1V_1^2$

$F = 3.89 \times 0.05 \times 60^2 - 3.89 \times 0.1 \times 30^2$

$F = 350.1N$

834. Bernoulli's theorem has been derived under the assumption that no external force acts on the liquid except

(a) Atmospheric pressure

(b) Force due to gravity

(c) Wind pressure

(d) None of the above

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Ans. (b) : Bernoulli's theorem has been derived under the assumption that no external force acts on the liquid except force due to gravity.

835. Water flows steadily down a vertical pipe of constant cross section. Neglecting friction, according to Bernoulli's equation

(a) Pressure is constant along the length of the pipe
(b) Velocity decreases with height
(c) Pressure decreases with height
(d) Pressure increases with height

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Ans. (c) : Water flows steadily down a vertical pipe of constant cross section.
According to Bernoulli's equation,

$$\frac{P_1}{\rho g}+\frac{V_1^2}{2g}+z_1=\frac{P_2}{\rho g}+\frac{V_2^2}{2g}+z_2$$

For constant cross-section, $V_1 = V_2$

$$\frac{P}{\rho g}+z=\text{Constant}$$

As water flows steadily downwards, pressure decreases with height.

836. Specific energy of a flowing liquid per unit weight is given by

(a) $E=y^2+\frac{v}{2g}$ (b) $E=y+\frac{v^2}{2g}$

(c) $E=\frac{v^2}{2g}$ (d) $E=y+\frac{v}{2g}$

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Ans. (b) : Specific energy of the flowing liquid per unit weight is given by;

$$\boxed{E=y+\frac{v^2}{2g}}$$

837. Water is flowing with a velocity of 15 m/s and under a pressure of 300 kPa. If the height above the datum is 30 m, the total energy per unit weight of water will be

(a) 11.42 Nm/N (b) 30.58 Nm/N
(c) 30 Nm/N (d) 72 Nm/N
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : As we know, total energy per unit mass is

$$gH=\frac{P}{\rho}+gz+\frac{V^2}{2} \qquad \therefore V=\frac{Q}{A}$$

$$=\frac{300\times1000}{1000}+9.8\times30+\frac{1}{2}\times15^2$$

$$=300+294+\frac{225}{2}$$

$$gH = 706.5$$

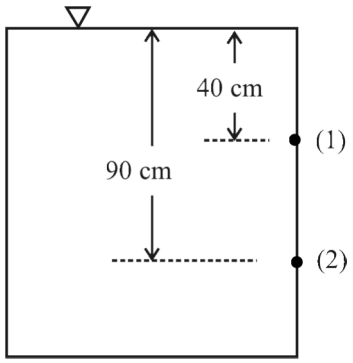
$$H=\frac{706.5\,\text{Nm}}{9.8\,\text{N}}=72\,\text{Nm/N}$$

838. A tank containing water has two orifices of the same size of depths of 40 cm and 90 cm below the free surface of water. The ratio of discharge through these orifices is :

(a) 2 : 3 (b) 1 : 1
(c) 4 : 8 (d) 16 : 81

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



$$\frac{Q_1}{Q_2}=\frac{A_1V_1}{A_2V_2}$$

$A_1 = A_2 = A$

$$V_1=\sqrt{2g\times40}$$

$$V_2=\sqrt{2g\times90}$$

$$\frac{Q_1}{Q_2}=\frac{\sqrt{40}}{\sqrt{90}}=\frac{2}{3}=0.667$$

839. A fluid jet discharging from 150 mm diameter orifice has a diameter of 100 mm at its vena contracta. The coefficient of contraction is-

(a) 0.44 (b) 1.24
(c) 0.24 (d) 0.64

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (a) Coefficient of contraction $(C_c)=\frac{A_c}{A}$

$$=\frac{\text{cross-sectional area of vena contracta}}{\text{cross-sectional area of orifice}}$$

$$C_c=\frac{\frac{\pi}{4}\times(100)^2}{\frac{\pi}{4}\times(150)^2}$$

$$\boxed{C_c = 0.44}$$

840. A jet of water discharging from a 40 mm diameter orifice has minimum area at its vena contracta. The coefficient of contraction is

(a) 0.46 (b) 0.61
(c) 0.78 (d) 0.87

ISRO Scientist/Engineer 11.10.2015

Ans. (b) : "Coefficient of contraction for orifice falls in the correct range".
In an orifice meter
Coefficient of contraction ranges from 0.61 to 0.69
Coefficient of velocity ranges from 0.95 to 0.99.

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841. When a fluid flows in concentric circles, it is known as

(a) Free circular motion
(b) Free rotational motion
(c) Free spiral vortex flow
(d) Free cylindrical vortex flow

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Ans. (d) : When a fluid flows in concentric circles it is known as free cylindrical vortex flow.

842. A right circular cylinder is filled with liquid upto its top level. It is rotated about is vertical axis at such a speed that half the liquid spills out, then the pressure at the point of intersection of the axis and bottom surface is

(a) same as before rotation
(b) half of the value before rotation
(c) quarter of the value before rotation
(d) equal to atmospheric pressure

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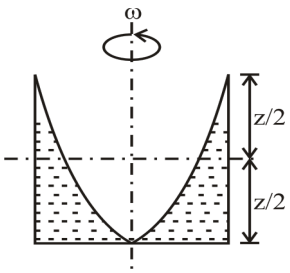
Ans. (d) :



■ Pressure at the point of intersection of the axis and bottom surface is equal to atmospheric pressure.

843. A cylindrical vessel having its height equal to its diameter is filled with liquid and moved horizontality at an acceleration equal to acceleration due to gravity. The ratio of the liquid left in the vessel to the liquid at static equilibrium condition is

(a) 0.2 (b) 0.4
(c) 0.5 (d) 0.75

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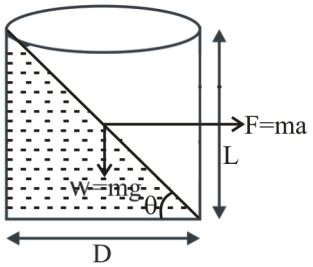
Ans. (c) : Given, a = g, L = D



Let height and diameter of cylinder be L and D respectively.

$$L = D$$

Given, $\tan\theta = \frac{L}{D} = \frac{L}{L} = 1 = 0.5$

Hence the ratio of liquid left in the vessel to the liquid if static equilibrium condition solution is 0.5.

844. Water is flowing through a pipe of 5 cm diameter under a pressure of 0.5 MPa and mean velocity of 2.0 m/s. What is total head of water at a cross section, Which is 4 m above the datum level? (Assume g = 10 m/s^2)

(a) 45.5 m
(b) 47.2 m
(c) 52.4 m
(d) 55.2 m

TSGENCO AE 14.11.2015

Ans. (d) : Given,

Diameter (D) = 5 cm = 5×10^{-2} m

Pressure (P) = 0.5 MPa, Velocity (V) = 2 m/s

$= 0.5 \times 10^6$ Pa

g = 10 m/sec^2

$$\text{Total head} = \frac{P}{\rho g} + \frac{V^2}{2g} + Z_1$$

$$= \frac{0.5\times10^6}{1000\times10} + \frac{(2)^2}{2\times10} + 4$$

$$= 50 + \frac{4}{2\times10} + 4$$

$$= 50 + 0.2 + 4 = 54.2 \text{ m}$$

$$= 54.2 \text{ m}$$

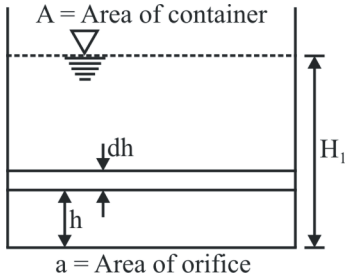
845. A tank of uniform cross section, sectional cross area (A) containing liquid up to height (H_1) has an orifice of cross sectional area (a) at its bottom. The time required to empty the tank completely will be

(a) $\frac{2A\sqrt{H_1}}{C_d a\sqrt{2g}}$ (b) $\frac{2AH_1}{C_d a\sqrt{2g}}$

(c) $\frac{2AH_1^{3/2}}{C_d a\sqrt{2g}}$ (d) $\frac{2AH_1^2}{C_d a\sqrt{2g}}$

ISRO Scientist/Engineer 11.10.2015

Ans. (a) : Given,



Let at any instant the height of water in the container be h and in dt time height changes by dh.

$$-A\,dh = \sqrt{2gh} \times a \times dt$$

$$\therefore \quad -A\,dh = C_d\sqrt{2gh}\cdot a\cdot dt$$

$$A\int_{H_1}^{H_2} \frac{-dh}{\sqrt{2gh}} = C_d a\int_0^t dt$$

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$$-A\left[\frac{1}{\sqrt{2g}}\cdot\frac{h^{-\frac{1}{2}+1}}{-\frac{1}{2}+1}\right]_{H_1}^{H_2} = C_d\, a\cdot t$$

$$-\frac{A}{\sqrt{2g}}\cdot 2\left[\sqrt{H_2}-\sqrt{H_1}\right] = C_d\, a\cdot t$$

When tank is empty $H_2 = 0$

$$\frac{2A\cdot\sqrt{H_1}}{\sqrt{2g}} = C_d\cdot a\cdot t$$

$$t = \frac{2A\sqrt{H_1}}{aC_d\sqrt{2g}}$$

846. The mean velocities of water at smaller end and larger end of a pipe are 20m/s and 5 m/s respectively. If the area of the smaller end is 2500 mm^2, then the area of larger end is

(a) 10.000 mm^2 (b) 1,000 mm^2
(c) 1,00,000 mm^2 (d) 100 mm^2

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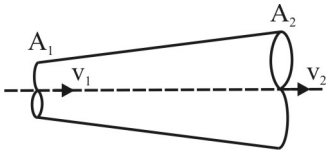
Ans. : (a)



Equation of

$A_1V_1 = A_2V_2$

$2500 \times 20 = A_2 \times 5$

$A_2 = 10000\text{mm}^2$

847. The line joining the sum of pressure heads and datum heads is called as

(a) Datum head line (b) Total energy line
(c) Piezometric line (d) pressure head line

TSPSC AEE 2015

Ans. : (c)

Piezometric head = Datum head + pressure head

$$\text{Piezometric head} = Z + \frac{P}{\rho g}$$

The line joining the sum of pressure heads and datum head is called piezometric line.

848. A vortex flow is

(a) rotational flow (b) irrotational flow
(c) viscous flow (d) free shear flow

TANGEDCO AE 2015

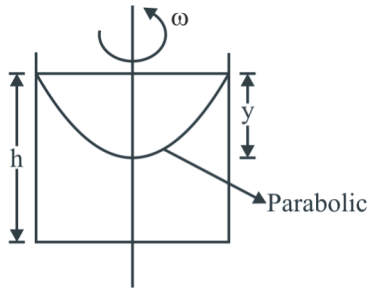
Ans. (a) : In vortex flow fluid mass rotates without any external compressed contact force as called rotational flow.

849. What is the type of flow when a cylindrical vessel containing some liquid is rotated about its vertical axis?

(a) steady flow (b) turbulent flow
(c) vortex flow (d) non-uniform flow

ISRO Scientist/Engineer 11.10.2015

Ans. (c) :



Vortex Flow– When a cylindrical vessel containing some liquid is rotated about its vertical axis.

850. A container carrying water is moved in a horizontal direction with an acceleration of 2.5 m/s^2. The approximate angle of indication of free water surface to the horizontal will be

(a) 65° (b) 45°
(c) 14° (d) 0°

GPSC AE 26.07.2015

Ans. (c) : $\tan\theta = \frac{a}{g} = \frac{2.5}{g}$

$$\theta = \tan^{-1}\left(\frac{2.5}{g}\right)$$

$\theta = 14°$

851. In a short cylindrical external mouthpiece, the vena-contracts occurs at a place which is at a distance equal to

(a) Diameter of the orifice from the outlet of orifice
(b) One-fourth the diameter of the orifice from the outlet of orifice
(c) One-third the diameter of the orifice from the outlet of orifice
(d) Two-third the diameter of the orifice from the outlet of orifice

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : In a short cylindrical external mouth-piece the vena contracta occurs at a distance one fourth the diameter of the orifice from the outlet of orifice.

852. In a free vortex motion, the radial component of velocity everywhere is

(a) Maximum (b) Minimum
(c) Zero (d) Non-zero and finite

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Ans. (c) : Free vortex motion the radial component of velocity everywhere is zero.

Free vortex flow–

Moment of momentum = Constant

$v \times r$ = Constant

Example- A whirlpool in a river.

Forced vortex–

Angular velocity, $\omega = \frac{v}{r}$ = constant

Example-Rotation of water through the runner of a turbine.

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853. Each term of the Bernoulli's equation stated in the form $P + wz + \rho V^2/2$ = constant has units of total energy per unit

(a) Mass (b) Weight
(c) Area (d) Volume

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (d)

$$p + wz + \frac{\rho v^2}{2} = \text{constant}$$

Bernoulli's Equation each term represent, energy/volume.

854. Bernoulli's equation cannot be applied when the flow is

(a) Rotational
(b) Turbulent
(c) Unsteady
(d) All options are correct

MPPSC State Forest Service Exam, 2014

Ans. (d) : Bernoulli's equation will be applied when the flow is steady state flow. It cannot be applied on rotational flow, turbulent and unsteady flow.

$$\frac{P}{\rho g} + \frac{v^2}{2g} + z = \text{Constant}$$

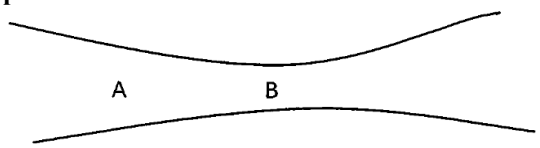
855. The locus of elevation that water will rise in a series of pitot tube is called

(a) Hydraulic grade line (b) Pressure head
(c) Energy grade line (d) Head loss

MPPSC State Forest Service Exam, 2014

Ans. (c) : The locus of elevation where the water rises in a series of pitot tube is called the energy grade line.

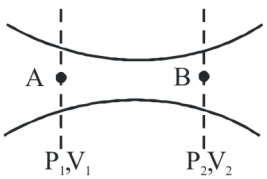
856. A liquid of density 1200 kg/m^3 is flowing steadily through a tube of varying cross section as shown below. The cross section at point A is 1 cm^2 and at B is 20 mm^2. The points A and B in the same horizontal plane. The speed of liquid at point A is 10 cm/sec. The difference of pressure at A and B is



(a) 100 Pa (b) 170 Pa
(c) 144 Pa (d) 180 Pa

ISRO Scientist/Engineer 24.05.2014

Ans. (c) :



From continuity Equation

$A_1v_1 = A_2v_2$

Or $v_2 = \frac{A_1}{A_2}v_1$

Or $v_2 = \frac{1\times10^{-4}}{20\times10^{-6}}\times10\times10^{-2}$

$= 0.52\,\text{m/s}$

From Bernoulli's Equation

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + Z_2$$

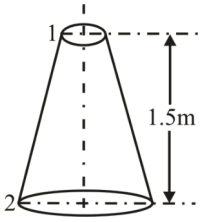
or $\frac{P_1 - P_2}{\rho g} = \frac{v_2^2 - v_1^2}{2g} (\because Z_1 = Z_2)$

$$P_1 - P_2 = \frac{\rho(v_2^2 - v_1^2)}{2}$$

$$= \frac{1200\left(0.5^2 - 0.1^2\right)}{2}$$

$= 144\,\text{Pa}$

857. A conical tube is fixed vertically with its smaller and upwards and it forms a part of pipeline. The velocity at the smaller end is 4.5 m/s and at the large and 1.5 m/s. Length of conical tube is 1.5 m. The pressure at the upper end is equivalent to a head of 10 m of water. Neglecting friction, determine the pressure at the lower and of the tube.



(a) 1.118 bar (b) 1.218 bar
(c) 1.318 bar (d) 1.418 bar
(e) 1.518 bar

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Given as,

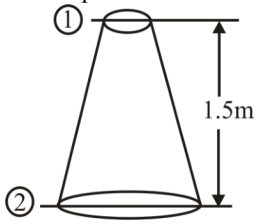
$V_1 = 4.5$ m/s, $V_2 = 1.5$ m/s

$\frac{P_1}{\rho g} = 10\,\text{m of water}$

$h = 1.5$

$P_2 = ?$

Applying Bernoulli's Equation



$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

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$$10 + \frac{(4.5)^2}{2\times 9.81} + 1.5 = \frac{P_2}{\rho \times g} + \frac{(1.5)^2}{2\times 9.81} + 0$$

$$\frac{P_2}{\rho \times g} = 12.4174 \text{ m}$$

$P_2 = 12.4174 \times 1000 \times 9.81$

$\boxed{P_2 = 1.218 \text{ bar}}$

858. The range of coefficient of discharge for orifice meter is

(a) 0.6 to 0.7 (b) 0.7 to 0.85
(c) 0.85 to 0.92 (d) 0.92 to 0.98
(e) 0.5 to 0.6

CGPSC AE 26.04.2015 SHIFT-II

Ans. (a) :
- The range of C_d for orifice meter (C_d) 0.6 to 0.7
- The range of C_d for venturimeter C_d - 0.95 to 0.99

859. Match list I with List II and select the correct answer:

List I		List II	
A	Bernoulli	1	Potential function
B	Continuity equation	2	Stream line
C	Eulers	3	Total head
D	Laplace equation	4	Conservation of momentum
		5	Conservation of mass.

(a) A-3, B-2, C-5, D-1 (b) A-5, B-3, C-2, D-4
(c) A-3, B-2, C-5, D-4 (d) A-3, B-5, C-2, D-1
(e) A-3, B-1, C-4, D-5

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) :
1. Bernoulli equation → Total head of energy

$$\left[\frac{P}{\rho g} + \frac{V^2}{2g} + Z = C\right]$$

2. Continuity equation → Conservation mass

$\rho_1 V_1 A_1 = \rho_1 V_2 A_2$ [For compressible fluid]

$V_1 A_1 = V_2 A_2$ [For Incompressible fluid]

3. Euler equation → Stream line
4. Laplace equation → Potential function

$$\left[\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0\right]$$

860. Consider the following flow.

1. Developed flow in a pipe
2. Flow of water over a long weir
3. Flow in a long, straight canal
4. The flow of exhaust gases
5. Flow of blood in a artery
6. Flow of air around a bullet
7. Flow of air in a tornado

Of the above, which could be modeled as in viscid flows?

(a) 4,5 (b) 4,7
(c) 2,5 (d) 2,7
(e) 2,4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : Flow of water over a long weir and flow of air in tornado could be modeled as inviscid flows whereas developed flow in a pipe, flow in long straight canal, the flow of exhaust gases exiting a rocked, the flow of blood in an artery and flow of air around a bullet could be modeled as viscid flows.

861. Select the false statement for Bernoulli's equation

(a) It can be applied to an inertial coordinate system.
(b) It can be applied to an unsteady flow.
(c) It can be applied in a in viscid flow.
(d) It can be applied between two points along a streamline.
(e) It can be applied to a incompressible fluid

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Ans. (b) : Bernoulli's equation can be applied to an internal coordinate system, in an inviscid flow, between two points along a streamline and incompressible fluid because inviscid flow, the viscosity is zero and hence no viscous force acts on the body also incompressible flow means the density remain constant whereas this equation can not be applied to unsteady flow, rotational flow and inside the boundary layer of fluid.

862. The Bernoulli's equation can take the form

(a) $\frac{P_1}{\rho_1} + \frac{V_1^2}{2g} + Z_1 = \frac{P^2}{\rho_2} + \frac{V_2^2}{2g} + Z^2$

(b) $\frac{P_1}{\rho_1 g} + \frac{V_1^2}{2} + Z_1 = \frac{P^2}{\rho_2 g} + \frac{V_2^2}{g} + Z^2$

(c) $\frac{P_1}{\rho_1 g} + \frac{V_1^2}{2g} + Z_1 g = \frac{P^1}{\rho_2 g} + \frac{V_2^2}{2g} + Z_2 g$

(d) $\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$

TNPSC AE 2014

Ans. (d) : $\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1$

$= \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$

863. Consider the following statements

1. In a given duct system, the total pressure drop varies linearly with flow rate
2. In a given duct system, the total pressure drop varies in a parabolic manner with flow rate
3. For a given flow rate. the total pressure drop of a duct increases as the dampers are opened more
4. For a given flow rate, the total pressure drop of a duct is less when the air filters are new

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Of the above, which is correct?

(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 3
(e) 2 and 4

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Ans. (e) : (1) In a given duct system, the total pressure drop varies in a parabolic manner with flow rate.
(2) For a given flow rate, the total pressure drop of a duct is less when the air filter are new.

864. In a forced vortex

(a) the fluid velocity is inversely proportional to the radius
(b) the fluid rotates with relative velocity
(c) the rice depends on the specific weight
(d) the rise is proportional to the cube of angular velocity
(e) the fluid rotates without any relative velocity

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Ans. (e) : Forced Vortex (Rotational Vortex)–It is defined as the type of vortex flow where fluid mass rotates about a vertical axis such that every particle has some angular velocity. Because of constant angular velocity the fluid rotates without any relative velocity.

ω = Constant

$$\frac{v}{r} = \text{Constant}$$

$$\boxed{v \propto r}$$

865. A Rotameter can be used

(a) Only in vertical orientation (direction)
(b) Only in horizontal orientation (direction)
(c) In any orientation (direction)
(d) For zero orientation (direction)

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Ans. (a) : A Rotameter is a device that measures the volumetric flow rate of fluid in a closed tube. A Rotameter can be used only in vertical orientation.

866. A hot wire anemometer is used to measure

(a) Mean flow velocity
(b) Fluctuating component of velocities
(c) Both mean and fluctuating component of velocities
(d) Constant velocity

TNPSC AE 2014

Ans. (c) : Hot wire Anemometer– The hot wire anemometer is a device used for measuring the both mean and fluctuating component of velocities and direction of the fluid.

867. The head of the water over the centre of orifice of diameter 20 mm is 1 m. The co-efficient of discharge is 0.61. What will be the actual discharge through orifice?

(a) 51 litre/min (b) 100 lite/ min
(c) 83.5 litre/min (d) 60 litre/min

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Theoretical velocity

$$V_{in} = \sqrt{2 \times 9.81 \times 1}$$

$$V_{in} = 4.43 \text{ m/s}$$

Theoretical discharge

$$Q_{in} = A \times V_{in}$$

$$= \frac{\pi}{4}(.02)^2 \times 4.43$$

$$Q_{in} = 1.39 \times 10^3 \text{ m}^3/\text{s}$$

Actual discharge

$$Q_a = C_d . Q_{in}$$

$$= 0.61 \times 1.391 \times 10^3$$

$$Q_a = 8.488 \times 10^{-4} \text{ m}^3/\text{s}$$

$$= 8.488 \times 10^{-3} \times 60 \times 10^3 \text{ lit/min}$$

$$Q_a = 50.93 \text{ lit/min}$$

$$Q_a = 51 \text{ lit/min}$$

868. A tank 1.5m stands on a trolly and is full of water. It has an orifice of diameter 0.1 m at 0.3m from the bottom of the tank. If the orifice is suddenly opened and coefficient of discharge of the orifice is 0.60, then the propelling force on the trolley will be

(a) 69.37 N (b) 67.39 N
(c) 63.79 N (d) 65.39 N

ISRO Scientist/Engineer 12.05.2013

Ans. (b) : Tank height h = 1.5m
Orifice of diameter d = 0.1m
at 0.3 from the bottom of the tank.
So, h = 1.5 - 0.3 = 1.2m
Coefficient of discharge $C_d = 0.60$

$$V = C_d \sqrt{2gh}$$

$$V = 0.6\sqrt{2 \times 10 \times 1.2} \qquad [\because g = 10 \text{m/s}^2]$$

V = 2.94 m/s

$$F = \rho a v^2 \qquad (\text{Area} = \pi/4 \times d^2 = 0.785 \times 0.01)$$

$$= 1000 \times 0.785 \times 0.01 \times (2.94)^2$$

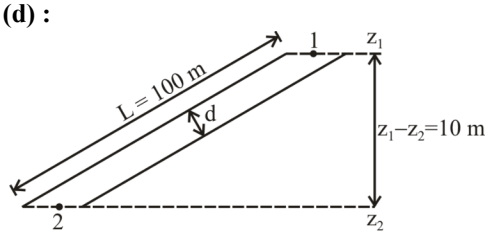
= 67.824 N

869. Water is flowing in an inclined pipe of uniform diameter and 100m length with its outlet being 10m above the inlet. If the pressure at the outlet is 100 kPa, assuming the specific weight of water as 10,000 N/m^3, what is the pressure at the inlet?

(a) 150 kPa (b) 50 kPa
(c) 75 kPa (d) 200 kPa

RPSC AEN Pre-2013

Ans. (d) :



Given – Diameter same = $d_1 = d_2 = d$

$A_1 = A_2 = A$

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So, velocity is also same $V_1 = V_2 = V$

Specific weight (γ) = (ρg) =

$= 10000\ N/m^3$

$P_1 = 100kPa, \quad P_2 = ?$

$Z_1 = 10, \quad Z_2 = 0$

According Bernoulli's equation

$$\frac{P_1}{Pg}+\frac{V_1^2}{2g}+Z_1=\frac{P_2}{Pg}+\frac{V_2^2}{2g}+Z_2$$

For uniform diameter of pipe

$$\frac{P_1}{pg}+Z_1=\frac{P_2}{pg}+Z_2$$

$$\frac{100\times10^3}{10,000}+10=\frac{P_2}{10,000}+0$$

$$10+10=\frac{P_2}{10,000}$$

$$20=\frac{P_2}{10,000}$$

$$\boxed{P_2 = 200kPa}$$

870. A venturimeter is preferred to an orifice plate because:

(a) it is cheaper (b) it is easy to install

(c) energy loss is less (d) it has very high life

SJVN ET 2013

Ans. (c) : A venturimeter is preferred to an orifice plate because energy loss is less.

871. A venturimeter is generally used to measure the rate of flow of

(a) Air (b) Steam

(c) Water (d) Hydrogen

TNPSC AE 2013

Ans. (c) : A venturimeter is generally used to measure the rate of flow of water.

872. Flow over an elliptical body may be idealized by superimposing a free stream and:-

(a) A source (b) A doublet

(c) A source and sink (d) A free vortex

UKPSC AE-2013, Paper-II

Ans. (c) : Flow over an elliptical body may be idealized by superimposing a free stream and a source and sink.

873. Flownet is used to analyse the flow problems related to :

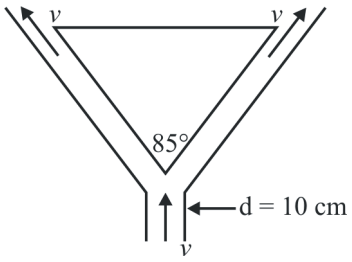
(a) Rotational flow (b) Irrotational flow

(c) Stream flow (d) Velocity flow

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Flownet is used to analyse the flow problems related to irrotational flow.

874. Air at 20°C and 10^5 Pa enters the bottom of an angle of 85° conical flow meter duct at a mass flow rate of 0.3 kg/s, as shown in the figure. It supports the centered conical body by steady annual flow around the cone and exists at the same velocity as it enters. Estimate the weight of the body is Newton's



(a) 2.5N (b) 1N

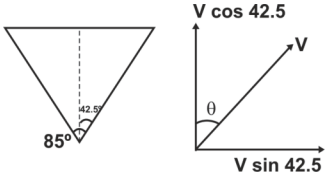
(c) 3.5N (d) 4N

ISRO Scientist/Engineer 2012

Ans. (a) : Pressure $p = 10^5$ Pa

mass rate $\dot{m} = 0.3$ kg/sec

diameter d = 10 cm



Weight of Body = change in linear momentum of air in vertical direction

$\dot{m} = 0.3$

$\dot{m} = \rho AV$

$$0.3 = 1.22\times\frac{\pi}{4}\times(0.1)^2\times V$$

$\therefore$ V = 31.31 m/s

$\therefore$ Weight of Body

= mv - mv cosθ

= mv (1 - cosθ)

= 0.3 × 31.3 (1 - cos 42.5)

= 2.47 N

W ≃ 2.5 N

875. Pressure drop of water in a pipe is measured by a manometer using a liquid of density 2000 kg/m³. The difference in height of liquid in the limbs is 10 cm. Then the pressure drop is

(a) 98.1 N/m² (b) 981 N/m²

(c) 1692 N/m² (d) 1620 N/m²

TRB Poly. Lect., 2012

Ans. (b) : Given, x = 10 cm = 0.1 m

$$\frac{P_1-P_2}{\rho g}=x\left(\frac{S_m}{s}-1\right)$$

$$P_1-P_2=1000\times9.81\times0.1(2-1)=981N/m^2$$

876. The discharge in (m^3/s) for laminar flow through a pipe of diameter 0.04 m bearing a centreline velocity of 1.5 m/s is

(a) $\frac{3\pi}{59}$ (b) $\frac{3\pi}{2500}$

(c) $\frac{3\pi}{5000}$ (d) $\frac{3\pi}{10000}$

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Ans. : (d) Given, Pipe diameter (d) = 0.04m
Maximum velocity = 1.5m/sec.
Average velocity = 0.75m/sec.

$$\text{Discharge (Q)} = AxV = \frac{\pi}{4}(0.04)^2 \times 0.75$$

$$Q = \frac{3\pi}{10000} m^3/\text{sec}$$

877. The Bernoulli's constants for points lying on the same stream line and those which lie on other stream lines will have the same value, if the flow is
(a) incompressible (b) steady
(c) irrotational (d) uniform

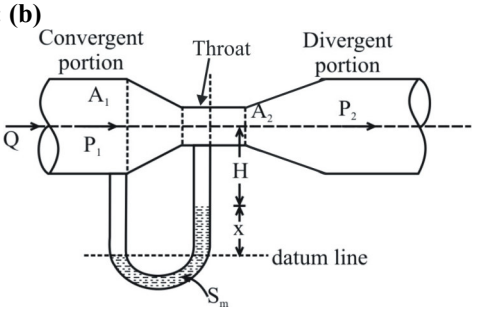
APPSC AEE 2012

Ans. : (c) The Bernoulli's constant for points lying on the same stream line and those which lie on other stream lines will have the same value, if the flow is irrotational.
Rotational flow is that type of flow in which the fluid particles which flowing along stream lines, also rotate about their own axis and if the fluid particles while flowing along stream lines do not rotate their own axis then the type of flow is called irrotatinal flow.

878. The velocity of liquid flowing through the divergent portion of a venturimeter
(a) remains constant (b) decreases
(c) increases (d) no relationship

APPSC AEE 2012

Ans. : (b)



The velocity of liquid flowing through the divergent portion of a venturimeter decreases.

879. The head loss is more in a
(a) nozzle meter
(b) venturimeter
(c) inclined venturimeter
(d) orifice meter

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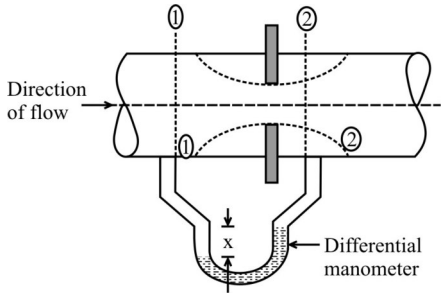
Ans. : (d) The head loss is more in a orificemeter.
orifice meter :-



It is a device used for measuring the rate of flow of a fluid through a pipe.

$$\text{Discharge (Q)} = \frac{C_d a_0 a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_0^2}}$$

C_d = Co-efficient of discharge for orifice meter = 0.62
The co-efficient of discharge for orifice meter is much smaller than that for a venturimeter.

880. If successive measurements with a Prandtl-pitot tube indicate that the tip piezo meter reading varies only across the flow, and the size piezometer reading varies only in the direction of flow, the flow is evidently
(a) uniform, irrotational
(b) non-uniform, rotational
(c) uniform, rotational
(d) non-uniform, irrotational

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Ans. : (b) If successive measurements with a prandtl-pitot tube indicate that the tip piezo meter reading varies only across the flow, and the size piezometer reading varies only in the direction of flow the flow is evidently non-uniform and rotational.

881. Momentum equation is completely independent of
(a) compressibility effects (b) frictional effects
(c) viscous effects (d) momentum flux

APPSC AEE 2012

Ans. : (c) Momentum equation is completely dependent of
(i) Compressibility effects
(ii) Frictional effects
(iii) Momentum flux

882. A mechanical device, which has rotation elements, the speed of rotation of which is a function of velocity of flow, and which is used measure the velocity of liquid in open channels, is
(a) Anemometer (b) Orifice meter
(c) Current meter (d) Rotameter

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Ans. : (c) A mechanical device, which has rotation elements, the speed of rotation of which is a function of velocity of flow, and which is used measure the velocity of liquid in open channels is current meter.

883. A mouthpiece and an orifice, both of the same diameter 'd', are discharging under the same head 'H'. The discharge through the mouthpiece will be
(a) the same as that of the orifice
(b) less than that of the orifice
(c) more than that of the orifice
(d) no relationship

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Ans. : (c) A mouth piece and an orifice, both of the same diameter 'd' are discharging under the same head 'H' the discharge through the mouth piece will be more than that of the orifice.

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884. Which is the flow measuring device through which fluid does not flow ?
(a) Orifice plate
(b) Venturimeter
(c) Pitot tube
(d) Elbow meter

UKPSC AE 2012 Paper–II

Ans. (c) : Pitot tube

885. The flow rate through a circular pipe is measured by
(i) pitot-tube
(ii) venturimeter
(iii) orificemeter
(iv) None of the above

Code–
(a) i
(b) i, ii, and iii
(c) ii and iii
(d) All of the above

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Ans. (c) : Flow rate through a circular pipe is measured by venturimeter & orificemeter both.

886. Measurement of velocity without any obstruction to the flow in a pipe may be made by
(a) pitot-static probe
(b) hot-wire anemometer
(c) hot-film anemometer
(d) laser-doppler anemometer

APPSC AEE 04.12.2012

Ans. (a) : Measurement of velocity without any obstruction to the flow is a pipe may be made by pitot static probe.

887. Calculate the maximum allowable discharge of water through a venturimeter throat 5 cm, fitted in a 10 cm diameter line with its inlet at an open channel Assume $c_d = 0.95$
(a) 0.0224 m^3/s
(b) 0.0448 m^3/s
(c) 0.8 m^3/s
(d) 0.04 m^3/s

APPSC AEE 2012

Ans. (d) : Data given, $d_i = 5$ cm, $d_o = 10$ cm, $c_d = 0.95$

$$Q_{max} = \frac{c_d . a_v a_i}{\sqrt{a_D^2 - a_i^2}} \sqrt{2gh}$$

$$Q_{max} = \frac{0.95 \times \left(\frac{\pi}{4} d_0^2\right) \times \left(\frac{\pi}{4} \times d_i^2\right)}{\sqrt{\left(\frac{\pi}{4} d_0^2\right) - \left(\frac{\pi}{4} d_i^2\right)^2}} \sqrt{2 \times 9.81 \times 25}$$

$Q_{max} = 0.04266$ m^3/sec

888. The term $v^2/2g$ is known as
(a) kinetic energy
(b) pressure energy
(c) kinetic energy/unit weight
(d) None of the above

APPSC AEE 2012

Ans. (c) : $\frac{V^2}{2g}$ is the term used for $= \frac{\frac{1}{2}mv^2}{mg}$

$\frac{V^2}{2g}$ = K.E./ weight

889. A current meter is a device used for measuring
(a) velocity (b) viscosity
(c) current (d) pressure

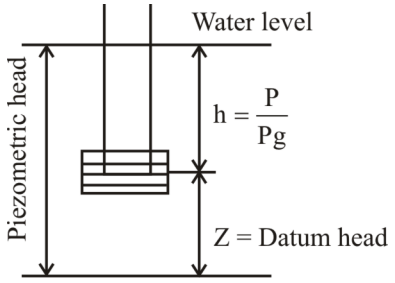
APPSC AEE 2012

Ans. (a) : Current meter- A current meter is oceanographic device for flow measurement by mechanical, tiet, acoustical or electrical means. It is an instrument for measuring the velocity of flow of fluid (as water) in a stream.

890. Piezometric head of a fluid is defined as
(a) the sum of absolute pressure head and datum head
(b) the stagnation pressure head
(c) the sum of stagnation head and datum head
(d) the sum of gauge pressure head and datum head

APPSC AEE 2012

Ans. (d) : Piezometric head $= \frac{P_{gauge}}{\rho g} + Z$



891. According to the King's Laws for a constant temperature hot wire anemometer
(a) $E \propto U$ (b) $E^2 \propto U$
(c) $E^2 \propto \sqrt{U}$ (d) $E \propto U^2$

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Ans. (c) : In practice, hot-wires are calibrated in the form of u = f(E) rather than the more conventional form of E = f(u)
The constant to and P version as King's law for a CTA is

$$E^2 = A + B\sqrt{U}$$

$$E^2 \propto \sqrt{U}$$

892. The integral momentum equation requires the assumption that
(a) the flow is uniform
(b) the flow is unidirectional
(c) the fluid is unidirectional
(d) the flow is steady

APPSC AE 04.12.2012

Ans. (d) : The integral momentum equation requires the assumption that the flow is steady.

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893. Observation of a flow net enables us to :

(a) determine the velocity at all points
(b) estimate the pressure at all points
(c) estimate the velocity variations
(d) determine the energy loss in flow

APPSC AE 04.12.2012

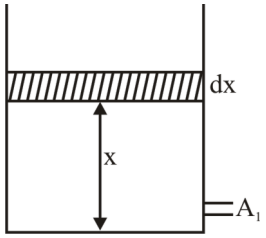
Ans. (c) : Observation of a flow net enables us to estimate the velocity variations.

894. For a given orifice time taken for the water level to fall down 8 to 4m is 10 seconds. The time taken for water level to fall down from 4m to zero is:

(a) 15 sec
(b) 24.14 sec
(c) 25.3 sec
(d) 20 sec

APGENCO AE 2012

Ans. (b) :



$$Q = \frac{dv}{dt}$$

$$Q = A_1 \times V$$

$$Q = A_1\sqrt{2gx}$$

$$A_1 \times \sqrt{2gx} \times dt = A_2 dx$$

$$\frac{A_1}{A_2} \times dt = \frac{dx}{\sqrt{2gx}}$$

$$\frac{A_1}{A_2}[t] = \frac{1}{\sqrt{2g}}\left[2\sqrt{x}\right]$$

$$\left(\frac{A_1}{A_2} = \frac{\sqrt{8}-\sqrt{4}}{10} \times \frac{\sqrt{2}}{\sqrt{g}}\right)$$

$$\frac{\sqrt{8}-\sqrt{4}}{10} \times \sqrt{\frac{2}{g}} \times t = \sqrt{\frac{2}{g}} \times \sqrt{4}$$

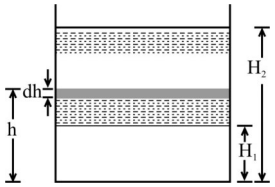
$$t = 24.14 \text{ sec}$$

995. The time taken for a tank, filled to a height 'h' above its flat base, to empty through an orifice in the base varies as the following power of 'h'.

(a) 1
(b) $\frac{1}{2}$
(c) $-\frac{1}{2}$
(d) $\frac{1}{3}$

APPSC AEE 2012

Ans. : (b) Time of emptying a tank through an orifice at its bottom :-



$$T = \frac{2A\left[\sqrt{H_1} - \sqrt{H_2}\right]}{C_d.a.\sqrt{2g}}$$

for emptying the tank Completely $H_2 = 0$

$$T = \frac{2A\sqrt{H_1}}{C_d.a.\sqrt{2g}}$$

896. Water flowing through variable cross section pipe has velocity of 2 m/s at a cross sectional area 0.4 m^2. The velocity of water at a cross sectional area of 0.1 m^2 is :

(a) 4 m/s (b) 2 m/s
(c) 1 m/s (d) 8 m/s

OPSC Civil Services Pre. 2011

Ans. (d) :
Given as,
$v_1 = 2$ m/s
$A_1 = 0.4\ m^2$
$A_2 = 0.1\ m^2$
By using continuity equation,
$A_1v_1 = A_2v_2$
$0.4 \times 2 = 0.1 \times v_2$
$v_2 = 8$ m/s

897. Which of the following is not an application of Bernoulli's equation ?

(a) Venturimeter (b) Orifice
(c) Anemometer (d) Pitot tube

OPSC Civil Services Pre. 2011

Ans. (c) :
(a) Venturimeter – It is a converging diverging tube to measure low velocity or discharge.
(b) Orificemeter – It is a holed plate in pipe carrying fluid. The small cross section of area is less than hole area plate and found a little downstream of plate.
(c) Anemometer – It measures the fluctuation of velocity in turbulent flow using the concept of convection.
(d) Pitot tube – It measure the flow velocity in open channels generally.

898. Which of the following surge tank is also called throttle surge tank ?

(a) Inclined surge tank
(b) Expansion chamber surge tank
(c) Restricted orifice surge tank
(d) None of the above

J&K PSC Civil Services Pre, 2010

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Ans. (c) : Surge tank are required to prevent sudden and potentially damaging change in pressure due to sudden changes or surger in the flow.

899. Water flows through a pipe of 200 mm diameter at a discharge of 62.8 Lps. At a section which is 5 m above the datum, the water pressure is 150 kN/m^2. In such a case, the total energy of the flow in terms of water head will be

(a) 12.5 m (b) 15.5 m
(c) 17.5 m (d) 20.5 m

UPSC JWM Advt. No.-50/2010

Ans. (d) : Given, d = 0.2m, Q = 62.8 × 10^{-3} m^3/s
z = 5 m, P = 150 kN/m^2

$$\text{Total energy} = \frac{P}{\rho g} + \frac{V^2}{2g} + z$$

$$V = \frac{Q}{A} = \frac{62.8\times10^{-3}}{\frac{\pi}{4}\times0.2^2} = 2\,m/s$$

$$= \frac{150\times10^3}{9810} + \frac{2^2}{2\times9.81} + 5$$

= 20.5 m of water

900. At a point A in a pipe line carrying water, the diameter is 1m, the pressure 98 kPa and the velocity 1m/s. At a point B, 2 m higher than A, the diameter is 0.5m and the pressure 20 kPa. The direction of flow would be

(a) A to B
(b) B to A
(c) Cannot be ascertained from data
(d) None of these

ISRO Scientist/Engineer 2010

Ans. (a) : At a point A,
d_A (diameter) =1m
P_A = 98 kPa
V_A = 1 m/s
z_A = 0

At point B,
2m higher than A
z_B = 2m
d_B = 0.5m
P_B = 20 kPa

$$E_A = \frac{P_A}{\rho g} + Z_A + \frac{V_A^2}{2g}$$

Let V_A =1m/s

$$E_A = \frac{98\times10^3}{1000\times10} + 0 + \frac{1^2}{20}$$

E_A = 9.8m
$A_AV_A = A_BV_B$

$$V_B = \frac{A_A}{A_B}V_A$$

$$= \frac{\frac{\pi}{4}(1)^2}{\frac{\pi}{4}\times(0.5)^2}\times1$$

V_B = 4 m/s

Then,

$$E_B = \frac{20\times10^3}{1000\times10} + 2 + \frac{4^2}{20}$$

E_B = 4.8m
$E_A > E_B$

"Direction of flow will be form high energy to low energy", i.e. from A to B.

901. For the flow of a Newtonian fluid with constant density and viscosity; the governing equation is the

(a) Navier-Stokes equation
(b) Laplace equation
(c) Euler equation
(d) Bernoulli equation

DRDO Scientists 2009

Ans. (a) :

902. Both free vortex and forced vortex can be expressed mathematically in terms of tangential velocity V at the corresponding radius r. Choose the correct combination

	Free vortex	Forced vortex
(a)	V = r × const.	V × r = const.
(b)	V^2 × r = const.	V = r × const.
(c)	V × r = const.	V^2 = r × const.
(d)	V × r = const.	V = r × const.

ISRO Scientist/Engineer 2009

Ans. (d) :

903. In a two-dimensional flow of a viscous fluid. Couette flow is defined for :

(a) pressure-gradient-driven laminar flow between fixed parallel plates
(b) pressure-gradient-driven laminar flow through non-circular duct
(c) pressure-gradient-driven laminar flow through pipe
(d) laminar flow between a fixed and a moving plate

DRDO Scientists 2009

Ans. (d) : Couette flow is defined for laminar flow between a fixed and a moving plate.

904. A cylindrical vessel of large diameter is filled with water to the brim. If a small hole is made in the wall of the vessel at a depth of 1.25 m, the velocity of outflow through the orifice is (take g = 10 m/s^2)

(a) 3.5 m/s (b) 5.0 m/s
(c) 15.0 m/s (d) 25.0 m/s

DRDO Scientists 2008

Search on Telegram @apna_pdf

Ans. (b) : $V = \sqrt{2gh}$

$= \sqrt{2 \times 10 \times 1.25}$

$= 5 m/s$

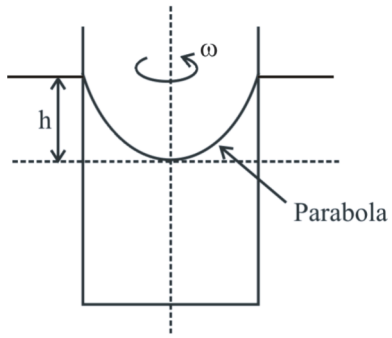
905. For a forced vortex flow, the height of paraboloid formed is

(a) $\frac{P}{\omega} + \frac{V^2}{2g}$ (b) $\frac{V^2}{2g}$

(c) $\frac{V^2}{r^2 \times 2g}$ (d) $\frac{\omega r^2}{2g}$

TNPSC AE, 2008

Ans. (b) :



(Forced Vortex flow)

Height of parabola (h) = $\frac{V^2}{2g}$ or $\frac{r^2\omega^2}{2g}$ $\{\because V = r\omega$

906. Which one of the following is a mis-match?

(a) velocity – anemometer
(b) flow – rotameter
(c) power – load cell
(d) level – float

TNPSC AE, 2008

Ans. (c) : A load cell is a transducer that is used to create an electrical signal whose magnitude is directly proportional to the force being measured. The various types of load cells include hydraulic load cells, pneumatic load cells and strain gauge load cells.
The most common type is a strain gauge load cell.

907. A pitot static tube is to measure the velocity of air-stream in a pipe. If the difference between the stagnation and static pressure as indicated by a vertical tube manometer is 12.5 cm of water, the velocity of the air-stream is (take $\rho_{water} = 1000 kg/m^3$ and $g = 10 m/s^2$)

(a) $5\sqrt{10}$ m/s (b) 50 m/s
(c) 400 m/s (d) 500 m/s

DRDO Scientists 2008

Ans. (b) : $h = x\left[\frac{\rho_{water}}{\rho_{air}} - 1\right] = 0.125 \times 999 m$

$V = \sqrt{2gh}$

$= \sqrt{2 \times 10 \times 0.125 \times 999}$

$= 49.975$ m/s

908. The SI unit of the term $u^2/2g$ that appears in the Bernoulli equation is

(a) N/m^2 (b) m
(c) J/m^3 (d) bar

DRDO Scientists 2008

Ans. (b) :

909. In the converging part of a horizontal venturi meter

(a) static pressure gradually decreases
(b) static pressure gradually increases
(c) axial velocity gradually decreases
(d) both the static pressure and axial velocity gradually decreases

DRDO Scientists 2008

Ans. (a) : In converging part of a horizontal venturimeter static pressure gradually decreases and axial velocity gradually increases.

910. In the fully developed laminar flow through a horizontal pipe of constant cross-section

(a) static pressure remains constant in the direction of flow
(b) Bernoulli equation holds along the axis of the pipe
(c) static pressure increases in the direction of flow
(d) static pressure decreases in the direction of flow

DRDO Scientists 2008

Ans. (d) : In the fully developed laminar flow through a horizontal pipe of constant cross-section static pressure decreases in the direction of flow.

911. A fluid jet discharging from a 100 mm diameter orifice has a diameter of 80 mm at its vena contracta. The coefficient of contraction is

(a) 0.8 (b) 1.25
(c) 0.2 (d) 0.64

ISRO Scientist/Engineer 2007

Ans. (d) : Dia of orifice = 100 mm
Dia at vena contracta = 80 mm
= Coefficient of contracted area

$$C_c = \frac{\text{Area of venacontracta}}{\text{Area of orifice}}$$

$$= \frac{\pi(80)^2/4}{\pi(100)^2/4}$$

$= 0.64$

912. The maximum height of a siphon for a fluid of specific gravity ρ under atmospheric conditions is

(a) ρ /10 meters (b) $\frac{10(1+\rho)}{\rho}$ meters
(c) 10/(1– ρ) meters (d) 10/ρ meters

ISRO Scientist/Engineer 2007

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Ans. (d) : We are assuming initially the tank has water in it

$$V_1 = 0, Z_1 = 0$$

$$\frac{P_0}{\rho g} + \frac{0^2}{2g} + 0 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

for max. height, the point 2 is so high that the pressure at the highest point is zero and the velocity is assumed to be zero as well. So

$$\frac{P_0}{\rho g} = Z_2 = y \qquad P_0 = P_{atm}$$

$$y = \frac{100000}{1000 \times 10} = 10 \text{ m of water}$$

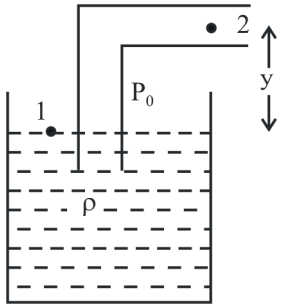
Now, it is said that another fluid

$$\rho_f gh = \rho_w \times g \times y$$

$$h = \frac{10}{(\rho_f / \rho_w)}$$

$$h = \frac{10}{\rho} \text{ m where } \rho \text{ is specific density}$$

where ρ_f, ρ_w are density of any fluid and density of standard fluid water



913. For flow through a horizontal pipe, the pressure gradient $\frac{dp}{dx}$ in the flow direction is-

(a) +ve (b) 1
(c) Zero (d) -ve

ISRO Scientist/Engineer 2006

Ans. (d) : According to Bernoulli's Equation–

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_f$$

$(v_1 = v_2, z_1 = z_2)$

$$\therefore \quad \boxed{\frac{p_1 - p_2}{\rho g} = h_f} \quad [\because \text{pressure decreases}]$$

$\therefore$ Pressure gradient is negative.

914. Bernoulli's equation is applicable to

(a) Compressible fluids
(b) Incompressible fluids
(c) Newtonian fluids
(d) Non-Newtonian fluids

UKPSC AE 2007 Paper -II

Ans. (b) : $\frac{P}{\rho g} + \frac{v^2}{2g} + z = \text{ constant}$

This is Bernoulli's equation and also known as Mechanical energy balance equation.

This equation is valid under the conditions –
(i) Steady flow
(ii) Fluid in non-viscous
(iii) Fluid is incompressible
(iv) Flow takes place along stream line

915. The actual velocity at vena contracta for flow through an orifice from a reservoir of height H=

(a) $\sqrt{2gH}$ (b) $C_V\sqrt{2gH}$
(c) $\sqrt{2gH/C_v}$ (d) $C_d\sqrt{2gH}$

ISRO Scientist/Engineer 2006

Ans. (b) : Actual velocity at vena-contracta for through an orifice from a reservoir of height H,

$$\boxed{V_{actual} = C_V\sqrt{2gH}}$$

C_V = Coefficient of velocity

916. A constant-head water tank has, on one of its vertical sides two identical small orifices issuing two horizontal jets in the same vertical plane. The vertical distance between the centers of orifices is 1.5 m and the jet trajectories intersect at a point 0.5 m below the lower orifice. What is the approximate height of water level in the tank above the point of intersection trajectories?

(a) 1.0 m (b) 2.5 m
(c) 0.5 m (d) 2.0 m

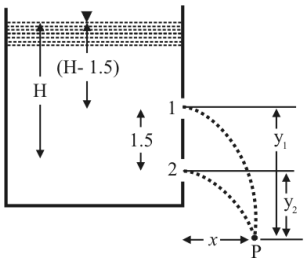
ISRO Scientist/Engineer 2006

Ans. (b) : data given,

$y_2 = 0.5$ m

$y_1 = 1.5 + 0.5 = 2$ m

orifice 1 and orifice 2 is same then co-efficient of velocity



$$C_V = \frac{x}{\sqrt{4yH}}$$

so, $C_{V_1} = C_{V_2}$

$$\frac{x}{\sqrt{4 \times y_1 \times H_1}} = \frac{x}{\sqrt{4 \times y_2 \times H_2}}$$

$$y_1 \times H_1 = y_2 \times H_2$$

$$2 \times (H - 1.5) = 0.5 \times H$$

H = 2m

then height of water level in tank above the point of intersection of trajectories is

$(H + y_2) = 2 + 0.5 = 2.5$ m

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7. Flow Through Pipes

917. Power transmitted through pipes will be maximum when–

(a) head lost due to friction $= \frac{1}{2}\times$ (total head at inlet of the pipe)

(b) head lost due to friction $= \frac{1}{4}\times$ (total head at inlet of the pipe)

(c) head lost due to friction = total head at inlet of the pipe

(d) head lost due to friction $= \frac{1}{3}\times$ (total head at inlet of the pipe)

ISRO Scientist/Engineer (RAC) 22.04.2018
Kerala PSC Poly. Lect., 2017
(BPSC POLY. TEACH 2016, UJVNL AE 2016)
RPSC IOF & Boiler 20.09.2017
VIZAG STEEL MT 2015
ISRO Scientist/Engineer 11.10.2015
J&K PSC Civil Services Pre, 2013
TNPSC AE, 2008, ESE 2001

Ans. : (d) For maximum power transmission,

H_L = Total head supplied

h = Head loss due to friction

$$h = \frac{H_L}{3}$$

918. For flow through pipes, the maximum transmission efficiency that can achieve is:

(a) 97% (b) 57%

(c) 47% (d) 67%

TSPSC Manager (Engg.) HMWSSB 12.11.2020
OPSC AEE 2019 PAPER - II
ESE 2019
Kerala PSC Poly. Lect. 09.01.2014
APPSC AEE 2012, APPSC AE 04.12.2012
RPSC AE GWD, 2011, TNPSC AE, 2008

Ans. : (d) : Efficiency of power transmission through pipe is given by equation

$$\eta = \frac{H - h_f}{H}$$

For maximum power transmission through pipe is given by equation,

$$h_f = \frac{H}{3}$$

Substituting the value of h_f in efficiency, we get maximum η,

$$\eta_{max} = \frac{H - H/3}{H} = 1 - \frac{1}{3}$$
$$= 66.7\%$$
$$\approx 67\%$$

919. Chezy's formula is given by

(a) $v = i\sqrt{mC}$ (b) $v = C\sqrt{mi}$

(c) $v = m\sqrt{Ci}$ (d) None of the above

VIZAG MT, 14.12.2020
TSPSC AEE 2015, TNPSC AE 2013
APPSC AEE 2012, APPSC AE 04.12.2012

Ans. (b) : Chezy's formula velocity of flow

$$V = C\sqrt{mi} \text{ m/sec}$$

Where, C = Chezy's constant, m = Hydraulic mean depth

$$m = \frac{A}{\rho} = \frac{\frac{\pi}{4}d^2}{\pi d} = \frac{d}{4} \text{ meter.}$$

920. Maximum velocity in fully developed laminar pipe flow is

(a) Half of average velocity
(b) Two-third of average velocity
(c) Twice of average velocity
(d) Equal of average velocity

ESE 2020
RPSC Lecturer 16.01.2016
UKPSC AE 2012 Paper–II
OPSC Civil Services Pre. 2011

Ans. (c) : Maximum velocity in fully developed laminar pipe flow is twice of average velocity.

$$U_{max} = \frac{R^2}{4\mu}\left(-\frac{\partial P}{\partial x}\right)$$

$$u = U_{max}\left(1 - \frac{r^2}{R^2}\right)$$

Average velocity

$$\bar{u} = \frac{R^2}{8\mu}\left(-\frac{\partial P}{\partial x}\right)$$

So,

$$\frac{U_{max}}{\bar{u}} = 2$$

$$U_{max} = 2\bar{u}$$

921. The hydraulic gradient line is:

(a) always moving up
(b) always moving down
(c) the velocity head below the energy grade line
(d) always above the energy grade line

Kerala PSC AE (Irrigation) 05.04.2017
APPSC AEE Mains 2016 (Civil Mechanical)
GWSSB DEE 07.07.2016
TSPSC AEE 2015

Ans. (c) : The line representing the sum of pressure heat (P/W) and potential head or datum head (z) with respect to some reference line is hydraulic gradient line.

The hydraulic gradient line is the velocity head below the energy grade line.

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922. What is hydraulic diameter used in place of diameter for non-circular ducts equal to?

(a) A/m (b) 4A/m
(c) A/(4m) (d) 4m/A
(e) None of the above

Where A = Area of flow and m = perimeter

APPSC-AE-2019, TNPSC AE 2018
CGPSC Poly. Lect. 22.05.2016
ESE 2008

Ans. (b)

For Non-circular, Hydraulic diameter (D) = $\frac{4A}{m}$

where,
A = Area
m = perimeter

923. Water at 25°C is flowing through a 1.0 km long GI pipe of 200 mm diameter at the rate of 0.07 m^3/s. If value of Darcy friction factor for this pipe is 0.02 and density of water is 1000 kg/m^3, the pumping power (in kW) required to maintain the flow is

(a) 1.8 (b) 17.4
(c) 20.5 (d) 41.0

ISRO Scientist/Engineer 11.10.2015
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2009

Ans. (b) : Given, ℓ = 1.0 km = 1000 m
D = 200 mm = 0.2 m
Q = 0.07 m^3/sec
f = 0.02
ρ = 1000 kg/m^3

Pumping power = $\rho g Q h_f$

But, $\boxed{h_f = \frac{f \ell Q^2}{12D^5}}$

$$= \frac{0.02 \times 1000 \times 0.07^2}{12 \times (0.2)^5}$$

h_f = 25.52 m

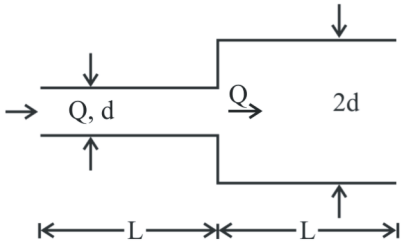
Pumping power = $\rho g Q h_f$
= 1000 × 9.81 × 0.07 × 25.52
= 17524.58W = 17.52 kW

924. Two pipe lines of equal length are connected in series. The diameter of second pipe is two times that of first pipe, the ratio of head loss between first and second pipe is

(a) 1:32 (b) 32:1
(c) 1:8 (d) 1:4

PPSC Asstt. Municipal Engineer 15.06.2021
APPSC AE Subordinate Service Civil/Mech. 2016
ISRO Scientist/Engineer 24.05.2014
CSE 1996

Ans. (b) :
Given - "Two pipe line are connected in series."
Let, Pipe first length = L_1
Pipe second length = L_2
i.e. L_1, L_2 = L [given]
And First pipe diameter = d_1
Second pipe diameter = $d_2 = 2d_1$ [∵ given]



$$\frac{h_{f_1}}{h_{f_2}} = \frac{V_1^2}{d_1} \times \frac{d_2}{V_2^2}$$

$$= \left(\frac{Q}{A_1}\right)^2 \frac{1}{d_1} \times d_2 \times \left(\frac{A_2}{Q}\right)^2$$

$$= \left(\frac{d_2}{d_1}\right)^5$$

$$\frac{h_{f_1}}{h_{f_2}} = 2^5 = 32$$

925. Oil flows through 200 mm diameter horizontal cast iron pipe with friction factor of 0.0225 of length 500 mtr. The volumetric flow rate is 0.2 m^3/sec. The head loss in mtr. due to friction will be

(a) 232.36 (b) 0.116
(c) 116.18 (d) 18.22

ISRO Scientist/Engineer (RAC) 07.05.2017
HPPSC Asstt. Prof. 18.11.2016
GATE 2012

Ans. (c) : D = 200 mm = 0.2m
f = 0.0225
L = 500 m
Q = 0.2 m^3/sec

$$h = \frac{f.L.V^2}{D \times 2g}$$

$$h = \frac{4^2.f.L.Q^2}{\pi^2.D^5.2.g}$$

$$= \frac{0.0225 \times 500 \times (0.2)^2 \times 4^2}{\pi^2 \times (0.2)^5 \times 2 \times 9.81}$$

h = 116.194 m

926. For steady, fully developed flow inside a straight pipe of diameter D, neglecting gravity effects, the pressure drop ΔP over a length L and the wall shear stress τ_W are related by

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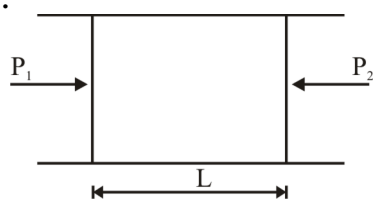
(a) $\tau_W = \frac{\Delta PD}{4L}$ (b) $\tau_W = \frac{\Delta PD^2}{4L^2}$

(c) $\tau_W = \frac{\Delta PD}{2L}$ (d) $\tau_W = \frac{4\Delta PL}{D}$

BPSC AE Mains 2017 Paper - V
RPSC VPITI 14.02.2016, GATE 2013

Ans. : (a) :



Considering equilibrium of a section of length L.

$\tau_w \times \pi DL = \Delta P \frac{\pi D^2}{4}$

$\Rightarrow \tau_w = \frac{\Delta PD}{4L}$

927. The head loss in a sudden expansion from 6 cm diameter pipe to 12 cm diameter pipe, in terms of velocity V_1 in the smaller diameter pipe is

(a) $\frac{3}{16}\frac{V_1^2}{2g}$ (b) $\frac{5}{16}\frac{V_1^2}{2g}$

(c) $\frac{7}{16}\frac{V_1^2}{2g}$ (d) $\frac{9}{16}\frac{V_1^2}{2g}$

BPSC AE Mains 2017 Paper - V
Nagaland PSC (CTSE) 2016, Paper-II
ESE 2010

Ans. : (d) : Head loss in a sudden expansion

$$h_L = \frac{(V_1 - V_2)^2}{2g} = \frac{V_1^2}{2g}\left(1 - \frac{V_2}{V_1}\right)^2$$

$A_1V_1 = A_2V_2$

$$\frac{V_2}{V_1} = \frac{A_1}{A_2}$$

$$\frac{V_2}{V_1} = \frac{d_1^2}{d_2^2}$$

$$= \frac{V_1^2}{2g}\left(1 - \frac{d_1^2}{d_2^2}\right)^2$$

$$= \frac{V_1^2}{2g}\left(1 - \left(\frac{6}{12}\right)^2\right)^2$$

$$= \frac{9}{16}\frac{V_1^2}{2g}$$

928. Head loss at entrance to pipe is:

(a) $\frac{3v^2}{2g}$ (b) $0.5\frac{v^2}{2g}$

(c) $\frac{2v^3}{g}$ (d) $\frac{v^2}{2g}$

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I
APGCL AM, 2021, OPSC AEE 2015 PAPER - II

Ans. : (b) Head loss at entrance to pipe is $0.5v^2/2g$.

Condition	Head loss
Losses due to sudden expansion	$\frac{(v_1 - v_2)^2}{2g}$
Losses due to sudden contraction	$\frac{v_2^2}{2g}\left(\frac{1}{C_c} - 1\right)^2$ OR $\frac{0.5v_2^2}{2g}$
Losses at exit of pipe	$v^2/2g$
Losses at entrance to pipe	$\frac{0.5v^2}{2g}$
Losses due to bends	$\frac{K.v^2}{2g}$, K = f(R, θ, r) R → Radius of curvature of pipe, r → Radius of pipe from horizontal, θ → Inclination of pipe.

929. The value of momentum correction factor (β) for the viscous flow through a circular pipe is

(a) 1.33 (b) 1.50

(c) 2.0 (d) 1.25

TSPSC AEE 2015, APPSC Poly. Lect. 2013
APPSC AE 04.12.2012

Ans. (a) : Momentum correction factor (β) for the viscous laminar flow through a circular pipe is 1.33 or (4/3) while energy correction factor is equal to 2.

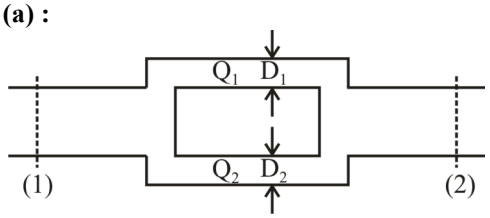
930. In order to replace a pipe of diameter D by n parallel pipes of diameter d the relation used is

(a) $d = \frac{D}{n^{2/5}}$ (b) $d = \frac{D}{n^2}$

(c) $d = \frac{D}{n^{3/2}}$ (d) $d = \frac{D}{n}$

BPSC Poly. Lect. 2016
APPSC AE Subordinate Service Civil/Mech. 2016
VIZAG STEEL MT 2015

Ans. (a) :



We know that, in parallel connection head loss in each pipe remains same.

$h_{f_1} = h_{f_2} = h_{f_3} = h_{f_n}$

$Q_n = Q_1 + Q_2 + Q_3 + ...$

$$h_f = \frac{fLQ^2}{12.1D^5}$$

$$Q = \left(\frac{12.1D^5}{fL}\right)^{1/2} = n\left(\frac{12.1d^5}{fL}\right)^{1/2}$$

$D^5 = n^2d^5 \Rightarrow d = \frac{D}{n^{2/5}}$

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931. Two pipe systems can be said to be equivalent, when the following quantities are same :

(a) friction loss and flow
(b) length and diameter
(c) flow and length
(d) friction factor and diameter

Vadodara Muncipal Corp. DEE, 2018
Kerala PSC AE (Irrigation) 05.04.2017
CSE 2007

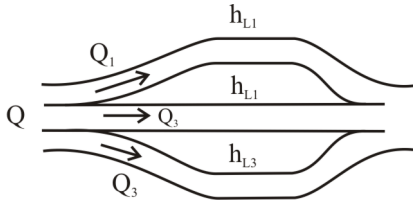
Ans. (a) : Two pipe system can be said to be equivalent when friction loss (h_f) and flow (Q) are same.

932. Three parallel pipes connected at the two ends have flow-rates Q_1, Q_2 and Q_3 respectively, and the corresponding frictional head losses are h_{L1}, h_{L2} and h_{L3} respectively. The correct expressions for total flow rate (Q) and frictional head low across the two ends (h_L) are

(a) $Q = Q_1 + Q_2 + Q_3$; $h_L = h_{L1} + h_{L2} + h_{L3}$
(b) $Q = Q_1 = Q_2 = Q_3$; $h_L = h_{L1} = h_{L2} = h_{L3}$
(c) $Q = Q_1 + Q_2 + Q_3$; $h_L = h_{L1} = h_{L2} = h_{L3}$
(d) $Q = Q_1 = Q_2 = Q_3$; $h_L = h_{L1} + h_{L2} + h_{L3}$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (c) : In parallel connection frictional head loss will be same.



$Q = Q_1 + Q_2 + Q_3$
$h_L = h_{L1} = h_{L2} = h_{L3}$

933. Which one of the following statements is correct for pipes connected in series?

(a) Same flow takes place through each pipe and the head loss across them is also same.
(b) Different flow takes place through each pipe but the head loss across them is same
(c) Same flow takes place through each pipe and head loss across the entire pipe is equal to the sum of head losses in the constituent pipes
(d) Different flow takes place through each pipe but the head loss across the entire pipe is equal to the sum of head losses in the constituent pipes.

Nagaland PSC CTSE 2017 Paper-2
RPSC AE GWD, 2011

Ans. (c) :
Pipes in series –

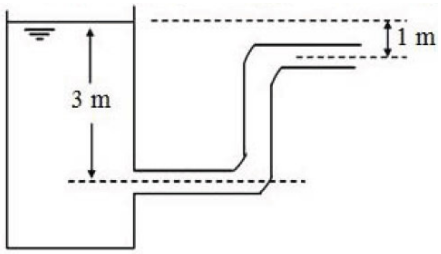
- $Q = Q_1 = Q_2 = Q_3 = Q_4$
- $h_L = h_{L_1} + h_{L_2} + h_{L_3} + h_{L_4}$

$$\left.\begin{matrix} Q = \text{discharge or flow rate} \\ h_L = \text{head loss} \end{matrix}\right\}$$

Pipes in Parallel –

- $Q = Q_1 + Q_2$
- $h_{L_1} = h_{L_2}$

934. What is the discharge velocity at the pipe exit in the given figure?



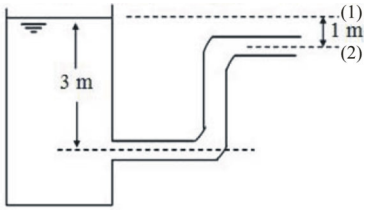
(a) $\sqrt{2\times9.81\times1}$ (b) $\sqrt{2\times9.81\times2}$
(c) $\sqrt{2\times9.81\times3}$ (d) $\sqrt{2\times9.81\times4}$
(e) $\sqrt{2\times9.81\times5}$

CGPSC Asstt. Workshop Supt., 17.07.2016
GATE ME 1998

Ans. (a) : Given,

$P_1 = P_2 = P_{atm}$
$V_1 = 0$, $V_2 = ?$

By applying Bernoulli's at section (1) & (2)



$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

$$\frac{P_1}{\rho g} = \frac{P_2}{\rho g} = \frac{P_{atm}}{\rho g}$$

For free surface, $v_1 = 0$

$$z_1 - z_2 = \frac{v_2^2}{2g}$$

$$v_2 = \sqrt{2g(z_1 - z_2)}$$

$$= \sqrt{2\times9.81\times1}$$

935. A pipe having a length 200 m and 200 mm diameter with friction factor 0.015, is to be replaced by a 400 mm diameter pipe of friction factor 0.012 to convey the same quantity of flow. The equivalent length of the new pipe for the same head loss will be

(a) 8300 m (b) 8240 m
(c) 8110 m (d) 8000 m

RPSC IOF, 2020, ESE 2019

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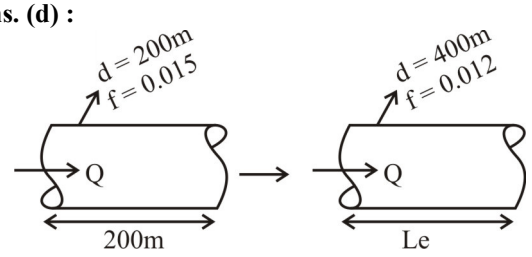
Ans. (d) :



$$h_{f_1} = \frac{f_1 L_1 Q_1^2}{12.1 d_1^5} \qquad h_{f_2} = \frac{f_2 L_2 Q_2^2}{12.1 d_2^5}$$

Given- $Q_1 = Q_2$

$h_{f_1} = h_{f_2}$

So, $\frac{f_1 L_1}{12.1 d_1^5} = \frac{f_2 L_2}{12.1 d_2^5}$

Where $L_2 = L_e$ equivalent length of new pipe

$$L_e = \frac{f_1}{f_2} \times \left(\frac{d_2}{d_1}\right)^5 \times L_1$$

$$L_e = \frac{0.015}{0.012} \times \left(\frac{400}{200}\right)^5 \times 200$$

$L_e = 8000m$

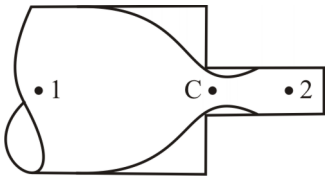
936. The loss of head due to sudden contraction of a pipe is equal to

(a) $\left(\frac{1}{C_C} - 1\right)^2 \frac{V_2^2}{2g}$ (b) $\left(1 - \frac{1}{C_C}\right)^2 \frac{V_2^2}{2g}$

(c) $\frac{1}{C_C}\left(1 - \frac{V_2^2}{2g}\right)$ (d) None of these

RPSC IOF, 2020, TNPSC AE 2018

Ans. (a) : Losses due to sudden contraction (h_c)



$$h_c = \frac{(V_c - V_2)^2}{2g} = \left[\frac{1}{C_c} - 1\right]^2 \frac{V_2^2}{2g}$$

Coefficient of contraction $C_c = \frac{V_2}{V_c} = \frac{A_c}{A_2}$

If C_c is not given

$$h_c = \frac{0.5 V_2^2}{2g}$$

937. Consider the information given below :
A smooth pipe of diameter 200 mm carries water. The pressure in the pipe at section S_1 (elevation : 10 m) is 50 kPa. At section S_2 (elevation : 12 m), the pressure is 20 kPa and the velocity is 2 m/sec. Density of water is 1000 kg/m^3 and acceleration due to gravity is 9.8 m/s^2.
Which of the following is true?

(a) Flow is from S_1 to S_2 and head loss is 0.53m
(b) Flow is from S_2 to S_1 and head loss is 0.53m
(c) Flow is from S_1 to S_2 and head loss is 1.06 m
(d) Flow is from S_2 to S_1 and head loss is 1.06 m

UPRVUNL AE 2014, GATE 2010

Ans. : (c) From Bernouli's Equation

$$\frac{P_1}{\rho g} + \frac{V_1^2}{\rho g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2 + h_f$$

$\Rightarrow V_1 = V_2$, because diameter of the pipe is same at the both section

$$\frac{50 \times 10^3}{1000 \times 9.81} + 10 = \frac{20 \times 10^3}{1000 \times 9.81} + 12 + h_f$$

$$h_f = \frac{50 \times 10^3 - 20 \times 10^3}{1000 \times 9.81} - 2$$

$h_f = 1.06m$ (+ve value of h_f)

So, fluid flow from S_1 to S_2 and head loss 1.06m.

938. For Viscous flow through pipes the friction factor is given as :

(a) 1600/Re (b) 1000/Re
(c) 16/Re (d) 32/Re

TSPSC Managers, 2015
UKPSC AE-2013, Paper-II

Ans. (c) : For viscous flow through pipes the friction

Factor (f) $= \frac{64}{Re}$

f = friction factor

f '= coefficient of friction

$\boxed{f = 4f'}$

$f' = \frac{16}{Re}$

939. The loss of head (or energy) due to velocity of liquid at the outlet of the pipe is given as :

(a) $V^4/4g$ (b) $V^4/2g$
(c) $V^2/4g$ (d) $V^2/2g$

Nagaland PSC (CTSE) 2018, Paper-II
TSPSC Managers, 2015

Ans. (d) : The head loss (or energy) due to velocity at liquid.

Velocity head = $\frac{V^2}{2g}$

(or Dynamic head)

940. For a fully developed flow of water in pipe having diameter 10 cm, velocity 0.1 m/s and kinematic viscosity 10^{-5} m^2/s, the value of Darcy friction factor is

(a) 0.064 (b) 0.016
(c) 0.008 (d) 0.004

GPSC Executive Engineer 23.12.2018
GATE 2014

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Ans. (a) : Given as,

$d = 10\ cm = 0.1\ m$

$V = 0.1\ m/s$

$\nu = 10^{-5}\ m^2/s$

$$R_e = \frac{\rho vd}{\mu} = \frac{vd}{\nu}$$

$$= \frac{0.1 \times 0.1}{10^{-5}} = 1000$$

Hence, flow is laminar.

Now, Darcy friction factor for laminar flow is given by,

$$f = \frac{64}{R_e} = \frac{64}{1000} = 0.064$$

941. Piezometric head is the sum of:

(a) Pressure head, datum head and velocity head

(b) Datum head and velocity head

(c) Pressure head and velocity head

(d) Pressure head and datum head

PTCUL AE 25.06.2017, ESE 2012

Ans. (d) : Pressure head and datum head

942. A pipe is said to be a siphon if

(a) it can operate without external power

(b) it can discharge is liquid at a fast rate

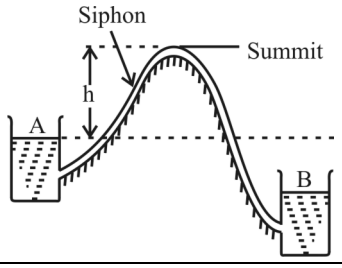
(c) it has sub-atmospheric pressure in it

(d) the exit is at a lower level than the inlet

APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (c) : A pipe is said to be a siphon if it has sub-atmospheric pressure in it.

Siphon is a long bent pipe which is used to transfer liquid from a reservoir at a higher elevation to another reservoir at a lower level when the two reservoirs are separated by a hill or high level ground as shown in figure–



943. If H is the total head at inlet and h is the head lost due to friction, the efficiency of power transmission trough a straight pipe is given by:

(a) $\frac{H-h}{H}$

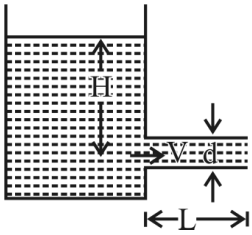
(b) $\frac{H}{H+h}$

(c) $\frac{H-h}{H+h}$

(d) $\frac{H}{H-h}$

UJVNL AE 2016, ESE 1995

Ans : (a)



L = length of the pipe

d = diameter of the pipe

H = Total head available at the inlet of pipe

V = Velocity of flow in pipe.

h = Loss of head due to friction.

Head available at the outlet of the pipe = (Total head at inlet- Head loss) = H-h

Efficiency of power transmission $(\eta) = \frac{H-h}{H}$

944. The hydraulic diameter, D_h of a rectangular duct with sides 'a' and 'b' is

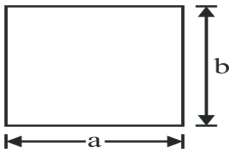
(a) $D_h = \frac{4ab}{2a+b}$ (b) $D_h = \frac{2ab}{a+b}$

(c) $D_h = \frac{2ab}{2a+b}$ (d) $D_h = \frac{4ab}{a+b}$

ISRO Scientist/Engineer 03.07.2016

CSE Pre-2007

Ans. (b) :



Hydraulic diameter $D_h = \frac{4A}{P}$

$$D_h = \frac{4 \times a \times b}{2(a+b)} = \frac{2ab}{a+b}$$

945. Consider fully developed flow in a circular pipe with negligible entrance length effects. Assuming the mass flow rate, density and friction factor to be constant, if the length of the pipe is doubled and the diameter is halved, the head loss due to friction will increase by a factor of :

(a) 2 (b) 16

(c) 32 (d) 64

UP Jal Nigam AE 2016, GATE 2015

Ans. (d) : Given :

Diameter of first pipe (d_1) $= d\ m$

Length of first pipe (ℓ_1) $= \ell\ m$

Diameter of second pipe (d_2) $= \frac{d}{2}\ m$

Length of second pipe (ℓ_2) $= 2\ell\ m$

Mass flow rate is given by,

$$m = \rho av = \rho \frac{\pi}{4} d^2 v$$

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$$v = \frac{4m}{\rho\pi d^2}$$

Head loss due to friction is given by,

$$h_f = \frac{f\ell v^2}{2gd} = \frac{f\ell}{gd} \times \frac{8m^2}{\rho^2\pi^2 d^4}$$

$$h_f \propto \frac{\ell}{d^5}$$

So, $$h_{f1} = \frac{\ell_1}{d_1^5} = \frac{\ell}{d^5} \quad ...(i)$$

Similarly, $$h_{f_2} = \frac{\ell_2}{d_2^5} = \frac{2\ell}{\left(\frac{d}{2}\right)^5} \quad ...(ii)$$

Hence taking the ratio of two different pipes
Now from equation (i) and (ii),

$$\frac{h_{f1}}{h_{f_2}} = \frac{\ell_1}{\ell_2} \times \frac{d_2^5}{d_1^5} = \frac{\ell}{2\ell} \times \left(\frac{d}{2}\right)^5 \times \frac{1}{d^5} = \frac{1}{64}$$

$$\boxed{h_{f_2} = 64 h_{f1}}$$

946. For laminar flow in a round pipe, the energy correction factor is

(a) 0.50 (b) 1.00
(c) 1.33 (d) 2.0

TSPSC AEE 2015, JPSC AE 2013 Paper-V

Ans. (d) : In Laminar flow,
Energy correction factor = 2
Momentum correction factor = 1.33
In Turbulent flow
Energy correction factor = 1.33
Momentum correction factor = 1.2

947. The ratio of average velocity to maximum velocity for steady laminar flow in circular pipes is

(a) 1/2 (b) 2/3
(c) 3/2 (d) 2
(e) 3

RPSC IOF & Boiler 20.09.2017
CGPSC AE 26.04.2015 SHIFT-II

Ans. (a) : Average velocity ($U_{avg.}$) for laminar steady flow in circular pipe is given as

$$U_{avg.} = \frac{U_{max}}{2}$$

Average velocity (U_{avg}) for laminar steady flow between two stationary plate is given as

$$U_{avg.} = \frac{3}{2} U_{max.}$$

948. A pipeline connecting two reservoirs has its diameter reduced by 20% due to deposition of chemicals. For a given head difference in the reservoirs with unaltered friction factor, this would cause a reduction in discharge of

(a) 42.8% (b) 20%
(c) 17.8% (d) 10.6%

ISRO Scientist/Engineer 2007
ESE 2000

Ans. (a) : Head of loss, n pipes connected in series.

$$h = h_1 + h_2 + ...h_n$$

$$h = \frac{8fLQ^2}{\pi^2 g D^5}$$

$$\frac{Q_1^2}{D_1^5} = \frac{Q_2^2}{D_2^5}$$

(for same head difference)

$$\frac{Q_2}{Q_1} = \left(\frac{D_2}{D_1}\right)^{\frac{5}{2}}$$

$$\frac{Q_2}{Q_1} = (0.8)^{\frac{5}{2}}$$

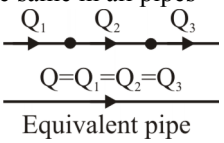
Q_2 = 57.2% of Q_1
Reduction in discharge = 42.8%

949. For pipes arranged in series:

(a) the head loss must be same in all pipes
(b) the flow rate must be the same in all pipes
(c) the velocity must be the same in all pipes
(d) the flow rate may be different in different pipes

Oil India Ltd. Sr. Engg. (Production) 30.11.2019
TNPSC ACF 2012

Ans. (b) : For pipe arranged in series-
(i) The flow rate be same in all pipes

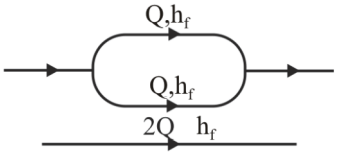


Equivalent pipe

(ii) Friction factor is same in all pipes
(iv) Minor losses neglected

$$\frac{L}{d^5} = \frac{L_1}{d_1^5} + \frac{L_2}{d_2^5} + \frac{L_3}{d_3^5} + ...$$

For pipes arranged in parallel



Equivalent pipeline

(i) Head loss must be same in all pipes
(ii) Friction factor is same in all pipes

950. Head loss in sudden expansion is given by

(a) $\frac{v_1^2 - v_2^2}{2g}$ (b) $\frac{(v_1 - v_2)^3}{2g}$

(c) $\frac{(v_1 - v_2)^2}{2g}$ (d) $\frac{2(v_1^2 - v_2)^2}{g}$

(Notations used have usual meanings)

ISRO Scientist/Engineer (RAC) 29.11.2015
UKPSC AE 2012 Paper–II

Ans. (c) : $\frac{(v_1 - v_2)^2}{2g}$

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951. In order to have a continuous flow through a siphon, no portion of the pipe be higher than _____ measured above the hydraulic grade line:

(a) 10m (b) 10.33m
(c) 5.5m (d) 7.75m

APPSC AEE Mains 2016 (Civil Mechanical)
UKPSC AE 2012 Paper–II

Ans. (d) : 7.75m

952. The Darcy - Weisbach friction factor 'f' which is a direct measure of resistance to flow in pipes is dependant on

(a) roughness height, diameter and velocity
(b) relative roughness, diameter and viscosity
(c) relative roughness, velocity and viscosity
(d) roughness height, diameter, velocity and kinematic viscosity

APPSC AEE 2012
UKPSC AE 2012 Paper–II

Ans. : (d) friction factor (f) = $\frac{64}{R_e}$, $f' = \frac{f}{4}$

for laminar flow.

$$f' = \frac{16}{R_e}$$

f′ = Fanning friction factor (or) friction co-efficient.

Friction factor depend upon :-

(i) Diameter
(ii) velocity
(iii) Kinematic viscosity
(iv) Relative height.

953. Which is the most commonly used following equation for velocity of distribution for laminar flow through pipes?

(a) $U = U_{max}\left[1-\frac{r}{R}\right]$

(b) $U = U_{max}\left[1-\left(\frac{r}{R}\right)^2\right]$

(c) $U = U_{max}\left[1-\left(\frac{r}{R}\right)^3\right]$

(d) $U = U^2_{max}\left[1-\left(\frac{r}{R}\right)^2\right]$

where U_{max} = centre line velocity
R = Radius of pipe

TNPSC AE 2018
UKPSC AE 2007 Paper -II

Ans. (b) : For laminar flow through pipes, velocity distribution equation,

$U = U_{max}\left[1-\left(\frac{r}{R}\right)^2\right] \Rightarrow$ Parabolic distribution

& $U_{max} = \frac{1}{4\mu}\left(\frac{-\partial P}{\partial x}\right)R^2$

* Shear stress equation,

$$\tau = \left(\frac{-\partial P}{\partial x}\right)\times\frac{r}{2} \text{ (linear distribution)}$$

954. The flow of a fluid in a pipe takes place from

(a) Higher level to lower level
(b) Small end to large end
(c) Higher pressure to lower pressure
(d) Higher energy to lower energy

UPPSC AE 13.12.2020, Paper-2
ISRO Scientist/Engineer 2008

Ans. (d) : Fluid flows from region of higher energy to lower energy and not from higher pressure to lower pressure.

Flow from a low pressure region to a high pressure region is possible if the sum of potential, pressure and kinetic energy at the low pressure region is greater than the high pressure region.

955. If V is the mean velocity of flow, then according to Darcy-Weisbach equation for pipe flow energy loss over a length of pipeline is proportional to

(a) V (b) 1/V
(c) $1/V^2$ (d) V^2

APGENCO AE, 2017
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (d) Head or Energy loss over a length of pipeline

$h = \frac{f.L.V^2}{d\times 2g}$

$$\left[\frac{h}{L}\right] \propto V^2$$

956. From a reservoir, water is drained through two pipes of 10 cm and 20 cm diameter respectively. If frictional head loss in both the pipes is same, then the ratio of discharge through the larger pipe to that through the smaller pipe will be

(a) $\sqrt{2}$ (b) $2\sqrt{2}$
(c) 4 (d) $4\sqrt{2}$

APPSC AE Subordinate Service Civil/Mech. 2016
CSE Pre-1998

Ans. (d) : $h_{f_1} = h_{f_2}$

d_1 = dia of largest pipe = 20 cm
d_2 = dia of smaller pipe = 10 cm
Q_1 = discharge through larger pipe
Q_2 = discharge through smaller pipe

$$\frac{fLV_1^2}{2gd_1} = \frac{fLV_2^2}{2gd_2}$$

$$\frac{fLQ_1^2}{12.1d_1^5} = \frac{fLQ_2^2}{12.1d_2^5}$$

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$$\left(\frac{Q_1}{Q_2}\right)^2 = \left(\frac{d_1}{d_2}\right)^5$$

$$\left(\frac{Q_1}{Q_2}\right) = \left(\frac{20}{10}\right)^{5/2} = (2)^{5/2}$$

$$\frac{Q_1}{Q_2} = \sqrt{32} = 4\sqrt{2}$$

957. The power available at the outlet of the pipe carrying water with a rate of flow of Q and head of H supplied at the entrance of the pipe is

(γ is the specific weight : and h_f is the loss of head due to friction)

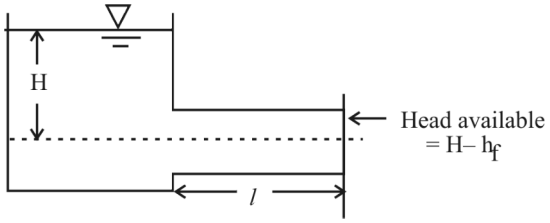
(a) $\gamma Q(H-h_f)$ (b) γQH

(c) $\gamma Q(H+h_f)$ (d) $\gamma QH(H+h_f)$

Sikkim PSC (Under Secretary), 2017

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



Power supplied at inlet of the pipe

= Weight of water per sec × Head at outlet

$\boxed{P_{inlet} = W \times H = \gamma QH}$

Power available at outlet of pipe

= Weight of water per sec × Head at outlet

= W × $(H-h_f)$

$\boxed{P_{outlet} = \gamma Q(H-h_f)}$

958. Due to which of the following phenomenon water hammer is caused?

(a) incompressibility of fluid

(b) sudden opening of a valve in a pipeline

(c) the material of the pipe being elastic

(d) sudden closure of a valve in pipe flow

TSPSC Manager (Engg.) HMWSSB 12.11.2020

ESE 1995

Ans. (d) : Phenomenon of sudden rise in pressure is known as water hammer or hammer blow.

*** Factors affecting water hammer**

- Speed at which valve is closed.
- Length of pipe
- Elastic properties of pipe material as well as that flowing fluid.

959. Which one of the following statements is true for fully developed flow through pipes?

(a) The flow is parallel, has no inertia effects, the pressure gradient is of constant value and the pressure force is balanced by the viscous force

(b) The flow is parallel, the pressure gradient is proportional to the intertia force and there is no viscous effect

(c) The flow is parallel, the pressure gradient is negligible and the inertia force is balanced by the viscous force

(d) The force is not parallel, the core region accelerates and the viscous drag is far too less than the inertia force

Nagaland PSC CTSE 2015 Paper-II

ESE 1997

Ans. (a) : For fully developed flow through pipes, the flow is parallel, has no inertia effects, the pressure gradient is of constant value and the pressure force is balanced by the viscous force.

960. Two identical pipes of length 'L', diameter 'D' and friction factor 'f' are connected in parallel between two points. For the same total volume flow rate with pipe of same diameter 'd' and same friction factor 'f', the single length of the pipe for a constant pressure drop will be

(a) $\frac{L}{2}$ (b) $\frac{L}{\sqrt{2}}$

(c) $\sqrt{2}L$ (d) $\frac{L}{4}$

Punjab PSC (Lect.) 06.08.2017

CSE 1999

Ans. (d) : For parallel flow

Equivalent Length of Pipe is given by

$$\sqrt{\frac{D_e^5}{L_e}} = \sqrt{\frac{D_1^5}{L_1}} + \sqrt{\frac{D_2^5}{L_2}}$$

For same Diameter and Length

$$\sqrt{\frac{D^5}{L_e}} = \sqrt{\frac{D^5}{L}} + \sqrt{\frac{D^5}{L}}$$

$$\sqrt{\frac{1}{L_e}} = 2\sqrt{\frac{1}{L}}$$

$$\boxed{L_e = \frac{L}{4}}$$

961. If coefficient of contraction at the vena contracta is equal to 0.62, then what will be the dynamic loss coefficient in sudden contraction in air-conditioning duct?

(a) 0.25 (b) 0.375

(c) 0.55 (d) 0.65

HPPSC Asstt. Prof. 20.11.2017

ESE 2004

Ans. (b) : Given, coefficient of contraction $(C_C) = 0.62$

Dynamic loss coefficient $(C_{DL}) = \left(\frac{1}{C_C} - 1\right)^2$

$$C_{DL} = \left(\frac{1}{0.62} - 1\right)^2 = 0.37565$$

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962. The head loss caused due to sudden expansion from A_1 to area A_2 causing velocity to change from V_1 to V_2 is

(a) $\left(1-\frac{A_1}{A_2}\right)^2\left(\frac{V_1^2}{2g}\right)$ (b) $\left(1-\frac{A_1}{A_2}\right)^2\left(\frac{V_2^2}{2g}\right)$

(c) $\left(1+\frac{A_1}{A_2}\right)^2\left(\frac{V_2^2}{2g}\right)$ (d) $\left(1-\frac{A_1}{A_2}\right)^2\left(\frac{V_2}{2g}\right)$

Nagaland PSC (CTSE) 2016, Paper-II
APPSC AE 04.12.2012

Ans. (a) : Head loss caused due to sudden expansion from A_1 to area A_2 causing velocity to change from V_1 to V_2 is,

$$(h_L)_{S.E.} = \frac{(V_1 - V_2)^2}{2g} = \frac{V_1^2}{2g}\left(1-\frac{V_2}{V_1}\right)^2$$

And, $A_1V_1 = A_2V_2 \Rightarrow \frac{V_2}{V_1} = \frac{A_1}{A_2}$

$$(h_L)_{S.E.} = \frac{V_1^2}{2g}\left[1-\frac{A_1}{A_2}\right]^2$$

963. The hydraulic mean depth for a circular pipe is given by (where D is the diameter)

(a) D/2 (b) D/4

(c) D/8 (d) D/16

Nagaland PSC (CTSE) 2017, 2016 Paper-II
OPSC AEE 2015 PAPER - II

Ans. (b) : The hydraulic mean depth for a circular pipe of diameter (D) is D/4.

964. The loss of head due to sudden enlargement is attributed to

(a) viscosity of fluid
(b) roughness of pipe
(c) generation of heat
(d) production and dissipation of turbulent energy

WBPSC AE 2003, ESE 2011

Ans. (d) : The loss of head due to sudden enlargement is attributed to production and dissipation of turbulent energy.
Due to sudden enlargement turbulent eddies at the corner of suction of pipe the formation of eddies cause loss of energy.

965. A fluid of viscosity 1.5 Pa-s and relative density 0.9 flows through a circular pipe of diameter 5 cm with a maximum velocity of 1.2 m/s. The shear stress at the wall in Pa is :

(a) 360 (b) 288

(c) 144 (d) 72

RPSC ACF & FRO, 26.02.2021

Ans. (c) : Given,
Viscosity (μ) = 1.5 Pa-s
Relative density (S_g) = 0.9
d = 5 cm
v_{max} = 1.2 m/s
Shear stress at wall

$$\tau = \frac{-\partial P}{\partial x}\cdot\frac{R}{2}$$

Pressure drop per unit length

$$\Delta P = \frac{32\mu \bar{V} L}{D^2}$$

$$\frac{\Delta P}{L} = \frac{32\mu\bar{V}}{D^2}$$

$$\frac{-\partial P}{\partial x} = \frac{\Delta P}{\Delta L} = \frac{32\times1.5\times(1.2/2)}{(0.05)^2} = 11520$$

$$\tau = -\frac{\partial P}{\partial x}\cdot\frac{R}{2}$$

$$= 11520\left(\frac{D}{4}\right) = 11520\times\frac{0.05}{4}$$

= 144 Pa

966. Oil is being pumped through a straight pipe, the pipe length, diameter and volumetric flow rate are all doubled in a new arrangement. The pipe friction factor, however, remains constant. The ratio of pipe frictional losses in the new arrangement to that of in the original configuration would be :

(a) 1/4 (b) 1/2

(c) 2 (d) 4

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : $\frac{h_{f_1}}{h_{f_2}} = \frac{\frac{fLQ^2}{12.1d^5}}{\frac{f(2L)(2Q)^2}{12.1\times(2d)^5}} = \frac{2^5}{2\times2^2}$

$$\boxed{\frac{h_{f_1}}{h_{f_2}} = 4}$$

$$\boxed{h_{f_2} = \frac{h_{f_1}}{4}}$$

967. The Reynold's number for the flow of a fluid in a horizontal circular tube of constant diameter is 1200. If the diameter of the tube and the kinematic viscosity of the fluid are doubled and that discharged at the pipe exit is unchanged, then the new Reynold's number for the flow in the tube will be :

(a) 4800 (b) 300

(c) 1200 (d) 600

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : Given, Re = 1200, $D_2 = 2D_1$, $v_2 = 2v_1$
Reynold's number,

$$Re_2 = \frac{\rho_1 V_1 D_1}{\mu_1} = \frac{V_1 D_1}{\nu_1}$$

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$$= \frac{\frac{Q}{(\pi/4)D_1^2} \times D_1}{v_1} = \frac{Q}{\frac{\pi}{4} D_1 \times v_1} = 1200$$

$$Re_2 = \frac{Q}{\frac{\pi}{4} D_2 v_2}$$

$$\frac{R_{e_1}}{R_{e_2}} = \frac{Q/\frac{\pi}{4} D \times v}{Q/\frac{\pi}{4}(2D)2v}$$

$$\frac{R_{e_1}}{R_{e_2}} = \frac{1}{4}$$

$$R_{e_2} = \frac{1}{4} R_{e_1} = \frac{1}{4} \times 1200$$

$$\boxed{R_{e_2} = 300}$$

968. Head loss due to friction in water flow through penstock can be minimised by :

(a) decreasing the diameter penstock
(b) increasing the diameter penstock
(c) increasing the length of penstock
(d) increasing the velocity of flow

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : Head loss due to friction =

$$h_f = \frac{fLV^2}{2gD}$$

$$Q = A.V$$

$$V = \frac{Q}{A} = \frac{4Q}{\pi D^2}$$

$$h_f = \frac{fLQ^2}{12.1 \times d^5}$$

$$\boxed{h_f \propto \frac{1}{d^5}} \qquad d \uparrow \rightarrow h_f \downarrow$$

Increasing the diameter of penstock minimises the head loss due to friction.

969. Mark the wrong statement in the context of laminar flow through a pipeline :

(a) Head loss is proportional to square of average flow velocity
(b) The friction fraction varies inversely with flow Reynolds number
(c) Shear stress is zero at the centre and varies linearly with pipe radius
(d) No dispersion of dye injected into the flow stream

CGPSC AE 15.01.2021

Ans. (a) : In laminar flow, head loss is proportional to the average flow velocity.
In turbulent flow, head loss proportional to square of the average flow velocity.

970. Water is flowing through a pipe of diameter 200 mm with a velocity of 3 m/s. What is the head loss due to friction for a length of 5 m if the coefficient of friction is given by $f = 0.02 + \frac{0.09}{Re^{0.3}}$, where Re is Reynolds number?
(Take the kinematic viscosity of water as 0.01 stoke, g = 9.81 m/s² and $(6 \times 10^5)^{0.3} = 54.13$)

(a) 0.993 m of water (b) 0.783 m of water
(c) 0.685 m of water (d) 0.552 m of water

ESE 18.07.2021

Ans. (a) :

$$Re = \frac{vD}{\nu} = \frac{3 \times 0.2}{0.01 \times 10^{-4}} = 6 \times 10^5$$

$$\Rightarrow f = 0.02 + \frac{0.09}{(6 \times 10^5)^{0.3}} = 0.02 + \frac{0.09}{54.13}$$

$$f = 0.02166$$

$\Rightarrow$ Friction factor = 4f = 4 × 0.02166 = 0.08664

Head loss due to friction = $h_f = \frac{(4f)\ell v^2}{2gD}$

$$\Rightarrow h_f = \frac{0.08664 \times 5 \times (3)^2}{2 \times 9.81 \times 0.2} = 0.993 \text{ m of water}$$

971. Water is flowing through a horizontal pipe of diameter 200 mm at a velocity of 3 m/s. A circular solid plate of diameter 150 mm is placed in the pipe to obstruct the flow. What is the loss of head due to obstruction in the pipe if $C_c = 0.62$? (Take g = 9.81 m/s²)

(a) 3.311 m (b) 4.211 m
(c) 5. 211 m (d) 6.211 m

ESE 18.07.2021

Ans. (a) : Given, diameter of pipe, d = 200 mm = 0.2m
diameter of solid plate, d_o = 150 mm = 0.15 m
Head loss due to obstruction in pipe

$$h_L = \frac{v^2}{2g}\left[\frac{A}{C_c(A-a)} - 1\right]^2$$

v = Velocity in pipe
A = Area of pipe
a = Max. area of obstruction
C_c = Coefficient of contraction

$$h_L = \frac{v^2}{2g}\left[\frac{\frac{\pi}{4}d^2}{C_c \frac{\pi}{4}(d^2 - d_o^2)} - 1\right]$$

$$h_L = \frac{v^2}{2g}\left[\frac{d^2}{C_c(d^2 - d_o^2)} - 1\right]$$

$$\Rightarrow h_L = \frac{(3)^2}{2 \times 9.81}\left[\frac{(0.2)^2}{0.62\left[(0.2)^2 - (0.15)^2\right]} - 1\right]^2$$

$$\boxed{h_L = 3.311 \text{ m}}$$

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972. Three pipes of length 800m, 500 m and 400 m and of diameters 500 mm, 400 mm and 300 mm respectively are connected in series. These pipes are to be replaced by a single pipe of length 1700 m. What is the diameter of the single pipe?

(a) $(0.007118)^{0.2}$ m (b) $(0.003609)^{0.3}$ m

(c) $(0.003609)^{0.2}$ m (d) $(0.007118)^{0.3}$ m

ESE 18.07.2021

Ans. (a) :

$l_1 = 800$ m $l_2 = 500$ m $l_3 = 400$ m

$d_1 = 500$ mm $d_2 = 400$ mm $d_3 = 300$ mm

$\rightarrow Q$

$l = 1700$ m

$d = ?$

Since the two pipes are two be equivalent, hence the discharge and head loss must be same in the two. Assuming same friction factor in the two sets of pipes.

$$\frac{f\ell Q^2}{12.1d^5} = \frac{f\ell_1 Q^2}{12.1d_1^5} + \frac{f\ell_2 Q^2}{12.1d_2^5} + \frac{f\ell_3 Q^2}{12.1d_3^5}$$

$$\frac{\ell}{d^5} = \frac{\ell_1}{d_1^5} + \frac{\ell_2}{d_2^5} + \frac{\ell_3}{d_3^5}$$

$$\Rightarrow \quad \frac{1700}{d^5} = \frac{800}{(0.5)^5} + \frac{500}{(0.4)^5} + \frac{400}{(0.3)^5}$$

$$= 239037.1785$$

$$\Rightarrow \quad d^5 = 0.0071118$$

$$d = (0.0071118)^{0.2}\,m$$

973. The head of water at the inlet of a pipe 2000 m long and 500 mm diameter is 60 m. A nozzle of diameter 100 mm at its outlet is fitted to the pipe. What is the velocity of water at the outlet of the nozzle if f = 0.01 for the pipe ? (Take g = 9.81 m/s^2)

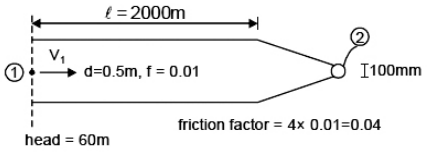
(a) 30.61 m/s (b) 34.81 m/s

(c) 36.52 m/s (d) 38.36 m/s

ESE 18.07.2021

Ans. (a) : Given, head loss, $h_f = 60$ m

$\ell = 2000$ m, $f = 0.01$



Head loss in 200 m length of pipe $= \dfrac{f\ell Q^2}{12.1d^5}$

Applying energy equation between (1) & (2)

$$60 = \overset{0}{\frac{P_2}{\gamma}} + \overset{0}{Z_2} + \frac{V_2^2}{2g} + h_f$$

[Neglecting loss in nozzle]

$$60 = \frac{v_2^2}{2g} + \frac{f\ell v_1^2}{2gd}$$

$$v_1 \times \frac{\pi}{4}(0.5)^2 = v_2 \times \frac{\pi}{4}(0.1)^2 \quad \text{(Continuity eqn.)}$$

$$v_1 = \frac{v_2}{25}$$

$$\Rightarrow \quad 60 = \frac{v_2^2}{2\times 9.81}\left[1 + \frac{f\ell}{d\times 625}\right]$$

$$= \frac{v_2^2}{2\times 9.81}\left[1 + \frac{0.04\times 2000}{0.5\times 625}\right]$$

$$\Rightarrow \quad \boxed{v_2 = 30.61\ m/s}$$

974. Water is flowing with a velocity of 1.5 m/s in a pipe of length 2500 m and of diameter 500 mm. At the end of the pipe, a valve is provided. What is the rise in pressure if the valve is closed in 25 seconds ? (Take the value of C as 1460 m/s)

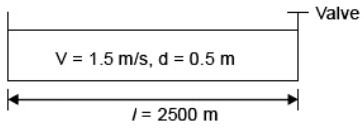
(a) 12 N/cm^2 (b) 15 N/cm^2

(c) 16 N/cm^2 (d) 18 N/cm^2

ESE 18.07.2021

Ans. (b) : Given, d = 500 mm = 0.5 m

$\ell = 2500$ m, $v = 1.5$ m/s



Time of closure of valve = $t_0 = 25$ sec

Critical time $(t_c) = \dfrac{2\ell}{C} = \dfrac{2\times 2500}{1460}$ sec $= 3.42$ sec

$\Rightarrow t_o > t_c$

$\Rightarrow$ It is a slow closure case

$$\Rightarrow \Delta P = \frac{\rho v L}{t_0} = \frac{1000\times 1.5\times 2500}{25}\frac{N}{m^2}$$

$$\Rightarrow \Delta P = 15 \times 10^4\ \frac{N}{m^2} = 15\frac{N}{cm^2}$$

975. In a 100 mm diameter pipe, water is flowing with a mean velocity of 4 m/s and a gauge pressure of 400 kN/m^2. The pipe is 10 metres above the ground level. The total head of the water neglecting friction is _____ (assume g = 10 m/s^2)

(a) 38.8 m (b) 48.8 m

(c) 30.8 (d) 50.8

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

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Ans. (d) : Diameter, d = 100 mm
Velocity, v = 4 m/s
Pressure, P = 400 kN/m^2
z = 10 m

$$\text{Total head} = \frac{P}{\rho g} + \frac{v^2}{2g} + z$$

$$= \frac{400\times10^3}{10^3\times10} + \frac{(4)^2}{2\times10} + 10$$

$$= \frac{400}{10} + \frac{8}{10} + 10$$

$$= 40 + 0.80 + 10$$

$$= 50.80 \text{ m}$$

976. If 'v' is the velocity of fluid entering the pipe entrance is equal to _____.

(a) v^2/g (b) $v^2/4g$
(c) $v^2/2g$ (d) $v^2/3g$

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (b) :

$$\text{The loss of head at entrance in a pipe} = \frac{0.5v^2}{2g} = \frac{v^2}{4g}$$

977. Water flows through a pipe of diameter 300 mm and length 50 m. The hydraulic mean depth (hydraulic radius) will be

(a) 0.3 m (b) 50 m
(c) 0.75 m (d) 1.2 m

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Hydraulic mean depth $= \frac{d}{4} = \frac{0.3}{4} = 0.75$

978. Which phenomenon will occur when the valve at the discharge end of a pipe connected to a reservoir is suddenly closed?

(a) Cavitation (b) Erosion
(c) Hammering (d) Surging

JPSC AE, 10.04.2021, Paper-I

Ans. (c) : Water hammer is the result of a pressure surge, or high pressure shockwave that propagates through a piping system when a fluid motion is forced to change direction or stop abruptly.

979. A smooth pipe of diameter 100 mm and length 100 m carries crude oil (Kinematic viscosity = 0.4 stokes and density = 900 kg/m^3) at velocity of 2 m/s. What will be the power (in watts) required to maintain the flow? [Friction factor = 0.03, g=10 m/s^2]

(a) 282.7 (b) 565.4
(c) 848.2 (d) 1130.9

APPSC Poly Lect. 13.03.2020

Ans. (c) : We know,
Power (P) required to maintain the flow is –
$P = \rho g Q h_f$
Where,
h_f = head loss due to friction
Q = Discharge = AV
f = friction factor = 0.03
D = 100 mm = 0.1 m
L = 100 m, V = 2 m/s

$$Q = AV = \frac{\pi}{4}\times(0.1)^2\times2$$

$$Q = 1.57\times10^{-2} \text{ m}^3/\text{s}$$

$\rho = 900$ kg/m^3, g = 10 m/s^2

$$h_f = \frac{fLV^2}{2gD}$$

$$= \frac{0.03\times100\times(2)^2}{2\times10\times0.1} = \frac{3\times4}{2} = 6\text{m}$$

$\therefore$ Power (P) $= \rho g Q h_f = 900\times10\times1.57\times10^{-2}\times6$

$\boxed{P = 848.2 \text{ Watts}}$

980. The relation between the skin friction coefficient C_f and the Darcy friction factor f for flow through a pipe are related as

(a) $C_f = 4_f$ (b) $4C_f = f$
(c) $BC_f = f$ (d) $C_f = 8f$

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) : Head loss $h_f = \frac{4flv^2}{2gD}$

981. When the velocity profile in a pipe is fully developed.

(a) the pressure gradient becomes invariant in the flow direction
(b) the pressure becomes invariant in the flow direction
(c) $\frac{\partial u}{\partial x}$ becomes linear with flow direction
(d) none of the above

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : For fully developed velocity profile in pipe flow $\frac{\partial p}{\partial x}$ pressure gradient remains constant (or invariant)

982. To maintain 0.08 m^3/s flow of petrol with a specific gravity of 0.7, through a steel pipe of 0.3 m diameter and 800 m length, with coefficient of friction of 0.0025 in the Darcy relation, the power required will be nearly

(a) 0.6 kW (b) 1.0 kW
(c) 2.6 kW (d) 2.6 kW

ESE 2020

Ans. (B) :
Given:
Coefficient of friction, f = 0.0025
Discharge, Q = 0.08 m^3/s
Length, ℓ = 800 m
Diameter, d = 0.3 m
Density, $\rho = 0.7\times1000 = 700$ kg/m^3

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Head loss to friction in pipe, $h_f = \frac{(4f)\ell V^2}{2dg}$

We know that friction factor is four times to that of coefficient of friction.

$\because V = \frac{4Q}{\pi d^2}\left(\text{for pipe, } A = \frac{\pi d^2}{4}\right)$

So, power required, $P = \rho g Q h_f$

$= \frac{\rho g Q \times (4f)\ell \times 16Q^2}{2 \times g \times \pi^2 \times d^5}$

$= \frac{\rho (4f)\ell \times 8 \times Q^3}{\pi^2 \times d^5}$

$P = \frac{700 \times (4 \times 0.0025) \times 800 \times 8 \times (0.08)^3}{\pi^2 \times (0.3)^5 \times 1000}$

$P = 0.956$ kW

≈ 1 kW

983. Two pipes of lengths 2500 m each and diameters 80 cm and 60 cm respectively, are connected in parallel. The coefficient of friction for each pipe is 0.006 and the total flow is 250 litres/s. The rates of flow in the pipes are nearly

(a) 0.17 m^3/s and 0.1 m^3/s
(b) 0.23 m^3/s and 0.1 m^3/s
(c) 0.17 m^3/s and 0.4 m^3/s
(d) 0.23 m^3/s and 0.4 m^3/s

ESE 2020

Ans. (a) : Given,
$f_1 = f_2 = 0.006$
$Q = 250$ liter/s $= 0.25$ m^3/s
Pipes are in parallel so head loss is equal

$h_{f_1} = h_{f_2}$

$\frac{f_1 L_1 Q_1^2}{12 d_1^5} = \frac{f_2 L_2 Q_2^2}{12 d_2^5}$

$\Rightarrow \frac{Q_1}{Q_2} = \left(\frac{d_1}{d_2}\right)^{5/2} = \left(\frac{0.8}{0.6}\right)^{2.5} = (1.33)^{2.5}$ ...(i)

$Q_1 + Q_2 = 0.25$ ...(ii)

From equation (i) and (ii)

$Q_1 = 0.17 m^3/sec$

and $Q_2 = 0.08 m^3/sec$

$\approx 0.1 m^3/sec$

984. A fluid of mass density 1790 kg/m^3 and viscosity 2.1 Ns/m^2 flows at a velocity of 3 m/s in a 6 cm diameter pipe. The head loss over a length of 12 m pipe will be nearly

(a) 62.0 m
(b) 54.0 m
(c) 46.5 m
(d) 38.5

ESE 2020

Ans. (d) : Given,
$\rho = 1790$ kg/m^3
$\mu = 2.1$ Ns/m^2
$V = 3$ m/s
$d = 0.06$ m
Head loss = ?

Major head loss is frictional head loss $= \frac{fLV^2}{2gd}$

$h_f = \frac{32\mu VL}{\rho g d^2}$

$= \frac{32 \times 2.1 \times 3 \times 12}{1790 \times 9.81 \times (0.06)^2}$

$= \frac{32 \times 12 \times 21 \times 3 \times 1000}{1790 \times 9.81 \times 36}$

$= 38.26$

≈ 38

985. Oil at 20°C (density = 870 kg/m^3 and viscosity = 0.104 kg/m-s) flows (laminar flow) at 1.1 m^3/hr through a horizontal pipe with diameter d = 2 cm and length L = 12 m. Find the power required to cause the flow

(a) 29.7 W
(b) 35.8 W
(c) 15.7 W
(d) 49.8 W

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : Given, $T_o = 20°C$, $\rho = 870$ kg/m^3

$\mu = 0.104$ kg/m-s

$Q = 1.1$ m^3/m, $d = 2$ cm $= 2 \times 10^{-2}$ m

$Q = \frac{1.1}{3600} m^3/sec$ $\quad L = 12$ m

$P = ?$

Minimum power required to maintain the flow (P) = $\rho g Q h_f$

h_f = Friction head loss

$h_f = \frac{32\mu \bar{u}^2}{\rho g D^2}$ (for laminar)

$\boxed{Q = \frac{\pi}{4} d^2 \times \bar{u}}$

$P = \frac{32\mu Q^2 L}{\left(\frac{\pi d^2}{4}\right) d^2} = \frac{32 \times 0.104 \times 1.1^2 \times 12}{\frac{\pi}{4} \times (0.02)^4 \times (3600)^2} = 29.7 W$

986. A liquid of specific weight (ρg) = 9000 N/m^3 flows by gravity from a tank of 0.3 m height through a 1.2 mm diameter capillary tube at a rate of 0.004 m^3/hr. as shown in figure. Sections 1 and 2 are at atmospheric pressure. Neglecting entrance effects and friction in the large tank, compute the viscosity of the liquid. Assume kinetic energy correction factor $\alpha = 2.0$.

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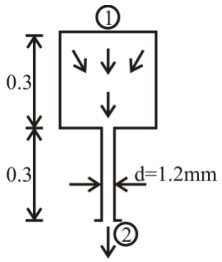


Q=0.004 m³/hr

(a) 8.9×10^{-4} kg/m-s (b) 10×10^{-4} kg/m-s

(c) 6.9×10^{-4} kg/m-s (d) 3×10^{-3} kg/m-s

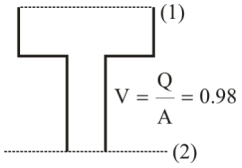
ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (c) : $V = \frac{Q}{A} = \frac{0.004}{3600 \times \frac{\pi}{4} \times 1.2^2 \times 10^{-6}} = 0.98\,m/sec$

Applying energy equation between (1) and (2)

$$\frac{P_{atm}}{\rho g} + 0 + z_1 = \frac{P_{atm}}{\rho g} + \frac{\alpha V_2^2}{2g} + z_2 + h_f$$

Assuming laminar flow in capillary tube,



$$\frac{32\mu VL}{\rho g d^2} = (z_1 - z_2) - \frac{\alpha V_2^2}{2g}$$

$$\frac{32 \times \mu \times 0.3}{9000 \times (0.0012)^3} = 0.6 - \frac{2 \times 0.98^2}{2 \times 10}$$

$\mu = 6.9 \times 10^{-4}$ kg/m-s

987. Gasoline at 20°C (density = 680 kg/m³) is pumped through a smooth 12 cm diameter pipe, 10 km long, at a flow rate of 75 m³/hr. The inlet is fed by a pump at an absolute pressure of 24 atm. The exit is at standard atmospheric pressure and is 150 m higher. Estimate frictional head loss h_f.

(a) 199 m (b) 299 m

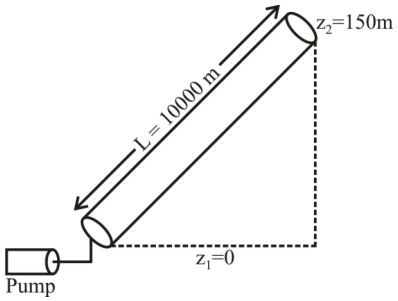
(c) 99 m (d) 399 m

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : $P_1 = 24$ atm $= 24 \times 1.01325 \times 10^5$

$\rho = 680$ kg/m³

$P_2 = 1$ atm $= 1.01325 \times 10^5$



Applying Bernoulli's equation,

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_f$$

Smooth pipe with pump,

$V_1 = V_2$ ($A_1V_1 = A_2V_2$)

$$h_f = \frac{P_1 - P_2}{\rho g} + (z_1 - z_2)$$

$$= \frac{(24-1) \times 1.01325 \times 10^5}{680 \times 9.81} - 150$$

$= 199.35 = 199$ m

988. A pipe 500 mm in diameter is 981 m long. It delivers water at a velocity of 2 m/s. The loss of head when f = 0.008 is

(a) 6.4 m (b) 12.8 m

(c) 20.2 m (d) 35.6 m

CIL MT 27.02.2020

Ans. (b) : Head loss due to friction in pipe

$$h_f = \frac{4f\ell v^2}{2gd}$$

$$= \frac{4 \times 0.008 \times 981 \times (2)^2}{2 \times 9.81 \times 0.5}$$

$h_f = 12.8$ m

989. In a free cylindrical vortex flow of air (density = 1.2 kg/m³), point A is located at a radius of 350 mm from the axis of rotation and at a height of 200 mm from the vessel bottom. Point B is however located at a radius of 500 mm and height 300 mm. If the velocity at point A is 20 m/s then the pressure difference between the points A and B is:

(a) 121.22 Pa (b) 10.29 Pa

(c) 12.35 Pa (d) 25.62 Pa

HPPSC ADF 05.03.2019

Ans. (a) : Given,

Point-A	**Point-B**
$r_A = 350$ mm	$r_B = 500$ mm
$z_A = 200$ mm	$z_B = 300$ mm
$v_A = 20$ m/sec	

Now, $v \times r =$ Constant

$v_A r_A = v_B r_B$

$20 \times 0.350 = v_B \times 0.5$

$v_B = 14$ m/sec

Now, $$\frac{P_A}{\rho g} + \frac{v_A^2}{2g} + z_A = \frac{P_B}{\rho g} + \frac{v_B^2}{2g} + z_B$$

$$\frac{P_A - P_B}{\rho g} = \left(\frac{v_B^2}{2g} - \frac{v_A^2}{2g}\right) + z_B - z_A$$

$$(P_A - P_B) = \frac{\rho}{2}(v_B^2 - v_A^2) + (z_B - z_A)\rho g$$

$$= \frac{1.2}{2}((14)^2 - (20)^2) + (0.300 - 0.200) \times 1.2 \times 9.81$$

$= -122.4 + 1.1772$

$= -121.22$ Pa

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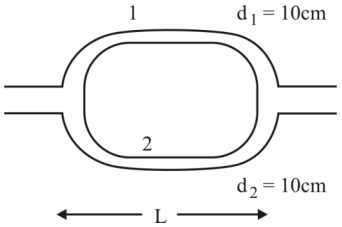
990. Two pipe lines of equal length and diameters of 10cm and 40 cm are connected in parallel between two reservoirs. If friction factor if is same for both the pipes, the ratio of the discharges in the larger to the smaller pipe is :

(a) 4 (b) 16
(c) 32 (d) 64

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) : $h_f = \frac{flQ^2}{12.1d^5}$

$$Q^2 = \frac{h_f \times 12.1d^5}{f_L}$$



$$Q^2 = \sqrt{\frac{h_f \times 12.1 \times d^5}{f_L}}$$

$$Q \propto d^{5/2}$$

$$\frac{Q_2}{Q_1} = \frac{40^{5/2}}{10^{5/2}} = 32$$

991. Which of the following drawing shows two lines representing the pipe diameter?

(a) Single-line (b) Double-line
(c) Standard piping (d) Centerline piping

MECON MT 2019

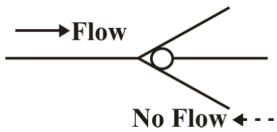
Ans. (b) : Double-line

992. Check valve allows the fluid flow

(a) Only in one direction
(b) In both the directions
(c) To change the direction
(d) To stop

APPSC-AE-2019

Ans. (a) : A check valve used in hydraulic circuits. It allows hydraulic liquid flow in one direction and prevents reversible (i.e. back flow) when the liquid in the line reverse direction. It is non-return (or) one-way valve pressurized liquid passing through a line opens the valve while, reversal of flow will close the valve represented as follows



993. The power transmission through nozzle is maximum if the ratio of total head supplied and head loss due to friction is:

(a) 1 (b) 1.5
(c) 2 (d) 3

OPSC AEE 2019 PAPER - II

Ans. : (d) : Total head = H

Head loss due to friction $h_L = \frac{H}{3}$ [for maximum power]

Then Ratio will be,

$$= \frac{H}{\left(\frac{H}{3}\right)} = 3$$

994. Major losses in the pipe are due to the

(a) Entrance effects (b) Frictional effects
(c) Valve and fittings (d) Pipe bends

APPSC-AE-2019

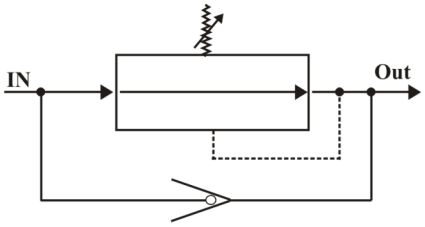
Ans. (b) : Loss due to friction is called major loss because its magnitude is generally high as compared to other losses.

995. Pressure-reducing valve is actuated by the fluid pressure at

(a) Upstream (b) Downstream
(c) Within the valve (d) Drain line

APPSC-AE-2019

Ans. (b) : Pressure reducing valve is actuated by the fluid pressure at downstream. It is normally an open type 2-way valve that allows the liquid pressurised system to flow through it until a required/desired set pressure is reached downstream.



It controls pressure pipe line at its outlet i.e. downstream pressure only. It is a pressure regulating at downstream of line.

996. In a laminar flow through pipe, the point of maximum instability exists at a distance of y from the wall which is

(a) $\frac{3}{2}$ of pipe radius R (b) $\frac{2}{3}$ of pipe radius R
(c) $\frac{1}{2}$ of pipe radius R (d) $\frac{1}{3}$ of pipe radius R

ESE 2019

Ans. (b) : A criterion for stability of laminar flow of any point,

$$\chi = \frac{y^2 \times \rho \times \frac{du}{dy}}{\mu}$$

For instability, $\frac{\partial \chi}{\partial y} = 0$

The velocity distribution in the pipe is,

$$u = v_{max}\left(1 - \frac{r^2}{R^2}\right)$$

$$\frac{du}{dr} = \frac{-du}{dy}$$

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$$\frac{du}{dy} = \frac{2v_{max}r}{R^2} = \frac{2v_{max}(R-y)}{R^2}$$

$$\chi = \frac{y^2 \times \rho \times \frac{du}{dy}}{\mu}$$

$$\Rightarrow \quad \chi = \frac{y^2 \times \rho \times \left(\frac{2\times(R-y)v_{max}}{R^2}\right)}{\mu} \qquad .....(i)$$

Differentiating equation (i)

$$\frac{d\chi}{dy} = \frac{\rho \times 2 \times v_{max}}{\mu \times R^2} \times (2yR - 3y^2)$$

$$0 = \frac{\rho \times 2 \times v_{max}}{\mu \times R^2} \times (2yR - 3y^2)$$

Here, ρ, v_{max}, μ cannot be zero,

So, equating $(2yR - 3y^2) = 0$

$$\boxed{y = \frac{3R}{3}}$$

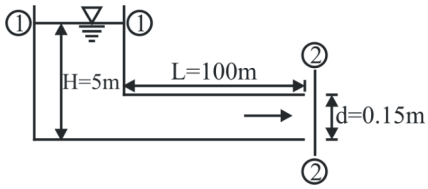
997. Water is discharged from a tank maintained at a constant head of 5 m above the exit of a straight pipe 100m long and 15 cm in diameter. If the friction coefficient for the pipe is 0.01 the rate of flow will be nearly

(a) 0.04 m^3/s (b) 0.05 m^3/s

(c) 0.06 m^3/s (d) 0.07 m^3/s

ESE 2019

Ans. (d) : Applying energy equation between points 1 and 2.



$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2 + h_f \qquad .....(1)$$

Here $P_1 = P_2 = P_{atm} = 0$

$V_1 = 0$, $Z_1 = 5m$ and $Z_2 = 0$

From equation (1)

$$0 + 0 + 5 = 0 + \frac{V_2^2}{2g} + 0 + h_f$$

$$5 = \frac{V_2^2}{2g} + \frac{fLV_2^2}{2gd}$$

$$5 = \frac{V_2^2}{2g}\left(1 + \frac{fL}{d}\right)$$

$V_2 = 3.576$ m/s

Discharge (Q) = AV_2

$$= \frac{\pi}{4}d^2 \cdot V_2$$

$$= \frac{\pi}{4}(0.15)^2 \times 3.576 = 0.063 \text{ m}^3/\text{s}$$

998. Water is flowing through a pipe of diameter 15 cm. At a section, the gauge pressure is 40 kPa and mean velocity is found to be 2 m/s. If the centerline of the pipe is 10 m above the datum, the total head in the flow will be given by [Kinetic energy correction factor = 1.2, g = 10 m/s^2]

(a) 12.80 m (b) 14.24 m

(c) 11.80 m (d) 13.20 m

SJVN ET 2019

Ans. (b) :

Total head = Energy correction factor $\times \frac{v^2}{2g} + z + \frac{p}{\rho g}$

$$= 1.2 \times \frac{2^2}{2\times 10} + 10 + \frac{40\times 10^3}{1000\times 10}$$

$$= 14.24\,m$$

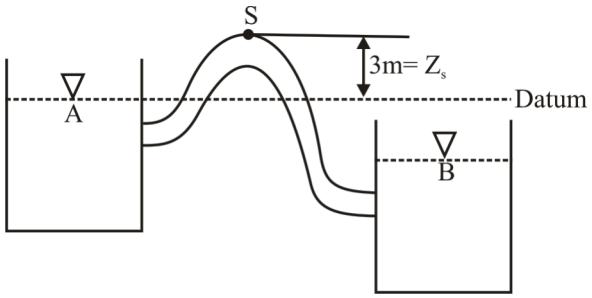
999. In a siphon, the summit is 3 m above the water level in the reservoir from which the flow is being discharged. If the head loss from the inlet of the siphon to the summit is 1.5 m and the velocity head at the summit is 0.5 m, the absolute pressure at the summit is :

(a) 5.3 m of water (b) 9.3 m of water

(c) 6.3 m of water (d) 5 m of water

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) :



1 atm = 760 mm of Hg = 10.3 m of water

$\frac{v^2}{2g} = 0.5m$ $\quad Z_s = Z_2 = 3m$ $\quad h_L = 1.5m$

Apply Bernoulli's equation between (A) and (S) points,

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + Z_2 + h_L$$

Let datum is at A $\Rightarrow Z_A = 0$

$$0 + 0 + 0 = \frac{P_s}{\rho g} + 0.5 + 3 + 1.5$$

$$\frac{P_s}{\rho g} = -5m \text{ of water}$$

(–ve sign means that pressure at S is below atmospheric pressure hence it is the gauge pressure head.)

$H_{Abs} = H_{Atm} + H_{Guage}$

$H_{Abs} = 10.3 + (-5) = 5.3$ m of water.

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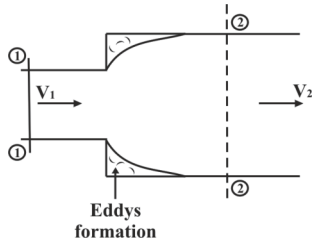
1000. A pipe of 0.1 m^2 suddenly changes to 0.5 m^2 area. The quantity of water flowing in the pipe is 0.5 ms^{-1}. Head loss due to sudden enlargement is nearly

(a) 0.025 m (b) 0.05 m
(c) 0.8 m (d) 1.2 m

ISRO Scientist/Engineer 22.04.2018

Ans. (c) : We know that head loss due to sudden enlargement



$$h = \frac{(V_1 - V_2)^2}{2g} \qquad (Q_1 = Q_2 = Q)$$

$$V_2 = \frac{Q}{A_2} = \frac{0.5}{0.5} = 1\ m/s$$

$$V_1 = \frac{Q}{A_1} = \frac{0.5}{0.1} = 5\ m/s$$

$$h = \frac{(5-1)^2}{2g} = \frac{16}{2\times 10}$$

$h = 0.8$ m

1001. The velocity along the radius of a pipe of 0.1 m radius varies as $u = 10 \times [1 - (r/0.1)^2]$ m/s. The viscosity of the fluid is 0.02 Ns/m^2. The maximum value shear stress in N/m^2 is

(a) –4 (b) –4.5
(c) –6 (d) –8

RPSC Vice Principal ITI 2018

Ans. (a) : Given,

$r = 0.1$ m

$\mu = 0.02\ Ns/m^2$

$$u = 10 \times \left[1 - \left(\frac{r}{0.1}\right)^2\right] m/s$$

$$= 10 - \frac{10 \times r^2}{0.1 \times 0.1}$$

$$= 10 - 1000r^2$$

$$\frac{du}{dr} = 0 - 2000r$$

$$\left(\frac{du}{dr}\right)_{r=0.1} = -200$$

By using Newton's Law of viscosity,

$$\tau = \mu \frac{du}{dy}$$

$$= 0.02 \times (-200)$$

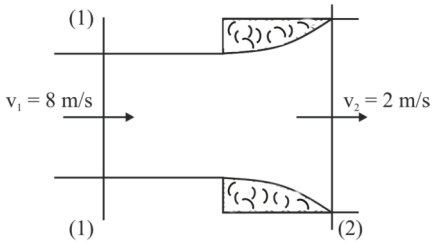
$\boxed{\tau = -4\ N/m^2}$

1002. If the pipe diameter is suddenly enlarged such that the velocity of flow is decreased from 8 m/s to 2 m/s. The head loss due to this enlargement will be : [$g = 10\ m/s^2$]

(a) 1.8 m of water (b) 0.18 m of water
(c) 18 m of water (d) 0.018 m of water
(e) 9 m of water

CGPSC AE 25.02.2018

Ans. (a) : Head loss due to sudden enlargement



$$h_{S.E.} = \frac{(V_1 - V_2)^2}{2g} = \frac{(8-2)^2}{2\times 10} = \frac{36}{2\times 10}$$

$\boxed{h_{S.E.} = 1.8\ m}$

1003. Pressure drop (ΔP) in a viscous flow through circular pipe of radius (R), length (L) with fluid of density (ρ), viscosity (μ) and flowing with average velocity (V) is given by:

(a) $\frac{32\mu VL}{\rho g R^2}$ (b) $\frac{8\mu VL}{R^2}$
(c) $\frac{32\mu VL}{R^2}$ (d) $\frac{8\mu VL}{\rho g R^2}$
(e) $\frac{16\mu VL}{\rho g R^2}$

CGPSC AE 25.02.2018

Ans. (b) : Pressure loss (ΔP) in circular pipe in a viscous flow

$$\Delta P = \frac{32\mu VL}{D^2}, \qquad D = 2R$$

$$\Delta P = \frac{8\mu VL}{R^2}$$

1004. The shear stress τ_0 for steady, fully developed flow inside a uniform horizontal pipe with coefficient of friction f, density ρ and velocity v, is given by

(a) $\frac{f\rho v^2}{2}$ (b) $\frac{f\rho^2 v}{2}$
(c) $\frac{\rho^2 v}{2f}$ (d) $\frac{\rho v^2}{2f}$

ESE 2018

Ans. (a) : The shear stress at wall is given by

$$\tau_0 = \frac{f}{2}\rho v^2$$

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1005. Statement I: In a pipeline, the nature of the fluid flow depends entirely on the velocity.
Statement II: Reynolds number of the flow depends on the velocity, the diameter of the pipe and the kinematic viscosity of the fluid.
ESE 2018

Ans. (d) : In pipeline the nature of flow depends on Reynolds number $\left(\frac{\rho VD}{\mu}\right)$ not entirely on velocity.
Hence statement-I is wrong.
The Reynolds number of flow is given by

$$R_e = \frac{\rho VD}{\mu} = \frac{VD}{v}$$

where R_e = Reynold Number
V = Velocity
D = Diameter of pipe
v = Kinematic viscosity
Statement I : wrong
Statement I : correct

1006. Darcy-Weisbach equation gives relation between_____
(a) Pressure and temperature
(b) Mass, volume and pressure
(c) Head loss and pressure loss
(d) Pressure loss only
GPSC Executive Engineer 23.12.2018

Ans. (c) : Darcy-Weisbach equation gives relation between.

$$h_f = \frac{flv^2}{2gd} = \frac{32\mu vl}{\rho gd^2}$$

- Pressure drop (ΔP) in laminar flow through pipe due to friction is given by,

$$\Delta P = \frac{32\mu vL}{D^2}$$

1007. When a liquid is flowing through a pipe, the velocity of the liquid is
(a) Maximum at the center and minimum near the walls
(b) Minimum at the center and maximum near the walls
(c) Zero at the center and maximum near the walls
(d) Maximum at the center and zero near the walls
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : Liquid is flowing through a pipe the velocity of liquid is maximum at the centre and minimum near the walls.

1008. The pressure of fluid due to hammer blow is
(a) Directly proportional to density of liquid
(b) Inversely proportional to density of liquid
(c) Directly proportional to (density) 1/2 of liquid
(d) Inversely proportional to (density) 1/2 of liquid
Nagaland PSC (CTSE) 2018, Paper-II

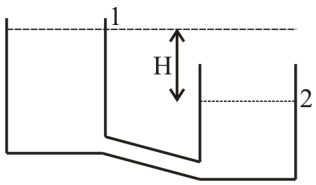
Ans. (c) : In water hammer pressure surge or wave caused when a fluid, usually a liquid but sometimes also a gas, in motion is forced to stop or change direction suddenly a momentum change.

$$\rho = (\rho k)^{1/2} \cdot V$$

So, pressure of fluid due to hammer blow is directly proportional to (density)$^{1/2}$ of fluid.

1009. The difference in water surface elevation in two reservoirs A and B is 10 m and the gauge pressure of air space in A is 50 kN/m^2. They are connected by a pipe 200 m long and 20 cm in diameter. If the friction factor for the pipe is 0.02 then the discharge is
(a) 0.118 m^3/s (b) 0.23 m^3/s
(c) 2.5 m^3/s (d) 1.5 m^3/s
Karnataka PSC Lect., 27.05.2017

Ans. (a) :



$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_L$$

$$\frac{V_1^2}{2g} = \frac{V_2^2}{2g} = 0$$

$$\left(\frac{P_2}{\rho g}\right)_{gauge} = 0$$

$$\frac{50\times10^3}{10^3\times9.81} + 10 = 0 + \frac{8fLQ^2}{\pi^2 D^5 g}$$

$$5.096 + 10 = \frac{8\times0.02\times200\times Q^2}{\pi^2\times(0.2)^5\times9.81}$$

$$15.096 = 1032.835\ Q^2$$

$$Q = 0.12\ m^3/s$$

1010. Water at 15ºC flow through a 200 m long galvanized steel pipe to diameter 250 mm and at 0.2 m^3/s. Average surface roughness for galvanized steel = 0.15 mm & loss of head due to friction is 10 m (Take density of water = 1000 kg/m^3)
(a) 1.962 kW (b) 19.62 kW
(c) 4.905 kW (d) 5.88 kW
GPSC EE Pre, 28.01.2017

Ans. (b) : Given, Length (L) = 200 m
Diameter (d) = 250 mm
h_f = 10 m, Q = 0.2 m^3/sec
Since, h_f is the head loss per unit weight of fluid, power required to maintain a flow at the rate of Q under a loss of head h_f.
$P = \rho g h_f Q$
$= 1000 \times 9.81 \times 10 \times 0.2$
P = 19620W
P = 19.620 kW

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1011. In a steady flow along a stream line at a location in the flow, the velocity head is 6m, the pressure head is 3m, and the potential head is 4m, the height of hydraulic gradient line at this location will be :

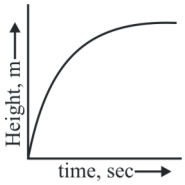
(a) 13 m (b) 09 m
(c) 10 m (d) 07 m

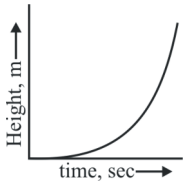
HPPSC Asstt. Prof. 20.11.2017

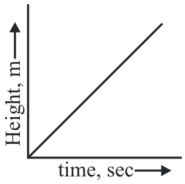
Ans. (d) : Hydraulic Gradient Line (HGL)- It is a graphical representation of piezometric head (ie. pressure head + potential head) at different points along the length of a pipe containing a fluid.

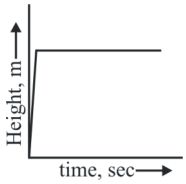
HGL = 4 + 3 = 7m

1012. Water flows into the top of an open drum at a constant mass flow rate. Water exits through a pipe near the base with a mass flow rate proportional to the height of the liquid. If the drum is initially empty, which of the following gives the plot of liquid height with time?

(a) 

(b) 

(c) 

(d) 

ISRO Scientist/Engineer 07.05.2017

Ans. (a) : Initially drum is empty with zero height so no discharge, but inlet has a constant mass flow rate. Hence height increases, along with which discharge also starts increasing and after attaining a certain height, discharge rate becomes equal to inlet rate, so after certain time, height becomes constant as depicted by option (a).

1013. Which of the following is not a minor energy loss?

(a) Loss due to sudden enlargement
(b) Loss due to friction
(c) Loss due to entrance of pipe
(d) Loss due to bend in pipe

TRB Polytechnic Lecturer 2017

Ans. (b) : Loss due to friction is a major energy loss in pipe flow

$$h_L = \frac{4f'LV^2}{2gD}$$

$$h_L = \frac{fLV^2}{2gD} \qquad [f = 4f']$$

f = friction factor
f ' = coefficient of friction (fanning factor)

1014. Engine oil flows in a 15 cm diameter horizontal tube with a velocity of 1.3 m/s, experiencing a pressure drop of 12 kPa. The pumping power requirement to overcome this pressure drop is,

(a) 190 W (b) 276 W
(c) 407 W (d) 655 W

TRB Polytechnic Lecturer 2017

Ans. (b) : Given as,
d = 15 cm = 0.15 m
V = 1.3 m/s
ΔP = 12 kPa
We know that power (P)
P = Force × Velocity

$$P = 12\times10^3 \times \frac{\pi}{4} \times (0.15)^2 \times 1.3$$

$$\boxed{P = 275.535\,W}$$

1015. The centre – line velocity in a pipe flow is 2m/s. What is the average flow velocity in the pipe if the Reynolds number of the flow is 800?

(a) 2 m/s (b) 1.5 m/s
(c) 1 m/s (d) 0.5 m/s

ESE 2017

Ans. (c) : Given, Centre line velocity = 2 m/s
i.e. V_{max} = 2 m/s
R_e = 800 < 2000 (Laminar flow)
The centre line velocity is maximum so average velocity in the pipe

$$V_{avg} = \frac{V_{max}}{2} = \frac{2}{2}$$

V_{avg} = 1 m/s

1016. In a steady laminar flow of a given discharge through a circular pipe of diameter D, the head loss is proportional to

(a) D^{-1} (b) D^{-2}
(c) D^{-3} (d) D^{-4}

ESE 2017

Ans. (d) : Head loss in steady laminar flow for a given discharge

$$h_f = \frac{128\mu QL}{\pi D^4}$$

$$h_f \propto D^{-4}$$

1017. An oil flows through a pipe at a velocity of 1.0 m/s. The pipe is 45m long and has 150mm diameter. What is the head loss due to friction, if ρ = 869kg / m^3 and μ = 0.0814kg / m s?

(a) 0.61 m (b) 0.51 m
(c) 0.41 m (d) 0.31 m

ESE 2017

Ans. (a) : Given, V = 1 m/s
L = 45 m
D = 150 mm = 0.15 m
ρ = 869 kg/m^3
μ = 0.0814 kg/ms

$$\text{Reynolds number } (R_e) = \frac{V\rho D}{\mu} = \frac{1\times869\times0.15}{0.0814} = 1601$$

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$R_e < 2000$ laminar flow

Head loss due to fluid friction in pipe

$$h_f = \frac{32\mu VL}{D^2 \cdot \rho \cdot g}$$

$$= \frac{32 \times 0.0814 \times 1 \times 45}{(0.15)^2 \times 869 \times 9.81} = 0.61 \text{ m}$$

1018. An air vessel is provided at the summit in a siphon to:

(a) avoid interruption in the flow
(b) increase discharge
(c) increase velocity
(d) maintain pressure difference

HPPSC Asstt. Prof. 18.09.2017

Ans. (a) : avoid interruption in the flow

1019. Pressure coefficient is the ratio of pressure force to:

(a) inertia force (b) viscous force
(c) gravity force (d) elasticity

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (a) : Pressure coefficient is the ratio of pressure force to inertial forces.

The pressure coefficient is important in most fluid flow application.

- Viscous force in a fluid are proportional to the rate at which the fluid velocity is changing in space, the proportionality constant is the viscosity.

1020. For pipes, transition flow occurs when Reynolds number is:

(a) less than 2000
(b) in between 1800 and 2200
(c) in between 2300 and 4000
(d) more than 4000

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : The dynamic viscosity of water is higher than that of air in keeping with our intuitive motion while the transition from laminar to turbulent flow occurs at a Reynolds number of approximately 2300 in a pipe, the precise value depends on weather any small disturbances are present.

For laminar flow $f = \frac{64}{Re}$ $Re < 2000$

For turbulent flow $f = \frac{0.316}{Re^{1/4}}$ $4000 < Re < 10^6$

Transition flow Reynold's number in between 2300 to 4000.

1021. Water is flowing with a flow rate of $0.003 m^3/s$. What is the average velocity at the outlet of pipe, whose area of cross section is 5 cm^2?

(a) 50 m/s (b) 45 m/s
(c) 30 m/s (d) 15 m/s
(e) 6 m/s

CGPSC ARTO & TSI 07.01.2017

Ans. (e) :

Given,

$Q = 0.003 \text{ m}^3/\text{sec}$

$V = ?$

$A = 5 \text{ cm}^2 = 5 \times 10^{-4} \text{ m}^2$

$Q = AV$

$$V = \frac{Q}{A}$$

$$V = \frac{0.003}{5 \times 10^{-4}} = 6 \text{m/s}$$

1022. Fluid is flowing in a curved path without any external impressed contact force. This flow is known as

(a) Free vortex flow (b) Forced vortex flow
(c) Radial flow (d) Spiral flow

UPPSC AE 12.04.2016 Paper-II

Ans. : (a) Fluid is flowing in a curved path without any external impressed contact force. This flow is known as free vortex flow.

Forced vortex flow : In this type of flow, the vessel containing the liquid is forced to rotate about the fixed vertical axis with the help of some external torque.

Free vortex flow : In this type of flow, the liquid particles describe circular paths about a fixed vertical axis, without any external torque acting on the particles. The flow of water through the hole in the bottom of a wash basin is an example of free vortex flow.

1023. A 1000 m long pipe discharges 1 m^3/sec of water with the head of 99 m at the inlet. The maximum power transmitted by pipe is (specific weight of water = 10 kN/m^3)

(a) 990 kW (b) 660 kW
(c) 445 kW (d) None

RPSC ADE 2016

Ans. (b) : Given specific weight

$w = 10 \text{ kN/m}^3$

$H = 99 \text{ mm}$, $Q = 1 \text{ m}^3/\text{s}$

For maximum power transmitted head less $h_f = H/3$

$$= \frac{99}{3} \text{m}$$

So, Working head at outlet $= 99 - \frac{99}{3} = 66 \text{m}$

$P = wQ(H - h_f)$

$= 10 \times 1 \times 66 = 660 \text{ kW}$

1024. For a steady compressible flow, the hydraulic grade line is represented by sum of the following heads :

(a) Pressure head and elevation head
(b) Velocity head and pressure head
(c) Velocity head and elevation head
(d) Velocity head, pressure head and elevation head

RPSC AE (GWD) 06.05.2016

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Ans. (a) : Hydraulic grade line is the line formed by joining the piezometric head $\left(\frac{p}{w}+z\right)$ at various point in a fluid.

$HGL = \frac{p}{\rho g}+z$ = pressure head + elevation head

1025. Simple or Plain Couette flow is characterized by

(a) Positive pressure gradient and linear velocity distribution.

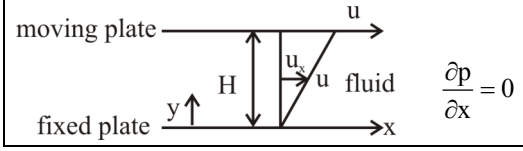
(b) Negative pressure gradient and parabolic velocity distribution.

(c) Negative pressure gradient and linear velocity distribution.

(d) Zero pressure gradient and linear velocity distribution.

RPSC AE (GWD) 06.05.2016

Ans. (d) : Couette flow i.e. the flow between the two plate. One plate is moving and other plate is fixed.



$\frac{\partial p}{\partial x}=0$

1026. A fluid is flowing at velocity of 2 m/s in a 1 km long pipe of 0.2 m diameter. The Darcy's coefficient of friction is 0.0025. What will be the head lost in friction in the pipe? (Take g = 10 m/s^2)

(a) 2.5m (b) 5 m

(c) 7 m (d) 9 m

RPSC AE (GWD) 06.05.2016

Ans. (*) : Given-

V = 2 m/sec.

L = 1 km = 1000m

D = 0.2 m

Friction coefficient f ' = 0.0025

g = 10 m/sec.

f = 4f'

According to Darcy's Weisbach equation-

Head loss $h_f = \frac{4f'LV^2}{2gD} = \frac{4\times0.0025\times1000\times(2)^2}{2\times10\times0.2}$

$h_f = \frac{4\times0.0025\times1000\times4}{2\times10\times0.2}$

h_f = 10 m

Note- Official answer given by commission is (a).

1027. A large Reynolds number is the indication of

(a) Highly turbulent flow

(b) Steady flow

(c) Laminar flow

(d) Smooth and streamline flow

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : For pipe flow

$Re \le 2300 \Rightarrow$ Laminar flow

$2300 \le Re \le 4000 \Rightarrow$ Transition flow

$Re \ge 4000 \Rightarrow$ Turbulent flow

So, the large Reynolds number is the indication of highly turbulent flow.

1028. Water flows through a horizontal contraction from a pipe of diameter d to another of diameter d/2. The flow velocity at the inlet to the contraction is 1 m/s and pressure is 150 KN/m^2. If the length of the contraction is 2 m, then pressure at the exit of the contraction will be nearly:

(a) 142.5 KN/m^2 (b) 150 KN/m^2

(c) 157.5 KN/m^2 (d) 162.5 KN/m^2

(e) 175 KN/m^2

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : Given,

v_1 = 1 m/s

From continuity equation,

$A_1V_1 = A_2V_2$

$\frac{\pi}{4}d^2\times1=\frac{\pi}{4}\left(\frac{d}{2}\right)^2\times v_2$

v_2 = 4 m/s

Now Bernoulli,

$\frac{P_1}{w}+\frac{v_1^2}{2g}+z_1=\frac{P_2}{w}+\frac{V_2^2}{2g}+z_2$

$\frac{150\times10^3}{9810}+\frac{1^2}{2\times9.81}+0=\frac{P_2}{9810}+\frac{4^2}{2\times9.81}+z$

P_2 = 142.5 kN/m^2

1029. Which of the following may be employed for measuring flow in a closed conduit (i.e., continuous pipe)?

(a) Orifice (b) Mouthpiece

(c) Venturimeter (d) Internal mouthpiece

RPSC AE (GWD) 06.05.2016

Ans. (c) : Venturimeter used for measuring flow in a closed conduit (i.e. continuous pipe). It basically works on the principle of Bernoulli's theorem.

1030. A jet of water initially 12 cm diameter when directed vertically upwards, reaches a maximum height of 20 meter. Assuming the jet remains circular, the flow rate will be :

(a) 0.224 m^3/sec (b) 0.112 m^3/sec

(c) 0.056 m^3/sec (d) 0.448 m^3/sec

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : D = 12 cm, h = 20 m

$v = \sqrt{2gh}$ $A = \frac{\pi}{4}D^2$

$Q = V.A = \sqrt{2gh}\times\frac{\pi}{4}D^2$

$= \boxed{\sqrt{2\times9.81\times20}\times\frac{\pi}{4}\times(0.12)^2}$

$\boxed{Q = 0.224\ m^3/sec}$

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1031. In pipe flow, the momentum correction factor is:

(a) Kinetic energy based on actual velocity/momentum based on average velocity

(b) Kinetic energy based on actual velocity/kinetic energy based on average velocity

(c) Momentum based on average velocity/Kinetic energy based on actual velocity

(d) Momentum based on actual velocity/Momentum based on average velocity

HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : In pipe flow, the momentum correction factor (β) is the ratio of actual momentum crossing per second through section to the momentum crossing per second based on mean velocity.

- $\beta \geq 1$, always
- $\beta = \frac{4}{3} \Rightarrow$ for laminar flow
- $\beta = 1.02$ to $1.05 \Rightarrow$ for turbulent flow.

1032. Hydraulic grade line for flow in a pipe of constant diameter is

(a) Always above the centre line of the pipe

(b) Always above the energy great line

(c) Always sloping downward in the direction of the flow

(d) Coincides with the pipe centerline

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Hydraulic gradient line which gives the sum of $\left(\frac{p}{w}\right)$ and datum head (z) of a flowing fluid in a pipe w.r.t. some reference line, which is obtained by joining the top of all vertical ordinates, showing the pressure head $\left(\frac{p}{w}\right)$ of a flowing fluid in a pipe from the centre of the pipe. Hydraulic gradient for flow in pipe of constant diameter is always sloping downward in the direction of the flow.

1033. The minor loss due to sudden contraction is due to

(a) flow contraction

(b) Expansion of flow after sudden contraction

(c) Boundary friction

(d) Cavitations

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : Minor loss due to sudden contraction is due to expansion of flow after sudden contraction.

1034. In a pipe of 300 mm diameter and 200 m length water is flowing at 2 m/s. The coefficient of friction is assumed to be negligible. What will be head loss due to friction

(a) 100 m (b) 415.9 m

(c) 533.3 m (d) 605 m

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (*) : Given,

D = 300 mm = 0.3 m

ℓ = 200 m

v = 2 m/sec

f′ = Negligible

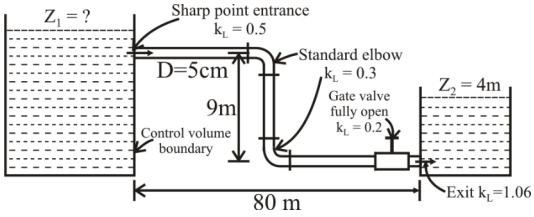
Suppose, f′ = 1

$$h_f = \frac{4f'\ell v^2}{2gD}$$

$$= \frac{4\times1\times200\times(2)^2}{2\times9.81\times0.3}$$

$$\boxed{h_f = 543.66\,\text{mm}}$$

1035. Water at 10°C flows from a large reservoir to a smaller one through a 5 cm diameter cast iron piping system, as shown in Fig. Determine the elevation z_1 for a flow rate of 6 L/s.



(a) 31.9 m (b) 29.9 m

(c) 35.9 m (d) 30.9 m

(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : As we know,

$$\frac{P_1}{\rho g} + \alpha_1\frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_L$$

$\therefore [V_1 = V_2 = 0]$

$[P_1 = P_2 = P_{atm}]$

So, $h_L = h_{Ltotal} = h_{Lmajor} + h_{Lminor}$

$$= \left(f\frac{L}{D} + \Sigma k_L\right)\frac{V^2}{2g}$$

$$V = \frac{Q}{A_c} = \frac{Q}{\pi\frac{D^2}{4}} = \frac{0.006}{\pi\left(\frac{0.05}{4}\right)^2} = 3.06\,\text{m/s}$$

$$R_e = \frac{\rho VD}{\mu} = \frac{999.7\times3.06\times0.05}{1.307\times10^{-3}}$$

$= 117{,}000$ $\quad \therefore R_e > 4000$

$\Sigma k_L = k_{Lentrance} + 2k_{Lelbow} + k_{Lvalve} + k_{Lexit}$

$= 0.5 + 2 \times 0.3 + 0.2 + 1.06 = 2.36$

$$h_L = \left(f\frac{L}{D} + \Sigma k_L\right)\frac{V^2}{2g}$$

$$= \left(0.0315\times\frac{89}{0.05} + 2.36\right)\frac{3.06}{2\times9.81} = 27.9$$

$Z_1 = Z_2 + h_L = 4 + 27.9 = 31.9$ m

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1036. In flow through a pipe, the transition from Laminar to turbulent flow does not depend on

(a) Length of the pipe
(b) Velocity of the flow
(c) Density of the fluid
(d) Diameter of the pipe

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : In flow through pipe the transition from laminar to turbulent flow depends upon.
(i) Velocity of the flow
(ii) Density of the fluid
(iii) Diameter of the pipe
(iv) Reynolds Number

$$Re = \frac{\rho v D}{\mu}$$

if $Re < 2000 \rightarrow$ Laminar flow
$Re = 2000$ to $4000 \rightarrow$ Transitional flow
$Re > 4000 \rightarrow$ Turbulent flow

1037. To avoid vaporization pipe lines are laid over the ridge so that these are above the hydraulic gradient line, not more than

(a) 6.4 m (b) 2.4 m
(c) 5.0 m (d) 10.0 m

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :

1038. The Darcy's friction factor in the pipe of uniform diameter in which a viscous flow is taking place is given in terms of Reynolds number (R_e)

(a) $\frac{64}{R_e}$ (b) $\frac{1}{R_e}$
(c) $\frac{4}{R_e}$ (d) $\frac{16}{R_e}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Loss of head due to friction in viscous flow

$$h_f = \frac{32\mu\bar{u}L}{\rho g D^2} = \frac{4fL\bar{u}^2}{2gD}$$

$$f = \frac{16\mu}{\bar{u}\rho D} = 16 \times \frac{\mu}{\rho v D}$$

$\boxed{f = \frac{16}{Re}}$ f = coefficient of friction

Friction factor (f ') = 4f

$$f' = \frac{64}{Re}$$

For turbulent flow –

$$f = \frac{0.079}{Re^{1/4}}$$

Friction factor = $f' = \frac{0.316}{Re^{1/4}}$

1039. In case of fluid flow through pipes, cavitation is caused by

(a) High pressure
(b) High velocity
(c) Low pressure below a limit
(d) Weak material of pipe

UPPSC AE 12.04.2016 Paper-II

Ans. : (c) In case of fluid flow through pipes, cavitation is caused by low pressure below a limit.
Cavitation : If the pressure at any point in this flowing liquid becomes equal to or less than vapour pressure, the vaporization of the liquid starts. The bubbles of these vapours are carried by the flowing liquid into the region of high pressure where they collapse, giving rise to high impact pressure. The pressure developed by the collapsing bubbles is so high that the material from the adjoining boundaries gets eroded and cavities formed of them. This phenomenon is known as cavitation.

1040. Energy grade line (EGL) represents the

(a) Elevation head
(b) Pressure head + Elevation head
(c) Pressure head+Velocity head+Elevation head
(d) Pressure head + Velocity head

RPSC LECTURER 16.01.2016

Ans. (c) : Energy grade line (EGL)
Pressure head + Velocity head + Elevation head

$$EGL = \frac{P}{\rho g} + \frac{V^2}{2g} + Z$$

Hydraulic gradient line HGL

HGL = Pressure head + elevation head $= \frac{P}{\rho g} + Z$

$$EGL - HGL = \frac{V^2}{2g}$$

1041. Hydraulic diameter of square duct is given by

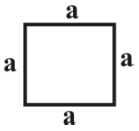
(a) Side of the square
(b) 2 × Side of the square
(c) 1.5 × Side of the square
(d) 0.5 × Side of the square

RPSC LECTURER 16.01.2016

Ans. (a) : We know that Hydraulic diameter

$$D_h = \frac{4A_c}{P}$$

$A_c = a^2$



$P = 2(a + a) = 4a$

$$D_h = \frac{4 \times a^2}{4a}$$

$D_h = a$ = side of the square

1042. The velocity of head of water flowing through the pipe of 0.3 m dia for the conditions shown in the figure below is

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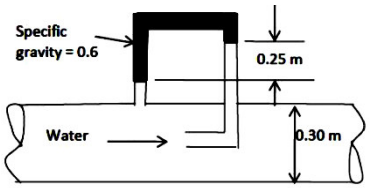

(a) 0.25 m (b) 0.15 m
(c) 0.44 m (d) 0.1 m

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : Velocity head $= x\left(1-\frac{s_m}{s_w}\right)$

$= 0.25\left(1-\frac{0.6}{1}\right)$

$= 0.1$ m

1043. The pressure drop in a laminar flow through circular pipe is dependent on

(a) velocity of flow
(b) viscosity of the fluid
(c) diameter of the pipe
(d) All the given answers

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : The pressure drop in a laminar flew through circular pipe is depend on
(i) Velocity of flow
(ii) Viscosity of the fluid
(iii) Diameter of the pipe

$$h_f = \lambda\frac{L}{D}\times\frac{V^2}{2g}$$

$\boxed{P_1 - P_2 = \frac{128\mu QL}{\pi D^4}}$ in a laminar flow.

1044. The deformation drag experienced by a sphere of d dia, moving at V velocity through a fluid of μ dynamic viscosity is given by

(a) $3\pi Vd/\mu$ (b) $3\pi\mu Vd$
(c) $3\pi\mu Vd^2$ (d) $3\pi\mu V^2d$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : We know,

$$R_e = \frac{\rho vd}{\mu}$$

for sphere, drag coefficient $\Rightarrow C_d = \frac{2D}{\rho v^2 A} = \frac{24}{R_e}$

$$\Rightarrow \frac{2D}{\rho v^2 A} = \frac{24\times\mu}{\rho vd}$$

$\boxed{D = 3\pi\mu vd}$

1045. Match List I and List II and select the correct answer using the codes given below the lists

List I	List II
P. Moment of momentum equation	1. Equation to find energy loss in pipeline having laminar flow.
Q. Bernoulli's equation	2. Equation of motion for 1-D steady flow
R. Euler's equation	3. Equation based on momentum conservation principle.
S. Hagen-poiseulle equation	4. 3-D equation of motion based on momentum conservation.

(a) P Q R S
2 3 4 1
(b) P Q R S
3 2 1 4
(c) P Q R S
2 3 1 4
(d) P Q R S
3 2 4 1

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : → Moment of momentum equation is equation based on momentum conservation principle.
→ Equation of motion for 1-D steady flow is Bernoulli's equation.
→ 3-D equation of motion based on momentum conservation is Euler's equation.
→ Equation to find energy loss in pipeline having laminar flow given by Hagen-poiseulle equation.

1046. A pipeline of diameter 10cm carries a discharge at a velocity of 1m/s. It branches into two pipes each of diameter 5cm. Consider the following statements

(1) velocity in each branch is 2m/s
(2) discharges in the two branched pipes are equal
(3) rate of flow before and after branching is same
(4) velocity in 5cm pipe is half of that in 10cm pipe

Select the correct answer using codes given below:

(a) 1 and 2 (b) 1 and 4
(c) 2,3 and 4 (d) 1,2 and 3

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) :

1047. The water pressure at the summit point of a siphon is

(a) minimum
(b) equal to atmospheric pressure
(c) maximum
(d) zero

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : The highest point of syphon is called as summit and a syphon can be defined as a long bend pipe, used to carry water from a reservoir at a higher level to another reservoir at lower level two reservoirs are separated by a hill.
Pressure at summit is below atmospheric as it lies above the free water surface in the reservoir at higher level.
So, the pressure at summit point of a siphon is minimum.

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1048. In practice the formation of air lock occurs at _____ water pressure at the summit of a siphon

(a) 2.5 m of water absolute
(b) 7.8 m of water absolute
(c) 10.3 m of water
(d) 13.6 m of water

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : A pipe which raises above its hydraulic gradient line has negative pressure and is known as siphon. Theoretically this negative pressure may be reduced to – 10.3 m of water but in actual practice this pressure is only –7.6 m of water or 10.3 – 7.6 = 2.7 m of water absolute.

If pressure at summit becomes less than 2.7 m of water absolute, the dissolved air and other gases would come out from water and collect at the summit and flow of water will be obstructed.

1049. Minor losses in a piping system are

(a) less than the friction losses $f\frac{L}{d^2}\frac{v^2}{g}$
(b) due to the viscous stresses
(c) assumed to vary linearly with the velocity
(d) found by using loss coefficients

VIZAG STEEL MT 2015

Ans. (d) : Following are some minor losses which occur in pipe flow.

Loss of energy due to sudden enlargement.

Loss of energy due to sudden contraction.

Loss of energy in bends and pipe fitting.

1050. The shear stress in a fluid flowing in a round pipe

(a) Is constant over the cross-section
(b) Is zero at the wall and increases linearly to the centre
(c) Is zero at centre and varies linearly with the radius
(d) Varies parabolically across the section

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Shear stress in a fluid flowing in a round pipe is zero at centre and varies linearly with the radius.

1051. The diameter of a pipe at two sections (section 1 and section 2) is 2 cm and 4 cm respectively. The velocity through the point at section 1 is 5 m/s. The velocity at section 2 will equal-

(a) 1.0 m/s (b) 1.25 m/s
(c) 2.5 m/s (d) 5.0 m/s

TSGENCO AE 14.11.2015

Ans. (b) : Given,

Two section –

Section -1	Section - 2
$d_1 = 2$ cm	$d_2 = 4$ cm
$v_1 = 5$ m/s	$v_2 = ?$

Applying continuity equation –

$A_1V_1 = A_2V_2$

$$\frac{\pi}{4}d_1^2V_1 = \frac{\pi}{4}d_2^2V_2$$

$$V_2 = \frac{d_1^2 \times V_1}{d_2^2} = \frac{2\times 2\times 5}{4\times 4} = 1.25 \text{ m/sec}$$

1052. The flow of fluid through a pipe is laminar, when

(a) the fluid is ideal
(b) the fluid is viscous
(c) the Reynold's number is less than 2000
(d) the Reynold's number is more than 3000

MPPSC AE 08.11.2015

Ans. (c) : The flow of fluid through a pipe is depend on the Reynold's number (R_e).

- If the Reynold's number (R_e) < 2000 : then the flow is said to be Laminar flow.
- If, $2000 < R_e < 4000$; Transition flow.
- If, $R_e > 4000$, Turbulent flow.

1053. The supply head of a pipe is 21 kPa. for maximum power transmission the loss of head due to friction is equal to (if w = 10 kN/m^3)

(a) 2 m (b) 1m
(c) 9.81 m (d) 0.7 m

TSPSC AEE 2015

Ans. : (d) Supply head of a pipe = 21 kPa.

$P = \rho gh$

$$h = \frac{P}{\rho g}$$

$$h = \frac{21\times 10^3}{1000\times 10}$$

$h = 2.1$ m

for maximum power transmission

$$\text{head loss} = \frac{\text{Total head}}{3}$$

$h_f = 2.1/3$

$h_f = 0.7$m

1054. Hydraulic Diameter D_h of circular tubes

(a) Equal to diameter of the tube
(b) Half of the diameter of the tube
(c) Twice the diameter of the tube
(d) Four times the diameter of the tube

TNPSC AE 2014

Ans. (a) : Hydraulic diameter

$$D_h = \frac{4A_C}{P}$$

For circular tube of diameter d

$$D_n = \frac{4\times\frac{\pi}{4}.d^2}{\pi d}$$

$$\boxed{D_h = d}$$

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1055. The slope of the total energy line is referred to as

(a) Potential Line (b) Gradient Line
(c) Energy Gradient (d) Stream Line
(e) Flow Line

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : The slope of the total energy line is referred to as energy gradient line where as the slope of the piezometric head is referred to as hydraulic gradient line

$$EGL = \frac{P}{\rho g} + \frac{V^2}{2g} + z$$

$$HGL = \frac{P}{\rho g} + z$$

$$EGL - HGL = \frac{V^2}{2g}$$

1056. Water flows in a 2×4 cm rectangular duct at 16 m/s. The duct undergoes a transition to a 6 cm diameter pipe. Calculate the velocity in the pipe

(a) 2.76 m/s (b) 3.14 m/s
(c) 3.95 m/s (d) 4.53 m/s
(e) 4.85 m/s

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : We know that continuity equation,

$$Q_1 = Q_2$$

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad [\rho_1 = \rho_2 \text{ For Liquid}]$$

So,

$$A_1 V_1 = A_2 V_2$$

$$2\times4\times10^{-4} \times 16 = \frac{\pi}{4}\times(6)^2 \times 10^{-4} \times V_2$$

$$\boxed{V_2 = 4.529 \text{m/s}}$$

1057. In Hagen – Poisuelle's flow of viscous fluid, one of the following pairs of forces are balanced

(a) Inertia and viscous force
(b) Pressure and viscous force
(c) Gravity and viscous force
(d) Inertia and gravity force

TNPSC AE 2014

Ans. (a) : In Hagen - Poisuelle's flow of viscous fluid inertia and viscous forces and balanced.

1058. Loss of head due to friction to maintain 0.05 m^3/s of discharge of petrol (sp. gravity 0.7) through a steel pipe 0.2 m diameter and 1000 m long, taking coefficient of friction 0.0025 will be

(a) 0.644 m (b) 6.44 m
(c) 64.4 m (d) 644 m

UKPSC AE-2013, Paper-II

Ans. (b) : Given as,

$Q = 0.05\ m^3/s$
$\rho_p = 700\ kg/m^3$
$D = 0.2$ m
$L = 1000$ m
$f = 0.0025$ m

$$h_L = \frac{4f \cdot LV^2}{2gD}$$

$$V = \frac{Q}{A} = \frac{0.05}{\frac{\pi}{4}\times(0.2)^2}$$

$$\boxed{V = 1.5915 \text{m/s}}$$

$$h_L = \frac{4\times0.0025\times1000\times(1.591)^2}{2\times9.81\times0.2}$$

$$\boxed{h_L = 6.45 \text{m}}$$

1059. When a fluid flows through a pipeline under viscous flow conditions, the ratio of velocity at the axis of the pipe to the mean velocity of flow is:-

(a) 0.5 (b) 1.0
(c) 1.67 (d) 2.0

UKPSC AE-2013, Paper-II

Ans. (d) : $u_{maximum} = 2v_{average}$

1060. Loss of head in the pipe due to various pipe fittings such as valves, coupling is expressed as

(a) $\frac{KV^2}{2g}$ (b) $\frac{V^2}{2g}$
(c) $0.5\frac{V^2}{2g}$ (d) $\frac{V^2}{2kg}$

V - velocity of flow
K = coefficient of pipe fitting

TNPSC AE 2013

Ans. (a) : Loss of head in the pipe due to various pipe fittings such as valves, coupling is expressed as

$$h_2 = \frac{KV^2}{2g}$$

1061. Loss of head (h_o) due to sudden contraction of a pipe is given by :
V_1 – Velocity of Vena contracta
V_2 – Velocity at exit

(a) $h_a = \frac{V_1^2 - V_2^2}{2g}$ (b) $h_a = \frac{0.5\,V_1^2}{2g}$
(c) $h_a = \frac{(V_1 - V_2)^2}{2g}$ (d) $h_a = \frac{V_1^2 - V_2^2}{4g}$

TNPSC AE (Industries) 09.06.2013

Ans. (c) : Loss of head (h_o) due to sudden contraction of a pipe is given by–

$$h_a = \frac{(V_1 - V_2)^2}{2g}$$

Where, V_1 – Velocity of Vena contracta
V_2 – Velocity at exit

1062. The purpose of surge tank in a pipe line is to

(a) smoothen the flow of water
(b) minimize friction losses in pipe
(c) prevent occurrence of hydraulic jump
(d) relieve pressure due to water hammer

ISRO Scientist/Engineer 12.05.2013

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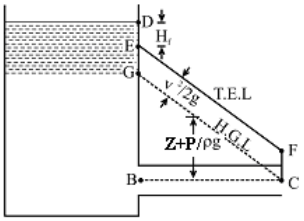
Ans. (d) : Surge tanks are installed on large pipelines to relieve excess pressure caused by water hammer surge tank in water storage device used as pressure neutralizer in feeder, dam, barrage pipe to resist sudden excess pressure rise and pressure drop conditions. Water hammer is a pressure surge or change direction suddenly. A water hammer occurs when a valve closes suddenly at an end of a pipeline system, and a pressure wave propagates in the pipe.

1063. The total energy line is always higher than the hydraulic gradient line, and the vertical distance between the two represents

(a) the datum head (b) the pressure head
(c) the velocity head (d) the piezometric head

APPSC AEE 2012

Ans. : (c) The total energy line is always higher than then hydraulic gradient line, and the vertical distance between the two represent the velocity head.



1064. Compare the cost of pumping the same fluid at the same flow rate through a 150 mm pipe and through 200 mm pipe both having the same roughness factor = 0.03 (negligible minor loss).

(a) 4.2 times (b) 2.1 times
(c) 8.4 times (d) 10 times

APPSC AE 04.12.2012

Ans. (a) : Frictional loss–

$$h = \frac{f\ell Q^2}{12d^5}$$

$d_1 = 150\ mm = 0.15\ m$

$d_2 = 0.20\ m$

The ratio of head loss in two pipe, $\frac{h_1}{h_2} = \frac{f\ell Q^2}{12d_1^5} \times \frac{d_2^5}{f\ell Q^2}$

$$\frac{h_1}{h_2} = \left(\frac{d_2}{d_1}\right)^5$$

$$= \left(\frac{0.20}{0.15}\right)^5 = 4.2$$

Hence the cost of pumping the same fluid at the same discharge through a 150 mm pipe is, therefore, 4.2 times the cost of pumping it through a 200 m pipe.

1065. To replace a compound pipe by a new pipe, the pipe will be equivalent when both the pipes have same

(a) length and flow
(b) diameter and flow
(c) loss of head and flow
(d) length and loss of head

TRB Asstt. Prof., 2012

Ans. (d) : To replace a compound pipe by a new pipe, the pipe will be equivalent when both the pipes have same length and loss of head.

1066. The hydraulic mean depth of the pipe 1m diameter flowing full is :

(a) 4m (b) 0.50 m
(c) 0.125 m (d) 0.25 m

APGENCO AE 2012

Ans. (d) : Given, Diameter (d) = 1m

Hydraulic mean depth (m) $= \frac{A}{P} = \frac{\frac{\pi}{4}d^2}{\pi d}$

$$= \frac{\pi}{4}d^2 \times \frac{1}{\pi d} = \frac{d}{4} = \frac{1}{4} = 0.25m$$

1067. Find the loss of head due to friction in a smooth pipe of rectangular cross section of 1m × 0.5 m and 10m long. Velocity of flow is 10 m/s and the friction coefficient is 0.01962.

(a) 4m (b) 6m
(c) 8m (d) 10m

ISRO Scientist/Engineer 2012

Ans. (b) : Rectangular cross-section = 1m × 0.5m

and Length l = 10m

Velocity v = 10 m/s

Friction coefficient f = 0.01962

Accordingto Darcy-Weisbach equation

$$h_f = \frac{f'Lv^2}{2gm}$$

f = friction coefficient

m = hydraullic mean depth

$$m = \frac{wetted\ area}{wetted\ perimeter} = \frac{A}{P}$$

$$m = \frac{1 \times 0.5}{2(1+0.5)} = 0.167m$$

$$\therefore \quad h_f = \frac{0.01962 \times 10 \times 10^2}{2 \times 9.81 \times 0.167}$$

$h_f = 6m$

1068. The extent of pressure produced due to water hammer depends on

(a) Velocity of flow of fluid in the pipe
(b) length of the pipe
(c) time taken to close the valve ; gradual or quick closure of the valve
(d) All the above

ISRO Scientist/Engineer 2012

Ans. (d) : The extent of pressure produced due to water hammer depends on velocity of flow of fluid in the pipe, length of the pipe and time taken to close the valve ; gradual or quick closure of the valve. So all options are correct.

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1069. Which of the following head loss is significant in a pipe flow ?

(a) Loss of head due to gradual contraction
(b) Loss of head due to friction
(c) Loss of head due to sudden enlargement
(d) Loss of head due to sudden contraction

UKPSC AE 2012 Paper–II

Ans. (b) : Loss of head due to friction

1070. A fully developed pipe flow implies that the

(a) flow should be laminar
(b) flow should be turbulent
(c) velocity profile should be uniform
(d) velocity profile should not change in the direction of flow.

UKPSC AE 2012 Paper–II

Ans. (d) : velocity profile should not change in the direction of flow.

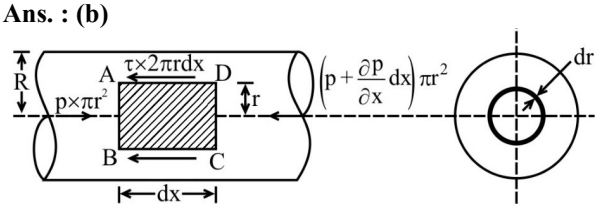
1071. The shear stress distribution in pipe flow is expressed by

(a) $\tau = \frac{1}{r}\left(\frac{dp}{dx}\right)$ (b) $\tau = -\left(\frac{dp}{dx}\right)\frac{r}{2}$

(c) $\tau = -2r\left(\frac{dp}{dx}\right)$ (d) $\tau = \frac{r^2}{L}\left(\frac{dp}{dx}\right)$

APPSC AEE 2012

Ans. : (b)



The force acting on the fluid elements are

(i) The pressure force $p \times \pi r^2$ on face AB

(ii) The pressure force $\left(p + \frac{\partial p}{\partial x}.dx\right)\pi r^2$ on force CD

(iii) The shear force. $\tau \times 2\pi r.dx$ on the surface of fluid elements.

$$p \times \pi r^2 - \left(p + \frac{\partial p}{\partial x}.dx\right)\pi r^2 - \tau \times 2\pi r.dx = 0$$

$$\tau = \frac{-\partial p}{\partial x}\left(\frac{r}{2}\right)$$

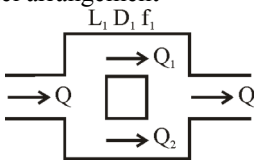
$$\tau = -\left(\frac{\partial p}{\partial x}\right)\frac{r}{2}$$

1072. Two pipes of diameters 0.1 m and 0.2 m are connected in parallel. If the head loss in smaller pipe is 16 m, the head loss in the other pipe is

(a) 4m
(b) 8m
(c) 16m
(d) 64m

TNPSC ACF 2012

Ans. (c) : Parallel arrangement



(1) Head loss = $h_L = h_{L_1} = h_{L_2}$ = 16 m

(2) Discharge $Q = Q_1 + Q_2$

(3) Equivalent pipe

$$\frac{D_{eq}^{5/2}}{\sqrt{L_{eq}}} = \frac{D_1^{5/2}}{\sqrt{L_1}} = \frac{D_2^{5/2}}{\sqrt{L_2}}$$

In given question arrangement are parallel so head loss is same.

1073. Compare the cost of pumping the same fluid at the same flow rate through a 150 mm pipe and ;through 200 mm pipe both having the same roughness factor = 0.03. (negligible minor loss)

(a) 4.2 times (b) 2.1 times
(c) 8.4 times (d) 10 times

APPSC AEE 2012

Ans. (a) : $h_f \propto \frac{1}{d^5}$

$$\frac{h_{f1}}{h_{f2}} = \left(\frac{d_2}{d_1}\right)^5 = \left(\frac{200}{150}\right)^5 = 4.16$$

$h_{f1} = 4.16\ h_{f2}$

$$\therefore \quad \frac{h_{f1}}{h_{f2}} \simeq 4.2 \text{ times}$$

1074. For the same discharge in a pipe with laminar flow, reduction of radius to half will increase the pressure gradient by a factor of

(a) 2 (b) 4
(c) 8 (d) 16

APPSC AE 04.12.2012

Ans. (d) : According to Hagen – Poiseuille equation–

Discharge $(Q) = \frac{\pi}{8\mu}\left(-\frac{\partial P}{\partial x}\right).R^4$

Where : R = Radius of pipe

For same discharge–

Pressure gradient $\left(-\frac{\partial P}{\partial x}\right) \propto \frac{1}{R^4}$

Where, negative sign shows the pressure decreases in the pipe along the length.

Let, Pressure gradient $\left(-\frac{\partial P}{\partial x}\right) = P$

$$\therefore \quad \frac{P_1}{P_2} = \left(\frac{R_2}{R_1}\right)^4$$

Given; $R_1 = R$ and $R_2 = R/2$

$$\therefore \quad \frac{P_1}{P_2} = \left(\frac{R/2}{R}\right) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

$$\Rightarrow \quad \boxed{P_2 = 16P_1}$$

$$\Rightarrow \text{Factor of pressure gradient } \boxed{\frac{P_2}{P_1} = 16}$$

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1075. If from energy line, $V^2/2g$ is subtracted at every point, we get:

(a) Head Line
(b) Hydraulic Grade Line
(c) Power line
(d) Streamline

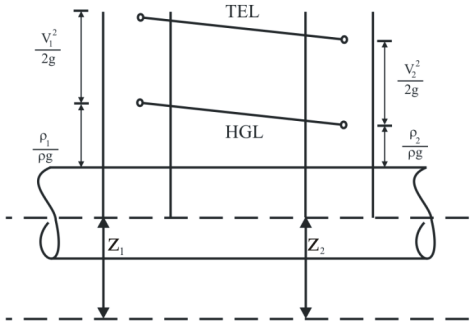
RPSC AE GWD, 2011

Ans. (b) : Total Energy Line (TEL)- It joins total energy head at various points.

$$\left(\frac{P}{\rho g}+\frac{V^2}{2g}+z\right)$$

Hence, If $\frac{V^2}{2g}$ is subtracted from energy line we get HGL at every points.

$$HGL = TEL - \frac{V^2}{2g}$$



1076. The loss of pressure head for the laminar flow through pipe varies :

(a) as the square velocity
(b) directly as the velocity
(c) as the inverse of the velocity
(d) none of the above

J&K PSC Civil Services Pre, 2010

Ans. (b) : Loss of pressure head for the laminar flow

Pressure head $h = \frac{P_1 - P_2}{\rho g} = \frac{32\mu VL}{\rho g D^2}$

Pressure head $\boxed{h \propto V}$

1077. A piping system consists of three pipes arranged in series, the lengths of the pipes are 1200 m, 750 m and 600 m and diameter 750mm, 600 mm, 450 mm respectively. Transform the system to an equivalent 450 mm diameter pipe

(a) 671.3 m (b) 771.3m
(c) 871.3m (d) 971.3m

ISRO Scientist/Engineer 2010

Ans. (c) : Three pipes arranged in series.

$L_1 = 1200m$, $L_2 = 750m$, $L_3 = 600m$
$d_1 = 750mm$, $d_2 = 600mm$, $d_3 = 450mm$
$d_e = 450mm$
$L_e = ?$

Pipe in series

(1) $Q_1 = Q_2 = Q_3 = Q = \text{Constant}$

(2) $h_e = h_1 + h_2 + h_3 + ...$

So, $\frac{fL_eQ^2}{12d_e^5} = \frac{fL_1Q^2}{12d_1^5} + \frac{fL_2Q^2}{12d_2^5} + \frac{fL_3Q^2}{12d_3^5}$

$$\frac{L_e}{d_e^5} = \frac{L_1}{d_1^5} + \frac{L_2}{d_2^5} + \frac{L_3}{d_3^5}$$

$$\frac{L_e}{(450)^5} = \frac{1200}{(750)^5} + \frac{750}{(600)^5} + \frac{600}{(450)^5}$$

$L_e = 871.25m \approx 871.3m$

1078. Consider the following statements :

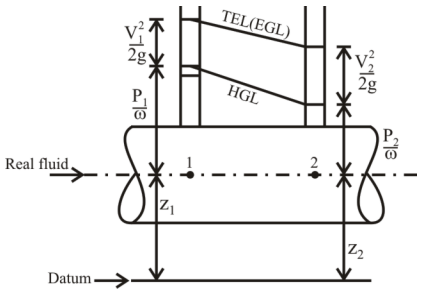
1. Energy gradient line always drops in the direction of flow.
2. Hydraulic gradient line always rises in the direction of flow.
3. Vertical intercept between the energy gradient line and hydraulic gradient line is equal to the velocity head.
4. For a pipe of uniform section, slope of hydraulic gradient line is equal to the slope of energy gradient line.

Which of the above statements is/are correct ?

(a) 1 only (b) 2 and 3 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

UPSC JWM Advt. No.-50/2010

Ans. (c) :



Energy gradient lines are obtained by plotting total head at various points along the axis of pipe, since the total head decrease in the direction of flow, so EGL will always drop in that direction.

Hydraulic gradient line is the sum of pressure head & datum head at a point, so the difference between the EGL & HGL is of velocity head.

1079. Pipes of lengths l_1, l_2 etc and diameters d_1, d_2, etc are connected in series. A single pipe of length $l_1 + l_2 + l_3 + ...$ having same pressure loss as all pipes in series will have diameter equal to

(a) $\left(\frac{l_1 + l_2 + l_3 + ...}{l_1/d_1^4 + l_2/d_2^4 + ...}\right)^{1/4}$

(b) $\left(\frac{l_1 + l_2 + l_3 + ...}{l_1/d_1^5 + l_2/d_2^5 + ...}\right)^{1/5}$

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(c) $\left(\frac{l_1+l_2+l_3+...}{l_1/d_1^3+l_2/d_2^3+...}\right)^{1/3}$

(d) $\left(\frac{l_1+l_2+l_3+...}{l_1/d_1^2+l_2/d_2^2+...}\right)^{1/2}$

ISRO Scientist/Engineer 2008

Ans. (b) : When pipe is connected in series

$$\frac{l}{d^5}=\frac{l_1}{d_1^5}+\frac{l_2}{d_2^5}+\frac{l_3}{d_3^5}+.....$$

$$d=\left[\frac{l_1+l_2+l_3}{\left[\frac{l_1}{d_1^5}+\frac{l_2}{d_2^5}+\frac{l_3}{d_3^5}+.....\right]}\right]^{\frac{1}{5}}$$

1080. There is a sudden increase in the diameter of a pipe from d_1 to d_2. The ratio d_2/d_1 for minor loss to be independent of the direction of flow is

(a) $\sqrt{3}$ (b) $\sqrt{2}$

(c) 2 (d) 3

WBPSC AE, 2007

Ans. (b) : Loss due to sudden expansion = Loss at exit

$$\frac{K(v_1-v_2)^2}{2g}=\frac{Kv_2^2}{2g}$$

$$v_1^2+v_2^2-2v_1v_2=v_2^2$$

$$v_1^2=2v_1v_2$$

$$\frac{v_1}{v_2}=2$$

$$\left(\frac{d_2}{d_1}\right)^2=2$$

$$\frac{d_2}{d_1}=\sqrt{2}$$

1081. If is the Fanning friction factor, loss of head due to friction in the pipe is $\frac{KfLU^2}{gD}$ where K is equal to-

(a) 4 (b) 1

(c) 2 (d) $\frac{1}{4}$

WBPSC AE 2003

Ans. (c) : ∵ Fanning friction equation.

$$H_f=\frac{4f_fL}{D}.\frac{U^2}{2g}$$

∵ Given, $H_f=\frac{KfLU^2}{gD}$

Where,

We get, $\boxed{K=2}$

8. Laminar Flow

1082. Laminar flow occurs in pipes, when Reynolds number is

(a) Lies between 2000-3000

(b) Lies between 3000-4000

(c) Less than 2000

(d) Less than 1000

Assam PSC AE (IWT) 14.03.2021
ISRO Scientist/Engineer (RAC) 07.05.2017
Vizag Steel (MT) 2017, TSPSC Managers, 2015
TRB Poly. Lect., 2012, RPSC AE GWD, 2011
WBPSC AE 2008
UKPSC AE 2007 Paper -II

Ans. (c) : If Re < 2000 → For Laimar flow

2000 ≤ Re ≤ 4000 → For Transition flow

Re > 4000 → Flow become Turbulent

1083. A fully developed laminar viscous flow through a circular tube has the ratio of maximum velocity to average velocity as

(a) 3.0 (b) 2.0

(c) 2.5 (d) 1.5

GPSC Executive Engineer 23.12.2018
MPPSC State Forest Service Exam, 2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
ISRO Scientist/Engineer 2011
J&K PSC Civil Services Pre, 2010
ESE 1994, GATE 1994

Ans. (b) : For fully developed laminar viscous flow through circular pipe

$V_{max}=2\,V_{avg}$

$$\frac{V_{max}}{V_{Avg}}=2$$

For two parallel fixed plate

$$\frac{V_{max}}{V_{Avg}}=\frac{3}{2}$$

1084. For the viscous flow the coefficient of friction is given by

(a) $f=\frac{8}{R_e}$ (b) $f=\frac{16}{R_e}$

(c) $f=\frac{32}{R_e}$ (d) $f=\frac{60}{R_e}$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
TNCSC AE 2020, RPSC IOF, 2020
OPSC AEE 2019 PAPER - II
Kerala PSC Poly. Lect. 09.01.2014
UKPSC AE 2012 Paper–II
WBPSC AE 2008

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Ans. (b) : For laminar/viscous flow :

$f = \frac{64}{R_e}$ where f = friction factor

$f = 4f'$ where f' = coefficient of friction

$$\boxed{f' = \frac{16}{R_e}}$$

1085. Reynolds number which quantifies the role of viscous effect expressed as

(a) $\frac{V}{\sqrt{gd}}$ (b) $\rho\frac{Vd}{\mu}$

(c) $\rho\frac{V^2L}{\sigma}$ (d) $V\sqrt{\frac{\rho}{E}}$

GPSC Asstt. Prof. 28.08.2016
ISRO Scientist/Engineer 24.05.2014
APPSC AEE 2012, TRB Asstt. Prof., 2012
UKPSC AE 2007 Paper -II

Ans : (b) Reynolds Number (R_e)

$$R_e = \frac{\text{Inertia force}}{\text{Viscous force}}$$

$$R_e = \frac{\rho vd}{\mu}$$

Here ρ = Density
d = Characteristics length
v = Average velocity

1086. In laminar flow through a circular tube, the Darcy - Weisbach friction is related to the Reynolds number by

(a) $f = \left(\frac{1}{R}\right)$ (b) $f = \left(\frac{16}{R}\right)$

(c) $f = \left(\frac{64}{R}\right)$ (d) $f = \left(\frac{0.316}{R^{1/4}}\right)$

Assam PSC AE (PHED), 18.10.2020
CIL MT 27.02.2020, APPSC AEE 2012
OPSC Civil Services Pre. 2011
UKPSC AE 2007 Paper -II

Ans. : (c) According to Darcy – Weisbach equation

$$\frac{p_1 - p_2}{\rho g} = h_f = \frac{4f'\ell V^2}{2gd} = \frac{32\mu VL}{\rho gd^2}$$

f' = Fanning friction factor

$$f' = \frac{16}{\left(\frac{\rho dV}{\mu}\right)} \qquad \frac{\rho dV}{\mu} = R_e$$

$$\boxed{f' = \frac{16}{R_e}}$$

But we know that $f = 4f'$

So, $\boxed{f = \frac{64}{R_e}}$ where f = friction factor

1087. The maximum velocity of a one-dimensional incompressible fully developed viscous flow, between two fixed parallel plates, is 6 m s^{-1}. The mean velocity (in m s^{-1}) of the flow is

(a) 2 (b) 3
(c) 4 (d) 5

GPSC Executive Engineer 23.12.2018
TSPSC AEE 2017
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2010

Ans. (c) : For fully developed in compressible viscous flow between two fixed parallel plate–

$$V_{mean} = \frac{2}{3}V_{max} = \frac{2}{3}\times 6$$

$$\boxed{V_{mean} = 4 \text{ m/sec.}}$$

1088. For a laminar flow through a pipe, the discharge varies–

(a) as the square of the diameter
(b) as the inverse of viscosity
(c) inversely as the pressure gradient
(d) as the square of viscosity

RPSC IOF & Boiler 20.09.2017
APPSC AEE Mains 2016 (Civil/Mechanical)
ESE 2011
OPSC Civil Services Pre 2006

Ans : (b) Discharge through pipe having Laminar flow

$$Q = \bar{u}.A$$

$$= \frac{R^4}{8\mu}\left(-\frac{\partial P}{\partial x}\right).\pi$$

$$Q \propto \frac{1}{\mu}$$

Hence discharge varies as inverse proportional of viscosity. Discharge increases as viscosity reduces or vice-versa.

1089. The Reynolds number for flow of a certain fluid in a circular tube is specified as 2500. What will be the Reynolds number when the tube diameter is increased by 20% and the fluid velocity is decreased by 40% keeping fluid the same?

(a) 1200 (b) 1800
(c) 3600 (d) 200

Vadodara Muncipal Corp. DEE, 2018
RPSC AE 2018
Karnataka PSC AE, 10.09.2017
GATE 1997

Ans. (b) : We know that Reynold's number

$$R_e = \frac{\rho VD}{\mu} \text{ (For pipe flow)}$$

$$R_e' = \frac{\rho}{\mu}\times 1.2D \times 0.6 \times V$$

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$$R_e' = \frac{\rho DV \times 1.2 \times 0.6}{\mu}$$

$$R_e' = 2500 \times 1.2 \times 0.6$$

$$\boxed{R_e' = 1800}$$

1090. For laminar flow through a long pipe, the pressure drop per unit length is

(a) In direct proportion to the cross sectional area

(b) In proportion to diameter of pipe

(c) In inverse proportion to the square of cross sectional area

(d) In inverse proportion to cross sectional area

GPSC Executive Engineer 23.12.2018
ISRO Scientist/Engineer (RAC) 07.05.2017
CGPSC Asstt. Workshop Supt., 17.07.2016
GATE 1996

Ans. (d) : Pressure drop (Δp) in laminar flow through pipe due to friction is given by,

$$\Delta p = \frac{32\mu VL}{D^2}$$

$$\frac{\Delta p}{L} = \frac{32\mu V}{D^2} = \frac{32\mu V}{(4/\pi) \times A}$$

Since, $\frac{\Delta p}{L} \propto \frac{1}{A}$

So, $\frac{\Delta p}{L}$ increases when cross sectional area decreases i.e., inversely proportional.

1091. The discharge in m^3/s for laminar flow through a pipe of diameter 0.04 m having centerline velocity of 1.5 m/s is:

(a) $\frac{3\pi}{10000}$

(b) $\frac{3\pi}{2500}$

(c) $\frac{3\pi}{5000}$

(d) $\frac{3\pi}{7500}$

VIZAG STEEL MT 2015, APGENCO AE 2012
CSE Pre-2004, GATE 1988

Ans. (a) : Given,

$d = 0.04$ m

$V_{max} = 1.5$ m/s

$V_{avg} = V_{max}/2$

$V_{avg} = 0.75$ m/s

$Q = A \times V_{avg}$

$$= \frac{\pi}{4} \times (0.04)^2 \times 0.75$$

$$= \frac{3\pi}{10000}$$

1092. For a laminar flow the loss of pressure head is given by

(a) Pressure head $\propto$ Velocity^2

(b) Pressure head $\propto$ Velocity

(c) Pressure head $\propto$ Velocity^3

(d) Pressure head^2 $\propto$ Velocity

Nagaland PSC (CTSE) 2017, 2016 Paper-II
ESE 2013, APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (b) : For a Laminar flow,
The loss of pressure head = Pressure head $\propto$ Velocity

1093. The pressure drop for laminar flow of a liquid in a smooth pipe at normal temperature and pressure is :

(a) Directly proportional to density

(b) Inversely proportional to density

(c) Independent of density

(d) Proportional to (density)$^{0.75}$

GPSC Executive Engineer 23.12.2018
HPPSC Asstt. Prof. 18.11.2016
GATE 2009

Ans. (c) : The pressure drop for laminar flow of a liquid in a smooth pipe at normal temperature and pressure is–

$$\Delta P = \frac{32\mu VL}{D^2}$$

Independent of density.

1094. Flow takes place at Reynolds Number of 1500 in two different pipes with relative roughness of 0.001 and 0.002. The friction factor

(a) Will be higher for the pipe with relative roughness of 0.001

(b) Will be higher for the pipe having relative roughness of 0.002

(c) Will be the same in both the pipes

(d) In the two pipes cannot be compared on the basis of data given

ISRO Scientist/Engineer 2011
ESE 2011, CSE Pre-1994

Ans. (c) : When Reynolds number is below 2000 then the flow is Laminar.
In case of Laminar fully developed flow through pipes, the friction factor, f is given by

$$f = \frac{64}{R_e}$$

It is not depends on roughness but for turbulent flow it will be higher for higher relative roughness in case of rough pipe.
If the flow is Laminar, friction factor will be same in both the pipes.

1095. At what distance (r) from the centre of a pipe of radius (R) does the average velocity occur in laminar flow:-

(a) $r = 0.33\ R$ (b) $r = 0.707\ R$

(c) $r = 0.50\ R$ (d) $r = 0.59\ R$

RPSC AE (GWD) 06.05.2016
UKPSC AE 2013, 2007 Paper-II

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Ans. (b) : We know that,

Velocity for laminar flow, $u = \frac{R^2}{4\mu}\left[\frac{-\partial P}{\partial x}\right]\left[1-\left(\frac{r}{R}\right)^2\right]$

Average velocity for laminar flow, $\bar{u} = \frac{R^2}{8\mu}\left[\frac{-\partial P}{\partial x}\right]$

At a distance r, $\bar{u} = u$

$$\frac{R^2}{8\mu}\left[\frac{-\partial P}{\partial x}\right] = \frac{R^2}{4\mu}\left[\frac{-\partial P}{\partial x}\right]\cdot\left[1-\left(\frac{r}{R}\right)^2\right]$$

$$\frac{r}{R} = 0.707$$

$$\boxed{r = 0.707\,R}$$

1096. The velocity distribution at any section of pipe for steady laminar flow is

(a) Laminar (b) Exponential
(c) Parabolic (d) Hyperbolic

CGPSC AE 16.10.2016
UKPSC AE 2012, 2007 Paper -II
CSE Pre-1996

Ans. (c) : At any section of pipe for steady laminar flow, velocity $u = \frac{1}{4\mu}\frac{-\partial P}{\partial x}R^2\left[1-\frac{r^2}{R^2}\right]$

From this equation, velocity profile or velocity distribution is parabolic.

Note : In turbulent fluid flow, velocity profile is logarithmic.

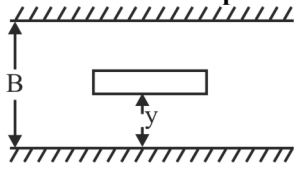
1097. Which two forces are most important in laminar flow between parallel plates?

(a) Inertial and viscous
(b) Viscous and pressure
(c) Gravity and pressure
(d) Pressure and inertial

UPPSC AE 12.04.2016 Paper-II
ESE 2014, GATE 1990

Ans. : (b) Viscous forces and pressure forces are most important in laminar flow between parallel plates.

Laminar flow between two fixed parallel plate:-



(i) Velocity $(\mu) = -\frac{1}{2\mu}\frac{\partial P}{\partial x}\left(By - y^2\right)$

(ii) pressure drop in given length, $(p_1 - p_2) = \frac{12\mu u l}{B^2}$

1098. The velocity at which the flow changes from laminar to turbulent for a given fluid at a given temperature is known as:-

(a) Maximum velocity (b) Minimum velocity
(c) Average velocity (d) Critical velocity

Nagaland PSC (CTSE) 2018, Paper-II
Sikkim PSC (Under Secretary), 2017
UKPSC AE-2013, Paper-II

Ans. (d) : The velocity at which the flow changes from laminar to turbulent for a given fluid at a given temperature is known as critical velocity.

1099. A fluid flowing through a pipe of diameter d with average velocity V. Over a length L the head loss is given by $\frac{FLV^2}{2gD}$. The friction factor, f for laminar flow in terms of Reynolds number (Re) is

(a) $f = 0.0014 + \frac{0.125}{Re^{0.32}}$ (b) $f = \frac{0.0791}{Re^{0.25}}$
(c) $f = \frac{14.227}{Re}$ (d) $f = \frac{64}{Re}$

GPSC Executive Engineer 23.12.2018
Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : For laminar flow, friction factor is given by

$\boxed{f = \frac{64}{R_e}}$ for laminar flow.

Friction factor (f) = 4 × friction of coefficient (f')

Friction factor and Reynold's number relationship for different conditions.

$$f = \frac{0.316}{(R_e)^{1/4}}$$

(For turbulent flow in smooth pipe $4000 < R_e < 10^5$)

1100. The momentum correction factor for laminar flow in pipes is _____.

(a) 1 (b) 1.33
(c) 1.05 (d) 0.98

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I
APPSC AEE 2012

Ans. (b) : For Laminar flow

(1) Momentum correction factor (β)

$\beta = 1.33$

(2) Kinetic energy correction factor (α) = 2

For turbulent flow–

(1) Momentum correction factor (β)

$\beta = 1.015$

(2) Kinetic energy connection factor (α) ≈ 1.04 to 1.11

1101. An oil of kinematic viscosity 0.25 stokes flows through a pipe of diameter 10 cm. The flow is critical at the velocity of :

(a) 7.2 m/sec (b) 5.0 m/sec
(c) 0.5 m/sec (d) 0.72 m/sec

RPSC ADE 2016
UKPSC AE-2013, Paper-II

Ans. (c) : For pipe flow, flow is laminar upto Reynolds's no. of 2000.

$$Re = \frac{\rho VD}{\mu} = \frac{VD}{\nu}$$

$$2000 = \frac{V \times 0.1}{0.25 \times 10^{-4}} \text{ or } V = 0.5 \text{ m/s}$$

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1102. In fluid flow through pipes, transition from laminar to turbulent flow, does not depend on

(a) Length of pipe (b) Density of fluid
(c) Diameter of pipe (d) Velocity of flow

UPPSC AE 12.04.2016 Paper-II
GATE 1996

Ans : (a) Nature of flow depend upon Reynolds Number.

$$\text{Reynolds Number (Re)} = \frac{\text{Inertia force}}{\text{viscous force}} = \frac{\rho v d}{\mu}$$

ρ = Density
d = Characteristic length (Diameter in case of pipe)
v = Average velocity.

1103. The pressure drop for a relatively low Reynolds number flow in a 600 mm, 30 m long pipeline is 70 kPa. What is the wall shear stress?

(a) 0 Pa (b) 1400 Pa
(c) 700 Pa (d) 350 Pa

ISRO Scientist/Engineer 2009
ESE 2004

Ans. (d) : D = 600 mm
l = 30 m
P = 70 kPa, τ = ?

For flow through circular pipe

$$\tau = -\frac{r}{2}\frac{dP}{dx}$$

$$= \frac{0.3}{2}\frac{(70\times10^3)}{30}$$

$$\tau = 350 \text{ Pa}$$

1104. The flow in the capillary tube is laminar because:-

(a) The capillary tube is made of glass
(b) The surface tension and capilarity promotes laminarisation
(c) Only inviscid liquid can flow through capillary tube
(d) The diameter of capillary tube is very small

UKPSC AE-2013, Paper-II
APPSC AEE 2012

Ans. (d) : The flow in the capillary tube is laminar because the diameter of capillary tube is very small.

1105. When various fluid particles move in layers with one layer of fluid sliding smoothly over the adjacent layer, then the flow is said to be

(a) Laminar flow
(b) Uniform flow
(c) Steady flow
(d) Turbulent flow

Nagaland PSC CTSE 2017, 2016 Paper-II

Ans. (a) : When various fluid particles move in layer with one layer of fluid sliding smoothly over the adjacent layer, then the flow is said to be, laminar flow.

1106. The velocity profile in fully developed laminar flow in a pipe of diameter D is given by $u = u_0\left[1-\left(\frac{4r^2}{D^2}\right)\right]$, where r is the radial distance from the centre. If the viscosity of the fluids is μ, the pressure drop across a length 'L' of the pipe is

(a) $\frac{u_o\mu L}{D^2}$ (b) $\frac{4u_o\mu L}{D^2}$
(c) $\frac{8\mu u_o L}{D^2}$ (d) $\frac{16\mu u_o L}{D^2}$

RPSC ADE 2016, GATE 2006

Ans. (d) : $u = u_0\left[1-\frac{4r^2}{D^2}\right]$

At, $r = 0, u = u_{max} = u_0$

$$u_{avg} = \frac{u_0}{2}$$

$$\Delta P = \frac{32\mu u_{avg} L}{D^2} = \frac{32\mu\times\frac{u_0}{2}\times L}{D^2} = \frac{16\mu u_0 L}{D^2}$$

1107. Water is flowing through a horizontal pipe on constant diameter and the flow is laminar. If the pipe diameter is increased by 50% keeping the volume flow rate constant, then the pressure drop in the pipe due to friction will be decreased by :

(a) 33% (b) 50%
(c) 70% (d) 80%

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
GATE 2011

Ans. (d) : Pressure drop in pipe $(\Delta P) = \frac{32\mu\bar{V}L}{D^2}$

$D_2 = 1.5D$

$$\Delta P = \frac{32\mu\left(\frac{Q}{\frac{\pi}{4}D^2}\right)L}{D^2}$$

$$\Delta P = \frac{32\mu QL}{\frac{\pi}{4}D^4} = \frac{128\mu QL}{\pi D^4}$$

$$\Delta P \propto \frac{1}{D^4}$$

$$\Delta P_1 = \frac{1}{D^4}, \Delta P_2 = \frac{1}{(1.5D)^4} = \frac{0.1975}{D^4}$$

% decrease,

$$= \frac{\Delta P_1 - \Delta P_2}{\Delta P_1} = \frac{1-0.1975}{1} = 0.8024 = 80.24\%$$

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1108. The shear stress between two fixed parallel plates with a laminar flow between them

(a) a constant across the gap
(b) varies parabolically as the distance from the mid plane
(c) varies inversely as the distance from the mid plane
(d) varies directly as the distance from the mid plane

UPPSC AE 13.12.2020, Paper-2
TNPSC AE, 2008

Ans. (d) : The shear stress between two fixed parallel plates with a laminar flow between them varies directly as the distance from the mid plane.

$$\tau_{max} = -\left(\frac{\partial P}{\partial x}\right).\frac{y}{2}$$

$$\tau \propto y$$

Where, y = distance between two fixed parallel plates

1109. Consider steady laminar incompressible axisymmetric fully developed viscous flow through a straight circular pipe of constant cross-sectional area at a Reynold's numbers of 5. The ratio of inertia force to viscous force on a fluid particle is :

(a) 5 (b) 1/5
(c) 0 (d) Infinity

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
GATE 2007

Ans. (a) : Reynolds number $= \frac{\text{Inertia Force}}{\text{Viscous Force}} = 5$

1110. When fluid is passing through a circular conduit, the flow will laminar when its Reynolds number is

(a) less than 5×10^5 (b) more than 5×10^5
(c) less than 2300 (d) more than 2300

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : When fluid is passing through a circular conduit, the flow will laminar when its Reynolds number is less than 2300.

1111. In Hagen Poiseuille flow the ratio of maximum velocity to average velocity will be

(a) 2 (b) 1.5
(c) 0.5 (d) 1.75

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : In Hagen Poiseuille flow the ratio of maximum velocity to average velocity will be 2.

1112. Water flows through a pipe of diameter 200 mm and length 65 m with a velocity of 2 ms^{-1}. If kinematic viscosity of water is 0.02×10^{-4}, the Reynolds number for the flow will be

(a) 9×10^5 (b) 2×10^5
(c) 3×10^5 (d) 5×10^5

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Given, d = 200 mm = 0.2 m
L = 65 m
v = 2m/sec
$v = 0.02\times10^{-4}$

$$Re = \frac{vd}{\nu} = \frac{2\times0.2}{0.02\times10^{-4}}$$

$Re = 2\times10^5$

1113. Which of the following applications regarding Navier-Stokes equations are correct?

1. Laminar unidirectional flow between stationary parallel plates.
2. Laminar unidirectional flow between parallel plates having no relative motion.
3. Laminar flow in circular pipes.
4. Laminar flow between concentric rotating cylinders.

(a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 1, 2 and 4 only (d) 2, 3 and 4 only

ESE 2020

Ans. (a) :
1. Laminar unidirectional flow between stationary parallel plates.
2. Laminar unidirectional flow between parallel plates having no relative motion.
3. Laminar flow in circular pipes.

1114. A fluid of viscosity 7 poise and density 1300 kg/m^3 is flowing through a circular pipe of diameter 200 mm. The maximum shear stress at the pipe wall is 200 N/m^2. What will be that pressure loss in one meter of pipe length?

(a) 400 N/m^2 (b) 4000 N/m^2
(c) 6000 N/m^2 (d) 8000 N/m^2

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given data :
Viscosity (μ) = 7 poise = 0.7 Ns/m^2
Density of fluid (ρ) = 1300 kg/m^3
Dia of pipe (D) = 200 mm
$\therefore$ Radius (R) = D/2 = 100 mm = 0.1 m
Maximum shear stress $(\tau_{max}) = 200\ N/m^2$

$$\therefore \tau_{max} = \left(\frac{-\partial P}{\partial x}\right).\frac{R}{2}; \qquad \frac{\partial P}{\partial x} = \text{Pressure gradient}$$

(–)ve sign indicates that pressure decreases along the length of pipe.

$$\frac{\partial P}{\partial x} = \frac{2\tau_{max}}{R} = \frac{2\times200}{0.1} = 4000$$

Pressure loss in one metre of pipe length
$(\Delta P) = 4000\ N/m^2$

1115. Oil (SG = 0.9, Dynamic viscosity = 1 poise) is flowing with a mean velocity of 1 m/s between two fixed parallel plates, which are 1 cm apart. What will be shear stress at the surface of plate?

(a) 40 N/m^2 (b) 50 N/m^2
(c) 60 N/m^2 (d) 70 N/m^2

APPSC Poly Lect. 13.03.2020

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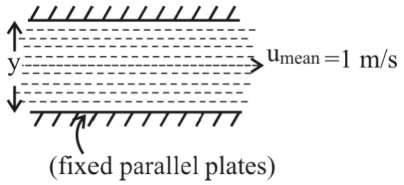
Ans. (c) :
For fixed parallel plates –
Mean speed velocity $(\bar{u})$ is –

$$\bar{u} = \frac{1}{12\mu}\left(\frac{-\partial P}{\partial x}\right).y^2$$

$$\Rightarrow \left(\frac{-\partial P}{\partial x}\right) = \frac{12\mu\bar{u}}{y^2}$$



(fixed parallel plates)

Where,
y = Distance between plates (y)
$= 1 \text{ cm} = 1 \times 10^{-2}$ m
$\bar{u} = 1$ m/s
$\mu = 1$ poise $= 0.1$ N-s/m^2

$$\left(-\frac{\partial P}{\partial x}\right) = \frac{12 \times 0.1 \times 1}{\left(10^{-2}\right)^2}$$

$$\left(\frac{-\partial P}{\partial x}\right) = 12000 \text{ N/m}^3$$

$$\because \tau_{max} = \left(\frac{-\partial P}{\partial x}\right).\frac{y}{2}$$

$$= \frac{12000 \times 10^{-2}}{2} = \frac{120}{2}$$

$$\tau_{max} = 60 \text{ N/m}^2$$

1116. The friction factor f in a laminar pipe flow was found to be 0.04. The Reynolds number of the flow was
(a) 2000 (b) 1600
(c) 1000 (d) 800
Assam PSC AE (PHED), 18.10.2020

Ans. (b) : Given-
Friction factor = 0.04

Reynold's number, $Re = \frac{64}{\text{Friction factor}}$ $\left[\because f = \frac{64}{Re}\right]$

$$Re = \frac{64}{0.04}$$

Re = 1600

$\Rightarrow$ Friction factor = 4 × Co-efficient of friction
$\Rightarrow$ For laminar flow, the friction factor is inversely proportional to the diameter of pipe,

1117. In a circular tube of diameter 100 mm and length 13 m with laminar flow, the friction factor is estimated to be 0.05. Calculate the Reynolds number?
(a) 950 (b) 2300
(c) 1280 (d) None of the above
ISRO Scientist/Engineer 12.01.2020

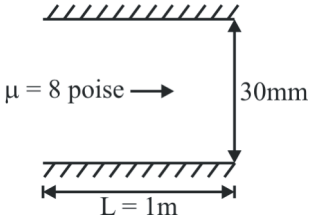
Ans. (c) : D = 100 mm, L = 13 m, f = 0.05
Re = ?

For laminar flow, $f = \frac{64}{Re}$

$$\Rightarrow Re = \frac{64}{f} = \frac{64}{0.05} = 1280$$

1118. An oil of viscosity 8 poise flows between two parallel fixed plates, which are kept at a distance of 30 mm apart. If the drop of pressure for a length of 1m is 0.3×10^4 N/m^2 and width of the plates is 500 mm, the rate of oil flow between the plates will be
(a) 4.2×10^{-3} m^3/s (b) 5.4×10^{-3} m^3/s
(c) 6.6×10^{-3} m^3/s (d) 7.8×10^{-3} m^3/s
ESE 2019

Ans. (a) :



For laminar flow through parallel plates separated by distance h, Δp is given as

$$\Delta p = \frac{12\mu VL}{h^2}$$

$$\frac{\Delta p}{L} = \frac{12\mu V}{h^2}$$

$$0.3 \times 10^4 = \frac{12 \times (8 \times 0.1) \times V}{(0.03)^2}$$

V = 0.281 m/s
So, discharge will be
$Q = V \cdot A$
$= V \cdot \text{width} \times h$
$= 0.281 \times 0.5 \times 0.03$
$= 4.22 \times 10^{-3}$ m^3/s

1119. The flow in open channel remains laminar if Reynold's number is less than:
(a) 5000 (b) 4000
(c) 2000 (d) 500
OPSC AEE 2019 PAPER - II

Ans : (d) : For open channel flow-
Laminar , transitional and turbulent flow characterized by Reynolds number,

$$Re = \frac{uL}{\nu}$$

Re < 500	(laminar flow)
500 < Re < 2000	(Transitional flow)
2000 < Re	(Turbulent flow)

1120. For the laminar flow through a pipe, the shear stress over the cross-section
(a) Varies inversely as the distance from the centre of the pipe

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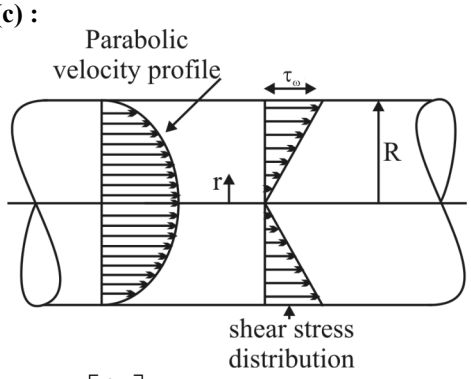
(b) Varies inversely as the distance from the surface of the pipe
(c) Varies directly as the distance from the centre of the pipe
(d) Remains constant over the cross-section

TNPSC AE 2019

Ans. (c) :



$$\tau = -\left[\frac{\partial P}{\partial x}\right].\frac{r}{2}$$

$$\tau \propto r$$

So, For the laminar flow through a pipe, the shear stress over the cross-section varies directly as the distance from the centre of the pipe.

1121. The laminar flow is characterised by
(a) Existence of eddies
(b) Irregular motion of fluid particles
(c) Fluid particles moving in layers parallel to the boundary surface
(d) Its space coordinates

TNPSC AE 2019

Ans. (c) : The laminar flow is characterised by fluid particles moving in layers parallel to the boundary surface

1122. When the fluid flows over the flat plate with a sharp leading edge is examined, laminar flow is generally observed for Reynolds number less than
(a) 2300 (b) 1500
(c) 5×10^5 (d) 1000

Nagaland PSC CTSE 2017 Paper-2

Ans. (c) : 5×10^5

1123. Which of the following is an example of laminar flow?
(a) underground flow
(b) flow past tiny bodies
(c) flow of oil measuring instruments
(d) all of these

Sikkim PSC (Under Secretary), 2017

Ans. (d) : Laminar flow- Laminar flow is defined as the type of flow in which the fluid particle move along well defined path or stream line and type of flow is also called stream line flow or viscous flow.

Ex.-
(i) Underground flow
(ii) Flow past tiny bodies
(iii) Flow of oil in measuring instrument.

1124. The shear stress at the wall of a 16 cm diameter pipe in laminar flow is 36 N/m^2. The shear stress at a radius of 4 cm in N/m^2 is :
(a) 9 (b) 18
(c) 6 (d) 72

HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : Shear stress in pipe in laminar flow

$$\tau = \frac{r}{2}\left(-\frac{\partial P}{\partial x}\right)$$

$$(\tau \propto r)$$

$$\frac{\tau_{r=8}}{\tau_{r=4}} = \frac{R}{r}$$

$$\frac{36}{\tau_{r=4}} = \frac{8}{4}$$

$$\tau_{r=4} = 18 N/m^2$$

1125. In a laminar flow through a circular pipe of diameter 20 cm, the maximum velocity is found to be 1 m/s. The velocity at a radial distance 5 cm from the axis of the pipe will be
(a) 0.25 m/s (b) 0.50 m/s
(c) 0.75 m/s (d) 1.0 m/s

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : $V = V_{max}\left(1-\frac{r^2}{R^2}\right) = 1\left(1-\frac{5^2}{10^2}\right) = 0.75\ m/s$

1126. The momentum correction factor is used to account for
(a) Change in direction of flow
(b) Change in pressure
(c) Non uniform distribution of velocity
(d) Unsteady flow

APPSC AEE Screening Test 2016

Ans. (c) : The momentum correction factor is defined as the ratio of momentum of the flow per second based on actual velocity to the momentum of the flow per second based on average velocity across a section.
It is considered to take into the effect of the non-uniform distribution of flow through pipes.
For turbulent flow, the momentum correction factor is 1.02 and for laminar flow, its value is 1.33.

1127. For laminar flow between two fixed parallel plates the flow velocity
(a) Is constant over the cross-section
(b) Varies parabolically across the section
(c) Varies as three halves power of distance from the mid-point
(d) Is maximum at the centres, zero at the plate and the variation in between is linear
(e) None of these

CGPSC AE 16.10.2016

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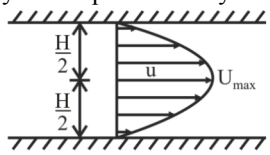
Ans. (b) : For laminar flow between two parallel plates the flow velocity varies parabolically across the section.



$$u = \frac{1}{2\mu}\left(-\frac{\partial p}{\partial x}\right)\left(Hy - y^2\right)$$

Where, H is the distance between the parallel plates.

1128. Which of the following does NOT hold good for fluid friction in case of steady stream line flow? The frictional resistance is

(a) Proportional to velocity
(b) Independent of the pressure
(c) Proportional to the surface area of contact
(d) Proportional to square of the velocity
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : The frictional head loss in a steady stream line flow, is

$$h_f = \frac{32\mu VL}{\rho g D^2}$$

$h_f \propto V$

$h_f \propto L$

$h_f \propto Q$

$$h_f \propto \frac{1}{D^4}$$

1129. Mark the wrong statement in the context of laminar flow through a pipeline.

(a) Shear stress is zero at the centre and varies linearly with pipe radius
(b) Head loss is proportional to square of the average flow velocity
(c) The friction factor varies inversely with flow Raynolds number
(d) All of these
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) : Laminar flow through a pipeline, these are the following properties–

(1) Shear stress is zero at the centre & varies linearly with pipe radius.
(2) Head loss is proportional to the average flow velocity.
(3) The friction factor varies inversely with flow reynolds number.

Note-Official answer given as option (e).

1130. While water passes through a given pipe at a mean velocity 'V' the flow is found to change from laminar to turbulent. If another fluid of specific gravity 0.8 and coefficient of viscosity 20% of that of water is passed through the same pipe the transition of flow from laminar to turbulent is expected if the flow velocity is

(a) V/4 (b) V
(c) 2V (d) V/2

UPRVUNL AE 21.08.2016

Ans. : (a) Let ρ_1, V_1 & μ_1 is for water

Given, $\rho_2 = 800$

$\mu_2 = 0.2\mu_1$

We know that,

For laminar critical value of Reynold's no. for pipe flow is 2000

$\therefore$ Re = 2000

$$\frac{Re_1}{Re_2} = 1$$

$$\frac{\rho_1 V_1 D_1/\mu_1}{\rho_2 V_2 D_2/\mu_2} = 1$$

$$\frac{1000 \times 0.2 V_1}{800 V_2} = 1$$

$$\boxed{V_2 = V/4}$$

1131. When the relationship between Reynolds number and the friction factor is represented by a straight line, the flow is said to be

(a) Isentropic (b) Laminar
(c) Turbulent (d) Vortex

UPPSC AE 12.04.2016 Paper-II

Ans. : (b) When the relationship between Reynolds number and the friction factor is represented by a straight line, the flow is said to be Laminar.

For laminar flow:-

$$f = \frac{64}{R_e}$$

Thus, the friction factor is inversely proportional to Reynolds no. so, after plotting in the log plane it is like a straight line coming downwards.

1132. In case of laminar flow through pipe, the ratio of total kinetic energy of fluid passing per second to the energy value obtained on the basis of average velocity is

(a) 1.2 (b) 1.54
(c) 2.0 (d) 2.37

UPPSC AE 12.04.2016 Paper-II

Ans : (c) In case of laminar flow through pipe, the ratio of total kinetic energy of fluid passing per second to the energy value obtained on the basis of average velocity is 2.0.

1133. Relation between shear and pressure gradients in a two dimensional laminar flow under steady and uniform conditions

(a) $\partial\tau/\partial x = \partial p/\partial x$ (b) $\partial\tau/\partial y = \partial p/\partial x$
(c) $\partial^2\tau/\partial x^2 = \partial^2 p/\partial x^2$ (d) $\partial^2\tau/\partial x^2 = \partial^2 p/\partial y^2$

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (b) : We know,

$$\tau + \frac{\partial\tau}{\partial y}\delta y = \text{shear stress on plane}$$

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Net shear force

$$\left[\left(\tau+\frac{\partial\tau}{\partial y}\delta y\right)\delta x\delta z-\tau\delta x.\delta z\right]$$

$$=\frac{\partial\tau}{\partial y}\delta x\delta y\delta z$$

Pressure intensity or

Net pressure force acting

$$=p\delta y\delta z-\left(p+\frac{\partial p}{\partial x}\delta x\right)\delta y\delta z$$

for steady & uniform flow,

The sum of all force is equal to zero.

So, $\boxed{\frac{\partial\tau}{\partial y}=\frac{\partial p}{\partial x}}$

1134. In laminar flow of a liquid down in an inclined plane, the surface velocity is found to be 30cm/s. The average velocity of the flow, in cm/s is

(a) 20 (b) 30

(c) 15 (d) 10

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (a) : As we know, for laminar flow

$$V_{avg}=V_{surface}\times\frac{2}{3}$$

$$V_{avg}=30\times\frac{2}{3}=20m/s$$

1135. If the velocity is zero over half of the cross-sectional area and is uniform over the remaining half, then the momentum correction factor is

(a) 2 (b) 1

(c) 4/3 (d) 4

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Momentum correction factor (β)

$$\beta=\frac{\text{Acutal momentum per sec.}}{\text{Average momentum per sec.}}$$

$$\beta=\frac{1}{A}\int_{A/2}^{A}\left(\frac{V}{V/2}\right)^2dA$$

$$\beta=\frac{4}{A}\int_{A/2}^{A}dA$$

$$\beta=\frac{4}{A}\times\frac{A}{2}=2$$

1136. The velocity at which the laminar flow stops is known as

(a) velocity of approach

(b) lower critical velocity

(c) higher critical velocity

(d) none of the above

ISRO Scientist/Engineer 11.10.2015

Ans. (b)

- A laminar flow is one in which fluid moves in the form of layer and there is no intermixing of fluid particles or molecular momentum transfer in one layer to another layer.
- A laminar flow generally occurs at high viscosity and low velocity.

The velocity at which the laminar flow stops is known as lower critical velocity.

1137. The pressure gradient along any free surface is

(a) Always negative

(b) Always positive

(c) May be positive or negative depending upon acceleration

(d) zero

(e) free from acceleration

CGPSC AE 26.04.2015 PAPER-II

Ans. (d) The pressure gradient along any free surface zero

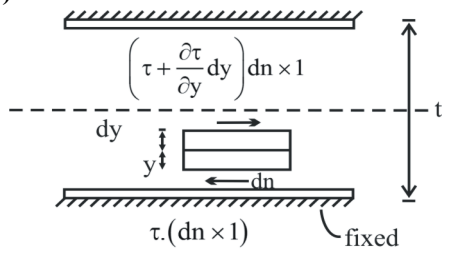
1138. In case of steady incompressible laminar viscous flow between two stationary parallel plates, the velocity distribution is

(a) Linear (b) Quadratic

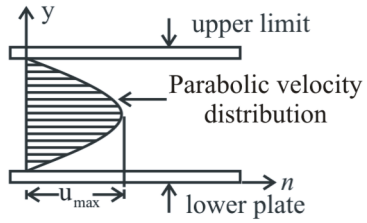
(c) Parabolic (d) Hyperbolic

ISRO Scientist/Engineer 24.05.2014

Ans. (c) :



$$\boxed{u_{max}=-\frac{1}{8\mu}\left(\frac{\partial P}{\partial x}\right)t^2}$$



Velocity distribution,

$$\boxed{u_{Avg}=\frac{2}{3}u_{max}}$$

1139. An oil of kinematic viscosity of 0.5 stokes flow though a pipe of 20cm diameter. What is the velocity if the flow is critical?

(a) 0.5 m/s

(b) 0.9 m/s

(c) 1.5 m/s

(d) 2 m/s

ISRO Scientist/Engineer 24.05.2014

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Ans. (a) : kinematic viscosity = 0.5 stokes

d = 20 cm

For pipe flow,-

$(Re)_{critical} = 2000$

$$\frac{VD}{\upsilon} = 2000$$

$$\text{or } V = \frac{2000 \times 0.5 \times 10^{-4}}{0.2}$$

$$= 0.5\,m/s$$

1140. If the free-stream fluid velocity (V) is 20 m/s and the pipe diameter (D) is 1 m, calculate the Reynolds number (R), if the dynamic density (ρ) is given by 0.150 kg/m^3 and the fluid viscosity is 0.0000122.

(a) 224201 (b) 232502

(c) 242602 (d) 245902

UPRVUNL AE 2014

Ans : (d) Reynolds Number = $\frac{\rho VD}{\mu}$

Denisty (ρ) =0.150 kg/m^3

Velocity (V) = 20m/sec

Diameter (D)= 1m

Viscosity (μ) = 0.0000122

$$R_e = \frac{0.150 \times 20 \times 1}{0.0000122}$$

Re = 245902

1141. An oil with specific gravity 0.85 and viscosity 3.8 poise flows in a 5cm diameter horizontal pipe at 2.0 m/s. The Reynolds number will be approximately:-

(a) 224 (b) 2240

(c) 22.4 (d) 22400

UKPSC AE-2013, Paper-II

Ans. (a) : Given as,

Specific gravity = 0.85

μ = 3.8 poise

d = 5 cm

V = 2.0 m/s

$$Re = \frac{\rho VD}{\mu} = \frac{0.85 \times 1000 \times 2 \times 0.05}{3.8 \times 10^{-1}}$$

Re = 223.68 $\simeq$ 224

1142. Which of the following is not correct for laminar flow in circular pipes?

(a) Velocity is maximum at the centre of the pipe.

(b) The variation of velocity from the pipe surface to the centre of pipe is parabolic.

(c) The average velocity is half the maximum velocity.

(d) The velocity equals the average velocity at a distance of 0.75 times radius from the centre of the pipe.

RPSC AEN Pre-2013

Ans. (d) : For Laminar flow in circular pipe –

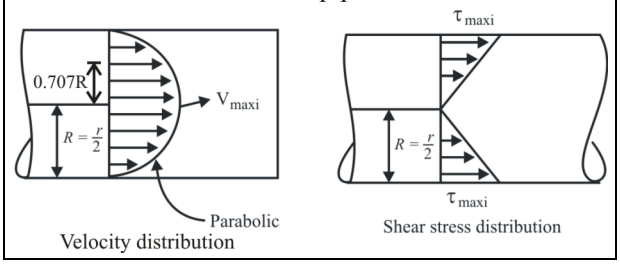
- Velocity is maximum at the centre of the pipe.
- The variation of velocity from the pipe surface to the centre of pipe is parabolic.
- The average velocity is half the maximum velocity.

$$V_{avg} = \frac{1}{2} V_{max}$$

- The velocity (local velocity) is equal to average velocity is given by

$$r = \frac{R}{\sqrt{2}} = 0.707R$$

So average velocity of flow occurs at a radial distance of 0.707R from the centre of pipe.



1143. In a laminar flow, the velocity at any point is:-

(a) Constant in magnitude but varying in direction

(b) Constant in magnitude as well as in direction

(c) Constant in direction but varying in magnitude

(d) Variable in magnitude as well as in direction

UKPSC AE-2013, Paper-II

Ans. (b) : In a laminar flow, the velocity at any point is constant in magnitude as well as in direction.

1144. Laminar flow changes to Turbulent flow when

(a) Diameter of pipe is decreased

(b) Velocity is increased

(c) Viscosity of fluid is increased

(d) Velocity is decreased

TNPSC AE 2013

Ans. (b) : Laminar flow changes to Turbulent flow when velocity is increased.

For pipe flow, if Reynolds number (Re),

Re $\leq$ 2000 [Flow will be laminar flow]

2000 $\leq$ Re $\leq$ 4000 [Flow will be transition]

Re > 4000 [Flow will be turbulent]

$$Re = \frac{\rho DV}{\mu}$$

Then $V \uparrow \rightarrow Re \uparrow \rightarrow$ Flow will be tends to turbulent flow.

1145. An oil of kinematic viscosity 0.5 stokes flows through a pipe of 5 cm diameter. The flow is critical at a velocity of about

(a) 0.2 m/s (b) 2 m/s

(c) 2.5 m/s (d) 4 m/s

UKPSC AE 2012 Paper–II

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Ans. (b) : For critical velocity in pipe flow Reynold's number is 2000.

$$Re = \frac{\rho VD}{\mu} = \frac{VD}{\nu}$$

$$2000 = \frac{V \times 0.05}{0.5 \times 10^{-4}}$$

$$V = 2 \text{ m/s}$$

1146. The lower limit of the critical Reynolds number below which all disturbances in pipe flow are damped out by viscous action has a value approximately equal to

(a) 1 (b) 500
(c) 1000 (d) 2000

APPSC AEE 2012

Ans : (d) The limiting values of Reynolds's Number corresponding to which flow is laminar is given by

Flow condition	Pipe flow	Open channel flow
Laminar flow	$R_e \le 2000$	$R_e \le 500$
Transitional flow	$2000 < R_e < 4000$	$500 < R_e < 2000$
Turbulent flow	$R_e > 4000$	$R_e > 2000$

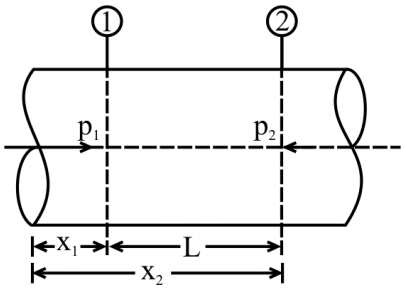
1147. The Hagen–Poiseuille equation which governs the velocity distribution in laminar flow through pipes may be expressed as

(a) $Q = \frac{\pi d^4 \Delta p}{128 \mu L}$ (b) $Q = \frac{\pi d^2 \Delta p}{128 \mu L}$

(c) $Q = \frac{128 \mu L}{\pi d^2 \Delta p}$ (d) $Q = \frac{128 \mu L}{\pi d^4 \Delta p}$

APPSC AEE 2012

Ans : (a)



Drop pressure for a given lenth (L) of a pipe :-

$$h = \frac{p_1 - p_2}{\rho g} = \frac{32 \mu \bar{u} L}{\rho g d^2} = \frac{128 \mu Q L}{\rho g \pi d^4} \quad \left(\bar{u} = \frac{Q}{A}\right)$$

$$p_1 - p_2 = \frac{128 \mu Q L}{\pi d^4}$$

$$\boxed{Q = \frac{\Delta p . \pi d^4}{128 \mu L}}$$

1148. The pressure drop per unit length of pipe in laminar flow is equal to

(a) $\frac{d^2}{32 \mu V}$ (b) $\frac{32 \mu VL}{\gamma d^2}$

(c) $\frac{32 \mu V}{d^2}$ (d) $\frac{8 \mu V}{d^2}$

APPSC AEE 2012

Ans : (c) $\frac{p_1 - p_2}{\rho g} = \frac{32 \mu \bar{u} L}{\rho g d^2}$

$$p_1 - p_2 = \frac{32 \mu \bar{u} L}{d^2}$$

For per unit length pressure drop, L = 1–

Pressure drop, $p_1 - p_2 = \frac{32 \mu \bar{u} L}{d^2}$ OR $\frac{32 \mu V}{d^2}$

$\bar{u} = V =$ Average velocity

1149. Steel and cast iron pipes carrying fluid under pressure are regarded as hydraulically smooth when

(a) the boundary surface is relatively smooth
(b) the roughness projections are of low height
(c) the roughness elements are completely covered by the laminar sub-layer
(d) the laminar layer is thin as compared to the a average height of roughness elements

APPSC AEE 2012

Ans : (c) Steel and cast iron pipes carrying fluid under pressure are regarded as hydraulically smooth when the roughness elements are completely covered by the laminar sub layer.

1150. The momentum correction factor β is used to account for

(a) change in pressure
(b) change in mass rate of flow
(c) change in total energy
(d) non-uniform distribution of velocities at inlet and outlet sections

APPSC AEE 2012

Ans : (d) Momentum correction factor (β) :-

It is defined as the ratio of momentum/sec based on actual velocity to the momentum/sec based on average velocity.

$$\beta = \frac{1}{AV^2} \int u^2 . dA$$

v = Average velocity
u = local velocity at distance r
For laminar flow $(\beta) = 1.33 = 4/3$

1151. Laminar flow through a circular tube studied experimentally by

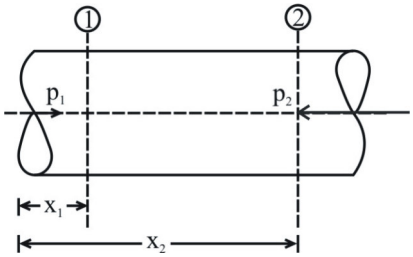
(a) Newton (b) Pascal
(c) Hagen and Poiseuille (d) Prandtl

APPSC AEE 2012

Ans : (c) Laminar flow through a circular tube studies experimentally by Hagen and Poiseuille.

Hagen Poiseuille formula :-

$$\frac{p_1 - p_2}{\rho g} = H_f = \frac{32 \mu \bar{u} L}{\rho g D^2 .}$$



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1152. The friction factor in laminar flow through a round tube

(a) varies inversely as Reynolds' number
(b) varies directly as Reynolds's number
(c) is a logarithmic function of viscosity
(d) varies as square of height of roughness

TNPSC ACF 2012

Ans. (a) : Friction factor for laminar flow

$$f = \frac{64}{Re}$$

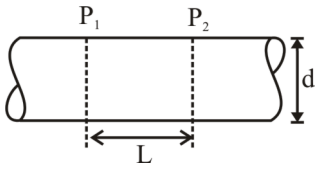
$$\boxed{f \alpha \frac{1}{Re}}$$

1153. For the same discharge in a pipe with laminar flow, reduction of radius to half will increase the pressure gradient by a factor of

(a) 2 (b) 4
(c) 8 (d) 16

APPSC AEE 2012

Ans. (d) :



Laminar flow through circular pipes

$$Q = \left(\frac{-\partial p}{\partial x}\right)\frac{\pi R^4}{8\mu}$$

Given, $Q_1 = Q_2$, $R_2 = R/2$

Then $$\frac{Q_1}{Q_2} = \frac{\left(\frac{-\partial p}{\partial x}\right)_1}{\left(\frac{-\partial p}{\partial p}\right)_2} \times \frac{\frac{\pi \times R^4}{8\mu}}{\frac{\pi (k/2)^4}{8\mu}}$$

$$\left(\frac{\partial p}{\partial x}\right)_2 = 16 \times \left(\frac{\partial p}{\partial x}\right)_1$$

1154. Which one of the following statements does not characterize the fully developed laminar flow in a pipe?

(a) Velocity profile does not change in the longitudinal direction
(b) Velocity profile across the pipe section is parabolic and maximum velocity occurs at the centre
(c) pressure gradient in the direction of flow is constant and positive
(d) Pressure gradient in the direction of flow is constant and negative.

RPSC AE GWD, 2011

Ans. (c) : For Fully developed laminar flow in pipes-

(1) Velocity profile across the pipe section is parabolic and maximum velocity occurs at centre.
(2) Pressure gradient in the direction of flow is constant and negative.

$$\tau = -\frac{\partial P}{\partial x} \cdot \frac{r}{2}$$

(3) Velocity profile does not change in the longitudinal direction.

1155. Match List I correctly with List II and select your answer using the codes given below :

	List I		List II
(A)	Navier-Stokes equation is useful in analysis of	1.	Momentum
(B)	Shear stress in turbulent flow is mainly due to	2.	Hydraulic gradient line
(C)	Pressure gradient is linear for developed flow obeys	3.	Eddy viscosity
(D)	Vapour lock in water pipeline may occur if ..... goes below conduit	4.	Viscous

	A	B	C	D
(a)	1	3	4	2
(b)	3	4	1	2
(c)	2	3	4	1
(d)	4	2	3	1

TNPSC AE, 2008

Ans. (a) :

Navier-Stokes equation is useful in analysis of	– Momentum
Shear stress in turbulent flow is mainly due to	– Eddy viscosity
Pressure gradient is linear for developed flow obeys	– Viscous
Vapour lock in water pipeline may occur if ..... goes below conduit	– Hydraulic gradient line

1156. The value of the kinetic energy correction factor (α) for the viscous flow through a circular pipe is

(a) 1.33 (b) 1.50
(c) 2.0 (d) 1.25

TNPSC AE, 2008

Ans. (c) : Kinetic energy correction factor (α)

$\boxed{\alpha = 2.0}$, for viscous flow through a circular pipe.

$\boxed{\alpha = 1.33}$, for turbulent flow through a circular pipe.

9. Turbulent Flow

1157. The head loss due to friction in turbulent flow through a pipe is:

(a) Directly proportional to velocity
(b) Inversely proportional to square of velocity
(c) Inversely proportional to velocity
(d) Directly proportional to square of velocity

JPSC AE, 10.04.2021, Paper-I
CIL MT 26.03.2017, UJVNL AE 2016

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APPSC AEE Screening Test 2016
CGPSC AE 16.10.2016, TRB Asstt. Prof., 2012
ISRO Scientist/Engineer 12.05.2013
ESE 2009, 2007, 1999, CSE Pre-2007

Ans. (d) : Head loss in turbulent flow $h_f = \frac{flV^2}{2gD}$

So head loss is directly proportional to square of velocity.

1158. Velocity distribution in a turbulent boundary layer follows :

(a) Logarithmic law (b) Parabolic law
(c) Linear law (d) Cubic law

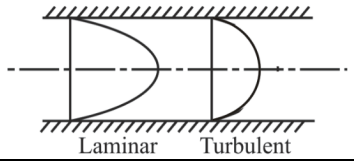
JPSC AE, 10.04.2021, Paper-I
GPSC Executive Engineer 23.12.2018
Vizag Steel (MT) 2017
Nagaland PSC CTSE 2015 Paper-II
MPPSC State Forest Service Exam, 2014
TNPSC ACF 2012, APPSC AEE 2012
UKPSC AE 2012, 2007 Paper–II
CSE Pre-2003

Ans. (a) : Velocity distribution in a turbulent boundary layer follows logarithmic law or power law.

$u = u_{max} + 2.5\ v^* \ln(y/R)$

For laminar = Parabolic law

$$u = u_{max}\left(1 - \frac{r^2}{R^2}\right)$$



1159. In the question ε denotes the average height of surface irregularities, y_w is the thickness of laminar sub-layer and y is the thickness of turbulent boundary layer. A pipe is termed as hydraulically smooth, if

(a) $\frac{\varepsilon}{y} < 0.35$ (b) $\frac{\varepsilon}{y} < 0.05$

(c) $\frac{\varepsilon}{y_w} < 0.35$ (d) $\frac{\varepsilon}{y_w} < 0.25$

OPSC AEE 2019 PAPER - II
TRB Polytechnic Lecturer 2017
APPSC AE Subordinate Service Civil/Mech. 2016
RPSC AE (GWD) 06.05.2016
APPSC AEE Mains 2016 (Civil/Mechanical)
APPSC AEE 2012
J&K PSC Civil Services Pre, 2010

Ans. (d) : $\frac{\varepsilon}{y_w} < 0.25$ – smooth

$0.25 < \frac{\varepsilon}{y_w} < 6$ – Transition

$\frac{\varepsilon}{y_w} > 6$ - Rough

1160. The parameters which determine the friction factor for turbulent flow in a rough pipe are

(a) Froude number and relative roughness
(b) Froude number and Mach number
(c) Reynolds number and relative roughness
(d) Mach number and relative roughness

GPSC Executive Engineer 23.12.2018
APPSC AEE 2012
UKPSC AE 2012 Paper–II
CSE Pre-2010, ESE 2010, GATE 1988

Ans. : (c) Friction factor :-
Friction factor is dimensional terms it is depends two terms.
(i) Reynolds's number
(ii) Relative Roughness of pipe surface

$f = \phi(R_e, k/d)$

1161. The intensity of turbulence refers to :

(a) correlation 'μ' and 'V'
(b) average kinetic energy of turbulence per unit mass
(c) root mean square value of turbulent velocity fluctuations
(d) the Reynolds stresses

TNPSC AE 2019, RPSC ADE 2016
APPSC AEE Mains 2016 (Civil/Mechanical)
TNPSC ACF 2012

Ans. (c) : The turbulence intensity, also often referred to as turbulence level, is defined as

$$I = \frac{u'}{U}$$

Where u' is the root mean square of the turbulent velocity fluctuations and U is the mean velocity.

$$u' = \sqrt{\frac{1}{3}\left[\left(u'_x\right)^2 + \left(u'_y\right)^2 + \left(u'_z\right)^2\right]}$$

1162. The Reynolds's number for turbulent flow is

(a) less than 2000
(b) greater than 4000
(c) between 2000 and 4000
(d) equal to 4000

APGENCO AE, 2017, Vizag Steel (MT) 2017
Rajasthan AE (Nagar Nigam) 2016 Shift-3
TSPSC AEE 2015

Ans : (b)

Flow condition	Pipe flow	open channel flow
Laminar flow	Re ≤ 2000	Re ≤ 500
Transitional flow	2000< Re<4000	500 <Re < 2000
Turbulent flow	Re > 4000	Re > 2000

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1163. Consider the turbulent flow of a fluid through a circular pipe of diameter, D. Identify the correct pair of statements.

I. The fluid is well-mixed
II. The fluid is unmixed
III. $Re_D < 2300$
IV. $Re_D > 2300$

(a) I, III (b) II, IV
(c) I, IV (d) II, III

Assam Engg. College AP/Lect. 18.01.2021
CGPSC Poly.Lect. 22.05.2016
GATE 2014

Ans. (c) : Fluid is well mixed when $Re_D > 2300$ means in turbulent flow.

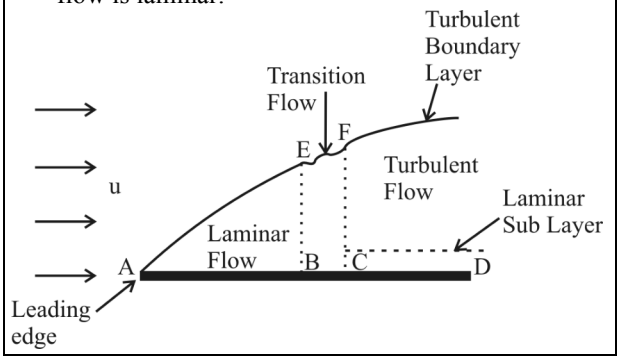
1164. Laminar sub layer may develop for a two dimensional flow over a flat plate. It exists in :

(a) Turbulent Zone
(b) Transition Zone
(c) Laminar Zone
(d) Laminar and Transition Zone

JPSC AE 10.04.2021 Paper-I
UPRVUNL AE 21.08.2016
OPSC Civil Services Pre. 2011

Ans. (a) : Laminar sub layer may develop for a two dimensional flow over a flat plate it exists in Turbulent zone.

- The laminar sublayer, also called the viscous sublayer, is the region of a mainly turbulent flow that is near a no-slip boundary and in which the flow is laminar.



1165. In Moody's diagram the abscissa and ordinate are respectively

(a) Reynolds number and relative roughness
(b) Friction factor and Reynolds number
(c) Relative roughness and Reynolds number
(d) Reynolds number and friction factor

GPSC DEE, Class-2 (GWSSB) 04.07.2021
OPSC Civil Services Pre 2006

Ans. (d) : In Moody's diagram the abscissa and ordinate are respectively Reynolds number and friction factor.

1166. If the fluid particles move in a zig-zag way, the flow is called :

(a) unsteady (b) non-uniform
(c) turbulent (d) incompressible

APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (c) : If the fluid particles move in a zig-zag way then the flow is called turbulent flow, i.e., the direction of fluid particles in randomly.

1167. The most essential feature of a turbulent flow is

(a) large discharge
(b) High velocity
(c) velocity and pressure at a point exhibit irregular fluctuations of high frequency
(d) velocity at a point remains constant with time

APPSC AEE 2012
UKPSC AE 2007 Paper -II

Ans : (c) The most essential feature of a turbulent flow is velocity and pressure at a point exhibit irregular fluctuation of high frequency.
Turbulent flow is that type of flow in which the fluid particles move in zig-zag way. Due to the movement of fluid particles in zig-zag way, the eddies formation take place which are responsible for high energy loss.

1168. In case of turbulent flow of a fluid through a circular tube, as compared to the case of laminar flow, at the same flow rate, the maximum velocity is _____, shear stress at the wall is _____, and the pressure drop across a given length is _____.

(a) higher, higher, higher
(b) higher, lower, lower
(c) lower, higher, higher
(d) lower, higher, lower

APPSC AEE 2012, GATE 1987

Ans : (c) In case of turbulent flow of a fluid through a circular tube, as compared to the case of laminar flow, at the same flow rate the maximum velocity is lower, shear stress at the wall is higher and the pressure drop across a given length in higher.

Laminar flow	Turbulent flow
(i) Flow is smooth	(i) Flow is Randomly
(ii) Energy loss $\propto$ velocity	(ii) Energy loss $\propto$ $(\text{velocity})^2$
(iii) velocity is small	(iii) velocity high
(iv) $R_e < 2000$	(iv) $Re > 4000$
(v) High viscosity	(v) Low viscosity

1169. A turbulent flow is considered steady when

(a) the algebraic sum of velocity fluctuation is zero
(b) the velocity at a point does not change with time
(c) temporal mean velocity at a point remains constant with time
(d) the discharge remains constant

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Mizoram PSC AE/SDO 2014, Paper-I

Ans. (c) : Turbulent flow is always unsteady. Turbulence is on inherently unsteady process since it involves rapid variation of the thermofluid properties. Turbulent flow can be strong when temporal mean velocity at a point remain constant with time.

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1170. The shear stress in turbulent flow is:

(a) Linearly proportional to the velocity gradient
(b) Proportional to the square of the velocity gradient
(c) Dependent on the mean velocity of flow
(d) Due to the exchange of energy between the molecules

HPPSC Asstt. Prof. 18.09.2017
CSE Pre-1994

Ans. (b) : Shear stress in turbulent flow due to Prandtl is $\overline{\tau} = \rho \ell^2 \left(\frac{du}{dy}\right)^2$, where $\ell \rightarrow$ mixing length.

1171. The turbulent flow has

(a) Streak line motion
(b) Parabolic velocity distribution
(c) Random orientation of fluid particles
(d) Small slope of velocity profile at the wall

MPPSC AE 08.11.2015, TSPSC AEE 2015

Ans. (c) : The turbulent flow has Random orientation of fluid particles i.e., zig-zag motion of fluid particles.

1172. Shear stress in a turbulent flow is due to

(a) the viscous property of the fluid
(b) the fluid density
(c) fluctuation of velocity in the direction of flow
(d) fluctuation of velocity in the direction of flow as well as transverse to it

JPSC AE, 10.04.2021, Paper-I, ESE 1997

Ans. (d) : Shear stress in a turbulent flow is due to fluctuation of velocity in the direction of flow as well as transverse to it.

1173. Turbulence in a flow implies

(a) random component of velocity superimposed on the mean flow
(b) unsteadiness of flow
(c) non-uniformity of flow
(d) unsteadiness and non-uniformity of flow

UPPSC AE 13.12.2020, Paper-2
J & K PSC Screening, 2006

Ans. (a) : Turbulence in a flow implies–

- random component of velocity superimposed on the mean flow.
- flow of layers of fluids with different velocities over one another.
- Frictional forces at the confining solid walls.

1174. The flow in which the particles of the fluid travels in different magnitudes and directions as well as from instant to instant is known as

(a) One dimensional flow
(b) Uniform flow
(c) Steady flow
(d) Turbulent flow

Rajasthan Nagar Nigam AE 2016, Shift-II
MPSC Poly. Lect. 2014

Ans. (d) : When fluid particles move in highly disorganized manner leading to rapid mixing of fluid particles then that flow is known as turbulent flow.

- Turbulent flow generally occurs in high velocity.

Uniform flow– Velocity at any given time does not change with respect to space $\left|\frac{dv}{ds}\right|_{t=0} = 0$

Steady flow– Fluid properties do not change with time

$$\frac{dv}{dt} = 0, \frac{dP}{dt} = 0 \ \frac{d\rho}{dt} = 0$$

One dimensional flow– All the flow parameter may be expressed as function of time and one space co-ordinate only.

1175. The friction factor ____ when the flow becomes fully turbulent.

(a) reaches maximum value
(b) reaches minimum value
(c) becomes zero
(d) becomes negative

UPRVUNL AE 05.07.2021

Ans. (a) : The friction factor reaches maximum value when the flow becomes fully turbulent.

1176. In rough turbulent-flow in a pipe, the friction factor would depend upon

(a) Velocity of flow
(b) Pipe diameter
(c) Type of fluid flowing
(d) Pipe condition and pipe diameter

JPSC AE, 10.04.2021, Paper-I

Ans. (d) : Friction factor for rough pipe,

$$\boxed{\frac{1}{\sqrt{f}} = 2\log_{10}\left(\frac{R}{k}\right) + 1.75}$$

This is independent from Reynold no.
Where, k = Average roughness height
In rough turbulent flow in a pipe, the friction factor would depend upon $\Rightarrow$ pipe diameter and pipe condition.

1177. At very high Reynolds numbers exceeding 10^6, variation of friction factor with Reynolds number is best described by the following statement :

(a) increase linearly with Reynolds number
(b) decrease linearly with Reynolds number
(c) is nearly constant with Reynolds number
(d) decreases logarithmically with Reynolds number

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Nearly constant with Reynolds number

1178. The entry length in turbulent flow through the circular pipe is :

(a) 5 to 50 times of Reynolds number
(b) 3 to 30 times of Prandtl number
(c) 10 to 60 times of Diameter of pipe
(d) 5 to 55 times of Nusselt number

NLC GET 17.11.2020, Shift-II

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Ans. (c) : The entry length in turbulent flow is about 10 to 60 times of the diameter remains constant and is generally more flat compared to laminar flow in which it is parabolic.

For turbulent flow the entrance length L_e can be estimated from the equation

$$\frac{L_e}{D} = 4.4(\text{Re})^{1/6}$$

1179. $Q = \frac{\partial u'}{\partial x} = \frac{-\partial v'}{\partial y}$ **for a turbulent flow signifies**

(a) Conservation of bulk momentum transport
(b) Increase in u' in x-direction followed by increase in v' in negative y-direction
(c) Turbulence is anisotropic
(d) Turbulence is isotropic

ESE 2019

Ans. (b) : $Q = \frac{\partial u'}{\partial x} = -\frac{\partial v'}{\partial y}$ for turbulent flow signifies that increase in u' in x-direction is followed by increase in v' in negative y-direction.

1180. Air flows past a cricket ball of 40 mm diameter. It is observed that the boundary layer becomes turbulent at Reynolds number of 200000. If the kinematic viscosity of air is 1.5×10^{-5} m²/s, then the speed of air when the flow becomes turbulent is:

(a) 25 m/s (b) 50 m/s
(c) 75 m/s (d) 100 m/s

AAI Jr. Executive 29.11.2018

Ans. (c) : d = 40 mm

$R_e = 200000$

$v = 1.5 \times 10^{-5}$ m²/sec

$$R_e = \frac{\rho v d}{\mu}$$

$$R_e = \frac{Vd}{v}$$

$$200000 = \frac{V \times 40 \times 10^{-3}}{1.5 \times 10^{-5}}$$

$$V = \frac{3 \times 10^3}{40}$$

V = 75 m/sec

1181. What is the value of momentum correction factor for a turbulent flow in a pipe?

(a) greater than 1 (b) 1.02-1.05
(c) 1.33 (d) 2.00

Kerala PSC Poly. Lect., 2017

Ans. (b) : Momentum Correction Factor (β)– It is defined as the ratio of momentum of the flow per second based on actual velocity to the momentum of the flow per second based on average velocity across a section. It is denoted by β.

$$\boxed{\beta = \frac{1}{AV^2}\int u^2 \cdot dA}$$

$\beta = \frac{4}{3}$, for laminar

β = 1.02-1.05, for turbulent

1182. The friction factor f at a given Reynolds number, in the turbulent region, depends on the ratio of average roughness to the diameter of the pipe. This ratio is known as

(a) Standard roughness (b) Absolute roughness
(c) Average roughness (d) Relative roughness

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : Relative roughness is the ratio of average roughness to the diameter of pipe at the friction factor at given Reynolds number in turbulent region.

1183. In fully developed turbulent flow, if the diameter is halved without changing the flow rate, the frictional drop will change by the factor

(a) 32 times (b) 16 times
(c) 8 times (d) 4 times

TNPSC AE 2017

Ans. (a) : We know that,

$$h_L = \frac{8 f \ell Q^2}{\pi^2 . g . D^5}$$

$$h_L \propto \frac{1}{D^5}$$

then,

$$\frac{h_{L_2}}{h_{L_1}} = \frac{D_1^5}{\left[\frac{D_1}{2}\right]^5} = 32$$

$$h_{L_2} = 32 h_{L_1}$$

1184. In a turbulent flow through a pipe the centre line velocity is 3.61 m/sec and the friction factor f = 0.02, the mean velocity of flow in m/sec is:

(a) 4.80 (b) 3.00
(c) 2.21 (d) 0.96

RPSC ADE 2016

Ans. (b) : For turbulent flow through pipe

$$\frac{V_{max}}{\bar{V}} = 1 + 1.33 \times \sqrt{f}$$

$$\frac{3.61}{\bar{V}} = 1 + 1.33 \times \sqrt{0.02}$$

$\bar{V} = 3.03$ m/s

1185. According to Rouse the distance required for the establishment of a fully developed turbulent flow is given as
(where x is the length of the pipe from the entrance of the pipe upto the section where the flow is established and d is the diameter of the pipe).

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(a) (x/d) = 50 (b) (x/d) = 20
(c) (x/d) = 30 (d) (x/d) = 40

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : A fluid flow need some length to fully develop the velocity profile after passing through components like bends, valves, pumps, turbine etc, the length after obstruction until the flow velocity profile is fully developed is called establishment length or entrance length.

For laminar flow $\frac{x}{d} = 0.07\ Re$

For turbulent flow, $\frac{x}{d} = 50$

1186. For the turbulent flow in a pipe of radius R, what will be the value of 'y' at which the point of velocity is equal to the mean velocity?

(a) 0.223 R (b) 0.45 R
(c) 0.523 R (d) 0.67 R

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :

Point velocity $\left(\frac{u}{u_*}\right)$ = Average Velocity $\left(\frac{\bar{u}}{u_*}\right)$

$$5.75 \log_{10}\left(\frac{y}{R}\right) + 8.5 = 5.75 \log\left(\frac{R}{K}\right) + 4.75$$

$$\left(\frac{u-\bar{u}}{u_*}\right) = 5.75 \log_{10}\left(\frac{y}{R}\right) + 3.75 = 0$$

$$\log\left(\frac{y}{R}\right) = -0.6521$$

$$\frac{y}{R} = 0.2228$$

$$\boxed{y = 0.223\ R}$$

1187. For fully developed turbulent flow through rough pipe at very large Reynolds numbers, friction factor is

(a) Dependent on Reynolds numbers only
(b) Dependent on relative roughness of pipe only
(c) Dependent on density of fluid only
(d) Independent of both Reynolds numbers and relative roughness of pipe

UPRVUNL AE 2016

Ans. (b) : For fully developed turbulent flow through rough pipe at very large Reynolds numbers, friction factor dependent on relative roughness (k) of pipe only

f = F (k only)

1188. The average coefficient of friction drag for turbulent boundary layer is expressed by

(a) $0.664/\sqrt{(Re)}$ (b) $0.664/(Re)^{1/5}$
(c) $0.074/(Re)^{1/4}$ (d) $0.074/\sqrt{(Re)}$

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (c) : The laminar boundary layer drag coefficient at $Re = 1 \times 10^6$ is 0.00135, while it is approximately 0.00445 for turbulent boundary layer.
So the average coefficient of friction drag for turbulent Boundary layer is expressed by: $0.074/(Re)^{1/4}$

1189. The entrance length or length of establishment of turbulent flow in pipes is

(a) Inversely proportional to pipe dia
(b) proportional to Reynolds number (Re)
(c) proportional to $Re^{0.25}$
(d) proportional to $Re^{0.6}$

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (c) : For turbulent flow the entrance length (Le) can be estimated from the equation

$$\frac{Le}{D} = 4.4\,Re^{1/4}$$

$$\boxed{Le \propto Re^{0.25}}$$

For laminar

$$\boxed{\frac{Le}{D} = 0.06\,Re}$$

1190. When the fluid particles are in extreme state of disarray, their velocity fluctuations are particularly violent and erratic then flow will be called as

(a) Rotational flow (b) Unsteady flow
(c) Turbulent flow (d) All the above

Nagaland PSC (CTSE) 2016, Paper-II

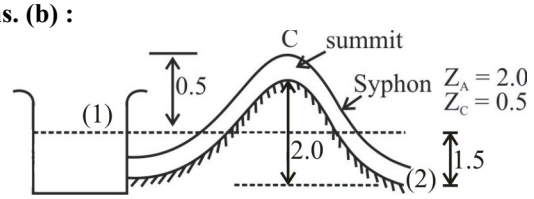
Ans. (c) : When the fluid particles are in extreme state of disarray their velocity fluctuations are particularly violent and erratic then flow will be called as turbulent flow.

1191. A siphon used to empty a tank consists essentially of pipe with its summit 0.5 m above the water surface of the tank and its outlet at 2.0 m below the summit. Neglecting friction and other losses, the velocity in the siphon is

(a) 4.4 m/s (b) 5.4 m/s
(c) 3.1 m/s (d) 3.8 m/s

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) :



$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$Z_1 - Z_2 = \frac{V_2^2}{2g}$$

$$1.5 = \frac{V_2^2}{2g}$$

$$V_2^2 = 1.5 \times 9.8 \times 2$$

$$V_2 = 5.42\,m/sec$$

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1192. In turbulent flow the friction factor 'f' for smooth pipe depends upon :

(a) The Reynolds number and the relatives roughness
(b) The relative roughness only
(c) The size of the pipe and the discharge only
(d) The Reynolds number only

TSPSC Managers, 2015

Ans. (d) : Turbulent Flow–

Friction factor, $f = \frac{0.3164}{(Re)^{1/4}}$ (for smooth pipe)

In rough pipe, $\frac{1}{\sqrt{f}} = 2.0 \log \frac{R}{K} + 1.74$

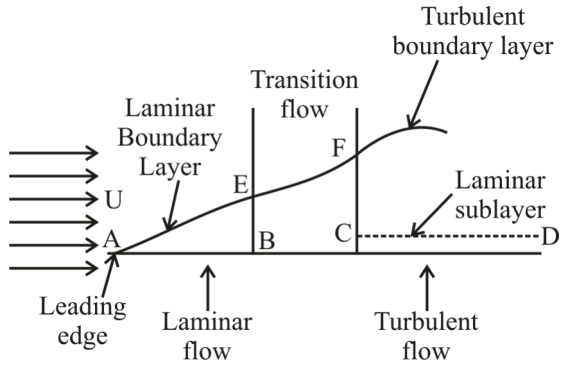
It depends upon both Reynolds number & relative roughness in turbulent rough pipe.

1193. Laminar sub layer acts as:

(a) An insulating medium
(b) A good conductor of heat
(c) A refractory substance
(d) A heat absorber

NPCIL ET, 2015

Ans. (a) :



The laminar sub layer, also called viscous sub layer is the region of a mainly-turbulent flow that is near a no-slip boundary and in which the flow is laminar. Because of this the Reynolds number decreases until at some point the flow crosses the threshold from turbulent to laminar.

1194. The head loss due to turbulent flow as compared to laminar flow is:-

(a) More (b) Less
(c) Equal (d) Unpredictable

UKPSC AE-2013, Paper-II

Ans. (a) : The head loss due to turbulent flow as compared to laminar flow is more.

1195. The frictional resistance in case of turbulent flow is independent of :-

(a) Pressure of flow
(b) Density of fluid
(c) Temperature of fluid
(d) Area of surface in contact

UKPSC AE-2013, Paper-II

Ans. (a) : Frictional resistance also depends on velocity and temperature of flow. It is proportional to the surface area of contact for both laminar and turbulent flows. It is independent of the pressure for both laminar and turbulent flows.

1196. In a turbulent flow through a pipe, the shear stress is:-

(a) Maximum at the centre and decreases linearly towards the wall
(b) Maximum at the centre and decreases logarithmically towards the wall
(c) Maximum mid–way between the centre line and the wall
(d) Maximum at the wall and decreases linearly to zero at the centre

UKPSC AE-2013, Paper-II

Ans. (d) : In a turbulent flow through a pipe, the shear stress is maximum at the wall and decreases linearly to zero at the centre.

1197. The ..... known as hydrodynamically smooth :

(a) $\frac{k}{\delta'} > 0.25$ (b) $\frac{k}{\delta'} > 0.3$
(c) $\frac{k}{\delta'} < 0.25$ (d) $\frac{k}{\delta'} = 6.0$

APPSC AE 04.12.2012

Ans. (c) : $\boxed{\frac{k}{\delta'} < 0.25}$

k = Average height of roughness projections
δ' = Laminar sub layer thickness
i.e., The flow is hydrodynamically smooth.

1198. Using Blasius Equation, the friction factor for turbulent flow through pipes varies as

(a) 1/Re (b) $1/Re^{0.5}$
(c) $1/Re^{0.33}$ (d) $1/Re^{0.25}$

UKPSC AE 2012 Paper–II

Ans. (d) : Friction coefficient (f) $= \frac{0.0791}{(Re)^{1/4}}$

This is given by Blasius equation for turbulent flow in smooth pipe.

1199. Which of the following is not a turbulent model?

(a) Prandtl mixing length model
(b) K-ε model
(c) Large eddy simulation
(d) Steady flow model

OPSC Civil Services Pre. 2011

Ans. (d) : Steady flow model is not a turbulent model.
K- epsilon model (K–ε) turbulence model is the most common model used in computational fluid dynamics to simulate mean flow characteristics for turbulent flow conditions. It is a two equation model which gives a general description of turbulence by means of two transport equations.

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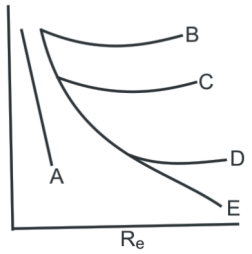
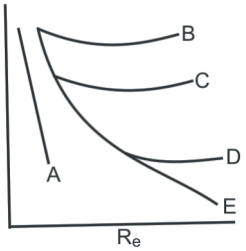
1200. The Moody diagram is shown in Fig. The friction factor for turbulent flow in a smooth pipe is given by curve



(a) A (b) B
(c) E (d) C

ISRO Scientist/Engineer 2010

Ans. (c) : Curve A - For Laminar
Curve B, C, D are for turbulent flow in rough pipe but curve E is for turbulent flow in smooth pipe.



1201. The ratio of shear stress to the rate of shear strain is not constant throughout the flow field if the flow is

(a) potential (b) laminar
(c) turbulent (d) both (a) and (b) above

WBPSC AE, 2007

Ans. (c) : The ratio of shear stress to the rate of shear strain is not constant throughout the flow field if the flow is turbulent.

1202. Prandtl's universal equation is given as

(a) $U = U_{max} + 2.5U_f \log_e\left(\frac{y}{R}\right)$

(b) $U = U_{max} + 3.5U_f \log_e\left(\frac{y}{R}\right)$

(c) $U = U_{max} + 4.5U_f \log_e\left(\frac{y}{R}\right)$

(d) $U = U_{max} + 5.5U_f \log_e\left(\frac{y}{R}\right)$

where U_{max} = centre line velocity
y = distance of pipe from wall
R = radius of pipe

U_f = shear friction velocity $= \sqrt{\frac{\tau_0}{\rho}}$

UKPSC AE 2007 Paper -II

Ans. (a) : Prandtl's universal equation is given as

$$U = U_{max} + 2.5U_f \log_e\left(\frac{y}{R}\right)$$

* This equation is a velocity distribution for turbulent flow in pipes.
* This equation is valid for both smooth and rough pipe boundaries.

10. Boundary Layer Theory

1203. Flow separation in flow past a solid object is caused by

(a) a reduction of pressure to vapour pressure
(b) a negative pressure gradient
(c) a positive pressure gradient
(d) the boundary layer thickness reducing to zero

JPSC AE, 10.04.2021, Paper-I
GPSC Executive Engineer 23.12.2018
TNPSC AE 2017
ISRO Scientist/Engineer 07.05.2017
OPSC AEE 2015 PAPER - II
NPCIL ET, 2015
Nagaland PSC CTSE 2015 Paper-II
APPSC Poly. Lect. 2013
Vizag Steel MT (Re-Exam) 24.11.2013
UKPSC AE 2013, 2012 Paper-II
J&K PSC Civil Services Pre, 2010
TNPSC AE, 2008
OPSC Civil Services Pre 2006
GATE 2002

Ans. (c) : Separation of Boundary layer- It has been observed that the flow is reversed at the vicinity of the wall under certain condition. This phenomenon is termed as separation of boundary layer. Separation takes place due excessive momentum loss near the wall in a boundary layer trying to move downstream against increasing pressure, i.e., $\frac{\partial P}{\partial x} > 0$, which is called adverse pressure gradient.

Condition for flow separation-

1. $\frac{\partial P}{\partial x} > 0$ 2. $\frac{\partial u}{\partial y} < 0$

1204. If x is the distance measured from the leading edge of a flat plate, then laminar boundary layer thickness varies as-

(a) $\frac{1}{x}$ (b) $x^{\frac{4}{5}}$
(c) x^2 (d) $x^{\frac{1}{2}}$

Assam PSC AE (IWT) 14.03.2021
JPSC AE, 10.04.2021, Paper-I
Oil India Ltd. Sr. Engineer (Production) 30.11.2019
RPSC AE 2018, TNPSC AE 2017
CIL MT 26.03.2017
BPSC AE Mains 2017 Paper - V
MPPSC State Forest Service Exam, 2014
UKPSC AE 2013, 2012 Paper-II
ISRO Scientist/Engineer 2012

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APPSC AE 04.12.2012
APPSC AEE 2012, GATE 2002

Ans. (d) : Laminar boundary layer thickness δ (HBL)

$$\delta = \frac{5.0x}{\sqrt{\mathrm{Re}_x}} \qquad \text{(Blasius equation)}$$

$$\delta = \frac{5.0x}{\sqrt{\frac{\rho V x}{\mu}}} = \frac{5.0x^{1/2}}{\sqrt{\frac{\rho V}{\mu}}}$$

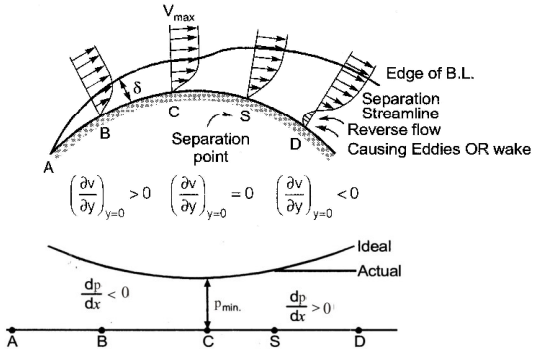
$$\delta \propto x^{1/2}$$

1205. At the point of boundary layer separation:

(a) Shear stress is maximum
(b) Shear stress is zero
(c) Velocity is negative
(d) Density variation is maximum

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II
Gujarat PSC AE 2019
TSPSC AEE 2017
UPPSC AE 12.04.2016 Paper-II
UJVNL AE 2016
APPSC AEE Mains 2016 (Civil/Mechanical)
Assam PSC CCE Pre 2015
GPSC AE 26.07.2015
APPSC AEE 2012, ESE 1996

Ans : (b) At the point of boundary layer separation shear stress is zero.



1206. The nominal thickness of the hydrodynamic boundary layer is generally taken equal to the distance from the surface to a point

(a) where the shear stress becomes constant.
(b) where the velocity of fluid attains value equal to the average velocity.
(c) where the velocity of fluid attains 99% value of the free stream velocity.
(d) of the centroid of the velocity profile.

RPSC AE (GWD) 06.05.2016
KPSC AE. 2015
Kerala PSC AE 06.08.2015
UKPSC AE-2013, Paper-II
RPSC AE GWD, 2011, ESE 1999

Ans. (c) : The boundary layer thickness is taken to be at a distance from the plate surface to a point at which the velocity is given by the velocity of fluid attains 99% value of the free stream velocity.

$u = 0.994\ u_\infty$

1207. Momentum thickness is given by which of following equations?

(a) $\int_0^\delta \left(1-\frac{u}{U}\right)dy$ (b) $\int_0^\delta \frac{u}{U}\left(1-\frac{u}{U}\right)dy$

(c) $\int_0^\delta \frac{u}{U}\left(1-\frac{u^2}{U^2}\right)dy$ (d) none of above

CGPSC AE 25.02.2018
RPSC LECTURER 16.01.2016
UKPSC AE 2007 Paper -II
CSE Pre-2008
ISRO Scientist/Engineer 2006, ESE 1997

Ans. (b) :

Momentum Thickness (θ) = $\int_0^\delta \frac{u}{U}\left(1-\frac{u}{U}\right)dy$

* Energy Thickness (δ_e) = $\int_0^\delta \frac{u}{U}\left(1-\frac{u^2}{U^2}\right).dy$

* Boundary layer thickness (δ), y = δ
when, u = 0.99 U

* Displacement Thickness (δ^*) = $\int_0^\delta \left(1-\frac{u}{U}\right).dy$

These equations associated with Boundary layer theory.

1208. Boundary layer exists in which of the following?

(a) flow of real fluids
(b) flow of ideal fluids
(c) flow over flat surfaces only
(d) pipe flow only

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Mizoram PSC AE/SDO 2014, Paper-I
APPSC Poly. Lect. 2013
APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (a) : When a real fluid flows over a solid body, the condition of no slip occurs at boundary. If boundary is stationary, the velocity of fluid at the boundary will be zero and as we move away from boundary in perpendicular direction, velocity increases to the free stream velocity. It means velocity gradient $\left(\frac{du}{dy}\right)$ will exist.

1209. The predominant forces acting on an element of fluid in the boundary layer over a flat plate placed in a uniform flow are

(a) Inertia and pressure forces
(b) Viscous and pressure forces
(c) Viscous and body forces
(d) Viscous and inertia forces

GPSC Executive Engineer 23.12.2018
CGPSC Asstt. Workshop Supt., 17.07.2016

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Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
ESE 2014, 1996, GATE 1996

Ans. (d) : The predominant forces acting on an element of fluid in the boundary layer over a flute plate in a uniform flow are viscous and inertia forces.

1210. Which of the following the condition for detached flow?

(a) $\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$ (b) $\left(\frac{\partial u}{\partial y}\right)_{y=0} > 0$

(c) $\left(\frac{\partial u}{\partial y}\right)_{y=0} < 0$ (d) $\left(\frac{\partial u}{\partial y}\right)_{y=0} = \infty$

TNPSC AE 2018, CSE Pre-2010
ESE 2011, 2000, GATE 2002

Ans. (c) : Flow separation occurs when the boundary layer travels for enough against an adverse pressure gradient $\left[\frac{\partial P}{\partial x} > 0\right]$ that the speed of the boundary layer relative to the object falls almost to zero. The fluid flow becomes detached from the surface of the object, and instead takes the forms of eddies and vortices.

Thus, condition for detached flow -

$$\frac{\partial P}{\partial x} > 0$$

$$\left.\frac{\partial u}{\partial y}\right|_{y=0} < 0$$

1211. Thickness of turbulent boundary layer at a distance X from the leading edge over flat plate varies as

(a) $X^{4/5}$ (b) $X^{5/4}$

(c) $X^{2/5}$ (d) $X^{5/3}$

OPSC AEE 2019 PAPER - II
MPPSC State Forest Service Exam, 2014
CSE Pre-2007, 2004, 2003
ESE 2000, 1997, 1996

Ans. (a) : We know,

$$\delta = \frac{0.371 \times x}{Re_x^{1/5}} = \frac{0.371 \times x}{\left(\frac{\delta v x}{\mu}\right)^{1/5}}$$

$$\delta \propto \frac{x}{x^{1/5}} \propto x^{4/5}$$

1212. In the boundary layer, the flow is

(a) viscous and rotational
(b) inviscid and irrotational
(c) inviscid and rotational
(d) viscous and irrotational

JPSC AE, 10.04.2021, Paper-I
Assam PSC CCE Pre 2015
TRB Asstt. Prof., 2012, ESE 2006

Ans. (a) : The boundary layer forms due to viscosity so the flow is viscous.

Assuming one dimensional flow along x – axis in x – y plane the rotation.

$$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

$$= \frac{1}{2}\left(0 - \frac{\partial u}{ry}\right)$$

$$= \frac{1}{2}\left(\frac{\partial u}{\partial y}\right) \neq 0$$

It exist in boundary layer and is main reason behind high shear stresses. So in boundary layer the flow is viscous and rotational.

1213. The velocity distribution in the boundary layer is given by $\frac{u}{U} = \frac{y}{\delta}$, where u is the velocity at a distance y from the plate and u = U at y = δ, δ being boundary layer thickness. The displacement thickness is given by

(a) δ (b) $\frac{\delta}{2}$

(c) $\frac{\delta}{3}$ (d) $\frac{2\delta}{3}$

UPPSC AE 13.12.2020, Paper-2
UPSC JWM 2017
UPSC JWM Advt. No.-50/2010
CSE Pre-2005

Ans. (b) : Velocity distribution in layer,

$$\frac{u}{U} = \frac{y}{\delta}$$

Displacement thickness,

$$\delta^* = \int_0^\delta \left(1 - \frac{u}{U}\right) dy = \int_0^\delta \left(1 - \frac{y}{\delta}\right) dy$$

Displacement thickness, $\Rightarrow \left[y - \frac{y^2}{2\delta}\right]_0^\delta \Rightarrow \frac{\delta}{2}$

1214. Separation of boundary layer take place when

(a) pressure and velocity gradient both positive
(b) pressure and velocity gradient both negative
(c) positive pressure gradient and negative velocity gradient.
(d) negative pressure gradient and positive velocity gradient.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper - II
GPSC EE Pre, 28.01.2017

Ans. (c) : Separation of boundary layer takes place when–

(1) Positive pressure gradient $\left(\frac{\partial P}{\partial x} > 0\right)$

(2) Negative velocity gradient $\left(\frac{\partial u}{\partial y} < 0\right)$

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1215. Hydrodynamic and thermal boundary layer thickness is equal when Prandtl number

(a) = 0 (b) < 1

(c) = 1 (d) > 1

RPSC AE (GWD) 06.05.2016
RPSC Lect. (Tech. Edu. Dept.) 2011
ESE 1992

Ans. (c) : Hydrodynamic & thermal Boundary layer thickness is defined as-

$$(Pr)^{\frac{1}{3}} = \frac{\delta}{\delta_t}$$

$\Rightarrow$ If Prandtl number of fluid, Pr = 1, then HBL and TBL thickness are identical.

$$(1)^{\frac{1}{3}} = \frac{\delta}{\delta_t}$$

$$\Rightarrow \delta = \delta_t$$

$\Rightarrow$ If Prandtl no > 1, the TBL thickness is less than the HBL thickness at a given location.

1216. The Blasius equation related to boundary layer theory is a:

(a) third-order linear partial differential equation

(b) third-order nonlinear partial differential equation

(c) third-order nonlinear ordinary differential equation

(d) second-order nonlinear ordinary differential equation

Assam Engg. College AP/Lect. 18.01.2021
DRDO Scientists 2009, GATE 2015

Ans. (c) : Blasius equation

$$\frac{d^3f}{d\eta^3} + \frac{1}{2} f \frac{d^2f}{d\eta^2} = 0$$

It is third order non linear differential equation.

1217. For a linear distribution of velocity in the boundary layer on a flat plate, the ratio of displacement thickness to nominal thickness is

(a) 1/2

(b) 1/3

(c) 1/4

(d) 2/3

UPPSC AE 12.04.2016 Paper-II
MPPSC AE 08.11.2015, ESE 2005

Ans. (a) : Let, δ = Nominal thickness or boundary layer thickness

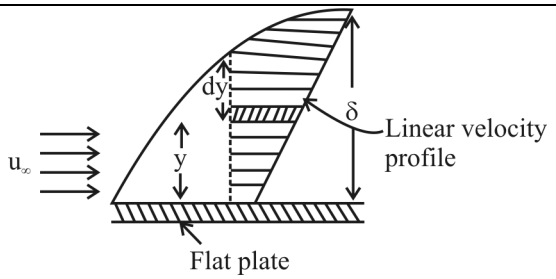
δ^* = Displacement thickness

Given, For a linear distribution of velocity in the boundary layer on a flat plate-

$$\frac{u}{u_\infty} = \frac{y}{\delta},$$

Where u = velocity of fluid at elemental strip

u_∞ = Free stream velocity



$$\because \quad \delta^* = \int_0^\delta \left(1 - \frac{u}{u_\infty}\right) dy$$

$$= \int_0^\delta \left(1 - \frac{y}{\delta}\right) dy$$

$$\delta^* = \delta - \frac{\delta^2}{2\delta} = \delta - \frac{\delta}{2} = \frac{\delta}{2}$$

Therefore; $\frac{\delta^*}{\delta} = \frac{\delta/2}{\delta} = \frac{1}{2} \quad \Rightarrow \frac{\delta^*}{\delta} = \frac{1}{2}$

1218. The transition Reynolds number for flow over a flat plate is 5×10^5. What is the distance from the leading edge at which transition will occur for flow of water with a uniform velocity of 1 m/s? (For water, the kinematic viscosity, $\upsilon = 0.86 \times 10^{-6} m^2/s$)

(a) 0.43m (b) 1m

(c) 43m (d) 103m

ISRO Scientist/Engineer 2009, 2008, 2006
UKPSC AE 2012 Paper–II
ESE 1994

Ans. (a) : Reynold number

$Re = 5 \times 10^5$

$V = 1$ m/s

$\upsilon = 0.86 \times 10^{-6}$ m^2/s

$$Re = \frac{VD}{\upsilon} = 5 \times 10^5$$

$$D = \frac{5 \times 10^5 \times 0.86 \times 10^{-6}}{1}$$

D = 0.43 m

1219. The drag coefficient for laminar flow varies with Reynolds Number (Re) as

(a) $Re^{1/2}$ (b) Re

(c) Re^{-1} (d) $Re^{-1/2}$

JPSC AE, 10.04.2021, Paper-I
CSE Pre-2003

Ans. (d) : • Drag coefficient for laminar flow,

$$C_D = 1.328\, Re^{-1/2}$$

$$\boxed{C_D \propto Re^{-1/2}}$$

• Drag coefficient for turbulent flow,

$$C_D = 0.074\, Re^{-1/5}$$

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1220. For flow along flat plate critical Reynolds number at which the transition from Laminar to Turbulent flow takes place is :

(a) $Re = 5 \times 10^5$ (b) $Re = 2000$

(c) $Re = 4 \times 10^5$ (d) $Re = 5 \times 10^{-5}$

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

UKPSC AE 2012 Paper–II

Ans. (a) : For flat plate –

$$\boxed{Re_x = \frac{\rho U x}{\mu}}$$

Critical Reynolds number

$Re < 5 \times 10^5$ Laminar

$Re > 5 \times 10^5$ Turbulent

For pipe-

$$\boxed{Re = \frac{\rho V D}{\mu}}$$

$Re < 2000 \rightarrow$ Laminar

$2000 < Re < 4000 \rightarrow$ Transition

$Re > 4000 \rightarrow$ Turbulent.

1221. The maximum thickness of boundary layer in case of a pipe of radius R is:

(a) R (b) 1/3R

(c) 0.1 R (d) 2/3 R

OPSC AEE 2015 PAPER - II

ISRO Scientist/Engineer 2010

Ans. : (a) The maximum thickness of boundary layer in case of a pipe of radius is R.

1222. For laminar flow over a flat plate, the thickness of the boundary layer at a distance from the leading edge is found to be 5 mm. The thickness of the boundary layer at a downstream section, which is at twice the distance of the previous section from the leading edge, will be-

(a) 10 mm (b) $5\sqrt{2}$ mm

(c) $5/\sqrt{2}$ mm (d) 2.5 mm

ISRO Scientist/Engineer 2006

ESE 1994

Ans. (b) : Laminar flow over a flat plate–

Thickness of the boundary layer at distance from the leading edge is found to be 5 mm.

$$\boxed{\delta \propto \sqrt{x}} \qquad [\because \delta_1 = 5 \text{ mm}]$$

$$\frac{\delta_1}{\delta_2} = \sqrt{\frac{x_1}{x_2}} = \sqrt{\frac{x_1}{2x_1}}$$

$$\frac{\delta_1}{\delta_2} = \frac{1}{\sqrt{2}} \Rightarrow \boxed{\delta_2 = \delta_1\sqrt{2}}$$

$$\delta_2 = 5\sqrt{2} \text{ mm}$$

1223. What will be the momentum thickness for the velocity distribution in boundary layer of :

$\frac{u}{U} = \frac{y}{\delta}$, **where u = velocity at a distance y from the plate and u = U at y = δ (Boundary layer thickness)**

(a) $\frac{\delta}{5}$ (b) $\frac{\delta}{3}$

(c) $\frac{\delta}{6}$ (d) $\frac{\delta}{2}$

SJVN ET 2019, CSE Pre-2005

Ans. (c) : $\frac{u}{U} = \frac{y}{\delta}$, momentum thickness

$$\theta = \int_0^\delta \frac{u}{U}\left(1 - \frac{u}{U}\right)dy$$

$$\theta = \int_0^\delta \frac{y}{\delta}\left(1 - \frac{y}{\delta}\right)dy$$

$$\theta = \left(\frac{y^2}{2\delta} - \frac{y^3}{3\delta^2}\right)\Bigg|_0^\delta = \left(\frac{\delta}{2} - \frac{\delta}{3}\right) = \frac{\delta}{6}$$

1224. The boundary layer is called turbulent boundary layer if.

(a) Reynolds number is more than 2000

(b) Reynolds number is more than 4000

(c) Reynolds number is more than 5×10^5

(d) Reynolds number is more than 6×10^5

TNPSC AE 2018

UKPSC AE-2013, Paper-II

Ans. (c) : For flow past a flat plate, the transition from laminar to turbulent begins when the critical Reynolds number reaches 5×10^5. The boundary layers changes from laminar to turbulent at this point.

1225. For a fluid flow over a flat pipe with zero pressure gradient, the boundary layer thickness $\left(\frac{\delta}{x}\right)$ is proportional to

(a) $\sqrt{Re_x}$ (b) $\frac{1}{\sqrt{Re_x}}$

(c) Re_x (d) $\frac{1}{Re_x}$

APPSC-AE-2019

APPSC AEE Screening Test 2016

Ans. (b) :

$$\delta(x) = \frac{5x}{\sqrt{Re_x}} \text{ for laminar boundary layer}$$

i.e. $\frac{\delta(x)}{x} \propto \frac{1}{\sqrt{Re_x}}$

1226. In the region of boundary layer on a flat plate surface where velocity is not zero, the viscous force is

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(a) Less than inertial force
(b) More than inertial force
(c) Equal in magnitude
(d) Not predictable

UPPSC AE 12.04.2016 Paper-II
ESE 1995

Ans : (b In the region of boundary layer on a flat plate surface where velocity is not zero, the viscous force is more than inertial force.

At nearest to the wall viscous forces are more than inerita forces.

1227. The velocity distribution in a turbulent boundary layer is given by $u/v = (y/\delta)^{1/7}$. What is the displacement thickness δ^*

(a) δ (b) $\delta/7$
(c) $7\delta/8$ (d) $\delta/8$

ISRO Scientist/Engineer 24.05.2014
ESE 2008

Ans. (d) : given- Turbulent Boundary

$$\frac{u}{v}=\left(\frac{y}{\delta}\right)^{1/7}$$

Displacement Thickness

$$\delta^*=\int_0^{\delta}\left(1-\frac{u}{v}\right)dy$$

$$=\int_0^{\delta}\left(1-\left(\frac{y}{\delta}\right)^{1/7}\right)dy$$

$$=\left[y-\frac{7}{8}\cdot\left(\frac{y^8}{\delta}\right)^{\frac{1}{7}}\right]_0^{\delta}$$

$$=\left[\delta-\frac{7}{8}\cdot\left(\delta^8\times\frac{1}{\delta}\right)^{\frac{1}{7}}\right]$$

$$=\delta-\frac{7}{8}\delta$$

$$\boxed{\delta^*=\frac{\delta}{8}}$$

1228. A laminar boundary layer occurs over a flat plate at zero incidence of the flow. The thickness of boundary layer at a section 2 m from the leading edge is 2 mm. The thickness of boundary layer at a section 4 m from the leading edge will be

(a) $2\times(2)^2$ mm (b) $2\times(2)^{1/2}$ mm
(c) $2\times(2)^{4/5}$ mm (d) $2\times(2)^{1/5}$ mm

Nagaland PSC CTSE 2015 Paper-II
ESE 1995

Ans. (b) : We know that,

$$\delta=\frac{5x}{\sqrt{Re_x}}=\frac{5x}{\sqrt{\rho vx/\mu}}=\frac{5\sqrt{x}}{\sqrt{\rho v/\mu}}$$

$$\delta\propto\sqrt{x}$$

Therefore,
Thickness of boundary layer at 4 mm

$$\text{From leading edge}=2\times\left(\frac{4}{2}\right)^{1/2}$$

$$=2\times2^{1/2}$$

1229. Flow separation is likely to take place when the pressure gradient in the direction of flow is

(a) Zero
(b) Adverse
(c) Slightly favourable
(d) Strongly favourable

JPSC AE, 10.04.2021, Paper-I
CSE Pre-1998

Ans. (b) : Flow separation is likely to take place when the pressure gradient in the direction of flow is adverse.

1230. The region between the boundary surface of the solid body and the separation streamline is:

(a) Crest (b) Wake
(c) Boundary Layer (d) Nappe

OPSC AEE 2019 PAPER - II
CGPSC AE 26.04.2015 PAPER-II

Ans : (b) : The region between the boundary surface of the solid body and the separation streamline is called wake.

1231. The concept of _____ thickness is used in the design of ducts, intakes of air-breathing engines, wind tunnels, etc. by first assuming a frictionless flow and then enlarging the passage walls by this thickness so as to allow the same flow rate.

(a) u = 0.99 U (b) Displacement
(c) Momentum (d) Friction

RPSC ACF & FRO, 26.02.2021

Ans. (b) : Displacement thickness– It is a distance measured normal to boundary by which solid boundary should displace in order to compensate reduction in mass flow rate due to lifting of plate by distance.

$$\boxed{\delta^*=\int_0^{\delta}\left(1-\frac{u}{U_\infty}\right)dy}$$

1232. The ratio of thickness of thermal boundary layer to the thickness of hydrodynamic boundary layer is equal to $(Pr)^n$, where (n) is equal to:

(a) $-\left(\frac{1}{3}\right)$
(b) $-\left(\frac{2}{3}\right)$
(c) $\left(\frac{1}{3}\right)$
(d) $-(1)$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

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Ans. (a) : $\delta \rightarrow$ Hydrodynamic boundary layer

$\delta_{th} \rightarrow$ Thermal boundary layer

$$\boxed{\frac{\delta}{\delta_{th}} = (Pr)^{1/3}}$$

$$\boxed{\frac{\delta_{th}}{\delta} = (Pr)^{-1/3}} \quad \left\{Pr = \frac{\mu Cp}{k}\right\}$$

1233. Which one of the following is not the methodology of control separation of flow from boundary in the application of aerofoils?

(a) Streamlining of blunt body shapes
(b) Fluid ejection from the boundary layer
(c) Suction of fluid from the boundary layer
(d) Creating a motion of the boundary wall

ESE 18.07.2021

Ans. (b) : The methodology of control separation of flow in the application of aerofoils-

1. Streamlining of blunt objects.
2. Fluid injection into Boundary layer.
3. Suction of retarded fluid form Boundary layer.
4. Creating a motion of boundary wall thereby giving energy to the retarded fluid.

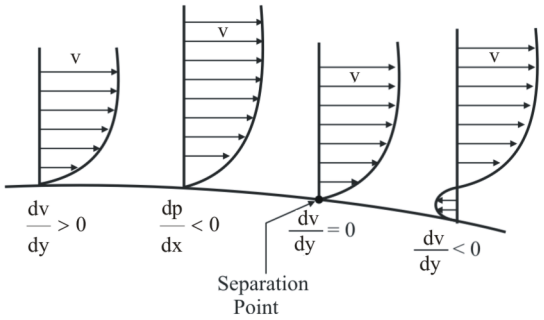
Hence answer should be (b) in which in place of injection it is given as ejection.

1234. The flow is said to have reached a separation point, when the velocity gradient at the surface becomes :

(a) positive (b) non-zero
(c) zero (d) negative

NLC GET 17.11.2020, Shift-II

Ans. (c) : Separation of flow occurs in a region of adverse pressure gradient. The separation point S is determined from the condition $\left(\frac{dv}{dy}\right)_{y=0} = 0$



If $\frac{dv}{dy} = +ve \Rightarrow$ separation has not occurred

If $\frac{dv}{dy} = 0$ Flow is at the verge of separation.

If $\frac{dv}{dy} = -ve$ Flow has separated.

1235. In liquid metal heat transfer, the thickness of the thermal boundary layer in comparison with the thickness of the momentum boundary is

(a) very small
(b) very large
(c) of the same order
(d) dependent on the thermal conductivity of the surface

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : In liquid metal heat transfer, the thickness of the thermal boundary layer in comparison with the thickness of the momentum boundary is very large.

1236. An experiment is conducted with a fluid of density 1 kg/m^3 at 10 m/s velocity. The free stream static pressure is 100 kPa and the local static pressure is 101 kPa. What is the pressure coefficient at the location?

(a) 70 (b) 80
(c) 20 (d) 50

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Pressure coefficient

$$C_P = \frac{\Delta P}{\frac{1}{2}\rho u_\infty^2} = \frac{P_{local} - P_{freestream}}{\frac{1}{2}\rho u_\infty^2}$$

$$C_P = \frac{(101-100)\times 10^3}{\frac{1}{2}\times 1\times 10^2} = 20$$

$C_P = 20$

1237. What is the energy thickness for the velocity distribution in the boundary layer given by uU = yδ. Where, u is velocity at a distance y from the plate and u = U at y = δ, where δ = boundary layer thickness.

(a) δ/2 (b) δ/4
(c) δ/3 (d) δ/6

Oil India Senior Officer 23.12.2020

Ans. (b) : Given, Boundary layer profile,

$$\frac{u}{U} = \frac{y}{\delta}$$

Energy thickness,

$$\theta^* = \int_0^\delta \frac{u}{U}\left[1 - \frac{u^2}{U^2}\right]dy$$

$$= \int_0^\delta \frac{y}{\delta}\left[1 - \left(\frac{y}{\delta}\right)^2\right]dy$$

$$= \left[\frac{y^2}{2\delta} - \frac{y^4}{4\delta^3}\right]_0^\delta$$

$$\Rightarrow \quad \frac{\delta}{2} - \frac{\delta}{4} = \frac{\delta}{4}$$

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1238. In liquid metals, thermal boundary layer develops much faster than velocity boundary layer due to

(a) Lower value of Nusselt number
(b) Higher value of Prandtl number
(c) Lower value of Prandtl number
(d) Higher value of Nusselt number

ESE 2019

Ans. (c) : Prandtl number is given by

$$\frac{\delta}{\delta_{th}} = (Pr)^{1/3}$$

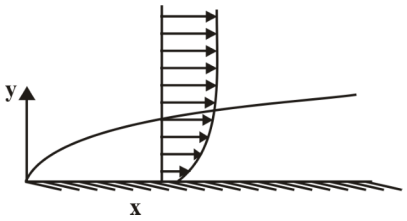
Where, δ = Hydrodynamic boundary layer thickness
δ_{th} = Thermal boundary layer thickness
For liquid metal, Prandtl number is very low since they have high thermal conductivity
When $Pr < 1$; $\delta_{th} > \delta$

1239. For a boundary layer flow over a flat plate with zero pressure gradient, if U is the free stream velocity, y is the coordinate perpendicular to the flow direction, and δ is the boundary layer thickness, at $y = \delta$,

(a) $\frac{du}{dy} = 0$ (b) $u = U$
(c) Both (a) and (b) (d) None of these

APPSC-AE-2019

Ans. (c) :



As per definition of boundary layer thickness
at $y = \delta$, $u = 0.99\ U \approx U$
As there is no variation in velocity beyond 'δ'

$$\text{at } y = \delta,\ \frac{du}{dy} = 0$$

1240. Within a boundary layer :

(a) The order of viscous force is much higher than that of the inertia force
(b) The order of viscous force is equal to that of the inertia force
(c) The order of viscous force is much less than that of the inertia force
(d) Both the forces are zero.

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (b) : Within Boundary layer the order of viscous force is equal to that of inertia force.

1241. For flow over a plate the hydrodynamic boundary layer thickness is 0.5 mm. The dynamic viscosity is 25×10^{-6} kg/m, specific heat is 2.0 kJ/kg and thermal conductivity is 0.05 W/mK. The thermal layer thickness would be:

(a) 0.1 mm
(b) 0.5 mm
(c) 1 mm
(d) 2 mm

HPPSC ADF 05.03.2019

Ans. (b) : Prondtl Number, $Pr = \frac{\mu C_p}{k}$

$$= \frac{25\times10^{-6}\times2.0\times10^{3}}{0.05} = 1$$

$$(Pr)^{1/3} = \frac{\delta}{\delta_t} = 1$$

$\therefore \quad \delta_t = \delta \Rightarrow \delta_t = 0.5$ mm

1242. A fluid near a solid wall has approximated velocity profile given by $u(y) = U_\infty \sin\left(\frac{\pi y}{2\delta}\right), 0 \le y \le \delta.$

The walls shear stress is given by :

(a) $\tau_w = \frac{\pi\mu U_\infty}{2\delta}$

(b) $\tau_w = \frac{3\pi\mu U_\infty}{\delta}$

(c) $\tau_w = \frac{2\pi\mu U_\infty}{\delta}$

(d) $\tau_w = \frac{\pi\pi U_\infty}{\delta}$

BHEL ET 2019

Ans. (a) : Given - velocity profile

$$u(y) = U_\infty \sin\left(\frac{\pi y}{2\delta}\right), 0 \le y \le \delta$$

$$\frac{du}{dy} = U_\infty \times \frac{\pi}{2\delta} \times \cos\left(\frac{\pi y}{2\delta}\right)$$

from $0 \le y \le \delta$

so, at $y = 0$

$$\frac{du}{dy} = U_\infty \times \frac{\pi}{2\delta}\cos\left(\frac{\pi\times0}{2\delta}\right)$$

$$\frac{dy}{du} = U_\infty \frac{\pi}{2\delta}\times1$$

Wall shear stress $\tau_W = \mu\frac{du}{dy} = \mu\times\frac{U_\infty\pi}{2\delta}$

$$\tau_W = \frac{\pi\mu U_\infty}{2\delta}$$

1243. Which statement is correct for outside regime of the boundary layer of fluid flow?

(a) Velocity is zero
(b) Shear stress is zero
(c) Velocity is not constant
(d) Shear stress is proportional to velocity gradient

OPSC AEE 2019 PAPER - II

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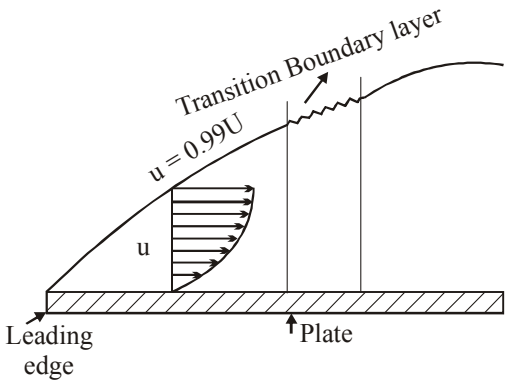
Ans : (b) : At the outer regime of the boundary layer of fluid shear stress will be zero, because at outer regime

$$\left.\frac{du}{dy}\right|_{y=\delta} = 0, \qquad (\tau = 0)$$



1244. In the laminar flow of air (Pr = 0.7) over a heated plate, if δ and δ_τ denote, respectively, the hydrodynamic and thermal boundary layer thickness, then

(a) $\delta = \delta_\tau$ (b) $\delta < \delta_\tau$

(c) $\delta > \delta_\tau$ (d) $\delta = 0$ but $\delta_\tau = \neq 0$

GPSC Executive Engineer 23.12.2018

Ans. (b) : The hydrodynamic and thermal boundary layer thickness can be related as,

$$\text{Prandtl Number, } Pr = 0.7 = \left(\frac{\delta}{\delta_\tau}\right)^3$$

$$(0.7)^{1/3} = 0.887 = \left(\frac{\delta}{\delta_\tau}\right)$$

$$\frac{\delta}{\delta_\tau} < 1$$

$$\boxed{\delta < \delta_\tau}$$

1245. The boundary layer thickness in laminar flow is proportional to

(a) $Re^{0.2}$ (b) $Re^{0.5}$

(c) $Re^{-0.5}$ (d) $Re^{-0.2}$

RPSC Vice Principal ITI 2018

Ans. (c) : The boundary layer thickness in laminar flow,

$$\delta = \frac{5x}{\sqrt{R_e}}$$

$$\delta \propto \frac{1}{\sqrt{R_e}}$$

But in turbulent flow,

$$\delta = \frac{0.377x}{R_e^{0.2}}$$

$$\delta \propto \frac{1}{R_e^{0.2}}$$

1246. Within boundary layer for a steady incompressible flow, the Bernoulli equation

(a) holds because the flow is steady

(b) holds because the flow is transitional

(c) holds because the flow is transitional

(d) does not hold because the flow is frictional

GPSC Executive Engineer 23.12.2018

Ans. (d) : Within boundary layer for a steady incompressible flow, the Bernoulli's equation does not hold because the flow is frictional.

1247. The velocity corresponding to Reynold's number of 2800, is called

(a) Sub-sonic velocity

(b) Super-sonic velocity

(c) Lower critical velocity

(d) Higher critical velocity

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : When Reynold's number is between 2000 and 2800, the flow is neither laminar nor turbulent. The velocity corresponding to Reynold's number of 2000 is called lower critical velocity and the velocity corresponding to Reynold's number of 2800 is called higher critical velocity.

1248. The celerity (velocity) of a pressure wave in a fluid is given by

(a) $\sqrt{(K.p)}$ (b) $\sqrt{(K/p)}$

(c) $\sqrt{(p/K)}$ (d) K.p

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : The celerity (velocity) of a pressure wave in a fluid is given by $\sqrt{K/\rho}$.

1249. In laminar flow over a flat plate.

(a) The thermal boundary layer and hydrodynamic boundary layers are of equal thickness

(b) The thermal boundary layer is thicker if the Prandtl number is greater than one.

(c) The thermal boundary layer is thicker if the Prandtl number is less than one.

(d) The thermal boundary layer is always thinner in the laminar region.

HPPSC Asstt. Prof. 20.11.2017

Ans. (c) : In laminar flow over a flat plate the thermal boundary layer is thicker if the prandtl number is less than one–

$$\boxed{\frac{\delta}{\delta_t} = (P_r)^{1/3}}$$

$$P_r < 1 \quad \Rightarrow \quad \boxed{\delta < \delta_t}$$

1250. Separation occurs when

(a) Velocity of sound is reached

(b) Boundary layer comes to rest

(c) Cross-section of a channel reduced

(d) Pressure reaches a minimum

Nagaland PSC CTSE 2017 Paper-2

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Ans. (b) : Separation occurs when, boundary Layer comes to rest.
Explanation:- Flow, separation occurs, when boundary layer travels for enough against an adverse pressure gradient, that the speed of the boundary layer relative of the objects falls almost to zero.

1251. The transition Reynolds number for flow over a flat plate is 6.5×10^5. What is the distance from the leading edge at which transition will occur for flow of water with a uniform velocity of 1.5 m/s? [For water the kinematic viscosity $v = 0.826\times10^{-6}$ m^2/s]?
(a) 4.3 m (b) 1 m
(c) 3.3 m (d) 0.36 m

UPRVUNL AE 21.08.2016

Ans. : (d) $Re = 6.5 \times 10^5$
$V = 1.5$ m/sec
$\nu = 0.826 \times 10^{-6}$ m^2/sec

$$Re = \frac{\rho V x}{\mu}$$

$$Re = \frac{Vx}{(\mu/\rho)}$$

$$Re = \frac{Vx}{\nu}$$

$$6.5 \times 10^5 = \frac{1.5\times x}{0.826\times10^{-6}}$$

$x = 0.36$ m

1252. In boundary layer theory the energy thickness (δ_e) is given by
(a) $\int_0^\infty \frac{u}{U}\left(1-\frac{u}{U}\right)dy$ (b) $\int_0^\infty \left(1-\frac{u}{U}\right)dy$
(c) $\int_0^\infty \frac{u}{U}\left(1-\frac{u}{U}\right)^2 dy$ (d) $\int_0^\infty \frac{u}{U}\left(1-\frac{u^2}{U^2}\right)dy$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : It is the vertical distance by which the solid plate must be shifted, in case of ideal to compensate for the loss of energy thickness, due to boundary layer formation.

$$\delta_e = \int_0^\infty \frac{u}{U}\left(1-\frac{u^2}{U^2}\right)dy$$

1253. The displacement thickness at section for an air stream (ρ = 1.4 kg/m^3) moving with a velocity of 12m/s over a flat plate is 0.6 mm what is the loss of mass rate of flow of air due to boundary layer formation in kg per meter width of plate per second?
(a) 10.08×10^{-3} (b) 6×10^{-3}
(c) 8.16×10^{-3} (d) 6.08×10^{-3}

UPRVUNL AE 21.08.2016

Ans. : (a) Mass flow rate =
$\rho Q = \rho \times \delta^* \times$ velocity per meter width of the plate

$$= 1.4\times\left(\frac{0.6}{1000}\right)\times12$$

$= 10.08 \times 10^{-3}$ kg/s

1254. Thickness of laminar boundary layer on flat plate:
(a) Decreases with increase in distance from leading edge
(b) Increases with increase in distance from leading edge
(c) Increases with increase in Reynold number at given location
(d) Constant with increase in distance from leading edge

UPRVUNL AE 2016

Ans. (b) : Thickness of laminar boundary layer on flat plate increases with increase in distance from leading edge.
We know that boundary layer thickness

$$\delta = \frac{5.x}{\sqrt{Re_x}}$$

$$\delta = \frac{5.x}{\sqrt{\frac{\rho\times u\times x}{\mu}}}$$

$$\delta \propto x^{1/2}$$

Where x is distance from leading edge of plate.

1255. The concept of boundary layer is introduced by:
(a) Prandtl (b) Reynolds
(c) Euler (d) Nusselt

TSPSC Managers, 2015

Ans. (a) : The concept of boundary layer is introduced by prandtl.

$$(Pr) = \frac{\mu C_p}{k}$$

or $$\boxed{(Pr)^{1/3} = \frac{\delta}{\delta_t}}$$

Pr = Prandtl number
δ = Thermal boundary layer thickness
δ_t = Hydrodynamic boundary layer thickness

1256. The non dimensional number associated with hydrodynamic boundary layer is
(a) Reynolds number (b) Prandtl number
(c) Boit number (d) Nussel number

TNPSC AE 2014

Ans. (a) : Reynolds number associated with hydrodynamic boundary layer.

1257. Separation of flow occurs due to reduction in pressure gradient to
(a) Zero
(b) Negligibly low value
(c) The extent such that vapour formation starts
(d) None of these

MPPSC State Forest Service Exam, 2014

Ans. (c) : Flow separation occurs when the boundary layer travels for enough against an adverse pressure gradient that the speed of boundary layer is zero, reduction in pressure gradient to extent the formation of vapours initially.

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1258. The distance from the surface at which the local velocity reaches 99 percent of the external velocity is

(a) Momentum thickness
(b) Free stream velocity
(c) Boundary layer thickness
(d) Momentum diffusivity

TNPSC AE 2013

Ans. (c) : The distance from the surface at which the local velocity reaches 99 percent of the external velocity is known as boundary layer thickness.

1259. The flow within the boundary layer is:-

(a) Only turbulent (b) Only laminar
(c) Either (a) or (b) (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) :

1260. If $A = u\frac{\partial u}{\partial x}, B = v\frac{\partial u}{\partial y}$ and $C = \upsilon\frac{\partial^2 u}{\partial y^2}$ then momentum equation of the boundary layer can be given as follows:-

(a) A + B + C = 0 (b) A + B – C = 0
(c) $A + B - C^2 = 0$ (d) A.B = C

UKPSC AE-2013, Paper-II

Ans. (b) : We know that momentum equation of the boundary layer is given as,

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \upsilon\frac{\partial^2 u}{\partial y^2}$$

1261. The velocity profile is approximated by a cubic parabola $\frac{u}{U} = \frac{3}{2}\left(\frac{y}{\delta}\right)^2 - \frac{1}{2}\left(\frac{y}{\delta}\right)^3$, where the displacement thickness for the profile is

(a) $\frac{3}{8}\delta$ (b) $\frac{5}{8}\delta$
(c) $\frac{11}{8}\delta$ (d) None of the above

UKPSC AE 2012 Paper–II

Ans. (b) : We know that displacement thickness of boundary layer is given as,

$$\delta^* = \int_0^\delta \left(1 - \frac{u}{U_\infty}\right) dy \qquad .....(1)$$

Given,

$$\frac{u}{U_\infty} = \frac{3}{2}\left(\frac{y}{\delta}\right)^2 - \frac{1}{2}\left(\frac{y}{\delta}\right)^3$$

Putting this value in equation (1)

$$\delta^* = \int_0^\delta \left[1 - \left(\frac{3}{2}\left(\frac{y}{\delta}\right)^2 - \frac{1}{2}\left(\frac{y}{\delta}\right)^3\right)\right] dy$$

$$= [y]_0^\delta - \left[\frac{3}{2}\times\frac{y^3}{3\cdot\delta^2} - \frac{1}{2}\cdot\frac{y^4}{4}\cdot\frac{1}{\delta^3}\right]_0^\delta$$

$$= \left[\delta - \left\{\frac{3}{2}\cdot\frac{\delta^3}{3\cdot\delta^2} - \frac{1}{8}\cdot\frac{\delta^4}{\delta^3}\right\}\right]$$

$$= \left[\delta - \frac{1}{2}\delta + \frac{1}{8}\delta\right]$$

$$= \left[\frac{8\delta - 4\delta + \delta}{8}\right]$$

$$= \frac{5\delta}{8}$$

1262. Which one of the following statement does not characterize the thermal boundary layer?

(a) Thermal boundary layer develops when a fluid at a specified temperature flows over a surface that is at different temperatures
(b) Temperature variation in the direction normal to the surface is significant
(c) The thickness of the thermal boundary layer increases in the flow direction
(d) The relative thickness of the velocity and the thermal boundary layers are always equal to 1.

RPSC AE GWD, 2011

Ans. (d) : Hydrodynamic boundary layer thickness δ
Thermal boundary layer thickness = δ_{th}
when Pr = 1
then $\frac{\delta}{\delta_{th}} = (Pr)^{1/3}$

$\boxed{\delta = \delta_{th}}$ only when Pr = 1 not always

1263. For laminar boundary layer over an entire length of plate, the drag coefficient C_D is given by :

(a) $\frac{1.328}{\sqrt{Re_L}}$ (b) $\frac{0.074}{(Re_L)0.25}$
(c) $\frac{0.37}{(Re_L)^{1.5}}$ (d) $\frac{0.455}{(Re_L)^2}$

J&K PSC Civil Services Pre, 2010

Ans. (a) : For laminar boundary layer, drag coefficient (C_D)

$$(C_D) = \frac{1.328}{\sqrt{Re_L}}$$

For turbulent boundary layer

Drag coefficient $(C_D) = \frac{0.074}{(Re_L)^{1/5}}$

1264. With increase in flow velocity of fluid the boundary layer thickness will :

(a) decrease
(b) increase
(c) does not change
(d) first increase then decrease

J&K PSC Civil Services Pre, 2010

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Ans. (a) : From Blasius, boundary layer thickness is given by–

$$\delta = \frac{5x}{\sqrt{Re_x}} = \frac{5x}{\sqrt{\frac{\rho V x}{\mu}}}$$

$$\delta \propto \frac{1}{\sqrt{V}}$$

1265. Select the correct option for the velocity profile

$$\frac{u}{v} = \frac{5}{2}\left(\frac{y}{\delta}\right) - \frac{1}{2}\left(\frac{y}{\delta}\right)^2 + \frac{2}{3}\left(\frac{y}{\delta}\right)^3$$

(a) Flow has separated
(b) Flow is on the verge of a separation
(c) Flow will not separate
(d) All of the above

UPSC JWM Advt. No.-50/2010

Ans. (c) : As we know,

$$\frac{u}{v} = \frac{5}{2}\left(\frac{y}{\delta}\right) - \frac{1}{2}\left(\frac{y}{\delta}\right)^2 + \frac{2}{3}\left(\frac{y}{\delta}\right)^3$$

Let, $\frac{u}{v} = U$

Differential w.r.t, y

$$\frac{\partial U}{\partial y} = \frac{5}{2}\left(\frac{1}{\delta}\right) - \frac{1}{2}\frac{2y}{\delta^2} + \frac{2}{3}\left(\frac{3y^2}{\delta^3}\right)$$

At $y = 0$

$$\frac{\partial U}{\partial y} = \frac{5}{2\delta} > 0$$

So, flow will not separate.

1266. Air flows past a golf ball of 20 mm radius. It is observed that the boundary layer becomes turbulent at Reynolds number of 2×10^5. If the kinematic viscosity of air is 1.5×10^{-5} m^2/s, then the speed of air when the flow becomes turbulent is :

(a) 150 m/s (b) 90 m/s
(c) 75 m/s (d) 62.5 m/s

DRDO Scientists 2009

Ans. (c) : Reynold's number, $Re = \frac{Vd}{v}$

$$2\times10^5 = \frac{V\times40\times10^{-3}}{1.5\times10^{-5}}$$

or $V = 75$ m/sec

1267. The displacement thickness for a boundary layer

(a) may be greater than the boundary layer thickness
(b) must be less than momentum thickness
(c) represents momentum deficit in a flow
(d) represents mass deficit in a flow

UKPSC AE 2007 Paper -II

Ans. (d) : Represents mass deficit in a flow

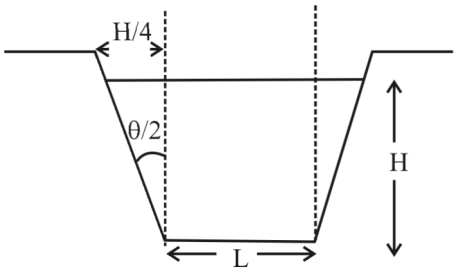
11. Flow over Notches and Weirs

1268. A Cipoletti notch is a trapezoidal notch having inclination of slopping sides as

(a) 1 horizontal to 4 vertical
(b) 1 horizontal to 2 vertical
(c) 1 horizontal to 3 vertical
(d) 1 horizontal to 1 vertical

OPSC AEE 2019 PAPER - II
TSPSC AEE 28.08.2017 (Civil/Mechanical)
APPSC AE Subordinate Service Civil/Mech. 2016
Nagaland PSC (CTSE) 2016, Paper-II
APPSC AEE 2012

Ans. (a) : A Cipoletti notch is a trapezoidal notch having inclination of sloping sides as 1 horizontal to 4 vertical.



$$\tan\frac{\theta}{2} = \frac{1}{4}$$

$$\boxed{\frac{\theta}{2} = 14^0 2'}$$

Discharge through Cipoletti weir–

$$\boxed{Q = \frac{2}{3}C_d \times L \times \sqrt{2g}\ H^{3/2}}$$

1269. The hydraulic mean depth is given by :

(a) $\frac{P}{A}$ (b) $\frac{P^2}{A}$
(c) $\frac{A}{P}$ (d) $\sqrt{\frac{A}{P}}$

Where A – Area, P = wetted perimeter

APPSC Poly. Lect. 2013, APPSC AE 04.12.2012
APPSC AEE 2012, ESE 2002

Ans. (c) : Hydraulic mean depth (m) = $\frac{A}{P}$

Where; A = Area of cross-section of pipe
P = Wetted perimeter of pipe

The ratio of $\frac{A}{P} = \left(\frac{\text{Area of flow}}{\text{Wetted perimeter}}\right)$ is called hydraulic mean depth or hydraulic radius and is denoted by 'm'.

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1270. Which of the following is/are related to measure the discharge by a triangular notch? (where θ is the central angle; H is the head; C_d is coefficient of discharge; and g is acceleration due to gravity.)

(a) $\frac{8}{15}C_d\sqrt{2g}\tan\frac{\theta}{2}H^{5/2}$

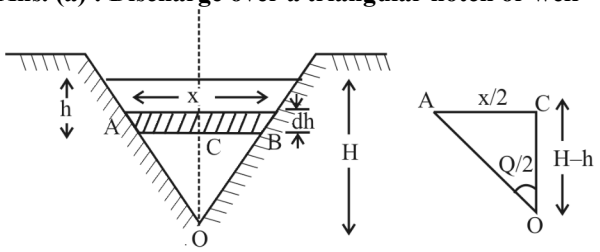
(b) $\frac{4}{15}C_d\sqrt{2g}\tan\frac{\theta}{2}H^{5/2}$

(c) $\frac{8}{15}C_d\sqrt{2g}\tan\frac{\theta}{2}H^{3/2}$

(d) $\frac{2}{3}C_d\sqrt{2g}\tan\frac{\theta}{2}H^{3/2}$

Nagaland PSC (CTSE) 2016, Paper-II
APPSC AE Subordinate Service Civil/Mech. 2016
APPSC AEE 2012

Ans. (a) : Discharge over a triangular notch or weir–



$$\tan\frac{\theta}{2}=\frac{x}{2(H-h)}\Rightarrow\frac{x}{2}=(H-h)\tan\frac{\theta}{2}$$

Area of strip = x.dh

Theoretical velocity of water through strip = $\sqrt{2gh}$

Discharge through strip,

$dQ = C_d \times$ Area of strip × then velocity

$= C_d \times (xdh)\sqrt{2gh}$

$= C_d \times 2\,(H-h)\tan\frac{\theta}{2}\,dh\,\sqrt{2gh}$

$$\therefore \text{ total discharge } = Q\int_0^H 2C_d(H-h)\tan\frac{\theta}{2}\times\sqrt{2gh}\times dh$$

$$Q = 2C_d\times\tan\frac{\theta}{2}\times\sqrt{2g}\int_0^H (H-h)h^{1/2}\,dh$$

$$= 2C_d\times\tan\frac{\theta}{2}\times\sqrt{2g}\int_0^H\left(H\times h^{1/2}-h^{3/2}\right)dh$$

$$= 2C_d\times\tan\frac{\theta}{2}\times\sqrt{2g}\left[\frac{Hh^{3/2}}{3/2}-\frac{h^{5/2}}{5/2}\right]^H$$

$$= 2C_d\times\tan\frac{\theta}{2}\times\sqrt{2g}\left[\frac{2}{3}H^{5/2}-\frac{2}{5}H^{5/2}\right]^H$$

$$= 2C_d\times\tan\frac{\theta}{2}\times\sqrt{2g}\left[\frac{4}{15}H^{5/2}\right]$$

$$\boxed{Q=\frac{8}{15}C_d\times\tan\frac{\theta}{2}\times\sqrt{2g}\times H^{5/2}}$$

Note : Discharge through rectangular notch.

$$\boxed{Q=\frac{2}{3}C_d\times L\times\sqrt{2g}\times H^{3/2}}$$

1271. A rectangular channel of 4 m width conveys water at 8 m^3/sec under critical condition. The specific energy for this flow is

(a) 1.1123 m (b) 1.4830 m
(c) 0.3703 m (d) 0.7416 m

JPSC AE 2013 Paper-V
APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (a) : Given, b = 4, Q = 8 m^3/s

$$q=\frac{Q}{b}=\frac{8}{4}=2m^2/s$$

$$h_c=\left(\frac{q^2}{g}\right)^{1/3}=\left(\frac{4}{10}\right)^{\frac{1}{3}}=(0.4)^{0.33}$$

$$h_c=0.73906$$

$$E_{min}=\frac{3}{2}h_c$$

$$=\frac{3}{2}\times0.7396$$

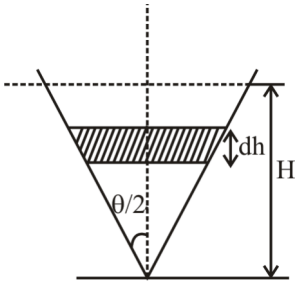
$$=1.1123\ m$$

1272. The rate of flow through a V-notch varies as

(a) H (b) $\sqrt{H}$
(c) $H^{3/2}$ (d) $H^{5/2}$

GPSC AE 26.07.2015, APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (d) :



The rate of flow through a V-notch varies.

$$Q=C_f\frac{8}{15}H^{5/2}\sqrt{2g}\tan(\theta/2)$$

$Q \propto H^{5/2}$

1273. The discharge through a rectangular channel is maximum when :

(a) m = d/3 (b) m = d/2
(c) m = 2d (d) m = 3d/2

TNPSC AE (Industries) 09.06.2013
APPSC AEE 2012, APPSC AE 04.12.2012

Ans. (b) : Maximum discharge through a rectangular channel if - $\boxed{m=\frac{d}{2}}$

Where, m = hydraulic mean depth

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1274. For a circular channel, the wetted perimeter is given by

(a) $\frac{R\theta}{2}$ (b) $3R\theta$

(c) $2R\theta$ (d) $R\theta$

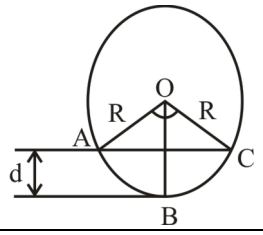
APPSC AEE 2012, APPSC AE 04.12.2012
ESE 2003

Ans. (c) : Wetted perimeter of a circular channel having radius R, depth d, and subtended angle θ = half the angle subtended by the water surface at the centre.

Wetted perimeter = Arc ABC

$P = R(2\theta)$

$P = 2R\theta$



1275. The discharge through a trapezoidal channel is maximum when

(a) half of top width = sloping side

(b) top width =half of sloping side

(c) top width = 1.5 $\perp$ 5 sloping side

(d) None of the above

APPSC AEE 2012, APPSC AE 04.12.2012

Ans. (a) : Maximum discharge for a trapezoidal channel section is the sloping side is equal to half of the width at the top.

Sloping side $= \frac{1}{2} \times$ Top width

1276. The head over a 90^0 V-notch increases from 0.15 to 0.3m. The ratio of the new discharge to the original discharge is :

(a) 1.414 (b) 2.25

(c) 4.00 (d) 5.657

KPSC AE. 2015
Kerala PSC AE 06.08.2015

Ans : (d) Discharge over . V –notch

$Q = 1.417\ H^{5/2}$

case 1st

$H_1 = 0.15$ m

$Q_1 = 1.417(0.15)^{5/2}\ m^3/sec$

$Q_1 = 0.0123\,m^3/sec$

case 2nd

$H_2 = 0.3$ m

$Q_2 = 1.417(0.3)^{5/2}$

$Q_2 = 0.0698\,m^3/sec$

$\frac{Q_2}{Q_1} = \frac{0.0698}{0.0123}$

Ratio of New discharge to the original discharge = 5.657

1277. In a hydraulic Jump the loss due to jump is given by

(a) $h = \frac{y_2 - y_1}{4y_1y_2}$ (b) $h = \frac{(y_2 - y_1)^2}{4y_1y_2}$

(c) $h = \frac{(y_2 - y_1)^3}{4y_1y_2}$ (d) $h = \frac{(y_2 - y_1)^4}{4y_1y_2}$

Nagaland PSC (CTSE) 2016, Paper-II
TNPSC AE 2013

Ans. (c) : In a hydraulic jump the energy head loss due to jump is given by,

$$h_L = \frac{(y_2 - y_1)^3}{4y_1y_2}$$

1278. The most economical section of a rectangular channel of maximum discharge is obtained when its depth is equal to

(a) Half the breadth

(b) Twice the breadth

(c) Same as the breadth

(d) 3/4th the breadth

Nagaland PSC CTSE 2015 Paper-II
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. (a) : Most economical two condition

1. Width is equal to two times of depth.

(b = 2d)

$\left(\because d = \frac{b}{2}\right)$

2. Hydraulic mean depth is equal to half the depth

$\left(M = \frac{d}{2}\right)$

1279. The critical depth of a rectangular channel of width 4.0 m for a discharge of 12 m^3/s is nearly,

(a) 300 mm (b) 30 mm

(c) 0.972 m (d) 0.674 m

Kerala PSC Poly. Lect., 2017
Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Given, b = 4 m, Q = 12 m^3/sec

Discharge per unit width

$q = \frac{Q}{b}$

$q = \frac{12}{4} = 3\,m^2/sec$

For rectangular channel

Critical depth $(h_c) = \left(\frac{q^2}{g}\right)^{1/3}$

$= \left(\frac{3^2}{9.81}\right)^{1/3}$

$h_c = 0.9721$ m

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1280. The hydraulic radius is given by

(a) Wetted perimeter divided by area
(b) Area divided by square of wetted perimeter
(c) Area divided by wetted perimeter
(d) Square root of area

Nagaland PSC CTSE 2015 Paper-II
UKPSC AE-2013, Paper-II

Ans. (c) : Hydraulic radius is defined as the cross-sectional area of flow divided by the wetted perimeter.

$$\text{Hydraulic radius (Rh)} = \frac{\text{Area of flow (A)}}{\text{Wetted perimeter (P)}}$$

1281. In a triangular notch there is an error of 4% in observing the head. The error in the computed discharge is

(a) 4% (b) 10%
(c) 6% (d) 2.5%

TSPSC AEE 28.08.2017 (Civil/Mechanical)
Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : For Triangular notch

$$Q = \frac{8}{15}\ C_d\sqrt{2g}\ \tan\frac{\theta}{2}H^{5/2}$$

Assumption only H to be the variable

$$dQ = \frac{8}{15}\ C_d\sqrt{2g}\ \tan\frac{\theta}{2}\times\frac{5}{2}H^{3/2}dH$$

$$\frac{dQ}{Q} = \frac{dH}{H}\times\frac{5}{2}$$

$$= 0.04 \times 2.5 = 0.1$$

i.e. $\frac{dQ}{Q}\times 100 = 10\%$

1282. In a 90° triangular notch, for a given head, the error in the estimated discharge due to a 2% error in the measurement of the vertex angle is

(a) π% (b) 5.0%
(c) 3.0% (d) π/2%

TSPSC AEE 28.08.2017 (Civil/Mechanical)
Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : For triangular notch

$$Q = \frac{8}{15}\ C_d\sqrt{2g}\ \tan\frac{\theta}{2}H^{5/2}$$

Assumption only θ to be the variable-

$$dQ = \frac{8}{15}\ C_d\sqrt{2g}\ \sec^2\frac{\theta}{2}\times\frac{d\theta}{2}\times H^{5/2}$$

$$\text{i.e. } \frac{dQ}{Q} = \frac{\text{Sec}^2\frac{\theta}{2}}{\tan\frac{\theta}{2}}\times\frac{d\theta}{2}$$

$$= \frac{1}{2}\left(\frac{\text{Sec}^2\frac{\theta}{2}}{\tan\frac{\theta}{2}}\right)\times\frac{d\theta}{\theta}\times\theta$$

taking $\theta = \frac{\pi}{2}$

$$\frac{dQ}{Q} = \frac{1}{2}\times\frac{\sec^2\left(\frac{\pi}{4}\right)}{\tan\left(\frac{\pi}{4}\right)}\times 0.02\times\frac{\pi}{2}$$

$$= 0.01\pi$$

$$\frac{dQ}{Q}\times 100 = \pi\%$$

1283. Which of the following is used for the measurement of rate of flow in open channel such as flow of water in river?

(a) Notches (b) weir
(c) orifices (d) Mouthpieces

APPSC AEE Mains 2016 (Civil/Mechanical)
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : Weir is used for the measurement of rate of flow in open channel such as flow of water in river.

1284. Notch is a device used for measuring :

(a) rate of flow through pipes
(b) rate of flow through a small channel
(c) velocity through a pipe
(d) velocity through a small channel

APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (b) : A notch is a device used for measuring the rate of flow of a liquid through a small channel or a tank. It may be defined as an opening in the side of a tank or a small channel in such a way that the liquid surface in the tank or channel is below the top edge of the opening.

1285. A submerged weir is one in which the water level on the downstream of the weir is

(a) Just at the crest level
(b) Below the crest level
(c) Is above the crest level
(d) At the same elevation as the upstream water surface

TSPSC AEE 28.08.2017 (Civil/Mechanical)
Nagaland PSC CTSE 2017, 2016 Paper-II

Ans. (c) : A submerged weir is one in which the water level on the downstream side of weir is above the top surface of weir it is known as submerged or drowned weir. Total discharge, over such a weir is found out by splitting up the height of water, above the sill of the weir, in two portions.

1286. The unit of Chezy's constant C in the Chezy's formula is :

(a) None (b) m/s
(c) m/s^2 (d) $m^{1/2}/s$

OPSC AEE 2019 PAPER - II
Kerala PSC Poly. Lect. 09.01.2014

Ans. : (d) : $V = C\sqrt{mi}$
C = Chezy's constant
The unit of C is $m^{1/2}/s$

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1287. The critical depth meter is used to measure

(a) Velocity of flow in an open channel
(b) Depth of flow in an open channel
(c) Depth of channel
(d) Hydraulic jump

RPSC Vice Principal ITI 2018
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : The critical depth meter is used to measure hydraulic jump.

1288. An error of 1% in measuring the head of water over the crest of a triangular notch, produces an error in the discharge which is equal to

(a) 2.5% (b) 1.25%
(c) 1.5% (d) 2.0%

APPSC AE Subordinate Service Civil/Mech. 2016
Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Effect of discharge due to error's in head measurement.

If there is x% error in measurement of head, error in measurement of discharge is given by $\frac{dQ}{Q}\times 100$

$Q = KH^n$

$dQ = KnH^{n-1}\, dH$

$\frac{dQ}{Q} = n \times \frac{dH}{H}$

$\left(\frac{dQ}{Q}\times 100\right) = n\left(\frac{dH}{H}\times 100\right)$

$\frac{dQ}{Q}\times 100 = \%$ error in discharged

$\frac{dH}{H}\times 100 = \%$ error in head measurement

% error in discharge = n×% error in head measurement

$n = \frac{3}{2} \rightarrow$ for rectangular weir

$n = \frac{5}{2} \rightarrow$ for triangular weir

$n = 1 \rightarrow$ for proportial weir

Hence an error of 1% in measuring the head of water over the crest of a triangular notch, produces an error in the discharge equal to 2.5%

1289. In an open channel flow :

(a) the hydraulic gradient line can never rise
(b) the hydraulic gradient line coincides with the energy gradient line
(c) the hydraulic gradient line and the free surface coincide
(d) the hydraulic gradient line, the energy line and the bed line are always parallel

CGPSC AE 15.01.2021, 16.10.2016

Ans. (c) : In open channel flow, the hydraulic gradient line and the free surface coincides.
A line joining the top of the liquid surface (free surface) in open channel is called Hydraulic gradient line.

1290. Statement (I): Gravity is the driving force behind flows through open channels.
Statement (II): Gravity stands to reason that the ratio of inertial to gravitational forces will play a major role in open channel flow analysis.

ESE 18.07.2021

Ans. (b) :

1291. _____ is a geometrical opening in the side of reservoir with upstream liquid level below the upper edge of an open

(a) Orifice (b) Venturi
(c) Mesh (d) Weir

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) :

1293. Hydraulic diameter for a duct of wetted perimeter P and cross-sectional area A is normally defined as

(a) 4 P/A (b) A/4 P
(c) 4A/P (d) P/4A

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Hydraulic diameter (m) = $\frac{4A}{P}$

1294. Uniform flow in an open channel exists when the flow is steady and

(a) the channel is frictionless
(b) the channel is non-prismatic
(c) the channel is prismatic
(d) the channel is prismatic and the depth of flow is constant along the channel

Assam PSC AE (PHED), 18.10.2020

Ans. (d) : Uniform flow in an open channel- If for a given length of the channel, the velocity of flow, depth of flow, slope of the channel and cross-section remain constant, the flow is said to be uniform. On the other hand, if for a given length of the channel the velocity of flow depth of flow do not remain constant the flow is said to be non uniform flow.

For uniform flow $\frac{\partial u}{ds} = 0, \frac{\partial v}{\partial s} = 0$

For non uniform flow $\frac{\partial u}{\partial s} \neq \frac{\partial v}{\partial s} \neq 0$

So, uniform flow in an open channel exists when the flow is steady and the channel is prismatic and depth of flow is constant along the channel.

1295. The whole area behind the dam draining into stream or river across which the dam is constructed is known as:

(a) Surge tank (b) Catchments area
(c) Spillways (d) Reservoir

SJVN ET 2019

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Ans. (b) : Catchment area- The whole area behind the dam draining into a stream or river across which the dam is constructed is known as catchment area.

1296. The safeguarding structure provided in the dam to relieve the reservoir of excess water when the water level in the reservoir rises, is known as:

(a) Spillways (b) Draft tube
(c) Catchment area (d) Surge tank

SJVN ET Mechanical 2019 paper

Ans. (a) :

- A spillway is an integral part of a Dam.
- A spillway is usually used to remove water from a reservoir to prevent overflow and to release pressure on a dam from increasing quantities of water. It is normally composed of three major components:

(1) The approach facility admits flow to the spillway.
(2) The discharging conduit evacuates the flow from the approach facility to an outlet structure.
(3) The outlet structure (tailwater channel) dissipates the excessive energy of the flow from the discharging conduits and conveys tranquil flow to the downstream.

Draft tube- Use in tail stock of reaction turbine to convert kinetic energy in pressure energy.
Catchment area- The area from which rain flows in to a particular river or lake.
Surge tank- It is a water storage device used as pressure neutralizer in hydropower water conveyance system to resist excess pressure rise and pressure drop conditions.

1297. Discharge through rectangle channel is maximum when the ratio of hydraulic mean depth and depth of flow is:

(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

OPSC AEE 2019 PAPER - II

Ans. : (a) : Discharge through rectangle channel is maximum when the ratio of hydraulic mean depth and depth of flow is 0.5.

1298. The most economic section of channel has:

(a) Minimum Wetted Perimeter
(b) Minimum Discharge
(c) Both Wetted Perimeter and Discharge Minimum
(d) Both Wetted Perimeter and Discharge Maximum

OPSC AEE 2019 PAPER - II

Ans. : (a) : A section of a channel is said to be most economical when the cost of construction of the channel is minimum. But the cost of construction of channel depends on excavation and the lining to keep the cost minimum the wetted perimeter should be minimum for a given discharge.

1299. The hydraulic mean depth for a circular pipe of diameter 100 mm when running full is:

(a) 75 mm (b) 100 mm
(c) 50 mm (d) 25 mm

AAI Jr. Executive 29.11.2018

Ans. (d) : For circular pipe, d = 100 mm

Cross section area = $\pi r^2 = \frac{\pi}{4}d^2$

Perimeter of pipe = $2\pi r = \pi d$

Hydraulic mean depth (m) = $\frac{A}{P} = \frac{\frac{\pi}{4}d^2}{\pi d} = \frac{d}{4}$

$\because$ d = 100

$\therefore \quad = \frac{100}{4}$

= 25 mm

1300. Time required to empty a tank form height h_1 to h_2 for rectangular notch is : [A = cross sectional area of reservoir, C_d = coefficient of discharge, L = length of crest of notch]

(a) $\frac{A}{C_d L\sqrt{2g}}\left[\frac{1}{h_2}-\frac{1}{h_1}\right]$

(b) $\frac{3A}{C_d L\sqrt{2g}}\left[\frac{1}{h_2^{3/2}}-\frac{1}{h_1^{3/2}}\right]$

(c) $\frac{5A}{C_d L\sqrt{2g}}\left[\frac{1}{\sqrt{h_2}}-\frac{1}{\sqrt{h_1}}\right]$

(d) $\frac{3A}{C_d L\sqrt{2g}}\left[\frac{1}{\sqrt{h_2}}-\frac{1}{\sqrt{h_1}}\right]$

(e) $\frac{A}{C_d L\sqrt{2g}}\left[\frac{1}{\sqrt{h_2}}-\frac{1}{\sqrt{h_1}}\right]$

CGPSC AE 25.02.2018

Ans. (d) : The time required to empty a tank from height h_1 to h_2 for rectangular notch is given as

$$t = \frac{3A}{C_d L\sqrt{2g}} \times \left[\frac{1}{\sqrt{h_2}}-\frac{1}{\sqrt{h_1}}\right]$$

1301. If the depth of water in an open channel is greater than the critical depth, the flow is called

(a) Critical flow (b) Turbulent flow
(c) Tranquil flow (d) Torrential flow

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Tranquil flow– Streaming flow or tranquil flow is the flow when depth of flow in an open channel is grater than the critical depth.

Turbulent flow– When fluid particles move in highly disorganized manner leading that flow is known as turbulent flow generally occur at high velocities.

Torrential flow– Torrential flow is the highest velocity flow that can occur in a given reach of a natural channel when slope is higher than a given value.

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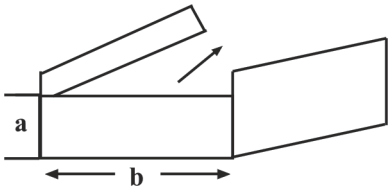
1302. The hydraulic diameter, D_h of a rectangular channel with sides 'a' and 'b' is,



(a) $D_h = \frac{4ab}{2a+b}$ (b) $D_h = \frac{2ab}{a+b}$

(c) $D_h = \frac{2ab}{2a+b}$ (d) $D_h = \frac{4ab}{a+b}$

ISRO Scientist/Engineer 22.04.2018

Ans. (a) : We know that hydraulic diameter in rectangular channel is given by

$$D_h = \frac{4A_c}{P}$$

where,

A_c = cross-sectional area

= a × b

P = wetted perimeter

= (2a + b)

$$\boxed{D_h = \frac{4(a\times b)}{(2a+b)}}$$

1303. Highest efficiency is obtained with following channel section-

(a) Circular (b) Triangular

(c) Rectangular (d) Trapezoidal

GMB AAE 25.06.2017

Ans. (d) : Highest efficiency is obtained with trapezoidal channel section.

1304. The discharge through a semi-circular weir is proportional

(a) $H^{1/2}$ (b) $H^{3/2}$

(c) $H^{5/2}$ (d) H^2

GMB AAE 25.06.2017

Ans. (b) : The discharge through a semi-circular weir is proportional to $H^{3/2}$.

1305. The type of jump that forms when the initial Froude number lies between 2.5 and 4.5 is known as

(a) A oscillating jump (b) Weak jump

(c) Steady jump (d) Undular jump

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : An oscillating jump that take place when the Froude's number is in between 2.5 to 4.5. During this jump, the jet water at the entrance of the jump fluctuates.

1306. The discharge over a 90° V-notch is written as $Q = 1.37\ H^{5/2}$ where Q is in $m^{3/4}$ and H is in metres. The C_d of this notch is

(a) 0.611 (b) 0.580

(c) 0.464 (d) 0.710

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : For Triangular notch

$$Q = \frac{8}{15} C_d \sqrt{2g}\ \tan\frac{\theta}{2} H^{5/2}$$

But, $Q = 1.37\,H^{5/2}$

$$\therefore \quad \frac{8}{15} C_d \sqrt{2g}\ \tan\frac{\theta}{2} = 1.37$$

$$\text{i.e. } \frac{8}{15}\times C_d \times \sqrt{2\times 9.81}\times \tan\frac{\pi}{4} = 1.37$$

$$C_d = 0.58$$

1307. In the open channels, the specific energy is the

(a) total energy per unit discharge

(b) total energy measured with respect to the datum passing through the bottom of the channel

(c) total energy measured above the horizontal datum

(d) total energy per velocity of flow

Karnataka PSC AE, 10.09.2017

Ans. (b) : In open channel flow the specific energy is the total energy measurement with respect to the datum passing through the bottom of the channel.

1308. The critical flow condition in a channel is given by:

(a) $\frac{\alpha Q T^{-2}}{gA^3} = 1$ (b) $\frac{\alpha Q^2 T^2}{gA^3} = 1$

(c) $\frac{\alpha Q T^2}{gA^3} = 1$ (d) None of these

PTCUL AE 25.06.2017

Ans. (c) :

1309. In the context of fluid mechanics, the ratio of the loss of head of fluid, issuing from an orifice or passing over a weir, to the remaining head is known as_____

(a) Head ratio

(b) Coefficient of orifice

(c) Coefficient of resistance

(d) None of these

KPCL AE 2016

Ans. (c) : In the context of fluid mechanics, the ratio of a loss of head of fluid, issuing from an orifice or passing over a weir, to the remaining head is known as coefficient of resistance.

1310. The most economical section of a trapezoidal channel is one which has hydraulic mean depth equal to

(a) 0.5 depth (b) 0.5 sloping side

(c) 0.5 width (d) 0.5 (width + depth)

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (a) : The most economical section of a trapezoidal channel is one which has hydraulic mean depth equal to half the depth of flow.

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1311. If weir is used as a spillway it will have a crest

(a) Ogee shaped (b) Broad and flat
(c) Narrow (d) Cippolet

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : If weir is used as a spillway it will have a crest in ogee shaped.

1312. In a tank or channel, notch is provided to measure

(a) Velocity (b) Pressure
(c) Discharge (d) Static energy

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : A notch is an opening in the side of a measuring tank or reservoir extending above the free surface. These notches are used to measure discharge of open channel flow by passing or placing or constructing them across the stream.

1313. Weir is also be used to measure

(a) Velocity to flow
(b) Pressure
(c) Discharge in a river of channel
(d) Kinetic energy

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Weir is also be used to measure discharge or flow rate and calculated by analyzing a weir. It is given by $Q = CLH^n$, where
C – Flow Coefficient
L – Width of Crest
H – Height of head
n – is different for different structure

1314. The sheet of water flowing through a notch or over a weir is known as

(a) Nappe (b) Crest
(c) Height of weir/notch (d) Sill

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : The sheet of water flowing over a notch (or a weir) is known as nappe or vein

1315. For a suppressed rectangular weir an arrangement for aeration of nappe is necessary.

(a) To maintain water quality
(b) To prevent submergence of the weir
(c) To have the highest value of C_d
(d) To have a constant head discharge relationship which is independent of time

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d): For a suppressed rectangular weir an arrangement for aeration of nappe is necessary to have a constant head discharge relationship which is independent of time.

1316. In a suppressed rectangular weir the computed discharge was found to be 3% in excess of the actual discharge. If this discrepancy was due to an error in reading the head, the measured head was

(a) 5% less (b) 2% excess
(c) 2% less (d) 1.5% excess

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : Discharge over a rectangular weir,

$$Q = \frac{2}{3} C_d \times L \times \sqrt{2g}\,[H]^{3/2}$$

$$Q = K \cdot [H]^{3/2}$$

$$\left(\frac{\Delta Q}{Q} \times 100\right) = \frac{3}{2}\left(\frac{\Delta H}{H} \times 100\right)$$

$$3 = \frac{3}{2}\left(\frac{\Delta H}{H} \times 100\right)$$

Percent error in measured head,

$$\left(\frac{\Delta H}{H} \times 100\right) = 2\% \quad \text{excess} \quad \left(\because Q \propto H^{3/2}\right)$$

1317. A cippoletti weir discharges water with the head of water above the crest being 250 mm. If the head due to velocity of approach is 0.01 m, what will be the excess percentage of discharge, as compared to when not so corrected?

(a) 3.5% (b) 6%
(c) 3% (d) 5.3%

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : Cippoletti weir discharge,

$$Q = 1.859\ BH^{3/2}$$

$$\left(\frac{\Delta Q}{Q} \times 100\right) = \frac{3}{2}\left(\frac{\Delta H}{H} \times 100\right)$$

$$= \frac{3}{2} \times \left(\frac{0.01}{250 \times 10^{-3}} \times 100\right) = 6\%$$

1318. The sequent depths in a hydraulic jump formed in a horizontal rectangular channel are 0.2 m and 2.0 m. The length of the jump is about

(a) 50 m (b) 25 m
(c) 8 m (d) 12 m

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : $\frac{h_2}{h_1} = \frac{1}{2}\left(\sqrt{1 + 8F_r^2} - 1\right)$

$$\frac{2}{0.2} = \frac{1}{2}\left(\sqrt{1 + 8F_r^2} - 1\right)$$

$F_r = 7.42$

If $3 < F_r < 400$ in a rectangular channel, we may use Ivadian's formula for length of hydraulic jump,

$$L_j = \frac{8\left(10 + \sqrt{F_r}\right)}{F_r}\left(\frac{(h_2 - h_1)^3}{4h_1h_2}\right)$$

$$L_j = \frac{8\left(10 + \sqrt{7.42}\right)}{7.42}\left(\frac{(2 - 0.2)^3}{4 \times 2 \times 0.2}\right)$$

$L_j = 50$ m

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1319. In a horizontal rectangular channel a hydraulic jump with a sequent depth ratio of 5.0 is formed. This jump can be classified as

(a) Steady jump (b) Strong jump
(c) Weak jump (d) Normal jump

Nagaland PSC CTSE 2017, 2016 Paper-II

Ans. (a) : In a horizontal rectangular channel a hydraulic jump with a sequent depth ratio of 5.0 is formed. Then this jump can be classified as Steady jump.

1320. An open channel may be defined as a passage flow in which

(a) A closed conduit is full of flowing fluid
(b) The solid boundary confining the flow an open at the top
(c) Liquid flow with its upper surface exposed to atmosphere
(d) Flow with mixed deposit at boundary

Nagaland PSC (CTSE) 2016, Paper-II

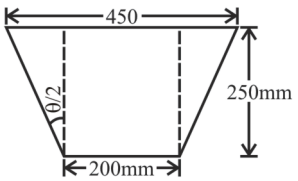
Ans. (c) : An open channel may be defined as a passage flow in which liquid flow with its upper surface exposed to atmosphere.

1321. A sharp edged weir is in the form of a symmetrical trapezium. The horizontal base is 200 mm. The top is 450 mm and the depth is 250 mm. If the head of water over the crest is 150 mm and $C_d = 0.62$, the discharge over the weir would be

(a) 0.0213 m^3/sec (b) 0.0064 m^3/sec
(c) 0.277 m^3/sec (d) 0.0277 m^3/sec
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) :



$$Q = Q_1 + Q_2$$

$$= \frac{2}{3} C_d \times L \times \sqrt{2g} \times H^{3/2} + \frac{8}{15} C_d \times \sqrt{2g} \times \tan\frac{\theta}{2} \times H^{5/2}$$

$$= \frac{2}{3} \times 0.62 \times 0.2 \times \sqrt{2 \times 9.81} \times 0.25^{3/2} + \frac{8}{15} \times 0.62 \times \sqrt{2 \times 9.81} \times \frac{125}{250} \times 0.25^{5/2}$$

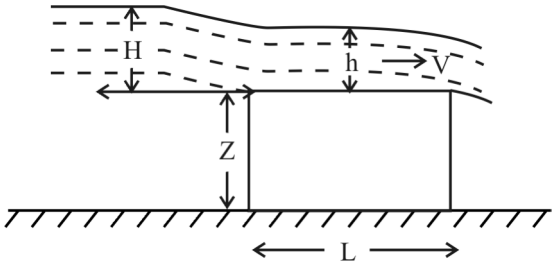
$$\boxed{Q = 0.0277\, m^3/sec}$$

1322. The maximum discharge over a broad crested weir is
(Where H is the height of water surface above the crest of the weir on the upstream side; L is the length of the crest of the weir, and C_d is the coefficient of discharge)

(a) 1.7 $C_d LH^{3/2}$ (b) 1.7 $C_d LH^{2/3}$
(c) 1.7 $LH^{3/2}$ (d) 1.98 $C_d LH^{3/2}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : A weir having a wide crest is known as broad crested weir.



If 2 L > H, the weir is called broad crested.
if 2L < H, the weir is called narrow crested.

Discharge over a broad crested weir –

H= Height of water above the crest
L = Length of crest
h = Head of water at the middle of weir.
v = velocity of flow over the weir.

Applying Bernoulli's equation to the still water surface on the upstream side and running water at the end of weir.

$$O + 0 + H = 0 + \frac{V^2}{2g} + h$$

$$\frac{V^2}{2g} = H - h$$

$$V = \sqrt{2g(H-h)}$$

$$Q = C_d \times \text{Area of flow} \times \text{Velocity}$$

$$= C_d \times L \times h \times \sqrt{2g(H-h)}$$

$$= C_d \times L \times \sqrt{2g\left(Hh^2 - h^3\right)}$$

The discharge will be maximum, if $(Hh^2 - h^3)$ is max.

$$\frac{d}{dh}\left(Hh^2 - h^3\right) = 0$$

$$2h \times H - 3h^2 = 0$$

$$2H = 3h$$

$$\boxed{h = \frac{2}{3}H}$$

$$Q_{max} = C_d \times L \times \sqrt{2g\left[H \times \left(\frac{2}{3}H\right)^2 - \left(\frac{2}{3}H\right)^2\right]}$$

$$= C_d \times L \times \sqrt{2g}\sqrt{\left[H \times \frac{4}{9}H^2 - \frac{8}{27}H^3\right]}$$

$$Q_{max} = C_d \times L \times \sqrt{2g}\sqrt{\frac{4}{27}H^3}$$

$$= C_d \times L \times \sqrt{2g} \times 0.3849 \times H^{3/2}$$

$$= 0.3849 \times \sqrt{2 \times 9.81} \times C_d \times L \times H^{3/2}$$

$$Q_{max} = 1.705 \times C_d \times L \times H^{3/2}$$

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1323. The error in discharge due to error in the measurement of head over a rectangular notch is given by

(a) $\frac{dQ}{Q} = \frac{1}{2}\frac{dH}{H}$ (b) $\frac{dQ}{Q} = \frac{3}{2}\frac{dH}{H}$

(c) $\frac{dQ}{Q} = \frac{3}{4}\frac{dH}{H}$ (d) $\frac{dQ}{Q} = \frac{5}{2}\frac{dH}{H}$

APPSC AEE Screening Test 2016

Ans. (b) : Discharge over a rectangular weir,

$$Q = \frac{2}{3} C_d \times L \times \sqrt{2g}\,[H]^{3/2}$$

$$Q = K \cdot [H]^{3/2}$$

$$\left(\frac{dQ}{Q}\right) = \frac{3}{2}\left(\frac{dH}{H}\right)$$

1324. A triangular notch ..... .

(a) has no end contractions

(b) is more accurate than rectangular notch for low discharge

(c) must have an inclined angle of 90

(d) is more accurate than rectangular notch for all flow rates

GWSSB DEE 07.07.2016

Ans. (b) : A triangular notch is also called a V-notch. Consider a triangular notch, in one side of a tank, over which water is flowing. It gives more accuracy when discharge charge is small.

The equation of discharge over a triangular notch,

$$Q = \frac{8}{15} \quad C_d\sqrt{2g} \quad \tan\frac{\theta}{2} H^{5/2}$$

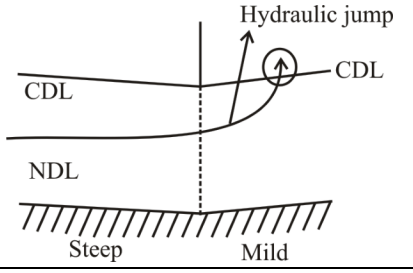
1325. Hydraulic jump is expected when slope of a channel changes from

(a) steep to steeper (b) steep to mild

(c) mild to critical (d) mild to steep

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (b) : A hydraulic jump is expected when slope of a channel changes from steep to mild.



1326. Parshall flume is used to measure discharge in

(a) pipes (b) wells

(c) canals (d) spillways

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (c) : Parshall flume is used to measure volumetric flow rate in Industrial discharges, municipal sewer lines, & influent/ effluent flows in wastewater treatment plants. It is also used in canals.

1327. In a rectangular channel of 1 m width the critical depth for a flow of 3.132 cumecs is approximately equal to

(a) 2 m (b) 1 m

(c) 0.1 m (d) 3.14 m

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (b) : As we know,

for critical depth,

$$h_c = \left(\frac{q^2}{g}\right)^{1/3}$$

$$h_c = \left[\frac{(3.132)^2}{g}\right]^{1/3}$$

$$h_c = \left[\frac{9.809}{9.81}\right]^{1/3} = 1\text{m}$$

1328. What is a condition for a hydraulically efficient channel

(a) Minimum flow rate

(b) Maximum wetted perimeter

(c) Constant velocity

(d) Minimum wetted perimeter

APPSC AEE Screening Test 2016

Ans. (d) : If we take the area as constant, hydraulic radius is maximum if the wetted perimeter is minimum. A channel section is considered as the most economical or most efficient when it passes a maximum discharge for given cross-section area, resistance coefficient and bottom slope.

1329. If the space under the nappe is only partially ventilated, then

(a) the discharge will be less than the theoretical discharge.

(b) the discharge will be more than the theoretical discharge.

(c) the discharge will be same as the theoretical discharge but there will be problems in taking measurements.

(d) the discharge will be completely shutoff due to vacuum.

RPSC AE (GWD) 06.05.2016

Ans. (b) : The height of the discharge over crest is called the nappe. The air space under the nappe, downstream from the weir is called the ventilation. If the space under the nappe is only partially ventilated. then the discharge will be more than the theoretical discharge.

1330. What is hydraulic diameter used in place of diameter for non-circular ducts equal to? Where A = area of flow and m = perimeter

(a) A/m (b) A/4 m

(c) 4m/A (d) 4A/m

UPRVUNL AE 21.08.2016

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Ans. : (d) Hydraulic diameter for non circular duct is equal to $\frac{4A}{m}$
where
A = Area of flow
m = Perimeter

1331. A 6m long rectangular weir is divided into 3 bays by two vertical posts each 0.3 m wide. What will be discharge through the weir for head of 0.45m?
(a) 2.85 m^3/s (b) 2.5 m^3/s
(c) 3.85 m^3/s (d) 1.85 m^3/s

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a)
Discharge through rectangular weir with end contraction–

$$Q = \frac{2}{3} \times C_d \times \left[L - 0.1 \times n \times H\right] \times \sqrt{2g} H^{3/2}$$

n = 6 = no. of end contractions, H = 0.45m
$C_d = 0.62$
Effective length, L = 6 – 2 × 0.3 = 5.4 m

$$Q = \frac{2}{3} \times 0.62\left[5.4 - 6 \times 0.1 \times 0.45\right] \times \sqrt{2g} \times (0.45)^{3/2}$$

$Q = 2.85\ m^3/s$

1332. Instrument used to measure the flow through open channel :
(a) Orifice plate (b) Venturi meter
(c) Pitot tube (d) V notch method

RPSC ADE 2016

Ans. (d) : V notch method – V notch method used to measure the flow through open channel.
Orifice plate and venturimeter used to measure the flow rate or discharge in pipe.
Pitot tube used to measure the velocity in the pipe.

1333. While dealing with flow in open channels one of the equations used is .....
(a) Sterling's equation
(b) Wilson's equation
(c) Tchebychev's equation
(d) Bazin's equation

HPPSC AE (PWD) 10.09.2016

Ans. (d) : While dealing with flow in open channels one of the equations used is Bazin's equation.

1334. The Francis formula for discharge over a rectangular weir without end contraction is
(a) $Q = (1/2)\ C_d \cdot b\sqrt{2} H^{3/2}$
(b) $Q = (1/2)\ C_d \cdot b\ H^{3/2}$
(c) $Q = 1.84\ C_d \cdot b\ H^{3/2} \times \sqrt{2}$
(d) $Q = 1.84\ C_d \cdot bH^{3/2}$

VIZAG STEEL MT 2015

Ans. (d) : The Francis formula for discharge over a rectangular weir without end contraction is–

$$Q = 1.84 C_d . bH^{3/2}$$

1335. When the water level on the downstream side of a weir is above the top surface of a weir, the weir is known as:
(a) Narrow – crested weir
(b) Broad-crested weir
(c) Ogee weir
(d) Submerged weir

NPCIL ET, 2015

Ans. (d) : When the water level on the downstream side of a weir is above the top surface of a weir, the weir is known as submerged weir.

1336. Manning formula is used to determine
(a) Head loss due to friction in pipes flowing full under pressure
(b) Head loss due to friction in open channels
(c) Flow in open channels
(d) Flow in pipes

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : The Manning equation is widely used and very versatile formula in water resources. It is used in open channel flow. The friction losses in channel, derive the capacity of pipe, check the performance of an area velocity flow meter.

$$\text{Manning formula} = \frac{1}{n} R^{2/3} S^{1/2}$$

$n \rightarrow$ Manning roughness
$R \rightarrow$ Hydraulic radius
$S \rightarrow$ Slope

1337. If flow in an open channel is gradually varied, then the flow will be
(a) Steady uniform flow
(b) Unsteady uniform flow
(c) Steady non-uniform flow
(d) Unsteady non-uniform flow

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Open channel when the depth of the open channel is increasing.
Gradually varied flow is called a "Steady non-uniform flow" inside a channel where there is a gradual change in the water depth.

1338. The ratio of the hydraulic radius of a pipe running full of water to the hydraulic radius of a square section of a channel whose side is equal to the diameter of the pipe, is
(a) 1 (b) 1/2
(c) 1/3 (d) 3/4

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Hydraulic radius also known as hydraulic mean depth.
For circular pipe hydraulic mean depth

$$= \frac{A}{P} = \frac{\pi d^2/4}{\pi d} = \frac{d}{4}$$

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For the square section of channel hydraulic mean depth

$$= \frac{A}{P} = \frac{d^2}{4d} = \frac{d}{4}$$

Ratio of the circle to square $= \frac{d/4}{d/4}$

$= 1$

1339. Hydraulic jump is used for

(a) Increasing the flow rate
(b) Reducing the flow rate
(c) Reducing the velocity of flow
(d) Reducing the energy of flow

TNPSC AE 2014

Ans. (d) : Hydraulic Jump–A Hydraulic jump is a phenomenon in the science of hydraulics which is frequently observed in open channel flow such as rivers and spillways. When liquid at high velocity discharges into a zone of lower velocity, rather abrupt rise occurs in the liquid surface.

1340. A quantity of 90 m^3/sec of water is smoothly flowing in a rectangular 16 m broad channel. If the depth of flow is 2 m and jump is established, then what would be the depth downstream of the jump?

(a) 3.50 m (b) 3.59 m
(c) 3.89 m (d) 3.98 m

UPRVUNL AE 2014

Ans. : (c) $y_2 = \frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2.a^2}{g.y_1}}$

Where $a = \frac{Q}{b}$

$b = 16$ m

$y_1 = 2$m

Depth downstream the jump (y_2)= ?

$$y_2 = \frac{2}{2} + \sqrt{\frac{(2)^2}{4} + \frac{2 \times \left(\frac{90}{16}\right)^2}{10 \times 2}}$$

$$y_2 = 1 + \sqrt{1 + \frac{405}{128}}$$

$$y_2 = 1 + \sqrt{\frac{533}{128}}$$

$$y_2 = 1 \pm 2.89 = 3.89\,\text{m}$$

1341. An ogee weir 5 meters long has a head of 40 cm of water. If $c_d = 0.6$, find the discharge over the weir.

(a) 2.2409 m^3/s (b) 1.120.45 m^3/s
(c) 4.4818 m^3/s (d) 1 m^3/s

APPSC Poly. Lect. 2013

Ans. (a) : $Q = 0.674\, C_d \times L \times \sqrt{2}\, g \times H^{1.5}$

$= 0.67 \times 0.6 \times 5 \times \sqrt{2 \times 9.81} \times (0.4)^{1.5}$

$= 2.2523\ m^3/sec$

1342. A reservoir having a surface area of 500 m^2 is emptied by a 0.5 m wide rectangular weir. How long should it take to empty the reservoir from a height of 3.2 m to 0.1 m above the sill? [Take : $C_d = 0.65$]

(a) 16 minutes and 5 seconds
(b) 8 minutes and 5 seconds
(c) 32 minutes and 5 seconds
(d) 10 seconds

JPSC AE 2013 Paper-V

Ans. (a) : $A = 500\ m^2$

$b = 0.5$ m

$h_1 = 3.2$ m

$h_2 = 0.1$ m

$C_d = 0.65$

$l = 1$ m

$$T = \frac{2A\left(\sqrt{h_1} - \sqrt{h_2}\right)}{C_d.b\sqrt{2g}}$$

$$= \frac{2 \times 500\left(\sqrt{3.2} - \sqrt{0.1}\right)}{0.65 \times 0.5 \times \sqrt{2 \times 10}} = 16\,\text{minutes and 5 seconds}$$

1343. Function of 'Spillways' is to:

(a) Provide structural stability to dam during reservoir flooding
(b) Provide safety during earthquakes
(c) Provide extra water during droughts
(d) All of these

SJVN ET 2013

Ans. (a) : Spillways are provided for storage and detention dams to release surplus or flood water which cannot be contained in the allotted storage space, and diversion dams to by pass flows exceeding those which are turned into the diversion system.

1344. The rise of water level, which takes place due to the transformation of the unstable shooting flow to the stable streaming flow is called :

(a) Critical flow (b) Torrential flow
(c) Tranquil flow (d) Hydralic jump

TNPSC AE (Industries) 09.06.2013

Ans. (d) :

1345. Match List I with List II and select the correct answer using the codes given below the Lists:

List-I		List-II	
A	Oscillating jump	1	Less than 2
B	Strong jump	2	between 2 and 3
C	Weak jump	3	around 4
D	Undular jump	4	around 10

Code:-

	A	B	C	D
(a)	3	4	2	1
(b)	1	2	3	4
(c)	4	3	1	2
(d)	2	1	3	4

APPSC AE 04.12.2012

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Ans. (a) : Oscillating jump → around 4
Strong jump → around 10
Weak jump → Between 2 and 3
Undular jump → Less than 2

1346. A 3 metre wide rectangular channel flowing at its normal depth of 0.8 m carries a discharge of 5.5 m^3/s. The bed slope of the channel is
(a) Steep
(b) Critical
(c) Mild
(d) Adverse

APPSC AEE 2012

Ans. (c) : $V = \frac{\theta}{y.b} = \frac{5.5}{3 \times 0.8} = 2.2917$

Froude No. $(F_r) = \frac{V}{\sqrt{gy}} = \frac{2.2917}{\sqrt{9.81 \times 0.8}}$

$F_r = \frac{2.2917}{2.8014} = 0.818$

$F_r < 1$ Subcritical flow.

So slope of rectangular channel- mild slope.

1347. A 3 metre wide rectangular channel flowing at its normal depth of 0.8 m carries a discharge of 5.5 m^3/s. The bed slope of the channel is
(a) Steep
(b) Critical
(c) Mild
(d) Adverse

APPSC AE 04.12.2012

Ans. (a) :

1348. When is the flow in a pipe or channel said to be uniform?
(a) The liquid particles at all section have the same velocities
(b) The liquid particles at different sections have different velocities
(c) The quantity of liquid flowing per second is constant
(d) Each liquid particle has a definite path

VIZAG STEEL MT 2011

Ans. (a) : Uniform Flow–A flow in which the liquid particles at all section of a pipe or channel have the same velocities is called a uniform flow.

1349. A most economical channel section is one which for a given cross-sectional area:
(a) has maximum velocity of fluid
(b) has maximum discharge
(c) has maximum depth of fluid flowing
(d) has maximum wetted perimeter

J & K PSC Screening, 2006

Ans. (b) : A channel section is considered as the most economical or most efficient when it passes a maximum discharge for given cross-section area, resistance coefficient and bottom slope.

12. Dimensional and Model Analysis

1350. The ratio between the inertia force of the flowing fluid and the pressure force is known as:
(a) Euler's number
(b) Mach number
(c) Webber's number
(d) Froude's number

NLC GET 17.11.2020, Shift-II
CIL MT 27.02.2020, MECON MT 2019
TNPSC AE 2018
ISRO Scientist/Engineer 22.04.2018
WBPSC AE, 2017, MPPSC AE 2016
NPCIL ET, 2015
MPPSC State Forest Service Exam, 2014
APPSC Poly. Lect. 2013
J&K PSC Civil Services Pre, 2013
UKPSC AE 2013 Paper-II
UKPSC AE 2007 Paper -II

Ans. (a) : Euler's number– The ratio between Inertia Force of the flowing fluid and the pressure force is known as Euler's number
(a) Reynolds number – It is the ratio of inertia force to the viscous Force.
(b) Froude's number– It is the ratio of inertia force to the gravity force.
(c) Weber's number – It is the ratio of the inertia Force to the surface tension Force.
(d) Mach's number or Cauchy's number – It is the ratio of the inertia Force to the elastic Force.

1351. Reynold's number is the ratio of the inertia force to the.....
(a) surface tension force
(b) viscous force
(c) gravity force
(d) pressure force

UPRVUNL AE 05.07.2021
NLCIL GET 17.11.2020, Shift-II
ISRO Scientist/Engineer (RAC) 22.04.2018
GPSC Executive Engineer 23.12.2018
GPSC Lect. 23.10.2016
APPSC AEE Screening Test 2016
KPCL AE 2016
TSPSC Managers, 2015
BPSC Asstt. Prof. 29.11.2015
MPSC Poly. Lect. 2014
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
TNPSC AE 2013
UKPSC AE 2012 Paper–II
OPSC Civil Services Pre 2006

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Ans. (b) : Reynold's number is the ratio of the inertia force to the viscous force

$$Re = \frac{\text{Inertia force}}{\text{Viscous force}}$$

$$Re = \frac{\rho v d}{\mu}$$

1352. The square root of the ratio of inertia force to gravity force is called :

(a) Reynolds number (b) Froude number
(c) Mach number (d) Euler number

JPSC AE 10.04.2021 Paper-I
OPSC AEE 2019 Paper-II
CGPSC AE 25.02.2018
Nagaland PSC (CTSE) 2018, Paper-II
HPPSC Asstt. Prof. 18.09.2017
RPSC Lect. 16.01.2016
Nagaland PSC (CTSE) 2016 Paper-II
ISRO Scientist/Engineer (RAC) 29.11.2015
TNPSC AE Industries 09.06.2013
UKPSC AE 2012 Paper-II
APPSC AE 04.12.2012
APPSC AEE 2012
OPSC Civil Services Pre-2006
CSE Pre-2003, GATE 1994

Ans. (b) : The square root of the ratio of inertia force to gravity force is called Froude number.

Froude number $(F_r) = \sqrt{\frac{F_i}{F_g}}$;

Where; F_i = Inertia force
F_g = Gravity force

Or $\frac{V}{\sqrt{gL}}$

1353. Mach number is the ratio of

(a) Elastic force to gravity force
(b) Viscous force to elastic force
(c) Inertial force to surface tension
(d) Inertial force to elastic force

APGCL AM 2021
HPPSC Workshop Suptd. 08.07.2021
Nagaland PSC (CTSE) 2018, Paper-II
WBPSC AE 2017, TRB Poly. Lect. 2017
UPPSC AE. 2016, TNPSC AE 2013
TNPSC ACF 2012
UKPSC AE 2012 Paper-II

Ans : (d) Mach number is the ratio of Inertia force to elastic force.

(i) Mach Number $= \sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}}$

(ii) Reynolds No $= \frac{\text{Inertia force}}{\text{Viscous force}}$

(iii) Euler's No. $= \sqrt{\frac{\text{Inertia force}}{\text{Pressure force}}}$

(iv) Weber No. $= \sqrt{\frac{\text{Inertia force}}{\text{Surface tension}}}$

(v) Froude No. $= \sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}}$

1354. Ratio of inertia force to surface tension is known as :

(a) Weber's number (b) Mach number
(c) Froude number (d) Euler number

OPSC AEE 2019 Paper-II
WBPSC AE 2017, GMB AAE 25.06.2017
Nagaland PSC (CTSE) 2016 Paper-II
Nagaland PSC CTSE 2015 Paper-II
OPSC AEE 2015 PAPER - II
SJVN ET 2013, ESE 2011

Ans : (a) (i) Reynold's Number $= \frac{\text{Inertia force}}{\text{viscous force}}$

(ii) Froude's Number $= \sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}}$

(iii) Weber's Number $= \sqrt{\frac{\text{Inertia force}}{\text{Surface tension force}}}$

(iv) Euler's Number $= \sqrt{\frac{\text{Inertia force}}{\text{Pressure force}}}$

(v) Mach's Number $= \sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}}$

1355. A dimensionless group formed with the variables ρ, ω, μ and D is

(a) $\rho\,\omega\,\mu / D^2$ (b) $\rho\,\omega\,D^2 / \mu$
(c) $\mu\,D^2\,\rho\,\omega$ (d) $\rho\,\omega\,\mu\,D$

UPPSC AE 13.12.2020, Paper-2
Oil India Ltd. Sr. Engineer (Production) 30.11.2019
APPSC AE 04.12.2012
APPSC AEE 2012, ESE 1995

Ans. (b) : density $(\rho) = kg/m^3$
ω = rad/sec
D = metre

dynamic viscosity,

$$\mu = \text{Pa-sec} = \frac{kg-m}{s^2}.\frac{1}{m^2}.\text{sec} = \frac{kg}{m-\text{sec}}$$

$$\frac{\rho\omega D^2}{\mu} = \frac{\frac{kg}{m^3}\times\frac{1}{\text{sec}}\times m^2}{\frac{kg}{m-\text{sec}}} = \text{Dimensionless group}$$

1356. Kinematic similarity between model and prototype is the

(a) similarity of discharge
(b) similarity of shape
(c) similarity of streamline pattern
(d) none of these

CGPSC AE 2014 Paper-II
Mizoram PSC AE/SDO 2014, Paper-I

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UKPSC AE 2012, Paper-II
J&K PSC Civil Services Pre, 2010
ESE 1996

Ans. (a) : Similitude–Similitude is defined as the similarity between the model and its prototype in every respect, which means that the model and prototype have similar properties or model and prototype are completely similar. Three type of similarities must exist between the model and prototype.

1. Geometric similarity
2. Kinematic similarity
3. Dynamic similarity

1. Geometric Similarity–It means similarity in shape, size may differ

Breadth Length Diameter

$$\frac{b_p}{b_m}=\frac{L_p}{L_m}=\frac{D_p}{D_m}=L_r \leftarrow \text{Scale ratio}$$

2. Kinematic Similarity–It means similarity in motion

$$\frac{V_p}{V_m}=V_r \leftarrow \text{Velocity ratio}$$

$$\frac{a_p}{a_m}=a_r \leftarrow \text{Acceleration ratio}$$

3. Dynamic Similarity–It means the similarity of forces between the model and prototype.

$$\frac{(F_i)_p}{(F_i)_m}=\frac{(F_v)_p}{(F_v)_m}=\frac{(F_g)_p}{(F_g)_m}=F_r \leftarrow \text{Force ratio}$$

Kinematic similarity between model and prototype is the similarity of discharge.

1357. Principle of similitude forms the basis of

(a) Comparing two identical equipments
(b) Designing models so that the result can be converted to prototypes
(c) Comparing similarity between design and actual equipment
(d) Hydraulics designs

CGPSC AE 15.01.2021
Nagaland PSC CTSE 2017, 2016 Paper-2
RPSC VPITI 14.02.2016
ESE 2012

Ans. (b) : Principle of similitude forms the basis of designing models so that the result can be convert to prototypes.

1358. If there 'm' physical quantities and 'n' fundamental dimensions in a particular process, the number of non-dimensional parameters is

(a) m + n (b) m × n
(c) m – n (d) m/n

RPSC AE 2018
GPSC Executive Engineer 23.12.2018
TSPSC AEE 2017, GATE 2002

Ans. (c) : We know from Buckingham π- Theorem.
m = physical quantities
n = fundamental dimensions
Non- dimensional parameters = m – n

1359. A ship whose full length is 100 m is to travel at a speed of 10 m/sec. For dynamic similarity, at what velocity should a 1 : 25 model be towed through water?

(a) 10 m/sec
(b) 25 m/sec
(c) 2 m/sec
(d) 50 m/sec

MPSC Poly. Lect. 2014
WBPSC AE 2003, ESE 2004, 2001

Ans. (c) : Given,

$L_p = 100m$ 1 : 25 model

$U_p = 10m/s$ $L_p = 100 \Rightarrow L_m = \frac{100}{25} = 4$

Using Froude's equation

$$\frac{V}{\sqrt{gL}} = \text{const.}$$

$$\frac{V_m}{\sqrt{L_m}} = \frac{V_p}{\sqrt{L_p}}$$

$$V_m = V_p \times \sqrt{\frac{L_m}{L_p}}$$

$$= 10 \times \sqrt{\frac{4}{100}}$$

$$\boxed{V_m = 2m/s}$$

1360. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Viscous force	1. Reynolds number
B. Elastic force	2. Froude number
C. Surface tension	3. Weber number
D. Gravity	4. Mach number

	A	B	C	D
(a)	1	4	2	3
(c)	3	4	1	2

	A	B	C	D
(b)	1	2	4	3
(d)	1	4	3	2

Nagaland PSC CTSE 2015 Paper-II
ISRO Scientist/Engineer 2007

Ans. (d) :

Viscous force	–	Reynolds number
Elastic force	–	Mach number
Surface tension	–	Weber number
Gravity	–	Froude number

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1361. A model test is to be conducted for an underwater structure, which is likely to be exposed to strong water currents. The significant force are known to be dependent on structure geometry, fluid velocity, fluid density and viscosity, fluid depth and acceleration due to gravity. Choose from the codes given below which of the following numbers must match for the model with that of the prototype:

1. Mach number
2. Weber number
3. Froude number
4. Reynolds number

Codes:

(a) 3 (b) 1, 2, 3 and 4
(c) 1 and 2 (d) 3 and 4

Nagaland PSC CTSE 2015 Paper-II
ESE 2002

Ans. (d) : Reynolds Number (R_e)– It is defined as ratio of inertia force and viscous force of flowing fluid.

$$R_e = \frac{\rho VL}{\mu} \text{(Plate)}$$

or $$R_e = \frac{\rho VD}{\mu} \text{(Pipe)}$$

Application–Laminar viscous flow in confined passages (where viscous effects are significant).

Froude Number (F_r)– Froude number is defined as square root of ratio of inertia force flowing fluid to gravity force.

$$F_r = \frac{V}{\sqrt{Lg}}$$

Application–Free surface flow (where gravity effects are important).

1362. Volumetric flow rate Q, acceleration due to gravity g and head H form a dimensionless group, which is given by

(a) $\sqrt{\frac{gH^5}{Q}}$ (b) $\frac{Q}{\sqrt{gH}}$

(c) $\frac{Q}{\sqrt{g^3H}}$ (d) $\frac{Q}{\sqrt{g^2H}}$

GPSC Asstt. Director of Transport 05.03.2017
ESE 2002

Ans. (a) : Given, 'Q' unit is m^3/sec
'g' unit is m/s^2
'H' unit is m

Substitute these in options and you will get dimensionless group.

$$\Rightarrow \quad \frac{\sqrt{gH^5}}{Q} = \frac{\sqrt{m/s^2 \times m^5}}{m^3/sec} = \frac{m^3/sec}{m^3/sec}$$

= No unit

So, option (a) is correct.

1363. The model of a propeller 3m diameter, cruising at 10m/sec in air, is tested in a wind tunnel on a 1:10 scale model. If thrust of 50N is measured on the model at 5 m/sec wind speed, then thrust on prototype will be

(a) 20000 N (b) 2000 N
(c) 5000 N (d) 500 N

ISRO Scientist/Engineer 24.05.2014
ESE 1995

Ans. (a) : $\frac{\text{Thrust to Model}}{\text{Thrust to Prototype}}$

$$= \left(\frac{v_m}{v_p}\right)^2 \left(\frac{d_m}{d_p}\right)^2$$

$$\frac{F_M}{F_P} = \left(\frac{5}{10}\right)^2 \times \left(\frac{1}{10}\right)^2$$

$$= \frac{1}{4} \times \frac{1}{100} = \frac{1}{400}$$

$F_P = 400\ F_M$
$= 400 \times 50$
$= 20{,}000$ N

1364. Select statement from List II matching the processes in List I. Enter your answer as A, B if the correct choice for (1) is (A) and that for (2) is (B)

	List I		List II
(A)	Fourier number	(1)	Surface tension
(B)	Weber number	(2)	Forced convention
(C)	Grashoff number	(3)	Natural convention
(D)	Schmidt number	(4)	Radiation
		(5)	Transient heat conduction
		(6)	Mass diffusion

(a) A-2, B-1, C-3, D-5 (b) A-5, B-1, C-3, D-6
(c) A-5, B-2, C-3, D-1 (d) A-5, B-1, C-3, D-4

ISRO Scientist/Engineer 12.05.2013
GATE 1996

Ans. (b) :

1365. List-I gives 4 dimensionless numbers and List-II gives the types of forces, which are one of the constituents describing the numbers. Match List I with List II and select the correct answer using the codes given below the lists:

List I	List II
A. Euler number	1. Pressure force
B. Froude number	2. Gravity force
C. Reynolds number	3. Elastic tension
D. Weber number	4. Surface tension
	5. Viscous force

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Code:

	A	B	C	D
(a)	2	3	4	5
(b)	3	2	4	5
(c)	1	2	5	4
(d)	2	1	5	4

ISRO Scientist/Engineer 2006
ESE 1993

Ans. (c) : Reynolds number $= \dfrac{\text{Inertia Force}}{\text{Viscous Force}}$

Froude number $= \sqrt{\dfrac{\text{Inertia Force}}{\text{Gravity Force}}}$

Euler number $= \sqrt{\dfrac{\text{Inertia Force}}{\text{Pressure Force}}}$

Weber number $= \sqrt{\dfrac{\text{Inertia Force}}{\text{Surface Tension}}}$

Mach number $= \sqrt{\dfrac{\text{Inertia Force}}{\text{Elastic Force}}}$

1366. For dimensional analysis the Buckinghm's π-theorem finds the resulting equation in form of.
(a) (n-m) dimensionless para-meters
(b) n dimensionless parameters
(c) Geometric and kinetic variables
(d) Mass, length and time

OPSC AEE 2015 PAPER - II
APPSC AEE 2012

Ans. : (a) For dimensional analysis the Buckingham's π-theorem finds the resulting equation in form of (n-m) dimensionless parameters.

Buckingham's π theorem:-
The Rayleigh's method of dimensional analysis becomes more laborious if the variable are more than the number of fundamental dimensions (M,L,T). This difficulty is overcome by the using Buckingham's π-theorem, which states "If there are n variables in a physical phenomenon and if variables contain m fundamental dimensional (M,L,T) then the variable are arranged into (n-m) dimensionless terms. Each terms is called π-term".

1367. Dynamic similarity means the ratio of:
(a) Linear dimension between model and prototype is equal
(b) Velocity between model and prototype at certain point is equal
(c) Forces between model and prototype is equal
(d) All of these

OPSC AEE 2019 PAPER - II
WBPSC AE 2008

Ans : (c) : Dynamic similarity means the similarity of forces between model and prototype. Thus dynamic similarity is said to exist between the model and the prototype if the ratios of the forces acting at the corresponding points in the model and prototype are the same in magnitude the direction also should be parallel.

1368. Which of the following dimensionless numbers is significant when studying surface tension phenomena in a flow problem?
(a) The Euler number (b) The Froude number
(c) The Mach number (d) The Weber number

APPSC Poly Lect. Automobile Engg., 2020
CIL MT 27.02.2020

Ans. (d) : Weber's number (We) – It is defined as the square root of the ratio of the inertia force of a flowing fluid to the surface tension force. It is expressed as

$$\text{Weber's Number We} = \sqrt{\frac{F_i}{F_s}}$$

where

F_i = Inertia force = ρAV^2

and F_s = Surface tension force

= Surface tension per unit length × length

$= \sigma \times L$

$$We = \sqrt{\frac{\rho AV^2}{\sigma \times L}} = \sqrt{\frac{\rho \times L^2 \times V^2}{\sigma \times L}}$$

$$= \sqrt{\frac{\rho L \times V^2}{\sigma}} = \sqrt{\frac{V^2}{\sigma/\rho L}} = \frac{V}{\sqrt{\sigma/\rho L}}$$

1369. Geometric similarity between model and prototype means
(a) the similarity discharge
(b) the similarity of linear dimensions
(c) the similarity of motion
(d) the similarity of forces

APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (b) : Geometric similarity between model and prototype means similarity of linear dimensions.

1370. Geometric similarity is said to exist between the model and the prototype, if both of them
(a) Have identical velocities
(b) Are equal in size and shape
(c) Are identical in shape, but differ only in size
(d) Have identical forces

Nagaland PSC (CTSE) 2018, Paper-II
UKPSC AE 2013 Paper-II

Ans. (c) : Geometric similarity is said to exist between the model and the prototype, if both of them are identical in shape, but differ only in size, and the ratio at the velocity and acceleration and prototype are the same in magnitude the direction also should be parallel.

1371. If there are n variables (dependent and independent) in a dimensionally homogeneous equation and if these variables contain m fundamental dimensions, then the variables are arranged in to _____ dimensionless terms.
(a) n + m (b) (n + m)/2
(c) n times m (d) n – m

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I
ESE 1998

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Ans. (d) : n = variable (dependent and independent)
m = fundamental dimensions
(n – m) = non-dimensionless parameter

1372. Which one of the following is dimensionless?

(a) $\frac{\partial p}{\partial x}\cdot\frac{D^4}{\mu Q^2}$ (b) $\frac{\partial p}{\partial x}\cdot\frac{D^3}{\mu Q}$

(c) $\frac{\partial p}{\partial x}\cdot\frac{D^4}{\mu Q}$ (d) $\frac{\partial p}{\partial x}\cdot\frac{\mu Q}{D^4}$

APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (c) : $\frac{\partial P}{\partial x}\cdot\frac{D^4}{\mu Q}$ is dimension less.

Now, Dimension of

$$\frac{\partial P}{\partial x}=\left[ML^{-2}T^{-2}\right],\quad D^4=\left[L^4\right]$$

$$\mu=\left[ML^{-1}T^{-1}\right],\quad Q=\left[L^3T^{-1}\right]$$

$$\Rightarrow\quad \frac{\partial P}{\partial x}\cdot\frac{D^4}{\mu Q}=\frac{\left[ML^{-2}T^{-2}\right]\times\left[L^4\right]}{\left[ML^{-1}T^{-1}\right]\times\left[L^3T^{-1}\right]}=\left[M^oL^oT^o\right]$$

1373. Which one of the dimensionless number identifies the compressibility effect of a fluid?

(a) Euler number (b) Froude number
(c) Mach number (d) Weber number

JPSC AE, 10.04.2021, Paper-I
Assam PSC AE (PHED) 18.10.2020

Ans. (c) :

Dimensionless Number		**Effect of Fluid**
Mach number	→	Compressible flow
Euler number	→	Cavitation
Froude number	→	Free surface

1374. In compressible fluid flow problems at high velocities, which of the following non-dimensional number is used?

(a) Euler number (b) Weber number
(c) Froude number (d) Mach number

VIZAG MT, 14.12.2020, ESE 2005

Ans. (d) : Mach number → Compressible fluid flow
Euler number → Cavitation studies
Weber number → Capilarity studies
Froude number → Free surface, open channel, weir, spilway
Reynold's number → Closed conduit pipe, submerged body, submarine

1375. The ratio of momentum diffusivity to thermal diffusivity is termed as _____.

(a) Grashoff number (b) Prandtl number
(c) Biot number (d) Schmidt number

VIZAG Steel MT 24.01.2021 Shift-II
UPSC JWM Advt. No.-52/2010

Ans. (b) : Prandtl number (Pr)

$$=\frac{\text{Momentum diffusivity (kinematic viscosity)}}{\text{Thermal diffusivity}}$$

$$\Pr=\frac{\nu}{\alpha}=\frac{\mu C_p}{k}$$

1376. Peclet number is the product of :

(a) Grashof and Rayleigh number
(b) Reynolds and Stanton number
(c) Reynolds and Prandtl number
(d) Grashof and Prandtl number

NLC GET 17.11.2020, Shift-II
TSPSC Managers 2015

Ans. (c) : The peclet number is equivalent to the product of Reynolds number (R_e) and the prandtl number (P_r).

Peclet number representing the ratio of heat transfer by motion of a fluid to heat transfer by thermal conduction.

$$Pe=\frac{uL}{\alpha}=\text{Re}\cdot\text{P}_r$$

Where, u is Fluid velocity, L is a characteristic length and α is thermal diffusivity of the fluid. Re is Reynolds number and Pr is prandtl number

• Peclet number for mass transfer has corresponding

From $P_{e_m}=\frac{uL}{\delta}=\text{R}_e\,S_c$

Where δ diffusivity and S_c is Schmidt number.

1377. For buoyancy-induced fluid flow and heat transfer, which of the following dimensionless numbers is significant?

(a) Reynolds number (b) Prandtl number
(c) Grashoff number (d) Nusselt number

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : For buoyancy induced fluid flow and heat transfer, Grashoff number is significant.
It is the ratio of buoyancy to viscous force acting on a fluid.
Reynolds Number–It is the force associated with momentum to force associated with viscous shear.
Prandtl Number–The prandtl number is a dimensionless number. It is defined as the ratio of momentum diffusivity to thermal diffusivity.
Nusselt Number–Nusselt number is the ratio of heat transfer by convection and heat transfer by conduction.

1378. Which is the correct option for Froude Number?

(a) Viscous force
(b) Elastic force
(c) Gravity force
(d) Surface tension force

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : Viscous force → Reynold number
Elastic force → Mach number
Gravity force → Froude number
Surface tension force → Weber number
Pressure force → Euler number

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1379. In a derivation, the relevant parameters are volume flow rate (V), density (ρ), viscosity (μ), bulk modulus (K), pressure difference (P), power consumption (E), rotational speed (ω) and characteristic dimension (D). Using the Buckingham's π-theorem, find the number of independent non-dimensional groups.

(a) 1 (b) 5
(c) 2 (d) 7

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (b) : Given, V, ρ, μ, K, P, E, ω, D
No. of relevant variable, n = 8
No. of independent dimension m = 3
(M, L and T)
No. of non-dimensional groups (π_s) = n – m
= 8 – 3
= 5

1380. The product of Grashoff number and Prandtl number is termed as _____.

(a) Rayleigh number (b) Peclet number
(c) Fourier number (d) Reynolds number

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (a) : $\boxed{Ra = Gr \times Pr}$

1381. If ratios of all forces acting on corresponding fluid particles and boundary surfaces in the two systems under comparison are same, then the similitude is called

(a) Geometric similitude
(b) Kinematic similitude
(c) Conditional similitude
(d) Dynamic similitude

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : If ratios of all forces acting on corresponding fluid particles and boundary surfaces in the two systems under comparison are same, then the similitude is called dynamic similitude.

1382. Oil at 20°C (density = 870 kg/m^3 and viscosity = 0.104 kg/m-s) flows at 1.1 m^3/hr through a horizontal pipe with diameter d = 2 cm and length L = 12 m. Find the Reynolds number for the flow.

(a) 104 (b) 163
(c) 208 (d) 78

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) : $\rho = 870\ kg/m^3$
$v = 0.104$ kg/m-s
$Q = 1.1\ m^3/hr$
$= \frac{1.1}{36000} = 3.055 \times 10^{-4}\ m^3/sec$
d = 2 cm, L = 12 m
$Q = AV = \frac{\pi}{4} d^2 \times V$
$V = \frac{Q \times 4}{\pi \times d^2} = \frac{3.055 \times 10^{-4} \times 4}{\pi \times (0.02)^2}$
V = 0.9727 m/sec
$R_e = \frac{\rho V d}{\mu} = \frac{Vd}{\nu}$
$R_e = 162.72$

1383. The Grashoff number is defined as :

(a) $\frac{\text{viscous force}}{\text{buoyancy force}}$ (b) $\frac{\text{buoyancy force}}{\text{inertia force}}$
(c) $\frac{\text{buoyancy force}}{\text{viscous force}}$ (d) $\frac{\text{inertia force}}{\text{buoyancy force}}$

APPSC Poly Lect. 13.03.2020

Ans. (c) : The Grashoff number is defined as the ratio of
buoyancy force to viscous force.
Grashoff Number (Gr) = $\frac{\text{buoyancy force}}{\text{viscous force}}$

1384. Lewis number is defined as :

(a) the ratio of momentum diffusivity to thermal diffusivity.
(b) the ratio of thermal diffusivity to mass diffusivity
(c) the ratio of mass diffusivity to thermal diffusivity.
(d) the ratio of momentum diffusivity to mass diffusivity.

NLC GET 17.11.2020, Shift-II

Ans. (b) : The Lewis number is defined as the ratio of thermal diffusivity and molecular diffusivity. It is given by,

$$Le = \frac{Sc}{Pr} = \frac{\upsilon}{D} \times \frac{\alpha}{\upsilon} = \frac{\alpha}{D}$$

where $\alpha = k/\rho C_p$ is the thermal diffusivity.
Lewis number is important in determining the relationship between mass and heat transfer coefficient.

1385. If 'a' is diffusivity, 't' is time and 'L' is the characteristic length, Fourier number is :

(a) $\frac{L^2}{(at)}$ (b) $\frac{a}{(tL^2)}$
(c) $\frac{(at)}{L^2}$ (d) $\frac{t}{(aL^2)}$

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : The Fourier number or Fourier modulus is a dimensionless number that characterize transient heat conduction. It is the ratio of diffusive or conductive transport rate to the quantity storage rate where the quality may be either heat or matter.

The Fourier number $= \frac{at}{L^2}$

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1386. In which of the following methods of dimensional analysis a functional relationship of some variables is expressed in the form of an exponential equation, which must be dimensionally homogeneous?

(a) Buckingham's π-method
(b) Rayleigh's method
(c) Bridgman's method
(d) Matrix-tensor method

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : Rayleigh's method- Lex x is a variable, which depends on x_1, x_2, x_3 variables. Then according to Rayleigh's method, x is function of x_1, x_2, x_3 and mathematically.

$$x = f[x_1, x_2, x_3]$$

This can be written as,

$$x = k\left[x_1^a \times x_2^b \times x_3^c\right]$$

Where k is constant and a, b and c are arbitrary powers. The values of a, b and c are obtained by comparing the power of the fundamental dimension on both sides. Thus the expression, for dependent variable is obtained.

1387. According to Euler's law, a model and prototype will be dynamically similar with respect to the ratio of inertial force to :

(a) gravity force (b) pressure force
(c) viscous force (d) buoyancy force

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : Euler's Number (E_u)– It is defined as the square root of the ratio of the inertia force of a flowing fluid to the pressure force

$$E_u = \sqrt{\frac{F_i}{F_p}}$$

F_p = Intensity of pressure × Area = P × A

$F_i = \rho AV^2$

$$E_u = \sqrt{\frac{\rho AV^2}{P \times A}} = \sqrt{\frac{V^2}{P/\rho}} = \frac{V}{\sqrt{P/\rho}}$$

1388. For froude model law, scale ratio of force and scale ratio of length are related as : [Assuming the experiment is conducted on same place and in same fluid]

(a) Scale ratio of force = Scale ratio of length
(b) Scale ratio of force = (Scale ratio of length)^4
(c) Scale ratio of force = (Scale ratio lf length)^2
(d) Scale ratio of force = (Scale ratio of length)^3

SJVN ET 2019

Ans. (d) : Scale ratio of force = (Scale ratio of length)^3

1389. A model of torpedo is tested in a towing tank at a velocity of 36 m/sec. The prototype is assumed to attain a velocity of 6 m/sec. What model scale should be used?

(a) 1:3 (b) 1:4.5
(c) 1:6 (d) 1:(6)^{½}

HPPSC ADF 05.03.2019

Ans. (c) : Given $V_m = 36$ m/sec

$V_p = 6$ m/sec

Reynolds model law is applicable for a model of torpedo which is completely submerged in water.

Reynolds model,

$$\frac{\rho_m V_m L_m}{\mu} = \frac{\rho_P V_P L_P}{\mu}$$

$$\frac{L_m}{L_P} = \frac{V_P}{V_m} = \frac{6}{36} = \frac{1}{6}$$

1390. Problems of fluid flow form an orifice or nozzle can be solved by:

(a) Froude Model (b) Mach Model
(c) Euler Model (d) Weber Model

OPSC AEE 2019 PAPER - II

Ans : (c) : Problems of fluid flow form an orifice or nozzle can be solved by Euler model.

1391. The pressure difference in a turbulent flow through the pipe of diameter (D) and length (L) depends on velocity, viscosity, density and roughness. The number of π-terms formed will be:

(a) 2 (b) 3
(c) 4 (d) 5
(e) 6

CGPSC AE 25.02.2018

Ans. (c) : Total no. of variable (n) = 7
Total no. of fundamental units (m) = 3
Then total no. of π term = (n – m)
= 7 – 3
= 4

1392. Position of a body with acceleration 'a' is given by $x = Ka^mt^n$, where t is time. The value of m and n are:

(a) m = 1, n = 1 (b) m = 2, n = 2
(c) m = 1, n = 3 (d) m = 1, n = 2

AAI Jr. Executive 29.11.2018

Ans. (d) : $x = Ka^mt^n$

Dimension,

$x = [L]$

Acceleration $a = [LT^{-2}]$

Time → $t = [T]$

$x = Ka^mt^n$

$[L] = K[LT^{-2}]^m [T]^n$

$[LT^0] = K[L]^m [T]^{(-2m+n)}$

Comparing both side,

$m = 1$

$-2m + n = 0$

$n = 2$

$m = 1, n = 2$

1393. If there are 4 number of fundamental dimensions and 7 number of variable, then the number of pi (π) terms for dimensional analysis will be:

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(a) 4 (b) 3
(c) 7 (d) 11

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : Bucklingham π theorem is used to determine the number of dimensional groups required to describe a phenomena. According to this group to define a problem equate the total number of variables n (like density viscosity etc.) minus the fundamental dimensions p (like length, time etc)

Given, $n = 7$

$p = 4$

Dimensional analysis $= n - p$

$= 7 - 4 = 3$

$F(\pi_1, \pi_2, \pi_3 .....) = 0$

1394. Consider steady laminar incompressible axis symmetric fully developed viscous flow through a straight circular pipe of constant cross sectional area at a Reynolds' number of 5. The ratio of inertia force to viscous force on a fluid particle is

(a) 5 (b) 1/5
(c) 0 (d) ∞

TSPSC AEE 2017

Ans. (a) : We know that,

$$\text{Reynolds No} = \frac{\text{Inertia force}}{\text{Viscous force}}$$

So, Reynolds No. = 5

1395. If the force due to inertia, gravity and frictional resistance are significant, the design of a channel may be done by comparing

(a) Weber number (b) Reynolds number
(c) Froude number (d) Prandtl number

Karnataka PSC Lect., 27.05.2017

Ans. (c) : Froude's Number-

$$F_e = \sqrt{\frac{\text{Inertia Force}}{\text{Gravity Force}}}$$

$$= \sqrt{\frac{\rho.L^2V^2}{\rho.L^3.g}} = \frac{V}{\sqrt{g.L}}$$

Applications-

(i) Open channels
(ii) Free surface flow such as flow over spillways, weirs, sluices.
(iii) Flow of jet from an orifice or nozzle.
(iv) Flow of different density fluids one above the other.

1396. Statement (I) : When variables are a few, then Rayleigh method is used in dimensional analysis.
Statement (II) : When variables are more, then Buckigham π-method is used since the Rayleigh method becomes cumbersome in dimensional analysis.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I).
(c) Statement (I) is true but Statement (II) is false.
(d) Statement (I) is false but Statement (II) is true.

UPSC JWM 2017

Ans. (a) : Buckingham's π-theorem- "If there are n variable (dependent and independent) in a dimensionally homogeneous equation and if these variable contain m fundamental dimensions, then the variables are arranged into (n - m), dimensionless terms."

Rayleigh's Method- This method gives a special form of relationship among the dimensional group. In this method a functional relationship of some variables is expressed in the form of an exponential equation which must be dimensionally homogeneous.

1397. Consider the following statements:

1. In model testing, the same fluid must be used in prototype and also the model
2. Models should be smaller than prototype for geometric similarity
3. Geometrically similar boundaries do not necessarily imply kinematically homogeneous flows
4. Existence of geometric similarity does not imply dynamic similarity

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 and 4 only (d) 1 and 4 only

UPSC JWM 2017

Ans. (c) : Geometrically similar boundaries do not necessarily imply kinematically homogenous flow.

- Existence of geometric similarity does not imply dynamic similarity.

1398. The advantages of dimensional analysis are

(a) It is used to change the theoretical equation into dimensionless form
(b) In model testing, it reduces the number of variables into three numbers
(c) It helps to convert the units of quantities from system to another
(d) All of these

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : The advantage of dimensional analysis are;

i) It is used to change the theoretical equation into dimensionless form.
ii) In model testing, it reduces the number of variables into three numbers.
iii) It helps to converts the units of quantities from one system to another system.

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1399. A phenomenon is modeled using n dimensional variables with k primary dimensions. The number of non-dimensional variables is

(a) k (b) k-n
(c) n/k (d) n-k

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : n–k

1400. A dimensionless combination of pressure drop 'ΔP', dynamic viscosity 'μ', velocity 'V' and length 'L' is :

(a) $\Delta P/(V^2.L)$ (b) $(V.L.)/\mu$
(c) $\Delta P/(\mu.V.L)$ (d) $(\Delta P.L)/(\mu.V)$

RPSC ADE 2016

Ans. (d) : Unit of viscosity 'μ'

$$\tau = \mu\frac{du}{dy} = \mu \times \frac{V}{l}$$

$$\Delta P \times L = \mu \times V$$

$$\Delta P = \frac{\mu V}{L}$$

Shear stress has also unit of pressure.

1401. Biot (Bi) Number is defined as ratio at :

(a) The convective thermal resistance to the conduction thermal resistance
(b) the conduction thermal resistance to the convective thermal resistance.
(c) The thermal energy storage capacity to the conduction thermal resistance.
(d) The thermal energy storage capacity to the convection thermal resistance.

GPSC Lect. (Auto) 16.10.2016

Ans. (b) :

$$\text{Biot number (Bi)} = \frac{\text{Internal Conductive Resistance[ICR]}}{\text{External Convective Resistance[ECR]}}$$

$$\boxed{B_i = \frac{hL_c}{k}}$$

Where, h = Convective heat transfer coefficient
L = Characteristic length of the geometry considered
k = Thermal conductivity of body.

1402. The dimensional number driving natural convection is :

(a) Grashoff number (b) Reynolds number
(c) Biot number (d) Prandtl number

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Grashoff number- The dimensional number driving natural convection is Grashoff number (Gr)

Grashoff number (Gr)

$$= \frac{\text{Inertia Force} \times \text{Buoyancy Force}}{(\text{Viscous Force})^2}$$

$$\boxed{Gr = \frac{L^3 g\beta\Delta T}{v^2}}$$

Where, v = Kinematic viscosity
β = Coefficient of thermal expansion

1403. Which of the following statements are true for dimensionless analysis?

(1) The functional relationship between dependent and non-dependent variables
(2) In model testing, it reduces the number of variables into three numbers
(3) It is used to change the theoretical equation into dimensionless form
(4) It helps to convert the units of quantities from one system to another system

(a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 3, 4 (d) 1, 2, 3, 4

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) The advantage of dimensionless analysis are:-

(i) It is used to change the theoretical equation into dimensionless form.
(iii) In model testing, it reduces the number of variable into three numbers.
(iii) It help to converts the unit of quantities from one system to another system.
(iv) Functioned relationship between dependent and non-dependent variables.

1404. The value of bulk modulus of a fluid is required to determine

(a) Reynold's number (b) Froude's number
(c) Mach number (d) Euler's number

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : The value of Bulk modulus of a fluid is required to determine, Mach number

→ Ma < 0.33, Value for water (incompressible fluid)
→ If fluid velocity (V) < 0.33C, then fluid treat as incompressible fluid

$$Ma = \frac{V}{C},\ C = \sqrt{\frac{K}{\rho}} \text{ for fluid}$$

K = Bulk modulus
V = Velocity of fluid
ρ = density of fluid

1405. The Froude number at the end of a hydraulic jump in a rectangular channel is 0.25. The sequent depth ratio of this jump is

(a) 2.5 (b) 5.2
(c) 8.9 (d) 9.8

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : For a given Froude number F_r, the sequent depth ratio of hydraulic jump,

$$\frac{h_2}{h_1} = \frac{1}{2}\left(\sqrt{1+8F_r^2} - 1\right)$$

$$\frac{h_2}{h_1} = \frac{1}{2}\left(\sqrt{1+8(0.25)^2} - 1\right)$$

$$\frac{h_2}{h_1} = 0.11 \Rightarrow \frac{h_1}{h_2} = 8.9$$

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1406. The type of jump that forms when the initial Froude number lies between 2.5 and 4.5 is known as

(a) A oscillating jump (b) Weak jump
(c) Steady jump (d) Undular jump

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : The oscillating type of jump that forms when the initial Froude number lies between 2.5 and 4.5.

1407. Which of the following is not used for model analysis

(a) Reynold's Model Law
(b) Froude Model Law
(c) Newton Model Law
(d) Mach Model Law

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) : Newton Model Law is not used for model analysis among the given option.

1708. Froude number is important for:

(a) Unsteady flow (b) Viscous flow
(c) Free surface flow (d) Compressible flow

UPRVUNL AE 2016

Ans. (c) : Froude number is important for free surface flow.

$$\text{Froude number} = \sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}}$$

1709. The scale ratio in model of spillway is 1:9. If the discharge in the prototype is 2430 cumecs, the discharge in the model is (in cumecs)

(a) 270 (b) 90
(c) 30 (d) 10

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (d) : As, we know,
Froude model Law will be applicable in this case.

$$\frac{Q_m}{Q_p} = \left(\frac{L_m}{L_p}\right)^{5/2}$$

$$Q_m = 2430 \times \left(\frac{1}{9}\right)^{5/2}$$

$$= 2430 \times \left(\frac{1}{3}\right)^{2\times\frac{5}{2}}$$

$$= 2430 \times \left(\frac{1}{3}\right)^{5}$$

$$Q_m = 10 \text{ cumecs}$$

1410. If a sphere of 10 mm dia is falling in a fluid medium of kinematic viscosity of 10 stokes with terminal velocity of 12 mm/s, then coefficient of drag on the sphere will be

(a) 12 (b) 24
(c) 100 (d) 200

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (d) : As We know,
V = 12 mm/s
$v = 10 \times 10^{-4}$ N-m/s
D = 10 mm

$$C_d = \frac{24}{Re}$$

$$C_d = \frac{24 \times 10^2 \times 10}{10 \times 12} \qquad \therefore Re = \frac{VD}{\upsilon}$$

$$\boxed{C_d = 200}$$

1411. Which of the following is dimensionless parameter

(a) Specific mass (b) Specific weight
(c) Specific gravity (d) Specific volume

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : Specific gravity(s)

- It is the ratio of specific weight (or mass density) of fluid to the specific weight (or mass density) of a standard fluid at a specified temperature.

$$\text{Specific gravity (S.G.)} = \frac{\rho_{fluid}}{\rho_{water}}$$

- Standard fluid for solid & liquid is water.
- Standard fluid for gases is air.

1412. Reynold's number for non- circular cross-section is

(a) $\frac{V4\rho}{\vartheta}$ (b) $\frac{V\delta}{\vartheta}$
(c) $\frac{V2\delta}{\vartheta}$ (d) $\frac{V\delta}{4\vartheta}$

VIZAG STEEL MT 2015

Ans. (a) : $\frac{V4\rho}{\vartheta}$

1413. For similarity, in addition to models being geometrically similar to prototype, the following in both cases should also be equal

(a) Ratio of inertial force to force due to viscosity
(b) Ratio of inertia force to force due to gravitation
(c) Ratio of inertial force to force due to surface tension
(d) All the four ratios of inertial force to force due to viscosity, gravitation, surface tension, and elasticity

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : For similarity, in addition to models being geometrically similar to prototype, in both cases should also be equal to all the four ratios of inertial force to force due to viscosity, gravitation, surface tension, and elasticity.

1414. A dimensionless number associated with transient conduction heat transfer is

(a) Reynolds number (b) Nusselt number
(c) Froude number (d) Biot number

TANGEDCO AE 2015

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Ans. (d) : The non-dimensional factor $\frac{hL_c}{k}$ is called Biot number (Bi). It gives an indication of the ratio of internal (Conduction) resistance to surface (Convection) resistance. If Bi < 0.1, the lumped heat capacity approach is used.

$$\text{Biot No.} = \frac{R_{cond.}}{R_{conv.}} = \left(\frac{L/kA}{\frac{1}{hA}}\right)$$

$$= \frac{hL_c}{K}$$

Dimensionless number used for transient heat transfer.

1415. A 1:20 model of a naval ship has a submerged area of 1.5m^3 and length 8 m with a total drag of 2 kgf when towed through water at a velocity of 1.4 m/s. Calculate the Reynold's number.

(a) 1.0×10^5 (b) 1.1×10^7
(c) 1.2×10^7 (d) 1.5×10^7

UPRVUNL AE 2014

Ans. : (c) : Given,
$\rho_w = 1000$ kg/m^3
v = 1.4 m/sec.
Characteristic dimension = D = L = 8m
Assume $\mu_w = 9.34 \times 10^{-4}$ Ns/m^2
Now,

$$R_e = \frac{\rho vD}{\mu} = \frac{1000 \times 1.4 \times 8}{9.34 \times 10^{-4}}$$

$$= 1.199 \times 10^7$$

$$\boxed{R_e = 1.2 \times 10^7}$$

1416. Water having kinematic viscosity of 0.01 stoke flows at a velocity of 2m/sec in a pipe of 15 cm diameter. For dynamic similarity, the velocity of oil of kinematic viscosity of 0.03 stoke in a pipe of same diameter will be

(a) 0.33 m/sec (b) 0.66 m/sec
(c) 2 m/sec (d) 6 m/sec

ISRO Scientist/Engineer 24.05.2014

Ans. (d) :
given,- for water,-
kinematic viscosity
$\upsilon_1 = 0.01$ Stoke
$V_1 = 2$ m/s
d = 15cm
For oil, -
$\upsilon_2 = 0.03$ Stoke
$V_2 = ?$
d = 15 cm (same)
Dynamic Similarity- Reynold's number
would be same $\left(\frac{VD}{\upsilon}\right)_1 = \left(\frac{VD}{\upsilon}\right)_2$

or $V_2 = \frac{\upsilon_2}{\upsilon_1}$ $V_2 = \frac{0.03}{0.01} \times 2 = 6\text{m/s}$

1417. The Reynolds's number for flow of fluid in a circular tube is 4000. If the diameter of the tube is increased by 25% and the fluid velocity is decreased by 20%, the Reynolds's number for the same fluid will be

(a) 4000 (b) 4200
(c) 3800 (d) 3000

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a)

$$\boxed{R_e = \frac{\rho vD}{\mu}}$$

where
v = Velocity
D = Diameter of Pipe
ρ = Density
μ = Dynamic Viscosity

$R_e \propto vD$

$$\frac{(R_e)_1}{(R_e)_2} = \frac{V_1 D_1}{V_2 D_2}$$

$(R_e)_1 = 4000$
$D_2 = 1.25D_1$
$V_2 = .8V_1$

$$\frac{4000}{(R_e)_2} = \frac{V_1 \times D_1}{1.25D_1 \times .8V_1}$$

$$\frac{4000}{(R_e)_2} = \frac{100 \times 10}{125 \times 8}$$

$(R_e)_2 = 4000$

1418. A proposed pier design is studied in a water channel to simulate forces due to hurricanes. Using a 1:10 scale model, what velocity should be selected in the model study to simulate a water speed of 12 m/s?

(a) 3.79 m/s (b) 4.28 m/s
(c) 5.91 m/s (d) 6.70 m/s
(e) 6.90 m/s

CGPSC AE 26.04.2015 PAPER-II

Ans. (a) :
In this case gravity force is only predominant force. Which controls the flow in addition to the force of inertia.
According to Froude model law.

$$(F_r)_m = (F_r)_s$$

$$\frac{V_m}{\sqrt{L_m g}} = \frac{V_s}{\sqrt{L_s g}}$$

$$V_m = \sqrt{\frac{L_m}{L_s}} \times V_s \qquad \text{Model Scale} = 1 : 10$$

$$V_m = \sqrt{\frac{1}{10} \times 12} \qquad \left[\frac{L_m}{L_m} = \frac{1}{10}\right]$$

$$\boxed{V_m = 3.794\,\text{m/s}}$$

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1419. Which of the following non-dimensional numbers is not correctly define?

(a) Reynolds number = inertia force /viscous force

(b) Froude number = (inertia force/gravity force)$^{1/2}$

(c) Weber number = inertia force/pressure force

(d) Euler number = pressure force/inertial force.

RPSC AEN Pre-2013

Ans. (*) : Deleted by official

$$\text{Reynolds Number} = \frac{\text{Inertia force}}{\text{Viscous force}}$$

$$\text{Mach Number} = \sqrt{\frac{\text{Inertia force}}{\text{Compressibility force}}}$$

$$\text{Froude Number} = \sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}}$$

$$\text{Weber Number} = \sqrt{\frac{\text{Inertia force}}{\text{Surface tension}}}$$

$$\text{Eulers Number} = \sqrt{\frac{\text{Inertia force}}{\text{Pressure force}}}$$

1420. Froude number is significant in:

(a) Supersonics, as with projectile and jet propulsion

(b) Full immersion or completely flow, as with pipes, air crafts wings, nozzles, etc.

(c) Simultaneous motion through two fluids where there is a surface discontinuity, gravity forces and wave making effect, as with slip's hulls

(d) All of these

ISRO Scientist/Engineer 12.05.2013

Ans. (c) : Froude number is significant in Simultaneous motion through two fluids where there is a surface discontinuity, gravity forces and wave making effect, as with slip's hulls.

1421. Match the following :

P: Compressible flow	U: Reynolds number
Q: Free surface flow	V: Nusselt number
R: Boundary layer flow	W: Weber number
S: Pipe flow	X: Froude number
T: Heat convection	Y: Mach number
	Z: Skin friction coefficient

(a) P-Q; Q-X; R-V; S-Z; T-W

(b) P-W; Q-X; R-Z; S-U; T-V

(c) P-Y; Q-W; R-Z; S-U; T-X

(d) P-Y; Q-W; R-Z; S-U; T-V

ISRO Scientist/Engineer 12.05.2013

Ans. (d)

1422. Lewis number is given by the following relation: (f_g = heat transfer coefficient, K_w is mass transfer coefficient and Cp is specific heat):-

(a) $\frac{C_p}{f_g \cdot K_w}$ (b) $\frac{K_w}{f_g \cdot C_p}$

(c) $\frac{f_g}{C_p - K_w}$ (d) $\frac{f_g}{K_w \cdot C_p}$

UKPSC AE-2013, Paper-II

Ans. (d) :

1423. Nature of flow is determined by:

(a) Mach number (b) Froude number

(c) Reynolds number (d) Weber's number

SJVN ET 2013

Ans. (c) : Nature of flow is determined by Reynolds number.

1424. A model of a golf ball is to be studied to determine the effects of the dimples. A sphere 10 times larger in diameter than an actual golf ball is used in the wind tunnel study. What speed should be selected for the model to simulate a prototype speed of 50 m/s?

(a) 0.2m/s (b) 5m/s

(c) 50m/s (d) 10m/s

ISRO Scientist/Engineer 2012

Ans. (b) : $(Re)_{model} = (Re)_{prototype}$

$$\left(\frac{\rho VD}{\mu}\right)_m = \left(\frac{\rho VD}{\mu}\right)_p$$

$V \times (10D) = 50 \times D$

$V = 5$ m/s

1425. Match the following and choose the correct alternatives :

List – I	List – II
A. Froude number	1. $p/\rho U^2$
B. Mach number	2. $U/(gd)$
C. Webber number	3. $U/\sqrt{gd}$
D. Euler number	4. $\rho LU^2/\sigma$
	5. $\frac{p}{\rho^2 U^2}$
	6. U/c (c = velocity of sound)

(Notations used have usual meaning)

Codes :

	A	B	C	D
(a)	3	6	3	5
(b)	3	6	4	1
(c)	2	4	5	2
(d)	2	6	1	5

UKPSC AE 2012 Paper–II

Ans. (b) : A-3, B-6, C-4, D-1

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1426. Rayleigh's method of dimensional analysis is used for determining the expression for a variable which depends on maximum following number of variables.

(a) 4 (b) 8
(c) 2 (d) 6

UKPSC AE 2012 Paper–II

Ans. (a) : 4

1427. Which dimensionless number has significant role in defining laminar and turbulent flow pattern ?

(a) Prandtl number (b) Reynolds number
(c) Nusselt number (d) Froude's number

OPSC Civil Services Pre. 2011

Ans. (b) : Reynolds play important role in laminar and turbulent flow.

1428. A flowmeter when tested in a laboratory gives a pressure drop of 100 kN/m^2 for a discharge of 0.1 m^3/s in a 150-mm diameter pipe. If a geometrically similar model is tested in 600-mm diameter pipe at identical conditions of fluid, the corresponding discharge will be

(a) 0.1 m^3/s (b) 0.2 m^3/s
(c) 0.3 m^3/s (d) 0.4 m^3/s

UPSC JWM Advt. No.-50/2010

Ans. (d) : $Re_1 = Re_2$

$$\frac{\rho v D_1}{\mu} = \frac{\rho v D_2}{\mu}$$

$$\frac{\rho (Q_1/A_1) D_1}{\mu} = \frac{\rho (Q_2/A_2) D_2}{\mu}$$

$$\frac{Q_1}{D_1} = \frac{Q_2}{D_2}$$

$$Q_2 = \frac{D_2}{D_1} \times Q_1$$

$$Q_2 = \frac{600}{150} \times 0.1$$

$$\boxed{Q_2 = 0.4\,m^3/sec}$$

1429. Reynolds model law is applicable for :

(a) flow over the spill way of a dam
(b) water hammer created in penstocks
(c) flow through low speed turbo machines
(d) flow of blood in veins and arteries

APPSC IOF, 2009

Ans. (c) : Reynolds law has applications –
(i) Incompressible fluid flow in closed pipes
(ii) Motion of a body fully submerged in a fluid
(iii) Motion of aeroplanes
(iv) Flow of a fluid very close to bodies fully submerged.

1430. The principle of dimensional homogeneity implies that :

(a) only the variables with the same dimensions may be multiplied or divided
(b) only the variables with the same dimensions may be added or subtracted
(c) only the dimensional quantities may be added or multiplied
(d) dimensions of the two sides of the equation may not be the same

APPSC IOF, 2009

Ans. (b) : Principle of homogeneity states that dimensions of each of the terms of a dimensional equation on both sides should be same. The principle is helpful because it helps to convert the units from one form to another.

1431. A given fluid flow problem involves the variables such as volume flow rate Q, dynamic viscosity μ, density ρ, characteristic length l and acceleration due to gravity g, Which of the following combination of variables is dimensionless?

(a) $\frac{Q^2}{gl^2}$ (b) $\frac{\rho g l^2}{Q^2}$

(c) $\frac{\rho Q l}{\mu}$ (d) $\frac{\rho Q}{\mu l}$

DRDO Scientists 2009

Ans. (d) : Dimension of $\frac{Q^2}{gL^2} = \frac{(m^3/s)^2}{\frac{m}{s^2} \times m^2} = m^3$

Dimension of $\frac{\rho g L^2}{Q^2} = \frac{\frac{kg}{m^3} \times \frac{m}{s^2} \times m^2}{\left(\frac{m^3}{s}\right)^2} = \frac{kg}{m^6}$

Dimension of $\frac{\rho Q L}{\mu} = \frac{\frac{kg}{m^3} \times \frac{m^3}{s} \times m}{\frac{kg}{ms}} = m^2$

Dimension of $\frac{\rho Q}{\mu L} = \frac{\frac{kg}{m^3} \times \frac{m^3}{s}}{\frac{kg}{ms} \times m}$

= No dimension

1432. Match List-I and List-II.

List-I	List-II
M. Stanton number	**R. Surface tension**
N. Mach number	**S. Mass transfer**
O. Weber number	**T. Free convection**
P. Grashoff number	**U. Forced convection**
Q. Schmidt number	**V. Compressibility**

(a) M-T; N-V; O-S; P-U; Q-R
(b) M-U; N-V; O-R; P-T; Q-S
(c) M-U; N-V; O-S; P-T; Q-R
(d) M-T; N-V; O-U; P-S; Q-R

DRDO Scientists 2009

Ans. (b) :

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1433. The flight Mach number of an aircraft flying at an altitude where the temperature is –48°C is 0.7. The speed of the aircraft is (take $\gamma = 1.4$, R = 287 J/kgK)

(a) 100 m/s (b) 140 m/s
(c) 210 m/s (d) 250 m/s

DRDO Scientists 2008

Ans. (c) : Mach number $= \frac{V}{\sqrt{\gamma RT}} = 0.7$

$= \frac{V}{\sqrt{1.4 \times 287 \times 225}} = 0.7$

V = 210.472 m/s

1434. Which of the following is not a dimensionless number?

(a) the coefficient of lift
(b) the pipe friction factor
(c) the Manning's coefficient
(d) the coefficient of discharge

UKPSC AE 2007 Paper -II

Ans. (c) : The Manning's coefficient is not a dimensional number. It represents the roughness or friction applied to the flow by the channel. According to Mannings formula, discharge through an open channel,
$Q = A\, Mm^{2/3}\, i^{1/2}$
where, M = Manning's coefficient
It's unit $\Rightarrow \frac{\sec}{m^{1/3}}$

13. Fluid Flow Around Submerged Bodies

1435. The critical angle of attack of an aerofoil is that, where

(a) the lift becomes zero
(b) the drag becomes zero
(c) the drag begins to rise
(d) the lift begins to drop

UKPSC AE-2013, Paper-II
JPSC AE 2013 Paper-V
APPSC AE 04.12.2012

Ans. (d) : The critical angle of attack of an aerofoil at which the flow of air about the airfoil changes abruptly so that lift is sharply reduced and drag is sharply increased. Also known as stalling angle of attack.

1436. A 2 m diameter cylinder rotates at 1800 r.p.m. in a stream (ρ = 1.225 kg/m^3) of 25 m/s. Estimate the lift per unit length of the cylinder

(a) L = 11.55kN (b) L = 9.22kN
(c) L = 24kN (d) L = 5.99kN

APPSC AEE 2012, APPSC AE 04.12.2012

Ans. (*) : Given, Diameter (d) = 2m, N = 1800 rpm.
$\rho = 1.225$ kg/m^3
V = 25 m/sec
$L = (4\pi^2 r^2 N \rho V)/60$

$= \frac{4\pi^2 \times 1^2 \times 1800 \times 1.225 \times 25}{60}$

= 36270.796
L = 36.27 kN

1437. A trip wire is mounted near the leading edge of a body

(a) to increase the lift
(b) to decrease the lift
(c) to advance the point of separation
(d) to delay the point of separation

APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (d) : To delay the point of separation, a trip wire is mounted near the leading edge of a body.

1438. Which one of the following causes lift on an immersed body in a fluid stream?

(a) Buoyant forces
(b) Resultant fluid force on the body
(c) Dynamic fluid force component exerted on the body parallel to the approach velocity
(d) Dynamic fluid force component exerted on the body perpendicular to the approach velocity

Vizag Steel MT (Re-Exam) 24.11.2013
ESE 2005

Ans. (d) : Lift–A fluid flowing past the surface of a body exerts a force on it. Lift is the component of this force that is perpendicular to the on coming flow direction.
If contracts with the drag force, which is the component of the force parallel to the flow direction.
Lift conventionally acts in an upward direction in order to counter the force of gravity, but it can act in any direction at right angles to the flow.

1439. A parachutist has a mass of 90 kg and a projected frontal area of 0.30 m^2 in free fall. The drag coefficient based on frontal area is found to be 0.75. If the air density is 1.28 kg/m^3, the terminal velocity of the parachutist will be

(a) 104.4 m/s (b) 78.3 m/s
(c) 25 m/s (d) 18.5 m/s

Nagaland PSC CTSE 2015 Paper-II
ESE 1999

Ans. (b) : Total drag (F_D) = Weight (W)

$C_D \times \frac{\rho V^2}{2} \times A = mg$

$V = \sqrt{\frac{2mg}{C_D \times \rho \times A}}$

$V = \sqrt{\frac{2 \times 90 \times 9.81}{0.75 \times 1.28 \times 0.3}}$

V = 78.3 m/sec

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1440. An automobile moving at a velocity of 40 km/hr is experiencing a wind resistance of 2 kN. If the automobile is moving at a velocity of 50 km/hr, the power required to overcome the wind resistance is

(a) 43.4 kW (b) 3.125 kW
(c) 2.5 kW (d) 27.776 kW

Nagaland PSC CTSE 2015 Paper-II
ESE 2000

Ans. (a) : Given, V_1 = 40 km/hr, V_2 = 50 km/hr, $F_{D_1} = 2\,kM$

$$\text{Power} = F_D \times V = C_D \times \frac{\rho V^2}{2} \times A \times V = C_d \frac{\rho A V^2}{2}$$

$$P \propto V^3$$

$$\frac{P_2}{P_1} = \left(\frac{V_2}{V_1}\right)^3 \text{ or } P_2 = P_1\left(\frac{V_2}{V_1}\right)^3$$

$$P_2 = \left(F_{D_1} \times V_1\right) \times \left(\frac{V_2}{V_1}\right)^3$$

$$P_2 = \left(2 \times 40 \times \frac{5}{18}\right) \times \left(\frac{50}{90}\right)^3 = 43.4\,kW$$

1441. A streamlined body is defined as a body about which

(a) the drag is zero
(b) the flow is laminar
(c) the flow is along streamlines
(d) the flow separation is suppressed

UKPSC AE 2012, 2007 Paper–II

Ans. (d) : A streamlined body is defined as a body about which the flow separation is suppressed.

1442. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Stoke's law	1. Strouhal number
B. Bluff body	2. Creeping motion
C. Streamline body	3. Pressure drag
D. Karman vortex street	4. Skin friction drag

	A	B	C	D
(a)	2	3	1	4
(c)	2	3	4	1

	A	B	C	D
(b)	3	2	4	1
(d)	3	2	1	4

Nagaland PSC CTSE 2015 Paper-II
ESE 2001

Ans. (c) :

Stoke's law	–	Creeping motion
Bluff body	–	Pressure drag
Streamline body	–	Skin friction drag
Karman vortex street	–	Strouhal number

1443. What will be the drag force on an object, if object speed is doubled in fluid?

(a) Doubled (b) Four times
(c) Remains same (d) Becomes zero.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper-II

Ans. (b) : Drag force $(F_d) = \frac{1}{2}\rho v^2 C_d A$

$$F_d \propto v^2$$

$$v_2 = 2v$$

$$\frac{F_{d_2}}{F_{d_1}} = \frac{(2v)^2}{v^2} = 4$$

$$\boxed{F_{d_2} = 4F_{d_1}}$$

1444. The drag due to tangential stresses on the surface is :

(a) Pressure drag (b) Friction drag
(c) Induced drag (d) Form drag

RPSC ACF & FRO, 26.02.2021

Ans. (b) : The drag due to tangential stresses on the surface is friction drag.

1445. The lift force on an airfoil is due to

(a) the circulation of air around it
(b) the pressure difference of the top and bottom surface
(c) the formation of tip vortices
(d) the angle of attach

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : The lift force on an airfoil is due to the pressure difference of the top and bottom surface.

1446. The body is called bluff body if the surface of the body.

(a) Very rough
(b) Very smooth
(c) Coincides with streamlines
(d) Does not coincides with streamlines

OPSC AEE 2019 PAPER - II

Ans : (d) : The bluff body can be defined as the body whose surface does not coincide with the streamline when places in a flow. As the pressure on downstream falls substantially and produces the pressure difference on the upstream and downstream side of body, a very wide wake is developed.

1447. The skin friction drag on a sphere for $R_e < 0.2$ is equal to:

(a) Equal to total drag
(b) Two-third of total drag
(c) One-third of total drag
(d) Double of total drag

OPSC AEE 2019 PAPER - II

Ans : (b) : The skin friction drag on a sphere for $R_e < 0.2$ is equal to two-third of total drag.

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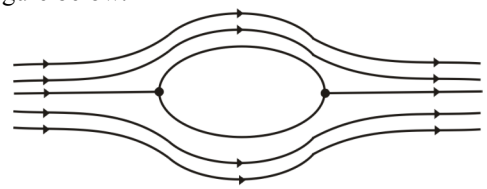
1448. A Rankine oval body has stagnation point (s) at

(a) $\theta = 0$ only (b) $\theta = \pi$ only

(c) $\theta = 0$ and $\theta = \frac{\pi}{2}$ (d) $\theta = 0$ and $\theta = \pi$

APPSC-AE-2019

Ans. (d) : For Rankine oval there are two stagnation points one on front and other at back side as shown in the figure below.



1449. Which object generates less drag when subjected to a flow field?

(a) Cylinder (b) Sphere

(c) Aerofoil (d) Square

APPSC-AE-2019

Ans. (c) : Aerofoil does not have pressure drag as flow doesn't separate on aerofoil.
Pressure drag is the dominant drag at moderate Reynolds number. Hence, aerofoil has least drag as compared to other blunt objects.

1450. The major drag force experienced by the body at very small velocity is due to

(a) pressure drag (b) skin friction drag

(c) both (a) and (b) (d) wakes

APPSC-AE-2019

Ans. (b) : When velocity is very small the flow falls into stokes regime i.e. creeping flow (Re < 1)
In creeping flow, there is no flow separation and pressure drag is less as compared to frictional drag.

1451. What will be the drag force exerted on a plate of size 2 m × 2 m, when the plate is moving at speed of 4 m/s normal to its plane in water? [Consider, Coefficient of drag = 1.2]

(a) 38.4 kN (b) 32.2 kN

(c) 45.7 kN (d) 40.2 kN

SJVN ET 2019

Ans. (a) : $F_D = \frac{1}{2} C_D \rho A V^2$

$= \frac{1}{2} \times 1.2 \times 1000 \times 2 \times 2 \times 4^2 = 1.2 \times 1000 \times 2 \times 16$

$= 38400 = 38.4 \text{ kN}$

1452. A sphere is settling down in a viscous fluid at a Reynolds number 0.24. Its drag coefficient is:

(a) 20 (b) 200

(c) 10 (d) 100

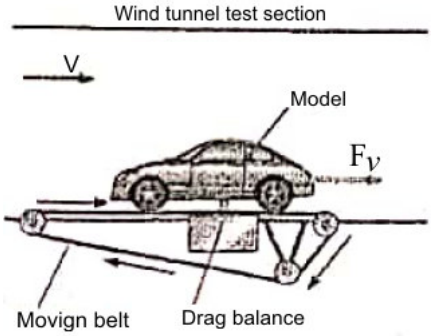
AAI Jr. Executive 29.11.2018

Ans. (d) : Drag in the sphere $= \frac{24}{R_e}$

$C_d = \frac{24}{0.24}$

$C_d = 100$

1453. The aerodynamic drag of a new sports car is to be predicted at a speed of 50.0 mi/h at an air temperature of 25°C. Automotive engineers build a one-fifth scale model of the car to test in a wind tunnel. It is winter and the wind tunnel is located in an unheated building; the temperature of the wind tunnel air is only about 5°C. Determine how fast the engineers should run the wind tunnel in order to achieve similarity between the model and the prototype



(a) 221 mi/h (b) 321 mi/h

(c) 211 mi/h (d) 121 mi/h

(e) None of these

CGPSC AE 16.10.2016

Ans. (a) :

Prototype (P)	**Model (M)**
$V_p = 50$ mi/h	$\frac{L_M}{L_P} = \frac{1}{5}$
Air at 25°C	At 5°C
$\rho_p = 1.184$ kg/m^3	$\rho_M = 1.269$ kg/m^3
$\mu_p = 1.849 \times 10^{-5}$ kg/ms	$\mu_M = 1.754 \times 10^{-5}$kg/m
& $v_m = ?$	

$$Re = \frac{\rho V L}{\mu} \Rightarrow Re_M = Re_P$$

$$\left(\frac{\rho_M V_M L_M}{\mu_M}\right) = \left(\frac{\rho_P V_P L_P}{\mu_P}\right)$$

$$V_M = \left(\frac{\rho L}{\mu}\right)_P \left(\frac{\mu}{\rho L}\right)_M \times V_P$$

$$= \left(\frac{1.184}{1.269}\right)\left(\frac{5}{1}\right)\left(\frac{1.754 \times 10^{-5}}{1.849 \times 10^{-5}}\right) \times 50$$

$$\boxed{V_M = 221 \text{mi/h}}$$

1454. A point, in a flow field where the local velocity of fluid is zero, is called

(a) critical point (b) vena contracta

(c) stagnation point (d) none of these

GPSC Lect. 23.10.2016

Ans. (c) : Stagnation point is a point in a flow field where the local velocity of the fluid is zero. It exists at the surface of objects in the flow field, where the fluid is brought to rest by the object.

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1455. Drag on cylinders and spheres decrease when the Reynolds number is in the region of $2x10^5$ since.

(a) Flow separation is advanced due to transition to turbulence
(b) Flow separation is delayed due to onset of turbulence
(c) Flow reattachment occurs
(d) Separation occurs due to transition to turbulence

UPRVUNL AE 21.08.2016

Ans. : (c) Drag on cylinder and sphere decrease when the Reynolds number is in the region of 2×10^5 since flow reattachment occurs.

1456. Drag force on a body is result of:

(a) Fluid dynamic force acting on the object
(b) A component of resultant fluid dynamic force in the flow direction
(c) Pressure variation over the surface of object due to horizontal force
(d) None of the above

OPSC AEE 2015 PAPER - II

Ans : (b) Drag force on a body is result of a component of resultant fluid dynamic force in the flow direction.

1457. A body is said to be provided optimum amount of streamlining when

(a) Friction drag is minimum
(b) Pressure drag is minimum
(c) Profile drag (i.e., sum of friction drag and pressure drag) is minimum
(d) Product of friction and pressure drag is minimum

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : A body is said to be provided optimum amount of streamlining when Profile drag (i.e., sum of friction drag and pressure drag) is minimum.

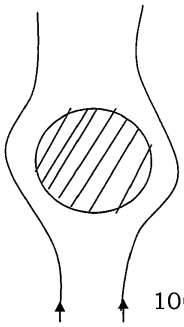
1458. In fluid modeling of flow pattern around a body submerged in a fluid, the non-dimensional number that has to be kept the same in the model and prototype is

(a) Weber number (b) Froude number
(c) Mach number (d) Reynolds number

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : For dynamic similarities between models and prototype, fully submerged body (e.g. aerofoil), it is Reynold's no. equality.
But in case of partially submerged bodies (e.g. ships) i.e. having free flow surfaces, it is Froude's no. equality.
In fluid modeling of flow pattern around a body submerged in a fluid, the non-dimensional number that has to be kept the same in the model and prototype is Reynold's number.

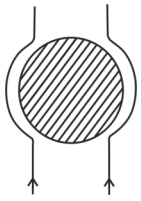
1459. A vertical air stream flowing at a velocity of 100 m/s supports a ball of 60 mm in diameter. Taking the density of air as 1.2 kg/m^3 and kinematic viscosity as 1.6 stokes, the weight of the ball that is supported is (if coefficient of drag $C_D = 0.8$)



(a) 20.2 N (b) 13.5 N
(c) 30.1 N (d) 25.2 N

ISRO Scientist/Engineer 24.05.2014

Ans. (b) :



$V = 100$ m/s

$d = 60$ mm $= 60 \times 10^{-3}$ m $= 0.06$ m

$\rho_{air} = 1.2$ kg/m^3

$\upsilon = 1.6$ stokes

coefficient of drag $C_D = 0.8$

Drag Force $F_D = C_D A \dfrac{\rho V^2}{2}$

$$F_D = 0.8 \times \frac{\pi}{4} \times (0.06)^2 \times \frac{1.2 \times 100 \times 100}{2}$$

$F_D = 13.5648$ N

1460. The total drag on a plate held normal to the flow is equal to:-

(a) Pressure drag
(b) Viscous drag
(c) Pressure drag × viscous drag
(d) $\dfrac{\text{Viscous drag}}{\text{Pressure drag}}$

UKPSC AE-2013, Paper-II

Ans. (a) : The total drag on a plate held normal to the flow is equal to pressure drag.

1461. The coefficient of drag and lift are functions of:-

(a) Frouds number (b) Reynolds number
(c) Weber number (d) Euler number

UKPSC AE-2013, Paper-II

Ans. (b) : The coefficient of drag and lift are functions of reynolds number.

1462. The aspect ratio of a wing is expressed as:-

(a) l/A (b) l/A^2
(c) l^2/A (d) l^2/A^2

UKPSC AE-2013, Paper-II

Ans. (c) :

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1463. Beyond the stall point, the drag coefficient:-

(a) Decreases rapidly (b) Increases rapidly
(c) Remains same (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (b) :

1464. The mathematical expression for lift force is given by

(a) $F_L = C_L \cdot \rho AU$

(b) $F_L = C_L \cdot \frac{\rho U^2}{2} \cdot A$

(c) $F_L = C_L \cdot \rho U^2 \cdot A$

(d) None of the above

(Notations used have usual meaning.)

UKPSC AE 2012 Paper–II

Ans. (b) : $F_L = C_L \cdot \frac{\rho U^2}{2} \cdot A$

1465. Match List I with List II and select the correct answer using the codes given below the Lists:

List-I		List II	
A	a sphere	1	0.2
B	an aerofoil	2	0.3
C	a bluff body	3	1.6
D	a racing car	4	0.1

Code:

	A	B	C	D
(a)	1	4	3	2
(b)	1	4	2	3
(c)	4	1	3	2
(d)	4	1	2	3

APPSC AEE 2012

Ans. (b) :

List-I		List- II
A sphere	-	0.2
An aerofoil	-	0.1
A bluff body	-	0.3
A racing car	-	1.6

1466. Which one of the following conditions renders the aerofoil inoperative?

(a) When angle of attack is zero
(b) When angle of attack is less than critical angle of attack.
(c) When angle of attack is equal to critical angle of attack.
(d) When angle of attack is slightly in excess to critical angle of attack

RPSC AE GWD, 2011

Ans. (a) : Lift is an upward force which is generated due to pressure difference between upper and lower surface. When angle of attack is 0° there will be no pressure difference due to which lift force will be zero and aerofoil become inoperative.

1467. Friction drag is generally larger than the pressure drag in

(a) Flow past a sphere
(b) Flow past a thin sheet
(c) Flow past an airfoil
(d) Flow past a cylinder

ISRO Scientist/Engineer 2010

Ans. (c) : Friction drag is generally larger than the pressure drag in flow past an airfoil. Pressure drag arises due to the shape of the object and depends on the flow separation point. Friction drag arises due to the friction between the fluid and surface of the object around which the flow is occurring.

1468. The value of the lift coefficient of a flat plate aerofoil for an angle of attack α is

(a) $2\pi \sin\alpha$ (b) $2\pi \cos\alpha$
(c) $\pi \sin\alpha$ (d) $\pi \cos\alpha$

WBPSC AE, 2007

Ans. (a) : From Kutta-Joukowsla, lift force F_L is given by

$F_L = \rho v L \tau$

where τ = circulation

$\tau = \pi\, cv \sin\alpha$

c = chord length

α = angle of attack

v = velocity of flow

We also have

$$C_L = \frac{F_L / A}{\frac{1}{2}\delta v^2}$$

$F_L = \rho VL \times \pi\, cv \sin\alpha$

$A = cL$

$$C_L = \frac{\rho vL \times \pi \times c \times v \times \sin\alpha \times 2}{cL \times \rho v^2}$$

$$\boxed{C_L = 2\pi \sin\alpha}$$

14. Flow of Compressible Fluids

1469. The flow is said to be subsonic when Mach number is

(a) Equal to unity (b) Less than unity
(c) Greater than unity (d) None of above

Sikkim PSC Under Secretary 2017
GPSC Asstt. Prof. 28.08.2016
TNPSC AE 2013, TNPSC ACF 2012
UKPSC AE 2007 Paper - II

Ans. (b) :

$$\text{Mach number (M)} = \frac{\text{velocity object in a medium}}{\text{velocity of sound in a medium}}$$

If $M < 1 \Rightarrow$ Subsonic flow

$M = 1 \Rightarrow$ Sonic flow

$M > 1 \Rightarrow$ Super sonic flow

$M > 6 \Rightarrow$ Hyper sonic flow

1470. Across a normal shock

(a) the pressure and temperature rise
(b) the velocity and pressure increase
(c) the velocity and pressure decrease
(d) the velocity and density decrease

ISRO Scientist/Engineer 07.05.2017
UKPSC AE 2007 Paper -II
ESE 2011

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Ans. (a) : Normal shock are a fundamental type of shock wave. The waves, which are perpendicular to the flow are called "Normal shocks".

* Across a normal shock pressure and temperature rise.
* Normal shocks only happen when the flow is supersonic.

1471. A Mach number (M) for supersonic fluid flow is:

(a) $M > 1$ (b) $M < 1$
(c) $M \geq 1$ (d) $M \leq 1$
(e) $M \approx 1$

CGPSC AE 15.01.2021
CGPSC Polytechnic Lecturer 2017
VIZAG Steel MT 2017

Ans. (a) : Mach Number (M) $= \dfrac{\text{Object Speed}}{\text{Speed of Sound}}$

$M < 1.0 \rightarrow$ Subsonic
$M = 1.0 \rightarrow$ Transonic flow or sonic flow
$M > 1.0 \rightarrow$ Supersonic flow
$M > 6.0 \rightarrow$ Hypersonic flow

1472. Gas flows can often be approximated as incompressible if the mach number is :

(a) equal to 1
(b) greater than 1
(c) less than 1 but greater than 0.3
(d) less than 0.3

RPSC ACF & FRO, 26.02.2021
Kerala PSC Poly. Lect. 2017

Ans. (d) : Compressible Flow– Compressible flows are usually high speed flow with mach number greater than about 0.3.

Incompressible Flow- Gas Flow approximated as incompressible if mach number is less than 0.3.

1473. For hypersonic flow, the mach number is

(a) Between 2-3 (b) Between 1-2
(c) Unity (d) Greater than 4

Rajasthan Nagar Nigam AE 2016, Shift-II
TNPSC AE 2008

Ans. (d) : In hypersonic flow mach Numbers is 5 to 10.

Supersonic mach no. –	1.3 – 5
Transonic Mach no. –	0.8 –1.3
Subsonic mach no. –	< 0.8

1474. Mach angle α and Mach number M are related as

(a) $M = \sin^{-1}\left(\dfrac{1}{\alpha}\right)$

(b) $\alpha = \cos^{-1}\left(\sqrt{\dfrac{M^2-1}{M^2}}\right)$

(c) $\tan\alpha = \left(\sqrt{M^2-1}\right)$

(d) $\alpha = \text{cosec}^{-1}\left(\dfrac{1}{M}\right)$

UPPSC AE 13.12.2020, Paper-2
ESE 1999

Ans. (b) : Relation between mach angle and mach number is given by –

$$\sin\alpha = \frac{1}{M}$$

$\alpha \rightarrow$ Mach angle

$M \rightarrow$ Mach number

$$\sqrt{1-\cos^2\alpha} = \frac{1}{M}$$

$$1-\cos^2\alpha = \frac{1}{M^2}$$

$$M^2 - M^2\cos^2\alpha = 1$$

$$\cos^2\alpha = \frac{M^2-1}{M^2}$$

$$\cos\alpha = \sqrt{\frac{M^2-1}{M^2}}$$

$$\alpha = \cos^{-1}\sqrt{\frac{M^2-1}{M^2}}$$

1475. For supersonic flow, increase in flow area

(a) Decreases velocity
(b) Does not cause any change
(c) Increases velocity
(d) None of these

MPPSC State Forest Service Exam, 2014
APPSC AE 04.12.2012

Ans. (c) : As we know, a supersonic flow that is turned while there is an increase in flow area is also isentropic, since there is an increase in area, therefore we call this an isentropic expansion. If a supersonic flow is turned abruptly the flow area decreases, the flow is irreversible due to the generation of shock waves.

1476. The Mach number for nitrogen flowing at 195 m/s when the pressure and temperature in the undisturbed flow at 690 kN/m and 93^0C respectively will be :

(a) 0.25 (b) 0.50
(c) 0.66 (d) 0.75

PSPCL AE, 2012, ESE 1992

Ans. (b) : V = 195 m/s
P = 690 kN/m
T = 273 + 93 = 366 K
C = sonic velocity in medium

$= = \sqrt{\gamma RT}$

γ for nitrogen = 1.4

For nitrogen $R = \dfrac{\bar{R}}{\text{Molecular mass}} = \dfrac{8.314}{28} = 0.296$

$$M = \frac{V}{C} = \frac{195}{\sqrt{1.4\times 296\times 366}} = 0.5$$

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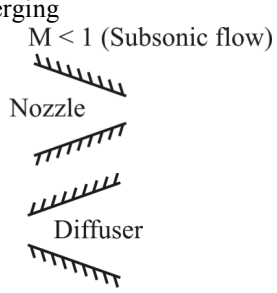
1477. The Mach number at inlet of gas turbine diffuser is 0.3. The shape of the diffuser would be :

(a) Converging
(b) Diverging
(c) Converging - diverging
(d) Diverging - converging

PSPCL AE, 2012, ESE 1992

Ans. (b) : Diverging

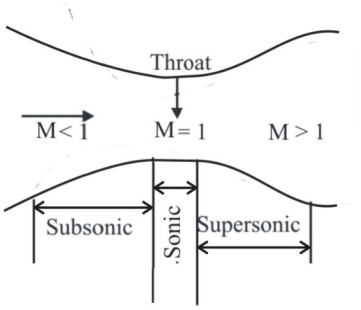


1478. If the cross-section of a nozzle is increasing in the direction of flow in supersonic flow, then in the downstream direction

(a) both pressure and velocity will increase
(b) both pressure and velocity will decrease
(c) pressure will increase but velocity will decrease
(d) pressure will decreases but velocity will increase

Nagaland PSC CTSE 2015 Paper-II
ESE 2005

Ans. (d) :



■ Pressure will decrease but velocity will increase.

1479. The prime parameter causing change of state in a Fanno flow is

(a) Heat transfer (b) Area change
(c) Friction (d) Buoyancy
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
ESE 1998

Ans. (c) : Fanno flow is the adiabatic flow through a constant area duct where the effect of friction is considered. Compressibility effects often come into consideration although the fanno flow model certainly also applies to incompressible flow.

The fanno flow model is considered an irreversible process due to viscous effects. The viscous friction causes the flow properties to change along the duct.

1480. At the eye tip of a centrifugal impeller, blade velocity is 200 m/s while the uniform axial velocity at the inlet is 150 m/s. If the sonic velocity is 300 m/s, then the inlet Mach number of the flow will be

(a) 0.75 (b) 0.66
(c) 0.90 (d) 0.83

ISRO Scientist/Engineer 2009
ESE 1998

Ans. (d) : Given, Blade velocity = 200 m/s
Axial velocity at Inlet = 150 m/s
Sonic velocity = 300 m/s
Mach Number

$$M = \frac{V}{C}$$

$$V = \sqrt{(\text{blade velocity}^2 + (\text{Axial velocity})^2}$$

$$= \sqrt{(200)^2 + (150)^2}$$

$$V = \sqrt{(200)^2 + (150)^2}$$

$$V = 250 \text{ m/s}$$

$$C = 300 \text{ m/s}$$

Sonic velocity

$$M = \frac{250}{300} = \frac{25}{30}$$

$$M = 0.83$$

1481. For one-dimensional isentropic flow in a diverging passage, if the initial static pressure is P_1, and the initial Mach number is M_1 ($M_1 < 1$), then for the downstream flow

(a) $M_2 < M_1$; $P_2 < p_1$ (b) $M_2 > M_1$; $P_2 > p_1$
(c) $M_2 < M_1$; $P_2 > p_1$ (d) $M_2 > M_1$; $P_2 < p_1$

ISRO Scientist/Engineer 2009
ESE 1993

Ans. (c) : One-dimensional Isentropic flow,
Static pressure = P_1
Initial Mach number is M ($M_1 < 1$)

$$\frac{dA}{A} = \frac{dv}{v}(M^2 - 1)$$

$$\frac{dA}{A} = -\frac{dv}{v}$$

($\therefore$ M < 1) and for diverging passage

$$\frac{dA}{A} = +ve$$

$$\therefore \frac{dv}{v} = -ve,$$

Velocity decreases and pressure increases.

1482. In supersonic flow, a diverging passage results in

(a) increase in velocity and pressure
(b) decrease in pressure and density
(c) increase in velocity and density
(d) decrease in velocity and pressure

APPSC AEE 2012, APPSC AE 04.12.2012

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Ans. (b) : In the supersonic flow, the density drops quite a lot faster than the pressure does. It drops so fast that the wall have to get farther apart to allow the density to decreases.

1483. An air stream with a velocity of 300 m/s at 10 kN/m^2 vacuum and temperature of 320 K passes through a section where its gauge is 20 kN/m^2. compute its stagnation properties (M, T, P)

(a) 0.84, 365 K, 142 kN/m^2
(b) 0.96, 385 K, 284 kN/m^2
(c) 0.42, 255 K, 71 kN/m^2
(d) None of the above

APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (a) : Given, $P_1 = 100 - 10 = 90$ kN/m^2, $\gamma = 1.4$, $u_1 = 300$ m/sec.

R = 287 J/kg k, $T_1 = 320$ k

$$C_p = \frac{\gamma}{(\gamma-1)}.R = \frac{1.4}{1.4-1} \times 287 = 1004.5$$

$$C_1 = \sqrt{\gamma RT_1} = \sqrt{1.4 \times 287 \times 320} = 358\,\text{m/sec}.$$

(1) $M_1 = u_1/C_1 = 300/358 = 0.84$

(2) $$T_o = T_1\left[1+\left(\frac{\gamma-1}{2}\right)M_1^2\right] = 320\,(1 + 0.2\times 0.84^2)$$

$T_0 = 365$ k

(3) $P_0 = 90 \times 1.14^{3.5} = 142$ kN/m^2

1484. How does the sonic velocity in air at STP condition change with the altitude in the standard atmosphere?

(a) 340.3 m/s (b) 17.15 m/s
(c) 0 (d) 680.6 m/s

APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (a) : The sonic velocity in air at STP condition is 340.3 m/s, changes with the altitude in the standard atmosphere.

1485. Compressibility effect can be treated as negligible when Mach number is

(a) Upto 0.2 (b) Upto 0.5
(c) Less than 1 (d) 1

UPPSC AE 12.04.2016 Paper-II
UKPSC AE 2007 Paper -II

Ans : (a) Compressibility effect can be treated as negligible when Mach number is upto 0.2.

$$\text{Mach Number} = \sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}} = \sqrt{\frac{\text{Inerita foce}}{\text{Compressible force}}}$$

(i) Incompressible flow
$M < 0.3$
(ii) Subsonic flow
$0.3 < M < 1.0$
(iii) Sonic flow
$M = 1$
(iv) Supersonic flow
$1.0 < M < 6.0$
(v) Hypersonic flow
$M > 6.0$

1486. What is the speed of sound in neon gas at a temperature of 500 K (Gas constant of neon is 0.4210 kJ/kgK)

(a) 492 m/s (b) 460 m/s
(c) 592 m/s (d) 543 m/s

VIZAG STEEL MT 18.08.2013, 10.06.2012
GATE 2002

Ans. (c) : Given, R = 0.421 kJ/kg-k, T = 500K

Speed of sound in terms as temperature s given by.

$$V = \sqrt{\gamma RT} \qquad \{\gamma = 1.67 \text{ for neon}\}$$

$$V = \sqrt{1.67 \times 0.421 \times 10^3 \times 500}$$

$$\boxed{V = 592.9\ \text{m/sec}}$$

1487. A shock wave is produced when

(a) a sonic flow changes to supersonic flow
(b) a subsonic flow changes to sonic flow
(c) a supersonic flow changes to subsonic flow
(d) none of the above

UKPSC AE 2013 Paper-II
UKPSC AE 2007 Paper -II

Ans. (c) : Shock wave is produced when a supersonic flow changes to subsonic flow. A shock wave is a type of propagating disturbance that moves faster than the local speed of sound in the medium.

1488. A compressible flow is that flow in which the _____of the fluid changes during flow.

(a) density (b) mass/weight ratio
(c) area/volume ratio (d) none of the above

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) :

1489. The pressure of fluid due to hammer blow is proportional to:

(a) Directly to $\sqrt{(\text{density})}$ of fluid
(b) Direct to density of fluid
(c) Inversely to density of fluid
(d) Inversely to $\sqrt{(\text{density})}$ of fluid

CIL MT 27.02.2020

Ans. (a) : Water hammer pressure,

$$P = V\sqrt{K\rho}$$

Where, K – Bulk modulus of water
V – Velocity of flow of water through pipe
ρ – Density of fluid flow in pipe

Pressure rise due to water hammer blow is directly proportional to $\sqrt{\text{Density of fluid}}$.

1490. Speed of sound in an ideal gas, eg. Oxygen, depends on

(a) Its pressure only
(b) Its temperature only
(c) Its viscosity
(d) all the above properties

Haryana PSC AE (PHED) 04.09.2020, Paper-I

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Ans. (b) : Velocity of sound in ideal gas = $\sqrt{\gamma RT/M}$ i.e. depends only on temperature.

1491. Pressure _____ and velocity _____ moving into a shock wave

(a) Reduces, increase (b) increases, increases
(c) increases, reduces (d) reduces, reduces

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : In shock wave.
(i) Shock occures only in supersonic flow
Pressure increases
velocity decreases from supersonic to subsonic
Temperature increase
Entropy increase
Stagnation temperature remain constant
Stagnation pressure decreases
density – increases
Stagnation density – decrease

1492. Mach number for subsonic flow

(a) $M < 0.8$ (b) $M > 1$
(c) $M > 2$ (d) $M > 2.5$

Gujarat PSC AE 2019

Ans. : (a) :
- $M < 0.3$: incompressible flow
- $0.3 < M < 0.8$: subsonic flow
- $0.8 < M < 1.2$: transonic flow
- $1.2 < M < 3.0$: supersonic flow
- $3.0 < M$: hypersonic flow

1493. Bernoulli's equation for compressible flows undergoing adiabatic process is given by:
[where γ = Ratio of specific heat, P = Pressure, v = velocity, Z = datum head, g = acceleration due to gravity]

(a) $\frac{p}{\rho g}\ln P + \frac{v^2}{2g} + z = \text{constant}$

(b) $\left(\frac{\gamma}{\gamma-1}\right)\frac{P}{\rho g} + \frac{v^2}{2g} + z = \text{constant}$

(c) $\frac{P}{(\gamma-1)\rho g} + \frac{v^2}{2g} + z = \text{constant}$

(d) $\frac{dv}{v} + \frac{dA}{A} + \frac{dP}{\rho} = 0$

SJVN ET 2019

Ans. (b) : Bernoulli's equation for compressible flows undergoing adiabatic process is given by :

$$\left(\frac{\gamma}{\gamma-1}\right)\frac{P}{\rho g} + \frac{v^2}{2g} + z = \text{Constant}$$

1494. A flow is said to be supersonic, if the

(a) If the velocity of flow is very high
(b) Mach number is between 1 and 6
(c) Mach number is more than 6
(d) If the discharge is difficult to measure

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (b) :

Type of flow	Mach. No. M (Range)
Subsonic	0.4 - 0.8
Transonic	08. - 1.2
Sonic	1
Supersonic	1 - 6
Hypersonic	> 6

1495. In a normal shock in a gas

(a) the upstream flow is supersonic
(b) the upstream flow is subsonic
(c) the downstream flow is sonic
(d) the downstream flow is supersonic

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : Normal shock wave is characterized by upstream super sonic flow and down stream is subsonic flow.

1496. In isentropic flow between two points

(a) stagnation pressure and stagnation temperature may vary
(b) the stagnation pressure decreases in the direction of the flow
(c) the stagnation temperature and stagnation pressure decrease with increase in the velocity
(d) the stagnation temperature and stagnation pressure remain constant

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : The stagnation temperature and stagnation pressure remain constant

1497. In a normal shock occurring in air (k = 1.4), if the upstream Mach number is 3.52, the Mach number after the shock is

(a) 0.61 (b) 0.45
(c) 0.13 (d) 0.28

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Mach number relations is normal shock wave pattern.
Given,

$m_1 = 3.52, \quad k = 1.4$

$$m_2 = \sqrt{\frac{2+(k-1)m_1^2}{2k\,m_1^2-(k-1)}}$$

$$m_2 = \sqrt{\frac{2+(1.4-1)\,(3.52)^2}{2\times1.4\times3.52^2-(1.4-1)}}$$

$$m_2 = \sqrt{\frac{6.9562}{34.293}} = \sqrt{0.203} = 0.45$$

1498. In a normal shock wave in a gas with k = 1.4, one of Mach numbers is 0.5. The other Mach number is

(a) 2.65 upstream of the shock
(b) 0.06 downstream of the shock
(c) 0.02 upstream of the shock
(d) 3.75 upstream of the shock

TSPSC AEE 28.08.2017 (Civil/Mechanical)

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Ans. (a) :

$$m_1 = \sqrt{\frac{2+(k-1)m_2^2}{2k\ m_2^2-(k-1)}}$$

Given,

$m_2 = 0.5$

(After normal shock wave, subsonic flow)

$$m_1 = \sqrt{\frac{2+(1.4-1)\ (0.5)^2}{2\times1.4\times0.5^2-(1.4-1)}}$$

$$m_1 = \sqrt{\frac{2.1}{0.3}} = \sqrt{7}$$

$m_1 = 2.65$ (upstream)

1499. Identify the incorrect statement:
A flow nozzle

(a) has a contraction coefficient of unity
(b) is less costly than a venturimeter
(c) is more efficient than an orifice meter
(d) has overall losses much smaller than in a venturimeter

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : Losses in flow nozzle (nozzle meter) are more as compared to venturimeter but less as compared orifice meter.

1500. The Mach number of an aircraft when the dynamic (velocity) temperature of air at entry to the engine equals the static temperature is

(a) 1.236 (b) 1.736
(c) 2.236 (d) 2.736

GPSC EE Pre, 28.01.2017

Ans. (c) : The Mach number of aircraft when the dynamic (velocity) temperature of air at entry to the engine equals the static temperature is 2.236.

1501. The power transmitted through the nozzle is maximum when the head lost due to friction in the pipe is :

(a) equal to the total supply head
(b) one-third of the total supply head
(c) one-half of the total supply head
(d) tow-third of the total supply head

Punjab PSC SDE 12.02.2017

Ans. (b) : For maximum power transmitted by a nozzle which is end of pipe fitted, the head loss due to friction should be $\frac{1}{3}$ of the supply head 'H'

• Diameter of nozzle for transmitting maximum power.

$$d = \left(\frac{D^5}{2fl}\right)^{\frac{1}{4}}$$

1502. Statement (I) : Both oblique shock and normal shock can be viewed as two different regions of a flow over a plate with a deflection section.
Statement (II) : Variation of the deflection angle from a zero to a positive value results in oblique shock. Further changing the deflection angle to a negative value results in expansion waves.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I).
(c) Statement (I) is true but Statement (II) is false.
(d) Statement (I) is false but Statement (II) is true.

UPSC JWM 2017

Ans. (b) : Normal shock wave is normal to the flow direction where as oblique shock wave is inclined at some angle.
In normal shock, flow direction does not change (flow deflection angle = 0)
In oblique shock, flow direction changes (flow deflection angle ≠ 0)

1503. A light plane flies at 720 km/hr in standard air at an altitude of 1000 m. Determine the stagnation pressure at the leading edge of the wing. Take, $P_{air} = 9.0 \times 10^4$ N/m^2, $\rho = 1.1$ kg/m^3.

(a) 90 kPa (b) 119 kPa
(c) 112 kPa (d) 100 kPa

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : Given, V = 720 km/hr = 200 m/s
Altitude = 1000 m
$P_{air} = 9.0 \times 10^4$ N/m^2, $\rho = 1.1$ kg/m^3

$$\boxed{P_s = P_0 + \rho\left(\frac{V_0^2}{2}\right)}$$

P_s = stagnation pressure
P_0 = static pressure

$$P_s = 9.0\times10^4 + 1.1\times\frac{200^2}{2}$$

$P_s = 112000$ Pa
$P_s = 112$ kPa

1504. Compressibility of fluid is insignificant if Mach number is

(a) greater than 1 (b) equal to 1
(c) between 0.4-1 (d) less than 0.4

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (d) : Usually compressibility effects can be neglected if the sarrau - mach number is less than 0.3^2. For air at atmospheric pressure & 20° C, this limit implies velocities less than 103 m/s or mach number less than 0.4.

1505. The velocity of sound in winter compared to summer will be

(a) higher (b) lower
(c) same (d) cannot be predicted

ISRO Scientist/Engineer 03.07.2016

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Ans. (b) : Velocity of sound $C = \sqrt{\gamma RT}$

In winter temperature is less hence velocity of sound is lower.

1506. The drag coefficient at supersonic velocities depends on

(a) Reynolds number only
(b) Mach number only
(c) Weber number only
(d) Euler number only

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (a) : The drag coefficient at supersonic velocities depends on Reynolds number only.

1507. A rigid pipe of length L, diameter D, wall thickness t and modulus of elasticity E is conveying water with a mean velocity of V. If mass density and bulk modulus of water are ρ and K respectively, then water hammer due to sudden closure in the pipe will cause increase in pressure equal to

(a) $\rho LV/t$ (b) $V\sqrt{\rho K}$

(c) $V\sqrt{\dfrac{\rho}{\dfrac{1}{K}+\dfrac{D}{Et}}}$ (d) None of the above

APPSC AEE Screening Test 2016

Ans. (b) : Water hammer due to sudden closure in the pipe–

Pressure head–

$$\frac{P}{\rho g} = \frac{VC}{g} \quad \text{[Allievi formula]}$$

$$\frac{P}{\rho} = VC$$

$$P = VC \times \rho$$

we know that,

$$C = \sqrt{\frac{K}{\rho}}$$

$$\therefore \quad P = \rho V \times \sqrt{\frac{K}{\rho}} \Rightarrow V\sqrt{\frac{K}{\rho} \times \rho^2}$$

$$\boxed{P = V\sqrt{\rho K}}$$

1508. Sonic velocity will have a low value in the medium having

(a) Low value of coefficient of compressibility
(b) High value of coefficient of compressibility
(c) High bulk modulus of elasticity
(d) Homogeneous composition

UPPSC AE 12.04.2016 Paper-II

Ans : (b) Sonic velocity will have a low value in the medium having high value of coefficient of compressibility.

Compressibility, $\beta = \dfrac{1}{K}$

Sonic velocity $C = \sqrt{\dfrac{K}{\rho}} \Rightarrow C = \sqrt{\dfrac{1}{\beta\rho}}$

$$C \propto \frac{1}{\sqrt{\beta}}$$

1509. If the valve in water pipe is instantaneously closed and the pipe is considered rigid, the intensity of pressure rise due to water hammer is given by

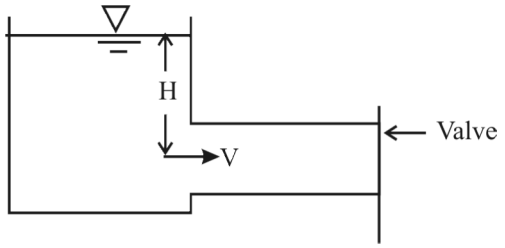
(Where V is the velocity of flow; γ is the specific weight of water, t is the time taken close the value; K is the bulk modulus of water and g is the acceleration due to gravity)

(a) $V\sqrt{\dfrac{\gamma K}{g}}$ (b) $Vt\sqrt{\dfrac{\gamma K}{g}}$

(c) $Vt\sqrt{\dfrac{K}{g}}$ (d) $V\gamma\sqrt{\dfrac{K}{g}}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Sudden closure of valve and pipe is rigid–



When valve is closed suddenly, the K.E. of flowing water is converted into strain energy of water if the effect of friction is neglected and pipe wall is assumed perfectly rigid.

Loss of kinetic energy = $\dfrac{1}{2}\times$ mass of water in pipe $\times V^2$

$$= \frac{1}{2} \times \rho AL \times V^2$$

Gain of strain energy = $\dfrac{1}{2}\left(\dfrac{P^2}{K}\right) \times$ Volume

$$= \frac{1}{2}\frac{P^2}{K} \times AL$$

Equating loss of K.E. to gain of strain energy

$$\frac{1}{2}\rho AL \times V^2 = \frac{1}{2}\frac{P^2}{K} \times AL$$

$$P^2 = \frac{1}{2}\rho AL \times V^2 \times \frac{2K}{AL}$$

$$P^2 = \rho K V^2$$

$$P = V\sqrt{K\rho}$$

$$P = V\sqrt{\frac{\gamma K}{g}} \qquad (\because \gamma = \rho g)$$

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1510. Compressibility force in fluid motion is due to

(a) Elastic property of fluid
(b) Cohesive property of fluid
(c) Adhesive property of fluid
(d) Viscosity of fluid

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Compressibility force in fluid motion is due to elastic property of fluid. It is reciprocal of bulk modulus of elasticity.

$$\beta = \frac{1}{K} = -\frac{dV/V}{dP}$$

1511. The Fanno line equation is given by

(a) $\frac{\gamma}{\gamma-1}\left(\frac{P_1}{P_2}\right)+\frac{(\rho_1 v_1)}{2P_1}=\frac{\gamma}{\gamma-1}\left(\frac{P_1}{P_2}\right)+\frac{(\rho_2 v_2)}{2P_2}$

(b) $\frac{\gamma}{\gamma-1}\left(\frac{P_1}{P_2}\right)+\frac{(\rho_1 v_1)}{2P_1}=\frac{\gamma}{\gamma-1}\left(\frac{P_1}{P_2}\right)+\frac{(\rho_2 v_2)^2}{2P_2}$

(c) $P_1 - P_2 = \rho_2 v_2^2 - \rho_1 v_1^2$

(d) $P_1 + P_2 = \rho_2 v_2^2 + \rho_1 v_1^2$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (c) :

1512. The strength of shock wave is defined as

(a) $\frac{\text{Ratio of pressure rise across the shock}}{\text{Upstream pressure}}$

(b) $\frac{\text{Upstream pressure}}{\text{Ratio of pressure rise across the shock}}$

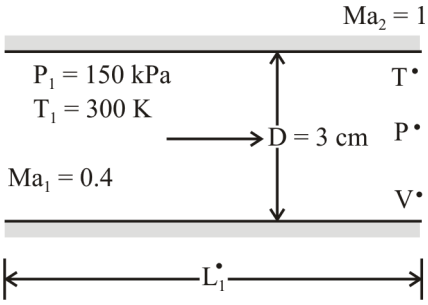
(c) $\frac{\text{Downstream pressure}}{\text{Upstream pressure}}$

(d) $\frac{\text{Upstream pressure}}{\text{Downstream pressure}}$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : The strength of shock wave is defined as, the ratio of pressure rise across the shock to the upstream pressure (static pressure).

1513. Air enters a 3 cm diameter smooth adiabatic duct at Ma_1 (Mach number) = 0.4, T_1 = 300 K, P_1 = 150 kPa. If the Mach number at the duct exit is 1, determine the duct length and the percentage of stagnation pressure lost in the duct



(a) 4.68 m and 37.1%
(b) 3.68 m and 37.1%
(c) 4.68 m and 35.1%
(d) 3.68 m and 35.1%
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) :

1514. For a normal shock wave the Rankine line equation is given by

(a) $P_1 + \frac{(\rho_1 + v_1)^2}{\rho_1} = P_2 + \frac{(\rho_2 + v_2)^2}{\rho_2}$

(b) $P_1 - \frac{(\rho_1 v_1)^2}{\rho_1} = P_2 - \frac{(\rho_2 v_2)^2}{\rho_2}$

(c) $P_1 + \frac{(\rho_1 v_1)^2}{\rho_1} = P_2 + \frac{(\rho_2 v_2)^2}{\rho_2}$

(d) $P_1 - \frac{(\rho_1 v_1)^2}{\rho_1} = P_2 - \frac{(\rho_2 v)}{\rho_2}$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (a) : Normal Shock wave, the Rankine line equations is given by,

$$p_1 + \frac{(\rho_1 + V_1)^2}{\rho_1} = p_2 + \frac{(\rho_2 + V_2)^2}{\rho_2}$$

1515. Pressure at a point in compressible fluid for an isothermal process is given by relation

(a) $P = P_0 e^{\left(-\frac{RT}{gz}\right)}$

(b) $P = P_0\left[1-\frac{\gamma-1}{\gamma}\times\frac{gz}{RT_0}\right]^{\frac{\gamma}{\gamma-1}}$

(c) $P = P_0\left[1-\frac{\gamma-1}{\gamma}\times\frac{gz}{RT_0}\right]$

(d) $P = P_0 e^{\left(-\frac{gz}{RT}\right)}$

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : Pressure at a point in compressible fluid for an isothermal process is given by relation of :-

$dp = -\rho g dz$ & $P = \rho RT$

$dp = -\frac{P}{RT} g.dz$ $\qquad \rho = \frac{P}{RT}$

$$\int_{P_0}^{P}\frac{dp}{P} = -\frac{g}{RT}\int_0^z dz$$

$$\ln\left(\frac{P}{P_0}\right) = -\frac{gz}{RT}$$

$$P = P_0 e^{\left(-\frac{gz}{RT}\right)}$$

1516. Mach number for a moving source at sonic speed will be :

(a) zero
(b) greater than 1
(c) less than 1
(d) equal to 1

HPPSC Asstt. Prof. 29.10.2016

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Ans. (d) :

Speed	Mach no. (M)
Subsonic	$M < 1$
Sonic	$M = 1$
Super sonic	$6 > M > 1$
Hypersonic	$M \geq 6$

1517. The flow velocity in the convergent section of a nozzle, designed for maximum discharge is:

(a) Subsonic (b) Sonic
(c) Supersonic (d) None of these

RPSC VPITI 14.02.2016

Ans. (a) : Convergent nozzles accelerate subsonic fluids. If the nozzle pressure ratio is high enough, then the flow will reach sonic velocity at the narrowest point.

1518. Pick the wrong statement about the normal shock :

(a) The plane of the shock wave is normal to the direction of the flow.
(b) The flow changes from supersonic upstream the shock to subsonic downstream the shock.
(c) The pressure decreases downstream the shock.
(d) The temperature increases downstream the shock.

RPSC AE (GWD) 06.05.2016

Ans. (c) : Across a shock wave, the static pressure temperature and gas density increases almost instantaneously. But because the flow is non-isentropic, the total pressure downstream of the shock is always less than the total pressure upstream of the shock. There is a loss of total pressure associated with a shock wave.

1519. In a normal shock wave in one-dimensional flow

(a) pressure, density and temperature increase
(b) velocity, temperature and density increase
(c) pressure, density and temperature decrease
(d) velocity, pressure and density decrease

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : In a normal shock wave in one-dimensional flow pressure, density and temperature increase.

1520. Which one of the following is the shock that occurs for an airplane travelling at supersonic speed?

(a) Normal shock (b) Oblique shock
(c) Bow shock (d) Air shock

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Bow shock.

1521. The area-velocity relationship for compressible fluid flow is
Note: M is the Mach number and C is the sonic speed.

(a) $\frac{dA}{A} = \frac{dV}{V}\left(M^2 - 1\right)$ (b) $\frac{dA}{A} = \frac{dV}{V}\left(C^2 - 1\right)$
(c) $\frac{dA}{A} = \frac{dV}{V}\left(1 - M^2\right)$ (d) $\frac{dA}{A} = \frac{dV}{V}\left(1 - V^2\right)$

MPPSC AE 08.11.2015

Ans. (a) : The area-velocity relationship for compressible fluid flow is-

$$\frac{dA}{A} = \frac{dV}{V}\left(M^2 - 1\right)$$, where, M = Mach number

1522. Sonic velocity C for adiabatic is mentioned as:

(a) $C = \sqrt{\gamma RT}$ (b) $C = \gamma RT$
(c) $C = \sqrt{\gamma RT^2}$ (d) $C = \left[\gamma RT\right]^{1/3}$

OPSC AEE 2015 PAPER - II

Ans : (a) velocity of sound wave is given by

$C = \sqrt{\frac{dp}{d\rho}} = \sqrt{\frac{k}{p}}$ ..... in term of bulk modulus

$C = \sqrt{\frac{p}{\rho}} = \sqrt{RT}$ ..... for isothermal process

$C = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\gamma RT}$ ..... for adiabatic process.

1523. Velocity of sound at 400 K is given by :

(a) 12.67 m/s (b) 400 m/s
(c) 2157.7 m/s (d) 225.8 m/s

TNPSC AE (Industries) 09.06.2013

Ans. (b) : $C = \sqrt{\gamma RT}$

$C = \sqrt{1.4 \times 287 \times 400}$

$\boxed{C = 400 \text{ m/s}}$

1524. Which one of the following is the expression of shock strength?

(a) $\frac{2v}{v+1}\left[m_1^2 - 1\right]$ (b) $\frac{v+1}{2v}\left[m_1^2 - 1\right]$
(c) $\frac{2v}{v-1}\frac{1}{\left[m_1^2 - 1\right]}$ (d) $\frac{v+1}{2v}\frac{1}{\left[m_1^2 - 1\right]}$

JPSC AE 2013 Paper-V

Ans. (a) : The expression of shock strength is-

$$\frac{2v}{v+1}\left[m_1^2 - 1\right]$$

1525. The flow of steam is supersonic

(a) at the entrance to the nozzle
(b) at the throat of the nozzle
(c) in the convergent portion of the nozzle
(d) in the divergent portion of the nozzle

JPSC AE 2013 Paper-V

Ans. (d) The flow of steam is supersonic in the divergent portion of the nozzle.

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1526. The ratio of stagnation pressure and static pressure

(a) $\left[1+\frac{\gamma-1}{2}M^2\right]^{\frac{\gamma}{\gamma-1}}$ (b) $\left[1+\frac{\gamma-1}{2}M^2\right]$

(c) $\left[1+\frac{\gamma-1}{2}M^2\right]^{\frac{\gamma-1}{\gamma}}$ (d) $\sqrt{\gamma RT}$

TNPSC AE 2013

Ans. (a) : The ratio of stagnation pressure and static pressure is given by

$$= \left[1+\frac{\gamma-1}{2}M^2\right]^{\frac{\gamma}{(\gamma-1)}}$$

1527. The angle which the oblique shock wave makes with the initial direction of flow is

(a) Mach angle (b) Wave angle

(c) Deviation angle (d) Angle of deflection

TNPSC AE 2013

Ans. (b) : The angle which the oblique shock wave makes with the initial direction of flow is known as wave angle.

1528. Shock waves are

(a) Infinitesimal pressure waves

(b) Non-stop pressure waves

(c) Expansion waves

(d) Steep pressure waves

TNPSC AE 2013

Ans. (d) : Shock waves are steep pressure waves.

1529. Prandtly-Meyer relation gives the relationship between

(a) stagnation temperature and initial temperature

(b) density before and after shock

(c) gas velocity before and after shock

(d) velocity fluid and sound

TNPSC AE 2013

Ans. (c) : Prandtly-Meyer relation gives the relationship between gas velocity before and after shock.

1530. The value of $\frac{4f\,L_{max}}{D}$ for sonic velocity is

(a) Less than one (b) One

(c) Zero (d) Greater than one

TNPSC AE 2013

Ans. (c) : Zero

1531. The temperature of the gas when it is adiabatically decelerated to zero velocity at zero elevation is called :

(a) Stagnation temperature

(b) Constant temperature

(c) Iso thermal temperature

(d) Adiabatic temperature

TNPSC AE (Industries) 09.06.2013

Ans. (a) : The temperature of the gas when it is adiabatically decelerated to zero velocity at zero elevation is called Stagnation temperature.

1532. Velocity of sound wave in fluid is :

(a) Square root of ratio of change of pressure the change of density of fluid

(b) Ratio of change of pressure to the change of density of fluid

(c) Square of ratio of change of pressure to the change of density of fluid

(d) Square of ratio of change of density of fluid to change of pressure

TNPSC AE (Industries) 09.06.2013

Ans. (a) :

1533. Air at 300 °C and 7 bar is expanded isentropically until the velocity is 300 m/s. Calculate the static temperature of the air at high velocity condition if Cp of air is 1.00 kJ/kg-°C:-

(a) 255 °C (b) 295 °C

(c) 300 °C (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (a) : $C_p(T_1 - T_2) = \frac{u_2^2}{2}$

$$1000(300 - T_2) = \frac{300^2}{2}$$

$$\boxed{T_2 = 255°C}$$

1534. For a convergent divergent nozzle, the mass flow rate remains constant if the ratio of exit and inlet pressures :

(a) is less than critical pressure ratio

(b) is equal to the critical pressure ratio

(c) is more than critical pressure ratio

(d) is infinity

J&K PSC Civil Services Pre, 2013

Ans. (c) : For a convergent divergent nozzle, the mass flow rate remains constant if the ratio of exit and inlet pressures is more than critical pressure ratio.

1535. Effect of compressibility of a fluid can neglected if Mach number is

(a) equal to 1

(b) greater than 1

(c) less than 1 but greater than 0.4

(d) less than 0.4

APPSC AEE 2012

Ans : (d) Effect of compressibility of a fluid can neglected if Mach number is less than 0.4

Compressibility effect can be treated as negligible when Mach number is upto 0.33

$$\text{Mach Number} = \sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}} = \frac{\text{Inerita force}}{\text{Compressible force}}$$

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(i) Incompressible flow	$M < 0.3$
(ii) Subsonic flow	$0.3 < M < 1.0$
(iii) Sonic flow	$M = 1$
(iv) Supersonic flow	$1.0 < M < 6.0$
(v) Hypersonic flow	$M > 6.0$

1536. The differential form of continuity equation for one dimensional steady flow compressible fluids with usual terms is

(a) $\frac{d\rho}{\rho}+\frac{dA}{A}=0$ (b) $\frac{dA}{A}=\frac{d\rho}{\rho}+\frac{dV}{V}$

(c) $\frac{dA}{\rho}=\frac{dV}{V}+\frac{d\rho}{\rho}$ (d) $\frac{dA}{A}=\frac{-d\rho}{\rho}-\frac{dV}{V}$

APPSC AEE 2012

Ans : (d) Continuity equation :- This is based on law of conservation of mass which states that matter cannot be created nor destroyed. For one-dimensional steady flow,

The mass per second = ρAV.

ρ = mass density. A = Area of cross - section, V = velocity

$$\rho AV = C$$

$$d(\rho AV) = 0$$

$$\rho d(AV) + AVd\rho = 0$$

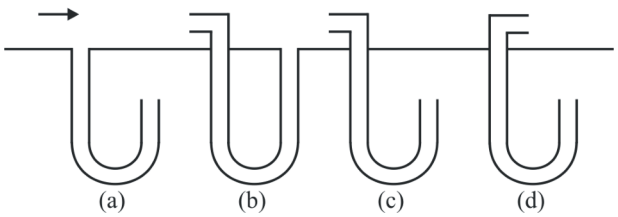
$$\rho[A.dV + V.dA] + AVd\rho = 0$$

$$\rho AdV + \rho VdA + AVd\rho = 0$$

dividing by ρAV

$$\boxed{\frac{dV}{V}+\frac{dA}{A}+\frac{d\rho}{\rho}=0}$$

1537. Which configuration measures the stagnation pressure?



ISRO Scientist/Engineer 2012

Ans. (c)

1538. Which of the following is conserved across a normal shock wave?

(a) Temperature (b) Pressure

(c) Momentum (d) All of the above

APPSC AEE 2012

Ans. (c) : Momentum is conserved across a normal shock wave.

1539. Assertion (A): In a supersonic flow, the velocity increases as the area increases.
Reason (R): Conservation of masses, momentum and energy require that dA/A should increase with dM/M for M greater than unity

(a) A is true; R is true, and it explains A
(b) A is true; R is true, but it does not explain A
(c) A is true; R is false
(d) A is false; R is true

APPSC AEE 2012

Ans. (a) : A is true, R is true and it explains A.

1540. Assertion (A): In a supersonic flow, the velocity increase as the area increases.
Reason (R): Conservation of masses, momentum and energy require that dA/A should increase with dM/M for M greater than unity

(a) A is true; R is true, and it explains A
(b) A is true; R is true, but it does not explain A
(c) A is true; R is false
(d) A is false; R is true

APPSC AE 04.12.2012

Ans. (a) : In supersonic flow, the velocity increases as the area increases.
Reason (R): Conservation of masses, momentum and energy require that dA/A should increase with dM/M for M greater than unity.
In the supersonic flow, velocity of flow exceed the velocity of second. for supersonic flow divergent portion of the nozzle is used. And mach number is greater than unity. (M > 1). Hence R is true, A is true, and R explains A.

1541. The velocity of elastic pressure wave in a fluid medium is equal to

(a) sonic velocity
(b) half of sonic velocity
(c) square root of sonic velocity
(d) Bulk Modulus

APPSC AEE 2012

Ans : (a)

$$\text{Mach number} = \frac{\text{velocity of fluid or body moving in fluid}}{\text{Velocity of sound in the fluid}}$$

The velocity of elastic pressure wave in a fluid medium is equal to sonic velocity.

1542. Mach number is given by

(a) $\frac{\text{Acoustic speed}}{\text{Stream speed}}$

(b) $\frac{\text{Stream speed}}{\text{Acoustic speed}}$

(c) product of gas constant and temperature
(d) half of Bulk Modulus

APPSC AEE 2012

Ans : (b) Mach number = $\frac{\text{stream speed}}{\text{Acoustic Speed}}$

Subsonic flow :- $M < 1$
Sonic flow :- $M = 1$
Supersonic flow : - $M > 1$

1543. Mach cone is possible in

(a) Stationary fluids (b) Subsonic flow
(c) Transonic flow (d) Supersonic flow

APPSC AEE 2012

Ans : (d) Mach cone is possible in Supersonic flow.

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1544. The range of Mach number for a subsonic flow is

(a) $0 < M < 1$ (b) $0.3 < M < 1$

(c) $0.8 < M < 1.2$ (d) $M > 1$

APPSC AEE 2012

Ans : (b) (i) Subsonic flow - $M < 1$

(ii) Sonic flow - $M = 1$

(iii) Supersonic flow - $M > 1$

(iv) Hypersonic flow - $M > 6$

M = Mach number

1545. The static pressure and temperature in air flow through a nozzle at a Mach number of 2 are 0.5 bar and 200 K. The stagnation temperature is recorded as

(a) 0ºC (b) 360ºC

(c) 360K (d) 273K

TNPSC ACF 2012

Ans. (c) : We know that,

$$\frac{T_1}{T_2} = 1 + \frac{\gamma - 1}{2} M^2$$

$$= 1 + \frac{0.4 \times 2 \times 2}{2} = 1.8$$

$$T_1 = 1.8 \times 200 = 360$$

1546. For adiabatic expansion with friction through a nozzle, the following remains constant :

(a) Entropy

(b) Static enthalpy

(c) Stagnation enthalpy

(d) Stagnation pressure

PSPCL AE, 2012

Ans. (c) : Stagnation enthalpy – The declaration of the gas stream of velocity, pressure and temperature to almost zero velocity in a infinite large setting chamber.

$$h_0 = h + \frac{v^2}{2}$$

Stagnation pressure – $\frac{P_0}{P} = \left[1 + \frac{\gamma - 1}{2} M^2\right]^{\frac{\gamma}{\gamma - 1}}$

M = Mach number

γ = adiabatic index

P = pressure

P_0 = Stagnation pressure

1547. The velocity of fluid particle at the centre of pipe section is ..... .

(a) zero

(b) minimum

(c) maximum

(d) average of full section

GWSSB DEE 07.07.2016

Ans. (c) : The velocity of fluid particle at the centre of pipe section is maximum.

1548. For flow in a nozzle discharging from a top "choking" condition occurs, when the flow through nozzle exit is

(a) subsonic (b) supersonic

(c) critical (d) transonic

APPSC AEE 2012

Ans. : (c) For flow in a nozzle discharging from a top choking condition occurs, when the flow the nozzle exit is critical.

1549. In subsonic Fanno flow of gas in a constant area duct with friction, pressure (P), temperature (T) and Mach number (M) varies in which of the following orders in flow direction?

(a) P decreases, T increases, M increases

(b) P decreases, T decreases, M increases

(c) P increases, T increases, M decreases

(d) P increases, T decreases, M increases

TNPSC ACF 2012

Ans. (b) : Fanno flow behavior

Parameter	–Subsonic flow	–Super flow
(1)Stagnation temperature	constant	constant
(2) Mach number	Increases	Decreases
(3) Pressure	Decreases	Increases
(4) Temperature	Decreases	Increases

1550. Velocity of sound in air at –40ºC, at high altitude in m/s is

[Take γ = 1.4, R = 287 J/kg K]

(a) 354.6 (b) 306

(c) 400 (d) 331

TNPSC ACF 2012

Ans. (b) : $T = -40^\circ C = 273 - 40 = 233$ K

$$C = \sqrt{\gamma RT}$$

$$C = \sqrt{1.4 \times 287 \times 233} = 305.972 \text{ m/s}$$

C = 306 m/s

1551. For supersonic flow, if the area of flow increases then

(a) velocity decreases (b) velocity increases

(c) velocity is constant (d) None of the above

APPSC AEE 2012

Ans. (b) : For supersonic flow, if the area of flow increases then velocity increases and pressure decreases.

1552. Which of the following statements is correct regarding the presence of the normal shock?

(a) It exists in the convergent passage

(b) It always exists at the entry of the divergent passage

(c) It always exists at the exit of the divergent passage

(d) It exists anywhere in the divergent passage.

RPSC AE GWD, 2011

Ans. (c) :

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1553. Mach number under choked condition in a steam nozzle is :

(a) Greater than 1 (b) Less than 1

(c) Less than 0.3 (d) Equal to 1

OPSC Civil Services Pre. 2011

Ans. (d) :

- Mach number under choked condition in a steam nozzle is equal to one.
- In the chocked condition of convergent divergent nozzle, a constant and maximum mass flow rate of air passes through nozzle in this condition.
- Critical pressure at throat and after that pressure reduces.
- Mach number at throat is one i.e. velocity of flow is sonic.

1554. The stagnation temperature of an isentropic flow of air (k = 1.4) is 360K. If the temperature is 200K at a section, then the Mach number of the flow will be

(a) 1.0 (b) 1.5

(c) 3.0 (d) 2.0

ISRO Scientist/Engineer 2011

Ans. (d) : Formula in temperature form

$$\frac{T_0}{T} = \left[1 + \frac{k-1}{2} M^2\right] \qquad [\text{Given, } k = \gamma]$$

Or $$\frac{T_0}{T} - 1 = \frac{k-1}{2} M^2$$

Or $$\frac{360}{200} - 1 = \frac{1.4-1}{2} M^2$$

Or $0.8 = 0.2\ M^2$

$M = 2$

Here,

T_0 = stagnation temperature

T = Temperature

M = Mach number

1555. The effect of supersaturation, in a nozzle is :

(a) to increase the heat drop

(b) to increase entropy

(c) to decrease dryness fraction of steam

(d) to decrease specific volume of steam

J&K PSC Civil Services Pre, 2010

Ans. (b) : Effect of Supersaturated Flows in Nozzle–

- The Supersaturation increases the entropy and specific volume of the steam.
- The Supersaturation reduces the heat drop (for the same pressure limits) below that for thermal equilibrium. Hence the exit velocity of the steam is reduced.
- The Supersaturation increases dryness fraction of steam.
- Since the condensation does not take place during supersaturated expansion, so the temperature at which the Supersaturation occurs will be less than the saturation temperature corresponding to the pressure. Therefore, the density of supersaturated steam will be more than for the equilibrium conditions, which gives the increase in the mass of steam discharged.

1556. Which one of the following properties drops across a normal shock ?

(a) Static pressure

(b) Stagnation temperature

(c) Stagnation pressure

(d) Entropy

DRDO Scientists 2009

Ans. (c) : Stagnation pressure

1557. Choose the correct statement for a quasi-one dimensional supersonic flow through a variable area duct :

(a) Velocity decreases when area increases

(b) Velocity increases when area increases

(c) Velocity increases when area decreases

(d) Velocity is independent of area

DRDO Scientists 2009

Ans. (b) : $\frac{dA}{A} = \frac{dV}{V}\left[M^2 - 1\right]$

$\because$ $M > 1$, $\frac{dA}{A} = \frac{dV}{V}$

$\because$ If $A\uparrow$, $V\uparrow$ and $A\downarrow$, $V\downarrow$

1558. In passing through a standing normal shock-wave, a subsonic flow cannot turn supersonic as it will not satisfy the

(a) continuity equation

(b) momentum equation

(c) energy equation

(d) second law of thermodynamics

DRDO Scientists 2008

Ans. (d) : second law of thermodynamics

1559. A steady flow of fluid crossing a normal shock-wave undergoes a sudden rise in

(a) stagnation pressure

(b) stagnation temperature

(c) static pressure accompanied by a fall in static temperature

(d) both static pressure and static temperature

DRDO Scientists 2008

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Ans. (d)

Property change summary across a normal shock

Property	Effect
Stagnation temperature	No change
Entropy	↑
Stagnation Pressure	↓
Temperature	↑
Velocity	↓
Density	↑
Pressure	↑
Mach number	↑

1560. Stagnation temperature in terms of Mach number is given by

(a) $\frac{T_0}{T} = \frac{K-1}{2}.M^2$ (b) $\frac{T_0}{T} = \frac{1+K^2}{2}.M^2$

(c) $\frac{T_0}{T} = 1-\frac{K-1}{2}.M^2$ (d) $\frac{T_0}{T} = 1+\frac{K-1}{2}.M^2$

TNPSC AE, 2008

Ans. (d) : Stagnation temperature in terms of Mach Number

$$\frac{T_0}{T} = 1+\left(\frac{K-1}{2}\right)M^2$$

Where,

T_0 = Stagnation temperature

T = Static temperature

K = Ratio of specific heat

1561. An aeroplane is flying at an height of 14km where temperature is -45°C. The speed of the plane is corresponding to M=2. Find the speed of the plane if R=287J/kg K and γ = 1.4.

(a) 2179 km/h (b) 3010 km/h

(c) 3250 km/h (d) 3379 km/h

ISRO Scientist/Engineer 2008

Ans. (a) : $v = \sqrt{\gamma RT} \times M$

$v = 2\sqrt{\gamma RT}$ [$\because M = 2$]

$\left(C = \frac{V}{M}, M = \text{Mach number } M = 2\right)$

$= 2\sqrt{1.4 \times 287 \times (273-45)}$

= 605.344 m/s

= 605. 344 × 3.6 km/hr

= 2179.24 km/hr.

1562. The normal shock wave in compressible flow is analogous to

(a) surges in open channel

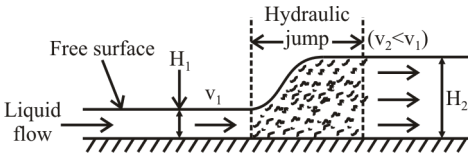
(b) vortex formation in centrifugal pump

(c) hydraulic bore in tidal rivets

(d) hydraulic jump in channel flow

TNPSC AE, 2008

Ans. (d) : A shock wave is extremely thin, the shock wave normally only being a few mean free path thick. A shock-wave is analogous in many wave to a "hydraulic jump" that occurs in free surface liquid flow. A hydraulic jump occurs for example in the flow down stream of a weir.



1563. The compressor performance at high altitude as compared to sea level will be

(a) same

(b) higher

(c) lower

(d) may be higher or lower depending on other factors

UKPSC AE 2007 Paper -II

Ans. (c) : The compressor performance at high altitude as compared to sea level will be lower.

1564. A convergent duct produces

(a) deceleration in supersonic and acceleration is subsonic flow

(b) acceleration in both supersonic and subsonic flow

(c) deceleration in both supersonic and subsonic flow

(d) acceleration in supersonic and deceleration in subsonic flow

UKPSC AE 2007 Paper -II

Ans. (a) : Deceleration in supersonic and acceleration is subsonic flow

1565. A shock wave which occurs in a supersonic flow represents a region in which

(a) a zone of silence exists

(b) there is no change in pressure, temperature and density

(c) there is sudden change in pressure, temperature and density

(d) velocity is zero

UKPSC AE 2007 Paper -II

Ans. (c) : There is sudden change in pressure, temperature and density

1566. The sonic velocity in a fluid medium is directly proportional to

(a) Mach number

(b) Density

(c) Pressure

(d) Square root of temperature

UKPSC AE 2007 Paper -II

Ans. (d) : For sonic velocity, Ma = 1

In a fluid medium is directly proportional to square root of temperature.

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06.

Fluid Machinery

1. Impact of Jets

1. A jet of water issues from a nozzle with a velocity of 20 m/s and it impinges normally on a flat plate moving away from it at 10 m/s. If the cross-sectional area of the jet is 0.02 m^2 and the density of water is taken as 1000 kg/m^3, then the force developed on the plate will be

(a) 2000 N (b) 10 N
(c) 100 N (d) 1000 N

APPSC AE Subordinate Service Civil/Mech. 2016
GATE 1990, CSE Pre-1994

Ans. (a) : $\rho = 1000\ kg/m^3$
Velocity of jet (V) = 20 m/s
Velocity of plate (u) = 10 m/s
Cross-sectional area of jet = (A) = 0.02 m^2
Force exerted by the jet on a moving flat vertical plate is given by

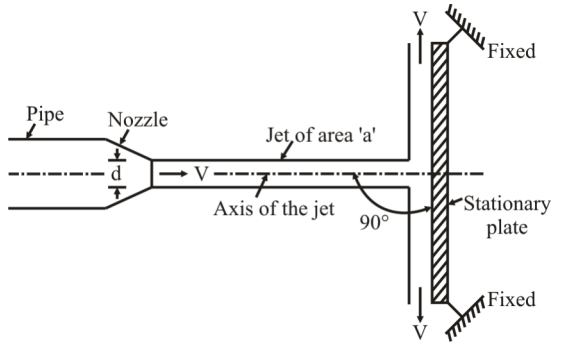
$$F_x = \rho A (V-u)^2$$
$$F_x = 1000 \times 0.02 (20 - 10)^2$$

2. The force exerted by a jet of water on a stationary vertical plate in the direction of jet is given by:

(a) $\frac{\rho AV^2}{2}$ (b) ρAV^2
(c) $2\rho AV^2$ (d) $4\rho AV^2$

TNPSC AE 2019
GPSC Asstt. Prof. 28.08.2016

Ans. (b) :



The force exerted by the jet on the plate (assuming it smooth) in the direction of jet (X-direction),
F_x = Rate of change of momentum (in the direction of force)
= (Initial momentum – final momentum)
= (Mass/sec) × [velocity of jet before striking the plate – velocity of jet after striking the plate]
= $\rho aV(V - 0)$ [$\because$ Mass/second = ρaV]
$F_x = \rho aV^2$
(Where, ρ = mass density of liquid; a = area of jet $= \frac{\pi}{4}d^2$, d being diameter of the jet).

3. A fluid jet of cross-sectional area A and velocity V strikes a flat plate moving with a velocity u. If ρ is the density of the fluid, the mass of fluid per second striking the plate is :

(a) $\rho A (V + u)$ (b) $\rho A (V - u)$
(c) ρAV (d) $\rho A (u - V)$

APPSC AEE Mains 2016 (Civil/Mechanical)
APGENCO AE 2012

Ans. (b) : The mass of fluid per second striking the plate is ($\dot{m}$) = $\rho A (V - u)$
Where, A = Area
ρ = density
V = strikes plate or jet velocity
u = moving plate
$F_x = 2000$ N

4. A jet of water issues from a nozzle with a velocity 20 m/s on a flat plate moving away from it at 10 m/s. The cross-sectional area of the jet is 0.01 m^2 and the density of water = 1000 kg/m^3. The force developed on the plate is

(a) 1000 N (b) 100 N
(c) 10 N (d) 2000 N

ISRO Scientist/Engineer 2012
ESE 2015

Ans. (a) : Velocity nozzle of water jet V_w = 20 m/s
plate, velocity V_p = 10 m/s
A = 0.01 m^2
ρ = 1000 kg/m^3
Force on plate = change in momentum of water

$$F = \rho A \times (V_w - V_p)^2$$

V_w = velocity of water jet
V_p = velocity of plate
$F = (1000 \times 0.01) \times (20 - 10)^2$
F = 1000 N

5. A jet of water of diameter 60 mm strikes a stationary flat plate with a velocity of 30 ms^{-1}. The force exerted by jet on plate will be

(a) 84.8 N (b) 2544.7 N
(c) 1766.8 N (d) 24963.5 N

GPSC DEE, Class-2 (GWSSB) 04.07.2021

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Ans. (b) : V = 30 m/s, d = 60 mm (dia of jet)
$= 60 \times 10^{-3}$ m
Force exerted by jet of water stationary flat plate,

$$\boxed{F = \rho a V^2}$$

Where, a = Area of jet of water
ρ = Density of water = 1000 kg/m^3
V = Velocity of water jet

$$F = \rho a V^2$$

$$= 1000 \times \frac{\pi}{4} \times 60^2 \times 10^{-6} \times 30^2$$

$$= 2{,}544.69 \simeq 2544.7 \text{ N}$$

6. A jet of water issuing from a nozzle strikes a flat plate normally (plane of plate makes an angle 90° with the axis of the jet). The plate moves towards the nozzle with a velocity of 10 m/s while the jet issuing from the nozzle has a velocity of 15 m/s. The absolute velocity components of the jet (in m/s) as it leaves the vane, parallel (x component) and normal component (y component) to the undisturbed jet are. (Neglect friction on the surface).

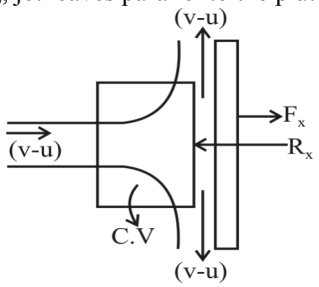
(a) $V_x = 5, V_y = 5$ (b) $V_x = -5, V_y = 5$
(c) $V_x = 10, V_y = 10$ (d) $V_x = -10, V_y = 25$

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) : Concept–The relative velocity of the jet with respect to the plate.

- When the plate is moving with velocity u in the direction of the jet then the relative velocity of jet:
  $V_r = V - u$
- When the plate is moving with velocity u in the opposite direction of the jet then the relative velocity of jet:
  $V_r = V + u$

After striking, jet leaves parallel to the plate



Calculation–
Given,
Velocity of plate, u = 10 m/s, velocity of jet, v = 15 m/s
V_x = absolute velocity component of jet after striking of plate in the x-direction.
V_y = absolute velocity component of the jet after striking to plate
The plate is moving away from the nozzle.
By the concept of relative velocity, the velocity of the jet along with the plate after jet strikes:

$$V_y = 15 - 10 = 5 \text{ m/s}^2$$

By the concept of relative velocity, the velocity normal to plate after jet strikes:

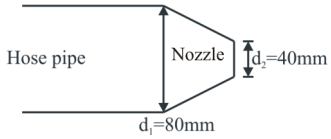
$$V_x = 5 \text{ m/s}^2$$

7. A nozzle at the end of an 80 mm hose pipe produces a jet 40 mm in diameter. When it is discharging the water 1200 Lpm, the force on the joint at the base of the nozzle will be

(a) 180 N (b) 200 N
(c) 220 N (d) 240 N

ESE 2019

Ans. (d) :



Given,

$d_1 = 80$ mm
$d_2 = 40$ mm

$$Q = 1200 \text{ Lpm} = \frac{1200 \times 10^{-3}}{60} = 0.02 \text{ m}^3/\text{s}$$

$$V_1 = \frac{Q}{A_1} = \frac{0.02}{\frac{\pi}{4} \times (0.08)^2} = 3.978 \text{ m/s}$$

$$V_2 = \frac{Q}{A_2} = \frac{0.02}{\frac{\pi}{4} \times (0.04)^2} = 15.915 \text{ m/s}$$

Force at joint (F) = $\rho Q (V_2 - V_1)$
$= 1000 \times 0.02 (15.915 - 3.978)$
$= 238.74$
$\simeq 240$ N

8. A jet of water is striking at the center of a curved vane moving with a uniform velocity in the direction of jet. For the maximum efficiency, the vane velocity is ..... of the jet velocity.

(a) One-half (b) One-third
(c) Two-third (d) Three-fourth

Nagaland PSC (CTSE) 2018, Paper-II

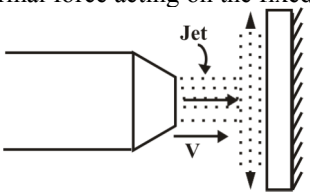
Ans. (b) : A jet of water is striking at the centre of a curved vane moving with a uniform velocity in the direction of jet for the maximum efficiency, the vane velocity is one third of the jet velocity.

9. If w = Specific weight of water in N/m^3, a = Cross sectional area of jet in m^2 and V = velocity of jet in m/s, the force exerted (in N) by a jet of water impinging normally on a fixed plate is .....

(a) waV/2g (b) waV/g
(c) waV^2/2g (d) waV^2/g

RPSC AE 2018

Ans. (d) : Normal force acting on the fixed plate is F_n.



$$F_x = F_n = \dot{m} \ [V_{jet} - V_{plate}]$$

$V_{jet} = V$, $V_{plate} = 0$ (in x-direction)

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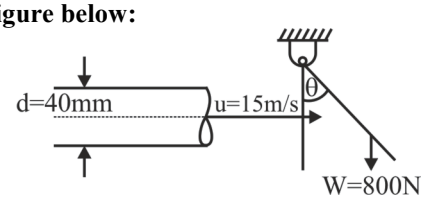
$$F_n = \rho \times a \times V[V-0]$$
$$= \rho \times a \times V^2$$
$$F_n = \rho \times g \times a \times \frac{V^2}{g}$$
$$F_n = \frac{w \times a \times V^2}{g}$$

10. A 40 mm diameter water jet strikes a hinged vertical plate of 800 N weight normally at its surface at its centre of gravity as shown in the figure below:

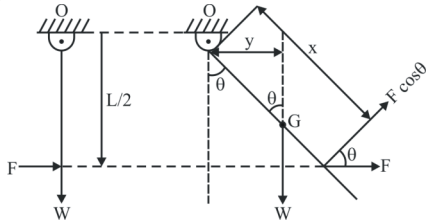


The angle of deflection is nearly

(a) $\sin^{-1} 0.353$ (b) $\sin^{-1} 0.321$
(c) $\tan^{-1} 0.353$ (d) $\tan^{-1} 0.321$

ESE 2018

Ans. (a) :



Taking moment about O at equilibrium position

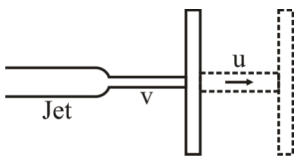
$$F\cos\theta \cdot x = W \cdot y$$
$$F\cos\theta \cdot \frac{L/2}{\cos\theta} = W\frac{L}{2}\sin\theta$$
$$F = W\sin\theta$$
$$\rho A V^2 = W\sin\theta$$
$$10^3 \times \frac{\pi}{4}(0.04)^2 \times 15^2 = 800 \times \sin\theta$$
$$\sin\theta = 0.353$$
$$\theta = \sin^{-1}(0.353)$$

11. What is the maximum efficiency of a jet striking a single plate moving in the direction of jet is :

(a) $\frac{4}{27}$ (b) $\frac{6}{27}$
(c) $\frac{1}{2}$ (d) $\frac{8}{27}$

Kerala PSC Poly. Lect., 2017

Ans. (d) :



Jet velocity = v
Plate velocity = u

Force exerted by jet $= \dot{m}[(v-u)-0]$
$= \rho a(v-u)(v-u)$
$= \rho a(v-u)^2$

Work done $= \rho a(v-u)^2.u$

$\dot{m} = \rho a V$

K.E of jet $= \frac{1}{2}\dot{m}v^2 = \frac{1}{2}\rho a v^3$

Efficiency $\eta = \frac{\text{Work done}}{\text{Energy supplied}}$

$$= \frac{2\rho a(v-u)^2 \cdot u}{\rho a v^3}$$
$$\eta = \frac{2(v-u)^2 \cdot u}{v^3}$$

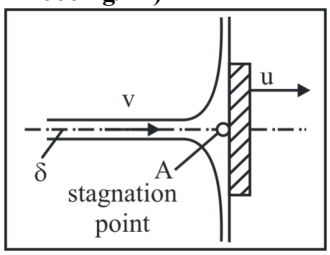
For maximum efficiency $\frac{d\eta}{du} = 0$

$$0 = \frac{d}{du}\left(\frac{2(v-u)^2 \cdot u}{v^3}\right)$$
$$\frac{d}{du}\left[2u(v^2+u^2-2vu)\right] = 0$$
$$\frac{d}{du}\left[2v^2u + 2u^3 - 4vu^2\right] = 0$$
$$2v^2 + 6u^2 - 8vu = 0$$
$$v^2 + 3u^2 - 4vu = 0$$
$$(v-3u)(v-u) = 0$$
$$(v-u) \neq 0$$
$$v = 3u \Rightarrow \boxed{u = \frac{v}{3}}$$

(Condition for maximum efficiency in case of a single moving plate)

$$\eta_{max} = \frac{2[3u-u]^2 \cdot u}{27u^3}$$
$$= \frac{2 \times 4u^3}{27u^3} = \frac{8}{27}$$
$$\boxed{\eta_{max} = \frac{8}{27}}$$

12. Calculate the dynamic head ($P_A - P_{static}$) when a jet of water with a velocity, v strikes a flat plate moving with a velocity, u in the same direction as jet. (Take u = 4 m/s, v = 10 m/s and $\rho = 1000$ kg/m^3)



(a) 8000 Pa (b) 50000 Pa
(c) 18000 Pa (d) 98000 Pa

ISRO Scientist/Engineer 07.05.2017

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Ans. (c) : $\boxed{P_s = P_0 + \rho\left(\frac{V_0^2}{2}\right)}$

$V_0 = V - u$

$V = 10$ m/sec., $u = 4$ m/sec

$V_0 = (10 - 4)$ m/s

$V_0 = 6$ m/s

$P_A = P_s$ = Stagnation pressure

$P_0 = P_{static}$ = Static pressure

So, $P_A = P_{static} + \rho\left[\frac{(V-u)^2}{2}\right]$

$(P_A - P_{static}) = \rho\left[\frac{(V-u)^2}{2}\right]$

$= 1000 \times \frac{(6)^2}{2}$

$= 1000 \times \frac{36}{2}$

$P_A - P_{static} = 18000$ Pa

13. A water jet with a velocity 20 m/sec is directed upwards at an angle of 45° to the horizontal. If air resistance is neglected, it will reach a maximum elevation at a distance of 'x' from nozzle. The value of 'x' in meters is:

(a) 20.38 (b) 40.77

(c) 10.19 (d) 14.41

RPSC ADE 2016

Ans. (c) : We know that,

Maximum elevation or height (x) =

$$\frac{u^2 \sin^2\theta}{2g} = \frac{(20)^2\left(\sin 45^0\right)^2}{2 \times 9.81}$$

= 10.193 meter

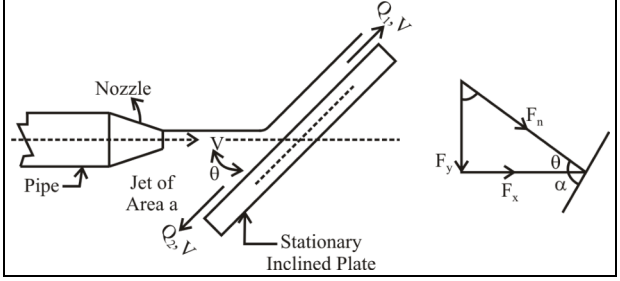
14. The normal force of thrust exerted by a jet of water of cross sectional area 'a' and velocity 'V' on a flat plate inclined at an angle θ to the jet is given by (where Q is the discharge)

(a) $QV \sin^2\theta/g$ (b) $QV \cos\theta/g$

(c) $QV \sin\theta/2g$ (d) $QV (1+\sin\theta)/g$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



15. Impulse momentum equation gives relationship between force (F), mass (m) and velocity (v) as

(a) $F = v\frac{dm}{dt}$ (b) $F = \frac{1}{m}\frac{dv}{dt}$

(c) $F = \frac{d(mv)}{dt}$ (d) $F = v\frac{dv}{dt}$

APPSC AEE Screening Test 2016

Ans. (c) : Impulse momentum equation gives relationship between force (F), mass (m) and velocity (v)

$$F = \frac{dp}{dt}$$

$$F = \frac{d}{dt}(mv)$$

$$F = m \cdot \frac{dv}{dt}$$

$$\boxed{F = m \cdot a}$$

16. A liquid jet of 30 m/s velocity is striking a single symmetrical moving curved vane at the centre. For maximum efficiency, the velocity of the vane should be

(a) 30 m/s (b) 20 m/s

(c) 15 m/s (d) 10 m/s

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (d) : As we know,

$$\text{Efficiency of jet} = \left(1 - \frac{V_2^2}{V_1^2}\right)$$

So, for maximum hydraulic efficiency

$$\eta = \frac{2u(V-u)^2(1+\cos\phi)}{V^3}$$

So, for maximum efficiency

$\therefore V = 3u$

$$u = \frac{V}{3} = \frac{30}{3} = 10\,m/s$$

17. The rate of change of linear momentum equals

(a) Active force (b) Reactive force

(c) Torque (d) Work done

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : According to Newton's second law of motion rate of change of linear momentum is directly proportional to the force applied.

$$\frac{\text{Change in momentum}}{\text{Change in time}} = \text{Force}$$

$$F = \frac{dP}{dt} = \frac{d}{dt}(mv)$$

$$= \frac{mdv}{dt}$$

$F = m.a$

18. The force exerted by a jet of water of 40 mm diameter on a flat stationary plane, when the jet strike the plate normally with 20 m/s velocity.

(a) 40 πN (b) 80 πN

(c) 120 πN (d) 160 πN

TSGENCO AE 14.11.2015

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Ans. (d) : Given,
Diameter (D) = 40 mm
$= 40 \times 10^{-3}$ m
Velocity (V) = 20 m/sec
Force exerted by jet of water on a flat stationary plate.

$F = \rho a v^2$

$$= 1000 \times \frac{\pi}{4} \times \left(40 \times 10^{-3}\right)^2 \times 20 \times 20$$

$$= 1000 \times \frac{\pi}{4} \times 1600 \times 10^{-6} \times 400$$

$= 160\ \pi N$

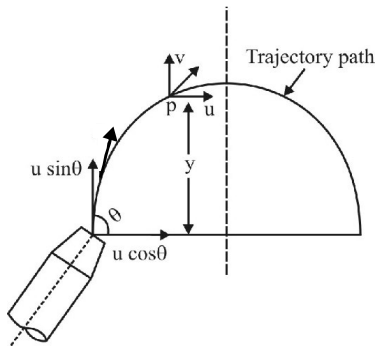
19. For a liquid coming out of a nozzle, maximum height attained is:

(a) $\frac{u^2 \sin^2\theta}{g}$ (b) $\frac{u^2 \sin^2 2\theta}{g}$

(c) $\frac{u^2 \cos^2\theta}{2g}$ (d) $\frac{u^2 \sin^2\theta}{2g}$

OPSC AEE 2015 PAPER - II

Ans. : (d)



free liquid jet is defined as the jet of water coming out from the nozzle in atmosphere. The travelled path by the free jet is parabolic.

(i) Maximum height attained by the jet:-

$$h = \frac{u^2 \sin^2\theta}{2g}$$

(ii) Time of flight :-

$$T = \frac{2u \sin\theta}{g}$$

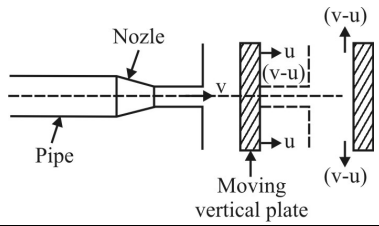
20. Force exerted by a jet of water impinging on a moving flat plate is

(a) $\frac{W}{g} a (v-u)^2$ (b) $\frac{W}{g} a (v-u)$

(c) $\frac{W}{2g} a (v-u)^2$ (d) $\frac{W}{2g} a (v-u)$

TSPSC AEE 2015

Ans : (a)



Let u = plate velocity, v = jet velocity
Force exerted by jet on moving flat plate normal to jet

$F_n = \rho a (v-u)^2$

$F_n = \frac{W}{g} a (v-u)^2 \qquad (W = \rho g)$

work done per second ($\dot{W}$) $= F_n \times u = \rho a (v-u)^2 u$

21. A 6-cm diameter horizontal stationary water jet having a velocity of 40 m/s strikes a vertical plate. The force needed to hold the plate if it moves away from the jet at 20 m/s is nearly

(a) 1365 N (b) 1270 N
(c) 1130 N (d) 1080 N
(e) 1150 N

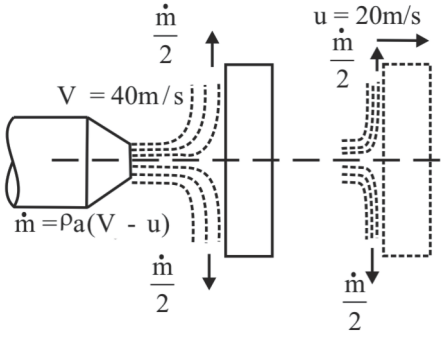
CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : Jet of water strikes on a moving plate
So
Jet of water strike with moving plate with relative velocity = (V - u)
So
Force acting in x - direction will be

$F_{x\text{-}x} = \dot{m}\,[V-u] = \rho \times a \times (V-u)^2$

$F_{x\text{-}x} = 1000 \times \frac{\pi}{4} \times 36 \times 10^{-4} \times [40-20]^2$

$F_{x\text{-}x} = 1130.4\ N \approx 1130\ N$

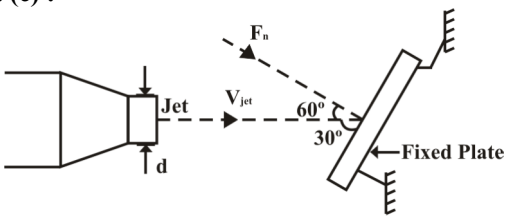


22. A 75 mm diameter water jet having a velocity of 12m/s impinges on a plane, smooth plate at an angle to 60° to the normal to the plate. The impact on the plate (when plate is stationary) is.

(a) 552.66N (b) 550.66N
(c) 317.93N (d) 319.93N
(e) 319.93N

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) :



We know that normal force acting on a inclined plate

$F_n = \rho\, a\, V^2 \sin\theta$

where, θ = angle made by jet with inclined plate

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$$a = \frac{\pi}{4}d^2$$
$$= \frac{\pi}{4} \times (75)^2$$
$$a = 4.415 \text{ mm}^2$$
$$F_n = 4.415 \times (12)^2 \times \sin(30^0)$$
$$\boxed{F_n = 317.925 \text{N}}$$

23. Maximum efficiency of a series of vertical plates is

(a) 66.67% (b) 33.33%
(c) 50% (d) 80%

APPSC Poly. Lect. 2013

Ans. (c) : Maximum efficiency of a series of vertical plates is 50%.

24. The internal momentum equation requires the assumption that

(a) the flow is uniform
(b) the flow is unidirectional
(c) the fluid is incompressible
(d) the flow is steady

APPSC AEE 2012

Ans. (c) : The fluid is incompressible.

25. A jet of water (density = ρ) with cross-sectional area a, moving with a velocity V strikes a hinged square plate of weight W at the centre of the plate. The plate is of uniform thickness. The angle through which the plate will swing is:

(a) $\cos\theta = \frac{\rho a V^2}{W}$ (b) $\sin\theta = \frac{\rho a V^2}{W}$
(c) $\cos\theta = \frac{\rho a V^2}{2W}$ (d) $\sin\theta = \frac{\rho a V^2}{2W}$

OPSC Civil Services Pre. 2011

Ans. (b) : The angle through which the plate will swing is :

$$\sin\theta = \frac{\rho a V^2}{W}$$

26. If a jet of water of area 'A' strikes with velocity 'V' a series of flat plates mounted on a wheel of mean diameter D rotating at N rpm, then force exerted will be equal to

(a) $\frac{\rho A V^2}{g}$ (b) $\frac{\rho A V}{g}\left(V - \frac{\pi DN}{60}\right)$
(c) $\frac{\rho A}{g} \times \frac{\pi DN}{60}$ (d) $\frac{\rho A}{g}\left(V - \frac{\pi DN}{60}\right)$

ISRO Scientist/Engineer 2007

Ans. (b) : Force Exerted

$$= \frac{\rho A V (V - u)}{g} \text{ [Force per unit weight]}$$

where $u = \frac{\pi DN}{60}$

$$= \frac{\rho A V}{g}\left[V - \frac{\pi DN}{60}\right]$$

27. A series of normal flat vanes are mounted on the periphery of a wheel, the vane speed being V. For maximum efficiency the speed of the liquid jet striking the vanes should be

(a) $\frac{V}{3}$ (b) $\frac{V}{2}$
(c) V (d) 2V

UKPSC AE 2007 Paper -II

Ans. (d) : For maximum efficiency,

Blade velocity $= \frac{\text{Jet velocity}}{2}$

Jet velocity = 2 × blade velocity
= 2V

2. Hydraulics Turbines

28. The specific speed of turbine is _____ where N is the speed of the turbine, P is power generated, Q is the rate of flow and H is the head available.

(a) $N_s = \frac{(N\sqrt{P})}{H^{(5/4)}}$ (b) $N_s = \frac{(N\sqrt{Q})}{H^{(3/4)}}$
(c) $N_s = \frac{(N\sqrt{Q})}{H^{(5/4)}}$ (d) $N_s = \frac{(N\sqrt{P})}{H^{(3/4)}}$

NLCIL GET 17.11.2020, Shift-II
VIZAG MT, 14.12.2020, TNPSC AE 2018
Nagaland PSC (CTSE) 2018, Paper-II
GPSC EE Pre- 28.01.2017
Punjab PSC Lect. 06.08.2017
TNPSC AEE 28.08.2017 (Civil/Mechanical)
MPPSC AE 2016
APPSC AE Subordinate Service Civil/Mech. 2016
OPSC AEE 2015, KPSC ADF 2015
Mizoram PSC AE/SDO 2014, Paper-I
UKPSC AE 2007, 2012 Paper-II
APPSC AE 2012
OPSC-Civil service Pre- 2011
WBPSC AE 2007, CSE Pre-1996
ESE 2007, 1997

Ans. (a) : Specific speed of turbine

$$\boxed{N_s = \frac{N\sqrt{P}}{H^{5/4}}}$$

Specific speed of pump

$$\boxed{N_s = \frac{N\sqrt{Q}}{H^{3/4}}}$$

29. In order to have maximum power from a Pelton turbine, the bucket speed must be :

(a) Equal to the jet speed
(b) Equal to half of the jet speed
(c) Equal to twice the jet speed
(d) Independent of the jet speed

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
APPSC AEE SCREENING 17.02.2019
GPSC Executive Engineer 23.12.2018
GPSC Asst. Director of Transport 05.03.2017

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Vizag Steel (MT)-2017
BPSC AE Mains 2017 Paper V
UP Jal Nigam AE 2016, APSC AEE 2016
Nagaland PSC CTSE 2015 Paper-II
GATE 2013

Ans. (b) : In order to have maximum power from a pelton turbine, the bucket speed must be equal to half of the jet speed.

30. A Kaplan turbine is

(a) A high head mixed flow turbine
(b) A low head axial flow turbine
(c) An outward flow reaction turbine
(d) An impulse inward flow turbine

OPSC AEE 2019 Paper-II
APPSC AEE SCREENING 17.02.2019
CIL MT 26.03.2017
Karnataka PSC AE 10.09.2017
TSPSC AEE 2017
CGPSC AE 26.04.2015 SHIFT-II
ESE 2011, WBPSC AE 2008
GATE 1997

Ans. (b) :

	Type of turbine	Head	Discharge	Flow	Specific speed (MKS unit)
1.	Pelton (impulse)	High (above 250 m)	Small	Tangential	10-35
2.	Francis (reaction)	Medium (60 m -250m)	Medium	Radial flow or mixed flow : Radial and axial	60-300
3.	Kaplan (reaction)	Low (below 60 m)	Large	Axial	300-1000

31. The specific speed of a turbine is speed of an imaginary turbine, identical with the given turbine, which

(a) delivers unit discharge under unit load
(b) delivers unit discharge under unit speed
(c) develops unit H.P. under unit head
(d) develops unit H.P. under unit speed

TNPSC AE 2019, ESE 2018
Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)
TNPSC AE (Industries) 09.06.2013
APPSC AEE 2012
UKPSC AE 2012 Paper–II, ESE 1996

Ans : (c) The specific speed of a turbine is speed of an imaginary turbine, identical with the given turbine which develop per unit H.P. under unit head.

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

Significance of specific speed :- Specific speed plays an important role for selecting the type of turbine. Also the performance of a turbine can be predicted by knowing the specific speed of the turbine.

32. A Pelton wheel is ideally suited for :

(a) High head and low discharge
(b) High head and high discharge
(c) Low head and low discharge
(d) Medium head and medium discharge

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Gujarat PSC AE 2019, RPSC AE 2018
UJVNL AE. 2016, KPSC AE 2015
TSPSC Managers, 2015, TNPSC AE 2013
UKPSC AE 2007, 2013 Paper -II
ESE 1998

Ans. : (a) A pelton wheel is ideally suited for high head and low discharge

Pelton wheel :-
i) High head
ii) Low discharge
iii) Low specific speed
iv) Horizontal shaft turbine
v) Tangential flow

33. A hydraulic turbine develops 1000 kW power for a head of 40 m. If the head is reduced to 20 m, the power developed (in kW) is

(a) 177 (b) 354
(c) 500 (d) 707

RPSC ACF & FRO, 26.02.2021
ISRO Scientist/Engineer (RAC) 07.05.2017
Kerala PSC Asstt. Prof. 27.09.2016
NPCIL ET, 2015
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2010

Ans. (b) : $P_u = \frac{P}{H^{3/2}}$

$$\therefore \quad \frac{P_1}{H_1^{3/2}} = \frac{P_2}{H_2^{3/2}}$$

$$P_2 = P_1\left(\frac{H_2}{H_1}\right)^{3/2} = 1000\left(\frac{20}{40}\right)^{3/2}$$

$$\boxed{P_2 = 353.55 \text{ kW}}$$

34. In a Pelton wheel the bucket peripheral speed is 10m/s, the water jet velocity is 25 m/s and volumetric flow rate of the jet is 0.1 m^3/s. If the jet deflection angle is 120° and the flow is ideal, the power developed is :

(a) 7.5 kW (b) 15.0 kW
(c) 22.5 kW (d) 37.5 kW

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
TSPSC AEE 2017
OPSC AEE 2015 PAPER - II
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2006

Ans. (c) : $u = 10 m/s$
$V_1 = 25$ m/s
$Q = 0.1$ m^3/s
$\phi = 60°$

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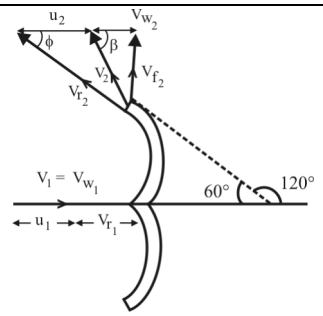

$$V_{r_1} = V_1 - u_1$$
$$= 25 - 10$$
$$= 15 m/s$$
$$V_{r_1} = V_{r_2} = 15 m/s$$
$$u_1 = u_2 = u = 10 m/s$$
$$\text{Power} = \rho Q\,[V_{w_1} + V_{w_2}]u$$
$$= 1000 \times 0.1[V_1 + V_{r_2}\cos\phi - u_2]u$$
$$= 1000 \times 0.1 \times [25 + 15\cos 60° - 10] \times 10$$
$$= 22500 \text{watt}$$
$$\boxed{P = 22.5 kW}$$

35. The purpose of surge tank in a pipeline is to

(a) remove friction
(b) prevent turbulence in the flow
(c) prevent flow loss
(d) relieve pressure due to water hammer

RPSC AE GWD 2019, Gujarat PSC AE 2019
TNPSC AE 2019, MPPSC AE 2016
TRB Poly Lect. 2012
J & K PSC Screening 2006

Ans. : (d) : Surge tank is a reservoir fitted on the penstock near the turbine to receive the rejected water when the pipe line is suddenly closed by the governing mechanism. It helps to reduce water hammer effect in the penstock pipe line.

36. Kaplan turbine is the reaction turbine of which of the following the types?

(a) Axial flow type
(b) Radial flow type
(c) Mixed flow type
(d) Tangential flow type

OPSC AEE 2019 Paper - II
Karnataka PSC AE 10.09.2017
TSPSC AEE 2017
GPSC Asstt. Prof. 28.08.2016
UPRVUNL AE 2016, TNPSC ACF 2012

Ans. (a) : Classification of Hydraulic turbine according to the type of energy at inlet

(I) Impulse Turbine- Ex Pelton turbine
(II) Reaction Turbine- Ex Francis turbine, Kaplan turbine, propeller turbine.

(2) According to the head at the inlet of turbine
(I) Tangential flow turbine- Ex Pelton turbine
(II) Radial flow turbine- Ex Francis turbine
(III) Axial flow turbine- Ex Kaplan turbine propeller turbine.

37. Cavitation in a hydraulic reaction turbine is most likely to occur at the turbine

(a) Rotor exit (b) Exit
(c) Stator exit (d) entry

GPSC Executive Engineer 23.12.2018
TSPSC AEE 2017
APPSC AEE MAINS 2016 PAPER-III
KPSC ADF 2015, GATE 1993

Ans. (a) : The formation, growth and collapse of vapour filled cavities or bubbles in a flowing liquid due to load fall in fluid pressure is called cavitation.

Cavitation in Turbines:- In turbines, only reaction turbines are subjected to cavitation, the cavitation may occur at the runner exit or at the draft tube inlet where the pressure is considerably reduced.

Caviation in centrifugal pumps:- In centrifugal pumps, the cavitation may occur at the inlet of the impeller of the pump or at the suction side of the pump, where the pressure is considerably reduced.

38. An impulse turbine is used for

(a) Low head of water
(b) High head of water
(c) Medium head of water
(d) High discharge

JPSC AE, 10.04.2021, Paper-I
Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)
ISRO Scientist/Engineer (RAC) 07.05.2017
GWSSB DEE 07.07.2016

Ans. (b) : An impulse turbine is used for high head of water.
In an impulse turbine the pressure energy of water is converted into kinetic energy when passed through the nozzle and form the high velocity jet of water.

39. The maximum hydraulic efficiency of an impulse turbine is

(a) $(1 + \cos\phi)/2$ (b) $(1 - \cos\phi)/2$
(c) $(1 + \sin\phi)/2$ (d) $(1 - \sin\phi)/2$

APPSC AEE SCREENING 17.02.2019
Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)
OPSC AEE 2015 Paper - II
ESE 2005

Ans. (a) : Hydraulic efficiency is the ratio of runner power to the kinetic energy per second.

$$\eta_{hyd} = \frac{2(v_1 - u)(1 + k\cos\phi)}{v_1^2}$$

$$(\eta_{hyd})_{max} = \frac{1 + k\cos\phi}{2}$$

When $u = \frac{v_1}{2}$

K = Friction factor for blades.

40. If D and d are the diameters of the runner and jet of the Pelton wheel respectively, the number of buckets on the runner is given by

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(a) $30+\frac{D}{d}$ (b) $30+\frac{D}{2d}$

(c) $15+\frac{D}{d}$ (d) $15+\frac{D}{2d}$

APPSC AEE SCREENING 17.02.2019
Kerala PSC Poly Lect. 2017
HPPSC W.S. Poly. 2016
TNPSC AE (Industries) 09.06.2013
APPSC AEE 2012

Ans. : (d) Number of Buckets on wheel are given as per Taygun's empirical relation.

$z=15+\frac{D}{2d}$ D = Dia of runner

$z=15+0.5m$ d = dia of jet

$m=\frac{D}{d}$ Jet ratio (12 for most case)

41. Water hammer is developed in

(a) penstock (b) draft tube

(c) turbine (d) surge tank

RPSC AE 2016, TNPSC AE 2014
TNPSC ACF 2012, RPSC ACF 2011
UKPSC AE 2007 Paper -II

Ans. : (a) Water hammer is developed in penstock.
The pressure rise due to water hammer depends upon
(i) The velocity of flow of water in pipe
(ii) The length of pipe
(iii) Time taken to close the valve
(iv) Elastic properties of the material of the pipe.

42. Which is the correct option for specific speed of pumps and turbines respectively?

(a) $\frac{NQ^{\frac{1}{2}}}{H^{\frac{3}{4}}}.\frac{NP^{\frac{1}{2}}}{H^{\frac{5}{4}}}$ (b) $\frac{NQ^{\frac{1}{2}}}{H^{\frac{2}{4}}};\frac{NP^{\frac{1}{2}}}{H^{\frac{5}{4}}}$

(c) $\frac{NP^{\frac{1}{2}}}{H^{\frac{5}{4}}};\frac{NQ^{\frac{1}{2}}}{H^{\frac{5}{4}}}$ (d) $\frac{NP^{\frac{1}{2}}}{H^{\frac{3}{4}}};\frac{NQ^{\frac{1}{2}}}{H^{\frac{5}{4}}}$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
TSPSC Managers, 2015
CGPSC Poly Lect. 22.05.2016
ISRO Scientist/Engineer 2007, ESE 1994

Ans. (a) : For pump,

Specific speed $N_s=\frac{N\sqrt{Q}}{H^{3/4}}=\frac{NQ^{1/2}}{H^{3/4}}$

For Turbine,

Specific speed $N_s=\frac{N\sqrt{P}}{H^{5/4}}=\frac{NP^{1/2}}{H^{5/4}}$

43. The draft tube in a turbine serves the following purpose :

(a) The scroll casing has to be fitted properly
(b) It supports the turbine as a strong base
(c) It reduces the working head thereby increasing the efficiency of turbine
(d) It increases the working head thereby increasing the efficiency of turbine.

CGPSC AE 15.01.2021
Punjab PSC SDE 12.02.2017
WBPSC AE, 2017, APPSC AEE 2012

Ans. (d) : Draft tube in a turbine serves the following purpose are –

- To reduce velocity/kinetic energy of water at exit thereby converting kinetic energy to pressure head allowing turbine to be installed above tail race level.
- It increases the working head thereby increasing the efficiency of turbine.

44. In a reaction turbine the function of a draft tube is to

(a) provide safety to turbine
(b) prevent air from entering
(c) reconvert the kinetic energy to flow energy
(d) increase the rate of flow

TSPSC Manager (Engg.) HMWSSB 12.11.2020
RPSC AE 2016, TSPSC Manager 2015
OPSC AEE 2015

Ans. (c) : The draft tube is a conduit which connects the runner exit to the tail race where the water is being finally discharged from the turbine. The primary function of the draft tube is to reduce the velocity of the discharged water to minimize the loss of kinetic energy at the outlet.

45. The condition for maximum efficiency of the Pelton wheel is that

(a) blade speed is one fourth of jet speed.
(b) blade speed is one-third of jet speed.
(c) blade speed is one half of jet speed.
(d) blade speed is equal to jet speed.

Nagaland PSC (CTSE) 2018, Paper-II
APPSC AEE SCREENING 17.02.2019
TSPSC AEE 2015, APPSC AEE 2012

Ans. : (c) Hydraulic efficiency

$$\eta_{hyd}=\frac{\text{Runner power}}{\text{Kinetic Energy per Second}}$$

$$\eta_{hyd}=\frac{2(V-u)(1+k\cos\phi)u}{V^2}$$

$$\eta_{hyd}=\left(\frac{1+K\cos\phi}{2}\right)$$

Hydraulic efficiency is maximum when blade speed is one half of jet speed.

$u=\frac{V}{2}$

u = Blade speed
V = Jet speed.

46. In an inward flow reaction turbine

(a) the water flows parallel to the axis of the wheel
(b) the water enters at the centre of the wheel and from there flows towards the outer periphery of the wheel
(c) the water enters the wheel at outer periphery, and then flows towards the centre of the wheel
(d) the flow of water is partly radial and partly axial

JPSC AE 10.04.2021 Paper I

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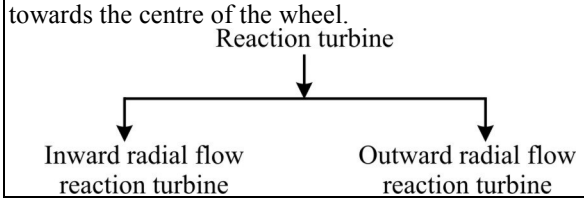
Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)
APPSC AEE 2012

Ans. : (c) In an inward flow reaction turbine the water enter the wheel at outer periphery, and then flow towards the centre of the wheel.



47. The unit speed of the turbine runner is

(a) $\frac{N}{\sqrt{H}}$ (b) $\frac{N}{H}$

(c) $N/H^{3/2}$ (d) $\frac{N}{H^2}$

Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)
TNPSC ACF 2012
OPSC Civil Services Pre-2006

Ans. (a) : $u \propto V \propto \sqrt{H}$

$\frac{\pi DN}{60} \propto \sqrt{H}$

$\frac{DN}{\sqrt{H}} = \text{constant}$

unit quantities are designed to predict performance of same hydraulic machine under different operating conditions.

$\therefore$ D = constant

or $\frac{N}{\sqrt{H}} = \text{constant} = N_u$

i.e. $N_u = \frac{N}{\sqrt{H}}$

48. In a reaction turbine the draft tube is used to

(a) transport water to downstream without eddies
(b) reconvert kinetic energy to flow energy by a gradual expansion of the flow cross-section
(c) increase the effective head
(d) prevent air from entering

UJVNL AE 2016, TRB Poly. Lect., 2012
APPSC AEE 2012, ESE 1996

Ans. (b) : In reaction turbine draft tube is used to convert the K.E to flow energy by a gradual expansion of the flow cross- section.

49. An impulse turbine

(a) Always operates submerged
(b) Makes use of draft tube
(c) Is most suited for low head installations
(d) Operates by initial complete conversion of kinetic head

RPSC ADE 2016, GPSC AE 16.10.2016
Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) :

	Impulse turbine	Reaction turbine
1.	Suitable for high head and low discharge	Suitable for maintain low head and large discharge
2.	Total potential energy converted into K.E. in nozzle.	Pressure energy converted into kinetic energy at the stage in fixed blade
3.	No draft tube required	draft tube required at pressure at outlet is less than atmospheric pressure
4.	No required to submerged as pressure at displaces remains constant and equal to atmospheric pressure	Should be submerged as pressure at all place done not equal and at exit it is less than atmospheric pressure .

50. Main characteristic curve of a turbine means

(a) Curves at constant speed
(b) Curves at constant efficiency
(c) Curves at constant head
(d) None of these

RPSC IOF, 2020
Nagaland PSC CTSE 2017 2nd Paper
J & K PSC Civil Service Pre 2010

Ans. (c) : Main characteristic curves of a turbine are also called curves at constant head. It helps to determine overall efficiency of the turbine by drawing curves with different set of parameters that play a major role in determining the performance of the turbine.

51. In general, the maximum number of jets employed in an impulse turbine without jet interference is:

(a) 2 (b) 1
(c) 6 (d) 4

VIZAG Steel MT 24.01.2021 Shift-II Shift-I
Nagaland PSC (CTSE) 2018, Paper-II
AAI Jr. Executive 29.11.2018

Ans. (c) : The maximum number of jets generally employed in an impulse turbine without jet interference are 6.

52. Which of the following turbines is most efficient at part load operation?

(a) Kaplan (b) Propeller
(c) Francis (d) Pelton wheel

RPSC ACF & FRO 26.02.2021
TSPSC Manager (Engg.) HMWSSB 12.11.2020
APPSC AEE Screening Test 2016

Ans. (a) : Kaplan Turbine-

- It is axial flow reaction turbine.
- Runner of Kaplan turbine has four to six blades.
- Runner blades of propeller turbines are fixed but Kaplan turbine blades are adjustable.
- At part load high efficiency can be obtained in the Kaplan turbine

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53. The hydraulic efficiency of a reaction turbine, is the ratio of

(a) power produced by the turbine to the energy actually supplied by the turbine
(b) actual work available at the turbine to energy imparted to the wheel
(c) work done on the wheel to the energy (or head of water) actually supplied to the turbine
(d) none of these

TNPSC AE 2018
Punjab PSC SDE 12.02.2017
GPSC Lect. 23.10.2016

Ans. (c) : Hydraulic efficiency of a reaction turbine, is the ratio of

$$\eta_H = \frac{\text{Work done on the wheel}}{\text{Energy actually supplied}}$$

$$\eta_H = \frac{Vw \times u}{gH}$$

54. Which turbine gives constant efficiency with high load condition ?

(a) Propeller turbine (b) Francis turbine
(c) Pelton wheel (d) Kaplan turbine

RPSC ACF and FRO 26.02.2021
APPSC AEE Screening test 2016
OPSC Civil Services Pre. 2011

Ans. (d) : The only turbine that can be regulated for best efficiency condition is Kaplan. Because its blades can be adjusted for optimum operation. This optimum adjustment mechanism is housed in the hub of turbine. so the inlet and outlet blade angles are adjusted as head or poor (load) varies.

55. What does the specific speed of a hydraulic turbine depend upon?

(a) Speed and power developed
(b) Discharge and power developed
(c) Speed and head of water
(d) Speed power developed and head of water

Nagaland PSC CTSE 2015 Paper-II
ESE 2018, 2012, VIZAG STEEL MT 2011

Ans. (d) : Specific speed of turbine $= \frac{N\sqrt{P}}{H^{5/4}}$

Hence, specific speed of turbine depends on– speed, power develop and head of the water.

56. Pelton turbine has ____ flow direction

(a) Tangential (b) Radial
(c) Axial (d) Universal

UPRVUNL AE 05.07.2021
Oil India Senior Officer 23.12.2020
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : Pelton turbine has tangential flow direction.

57. A model of a hydraulic turbine is tested at a head of 1/4th of that under which the full scale turbine works. The diameter of the model is half of that of the full scale turbine. If N is the RPM of the full scale turbine, then RPM of the model will be

(a) N/4 (b) N/2
(c) N (d) 2N

TSPSC AEE 2017
ISRO Scientist/Engineer 12.05.2013
GATE 2007

Ans. (c) : $H_m = \frac{H_p}{4}$

$$D_m = \frac{D_p}{2}$$

$$\frac{\sqrt{H}}{DN} = \text{Constant}$$

Let prototype of diameter D_p and N_p

So, $$\frac{\sqrt{H_m}}{D_m N_m} = \frac{\sqrt{H_p}}{D_p N_p}$$

$$N_m = N_p \left(\frac{D_p}{D_m}\right)\sqrt{\frac{H_m}{H_p}}$$

$$N_m = N \times \left(\frac{D_p}{\frac{D_p}{2}}\right)\sqrt{\frac{\frac{H_p}{4}}{H_p}}$$

$$= N \times 2\sqrt{\frac{1}{4}}$$

$$= N \times 2 \times \frac{1}{2} = N$$

58. Euler equation of water turbine is derived on the basis of

(a) conservation of mass
(b) rate of change of linear momentum
(c) rate of change of angular momentum
(d) rate of change of velocity

APPSC AEE SCREENING 17.02.2019
APPSC AEE MAINS 2016 PAPER-III
ESE 1995

Ans. (c) : Euler's equation for turbine is derived from Newton's second law related to angular momentum.

i.e. $$T = \frac{d}{dt}(\vec{r} \times m\vec{v})$$

or $$T = \rho Q\left[V_{w1}r_1 \pm V_{w2}r_2\right]$$

59. A pelton wheel is

(a) axial flow impulse turbine
(b) radial flow impulse turbine
(c) inward flow impulse turbine
(d) outward flow impulse turbine

JPSC AE, 10.04.2021, Paper I
GPSC Lect. 23.10.2016

Ans. (*) : The pelton wheel is an impulse turbine with tangential flow.
Note-No option is given by the commission.

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60. Match List I with List II and select the correct answer

	List I		List II
A.	Pelton wheel (single jet)	1.	Medium discharge, low head
B.	Francis Turbine	2.	High discharge, low head
C.	Kaplan Turbine	3.	Medium discharge, medium head
		4.	Low discharge, high head

	A	B	C
(a)	4	3	2
(b)	1	3	4
(c)	4	1	3
(d)	1	2	3

TNPSC. Manager 2015
ISRO Scientist/Engineer 2011

Ans. (a) : Pelton Wheel (single jet)
Single jet pelton turbine with slow speed, low discharge and high head
Francis turbine
Its inward flow reaction turbine
It have medium discharge medium head
Kaplan Turbine
Its axial flow turbine
Its have high discharge, high speed and low head.

61. Run away speed of a Pelton wheel means
(a) Full load speed
(b) No load speed
(c) No load speed with no governor
(d) Full load speed with governor mechanism

Nagaland PSC CTSE 2017 2nd Paper
UKPSC AE 2007 Paper -II

Ans. (c) : The maximum unsafe speed of the runner (without governor control) due to sudden decrease in load on turbine is called runaway speed of the runner of pelton wheel.

62. A Draft tube is used with:
(a) Impulse Turbine
(b) Reaction Turbine
(c) Reciprocating Pump
(d) Centrifugal pump

OPSC AE-II 2019, TNPSC AE 2014

Ans. : (b) : Draft Tube- The draft tube is a pipe of gradually increasing area used for discharging water from the exit of a reaction turbine. It is an integral part of mixed and axial flow turbines. Because of the draft tube, it is possible to have the pressure at runner outlet much below the atmospheric pressure.

63. If the number of jets in a Pelton wheel installation are n, its specific speed is proportional to
(a) n (b) $n^{1/2}$
(c) $n^{3/4}$ (d) $n^{5/4}$

UKPSC AE 2007 Paper -II, ESE 2008

Ans. (b) : If the number of jets in a Pelton wheel installation are n, its specific speed is proportional to $n^{1/2}$

64. Francis and Kaplan turbines fall under the category of
(a) Impulse turbines
(b) Reaction turbines
(c) Axial flow turbines
(d) Mixed flow turbines

CGPSC AE 16.10.2016, TNPSC AE 2014

Ans. (b) : Francis and Kaplan turbine are reaction turbines whereas Pelton wheel turbine is impulse turbine.

65. High head hydro-power plant uses
(a) Kaplan turbine
(b) Francis turbine
(c) Pelton turbine
(d) The type of turbine used does not depend on head of hydro-power plant

Jharkhand JPSC AE Pre 2019
RPSC Lecturer 16.01.2016

Ans. (c) :

Head	Turbine
(1) High and low discharge	Pelton turbine
(2) Medium head medium discharge	Francis turbine
(3) Low head and maximum discharge	Kaplan turbine

66. Turbine is a machine which converts:
(a) mechanical energy to hydraulic energy
(b) Hydraulic energy to mechanical energy
(c) Electrical energy to mechanical energy
(d) Mechanical energy to electrical energy

(HPPSC LECT. 2016)
UKPSC AE-2013, Paper-II

Ans. : (b) Turbines:- Turbines are defined as the hydraulic machines which convert hydraulic energy into mechanical energy. This mechanical energy is used in running on electric generator which is directly coupled to the shaft of the turbine.

67. Consider the following statements with respect to Kaplan Turbine:
(1) It is a reaction turbine
(2) It is an impulse turbine
(3) It has adjustable blades
(a) 1, 2 and 3 (b) 2 and 3 only
(c) 1 and 2 only (d) 1 and 3 only

MPPSC AE 2016
Vizag Steel MT (Re-Exam) 24.11.2013

Ans. : (d) Kaplan Turbine
i. Reaction Turbine ii. Adjustable Blade
iii. Low head iv. High discharge
v. Axial flow turbine

68. The head loss due to friction for the flow of water through penstocks can be minimized by:
(a) Decreasing the diameter of penstock
(b) Increasing the diameter of penstock
(c) Increasing the length of penstock
(d) Increasing the velocity of low

OPSC AEE 2019 Paper - II
Kerala PSC Poly. Lect. 09.01.2014

Ans. : (b) : The head loss due to friction for the flow of water through penstocks can be minimized by increasing the diameter of penstock.

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69. The ratio of power available at the shaft of the turbine to power delivered by water to the runner is known as

(a) Hydraulic efficiency
(b) Mechanical efficiency
(c) Volumetric efficiency
(d) Overall efficiency

Oil India Senior Officer 23.12.2020
RPSC LECTURER 16.01.2016

Ans. (b) : Hydraulic efficiency—Hydraulic efficiency is defined as the ratio between power given by water to runner of turbine and to the power supplied by water at inlet of the turbine.

$$\eta_H = \frac{\text{Power given by water to runner of turbine}}{\text{Power supplied by water at inlet}}$$

Mechanical efficiency—The ratio of power available at the shaft of the turbine to power delivered by water to the runner is known as mechanical efficiency.

$$\eta_M = \frac{\text{Runner Power}}{\text{Power delivered by water to the runner inlet}}$$

Volumetric efficiency—Volumetric efficiency is the ratio between volume of water actually striking the runner of turbine and volume of water supplied to the turbine.

Overall efficiency—It is defined as the ratio of runner power to power of water at inlet of turbine

$$\eta_O = \frac{\text{Runner Power}}{\text{Water Power}}$$

$$\boxed{\eta_O = \eta_H \times \eta_M}$$

70. For high head and low discharge, the hydraulic machinery selected is

(a) Francis turbine (b) Kaplan turbine
(c) Pelton turbine (d) none of these

Kerala PSC IOF 19.04.2016
Mizoram PSC AE/SDO 2014, Paper-I

Ans. (c) : Based on heads and discharge, turbine are classified into Pelton, Francis or Kaplan. Pelton has high head and low discharge.

71. A high specific Francis is turbine is

(a) axial flow turbine
(b) tangential flow turbine
(c) mixed flow turbine
(d) radial flow turbine

Mizoram PSC AE/SDO 2014, Paper-I
UKPSC AE 2012 Paper–II

Ans. (c) : The Francis turbine is a mixed flow turbine in which the water passes through the curve guide vanes and create curved guide vanes and create a high curved rotational flow at the outlet.

72. Specific speed of Kaplan turbine ranges between

(a) 600 to 900 (b) 300 to 600
(c) 100 to 300 (d) 30 to 60

GPSC Executive Engineer 23.12.2018
GATE 1993

Ans. (a)

S.No.	Specific Speed	Type of Turbine
1.	10 to 35	Pelton wheel with single jet
2.	35 to 60	Pelton wheel with multi jets
3.	60 to 300	Francis Turbine
4.	300 to 600	Propeller Turbine
5.	600 to 1000	Kaplan Turbine

73. Kaplan turbine is

(a) A high head mixed flow turbine
(b) a low head axial flow turbine
(c) an impulse inward flow turbine
(d) an outward flow reaction turbine

GPSC Executive Engineer 23.12.2018
GATE 1997

Ans. (b) : Kaplan and propeller turbine:-

- Low head, high discharge turbine.
- They are impulse reaction turbine.
- Axial flow turbine
- Minimum pressure is less than that of atmospheric pressure at runner exit or draft tube inlet.
- Chance of cavitation takes place.

74. Two hydraulic turbines having identical specific speed and effective head at the inlet. If the speed ratio (N1/N2) of the two turbines is 2, then the respective power ratio (P1/P2) is

(a) 1/2 (b) 1/4
(c) 1/8 (d) 1/16

GPSC Executive Engineer 23.12.2018
GATE 2016

Ans. (b) : Given: Turbines having equal specific speed

$$\left(N_{s_1}\right) = N_{s_2}$$

Turbine having equal head

$$(H_1) = H_2$$

Speed ratio, $\left(\frac{N_1}{N_2}\right) = 2$

$\therefore$ (Specific speed)$_1$ = (Specific Speed)$_2$

$$\frac{N_1\sqrt{P_1}}{H_1^{5/4}} = \frac{N_2\sqrt{P_2}}{H_2^{5/4}}$$

$$N_1\sqrt{P_1} = N_2\sqrt{P_2}\left(\because H_1 = H_2\right)$$

$$\frac{P_1}{P_2} = \left(\frac{N_2}{N_1}\right) - \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{4}$$

Hence, the respective power ratio

$\left(\frac{P_1}{P_2}\right)$ is $\frac{1}{4}$

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75. The ratio of the pitch diameter of Pelton wheel to the diameter of the jet is called :

(a) Jet ratio (b) Speed ratio
(c) Wheel ratio (d) None of the above

OPSC AEE 2019 Paper - II
Kerala PSC Poly. Lect. 09.01.2014

Ans. (a) : Jet ratio $= \frac{D}{d}$
D = Pitch diameter
d = Diameter of jet

76. Guide angle as per the aerofoil theory of Kaplan turbine blade design is defined as the angle

(a) lift and resultant force
(b) drag and resultant force
(c) lift and tangential force
(d) lift and drag

ESE 2018
Kerala PSC AE (Irrigation) 05.04.2017

Ans. (a) : Guide angle as per the aerofoil theory of Kaplan turbine blade design is defined as the angle between lift and resultant force.

77. The ratio of the actual work available at the turbine to energy imparted to the wheel is called:

(a) hydraulic efficiency
(b) mechanical efficiency
(c) overall efficiency
(d) turbine efficiency

Nagaland PSC (CTSE) 2018, Paper-II
Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : Mechanical efficiency–The ratio of actual work available at the turbine to energy imparted to the wheel is called mechanical efficiency.
Hydraulic efficiency–Hydraulic efficiency is the ratio between power given by water to runner of turbine and to power supplied by water at inlet of turbine.
Overall efficiency–Ratio of power available at shaft to power supplied at inlet of turbine which is also product of mechanical and hydraulic efficiency.
Turbine efficiency–It is the ratio of actual work output of the turbine to the net energy input supplied in the form of fuel.

78. In a reaction turbine, draft tube is used because

(a) the pressure at exit of the runner is more than atmospheric pressure
(b) The pressure at exit of the runner is equal to atmospheric pressure
(c) the pressure at exit of the runner is less than atmospheric pressure
(d) the pressure at inlet of the runner is less than atmospheric pressure

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC AE (Industries) 09.06.2013

Ans. (c) : The pressure at exit of the runner is less than atmospheric pressure

79. For pelton wheel turbine having a jet ratio at 10, number of buckets to be used are :

(a) 16 (b) 22
(c) 20 (d) 17

KPSC ADF 2015
TNPSC AE (Industries) 09.06.2013

Ans. (c) : No. of buckets $= \frac{D}{2d} + 15$

$= \frac{10}{2} + 15 \quad \left(\because \frac{D}{d} = 10\right)$

$= 20$

80. Match List I with List II and select the correct answer using the codes given below the Lists:

List-I		List-II	
A	**Kaplan turbine**	**1**	**works at atmospheric pressure**
B	**Pelton wheel**	**2**	**high-part load efficiency**
C	**Axial flow pumps**	**3**	**pressure head recovery**
D	**Draft tube**	**4**	**high value of N_S**

Code:-

	A	B	C	D
(a)	2	1	4	3
(b)	1	2	4	3
(c)	2	1	3	4
(d)	1	2	3	4

APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (a) : Kaplan turbine – High part load efficiency
Pelton wheel – Work at atmospheric pressure
Axial flow pump – High value of specific speed (N_s)
Draft tube – Pressure head recovery.

81. A double overhung Pelton wheel has

(a) Two jets (b) Two runners
(c) Four jets (d) Four runners

JPSC AE, 10.04.2021, Paper-I
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : A double overhung pelton wheel has two runner. In general, Number of jets are limited to two in case of vertical runner and six in case of horizontal runner.

82. In a Kaplan turbine runner, the number of blades is generally between

(a) 2 to 4 (b) 4 to 8
(c) 8 to 16 (d) 16 to 24

Nagaland PSC (CTSE) 2018, Paper-II
APPSC AEE 2012

Ans. (b) : Kaplan turbine is a axial flow reaction turbine.
Runner of Kaplan turbine has four to eight blades.
Runner blades of propeller turbines are fixed but of Kaplan turbine can be turned about axis.

83. Which of the following turbine is preferred for 0 to 25 m head of water?

(a) Pelton wheel (b) Kaplan turbine
(c) Francis turbine (d) None of these

Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)

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Ans. (b)

Turbine	Head	Discharge
Pelton turbine	Above 250 m	Low
Francis turbine	60 m – 250 m	medium
Kaplan and propeller turbine	Below 60 m	High

So, Kaplan turbine is preferred for 0 to 25 m head of water.

84. Cavitation in hydraulic turbine result in

(a) Noise and vibration
(b) Reduction of discharge
(c) Drop in output and efficiency
(d) Rough surfaces

Nagaland PSC CTSE 2015 Paper-II
APPSC AEE 2012

Ans. (d) : Cavitation in hydraulic turbine result in rough surfaces.

When the absolute local pressure at any point in a conduct carrying a liquid approaches the vapour pressure of liquid, the dissolved gases & liquid vapour emerge out of the liquid as bubbles. The phenomenon of formation, travel and collapse of vapour is known as cavitation.

85. An impulse turbine produces 50 kW of power when the blade mean speed is 400 m/s. What is the rate of change of momentum tangential to the rotor?

(a) 200 N (b) 175 N
(c) 150 N (d) 125 N

RPSC IOF 2020, IES 1997

Ans. (d) : Given, Power = 50 kW

$V_b = 400$ m/sec

$$\text{Power} = \frac{m\left(V_{w_1} - V_{w_2}\right)V_b}{1000}$$

$$\frac{50 \times 1000}{400} = m\left(V_{w_1} - V_{w_2}\right)$$

Where $m\left(V_{w_1} - V_{w_2}\right)$ = Rate of change of momentum

$$m\left(V_{w_1} - V_{w_2}\right) = 125 \text{ N}$$

86. Cavitation damage in the turbine runner occurs near the

(a) inlet on the concave side of the blades
(b) inlet on the convex side of the blades
(c) outlet on the convex side of the blades
(d) outlet on the concave side of the blades

Harayana PSC Civil Service Pre 2014
TNPSC AE, 2008

Ans. (c) : The cross-section of blades of Francis turbine is of thin airfoils. It is so shaped that when water flows over it a low pressure will be induced on convex side and high pressure on concave side. Cavitation is likely to occur on low pressure side so the chances of cavitation is near the outlet on the convex side of blades.

87. For a Pelton wheel with a given water jet velocity, the maximum output power from the Pelton wheel is obtained when the ratio of the bucket speed to the water jet speed is _____ (correct to two decimal places)

(a) 0.75 (b) 0.5
(c) 0.8 (d) 0.95

Assam Engg. College AP/Lect. 18.01.2021
Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : Power developed in case of pelton wheel

$$P = \rho a v (v - u)(1 + \cos\phi).u$$

For max power,

$$\frac{dP}{du} = 0$$

$$\frac{d}{du}\left[v^2u - vu^2\right] = 0$$

$$v^2 - 2vu = 0$$

$$v(v - 2u) = 0$$

$$v \neq 0$$

$$v = 2u \Rightarrow \boxed{u = \frac{v}{2}}$$

88. A hydraulic reaction turbine working under a head of 16 m develops 640 kW of power. What is the unit power of the turbine?

(a) 10 kW (b) 40 kW
(c) 60 kW (d) 160 kW

RPSC ACF & FRO, 26.02.2021
ESE 2009

Ans. (a) : Unit power $P_u = \frac{P}{(H)^{3/2}}$

$$= \frac{640}{(16)^{3/2}}$$

$$\boxed{P_u = 10\text{kW}}$$

89. Two Pelton wheels A and B have the same specific speed and are working under the same head. Wheel A produces 400 kW at 1000 rpm. If B produces 100 kW, then its speed is

(a) 4000 rpm (b) 2000 rpm
(c) 1500 rpm (d) 1250 rpm

TRB Poly. Lect., 2012, ESE 2003

Ans. (b) Specific speed of pelton wheel , $N_s = \frac{N\sqrt{P}}{H^{5/4}}$

→ Two pelton wheels A and B have the same Specific speed.

$$(N_S)_A = (N_S)_B$$

$$\frac{N_A\sqrt{P_A}}{(H_A)^{5/4}} = \frac{N_B\sqrt{P_B}}{(H_B)^{5/4}}$$

$$\frac{N_A}{N_B} = \frac{\sqrt{P_B}}{\sqrt{P_A}}$$

Given- $N_A = 1000$ rpm
$P_A = 400$ kW
$P_B = 100$ kW

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$N_B = ?$

$$\frac{1000}{N_B} = \sqrt{\frac{100}{400}}$$

$N_B = 2000$ rpm

90. Consider the following energies connected with a Pelton turbine:

I. Mechanical energy

II. Kinetic energy

III. Potential energy

The correct sequence of energy conversion from the entry of the fluid is

(a) I, II, III (b) II, III, I

(c) III, II, I (d) I, III, II

TRB Poly. Lect., 2012, ESE 2003

Ans. (c) The sequence of energy conversion from the entry of the fluid in (Pelton turbine) is,

i) Potential energy

ii) Kinetic energy

iii) Mechanical energy

91. Water hammer in pipe line takes place when:

(a) Fluid is flowing with high velocity

(b) Fluid is flowing with high pressure

(c) Flowing fluid is suddenly brought to rest by closing a valve

(d) Flowing fluid is brought to rest by gradually closing a valve

RPSC VPITI 14.02.2016

RPSC Lect. (Tech. Edu. Dept. 2011)

Ans. (c) : Water hammer in pipe due to flowing fluid is suddenly brought to rest by closing a valve

If the valve is suddenly closed, the momentum of the flowing water will be destroyed and consequently. A wave of high pressure will be setup. This wave of high pressure will be transmitted along the pipe with a velocity equal to the velocity of sound wave and may create noise called knocking. This wave of high pressure has the effect of hammering action an pipe wall.

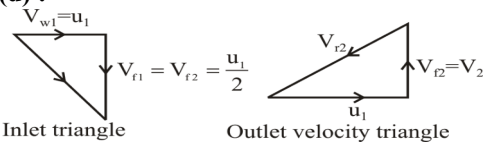
92. The inlet angle of runner blades of a Francis turbine is 90°. The blades are so shaped that the tangential component of velocity at blade outlet is zero. The flow velocity remains constant throughout the blade passage and is equal to half of the blade velocity at runner inlet. The blade efficiency of the runner is

(a) 25% (b) 50%

(c) 80% (d) 89%

TSPSC AEE 2017, GATE 2007

Ans. (d) :



Blade angle at inlet $(\beta_1) = 90°$

Tangential component of velocity at outlet $(V_{w_2}) = 0$

Given, flow velocity, $V_{f_1} = V_{f_2} = \frac{u_1}{2}$; $V_{f_2} = V_2$

Bade efficiency of runner,

$$\eta_H = \frac{V_{w_1} u_1}{gH} = \frac{u_1^2}{gH} \qquad (\because V_{w_1} = u_1)$$

By balancing energy equation,

$$\rho gQH = \rho Q V_{w_1} u_1 + \frac{\rho Q V_2^2}{2}$$

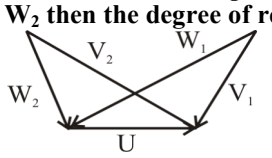
$$gH = u_1^2 + \frac{V_{f_2}^2}{2} = u_1^2 + \frac{\left(\frac{u_1}{2}\right)^2}{2}$$

$$gH = u_1^2 + \frac{u_1^2}{8} = \frac{9}{8}u_1^2$$

$$\eta_H = \frac{u_1^2}{\frac{9}{8}u_1^2} = \frac{8}{9} = 0.88$$

$\boxed{\eta_H = 89\%}$

93. The velocity triangles at the inlet and exit of the rotor of a turbo machine are shown below. V denotes the absolute velocity of the fluid. W denotes the relative velocity of the fluid and U denotes the blade velocity. Subscripts 1 & 2 refer to inlet and outlet respectively. If $V_2 = W_1$, $V_1 = W_2$ then the degree of reaction is

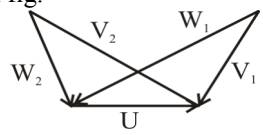


(a) 0 (b) 1

(c) 0.5 (d) 0.25

TSPSC AEE 2017, GATE 2012

Ans. (c) : From fig.



$V_1, V_2 \rightarrow$ Absolute velocity

$W_1, W_2 \rightarrow$ Relative velocity

$V_1 = V_2 = V$

$W_2 = V_1$, $W_1 = V_2$, $U_1 = U_2$

Degree of reaction (R)

$$R = 1 - \frac{V_1^2 - V_2^2}{\left(U_1^2 - U_2^2\right) + \left(V_1^2 - V_2^2\right) + \left(W_2^2 - W_1^2\right)}$$

$$= 1 - \frac{V_1^2 - V_2^2}{2\left(V_1^2 - V_2^2\right)} = 1 - \frac{1}{2}$$

$$R = \frac{1}{2} = 0.5$$

94. The degree of reaction of an impulse turbine :

(a) is less than zero

(b) is greater than zero

(c) is equal to zero

(d) increase with steam velocity at the inlet

GPSC ARTO Pre 30.12.2018

WBPSC AE 2007

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Ans. (c) : Degree of Reaction is a ratio of change of pressure energy inside turbine to the change of total energy inside turbine.
For pure impulse turbine, there is no change in static pressure energy and thus degree of reaction for pure impulse turbine is zero.

95. The hydraulic efficiency of an impulse turbine is the:
(a) ratio of the actual power produced by the turbine to the energy actually supplied by the turbine
(b) ratio of actual work available at the turbine to the energy imparted to the wheel
(c) ratio of the work done on the wheel to the energy of the jet
(d) none of the above

Punjab PSC SDE 12.02.2017
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : The hydraulic efficiency of an impulse turbine is the ratio of the work done on the wheel to the energy of the jet.
Volumetric efficiency (η_v) – The volumetric efficiency is the ratio of the quantity of water actually striking the runner and the quantity of water supplied to the turbine. These two quantities differ by the amount of water that slips directly to the tail race without striking the runner.
Mechanical efficiency (η_m) –

$$\eta_m = \frac{\text{Power available at turbine shaft}}{\text{Power developed by runner}} = \frac{S.P.}{R.P.}$$

- **Overall efficiency–** The overall efficiency of the turbine is the ratio of the power available at the turbine shaft to the power supplied by the water at the entrance to the turbine.

$$\eta_0 = \frac{\text{Power available at the turbine shaft}}{\text{Net power supplied at the turbine entrance}}$$

96. Cavitation can not occur in :
(a) Francis turbine (b) Centrifugal pump
(c) Piston pump (d) Pelton wheel

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (d) : Cavitation is the phenomenon of formation of bubbles on the low pressure area and collapsing at the high pressure area which creates shockwaves due to bursting of bubble.
The main reason for cavitation is to take place is the low pressure area. Since the pelton turbine works at the atmospheric pressure i.e. water coming from the penstocks strikes the buckets at atmospheric pressure. So there is no chance of low pressure area to be created that can generate water bubbles.
Since, there is no any bubble formation takes place. So, there is no any chance of cavitation in pelton turbine.

97. Runaway speed of a turbine (hydraulic) is the speed
(a) at which there would be no damage to the turbine runner
(b) at which the turbine runner can be allowed to run freely without load and wicket gates wide open
(c) corresponding to maximum overload permissible
(d) at full load

GPSC EE Pre 28.01.2017, WBPSC AE 2008

Ans. (b) : Runaway speed for different turbine
For Pelton → 1.8–1.9 of normal seed
For Francis → 2–2.2 of normal speed
For Kaplan → 2.5–3 of normal speed

98. The term.....is used to specify what portion of the total energy transfer across a rotor is related to changes of static pressure enthalpy across the rotor.
(a) Degree of freedom (b) Degree of transfer
(c) Degree of conversion (d) Degree of reaction

HPPSC AE (PWD) 10.09.2016
MPPSC AE 2016

Ans. (d) : Degree of reaction or reaction ratio (R) is defined as the ratio of the static pressure drop in the rotor to the static pressure drop in the stage or as the ratio of static enthalpy drop in the rotor to the static enthalpy drop in the stage.
Degree of reaction of a turbine is defined as a ratio of static pressure drop to total energy

$$R = \frac{\frac{V_{r_2}^2 - V_{r_1}^2}{2g} + \frac{u_1^2 - u_2^2}{2g}}{H}$$

99. In pelton wheel, maximum efficiency takes place when
(a) velocity of wheel = 0.46 × velocity of jet
(b) velocity of wheel = 0.64 × velocity of jet
(c) velocity of wheel = 0.50 × velocity of jet
(d) velocity of wheel = 0.70 × velocity of jet

APPSC AEE MAINS 2016 Paper-III
Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Velocity of wheel must be equal to half the jet speed.

$$u = \frac{V}{2}$$

Velocity of wheel (u) = 0.5 × Velocity of jet

100. Which of the following sequence of hydraulic turbines is in the descending order of specific speed
(a) Kaplan, Francis, Pelton
(b) Francis, Kaplan, Pelton
(c) Pelton, Francis, Kaplan
(d) Francis, Pelton, Kaplan

Nagaland PSC CTSE 2015 Paper-II
ESE 1999

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Ans. (a) : $(Ns)_{kaplan} > (Ns)_{francis} > (Ns)_{pelton}$

Turbine	Specific speed	Head
Pelton	10 – 35	250 m above
Francis	60 – 300	60 m – 250m
Kaplan & propeller	300 – 1000	Below 60m

101. In a Francis turbine, the maximum hydraulic efficiency is obtained when :

(a) absolute velocity at exit is radial
(b) relative velocity at exit is radial
(c) relative velocity at inlet is radial
(d) whirl velocity at exit is zero

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : In a Francis turbine, the maximum hydraulic efficiency is obtained when absolute velocity at exit is radial.

102. Pelton wheel is a:

(a) tangential flow impulse turbine and suitable for high head applications
(b) radial flow impulse turbine and suitable for low head applications
(c) radial flow impulse turbine and suitable for high head applications
(d) tangential flow impulse turbine and suitable for low head applications

NLCIL GET 17.11.2020, Shift-II
SJVN ET 2019

Ans. (a) : Pelton wheel is a tangential flow impulse turbine and it has high head low discharge turbine.

- The bucket of pelton wheel is double semi-ellipsoidal in shape. The jet of water impinges at the centre of the bucket and deflects through 160–170°.

103. Consider the following statements :

(1) Pelton wheel is a tangential flow impulse turbine
(2) Francis turbine is an axial flow reaction turbine
(3) Kaplan turbine is a radial flow reaction turbine

Which of the above statements is /are correct?

(a) 1 alone (b) 1 and 3
(c) 2 alone (d) 3 alone

APPSC AE Subordinate Service Civil/Mech. 2016
ESE 2002

Ans. (a) : Pelton wheel is a tangential flow impulse turbine.
Francis turbine is an mixed flow reaction turbine.
Kaplan and propeller turbines are axial flow reaction turbine.

104. The hydraulic efficiency of Francis turbine is calculated using which of the following equation
(Where V_w is the velocity of whirl at inlet : u is the tangential velocity at the inlet. H is head and g is the acceleration due to gravity.)

(a) $V_w u/gH$ (b) $V_w H/gu$
(c) $V_w u/2gH$ (d) $V_w u^2/gH$

APPSC AE Subordinate Service Civil/Mech. 2016
TSPSC AEE 2015

Ans. (a) : Hydraulic efficiency of Francis turbine –

$$\eta_h = \frac{\text{R.P.}}{\text{W.P.}} = \frac{\frac{W}{1000g}\left[V_{w_1}u_1 \pm V_{w_2}u_2\right]}{\frac{W \times H}{1000}}$$

$$\eta_h = \frac{V_{w_1}u_1 \pm V_{w_2}u_2}{gH}$$

If discharge at outlet in radial direction then $V_{w_1} = 0$

$$\boxed{\eta_h = \frac{V_{w_1}u_1}{gH}}$$

105. The value of speed ratio in case of a Francis turbine ranges from

(a) 0.2 to 0.3 (b) 0.4 to 0.5
(c) 0.6 to 0.9 (d) 1.0 to 2.0

RPSC IOF 2020
TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : • The speed ratio $= \frac{u_1}{\sqrt{2gH}}$ varies from 0.6 to 0.9.

• Flow ratio $= \frac{v_{f1}}{\sqrt{2gH}}$ and varies from 0.15 to 0.30.

106. A single-stage impulse turbine with a diameter of 120 cm runs at 3000 rpm. If the blade speed ratio is 0.42, then, the inlet velocity of steam will be

(a) 79 m/s (b) 188 m/s
(c) 450 m/s (d) 900 m/s

ISRO Scientist/Engineer 2009, 2006

Ans. (c) : For single stage impulse turbine diameter
D = 1.20 m
N = 3000 rpm
Blade speed Ratio = 0.42

$$u = \frac{\pi DN}{60}$$

$$u = \frac{\pi \times 1.20 \times 3000}{60}$$

= 188.49 m/sec
Inlet velocity is given

$$V = \frac{u}{0.42} = \frac{188.49}{0.42}$$

= 448.79 m/sec.

107. In hydraulic turbines, if the energy available at inlet is only kinetic energy, then that type of turbine is

(a) Reaction turbine (b) Impulse turbine
(c) Francis turbine (d) Kaplan turbine

ESE 18.07.2021

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Ans. (b) : In Impulse turbines, the energy at the entrance to the runner is in the form of kinetic energy only. The pressure remains atmospheric throughout the runner.

108. Which is/are not the correct statements for hydroelectric power plant?

(1) Cost of electricity generation is affected by load factor.

(2) Useful life is lower than thermal power plant.

(3) Operating cost is low.

(4) Quickly respond to change in load.

(a) (2), (4) (b) (1), (2)

(c) (1), (3) (d) only (4)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : Life of hydroelectric power plant is more as compared to thermal power plant & it does not quickly respond to the change in load.

Hydroelectric Power plants are mainly used in hilly areas.

109. Catchment area of hydroelectric power plant is 100 sq. km and annual average rainfall is 100 cm. The head drop available at power house site is 120 m. How much approximate power can be developed by the plant if evaporation loss is 20% and load factor as unity?

(a) 0.30 MW (b) 3.0 MW

(c) 30 MW (d) 300 MW

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : Discharge in 1 sec $= \dfrac{100\times10^6\times1}{52\times7\times24\times60\times60}$

$= 3.179\ m^3/sec$

Discharge after evaporation loss(Q) $= 3.179 \times 0.8$

$Q = 2.54\ m^3/sec$

Power (P) $= \rho gQH$

$= 1000 \times 9.81 \times 2.54 \times 120$

$= 2.99 \times 10^6$

$\boxed{P = 3MW}$

110. In impulse turbines :

(a) Water remains at atmospheric pressure before and after making contact with the runner blades.

(b) water remains at atmospheric pressure before entering the runner blades but the pressure reduces thereafter.

(c) water enters at low pressure, which increases after making contact with the runner blades.

(d) water pressure remains below atmospheric before and after making contact with the runner blades.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : In impulse turbines water remains at atmospheric pressure before and after making contact with the runner blades.

111. For Pelton wheel turbine correct relation between nozzle efficiency (η_n) & nozzle velocity coefficient (C_v) is :

(a) $\eta_n = C_v^2$ (b) $\eta_n = C_v$

(c) $\eta_n = \dfrac{C_v}{2}$ (d) $\eta_n^2 = C_v$

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Ans. (a) : Nozzle efficiency (η_n)

Nozzle velocity coefficient C_v

$\boxed{\eta_n = C_v^2}$

$$\text{Nozzle efficiency} = \frac{\text{KE of jet per sec}}{\text{Water power at the base nozzle}}$$

112. For a hydraulic turbine working of two different head, the ratio of discharges through the turbine is determined to be 1 : $\sqrt{7}$, for the two head, the ratio of the shaft powers developed by the turbine will be :

(a) $1 : \sqrt{7}$ (b) 1 : 49

(c) 1 : 7 (d) $1 : 7\sqrt{7}$

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Ans. (d) : $\dfrac{Q_1}{Q_2} = \dfrac{1}{\sqrt{7}}$ $\qquad \dfrac{Q}{ND^3} = C_Q\,(\text{flow coefficient})$

$Q \propto N$ $\qquad Q \propto N$

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2}$$

$$\frac{N_1}{N_2} = \left(\frac{Q_1}{Q_2}\right) = \left(\frac{1}{\sqrt{7}}\right)$$

$$\boxed{\frac{N_1}{N_2} = \frac{1}{\sqrt{7}}}$$

$P \propto N^3$ $\qquad \dfrac{P}{N^3D^5} = C_p\,(\text{Power coefficient})$

$$\frac{P_1}{P_2} = \left(\frac{N_1}{N_2}\right)^3$$

$$= \left(\frac{1}{\sqrt{7}}\right)^3$$

$$\boxed{\frac{P_1}{P_2} = \frac{1}{7\sqrt{7}}}$$

113. A hydraulic turbine is working under a head of 20m. If the head available to the turbine is increased by 700%, then the speed of the turbine will:

(a) Increase by 700% (b) Increase by 2162%

(c) Decrease by 64.4% (d) Increase by 182.8%

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : $H_1 = 20m$

$H_2 = 20\ (1 + 7)$

$= 20 \times 8$

$H_2 = 160m$

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$$\frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$$

$$N_2 = N_1 \times \frac{\sqrt{H_2}}{\sqrt{H_1}}$$

$$= N_1 \times \left[\frac{160}{20}\right]^{1/2}$$

$$= N_1 \times [8]^{1/2}$$

$$= N_1 \times 2.828$$

$$N_2 = N_1\,[1 + 1.828]$$

Speed is increased by 182.8%

114. The head available to both the Francis turbines A and B is 80m. The mean atmospheric pressure is 101.043 kPa and the vapour pressure for water is 2.943 kPa. The height of the runner of turbine A above the tail water level is 0.5 m and for turbine B it is 0.6 m. If the critical cavitation factor is 0.1144, then which one of the following statements is only correct?

(a) Cavitation does not occur in the turbine.
(b) Cavitation occurs in both the turbines
(c) Cavitation occurs only in turbine A.
(d) Cavitation occurs only in turbine B.

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Ans. (a) : Given,

Head available for both turbines (H_m) = 80 m

Mean atmospheric pressure, P_{atm} = 101.043 kPa

$$H_{atm} = \frac{101.043 \times 10^3}{10^3 \times 9.81}$$

$$H_{atm} = 10.3 \text{ m}$$

Vapour pressure for water, P_v = 2.943 kPa

$$H_v = \frac{2.943 \times 10^3}{10^3 \times 9.81}$$

$$H_v = 0.3 \text{ m}$$

Thomas cavitation factor (σ)

For turbine A, $\sigma_A = \dfrac{(H_{atm} - H_v) - H_s - h_{fs}}{H_m}$

$$= \frac{10.3 - 0.3 - 0.5}{80} = 0.11875$$

Similarly,

For turbine B, $\sigma_B = \dfrac{(H_{atm} - H_v) - H_s - h_{fs}}{H_m}$

$$= \frac{10.3 - 0.3 - 0.6}{80} = 0.1175$$

Critical cavitation factor for turbines (σ_c) = 0.1144

$\sigma_A > \sigma_C$

$\sigma_B > \sigma_C$

Therefore, cavitation does not occurs in any turbine.

115. For a Pelton wheel turbine theoretical maximum efficiency and the speed factor for maximum efficiency can be obtained from -

(a) $\frac{1}{2}\left(1+\frac{\cos\beta_2}{2}\right); \frac{U}{V_1}$ (b) $\frac{1}{2}(1-\cos\beta_2); \frac{U}{V_{r_1}}$

(c) $\frac{1}{2}(1+\cos\beta_2); \frac{U}{V_1}$ (d) $\frac{1}{2}(1+\cos\beta_2); \frac{U}{V_{r_1}}$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : Hydraulic efficiency of Pelton wheel.

$$\eta_h = \frac{\text{Works done per second}}{\text{KE of jet per second}} = \frac{\rho A V_1 (Vw_1 + Vw_2) \times u}{\frac{1}{2}\rho A V_1 \times V_1^2}$$

$$= \frac{2(V_1 - u)(1 + \cos\phi)u}{V_1^2}$$

For max. efficiency $u = \frac{V_1}{2}$

$$(\eta_n)_{max} = \frac{1+\cos\phi}{2} = \frac{1}{2}[1+\cos\phi]$$

$$\boxed{(\eta_n)_{max} = \frac{1}{2}(1+\cos\beta_2)} \qquad (\text{here } \phi = \beta_2)$$

116. Approximate head gain due to installation of a tapered draft tube after a turbine, if discharge is 9.42 m^3/s, draft tube inlet and outlet diameter respectively is 2m and 4m with efficiency of 90%.

(a) 0.86 m (b) 0.38 m
(c) 0.42 m (d) 0.95 m

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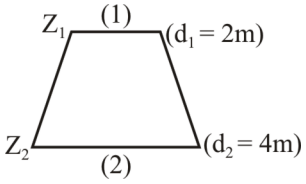
Ans. (b) :



Given,

$Q = 9.42 \text{ m}^3/\text{sec}$

At inlet, $\frac{\pi}{4}(d_1)^2 \times V_1 = 9.42$

$$\frac{\pi}{4}(2)^2 \times V_1 = 9.42$$

$$V_1 = 3 \text{ m/s}$$

At outlet,

$$\frac{\pi}{4}(d_2)^2 \times V_2 = 9.42$$

$$\frac{\pi}{4} \times (4)^2 \times V_2 = 9.42$$

$$V_2 = 0.75 \text{ m/s}$$

Applying Bernoulli's equation at (1) & (2)

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$\frac{P_1 - P_2}{\rho g} + Z_1 - Z_2 = \frac{V_2^2 - V_1^2}{2g}$$

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$$\frac{\Delta P}{\rho g} + \Delta Z = \frac{(3)^2 - (0.75)^2}{2 \times 9.81}$$

$$\Delta Z = 0.430 \text{ m}$$

At efficiency $\eta = 0.9$

Approximate head gain = $0.430 \times 0.9 = 0.38$ m

117. An inward flow reaction turbine running at 250 rpm has $D_1 = 1.0$m. $b_1 = 0.25$ m, $D_2 = 0.50$ m and $b_2 = 0.30$ m. If water enters the wheel radially at 3.5 m/s velocity, the discharge and velocity of flow at outlet (assume vane thickness at extremities as negligible) are :

(a) 2.199m^3/sec, 4.666 m/s
(b) 14.728m^3/sec, 10.148 m/s
(c) 10,000 m^3/sec, 48.100 m/s
(d) 5.211 m^3/sec, 9.121 m/s

CGPSC AE 15.01.2021

Ans. (a) : Discharge = $\pi D_1 b_1 V_{f_1} = \pi D_2 b_2 V_{f_2}$

$Q = \pi \times 1 \times 0.25 \times 3.5$

$Q = 2.748$

$$V_{f_2} = \frac{D_1 b_1}{D_2 b_2} V_{f_1} = \frac{1 \times 0.25}{0.5 \times 0.3} \times 3.5 = 5.834 \text{ m/sec}$$

So nearest answer is option (a).

118. Which one of the following is not a reaction turbine?

(a) Radial flow turbine
(b) Mixed flow turbine
(c) Axial flow turbine
(d) Pelton wheel turbine

CGPSC AE 15.01.2021

Ans. (d) : Types of Reaction Turbine-
(i) Francis Turbine (Inward Radial Flow)
(ii) Modern Francis (Mixed Flow)
(iii) Kaplan Turbine (Axial Flow)
• Pelton wheel turbine is a tangential flow impulse turbine.

119. A turbine develops 9000 kW when running at 100 rpm. The head on the turbine is 25 m. If the head on the turbine is reduced to 16 m. Determine the speed.

(a) 90 rpm (b) 84 rpm
(c) 80 rpm (d) 87 rpm

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (c) : According to unit head–

$$N_u = \frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$$

So, $\frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$

Given, $N_1 = 100$ rpm, $H_1 = 25$, $H_2 = 16$

$$\frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}} \Rightarrow \frac{100}{\sqrt{25}} = \frac{N_2}{\sqrt{16}}$$

$$\Rightarrow \quad \frac{100}{5} = \frac{N_2}{4} \Rightarrow N_2 = \frac{400}{5}$$

$$\boxed{N_2 = 80 \text{ rpm}}$$

120. In a Francis turbine, runner, the number of blades are generally

(a) 2 to 4 (b) 4 to 8
(c) 8 to 16 (d) 16 to 24

JPSC AE, 10.04.2021, Paper-I

Ans. (d) :

Turbine		No. of blades
Francis	→	16 to 24
Kaplan	→	4 to 8

121. The power produced by a reaction turbine is

(a) directly proportional to H
(b) inversely proportional to H
(c) directly proportional to $\sqrt{H}$
(d) inversely proportional to $\sqrt{H}$

JPSC AE, 10.04.2021, Paper-I

Ans. (a) : The power produced by reaction turbine = $\rho Q g H$

Power $\propto$ Head

$\boxed{P \propto H}$

122. In Francis turbine, as the water discharge is radial at the outlet, the velocity whirl at the outlet becomes

(a) 1 (b) 0
(c) ∞ (d) 0.5

ESE 18.07.2021

Ans. (b) : In Francis turbine,
Discharge is radial at the outlet and hence whirl velocity (Tangential component of absolute velocity) at the outlet is zero.

123. A turbine develops 800 kW when running at 1000 rpm. The head on the turbine is 30 m. If the head is reduced to 18 m, what is the speed developed by the turbine?

(a) 67.46 rpm (b) 95.24 rpm
(c) 54.67 rpm (d) 77.46 rpm

ESE 18.07.2021

Ans. (*) : $P_1 = 800$ kW
$N_1 = 1000$ rpm, $H_1 = 30$ m
$H_2 = 18$ m

$$\frac{N_2}{N_1} = \sqrt{\frac{H_2}{H_1}} = 1000 \times \sqrt{\frac{18}{30}}$$

$= 774.60$ rpm

None of the given option is correct

124. The phenomenon of water hammer in pipe flow originates from:

(a) the compressibility of water when subjected to sudden applied high pressure
(b) the microscopic form of all matter
(c) the non-Newtonian behaviour of water
(d) the critical point singularity of the phase diagram

AAI Jr. Executive 26.03.2021

Ans. (a) : The phenomenon of water hammer in pipe flow originates from the compressibility of water when subjected to sudden applied high pressure.

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125. The energy possessed by water at inlet of reaction turbine will be

(a) kinetic energy only
(b) pressure energy only
(c) both kinetic energy and pressure energy
(d) static energy

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : The energy possessed by water at inlet of reaction turbine will be both kinetic energy and pressure energy.

126. In a reaction turbine, speed ratio is 0.6 while head developed is 10 m. The tangential velocity of wheel at inlet will be

(a) $10.28 ms^{-1}$ (b) $8.4 ms^{-1}$
(c) $6.356 ms^{-1}$ (d) $10\ ms^{-1}$

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Speed ratio, $\phi = \frac{V_f}{\sqrt{2gH}}$

$$0.6 = \frac{V_f}{\sqrt{2 \times 9.81 \times 10}}$$

$$V_f = 8.4\ ms^{-1}$$

127. The spillway which has a crest in the form of an ogee or S-shape is

(a) Chute spill way
(b) free over fall spill way
(c) siphon spill way
(d) overflow spill way

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) :

128. The structure provided in the body of a dam or near dam to relieve the reservoir of excess water is called

(a) penstock (b) spillways
(c) surge tank (d) draft tube

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :

129. Kaplan turbine is used

(a) when head of the water is high and volume of water is high
(b) when head of the water is low and volume of water is high
(c) when head of the water is low and volume of water is low
(d) when head of the water is high and volume of water is low

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :

130. Which one of the following statements is true?

(a) Both Pelton and Francis turbines are impulse turbines.
(b) Francis turbine is a reaction turbine but Kaplan turbine is an impulse turbine.
(c) Francis turbine is an axial-flow reaction turbine.
(d) Kaplan turbine is an axial-flow reaction turbine.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d) :
Kaplan turbine – Axial flow reaction turbine
Francis turbine – Inward radial flow reaction turbine
Pelton turbine – Tangential flow impulse turbine

131. Flow a hydraulic turbine operating at two different heads, the ratio of their speeds will be 3 : 1. For the two heads, the ratio of the shaft power developed by the turbine will be :

(a) 9 : 1 (b) 243 : 1
(c) 3 : 1 (d) 27 : 1

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : $\frac{P}{D^5 N^3} = \text{constant}$

$$P \propto N^3$$

$$\frac{P_1}{P_2} = \left(\frac{N_1}{N_2}\right)^3$$

$$\frac{P_1}{P_2} = \left(\frac{3}{1}\right)^3$$

$$\boxed{\frac{P_1}{P_2} = \frac{27}{1}}$$

132. The gross head available to a hydraulic power plant is 100 m. The utilized head in the runner of the hydraulic turbine is 72 m. If the hydraulic efficiency of the turbine is 90%, then the estimated pipe friction head is _____.

(a) 18 m (b) 20 m
(c) 16.6 m (d) 18.4 m

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (b) : Given,
Gross head, $H_G = 100$ m
Utilized head, $H_u = 72$ m
Hydraulic efficiency $(\eta_h) = 90\%$

$$\because \text{Net head} = \frac{\text{Utilized head}}{\eta_h} = \frac{72}{0.90} = 80 \text{ m}$$

$\therefore$ Friction head = Gross head – Net head
= 100 – 80
= 20 m

133. For the two straight conical draft tubes A and B, the head loss due to friction in the tube is equal to 0.4 times the velocity head at outlet. If the inlet to outlet diameter ratio for draft tube A is 0.56 while for draft tube B is 0.71, then which one of the following statements is only correct for the same discharge through the tubes?

(a) Draft tube B is more efficient.
(b) Draft tube A is more efficient.
(c) Both the draft tubes have the same efficiency.
(d) Efficiency of the draft tube cannot be determined.

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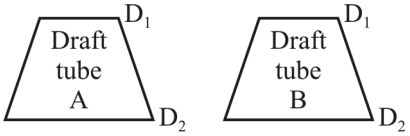
Ans. (b)



Given,

$$\left(\frac{D_1}{D_2}\right)_A = 0.56 \qquad \left(\frac{D_1}{D_2}\right)_B = 0.71$$

Head loss, $(h_f)_A = (h_f)_B = 0.4\frac{V_2^2}{2g}$

$$(\eta_A)_{\text{Draft tube}} = \frac{\frac{V_1^2}{2g} - \frac{V_2^2}{2g} - h_f}{\frac{V_1^2}{2g}} = \frac{\frac{V_1^2}{2g} - \frac{V_2^2}{2g} - 0.4\frac{V_2^2}{2g}}{\frac{V_1^2}{2g}}$$

$$= 1 - 1.4\frac{V_2^2}{V_1^2} \qquad \left\{\therefore \frac{V_2}{V_1} = \frac{d_1^2}{d_2^2}\right\}$$

$$= 1 - 1.4\left(\frac{d_1}{d_2}\right)^4 = 1 - (1.4)(0.56)^4$$

$\eta_A = 86.23\%$

$$(\eta_B)_{\text{Draft tube}} = \frac{\frac{V_1^2}{2g} - \frac{V_2^2}{2g} - h_f}{\frac{V_1^2}{2g}} = \frac{\frac{V_1^2}{2g} - \frac{V_2^2}{2g} - 0.4\frac{V_2^2}{2g}}{\frac{V_1^2}{2g}}$$

$$\Rightarrow \quad 1 - 1.4\frac{V_2^2}{V_1^2} = 1 - 1.4\frac{d_1^4}{d_2^4} = 1 - 1.4(0.76)^4$$

$\eta_B = 53.29\%$

$\therefore \quad (\eta_A)_{\text{Draft tube}} > (\eta_B)_{\text{Draft tube}}$

134. A water turbine delivering 10 MW power is to be tested with the help of a geometrically similar 1:10 model which runs at the same speed and efficiency. The power developed by the model will be

(a) 10 W (b) 100 W
(c) 1000 W (d) 10000 W

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) : Given, $P_p = 10$ MW, $\frac{D_m}{D_p} = \frac{1}{10}$

$N_p = N_m$, $\eta_p = \eta_m$, $P_m = ?$

From model relationship,

$$\left(\frac{P}{N^3D^5}\right)_m = \left(\frac{P}{N^3D^5}\right)_p$$

$$P_m = \left(\frac{D_m}{D_p}\right)^5 \times P_p$$

$$P_m = 10 \times 10^6 \times \frac{1}{10^5}\text{ W}$$

$P_m = 100$ W

135. A fan running at 720 rpm develops static pressure of 12 mm. If the speed of the fan is doubled the static pressure developed will be

(a) 24 mm (b) 12 mm
(c) 48 mm (d) 96 mm

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (c) : Head coefficient $C_H = \frac{H_m}{N^2D^2}$

$$\left(\frac{H_m}{N^2D^2}\right)_1 = \left(\frac{H_m}{N^2D^2}\right)_2$$

$D_1 = D_2$, $N_1 = 720$ rpm, $H_1 = 12$ mm

$N_2 = 2\,N_1 = 1440$ rpm, $H_2 = ?$

$$\frac{12}{720^2} = \frac{H_{m_2}}{1440^2}$$

$$H_{m_2} = 48\text{cm}$$

136. A turbine is to operate under a head of 25 m at 200 rpm. The discharge is 9 m^3/s. If the overall efficiency is 90% then the power generated by the turbine will be

(a) 1968.5 kW (b) 1986.5 kW
(c) 1998.5 kW (d) 1960 kW

RPSC IOF, 2020

Ans. (b) : Head $H = 25$ m

$N = 200$ rpm

$Q = 9\ m^3/\text{sec}$

$\eta_o = 90\% = 0.90$

Power generated by turbine-

$$\eta_o = \frac{P}{\rho g Q H}$$

$$0.9 = \frac{P}{10^3 \times 9.81 \times 9 \times 25 \times 10^3}$$

$P = 1986.5$ kW

137. Mechanical efficiency of a turbine is the ratio of

(a) Power at the inlet to the power at the shaft of turbine
(b) Power at the shaft to the power given to the runner
(c) Power at the shaft to the power at the inlet of turbine
(d) None of these

RPSC IOF, 2020

Ans. (b) : Mechanical efficiency

$$= \frac{\text{Power available at the turbine shaft}}{\text{Power developed by the runner}}$$

Hydraulic efficiency η_h

$$= \frac{\text{Power developed by the runner}}{\text{Net power supplied at turbine entrance}}$$

Volumetric efficiency (η_v)

$$= \frac{\text{Quantity of water actually striking the runner}}{\text{Quantity of water supplied to the turbine}}$$

138. Which of the following is most suitable for Kaplan turbine?

(a) It works under the head between 500–100 m
(b) It works under the head between 1000–2000 m
(c) It works under the head between 5–70 m
(d) It works under the head between 200–500 m

APPSC Poly Lect. 13.03.2020

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Ans. (c) : Kaplan turbine works under low head that means it is suitable for working under the head between 5–70 m. It is axial flow reaction turbine.

139. If the flow deflection angle is 120 degree, the hydraulic efficiency of impulse hydraulic turbine is :

(a) 0.10 (b) 0.66
(c) 0.45 (d) 0.25

NLC GET 17.11.2020, Shift-II

Ans. (d) : The hydraulic efficiency of an impulse turbine is the ratio of the work done on the wheel to the energy of the jet.

The hydraulic efficiency of an impulse turbine is maximum when the velocity of wheel is one half the velocity of jet of water at inlet.

$$\text{Hydraulic efficiency } (\eta_H) = \frac{1+\cos\phi}{2}$$

$$= \frac{1+\cos 120}{2} = \frac{1-1/2}{2} \qquad (\phi = 120°)$$

$$= \frac{1}{4} \Rightarrow \eta_{\text{Hydraulic}} = 0.25 \text{ or } 25\%$$

140. A modern Francis turbine is a :

(a) inward flow reaction turbine
(b) mixed flow reaction turbine
(c) outward flow reaction turbine
(d) axial flow reaction turbine

NLC GET 17.11.2020, Shift-II

Ans. (b) : Modern Francis Turbine- It is a mixed flow reaction turbine. An inward Flow, reaction-type water turbine in which the runner vanes are so shaped that they are acted on by the water pressure both axially and radially.

141. The order in which specific speed for best efficiency increases in hydraulic turbines is :

(a) Pelton, Francis, Kaplan turbines
(b) Francis, Pelton Kaplan turbines
(c) Kaplan, Francis, Pelton turbines
(d) Pelton, Kaplan, Francis turbines

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : Specific speed of turbine

$$N_S = \frac{N\sqrt{P}}{H^{5/4}}$$

Pelton wheel works on high head and low discharge Kaplan turbine work on low head and high discharge.
Increasing order of specific speed,
Pelton < Francis < Kaplan turbine.

142. In a Kaplan turbine, the velocity of flow through the rotor:

(a) decreases (b) increases and then decreases
(c) increases (d) remains the same

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : The Kaplan turbine is also called as propeller turbine. Consequently, the specific speed of these turbines are high. Kaplan turbine is an axial flow reaction turbine which means that the flow direction does not change as it crosses the rotor. So, the velocity of flow through rotor remains the same.

143. _____ turbine is an inward flow reaction turbine with the radial discharge at outlet.

(a) Curtis (b) Francis
(c) Pelton (d) Kaplan

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Pelton turbine is a impulse turbine, Kaplan turbine is a axial flow turbine and Francis is inward flow reaction turbine with the radial discharge at outlet. Discharge of radial flow reaction turbine is product of perimeter of runner, thickness whirl velocity at inlet as well as outlet runner vanes.

144. In an impulse turbine, if the blade losses are neglected, then for maximum efficiency, the inlet absolute fluid flow velocity is:

(a) half of the blade velocity
(b) twice as that of the blade velocity
(c) equal to that of the blade velocity
(d) thrice as that of the blade velocity

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Hydraulic efficiency is defined as ratio between power given by water to runner of turbine and to the power supplied by water at inlet of turbine. This efficiency will be maximum when differentiating with u and it is obtained as v = 2u
Where, u = Blade velocity
v = Fluid flow velocity

145. Specific speed of Pelton Wheel ranges from

(a) 5 to 10 (b) 10 to 50
(c) 60 to 300 (d) 350 to 950

TSPSC Manager (Engg.) HMWSSB 12.11.2020

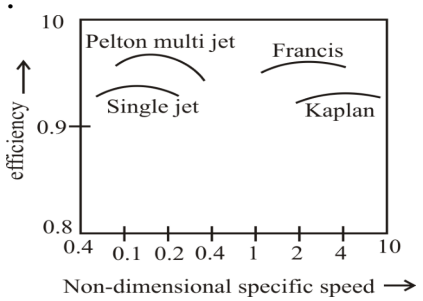
Ans. (b) : Turbine	**Specific speed**
• Pelton turbine	10 to 60
• Francis turbine	60 to 300
• Kaplan and propeller turbine	300 to 1000

146. For a non-dimensional specific speed value of 1, for maximum efficiency, which of the following turbines is preferred?

(a) Pelton wheel (b) Francis turbine
(c) Kaplan turbine (d) Tyson wheel

ISRO Scientist/Engineer 12.01.2020

Ans. (b) :



So for a non-dimensional specific speed value of 1, for maximum efficiency, Francis turbine is preferred.
Non-dimensional specific speed is given by –

$$N_S = \frac{N\sqrt{P}}{\sqrt{\rho}(gH)^{5/4}}$$

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147. When blade speed ratio is zero, no work is done because the distance travelled by the blade is zero even if the torque on the blade

(a) is minimum (b) is zero
(c) is maximum (d) remains the same

ESE 2020

Ans. (c) : When blade speed ratio is zero, no work is done because the distance travelled by the blade is zero even if the torque on the blade is maximum

148. In an axial flow turbine, the utilization factor has an absolute maximum value of unity, for any degree of reaction if the value of nozzle angle α is

(a) 270° (b) 180°
(c) 90° (d) 0°

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Ans. (d) : Utilization factor

$$= \frac{2\cos^2\alpha}{1+\cos^2\alpha} = \frac{2\times1}{1+1} = 1 \quad [\because \alpha = 0]$$

149. An inward flow reaction turbine has an external diameter of 1 m and its breadth at inlet is 250 mm. If the velocity of flow at inlet is 2 m/s and 10% of the area of flow is blocked by blade thickness, the weight of water passing through the turbine will be nearly

(a) 10 kN/s (b) 14 kN/s
(c) 18 kN/s (d) 22 kN/s

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Ans. (b) : Given, $d_1 = 1$ m
$b_1 = 0.25$ m
$V_{f1} = 2$ m/s
$k = 0.9$ = coefficient of blade thickness
$Q = k\pi d_1 b_1 V_{f1}$
$Q = 0.9 \times \pi \times 1 \times 0.25 \times 2 = 1.413\ m^3/sec$
Weight of water passing through turbine = ρgQ
$= 1000 \times 9.81 \times 1.413$
$= 14$ kN/s

150. While designing a Pelton wheel, the velocity of wheel 'u' is

(a) $K_u\sqrt{gH}$ (b) $2K_u\sqrt{gH}$
(c) $K_u\sqrt{2gH}$ (d) $2K_u\sqrt{2gH}$

where, K_u = speed ratio; H = net head on turbine; g = gravity

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Ans. (c) : Speed ratio, $K_u = \frac{u}{\sqrt{2gH}}$

151. The turbines of the same shape will have the same

(a) Thomas number (b) Reynolds number
(c) Specific speed (d) Rotational speed

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Ans. (c) : The specific speed remains same for all the similar turbine.

152. In reaction turbine, water at the inlet of turbine possesses.

(a) Kinetic energy
(b) Kinetic and potential energy
(c) Potential energy
(d) None of these

Oil India Senior Officer 23.12.2020

Ans. (b) :

153. Draft tube is pipe of,

(a) Decreasing area
(b) Increasing area
(c) Uniform circular area
(d) Connecting inlet of the runner

Oil India Senior Officer 23.12.2020

Ans. (b) :

154. A large power generation unit uses a hydraulic turbine to generate 300 kW at 1000 rpm under a head of 40 m. For initial testing, a 1 : 4 scale model of the turbine operates under a heat of 10 m. The power generated by the model (in kW) will be:

(a) 234 (b) 2.34
(c) 23.4 (d) 0.234

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Ans. (b) : Power (P_2) = 300 kW
N = 1000 rpm
Head (H)= 40 m
Initial scale model scale = 1 : 4
Head (H) = 10 m
power (P_2) = ?

$$\text{Scale ratio } (L_r) = \frac{D_p}{D_m}\left(\frac{4}{1}\right)$$

For similar turbines $\frac{P}{H^{3/2}D^2}$ equal.

$$\left(\frac{P}{H^{3/2}D^2}\right)_p = \left(\frac{P}{H^{3/2}D^2}\right)_m$$

$$\left(\frac{D_p}{D_m}\right) = (L_r) = \sqrt{\frac{P_p}{P_m}\times\left(\frac{H_m}{H_p}\right)^{3/2}}$$

Squaring both side

$$\left(\frac{D_p}{D_m}\right)^2 = \frac{P_p}{P_m}\times\left(\frac{H_m}{HP}\right)^{3/2}$$

$$\left(\frac{4}{1}\right)^2 = \frac{300}{P_m}\times\left(\frac{10}{40}\right)^{3/2}$$

$$= \frac{300}{P_m}\times\frac{1}{8}$$

$$\frac{16}{1} = \frac{300}{P_m\times8}$$

$$P_m = \frac{300}{16\times8}$$

$$\boxed{P_m = 2.34kW}$$

155. Cavitations gives damage to turbine on:

(a) Outlet on the convex side
(b) Exit side of casing
(c) Inlet on concave side of blades
(d) Inlet on the convex side of blades

HPPSC ADF 05.03.2019

Ans. (a, b) : The cross-section of blades of Francis Turbine is of thin airfoils. It is so shaped that when water flows over it a low pressure will be induced on convex side and high pressure on the concave side. Cavitation is likely to occur on low pressure side so the chances of cavitation is near the outlet on the convex side blades.

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156. A hydro-electric site has a head of 100 m and an average discharge of $10m^3/s$. Assume 92% efficiency, for a greater speed of 6000 rpm, the specific speed of turbine would be:

(a) 1500 (b) 1000
(c) 1900 (d) 1802.4

HPPSC ADF 05.03.2019

Ans. (d) : Given-

$H = 100$ m

$Q = 10^3$ m/sec.

$N = 6000$ rpm, $\eta = 92\%$

Power (p) = pg QH × η

$= 1000 \times 9.81 \times 10 \times 100 \times 0.92$

$P = 9025200$

$= 9025.2$ kW

Specific speed of turbine

$$(N_s) = \frac{N\sqrt{P}}{H^{5/4}} = \frac{6000\sqrt{9025200}}{(100)^{5/4}}$$

$$= \frac{6000 \times 95.0010}{(100)^{1.25}}$$

$= 1802.517$ rpm

157. Which of the following is the highest specific speed turbine?

(a) Pelton wheel with single jet
(b) Francis turbine
(c) Pelton wheel with double jet
(d) Kaplan turbine

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Ans. (d) :

	Type of turbine	Specific speed (Ns)	Head (H)
1	Single jet pelton turbine	10-30	High Head (300-2000 m)
2.	Double jet pelton turbine	30-60	High Head (300-1500 m)
3.	Francis turbine	60-300	Medium Head (30-500 m)
4.	Kaplan turbine	300-1000 (Adjustable runner vanes)	Low Head (3-70 m)
5.	Propeller turbine	300-1000 (Fixed runner vanes)	Low Head (3-70 m)

158. Regulation of flow in Pelton Turbine is done by changing:

(a) The vane angle
(b) The velocity of flow from nozzle
(c) The head available at the nozzle
(d) The annular area of the nozzle

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

Ans. (d) : In order to control the quantity of water striking the runner, the nozzle fitted at the end of the penstock is provided with a spear and needle having streamlined head which is fixed to the spear rod support. Spear regulates the flow in pelton turbine by changing the annular area of the nozzle.

159. Cavitations happens in turbo machines primarily because :

(a) the temperature increases above a critical value
(b) the velocity attains very high value
(c) the pressure falls below the vapour pressure
(d) the thomas cavitation parameter exceeds a certain limit

BHEL ET 2019

Ans. (c) Cavitation happens in turbomachines primarily because the pressure falls below the vapour pressure.

160. A Pelton wheel operates under a head of 900 m. The speed ratio of the wheel is 0.45. The peripheral speed of the turbine wheel is :

(a) 60 m/s (b) 45 m/s
(c) 75 m/s (d) 90 m/s

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Ans. (a) : Given -

Head (H) = 900 m

speed ratio (ϕ) = 0.45

velocity of Pelton wheel $u = \phi\sqrt{2gH}$

$= 0.45\sqrt{2 \times 9.81 \times 900}$

$= 59.797$

$\boxed{u \approx 60 \text{ m/sec.}}$

161. If the outlet angle of the bucket for pelton wheel is 60^0, the maximum efficiency in the case of Pelton wheel neglecting friction in the bucket is :

(a) 75% (b) 80%
(c) 50% (d) 90%

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Ans. (a) : Given -

Outlet angle of bucket pelton wheel $\phi = 60^o$

maximum hydraulic efficiency

$$\eta_H = \left(\frac{1+\cos\phi}{2}\right) = \frac{1+\cos 60^o}{2}$$

$$= \frac{1+0.5}{2} \times 100 = 75\%$$

162. If D, D_0 and D_b refer to pelton wheel diameter, d refers to jet diameter, B refers to breadth the wheel V_f refers to the velocity of the jet and H refers to net head, discharge through Pelton turbine is given by one of the following equation

(a) $Q = \pi DBV_f$

(b) $Q = \frac{\pi}{4} d^2 \times \sqrt{2gH}$

(c) $Q = \frac{\pi}{4}\left[D_0^2 - D_b^2\right] \times V_f$

(d) $Q = \frac{\pi}{4}\left[D_b^2 - D_0^2\right] \times V_f$

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Ans. (b) : Turbine	**Discharge**
Pelton wheel turbine	$\frac{\pi}{4}d^2 \times \sqrt{2gh}$
Francis Turbine	πDBV_f
Kaplan turbine	$\frac{\pi}{4}\left[D_0^2 - D_b^2\right] \times V_f$

163. A turbine develops 10 MW when running at 100 rpm under the head of 30 m. If the head is reduced to 20 m, the speed and power developed will be respectively given as
[given $\sqrt{1.5} = 1.22$]
(a) 97.62 rpm, 5.46 MW
(b) 81.96 rpm, 4.82 MW
(c) 97.62 rpm, 4.82 MW
(d) 81.96 rpm, 5.46 MW

SJVN ET 2019

Ans. (d) : Given,
$P_1 = 10 \text{ MW} = 10 \times 10^6 \text{ W}$
$N_1 = 100 \text{ rpm}$
$N_2 = ?$
$H_1 = 30 \text{ m}$
$H_2 = 20 \text{ m}$

$$\frac{N_1}{(H_1)^{1/2}} = \frac{N_2}{(H_2)^{1/2}}$$

$$\frac{100}{(30)^{1/2}} = \frac{N_2}{(20)^{1/2}}$$

$$N_2 = 81.65 \text{ rpm}$$
$$\approx 81.96 \text{ rpm}$$

$$\frac{P_1}{(H_1)^{3/2}} = \frac{P_2}{(H_2)^{3/2}}$$

$$P_2 = \frac{10 \times 10^6 \times (20)^{3/2}}{(30)^{3/2}}$$

$$P_2 = 5.46 \times 10^6 \text{ W} = 5.46 \text{ MW}$$

164. Which of the following is the best suitable range of working head of Kaplan turbine?
(a) 5 - 70 m (b) 500 - 2000 m
(c) 300 - 1500 m (d) 100 - 300 m

SJVN ET 2019

Ans. (a) : High head 500 to 2000 m - Pelton wheel - Impulse turbine tangential flow
Medium head 60 m to 250 m - Francis turbine - mixed flow
Low head - head less than 60 m - Kaplan & propeller - axial flow

165. Which of the following is not open conduit?
(a) Tunnel (b) Flume
(c) Penstock (d) Pipeline

SJVN ET 2019

Ans. (b : Flume is a open channel or conduit. Tunnel penstock and pipe lines are not open channel.

166. An inward flow reaction turbine has following data:
Head available = 25 m
Velocity of flow = 2.5 m/s (constant)
Guide blade angle = 10^o
Runner vanes are radial at inlet and discharge at outlet is radial. What will be the approximate hydraulic efficiency of turbine?
[give $\tan 10^o = 0.175$, $g = 10 \text{ m/s}^2$]
(a) 95.34% (b) 70.23%
(c) 61.32% (d) 81.63%

SJVN ET 2019

Ans. (d) : Given
$\tan\alpha = \tan 10$
$= 0.175$

$$RP = \rho Q\left(V_{w_1}u_1 + V_{w_2}u_2\right)$$

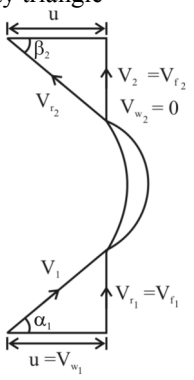
$$V_{w_2} = 0$$

$$u_1 = u_2 = u$$

$$RP = \rho Q V_w . u$$

$$SP = \rho Q g H$$

From - Inlet velocity triangle



$$\tan\alpha = \frac{V_{f_1}}{V_{w_1}}$$

$$V_{w_1} = \frac{2.5}{0.175} = 14.28$$

$$\eta_H = \frac{RP}{WP} = \frac{V_{w_1}u}{gH} = \frac{(14.28)^2}{10 \times 25} = 81.5$$

$$\approx 81.63\%$$

167. Consider the following statements regarding reaction turbine:
1. Blade shape is aerofoil type and its manufacturing is difficult.
2. It is suitable for small power.
3. Leakages losses are less compared to friction losses.

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

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Ans. (a) : Reaction turbine blades are asymmetrical and the design of these blades are aerofoil type hence it's manufacturing is difficult.

168. The hydraulic efficiency of a turbine is the ratio of
(a) mechanical energy in the output shaft at coupling and hydrodynamic energy available from the fluid
(b) mechanical energy supplied by the rotor and hydrodynamic energy available from the fluid
(c) useful hydrodynamic energy in the fluid at final discharge and mechanical energy supplied to the rotor

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(d) useful hydrodynamic energy in the fluid at final discharge and mechanical energy supplied to the shaft and coupling

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Ans. (b) : Hydraulic efficiency

$$\eta_H = \frac{\text{Runner power}}{\text{Water power at inlet of turbine}}$$

$$= \frac{\text{Mechanical energy supplied by the rotor}}{\text{Hydrodynamic energy available from fluid}}$$

169. In a lawn sprinkler, water leaves the jet with an absolute velocity of 2m/s and the sprinkler arms are 0.1 m in length. The sprinkler rotates at a speed of 120 rpm. The utilization factor of this sprinkler will be nearly

(a) 0.72 (b) 0.64
(c) 0.56 (d) 0.49

ESE 2019

Ans. (c) : Given, Velocity of water (v) = 2 m/s
Length of sprinkler arm = 0.1 m
N = 120 rpm

$$u = \frac{\pi DN}{60} = \frac{\pi \times 0.2 \times 120}{60} = 1.25 \text{ m/s}$$

$$\text{Utilization factor } (\in) = \frac{\text{Ideal work output}}{\text{Work output + kinetic energy}}$$

$$= \frac{\frac{vu}{g}}{\frac{vu}{g} + \frac{v^2}{2g}} = \frac{vu}{vu + \frac{v^2}{2}}$$

$$= \frac{2 \times 1.25}{2 \times 1.25 + \frac{2^2}{2}} = 0.56$$

170. For a given design of bucket, if U is the velocity of the bucket the V_1 is the velocity of the jet, then the ratio U/V_1 for the efficiency of a Pelton wheel is theoretically maximum :

(a) 0.75 (b) 0.25
(c) 0.50 (d) 1

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Ans. (c) : Given -
U = velocity of bucket
V_1 = velocity of jet

$$\frac{\text{velocity of the bucket}}{\text{velocity of jet}} = \frac{U}{V_1} = 0.5$$

171. A single jet impulse turbine of 10 MW capacity works with a head of 500 m. If the specific speed of the turbine is 10, the actual speed of the turbine is

(a) 106.9 rpm (b) 236.4 rpm
(c) 392.5 rpm (d) 50 rpm

APPSC AEE SCREENING 17.02.2019

Ans. (b) : $N_s = \frac{N\sqrt{P}}{H^{5/4}}$

$$10 = \frac{N\sqrt{10 \times 10^3}}{500^{5/4}}$$

$\Rightarrow$ N = 236.4

Note: Power is consider in kW to calculate specific speed.

172. Basic method to measure the flow rates in hydro power plants is

(a) velocity integration method
(b) Pitot-tube method
(c) Pressure-time method
(d) Notch method

APPSC AEE SCREENING 17.02.2019

Ans. (c) : One of the common method used in flow rate measurement in hydro power plant is pressure time method also called Gibson method. This method is based on water hammer principle.

173. If a turbine generates 10000 kW under the head of 10m at the speed 100 rpm, the specific speed of the turbine is

(a) 177.8 rpm (b) 562.3 rpm
(c) 1052 rpm (d) 2056 rpm

APPSC AEE SCREENING 17.02.2019

Ans. (b) : $N_s = \frac{N\sqrt{P}}{H^{5/4}} = \frac{100\sqrt{10000}}{10^{5/4}} = 562.3$

174. If V_w and u are the whirl and tangential velocity at the impeller, work done by the impeller on the water per second per unit weight of water striking per second is

(a) $V_w u/g$ (b) $V_w u$
(c) $\rho Q V_w u$ (d) $\rho Q V_w V_r$

APPSC AEE SCREENING 17.02.2019

Ans. (a) : Work done by impeller per unit weight per unit time is represent by Euler's head which is given as

$$H_e = \frac{V_{w1} u_1}{g} \quad \{\text{Assuming } V_{w2} = 0\}$$

175. Which of the condition leads to Cavitaion process?

(a) Too low local temperature
(b) Very high local pressure
(c) Local pressure falls below vapour pressure
(d) Thoma cavitation factor is higher than safe limit

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Ans. : (c) : When local pressure of flowing liquid falls below the vapor pressure then cavitation will take place.

176. Draft tube is used for discharging water from the exit of

(a) Impulse turbine (b) Francis turbine
(c) Kaplan turbine (d) Pelton wheel

TNPSC AE 2019

Ans. (b, c) : Draft tube is used for discharging water from the exit of Kaplan turbine and also Francis Turbine.
Note - Option (c) is given by commission.

177. The overall efficiency for a Pelton wheel lies between

(a) 0.50 to 0.65 (b) 0.65 to 0.75
(c) 0.75 to 0.85 (d) 0.85 to 0.90

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : The overall efficiency for a pelton wheel lies between 0.85 to 0.90.

178. In a reaction turbine, the draft tube is used

(a) To run the turbine full
(b) To prevent air to enter the turbine

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(c) To increase the head of water by an amount equal to the height of the runner outlet above the tail race
(d) To transport water to downstream

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : It is a pipe of gradually increasing area, which connects the turbine exhausted from the runner, flows to the tail race.
It makes the installation possible above the tail race level without the loss of head. Draft tube prevents the splashing at water coming out of the runner.

179. The Furneyron turbine is ..... reaction turbine
(a) An axial flow (b) An inward flow
(c) An outward flow (d) A mixed flow

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The furneyron turbine is an outward flow reaction turbine. In this water enters at outer periphery of runner and flows radially inwards toward the centre. Here the flow is radially outwards and the fluid get discharged at the outer periphery of the runner.

180. The flow ratio in case of Francis turbine varies from
(a) 0.15 to 0.3 (b) 0.4 to 0.5
(c) 0.6 to 0.9 (d) 1 to 1.5

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : Flow ratio $\psi = \frac{v_{f1}}{\sqrt{2gH}} = 0.15 \text{ to } 0.30$
So the flow ratio in Francis turbine varies from 0.15 to 0.3.

181. The unit power developed by a turbine is
(a) $P/\sqrt{H}$ (b) P/H
(c) $P/H^{3/2}$ (d) P/H^2

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The power developed by a turbine working under a head of one meter is known as unit power.
$P_u = \frac{P}{H^{3/2}}$

182. The discharge through a turbine is
(a) Directly proportional to $H^{1/2}$
(b) Inversely proportional to $H^{1/2}$
(c) Directly proportional to $H^{3/2}$
(d) Inversely proportional to $H^{3/2}$

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : The discharge through a turbine is directly proportional to $H^{1/2}$.

183. A pelton wheel working under a constant head and discharge, has maximum efficiency when the speed ratio is
(a) 0.26 (b) 0.36
(c) 0.46 (d) 0.56

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Speed ratio of pelton wheel is Ku = 0.43 to 0.47
So, A pelton wheel working under a constant head and discharge, has maximum efficiency when speed ratio is 0.46.

184. A pelton wheel develops 1750 kW under a head of 100 metres while running at 200 rpm and discharging 2500 litres of water per sec. The unit power of wheel will be
(a) 3.75 kW (b) 0.75 kW
(c) 1.75 kW (d) 0.25 kW

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (c) : We know that unit power of wheel
$$P_u = \frac{P}{H^{3/2}} = \frac{1750}{100^{3/2}} = 1.75 \text{ kW}$$

185. An impulse turbine produces 125 hp under a head of 25 meters. By what percentage should the speed be increased for a head of 100 metres
(a) 25% (b) 50%
(c) 75% (d) 100%

ISRO Scientist/Engineer 22.04.2018

Ans. (d) : We know that unit speed of turbine—
$$\frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}} = N_u$$
$$\frac{N_1}{\sqrt{25}} = \frac{N_2}{\sqrt{100}}$$
$N_2 = 2N_1$
So, speed should be increased by 100%.

186. If the head on the turbine is more than 300m, the type of turbine used should be
(a) Kaplan (b) Francis
(c) Pelton wheel (d) Propeller

TNPSC AE 2018

Ans. (c)

Turbine	Head (m)
Kaplan	10 - 30 (High discharge)
Propeller	10 - 30 (High discharge)
Francis	30 - 250
Pelton wheel	Above 300 m with low discharge

187. The critical speed of a turbine is
(a) same as the runaway speed
(b) the speed that will lead to mechanical failure of the shaft
(c) the speed which equals the natural frequency of the rotor
(d) the speed equal to the synchronous speed of the generator

ESE 2018

Ans. (c) : Critical speed of a turbine is the speed of the runner at which natural frequency of the rotor unit equal to the operating speed.

188. Consider the following statements:
1. The distinguishing features of a radial flow reaction turbine are – (i) only a part of the total head of water is converted into velocity head before it reaches the runner and (ii) the flow through water completely fills all the passages in the runner.
2. Kaplan turbine is essentially a propeller working in reverse, and its blades are so mounted that all the blade angles can be adjusted simultaneously by means of suitable gearing even as the machine is in operation.

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3. A draft tube is pipe of gradually increasing cross-sectional area which must be airtight, and under all conditions of operation, its lower end must be submerged below the level of the discharged water in the tailrace.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d)
(i) In reaction turbine, head is partially converted into velocity head and partially into pressure head.
(ii) In kaplan turbine, the orientation of moving angle can also be adjusted.
(iii) The draft tube is always immersed in tail race.

189. The maximum hydraulic efficiency of Pelton turbine and ratio of the velocity of jet to the velocity of bucket at this efficiency will be respectively: [Neglecting the bucket friction]

(a) $(1+\cos\phi);2$ (b) $\left(\frac{1+\cos\phi}{2}\right);0.5$
(c) $(1-\cos\phi);0.5$ (d) $(1-\cos\phi);0.2$
(e) $\left(\frac{1+\cos\phi}{2}\right);2$

CGPSC AE 25.02.2018

Ans. (e) : Maximum hydraulic efficiency of Pelton turbine

$$\eta_{max}=\frac{1+\cos\phi}{2}$$

and ratio of the velocity of jet (v) to the velocity of bucket (u) at maximum efficiency

$$r=\frac{v}{u}=2$$

190. Cavitation may occur at :

(a) the exit of reaction turbine
(b) the inlet of reaction turbine
(c) discharge side of centrifugal pump
(d) the inlet of penstock of hydroelectric power plant
(e) at the bucket of Pelton turbine

CGPSC AE 25.02.2018

Ans. (a) : Cavitation– Cavitation is a phenomenon in which rapid changes of pressure in a liquid lead to the formation of small vapor-filled cavities, in place where the pressure is relatively low. When subjected to higher pressure, these cavities called "bubbles" or "voids". Collapse and can generate an intense shock wave.
In Turbine cavitation may occur at exit of reaction turbine and at inlet of centrifugal pump.

191. If V_1 is inlet jet velocity, u is blade velocity and ϕ = outlet blade angle (bucket angle), the hydraulic efficiency of a Pelton wheel is expressed as .....

(a) $\eta_h=\frac{(2u)(V_1-u)(1+\cos\phi)}{V_1^2}$

(b) $\eta_h=\frac{(2u)(V_1+u)(1+\cos\phi)}{V_1^2}$

(c) $\eta_h=\frac{(2u)(V_1+u)(1-\cos\phi)}{V_1^2}$

(d) $\eta_h=\frac{(2u)(V_1-u)(1-\cos\phi)}{V_1^2}$

RPSC AE 2018

Ans. (a) : $\eta_h=\frac{(2u)(V_1-u)(1+\cos\phi)}{V_1^2}$

192. For the medium head of the water (24-180m), which hydraulic turbine is best suited

(a) Pelton turbine (b) Francis turbine
(c) Kaplan turbine (d) Any one

HPPSC AE 2018

Ans. (b) :

	Head	Turbine
Low head Turbine	2m–15m	Kaplan/ Propeller
Medium head Turbine	16m–70m	Kaplan/ Francis
High head Turbine	71m–500m	Francis/ Pelton
Very high head Turbine	> 500 m	Pelton turbine

193. The discharge through Kaplan turbine is given by

(a) $Q=\pi DBV_f$

(b) $Q=\frac{\pi}{4}d^2\times\sqrt{2gH}$

(c) $Q=\frac{\pi}{4}\left[D_0^2-D_b^2\right]\times V_f$

(d) $Q=0.9\pi DBV_f$

TNPSC AE 2018

Ans. (c) : The discharge through Kaplan turbine is given as

$$Q=\frac{\pi}{4}\left[D_0^2-D_b^2\right]\times V_f$$

where $V_f \rightarrow$ flow velocity

194. Which one of the following is impulse turbine?

(a) Francis turbine (b) Kaplan turbine
(c) Pelton turbine (d) none of these

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Pelton turbine is an impulse turbine while Francis and Kaplan turbines are reaction turbine.

195. The specific speed of the Kaplan turbine is

(a) 2.0–5.0 (b) < 0.3
(c) 0.3–2.0 (d) none of these

ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Dimensionless specific speed range
Pelton wheel : 10-35
Francis turbine : 60-300
Propeller turbine : 300-600
Kaplan turbine : 600-1000

196. A turbine develops 500 kW power under a head of 100 m at 200 rpm. What would be its normal speed under a head of 81 m?

(a) 100 rpm (b) 180 rpm
(c) 1800 rpm (d) 162 rpm

ISRO Scientist/Engineer 07.05.2017

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Ans. (b) : $P_1 = 500$ kW $H_1 = 100$ m $N_1 = 200$ rpm
$H_2 = 81$ m $N_2 = ?$

$$\boxed{H \propto N^2}$$

$$\frac{H_1}{N_1^2} = \frac{H_2}{N_2^2}$$

$$N_2^2 = N_1^2 \times \frac{H_2}{H_1} = (200)^2 \times \frac{81}{100}$$

$$= 200 \times 200 \times \frac{81}{100}$$

$$N_2^2 = 400 \times 81$$

$$N_2 = \sqrt{400 \times 81}$$

$$N_2 = 20 \times 9 = 180 \text{ rpm}$$

197. The valve closure is said to be gradual if the time required to close the valve is
(a) $t = 2L/C$ (b) $t \le 2L/C$
(c) $t \ge 2L/C$ (d) $t > 2L/C$
Where, L = length of pipe, C = velocity of the pressure wave
Nagaland PSC CTSE 2017 2nd Paper

Ans. (d) : The valve closure is said to be, sudden or gradual, the time required to close the Valve is "$t > 2L/C$"

198. The speed ratio of a Pelton wheel varies from
(a) 0.6 to 0.7
(b) 0.8 to 0.9
(c) 0.45 to 0.5
(d) It cannot be defined
Nagaland PSC CTSE 2017 2nd Paper

Ans. (c) : The Speed ratio of Pelton wheel Varies from 0.45 to 0.5.
$u = \phi\sqrt{2gh}$, $\phi =$ speed ratio

199. Statement (I): The speed of a governed water turbine will remain constant irrespective of load.
Statement (II): In governing, the water supply is regulated to maintain the speed constant.
ESE 2017

Ans. (a) : The primary objective in turbine operation is to maintain a constant speed of rotation irrespective of the varying load. This is achieved by means as governing in a turbine.

200. The pressure inside a Pelton turbine casing during working _____.
(a) Increases
(b) Remains constant
(c) Decreases
(d) First decreases and then increases
CIL MT 26.032017

Ans. (b) : Pelton wheel is a kind of impulse turbine. As in case of impulse turbine pressure throughout turbine is constant and equal to atmospheric pressure, so only energy available for turbine is kinetic energy of fluid. There is no change in pressure energy as it takes place in reaction turbine.

201. A turbine develops 9000 kW when running at 100 rpm. Head on turbine is 30 m. If the head on the turbine is reduced to 18 m, speed of turbine under 18 m is:
(a) 77 rpm (b) 18 rpm
(c) 180 rpm (d) 30 rpm
TRB Polytechnic Lecturer 2017

Ans. (a) : Given,
$H_1 = 30$ m
$H_2 = 18$ m
$N_1 = 100$ rpm
$N_2 = ?$
We know that
$H \propto N^2$
Then

$$\frac{N_2}{N_1} = \sqrt{\frac{H_2}{H_1}}$$

$$N_2 = 100 \times \sqrt{\frac{18}{30}}$$

$N_2 = 77.45 \simeq 77$ rpm.

202. Which of the following statements are correct?
1. The specific speed of a turbine is the speed at which a homologous turbine develops 1 kW under unit head at its maximum efficiency.
2. The specific speed is a dimensionless parameter used for the selection of turbines.
3. The function of guide vanes in reaction turbines is to minimize shock at entry of the fluid onto the runner blades.

Select the correct answer using the code given below.
(a) 1, 2 and 3 (b) 2 and 3 only
(c) 1 and 2 only (d) 1 and 3 only
ESE 2017

Ans. (a) : All the statements are correct. The specific speed may be dimensional or non-dimensional and is an important parameter in turbine selection. Specific speed is defined as the speed of a similar turbine working under a head of 1 m to produce a power output of 1 kW.

Specific speed $N_S = \frac{N\sqrt{P}}{H^{5/4}}$
N = Speed in rpm
P = Power
H = Head available

203. A turbine is to operate under a head of 30 m while running at 120 rpm using 12 m^3/s discharge at 90% efficiency. Which one of the following turbine types would suit the conditions?
(a) Pelton (b) Kaplan
(c) Francis (d) Turgo
JWM 2017

Ans. (c) : Specific speed, $N_S = \frac{N\sqrt{P}}{H^{5/4}}$

Power, $P = \eta\rho Qhg$
$P = 0.9 \times 1000 \times 9.81 \times 12 \times 30$
$= 3178.44$ kW

$$N_S = \frac{120\sqrt{3178.44}}{(30)^{5/4}} = 96.35$$

For Francis turbine specific speed should lies b/w 50 – 250

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204. The speed of a reaction turbine is 250 rpm when working under a head of 120 m. The speed of the turbine when the head is changed to 90 m will be

(a) 167 rpm (b) 197 rpm
(c) 217 rpm (d) 237 rpm

JWM 2017

Ans. (c) : Given, speed of Reaction turbine, $N_1 = 250$ rpm

Working head, $H_1 = 120$ m

Working head for N_2 speed is 90 m

$$\frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$$

$$N_2 = N_1\sqrt{\frac{H_2}{H_1}} = 250\sqrt{\frac{90}{120}}$$

$$N_2 \simeq 217\text{rpm}$$

205. 'Shockless entry' of water in hydraulic turbine is of much significance from the standpoint of

(a) Hydraulic efficiency
(b) Mechanical efficiency
(c) Both hydraulic efficiency and mechanical efficiency
(d) Neither hydraulic efficiency nor mechanical efficiency

JWM 2017

Ans. (a) :
Hydraulic efficiency

$$\eta_H = \frac{\text{Power developed by the runner}}{\text{Net power supplied at the turbine entrance}}$$

Mechanical efficiency,

$$\eta_{mech} = \frac{\text{Power available at the turbine shaft}}{\text{Power developed by the runner}}$$

The hydraulic efficiency of hydraulic turbine is very much effected by shock less entry, shock less entry in turbine ensure that the power loss in the turbine is less.

206. When water glides over the runner blades of a hydraulic reaction turbine:

(a) Pressure remains constant
(b) Pressure decreases
(c) Pressure first increases and then decreases
(d) Pressure increases

CIL MT 26.032017

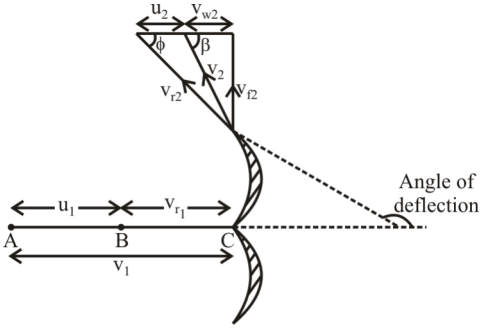
Ans. (b) : Reaction turbine means that water at the inlet of the turbine possesses kinetic energy as well as pressure energy. As the water flows through the runner, a part of pressure energy is converted into kinetic energy. So pressure decreases.

207. Inlet velocity triangle of Pelton wheel is :

(a) Straight line (b) Triangle
(c) Inverted triangle (d) None of the above

TRB Polytechnic Lecturer 2017

Ans. (a) : Inlet velocity triangle of Pelton wheel is straight line.



208. Hydraulic efficiency of Francis turbine whose vanes are radial at inlet is (α - guide blade angle)

(a) $\frac{2+\tan^2\alpha}{2}$ (b) $\frac{\tan^2\alpha}{2}$
(c) $\frac{2}{2+\tan^2\alpha}$ (d) $\frac{2}{\tan^2\alpha}$

TRB Polytechnic Lecturer 2017

Ans. (c) : Hydraulic efficiency of Francis turbine whose vanes are radial at inlet is (α - guide blade angle)

$$\eta_H = \frac{2}{2+\tan^2\alpha}$$

209. Turbine having adjustable blade impeller is called:

(a) Pelton turbine (b) Francis turbine
(c) Kaplan turbine (d) Deriaz turbine

CGPSC ARTO & TSI 07.01.2017

Ans. (c) : The kaplan Turbine is a propeller type water turbine which has adjustable blades.
It was developed in 1913 by Austrian professor Viktor kaplan who combined automatically adjusted propeller blades with automatically adjusted wicket to achieve efficiency over a wide range of flow and water level.

210. Two turbines A and B are having same specific speed and working under the same head. Turbine A produces 400 kW at 800 rpm. If Turbine B produces 100 kW, then what is its rpm?

(a) 200 (b) 320
(c) 800 (d) 1600
(e) 3200

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : $N_s = \frac{N\sqrt{P}}{H^{5/4}}$

Given $(N_s)_A = (N_s)_B$

and here $h_A = h_B$

$N_A = 800$ rpm, $P_A = 400$ kw, $N_B = ?$

$$\frac{N_A\sqrt{P_A}}{H_A^{5/4}} = \frac{N_B\sqrt{P_B}}{H_B^{5/4}}$$

$$\frac{N_A}{N_B} = \sqrt{\frac{P_B}{P_A}} \qquad (\because H_A = H_B)$$

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$$\frac{800}{N_B}=\sqrt{\frac{100}{400}}$$

$$\frac{800}{N_B}=\frac{10}{20}$$

$$N_B=1600\text{rpm}$$

211. In a simple impulse turbine, the nozzle angle at the entrance is 30º. What will be the blade-speed ratio for maximum diagram efficiency?

(a) 0.433 (b) 0.25
(c) 0.5 (d) 0.75

HPPSC Asstt. Prof. 18.09.2017

Ans. (a) : Nozzle angle $\alpha = 30°$
For maximum efficiency,

$$\text{Blade speed ratio} = \frac{u}{V}=\frac{\cos\alpha}{2}$$

$$=\frac{\cos 30}{2}$$

$$= 0.433$$

212. A draft tube is used in:

(a) Pelton wheel turbine
(b) impulse turbine
(c) Francis turbine
(d) very high specific speed turbines

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : The draft tube is an important component of a Francis turbine which influences the hydraulic performance. It is located just under the runner and allowed to decelerate the flow velocity existing the runner, thereby converting the excess of kinetic energy into static pressure.

213. Choose the wrong statement:

(a) impulse type turbine is suited for high heads
(b) propeller type turbine is suited for low heads
(c) Francis turbine is a medium head turbine
(d) impulse type turbine is a high specific speed turbine

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) :

	Turbine	Specific speed (MKS unit)
1	Pelton Turbine (Impulse turbine)	10-30
2.	Francis turbine	60-300
3.	Kaplan and Propeller turbine	300-1000

Value of head and discharge

Turbine	Head	Discharge
1. Pelton turbine	above 25 m	Low
2. Francis turbine	60m-250m	Medium
3. Kaplan and Propeller turbine	below 60 m	High

So, The wrong statement is impulse type turbine is a high specific speed turbine.

214. Reaction turbine requires

(a) High head and low discharge
(b) High head and high discharge
(c) Low head and low discharge
(d) Low head and high discharge

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : Francis and Kaplan turbines are reaction turbines Francis turbine works on medium head and medium discharge and Kaplan turbine works on low head and high discharges.

215. In mixed flow turbines, the water enters the blades _____ and comes out _____.

(a) radially, axially (b) radially, radially
(c) axially, radially (d) axially, axially

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : In mixed flow turbines like modern Francis turbines water enters radially and comes out axially.

216. Which of the following statement is correct?

(a) In an impulse turbine, the water impinges on the buckets with pressure energy
(b) In a reaction turbine, the water glides over the moving vanes with kinetic energy
(c) In an impulse turbine, the pressure of the flowing water remains unchanged and is equal to atmospheric pressure
(d) In a reaction turbine, the pressure of the flowing water increase after gliding over the vanes

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : In impulse turbine pressure of water remains constant as it flows over the blade. As the where is exposed to atmosphere, the pressure is equal to atmospheric pressure.
In reaction turbine the pressure continuously decreases as water flows guide vanes as well as runners blades.

217. The flow ratio of Francis turbine is defines as the ratio of the

(a) velocity of flow at inlet to the theoretical jet velocity
(b) theoretical velocity of jet to the velocity of flow at inlet
(c) velocity of runner at inlet to the velocity of flow at inlet
(d) velocity of runner at outlet to the velocity of flow at inlet

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : $k_f = \frac{V_{fi}}{\sqrt{2gH}} = \frac{\text{Velocity of flow at Inlet}}{\text{Theoretical jet speed}}$

218. A Pelton wheel develops 1750 kW under a head of 100 metres while running at 200 rpm and discharging 2500 litres of water per second. The unit power of the wheel is

(a) 0.25 kW (b) 0.75 kW
(c) 1.75 kW (d) 3.75 kW

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : $P_u = \frac{P}{H^{3/2}} = \frac{1750\,\text{kW}}{(100)^{3/2}} = 1.75\,\text{kW}$

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219. The speed of an imaginary turbine, identical with the given turbine, which will develop a unit power under a unit head, a known as

(a) normal speed (b) unit speed
(c) specific speed (d) pitot speed

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : $N_s = \frac{N\sqrt{P}}{H^{5/4}}$

When = P = 1 kW, H = 1 m then N = N_S

220. The function of guide vanes in a reaction turbine is to

(a) allow the water to enter the runner without shock
(b) allow the water to flow over them, without forming eddies
(c) allow the required quantity of water to enter the turbine
(d) allow the water to flow to the draft tube

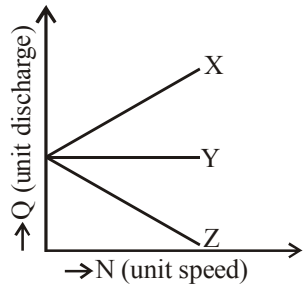
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : The Primary function of guide vanes is to direct water towards the runner in such a way that the flow enters into the runner without shock.

Note- By changing the flow area between the guide vanes the discharge entering into the turbine can also be controlled.

Hence both (1) and (3) are correct but (1) is most appropriate were as it is the main function.

221. The unit discharge (Q) and unit speed (N) curves for different turbines are given in figure. Which curve represent to Francis turbine:



(a) X (b) Y
(c) Z (d) None of these

PTCUL AE 25.06.2017

Ans. (c) :

222. In the impulse turbine if friction is neglected the:

(a) Relative velocity at inlet and outlet is of the same magnitude
(b) Passage of the steam over the moving blades provides the motive force
(c) Pressure increases as it flows over the moving blades
(d) Work done is zero

PTCUL AE 25.06.2017

Ans. (c) :

223. For Francis Turbine Thoma's critical cavitation factor 'σ_c' can be related to specific speed as :

(a) $0.30 + 0.0036\left(\frac{N_s}{100}\right)^{2.73}$

(b) $0.30 + 0.0036\left(\frac{N_s}{100}\right)$

(c) $0.0431\left(\frac{N_s}{100}\right)^2$

(d) $0.0431\left(\frac{N_s}{100}\right)$

Kerala PSC Poly. Lect., 2017

Ans. (c) : For pump and turbine,
Thoma's critical cavitation factor (σ) > Critical cavitation number (σ_c)
Then cavitation will not occur,

	Turbine	Pumps
Francis turbine	$\sigma_c = 0.625\left(\frac{N_s}{380.78}\right)^2 = 0.0431\left(\frac{N_s}{100}\right)^2$	$\sigma_c = 0.103\left(\frac{N_s}{1000}\right)^{4/3}$
Propeller turbine	$\sigma_c = 0.28\left[\frac{1}{7.5}\left(\frac{N_s}{380.78}\right)^2\right]$	

224. A single stage impulse turbine with a diameter of 120 cm runs at 3000 rpm. If the blade speed ratio is 0.42, then the inlet velocity of steam will be

(a) 79 m/s (b) 450 m/s
(c) 188 m/s (d) 900 m/s

TSPSC AEE 2017

Ans. (b) : We know,

$$\text{Blade speed ratio} = \frac{\text{blade speed}}{\text{velocity of steam at entry}}$$

$$V_b = \frac{\pi DN}{60} = \frac{\pi \times 1.2 \times 3000}{60}\text{m/s} = 50 \times \pi \times 1.2\,\text{m/s}$$

$$\Rightarrow \quad V_i = \frac{\text{blade speed}}{\text{blade speed ratio}}$$

$$V_i = \frac{50 \times \pi \times 1.2}{0.42}$$

$$V_i = = 448.97 \approx 450\ \text{m/s}$$

225. What is the head of water available at turbine inlet in hydro-electric power plant?

(a) Head race (b) Tail race
(c) Gross head (d) Net head

WBPSC AE, 2017

Ans. (d) : Gross head-The difference between the head race level and tail race level when no water is flowing is known as gross head.

Net head- It is the effective head available at the inlet of turbine.

$$H = H_{net} = H_g - h_f$$

h_f is head loss due to friction between penstock and water.

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226. Which component of fluid velocity is responsible for transmission of power through roto dynamic machine?

(a) Radial (b) Axial
(c) Tangential (d) None of the above

WBPSC AE, 2017

Ans. (c) : Euler Equation- Work done by the jet on the runner per second

$W = F_x \times u = \rho A V_1 \left[V_{w_1} + V_{W_2} \right] \times u$ N/m/sec.

V_{W_1} = Whirl velocity component/tangential velocity component at inlet.

V_{W_2} = Whirl velocity component/tangential velocity component at outlet.

So tangential component of fluid velocity is responsible for power transmission.

227. In reaction turbine work is done_____

(a) totally due to change in pressure head.
(b) mostly due to change in pressure head and small amount of work is due to change in kinetic energy.
(c) mostly due to change in kinetic energy and small amount of work is due to change in pressure head.
(d) totally due to change in kinetic energy.

WBPSC AE, 2017

Ans. (b) : In reaction turbine work is done mostly due to change in pressure head and small amount of work is due to change in kinetic energy.

228. In which type of turbine, the runner need to be run full of water?

(a) Impulse turbine (b) reaction turbine
(c) Both (A) and (B) (d) None of the above

WBPSC AE, 2017

Ans. (b) : Reaction turbine:- In this turbine the water at the inlet of the turbine possesses Kinetic energy as well as pressure energy. As the water flows through the runner, a part of pressure energy goes on changing into kinetic energy. The water through the runner in under pressure. The runner is completely enclosed in an air- tight casing and the runner is always full of water.

229. Generally, the value of specific speed of Kaplan turbine, in comparison to that of Pelton turbine, is

(a) lower (b) higher
(c) same (d) unpredictable

WBPSC AE, 2017

Ans. (b) :

S.N.	Specific speed	Types of Turbine
1	10-35	Pelton wheel with single jet
2	35-60	Pelton wheel with two or more jets
3	60-300	Francis Turbine
4	300-1000	Kaplan or Propeller Turbine

230. The water from runner in reaction turbine is discharged into tail race through a draft tube, the cross sectional area of which is

(a) constant
(b) gradually decreasing
(c) gradually increasing
(d) uneven

WBPSC AE, 2017

Ans. (c) : Draft tube:- The draft- tube is a pipe of gradually increasing area which connects the outlet of the runner to the tail race. It is used for discharging water from the exit of the turbine to the tail race. This pipe of gradually increasing cross-sectional area is called a draft tube. one end of the draft- tube is connected to the outlet of the runner while the other end is sub-merged below the level of water in the tail race.

231. The energy of water entering the reaction turbine is

(a) fully kinetic energy
(b) fully pressure energy
(c) partly pressure and partly kinetic energy
(d) unpredictable

WBPSC AE, 2017

Ans. (c) : Reaction turbine:- Reaction turbine means that the water at the inlet of the turbine possesses Kinetic energy as well as pressure energy.

232. Which of the following statements are false for reaction turbines?

a. Flow of water can be regulated without any losses.
b. It works on the principle of impulse and reaction.
c. They are suitable for low heads.
d. Water flows at constant atmospheric pressure over the runner.

Select the code for the correct, answer for the options given below:

(a) Only a, b and c (b) Only b and d
(c) Only a and d (d) a, b, c and d

Karnataka PSC Lect., 27.05.2017

Ans (c) : In reaction turbine water is not actually hitting the blade rather change in pressure and change in momentum of flow causes the reaction force in the system which results in torque about the axis of the rotation of the shaft.

- In reaction turbine only a part of total energy at entry is in the form of kinetic energy the remaining energy is in the form of pressure energy the flow entering the casing is turned towards the runners by stay vanes and then it passes through the guide vanes and then through the runner blade while passing through the runner blade, both pressure and velocity changes and at the exit of the runner blade, pressure becomes negative and velocity becomes very small.

233. What is the formula for the velocity of water jet at the inlet of a turbine?

(a) $V = C_V\sqrt{gH}$ (b) $V = 2C_V\sqrt{gH}$
(c) $V = C_V\sqrt{2gH}$ (d) $V = 2C_V\, g\sqrt{H}$

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Where,
H = Net head acting on Pelton wheel
C_V = Coefficient of velocity of the jet

Karnataka PSC Lect., 27.05.2017

Ans. (c) : The velocity of jet at inlet is given by

$$V_1 = C_V\sqrt{2gH}$$

C_V = Coeff. of velocity = (0.98 to 0.99)
H = Net head on turbine

- The velocity of wheel (u) is given by

$$u = K_u\sqrt{2gH}$$

K_u = speed ratio = (0.43 to 0.48)

- The mean diameter or the pitch diameter (D) of the Pelton wheel is given by

$$u = \frac{\pi DN}{60}$$

$$D = \frac{60u}{\pi N}$$

234. A turbine develops 1200 kW while running at 300 rpm under 8-meter head. The turbine should be

(a) Francis turbine (b) Kaplan turbine
(c) Pelton wheel (d) Any of the above

Karnataka PSC AE, 10.09.2017

Ans. (b) : Specific speed of turbine,

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

$$N_s = \frac{300\sqrt{1200}}{(8)^{5/4}} = 772.4$$

$N_s > 600$ – Kaplan turbine
So, It is Kaplan turbine.

235. Propeller turbine is suitable for

(a) low head installation upto 30 m
(b) high head installation above 180 m
(c) medium head installations between 20 m and 180 m
(d) all types of heads

APGENCO AE, 2017

Ans. (a) :

Turbine	**Head**
Propeller	upto 10 m
Kaplan	10 to 60 m
Francis	60 to 30 m
Pelton	300 m to above

236. The cavitation in reaction turbine is avoided, to a great extent by :

(a) installing the turbine below the tail race level
(b) using stainless steel runner of the turbine
(c) providing highly polished blades to the runner
(d) all of the above

Punjab PSC SDE 12.02.2017

Ans. (d) : The cavitation in reaction turbine is avoided to a great extent by .
(i) Installing the turbine below the tail race level.
(ii) Using stainless steel runner of the turbine.
(iii) Providing highly polished blades to the runner.

237. A turbine develops 100 kW at a head of 25 m. What will be the power output of the same turbine when the head is reduced to 16 m?

(a) 51.2 kW (b) 64 kW
(c) 80 kW (d) 26.2 kW

RPSC AE (GWD) 06.05.2016

Ans. (a) : $P_1 = 100$ kW
$H_1 = 25$ m
$P_2 = ?$
$H_2 = 16$ m
Same unit quantity-

$$\frac{P_1}{H_1^{3/2}} = \frac{P_2}{H_2^{3/2}}$$

$$\frac{100}{(25)^{3/2}} = \frac{P_2}{(16)^{3/2}}$$

$$\frac{100}{(5^2)^{3/2}} = \frac{P_2}{(4^2)^{3/2}} \Rightarrow \frac{100}{5^3} = \frac{P_2}{4^3}$$

$$P_2 = \frac{100\times(4)^3}{(5)^3}$$

= 51.2 kW

238. Pick the wrong statement about the Kaplan turbine :

(a) It is purely axial flow turbine.
(b) It is mostly employed with the shaft kept in vertical position only.
(c) In this only guide vanes are adjustable.
(d) It has smaller number of vanes as compared to a Francis turbine.

RPSC AE (GWD) 06.05.2016

Ans. (c) : Guide vanes are used as governor in Francis and Kaplan turbines. These guide vanes controls the flow of water to the turbine and hence governs the speed of rotation of turbine under different loading conditions the speed of rotation of turbine is needed to be controlled for constant frequency output.

239. The term 'run a way speed' for a Pelton turbine means that speed of the turbine at which

(a) the turbine just starts to rotate.
(b) the velocity of bucket at mean radius equals to the velocity of the jet.
(c) the buckets will break off due to centrifugal force.
(d) the efficiency of the turbine will be maximum.

RPSC AE (GWD) 06.05.2016

Ans. (b) : The maximum unsafe speed of runner due to sudden decrease in load on turbine is called runaway speed. In a runway speed, the velocity of bucket at mean radius equals to velocity of jet.

240. Which of the following types of draft tubes is most efficient?

(a) Straight conical type draft tube
(b) Simple elbow type draft tube with circular cross section

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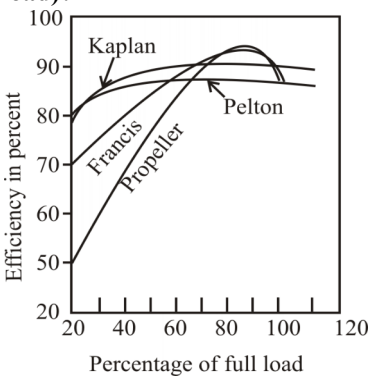
(c) Elbow type draft tube with circular cross section at entrance and square cross section at exit.
(d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (a) : Draft tube is one of the most commonly device used in the Reaction turbine. It works as an outlet at the Reaction turbine. Draft tube is also called straight conical type draft tube. Draft tube of conical with angles of 10 degree with respect to its position, Draft tubes are situated at the outlet of the turbine.

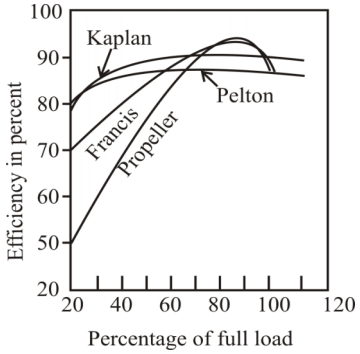
241. Which of the following turbines is more efficient at highly part load (upto 20-25% of full load)?



(a) Francis turbine
(b) Propeller Turbine
(c) Pelton turbine
(d) Both Pelton and Francis turbines

RPSC AE (GWD) 06.05.2016

Ans. (c)



From figure it is clear that at part load (upto 20-25%) of full load Kaplan is most efficient but in option Kaplan is not given so best suitable will be pelton turbine.
Hence option (c) is correct.

242. A turbine is working under a head of 20 m. It has been installed at a power plant where the atmospheric pressure is 10 m of water and vapour pressure is 0.20 m of water. If the Thoma's cavitation factor is 0.25, what can be the maximum height of the discharge end of the runner from the tail race?

(a) 9.8 m (b) 4.8 m
(c) 2.4 m (d) 4.4 m

RPSC AE (GWD) 06.05.2016

Ans. (b) : Given,
$H_{atm} = 10$ m
$H_{vapour\ head} = 0.20$ m
h_s = suction head
H = working head = 20 m
Thomas cavitation factor (σ) = 0.25

$$\sigma = \frac{(H_{atm} - h_s) - H_{vapour}}{H}$$

$$0.25 = \frac{(10 - h_s) - 0.20}{20} \Rightarrow 5 = (10 - h_s) - 0.20$$

$$10 - h_s = 5.2$$

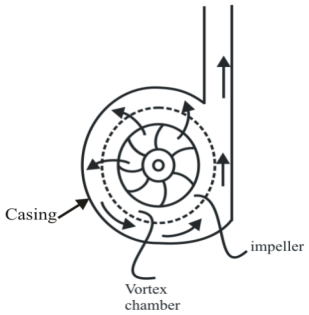
$$h_s = 4.8 \text{ m}$$

243. In vortex casing ….. type of chamber is introduced between the casing and the impeller.

(a) Semi-elliptical (b) Circular
(c) Triangle (d) Hexagonal

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : Vortex casing is a casing in which circular chamber is provided b/w the casing and the impeller vortex casing will increase pump efficiency by reducing eddies formation to a considerable extents.



244. A pressure surge caused when a valve closes suddenly at an end of pipeline system, leading to a pressure wave propagating in the pipe is called_____

(a) fluid pressure
(b) water hammer or fluid hammer
(c) water wave or fluid wave
(d) none of these

KPCL AE 2016

Ans. (b) : A pressure surge caused when a valve closes suddenly at an end of a pipeline system, leading to a pressure wave propagating in the pipe is called water hammer or fluid hammer.

245. In a two-way spool valve in hydraulic systems, the moving member is called :

(a) land (b) groove
(c) spool (d) shuttle

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : In a two way spool valve used in hydraulic systems, the moving member is called spool.

246. In Francis turbine, the spiral casing surrounds the ..... completely.

(a) shaft (b) runner
(c) penstock (d) governor

HPPSC Asstt. Prof. 29.10.2016

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Ans. (b) : In Francis turbine, the spiral casing surrounds the runner. Casing is always full of water. Spiral casing is reducing in cross-section in flow direction.

247. While dealing with fluid-rotor energy transfer related cases, the term "degree of reaction" for a given machine:

(a) is always positive
(b) is always negative
(c) is always zero
(d) may be positive or negative

HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : The degree of reaction (R) can be positive, negative or zero.

248. Draft tube.....the available head in the case of reaction turbines.

(a) Increases (b) Decreases
(c) Maintains constant (d) None of these

HPPSC AE (PWD) 10.09.2016

Ans. (a) : By a placing a draft tube at the exit of the turbine pressure head is increased by decreasing the exit velocity and both the overall efficiency and the output of the turbine can be improved.

249. The diameter of a pelton wheel operating with a speed of 600 rpm, speed ratio of 0.44 and a net head of 300 m, is

(a) 0.82 m (b) 1.07 m
(c) 1.51 m (d) 2.14 m
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) : $u = \frac{\pi DN}{60}$, $\frac{u}{v} = 0.44$

H = 300 m

$v = \sqrt{2gH} = \sqrt{2 \times 9.81 \times 300} = 76.72\,m/s$

$u = 76.72 \times 0.44$

$76.72 \times 0.44 = \frac{\pi \times D \times 600}{60}$

D = 1.07 m

250. The mean diameter of an impulse turbine is 85 cm and the speed is 3200 rpm. The nozzle angle is 20° and the ratio of blade speed to steam speed is 0.45. The blade velocity coefficient is 0.85. The outlet angle of blade is 2° less than the blade angle at the inlet. The steam flow is 9 kg/s. Tangential, axial and resultant thrusts on the blades will be respectively

(a) 260.856 N, 188.05 N and 261.532 N
(b) 2608.56 N, 188.05 N and 2615.32 N
(c) 2608.56 N, 1880.5 N and 2615.32 N
(d) 260.856 N, 1880.5 N and 261.532 N
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) :

Given, D = 85 cm = 0.85 m k = 0.85

N = 3200 rpm $\frac{u}{\sqrt{2gH}} = 0.45$

$\alpha = 20°$ $\dot{m} = 9\,kg/sec$

$\phi = \theta - 2°$

$u = \frac{\pi DN}{60} = \frac{\pi \times 0.85 \times 3200}{60} = 142.41\,m/s$

$\frac{u}{v_1} = 0.45 \Rightarrow v_1 = 316.466\,m/s$

$v_{w1} = v_1 \cos 20° = 297.38\,m/s$

$\tan\alpha = \frac{v_{f1}}{v_{w1}} \Rightarrow \frac{v_{f1}}{297.38} = \tan 20°$

$v_{f1} = 108.23$ m/s

$\tan\theta = \frac{v_{f1}}{v_{w1} - u_1} = \frac{108.23}{297.38 - 142.41}$

$\theta = 34.932°$ (Inlet blade angle)

$\phi = 32.932°$ (Outlet blade angle)

$v_{r1} = \frac{v_{f1}}{\sin\theta} = \frac{108.23}{\sin 34.932}$

$v_{r1} = 189.01$ m/s

$v_{r2} = k v_{r1}$

$= 0.85 \times 189.01$

$v_{r2} = 160.66$ m/s

$v_{f2} = v_{r2}\sin\phi$

$v_{f2} = 87.34$ m/s

$v_{r2}\cos\phi = u_2 + v_{w2}$

$v_{w2} = -7.565$ m/s

Thus, $F_T = \dot{m}(v_{w1} + v_{w2})$

$= 9(297.38 - 7.565)$

$F_T = 2608.33$ N

$F_{axial} = \dot{m}(v_{f1} - v_{f2})$

$= 9(108.23 - 87.34)$

$= 188.01$ N

Therefore resultant thrust on the blade,

$F_R = \sqrt{F_A^2 + F_T^2}$

$= \sqrt{(188.01)^2 + (2608.33)^2}$

$F_R = 2615.09$ N

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251. A turbine where in entire total energy of water available is converted to kinetic energy is called

(a) Reaction turbine (b) Mixed flow turbine
(c) Impulse turbine (d) Axial flow turbine
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : Impulse turbine change the velocity a water jet, in impulse turbine pressure of fluid flowing over the rotor blades is constant & all the work output is due to change in K.E. of the fluid.

252. Principles of similitude form the basis of

(a) Performing acceptance tests
(b) Comparing two identical equipment
(c) Comparing similarity between design ad actual equipment
(d) Designing and testing models so that the results can be worked out for the prototype
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : The principle of similitude is a supplement to the scientific method advocated by Rayleigh, that any suggested law be examined for its relationship to the similar laws. Designing & testing models, can be worked out for the prototype.

253. Match List-I (Type of Turbine) with List-II (Specific Speed) and select the correct answer using the code given below the lists

(List-I) (Type of Turbine)	(List-II) (Specific Speed)
A. Propeller	1. 1.0
B. Pelton	2.4.0
C. Kaplan	3. 3.0
D. Francis	4. 0.1

(a) A-1,b-4,C-2,D-3 (b) A-1,B-2,C-4,D-3
(c) A-3,B-4,C-2,D-1 (d) A-3,B-2,C-4,D-1

UPRVUNL AE 21.08.2016

Ans. : (c)

Type of turbine	Specific Speed
Propeller	3.0
Pelton	0.1
Francis	1.0
Kaplan	4.0

254. Consider the following statements

1. **Runner blades of Francis turbine are adjustable**
2. **Draft tube is used invariably in all reaction turbine installations**
3. **Surge tank is a protective device on penstock**
4. **A reverse jet protects Pelton turbine from over speeding**

(a) 1 and 2 are correct
(b) 2 and 3 are correct
(c) 3 and 4 are correct
(d) 1 and 4 are correct

UPRVUNL AE 21.08.2016

Ans. : (b) (i) Draft tube is used invariably in all reaction turbine installations
(ii) Surge tank is a protective device on penstock.

255. A reaction turbine discharge 30 m^3/s of water under a head of 10 m with an overall efficiency of 92%. The power developed is :

(a) 2707 kW (b) 2952 kW
(c) 2870 kW (d) 2652 kW

APPSC AE Subordinate Service Civil/Mech. 2016
ESE 1997

Ans. (a) : Given,
$Q = 30\ m^3/s$, $H = 10m$, $\eta_0 = 92\%$
Power developed $= \rho g Q H \eta_0$

$$P = 1000 \times 9.81 \times 30 \times 10 \times 0.92$$
$$= 2707.56\ kW$$
$$P = 2707\ kW$$

256. A kaplan turbine with a runner diameter of 9m is operating under a head of 10m. At optimum regime the turbine discharge 500 cumecs of water to develop the power. What is the unit discharge of the turbine?

(a) 1.952 (b) 0.617
(c) 1.667 (d) 2.68

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (*) : Given, D = 9 m, H = 10 m, Q = 500 m^3/sec

$$Q_u = \frac{Q}{\sqrt{H}} = \frac{500}{\sqrt{10}} = 158.11 \text{cumecs}$$

No answer is correct.

257. The relationship between various efficiencies of turbine is expressed as (where ε_0 is the overall efficiency. ε_m is the mechanical efficiency, ε_h is the hydraulic efficiency and ε_v is the volumetric efficiency.)

(a) $\varepsilon_0 = \varepsilon_h \times \varepsilon_v \times \varepsilon_m$ (b) $\varepsilon_0 = \varepsilon_h + \varepsilon_v + \varepsilon_m$
(c) $\varepsilon_h = \varepsilon_0 - \varepsilon_0/(\varepsilon_v\ \varepsilon_m)$ (d) $\varepsilon_m = \varepsilon_h - \varepsilon_0/(\varepsilon_v\ \varepsilon_h)$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Efficiencies of turbines –

(i) Hydraulic efficiency (η_h)

$$\eta_h = \frac{\text{Power delivered to runner}}{\text{Power supplied at inlet}} = \frac{\text{R.P.}}{\text{W.P.}}$$

(ii) Mechanical efficiency (η_m)

$$= \frac{\text{Power at the shaft of the turbine}}{\text{Power delivered by water to the runner}} = \frac{\text{S.P.}}{\text{R.P.}}$$

(iii) Volumetric efficiency (η_v)

$$\eta_v = \frac{\text{Volume of water actually striking the runner}}{\text{Volume of water supplied to the turbine}}$$

(iv) Overall efficiency (η_0)

$$\eta_0 = \frac{\text{Volume available of the shaft of the turbine}}{\text{Power supplied at the inlet of the turbine}}$$

$$\boxed{\eta_0 = \eta_h \times \eta_m \times \eta_v}$$

258. A Pelton wheel turbine operates under a head of 125 m at a speed of 2000 rpm. To operate very close to the maximum efficiency, the mean diameter of the Pelton wheel is nearly

(a) 24 cm (b) 48 cm
(c) 12.5 cm (d) 36 cm

ISRO Scientist/Engineer 03.07.2016

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Ans. (a) : Given Data–

H = 125 m, N = 2000 rpm

Assume $C_v = 1$

Jet velocity

$$V = C_v\sqrt{2gh}$$

$$= 1\times\sqrt{2\times 9.81\times 125}$$

$$= 49.5227 \text{ m/sec}$$

For maximum efficiency

$$u = \frac{v}{2} = \frac{49.52}{2} = 24.76 \text{ m/s}$$

Also $u = \frac{\pi DN}{60}$

$\therefore$ $24.76 = \frac{3.14\times D\times 2000}{60}$

Or $D = 0.2365 \text{ m} = 23.65 \text{ cm}$

$D \approx 24 \text{ cm}$

259. Draft tube for a reaction turbine is arranged for

(a) Safety purpose only
(b) Increasing velocity head
(c) Converting kinetic head into pressure head
(d) Diverting water only

APPSC AEE Screening Test 2016

Ans. (c) : Draft tube used in reaction turbine for main purpose i.e. to convert the excess velocity head available at the exit of the runner into pressure head.

260. The magnitude of water hammer in the flow of a liquid through a pipe does not depend upon

(a) Length of pipe
(b) Elastic properties of pipe material
(c) Temperature of liquid
(d) Time of valve closure

UPPSC AE 12.04.2016 Paper-II

Ans. : (c) The pressure rise due to water hammer depends upon
(i) The velocity of flow of water in pipe
(ii) The length of pipe
(iii) Time taken to close the valve
(iv) Elastic properties of the material of the pipe
The following cases of water hammer in pipe will be considered
(i) Gradual closure of valve
(ii) Sudden closure of valve and considering pipe rigid
(iii) Sudden closure of valve and considering pipe elastic

261. Impulse turbines work on the principle of

(a) Newton's first law
(b) Newton's second law
(c) Newton's third law
(d) Conservation of mass

RPSC LECTURER 16.01.2016

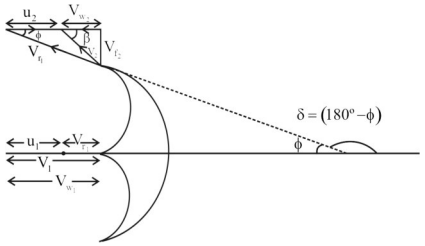
Ans. (b) : Impulse turbines work on the principle of Newton's second law.

262. Which of the following statements is correct for Pelton wheel (where, V_W = whirl velocity at inlet tip of bucket, V = absolute velocity of jet before striking of bucket, u = absolute velocity of bucket)?

(a) $V_W = V$ (b) $V_W = V + u$
(c) $V_W = u$ (d) $V_W = V - u$

UPRVUNL AE 2016

Ans. (a) : For pelton wheel turbine
Whirl velocity at input tip V_w= absolute velocity of jet before striking of bucket V.



263. Francis turbine is usually used for

(a) law head installation up to 30 m
(b) medium head installation from 30 m to 180 m
(c) high head installation above 180
(d) for all heads

RPSC AE 2016

Ans. : (b)

Turbine	Head
Pelton turbine	High head installation above 180 m
Francis turbine	medium head installation from 30m to 180 m.
Kaplan turbine	low head installation up to 30 m.

264. Operating charges for same power output are minimum for

(a) gas turbine plant (b) hydroelectric plant
(c) thermal plant (d) nuclear plant

RPSC AE 2016

Ans. : (b) Operating charges for same power output are minimum for hydro electric plant.

265. Load centre in a power station is

(a) centre of coal fields
(b) centre of maximum load of equipments
(c) centre of gravity of electrical load
(d) centre of power station

RPSC AE 2016

Ans. : (c) Load centre in a power station is centre of gravity of electrical load.

266. An impulse turbine produces 50 kW of power when the blade mean speed is 400 m/s. what is the rate of change of momentum tangential to the rotor?

(a) 200 N (b) 175 N
(c) 150 N (d) 125 N

RPSC IOF & Boiler 20.09.2017

Ans. : (d) $P = F\times v$

$50\times 1000 = F\times 400$

(F = rate of change of momentum)

$F = \frac{50000}{400}$

$\boxed{F = 125 \text{ N}}$

267. Which of the following components of reaction turbine increases the head on the turbine by an amount equal to the height of runner outlet above the tail race?

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(a) Draft tube (b) Guide vanes
(c) Scroll casing (d) Moving blades

APPSC AEE MAINS 2016 PAPER-III

Ans. (a) : Draft tube is the components of reaction turbine increases the head on the turbine by an amount equal to the height of runner outlet above the tail race.

268. Which of the following is lowest specific speed turbine?
(a) Pelton wheel with double jet
(b) Francis turbine
(c) Pelton wheel with single jet
(d) Kaplan turbine

UPRVUNL AE 2016

Ans. (c) : Turbine specific speed

Pelton wheel (with single jet)	0 - 30
Pelton wheel (with double jet)	30 - 60
Francis turbine	60 - 300
Propeller turbine	300 - 600
Kaplan turbine	600 - 1000

269. Which of the following hydraulic turbine has lowest specific speed
(a) Kaplan turbine (b) Propeller turbine
(c) Impulse turbine (d) Francis turbine

RPSC LECTURER 16.01.2016

Ans. (c) :

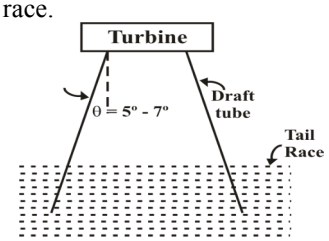
Turbine	Specific Speed
1. Single jet impulse turbine	10 - 30
2. Double jet impulse turbine	30 - 60
3. Francis turbine	60 - 300
4. Propeller turbine	300 - 600
5. Kaplan turbine	600 - 1000

270. Which of the following statements is not correct for draft tube
(a) It allows the negative head at the outlet of runner
(b) It is of gradually increasing area of cross-section
(c) It increases the efficiency of turbine
(d) It converts the pressure energy of the outlet of turbine into useful kinetic energy

RPSC LECTURER 16.01.2016

Ans. (d) : Functions of draft tube—
1. It allows the negative head at the outlet of runner
2. It converts the kinetic energy at the outlet of the turbine into useful pressure energy.
3. Draft tube increase total net head and efficiency of turbine.

Draft tube— The draft is a gradually increasing area of cross-section which connects between the water turbine exit and tail race.



271. A model of Francis turbine one fifth of full size, develops 3 kW at 306 rpm under a head of 1.77 m. Speed of the full size turbine operating under head of 1.77 m if efficiency of the model and full size turbine are same will be
(a) 109.8 rpm (b) 209.8 rpm
(c) 259 rpm (d) 300.8 rpm

GPSC ARTO 01.05.2016

Ans. (a) : $P_1 = 3kW$, $N_2 = 306$ rpm, $H_1 = 1.77$ m
$D_m/D_p = 1/5$, $H_2 = 5.7$ m. $N_2 = 1$
For the same efficiency of the model and prototype

$$\frac{D_m N_m}{\sqrt{H_m}} = \frac{D_p N_p}{\sqrt{H_p}} \Rightarrow N_p = \frac{D_m}{D_p}\frac{N_m}{\sqrt{H_m}} \times \sqrt{H_p}$$

$$\boxed{N_p = \frac{1}{5} \times \frac{306}{\sqrt{1.77}}\sqrt{5.7} = 109.8\,\text{rpm}}$$

272. Cavitation in water turbine results in :
(a) Rough surfaces (b) Noisy operation
(c) Drop in discharge (d) All the above

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (d) : The effects of cavitation are hydraulic (low efficiency due to flow instability) and mechanical (surface damage, noise and vibration). It may also lead to surface erosion. It is difficult to avoid cavitation in turbines, which cannot be avoided completely but can be reduced.

273. Under part load operation, the turbine which is least efficient is :
(a) Pelton turbine (b) Francis turbine
(c) Kaplan turbine (d) Propeller turbine

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (d) : Part load operation occurs when a system or device is operating below its maximum rated capacity.
- Under part load operation propeller turbine is least efficient.
- Under part load operation Kaplan turbine is most efficient because Kaplan turbine blades are arjustable.

274. Francis turbine is a :
(a) radial flow turbine
(b) axial flow turbine
(c) mixed flow turbine
(d) inward flow radial turbine

Kerala PSC IOF 19.04.2016

Ans. (c)

275. The magnitude of water hammer depends upon ..... .
(a) elastic properties of pipe material only
(b) length of pipe only
(c) rate of stoppage of flow only
(d) all three mentioned here

GWSSB DEE 07.07.2016

Ans. (d) : The magnitude of water hammer depends upon–
(1) Elastic properties of pipe material
(2) Length of pipe
(3) Rate of stoppage of flow

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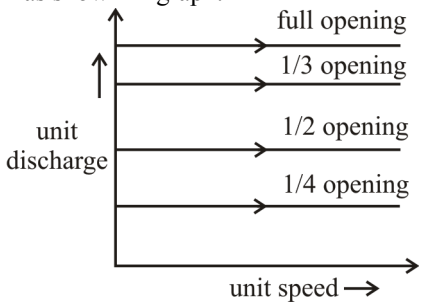
276. Performance characteristic curves indicate that unit discharge is independent of unit speed for

(a) Pelton wheel
(b) Low specific speed Francis turbine
(c) High specific speed Francis turbine
(d) Kaplan turbine

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (a) : For pelton turbine characteristics Graph between unit speed Vs unit discharge shows straight line. Such as shown in graph.



277. The speed of a geometrically similar turbine such that when it is developing 1 kW the total suction head is equal to 1 meter. The speed is known as

(a) specific speed
(b) unit speed
(c) suction specific speed
(d) critical speed

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (a) : The specific speed of a hydraulic turbine is define as the speed of a geometrically similar hydraulic turbine that would develop 1 kW power under 1m head of water column.

278. A penstock is 3000m long. Pressure wave travels in it with a velocity of 1500m/s. If the turbine gates are closed uniformly and completely in a period of 4.5 seconds, then it is called

(a) rapid closure (b) slow closure
(c) sudden closure (d) uniform closure

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (b) : We know, that critical time is*

$$T_0 = \frac{2L}{C} \qquad T_0 = \frac{2\times 3000}{1500}$$

$T_0 = 4$ Sec.

Actual time for the closure is 4.5 sec. So a slow closure will occur.

279. A reaction turbine discharge $35m^3/s$ under a head of 9m and with an overall efficiency of 91%. The power developed in kW is

(a) 286.65 (b) 37.49
(c) 3.822 (d) 28.665

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (c) : We know,

$$\eta = \frac{P_0}{P_i}$$

$P_i = \rho g Q h \qquad \Rightarrow P_0 = ?$

$\rho = 1000\ kg/m^3, \qquad h = 9m, \qquad \eta = 91\%$

$P_i = 1000 \times 9.81 \times 35 \times 9m$

$$0.91 = \frac{P_0}{1000\times 9.81\times 35\times 9}$$

$$\boxed{P_0 = 3.822\ kW}$$

280. A draft tube is fitted to the exit of a reaction turbine, located at 1.82m above tail race level. The pressure at the entrance section of draft tube is

(a) –5 (b) –3.82m
(c) 10.2m (d) 1.82m

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (a) : We know,

$$\eta_a = \frac{\left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g}\right)}{\left(\frac{V_1^2}{g}\right)} \qquad [\because V_1 = 10\ m/s]$$

$\eta_d = 75\%$

We get V_2

So, for entry pressure-

$$\frac{P_1}{\rho g} = \frac{P_a}{\rho g} - H_s - \left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g} - h_f\right)$$

$$\boxed{\frac{P_1}{\rho g} = -5m}$$

281. A Pelton wheel has a mean bucket speed of 10m/s with a jet of water flowing at the rate of 700 liter/sec under a head of 30m. The bucket deflects the jet through an angle of 160°. Theoretical velocity of jet is 24.26 m/s. Assuming the coefficient of velocity 0.98, the velocity of whirl at inlet in cm/sec is

(a) 2377 (b) 23.77
(c) 237.7 (d) 23770

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (a) : Given,

$u = 10$ m/s, $\quad Q = 0.7\ m^3/s, \quad \phi = 180 - 160 = 20^0$

$H = 30m, \quad V_v = 0.98$

$g = 10\ m/s^2$

$V = C_v\sqrt{2gH} \quad = 0.98 \times \sqrt{2\times 10\times 30}$

$= 24$ m/s $= 2400$ cm/s

$(\because$ due to g = 10 m/s)

282. Two jets strikes the buckets of a Pelton wheel, which is having shaft as 15450kW. The diameter of each jet is given as 200 mm. If the net head on the turbine is 400m, the power at the inlet of turbine is (Take $C_v = 1.0$)

(a) 21817.44 kW (b) 2181.17 kW
(c) 21.81 kW (d) 21817.44 W

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (a) : We know-

$$\therefore\ \eta_0 = \frac{P_0}{P_i}$$

$P_i = \rho g h \quad Q = 1000 \times 9.81 \times 400 \times Q$

$$Q = A_1V_1 \Rightarrow A_1 = \pi/4d^2 = \frac{\pi}{4}\times(0.2)^2$$

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$A_1 = 0.031\ m^2$

$V_1 = C_v\sqrt{2gH} = 1\times\sqrt{2\times9.81\times400}$

$V_1 = 88.58$ m/s

$P_i = 1000 \times 9.81 \times 400 \times 0.031 \times 88.58$

$\boxed{P_i = 21817.44\ kW}$

283. Windage is :

(a) Effect of runner speed on wind
(b) Winding of electrical wire on motor
(c) Friction between pelton wheel and atmosphere
(d) Cavitation

RPSC ADE 2016

Ans. (c) : Windage is air resistance of a moving object such as a vessel or a rotating machine part or the force of the wind on a stationary object.

284. The function of draft tube is :

(a) Recuperation of energy
(b) To make it possible to establish the turbine above tail race
(c) Both recuperation of energy and to make it possible to establish the turbine above tail race
(d) None of these

RPSC ADE 2016

Ans. (c) : Draft tube is a conical passage which convert K.E. into pressure energy used in reaction turbine. It also make possible to install the turbine above tail race.

285. A reaction turbine discharges 50 m³/sec of water under a head of 7.5 m with an overall efficiency of 80%. The H.P developed is :

(a) 5000 (b) 300
(c) 4000 (d) None of these

RPSC ADE 2016

Ans. (c) :

$$\text{Overall efficiency} = \frac{\text{output (shaft power)}}{\text{input (power of water)}} = \frac{P_{output}}{\rho g h Q}$$

$$0.80 = \frac{\text{output power}}{(50\times1000)\times9.81\times7.5}$$

Power output = 2943 kW

$$= \frac{2943}{0.746}\text{HP} = 3945\ \text{HP} \approx 4000\ \text{HP}$$

286. A large hydraulic turbine is to generate 300 kW at 1000 rpm under a head of 40 m. For initial testing, a 1:4 scale model of the turbine operates under a head of 10 m. The power generated by model (in kW) will be :

(a) 2.34 (b) 4.68
(c) 9.38 (c) 18.75

RPSC ADE 2016, GATE 2006

Ans. (a) : $u \propto \sqrt{2gH}$

$$u \propto \frac{\pi DN}{60}$$

$\sqrt{H} \propto DN$

$$\left(\frac{D^2N^2}{H}\right)_P = \left(\frac{D^2N^2}{H}\right)_M$$

$$\frac{4^2\times100^2}{40} = \frac{1^2\times N_M^2}{10} \quad \text{or } N_M = 2\times100 = 200$$

$$\text{Specific speed } \left(\frac{N\sqrt{P}}{H^{5/4}}\right)_P = \left(\frac{N\sqrt{P}}{H^{5/4}}\right)_M$$

$$= \frac{1000\sqrt{300}}{40^{5/4}} = \frac{200\sqrt{P}}{10^{5/4}}$$

or P = 2.37 kW

287. Gross head of a hydro-power station is:

(a) The height of water level in the river where the storage is provided
(b) The height of the water level in the river tail race is provided
(c) The difference between maximum and minimum storage level of river
(d) The difference of water level between the level in the storage and tail race

RPSC VPITI 14.02.2016

Ans. (d) : Gross head of a hydro-power station is the difference of water level between the level in the storage and tail race is called gross head.

Net head (H) = Gross head (H_G) – Head loss (H_F)

$$H_F = \frac{fLV^2}{2gd}$$

f = Darcy friction factor

288. A Francis turbine working at 400 rpm has a unit speed of 50 rpm and develops 500 KW of power. What is the effective head under which this turbine operates?

(a) 49 m (b) 64 m
(c) 81 m (d) 100 m

RPSC VPITI 14.02.2016, ESE 2009

Ans. (b) : Given-

Unit speed Nu = 50 rpm

Turbine speed N = 400 rpm

$$\text{Effective head } H = \left(\frac{N}{Nu}\right)^2$$

$$H = \left(\frac{400}{50}\right)^2$$

H = 64 m

289. Consider the following turbines

(1) Francis turbine
(2) Pelton wheel with 2 or more jets
(3) Pelton wheel with single jet
(4) Kaplan

The correct sequence of these turbines in increasing order of their specific speed is

(a) 2, 3, 1, 4 (b) 3, 2, 1, 4
(c) 2, 3, 4, 1 (d) 3, 2, 4, 1

TSPSC AEE 2015

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Ans. (b)

Turbine	Specific speed
Pelton wheel with single jet	10 - 30
Pelton wheel with 2 or more jets	30 - 60
Francis turbine	60 - 300
Propeller turbine	300 - 600
Kaplan turbine	600 - 1000
Turbo turbine	20 - 80

290. In reaction turbine:

(a) The vanes are partly filled
(b) Total energy of fluid is converted to kinetic energy in the runner
(c) It is exposed to atmosphere
(d) It is not exposed to atmosphere

OPSC AEE 2015 Paper - II

Ans : (d) Reaction turbine:- Reaction turbine means that the water at the inlet of the turbine possesses Kinetic energy as well as pressure energy. As the water flows through the runner, a part of pressure energy goes on changing into kinetic energy. The water through the runner in under pressure. The runner is completely enclosed in an air- tight casing and the runner is always full of water.

291. Capacity of a hydroelectric plant in service in excess of the peak load is known as:

(a) Operating reserve (b) Spinning reserve
(c) Peak reserve (d) Cold reserve

OPSC AEE 2015 Paper - II

Ans : (d) Capacity of a hydroelectric plant in service in excess of the peak load is known cold reserve.

292. A hydraulic turbine generated 500 KW at 1500 rpm under a head of 60 m. During testing conditions, a scale model of 5 works under the head of 15m. the power generated by the model will be.

(a) 2 kW (b) 1.6 kW
(c) 3 kW (d) 2.5 kW

OPSC AEE 2015 Paper - II

Ans. : (d) Given data for prototype power generated

$P_P = 500$ KW
$N_P = 1500$ rpm
$H_P = 60$ m
for model
Power generated
$P_m = ?$
$H_m = 15$ m

Scale ratio : $\frac{D_P}{D_m} = 5$

We know that the head coefficient

$$C_H = \frac{gH}{N^2D^2}$$

$$(C_H)_m = (C_H)_P$$

$$\left(\frac{gH}{N^2D^2}\right)_m = \left(\frac{gH}{N^2D^2}\right)_P$$

$$\frac{H_m}{N_m^2D_m^2} = \frac{H_P}{N_P^2D_P^2}$$

$$\frac{H_m}{H_P}\left(\frac{D_P}{D_m}\right)^2 = \left(\frac{N_m}{N_P}\right)^2$$

$$\frac{15}{60}(5)^2 = \frac{N_m^2}{(1500)^2}$$

$$N_m = 3750\,\text{rpm}$$

Power coefficient

$$C_P = \frac{P}{\rho N^3D^5}$$

$$(C_P)_m = (C_P)_\rho$$

$$\left(\frac{P}{\rho N^3D^5}\right)_m = \left(\frac{P}{\rho N^3D^5}\right)_\rho$$

$$\frac{P_m}{N_m^3D_m^5} = \frac{P_\rho}{N_P^3D_P^3}, \rho = \text{constant}$$

$$P_m = P_P\left(\frac{N_m}{N_P}\right)^3\left(\frac{D_m}{D_P}\right)^5$$

$$P_m = 500\left(\frac{3750}{1500}\right)^3\left(\frac{1}{5}\right)^5$$

$$\boxed{P_m = 2.5\,\text{KW}}$$

293. A hydro plant is having a head of 150 metres. The turbine that best suits this height is:

(a) Pelton (b) Francis
(c) Kaplan (d) Turgo wheel

NPCIL ET, 2015

Ans. (b) :

Turbine	Head
Pelton Turbine	above 250 m
Francis	60 to 250 m
Kaplan	10 to 60 m

294. Any change in load is adjusted by adjusting following parameter on turbine

(a) Net head (b) Absolute velocity
(c) Blade velocity (d) Flow

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : Any change in load is adjusted by adjusting following parameter on turbine flow.

295. A draft tube is a must in :

(a) Francis turbine (b) Pelton turbine
(c) axial pump (d) Centrifugal pump

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Draft tube must in Francis turbine.
A draft tube is a conduit attachment to the reaction turbine exit to achieve the followings benefits–

(i) To enable the turbine to be set up at an elevation higher than the tail water level.
(ii) To utilize a major part of the kinematic energy of the water exiting the turbine.
(iii) Draft tube in reaction turbine to convert residual kinetic energy in to pressure energy.

296. The speed of a turbine and discharge through turbine are proportional to

(a) Head H (b) $\sqrt{H}$
(c) H^2 (d) $H^{3/2}$

Nagaland PSC CTSE 2015 Paper-II

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Ans. (b) : Discharge through turbine is directly proportional to $\sqrt{H}$.

For model relationship,

$\frac{H}{N^2D^2}$ = Constant; $\frac{Q}{ND^3}$ = Constant, $\frac{P}{N^3D^5}$ = Constant

$Q \propto N$, So, $Q \propto \sqrt{H}$

297. A turbine generates 10 MW under a head of 100 m while working with a speed of 500 rpm. The turbine selected is :

(a) Francis (b) Kaplan
(c) Pelton (d) Propeller

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : We know that,

Specific speed $N_s = \frac{N\sqrt{P}}{H^{5/4}}$

$$N_s = \frac{500\sqrt{10\times1000}}{100^{5/4}}$$

$N_s = 158.11$

Francis turbine selected.

298. For a free vortex design of blade in the rotor or a reaction axial turbine, the specific work along the blade height is

(a) higher at the blade hub and lower at the blade tip
(b) constant from hub to tip
(c) lower at the hub and tip but different from the mean section
(d) same at the hub and tip but different from the mean section

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : $\because$ For free vortex $V_w \propto \frac{1}{r}$

$\therefore$ Free vortex design of blade in the rotor or a reaction axial turbine, the specific work along the blade height is higher at the blade hub and lower at the blade tip.

299. Which one of the following is false

(a) Pelton wheel turbine is an axial flow turbine
(b) If at the inlet of the turbine, the energy available is only kinetic energy, the turbine is known as impulse turbine
(c) Radial flow turbines are those in which water flows in the radial direction
(d) In Francis turbine, the water leaves in the radial direction

TSGENCO AE 14.11.2015

Ans. (a) : Impulse turbine – These have only kinetic energy at inlet.

e.g. Pelton wheel

Reaction turbine – These have kinetic energy and pressure both at inlet.

e.g. – Francis turbine, propeller turbine, Kaplan turbine.

Pelton wheel is a tangential flow turbine, Francis is inward radial flow turbine and propeller and Kaplan turbine is axial flow turbine.

300. A turbine develops 8000 kW when running at 100 rpm. The head on the turbine is 36 m. If the head is reduced to 9m, the power developed by the turbine will be

(a) 16000 kW (b) 4000 kW
(c) 1414 kW (d) 1000 kW

MPPSC AE 08.11.2015

Ans. (d) : $\frac{P_1}{(H_1)^{3/2}} = \frac{P_2}{(H_2)^{3/2}}$

$$\Rightarrow \frac{8000\text{kW}}{(36)^{3/2}} = \frac{P_2}{(9)^{3/2}} \Rightarrow P_2 = \frac{8000\text{kW}\times(3)^3}{(6)^3}$$

$$\Rightarrow P_2 = \frac{8000\times27}{216}$$

$\Rightarrow P_2 = 1000$ kW

301. Specific speed of an impulse turbine (Pelton wheel) ranges from

(a) 10-40 (b) 50-100
(c) 60-250 (d) 300-800

MPPSC AE 08.11.2015

Ans. (a) : Range of specific speed (N_s) of an Impulse turbine (Pelton wheel) is-

10-40 rpm

302. In a reaction turbine:

(a) Flow can be regulated without loss
(b) There is only partial conversion of available head to velocity head before entry to runner
(c) The outlet must be above the tail race
(d) Water may be allowed to enter a part or whole of wheel circumference

MPPSC AE 08.11.2015

Ans. (b) : There is only partial conversion of available head to velocity head before entry to runner in a reaction turbine.

303. The power obtained from an impulse an impulse turbine is proportional to the number of nozzles used. A pelton turbine with six nozzles has a specific speed of 8.1. The specific speed per nozzle is :

(a) 1.35 (b) 2
(c) 3.3 (d) 8.1

KPSC AE 2015, Kerala PSC AE 06.08.2015

Ans. (c) : $N_S = \frac{N\sqrt{P}}{H^{5/4}}$

$N_S \propto \sqrt{P}$ $\{\because P \propto n\}$ where, n = no. of nozzles

$N_S \propto \sqrt{P} \propto \sqrt{n}$

$\frac{N_S}{\sqrt{n}}$ = constant

$$\frac{x}{8.1} = \frac{1}{\sqrt{6}} \Rightarrow x = \frac{8.1}{\sqrt{6}} = 3.3$$

304. The impulse turbine among the following is

(a) Pelton wheel (b) Francis turbine
(c) Kaplan turbine (d) Propeller turbine

Kerala LBS Centre for Sci. and Tech. Asstt. Prof. 2014

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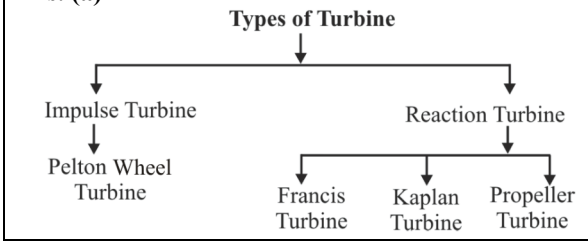
Ans. (a)



305. The Francis turbine is suited for
(a) High head and low discharge
(b) High head and high discharge
(c) Low head and low discharge
(d) Medium head and medium discharge

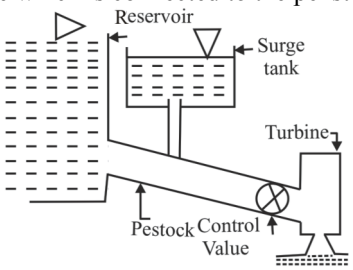
Kerala LBS Centre for Sci. and Tech. Asstt. Prof. 2014

Ans. (d) : Francis Turbine is a reaction turbine which suited for radial flow, medium specific speed, medium head and medium discharge.

306. The surge tanks are normally integral part of power plant as they prevent immediate damage to:
(a) Tail race (b) Turbine
(c) Penstocks (d) Draft tube
(e) Water hammer

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : Surge tank - A Simple surge tank is like vertical pipe which is connected to the penstock.



Function of surge tank–
1. Upon the rapid closure of the turbine, water masses moving in the pressure tunnel and in the penstock are suddenly decelerated .
2. The surge provides protection to the penstock against damage of water hammer.
3. The third purpose of the surge tank is to provide water supply to the turbines in case of starting up.

307. Consider the following statements:
1. A reverse jet protects Pelton turbine form over-speeding
2. Runner blades of Francis turbine are adjustable.
3. Draft tube is used invariably in all reaction turbine installation.
4. Pelton wheel is a tangential flow impulse turbine
5. The peripheral velocity at inlet and outlet are same in Kaplan turbine

Which of the following statements are correct.
(a) 1,2 and 5 are correct (b) 2 and 3 are correct
(c) 3,4 and 5 are correct (d) 3 and 4 are correct
(e) 1,2 and 3 are correct

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c)
1. Draft tube is used invariably in all reaction turbine installations.
2. Pelton wheel is a tangential flow impulse turbine.
3. The peripheral velocity at inlet and outlet are same in Kaplan turbine

308. In the case of reaction radial flow turbine, velocity ratio is defined as $\frac{V}{\sqrt{2gh}}$ where H is the available head and V is the
(a) Absolute velocity at the draft tube inlet
(b) Mean velocity of flow in the turbine
(c) Absolute velocity at the guide vane inlet
(d) Tangential velocity of wheel at inlet
(e) Tangential velocity of wheel at outlet

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : In the case of reaction radial flow turbine, velocity ratio is defined as $\frac{V}{\sqrt{2gh}}$ where H is the available head and V is the tangential velocity of wheel at inlet.

309. The draft tube in a reaction turbine is used
(a) To minimize the whirl losses at the turbine outlet
(b) To change a large part of pressure energy at turbine outlet into kinetic energy
(c) To enable vertical operation of the draft
(d) To minimize the loss of kinetic energy at the turbine outlet
(e) To increase the potential energy of the fluid at turbine outlet

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : Draft tube - A draft tube is a kind of tube which connects between the water turbine exit and tail race.
Due to high K.E. at exit of reaction turbine, there would be a loss of overall reduction in K.E. So to prevent this loss, the draft tube is provided which produces net efficiency.
Functions of Draft tube -
- It is to increase the pressure from inlet to outlet of the draft tube as it flows through it and hence increase it more than atmospheric pressure.
- To safely discharge the water that has worked on the turbine to the tail race.

310. The range of speed factor for Francis turbine is
(a) 0.20-0.30 (b) 0.30-0.45
(c) 0.45-0.70 (d) 0.70-0.85
(e) 0.85-0.95

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) : The range of speed factor for Francis turbine is 0.70 - 0.85

311. Flow ratio is
(a) Flow velocity/blade velocity
(b) Flow velocity/Relative velocity
(c) Flow velocity/$\sqrt{2}$gH
(d) Flow velocity/ whirl velocity
(e) Flow velocity/ absolute velocity

CGPSC AE 26.04.2015 SHIFT-II

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Ans. (c)

$$\text{Flow Ratio} = \frac{\text{Flow Velocity}}{\sqrt{2gH}}$$

312. If the specific speed of a turbine is the range of 300 – 1000 then the turbine is

(a) Pelton (b) Francis
(c) Kaplan (d) Mixed

TNPSC AE 2014

Ans. (c) : If the specific speed of a turbine is the range of 300 – 1000 then the turbine is Kaplan.

Turbine	Specific speed (N_s)
Impulse turbine	10 - 60
Francis turbine	60 - 300
Propeller turbine	300 - 600
Kaplan turbine	600 - 1000

313. For a given head the discharge through a pelton turbine with increase in speed

(a) Decreases
(b) Increases
(c) Does not change
(d) First increases then decreases

TNPSC AE 2014

Ans. (b) : We know that for a given head for turbine, speed of turbine N is directly proportional to discharge (Q).

$$Q \propto N$$

314. The hydraulic unit used to minimize frictional loss in pipe line is called as

(a) safety tank (b) surge tank
(c) relief tank (d) pressure tank

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : The hydraulic unit used to minimize frictional loss in pipe line is called as pressure tank.

315. Pelton turbine is a

(a) Reaction turbine (b) Impulse turbine
(c) Radial flow turbine (d) Axial flow turbine

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (b) : Impulse turbine–If at the inlet of the turbine, the energy available is only kinetic energy then the turbine is known as an impulse turbine e.g. pelton wheel turbine.

316. The ratio of power produced by the turbine to the energy actually supplied by the turbine is called

(a) mechanical efficiency
(b) hydraulic efficiency
(c) overall efficiency
(d) turbine efficiency

Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : The ratio of power produced by the turbine to the energy actually supplied by the turbine is called turbine efficiency.

317. The hydraulic efficiency of a turbine with mean bucket diameter 1m and running at 1000 rpm with net head 700m when the side clearance angle is 15 degree and discharge of 100 liters/s is :

(a) 60% (b) 70%
(c) 85% (d) 97%

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) :

318. The specific speed of a Francis turbine which develops 7225 kW power under a head of 25 meters at 135 r.p.m is :

(a) 150 (b) 180
(c) 195 (d) 205

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) :

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

$$= \frac{135\sqrt{7225}}{25^{5/4}} = 205.27 \text{ rpm}$$

319. An example of a turbine with high head and low specific speed is :

(a) Kaplan (b) Francis
(c) Pelton (d) None

Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : Pelton turbine

320. The overall efficiency of a Pelton turbine is 70%. If the mechanical efficiency is 85%, what is its hydraulic efficiency?

(a) 82.4% (b) 59.5%
(c) 72.3% (d) 81.5%

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Given, $\eta_o = 70\%$, $\eta_m = 85\%$, $\eta_H = ?$

$$\eta_o = \eta_m \times \eta_H$$

$$\eta_H = \frac{\eta_o}{\eta_m} = \frac{70}{85} \times 100$$

$$\eta_H = 82.35\%$$

$$\eta_H = 82.4\%$$

321. In axial flow fans and turbines, fluid enters and leaves as follows :

(a) radially, axially (b) axially, radially
(c) axially, axially (d) radially, radially

J&K PSC Civil Services Pre, 2013

Ans. (c) : Axial flow means axial inlet and axial outlet. Example – exhaust, ceiling fan.

322. Which of the following is a function of surge tank provided in hydroelectric power plants?

(a) To provide backup storage of water to meet demand in the months when the water level in the dam reduces.
(b) To increase the efficiency of the turbine.
(c) To avoid cavitation in the turbine.
(d) To reduce water hammer effect due to sudden closing of water supply to turbine in the event of reduced load.

RPSC AEN Pre-2013

Ans. (d) : In hydroelectric power plants the function of surge tank is to reduce water hammer effect due to sudden closing of water supply to turbine in the event of reduced load.

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323. In which of the following methods, the available energy is directly converted to electrical energy without the aid of a turbine or engine?

(a) Magneto Hydrodynamic system
(b) Tidal power plant
(c) Geothermal power plant
(d) Solar thermal power plant

RPSC AEN Pre-2013

Ans. (a) : Magneto Hydro Dynamic System- It is a direct energy conversion system which converts the available energy directly into electrical energy without any intermediate mechanical energy conversion unlike in all other power generating plant.

324. Which relation is incorrect?

(a) Francis Turbine - Impulse Turbine
(b) Pelton Turbine - Impulse Turbine
(c) Kaplan Turbine - Reaction Turbine
(d) Francis Turbine - Reaction Turbine

SJVN ET 2013

Ans. (a) : Pelton Turbine - Impulse Turbine
Francis Turbine - Reaction Turbine
Kaplan Turbine - Reaction Turbine

325. In hydro turbines, cavitation results in:

(a) Pitting on metallic surfaces of runner blades
(b) Long life for blades
(c) More power generation efficiency
(d) All of these

SJVN ET 2013

Ans. (a) : Cavitation in hydraulic machine occurs when static pressure at any location is close to or lower than vapour pressure and we know that sum of velocity head and pressure head is constant at constant elevation. So the location where velocity is maximum, pressure will be minimum or close to vapour pressure. Hence maximum velocity location in machines are probable points of cavitation. In pump it is occurs at the entry to impeller and in turbine it is exit from runner or entry to draft tube.

326. The cavitation in reaction type hydraulic turbine is avoided by

(a) Using high polished blades
(b) Using stainless steel runner
(c) Running turbine at designed speed
(d) All of above

SJVN ET 2013

Ans. (d) : The cavitation in reaction type hydraulic turbine is avoided by
- Using high polished blades.
- Using stainless steel runner.
- Running turbine at designed speed.

327. Running cost of the hydroelectric power plant is

(a) more than the running cost of steam power plant
(b) less than the running cost of steam power plant
(c) equal to running cost of steam power plant
(d) equal to running cost of nuclear power plant

TNPSC AE 2013

Ans. (b) : Running cost of the hydroelectric power plant is less than the running cost of steam power plant.

328. The velocity ratio of a fluid whose velocity is 330 m/s and blade velocity is 150 m/s

(a) 1 (b) 2.2
(c) 0.45 (d) 1.5

TNPSC AE 2013

Ans. (c) : Velocity ratio of a fluid is given as

$$\text{V.R.} = \frac{\text{Velocity of blade}}{\text{Fluid velocity}}$$

$$= \frac{150}{330} = 2.2$$

329. A pump lifts 50 m^3 of water to a tank at a height of 50 m. What is the work done to lift the water?

(a) 24.525 MJ (b) 20.525 MJ
(c) 32.525 MJ (d) 18.525 MJ

TNPSC AE 2013

Ans. (a) : Work done to the lift the water by pump is given by

$$W = \rho \times g \times Q \times H$$
$$= 1000 \times 9.81 \times 50 \times 50$$
$$= 24.525 \text{ MJ}$$

330. Run away speed for Francis turbine ranges:-

(a) 1.8 to 1.9 times normal speed
(b) 2.5 to 3 times normal speed
(c) 3 to 3.5 times normal speed
(d) 2 to 2.2 times normal speed

UKPSC AE-2013, Paper-II

Ans. (d) : Run away speed for Francis turbine ranges of order 2 to 2.2 times normal speed.

331. A Pelton turbine works under a head of 405 m and runs at 400 rpm. What will be diameter of its runner? ($k_n = 0.45$):-

(a) 1.93m (b) 3.80m
(c) 2.10m (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (a) : Given as,
h = 405 m
N = 400 rpm
$k_n = 0.45$
We know that,

$$\text{Speed ratio, } (k_n) = \frac{u}{V}$$

$$u = \sqrt{2gh} \cdot k_n$$

$$\frac{\pi DN}{60} = \sqrt{2gh} \cdot k_n$$

$$D = \frac{\sqrt{2 \times g \times 405} \times 0.45 \times 60}{\pi \times 400}$$

$$\boxed{D = 1.9162 \text{ m}}$$

332. A Kaplan turbine develops 3000 kW power under a head of 5m and a discharge of 75 m^3/s. Determine the overall efficiency:-

(a) 0.79 (b) 0.82
(c) 0.90 (d) None of the above

UKPSC AE-2013, Paper-II

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Ans. (b) : Given as,

$P = 3000$ kW

$h = 5$ m

$Q = 75\ m^3/s$

$\eta_0 = ?$

Overall efficiency of Kaplan turbine,

$$\eta_0 = \frac{\text{Shaft Power}}{\text{Water Power}(\rho_w Qgh)}$$

$$= \frac{3000 \times 1000}{1000 \times 75 \times 9.81 \times 5}$$

$$\eta_0 = 0.8154 \simeq 0.82$$

333. If the jet ratio in a Pelton turbine wheel is 18, the number of buckets will be about:-

(a) 24 (b) 21

(c) 26 (d) 18

UKPSC AE-2013, Paper-II

Ans. (a) : Number of buckets

$$N = 15 + \frac{\text{Jet Ratio}}{2} = 15 + \frac{18}{2}$$

$N = 24$

334. The specific speed of a turbine is:-

(a) Its maximum speed

(b) Its minimum speed

(c) Its speed at maximum output

(d) The speed of a geometrically similar turbine that would develop one metric horse power under a head of one meter

UKPSC AE-2013, Paper-II

Ans. (d) : The specific speed of a turbine is the speed of a geometrically similar turbine that would develop one metric horse power under a head of one meter.

335. The type of turbine recommended for a head of 10 meter is:-

(a) Francis turbine (b) Kaplan turbine

(c) Pelton turbine (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (b) : Kaplan turbine.

336. Which phenomenon will occur when the value at the discharge end of a pipe connected to a reservoir is suddenly closed?

(a) Cavitation (b) Erosion

(c) Hammering (d) Surging

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : When a liquid flowing through a long pipe is suddenly brought to rest by closing the valve at the end of a pipe, then a pressure wave of high intensity is produced behind the valve.

This pressure wave of high intensity has the effect of hammering action on the walls of the pipe. This phenomenon is known as water hammer or hammer blow.

337. Water hammer is developed in :

(a) Penstock (b) Draft tube

(c) Turbine (d) Surge tank

TNPSC AE (Industries) 09.06.2013

Ans. (a) : Water hammer is developed in penstock.

338. In impulse turbines, the energy transfer takes place at :

(a) Rotor (b) Nozzle

(c) Stator blades (d) Casing

TNPSC AE (Industries) 09.06.2013

Ans. (b)

339. The gross head at a turbine working with a specific speed at 200 rpm, having frictional head at 10 m and net head as 150 m is :

(a) 162.5 m (b) 160 m

(c) 172.5 m (d) 212 m

TNPSC AE (Industries) 09.06.2013

Ans. (b)

340. Cavitation depends upon

(a) vapour pressure which is function of temperature

(b) absolute pressure or barometric pressure

(c) suction pressure (H_s) which is height of runner outlet above tail race level.

(d) all the above

UKPSC AE 2012 Paper–II

Ans. (d) : all the above

341. Higher specific speed (300 to 1000) of turbine indicates that the turbine is :

(a) Pelton wheel (b) Kaplan

(c) Francis (d) none of these

TRB Asstt. Prof., 2012

Ans. (b)

S.N.	Specific speed	Types of Turbine
1	10-35	Pelton wheel with single jet
2	35-60	Pelton wheel with two or more jets
3	60-300	Francis Turbine
4	300-1000	Kaplan or Propeller Turbine

342. Reaction turbines are used for :

(a) low head

(b) high head

(c) high head and low discharge

(d) high head and high discharge

TRB Asstt. Prof., 2012

Ans. (a) : Reaction Turbine is used for low head, large discharge and high specific speed of turbine while impulse turbine is used for high head, small discharge and low specific speed.

343. A Francis turbine running at 200 rpm develops a power of 5000 kW under a head of 25m. The power output under a head of 100m will be

(a) 20, 000 kW (b) 30,000 kW

(c) 40,000 kW (d) 50,000 kW

ISRO Scientist/Engineer 2012

Ans. (c) : Francis Turbine

$N_1 = 200$ rpm

$P_1 = 5000$ kW

$H_1 = 25$ m

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$P_2 = ?$

$H_2 = 100$ m

Model and prototype,

$$\left(\frac{H}{N^2D^2}\right)_m = \left(\frac{H}{N^2D^2}\right)_p$$

Or $$\left(\frac{H}{N^2D^2}\right)_1 = \left(\frac{H}{N^2D^2}\right)_2 \quad [\because D \text{ are same}]$$

$$\left(\frac{H}{N^2}\right)_1 = \left(\frac{H}{N^2}\right)_2$$

$$\frac{H_1}{N_1^2} = \frac{H_2}{N_2^2}$$

$$N_2^2 = \frac{N_1^2 \times H_2}{H_1}$$

$$= \frac{100 \times (200)^2}{25}$$

$N_2 = 400$ rpm

Now, $$\left(\frac{P}{N^3D^5}\right)_1 = \left(\frac{P}{N^3D^5}\right)_2$$

$$\frac{P_1}{N_1^3} = \frac{P_2}{N_2^3}$$

$$P_2 = \frac{N_2^3 \times P_1}{(N_1)^3}$$

$$P_2 = 5000 \times \left(\frac{400}{200}\right)^3$$

$P_2 = 40{,}000$ kW

344. The power developed by a turbine is

(a) Directly proportional to $H^{1/2}$

(b) Inversely proportional to $H^{1/2}$

(c) Directly proportional to $H^{3/2}$

(d) Inversely proportional to $H^{3/2}$

APPSC AEE 2012

Ans. : (c) $N_u = \frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$

$N \propto H^2$

speed $\propto H^2$ ……..(i)

$Q_u = \frac{Q_1}{\sqrt{H_1}} = \frac{Q_2}{\sqrt{H_2}}$

$Q \alpha H^2$

Discharge $\propto H^2$ …….. (ii)

$P_u = \frac{P_1}{H_1^{3/2}} = \frac{P_2}{H_2^{3/2}}$

Power $\propto H^{3/2}$ …….. (iii)

345. The turbine to be used for 450 m head of water is

(a) Pelton wheel (b) Francis turbine

(c) Kaplan turbine (d) None of these

APPSC AEE 2012

Ans. : (a)

Turbine	Head
Pelton turbine	High head (above 250m)
Kaplan turbine	Low head (below 60m)
Francis turbine	Medium head (60-250m)

The turbine to be used for 450 m head of water is pelton wheel.

346. Hammer blow in pipes occurs due to

(a) sudden sharp bends

(b) sudden contraction

(c) sudden stoppage of flow

(d) sudden release of fluid from pipe

UKPSC AE 2012 Paper–II

Ans. (c) : sudden stoppage of flow

347. For a reaction turbine, specific flow is given by following expression :

(a) $\frac{Q}{D_1^2}H$ (b) $\frac{Q}{D_1}\sqrt{H}$

(c) $\frac{Q}{D_1^2}H^{3/2}$ (d) $\frac{Q}{D_1^2}\sqrt{H}$

UKPSC AE 2012 Paper–II

Ans. (d) : $\frac{Q}{D_1^2}\sqrt{H}$

348. Mean diameter of runner of a pelton turbine is 200mm and least diameter of jet is 1 cm. Calculate the jet ratio and number of buckets.

(a) 20, 25 (b) 200, 115

(c) 20, 40 (d) 20, 45

UKPSC AE 2012 Paper–II

Ans. (a) : Mean diameter (D) = 200 mm

Jet diameter (d) = 1 cm = 10 mm

Jet ratio $= \frac{D}{d} = \frac{200}{10} = 20$

No. of buckets $(n) = 15 + \frac{D}{2d} = 15 + \frac{200}{2 \times 10} = 15 + 10$

$(n) = 25$

349. A water turbine is usually designed for the given values of

(a) N, T and Q (b) P, T and Q

(c) P, H and Q (d) P, H and N

UKPSC AE 2012 Paper–II

Ans. (d) : P, H and N

350. Hydraulic efficiency of water turbine is the ratio of

(a) power from runner to water power at turbine entrance

(b) power of shaft to water power at turbine entrance

(c) power of shaft to power from runner

(d) power from runner to shaft power

TNPSC ACF 2012

Ans. (a) : Hydraulic efficiency η_h

$= \frac{\text{Power delivered to runner}}{\text{Power supplied at inlet}}$

$= \frac{\text{Runner Power}}{\text{Water Power}}$

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351. A typical pelton wheel bucket deflects the water through an angle of

(a) 150º (b) 159º
(c) 170º (d) 179º

TNPSC ACF 2012

Ans. (c) : 170º

352. In which type of turbine water enters the runner radially and leaves axially?

(a) Pelton (b) Propeller
(c) Kaplan (d) Francis

TNPSC ACF 2012

Ans. (d) : In Francis turbine water enters the runner radially and leave axially. In case of Francis turbine, the discharge at outlet is radial thus, velocity of whirl at outlet valve is zero.

353. Francis turbine usually used for

(a) high heads (b) medium heads
(c) low heads (d) very high heads

TNPSC ACF 2012

Ans. (b) :

Type of Turbine	**– Head**	**– Discharges**
Pelton (Impulse)	– High	– Small
Francis/Reaction	– Medium	–Medium
Kaplan/Reaction	– Low	– Large

354. The speed ratio for Pelton wheel varies from

(a) 0.45 to 0.50 (b) 0.6 to 0.7
(c) 0.3 to 0.4 (d) 0.8 to 0.9

APPSC AEE 2012

Ans. (a) : Speed ratio (K_u) of a pelton wheel is designed as the ratio of tangential velocity of wheel u to the theoretical velocity of jet

$$K_u = \frac{u}{\sqrt{2gh}} = 0.45 \text{ to } 0.50$$

355. Turbine gives best performance (i.e. work at peak efficiency) when they are operated at full or design load. The performance of many turbines deteriorate considerably at part loads. Which of the following turbines is best suited for operation at part loads. ?

(a) Pelton Turbine (b) Francis Turbine
(c) Propeller Turbine (d) Kaplan Turbine

UKPSC AE 2012 Paper–II

Ans. (d) : Kaplan Turbine

356 If a draft tube is used with a Francis turbine (installed above tail race level), the pressure at the runner outlet

(a) is equal to atmospheric pressure
(b) is above atmospheric pressure
(c) is below atmosphere pressure
(d) depends upon turbine speed

UKPSC AE 2012 Paper–II

Ans. (c) :

357. An impulse turbine operating with a single nozzle has a specific speed of 5. What will be the approximate specific speed of the turbine if the turbine is operated with one more additional nozzle of the same size?

(a) 4 (b) 6
(c) 7 (d) 10

TRB Poly. Lect., 2012

Ans. (c) : Power will be double with additional nozzle.

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}, \; N_s \propto \sqrt{P} \text{ or } \frac{(N_s)_2}{(N_s)_1} = \sqrt{\frac{P_2}{P_1}}$$

$$(N_s)_2 = (N_s)_1 \times \sqrt{P_2/P_1}$$

N & H are constant $(N_s)_2 = 5 \times \sqrt{2P/P}$

$$(N_s)_2 = 5\sqrt{2} = 7.071 \approx 7$$

358. In hydraulic reaction turbine, function of the draft tube is to :

(a) increase the flow rate
(b) reduce water hammer effect
(c) convert kinetic energy of water to potential energy by a gradual expansion in divergent part
(d) increase efficiency of the turbine

RPSC ACF-2011

Ans. (c) : In hydraulic reaction turbine, function of the draft tube is to convert kinetic energy of water to potential energy by a gradual expansion in divergent part. In reaction turbine changes in both pressure and velocity as the steam flows through the moving blades.

359. Francis turbine is usually used for :

(a) low head installation upto 30 m
(b) medium head installation from 30m to 180m
(c) high head installation above 180 m
(d) for all heads

RPSC ACF-2011

Ans. (b) : Francis turbine is usually used for medium head installation from 30m to 180m.

360. Pelton wheels are available in the following range of specific speed (N_S)

(a) 8 – 30 (b) 380 – 950
(c) 4 – 70 (d) 40 – 420

RPSC AE GWD, 2011

Ans. (a) :

Turbines		**Specific Speed**
Pelton	→	0 – 60
Francis	→	60 – 300
Propeller	→	300 – 600
Kaplan	→	600 – 900

361. In a Francis turbine, the discharge leaves the runner radially at the exit. For this turbine.

(a) The blade tip is radial at the outlet
(b) The blade tip is radial at the inlet
(c) The guide vane angle is 90°
(d) The absolute velocity is radial at the outlet

RPSC AE GWD, 2011

Ans. (d) : In a Francis turbine, the discharge leaves the runner radially at the exit for this the absolute velocity is radial at the outlet.

362. Which is not a part of a Pelton turbine ?

(a) Nozzle (b) Spear
(c) Draft tube (d) Breaking jet

OPSC Civil Services Pre. 2011

Ans. (c) : Draft tube is used only in Reaction Turbine.

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363. Which of the following statements is wrong?

(a) The reaction turbines are used for low head and high discharge

(b) The angle of taper on draft tube is less than 8°

(c) An impulse turbine is generally fitted slightly above tail race

(d) A Francis turbine is an impulse turbine

VIZAG STEEL MT 2011

Ans. (d) :

364. Consider the following statements :

The amount of water striking the vanes of the runner of a turbine is controlled by a spear which is

1. Provided in the nozzle
2. Conical needle shaped

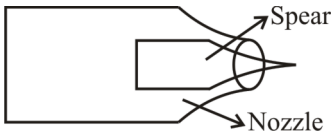
Which of the above statements is/are correct?

(a) Neither 1 nor 2 (b) Both 1 and 2

(c) 1 only (d) 2 only

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know, the amount of water striking the values of the runner of a turbine is controlled by a spear which is conical needle shaped & it is provided in nozzle.



365. If the specific speed of a turbine is more than 300, the type of turbine is :

(a) Pelton (b) Kaplan

(c) Francis (d) Pelton with more jets

J&K PSC Civil Services Pre, 2010

Ans. (b) :

Turbine	Specific speed
Pelton turbine	10 to 25
Francis turbine	60 to 300
Kaplan and propeller	300 to 1000

366. Match List I with List II and select the correct answer using the code given below the lists :

	List I		List II
A.	Degree of reaction	1.	Power given by water to runner
B.	Net head	2.	Inverse of $\sqrt{H}$, where H is head on turbine
C.	Flow ratio	3.	Change of total energy inside the runner
D.	Hydraulic efficiency	4.	Friction between water and penstock

Code:

	A	B	C	D
(a)	3	2	4	1
(c)	3	4	2	1

	A	B	C	D
(b)	1	2	4	3
(d)	1	4	2	3

UPSC JWM Advt. No.-50/2010

Ans. (c) :

- Degree of reaction or reaction ratio (R) is defined as the ratio of the static pressure drop in rotor to the static pressure drop in the stage or as the ratio of static enthalpy drop in the rotor to static enthalpy drop in stage.
- Net head is nothing but it is the friction between water & penstock.
- Flow ratio is inverse of $\sqrt{H}$, where H is head of turbine.

$$\text{Flow ratio } (\psi) = \frac{v_r}{\sqrt{2gH}}$$

- Hydraulic efficiency is also referred to as vane efficiency, is defined as ratio between power given by water to runner of turbines to power supplied by water at inlet of the turbine.

367. Water enters a turbine at 900 kPa with negligible velocity. What maximum speed, in m/s, can it reach before it enters the turbine rotor?

(a) 42 (b) 47

(c) 45 (d) 52

ISRO Scientist/Engineer 2010

Ans. (a) : P = 900 kPa

For maximum speed, pressure energy completely converted into the kinetic energy.

v = m/s

$$\frac{P}{\rho g} = \frac{V^2}{2g}$$

$$\frac{900\times10^3}{1000\times10} = \frac{V^2}{2\times10}$$

V = 42 m/s

368. Assertion (A) : A surge tank is provided in high-head power plant between main reservoir and penstock.

Reason (R) : In high-head power plants, the water hammer affects penstock.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true and R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (a) : The main function of a surge tanks are–

(1) When the load decreases, the water moves backward & gets, stored in it, which is provided in high head power plant.

(2) When load increases, additional supply of water will be provided by the surge tank.

- When water moves backward cause water hammering.

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369. **Assertion (A) :** A surge tank is provided in high-head power plant between main reservoir and penstock.

Reason (R) : In high-head power plants, the water hammer affects penstock.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A

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UPSC JWM Advt. No.-50/2010

Ans. (a) : The main function of a surge tanks are–

(1) When the load decreases, the water moves backward & gets stored in it, which is provided in high head power plant.

(2) When head increases, additional supply of water will be provided by the surge tank.

- When water moves backward cause water hammering.

370. **Specific speed of impulse wheels ranges from**

(a) 0 to 4.5 (b) 10 to 100

(c) 80 to 200 (d) 250 to 300

TNPSC AE, 2008

Ans. (b) : Low Specific Speed Turbine–The specific speed is less than 60. Varying from 10 to 35 for single jet and upto 60 for a double jet.

e.g.– Pelton wheel turbine

Medium-Specific Turbine–The specific speed varies from 60 to 300.

e.g.– Francis turbine

High Specific Turbine–The specific speed is more than 300.

e.g. Kaplan or propeller turbine

- Reaction turbines are used for low and medium heads (usually less than 300 m) and for high discharge.

371. **A turbine runs at 240 rpm under a head of 9 m. The speed of the turbine operating under a head of 16 m will be**

(a) 420 rpm (b) 320 rpm

(c) 240 rpm (d) 120 rpm

TNPSC AE, 2008

Ans. (b) : Given,

turbine speed (N_1) = 240 rpm; Head (H_1) = 9 m

N_2 = ?; H_2 = 16m

For similar turbine unit quantities must be same

Hence equating unit speed

$$\because \frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}$$

$$\therefore \frac{240}{\sqrt{9}} = \frac{N_2}{\sqrt{16}} \Rightarrow \frac{240}{3} = \frac{N_2}{4}$$

$N_2 = 80 \times 4$

$N_2 = 320$ rpm

372. **The only correct statement related to hydraulic turbines among the following is**

(a) Pelton wheel and Francis turbine are impulse turbines

(b) The adjustable blade Kaplan turbine has a higher part-load efficiency compared to a Francis turbine

(c) A draft tube can improve the performance of a Pelton wheel

(d) Cavitation occurs at the entry to a Francis turbine

DRDO Scientists 2008

Ans. (b) :

373. **The main function of the draft tube in a hydraulic reaction turbine is to**

(a) reduce the effective head across the turbine

(b) increase the discharge and hence the work

(c) decrease the discharge to prevent flow separation

(d) increase the effective head across the turbine

DRDO Scientists 2008

Ans. (c) :

374. **For an axial-flow turbine rotor with mass flow rate 10 kg/s a change in whirl of 60 m/s and a blade speed of 30 m/s, the Euler equation of turbo machine gives that the specific work done by**

(a) the fluid on the rotor is 1800 J/kg

(b) the fluid on the rotor is 18000 W

(c) the turbine rotor on the fluid is 18000 J/kg

(d) the turbine rotor on the fluid is 18000 W

DRDO Scientists 2008

Ans. (b) :

$\dot{m} = 10$ kg/s

$V_w = 60$ m/s

$u = 30$ m/s

Work done by fluid on rotor $= \dot{m} V_w . u$

$= 10 \times 60 \times 30$

$= 18000$ W

375. **The most probable value of speed ratio of Kaplan turbine is**

(a) 0.45 (b) 0.75

(c) 1.15 (d) 2.0

UKPSC AE 2007 Paper -II

Ans. (d) : Kaplan turbine is a axial flow, low head and high speed ratio reaction turbine.

The most probable value of speed ratio of Kaplan Turbine is, $k_u = 2.0$

$$\text{where } k_u = \frac{u}{\sqrt{2gh}}$$

376. **Which of the following turbines is suitable for specific speed ranging from 300 to 1000 and heads below 30 m?**

(a) Francis (b) Kaplan

(c) Propeller (d) Pelton

UKPSC AE 2007 Paper -II

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Ans. (b) : Kaplan turbine is suitable for specific speed ranging from 300 to 1000 and heads below 30m.

Turbines	Head	Specific speed (M.K.S. Unit)
1. Pelton	above 250 m	10-35
2. Francis	30-250 m	60-300 m
3. Kaplan	below 30	300 - 1000

Note : At part load, high efficiency can be obtained in Kaplan turbine.

377. Unit speed of a turbine can be determined by dividing the normal speed by

(a) $\sqrt{P}$ (b) $\sqrt{H}$
(c) $P^{3/2}$ (d) $H^{3/2}$

OPSC Civil Services Pre 2006

Ans. (b) : Unit speed $(N_u) = \frac{N}{\sqrt{H}}$

Unit discharge $(Q_u) = \frac{Q}{\sqrt{H}}$

Unit power $(P_u) = \frac{P}{H^{3/2}}$

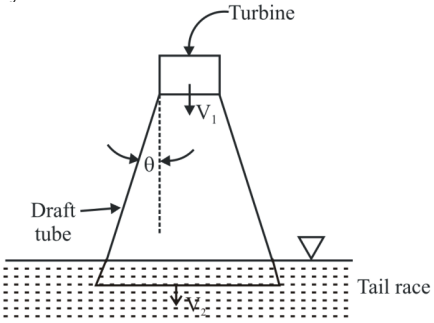
378. Efficiency of a draft tube is the ratio of actual regain of pressure head and

(a) velocity head at the entrance
(b) pressure head at the entrance
(c) theoretical regain of pressure head
(d) none of the above

OPSC Civil Services Pre 2006

Ans. (a) : Efficiency of the draft tube is defined as the ratio of actual conversion of kinetic head into pressure head in the draft tube to the kinetic head at the inlet of the draft tube.

In a reaction turbine, the draft tube is used to increase the effective head of water.
Hence by using draft tube, the net head on the turbine increases The turbine develops more power and also the efficiency of the turbine increases.



$\theta = 5° - 7°$ (This is limit of angle to avoid separation of flow)

Efficiency of draft tube (η_d)

$$= \frac{\text{Actval conversion of kinetic head in to pressure head}}{\text{kinetic head at the inlet of draft tube}}$$

$$\eta_d = \frac{\left(\frac{v_1^2}{2g} - \frac{v_2^2}{2g}\right) - H_f}{v_1^2/2g}$$

Where H_f = Head loss in draft tube

379. Specific speed of a hydraulic turbine is inversely proportional to

(a) $P^{5/4}$ (b) $H^{5/4}$
(c) $P^{3/4}$ (d) $H^{3/4}$

OPSC Civil Services Pre 2006

Ans. (b) : Specific speed of turbine $N_S = \frac{N \times \sqrt{P}}{H^{5/4}}$

380. The pipe carrying water from reservoir to the turbine house in a hydraulic power plant is called

(a) supply pipe (b) inlet pipe
(c) pen stock (d) draft tube

OPSC Civil Services Pre 2006

Ans. (c) : Pipe carrying water from reservoir to turbine house - Pen stock. Pipe carrying water from turbine to tailrace- Draft tube (always used in reaction turbine)

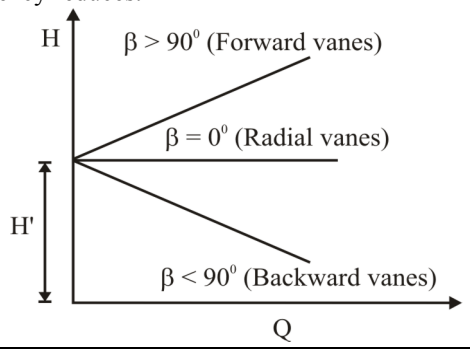
3. Reciprocating Pumps

381. The efficiency of a centrifugal pump is maximum when its blades are

(a) Bent forward
(b) Bent forward first and then straight
(c) Bent backward
(d) Straight

ISRO Scientist/Engineer 22.04.2018
Rajasthan Nagar Nigam AE 2016, Shift-I
GPSC Lect. 23.10.2016
RPSC VPI TI 14.02.2016
J & K PSC Civil Service pre 2013
ISRO Scientist/Engineer 2008
ESE 2002

Ans. (c) : Backward bent blades give less absolute velocity at the exit of impeller. Higher velocity at impeller exit is difficult to convert into pressure hence efficiency reduces.



381. What is the need of air vessel used in a reciprocating pump?

(a) To obtain a continuous supply of water at uniform rate
(b) To reduce suction head
(c) To increased the delivery head
(d) To reduce Cavitation

Nagaland PSC CTSE 2017 2nd paper
APPSC AEE MAINS 2016 PAPER-III
PSPCL AC 2012, TRB Poly Lect. 2012
GPSC Astt. Prof. 28.08.2011, ESE 2008

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Ans. (a) : The air vessel, in a reciprocating pump, is a cast iron closed chamber having an opening at its base. These are fitted to the suction pipe and delivery pipe close to the cylinder of the pump. The vessels are used to get continuous supply of liquid at a uniform rate and to save the power required to drive the pump.

382. The cavitation in a hydraulic machine is mainly due to

(a) Low velocity (b) High velocity
(c) Low pressure (d) High pressure

CIL MT 27.02.2020
Nagaland PSC (CTSE) 2018, Paper-II
GPSC Lect. 23.10.2016, TNPSC AE, 2008

Ans. (c) : Cavitation is the formation of bubbles in the liquid flowing through any hydraulic turbine, cavitation occurs when the static pressure of the liquid falls below its vapour pressure. Cavitation is most likely to occur near the fast moving blades of the turbines and in the exit region of the turbines.

384. The power saved by fitting air vessel to a single –acting reciprocating pump is :

(a) 88.4% (b) 84.8%
(c) 39.2% (d) 48.8%

Kerala PSC AE (Irrigation) 05.04.2017
APGENCO AE 2012, APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (b) : Percentage of work saved is 84.8% when single acting pump air vessel is used while this saving is only 39.2% when air vessel is used in double acting pump.

385. The discharge through a single acting reciprocating pump is

(a) Q = ALN (b) Q = ALN/60
(c) Q = 2 ALN/60 (d) Q = 2ALN

RPSC IOF, 2020, APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (b) : The discharge through a single acting reciprocating pump or volume of water discharged per second.

$$Q = \frac{ALN}{60} m^3/sec$$

Where, A- Area of cylinder (m^2)
L- Length of cylinder (m)
N- Crank speed in rpm

To increase discharge and to maintain it more uniform by double acting reciprocating pump are used

$$Q \approx \frac{2ALN}{60} m^3/sec$$

386. Which pump is more suitable for an application where high pressure is required to be developed at moderate discharge?

(a) Centrifugal pump
(b) Reciprocating pump
(c) Turbine
(d) None of above

WBPSC AE, 2017, GWSSB DEE 07.07.2016
Mizoram PSC AE/SDO 2014, Paper-I

Ans. (b) : Reciprocating pump is more suitable for an application where high pressure is required to be developed at moderate discharge.

387. The specific speed of a hydraulic pump is the speed of geometrically similar pump working against a unit head and

(a) delivering unit quantity of water
(b) consuming unit power
(c) having unit velocity of flow
(d) having unit radial velocity

Gujarat PSC AE 2019
APPSC AEE MAINS 2016 PAPER-III
UKPSC AE 2007 Paper -II

Ans. (a) : The specific speed of a hydraulic pump is the speed of geometrically similar pump working against a unit head and delivering unit quantity of water

388. Which one of the following pumps is not a positive displacement pump?

(a) Reciprocating pump (b) Centrifugal pump
(c) Vane pump (d) Lobe pump

APPSC AEE MAINS 2016 PAPER-III
Vizag Steel MT (Re-Exam) 24.11.2013, ESE 2007

Ans. (b) : Positive Displacement Pump–These are those pumps in which the liquid is sucked and then it is pushed or displaced to the thrust exerted on if by a moving member, which results in lifting the liquid to the required height.

Reciprocating Pump–Vane pump, lobe pump are the examples of positive displacement pump whereas the centrifugal pump is non-positive displacement pump.

389. Slip of a reciprocating pump is defined as the

(a) Ratio of actual discharge to the theoretical discharge
(b) Sum of actual discharge and the theoretical discharge
(c) Difference of theoretical discharge and the actual discharge
(d) Product of theoretical discharge and the actual discharge

Nagaland PSC (CTSE) 2018, Paper-II
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : Slip of a reciprocating pump is defined as the difference of theoretical discharged and the actual discharge.

$Slip = Q_{theoretical} - Q_{actual}$

The value of slip is generally positive

$$Slip = \frac{\text{Swept volume/stroke}}{\text{Actual discharge/stroke}}$$

390. Slip of a reciprocating pump is negative when

(a) Suction pipe is short and pump is running at low speeds
(b) Delivery pipe is long and pump is running at high speeds
(c) Suction pipe is short and delivery pipe is long and the pump is running at low speeds
(d) Suction pipe is long and delivery pipe is short and the pump is running at high speeds

GPSC DEE Class-II (GWSSB) 04.07.2021
Nagaland PSC (CTSE) 2018, Paper-II

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Ans. (d) : Slip of a reciprocating pump is negative when suction pipe is long and delivery pipe is short and the pump is running at high speeds.
However in practice sometimes delivery valve opens before suction stroke is completed thus delivering a greater volume of water than actually swept by the piston. Hence the slip will be negative in such a case.

391. In a reciprocating pump, air vessels are used to
(a) Smoothen the flow
(b) reduce suction head
(c) Increase delivery head
(d) Reduce acceleration head

Nagaland PSC (CTSE) 2018, Paper-II
VIZAG STEEL MT 2011

Ans. (d) : Air vessel is a closed chamber containing compressed air at the top and liquid at the bottom.
In reciprocating pump air vessels are used to reduced acceleration head.
It provides uniform discharge from pump. The pump can run at higher speed and provide higher discharge.

392. If a reciprocating pump having a mechanical efficiency of 80%, delivers water at the rate of 80 kg/s with a head of 30 m, the brake power of the pump is _____ .
(a) 20.8 kW (b) 15.4 kW
(c) 29.4 kW (d) 40.8 kW

VIZAG MT, 14.12.2020, ESE 2001

Ans. (c) : $\eta_{mech} = 80\%$
$Q = 80$ kg/sec
$H = 30$ m
$P = \rho gQH/\eta$
$= \dfrac{1000 \times 9.81 \times 80 \times 30}{0.80}$
$\boxed{P = 29.43\,kW}$

393. Given power 'P' of a pump, the head 'H', the discharge 'Q' and specific weight 'w' of the liquid, dimensional analysis would lead to the result that 'P' is proportional to
(a) $H^{1/2}Q^2 w$ (b) $H^{1/2} Qw$
(c) $H Q^{1/2} w$ (d) HQw

UPPSC AE 13.12.2020, Paper-2, ESE 1998

Ans. (d) : Power,
$P = \rho g\, QH$
$P \propto \rho g\, QH$
$\boxed{P \propto wQH}$
Where,
w = specific weight of liquid
Q = discharge
H = head

394. Which of the following pumps is NOT a positive displacement pump?
(a) Reciprocating pump (b) Jet pump
(c) Sliding –vane pump (d) Lobe pump

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
ESE 2007

Ans. (b) : positive displacement pumps are –
(1) Reciprocating pump
(2) Sliding vane pump
(3) Lobe pump
Jet pump is a dynamic displacement pump

395. The preferred type of pump for small discharge and high head is:
(a) Centrifugal type (b) Reciprocating type
(c) Axial flow type (d) Radial flow type

TSPSC AEE 28.08.2017 (Civil/Mechanical)
RPSC VPITI 14.02.2016

Ans. (b) : Reciprocating pumps are positive displacement pump.
Reciprocating pumps preferred for small discharge and high head.

396. Which of the following is not a turbo machine
(a) Centrifugal pump
(b) Kaplan Turbine
(c) Reciprocating pump
(d) Fans

RPSC ADE 2016, KPSC ADF 2015

Ans. (c) : Turbo machinery describes the machines that transfer energy between a rotor and a fluid, including both turbines and pumps.

397. Water (density 1000 kg/m^3) at ambient temperature flows through a horizontal pipe of uniform cross-section at the rate of 1 kg/s. If the pressure drop across the pipe is 100 KPa, the minimum power required to pump the water across the pipe, in watts, is
(a) 10 W (b) 100 W
(c) 200 W (d) 20 W

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : Power is required to overcome losses h_f,
Power = ρQgh
$h_f = \dfrac{\Delta P}{\rho g}$
$\dot{m} = \rho Q$
$Q = \dfrac{\dot{m}}{\rho} = \dfrac{1}{10^3} = 10^{-3}\,m^3$
$h = \dfrac{\Delta P}{\rho g}$
$h = \dfrac{100 \times 10^3}{10^3 \times 9.81} = 10.19\,m$
Power = ρQgh
$= 9810 \times 10^{-3} \times 10.19$
$= 100$ W

398. Which of the following is not a correct statement for a reciprocating pump?
(a) The flow from a reciprocating pump may be evened out by employing properly phased out multi cylinder units or by fitting air vessels close to the cylinder.
(b) Percentage of power saved by fitting air vessel is more in a double acting pump than in a single acting pump.

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(c) The reciprocating pump is essentially a low speed machine and that speed is limited due to considerations of flow separation.
(d) The reciprocating pumps can handle only low viscosity liquids free from impurities.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : By using air vessel to reciprocating pump, the saving of work and subsequently the power is about 84.8% in case of single acting reciprocating pump and 39.2% in case of double acting reciprocating pump.
So, percentage of power saving in case of single acting reciprocating pump is more as compared to double acting reciprocating pump.

399. If Net Positive Suction Head (NPSH) requirement of a pump is not satisfied, then correct statements are:
(a) It will not develop sufficient head to raise water.
(b) Its efficiency will be low.
(c) It will deliver very low discharge.
(d) It will be cavitated.
(a) (a), (b) and (c) (b) (b), (c) and (d)
(c) (a) and (d) (d) (a), (b), (c) and (d)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : If Net positive suction head (NPSH) requirement of a pump is not satisfied, then correct statements are :
(1) It will not develop sufficient head to raise water.
(2) It will be cavitated.

400. A single-acting reciprocating pump, running at 50 rpm delivers 0.00736 m^3/s of water. The diameter of the piston is 200 mm and stroke length is 300 mm. What is the percentage slip of the pump?
(a) 5.29% (b) 6.29%
(c) 7.29% (d) 8.29%

ESE 18.07.2021

Ans. (b) : Given,
N = 50 rpm
Q = 0.00736 m^3/s
D = 200 mm
L = 300 mm

$$\% \text{ slip} = \frac{Q_{th} - Q_{act}}{Q_{th}} \times 100$$

$$Q_{th} = \frac{ALN}{60} = \frac{\pi}{4}(0.20)^2 \times 0.30 \times \frac{50}{60}$$

$$= 0.00758 \text{ m}^2/\text{s}$$

$$\% \text{ slip} = \frac{0.00785 - 0.00736}{0.00785} \times 100$$

$$= 6.25\%$$

401. A pump discharges a liquid into a tank at the rate of 0.032 m^3/s. The tank 1.5 m in diameter and 4.20 m in height, can hold 3500 kg of liquid. The density of the liquid and mass flow rate of the liquid handled by the pump are respectively,
(a) 471.57 kg/m^3 and 16 kg/sec.
(b) 471.57 kg/m^3 and 15 kg/sec.
(c) 481.57 kg/m^3 and 16 kg/sec.
(d) 481.57 kg/m^3 and 15 kg/sec.

ESE 18.07.2021

Ans. (b) : Given, d = 1.5m, h = 4.20

$$\text{Volume of the tank} = \frac{\pi}{4} d^2 \times h$$

$$= \frac{\pi}{4} \times (1.5)^2 \times 4.20 = 7.418 \text{ m}^3$$

$$\text{Density of liquid handled by pump } (\rho) = \frac{\text{Mass}}{\text{volume}}$$

$$= \frac{3500}{7.418} = 471.80 \text{ kg/m}^3$$

Mass flow rate of liquid handled by the pump
= 0.032 × 471.80
= 15.097 kg/s

402. A pump impeller is 375 mm in diameter and it discharges water with velocity components of 2 m/s and 12 m/s in the radial and tangential directions respectively. The impeller is surrounded by a concentric cylindrical chamber with parallel sides, the outer diameter being 450 mm. If the flow in the chamber is a free spiral vortex, what are the tangential velocity and radial velocity at the outlet of the chamber respectively?
(a) 12 m/s and 1.67 m/s
(b) 10 m/s and 1.67 m/s
(c) 12 m/s and 1.76 m/s
(d) 10 m/s and 1.76 m/s

ESE 18.07.2021

Ans. (b) : Given,

$d_2 = 375$ mm $\quad r_2 = \frac{375}{2}$

$v_{w_2} = 12$ m/s

$v_{f_2} = 2$ m/s

Free vortex motion;

$v_w.r$ = constant

$v_{w_2} r_2 = v_{w_3} r_3$

$$\Rightarrow 12 \times \frac{375}{2} = v_{w_3} \times \frac{450}{2}$$

$\Rightarrow v_{w_3} = 10$ m/s

Further;

$$\frac{v_{w_2}}{v_{f_2}} = \frac{v_{w_3}}{v_{f_3}}$$

$$\Rightarrow v_{f_3} = v_{f_2} \times \frac{v_{w_3}}{v_{w_2}}$$

$$= 2 \times \frac{10}{12} = 1.67 \text{ m}$$

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403. Which one of the following types of impellers is used to handle highly solid laden liquids like concrete and slurry?

(a) Fully open impeller
(b) Semi-Enclosed Impeller
(c) Fully-Enclosed Impeller
(d) Quarter Open Impeller

ESE 18.07.2021

Ans. (a) : Fully open impeller is used to handle high solid laden liquids like concrete and slurry.

404. In a single reciprocating pump without air vessel, the ratio of the average frictional head to the maximum frictional head in the delivery pipe is

(a) 1/2 (b) 1/3
(c) 2/3 (d) 3/4

ESE 18.07.2021

Ans. (c) : In single reciprocating pump without air vessel; the ratio of average frictional head to maximum frictional head in delivery pipe is 2/3.

405. The theoretical discharge of a single acting reciprocating pump running at 60 rpm delivering 0.02 m^3s^{-1} of water when the piston diameter is 200 mm and stroke length is 400 mm is

(a) 0.025 m^3s^{-1} (b) 1.508 m^3s^{-1}
(c) 0.754 m^3s^{-1} (d) 0.0125 m^3s^{-1}

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : Area, $A = \frac{\pi}{4} \times d^2$

$= \frac{\pi}{4} \times (0.2)^2$

$= 0.0314\ m^2$

$Q_{th} = \frac{\ell AN}{60} m^3/s$

$= \frac{0.4 \times 0.0314 \times 60}{60}$

$= 0.0125\ m^3/s$

406. A pump having an efficiency of 68% delivers 3 L/s of water against a head of 22 m. Compute the power requirement of the pump in W

(a) 952 (b) 850
(c) 1050 (d) 752

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : $\eta_{overall} = 68\%$, $H_m = 22$ m

$\eta_{overall} = \frac{\text{Water power}}{\text{Shaft power}}$

$\text{Shaft power} = \frac{\text{Water power}}{\eta_{overall}}$

$SP = \frac{\rho g Q H_m}{\eta_0} = \frac{1000 \times 9.81 \times 0.003 \times 22}{0.68}$

$\boxed{SP = 952\ W}$

407. Delivery head of a centrifugal pump is

(a) Directly proportional to N
(b) Inversely proportional to N
(c) Directly proportional to N^2
(d) Inversely proportional to N^2

RPSC IOF, 2020

Ans. (c) : Delivery head of centrifugal pump is directly proportional to N^2

$\left(\frac{\sqrt{H}}{DN}\right)_m = \left(\frac{\sqrt{H}}{DN}\right)_p$

$\sqrt{H} \propto N$

$H \propto N^2$

408. Which of the following statements regarding hydraulic pumps are correct?

1. **The gear pump consists of two close-meshing gear wheels which rotate in opposite directions.**
2. **In vane pump, as the rotor rotates, the vanes follow the contours of the casing.**
3. **This leakage is more in vane pump compared to gear pump.**

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

ESE 2020

Ans. (b) :
1. The gear pump consists of two close- meshing gear wheels which rotate in opposite directions.
2. In vane pump, as the rotor rotates, the vanes follow the contours of the casing.
3. The leakage is less in vane pump than with the gear pump.

409. In reciprocating pumps, the acceleration pressure head is maximum at the :

(a) beginning of the stroke only
(b) mid of the stroke
(c) beginning and end of the stroke
(d) end of the stroke only

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Velocity of water (Due to acceleration of piston)

$V_p = \frac{A}{(a)_{s,d}} \cdot \omega r \sin\theta$

Acceleration $\rightarrow$ max, $V_p = 0$

$V_p = 0$ at $\theta = 0°, 180°$

So, In reciprocating pumps, the acceleration pressure head is maximum at the beginning and end of the stroke.

410. A single acting reciprocating pump having theoretical 9×10^{-3} m^3/s runs at 40 rpm. If the actual discharge is 300 litre/m, then the percent slip in this pump is :

(a) 16.6% (b) 26.2%
(c) 8.3% (d) 13.1%

APPSC Poly Lect. Automobile Engg., 2020

Ans. () : Given that

$Q_{th} = 9 \times 10^{-3}$ m^3/sec

$Q_{act} = 300$ litre/min $= \frac{300}{60} \times 10^{-3}$

$Q_{act} = 5 \times 10^{-3}$

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$$\% \text{ slip} = \frac{Q_{th} - Q_{act}}{Q_{th}}$$
$$= \frac{9\times10^{-3} - 5\times10^{-3}}{9\times10^{-3}} \times 100$$
$$= \frac{10^{-3}(9-5)}{9\times10^{-3}}$$
$$= \frac{4}{9} \times 100$$
$$= 44.4\%$$

Note-Solution does not matches.

411. Two identical pumps having the same discharge (Q) and working against the same head (H). If they are connected in parallel, which of the following is correct?

(a) Combined discharge = 2Q
Combined head = h
(b) Combined discharge = Q
Combined heat = 2H
(c) Combined discharge = 4Q
Combined head = H/2
(d) Combined discharge = 2Q
Combined head = 2H

SJVN ET 2019

Ans. (a) :
• If they are connected in parallel,
$Q = Q_1 + Q_2$
$H = H_1 = H_2$
• If they are connected in series,
$Q = Q_1 = Q_2$
$H = H_1 + H_2$

412. Which type of the pump is different from others in the same group?

(a) Screw pump
(b) Gear pump
(c) Cam and piston pump
(d) Plunger pump

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Screw pump, gear pump, vane pump axial pump, radial pump are rotary pump or centrifugal pump. Whereas plunger pump is reciprocating pump So, plunger pump is different from others in the same group.

413. The component fitted in the suction and delivery pipe to maintain the uniform velocity of flow of liquid in single acting reciprocating pump is :

(a) Air vessel
(b) Hydraulic coupling
(c) Hydraulic press
(d) Hydraulic intensifier
(e) Supercharge

CGPSC AE 25.02.2018

Ans. (a) : Air vessel fitted in the suction and delivery pipe to maintain the uniform velocity of flow of liquid in single acting reciprocating pump. This pump use for high pressure water with low discharged liquid.

414. The service pump in a water supply system has to maintain a net static head lift of 5 m at the tank to which it delivers freely through a 4 km long pipe, wherein all minor losses can be neglected. The diameter of the pipe is 0.2m and its friction factor f = 0.01. The pumped water is discharged at 2 m/s. The absolute pressure differential developed by the pump is nearly (taking atmospheric pressure as 10.3 m of water)

(a) 4.5 bar (b) 5.5 bar
(c) 45 bar (d) 55 bar

ESE 2018

Ans. (b) : Given,
Static head (H_{st}) = 5 m
Length of pipe (ℓ) = 4 km = 4000 m
Dia of pipe (d) = 0.2 m
Friction factor (f) = 0.01
Discharge velocity (v) = 2 m/s
h_{atm} = 10.3 m of water
Loss of head due to friction

$$(h_f) = \frac{f\ell v^2}{2gd} = \frac{0.01\times4000\times(2)^2}{2\times9.81\times0.2} = 40.77 \text{ m}$$

We know that the pressure difference developed by pump depends on manometric head produced by the pump and is given by

$$H_m = H_{st} + h_{fs} + h_{fd} + \frac{v^2}{2g}$$

(minor losses can be neglected)

Where H_m = Manometric head
H_{st} = Static head
h_{fs}, h_{fd} = loss of head due to friction in suction and delivery pipes.
By the given condition (according to question)

$$H_m = H_{st} + h_f + \frac{v^2}{2g} + h_{atm}$$
$$= 5 + 40.77 + \frac{2^2}{2\times9.81} + 10.3 = 56.27 \text{ m}$$

Pressure rise is given by
$\Delta P = \rho g H_m = 1000 \times 9.81 \times 56.27$
$= 552008.7 \text{ N/m}^2 = 5.520 \text{ bar}$

415. When an air vessel is fitted in a double acting Reciprocating pump, percentage work done per stroke saved against friction is :

(a) 84.8% (b) 39.2%
(c) 46.4% (d) 76.4%

Kerala PSC Poly. Lect., 2017

Ans. (b) : Air vessel is used to obtain a continuous supply of water at uniform rate, to save a considerable amount of work to run the pump at a high speed separation.
Work saved by fitting air vessels in single acting reciprocating pump is 84.8%, while in double acting reciprocating pump, the work saved is 39.2%.

416. Rotary displacement pumps are suitable for handling_____

(a) Oils (b) Gritty liquids
(c) Both (A) and (B) (d) none of the above

WBPSC AE, 2017

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Ans. (c) : Rotary displacement pumps are suitable for handling Oils and Gritty liquids.

417. Pump transfers the mechanical energy of motor or an engine into _____ of a fluid.

(a) Pressure energy
(b) kinetic energy
(c) (A) or (B)
(d) pressure energy, kinetic energy or both

WBPSC AE, 2017

Ans. (d) : Pump transfers the mechanical energy of motor or an engine into pressure energy, kinetic energy or both of a fluid.

418. _____ pump is also called as velocity pump.

(a) Reciprocating (b) Rotary displacement
(c) Centrifugal (d) Screw

WBPSC AE, 2017

Ans. (c) : Centrifugal pump is also called as velocity pump.

419. Which of the following is/are example/s of rotary displacement pump?

(a) Gear pump (b) Vane pump
(c) Rotary piston pump (d) All of the above

WBPSC AE, 2017

Ans. (d) : Rotary displacement pumps are –
(1) Gear pump
(2) Vane pump
(3) Rotary piston pump

420. What is the difference between vane pump and radial piston pump?

(a) In radial piston pump, radial slots in vane pumps are replaced by radial bores which accommodate pistons.
(b) In radial piston pump, radial slots in vane pumps are replaced by radial bores which accommodate swash plate.
(c) In radial piston pump, radial slots in vane pumps are replaced by radial bores which accommodate both swash plate and pistons.
(d) None of the above

Karnataka PSC Lect., 27.05.2017

Ans. (a) : In radial piston pump, radial slots in vane pumps are replaced by radial bores which accommodate pistons.
Hydraulic vane pump are used in die casting, injection moulding machine industry plus land and road construction machinery.
Piston pump handle large flow at high hydraulic system pressure delivering optimum efficiency and reliability while maintaining a compact size with a high power density.

421. Consider the following statements regarding reciprocating pump:

1. Frictional losses are maximum at the middle of the stroke
2. Maximum inertia effect occurs in place with zero frictional losses
3. Negative slip may occur when the delivery head is high.

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 and 3 only
(c) 2 and 3 only (d) 1 and 2 only

JWM 2017

Ans. (d) : Reciprocating pump is a positive displacement pump where certain volume of liquid is collected in enclosed volume and is discharge using pressure to the required application. This pump is more suitable for low volumes of flow at high pressure. In reciprocating pump frictional losses are maximum at the middle of the stroke and maximum inertia effect occur in place with zero frictional losses.

422. A pump handling a liquid raises its pressure from 1 bar to 30 bar. Take the density of the liquid as 990 kg/m^3. The isentropic specific work done by the pump in kJ/kg is

(a) 0.10 (b) 0.30
(c) 2.50 (d) 2.93

TNPSC AE 2017

Ans. (d) : Data given, $P_1 = 1$ bar $= 100$ kPa
$P_2 = 30$ bar $= 3000$ kPa
$\rho = 990$ kg/m^3 Then, isentropic specific work done by pump (W)

$$W = \frac{(P_2 - P_1)}{\rho} = \frac{(3000-100)\times 10^5}{990}$$

$$= \frac{2900}{990} = 2.929\,\text{kJ/kg}$$

$$\simeq 2.93\,\text{kJ/kg}$$

423. In a single casing, multi stage pump running at constant speed, how the capacity rating is to be lowered:

(a) coating a new material on the impeller
(b) trimming the impeller size to the required size by machining
(c) impossible
(d) none of the above

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : Trimming the impeller size to the required size by machining.

424. Negative slip in a reciprocating pump occurs when:

(a) the speed of the pump is too high
(b) actual discharge is greater than theoretical discharge
(c) all of the above
(d) none of the above

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : Slip of reciprocating pump defines as the difference between the theoretical and actual discharge of the pump. Negative slip occurs when delivery pipe is sort, suction pipe is long, pump running at high speed and actual discharge is greater than theoretical discharge.

425. Which of the following parameters are true when the cavitations parameter $\sigma = 0$?

i. The local pressure is reduced to vapour pressure
ii. Cavitation starts
iii. Boiling of liquid starts
iv. Cavitation stops

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(a) (i), (ii) and (iv) (b) only (ii) and (iii)
(c) only (i) and (iii) (d) (i), (ii) and (iii)

RPSC ADE 2016

Ans. (d) :

Cavitation factor $(\sigma) = \frac{H_{atm} - H_v - H_s - H_{LD}}{H_m}$

H_m = Head developed by pump
H_{atm} = Atmospheric pressure head
H_v = Vapour pressure head in m of water
H_s = suction head of water
H_{LD} = Head loss in suction side.

If value of σ is less than the critical value cavitation occur in the pump and pump stops flowing water.

If $\sigma = 0$, local pressure is reduced to vapour pressure and Cavitation starts. During cavitation liquid starts boiling at the atmospheric temperature.

426. Which of the following is a wrong statement about the effect of inertia of reciprocating parts in a reciprocating pump?

(a) It causes variation in pressure.
(b) It limits the suction height.
(c) It puts an upper limit on the speed at which the pump can be run to obtain higher discharge.
(d) It increases the input work required.

RPSC AE (GWD) 06.05.2016

Ans. (d) :

427. ..... is an example for rotary displacement pump.

(a) Slider pump (b) Piston pump
(c) Lobe pump (d) Quadra pump

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Example of rotary positive displacement pumps are gear pumps. Screw pump, progressive cavity pumps, rotary piston pumps, water ring pumps, rotary vane pumps, roller vane pump and peristaltic pump.

428. Reciprocating pumps are best suited

(a) Where operating speeds are much high
(b) For pumping large liquid flows for smaller heads
(c) Where constant heads are required despite fluctuation in discharge
(d) Where constant supplies are required regardless of pressure fluctuation
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : Reciprocating pumps are best suited for high pressure & volume applications. They frequently require pulsation dampers because of pulsating nature of the flow. It's supply is constant.
It is also high head and low discharge.

429. Air vessels in a reciprocating pump are used to

(a) Increase the pump head
(b) Increase the pump efficiency
(c) Reduce accelerating heads to minimum
(d) Save pump from cavitation
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : The air vessels in a reciprocating pump is a cast iron closed chamber having an opening at its base, it helps to get continuous supply of liquid at uniform rate & to save the power required to drive the pump, the acceleration & friction heads are reduced.

430. Match List-I with List-II and select the correct answer using the code given below the lists:

List - I	List -II
A. Draft Tube	**1. Impulse Turbine**
B. Surging	**2. Reciprocating Pump**
C. Air Vessel	**3. Reaction Turbine**
D. Nozzle	**4. Centrifugal Pump**

(a) A-1, B-2, C-4, D-3
(b) A-3, B-2, C-4, D-1
(c) A-3, B-4, C-2, D-1
(d) A-1, B-4, C-2, D-3

UPRVUNL AE 21.08.2016

Ans. : (c)

List-I	List-II
Draft tube	Reaction turbine
Surging	Centrifugal pump
Air vessel	Reciprocating pump
Nozzle	Impulse turbine

431. Air vessels used in reciprocating pumps are initially filled with:

(a) Water (b) Vacuum
(c) Compressed air (d) None of the above

OPSC AEE 2015 PAPER - II

Ans. : (c) Air Vessels:-The air vessel, in a reciprocating pump, is a cast iron closed chamber having an opening at its base. These are fitted to the suction pipe and delivery pipe close to the cylinder of the pump. The vessels are used for the following purposes:
(i) To get continuous supply of liquid at a uniform rate
(ii) To save the power required to drive the pump. This is due to the fact that by using air vessels, the acceleration and friction heads are reduced. Thus the work is also reduced.

432. In reciprocating pump, the percentage of slip is equal to

(a) 1 - coefficient of discharge
(b) 1 + coefficient of discharge
(c) $\frac{1}{\text{cofficient of discharge}}$
(d) $\frac{\text{cofficient of discharge}}{\text{theoretical discharge}}$

TSPSC AEE 2015

Ans. : (a) Slip in percentage is given by

$$\%\,\text{slip} = \frac{(Q_{th} - Q_{act}) \times 100}{Q_{th}}$$

$$= \left(1 - \frac{Q_{act}}{Q_{th}}\right) \times 100$$

$$\%\ \text{Slip} = (1 - C_d) \times 100$$

Where C_d = Coefficient of discharge
Slip is negative when delivery pipe is small and suction pipe is long and pump is running at very high speed.

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433. For pumping viscous oil, pump used will be

(a) low speed (b) air leakage
(c) air in water (d) heavy liquid

GPSC AE 26.07.2015

Ans. (d)

434. Due to rotation of the impeller of centrifugal pump in liquid surroundings

(a) a momentum acts on the liquid
(b) a torque acts on the liquid in the direction of rotation
(c) a torque acts on the liquid in the direction opposite to the direction of impeller rotation
(d) just loss of energy alone takes place

Haryana PSC Civil Services Pre, 2014

Ans. (c) : Due to rotation of the impeller of a centrifugal pump in liquid a torque exist on the liquid in the direction opposite to the direction of impeller rotation.

435. The slip of a single acting reciprocating pump, running at 50 rpm, delivers 10 liter/s when the diameter of piston and stroke length are 200 mm and 400 mm respectively is :

(a) 0.47 m^3/sec (b) 0.047 m^3/sec
(c) 0.00047 m^3/s (d) 4.7 m^3/s

Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : Given,

Speed, N = 50 rpm, dia (d) = 200 mm

Q_{act} = 10 liter/s = 0.2 m

= 0.01 m^3/s ℓ = 400 mm

= 0.4 m

$$Q_{th} = \frac{\ell AN}{60} m^3/s \qquad A = \frac{\pi}{4} \times d^2$$

$$= \frac{0.4 \times 0.0314 \times 50}{60} \qquad = \frac{\pi}{4} \times (0.2)^2$$

= 0.01047 m^3/s = 0.0314 m^2

Q_a = 0.01 m^3/s

∵ We know,

Slip = Theoretical discharge – Actual discharge

$= Q_{th} - Q_{act}$

= 0.01047 – 0.01

= 0.00047 m^3/s

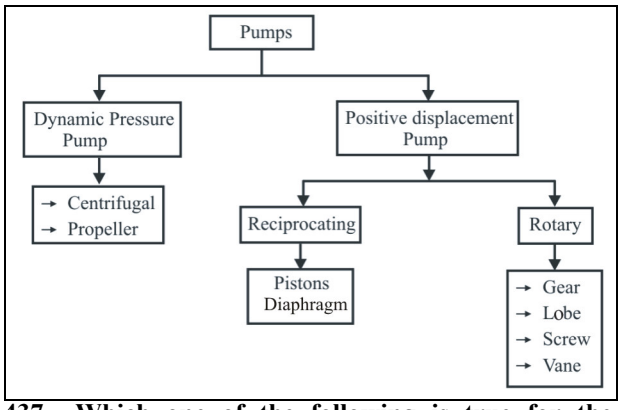
= 4.7×10^{-4} m^3/s

436. The gear pump is of which type ?

(a) Rotodynamic
(b) Positive displacement
(c) Negative displacement
(d) Hydrodynamic

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Gear pump is positive displacement type pump.



437. Which one of the following is true for the positive displacement pumps

(a) They can generate high pressure
(b) They have low power to weight ratio
(c) They have low volumetric efficiency
(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Positive displacement pump- It is used for higher pressure and low discharge flow rates.
Centrifugal pump is used for low pressure and high discharge.

438. For only hydraulic pump, delivery of a pump reduces with.

(a) Increase in pressure at constant speed
(b) Increase in speed at constant pressure
(c) Reduction in pressure
(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : In hydraulic pump delivery of a pump reduces with increase in pressure at constant speed.

439. Rotary actuator is a device that converts hydraulic energy into:

(a) Pressure (b) Velocity
(c) Discharge (d) Rotational motion

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Rotary actuator is a device that converts hydraulic energy into rotational motion.

440. The torque developed by hydraulic motor is a function of :

(a) Displacement of hydraulic fluid
(b) Velocity of hydraulic fluid
(c) Acceleration of hydraulic fluid
(d) Pressure of hydraulic fluid

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : The torque developed by hydraulic motor is a function of pressure of hydraulic fluid.

Performance of Hydraulic motors.

$$\eta_{mech} = \frac{\text{Actual Torque Produced}}{\text{Theroretical Torque}} \times 100$$

$$T_{act} = \frac{P}{2\pi N} \qquad P = \text{Pressure},$$

$$T_{th} = \frac{V_D \times P}{2\pi}$$

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441. Theoretical power required to drive a reciprocating pump is

(a) $\frac{WQH_s}{60}$ (b) $\frac{WQH_s}{75}$

(c) $\frac{WQH_d}{60}$ (d) $\frac{270}{\pi}$

APPSC AEE 2012

Ans. : (b) Theoretical power required to drive a reciprocating pump is $\frac{\rho gQH_s}{75}$

$$P = \frac{\rho gQH_s}{75}$$

$$P = \frac{WQH_s}{75}$$

ρ = Density of liquid

Q = Discharge

H_s= static head = suction head + delivery head

442. Match List I with List II and select the correct answer using the codes given below the Lists:

List-I		List II	
A	Oscillating jump	**1**	less than 2
B	Strong jump	**2**	between 2 and 3
C	Weak jump	**3**	around 4
D	Undular jump	**4**	around 10

Code:

	A	**B**	**C**	**D**
(a)	3	4	2	1
(b)	1	2	3	4
(c)	4	3	1	2
(d)	2	1	3	4

APPSC AEE 2012

Ans. (a) : Undular Jump - 1-1.7 (less than 2)
Weak Jump - 1.7 - 2.5 (2 to 3)
Oscillating Jump - 2.5 - 4.5 (around 4)
Strong Jump - > 9.0 (around 10)

443. If h_a = height of inlet of pump from the sump water level h_f = loss of head in the foot valve strainer and suction pipe,
H_a = Atmospheric pressure head
H_v = Vapour pressure head
Then net positive suction head (NPSH) is expressed as :

(a) $H_a - H_v - h_s - h_{fs}$ (b) $H_a - H_v + h_a + h_{fs}$

(c) $H_a - H_v + h_s - h_{fs}$ (d) $H_a + H_v - h_s - h_{fs}$

RPSC AE GWD, 2011

Ans. (a) : Net positive suction Head (NPSH)
$= H_a - H_v - h_s - h_{fs}$

H_a = atmospheric pressure head

H_v = vapour pressure head

H_{fs} = Loss of head in the foot valve strainer and suction pipe.

H_s = Suction head

444. For a reciprocating water pump having cylinder dia d and crank radius r, if W is the weight of water lifted, the coefficient of discharge is

(a) $\frac{W}{\pi d^2 r}$ (b) $\frac{2W}{\gamma \pi d^2 r}$

(c) $\frac{W}{\gamma d^2 r}$ (d) $\frac{W}{\gamma \pi d^2}$

ISRO Scientist/Engineer 2009

Ans. (b) : Cylinder diameter = d
Crank radius = r
Water weight = W
Weight of water delivered per second
$W = \rho gQ$

$$= \frac{\rho g C_d ALN}{60}$$

$$C_d = \frac{60W}{\gamma ALN}$$

(weight density) = $w = \rho g$

$$= \frac{60W}{\gamma \frac{\pi d^2}{4} \times 2r \times N}$$

$$\frac{60 \times 2W}{N \times \gamma \times \pi \times d^2 \times r}$$

$$C_d = \frac{2W}{\gamma \pi d^2 r}$$ (for one revolution)

4. Centrifugal Pumps

445. In a centrifugal pump, cavitation is reduced by

(a) Increasing the flow velocity
(b) Reducing the discharge
(c) Reducing the suction head
(d) Throttling the discharge

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TSPSC AEE 28.08.2017 (Civil/Mechanical)
NPCIL ET, 2015, (KPSC AE 2015)
GPSC Lect. (Poly.) 07.12.2014
UKPSC AE 2007 Paper -II

Ans. : (c) Cavitation in centrifugal pumps reduced by reducing the suction head.

Cavitation in centrifugal Pumps:- In centrifugal pumps the cavitation may occur at the inlet of the impeller of the pump, or at the suction side of the pumps, where the pressure is considerably reduced. Hence if the pressure at the suction side of the pump drops below the vapour pressure of the liquid then the cavitation may occurs.

Cavitation factor for centrifugal pumps.

Thoma's cavitation factor

$$\sigma = \frac{NPSH}{H} = \frac{H_{atm} - h_s - h_v - h_F}{H}$$

Note : $H = H_m$ [Monometric head]

For centrifugal pump

$H = H_N$ [Net head for turbine]

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446. The specific speed (N_S) of a centrifugal pump is given by

(a) $\frac{N\sqrt{Q}}{H^2}$ (b) $\frac{N\sqrt{Q}}{H^{3/4}}$

(c) $\frac{N\sqrt{Q}}{H^{1/2}}$ (d) $\frac{N\sqrt{Q}}{H^{5/4}}$

OPPSC AEE 2019 Paper II
TSPSC AEE 28.08.2017 (Civil/Mechanical)
APPSC AEE Mains 2016 (Civil/Mechanical)
Vizag STEEL MT 2015
MPSC Dy. Engineer 2014, APPSC AEE 2012

Ans. (b) : for pump

$N_S = \frac{N\sqrt{Q}}{H^{3/4}}$

Where,

Q = discharge (m^3/sec)

N = Actual speed of pump in rpm

H = delivery head total manomentric in (meters)

447. The delivery valve, while starting centrifugal pump is kept

(a) fully closed (b) fully open

(c) half open (d) in any position

GPSC Executive Engineer 23.12.2018
UJVNL AE 2016, KPSC AE 2015
RPSC Lect. (Tech Edu. Dept.) 2011
WBPSC AE 2008, ESE 1997

Ans. (a) : The delivery valve while starting centrifugal pump is kept fully closed.

The delivery valve while starting axial pump is kept fully open.

448. A surge tank is provided in hydropower schemes to-

(a) Avoid reversal of flow

(b) Reduce water hammer pressure

(c) Strengthen the penstock

(d) Reduce friction losses in the system.

Gujarat PSC AE 2019, TNPSC AE 2019
MPPSC AE 2016, MPSC Dy. Engineer 2014
TRB Poly Lect. 2012, RPSC AE GWD 2011
J & K PSC Screening 2006

Ans. (b) : A surge tank is provided in hydropower schemes to reduce water hammer pressure.

Surge tank is a small storage tank or reservoir required in the hydro power plants for regulating the water flow during load reduction and sudden increases in the load on the hydro generator (water flow transients in penstock) and thus reducing the pressure on the penstock.

449. If two pumps identical in all respects and each capable of delivering a discharge Q against a head H are connected in series, the resulting discharge is :

(a) 2 Q against a head 2H

(b) 2Q against a head H

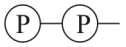
(c) Q against a head 2H

(d) Q against a head H.

SJVN ET Mechanical 2019 paper
RPSC VPI TI 14.02.2016, KPSC AE 2015
Kerala PSC AE 06.08.2015
MPSC AE 08.11.2015, RPSC AE GWD, 2011

Ans. (c) : Two pumps indentical in all respects and each capable

(i) When pump connected in series.



Q = discharge

H = head

Net discharge $[Q_{net} = Q_1 = Q_2] = Q$

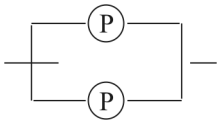
$Q_1 = Q_2 = Q$ resulting discharge Q

total head $[H_{total} = H_1 + H_2]$

$\because$ $H_1 = H_2 = H$

$\boxed{H_{total} = 2H}$

(ii) when connected in parallel



$Q_{net} = Q_1 + Q_2$

$H_{total} = H_1 = H_2$

450. The cavitation parameter σ is defined in terms of net positive suction head (NPSH) and net head H as

(a) NPSH/√H (b) H/NPSH

(c) H/√NPSH (d) NPSH/H

GPSC EE Pre 12.01.2017
GPSC EE Pre, 28.01.2017
APPSC AEE Screening Test 2016
MPSC Dy. Engineer 2014
TNPSC AE 2008

Ans. (d) : Cavitation in Pump- A cavitation parameter (σ) called as thoma's cavitation factor

$$\sigma = \frac{\frac{P_s(\text{Absolute})}{\rho g} + \frac{V_s^2}{2g} - \frac{P_v(\text{absolute})}{\rho g}}{H_m} = \frac{NPSH}{H_m}$$

$\sigma > \sigma_C$ to avoid cavitation,

The following factors Contribute towards onset of cavitation.

- High runner speed
- High temperature
- Less available NPSH.

451. In order to avoid cavitation in centrifugal pumps

(a) The suction pressure should be high

(b) The delivery pressure should be high

(c) The suction pressure should be low.

(d) The delivery pressure should be low

Nagaland PSC (CTSE) 2018, Paper-II
HPPSC AE 2018
MPSC Dy. Engineer 2014
UKPSC AE 2013 Paper II
APPSC AEE 2012

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Ans. (a) : In order to avoid cavitation in centrifugal pumps.
(1) The suction pressure should be high
(2) Lower temperature
(3) Reduce motor speed (rpm)
(4) Increase diameter of the eye of impeller.

452. If the net positive suction head (NPSH) requirement for the pump is not satisfied, then:
(a) no flow will take place
(b) cavitation will be formed
(c) efficiency will be low
(d) excessive power will be consumed

Nagaland PSC (CTSE) 2018, Paper-II
ISRO Scientist/Engineer (RAC) 01.05.2017
Punjab PSC SDE 12.02.2017
GPSC Lect. 23.10.2016
OPSC AEE 2015 PAPER - II, TNPSC AE 2014

Ans. (b) : Cavitation is defined as the phenomenon of formation of vapour bubbles and sudden collapsing of the vapour bubbles. Thoma suggested a dimensionless number called after his name Thoma's cavitation factor σ (sigma), Which can used for determining the region where cavitation takes place in reaction turbine.

$$\sigma = \frac{P_S(\text{Absolute}) + \frac{V_s^2}{2g} - \frac{P_V(\text{Absolute})}{\rho g}}{H_m}$$

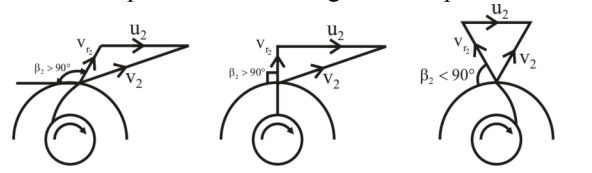
$$\sigma = \frac{NPSH}{H_m} \quad (\sigma > \sigma_C \text{ to avoid cavitation})$$

$$NPSH = (H_a - h_s - h_{fs}) - H_v$$

453. In general, the vanes of a centrifugal pump are
(a) Curved forward (b) Radial
(c) Curved backward (d) Spiral

CGPSC AE 05.10.2021, RPSC IOF, 2020
MPPSC AE 2016, CGPSC AE 16.10.2016

Ans. (c) : The vanes of centrifugal pumps are curved backward to prevent overloading of the impeller motor.



Forward vanes Radial vanes Backward vanes

454. The correct sequence of the centrifugal pump components through which the fluid flows is
(a) Impeller, Suction Pipe, Foot Valve and Strainer, Delivery Pipe.
(b) Foot Valve and strainer, Suction Pipe, Impeller, Delivery Pipe
(c) Impeller, Suction Pipe, Delivery Pipe, Foot Valve Strainer
(d) Suction Pipe, Delivery Pipe, Impeller, Foot Valve and Strainer

GPSC Asstt. Director of Transport 05.03.2017
ISRO Scientist/Engineer 2009
ESE 2002

Ans. (b) : In a centrifugal pump, the sequence of flow is 1st a foot valve and strainer is provided at the lowest point on suction pipe, then fluid start flowing through suction pipe, after that fluid enter into the impeller, which is inside casing, fluid then passes through eye of pump to the delivery valve to delivery pipe.

455. In a centrifugal pump casing, the flow of water leaving the impeller is
(a) rectilinear flow (b) free vortex motion
(c) radial flow (d) forced vortex

ISRO Scientist/Engineer 22.04.2018
APPSC AE Subordinate Service Civil/Mech. 2016
OPSC AEE 2015, Paper II

Ans. (b) : Flow inside the impeller of centrifugal pump is forced vortex motion and the flow of water leaving the impeller is free vortex motion.

456. Multi stage centrifugal pumps are used for
(a) high discharge
(b) high head and high discharge
(c) high pressure
(d) high efficiency

OPSC AEE 2019 PAPER - II
ISRO Scientist/Engineer 03.07.2016
TNPSC AE 2014, 2008

Ans. (c) : Generally in multi-stage centrifugal pumps are arranged in series. In case of multistage series arrangement head is increased, we get high pressure.

457. Multistage centrifugal pumps are used to
(a) give high discharge (b) pump viscous fluids
(c) produce high heads (d) None of these

APPSC AEE 2012
J & K PSC Screening 2006
ESE 2008

Ans. : (c) Multistage centrifugal pumps are used to produced high head.
Multistage Centrifugal pumps : - If a centrifugal pump consists of two or more impeller, the pump is called a multistage centrifugal pump. The impellers may be mounted on the same shaft or on different shaft. A multistage pump is having the following two important functions
(i) To produced a high head
(ii) To discharge a large quantity of liquid
If a high head is to be developed, the impellers are connected in series, while for discharging large quantity of liquid, the impeller are connected in parallel.

458. In a centrifugal pump, the liquid enters the pump
(a) At the centre (b) At the bottom
(c) At the top (d) From side

APPSC AE Subordinate Service Civil/Mech. 2016
MPPSC Dy. Engg. (GWD) 19.05.2013
APPSC AEE 2012

Ans. (a) : The centrifugal pump acts as a reverse of an inward radial flow reaction turbine. This means that the flow in centrifugal pumps is in the radial outward direction. The liquid enters the pump at the centre and leaves through casing.

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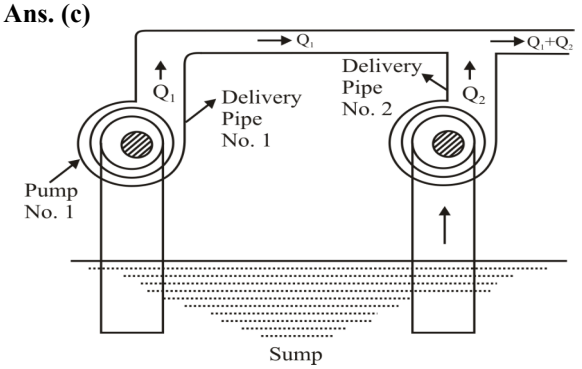
459. Multistage centrifugal pumps with impellers in series are used to :

(a) distribute the power among pumps
(b) increase the discharge of liquid
(c) increase the head of liquid
(d) decrease the head of liquid

APPSC Poly Lect. Automobile Engg., 2020
Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c)



The function of a multistage centrifugal pump is two types. When impeller is corrected in series with the same shaft of a centrifugal pump. It produced high head and to discharge enlarge quantity of liquid.

460. If Q is the volume flow rate, H is the net head and N is the rotational speed of a centrifugal pump, and if g denotes the acceleration due to gravity, then the dimensionless speed of the pump is given by :

(a) $\frac{N\sqrt{Q}}{gH^{0.7}}$ (b) $\frac{N\sqrt{Q}}{H^{0.75}}$

(c) $\frac{N\sqrt{Q}}{(gH)^{0.75}}$ (d) $\frac{N\sqrt{Q}}{H^{0.5}}$

Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017, DRDO Scientists 2009

Ans. (b) :

461. Overall efficiency of a centrifugal pump is the ratio of

(a) energy available at the impeller to the energy supplied to the pump by the prime mover
(b) actual work done by the pump to the energy supplied to the pump by the prime mover.
(c) energy supplied to the pump to the energy available at the impeller
(d) manometric head to the energy supplied by the impeller per kN of water

TSPSC AEE 28.08.2017 (Civil/Mechanical)
Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017

Ans. (b) : Overall efficiency of a centrifugal pump is the ratio of actual work done by the pump to the energy supplied to the pump by the prime mover

$$\eta_0 = \frac{\text{Actual energy added by pump to water}}{\text{Work done by prime mover}}$$

462. What is the slip responsible for in a centrifugal pump?

(a) Increases Cavitation
(b) Reduces the speed
(c) Reduces the flow rate
(d) Reduces the energy transfer.

TNPSC AE 2017
Mizoram PSC AE/SDO 2014, Paper-I
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Slip–In a centrifugal pump slip is responsible for reduces the energy transfer.
Slip is the internal backflow of a fluid in a pump returning from the discharge side to the suction side. In a centrifugal pump, slip is the loss of efficiency from a theoretical ideal arising from design constraint and turbulence at the impeller vanes.

463. A centrifugal hydraulic pump is used to force water to an open tank through a pipe having a diameter of 2 decimeters. Given that the pump is 4 km away from the tank, the average speed of the water in the pipe is 2 m/s. After neglecting the other minor losses, evaluate the absolute discharge pressure at the pump exit if it is to maintain a constant head of 5 m in the tank.
(Assume Darcy's friction factor of 0.01 for the pump).

(a) 0.449 bar (b) 5.503 bar
(c) 44.911 bar (d) 55.203 bar

HPPSC ADF 05.03.2019
UPRVUNL AE 2014, GATE 2004

Ans. : (b) Atmospheric pressure = 1.01325×10^5 bar
= 101325 Pa

Now atmospheric head $h = \frac{P}{\rho g}$

$= \frac{101325}{1000 \times 9.81}$

= 10.328 m

$h_f = \frac{f\ell v^2}{2gD} = \frac{0.01 \times 4000 \times 2^2}{2 \times 9.81 \times 0.2} = 40.77\,m$

Datum head required at the pump exit = 5 m
∴ Absolute head at the exit of pump
= 10.328 + 40.77 + 5
= 56.095 m
Absolute discharge pressure = ρgh
= 1000 × 9.81 × 56.095
= 5.503 bar

464. Which of the following pump is used for pumping viscous fluids?

(a) Reciprocating pump (b) Centrifugal pump
(c) Screw pump (d) Jet Pump

Nagaland PSC (CTSE) 2018, Paper-II
APGENCO AE 2017, TNPSC AE 2014

Ans. (c) : Screw pumps are used for transport of viscous fluids with lubricating properties. They are suited for a variety of application such as fuel injection, oil burners, boosting fuel, lubrication, feed so on.

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465. The specific speed $N_s = \left(N\sqrt{Q}\right)/H^{3/4}$ for a double suction pump is to be evaluated. The discharge would be taken

(a) half the actual discharge
(b) actual discharge
(c) double the actual discharge
(d) square of the actual discharge

(KPSC AE. 2015)
Kerala PSC AE 06.08.2015, ESE 2012

Ans. : (a) The specific speed $Ns = \left(N\sqrt{Q}\right)/H^{3/4}$ for a double suction pump is to be evaluated. The discharge would be taken half the actual discharge.

466. A pump running at 1000 rpm consumes 1 kW and generates head of 10m of water. When it is operated at 2000rpm, its power consumption and head generated would be

(a) 4 kW, 50m of water
(b) 6 kW, 20m of water
(c) 3 kW, 30m of water
(d) 8 kW, 40m of water

ISRO Scientist/Engineer 24.05.2014
ESE 2002

Ans. (d) : In Pump,-

$N_1 = 1000$ rpm
$H_1 = 10$ m
$P_1 = 1$kW
$N_2 = 2000$ rpm
$H_2 = ?$
$P_2 = ?$

$$\frac{\sqrt{H}}{DN}$$

For, Head-

$$\frac{\sqrt{H_1}}{D_1N_1} = \frac{\sqrt{H_2}}{D_2N_2}$$

$$\sqrt{\frac{H_2}{H_1}} = \frac{N_2}{N_1}$$

$$\frac{H_2}{H_1} = \left(\frac{N_2}{N_1}\right)^2$$

$$\frac{H_2}{10} = \left(\frac{2000}{1000}\right)^2$$

$$\boxed{H_2 = 40\,\text{m}}$$

For power-

$$\frac{P}{D^5N^3} = C$$

$$\frac{P_1}{D_1^5N_1^3} = \frac{P_2}{D_2^5N_2^3}$$

$$\frac{P_1}{P_2} = \left(\frac{N_1}{N_2}\right)^3 = 2^3 = 8$$

or $\boxed{P_2 = 8\,\text{kW}}$

467. The comparison between pumps operating in series and in parallel is :

(a) Pumps operating in series boost the head, whereas pumps in parallel boost the discharge.
(b) Pumps operating in series boost the discharge, whereas pumps in parallel boost the head.
(c) In both the cases there would be a boost in head only
(d) In both the cases there would be a boost in discharge only.

ISRO Scientist/Engineer (RAC), 10.03.2019
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : <u>Pumps in series</u>–

1. $Q = Q_1 = Q_2 = Q_3 = Q_4$ (Discharge is same)
2. $h = h_1 + h_2 + h_3 + h_4$ (Head is added)

Pumps in parallel

1. $h = h_1 = h_2 = h_3 = h_4$ (Head is same)
2. $Q = Q_1 + Q_2 + Q_3 + Q_4$ (Discharge is added)

468. A centrifugal pump driven by a directly coupled 3 kW motor of 1450 rpm speed, is proposed to be connected to a motor of 2900 rpm speed. The power of the motor should be

(a) 6 kW (b) 12 kW
(c) 18 kW (d) 24 kW

UPPSC AE 13.12.2020, Paper-2, ESE 2002

Ans. (d) : Given,

$P_1 = 3$ kW | $P_2 = ?$
$N_1 = 1450$ rpm | $N_2 = 2900$ rpm

$$\frac{P}{D^5N^3} = \text{constant}$$

$$\left(\frac{P_1}{N^3}\right)_1 = \left(\frac{P_2}{N^3}\right)_2 \Rightarrow \frac{3}{(1450)^3} = \frac{P_2}{(2900)^3}$$

$$\Rightarrow \boxed{P_2 = 24\text{kW}}$$

469. For a given Centrifugal pump

(a) Head varies inversely as square of speed
(b) Discharge varies directly as speed
(c) Discharge varies directly as square of speed
(d) Power varies directly as fifth power of speed

RPSC VPITI 14.02.2016, ESE 2005

Ans. (b) : In centrifugal pump discharge varies directly as speed.
It can handle large quantity of liquid.
It is used for large discharge and low head.

470. Priming is necessary in

(a) Centrifugal pumps to lift water from a greater depth
(b) Centrifugal pumps to remove air in the suction pipe and casing
(c) Hydraulic turbine to remove air in the turbine casing
(d) Hydraulic turbine to increase the speed of the turbine and to generate more power

RPSC VPITI 14.02.2016, ESE 2003

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Ans. (b) : Priming in which suction pipe casing of the pump and the portion of delivery pipe up to delivery valve is completely filled up form out side source with the liquid to be raised by the pump before the starting. Thus the air from these parts of the pump is removed and these parts are filled with the liquid to be pumped.

471. A centrifugal pump will start delivering liquid only when the pressure rise in the impeller is equal to the

(a) kinetic head
(b) velocity head
(c) manometric head
(d) static head

Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : A centrifugal pump will start delivering liquid only when the pressure rise in the impeller is equal to the manometric head.

472. Discharge of a centrifugal pump is

(a) directly proportional to diameter of its impeller
(b) inversely proportional to diameter of its impeller
(c) directly proportional to $(\text{diameter})^2$ of its impeller
(d) inversely proportional to $(\text{diameter})^2$ of its impeller

Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d)

473. For centrifugal pump impeller, the maximum value of the vane exit angle is

(a) 10º to 15º
(b) 15º to 20º
(c) 20º to 25º
(d) 25º to 30º

Nagaland PSC (CTSE) 2018, Paper-II
APPSC AEE 2012

Ans. (c) : For a centrifugal pump impeller, the maximum value of the none exit angle is 20ºC to 25º.

474. The centrifugal pump preferred for a specific speed between 80 to 160 r.p.m. is

(a) Slow speed with radial flow at outlet
(b) Medium speed with radial flow at outlet
(c) High speed with radial flow at outlet
(d) Highs speed with mixed flow at outlet

Nagaland PSC (CTSE) 2018, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) :

Pump	Specific speed
Radial flow	10 to 80
Mixed flow	80 to 160
Axial flow	160 to 450

Centrifugal pump preferred for a specific speed between 80 to 160 r.p.m. is high speed with mixed flow at outlet.

475. A minimum Net Positive Suction Head (NPSH) is required for a hydraulic pump to

(a) increase the delivery head
(b) increase the suction head
(c) prevent the cavitation
(d) increase the efficiency

APPSC AEE SCREENING 17.02.2019
ESE 2005

Ans. (c) : NPSH is used to study possibility cavitation in centrifugal pump.
To avoid cavitation NPSH > $(\text{NPSH})_{\text{min}}$

476. Mechanical efficiency of a centrifugal pump is the ratio of

(a) energy available at the impeller of the energy supplied to the pump by the prime mover
(b) actual workdone by the pump to the energy supplied to the pump by the prime mover
(c) energy supplied to the pump to the energy available at the impeller
(d) manometric head to the energy supplied by the the impeller per kN of water

TSPSC AEE 28.08.2017 (Civil/Mechanical)
ESE 2018

Ans. (a) : Mechanical efficiency (η_m) of a centrifugal pump is the ratio of energy available at the impeller to the energy supplied to the pump by the prime mover.

$$\eta_m = \frac{\text{Rotar Power}}{\text{Shaft Power}}$$

477. When the speed of the centrifugal pump is doubled the power required to drive the pump will

(a) increases 4 times (b) increases 8 times
(c) remains same (d) doubles

GPSC AE 26.07.2015, GATE 2000

Ans. (b) : $P \propto N^3$

478. Why is a minimum of Net Positive Suction Head required for a hydraulic pump?

(a) To prevent cavitations
(b) To increase discharge
(c) To increase suction head
(d) To increase efficiency

Vizag Steel MT (Re-Exam) 24.11.2013
ESE 2005

Ans. (a) : Net Positive Suction Head (NPSH)–The Net Positive Suction Head (NPSH) is define as the absolute pressure head at the inlet to the pump, minus the vapour pressure head (in absolute units) plus the velocity head.

$$\text{NPSH} = \frac{P_l}{\rho g} - \frac{P_v}{\rho g} + \frac{v_s^2}{2g}$$

$P_l \rightarrow$ Absolute pressure at the inlet of the pump
$P_v \rightarrow$ Vapour pressure at the inlet
$v_s \rightarrow$ Velocity of the fluid in the suction pipe

- Vapour pressure is the amount of pressure required for a specific liquid to stay in liquid form.
- When the pressure of the fluid falls below the vapour pressure then cavitation occurs.
- NPSH must be positive to prevent cavitation.

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479. A centrifugal pump running at 500 rpm and at its maximum efficiency is delivering a head of 30 m at a flow rate of 60 L/min. If the rpm is changed to 1000, then the head 'H' in meter and flow rate 'Q' in L/min at maximum efficiency are estimated to be

(a) H = 60, Q = 120 (b) H = 120, Q = 120
(c) H = 60, Q = 480 (d) H = 120, Q = 30

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2003

Ans. (b) : Given, $H_1 = 30$ m, $Q_1 = 60$ L/min, $N_1 = 500$ rpm

$N_2 = 1000$ rpm, $H_2 = ?$, $Q_2 = ?$

$$\frac{H_2}{H_1} = \left(\frac{N_2}{N_1}\right)^2 \Rightarrow \frac{H_2}{30} = \left(\frac{1000}{500}\right)^2 = 4$$

$H_2 = 120$ m

$$\frac{Q_1}{D_1^3 N_1} = \frac{Q_2}{D_2^3 N_2}$$

$$Q_2 = \left(\frac{1000}{500}\right) \times 60 = 120 \text{ litre/min}$$

480. Which of the followings is the appropriate equation used to compute minimum speed for starting a centrifugal pump?
(Where H_m is the manometric head, g is acceleration due to gravity, D_1 is the outside diameter of the impeller and D_2, is the inter diameter of the impeller.

(a) $\frac{60\sqrt{2gH_m}}{\pi\sqrt{D_1^2 - D_2^2}}$ (b) $\frac{12\sqrt{gH_m}}{\pi\sqrt{D_1^2 - D_2^2}}$

(c) $\frac{60\sqrt{2H_m}}{\pi\sqrt{D_1^2 - D_2^2}}$ (d) $\frac{48\sqrt{2gH_m}}{\sqrt{D_1^2 - D_2^2}}$

APPSC AE Subordinate Service Civil/Mech. 2016
MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Minimum speed for starting a centrifugal pump– When a centrifugal pump is started, it will start delivering liquid only if the pressure rise in the impeller is more than or equal to monometric head flow will commence only if, $\frac{u_2^2 - u_1^2}{2g} \geq H_m$

For minimum speed

$$\frac{u_2^2 - u_1^2}{2g} = H_m \quad \text{..... (i)}$$

also, $\eta_{man} = \frac{gH_m}{v_{w_2} u_2} \Rightarrow H_m = \eta_{man.} \times \frac{v_{w_2} u_2}{g}$ ..... (ii)

equating equation (i) and (ii)

$$\frac{u_2^2 - u_1^2}{2g} = \eta_{man.} \times \frac{v_{w_2} u_2}{g} \quad \text{..... (iii)}$$

$$u_2 = \frac{\pi D_2 N}{60} \text{ and } u_1 = \frac{\pi D_1 N}{60}$$

Substituting the values of u_2 and u_1 in equation (iii)

$$\left(\frac{\pi D_2 N}{60}\right)^2 - \left(\frac{\pi D_2 N}{60}\right)^2 = 2\eta_{man.} \times v_{w_2} u_2$$

$$\eta_{man} \times v_{w_2} \times u_2 = H_m \times g$$

$$\left(\frac{\pi N}{60}\right)^2 \left[D_2^2 - D_1^2\right] = 2gH_m$$

$$N = \frac{60\sqrt{2gH_m}}{\pi\sqrt{D_2^2 - D_1^2}}$$

481. Discharge (Q) of a centrifugal pump is given by

(a) $Q = \pi D V_f$ (b) $Q = \pi b V_f$
(c) $Q = \pi D b V_f$ (d) $Q = D b V_f$

(Where D = diameter of impeller at inlet
b = Width of the impeller at inlet
V_f = Velocity of flow at inlet)

TSPSC AEE 28.08.2017 (Civil/Mechanical)
TSPSC AEE 2015

Ans. : (c) Discharge (Q) of a centrifugal pump
$Q = \pi D b V_f$
D = Diameter of impeller at inlet
b = width of the impeller at inlet
v_f = Velocity of flow at inlet.

482. A centrifugal pump lift water through a height h and delivers it at a velocity v_d. The loss of head through piping is h_f. The gross lift is-

(a) $h + h_f$ (b) $h_f + \frac{v_d^2}{2g}$

(c) $h + h_f + \frac{v_d^2}{2g}$ (d) $h + \frac{v_d^2}{2g}$

RPSC AE 2018, UKPSC AE 2012 Paper–II

Ans. (c) : The gross lift by centrifugal pump = total height of lifts water + discharge velocity head + loss of head through piping

$$= h + \frac{V_d^2}{2g} + h_f$$

483. Power need to drive the centrifugal pump is proportional to

(a) N (b) N^2
(c) N^3 (d) N^{5-2}

HPPSC AE 2018, Vizag STEEL MT 2015

Ans. (c) : We know that

$$\left.\frac{P}{D^5 N^3}\right|_m = \left.\frac{P}{D^5 N^3}\right|_P = \text{constant}$$

So From this relation–

$P \propto N^3$

484. Two centrifugal pumps 'A' operate at their maximum efficiencies at 2000 rpm and 1000 rpm respectively. Against the same delivery head, pump 'A' discharge 2 m^3/s and pump B discharge 8m^3/s respectively. What is the ratio of specific speeds $(N_s)_A : (N_s)_B$?

(a) 1:1 (b) 1:2
(c) 1:4 (d) 4:1
(e) 2:1

CGPSC AE 26.04.2015 SHIFT-II, ESE 2004

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Ans. (a) : Data given as:
N_A= 2000 rpm N_B = 1000 rpm
$Q_A = 2m^3/s$ $Q_B = 8\ m^3/s$
$H_A = H_B$
We know that,

$$N_S \propto N\sqrt{Q}$$

$$\frac{(N_s)_A}{(N_s)_B} = \frac{2000\times\sqrt{2}}{1000\times\sqrt{8}} = 1:1$$

485. A centrifugal pump is required to lift 0.0125 m^3/s of water from a well with depth 30 m. If rating of the pump motor is 5 kW, and the density of water is 1000 kg/m^3, the efficiency of the pump will be nearly
(a) 82% (b) 74%
(c) 66% (d) 58%

ESE 2020
RPSC IOF & Boiler 20.09.2017

Ans. (b) : Given,
$Q = 0.0125\ m^3/s$
H = 30 m
Actual power,
$P_a = \rho QgH$
$= 1000\times0.0125\times9.81\times30$
= 3.678 kW
Pump motor power,
P_m = 5 kW
Efficiency of pump,

$$\eta = \frac{P_a}{P_m}\times100\% = \frac{3.678}{5}\times100\% = 73.57\%$$

486. Chances of occurrence of cavitation are high if the
(a) Local pressure becomes very high
(b) Local pressure falls below the vapour pressure
(c) Thoma cavitation parameter exceeds a certain limit
(d) Local temperature becomes low

UPPSC AE 13.12.2020 Paper II
ISRO Scientist/Engineer 2007

Ans. (b) : Chances of occurrence of cavitation are high if the local pressure falls below the vapour pressure.

487. If D is the diameter of impeller at inlet, w is the width of impeller at inlet and V_f is the velocity of flow at inlet, then discharge through a centrifugal pump is equal to
(a) πDV_f (b) $DV_f w$
(c) $\pi DV_f w$ (d) πDw

OPSC AEE 2015 Paper II
ISRO Scientist/Engineer 2010

Ans. (c) : Impeller diameter= D at Inlet
width of Impeller at Inlet = w
velocity of flow at Inlet = V_f
Discharge through a centrifugal pump is equal to $\pi DV_f w$

$$Q = \pi DV_f w$$

488. Motion of liquid in volute casing of a centrifugal pump is an example of:
(a) spiral vortex flow
(b) radial flow
(c) forced cylindrical vortex flow
(d) forced spiral vortex flow

Kerala PSC AE (Irrigation) 05.04.2017
TNPSC AE, 2008

Ans. (a) : The volute casing is designed to guide the flow out of the impeller in order to convert the fluid flow. Kinetic energy into static pressure, it serves to collect the fluid discharges from the impeller and route it to the discharge nozzle.
Motion of liquid in volute casing of a centrifugal pump is an example of spiral vortex flow.

489. The manometric efficiency of a centrifugal pump whose overall efficiency is 30% while its mechanical efficiency is 60% is
(a) 90% (b) 0.18%
(c) 50% (d) 30%

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : $\eta_0 = \eta_m \times \eta_H$

$$\eta_m = \frac{30}{60} = 50\%$$

490. To discharge large quantity of water in centrifugal pumps
(a) multistage pumps with impellers connected in parallel should be used
(b) manometric efficiency should be high
(c) delivery head should be high
(d) multistage pumps with impellers connected in series should be used

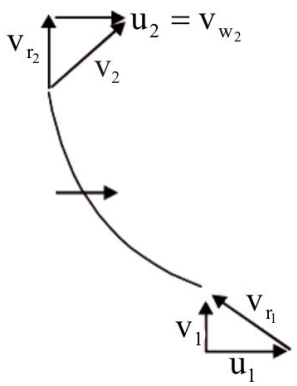
GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) :

491. A centrifugal pump has an impeller of 30 cm outer diameter. The vane tips are radial at the outlet. For a rotating speed of 1450 rpm, what is the manometric head developed? (Assume a manometric efficiency of 82% and take g = 9.81 m/s_2)
(a) 37.24 m (b) 43.38 m
(c) 29. 46 m (d) 32.88 m

ESE 18.07.2021

Ans. (b) :



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D2 = 30 cm
Vane tips are radial at outlet
N = 1450 rpm
$\eta_{mano} = 82\%$
$H_m = ?$

$$u_2 = \frac{\pi D_2 N}{60} = \frac{\pi \times 0.30 \times 1450}{60} = 22.765\,m/s$$

$$\eta_{mano} = \frac{gH_m}{v_{w_2} u_2} = \frac{gH_m}{u_2^2}$$

$$= 0.82 = \frac{9.81 \times H_m}{(22.765)^2} \qquad (\because v_{w_2} = u_2)$$

Manometric head $(H_m) = 43.32\,m$

492. Correct combinations for centrifugal pump efficiencies are :

(1) Manometric efficiency (I) $\frac{\text{Water Power}}{\text{Shaft Power}}$

(2) Mechanical efficiency (II) $\frac{\text{Water Power}}{\text{Power of Impeller}}$

(3) Hydraulic efficiency (III) $\frac{\text{Power in fluid}}{\text{Impeller Power}}$

(4) Overall efficiency (IV) $\frac{\text{Impeller Power}}{\text{Shaft Power}}$

(a) (b)
(c) (d)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : Manometric efficiency = $\frac{\text{Water Power}}{\text{Power of Impeller}}$

Mechanical efficiency = $\frac{\text{Impeller Power}}{\text{Shaft Power}}$

Hydraulic efficiency = $\frac{\text{Power in fluid}}{\text{Impeller Power}}$

Overall efficiency = $\frac{\text{Water Power}}{\text{Shaft Power}}$

493. The specific speed of a pump is 25 and rated capacity of the pump is 0.16 m^3/s. Find head developed by the pump at 800 RPM.

(a) 25.6 m (b) 29.94 m
(c) 12.8 m (d) 59.88 m

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (b) : Given, $N_s = 25$, $Q = 0.16\ m^3/sec$, $N = 800$ rpm

Specific speed of pump $(N_s) = \frac{N\sqrt{Q}}{H^{3/4}}$

$$25 = \frac{800\sqrt{0.16}}{H^{3/4}}$$

$$25 = \frac{800 \times 0.4}{H^{3/4}} = \frac{320}{H^{3/4}}$$

$$H^{3/4} = \frac{320}{25} = 12.8$$

$$H = (12.8)^{4/3}$$

$$H = 29.94\ m$$

494. Two homologous pumps A and B are to run at the same speed. Pump A has an impeller diameter of 50 cm and discharge 0.4 m^3/s under the head of 50 m. What will be the diameter of pump B, if its discharge is 0.2 m^3/s? [Assuming $2^{1/3} = 1.26$]

(a) 63.0 cm (b) 52.31 cm
(c) 47.17 cm (d) 39.68 cm

APPSC Poly Lect. 13.03.2020

Ans. (d) : Given, speed is constant for the both pump

Given, $Q_A = 0.4\ m^3/s$
$Q_B = 0.2\ m^3/s$
$D_A = 50\ cm = 0.5\ m$
$D_B = ?$

In case of the pump–

$$\frac{Q}{D^3 N} = \text{constant} \quad \text{or} \quad \frac{Q}{D^3} = \text{constant}$$

$$\Rightarrow \left(\frac{Q}{D^3}\right)_A = \left(\frac{Q}{D^3}\right)_B$$

$$\Rightarrow D_B^3 = \frac{Q_B}{Q_A} \times (D_A)^3$$

$$\Rightarrow D_B^3 = \frac{0.2}{0.4} \times (0.5)^3$$

$$\Rightarrow D_B = \frac{0.5}{(2)^{1/3}} = \frac{0.5}{1.26} \qquad [\text{Given; } 2^{1/3} = 1.26\ cm]$$

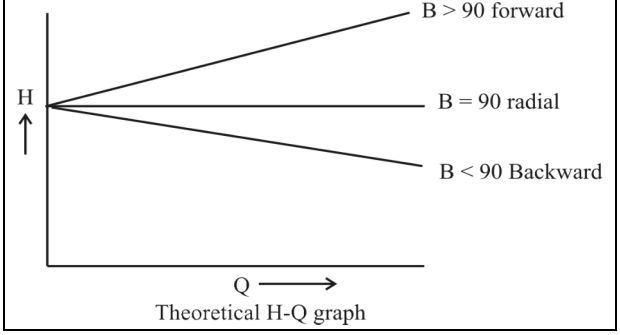
$$\Rightarrow D_B = 0.3968\ m \quad \text{or} \quad \boxed{D_B = 39.68\ cm}$$

495. The slope of ideal head-discharge line is [positive, zero, negative] corresponding to the following vane curvature profiles in centrifugal pumps:

(a) backward, radial forward
(b) forward, backward, radial
(c) backward, forward, radial
(d) forward, radial, backward

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Forward has positive slope & backward has negative slope.



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496. A centrifugal pump having an impeller diameter of 127 mm, delivers a power of 12 hp. If the impeller diameter is changed to 254 mm, what is the power, if other parameters are kept constant?

(a) 48 hp (b) 192 hp
(c) 24 hp (d) 96 hp

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : The main output of centrifugal pump is manometric head.

Take same manometric head (H).

$P_{output} = P = \eta_o \times mg\,H$

$P = \eta_o \times \rho Q g H$

$Q = A_{flow} \times V_{flow}$ $\quad \left[\begin{matrix} A_{flow} \propto D^2 \\ V_{flow} \propto \sqrt{H} \end{matrix}\right]$

$Q \propto D^2\sqrt{H}$

$P \propto D^2\sqrt{H} \times H$

$P \propto D^2(H)^{3/2}$

$\dfrac{P}{D^2(H)^{3/2}} = \text{constant}$ (which is specific quantity)

$\dfrac{P_1}{D_1^2(H_1)^{3/2}} = \dfrac{P_2}{D_2^2(H_2)^{3/2}}$ $\quad \left[\text{Here } H_1 = H_2\right]$

$\dfrac{P_2}{P_1} = \left(\dfrac{D_2}{D_1}\right)^2 = \left(\dfrac{254}{127}\right)^2$

$P_2 = 4P_1 = 4 \times 12 = 48$ hp

Option (a) is correct.

Note-But ISRO consider option (d) is correct.

497. Impellers of a rotary pump for high heads usually have :

(a) medium specific speed
(b) high specific speed
(c) low specific speed
(d) variable specific speed

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Impellers of a rotary pump for high heads usually have low specific speed.

498. The fixed guide blades in a centrifugal pump are used to :

(a) increase the kinetic energy
(b) increase the pressure energy
(c) simply direct the fluid
(d) decrease the pressure energy

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The fixed guide blades in a centrifugal pump are used to increase the pressure energy.

499. The pressure rise in the impeller of a centrifugal pump is :

(a) $[V_f^2 + v_1^2 + v_{fl}^2\ \text{Cosec}^2\ \Phi]$
(b) $[V_f^2 + v_1^2 - v_{fl}^2\ \text{Cosec}^2\ \Phi]/(2g)$
(c) $[V_f^2 + v_1^2 - v_{fl}^2\ \text{Cosec}^2\ \Phi]/g$
(d) $[V_f^2 + v_1^2 + v_{fl}^2\ \text{Cosec}^2\ \Phi]/(2g)$

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The pressure rise in the impeller of a centrifugal pump is

$V_f^2 + v_1^2 - v_{fl}^2\ \text{Cosec}^2\ \Phi/2g$

500. A centrifugal pump runs at a speed of 1450 rpm and discharges 0.118 m^3/s of water. The pump works against a head of 25m. The impeller radius is 12.5 cm. its width at the outlet is 5 cm and manometric efficiency is 75%. The value angle in degrees at the outlet periphery of the impeller is :

(a) 79.77 (b) 39.77
(c) 49.77 (d) 59.77

BHEL ET 2019

Ans. (d) : Given -

Speed N = 1450 rpm

$Q = 0118\ m^3/sec.$

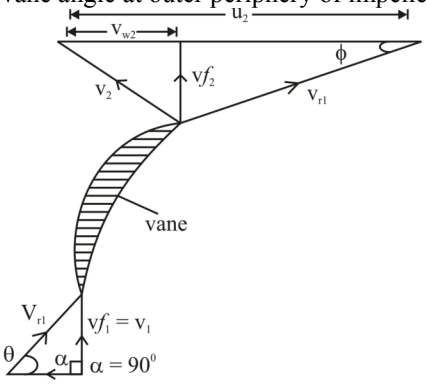
Head (H) = 25 m

Impeller radius (r) = 12.5 cm = D = 0.25 m

width at outlet (w) = 5 cm = 0.05 m

manometric efficiency (η_m) = 75%

vane angle at outer periphery of impeller.



vane angle at outlet (ϕ)

Tangential velocity of impeller at outlet,

$$u_2 = \frac{\pi D_2 N}{60}$$

$$= \frac{\pi \times 0.25 \times 1450}{60}$$

$$= 18.98 \text{ m/s}$$

Discharge $Q = \pi D_2 B_2 \times v_{f2}$

$$v_{f2} = \frac{Q}{\pi D_2 B_2} = \frac{0.118}{\pi \times 0.25 \times 0.05}$$

$v_{f2} = 3.0$ m/s

manometric efficiency = $\eta_{mano} = \dfrac{gH_{mano}}{v_{W2}u_2}$

$$0.75 = \frac{9.81 \times 25}{v_{w2} \times 18.98}$$

or $$v_{W2} = \frac{9.81 \times 25}{0.75 \times 18.98}$$

$v_{W2} = 17.23$ m/s

from velocity triangle at outlet,

$$\tan\phi = \frac{v_{f2}}{u_2 - v_{W2}}$$

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$$= \frac{3}{18.98 - 17.23} = 1.7143$$

$$\tan\phi - = 1.7143$$

$$\phi = \tan^{-1}(1.7143)$$

$$\phi = 59.74^0$$

501. Impeller of two geometrical similar centrifugal pump operates at same speed, how power, discharge and head will vary with diameter ratio, d, respectively as:

(a) d^5, d^3, d^2 (b) d, d^3, d^2

(c) d^5, d^3, d (d) d^3, d^2, d

OPSC AEE 2019 Paper - II

Ans. : (a) : We know that,

$$\frac{H}{D^2N^2} = \text{Constant}$$

then, $H \propto D^2$

$$\frac{Q}{D^3N} = \text{constant}$$

$Q \propto D^3$

$$\frac{P}{D^5N^3} = \text{constant}$$

$P \propto D^5$

502. The head against which a centrifugal pump has to work is known as:

(a) Static head

(b) Total load

(c) Net positive suction head

(d) Manometric head

OPSC AEE 2019 Paper - II

Ans. : (d) : Manometric head- The manometric head is defined as the head against which a centrifugal pump has to work.

503. For high discharge and low head such as irrigation, the type of pump preferred is

(a) centrifugal pump (b) reciprocating pump

(c) propeller pump (d) gear pump

Gujarat PSC AE 2019

Ans. : (c) : For high discharge and low head such as irrigation, the propeller type of pump is preferred.
Axial flow type is also used for very high discharge at low pressure such as flood control and irrigation applications.

504. Axial thrust is zero in

(a) Screw pump (b) Axial flow pump

(c) Vane pump (d) Lobe pump

Gujarat PSC AE 2019

Ans. : (a) : Axial thrust is zero in screw pump.

505. A centrifugal pump has an impellor of outer diameter of 30 cm. Water enters the impellor radially. Vane tips are radial at outlet. The rotative speed is 1500 rpm and manometric efficiency is 80%. What is the net head developed (in meter)? [Assuming acceleration due to gravity = g = π]

(a) 45 (b) 60

(c) 70 (d) 100

SJVN ET 2019

Ans. (a) : Manometric efficiency of centrifugal pump

$$\eta_{man} = \frac{gH}{V_w u}$$

For radial tip, whirl velocity at outlet is equal to peripheral speed at outlet,

$$u = v_w = \frac{\pi DN}{60}$$

$$= \frac{\pi \times 0.3 \times 1500}{60}$$

$$= 23.56 \text{ m/sec}$$

$$H = \frac{\eta_{max} V_w u}{g} = \frac{0.8 \times 23.56 \times 23.56}{9.81}$$

$$= 45.26 \text{ m}$$

$$\approx 45 \text{ m}$$

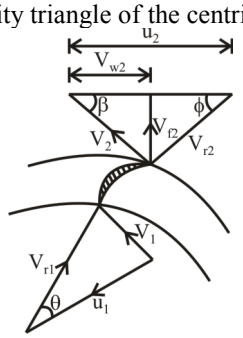
506. A pump works on the principle of centrifugal theory, running at 900 rpm is working against a head of 16 m. The external diameter of the impeller is 360 mm and the outlet width is 40 mm. If the vane angle at outlet is 30° and the manometric efficiency is 80%, the discharge of the pump will be :

(a) 0.22 m^2/s (b) 0.24 m^2/s

(c) 0.18 m^2/s (d) 0.14 m^2/s

BHEL ET 2019

Ans. (d) : Velocity triangle of the centrifugal pump–



Given, $H_m = 16$ m, $N = 900$ rpm

$\phi = 30°$, $D_2 = 360$ mm $= 0.36$ m

$B_2 = 40$ mm $= 0.04$ m

$\eta_{mano} = 80\% = 0.8$

$$u_2 = \frac{\pi D_2 N}{60} = \frac{\pi \times 0.36 \times 900}{60} = 16.96 \text{ m/s}$$

$$\eta_{mano} = \frac{gH_m}{V_{w2} u_2}$$

$$0.8 = \frac{9.81 \times 16}{V_{w2} \times 16.96}$$

$V_{w2} = 11.56$ m/s

$$\tan\phi = \frac{V_{f2}}{(u_2 - V_{w2})}$$

$$\tan 30° = \frac{V_{f2}}{16.96 - 11.56} \Rightarrow V_{f2} = 5.4 \times \tan 30° = 3.117$$

Discharge $Q = \pi D_2 B_2 V_{f2}$

$= \pi \times 0.36 \times 0.04 \times 3.117 = 0.14 \text{ m}^3/\text{sec}$

507. A centrifugal pump driven by a directly coupled 3 kW motor of 1450 rpm speed is proposed to be connected to another motor of 2900 rpm. The power of the motor should be

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(a) 24 kW (b) 18 W
(c) 12 kW (d) 6 kW

APPSC AEE SCREENING 17.02.2019

Ans. (a) : We know that,

$$P \propto D^2 H^{3/2} \quad \& \quad ND \propto \sqrt{H}$$

$$\therefore \quad P \propto D^2 (ND)^3$$

$$\propto D^5 N^3$$

$$\therefore \quad \frac{P_2}{P_1} = \left(\frac{D_2}{D_1}\right)^5 \times \left(\frac{N_2}{N_1}\right)^3$$

$$\frac{P_2}{3} = 1^5 \times \left(\frac{2900}{1450}\right)^3$$

$$P_2 = 2^3 \times 3 = 24kW$$

508. A diffuser blade in the centrifugal pump is used to

(a) convert mechanical energy to kinetic energy
(b) convert kinetic energy to pressure energy
(c) convert pressure energy to kinetic energy
(d) convert potential energy to kinetic energy

APPSC AEE SCREENING 17.02.2019

Ans. (b) : In diffuser, flow area increases, velocity decreases and as per Bernoulli's equation pressure increases.

509. If H is the unit head, the unit power of the turbine is proportional to

(a) H (b) H^{-2}
(c) $H^{-(1/2)}$ (d) $H^{-(3/2)}$

APPSC AEE SCREENING 17.02.2019

Ans. (d) : Unit Power—It is defined as the power developed by a turbine, working under a unit head (i.e. under a head or 1m). It is denoted by the symbol 'P_u'. The expression for unit power is obtained as

$$P_u = \frac{P}{H^{3/2}}$$

510. Net positive suction head is

(a) pressure head + kinetic head
(b) pressure head + vapour pressure + kinetic head
(c) pressure head - vapour pressure + kinetic head
(d) pressure head - vapour pressure - kinetic head

APPSC AEE SCREENING 17.02.2019

Ans. (c) : The net positive suction head is defined as

$$NPSH = \frac{(P_{stag})_{inlet\,(or)\,suction} - P_v}{\rho g}$$

$$= \left(\frac{P}{\rho g} + \frac{V^2}{2g}\right)_{inlet} - \frac{P_v}{\rho g}$$

511. In axial flow pumps and compressors, the combined velocity diagram with common base is used to determine change in

(a) absolute velocity $(V_2 - V_1)$
(b) relative velocity $\left(V_{r_2} - V_{r_1}\right)$
(c) tangential velocity $(U_2 - U_1)$
(d) whirl velocity $\left(V_{u_2} - V_{u_1}\right)$

ESE 2019

Ans. (d) The whirl velocity is the tangential component of absolute velocity at the blade inlet and outlet.
This component of velocity is responsible for the whirling or rotating of the turbine rotor.

512. Manometric head, in case of a centrifugal pump, is equal to

(a) Suction lift + Loss of head in suction pipe due to friction + delivery lift + Loss of head in delivery pipe due to friction + Velocity head in the delivery pipes
(b) Workdone per k N of water – Losses within the impeller
(c) Energy per k N at outlet of impeller
Energy per k N at inlet of impeller
(d) All of the above

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : The manometric head is defined as the head against which a centrifugal pump has to work. Manometric head is the head measured across the pump inlet and outlet flanges.
In case of centrifugal pump manometric head = suction lift + Loss of head in suction pipe due to friction + delivery lift + Loss of head in delivery pipe due to friction + velocity head in the delivery pipes
Manometric head = workdone Per KN of water – Losses within the impeller.
And Energy per KN at outlet of impeller – energy
So, option 4 is correct.

513. In a centrifugal pump, the regulating valve is provided on the

(a) Casing (b) Delivery pipe
(c) Suction pipes (d) Impeller

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : A delivery valve is a regulating valve which is of sluice type and is required to provide in order to control the flow from the pump into delivery pipe.

514. Power required to drive a centrifugal pump is

(a) Directly proportional to speed of its impeller
(b) Directly proportional to square of speed of its impeller
(c) Directly proportional to cube of the speed of its impeller
(d) Inversely proportional to cube of the speed of its impeller

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (c) : We know that, for centrifugal pump

$$\frac{P}{D^5 N^3} = \text{constant}$$

$$P \propto N^3$$

$$P \propto D^5$$

515. The specific speed of a centrifugal pump delivering 810 litres of water per sec. against head of 16 metres. at 725 RPM is

(a) 81 RPM (b) 810 RPM
(c) 40 RPM (d) 16 RPM

ISRO Scientist/Engineer (RAC) 22.04.2018

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Ans. (*) : No answer given by ISRO

$$N_s = \frac{N\sqrt{Q}}{H^{3/4}}$$

$$= \frac{725 \times \sqrt{0.81}}{(16)^{3/4}}$$

$N_s = 81.56$

Closed answer is (a)

516. A centrifugal pump delivers 1000 litres per minute at 2000 RPM against total head of 50 metres. and requires 32 BHP for its operation. If speed is reduced to 1000 RPM, the discharge and head developed will be

(a) 500 litres/min and 25 metre
(b) 500 litres/min and 12.5 metre
(c) 250 litres/min and 12.5 metre
(d) 250 litres/min and 25 metre

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (b) : $Q_1 = 1000$ litres/min., $Q_2 = ?$

$N_1 = 2000$ RPM, $N_2 = 1000$ RPM

$H_1 = 50$ m, $H_2 = ?$

$P_1 = 32$ BHP

We know that unit speed,

$$N_u = \frac{N_1}{H_1^{1/2}} = \frac{N_2}{H_2^{1/2}} = \text{constant}$$

$$H_2 = \left(\frac{N_2}{N_1}\right)^2 \times H_1$$

$$= \left(\frac{1000}{2000}\right)^2 \times 50$$

$H_2 = 12.5$ m

and unit discharge Q_u

$$Q_u = \frac{Q_1}{H_1^{1/2}} = \frac{Q_2}{H_2^{1/2}}$$

$$Q_2^2 = \frac{H_2}{H_1} \times Q_1^2 = \frac{12.5}{50} \times (1000)^2$$

$Q_2 = 500$ litres/min

517. The specific speed of a centrifugal pump, delivering 1000 litres of water per second against a head of 16 metre at 800 rpm

(a) 50 rpm (b) 100 rpm
(c) 150 rpm (d) 200 rpm

ISRO Scientist/Engineer 22.04.2018

Ans. (b) : Sp. speed of centrifugal pump

$$N_S = \frac{N\sqrt{Q}}{H^{3/4}} = \frac{800 \times \sqrt{1000 \times 10^{-3}}}{16^{3/4}}$$

$N_S = 100$ rpm

518. Which of the following is correct for centrifugal pumps?

(a) Constant efficiency curve is obtained by using manometric head versus discharge curve for different speed and overall efficiency versus discharge curve for different speed
(b) Constant efficiency curve is obtained by using input power versus discharge curve for different speed and overall efficiency versus discharge curve for different speed
(c) Constant efficiency curve is obtained by combining input power versus discharge curve for different speed and head versus discharge curve for different speed
(d) For pumps connected in series, the total discharge capacity will be the sum of individual discharge capacity
(e) For pumps connected in parallel, the total head developed will be the sum of individual head developed by each pump

CGPSC AE 25.02.2018

Ans. (a) : In centrifugal pump, constant efficiency curve is obtained by using manometric head versus discharge curve for different speed and overall efficiency versus discharge curve for different speed.

519. Consider the following advantages of rotary pumps compared to reciprocating pumps:

1. Steady discharge which increases with decrease in head
2. Suitable for handling fluids with suspended solid particles
3. Less bulky than positive displacement pumps
4. Can be started with open delivery with least load

Which of the above advantages are correct?

(a) 1, 2 and 3 (b) 1, 2 and 4
(c) 1, 3 and 4 (d) 2, 3 and 4

ESE 2018

Ans. (c) : Rotary pumps can handle all types of liquids that do not have any solid content and are poor is transporting thick and viscous while reciprocating pump is suitable for fluid having solid particles. Rotary pump gives steady discharge and discharge increase with decrease in head.

Note : The answer given by UPSC is option (a).

520. In the casing of a centrifugal pump, the kinetic energy of the water is converted into _____ energy before the water leaves the casing.

(a) Pressure (b) Magnetic
(c) Electronic (d) Dynamic

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : Pump casing has gradually increasing area it performs similar to deffuser. As area increases, velocity reaches and pressure increases. Thus the pump casing converts kinetic energy of water partially into energy.

521. In a centrifugal pump with usual notations with radial entry of liquid, the manometric efficiency (η_H) is given by

(a) $\frac{u_2 v_{w_2}}{gH}$ (b) $\frac{gH}{u_2 v_{w_2}}$

(c) $\frac{H}{g u_2 v_{w_2}}$ (d) $\frac{2gH}{u_2 v_{w_2}}$

TSPSC AEE 28.08.2017 (Civil/Mechanical)

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Ans. (b) : Hydraulic (or) manometric efficiency (η_{meno}) : it is ratio of manometric head to head imparted to the fluid to the mark done by the impeller per unit weight to fluid.

$$\eta_{mano} = \frac{H}{H_e} = \frac{H}{\frac{u_2 v_{w_2}}{g}} = \frac{gH}{u_2 V_{w_2}}$$

522. The specific speed of pump has following dimensions :

(a) $L^{3/4}T^{3/2}$ (b) $M^0L^0T^0$

(c) $M^{-1/2}L^{1/2}T^{1/4}$ (d) $L^{3/4}T^{1/2}$

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : For pumps,

$$N_S = \frac{N\sqrt{Q}}{H^{3/4}} = \frac{T^{-1}\sqrt{L^3T-1}}{L^{3/4}}$$

$$= M^oL^{3/4}T^{-3/2}$$

Note: (–) Minus sign is missing in the dimension of T in.

523. Head developed by a centrifugal pump is

(a) Proportional to diameter of impeller

(b) Proportional to speed of impeller

(c) Proportional to diameter and speed of impeller

(d) Proportional to diameter of delivery pipe

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : Head developed by the centrifugal pump is proportional to impeller diameter and impeller speed.

524. Power required to drive a centrifugal pump is directly proportional to _____ of its impeller.

(a) diameter

(b) square of diameter

(c) cube of diameter

(d) fourth power of diameter

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : $P \propto \rho gQh$

$\propto (D^2\sqrt{H})H$

$\propto D^2 H^{3/2}$

$\therefore$ $P \propto D^2$ if head is constant

Note : $u \propto v \propto \sqrt{H}$

$\frac{\pi DN}{60} \propto \sqrt{H}$

$ND \propto \sqrt{H}$

$\therefore$ $P \propto D^2 \times (ND)^3$

$\propto D^5N^2$

$\therefore$ ($P \propto D^5$ if speed is kept constant)

525. The specific speed from 160 to 500 rpm of a centrifugal pump indicates that the pump is

(a) slow speed with radial flow at outlet

(b) medium speed with radial flow at outlet

(c) high speed with radial flow at outlet

(d) high speed with axial flow at outlet

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : Range of specific speeds for different types of centrifugal pumps is given below.

N_S	Type of pump
10 to 30	Low speed radial flow
30 to 50	Medium speed radial flow
50 to 100	High speed radial flow
75 to 220	Mixed flow pumps
180 to 450	Axial flow pumps

Based on above table axial flow pumps are most suitable for given range of specific speed.

526. Which of the following centrifugal pumps has the maximum specific speed?

(a) Axial flow

(b) Radial flow

(c) Mixed flow

(d) All have same specific speed

WBPSC AE, 2017

Ans. (a) : Axial flow centrifugal pumps has the maximum specific speed.

527. Power in kW required to drive Centrifugal pump with discharge of Q m^3/sec. Head in H mtr, specific weight of fluid as ω N/m^3 and overall efficiency of pump as η shall be

(a) ωH/Q η (b) ωQH/ η

(c) ωQ/H η (d) Qωη/H

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (b) : We know that overall efficiency of centrifugal pump is

$$\eta = \frac{\omega \times Q \times H}{P}$$

So, $$P = \frac{\omega \times Q \times H}{\eta}$$

528. Consider the following statements:

1. The specific speed of a two stage centrifugal pump is twice that of a single stage pump
2. The specific speed of a double end suction pump is half that of a single end suction pump
3. The specific speed is not a dimensionless number

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 only

(c) 2 only (d) 3 only

JWM 2017

Ans. (d) : The specific speed is a dimensional number as well as dimensionless number

$$N_S = \frac{N\sqrt{P}}{H^{5/4}} \rightarrow \text{Dimensioned specific speed}$$

$$N_S = \frac{N\sqrt{P}}{\sqrt{\rho}(gH)^{5/4}} \rightarrow \text{Dimensionless specific speed}$$

Where P = Power; g = Acceleration due to gravity

ρ = Density; H = Head (m)

N= Rotation per minute

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529. The torque transmitting capacity of fluid coupling [T] for a given slip varies with impeller internal diameter 'D' and its speed 'N' as

(a) $T \propto D^3 N^2$ (b) $T \propto D^3 N^3$
(c) $T \propto D^5 N^5$ (d) $T \propto D^5 N^2$

TNPSC AE 2017

Ans. (d) :

$$P = T\omega$$
$$= T.\frac{2\pi N}{60}$$
$$T = \frac{60 \times P}{2\pi N}$$
$$T \propto \frac{P}{N}$$
$$\frac{P}{D^5 N^3} = \text{constant}$$
$$P \propto D^5 N^3$$
$$T \propto \frac{D^5 N^3}{N}$$
$$T \propto D^5 N^2$$

The torque transmitting capacity of fluid coupling [T] for a given slip varies with impeller internal diameter 'D' and its speed 'N' as $T \propto D^5 N^2$.

530. A centrifugal pump lifts 0.0125 m^3/s of water from a well with a static lift of 30m. If the brake power of the driving electric motor is 5 kW, what is the overall efficiency of the pump-set?

(a) 57.6% (b) 63.9%
(c) 65.3% (d) 73.6%

ESE 2017

Ans. (d) : Given,

$Q = 0.0125\ m^3/s$
$H = 30\ m$
$P = 5\ kW$

$$\text{Overall efficiency } (\eta_0) = \frac{\rho g Q H}{P}$$
$$= \frac{1000 \times 9.81 \times 0.0125 \times 30}{5 \times 1000}$$
$$= 0.736 = 73.6\%$$

531. For backward curved vane of centrifugal pump:

(a) ideal head developed decreases linearly with increase in flow rate.
(b) ideal head developed increases linearly with increase in flow rate.
(c) ideal head developed remains constant with increase in flow rate.
(d) ideal head developed increases exponentially with increase in flow rate.
(e) ideal head developed decreases exponentially with increase in flow rate.

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : For backward curved vane of centrifugal pump ideal head developed decrease linearly with increase in flow rate.

532. A centrifugal pump driven by a directly coupled 1.5kW motor of 1150 of 1150 rpm speed is proposed to be connected to another motor of 2300 rpm speed. What should be the power of the motor?

(a) 8 kW (b) 12 kW
(c) 16 kW (d) 18 kW
(e) 20 kW

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Given,

$P_1 = 1.5$ kw, $N_1 = 1150$, $N_2 = 2300$, $P_2 = ?$

We know

$$\frac{P_1}{P_2} = \left(\frac{N_1}{N_2}\right)^3$$
$$\frac{1.5}{P_2} = \left(\frac{1150}{2300}\right)^3$$
$$\boxed{P_2 = 12.0 kw}$$

533. If Thoma's cavitation factor is less than the critical cavitation factor, then the overall efficiency of a reaction will have

(a) abnormal increase (b) no change
(c) sharp fall (d) gradual increase

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (c) : Thoma's Cavitations factor,

$$\sigma = \frac{H_b - H_s}{H} = \frac{(H_{atm} - H_r) - H_s}{H}$$

If the value of Thoma's cavitations is more than the critical cavitations factor (σ_C). Cavitations will not occur in reaction turbine, but if 'σ' is less than 'σ_C' cavitations will occur, caused sharp fall in overall efficiency of the reaction turbine.

534. Which of the following statement is correct?

(a) centrifugal pump convert mechanical energy into hydraulic energy by sucking liquid into chamber
(b) reciprocating pumps convert mechanical energy into hydraulic energy by means of centrifugal energy
(c) centrifugal pumps convert mechanical energy into hydraulic energy by means of centrifugal force
(d) reciprocating pumps convert hydraulic energy into mechanical energy

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (c) : Centrifugal pumps converts mechanical. Energy into Hydraulic energy by means of centrifugal force.

535. The internal and external diameter of the impeller of a centrifugal pump are 200mm and 400 mm respectively. The pump is running at 1200r.p.m. The vane angles of the impeller at inlet and outlet are 20° and 30° respectively. The water enters the impeller radially and velocity of flow is constant. The tangential velocity at the inlet and outlet in m/s is

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(a) 125.6, 25.13 (b) 12.56, 25.13
(c) 12.56, 2.513 (d) 1.256, 25.13

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (b) : Tangential velocity at inlet $u_1 = \frac{\pi D_1 N}{60}$

$$\tan\theta = \frac{vf_1}{u_1}$$

$$u_1 = \frac{\pi \times 0.2 \times 1200}{600} = 12.56 \text{ m/s}$$

$$\tan\theta = \frac{vf_1}{u_1} = \frac{vf_1}{12.56} \qquad (\because \theta = 20^0)$$

$vf_1 = \tan 20^0 \times 12.56 = 4.57$ m/s

$$u_2 = \frac{\pi D_2 N}{60} = 25.13 \text{ m/s}$$

536. A centrifugal pump having outer diameter equal to two times the inner diameter and running at 1000 rpm. Works against a total head of 40m. The velocity of flow through the impeller is constant and equal to 2.5m/s. The vanes are set back at an angle 40º at outlet. If the outer diameter of the impeller is 500 mm and width at outlet is 50mm, the discharge is.

(a) 1.963 m^3/sec (b) 0.1963 m^3/min
(c) 0.1963 m^3/sec (d) 0.1963 cm^3/sec

APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (c) : We know, $Q = \pi D_2 \ B_2 \times V_{f_2}$

$$u_1 = \frac{\pi D_1 N}{60} \quad \& \quad u_2 = \pi D_2 N/60$$

$Q = \pi \times 0.05 \times 2.5$ m/s $\times D_2$

$u_1 = 242 \qquad [\because D_2 = 2D_1)$

$D_2 = 0.5$m

$Q = \pi \times 0.05 \times 2.5 \times 0.5$

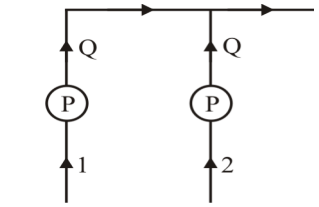
$Q = 0.1963$ m^3/sec.

537. If two pumps, identical in all respects and each capable of delivering a discharge Q against a head H are connected in parallel, the resulting discharge is

(a) Q against a head 2H
(b) 2Q against a head H
(c) 2Q against a head 2H
(d) 2Q against a head $H^{1/2}$

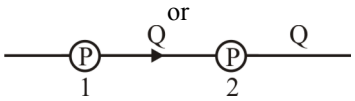
APPSC AEE Mains 2016 (Civil/Mechanical)

Ans. (b) :



$Q_1 = Q + Q = 2Q$

$H_1 = H_2 = H$



$Q_1 = Q_2$

$H = H_1 + H_2 = 2H$

538. Specific speed of a pump is the speed at which a pump will run when the:

(a) Discharge is one cubic meter and head developed is one meter
(b) Power is one unit and head developed is one meter
(c) Power is one unit and discharge is one cubic meter
(d) Power is one unit and speed is one unit

GPSC Asstt. Prof. 28.08.2016

Ans. (a) : We know that,
Specific speed for pump

$$N_s = \frac{N\sqrt{Q}}{H^{3/4}}$$

If, Q = 1 m^3/s, H = 1m

So, $\boxed{N_s = N}$

539. In a system of two identical pumps in series:

(a) The head will be the same and the flow rate will be twice that of a single pump
(b) The head will be the same and the flow rate will be half that of a single pump
(c) The head will be half that of a single pump and the flow rate will be the same
(d) The head will be twice that of a single pump and the flow rate will be the same

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : Flow Through Pipes in Series–
Discharge (Q) = $Q_1 = Q_2 = Q_3 = Q_n$ = Constant
Head loss, $(h_L) = h_{L1} + h_{L2} + h_{L3} + \ldots\ldots + h_{Ln}$

$$\left[\because \text{Where}, h_L = \frac{fLQ^2}{12d^5}\right]$$

For two identical pipes in series–
Discharge, $Q = Q_1 = Q_2$
Head loss $(h_L) = h_{L1} + h_{L2}$
$= h_{L1} + h_{L1} = 2h_{L1}$

540. Axial flow pumps can handle:

(a) very large volumes at low pressure
(b) very low volumes at high pressure
(c) very low volumes at medium pressure
(d) none of the above

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : Axial flow pumps can handle very large volume at low pressure.

541. The hydraulic pump efficiency is defined as :

(a) rotor specific energy/specific energy ideally available from fluid
(b) useful specific energy transferred to the fluid/rotor specific energy
(c) useful specific energy transferred to the fluid/specific energy ideally available from fluid
(d) rotor specific energy/useful specific energy transferred to the fluid

HPPSC Asstt. Prof. 29.10.2016

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Ans. (b) : Hydraulic pump efficiency = $\frac{\text{Useful specific energy transferred to the fluid}}{\text{Rotor specific energy}}$

542. In order of avoid Cavitation in centrifugal pumps, Thoma's equation for determination of cavitation factor is
(Which N_s is the specific speed of the pump; N is speed of the pump; and Q is the rate of flow)
(a) $0.103\ (N_s/1000)^{4/3}$
(b) $0.103\ (QN_s/1000)^{4/3}$
(c) $0.103\ (QN_s/1000)^{3/4}$
(d) $0.103\ (N_s/1000)^{3/4}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Cavitation in centrifugal pumps – In centrifugal pumps the Cavitation may occur at the inlet of the impeller of the pump or at the suction side of the pumps. where the pressure is considerably reduced. Hence if the pressure at the suction side of the pump drops below the vapour pressure of the liquid then the cavitation may occur. The cavitation in a pump can be noted by sudden drop in efficiency and head. In order to determine whether cavitation will occure in any portion of the suction side of the pump, the critical value of Thoma's cavitation factor (σ) is calculated.

Thomos cavitation factor for centrifugal pumps –

$$\sigma = \frac{(H_b) - H_s - h_{LS}}{H} = \frac{(H_{atm} - H_v) - H_s - H_{LS}}{H}$$

If in terms of specific speed –

$$\sigma = 0.103\left(\frac{N_s}{1000}\right)^{4/3}$$

543. The discharge from a centrifugal pump is 100 ℓpm and the leakage is 11 ℓpm. Then the volumetric efficiency of the pump is
(a) 89% (b) 90%
(c) 99% (d) 80%

APPSC AEE Screening Test 2016

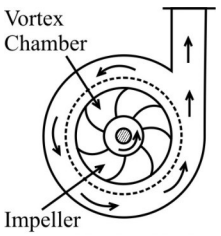
Ans. (a) : Given,
Discharge,
$Q_T = 100\ \ell pm$
Leakage is 11 ℓpm
So,
Actual discharge $Q_A = 100 - 11$
$Q_A = 89$
volumetric efficiency

$$\eta_{vol} = \frac{Q_A}{Q_T} = \frac{89}{100} \times 100 = 89\%$$

544. If a circular chamber is introduced between the casing and the impeller, then is known as:
(a) guide blades casing (b) vortex casing
(c) volute casing (d) none of these

(HPPSC LECT. 2016)

Ans. : (b) If a circular chamber is introduced between the casing and the impeller, then casing is known as vortex casing.

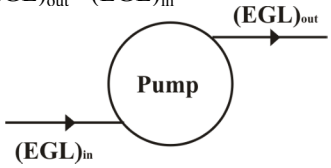


Vortex Casing

545. Net head provided by the liquid pump is given by [where, EGL = Energy grade line, HGL = hydraulic grade line, out = outlet of pump, in = inlet of pump]
(a) $(EGL)_{out} + (EGL)_{in}$ (b) $(HGL)_{out} + (HGL)_{in}$
(c) $(EGL)_{out} - (EGL)_{in}$ (d) $(HGL)_{out} - (HGL)_{in}$

RPSC LECTURER 16.01.2016

Ans. (c) : Net head provided by the liquid pump
$= (EGL)_{out} - (EGL)_{in}$



546. Maximum flow rate through pump occurs when
(a) Net head is maximum
(b) Net head is zero
(c) Pump operates at best efficiency point
(d) Net head is greater than zero but less than maximum

RPSC LECTURER 16.01.2016

Ans. (b) : Maximum volume flow rate through pump occurs when net head is zero.

547. Cavitation in the pump occurs when
(a) Vapour pressure of liquid is greater than local pressure of liquid inside the pump
(b) Vapour pressure of liquid is lower than local pressure of liquid inside the pump
(c) Vapour pressure of liquid is lower than that of atmospheric pressure
(d) local pressure of liquid inside the pump is greater than atmospheric pressure

RPSC LECTURER 16.01.2016

Ans. (a) : Cavitation—Cavitation in a centrifugal pump has a significant effect on pump performance. Cavitation degrade the performance of a pumps, resulting in a fluctuating flow rate and discharge pressure. Cavitation can also be destructive to pumps internal components. When a pump cavitates, vapour bubbles form in the low pressure region the rotating impeller vanes. Where they collapse and create a physical shock to leading edge of the impeller vane.

548. Slip of reciprocating pump becomes negative when
(a) Theoretical discharge is more than actual discharge
(b) Actual discharge is more than theoretical discharge

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(c) Theoretical discharge is equal to actual discharge
(d) Pump is running at low speed

RPSC LECTURER 16.01.2016

Ans. (b) : Slip-Slip of a reciprocating pump is defined as the difference between theoretical and actual discharge.

$$\text{Slip} = Q_{th} - Q_{act} \quad \text{(Positive Slip)}$$

$$= Q_{th}\left[1 - \frac{Q_{act}}{Q_{th}}\right]$$

$$= Q_{th}\,[1 - C_d]$$

Percentage Slip—

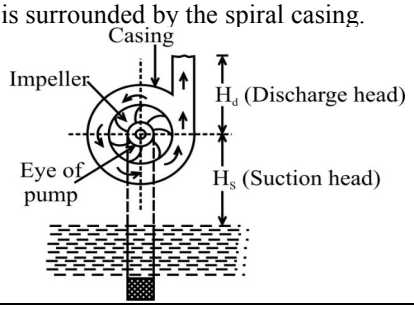
$$= \frac{Q_{th} - Q_{act}}{Q_{th}} \times 100 = [1 - C_d) \times 100$$

Negative Slip—It has been found in some cases that $Q_{act} > Q_{th}$, due to operating conditions. In this case the slip is called negative slip. When the delivery pipe is short or delivery head is small and the accelerating head in the suction side is high, the delivery valve is found to open before the end of suction stroke and the water passes directly into the delivery pipe. Such a situation leads to negative slip.

549. In.....type of centrifugal pump the impeller is surrounded by the spiral casing.
(a) involute (b) bevel
(c) volute (d) spur

(HPPSC LECT. 2016)

Ans. : (c) In volute type of centrifugal pump the impeller is surrounded by the spiral casing.



550. Is it possible to pump water available at around 100°C under atmospheric condition using centrifugal pump placed near the tank:
(a) No
(b) Yes
(c) Yes, if pump is selected properly
(d) None of these are correct?

MPPSC AE 2016

Ans. : (a) Centrifugal Pumps:-A centrifugal pump is a machine which converts the kinetic energy of the water into pressure energy before the water leaves its casing. The flow of water leaving the impeller is free vortex. The impeller of a centrifugal pump may have volute casing, vortex casing and volute casing with guide blades.

551. Axial flow pumps are suitable for
(a) Large discharge with low head
(b) Large discharge with high head
(c) Low discharge with low head
(d) Low discharge with high head

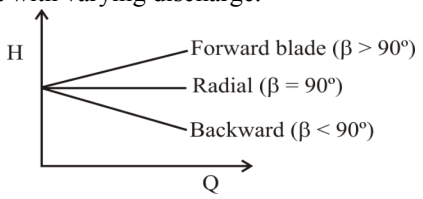
RPSC AE (GWD) 06.05.2016

Ans. (a) : Axial flow pumps are most suitable for providing large discharge and low heads. In axial flow pumps, the size of inlet diameter is greater than that of impeller diameter.

552. In which of the following types of impeller blades or vanes, theoretically, the head remains constant with varying discharge?
(a) Radial blades
(b) Backward curved blades
(c) Forward curved blades
(d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (a) : For both radial and forward facing blades, the power is rising monotonically as the flow rate is increased. In the case of backward curved blade, the maximum efficiency occurs in the region of maximum power. Radial blades or vanes, the head remains constant with varying discharge.



553. In a centrifugal pump casing, the flow of water leaving ..... .
(a) radial (b) centripetal
(c) rectilinear (d) vortex

GWSSB DEE 07.07.2016

Ans. (d)

554. Casting of a centrifugal pump is designed so as to minimise ..... .
(a) friction loss (b) cavitation
(c) static head (d) loss of kinetic energy

GWSSB DEE 07.07.2016

Ans. (d) :

555. A centrifugal pump takes too much power due to
(a) low speed (b) air leakage
(c) air in water (d) heavy liquid

GPSC AE 26.07.2015

Ans. (d) :

556. To discharge a large quantity of liquid by multi-stage centrifugal pump, the impellers are connected
(a) in parallel
(b) in series
(c) in parallel and in series
(d) none of the above

GPSC AE 26.07.2015

Ans. (a) :

557. As pump speed increases, its NPSH (net positive suction head) requirement
(a) Increases
(b) Decreases
(c) Remains unaffected
(d) May increase/decrease depending on other considerations

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) :

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558. In mixed flow centrifugal pump

(a) Two fluids are mixed up at suction
(b) Two fluids are pumped separately and then both mix up at delivery
(c) Two impellers, radial and axial type are employed
(d) Flow through the impeller is a combination of radial and axial flows

Nagaland PSC CTSE 2015 Paper-II

Ans. (d) : In mixed flow centrifugal pump employes such an impeller, through which the flow is a combination of radial and axial flow.

559. The specific speed of a turbo-pump is given by

(a) $\frac{\omega(Q)^{1/2}}{(gH)^{3/4}}$ (b) $\frac{\omega(Q)^{1/2}}{(gH)^{5/4}}$

(c) $\frac{\omega(Q)^{1/2}}{(gH)^{1/2}}$ (d) $\frac{\omega(Q)^{1/2}}{(gH)^{2}}$

Nagaland PSC CTSE 2015 Paper-II

Ans. (a) : Specific Speed of Pump–It is defined as the speed in revolutions per minute of a geometric similar pump of such a size under a head of 1 metre. It represented by N_s

Specific speed may be determined independent of pump size.

$$N_s = \frac{(Q/\omega D^3)^{1/2}}{(gH/\omega^2D^2)^{3/4}} = \frac{\omega\sqrt{Q}}{(gH)^{3/4}}$$

Specific speed of turbo pump $= \frac{\omega(Q)^{1/2}}{(gH)^{3/4}}$

560. The rotating part of the centrifugal pump is called -

(a) impeller (b) rotor
(c) compressed (d) centrifugal

TSGENCO AE 14.11.2015

Ans. (a) : The rotating part of the centrifugal pump is called impeller.

561. When a centrifugal pump is started, there will be no flow of water until the pressure rise in the impeller is large enough to overcome the:

(a) Manometric head (b) Total head
(c) Static head (d) Friction head

OPSC AEE 2015 PAPER - II

Ans. : (a) Manometric head:- The manometric head is defined as the head against which a centrifugal pump has to work. It is denoted by H_m.

H_m = Head imparted by the impeller to the water- Loss of head in the pump.

$H_m = \frac{V_{w_2}u_2}{g}$ - Loss of head in impeller and casing

$$H_m = h_s + h_d + h_{fs} + h_{fd} + \frac{v_d^2}{2g}$$

h_s = Suction head
h_d = Delivery head
h_{fs} = frictional head loss in suction pipe
h_{fd} = frictional head loss in delivery pipe
v_d = velocity of water in delivery pipe.

562. Slip in the case of a centrifugal pump.

(a) Reduces the flow rate
(b) Reduces the tangential component of fluid velocity at impeller outlet
(c) Reduces the speed
(d) Increases cavitation
(e) Decreases cavitation

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : Slip in the case of a centrifugal pump reduces the tangential component of fluid velocity at impeller outlet.

563. When the speed of a centrifugal pump is tripled, the power required to drive the pump will:

(a) Triple (b) Increase 9 times
(c) Increase 27 times (d) Remain the same
(e) not depend upon peed

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : We know that,

$P \alpha N^3$

that

$$\frac{P_2}{P_1} = \left(\frac{N_2}{N_1}\right)^3$$

$$P_2 = \left(\frac{3N}{N}\right)^3 \times P_1$$

$$\boxed{P_2 = 27 \times P_1}$$

564. At the eye tip of a centrifugal impeller, blade velocity is 250 m/s while the uniform axial velocity at the inlet is 200 m/s. If the sonic velocity is 300 m/s, then the inlet Mach number of the flow will be:

(a) 0.81 (b) 1.06
(c) 1.91 (d) 0.16
(e) 0.98

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) :

$$\text{Velocity (V)} = \sqrt{(\text{Blade velocity})^2 + (\text{Axial velocity})^2}$$

$$= \sqrt{(250)^2 + (200)^2} = 320.15$$

$$\text{Mach no.} = \frac{V}{C} = \frac{320.15}{300} = 1.067$$

565. Two geometrically similar pumps are running at the same speed of 1000 RPM. One pump has an impeller diameter of 300 mm and lifts water at the rate of 0.02m^3/s against a head of 15 m. The head of the other pump to deliver half the discharge.

(a) 9.15m (b) 9.25m
(c) 9.35m (d) 9.45m
(e) 9.55m

CGPSC AE 26.04.2015 SHIFT-II

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Ans. (d) : We know that

$$\frac{h_1}{D_1^2N_1^2} = \frac{h_2}{D_2^2N_2^2} \quad \text{.....(1)} \quad [N_1 = N_2 \text{ given}]$$

$$\frac{Q_1}{D_1^3\ N_1^2} = \frac{Q_2}{D_2^3\ N_2^2} \quad \text{.....(2)}$$

then $D_2^3 = \left[\frac{Q_2}{Q_1}\right] \times D_1^3$

$Q_1 = 0.02\ m^3$, $Q_2 = \frac{Q_1}{2}$, $D_1 = 0.3m$, $h_1 = 15m$

$$D_2 = \left[\left(\frac{Q_1/2}{Q_1}\right) \times (0.3)^3\right]^{\frac{1}{3}}$$

$$\boxed{D_2 = 0.2381m}$$

From Equation (1)

$$\frac{15}{(0.3)^2} = \frac{h_2}{(0.2381)^2}$$

$$h_2 = \left(\frac{0.2381}{0.300}\right)^2 \times 15$$

$$\boxed{h_2 = 9.449m}$$

566. Which of the following is not a rotary pump?

(a) Gear (b) Vane
(c) Axial (d) Screw

OPSC AEE 2015 PAPER - II

Ans. : (c) Axial flow pump is a common type of pump that essentials consist of a propeller in a pipe.
The propeller can be driven directly by a sealed motor in the pipe or by electric motor or petrol/diesel engine mounted to the pipe from the outside or by a right angle drive shaft that pierce the pipe.

567. The function of the foot valve in a centrifugal pump is

(a) not to permit the liquid to move downward back to the sump when the pump is not working.
(b) to permit the liquid to move downwards back to sump when the pump is not working.
(c) not to permit the liquid to move upwards from the sump when the pump is working.
(d) to permit the liquid to move downwards back to the sump when the pump is working

TSPSC AEE 2015

Ans. : (a) The function of the foot valve in a centrifugal pump is not to permit the liquid to move downwards back to the sump when the pump is not working.

568. Which of the following hydraulic machines are subjected to cavitation?

(a) Impulse turbines only
(b) Reciprocating pumps only
(c) Reaction turbines and centrifugal pumps
(d) Impulse turbines and reciprocating pumps

TSPSC AEE 2015

Ans. : (c) Reaction turbines and centrifugal pumps are subjected to cavitation.

Caviation:- The formation, growth and collapse of vapour bubbles in a flowing liquid due to local fall in fluid pressure is called cavitation.
(a) It causes noise and vibration of various parts
(b) It makes surface rough
(c) It reduces the discharge of a turbine
(d) It causes sudden drop in power output and efficiency

569. The head against which the centrifugal pump has to work is called :

(a) Static head (b) Suction head
(c) Delivery head (d) Manometric head

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) :

570. The diameter of an impeller of a centrifugal pump at inlet and outlet are 0.2 m and 0.4 m respectively. Determine the minimum starting speed of the pump if it works against a head of 25m.

(a) 1125.57 rpm (b) 1225.57 rpm
(c) 1335.57 rpm (d) 1355.57 rpm

MPSC Dy. Engineer 2014

Ans. (b) : Flow will commence only if

$$\frac{u_2^2 - u_1^2}{2g} \geq H_m$$

For minimum speed,

$$\frac{u_2^2 - u_1^2}{2g} = H_m$$

$$\boxed{N = \frac{60}{\pi}\sqrt{\frac{2gH_m}{D_2^2 - D_1^2}}}$$

$$N = \frac{60}{\pi}\sqrt{\frac{2 \times 9.81 \times 25}{0.4^2 - 0.2^2}}$$

$$\boxed{N = 1221.04\,rpm}$$

Nearest option is (b) i.e 1225.57 rpm.

571. Consider the following forces to determine hoist capacity required for the operation of hydraulic gates-

A. Weight of gate
B. Frictional force between roller wheel and track plate
C. Frictional force between roller pin and bushing
D. Frictional force between rubber seal and seal plates.

Which of the following is correct?

(a) A and B (b) C and D
(c) A, B and D (d) A, B, C and D

MPSC Dy. Engineer 2014

Ans. (d) : Forces required to determine hoist capacity required for the operation of hydraulic gates-
(1) weight of gate
(2) Frictional force between roller wheel and track plate.
(3) Frictional force between roller pin and bushing.
(4) Frictional force between rubber seal and seal plates.

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572. For estimating the flood discharge by the use of fanning formula ($Q = C A^{5/6}$), an average value of 'C' is taken as equal to :

(a) 2.34 (b) 2.54
(c) 3.24 (d) 3.54

MPSC Dy. Engineer 2014

Ans. (b) : Flood discharge empirical formula

Where, $Q = CA^n$

Q = Flood discharge
A = Catchment area
C = Flood co-efficient
n = flood index

Fanning Formula $Q = C(A)^{5/6}$

Average value of C is 2.54

573. A sluice gate controlling flow in a canal is suddenly lowered by an amount to cause partial closure. This will produce.

(a) A negative wave on the upstream
(b) A positive wave on the upstream
(c) A positive surge on the downstream
(d) A standing wave on the downstream

MPSC Dy. Engineer 2014

Ans. (b) : A sluice gate controlling flow in a canal is suddenly lowered by an amount to cause partial closure. This will produce a positive wave on the upstream.

574. The correct sequence in the direction of flow of water for installation in a hydropower plant is-

(a) Reservoir, surge tank, turbine, low pressure penstock
(b) Reservoir, penstock, surge tank, turbine
(c) Surge tank, reservoir, high pressure penstock, turbine.
(d) Reservoir, turbine, surge tank, high pressure penstock

MPSC Dy. Engineer 2014

Ans. (b) : Sequence in direction of flow of water in hydropower plant is-

Reservoir → Penstock → Surge tank → Turbine.

575. The maximum practical suction lift for a self-priming centrifugal pump is-

(a) 35 ft (b) 25 ft
(c) 45 ft (d) 50 ft

MPSC Dy. Engineer 2014

Ans. (b) : The maximum practical suction lift for a self priming centrifugal pump is 25 ft.

Suction lift– It is vertical distance that the pump can be above the liquid source. Typically atmospheric pressure limit for vertical suction lift of pumps to 25 feet at sea level.

576. A diaphragm pump is classified an operates as which type of pump-

(a) Rotary
(b) Centrifugal
(c) Positive displacement
(d) Multi-stage centrifugal

MPSC Dy. Engineer 2014

Ans. (c) : A diaphragm pump is a positive displacement pump that uses a combination of the reciprocating action of a rubber, thermoplastic or Teflon diaphragm and suitable valves on either side of the diaphragm (butterfly valves, flap valves or any other form of shut off valves) to pump fluid.

577. For small lift irrigation schemes type of pump used is

(a) Centrifugal pump
(b) Rotary pump
(c) Dynamic pump
(d) Reciprocating pump

MPSC Dy. Engineer 2014

Ans. (a) : For small lift irrigation schemes centrifugal type of pump is used.

578. A centrifugal pump draws too much power compared to design value. It could be due to :

(a) Air leakage
(b) Presence of air in water
(c) Hot liquid
(d) Heavy liquid

UPRVUNL AE 2014

Ans. : (a) A centrifugal pump draws in too much power compared to design value due to its heavy liquid.

Note :– The commission has given option (d).

579. A centrifugal pump has the following specifications

Speed – 1000 rpm
Flow – 1200 cubic meters/second
Head – 20 m
Power – 5 H.P.

If speed is increased to 1500 rpm, new flow will be

(a) 1800 Cubic meter/second
(b) 2700 Cubic meter/second
(c) 1200 Cubic meter/second
(d) 4500 Cubic meter/second

UPRVUNL AE 2014

Ans. : (a) Centrifugal pump (Q) ∝ speed (N)

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2} = \frac{1200}{Q_2} = \frac{1000}{1500}$$

$$Q_2 = 1800\,m^3/sec$$

580. Impellers for high heads usually have

(a) High specific speed
(b) Low specific speed
(c) Constant specific speed
(d) Speed independent of head

TNPSC AE 2014

Ans. (b) : Impellers for high heads usually have low specific speed.

581. The work done by impeller of a centrifugal pump on water per second per unit weight of water is given by

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(a) $\frac{V_{w_1}.u_2}{g}$

(b) $\frac{(V_{w_1}.u_2 - V_{w_2}.u_1)}{g}$

(c) $\frac{V_{w_2}.u_2}{g}$

(d) $\frac{(V_{w_1}.u_2 + V_{w_2}.u_2)}{g}$

GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : The work done by impeller pump on water per second per unit weight of water is given by

$$= \frac{(V_{w_2}u_2 - V_{w_1}u_1)}{g}$$

for radial inlet $V_{w_1} = 0$

$x = 90^\circ$

$$\text{W.D./wt} = \frac{V_{w_2}u_2}{g}$$

582. In case of gear pumps.

(a) Only spur gears are used.
(b) Only helical gears are used
(c) Both spur and helical gears are used.
(d) Spiral gears are used.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : In case of gear pumps both spur and helical gears are used.

The gear pump uses the meshing of gears to pump fluid by displacement. They are one of the most common types of pumps for hydraulic fluid power applications.

External gear pump designs can utilise spur, helical or herring bone gears.

583. Following is not the characteristic feature of a gear pump -

(a) Self priming
(b) Bidirectional
(c) No pulsation in the flow
(d) Long-time dry run will not harm the pump.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Characteristic features of a gear pumps are-
(1) Self priming
(2) Bidirectional
(3) Pulse free flow
Gear pump is a type of positive displacement pump. It moves a fluid by repeated by endorsing a fixed volume using interlocking cogs or gears, transferring it mechanically using a cyclic pumping action. It delivers a smooth pulse free flow proportional to the rotational speed of gears.

584. In vane type pumps, vane throw is adjusted by adjusting the eccentricity between the rotor and the stator. More the eccentricity.

(a) Maximum is the discharge
(b) Minimum is the discharge
(c) Same is the discharge
(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : In vane type pumps, vane throw is adjusted by adjusting the eccentricity between the rotor and stator. As eccentricity increases discharge also increases.

585. How is the pump quality not rated in efficiency?

(a) Mechanical efficiency
(b) Overall efficiency
(c) Volumetric efficiency
(d) Pressure efficiency

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Pump quality rated in –

(1) Mechanical efficiency $(\eta_{mech}) = \frac{\text{Impeller power}}{\text{Shaft power}}$

(2) Overall efficiency $(\eta_o) = \eta_{mech} \times \eta_{vol} \times \eta_{mano}$

(3) Volumetric efficiency $(\eta_v) = \frac{Q}{Q + \Delta Q}$

(4) Manometric efficiency $(\eta_{mano}) = \frac{gHm}{Vw_2u_2}$

586. Which of the following is not a type of casing or chamber of a centrifugal pump?

(a) Volute or spiral casing
(b) Vortex casing
(c) Volute casing with guide blades
(d) CUBIC casing.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Type of casing of centrifugal pumps are-
(1) Volute or spiral casing
(2) Vortex casing
(3) Volute casing with guide blades

587. In a centrifugal pump, the water

(a) enters the impeller axially and leaves the vanes axially.
(b) enters the impeller radially and leaves the vanes axially
(c) enters the impeller radially and leaves the vanes radially
(d) enters the impeller axially and leaves the vanes radially

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Centrifugal pump is a reverse of inward flow reaction turbine. It works on forced vortex motion principle. The water enters the impeller axially and leaves the vanes radially. It has high discharge capacity and can be used for drifting highly viscous liquid e.g. sewage water, chemicals etc.

588. A four stage centrifugal pump with identical impellers develops a total head of 80m when running at 500 rpm. The outer diameter of impeller is 65 cm. What will be the head per stage?

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(a) 320 m (b) 20 m
(c) 60.5 m (d) 20 m

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : In multistage pump

$$\text{Head per stage } (H_m) = \frac{\text{Total head}}{\text{No. of stage}}$$

$$H_m = \frac{80}{4}$$

$$\boxed{H_m = 20\,m}$$

589. In the casing of a centrifugal pump, the kinetic energy of water is converted into…. before the water leaves the casing.

(a) heat energy (b) Potential energy
(c) Pressure energy (d) All of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : In the casing of a centrifugal pump, the kinetic energy of water is converted into pressure energy before the water leaves the casing.

590. Which of the following centrifugal pumps will be utilized for high head and low discharge?

(a) Axial flow pump (b) Mixed flow pump
(c) Radial flow pump (d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : For high head and low discharge Radial flow centrifugal pump is used.

Centrifugal pump →
- Axial flow → High flow rate and very low head
- Radial flow → Low discharge and high head
- Mixed flow → Medium flow rate and medium delivery head

591. If a centrifugal pump works at high suction head or high vapour pressure then.

(a) cavitation will be formed.
(b) Efficiency will be high
(c) No flow will take place
(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : If a centrifugal pump works at high suction head or high vapour pressure then cavitation will be formed.

Cavitation is defined as phenomenon of formation of vapour bubbled and sudden collapsing of the vapour bubbles this occurs when suction head becomes equal to the head corresponding to vapour pressure. So to avoid cavitation suction head should be as low as possible.

Factor contribute towards on set of cavitation

1. High runner speed.
2. High temperature
3. Less available NPSH (Net Positive Suction Head)

$$\boxed{NPSH = (H_a - h_s - h_{fs})}$$

592. The forward curved blading in case of centrifugal pump leads to

(A) Increase of power input with increase of discharge.
(B) Increase of power input with decrease of discharge.
(C) The power curve is not self limiting and damage to motor is possible
(D) Decrease of power input with increase of discharge

Which of the following statements given above is / are correct.

(a) A only (b) B & C only
(c) A & C only (d) C & D

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : The forward curved blading in case of centrifugal pump leads to
(1) Increase of power input with increase of discharge.
(2) The power curve is not self limiting and damage to motor is possible.

593. The unit speed of pump

(a) N/H (b) $\frac{\sqrt{H}}{N}$
(c) $\frac{N}{\sqrt{H}}$ (d) $\frac{Q}{\sqrt{H}}$

TNPSC AE 2013

Ans. (c) : Unit speed of pump

$$N_u = \frac{N}{\sqrt{H}}$$

Unit discharge of pump

$$Q_u = \frac{Q}{\sqrt{H}}$$

Unit power of pump

$$P_u = \frac{P}{H^{3/2}}$$

594. A centrifugal pump lifts 0.013 m^3/s water from a depth of 32 m. If the pump motor consumes 6 kW, what is the overall efficiency of pump. Given density of water = 1000 kg/m^3:-

(a) 88% (b) 75%
(c) 69% (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) : Given as,

$Q = 0.013\ m^3/s$
$h = 32\ m$
$P = 6\ kW$
$\eta_0 = ?$
$\rho_w = 1000\ kg/m^3$

Overall efficiency of centrifugal pump is given as,

$$\eta_0 = \frac{\text{Water Power}[\rho_w Qgh]}{\text{Shaft Power}}$$

$$= \frac{1000 \times 0.013 \times 9.81 \times 32}{6000}$$

$$\eta_0 = 0.68 \simeq 0.69$$

$$\boxed{\eta_0 = 69\%}$$

595. The vanes of a centrifugal pump move due to:-

(a) Pressure energy of water
(b) Kinetic energy of water
(c) Both pressure and kinetic energy of water
(d) Power supplied by prime mover

UKPSC AE-2013, Paper-II

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Ans. (d) : The vanes of a centrifugal pump move due to power supplied by prime mover.

596. The maximum lift in a centrifugal pump would be close to 'X' if the pump is handling cold water (25°C) from an open tank at sea level with no friction and turbulence losses, where 'X' is equal to

(a) 10 m (b) 1.03 m
(c) 100 m (d) 30 m

ISRO Scientist/Engineer 2012

Ans. (a) : Maximum lift is when the pressure at the pump impeller reaches absolute zero.

$\therefore$ Drop in pressure for x lift.

ρgx = Pressure

$\rho gx = 1.01 \times 10^5$

$$x = \frac{1.01\times10^5}{9810}$$

= 10 m.

597. Open type impeller centrifugal pump is used to handle

(a) water
(b) mixture of water, sand, pebbles and clay
(c) sewage
(d) liquids lighter than water

UKPSC AE 2012 Paper–II

Ans. (b) : Open impellers are usually used in small expensive pumps or pumps handling abrasive liquid or suspended solid (in which the impeller rotates between two side plate between the casting wall of volute or between the stuffing box head and the section head.)

598. The impeller of a centrifugal blower

(a) decreases the velocity and increases the pressure of gas
(b) increases the velocity and decreases the pressure of gas
(c) increases both velocity and pressure of the gas
(d) none of these

TNPSC ACF 2012

Ans. (c) : The impeller of centrifugal blower increases cavity and pressure of gases.

599. Foot valves are provided in the suction line of a centrifugal pump to

(a) avoid cavitation
(b) remove the contaminants present in the liquid
(c) avoid priming every time the pump is started
(d) control the liquid discharge

APPSC AEE 2012

Ans. (c) : The foot value is a lift check valve which is usually installed together with section strainers in the suction line. In prevents the section line from running empty after the pump is stopped it avoid priming every time the pump is started.

600. Cavitation will take place if the pressure of the following fluid at any point is

(a) more than vapour pressure of the fluid
(b) equal to vapour pressure of the fluid
(c) less than vapour pressure of the fluid
(d) None of the above

APPSC AEE 2012

Ans. (c) : Cavitation will take place if the pressure of the flowing fluid at any point is less than vapour pressure of the fluid.
To avoid cavitation $P_{min} > P_{vapour}$

601. Where is the regulating valve provided in a centrifugal pump?

(a) Casing (b) Delivery pipe
(c) Suction pipe (d) Impeller

VIZAG STEEL MT 2011

Ans. (c) :

602. Pick up the incorrect statement for centrifugal pumps

(a) Discharge $\propto$ diameter
(b) Head $\propto$ (speed)^2
(c) Head $\propto$ (diameter)^2
(d) Discharge $\propto$ speed

ISRO Scientist/Engineer 2011

Ans. (a) : We know that

$$\left.\frac{H}{D^2N^2}\right|_P = \left.\frac{H}{D^2N^2}\right|_m = \text{constant}$$

So,

$$H \propto D^2$$

$$H \propto N^2$$

and also,

$$\left.\frac{Q}{D^3N}\right|_P = \left.\frac{Q}{D^3N}\right|_m = \text{constant}$$

$$Q \propto N$$

$$Q \propto D^3$$

So, $Q \propto D$ is incorrect statement about centrifugal pump.

603. If the discharge of a centrifugal pump is throttled then it suction lift

(a) Decreases
(b) First increases and then decreases
(c) Remains unchanged
(d) Increases

ISRO Scientist/Engineer 2011

Ans. (d) : If the discharge of a centrifugal pump is throttled then its suction lift **increases**. The "Suction Lift" is known as Net positive suction head available (NPSH). Throttling the discharge of centrifugal pump is common method of stopping a cavitation problem.

604. Cavitation can take place in case of :

(a) Pelton wheel
(b) Centrifugal pump
(c) Reciprocating pump
(d) All of these

J&K PSC Civil Services Pre, 2010

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Ans. (b) : Cavitation in Centrifugal Pumps–

- Cavitation is likely to occur at the inlet to the pump.
- Since the pressure there is the minimum and is lower than the atmospheric pressure the formation of bubbles or cavities in liquid, developed in areas of relatively low pressure around an impeller.
- The imploding or collapsing of these bubbles trigger entense shockwaves inside the pump, causing significant damage to the impeller and or the pump housing.

605. Consider the following statements regarding multistage centrifugal pumps :

1. For discharging large quantity of liquid, the impellers are connected in parallel.
2. At the outlet of 2nd impeller, the pressure of water will be less than the pressure of water at the outlet of 1st impeller, when the impellers are connected in series.

Which of the above statements is/are correct?

(a) Both 1 and 2 (b) 1 only
(c) 2 only (d) Neither 1 nor 2

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know, in a multi stage centrifugal pumps, for discharging large quantity of liquid, the impellers are connected in parallel.

606. Cavitation parameter is defined by

(a) $\frac{P_v - P}{\rho V^2/2}$ (b) $\frac{P_{atm} - P_v}{\rho V^2/2}$

(c) $\frac{P - P_{atm}}{\frac{\rho V^2}{2}}$ (d) $\frac{P - P_v}{\frac{\rho V^2}{2}}$

ISRO Scientist/Engineer 2010

Ans. (d) : Cavitation parameter

$$\frac{P - P_V}{\frac{\rho V^2}{2}}$$

where, P = pressure
P_v = Vapour pressure
ρ = density
V = velocity

607. A necessary precaution in selection of pumps for parallel operations is that:

(a) H-Q characteristics of both should be identical
(b) Both pumps should be centrifugal type
(c) Both pumps should be identical
(d) There should not be change from positive to negative slope in H-Q curve

ISRO Scientist/Engineer 2009

Ans. (a) : A necessary precaution in selection of pumps for parallel operations is H-Q characteristics of both should be identical.

608. In a fluid machinery, the relationship between saturation temperature and pressure decides the process of

(a) Flow separation (b) Cavitation
(c) Turbulent mixing (d) Water hammer

ISRO Scientist/Engineer 2009

Ans. (b) : In a fluid machinery, the relationship between saturation temperature and pressure decides the process of cavitation.

609. In case of centrifugal fan or blower the gas capacity varies directly with

(a) speed (b) $(\text{speed})^2$
(c) $(\text{speed})^{3/2}$ (d) $(\text{speed})^3$

TNPSC AE, 2008

Ans. (a) : $\frac{Q}{ND^3} = \text{Constant}$

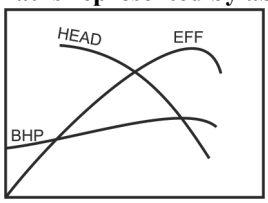
610. Which of the following types of impeller is used for centrifugal pumps dealing with mud's?

(a) One side shrouded
(b) Two side shrouded
(c) Double suction
(d) Open type impeller

WBPSC AE 2008

Ans. (d) : An open impeller is one whose vanes have neither the crown plate nor the base plate such impellers are useful in the pumping of liquids containing suspended solid matter such as paper pulp, sewage and water containing sand or grit.
Shrouded impeller are those whose vanes are provided with metal cover plates.

611. If the characteristics of a pump are as shown in fig. What is represented by abscissa



(a) Head (b) RPM
(c) Discharge (d) Power

ISRO Scientist/Engineer 2008

Ans. (c) :

612. Cavitation is a phenomenon observed

(a) only in centrifugal pumps
(b) only in hydraulic reaction turbines like Francis and Kaplan
(c) only in marine propellers
(d) in all of centrifugal pumps, hydraulic reaction turbines and marine propellers

DRDO Scientists 2008

Ans. (d) :

613. A centrifugal pump with peripheral speed 'V' was selected for a total lift of 100 m. Actually pump was to be operated for a total lift of 400m. The peripheral speed should actually be (other conditions remaining same)

(a) 2V (b) 4V
(c) 8V (d) 1V

ISRO Scientist/Engineer 2007

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Ans. (a) : $V_1 = V$, $H_1 = 100m$, $V_2 = ?$, $H_2 = 400m$

$$\frac{H_1}{N_1^2} = \frac{H_2}{N_2^2}$$

$$N \propto V$$

$$\frac{100}{V^2} = \frac{400}{V_2^2}$$

$$V_2^2 = 4V^2$$

$$V_2 = 2V$$

614. The best suited pump for lifting water having solids in suspension is

(a) air lift pump (b) jet pump
(c) centrifugal pump (d) reciprocating pump

OPSC Civil Services Pre 2006

Ans. (c) : Centrifugal pump with open type impeller is used for mud/solid suspension.

615. The maximum depth from which a centrifugal pump can draw water is-

(a) Dependent on the speed N of the pump
(b) Dependent on the power of the pump
(c) Around 10 m
(d) Dependent on N^2

ISRO Scientist/Engineer 2006

Ans. (c) : The maximum depth from which the centrifugal pump can draw water in around 10 m.

we know that

$P_{atm} = \rho \times g \times h$

$101.3 \times 10^3 = h \times 1000 \times 10$

$$\boxed{h = 10.13m}$$

This is the maximum height to which water can be raised by suction pump.

5. Miscellaneous Hydraulic Machines

616. For flood control and irrigation applications, the pump generally used is

(a) Centrifugal type (b) Reciprocating type
(c) Axial flow type (d) Radial flow type

Nagaland PSC (CTSE) 2018, Paper-II
Nagaland PSC CTSE 2015 Paper-II
APPSC AEE 2012

Ans. (c) : Axial flow pump is preferred for flood control and irrigation application.

We know that,

$$(N_s)_{pump} = \frac{N\sqrt{Q}}{H^{3/4}}$$

So, $(N_s)_{pump} \propto Q^{1/2}$

Specific speed of axial flow pump higher than any other pump which is $N_s = 200 - 300$.

So, it has highest discharge capacity due to this reason axial flow pump used in flood control and irrigation applications.

617. A hydraulic press has a ram of 20 cm diameter and a plunger of 5 cm diameter. The force required at the plunger to lift a weight of 16×10^4 N shall be:

(a) 256×10^4 N (b) 64×10^4 N
(c) 4×10^4 N (d) 1×10^4 N

GPSC Asstt. Director of Transport 05.03.2017
KPSC ADF 2015, ESE 2002

Ans. (d) : Given, Dia of press (D_p) = 20 cm

Dia of plunger (d_p) = 5 cm

Weight (W) = 16×10^4 N = 160 kN

Equating pressures,

$$P_1 = P_2$$

$$\frac{F}{a} = \frac{W}{A}$$

$$'a' = \text{Area of plunger} = \pi\frac{dp^2}{4}$$

$$'A' = \text{Area of press} = \pi\frac{Dp^2}{4}$$

$$F = \frac{W}{A} \times a = \frac{160 \times 3.14 \times 0.05 \times 0.05 \times 4}{3.14 \times 0.2 \times 0.2 \times 4}$$

$$\boxed{F = 10000\,N}$$

618. Which one of the following is NOT a rotating machine?

(a) Centrifugal pump (b) Gear pump
(c) Jet pump (d) Vane pump

Assam Engg. College AP/Lect. 18.01.2021
GATE 2017

Ans. (c) : Jet pump

Jet pump is not a rotating machine while centrifugal pump, gear pump and vane are rotating machine.

Jet pump utilizes ejector principle which have nozzle and diffuser which has not a rotating parts.

619. Hydraulic transmission through fluid coupling is suitable for :

(a) Unsteady operation and increasing torque
(b) Unsteady operation and increasing speed
(c) Unsteady operation and low starting torque
(d) Increasing torque and low starting load

PSPCL AE, 2012, ESE 1992

Ans. (c) : Unsteady operation and low starting torque.

620. A hydraulic coupling belongs to the category of

(a) power absorbing machines
(b) power developing machines
(c) energy generating machines
(d) energy transfer machines

TRB Poly. Lect., 2012, CSE Pre-1994

Ans. (d) : A hydraulic coupling belongs the category of energy transfer machines.

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621. Which among the following is non structural flood mitigation measure

(a) Embankments, flood walls, sea walls
(b) Flood plain management and zoning
(c) Dams and reservoir
(d) Diversion of flood waters

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :

622. The function of master cylinder hydraulic system is :

(a) To increase pressure equally in all cylinders
(b) To increase pressure unequally in all cylinders
(c) To increase power equally in all cylinders
(d) To decrease power equally in all cylinders

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : The function of master cylinder Hydraulic system is to increase pressure equally in all cylinder.

623. Which of the following statements represents the effect of cavitation?

(a) Considerable cracking or rattling may be produced leading to objectionable vibrations
(b) The pressure velocity distribution in this affected region is greatly distorted
(c) The metallic or concrete walls containing the water stream may be severely damaged
(d) All of the above

CGPSC AE 15.01.2021

Ans. (d) : Effects of Cavitation-
(i) Considering cracking or rattling may be produced leading to objectionable vibrations.
(ii) The pressure velocity distribution in this affected region is greatly distorted.
(iii) The metallic or concrete walls containing the water stream may be severely damaged.

624. Pumping power is equal to

(a) Total power produced
(b) Net power produced
(c) (a) + (b)
(d) None of the above

JPSC AE, 10.04.2021, Paper-I

Ans. (c) : Pumping power is equal to net power produced and total power produced.

625. A hydraulic fuse is incorporated in the system for safety against :

(a) velocity of fluid (b) pressure of fluid
(c) temperature of fluid (d) viscosity of fluid

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : A hydraulic systems a fuse (or velocity fuse) is a component which prevent the sudden loss of hydraulic fuses help guard against catastrophic failure of a hydraulic system by automatically Isolating the defecting branch.

626. The component of torque converter that allows torque multiplication is

(a) Turbine (b) Impeller
(c) Stator (d) Flywheel

RPSC IOF, 2020

Ans. (c) : Stator torque multiplication a torque converter differs from a simple fluid coupling by the addition of a stator a disc with fan-like blade connected to the transmission via a fixed shaft with a one-way clutch that allows it to rotate only in the opposite direction of a fluid's radial motion.

627. The order in which the specific speed for best efficiency increase in pumps is :

(a) axial, mixed, centrifugal, piston pumps
(b) piston, centrifugal, mixed, axial flow pumps
(c) piston, axial, mixed, centrifugal pumps
(d) mixed, axial, centrifugal, piston pumps

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (b) :

628. A hydraulic ram is a device used to

(a) Store the energy of water
(b) Increase the pressure of water
(c) To lift water from deep wells
(d) To lift small quantity of water to a greater height when a large quantity of water is available at a smaller height

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : A hydraulic ram is a device used to lift small quantity of water to a greater height when a large quantity of water is available at a smaller height. A hydraulic ram pump is water pump powered by water with a height difference. In areas where natural flows exist with a height difference of the water over a small distance.

629. If a work piece is moved by 50 mm in 10 s by a piston of diameter 100 mm, the hydraulic liquid flow rate is nearly

(a) 3.00×10^{-5} m^3/s (b) 3.93×10^{-5} m^3/s
(c) 4.74×10^{-5} m^3/s (d) 5.00×10^{-5} m^3/s

ESE 2018

Ans. (b) : Given,

$$\text{Velocity (v)} = \frac{50 \times 10^{-3}}{10} \text{m/s} = 5 \times 10^{-3} \text{ m/s}$$

$$\text{Area (A)} = \frac{\pi}{4}(0.1)^2 = 0.0078 \text{ m}^2$$

We know that

$$\text{discharge (Q)} = A \cdot v$$
$$= 0.0078 \times 5 \times 10^{-3}$$
$$= 0.0392 \times 10^{-3}$$
$$= 3.92 \times 10^{-5} \text{ m}^3/\text{s}$$

630. Which of the following uses the water hammer effect to lift water?

(a) Hydraulic accumulator
(b) Hydraulic intensifier
(c) Hydraulic ram
(d) Hydraulic pump

APGENCO AE, 2017

Ans. (c) : Hydraulic accumulator – Used for storing energy of liquid in the form of pressure energy.
Hydraulic intensifier – Used to increase of pressure of water

Search on Telegram @apna_pdf

Hydraulic ram – used to raise water without any external prior for its operation and work on principle of water hammer.

631. What will be the velocity of piston movement for a single-acting hydraulic actuator, when the fluid pressure is 100 bar, the diameter of the piston is 50 mm and the flow rate is 0.3 m^3/min?

(a) 2.41 m/s (b) 2.55 m/s
(c) 2.67 m/s (d) 2.84 m/s

ESE 2017

Ans. (b) : $D = 0.05$ m

$Q = 0.3\ m^3/min$

$\because \quad Q = A \cdot v$

$$Q = \frac{\pi}{4} D^2 \cdot v$$

$$v = \frac{4 \times 0.3}{\pi \times (0.05)^2}$$

$= 2.55$ m/s

632. Consider the following statements:

1. The head produced in the primary of a fluid coupling is more than the corresponding centrifugal head
2. When the two speeds are equal, the torque transmitted is zero

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

JWM 2017

Ans. (a) : The flow in the primary is radially outwards and radially inwards in the secondary flow occur in this direction because the speed of primary exceeds that of the secondary. The heat produced in the primary is this greater than the centrifugal head resisting flow through the secondary.
If the two speeds same, head balance and no flow will occur and no torque will be transmitted.

633. The operation of removing trapped air from the hydraulic braking system is known as

(a) Trapping (b) Tapping
(c) Bleeding (d) Cleaning

TNPSC AE 2017

Ans. (c) : The operation of removing trapped air from the hydraulic braking system is known as bleeding.

634. In hydraulic Brakes "Bleeding" refers to

(a) The process of removing air out from the braking system
(b) The process of filling the brake fluid in the brake cylinders
(c) The leakage of brake fluid in the brake system
(d) The process of emptying the brake fluid from the brake system

TNPSC AE 2017

Ans. (a) : In hydraulic Brakes "Bleeding" refers to The process of removing air out from the braking system.

635. Hydraulic accumulator is a device used for

(a) Lifting heavy weights
(b) Storing the energy of a fluid in the form of pressure energy
(c) Increasing the pressure intensity of the fluid
(d) Transferring kinetic energy of the fluid to the shaft

Nagaland PSC CTSE 2017 2nd Paper

Ans. (b) : Hydraulic accumulators are energy storage device. Analogous to rechargeable batteries in electrical System, they store and discharge energy in the form pressurized fluid and are after used to improve hydraulic-system efficiency.

636. Hydraulic coupling is a device used for…..

(a) Transmitting same torque to the driven shaft
(b) Transmitting increased torque to the driven shaft
(c) Transmitting decreased torque to the driven shaft
(d) Transmitting increased or decreased torque to the driven shaft depending on the load

GPSC Asstt. Prof. 28.08.2016

Ans. (a) : A fluid coupling or hydraulic coupling is a hydrodynamic or hydro kinetic device used to transmit same rotating mechanical power. It has been used in automobile transmissions as an alternative to a mechanical clutch transmitting some torque to driven shaft.

637. At constant efficiency, the horse power of a fan is

(a) Proportional to rpm
(b) Proportional to $(rpm)^2$
(c) Proportional to $(rpm)^3$
(d) A polynomial function of rpm
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (c) : At constant efficiency, the horse power of a fan is proportional to $(rpm)^3$.

$$\frac{PN^3}{D^5} = \text{Constant}$$

638. Which of the followings is not the element of hydroelectric power plant?

(a) Catchment area (b) Dam
(c) Draft tube (d) Condenser

RPSC LECTURER 16.01.2016

Ans. (d) : Hydroelectric power plant—Hydropower plant uses hydraulic energy of water to produce electricity. The power obtained from this plant is termed as hydro-electric power. Nearly 16% of total power used by the world.

Components of hydropower plant—

1. Forebay
2. Catchment area
3. Dam
4. Intake structure
5. Penstock
6. Surge chamber
7. Hydraulic turbine
8. Draft tube
9. Tail race

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639. Consider the following statements: Cavitation in hydraulic machines occurs at the 1. Exit of the pump 2. Entry of the pump 3. Exit of the turbine 4. Entry of the turbine of these the correct statements are:

(a) 1 and 2 (b) 1 and 3
(c) 2 and 4 (d) 2 and 3

RPSC VPITI 14.02.2016

Ans. (d) : Cavitation is the formation of the bubbles inside the liquid being pumped. The bubble when vaporizes phase change from liquid to vapour and then bubbles collapse.
Cavitation in hydraulic machine occur-
1. Entry of the pump
2. Exit of the turbine

640. Which of the following statements represent the effects of cavitation?

(a) The metallic or concrete walls containing the water stream may be severely damaged
(b) Considerable cracking or rattling may be produced leading to objectionable vibrations
(c) The pressure velocity distribution in this affected region is greatly distorted
(d) All of the above
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : Cavitations, formation of vapour bubbles within a liquid at low pressure regions that occur in places where the liquid has been accelerated to high velocities. Cause damage to metallic walls containing water stream, cracking or rattling may be produced & pressure velocity distribution in this affected region is greatly distorted.

641. Axial flow pump is started with its delivery valve

(a) kept fully closed
(b) kept fully open
(c) irrespective of any position
(d) kept 50% open

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : For starting an axial flow pump, its delivery valve should be open.

642. When a piping system is made up primarily of friction head and very little of vertical lift, then pump characteristics should be

(a) horizontal
(b) nearly horizontal
(c) Steep
(d) First rise and then fall

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Piping system is made up primarily of vertical lift and very little pipe friction, the pump characteristics should be steep.

643. According to fan laws, for fans having consultant wheel diameter, the pressure varies

(a) Directly as fan speed
(b) Square of fan speed
(c) Cube of fan speed
(d) Square root of fan speed

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : According to the Fan Laws– Pressure varies with the square of the airflow rate and therefore the system curve appears as a parabola.
The pressure developed by fan and the pressure drop in a system varies with the square of change of the speed of the impeller or volume passing through the system.

$$P \propto v^2$$

644. A hydraulic intensifier normally consists of

(a) Two cylinders, two rams and a storage device
(b) A cylinder and a ram
(c) Two co-axial rams and two cylinders
(d) A cylinder, a piston, storage tank and control valve

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Two co-axial rams and two cylinders.
A hydraulic intensifier is component that converts the low pressure from a cylinder into high pressure in a smaller cylinder.
Intensifiers operates on the ratio of areas principle in inter connected cylinders. A common rod connects the pistons of two cylinders of different bore.

645. For incompressible flow, the pumping power is given by the expression

(a) (Pressure drop) × (volumetric flow rate)
(b) (Average pressure) × (volumetric flow rate)
(c) 1/2 × (Average pressure) × (volumetric flow rate)
(d) 3/4 × (Pressure drop) × (volumetric flow rate)

TSGENCO AE 14.11.2015

Ans. (a) : For incompressible flow, the pumping power.
$(P) = \rho gQH$
$= (\rho gH)Q$
= (Pressure drop) × (Volumetric flow rate)

646. How does pressure intensity varies when liquid rotates at an angular velocity of a vertical axis?

(a) Directly with radial distance
(b) Directly with square of the radial distance
(c) Inversely with square of the radial distance
(d) None of the above

OPSC AEE 2015 PAPER - II

Ans. : (b) Pressure intensity varies directly with square of the radial distance when liquid rotates at an angular velocity of a vertical axis.

647. If a hydraulic press has a ram of 32.5 cm diameter and plunger of 3.25 cm diameter, what force would be required on the plunger to raise a mass of 500kg on the ram?

(a) 98.1N (b) 9.81N
(c) 49.05N (d) 4.905N
(e) 19.62N

CGPSC AE 26.04.2015 SHIFT-II

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Ans. (c) : Given,

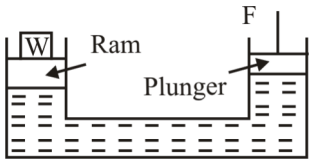
$d_R = 3.5$ cm

$d_P = 3.25$ cm

$m \times g = 500 \times 9.81$

$= 4905$ N

According to Pascal law -



$$\frac{F}{A_P} = \frac{W}{A_R}$$

$$F = 500 \times 9.81 \times \left[\frac{\frac{\pi}{4}(3.25)^2}{\frac{\pi}{4}(32.5)^2}\right]$$

$$\boxed{F = 49.5\,N}$$

648. Which of the following pumps is successfully used for lifting water from deep wells?

(a) Reciprocating pump (b) Centrifugal pump

(c) Gear pump (d) Air lift pump

MPSC Dy. Engineer 2014

Ans. (d) : Airlift pump is a pump that has low suction and moderate discharge of liquid and entrained solids. The pump injects compressed air at the bottom of the discharge pipe which is immersed in the liquid.

Air lift pumps are often used in deep dirty wells where sand would quickly abrade mechanical parts.

649. Consider the following pumps-

A. Centrifugal pump – single stage

B. Centrifugal pump – multi stage

C. Reciprocating pump

D. Jet pump

The pump (s) which could be used to lift water through a suction head of 12 m from a well would include.

(a) B only (b) A, C and D

(c) D only (d) A and C

MPSC Dy. Engineer 2014

Ans. (c) : Jet pump is used to lift water through a suction head of 12 m from a well.

650. A suction hose sized larger than the pump size will -

(a) Increase capacity (b) Decrease capacity

(c) Early pump failure (d) Cause cavitation

MPSC Dy. Engineer 2014

Ans. (a) : If suction have sized larger than the pump size will increases capacity.

651. Capacity of a hydraulic accumulator is generally specified as

(a) Quantity of liquid accumulated

(b) Maximum pressure developed

(c) Maximum energy stored

(d) Maximum Quantity of discharge allowed

TNPSC AE 2014

Ans. (c) : Capacity of a hydraulic accumulator is generally specified as maximum energy stored.

652. Which of the following is a positive displacement device?

(a) Ultrasonic flow meter

(b) Turbine flow meter

(c) Laser Doppler Anemometer

(d) Hot wire Anemometer

TNPSC AE 2014

Ans. (b) : Turbine flow meter is a positive displacement device.

653. Which of the following geometric characteristics does not require datum surface.

(a) Angularity (b) Straightness

(c) Run out (d) Concentricity

GPSC Lect. (Poly.) 07.12.2014

Ans. (b) : The geometric require datum surface are

(i) Angularity

(ii) Run out

(iii) Concentricity

Straightness doesn't required datum surface.

654. Axial pumps are bidirectional and are best suited for.

(a) High –pressure-high-volume

(b) High-pressure-low volume

(c) Low-pressure-high-volume

(d) Low-pressure-low-volume

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Axial pumps are bidirectional and best suited for high-pressure and low-volume.

655. Which type of pump stops pumping of liquid further into the system when the predetermined pressure is reached?

(a) Balanced vane pump

(b) Axial flow pump

(c) Unbalanced vane pump

(d) Pressure compensated vane pump

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Pressure compensator is a device built into some pumps for the purpose of automatically reducing (or stopping) pump flow if system pressure sensed on the pump outlet port, should rise above a pre-set desired maximum pressure (sometimes called the "firing" pressure).

656. Evaluate the statement given below and choose the correct option. The flow rate capacity of any pump is dependent mainly on

(A) The geometric size of the pumping chamber.

(B) Rotational speed of the pump

(a) Both A & B (b) Only A

(c) Only B (d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

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Ans. (a) : The flow rate capacity of pump depends on-
(1) The geometric size of the pumping chamber
(2) Rotational speed of the pump

657. Pressure Relief Valve is used in case of…

(a) Gear Pump (b) Vane Pump
(c) Piston Pump (d) All of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Pressure relief valve is used in case of-
(1) Gear pump
(2) Vane pump
(3) Piston pump.

658. Efficiency of which pump is the maximum?

(a) Gear Pump (b) Lobe Pump
(c) Vane Pump (d) Piston Pump

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Efficiency of piston pump is maximum as compare to gear pump, lobe pump and vane pump.

659. Noise level of which pump is the maximum?

(a) Gear Pump (b) Vane Pump
(c) Piston Pump (d) Screw Pump

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Noise level of gear pump is the maximum.

660. Following is the most widely used actuator of hydraulic system.

(a) Vane (b) Gear
(c) Pump (d) Hydraulic cylinder

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : The most widely used actuator of hydraulic system is hydraulic cylinder.

A hydraulic cylinder is a mechanical actuator that is used to give a unidirectional force through a unidirectional stroke.

661. In case of vane type limited rotary actuator, what creates a differential force on vane causing the vane to turn and thus rotate the shafts?

(a) Velocity difference in two chambers
(b) Volume difference in two chambers.
(c) Pressure difference in two chambers
(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : In Vane type pump pressure difference in two cylinder creates a differential force on vane causing the vane to turn and thus rotates the shaft.

662. Which force contributes to driving the rotor in turbo machines ?

(a) Axial force (b) Tangential force
(c) Radial force (d) Centrifugal force

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

663. Muschel curve means :

(a) curve at constant head
(b) curve at constant speed
(c) curve at constant efficiency
(d) None of the above

APPSC AE 04.12.2012

Ans. (c) : Muschel curve means the curve at constant efficiency.
The constant efficiency curves or muschel curves help to determine the range of pump operation for a particular efficiency.

664. In high speed billet forming machines, the impact speed of hydraulic press is of the order of:

(a) 3 to 30 m/s (b) 0.03 to 0.3 m/s
(c) 0.3 to 3 m/s (d) 0.003 to 0.03 m/s

APPSC IOF, 2009

Ans. (a) : In high speed billet forming machines, the impact speed of hydraulic press is of the order of 3 to 30 m/s.

665. Consider the following statements :

I. Run-of the river plant is cheaper than the storage plant of equal capacity
II. Run-of-the river plant serves base load, when located upstream of the storage plant
III. Run of the river plant output depends on what storage plants pass on, when located between storage plants

Of the statements :

(a) (I) and (II) are correct
(b) all are correct
(c) (II) and (III) are correct
(d) (I) and (III) are correct

TNPSC AE, 2008

Ans. (b) : Hydropower schemes can be classified into four main types, run-of-river (ROR), storage (reservoir-based), pumped storage hydro (PSH) and in-stream (hydrokinetic) technologies.
(i) Run-of-river
(ii) Storage hydropower
(iii) Pumped storage
(iv) Hydrokinetic
(v) Underground power plants
(vi) Large and small hydro

666. Hydel plants having capacity of the order of 100 kW are called

(a) pico hydel plants
(b) micro hydel plants
(c) mini hydel plants
(d) none of the above

OPSC Civil Services Pre 2006

Ans. (b) : Micro hydel power plant - upto 100 kW
Pico hydel plant - upto 5 kW

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07.

Heat and Mass Transfer

1. Steady and Unsteady Heat Conduction

1. What is the critical radius of insulation for a sphere equal to?
k = thermal conductivity in W/m-K
h = heat transfer coefficient in W/m^{2}K
(a) 2kh (b) 2k/h
(c) k/h (d) $\sqrt{2}$kh

GPSC DEE Class-2 (GWSSB) 04.07.2021
JPSC AE, 10.04.2021, Paper-I
VIZAG MT, 14.12.2020
RPSC IOF, 2020, TNPSC AE 2018
ISRO Scientist/Engineer 22.04.2018
Kerala PSC Poly. Lect., 2017
TSPSC AEE 2017
UPPSC AE 12.04.2016 Paper-II
Nagaland PSC (CTSE) 2018, 2015 Paper-II
ISRO Scientist/Engineer (RAC) 29.11.2015
Kerala PSC Poly. Lect. 09.01.2014
CGPSC AE 2014 -II
APPSC Poly. Lect. 2013
UKPSC AE 2007, 2012 Paper -II
J&K PSC Civil Service Pre-2010
TNPSC AE, 2008, ESE 2008

Ans. (b) : Critical Radius of insulation

For sphere $(r_c)_s = \frac{2k}{h}$

For cylinder $(r_c)_c = \frac{k}{h}$

2. If K is the thermal conductivity, ρ is the mass density and C is the specific heat of a substance, then its thermal diffusivity is
(a) ρC/K (b) K/ρC
(c) KC/ρ (d) Kρ/C

GPSC DEE Class-2 (GWSSB) 04.07.2021
Assam PSC AE (IWT) 14.03.2021
Nagaland PSC (CTSE) 2018 Paper-II
GPSC ARTO Pre 30.12.2018
Karnataka PSC AE 10.09.2017
UPRVUNL AE 21.08.2016
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
APGENCO AE 2012
ISRO Scientist/Engineer 2008
UKPSC AE 2007 Paper-II, ESE 2006

Ans. (b) : Thermal diffusivity is the thermal conductivity divided by density and specific heat capacity at constant pressure. It measures the rate of transfer of heat of the material from the hot end to the cold end. It has the SI derived unit of m^2/sec.

Thermal diffusivity $\alpha = \frac{k}{\rho C}$

3. A copper pipe carrying refrigerant at T°C is covered by cylindrical insulation of thermal conductivity k W/mK. The heat transfer co-efficient over the insulation surface is h W/m^{2}K. The critical thickness of insulation would be.
(a) k/h (b) 2 k/h
(c) 2h /k (d) h/k

TNPSC AE 2020
Haryana PSC AE (PHED) 04.09.2020, Paper-I
UPPSC AE 13.12.2020, Paper-2
MECON MT 2019, SJVN ET 2019
CIL MT 26.03.2017, TSPSC AEE 2015
Kerala PSC Poly Lect. 09.01.2014
UKPSC AE 2012 Paper–II
UPSC JWM Advt. No. 52/2010
CSE Pre-2001

Ans. (a) : For sufficient thin wires, putting the insulation around the wire may result in increase of heat transfer rate from the wire as compare to the case where there is no insulation.

For cylindrical wire, critical radius of insulation is given as

$r_c = \frac{k}{h}$

For spherical wire

$r_c = \frac{2k}{h}$

4. Upto the critical radius of insulation
(a) Adding insulation to a wall decreases heat loss
(b) Adding insulation to a wall increases heat loss
(c) Heat flux of the wall decreases
(d) Convection heat loss is less than conduction heat loss in the wall

GPSC DEE Class-2 (GWSSB) 04.07.2021
TNPSC AE 2019
RPSC Vice Principal ITI 2018
HPPSC Asst. Prof. 18.09.2017
(KPSC AE. 2015), Kerala PSC AE 06.08.2015
TRB Poly. Lect., 2012, RPSC AE GWD, 2011
ESE 2010, 2005

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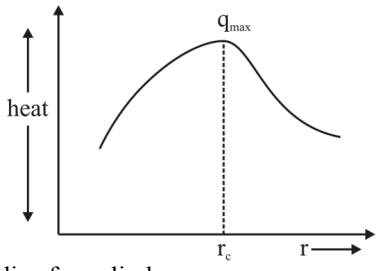
Ans. (b) : Upto the critical radius of insulation adding insulation to a wall increases heat loss.



Critical radius for cylinder

$$r_c = \frac{k}{h}$$

Critical radius for sphere

$$r_c = \frac{2k}{h}$$

5. For a given heat flow and for the same thickness, the temperature drop across the material will be maximum for

(a) copper (b) steel
(c) glass-wool (d) refractory brick

Gujarat PSC AE 2019
GPSC Executive Engineer 23.12.2018
TNPSC AE 2018, TSPSC AEE 2017
ISRO Scientist/Engineer (RAC) 07.05.2017
MPPSC State Forest Service Exam, 2014
ISRO Scientist/Engineer 12.05.2013
UKPSC AE 2012 Paper–II, GATE 1996

Ans. (c) : We know that

$$\frac{dT}{dx} \propto \frac{1}{k}$$

$$k \downarrow \rightarrow \frac{dT}{dx} \uparrow$$

So, form the given material glass-wool [0.075 W/mK] have low (value of) thermal conductivity (K).

$$\text{Copper} = 385\frac{W}{mK}$$

$$\text{Steel} = 17 - 55\frac{W}{mK}$$

$$\text{Refractory brick} = 0.9\frac{W}{mK}$$

Hence for glass wool temperature drop across the material will be maximum.

6. The thermal conductivity is expressed as :

(a) W/mK (b) W/m^2K
(c) W/hmK (d) W/h^2m^2K

GPSC DEE Class-2 (GWSSB) 04.07.2021
Nagaland PSC (CTSE) 2018 Paper-II
CGPSC ARTO & TSI 07.01.2017
Jharkhand Urja Vikas Nigam Ltd. AE 2017
GMB AAE 25.06.2017
(KPSC AE 2015), TSPSC AEE 2015
Kerala PSC AE 06.08.2015
UKPSC AE 2007 Paper -I, II

Ans : (a) Fourier's law of heat Conduction
It is also an important law in heat Conduction, which is represented by the equation

$$\dot{Q} = KA\frac{dT}{dx}$$

$\dot{Q}$ = Amount of heat flow through the body in a unit time (in watt)
A = Surface area perpendicular to heat flow (in m^2)
dT = Temperature difference on the two faces of the body (in K)
dx = Thickness of the body through which the heat flows (in m)
k = Constant of Proportionality known as thermal Conductivity

$$Q = KA\frac{dT}{dX}$$

$$K = \frac{Q \times dx}{A \times dT}$$

$$K = \frac{\text{watt} \times \text{meter}}{\text{meter}^2 \times K}$$

$$\boxed{K = W/m - K}$$

7. Fins are provided on heat transferring surface to:

(a) Increase surface area to promote the rate of heat transfer
(b) increase temperature gradient so as to enhance heat transfer
(c) Increase turbulence in flow for enhancing heat transfer
(d) Decrease the pressure drop of fluid

JPSC AE, 10.04.2021, Paper-I
Assam PSC AE (IWT) 14.03.2021
ESE 2020
GPSC Executive Engineer 23.12.2018
UPPSC AE 12.04.2016 Paper-II
Nagaland PSC (CTSE) 2016, Paper-II
OPSC AEE 2015 PAPER - II
UPSC JWM Advt. No.-52/2010
ESE 2008

Ans : (a) Fins are provided on heat transferring surface to increase surface area to promote the rate of heat transfer.
Heat transfer from extended surface (FINS):- Finned surfaces are manufactured by extruding, welding or wrapping a thin metal sheet on a surface. Fins enhance heat trnsfer from a surface by exposing a larger surface area to convection.

8. The ratio of heat flow Q_1/Q_2 from two walls of same thickness having their thermal conductivities as $k_1 = 2k_2$ will be

(a) 1 (b) 0.5
(c) 2 (d) 0.25

ISRO Scientist/Engineer (RAC) 22.04.2018
GPSC Asstt. Director of Transport 05.03.2017
ISRO Scientist/Engineer (RAC) 29.11.2015
TNPSC AE 2013, SJVN ET 2013
J&K PSC Civil Services Pre, 2013
BPSC AE 2012 Paper - V
J & K PSC Screening, 2006, ESE 2008

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Ans : (c) : $Q = -kA\frac{dT}{dx}$

$$Q_1 = -k_1 A\frac{dT}{dx}$$

$$Q_2 = -k_2 A\frac{dT}{dx}$$

$$k_1 = 2k_2$$

$$\frac{Q_1}{Q_2} = \frac{-2k_2 A\frac{dT}{dx}}{-k_2 A\frac{dT}{dx}} = 2$$

$$\boxed{\frac{Q_1}{Q_2} = 2}$$

9. Thermal diffusivity of a substance is

(a) Inversely proportional to thermal conductivity
(b) Directly proportional to thermal conductivity
(c) Directly proportional to the square of thermal conductivity
(d) Inversely proportional to the square of thermal conductivity

TNPSC AE 2019
GPSC Executive Engineer 23.12.2018
Sikkim PSC (Under Secretary), 2017
Punjab PSC SDE 12.02.2017
Nagaland PSC (CTSE) 2016, Paper-II
RPSC VPITI 14.02.2016
OPSC AEE 2015 PAPER - II
ESE 2006

Ans. (b) : Thermal diffusivity- Thermal diffusivity is the thermal conductivity divided by density and specific heat capacity at constant pressure.

It measure the rate of heat transfer of heat of a material from the hot end to the cold end. It has the SI derived unit of m^2/s.

$$\alpha = \frac{K}{\rho C_p}$$

$$\alpha \propto K$$

10. Which of the following has the lowest thermal conductivity

(a) Air (b) Water
(c) Brick (d) Copper

GPSC ARTO Pre 30.12.2018
Nagaland PSC CTSE 2017 Paper-2
HPPSC Asstt. Prof. 18.11.2016
UPPSC AE 12.04.2016 Paper-II
Rajasthan Nagar Nigam AE 2016, Shift-I
APPSC Poly. Lect. 2013, ESE 2011
UKPSC AE 2007 Paper -II

Ans. (a) : Air has the lowest thermal conductivity among the all which given in the question.

Air (at 0°C) – 0.024 (W/mK)
Brick – 0.15 (W/mK)
Water (at 20°C) – 0.6 (W/mK)
Brass – 109 (W/mK)

11. Usually fins are provided to increase the rate of heat transfer. But fins also act as insulation. Which one of the following non-dimensional numbers decides this factor?

(a) Eckert number (b) Biot number
(c) Fourier number (d) Peclet number

Assam PSC AE (PHED), 18.10.2020
Oil India Ltd. Sr. Engineer (Production) 30.11.2019
RPSC Vice Principal ITI 2018
HPPSC Asst. Prof. 18.09.2017
UPRVUNL AE 2016, 21.08.2016, ESE 2007

Ans. (b) : The effectiveness of fin

$$\varepsilon = \sqrt{\frac{kP}{hA_c}}$$

Since ratio of volume 'V' and cross section 'A_c' has dimension of length. So, characteristic dimension can be define as.

$$L_c = \frac{V}{A_c}$$

$$\therefore \varepsilon \propto \sqrt{\frac{k}{h.L_c}} \propto \sqrt{\frac{1}{Bi}}$$

where 'Bi' is Biot Number

$$\therefore Bi \propto \frac{1}{\varepsilon^2}$$

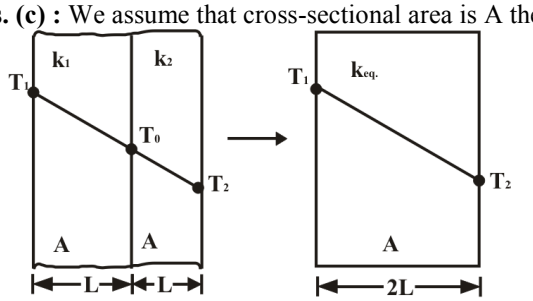
It Bi is high, the effectiveness will be low or vice versa.

12. A composite wall has two layers of different materials having thermal conductivities of k_1 and k_2. If each layer has the same thickness, the equivalent thermal conductivity of the wall is .....

(a) k_1k_2 (b) $\frac{(k_1+k_2)}{k_1k_2}$
(c) $\frac{(2k_1k_2)}{k_1+k_2}$ (d) k_1k_2

ISRO Scientist/Engineer (RAC) 22.04.2018
RPSC AE 2018
RPSC IOF & BOILER 20.09.2017
UJVNL AE 2016
Assam PSC CCE Pre 2015
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
ESE 2010, 2000, 2009, 1997

Ans. (c) : We assume that cross-sectional area is A then



Thermal resistance:

$$R_{th} = \frac{(T_1 - T_2)}{\left[\frac{L}{k_1A} + \frac{L}{k_2A}\right]} = \frac{(T_1 - T_2)}{\frac{2L}{k_{eq} \times A}}$$

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$$\frac{2}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{(k_1 + k_2)}{k_1 k_2}$$

$$k_{eq} = \frac{2k_1 k_2}{k_1 + k_2}$$

13. The concept of overall coefficient of heat transfer is used in heat transfer problems of:

(a) Fin conduction
(b) Convection
(c) Radiation
(d) Conduction and convection

RPSC IOF, 2020
OPSC AEE 2019 PAPER - II
Kerala PSC AE (Irrigation) 05.04.2017
ISRO Scientist/Engineer (RAC) 07.05.2017
TRB Poly. Lect., 2012, TNPSC AE, 2008

Ans : (d) : It is used in calculating the heat transfer typically by convection or phase transition between a fluid and a solid. The heat transfer coefficient has SI units watts per square meter Kelvin (W/m^2K). The heat transfer coefficient is reciprocal of thermal insulation.
The concept of overall coefficient of heat transfer is used in heat transfer problems of conduction and convection.

14. Thermal conductivity of air with rise in temperature

(a) Increases
(b) decreases
(c) remains constant
(d) May increases or decrease depending on temperature

JPSC AE 10.04.2021 Paper- I
Nagaland PSC CTSC 2017 Paper-II
Vizag Steel (MT) 2017, SJVN ET 2013
TRB Asstt. Prof., 2012
ISRO Scientist/Engineer 2007

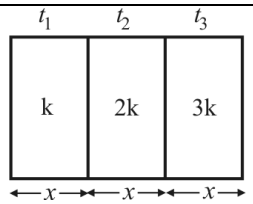
Ans. (a) : As the temperature of air increases the molecular diffusion also get increases and in case of air, thermal conductivity is directly proportional to the lattice vibration and molecular diffusion and hence as the temperature increases the thermal conductivity of air increases.

15. Three metal walls of the same cross-sectional area having thermal conductivities in the ratio 1:2:4 transfers heat at the rate of 3000 kJ/hr. For the same wall thickness, the temperature drops will be in the ratio.

(a) 1:2:4 (b) 1:1/2:1/4
(c) 1/4:1/2:1 (d) 1:1:1

ISRO Scientist/Engineer (RAC), 10.03.2019
GPSC ARTO Pre 30.12.2018
GPSC ARTO 01.05.2016
ISRO Scientist/Engineer 11.10.2015
MPPSC State Forest Service Exam, 2014
APPSC AEE 2012

Ans. (b) : Q = 3000 kJ/hr
$A_1 = A_2 = A_3 = A$ (same cross-section)
$x_1 = x_2 = x_3 = x$ (same thickness)



$$Q = \frac{t_1 - t_2}{\frac{x}{KA}} = \frac{t_2 - t_3}{\frac{x}{2KA}} = \frac{t_3 - t_4}{\frac{x}{4KA}}$$

$$\begin{cases} (t_1 - t_2) = k \\ (t_2 - t_3) = k/2 \\ (t_3 - t_4) = k/4 \end{cases}$$

$(t_1 - t_2) = 2\ (t_2 - t_3) = 4\ (t_3 - t_4) = K$

Ratio of temperature drop will be $1 : \frac{1}{2} : \frac{1}{4}$

1 : 1/2 : 1/4

16. Which one of the following configurations has the highest fin effectiveness?

(a) Thin, closely spaced fins
(b) Thin, widely spaced fins
(c) Thick, widely spaced fins
(d) Thick, closely spaced fins

APGCL AM, 2021
UPPSC AE 13.12.2020, Paper-2
GPSC ARTO Pre 30.12.2018, TSPSC AEE 2017
UPPSC AE 12.04.2016 Paper-II, GATE 2012

Ans. (a) : Effectiveness ($\in$) of fin $= \sqrt{\frac{KP}{hA_C}}$

Effectiveness ($\in$) of fin $\propto \sqrt{\frac{P}{A}}$

For better fin effectiveness ($\in$) P/A_C must be large i.e. $\in$ will be high if A_C is small therefore thin fins are preferred and fins must be closed (Not too closed).
so to have high effectiveness of fin, fin must be thin and closely spaced.

17. In a long cylindrical rod of radius R and a surface heat flux of q_0, the uniform internal heat generation rate is

(a) $2q_0/R$ (b) $2q_0$
(c) q_0/R (d) q_0/R^2

UPPSC AE 13.12.2020, Paper-2, UJVNL AE 2016
Mizoram PSC 2014 Paper- I
APPSC Poly. Lect. 2013, ESE 1998

Ans. (a) : Uniform internal heat generation (q) is define as heat generation per unit volume.

$$\therefore \quad q = \frac{Q}{V} = \frac{Q}{\pi R^2 H} \quad .....(1)$$

$$\text{Surface heat flux } (q_0) = \frac{Q}{\text{Surface area}} = \frac{Q}{2\pi RH}$$

$$\therefore \quad Q = q_0 \cdot 2\pi RH \quad .....(2)$$

Put the value of Q from equation (2) to equation (1)

$$\because \quad q = \frac{q_0 \times 2\pi RH}{\pi R^2 H}$$

$$\boxed{q = \frac{2q_0}{R}}$$

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18. The dimensional number relevant in transient heat conduction is :

(a) Grashof number (b) Weber number
(c) Fourier number (d) Reynold number

JWM 2017
Kerala LBS Centre for Sci. & Tech. Asst. Prof. 2014
ESE 2011
J&K PSC Civil Services Pre, 2010
UKPSC AE 2007 Paper -II

Ans. (c) : Biot and Fourier number simultaneously are applied when transient (unsteady) heat transfer.

$$\text{Biot number (Bi)} = \frac{\text{Internal resistance of conduction}}{\text{External resistance of convection}}$$

- Biot number is a dimensionless quantity.

Fourier number at time t can be viewed as the ratio of the rate of heat conducted to the rate of heat stored at that time.

19. For simple shapes such as plates, cylinders, sphere and cubes, the lumped heat capacity approach can be used if

(a) $Bi < 0.1$ (b) $Bi = 0.1$
(c) $Bi > 1.0$ (d) $Bi > 0.1$

NLCIL GET 17.11.2020, Shift-II
TNPSC AE 2018
UPPSC AE 12.04.2016 Paper-II
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
UKPSC AE-2013, Paper-II

Ans. (a) : The lumped heat capacity approach can be used when internal conductivity thermal resistance of material become zero ($k = \infty$) and Biot number should be less than 0.1

20. The plastic sleeve of outer radius r_0 = 1 mm covers a wire (radius = 0.5 mm) carrying electric current. Thermal conductivity of the plastic is 0.15 W/mK. The heat transfer coefficient on the outer surface of the sleeve exposed to air is 25 W/m^{2}K. Due to the addition of the plastic cover, the heat transfer from the wire to the ambient will.

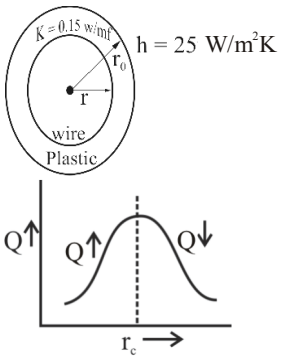
(a) increase (b) remain the same
(c) decrease (d) be zero

Assam Engg. College AP/Lect. 18.01.2021
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
GPSC Asstt. Director of Transport 05.03.2017
RPSC IOF & Boiler 20.09.2017
GATE 2016

Ans : (a)

Outer radius of Plastic sleeve (r_0) = 1 mm
Electric current carrying wire radius
(r) = 0.5 mm
Thermal conductivity of plastic
(k) = 0.15 mm
Convective heat transfer coefficient
(h) = 25 W/m^{2}K

According to question it is said that the wire is insulated by a plastic sleeve which shown in figure below.



critical radius $r_C = \frac{k}{h} = \frac{0.15}{25} = 6\times10^{-3}\,m$

$r_C = 6\,mm$

Here, we can see that the critical radius is greater than the radius of coating ($r_c > r_0$) therefore addition of plastic sleeve will increase The heat transfer.

21. With rise in temperature, thermal conductivity of solid material

(a) Decreases
(b) Increases
(c) Remains constant
(d) Cannot be predicted

Kerala PSC Asst. Prof. 27.09.2016
Rajasthan Nagar Nigam AE 2016, Shift-II
J&K PSC Civil Services Pre, 2013
TNPSC AE, 2008
UKPSC AE 2007 Paper -II

Ans. (a) : With rise in temperature, thermal conductivity of solid material decreases but for insulators it is not true.

For solids : (i) $\boxed{k \propto \frac{1}{T}} \Rightarrow$ For metals

(ii) $\boxed{k \propto T} \Rightarrow$ For non-metals and semi-conductors

where k = Thermal conductivity
T = Absolute Temperature

Thermal conductivity of metal decreases with increase in temperature except mercury, Aluminium and Uranium.

- Thermal conductivity is independent of pressure except in extreme cases as for e.g. vacuum.
- Thermal conductivity of gas increases with increasing temperature and decreasing molecular weight.

$$k \propto \sqrt{\frac{T}{m}}$$

- Thermal conductivity of liquid can decrease with temperature except water, glycerin.

22. Which of the following is expected to have highest thermal conductivity?

(a) Water (b) Melting ice
(c) Solid ice (d) Steam

ISRO Scientist/Engineer (RAC), 10.03.2019
Rajasthan Nagar Nigam AE 2016, Shift-I
UPSC JWM Adv. No.-16/2009
UKPSC AE 2007 Paper -II

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Ans. (c) : In given option thermal conductivity of solid ice is higher than other.

$k_{ice} > k_{melting\ ice} > k_{water} > k_{steam}$

Note– Thermal conductivity of diamond is highest (2300 W/m-k).
Where Freon-12 is lowest (0.0083 W/m-k)

23. Basic law of heat conduction is

(a) Fourier's law
(b) Stefan's law
(c) Newton's law
(d) First law of thermodynamics

Nagaland PSC CTSE 2017 Paper-II
Kerala PSC IOF 19.04.2016
MPPSC AE 08.11.2015, TRB Asstt. Prof., 2012

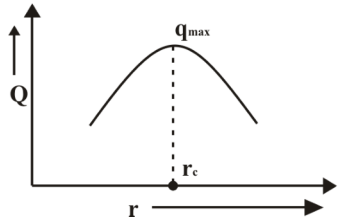
Ans. (a) : • Fourier's law - Conduction
• Newton's law – Convection
• Stefan's law – Radiation.

24. The critical radius is a insulation radius at which the resistance to heat flow is .....

(a) Minimum (b) Zero
(c) Maximum (d) Double

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 PAPER - II
RPSC AE 2018, ESE 2005, 1993

Ans. (a) : The critical radius is the insulation radius at which the resistance to heat flow is minimum and heat flow is maximum.



$r_c = \frac{k}{h}$ (For cylinder)

$r_c = \frac{2k}{h}$ (For spherical)

25. Various modes of heat transfer are:

(a) Conduction (b) Radiation
(c) Convection (d) All of these

CGPSC AE 15.01.2021, 16.10.2016
SJVN ET 2013, ESE 1993

Ans. (d) : Various modes of heat transfer are
• Conduction
• Convection
• Radiation

26. Addition of fin to the surface increases the heat transfer if $\sqrt{\frac{hA}{kP}}$ is

(a) equal to 1
(b) greater than 1
(c) less than 1
(d) greater than 1 but less than 2
(Notations used have usual meaning)

UPRVUNL AE 2014, APPSC Poly. Lect. 2013
UKPSC AE 2012 Paper–II, ESE 1996

Ans. (c) : Effectiveness $\varepsilon_{fin} = \frac{\tanh(mL)}{\sqrt{\frac{hA}{kP}}}$

$= \sqrt{\frac{kP}{hA}} \times \tanh(mL)$

for increased heat transfer, ε_{fin} should be higher. Hence effectiveness of fin 'ε_{fin}' should be greater than one.

So $\sqrt{hA/kP} \propto \frac{1}{\varepsilon_{fin}}$ should be less than one.

27. It is desired to increase the heat dissipation rate from the surface of an electronic device of spherical shape of 5 mm radius exposed to convection with h = 10 W/m^2 K, by encasing it in a spherical sheath of conductivity 0.04 W/m K. For maximum heat flow, the diameter of sheath should be

(a) 18 mm (b) 12 mm
(c) 8 mm (d) 16 mm

ESE 2018, 1996
RPSC IOF & Boiler 20.09.2017
APPSC AEE MAINS 2016 PAPER-III
MPPSC AE 2016

Ans. (d) : We know that, critical radius for spherical shape,

$r_c = \frac{2k}{h} = \frac{2 \times 0.04}{10}$

$r_c = 0.008$ m

for maximum heat flow, the diameter of spherical sheath should be

$d_c = 2 \times r_c = 2 \times 0.008 \times 1000$ mm

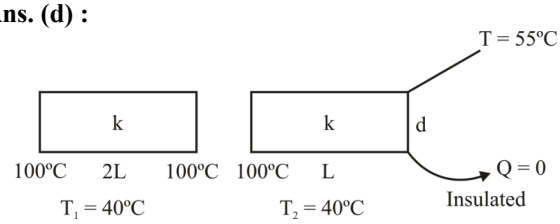
$d_c = 16$ mm

28. Two rods, one of length L and the other of length 2L, are made of the same material and have the same diameter. Both ends of the longer rod are maintained at 100 °C. One end of the shorter rod is maintained at 100 °C, while the other end is insulated. Both the rods are exposed to the same environment at 40 °C. The temperature at the insulated end of the shorter rod is measured to be 55 °C. What would be the temperature at the mid-point of the longer rod?

(a) 40 °C (b) 45 °C
(c) 50 °C (d) 55 °C
(e) 75 °C

ESE 2017
CGPSC Asstt. Workshop Supt., 17.07.2016
ISRO Scientist/Engineer 2010
GATE 1992

Ans. (d) :



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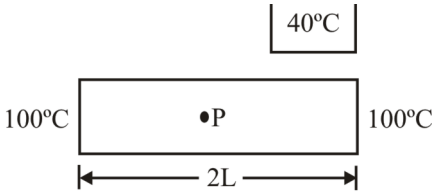
The longer rod has symmetric boundary conditions i.e. temperature at both the ends are the same, it is clear that the minimum temperature will occur at the midpoint, i.e. at midpoint dT/dx = 0 i.e. heat flux is zero.
For shorter rod, at dT/dx = 0, the temperature is 55ºC
If we assume the longer rod as two parts, then first part will be similar to the shorter rod, where base is at 100ºC and at tip dQ/dx = 0
∴ temperature at the mid point = 55ºC

Alternate Method

Mathematically-
Longer rod like a finite length fin exposed to ambient condition.
Temperature distribution for finite length is given as:



Longer fin

29. A composite wall consists of three different materials having thermal conductivity k, 2k and 4k respectively. The temperature drop across different materials will be in the ratio :

(a) 1 : 1 : 1 (b) 1 : 2 : 4
(c) 4 : 2 : 1 (d) 2 : 4 : 1

HPPSC W.S. Poly. 2016
UKPSC AE 2013, 2007 Paper-II, ESE 2009

Ans : (c) Fourier's law of heat conduction :-

$$Q = KA\frac{dT}{dx}$$

Thermal conductivity of different material K, 2K and 4K.

$$Q = K_1A_1\frac{dT}{dx}$$

$$dT_1 = \frac{Q \times dx}{K_1A_1}$$

$$dT_1 \propto \frac{1}{K_1}$$

$$dT_1 : dT_2 : dT_3 = \frac{1}{K_1} : \frac{1}{K_2} : \frac{1}{K_3}$$

$$dT_1 : dT_2 : dT_3 = \frac{1}{K} : \frac{1}{2K} : \frac{1}{4K}$$

$$dT_1 : dT_2 : dT_3 = 4 : 2 : 1$$

30. In the non-dimensional Biot number, the characteristic length is the ratio of

(a) Volume of solid to its surface area
(b) Surface area to volume of solid
(c) Perimeter to surface area of solid
(d) Surface area to perimeter of solid

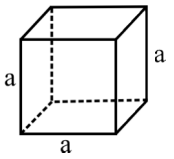
CGPSC AE 26.04.2015 SHIFT-II
MPPSC State forest service Exam 2014
UKPSC AE 2007 Paper -II

Ans. (a) :

$$\text{Biot No. (Bi)} = \frac{\text{Conductive Resistance}}{\text{Convective Resistance}} = \frac{hL_c}{k}$$

where,

$$\text{Characteristic Length } (L_c) = \frac{\text{volume of body}}{\text{surface area of contact}}$$



For cube-

$$\text{characteristic length, } L_c = \frac{a \times a \times a}{2a^2 + 2a^2 + 2a^2}$$

$$L_c = \frac{a^3}{6a^2} = \frac{a}{6}$$

* For sphere–

$$\text{Characteristic length, } L_c = \frac{R}{3} \quad (R = \text{Radius})$$

31. A 0.5 m thick plane wall has its two surfaces kept at 300^0C and 200^0C. The thermal conductivity of the wall varies linearly with temperature and its value at 300^0C and 200^0C are 25 W/mK and 15 W/mk, respectively. Then the steady heat flux through the wall is

(a) 8 kW/m^2 (b) 5 kW/m^2
(c) 4 kW/m^2 (d) 3 kW/m^2

TRB Asstt. Prof., 2012
UKPSC AE 2012 Paper–II, ESE 2002

Ans. (c) : Given,

$t = 0.5$ m
$T_1 = 300°C$
$T_2 = 200°C$
$K_{300} = 25$ W/mK
$K_{200} = 15$ W/mK

$$K_{avg} = \frac{25+15}{2} = 20 W/mK$$

$$Q = \frac{K\Delta T}{t} = \frac{20 \times (300-200)}{0.5} = 4 kW/m^2$$

32. It is proposed to coat a 1 mm diameter wire with enamel paint (k = 0.1 W/mK) to increase the heat transfer with air. If the air side heat transfer coefficient is 100 W/m^{2}K, the optimum thickness of enamel paint should be

(a) 0.25 mm (b) 0.5 mm
(c) 1 mm (d) 2 mm

HPPSC Asstt. Prof. 18.11.2016
ISRO Scientist/Engineer 11.10.2015
GATE 1999

Ans. (b) : Given, r = 1 mm, k = 0.1 W/mK, h = 100 W/m^{2}K

For cylinder,

$$r_{critical} = \frac{k}{h}$$

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$= \frac{0.1}{100} = 1 \times 10^{-3}$ m

$= 1$mm

Thickness of enamel point

$= r_{critical} - r_{wire}$

$= 1 - 0.5 = 0.5$ mm

33. According to Fourier's law, amount of heat flow (Q) through the body in unit time is equal to

(a) $kA\frac{dT}{dx}$ (b) $kA\frac{dT^2}{dx^2}$

(c) $k\frac{dT}{dx}$ (d) $kA\frac{dx}{dT}$

GPSC DEE Class-2 (GWSSB) 04.07.2021
UKPSC AE 2013 Paper-II
ISRO Scientist/Engineer 2011

Ans. (a) : According to Fourier's law,

Amount of heat flow (Q) = $kA\frac{dT}{dx}$

k = thermal conductivity

A = cross-section area

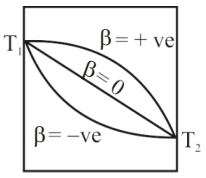
$\frac{dT}{dx}$ = Temperature gradient.

34. In steady state conduction with thermal conductivity given by $k = k_0 (1 + \beta T)$ where β, is positive, a slab of given thickness and given temperature drop will conduct

(a) more heat at lower temperature levels
(b) more heat at higher temperature levels
(c) same heat as flow depends on the temperature drop
(d) same heat as flow depends on the thickness only

RPSC Vice Principal ITI 2018
HPPSC Asst. Prof. 20.11.2017
ISRO Scientist/Engineer 2008

Ans. (b) : $k = k_0 (1 + \beta t)$



More heat at higher temperature levels.

35. What is the expression for the thermal conduction resistance to heat transfer through a hollow sphere of inner radius r_1, and outer radius r_2, and thermal resistance?

(a) $\frac{(r_2 - r_1) r_1 r_2}{4\pi K}$ (b) $\frac{4\pi K (r_2 - r_1)}{r_1 r_2}$

(c) $\frac{r_2 - r_1}{4\pi K\ r_1 r_2}$ (d) None of these is correct

RPSC IOF, 2020, MPPSC AE 2016, ESE 2007

Ans : (c)

Heat transfer through hollow sphere –

$$Q = \frac{4\pi r_1 r_2 K (T_1 - T_2)}{(r_2 - r_1)}$$

$$Q = \frac{(T_1 - T_2)}{\frac{(r_2 - r_1)}{4\pi r_1 r_2 K}}$$

We know that

$$Q = KA\frac{(T_1 - T_2)}{dx} = \frac{(T_1 - T_2)}{R_{th}}$$

$$\text{Thermal Resistance } (R_{th}) = \frac{(r_2 - r_1)}{4\pi r_1 r_2 K}$$

36. The MLTθ system the dimension of thermal conductivity is:

(a) $ML^{-1} T^{-1} \theta^{-3}$ (b) $MLT^{-1} \theta^{-1}$
(c) $ML\theta^{-1} T^{-3}$ (d) $ML\theta^{-1} T^{-2}$

Assam PSC AE (IWT) 14.03.2021
OPSC AEE 2015 PAPER - II
ESE 2009

Ans : (c) Fourier law of heat conduction:-

$$Q = -KA\frac{dt}{dx}$$

$$k = \frac{Q \times dx}{A \times dT.}$$

Unit of thermal conductivity = W/mK

$$K. = \frac{J}{sec - mK} = \frac{N - m}{sec - m.K} = \frac{N}{sec.K}$$

Dimension of thermal conductivity = $[MLT^{-3} \theta^{-1}]$

37. In lumped capacity heat transfer model, the variation of temperature with time is

(a) Linear (b) Parabolic
(c) Exponential (d) Hyperbolic

JPSC AE PRE 2019
Oil India Limited Sr. Engineer (Drilling) 30.11.2019
UKPSC AE 2012 Paper–II

Ans. (c) :

$$\frac{T - T_\infty}{T_i - T_\infty} = e^{-\frac{hA\tau}{\rho C_p V}}$$

Where, T_i = Temperature of body at the instant of time t = 0 second i.e. when the body is just exposed to ambient fluid

T = Temperature of body at any instant of time t (sec.)

T_∞ = Ambient fluid temperature

Hence, temperature variation will be exponential.

38. In unsteady-state heat conduction for bodies with negligible temperature gradients, the time temperature variation curve is

(a) linear (b) parabolic
(c) sinusoidal (d) exponential

JPSC AE, 10.04.2021, Paper-I
GPSC EE Pre, 28.01.2017, ESE 2012

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Ans. (d) : According to time temperature variation curve-

$$\frac{T - T_\infty}{T_1 - T_\infty} = e^{-Bi\,\tau_0}$$

Where, τ_o = time

B_i = Biot Number

39. A very low value of the Biot number indicates that:

(a) the internal-conduction resistance is considerable in comparison with surface convection resistance.

(b) unsteady state conduction is dominating over the steady state conduction steady state conduction is dominating over the unsteady state conduction.

(c) steady state conduction is dominating over the unsteady state conduction.

(d) the internal-conduction resistance is negligible in comparison with surface convection resistance.

RPSC ACF & FRO, 26.02.2021
NLC GET 17.11.2020, Shift-II
ISRO Scientist/Engineer 22.04.2018

Ans. (d) : Biot No. (Bi)

$$= \frac{\text{Internal conductive Resistant } (R_{cond})}{\text{External convective Resistant } (R_{conv})}$$

$$= \frac{hL}{k_{solid}} \qquad B_i < 0.1, \quad ICR <<< FCR$$

If Bi is very low then internal conductive resistant is negligible in comparison with surface convection resistance.

Value of the Biot number smaller than 0.1 imply that the heat conduction inside the body is much faster than the heat convection away from its surface and temperature gradients are negligible inside of it.

40. Which of the following has maximum value of thermal conductivity?

(a) Steel (b) Brass
(c) Copper (d) Aluminum

ISRO Scientist/Engineer 22.04.2018
Rajasthan Nagar Nigam AE 2016, Shift-I
Nagaland PSC CTSE 2015 Paper-II

Ans. (c)

Material	Thermal Conductivity (W/mK)
(i) Silver	429
(ii) Copper	393
(iii) Aluminium	247
(iv) Cast iron	55-65
(v) Steel	20-45
(vi) Concrete	1.20
(vii) Glass (window)	0.75
(viii) Asbestos sheet	0.17
(ix) Ash	0.12
(x) Cork, felt	0.05 – 0.10
(xi) Saw dust	0.07
(xii) Glass wool	0.03
(xiii) Water	0.55 – 0.7
(xiv) Freon	0.0083

41. Fourier law of heat conduction gives the heat flow for

(a) irregular surfaces
(b) non-uniform temperature surfaces
(c) one dimensional case only
(d) two dimensional case only

RPSC IOF, 2020, MECON MT 2019
Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : Fourier's law of heat conduction –

$$Q = kA \times \frac{dT}{dx}$$

Fourier's law of heat conduction is valid for one dimensional case only.

Fourier equation is valid for all matter solid liquid or gas. The vector expression indicating that heat flow rate is normal to an isothermal and is in the direction of decreasing temperature.

42. A steam pipe is covered with two layers of insulating materials, with the better insulating material forming the outer part. If the two layers are interchanged, the heat conducted:

(a) Will decrease
(b) Will increase
(c) Will remain unaffected
(d) may increase or decrease depending upon the thickness of each layer

HPPSC Astt. Prof. 18.09.2017
RPSC ADE 2016, APPSC AE 2012

Ans. (a) : We should select a material which have low heat transfer capacity. So we should select a material having low thermal conductivity. So that heat will pass through material is minimum. We should use low thermal conductivity material both sides because we need to decrease the rate of heat transfer.

43. A steel steam pipe 10 cm inner diameter and 11 cm outer diameter is covered with an insulation having a thermal conductivity of 1 W/mK. If the convective heat transfer coefficient between the surface of insulation and the surrounding air is $8 W/m^2K$, the critical radius of insulation is :

(a) 100 mm (b) 110 mm
(c) 125 mm (d) 150 mm

Vadodara Municipal Corp. DEE, 2018
GPSC ARTO Pre 30.12.2018
GATE 2000

Ans. (c) For cylinder,

$$\text{Critical radius of insulation } (r_c) = \frac{k_{insulation}}{h_{surrounding}}$$

$$r_c = \frac{1}{8}$$

$$r_c = 0.125m = 125mm$$

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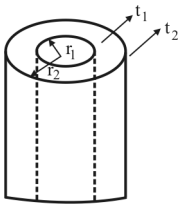
44. The heat flow equation through a cylinder of inner radius r_1 and outer radius r_2 is desired to be written in the same form as that for heat flow through a plane wall. For wall thickness $(r_2 - r_1)$ the equivalent area A_m would be :

(a) $\frac{A_1 + A_2}{2}$ (b) $\frac{A_1 + A_2}{2\log\left(\frac{A_2}{A_1}\right)}$

(c) $\frac{A_2 - A_1}{\log\left(\frac{A_2}{A_1}\right)}$ (d) $\frac{A_2 + A_1}{2\log\left(\frac{A_2}{A_1}\right)}$

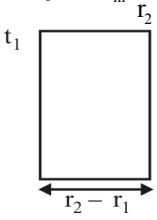
J & K PSC Screening, 2006

Ans. (c) : Heat flow through cylinder $= \frac{2\pi kL(t_1 - t_2)}{\ln\frac{r_2}{r_1}}$



Thermal Resistance cylinder $= \frac{1}{2\pi kL}\ln\left(\frac{r_2}{l_1}\right)$ ..(i)

In case of plane wall $Q = kA_m\frac{t_1 - t_2}{r_2 - r_1}$



Thermal resistance wall $= \frac{r_2 - r_1}{kA_m}$ ..(ii)

equat.. (i) & (ii)

$$\frac{1}{2\pi kL}\ln\left(\frac{r_2}{r_1}\right) = \frac{r_2 - r_1}{kA_m}$$

$$\frac{1}{2\pi kL}\ln\frac{r_2}{r_1} = \frac{r_2 - r_1}{k \times 2\pi r_m}$$

$$r_m = \frac{r_2 - r_1}{\ln\frac{r_2}{r_1}}$$

or $\boxed{A_m = \frac{A_2 - A_1}{\ln\frac{A_2}{A_1}}}$

45. The inner surface of a plain brick wall is at 50°C and the outer surface is at 25°C. The rate of heat transfer per m^2 of surface area of the wall which is 220 mm thick will be (Take thermal conductivity of the brick as 0.51 W/m-k) :

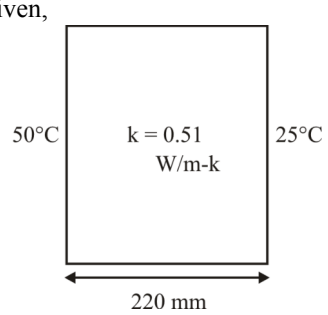
(a) 57.95 W/m^2 (b) 579.5 W/m^2

(c) 5795 W/m^2 (d) 5.795 W/m^2

CGPSC AE 15.01.2021, 16.10.2016

TSPSC AEE 2017

Ans. (a) : Given,



Thermal conductivity

(k) = 0.51 W/m-K

Thickness (L) = 220 mm = 0.22 m

Rate of Heat transfer per m^2

$$q = \frac{T_1 - T_2}{\left(\frac{L}{k}\right)} = \frac{50 - 25}{\left(\frac{0.22}{0.51}\right)}$$

$$q = 57.95\,W/m^2$$

46. Fins are made as thin as possible to

(a) reduce the total weight

(b) accommodate more number of fins

(c) increase the width for the same profile area

(d) improve flow of coolant around the fins

RPSC VPITI 14.02.2016

Assam PSC CCE Pre 2015, ESE 2010

Ans. (b) : For more heat transfer from fin, the effectiveness should be high

i.e. $\in = \sqrt{\frac{KP}{hA_C}}$

For better effectiveness ($\in$) of fins $\frac{P}{A_c}$ must be large i.e., effectiveness will be high if A_c is small. So more number of thin fins are preferred and fins must be close (Not too closely) space.

Note : Generally fins are effective if $\in > 2$.

47. Two walls of the same thickness and cross-sectional area have thermal conductivities in the ratio of 1 : 2. If the ratio of the temperature drop across the two walls is 2 : 3, what is the ratio of the heat flow?

(a) 3 : 1 (b) 1 : 3

(c) 1 : 2 (d) 2 : 1

CIL MT 27.02.2020

APPSC Poly Lect. Automobile Engg., 2020

ESE 2008

Ans. (b) : given that

$A_1 = A_2 = A$ $K_1 : K_2 = 1:2$

$\Delta x_1 = \Delta x_2 = dx$ $dt_1 : dt_2 = 2:3$

$$Q = kA\frac{dt}{dx}$$

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$$Q_1 = k_1 A \frac{dt_1}{dx_1} ----(I)$$

$$Q_g = k_2 A \frac{dt_2}{dx_2} ----(II)$$

$$\frac{Q_1}{Q_2} = \frac{K_1 A dt_1}{dx} \times \frac{dx}{k_2 A dt_2}$$

$$\frac{Q_1}{Q_2} = \frac{k_1}{k_2}\frac{dt_1}{dt_2} = \frac{1}{2} \times \frac{2}{3}$$

$$\frac{Q_1}{Q_2} = \frac{1}{3}$$

$$\boxed{Q_1 : Q_2 = 1:3}$$

48. When heat is transferred by molecular collision, it is referred to as heat transfer by

(a) conduction (b) convection

(c) radiation (d) radiation and convection

ISRO Scientist/Engineer 03.07.2018
Mizoram PSC AE/SDO 2014, Paper-I
JPSC AE 2013 Paper- V

Ans. (a) : Convection–Heat transfer by motion of particles.
Conduction–Heat transfer by molecular collision actually convection involves conduction and convection both so convection can also correct but the most appropriate option will be conduction.

49. In case of one dimensional heat conduction in a medium with constant properties, T is a temperature at position x, at time t. Then $\frac{\partial T}{\partial t}$ is proportional to

(a) $\frac{T}{x}$ (b) $\frac{\partial t}{\partial x}$

(c) $\frac{\partial T}{\partial x \partial t}$ (d) $\frac{\partial^2 T}{\partial x^2}$

VIZAG STEEL MT 18.08.2013, 10.06.2012
GATE 2005

Ans. (d) : One dimensional heat conduction equation.

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha}\frac{\partial T}{dt}$$

So, $$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$\boxed{\frac{\partial T}{\partial t} \propto \frac{\partial^2 T}{\partial x^2}}$$

50. Lumped heat transfer analysis of a solid object suddenly exposed to a fluid medium at a different temperature is valid when

(a) Biot number < 0.1
(b) Biot number > 0.1
(c) Fourier number < 0.1
(d) Fourier number > 0.1

VIZAG STEEL MT 18.08.2013, 10.06.2012
GATE 2001

Ans. (a) : Lumped heat transfer analysis of a solid object is valid.
When,

$$\boxed{\text{Biot Number} < 0.1}$$

51. General conduction equation in its simple form, $\nabla^2 T=0$ (where T is temperature), is called:-

(a) Poisson's equation (b) Fourier's equation

(c) Laplace equation (d) Kirchoff's equation

TNPSC AE (Industries) 09.06.2013
UKPSC AE-2013, Paper-II
J&K PSC Civil Services Pre, 2013

Ans. (c) : General conduction equation in its simple form, $\nabla^2 T=0$ (where T is temperature), is called Laplace equation.

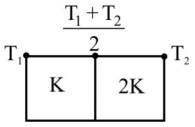
52. Two plane slabs of equal area but with thermal conductivities in the ratio 1 : 2 are held together with temperature between the two outer surfaces being T_1 and T_2. If the juction temperature between the two surfaces is desired to be $\frac{T_1+T_2}{2}$, then their thicknesses should be in the ratio of :

(a) 1 : 2 (b) 2 : 1

(c) 1 : 1 (d) 3 : 1

OPSC AEE 2015 PAPER - II
ISRO Scientist/Engineer (RAC) 29.11.2015
HPPSC Asstt. Prof. 2014

Ans : (a)



for first slab.

$$Q = KA\frac{dt}{dx}$$

$$dx_1 = \frac{KA}{Q}\left[T_1 - \left(\frac{T_1+T_2}{2}\right)\right]$$

$$dx_1 = \frac{KA}{Q}\left(\frac{T_1-T_2}{2}\right) \quad \ldots\ldots\ldots (i)$$

for second slab

$$dx_2 = \frac{2K.A}{Q}\left(\frac{T_1+T_2}{2} - T_2\right)$$

$$= 2\frac{K.A}{Q}\left(\frac{T_1-T_2}{2}\right) \ldots\ldots\ldots (ii)$$

Eq$^{n.}$ (i) divided by eq$^{n.}$ (ii)
so

$$\boxed{\frac{dx_1}{dx_2} = \frac{1}{2}}$$

53. The radial heat transfer rate through hollow cylinder increases as the ratio of outer radius to inner radius-

(a) decreases (b) increases

(c) constant (d) decreases as well as increases

RPSC AE 2018, UKPSC AE 2012, 2007 Paper -II

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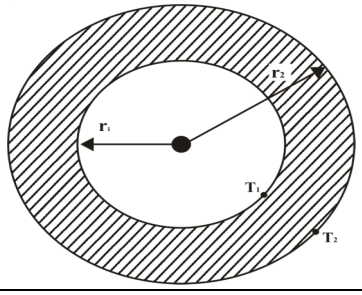
Ans. (a) : We know that heat transfer through hollow cylinder

$$Q = \frac{2\pi kL(T_1 - T_2)}{\ln\left(\frac{r_2}{r_1}\right)}$$

If $\left(\frac{r_2}{r_1}\right) \uparrow \rightarrow Q \downarrow$



54. For a current wire of 20 mm diameter exposed to air (h = 25 W/m²K), maximum heat dissipation occurs when thickness of insulation (k = 0.5 W/mK) is,

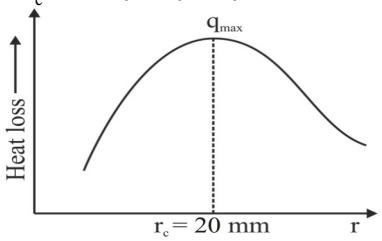
(a) 20 mm (b) 25 mm
(c) 28 mm (d) 10 mm

TNPSC AE 2017
ISRO Scientist/Engineer 11.10.2015
GATE 1996, 1993

Ans. (d) : Data given,
D = 20 mm
h = 25 W/m²K
k = 0.5 W/mK

$$r_c = \frac{0.5}{25} \times 1000 = 20\text{mm}$$

Thickness = $r_c - r = 20 - 10 = 10$ mm



55. Which of the following material has highest thermal conductivity at room temperature

(a) Gold (b) Diamond
(c) Iron (d) Aluminium

APPSC Poly. Lect. Automobile Engg. 2020
CIL MT 27.02.2020
RPSC Lecturer 16.01.2016

Ans. (b) :

Metal	Thermal conductivity $\left[\frac{W}{mK}\right]$
Gold	310
Diamond	2300
Iron	80.2
Aluminium	200
Silver	410
Copper	385

56. Consider following facts about Thermal diffusivity

1. Its unit is m^2/s
2. It is the relation of k, ρ and Cp
3. depend upon temperature

Of these, which are correct

(a) 1 (b) 1 and 2
(c) 1 and 3 (d) 2 and 3
(e) 1, 2 and 3

SJVN ET 2019
Nagaland PSC CTSE 2017 Paper-2
CGPSC AE 2014 -II

Ans. (e) : Thermal diffusivity– Thermal diffusivity is the thermal conductivity divided by density and specific heat capacity at constant pressure.

It measure the rate of transfer of heat of a material from the hot end to the cold end. Unit of thermal diffusivity is m^2/s.

1. Unit of thermal diffusivity is m^2/s
2. It is the relation of K , ρ , and C_P

$$\alpha = \frac{K}{\rho C_P}$$

3. It depends upon temperature.

57. Transient conduction means

(a) very little heat transfer
(b) heat transfer for a short time
(c) heat transfer with a very small temperature difference
(d) conduction when the temperature at a point varies with time

GPSC DEE, Class-2 (GWSSB) 04.07.2021
JPSC AE, 10.04.2021, Paper-I
TSPSC AEE 2015

Ans. (d) : Transient conduction means conduction when the temperature at a point varies with time.

58. For steady flow and constant value of conductivity, the temperature distribution for a plane wall is

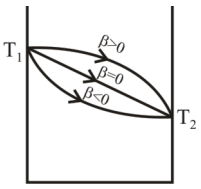
(a) Parabolic (b) Linear
(c) Logarithmic (d) Cubic

GPSC Engineer, Class-II Pre-19.01.2020
JPSC AE PRE 2019, NPCIL ET, 2015
MPPSC State Forest Service Exam, 2014

Ans. (b) : $Q = k_m A\left(\frac{T_1 - T_2}{L}\right)$

$k_m = k_0(1 + \beta T)$

k_0 = Thermal conductivity at zero temperature
k_m = Mean thermal conductivity
β = Slope of temperature variation line



Variation of thermal conductivity with temperature.

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59. Time constant of a thermocouple is the time taken to attain

(a) The final value to be measured
(b) 50% of value of the initial temperature difference
(c) 63.2% of value of the initial temperature difference
(d) 98.8% of value of the initial temperature difference.

CGPSC ARTO & TSI 07.01.2017
Sikkim PSC (Under Secretary), 2017
UKPSC AE 2012 Paper-II
UPSC JWM Advt. No.-52/2010, ESE 2010, 1997

Ans. (c) : Explanation– The time constant of a thermocouple is defined as the time required for the sensor to respond to 63.2% of its total output signal when subjected to a step change in temperature.

60. Thermal diffusivity is:

(a) a dimensionless parameter
(b) function of temperature
(c) a physical property of the material
(d) useful in case of heat transfer by radiation

TANGEDCO AE 2015, Mizoram PSC 2014 Paper-I
HPPSC Asstt. Prof. 2014

Ans. (c) :

$$\text{Thermal diffusivity } (\alpha) = \frac{k}{\rho \cdot C_p} = \frac{\text{Heat conductivity}}{\text{Heat capacity}}$$

Unit = m^2/sec

Thermal diffusivity is a physical property of the material.

61. Consider the following statements pertaining to heat transfer using fins :

1. Fins should be used on the side where heat-transfer coefficient is small.
2. Long and thick fins should be used.
3. Short and thin fins should be used.
4. Thermal conductivity of the fin material should be large.

Which of the above statements are correct ?

(a) 1, 2 and 4 (b) 1 and 3 only
(c) 3 and 4 only (d) 1, 3 and 4

UPRVUNL AE 21.08.2016
TRB Poly. Lect., 2012
UPSC JWM Advt. No.-50/2010, ESE 2002

Ans. (d) : As we know,

$$\text{Effectiveness } (\varepsilon) = \sqrt{\frac{Pk}{hA}}$$

Heat transfer using fins, increase, when k is increases Q 'h' is decreases, and long & thick fins have less effective area than short & thin fins.

62. A furnace wall has a thickness of 50cm and thermal conductivity of 0.7 W/m-K. What will be the required thickness of a new material having thermal conductivity equal to 0.14 W/m-K, if the heat loss and temperature drop across the wall are the same as before ?

(a) 5 cm (b) 10 cm
(c) 20 cm (d) 50 cm

GPSC ARTO Pre 30.12.2018
ISRO Scientist/Engineer (RAC) 07.05.2017
UPSC JWM Advt. No.-50/2010

Ans. (b) : Given, $\ell_1 = 50$ cm, $k_1 = 0.7$ W/mK, ΔT_1
$k_2 = 0.14$ W/mK, ΔT_2

Given that,

$$q_1 = q_2$$
$$\Delta T_1 = \Delta T_2$$
$$q = \frac{kA\Delta T}{\ell}$$

So, $$\frac{k_1 A_1 \Delta T_1}{\ell_1} = \frac{k_2 A_2 \Delta T_2}{\ell_2}$$ [∴ Same heat loss and temperature drop]

or $$\frac{k_1}{\ell_1} = \frac{k_2}{\ell_2} \Rightarrow \ell_2 = \frac{k_2 \times \ell_1}{k_1}$$

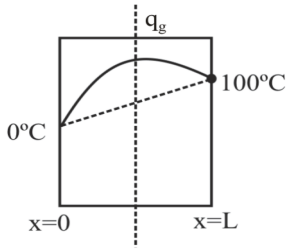
$$\ell_2 = \frac{0.14}{0.7} \times 50 = 10\,\text{cm}$$

63. Consider one-dimensional steady state heat conduction along x-axis through a plane wall with the boundary surfaces (x = 0 and x = L) maintained at temperatures 0°C and 100°C. Heat is generated uniformly throughout the wall. Choose the correct statement.

(a) The direction of heat transfer will be from the surface at 100°C to surface at 0°C.
(b) The temperature distribution is linear within the wall.
(c) The maximum temperature inside the wall must be greater than 100°C.
(d) The temperature distribution is symmetric about the mid-plane of the wall.

Assam Engg. College AP/Lect. 18.01.2021
GPSC Asstt. Director of Transport 05.03.2017
GATE 2013

Ans. (c) :



Heat generation per unit volume.

1-D steady state heat conduction equation.

$$\frac{d^2T}{dx^2} + \frac{q_g}{k} = 0$$

$$T = \frac{-q_g}{2k}x^2 + C_1 x + C_2$$

This equation shows that temperature distribution will be parabolic from figure it is clear that maximum temperature will be more than 100°C inside the wall.

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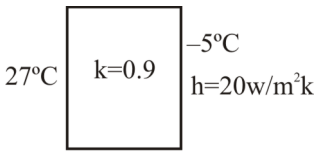
64. **A brick wall (k = 0.9 W/m–K) of thickness 0.18 m separates the warm air in a room from the cold ambient air. On a particular winter day, the outside air temperature is –5°C and the room needs to be maintained at 27°C. The heat transfer coefficient associated with outside air is 20 W/m^2-K. Neglecting the convective resistance of the air inside the room, the heat loss, in (W/m^2) is_____**

(a) 88 (b) 110
(c) 160 (d) 128

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (d) :



Heat loss (W/m^2) $q = \dfrac{T_1 - T_2}{\Sigma R_{Th}}$

$$q = \frac{T_1 - T_2}{\frac{t}{k} + \frac{1}{h}}$$

$$= \frac{27 - (-5)}{\frac{0.18}{0.9} + \frac{1}{20}}$$

$q = 128$ W/m^2

65. **The ratio of heat flow from two walls of thickness ratio 1 : 2 and thermal conductivity ratio 3 : 1 for the same temperature difference on the two sides is :**

(a) 5 : 1 (b) 6 : 1
(c) 2 : 3 (d) 3 : 2

Nagaland PSC CTSE 2017 Paper-2
J&K PSC Civil Services Pre, 2010

Ans. (b) : Given,
Thickness ratio of two walls

$$\frac{x_1}{x_2} = \frac{1}{2}$$

and same temperature difference

$$(\Delta T)_1 = (\Delta T)_2$$

Thermal conductivity ratio

$$\frac{K_1}{K_2} = \frac{3}{1}$$

According to Fourier's law =

$$\frac{Q_1}{Q_2} = \frac{K_1 A_1 \left(\frac{dT}{dx}\right)_1}{K_2 A_2 \left(\frac{dT}{dx}\right)_2}$$

$$\frac{Q_1}{Q_2} = \frac{K_1}{K_2} \times \frac{x_2}{x_1}$$

$$Q_1 : Q_2 = \frac{3}{1} \times \frac{2}{1} = \frac{6}{1} = 6:1$$

$$\boxed{Q_1 : Q_2 = 6:1}$$

66. **A copper wire of radius 0.5 mm is insulated with a sheathing of thickness 1 mm having a thermal conductivity of 0.5 W/mK. The outside surface convective heat transfer coefficient is 10 W/m^{2}K. If thickness of insulation sheathing is raised by 10 mm, then the electrical current carrying capacity of the wire will :**

(a) increase
(b) decrease
(c) remains same
(d) vary depending upon the electrical conductivity of the wire

APPSC IOF, 2009, ESE 2000

Ans. (a) : Given, $r_i = 0.5$ mm
$r_0 = 0.5 + 1 + 10 = 11.5$ mm
$k = 0.5$ W/mK, $h = 10$ W/m^{2}K

$$r_c = \frac{k}{h} = \frac{0.5}{10} = 0.05\,\text{m} = 50\,\text{mm}$$

Since $r_c > r_o$, there will be increase in rate of heat transfer, current carrying capacity increase.

67. **Two walls of same thickness and cross-sectional area have thermal conductivities in the ratio 1:2. If the same temperature difference is maintained across the wall faces, the ratio of heat flow Q_1/Q_2 will be**

(a) 1/2 (b) 1
(c) 2 (d) 4

Assam PSC AE (PHED), 18.10.2020
UKPSC AE 2012 Paper–II, ESE 2008

Ans. (a) : Given $\dfrac{k_1}{k_2} = \dfrac{1}{2}$ Conductivity ratio

heat transferred $Q = -kA\dfrac{\Delta T}{\Delta X}$

First wall face heat $Q_1 = -k_1 A_1 \dfrac{\Delta T_1}{\Delta X}$

$$Q_2 = -k_2 A_2 \frac{\Delta T_2}{\Delta X}$$

Ratio of heat flow $\dfrac{Q_1}{Q_2} = \dfrac{k_1 A_1 \frac{\Delta T_1}{\Delta X}}{k_2 A_2 \frac{\Delta T_2}{\Delta X}}$

For same temperature difference, same thickness and cross sectional area

$A_1 = A_2$, $\Delta T_1 = \Delta T_2$

$$\frac{Q_1}{Q_2} = \frac{k_1}{k_2}$$

$$\frac{Q_1}{Q_2} = \frac{1}{2}$$

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68. A furnace is made of a red brick wall of thickness 0.25 m and conductivity 0.4 W/m-K. For the same heat loss and temperature drop this can be replaced by layer of diatomite earth of conductivity 0.16 W/m-K and thickness

(a) 0.05 m (b) 0.1 m
(c) 0.5 m (d) 0.2m

UPRVUNL AE 21.08.2016, ESE 1993

Ans. : (b) $Q = kA\frac{dT}{dx}$

$$\frac{k_1}{dx_1} = \frac{k_2}{dx_2}$$

$$\frac{0.40}{0.25} = \frac{0.16}{dx_2}$$

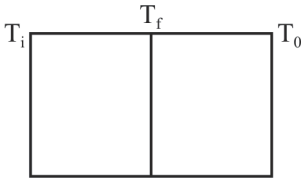
$$dx_2 = 0.1 \text{ m}$$

69. An iron sheet of 20 mm thickness is brought in contact with an aluminium sheet of 10 mm thickness. The temperature at the outer surface of the aluminium and iron is 100 °C and 0 °C respectively. Find the interface temperature if the ratio of thermal conductivity of aluminium and iron is 3 : 1:

(a) 80 °C (b) 45 °C
(c) 8.57 °C (d) 85.7 °C

AAI Jr. Executive 29.11.2018
DRDO Scientist 2009

Ans. (d) :



$$\frac{0 - T_f}{\frac{20\times10^{-3}}{1}} = \frac{T_f - 100}{\frac{10\times10^{-3}}{3}}$$

$$-T_f = 6T_f - 600$$

$$7T_f = 600$$

$$T_f = 85.7 \text{ °C}$$

70. In a furnace the wall thickness is 60 cm and is 100 cm wide and 150 cm high out of material with thermal conductivity of 0.4 W/m K. The temperature inside and outsides are 1000° and 4° C respectively. The thermal resistance is

(a) 1 K/W (b) 2 K/W
(c) 18 K/W (d) 15K/W

ISRO Scientist/Engineer (RAC), 10.03.2019
GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : Given, Length = Thickness of wall = 0.6 m

Thermal resistance $R_{th} = \frac{L}{kA}$

$$R_{th} = \frac{0.6}{0.4\times1\times1.5} \quad \therefore \text{ Area} = \text{Height} \times \text{Width}$$

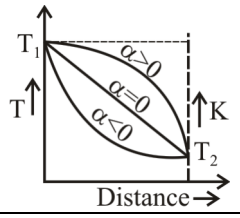
$$R_{th} = 1 \text{ K/W}$$

71. If the thermal conductivity of a material of wall varies as $k_0(1-\alpha_0)$, the temperature at the centre of the wall will be

(a) Some as in case of constant thermal conductivity
(b) More than that in case of constant thermal conductivity
(c) Less than that in case of constant thermal conductivity
(d) Temperature will depend on other factors

GPSC Asstt. Director of Transport 05.03.2017
ISRO Scientist/Engineer 2010

Ans. (c) : If the thermal conductivity of a material of wall varies as $k_o(1 - \alpha_o)$, the temperature at the centre of wall will be less than that in case of constant thermal conductivity.



72. It is required to insulate a kitchen oven with cork board (K=0.043 W/(m K)) so that the heat losses from the oven does not exceed 400 W/m^2 when the inner surface of the oven is at 225°C and the outer surface of the oven is at 40°C. The thickness of insulation required is nearly

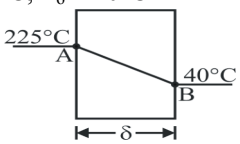
(a) 1 cm (b) 2 cm
(c) 3 cm (d) 4 cm

GPSC ARTO Pre 30.12.2018
ISRO Scientist/Engineer 03.07.2016

Ans. (b) : K = 0.043 W/m-K

$$\frac{Q}{A} = 400 \text{ W/m}^2$$

$T_i = 225°C$, $T_o = 40°C$



Applying Fourier Low–

$$\boxed{Q = kA\frac{dT}{dx}}$$

$$\frac{Q}{A} = k\frac{dT}{dx}$$

$$400 = \frac{0.043}{\delta}(225 - 40)$$

$$\delta = 0.01988 \text{ m}$$

$$\delta \approx 2 \text{ cm}$$

73. In the heat flow equation $Q = \frac{kA(t_1 - t_2)}{x}$, the term $\frac{t_1 - t_2}{x}$ is known as

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(a) Thermal resistance
(b) Thermal coefficient
(c) Temperature gradient
(d) Thermal conductivity

Nagaland PSC CTSE 2015 Paper-II
UKPSC AE 2007 Paper -II

Ans. (c) : According to Fourier's law of heat transfer by conduction-

$$Q = kA\frac{dT}{dx} \quad .....(1)$$

where, $\frac{dT}{dx}$ = temperature gradient

According to question $Q = \frac{kA(t_1 - t_2)}{x}$ .....(2)

Comparing equation (1) & (2)

$$\frac{dT}{dx} = \frac{t_1 - t_2}{x}$$

Term $\frac{t_1 - t_2}{x}$ is known as temperature gradient.

74. The heat flow equation $Q = \frac{kA(t_1 - t_2)}{x}$ the term $\frac{x}{kA}$ is known as

(a) Thermal resistance
(b) Thermal coefficient
(c) Thermal gradient
(d) Thermal conductivity
(e) None of the above

ISRO Scientist/Engineer (RAC) 07.05.2017
CGPSC Poly. Lect. 22.05.2016

Ans. (a) : According to Fourier's law heat transfer –

$$Q = kA\left(\frac{t_1 - t_2}{x}\right)$$

where,
Thermal resistance

$$(R_{th}) = \frac{x}{kA}$$

- Unit of 'R_{th}' is $\left(\frac{°C}{W}\right)$

75. The thermal resistance of a rectangular slab of thickness L, area A and thermal conductivity K is given as:

(a) K/(LA) (b) L/(AK)
(c) A/(LK) (d) 1/LAK

NLC GET 17.11.2020, Shift-II
NLCIL GET 17.11.2020, Shift-II

Ans. (b) : According to Fourier's law of heat conductor

$$Q = -KA\frac{dT}{dx} \quad \text{or} \quad Q = -KA\frac{(T_2 - T_1)}{L}$$

$$(T_1 - T_2) = Q \times \left[\frac{L}{KA}\right] \quad .....(i)$$

T_1 —Q→ —R_{th}→ T_2

$(T_1 - T_2) = Q \times R_{th}$ .....(ii)

Comparing both equation

$$R_{th} = \left[\frac{L}{KA}\right]$$

So, The thermal resistance of a rectangular slab of thickness L, area A and thermal conductivity K is given as L/(AK).

76. Film coefficient is defined as the ratio of :

(a) thermal conductivity to the equivalent thickness of the film of fluid
(b) temperature drop through the films of fluids to the thickness of film of fluid
(c) thickness of film of fluid to the thermal conductivity
(d) thickness of film of fluid to the temperature drop through the films of fluids

Punjab PSC SDE 12.02.2017
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. (a) : Film coefficient is defined as the ratio of thermal conductivity to the equivalent thickness of the film of fluid.

77. A solid sphere and a hollow sphere of the same material and size are heated to the same temperature and allowed to cool in the same surrounding. If the temperature difference between the body and that of the surroundings is T, then.

(a) Both spheres will cool at the same rate for small values of T
(b) Both spheres will cool at the same rate for small values of T
(c) The hollow sphere will cool at a faster rate for all the values of T
(d) The solid sphere will cool at a faster rate for all the values of T

GPSC ARTO Pre 30.12.2018, GATE 1992

Ans. (c) : The hollow sphere will cool at a faster rate for all values of T because hollow sphere has more area to transfer heat.

78. A metal rod of 2 cm diameter has conductivity of 40 W/mK, which is to be insulated with an insulating material of conductivity of 0.1 W/mK. If the convective heat transfer coefficient with the ambient atmospheric is 5 W/m^{2}K, the critical thickness of insulation will be –

(a) 1 cm (b) 2 cm
(c) 7 cm (d) 8 cm

TRB Asstt. Prof., 2012, ESE 2003

Ans. (a) : Given,
Diameter (d) = 2 cm
radius (r) = 1 cm
Thermal conductivity of metal = 40 W/mK
Thermal conductivity of insulating material = 0.1 W/mK
Heat Transfer co-efficient = 5W/m^{2}K

$$\text{Critical Radius } (r_c) = \frac{k}{h} = \frac{0.1}{5} = 0.02\text{m or 2 cm}$$

Critical thickness (t_c) = r_c – r = 2 – 1 = 1 cm

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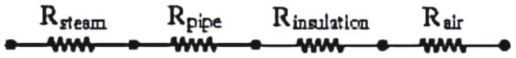
79. Heat pipe is widely used now-a-days because it acts as

(a) an insulator
(b) conductor and insulator
(c) a superconductor
(d) a fin

TRB Asstt. Prof., 2012, ESE 1995

Ans. (c) : Heat pipe is widely used now-a-days because it acts as superconductor.

80. The pipe carrying saturated steam is covered with a layer of insulation and exposed to ambient air. The thermal resistance are as shown in figure.

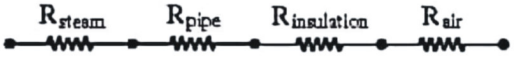


Which one of following statements is correct in this regard :

(a) R_{steam} and R_{pipe} negligible as compared to $R_{insulation}$ and R_{air}
(b) R_{pipe} and R_{air} are negligeable as compared to $R_{insulation}$ and R_{steam}
(c) R_{steam} and R_{air} are negligible as compared to R_{pipe} and $R_{insulation}$
(d) No quantitative data is provided, therefore no comparison is possible

KPSC ADF 2015, ESE 1996

Ans. (a) : The thermal resistance are as shown in figure



- The resistance due to steam film and pipe material are negligible in comparison to resistance of insulation material and resistance due to air film.

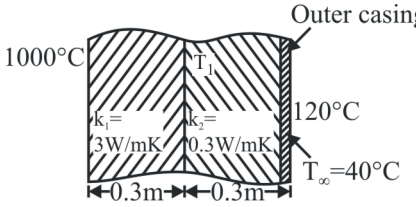
81. Heat conduction in gases is due to

(a) motion of electrons
(b) elastic impact of molecules
(c) mixing motion of the different layers of the gas
(d) electromagnetic waves

JPSC AE, 10.04.2021, Paper-I
UKPSC AE 2013, Paper-II

Ans. (b) : Heat conduction in gases is due to elastic impact of molecules.

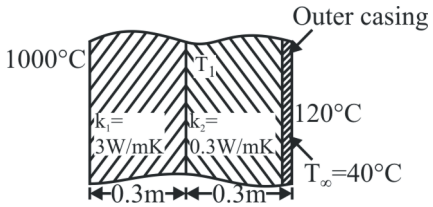
82. A furnace wall is constructed as shown in the figure. The interface temperature T_1 will be



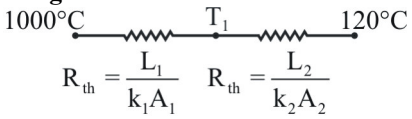
(a) 560°C (b) 200°C
(c) 920°C (d) 1120°C

ISRO Scientist/Engineer 11.10.2015
ESE 1998

Ans. (c) :



Network Diagram–



For steady state heat transfer

$$\frac{\Delta T_1}{\frac{L_1}{k_1 A_1}} = \frac{\Delta T_2}{\frac{L_2}{k_2 A_2}}$$

$$\frac{1000 - T_1}{\frac{0.3}{3}} = \frac{T_1 - 120}{\frac{0.3}{0.3}}$$

$$\frac{1000 - T_1}{0.1} = T_1 - 120$$

$$1000 + 12 = T_1 + 0.1\ T_1$$

$$T_1 = \frac{1012}{1.1} = 920°C$$

83. A spherical thermometer junction of diameter 0.706 mm is to be used for the measurement of temperature of a gas stream. The convective heat transfer co-efficient on the bead surface is 400 W/m^{2}K. Thermo-physical properties of thermocouple material are k = 20W/mK, C = 400 J/kg K and r = 8500 kg/m^3. If the thermocouple initially at 30°C is placed in a hot stream of 300°C, the time taken by the bead to reach 298°C is

(a) 2.35 s (b) 4.9 s
(c) 14.7 s (d) 29.4 s

ISRO Scientist/Engineer 12.05.2013
GATE 2004

Ans. (b) : Given,

$h = 400\ W/m^2K$
$d = 0.706\ mm$
$k = 20\ W/mK$
$C = 400\ J/kgK$
$\rho = 8500\ kg/m^3$
$T = 298°C$
$T_i = 30°C$
$T_\infty = 300°C$

Characteristic length

$$L = \frac{V}{A}$$

$$L = \frac{\frac{4}{3}\pi r^3}{4\pi r^2} = \frac{r}{3} = \frac{d}{6}$$

$$L = \frac{0.706}{6} = 0.11767 mm$$

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Biot Number

$$Bi = \frac{hL}{k}$$

$$= \frac{400 \times 0.11767 \times 10^{-3}}{20}$$

$$= 2.3533 \times 10^{-3} << 1$$

$$\frac{T - T_\infty}{T_i - T_\infty} = exp\left[\frac{-hA}{\rho VC}t\right]$$

or $$exp\left[\frac{-hA}{\rho VC}t\right] = \frac{T - T_\infty}{T_i - T_\infty}$$

$$exp\left[\frac{-hA}{\rho VC}t\right] = \frac{298 - 300}{30 - 300}$$

$$= \frac{2}{270} = 7.4074 \times 10^{-3}$$

or $$\frac{-hA}{\rho VC}t = -4.9053$$

or $$t = 4.9053 \times \frac{V}{A} \times \frac{\rho C}{h}$$

$$= 4.9053 \times 0.11767 \times 10^{-3} \times \frac{8500 \times 400}{400}$$

$$= 4.9062 \text{ s.}$$

84. The heat flow rate through parallel walls of thickness L_1, L_2 and L_3 and having surface areas A_1, A_2 and A_3, thermal conductivities k_1, k_2 and k_3, respectively and first and last walls maintained at temperature t_1 and t_2 will be

(a) $$\frac{t_1 - t_2}{\frac{L_1}{A_1k_1} + \frac{L_2}{A_2k_2} + \frac{L_3}{A_3k_3}}$$

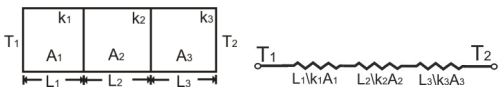
(b) $$\frac{t_1 - t_2}{\frac{k_1}{A_1L_1} + \frac{k_2}{A_2L_2} + \frac{k_3}{A_3L_3}}$$

(c) $$\frac{t_1 - t_2}{\frac{k_1A_1}{L_1} + \frac{k_2A_2}{L_2} + \frac{k_3A_3}{L_3}}$$

(d) $$\frac{t_1 - t_2}{\frac{L_1A_1}{k_1} + \frac{L_2A_2}{k_2} + \frac{L_3A_3}{k_3}}$$

ISRO Scientist/Engineer 2010, 2007

Ans. (a)



$$R_{th} = \frac{L_1}{k_1A_1} + \frac{L_2}{k_2A_2} + \frac{L_3}{k_3A_3}$$

$$Q = \frac{\Delta T}{R_{th}}$$

$$Q = \frac{T_1 - T_2}{\frac{L_1}{k_1A_1} + \frac{L_2}{k_2A_2} + \frac{L_3}{k_3A_3}}$$

85. In a furnace made of bricks,
Inside heat transfer coefficient = 35 W/m^{2}K
Outside heat transfer coefficient = 25 W/m^{2}K
Thermal conductivity of bricks (15 cm thick) = 0.15 W/mK
The overall heat transfer coefficient (in W/m^{2}K) will be closer to the

(a) Inside heat transfer coefficient
(b) Outside heat transfer coefficient
(c) Thermal conductivity of bricks
(d) Heat transfer coefficient based on the thermal conductivity of the bricks alone.

CGPSC AE 2014 -II
ISRO Scientist/Engineer 2009

Ans. (d) : Inside heat transfer h_i = 35 W/m^{2}K
Outside heat transfer coefficient = h_o = 25 W/m^{2}K
Thermal conductivity of bricks = (15 cm thick)
= 0.15 W/mK
Overall heat transfer Coefficient U

$$\frac{1}{U} = \frac{1}{h_{in}} + \frac{1}{h_o} + \frac{t}{k}$$

$$= \frac{1}{35} + \frac{1}{25} + \frac{15 \times 10^{-2}}{0.15}$$

$$= \frac{1}{35} + \frac{1}{25} + \frac{0.15}{0.15}$$

$$\frac{1}{U} = \frac{1}{35} + 1 + \frac{1}{25}$$

U = 0.9358 W/m^{2}K $\simeq 1 W/m^2K$

Heat transfer coefficient based on thermal conductivity of bricks alone,

$$\frac{1}{h} = \frac{t}{k} = \frac{0.15}{0.15}$$

h = 1 W/m^{2}K

This value is close to overall heat transfer coefficient (U).

86. A spherical shaped vessel of 1.4m outer diameter is 90 mm thick. Find the rate of heat leakage, if the temperature difference between the inner and outer surfaces is 220ºC. Thermal conductivity of the material of the sphere is 0.083 W/mK.

(a) 0.2 kW (b) 0.5 kW
(c) 1.088 kW (d) 1.6 kW

ISRO Scientist/Engineer 2008, 2007

Ans. (c) : r_2 = 0.7 m
r_1 = (0.7 - 0.09)m
= 0.61 m
dT = 220ºC
k = 0.083 W/mk
Rate of heat leakage

$$Q = \frac{dT}{R}$$

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For spherical,

$$R_{th} = \frac{r_2 - r_1}{4\pi k r_1 r_2}$$

$$= \frac{(0.7) - (0.61)}{(4\pi \times 0.083 \times 0.7 \times 0.61)}$$

$$R_{th} = 0.2020$$

$$Q = \frac{220}{0.2020}$$

$$= 1.088 \text{ kW}$$

87. The temperature distribution at a certain input of time in concrete slab during occurring is given by $T = 3x^2 + 3x + 16$ where x is in cm and T is in K. The rate of change of temperature with time is given (α = 0.003 cm^2/sec)

(a) 0.009 K/sec (b) 0.0048 K/sec
(c) -0.0012 K/sec (d) -0.0018 K/sec

ISRO Scientist/Engineer 2007, ESE 2010

Ans. (d) : $T = 3x^2 + 3x + 16$

$\alpha = 0.003$ cm^2/sec

$$\frac{dT}{dt} = ?$$

From, $$\frac{d^2T}{dx^2} = \frac{1}{\alpha}\frac{dT}{dt} \quad ...(1)$$

$T = 3x^2 + 3x + 16$

$$\frac{dT}{dx} = 6x + 3$$

$$\frac{d^2T}{dx^2} = 6$$

From equation (1)

$$6 = \frac{1}{0.003}\frac{dT}{dt}$$

$$\frac{dT}{dt} = 0.0018\,K/s$$

88. A composite wall of a furnace has 2 layers of equal thickness having thermal conductivities in the ratio of 3 : 2. What is the ratio of the temperature drop across the two layers?

(a) 2 : 3 (b) 3 : 2
(c) 1 : 2 (d) $\log e^2 : \log e^3$

TNPSC AE 2019
Kerala PSC Poly. Lect. 09.01.2014
ESE 2008

Ans. (a) : Given, $k_1:k_2 = 3:2$

$$k_1 : k_2 = \frac{1}{\left(\frac{dT}{dx}\right)_1} : \frac{1}{\left(\frac{dT}{dx}\right)_2}$$

$$\left(\frac{dT}{dx}\right)_1 : \left(\frac{dT}{dx}\right)_2 = \frac{1}{k_1} : \frac{1}{k_2}$$

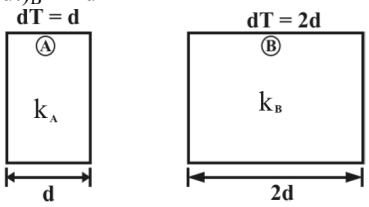
$$= 2 : 3$$

89. Two walls of thickness 'd' and '2d' called A and B are made of materials such that their thermal conductivities are $K_A = 2K_B$. If the difference of temperature on two sides is proportional to thickness the ratio of heat transfer through A to that through B is–

(a) 6 (b) 4
(c) 2 (d) 1

RPSC IOF & Boiler 20.09.2017
UKPSC AE 2012 Paper–II

Ans : (c) Given, $k_A = 2k_B$, $(dx)_A = d$, $(dx)_B = 2d$, $(dt)_A = d$, $(dt)_B = 2d$



Heat transfer through wall A (Q_A) $= k_A . A . \left(\frac{dT}{dx}\right)_A$

$$= 2k_B . A . \left(\frac{d}{d}\right) = 2\,k_B \,.\, A$$

Heat transfer through wall B (Q_B) $= k_B . A . \left(\frac{dT}{dx}\right)_B$

$$= k_B . A . \left(\frac{2d}{2d}\right) = k_B \,.\, A$$

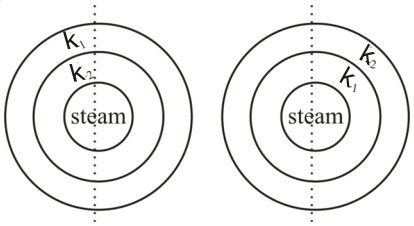
$$\frac{Q_A}{Q_B} = \frac{2k_B . A}{k_B . A} = 2$$

90. A steam pipe is to be insulated by two insulating materials put over each other. For best results

(a) better insulation should be put over pipe and better one over it
(b) inferior insulation should be put over pipe and better one over it
(c) both may be put in any order
(d) Whether to put inferior one over pipe or the better one would depend on steam temperature

ISRO Scientist/Engineer 22.04.2018
BPSC AE 2012 Paper - V

Ans : (a) :



$k_1 < k_2$ [k_1 better insulator]

$$Q = kA\frac{dT}{dx}$$

This means that the material having better insulating material should be inside to minimize the heat flow hence if better insulating material is placed inside it will reduce the heat transfer.

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91. The temperature distribution at a certain instant of time in a slab during a process is given by $T = 2x^2 + x + 5$, where x is in cm and T is in K. If the thermal diffusivity is 0.0002 cm^2/s, the rate of change of temperature with time is given by:

(a) 0.0008 K/s (b) 0.0004 K/s
(c) -0.0004 K/s (d) -0.0008 K/s

CIL MT 26.03.2017, ESE 1994

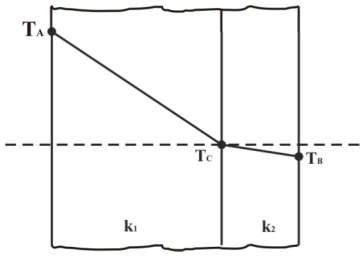
Ans. (a) : $T = 2x^2 + x + 5$

General equation without generation (q = 0) and in x-direction only,

$$\frac{1}{\alpha}\frac{\partial T}{\partial t} = \frac{\partial^2 T}{\partial x^2}$$

$$\frac{\partial T}{\partial t} = 4 \times \alpha = 4 \times 0.0002 = 0.0008 \text{ K/s}$$

92. The temperature variation under steady heat conduction across a composite slab of two materials of conductivities k_1 and k_2 is shown in the figure. Then which one of the following statements holds?

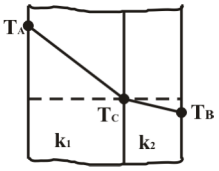


(a) $k_1 > k_2$ (b) $k_1 = k_2$
(c) $k_1 = 0$ (d) $k_1 < k_2$

RPSC AE 2018, GATE 1998

Ans. (d) : We know that

$$Q = -KA\frac{dT}{dx}$$



$$K \propto \frac{1}{\left(\frac{dT}{dx}\right)}$$

materials 1 has more temperature gradient than material 2.
So,

$K_2 > K_1$

93. Three dimensional steady state heat conduction equation with internal heat generation and constant thermal conductivity is known as

(a) Laplace equation (b) Poisson equation
(c) Fourier equation (d) Diffusion equation

RPSC LECTURER 16.01.2016
RPSC AEN Pre-2013

Ans. (b) : General heat conduction equation in cartesian coordinates—

$$\frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2} + \frac{q_{gen}}{k} = \frac{\rho C}{k}\cdot\frac{\partial t}{\partial \tau} = \frac{1}{\alpha}\frac{\partial t}{\partial \tau} \quad ...(1)$$

1. For the case when no internal source of heat is present

$$q_{gen} = 0$$

$$\frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2} = \frac{1}{\alpha} \times \frac{\partial t}{\partial \tau} \quad ...(2)$$

(called Fourier's equation)

2. In steady state

$$\frac{\partial t}{\partial \tau} = 0$$

$$\frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2} + \frac{q_{gen}}{k} = 0 \quad ...(3)$$

(called Poisson Equation)

3. If no internal heat source and in steady state then

$$\frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2} = 0 \quad ...(4)$$

(called Laplace equation)

94. A pipe of 25 mm outer diameter carries steam. The heat transfer coefficient between the cylinder and surroundings is 5 W/m^2K. It is proposed to reduce the heat loss from the pipe by adding insulation having a thermal conductivity of 0.05 W/mK. Which one of the following statements is TRUE?

(a) The outer radius of the pipe is equal to the critical radius
(b) The outer radius of the pipe is less than the critical radius
(c) Adding the insulation will reduce the heat loss
(d) None of the above

GPSC Asstt. Director of Transport 05.03.2017
UP Jal Nigam AE 2016

Ans. (c) : given as,
D = 25 mm $\Rightarrow$ R = 12.5 mm
h = 5 W/m^2K
k = 0.05 W/mK
We know that
Critical radius for cylinder,

$$r_c = \frac{k}{h} = \frac{0.05 \times 1000}{5}$$

$$r_c = 10 \text{ mm}$$

if

$$r_c < R$$

So adding the insulation will reduce the heat loss.

95. For a current wire of 25 mm diameter exposed to air (h = 15 W/m^2 K), maximum heat dissipation occurs when thickness of insulation (k = 0.6 W/m K) is

(a) 20 mm (b) 25 mm
(c) 30 mm (d) 40 mm
(e) 50 mm

CGPSC AE 2014 -II, GATE 1996

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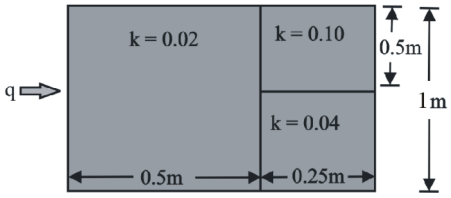
Ans. (d) : Maximum heat dissipation occurs upto critical Radius (R_c)

$$r_c = \frac{k}{h} = \frac{0.6}{15} \times 1000 \qquad \text{[for cylinder]}$$

$$r_c = 40\text{mm}$$

96. Heat flows through a composite wall, as shown in Figure. The depth of the slab is 1m. The values of k are in SI unit. The overall thermal resistance in K/W is



(a) 17.2 (b) 28.6
(c) 39.4 (d) 41.3
(e) 43.5

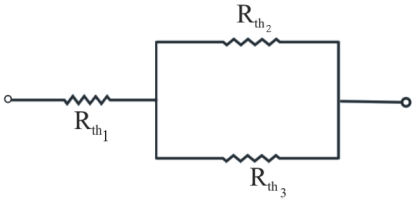
CGPSC AE 2014 -II, GATE 2005

Ans. (b) : $R_{th_1} = \frac{0.5}{0.02 \times 1}$

$R_{th_1} = 25$ K/W

$R_{th_2} = \frac{0.25}{0.10 \times 0.5 \times 1} = 5$ K/W

$R_{th_3} = \frac{0.25}{0.04 \times 0.5 \times 1} = 12.5$ K/W



Equivalent thermal resistance between R_2 and R_3 is

$$\frac{1}{R_{th}} = \frac{1}{R_{th_2}} + \frac{1}{R_{th_3}} = \frac{1}{12.5} + \frac{1}{5} = 0.28$$

$R_{th} = 3.57142$

Then total thermal resistance will be

$R_{th_{Total}} = R_{th_1} + R_{th} = 25 + 3.357142$

$= 28.57142$ K/W

97. Thermal diffusivity of a substance is inversely proportional to

(a) specific heat
(b) density of substance
(c) both (a) and (b)
(d) none of the above

UKPSC AE 2007 Paper -II, ESE 2006

Ans. (c) : Thermal diffusivity (α) is given by

$$\alpha = \frac{k}{\rho C_p}$$

$$\alpha \propto \frac{1}{\rho C_p}$$

$\alpha \propto k$

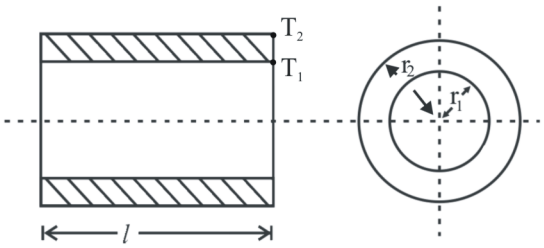
Hence, thermal diffusivity of a substance is inversely proportional to specific heat and density of substance.

98. If R_1 and R_2 are the inner and outer radii of a cylinder, the heat conduction through a cylinder is proportional to

(a) $(R_1 - R_2)$ (b) $(R_1 \times R_2)$
(c) $\frac{1}{(R_1 - R_2}$ (d) $\frac{1}{\log_e\left\{\frac{R_1}{R_2}\right\}}$

BPSC Poly. Lect. 2016, VIZAG STEEL MT 2015

Ans : (d) Radial heat transfer by Conduction through a thick cylinder:-



T_1 = Inside temperature of liquid
T_2 = Outside temperature of liquid
r_1 = Inside radius of the pipe
r_2= Outside radius of the pipe
(r_2-r_1) = Thickness of the pipe

$$Q = \frac{2\pi K \ell (T_1 - T_2)}{\ln\left(\frac{r_2}{r_1}\right)} \text{ or } \frac{2\pi K \ell (T_2 - T_1)}{\ln(r_1 / r_2)}$$

99. A wall of 0.6m thickness has normal area of $1.5m^2$ and is made up of material of thermal conductivity 0.4W/m K. If the temperatures on the two sides of the wall are 800°C and 1000°C, the thermal resistance of the wall is

(a) 1.8 K/W (b) 1.8 W/K
(c) 1 K/W (d) 1 W/K

UPPSC AE 13.12.2020, Paper-2
ESE 2017, 2007

Ans. (c) : Given, Thickness (L) = 0.6 m
Area (A) = 1.5 m^2
Thermal conductivity (k) = 0.4 W/mK
Temperature difference (ΔT) = 200°C

Thermal resistance $(R_{th}) = \frac{L}{kA}$

$$= \frac{0.6}{0.4 \times 1.5} = 1 \text{ K/W}$$

100. The overall heat transfer coefficient for a heat exchanger is the

(a) Sum of all conductance
(b) Sum of all resistances
(c) Sum of all convective coefficients
(d) Sum of thermal conductivities of all surfaces

Nagaland PSC CTSE 2017 Paper-2
CGPSC AE 2014 -II

Ans. (b) : Overall heat transfer coefficient for a heat exchanger is the Sum of all resistances.

$$\boxed{\frac{1}{U_0} = \frac{1}{h_1} + \frac{b_1}{k_1}}$$

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101. Cork is a good thermal insulator because

(a) Its density is low (b) It is porous

(c) It can be powdered (d) It is flexible

UPPSC AE 12.04.2016 Paper-II, ESE 2009

Ans : (b) Cork is a good thermal insulator because it is porous.

Porous material :- Porous materials are material with pores (cavities, channels or interstics). The characteristics of a porous material vary depending on the size, arrangement and shape of the pores, as well as the porosity (The ratio of the total pore volume relative to apparent volume of the material) and composition of material it self.

102. It is desired to increase the heat dissipation rate over the surface of an electronic device of spherical shape of 5 mm radius exposed to convection with h = W/m^2 K encasing it in a spherical sheath of conductivity 0.04 W/mK. For maximum heat flow, the diameter of the sheath should be

(a) 18 mm (b) 16 mm

(c) 12 mm (d) 8 mm

APPSC AEE 2012, ESE 1996

Ans. (b) : Data given h = 10 w/m^2, K = 0.4 w/mk

The critical radius of insulation for insuring max head transfer by conduction $(r_c) = \frac{2k}{h} = \frac{2\times 2.04}{10}$

r_c = 8 mm

Therefore = diameter should be 16 mm

103. An electric cable of Aluminum conductor of thermal conductivity 240 W/mK is to be insulated with rubber of thermal conductivity 0.15 W/mK. The cable is to be located in air of heat transfer coefficient of 6 W/m^{2}K. The critical thickness of insulation will be

(a) 25 mm (b) 40 mm

(c) 150 mm (d) 160 mm

ISRO Scientist/Engineer (RAC) 12.01.2020

ESE 1992

Ans. (a) : k_{Ae} = 240 W/mK

$k_{rubber} = k_{insulasion}$ = 0.15 W/mK

$h_{surroudning}$ = 6 W/m^{2}K

Critical radius of insulation for wire $= \frac{k_{insulasion}}{h_{surroudning}}$

$$r_c = \frac{0.15}{6} = 0.025\,m$$

Although this is critical radius of insulation, critical thickness

= Critical radius of thickness – Radius of wire

So, critical thickness will be less than 25 cm as no option is less than 25 so most appropriate is option (a).

104. Usually fins are provided to increase the rate of heat transfer. But fins also acts as insulation. Which one of the following non-dimensional number decides this factor?

(a) Eckert number (b) Biot number

(c) Fourier number (d) Peclet number

Vizag Steel MT (Re-Exam) 24.11.2013

ESE 2007

Ans. (b) : For long fins, the rate of heat loss from the fin is given by $\theta = \sqrt{hpkA}\times\theta_o = kA\sqrt{\frac{hp}{kA}}\,\theta_o = kAm\theta_o$

When $\frac{h}{mk} = 1$ or h = mk

$Q = hA\theta_o$

Which is equal to the heat loss from the primary surface with no extended surface.

Thus, when h = mk, an extended surface will not increase the heat transfer rate from the primary surface whatever be the length.

When $\frac{h}{mk} > 1$, Q < $hA\theta_o$ and hence adding secondary surfaces reduces the heat transfer and the added surface will act as insulation.

When $\frac{h}{mk} < 1, Q > hA\theta_o$ and the hence extended surface will increase the heat transfer.

- The term 'm' can also remodified and rewritten as 'Bi number', it can give an idea of the fin as an insulator.

105. A furnace is made of a red brick wall of thickness 1 m and conductivity 0.8 W/mK. For the same heat-loss and temperature drop, this can be replaced by a layer of diatomite earth of conductivity 0.16 W/mK and thickness.

(a) 1 m (b) 0.5 m

(c) 5 m (d) 0.2 m

GPSC ARTO 01.05.2016, ESE 1993

Ans. (d) :

Brick wall thickness = 1m, $K_{brick\ wall}$ = 0.8 W/mK

$K_{diatomite\ earth\ surface}$ = 0.16 W/mK, $dx_{Diatomite\ e.s}$ = ?

Heat loss and temperature drop are equal in both cases and assuming cross-section also same. Heat loss for conduction $Q = \frac{kAdT}{dx}$

$$Q = \frac{k_1AdT}{dx_1} = \frac{k_2AdT}{dx_2}$$

$$dx_2 = \frac{k_2}{k_1}\times dx_1 = \frac{0.16}{0.8}\times 1 = 0.2$$

Hence thickness as the diatomite earth surface will be 0.2 m.

106. For maximum heat dissipation in fins, the geometry of fins should have the lateral surfaces:

(a) maximum near the root

(b) maximum near the tip

(c) constant throughout

(d) minimum near the centre

APPSC Poly Lect. Automobile Engg., 2020

ESE 2005

Ans. (a) : For maximum heat dissipation in fins the geometry of fins should have the lateral surfaces fins are so designed that lateral surfaces at the root side of the fin is maximum. This aspect results into higher heat dissipation.

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107. A fin will be necessary and effective only when (Where k = thermal conductivity of fin material, h = convective heat transfer coefficient between the fin surface and environment temperature.)

(a) k is small and h is large
(b) k is large and h is also large
(c) k is small and h is also small
(d) k is large and h is small

GPSC EE Pre, 28.01.2017, ESE 2012

Ans. (d) : Effectiveness of fin–

$$\varepsilon_{fin} = \frac{\tan h\, mL}{\sqrt{\frac{hA}{kP}}}$$

$$\varepsilon_{fin} = \sqrt{\frac{kP}{hA}} \times \tan hmL$$

$$\varepsilon_{fin} = \propto \sqrt{\frac{kP}{hA}}$$

$$\varepsilon_{fin} \propto \sqrt{k}$$

$$\varepsilon_{fin} \propto \frac{1}{\sqrt{h}}$$

Fin is effective when k is large and h is small.

108. With an increase in the thickness of insulation around a circular pipe, heat loss the surroundings due to

(a) Convection and conduction increases
(b) Convection and conduction decreases
(c) Convection increases, while that due to conduction decreases
(d) Convection decreases, while that due to conduction increases

GPSC Executive Engineer 23.12.2018
ISRO Scientist/Engineer (RAC) 07.05.2017
GATE 2006

Ans. (c) : Heat loss to surrounding in a circular pipe,

$$Q = \frac{2\pi L(\Delta T)}{\frac{1}{Lr_o} + \frac{\ln[r_o/r_i]}{K}}$$

The addition of insulation to a cylindrical piece or spherical shell increase the conduction resistance of the insulation layer but decrease the convection resistance of the surface because of the increase in the outer surface area for convection. The heat transfer from the pipe may increase or decrease, depending on which effect dominates.

109. The heat transferred by conduction, convection and radiation is:

(a) Melting of ice
(b) Boiler furnaces
(c) Condensation of steam in condenser
(d) None of the above

HPPSC Asstt. Prof. 18.09.2017, ESE 1998

Ans. (b)

110. Which of the following is a case of steady state heat transfer?

(a) I.C. engine
(b) Air preheaters
(c) Heating of building in winter
(d) None of these

OPSC AEE 2019 PAPER - II
RPSC AE GWD, 2011

Ans : (d) : I.C. engine, air preheaters and heating of building in winter are the example of transient heat conduction. whereas in the case of steady state heat transfer not a function of time it means temperatures will remain constant.

111. Two insulating materials of thermal conductivity K and 3K are available for lagging a pipe carrying a hot fluid. If the radial thickness of each material is same,

(a) Material with higher thermal conductivity should be used for the inner layer and one with lower thermal conductivity for the outer
(b) Material with lower thermal conductivity should be used for the inner layer and one with higher thermal conductivity for the outer
(c) It is immaterial in which sequence insulating materials are used
(d) It is not possible to judge unless numerical value of dimensions are given

Gujarat PSC AE 2019
ISRO Scientist/Engineer 12.05.2013

Ans : (b) : Two insulating materials of thermal conductivity K and 3K are available for lagging a pipe carrying a hot fluid here the critical radius for insulation must be

(i) For first material $(r_c)_1 = \frac{k}{h}$

(ii) For second material $(r_c)_2 = \frac{3k}{h}$

It is apparent that $\boxed{(r_c)_2 = 3(r_c)_1}$

Hence, the first material is to be used inside and second material is to be used for outside as it has bigger critical radius. Hence, the correct option is B.

112. For a given amount of heat transfer rate during steady one-dimensional heat conduction through a plane wall – with constant thermal conductivity and no internal heat generation – the higher the value of thermal conductivity of the wall, the temperature gradient within the wall will be :

(a) higher
(b) lower
(c) same
(d) dependent on ambient conditions

APPSC Poly Lect. 13.03.2020, BHEL ET 2019

Ans. (b) : According to Fourier's law for conduction –

$$Q = -kA\left(\frac{dT}{dx}\right)$$

Where; k = thermal conductivity

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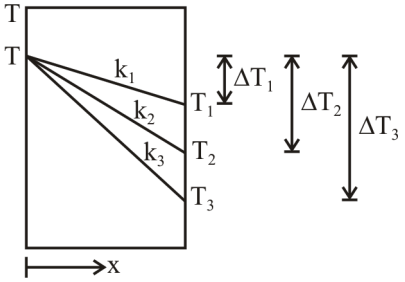
$$\frac{dT}{dx} = \text{temperature gradient}$$

$$\boxed{\frac{dT}{dx} \propto \frac{1}{k}}$$

If k increase then temperature gradient $\left(\frac{dT}{dx}\right)$ decreases.



$k_1 > k_2 > k_3$;

$$\frac{dT_1}{dx} < \frac{dT_2}{dx} < \frac{dT_3}{dx}$$

113. A copper plate of 2mm thickness is heated up to 400°C and then quenched into water at 20°C. Assuming h (heat transfer coefficient on the surface of the plate) = 80 W/m^2-K, plate dimension 30 cm × 30 cm, C_p (specific heat of copper) = 0.4 kJ/kg-K, ρ (density of copper) = 8800 kg/m^2, the time required for the plate to reach the temperature of 40°C will be :

(a) 1.29 minutes (b) 12.9 minutes
(c) 2 minutes 9 seconds (d) 12.9 seconds

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (c) : Given,

heat transfer coefficient of plate = 80 W/m^{2}k
density of copper = 8800 kg/m^3
specific heat of copper (C_p) = 0.4 kJ/kg-k
= 400 J/kg k

$$\boxed{\frac{T - T_\infty}{T_0 - T_\infty} = e^{-\left(\frac{hA}{\rho V C_p}\right)t}}$$

As copper plate quenched into water, the heat transfer takes place from both side surface,

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-\left(\frac{hA}{\rho V C_p}\right)t}$$

$$\frac{40 - 20}{400 - 20} = e^{-\left(\frac{80 \times 2 \times 10^3}{8800 \times 2 \times 400}\right)t}$$

Taking log both sides,

$$\ln\left(\frac{20}{380}\right) = -(0.02272)t$$

$$t = \frac{2.944}{0.02272} = 129.536 \text{ seconds}$$

$$\boxed{t = 2.158 \text{ minutes} = 2 \text{ min } 9 \text{ sec}}$$

114. A coolant fluid at 30° C flows over a heated flat plate maintained at a constant temperature of 100°C. The boundary layer temperature distribution at a given location on the plate may be approximated as T = 30 + 70 exp (–y), where y is the distance normal to the plate in meters and T is in °C. If thermal conductivity of the fluid is 1.0 W/mK, then the local convective heat transfer coefficient in W/m^{2}K at that location will be

(a) 0.2 (b) 1
(c) 5 (d) 10

ISRO Scientist/Engineer 11.10.2015
GATE 2009

Ans. (b) : Given,

$T_\infty = 30°C$, $T_w = 100°C$
$T = 30 + 70e^{-y}$
k = 1 W/mk
h = ?

Condition–

$$h(T_w - T_\infty) = -k\left.\frac{dT}{dy}\right|_{y=0}$$

$$h(T_w - T_\infty) = -1 \times \frac{d}{dy}(30 + 70e^{-y})\Big|_{y=0}$$

$$= -1\left[0 + 70 \times e^{-y} \times -1\right]\Big|_{y=0} = 70$$

$$h = +\frac{70}{T_w - T_\infty}$$

$$= \frac{70}{100 - 30} = \frac{70}{70} = 1 \text{ W/m}^2\text{K}$$

115. A solid copper ball of mass 500 gm when quenched in a water bath at 30^{0}C, cools from 530^{0}C to 430^{0}C in 10 sec. What will be the temperature of the ball after the next 10 seconds?

(a) 300^{0}C
(b) 320^{0}C
(c) 350^{0}C
(d) Cannot be determine

UPPSC AE 13.12.2020, Paper-2
RPSC ADE 2016, ESE 1997

Ans. (c) : Given,

m = 500 gm = 0.5 kg
$T_\infty = 30^0$C
T = 430^{0}C
$T_0 = 530^0$C

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-\left(\frac{hA}{\rho V C_p}\right)t}$$

$$\frac{430 - 30}{530 - 30} = e^{-\left(\frac{hA}{\rho V C_p}\right) \times 10} \quad (\therefore t = 10 \text{ seconds})$$

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Taking log both sides

$$0.223 = \left(\frac{hA}{\rho VC_p}\right)10 \Rightarrow \frac{hA}{\rho VC_p} = 0.0223$$

Now, after 20 seconds

$$\frac{T-30}{530-30} = e^{-\left(\frac{hA}{\rho VC_p}\right)\times 20}$$

$$\frac{T-30}{530-30} = e^{-(0.0223\times 20)}$$

$$\boxed{T = 350^0 C}$$

116. Biot number signifies

(a) the ratio of heat conducted to heat convicted
(b) the ratio of heat convicted to heat conducted
(c) the ratio of external convective resistance to internal conductive resistance
(d) the ratio of internal conductive resistance to external convective resistance

UPPSC AE 13.12.2020, Paper-2
GPSC Executive Engineer 23.12.2018
GATE 2014, 1991

Ans. (d)

$$\text{Biot No.} = \frac{\text{Internal Conductive Resistance}}{\text{External Convective Resistance}}$$

117. A steam pipe is to be lined with two layers of insulating materials of different thermal conductivities. For the minimum heat transfer

(a) the better insulation must be put inside
(b) the better insulation must be put outside
(c) one could place either insulation on either side
(d) one should take into account the steam temperature before deciding as to which insulation is put where

Nagaland PSC CTSE 2015 Paper-II
UKPSC AE 2007 Paper -II
ESE 1992, 1994, 1997

Ans. (a) : A steam pipe is to be lined with two layers of insulating materials of different thermal conductivities for the minimum heat transfer, the better insulation must be put inside.

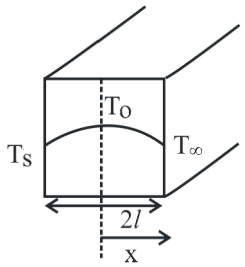
118. A plane wall of thickness 2L has a uniform volumetric heat sources $q^*(W/m^3)$. It is exposed to local ambient temperature T_a at both the ends ($x = \pm L$). The surface temperature T_s of the wall under steady-state condition (where h and k have their usual meanings) is given by :

(a) $T_s = T_\infty + \frac{q^*L}{h}$ (b) $T_s = T_\infty + \frac{q^*L^2}{h}$

(c) $T_s = T_\infty + \frac{q^*L^2}{h}$ (d) $T_s = T_\infty + \frac{q^*L^3}{2k}$

RPSC ADE 2016, ESE 2001

Ans. (a) :



q^* = uniform volumetric heat source (W/m^3)
T_∞ = Local ambient temperature
T_s = Surface temperature
T_0 = Temperature generated in slab
h = convection heat transfer coefficient

$$T_s - T_\infty = \frac{q^*L^2}{2k} = \left[1-\left(\frac{x}{L}\right)^2\right] + \frac{q^*L}{h}$$

at x = L

$$T_s - T_\infty = \frac{q^*L}{h}$$

$$T_s = T_\infty + \frac{q^*L}{h}$$

119. Determine the heat transfer through a plane of length 4m, height 3m and thickness 0.2m. The temperature of inner and outer surfaces are 150ºC and 90ºC respectively. Thermal Conductivity of the wall is 0.5 W/mK.

(a) 1800W (b) 2000W
(c) 2200W (d) 2400W

RPSC VPITI 14.02.2016, ESE 2015

Ans. (a) : Heat transfer through plane wall

$$Q = \frac{kA\Delta T}{L}$$

$$= \frac{0.5\times 4\times 3[150^\circ - 90^\circ]}{0.2}$$

Q = 1800W

120. A 10 mm diameter electrical conductor is covered by an insulation of 2mm thickness. The conductivity of the insulation is 0.8 W/mK and the convection coefficient at the insulation surface is 10 W/m^2K. Addition of further insulation of the same materials will be :

(a) Increase heat loss continuously
(b) Decrease heat loss continuously
(c) Increase heat loss to a maximum and then decrease heat loss
(d) Decrease heat loss to a minimum and then increase heat loss.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
GATE 2015

Ans. (c) : Given,
d = 10 mm
thickness of insulation = 2mm
k = 0.8 W/mK

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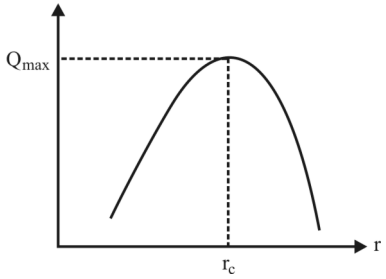
$h = 10\ W/m^2K$

$$r_c = \frac{k}{h} = \frac{0.8}{10} = 0.08m. = 80mm$$



So, If more insulation is added the heat loss increases upto maximum and then decreases.

121. The heat flow equation through a sphere of inner radius r_1 and outer radius r_2 is to be written in the same form as that for heat flow through a plane wall. For wall thickness ($r_2 - r_1$), the equivalent mean radius for the spherical shell is

(a) $\frac{r_1 + r_2}{2}$ (b) $r_1 r_2$

(c) $\sqrt{r_1 r_2}$ (d) $\frac{r_1 + r_2}{\log_e (r_2 / r_1)}$

RPSC IOF, 2020, ESE 2004

Ans. (c) : The heat flow equation through a sphere of inner radius r_1 and outer radius r_2 is to written in the same form as that for heat flow through a plane wall.
For wall thickness ($r_2 - r_1$)
Equivalent mean radius for the spherical shell is equal to $\sqrt{r_1 r_2}$

122. Which substance has the maximum value of thermal conductivity?

(a) Glass (b) Steel
(c) Cork (d) Freon
(e) Silver

JPSC AE 10.04.2021 Paper-I
CGPSC ARTO & TSI 07.01.2017

Ans. (e) : Value of different thermal conductivity
Diamond = 2300 W/m°C
Silver = 430 W/m°C
Copper = 400 W/m°C
Gold = 320 W/m°C
Glass = 0.8 W/m°C
Steel = 14 W/m°C
Freon- 12= 0.0083 W/m°C

123. In descending order of magnitude, the thermal conductivity of (a) pure iron, (b) liquid water, (c) saturated water vapour and (d) aluminium can be arranged as

(a) a b c d (b) b c a d
(c) d a b c (d) d c b a

ISRO Scientist/Engineer 12.05.2013
GATE 2001

Ans. (c) : Thermal conductivity (k) = W/mK
In descending order—
Aluminium > Pure Iron > liquid water > Saturated water vapour.

124. Which of the following is the best thermal insulator material?

(a) Saw dust (b) Glass wool
(c) Cork (d) Asbestos sheet

JPSC AE, 10.04.2021, Paper-I
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b)

	Materials	**Thermal Conductivity**
1.	Saw Dust	0.324 w/mk
2.	Glass Wool	0.04 w/mk
3.	Cork	0.37 – 0.04 w/mk
4.	Asbestos shul	0.08 w/mk

125. Arrange the metals Aluminium, Copper, Iron & Silver in increasing order of thermal conductivity

(a) Iron, Aluminium, Copper, Silver
(b) Aluminium, Iron, Copper, Silver
(c) Copper, Iron, Silver, Aluminium
(d) Iron, Copper, Aluminium, Silver

ISRO Scientist/Engineer 17.12.2017
UKPSC AE 2012 Paper-II

Ans. (a) : Increasing order of thermal conductivity
Iron < Aluminium < Copper < Silver

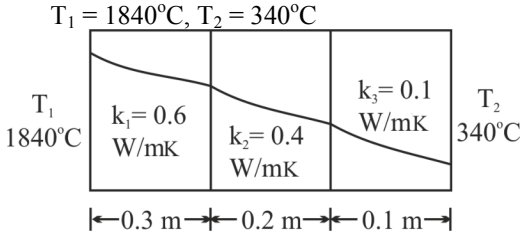
Metal	**Thermal Conductivity**
Silver	410
Copper	385
Aluminium	210
Iron	55-35

126. A composite wall of surface area 1 m^2 has three layers of thickness 0.3 m, 0.2 m and 0.1 m has thermal conductivities 0.6, 0.4 and 0.1 W/m. ^{0}C, respectively. There is no generation of thermal energy within the wall. If the inner and outer temperature of the composite wall are 1840^0 C and 340^0 C, respectively, the rate of heat transfer through this wall is :

(a) 0.75 kW (b) 0.150 kW
(c) 7.5 kW (d) 1.5 kW

BHEL ET 2019, ESE 2007

Ans. (a) : Given - Area = 1 m^2
Three layer of thickness
$l_1 = 0.3$ m
$l_2 = 0.2$ m
$l_3 = 0.1$ m
thermal conductivities = $k_1 = 0.6$ W/mK
$k_2 = 0.4$ W/mK
$k_3 = 0.1$ W/mK
$T_1 = 1840^oC$, $T_2 = 340^oC$



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Rate of heat transfer $Q = \dfrac{T_1 - T_2}{\dfrac{L_1}{k_1 A} + \dfrac{L_2}{k_2 A} + \dfrac{L_3}{k_3 A}}$

Equivalent thermal resistance R_{Th}

$$= \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A}$$

$$= \frac{0.3}{0.6 \times 1} + \frac{0.2}{0.4 \times 1} + \frac{0.1}{0.1 \times 1}$$

$$= 0.5 + 0.5 + 1 = 2$$

Heat transfer $(Q) = \dfrac{1840 - 340}{2} = \dfrac{1500}{2}$

$= 750\ W = 0.75\ kW$

127. If the radius of any current carrying wire is less than the critical radius, then the addition of electrical insulation will enable the wire to carry a higher current because the :

(a) Thermal resistance of the insulator is reduced
(b) Thermal resistance of the insulator is increased
(c) Heat loss from the wire is reduced
(d) Heat loss from the wire is increased

UPPSC AE 12.04.2016 Paper- II
OPSC AEE 2015 PAPER - II
ESE 2011

Ans. : (d) If the radius of any current carrying wire is less than the critical radius then the addition of electrical insulation will enable the wire to carry a higher current because the heat loss from the wire is increase.

128. What happens when the thickness of insulation on a pipe exceeds the critical value?

(a) Heat transfer rate increases
(b) Heat transfer rate decreases
(c) Heat transfer rate remains constant
(d) None of these

UKPSC AE 2007 Paper- II
UPPSC AE 12.04.2016 Paper-II

Ans. : (b) Heat transfer rate decreases

129. For effective working of fins, the thickness of the fins should be:

(a) Large
(b) Small
(c) Unpredictable
(d) Thickness of fin doesn't affect fin effectiveness

OPSC AEE 2019 PAPER - II
APPSC AEE MAINS 2016 PAPER-III

Ans. : (b) : Effectiveness 'ε'

$$\varepsilon = \sqrt{\frac{k.P.}{h.A_c}}$$

For, higher 'ε', 'h' and 'A_c' should be low 'k', and 'P' should be high. For low 'A_c' and high 'P', the fins should be thin.

130. In the process of heat transfer through extended surfaces or fins, the entire surface area is at

(a) the same constant temperature
(b) different temperatures
(c) maximum base temperature
(d) minimum temperature

OPSC AEE 2019 Paper - II
Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (b) : In the process of heat transfer through extended surfaces or fins, the entire surface area is at different temperature.

Heat transfer, $Q_{fin} = \sqrt{hPkA}\,(T_0 - T_\infty)$

131. The efficiency of a pin fin with insulated tip is:

(a) $\dfrac{\tan h mL}{\sqrt{hA/kP}}$ (b) $\dfrac{\tan h mL}{mL}$

(c) $\dfrac{mL}{\tan h mL}$ (d) $\dfrac{\sqrt{hA/kP}}{\tan h mL}$

UPPSC AE 13.12.2020 Paper-2
TRB Polytechnic Lecturer 2017
ESE 2001

Ans. (b) : Efficiency of Fin–

$$\eta_{fin} = \frac{\text{Actual heat transfered by the fin } (Q_{fin})}{\text{Maximum heat transfered by the fin if the entire surface of fin at base temperature}}$$

For a fin which insulated at the tip–

$$\eta_{fin} = \frac{\sqrt{Phk A_{cs}}\,(T_o - T_\infty) \tan h\, mL_c}{hPL_c (T_o - T_\infty)}$$

$$\eta_{fin} = \frac{\tan h (mL_c)}{mL_c} \quad \text{Where, } m = \sqrt{\frac{hP}{kA}}$$

Where, L_c = Corrected length

For rectangular fin, $L_c = \left(L + \dfrac{t}{2}\right)$

For pin fin, $L_c = \left(L + \dfrac{d}{4}\right)$

132. The purpose of using fins in a particular heat transfer system is

(a) to decrease rate of heat transfer
(b) to increase rate of heat transfer
(c) to maintain rate of heat transfer at a constant rate
(d) cannot say

GWSSB DEE Class-2 (GWSSB) 04.07.2021
Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : Fins are surfaces that extended from a object to increase the rate of heat transfer to on from the environment by increasing convection.

133. Log mean area 'A' can be given as

(a) $\dfrac{\log A_2 - \log A_1}{A_2 - A_1}$ (b) $\dfrac{A_2 - A_1}{\log A_2 - \log A_1}$

(c) $\dfrac{\log A_2 - \log A_1}{\log (A_2 / A_1)}$ (d) None

ISRO Scientist/Engineer 2012, ESE 1999

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Ans. (b) : Logarithmic mean Area 'A'.

$$A_m = \frac{A_2 - A_1}{ln\left(\frac{A_2}{A_1}\right)}$$

134. The unit of overall coefficient of heat transfer is:

(a) W/m^2K (b) W/m^2

(c) W/mK (d) W/m

ISRO Scientist/Engineer (RAC) 22.04.2018

Nagaland PSC (CTSE) 2018 Paper-II

Ans. (a) : We know that

$$Q = UA\Delta T$$

So, $$U = \frac{Q}{A\Delta T} = \frac{W}{m^2K}$$

where U is overall heat transfer coefficient of heat transfer.

135. A thermocouple in a thermowell measures the temperature of hot gas flowing through the pipe. For the most accurate measurement of temperature the thermowell should be made of

(a) Copper (b) brass

(c) Steel (d) aluminium

UPRVUNL AE 21.08.2016

GPSC AE 26.07.2015

Ans. : (c) A thermocouple in a thermowell measures the temperature of hot gas flowing through the pipe. For the most accurate measurement of temperature the thermowell should be made of steel.

136. Lumped system analysis of transient heat conduction situation is valid when the biot number is–

(a) very small

(b) very large

(c) approximately one

(d) cannot say unless the Fourier

PPPSC Asst. Municipal Engineer 15.06.2021

RPSC IOF & Boiler 20.09.2017

Ans : (a) For Lumped capacity method to be applicable, B_i should be less than 0.1 i.e., $B_i < 0.1$.

137. In solving a lumped parameter problem, which one of the following pair of non-dimensional numbers is used?

(a) Grashoff number and Biot number

(b) Prandtl number and Nusselt number

(c) Biot number and Fourier number

(d) Fourier number and Reynolds number

AAI Junior Executive 26.03.2021

JWM 2017

Ans. (c) : Biot number and Fourier number used while solving a lumped parameter problem.

$$\text{Biot number}(Bi) = \frac{\text{Conduction resistance}}{\text{Convection resistance}} = \frac{h \times L_c}{k_{solid}}$$

$$\text{Fourer Number}(F_o) = \frac{\text{Heat conduction}}{\text{Rate of Heat storage}} = \frac{\alpha \times t}{L^2}$$

138. Fin ____ is termed as the ratio of heat transfer rate with fin to without fin.

(a) effectiveness (b) conductance

(c) efficiency (d) resistance

UPRVUNL AE 05.07.2021

Ans. (a) :

139. The outer surface of a long cylinder is maintained at constant temperature. The cylinder does not has any heat source. The temperature in the cylinder will :

(a) increase linearly with radius

(b) decrease linearly with radius

(c) be independent of radius

(d) vary logarithmically with radius

RPSC ACF & FRO, 26.02.2021

ESE 2000

Ans. (d) : In long cylinder the temperature will vary logarithmically with radius.

Heat transfer through cylinder

$$Q = \frac{2\pi kL(T_1 - T_2)}{\ln\left[\frac{r_0}{r_i}\right]}$$

Temperature distribution –

$$\boxed{\frac{T - T_1}{T_2 - T_1} = \frac{\ln\left[\frac{r}{r_1}\right]}{\ln\left[\frac{r_2}{r_1}\right]}}$$

140. For steady flow and constant value of conductivity, the temperature distribution for a hollow cylinder or radii r_1 and r_2 is

(a) Parabolic

(b) Linear

(c) Logarithmic function of radii

(d) Cubic

APGCL AM, 2021

Ans. (c) : For hollow cylinder, rate of heat transfer

$$(q) = \frac{2\pi kL(T_1 - T_2)}{\ln\left(\frac{r_2}{r_1}\right)}$$

so $$\boxed{(T_1 - T_2) \propto \ln\left(\frac{r_2}{r_1}\right)}$$

$$\Delta T = f\left\{\ln\left(\frac{r_2}{r_1}\right)\right\}$$

So we can clearly say that the temperature distribution is Logarithmic function of radii

141. In the process of heat transfer through fins, the entire surface area is considered at :

(a) Equal to fin base temperature

(b) Equal to surrounding temperature.

(c) Any temperature between base and surrounding temperature that remain constant throughout the length of fin.

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(d) Any temperature between base and surrounding temperature that varies throughout the length of fin.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : In the process of heat transfer through fins the entire surface area is considered at any temperature between base and surrounding temperature that varies throughout the length of fin.

142. The dimension of a wall are 5m long, 10 m wide & 0.25 m thick, made of material having thermal conductivity at 1 W/mK. The temperature of inner and outside wall are 25°C and 15°C respectively for 10 hrs. If cost of electricity is 10 Rs/kWh, the cost of heat loss will be in rupees.

(a) 20 (b) 200
(c) 10 (d) 100

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : Given

$L = 5m$, $b = 10m$
$h = 0.25m$, $k = 1W/mK$
$T_2 = 25°C$, $T_1 = 15°C$
Time = 10 hours, cost = 10 Rs/kWh.

$$Q = -k.A\frac{dT}{dx}$$

$$Q = 1\times(5\times10)\times\left[\frac{25-15}{0.25}\right]$$

$Q = 2000$ watt
$Q = 2$ kW

Energy = 2×10 [For 10 hours]
= 20 kWh

Cost of heat loss = energy × Rupees $\left[\text{kWh}\times\frac{\text{Rs}}{\text{kWh}}\right]$

$= 20 \times 10$
= 200 Rs.

143. The measure of heat conducted through a body relative to the rate of heat storage is called as :

(a) Thermal diffusivity (b) Biot number
(c) Fourier number (d) Nusselt number.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) : Fourier number is a measurement heat conduction through a body relative to heat stored. Thus a large value of Fourier number indicates faster propagation of heat through a body.

$$\text{Fourier number (Fo)} = \frac{\alpha t}{L^2} = \frac{Q_{\text{conducted}}}{Q_{\text{stored}}}$$

Fourier number represents how during the time the heat is transporting in the solid body from hot to cold spots and is the ratio of thermal diffusivity per rate of storage.

144. For which of these solid materials at room temperature, the thermal diffusivity is highest?

(a) Copper (b) Stainless Steel
(c) Alluminium (d) Glass

RPSC ACF & FRO, 26.02.2021

Ans. (a) :

Material	Thermal Diffusivity (m^2/sec)
Copper	1.15
Stainless steel	0.042
Aluminium	0.97
Glass	0.34

145. The condition to be satisfied in lumped heat transfer analysis is:

(a) Zero thermal conductivity
(b) Large value of prandtl number
(c) Infinite thermal conductivity
(d) Large value of biot number

VIZAG Steel MT 24.01.2021

Ans. (c) : The condition to be satisfied in lumped heat transfer analysis is infinite thermal conductivity.

146. The Laplace heat conduction equation satisfies the following condition:

(a) steady state without internal heat generation
(b) steady state with internal heat generation
(c) unsteady state with internal heat generation
(d) unsteady state without internal heat generation

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Ans. (a) : The Laplace heat conduction equation satisfies the steady state without thermal heat generation.

$$\boxed{\frac{\partial^2T}{\partial x^2}+\frac{\partial^2T}{\partial y^2}+\frac{\partial^2T}{\partial z^2}=0}$$

$$\Rightarrow \quad \boxed{\nabla^2T=0}$$

147. A metal ball of diameter 60 mm is initially at 220°C. The ball is suddenly cooled by an air jet of 20°C. The heat transfer coefficient is 200 W/m^2K and 9000 kg/m^3, respectively, The ball temperature in (°C) after 90 seconds will be approximately_____.

(a) 210 (b) 163
(c) 189 (d) 141

Assam Engg. College AP/Lect. 18.01.2021
GATE 2017

Ans. (d) : 141

$$\frac{Q}{Q_i} = e^{-\frac{hA_S}{\rho vc}\tau} \qquad C_p = 400$$

$$\frac{t-t_a}{t_i-t_a} = e^{-\frac{hA_S}{\rho vc}\tau}$$

$$\frac{t-20}{220-20} = e^{-\frac{200}{9000\times\left(\frac{0.060}{6}\right)\times400}\times90}$$

$t = 141.30°C$

$$\left(\frac{V}{A_s}=\frac{\frac{4}{3}\pi r^3}{4\pi r^2} = r/3 = d/6\right)$$

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148. Finned surfaces have improved rate of heat dissipation due to

(a) decrease in ambient temperature
(b) increase in the surface area exposed to the surrounding
(c) increase in the corrective film coefficient
(d) All of the above

JPSC AE, 10.04.2021, Paper-I
TNPSC AE (Industries) 09.06.2013
ESE 2008

Ans. (b) : Finned surfaces have improved rate of heat dissipation due to increase in the surface area exposed to the surrounding.

149. An increase in fin effectiveness is caused by high value of

(a) convective coefficient
(b) thermal conductivity
(c) circumference
(d) (b) and (c)

JPSC AE, 10.04.2021, Paper-I

Ans. (d) : Fin effectiveness $= \dfrac{q_{\text{with fin}}}{q_{\text{without fin}}}$

$= \sqrt{\dfrac{PK}{hA}} \times \tanh(mL)$

An increase in fin effectiveness is caused by high valve of –
(a) Thermal conductivity
(b) Circumference

150. What is the critical radius of insulation of asbestos (thermal conductivity = 0.17 $W.m^{-1}°C^{-1}$) surrounding a circular pipe and exposed to room air at 20°C with heat transfer coefficient 3 $W.m^{-2}°C^{-1}$?

(a) 7.21 cm (b) 6.37 cm
(c) 5.67 cm (d) 6.93 cm

ESE 18.07.2021

Ans. (c) : Given,

$h = 3\ Wm^{-2}°C^{-1}$

$k = 0.17\ Wm^{-1°}C^{-1}$

Critical radius of insulation $= \dfrac{k}{h}$

$= \left(\dfrac{0.17}{3} \times 1000\right) mm$

$= 56.67\ mm \simeq 5.667\ cm$

151. Which of the following CANNOT be considered as mode of heat transfer?

(a) Radiation (b) Convection
(c) Advection (d) Induction

AAI Jr. Executive 26.03.2021

Ans. (d) : Induction cannot be considered as mode of heat transfer.

152. For pipes, the radius of the pipe is taken higher than the critical radius, so that any insulation added will _____.

(a) keep loss unaltered
(b) decrease heat loss
(c) enable heat gain
(d) increase heat loss

UPRVUNL AE 05.07.2021

Ans. (b) :

153. Fins must be arranged ____ to the direction of fluid flow

(a) inclined at 60° (b) inclined at 30°
(c) parallel (d) perpendicular

UPRVUNL AE 05.07.2021

Ans. (d) :

154. The fin effectiveness can be enhanced by selecting _____ value of heat transfer co-efficient.

(a) low (b) zero
(c) high (d) positive

UPRVUNL AE 05.07.2021

Ans. (a) :

155. Heat conduction in solids in which the faster moving molecules or atoms in the hottest part of a body transfer heat by impact as _____

(a) lattice vibration
(b) asymptotic vibration
(c) Scotty vibration
(d) dissipation

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : Heat conduction in solids in which the faster moving molecules or atoms in the hottest part of a body transfer heat by impact as lattice vibration.

156. Heat transmission is directly linked with the transport of medium itself, i.e., there is actual motion of heated particles during

(a) Conduction only
(b) Convention only
(c) Radiation only
(d) Conduction as well as radiation

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Convention only

157. Which of the following is anisotropic, i.e., exhibits change in thermal conductivity due to directional preferences

(a) Wood (b) Glass wool
(c) Concrete (d) Masonary brick

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Wood

158. The heat loss from a fin is 6 W. The effectiveness and efficiency of the fin are 3 and 0.75, respectively. The heat loss (in W) from the fin, keeping the entire fin surface at base temperature is

(a) 2 W (b) 3 W
(c) 4 W (d) 8 W

PPSC Asstt. Municipal Engineer 15.06.2021

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Ans. (d) : Given, $Q_A = 6$ W, $\in = 3$, $\eta = 0.75$

We know that,

$$\eta = \frac{Q_A}{Q_{max}}$$

$$Q_{max} = \frac{Q_A}{\eta} = \frac{6}{0.75}$$

$$Q_{max} = 8 \text{ W}$$

159. Lumped system analysis of transient heat conduction situation is valid when the Biot number is

(a) very small.
(b) very large.
(c) approximately one.
(d) cannot say unless the Fourier number is also known.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : Lumped system analysis of transient heat conduction situation is valid when the Biot number is very small (less than 0.1).

160. A spherical steel ..... diameter is initially at 1000 K. It is slowly cooled in a surrounding of 300 K. The heat transfer coefficient between the steel ball and the surrounding is 5 W/m^2 K. The thermal conductivity of steel is 20 W/mK. The temperature difference between the centre and the surface of the steel ball is

(a) large because conduction resistance is far higher than the convective resistance.
(b) large because conduction resistance is far less than the convective resistance.
(c) small because conduction resistance is far higher than the convective resistance.
(d) small because conduction resistance is far less than the convective resistance.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d) : Small because conduction resistance is far less than the convective resistance.

161. In transient heat conduction systems, the degree of penetration of heating or cooling effect is represented by a non-dimensional number called _____.

(a) Prandtl number (b) Heisler Number
(c) Fourier Number (d) Biot Number

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : Fourier Number–In transient heat conduction systems, the degree of penetration of heating or cooling effect is represented by a non-dimensional number called Fourier number.

162. Steady state have three dimensional conductive heat transfer equation without internal heat generation is also called _____.

(a) Fourier Equation (b) Gauss Equation
(c) Poisson Equation (d) Laplace Equation

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : Steady state, 3-D conductive heat transfer equation without internal heat generation is called Laplace equation.

$$\boxed{\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0}$$

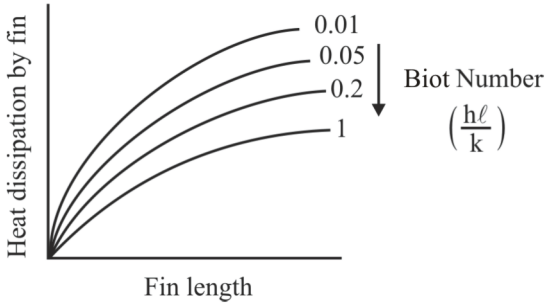
163. In case of fins, the values of the Biot number for effective functioning of fins is :

(a) zero (b) less than one
(c) equal to one (d) more than one

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The Biot number is the ratio of the internal resistance of a body to heat conduction to its external resistance to heat convection. Therefore a small Biot represents small resistance to heat conduction and thus small temperature gradients within the body.

We know that fin effectiveness is inversely proportional to Biot number. So a fin will be more effective when Biot number is less than 1



From the graph it is clear that when the Biot number increases. Heat dissipation by fin effectiveness decreases.

164. A wall of thickness 0.6 m and thermal conductivity 0.6 W/m-K has a surface area of 1 m^2. If the inner and outer temperature of the wall are 1840^{0}C and 340^{0}C, respectively, the rate of heat transfer will be:

(a) 150 W (b) 75 W
(c) 750 W (d) 1500 W

APPSC Poly Lect. 13.03.2020

Ans. (d) : Given; $T_i = 1840^0$C, $T_0 = 340^0$C, x = 0.6 m
k = 0.6 W/m-K, A = 1 m^2

$$\text{Rate of heat transfer (Q)} = -kA\frac{dT}{dx} = \frac{kA(T_i - T_o)}{x}$$

$$= \frac{0.6 \times 1(1840 - 340)}{0.6} = 1500$$

$$\Rightarrow \boxed{Q = 1500 \text{ Watt}}$$

165. During unsteady state conduction analysis, the characteristic length of cube of side L is taken as:

(a) L/4 (b) L/6
(c) L (d) 4L

NLC GET 17.11.2020, Shift-II

Ans. (b) : If the temperature of a body varies with time and body is said to be in the unsteady or transient heat conduction characteristic length for cube

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$$L_c = \frac{Volume}{Surface\ Area\ (A_s)} = \frac{L^3}{6L^2} = \frac{L}{6}$$

Characteristic length for sphere $L_c = \frac{R}{3}$

Characteristic length for solid cylinder $l_c = \frac{R}{2}$

Characteristic length for rectangular plate $l_c = \frac{lbt}{2lb} = \frac{t}{2}$

166. Which of the following is defined as the amount of heat transmitted for a unit temperature difference between the fluid and unit area of surface in unit time?

(a) Fourier law
(b) Coefficient of convective heat transfer
(c) Thermal conductivity
(d) Thermal resistance

VIZAG MT, 14.12.2020

Ans. (b) : In a fluid heat transfer in convection medium dominates.

Heat transfer in convection medium,

$Q = hA(T_s - T_f)$

If $A = 1$

$T_s - T_f = 1$

$\boxed{Q = h}$

Hence coefficient of convective heat transfer is defined as amount of heat transferred for a unit temperature difference between the fluid and unit area of surface in unit time.

167. Uniform heat generation takes place in a symmetric slab so that heat flows towards both side in contact with fluid. The zero gradient boundary condition $\frac{\partial T}{\partial x} = 0$ occurs at

(a) centre line of slab (b) left wall of slab
(c) right wall of slab (d) nowhere in slab

UPPSC AE 13.12.2020, Paper-2, ESE 2013

Ans. (a) : Uniform heat generation takes place in a symmetric slab. So that heat flows towards both side in contact with fluid. The zero gradient boundary condition $\frac{\partial T}{\partial x} = 0$ occurs at centre line of slab.

168. A flat plate has thickness 5 cm, thermal conductivity 1 W/mk, convective heat transfer coefficients on its two flat faces are 10 W/m^{2}K and 20 W/m^{2}K. The overall heat transfer coefficient for such a plate is

(a) 5 W/m^{2}K (b) 6.33 W/m^{2}K
(c) 20 W/m^{2}K (d) 30 W/m^{2}K

UPPSC AE 13.12.2020, Paper-2

Ans. (a) : Given,

$t = 5\ cm = 0.05\ m$

$k = 1\ W/m–K$

$h_i = 10\ W/m^2K$

$h_0 = 20\ W/m^2K$

Overall heat transfer coefficient,

$$\frac{1}{U} = \frac{1}{h_i} + \frac{L}{k} + \frac{1}{h_0} = \frac{1}{10} + \frac{0.05}{1} + \frac{1}{20} = \frac{1}{5}$$

$\boxed{U = 5W/m^2K}$

169. Heat is conducted through a 10 cm thick wall at the rate of 30 W/m^2, when the temperature difference across the wall is 10°C. What is the thermal conductivity of the wall material?

(a) 0.03 W/m-k (b) 0.3 W/m-k
(c) 3.0 W/m-k (d) 30.0 W/m-k

Assam PSC AE (PHED), 18.10.2020
ESE 2005

Ans. (b) : Given- $\Delta x = 10\ cm = 10\times10^{-2}\ m$

$Q = 30\ W/m^2$

$\Delta T = 10°C$

$k = ?$

$$Q = kA\frac{\Delta T}{\Delta x} \Rightarrow \frac{Q}{A} = k\frac{\Delta T}{\Delta x}$$

$$30 = k \times \frac{10}{10\times10^{-2}}$$

$k = 30 \times 10^{-2}$

$k = 0.30\ W/mk$

170. In the thermal conductivity is 0.5 W/mK and convective heat transfer coefficient is 8 W/m^{2}K, then the critical thickness of wire is about:

(a) 60 mm (b) 50 mm
(c) 10 mm (d) 80 mm

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : Thermal conductivity $k = 0.5\ W/mK$

Convective heat transfer coefficient (h) = 8 W/m^{2}K

Critical radius of wire

$$r_c = \frac{k}{h}$$

$$= \frac{0.5}{8} = 0.0625\,m$$

$r_c = 62.5\ mm$

$\boxed{r_c \approx 60\,mm}$

171. The effectiveness, $\in$, for a very long fin
(i) Increases with increasing length of the fin
(ii) Is equal to the efficiency when the surface area of the fin is equal to its cross-section area.
(iii) should always be greater than 1.
Of these statements:

(a) only (i) is correct
(b) only (i) and (ii) are correct
(c) only (i) and (iii) are correct
(d) (i), (ii) and (iii) are correct

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : We know the effectiveness of the fin can be written as

$$\in \propto \frac{1}{\sqrt{\frac{hA}{kP}}}$$

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The effectiveness of fin must be greater than 1.

Therefore; $\sqrt{\frac{hA}{kP}}$ must be less than 1 to increases the heat transfer.

172. A solid metallic plate with thermal conductivity of 200 W/m^{0}C and 100 mm thick is exposed to surface temperatures of 100 to 75^{0}C. The heat transfer rate per unit area is :

(a) 50 kW (b) 500 kW
(c) 250 kW (d) 100 kW

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : given that

$K = 200\ W/m^0C$

$\Delta t = 100 - 75 = 25^0C$

$\Delta x = 100\ mm$

$$Q = kA\frac{dt}{dx}$$

$$(Q/A) = k\frac{dt}{dx}$$

$$= \frac{200 \times 25^0C}{100} = 50kW$$

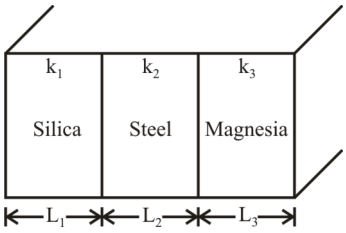
$= 50$ kW

173. A furnace wall made of steel plate 10 mm thick and thermal conductivity 15 kcal/m-hr-°C is lined inside with silica brick of 150 mm thick of thermal conductivity 1.75 kcal/m-hr-°C and on outside with magnesia bricks of 200 mm thick with thermal conductivity of 4.5 kcal/m-hr-°C. The total resistance of the composite wall will be

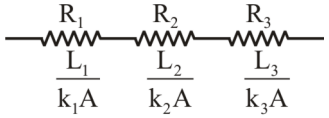
(a) 0.13 °C-hr/kcal (b) 0.23 °C-hr/kcal
(c) 0.03 °C-hr/kcal (d) 0.33 °C-hr/kcal

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) :



Resistance $(R) = \frac{L}{kA}$



Assume, $A = 1\ m^2$

For series, total resistance,

$$R_{total} = R_1 + R_2 + R_3$$

$$R_{total} = \frac{0.15}{1.75 \times 1} + \frac{0.01}{15 \times 1} + \frac{0.2}{4.5 \times 1}$$

$R_{total} = 0.13°$ C-hr/kcal.

174. A hot fluid is flowing through a long pipe of 4 cm outer diameter and covered with 2 cm thick insulation. It is proposed to reduce the conduction heat loss to the surroundings to one third of the present rate by increasing the same insulation thickness. The additional thickness of insulation required will be

(a) 2 cm (b) 6 cm
(c) 9 cm (d) 12 cm

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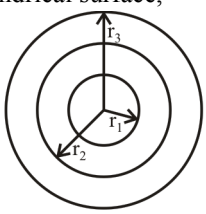
Ans. (d) : For cylindrical surface,



(Cross-section view of cylinder)

$$Q = \frac{\Delta T}{R_{th}} = \frac{2\pi k\ell \cdot \Delta T}{\ln(r_2/r_1)}$$

$$Q \propto \frac{1}{R_{th}}$$

$r_1 = 2$ cm, $r_2 = 2 + 2 = 4$ cm

Assume, r_3 radius after the additional thickness of insulation,

$$\frac{Q_2}{Q_1} = \frac{\ln(r_2/r_1)}{2\pi k\ell} \cdot \frac{2\pi k\ell}{\ln(r_3/r_1)}$$

$$\frac{Q_2}{Q_1} = \frac{1}{3}$$

$$\frac{1}{3} = \frac{\ln(r_2/r_1)}{\ln(r_3/r_1)}$$

$$\frac{r_3}{r_1} = \left(\frac{r_2}{r_1}\right)^3 = 2^3 = 8$$

$r_3 = 8 \times 2 = 16$ cm

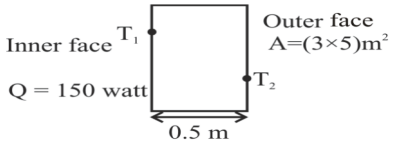
Increment, $r_3 - r_2 = 16 - 4 = 12$ cm

175. The external surface of a wall of 3 m height, 5 m width and 0.5 m thickness is at a temperature of 2ºC. If a heat loss of 150 W from the room is measured across the wall, find the inner wall temperature? The thermal conductivity of wall material can be taken as 1 W/m.K

(a) 280 K (b) 285 K
(c) 268 K (d) 282 K

ISRO Scientist/Engineer 12.01.2020

Ans. (a) :



$$Q = \frac{KA\Delta T}{\Delta x} \Rightarrow 150 = \frac{1 \times (3 \times 5)(T_1 - 2)}{0.5}$$

$T_1 = 7$ºC $= 280$ K

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176. A hot plate of 2 m × 2 m is maintained at 300°C. Air at 20°C flows over the plate. The convective heat transfer coefficient is 20 W/m^2 °C. The rate of heat transfer is ____.

(a) 89.6 kW (b) 11.2 kW
(c) 44.8 kW (d) 22.4 kW

VIZAG MT, 14.12.2020

Ans. (d) : $Q = hA\Delta T$
$= 20 \times 2 \times 2 \times (300 - 20)$
$= 22400$ Watt
$\boxed{Q = 22.4\,kW}$

177. Heat conduction does not occur

(a) if a physical body is impermeable to any kind of rays
(b) if the parts of a body are not in motion relative to one another
(c) if the bodies are kept in vaccum
(d) if the temperature difference between the bodies does not exist

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Heat conduction does not occur if the temperature difference between the bodies does not exist.

178. Which of the following relation is correct for fin effectiveness? [Where k = thermal conductivity of fin material, p = perimeter of the fin, A = cross sectional area of fin, h = convective heat transfer coefficient]

(a) $\sqrt{\frac{kA}{hp}}$ (b) $\sqrt{\frac{kph}{A}}$
(c) $\sqrt{\frac{kp}{hA}}$ (d) $\sqrt{\frac{kh}{pA}}$

SJVN ET 2019

Ans. (c) : $\varepsilon_{fin} = \frac{\tanh mL}{\sqrt{\frac{hA}{kP}}}$

Fin effectiveness = $\sqrt{\frac{kp}{hA}}$

Where,
k = Thermal conductivity
p = Perimeter
A = Cross section area of fin
h = Convective heat transfer coefficient

179. For an infinitely long fin, efficiency is given by

(a) $\frac{1}{mL}$ (b) $\frac{2}{mL}$
(c) $\frac{1}{2mL}$ (d) $\frac{3}{mL}$

JPSC AE PRE 2019

Ans. (a) : • Fin efficiency for long fin (infinite)

$$\eta_{\text{long fin(infinite)}} = \frac{q_{actual}}{q_{max}} = \frac{1}{mL}$$

• The efficiency of a fin with insulated tip is

$$\eta_{fin} = \frac{q_{actual}}{q_{max}} = \frac{\tanh(mL_c)}{mL_c}$$

Where, L_c = Corrected length

180. What is the affect of thermal conductivity k on fin effectiveness?

(a) Fin is effective for smaller value of thermal conductivity
(b) Fin is effective for larger value of thermal conductivity
(c) Thermal conductivity does not affect fin effectiveness
(d) Initially Increases and after attaining peak reduces

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Ans. : (b) : The effectiveness 'ε' of fins

$\varepsilon = \sqrt{\frac{k.P.}{h.A_c}}$

Where, k - Thermal conductivity
P – Perimeter of fin cross - section
h - Convective heat transfer coefficient
A_c - Cross-section of fin.
ε - For higher effectiveness

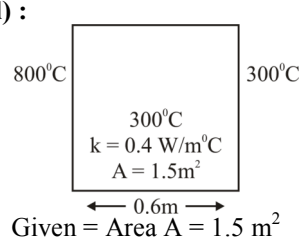
'k' Should be high and 'h' should be small

181. Consider steady-state heat conduction across the thickness in a plane wall of thickness 0.6 m. The wall has a normal area 1.5 m^2 and is made up of material of thermal conductivity 0.4 W/m^{0}C. There is no generation of thermal energy within the wall. The temperature on the two sides are 800^{o}C and 300^{o}C. The thermal resistance of the wall is :

(a) 1.5 W/^{o}C (b) 1 W/^{o}C
(c) 1.5 ^{o}C/W (d) 1 ^{o}C/W

BHEL ET 2019

Ans. (d) :



Given = Area A = 1.5 m^2
Thickness of wall = t = 0.6 m
Thermal conductivity (k) = 0.4 W/m^{0}C
$T_1 = 800^0$C, $T_2 = 300^0$C

Thermal resistance (R_{th}) = $\frac{t}{kA}$

$= \frac{0.6}{0.4 \times 1.5} = \frac{0.6}{0.6}$

$\boxed{R_{th} = 1^0 C/W}$

182. Conduction takes place due to temperature gradients:

(a) in a medium which is not vacuum
(b) in a fluid medium only
(c) in solid only
(d) only in stationary medium

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Ans. (a) : Conduction– Its the mode of heat transfer which generally occurs in solids, gas as liquid due to temperature difference associated with molecular lattices vibrational energy transfer and also by free electron transfer conduction take place due to medium but in vacuum not any medium presents.

183. Physical signification of the thermal diffusivity is :

(a) the ability of a substance to conduct relative to its ability to store thermal energy
(b) the ability of a substance to store thermal energy
(c) the ratio of rate of diffusion of momentum through a substance to the diffusion of heat in that substance
(d) the ability of a substance to conduct heat

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Ans. (a) : Thermal Diffusivity (α)– A thermo physical property of material as the ratio between thermal conductivity of material and its thermal capacity i.e.,

$$\alpha = \frac{k}{\rho C_p}$$

unit $\alpha = m^2 / sec.$

- Thermal diffusivity (α) of a medium signifies the ability of a material or medium to allow the heat energy to get diffused through the material more rapidly or quickly.

184. 'Fourier's law of heat conduction defines the rate of heat transfer through a body as :

(a) only dependent upon the material of the body
(b) proportional to the area normal to the direction of the heat flow and the temperature gradient along the direction of the heat flow, and is dependent upon the material of the body
(c) proportional only to the area normal to the direction of the heat flow
(d) proportional only to the temperature gradient along the direction of the heat flow

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Ans. (b) : According to Fourier's Law of Heat Conduction–Fourier's law is defined as head flow directly proportional to the area normal to the direction of the heat flow and the temperature gradient along the direction of the heat flow, and is dependent upon the material of the body.

$$Q \propto A$$

$$Q \propto A\frac{dT}{dx}$$

$$\boxed{Q = kA\frac{dT}{dx}}$$

Where, A = Area

$\frac{dT}{dx}$ = Temperature gradient

k = Thermal conductivity

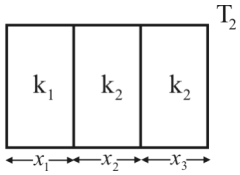
185. The overall heat transfer coefficient (U) for a composite wall of thickness x_1, x_2, x_3 and of corresponding thermal conductivities k_1, k_2, k_3 is given by equation.

(a) $1/U = k_1/x_1 + k_2/x_2 + k_3/x_3$
(b) $U = k_1x_1 + k_2x_2 + k_3x_3$
(c) $1/U = x_1/k_1 + x_2/k_2 + x_3/k_3$
(d) $1/U = x_1/k_1 + x_2/k_2 + k_3/x_3$

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) : $Q = UA\Delta T = \dfrac{A\Delta T}{\frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3}}$

$\boxed{\frac{1}{U} = \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3}}$ U = overall heat transfer coefficient



186. The rate of heat flow through a composite wall of three layers of thickness 0.3 m, 0.2 m, 0.15 m and of corresponding thermal conductivities 1.2, 0.8 and 0.6 kJ/hr-m°C is 1280 kJ/hr. If the surface area normal to the direction of heat flow is 1 sq. mtr. and inner surface temperature is 1000°C, then the interface temperature at the end of first layer will be :

(a) 700°C (b) 680°C
(c) 500°C (d) 360°C

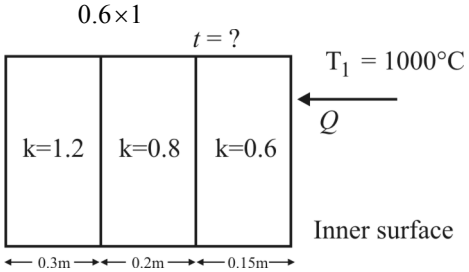
ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (b) : Given,

Area = $1m^2$

Q = 1280 kJ/hr

$$Q = \frac{\frac{T_1 - t}{0.15}}{0.6 \times 1}$$



$$1280 = \frac{\frac{1000 - t}{0.15}}{0.6}$$

$$320 = 1000 - t$$

$$\boxed{t = 680°C}$$

187. Thermal conductivity of non-metallic amorphous solids _____ with decrease in temperature.

(a) Increases
(b) Decreases
(c) Remains constant
(d) May increase or decrease depending on temperature

MECON MT 2019

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Ans. (b) : Thermal conductivity of non-metallic amorphous solids decreases with decrease in temperature.

188. A solid spherical ball taken of a furnace at 1200 K is allowed to cool in air. Given the following radius of the ball = 2 cm
Density of the ball = 7800 kg/m^3
Specific heat of the ball = 420 kJ/kg emissivity = 0.85
Stefan-Boltzman constant = 5.67 × 10^{-8} W/m^{2}K^4
Ambient temperature = 300 K Convection coefficient to air = 350 W/m^{2}K
The differential equation governing the temperature θ of the ball as a function of time t is given by:

(a) $\frac{d\theta}{dt} = -1.6026\times10^{-2}(\theta-300)$

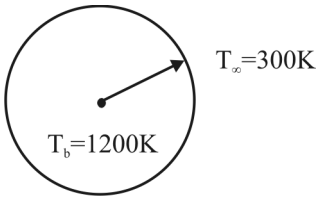
(b) $\frac{d\theta}{dt} = -2.2067\times10^{-12}(\theta^4-81\times10^8)$

(c) $\frac{d\theta}{dt} = -2.2067\times10^{-12}(\theta^4-81\times10^8)$
$-1.6026\times10^{-2}(\theta-300)$

(d) $\frac{d\theta}{dt} = -2.2067\times10^{-12}(\theta^4-81\times10^8)$
$+1.6026\times10^{-2}(\theta-300)$

Oil India Ltd Sr. Engineer (Production) 30.11.2019

Ans. (c) :



Given,
Radius of ball (r) = 2 cm = .02m
Density of ball (ρ) = 7800 kg/m^3
Specific heat of ball (Cp) = 420 kJ/kgK
Emissivity (ε) = 0.85
Stefan- Boltzman constant (σ) = 5.67×10^{-8} W/m^{2}K^4
Ambient temperature (T_∞) = 300k
Convection coefficient of air (h) = 350 W/m^{2}K
Furnace temperature (T_F) = 1200K
Find differential governing equation $\frac{d\theta}{dt} = ?$
Now, During cooling, the ball heat will transfer to air by convection and radiation from body to environment let ball temperature is θ
Energy balance equation,
$Q_{body} + Q_{conv.} + Q_{rad.}$ = constant

$$\frac{d}{dt}\left[Q_{body}+Q_{conv}+Q_{rad.}\right]=0$$

$$\frac{d}{dt}\left[mc(\theta-\theta_\infty)+hA(\theta-\theta_\infty)+\varepsilon\sigma A(\theta^4-\theta_\infty^4)\right]=0$$

$$\frac{d}{dt}[\theta-\theta_\infty] = -\frac{d}{dt}\left[\frac{hA}{mc}(\theta-300)+\frac{\varepsilon\sigma A}{mc}(\theta^4-300^4)\right]$$

$$\frac{d\theta}{dt} = -\left[\frac{h\times4\pi r^2}{\rho vc}(\theta-300)+\frac{\varepsilon\sigma\times4\pi r^2}{\rho vc}(\theta^4-300^4)\right]$$

$$= -\frac{350\times4\pi\times(0.02)^2}{7800\times\frac{4}{3}\pi(0.02)^3\times420\times10^3}(\theta-300)$$
$$+\frac{0.85\times5.67\times10^{-8}\times4\pi\times(0.02)^2}{7800\times\frac{4}{3}\pi(0.02)^3\times420\times10^3}(\theta^4-300^4)$$

$$= -1.6026\times10^{-5}(\theta-300^o)-2.2067\times10^{-15}(\theta^4-81\times10^8)$$

In question specific heat of ball
C = 420 kJ/kgK

$$\therefore \left|\frac{d\theta}{dt} = -2.2067\times10^{-15}(\theta^4-81\times10^8)\right.$$
$$-1.6026\times10^{-5}(\theta-30)$$

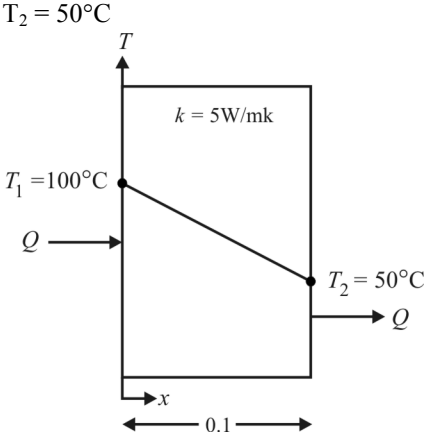
If C = 420J/kgK
Then,

$$\frac{d\theta}{dt} = -2.2067\times10^{-12}(\theta^4-81\times10^8)$$
$$-1.6026\times10^{-2}(\theta-300)$$

189. The heat flow across a plane wall (area 2m^2 and thickness 0.1m) with a constant thermal conductivity of 5W/mk and steady state temperature at 100°C and 50°C is:
(a) 5 kW (b) 50 kW
(c) 60 kW (d) 6 kW

Oil India Ltd. Sr. Engineer (Drilling) 30.11.2019

Ans. (a) : Given, A = 2m^2
t = 0.1 m
k = 5 W/mk
T_1 = 100°C
T_2 = 50°C



∴ According to Fourier's Law of heat transfer by conduction.

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$$Q = -kA\frac{dT}{dx}$$

or $$Q = kA\frac{T_1 - T_2}{t} = \frac{5\times 2\times(100-50)}{0.1}$$

$$= \frac{5\times 2\times 50}{0.1} = 5000 = 5\text{ kW}$$

190. Which of the following statement is correct with respect to critical radius of insulation?

(a) The radius up to which heat flow decreases and after which heat flow increases is known as critical radius of insulation

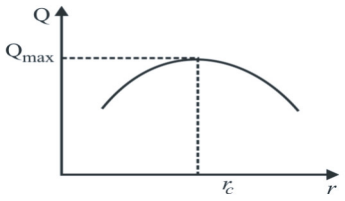
(b) The radius up to which heat flow increases and after which heat flow remains constant is known as critical radius of insulation

(c) The radius up to which heat flow increases and after which heat flow decreases is known as critical radius of insulation

(d) The radius up to which heat flow is constant and after which heat flow decreases is known as critical radius of insulation.

Oil India Ltd. Sr. Engineer (Drilling) 30.11.2019

Ans. (c) : Critical radius of Insulation (r_c)– It is such a radius of insulation at which the total resistance to the heat flow is minimum or heat transfer is maximum



For cylinder; $(r_c) = \frac{k_{insulation}}{h}$

For sphere; $(r_c) = \frac{2k_{insulation}}{h}$

So, the radius up to which heat flow increases and after which heat flow decreases is known as critical radius of insulation.

191. For a given set of parameters, which of the following cross section of the fin has highest fin efficiency?

(a) Convex parabolic (b) Concave parabolic

(c) Trapezoidal (d) Rectangular

Oil India Ltd. Sr. Engineer (Drilling) 30.11.2019

Ans. (b) : We know that

Fin efficiency $(\eta) = \frac{Q_{act.}}{Q_{max.}} = \frac{\sqrt{hPkA_C}\theta_0}{hPL\theta_0}$

$$\eta = \sqrt{\frac{KA_C}{hP}}.\frac{1}{L}\ ;\ m = \sqrt{\frac{hP}{KA_C}} \qquad \text{if } A_C\uparrow - m\downarrow$$

$$\boxed{\eta = \frac{1}{mL}}$$

$\eta \uparrow \rightarrow m\downarrow$

Concave parabolic has highest cross-section area compared to other cross-section for given parameter so it provides larger heat transfer.

192. $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0$ is an equation for : [where, T = temperature]

(a) 3-D transient heat conduction equation with no heat generation with constant thermal conductivity in Cartesian

(b) 3-D steady state heat conduction equation with heat generation and with constant thermal conductivity in Cartesian

(c) 3-D steady state, heat conduction equation with no heat generation and with temperature dependent thermal conductivity in Cartesian coordinates

(d) 3-D steady state heat conduction equation with no heat generation and with constant thermal conductivity in Cartesian coordinates

SJVN ET 2019

Ans. (d) : 3-D steady state heat conduction equation with no heat generation and with constant thermal conductivity in Cartesian coordinates

193. If r_1 and r_2 are respectively inner and outer diameter of a pipe and k is the constant, R-value for pipe insulation is determined from

(a) $R - value = \frac{r_1}{k} l_n \frac{r_2}{r_1}$

(b) $R - value = \frac{r_1}{k} l_n \frac{r_1}{r_2}$

(c) $R - value = \frac{r_2}{k} l_n \frac{r_1}{r_2}$

(d) $R - value = \frac{r_2}{k} l_n \frac{r_2}{r_1}$

TNPSC AE 2019

Ans. (d) : R-value for pipe insulation is given as

$$R - \text{value} = \frac{r_2}{k}\ln\left(\frac{r_2}{r_1}\right)$$

194. Among the following, the best insulator is

(a) Air (b) Water

(c) Ash (d) Aluminium

JPSC AE PRE 2019

Ans. (a) : Air

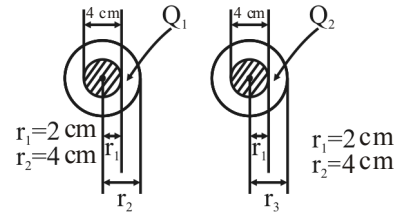
195. A long conduit of 4 cm outer diameter is carrying steam. Currently it is insulated with 20 mm thick insulation. Additional insulation required to reduce the heat loss by two-third is

(a) 90 mm (b) 110 mm

(c) 120 mm (d) 140 mm

JPSC AE PRE 2019

Ans. (c) :



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Given that,

$$Q_2 = \frac{1}{3}Q_1 \qquad \left[\because Q = \frac{2\pi k l(\Delta T)}{ln\left(\frac{r_o}{r_i}\right)}\right]$$

$$\frac{1}{\ell n\left(\frac{r_3}{r_1}\right)} = \frac{1}{3}\frac{1}{\ell n(r_2/r_1)}$$

$$\frac{r_3}{r_1} = \left(\frac{r_2}{r_1}\right)^3$$

$$\Rightarrow \quad r_3 = \left(\frac{4}{2}\right)^3 \times 2$$

$r_3 = 16$ cm

Extra insulation $= r_3 - r_2$

$= 16 - 4 = 12$ cm $= 120$ mm

196. A wire of 8 mm diameter at a temperature of 60°C is to be insulated by a material having k = 0.174 W/mK. The heat transfer coefficient on the outside h_a = 8W/m²K and ambient temperature T_a = 25°C. The maximum thickness of insulation for maximum heat loss will be

(a) 15.25 mm (b) 16.50 mm

(c) 17.75 mm (d) 18.25 mm

ESE 2019

Ans. (c) : Given,

$d = 8$ mm, $r = 4$ mm

$k_{ins} = 0.174$ W/m·k

$h_0 = 8$ W/m²k

Critical radius of insulation $(r_c) = \frac{k_{ins}}{h_0}$

$= \frac{0.174}{8} = 0.02175$ m

$= 21.75$ mm

Maximum thickness $= r_c - r$

$= 21.75 - 4 = 17.75$ mm

197. The differential governing equation for the fin of uniform cross-sectional area (A_c) and constant thermal conductivity (k) is written as $\frac{d^2\theta}{dx^2} - m^2\theta = 0$. Which of the following is correct for this equation? [Where, P = Perimeter of fin, h = convective heat transfer coefficient]

(a) θ = Temperature of fin; $m = \sqrt{\frac{hP}{kA_c}}$

(b) θ = Excess temperature; $m = \frac{hP}{kA_c}$

(c) θ = Excess temperature; $m = \sqrt{\frac{hP}{kA_c}}$

(d) θ = Temperature of fin; $m = \frac{hP}{kA_c}$

(e) θ = Excess Temperature; $m = \left(\frac{hP}{kA_c}\right)^{3/2}$

CGPSC AE 25.02.2018

Ans. (c) : θ = Excess temperature

$$m = \sqrt{\frac{hP}{kA_c}}$$

198. Non dimensional temperature distribution for lumped system analysis is given by [where, B_i = Biot number, F_0 = Fourier number, T_0 = Initial temperature, T_∞ = Environmental temperature]

(a) $\frac{T - T_\infty}{T_0 - T_\infty} = e^{-(B_i/F_0)}$ (b) $\frac{T - T_\infty}{T_0 - T_\infty} = e^{-(B_i^2 \times F_0)}$

(c) $\frac{T - T_\infty}{T_0 - T_\infty} = e^{-(B_i \times F_0)}$ (d) $\frac{T - T_\infty}{T_0 - T_\infty} = e^{-(B_i^2 \times F_0)}$

(e) $\frac{T - T_\infty}{T_0 - T_\infty} = \ln\left(B_i^2 \times F_0\right)$

CGPSC AE 25.02.2018

Ans. (c) : $\frac{T - T_\infty}{T_o - T_\infty} = e^{-(B_i \times F_o)}$

where $B_i = \frac{hL_c}{k}$

$L_c = \frac{\text{volume of solid}}{\text{surface area of the solid}}$

$F_o = \frac{\alpha\tau}{L_c^2}$

For Lumped system $B_i < 0.1$

199. What is the advantage of using the spherical vessel to store fluids at low temperature?

(a) heat transfer in sphere is restricted in all direction by insulation

(b) sphere has the smallest volume per unit surface area

(c) sphere has the largest volume per unit surface area

(d) none of the above

Vadodara Municipal Corp. DEE, 2018

Ans. (c) : An advantage of spherical storage vessels to store fluid at low temperature is that they have largest volume per unit surface area than any other transfer from warmer surroundings to the liquid in the sphere, will be less than that for cylindrical or rectangular storage vessels.

200. A plane wall is 25 cm thick with an area of 1 m², and has a thermal conductivity of 0.5 W/mK. If a temperature difference of 60°C is imposed across it, what is the heat flow ?

(a) 120 W (b) 140 W

(c) 160 W (d) 180 W

GPSC ARTO Pre 30.12.2018, ESE 2005

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Ans. (a): Given,
Thickness, L = 25 cm = 0.25 m
Area, $A = 1\ m^2$
Thermal conductivity, K = 0.5 W/m-K
Temperature difference, $\Delta T = 60°C$

$$\text{Heat flow, } Q = \frac{\Delta T}{\left(\frac{L}{KA}\right)} = \frac{60}{\left(\frac{0.25}{0.5\times1}\right)} = 120\ W$$

201. The heat transfer by conduction through a thick cylinder of inner radius r_1, outer radius r_2 Higher temperature T_1, lower temperature T_2, length of cylinder l and thermal conductivity k is given by

(a) $\dfrac{2\pi lk(T_1 - T_2)}{2.3\log\left(\frac{r_2}{r_1}\right)}$

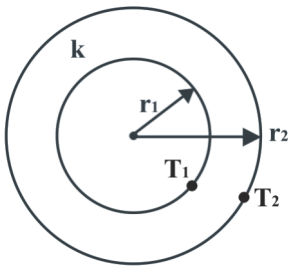
(b) $\dfrac{2.3\log\left(\frac{r_2}{r_1}\right)}{2\pi lk(T_1 - T_2)}$

(c) $\dfrac{2\pi(T_1 - T_2)}{2.3lK\log\left(\frac{r_2}{r_1}\right)}$

(d) $\dfrac{2\pi lk}{2.3(T_1 - T_2)\log\left(\frac{r_2}{r_1}\right)}$

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (a) :



Heat transfer through cylinder is given as

$$Q = \frac{2\pi lk(T_1 - T_2)}{2.3\log\left(\frac{r_2}{r_1}\right)}$$

202. Thermal conductivity of copper with rise in temperature

(a) Decrease (b) Increase
(c) Remains constant (d) None of these

RPSC Vice Principal ITI 2018

Ans. (a) : Copper has a fairly high conductivity at room temperature. This is because of the large number of valence electrons which are free to move in the lattice. Thus the electrical conductivity will decreases with increases in temperature.

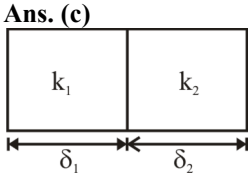
203. What is the equivalent thermal conductivity of a composite wall made up of two layers of different materials of thermal conductivities k_1 and k_2 and having thicknesses of δ_1 and δ_2 respectively?

(a) $\dfrac{2(k_1 + k_2)}{\delta_1 + \delta_2}$ (b) $\dfrac{(k_1 + k_2)}{k_1k_2} \times \dfrac{\delta_1\delta_2}{\delta_1 + \delta_2}$

(c) $\dfrac{k_1k_2(\delta_1 + \delta_2)}{k_1\delta_2 + k_2\delta_1}$ (d) $\left(\dfrac{k_1}{\delta_1}\right) + \left(\dfrac{k_2}{\delta_2}\right)$

RPSC Vice Principal ITI 2018

Ans. (c)



$$\Rightarrow R_{eq} = R_1 + R_2$$

$$\Rightarrow \frac{(\delta_1 + \delta_2)}{k_{eq}A} = \frac{\delta_1}{k_1A} + \frac{\delta_2}{k_2A}$$

$$\Rightarrow \frac{\delta_1 + \delta_2}{K_{eq}} = \frac{\delta_1}{k_1} + \frac{\delta_2}{k_2}$$

$$\Rightarrow \frac{\delta_1 + \delta_2}{k_{eq}} = \frac{\delta_1k_2 + k_1\delta_2}{k_1k_2}$$

$$\boxed{k_{eq} = \frac{k_1k_2(\delta_1 + \delta_2)}{k_1\delta_2 + k_2\delta_1}}$$

204. For steady 1-D conduction, with no heat generation temperature distribution in the cylinder along the radius is : [T = temperature, r = radius, i = inside, o = outside]

(a) $\dfrac{T(r) - T_i}{T_o - T_i} = \dfrac{\ln r}{\ln r_i}$

(b) $\dfrac{T(r) - T_i}{T_o - T_i} = \dfrac{\ln r}{\ln r_o}$

(c) $\dfrac{T - T_i}{T_o - T_i} = \dfrac{\ln(r/r_o)}{\ln(r/r_i)}$

(d) $\dfrac{T(r) - T_i}{T_o - T_i} = \dfrac{\ln(r/r_i)}{\ln(r_o/r_i)}$

(e) $\dfrac{T(r) - T_i}{T_o - T_i} = \dfrac{r}{r_i}$

CGPSC AE 25.02.2018

Ans. (d) : For steady, 1-D conduction, with no heat generation temperature distribution in the cylinder along the radius is given as

$$\frac{T(r) - T_i}{T_o - T_i} = \frac{\ln(r/r_i)}{\ln(r_o/r_i)}$$

205. A glass window 3m high, 5 m wide, 0.3 m thick of thermal conductivity k = 0.9 W/mK, having inner and outer surface temperature at 16^0C and 2^0C respectively. The rate of heat loss and thermal resistance of the glass will be respectively given by:

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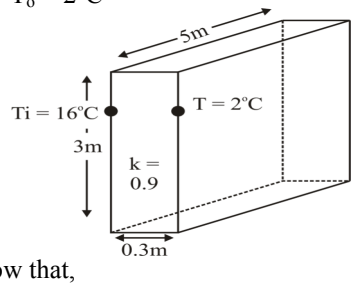
(a) 63W, 0.022 K/W (b) 630 W, 0.22 K/W
(c) 630 W, 0.022 K/W (d) 63 W, 0.22 K/W
(e) 6.3 W, 2.2 K/W

CGPSC AE 25.02.2018

Ans. (c) : Data given-

$t = 0.3$ m
$\lambda = 3$ m
$w = 5$ m
$T_i = 16^oC$
$T_o = 2^oC$



We know that,

$$Q = k.A.\frac{\Delta T}{t} \qquad (A = 3\times 5\ m^2)$$

$$= 0.9\times 3\times 5\times \frac{[16-2]}{0.3}$$

$$\boxed{Q = 630W}$$

Thermal resistance

$$R_{th} = \frac{t}{kA} = \frac{0.3}{0.9\times 3\times 5} = 0.022\ K/W$$

206. Which of the following is correct regarding critical radius of insulation (r_c)?

(a) The rate of heat transfer is minimum at critical radius of insulation.
(b) Rate of heat transfer is maximum at critical radius of insulation.
(c) Rate of heat transfer is zero at critical radius of insulation.
(d) Resistance to heat flow is maximum at critical radius of insulation.

(d) $(r_c)_{\text{cylindrical pipe}} = \dfrac{4\times \text{thermal conductiviy of insulating material}}{\text{heat transfer coefficient between insulation surface and air}}$

CGPSC AE 25.02.2018

Ans. (b) : Rate of heat transfer is maximum at critical radius of insulation.

$$r_c = \frac{k}{h}\ [\text{for cylinder}]$$

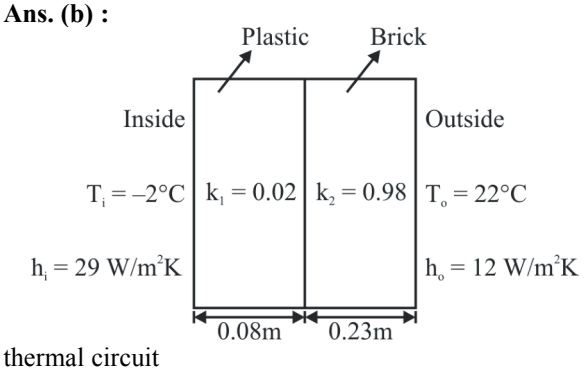
$$r_c = \frac{2k}{h}\ [\text{for sphere}]$$

207. A cold storage has 23 cm brick wall on the outside and 8 cm plastic foam on the inside. The inside and outside temperatures are – 2°C and 22°C, respectively. If the thermal conductivities of brick and foam are 0.98 W/ m-K and 0.02 W/m-K, and the inside and outside heat transfer coefficients are 29 W/ m^2-K and 12 W/m^2-K, respectively, then the rate of heat removal for a (projected) wall area of 90 m^2 will nearly be

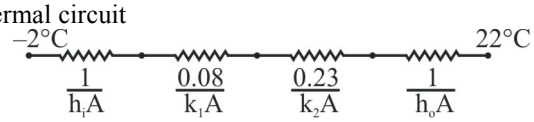
(a) 503 W (b) 497 W
(c) 490 W (d) 481 W

ESE 2018

Ans. (b) :



thermal circuit



$$\text{Heat transfer rate} = \frac{\Delta T}{\Sigma R_{th}}$$

$$= \frac{22-(-2)}{\frac{1}{29\times 90}+\frac{0.08}{0.02\times 90}+\frac{0.23}{0.98\times 90}+\frac{1}{12\times 90}}$$

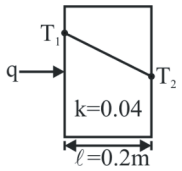
$$= 496.26\ W \simeq 497\ W$$

208. The refractory lining of a furnace has a thickness of 200 mm. The average thermal conductivity of the refractory material is 0.04 W/m-K. The heat loss is estimated to be 180kJ/hr/m^2. The temperature difference across the lining will be

(a) 280ºC (b) 250ºC
(c) 240ºC (d) 220ºC

ESE 2018

Ans. (b) : Given,



$\ell = 0.2$ m
$k = 0.04$ W/mk
$q = 180$ kJ/hr/m^2

$$= \frac{180\times 1000}{3600}\ J/s/m^2 = 50\ W/m^2$$

$$\text{Heat flux } (q) = \frac{k\Delta T}{\ell}$$

$$50 = \frac{0.04\times \Delta T}{0.2}$$

$$\Delta T = 250°C$$

209. A wall surface of 200 mm thickness has an outside temperature of 50°C and inside temperature of 25°C with thermal conductivity of 0.51 W/m-K. The heat transfer through this wall will be

(a) 63.75 W/m^2 (b) 65.75 W/m^2
(c) 70.25 W/m^2 (d) 73.25 W/m^2

ESE 2018

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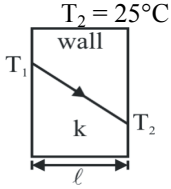
Ans. (a) : $\ell = 200$ mm k = 0.51 W/mK

$T_1 = 50°C$, $T_2 = 25°C$



Heat transfer through this wall i.e. heat flux is given by

$$q = k\left(\frac{T_1 - T_2}{\ell}\right) = 0.51\left(\frac{50-25}{0.2}\right) = 63.75 \text{ W/m}^2$$

210. Find efficiency of a plate fin of length 1.5 cm and thickness 2 mm (k = 210 W/mK and h = 285 W/m^{2}K).

(a) 89.9% (b) 92.4%
(c) 92.9% (d) 74%

AAI Jr. Executive 29.11.2018

Ans. (a) : $\eta = \frac{\tanh(mL_c)}{mL_c}$, For plate $L_c = c + \frac{t}{2}$

$$L_c = 1.5 + \frac{0.2}{2}$$
$$= 1.6 \text{ cm}$$
$$m = \sqrt{\frac{hP}{kA}}$$
$$mL_c = 1.6\times10^{-2}\sqrt{\frac{2\times285}{210\times2\times10^{-3}}}$$
$$= 0.589$$
$$\eta = \frac{\tanh(0.589)}{0.589}$$
$$= 0.899$$
$$= 89.9\%$$

211. Three metal walls of same wall thickness and cross-sectional area have thermal conductivities k, 2k and 3k respectively. For the same heat transfers, the temperature drops across the wall will be in the ratio

(a) 1 : 2 : 3
(b) 3 : 2 : 1
(c) 1 : 1 : 1
(d) temperature drop ratios cannot be worked out as the given data is insufficient

GPSC Executive Engineer 23.12.2018

Ans. (b) : $K_1 : K_2 : K_3 = 1 : 2 : 3$

$$Q_2 = Q_1$$
$$\Rightarrow \frac{K_2 A\Delta T_2}{\delta_2} = \frac{K_1 A\Delta T_1}{\delta_1}$$
$$\Rightarrow \frac{\Delta T_2}{\Delta T_1} = \frac{K_1}{K_2} = \frac{1}{2}$$
$$\Rightarrow Q_3 = Q_1$$
$$\Rightarrow \frac{K_3 A\Delta T_3}{\delta} = \frac{K_1 A\Delta T_1}{\delta_1}$$
$$\frac{\Delta T_3}{\Delta T_1} = \frac{K_1}{K_3} = \frac{1}{3}$$
$$\Delta T_1 : \Delta T_2 : \Delta T_3 = \Delta T_1 : 0.5\Delta T_1 : 0.33\Delta T_1$$
$$= 3 : 2 : 1$$

or

Higher the thermal conductivity (K), lower the temperature drop.

212. The process of heat transfer from one particle of the body to another is called conduction, when the particles of the body

(a) move actually
(b) do not move actually
(c) affect the intervening medium
(d) does not effect the intervening medium

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : The process of heat transfer from one particle of the body to another is called conduction, when the particles of the body do not move actually.

213. The heat transfer from a hot body to a cold body is directly proportional to the surface area and difference of temperatures between the two bodies. This statement is called

(a) First law of thermodynamics
(b) Newton's law of cooling
(c) Newton's law of heating
(d) Stefan's law

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : The heat transfer from a hot body to a cold body is directly proportional to the surface area and difference of temperature between the two bodies. This statements is called Newton's law of cooling.

214. Thermal conductivity of water ..... with rise in temperature

(a) remains same
(b) decreases
(c) increases
(d) may increase or decrease depending upon temperature

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Thermal conductivity of water may increase or decrease with rise in temperature.

215. One dimensional wave equation is

(a) $\frac{\partial^2 y}{\partial t^2} = \alpha^2 \frac{\partial^2 y}{\partial x^2}$ (b) $\frac{\partial y}{\partial t} = \alpha^2 \frac{\partial^2 y}{\partial x^2}$

(c) $\frac{\partial^2 y}{\partial t^2} + \frac{\partial^2 y}{\partial x^2} = 0$ (d) none of these

ISRO Scientist/Engineer 22.04.2018

Ans. (a) : Wave equation-

$$\frac{\partial^2 y}{\partial t^2} = \alpha^2 \frac{\partial^2 y}{\partial x^2}$$

Heat equation-

$$\frac{\partial y}{\partial t} = \alpha^2 \frac{\partial^2 y}{\partial x^2}$$

Laplace equation

$$\frac{\partial^2 y}{\partial t^2} + \frac{\partial^2 y}{\partial x^2} = 0$$

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216. The unit of thermal diffusivity is

(a) m/h K (b) m/h

(c) m^2/h (d) m^2/hK

JPSC AE, 10.04.2021, Paper-I
Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : Thermal diffusivity is the thermal conductivity divided by density and specific heat capacity at constant pressure. It measures the rate of transfer at heat of a material from the hot end to the cold end it has the SI derived unit of m^2/hr.

217. The lowest thermal diffusivity is of

(a) iron (b) lead

(c) aluminium (d) rubber

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : The lowest thermal diffusivity is of rubber.

Materials		**Thermal diffusivity (mm^2/sec.)**
Pyrolytic graphite	–	1220
Corbon	–	216.5
Silver	–	165.63
Aluminium	–	97
Tin	–	40
Iron	–	23
Lead	–	0.34
Rubber	–	0.089 – 0.13
Wood	–	0.082
Oil, engine	–	0.0738

218. Which of the following statement is wrong?

(a) the heat transfer in liquid and gases takes place according to convection

(b) the amount of heat flow through a body is dependent upon the material of the body

(c) the thermal conductivity of solid metals increases with rise in temperature

(d) logarithmic mean temperature difference is not equal to the arithmetic mean temperature difference

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : For metals, the thermal conductivity is mainly a function of the motion of free electrons. As the temperature increases the molecular vibrations increase so, they obstruct the flow of free electrons, thus reducing, the conductivity.

219. A platinum resistance thermometer has a resistance of 2Ω at 0°C and 3Ω at 100°C. What will be the temperature when the resistance indicates 5Ω?

(a) 300°C (b) 200°C

(c) 350°C (d) 400°C

Karnataka PSC Lect., 27.05.2017

Ans. (a) : $R = R_0(1 + \alpha T)$

$T = 0°C, R = 2\Omega$

$2\Omega = R_0(1 + 0)$

$\boxed{R_0 = 2\Omega}$

At $T = 100°C, R = 3\Omega$

$3 = 2(1 + 100\alpha)$

$\frac{3}{2} = 1 + 100\alpha$

$\frac{1}{200} = \alpha$

$\boxed{\alpha = 0.005}$

At $R = 5\Omega, T = ?$

$R = R_0(1 + \alpha T)$

$5 = 2(1 + 0.005\ T)$

$\frac{5}{2} - 1 = \frac{5T}{1000}$

$\frac{3000}{2 \times 5} = T$

$\boxed{T = 300°C}$

220. When the thickness of insulation is less than the critical thickness of the insulation, then the heat transfer rate of insulated pipe

(a) will be more than the heat transfer rate from the same pipe without insulation

(b) will be less than the heat transfer rate from the same pipe without insulation

(c) will become equal to the heat transfer rate from the same pipe without insulation

(d) None of the above

Karnataka PSC Lect., 27.05.2017

Ans. (a) : The thickness of insulation is less than the critical thickness of the insulation, then the heat transfer rate of insulate of pipe will be more than the heat transfer rate from the same pipe without insulation.

221. Which of the following would not increase the rate of heat transferred from a heater pipe?

(a) Insulating with materials whose thickness is below that of critical thickness for insulation.

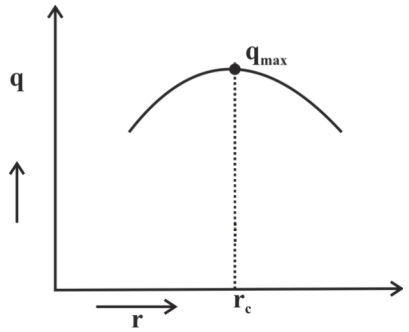
(b) Providing cooling jackets over the heater pipe surface.

(c) Providing heat exchangers.

(d) Covering heater pipe with another pipe whose thermal conductivity is smaller in number and 50 mm thick.

Karnataka PSC AE, 10.09.2017

Ans. (d) :



For cylindrical,

Critical radius $(r_c) = \frac{k}{h}$

For spherical,

Critical radius $(r_c) = \frac{2k}{h}$

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222. A furnace has a 20 cm thick wall with thermal conductivity 0.8 W/m-K. For the same heat loss from the furnace, what will be the thickness of the wall if the thermal conductivity of the material is 0.16 W/m-K?

(a) 4 cm (b) 6.3 cm
(c) 10 cm (d) 40 cm

GPSC EE Pre, 28.01.2017, ESE 2012

Ans. (a) : Given-

Thickness $(x_1) = 20$ cm

$k_1 = 0.8$ W/m-K

Thickness $(x_2) = ?$

$k_2 = 0.16$ W/m-K

According to Fourier's equation-

$$Q = kA\frac{dT}{dx}$$

$$\therefore \quad Q \propto \frac{k}{dx}$$

In question, heat (Q) is same in both cases.

$$\therefore \quad \frac{k_1}{(dx)_1} = \frac{k_2}{(dx)_2}$$

$$\frac{0.8}{20} = \frac{0.16}{(dx)_2}$$

$(dx)_2 = 4$ cm

223. For quick response of a thermocouple to observe varying temperatures of fluids

(a) wire diameter must be large
(b) wire material density must be large
(c) wire material specific heat must be large
(d) wire surface heat transfer coefficients must be large

GPSC EE Pre, 28.01.2017

Ans. (d) : For quick, response of a thermocouple 'time constant'

$$t = \frac{\rho VC}{hA}$$

$$t \propto \frac{1}{h}$$

- If h↑ will large t↓ will be small therefore thermocouple set the temperature fast.

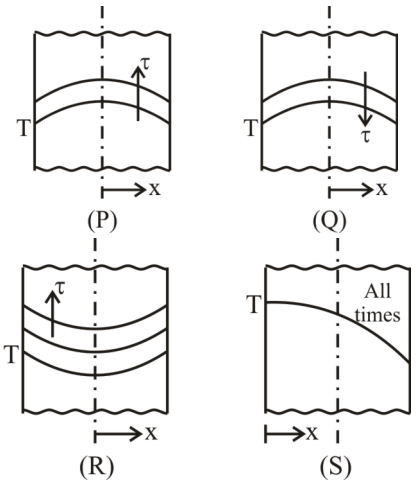
224. If a square section fin is split longitudinally and used as two fins.

(a) The total heat flow will decrease
(b) The total heat flow will increase
(c) The total heat flow will remain constant
(d) Heat flow many increase or decrease depending on the material used

HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : If a square section fin is split longitudinally and used as two fins, the total heat flow will increase. Because of increase in total surface area.

225. Match the figures and the statements under transient conditions :



Statements–

(1) Cooling **(2) Heating**
(3) Steady state **(4) Heat generation.**

(a) 1-Q, 2- R, 3-S, 4-P (b) 2-Q, 1- R, 3-S, 4-P
(c) 3-Q, 2- R, 1-S, 4-P (d) 1-Q, 2- R, 4-S, 3-P

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : Figure (P)→ Maximum temperature at mid point and heat is releasing to the surrounding it means heat generation.

Figure (Q)→ Wall temperature is decreasing. It means cooling occurs.

Figure (R)→ Wall temperature is increasing It means heating occurs.

Figure (S)→ Temperature is decreasing in the direction of length. It means steady state condition.

226. A plane wall of thickness 100 mm and wall area 4 m^2 has conductivity of 9.35 W/m °C. The surface temperatures are steady at 130 °C and 30 °C. The heat flow across the wall will be

(a) 37400 W (b) 37200 W
(c) 37300 W (d) 37500 W

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : L = 100 mm = 0.1 m

$A = 4\ m^2$

$K = 9.35\ W/m^0c$

$T_1 = 130^0C$

$T_2 = 30^0C$

Heat transfer,

$$Q = \frac{KA(T_1 - T_2)}{L}$$

$$= \frac{9.35 \times 4 \times (130 - 30)}{0.1}$$

Q = 37400 watt

227. Among the following materials, which has the lowest resistivity?

(a) Tungsten (b) Gold
(c) Iron (d) Brass

Jharkhand Urja Vikas Nigam Ltd. AE 2017

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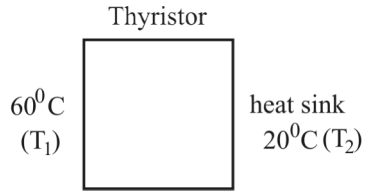
Ans. (b) : Resistivity electrical resistance at a conductor of unit cross sectional area and unit length.
High resistivity designates poor conductors
(A) Tungesten – 5.6×10^{-8} ohm-m
(B) Gold – 2.4×10^{-8} ohm-m
(C) Iron – 0.71×10^{-8} ohm-m
(D) Brass – 60×10^{-8} ohm-m

228. In a thyristor the junction temperature is 60 degree C. Thermal power flow between the junction and heat sink is 4W. What will be the thermal resistance between junction and heat sink, if the latter is at a temperature of 20 degree C?
(a) 20 degree C/Watt (b) 10 degree C/Watt
(c) 5 degree C/Watt (d) 15 degree C/Watt

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (b) :



The heat conduction (heat rate)

$$q = \frac{k_A (T_1 - T_2)}{L}$$

$$q = \frac{(k_A)}{L}(\Delta T) \quad \text{..... (i)}$$

Compare with V = iR

$$\text{or } i = \frac{V}{R} \quad \text{..... (ii)}$$

Compare (i) and (ii)

$$\frac{1}{R_H} = \frac{k_A}{L}$$

$$R_H = \frac{L}{k_A}$$

$$q = \frac{1}{R_H}\Delta T$$

or $$R_H = \frac{\Delta T}{q}$$

$$R_H = \frac{60-20}{4} = 10^0 \text{C/watt}$$

229. The chart solutions for temperature distributions and heat flows in plane walls, long cylinders and spheres with finite internal and surface resistance is present in
(a) Compressibility chart
(b) Mollier chart
(c) Heisler chart
(d) Psychrometric chart

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) : Heisler charts are a graphical analysis tool for the evaluation of heat transfer in thermal engineering. They are set of two charts for included geometry.

Heisler charts permit evaluation at the central temperature for transient heat conduction through an infinitely long plane wall of thickness 2L, and infinitely long cylinder at radius r_o and sphere of radius r_o.

230. Heat flows one body to other when they have
(a) Different heat contents
(b) Different specific heat
(c) Different atomic structure
(d) Different temperature

Vizag Steel (MT) 2017

Ans. (d) : Heat flows from one body to other when they have different temperature.

231. A 3 cm long, 2 mm × 2mm rectangular cross-section aluminium fin [k = 237 k/m^{0}c] is attached to a surface. If the fin efficiency is 65%, the effectiveness of this single fin is :
(a) 30% (b) 24%
(c) 8% (d) 39%

TRB Polytechnic Lecturer 2017

Ans. (d) : Given as,
$\eta_{fin} = 65\%$
$A_c = 2 \times 2$ mm^2
L = 30 mm
We know that

$$\varepsilon = \eta_{fin} \times \frac{PL}{A_c} = 0.65 \times \frac{2 \times [2+2] \times 30}{2 \times 2}$$

$$\boxed{\varepsilon = 39\%}$$

232. A long glass cylinder of inner diameter 0.03m and outer diameter 0.05m carries hot fluid inside. If the thermal conductivity of glass is 1.05 W/mK, the thermal resistance per unit length of the cylinder is
(a) 0.031 (b) 0.077
(c) 0.017 (d) 0.77

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (b) : Given, Inner dia = 0.03 m and outer dia = 0.05 m. Thermal conductivity k = 1.05 W/mK
Thermal resistance per unit length of the cylinder is

$$R_{th} = \frac{\ln(d_2/d_1)}{2\pi k.l}, \; l = 1$$

$$R_{th} = \frac{\ln(0.05/0.03)}{2 \times 3.14 \times 1.05}$$

$R_{th} = 0.077$

233. The distribution of temperature T, at a particular instant, across a large concrete wall, 1 m thick, which is heated from one side is given by the equation $T = 150 - 80x + 16x^2$, where the distance x is in meters and T is in degree Celsius. Thermal conductivity of wall is 0.6 W/mK, area of cross section is 10 m^2. The heat accumulated in unit time at this instant in the concrete wall is
(a) 768 J/sec (b) 384 J/sec
(c) 96 J/sec (d) 192 J/sec

ISRO Scientist/Engineer 17.12.2017

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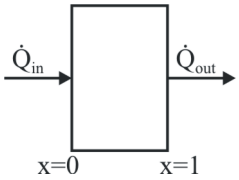
Ans. (d) : $\Delta x = 1$ m

$A = 10\ m^2$

$k = 0.6$ W/mK

$T = 150 - 80x + 16x^2$

$$\frac{dT}{dx} = -80 + 32x$$



Heat accumulated

= Heat transfer at inlet – Heat transfer at outlet

$$= \left[-kA\frac{dT}{dx}\bigg|_{x=0}\right] - \left[-kA\frac{dT}{dx}\bigg|_{x=1}\right]$$

$= [-0.6 \times 10\ (-80)] - [-0.6 \times 10\ (-80 + 32)]$

$= 480 - 288$

$= 192$ J/sec

234. For heat transfer across a composite slab with materials having different thermal conductivity, study the following statements.

I. Temperature is continuous always.

II. Temperature gradient is not continuous.

III. Heat flow is not continuous.

(a) Statement I alone is correct

(b) Statement I and II are correct

(c) Statement II alone is correct

(d) All the statements are correct

ISRO Scientist/Engineer 07.05.2017

Ans. (b) : For heat transfer across a composite slab with materials having different thermal conductivity—

- Temperature is continuous always
- Temperature gradient is not continuous

I. Temperature is continuous always—

In a single slab temperature has linear profile.

$$Q = -kA\frac{dT}{dx}$$

$$Qdx = -kA\ dT$$

$$dT = -\frac{Q}{kA}dx$$

$$T = -\frac{Q}{kA}x + C_1$$

So, we can say that temperature profile in any single slab is linear and continuous.

II. Temperature gradient is not continuous—

Temperature gradient in single slab is

$$Q = -kA\frac{dT}{dx}$$

$$\frac{dT}{dx} = -\frac{Q}{kA}$$

If Q and A both are constant for both slab then

$$\frac{dT}{dx} \propto \frac{1}{k}$$

Therefore, temperature gradient is constant in a single slab having constant thermal conductivity which is the usual case. If there are 2 slabs having different thermal conductivity they will have different gradient and hence at the interface gradient will be non-continuous.

235. An electric heater is sandwiched between two plates each 0.3 m long and 0.1 m wide with a thickness of 30 mm. At steady state condition the heater is maintained at a temperature of 100°C, with a current of 0.25 A and voltage of 200 V. Assume the plates are perfectly insulated at the edges, and the heater is having perfect contact with the plates to given a temperature of 50°C on the outside of the plate surface. What is the thermal conductivity of the plate material?

(a) 1.0 W/m-K (b) 0.5 W/m-K

(c) 0.015 W/m-K (d) 0.3 W/m-K

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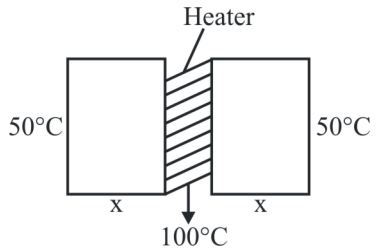
Ans. (b) :



$$q = kA\frac{\Delta T}{\Delta x}$$

$V.I = H_{gen.}$

$= 200 \times 0.25$

$= 50$ W

$$\frac{50}{2} = k \times (0.3 \times 0.1) \times \frac{(100-50)\times 1000}{30}$$

$$k = \frac{50 \times 30 \times 100}{2 \times 3 \times 1 \times (100-50)}$$

$$k = \frac{1}{2} = 0.5 \text{ W/m-K}$$

236. Thermal conductivity through walls of cylinder of inner and outer radii r_1 and r_2, respectively is inversely proportional to

(a) $(r_1 - r_2)$ (b) $\dfrac{1}{r_2 - r_1}$

(c) $\ell n\left(\dfrac{r_2}{r_1}\right)$ (d) $\dfrac{1}{\ell n\left(\dfrac{r_2}{r_1}\right)}$

BPSC AE Mains 2017 Paper - V

Ans : (c) : Thermal conductivity is a material property and it does not depends on the dimension.

$$Q = -kA\frac{dT}{dx}$$

$$Q = -k2\pi r\ell\frac{dT}{dr}$$

$$\int_1^2 Q\frac{dr}{r} = \int_{T_1}^{T_2} -2\pi k\ell dT$$

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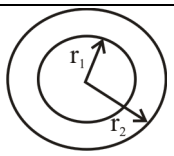

$$Q\ell n\left(\frac{r_2}{r_1}\right) = -2\pi k\ell(\Delta T)$$

$$Q = \frac{-2\pi k\ell(\Delta T)}{\ell n\left(\frac{r_2}{r_1}\right)}$$

237. A pipe carrying hot gasses is insulated with a material having a thermal conductivity of 0.15 W/mK. The average local heat transfer coefficient by convection to the ambient air is 3 W/m^2K. The critical diameter of the pipe under these conditions is

(a) 100 mm (b) 130 mm
(c) 160 mm (d) 200 mm

JWM 2017

Ans. (a) : Thermal conductivity k = 0.15 W/mK

Heat transfer coefficient, h = $3 W/m^2K$

Critical radius, $r_c = \frac{k}{h} = \frac{0.15}{3} = 0.05$ m

r_c = 50 mm

Critical diameter, d = 2 × r_c = 2 × 50 = 100 mm

Note : Critical radius is the radius for which the heat transfer rate is maximum.

238. In a furnace the thickness of the refractory wall lining is 150 mm and the average thermal conductivity of the refractory material is 0.05 W/mK. If the temperature difference between the inner and outer sides of the furnace is 150°C, the heat loss per unit area to the surroundings will be:

(a) 50 W/m^2 (b) 70 W/m^2
(c) 90 W/m^2 (d) 120 W/m^2

JWM 2017

Ans. (a) : Refractory wall thickness, x = 150 mm = 0.15m

Thermal conductivity, k = 0.05 W/mK

Temperature difference, $\Delta T = T_1 - T_2 = 150^oC$

According to Fourier's law of heat conduction

$$= -kA\left(\frac{T_2 - T_1}{x}\right)$$

Heat loss per unit area, $q = \frac{Q}{A} = -k\frac{(T_2 - T_1)}{x}$

$$= 0.05 \times \frac{150}{0.15} = 50 W/m^2$$

239. Heat is lost steadily through a 0.5 cm thick 2m × 3m window glass whose thermal conductivity is 0.7 W/m^oc. The inner and outer surface temperature of the glass are measured to be 12^oC to 9^oC. The rate of heat loss by conduction through the glass is :

(a) 420 W (b) 5040 W
(c) 1256 W (d) 2520 W

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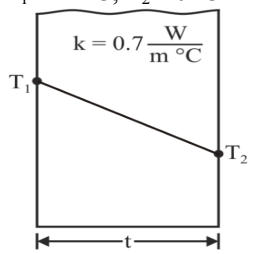
Ans. (d) : Data given as,

t = 0.5 cm = 0.5×10^{-2} m

A = 2×3 m^2

$T_1 = 12°C$, $T_2 = 9°C$



Heat loss through the glass

$$Q = kA\frac{dT}{dt}$$

$$= \frac{0.7 \times 2 \times 3 \times (12 - 9)}{0.5 \times 10^{-2}} = 2520 W$$

240. For a given volume and specified heat input which will have the smallest temperature rise

(a) copper (b) mild steel
(c) water (d) None of these

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : For a given volume and specified heat input, mild steel have the smallest temperature rise.

241. Thermal conductivity of a substance is

(a) Temperature dependent
(b) Independent of temperature
(c) A strong function of the pressure
(d) Independent of pressure

Nagaland PSC CTSE 2017 Paper-2

Ans. (a) : Thermal conductivity of a substance is temperature dependent, as the temperature increases, the molecular vibrations increase.

242. A furnace is provided with an insulating refractory lining. The overall thermal conductivity of the material is 0.03W/m K. The thickness of the lining is 100mm. The inner and outer temperatures are 250°C and 50°C, respectively. The heat loss the surroundings will be

(a) 30 $J/m^2/s$ (b) 60 $J/m^2/s$
(c) 60 J/s (d) 30 J/s

ESE 2017

Ans. (b) : Overall thermal conductivity

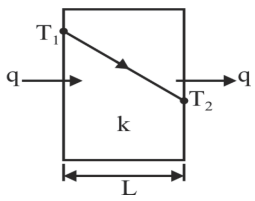
(k) = 0.03W/mK

Thickness of the lining (L) = 100 mm = 0.1 m

$T_1 = 250°C$

$T_2 = 50°C$

Heat flux $q = \frac{Q}{A}$



$$= \frac{k(T_1 - T_2)}{L} = \frac{0.03(250 - 50)}{0.1} = 60 J/m^2\text{-}s$$

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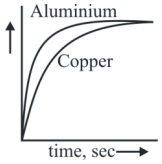
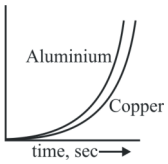
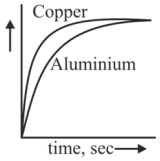
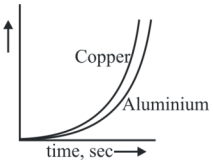
243. A packed bed of solid spheres acts as a thermal energy storage system. In a charging process, heat transfer from the hot gas increases thermal energy stored within the spheres. Consider two beds, one with aluminium spheres (ρ = 2700 kg/m^3, C = 950 J/kg-K, k = 240 W/m-K) and other with copper spheres (ρ = 8900 kg/m^3, C = 400 J/kg-K, k = 380 W/m-K). Select the correct graph for the temperature rise of a sphere at the inlet of the bed

(a) 

(b) 

(c) 

(d) 

ISRO Scientist/Engineer 07.05.2017

Ans. (a) : Thermal diffusivity $= \dfrac{\text{Heat Conducted}}{\text{Heat Stored}}$

$$\boxed{\alpha = \frac{k}{\rho C_p}}$$

α = Thermal diffusivity

Its represent propagation of heat in to the medium

C_P = Heat capacity represents amount of heat stored per unit volume.

$(C_P)_{Al} = 950 \times 2700 = 2565000$ J/m^3-k

$(C_P)_{Cu} = 400 \times 380 = 152000$ J/m^3-k

Heat stored in 'Al' > Heat stored in 'Cu'

$k_{Al} = 240$, $k_{Cu} = 380$ W/m-k

Heat conducted 'Al' < Heat conducted in 'Cu' then

$$\alpha_{Al} = \frac{240}{2700 \times 950} = 9.35 \times 10^{-5}$$

$$\alpha_{Cu} = \frac{365}{8900 \times 400} = 10.11 \times 10^{-5}$$

$$\boxed{\alpha_{Al} < \alpha_{Cu}}$$

Therefore, from above values we can conclude that aluminium has more heat storing capacity then copper and it will have higher temperature then copper at any given point of times as it allows less heat to transfer through it and rather stores that heat increasing its temperature, hence option (c) and (d) can be rejected.

Between option (a) and (b), option (a) is correct because at the starting of charging due to high temperature difference the heat flow will be large and more heat will be stored by aluminium which in turn will increase the temperature drastically and as the temperature difference decrease, there will be less rise in temperature for the same amount of time. In duration of time the temperature of storage bed becomes approximately equal to the hot gas and hence achieves a constant temperature.

244. The instrument for measuring high temperatures is

(a) Pycnometer (b) Barometer

(c) Pyrometer (d) Manometer

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) : Pyrometer measures high temperatures it is used to measure the temperature of distance objects.

- Barometer –measures atmospheric pressure
- Manometer measures gas pressure
- Pycnometer used for measuring the density or volume of the solid.

245. Temperature at the end tip of the fin having uniform cross-sectional area is

(a) maximum

(b) similar to the heat generation temperature

(c) unpredictable

(d) minimum

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (d) : Temperature at the end tip of the fin having uniform cross-sectional area is minimum.

246. Temperature distribution $\dfrac{T - T_\infty}{T_o - T_\infty} = e^{-mx}$ is valid for :

(a) Fin of finite length with heat loss by convection at end

(b) Fin of finite length with insulated end

(c) Very long fins

(d) Fin of finite length with specified temperature at end

UPRVUNL AE 2016

Ans. (c) : Temperature distribution—

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-mx} \quad \text{(For very long fin)}$$

$$\frac{T - T_\infty}{T_0 - T_\infty} = \frac{\cos h\, m(L - x)}{\cos h\, m L}$$

(For fin of finite length with insulated end)

$$\frac{T - T_\infty}{T_0 - T_\infty} = \frac{\cos h\, m(L_c - x)}{\cos h\, m L_c}$$

(For fin of finite length with heat loss by convection at end)

where L_c = Corrected length of fin

$$L_c = L + \frac{t}{2} \text{ (for rectangular fin)}$$

$$L_c = L + \frac{d}{4} \text{ (for Pin fin)}$$

247. Lumped heat capacity approach of transient heat conduction analysis is used:

(a) For bodies of very high conductive resistance

(b) When temperature variation throughout the body is very high

(c) When temperature variation throughout the body is very low

(d) For Biot number > 10

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Ans. (c) : Lumped heat capacity approach of transient heat conduction analysis is used when heat conduction in solids having infinite thermal conductivity (negligible internal resistance) or when temperature variation throughout the body is very low.

248. A heat flux of 40 kW/m^2 is passing through a plate of thickness 0.2 m and thermal conductivity 100 W/m.K. If the temperature of the hot side is 100°C, what will be the temperature of the cooler side ?

(a) 20°C (b) 40°C
(c) 10°C (d) 0°C

RPSC AE (GWD) 06.05.2016

Ans. (a) : Given- $\frac{Q}{A} = q = 40\,kW/m^2$

Thickness (t) = 0.2 m
k = 100 W/mK
$T_H = 100°C$
According to heat conduction of Fourier low

$$Q = kA\frac{dT}{dx}$$

$$q = \frac{Q}{A} = k\left(\frac{T_H - T_L}{t}\right)$$

$$40 \times 10^3 = 100\left(\frac{100 - T_L}{0.2}\right)$$

$$8\times10^3 = 100\,(100 - T_L)$$

$$8\times10^3 = 100[100 - T_L]$$

$$8\times10 = 100 - T_L$$

$$T_L = 100 - 80 = 20°C$$

249. A 0.6 m thick wall is made of a material with thermal conductivity of 1.2 W/m.K. This wall is to be insulated with a material of thermal conductivity 0.2 W/m. K so that the heat loss per m^2 area of the wall does not exceed 1200 W. If the inner and outer temperatures of the composite wall are 930°C and 30°C respectively, determine the thickness of the insulation required?

(a) 7.5 cm (b) 10 cm
(c) 5 cm (d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (c) : Given-
$l_1 = 0.6$ m
$k_1 = 1.2$ W/m-K
$l_2 = ?$
$k_2 = 0.2$ W/m-K

$\frac{Q}{A} = 1200\,W/m^2$, $T_1 = 930°\,C$

$T_2 = 30°C$

According to heat conduction of Fourier's law-

$$Q = -kA\frac{dT}{dx}$$

We know that $R_{th} = \frac{l_1}{k_1 A} + \frac{l_2}{k_2 A}$

$$\frac{\Delta T}{q} = \frac{l_1}{k_1} + \frac{l_2}{k_2}$$

$$\frac{930 - 30}{1200} = \frac{0.6}{1.2} + \frac{l_2}{0.2}$$

$$\frac{900}{1200} = \frac{1}{2} + \frac{l_2}{0.2}$$

$$\frac{3}{4} - \frac{1}{2} = \frac{l_2}{0.2} \Rightarrow \frac{1}{4} = \frac{l_2}{0.2}$$

$$4l_2 = 0.2$$

$$l_2 = \frac{0.2}{4}$$

$[\because 1m = 100cm]$ $$l_2 = \frac{0.2\times100}{4}$$

$$l_2 = 5\text{ cm}$$

250. A pipe of external diameter of 20 mm is carrying hot fluid. It is required to provide a lagging of insulating material of thermal conductivity 0.1 W/mK to reduce the heat loss from the pipe. The heat transfer coefficient with the ambient air is assumed to remain same at 10 W/m^{2}K with or without the lagging. Pick the correct statement :

(a) The insulation will increase heat loss in this case.
(b) Any thickness of insulation will reduce heat loss.
(c) The heat loss will be reduced only if the thickness of insulation is more that the critical thickness of insulation.
(d) For reducing heat loss, the thickness of insulation will have to be less than critical thickness.

RPSC AE (GWD) 06.05.2016

Ans. (b) : Given-
External diameter
(D) = 20 mm
R = 10 mm
Insulated material thermal conductivity
k = 0.1 W/m-K
h = 10 W/m^2-K

Critical radius (r_c) = $\frac{k}{h}$ $[\because 1m = 1000mm]$

$$= \frac{0.1}{10} = \frac{1}{100}\times1000$$

$$r_c = 10\text{ mm}$$

$\therefore$ Here, pipe radius is equal to critical radius ($r_c = R$), then added insulation shall decreases the heat transfer rate.
If pipe radius is lesser than critical radius of insulation any insulation put up around it shall result in increases of heat transfer rate.

251. The condition for heat dissipation in a parabolic fin to be independent of the distance from the base of the fin is (where δ is the thickness of the fin, b is the width of the fin at the base, L is the length of the fin and B_i is the Biot number)

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(a) $\frac{\delta/2}{L}=\frac{B_i}{2}$ (b) $\frac{\delta/2}{L}=\sqrt{\frac{B_i}{2}}$

(c) $\frac{\delta/2}{b}=\sqrt{\frac{B_i}{2}}$ (d) $\frac{b}{L}=\sqrt{B_i}$

RPSC AE (GWD) 06.05.2016

Ans. (b) : $\frac{\delta/2}{L}=\sqrt{\frac{B_i}{2}}$

252. The effectiveness of an infinitely long rectangular fin is NOT increased by

(a) increasing thermal conductivity of the fin material
(b) increasing heat transfer coefficient
(c) decreasing heat transfer coefficient
(d) decreasing area of cross section of the fin

RPSC AE (GWD) 06.05.2016

Ans. (b) : To increase fins effectiveness, one can conclude

- The thermal conductivity of the fin material must be as high as possible.
- The ratio of perimeter to the cross-section area (P/A) should be as high as possible
- To use of fin is most effective in applications that involve low convection heat transfer coefficient i.e. natural convection.

For infinitely long fin effectiveness $\varepsilon_{fin}=\sqrt{\frac{Pk}{hA}}$

$$\varepsilon_{fin}\propto\frac{1}{\sqrt{h}}$$

$$\varepsilon_{fin}\propto\sqrt{k}$$

253. Thermal conductivity of gases varies as

(a) First power of absolute temperature
(b) Square root of absolute temperature
(c) Zeroth power of absolute temperature
(d) Square of absolute temperature

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : Thermal conductivity of Gases proportional to the square root of temperature.

$K\propto T^{1/2}$

$K\propto\sqrt{T}$

254. Thermal conductivity of water at 20 degree C is of the order of

(a) 0.1 W/m K (b) 0.32 W/m K
(c) 0.6 W/m K (d) 0.23 W/m K

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : Thermal conductivity of water at 20 degree C is of the order of 0.6 W/m K

255. Thermal conductivity of air at room temperature in kcal/m hr degree C is –

(a) 0.02 (b) 0.01
(c) 0.1 (d) 0.002

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (a)

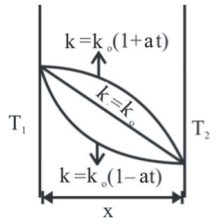
Material	Thermal Conductivity (W/mK)
(i) Diamond	2300
(ii) Silver	429
(iii) Copper	401
(iv) Gold	317
(v) Aluminum	237
(vi) Iron	80.2
(vii) Mercury	8.54
(viii) Glass	0.78
(ix) Brick	0.72
(x) Water	0.613
(xi) Human skin	0.37
(xii) Wood	0.17
(xiii) Helium	0.152
(xiv) Soft rubber	0.13
(xv) Glass fiber	0.043
(xvi) Air	0.026

256. Thermal conductivity of material of a plane wall varies linearly with temperature as $k = k_0 (1 + aT)$ where k_0 and a are positive constant. Considering one dimensional heat transfer in x direction through the wall, how the temperature gradient $\left(\frac{dt}{dx}\right)$ changes in the direction of heat transfer?

(a) Remain constant (b) Increases
(c) Decreases (d) Equal to zero

GPSC Asstt. Prof. 28.08.2016

Ans. (b) :



$k = k_0(1 + aT)$

a and k_0 are positive constant

As T increases k also increases.

$$Q=-kA\frac{dT}{dx}$$

$$\therefore\quad\frac{dT}{dx}\propto-\frac{1}{k}$$

Therefore, As k increases $\frac{dT}{dx}$ also increases.

257. For a current carrying wire of 25 mm diameter and k = 0.45 w/m-K is exposed to air (h = 17 w/m^2-K), maximum heat dissipation occurs when thickness of insulation is:

(a) 12 mm (b) 14 mm
(c) 16 mm (d) 19.8 mm
(e) 25.3 mm

CGPSC Asstt. Workshop Supt., 17.07.2016

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Ans. (b) : For maximum heat dissipation of cylindrical surface,

$$r_c = \frac{k}{h}$$

Where, $k = 0.45$ w/mK $\quad \because d = 25$ mm

$h = 17$ w/m^{2}K $\quad r = 12.5$ mm

$$r_c = \frac{0.45}{17} = 26.47\,\text{mm}$$

Insulation thickness $= r_c - r$

$= 26.47 - 12.5$

$= 13.97$

≈ 14 mm

258. In the.....heat flow problems, the temperature at any point within the system varies non-linearly with time.

(a) Periodic
(b) Non-periodic
(c) Non-transient counter
(d) radiative counter

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : Non-periodic- The temperature at any point within the system changes as a non-linear function with time.

Periodic Heat flow system– The temperature within the system undergoes periodic changes which may be regular or irregular but definitely cyclic.

259. The boundary condition which describes the process of conduction between two bodies in perfect contact is known as:

(a) Boundary condition of the first kind
(b) Boundary condition of the second kind
(c) Boundary condition of the third kind
(d) Boundary condition of the fourth kind

HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : The boundary condition which describes the process of conduction between two bodies in perfect contact is known as boundary condition of the fourth kind.

260. A 2-m-long, 0.3-cm-diameter electrical wire extends across a room at 15°C. Heat is generated in the wire as a result of resistance heating, and the surface temperature of the wire is measured to be 152°C in steady operation. Also, the voltage drop and electric current through the wire are measured to be 60 V and 1.5 A, respectively. Disregarding any heat transfer by radiation, determine the convection heat transfer coefficient for heat transfer between the outer surface of the wire and the air in the room.

(a) 34.9 W/m^2.°C (b) 44.3 W/m^2.°C
(c) 72.8 W/m^2.°C (d) 81.5 W/m^2.°C
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : $\dot{Q} = E_{generated} = VI = 60 \times 1.5 = 90$ W

$A = \pi DL = \pi(0.003) \times (2)$

$= 0.0204$ m^2

$\dot{Q} = hA(T_s - T_\infty)$

$$h = \frac{Q}{A(T_s - T_\infty)} = \frac{90}{(0.0204) \times (152 - 15)}$$

$$\boxed{h = 34.9\,\text{w/m}^2\,^\circ\text{C}}$$

261. Heat is being transferred from fluid to fluid across a metal boundary of thickness 'x'. If h_a and h_b are heat transfer coefficients from hot fluid to metal surface and from metal surface to cold fluid respectively and R = thermal conductivity of metal wall, the overall heat transfer coefficient will be

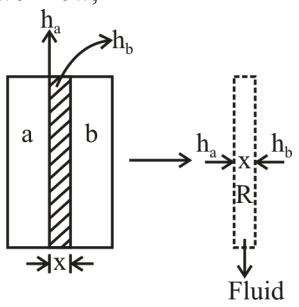
(a) $\dfrac{1}{\frac{1}{h_a} + \frac{x}{R} + \frac{1}{h_b}}$ (b) $\dfrac{1}{\frac{1}{h_a} + \frac{R}{x} + \frac{1}{h_b}}$

(c) $\dfrac{(t_a - t_b)}{\frac{1}{h_a} + \frac{x}{R} + \frac{1}{h_b}}$ (d) $\dfrac{(t_a - t_b)}{\frac{1}{h_a} + \frac{R}{x} + \frac{1}{h_b}}$

(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : As we know,



Overall heat transfer coefficient,

$$U = \frac{1}{\frac{1}{h_a} + \frac{x}{R} + \frac{1}{h_b}}$$

$$U = \frac{1}{R_{th}}$$

Overall thermal resistance

$$\therefore \quad R_{th} = \frac{1}{h_a A_s} + \frac{1}{h_b A_s} + \frac{x}{R A_s}$$

Where, R = Thermal conductivity.

262. A composite hollow sphere with steady internal heating is made of 2 layers of materials of equal thickness with thermal conductivities in the ratio of 1 : 3 for inner to outer layers. Ratio of inside to outside diameter is 0.75. What is ratio of temperature drop across the inner and outer layers?

(a) 1.2 (b) 4
(c) 0.8 (d) 2 In 0.4

UPRVUNL AE 21.08.2016

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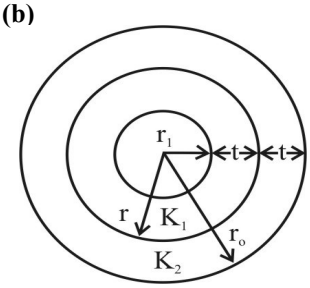
Ans. : (b)



$$R_{sp} = \frac{r_2 - r_1}{4\pi k r_1 r_2}$$

$$Q = \frac{\Delta T}{R_{sp}}$$

$$\frac{d_i}{d_o} = 0.75$$

$$\frac{r_i}{r_o} = 0.75$$

$$r_o - r_1 = 3t$$

$$3t = r_o - 0.75r_o$$

$$t = \frac{0.25r_o}{3}$$

$$t = 0.0833$$

$$\frac{k_1}{k_2} = \frac{1}{3}(\text{given})$$

$$\frac{(\Delta T)_{inner}}{(\Delta T)_{outer}} = \frac{R_{inner}}{R_{outer}}$$

$$\frac{k_2}{k_1} \times \frac{1}{0.75} = \frac{3}{0.75} = 4$$

263. Heat is conducted through a 20 cm thick wall at the rate of 40 W/m^2 when the temperature difference across the wall is 20^{0}C. What is the thermal conductivity of the wall?

(a) 4.0 W/mK (b) 0.4 W/mK
(c) 0.04 W/mK (d) 40.0 W/mK

UPRVUNL AE 21.08.2016

Ans. : (b) given data

$t = 20 \text{ cm} = 20 \times 10^{-2} \text{ m}$

$$\frac{\dot{Q}}{A} = 40 \text{w/m}^2$$

$$dT = 20^0\text{C}$$

$$\dot{Q} = KA\frac{dT}{dx}$$

$$\frac{\dot{Q}}{A} = \frac{K \times 20}{20 \times 10^{-2}}$$

$$K = \frac{40 \times 20 \times 10^{-2}}{20}$$

$$K = 0.4 \text{ W/mK}$$

264. If two ends of rods of length L and radius r, made of same material are kept at the same temperature difference, which of the following rods conduct most heat per unit time?

(a) L = 50 cm, r = 1 cm
(b) L = 2 cm, r = 0.5 cm
(c) L = 100 cm, r = 2 cm
(d) L = 3 cm, r = 1 cm

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Fourier Law–

$$\boxed{q = -kA\frac{dT}{dx}}$$

$$\boxed{q \propto \frac{A}{dx}}$$

So, option check,

(a) $q \propto \left(\frac{r^2}{50}\right)$

$$q \propto \frac{(1)^2}{50} = 0.02$$

(b) $q \propto \left[\frac{(0.5)^2}{2}\right]$

$= 0.125$

(c) $q \propto \left[\frac{(2)^2}{100}\right]$

$= 0.04$

(d) $q \propto \left(\frac{(1^2)}{3}\right)$

$= 0.33$

265. Critical thickness rate from a yields

(a) No heat transfer rate from a pipe
(b) Minimum heat transfer rate from a pipe
(c) Maximum heat transfer rate from a pipe
(d) None of the above

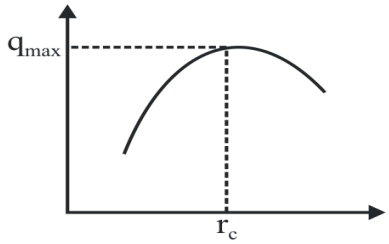
ISRO Scientist/Engineer 03.07.2016

Ans. (c) : When the total thermal resistance is made of conductive thermal resistance ($R_{cond.}$) and convective thermal resistance ($R_{conv.}$) the addition of insulation in some cases, may reduces the convective thermal resistance due to increase in surface area, as in the case of cylinder and sphere, and the total thermal resistance may actually decreases resulting in increased heat flow.

"Critical thickness upto which heat flow increases and after which heat flow decreases is termed as critical thickness.

For cylinder $\boxed{r_c = \frac{k}{h}}$

For sphere $\boxed{r_c = \frac{2k}{h}}$



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266. The rate of heat flow through a hollow sphere of inner radius 0.25 m and outer radius 0.35m, whose thermal conductivity is 5 W/mK, maintained at temperatures of 400°C and 300°C respectively equal to

(a) 2425 W (b) 5495 W
(c) 2747.5 W (d) 4850 W

APPSC AEE MAINS 2016 PAPER-III

Ans. (b) : Data given as,
$r_i = 0.25m$, $r_0 = 0.35m$ $k = 5$ w/mk,
$T_i = 400°C$ $T_0 = 300°C$
We know that
The rate of heat flow through hollow sphere,

$$Q = \frac{4\pi r_i r_\circ k(T_i - T_\circ)}{r_\circ - r_i}$$

$$Q = \frac{4 \times \frac{22}{7} \times 0.25 \times 0.35 \times 5 \times [400 - 300]}{0.35 - 0.25}$$

$$Q = 5500W \approx 5495\,W$$

267. The maximum temperature raised in the solid sphere with internal heat generation is given by

(a) $T_w + Q_g R^2/6K$ (b) $T_w - Q_g R^2/6K$
(c) $T_w + Q_g/6K$ (d) $T_w - Q_g/6K$

APPSC AEE MAINS 2016 PAPER-III

Ans. (a) : The maximum temperature raised in the solid sphere with internal heat generation is given by
$T_w + Q_g R^2/6K$.

268. Thermal resistance for hollow cylinder is given as (where A_m = logarithmic mean area of cylinder. r_o, r_i, L = Outer radius, inner radius and length of cylinder respectively and k = Thermal conductivity):

(a) $\frac{r_o - r_i}{kA_m}$ (b) $\frac{r_o - r_i}{4\pi k(r_o \times r_i)}$
(c) $\frac{L}{kA}$ (d) $\frac{r_o - r_i}{k}$

UPRVUNL AE 2016

Ans. (a) : Thermal resistance for hollow cylinder is given as

$$R_{th} = \frac{(r_o - r_i)}{kA_m}$$

269. Match list-I with list-II select the correct Answer using the codes given below the list:

List -I	List -II
(a) Momentum transfer	1. Thermal diffusivity
(b) Mass transfer	2. Kinematic viscosity
(c) Heat transfer	3. Diffusion coefficient

Code

	A	B	C
(a)	2	3	1
(b)	1	3	2
(c)	3	2	1
(d)	1	2	3

UJVNL AE 2016

Ans : (a) List -I	**List -II**
i) Momentum transfer	Kinematic viscosity
ii) Mass transfer	Diffusion coefficient
ii) Heat transfer	Thermal diffusivity

270. A high value of thermal diffusivity represents

(a) High storage, less conduction of heat
(b) Less storage, more conduction of heat
(c) There is always equal amount of conduction and storage since it is a property
(d) It has no relevance

UPPSC AE 12.04.2016 Paper-II

Ans : (b) A high value of thermal diffusivity represent less storage, more conduction of heat.
Thermal diffusivity:- Thermal diffusivity indicates the ease at which energy get diffused in the volume of the substance. It is defined as the ratio of thermal conductivity to the heat capacity of a substance

$$\alpha = \frac{K}{\rho . C_p}$$

Hence it is directly proportional to the thermal conductivity.

271. Unsteady state of heat flow occurs in

(a) Flow of heat through furnace walls
(b) Flow of heat through insulated pipe with constant surface temperature
(c) Annealing of castings
(d) Flow of heat through refrigerator walls

UPPSC AE 12.04.2016 Paper-II

Ans : (c) Unsteady state of heat flow occur in annealing of casting

272. A 30 mm OD pipe is to be insulated with asbestos having a thermal conductivity of 0.1 W/mK. The convective heat transfer coefficient is 5 W/m^2/K. The critical radius of insulation for this pipe would be :

(a) 10 mm (b) 20 mm
(c) 40 mm (d) 60 mm

HPPSC W.S. Poly. 2016

Ans : (b) Thermal Conductivity (K)= 0.1W/mK
Heat transfer coefficient (h) = 5W/m^{2}K.
Critical thickness of insulation for cylinder

$$r_c = \frac{K}{h}$$

$$r_c = \frac{0.1}{5}\,m, \qquad \boxed{r_c = 20mm}$$

273. With the rise in temperature, the thermal conductivity of solid metals :

(a) Increases (b) Decreases
(c) Remains the same (d) None of these

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (b) : Thermal conductivity of metal decreases with increase in temperature except mercury, Aluminium and Uranium.

- Thermal conductivity is independent of pressure except in extreme cases as for e.g. vacuum.
- Thermal conductivity of gas increases with increasing temperature and decreasing molecular weight.

$$k \propto \sqrt{\frac{T}{m}}$$

- Thermal conductivity of liquid can decrease with temperature except water, glycerin.

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274. In a cylinder under steady state conduction with uniform heat generation, the temperature gradient at half the radius location will be

(a) One half of that at surface
(b) One fourth of that at surface
(c) Twice that at surface
(d) Four times that at surface

UPPSC AE 12.04.2016 Paper-II

Ans : (a) In a cylinder under steady state conduction with uniform heat generation the temperature gradient at half the radius location will be one half of that at surface.

275. A beaker filled with hot water in a room cools from 70°C to 65°C in t_1 minutes, 65°C to 60°C in t_2 minutes and from 60°C to 55°C in t_3 minutes. Then,

(a) $t_1 > t_2 > t_3$ (b) $t_1 = t_2 = t_3$
(c) $t_1 < t_2 < t_3$ (d) cannot be concluded

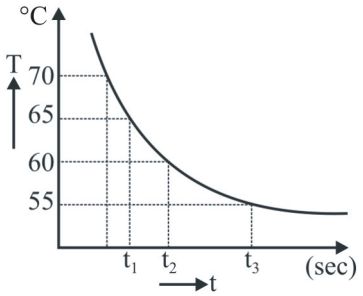
ISRO Scientist/Engineer 03.07.2016

Ans. (c) : "Newton Cooling Law"–

$\Delta T \propto e^{-kt}$

t = time

$t_1 < t_2 < t_3$



276. For the quick response of a thermocouple

(a) Its wire diameter should be large
(b) The convective heat transfer coefficient should be high
(c) The specific heat should be high
(d) The density should not be very small

UPPSC AE 12.04.2016 Paper-II

Ans. : (b) For the quick response of a thermocouple the convective heat transfer coefficient should be high.

277. A composite walls of two layers of thickness Δx_1, Δx_2 and thermal conductivities k_1 and k_2 having cross-sectional area A normal to the path of heat flow. If the wall surface temperatures are at t_1 and t_3, the rate of heat flow Q is equal to

(a) $\dfrac{(t_1 - t_3)}{\left(\dfrac{\Delta x_1}{k_1}\right) + \left(\dfrac{\Delta x_2}{k_2}\right)}$

(b) $\dfrac{A(t_1 - t_3)}{\left(\dfrac{\Delta x_1}{k_1}\right) + \left(\dfrac{\Delta x_2}{k_2}\right)}$

(c) $\left(\dfrac{Ak_1k_2(t_1 - t_3)}{\Delta x_1 + \Delta x_2}\right)$

(d) $\left(\dfrac{Ak_1 + Ak_2)(t_1 - t_3)}{\Delta x_1 + \Delta x_2}\right)$

APPSC AEE Screening Test 2016

Ans. (b) : A composite walls of two layers of thickness (Δx_1 and Δx_2), the rate of heat flow (Q) = $\dfrac{A(t_1 - t_3)}{\left[\dfrac{\Delta x_1}{k_1} + \dfrac{\Delta x_2}{k_2}\right]}$

278. Insulation in cavity walls is

(a) Cork (b) Wood
(c) Air (d) Water

APPSC AEE Screening Test 2016

Ans. (c) : Cavity wall insulation is used to reduce heat loss through a cavity wall by filling the air space with material that inhibits heat transfer.
This immobilises the air within the cavity, preventing convection, and conduction substantially reduce space heating costs.

279. What should be the shape of an ideal thermometer bulb?

(a) cylindrical (b) elliptical
(c) spherical (d) circular

GWSSB DEE 07.07.2016

Ans. (c) :

280. Which of the following does not pertain to transient heat conduction.

(a) Biot number (b) Fourier number
(c) Heisler chart (d) Interchange factor

TSPSC Managers, 2015

Ans. (d) : Biot Number (Bi), Fourier number (Fo) and Heisler chart is a transient (or unsteady state) heat conduction but interchange factor is not a transient heat conduction.

281. Match List-I with List-II and select the correct answer :

	List-I		List-II
A.	Regenerative heat exchanger	1.	Water cooling tower
B.	Direct contact heat exchanger	2.	Lungstrom air heater
C.	Conduction through a cylindrical wall	3.	Hyperbolic curve
D.	Conduction through a spherical wall	4.	Logarithmic curve

	A	B	C	D
(a)	1	4	2	3
(b)	3	1	4	2
(c)	2	1	3	4
(d)	2	1	4	3

Nagaland PSC CTSE 2015 Paper-II
ESE 1995

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Ans. (d) :

Regenerative heat exchanger	–	Lungstrom air heater
Direct contact heat exchanger	–	Water cooling tower
Conduction through a cylindrical wall	–	Logarithmic curve
Conduction through a spherical wall	–	Hyperbolic curve

282. Which of the following is not a desirable property of good insulating material

(a) Light weight (b) High heat conductivity
(c) Odourless (d) Low initial cost

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) : High heat conductivity is not a desirable property of good insulating material.
Desirable properties of insulating materials–
(1) Low thermal conductivity
(2) Chemical stability
(3) Non-combustible
(4) Low cost
(5) Light weight
(6) Fire proof
(7) Water repellent

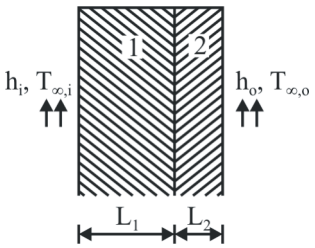
283. Fundamental laws used in heat transfer are

(a) The laws of conservation of mass
(b) Newton's laws of motion
(c) Laws of thermodynamics
(d) All of the above

TSGENCO AE 14.11.2015

Ans. (d) : All of the above

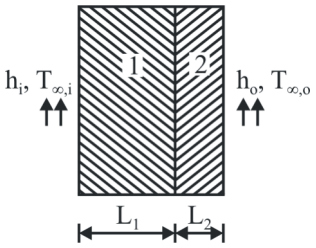
284. Consider steady state heat conduction across the thickness in a plane composite wall as shown in the figure exposed to convection conditions on both sides. Given : h_i = 20 W/m^{2}K; h_0 = 50 W/m^{2}K; $T_{\infty,i}$ = 20°C; $T_{\infty,0}$ = – 2°C; k_1 = 20 W/mK; k_2 = 50 W/mK; L_1 = 0.30 m and L_2 = 0.15 m. Assuming negligible contact resistance between the wall surfaces, the interface temperature, T (in °C) of the two walls will be



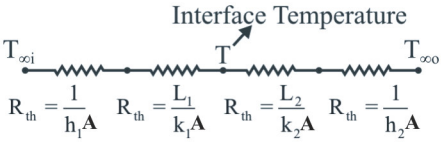
(a) 0.5 (b) 2.75
(c) 3.75 (d) 4.5

ISRO Scientist/Engineer 11.10.2015

Ans. (c) :



Network Diagram–



$$\therefore \quad \frac{T_{\infty i} - T}{\frac{1}{h_1} + \frac{L_1}{k_1}} = \frac{T - T_{\infty o}}{\frac{L_2}{k_2} + \frac{1}{h_2}}$$

$$\frac{20 - T}{\frac{1}{20} + \frac{0.3}{20}} = \frac{T + 2}{\frac{0.15}{50} + \frac{1}{50}}$$

$$\frac{400 - 20T}{1.3} = \frac{50T + 100}{1.15}$$

$$460 - 26T = 65T + 130$$

$$330 = 88T$$

$$T = 3.75\ °C$$

285. The radius of a bare cable carrying electric current before providing insulation should be _____.

(a) Equal to r_c (b) Less than r_c
(c) Greater then r_c (d) Twice of r_c

TSPSC AEE 2015

Ans. (b) : The radius of a bare cable carrying electric current before providing insulation should be less than r_c.

286. A body cooling from 80°C to 70°C takes 10 minutes when left exposed to environmental conditions. If the body is to cool further from 70°C to 60°C under the same external conditions, it will take

(a) more than 10 minutes
(b) same time of 10 minutes
(c) less than 10 minutes
(d) none of these

(KPSC AE 2015)

Ans : (a) According to Newton's law of cooling:-
Rate of heat loss ∝ Temperature difference

287. Fin effectiveness equals

(a) $\dfrac{\text{heat transfer rate from fin surface}}{\text{heat transfer rate from a standard fin but of the same size}}$

(b) $\dfrac{\text{heat transfer rate from fin surface}}{100}$

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(c) $\dfrac{\text{heat transfer rate from fin surface}}{\text{heat transfer rate from an identical fin of infinite thermal conductivity}}$

(d) $\dfrac{\text{heat transfer rate from fin surface}}{\text{heat transfer rate from an identical made of copper}}$

TSGENCO AE 14.11.2015

Ans. (*) : Fin effectiveness

$$\in_{fin} = \frac{q_{withfin}}{q_{without\ fin}}$$

288. Your finger sticks to an ice tray just taken from the refrigerator. Which factor has more effect on this phenomenon?

(a) The inside temperature of the freezer
(b) The humidity of the air
(c) The heat capacity of both your finger and the tray
(d) The thermal conductivity of the tray

ISRO Scientist/Engineer 11.10.2015

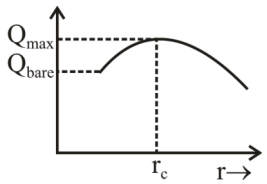
Ans. (d) : Heat is transferred between a system and its environment when their temperature differs.
Energy is transferred by thermal conduction. Metal is good conductor of heat. When you touch the metal ice tray, heat will transfer from your fingers to the bit of ice and the ice melts results a thin film of water between the finger and the metal tray.

289. The critical radius of insulation signifies

(a) The insulation thickness above which the heat transfer is increased
(b) The insulation thickness above which the heat transfer remains unaltered
(c) The insulation thickness below which the electric cable gets overheated and rails
(d) The insulation thickness above which the heat transfer is reduced

TANGEDCO AE 2015

Ans. (d) :



Increasing radius of insulation after critical thickness or radius decreases heat transfer.

290. If the diameter of a conductor is doubled then its resistance will be reduced to

(a) Half (b) One-fourth
(c) One-eighth (d) $1/\sqrt{2}$ time

GPSC AE 26.07.2015

Ans. (b) : $R = \rho\dfrac{l}{A}$

$$R \propto \frac{1}{A} \propto \frac{1}{d^2}$$

$$\frac{R_2}{R_1} = \left(\frac{d_1}{d_2}\right)^2$$

$$R_2 = \left(\frac{1}{2}\right)^2 \times R_1$$

$$R_2 = \frac{R_1}{4}$$

291. All of the following are units of thermal conductivity except :

(a) kcal/m-hr-^{0}C (b) kj/m-hr-k
(c) W/m-s-k (d) cal/cm-s-^{0}C

Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) :

292. A steam engine intakes 100 gm of steam at 100ºC per minute and cools it down to 20ºC. Calculate the heat rejected by the steam engine per minute (consider Latent Heat of vaporization of steam = 540 cal/gm and c_p = 1 cal/gmºC)

(a) 6.2×10^4 kcal (b) 62×10^4 cal
(c) 6.2×10^4 cal (d) 620×10^4 kcal

ISRO Scientist/Engineer 24.05.2014

Ans. (c) : Heat rejected by steam per minute:

$$Q_R = mc_p\Delta T + mL$$
$$= 100 \times 1 \times (100 - 20) + 540 \times 100$$
$$= 100 \times 80 + 54000$$
$$= 62000\,cal = 6.2 \times 10^4\,cal.$$

293. The highest thermal diffusivity is of

(a) Iron (b) Lead
(c) Concrete (d) Wood

APGENCO AE, 2017
GPSC Lect. (Poly.) 07.12.2014

Ans. (b) : Thermal diffusivity of substance

$$\alpha = \frac{k}{\rho c_p}\ m^2/s$$

k = Thermal conductivity of substance (w/m.k)
ρ = density (kg/m^3)
c_p = sp. heat of substance (J/kg.k)

i.e higher the thermal conductivity and lower sp. heat capacity storage has higher thermal diffusivity.

for iron, $\alpha = \dfrac{46.33}{26 \times 0.12} = 14.8493$

for lead, $\alpha = \dfrac{20.39}{41 \times 0.032} = 15.5412$

hence lead has high thermal diffusivity

294. The fin efficiency is defined as the ratio of actual heat transfer from the fin to

(a) The heat transfer from the same fin with an adiabatic tip
(b) The heat transfer from an equivalent fin which is infinitely long
(c) The heat transfer from the same fin if the temperature along the entire length of the fin is the same as the base temperature

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(d) The heat transfer through the base area of the same fin

TNPSC AE 2014

Ans. (c) : The fin efficiency is defined as the ratio of actual heat transfer from the fin to the heat transfer from the same fin if the temperature along the entire length of the fin is the same as the base temperature.

$$\eta_{fin} = \frac{\sqrt{PhkA_{CS}}\,(T_o - T_a)}{hPl(T_o - T_a)}$$

$$\eta_{fin} = \frac{1}{ml}$$

295. The term 'lumped systems' in heat transfer refers to system having

(a) heat transfer varying with time and position
(b) heat transfer varying with time but not with position
(c) heat transfer varying with position but not with time
(d) heat transfer varying with position only
(e) heat transfer varying with time only

CGPSC AE 2014 -II

Ans. (b) : The term 'Lumped systems' in heat transfer refers to system having heat transfer varying with time but not with position.

296. A finned surface consists of root or base area of 2 W/m^2 and fin surface area of 2m^2. The average heat transfer coefficient for finned surface is 32 W/m^{2}K, effectiveness of fins provided is 0.65. If finned surface with root or base temperature of 55^{0}C is transferring heat to a fluid at 35^{0}C, then what is the rate of heat transfer?

(a) 448 W (b) 825 W
(c) 1050 W (d) 1280 W

UPRVUNL AE 2014

Ans. : (b) Given that Root area = 2m^2, and h = 32 W/m^{2}K,

Heat transfer without fin = $hA(T_0 - T_\infty) = 32 \times 2 \times (55–35) = 1280$

$$0.65 = \frac{Q_{Fin}}{1280}$$

$$Q_{Fin} = 1280 \times 0.65 = 832W$$

Here, which is close to option (b).

Note :– The commission has given answer option (d).

297. For which of the given conditions heat transfer from the insulated tip can be considered the case of fin of finite length?

(where m is slope of differential equation and L is length of fin).

(a) m = 0.75, L = 3 (b) m = 1, L = 3
(c) m = 3, L = 0.72 (d) m = 2, L = 1.2

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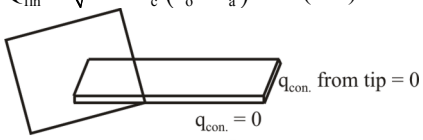
Ans. : (b) If that loss is assumed to be negligible at its length and tip at tin insulated.

at x = L

$$Q = 0 \Rightarrow \frac{dQ}{dx} = m$$

$$Q_{fin} = \sqrt{PhkA_c}\,(t_o - t_a)\tanh(mL)$$



We will check the value of tanh(mL) for tall the given condition & the condition which is having value will have the higher heat transfer.

(i) m = 0.75, L = 3, tanh(mL) = 0.9780
(ii) m = 1, L = 3, tanh(mL) = 0.9550
(iii) m = 3, L = 0.72, tanh(mL) = 0.9737
(iv) m = 3, L = 1.2, tanh(mL) = 0.9836

Therefore the condition having m = 1 & L = 3 will have the higher heat transfer.

298. A copper wire of radius 0.65 mm is insulated with a sheathing of thickness 1.25 mm having a thermal conductivity of 0.7 W/mK. The outside surface convective heat transfer coefficient is 10 W/m^{2}K. If the thickness of insulation sheathing is raised by 15 mm, then what will be the result on electrical current-carrying capacity of the wire?

(a) It will increase
(b) It will decrease
(c) It will remain same
(d) will vary depending upon the electrical conductivity of the wire

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Ans : (a) We know

for cylinder Critical Radius (r_c) = $\frac{k}{h}$

$$= \frac{0.7}{10} = 0.07m = 70mm$$

Hence, heat transfer rate increases up to the r_c.

299. Which of the following phenomena is responsible for the heat transfer through the walls of steel radiator?

(a) Conduction only
(b) Convection only
(c) Radiation only
(d) Conduction and convection

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Ans : (a) The Heat transfer through wall of steel radiator through conduction.

Conduction heat transfer take palce from solid surface to solid surface

$$Q = \frac{kA\,dt}{dx}$$

300. A flat plate has thickness 6 cm, thermal conductivity 1 W/m-K, convective heat transfer coefficient on its two flat faces are of 50 W/m^2-K and 20 W/m^2-K determine it's overall heat transfer coefficient.

(a) 6.33 W/m^2 – K (b) 7.69 W/m^2 – K
(c) 20 W/m^2 – K (d) 30 W/m^2 – K

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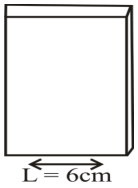
Ans : (b) U = Overall heat transfer coefficient

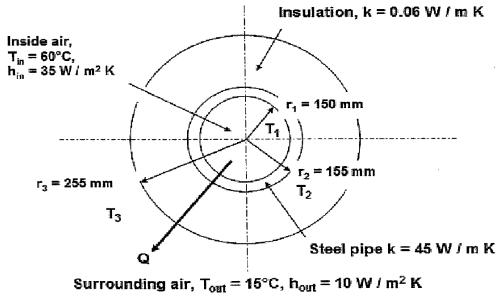


$$U = \frac{1}{\frac{1}{h_{hf}} + \frac{L}{k} + \frac{1}{h_{cf}}}$$

$$U = \frac{1}{\frac{1}{50} + \frac{6}{100} + \frac{1}{20}} = \frac{100}{13} = 7.69\ W/m^2K$$

$$U = 7.69\ W/m^2K$$

301. The figure below shows a cross section through an insulated heating pipe which is made from steel (k = 45 W/m K) with an inner radius of 150mm and outer radius of 155mm. The pipe is coated with 100mm thickness of insulation having thermal conductivity of k = 0.06 W/m K. Air at inlet temperature T_i = 60°C flow through the pipe has a value of convective coefficient h_i = 35 W/m²k. The outside surface of the pipe is surrounded by air which is at 15°C and convective heat transfer coefficient on this surface has a value of ho = 10 W/m²K. Find the critical insulation radius?



(a) 14.00 mm (b) 3.00 mm
(c) 20.00 mm (d) 6.0 mm

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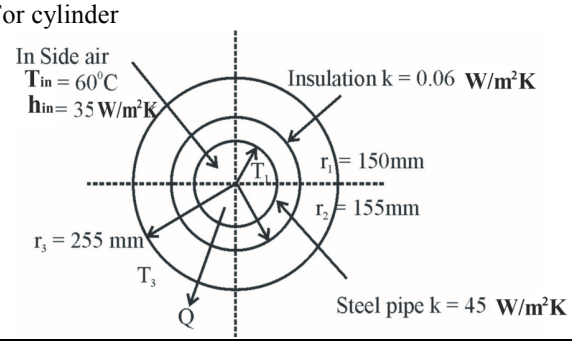
Ans. (d) : If two or more Insulated material have joined i.e. one insulated layer to another insulated layer then heat transfer rate considering or critical radius has calculated from other layer of Insulated material.

Surrounding air, T_{out} = 150°C, h_{out} = 10 W/m²K

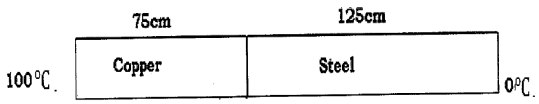
Critical Insulation Radius

For cylinder



$$r_c = \frac{k}{h} = \frac{k_{outer}}{h_{outer}} = \frac{0.06}{10} = 0.006m = 6mm$$

302. A bar of Cu of length 75 cm and a steel bar of 125 cm length are joined together end to end. Both the bars are having same diameter of 2 cm. The free ends of the Cu and the steel bars are maintained at 100°C and 0°C respectively. The curved surfaces of the bars are thermally insulated, what is the approximate temperature at Cu-steel junction? (consider conductivity K of Cu = 400 $Js^{-1}m^{-1}{}^oC^{-1}$ and K for steel 50 $Js^{-1}m^{-1}{}^oC^{-1}$)



(a) $\approx 95.0^oC$ (b) $\approx 93.0^oC$
(c) $\approx 90.0^oC$ (d) $\approx 89.0^oC$

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Ans. (b) :

Copper	Steel
T = 100°C	T = 0°C
L = 75cm	L = 125cm

Let Interface temperature 'T'

or $\frac{k_{cu}(100-T)}{75} = \frac{k_{st}(T-0)}{125}$

or $\frac{400(100-T)}{75} = \frac{50(T)}{125}$

or 5.33 (100-T) = 0.4 T

533 = (5.33 + 0.4) T

T = 93°C

303. $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ **represents a :**

(a) Wave equation
(b) One-dimensional heat flow equation
(c) Two dimensional heat flow equation
(d) Radio equation

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ represents one – dimensional heat flow equation.

304. An aluminium (k = 240 W/mK) electric cable of diameter 5 mm is to be insulated with a material of k = 0.04 W/mK. If the cable is located in air (h = 10 W/m²K), the optimum thickness of insulation will be

(a) 1.5 mm (b) 8 mm
(c) 3 mm (d) 4 mm

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Ans. (a) : Given

$$\text{Critical radius} = \frac{k}{h}$$

where

k = insulation thermal conductivity

h = heat transfer coefficient

$$r_c = \frac{0.04}{10} = 0.004 = 4\text{mm}$$

d = 5mm or r = 2.5 mm

Critical thickness $t_c = r_c - r = 4 - 2.5 = 1.5\text{mm}$

305. Which of the following statements about fins is wrong?

(a) Fin effectiveness is high with a material of high thermal conductivity.

(b) Use of thin, but closely spaced fins is preferred in most engineering applications.

(c) Use of fins can be better justified under conditions for which the convection coefficient is large.

(d) The need for fin is stronger when the surface heat transfer is by free convection.

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Ans. (c)

$$\text{Effectiveness } (\varepsilon) = \sqrt{\frac{Pk}{hA}}$$

- $\varepsilon \propto \sqrt{k}$ – if we select high thermal conductivity material they are highly effective
- $\varepsilon \propto \frac{1}{\sqrt{h}}$ – we should keep small value as much as possible
- $\varepsilon \propto \sqrt{\frac{P}{A}}$ – Therefore thin & closely spaced fins are selected.

306. Consider following units

1. J/m^2s
2. J/mKs
3. W/mK
4. W/m^2K
5. Kcal/hr.m.^{0}C

Of the above, which represent Thermal Conductivity in S.I. unit

(a) 1 and 5 (b) 2 and 5

(c) 1 and 2 (d) 2 and 3

(e) 3 and 4

CGPSC AE 2014 -II

Ans. (d) : We know that,

unit of thermal conductivity in S.I. unit

$$= \frac{W}{m.K} = \frac{J/s}{m.K} = \frac{J}{m.K.s}$$

307. A wall has two layers of materials A and B; each made of a different material. Both the layers have the same thickness. The thermal conductivity of material A is twice that of B. Under the equilibrium, the temperature difference across the wall is 36^{0}C. The temperature difference across the layer B is

(a) 6^{0}C (b) 12^{0}C

(c) 14^{0}C (d) 18^{0}C

(e) 24^{0}C

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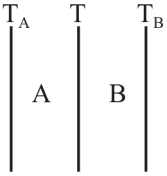
Ans. (e) : Given, $T_A - T_B = 36°C$

$k_A = 2k_B$, $L_A = L_B = L$



Heat conducted through B layer = Heat conducted across the wall

$$\frac{T - T_B}{\frac{L}{k_B}} = \frac{T_A - T_B}{\frac{L}{k_A} + \frac{L}{k_B}}$$

$$\frac{T - T_B}{\frac{L}{k_B}} = \frac{36}{\frac{L}{2k_B} + \frac{L}{k_B}} = \frac{36}{\frac{3L}{2k_B}}$$

$$\frac{T - T_B}{\frac{L}{k_B}} = \frac{36}{\frac{3L}{2k_B}}$$

$$T - T_B = \frac{36 \times 2}{3}$$

Temperature difference across layer B

$$T - T_B = 24°C$$

308. Following ways have been suggested to increase the effectiveness of an infinitely long fin.

A. Using fin material of high thermal conductivity

B. Increasing the ratio of area to perimeter of cross section of the fin.

C. Increasing the convective heat transfer coefficient.

D. Increasing the ratio of perimeter to area of cross section of the fin.

Which of the above actions will all increase the effectiveness of the fin?

(a) (A) and (D) only (b) (A), (B) an (C)

(c) (B) and (C) only (d) (A) and (B) only

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Ans. (a) : Increasing the effectiveness of an infinitely long fin.

$$\varepsilon_{fin} = \frac{\tanh(mL)}{\sqrt{\frac{hA}{Pk}}}$$

$$\boxed{\varepsilon_{fin} \propto \sqrt{k}}$$

$$\boxed{\varepsilon_{fin} \propto \frac{1}{\sqrt{h}}}, \quad \boxed{\varepsilon_{fin} \propto \sqrt{\frac{P}{A}}}$$

- $\varepsilon_{fin} \uparrow$ with $k \uparrow$
- $\varepsilon_{fin} \uparrow$ with $\left(\frac{P}{A}\right) \uparrow$
- $\varepsilon_{fin} \uparrow$ with $h \downarrow$

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309. Following ways have been suggested to increase the effectiveness of an infinitely long fin.

A. Using fin material of high thermal conductivity

B. Increasing the ratio of area to parameter of cross section of the fin.

C. Increasing the convective heat transfer coefficient.

D. Increasing the ratio of parameter to area of cross section of the fin.

Which of the above actions will all increase the effectiveness of the fin?

(a) (A) and (D) only (b) (A), (B) an (C)

(c) (B) and (C) only (d) (A) and (B) only

RPSC AEN Pre-2013

Ans. (a) : Increasing the effectiveness (ε) of an infinitely long fin.

$$\varepsilon_{fin} = \frac{\tan h\, mL}{\sqrt{\frac{hA}{Pk}}}$$

$$\boxed{\varepsilon_{fin} \propto \sqrt{k}}$$

$$\boxed{\varepsilon_{fin} \propto \frac{1}{\sqrt{h}}}, \quad \boxed{\varepsilon_{fin} \propto \sqrt{\frac{P}{A}}}$$

- $\varepsilon_{fin} \uparrow$ with $k \uparrow$
- $\varepsilon_{fin} \uparrow$ with $\left(\frac{P}{A}\right) \uparrow$
- $\varepsilon_{fin} \uparrow$ with $h \uparrow$

310. With the help of fins, the heat transfer can be increased by about:-

(a) 4 times (b) 6 times

(c) 8 times (d) 10 times

UKPSC AE-2013, Paper-II

Ans. (d) :

311. With the increase in the length of fin, fin efficiency:-

(a) Decreases

(b) Increases

(c) Remains unaffected

(d) First increases and then decreases

UKPSC AE-2013, Paper-II

Ans. (a) : With the increase in the length of fin, fin efficiency decreases.

$$\boxed{\eta_{fin} = \frac{\tanh(mL_c)}{mL_c}}$$

312. A pipe carrying hot fluid is to be insulated by two layers of insulators, A and B, having thermal conductivities k and 2k respectively an of equal radial thickness. If the heat transfer coefficient from the outer surface to the air remains constant, how the two insulators must be placed to minimise the heat transfer?

(a) Material A must be outside.

(b) Material B must be outside

(c) The heat transfer will be same regardless of their relative placement.

(d) The order of placement will depend upon the outside heat transfer coefficient.

RPSC AEN Pre-2013

Ans. (b) : Given $k_A = k$

$k_B = 2k$

According Fourier's law of equation –

$$Q_A = kA\frac{dT}{dx}$$

$$Q_B = 2kA\frac{dT}{dx}$$

Here, $Q_B > Q_A$, because $k_B > k_A$ then material B must be outside.

313. A pipe of 40 mm outside radius carrying hot fluid in a factory is to be insulated by a layer of insulator having thermal conductivity 0.5 W/m°C. The heat transfer coefficient for convection from pipe surface (bare as well as insulated) to the air is 10 W/m²°C. The objective of providing insulation is to reduce the heat loss from the pipe. Choose the correct statement about the thickness of insulation to be provided.

(a) Any thickness of insulator will always reduce heat loss.

(b) Any thickness of insulator will always reduce heat loss.

(c) Heat loss will be minimum if the outside radius of insulation is 50 mm.

(d) In order to achieve a heat loss less than that obtained with the bare pipe, the outside radius of insulation has to be more than a certain value that requires calculations but it will be definitely greater than 50 mm

RPSC AEN Pre-2013

Ans. (d) : Given,

pipe radius

$r_p = 40$ mm

$k = 0.5$ w/m°C

$h = 10$ w/m²°C

Critical radius $(r_c) = \frac{k}{h} = \frac{0.5}{10}$

$= 0.05 \times 1000$

$= 50$ mm

Here, pipe radius (40mm) less than critical radius (50 mm) of insulation. In order to achieve a heat loss less than that obtained with the bare pipe. The outside radius of insulation has to be more than a certain value that requires calculation but it will definitely greater than 50 mm.

314. Which one of the following is electrically most conductive?

(a) Copper (b) Silver

(c) Aluminium (d) Gold

UKPSC AE-2013, Paper-I

Ans. (b) : Silver is electrically most conductive.

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315. For given combined radioactive and convective heat transfer coefficient 'h_t' and given thermal conductivity k. Critical thickness of insulation for cylinder and sphere is given as

(a) $\frac{k}{h_t}$ and $\frac{k}{h_l^2}$ (b) $\frac{k}{h_t}$ and $\frac{2k}{h_l}$

(c) $\frac{2k}{h_t}$ and $\frac{k}{h_l^2}$ (d) $\frac{2k}{h_t}$ and $\frac{k}{h_t}$

ISRO Scientist/Engineer 12.05.2013

Ans. (b) : Convective heat transfer coefficient = h_t

Thermal conductivity = k

For, cylinder

Critical thickness of Insulation

$$t_c = \frac{k}{h_t}$$

For, sphere

Critical thickness of Insulation

$$t_c = \frac{2k}{h_t}$$

316. In heat transfer, conductance equals conductivity (Kcal/hr/m^2/^{o}C/cm) divided by

(a) hr (Time) (b) m^2 (Area)

(c) ^{o}C (Temp) (d) cm (Thickness)

SJVN ET 2013

Ans. (d) : cm (Thickness)

317. The thermal diffusivities for solids are generally.

(a) Less than those for gases

(b) Less than those for liquids

(c) More than those for liquid and gases

(d) Same as for gases

SJVN ET 2013

Ans. (c) : Thermal diffusivity $(\alpha) = \frac{k}{\rho c}$

$$(k)_{solid} > (k)_{liq} > (k)_{gas}$$

$$\alpha_{solid} > \alpha_{liq} > \alpha_{gas}$$

Thermal diffusivity for solids are generally more than those for liquid and gases.

318. A body cools from 90°C to 70°C in 5 minutes. The time required by body for further cooling to 50°C will be:-

(a) 5 minutes

(b) Less than 5 minutes

(c) More than 5 minutes

(d) 10 minutes

UKPSC AE-2013, Paper-II

Ans. (c) :

319. Which of the following statements is not true for diamond?

(a) It is hardest known material

(b) Diamond is non-metallic

(c) It has high thermal conductivity

(d) It has a very high electrical conductivity

UKPSC AE-2013, Paper-I

Ans. (d) : It has a very low electrical conductivity.

320. Consider a cylinder of radius R with uniformly distributed heat source of heat generation rate and constant thermal conductivity K. Assume temperature as a function of radius (r) only. The boundary conditions are; T = T_w at r = R and heat generated equals heat lost at surface. The temperature distribution in the cylinder is given by:-

(a) $T = T_w + \frac{\dot{q}}{2K}\left(1 - \frac{r^2}{R^2}\right)$

(b) $T = T_w + \frac{\dot{q}}{4K}\left(1 - \frac{r^2}{R^2}\right)$

(c) $T = T_w + \frac{\dot{q}}{2K}\left(R^2 - r^2\right)$

(d) $T = T_w + \frac{\dot{q}}{4K}\left(R^2 - r^2\right)$

UKPSC AE-2013, Paper-II

Ans. (d) : $T = T_w + \frac{\dot{q}}{4K}\left(R^2 - r^2\right)$

321. A metal plate has a surface area of 2m^2 thickness 10 mm and a thermal conductivity of 200 W/mK. What is the thermal resistance of the plate?

(a) 4×10^4 K/W (b) 2.5×10^{-3} K/W

(c) 1.5×10^{-4} K/W (d) 2.5×10^{-5} K/W

TRB Poly. Lect., 2012, ESE 2006

Ans. (d) : A=2m^2, L (thickness) = 10 mm K= 200 W/mK

$$R_{thermal} = \frac{\ell}{KA} = \frac{10 \times 10^{-3}}{200 \times 2}$$

$R_{th} = 2.5 \times 10^{-5}$ K/W

322. Which of the following would not increase the rate of heat transferred from a heater pipe?

(a) Insulation with materials whose thickness is below critical thickness

(b) Blowing air over it

(c) Providing fins

(d) Putting heater pipe within another whose thermal conductivity is smaller in number and 2 inches thick

TRB Poly. Lect., 2012

Ans. (d) : Putting heater pipe with another whose thermal conductive is smaller in number and 2 inches thick.

323. One face of a copper plate, 3 cm thick, is maintained at 300°C, while the other face is at 100°C. If thermal conductivity of copper at the average temperature is 374 W/m-°C, the amount of heat transferred through the plate, in MW/m^2, is:

(a) 3.74 (b) 2.49

(c) 2.74 (d) 3.49

APGENCO AE 2012

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Ans. (b) : Given, Thickness (t) = 3 cm = 3×10^{-2}m

$T_1 = 300°C$, $T_2 = 100°C$

k = 374 W/m°C

$$Q = kA \times \frac{dT}{t}$$

$$\frac{Q}{A} = k \times \frac{dT}{t} \Rightarrow 374 \times \frac{(300-100)}{3\times10^{-2}}$$

$$= 374 \times \frac{200\times100}{3}$$

$$= 2493333.333 \text{ W/m}^2$$

$$= 2.49 \text{ MW/m}^2$$

324. The insulating ability of an insulator with the presence of moisture would

(a) increase (b) decrease

(c) remain unaffected (d) none of the above

BPSC AE 2012 Paper - V

Ans : (b) : The insulating ability of an insulator with the presence of moisture would decrease.

325. Metals are good thermal conductors since

(a) they have free electrons.

(b) their atoms are relatively closer.

(c) their surfaces reflect.

(d) their atoms are of larger size.

UKPSC AE 2012 Paper–II

Ans. (a) : they have free electrons.

326. If V = volume, A = surface area, h = surface film conductance, ρ = density and C = specific heat, then the time constant of a thermocouple is equal to

(a) $\frac{V\rho C}{Ah}$ (b) $\frac{V\rho}{CAh}$

(c) $\frac{Ah}{V\rho C}$ (d) $\frac{VC}{\rho Ah}$

UKPSC AE 2012 Paper–II

Ans. (a) : Time constant of thermo couple–

$$\frac{V\rho C}{Ah}$$

327. The rate of heat transfer through a hollow cylinder of inner and outer radii r_1 and r_2, respectively, depends on

(a) difference of radii, $(r_2 - r_1)$

(b) sum of radii, $(r_2 + r_1)$

(c) product of radii, $(r_1 r_2)$

(d) ratio of radii, $\left(\frac{r_2}{r_1}\right)$

UKPSC AE 2012 Paper–II

Ans. (d) : The rate of heat transfer through a hollow cylinder of inner and outer radii r_1 and r_2, respectively, depends on ratio of radii, $\left(\frac{r_2}{r_1}\right)$.

328. Which one of the following techniques will not improve effectiveness?

(a) By selecting the fin material which has high thermal conductivity

(b) By selecting the shape and size of fin section which has high ratio of the perimeter to the cross-section area.

(c) By selecting the medium and mode which results in high convective heat transfer coefficient.

(d) By selecting the medium and mode which results in low convective heat transfer coefficient.

RPSC AE GWD, 2011

Ans. (c) : Fin effectiveness $(\varepsilon) = \sqrt{\frac{Pk}{hA}}$

$$\varepsilon\uparrow \rightarrow \left(\frac{P}{A}\right)\uparrow \quad k\uparrow \quad h\downarrow$$

By selecting medium which results in high convective heat transfer coefficient fin effectiveness decreases.

329. Which one of the following techniques will not increase the rate of heat conduction through a plane wall?

(a) By using the material of wall having high thermal conductivity.

(b) By increasing the surface area of the wall

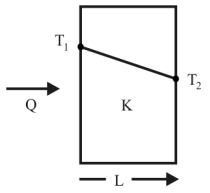
(c) By making the wall thinner

(d) By making the wall thicker

RPSC AE GWD, 2011

Ans. (d) : Rate of heat conduction through plane wall.

$$Q = \frac{kA[T_1 - T_2]}{L}$$



$$Q \propto k$$

$$Q \propto A$$

$$\downarrow Q \propto \frac{1}{L\uparrow}$$

If thickness of wall increases then rate of heat conduction decreases.

330. Thermal conductivity of a material may be defined as the :

(a) Quantity of heat flowing in one second through one centimeter cube of material when opposite faces are maintained at a temperature difference of 1°C.

(b) Quantity of heat flowing in one second through a slab of material of area one centimeter square, thickness 1 cm when its faces differ in temperature by 1°C

(c) Heat conducted in unit time across unit area through unit thickness, when a temperature difference of unity is maintained between opposite faces.

(d) All the above.

RPSC AE GWD, 2011

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Ans. (d) : From Fourier law of heat conduction

$$Q = -kA\left(\frac{dT}{dX}\right)$$

$$k = \frac{QdX}{AdT}$$

If $dT = 1$

$A = 1$ $\boxed{k = Q}$

$dX = 1$

331. Temperature of steam at around 540°C can be measured by:

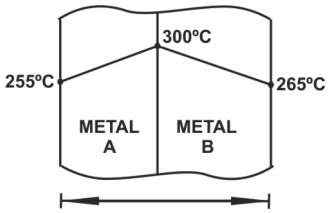
(a) Thermometer (b) Radiation pyrometer

(c) Thermocouple (d) Thermistor

RPSC AE GWD, 2011

Ans. (c) : Thermocouple can easily sense temperature of 540°C or even up to 1200°C. For measuring any temperature above 1200°C a radiation pyrometer type thermometer is used because it is contactless thermometer.

332. The temperature profile between two metal wall joined together is shown in Fig. From the figure it can be concluded that



(a) Heat flows from A to B

(b) Heat flows from B to A

(c) Heat is generated at the interface

(d) A is bad conductor of heat

ISRO Scientist/Engineer 2011

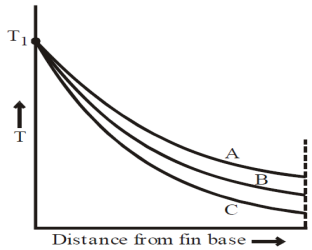
Ans. (c) : The figure shows that the maximum temperature is at the interface of the two metal walls.

$$Q = -kA\frac{dt}{dx}$$

And from the maximum temperature, the heat is transferred to both side.

Therefore, at the interface, we can conclude that heat is generated at the interface.

333.



Three fins of identical shape and size, each 10 cm long, are attached to a source at temperature T_1, while the surrounding temperature is T_0. A, B and C are the temperature profiles observed along the fins as shown in the figure. The fin materials are aluminium, copper and fibre glass. The correct sequence of A, B and C is

(a) Aluminium, fibreglass and copper

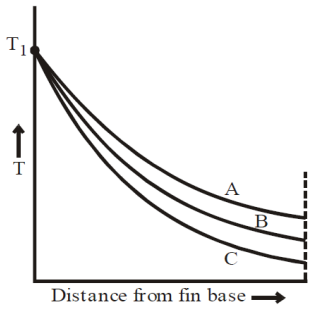
(b) Fibreglass, aluminium and copper

(c) Copper, aluminium and fibreglass

(d) Copper, fibreglass and aluminium

UPSC JWM Advt. No.-50/2010

Ans. (c) :



As we know,

Heat transfer, $Q = kA\frac{dT}{dx}$

or $\boxed{\frac{dT}{dx} \propto \frac{1}{k}}$

From above formula,

$$k_{cu} > k_{A\ell} > k_{\text{fibre glass}}$$

So, temperature difference

$$\therefore \quad k \propto \frac{1}{\Delta T}$$

$$T_{fb} > T_{A\ell} > T_{cu}$$

334. Which liquid metal can be taken as the best conductor ?

(a) Tin (b) Mercury

(c) Bismuth (d) Sodium

J&K PSC Civil Services Pre, 2010

Ans. (b) : Mercury is liquid at room temperature, It can be taken as best conductor.

Bismuth is the poor conductor of heat due to less number of force electrons.

335. A decrease in heat transfer coefficient over the surface of a fin :

(a) Decreases its effectiveness

(b) Increases its effectiveness

(c) Does not affect its effectiveness

(d) Its effectiveness will first increase and then decrease

J&K PSC Civil Services Pre, 2010

Ans. (b) : Fin effectiveness –

$$\varepsilon_{fin} = \frac{q_{\text{with fin}}}{q_{\text{without fin}}}$$

$$\varepsilon_{fin} = \frac{\tan h\, mL}{\sqrt{\frac{hA}{kP}}}$$

$$\boxed{\varepsilon_{fin} \propto \frac{1}{\sqrt{h}}}$$

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336. A composite slab has 3 layers of insulation of equal thickness with thermal resistances in the ratio of 1 : 2 : 3. What will be the ratio of respective temperature drops across the layers?

(a) 1 : 2 : 3 (b) 3 : 2 : 1
(c) 1 : 1.5 : 3 (d) 3 : 1.5 : 1

UPSC JWM Advt. No.-50/2010

Ans. (a) : As we know,

$$\text{Heat } (Q) = \frac{\Delta T}{\Sigma R_{th}}$$

$\Delta T \propto \sum R_{th}$ $\quad\therefore$ Q = Constant

So, for R_{th}, 1:2:3 then the change temperature is also is in form of 1:2:3.

337. Critical radius of insulation for a cylinder of diameter = r is given by :

(a) $\frac{h_0}{K}$ (b) $\frac{h_0}{K} - r$
(c) $\frac{K}{h_0}$ (d) $\frac{K}{h_0} - r$

UPSC JWM Advt. No.-52/2010

Ans. (c) : Solution- Critical radius (r) for a cylinder –

$$r = \frac{k}{h_0}$$

where, K = Thermal conductivity of insulating material

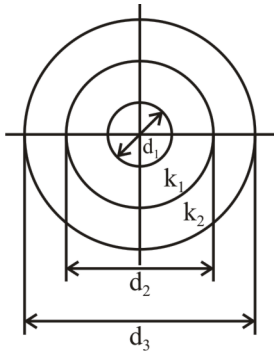
h_0 = Heat transfer coefficient of surrounding to the cylinder.

338. A composite hollow sphere with radial outflow of heat is made of two layers of materials of equal thickness with thermal conductivities in the ratio of 2 : 3 for inner and outer layers. Ratio of outer to inner diameter is 1.25. What will be the ratio of temperature drops across the inner layer to the outer layer?

(a) 0.833 (b) 1.2
(c) 1.875 (d) 2.34

UPSC JWM Advt. No.-50/2010

Ans. (c) : As we know,



$$\frac{k_1}{k_2} = \frac{2}{3}$$

$d_3 = 1.25\ d_1$

$d_2 - d_1 = d_3 - d_2$

So, $q = \dfrac{\Delta T}{\left(\dfrac{R_2 - R_1}{4\pi k R_1 R_2}\right)}$

$$\frac{\Delta T_1}{\Delta T_2} = \frac{q \times \dfrac{(R_2 - R_1)}{4\pi k_1 R_1 R_2}}{q \times \left(\dfrac{R_3 - R_2}{4\pi k_2 R_2 R_3}\right)}$$

$$\frac{\Delta T_1}{\Delta T_2} = \frac{k_2 R_3}{k_1 R_1} = \frac{3}{2} \times 1.25 = 1.875$$

339. Consider the following statements related to transient heat conduction :

1. Biot number and Fourier number play a vital role in unsteady heat conduction.
2. Schmidt plot is used in one-dimensional transient heat conduction problems.

Which of the above statements is/are correct ?

(a) 1 only (b) 2 only
(c) Neither 1 nor 2 (d) Both 1 and 2

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Ans. (d) : As we know, for an unsteady heat transfer,

- Biot number & Fourier number plays vital role

$$\Rightarrow \quad B_i = \frac{hL}{k}$$

- Schmidt plot is used in 1-D transient heat conduction problems. It is the ratio of kinematic viscosity (momentum diffusivity) to mass diffusivity.

$$Sc = \frac{(\text{Kinematic viscosity})}{\text{Mass diffusivity}} = \frac{\nu}{\alpha}$$

340. Surface temperatures on the two sides of a flat wall are measured to be 600°C and 400° C respectively. Thermal conductivity of the wall material is 0.1 kW/m-K. If the wall thickness is 0.2 m, the heat flux through the wall will be

(a) 1 kW/m^2 (b) 100 kW/m^2
(c) 236.5 kW/m^2 (d) 1000 kW/m^2

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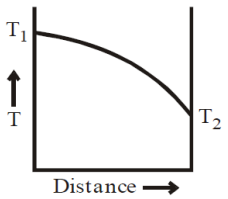
Ans. (b) : As we know,

$$\text{Heat transfer } (Q) = -\frac{kAdT}{dx}$$

$$\text{Heat flux } (q) = \frac{Q}{A} = \frac{0.1 \times 200}{0.2} = 100\,kW/m^2$$

341.



Temperature distribution in a furnace wall is shown in the figure. Which of the following expressions would represent the correct variation of thermal conductivity of this material with respect to temperature ?

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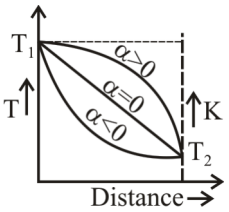
(a) $\frac{K_0}{1+\alpha T}$ (b) $K_0(1+\alpha T)$

(c) $\frac{K_0}{\alpha T}$ (d) $K_0(1-\alpha T)$

Where K_0 is thermal conductivity at 0°C. T is in °C and α is a constant.

UPSC JWM Advt. No.-50/2010
ESE 1998

Ans. (b) :



As we know from figure,

$K = K_0(1 + \alpha T)$

So,

$\therefore$ $\alpha > 0$ for the given problem.

342. Match List I with List II and select the correct answer using the code given below the Lists :

List I

(Type of wall material)

A. With internal heating and constant thermal conductivity

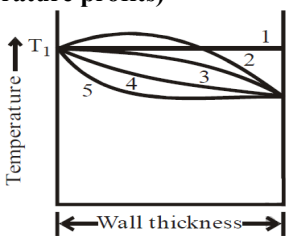
B. Constant thermal conductivity

C. Infinite thermal conductivity

D. Thermal conductivity increasing with temperature

List II

(Temperature profits)



Code:

	A	B	C	D		A	B	C	D
(a)	1	4	2	5	(b)	2	4	1	3
(c)	1	4	3	5	(d)	2	1	4	5

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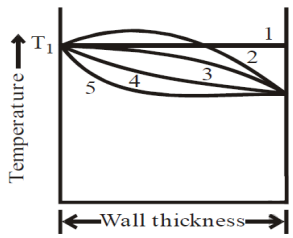
Ans. (b) :



According to Fourier's law –

$$\boxed{\frac{dT}{dx} \propto \frac{1}{k}}$$

From the figure,

- For infinite thermal conductivity temperature will be zero, so curve 1 shows the correct option.
- Temperature first increase & then decrease in case of internal heating & constant thermal conductivity.
- For a constant thermal conductivity drop linearly.
- For substance in which thermal conductivity increasing with temperature, $k = k_0(1 + \alpha T)$ show the nature of above equation.

343. A body cools from 90ºC to 80ºC in 5 minutes. Under the same external conditions to cool from 80 ºC to 70 ºC the body will take

(a) 5 minutes

(b) 4 minutes

(c) 2.5 minutes

(d) More than 5 minutes

ISRO Scientist/Engineer 2010

Ans. (d) : Case - I

$$\frac{80-T_\infty}{90-T_\infty} = e^{-5\times 60\times b}$$

Case - II

$$\frac{70-T_\infty}{80-T_\infty} = e^{-bt}$$

Let $T_\infty = 0^\circ C$

From case I

$$\frac{8}{9} = e^{-300t}$$

$\Rightarrow$ $b = 3.92 \times 10^{-4}$

From case - II

$$\frac{7}{8} = e^{-bt}$$

$$\ln\left(\frac{7}{8}\right) = -3.92\times 10^{-4}\times t$$

t = 340 sec

= 5.668 min.

344. A metal wall has an area of $10m^2$, thickness = 5 cm and thermal conductivity = 200 W/mK. What is the thermal resistance of the wall in K/W?

(a) 2.5×10^{-5} (b) 2.5×10^{-4}

(c) 2.5×10^{-3} (d) 2.5×10^{-2}

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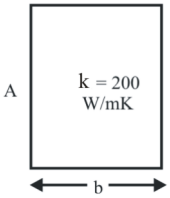
Ans. (a) : Given data :

Cross-section area of wall (A) = $10m^2$

Wall thickness (b) = 5 cm = 0.05 m

Thermal Conductivity (k) = 200 W/mK



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Thermal resistance $(R_{Th}) = \frac{b}{kA} = \frac{0.05}{200\times10} = \frac{5}{2}\times10^{-5}$

$\boxed{R_{Th} = 2.5\times10^{-5}} K/W$

345. An fin of length 'L' protrudes from a surface held at temperature t_0 greater than the ambient temperature t_a. The heat dissipation from the free end of the fin is assumed to be negligible. The temperature gradient at fin tip is :

(a) zero (b) $(t_1 - 1_a)/(t_0 - t_a)$

(c) $h(t_0 - t_1)$ (d) $(t_0 - t_1)L$

APPSC IOF, 2009

Ans. (a) : We know, $\left.\frac{dT}{dx}\right|_{x=l}$

So, temperature gradient at fin tip is zero.

346. Four rods with different radii r and length l are connected to two reservoirs at different temperature. Which one of them will conduct most heat?

(a) r = 2 cm & l = 2 m

(b) r = 1 cm & l = 1 m

(c) r = 2 cm & l = $\frac{1}{2}$ m

(d) r = $\frac{1}{2}$ cm & l = $\frac{1}{2}$ m

ISRO Scientist/Engineer 2009

Ans. (c) : According Fourier's Law

$$Q = -KA\frac{dT}{dx}$$

$$Q \propto A\frac{dT}{dx}$$

$$Q \propto \frac{r^2}{l}$$

length will be minimum and cross section area will be maximum for maximum heat transfer

347. As thickness of insulation around a heated cable gradually increases from zero, heat transfer from the conductor

(a) goes on decreasing monotonically

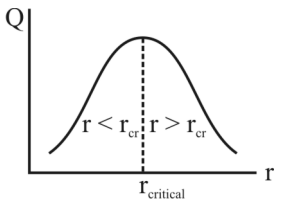
(b) goes on increasing monotonically

(c) first increases and then decreases

(d) first decreases and then increases

DRDO Scientists 2008

Ans. (c) :



348. In transient heat conduction, two significant dimensionless parameters are

(a) Reynolds and Fourier

(b) Fourier and Prandtl

(c) Biot and Fourier

(d) Biot and Nusselt

TNPSC AE, 2008

Ans. (c) : We know that,

According to transient heat transfer,

$$\frac{T - T_\infty}{T_i - T_\infty} = e^{-\frac{hA}{\rho VC}\cdot\tau} = e^{-\frac{hL}{K}\times\frac{\alpha t}{L^2}}$$

$$\left(Bi = \frac{hL}{K}, \quad Fo = \frac{\alpha t}{L^2}\right)$$

$$\frac{T - T_\infty}{T_i - T_\infty} = e^{-Bi\times Fo}$$

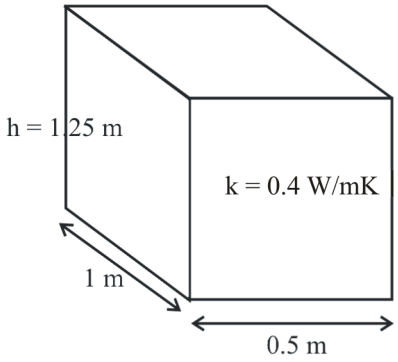
349. Thermal resistance (K/W) of a 0.5 m thick, 1 m wide and 1.25 m high furnace wall having a thermal conductivity 0.4 W/mK is

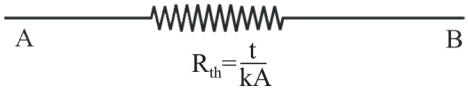
(a) 2 (b) 10

(c) 20 (d) 1

TNPSC AE, 2008

Ans. (d) :





Let; R_{th} = thermal resistance, unit of $\left(\frac{K}{W}\right)$

A = cross section area; Height of wall (h) = 1.25 m

Width of wall (b) = 1 m

Cross-section area (A) = $1.25 \times 1 = 1.25\ m^2$

thickness (t) of wall = 0.5 m

$$R_{th} = \frac{t}{kA} = \frac{0.5}{1.25\times0.4}$$

$$= \frac{0.5}{0.5} = 1\ K/W$$

350. The variation of temperature inside a wall is shown by three different curves. In which case thermal conductivity least varies with temperature

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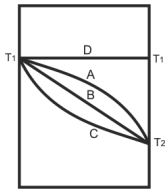

(a) A
(b) B
(c) C
(d) Cannot be ascertained on the basis of information provided

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Ans. (b) : For B, the thermal conductivity is not the function of temperature

351. A thin copper wire at 300º is suddenly immersed in water at 30ºC. It cools down to 150ºC in 70 seconds. It is then reheated to the initial temperature of 300ºC and suddenly exposed to air at 30ºC where it cools down to 150ºC in 200 seconds. This difference in cooling time is due to

(a) Larger specific heat of water
(b) Larger heat transfer coefficient in water
(c) Smaller heat transfer coefficient in water
(d) None of the above

ISRO Scientist/Engineer 2008

Ans. (b) : The convection heat transfer coefficient depend on the type of fluid flow properties coefficient of air. (Free convective) = 6 - 30 W/m^2K and water (free convection) = 20 - 100 W/m^2K.

352. In which of the following cases, transmission of heat is smallest?

(a) Solids (b) Alloys
(c) Gases (d) Liquids

UKPSC AE 2007 Paper -II

Ans. (c) : In gases, transmission of heat is smallest because its thermal conductivity is very low.

$k_{gas} < k_{liquid} < k_{soild}$

where, k = Thermal Conductivity

Thermal conductivity (k) of different material –

1. Diamond	2300 W/m-K	(Highest)
2. Silver	405 W/m-K	
3. Copper	385 W/m-K	
4. Steels	15-55 W/m-K	
5. Air	0.024W/m-K	
6. Water	0.6 W/m-K	
7. Ice	2.25 W/m-K	

353. A thin copper wire at 300°C is suddenly immersed in water at 30°C. It cools down to 150°C in 70 seconds. It is then reheated to the initial temperature of 300°C and suddenly exposed to air at 30°C where it cools down to 150°C in 200 seconds. This difference in cooling time is due to

(a) Larger specific heat of water
(b) Larger heat transfer coefficient in water
(c) Smaller heat transfer coefficient in water
(d) None of the above

ISRO Scientist/Engineer 2007

Ans. (b) : Heat transfer coefficient is the proportionality constant between the heat flux and the temperature difference ΔT. So larger the heat transfer coefficient lessen the time the body will take to cover the temperature difference.

354. The heat transfer equation $\frac{\partial^2 t}{\partial x^2}+\frac{\partial^2 t}{\partial y^2}+\frac{\partial^2 t}{\partial z^2}=0$ is known as

(a) General heat transfer equation
(b) Poisson's equation
(c) Fourier's equation
(d) Laplace's equation

UKPSC AE 2007 Paper -II

Ans. (d) : From Fourier-Biot equation,

$$\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}+\frac{q_g}{k}=\frac{1}{\alpha}\left(\frac{\partial T}{\partial \tau}\right)$$

In case of steady state $\left(\frac{1}{\alpha}\frac{\partial T}{\partial \tau}=0\right)$ and no heat generation $\left(\frac{q_g}{k}=0\right)$, this equation is called Laplace's equation.

ie. $\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}=0 \Rightarrow$ Lalplace's equation

355. For conduction through thick walled tube, the value of mean radius, r_m used in heat conduction (with r_1 inner radius and r_2 outer radius) is given by

(a) $\frac{(r_2-r_1)}{2}$ (b) $\frac{(r_2+r_1)}{2}$
(c) $\frac{(r_2-r_1)}{\ln\frac{r_2}{r_1}}$ (d) $\frac{(r_2+r_1)}{\ln\frac{r_2}{r_1}}$

UKPSC AE 2007 Paper -II

Ans. (c) : For conduction through thick walled tube, the value of mean radius,

$$r_m=\frac{r_2-r_1}{\ln(r_2/r_1)}$$

* In this case, temperature profile is logarithmic

* Thermal Resistance $R_{th}=\frac{\ln(r_2/r_1)}{2\pi KL}$

356. Thermal conductivity of water

(a) First increases with temperature then decreases with temperature
(b) Increases steadily with temperature
(c) Decreases with temperature
(d) Does not depend on temperature

UKPSC AE 2007 Paper -II

Ans. (a) : Thermal conductivity of most of the liquid decrease with increasing temperature, water is exception. In case of water, thermal conductivity first increase with temperature, then decreases with temperature.

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357. Which instrument is used to measure the temperature inside furnace?

(a) Gas thermometer
(b) Optical pyrometer
(c) Alcohol thermometer
(d) Mercury thermometer

UKPSC AE 2007 Paper -II

Ans. (b) : Optical pyrometer is used to measure the temperature inside furnace.

358. An increase in convective coefficient over a fin

(a) increases effectiveness
(b) decreases effectiveness
(c) does not influence effectiveness
(d) influence only the fin efficiency

J & K PSC Screening, 2006

Ans. (b) : Fin effectiveness $\varepsilon = \sqrt{\frac{Pk}{hA_c}}$

$$\downarrow \varepsilon \propto \frac{1}{\sqrt{h}\uparrow}$$

359. Which of the following has maximum thermal conductivity.

(a) aluminium (b) steel
(c) brass (d) glass

J & K PSC Screening, 2006

Ans. (a) : Aluminium has maximum thermal conductivity among steel, brass, glass and aluminium.

Material	Thermal conductivity
Aluminium	200W/m-K
steel	15-35 W/m-K
brass	109 W/m-K
glass	1.2 W/m-K

360. Thermal conductivity of glass wool depends on:

(a) porosity (b) density
(c) both (a) & (b) (d) none of the above

J & K PSC Screening, 2006

Ans. (c) : Thermal conductivity of glass wool depends on Composition, Density, Porosity and structure.

361. If the inner and outer surfaces of a hollow cylinder (having radi r_1 and r_2 and length L) are at temperatures t_1 and t_2 then rate of radial heat flow will be

(a) $\frac{k}{2\pi L}\frac{t_1 - t_2}{log\frac{r_2}{r_2}}$ (b) $\frac{k}{2\pi Lk}\frac{t_1 - t_2}{log\frac{r_2}{r_1}}$

(c) $\frac{2\pi L}{k}\frac{t_1 - t_2}{log\frac{r_2}{r_2}}$ (d) $2\pi Lk\frac{t_1 - t_2}{log\frac{r_2}{r_2}}$

ISRO Scientist/Engineer 2006

Ans. (d) : Hollow cylinder–Inner and outer radii r_1 and r_2 and length L are at temperatures t_1 and t_2.

Thermal resistance $R_{th} = \frac{\log\frac{r_2}{r_1}}{2\pi kL}$

Heat transfer, $Q = \frac{\Delta T}{R_{th}} = \frac{(t_1 - t_2)2\pi kL}{\log\frac{r_2}{r_1}}$

362. A mild steel tank of wall thickness 12 mm contains water at 100^0C. The atmospheric temperature is 20^0C. The thermal conductivity of mild steel is 50 W/mK, and the heat transfer co-efficients for inside and outside the tank are 2850 and 10 W/m^{2}K, respectively. Calculate the rate of heat loss per m^2 of tank surface and the temperature of the outside surface of the tank.

(a) 300.5 W/m^2, 45.5^0C
(b) 495.2 W/m^2, 67.6^0C
(c) 602.6W/m^2, 80.6^0C
(d) 795.2 W/m^2, 99.52^0C

ISRO Scientist/Engineer 2006

Ans. (d) : Given–

Thickness t = 12 mm

$\Delta T = 100°C - 20°C = 80°C$

Thermal Resistance $R_{th} = \frac{1}{h_i A} + \frac{t}{kA} + \frac{1}{h_o A}$

$= \frac{1}{h_i} + \frac{t}{k} + \frac{1}{h_o}$ (For unit area)

$R_{th} = \frac{1}{2850} + \frac{12\times10^{-3}}{50} + \frac{1}{10}$

$R_{th} = 0.10059\frac{Km^2}{W}$

Heat loss per unit area $(q) = \frac{\Delta T}{R_{th}}$

$= \frac{80}{0.10059} = 795.3\ W/m^2$

$\frac{100 - T}{\frac{1}{h_1} + \frac{t}{k}} = \frac{T - 20}{\frac{1}{h_2}}$

$\frac{100 - T}{\frac{1}{2850} + \frac{0.012}{50}} = \frac{T - 20}{\frac{1}{10}}$

$\boxed{T = 99.53°C}$

2. Free and Forced Convection

363. The thickness of thermal and hydrodynamic boundary layers is equal, if

(a) Pr = 1 (b) Pr > 1
(c) Pr < 1 (d) Pr = Nu

RPSC IOF, 2020
OPSC AE 2019 Paper-II
Sikkim PSC (Under Secretary), 2017
Nagaland PSC (CTSE) 2016 Paper-I
Assam PSC CCE Pre 2015
APPSC Poly. Lect. 2013

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JPSC AE 2013 Ist Paper
UKPSC AE 2013, 2012 2007 Paper-II
UPSC JWM Adv. No.- 52/2010

Ans. (a) :

$$\frac{\text{Hydrodynamic boundary layer thickness } \delta}{\text{Thermal boundary layer thickness, } \delta_t} = (\text{Pr})^{1/3}$$

If,

$\text{Pr} = 1$

$\frac{\delta}{\delta t} = 1$

$\boxed{\delta = \delta_t}$

364. In free convection heat transfer transition from laminar to turbulent flow is governed by the critical value of the :

(a) Reynold's number
(b) Grashoff's number
(c) Reynold's number, Grashoff's number
(d) Prandtl number, Grashoff's number

RPSC ACF & FRO, 26.02.2021
CGPSC AE 15.01.2021, RPSC IOF, 2020
Nagaland PSC (CTSE) 2018 Paper-II
TNPSC AE 2018, CIL MT 26.03.2017
APPSC AEE MAINS 2016 PAPER-III
OPSC AEE 2019, 2015 Paper - II
TSPSC Managers, 2015
GPSC Lect. (Poly) 07.12.2014
ESE 2007, 1992

Ans. (d) : In free convection heat transfer from laminar to turbulent is governed by the critical value of the Prandtl number & Grashoff's number.

Case I- Laminar Flow over flat plate-

$\text{Nu} = 0.59\ (\text{Ra})^{1/4}$

$\text{Nu} = 0.59\ [\text{Gr} \times \text{Pr}]^{1/4}$

Case II- Turbulent Flow over flat plate-

$\text{Nu} = 0.13\ (\text{Ra})^{1/3}$

$\text{Nu} = 0.13\ [\text{Gr} \cdot \text{Pr}]^{1/3}$

365. In free convection heat transfer, Nusselt number is the function of

(a) Grashof number and Prandtl number
(b) Grashof number and Reynolds number
(c) Prandtl number and Reynolds number
(d) Grashof number, Prandtl number and Reynolds number

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
UPRVUNL AE 05.07.2021
GPSC DEE Class-2 (GWSSB) 04.07.2021
Kerala PSC AE (Irrigation) 05.04.2017
Karnataka PSC Lect., 27.05.2017
KPSC AE. 2015
Kerala PSC AE 06.08.2015
UKPSC AE 2012 Paper–II
ESE 1992

Ans (a) : In free or natural convection heat transfer.

$\text{Nu} = f(\text{Gr}, \text{Pr})$

In forced convection heat transfer

$\text{Nu} = f(\text{Re}, \text{Pr})$

366. The Prandtl number value for air is

(a) 0.1 (b) 0.7
(c) 0.3 (d) 1.7

Nagaland PSC CTSE 2017 Paper-2
Rajasthan Nagar Nigam AE 2016, Shift-II
Nagaland PSC CTSE 2015 Paper-II
J&K PSC Civil Services Pre, 2013
UKPSC AE 2012 Paper–II
J&K PSC Civil Services Pre, 2010
ISRO Scientist/Engineer 2010

Ans. (b) : Prandtl number is a characteristic of fluid only, for air at room temperature P_r is 0.71.
Prandtl number of water at 17ºC is 7.56. Liquid in general have high Prandtl number.

367. Heat transfer in liquid and gases primarily takes place by:

(a) Conduction
(b) Convection
(c) Radiation
(d) Conduction and Radiation

JPSC AE, 10.04.2021, Paper-I
GPSC Asstt. Prof. 28.08.2016
KPCL AE 2016, GPSC Lect. 23.10.2016
Rajasthan AE (Nagar Nigam) 2016 Shift-3
GWSSB AAE, 27.12.2015
HPPSC Asstt. Prof. 2014
TNPSC AE (Industries) 09.06.2013
UKPSC AE 2012 Paper–II, TRB Poly. Lect., 2012

Ans. (b) : Convection heat transfer, often referred to simply as convection, is the transfer of heat from one place to another by the movement of fluids.

368. Reynold's analogy may expressed as: (where, S_t = Stanton Number, N_u = Nusselt number, P_r = Prandtl number, R_e = Reynolds number)

(a) $(S_t) = \frac{N_u}{R_e P_r}$ (b) $(S_t) = \frac{N_u P_r}{R_e}$

(c) $(S_t) = N_u R_e P_r$ (d) $(S_t) = \frac{R_e P_r}{N_u}$

APPSC AEE MAINS 2016 PAPER-III
BPSC Poly. Lect. 2016, UPRVUNL AE 2016
VIZAG STEEL MT 2015
CGPSC AE 26.04.2015 SHIFT-II
APGENCO AE 2012

Ans. (a) : Reynold's Analogy—Reynold's analogy is the inter-relationship between fluid friction and Newton's law of viscosity.

$\frac{N_u}{R_e P_r}$ is called the stanton number (S_t). It represents the Nusselt number divided by the product of the Reynolds and Prandtl numbers.

369. For forced convection, Nusselt number is a function of

(a) Reynolds and Prandtl numbers
(b) Grashoff number only
(c) Reynolds and Grashoff numbers
(d) Prandtl and Grashoff numbers

GPSC Engineer Class-II Pre 19.01.2020, ESE 2018

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RPSC IOF & Boiler 20.09.2017
Rajasthan AE (Nagar Nigam) 2016 Shift-3
MPPSC AE 08.11.2015
UKPSC AE 2012 Paper–II

Ans. (a) : In forced convection, the Nusselt number depends on the rate of heat transfer through a boundary layer from a surface hotter or cooler than the air passing over it,

We know that,

Overall nusselt No. throughout the plate for forced convection.

$Nu = 0.664\ (Re_L)^{1/2}\ (Pr)^{1/3}$

$\therefore\ \boxed{Nu = f(Re, Pr)}$

Reynolds number – It is the ratio of inertia force to viscous force.

$$Re = \frac{\rho V d}{\mu} = \frac{Vd}{\nu}$$

Prandtl No.– It is the ratio of kinematic viscosity (v) to thermal diffusivity (α).

$$\text{Prandtl No. } (Pr) = \frac{\nu}{\alpha} = \frac{\mu C_p}{K}$$

Prandtl Number is a measure of relative thickness of the velocity and thermal boundry layer.

$$Pr = \frac{\nu}{\alpha} = \frac{\mu C_p}{K}$$

370. The ratio of kinematic viscosity to thermal diffusivity is known as

(a) Prandtl number (b) Nusselt number
(c) Peclet number (d) Schmidt number

JPSC AE, 10.04.2021, Paper-I
GPSC Executive Engineer 23.12.2018
RPSC AE (GWD) 06.05.2016
GATE 2015, ESE 2005

Ans. (a) : The ratio of kinematic viscosity to thermal diffusivity is called Prandtl number.

$$\boxed{\text{Prandtl no.} = \frac{\text{Kinematic viscosity}}{\text{Thermal diffusivity}} = \frac{\nu}{\alpha}}$$

371. The ratio of energy transferred by convection to that by conduction is called-

(a) Stanton number (b) Nusselt number
(c) Biot number (d) Peclet number

Assam PSC AE (PHED), 18.10.2020
Assam PSC CCE Pre 2015
Kerala PSC Poly. Lect. 09.01.2014
TRB Asstt. Prof., 2012, ESE 1992

Ans. (b) : Nusselt Number– It is the ratio of convection heat transfer to the conduction heat transfer.

$$Nu = \frac{\text{Convection heat transfer}}{\text{Conduction heat transfer}}$$

$$= \frac{h}{k/L} = \frac{hL}{k}$$

Where, h = Convection heat transfer
L = Characteristic length
k = Thermal conductive of the fluid

Note– Nu for most of fluid greater than 1.

Stanton Number (St) – (Modified Nusselt No.)

$$\frac{Nu}{Pe}\frac{h}{\rho V_\infty} = \frac{NU_L}{Re_L\ pr}$$

- **Biot Number (Bi) –** Ratio of the internal conductive resistance to the boundary layer external convective resistance.

$$\boxed{Bi = \frac{ICR}{ECR} = \frac{hL}{Ks}}$$

- **Peclet No. (Pe) –** Ratio of convection to conduction heat transfer rates.

$$\boxed{Pe_L = \frac{uL}{\alpha} = Re_L\ pr}$$

372. Which one of the following factors has no influence on free convection?

(a) Specific gravity
(b) Coefficient of viscosity
(c) Gravitational force
(d) Velocity

JPSC AE, 10.04.2021, Paper-I
TNPSC AE 2017, KPSC AE 2015
MPPSC AE 08.11.2015
Mizoram PSC AE/SDO 2014, Paper-I

Ans. (d) : Grashof Number (Gr)–It indicates the relative strength of buoyant force to viscous forces.

$$Gr = \frac{\text{Buoyant force}}{\text{Viscous force}}$$

$$Gr = \frac{g\beta(T_S - T_\infty)\ell_c^3}{\upsilon^2}$$

$$Gr = \text{Buoyant force} \times \frac{\text{Inertia force}}{(\text{Viscous force})^2}$$

Where, g = Gravitational acceleration, β = Coefficient of volume expansion,

T_S = Temperature of surface, T_∞ = Temperature of fluid far from surface (°C)

ℓ_c = Characteristics length of surface and υ = Kinematic viscosity of fluid.

Significance of Gr–Grashof number provides the main criteria in determining whether the fluid flow is laminar or turbulent in natural convection.

Velocity has no influence on free convection.

- Free convection is caused by a change in density of a fluid due to temperature change or gradient.

373. For the fluid flowing over a flat plate with Prandtl number greater than unity, the thermal boundary layer for laminar forced convection :

(a) is thinner than hydrodynamic boundary layer
(b) has thickness equal to zero
(c) is of same thickness as hydrodynamic boundary layer
(d) is thicker than hydrodynamic boundary layer

GPSC Executive Engineer 23.12.2018
GPSC Asstt. Director of Transport 05.03.2017

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HPPSC Asstt. Prof. 18.11.2016
RPSC Lecturer 16.01.2016, GATE 1988

Ans. (a) : $\text{Pr} = \frac{\delta}{\delta_t} = \frac{\nu}{\alpha}$

$\text{Pr} > 1$

$\delta_t < \delta$

Hence thermal boundary layer is thinner than hydrodynamic boundary layer.

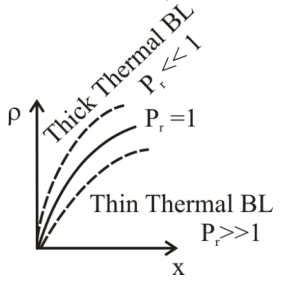
374. The non-dimensional number that gives the relationship between the thermal boundary layer and hydrodynamic boundary layer is :

(a) Rayleigh number (b) Peclet number
(c) Grashof number (d) Prandtl number

SJVN ET 2019, MECON MT 2019
GPSC ARTO Pre 30.12.2018, UPRVUNL AE 2014

Ans : (d) Prandtl number gives the relationship between the thermal boundary layer and hydrodynamic boundary layer

$$\frac{\text{Hydrodynaimc Boundary layer}(\delta)}{\text{Thermal Boundary layer}(\delta_t)} = (\text{Pr})^{\frac{1}{3}}$$



375. The non-dimensional parameter known as Stanton number (St) is used in

(a) Forced convection heat transfer in flow over flat plate
(b) Condensation heat transfer with laminar film layer
(c) Natural convention heat transfer over flat plate
(d) Unsteady heat transfer from bodies in which internal temperature gradients cannot be neglected

Gujarat PSC AE 2019
GPSC ARTO Pre. 30.12.2018
RPSC Vice Principal ITI 2018, ESE 2004

Ans : (a) : We know Stanton number is defined as,

$$S_t = \frac{N_u}{R_e . P_r} = \frac{C_f}{2}$$

The expression contains Reynolds number R_e. So the convection should be forced. The presence of friction factor signifies that the total friction should be due to skin friction not due to form friction. So forced convection over flat plate uses Stanton number 'S_t'.

376. Grashof number may be defined as

(a) Viscous force/buoyancy force
(b) Inertia force/pressure force
(c) Buoyancy force/viscous force
(d) Inertia force/gravity force

Oil India Ltd. Sr. Engineer (Production) 30.11.2019
GPSC Executive Engineer 23.12.2018

CGPSC AE 25.02.2018, GATE 2016
RPSC Lecturer 16.01.2016

Ans. (c) : Grashof number may be defined as it is the ratio of buoyancy force to viscous force.

$$Gr = \frac{\text{Inertia force} \times \text{Buoyancy force}}{(\text{Viscous force})^2}$$

$$= \frac{L^3 \times g \times \beta \times \Delta t}{\upsilon^2}$$

377. Which of the following is correct?
The effectiveness of a fin will be maximum in an environment with

(a) Free convection
(b) Forced convection
(c) Radiation
(d) Convection and radiation
(e) None of the above

GPSC ARTO Pre 30.12.2018, RPSC AE 2018
CGPSC Poly. Lect. 22.05.2016

Ans. (a) : The effectiveness of a fin will be maximum in an environment with force convection

$$\text{effectiveness } (\varepsilon) = \sqrt{\frac{kP}{hA}}$$

If h is small effectiveness will be more.

Conditions	h(W/m^{2}K)
Free convection in gases	3 to 25
Forced convection in gases	25 to 400
Free convection in liquid	250 to 600
Forced convection in liquid	600 to 4000
Condensation H.T. (Vapour to liquid)	3000 to 25,000
Boiling heat transfer	5000 to 50,000

378. Nusselt number is defined as

(a) Heat transfer by conduction/heat transfer by convection
(b) (Heat transfer by conduction/heat transfer by convection)^2
(c) (Heat transfer by convection/heat transfer by conduction)^2
(d) Heat transfer by convection/heat transfer by conduction

RPSC Lecturer 16.01.2016
KPCL AE 2016, SJVN ET 2013

Ans. (d) : Nusselt number is defined as heat transfer by convection to heat transfer by conduction.

$$N_U = \frac{hL}{K}$$

379. Heat is lost from a 100mm diameter steam pipe placed horizontally in ambient air at 30°C. If the Nusselt number is 25 and the thermal conductivity of the air is 0.03W/m K, then the heat transfer coefficient will be.

(a) 7.5 W/m^{2}K (b) 15 W/m^{2}K
(c) 25 W/m^{2}K (d) 35 W/m^{2}K

TNPSC AE 2019, ESE 2017, 1999
MPPSC AE 2016, RPSC ADE 2016

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Ans. (a) : Given,

$D = 100 \text{ mm} = 0.1 \text{ m}$

$T_\infty = 30°C$

$Nu = 25$

$k = 0.03 \text{ W/mK}$

Nusselt number is given by

$$Nu = \frac{hL_c}{k} = \frac{hD}{k} \quad (\because L_c = D)$$

$$25 = \frac{h \times 0.1}{0.03}$$

$h = 7.5 \text{ W/m}^2\text{k}$

380. Which dimensionless number has the significant role in forced convection:-

(a) Mach number (b) Reynold number

(c) Prandtl number (d) Pecklet number

GPSC DEE Class-2 (GWSSB) 04.07.2021
UKPSC AE-2013, Paper-II
UKPSC AE 2007 Paper -II

Ans. (b) : Reynold number has the significant role in forced convection.

381. Least value of Prandtl number can be expected in case of

(a) water (b) liquid metals

(c) salt solution (d) sugar solution

JPSC AE, 10.04.2021, Paper-I
JPSC AE 2013 Ist Paper
UKPSC AE 2012, 2007 Paper–II

Ans. (b) : Liquid Metals

Liquid Metals-	– 0.004 – 0.03
Gases	– 0.7-1.0
Water	– 1.7-13.7
Glycerin	$– 2 \times 10^3 – 10^5$

382. The free convection heat transfer is significantly affected by

(a) Reynolds number (b) Grashof number

(c) Prandtl number (d) Stanton number

GPSC DEE Class-2 (GWSSB) 04.07.2021
BPSC AE Mains 2017 Paper - V
UKPSC AE-2013, Paper-II

Ans : (b) : Grashof number = $\frac{\text{Buoyance force}}{\text{viscous force}}$

Buoyant force occurs due to free convection.

383. Air at 20^0C blows over a plate of 50 cm × 75 cm maintained at 250^0C. If the convection heat transfer coefficient is 25 W/m^2°C the heat transfer rate is

(a) 215.6 kW (b) 2156 kW

(c) 2.156 kW (d) 23 kW

MPSC Poly. Lect. 2014, TRB Asstt. Prof., 2012
ISRO Scientist/Engineer 2011

Ans. (c) : From Newton's law of cooling

$Q = hA\Delta T$

Given,

$h = 25 \text{W/m}^2\text{–°C}$

$T_\infty = 20°C$

$T_s = 250°C$

$A = 50\text{cm} \times 75 \text{ cm}$

$A = 0.5\text{m} \times 0.75 \text{ m}$

Heat transfer rate, $Q = hA(T_s – T_\infty)$

$Q = 25 \times 0.5 \times 0.75 \times (250–20)$

$\boxed{Q = 2156.25 \text{ W}}$

$= 2.156 \text{ kW}$

384. Given that Nu = Nusselt number, Re = Reynolds number, Pr = Prandtl number, Sh = Sherwood number, Sc = Schmidt number and Gr = Grashoff number. The function relationship for free convective heat transfer is given as,

(a) Nu = f(Gr, Pr) (b) Sh = f(Sc, Gr)

(c) Nu = f(Re, Pr) (d) Sh = f(Re, Sc)

NPCIL ET, 2015, DRDO Scientist 2009
ISRO Scientist/Engineer 2008

Ans. (a) : Nu = f (Gr, Pr)

Free convection heat transfer

385. The Prandtl number is the ratio of

(a) The momentum and mass diffusivities

(b) The momentum and thermal diffusivities

(c) The thermal and mass diffusivities

(d) The inertia and viscous forces

RPSC IOF & Boiler 20.09.2017
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
CGPSC AE 2014 -II

Ans. (b) : Prandtl Number (P_r) :– It is the ratio of momentum diffusivity and thermal diffusivity of fluid.

$$P_r = \frac{\text{Momentum diffusivity}}{\text{Thermal diffusivity}} = \frac{\nu}{\alpha}$$

$$\boxed{P_r = \frac{\mu C_P}{K}}$$

386. Nusselt number is given by :

(a) $\frac{\mu C_p}{K}$ (b) $\frac{\mu K}{C_p}$

(c) $\frac{hL}{K}$ (d) $\frac{hK}{L}$

Nagaland PSC (CTSE) 2018 Paper-II
J&K PSC Civil Services Pre, 2010
UKPSC AE 2007 Paper-II

Ans. (c) : Nusselt Number (Nu) = $\frac{hL}{k_{fluid}}$

l = length

h = heat transfer coefficient

387. For a flow over a flat plate, the hydrodynamic boundary layer thickness is 0.5 mm. The fluid viscosity is 25 × 10^{-6} Pa.s specific heat is 2.0 kJ/kgK and thermal conductivity is 0.05 W/m-K. The thermal boundary layer thickness would be

(a) 0.1 mm (b) 0.5 mm

(c) 1 mm (d) 1.5 mm

UPPSC AE 13.12.2020, Paper-2
GPSC Executive Engineer 23.12.2018
APPSC IOF, 2009, GATE 1992

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Ans. (b) : Given,

Hydrodynamic Boundary layer thickness, $\delta = 0.5$ mm

viscosity, $\mu = 25 \times 10^{-6}$ Pa-sec

specific heat, $C_p = 2.0$ kJ/kg K

$k = 0.05$ W/m–K

Prandtl No. $\frac{\mu C_p}{k} = \frac{25\times10^{-6}\times2\times10^{3}}{0.05}$

$\boxed{\text{Prandtl No.} = 1}$

Pr No. $\frac{\delta}{\delta_t} = \frac{0.5}{\delta_t} = 1$

Thermal boundary layer thickness, $\boxed{\delta_t = 0.5\text{mm}}$

388. The average Nusselt number in laminar natural convection from a vertical wall at 180^0C with still air at 20^0C is found to be 48. If the wall temperature becomes 30^0C, all other parameter remaining the same, average Nusselt number will be

(a) 8 (b) 16

(c) 24 (d) 32

UPPSC AE 13.12.2020, Paper-2, ESE 2002

Ans. (c) : For laminar natural convection

$$Nu \propto (Gr)^{1/4} \propto \Delta T$$

$$\frac{(Gr)_2}{(Gr)_1} \propto \frac{30-20}{180-20} = \frac{1}{16}$$

$$\frac{(Nu)_2}{(Nu)_1} = \left(\frac{1}{16}\right)^{1/4} = \frac{1}{2}$$

$$(Nu_2) = \frac{(Nu)_1}{2} = \frac{48}{2}$$

$$Nu_2 = 24$$

389. In case of turbulent flow through a horizontal isothermal cylinder of diameter 'D' free convection heat transfer coefficient for the cylinder will

(a) be independent of diameter

(b) vary as $D^{3/4}$

(c) vary as $D^{1/4}$

(d) vary as $D^{1/2}$

UPPSC AE 13.12.2020, Paper-2, ESE 1997

Ans. (a) : For turbulent flow,

$$Nu = f(Gr\,Pr)^{1/3}; \quad Gr = \frac{g\beta\Delta TL^3}{\nu^2}$$

For a horizontal isothermal cylinder,

$$Nu = f(L^3)^{1/3}$$

$$Nu \propto L \ (\because L = D)$$

$$\frac{hD}{K} \propto D$$

$\therefore$ Heat transfer coefficient h for cylinder will be independent of diameter.

390. The ratio of momentum diffusivity (v) to thermal diffusivity (α) is called_____

(a) Lewis number (b) Nusselt number

(c) Biot number (d) Prandtl number

PPSC Asst. Municipal Engineer 15.06.2021

Assam Engg. College AP/Lect. 18.01.2021

GATE 2015

Ans. (d) : Prandtl number.

Prandtl number $(P_r) = = \frac{\nu}{\alpha} = \frac{\mu C_p}{K}$

391. Nusselt number is the ratio of

(a) Convective and Conductive resistances

(b) Conductive and Convective resistances

(c) Wall heat transfer rate to mass heat flow rate

(d) Kinetic viscosity to thermal diffusivity

CGPSC AE 2014 Paper-II

ISRO Scientist/Engineer 2012

Ans. (b) : Nusselt number $(Nu) = \frac{hL}{K_{fluid}}$ or $\frac{hD}{K_{fluid}}$

$$= \frac{\frac{D}{kA}}{\frac{l}{ha}} \text{ or } \frac{\frac{L}{kA}}{\frac{l}{h}}$$

$$= \frac{\text{Conduction Resistance}}{\text{Convective Resistance}}$$

392. A flat plate of length 1 m and width 50 cm is placed in an air stream at 30°C blowing parallel to it. The convective heat transfer co-efficient is 30 W/(m^2K). The heat transfer if the plate is maintained at a temperature of 400°C is

(a) 5.55 kW (b) 4.05 kW

(c) 55.5 kW (d) 6 kW

GPSC ARTO Pre 30.12.2018

ISRO Scientist/Engineer 03.07.2016

Ans. (a) : L = 1 m, w = 50 cm

$T_1 = 30°C$, $T_2 = 400°C$

Convective heat transfer co-efficient

$h = 30$ W/m^2K

Heat transfer by convection = $hA\Delta T$

$= 30 \times 1 \times 0.50 \times (400 - 30)$

$= 30 \times 1 \times 0.50 \times 370$

$= 5550$ W $= 5.5$ kW

393. Prandtl number is expressed by-

(a) $\frac{\text{Inside diameter of tube}}{\text{equivalent thickness of film}}$

(b) $\frac{\text{specific heat} \times \text{viscosity}}{\text{thermal conductivity}}$

(c) $\frac{\text{thermal conductivity}}{\text{equivalent thickness of film}}$

(d) $\frac{\text{molecular diffusivity of momentum}}{\text{thermal diffusivity}}$

ISRO Scientist/Engineer (RAC) 29.11.2015

UKPSC AE 2007 Paper-II

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Ans : (b & d) Prandtl number is defined as the ratio of kinematic viscosity (υ) to the thermal diffusivity (α).

$$\Pr = \frac{\upsilon}{\alpha}$$

$\because$ $\alpha = \frac{k}{\rho C_p}$ **and** $\upsilon = \frac{\mu}{\rho}$

$$\Pr = \frac{\left(\frac{\mu}{\rho}\right)}{\left(\frac{k}{\rho C_p}\right)} = \frac{\mu C_p}{k}$$

$$\Pr = \frac{Vis\cos ity \times specific\ heat}{thermal\ conductivity}$$

394. For a fluid having Prandtl number equal to unity, the relation between hydrodynamic boundary layer thickness (δ_1) and thermal boundary layer thickness (δ_2).

(a) $\delta_1 > \delta_2$ (b) $\delta_1 < \delta_2$

(c) $\delta_1 = \delta_2$ (d) $\delta_2 = \frac{\delta_1}{3}$

ISRO Scientist/Engineer 22.04.2018
ESE 2007

Ans. (c) : $P_r^{1/3} = \frac{\delta_1}{\delta_2}$

If, $\Pr = 1$

So, $\delta_1 = \delta_2$

395. A thin flat plate 3 m × 2 m is hanging freely in air. The temperature of surrounding is 24^0C. Solar radiation is falling on one side of the plate at the rate of 432 W/m^2. What should the convective heat transfer coefficient (in W/m^2 – ^{0}C), if the temperature of plate is to remain constant at 30^0C?

(a) 25 (b) 36

(c) 100 (d) 200

UPRVUNL AE 2014, ESE 1993

Ans : (b) From Newton's law of cooling

$$Q = hA(T_s - T_\infty)$$

$$\frac{Q}{A} = 2h(T_s - T_\infty)$$

{since convection is taking place from both side}

$432 = 2 \times h(30 - 24)$

$h = 36$ W/m^2°C

396

(a) –1/3 (b) –2/3

(c) 1 (d) –1

Nagaland PSC (CTSE) 2018, Paper-II
ESE 1994

Ans. (a) : The ratio of the thickness of thermal boundary layer to the thickness of hydrodynamic boundary layer is equal to (Prandtl's number)^n, where n is equal to -1/3.

397. In laminar flow over a flat plate the convective heat transfer coefficient is proportional to

(a) x (b) $x^{-0.5}$

(c) $x^{0.5}$ (d) $x^{-0.2}$

Nagaland PSC CTSE 2017 Paper-2, ESE 2012

Ans. (b) : $Nu = 0.664 (R_e)^{\frac{1}{2}} (P_r)^{\frac{1}{3}}$

$$\frac{hx}{K} = \left(\frac{\rho v x}{\mu}\right)^{\frac{1}{2}}$$

$h \propto x^{\frac{1}{2}-1}$

$h \propto x^{-0.5}$

398. Peclet number s the product of:

(a) Reynold's number and Nusselt Number

(b) Reynold's number and Prandtl Number

(c) Prandtl number and Nusselt Number

(d) Poisson's number and Nusselt number

Nagaland PSC (CTSE) 2018 Paper-II
HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : Peclet number = Pe = Re.Pr.

399. The velocity parameters within the given fields are same when:

(a) Nusselt number are different

(b) Nusselt number are same

(c) Reynolds number are same

(d) Reynolds number are different

APPSC Poly Lect. Automobile Engg. 2020
CIL MT 27.02.2020

Ans. (c) : Reynold's number defined as the ratio of inertia force of flowing fluid and the viscous force of liquid.

$$R_e = \frac{\rho VL}{\mu}$$

Velocity is directly proportional to Reynolds number so when velocity parameter for a given field is same then Reynolds number are same.

400. Reynolds analogy states that

(a) $St = \frac{C_{fx}}{4}$ (b) $St = \frac{C_{fx}}{2}$

(c) $St = \sqrt{C_{fx}}$ (d) $St = 2C_{fx}$

BPSC AE Mains 2017 Paper - V
UKPSC AE 2012 Paper–II

Ans : (b) : Reynolds analogy states that

$$St = \frac{C_{fx}}{2}$$

401. Match list I with List II and select the correct answer:

List I		List II	
A	Fourier number	1	Surface tension
B	Weber number	2	Forced convection
C	Grashoff number	3	Natural convection
D	Schmidt number	4	Radiation
		5	Transient heat conduction
		6	Mass diffusion

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(a) A-6, B-5, C-2, D-3 (b) A-5, B-1, C-3, D-6
(c) A-2, B-4, C-5, D-3 (d) A-1, B-6, C-3, D-5
(e) A-5, B-3, C-1, D-6

CGPSC AE 2014 -II, GATE 1996

Ans. (b) : Fourier number → Transient heat conduction
Weber number → Surface tension
Grashoff number → Natural Convection
Schmidt number → Mass diffusion

402. Maximum heat transfer rate is attained in
(a) laminar flow (b) turbulent flow
(c) transition region (d) creeping flow
(e) steady flow

TSPSC Manager (Engg.) HMWSSB 12.11.2020
CGPSC AE 2014 -II, UKPSC AE 2012 Paper-II

Ans. (b) : Maximum heat transfer rate is attained in turbulent flow due to zig-zag motion of fluid particles with high velocity.

403. Air at 20°C blows over a hot plate of 50 × 60 cm^2 made of carbon steel maintained at 220°C. The convective heat transfer co-efficient is 25 W/m^2K. What will be the heat loss from the plate?
(a) 1500 W (b) 2500 W
(c) 3000 W (d) 4000 W

TRB Polytechnic Lecturer 2017, ESE 2009

Ans. (a) : $Q = h A_s (T_o - T_\infty)$
$= 25 \times 50 \times 60 \times [220 - 20] = 1500$ W

404. If Nusselt number is 390, Reynolds number is 39 and Prandtl number is 20, then Stanton will be
(a) 780 (b) 200
(c) 2 (d) 0.5

UPPSC AE 12.04.2016 Paper-II
UKPSC AE 2013 Paper-II

Ans : (d) Stanton number

$$= \frac{\text{Nusselt Number}}{\text{Reynolds number} \times \text{Prandtl Number}}$$

$$S_t = \frac{N_u}{R_e \times P_r}$$

$$S_t = \frac{390}{39 \times 20}$$

$$\boxed{S_t = 0.5}$$

405. According to Newton's law of cooling, the rate of heat transfer from a solid surface of area A, at a temperature t_1, to a fluid at temperature t_2 is given by:-
(a) $Q = hA(t_1 + t_2)$ (b) $Q = hA(t_1 - t_2)$
(c) $Q = h / A(t_1 - t_2)$ (d) $Q = A / h(t_1 + t_2)$

PTCUL AE 25.06.2017, UKPSC AE-2013, Paper-II

Ans. (b) : According to Newton's law of cooling-
$Q = hA(t_1 - t_2)$

406. Prandtl number is connecting link between:-
(a) Velocity field and temperature field
(b) Gravity field and temperature field
(c) Gravity field and velocity field
(d) None of the above

ESE 2014, UKPSC AE-2013, Paper-II

Ans. (a) : Prandtl number is connecting link between velocity field and temperature field.

407. The properties of mercury at 300 K are: density = 13529 kg/m^3, specific heat at constant pressure = 0.1393 kJ/kg-K, dynamic viscosity = 0.1523×10^{-2} Ns/m^2 and thermal conductivity = 8.540 W/m-K. The Prandtl number of mercury at 300 K is
(a) 0.0248 (b) 2.48
(c) 24.8 (d) 248

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2002

Ans. (a) : Given, $\rho = 13529$ kg/m^3, $C_p = 0.1393$ kJ/kg-K
$\mu = 0.1523 \times 10^{-2}$ Ns/m^2, $k = 8.540$ W/m-K

$$P_r = \frac{\mu C_p}{k} = \frac{0.1523 \times 10^{-2} \times 0.1393 \times 10^3}{8.540} = 0.0248$$

408. In a free convection heat transfer system, Rayleigh Number is represented as _____.
(a) product of Grashoff Number and Prandtl Number
(b) quotient of Grashoff Number and Prandtl Number
(c) difference of squares of Grashoff Number and Prandtl Number
(d) sum of squares of Grashoff Number and Prandtl Number

VIZAG Steel MT 24.01.2021, Shift-I
ISRO Scientist/Engineer 24.05.2014

Ans. (a) : In free convection transfer system–
Rayleigh Number, $(R_a) = G_r \times P_r$
= Grashoff number (G_r) × Prandtl Number (P_r)

409. Reynolds analogy expresses the relation between:
(a) the fluid friction and heat transfer for a laminar flow on a flat plate
(b) the momentum diffusivity and forced convection on a flat plate
(c) the natural convection and forced convection on a flat plate
(d) the fluid friction and boundary layer thickness over a flat plate

NLCIL GET 17.11.2020, Shift-II
UKPSC AE-2013, Paper-II

Ans. (a) : This relation is called the Reynolds Colburn analogy, expresses the relation between fluid friction and heat transfer for laminar flow on a flat plate. Three region can be observed.
(1) A very thin region near the plate surface is called laminar sublayer has the same character of laminar flow.

410. Which one of the following non-dimensional numbers is used for transition from laminar to turbulent flow in free convection?
(a) Reynolds number (b) Nusselt number
(c) Peclet number (d) Rayleigh number

RPSC AE 2018, ESE 2007

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Ans. (d) : Reynolds number— It is defined as the ratio of the inertia force to the viscous force.

$$\text{Re} = \frac{\rho UL}{\mu}$$

- The higher the value of Re the greater will be the relative contribution of inertia effect. The smaller the value of Re, the greater will be the relative magnitude of the viscous stresses.

Prandtl Number (Pr)—It is the ratio of kinematic viscosity (ν) to thermal diffusivity (α)

$$\text{Pr} = \frac{\mu C_p}{k} = \frac{\nu}{\alpha}$$

Kinematic viscosity indicates the impulse transport through molecular friction whereas thermal diffusivity indicates the heat energy transport by conduction process.

- Prandtl Number provides a measure of the relative effectiveness of the momentum and energy transport by diffusion.
- Prandtl Number is a connecting link between the velocity field and temperature field and its value strongly influences relative growth of velocity and thermal boundary layers.

Nusselt Number (Nu)—It is the ratio of heat flow rate by convection process under a unit temperature gradient to the heat flow rate by conduction process under a unit temperature gradient through a stationary thickness of L meters. Thus

$$Nu = \frac{Q_{conv.}}{Q_{cond.}} = \frac{h}{k/L} = \frac{hL}{k}$$

Rayleigh Number—The Rayleigh number (Ra) for a fluid is a dimensionless associated with buoyancy driven flow, also known as free or natural convection.

Ra = Grashof number × Prandtl number

Ra = Gr × Pr

411. When all the condition identical, in the case of flow through pipes with heat transfer, the velocity profile will be identical for :

(a) liquid heating and liquid cooling
(b) gas heating and gas cooling
(c) liquid heating and gas cooling
(d) heating and cooling of any fluid

RPSC ACF & FRO, 26.02.2021
ESE 1997

Ans. (a) : The velocity profile for flow through pipes with heat transfer is identical for liquid heating and liquid cooling.

412. Consider the following :

1. Density
2. Kinematic Viscosity
3. Specific heat at constant pressure
4. Thermal diffusivity

Which of these properties of air increase with temperature

(a) 1, 2, 3 and 4 (b) 2, 3 and 4 only
(c) 1, 2 and 3 only (d) 3 and 4 only

ESE 2012
UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation :

(1) Density– $\boxed{\rho = \frac{P}{RT}}$

as T increases, the density will be decreases.

(2) Kinematic Viscosity–

The Kinematic viscosity ratio (ν) is the ratio of dynamic viscosity to density.

$$\text{Kinematic viscosity } (\nu) = \frac{\text{Dynamic Viscosity } (\mu)}{\text{Density } (\rho)}$$

for gases, Dynamic Viscosity $(\mu) \propto \sqrt{T}$

density $(\rho) \propto \frac{1}{T}$

$$\nu \propto \frac{\sqrt{T}}{1/T}$$

$$\nu \propto T^{3/2}$$

So T increases, Kinematic viscosity will also be increases.

(3) Specific heat at constant Pressure (C_p)– For air, when we increase the temperature then the value of C_p increases.

(4) Thermal diffusivity- When we increase the temperature of the air, then the thermal diffusivity will be increase because molecular collision should be increased.

413. Forced convection in a liquid path is caused by:

(a) Density difference brought about by temperature gradients
(b) Molecular energy interaction
(c) Flow of electrons in a random fashion
(d) Intense stirring by an external agency

MECON MT 2019, TSPSC Managers, 2015

Ans. (d) : If the fluid motion involve in the process is induced by some external means then it is called forced convection.

414. A Prandtl number of a flowing fluid greater than unity indicates that hydrodynamic boundary layer thickness is:

(a) Greater than thermal boundary layer thickness
(b) Equal to thermal boundary layer thickness
(c) Smaller than thermal boundary layer thickness
(d) Independent of thermal boundary layer thickness

RPSC ADE 2016, ESE 2002

Ans. (a) Prandtl number, $\text{Pr} = \frac{\mu Cp}{K} = \frac{\nu}{\alpha}$, $\nu = \frac{\mu}{\rho}$

$$\frac{\delta}{\delta_{th}} = (\text{Pr})^{1/3} \qquad \alpha = \frac{K}{\rho c_P}$$

δ_{th} = thickness of thermal boundary layer
δ = thickness of hydrodynamic boundary layer
P_r value varies small, greater, equal to one
$P_r > 1$
$\delta > \delta_{th}$, means hydrodynamic boundary layer is greater than thermal boundary layer.

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415. For fully developed turbulent flow in a pipe with heating, the Nusselt number Nu, varies with Reynolds number Re and Prandtl number Pr as:

(a) $Re^{0.5}Pr^{1/3}$ (b) $Re^{0.8}Pr^{0.2}$
(c) $Re^{0.8}Pr^{0.4}$ (d) $Re^{0.8}Pr^{0.3}$

RPSC ADE 2016, ESE 2003

Ans. (c) : For fully developed turbulent flow in a pipe the Nusselt number Nu varies, with Reynold number Re & Prandtl number Pr

$Nu = 0.023\ Re^{0.8}\ Pr^{n}$

n = 0.4 for heating of fluid

n = 0.3 for cooling of fluid

But option not mention 0.023 then

$Nu = Re^{0.8}\ Pr^{0.4}$ [for heating]

$Nu = Re^{0.8}\ Pr^{0.3}$ [for cooling]

For turbulent flow [Re > 4000]

for laminar flow [Re < 2000]

416. A non-dimensional number generally associated with natural convection of heat transfer is

(a) Grashoff Number (b) Weber Number
(c) Reynold Number (d) Nusselt Number

RPSC AE 2018
Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : The Grashoff number (Gr) is dimensionless number in fluid dynamics and heat transfer which approximates the ratio of the buoyancy to the viscous force acting on a fluid.

It frequently arises in the study of situations involving natural convection and is analogous to Reynolds number.

$$\text{Weber Number} = \sqrt{\frac{\text{Inertia force}}{\text{Surface tension}}} \Rightarrow \text{In capillary tubes}$$

$$\text{Renolds Number} = \frac{\text{Inertia force}}{\text{Viscous force}} \Rightarrow \text{Laminar Viscous flow in confined passage}$$

$$\text{Nusselt Number} = \frac{Q_{convection}}{Q_{conduction}} = \frac{h\ell_c}{k}$$

$$\text{Grashoff Number (Gr)} = \frac{\text{Buoyant force}}{\text{Viscous force}}$$

417. A dimension heat transfer coefficient which gives a measure of the ratio of the heat transfer rate to the rate at which heat would be conducted within the fluid under temperature gradient is known as

(a) Reynold's number (b) Grashoff's number
(c) Nusselt number (d) Stanton number

GPSC Asstt. Director of Transport 05.03.2017
APPSC AEE 2012

Ans. (c) : The Nusselt number is the ratio of convective to conductive heat transfer across a Boundary. The convective and conduction heat flows are parallel to each other and to the surface normal of Boundary surface and are all perpendicular to mean fluid flow in the simple case. It is the ratio of the heat transfer rate to the rate at which heat would be conducted within the fluid under temperature gradient.

418. The process of heat transfer from one particle of the fluid to another by the actual movement of fluid particles due to difference of density caused by temperature of the particles is known as

(a) conduction (b) free convection
(c) forced convection (d) radiation

Nagaland PSC (CTSE) 2018 Paper-II
Assam PSC CCE Pre 2015

Ans. (b) : The process of heat transfer from one particle of the fluid to another by the actual movement of fluid particles due to difference of density caused by temperature of the particles is known as free convection.

419. The rate of heat transfer from a solid surface to a fluid is obtained from

(a) Newton's law of cooling
(b) Fourier's law
(c) Kirchhoff's law
(d) Stefan's law

JPSC AE 2013 Ist Paper
UKPSC AE 2012 Paper–II

Ans. (a) : The rate of heat transfer from a solid surface to a fluid is obtained from Newton's law of cooling.

420. In natural convection, the ratio of thermal energy liberated by buoyancy to the energy dissipated by heat conduction and viscous drag is known as :

(a) Nusselt number (b) Grashof number
(c) Rayleigh number (d) Prandtl number

RPSC ACF & FRO, 26.02.2021

Ans. (b) : In natural convection, the ratio of thermal energy liberated by buoyancy to the energy dissipated by heat conduction and viscous drag is known as Grashof number.

421. A case of natural convection is given by :

(a) Cooling of billets in atmosphere
(b) Cooling in IC engines
(c) Flow of water inside condensers
(d) Cooling of a hot plate in a steam of cold water

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) : Cooling of billets in atmosphere → Natural Convection cooling in IC engine

Flow of water inside condensers → Force convection

Cooling of hot plate in a stream of cold water → Force convection

422. _____ is a process involving mass movement of fluids during heat transfer.

(a) conduction (b) transition
(c) convection (d) radiation

UPRVUNL AE 05.07.2021

Ans. (c) :

423. Controlling heat transfer film co-efficient is the one, which offers—— ressistance to heat transfer.

(a) no (b) the least
(c) the largest (d) lower

APGCL AM, 2021

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Ans. (c) : From Newton's law of cooling (for convection)

$q_{conv.} = hA\,(T_w - T_0)$

To control heat transfer, the film coefficient or convective heat transfer coefficient should be less. That means persistence $\left(\frac{1}{hA}\right)$ becomes large.

424. Peclet number is defined as

(a) Kinematic viscosity/Thermal diffusivity
(b) Convective heat transfer/Conduction heat transfer
(c) Buoyancy force × Inertial force/Viscous force
(d) Wall heat transfer rate/ Convection heat transfer

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :

425. In forced convection heat transfer systems, the ratio of actual heat flux of the fluid to the heat flux capacity of the fluid flow is called _____.

(a) Reynold number (b) Prandtl number
(c) Nusselt number (d) Stanton number

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : In forced convection, heat transfer systems, the ratio of actual heat flux of the fluid to the heat flux of the fluid to heat flux capacity of the fluid flow is called Stanton number.

$$St = \frac{N_u}{Re.Pr} = \frac{h}{P_v C_p}$$

where, P = density of fluid
N = Speed of fluid
C_p = specific heat capacity of the fluid

426. As defined by Prandtl, in convective heat transfer, a thin layer next to the wall where shear stress is confined is called.

(a) radiative layer (b) diffusion layer
(c) boundary layer (d) compound layer

UPRVUNL AE 05.07.2021

Ans. (c) : According to Prandtl, in convective heat transfer, a thin layer next to the wall where shear stress is confined is called boundary layer.

427. If there are no externally induced flow velocities, then the Nusselt number (Nu) does not depend upon

(a) Prandtl number (Pr)
(b) Reynolds number (Re)
(c) Grashof number (Gr)
(d) None of the above

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) :

428. In a heat exchanger, higher flow Reynolds number can help to get

(a) higher heat transfer effectiveness
(b) higher mean temperature difference
(c) Lower pressure drop
(d) Lower Transfer units

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : $Nu = 0.662\,(Re)^{1/2}\,(Pr)^{1/3}$

$$\frac{hd}{k} = 0.662\left(Re\right)^{1/2}\left(Pr\right)^{1/3}$$

$$h \propto \left(Re\right)^{1/2}$$

429. A cylinder of outer surface area 10 m^2 loses heat on its surface at 85^0C by convection to a fluid at 25^0C. The average heat convection heat transfer coefficient is 35 W/m^2K. The heat transfer rate is :

(a) 25 kW (b) 21 kW
(c) 35 kW (d) 60 kW

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : Given that

$A = 10m^2$
$h = 35w/m^2k$
$\Delta T = 85 - 25 = 60^0C$

we know that

$Q = h\,A\,\Delta T$
$= 35 \times 10\,(85-25)$
$= 35 \times 10 \times 60$
21000
$= 21$ kW

430. A steel plate is exposed to solar heat flux of 800 W/m^2 on one side. The plate is exposed to air at 30°C on both sides. The convection coefficients are 10 W/m^2K on the back side and 15 W/m^2K on the front. Neglecting radiation loss, the equilibrium temperature is :

(a) 26°C (b) 62°C
(c) 38°C (d) 56°C

NLC GET 17.11.2020, Shift-II

Ans. (b) : Given, Solar heat Flux (Q) = 800 W/m^2
Exposed air temperature (T_∞) = 30°C
Convection coefficient on back side (h_b) = 10W/m^2K
Convection coefficient on front side (h_f) = 15 W/m^2K
In equilibrium condition → heat flux incident on plate = heat release by plate via cooling.

$800 \times A = h_b\,A\,(T_P - T_\infty) + h_f A\,(T_P - T_\infty)$
$800 = 10(T_P - 30) + 15(T_P - 30)$
$800 = 25(T_P - 30)$
$32 = T_P - 30$
$\boxed{T_P = 62°C}$

431. In natural convection of vertical plate, the ratio of the average heat-transfer coefficient to the local heat-transfer coefficient is :

(a) 3/4 (b) 3/2
(c) 2/3 (d) 4/3

NLC GET 17.11.2020, Shift-II

Ans. (d) : In natural convection of vertical plate, the ratio of the average heat transfer coefficient to the local heat transfer coefficient is 4/3.

432. The Reynolds-Colburn analogy :

(a) expresses the relation between convection and conduction heat transfer for turbulent flow on a flat plate.

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(b) is the criteria for the identification of the attainment of the fully developed flow.
(c) is the criteria for the identification of the attainment of the fully turbulent flow.
(d) expresses the relation between fluid friction and heat transfer for laminar flow on a flat plate.

NLC GET 17.11.2020, Shift-II

Ans. (d) : The Reynolds analogy is popularly known to relate turbulent momentum and heat transfer. That is because in a turbulent flow the transport of momentum and the transport of heat largely depends on the same turbulent eddies. The velocity and the temperature profile have the same shape.

The Reynolds-Colburn analogy expresses the relation between Fluid friction and heat transfer for laminar flow on a flat plate.

433. Heat transfer coefficient is related to thermal conductivity by

(a) Fourier number (b) Prandtl number
(c) Weber number (d) Nusselt number

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Nusselt No. (Nu) = $\frac{hd}{k}$

h = convective heat transfer co-efficient
k = conductive heat transfer co-efficient
d = diameter of pipe

434. In a flying aircraft, heat transfer between the atmosphere and the aircraft's outer wall surface is dominated by:

(a) forced convection (b) conduction
(c) natural convection (d) radiation

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : In an aircraft installations, electric fans and blowers are used to aid convection. An aircraft air-cooled piston engine is a good example of convection being used to transfer heat.

In a flying aircraft, heat transfer between the atmosphere and the aircraft's outer wall surface is dominated by forced convection.

435. If 'Pr' is the Prandtl number, in forced convection, the ratio of velocity and thermal boundary layer thickness is given as:

(a) $Pr^{2/3}$ (b) $Pr^{1/3}$
(c) $Pr^{5/3}$ (d) $Pr^{4/3}$

NLCIL GET 17.11.2020, Shift-II

Ans. (b) :

$$\frac{\text{Hydrodynamic boundary layer thickness}}{\text{Thermal boundary layer thickness}} = Pr^{1/3}$$

$$\frac{\delta}{\delta_t} = Pr^{1/3}, 0.6 \le Pr \le 50$$

436. Nusselt number in a thermal boundary layer is similar to:

(a) friction coefficient in the velocity boundary layer
(b) normalised momentum thickness in the velocity boundary layer
(c) normalised displacement thickness in the velocity boundary layer
(d) normalised energy thickness in the velocity boundary layer

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : The Nusselt number represents the enhancement of heat transfer through a fluid layer as a result of convection relative to conduction across the same fluid layer. The larger the Nusselt number, the more effective is the convection.

Nusselt number $Nu = \frac{q_{conv}}{q_{cond}} = \frac{hL}{k}$

Nusselt number in a thermal boundary layer is similar to the friction coefficient is to the velocity boundary layer.

437. In forced convection, over the flat plate, the average value of Nusset number is given as:

(a) $0.664\ Re^{0.3}\ Pr^{0.5}$ (b) $0.332\ Re^{0.5}\ Pr^{0.3}$
(c) $0.332\ Re^{0.3}\ Pr^{0.5}$ (d) $0.664\ Re^{0.5}\ Pr^{0.3}$

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : The average value of Nusset number

$$N_u = \frac{hL}{k} = 0.664\,Re^{1/2}.Pr^{1/3}$$

$$\boxed{N_u = 0.664\,Re^{0.5}\,Pr^{0.3}}$$

438. When we have free and forced convection simultaneously, Gr/Re^2 is

(a) greater than 1
(b) almost equal to 1
(c) less than 1
(d) equal to Peclet number

TSPSC Manager (Engg.) HMWSSB 12.11.2020
ISRO Scientist/Engineer (RAC) 2014

Ans. (b) : • $\frac{Gr}{R_e^2} = 1$, Effect of both forced convection and free convection should be considered.

• $\frac{Gr}{R_e^2} >> 1$, Forced convection effect neglected.

• $\frac{Gr}{R_e^2} << 1$, Free convection effect neglected.

439. Which of the following condition is correct for natural convection?

(a) $(Gr/ Re^2) = 1$ (b) $(Gr/ Re^2) << 1$
(c) $(Gr/ Re^2) >> 1$ (d) None of the above

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (c) : For natural convection : $(Gr/ Re^2) >> 1$

440. The typical range of Prandtl number for water is

(a) 0.004-0.300 (b) 1.7-13.7
(c) 50-500 (d) 2000-1000

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : • Typical range of prandtl no. for water : 1.7 to 13.7
• Typical range of prandtl no. for air : 0.65 to 0.71

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441. A fluid having a density of 1 g/cc is in a state with Grashof number 2×10^6 and prandtl number 0.7. Assuming acceleration due to gravity as 10 m/s^2, calculate the Rayleigh number?

(a) 1.4×10^6 (b) 2.86×10^6
(c) 3.7×10^5 (d) 8.4×10^6

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : $Ra = Gr \times Pr$
$= 2 \times 10^6 \times 0.7$
$= 1.4 \times 10^6$

442. The Nusselt number for a laminar flow is related to the Reynolds number as :

(a) $R_e^{-0.5}$ (b) $R_e^{0.8}$
(c) $R_e^{0.5}$ (d) $R_e^{0.33}$

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Empirical correlation for forced conversation:-

(A) Laminar flow.

(i) Plane surface : Flow part flat plates and walls

$$Nu_x = \frac{h_x.x}{k} = 0.332(Re_x)^{0.5}(Pr)^{0.333}$$

$$\overline{Nu} = \frac{\bar{h}\ell}{k} = 0.664(Re)^{0.5}(Pr)^{0.333}$$

(B) Turbulent flow

(i) Turbulent flow over flat surface

$Nux = 0.0282\ (Re)^{0.8}\ (Pr)^{0.33}$

$$\overline{Nu} = 0.036(Re)^{0.8}(Pr)^{0.33}$$

Nusselt number for a laminar flow is related to the Reynolds Number.

443. The convection heat transfer is a combination of :

(a) conduction and radiation
(b) conduction and advection
(c) conduction and work transfer
(d) advection and work transfer

APPSC Poly Lect. 13.03.2020

Ans. (b) : The convection heat transfer is a combination of conduction and advection.

444. Nusselt number may be characterized by :

(a) the temperature (dimensional) gradient at the surface
(b) the dimensionless temperature gradient at the surface
(c) the ratio of the convective to the conductive resistance
(d) the ratio of the rate of heat transfer due to convection to the rate of the store of the energy

APPSC Poly Lect. 13.03.2020

Ans. (b) : Nusselt number may be characterized by the dimensionless temperature gradient at the surface.

445. The black surface of a plate is heated at the rate of 50 W/m^2 and the front surface is cooled by air flow at 30^{0}C. It is given that the heat transfer coefficient between the air and the plate surface is 50 W/m^2 – K. Considering steady state heat transfer, what is the temperature (in ^{0}C) of the front surface of the plate.

(a) 21 (b) 31
(c) 41 (d) 51

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given :

Heat transfer coefficient (h) = 50 W/m^2–K

Ambient temperature (T_{air}) = 30^{0}C

Heat supplied to plate $\left(\frac{Q}{A}\right) = 50$ W/m^2

Let;

T = Temperature of the surface

$Q = hA\Delta T = hA\ (T - T_{air})$

$$\Rightarrow \frac{Q}{A} = h\left(T - T_{air}\right)$$

$$\Rightarrow 50 = 50\left(T - 30\right)$$

$$\Rightarrow T - 30 = 1$$

$$\boxed{T \quad 31^0C}$$

446. In free convection, motion of fluid is caused by

(a) Weight of fluid element
(b) Hydrostatic force
(c) Variable buoyancy forces
(d) None of the above

GPSC Engineer, Class-II Pre-19.01.2020

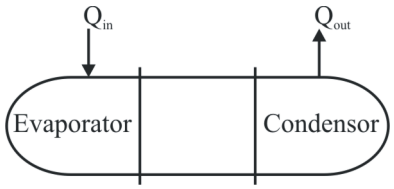
Ans. (c) :

447. The solar heat pipe works on the principle of

(a) heating and condensation cycle
(b) evaporation and condensation cycle
(c) cooling and condensation cycle
(d) heating and evaporation cycle

ESE 2019

Ans. (b) : Heat pipes employ evaporative cooling to transfer thermal energy from one point to another by the evaporation and condensation of a working fluid or coolant.



448. A electric heater of exposed surface area 0.09 m^2 and output 600 watts is designed to operate fully submersed in water. When the water is at 37°C and the surface coefficient of heat transfer is 285.3 W/m^2-deg C, surface temperature of the heater will be:

(a) 30.5°C (b) 60.5°C
(c) 90.5°C (d) 120.5°C

ISRO Scientist/Engineer (RAC), 10.03.2019

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Ans. (b) : Given,

$A = 0.09m^2$

$Q = 600$ watt $\quad Q = hA(t_s - t_1)$

$t_1 = 37°C \quad 600 = 285.3 \times 0.09 \times (t_s - 37°C)$

$h = 285.3$ w/m2°C $\quad \boxed{t_s = 60.367°C}$

$t_s = ?$

449. Air at 30° C flows over a surface at 80°C. The local heat flow was measured at a point as 500 W/m^2. The local value of convective heat transfer coefficient is :

(a) 45.4 W/m^2K (b) 10.0 W/m^2K

(c) 4.54 W/m^2K (d) 25.68 W/m^2K

Oil India Ltd. Sr. Engineer (Drilling) 30.11.2019

Ans. (b) : Given, $T_w = 80°C$

$T_\infty = 30°$

Heat flow per unit area

$Q = 500\ W/m^2$

$h = ?$

we know that,

$$Q = Ah(T_w - T_\infty) \quad \rightarrow \quad h = \frac{Q}{A}\cdot\left(\frac{1}{T_w - T_\infty}\right)$$

Convective heat transfer coefficient (h)

$$h = 500 \times \left(\frac{1}{80-30}\right) = 10.0\ W/m^2K$$

450. When a hot fluid is flowing over a cold flat plate, the temperature gradient is .....

(a) zero at the surface

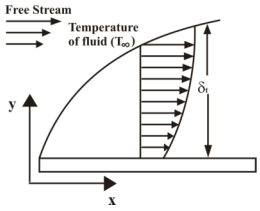
(b) negative at the surface

(c) zero at the edge of the thermal boundary layer

(d) positive at the edge of the thermal boundary layer

RPSC AE 2018

Ans. (c) : When $T_\infty > T_P$



The boundary conditions of thermal boundary layer—

At any distance x, measured from leading edge of plate.

At $y = 0, \quad T = T_p$

At $y = \delta_t, \quad T = T_\infty$

At $y = \delta_t, \quad \frac{\partial T}{\partial y} = 0$

At $y = 0, \quad \frac{\partial^2 T}{\partial y^2} = 0$

$\left(\frac{\partial T}{\partial y}\right) = f(y), \frac{\partial T}{\partial y}$ is maximum at y = 0.

451. Air is flowing over a hot plate at temperature 120 °C. If at a point Reynold number is increased by 4 times and Nusselt number is increased by 2 times, then new value of Prandtl number will be:

(a) 2 times (b) 4/9 times

(c) same (d) 1/2 times

AAI Jr. Executive 29.11.2018

Ans. (c) : Same

452. The velocity and thermal boundary layers coincide, and the non dimensional velocity and temperature profiles are identical for steady, incompressible, laminar flow over a flat plate when

(a) Pr = 1 (b) Pr < 1

(c) Pr > 2 (d) Pr ≥ 2.5

TNPSC AE 2018

Ans. (a) : When, $\delta_t = \delta_H$

Then, Pr = 1

where Pr = Prandtl Number

453. The ratio $\left(\frac{\text{Molecular diffusivity of momentum}}{\text{Molecular diffusivity of heat}}\right)$, is known as :

(a) Prandtl number (Pr)

(b) $\frac{1}{\text{Prandtl number}(\text{Pr})}$

(c) Nesselt Number (Nu)

(d) $\frac{1}{\text{Nusselt number (Nu)}}$

(e) Peclet number (Pc)

CGPSC AE 25.02.2018

Ans. (a) : Prandtl Number– It is defined as the ratio of molecular diffusivity of momentum (ν) to molecular diffusivity of heat (α).

$$\text{Pr} = \frac{\nu}{\alpha} = \frac{\mu C_p}{k}$$

454. A spherical steel ball of 12mm diameter is initially at 1000K. It is slowly cooled in a surrounding of 300K. The heat transfer coefficient between the steel ball and the surrounding is 5W/m^2 K. The thermal conductivity of steel is 20W/mK. The temperature difference between the centre and the surface of the steel ball is

(a) Large because conduction resistance is far higher than the convective resistance

(b) Large because conduction resistance is far less than the convective resistance

(c) Small because conduction resistance is far higher than the convective resistance

(d) Small because conduction resistance is far less than the convective resistance

GPSC Asstt. Director of Transport 05.03.2017

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Ans. (d) : We know, Biot number $\left(B_i\right)=\frac{hL_c}{k}$

$$\left\{L_c=\frac{d}{6}=\frac{0.012}{6}=0.002\right\}$$

$$=\frac{5\times0.002}{20}=0.0005$$

For given condition, biot number tends to zero, i.e conduction resistance is far less than convection resistance. Therefore temperature between centre and surface is very small.

455. In turbulent flow, one can estimate the Nusselt number using the analogy between heat and momentum transfer. (colburn analogy). This analogy relates the Nusselt number to the co-efficient of friction, C_f as.

(a) $(Nu) = 0.5\ C_f\ (Re)\ (Pr)^{1/3}$
(b) $(Nu) = 0.5\ C_f\ (Re)\ (Pr)^{2/3}$
(c) $(Nu) = C_f,\ Re.\ (Pr)^{1/3}$
(d) $(Nu) = C_f\ (Re)^{1/2}\ (Pr)^{1/3}$

TRB Polytechnic Lecturer 2017

Ans. (a) : Reynold's analogy –

$$St.Pr^{2/3} = C_f/2$$

$$\frac{Nu}{Re\cdot Pr}\cdot Pr^{2/3}=\frac{C_f}{2}$$

$$Nu = 0.5C_f\left(R_e\right)\left(P_r\right)^{\frac{1}{3}}$$

456. The prominent mode of heat transfer from electrical transmission lines is

(a) conduction (b) natural convection
(c) forced convection (d) radiation

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : Natural convection, the prominent modes of heat transfer from electrical transmission lines.

457. The thermal boundary layer thickness is thicker than the momentum boundary layer thickness when Prandtl number is

(a) 0 (b) 1
(c) Less than 1 (d) Greater than 1

Nagaland PSC CTSE 2017 Paper-2

Ans. (c) : When thermal boundary layer thickness is thicker than the momentum boundary layer thickness when Prandlt number is less than 1.
Similarly,
If thermal boundary layer thickness is thinner than the momentum boundary layer thickness, when Prandlt number is greater than 1.

458. The free convection heat transfer is influenced by the

(a) Buoyant and inertia forces only
(b) Viscous and buoyant forces only
(c) Inertia forces only
(d) Viscous, buoyant and inertia forces

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : Viscous, buoyant and inertia forces

459. Air at 1 atmospheric pressure and 27°C blows across a 12mm diameter sphere at a free stream velocity of 4 m/s. A small heater inside the sphere maintains the surface temperature at 77°C. With k = 0. 0.026 W/m (Kelvin) and with (Nu) = 31.4, the heat loss by the sphere would be

(a) 1.93J/s (b) 1.76J/s
(c) 1.65J/s (d) 1.54J/s

ESE 2017

Ans. (d) : Given,

$T_\infty = 27°C$ $D = 12\ mm = 0.012\ m$
$k = 0.026\ W/mK$ $R = 0.006\ m$
$Nu = 31.4$
$V_\infty = 4\ m/s$
$T_w = 77°C$

$$\text{Nusselt number (Nu)} = \frac{hL_c}{k}$$

$$31.4 = \frac{h\times0.012}{0.026}$$

$$h = 68.03\ W/m^2K$$

Heat lost by sphere = $hA\ \Delta T$

$$= h(4\pi R^2)\ (T_w - T_\infty)$$
$$= 68.03 \times 4 \times \pi\ (0.006)^2\ (77 - 27)$$
$$= 1.538\ Watt$$

460. The process of heat transfer from one particle of the body to another by the actual motion of the heated particles, is called.

(a) conduction (b) convection
(c) radiation (d) all of the above

Punjab PSC SDE 12.02.2017

Ans. (b) : The process of heat transfer from one particle of the body to another by the actual motion of the heated particles, is called convection.

461. The convective heat transfer coefficient in laminar flow over a flat plate.

(a) increases if a lighter fluid is used
(b) increases if a higher viscosity fluid is used
(c) increases if higher velocities are used
(d) increases with distance

HPPSC Asstt. Prof. 20.11.2017

Ans. (c) : Local Nusselt No. for laminar flow.

$$= Nu_x = \frac{h_x.x}{k} = 0.332\times(Re_x)^{1/2}.(P_r)^{1/3}$$

$$\frac{h_x.x}{k} = 0.332\times\left(\frac{\rho Vx}{\mu}\right)^{1/2}\left(P_r\right)^{1/3}$$

$h_x \propto \rho^{1/2}$ $\quad h_x \uparrow \Rightarrow \rho\uparrow$

$h_x \propto \frac{1}{(\mu)^{1/2}}$ $\quad h_x \uparrow \Rightarrow \mu\downarrow$

$h_x \propto v^{1/2}$ $\quad \boxed{h_x \uparrow \Rightarrow v\uparrow}$

$h_x \propto \frac{1}{x^{1/2}}$ $\quad h_x \uparrow \Rightarrow x\downarrow$

So option C is correct.

462. The ratios of the laminar hydrodynamic boundary layer thickness to thermal boundary layer thickness of flows of two fluids P and Q on a flat plate are 1/2 and 2 respectively. The Reynolds number based on the plate length for both the flows is 10^4. The Prandtl and Nusselt numbers for P are 11/8 and 35 respectively. The Prandtl and Nusselt numbers for Q are respectively

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(a) 8 and 140 (b) 8 and 70
(c) 4 and 70 (d) 4 and 35

GPSC Asstt. Director of Transport 05.03.2017
GATE 2011

Ans. (a) : We know,

$$\frac{\delta_t}{\delta} = P_r^{-1/3}$$

For fluid Q, $\frac{1}{2} = \frac{\delta_t}{\delta} = P_r^{-1/3}$

$P_r = 8$

For fluid P, Laminar flow over flat plate

$$Nu = 0.664\, Re_L^{1/2} \times Pr^{1/3} = 35$$

Similarly, fluid Q,

$$= 0.664 \times (10^4)^{1/2} \times (8)^{1/3} = 140$$

Pr No = 8, Nu No. = 140

463. Water (specific heat, c_p = 4.18 kJ / kgK) enters a pipe at a rate of 0.01 kg/s and a temperature of 20°C. The pipe, of diameter 50 mm and length 3m, is subjected to a wall heat flux qw in W/m^2 .If qw = 5000, and the convection heat transfer coefficient at the pipe outlet is 1000 W/m^2 K , the temperature in °C at the inner surface of the pipe at the outlet is

(a) 71 (b) 76
(c) 79 (d) 81

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Given,

$q_w = 5000\, w/m^2$, $h = 1000\ w/m^2k$

$Q = q_w \times$ surface area

$= 5000 \times \pi \times d \times \ell$

$= 5000 \times \pi \times 0.05 \times 3 = 2355$ W

$Q = mC_p(T_o - T_i)$

Amount of heat gained by the water.

$2355 = 0.01 \times 4180 \times (T_o - 20)$

$T_o = 76.33°C$

$q_w = h(T_{so} - T_o)$

$5000 = 1000(T_{so} - 76.33)$

$T_{so} = 81°C$

464. What is the Prandtl number for a gas at 27°C and 1 × 10^5 Pa, with kinematic viscosity equal to 1.8 × 10^{-5} m^2/s, specific heat capacity at constant pressure = 700 J/kg-K and thermal conductivity of 0.02 W/m-K? (Take Gas constant = 300 J/kg-K)

(a) 0.52 (b) 0.62
(c) 0.70 (d) 0.78

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : $\boxed{\text{Prandtl Number } Pr = \frac{\mu C_P}{K}}$

$T = 27°C = 27 + 273 = 300$ K

$P = 1 \times 10^5$ Pa

Kinematic viscosity $\upsilon = 1.8 \times 10^{-5}$ m^2/sec

$C_P = 700$ J/kg-K

$k = 0.02$ W/m-K

$R = 300$ J/kg-K

Ideal gas equation PV = mRT

$$P = \frac{m}{V}RT$$

$$P = \rho RT$$

$$\rho = \frac{P}{RT} = \frac{1\times10^5}{300\times300}$$

$$= \frac{1\times10^5}{9\times10^6}$$

$$\rho = \frac{1\times10^{-1}}{9} = \frac{1}{90}\ kg/m^3$$

$$\boxed{Pr = \frac{\text{Kinematic viscosity}}{\text{Thermal diffusivity}} = \frac{\upsilon}{\frac{k}{\rho C_P}}}$$

$$Pr = \frac{1.8\times10^{-5}}{\frac{0.02}{\frac{1}{90}\times700}} = \frac{1.8\times10^{-5}}{\frac{0.02\times9}{70}} = \frac{1.8\times10^{-5}\times70}{0.02\times9}$$

$$= \frac{18\times7\times10^{-5}}{0.02\times9} = \frac{2\times7\times10^{-5}}{0.02} = \frac{14\times10^{-3}}{2}$$

Pr = 0.70

465. Match List I with List II and select the correct answer using the codes given below.

List I	List II
(A) Momentum transfer	1. Thermal diffusivity
(B) Mass transfer	2. Kinematic viscosity
(C) Heat transfer	3. Diffusion co-efficient

	A	B	C
(a)	2	3	1
(b)	1	3	2
(c)	3	2	1
(d)	1	2	3

TNPSC AE 2017

Ans. (a) : Momentum transfer → Kinematic viscosity
Mass transfer → Diffusion co-efficient
Heat transfer → Thermal diffusivity

466. If Nusselt number is 4000 with corresponding Reynolds and Prandtl numbers as 40 and 20 respectively, the relevant Stanton number will be

(a) 40 (b) 20
(c) 10 (d) 5

JWM 2017

Ans. (d) : Stanton number

$$(St\ No.) = \frac{\text{Nusselt number}}{\text{Reynold No.} \times \text{Prandtl No.}}$$

$$St\ No. = \frac{4000}{40\times20} = 5$$

467. The convective heat transfer co-efficient from a hot cylindrical surface exposed to still air varies in accordance with

(a) $(\Delta T)^{0.5}$ (b) $(\Delta T)^{1.25}$
(c) $(\Delta T)^{0.25}$ (d) $(\Delta T)^{0.75}$

GPSC ARTO 01.05.2016

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Ans. (c) : Heater transfer co-efficient from a hot cylindrical surface varies with temperature.

$$h \propto (\Delta T)^{0.25}$$

468. Match List-I and List-II and select the correct answer :

List-I	List-II
A. Reynolds number	1. Film coefficient, pipe diameter, thermal conductivity
B. Prandtl number	2. Flow velocity, acoustic velocity
C. Nusselt number	3. Heat capacity, dynamic viscosity, thermal conductivity
D. Mach number	4. Flow velocity, pipe diameter, kinematic viscosity

Code :

	A	B	C	D
(a)	4	1	3	2
(b)	4	3	1	2
(c)	2	3	1	4
(d)	2	1	3	4

(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (b)	
A. Reynolds number $\left(R_e = \frac{V_D}{v}\right)$	1. Flow velocity, pipe diameter, kinematic viscosity
B. Prandtl number $\left(Pr = \frac{\mu C_P}{k}\right)$	2. Heat capacity, dynamic viscosity, thermal conductivity
C. Nusset number $\left(Nu = \frac{hD}{k}\right)$	3. Film coefficient, pipe diameter, thermal conductivity
D. Mach number $\left(M = \frac{V}{C}\right)$	4. Flow velocity, acoustic velocity

469. Consider the development of laminar boundary layer for a moving non-reacting fluid in contact with a flat plate of length 'l' along the flow direction. The average value of heat transfer coefficient can be obtained by multiplying the local heat transfer coeffieient at the trailing edge by the factor

(a) 1.0 (b) 2.0

(c) 1.5 (d) 0.75

UPRVUNL AE 21.08.2016

Ans. : (b)

Average value of heat transfer coefficient = h_{avg}

Local value of heat transfer coefficient = h_{local}

$$\frac{h_{avg}}{h_{local}} = 2$$

$$h_{avg} = 2 \times h_{local}$$

470. Combined natural and forced convection analysis is done only when (G_r = Grashoff number and R_e = Reynold number):

(a) $0.1 < \frac{G_r}{R_e^2} < 10$ (b) $0.1 < \frac{G_r}{R_e} < 10$

(c) $0.1 < \frac{G_r}{R_e^3} < 10$ (d) $0.1 < \frac{G_r}{R_e^4} < 10$

UPRVUNL AE 2016

Ans. (a) : Combined natural and forced convection analysis is done only when

$$0.1 < \frac{G_r}{R_e^2} < 10$$

471. By which of the following modes of heat transfer, Heat is mainly transferred from an insulated pipe to the surrounding still air.

(a) conduction (b) Radiation

(c) forced convection (d) natural convection

APPSC AEE MAINS 2016 PAPER-III

Ans. (d) : By natural convection, heat is mainly transferred from an insulated pipe to the surrounding still air.

472. Which of the following is correct statement for local heat transfer coefficient variation in boundary layer along isothermal plate?

(a) Local heat transfer coefficient increases in flow direction for laminar flow and increases in flow direction for fully developed turbulent flow

(b) Local heat transfer coefficient increases in flow direction for laminar flow and decreases in flow direction for fully developed turbulent flow

(c) Local heat transfer coefficient decreases in flow direction for laminar flow and decreases in flow direction for fully developed turbulent flow

(d) Local heat transfer coefficient decreases in flow direction for laminar flow and increases in flow direction for fully developed turbulent flow

UPRVUNL AE 2016

Ans. (c) : For local heat transfer coefficient variation in boundary layer along isothermal plate, local heat transfer coefficient decreases in flow direction for laminar flow and decreases in flow direction for fully developed turbulent flow.

473. Convective heat transfer coefficient is a function of:

(a) Configuration of body only

(b) Nature of the fluid flow, properties of fluid and configuration of body

(c) Nature of the fluid flow only

(d) Properties of fluid only

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Ans. (b) : Convective heat transfer coefficient—The convective heat transfer coefficient, h can be defined as the rate of heat transfer between a solid surface and a fluid per unit surface area per unit temperature difference. The convective heat transfer coefficient is developed upon the physical properties of the fluid and the physical situation.

474. For flow of fluid over a heated plate, the following fluid properties are known:
Viscosity = 0.001 Pa-s, sp. heat at constant pressure = 1 kJ/kg-K, thermal conductivity = 1W/mK
The hydrodynamic boundary layer thickness at a specified location on the plate if 1 mm, the thermal boundary layer thickness at the same location is
(a) 0.001 mm (b) 0.01 mm
(c) 1 mm (d) 10 mm

PPSC Asstt. Municipal Engineer 15.06.2021
UPPSC AE 12.04.2016 Paper-II, GATE 2008

Ans : (c) Given data:-

$$\mu = 0.001 Pa - sec$$

$$C_p = 1 kJ/kg K = 1000 J/kg - K$$

$$k_f = 1 W/mK$$

Hydrodynamic boundary layer

$$\delta = 1 mm$$

we know that

$$\text{Prandtl Number } (P_r) = \frac{\mu c_p}{k_f} = \frac{0.001 \times 1000}{1} = 1$$

Thermal boundary layer

$$\delta_t = \delta . P_r^{-\frac{1}{3}}$$

$$\delta_t = 1 \times (1)^{-\frac{1}{3}}$$

$$\boxed{\delta_t = 1 mm}$$

475. If in a heat transfer process $Nu \propto Re^{0.8}$, then heat transfer coefficient varies with D in the following pattern (symbols/notation carry their usual meaning)
(a) $h \propto 1/D^{1.6}$ (b) $h \propto 1/D^{0.8}$
(c) $h \propto 1/D^{0.4}$ (d) $h \propto 1/D^{0.2}$

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (d) The average Nusselt Number for flow across cylinders can be expressed as

$$Nu = \frac{hD}{k}$$

and Reynold Number $(Re) = \frac{\rho VD}{\mu}$

Given,

$$Nu \propto Re^{0.8},$$

$$\frac{hD}{k} \propto \left(\frac{\rho VD}{\mu}\right)^{0.8}$$

$$\boxed{h \propto \frac{1}{D^{0.2}}}$$

476. The thermo-physical properties of a fluid are: density = 13600 kg/m^3, dynamic viscosity = 0.152 × 10^{-2} Ns/m^2, thermal conductivity = 8.5 W/mK, Specific heat C_p = 0.14 kJ/kgK., Prandtl Number = 0.025. This liquid flows over a flat plate which is heated over its entire length. Which of the following relationship between hydrodynamic boundary layer (δ) and thermal boundary layer (δ_t) is true?
(a) $\delta_t = \delta$
(b) $\delta_t < \delta$
(c) $\delta_t > \delta$
(d) Cannot be predicted

GPSC Asstt. Prof. 28.08.2016

Ans. (c) : Prandtl number, $(Pr)^{1/3} = \frac{\delta}{\delta_t}$

$$\frac{\delta}{\delta_t} = (Pr)^{1/3}$$

$$Pr = 0.025 < 1$$

$$\frac{\delta}{\delta_t} < 1$$

$$\boxed{\delta_t > \delta}$$

477. What is Stanton number?
(a) It measures the ratio of specific volume of a fluid to the thermal capacity of the fluid
(b) It measures the ratio of specific gravity of a fluid to the thermal capacity of the fluid
(c) It measures the ratio of heat specific gravity of a fluid to specific volume of the fluid
(d) It measures the ratio of heat transferred into a fluid to the thermal capacity of the fluid

KPCL AE 2016

Ans. (d) : The Stanton number (St) is dimensionless number that measures the ratio of heat transferred into a fluid to the thermal capacity of fluid.

$$St = \frac{Nu}{Re . Pr} = \frac{h}{\rho \mu C_P}$$

u = speed of fluid , ρ = density of fluid, C_P= specific heat

478. The "Characteristic length" for horizontal plates while dealing with natural convection problems is given by :
(a) surface area of the plate/perimeter of the plate
(b) perimeter of the plate/surface area of the plate
(c) surface area of the plate/volume of the plate
(d) volume of the plate/surface area of the plate

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : For horizontal plates while dealing with natural convection.

$$\boxed{\text{Characteristic length} = \frac{\text{Surface area of the plate}}{\text{Perimeter of the plate}}}$$

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479. What is the relation between the rate of convection heat and the rate of mixing in turbulent fluid flow?

(a) The rate of convection heat transfer decreases with increase in the rate of mixing in turbulent fluid flow

(b) The rate of convection heat transfer increases with increase in the rate of mixing in turbulent fluid flow

(c) The rate of convection heat transfer does not affected by the change in the rate of mixing in turbulent fluid flow

(d) None of the above

Nagaland PSC (CTSE) 2016, Paper-II

Ans. (b) : The rate of convection heat transfer increases with increase in the rate of mixing in turbulent fluid flow.

480. Which one of the following is INCORRECT about convective heat transfer coefficient 'h'?

(a) It is a function of fluid properties

(b) It is a function of system geometry

(c) It is a function of temperature difference

(d) It is a constant

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (d) : According to Newton's law of cooling (for convection).

$Q_{conv.} = hA\Delta T = hA(T_s - T_\infty)$ Watt

$h = \frac{Q_{conv.}}{A\Delta T}$ Watt / m^2K

$h \rightarrow$ Convective heat transfer coefficient.

Unlike thermal conductivity, it is not a property of the material but it depends upon some of the thermophysical properties like ρ, μ, C_p, k

$(h)_{forced\ conv.} = f\left(\vec{v}, D, \rho, \mu, C_p, k\right)$

$(h)_{free\ conv.} = f(g, \beta, \Delta T, L, \mu, \rho, C_p, k)$

ρ, μ, C_p, k $\Rightarrow$ properties of fluid

Note- $(h)_{forced\ convection} > (h)_{free\ convection}$

So, h is not a constant because it varies with fluid properties, system geometry and temperature differences.

481. Which heat transfer method is comprised of two mechanisms: random molecular motion (diffusion) and energy transferred by bulk or macroscopic motion of the fluid?

(a) Scattering (b) Radiation

(c) Convection (d) Conduction

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : In convection process heat transfer method is comprised of two mechanism.

1- Random molecular motion.

2- Energy transferred by bulk or macroscopic motion.

482. For one-dimensional transient heat conduction problems the graphical analysis can be done by .....method.

(a) Graffer plot (b) Schmidt plot

(c) Sterling plot (d) Mohr's plot

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : For one-dimensional transient heat conduction problems the graphical analysis can be done by Schmidt plot method.

Schmidt plot is a graphical method for determining the temperature at any point in a body at a specified time during the transient heating or cooling.

483. For free convection in laminar region on vertical plates, the correlation is given by (Nu = Nusselt Number, Pr = Prandtl Number and Gr = Grashof Number) :

(a) $Nu = 0.13\ (Gr.\ Pr)$

(b) $Nu = 0.59\ (Gr.\ Pr)^{0.25}$

(c) $Nu = 0.15(Gr.\ Pr)^{0.33}$

(d) $Nu = 0.27\ (Gr.\ Pr)^{0.25}$

RPSC AE (GWD) 06.05.2016

Ans. (b) : $Nu = 0.59\ (Gr.\ Pr.)^{0.25}$

Empirical correlation for free convection.

1. Heated surface up or cooled surface down.

Laminar flow, $2 \times 10^5 < Gr \times Pr < 2 \times 10^7$

$Nu = 0.54\ (Gr \times Pr)^{0.25}$

Turbulent flow, $2 \times 10^7 < Gr \times Pr < 3 \times 10^{10}$

$Nu = 0.14 \times (Gr \times Pr)^{0.33}$

2. Heated surface down or cooled surface up–

Laminar flow, $3 \times 10^5 < Gr \times Pr < 7 \times 10^8$

$Nu = 0.27 \times (Gr \times Pr)$

Turbulent flow, $7 \times 10^8 < Gr \times Pr < 11 \times 10^{10}$

$Nu = 0.107 \times (Gr \times Pr)^{0.33}$

484. A thin flat plate is hanging freely in air at 27°C. Solar radiation falls in one of its sides at the rate of 500 W/m^2.For maintaining the temperature of the plate constant at 32°C, the value of heat transfer coefficient is:

(a) 25 W/m^2K (b) 50 W/m^2K

(c) 100 W/m^2K (d) 200 W/m^2K

OPSC AEE 2015 PAPER - II

Ans. : (c) $Q = hA(\Delta T)$

$500 = h(32 - 27)$

$h = 100$ W/m^2K

485. For natural convection flow over a vertical plate, the velocity at the edge of the boundary layer is:

(a) Equal to the ambient air velocity

(b) $0.99U_{max}$

(c) Zero

(d) $0.5\ U_{max}$

NPCIL ET, 2015

Ans. (c) : Natural convection on a surface depends on the geometry of the surface as well as orientation. It also depends on the variation of temperature on the surface and the thermophysical properties of the fluid.

The velocity and temperature distribution for natural convection over a hot vertical plate. Note that the velocity at the edge of the boundary layer becomes zero. Its expected the fluid beyond the boundary layer is stationary.

486. In a laminar flow over a flat plate, the average heat transfer coefficient is 8 W/m^2K. The corresponding local heat transfer coefficient is ..... W/m^2K.

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(a) 16 (b) 04
(c) 08 (d) 01

KPSC ADF 2015

Ans. (b) : Since, Relation between average and local heat transfer coefficient is :

Local heat transfer coefficient

$$h_L = \frac{1}{2}\bar{h}$$

$\bar{h} = 2h_L$ (for laminar flow)

where; h_L = local heat transfer coefficient

$$\Rightarrow \quad h_L = \frac{\bar{h}}{2} = \frac{8}{2}$$

$$\Rightarrow \quad \boxed{h_L = 4W/m^2K}$$

487. Match List-I with List-II and select the correct answer :

List-I	List-II
A. Stanton number	1. Natural convection for ideal gases
B. Grashoff number	2. Mass transfer
C. Peclet number	3. Forced convection
D. Schmidt number	4. Forced convection for small prandtl number

	A	B	C	D
(a)	2	4	3	1
(b)	3	1	4	2
(c)	3	4	1	2
(d)	2	1	3	4

Nagaland PSC CTSE 2015 Paper-II

Ans. (b) :

Stanton number	–	Forced convection
Grashoff number	–	Natural convection for ideal gases
Peclet number	–	Forced convection for small prandtl number
Schmidt number	–	Mass transfer

488. The ratio of mass heat flow rate by convection to the flow rate by conduction under a unit temperature gradient and through a characteristic length L is known as

(a) Prandtl number (b) Nusselt number
(c) Stanton number (d) Peclet number

MPPSC AE 08.11.2015

Ans. (d) : It is a dimensionless no representing the ratio of heat transfer by motion of a fluid to heat transfer by thermal conduction.

$$P_e = R_e \times P_r = \frac{VL_c}{\alpha}$$

Where, V = Flow velocity

α = Thermal diffusivity

L_c = Characteristic length.

489. In natural convection the significant number which decides whether the flow is laminar or turbulent is :

(a) Reynold's Number (b) Nusselt Number
(c) Grashoff's Number (d) Prandtl Number

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c & d) : In natural convection Rayleigh number decides whether the flow is laminar or turbulent.

Rayleigh Number (Ra) = Gr.× Pr.

In free or natural convection,

$10^4 < Ra < 10^9$ Laminar

$Ra > 10^9$ Turbulent

490. Match the following and select the correct answer from the options given below the lists :

Column A	Column B
P: Compressible flow	U: Reynolds number
Q: Free surface flow	V: Nusselt number
R: Boundary layer flow	W: Weber number
S: Pipe flow	X: Froude number
T: Heat convection	Y: Mach number
	Z: Skin friction coefficient

(a) P – U; Q – X; R – V; S – Z; T – W
(b) P – W; Q – X; R – Z; S – U; T – V
(c) P – Y; Q – W; R – Z; S – U; T – X
(d) P – Y; Q – W; R – Z; S – U; T – V

UPRVUNL AE 2014

Ans : (d) Compressible flow-Mach Number $\left(m = \frac{v}{c}\right)$

Free surface flow-weber Number $\left(\sqrt{\frac{F_I}{F_S}}\right)$

F_I = Inertia force

F_s = Surface Tension free

Boundary Layer flow = Skin friction co-efficient

Pipe flow = Reynolds Number $\left(Re = \frac{\rho vL}{\mu}\right)$

Heat convection $\left(Nu = \frac{h\ell}{k}\right)$

491. The Grashoff number, Gr is defined as (where g is gravitational acceleration, β is volumetric thermal expansion coefficient, ΔT is temperature difference, L is characteristic length, μ is the viscosity and υ is kinematic viscosity.

(a) $\frac{g\beta\Delta TL^3}{\mu^2}$ (b) $\frac{\rho^2 g\beta\Delta TL^3}{\mu^2}$

(c) $\frac{g\beta\Delta TL^3}{\upsilon^2}$ (d) $\frac{g\beta\Delta TL}{\upsilon^2}$

ISRO Scientist/Engineer 24.05.2014

Ans. (b & c) : Grashoff number

$$Gr = \frac{\text{buoyancy force}}{\text{viscous force}}$$

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$$= \frac{g\beta L^3 \Delta T}{\upsilon^2} \qquad \because \nu = \frac{\mu}{\rho}$$

$$= \frac{\rho^2 g\beta \Delta T L^3}{\mu^2}$$

492. The convective heat transfer coefficient h is defined by the relationship (m = mass, Cp = specific heat, k = conductivity, q = heat flow rate, Nu = Nusselt Number, L = Thickness, ΔT = Temperature difference)

(a) $h = mCp\Delta T$ (b) $h = \frac{k}{L}$

(c) $h = \frac{q}{\Delta T}$ (d) $h = \frac{NuK}{2L}$

ISRO Scientist/Engineer 24.05.2014

Ans. (c) : Convective heat transfer coefficient = h

mass = m, Specific heat = c_P

Conductivity = k , heat = q

Nu = Nusselt number

L = Thickness , Δt = Temperature difference

$$Nu = \frac{hL}{k}$$

and q convection = hΔT (for unit area)

$$\boxed{h = \frac{q}{\Delta T}}$$

493. The ratio of thermal conductivity to electrical conductivity is equal to?

(a) Prandtl number (b) Schmidt number
(c) Lorenz number (d) Lewis number
(e) Weber Number

CGPSC AE 2014 -II

Ans. (c) : The ratio of thermal conductivity to electrical conductivity is equal to Lorenz number.

494. The heat transfer rate in free convection in the turbulent region depends on

(a) $\Delta T^{1.33}$ (b) ΔT
(c) $\Delta T^{0.33}$ (d) $\Delta T^{1.25}$
(e) $\Delta T^{0.25}$

CGPSC AE 2014 -II

Ans. (a) : From turbulent flow in free convection

$$Nu = C(Gr \,.\, Pr)^{1/3}$$

$$Gr = \frac{g\beta.\Delta T.L^3}{\nu^2}$$

$$Nu = C\left(\frac{g.\beta.\Delta T.L^3}{\nu^2}.Pr\right)^{1/3}$$

$$Nu \propto \Delta T^{1/3}$$

$$Nu = \frac{hD}{k}$$

$$\frac{hD}{k} \propto (\Delta T)^{1/3}$$

$$h \propto (\Delta T)^{1/3}$$

$\because$ $Q = h.A.\Delta T$

$$Q \propto h.\Delta T$$

$$Q \propto (\Delta T)^{1/3}.\Delta T$$

$$Q \propto (\Delta T)^{4/3}$$

$$Q \propto (\Delta T)^{1.33}$$

The heat transfer rate in free convection in the turbulent region depends on $\Delta T^{1.33}$

495. Which dimensionless number has a significant role in natural convection?

(a) Grashoff Number (b) Peclet Number
(c) Mach Number (d) None

Kerala PSC Poly. Lect. 09.01.2014

Ans. (a) :

496. A 2 m long vertical pipe of 400 mm diameter is placed in a fluid with dynamic viscosity, thermal conductivity and specific heat of 1/1200 N.s/m², 100 W/m°C and 120 J/kg°C respectively. If the Grashoff number is 10^9 and the empirical relation governing the free convection in this case is known to be Nu = 0.1 $(Gr. Pr)^{1/3}$, where the symbols have standard or usual meaning. The curvature effect of pipe may be neglected. What will be the value of heat transfer coefficient due to free convection at the pipe surface?

(a) 125 W/m². K (b) 100 W/m². K
(c) 25 W/m². K (d) 500 W/m². K

RPSC AEN Pre-2013

Ans. (d) : Given, Grashoff Number (Gr) = 10^9

k = 100 W/m°C

$$\mu = \frac{1}{1200} N - s/m^2$$

$$C = 120 J/kg°C$$

$$\text{Prandtle number } (Pr) = \frac{\mu C}{k}$$

$$Pr = \frac{\frac{1}{1200} \times 120}{100}$$

$$Pr = \frac{\frac{1}{10}}{100} \Rightarrow \frac{1}{1000}$$

In case , $Nu = 0.1(Gr.Pr)^{\frac{1}{3}}$

$$\frac{hL}{k} = 0.1\left(10^9.\frac{1}{1000}\right)^{\frac{1}{3}}$$

$$\frac{h \times 2}{100} = 0.1(10^6)^{\frac{1}{3}}$$

$$\frac{h \times 2}{100} = (0.1)\,(10^2)$$

$$\frac{h}{50} = 100 \times 0.1 = 10$$

$$\boxed{h = 500 W/m^2 K}$$

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497. Forced convective heat transfer is predominant when :

(a) $\left(\frac{Gr}{Re^2}\right) << 1$

(b) $\left(\frac{Gr}{Re^2}\right) \approx 1$

(c) $\left(\frac{Gr}{Re^2}\right) >> 1$

(d) There is no appreciable change in temperature

TNPSC AE (Industries) 09.06.2013

Ans. (a) : For forced convective heat transfer, $\left(\frac{Gr}{Re^2}\right) << 1$

498. For mixed (free and forced) convection in horizontal tubes of diameter d and length L, Graetz number (Gz) is defined as:-

(a) $Re.Pr.\frac{d}{L}$ (b) $Gr.Pr.Re.\frac{d}{L}$

(c) $Gr.Re.\frac{d}{L}$ (d) $Gr.Pr.\frac{d}{L}$

UKPSC AE-2013, Paper-II

Ans. (a) :

499. The value of Stanton number for value of Nusselt number 400, Reynolds number 40 and Prandtl number 20 will be:-

(a) 800 (b) 200

(c) 2 (d) 0.5

UKPSC AE-2013, Paper-II

Ans. (d) : We know that,

$$\text{Stanton number} = \frac{N_u}{P_r \times R_e} = \frac{400}{40 \times 20} = 0.5$$

500. Dittus – Boelter equation for heating of a fluid in a tube is given as follows:-

(a) $Nu=0.023\ Re^{0.8}\ Pr^{0.4}$

(b) $Nu=0.023\ Re^{0.8}\ Pr^{0.3}$

(c) $Nu=C\ Re^n$ (where C is a constant and n is a real number)

(d) $Nu=C\ \log_e Re^n$

UKPSC AE-2013, Paper-II

Ans. (a) :

501. The measure of relative effectiveness of momentum is :

(a) Reynold's Number (b) Prandtl Number

(c) Nusselt Number (d) Grashof Number

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

502. A wire, 1mm in diameter and 10 cm long, is submerged in liquid water at atmospheric pressure. An electric current is passed through the wire. The current is increased until the water boils. For this situation, h = 4000 W/m^2–°C. The water temperature is 100°C. If the wire surface is to be maintained at 120°C. The electric power that need be supplied to the wire is :

(a) 28.142 W (b) 27.136 W

(c) 12.568 W (d) 25.136 W

APGENCO AE 2012

Ans. (d) : Given, h = 4000 W/m^2 °C

$T_\infty = 100°C$ $T_s = 120°C$

d = 1 mm, l = 10 cm

Electric power that need be supplied of the wire is

$Q = (\pi dL)h\ (T_s - T_\infty)$

$= \pi \times 10^{-3} \times (10 \times 10^{-2}) \times 4000 \times (120 - 100)$

Q = 25.136W36W

503. Match the List-I and List-II and select the correct answer by using the codes given below:

List-I	List-II
(A) Reynolds Number	(X) Dimensionless heat transfer coefficient.
(B) Nusselt Number	(Y) Governs relative thickness of the velocity and the thermal boundary layers
(C) Grashoff Number	(Z) Governs the flow regime in forced convection
(D) Prandtl Number	(T) Governs the flow regime in natural convection.

	(A)	(B)	(C)	(D)
(a)	(X)	(Y)	(Z)	(T)
(b)	(T)	(X)	(Y)	(Z)
(c)	(Z)	(X)	(T)	(Y)
(d)	(Z)	(Y)	(X)	(T)

RPSC AE GWD, 2011

Ans. (c) : Reynold's number– Governs the flow regime in forced convection.

Nusselt number– Dimensionless heat transfer coefficient.

Grashoff number – Governs the flow regime in natural convection.

Prandtl number – Governs relative thickness of velocity and thermal boundary layer.

504. The heat transfer coefficient for a fully developed laminar flow of a fluid (k = 0.173 W/mK) inside a 6.35 mm ID tube under uniform wall temperature boundary conditions is

(a) 99.7 W/m^{2}K (b) 88.5 W/m^{2}K

(c) 120 W/m^{2}K (d) 50 W/m^{2}K

APPSC IOF, 2009

Ans. (a) : Nu = 3.66 for uniform wall temperature

$$\frac{hD}{k} = 3.66$$

$$h = \frac{3.66 \times 0.173}{6.35 \times 10^{-3}} = 99.7\ W/m^2K$$

505. For the laminar flow of air (Prandtl number = 0.71) over a flat plate at zero incidence, the thicknesses of the velocity boundary layer δ and the thermal boundary layer δ_t are such that

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(a) $\delta_t > \delta$ (b) $\delta_t = \delta_t$
(c) $\delta_t < \delta_t$ (d) $0.5 \le \delta_t/\delta \le 1.5$

DRDO Scientists 2008

Ans. (a) : Prandtl law

$$Pr = \left(\frac{\delta}{\delta_T}\right)^3$$

$$\frac{\delta}{\delta_T} = (0.71)^{1/3} = 0.89$$

506. The value of convective heat transfer coefficient depends upon
(a) physical properties of fluid
(b) nature of fluid flow
(c) velocity of fluid flow
(d) all of these

TNPSC AE, 2008

Ans. (d) : The value of convective heat transfer coefficient (h) –
- Physical properties of fluid (ex : viscosity, density, specific heat etc.)
- Nature of fluid flow
- Velocity of fluid flow
- Geometry of the surface

507. $(Nu)_x$ for the forced convection in laminar flow over a flat plate is
(a) $0.116\ P_r^{1/3}\ R_e^{1/2}$ (b) $0.332\ P_r^{1/3}\ R_e^{1/2}$
(c) $0.037\ R_r^{0.8}\ R_e^{1/2}$ (d) $0.0.023\ R_e^{0.8}\ P_r^{0.3}$

TNPSC AE, 2008

Ans. (b) : $(Nu)_x$ for the forced convection in laminar flow over a flat plate is –
$(Nu)_x = 0.332\ (Pr)^{1/3}.\ (Re)^{1/2}$ $(Pr \ge 0.6)$
Pr = Prandtl Number,
Re = Reynolds Number
$(Nu)_x$ = Nusselt number at a location – x.

508. Prandtl number for water varies from
(a) 0.5 - 1.0 (b) 1.0 - 5.0
(c) 5.0 - 10.0 (d) $10.0 - 10^2$

UKPSC AE 2007 Paper -II

Ans. (c) : Prandtl number (Pr) for water varies from 5.0-10.0. It is approx, 7.56.
* For engine oil, Pr is varies from 100 – 40000.
* For Hg Pr << 1

509. What does a high value of Prandtl number indicate?
(a) rapid heat transfer by forced convection to natural convection
(b) rapid diffusion of momentum by viscous action compared to diffusion energy
(c) relative heat transfer by conduction to convection
(d) all the above

UKPSC AE 2007 Paper -II

Ans. (b) :

510. A body cools from 85ºC to 70ºC in 7 minutes. The time taken by the body for further cooling to 55ºC will be
(a) 7 minutes
(b) more than 7 minutes
(c) less than 7 minutes
(d) 0.7 minutes

UKPSC AE 2007 Paper -II

Ans. (b) : According to the question, during cooling process, the time taken by the body from 70^0C to 55^0C is more than 7 minutes, because –
Temperature difference between hot body of 85^0C and surrounding is more than that of 70^0C hot body and surroundings.

511. For gases Prandtl number is
(a) near unity (b) between 5 to 50
(c) between 60 to 100 (d) between 150 to 300

UKPSC AE 2007 Paper -II

Ans. (a) : Prandtl number is a dimensionless number, defined as the ratio of momentum diffusivity to thermal diffusivity.

$$P_r = \frac{\nu}{\alpha} = \frac{\mu C_p}{k}$$

Prandtl number for different matter –
1. For gases – near 1
2. Oxygen – 0.63
3. Water – 7.56

Note – The P_r number of gases are about 1, which indicates that both momentum and heat dissipate through the fluid at about the same rate.

512. The lowest value of Nusselt Number will be
(a) Less than one
(b) Always greater than one
(c) May be less than one or greater than one
(d) One

UKPSC AE 2007 Paper -II

Ans. (d) : One

513. $Nu = CR_e^m P_r^n$ represents heat transfer under-
(a) Free convection
(b) Forced convection
(c) Combined convection
(d) None of the above

ISRO Scientist/Engineer 2006

Ans. (b) : $N_u = CR_e^m \cdot P_r^n$ represent
Forced convection heat transfer.

514. The Newmann boundary condition on heat transfer surface means,
(a) $q/A = -k\,\partial T/\partial x$ at $w = 0$
(b) $k\,\partial T/\partial x = 0$ at $w = 0$
(c) $k\,\partial T/\partial x = h\,[T_w - T_\infty]$ at $w = 0$
(d) $k\,\partial T/\partial x = \sigma\,[T_1^4 - T_2^4]$ at $w = 0$

APPSC AEE MAINS 2016 PAPER-III

Ans. (c) : The Newmann boundary condition on heat transfer surface means,
$k\,\partial T/\partial x = h\,[T_w - T_\infty]$ at $w = 0$

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3. Heat Transfer by Radiation

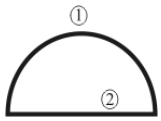
515. The shape factor of a hemispherical body placed on a flat surface with respect to itself is:

(a) 0 (b) 0.25
(c) 0.5 (d) 1.0

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
UPPSC AE 13.12.2020, Paper-2
NLCIL GET 17.11.2020, Shift-II
APPSC Poly Lect. Automobile Engg., 2020
CGPSC Poly. Lect. 22.05.2016
UPRVUNL AE 2014, RPSC AE GWD, 2011
APPSC IOF, 2009, ESE 2005, 2001

Ans. (c) : $F_{11} = ?$

$F_{22} + F_{21} = 1$



$F_{22} = 0$

$\boxed{F_{21} = 1}$

$A_2F_{21} = A_1F_{12}$

$$F_{12} = \frac{A_2}{A_1}F_{21}$$

$F_{11} + F_{12} = 1$

$$F_{11} = 1 - F_{12} = 1 - \frac{A_2}{A_1}F_{21}$$

$$F_{11} = 1 - \left(\frac{\pi r^2}{2\pi r^2}\right) \times 1$$

$$= 1 - \frac{1}{2}$$

$\boxed{F_{11} = 0.5}$

516. A grey body is one whose absorptivity

(a) varies with temperature
(b) varies with wavelength of incident ray
(c) is equal to emissivity
(d) does not vary with temperature and wavelength of incident ray

RPSC IOF, 2020
CGPSC AE 26.04.2015 SHIFT-II
HPPSC Asstt. Prof. 2014
SJVN ET 2013, TRB Asstt. Prof., 2012
UPSC JWM Adv. No. 16/2009
UKPSC AE 2007 Paper -II

Ans. (d) : A grey body is a physical body whose absorptivity of a surface does not vary with variation in temperature and wavelength of incident ray.

517. The total heat radiation from a black body per second per unit area is proportional to (Where T is an absolute temperature).

(a) T^4 (b) T^3
(c) T^2 (d) T

ISRO Scientist/Engineer (RAC) 22.04.2018
HPPSC AE 2018
Vadodara Municipal Corp. DEE, 2018
CGPSC AE 16.10.2016, GPSC Lect. 23.10.2016
UKPSC AE 2013 Paper-II
RPSC AE GWD, 2011
J & K PSC Screening, 2006

Ans. (a) : Stefan Boltzmann Law–According to Stefan-Boltzmann law, the amount of Radiation emitted per unit time from an area of a black body at absolute temperature T is directly proportional to the fourth power of the temperature.

$Q = \sigma AT^4$

$Q \propto T^4$

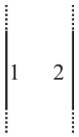
where σ is Stefan's constant = $5.67 \times 10^{-8} W/m^2K^4$

518. What is the value of shape factor for two infinite parallel surfaces separated by a distance X ?

(a) 0 (b) Infinite
(c) 1 (d) X

TNPSC AE 2019, JPSC AE PRE 2019
RPSC Vice Principal ITI 2018
BPSC AE Mains 2017 Paper - V
Kerala PSC Lect. 09.01.2014
UKPSC AE 2012 Paper–II
J&K PSC Civil Services Pre, 2010
ESE 2006

Ans. (c) : The infinite parallel planes with the assumption that leakage of radiation from space between them is zero.



$\therefore F_{11} + F_{12} = 1$

$\because$ Plane are flat so,

$F_{11} = 0$

$F_{12} = 1 = F_{21}$

519. If the temperature of a solid surface increases form 27°C to 627° C, its emissive power increases in the ratio:

(a) 1 : 3 (b) 1 : 9
(c) 1 : 27 (d) 1 : 81

TSPSC AEE 2017
Rajasthan Nagar Nigam AE 2016 Shift-I
RPSC IOF & Boiler 20.09.2017
UJVNL AE 2016
OPSC AEE 2015 PAPER - II
SJVN ET 2013, TRB Poly. Lect., 2012
ESE 2006, 1999

Ans : (d) $T_1 = 27 + 273 = 300$ K

T_2 627 + 273 = 900 K

Emissive power (E) = σT^4 W/m^2

$$\frac{E_1}{E_2} = \left(\frac{300}{900}\right)^4$$

$$\frac{E_1}{E_2} = \frac{1}{81}$$

$E_1 : E_2 = 1 : 81$

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520. At thermal equilibrium, the absorptivity of a body is equal to

(a) Emissivity (b) Reflectivity
(c) Transmissivity (d) Diffusivity

JPSC AE, 10.04.2021, Paper-I
APGENCO AE, 2017, NPCIL ET, 2015
TSPSC Managers, 2015, TNPSC AE 2013
TRB Asstt. Prof., 2012
UKPSC AE-2013, 2007 Paper-II

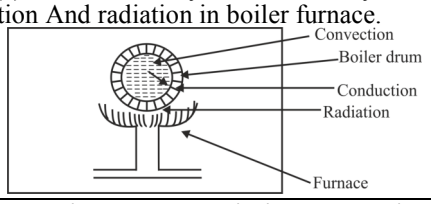
Ans. (a) : According to Kirchoff's Law of thermal radiation the emissivity and the absorptivity of a surface at a given temperature and wavelength are equal.

521. Heat is mainly transferred by conduction, convection and radiation in :

(a) Insulated pipes carrying hot water
(b) Refrigerator freezer coil
(c) Boiler furnaces
(d) Condensation of steam in a condenser

RPSC ACF & FRO, 26.02.2021
APPSC Poly Lect. Automobile Engg., 2020
RPSC IOF, 2020, TNPSC AEE 2017
Vizag Steel (MT) 2017
RPSC IOF & Boiler 20.09.2017
UJVNL AE 2016
UPPSC AE 12.04.2016 Paper-II
J&K PSC Civil Services Pre, 2013
UKPSC AE 2007 Paper-II

Ans : (c) Heat is mainly, transferred by Conduction, Convection And radiation in boiler furnace.



522. Depending on the radiation properties, a body will be opaque when,

(a) $\tau = 1; \rho = \alpha = 0$ (b) $\alpha = 0; (\tau + \rho) = 1$
(c) $\tau = 0; (\alpha + \rho) = 1$ (d) $\rho = 0; (\tau + \alpha) = 1$

GPSC Executive Engg. 23.12.2018
GPSC ARTO Pre. 30.12.2018
UKPSC AE 12.04.2016
Nagaland PSC CTSE 2017, 2015 Paper-II
CGPSC AE 2014- II, TNPSC AE 2013

Ans. (c) : Opaque body– When no radiation is transmitted through the body, it is called opaque body.

For opaque body

$$\tau = 0$$

And $\alpha + \rho = 1$

Gray body– A gray body is defined as a body whose absorptivity of a surface does not vary with variation in temperature and wavelength of the incident radiation.

523. The amount of radiation mainly depends on:

(a) Nature of body
(b) Temperature of body
(c) Type of surface of body
(d) All of above

CGPSC AE 16.10.2016, GPSC Lect. (Poly) 17.12.2014
Mizoram PSC 2014 Paper-I
SJVN ET 2013, RPSC AE GWD, 2011
UKPSC AE 2007 Paper -II

Ans. (d) : The amount of radiation mainly depends on
- Nature of body
- Temperature of body
- Type of surface of body.

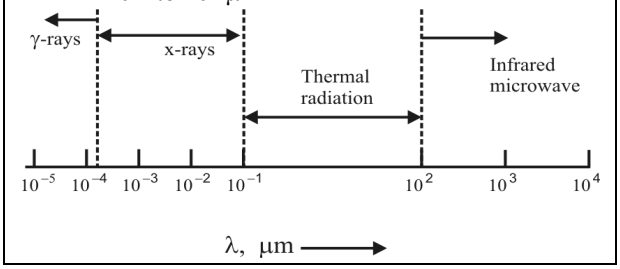
524. Thermal radiations occur in the portion of electromagnetic spectrum between the wavelengths

(a) 10^{-2} to 10^{-4} micron
(b) 10^{-1} to 10^{-2} micron
(c) 0.1 to 100 micron
(d) 100 micron onwards

RPSC ACF & FRO, 26.02.2021
BPSC AE Mains 2017 Paper - V
Nagaland PSC CTSE 2015 Paper-II
J&K PSC Civil Services Pre, 2013
UKPSC AE 2007 Paper -II

Ans. (c) : Thermal radiation occurs in the portion of electromagnetic spectrum between the wavelength –

0.1 to 100 μm
10^{-1} to 10^{2} μm



525. For the radiation between two infinite parallel planes of emissivity ε_1 and ε_2 respectively, which one of the following is the expression for emissivity?

(a) ε_1 and ε_2 (b) $1/\varepsilon_1 + 1/\varepsilon_2$
(c) $1/[1/\varepsilon_1 + 1/\varepsilon_2]$ (d) $1/[1/\varepsilon_1 + 1/\varepsilon_2 - 1]$

RPSC ADE 2016, MPPSC AE 2016
UKPSC AE-2013, Paper-II
UPSC JWM Advt. No.-52/2010
ESE 2007, 1993

Ans. (d) : $$\frac{1}{\left(\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1\right)}$$

ε_1 = Emissivity of first plate
ε_2 = Emissivity of second plate
In initially long parallel plate

$$\frac{Q_{1-2}}{A} = \frac{\sigma\left(T_1^4 - T_2^4\right)}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1}$$

$\frac{1-\varepsilon_1}{A_1\varepsilon_1}$ = Surface resistance of first plate

$\frac{1}{A_1F_{12}}$ = Space resistance

$\frac{1-\varepsilon_2}{A_2\varepsilon_2}$ = Surface resistance of second plate

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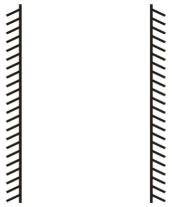
526. The radioactive heat transfer per unit area (W/m^2) between two plane parallel gray surfaces (emissivity = 0.9) maintained at 400 K and 300 K is

(a) 992 (b) 812
(c) 464 (d) 567

UPPSC AE 13.12.2020, Paper-2
MPPSC AE 08.11.2015
UKPSC AE 2012 Paper-II
ISRO Scientist/Engineer 2009, 2007
GATA 1993

Ans. (b) : Given,

$\epsilon_1 = \epsilon_2 = 0.9$
$T_1 = 400$ K
$T_2 = 300$ K



Heat transfer per unit area for two plane parallel gray surface –

$$Q = \frac{\sigma\left(T_1^4 - T_2^4\right)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$

$$Q = \frac{5.67\times10^{-8}\left(400^4 - 300^4\right)}{\frac{1}{0.9} + \frac{1}{0.9} - 1}$$

$Q = 812$ W/m^2

527. The ratio of the emissive power and absorptive power of all bodies is the same and is equal to the emissive power of a perfectly blackbody. This statement is known as

(a) Kirchhoff's law (b) Stefan's law
(c) Wien law (d) Planck's law

CGPSC AE 15.01.2021
Punjab PSC SDE 12.02.2017
HPPSC LECT. 2016, KPSC AE. 2015
BPSC AE 2012 Paper - V

Ans : (a) : Kirchhoff's law- The ratio of the emissive power to absorptive power of all bodies is the same and is equal to the emissive power of perfectly blackbody.

$$\frac{E}{E_b} = \text{Constant}$$

528. 40% of incident radiant energy on the surface of thermally transparent body is reflected back, if the transmissivity of the body be 0.15 then emissivity of the surface is–

(a) 0.45 (b) 0.55
(c) 0.40 (d) 0.75

RPSC IOF & Boiler 20.09.2017
GPSC Asstt. Prof. 28.08.2016
ISRO Scientist/Engineer (RAC) 29.11.2015
SJVN ET 2013, TRB Asstt. Prof., 2012

Ans : (a) Given,

$\rho = 0.4$, $\tau = 0.15$

We know that,

$\alpha + \rho + \tau = 1$
$\alpha + 0.4 + 0.15 = 1$
$\alpha = 0.45$ $[\alpha = \varepsilon]$

529. The intensity of solar radiation on Earth is

(a) 1 kW/m^2 (b) 2 kW/m^2
(c) 4 kW/m^2 (d) 8 kW/m^2

Gujarat PSC AE 2019
Kerala PSC Poly. Lect. 09.01.2014
APPSC Poly. Lect. 2013
UKPSC AE-2013, Paper-II

Ans. (a) : The intensity of solar radiation on earth is 1 kW/m^2 (1360 watt/m^2) as per recent NASA Satellite missions.

530. A perfect black body is one which :

(a) reflects all heat
(b) transmits all heat radiations
(c) absorbs heat radiations of all wave lengths falling on it :
(d) is black in colour.

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Sikkim PSC (Under Secretary), 2017
GWSSB AAE, 27.12.2015
TRB Poly. Lect., 2012

Ans. (c) : Absorbs heat radiations of all wave length falling on it.

Absorptivity (α) = 1.

$\alpha + \rho + z = 1$ Where, α = Absorptivity
ρ = Reflectivity
z = transmissivity.

531. The unit of Stefan Boltzmann constant is :

(a) Watt/cm^{2}K (b) Watt/cm^{4}K
(c) Watt/cmK^4 (d) Watt/cm^{2}K^4

ISRO Scientist/Engineer (RAC), 10.03.2019
Nagaland PSC (CTSE) 2018 Paper-II
APGENCO AE, 2017
TNPSC AE 2008

Ans. (d) :

Radiation Heat Transfer $Q = \sigma_A(T_1^4 - T_2^4)watt.$

$$\sigma = \frac{Q}{A(T_1^4 - T_2^4)} \text{ watt/m}^2\text{k}^4 \text{ or watt/cm}^2\text{k}^4$$

$\sigma = 5.67\times10^{-8} W/m^2k^4$

532. The wavelength of the radiation emitted by a body depends upon -

(a) the nature of the surface
(b) the area of its surface
(c) the temperature of its surface
(d) all the above factors

Nagaland PSC CTSE 2015 Paper-II
Mizoram PSC 2014 Paper-I
PSPCL AE, 2012, ESE 1992

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Ans. (c) : According to Wien's displacement law the wave length at which the intensity of radiation is maximum λ_{max} for a black body radiating at absolute temperature

$$\lambda_{max} \times T = 2.898 \times 10^{-3}\ mK$$

533. The emissive power of a body depends on:

(a) temperature (b) wavelength

(c) physical nature (d) all of these

ISRO Scientist/Engineer (RAC) 07.05.2017
GWSSB DE 07.07.2016
Nagaland PSC (CTSE) 2016, Paper-II
BPSC AE 2012 Paper - V

Ans. (d) : The emissive power of a body depends on temperature, wavelength and physical nature.

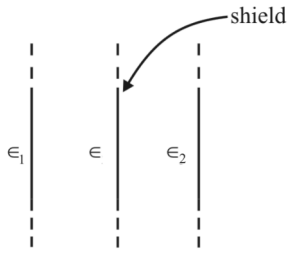
534. Two long parallel plates of same emissivity 0.5 are maintained at different temperatures and have radiation heat exchange between them. A radiation shield of emissivity 0.25 placed in the middle will reduce radiation heat exchange to

(a) 1 (b) $\frac{1}{4}$

(c) $\frac{3}{10}$ (d) $\frac{3}{5}$

RPSC Vice Principal ITI 2018
BPSC AE Mains 2017 Paper - V
TRB Polytechnic Lecturer 2017
ESE 2002

Ans : (c) :



Radiation heat exchange between parallel plates

$$Q_1 = \frac{\sigma(T_1^4 - T_2^4)}{\left(\frac{1}{\in_1} + \frac{1}{\in_2} - 1\right)}$$

$\in_1 = 0.5$

$$Q_1 = \frac{\sigma(T_1^4 - T_2^4)}{3}$$

Now a radiation shield of emissivity $\in = 0.25$ is inserted between the plates, so new radiation change.

$$Q_2 = \frac{\sigma(T_1^4 - T_2^4)}{\left(\frac{1}{0.5} + \frac{1}{0.25} - 1\right) + \left(\frac{1}{0.25} + \frac{1}{0.5} - 1\right)}$$

$$Q_2 = \frac{\sigma(T_1^4 - T_2^4)}{10}$$

$$\therefore \frac{Q_1}{Q_2} = \frac{10}{3}$$

$$\frac{Q_2}{Q_1} = \frac{3}{10}$$

535. For radiation shield which of the following parameters should be highest?

(a) Emissivity (b) Reflectivity

(c) Absorptivity (d) Transmissivity

JPSC AE, 10.04.2021, Paper-I
TSPSC AEE 2017
Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : A material with having a good or better reflectivity is best suited for radiation shield, because all the (most of) radiation will be reflected.

536. According to Kirchoff's law

(a) Radiant heat is proportional to fourth power of absolute temperature

(b) Emissive power depends of temperature

(c) Ratio of emissive power to absorptive power is maximum for perfectly black body

(d) Ratio of emissive power to absorptive power for all bodies is same and equal to the emissive power of perfectly black body

Punjab PSC SDE 12.02.2017
Karnataka PSC AE, 10.09.2017
RPSC VPITI 14.02.2016

Ans. (d) : According to Kirchoff's law Ratio of emissive power to absorptive power for all bodies is same and equal to the emissive power of perfectly black body.

537. In a black body, all the incident radiations are

(a) Reflected (b) Transmitted

(c) Absorbed (d) Refracted

Rajasthan AE (Nagar Nigam) 2016 Shift-3
Rajasthan Nagar Nigam AE 2016, Shift-I
MPPSC State Forest Service Exam, 2014

Ans. (c) : In a black body all the incident radiation are absorbed. Hence the black body is a perfect absorber for all incident radiation.

According to Kirchhoff's law of thermal radiation. A good absorber is always a good emitter at same temperature.

$\alpha_b = 1;\ \in_b = 1$

For a black body $\rho = 0;\ \tau = 0$

$\therefore \quad \alpha + \rho + \tau = 1 \Rightarrow \alpha = 1$

538. For a glass plate, transmittivity and reflectivity are specified as 0.86 and 0.08 respectively, the absorptivity of the plate is

(a) 0.86 (b) 0.08

(c) 1.00 (d) 0.06

UPPSC AE 13.12.2020, Paper-2
ISRO Scientist/Engineer (RAC) 22.04.2018
TSPSC AEE 2017

Ans. (d) : Given,

$\tau = 0.86$

$\rho = 0.08$

$\alpha = ?$

We know that,

$\alpha + \rho + \tau = 1$

$\alpha + 0.08 + 0.86 = 1$

$\alpha = 0.06$

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539. If a body transmits all the radiations incident on it, then it is known as

(a) black body (b) white body
(c) grey body (d) transparent body

Assam PSC AE (PHED), 18.10.2020
GPSC Engineer, Class-II Pre-19.01.2020
Nagaland PSC (CTSE) 2016, Paper-II

Ans. (d) : A transparent body is one that transmits all the radiation that reaches it materials like air, water and clear glass are called transparent. When light encounters transparent materials almost all of it passes directly through them.

Black body- A black body is completely absorbs all wavelengths of thermal radiation incident on it. Such bodies do not reflect light and therefore appears black.

White body- A white body is one with a rough surface that reflects all incident rays completely and uniformly in all directions.

Gray body- Bodies that emit less thermal radiation than a black body have surface emissivity (ε) less than 1.

540. The total emissive power of a surface is 500 W/m^2 at a temperature T_1 and 1200 W/m^2 at a temperature T_2 where the temperatures are in Kelvin. Assuming the emissivity of the surface to be constant, the ratio of the temperatures $\frac{T_1}{T_2}$ is _____

(a) 0.308 (b) 0.803
(c) 0.416 (d) 0.874

Assam Engg. College AP/Lect. 18.01.2021
MPPSC State Forest Service Exam, 2014
GATE 2005

Ans. (b) : 0.803

$E_1 = 500$ W/m^2, $E_2 = 1200$ W/m^2

$E \propto T^4$

$$\frac{E_1}{E_2} = \frac{T_1^4}{T_2^4}$$

$$\frac{T_1^4}{T_2^4} = \frac{500}{1200}$$

$$\frac{T_1}{T_2} = \left(\frac{500}{1200}\right)^{1/4}$$

$$\frac{T_1}{T_2} = 0.8034$$

541. Solar radiation of 1200 W/m^2 falls perpendicularly on a gray opaque surface of emissivity 0.5. If the surface temperature is 50°C and surface emissive power is 600 W/m^2, the radiosity of the surface will be :

(a) 600 W/m^2 (b) 1000 W/m^2
(c) 1200 W/m^2 (d) 1800 W/m^2

CGPSC AE 26.04.2015 SHIFT-II
APPSC IOF, 2009, ESE 2000

Ans. (c) : Given, Emissivity (ε) = 0.5
Surface temperature = 20°
Radiosity = 600 W/m^2
Solar radiation = 1200 W/m^2

$J = E + (1 - \varepsilon)G_s$

$J = 600 + (1 - 0.5)1200$

$= 1200$ w/m^2

542. Two long parallel surfaces each of emissivity 0.7 are maintained at different temperatures and accordingly have radiation heat exchange between them. It is desired to reduce 80% of this radiant heat transfer by inserting thin parallel shields of emissivity on both sides. The number of shields should be.

(a) 1 (b) 4
(c) 2 (d) 3

UPRVUNL AE 21.08.2016
OPSC AEE 2015 PAPER - II, ESE 2004, 1992

Ans. : (b) $q_{\text{with shield}} = \frac{1}{n+1} q_{\text{without shield}}$

$$1 - \frac{q_{\text{with shield}}}{q_{\text{without shield}}} = 1 - \frac{1}{n+1}$$

because $\frac{q_{\text{with shield}}}{q_{\text{without shield}}} = 80\% = \frac{4}{5}$

$$1 - \frac{4}{5} = 1 - \frac{1}{n+1}$$

$$1 - \frac{4}{5} = \frac{1}{n+1}$$

$$\frac{1}{5} = \frac{1}{n+1}$$

$$n = 4$$

543. In a radioactive heat transfer, a gray surface is one

(a) which appears gray to the eye
(b) whose emissivity is independent of wavelength
(c) which has reflectivity equal to zero
(d) which appears equally bright from all directions

Gujrat PSC AE 2019
ISRO Scientist/Engineer 12.05.2013
RPSC AE GWD, 2011

Ans. (b) : Gray body is defined as a substance whose monochromatic emissivity and absorptivity are independent of wavelength and temperature.

544. Sun's surface at 5800 K emits radiation at a wavelength of 0.5μ. A furnace at 300°C will emit through a small opening, radiation at a wavelength of

(a) 5μ (b) 0.5μ
(c) 2.5μ (d) 0.25μ

APPSC Poly. Lect. Automobile Engg. 2020
ISRO Scientist/Engineer 2010, ESE 1997

Ans. (a) : Given
$T_1 = 5800$ K
$T_2 = 300 + 273 = 573$ K
$\lambda_1 = 0.5\mu$
$\lambda_2 = ?$

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According to wien's displacement law

λT = constant

$\lambda_1 T_1 = \lambda_2 T_2$

$0.5\mu \times 5800 = 573 \times \lambda_2$

$$\lambda_2 = \frac{5800 \times 0.5}{573}\mu = 5.061\mu \approx 5\mu$$

$\lambda_2 = 5\mu$

545. What is the net radiant interchange per square meter for two very large plates at temperatures 800 K and 500 K respectively? (The emissivity of the hot and cold plates are 0.8 and 0.6 respectively. Stefan Boltzmann constant is 5.67×10^{-8} W/m^2 K^4).

(a) $10.26 \frac{kW}{m^2}$ (b) $1.026 \frac{kW}{m^2}$

(c) $102.6 \frac{kW}{m^2}$ (d) $1026 \frac{kW}{m^2}$

Nagaland PSC CTSE 2015 Paper-II
ISRO Scientist/Engineer 2008
ESE 2013, 1994

Ans. (a) : $\sigma = 5.67 \times 10^{-8}$ w/m^{2}K^4

$T_1 = 800$ K, $T_2 = 500$ K

$\varepsilon_1 = 0.8$, $\varepsilon_2 = 0.6$

Net Radiant heat exchange per unit area

$$\left[\frac{Q_{net}}{A}\right] = \frac{\sigma(T_1^4 - T_2^4)}{\left(\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1\right)}$$ [$\because$ where Q_{net} for two parallel plate]

$$= \frac{5.67 \times 10^{-8}(800^4 - 500^4)}{\left(\frac{1}{0.8} + \frac{1}{0.6} - 1\right)}$$

$= 10.26$ kW/m^2

546. Stefan Boltzmann law is expressed as

(a) $E_b = \sigma T^4$ (b) $E_b = \sigma(\Delta T)^4$

(c) $E_b = \sigma(\Delta T)^{1.4}$ (d) $E_b = \sigma T^{1.4}$

GPSC DEE Class-2 (GWSSB) 04.07.2021
UKPSC AE 2012, 2007 Paper-II

Ans. (a) : According to Stefan Boltzman's emissive power of black body is directly proportional to fourth power of absolute temperature of body

$E_b \propto T^4$ or $\boxed{E_b = \sigma T^4}$

where σ =Stefan Boltzman's constant

and $\sigma = 5.67 \times 10^{-8} \frac{w}{m^2 - k^4}$

547. Two spheres A and B of same material have radii 1 m and 4 m and temperature 4000 K and 2000 K respectively. Which one of the following statements is correct?
The energy radiated by sphere A is

(a) greater than that of sphere B
(b) less than that of sphere B
(c) equal to that of sphere B
(d) equal to double that of sphere B

Gujarat PSC AE 2019, TNPSC AE 2017
TRB Poly. Lect., 2012, ESE 2004

Ans : (c) : $E_{b_1} = A_1 \sigma_b T_1^4$

$E_{b_2} = A_2 \sigma_b T_2^4$

$$\frac{E_{b1}}{E_{b_2}} = \frac{4\pi \times 1^2 \times \sigma_b \times 4000^4}{4\pi \times 4^2 \times \sigma_b \times 2000^4}$$

$= 1$

$\boxed{E_{b1} = E_{b2}}$

548. The emissive power of a black body is P. If its absolute temperature is doubled, the emissive power becomes

(a) 2P (b) 4P
(c) 8P (d) 16P

PPSC Asstt. Municipal Engineer 15.06.2021
GATE 2017
BPSC AE Mains 2017 Paper - V

Ans : (d) : As emissive power, $E \alpha T^4$

So, $\frac{E_2}{E_1} = \frac{(2T)^4}{T^4}$

$= 16$

549. The value of wavelength for maximum emissive power is given by

(a) Wien's Displacement Law
(b) Fourier Law
(c) Planck's Distribution Law
(d) Kirchhoff's Law

OPSC AEE 2019 PAPER - II
MPPSC State Forest Service Exam, 2014
TNPSC AE 2013

Ans. (a) : The value of wavelength for maximum emissive power is given by Wien's Displacement Law.

$\lambda_{max} \propto \frac{1}{T}$

$\lambda_{max} T$ = constant

$\lambda_{max} T = 2898$ μmk

550. Which mode(s) of heat transfer does (do) not need a material medium?

(a) Conduction
(b) Convection
(c) Radiation
(d) Convection and Radiation

UPRVUNL AE 05.07.2021
JPSC AE, 10.04.2021, Paper-I
UKPSC AE 2013 Paper-II

Ans. (c) : Radiation mode of heat transfer does not require any material medium whereas other mode conduction and convection requires material medium.

551. Steafan Boltzman law is applicable for heat transfer by

(a) conduction
(b) convection
(c) radiation
(d) conduction and radiation combination

MECON MT 2019
GPSC Executive Engineer 23.12.2018
Mizoram PSC AE/SDO 2014, Paper-I

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Ans. (c) : Stefan-Boltzman law is applicable for radiation. it states that total radiant heat power emitted from a surface is proportional to the fourth power of its absolute temperature. The law applied only to black bodies theoretical surface that absorb all incident heat radiation.

552. When α is absorptivity, p is reflectivity and τ is transmittivity, then for a diathermanous body.

(a) $\alpha = 1$, $p = 0$ and $\tau = 0$
(b) $\alpha = 0$, $p = 1$ and $\tau = 0$
(c) $\alpha = 0$, $p = 0$ and $\tau = 1$
(d) $\alpha + p = 1$ and $\tau = 0$

GPSC DEE Class-2 (GWSSB) 04.07.2021
Nagaland PSC (CTSE) 2018, Paper-II
ESE 1992

Ans. (c) : Diathermanous body is a body which transmits all the incident radiation without absorbing or reflecting.
(A body with transmitivity 1 reflectivity 0 and absorptivity zero) air, up to certain extent can be considered as the diathermanous body because, it transmits all the radiation without hearing itself.

553. Radiation heat transfer occurs at a speed of

(a) sound (b) light
(c) 60000 km/h (d) 3000 m/h

APPSC Poly. Lect. 2013
UKPSC AE 2007 Paper -II

Ans. (b) : Energy transfer by radiation occurs at the speed of light and suffers no attenuation in Vacuum.

554. Gases have poor

(a) absorptivity (b) reflectivity
(c) transmissivity (d) (a) and (c)

APPSC Poly. Lect. 2013
UKPSC AE 2007 Paper -II

Ans. (b) : Gases have poor reflectivity because of low density.

555. A large spherical enclosure has a small opening. The rate of emission to radioactive flux through this opening is 7.35 KW/m^2. The temperature at the inner surface of the sphere will be about (assume Stefan Boltzman constant $\sigma = 5.67 \times 10^{-8}$ W/m^{2}k^4

(a) 600 K (b) 330^0C
(c) 373 K (d) 1000 K

RPSC ADE 2016, ESE 1998

Ans. (a) :
Rate of emission of radioactive flux = 7.35 kW/m^2
Stefan Boltzman constant $\sigma = 5.67 \times 10^{-8}$ W/m^{2}K^4
Temperature = ?
$E_b = \sigma T^4$
$7.35 \times 10^3 = 5.67 \times 10^{-8} \times T^4$

$$T = \left(\frac{7.35\times10^3}{5.67\times10^{-8}}\right)^{0.25}$$

Temperature = 600.034 K
Temperature = 600 K

556. The emissivity of a black body is

(a) zero (b) 0.5
(c) 0.75 (d) 1

Gujarat PSC AE 2019
Assam PSC CCE Pre 2015

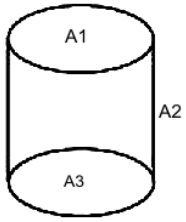
Ans. (d) : The emissivity of a surface is defined as the ratio of emissive power of the surface to the emissive power of the black body of the same temperature.

$$\boxed{\in = \frac{E}{E_b}}$$

$$\boxed{\in = E\sigma T^4}$$

Black body is perfect absorber as well as perfect emitter is known as Kirchoff's law.
For a black body
$\in = 1$, $\alpha = 1$

557. The circular tube of equal length and diameter is shown in the figure below. The view-factor F_{13} is 0.2.

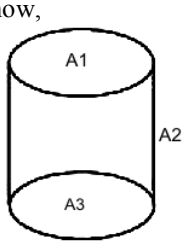


The view factor F_{12} in this case will be:

(a) 0.2 (b) 0.4
(c) 0.6 (d) 0.7
(e) 0.8

CGPSC Asstt. Workshop Supt., 17.07.2016
GATE 2001

Ans. (e) : As we know,



$F_{11} + F_{12} + F_{13} = 1$
$\downarrow$
$0 + F_{12} + 0.2 = 1$
$\boxed{F_{12} = 0.8}$

558. Two large diffuse gray parallel plates, separated by a small distance, have surface temperatures of 400 K and 300 K. If the emissivities of the surface are 0.8 and the Stefan-Boltzmann constant is 5.67 × 10^{-8} W/m^{2}K^4, the net radiation heat exchange rate in kW/m^2 between the two plates is_____.

(a) 0.66 (b) 0.79
(c) 0.99 (d) 3.96

Assam Engg. College AP/Lect. 18.01.2021
GPSC Asstt. Director of Transport 05.03.2017
GATE 2013

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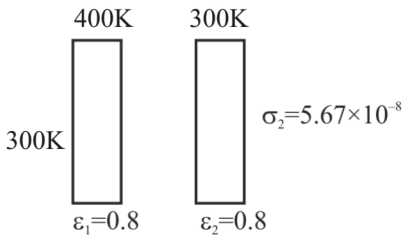
Ans. (a) : 0.66



$$q=\frac{\sigma\left(T_1^4-T_2^4\right)}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1}$$

$$=\frac{5.67\times10^{-8}\times\left(400^4-300^4\right)}{\frac{1}{0.8}+\frac{1}{0.8}-1}$$

$$q=661.5\frac{W}{m^2}$$

$q = 0.6615\ kW/m^2$

559. The intensity of radiation is obtained by multiplying the emissive power by a factor

(a) π (b) $\frac{1}{\pi}$

(c) $\frac{1}{\sqrt{2}}\pi$ (d) $\frac{\sqrt{2}}{\pi}$

Sikkim PSC (Under Secretary), 2017
UPSC JWM Advt. No.-52/2010

Ans. (b) : The relation between emissive power (E_b) and Intensity of radiation (I_b) is –

$\Rightarrow \quad E_b = \pi I_b$

$\Rightarrow \therefore \quad \boxed{I_b=\frac{1}{\pi}\times E_b}$

560. In grey body, rate of emission is

(a) Less than one (b) Equal to one
(c) More than one (d) None of these

UPRVUNL AE 05.07.2021
MPPSC State Forest Service Exam, 2014

Ans. (a) : A body with an emissivity less than unity and the same at all wavelengths is called a grey body, by comparison with a black body that has unit emissivity at all wavelengths. The emission from a grey body with ε = 0.80 is by lower grey emission curve. So, rate of emission is less than one.

561. Relation between emissive and absorptive power of a body is given by

(a) Wein's law (b) Stefan's law
(c) Kirchoff's law (d) Planck's law

JPSC AE, 10.04.2021, Paper-I
MPPSC State Forest Service Exam, 2014

Ans. (c) : Kirchoff's–The emissivity of a body is equal to its absorptivity when the body remains in the thermal equilibrium with the surroundings.

$\boxed{\alpha=\epsilon}$

Kirchoff's radiation law states that the ratio of emissive power to absorptive power,
This law accounts for the fact that for a given wavelength of radiation, good absorbers are also good emitters.

562. A satellite floats in deep space with very high velocity. It will continuously lose heat by

(a) Convection
(b) Conduction and Convection
(c) Radiation
(d) Radiation and Convection

GPSC DEG Class-2 (GWSSB) 04.07.2021
GPSC EE Pre, 28.01.2017

Ans. (c) : A satellite floats in deep space with very high velocity. It will continuously lose heat by radiation. As there will be no mass surrounding the satellite. It will lose heat by radiation only.

563. Assertion (A) : Heat transfer at high temperature is dominated by radiation rather than convection.
Reason (R) : Radiation depends on fourth power of temperature while convection depends on unit power relationship.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

Punjab PSC (Lect.) 06.08.2017
ESE 2002

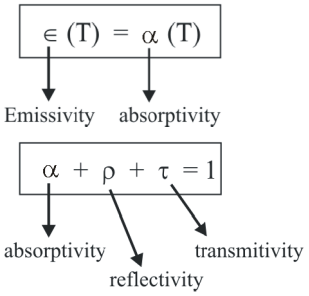
Ans. (a) : Both (A) and (R) are true and (R) is the correct explanation of (A)

564. The emissivity of a body is equal to absorptivity when the body remains in thermal equilibrium. This law refers to as :

(a) Planck's law
(b) Lambert's law
(c) Kirchoff's law
(d) Wien's displacement law

Kerala PSC AE 06.08.2015
RPSC AEN Pre-2013

Ans. (c) : Kirchhoff's law – The total hemispherical emissivity of a surface at a temperature T is equal to its total hemispherical absorptivity for radiation coming from a block body at the same temperature.



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→ for opaque surfaces, T = 0, and thus

$$\alpha + \rho = 1$$

→ By stefan-Boltzman law

$$E_b(T) = \sigma T^4 \quad W/m^2$$

where, $\sigma = 5.67 \times 10^{-8} W/m^2K^4$

σ = Stefan -Boltzman constant

total black body emissive power

565. Consider the following statements :

1. **Temperature of the surface**
2. **Emissivity of the surface**
3. **Temperature of air in the room**
4. **Length and diameter of the pipe**

Select parameter(s) responsible for loss of heat from at hot surface in a room would include

(a) 1 alone (b) 1 and 2
(c) 1, 2 and 3 (d) 1, 2, 3 and 4
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
ESE 1995

Ans. (d) : Heat loss $(Q) = \varepsilon \sigma A (\Delta T)^4$
where,
ε = emissivity of the surface

566. A spherical aluminium shell of inside diameter 2 m is evacuated and used as a radiation test chamber. If the inner surface is coated with carbon black and mainained at 500 K. the irradiation on a small test surface placed inside the chamber is (Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8}$ w/m^2 K)

(a) 1000 w/m^2 (b) 7348 w/m^2
(c) 3544 w/m^2 (d) 3400 w/m^2

UPRVUNL AE 21.08.2016, ESE 1999

Ans. : (c) According to Stefan-Boltzman law

$$E_b = \frac{\dot{Q}}{A} = \sigma T^4$$

σ = Stefan-Boltzman constant = $\left(5.67 \times 10^{-8}\ w/m^2k^4\right)$

T = 500 k

$E_b = 5.67 \times 10^{-8} \times (500)^4$

$E_b = 3544$ w/m^2.

567. For an opaque plane surface the radiosity, irradiation and emissive power are respectively 16, 24 and 12 W/m^2. Determine the emissivity of surface.

(a) 0.2 (b) 0.45
(c) 0.87 (d) 0.67

UPRVUNL AE 2014, ESE 2004

Ans : (c) Given,
Radiosity(J) = 16 w/m^2
Irradiation (G) = 24 w/m^2
Emissive power (E) = 12 w/m^2
$J = E + (1-\varepsilon)G$
$16 = 12 + (1-\varepsilon)24$

$$\varepsilon = 1 - \frac{4}{24} = \boxed{0.83}$$

Note-
$J = E + \rho G$
$J = \varepsilon E_b + \rho G$
E_b = Emissive power of a perfect black body.
Note: UPRVUNL gives (d) as correct answer.

568. Two long parallel plates of same emissivity 0.2 are maintained at different temperatures and have radiation heat exchange between them. To what amount will the radiation heat exchange get reduced by the radiation shield of emissivity 0.4 placed in the middle?

(a) 2/11 (b) 1/4
(c) 9/13 (d) 3/5

UPRVUNL AE 2014, ESE 2002

Ans : (c) Amount of radiation exchange

$$\frac{\frac{2}{\varepsilon_1} - 1}{2\left(\frac{1-\varepsilon_1}{\varepsilon_1}\right) + 2\left(\frac{1-\varepsilon_2}{\varepsilon_2}\right) + 2}$$

Given $\varepsilon_1 = 0.2$
$\varepsilon_2 = 0.4$

$$= \frac{\frac{2}{0.2} - 1}{2\left(\frac{1-0.2}{0.2}\right) + 2\left(\frac{1-0.4}{0.4}\right) + 2}$$

$$= \frac{9}{8+3+2}$$

$$= \frac{9}{13}$$

569. Which of the following law signifies the wave length maximum emissive power:

(a) Wien's law
(b) Stefan-Bolzmann law
(c) Newton's law
(d) Kirchhoff's law

HPPSC ADF 05.03.2019, MECON MT 2019

Ans. (a) : Wien's displacement low states that the block body radiation curve for different temperature peaks at a wavelength inversely proportional to the temperature.

λT = Constant

$\lambda_{peak} T = 2.898 \times 10^{-3}$ m-k

570. Two large parallel grey plates with a small gap exchange radiation at a rate of 1000 W/m^2, when their emissivities are 0.5 each. One plate is cooled such that its emissivity becomes 0.25. Find the new rate of heat transfer if the temperature difference remains same.

(a) 600 W/m^2 (b) 550 W/m^2
(c) 850 W/m^2 (d) 1000 W/m^2

AAI Jr. Executive 29.11.2018, ESE 2002

Ans. (a) : Given,
$Q_1 = 1000$ W/m^2

$$Q_1 = \frac{A\sigma\left(T_1^4 - T_2^4\right)}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1}$$

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$$Q_1 = \frac{A\sigma\left(T_1^4 - T_2^4\right)}{\frac{1}{0.5}+\frac{1}{0.5}-1} \quad \text{.... (i)}$$

$$Q_2 = \frac{A\sigma\left(T_1^4 - T_2^4\right)}{\frac{1}{0.5}+\frac{1}{0.25}-1} \quad \text{.... (ii)}$$

from equation (i)/(ii)

$$\frac{Q_1}{Q_2} = \frac{\frac{1}{0.5}+\frac{1}{0.25}-1}{\frac{1}{0.5}+\frac{1}{0.5}-1}$$

$$Q_2 = \frac{3}{5}Q_1$$

$$Q_2 = \frac{3}{5}\times 1000 \qquad \left[\therefore Q_1 = 1000 \ \text{W/m}^2\right]$$

$Q_2 = 600$ W/m^2

571. Two spherical balls of same material and surface finish have their diameters in the ratio of 2:1. Both are heated to same temperature and allowed to cool by radiation. Rate of cooling of big ball as compared to smaller one will be in the ratio of

(a) 1:2 (b) 2:1
(c) 1:1 (d) 4:1

Nagaland PSC CTSE 2017 Paper-II
ISRO Scientist/Engineer 2011

Ans. (d) : By Stefan Boltzman law

Rate of cooling = $\varepsilon\sigma A(T_1^4)$

Given, $\frac{d_1}{d_2} = \frac{2}{1}$

$d_1 = 2d_2$

$$\frac{Q_1}{Q_2} = \frac{A_1}{A_2} = \frac{\frac{\pi d_1^2}{4}}{\frac{\pi d_2^2}{4}}$$

$$= \left(\frac{d_1}{d_2}\right)^2$$

$$= \left(\frac{2d_2}{d_2}\right)^2$$

$$= \frac{4}{1} = 4:1$$

572. The spectral emissive power E_1 for a diffusely emitting surface is $E_1 = 0$ for $\lambda > 3$ μm ; $E_1 = 150$ W/m^2 μm for $3 < \lambda < 12$ μm ; $E_1 = 300$ W/m^2 μm for $12 < \lambda < 25$ μm ; $E_1 = 0$ for $\lambda > 25$ μm. The total emissive power of the surface over the entire spectrum is

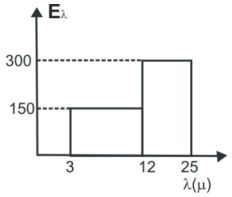
(a) 1250 W/m^2 (b) 2500 W/m^2
(c) 4000 W/m^2 (d) 5250 W/m^2

ISRO Scientist/Engineer 2010, ESE 1998

Ans. (d) : Total emissive power

$$= \int_o^\infty E_\lambda d\lambda = \int_0^3 0\times d\lambda + \int_3^{12} 150 d\lambda + \int_{12}^{25} 300 d\lambda + \int_{25}^\infty 0.d\lambda$$

$$= [\lambda]_0^3 + 150[\lambda]_3^{12} + 300[\lambda]_{12}^{25} + 0.[\lambda]_{25}^\infty$$



total emissive power

= 0 + 150 × (12 - 3) + 300 (25 - 12) + 0

= 5250 W/m^2

573. The temperature of sun can be measured using a

(a) Standard thermometer
(b) Radiation pyrometer
(c) Platinum resistance thermometer
(d) Mercury thermometer

ISRO Scientist/Engineer 2010
UKPSC AE 2007 Paper -II

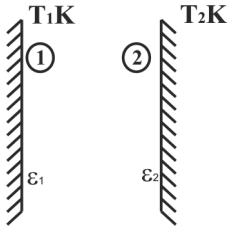
Ans. (b) : The temperature of sun can be measured using a radiation pyrometer.

574. For infinite parallel planes with emissivities ε_1 and ε_2, the interchange factor for radiation from surface 1 to surface 2 is given by-

(a) $\frac{\varepsilon_1\varepsilon_2}{\varepsilon_1+\varepsilon_2-\varepsilon_1\varepsilon_2}$ (b) $\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}$
(c) $\varepsilon_1+\varepsilon_2$ (d) $\varepsilon_1\varepsilon_2$

DRDO Scientist 2009
ISRO Scientist/Engineer 2006

Ans. (a) : For infinite parallel planes with emissivities ε_1 and ε_2, the interchange factor for radiation from surface 1 to surface 2.



$$F_{12} = \frac{1}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1}$$

$$= \frac{1}{\frac{\varepsilon_2+\varepsilon_1}{\varepsilon_1\varepsilon_2}-1} \Rightarrow \frac{1}{\frac{\varepsilon_2+\varepsilon_1-\varepsilon_1\varepsilon_2}{\varepsilon_1\varepsilon_2}}$$

$$\boxed{F_{12} = \frac{\varepsilon_1\varepsilon_2}{\varepsilon_2+\varepsilon_1-\varepsilon_1\varepsilon_2}}$$

For black plates $\varepsilon_1 = \varepsilon_2 = 1$

$\boxed{F_{1-2} = 1}$

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575. A grey body is defined such that

(a) Monochromatic emissivity of the body is independent of temperature

(b) Monochromatic emissivity of the body is dependent of temperature

(c) Monochromatic emissivity of the body is independent of wave length

(d) Monochromatic emissivity of the body is dependent of wave length

GPSC ARTO Pre 30.12.2018
ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Monochromatic Emissivity (E_λ)– It is defined as the ratio between monochromatic hemispherical emissive power of a non-block body and monochromatic hemispherical emissive power of a block body both being at the temperature and wave length.

Note : For a grey body – Monochromatic emissivity E_λ is independent of wave length (λ).

576. With usual notations, for black body

(a) $\alpha = 0, \tau = 0, \rho = 1$ (b) $\alpha = 1, \tau = 0, \rho = 0$

(c) $\alpha = 1, \tau = 1, \rho = 0$ (d) None of the above

Kerala PSC Poly. Lect. 09.01.2014
UKPSC AE 2007 Paper -II

Ans. (b) : $\alpha = 1, \tau = 0, \rho = 0$

Absorptivity (α)	Fraction of incident radiation absorbed.
Reflectivity (ρ)	Fraction of incident radiation reflected.
Transmittivity (τ)	Fraction of incident radiation transmitted.

577. Two long parallel surfaces each of emissivity 0.7 are maintained at different temperatures and accordingly have radiation heat exchange between them. It is desired to reduce 75% of the radiant heat transfer by inserting thin parallel shields of emissivity 1 on both sides. The number of shields should be:

(a) 2 (b) 1

(c) 3 (d) 4

OPSC AEE 2019 PAPER - II, ESE 2004, 1992

Ans : (c) : Since emissivity '$\in$' of both surfaces and shield are same and equal to 0.7 Radiation shields reduces 75% of radiation so radiation exchange in presence of n shield,

$$\therefore \frac{E_1 - E_2}{4} = \frac{E_1 - E_2}{n+1}$$

$\therefore n = 3$

Where E_1 and E_2 are emissive power of long parallel surfaces.

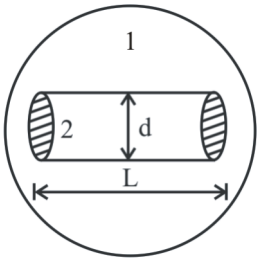
578. At the centre of hollow sphere (Surface 1) of 2 m diameter a solid cylinder of 1 m diameter and length each is placed (Surface 2), what will be the view factor F_{11}?

(a) 0.375 (b) 1

(c) 0.625 (d) 0.75

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
OPSC AEE 2019 Paper - II
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2005

Ans : (c) : Given : Diameter of sphere (D) = 1 m
Diameter of cylinder (d) = 0.5 m
Length of cylinder (L) = 0.5 m



$F_{21} = 1$

$\because$ (Cylinder is completely enclosed with sphere)

From reciprocity theorem

$A_1 F_{12} = A_2 F_{21}$

$$F_{12} = \frac{A_2}{A_1} F_{21} = \frac{\left(\pi dL + 2\frac{\pi}{4}d^2\right)}{4\pi\left(\frac{D}{2}\right)^2}$$

$$F_{12} = \frac{0.5 \times 0.5 + \frac{0.5 \times 0.5}{2}}{1 \times 1}$$

$F_{12} = 0.375$

According to the summation rule,

$F_{11} + F_{12} = 1$

$F_{11} = 1 - F_{12} = 1 - 0.375$

$\boxed{F_{11} = 0.625}$

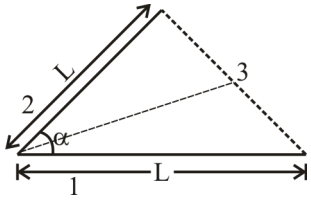
579. What is the value of the view factor for two inclined flat plates having common edge of equal width and with an angle of 20 degrees?

(a) 0.83 (b) 1.17

(c) 0.66 (d) 1.34

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2002

Ans. (a) : Imagine a surface (3) of equal width using geometry.



$$L_3 = 2L\sin\left(\frac{\alpha}{2}\right)$$

$$F_{12} = \frac{A_1 + A_2 - A_3}{2A_1} = \frac{L_1 + L_2 - L_3}{2L_1}$$

$$F_{12} = 1 - \sin\left(\frac{\alpha}{2}\right)$$

$$= 1 - \sin\left(\frac{20°}{2}\right)$$

$F_{12} = 1 - \sin(10°)$

$F_{12} = 0.83$

But official answer given is option (c) i.e. 0.66.

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580. Sun's surface at 5800 K emits radiation at a wavelength of 0.5μ. A furnace at 580^0C will emit through a small opening radiation at a wavelength of nearly

(a) 10μ (b) 0.1μ
(c) 3.4μ (d) 0.01μ
(e) 5.6μ

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : According to Wien's displacement law

$\lambda . T = \text{Constant}$

$\lambda_1 \times T_1 = \lambda_2 \times T_2$

$0.5\ \mu \times 5800 = \lambda_2 \times (580+273)$

$\boxed{\lambda_2 = 3.399\mu} \simeq 3.4\mu$

581. When a black body absorbs all falling radiations due to absorption, then

(a) A Black body shines
(b) The temperature of black body rises
(c) Black body radiates energy to other
(d) Black body becomes good conductor of heat

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (a) : A surface that absorbs all radiation that falling on it. The term arises because incident visible light will be absorbed rather than reflected, therefore this surface will appear black.

582. If the ratio of emission of a body at a given temperature is a constant for all wavelengths, the body is termed as

(a) Grey body (b) White body
(c) Qpaque body (d) Black body

Nagaland PSC CTSE 2017 Paper-2
UKPSC AE 2013 Paper-II

Ans. (a) : A gray body of which the monochromatic emissivity (ε) is constant for the entire wavelength spectrum.

583. The temperature of a solid surface is raised from 227°C to 727°C, the emissive power of the body will change from E_1 to E_2 such that E_2/E_1 is

(a) 400 (b) 16
(c) 4000 (d) 1600

HPPSC W.S. Poly. 2016
UPPSC AE 12.04.2016 Paper-II

Ans : (b) $T_1 = 227°C + 273$

$T_2 = 727°C + 273$

$T_1 = 500K$

$T_2 = 1000K$

According to Stefan - Boltzmann law:-

$E_b = \sigma T^4\ W/m^2$

σ = Stefan - Boltzmann constant

E_b = emissive power

$$\frac{(E)_1}{(E)_2} = \frac{\sigma T_1^4}{\sigma T_2^4}$$

$$\frac{(E)_1}{(E)_2} = \left(\frac{500}{1000}\right)^4$$

$$\frac{(E)_1}{(E)_2} = 1/16$$

$E_2/E_1 = 16$

584. Match list-I (law) with list-II (equation) select the correct Ans.wer using the codes given below the list:

List -I	List -II
(a) stefan-Boltzmann law	1. $q = hA(T_1 - T_2)$
(b) Newton's law of cooling	2. $E = \sigma E_0$
(c) Fourier's law	3. $q = \frac{KA(T_1 - T_2)}{L}$
(d) Kirchoff's law	4. $q = \sigma A(T_1^4 - T_2^4)$
	5. $q = kA(T_1 - T_2)$

Code

	A	B	C	D
(a)	4	1	3	2
(b)	1	2	4	3
(c)	1	4	2	3
(d)	3	2	4	1

UJVNL AE 2016, ESE 1999

Ans : (a)

List -I	List -I
i) stefan-Boltzmann law	$q = \sigma A(T_1^4 - T_2^4)$
ii) Newton's law of cooling	$q = hA(T_1 - T_2)$
iii) Fourier's law	$q = \frac{KA}{L}(T_1 - T_2)$
iv) Kirchoff's law	$E = \sigma E_0$

585. What is the equivalent emissivity for radiant heat exchange between a small body (emissivity = 0.4) in a very large enclosure (emissivity = 0.5) ?

(a) 0.5 (b) 0.4
(c) 0.2 (d) 0.1

UKPSC AE 2012 Paper–II, ESE 2008

Ans. (b) : When body 1 completely enclosed by body 2, body 1 is very small,

$$\varepsilon = \frac{1}{\frac{1}{\varepsilon_1} + \frac{A_1}{A_2}\left(\frac{1}{\varepsilon_2} - 1\right)}$$

Since A_1/A_2 is very less whole term is neglected

$\boxed{\varepsilon = 0.4}$

586. For an opaque plane surface, the irradiation, radiosity and emissive power are 20,12 and 10 W/m^2 respectively. The emissivity of the surface is

(a) 0.2 (b) 0.4
(c) 0.8 (d) 1.0

UKPSC AE 2012 Paper–II, ESE 2004

Ans. (c) : $J = \varepsilon E_b + (1-\varepsilon)G$

$12 = \varepsilon \times 10 + (1-\varepsilon) \times 20$

$12 = 10\varepsilon + 20 - 20\varepsilon$

$10\varepsilon = 8$

$\varepsilon = 8/10$

Emissivity of surface $\boxed{\varepsilon = 0.8}$

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587. The opaque body is that which:-

(a) Absorbs all radiations
(b) Reflects all radiations
(c) Transmits all radiations
(d) Partly reflects and partly absorbs the radiation

GPSC DE Class-2 (GWSSB) 04.07.2021
UKPSC AE-2013, Paper-II

Ans. (d) : A body which partly reflects and partly absorbs the radiation is known as the opaque body.

588. According to Wien's law, the wavelength corresponding to maximum energy is proportional to:-

(a) T^{-1} (b) T^{-2}
(c) T^{-3} (d) T^{-4}

UKPSC AE 2013, 2007 Paper-II

Ans. (a) : According to Wien's law

$T\,\lambda_{max} = \text{Constant}$

So, $\lambda_{max} \propto T^{-1}$

589. In case of black body

(a) Transmissivity is one
(b) Absorptivity is zero
(c) Reflectivity is one
(d) None of the above

APPSC Poly. Lect. 2013
UKPSC AE 2007 Paper -II

Ans. (d) : In case of black body absorptivity is one

(i) For black body – $\alpha = 1,\ \rho = 0,\ \tau = 0$

(ii) For perfectly white body $\alpha = 0,\ \rho = 0\ \&\ \tau = 0$

(iii) For opaque body $\alpha + \rho = 1,\ \tau = 0$

where, α = Absorptivity, ρ = Reflectivity
τ = Transmissivity

590. A body which reflects all the radiations falling on it is called as :

(a) White body (b) Opaque body
(c) Black body (d) Transparent body

Kerala PSC IOF 19.04.2016
UPSC JWM Adv. No-16/2009

Ans. (a) : White body is a body which reflects all the radiations falling on it.

$\alpha + \rho + \tau = 1$

for white body ; $\alpha = 0$

$\tau = 0$

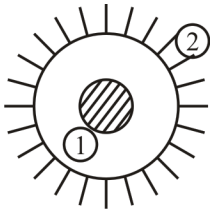
$\therefore$ $\boxed{\rho = 1}$

591. A hollow enclosure is formed between two infinitely long concentric cylinders of radii 1 m and 2 m respectively. Radiative heat exchange takes place between the inner surface of the larger cylinder (surface 2) and the outer surface of the smaller cylinder (surface 1). The radiating surfaces are diffuse and the medium in the enclosure is non-participating. The fraction of the thermal radiation leaving the larger surface and striking itself is

(a) 0.25 (b) 0.50
(c) 0.75 (d) 1

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2008

Ans. (b) :



Given, $r_1 = 1m,\ r_2 = 2m,\ F_{22} = ?$

$F_{12} + F_{11} = 1$

$F_{11} = 0$ (due to convex)

$\therefore$ $F_{12} = 1$

$A_1F_{12} = A_2F_{21}$

$$F_{21} = \frac{A_1}{A_2} = \frac{\pi r_1 \ell}{\pi r_2 \ell} = 0.5$$

$F_{21} = 0.5$

$F_{21} + F_{22} = 1$

$F_{22} = 1 - F_{21} = 0.50$

592. Fraction of radiative energy leaving one surface that strikes

(a) Radiative flux
(b) Emissive power of the first surface
(c) View factor
(d) Re-radiation flux

HPPSC Asstt. Prof. 18.09.2017, ESE 2003

Ans. (c) :

593. The ratio of total emissive power of the body to the total emissive power of a black body at the same temperature is called

(a) Absorptivity (b) Transmissivity
(c) Reflectivity (d) Emissivity

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GPSC Engg. Class-2 Pre. 19.01.2020

Ans. (d) :

594. A radiation shield should

(a) Have high transmissivity
(b) Abrob all the radiations
(c) Have high reflective power
(d) Partly absorb and partly transmit the incident radiation

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TANGEDCO AE 2015

Ans. (c) :

595. A perfect black body is one which

(a) is black in colour
(b) absorbs heat radiations of all wave lengths falling on it
(c) reflects all the heat radiations
(d) transmits the heat radiations

Nagaland PSC (CTSE) 2018, Paper-II
GWSSB DEE 07.07.2016

Ans. (b) : A perfect black body is one which absorbs all incoming light and does not reflect any. Such on item will tend to be completely black at room temperature. However, a black body can begin to glow with thermal radiation if heated to a high temperature.

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596. A thin shield of emissivity ε_3 (on both sides) is placed between two infinite parallel plates of emissivities ε_1 and ε_2, and temperatures T_1 and T_2 respectively. If $\varepsilon_1 = \varepsilon_2 = \varepsilon_3$ then the ratio of radiant energy transfer without shield to with shield will take the value.

(a) 0.25 (b) 0.5
(c) 0.75 (d) 2

RPSC ACF & FRO, 26.02.2021

Ans. (b) : $\frac{Q_{\text{with shield}}}{Q_{\text{without shield}}} = \frac{1}{n+1}$ n = no. of shield

$= \frac{1}{1+1}$

$= \frac{1}{2}$

$= 0.5$

597. In a typical central heating Radiator', approximately half the heat is transferred by :

(a) Conduction (b) Convection
(c) Radiation (d) Evaporation

RPSC ACF & FRO, 26.02.2021

Ans. (b) : Radiators are heat exchangers used to transfer thermal energy from one medium to another for the purpose of cooling and heating. Approximately half the heat transfer by convection.

598. Radiation heat transfer between two surfaces can be reduced greatly by inserting radiation shield with :

(a) Greater thickness (b) Low reflectivity
(c) High absorbivity (d) Low emissivity

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Radiation heat transfer between two surface can be reduced greatly by inserting a thin, high reflectivity (low emissivity) sheet of material between two surface. Such highly reflective thin plates or shields are called radiation shield.

If no. of shields, are 'n' then

$$\boxed{Q_{\text{with shield}} = \frac{Q_{\text{without shield}}}{n+1}}$$

599. The total emissive power of surface is 100 W/m^2 at temperature T_1 and 800 W/m^2 at a temperature T_2, where the temperature are in Kelvin. Assuming the emissivity of the surface to be constant, the ratio of the temperature T_2/T_1 is:

(a) 2 (b) 4
(c) 6 (d) 8

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (*) : $E_{b_1} = \sigma T_1^4$

$E_{b_2} = \sigma T_2^4$

$\frac{E_{b_2}}{E_{b_1}} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{T_2}{T_1} = \left(\frac{E_{b_2}}{E_{b_1}}\right)^{1/4}$

$\frac{T_2}{T_1} = \left(\frac{800}{100}\right)^{1/4}$

$\frac{T_2}{T_1} = (8)^{1/4}$

600. If the gray body emission is 90% of black body emission at 1200 K, then find the temperature required for the gray body.

(a) 1788 K (b) 1655 K
(c) 1830 K (d) 1754 K

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (*) : Emission of gray body = 90% of emission of black body

$E = 0.9\ E_b$

$$\boxed{\frac{E}{E_b} = 0.9}$$

Emissivity of gray body, $\epsilon = 0.9$

$\epsilon T_2^4 = \epsilon_b T_b^4$

$0.9T_2^4 = 1 \times (1200)^4$

$T_2 = 1232.028$ K

601. What is the value of reflectivity in an opaque body, when absorptivity is 0.4?

(a) 0 (b) 0.6
(c) 1 (d) 0.4

VIZAG STEEL MT 24.01.2021 Shift-II

Ans. (b) : Given,

$\alpha = 0.4$

We know that,

$\alpha + \rho + \tau = 1$

For opaque body,

$\tau = 0$

$\rho = 1 - \alpha$

$= 1 - 0.4$

$\rho = 0.6$

602. The gray body shape factor for radiant heat exchange between a small body (emissivity 0.4) in a large enclosure (emissivity 0.5) is

(a) 0.1 (b) 0.2
(c) 0.4 (d) 0.5

JPSC AE, 10.04.2021, Paper-I

Ans. (*) : Given, $\epsilon_1 = 0.4$, $\epsilon_2 = 0.5$

$F_{12} = \frac{1}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{1}{\frac{1}{0.4} + \frac{1}{0.5} - 1}$

$F_{12} = \frac{1}{2.5 + 2 - 1} = \frac{1}{3.5} = 0.285$

No correct answer is given

Note-Official answer is given option (c).

603. The emissivity is likely to be higher in case of

(a) Rubber (b) Paper
(c) Carbon (d) Iron oxide

JPSC AE, 10.04.2021, Paper-I

Ans. (b) : The emissivity is likely to be higher in case of paper.

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604. Heat transfer by Radiation is encountered least in

(a) Boiler furnace
(b) Insulated steam pipe
(c) Electric bulb
(d) Nuclear reactor

JPSC AE, 10.04.2021, Paper-I

Ans. (b) : When temperature is very high then radiation mode of heat transfer dominates over conduction or convection mode of heat transfer.

605. Surface finish factor for a mirror polished component is :

(a) 0.75 (b) 0.85
(c) 0.9 (d) 1

JPSC AE 10.04.2021, Paper-II

Ans. (d) : Surface finish factor for a polished component is 1.

606. The rate at which energy is radiated by a black body at an absolute temperature is given by

(a) Grashof law
(b) Prandtl law
(c) Nusselt law
(d) Stefan Boltzmann law

UPRVUNL AE 05.07.2021

Ans. (d) :

607. When radiant energy is incident upon the surface of the body, which of the following phenomena does NOT occur?

(a) Conversion (b) Reflection
(c) Absorption (d) transmission

UPRVUNL AE 05.07.2021

Ans. (a) :

608. At room temperature, the radiation heat transfer co-efficient to surface emissivity is about:

(a) 2 (b) 6
(c) 3 (d) 1.5

UPRVUNL AE 05.07.2021

Ans. (b) :

609. The substances which are in thermal equilibrium with their environment will bear a constant ratio between total emissive power to the total absorptivity at an temperature. This is law is called

(a) Stefan Boltzman Law
(b) Planck Law
(c) Wien Displacement Law
(d) Kirchhoff Law

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (d) : Kirchhoff Law–The substances which are in thermal equilibrium with their environment will bear a constant ratio between total emissive power to the total absorptivity at any temperature. This is low is called Kirchhoff Law.

610. How is the incident radiation absorbed by a gas layer?

(a) A gas layer absorbs incident radiation slowly
(b) A gas layer absorbs incident radiation rapidly
(c) Doesn't matter
(d) unpredictable

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (a) :

611. Total emissive power and absorptivity of a general and black bodies are connected by

(a) Wein's Law
(b) Plank's Law
(c) Kirchoff's Law
(d) Stephen Boltzment's Law

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (c) :

612. Ratio of total emissive power of a body to the total emissive power of black body is called:

(a) Absorptivity (b) Reflectivity
(c) Transmissivity (d) Emissivity

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) :

613. The overall heat transfer coefficient due to convection and radiation for a steam maintained at 200°C running in a large room at 30°C is 17.95 W/m^{2}K. If the emissivity of the pipe surface is 0.8; the value of $\sigma = 5.67 \times 10^{-8}$ W/ m^{2}K^4; the heat transfer coefficient due to radiation will be nearly

(a) 17 W/m^{2}K (b) 14 W/m^{2}K
(c) 11 W/m^{2}K (d) 8 W/m^{2}K

ESE 2020

Ans. (c) : Heat lost by pipe due to radiation to the room air

$= \sigma A_{pipe} \in_{pipe} \left(T_{pipe}^4 - T_{room}^4\right)$

$= h_{radiation} \times A_{pipe} \times \left(T_{pipe} - T_{room}\right)$

$\therefore$ Heat transfer coefficient due to radiation

$$h_{radiation} = \frac{\sigma \times \in_{pipe} \left(T_{pipe}^4 - T_{room}^4\right)}{T_{pipe} - T_{room}}]$$

$$= \frac{5.67 \times 10^{-8} \times 0.8\left[(200+273)^4 - (30+273)^4\right]}{(200-30)}$$

$= 11.1067\ W/m^2K$

614. A 40 cm diameter disk with emissivity of 0.65 is placed in a large enclosure at 30^{0}C and is effectively a black body. If the disc has a temperature of 55^{0}C, calculate the radiosity of its upper surface.

(a) 604 W/m^2 (b) 594 W/m^2
(c) 560 W/m^2 (d) 749 W/m^2

UPPSC AE 13.12.2020, Paper-2

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Ans. (b) : Given,

Emissivity (E) = 0.65

Radiosity, $J = \varepsilon E_b + (1-\varepsilon) G$

$= \varepsilon\sigma T_b^4 + (1-\varepsilon)\sigma T^4$

$= [0.65 \times (328)^4 + (1-0.65)(303)^4] 5.67 \times 10^{-8}$

$= 426.57 + 167.271$

$= 593.84 \text{ W/m}^2$

615. The radiation pyrometers work on the principle of

(a) Newton's law

(b) Third law of thermodynamics

(c) Wien's displacement law

(d) Stefan boltzmann's law

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (d) : Radiation pyrometer is used to measure temperature more than 1200^0C such as boiler - furnace, sun. It works on Stefan Boltzmann's law.

616. All substances emit radiation, the quality and quantity depending upon the absolute temperature and the properties of the material composing the radiating body. The above statement is known as the statement of

(a) Planck's law

(b) Kirchoff's law

(c) Electrical analogy

(d) Stefan-Boltzmann law

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Planck's Law- It gives satisfactory result in the whole wavelength range.

$(E_{b\lambda})$ = Spectral monochromatic black body emissive power.

$$E_{b\lambda} = 2\pi C_1^2 h \left[\frac{\lambda^{-5}}{\frac{C_2 h}{e\lambda K T^{-1}}}\right]$$

Kirchoff's Law - At any particular temperature the ratio of ε and α is constant for all the substance, which are in thermal equilibrium with their environment i.e.

$$\frac{\varepsilon}{\alpha} = \text{constant}$$

• If body is black body then $(\varepsilon = \alpha)$ i.e. if a body is in thermal equilibrium with its environment its absorptivity is equal to emissivity.

Stefan-Boltzman Law- This law states that the total hemispherical emissive power of a black body is directly proportional to the fourth power of the absolute temperature of black body.

$$E_b = \int_0^\infty E_\lambda . d\lambda = \sigma T^4$$

• The total emissive power of a diffuse surface is equal to π times its radiation intensity.

$E_b = \pi I$

Stefan- Boltzman constant = 5.67×10^{-8} W/m^{2}K^4

617. The temperature of a surface with 0.2 m^2 area is 17 deg C. Calculate the wavelength corresponding to maximum monochromatic emissive power

(a) 20 micrometers (b) 30 micrometers

(c) 10 micrometers (d) 40 micrometers

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Wein's displacement law:

$\lambda_{max} \times T = 2898$ μm–K (assuming body as black body)

$$\lambda_{max} = \frac{2898}{T} = \frac{2898}{17+273} = 9.993\mu m$$

$\lambda_{max} \simeq 10$ μm

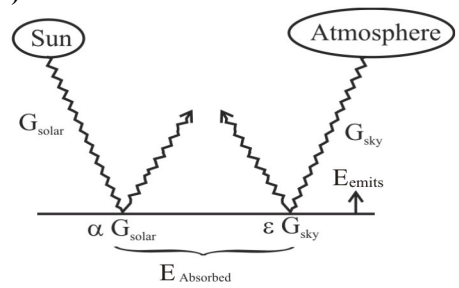
618. For a surface, the direct and diffuse components of the solar radiation are 400 and 300 W/m^2 and the direct radiation makes 60 deg angle with the normal. If the surface temperature is 300 K and effective sky temperature is 200 K, calculate net rate of radiation heat transfer. (Assume solar absorptivity and emissivity as 0.1 each; For calculation take Stefan Boltzmanns constant as 6×10^{-8} W/m^2.K^4)

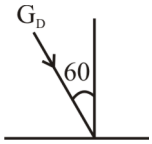
(a) 11 W/m^2 (b) 45 W/m^2

(c) 33 W/m^2 (d) 64 W/m^2

ISRO Scientist/Engineer 12.01.2020

Ans. (a) :





The total solar energy incident on the surface $G_{solar} = (G_{Direct} \times \cos\theta) + G_{diffuse}$

$= (400 \times \cos 60) + 300$

$= 500 \text{ W/m}^2$

Net rate of radiation heat transfer

$= (\alpha_{solar} \times G_{solar}) + \varepsilon\sigma \left(T_{sky}^4 - T_{surface}^4\right)$

$= (0.1\times500) + 0.1\times6\times10^{-8}\left(200^4 - 300^4\right)$

$= 50 - 39$

$= 11 \text{ W/m}^2 \qquad \left(\because \sigma = 6\times10^{-8} \text{ W/m}^2 - \text{K}^4\right)$

619. In which type of collector is solar radiation focused into the absorber from the top, rather than from the bottom ?

(a) Fresnel lens (b) Paraboloidal

(c) Concentrating (d) Compound parabolic

ESE 2020

Ans. (d) : Compound parabolic

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620. A flat plate collector is 150 cm wide and 180 cm high and is oriented such that it is perpendicular to the sun rays. Its active area is 90% of the panel size. If it is in a location that receives solar insulation of 1000 W/m^2 peak, the peak power delivered to the area of the collector will be

(a) 1.23 kW (b) 2.43 kW
(c) 4.46 kW (d) 6.26 kW

ESE 2020

Ans. (b) : Area of the flat plate
$= 150 \times 180 = 27000\ cm^2 = 2.7\ m^2$
Solar insolation = $1000\ W/m^2 = 1000 \times 2.7 \times 0.90$
$= 2700 \times 0.90 = 2.43\ kW$

621. A surface having high absorptance for shortwave radiation (less than 2.5 μm) and a low emittance of long-wave radiation (more than 2.5 μm), is called

(a) Absorber (b) Emitter
(c) Selective (d) Black

ESE 2020

Ans. (c) : Selective

622. In quantum theory, the value of Planck's constant in the equation E = hν is :

(a) 1.982×10^{-34} J-s (b) 6.476×10^{-8} J-s
(c) 6.626×10^{-34} J-s (d) 3.453×10^{-8} J-s

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Planck Radiation Law– Planck is considered the father of the quantum theory according to planck $E = h\nu = \frac{hc}{\lambda}$

where
E = Quantum Energy
ν = Frequency
h = Plancks constant
$h = 6.626 \times 10^{-34}$ J. sec

623. The intensity of solar radiation is maximum at a wavelength of 0.49 μm. Assuming the Sun as a black body, what is the approximate total emissive power of Sun? [Consider Wien's displacement constant = 2890 μm-K]

(a) 6.86×10^4 kW/m^2
(b) 6.86×10^7 kW/m^2
(c) 6.86 W/m^2
(d) 6.86 kW/m^2

SJVN ET 2019

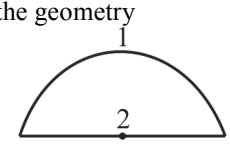
Ans. (a) :
$\lambda T = 2890\ \mu mK$
$T = \frac{2890}{0.49} = 5897.95\ K$
$E = \sigma A T^4$
$= 5.67 \times 10^{-8} \times 1 \times (5897.95)^4$
$= 6.861 \times 10^7\ W/m^2$
$= 6.861 \times 10^4\ kW/m^2$

624. A hemispherical furnace of radius 1.0 m has a roof temperature of T_1 = 800 K and emissivity ε_1 = 0.8 . The flat floor of the furnace has a temperature T_2 = 600K and emissivity ε_2 = 0.5 . The view factor F_{12} from surface 1 to 2 will be

(a) 0.3 (b) 0.4
(c) 0.5 (d) 0.6

ESE 2019

Ans. (c) : From the geometry



$F_{21} + F_{22} = 1$
$F_{21} + 0 = 1$
$F_{21} = 1$

From reciprocity theorem
$A_1 F_{12} = A_2 F_{21}$
$F_{12} = \frac{A_2}{A_1} F_{21}$
$= \frac{\pi R^2}{2\pi R^2} \times 1$
$F_{12} = 0.5$

625. The temperature of a body of area 0.1 m^2 is 900 K. The wavelength for maximum monochromatic emissive power will be nearly

(a) 2.3μm (b) 3.2μm
(c) 4.1μm (d) 5.0μm

ESE 2019

Ans. (b) : From Wien's displacement law
$\lambda_{max} \cdot T = 2898\ \mu mk$
$\lambda_{max} \cdot 900 = 2898$
$\lambda_{max} = 3.22\ \mu m$

626. The rate of energy emission from unit surface area through unit solid angle, along a normal to the surface, is known as:

(a) Emissivity (b) Transmissivity
(c) Reflectivity (d) Intensity of radiation

OPSC AEE 2019 Paper - II

Ans : (d) : The rate of energy emission from unit surface area through unit solid angle, along a normal to the surface, is known as intensity of radiation.

627. In what form solar energy is radiated from the Sun?

(a) Ultraviolet radiation
(b) Infrared radiation
(c) Electro-magnetic waves
(d) Transverse waves

JPSC AE PRE 2019

Ans. (c) : Electro-magnetic waves.

628. Which of the following is an example of transient heat flow?

(a) Current carrying conductors
(b) Spherical conductor carrying superheated steam
(c) Heating and cooling of enclosures due to solar radiations
(d) Cooling of water

HPPSC ADF 05.03.2019

Ans. (c) : Transient implies variation with time or time dependence. Transient heat flow in heating and cooling of enclosures due to solar radiations.

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629. Which of the following is the case of heat transfer by radiation?

(a) Touching a stove and being burned
(b) Ice cooling down your hand
(c) Heat from a light bulb
(d) Hot air rising, cooling and falling

MECON MT 2019

Ans. (c) : Which of the following is the case of heat transfer by radiation heat from a light bulb.

630. The total radiation leaving a surface per unit time and per unit area is called

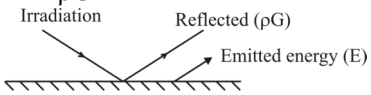
(a) radiosity (b) shape factor
(c) radiation intensity (d) black body radiation

TNPSC AE 2019

Ans. (a) : Radiosity (J)- The total thermal radiation energy leaving a surface per unit time per unt area is known as radiosity.

J = Emitted energy + Reflected part of incident.

$J = E + \rho G$



631. A good approximation of the measured solar spectrum is made by

(a) black-body energy distribution
(b) Planck's energy distribution
(c) inverse square law
(d) solar constant

ESE 2019

Ans. (a) : The blackbody model with a temperature of 5900 K is a good approximation to measured solar radiation.

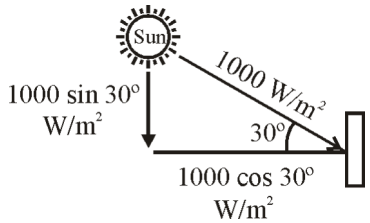
632. A room window (consisting of a vertical sheet of plane glass) is exposed to direct sunshine at a strength of 1000 W/m^2. The window is pointing due south, While the sun is in the southwest, 30^0 above the horizon. Estimate the amount of solar energy in W/m^2 reflected by the window: Assume glass to be gray with ρ (reflectivity) = 0.08.

(a) 49 (b) 490
(c) 612.3 (d) 61.2

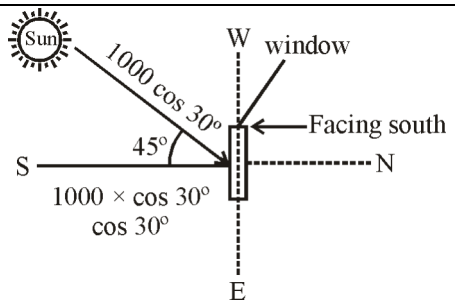
BHEL ET 2019

Ans. (a) : Solar radiation reflected by that surface is given by–

Reflected solar radiation = Reflectivity × incident solar radiation



This solar energy incidents on window in south west direction.



Strength of sunshine = 1000 W/m^2
Reflectivity (ρ) = 0.08
Solar radiation perpendicular to the plane of the window
= 1000 × cos 30° × cos 45°
The amount of solar energy reflected from the window therefore, Reflection = ρ × 1000 × cos 30°× cos 45°
= 0.08 × 1000 × cos 30° × cos 45°
= 48. 985
=49 W/m^2

633. A wave of radiation falls on a body. 35% of the radiation is reflected back. If transmissivity of the body is 0.25, then emissivity is :

(a) 0.35 (b) 0.45
(c) 0.40 (d) 0.25

BHEL ET 2019

Ans. (c) : Given– transmissivity $\tau = 0.25$
reflectivity $\rho = 0.35$
$\alpha + \rho + \tau = 1$
$\alpha + 0.35 + 0.25 = 1$
$\alpha = 1 - 0.60 = 0.40$ (according to Kirchoff's law of thermal radiation.)
$\boxed{\alpha = \epsilon = 0.40}$

634. An electric flat-plate square heater of sides 10 cm provides 100 W power from each side. If the heater is assumed black, its temperature is approximately:

(a) 648^0 C (b) 648 K
(c) 6480 K (d) 6480^0 C

BHEL ET 2019

Ans. (b) : Area = 10 cm × 10 cm
= 100 cm^2 = 100 × 10^{-4} m^2
Q = 100 W

$\boxed{E = \epsilon \sigma A T^4}$

for black body $\boxed{\epsilon = 1}$

$100 = 1 \times 5.67 \times 10^{-8} \times 100 \times 10^{-4} \times T^4$
$100 = 567 \times 10^{-12} \times T^4$

$$T^4 = \frac{100}{567 \times 10^{-12}}$$

$T^4 = 0.17636 \times 10^{12}$
$T = \sqrt[4]{0.17636 \times 10^{12}}$
$\boxed{T = 648\ K}$

635. Two very large parallel plates with emmisionfies 0.3 and 0.8 exchange heat. Find the percentage (%) reduction in heat transfer when a polished aluminium shield (ε = 0.04) is placed between them)

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(a) 85% (b) 88%
(c) 91% (d) 93%

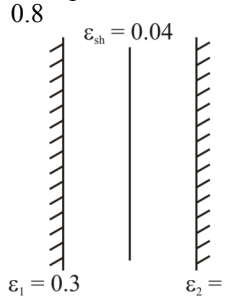
TNPSC AE 2019

Ans. (d) : Data given-

$\varepsilon_1 = 0.3, \varepsilon_2 = 0.8, \varepsilon_{sh} = 0.04$

$$\left[\frac{q}{A}\right]_{\text{net without shield}} = \frac{\sigma\left(T_1^4 - T_2^4\right)}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1}$$

$$= \frac{\sigma\left(T_1^4 - T_2^4\right)}{\frac{1}{0.3} + \frac{1}{0.8} - 1}$$



$$\left[\frac{q}{A}\right]_{\text{without shield}} = 0.279 \times \sigma\left(T_1^4 - T_2^4\right) W/m^2$$

$$\left[\frac{q}{A}\right]_{\text{with shield}} = \frac{\sigma\left(T_1^4 - T_2^4\right)}{\frac{1}{0.3} + \frac{2}{0.04} + \frac{1}{0.8} - 2}$$

$$= 0.019 \times \sigma \times \left(T_1^4 - T_2^4\right) W/m^2$$

$$\% \text{ Reduction} = \frac{q_{\text{without}} - q_{\text{with}}}{q_{\text{without}}} \times 100\%$$

% Reduction = 93.18%

636. The emissitivities for non-metallic surfaces generally

(a) increase with increase in temperature
(b) decrease with increase in temperature
(c) increase exponentially with temperature
(d) remain constant at all temperatures

TNPSC AE 2019

Ans. (b) : The emissitivities for non-metallic surfaces generally decrease with increase in temperature.

637. The ratio between emissive power and intensity of normal radiation is

(a) π (b) $\pi/2$
(c) $2/\pi$ (d) $\pi/3$

JPSC AE PRE 2019

Ans. (a) : The ratio between emissive power and intensity of normal radiation is π.

$$\frac{\text{Emissive Power}}{I_b} = \pi$$

638. An insulated pipe of 50 mm outside diameter with ε =0.8 is laid in a room at 30°C. If the surface temperature is 250°C and the convective heat transfer coefficient is 10W/ m^2K, the total heat loss per unit length of the pipe will be

(a) 896.6 W/m (b) 818.8 w/m
(c) 786.4W/m (d) 742.2 W/m

ESE 2019

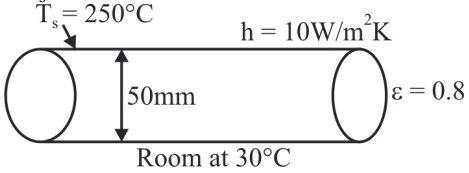
Ans. (b) : Given,

d = 50 mm = 0.05 m

$\in = 0.8$

$T_\infty = 30 + 273 = 303$ K

$T_s = 250 + 273 = 523$ K



Total heat loss per unit length

$= Q_{convection} + Q_{radiation}$

$= hA(T_s - T_\infty) + \in A\sigma\left(T_s^4 - T_\infty^4\right)$

$= h \times \pi d\ell\left(T_s - T_\infty\right) + \in \times \pi d\ell \times \sigma\left(T_s^4 - T_\infty^4\right)$

$= 10 \times \pi \times 0.05\ (250 - 30) + 0.8 \times \pi \times 0.05 \times 5.67 \times 10^{-8}\ (523^4 - 303^4)$

$= 345.57 + 473.03$

$= 818.6$ W/m

639. When absorptivity (α) = 1 reflectivity (p) = 0 and transmittivity (τ) = 0, then the body is said to be a

(a) black body (b) grey body
(c) opaque body (d) white body

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (a) : A black body surface completely absorbs external irradiation, regardless its surface absorptivity is always 1 so $\alpha\lambda = \alpha = 1$ the reflectivity of a blackbody is always zero.

640. The value of the wavelength for maximum emissive power is given by

(a) Kirchhoff's law (b) Stefan's law
(c) Wein's law (d) Planck's law

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The value of the wavelength for maximum emissive power is given by wen's law.
When the temperature of a blackbody radiator increases, the overall radiated energy increases and the peak of the radiation curve moves to shorter wavelengths. When the maximum is evaluated from the Planck radiation formula, the product of the peak wavelength and the temperature is found to be a constant.

641. If the temperature of a solid surface changes from 27ºC to 81ºC, then its emissive power changes in the ratio of nearly

(a) 81 (b) 9
(c) 1.49 (d) 1.94

ISRO Scientist/Engineer 22.04.2018

Ans. (d) : $T_2 = 81$ºC, $T_1 = 27$ºC

We know that Emissive Power (E)

$$E = \sigma T^4 \Rightarrow E \propto T^4$$

$$\frac{E_2}{E_1} = \left[\frac{T_2}{T_1}\right]^4 = \left(\frac{81+273}{27+273}\right)^4 = 1.938$$

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642. Two radiation surfaces $A_1 = 6\ m^2$ and $A_2 = 4\ m^2$ have the shape factor $F_{1-2} = 0.1$; the shape factor F_{2-1} will be :

(a) 0.18 (b) 0.15
(c) 0.12 (d) 0.10

RPSC Vice Principal ITI 2018

Ans. (b) :
By reciprocity theorem,
$A_1F_{1-2} = A_2F_{2-1}$
$6 \times 0.1 = 4 \times F_{2-1}$
$\boxed{F_{2-1} = 0.15}$

643. The intensity of the radiation emitted by the sun is maximum at a wavelength of 0.5 μm. Assuming the sun to a black body, the surface temperature of the sun will approximately be:

(a) 5000 K (b) 5780 K
(c) 6280 K (d) 6490 K
(e) 6600 K

CGPSC AE 25.02.2018

Ans. (b) : Wavelength (λ) = 0.5 μm
Surface temperature (T) = ?
We know that-

$$T \times \lambda = 2898\ \mu m - K$$

$$T = \frac{2898}{0.5} = 5796 = 5780\ K$$

644. The radiation heat transfer from an inner cylindrical surface of radius r_1 and emissivity ε_1 at temperature T_1 to concentric cylinder of radius r_2, emissivity ε_2 and at temperature T_2 is proportional to-

(a) $\dfrac{T_1^4 - T_2^4}{\left[\dfrac{1}{\varepsilon_1} + \left(\dfrac{1}{\varepsilon_2} - 1\right)\left(\dfrac{r_1}{r_2}\right)\right]}$

(b) $\dfrac{T_1^4 - T_2^4}{\left[\dfrac{1}{\varepsilon_1} - \left(\dfrac{1}{\varepsilon_2} - 1\right)\left(\dfrac{r_1}{r_2}\right)\right]}$

(c) $\dfrac{T_1^4 - T_2^4}{\left[\dfrac{1}{\varepsilon_1} + \left(\dfrac{1}{\varepsilon_2} - 1\right)\left(\dfrac{r_2}{r_1}\right)\right]}$

(d) $\dfrac{\left(\dfrac{r_2}{r_1}\right)T_1^4 - T_2^4}{\dfrac{1}{\varepsilon_1} + \left(\dfrac{1}{\varepsilon_2} - 1\right)}$

RPSC AE 2018

Ans. (a) : $Q \propto \dfrac{T_1^4 - T_2^4}{\left[\dfrac{1}{\varepsilon_1} + \left(\dfrac{1}{\varepsilon_2} - 1\right)\left(\dfrac{r_1}{r_2}\right)\right]}$

645. In solar flat-plate collectors, the absorber plate is painted with selective paints. The selectivity is the ratio of

(a) Solar radiation-absorption to thermal infrared radiation-emission
(b) Solar radiation emission to thermal infrared radiation-absorption
(c) Solar radiation reflection to thermal infrared radiation-absorption
(d) Solar radiation absorption to thermal infrared radiation-reflection

ESE 2018

Ans. (a) : In solar thermal collectors, a selective surface or selective absorber is a means of increasing its operation temperature.
The selectivity is defined as the ratio of solar radiation absorption to thermal infrared radiation emission.

646. The emissivity of a polished silver body is ..... as compared to black body

(a) same (b) low
(c) very low (d) high

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (c) : The emissivity of a polished silver body is very low as compared to black body.
The emissivity of a surface depends not only on the material but also on the nature of the surface. For example a clean and polished metal surface will have a low emissivity whereas a roughened and oxidized metal surface, black body will have high emissivity.

647. The emissivity for a black body is

(a) 0 (b) 0.5
(c) 0.75 (d) 1

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (d) : Real materials emit energy at a fraction called the emissivity of black body energy levels By definition a black body in thermal equilibrium has an emissivity of $\varepsilon = 1$.
A black body has an emissivity of 1 while a perfect reflector or white body has an emissivity of zero.

648. Absorptivity of a body will be equal to its emissivity:

(a) at all temperature
(b) at one particular temperature
(c) when system is in thermal equilibrium
(d) for a polished surface

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : Kirchhoff's law states that for an arbitray body emitting and absorbing thermal radiation in thermodynamic equilibrium. The emissivity is equal to the absorptivity.

649. An oxygen and nitrogen mixture has transmissivity for heat radiations that is practically equal to

(a) 1 (b) 0.5
(c) 2 (d) 0

GPSC EE Pre, 28.01.2017

Ans. (a) : An oxygen and nitrogen mixture has transmissivity for heat radiations that is practically equal to 1. This is very small quantity. The response of the body to incident radiations is, however, completely incident of and unaffected by the simultaneous emission from the body.

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650. A hot and cold body are kept separated from each other in perfect vacuum. Which of the following causes decreases in temperature of the hot body?

(a) Radiation
(b) Convection
(c) Conduction
(d) Temperature remains unchanged

GPSC EE Pre, 28.01.2017

Ans. (a) : As the two bodies are kept in vacuum and without contact, conduction and convection are not possible. So, the heat has travelled from one body to the other in the form of radiation.

651. The ratio of the energy absorbed by the body to the total energy falling on it, is called

(a) emissivity (b) emissive power
(c) absorbtivity (d) none of the above

Nagaland PSC (CTSE) 2018, Paper-II
Punjab PSC SDE 12.02.2017

Ans. (c) : Absorbtivity - The ratio of the energy absorbed by the body to the total energy falling on it, is called absorbtivity.

Emissivity- Ratio of emissive power of non-black body to the emissive power of black body.

$$\boxed{\in = \frac{E}{E_b}}$$

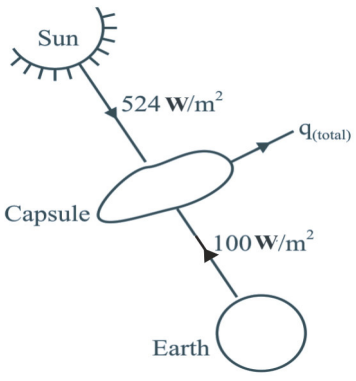
Emissive Power (E_b)- Is the energy emitted by the surface per unit time per unit area.

652. A radiator on a space capsule must dissipate heat generated inside the capsule, by radiating it into space. While on the bright side of earth, the capsule sees the earth's radiation and the solar radiation. The radiator surface has a solar absorptivity of 0.5 and an emissivity of 0.9. What is the maximum equilibrium surface temperature of the radiator, if the maximum heat generated in the capsule is 800 W/m^2?
Assume earth's radiation = 100 W/m^2 and solar irradiation = 524 W/m^2.
Take Stefan Boltzmann constant as 5×10^{-8} W/m^2-K^4

(a) 400°C (b) 100°C
(c) 127°C (d) 160°C

ISRO Scientist/Engineer 07.05.2017

Ans. (c) :



Heat generated in the capsule = 800 W/m^2
Earth's radiation = 100 W/m^2
Solar irradiation = 524 W/m^2
Absorptivity, $\alpha = 0.5$
Emissivity, $\varepsilon = 0.9$
Equating the net heat generated.

$$800 = \varepsilon\sigma T^4 - 0.5 \times 524 - 100$$
$$\Rightarrow \quad 800 = 0.9 \times 5.67 \times 10^{-8} \times T^4 - 0.5 \times 524 - 100$$
$$\Rightarrow \quad T^4 = 2.58 \times 10^{10}$$
$$\Rightarrow \quad T = 400.86K = 127.86°C$$

653. Two balls of same material and finish have their diameters in the ratio of 2 : 1 and both are heated to same temperature and allowed to cool by radiation. Rate of cooling by big ball as compared to smaller one will be

(a) 1 : 4 (b) 2 : 1
(c) 1 : 2 (d) 4 : 1

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : Use, Stefen Boltzman Law,

$$E \propto T^4$$
$$\frac{Q}{A} = \sigma T^4 \qquad \boxed{Q \propto A}$$
$$Q = \sigma T^4 A$$
$$\frac{Q_1}{Q_2} = \frac{A_1}{A_2} = \frac{d_1^2}{d_2^2} = \left(\frac{2}{1}\right)^2$$
$$\frac{Q_1}{Q_2} = \frac{4}{1}$$
$$Q_1 : Q_2 = 4 : 1$$

654. All grey bodies obey the

(a) Stefan-boltzmann's law
(b) Planck's
(c) Kirchhoff's law
(d) Law of inertia

Nagaland PSC CTSE 2017 Paper-2

Ans. (c) : A grey bodies obey the, **Kirchhoff's Law :-** It states that, whenever a body is in thermal equilibrium with it's surrounding, its emissivity to its absorptivity.

$$\boxed{\varepsilon = \alpha}$$

655. If G is irradiation and J is the radiosity, the net radiation leaving the surface is

(a) J (b) G
(c) G-J (d) J-G

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : J-G

656. For the calculation of the shape factor

(a) There should not be any intervening reflections between the surfaces
(b) There should be at least one intervening reflection between the surfaces
(c) Reflections do not affect the shape
(d) None of the above

Nagaland PSC CTSE 2017 Paper-2

Ans. (b) : There should be at least one intervening reflection between the surfaces.

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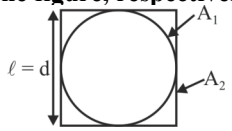
657. The view factors F_{12} and F_{21}, for the sphere of diameter d and a cubical box of length ℓ = d as shown in the figure, respectively, are



(a) 1 and $\frac{\pi}{3}$ (b) $\frac{\pi}{3}$ and 1

(c) 1 and $\frac{\pi}{6}$ (d) $\frac{\pi}{6}$ and 1

ESE 2017

Ans. (c) : Summation rule for sphere

$$F_{11} + F_{12} = 1$$

$$F_{12} = 1 \qquad (\because F_{11} = 0)$$

By reciprocity theorem

$$A_1F_{12} = A_2F_{21}$$

$$F_{21} = \frac{A_1}{A_2}F_{12}$$

$$F_{21} = \frac{\pi d^2}{6d^2}$$

$$F_{21} = \frac{\pi}{6}$$

658. Statement (I) : The radiation emitted by earth is of longer wavelength than that emitted by the sun.
Statement (II) : Earth is at a lower temperature than the sun.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I)
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I)
(c) Statement (I) is true but Statement (II) is false
(d) Statement (I) is false but Statement (II) is true

JWM 2017

Ans. (a) : The radiation emitted by sun is of shorter wave length then that emitted by earth. This is because that sun has high temperature than earth.

659. Which of the following metal has highest thermal conductivity at normal pressure and temperature

(a) brass (b) copper
(c) steel (d) silver

GPSC ARTO 01.05.2016

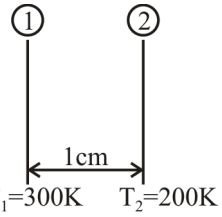
Ans. (d) : At normal pressure and temperature, highest. Thermal conductivity is silver (Ag.)

660. Consider steady heat transfer between two large parallel plates at constant temperatures of T_1 = 300 K and T_2 = 200 K that are L = 1 cm apart. Assuming the surfaces to be black (emissivity ε = 1), determine the rate of heat transfer between the plates per unit surface area assuming the gap between the plates is filled with super insulation that has an apparent thermal conductivity of 0.00002 W/m°C.

(a) 0.2 W (b) 0.4 W
(c) 0.7 W (d) 0.8 W
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) :



L = 1 cm = 0.01 m

$\epsilon_1 = \epsilon_2 = 1$

k_{ins} = 0.00002 W/m°C

As the gap between the plates is filled with super insulation thus, heat transfer between plates due to radiation is not possible.

$$\dot{q}_{cond.} = \frac{T_1 - T_2}{\left(\frac{L}{k}\right)}$$

$$= \frac{(300 - 200)}{\left(\frac{0.01}{0.00002}\right)}$$

$$\dot{q}_{cond.} = 0.2\,W$$

661. What is the radiation intensity in a particular direction?

(a) Radiation energy per unit time per unit area of the radiating surface
(b) Radiant energy per unit time per unit solid angle per unit area of the radiation surface
(c) Radiant energy per unit time per unit solid angle per unit projected area of the radiation surface in the given direction
(d) Radiant energy per unit time per unit projected area of the radiating surface in the given direction

RPSC ADE 2016

Ans. (c) : Radiant energy per unit time per unit solid angle per unit projected area of the radiation surface in the given direction.

662. Which instruments are used for measuring Solar radiation parameters :
1. Pyrheliometer
2. Pyranometer
3. Pyradiometer
4. Pyrometer

(a) 1, 2, 3 and 4 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) 1, 2 and 3

RPSC VPITI 14.02.2016

Ans. (d)

1. Pyrheliometer is an instrument for measurement of direct beam solar irradiance.
2. A pyranometer used for measuring solar irradiance on a planar surface.

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3. Pyradiometer is measuring net radiation at earth surface for meteorological application.
4. Pyrometer, device for measuring relatively high temperatures.

So, option (d) 1, 2, 3 are correct.

663. Cylindrical parabolic solar concentrator can give temperatures in the range of

(a) less than 50°C (b) 50 °C to 100°C
(c) 100°C to 200°C (d) 300°C to 400°C

RPSC AE (GWD) 06.05.2016

Ans. (d) : Temperature ranges 300ºC to 400ºC

664. Consider heat transfer by radiation for two concentric infinitely long cylindrical surfaces. If the diameter ratio of cylinders is 1:3, the shape factor for the outer cylinder with the inner cylinder is:

(a) 1/3 (b) 2/3
(c) 1 (d) 1/9

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : We know that,

$F_{12}A_1 = F_{21}A_2 \quad \left[\because A_1 = \frac{\pi}{4}D_1^2 \;\&\; A_2 = \frac{\pi}{4}D_2^2\right]$

$F_{12}D_1^2 = F_{21}D_2^2$

$D_1^2 = F_{21}D_2^2$

$F_{21} = \left(\frac{1}{3}\right)^2$

[$F_{12} = 1$ for concentric infinitely long cylinder]

For outer to inner, $F_{21} = \frac{1}{9}$

665. What is the blackbody radiation?

(a) Electrical currents emitted by a black body
(b) Electromagnetic radiation emitted by a black body
(c) Electromagnetic radiation emitted by a black hole
(d) None of these

KPCL AE 2016

Ans. (b) : Blackbody radiation is the thermal electromagnetic radiation within or surrounding of a body in thermodynamic equilibrium with its environment.

666. Hottel's crossed String method is used for determining .....which are infinite in one direction.

(a) surface factor (b) line factor
(c) shape factor (d) volume factor

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Hottel's crossed string method is used for determining shape factor which are infinite in one direction.

667. The ratio of 'actual radiation at T to black surface radiation at T' is known as:

(a) Radiance (b) Emittance
(c) Convectness (d) Irradiance

HPPSC AE (PWD) 10.09.2016

Ans. (b) : The ratio of actual radiation at T to the black surface radiation at T is known as emittance.

$$\text{Emittance} = \frac{\text{Actual radiation}}{\text{Black surface radiation}}$$

668. Consider a person standing in a breezy room at 20°C. Determine the total rate of heat transfer from this person if the exposed surface area and the average outer surface temperature of the person are 1.6 m^2 and 29°C, respectively, and the convection heat transfer coefficient is 6 $W/m^2.°C$. Given, the emissivity of a person is ε = 0.95.

(a) 86.4 W (b) 82.1 W
(c) 122.7 W (d) 168.1 W
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : $Q_{radiation} = \varepsilon\sigma A\left(T_1^4 - T_2^4\right)$

$= 0.95\times5.67\times10^{-8}\times1.6\times[(29+273)^4-(20+273)^4]$

$= 81.712$ W

$Q_{convection} = hA_s(T_s - T_\infty)$

$= 6 \times 1.6 \times (29 - 20)$

$= 86.4$ W

$Q_{total} = Q_{radiation} + Q_{convection}$

$= 81.712 + 86.4$

$= 168.112$ W

669. Identify the wrong statement.

(a) The shape factor is purely a function of geometrical parameters
(b) When only two bodies are radiating energy with each other the shape factor is given by $A_2F_1 = A_1F_2$
(c) The shape factor of a flat surface with the other surface enclosing the first is always unity
(d) A concave surface has a shape factor with itself because the radiation energy coming out from one part of the surface is intercepted by another part of the same surface
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) : When only two bodies are radiating energy with each other the shape factor is given by–

$A_1F_{12} = A_2F_{21}$

$F_{21} + F_{22} = 1$

670. If the temperature of the sun is doubled, the rate of energy received on earth will be increased by a factor of

(a) 2 (b) 4
(c) 8 (d) 16

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Rate of energy $E \propto T^4$

$\boxed{\text{OR} \quad E_1 = \sigma T^4}$

"If temperature of the sun 'T' is double"

Then, $E_2 = \sigma(2T)^4$

$E_2 = 16\,\sigma T^4$

$E_2 = 16\,\sigma T^4 \rightarrow E_1$

$E_2 = 16\,E_1$

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671. A radiator in a domestic heating system operates at a surface temperature of 55°C. Assuming the radiator behaves as a black body, the rate at which it emits the radiant heat per unit area is (assume $\sigma = 5.67 \times 10^{-8}$ W/(m^{2}K^4))

(a) 0.66 kW/m^2 (b) 0.0005 kW/m^2
(c) 0.5 kW/m^2 (d) 66 kW/m^2

ISRO Scientist/Engineer 03.07.2016

Ans. (a) : $T = 55°C$

$\sigma = 5.67 \times 10^{-8}$ W/m^{2}K^4

Radiation heat transfer

$Q = \varepsilon A \sigma T^4$

For black body $\boxed{\varepsilon = 1}$

$$\frac{Q}{A} = \varepsilon \sigma T^4$$
$$= 1 \times 5.67 \times 10^{-8} (55 + 273)^4$$
$$= 656.2637 \text{ W/m}^2$$
$$= 0.66 \text{ kW/m}^2$$

672. Which of the following is not the characteristics of Planck's black body radiation distribution

(a) As temperature increases, the peak of the curve shift towards higher wavelength
(b) Spectral emissive power varies continuously with the change in wavelength
(c) At a given wavelength, as temperature increases, emissive power also increases
(d) Total emissive power is proportional to T^4

RPSC LECTURER 16.01.2016

Ans. (a) : According to the characteristics of Planck's black body radiation distribution as temperature increases, the peak of the curve shift towards shorter wavelength.

673. The wave length at which the black body emissive power reaches its maximum value at 300 K is –

(a) 9.6 μm (b) 15.5μm
(c) 5.1μm (d) 38.0μm

RPSC IOF & Boiler 20.09.2017

Ans : (a)

Temperature of body = 300 K

Let the wavelength of maximum amount of radiation i.e., peak emissive power wavelength is λ_{max}.

From Wien's displacement law

$T.\lambda_{max} = 2898$ μmK

$$\lambda_{max} = \frac{2898}{300} = 9.66 \text{ μmK}$$

674. Two plates spaced 150 mm apart are maintained at 1000^{0}C & 70^{0}C. The heat transfer will take mainly by–

(a) convection (b) free convection
(c) forced convection (d) radiation

RPSC IOF & Boiler 20.09.2017

Ans : (d) $Q = \sigma(T^4 - T_\infty^4)$

At high temperature, radiation transfer will be high due to 4th power of temperature.

675. Thermal radiation includes:

(a) Entire visible radiation and entire infrared radiation
(b) Entire visible radiation, entire infrared radiation and entire UV radiation
(c) Entire visible radiation, entire infrared radiation and some portion of UV radiation
(d) Some portion of visible radiation and entire infrared radiation

UPRVUNL AE 2016

Ans. (c) : Thermal radiation includes entire visible radiation, entire infrared radiation and some portion of UV radiation.

676. If 5 radiation shields are placed between two parallel plates, the rate of radiation heat transfer is reduced to: (ε = 1 for all surfaces)

(a) 1/10 times (b) 1/5 times
(c) 1/4 times (d) 1/6 times

UPRVUNL AE 2016

Ans. (d) :

$$(q/A)_{\text{with 'n' shields}} = \frac{1}{(n+1)} \left(\frac{q}{A}\right)_{\text{with out any shield}}$$
$$= \frac{1}{6} \text{ times}$$

677. Incident radiation of 1000 W/m^2 fall on the object. The energy absorbed by the object is 400 W/m^2 and energy transmitted is 350 W/m^2. What will be the value of reflectivity?

(a) 0.40 (b) 0.35
(c) 0.75 (d) 0.25

RPSC LECTURER 16.01.2016

Ans. (d) : Incident radiation (G) $= 1000 \frac{W}{m^2}$

Energy absorbed by object $= 400 \frac{W}{m^2}$

Energy transmitted $= 350 \frac{W}{m^2}$

So, energy reflected by body $= 250 \frac{W}{m^2}$

We know that

$$\alpha + \rho + \tau = 1$$
$$\alpha = \frac{400}{1000} = 0.4$$
$$\rho = \frac{250}{1000} = 0.25$$
$$\tau = \frac{350}{1000} = 0.35$$

678. Radiation thermal resistance may be written as (where F, A, σ are shape factor, area and Stefan-Boltzmann constant respectively)

(a) $\frac{1}{FA\sigma(T_1 + T_2)(T_1^2 + T_2^2)}$

(b) $\frac{1}{FA\sigma(T_1 + T_2)(T_1^2 - T_2^2)}$

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(c) $\frac{1}{FA\sigma(T_1^4 - T_2^4)}$

(d) $\frac{1}{FA\sigma(T_1^4 + T_2^4)}$

RPSC LECTURER 16.01.2016

Ans. (a) : We know that

$$Q = FA\sigma(T_1^4 - T_2^4)$$

$$Q = FA\sigma(T_1^2 + T_2^2)(T_1 + T_2)(T_1 - T_2)$$

$$Q = \frac{(T_1 - T_2)}{\left[\frac{1}{FA\sigma(T_1^2 + T_2^2)(T_1 + T_2)}\right]}$$

$$Q = \frac{(T_1 - T_2)}{R_{th}}$$

So,

$$R_{th} = \frac{1}{FA\sigma(T_1^2 + T_2^2)(T_1 + T_2)}$$

679. If ε is the emissivity of surfaces and shields and n is the number of shields, introduced between the two surfaces, then overall emissivity is given by

(a) $\frac{1}{n\varepsilon}$ (b) $\frac{1}{n(2-\varepsilon)}$

(c) $\frac{1}{(n+1)(2-\varepsilon)}$ (d) $\frac{\varepsilon}{(n+1)(2-\varepsilon)}$

BPSC Poly. Lect. 2016

Ans : (d) overall emissivity = $\frac{\varepsilon}{(n+1)(2-\varepsilon)}$

680. Which non-metallic body is expected to have highest value of emissivity?

(a) Iron oxide (b) Carbon

(c) Ice (d) Paper

UPPSC AE 12.04.2016 Paper-II

Ans : (c)

Material	Emissivity
Ice	0.97
Paper	0.86
Aluminum foil	0.03
Brick	0.90
Glass	0.95
Silver	0.04

681. Terrestrial radiation has a wavelength in the range of

(a) 0.2μm to 4μm

(b) 0.2μm to 0.5μm

(c) 0.380μm to 0.760μm

(d) 0.29μm to 2.3μm

UPPSC AE 12.04.2016 Paper-II

Ans : (d) Terrestrial radiation has a wavelength in the range of 0.29μm to 2.3μm .

682. Visible light is a kind of electromagnetic wave whose wavelength is between:

(a) 0.40 and 0.76 μm (b) 0.76 and 1 μm

(c) 0.01 and 0.40 μm (d) 0.01 and 1 μm

GPSC Asstt. Prof. 28.08.2016

Ans. (a) : The visible spectrum is the segment of the electromagnetic spectrum that the human eye can view, the range of wavelength is called visible light have wavelengths from 380 to 700 nano meters.

Note-Option (b) is given by commission.

683. An ideal absorber of radiation is also an ideal emitter. It is known as :

(a) Kirchhoff's law (b) Wien's law

(c) Planck's law (d) Lambert's law

OPSC AEE 2015 PAPER - II

Ans : (a) An ideal absorber of radiation is also an ideal emitter. it is known as Kirchhoff's law.

Kirchhoff's law:- The emissivity of a body is equal to its absorptivity when the body remains in thermal equilibrium with its surrounding

ε = α (Kirchhoff's law).

684. The roof of a house has been given a coating of shining metal paint. Consequently the temperature inside the room will :

(a) Rise (b) No effect

(c) Cannot be decided (d) Fall

TSPSC Managers, 2015

Ans. (d) : The radiant energy emitted from a non-black body depends upon its emissivity. In general, the emissivity of the shining metallic paint is high. Consequently, the emitted radiant energy will increase and the temperature inside the room will fall.

685. If velocity of water inside a smooth tube is doubled, the turbulent flow heat transfer coefficient between the water and the tube wall

(a) Remains unchanged

(b) Increases to double its value

(c) Increases but will not reach double its value

(d) Increases to more than double its value

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) : Increases but will not double its value

686. How can radiation heat transfer between two surfaces be reduced?

(a) By introducing radiation shield between the surfaces

(b) By bringing surfaces closer together

(c) By polishing surfaces

(d) All options are correct

MPPSC State Forest Service Exam, 2014

Ans. (a) : As we know, the radiation heat transfer between two parallel plates can be reduced by placing obstruction like radiation shield having low thermal conductivity & low absorbitivity.

687. Sun's surface at 6400 K emits radiation at a wavelength of 0.75μ. A furnace at 327^0C will emit through a small opening, radiation at a wavelength of nearly :

(a) 12 μ (b) 8 μ

(c) 0.25 μ (d) 0.025 μ

UPRVUNL AE 2014

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Ans : (b) Form wien's displacement law,

$\lambda_{max} T = C$

$\lambda_1 T_1 = \lambda_2 T_2$

$6400 \times 0.75\mu = 600 \times \lambda_2$

$\lambda_2 = 8\mu$

λ= wavelength

$T_1 = 6400$ K

$T_2 = 327 + 273 = 600$ K

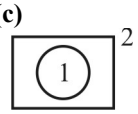
688. A small sphere of outer area 0.36 m^2 is totally enclosed by a large cubical hail. Shape factor of hail with respect to sphere is 0.008. The measure of internal side of cubical hail is :

(a) 4 m (b) 5.2 m

(c) 2.74 m (d) 10 m

UPRVUNL AE 2014

Ans : (c)



$A_1 = 0.36 m^2$

$A_2 = 6.a^2$

$F_{21} = 0.008$

$F_{11} = 0$

$F_{11} + F_{12} = 1$

$\therefore \quad F_{12} = 1$

$A_1 F_{12} = A_2 F_{21}$

$0.36 \times 1 = 6.a^2 \times 0.008$

$a = 2.74m$

689. Two bodies A and B having equal surface area are maintained at temperatures 10°C and 20°C. The energy of thermal radiations emitted in a given time by A and B are in the ratio:

(a) 1:1.15 (b) 1:2.5

(c) 1:4.2 (d) 1:5.6

ISRO Scientist/Engineer 24.05.2014

Ans. (a) : Let thermal radiations A and B are E_A and E_B

$E_A = \varepsilon_1 A_1 \sigma_1 T_1^4$

σ = Stefan constant

$\sigma = 5.67 \times 10^{-8}$ W/m^2-k^4

So, $E_A \propto T^4$

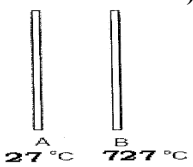
$\frac{E_A}{E_B} = \frac{(T_A)^4}{(T_B)^4}$

$T_A = 10°C = 273 + 10 = 283K$

$T_B = 20°C = 273 + 20 = 293K$

$\frac{E_A}{E_B} = \left(\frac{283}{293}\right)^4 = \frac{1}{1.149}$

690. Surface A in the figure is coated with white paint and maintained at temperature of 27°C. It is located directly opposite to surface B which can be considered a black body and maintained at temperature of 727°C. Calculate the amount of heat that needs to be removed from surface A per unit area to maintain its constant temperature (consider emissivity of the white paint = 1 and view factor = 1, σ = Stefan - Boltzmann constant)



(a) $9919 \times 10^8 \sigma W/m^2$ (b) $9999 \times 10^8 \sigma W/m^2$

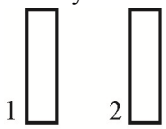
(c) $9990 \times 10^8 \sigma W/m^2$ (d) $9909 \times 10^8 \sigma W/m^2$

ISRO Scientist/Engineer 24.05.2014

Ans. (a) : $T_1 = 27^0$C at Surface 'A'

$T_2 = 727^0$C at Surface 'B'

emissivity $\varepsilon = 1$



$Q_{12} = \sigma A_1 F_{12} \left[T_2^4 - T_1^4\right]$

F_{12} = View factor

$Q_{12} = \sigma \times 1 \times 1 \left(1000^4 - 300^4\right)$

$Q_{12} = 9919 \times 10^{11} \sigma \quad W/m^2$

$= 9919 \times 10^{11} \sigma \quad W/m^2$

691. The temperature of a solid surface is changed for 127°C to 927°C. The emissive power shall increase in the ratio of

(a) 2 (b) 8

(c) 3 (d) 9

(e) 81

CGPSC AE 26.04.2015 SHIFT-II

Ans. (e) : We know that,

$E \propto T^4$

$T_1 = 127+273 = 400K$

$T_2 = 927 +273 = 1200K$

then,

$\frac{E_2}{E_1} = \left[\frac{T_2}{T_1}\right]^4 = \left[\frac{1200}{400}\right]^4 = [3]^4$

$\boxed{E_2 = 81 E_1}$

692. Shape factor of a plane surface about itself is :

(a) 1 (b) zero

(c) 0.5 (d) 0.75

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

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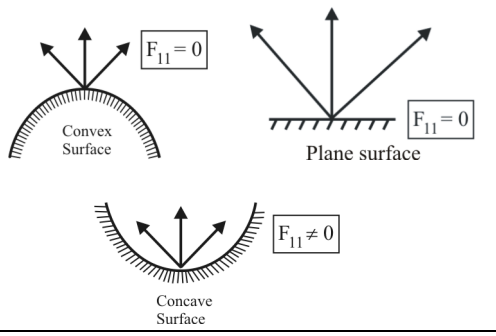
Ans. (b) :



693. The rate at which radiation is emitted per unit area at all possible wavelengths and in all possible directions is the

(a) Total emissive power
(b) Monochromatic emissive power
(c) Radiosity
(d) Irradiation

Kerala LBS Centre For Sci. And Tech. Asstt. Prof. 2014

Ans. (a)

- **Total Emissive Power–** It is defined as a radiation energy emitted/time–surface area in all possible hemispherical direction at all possible wavelength.
- **Monochromatic Emissive Power–** It is defined as a thermal radiation emitted per unit time per unit surface area for a particular wavelength.

694. A body whose reflectivity is unity is called a

(a) White body (b) Black body
(c) Transparent body (d) Gray body

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a)

$$\boxed{\alpha + \rho + \tau = 1}$$

Where,

α = Absorptivity
ρ = Reflectivity
τ = Transmissivity

- For Black body
  $\alpha = 1, \ \rho = 0, \ \tau = 0$
- For Opaque body
  $\tau = 0$
  $\alpha + \rho = 1$
- For White body
  $\alpha = 0, \ \tau = 0$
  $\boxed{\rho = 1}$

695. The value of Stefan Boltzmann constant is

(a) $6.67 \times 10^{-8}\ W/m^2K^4$ (b) $5.67 \times 10^{-8}\ W/m^2K^4$
(c) $6.67 \times 10^{-8}\ W/m^4K^2$ (d) $5.67 \times 10^{-8}\ W/m^4K^2$
(e) $5.67 \times 10^{-8}\ W/mK^4$

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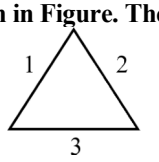
Ans. (b) : The value of Stefan Boltzmann constant (σ) is given as $\sigma = 5.67 \times 10^{-8}\ W/m^2K^4$

696. Consider an enclosure of three non-concave surfaces shown in Figure. The shape factor F_{23} is



(a) $\dfrac{A_1 + A_2 - A_3}{2A_1}$ (b) $\dfrac{A_2 + A_3 - A_1}{2A_2}$
(c) $\dfrac{A_3 + A_1 - A_2}{2A_3}$ (d) $\dfrac{A_1 + A_2 + A_3}{2A_1}$
(e) $\dfrac{A_1 + A_2 + A_3}{2A_2}$

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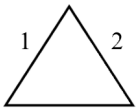
Ans. (b) : The shape factor F_{23} is given as

$$F_{23} = \frac{A_2 + A_3 - A_1}{2A_2}$$



697. A solid cylinder (surface 2) is located at the centre of a hollow sphere (surface 1). The diameter of the sphere is 2 m, while the cylinder has a diameter and length of 1.5 m each. The radiation configuration factor F_{11} is

(a) 0.225 (b) 0.8437
(c) 0.1562 (d) 0.4375
(e) 0.2969

CGPSC AE 26.04.2015 SHIFT-II

Ans. (c) : We know that,

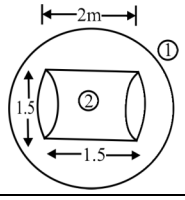
$F_{1-1} + F_{1-2} = 1$

and $A_1F_{1-2} = F_{2-1}\ A_2$ [Reciprocity theorem]

$$F_{1-2} = \frac{A_2}{A_1} = \frac{\left[\pi DL + 2 \times \frac{\pi}{4} D^2\right]_2}{\left[4\pi R^2\right]_1} \qquad F_{2-1} = 1\ ,\ F_{2-2} = 0$$

$$F_{1-2} = \frac{\left[\pi \times 1.5 \times 1.5 + 2 \times \frac{\pi}{4} \times (1.5)^2\right]}{\left[4 \times \pi \times (1)^2\right]} = 0.84375$$

$F_{1-1} = 1 - 0.84375 = 0.15625$



698. The radiation heat transfer through large plates separated by N radiation shields becomes, when the emissivities of all surfaces are equal

(a) Q_{12}, N shields $= \dfrac{1}{N+1} Q_{12}$, no shield

(b) Q_{12}, N shields $= \dfrac{1}{N+1} Q_{12}$

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(c) Q_{12}, N shields = $(N+1)Q_{12}$, no shield

(d) Q_{12}, N shields = $N(N+1)Q_{12}$, no shield

TNPSC AE 2014

Ans. (a) :

$$Q_{1-2 \text{ with N shields}} = \frac{1}{(N+1)} \times Q_{1-2 \text{ without shield}}$$

699. Radiosity (J) for black surface is

(a) Equivalent to emissive power E_b
(b) Greater than emissive power
(c) Less than emissive power
(d) None of the above

TNPSC AE 2014

Ans. (a) : We know that

J = Emitted energy + Reflected part of incident energy.

$J = E + \rho G$

For black body

$\rho = 0$

so, $\boxed{J = E_b}$

700. Consider a surface at -5°C in an environment at 25°C. The maximum rate of heat that can be emitted from this surface by radiation is

(a) 0 W/m^2 (b) 155 W/m^2
(c) 293 W/m^2 (d) 354 W/m^2

TNPSC AE 2014

Ans. (b) : $T_1 = 25^oC = 273 + 25$

$= 298$ K

$T_2 = -5^oC = 268$ K

then $\frac{Q}{A} = \sigma\left(T_1^4 - T_2^4\right)$

$= 5.67 \times 10^{-8} [(298)^4 - (268)^4]$

$\frac{Q}{A} = 154.64 \frac{W}{m^2}$

701. If the temperature of a hot body is increased by 50% the amount of radiation emitted by it would increase by nearly :

(a) 50% (b) 100%
(c) 200% (d) 500%

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) :

702. The sun is emitting radiation with maximum intensity at $\lambda = 0.5$ μm. Determine the surface temperature of the sun :

(a) 5796°K (b) 5796°C
(c) 2898°K (d) 2898°C

TNPSC AE (Industries) 09.06.2013

Ans. (a) :

703. The angular limit to which the incident sun ray may deviate from the normal to the aperture plane and still reach the absorber :

(a) Aperture angle (b) Acceptance angle
(c) Absorber angle (d) Incident angle

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

704. The energy distribution of an ideal reflector at higher temperature is largely in the range of :

(a) shorter wavelength
(b) longer wavelength
(c) remains same at all wavelengths
(d) wavelength has nothing to do with it

J&K PSC Civil Services Pre, 2013

Ans. (a) : The energy distribution of an ideal reflector at higher temperature is largely in the range of shorter wavelength.

705. Stefan-Boltzmann's law is applicable to

(a) gray body (b) white body
(c) black body (d) All of the above

APPSC Poly. Lect. 2013

Ans. (c) : Stefan- Boltzmann Law, state that the total hemispherical emissive power of a black body is directly proportional to the fourth power of the absolute temperature of the black body.

$E_b \propto T^4$

$E_b = \sigma T^4$ watt/m^2

σ = Stefan Boltzmann constant

$= 5.67 \times 10^{-8}$ watt/m^2-K^4

706. Loss of heat from unlagged steam pipe to the ambient air is by

(a) conduction (b) convection
(c) radiation (d) All of the above

APPSC Poly. Lect. 2013

Ans. (d) : Loss of heat from unlagged steam pipe to the ambient air is by- conduction, convection, radiation

707. A smaller sphere (1) of radius 50 mm is inside a larger (2) of radius 100 mm. What will be the value of the radiation shape factor F_{22}.

(a) 3/4 (b) 4/3
(c) 1.0 (d) Zero

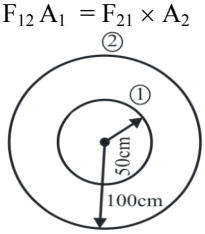
RPSC AEN Pre-2013

Ans. (a) : Given, $F_{12} = 1$

Smaller (1) radius = 50 mm

larger (2) radius = 100 mm

$F_{12} A_1 = F_{21} \times A_2$



$1 \times 4\pi (50)^2 = F_{21} \times 4\pi (100)^2$

$F_{21} = \frac{(50)^2}{(100)^2}$

$F_{21} = \frac{50 \times 50}{100 \times 100} = \frac{1}{4}$

$= 0.25$

As we know that $F_{21} + F_{22} + \ldots\ldots = 1$

$0.25 + F_{22} = 1$

$F_{22} = 1 - 0.25$

$= 0.75$, $\boxed{F_{22} = 0.75}$

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708. The equivalent emissivity of two parallel gray planes, whose emissivities are 0.2 and 0.5 is

(a) 0.1 (b) 0.166
(c) 0.4 (d) 2.5

TNPSC AE 2013

Ans. (b) : We know that interchanging factor for the radiation from surface 1 to surface 2

$$\varepsilon = \frac{\varepsilon_1 \varepsilon_2}{\varepsilon_1 + \varepsilon_2 - \varepsilon_1 \varepsilon_2}$$

$$\varepsilon = \frac{0.2 \times 0.5}{0.2 + 0.5 - 0.5 \times 0.2}$$

$$\boxed{\varepsilon = 0.166}$$

709. The degradation of plastics accelerated by

(a) Ultraviolet radiation
(b) Dampness
(c) Corrosive atmosphere
(d) None of these

SJVN ET 2013

Ans. (a) : The degradation of plastics is accelerated by ultraviolet radiation.

710. The process in which heat energy is transmitted by means of electromagnetic waves is known as:-

(a) Heat conduction (b) Heat convection
(c) Heat radiation (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) : The process in which heat energy is transmitted by means of electromagnetic waves is known as heat radiation.

711. Three radiation shields are placed between two infinite parallel plates. The emissivities of plates and shields are same. As compared to heat transfer without shields, the heat transfer with shield will become:-

(a) $\frac{1}{3}$ (b) $\frac{1}{9}$
(c) $\frac{1}{4}$ (d) None of the above

UKPSC AE-2013, Paper-II

Ans. (c) : $Q_{\text{with radiation shields}} = \frac{1}{(n+1)} Q_{\text{without radiation shields}}$

$$= \frac{1}{(3+1)} Q_{\text{without radiation shields}}$$

$$\frac{Q_{\text{with radiation shields}}}{Q_{\text{without radiation shields}}} = \frac{1}{4}$$

712. Most of the terrestrial solar radiations (received on the earth) lie within wavelength range:-

(a) 0.10μm to 0.29μm (b) 0.29μm to 2.5μm
(c) 3.8μm to 7.8μm (d) 10^2μm to 10^{10}μm

UKPSC AE-2013, Paper-II

Ans. (b) :

713. Flat plate solar collectors are used for temperature applications above ambient of about:-

(a) 20°C (b) 50°C
(c) 100°C (d) 1000°C

UKPSC AE-2013, Paper-II

Ans. (c) :

714. Assuming the Sun to be a black body emitting radiation with maximum intensity at , the surface temperature of the sun will be:-

(a) 491.4 K (b) 4914 K
(c) 49140 K (d) 491.4° C

UKPSC AE-2013, Paper-II

Ans. (b) :

715. A hot body will radiate heat most rapidly if its surface is

(a) white and polished
(b) grey
(c) black and rough
(d) black and polished

APPSC AEE 2012

Ans. (d) : A hot body will radiate heat most rapidly if its surface is black and polished.

716. If one radiation shield is placed between two infinite parallel radiating plane surfaces, then the amount of heat radiated becomes

(a) one third (b) one fourth
(c) half (d) none of the above

UKPSC AE 2012 Paper–II

Ans. (c) : half

717. A black plate (0.5 m × 1 m) is at a distance of 0.5 m from a second black plate, (1m × 2m). The shape factor F_{12} of the first plate is 0.48. What is the shape factor F_{21} of the second plate?

(a) 0.24 (b) 0.12
(c) 0.06 (d) 0.36

APGENCO AE 2012

Ans. (b) : Given, $A_1 = 0.5 \times 1$, $F_{12} = 0.48$

$A_2 = 1 \times 2$, $F_{21} = ?$

$F_{12} \times A_1 = F_{21} \times A_2$

$0.48 \times 0.5 = F_{21} \times 2$

$$F_{21} = \frac{0.240}{2} = 0.12$$

718. Two very large parallel plates are at 127°C and 27°C. Their emissivities are 0.4 and 0.5 respectively, Stefan-Boltzmann constant is 5.669×10^{-8} W/m^2-K^4 . Then, the net heat exchanged between them, in W/m^2, is

(a) 243 (b) 273
(c) 283 (d) 343

APGENCO AE 2012

Ans. (c) : Given $T_1 = 127°C$, $T_2 = 27°C$

$T_1 = 400k$, $T_2 = 300k$

$\sigma = 5.669 \times 10^{-8}$ W/m^2-k^4

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For parallel plate– $\epsilon_1 = 0.4, \quad \epsilon_2 = 0.5$

$$Q = \frac{\sigma_A (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$

$$\frac{Q}{A} = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} \Rightarrow \frac{5.669 \times 10^{-8}[(400)^4 - (300)^4]}{\frac{1}{0.4} + \frac{1}{0.5} - 1}$$

$$\frac{Q}{A} = \frac{5.669 \times 10^{-8} \times 10^{+8}[256 - 81]}{2.5 + 2 - 1}$$

$$= \frac{992.075}{3.5}$$

$$\frac{Q}{A} = 283.45 W/m^2$$

719. A thermal transparent body is characterised by

(a) absorptivity = 1
(b) reflectivity =1
(c) absorptivity = reflectivity =0
(d) none of the above

UKPSC AE 2012 Paper–II

Ans. (c) : absorptivity = reflectivity =0

720. The shape factor for radiation heat transfer of a long cylinder of radius r_1 enclosed by another concentric long cylinder of radius r_2 is

(a) 0.25 (b) 0.50
(c) 0.75 (d) 1.0

UKPSC AE 2012 Paper–II

Ans. (d) : Shape factor for radiation heat transfer of a long cylinder of radius r_1 enclosed by another concentric long cylinder of radius r_2

$\boxed{F_{12} = 1}$

721. _____ will radiate heat to a large extent.

(a) Black polished surface
(b) White rough surface
(c) White polished surface
(d) Black rough surface

UKPSC AE 2012 Paper–II

Ans. (c) : White polished surface

722. Which pair, out of the following alternatives, is not correctly matched ?

	List – I		List – II
(a)	Fourier's law	–	Conduction
(b)	Newton's law of cooling	–	Convection
(c)	Stefan-Boltz-man law	–	Radiation
(d)	Kirchoff's law	–	Radiation +Convection

UKPSC AE 2012 Paper–II

Ans. (d) : Kirchoff's law – Radiation +Convection

723. When the temperature of a solid surface changes from 227ºC to 1227 ºC, its total emissive power changes from E1 to E2. The ratio $\left(\frac{E_2}{E_1}\right)$ will be

(a) 3
(b) 9
(c) 81
(d) Cannot be determined on the basis of information provided

ISRO Scientist/Engineer 2010

Ans. (c) : $T_1 = 227°C = 273 + 227 = 500$ K

$T_2 = 1227$ °C $= 1227 + 273 = 1500$ K

$$E_1 = \sigma T_1^4,$$

$$E_2 = \sigma T_2^4$$

$$\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4$$

$$= \left(\frac{1500}{500}\right)^4$$

$= 81$

724. The absorptivity of a freshly white washed wall is close to :

(a) 0.1 (b) 0.3
(c) 0.5 (d) 0.9

J&K PSC Civil Services Pre, 2010

Ans. (d) : The absorptivity of a freshly white washed wall is close to 0.9.

725. A copper ball at 227°C cools by radiation to ambient atmosphere at 27°C. When the body has cooled down to 127°C, what will be the decreasing cooling rate as percentage of original cooling rate?

(a) 90.2 (b) 67.8
(c) 59 (d) 11

UPSC JWM Advt. No.-50/2010

Ans. (b) : Given,

$T_1 = 227°C + 273 = 500$ K

$T_0 = 27°C + 273 = 300$ K

$T_2 = 127°C + 273 = 400$ K

$$\frac{\dot{Q}_1}{\dot{Q}_2} = \frac{T_1^4 - T_0^4}{T_2^4 - T_0^4}$$

$$\frac{\dot{Q}_1}{\dot{Q}_2} = \frac{500^4 - 300^4}{400^4 - 300^4}$$

$$\frac{\dot{Q}_1}{\dot{Q}_2} = \frac{625 - 81}{256 - 81}$$

$$\frac{\dot{Q}_1}{\dot{Q}_2} = \frac{544}{175}$$

$$\frac{\dot{Q}_2}{\dot{Q}_1} = \frac{175}{544}$$

So, decrease in cooling rate

$$= \left(\frac{\dot{Q}_1 - \dot{Q}_2}{\dot{Q}_1}\right) \times 100 = 67.8\%$$

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726. A body at 500 K cools down to 400 K by radiation to atmosphere at 300 K. What will be the ratio of heat loss rate at 500 K to the heat loss rate at 400 K?

(a) 1.25 (b) 2.00
(c) 2.44 (d) 3.11

UPSC JWM Advt. No.-50/2010

Ans. (d) : As we know,

$$\Rightarrow \frac{\text{The rate of heat loss at 500K}}{\text{The rate heat loss at 400K}}$$

$$= \frac{500^4 - 300^4}{400^4 - 300^4} = \frac{544}{175} = 3.11$$

727. A man sits on the floor by a fire burning at some distance at the same level. The mode of heat transfer mainly responsible for the man receiving heat is

(a) conduction (b) convection
(c) radiation (d) advection

DRDO Scientists 2008

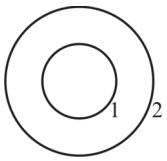
Ans. (c) :

728. A spherical body with surface A_1 is completely enclosed by another hollow body with inner surface A_2. The shape factor of A_2 with respect to A_2 is

(a) 0.0 (b) 0.5
(c) 1.0 (d) π

DRDO Scientists 2008

Ans. (b) :



$$A_1 F_{12} = A_2 F_{21}$$

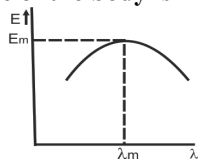
$$F_{21} = \frac{A_1}{A_2} F_{12} \qquad (\because F_{12} = 1)$$

$$F_{21} = \frac{A_1}{A_2}$$

As $A_2 > A_1$, $F_{21} < 1$ from the given option (b) is only less than one.

F_{21} can not be zero as this will lead to trivial solution.

729. The spectral energy distribution for a body at temperature T is shown in figure. Now if the temperature of the body is increased



(a) λ_m Will shift towards right
(b) λ_m Will shift towards left
(c) The total energy emitted by the body will decrease
(d) λ_m will remain the same but E_m will increase

ISRO Scientist/Engineer 2008

Ans. (b) : $\lambda_{max} T_{max} = 2898\ \mu mK$

730. A radiation shield is used around thermocouples in order to measure more accurately the temperature of

(a) Solid (b) Gases
(c) Freezing liquid (d) Boiling liquid

UKPSC AE 2007 Paper -II

Ans. (b) : A thermocouple is a sensor used to measure temperature. Thermocouple consist of two wire legs made from different metals. The wire's leg are welded together at one end, creating a junction. This junction is where the temperature is measured.
A radiation shield is used around thermocouples in order to measure more accurately the temperature of gases, because its volume is not fixed or its molecules are far away.

731. Planck's law holds good for

(a) Polished bodies (b) Black bodies
(c) All coloured bodies (d) None of above

UKPSC AE 2007 Paper -II

Ans. (b) : Black bodies

732. Which mode of heat transfer plays insignificant role in a cooling tower?

(a) Radiation (b) Evaporative cooling
(c) Convective cooling (d) All the above

UKPSC AE 2007 Paper -II

Ans. (a) : A cooling tower is a heat rejection device that rejects waste heat to the atmosphere through the cooling of a water stream to a lower temperature.
* In cooling tower, mode of heat transfer is radiation which plays insignificant rate.

733. A thin metal plate is insulated on the back and exposed to solar radiation at the front surface. The exposed surface of the plate has an absorptivity of 0.6 for solar radiation. If solar radiation is incident on the plate at a rate of 700 W/m^2 and the surrounding air temperature is 25°C, determine the surface temperature of the plate when the heat loss by convection and radiation equals the solar energy absorbed by the plate. Assume the combined convection and radiation heat transfer coefficient to be 50 W/m^2°C.

(a) 42.7°C (b) 38.9°C
(c) 33.4°C (d) 21.8°C
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : As we know,

$$\dot{q}_{absorption} = \dot{q}_{convection}$$

$$\alpha \cdot \dot{q}_{solar} = h(T_s - T_a)$$

Hence the plate surface temp is

$$T_s = T_a + \frac{\alpha \dot{q}_{solar}}{h}$$

$$= 25 + \frac{0.6 \times 700}{50}$$

$$T_s = 33.4°C$$

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734. Which of the following is FALSE?

(a) Heat flow by conduction is directly proportional to the area and temperature gradient

(b) Heat flow by convection is directly proportional to the area and temperature difference

(c) Heat flow by radiation is directly proportional to the cube of absolute temperature of the radiating body

(d) Emissivity of perfect black body is unity

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Heat flow by radiation is directly proportional to the fourth power of absolute temperature of radiating body.

4. Boiling and Condensation

735. Dropwise condensation occurs on the following surface:

(a) Oily (b) Smooth

(c) Glazed (d) Coated

JPSC AE, 10.04.2021, Paper-I
TRB Poly. Lect., 2012
UKPSC AE 2007 Paper -II
ESE 1992

Ans. (a) : Dropwise condensation is a surface condensation in which 90% of surface is covered by drops. It occurs on the oily surface.

* Droplet formation is superior to film formation in terms of maintaining high condensation and heat transfer rates.

736. In pool boiling, the highest heat transfer coefficient (HTC) occurs in:

(a) Film boiling zone

(b) Partial film boiling zone

(c) Nucleate boiling zone

(d) Sub-cooled boiling zone

(e) Both film boiling zone and partial film boiling zone

GPSC Executive Engineer 23.12.2018
TSPSC AEE 2017
CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : Nucleate Boiling region, is the most important boiling region, for technical applications. The temperature difference required to enter this region is approximately 3 K. Superheated liquid overcomes the surface fusion, forming instable. Vapour bubbles that collapse, in contact with sub-cooled refrigerant liquid.

737. Which of the following is not the regimes of pool boiling?

(a) Natural convection boiling

(b) Nucleate boiling

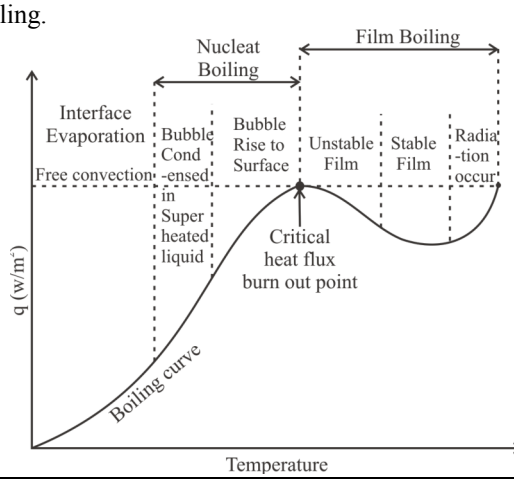
(c) Film boiling

(d) Flow boiling

(e) Transition boiling

CGPSC AE 25.02.2018

Ans. (d) : Flow boiling is not the regimes of pool boiling.



738. Consider the following statements for Nucleate boiling:

1. For water, the critical heat flux does not exceed 1 MW/m^2.

2. Nucleate boiling is the most desirable boiling regime in practice because of high heat transfer rates.

3. Heat flux increases at a higher rate with increase in temperature.

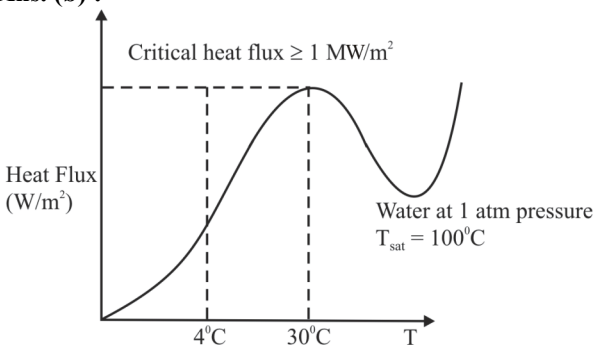
Which of the above statements is/are correct?

(a) 1 Only (b) 2 Only

(c) 1 and 3 only (d) 2 and 3 only

ESE 18.07.2021

Ans. (b) :



- For water, critical heat flux > 1 MW/m^2.
- From heat transfer point of view, nucleate boiling regime is the most desirable range to operate.
- Heat transfer rate increase at a lower rate as ΔT_e increases.

739. In drop-wise condensation, the heat transfer rate is

(a) 5 times less than that in film-wise condensation.

(b) 15 times less than that in film-wise condensation

(c) 25 times more than that in film-wise condensation.

(d) 10 times more than that in film-wise condensation.

ESE 18.07.2021

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Ans. (d) : In drop-wise condensations, the vapours condense on the surface on drops, which drips down the surface. A continuous films of liquid is not formed on the surface. Thus, more of the base area at temperature T_s is always exposed to the vapours. Therefore, heat transfer rate is higher (up to 10 times) in drop-wise condensation as compared to the value in film-wise condensation.

In film wise condensation, the sliding film will act as barrier to the heat transfer offering thermal resistance between vapour and wall thereby reducing the heat transfer rate as compared to that in drop wise condensation.

740. Which are the correct statements ?

(1) Film condensation usually occur on oily surface

(2) Temperature of condensate gradually increases towards the liquid vapour interface.

(3) Suitable coating is used to promote dropwise condensation.

(4) Dropwise condensation gives lower heat transfer rate than film condensation.

(a) (1), (4) (b) (3), (4)

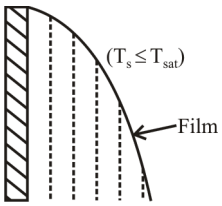
(c) (1), (2) (d) (2), (3)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : Surface Condensation–

(1) Film Wise Condensation–The liquid condensate wets the surface, spreads out and forms a continuous film over the entire surface.

Film condensation usually occurs when a vapour relatively free from impurities, is allowed to condense on a clean surface.



(2) Drop-Wise Condensation–This type of condensation occurs when the liquid doesnot wet the surface and droplets of liquid is formed in random fashion on the surface.

- Heat transfer coefficient of drop wise condensation is more as compared to film wise condensation.
- Coatings like silicon, teflon etc are used to promote droplet formation.

741. Large heat transfer coefficients for vapour condensation can be achieved by promoting

(a) Film condensation

(b) Dropwise condensation

(c) Cloud condensation

(d) Dew condensation

ESE 2020

Ans. (b) : Dropwise condensation.

742. Drop wise condensation is the liquid condensate collects in

(a) Droplets and does not wet the solid

(b) Droplets and wet solid surface

(c) Droplets do not wet and spreads out

(d) Droplets do not wet and do not spreads out

TNCSC AE 2020

Ans. (a) : Drop wise condensation is the liquid condensate collects in droplets and does not wet the solid. For non-metal vapours, drop wise condensation clean metal surfaces are wetted by non-metallic liquids and film condensation is the mode which normally occurs in practice.

743. A liquid is boiling in a airtight vessel. Using an exhaust tube the vapour is pumped out at a faster rate. What will happen to the liquid?

(a) Temperature will go down but boiling will continue

(b) Temperature will rise and boiling will continue

(c) Boiling will stop

(d) Vapour starts condensing

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) : Temperature will go down boiling will continue.

744. In transition boiling heat flux decreases due to which of the following?

1. Low value of film heat transfer coefficient at the surface during 100°C to 120°C surface temperature
2. Major portion of heater surface is covered by vapour film which has smaller thermal conductivity as compared to liquid
3. Nucleate boiling occurs very fast

Select the correct answer using the code given below.

(a) 1 only (b) 2 only

(c) 3 only (d) 1, 2 and 3

ESE 2019

Ans. (b) : The heat flux decreases in the transition zone of boiling because a large fraction of the heat surface is covered by a vapour film, which acts as an insulation due to the low thermal conductivity of the vapour relative to that of the liquid.

745. Consider the following statements:
For the laminar condensation on a vertical plate, the Nusselt theory says that

1. Inertia force in the film is negligible compared to viscosity and weight
2. Heat flow is mainly by conduction through the liquid film, convection in liquid film as well as in vapour is neglected
3. Velocity of vapour is very high

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Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

ESE 2019

Ans. (b) : Nusselt theory for the vertical plate

(i) The acceleration of condensate layer is negligible.
(ii) Heat transfer across the liquid film is by pure conduction (no convection current and in the liquid film and vapour)
(iii) The velocity of vapour is low so that it exerts no drag on the condensate (no viscous shear on the liquid vapour interface)

746. Nucleate boiling is promoted

(a) on polished surface
(b) on rough surfaces
(c) in the absence of agitation
(d) none of these

RPSC Vice Principal ITI 2018

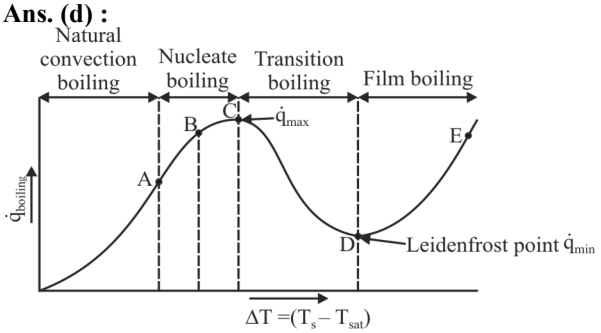
Ans. (b) : Nucleate boiling is a type of boiling that takes place when the surface temperature is hotter than the saturated fluid temperature by a certain amount. The heat transfer from surface to liquid is greater than that in film boiling.

747. Which one of the following regimes of boiling curve can be considered as reverse of condensation?

(a) Free convection boiling regime
(b) Nucleate boiling regime
(c) Transition boiling regime
(d) Film boiling regime

ESE 2018

Ans. (d) :



(i) Natural Convection Boiling (to point A on the boiling curve)–We do not see any bubbles forming on the heating surface.

(ii) Nucleate Boiling (Between Point A and C)–The first bubbles start forming at point A of the boiling curve. The nucleate boiling regime can be separated into two distinct regions i.e. AB and BC.

(iii) Transition Boiling (Between Point C and D)–As the ΔT is increases past point C, the heat flux decreases.

(iv) Film Boiling (Beyond Point D)–In this region the heater surface is completely covered by a continuous stable vapour film.

■ Boiling take heat and condensation release heat. Boiling and condensation is opposite phenomenon. The nucleate boiling exist up to $\Delta T \simeq 40°C$ while film boiling temperature is greater than nucleate boiling.

So for film regime, condensation and boiling can be considered as reverse phenomenon.

748. Which of the following is true?

(a) Heat transfer coefficient in dropwise condensation is very low in comparison to filmwise condensation
(b) Filmwise condensation is preferred over dropwise condensation on heat transfer surface
(c) Dropwise condensation required very high area of heat exchanger relative to filmwise condensation
(d) Dropwise condensation cannot be easily sustained for prolonged period of time

UPRVUNL AE 2016

Ans. (d) : Dropwise condensation cannot be easily sustained for prolonged period of time.

749. Nukiyama's Boiling curve is plotted between

(a) Boiling temperature vs excess temperature
(b) Boiling heat flux vs boiling temperature
(c) Boiling temperature vs boiling pressure
(d) Boiling heat flux vs excess temperature

RPSC LECTURER 16.01.2016

Ans. (d) : Nukiyama's Boiling curve is plotted between boiling heat flux vs excess temperature.

750. Nucleate boiling regime is formed approximately between (ΔT excess = excess temperature)

(a) $5°C \le \Delta T_{excess} \le 10°C$
(b) $50°C \le \Delta T_{excess} \le 80°C$
(c) $80°C \le \Delta T_{excess} \le 100°C$
(d) $5°C \le \Delta T_{excess} \le 50°C$

UPRVUNL AE 2016

Ans. (d) : When ΔT excess equal or greater than 5°C and equal or less than 50°C then Nucleate boiling regime is formed

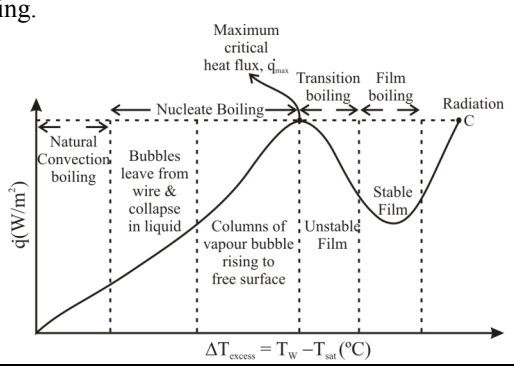
$$5°C \le \Delta T_{excess} \le 50°C$$

751. Which of the following is not the regimes of pool boiling?

(a) Film boiling regime
(b) Nucleate boiling regime
(c) Slug flow regime
(d) Natural convection boiling regime

UPRVUNL AE 2016

Ans. (c) : Slug flow regime is not the regimes of pool boiling.



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752. Consider the following phenomena

1. Boiling
2. Free convection in air
3. Forced convection
4. Conduction in air

Their correct sequence in the increasing order of heat transfer is

(a) 4, 2,3,1 (b) 4,1,3,2
(c) 4,3,2,1 (d) 3,4,1,2
(e) 4,2,1,3

CGPSC AE 26.04.2015 SHIFT-II

Ans. (a) : Current sequence in the increasing order of heat transfer is
Conduction in air < Free convection < forced convection < Boiling

753. Heat transfer rate is high in

(a) film boiling (b) nucleate boiling
(c) both (1) and (2) (d) simple convection

APPSC Poly. Lect. 2013

Ans. (b) : Heat transfer rate is high in nucleate boiling.

754. If the bubbles formed on a submerged hot surface get absorbed in the mass of liquid, the process of boiling is called

(a) nucleated boiling (b) film boiling
(c) pool boiling (d) normal boiling

VIZAG STEEL MT 18.08.2013

Ans. (c) : Pool boiling– Pool boiling is defined as boiling form a heated surface submerged in a large volume of stagnant liquid. The liquid may be at its boiling is employed or below its boiling point, when the term sub-cooled pool boiling is used.

755. The heat transfer coefficient in laminar film condensation on a vertical plate is maximum at:

(a) Upper edge of the plate
(b) Lower edge of the plate
(c) Middle of the plate
(d) It remains constant over the plate.

RPSC AEN Pre-2013

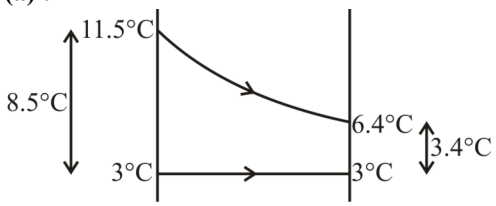
Ans. (a) : The heat transfer coefficient in laminar film condensation on a vertical plate is maximum at upper edge of the plate.

756. A fluid evaporates at 3°C and cools water from 11.5°C to 6.4°C. What is the heat transfer if it has a surface area of 420 m^2 and the thermal transmittance is 110 W/m^2 k?

(a) 257 kW (b) 324 kW
(c) 382 kW (d) 427 kW
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) :



$$LMTD = \frac{8.5 - 3.4}{\ln\left(\frac{8.5}{3.4}\right)} = 5.566°C$$

$Q = U \times A \times \Delta T$
$= 110 \times 420 \times 5.566$
$= 257149.2$ W
$\boxed{Q = 257.1\,kW}$

5. Heat Exchanger

757. The automobile radiator is heat exchanger of-

(a) counter flow type
(b) cross flow type
(c) none of three mentioned here.
(d) parallel flow type

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
Sikkim PSC (Under Secretary), 2017
GMBAAE 25.06.2017
GWSSB AAE, 27.12.2015
TRB Asstt. Prof., 2012
ISRO Scientist/Engineer 2008
UKPSC AE 2007 Paper-II

Ans. (b) : The automobile radiator is cross flow type heat exchanger in which atmospheric air flow out side the shell and water flow inside the tubes.

758. A cross flow type air heater has an area of 50 m^2, The overall heat transfer coefficient is 100 W/m^2K and heat capacity of both hot and cold stream is 1000 W/m^2K. The value of NTU is

(a) 1000 (b) 500
(c) 5 (d) 0.2

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Gujarat PSC AE 2019
GPSC ARTO 01.05.2016
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
UKPSC AE 2012 Paper–II, ESE 2016, 1999

Ans. (c) : Given-
$A = 50\ m^2$
$U = 100\ W/m^2K$
$C_h = C_c = 1000\ W/m^2K$
$C_{min} = C_{max}$

$$NTU = \frac{UA}{C_{min}} = \frac{100 \times 50}{1000}$$

$NTU = 5$

759. In a heat exchanger, the hot liquid enters with a temperature of 180ºC and leaves of 160ºC. The cooling fluid enters at 30ºC and leaves at 110ºC. The capacity ratio of the heat exchanger is

(a) 0.25 (b) 0.40
(c) 0.50 (d) 0.55

TNPSC AE 2018, Assam PSC CCE Pre 2015
Mizoram PSC 2014 Paper-I
ISRO Scientist/Engineer 12.05.2013, ESE 2010

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Ans. (a) : Capacity Ratio $= \frac{C_{min}}{C_{max}}$

Balance equation heat in heat exchanger

C_{min} (110 - 30) = C_{max} (180 - 160)

or $\frac{C_{min}}{C_{max}} = C = \frac{20}{80} = \frac{1}{4}$

C = 0.25

760. Fouling factor is used

(a) In heat exchanger design as a safety factor
(b) In case of Newtonian fluids
(c) When a liquid exchanges heat with a gas
(d) None of the above

OPSC AEE 2019 PAPER - II
ISRO Scientist/Engineer (RAC) 22.04.2018
APPSC AEE Screening Test 2016
GPSC Lect. (Poly) 07.12.2014
UKPSC AE-2013, Paper-II

Ans. (a) : Fouling factor represents the theoretical resistance to heat flow due to a build up of a layer of dirt or other fouling substance on the tube surface of the heat exchanger.

761. The mathematical expression of NTU in a heat exchanger is,
U = Overall heat transfer coefficient
C = Heat capacity
E = Effectiveness
A = Area of heat exchanger

(a) UA/C_{min} (b) UA/C_{max}
(c) UA/E (d) $\frac{UC_{max}}{C_{min}}$

VIZAG MT, 14.12.2020, TNPSC AE 2019
Nagaland PSC (CTSE) 2016 Paper-II
APPSC AEE Screening Test-2016
RPSC ADE 2016, TSPSC AEE 2015
TNPSC AE 2013
TNPSC AE (Industries) 09.06.2013
APGENCO AE 2012, TNPSC AE, 2008
ESE 2008

Ans. (a) : The mathematical expression of NTU in a heat exchanger is given as $NTU = \frac{UA}{C_{min}}$

Since, number of transfer units (NTU) is directly proportional to the area. It indicates the overall size or bulkiness of H.E. usely NTU values are practically 1.115

1.2, 1.5, 2, 2.5, 4

NTU $\not>$ 18

762. The equation of effectiveness $\varepsilon = 1 - e^{-NTU}$ of heat exchanger is valid (NTU) is number of transfer units) in the case of-

(a) boiler and condenser for parallel flow
(b) boiler and condenser for counter flow
(c) boiler and condenser for both parallel and counter flow
(d) gas turbine for both parallel and counter flow

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
TNPSC AE 2019, RPSC AE 2018
RPSC IOF & Boiler 20.09.2017
ESE 2006, 2000

Ans. (c) : The equation of effectiveness for parallel flow heat exchanger

$$\varepsilon_{parallel} = \frac{1 - e^{-(1+C)NTU}}{1 + C}$$

and

$$\varepsilon_{counter} = \frac{1 - e^{-(1-C)NTU}}{1 - C \times e^{-(1-C)NTU}}$$

If one of the fluid in heat exchanger is undergoing change of phase like in steam condenser or evaporator and boiler then C = 0.

So,

$\varepsilon_{parallel} = \varepsilon_{counter}$
$= 1 - e^{-NTU}$

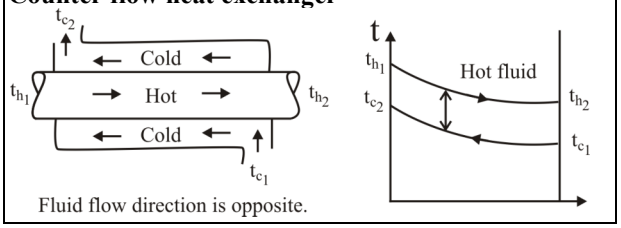
763. LMTD in case of counter flow heat exchanger as compared to parallel flow heat exchanger is

(a) higher
(b) lower
(c) same
(d) cannot be predicted

MECON MT 2019
ISRO Scientist/Engineer (RAC) 07.05.2017
Nagaland PSC CTSE 2017 Paper-2
CGPSC Poly. Lect. 22.05.2016
Nagaland PSC CTSE 2015 Paper-II
UKPSC AE-2013, Paper-II
TNPSC AE 2013
J&K PSC Civil Services Pre, 2010
UKPSC AE 2007 Paper -II
J & K Screening 2006

Ans. (a) : LMTD in case of case of counter flow heat exchanger as compared to parallel flow heat exchanger is higher so counter flow heat exchanger is more effective and more useful.

Counter flow heat exchanger -



764. In a certain Heat exchanger, both fluids have identical mass flow rate – sp. heat product. The hotfluid enters at 76°C and leaves at 47°C and cold fluid enters at 26°C and leaves at 55°C. The effectiveness of heat exchanger is :

(a) 0.16 (b) 0.58
(c) 0.72 (d) 1.0

UPPSC AE 13.12.2020, Paper-2
Kerala PSC Poly. Lect., 2017
RPSC ADE 2016
UKPSC AE 2012 Paper–II
GATE 1997

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Ans. (b) : Given $h_i = 76°C$ $h_o = 47°C$
$C_i = 26°C$ $C_o = 55°C$

Effectiveness, $\varepsilon = \frac{h_i - h_o}{h_i - c_i} = \frac{76-47}{76-26} = \frac{29}{50}$

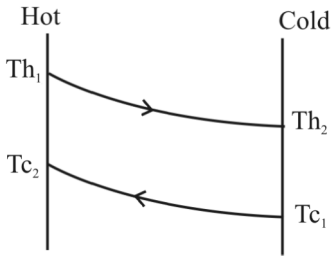
$\varepsilon = 0.58$

765. Which one of the following heat exchangers gives parallel straight line pattern of temperature distribution for both cold and hot fluids?

(a) Parallel flow with unequal heat capacities.
(b) Counter flow with equal heat capacities.
(c) Parallel flow with equal heat capacities.
(d) Counter flow with unequal heat capacities

CGPSC AE 15.01.2021, RPSC IOF, 2020
Punjab PSC (Lect.) 06.08.2017
RPSC ADE 2016, CGPSC AE 16.10.2016
RPSC AE GWD, 2011

Ans. (b) :



Counter flow with equal heat capacities

$$T_{h_1} - T_{c_2} = T_{h_2} - T_{c_1}$$

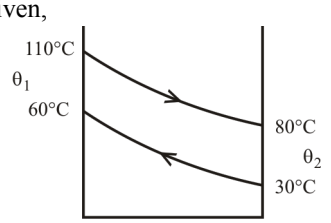
Parallel straight line pattern of temperature distribution for both hot and cold fluid is given counter flow with equal heat capacities.

766. In a counter flow heat exchanger, the hot fluid is cooled from 110°C to 80°C by a cold fluid which gets heated from 30°C to 60°C. LMTD for heat exchanger is :

(a) 50°C (b) 30°C
(c) 20°C (d) 80°C

CGPSC AE 15.01.2021, CGPSC AE 16.10.2016
HPPSC W.S. Poly. 2016
UKPSC AE-2013, Paper-II, ESE 2001

Ans. (a) : Given,



$T_{h_i} = 110°C$

$T_{h_e} = 80°C$

$T_{c_i} = 30°C$

$T_{c_e} = 60°C$

Inlet temp difference $\theta_1 = 110° - 60° = 50°C$
Outlet temp difference $\theta_2 = 80° - 30° = 50°C$

$\boxed{LMTD = 50°C}$

This is a case of balanced counter flow heat exchange.

767. In a counter flow heat exchanger, the product of specific heat and mass flow rate is the same for hot and cold fluids. If NTU is equal to 0.5, the effectiveness of the heat exchanger is

(a) 1.0 (b) 0.5
(c) 0.33 (d) 0.2

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
UPPSC AE 13.12.2020, Paper-2
Assam PSC AE (PHED), 18.10.2020
Oil India Ltd. Sr. Engg. (Production) 30.11.2019
TRB Asst. Prof. 2012
APPSC IOF, 2009, ESE 2001

Ans. (c) : $m_c(C_p)_c = m_h(C_p)_h$

For balanced counter flow heat exchanger

$$\varepsilon = \frac{NTU}{1+NTU} = \frac{0.5}{1+0.5} = \frac{0.5}{1.5} = 0.33$$

$\varepsilon = 0.33$

768. For evaporators and condensers, for the given condition, the logarithmic mean temperature difference (LMDT) of parallel flow compared to counter flow is

(a) more
(b) equal
(c) less
(d) may be equal or more

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Gujrat PSC AE 2019
GPSC Executive Engineer 23.12.2018
TNPSC AE 2017
Mizoram PSC AE/SDO 2014, Paper-I
ESE 1993

Ans. (b) : For evaporators and condensers, for the given condition, the logarithmic mean temperature difference (LMTD) of parallel flow compared to counter flow is equal.

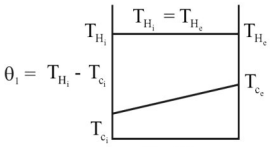


fig (1)

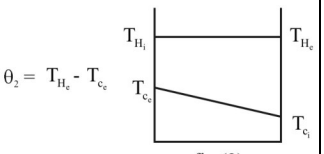


fig (2)

So, LMTD for parallel

$$LMTD = \frac{\theta_1 - \theta_2}{ln\left(\frac{\theta_1}{\theta_2}\right)}$$

for counter flow.

same for counter flow

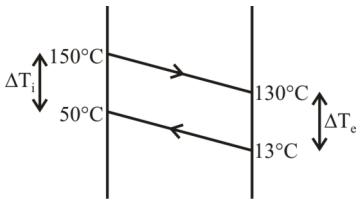
769. In a counter flow heat exchanger, cold fluid enters at 30 °C and leaves at 50 °C, whereas the hot fluid enters at 150 °C and leaves at 130 °C. The log mean temperature difference for this case is

Search on Telegram @apna_pdf

(a) 20 °C (b) 80 °C
(c) 100 °C (d) 120 °C

GPSC DEE Class-2 (GWSSB) 04.07.2021
ISRO Scientist/Engineer (RAC) 12.01.2020
RPSC Vice Principal ITI 2018
GPSC Executive Engg. 23.12.2018
Nagaland PSC CTSE 2015 Paper-II
TSPSC Managers, 2015, ESE 1994

Ans. (c) :



For counter flow heat exchanger,

If $\boxed{\Delta T_i = \Delta T_e}$

$100 = 100$

Then, $\boxed{(LMTD)_{counter} = \Delta T_i = \Delta T_e}$

So, $(LMTD)_{counter} = 100°C$

770. For the same terminal temperatures of the fluids, LMTD:

(a) of a recuperative heat exchanger has the maximum value

(b) of a counter flow heat exchanger is equal to that for a parallel flow heat exchanger

(c) of a counter flow heat exchanger is less than that for a parallel flow heat exchanger

(d) of a counter flow heat exchanger is more than that for a parallel flow heat exchanger

NLCIL GET 17.11.2020, Shift-II
CGPSC AE 26.04.2015 SHIFT-II
ISRO Scientist/Engineer 12.05.2013
GATE 2002

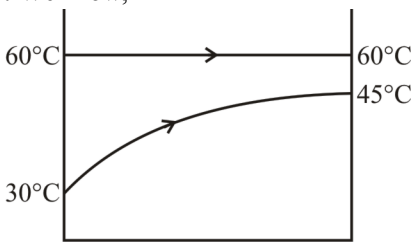
Ans. (d) : For the same terminal temperatures of the fluids logarithmic mean temperature difference of a counter flow heat exchanger is more than that for a parallel flow heat exchanger.

771. In a condenser of a power plant, the steam condenses at a temperature of 60° C The cooling water enters at 30° C and leaves at 45° C. The logarithmic mean temperature difference (LMTD) of the condenser is

(a) 16.2° C (b) 21.6° C
(c) 30° C (d) 37.5° C

ISRO Scientist/Engineer (RAC) 22.04.2018
GPSC Asstt. Director of Transport 05.03.2017
HPPSC Asstt. Prof. 18.11.2016
UPPSC AE 12.04.2016 Paper-II
GATE 2011

Ans. (b) : We know,



$\Delta T_1 = 30°C$

$\Delta T_2 = 15°C$

$$LMTD = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} = \frac{30-15}{\ln\left(\frac{30}{15}\right)} = \frac{15}{\ln(2)} = \frac{15}{0.693}$$

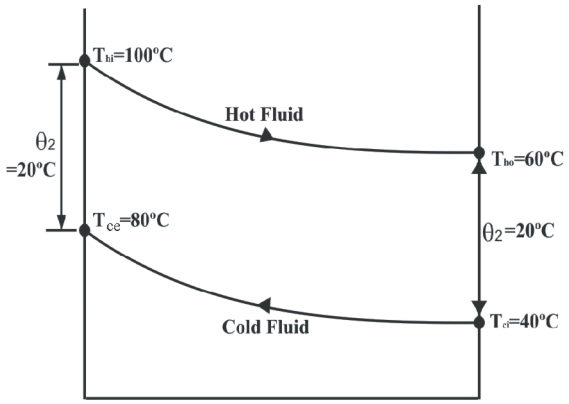
$LMTD = 21.6°C$

772. In a Counter flow heat exchanger design, fluid rates and specific heats were chosen in such a manner that heat capacities of both the fluids are same. A hot fluid enters at 100ºC and leaves at 60ºC. The cold fluid enters that exchanger at 40ºC. The mean temperature difference between the two fluids is

(a) 40ºC (b) 60ºC
(c) 20ºC (d) 36.6ºC

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II
ISRO Scientist/Engineer (RAC) 22.04.2018
Vadodara Municiapal Corp. DEE 2018

Ans. (c) :



From energy balance equation

$C_h (100 - 60)$

$= C_c (T_{ce} - 40)$

Equal heat capacity, $m_h C_{p_h} = m_c C_{p_c}$

$C_h = C_c$

$T_{ce} = 80°C$

So, mean temperature difference will be same throughout the length for same heat capacity of fluid

So, $\Delta T = 100 - 80 = 60 - 40 = 20° C$

773. During the process of boiling and condensation, only a phase change takes place and one fluid remains at constant temperature throughout the heat exchanger. In terms of number of transfer units (NTU), the effectiveness of such an exchanger would be -

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(a) $\frac{NTU}{1+NTU}$ (b) $1 - \exp(-NTU)$

(c) $\frac{1-\exp(-2NTU)}{2}$ (d) None of these

RPSC ACF & FRO, 26.02.2021
RPSC IOF, 2020
GPSC Lect. (Auto) 16.10.2016
UKPSC AE 2012 Paper–II

Ans. (b) : Effectiveness (ε)

(i) Parallel Flow –

$$\boxed{\varepsilon = \frac{1-\exp[-NTU(1+R)]}{1+R}} \quad \left\{R = \frac{C_{min}}{C_{max}}\right\}$$

(ii) Counter Flow –

$$\boxed{\varepsilon = \frac{1-\exp[-NTU(1-R)]}{1-R\exp[-NTU(1-R)]}}$$

Special cases

(i) For regenerator, gas turbine Recuperator

R = 1

$$\boxed{\varepsilon_{parallel} = \frac{1-\exp(-2NTU)}{2}}$$

$$\boxed{\varepsilon_{counter} = \frac{NTU}{NTU+1}}$$

(ii) For boiler/evaporator/condenser

R = 0

$$\boxed{\varepsilon_{parallel} = \varepsilon_{counter} = 1-\exp(-NTU)}$$

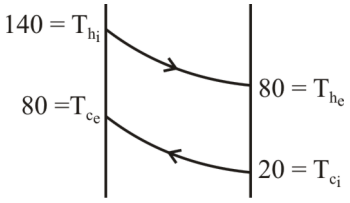
774. A counter flow heat exchanger is used to heat water from 20ºC to 80ºC by using heat exchanger gas entering at 140ºC and leaving at 80ºC. The log mean temperature difference for the heat exchanger is

(a) 80ºC (b) 60ºC

(c) 110ºC (d) 100ºC

TSPSC AEE 2017, MPPSC AE 08.11.2015
APPSC AEE 2012, ESE 1996

Ans. (b) : We know, If counter flow,



$$LMTD = \frac{(\Delta T_1 - \Delta T_2)}{ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

$\Rightarrow$ $\Delta T_1 = 80 - 20 = 60$

$\Delta T_2 = 140 - 80 = 60$

LMTD = Arithmetic mean temperature difference = $\frac{\Delta T_1 + \Delta T_2}{2} = 60$

775. In a shell and tube heat exchanger, baffles are mainly used to

(a) Increase the mixing of fluid
(b) Increase the heat transfer area
(c) Deflect the flow in desired direction
(d) Reduce fouling of the tube surface

APPSC Poly Lect. Automobile Engg., 2020
GPSC Executive Engg. 23.12.2018
TSPSC AEE 2017, GATE 1991

Ans. (b) : In shell and tube heat exchanger, baffles are mainly used to increase the heat transfer area.

776. Fouling factor in heat exchanger is written as:

(a) $R_f = \frac{1}{U_{dirty}} - \frac{1}{U_{clean}}$

(b) $R_f = U_{dirty} - U_{clean}$

(c) $\frac{1}{R_f} = U_{dirty} - U_{clean}$

(d) $\frac{1}{R_f} = \frac{1}{U_{dirty}} - \frac{1}{U_{clean}}$

KPSC ADF 2015, TSPSC AEE 2015
ESE 2012

Ans. (a) Fouling factor in heat exchanger is written as–

$$\boxed{R_f = \frac{1}{U_{dirty}} - \frac{1}{U_{clean}}}$$

777. The decreasing order of effectiveness for a given situation among types of heat exchangers is :

(a) parallel flow, cross flow, shell and tube, counter flow
(b) cross flow, counter flow, shell and tube, parallel flow
(c) counter flow, shell and tube, cross flow, parallel flow
(d) counter flow, cross flow, shell and tube, parallel flow.

TNPSC AE 2018
HPPSC Asstt. Prof. 20.11.2017, ESE 2012

Ans. (c) : The effectiveness of a heat exchanger can also be measured by heat transfer area for a given quantity of heat transferred. Out of given types of Heat exchangers, least area is of counter flow i.e. highly effective and maximum area in parallel heat exchanger i.e. least effective (from economical point of view).

Between shell and tube and cross flow, shell and tube is more effective than cross-flow.

So out of all four types, the correct sequence is.

$\in_{counter\ flow} > \in_{shell\ tube} > \in_{cross\ flow} > \in_{parallel\ flow}$.

778. Consider the following statements :
In a shell-and-tube heat exchanger, baffles are provided on the shell side to

1. Prevent the stagnation of shell-side fluid
2. Improve heat transfer
3. Provide support for tube
4. Prevent fouling of tubes

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Which of the above statements are correct ?

(a) 1, 2, 3 and 4 (b) 1 and 2 only

(c) 3 and 4 only (d) 2 and 3 only

RPSC IOF & Boiler 20.09.2017
UPSC JWM Advt. No.-50/2010
ESE 1994

Ans. (a) : As we know, in a shell and tube heat exchanger, the baffles are provided on the shell side have different functions such as –

(1) Prevent the stagnation of shell-side fluid.

(2) Improve heat transfer.

(3) Provide mechanical support for tube.

(4) Prevent fouling of tubes.

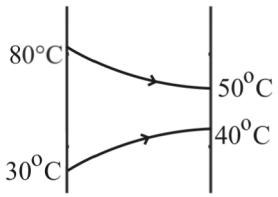
779. Air is cooled from 80 °C to 50 °C in a cooler. The coolant temperature rises from 30 °C to 40 °C. The LMTD value is 26 °C. Find the type of heat exchanger.

(a) Cross Flow (b) Parallel Flow

(c) Double Flow (d) Counter Flow

AAI Jr. Executive 29.11.2018
GPSC Executive Engg. 23.12.2018
GATE 2005

Ans. (a) :



Parallel Flow

$\theta_1 = 80 - 30$

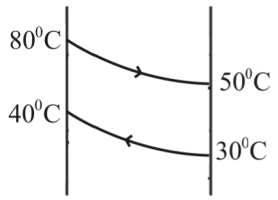
$\theta_1 = 50$

$\theta_2 = 50 - 40$

$\theta_2 = 10$

$$\theta_{LMTD} = \frac{\theta_1 - \theta_2}{\ln\left(\frac{\theta_1}{\theta_1}\right)} = \frac{50-10}{\ln\left(\frac{50}{10}\right)}$$

$$= \frac{40}{\ln 5} = 24.85\ ^\circ C$$



Counter Flow

$\theta_1 = 80 - 40$

$\theta_1 = 40$

$\theta_2 = 50 - 30$

$\theta_2 = 20$

$$\theta_{LMTD} = \frac{\theta_1 - \theta_2}{\ln\left(\frac{\theta_1}{\theta_2}\right)} = \frac{40-20}{\ln\left(\frac{40}{20}\right)}$$

$$= \frac{20}{\ln 2} = 28.85\ ^\circ C$$

As given value is between these two, so cross flow is the best option.

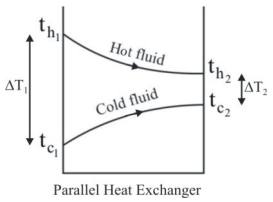
780. LMTD of a heat exchanger with ΔT_1 , and ΔT_2 being temperature differences between the hot and cold fluids at entrance and exit, respectively is:

(a) $\dfrac{\Delta T_2 - \Delta T_1}{\log\left[\dfrac{\Delta T_2}{\Delta T_1}\right]}$ (b) $\dfrac{\Delta T_2 - \Delta T_1}{\log\left[\dfrac{\Delta T_1}{\Delta T_2}\right]}$

(c) $\dfrac{\Delta T_2 - \Delta T_1}{\dfrac{\Delta T_2}{\Delta T_1}}$ (d) $\log\dfrac{\left[\Delta T_2 - \Delta T_1\right]}{\left[\dfrac{\Delta T_2}{\Delta T_1}\right]}$

OPSC AEE 2015 PAPER - II
Nagaland PSC CTSE 2015 Paper-II
UKPSC AE 2012 Paper–II

Ans : (a)



Parallel Heat Exchanger

$\Delta T_1 = t_{h_1} - t_{c_1}$

$\Delta T_2 = t_{h_2} - t_{h_1}$

LMTD for Parallel flow heat exhanger

$$LMTD = \frac{\Delta T_2 - \Delta T_1}{\log_e\left(\Delta T_2 / \Delta T_1\right)}$$

781. An adiabatic heat exchanger is used to heat cold water at 15ºC entering at a rate of 5 kg/s by hot air at 90ºC entering also at a rate of 5 kg/s. If the exit temperature of hot air is 20ºC, the exit temperature of cold water is

(a) 27ºC (b) 32ºC

(c) 52ºC (d) 85ºC

Gujarat PSC AE 2019
ISRO Scientist/Engineer 12.05.2013
ISRO Scientist/Engineer 2012

Ans : (b) : Applying heat balancing equation,

$$m_h C_{P_h}\left(T_{h_1} - T_{h_2}\right) = m_c C_{P_c}\left(T_{c_2} - T_{c_1}\right)$$

Here, $T_{h_1} = 90^\circ C$

$T_{h_2} = 20^\circ C$

$T_{c_1} = 15^\circ C$

$T_{c_2} = ?$

$m_h = 5\ kg/s$

$m_c = 5\ kg/s$

$C_{P_h} = 1.008\ kJ/kgK$

$C_{P_c} = 4.18\ kJ/kgK$

So, $5 \times 1.0085\ (90 - 20) = 5 \times 418\ (T_{c_2} - 15)$

$70.56 = 418.(T_{c_2} - 15)$

$T_{c_2} = 31.88^\circ C$

$$\boxed{T_{c_2} = 32^\circ C}$$

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782. Air enters in a counter flow heat exchanger at 70°C and leaves at 40°C. Water enters at 30°C and leaves at 50°C. The LMTD in °C is

(a) 5.65 (b) 14.43
(c) 19.52 (d) 20.17

MPSC Poly. Lect. 2014
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2000

Ans. (b) : Given, $T_{h_1} = 70°C$, $T_{h_2} = 40°C$,

$T_{c_1} = 30°C$ and $T_{c_2} = 50°C$

$\Delta T_1 = T_{h_1} - T_{c_2} = 70 - 50 = 20°C$

$\Delta T_2 = T_{h_2} - T_{c_1} = 40 - 30 = 10°C$

$$LMTD = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

$$LMTD = \frac{20-10}{\ln\left(\frac{20}{10}\right)} = 14.43°C$$

783. In heat exchanger degree of approach is defined as the difference between temperature of

(a) cold water inlet and outlet
(b) hot medium inlet and outlet
(c) hot medium outlet and cold water inlet
(d) hot medium outlet and cold water outlet

ISRO Scientist/Engineer (RAC) 07.05.2017
J&K PSC Civil Services Pre, 2013
TRB Poly. Lect., 2012

Ans. (d) : In heat exchanger, The degree of approach is defined as the difference between temperature of Hot medium outlet and cold water outlet.

784. If one of the two fluids flowing through a heat exchanger of NTU=2 remains at constant temperature throughout heat exchanger length the effectiveness of the heat exchanger will be

(a) $1- e^{-4}$ (b) $1-e^{-2}$
(c) $(1-e^{-2})/2$ (d) $(1-e^{2})/2$

BPSC Poly. Lect. 2016
VIZAG STEEL MT 2015
J&K PSC Civil Services Pre-2010

Ans. (b) : This is the case of boiling and condensation

$$C = \frac{C_{min}}{C_{max}} = 0 \qquad (C_{max} = \infty \text{ for phase change})$$

$NTU = 2$

$$\varepsilon_{parallel} = \frac{1 - e^{-NTU(1+C)}}{1+C}$$

$$= \frac{1 - e^{-NTU}}{1}$$

$$= 1 - e^{-2}$$

$$\varepsilon_{parallel} = \frac{1 - e^{-NTU(1-C)}}{1 - C\, e^{-NTU(1-C)}}$$

$$= \frac{1 - e^{-NTU}}{1} \qquad (C = 0, NTU = 2)$$

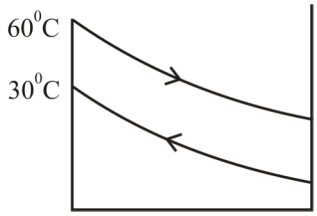
$$= 1 - e^{-2}$$

785. In a counter flow heat exchanger, hot fluid enters at 60^0C and cold fluid leaves at 30^0C. Mass flow rate of the hot fluid is 1 kg/s and that of the cold fluid is 2 kg/s. Specific heat of the hot fluid is 10 kJ/kgK and that of cold fluid is 5 kJ/kgK. The log mean temperature difference (LMTD) for the heat exchanger in ^{0}C is

(a) 15 (b) 30
(c) 35 (d) 45

UPPSC AE 13.12.2020, Paper-2
GPSC Executive Engineer 23.12.2018
GATE 2007

Ans. (b)



Given,

$(m_h) = 1$ kg/s
$(m_c) = 2$ kg/s
$(C_p)_h = 10$ kJ/kgK
$(C_p)_c = 5$ kJ/kgK

$(m_h)\,(C_p)_h = (m_c)\,(C_p)_C$

$1 \times 10 = 2 \times 5$

$10 = 10$

Thus, this is a case of balanced counter flow heat exchanger

LMTD = 60 – 30 = 30^0C

786. The log mean temperature difference (LMTD) is highest for _____ heat exchanger

(a) Parallel or co-flow (b) Cross- flow
(c) Mixed flow (d) Counter-flow

Haryana PSC AE (PHED) 04.09.2020, Paper-I
GATE 2007

Ans. (d) : Increasing order LMTD
Parallel flow < mixed flow < cross flow < counter flow

787. For boilers and condensers, the ratio of heat capacities $\frac{C_{min}}{C_{max}}$ is :

(a) Infinity (b) Zero
(c) One (d) Two

Sikkim PSC (Under Secretary), 2017
UPSC JWM Advt. No.-52/2010

Ans. (b) : For boilers and condensers, the ratio of heat capacities $\frac{C_{min}}{C_{max}}$ is zero.

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788. What does NTU indicate?

(a) Effectiveness of heat exchanger
(b) Efficiency of heat exchanger
(c) Size of the heat exchanger
(d) Temperature drop in heat exchanger

KPSC ADF 2015, ESE 2012

Ans. (c) : Number of transfer unit (NTU) indicates the size of the heat exchanger.

$$NTU = \frac{UA}{(mc_p)_{small}}$$

789. In a double-pipe counter flow heat exchanger, 1000 kg/h of oil having a specific heat 20 J/kg-K is cooled from 150°C to 125°C by 1250 kg/h of fluid of specific heat 16 J/kg-K. The fluid leaves the heat exchanger at 75°C. In such case, the temperature at which the cooling fluid is entering the heat exchanger and LMTD will be

(a) 25°C and 50°C (b) 50°C and 50°C
(c) 50°C and 75°C (d) 75°C and 50°C

GPSC EE Pre, 28.01.2017, ESE 2012

Ans. (c) : Given- $m_h = \dot{M}_{oil} = 1000$ kg/h

$C_{ph} = C_{poil} = 20$ J/kg-K

$T_{hi} = (T_{hi})_{oil} = 150°C$

$T_{he} = (T_{he})_{oil} = 125°C$

$\dot{M}_{fluid} = 1250$ kg/hr

$mc_{pc} = (C_p)_{fluid} = 16$ J/kg-K

$T_{ce} = (T_{ce})_{fluid} = 75°C$

We know that,

Heat lost by oil = Heat gained by water

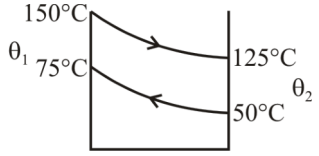
$$m_h C_{ph} (T_{hi} - T_{he}) = m_c C_{pc} (T_{ce} - T_{ci})$$

$1000 \times 20(150 - 125) = 1250 \times 16 (75 - T_{ci})$

$1000 \times 20 \times 25 = 1250 \times 16 (75 - T_{ci})$

$T_{ci} = 75 - 25$

$T_{ci} = 50°C$



$\theta_1 = 150 - 75 = 75°C$

$\theta_2 = 125 - 50 = 75°C$

$LMTD = \theta_1 = \theta_2 = 75°C$

790. A counter flow shell and tube heat exchanger is used to heat water with hot exhaust gases. The water (C_p = 4180 J/kg °C) flow at a rate 2kg/s while the exhaust gas (C_p = 1030 J/kg °C) flows at the rate of 5.25 kg/s. If the heat exchanger surface area is 32.5 m^2 and the over all heat transfer coefficient is 200 W/m^2 °C, what is the NTU for the heat exchanger?

(a) 1.2 (b) 2.4
(c) 4.5 (d) 8.6

MPPSC AE 08.11.2015
MPSC Poly. Lect. 2014

Ans. (a) : Given, A = 32.5 m^2, U = 200 W/m^2

For water	For exhaust gas
$c_p = 4180$ J/kg°C	$c_p = 1030$ J/kg°C
$\dot{m} = 2$ kg/s	$\dot{m} = 5.25$ kg/s

For water $(mc_p) = 2 \times 4180 = 8360$ J/sec°C

For exhaust gas $(mc_p) = 5.25 \times 1030$
$= 5407.5$ J/sec°C

Minimum value of $(mc_p) = 5407.5$

$$NTU = \frac{UA}{(mc_p)_{min}} = \frac{200 \times 32.5}{5407.5} = 1.2$$

791. Fouling significantly impacts the thermal and mechanical performance of heat exchangers. Which of the following is TRUE for Fouling?

(a) It reduces corrosion of heat exchangers
(b) It decreases the overall heat transfer coefficient
(c) It decreases the overall thermal resistance
(d) It decreases pressure drop across heat exchangers

Jharkhand Urja Vikas Nigam Ltd. AE 2017
UPPSC AE 2007 PAPER-II

Ans. (b) : The deposition of any undesired material on heat transfer surfaces is called fouling.
Fouling increases the overall thermal resistance and lowers the overall heat transfer coefficient of heat.

792. Match List-I with List-II and select the correct answer using the code given below the lists:

	(List-I)		(List-II)
A.	Direct contact heat Exchanger	1.	Lungstrom air heater
B.	Regenerative heat Exchanger	2.	Water cooling tower
C.	Conduction through a spherical wall	3.	Logarithmic curve
D.	Conduction through cylindrical wall	4.	Hyperbolic curve

(a) A-3, B-4, C-2, D-1 (b) A-2, B-4, C-1, D-3
(c) A-1, B-2, C-4, D-3 (d) A-3, B-2, C-4, D-1

UPRVUNL AE 21.08.2016, ESE 1995

Ans. : (*) No answer is correct.
The correct answer should be
A-2, B-1, C-4, D-3

793. For a compression or heating process what is the expression for effectiveness ∈

(a) $\in = \frac{\text{increase of availability of surroundings}}{\text{loss of availability of the system}}$

(b) $\in = \frac{\text{increase of availability of the system}}{\text{loss of availability of the surroundings}}$

(c) $\in = \frac{\text{loss of availability of the surroundings}}{\text{increase of availability of the system}}$

(d) $\in = \frac{\text{loss of availability of the system}}{\text{increase of availability of the surroundings}}$

TNPSC AE 2018, CSE Pre-2008

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Ans. (b) : $\in = \dfrac{\text{increase of availability of the system}}{\text{loss of availability of the surroundings}}$

794. For shell and tube heat exchanger, the corrosive liquid is normally passed through
(a) Shell side
(b) Tube side
(c) Either of (1) and (2) above
(d) None of the above

Nagaland PSC (CTSE) 2017, 2016 Paper-II

Ans. (b) : In a shell and tube exchanger, the corrosive liquid is normally passed through tube side.

795. A correction of L.M.T.D. is necessary in case of
(a) cross flow heat exchanger
(b) parallel flow heat exchanger
(c) counter flow heat exchanger
(d) all of the above

APPSC Poly. Lect. 2013
UKPSC AE 2007 Paper -II

Ans. (a) : Cross flow heat exchanger is a recuperater type heat exchanger in which hot fluids and cold fluid flow line intersect to each other.
* In this heat changer a correction of LMTD is necessary.
* LMTD for cross flow and multiplies heat exchanger
= Correction factor × $LMTD_{counter\ flow}$

796. In case of heat exchanger, the value of logarithmic mean temperature difference should be
(a) as small as possible (b) as large as possible
(c) constant (d) none of the above

Kerala PSC Asstt. Prof. 27.09.2016
UKPSC AE 2007 Paper -II

Ans. (b) : LMTD – Logarithmic mean temperature difference. It is a performance approach of heat exchangers.
* It is defined as the temperature different which, if constant would give the same rate of heat transfer as actually occurs under variable conditions of temperature difference.
* LMTD ↑ ⇒ Performance of Heat exchanger ↑

So, in case of heat exchanger, the value of logarithmic mean temperature difference (LMTD) should be as large as possible.

797. The log mean temperature difference is given by:
(a) $(\theta_1 - \theta_2) / \text{Log}_e (\theta_1/\theta_2)$
(b) $(\theta_1 + \theta_2) / \text{Log}_e (\theta_1/\theta_2)$
(c) $(\theta_1 - \theta_2) / \text{Log}_e (\theta_1-\theta_2)$
(d) $(\theta_1 - \theta_2) / \text{Log}_e (\theta_1+\theta_2)$

RPSC AE (GWD) 06.05.2016
TSPSC Managers, 2015

Ans. (a) : Log Mean Temperature Difference (LMTD)

$$(\Delta T)_{LMTD} = \frac{\theta_1 - \theta_2}{\ln\left(\frac{\theta_1}{\theta_2}\right)}$$

where, θ_1 = Inlet temperature difference.
θ_2 = Outlet temperature difference.

798. Air can be best heated by steam in a heat exchanger of
(a) plate type
(b) double pipe type with fins on steam side
(c) double pipe type with fins on air side
(d) shell and tube type

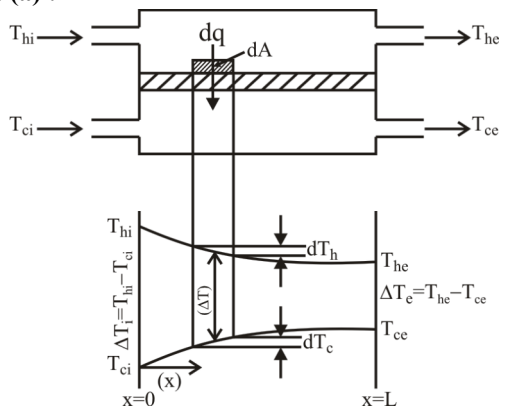
TRB Poly. Lect., 2012, ESE 2006

Ans. (c) : Air can be best heated by steam in a heat exchanger of double pipe type with fins on air side

799. In heat exchanger, the temperature of the hot fluid decreases while the temperature of the cold fluid increases. The increase and decrease follows:
(a) An exponential law (b) A quadratic law
(c) A linear law (d) A cubic law

RPSC VPITI 14.02.2016
UPPSC AE 12.04.2016 Paper-II

Ans. (a) :



Consider differential heat transfer area dA of the heat exchanger of length dx through which the differential heat transfer rate between hot and cold fluids is dq, Then

$$dq = U\Delta T dA$$

Where, $dA = Bdx$

And $\Delta T = T_h - T_c = f(x)$

At $x = 0, \Delta T = \Delta T_i = T_{hi} - T_{ci}$

At $x = L, \Delta T = \Delta T_e = T_{he} - T_{ce}$

Also, $dq = -\dot{m}_h C_{ph} dT_h = +\dot{m}_c C_{pc} dT_c$

$$\Delta T = T_h - T_c$$

$$d(\Delta T) = dT_h - dT_c$$

$$d(\Delta T) = \frac{-dq}{\dot{m}_h C_{ph}} - \frac{dq}{\dot{m}_c C_{pc}}$$

$$d(\Delta T) = -U\Delta T B dx\left(\frac{1}{\dot{m}_h C_{ph}} + \frac{1}{\dot{m}_c C_{pc}}\right)$$

$$\int_{\Delta T_i}^{\Delta T_e} \frac{-d(\Delta T)}{\Delta T} = \int_{x=0}^{L} UB\left(\frac{1}{\dot{m}_h C_{ph}} + \frac{1}{\dot{m}_c C_{pc}}\right) dx$$

$$\ell n\left(\frac{\Delta T_i}{\Delta T_e}\right) = UBL\left(\frac{1}{\dot{m}_h C_{ph}} + \frac{1}{\dot{m}_c C_{pc}}\right)$$

BL = A = Total heat transfer area
For inexact differential

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$$\ell n\left(T_h - T_c\right) = UA\left(\frac{1}{\dot{m}_h C_{ph}} + \frac{1}{\dot{m}_c C_{pc}}\right)$$

$$T_h - T_c = e^{UA\left(\frac{1}{\dot{m}_h C_{ph}} + \frac{1}{\dot{m}_c C_{pc}}\right)}$$

800. In a counter flow heat exchanger, the thermal capacities of both the hot and cold fluids are the same. If NTU of the heat exchanger is 1, then the effectiveness of the heat exchanger is

(a) 0.5 (b) 2

(c) 0.75 (d) None of these

RPSC AE (GWD) 06.05.2016, ESE 2009

Ans. (a) : For counter flow- When both hot and cold fluid have equal capacity rates i.e. $\dot{m}C_{ph} = \dot{m}C_{pc}$

$$\varepsilon_{\text{counter flow}} = \frac{NTU}{1+NTU}$$

[$\because$ given, NTU = 1]

$$\varepsilon_{\text{counter}} = \frac{1}{1+1} = \frac{1}{2}$$

$$= 0.5$$

801. In a two-fluid heat exchanger, the inlet and outlet temperatures of the hot fluid are 65 °C and 40 °C respectively. For the cold fluid, these are 15 °C and 43 °C. The heat exchanger is a

(a) parallel-flow heat exchanger

(b) counter-flow heat exchanger

(c) heat exchanger where both parallel-flow and counter-flow operations are possible

(d) None of the above

Assam PSC AE (IWT) 14.03.2021

ESE 2014

Ans. (b) : In case of parallel flow the temperature lines will cross each other and that is impossible. And in counter flow lines would not cross each other and hence is possible. So counter flow heat exchanger is true.

802. Water (C_p = 4.18 kJ/kgK) at 80°C enters counter flow heat exchanger with a mass flow rate of 0.5 kg/s. Air (C_p = 1kJ/kgK) enters at 30°C with a mass flow rate of 2.09 kg/s. If the effectiveness of the heat exchanger is 0.8, the LMTD in (in °C) is_____

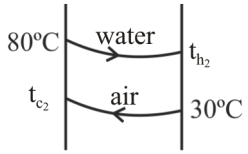
(a) 40 (b) 20

(c) 5 (d) 10

Assam Engg. College AP/Lect. 18.01.2021

GATE 2012

Ans. (d) :



Given,

$\dot{m}_w$ = 0.5 kg/s.

$(C_P)_{water}$ = 4.18 kJ/kgK

$\dot{m}_a$ = 2.09 kg/s

$(C_P)_{air}$ = 1 kJ/kg

$\dot{m}_w (C_P)_{water} = \dot{m}_a (C_P)_{air}$

$0.5 \times 4.18 = 2.09 \times 1$

$2.09 = 2.09$

Hence it is balanced heat exchanger.

Effectiveness of heat exchanger.

$$\varepsilon = \frac{80°C - t_{h2}}{80°C - 30°C}$$

$$0.8 = \frac{80 - t_{h2}}{50}$$

t_{h2} = 40°C

In case of balanced heat exchanger.

LMTD (θm) = $\theta_1 = \theta_2$

= 40°C – 30°C

= 10°C

803. In certain heat exchanger, both the fluids have identical mass flow rate specific heat product. The hot fluid enters at 81°C and leaves at 53°C, and the cold fluid entering at 25°C leave at 58°C, the effectiveness of the HE is

(a) 0.06 (b) 0.20

(c) 0.50 (d) 1.00

GPSC Executive Engineer 23.12.2018, GATE 1997

Ans. (c) : $\varepsilon = \frac{m_h C_h (T_{hi} - T_{ho})}{m_h C_h (T_{hi} - T_{ci})}$

$$= \frac{81-53}{81-25}$$

$$= \frac{28}{56} = 0.5$$

804. In a counter flow heat exchanger, for the hot fluid the heat capacity = 2kJ/kgK, mass flow rate = 5kg/s, inlet temperature = 150°C, outlet temperature = 100°C. For the cold fluid heat capacity = 4 kJ/kgK, mass flow rate = 10kg/s, inlet temp. = 20°C. Negleting the heat transfer to the surroundings, the outlet temperature of the cold fluid in °C is

(a) 7.5 (b) 32.5

(c) 45.5 (d) 70.0

GPSC Executive Engineer 23.12.2018, GATE 2003

Ans. (b) : $m_h C_h (T_{hi} - T_{ho}) = m_c C_c (T_{co} - T_{ci})$

$= 2 \times 5(150 - 100) = 4 \times 10\ (T_{co} - 20)$

$$T_{co} - 20 = \frac{50}{4} = 12.5$$

$T_{co} = 12.5 + 20$

= 32.5°C

805. An industrial gas (cp = 1kJ/kgK) enters a parallel flow heat exchanger at 250°C with a flow rate of 2kg/s to heat a water stream. The water stream (cp = 4kJ/kgK) enters the heat exchanger at 50°C with a flow rate of 1 kg/s. The heat exchanger has an effectiveness of 0.75. The gas stream exit temperature will be

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(a) 75ºC (b) 100ºC
(c) 125ºC (d) 150ºC

GPSC Executive Engineer 23.12.2018
GATE 2010

Ans. (b) : $\varepsilon = \frac{Q_{act.}}{Q_{max}} = \frac{m_g C_g (T_{hi} - T_{ho})}{m_g C_g (T_{hi} - T_{wi})}$

$0.75 = \frac{2 \times 1(250 - T_{ho})}{2 \times 1(250 - 50)}$

$150 = 250 - T_{ho}$

$T_{ho} = 100°C$

806. In a parallel flow gas turbine recuperator, the maximum effectiveness is
(a) 100%
(b) 75%
(c) 50%
(d) between 25% & 45%

JPSC AE, 10.04.2021, Paper-I, ESE 1992

Ans. (c) : In a parallel flow, gas turbine recuperator the maximum effectiveness is 50%.
For parallel flow configuration,
Effectiveness $(\in) = \frac{1 - \exp(-2NTU)}{2}$
Hence limiting value of effectiveness $(\in)$ is 1/2 or 50%.

807. After expansion from a gas turbine the hot exhaust gases are used to heat the compressed air from a compressor with the help of a cross flow compact heat exchanger of 0.75 effectiveness. What is the number of transfer units of the heat exchanger?
(a) 2 (b) 3
(c) 9 (d) 6

UPRVUNL AE 21.08.2016, ESE 2005

Ans. : (b) Since heating and cooling both sides are gases, so both heat capacity rates are equal
$\therefore C_{min} = C_{max}$
In counter flow design, effectiveness, expression in this situation is,

$\varepsilon = \frac{NTU}{1 + NTU}$

$0.75 = \frac{NTU}{1 + NTU}$

$1 + NTU = \frac{NTU}{0.75}$

$1 + NTU = \frac{4NTU}{3}$

$3 + 3\ NTU = 4\ NTU$

$3 = NTU$

808. Recuperators are a type of _____
(a) compressors (b) Solidifiers
(c) heat exchangers (d) accumulators

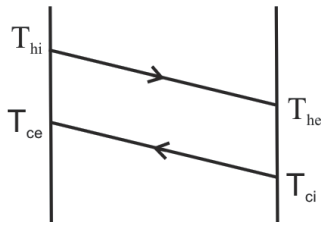
HPPSC Workshop Suptd. 08.07.02021

Ans. (c) :

809. In a heat exchanger, it is observed that where is the temperature difference between the two single phase fluid streams at one end and is the temperature difference at the other end. This heat exchanger is
(a) a condenser
(b) an evaporator
(c) a counter flow heat exchanger
(d) a parallel flow heat exchanger

APGCL AM, 2021

Ans. (c) :
$\Delta T_1 = T_{hi} - T_{ce}$
$\Delta T_2 = T_{he} - T_{ci}$
When $\Delta T_1 = \Delta T_2$, then Heat exchanger is known as balanced counter flow Heat exchanger and this is Happen when both hot and could fluids have equal capacity rate in a counter flow heat exchanger.



i . e. $\dot{m}_h C_{p_h} = \dot{m}_c C_{p_c}$

810. In regenerator type heat exchanger, heat transfer takes place by –
(a) Direct mixing of hot and cold fluids
(b) A complete separation between hot and cold fluids
(c) Flow of hot and cold fluids alternately over a surface
(d) Generation of heat again and again

APGCL AM, 2021

Ans. (c) : Regenerative type heat exchanger:
- The fluids flow through high heat capacity cellulose matrix
- Hot and cold fluids flow alternatively (only one fluid at a time through the heat exchanger)
- Hot fluid heats the cellulose matrix and cold fluid picks heat from it.
- No continuity of fluid flours.

811. In design of heat exchanger fouling factor largely depends on :
(1) Operating temperature
(2) Velocity of fluid
(3) Length of service
(4) Area of surface
(a) (1), (2) (4) (b) (2), (3) (4)
(c) (1), (3) (4) (d) (1), (2) (3)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (d) : The fouling factor depends on the operating temperature and velocity of the fluid as well as the length of service.
Fouling increases with increasing temperature and decreasing velocity.

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812. A heat exchanger is used to heat cold water (C_p = 4.18 kJ/kgK) entering at 12°C at a rate of 1.2 kg/s by hot air (C_p = 1.0 kJ/kgK) at 90°C at rate of 2.5 kg/s. The highest rate of heat transfer in the heat exchanger is :

(a) 82 kW (b) 195 kW
(c) 391 kW (d) None of these

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : Highest rate of heat transfer =

$Q_{max} = C_{min}\,\Delta T_{max}$

$C_{hot} = 2.5 \times 1.0 = 2.5$

$C_{cold} = 1.2 \times 4.18 = 5.016$

$Q_{max} = C_{min} \times (\Delta T)_{max}$

$= 2.5 \times (90 - 12)$

$\boxed{Q_{max} = 195 \text{ kW}}$

813. In a heat exchanger, 50 kg of water is heated per minute from 50°C to 110°C by hot gases which enter the heat exchanger at 250°C. The value of C_p for water is 4.186 kJ/kg.K and for air is 1 kJ/kg.K. If the flow rate of gases is 100 kg/min, the net change of enthalpy of air will be nearly

(a) 17.6 MJ/min (b) 15.0 MJ/min
(c) 12.6 MJ/min (d) 10.0 MJ/min

ESE 2020

Ans. (c) : $m_w c_{p_w} (T_1 - T_2) =$ Enthalpy change of air

$= \dfrac{50 \times 4.186 \times 60}{1000} = 12.558$ MJ/min

≈ 12.6 MJ/min

814. The purpose of using multiple pass heat exchangers is to

(a) reduce fluid flow friction losses
(b) reduce pressure drop
(c) increase pressure drop
(d) increase the rate of heat transfer

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : The purpose of using multiple pass heat exchangers is to increase the rate of heat transfer.
As by using multiple pass heat exchangers, heat transfer area increases. Thus, rate of heat transfer increases.

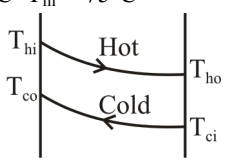
815. A cold liquid enters a counter flow heat exchanger at 15 deg C at a rate of 8 kg/s. A hot stream of the same liquid enters the heat exchanger at 75 deg C at 2 kg/s. Assuming the specific heat of the fluid as 4 kJ/kg. °C, determine the maximum heat transfer rate.

(a) 960 kW (b) 240 kW
(c) 1920 kW (d) 480 kW

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : $\dot{m}_c = 8 \text{kg/sec}, \quad \dot{m}_h = 2 \text{kg/sec}$

$T_{ci} = 15°C \quad T_{hi} = 75°C$



Maximum heat transfer rate in heat exchanger

$= (\dot{m}C)_{min} \times (T_{hi} - T_{ci})$

$= (2 \times 4)(75 - 15)$

$= 480$ kW

816. NTU refers to :

(a) the number of transfer units of pipes required for effective heat transfer
(b) the number of transfer units of passes required for effective heat transfer
(c) the compactness of heat transfer unit irrespective of heat transfer effectiveness.
(d) the closeness of heat exchanger with its thermodynamic limit of operation.

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : Number of transfer units (NTU)–

NTU refers to the closeness of heat exchanger with its thermodynamic limit of operation.

The number of transfer unit itself is a combination of overall heat transfer coefficients, transfer area, Fluid Flow rate and heat capacity.

NTU is also a measure of size of heat exchanger, it provides some indication of the size of heat exchanger.

$\text{NTU} = \dfrac{UA}{m_c c_c}$ when $m_h c_h > m_c c_c$

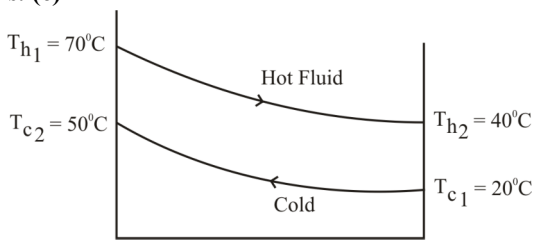
$\text{NTU} = \dfrac{UA}{m_h c_h}$ when $m_h c_h < m_c c_c$

817. In a heat exchanger, hot fluid enters at 70^0C and leaves at 40^0C, while cold fluid enters at 20^0C and leaves at 50^0C. If both hot and cold fluids have the same mass flow rate and specific heat, then the effectiveness of the heat exchanger is :

(a) 0.2 (b) 0.3
(c) 0.6 (d) 0.5

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c)



Given that

$T_{h_1} = 70^0$C

$T_{h_2} = 40^0$C

$T_{c_1} = 20^0$C

$T_{c_2} = 50^0$C

For same mass flow rate and specific heat the effectiveness (ε)

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$$\varepsilon = \frac{\left(T_{h_1} - T_{h_2}\right)}{\left(T_{h_1} - T_{c_1}\right)}$$

$$\varepsilon = \frac{70-40}{70-20}$$

$$= \frac{30}{50} = 0.6$$

818. For evaporation and condensation in a heat exchanger, the required surface area will be minimum for which type of flow?

(a) Cross (b) Counter
(c) Parallel (d) Same for all the cases

OPSC AEE 2019 PAPER - II

Ans : (d) : For evaporation and condensation in a heat exchanger, LMTD is same for both cases. Therefore, the required surface area is also same for both cases for all type of flow.

819. In a heat exchanger, the oil is cooled from 135°C to 15°C with the water, if water inlet temperature is 5°C and oil and water has the same mass flow rate $(C_p)_{oil}$ = 1.04 Joule/gram and $(C_p)_{water}$ = 4.16 Joule/gram, the water outlet temperature is:

(a) 150 (b) 135
(c) 30 (d) 35

Gujarat PSC AE 2019

Ans : (d) : Applying heat balancing equation,

$$m_h c_{P_h}\left(T_{h_1} - T_{h_2}\right) = m_c C_{P_c}\left(T_{c2} - T_{c1}\right)$$

Here, $T_{h_1} = 135^oC$
$T_{h_2} = 15^oC$
$T_{c_1} = 5^oC$
$T_{c_2} = ?$
$C_{P_h} = 1.04$ Joule/gram
$C_{P_c} = 4.16$ Joule/gram
$\dot{m}_h = \dot{m}_c$

$1.04\ (135 - 15) = 4.16\ (T_{c_2} - 5)$

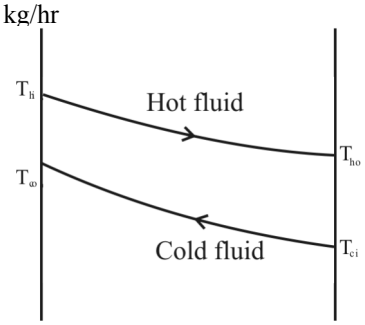
$$\boxed{T_{c2} = 35^oC}$$

820. In a double pipe counter flow heat exchanger, 10,000 kg/hr of oil [C_P = 2.09 kJ/kg-K] is cooled from 80°C to 50°C by water [C_P = 4.18 kJ/kg-K] of flow rate 8000 kg/hr entering at 25°C. What will be outlet temperature of water?

(a) 63.75°C (b) 52.55°C
(c) 48.15°C (d) 43.75°C

SJVN ET 2019

Ans. (d) : Given,
$T_{hi} = 80^oC$
$T_{ho} = 50^oC$
$C_{ph} = 2.09$ kJ/kg-K
$m_{ph} = 10000$ kg/hr
$T_{ci} = 25^oC$
$T_{co} = ?$
$C_{pc} = 4.18$ kJ/kgk
$m_c = 8000$ kg/hr



Applying heat balance equation,

$m_h C_{ph} (T_{hi} - T_{ho}) = m_c C_{pc} (T_{co} - T_{ci})$

$$\frac{10000}{3600} \times 2.09 \times (80-50) = \frac{8000}{3600} \times 4.18 \times (T_{co} - 25)$$

$T_{co} - 25 = 18.75$

$T_{co} = 43.75^oC$

821. In counter-current flow heat exchangers

(a) both the fluids at inlet are in their hottest state
(b) both the fluids at inlet are in their coldest state
(c) both the fluids at exit are in their hottest state
(d) one fluid is coldest and the other is hottest at inlet

Nagaland PSC (CTSE) 2018, Paper-II

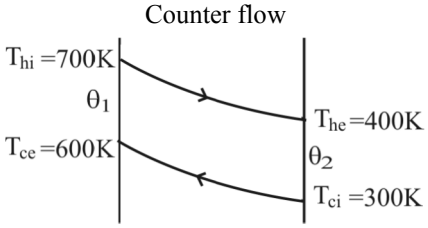
Ans. (a) : A counter-Flow heat exchanger is one in which the direction of the flow of one of the working fluids is opposite to the direction to the flow of the other fluid. In a parallel flow exchanger both Fluids in the heat exchange flow in the same direction.

822. In a single pass counter flow heat exchanger, air enters at 700 K and leaves at 400 K. Water enters at 300 K and leaves at 600 K. Find the LMTD for the heat exchanger.

(a) 200 K (b) 144 K
(c) 100 K (d) 150 K

AAI Jr. Executive 29.11.2018

Ans. (c) :



$\theta_1 = T_{hi} - T_{ce} = 700 - 600 = 100K$
$\theta_2 = T_{he} - T_{ci} = 400 - 300 = 100$ K
$\theta_1 = \theta_2$ then LMTD = 100 K

823. In shell and tube heat exchanger, baffles are mainly used for

(a) Increase the mixing of fluid
(b) Direct the flow in desired direction
(c) Reduce fouling of the tube surface
(d) Increase the heat transfer area

ISRO Scientist/Engineer (RAC) 22.04.2018

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Ans. (d) : In shell and tube heat exchanger, baffles are mainly used for increase the heat transfer area.
Note-ISRO excluded this question from evaluation.

824. In a counterflow heat exchanger, hot gases enter the system at 200°C and leave at 80°C. The temperature of the outside air entering the unit is 35°C. Its temperature at the exit is 90° C. The heat exchanger has an effectiveness of
(a) 0.35 (b) 0.34
(c) 0.33 (d) 0.32

ESE 2018

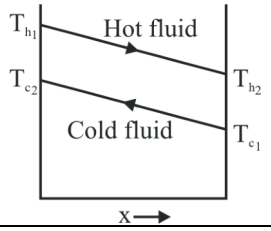
Ans. (*) : Given,

$T_{h_1} = 200°C$

$T_{h_2} = 80°C$

$T_{c_1} = 35°C$

$T_{c_2} = 90°C$



From energy balance

$$C_h\left(T_{h_1} - T_{h_2}\right) = C_c\left(T_{c_2} - T_{c_1}\right)$$

$$C_h\,(200 - 80) = C_c\,(90 - 35)$$

$$120\,C_h = 55\,C_c$$

$$\frac{C_h}{C_c} = \frac{55}{120}$$

$C_h = C_{min}$

$C_c = C_{max}$

$$\text{effectiveness } (\in) = \frac{C_h\left(T_{h_1} - T_{h_2}\right)}{C_{min}\left(T_{h_1} - T_{c_1}\right)}$$

$$= \frac{200 - 80}{200 - 35} = 0.727$$

825. After cooler is used to:
(a) cool the air
(b) reduce volume
(c) cause moisture and oil vapour to drop out
(d) increase pressure

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : An after cooler is a heat exchanger used to cool compressed air after leaving the compressor. Cooling reduces the air temperature and precipitates out water droplets which can be drained off.
After coolers incorporate a moisture separator for this purpose.
So, After cooler is used to cause moisture and oil vapour to drop out.

826. For a specified inlet and outlet temperatures, for which kind of heat exchanger will the log mean temperature difference, ΔT_m will be highest?
(a) Double pipe parallel flow heat exchanger
(b) Double pipe counter flow heat exchanger
(c) Cross flow heat exchanger
(d) Multi pass shell and tube heat exchanger

ISRO Scientist/Engineer 07.05.2017

Ans. (b) : Double pipe counter flow heat exchanger have large LMTD.

827. In a heat exchanger, hot gasses enter with a temperature of 250°C and leave at 50°C. On the other side, air enters at a temperature of 50°C and leaves at 90°C. The effectiveness of the exchanger is to be quoted as:
(a) 0.15 (b) 0.20
(c) 0.25 (d) 0.30

JWM 2017

Ans. (*) : Effectiveness of heat exchanger

$$= \frac{\text{Actual heat transfer}}{\text{Max. possible heat transfer}}$$

$$\in = \frac{Q}{Q_{max}} = \frac{C_h\left(t_{h_1} - t_{h_2}\right)}{C_{min}\left(t_{h_1} - t_{c_1}\right)}$$

Where C_{min} = Minimum of C_h and C_c
C_h = and C_c = heat capacity
t_{h1} = Temperature of hot fluid at inlet
t_{h2} = Temperature of hot fluid at exit
t_{c1} = Temperature of cold fluid at inlet
t_{c2} = Temperature of cold fluid at exit.

$$\in = \frac{C_h\left(250 - 50\right)}{C_{min}\left(250 - 50\right)} = 1$$

828. The NTU of a heat exchanger is an index of its
(a) Number of tubes (b) Number of passes
(c) Mode of operation (d) Performance

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : The NTU is a measure of heat transfer size of the exchanger, the larger the value of NTU the closer the heat exchanger approaches its thermodynamics limit.

829. The overall heat transfer coefficient in a fouled heat exchanger in comparison to the clean heat exchanger is
(a) Negligible (b) equal
(c) more (d) less

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : The overall heat transfer coefficient in a fouled heat exchanger in comparison to the clean heat exchanger is less, because it leaves trace/deposits on the surface of the separating wall.

830. Consider the following statements:
1. The efficiency of heat transfer in a condenser will improve by increase of the overall heat transfer coefficient.
2. The efficiency of heat transfer in a condenser will improve by increase of the velocity of flow of water in the tube.
3. The difference between the temperature of steam entering the condenser and the inlet water temperature should be maximum for maximum efficiency.

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Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (a) : $Q = UA\,\Delta T_m$

As velocity of flow of water increases the convective heat transfer co-efficient on water side increases. As a result U value increases.

$$\text{Condenses efficiency} = \frac{T_{wo} - T_{wi}}{T_s - T_{wi}}$$

If $(T_s - T_{wi})$ is maximum denominator increases hence condenser efficiency decreases.

Where

T_{wi} = temperature of cooling water at inlet

T_{wo} = temperature of cooling water at outlet

T_s = temperature of steam corresponding to the actual absolute pressure in the condenser.

831. In a counter flow heat exchanger, hot gases enter at 250°C and leave at 100°C. Atmospheric air enters at 50° C and leaves at 80°C. The effectiveness of the heat exchanger will be

(a) 0.20 (b) 0.25
(c) 0.30 (d) 0.75

ESE 2017

Ans. (d) : Given,

$T_{h_1} = 250°C$ $T_{c_1} = 50°C$

$T_{h_2} = 100°C$ $T_{c_2} = 80°C$

From energy balance equation

$$\dot{m}_h c_{ph}\left(T_{h_1} - T_{h_2}\right) = \dot{m}_c c_{pc}\left(T_{c_2} - T_{c_1}\right)$$

$$\dot{m}_h c_{ph}\,(250 - 100) = \dot{m}_c c_{pc}\,(80 - 50)$$

$$\dot{m}_h c_{ph} < \dot{m}_c c_{pc}$$

Hence effectiveness of heat exchanger

$$\in = \frac{T_{h_1} - T_{h_2}}{T_{h_1} - T_{c_1}} = \frac{250 - 100}{250 - 50}$$

$$= 0.75$$

832. A heat exchanger is used to cool the hot oil (C_p = 2kJ/kgK) at 120°C coming from the engine at the rate of 2kg/s by atmospheric air at 30°C entering at rate of 7 kg/s. The heat exchanger is not insulated and is losing heat at a rate of 30 kJ/s. If the exist temperature of air leaving heat exchanger is 60°C, determine exit temperature of hot oil.

(a) 60°C (b) 75°C
(c) 66.8°C (d) 48.84°C

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Heat loss by oil = Heat gain by air

$$mCp\left[T_{h_1} - T_{h_2}\right] = mCp\left[T_{c_2} - T_{c_1}\right] + 30 \text{ kJ/sec.}$$

$$2 \times 2\,[120° - T_{h_2}] = 7 \times Cp\ air \times [60°C - 30°C] + 30$$

$$2 \times 2\,[120 - T_{h_2}] = 30 + 7 \times 1.005 \times (30)$$

$$4 \times (120 - T_{h_2}) = 241.05$$

$$120 - T_{h_2} = 60.2625$$

$$\boxed{T_{h_2} = 59.7375 \approx 60°C}$$

833. In a heat exchanger, a pure fluid X condenses from a saturated vapor to saturated liquid, while exchanging heat with a fluid Y. The temperature of fluid X.....in the direction of flow.

(a) stays constant (b) increases
(c) decreases (d) none of these

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : Fluid 'X' condenses from a saturated vapor to saturated liquid. It means fluid 'X' undergoing phase change. So that there is no temperature change in phase change process.

834. Air circulates at the rate of 68 kg/s and is to be heated from 16°C to 34°C. Calculate the water mass flow for an air heater coil having hot water entering at 85°C and leaving at 74°C.

(a) 18 kg/s (b) 27 kg/s
(c) 32 kg/s (d) 48 kg/s
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) : We know,

$$m_{air} \times C_p \times \Delta T = m_{water} \times C_{pwater} \times \Delta T$$

$$68 \times 1.02 \times (34 - 16) = m_{water} \times 4.186 \times (85 - 74)$$

$$m_{water} = 27 \text{ kg/s}$$

835. Depending upon directions of two moving fluids, heat exchangers may be classified as

(a) Parallel flow (b) Counter flow
(c) Cross flow (d) All of the above
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : The classification of heat exchangers is depending upon directions of two moving fluids, are

(1) Parallel flow
(2) Counter flow
(3) Cross flow

836. When is arithmetic mean temperature difference used instead of LMTD in heat exchange?

(a) When temperature profiles of two fluids of heat exchanger are sloping downward with curve
(b) When the temperature profiles of two fluids of heat exchanger are sloping upward with curve
(c) When the temperature profiles of two fluids of heat exchanger are straight
(d) When one of the temperature profile for the fluid is straight

APPSC AEE MAINS 2016 PAPER-III

Ans. (c) : Arithmetic mean temperature difference used instead of LMTD in heat exchange when the temperature profiles of two fluids of heat exchanger are straight.

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837. Effectiveness of heat exchanger is function of:
(a) Heat capacity ratio only
(b) Surface area of heat exchanger only
(c) NTU and heat capacity ratio
(d) NTU only

UPRVUNL AE 2016

Ans. (c) : The effectiveness (ε) of a heat exchanger is defined as the ratio of the actual heat transfer to the maximum possible heat transfer.

$$\varepsilon = \frac{q}{q_{max}}$$

For heat exchanger

$$\varepsilon = f\left[NTU, \frac{C_{min}}{C_{max}}\right]$$

where the number of transfer unit, NTU

$$NTU = \frac{UA}{C_{min}}$$

where,
U - overall heat transfer coefficient and A is the heat transfer area.

838. Effectiveness (ε) and NTU relation for condenser may be written as
(a) NTU = ln(1 + ε) (b) NTU = ln(1 - ε)
(c) NTU = - ln(1 - ε) (d) $\varepsilon = \frac{NTU}{1+NTU}$

RPSC LECTURER 16.01.2016

Ans. (c) : In condenser and evaporator there will be phase change of fluid so heat capacity ratio will be zero.

$$\varepsilon = 1 - e^{-NTU}$$

(same for parallel and counter flow heat exchanger)

$$NTU = -\ln(1-\varepsilon)$$

839. Flat plate collectors are used to heat the water upto the temperature of
(a) 70-90ºC (b) 100-200ºC
(c) 200-300ºC (d) 300-400ºC

RPSC LECTURER 16.01.2016

Ans. (a) : Flat plate collectors—A Flat plate collector is a heat exchanger that converts the radiant solar energy from the sun into heat energy using the well known green house effect. It collects or capture, solar energy and uses that energy to heat water in the home for bathing, washing and heating etc.
A solar flate plate collector typically consists of a large heat absorbing plate, usually a large sheet of copper or aluminium as they are both good conductors of heat, which is painted or chemically etched black to absorb as much solar radiation as possible for maximum efficiency. This is used to heat the water upto the temperature of 70-90ºC.

840. Scale is formed on heat transfer surfaces when-
(a) water is acidic
(b) water is alkaline
(c) water contains dissolved gases
(d) water contains dissolved calcium and magnesium salts

RPSC IOF & Boiler 20.09.2017

Ans : (d) Water contains dissolved calcium and magnesium salts.

841. Cooling towers generally employ
(a) Single speed fan drive
(b) two speed fan drives
(c) Multi-speed fan drives in steps
(d) Infinitely variable multi-speed fan drives

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (b) : Cooling tower are a special type of heat exchanger that allows water and air to come in contact with each other to lower the temperature of the hot water. During this process, small volumes of water evaporate, lowering the temperature of the water that's being circulated throughout the cooling tower.

842. In a certain heat exchanger, both the hot fluid and cold fluid have identical heat capacity. $\left(\dot{m} \times Cp\right)$. The hot fluid enters at 80ºC and leaves at 50ºC. The cold fluid enters at 30ºC and leaves at 60ºC. The effectiveness of heat exchanger is:
(a) 0.4 (b) 0.6
(c) 0.667 (d) 1

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : Given,
Heat capacity are equal, $m_hC_{ph} = m_cC_{pc}$

$$\varepsilon = \frac{Q_{actual}}{Q_{max\,possible}} = \frac{m_hC_{ph}\left(T_{hi} - T_{he}\right)}{m_hC_{ph}\left(T_{hi} - T_{ci}\right)}$$

$$\varepsilon = \frac{T_{hi} - T_{he}}{T_{hi} - T_{ci}} = \frac{(80-50)}{(80-30)}$$

$$\varepsilon = 0.6$$

843. Oil is cooled from 125° C to 65° C with the help of cold water coming at 15° C in a counter flow heat exchanger. The mass flow rate of oil and water is in ratio 1 : 2 and specific heat capacity of oil and water is also in the ratio 1 : 2. Calculate the temperature of water at the outlet.
(a) 25° C (b) 35° C
(c) 40° C (d) 45° C
(e) 50° C

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : Heat transfer,

$$\frac{M_{oil}}{M_w} = \frac{1}{2} \,\&\, \frac{C_{oil}}{C_{pw}} = \frac{1}{2}$$

$$Q = mC_p\,\Delta T_{oil} = m_{oil} \times C_{oil}\,(125 - 65)$$
$$= m_{oil} \times C_{oil} \times 60$$

Outlet,

$$Q = mC_p\,\Delta T_{water}$$

$$\Rightarrow \quad T_{out} = T_{in} + \frac{Q}{mC_p}$$

$$T_{out} = 15° + \frac{M_{oil}C_{oil} \times 60}{M_wC_{pw}}$$

$$T_{out} = 15° + \frac{1}{2} \times \frac{1}{2} \times 60$$

$$= 30°C$$

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844. The effect of scales, deposits, rust formation etc. formed in heat exchanger can be taken care by introducing additional thermal resistance, called:

(a) caulking resistance
(b) peeling resistance
(c) fouling resistance
(d) dripping resistance

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Fouling factor is the factor which takes into account, the thermal resistance offered by any chemical deposits, scales, rust formation, that are formed heat transfer surface on both side and cold side.

$$f = \frac{1}{U_{dirty}} - \frac{1}{U_{clean}}$$

Generally,

$$f = 0.0005 \frac{m^2K}{W}$$

845. The steam condenser is a heat exchanger of which of the following types?

(a) Direct contact (b) Regenerator
(c) Recuperator (d) None of the above

Assam PSC CCE Pre 2015

Ans. (c) : The steam condenser is a heat exchanger of recuperator type.

846. Effectiveness of heat exchanger is defined as

(a) $\frac{t_{hi} - t_{he}}{t_{hi} - t_{ci}}$ (b) $\frac{t_{ce} - t_{ci}}{t_{hi} - t_{ci}}$

(c) Both (a) and (b) (d) $\frac{t_{hi} - t_{ce}}{t_{ho} - t_{hi}}$

Where

t_{hi} = temperature of hot fluid at inlet
t_{he} = temperature of hot fluid at exit
t_{ci} = temperature of cold fluid at inlet
t_{ce} = temperature of cold fluid at exit

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : Effectiveness of Heat Exchanger ($\in$)–It is the ratio of actual heat transfer to the max. possible heat transfer.

$$\in = \frac{Q_{actual}}{Q_{maximum}}$$

Case -1
When, $m_h\, Cp_h < m_c\, Cp_c$

$$\in = \frac{C_h(t_{hi} - t_{he})}{C_{min}(t_{hi} - t_{ci})}$$

$$\in = \frac{(t_{hi} - t_{he})}{(t_{hi} - t_{ci})}$$

Case -2
When, $m_c\, Cp_c < m_h\, Cp_h$

$$\in = \frac{C_c(t_{ce} - t_{ci})}{C_{min}(t_{hi} - t_{ci})}$$

$$\in = \frac{(t_{ce} - t_{ci})}{(t_{hi} - t_{ci})}$$

847. Approach of cooling tower means-

(a) difference in temperature of hot water entering the cold water leaving
(b) difference in temperature of the cold water and atmospheric temperature
(c) difference in temperature of the cold water leaving the cooling tower and the wet bulb temperature of surrounding air
(d) amount of heat thrown away by the cooling tower in kcal/hr

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (c) Approach of cooling tower means difference in temperature of the cold water leaving the cooling tower and the wet bulb temperature of surrounding air.

848. The effectiveness of a heat exchanger is the ratio of two heat transfer which is defined as the ratio of :

(a) Ideal to actual
(b) Ideal to maximum
(c) Actual to Maximum
(d) Maximum to actual

TSPSC Managers, 2015

Ans. (c) : The effectiveness (ε) of a heat exchanger is the ratio of two heat transfer which is defined as the ratio of actual to maximum.

$$\text{effectiveness } (\in) = \frac{Q_{Actual}}{Q_{Maximum}}$$

849. Cooling range in cooling tower is the-

(a) Difference in temperature of hot water entering and cold water leaving
(b) Amount of heat removed by the cooling tower in kcal/hr
(c) Difference in temperature of the cold water and atmospheric temperature
(d) Difference in temperature of the cold water leaving the cooling tower and the wet bulb temperature of surrounding air

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a) Cooling range in cooling tower is the difference in temperature of hot water entering and cold water leaving from the cooling tower.

850. In parallel flow heat exchangers, the two fluids i.e. hot and cold travel in the

(a) same direction (b) opposite direction
(c) cross direction (d) axial direction

Mizoram PSC AE/SDO 2014, Paper-I

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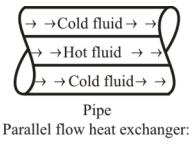
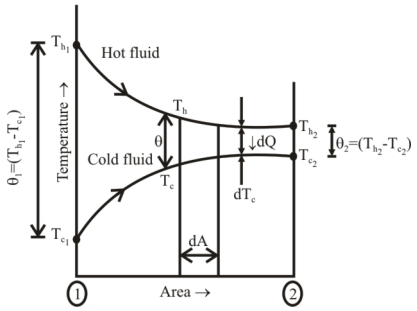
Ans. (a) :



Pipe

Parallel flow heat exchanger:



In parallel flow heat exchange the, two-fluids i.e. hot an cold travel in the same direction.

851. A hot fluid at 200^0C enters a heat exchanger at a mass flow rate of 10^4 kg/h. Its specific heat is 2000 J/kg-K. It is to be cooled by another fluid entering at 25^0C with a mass flow rate of 2500 kg/h and specific heat of 400 J/kg-K. The overall heat transfer coefficient based on outside area of 20 m^2 is 250 W/m^2-K. The exit temperature of hot fluid when fluids are in parallel flow is :

(a) 201^0C (b) 192^0C
(c) 309^0C (d) 167^0C

UPRVUNL AE 2014

Ans : (b) : Given,

$\dot{m}_h = 10^4 \text{ kg/h}, C_h = 2000 \text{J/kg-K},$

$\dot{m}_h C_h = 10^4 \times 2000 = 20 \times 10^6 \text{ J/kg-h} =$ 5555.6J/kg-s

$\dot{m}_c C_c = 2500 \times 400 = 1 \times 10^6 \text{ J/kg-h} =$ 277.78 J/kg-s

The minimum of the two is clearly $\dot{m}_c C_c$

$$\text{Now NTU} = \frac{UA}{C_{min}} = \frac{250 \times 20}{277.78} = 18$$

$$C = \text{Capacity ratio} = \frac{277.78}{5555.6} = 0.05$$

Now,

$$\varepsilon_{llel} = \frac{1 - e^{(1+C)NTU}}{1+C} = \frac{1 - e^{(1+0.05)_{18}}}{1+0.05} = 0.952$$

Now we know that $\varepsilon = \left(\frac{q_{act}}{q_{max\ possible}}\right)$

$$\varepsilon = \left(\frac{q_{act}}{q_{max\ possible}}\right) = \frac{\dot{m} C_c (T_{ce} - T_{ci})}{\left(\dot{m} C_p\right)_{smal} \times \left(T_{hi-T_{ci}}\right)}$$

$$0.952 = \frac{T_{ce} - 25}{200 - 25}$$

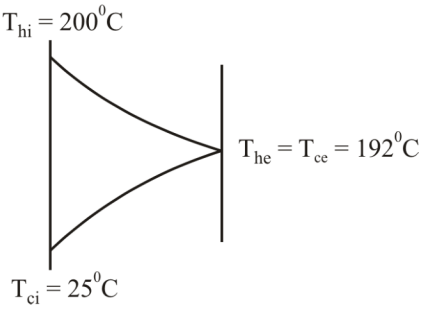
$T_{ce} = 191.6$

Now exit temperature of hot fluid can be found from this equation

$$\dot{m}_h C_h (T_{hi} - T_{he}) = \dot{m}_c C_c (T_{ce} - T_{ci})$$

$5555.6\ (200 - T_{he}) = 277.78\ (191.6 - 25)$

$T_{he} = 191.67 = 192^0$C



852. In a shell and tube heat exchanger, floating head is used for

(a) less corrosion of tubes
(b) large temperature differentials
(c) high heat transfer co-efficient
(d) low pressure drop
(e) small temperature differentials

CGPSC AE 26.04.2015 SHIFT-II

Ans. (b) : In a Shell and tube heat exchanger, floating head is used for large temperature differentials.

853. Consider following facts about fouling factor.

1. is a dimensionless quantity
2. accounts for additional resistance to heat flow
3. depend upon temperature
4. Its unit is m^2 K/W

Of these, which are correct

(a) 1 (b) 1 and 2
(c) 2 and 3 (d) 2, 3 and 4
(e) 1 and 4

CGPSC AE 26.04.2015 SHIFT-II

Ans. (d) 1. It is reciprocal of heat transfer coefficient (h)

$$F \propto \frac{1}{h}$$

2. Its unit is m^2 K/W
3. Depend upon temperature
4. Accounts for additional resistance to heat flow.

854. The unit of fouling factor is:

(a) W/m^2 -K (b) m-K/W
(c) K/W (d) m^2/K
(e) m^2-K/W

CGPSC AE 2014- II

Ans. (e) : We know that-

$$\text{unit of fouling factor} = \frac{1}{\text{unit of heat transfer coefficient } (h)}$$

$$= \frac{1}{\left[\frac{W}{m^2 - K}\right]} = \frac{m^2 - K}{W}$$

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855. In a parallel flow heat exchanger, the NTU is calculated to be 2.5. The lowest possible effectiveness for this heat exchanger is

(a) 10% (b) 27%
(c) 41% (d) 50%

TNPSC AE 2014

Ans. (d) : In a parallel flow heat exchanger, NTU = 2.5. Then the lowest possible effectiveness for this heat exchanger will be 50%.
For lowest possible effectiveness, C = 1

$$\in = \frac{1 - e^{-(1+C)NTU}}{1+C}$$

$$\in = \frac{1 - e^{-2\times 2.5}}{2}$$

$$\in = 0.49 \simeq 50\%$$

856. The correction factor of multipass counter flow heat exchanger depends on

(a) Fluid properties
(b) Geometry alone
(c) Temperature of Inlet and outlet fluid stream only
(d) Mass flow rates of hot and cold fluid streams

TNPSC AE 2014

Ans. (c) : The correction factor of multipass counter flow heat exchanger depends on temperature of Inlet and outlet fluid stream only.

857. Cold water (C_p = 4.18 kJ/kg°C) enters a heat exchanger at 15°C at a rate of 0.5kg/s, where it is heated by hot air (C_p = 1.0 kJ/kg°C) that enters the heat exchanger at 50°C at a rate of 1.8 kg/s. The maximum possible heat transfer rate in this heat exchange is

(a) 51.1 kW (b) 63.0 kW
(c) 66.8 kW (d) 73.2 kW

TNPSC AE 2014

Ans. (b) : For cold water
$C_P = 4.18$ kJ/kg°C
$T_{iw} = 15^0$C
$m_{cw} = 0.5$ kg/s
For hot air
$C_p = 1.0$ kJ/kg°C
$T_{iA} = 50$°C
$m_{HA} = 1.8$ kg/s
We know that, maximum possible heat transfer rate

$$\dot{Q}_{max} = C_{min}(T_{iA} - T_{iW})$$

$$C_W = \left[m\, c_p\right]_W = 0.5 \times 4.18$$

$C_W = 2.09$ kJ/°C
$C_{HA} = (m\, c_p) = 1.8 \times 1 = 1.8$ kJ/°C
$C_{min} = 1.8$ kJ/°C

$$\dot{Q}_{max} = 1.8 \times (50 - 15)$$

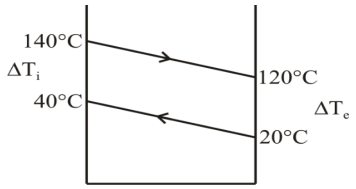
$$\boxed{\dot{Q}_{max} = 63\text{kW}}$$

858. In a counter flow heat exchanger, the cold fluid enters at 20°C and leaves at 40°C. The hot fluid enters at 140°C and leaves at 120°C. What is the heat transferred per unit area per unit heat transfer coefficient?

(a) 20 W (b) 80 W
(c) 100 W (d) 60 W

RPSC AEN Pre-2013

Ans. (c) :



Given,
For Counter flow heat exchanger –

$T_{h_i} = 140$°C $\quad \Delta T_i = T_{hi} - T_{ce} = 140 - 40$

$T_{h_e} = 120$°C $\quad \Delta T_e = T_{he} - T_{ci} = 120 - 20$

$T_{c_i} = 20$°C

$T_{c_e} = 40$°C

$(\Delta T)_{LMTD}$ of counter flow = ΔT_i or ΔT_e
$(\Delta T)_{LMTD} = (140 - 40)$ or $(120 - 20)$
$(\Delta T)_{LMTD} = 100$
We know that, $Q = UA\,(\Delta T)_{LMTD}$

$$\boxed{\frac{Q}{UA} = 100\text{W}}$$

859. In a counter flow heat exchanger with effectiveness of 0.75, the properties and flow rate are such that the both fluids have thermal capacity equal to 1000 W/°C. What will be the are of heat exchanger required, if the heat transfer coefficient is 100 W/m² °C?

(a) 40 m² (b) 30 m²
(c) 20 m² (d) None of the above

RPSC AEN Pre-2013

Ans. (b) : Given,
Efficiency ($\in$) = 0.75
(C) = 1000 W/°C
U = 100 W/m²°C

We know that, $\boxed{NTU = \frac{UA}{(C)}}$

For counter flow heat exchanger and having equal thermal capacity–

$$\in = \frac{NTU}{1 + NTU}$$

$$0.75 = \frac{NTU}{1 + NTU}$$

$$\frac{3}{4} = \frac{NTU}{1 + NTU} \Rightarrow 3 + 3NTU = 4NTU$$

$$3 = NTU$$

$$\therefore \quad NTU = \frac{UA}{(C)}$$

$$3 = \frac{100 \times A}{1000}$$

$$\boxed{A = 30\text{m}^2}$$

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860. For a double pipe, counter flow heat exchanger with C = 1 ($C = C_{min} / C_{max}$) the effectiveness is equal to:-

(a) $\frac{NTU}{NTU-1}$ (b) 1+1 / NTU

(c) $\frac{NTU}{NTU+1}$ (d) $\frac{NTU+1}{NTU-1}$

UKPSC AE-2013, Paper-II

Ans. (c) :

861. If Q = actual rate of heat transfer and Q_{max} = maximum possible rate of heat transfer then, heat exchanger effectiveness is equal to

(a) $Q_{max} - Q$ (b) Q/Q_{max}

(c) Q_{max}/Q (d) $\frac{Q_{max}+Q}{2}$

UKPSC AE 2012 Paper–II

Ans. (b) : Q/Q_{max}

862. Water at the rate of 60 Kg/min is heated from 35 to 85°C by an oil. The LMTD is given as 40°C. The overall coefficient is 300 W/m^2-°C. Specific heat of water is 4.2 kJ/kg°C. The heat exchanger area, in square meters, is:

(a) 18.5 (b) 15.5

(c) 17.5 (d) 16.5

APGENCO AE 2012

Ans. (c) : Given, m = 60kg/min = 1 kg/sec.

$\Delta T = 85 - 35 = 50°C$

LMTD (ΔT_m) = 40°C

U = 300 W/m^2-°C, C_w = 4.2 kJ/kg°C

$Q = mC_w\Delta T$

$= 1 \times 4.2 \times 10^3 \times 50$

Q = 210000W

$Q = UA\Delta T_m$

$$A = \frac{Q}{U\Delta T_m} = \frac{210000}{40 \times 300}$$

$$A = \frac{210}{12} = 17.5m^2$$

863. A list representing the thermal resistances of a heat exchanger is given below :

(1) Convection resistance of the surface of separating wall towards the hot fluid.

(2) Convection resistance of the surface of separating wall towards the cold fluid

(3) Convection resistance of the surface of separating wall.

Which of these will be used to determine the overall resistance of heat exchanger when the wall thickness of the separating wall is small and the thermal conductivity of wall material is very high?

(a) (1) and (2) (b) (1) ad (3)

(c) (2) and (3) (d) (3) only

RPSC AE GWD, 2011

Ans. (a) :

864. In cooling towers, the drift loss is around :

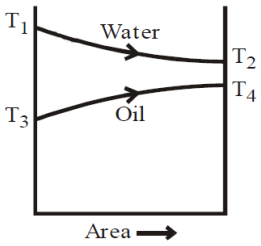
(a) 20% (b) 5 to 10%

(c) 10 to 20% (d) 20 to 30%

J&K PSC Civil Services Pre, 2010

Ans. (c) : In cooling towers, the drift loss is around 10 to 20%.

865.



In a heat exchanger used to heat up oil by hot water, the temperature profiles are as shown in the figure. If the rate of flow of oil is decreased, which of the following quantities will decrease?

(1) $(T_1 - T_2)$ (2) T_2

(3) $(T_1 - T_4)$ (4) $(T_4 - T_3)$

Select the correct answer using the code given below :

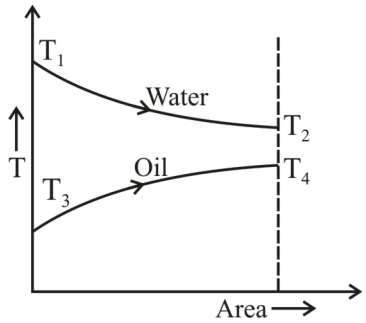
Code :

(a) 2 and 3 (b) 1 only

(c) 1 and 3 (d) 3 and 4

UPSC JWM Advt. No.-50/2010

Ans. (b) :



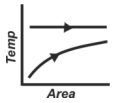
From energy balance,

$$Q = \dot{m}_h C_{ph}(T_1 - T_2) = \dot{m}_c C_{pc}(T_4 - T_3)$$

If $\dot{m}_c$ decrease, $(T_4 - T_3)$ increases to balance the equation.

So $(T_1 - T_2)$ decrease & due less removal of heat.

866. Choose the correct figure representing to gas heat transfer in parallel flow heat exchanger?

(a) 

(b)

(c)

(d)

ISRO Scientist/Engineer 2010

Ans. (c) : Heat transfer in parallel flow heat exchanger.

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867. The engine oil at 150°C is cooled to 80°C in a parallel flow heat exchanger by water entering at 25°C and leaving at 60°C. If the fluid flow rates and the inlet conditions remain unchanged the lowest temperature to which the oil may be cooled by increasing the length of the heat exchanger is :

(a) 60°C (b) 80°C
(c) 25°C (d) 66.6°C

APPSC IOF, 2009

Ans. (d) : Heat gain = Heat loss

$$m_c C_c (t_{c_2} - t_{c_1}) = m_h C_h (t_{h_1} - t_{h_2})$$

$$\frac{m_h C_h}{m_c C_c} = \frac{60-25}{150-80} = 0.5$$

$$\frac{C_h}{C_c} = \frac{C_{min}}{C_{max}} = 0.5 < 1$$

$(C_h) < (C_c)$

$$\varepsilon = \frac{1-\exp(-NTU(1+C))}{1+C}$$

$$NTU = \frac{UA}{C_{min}}$$

For increasing length of heat exchanger

$$NTU \to \infty$$

$$\varepsilon = \frac{1}{1+C} = \frac{1}{1+0.5} = 0.667$$

$$\varepsilon = \frac{C_h(150 - T_{h_2})}{C_h(150-25)}$$

$$0.667 \times (150 - 25) = 150 - T_{h_2}$$

$$\boxed{T_{h_2} = 66.62°C}$$

868. In a solar-assisted air-conditioning system, 1 kg/s of ambient air is to be preheated by the same amount of air leaving the system. A counter-flow heat exchanger having an area of 60 m^2 with overall heat transfer coefficient of 25 W/m^2K is used for this purpose. Assuming C_p for air as 1 kJ/kgK. the effectiveness of the heat exchange is :

(a) 0.4 (b) 0.6
(c) 1.6 (d) 2.5

DRDO Scientists 2009

Ans. (b) : $NTU = \frac{UT}{C_{min}} = \frac{25 \times 60}{1000} = 1.5$

$$\varepsilon = \frac{NTU}{1+NTU} = \frac{1.5}{2.5} = 0.6$$

869. A 10 kg solid at 100ºC with a specific heat of 0.8kJ/kgºC is immersed in 40 of 20ºC liquid with a specific heat of 40 kJ/kgºC. Estimate the temperature after a long time if the container is insulated?

(a) 30ºC (b) 28ºC
(c) 26ºC (d) 24ºC

ISRO Scientist/Engineer 2008

Ans. (d) : Let Intermediate temp. in 'T'.

$m_s = 10$ kg, $T_s = 100°C$
$c_s = 0.8$ kJ/kgºC
$m_c = 40$ kg, $T_C = 20°C$
$c_L = 4.0$ kJ/kgºC

$m_s c_s (T_s - T)$
$= m_L c_L (T - T_C)$
$10 \times 0.8 (373 - T)$
$= 40 \times 4 (T - 293)$
$T = 297 - 273$
$T = 24°C.$

870. In comparison to parallel flow heat exchangers, counter flow versions have

(a) lower LMTD
(b) larger heat transfer area
(c) smaller heat transfer area
(d) highest LMTD

TNPSC AE, 2008

Ans. (c & d) : In comparison to parallel flow heat exchanger, counter flow versions have highest LMTD and smaller heat transfer area.

$$Q = UA(\Delta T)_{LMTD}$$

$$\boxed{(\Delta T)_{LMTD} = \frac{Q}{UA}}$$

871. The function of a heat exchanger in gas turbine unit is

(a) to heat the compressed air before inlet to combustion chamber
(b) to heat the gas before inlet to gas turbine
(c) to exchange heat from hot gases of combustion chamber to the exhaust gases of gas turbine
(d) to heat the compressed air in between the stages of air compressor

TNPSC AE, 2008

Ans. (a) : The function of a heat exchanger in a gas turbine to increase the temperature of the air leaving to compressor prior to combustion. This reduce the amount of fuel needed to reach the desired turbine-inlet.

6. Mass Transfer

872. The functional relationship for free convective mass transfer is given as

(a) Nu = f(Gr, Pr) (b) Sh = f(Sc, Gr)
(c) Nu = f(Re, Pr) (d) Sh = f(Re, Sc)

TNPSC AE 2018, UJVNL AE 2016
APPSC Poly. Lect. 2013, ESE 1997

Ans. (b) : The functional relationship for free convection mass transfer is given as:

Sh = f(Sc, Gr)

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Sh = Sherwood no.	Nu = f(Gr, Pr)
Sc = Schmidt no.	For free convective heat
Pr = Prandtl no.	transfer
Gr = Grashof no.	
Nu = Nusselt no.	

873. Convective mass transfer coefficient depends only upon :

(a) dynamic characteristics of the flowing fluid
(b) geometry of the specific system
(c) properties of the fluid
(d) All of the above

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (d) : Convective mass transfer coefficient depends upon –
(i) dynamic characteristics of flowing fluid.
(ii) Geometry of the specific system.
(iii) Properties of the fluid.

874. Ratio of the molecular free path to characteristic length scale of flow is:

(a) Laplace number (b) Knudsen number
(c) Marangoni number (d) Galilei number

AAI Jr. Executive 26.03.2021

Ans. (b)
(i) Knudsen Number–The knudsen number (k_n) is a dimensionless number defined as the ratio of molecular mean free path to characteristic length of scale of flow.
(ii) Marangoni Number–The marangoni number (Ma) is as usually defined, the dimensionless number that compares the rate of transport due to marangoni flows, with the rate of transport of diffusion.
(iii) Galilei Number–Also referred as Galileo number (Ga) gravity forces divided by viscous forces. The Galilei number is used in viscous flow and thermal expansion.

$$Ga = R_e^2 R_i = \frac{gL^3}{v^2}$$

R_e = Reyonlds number
R_i = Richardson number
L = Characteristic length
v = Kinematic viscosity

875. Identity the wrong statement :

(a) Mass transfer ceases when concentration gradient is reduced to zero
(b) Mass transfer rate depends on a driving potential
(c) Rate of mass transfer does not depend on driving potential and resistance
(d) Mass is transferred towards the lower concentration

CGPSC AE 15.01.2021

Ans. (c) : Mass Transfer-
(i) Mass transfer rate depends upon the driving potential and resistance.
(ii) Mass is transferred from higher concentration to towards lower concentration.
(iii) Mass transfer increases when concentration gradient is reduced to zero.

876. Ratio of heat and mass transfer rates can be related by

(a) Nusselt number (b) Prandtl number
(c) Lewis number (d) Biot number

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (c) : Lewis number is a dimensionless number defined as the ratio of thermal diffusivity to mass diffusivity. It is used to characterized fluid flows where there is simultaneous heat and mass transfer.

877. Under steady state condition, Fick's first law is given as

$$J = -D\frac{dc}{dx}$$

Where, J = Diffusion flux,
D = Diffusion coefficient

$\frac{dc}{dx}$ **= Concentration gradient**

The unit of diffusion coefficient will be :

(a) m/s^2 (b) m/s
(c) m^2/s (d) kg/m^3

APPSC Poly Lect. 13.03.2020

Ans. (c) : Under steady state condition, Fick's first law

$J = -D\frac{dc}{dx}$; J = diffusion flux $kg/m^2.sec$

$\frac{dc}{dx}$ = Concentration gradient

D = diffusion coefficient, c = Concentration kg/m^3
Unit of diffusion coefficient (D) = m^2/sec
D is similar to kinematic viscosity.

878. Ratio of convective mass transfer to the mass diffusion rate is called?

(a) Sherwood number (b) Schmidt number
(c) Rayleigh number (d) Strouhal number

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : Sherwood number

$$= \frac{\text{Convective mass transfer}}{\text{mass diffusion rate}} = \frac{h_m.L}{D}$$

D = mass diffusivity
Sherwood number is similar to Nusselt number.

879. Analogy between momentum and heat transfer is known as

(a) Stanton-Prandtl analogy
(b) Crassoff-Meyer analogy
(c) Chilton-Colburn analogy
(d) None of the above

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Chilton-Colburn analogy: This is a successful and widely used analogy between heat, momentum and mass transfer.

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880. Thermal, velocity and concentration boundary layers become identical, when

(a) Prandtl number = Nusselt number = Schmidt number = 1
(b) Prandtl number = Sherwood number = Schmidt number = 1
(c) Prandtl number = Nusselt number = Lewis number = 1
(d) Prandtl number = Schmidt number = Lewis number = 1

TNPSC AE 2019

Ans. (d) : Thermal, velocity and concentration boundary layers become identical, when Prandtl number = Schmidt number = Lewis number = 1.

881. The dimensionless number in mass transfer which plays an equivalent role of Prandtl Number in heat transfer is

(a) Nusselt number (b) Lewis number
(c) Schmidt number (d) Grashof number

TNPSC AE 2018

Ans. (c) : Schmidt number- Schmidt number (Sc) is a dimensionless number defined as the ratio of momentum diffusivity (kinematic viscosity) and mass diffusivity and is used to characterize fluid flows in which there are simultaneous momentum and mass diffusion convection processes.

882. The driving potential in mass transfer process is

(a) concentration gradient
(b) temperature gradient
(c) pressure gradient
(d) velocity gradient

TNPSC AE 2018

Ans. (a) : Due to concentration gradient, mass transfer take place.

883. "The rate of compositional change is equal to the diffusivity times the rate of change of the concentration gradient" is

(a) Fick's first law of diffusion
(b) Fick's second law of diffusion
(c) Gibbs' phase rule
(d) Diffusion by vacancy migration

Karnataka PSC Lect., 27.05.2017

Ans (b) : "The rate of compositional change is equal to the diffusivity times the rate of change of the concentration gradient" is Fick's second law of diffusion.

884. The Lewis number which is a measure of relative thermal and concentration boundary layer thicknesses is defined as, (Re: Reynold's number, Pr: Prandtl number, Sc: Schmidt number)

(a) Re × Pr (b) Sc/Pr
(c) Re/Sc (d) Sc × Pr

ISRO Scientist/Engineer 07.05.2017

Ans. (b) : Lewis number = $\frac{Sc}{Pr}$

Sc = Schmidt number = $\frac{\upsilon}{D}$

Pr = Prandtl number = $\frac{\upsilon}{\alpha}$

where υ is molecular diffusivity (kinematic viscosity)
α is heat diffusivity
D is mass diffusivity

885. The ratio of thermal diffusivity to mass diffusivity is known as

(a) Schmidt number (b) Sherwood number
(c) Lewis number (d) Stanton number

Nagaland PSC CTSE 2017 Paper-2

Ans. (c) : The ratio of thermal diffusivity to mass diffusivity is known as lewis number (L_e). It is used to characterize fluid flow's where there is simultaneous heat and mass transfer. It is a dimension less number.

886. The dimensional number associated with mass transfer is

(a) Schmidt Number (b) Nusselt Number
(c) Biot Number (d) Mach Number

TANGEDCO AE 2015

Ans. (a) : The Schmidt Number (Sc) in convective mass transfer is similar to Prandtl no. in convective heat transfer. The Schmidt no. compares velocity and concentration boundary layer like Prandtl no. compares velocity and thermal boundary layer.

$$\text{Schmidt No.} = \frac{\text{Momentum diffusitivity}}{\text{Mass diffusitivity}} = \frac{\upsilon}{D}$$

887. If heat and mass transfer take place simultaneously, the ratio of heat transfer coefficient to the mass transfer coefficient is a function of the ratio of

(a) Schmidt and Reynolds numbers
(b) Schmidt and Prandtl numbers
(c) Nusselt and Lewis numbers
(d) Reynolds and Lewis numbers

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Heat and mass transfer take place simultaneously, the ratio of heat transfer coefficient to the mass transfer coefficient is a function of the ratio of Schmidt and Prandtl number.

$$\text{Schmidt no.}(Sc) = \frac{\nu}{D} = \frac{\mu}{\rho D} = \frac{\text{viscous diffusion rate}}{\text{mass diffusion rate}}$$

$$\text{Prandtl number (Pr)} = \frac{\nu}{\alpha} = \frac{\text{momentum diffusivity}}{\text{thermal diffusivity}}$$

888. Which of the following concepts do astronomers use to measure the mass of stars?

(a) Wein's law
(b) Spectral lines
(c) Generalized keplar's law
(d) Stefan-Boltzman law

WBPSC AE 2008

Ans. (c) : Astronomers uses Generalized Keplar's law to measure the mass of stars.

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08.

Industrial Engineering

1. Break Even Analysis

1. In break even analysis, total cost consists of :

(a) Fixed Cost + Sales Revenue
(b) Variable Cost + Sales Revenue
(c) Fixed Cost + Variable Cost
(d) Fixed Cost + Variable Cost + Profit

Nagaland PSC (CTSE) 2018, Paper-I
GPSC Lect. 23.10.2016
Rajasthan Nagar Nigam AE 2016, Shift-I
Kerala PSC IOF 19.04.2016
VIZAG STEEL MT 2015
GWSSB AAE, 27.12.2015
J&K PSC Civil Services Pre, 2013
OPSC Civil Services Pre. 2011
RPSC ACF-2011
OPSC Civil Services Pre 2006

Ans. (c) : In break even analysis, total cost consists of Fixed Cost + Variable Cost.

Total selling cost (S) = Fixed Cost (F) + Variable Cost (V) + Profit (P)

At BEP, Profit = 0

Then,

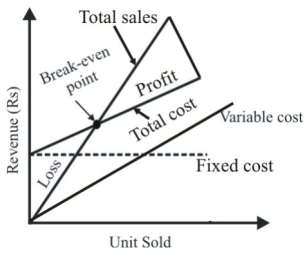
$$\left[\text{Total selling cost}\right]_{BEP} = \text{Fixed cost} + \text{Variable cost}$$

2. At break-even point :

(a) Fixed costs and variable costs are equal
(b) Sales revenue and total cost are equal
(c) Sales revenue is more than total cost
(d) Sales revenue is less than total cost

NLC GET 17.11.2020, Shift-II
Kerala PSC Asstt. Prof. 27.09.2016 (KPSC AE. 2015)
Kerala PSC AE 06.08.2015

Ans. (b) : A Break even point sales revenue and total cost are equal.



A breakeven analysis is used to determine how much sales value a business needs to start making a profit.

In business or economics the break-even point (BEP) is point at which the total of fixed and variable costs of business becomes equal to its total revenue. At this point, a business neither earns any profit nor suffers any loss. The above graph is used to find out the break even point.

3. A company produced 15,000 quantity of a product whose selling price is Rs. 300. At that quantity, if the fixed cost is Rs. 15.2 lakhs and total variable cost Rs. 21 lakhs, then the breakdown quantity for the product is

(a) 4,000 (b) 7,800
(c) 8,400 (d) 9,500

ESE 2000
Mizoram PSC (Insp. of Factories) 2019 Paper-III
GPSC Asstt. Director of Transport 05.03.2017
APGENCO AE, 2017

Ans. (d) : $x_{bep} = \dfrac{F}{(s-v)} = \dfrac{1520000}{4500000 - 2100000}$

$= \dfrac{152}{240}$

$x = 0.633$

Total quantity at break even point

$= 0.633 \times 15000$

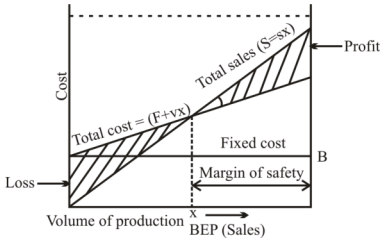
$= 9500$

4. Break-even analysis shows profit when–

(a) sales revenue is greater than total cost
(b) sales revenue equal to total cost
(c) sales revenue is less than total cost
(d) sales revenue is less than fixed cost

Nagaland PSC CTSE 2017, Paper-I
TSPSC AEE 2015
UKPSC AE-2013, Paper -I

Ans. (a) : Break-even analysis shows profit when sales is greater than total cost.



Total Cost (F') = F + vx

Total Sales (S) = sx

Break even point is the volume of production where the total cost of manufacturing equal to the total sales (also known as no profit no loss point.)

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5. In the production of a product, the fixed costs are Rs. 6,000 and the variable cost is Rs. 10 per product. If the sale price of the product is Rs. 12 the break even volume of the product to be made will be

(a) 2000 (b) 3000
(c) 4000 (d) 6000

TNPSC AE 2017, ESE 2008

Ans. (b) : Data given,
F = Rs. 60,000
v = Rs. 10/product
s = Rs. 12/product
We know that,
Total sale = Fixed costs + Total variable costs + Profit

$$x \times s = F + v \times x + P$$

For B.E.V. of products.

$$P = 0$$

Then,

$$x_{bev} \times s = F + v \times x_{bev}$$

$$x_{bev} = \frac{F}{(s-v)} = \frac{6000}{12-10}$$

$$x_{bev} = 3000 \text{ units}$$

6. If F is the fixed cost, V is the variable cost per unit (or total variable costs) and P is the selling price of each unit (or total sales value), then break-even point is equal to

(a) $\frac{F \times V}{P}$ (b) $\frac{F \times V}{V}$

(c) $\frac{F}{1+\frac{V}{P}}$ (d) $\frac{F}{P\left[1-\frac{V}{P}\right]}$

RPSC ACF & FRO, 26.02.2021
Nagaland PSC (CTSE) 2018, Paper-I
GPSC Asstt. Prof. (Prod.) 21.08.2016
JPSC AE 2013, Paper-V

Ans. : (d) :

$$x_{BEP} = \frac{F}{P\left[1-\frac{V}{P}\right]}$$

7. The variable cost per unit associated with automated assembly line (V_a), cellular manufacturing (V_b), and job production (V_c) will be such that

(a) $V_a > V_b > V_c$ (b) $V_b > V_a > V_c$
(c) $V_c > V_b > V_a$ (d) $V_c > V_a > V_b$

APPSC AEE 2012, CSE Pre-1995

Ans. (c) : Automated assembly line (V_a)
Cellular manufacturing (V_b)
Job production (V_c)
So variable cost per unit associated by $V_c > V_b > V_a$

8. Process I requires 20 units of fixed cost and 3 units of variable cost per piece. while process II requires 50 units of fixed cost and 1 unit of variable cost per piece. For a company producing 10 pieces per day.

(a) Process I should be chosen
(b) Process II should be chosen
(c) Either of the two processes could be chosen
(d) A combination of the two processes should be chosen

Punjab PSC (Lect.) 06.08.2017
APPSC AEE 2012, ESE 1997

Ans. (a) : for I items
Sales = F + V × Q
= 20 + 3 × 10
= 50
Where,
F = Fixed cost
V= Variable Cost/ Quantity
Q = No. of items
For II Items –
Sales = F + V × Q = 50 + 10 × 1 = 60
So sales price of Ist items are lesser them IInd items therefore I should be chosen

9. Production cost per unit can be reduced by :

(a) Eliminating idle time.
(b) Producing more with increased input
(c) Producing more with same inputs
(d) Minimizing resource waste

RPSC AE GWD, 2011
UPSC JWM Advt. No.-52/2010
CSE Pre-1995

Ans. (c) : Total product costs can be determined by adding together the total direct materials and labour costs as well as the total manufacturing overhead costs. To determine the product cost per unit of product, divide this sum by the number of units manufactured in the period covered by those costs. The production cost per unit can be reduced by producing more with the same inputs.

10. At Break Even Point there is :

(a) Profit (b) Loss
(c) No profit or no loss (d) None of the above

JPSC AE 10.04.2021, Paper-II
KPCL AE 2016

Ans. (c) : At Break evens point there is no profit and no loss.

11. Process X has fixed cost of Rs. 40,000/- and variable cost of Rs. 9 per unit whereas process Y has a fixed cost of Rs. 16,000/- and variable cost of Rs. 24 per unit. At what production quantity are the total costs of X and Y equal?

(a) 1200 units (b) 1600 units
(c) 2000 units (d) 2500 units

TRB Asstt. Prof., 2012, ESE 2009, 2004

Ans. (b) : Total cost = Fixed cost + variable cost
For X
Fixed cost = Rs.40,000
Variable Cost = Rs. 9/unit.
For Y
Fixed cost = Rs.16000
Variable cost = Rs.24/unit

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Let, at x quantity total cost of x and y will be same

$40000 + x \times 9 = 16000 + 24 \times x$

$15x = 24000$

$\boxed{x = 1600}$

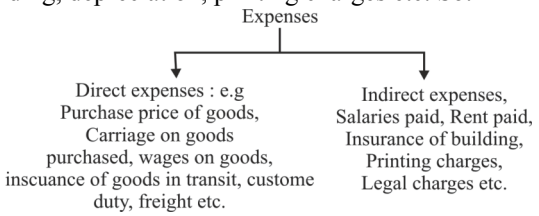
12. Direct Expenses include:

(a) Factory Expenses
(b) Selling Expenses
(c) Administrative Expenses
(d) None of these

OPSC AEE 2015 Paper-I
J&K PSC Civil Services Pre, 2013

Ans. : (d) Direct Expenses : Expenses connected with purchases of goods are known as direct expenses. For example, freight, insurance of goods in transit, carriage, wages, custom duty, import duty, etc. Without incurring these expenses, it is not possible to bring the goods from the purchase point to the go down of the business. such expenses are collectively known as direct expenses.

Indirect Expenses : All expenses other than direct expenses are assumed as indirect expenses. Such expense have no relationship with purchase of goods. Examples of indirect expenses include rent of building, salaries to employees, legal charges, insurance of building, depreciation, printing charges etc. So.



13. The break-even point can be lowered by

(a) increasing the fixed cost
(b) increasing the variable cost
(c) decreasing the slope of the income line
(d) reducing the variable cost

Assam PSC AE (PHED), 18.10.2020
ESE 2011

Ans. (d) : The break-even point can be lowered by

(1) Reducing the amount of fixed cost.
(2) Reducing the variable cost per unit thereby increasing the unit's contribution margin.
(3) Improving the sales mix by selling a greater proportion of the products having longer contribution margin.
(4) Increasing selling price so long as the number of units sold will not decline significantly.

14. Identify the chart which is used to determine the profit potential under varying conditions of output and cost :

(a) Gantt chart (b) Travel chart
(c) Break even chart (d) Emerson chart

TRB Polytechnic Lecturer 2017
J&K PSC Civil Services Pre, 2010

Ans. (c) : A graphical device used to determine the break-even point and profit potential under varying conditions of output and cost is known as break even chart.

15. For a small scale industry, the fixed cost per month is 5000. The variable cost per product is 20 and sales price is 30 per piece. The break even production per month will be :

(a) 300 (b) 460
(c) 500 (d) 1000

RPSC ACF & FRO, 26.02.2021
ESE 2016, 1995

Ans. (c) : Break even production per month

$$x = \frac{F}{P - V}$$

$$x = \frac{5000}{30 - 20}$$

$\boxed{x = 500}$

16. At breakeven point, slope of sales line is equal to :

(a) $\frac{\text{Variable expenses} + \text{Constant expenses}}{\text{Total sales}}$

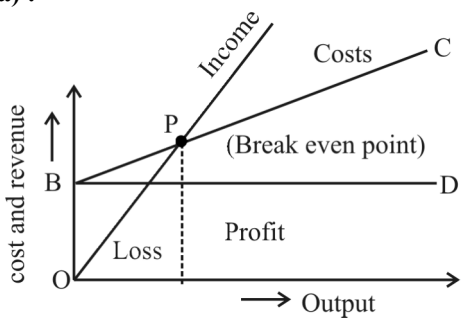
(b) Slope of total expenses line

(c) $\frac{\text{Total sales - Profit}}{\text{Variable expenses} + \text{Profit}}$

(d) $\frac{\text{Variable expenses} - \text{Constant expenses}}{\text{Total sales}}$

RPSC ADE 2016, VIZAG STEEL MT 2015

Ans. (a) :



Slope of sales line

$$= \frac{\text{Variable expenses} + \text{Constant expenses}}{\text{Total sales}}$$

17. A company sells 14000 units of its products. It has a variable cost of Rs. 15 unit. Fixed cost is Rs. 47000 and total required profit is Rs. 23000. Per unit product price (in Rs) will be:

(a) 60 (b) 40
(c) 30 (d) 20

RPSC VPITI 14.02.2016, CSE Pre-1999

Ans. (d) : Given-

Fixed cost F = Rs. 47000
Quantity Q = 14000 unit product
Variable cost V = 15 per unit
Profit P = Rs. 23000

Formula

$F + QV + P = Q.S.$

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Product price $S = \frac{F + QV + P}{Q}$

$= \frac{47000 + 14000 \times 15 + 23000}{14000}$

S = 20 Per unit

18. The Break Even quantity for an industry selling a product for Rs. 10/ unit, with fixed cost for assets Rs. 40,000 and variable cost of Rs. 6 per unit will be :

(a) 8,000 (b) 10,000
(c) 12,000 (d) 14,000

JPSC AE 10.04.2021, Paper-II

Ans. (b) : Given,
Selling price (S) = Rs.10/unit
Fixed cost (F) = Rs. 40,000
Variable cost (V) = Rs. 6/unit

Break even quantity $(x) = \frac{F}{S - V}$

$= \frac{40,000}{10 - 6}$

$= \frac{40,000}{4}$

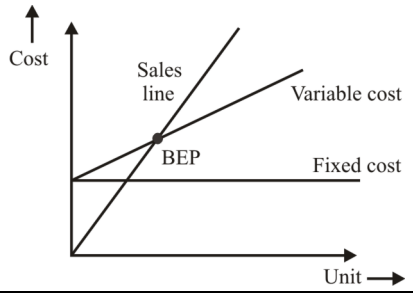
$\boxed{x = 10,000}$ units

19. A company spends considerable amount on publicity to promote sales. This expenditure in break-even chart is shown below the :

(a) fixed cost line (b) sales revenue line
(c) total cost line (d) variable cost line

CGPSC AE 15.01.2021

Ans. (d) : A company spends considerable amount on publicity to promote sales. This expenditure in break-even point chart is shown by variable cost line.



20. Fixed cost of an equipment is Rs. 6,000, if variable cost of an item it produces is Rs. 2 per item and sells it for Rs. 7 per item, what is the break-even point?

(a) 1200 items (b) 3000 items
(c) 7000 items (d) 6500 items

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : BEQ $= \frac{\text{Fixed cost}}{\text{Selling price} - \text{Variable cost}}$

$= \frac{6000}{7 - 2} = \frac{6000}{5}$

BEQ = 1200 items

21. If the fixed cost of the assets for a given period tripled, then how much will the break even quantity becomes.

(a) Half the original value
(b) Same as the original value
(c) Thrice the original value
(d) Twice the original value

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : We know,

$BEQ = \frac{F.C}{S.P. - VC}$

F.C.= Fixed cost
S.P.= Sales Price
V.C.= Variable cost

$BEQ \propto F.C.$

If F.C. is 3 times
Then BEQ is become 3 times

22. A company has the data of a product as:
Fixed cost/month = Rs. 60,000
Variable cost/unit = Rs. 210
Selling price/unit = Rs. 320
Production capacity = 1600 unit/month.
If the production is carried out at 80% of the rated capacity. What will be the monthly profit?

(a) Rs. 80,800 (b) Rs. 90,900
(c) Rs. 85,850 (d) Rs. 75,750

SJVN ET 2019

Ans. (a) : Fixed cost/month = Rs. 60,000
Variable cost/unit = Rs. 210
Selling price/unit = Rs. 320
Production capacity = 1600 unit/month
Rated capacity of production (r) = 80%
Actual production,
$= 0.8 \times 1600$
$= 1280$ units
Total cost = FC + Variable cost
$= 60000 + 210 \times 1280$
$= 328800$
Total Selling price (sales)
$= 320 \times 1280$
$= 409600$
Profit = Sales – Total cost
$= 409600 - 328800$
= Rs. 80800

23. The fixed cost per month is Rs. 5,000, variable cost is Rs. 2 per unit, selling price is Rs. 7 per unit. To achieve a profit of Rs. 1,000, the quantity produced per month will be:

(a) 800 (b) 1000
(c) 1200 (d) 1400
(e) 1600

CGPSC AE 25.02.2018

Ans. (c) : Data given,
F.C. = Rs.5000
v = Rs. 2 per unit
s = Rs. 7 per unit

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P = Rs. 1000

x = ?

Let us assume total quantity of product produced in one month is x

We know that,

Total sell = F.C. + V.C. + Profit

$x \times s = F.C. + x \times v + P$

$x[s - v] = F.C. + P$

$$x = \frac{F.C. + P}{(s - v)} = \frac{5000 + 1000}{(7 - 2)} = \frac{6000}{5}$$

$$\boxed{x = 1200 \text{ units}}$$

24. A standard machine tool and an automatic machine tool are being compared for the production of a component. The following data refers to the two machines. Setup time, machining time and machine rate for standard machine tool are : 30 min, 22 min and Rs.200 per hour, and those for automatic machine tool are 2 hours, 5 min and Rs. 800 per hour, respectively. The break-even production batch size above which the automatic machine tool will be economical to use, will be :

(a) 4 (b) 5

(c) 24 (d) 225

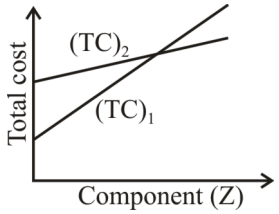
GPSC ARTO Pre 30.12.2018

Ans. (d) : Total cost of Z_1 component by using standard machine tool,

$$(TC)_1 = \left[\frac{30}{60} + \frac{22 \times Z_1}{60}\right] \times 200 = 100 + \frac{2200}{30} Z_1$$

Total cost of Z_2 component by using automatic machine tool,

$$(TC)_2 = \left[2 + \frac{5}{60} \times Z_2\right] \times 800 = 1600 + \frac{2000}{30} Z_2$$



Let breakeven point be Z number of components.

$$\therefore \quad 100 + \frac{2200}{30} Z = 1600 + \frac{2000}{30} Z$$

or $$\frac{200}{30} Z = 1500$$

$$\boxed{Z = \frac{1500 \times 30}{200} = 225}$$

25. Break-even analysis can be used for

(a) Short run analysis

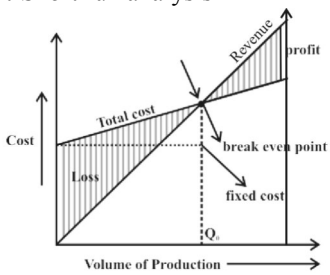
(b) Long run analysis

(c) Average of above two run analysis

(d) There is no such criterion

Vizag Steel (MT) 2017

Ans. (a) : Short run analysis



Break even analysis– It is the technique to study the total cost, total revenue and output relationship.

– It determines the level of operation/production where the firm neither earn profit nor suffers loss.

– At break even point total revenue equals to total cost.

26. A manufacturer has the following data regarding a product. Fixed cost per month = Rs. 50,000, variable cost per unit = Rs. 200, selling price per unit = Rs. 300, production capacity = 1500 units per month. If the production is carried out 80% of the rated capacity, then monthly profit (in Rs.) is

(a) 50,000 (b) 70,000

(c) 80,000 (d) 60,000

TSPSC AEE 2017

Ans. (b) : FC = Rs. 50,000, VC = Rs. 200 per unit

SP = Rs. 300 per unit

Production capacity = 1500 per month.

At 80% rate capacity = 0.80×1500

= 1200 unit/month

Total cost = FC + VC = 50000 + (1200 × 200)

= 290000

Sales = Selling price × production capacity

= 300 × 1200 = 360000 Rs.

Profit = Sales –T.C. = 360000–290000= Rs. 70000

27. The Break-Even Point represents

(a) The most economical level of operation of any industry

(b) The time when an industrial unit can run without loss and profit

(c) The time for maintenance of the plant

(d) The time when industry will undergo loss

Karnataka PSC Lect., 27.05.2017

Ans. (b) : The Break-even point represents the time when an industrial unit can run without loss and profit.

28. If excavation of earth is done manually then it costs Rs. 100 per cu.m. A hydraulic excavator can excavate at the fixed cost of Rs. 40000 plus a variable cost of Rs. 20 per cu.m. The quantity of earth work for which the cost of excavation by machine will be equal to the cost of manual excavation is

(a) 400 cu.m (b) 500 cu.m

(c) 1000 cu.m (d) 1200 cu.m

GPSC EE Pre, 28.01.2017

Ans. (b) : Given,

F.C = Rs. 40000

V.C. = 20 per cu.m.

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Manual costs = 100 per cu.m.
fixed cost + variable cost = manual costs
40000 + 20x = 100x
40000 = 80x
x = 500.0 per cu.m.

29. For any manufacturing setup the fixed cost may include the following costs ____.

(a) Overhead cost (b) Tooling cost
(c) Part material cost (d) None of the above

Punjab PSC SDE 12.02.2017

Ans. (a) : Fixed costs– Costs incurred by the enterprise which do not vary with increase or decrease of its production and sales activity. They include capital assets which however are a function of time, which depreciate irrespective of intensity of utilization

30. If F is the fixed cost of the assets for a given period, S is the unit selling price, V is variable cost per unit & P = Profit. Break Even Point is calculated by :

(a) F/(S-V) (b) (F-V)/S
(c) (F-S)/V (d) F-S+V

Punjab PSC SDE 12.02.2017

Ans. (a) : Break Even Point

$$= \frac{F}{S-V}$$

F = Fixed cost per period
V = Variable cost per unit
S = Selling price per unit

31. Initial cost for machine A is Rs. 1.5 lakh and the unit production cost on the machine is Rs. 75 each. For another machine B, the initial cost in Rs. 6 lakh, and the unit production cost on this machine is Rs. 15 only. The volume of production at the breakeven point is

(a) 6000 units (b) 9000 units
(c) 7500 units (d) 10500 units

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : At value of x the total cost of A and B are equal
150000 + 75x = 600000 + 15x
x = 7500 units

32. If the fixed expenses of a company are Rs. 1,38,00,000 variable expenses are Rs. 270,00,000 total sales volume is Rs. 500,00,000. Break even sales volume would be

(a) Rs. 330,00,000 (b) Rs. 300,00,000
(c) Rs. 30,00,000 (d) Rs. 3,00,000

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Given,
FC = Rs. 1,38,00,000
VC = Rs. 2,70,00,000
SV = Rs. 5,00,00,000
Break even sales volume

$$= \frac{\text{fixed cost} \times \text{sales}}{\text{sales} - \text{Variable cost}} = \frac{1,38,00,000 \times 500,00,000}{500,00,000 - 2,70,00,000}$$

= Rs. 300,00,000

33. Pick the wrong statement in the context of break-even analysis:

(a) Large angle of incidence in break-even chart indicates more profit rate
(b) Contribution = Selling price per unit-Total cost per unit
(c) Contribution per unit = Selling price per unit- Variable cost per unit
(d) Margin of safety = Profit/(P/V) ratio

RPSC AE (GWD) 06.05.2016

Ans. (b) : In a break-even analysis
- Large angle of incidence in break-even chart indicates more profit rate.
- Contribution per unit = selling price per unit – variable cost per unit
- Margin of safety = profit/$\left(\frac{P}{V}\right)$ ratio

34. A company is manufacturing a product which is sold for Rs. 15 per unit. The fixed cost of production is Rs. 50,000 and the variable cost is Rs. 10 per unit. How many units must be produced to break-even?

(a) 10000 (b) 5000
(c) 3333 (d) 2000

RPSC AE (GWD) 06.05.2016

Ans. (a) : Given-
Selling cost of product
s = Rs. 15 per unit
Fixed cost (F) = Rs. 50,000
Variable cost (v) = Rs. 10 per unit
At break even point analysis-
∵ Unit at BEP 'x_{BEP}' with profit (p) = 0

$$x_{BEP} = \frac{F}{s-v}$$
$$= \frac{50,000}{15-10}$$
$$= \frac{50,000}{5}$$

x_{BEP} = 10,000 unit

35. For a typical manufacturing setup, the fixed cost may include the following

(a) over head cost
(b) part material cost
(c) running cost
(d) initial investment on equipments and tools

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (d) : Fixed cost do not change for a given period with change in volume of production. Ex- rent, taxes, investment, increment on equipments and tools etc.

36. The fixed cost of the firm is Rs. 60,000/- per month. The variable cost is Rs. 10/- per unit and selling price is Rs. 50 per unit. The break even quantity will be

(a) 1300 (b) 1400
(c) 1500 (d) 1600

RPSC Lecturer 16.01.2016

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Ans. (c) : F.C. = 60000/month

Variable cost (v) = 10/unit

Selling price (s) = 50/unit

We know that,

Total sale = Total cost + Profit

$x \times s = F.C. + x \times v + Profit$

For break even quantity, profit = zero

$$x_{beq} \times s = FC + x_{beq} \times v$$

$$x_{beq} = \frac{FC}{s - v}$$

$$x_{beq} = \frac{60000}{50 - 10}$$

$x_{beq} = 1500$ unit

37. Break even point is the point where :

(a) fixed and variable cost line will intersect

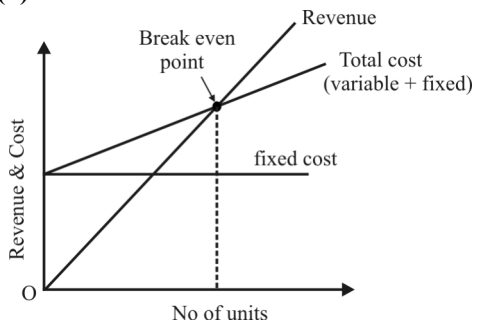
(b) variable and total cost lines intersect

(c) total cost and fixed cost lines intersect

(d) total cost and sales revenue lines intersect

Kerala PSC AE 06.08.2015

Ans. (d) : Break Evan Point–



Break even point in economics, is the point of which total cost and total revenue are equal. There is no net loss or gain.

38. Process X has a fixed cost of Rs. 30,000 per month and a variable cost of Rs. 8 per unit. Process Y has a fixed cost of Rs. 12,000 per month and a variable cost of Rs. 17 per unit. At which value, total cost of processes X and Y will be equal?

(a) 800 (b) 1200

(c) 1600 (d) 2000

(e) 2100

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : At value of x, the total cost of X and Y will be equal then.

$30000 + 8 \times x = 12000 + 17x$

$9x = 18000$

$x = 2000$ units.

39. Total unit cost is given by :

(a) $\frac{\text{Set up costs} + \text{Tooling costs}}{\text{Volume}} + \text{Variable cost}$

(b) $\frac{\text{Set up costs} + \text{Tooling costs}}{\text{Volume}} - \text{Variable cost}$

(c) $\frac{\text{Set up costs} - \text{Tooling costs}}{\text{Volume}} + \text{Variable cost}$

(d) $\frac{\text{Volume}}{\text{Set up costs} + \text{Tooling costs}} + \text{Variable cost}$

VIZAG STEEL MT 2015

Ans. (a) : Total unit cost is given by

$$= \frac{\text{Set up costs} + \text{Tooling Costs}}{\text{Volume}} + \text{Variable cost}$$

40. A machine costing Rs. 1,20,000/- was purchased in Dec. 2008. The installation and erection charges were Rs. 5000/- and the cost of labour for operating the machine is Rs. 6,000/- per month. The useful life of the machine is expected to be 12 years and its scrap value to be Rs. 5000/-. The yearly depreciation by straight line method will be

(a) 10000/- (b) 12000/-

(c) 6000/- (d) 1000/-

GPSC AE 26.07.2015

Ans. (a) :

41. Which of the following statements is not true for break even analysis?

(a) fixed cost does not vary with the total production

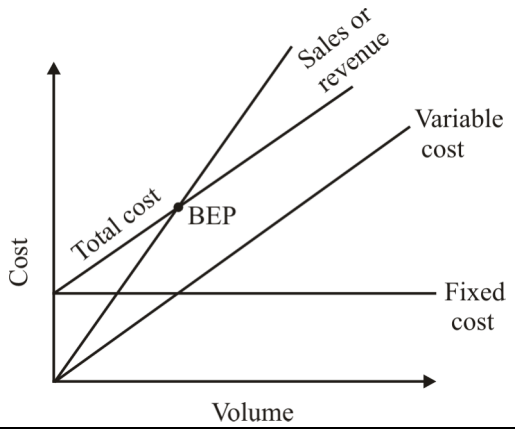
(b) total cost varies with the total production

(c) variable cost is dependent on total production

(d) break – even point is the point where fixed cost line and variable cost line intersect with each other on a volume cost graph.

Haryana PSC Civil Services Pre, 2014

Ans. (d) : Break even point is the point where total cost line cuts the total sales/revenue line on cost volume graph.



42. Break-even point does not depend on

(a) Fixed cost (b) Variable cost

(c) Overheads (d) Revenue

MPPSC State Forest Service Exam, 2014

Ans. (c) : Break even point is determined by total cost revenues do not directly affect the break even point. Sales revenue do to reaches its BEP. Overheads does not effect BEP.

$$BEP = \frac{FC}{SP - VC}$$

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43. Two alternatives can produce a product first has a fixed cost of Rs. 3000 and a variable cost of Rs. 20 per price. The second method has a fixed cost of Rs. 1500 and a variable cost of Rs. 30 per price. The break even quantity between the two alternative is

(a) 150 units (b) 25 units
(c) 75 units (d) 100 units

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Total cost = F.C + V.C

$T_{c1} = 3000 + 20.x$

$T_{c2} = 1500 + 30\ x$

at BEP $T_{c1} = T_{c2}$

$3000 + 20\ x = 1500 + 30\ x$

$10\ x = 1500$

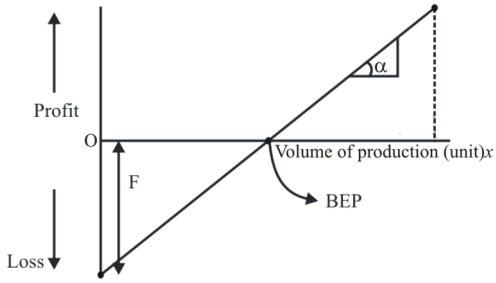
$x = 150$ units

44. Which of the following actions will not reduce break-even point?

(a) Reducing fixed cost
(b) Reducing variable cost
(c) Increasing sales price
(d) Decreasing sales price

RPSC AEN Pre-2013

Ans. (d) :



Total sale (S) = F + P + V

$sx = F + P + vx$

$P = (s - v)\,x - F$

At BEP, Profit (P) = 0

$$\Rightarrow \quad \boxed{x_{BFP} = \frac{F}{s - v}}$$

As the above relation shows, x_{BEP} is inversely proportional to sale price of unit. So, decreasing sale price will not reduce break even point.

45. Production cost refers to prime cost plus :

(a) factory overheads
(b) factory and administration overheads
(c) factory, administration and sales overheads
(d) factory, administration, sales overheads and profit

J&K PSC Civil Services Pre, 2013
ESE 1995

Ans. (a) : Production cost includes only prime cost and factory overheads (total direct cost including raw material and labour.)

46. Which of the following is not included in prime cost?

(a) Direct overheads
(b) Factory overheads
(c) Direct labour cost
(d) Direct materials cost

RPSC AEN Pre-2013

Ans. (b) : Prime (or) Direct Cost- Direct materials, direct labour, direct expenses.
Factory Overhead or Indirect Cost- Indirect material, indirect labour indirect expenses.

47. Fixed investments for manufacturing a product in a particular year is ` 60,000. The estimated sales during the year is ` 2,00,000. The variable cost per unit for this product is ` 4. If each unit is sold at ` 14, then the break even point will be

(a) 6000 units (b) 15000 units
(c) 9000 units (d) 10000 units

APPSC AEE 2012

Ans. (a) : Fixed cost = 60000 rupees
variable cost per unit = 4 rupees
sales price per unit = 14 rupees
Break even point = fixed cost
Break even point

$$= \frac{\text{fixed cost}}{\text{sales price per unit} - \text{Variable cost per unit}}$$

$$BEP = \frac{60000}{14 - 4}$$

$$\boxed{BEP = 6000 \text{ unit}}$$

48. Break-even analysis chart is drawn between

(a) overhead cost and fixed cost
(b) volume of production and income
(c) material cost and labour cost
(d) none of these

TRB Poly. Lect., 2012

Ans. (b) : Break-even analysis chart is drawn between volume of production & income. Graphical device used to determine the break-even point and profit potential under varying condition of output & costs i.e. known as break-even chart.

49. BEP indicates the recovery of

(a) variable costs only
(b) both fixed and variable costs
(c) fixed cost only
(d) both fixed and variable costs along with margin of profit

UJVNL AE 2016
UKPSC AE 2012 Paper-I

Ans. (b) : BEP indicates the recovery of both fixed and variable cost. In very simple terms, the breakeven point is the moment where your business has made enough money to recover cost.

50. What is the break-even point for the following situations: Fixed cost of product Rs. 2000; variable cost of product Rs. 5; price per unit Rs. 10?

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(a) 4 units (b) 40 units
(c) 400 units (d) 1000 units

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : Break even point (BEP)

$$= \frac{\text{Fixed cost}(F)}{\text{Price per unit}(S) - \text{Variable cost}(V)} = \frac{F}{S-V}$$

Given, Variable cost (V) = Rs. 5
Total sale or Revenue cost (S) = Rs. 10
At BEP,

$$X_{BEP} = \frac{2000}{10-5} = 400 \text{ units}$$

51. Which one of the following constitutes a fixed capital?
(a) Purchase of raw material and supplies
(b) Equipment and plant maintenance cost
(c) Purchase of equipment and selling expenses
(d) None of the above

RPSC AE GWD, 2011

Ans. (c) : Equipment in plant maintenance cost comes under a fixed capitals.

52. A list of various components of the total cost associated with the item transport is given below.
(1) Direct labour
(2) Direct material
(3) Manufacturing overheads
(4) Administrative and distribution overheads
Which of these forms the variable cost?
(a) (1) and (2) (b) (3) and (4)
(c) (2) and (3) (d) (1) (2) and (4)

RPSC AE GWD, 2011

Ans. (a) : Direct material and direct labour are variable cost and manufacturing overheads, administrative and distribution overheads are fixed costs.

53. A firm is operating at a fixed cost of Rs. 4,00,000/-, a revenue of Rs. 11/unit and a variable cost of Rs. 6/unit. If the number of units produced annually is 1,00,000 the annual profit is :
(a) Rs. 25,000/- (b) Rs. 5,00,000/-
(c) Rs.1,30,000/- (d) Rs.1,00,000/-

UPSC JWM Advt. No.-52/2010

Ans. (d) : Given;
Let, P = Annual Profit
Fixed Cost (F) = Rs. 400000
Selling Cost per unit (s) = Rs. 11/unit
Variable cost per unit (v) = Rs. 6 /unit
Number of units (x) = 100000 units

$$\because \quad x = \frac{F+P}{s-v}$$

$$\Rightarrow \quad 100000 = \frac{400000+P}{11-6}$$

$$400000 + P = 500000$$

$$P = 100000$$

$$\Rightarrow \quad \boxed{\text{Profit (P)} = \text{Rs. } 1,00000}$$

54. The annual fixed cost of a product is Rs. 2,00,000 and the annual net profit is Rs. 40,000 the average monthly sale being 820 units. A new design is contemplated, involving expenditure for preparations amounting to Rs. 80,000 to be returned in two years. It is expected that with the new method the profit-volume ratio may be increased by 5%. For the same net profit to be realized, the annual sales figure for the new design should be :
(a) 9840 (b) 9920
(c) 10920 (d) 11220

APPSC IOF, 2009

Ans. (c) : $\left(\frac{P}{V}\right)_{ratio} = \frac{F+P}{S} = \frac{F+P}{s \times x}$

Where, S = Total sales
s = Selling price per unit
For (1) case when,
F = 200,000
P = 40,000

$$\left(\frac{P}{V}\right)_{ratio} = \frac{240000}{s \times 820 \times 12}$$

For (2) case when,
F = 200,000 + 40,000 (for one year)

$$\left(\frac{P}{V}\right)_{ratio} = \frac{240000+40,000}{s \times x} = \frac{280,000}{s \times x}$$

According to the given condition,

$$\frac{240,000}{s \times 820 \times 12} \times \frac{105}{100} = \frac{280,000}{s \times x}$$

$$\boxed{x = 10,933.33 \text{ units}}$$

Nearest option is (c).

55. Two alternative methods can produce a product. First method has a fixed cost of Rs. 2,000 and variable cost of Rs. 20 per piece. The second method has a fixed cost of Rs. 1,500 and a variable cost of Rs. 30. The break-even quantity between two alternatives is :
(a) 25 (b) 50
(c) 75 (d) 100

APPSC IOF, 2009, ESE 1996

Ans. (b) : $\frac{\text{Fixed cost}}{\text{Selling cost} - \text{Variable cost}} = x_{BEP}$

$$\frac{2000}{s-20} = \frac{1500}{s-30}$$

$$4s - 120 = 3s - 60$$

$$\boxed{s = \text{Rs}60}$$

$$x_{BEP} = \frac{2000}{60-20} = \frac{2000}{40} = 50 \text{ unit}$$

56. The probability distribution for the sale of a product is as follows:

Sale quantity	Probability
1	0.3
2	0.4
3	0.3

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On the sale of an item the profit is Rs. 500. The expected profit is

(a) Rs. 1000 (b) Rs. 500
(c) Rs. 250 (d) Rs. 200

DRDO Scientists 2008

Ans. (a) : Expected profit = 500 [1 × 0.3 + 2 × 0.4 + 3 × 0.3]
= 500 × 2 = 1000

57. Difference of actual sales and break even point is called

(a) margin of safety (b) price-cost margin
(c) contribution (d) none of these

TNPSC AE, 2008

Ans. (a) : Margin of safety (MOS) is the difference between the actual sales and the sales at break-even level.
Margin of safety = Actual sales - Break even sales

$$MOS = x_{sales} - x_{BEP}$$

58. The input-output analysis is often called as

(a) cost benefit analysis (b) value analysis
(c) non-pricing analysis (d) none of these

TNPSC AE, 2008

Ans. (a) : The input-output analysis is often called as cost benefit analysis. Purpose of cost-benefit analysis to determine if the project is worthwhile financially (justification/feasibility).

59. The break-even point is least affected by:

(a) product mix (b) selling price
(c) fixed cost (d) volume of production

J & K PSC Screening, 2006

Ans. (a) : The break even point is least affected by product mix.

60. Break even point increases with :

(a) increase in fixed cost
(b) increase in variable cost for a given value of price
(c) decrease in unit contribution
(d) All of the above

J & K PSC Screening, 2006

Ans. (d) : Break even point increases with
- Increase in fixed cost.
- Increase in variable cost for a given value of price.
- Decrease in unit contribution

Break even point is a point at which there is neither profit nor loss occurs.

2. Inventory Control

61. If the demand for an item is doubled and the ordering cost is halved, the economic order quantity for the item will be :-

(a) A half of the earlier quantity
(b) Double of the earlier quantity
(c) Increased by a factor of
(d) Will remain unchanged

Vadodara Muncipal Corp. DEE, 2018
Punjab PSC (Lect.) 06.08.2017
TSPSC AEE 2017, RPSC VPITI 14.02.2016
GPSC Asstt. Prof. (Prod.) 21.08.2016
GPSC AE 26.07.2015, TSPSC Managers, 2015
UKPSC AE-2013, Paper-I, TRB Asstt. Prof., 2012
J&K PSC Civil Services Pre, 2010
ESE 2009, GATE 1995

Ans. (d) : $[EOQ]_1 = \sqrt{\frac{2DC_o}{C_h}}$

If $D_2 = 2D$

$$[EOQ]_2 = \sqrt{\frac{2\times[2D]\times\left[\frac{C_o}{2}\right]}{C_h}}$$

Then, $[EOQ]_1 = [EOQ]_2$

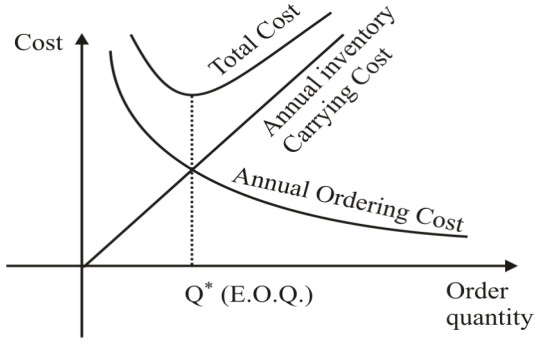
So it will remain unchanged.

62. In inventory control theory, the economic order quantity is :

(a) average level of inventory
(b) optimum lot size
(c) capacity of a warehouse
(d) lot size corresponding to break-even analysis

Mizoram PSC (Insp. of Factories) 2019 Paper-III
TNPSC AE 2019
Nagaland PSC (CTSE) 2018, Paper-I
Karnataka PSC AE, 10.09.2017
Sikkim PSC (Under Secretary), 2017
RPSC VPITI 14.02.2016
MPPSC AE 2016
J&K PSC Civil Services Pre, 2013
TNPSC ACF 2012, RPSC AE GWD, 2011
ESE 1995

Ans. (b) : Economic Order Quantity (E.O.Q.)
There are mainly two costs associated with inventory. Ordering cost and inventory carrying cost. Above two cost is opposite in nature so order quantity must be balanced quantity, to obtain optimum lot size or minimum total cost. The quantity is known as economic order quantity.



63. ABC analysis deals with

(a) analysis of process charts
(b) flow of material
(c) ordering schedule of jobs
(d) controlling inventory costs

Nagaland PSC CTSE 2017, Paper-I

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Vizag Steel (MT) 2017, TSPSC AEE 2015 (KPSC AE. 2015), APPSC Poly. Lect. 2013 TRB Asstt. Prof., 2012, UKPSC AE 2012 Paper-I TNPSC AE, 2008

Ans. (d) : ABC analysis deals with controlling inventory costs.

64. There are two products P and Q with the following characteristics

Product (Units)	Demand (Rs./order)	Order cost (Rs./unit /year)	Holding cost
P	100	50	4
Q	400	50	1

The Economic Order Quantity (EOQ) of products P and Q will be in the ratio:

(a) 1 : 1 (b) 1 : 2
(c) 1 : 4 (d) 1 : 8

Mizoram PSC (Insp. of Factories) 2019 Paper- III VIZAG STEEL MT 2015, APGENCO AE 2012 GATE 2004, ESE 1994

Ans. (c) : For Product 'P'–

Demand (D) = 100 units

C_o = Rs 50 / order

C_h = Rs.4/unit/order

$$(EOQ)_P = \sqrt{\frac{2DC_o}{C_h}}$$

$$= \sqrt{\frac{2\times 100\times 50}{4}}$$

$$(EOQ)_P = 50$$

For product 'Q' –

D = 400 units

C_o = Rs.50 /order

C_h = Rs.1/unit/order

$$(EOQ)_Q = \sqrt{\frac{2\times 400\times 50}{1}}$$

$$= 200$$

$$\frac{(EOQ)_P}{(EOQ)_Q} = \frac{50}{200} = \frac{1}{4}$$

65. A company uses 2555 units of an item annually. Delivery lead time is 8 days. The reorder point (in number of units) to achieve optimum inventory is

(a) 7 (b) 8
(c) 56 (d) 60

RPSC ACF & FRO, 26.02.2021 HPPSC Asstt. Prof. 18.11.2016 VIZAG STEEL MT 18.08.2013 VIZAG STEEL MT 10.06.2012 GATE 2009

Ans. (c) : D = Demand per year

L = Lead time in days

α = Demand per day

$$\alpha = \frac{D}{\text{Working days / year}}$$

$$= \frac{2555}{365}$$

$$= 7$$

Reorder point for optimum inventory.

$$R.O.P = \alpha.L$$

$$= 7 \times 8$$

$$\boxed{R.O.P = 56}$$

66. In the EOQ model, if the unit ordering cost is double, the EOQ

(a) is halved
(b) is doubled
(c) increases 1.414 times
(d) decreases 1.414 times

RPSC Vice Principal ITI 2018 GPSC Asstt. Director of Transport 05.03.2017 UKPSC AE 2012 Paper-I ESE 2007

Ans. (c) :

$$EOQ = \sqrt{\frac{2DC_o}{C_h}}$$

When C_o is doubled i.e. $2C_o$, the EOQ is

$$EOQ_{new} = \sqrt{2\left(\frac{2DC_o}{C_h}\right)}$$

$$= \sqrt{2\times (EOQ)_{old}}$$

$(EOQ)_{new} = 1.414\ (EOQ)_{old}$

Thus, EOQ increases by 1.414 times.

67. In an ideal inventory control system, the economic lot size for a part is 1,000. If the annual demand for the part is doubled, then new economic lot size required will be :

(a) 500 (b) 2,000
(c) 1,000/√2 (d) 1000√2

AAI Jr. Executive 26.03.2021 OPSC AEE 2019 Paper-I GPSC Asstt. Director of Transport 05.03.2017 GATE 1989

Ans : (d) :

Economic lot size = 1000

Economic order quantity,

$$EOQ = \sqrt{\frac{2C_oD}{C_h}}$$

Where,

D = Annual demand

C_o = Ordering cost per order

C_h = Unit holding cost per year

$$\Rightarrow 1000 = \sqrt{\frac{2C_oD_1}{C_h}} \quad \text{——— (i)}$$

$\because \quad D_2 = 2D_1 (given)$

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Again,

$$(EOQ)_2 = \sqrt{\frac{2\times 2C_oD_1}{C_h}} \quad \text{——————(ii)}$$

Dividing equation (ii) by (i), we get

$$\frac{(EOQ)_2}{1000} = \frac{\sqrt{\frac{2\times 2C_oD_1}{C_h}}}{\sqrt{\frac{2\times C_oD_1}{C_h}}}$$

$$(EOQ)_2 = 1000\sqrt{2}$$

68. In ABC analysis the C items are those which represent

(a) small percentage of total annual consumption value
(b) high percentage of total annual consumption value
(c) small percentage of closing inventory value
(d) high percentage of closing inventory value

Karnataka PSC AE, 10.09.2017
Kerala PSC IOF 19.04.2016
MPPSC AE 08.11.2015, TRB Poly. Lect., 2012

Ans. (a) : ABC analysis deal with controlling inventory cost It is divided into three categories, A,B,&C in descending Value. A has highest Value items, B is lower value then A, and C has the lowest value of consumption.

69. A-B-C analysis is used in-

(a) CPM (b) PERT
(c) Inventory Control (d) All of above

HPPSC Asstt. Prof. 20.11.2017
GMB AAE 25.06.2017
GWSSB DEE 07.07.2016
Assam PSC CCE Pre 2015

Ans. (c) : In material management, ABC analysis is an inventory categorization technique. ABC analysis divides an inventory into three categories.

(i) A 'items' with very tight control and accurate record.
(ii) B 'items' with less tightly controlled and good records.
(iii) C 'items' with the simplest controls possible and minimum records.

70. An item can be purchased for Rs. 100. The ordering cost is Rs. 200 and the inverse carrying cost is 10% of the item cost per annum. If the annual demand is 4000 units, the economic ordering quantity is

(a) 50 (b) 100
(c) 200 (d) 400

Karnataka PSC Lect., 27.05.2017
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2002

Ans. (d) : Ordering cost (C_o) = Rs. 200/order
Unit cost (C) = Rs. 100
Inventory carrying cost (C_h) = 10% of unit cost

$$\text{So, } C_h = \frac{10\times 100}{100} = \text{Rs.}10/\text{unit}/\text{year}$$

Annual demand (D) = 4000 units/year

$$EOQ = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2\times 4000\times 200}{10}}$$

EOQ = 400 units

71. The time period between placing an order and its receipt in stock is known as :

(a) Carrying time (b) Shortage time
(c) Over time (d) Lead time

JPSC AE 10.04.2021, Paper-II
UPRVUNL AE 05.07.2021
HPPSC Asstt. Prof. 18.11.2016
VIZAG STEEL MT 2011

Ans. (d) : Lead time – It is the time period between placing an order and its receipt in stock.

72. In the ABC method of inventory control. Group A constitutes costly items. What is the usual percentage of such items of the total items?

(a) 40 to 50% (b) 20 to 30%
(c) 30 to 40% (d) 10 to 20%

UPRVUNL AE 07.10.2016
UPRVUNL AE 21.08.2016
UKPSC AE 2007 Paper -I
ESE 2006

Ans. : (d) In the ABC method of inventory control. Group A constitutes costly items, 10–20% is the usual percentage of the total items.

'A' Items – Most costly and valuable items 10 to 20% of items accounts for 75% of the total capital invested in the inventory.
'B' Items – 20 to 30% of items account for 20 to 30% total invested capital.
'C' Items – 70 to 100% items accounts for 10% of invested capital.

73. Annual demand for a product costing Rs. 100 per piece is 900. Ordering cost per order is Rs. 100 and holding cost is Rs. 2 per unit per year. The economic lot size is then

(a) 200 (b) 300
(c) 400 (d) 500

TSPSC Manager (Engg.) HMWSSB 12.11.2020
KPSC AE, 2015
UKPSC AE 2012 Paper-I
ESE 1997

Ans. (b) : Given,
Demand (D) = 900
Ordering cost (C_o) = 100
Holding cost (C_h) = 2

$$\text{Economic lot size (EOQ)} = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2\times 900\times 100}{2}}$$

EOQ = 300

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74. Market demand for springs is 8 lakh per annum. A company purchases these springs in lots and sells them. The ordering cost is Rs. 1200/-. The storage cost is Rs. 120/- per stored piece per annum. The economic order quantity is:

(a) 400 (b) 2,828
(c) 4,000 (d) 8,000

HPPSC ADF 05.03.2019
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2003

Ans. (c) : Given,

$D = 8 \text{ lakh} = 8 \times 10^5$

$C_o = \text{Rs. } 1200$

$C_h = \text{Rs. } 120$

$$EOQ = \sqrt{\frac{2DC_o}{C_h}}$$

$$EOQ = \sqrt{\frac{2 \times 8 \times 10^5 \times 1200}{120}}$$

$$EOQ = 4000$$

75. When the ordering cost is increased to four times, the EOQ will be increased to

(a) Two times (b) Three times
(c) Eight times (d) Ten times

RPSC ADE 2016, ESE 2016
MPPSC AE 08.11.2015

Ans. (a) : $\because EOQ = \sqrt{\frac{2DC_o}{C_h}}$

Where, D = Annual demand
C_o = ordering cost
C_h = holding cost

$$\frac{(EOQ)_1}{(EOQ)_2} = \frac{\sqrt{\frac{2DC_o}{C_h}}}{\sqrt{\frac{2D \times 4C_o}{C_h}}}$$

$$= \sqrt{\frac{1}{4}} = \frac{1}{2}$$

$\Rightarrow$ $(EOQ)_2 = 2 \times (EOQ)_1$

Two times.

76. A manufacturer can produce 12000 bearings per day. The manufacturer received an order of 8000 bearings per day from a customer. The cost of holding a bearing in stock is Rs.0.20 per month. Setup cost per production run is Rs.500. Assuming 300 working days in a year, the frequency of production run should be -

(a) 4.5 days (b) 4.5 months
(c) 6.8 months (d) 6.8 days

GPSC Asstt. Director of Transport 05.03.2017
NPCIL ET, 2015, GATE 2014

Ans. (d) :

D = 8000 × 300 bearing/year = 2400000 bearing/year

P = 12000 bearing/day

C = 800 bearing/day

$C_h = 0.20 \times 12$/year

$C_0 = 500$/production

$$Q = \sqrt{\frac{2D.C_0}{C_h}} \times \left(\frac{p}{p-C}\right)$$

$$= \sqrt{\frac{2 \times 2400000 \times 500}{0.20 \times 12}} \times \left(\frac{12000}{12000-8000}\right)$$

$$= 54772.526$$

Time, $T = \frac{Q}{C}$

$$T = \frac{54772.526}{8000}$$

$$T = 6.84 \text{ days}$$

77. Economic order quantity is the quantity at which the cost of carrying is _____.

(a) minimum
(b) maximum
(c) Less than the cost of ordering
(d) Equal to the cost of ordering

VIZAG STEEL MT 24.01.2021 Shift-II
Mizoram PSC (Insp. of Factories) 2019 Paper-III
TNPSC AE 2013, ESE 2002

Ans. (d) : Economic Order Quantity (EOQ) is the quantity at which the cost of carrying is equal to the cost of ordering.

$$\boxed{EOQ = \sqrt{\frac{2DC_o}{C_h}}}$$

78. If the annual demand of an item becomes half, ordering cost double, holding cost one-fourth and the unit cost twice, then what is the ratio of the new EOQ and earlier EOQ?

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$
(c) 2 (d) $\sqrt{2}$

VIZAG STEEL MT 24.01.2021 Shift-II
GPSC ARTO Pre 30.12.2018
ESE 2006

Ans. (c) : Given,

$$EOQ = \sqrt{\frac{2DC_o}{C_h}}$$

If $D_1 = \frac{D}{2}$, $(C_o)_1 = 2C_o$

$$(C_h)_1 = \frac{C_h}{4}$$

$$(EOQ_1) = \sqrt{\frac{2D_1(C_o)_1}{(C_h)_1}} = \sqrt{\frac{2 \times \frac{D}{2} \times 2C_o}{\frac{C_h}{4}}}$$

$$= \sqrt{\frac{2DC_o \times 4}{C_h}} = 2\sqrt{\frac{2DC_o}{C_h}}$$

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$$\text{Ratio} = \frac{(EOQ)_1}{(EOQ)} = \frac{2\sqrt{\frac{2DC_o}{C_h}}}{\sqrt{\frac{2DC_o}{C_h}}} = 2$$

79. Annual demand for window frames is 10000. Each frame costs Rs 200 and ordering cost is Rs 300 per order. Inventory holding cost is Rs 40 per frame per year. The supplier is willing to offer 2% discount if the order quantity is 10000 or more and 4% if order quantity is 2000 or more. If the total cost is to be minimized, the retailer should.

(a) Order 200 frames every time
(b) Accept 4% discount
(c) Accept 2% discount
(d) Order economic order quantity

NPCIL ET, 2015
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2010

Ans. (b) : Given, D = 10,000 units, C_u = Rs. 200 each frame

C_o = Rs. 300/order, C_u = Rs. 40 per frame/year

$$EOQ = \sqrt{\frac{2DC_o}{C_u}} = 387.298$$

$$\text{Total cost, } TC = DC + \frac{Q}{2}C_u + \frac{D}{Q}\cdot C_o$$

At EOQ

$$TC = (10,000)(200) + \frac{387.298}{2}(40) + \frac{10,000}{387.298}(300)$$

= Rs. 2015492

At Q = 1000, discount of 2%

$$TC = (10,000)(200)(0.98) + \frac{1000}{2}(40) + \frac{10000}{1000}(300)$$

= Rs. 1983000

At Q = 2000, discount of 4%

$$TC = (10000)(200)(0.96) + \frac{2000}{2}(40) + \frac{10000}{2000}(300)$$

= Rs. 1961500

So, the minimum total cost is with a discount of 4%.

80. In computing Wilson's lot size for an item, by mistake the demand rate estimate used was 40% higher than the true demand rate. Due to this error in the lot size computation, the total cost of setups plus inventory holding per unit time would rise above the optimum by approximately

(a) 1.4% (b) 6.3%
(c) 18.3% (d) 8.7%

AAI Jr. Executive 26.03.2021
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 1999

Ans. (c) : Given, $D_1 = D$, $D_2 = 1.4\ D$

Actual cost, $TIC_1 = \sqrt{2DC_oC_h} = C$

Computed cost, $TIC_2 = \sqrt{2\times1.4\times D\times C_o\times C_h} = 1.183C$

$$\text{Error} = \frac{TIC_2 - TIC_1}{TIC_1}\times 100 = \frac{1.183-1}{1}\times 100$$

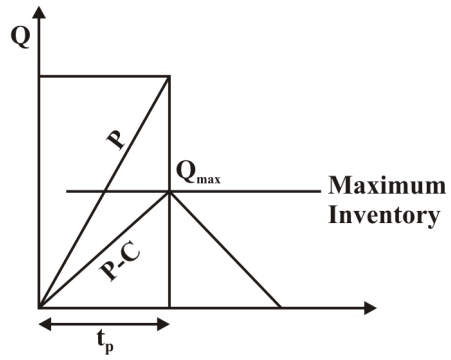
Error = 18.3%

81. In a machine shop, of 15 mm diameter are produced at rate of 1000 per month and the same is consumed at a rate of 500 per month. The production and consumption continue simultaneously till the maximum inventory is reached. Then inventory is allowed to reduce to zero due to consumption. The lot size of production is 1000. If backlog is not allowed, the maximum inventory level is

(a) 400 (b) 500
(c) 600 (d) 700

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2007

Ans. (b) :



$$Q_{max} = t_p(P\text{-}C) = \frac{Q}{P}(P-C) = Q(1-C/P)$$

$$Q_{max} = 1000\left(1-\frac{500}{1000}\right)$$

$\boxed{Q_{max} = 500}$ max inventory level

82. In machine shop, pins of 15 mm diameter are produced at a rate of 1000 per month and the same is consumed at 500 per month. The production and consumption continue simultaneously till maximum inventory is achieved. Then inventory is allowed to reach zero due to consumption. The lot size of production is zero. If backlog is not allowed, max inventory level is :

(a) 400 (b) 500
(c) 600 (d) 700

APGCL AM, 2021
HPPSC Asstt. Prof. 18.11.2016
GATE 2007

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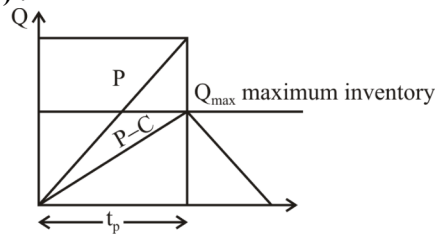
Ans. (b) :



$$Q_{max} = t_p(P-C) = \frac{Q}{P}(P-C)$$

$$= Q\left[1-\frac{C}{P}\right]$$

$$Q_{max} = 1000\left[1-\frac{500}{1000}\right]$$

$$= \frac{1000}{2}$$

$$Q_{max} = 500$$

83. Set-up costs do not include

(a) Labour costs of preparing the machines for the production of the item

(b) Ordering cost of raw material

(c) Maintenance cost of the machines

(d) Paperwork cost of processing the work order.

GPSC DEE, Class-2 (GWSSB) 04.07.2021
UPSC JWM Advt. No.-52/2010, GATE 1997

Ans. (d) : Set-up costs do not include paperwork cost of processing the work order.

84. The weekly sales for an item is A units. The ordering cost per or B rupees. The carrying cost per unit per month is C rupees. The EOQ will

(a) $\sqrt{\frac{8.67A}{BC}}$ (b) $\sqrt{\frac{8.67AB}{C}}$

(c) $\sqrt{\frac{8.67B}{AC}}$ (d) $\sqrt{\frac{8.67AC}{B}}$

TRB Asstt. Prof., 2012, ESE 2016

Ans. (b) : Weekly sales for an item = A
or yearly sales for an item = 52A
Ordering cost per order = Rs. B
Carrying cost per month = C
or Carrying cost per year = 12C

The $EOQ = \sqrt{\frac{2DC_O}{C_H}} = \sqrt{\frac{2\times 52A\times B}{12C}} = \sqrt{\frac{8.67AB}{C}}$

85. If rate of consumption of a particular item is 20 unit/year. The cost of procurement per order is Rs.40/-. The unit cost is Rs. 100/-. The inventory carrying cost is 0.16% & it depends upon the average stock. What would be Economic ordering quantity (E.O.Q.)?

(a) 8 items (b) 10 items

(c) 12 items (d) 14 items

Punjab PSC SDE 12.02.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Given, D = 20 unit /year
C_0 = 40 Rs./order
C_h = 0.16%C
C = unit cost price
= Rs. 100

$$EOQ = \sqrt{\frac{2DC_0}{C_h}}$$

$$EOQ = \sqrt{\frac{2\times 20\times 40}{0.16\times 100}}$$

$$= 10$$

86. A company requires 9000 units of a product annually. It costs Rs. 3 per unit the cost per purchase order is Rs. 300 and the inventory carrying cost per unit year is 20% of the unit cost. The economic order quantity (EOQ) for the company is :

(a) 4000 units (b) 3500 units

(c) 3000 units (d) 5000 units

HPPSC Asstt. Prof. 18.11.2016
RPSC AEN Pre- 2013

Ans. (c) : D = 9000 unit/year
C = 3, C_0 = Rs. 300/order
C_h = 20% C

$$Q = \sqrt{\frac{2DC_o}{C_h}}$$

$$Q = \sqrt{\frac{2\times 9000\times 300}{0.2\times 3}}$$

Q = 3000 Units

87. Which analysis is not related to inventory control ?

(a) ABC Analysis (b) VED Analysis

(c) FSN Analysis (d) FMS Analysis

APPSC Poly Lect. 13.03.2020
HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : ABC, VED and FSN is related to inventory control. FMS analysis is not related to inventory control.

ABC Analysis (Annual Usage Value)–
ABC analysis divides an inventory into three categories.
A items with very tight control and accurate records.
B items – with less tightly controlled
C items – with simples controls possible.

VED Analysis (Vital Essential Desirable)–
In this technique that classifies inventory based on its functional important.

FSN Analysis–(Fast moving slow moving and non moving). This analysis is helpful in identifying active items which need to be revised regularly and surplus items and non-moving items are examined.

FMS (Flexible Manufacturing System)–
It is a production method that is designed to quickly adapt to changes in the type and quantity of the product being manufactured.

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88. In ABC analysis of inventory control B, A and C category items respectively refer to

(a) High, medium, low value items
(b) Medium, high, low value items
(c) Low, medium, high value items
(d) Low, high, medium value items
(e) None of these

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (b) : In material management, ABC analysis is an inventory. Categorization technique ABC analysis divided an inventory into three.
Category 'A' items with very tight control and accurate records, 'B' items, with less tightly controlled and good records & 'C' items with simplest controls possible and minimal records.

89. The demand for a commodity is 100 units per day. Every time an order is placed, a fixed cost of Rs. 400 is incurred. Holding cost is Rs. 0.08 per unit per day. If the lead time is 13 days, then the economic lot size and the reorder point are in units

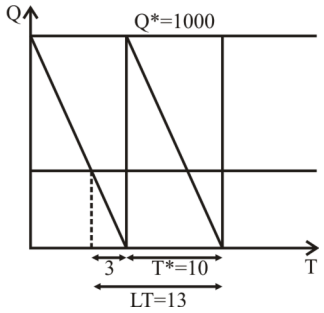
(a) 800 and 130 (b) 840 and 100
(c) 890 and 300 (d) 1000 and 300

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
NPCIL ET, 2015

Ans. (d) $EOQ = \sqrt{\frac{2DC_o}{C_h}}$

$= \sqrt{\frac{2\times100\times400}{0.08}} = \sqrt{10^6}$

= 1000 lot size



ROL is not more Q* so find effective lead time

$T^* = \frac{Q^*}{d} = \frac{1000}{100} = 10$

$\boxed{LT > T^*}$

Effective LT = LT – T*
= 13 – 10 = 3
ROL = L × d = 3 × 100 = 300

90. Vital-Essential-Desirable (VED) analysis is very useful for the control _____.

(a) of spare parts
(b) of dead stock
(c) of important items
(d) at department levels

VIZAG MT, 14.12.2020, RPSC ADE 2016

Ans. (a) : Vital-Essential-Desirable (VED) analysis is very useful for the control of spare parts.
VED analysis related to the classification of maintenance spare parts and denotes the essentiality of stocking spaces. The spares are classified into three categories in order of importance.

91. When ordering cost is increased to 64 times, EOQ will be increased to :

(a) 2 times (b) 4 times
(c) 8 times (d) Remains same

CGPSC AE 15.01.2021, UPRVUNL AE 2014

Ans. : (c) $EOQ = \sqrt{\frac{2AP}{C}}$

Where
A = Annual Demand
P = Cost per order
C= Annual Holding cost per unit

$EOQ = \sqrt{P}$
$EOQ = \sqrt{64}$
EOQ = 8

92. In ABC inventory control technique, maximum attention is given to those items which are

(a) surplus (b) more in demand
(c) costly (d) perishable in nature

TSPSC Manager (Engg.) HMWSSB 12.11.2020
GPSC Executive Engineer 23.12.2018

Ans. (c) : ABC (Always better control)- Annual usage value = (Annual Requirement) × Per unit cost)

- The 'ABC' Analysis is based on Pareto's Law .
- In 'ABC' analysis the items are classified in three main categories based on their respective usage value.
- **Category 'A' Items-** Most costly and valuable items.
- 10% of items accounts for 75% of total capital invested in inventory.
- **Category 'B' Items-** These items have average consumption value nearly 15% of the items in an inventory accounts for 15% for the total investment.
- **Category 'C' Items-** These items having low consumption value.
- Nearly 75% of inventory items account only for 10% of the total invested capital.

93. If orders are placed once a month to meet an annual demand of 6,000 units, then the average inventory would be :

(a) 200 (b) 250
(c) 300 (d) 500

Mizoram PSC (Insp. of Factories) 2019 Paper-III
CSE Pre-1994

Ans. (b)
Annual Demand = 6,000

Monthly Demand = $\frac{6,000}{12} = 500$

Monthly average Demand = $\frac{500}{2} = 250$

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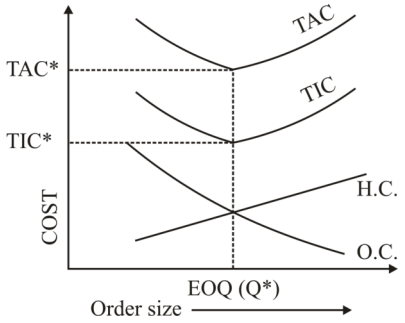
94. Increase in economic order quantity results in:-

(a) Increase in inventory carrying cost
(b) Decrease in ordering cost
(c) Decrease in total cost
(d) Total cost first decreases and then increases

UKPSC AE-2013, 2012 Paper-I

Ans. (d) : Increase in economic order quantity results in total cost first decreases and then increases.

If we change the EOQ (Q*), it can affect the different type of inventory costs in different ways. Large order size results in lower order costs because fewer orders need to be placed to cover the annual demand. This however results in higher holding costs because of the increase in inventory level.



Note- Option (c) is given by UKPSC.

95. The annual demand for an item is 3200 parts. Unit cost is Rs. 6 and the inventory carrying charges are estimated as 25% per annum. If the cost of one procurement is Rs. 150, what will be the number of orders per year?

(a) 4 (b) 6
(c) 8 (d) 10

NLC GET 17.11.2020 Shift-II
UKPSC AE-2013, Paper-I

Ans. (a) : Given as,

D = 3200 parts

c = Rs. 6/unit

$C_h = \frac{25}{100} \times 6 / \text{year}$

C_o = Rs. 150/year

Then,

$$[EOQ] = \sqrt{\frac{2DC_o}{C_h}}$$

$$= \sqrt{\frac{2 \times 3200 \times 150}{\left[\frac{25}{100}\right] \times 6}}$$

= 800 unit

Number of order per year

$$N = \frac{D}{[EOQ]} = \frac{3200}{800}$$

$$\boxed{N = 4}$$

96. In inventory control the optimum (EOQ) lot size is given by

(a) $\sqrt{\frac{2 \times \text{Carrying cost} \times \text{Annual demand}}{\text{Ordering cost}}}$

(b) $\sqrt{\frac{2 \times \text{Annual demand} \times \text{Ordering cost}}{\text{Carrying cost}}}$

(c) $\sqrt{\frac{4 \times \text{Demand} \times \text{Ordering cost}}{\text{Carrying cost}}}$

(d) $\sqrt{\frac{2 \times \text{Carrying cost} \times \text{Ordering cost}}{\text{Annual demand}}}$

TSPSC AEE 2015, VIZAG STEEL MT 2015

Ans : (b) $\sqrt{\frac{2 \times \text{Annual demand} \times \text{Ordering cost}}{\text{Carrying cost}}}$

97. A company has an annual demand of 1000 units, ordering cost of Rs. 100/order and carrying cost of Rs. 100/unit year. If the stock out costs are estimated to be nearly Rs. 400 each time the company runs out of stock, the safety stock justified by the carrying cost will be

(a) 4 (b) 20
(c) 40 (d) 100

VIZAG STEEL MT 18.08.2013, GATE 2004

Ans. (c) : Given, D = 1000 units, C_o = 100 Rs./order
C_h = 100/unit/order, C_s = 400 each time

$$Q^* = \sqrt{\frac{2DC_o}{C_n}\left(\frac{C_h + C_s}{C_s}\right)}$$

$$Q^* = \sqrt{\frac{2 \times 1000 \times 100}{100}\left(\frac{100 + 400}{400}\right)} = 50$$

$$S^* = Q^*\left(\frac{C_n}{C_s + C_n}\right) = 50\left(\frac{100}{500}\right) = 10$$

Safety stock justified by the carrying cost =

$$\boxed{\text{Maximum inventory level (m)} = Q^* - S^* = 50 - 10 = 40}$$

98. Buffer stock + Reserve stock + Safety stock equals

(a) Order quantity
(b) EOQ
(c) Reorder point
(d) Maximum inventory level

TNPSC AE 2017, GPSC AE 26.07.2015

Ans. (c) : Buffer stock + Reserve stock + Safety stock equals Reorder point.

99. In perpetual inventory control, the material is checked as it reaches its

(a) Minimum value (b) Maximum value
(c) Average value (d) Middle value

Vizag Steel (MT) 2017
APPSC AEE Mains 2016, Paper-III

Ans. (a) : In perpetual inventory control, the material is checked as it reaches its minimum value.

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100. Just in time manufacturing philosophy emphasizes on :

(a) man power (b) manufacturing
(c) profit (d) inventory

Kerala PSC AE 06.08.2015
(KPSC AE. 2015)

Ans. (d) : JIT manufacturing philosophy emphasizes on inventory-system

Just in time (JIT), associated with overproduction waiting and excess inventory. JIT production is to avoid the waste.

Benefits of JIT

- Reduced operating costs.
- Higher quality
- Shortened lead time
- Reduce inventory
- Increased flexibility and innovativeness
- greater performance.

101. Classifying items in A, B and C categories for selective control of inventory management is done by arranging items in the decreasing order of.

(a) Total inventory costs
(b) Value of item
(c) Annual usage value
(d) Demand of the item

UPPSC AE 13.12.2020, Paper-I
APPSC AEE 2012, ESE 1995

Ans. (c) : The ABC analysis suggests that inventories of an organization are not of equal value. Thus the inventory is grouped into three categories (A, B and C) in order of their estimated importance or annual usage value.

A Items– it is very important for organization.

B Items– It is important but less important than A items and more important than C items.

C Items– it is marginally important

102. A company offered the following price breaks for order quantity as follows:

1. Order quantity: < 500, price Rs 10.00

2. Order quantity: ≥ 500, price Rs 9.00

Order cost is Rs 100.00 per order, while the holding cost is 25% of the unit price. Determine the economic order quantity if the annual requirement is 2500.

(a) 447 (b) 471
(c) 500 (d) 600

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2006

Ans. (c) : D = 2500 units per year

C_o = Rs 100 per order

Inventory holding rate I = 0.25 of unit price

Case-1

Let EOQ is less than 500 for the inventory holding cost.

$$C_h = \frac{25}{100} \times 10 = 2.5$$

For this case,

$$EOQ = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2 \times 2500 \times 100}{2.5}} = 447.2$$

Total inventory cost (TIC) = $\sqrt{2DC_oC_h}$

$$= \sqrt{2 \times 2500 \times 0.25 \times 10 \times 100} = 1118.03$$

Case-2

Let EOQ is greater than 500, for this case

$$C_h = \frac{25}{100} \times 9 = 2.25$$

For this case, EOQ

$$\sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2 \times 2500 \times 100}{2.25}} = 471.4$$

This is against the assumption made, so correct EOQ = 447.2 = 447 unity.

When price break occurs, TIC at Q = 500

$$TIC = \frac{Q}{2} \times C_h + \frac{D}{Q} \times C_o$$

$$= \frac{500}{2} \times 9 \times 0.25 + \frac{2500}{500} \times 100$$

$$= 1062.5$$

Hence the quantity to be ordered is 500.

103. A shop owner with an annual constant demand of 'A' unit has ordering costs of Rs 'P' per order and carrying costs Rs 'I' per unit per year. The economic order quantity for a purchasing model having no shortage may be determined from

(a) $\sqrt{\frac{24P}{AI}}$ (b) $\sqrt{\frac{24AP}{I}}$
(c) $\sqrt{\frac{2AP}{I}}$ (d) $\sqrt{\frac{2AI}{P}}$

GPSC ARTO 01.05.2016, ESE 2002

Ans. (c) : Given, Demand (R) = A

Ordering Cost (C_o) = P/order

Carrying cost (C_c) = I per unit year.

Economic order quantity EOQ = $\sqrt{\frac{2RC_o}{C_c}} = \sqrt{\frac{2 \times A \times P}{I}}$

$$\boxed{EOQ = \sqrt{\frac{2AP}{I}}}$$

104. The maximum level of inventory of an item is 100 and it is achieved with infinite replenishment rate. The inventory becomes zero over one and half month due to consumption at a uniform rate. This cycle continues throughout the year. Ordering cost is Rs 100 per order and inventory cost is Rs 10 per item per month. Annual cost (in Rs) of the plan, neglecting material cost is

(a) 800 (b) 2800
(c) 4800 (d) 6800

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2007

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Ans. (d) : Number of cycle required per year

$$= \frac{\text{No. of months in year}}{\text{No. of months for inventory}}$$

Given,

Number of cycles a required per year

$$= \frac{12\,\text{month}}{1.5\,\text{month}} = 8\,\text{cycles}$$

Ordering cost = 8 × 100 = Rs. 800

Average inventory per cycle = 100/2 = 50

Inventory carrying cost = 50 × 12 × 10 = Rs. 6000

∴ Total cost = 800 + 6000 = 6800

105. Which principle serves as the baseline for ABC analysis?

(a) Samuelson (b) Aristotle

(c) Parker (d) Pareto

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (d) : ABC analysis is based on the Pareto principle. Its widely used in logistics and procurement for the purpose of optimising stock of goods as well as costs of keeping and replenishing that stock.

106. In a basic EOQ based inventory model, the total variable cost is the _____.

(a) quotient of carrying cost and ordering cost

(b) sum of carrying cost and ordering cost

(c) difference of carrying cost and ordering cost

(d) product of carrying cost and ordering cost

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (b) : In a basic EOQ based inventory model, the total variable cost is the sum of carrying cost and ordering cost.

Total variable cost = Carrying cost + Ordering cost

107. Reorder point is the _____.

(a) sum of demand rate per time period and lead time

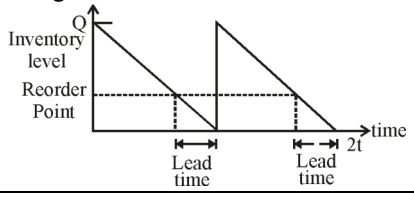
(b) quotient of demand rate per time and lead time

(c) product of demand rate per time period and lead time

(d) difference of demand rate per time period and lead time

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (c) : Reorder point is the product of demand rate per time period and lead time. Reorder point meets the demand during the lead time.



108. In E.O.Q. model, if holding cost is halved and ordering cost is doubled, the optimal order quantity will

(a) remain same (b) be doubled

(c) be halved (d) be one fourth

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : Given-

Q = Quantity, D = Annual Demand

C_o = Cost of placing order

C_h = Cost of holding $\quad \therefore C_{o\ new} = 2C_o$

$$\therefore C_{h\ new} = \frac{C_h}{2}$$

$$EOQ = \sqrt{\frac{2DC_o}{C_h}} \qquad \therefore D_{new} = D$$

$$\text{So, } EOQ_{new} = \sqrt{\frac{2 \times D \times C_o \times 2}{C_h / 2}}$$

$EOQ_{new} = 2 \times EOQ$

109. Given annual usage, U = 50 units, Procurement cost, P = Rs. 15 per order, cost per piece, C_1 = Rs. 100 per item and C_2, a percentage including insurance, taxes, etc = 10%, the economic order quantity EOQ will be:

(a) $\sqrt{15}$ (b) $\sqrt{100}$

(c) $\sqrt{150}$ (d) $\sqrt{105}$

JPSC AE 10.04.2021, Paper-II

Ans. (c) : Given,

D = 50

C_o = 15

C_h = 10% of Cost price

$$= \frac{10}{100} \times 100$$

$$EOQ = \sqrt{\frac{2DC_O}{C_h}}$$

$$= \sqrt{\frac{2 \times 50 \times 15}{10}}$$

$$= \sqrt{150}$$

110. Inventory carrying cost :

(a) Increases with quantity ordered

(b) Is a fixed cost with reference to quantity ordered

(c) Decreases with quantity ordered

(d) First increases then decreases

JPSC AE 10.04.2021, Paper-II

Ans. (a) : Inventory carrying cost is the sum of expenses associated with holding or storing goods. It includes ware housing, labor, insurance and rent. This cost increases with increase in quantity ordered.

111. ABC analysis helps segregating the items and helps the organization to control them. Amongst all items, A items are :

(a) low valued and constitute about 75% of total items.

(b) account for 15-20% of the total inventory and constitute about 15-20% of total items.

(c) account for 70-80% of the total inventory and constitute about 10% of total items.

(d) None of these

JPSC AE 10.04.2021, Paper-II

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Ans. (c) : ABC analysis is inventory characterization technique in which items are account for 70 – 80% of the total inventory and constitute about 10% of total items.

112. A bearing manufacturing company supplies 10,000 bearings per day to an automobile company. Company finds it can produce 25,000 per day when production run starts. Holding the stock for a year at Rs.2 per unit and setup cost of a production run is Rs.1,800. The economic lot size for each production run is:

(a) 265430 (b) 350765
(c) 124631 (d) 104642

AAI Jr. Executive 26.03.2021

Ans. (d) : Given, Demand d = 10,000
Production rate, p = 25000
C_h = Rs 2 per unit for a year
Demand = 10,000 × 365 = 3650000
Setup cost = 1800 Rs

$$EOQ,\ Q^* = \sqrt{\frac{2DC_o}{C_h}\left(\frac{p}{p-d}\right)}$$

$$= \sqrt{\frac{2\times 3650000\times 1800}{2}\left(\frac{25000}{15000}\right)}$$

Economic lot size, Q^* = 104642 units.

113. Which of the following is a type of indirect inventory?

(a) Raw material inventory
(b) Anticipation inventory
(c) Work in process inventory
(d) Finished goods inventory

UPRVUNL AE 05.07.2021

Ans. (b) : Anticipation Inventory–These inventory items are build up to meet anticipation demand in the future like big selling forecast government policy change price hike, strike shut down etc.

114. The 'E' in EOQ concept stands for _____.

(a) Economic (b) Early
(c) Essential (d) easy

UPRVUNL AE 05.07.2021

Ans. (a) : EOQ – Economic order quantity

115. Fixed quantity system are also termed as _____ systems of inventory control

(a) q (b) p
(c) qv (d) pq

UPRVUNL AE 05.07.2021

Ans. (a) :

Review system
- Q-system or fixed order sytem or Re-order level system or Two bin system
- P-system or fixed period sytem or Periodic review system

116. Inventory control is most closely related to _____.

(a) quality control
(b) quality management
(c) machining control
(d) materials management

UPRVUNL AE 05.07.2021

Ans. (d) :

117. fixed period systems are also termed as ____ systems of inventory control.

(a) q (b) pq
(c) qv (d) p

UPRVUNL AE 05.07.2021

Ans. (d) :

Review system
- Q-system or fixed order sytem or Re-order level system or Two bin system
- P-system or fixed period sytem or Periodic review system

118. Which equation among the following represents the economic batch quantity (EBQ)? (Consider R – annual requirement of products, S – preparation and set-up cost, C – constant per piece and I – inventory carrying charges.)

(a) $EBQ = \sqrt{\frac{RS}{2CI}}$ (b) $EBQ = \sqrt{\frac{RS}{CI}}$
(c) $EBQ = \sqrt{\frac{2RS}{CI}}$ (d) $EBQ = \sqrt{\frac{2CS}{RI}}$

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Economic bath quantity (EBQ) = $\sqrt{\frac{2RS}{CI}}$

Where,
R = annual requirement
S = preparation and set-up cost
C = constant per piece
I = Inventory carrying charges

119. The cost NOT associated with inventory is :

(a) Ordering cost (b) Building cost
(c) Carrying cost (d) Purchase cost

NLC GET 17.11.2020, Shift-II

Ans. (b) : Building cost is not inventory cost.
Inventory cost are-
(i) ordering costs (C_o)
(ii) carrying cost or holding cost (C_h)
(iii) Purchase cost (C)

120. The annual demand for an item is 3200 units. The unit cost is Rs 6 and inventory carrying charges are estimated as 25 percent per annum. If the cost of one procurement is Rs. 150, the EOQ is :

(a) 200 (b) 800
(c) 600 (d) 950

NLC GET 17.11.2020, Shift-II

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Ans. (b) : Given, Annual demand A = 3200
Cost of one procurement S = 150
Cost per unit C = 6
Inventory carrying cost I = 0.25

$$Q = EOQ = \sqrt{\frac{AS2}{CI}} = \sqrt{\frac{3200 \times 150 \times 2}{6 \times 0.25}} = \sqrt{320000 \times 2}$$

EOQ = $\sqrt{640000} = 800$ Units

121. Reorder point in inventory indicates the
(a) Quantity to be ordered
(b) Cost of the item
(c) Stock level of inventory when the order is to be placed
(d) None of these

RPSC IOF, 2020

Ans. (c) : Reorder point in inventory indicates the stock level of inventory when the order is to be placed.

122. A manufacturing company purchases 9000 parts of a machine for its annual requirement ordering for month usage at a time, each part costing Rs. 20. The ordering cost per order is Rs 15 and carrying charges are 15% of the average inventory per year. What should be the optimum order quantity?
(a) 200 (b) 300
(c) 400 (d) 500

APPSC Poly Lect. 13.03.2020

Ans. (b) : Optimum order quantity(Or) Economic order Quantity

$$(EOQ) = \sqrt{\frac{2DC_0}{C_h}}$$

Given data D = 9000 parts,
Ordering cost (C_0) = Rs 15/order
Carrying cost (OR) Holding Cost (C_h) = 15% of the average inventory/year
$\because$ Purchase cost of each part (C) = Rs. 20
$\therefore C_h = 20 \times 0.15 =$ Rs. 3/unit/year

$$\therefore \text{ EOQ} = \sqrt{\frac{2 \times 9000 \times 15}{3}} = \sqrt{90000}$$

EOQ = 300 units

123. ABC Analysis refers to :
(a) Always Be Closing
(b) Active Best Class
(c) Anyway Better Class
(d) Always Better Control

NLC GET 17.11.2020, Shift-II

Ans. (d) : ABC analysis refers to always better control. In materials management, ABC analysis is an inventory categorization technique. ABC analysis divides an inventory into three categories.
"A items" with less tightly controlled and good records and "C items" with the simplest controls possible and minimum record.

124. In a static inventory problem, the demand of an inventory item may be assumed to be under a number of conditions. They are _____
(a) risk and uncertainty
(b) certainty and uncertainty
(c) risk, certainty, uncertainty
(d) risk and certainty

VIZAG MT, 14.12.2020

Ans. (a) : In a static inventory problem, the demand of an inventory item may be assumed to be under a number of risk and uncertainty.

125. Consider a manufacturing system with a setup cost of Rs. 2000 per production run, a holding cost of 10 Rs unit/year, an annual demand of 14000 units, and a production rate of 16000 units year. What is the optimal number of production runs that minimize annual total cost for meeting the demand?
(a) 2.1 (b) 5.9
(c) 3.5 (d) 4.8

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Setup cost = 2000 per production run
Holding or inventory cost = Rs 10 / unit/year
Annual demand = 14000 unit.
Production rate = 16000 unit / year.

$$q_0 = \sqrt{\frac{2 \times \text{setup cost} \times \text{Annual demand}}{\text{Inventory cost}}} = \sqrt{\frac{2 \times 2000 \times 14000}{10}}$$

= 2366.43
Number of run per year =

$$\frac{\text{Annual demand}}{\text{Quantity per order}} = \frac{14000}{2366.43} = 5.916$$

126. What is the percentage increase in the total cost of inventory and setup computed under the EOQ model (from the optional total cost) if the actual amount ordered is 80%, of that of the economic order quantity?
(a) 1.20% (b) 3.75%
(c) 2.00% (d) 2.50%

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : C_h = Inventory cost or carrying cost
C_0 = ordering cost per order
D = Annual demand

$$TIC = \frac{D}{Q}.C_0 + \frac{Q}{2}.C_h$$

$$TIC_1 = \frac{D}{0.8Q}.C_0 + \frac{0.8Q}{2}.C_h$$

Since at EOQ $\frac{D}{Q}.C_0 = \frac{Q}{2}.C_h$

$$TIC = 2 \times \frac{Q}{2}.C_h \quad \text{.....(i)}$$

$$TIC_1 = \frac{1}{0.8} \times \frac{Q}{2}.C_h + 0.8 \times \frac{Q}{2}.C_h$$

$$TIC_1 = \left[\frac{1}{0.8} + 0.8\right].\frac{Q}{2}.C_h \quad \text{.....(ii)}$$

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Dividing (ii) by (i)

$$\frac{TIC_1}{TIC} = \frac{1.64}{0.8 \times 2} = 1.025$$

% change

$$\left(\frac{TIC_1 - TIC}{TIC}\right) \times 100 = 2.5\%$$

127. Which of the following in not a valid assumption made when working with the EOQ model?

(a) Shortage allowed
(b) Constant demand
(c) Infinite production rate
(d) Immediate delivery

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) :

128. The rate at which parts or products must be produced in order to meet customer demand is:

(a) just time (b) best time
(c) takt time (d) lead time

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : Takt Time–Takt time is the rate at which parts or product must be produced in order to meet customer demand.

129. Inventory carrying cost does NOT include:

(a) interest on investment
(b) cost of storage facility
(c) cost of inspection
(d) cost of insurance

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : Inventory carrying cost is the total of all expenses related to storing unsold goods. The total includes intangibles like depreciation and lost opportunity cost as well as warehousing cost. A business inventory carrying cost will generally 20% to 30% of its total inventory costs.

Four main component–

(1) Capital cost
(2) Storage space cost
(3) Inventory service cost
(4) Inventor risk cost

130. An oil engine manufacturer purchases lubricant cans at the rate of Rs 42 per piece from a vendor. The requirement of these lubricant cans is 1800 per year. If the cost per placement of an order is Rs. 16 and inventory carrying charges per rupee per year is 20 paise, the order quantity per order will be

(a) 91 cans (b) 83 cans
(c) 75 cans (d) 67 cans

ESE 2020

Ans. (b) : Given,

D = 1800 cans

C_o = Rs. 16/order

Carrying cost (C_c)= Rs. 0.2/ rupee/year

EOQ = ?

$$EOQ = \sqrt{\frac{2DC_o}{C_c}} = \sqrt{\frac{2 \times 1800 \times 16}{0.2 \times 42}}$$

$\simeq 83$ cans

131. Los Alamitos Store buys sets of steel at $40 per set from an outside vendor. It will sell 6,400 sets evenly throughout the year. The store's carrying cost is $8 per unit per year. The ordering cost is $ 100 per order. Total number of order per year is:

(a) 16 (b) 18
(c) 14 (d) 20

Oil India Limited Sr. Engineer (Production) 30.11.2019

Ans. (a) : Given-

D = 6400 sets

C_0 = $ 100 per order

C_h = $ 8 per unit

$$EOQ\ (Q^*) = \sqrt{\frac{2DC_0}{C_h}}$$

$$= \sqrt{\frac{2 \times 6400 \times 100}{8}}$$

$Q^* = 400$

$$\text{No. of orders } \frac{D}{Q^*} = \frac{6400}{400} = 16$$

132. Which of the following is not a direct inventory?

(a) Raw material (b) Purchased parts
(c) Finished goods (d) Tools

HPPSC ADF 05.03.2019

Ans. (d) :

133. If annual usage = 1000 pieces, expediting cost = Rs. 4 per order, Cost per piece = Rs. 250 inventory holding cost is 20% of average inventory cost of ordering = Rs. 6 per order, then the Economic ordering Quantity (EOQ) is:

(a) 40 (b) 35
(c) 30 (d) 20

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (d) : Economic ordering Quantity (EOQ)

$$= \sqrt{\frac{2DC_o}{C_h}} \qquad \text{Given, D = 1000}$$

C_o = Expenditing cost + Ordering cost

C_o = 4 + 6 = 10 Rs./order

C_h = 20% of cost per piece

$$C_h = \frac{1}{5} \times 250 = \text{ Rs. 50/unit}$$

$$EOQ = \sqrt{\frac{2 \times 1000 \times 10}{50}}$$

$$= \sqrt{400} = 20 \text{ unit}$$

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134. Demand for a product is 12,500,000 per annum, Company purchase this product in lots and sells them. The cost of purchase order is Rs. 1500 and cost of storage is Rs. 150 per piece per annum. EOQ will be:

(a) 6000 (b) 7000
(c) 8000 (d) 5000

SJVN ET 2019

Ans. (d) : Given,
D = 12,50,000
C_o= Rs. 1500
C_h= Rs.150

$$EOQ = \sqrt{\frac{2C_oD}{C_h}}$$

$$= \sqrt{\frac{2\times1250000\times1500}{150}}$$

$$= 5000$$

135. Inventory control in production, planning and control aims at

(a) Improving forecasting accuracy
(b) Cushioning production against market fluctuations
(c) Acceptable customer service at low capital investment in inventory
(d) Increasing the volume of transactions

TNPSC AE 2019

Ans. (c) : Inventory control in production, planning and control aims at acceptable customer service at low capital investment in inventory.

136. Calculate the total ordering cost incurred per year for the following inventory details connected to purchase of an item.
Annual consumption : 5000 units
Ordering cost per order : Rs. 100
Number of units ordered per order = 100 units

(a) Rs. 5,000 (b) Rs. 10,000
(c) Rs. 6,000 (d) Rs. 4,000

TNPSC AE 2019

Ans. (a) : D = 5000 units/year
Q = 100 units/order
C_o = Rs. 100/order
Then ordering cost per year

$$O.C. = \frac{D}{Q}\times C_o$$

$$= \frac{5000}{100}\times100$$

O.C. = Rs. 5,000/year

137. Just-in Time concept was developed by

(a) Taiichi Ohno (b) Kiichito Toyoda
(c) Ejji Toyoda (d) Demin

HPPSC AE 2018

Ans. (a) : Just in Time (JIT)–Just in Time (JIT) manufacturing, also known as just-in-time production or Toyota production system, is a methodology aimed primarily at reducing times within production system as well as response times from suppliers and to customers. its origin and development was in Japan by Taiichi Ohno, largely in the 1960s and 1970s and particularly at Toyota.

138. A cutting tool manufacturer purchases 2000 rack and pinion per year from a gear manufacturer. The ordering cost is Rs. 100 per order and carrying cost per unit per year is ` 10. The EOQ is:

(a) 150 (b) 200
(c) 250 (d) 100

AAI Jr. Executive 29.11.2018

Ans. (b) : Given, D = 2000
C_o = Rs. 100 / order
C_h = Rs 10/order/year

$$EOQ = \sqrt{\frac{2\times D\times C_o}{C_h}}$$

$$= \sqrt{\frac{2\times2000\times100}{10}}$$

EOQ = 200

139. In ABC analysis :

(a) item 'A' is large in number
(b) item 'B' is very expensive item
(c) item 'C' is very few in number
(d) item 'C' is low valued items
(e) item 'A' is low valued items

CGPSC AE 25.02.2018

Ans. (d) : ABC Analysis- ABC analysis is a type of inventory categorization method in which inventory is divided into three categories, A, B & C, in descending value. A has highest value items, B is lower value than A, and C has the lowest value.

140. A manufacturer has to supply 9,000 unit of product/per. The cost of one procurement is Rs. 10 and holding cost per unit is Rs. 2 per year. The replacement is instantaneous and no shortages are allowed. Number of orders per year will be:

(a) 10 (b) 20
(c) 30 (d) 40
(e) 50

CGPSC AE 25.02.2018

Ans. (c) : D = 9,000 unit
C_o = Rs. 10/order
C_h = Rs. 2/year
Then

$$EOQ = \sqrt{\frac{2\times D\times C_o}{C_h}} = \sqrt{\frac{2\times9000\times10}{2}}$$

EOQ = 300 unit
Then Number of order per year

$$N = \frac{D}{EOQ} = \frac{9000}{300} = 30$$

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141. A stocking finds that monthly demand for a particular ball pen is 2000. The price of each pen is 0.8 rupees and cost of placing an order is Rs. 20. The cost of stocking the pens per month is 10% of price of pen. What is EOQ?

(a) 1000 (b) 2000
(c) 500 (d) 750

RPSC AE 2018

Ans. (a) : Data given,

D = 2000/month

C_O = Rs. 20/order

$C_h = 0.8 \times \frac{10}{100}$

We know that

$$EOQ = \sqrt{\frac{2DC_o}{C_h}}$$

$$= \sqrt{\frac{2 \times 2000 \times 20 \times 100}{0.8 \times 10}}$$

EOQ = 1000 unit

142. The most suitable system for a retail shop is

(a) FSN Analysis (b) ABC Analysis
(c) VED Analysis (d) GOLF Analysis

RPSC Vice Principal ITI 2018

Ans. (a) : FSN stands for fast moving slow moving and non moving items. It looks at quantity consumption rate and how often the item is issued and used.

143. Losses due to deterioration, theft and pilferage come under :

(a) Inventory carrying charges
(b) Not any cost
(c) Losses due to theft
(d) Consumption cost

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : Holding cost– It is a cost associated with storing and maintaining inventory within production system. It include storage cost, handling cost, damage deterioration, theft pilferage, insurance cost, interest etc. This cost depends upon the quantity and period for which inventory is used.

144. The order cost per order of an inventory is Rs. 200 with annual carrying cost of Rs. 5 per unit. The Economic Order Quantity (EOQ) for an annual demand of Rs. 2000 unit is

(a) 100 (b) 400
(c) 480 (d) 500

GPSC EE Pre, 28.01.2017

Ans. (b) : Given-

C_o = Rs. 200 per order

C_h = Rs. 5 per unit

D = Rs. 2000 unit

EOQ (Economic Order Quantity) $= \sqrt{\frac{2DC_o}{C_h}}$

$$= \sqrt{\frac{2 \times 2000 \times 200}{5}}$$

$$= \sqrt{\frac{800000}{5}}$$

$$= \sqrt{160000}$$

= 400 unit

145. Low cost, higher volume items requires :

(a) non inspection (b) little inspection
(c) intensive inspection (d) 100% inspection

Kerala PSC Poly. Lect., 2017

Ans. (b) : Low cost, higher volume items requires little inspection.

146. Which inventory follows criticality of items control techniques?

(a) ABC analysis (b) VED analysis
(c) SDE analysis (d) MNG analysis

Karnataka PSC Lect., 27.05.2017

Ans. (b) : The VED analysis based on the critically of the items. If the items are arranged in the descending order of their criticality viz.

V → Vital items

E → Essential items

D → Desirable items, then more attention is paid to the V- type items.

147. The goal of basic EOQ model is to minimize

(a) order size
(b) ordering cost
(c) sum of purchasing and ordering cost
(d) sum of ordering and holding cost

Punjab PSC (Lect.) 06.08.2017

Ans. (d) : The goal of basic EOQ model is to minimize sum of ordering and holding cost.

148. We can reduce the materials cost by :

(a) Using systematic inventory control techniques
(b) Using the cheap material
(c) Reducing the use of materials
(d) Making hand to mouth purchase

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : We can reduce the materials cost by using systematic inventory control techniques.

149. In a quantity discount model O inventory control, the relevant costs are

(a) Annual purchase cost
(b) Annual order cost and annual carrying cost
(c) Annual purchase cost, annual order cost and annual carrying cost
(d) Annual order cost

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : In a quantity discount model 'O' inventory control, the relevant costs are, annual purchase cost, annual order cost and annual carrying cost.

150. For a given level of safety stock and EOQ ordering

(a) The Reorder point depends only on the rate of consumption
(b) The Reorder point is independent of the rate of consumption

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(c) The Reorder point depends only on the lead time

(d) The Reorder point depends upon the rate of consumption and lead time

Karnataka PSC AE, 10.09.2017

Ans. (d) : The Re-order point depends upon the rate of consumption and lead time for given level of safety stock and EOQ ordering.

151. The inventory model with finite rate of replenishment with shortages shows the minimum inventory cost as

(a) $C_{min} = \sqrt{\dfrac{2DC_o C_c C_s\left(1-\dfrac{d}{p}\right)}{C_s + C_c}}$

(b) $C_{min} = \sqrt{\dfrac{2DC_o C_c\left(1-\dfrac{d}{p}\right)}{C_s + C_c}}$

(c) $C_{min} = \sqrt{\dfrac{2DC_o\left(1-\dfrac{d}{p}\right)}{C_s + C_c}}$

(d) $C_{min} = \sqrt{\dfrac{2D\left(1-\dfrac{d}{p}\right)}{C_s + C_c}}$

Karnataka PSC AE, 10.09.2017

Ans. (a) :

$$C_{min} = \sqrt{\frac{2DC_o C_c C_s\left(1-\frac{d}{p}\right)}{C_s + C_c}}$$

Where,

p = Production for build up rate

d = Demand or consumption rate

C_c = Carrying cost

C_o = Ordering cost

C_s = shortage cost

152. The rent for the stores where materials are kept falls under :

(a) Set-up cost (b) Shortage cost

(c) Holding cost (d) Ordering cost

TRB Polytechnic Lecturer 2017

Ans. (c) : The rent for the stores where materials are kept is called holding cost.

153. In a fixed order quantity constant lead time inventory system, the safety stock is dependent on :

(a) Service level

(b) Demand variation during lead time

(c) Service level and demand variation during lead time

(d) Order quantity

TRB Polytechnic Lecturer 2017

Ans. (c) : In a fixed order quantity constant lead time inventory system, the safety stock is dependent on service level and demand variation during lead time.

154. A particular item has a demand of 9000 units/year. The cost of one procurement is Rs. 108 and the holding cost per unit is Rs. 2.40/year. The replacement is instantaneous and no shortages are allowed. What is the optimum number of orders/year?

(a) 7 orders/year (b) 8 orders/year

(c) 9 orders/year (d) 10 orders/year

ESE 2017

Ans. (d) : Annual demand (D) = 9000 units/year

Cost of procurement or ordering cost (C_o) = Rs. 108

Holding cost (C_h) = Rs. 2.4 / unit/ year

$$EOQ = \sqrt{\frac{2DC_o}{C_h}}$$

$$= \sqrt{\frac{2\times 9000\times 108}{2.4}}$$

= 900 units

Optimum No. of orders per year

$$N = \frac{D}{EOQ} = \frac{9000}{900} = 10 \text{ order/year}$$

155. An organization's sales during a financial year is Rs. 6,00,000 with 90% of it on credit. At the end of the year, the receivables turnover was found to be 5. Considering 365 days to a year, the average collection period and receivables are, respectively.

(a) 81 days and Rs. 1, 08,000

(b) 73 days and Rs. 1, 08,000

(c) 81 days and Rs. 1, 20,000

(d) 73 days and Rs. 1, 20,000

ESE 2017

Ans. (b) : Sales = Rs. 6,00,000

Credit = 0.9

Turnover = 5

$$\text{Turnover} = \frac{\text{Sales}}{\text{Inventory cost}}$$

$$\text{Inventory cost} = \frac{6,00,000}{5}$$

= Rs. 1,20,000

Received inventory cost = 1,20,000 × 0.9

= 1,08,000

$$\text{Days inventory} = \frac{1}{5}\times 365 = 73 \text{ days}$$

156. A firm's inventory turnover of Rs. 8,00,000 is 5 times the cost of goods sold. If the inventory turnover is improved to 8 with the cost of goods sold remaining the same, a substantial amount of fund is either released from, or gets additionally invested in, inventory. Which one of the following statements is correct?

(a) Rs. 1, 60,000 is released.

(b) Rs. 1, 60,000 is additionally invested.

(c) Rs. 60,000 is released.

(d) Rs. 60,000 is additionally invested.

ESE 2017

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Ans. (c) : Inventory turnover $= \dfrac{\text{Cost of goods sold}}{\text{Average inventory cost}}$

Old avg. inventory cost $= \dfrac{800000}{5} = 160000$

New avg. inventory cost $= \dfrac{800000}{8} = 100000$

$\therefore$ Old inventory – New inventory
= 1600000 – 100000
= Rs. 60,000

So Rs. 60,000 fund will be released.

157. A stockiest has to supply 400 units of a product every Monday to his customers. He gets the product at Rs. 50 per unit from the manufacturer. The cost of ordering and transportation from the manufacturer to the stockist's premises is Rs. 75 per order. The cost of carrying inventory is 7.5% per year of the cost of the product. What are the economic lot size and the total optimal cost (including capital cost) for the stockist?

(a) 989 units/order and Rs. 20,065.80/week
(b) 912 units/order and Rs. 20,065.80/week
(c) 989 units/order and Rs. 18,574.50/week
(d) 912 units/order and Rs. 18,574.50/week

ESE 2017

Ans. (b) : Given,
Weekly demand (d) = 400 units
Annual demand (D) = 400 × 52
= 20800 units
Unit cost (C_u) = Rs. 50
Ordering cost (C_o) = 75/orders
Carrying cost (C_h) = 7.5% of C_u = 7.5% of 50
= 3.75 / unit / year

$$EOQ = \sqrt{\frac{2DC_o}{C_h}}$$

$$= \sqrt{\frac{2 \times 20800 \times 75}{3.75}} = 912 \text{ units}$$

Total optimal cost/year $= \sqrt{2DC_oC_h} + DC_u$

$$= \sqrt{2 \times 20800 \times 75 \times 3.75} + 20800 \times 50$$

= 1043420.52

Total optimal cost/week = Rs. 20065

158. If T is the total item consumed per year, P is the procurement cost per order and A is the annual inventory carrying cost per item then the most Economic Order Quantity (EOQ) is given by:

(a) $\sqrt{\dfrac{2TP}{A}}$ (b) $\sqrt{\dfrac{2T}{AP}}$
(c) $\dfrac{AP}{\sqrt{2T}}$ (d) $\dfrac{\sqrt{2AP}}{T}$
(e) $\dfrac{2TR}{A}$

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : $EOQ = \sqrt{\dfrac{2DC_o}{C_H}}$

Where ,
D = Annual Demand
C_o = Ordering cost
C_H = Holding cost
Given -
T = Total item consumed year.
P = Procurement cost per order
A = Annual inventory carrying cost per item

$$EOQ = \sqrt{\frac{2TP}{A}}$$

159. ABC inventory control focuses on those

(a) Items not readily available
(b) Items which consume less money
(c) Items which have more demand
(d) Items which consume more money

ISRO Scientist/Engineer 07.05.2017

Ans. (d) : ABC inventory control focuses on those items which consume more money.

160. EOQ stands for :

(a) Economic Order Quantity
(b) Elimination for Quality Inspection
(c) Elements of Quality Control
(d) End of Quality Inspection Change

Kerala PSC IOF 19.04.2016

Ans. (a) Economic Order Quantity

161. ABC analysis :

(a) is meant for relative inventory control
(b) is a basic technique of materials management
(c) does not depend upon the unit cost of the item but on its annual consumption
(d) all of the above

Kerala PSC IOF 19.04.2016

Ans. (d)

162. If EOQ is within the range of the lowest discounted rate offered, then

(a) accept the discount offer and order for the minimum in the range
(b) consider the total costs of the ranges of discount before taking the decision
(c) reject the discount offer
(d) accept the discount offer and order at EOQ level

Kerala PSC IOF 19.04.2016

Ans. (d) :

163. A company requires 16000 units of raw material per year costing Rs. 2 per unit. The cost of placing an order is Rs. 45 and the carrying costs are 10% per year per unit of the average inventory. The economic order quantity, number of orders placed per year and total variable cost for managing the inventory are respectively

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(a) 2684, 6, 5384 (b) 268.4, 6, 538.4
(c) 2684, 8, 538.4 (d) 2684, 6, 538.4
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : Given,

D = 16000 units

C_h = 10% of 2 Rs per unit

$C_h = 0.1 \times 2 = 0.2$ Rs per unit

C_o = 45 Rs per order

We know,

$$EOQ = \sqrt{\frac{2 \times \text{Annual demand} \times \text{Ordering cost}}{\text{Storage cost}}}$$

$$= \sqrt{\frac{2 \times 16000 \times 45}{2 \times 0.1}} = 2684$$

$$\& \text{ number of order placed} = \frac{\text{Annual demand}}{EOQ}$$

$$= \frac{16000}{2684} \approx 6$$

$$\text{Total variable cost} = \sqrt{2DC_oC_h}$$

$$\text{Total variable cost} = \sqrt{2 \times 16000 \times 45 \times 2 \times 0.1} = 538.4$$

164. Which of the following cost is not considered in basic EOQ Model?

(a) Ordering Cost (b) Stockout Cost
(c) Holding Cost (d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (b) : EOQ (Economic order quantity)

$$= \sqrt{\frac{2DC_o}{C_h}}$$

D = Annual demand for the item

C_o = Cost of order or setup cost

C_h = Inventory cost per unit per year

165. A company has a monthly demand of 5000 units for certain item which costs Rs. 60 per unit. The ordering cost per order is Rs. 225 and the holding or carrying cost of the inventory per year is 5% of the unit price of the item. What must be the Economic Order Quantity (EOQ), considering instantaneous replenishment of items?

(a) 2600 (b) 3200
(c) 3000 (d) 3600

RPSC AE (GWD) 06.05.2016

Ans. (c) : Given,

A monthly demand (D) = 5000 unit

Item cost = Rs. 60 per unit

Ordering cost (C_o) = Rs. 225 per order (Rs./order)

Holding cost of inventory per year 5%

$C_h = 60 \times 5\%$

$$= 60 \times \frac{5}{100} = 3$$

C_h = 3 Rs./unit/year

Economic order quantity (EOQ)

$$Q^* = \sqrt{\frac{2DC_o}{C_h}}$$

$$\left[\because 12 \text{ month} = 1 \text{ year}, \quad 1 \text{ month} = \frac{1}{12} \text{ year}\right]$$

$$= \sqrt{\frac{2 \times 5000 \times 225 \times 12}{3}}$$

$$= \sqrt{\frac{27000000}{3}}$$

$$= \sqrt{9000000} = 3000 \text{ unit}$$

166. Consider these models of inventory Economic Ordering Quantity (EOQ) in all of which the inventory consumption is at a constant rate : (A) the replenishment is instantaneous but shortages are not permitted, (B) the replenishment is at a finite constant rate and no shortages are permitted, and (C) the replenishment is instantaneous but shortages are permitted. In which of these cases the total optimal inventory cost will be largest?

(a) model A
(b) model B
(c) model C
(d) model C or model A depending upon parameters

RPSC AE (GWD) 06.05.2016

Ans. (a) : Economic order quantity (Q*) = $\sqrt{\frac{2DC_o}{C_h}}$

Model (A)– The replenishment is instantaneous but shortages are not permitted.

Model (B)– Replenishment is at a finite constant rate no shortage are permitted.

Model (C)– Replenishment is instantaneous but shortage are permitted.

∴ In above three case the total optimal inventory cost will be largest in model (A).

167. If the total investment in stock is limited, then the best order quantity for each item will be:

(a) equal to EOQ
(b) greater than EOQ
(c) less than EOQ
(d) less than or equal to EOQ
(e) greater than or equal to EOQ

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : As we know, if the total investment in stock is limited, then the best order quantity for each item will be less than EOQ due to less total cost.

168. If EOQ is calculated but an order is then placed which is smaller than this, what will happen to the variable cost?

(a) It will increase
(b) It will decrease
(c) It will either increase or decrease
(d) It remains unchanged
(e) Cannot be determined

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (d) :

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169. If the annual demand of an item becomes double, ordering cost half, holding cost one fourth and unit cost twice, then what is the ratio of old EOQ and new EOQ?

(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$

(c) 2 (d) $\frac{1}{2}$

(e) 1

CGPSC Asstt. Workshop Supt., 17.07.2016
ESE 2006

Ans. (d) : Given, $D_2 = 2D_1$

$$C_{0_2} = \frac{1}{2}C_{0_1}$$

$$C_{H_2} = \frac{1}{4}C_{H_1}$$

$$\frac{(EOQ)_1}{(EOQ)_2} = \frac{\sqrt{2D_1C_{0_1}/C_{H_1}}}{\sqrt{\frac{2\times 2D_1\times C_{0_1}/2}{C_{H_1}/4}}}$$

$$\frac{(EOQ)_1}{(EOQ)_2} = \frac{1}{2}$$

170. Total inventory cost consists of :

(a) 3 cost components (b) 2 cost components

(c) 4 cost components (d) 5 cost components

HPPSC Asstt. Prof. 18.11.2016

Ans. (c) : There are four main components to the cost of inventory–
(1) Capital cost
(2) Storage space cost
(3) Inventory services costs
(4) Inventory risk costs

171. Just-in-time manufacturing concept is based on :

(a) Reduction of production
(b) Reduction of waste
(c) Reduction of inspection
(d) Reduction of defects

HPPSC Asstt. Prof. 18.11.2016

Ans. (b) : Just in time also known as JIT is an inventory management method where by labour material and goods (to be used in manufacturing) are unfilled or scheduled to arrive exactly when needed in the manufacturing process.

172. Consider the following costs:
1. Cost of insurance
2. Cost of scrap
3. Cost of obsolescence
4. Cost of negotiation with suppliers
5. Cost of inspection and return of goods
Which of these costs are related to inventory cost ?

(a) 1.3 and 4 (b) 2, 3 and 4

(c) 1, 2 and 3 (d) 2, 4 and 5

UPRVUNL AE 21.08.2016
ESE 1999

Ans. : (c) Cost of insurance, cost of scrap, cost of obsolescence these cost are related to inventory carrying cost.

173. Which one of the following is an inventory system that keeps a running record of the amount in storage and replenishes the stock when it drops to a certain level by ordering a fixed quantity?

(a) ABC (b) EOQ

(c) Peripheral (d) Periodic

UPRVUNL AE 21.08.2016
ESE 2006

Ans. : (b) EOQ inventory system that keeps a running record of the amount in storage and replenishes the stock when it drops to a certain level by ordering a fixed quantity.

174. Consider the following statements
1. Simulation can be used for inventory control
2. EOQ (Economic Order Quantity) formula ignores variations in demand pattern
3. ABC analysis is based on Pareto's principal
4 FIFO and LIFO policies can be used for material valuation in materials management of these statements

(a) 1 alone is correct
(b) 2, 3 and 4 are correct
(c) 1 and 3 are correct
(d) 1, 2, 3 and 4 are correct

UPRVUNL AE 21.08.2016
ESE 1995

Ans. : (d)
(i) Simulation can be used for inventory control
(ii) EOQ formula ignores variation in demand pattern
(iii) ABC analysis is based on Pareto's principal
(iv) FIFO and LIFO policies can be used for material valuation in materials management.

175. Which one of the following expressions give the rate depreciation by the straight line method? (C = initial cost. S = scrap value. N = life in years)

(a) D = (C –S)/N (b) D = N (C +S)

(c) D = (C +S)/N (d) D = N(C –S)

UPRVUNL AE 21.08.2016
CSE Pre-2005

Ans. : (a) Rate depreciation by the straight line method
D = (C – S)/N
C = initial cost
S = Scrap value
N = life in gear

176. Annual demand for a product Rs. 50 per piece is Rs. 300. Ordering cost per order is Rs. 200 and inventory holding cost is Rs. 3 per unit per year. The economic size is

(a) 300 (b) 200

(c) 500 (d) 400

UPRVUNL AE 21.08.2016
ESE 1997

Ans. : (b) A = Annual consumption in rupees
C_o = cost of placing an order
C_h = Inventory carrying cost per unit per year

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$$EOQ = \sqrt{\frac{2AC_o}{C_h}}$$

$$EOQ = \sqrt{\frac{2 \times 300 \times 200}{3}}$$

$$\boxed{EOQ = 200}$$

177. Which of the following is not a part of inventory carrying cost

(a) Cost of inwards goods inspection
(b) Cost of obsolescence
(c) Cost of insurance
(d) Cost of storage space

APPSC AEE Screening Test 2016

Ans. (a) : An incoming inspection also known as receiving inspection or material inspection validates the quality of purchased raw materials based on set acceptance criteria. It is performed by quality assurance personnel in the manufacturing facility to resolve quality issues during pre-production.

178. In 'ABC' analysis of inventory control, 'A' stands for:

(a) Medium valued and medium number of items
(b) Low valued and small number of items
(c) High valued and maximum number of items
(d) High valued and small number of items

UPRVUNL AE 07.10.2016

Ans. (d) : Meaning of ABC analysis—For selective inventory control, three classes of materials and not more than that have been found to be convenient. This is known in general as the ABC classification. The alphabets A, B and C stands for three different classes and it is popularly known as Always Better Controls.

- A items are good where annual consumption value is the highest.
- B items are interclass items. Their consumption values are lower than A items but higher than C items.
- C items have the lowest consumption value. This class has a relatively high proportion of the total number of lines but with low consumption value.

179. What will be economic order quantity for an item with annual usage of 50 units, procurement cost is Rs. 10 per order. Cost of one piece is Rs. 100 and cost of carry inventory per year is 10%?

(a) 9 (b) 10
(c) 8 (d) 11

UPRVUNL AE 07.10.2016

Ans. (b) : Data given,

$D = 50$ units

$C_o = 10$ per order

$C = Rs.\ 100$/piece

$$C_h = 100 \times \frac{10}{100} = Rs.10$$

We know that,

$$EOQ = \sqrt{\frac{2D \times C_o}{C_h}}$$

$$EOQ = \sqrt{\frac{2 \times 50 \times 10}{10}}$$

$$\boxed{EOQ = 10 \text{ units}}$$

180. The most economic order quantity in terms of total item consumed per year (A), procurement cost, (P) per order and the annual inventory carrying cost (C) per item is given by :

(a) $\frac{AP}{2C}$ (b) $\frac{2AP}{C}$
(c) APC (d) $\sqrt{\frac{2AP}{C}}$

HPPSC W.S. Poly. 2016

Ans : (d) Economic order quantity (EOQ)

Assumption:-

(i) Demand is known, deterministic and constant
(ii) Lead time is zero
(iii) Usage rate is constant
(iv) Price of material is fixed
(v) Holding cost per unit item per unit time is fixed.

$$EOQ = \sqrt{\frac{2AP}{C}}$$

A = Item consumed per year
P = Procurement Cost
C = Annual inventory carrying cost.

181. In case of 'VED' analysis of inventory control 'E' stands for

(a) Easily available items
(b) Essential items
(c) Extra-ordinary items
(d) Extra items

UJVNL AE 2016

Ans : (b) In case of 'VED' analysis of inventory control 'E' stands for Essential items. The degree of critically can be stated as whether the material is vital to the process of production, or essential to the process of production or desirable for the process of production. The classification is known as VED Analysis, V-stands for vital, E stands for essential and D stands for desirable items.

182. ABC analysis, as an input, requires

(a) Annual usage and cost of the items.
(b) Cost and criticality of the items.
(c) Criticality and availability of the items.
(d) Availability and annual usage of items.

UPPSC AE 12.04.2016 Paper-I

Ans : (a) ABC analysis, as an input, requires annual usage and cost of the items.

ABC (Always Better Control):-ABC analysis is a basic technique of materials management and is used in inventory control. The technique is based on annual consumption of an item rather on its unit cost.

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183. The finite production rate inventory model relaxes which of the following EOQ assumptions?

(a) Instantaneous replenishment
(b) Constant lead time
(c) Fixed deterministic demand
(d) No variation in unit time

UPPSC AE 12.04.2016 Paper-I

Ans : (a) Instantaneous replenishment
Assumption of the basic EOQ model:-
(i) Demand is known, deterministic and constant
(ii) Lead time is zero
(iii) Usage rate is constant
(iv) Price of material is fixed
(v) Holding cost per unit item per unit time is fixed

184. Which of the following is not an underlying assumption of the basic EOQ model?

(a) Stochastic demand
(b) Instant replenishment
(c) Fixed lead time
(d) No shortages

UPPSC AE 12.04.2016 Paper-I

Ans : (a) Stochastic demand is not an underlying assumption of the basic EOQ model:
(i) Demand is known, deterministic and constant
(ii) Lead time is zero
(iii) Usage rate is constant
(iv) Price of material is fixed
(v) Holding cost per unit item per unit time is fixed

185. Optimum reorder quantity is

(a) $\sqrt{\frac{2(\text{annual unit usage})\times(\text{cost per set - up or order})}{(\text{cost per unit})\times(\text{percent annual carrying cost})}}$

(b) $\sqrt{\frac{2(\text{cost of item})}{(\text{annual usage})\times(\text{percent annual carying cost})}}$

(c) $\sqrt{\frac{2(\text{cost per unit})\times(\text{percent annual carrying cost})}{(\text{annual unit storage})\times(\text{cost per set}-\text{up order})}}$

(d) $\sqrt{\frac{2(\text{cost per unit})\times(\text{annual carrying cost})}{(\text{cost per set up order})\times(\text{percent annual carrying cost})}}$

BPSC Poly. Lect. 2016

Ans : (a) Optimum reorder quantity

$$\sqrt{\frac{2(\text{annual unit usage})\times(\text{cost per set - up or order})}{(\text{cost per unit})\times(\text{percent annual carrying cost})}}$$

186. Service level in inventory is defined as

(a) % age of time consumed during order
(b) % age of time where there is stock out
(c) % of time during which inventory is available
(d) % age of time during which there is production

TSPSC AEE 2015

Ans. (c) : Service level in inventory is defined as % of time during which inventory is available.

187. Annual demand for a product costing Rs.100 is 400. Ordering cost per order is Rs. 100 and carrying cost is Rs. 2 per unit per year. The economic lot size is then :

(a) 200 (b) 300
(c) 400 (d) 500

Kerala PSC AE 06.08.2015, ESE 1997

Ans. (a) : Given demand = 400
Ordering cost $C_o = 100$`
holding cost $C_h = 2$`
EOQ = ?

$$EOQ = \sqrt{\frac{2C_oD}{C_h}} = \sqrt{\frac{2\times100\times400}{2}} = 200$$

188. Inventory control in production planning and control aims at :

(a) Achieving optimization
(b) Ensuring against market fluctuations
(c) Acceptable customer service at low capital investment in inventory
(d) Discounts allowed in bulk purchase.

TSPSC Managers, 2015

Ans. (c) : Inventory control in production planning and control aims at acceptable customer service at low capital investment in inventory.

189. One of the important reasons for carrying inventory is to-

(a) Improve customer service
(b) Get quantity discounts
(c) Maintain operational capability
(d) All of these

NPCIL ET, 2015

Ans. (d) : Important reasons for carrying inventory
- Improve customer service
- Get quantity discount
- Maintain operational capability

190. The demand of product is 25 units per month and the items are withdrawn uniformly. The setup cost for each time production run is Rs. 15. The inventory holding cost is Rs 0.30 per unit per month. Determine how often to make production run, if shortage is not allowed.

(a) 3 run/year (b) 6 run/year
(c) 9 run/year (d) 12 run/year

NPCIL ET, 2015

Ans. (b) : D = 25 units per month = 25 × 12 per year.
$C_h = 0.30$
$C_o = 15$

$$EOQ = \sqrt{\frac{2DC_o}{C_h}}$$

$$= \sqrt{\frac{2\times25\times15}{0.30}}$$

EOQ = 50

$$\text{No. of production run} = \frac{25\times12}{50} = 6 \text{ run per year}$$

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191. Inventory management consists of :

(a) state of merchandise methods of storing and maintenance etc.
(b) stock control system
(c) All of three mentioned here.
(d) effective running of stores.

GWSSB AAE, 27.12.2015

Ans. (c) : Inventory management consists of –
- State of merchandise methods of storing and maintenance etc.
- Stock control system
- Effective running of stores.

192. Consider the following statements

1. Interests of Materials manager
2. Interests of the top management
3. Pareto's 80-20 rule
Of the above, the basis for ABC analysis is

(a) 2 (b) 2 and 3
(c) 1 (d) 3
(e) 1 and 2

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : ABC analysis may be seen to share similar ideas as the Pareto principle, which states that 80% of overall consumption value comes from only 20% of items. Plainly, it means that 20% of your products will bring in 80% of your revenues.

193. The annual demand for an item is 4000 parts. The unit cost is Rs. 8/- and inventory carrying charges are estimated as 25 percent per annum. If the cost of one procurement is Rs. 150, the Economic order quantity is

(a) 675 units (b) 775 units
(c) 875 units (d) 2400 units
(e) 2200 units

CGPSC AE 26.04.2015 Shift-I

Ans. (b) : given,
D = 4000 units
C = Rs 8 / unit

$C_h = 8 \times \frac{25}{100} = \text{Rs. } 2 / \text{year}$

$C_o =$ Rs. 150
then,

$$EOQ = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2 \times 4000 \times 150}{2}}$$

$EOQ = 774.59 \simeq 775$ units

194. Which of the following does not pertain to inventory management?

(a) Production schedule
(b) State of merchandise
(c) Control of stock
(d) Effective running of stores

(KPSC AE 2015)

Ans : (a) production schedule does not pertain to inventory management.

195. The size of order which minimize total cost of carrying and cost of ordering is called

(a) minimum order quantity
(b) economic order quantity
(c) safety order quantity
(d) re-order quantity

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (b) : The size of order which minimizes total cost of carrying and cost of ordering is called economic order quantity.

196. VED analysis of inventory management stands for

(a) Vital – Essential – Desirable
(b) Valuable – Easy – Difficult to obtain
(c) Very essentially desired
(d) Valuable – Effective – Difficult to obtain

Haryana PSC Civil Services Pre, 2014

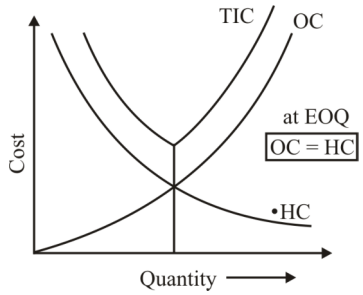
Ans. (a) VED = Vital – Essential – Desirable

197. Formula for calculating EOQ in inventory management is :

(a) $EOQ = (2 \times D \times K/H)^{1/2}$
(b) $EOQ = (D \times K/2H)^{1/2}$
(c) $EOQ = (2 \times K/D \times H)^{1/2}$
(d) $EOQ = (K/2 \times D \times H^{1/2}$

Haryana PSC Civil Services Pre, 2014

Ans. (a)



$$\boxed{EOQ = \sqrt{\frac{2DK}{H}}}$$

EOQ – Economic Order Quantity
D – Demand
K – Fixed Cost per Order (OC)
H – Holding Cost (HC)
TIC – Total Cost Inventory

198. The lead time consumption is 500 units. The annual consumption is 8000 units. The company has a policy of EOQ ordering and maintenance of 200 units as safety stock. The reorder point (ROP) is :

(a) 500 units (b) 700 units
(c) 200 units (d) None of these

MPSC Poly. Lect. 2014

Ans. (b) : Reorder point = LT × d + safety stock.
= 500 + 200
Reorder point = 700 units.

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199. The total annual inventory cost includes the following costs :

(a) Carrying cost, ordering cost and setup cost
(b) Carrying cost and setup cost
(c) Production cost, carrying cost, ordering cost and setup cost
(d) Production cost, carrying cost and ordering cost.

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : Total Annual inventory cost includes.
(1) Purchase cost
(2) Order cost or setup cost
(3) carrying cost or holding cost.
(4) shortage cost.

200. Which of the following manufacturing system keeps maximum inventory?

(a) Make to stock
(b) Make to order
(c) Assemble to order
(d) Engineer to order

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : Make to stock manufacturing systems keeps maximum inventory.

201. The formula for economic order quantity does not contain

(a) Order cost (b) Carrying cost
(c) Cost of the item (d) Annual demand

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) : Cost of the item

202. Cycle counting in inventory

(a) provides a measure of inventory turnover
(b) assumes that all inventory records must be verified with the same frequency
(c) is a process by which inventory records are periodically verified
(d) provides annual demand

TNPSC AE 2014

Ans. (c) : Cycle counting in inventory is a process by which inventory records are periodically verified.

203. Effect of stockout of a commodity is:-

(a) Loss of profit (b) Loss of customers
(c) Loss of goodwill (d) All of the above

UKPSC AE-2013, Paper-I

Ans. (d) : Effect of stockout- If an item is not available for manufacturing then it may be possible to change the production schedule, although there is a significant cost in this due to the changes in machine, teardown costs, resource changes, loss of profit, loss of customers and also loss of goodwill.

204. Which of the following statements is correct ?

(a) A-B-C analysis is based on Pareto's principle
(b) Simulation can be used for inventory control
(c) Economic order quantity formula ignores variations in demand pattern
(d) all of the above

J&K PSC Civil Services Pre, 2013

Ans. (d) : All sentences are correct.

205. In ABC analysis of inventories, 'A' items usually constitute:-

(a) 10% (b) 20%
(c) 30% (d) 70%

UKPSC AE-2013, Paper-I

Ans. (a) : In ABC analysis of inventories, 'A' items usually constitute 10%.

206. When ordering cost is increased to 16 times, the EOQ will be increased to:-

(a) 2 times (b) 4 times
(c) 8 times (d) None of the above

UKPSC AE-2013, Paper-I

Ans. (b) : $[EOQ]_1 = \sqrt{\frac{2DC_o}{C_h}}$

If C_o increased to 16 C_o

Then, $[EOQ]_2 = \sqrt{\frac{2D(16C_o)}{C_h}}$

$= 4 \times \sqrt{\frac{2DC_o}{C_h}}$

$[EOQ]_2 = 4[EOQ]_1$

207. In ABC analysis items are classified in three categories namely A, B, and C in accordance with their:-

(a) Values (b) Number
(c) Characteristics (d) Priorities

UKPSC AE-2013, Paper-I

Ans. (a) : In ABC analysis items are classified in three categories namely A, B, and C in accordance with their values.

208. Depreciation can be calculated by:

(a) Diminishing balance method
(b) Time series analysis
(c) ABC analysis
(d) All of these

SJVN ET 2013

Ans. (a) : Depreciation can be calculated by diminishing balance method.

209. In inventory control, EOQ depends on:

(a) Holding cost (b) Ordering cost
(c) Consumption rate (d) All of these

SJVN ET 2013

Ans. (d) : In inventory control, EOQ depends on-
- Holding cost
- Ordering cost
- Consumption rate

$$EOQ = \sqrt{\frac{2C_o D}{C_h}}$$

D = Demand
C_h = Holding cost
C_o = Ordering cost

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210. In inventory EOQ means :

(a) The quantity which when ordered will minimum the total inventory cost

(b) Average quantity will minimize the total inventory cost

(c) Maximum quantity which going to order

(d) Minimum quantity order

TNPSC AE (Industries) 09.06.2013

Ans. (a) : The quantity which when ordered will minimum the total inventory cost.

211. Group 'C' items constitute the following percentage of items in ABC analysis :

(a) 10 (b) 20

(c) 50 (d) 70

UKPSC AE 2012 Paper-I

Ans. (d) : 70

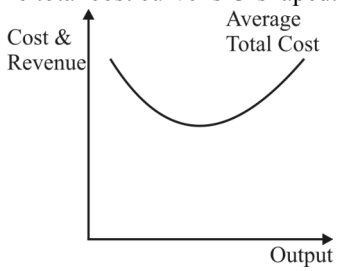
212. The total cost curve is

(a) L-shaped (b) O-shaped

(c) U-shaped (d) S-shaped

BPSC AE 2012 Paper - VI

Ans : (c) : The total cost curve is U-shaped.



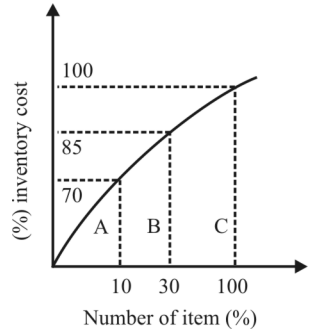
213. Based on ABC analysis, items requiring frequent ordering are

(a) A items (b) B items

(c) C items (d) none of these

TRB Asstt. Prof., 2012

Ans. (a) : ABC Analysis– This analysis follows the basic Pareto's principal few usage value items constitute a major part of capital invested in inventories.



214. Inventory is in the form of

(a) raw materials (b) finished goods

(c) in-process goods (d) all of the these

TRB Poly. Lect., 2012

Ans. (d) : Inventory is in the form of raw materials or finished goods of improves goods, all comes under inventory

215. In ABC analysis, 'A' items are responsible to share approximately the following percentage of cost :

(a) 80 (b) 60

(c) 40 (d) 20

UKPSC AE 2012 Paper-I

Ans. (a) :

216. The following is the general policy for A class items in ABC analysis :

1. Very strict control
2. Frequent review of their consumption
3. Safety stock kept

Which of these statement/s is/are correct ?

(a) 1 only (b) 1 and 2 only

(c) 2 only (d) 1, 2 and 3

UKPSC AE 2012 Paper-I

Ans. (b) : 1 and 2 only

217. What is the time period between two successive orders called.

(a) Order cycle (b) Order review

(c) Ordering process (d) Ordering period

VIZAG STEEL MT 2011

Ans. (a) :

218. According to Pareto principle, and effective man is one who

(a) can manage his boss

(b) can manage his subordinates

(c) can manages his colleagues

(d) pick up vital from the trivial many things

RPSC ACF-2011

Ans. (d) : The Pareto's principal states that 80% of outcomes come from 2% of all causes or possible factors in given event. For this reason, it is also known as 80/20 rule. Other names for this principle are the 80/20 rule, the law of the vital few or the principle of factor sparsity.

219. Bin cards are used in :

(a) machine loading

(b) accounts

(c) stores

(d) preventive maintenance

RPSC ACF-2011

Ans. (c) : Bin card is the record maintained under the perpetual inventory system by the store department and shows the quantities of materials received, issued and balance in hand after each receipt and issue. It is also known as stock card or bin tag. It records only quantity of materials not the value.

220. For a company that uses a periodic inventory system, which of the following is true?

(a) inventory is counted only at specific time intervals

(b) the order is for a fixed quantity which minimizes inventory costs

(c) the inventory system is also referred to as a perpetual inventory system

(d) the company is using a fixed-order-quantity system

RPSC Lect. (Tech. Edu. Dept.) 2011

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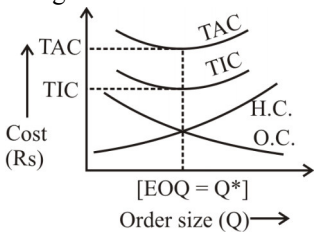
Ans. (a) : Periodic inventory is a system of inventory in which updates are made on a periodic basis.
In this, no effort is made to keep up-to-date records of either the inventory or the cost of goods sold.

221. Which of the following statements concerning the basic EOQ model is true?
(a) A decrease in demand will increase the EOQ value
(b) If an actual order quantity is smaller than the EOQ, the annual holding cost is less than the annual ordering cost.
(c) An increase in holding cost will increase the EOQ value.
(d) In the EOQ formula there is an inverse relationship between setup and carrying costs

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : EOQ model: If an actual order quantity is smaller than EOQ, the annual holding cost is less than the annual ordering cost.



At, EOQ ; HC = OC

222. For a company with an average daily demand of 9 units and a standard deviation of 3 units, a 9-day lead time, and a 95% service level, which of the following statements is true?
(a) The reorder point is about 81 units
(b) The safety stock is about 15 units
(c) The reorder point is about 96 days
(d) The reorder point is about 81 days

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (*) : Given,
Average demand, d = 9 units
$\sigma = 3$ units
Lead time, LT = 9 days
Service level = 95%
z = 1.645
(SS) safety stock = $z.\sigma$
$\Rightarrow 1.645 \times 3 = 4.935$ units
Reorder point, ROL = d × LT + SS
$= 9 \times 9 + 4.935$
= 85.935 units
Note- Option (b) is given by commission.

223. In Two Bin System
(a) Jobs move in batches of fixed quantity between the machines.
(b) products are delivered in batches of fixed quantity.
(c) Inspection for quality control is done in fixed quantity.
(d) Inventory is ordered in fixed quantity.

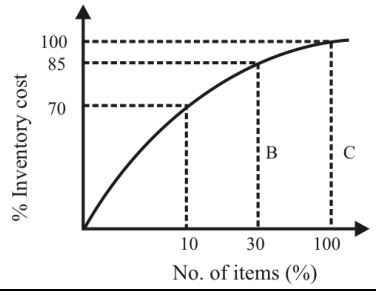
RPSC AE GWD, 2011

Ans. (b) : In Two Bin System products are delivered in batches of fixed quantity.

224. In ABC inventory control analysis :
(a) Careful control on high value and low volume items
(b) Careful control of high demand items
(c) Careful control on quality assurance o inventory
(d) Careful control on low value and high volume items.

RPSC AE GWD, 2011

Ans. (a) : ABC inventory control is based on Pareto law or (80-20) rule. It is based on usage value.
In ABC analysis few usage value items constitute a major part of capital investment in inventories.



225. In inventory management when the quantities needed with certainty over subsequent periods of time what can this be called as.
(a) Lead demand
(b) Probabilistic demand
(c) Deterministic demand
(d) Direct demand

VIZAG STEEL MT 2011

Ans. (c) :

226. Ordering costs do not include
(a) Obsolescence
(b) Issuing purchase orders
(c) Placing goods into the appropriate location in the stores
(d) Follow up of pending receipts

UPSC JWM Advt. No.-52/2010

Ans. (a) : Ordering Cost–Ordering cost also known as set-up costs which is incurred at the time of placing an order. Ordering costs include expenses for a purchase order, labour costs for the inspection of goods received, placing goods into the appropriate location in the stores and follow up of pending receipt.

227. Consider the following assumptions on EOQ model of inventory management :
1. Annual demand is known and constant.
2. Ordering cost is known.
3. Inventory carrying cost (holding cost) is known.

Which of these are correct?
(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-50/2010

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Ans. (a) : As we know, EOQ Model of inventory management consist of,

$$EOQ = \sqrt{\frac{2DC_o}{C_h}}$$

$C_o \rightarrow$ Ordering cost
$D \rightarrow$ Annual demand
$C_h \rightarrow$ Holding cost

228. In finding out the EOQ for an item, it is observed that the carrying cost is estimated as k C_c instead of C_c (i.e., the true carrying cost is C_c). There is no error in the estimate of ordering cost. The potential percentage savings when k = 0.5 is :

(a) 5.72 (b) 2.02
(c) 8.12 (d) 4.02

APPSC IOF, 2009

Ans. (b) : $EOQ_1 = \sqrt{\frac{2 \times S \times D}{C}}$

$$EOQ_2 = \sqrt{\frac{2 \times S \times D}{0.5C}}$$

$$\frac{EOQ_1}{EOQ_2} \Rightarrow 2.02$$

229. In ABC analysis, the items which represent a large group are :

(a) A group items
(b) B group items
(c) C group items
(d) All have the same proportion

APPSC IOF, 2009

Ans. (c) : For A group items usable value is high and quantity is less.
For C group items usable value is low and quantity is more.

230. In the application of EOQ to minimize inventory level without increasing purchase work load, average inventory is used as a surrogate measure for:

(a) Purchasing cost
(b) Annual demand
(c) Price per unit of the inventory
(d) Carrying costs

UPSC JWM Adv. No-16/2009

Ans. (d) : Average inventory is used as a surrogate measure for carrying costs.

231. We would expect it to be more difficult to justify the use of an EOQ model for production lot sizing, when cost information is not available, than for an inventory application under the same conditions, because, with the lot sizing model, we would normally expect a wide variations among stock keeping units in the :

(a) Holding costs (b) Production rate
(c) Daily demand rate (d) Cost per setup

UPSC JWM Adv. No-16/2009

Ans. (a) : Holding costs

232. A machine tool manufacturing industry requires to purchase 2000 helical gears/year from a gear manufacturing industry. The ordering cost is ₹ 100/-per order and the carrying cost per unit per year is ₹ 10/. The economic order quantity is :

(a) 2000 (b) 1000
(c) 200 (d) 100

DRDO Scientists 2009

Ans. (c) : $C_O = 100$/order, $D = 2000$ unit
$C_h = 10$ Rs/unit

$$EOQ = \sqrt{\frac{2DC_O}{C_h}}$$

$$= \sqrt{\frac{2 \times 2000 \times 100}{10}} = 200$$

233. In basic economic order quantity model, for the optimal order quantity,

(a) holding cost is more than ordering cost
(b) holding cost is less than ordering cost
(c) holding cost is equal to ordering cost
(d) holding cost is two times the ordering cost

DRDO Scientists 2008

Ans. (c) : holding cost is equal to ordering cost

234. The expected demand of a product is 150 per day. The lead time is 10 days. An order is placed when the inventory falls to 4000 units. The safety stock is

(a) 1500 units (b) 2500 units
(c) 3000 units (d) 4000 units

DRDO Scientists 2008

Ans. (b) : $\alpha = 150$/day
L.T. = 10 days
Reorder level = 4000 units
L.T. × d + S.S. = R.O.L.
10 × 150 + S.S. = 4000
S.S. = 2500
Safety stock = 2500 units

235. The following is not the guideline for controlling A class item:

(a) Very strict control
(b) Frequent ordering
(c) Maximum efforts to reduce lead time
(d) Higher safety stock

UKPSC AE 2007 Paper -I

Ans. (d) : Higher safety stock

3. Forecasting

236. Which one of the following forecasting techniques is not suited for making forecast for planning production schedules in the short range?

(a) Moving average
(b) Exponential moving average
(c) Regression analysis
(d) Delphi

RPSC ACF & FRO, 26.02.2021

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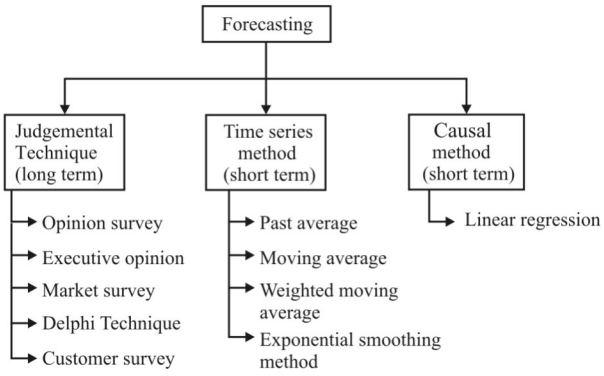
APPSC Poly Lect. 13.03.2020
VIZAG MT, 14.12.2020, TRB Asstt. Prof., 2012
CSE Pre-2005, GATE 1998

Ans. (d) : Delphi



237. For a product, the forecast and the actual sales for December 2002 were 25 and 20 respectively. If the exponential smoothing constant (α) is taken as 0.2, then forecast sales in January 2003 would be

(a) 21 (b) 24
(c) 23 (d) 27

GPSC ARTO Pre 30.12.2018
TSPSC AEE 2017
VIZAG STEEL MT 2015
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2004

Ans. (b) : $F_{t+1} = F_t + \alpha(D_t - F_t)$
$F_{Jan}(2003) = F_{Dec} + \alpha(D_{Dec} - F_{Dec})$
$\Rightarrow D_{Dec} = 20$ units
$F_{Dec} = 25$ units
$F_{Jan} = 25 + 0.2(20–25)$
$= 25 – 1 = 24$ units
$F_{Jan} = 24$ units

238. Which of the following forecasting methods takes a fraction of forecast error into account for the next period forecast?

(a) simple average method
(b) moving average method
(c) weighted moving average method
(d) exponential smoothening method

Mizoram PSC (Insp. of Factories) 2019 Paper-III
Nagaland CTSE 2017 Paper-I
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2009

Ans. (d) : Exponential Smoothening method-
Exponential Smoothening method is given by-
$f_t = f_{t-1} + \alpha(D_{t-1} - f_{t-1})$
$f_t = f_{t-1} + \alpha(\text{error})$
Where,
Error $= D_{t-1} - f_{t-1}$
The error takes into consideration the difference b/w the demand and the forecast as the previous period.

239. Which one of the following method can be used for forecasting the sales potential of a new product

(a) Time series analysis
(b) Jury of executive opinion method
(c) Sales force composite method
(d) Direct survey method

Mizoram PSC (Insp. of Factories) 2019 Paper-III
Kerala PSC Poly. Lect., 2017
RPSC VPITI 14.02.2016
MPPSC AE 2016

Ans. (d) : Direct survey method used for forecasting the sales potential of new product.
The direct survey method was developed in order to provide a fast, simple and accurate technique with quantifiable error. Particularly for underwater sites with limited visibility and those with extremely complex three dimensional architecture.

240. The demand and forecast for June are 12000 and 10275 respectively. If the smoothening coefficient = 0.25, the forecast for the month of July will be-

(a) 9431 (b) 9587
(c) 10706 (d) 11000

HPPSC ADF 05.03.2019
GPSC ARTO 01.05.2016
NPCIL ET, 2015, GATE 2010

Ans. (c) : Given, Demand and forecast
$D_{t-1} = 12000$
$F_{t-1} = 10275$
Smoothing coefficient (α) = 0.25
Forecast for month July
$F_t = F_{t-1} + \alpha (D_{t-1} - F_{t-1})$
$= 10275 + 0.25 (12000 – 10275)$
$= 10275 + 431.25$
$F_t = 10706.25$

241. The most commonly used criteria for measuring forecast error is:

(a) mean absolute deviation
(b) mean absolute deviation percentage error
(c) mean standard deviation error
(d) mean square error

GPSC Asstt. Prof. (Prod.) 21.08.2016
HPPSC LECT. 2016, TRB Poly. Lect., 2012
GATE 1997

Ans : (d) The most commonly used criteria for measuring forecast error is mean square error.

242. Which of the following is independent of sales forecast ?

(a) Productivity
(b) Inventory control
(c) Production planning
(d) Production control

APGCL AM, 2021, TSPSC Managers, 2015
GPSC Lect. (Poly.) 07.12.2014

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Ans. (a) **: Forecasting :-**

⇒ It helps in determine volume of production & production Rate.

⇒ It forms the basis for production budget, labour budget, material budget etc.

⇒ It is essential for project design and development.

⇒ It suggest the Need for plant expansion.

243. It is given that the actual demand is 61 units, a previous forecast 65 units and smoothening factor is 0.3. What will be the forecast for next period, using exponential smoothing?

(a) 59.8 (b) 62.2

(c) 63.8 (d) 66.2

RPSC ACF & FRO, 26.02.2021

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : Given,

$D_t = 61$

$F_t = 65$

$\alpha = 0.3$

$F_{t+1} = F_t + \alpha (D_t - F_t)$

$= 65 + 0.3 (61 - 65)$

$\boxed{F_{t+1} = 63.8}$

244. Regression method of forecasting is applicable mainly for :

(a) Causal models

(b) Qualitative forecasting

(c) Time series models

(d) Delphi method

MPSC Lect. (Prod.) 2018, RPSC ADE 2016

Ans. (a) : Regression analysis is widely used for prediction and forecasting where its use has substantial overlap with the field of machine learning. Regression can also be used to single caused relationship between independent and dependent variable.

245. Which one of the following forecasting techniques is most suitable for making long-range forecasts?

(a) Time series analysis

(b) Regression analysis

(c) Exponential smoothing

(d) Market surveys

Assam PSC AE (PHED), 18.10.2020

UJVNL AE 2016, ESE 2005

Ans. (d) : Market survey is another most widely used sales forecasting method which is used to gather information related to market that cannot be collected from the company's internal records or the external sources of data.

Most suitable for making long range forecasts is market surveys.

246. The demand for a two-wheeler was 900 units and 1030 units in April 2015 and May 2015, respectively. The forecast for the month of April 2015 was 850 units. Considering a smoothing constant of 0.6, the forecast for the month of June 2015 is_____

(a) 970 units (b) 850 units

(c) 927 units (d) 965 units

Assam Engg. College AP/Lect. 18.01.2021

GATE 2016

Ans. (a) $D_{April} = 900$

$\alpha = 0.6$

$D_{May} = 1030$

$F_{April} = 850$

$F_{June} = F_{May} + \alpha(D_{May} - F_{May})$

$F_{May} = F_{April} + \alpha(D_{April} - F_{April})$

$= 850 + 0.6(900 - 850)$

$F_{May} = 880$

$F_{June} = F_{May} + \alpha(D_{May} - F_{May})$

$= 880 + 0.6(1030 - 880)$

$F_{June} = 970$ units

247. In exponential smoothening method, which one of the following is true?

(a) $0 \le \alpha \le 1$ and high value of α is used for stable demand

(b) $0 \le \alpha \le 1$ and high value of α is used for unstable demand

(c) $\alpha \ge 1$ and high value of α is used for stable demand

(d) $\alpha \le 0$ and high value of α is used for unstable demand

(e) None of the above

CGPSC Poly. Lect. 22.05.2016, GATE 2014

Ans. (b) : $0 \le \alpha \le 1$ and high value of α is used for unstable demand.

248. Which of the following is the measure of forecast error?

(a) Mean absolute deviation

(b) Trend value

(c) Moving average

(d) Price fluctuation

Mizoram PSC (Insp. of Factories) 2019 Paper-III

MPPSC AE 2016, ESE 2009

Ans. (a) : Method of measure of forecast error–

1. Mean absolute deviation (MAD)
2. Mean forecast error (MFE) or Bias
3. Mean square error (MSE)
4. Mean absolute percentage error (MAPE)
5. Tracking signal (TS)

249. For sales forecasting pooling of expert opinions is made use of in

(a) Statistical correlation

(b) Delphi technique

(c) Moving average method

(d) Exponential smoothing

Mizoram PSC (Insp. of Factories) 2019 Paper-III

RPSC Vice Principal ITI 2018

Ans. (b) : For sales forecasting, pooling of expert opinions is made use of in Delphi technique.

250. When using a simple moving average to forecast demand, one would :

(a) Given equal weight to all the demand data

(b) Assign more weight to the recent demand data

(c) Include new demand data in the average after discarding some of the earlier demand data

(d) All of these

OPSC AEE 2019 Paper-I, GATE 2001

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Ans : (a & c) :
(a) Given equal weight to all the demand data
(c) Include new demand data in the average discarding some of the earlier demand data.
In simple moving average method to forecast demand, equal weight is given to all demand data and it include new demand data in the average after discarding some of the earlier demand data.

251. Which of the following is a technique for forecasting?
(a) Exponential smoothing (b) PERT
(c) CPM (d) Control charts

OPSC AEE 2019 Paper-I, GATE 1989

Ans. : (a) : Exponential smoothing is a technique for forecasting.

252. The sale of cycles in a shop in four consecutive months are given as 70, 68, 82, 95. Exponentially smoothing average method with a smoothing factor of 0.4 is used in forecasting. The expected number of sales in the next month is
(a) 59 (b) 72
(c) 86 (d) 136

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2003

Ans. (b) : $F_t = 95$, $F_{t-1} = 82$, $F_{t-2} = 68$, $F_{t-3} = 70$ and $\alpha = 0.4$
$F_{t+1} = \alpha F_t + \alpha(1-\alpha)F_{t-1} + \alpha(1-\alpha)^2 F_{t-2} + \alpha(1-\alpha)^3 F_{t-3}$
F_{t+1} = Forecast for next month
F_t = Forecast for the previous month and so on.
$F_{t+1} = (0.4 \times 95) + (0.4 \times 0.6 \times 82) + (0.4 \times 0.6^2 \times 68) + (0.4 \times 0.6^3 \times 70) = 73.2$
So expected no. as per given option is 72.

253. In a time series forecasting model, the demands for five time periods were 10, 13, 15, 18 and 22. A linear regression fit resulted in an equation F = 6.9 + 2.9t where F is the forecast for period t. The sum of absolute deviations for the five data is
(a) 2.3 (b) 0.2
(c) –1.2 (d) 24.3

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2000

Ans. (a) : The absolute deviation is $|D - F|$.
The forecast for each period can be calculated using the regression line equation.

Period	D_t	$F_t = 6.9 + 2.9t$	$\lvert D_t - F_t \rvert$
1	10	9.8	0.2
2	13	12.7	0.3
3	15	15.6	0.6
4	18	18.5	0.5
5	22	21.4	0.6

Sum of absolute deviation is
$= 0.2 + 0.3 + 0.6 + 0.5 + 0.6 = 2.2 \approx 2.3$

254. In simple exponential smoothing forecasting, the value of smoothening coefficient should be kept close to _____ in order to give lower weight age to past demand.
(a) 1.0 (b) 0
(c) –1 (d) 0.5

PPSC Asstt. Municipal Engineer 15.06.2021
GATE 2013

Ans. (a) : In exponential smoothing forecast
$F_{t+1} = F_t + \alpha (D_t - F_t)$
If $\alpha = 1$ then
$F_{t+1} = D_t$
So, if we want to give more weightage for recent demand then the value of α should higher. In order to give lower weightage to past demand $\alpha = 1$.

255. A TV seller predicts the demand at the rate of 600 units per month for the next four months. Actual demand is found 500, 680, 800 and 900 units, respectively. The mean absolute deviation (MAD) and the BIAS are _____ and _____, respectively.
(a) –170; 120 (b) 170; –120
(c) 170; 120 (d) 120; 170

AAI Jr. Executive 26.03.2021

Ans. (c) :

Demand	Fore cast	Error
500	600	–100
680	600	80
800	600	200
900	600	300
		ΣError = 480

$$BIAS = \frac{\Sigma Error}{n} = \frac{480}{4} = 120$$

$$MAD = \frac{\Sigma|D_i - F_i|}{n} = \frac{|-100| + |80| + |200| + |300|}{4}$$

$$\Rightarrow \frac{680}{4} = 170$$

256. Which forecasting method gives decreasing weightages to the demands for all the past periods?
(a) Weighted moving average
(b) Moving average
(c) Exponential Smoothening method
(d) Simple average method

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) The weighted moving average of forecasting is a technical indicator that assigns a greater weighting to most recent data points, but a less weighting to data points in the distant past periods. It is obtained by multiplying each no. in data set by a predetermined weight and adding up the results value.

257. Select the most sophisticated successful forecasting model.
(a) Statistical forecasting
(b) Qualitative Technique
(c) Causal model
(d) Time series analysis

VIZAG STEEL MT 24.01.2021 Shift-II

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Ans. (c) : A causal model is the most sophisticated kind of forecasting tool. The causal model takes into account every thing known of the dynamics of the flow system and utilizes predictions of related events such as competitive actions, strikes and promotions.

258. Short term regular variations related to the calendar or time of day is known as

(a) Seasonality (b) Trend
(c) Cycles (d) Random variations

JPSC AE 10.04.2021, Paper-II

Ans. (a) : Seasonality- Seasonality is a characteristics of a time series in which the data experiences regular and predictable changes that recur every calendar year. Any predicable fluctuations or pattern that recur or repeats over a one-year period is said to be seasonable.

Trend- The trend shows the general tendency of the data to increase or decrease during a long period of time. A trend is a smooth, general long term, average tendency.

Cycles- The variations in a time series which operate themselves over a span of more than one year are the cyclic variations. This oscillatory movement has a period of oscillation of more than one year.

Random variation- The tendency for the estimated magnitude of a parameter to deviate randomly from the true magnitude of that parameter.

259. If the sales has seasonal fluctuation, forecasting is better done by

(a) Linear forecaster
(b) constant forecaster
(c) Cyclic forecaster
(d) Linear-Cyclic forecaster

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : Cyclic forecaster

260. Which one of the following statements is not correct for the exponential smoothing method of demand forecasting?

(a) Demand for the most recent data is given more weightage
(b) This method requires only the current demand and forecast demand
(c) This method assigns weight to all the previous data
(d) This method gives equal weightage to all the periods

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (d) : Exponential smoothing of demand forecasting– This method requires only the current demand and the forecasted value for the current period to give next forecast. This method is modified form of weighted moving average which gives weight to all the previous data but in exponentially decreasing order.

The most recent data is given the highest weight and the weight assigned to older data decrease exponentially.

261. Delphi method is used for:

(a) Technological forecasting
(b) Developing new products
(c) Acquiring new capacity and penetrating new market
(d) All of the above

HPPSC ADF 05.03.2019

Ans. (d) :

262. Which of the following is not a type of predictable demand behavior?

(a) trend (b) random variation
(c) cycle (d) seasonal pattern

RPSC AE 2018

Ans. (b) : Random variation—Random variation occur without any known reason or explanation. They are the unforeseeable and unexpected change in demand.

Cycles—Cycles are wave like occurrence that repeat over longer periods of time, they are usually tied to economic conditions or the business cycle. Big-ticket items such as automobiles often display this type of pattern.

Seasonal factors—Seasonal factors are regular variations that occur over and over and are related to some particular event.

Products and services with significant seasonal demand include air conditioners and snow shovels and back to school clothing and supplies.

Trends—Trends are a long-term upward or downward movement in demand when products experience a growth or decline in the market, trends tends to emerge. Examples include the growth of organic products in the super market and the decline in sales of paper books in favour of the electronic variety.

263. The demand during the months of October, November and December are 10, 13 and 20 units and the weights assigned are 1, 2 and 3 respectively. The forecast for January is:

(a) 18 units (b) 12 units
(c) 14 units (d) 16 units

AAI Jr. Executive 29.11.2018

Ans. (d) : Given,
October demand = 10
November demand = 13
December demand = 20
Here, n = 3
n = number of period

$$\Sigma n = \frac{n(n+1)}{2} = \frac{3\times 4}{2} = 6$$

$$= 20\times\frac{n}{\Sigma n} + 13\times\frac{n-1}{\Sigma n} + 10\times\frac{n-2}{\Sigma n}$$

$$= 20\times\frac{3}{6} + 13\times\frac{2}{6} + 10\times\frac{1}{6}$$

January forecast = 20 × 0.5 + 13 × 0.33 + 10 × 0.17
=16 Unit

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264. Which one of the following forecasting models best predicts the turning point?

(a) Simple exponential smoothing model
(b) Brown's quadratic smoothing model
(c) Double exponential smoothing model
(d) Moving average model (using 5 data points)

ESE 2018

Ans. (d) : Moving average model (using 5 data points)

265. Delphi technique is used in

(a) Forecasting
(b) Inventory management
(c) Quality planning
(d) Material handling

HPPSC AE 2018

Ans. (a) : Delphi technique–The Delphi technique refers to the systematic forecasting method used together opinions of the panel of experts on the problem experts on the problem being encountered, through mail. In other words, a set of opinions pertaining to a specific problem, obtained in writing usually through questionnaires. From several experts in the specific field is called as a Delphi technique.

266. Which one of the following is not a causal forecasting method?

(a) Trend adjusted exponential smoothing
(b) Econometrics models
(c) Linear regression
(d) Multiple regression

TRB Polytechnic Lecturer 2017

Ans. (a) : Trend adjusted exponential smoothing is not a causal forecasting method.

267. In forecasting by exponential smoothing, if α is a smoothing constant, then :

(a) New forecast = α (least sales figure) + (1 – α) (old forecast)
(b) New forecast = α (least sales figure) - (1 + α) (old forecast)
(c) New forecast = α (least sales figure) + (1 - α) (old forecast)
(d) New forecast = α (least sales figure) - (1 + α) (old forecast)

TRB Polytechnic Lecturer 2017

Ans. (a) : New forecast = α (least sales figure) + (1 - α) (old forecast)

$F = \alpha \times D_n + (1 - \alpha)F_{n-1}$

268. For a simple moving average forecasting method, as the length of averaging period increases, the forecast sensitivity–

(a) Increases (b) Decreases
(c) Remains constant (d) Cannot be predicted

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Simple moving average forecasting method, as the length averaging period increases, the forecast sensitivity decreased.

269. If the actual and forecasted demands for the month of November 2016 are 721 and 706 respectively, what may be the forecast for December 2016 if the smoothening coefficient is 0.2?

(a) 721 (b) 719
(c) 709 (d) 701

Karnataka PSC AE, 10.09.2017

Ans. (c) : $F_T = F_{T-1} + \alpha [D_{T-1} - F_{T-1}]$
$\alpha = 0.2$
$F_{T-1} = 706$
$D_{T-1} = 721$
$\therefore F_T = 706 + 0.2 (721–706)$
$= 709$

270. Rolling horizon in forecast is used for

(a) Allowing same length of forecast horizon by easily adding a new period when one period is over
(b) Easy updating of changes and maintaining same length of forecast horizon by adding a new period when one period is over
(c) Easy updating of changes and there is no addition of a new period
(d) Different reasons other than the above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Rolling horizon are more agile than static forecasts, which project numbers based on a single time frame, say January through December, is used for easy updating of changes and maintaining same length of forecast horizon by adding a New period when one period is over.

271. In an exponentially weighted moving average, the weight of the demand of past periods

(a) Increases as age of the data increases
(b) Increases as age of the data decreases
(c) Decreases as age of the data increases
(d) Has no relationship with age of the data.

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : In an exponentially weighted moving average, the weight of the demand of past periods decreases as age of the data increases.

272. Adaptive response rate forecasting is related to

(a) Production Planning
(b) Production Scheduling
(c) Value analysis
(d) Inventory control

ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Adaptive response rate forecasting is a technique which is suitable for estimating mean demand in large scale inventory control system. It deals satisfactorily with fast moving items, slow moving items, erratic demand abnormal demand and step change in demand.

273. Delphi technique is a _____ method of forecasting.

(a) Time series (b) Judgmental
(c) Causal method (d) None of these

Kerala PSC Asstt. Prof. 27.09.2016

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Ans. (b) : Delphi Technique–Delphi is a method for obtaining an explicit judgmental forecast and can be evaluated only by comparing it to other judgmental forecasting methods that can be used when a forecast must be made despite insufficient data, inadequate models, and lack of time and resources of or through scientific study. It is suitable for long range forecasting.

274. The following data relates to annual sales in thousands of the product for a period of 16 years. Using 3 yearly moving averages, the last two moving averages, will be respectively

Year	Sales	Year	Sales
1	16	9	25
2	18	10	28
3	15	11	26
4	17	12	22
5	20	13	28
6	22	14	24
7	25	15	25
8	24	16	30

(a) 25.33 and 25.33 (b) 25.33 and 24.66
(c) 24.66 and 25.66 (d) 25.66 and 26.33
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : Moving averages for 3 years is given by–

$$= \frac{28+24+25}{3} = 25.66$$

Last two moving average

$$= \frac{24+25+30}{3} = 26.33$$

275. Match List I with List II and select the correct answer.

List-I	List-II
A. Decision making under complete certainty	1. Delphi approach
B. Decision making under risk	2. Maximax criterion
C. Decision making under complete uncertainty	3. Transportation model
D. Decision making based on expert opinion	4. Decision tree

	A	B	C	D
(a)	1	2	4	3
(b)	3	2	1	4
(c)	3	4	2	1
(d)	1	3	4	2
(e)	3	4	1	2

CGPSC AE 26.04.2015 Shift-I

Ans. (c) :

276. Forecasting which assumes a static environment in the future is
(a) Passive forecasting
(b) Active forecasting
(c) Long term forecasting
(d) Short term forecasting

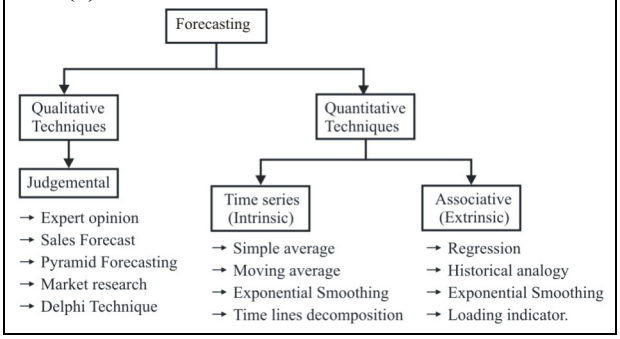
MPPSC AE 08.11.2015

Ans. (a) : Forecasting which assumes a static environment in the future is called passive forecasting.

277. Which of the following forecasting methods is also called "Extrinsic Method"?
(a) Regression Method
(b) Time Series Analysis Method
(c) Weighted Moving Average Method
(d) Exponential Smoothing Method

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) :



278. Current period forecast becomes equal to last period forecast for the value of smoothing constant equal to
(a) 1 (b) 2
(c) 0 (d) 0.5

BPSC AE 2012 Paper - VI

Ans : (c) : Current period forecast becomes equal to last period forecast for the value of smoothing constant equal to zero.

279. Which of the following is not a forecasting technique?
(a) Trend line estimate
(b) Delphi method
(c) Hungarian technique
(d) Judgemental technique

TRB Poly. Lect., 2012

Ans. (c) : Hungarian technique is not among these which is used for forecasting. This method is used for solving an assignment problem, to optimize the cost & maximize the profit.

280. Which of the following items has a seasonal demand?
(a) Torch light (b) Air conditioner
(c) Cooker (d) None of these

TRB Asstt. Prof., 2012

Ans. (b) :

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281. Which of the following is not connected with forecasting?

(a) Involves projection of past into future
(b) More objective
(c) Scientific
(d) Involves judgment by intuition

TRB Asstt. Prof., 2012

Ans. (b) :

282. Degree of relationship between two or more variables is expressed by

(a) seasonal index (b) correlation
(c) averages (d) none of these

TRB Asstt. Prof., 2012

Ans. (b) : Degree of relationship between two or more variables is expressed by correlation.

283. Which of the following is true concerning the weighted moving average?

(a) The oldest data will generally be given the greatest weight
(b) If the weighted moving average forecast is 57.3, then the final forecast must be rounded up to 58.
(c) If the most recent periods are too heavily weighted, the forecast might overreact
(d) The weighted moving average is usually more accurate than a simple moving average

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : The weighted moving average is more accurate than the simple moving average.

In weighted moving average, the most recent data is given the highest weight and the weight assigned to oldest will be least.

Simple moving average assigns identical weights to all the numbers in the data set.

Note- No answer is given by commission.

284. Assertion (A) : Forecasts for groups of items tend to be more accurate than forecasts for individual items.

Reason (R) : Opportunities for grouping arise if parts or raw materials are used for multiple products manufactured in the same organization.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know, forecasts for groups of items tend to more accurate than forecast for individual items, we can easily predict the real trend for forecast which will be difficult for individual. Opportunities for grouping arise when a parts or raw material are used for multiple product so we can use as a substitute of the materials.

285. A firm uses simple exponential smoothing with $\alpha = 0.2$ to forecast demand. The forecast for the month of June was 500 units, whereas the actual demand turned out to be 460 units. Assume the demand for the month of July turned out to be 480 units, the demand forecast for the month of August is :

(a) 490 units (b) 520 units
(c) 477 units (d) None of the above

APPSC IOF, 2009

Ans. (a) : $F_t = F_{t-1} + \alpha (D_{t-1} - F_{t-1})$

$= 500 + 0.2 (460 - 500)$

$F_t = 492$

$F_{t+1} = F_t + \alpha (D_t - F_t)$

$= 492 + 0.2(480 - 492)$

$F_{t+1} = 490$ units

286. The least squares forecasting attempts to find a fit to the linear equation $F_1 = a + bt$. The trend of the series is indicated by which value?

(a) Coefficient 'a'
(b) Coefficient 'b'
(c) Variable 't'
(d) The sum of the coefficient 'a' and 'b'

UPSC JWM Adv. No-16/2009

Ans. (c) : The trend of the series is indicated by the value of variable "t".

287. According to the simple exponential smoothing model of forecasting:

$S_t = \alpha D_t + (1 - \alpha) S_{t-1}$

where D_t is the demand in period t, S_t is the estimated mean demand including all data upto period t and the parameter α is the proportion of weight of the most recent observation. If we choose $\alpha = 0.03$ for the monthly forecast, then for the estimation of mean demand up to March 2008, the weightage of data of January 2008 is

(a) 0.3 (b) 0.09
(c) 0.027 (d) 0.147

DRDO Scientists 2008

Ans. (c) :

Period	Weight
Mar. 08	α
Feb. 08	$\alpha(1-\alpha)$
Jan. 08	$\alpha(1-\alpha)^2$

So, weight for January would be $\alpha(1-\alpha)^2$

Weight $= 0.03(1 - 0.03)^2$

$= 0.0282$

$\simeq 0.027$

4. Queuing Theory

288. Queuing theory is associated with

(a) Production time (b) Waiting time
(c) Planning time (d) Sales time

JPSC AE PRE 2019
Nagaland PSC (CTSE) 2018, Paper-I

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Vizag Steel (MT) 2017
Kerala PSC Poly. Lect., 2017
GPSC Lect. 23.10.2016
KPSC AE 2015, TSPSC AEE 2015
MPSC Poly. Lect. 2014
J&K PSC Civil Services Pre, 2013
RPSC ACF-2011
OPSC Civil Services Pre. 2011

Ans. (b) : Queuing theory—It is the quantitative technique which consists of constructing mathematical models for various types of queuing systems.
Mathematical models are constructed so that queue length and waiting times can be predicated which help in balancing the cost of service and the cost associated with customers waiting for services.

289. In a queuing problem, if the arrivals are completely random, then the probability distribution of number of arrivals in a given time follows :
(a) Poisson distribution
(b) Normal distribution
(c) Binomial distribution
(d) Exponential distribution

Haryana PSC AE (PHED) 05.09.2020, Paper-II
Mizoram PSC (Insp. of Factories) 2019 Paper-III
GPSC Executive Engineer 23.12.2018
MPPSC AE 2016
UKPSC AE-2013, Paper-I
ESE 2006

Ans. (a) : In a queuing problem, if the arrivals are completely random, then the probability distribution of number of arrivals in a given time follows Poisson distribution.

290. For a M/M/1 : ∞/FCFS queue, the mean arrival rate is equal to 10 per hour and the mean service rate is 15 per hour. The expected queue length is
(a) 1.33 (b) 1.53
(c) 2.75 (d) 3.20

VIZAG MT, 14.12.2020, RPSC IOF, 2020
UPPSC AE 12.04.2016 Paper-I
ESE 1998

Ans. (a) : $\rho = \dfrac{\text{mean arrival rate per hour}(\lambda)}{\text{mean service rate per hour}(\mu)}$

$$\rho = \frac{10}{15} = 0.66$$

$$L_s = \frac{\rho}{1-\rho} = \frac{0.66}{1-0.66} \approx 2$$

$$L_q = L_s - \frac{\lambda}{\mu}$$

$$= 2 - 0.66$$

$$L_q = 1.33$$

291. The utilization factor or traffic intensity in queuing theory is defined as :
(a) the ratio of mean service rate to mean arrival rate
(b) the product of mean arrival rate and mean service rate
(c) the sum of mean arrival rate and mean service rate
(d) the ratio of mean arrival rate to mean service rate

CGPSC AE 15.01.2021
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
UKPSC AE-2013, Paper-I
TRB Poly. Lect., 2012

Ans. (d) : The utilization factor or traffic intensity in queuing theory is defined as the ratio of mean arrival rate to mean service rate. It is represented by ρ.

$$\rho = \frac{\text{Mean arrival Rate}(\lambda)}{\text{Mean service Rate}(\mu)}$$

292. Cars arrive at a service station according to Poisson's distribution with a mean rate of 5 per hour. The service time per car is exponential with a mean of 10 minutes. At steady state, the average waiting time in the queue is
(a) 10 minutes (b) 20 minutes
(c) 25 minutes (d) 50 minutes

GPSC Asstt. Director of Transport 05.03.2017
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012, GATE 2011

Ans. (d) : We know, average waiting time of an arrival is given by $= \dfrac{\lambda}{\mu(\mu-\lambda)}$

Given,

Where, $\lambda = 5\text{hr}^{-1}$; $\mu = \dfrac{60}{10} = 6\text{hr}^{-1}$

$$= \frac{5}{6(6-5)} = \frac{5}{6}\text{hr} = 50\,\text{min}$$

293. The cost of providing service in a queuing system increases with :
(a) Increased mean time in the queue
(b) Increased arrival rate
(c) Decreased mean time in the queue
(d) Decreased arrival rate

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Gujarat PSC AE 2019
Mizoram PSC (Insp. of Factories) 2019 Paper-III
GATE 1997

Ans. (c) : The cost of providing service in a queuing system increases with decreased mean time in the queue.

294. Consider a single server queuing with poisons arrivals (λ = 4/hour) and exponential service (μ = 4/hour). The number in the system is restricted to a maximum of 10. The probability that a person who comes without joining the queue is

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(a) 1/11 (b) 1/10
(c) 1/9 (d) 1/2

TSPSC AEE 2017
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2005

Ans. (a) : Traffic intensity, $\rho = \frac{\lambda}{\mu} = \frac{4}{4} = 1$

If P_n is probability of n people in queue.

$$\sum_{\lambda=0}^{10} P_n = 1$$

$$P_0 + P_1 + P_2 + ...P_{10} = 1$$

$$P_0 + \rho P_0 + \rho^2 P_0 + ...+\rho^{10} P_0 = 1$$

$$P_0 (1 + \rho + \rho^2 + ...\rho^{10}) = 1$$

$$P_0(1+1.....1) = 1$$

$$P_0 = \frac{1}{11}$$

So, probability of a person who comes after the system.

$$P_{11} = \rho^{11} \times P_0 = 1^{11} \times \frac{1}{11} = \frac{1}{11}$$

295. In queue designation a/b/c : (d/e/f), what does c represent ?
(a) Arrival Pattern
(b) Service Pattern
(c) Number of service channels
(d) Capacity of the system

VIZAG STEEL MT 24.01.2021 Shift-II
GPSC ARTO Pre 30.12.2018
RPSC Vice Principal ITI 2018

Ans. (c) : Kendall Notation
a/b/c : d/e/f where,
a – is probability law for arrival
b – is probability law for service
c – is number of service channels
d – queue discipline
e – Size of system
f – calling source population

296. At a production machine, parts arrive according to a Poisson process at the rate of 0.35 parts per minute. Processing time for parts have exponential distribution with mean of 2 minutes. What is the probability that a random part arrival finds that there are already 8 parts in the system (in machine + in queue)?
(a) 0.0247 (b) 0.0576
(c) 0.0173 (d) 0.082

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 1999

Ans. (c) : Given, $\lambda = 0.35$/min, $n = 8$

$$\mu = \frac{1}{2}\text{/min} = 0.5\text{/min}$$

$$\rho = \frac{\lambda}{\mu} = \frac{0.35}{0.5}$$

$$\rho = 0.7$$

Probability of 'n' numbers in the system,

$$\rho_n(t) = \rho^n(1-\rho)$$

$$= (0.7)^8 (1–0.7)$$

$$\boxed{\rho_n(t) = 0.0173}$$

297. If average arrival rate in a queue is 6/hr and the average service rate is 10/hr, which one of the following is the average number of customers in the line, including the customer being served?
(a) 0.3 (b) 0.6
(c) 1.2 (d) 1.5

NPCIL ET, 2015, ESE 2007

Ans. (d) : Avg. Arrival rate in queue (λ) = 6/hr
Avg. service rate (μ) = 10/hr
Number of customer in the line including the customer being served.

$$= \left(\frac{\lambda}{\mu - \lambda}\right)$$

$$= \frac{6}{10-6} = \frac{6}{4} = 1.5$$

298. At a self-service store, a cashier can serve 10 customers in 5 minutes. On an average 15 customers arrive every 10 minutes. If the arrivals are as per Poisson distribution and services as per exponential distribution, the probability that the cashier would be idle is :
(a) 0.5 (b) 0.75
(c) 0.25 (d) 0

GPSC ARTO Pre 30.12.2018
ESE 2003

Ans. (c) : Given,
μ = 10 customers in 5 minutes
μ = 2 customers /min.
λ = 15 customers in 10 minutes
λ = 1.5 customers/min.

Utilization factor, $\rho = \frac{\lambda}{\mu} = \frac{1.5}{2} = 0.75$

The probability that the cashier would be idle is.
1 – 0.75 = 0.25

299. In Kendall's notation for representing queuing models :(a | b | c) : (d | e), d stands for
(a) probability law for inter-arrival time
(b) probability law for customers being served
(c) number of channels
(d) capacity of the system

OPSC Civil Services Pre 2006, ESE 2010

Ans. (d) : Kendall's Notation for Representing Queuing Models–
(a/b/c) : (d/e)
a = Probability law of inter arrival
b = Probability law for customer being served
c = No of parallel service channel
d = Capacity of the system
e = Queue discipline

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Kendall's and Lee Representation–
(a/b/c) : (d/e/f)
a = Probability law of inter arrival
b = Probability law for customer being served
c = No of parallel service channel
d = Service discipline like FCFS, FCLS, SIRO (service in random order) or GD (several service discipline)
e = Customer in system/capacity of system.
f = calling source

300. SIRO discipline is generally found in :
(a) Loading and unloading
(b) Office filing
(c) Lottery draw
(d) Train arrivals at platform

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (c) : SIRO (Service in Random Order) discipline is generally found in lottery draw.

301. When a doctor attends to an, emergency case leaving his regular service is called.
(a) Reneging
(b) Balking
(c) Pre-emptive queue discipline
(d) Non-Pre-emptive queue discipline

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (d) : When a doctor attends to an emergency case leaving his regular service is called Non-pre-emptive queue discipline.

302. In a M/M/1 queue, with utilization factor of 0.5, the probability of only one person waiting in the queue is
(a) 0 (b) 1.0
(c) 0.125 (d) 1.25

GPSC ARTO 01.05.2016, MPPSC AE 08.11.2015

Ans. (c) : $0 \le P \le 1$
$\because$ Probability is always less than or equal to one therefore option (d) is incorrect
Given, Utilization factor (ρ) = 0.5
When only one person waiting in the queue therefore total number of person in the system is two (n = 2).
$\therefore$ Probability (P) = $\rho^n . P_o$ $\quad \because P_o = 1 - \rho$

$$= (0.5)^2 \times (1 - 0.5)$$
$$= 0.125$$

303. In a single server queue customers are served at a rate of μ. If W and W_q represent the mean waiting time in the system and mean waiting time in the queue respectively then W will be equal to?
(a) $W_q + 1/\mu$ (b) $W_q - \mu$
(c) $W_q - 1/\mu$ (d) $W_q - \mu$

UPRVUNL AE 21.08.2016, ESE 1997

Ans. : (a) $W = \frac{L_s}{\lambda} = \frac{1}{\lambda}\left(\frac{\rho}{1-\rho}\right)$

$$W_q = \frac{L_q}{\lambda} = \frac{1}{\lambda}\left(\frac{\rho^2}{1-\rho}\right)$$

$$W = \frac{L_s}{\lambda} = \frac{L_q + \rho}{\lambda} = \frac{L_q}{\lambda} + \frac{\rho}{\lambda}$$
$$= \frac{L_q}{\lambda} + \frac{\lambda}{\mu \times \lambda} = \frac{L_q}{\lambda} + \frac{1}{\mu}$$
$$W = W_q + \frac{1}{\mu}$$

304. Little's law is a relationship between :
(a) Stock level and lead time in an inventory system
(b) Waiting time and length of the queue in a queuing system
(c) Number of machines and job due dates in a scheduling problem
(d) Uncertainty in the activity time and project completion time

Mizoram PSC (Insp. of Factories) 2019 Paper-III
GATE 2010

Ans. (b) : 'Little's law is a relationship between waiting and length of the queue in a queuing problem.

$$L_s = \lambda W_s \quad \Rightarrow W_s = \frac{L_s}{\lambda}$$

$$L_q = \lambda W_q \quad \Rightarrow W_q = \frac{L_q}{\lambda}$$

$$\boxed{W_s = W_q = \frac{1}{\mu}}$$

$L_s \Rightarrow$ Average no. of customer in the system
$L_a \Rightarrow$ Average no. of customer in the queue
$L_p \Rightarrow$ Avg. waiting time of customer in the system
$L_q \Rightarrow$ Avg. waiting time of customer in the queue

305. Service time in queuing theory is usually assumed to follow :
(a) Normal distribution
(b) Poisson's distribution
(c) Erlang distribution
(d) Exponential law

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Mizoram PSC (Insp. of Factories) 2019 Paper-III
ESE 1992

Ans. (d) : Service time in queuing theory is usually assumed to follow exponential law.
- Probability that more than 'T' time period is required to service a customer

$$\boxed{P(\text{time} > T) = e^{-\mu T}} \qquad \mu \Rightarrow \text{service rate}$$

306. If the number of arrivals in a queue follows the Poisson distribution, then the inner arrival time obeys which one of the following distributions?
(a) Poisson's distribution
(b) Negative exponential law
(c) Normal distribution
(d) Binomial

Mizoram PSC (Insp. of Factories) 2019 Paper-III
ESE 2007

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Ans. (b)

Arrival rate = $\lambda \Rightarrow$ Poisson's Distribution

Inter arrival rate = $\frac{1}{\lambda} \Rightarrow$ Exponential Distribution

Service rate = $\mu \Rightarrow$ Exponential Distribution

Inter service rate = $\frac{1}{\mu} \Rightarrow$ Poisson's Distribution

307. A single bay car wash with a Poisson arrival rate and exponential service time has cars arriving at an average rate of 10 min. apart and an average service time of 4 min. What is the system utilization?

(a) 1.00 (b) 0.67

(c) 0.40 (d) 0.24

GPSC DEE, Class-2 (GWSSB) 04.07.2021

ESE 2009

Ans. (c) : Arrival rate, λ = 10 min = 6/hr

Service rate, μ = 4 min = 15/hr

$\therefore \quad \frac{\lambda}{\mu} = \frac{6}{15} = \frac{2}{5} = 0.4$

308. A television repairman finds that the time spent on TV sets has an exponential distribution with mean of 20 minutes. If he repairs sets in the order in which they came in, and if the arrival of sets follows a Poisson distribution approximately with an average rate of 15 per 8-hour day, what is the repairman's expected idle time each day in hours?

(a) 2 (b) 5

(c) 4 (d) 3

AAI Jr. Executive 26.03.2021

Ans. (d) : Given,

Arrival rate, 15 per 8 hour day

$$\lambda = \frac{15}{8}$$

$$\mu = \frac{60}{20} = 3$$

Utilization factor, $\rho = \frac{\lambda}{\mu} = \frac{15}{8 \times 3} = \frac{5}{8}$

Idle time = $1 - \rho = 1 - \frac{5}{8} = \frac{3}{8}$

Idle time each day in hours,

$\Rightarrow \quad \frac{3}{8} \times 8 = 3$ hours

309. Vehicles pass through a toll gate at a rate of 90 per hour. The average time to pass the gate is 36 s. The arrival and service rate follow Poisson distribution. The number of vehicles waiting in the queue is:

(a) 7 (b) 8

(c) 10 (d) 9

AAI Jr. Executive 26.03.2021

Ans. (d) : Arrival rate for vehicle, λ = 90/hr

Service rate at toll, $\mu = \frac{3600}{36}$/hr

μ = 100/hr

$$\rho = \frac{\lambda}{\mu} = \frac{90}{100} = 0.9$$

No. of vehicles in queue, $\boxed{L_q = \frac{\rho^2}{1-\rho}}$

$$L_q = \frac{(0.9)^2}{1-0.9} = \frac{0.81}{0.1} = 8.1$$

So, 8 vehicles are in the queue and 9th vehicle remains in the queue for 10% of time.

So, answer will be 9.

310. Which of the following scheduling algorithms is suitable in a congested job shop?

(a) Shortest processing time

(b) Earliest Due date

(c) Minimum Slack method

(d) First Come first Serve

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : First come first serve (FCFS) is an operating system scheduling algorithm that automatically executes queued requests and processes in order of their arrival.

311. A workstation has a mean process time(in normal operating conditions) of 15 minutes, a mean time to failure of 120 minutes and a mean time to repair of 15 minutes. What is the effective process time of this workstation?

(a) 15 minutes (b) 150 minutes

(c) 16.875 minutes (d) 13.33 minutes

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : MTBF = MTTF + MTTR

= 120 + 15

= 135

Effective process time

= Mean process time + MTBF

= 135 + 15

= 150 min

312. Repairing of a machine consists of 5 steps that must be performed sequentially. Time taken to perform each of the 5 steps is found to have an exponential distribution with a mean of 5 minutes and is independent of other steps. If these machines break down in Poisson fashion at an average rate of 2/hour and if there is only one repairman, the average idle time for each machine that has broken down will be

(a) 120 minutes (b) 110 minutes

(c) 100 minutes (d) 90 minutes

ESE 2020

Ans. (c) : No. of phases service = k = 5

Machine break down rate (λ) = 2/hr

Service time = 25 min (5 × 5 = 25 min)

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Mean service rate $(\mu) = \frac{1}{25} \times 60$
$= 2.4/\text{hr}$
Average idle time of break down machine (W_s) = ?
Waiting time in the system = Idle time of the machine

$$W_s = \frac{k+1}{2k} \times \frac{\lambda}{\mu} \times \frac{1}{\mu - \lambda} + \frac{1}{\mu}$$

$$= \frac{5+1}{2\times 5} \times \frac{2}{2.4} \times \frac{1}{(2.4-2)} + \frac{1}{2.4} = \frac{5}{3}\text{hour} = \frac{5}{3} \times 60\,\text{min}$$

$= 100$ minutes

313. Which one of the following ways will be adopted to store the program counter contents?
(a) Last-in-First-out (LIFO)
(b) First-in-First-out (FIFO)
(c) Last-in-Last-out (LILO)
(d) First-in-Last-out (FILO)

ESE 2019

Ans. (a) : Last in first out (LIFO) is a method used to account for inventory, where the most recently produced items are recorded as sold first. It is necessary to store the program counter contents in last in first out (LIFO).

314. The expected waiting time in queue:
(a) $\mu/\lambda(\mu - \lambda)$ (b) $\lambda/\mu\,(\mu - \lambda)$
(c) $\mu/\lambda(\mu + \lambda)$ (d) $\lambda(\mu - \lambda)/\mu$

HPPSC ADF 05.03.2019

Ans. (b) : $W_q = \frac{L_q}{\lambda}$, Where $L_q = \frac{\rho^2}{1-\rho}$ and $\rho = \frac{\lambda}{\mu}$

$$= \frac{\rho^2}{(1-\rho)\times\lambda} = \frac{\lambda^2}{\mu^2\left(\frac{\mu-\lambda}{\mu}\right)\times\lambda}$$

$$= \frac{\lambda}{\mu(\mu-\lambda)}$$

315. In the notation (a/b/c) : (d/e/f) for summarizing the characteristics of queuing situation, the letter 'b' and 'd' stand respectively for:
(a) Service time distribution and queue discipline
(b) number of servers and size of calling source
(c) number of servers and queue discipline
(d) service time distribution and maximum number allowed in system.

HPPSC ADF 05.03.2019

Ans. (a) :

316. Arrivals at a bank counter are considered to be Poisson distribution, with an average time of 12 min between two successive arrivals. The time required to serve is distributed exponentially with a mean of 3 min. The probability that an arrival does not have to wait is before service:
(a) 0.25 (b) 0.75
(c) 0.9 (d) 0.5

SJVN ET 2019

Ans. (b) : Probability that customer will have to wait

$$\rho = \frac{\lambda}{\mu}$$

$$\lambda = \frac{60}{12} = 5/\text{hr}$$

$$\mu = \frac{60}{3} = 20/\text{hr}$$

$$\rho = \frac{5}{20} = \frac{1}{2}$$

$= 0.25$

The probability that an arrival does not have to wait is before service :
$= 1 - \rho$
$= 1 - 0.25$
$= 0.75$

317. The distribution of arrivals in a queuing system can be considered as a
(a) Death process (b) Pure birth process
(c) Pure live process (d) Sick process

RPSC Vice Principal ITI 2018

Ans. (b) : The distribution of arrivals in a queuing system can be considered as pure birth process.

318. In queuing theory with multiple servers, the nature of the waiting situation can be studied and analyzed mathematically, if
(a) the complete details of the items in the waiting lines are known
(b) the arrival and waiting times are known and can be grouped to form an appropriate waiting line model
(c) all the variables and constants are known and they may form a linear equation
(d) the laws governing arrivals, service times and the order in which the arriving units are taken into service are all known

ESE 2018

Ans. (d) : Arrival pattern, service pattern and Queue discipline must be known in Queuing theory.

319. The reasons which are basically responsible for the formation of a queue should be that-
(a) The average service rate is less than the average arrival rate
(b) Output rate is linearly proportional to input
(c) Output rate is constant and the input varies in a random manner
(d) All of the above

Vizag Steel (MT) 2017

Ans. (d) : All of the above

320. Which of the following stock valuation methods is based on a cost estimate made before the part is purchased :
(a) FIFO (b) WFO
(c) Standard costing (d) Average costing

Kerala PSC Poly. Lect., 2017

Ans. (c) : Standard costing is stock valuation methods based on a cost estimate made before the part is purchased.

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321. In the queuing theory, the relationship between expected number of customers in the system (or queue) and expected waiting time in the system (or queue) is known as :

(a) Kendall's formula (b) Little's formula
(c) Bellman's formula (d) Erlang formula

TRB Polytechnic Lecturer 2017

Ans. (b) : In the queuing theory, the relationship between expected number of customers in the system (or queue) and expected waiting time in the system (or queue) is known as little's formula.

322. The term 'jockeying' in queuing theory refers to

(a) not entering the long queue
(b) leaving the queue
(c) shifting from one queue to another parallel queue
(d) None of the above

BPSC AE Mains 2017 Paper - VI

Ans. : (c) : When a customer keep on changing queue in hope to get service faster is known as jockeying in queuing theory.

333. A queuing system, using Kendall's notation, is expressed in the symbolic from as (M/M/3); (FCFS/6). How many number of servers in the system?

(a) 6 (b) 3
(c) 2 (d) 1

TRB Polytechnic Lecturer 2017

Ans. (b) : In (M/M/3); (FCFS/6)
In this symbol number of servers is 3.

334. Monte Carlo simulation of queues is used when

(a) arrival time distributions are standard
(b) service time distributions are standard
(c) mathematical analysis cannot be adopted
(d) None of the above

BPSC Poly. Lect. 2016

Ans. : (c) Monte carlo simulation of queues is used when mathematical analysis cannot be adopted.

335. The calling population is assumed to be infinite when:

1. Arrivals are independent of each other
2. Capacity of system is infinite
3. Service rate is faster than the arrival rate

(a) Only (1) (b) Only (2)
(c) Only (3) (d) Both (1) and (2)
(e) (1), (2) and (3)

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : A calling population is considered to be infinite when arrivals are independent of each other. If the new customers arrival is independent of number of customers already in the system, the calling population is called unlimited or infinite.

336. If the queue length appears very large a customer he/she may not join the queue. This property is known as.....of customers.

(a) Jockeying (b) Balking
(c) Reneging (d) None of these

HPPSC AE (PWD) 10.09.2016

Ans. (b) : Balking- If the queue length appears very large a customer may not join the queue this property is known as balking.

337. Arrivals at a telephone booth are considered to be Poisson with an average time of 16 minutes between one arrival and the next. The length of a phone call is assumed to be distributed exponentially with mean of 4 minutes. The probability that a person arriving at the book will have to wait is?

(a) 0.25 (b) 0.75
(c) 0.5 (d) 0.35

UPRVUNL AE 21.08.2016
GATE 2002

Ans. : (a) λ = No of arrivals/hour = $\frac{60}{16}=3.75$

$$\mu = \frac{60}{4}=15$$

$$\rho = \lambda/\mu = \frac{3.75}{15}$$

$$\rho = 0.25$$

338. What type of example is a three-step repair service with one server at each step:

(a) Single-phase, single-server
(b) Multiple-phase, single-server
(c) Single-phase, multiple-server
(d) Multiple-phase, multiple-server

Nagaland CTSE 2016 Ist Paper

Ans. (b) : Three-step repair service with one server at each step is a example of multiple-phase, single-server.

339. Babies are born in a light populated state at the rate of one birth every 12 minutes. The times between births follows an exponential distribution. Find the probability that no births will occur in any one day

(a) 1 (b) 0
(c) 3 (d) 5
(e) 4

CGPSC AE 26.04.2015 Shift-I

Ans. (b) : The birth rate per day is computed as

$$\lambda = \frac{24\times 60}{12} = 120 \text{ birth/day}$$

The probability of number of birth in any one day is computed from the Poisson distribution as

$$P_o(1) = \frac{(120\times 1)^0 \times e^{-120\times 1}}{\lfloor 0} = 0$$

340. Which one of the following is not a queuing theory model?

(a) orders waiting to be processed
(b) machines waiting to be repaired
(c) machines in an assembly line to be operated
(d) batch of finished products waiting to be transported

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (c)

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341. Arrival to a system is Poisson with mean rate of 6 per hour, and service time of 3 minutes. The expected queue length will be :

(a) 0.3 (b) 0.129

(c) 1.29 (d) 12.9

MPSC Poly. Lect. 2014

Ans. (b) : Given,

mean arrival rate (λ) = 6 customer /hours.

$$\mu = 3 \text{ minute} \rightarrow \frac{60}{3} = 20 \text{customer/hr.}$$

$$\rho = \frac{\lambda}{\mu} = \frac{6}{20} = \frac{3}{10} = 0.3$$

$$L_q = \frac{\rho^2}{1-\rho} = \frac{0.3^2}{1-0.3} = 0.128$$

$$L_q = 0.128$$

342. Customers arrive at a reception counter at an average interval rate of 10 minutes and receptionist takes an average of 6 minutes for one customer. Determine average queue length.

(a) 9/10 (b) 7/10

(c) 11/10 (d) 3/10

UPRVUNL AE 2014

Ans. : (a) Average queue length $P = \frac{\rho}{1-\rho}$

Given

$$\text{Arrival Rate } (\lambda) = \frac{1}{10}$$

$$\text{Service Rate } (\mu) = \frac{1}{6}$$

$$\rho = \frac{\frac{1}{10}}{\frac{1}{6}} = \frac{6}{10} = \frac{3}{5}$$

Where $\rho = \frac{\lambda}{\mu}$

$$L_q = \frac{\rho^2}{1-\rho}$$

$$= \frac{25}{1-\frac{3}{5}} = \frac{9}{10}$$

343. As the ratio of mean arrival to mean service rate is increased. It is likely that

(a) customer moves faster in the system

(b) customer moves slowly in the system

(c) utilization is decreased because of added strain on the system

(d) the average number in the system decreases

TRB Poly. Lect., 2012

Ans. (b) : As the ratio of mean arrival to mean service rate is inverted, it is likely that customer moves slowly in the system.

344. Queuing theory is used for:

(a) Job shop scheduling

(b) Inventory problems

(c) Traffic congestion studies

(d) All of these

RPSC AE GWD, 2011

Ans. (c) : Queuing theory is used for traffic congestion studies.

345. For a facility with an average arrival rate of 45 units per hour and a service rate of 60 units per hour, with a maximum system size of 3 units, which of the following statements is true?

(a) The probability the system is full is .63

(b) The average number of units waiting to be served and being served is 1.15 units

(c) The average time a unit spends waiting to be served is 1.8 minutes

(d) The average number of units waiting to be served is .64 units

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (*) : Arrival rate λ = 45 unit/hr

Service rate μ = 60 unit/hr

$$\rho = \frac{\lambda}{\mu} = 0.75$$

Now the probability that the system is full can be calculated as we know the maximum quantity of the system i.e. K = 3

The probability that there are 3 customer in the system

$= \rho^3 (1 - \rho)$

$= (0.75)^3 (1 - 0.75)$

$= 0.1054$

The probability that there are 3 customer in the system

$= 0.1054$

Note-None correct option given.

346. A bank queue is having 3 counters with 30 system capacity cater to the whole campus. If arrival rate is 10/hr and service rate is 12/hr on FCFS basis the system can be represented as:

(a) M/M/30 : FCFS/3/∞

(b) M/M/3 : FCFS/∞/30

(c) M/M/3 : FCFS/30/∞

(d) M/M/3 : FCFS/10/12

TSPSC Managers, 2015

Ans. (c) : Queue = 3 counters

System capacity = 30

Arrival rate = 10/hr

Service rate 12/hr

System = 'σ' FCFS

Then, M/M/3 : FCFS/30/∞

347. Consider the following characteristics of the system under investigation regarding queuing model choice :

1. Number of servers
2. Arrival and service time patterns
3. Queue discipline (order of service)
4. VIP items and their number

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Queuing model choice is affected by which of these characteristics?

(a) 1, 2, 3 and 4 (b) 2, 3 and 4 only
(c) 1, 2 and 3 only (d) 1, 3 and 4 only

UPSC JWM Advt. No.-50/2010

Ans. (a) : Queuing model depends on these functions or characteristics.
(1) Number of servers
(2) Arrival & service time patterns
(3) Queue discipline (order of service)
(4) VIP items and their number.
According to Kendall and Lee Notation queuing model (a/b/c) : (d/e/f).
a → Probability distribution for arrival pattern
b → Probability distribution for service pattern
c → No. of server within the system
d → Service rule or service order
e → Size or capacity of system
f → Size or capacity of calling population or input

348. In an automobile manufacturing industry, for every 3 minutes a new motor cycle arrives for inspection and is attended by a quality control engineer on a first-come, first-serve basis. There is only one engineer on duty and it takes exactly 4 minutes to inspect each motor cycle. If there is no motor cycle awaiting inspection at the beginning of the shift, then the average number of motor cycle waiting for inspection in the first nine minutes is :

(a) 0 (b) 1/9
(c) 2/9 (d) 1

DRDO Scientists 2009

Ans. (d) : Average waiting time in queue :

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{\frac{1}{4}}{\frac{1}{3}\times\left(\frac{1}{3}-\frac{1}{4}\right)}$$

$\therefore$ Number of motor cycle in first nine minutes

$$\frac{9}{9} = 1$$

349. Patients arrive at a doctor's clinic according to the Poisson distribution. Check up time by the doctor follows an exponential distribution. If on an average, 9 patients/hr arrive at the clinic and the doctor takes on an average 5 minutes to check a patient, the number of patients in the queue will be

(a) 1 (b) 1.25
(c) 2.25 (d) 3.25

DRDO Scientists 2008

Ans. (c) : $\lambda = 9$ customer/hr

$$\mu = \frac{1}{5}\times 60 = 12 \text{ customer/hr}$$

$$\rho = \frac{\lambda}{\mu} = \frac{3}{4}$$

$$L_q = \frac{\rho^2}{1-\rho} = \frac{\left(\frac{3}{4}\right)^2}{1-\frac{3}{4}} = 2.25$$

350. In the model M/M/I:∞/FCFS the expected queue length is equal to

(a) $\frac{\rho}{1-\rho}$ (b) $\frac{\rho^2}{1-\rho}$
(c) $\frac{1}{1-\rho}$ (d) $1-\rho$

where ρ = utilisation factor

UKPSC AE 2007 Paper -I

Ans. (b) : Average number of customer in system,

$$L_s = \frac{\lambda}{\mu-\lambda} = \frac{\rho}{1-\rho}$$

Average number of customer (length) of queue,

$$L_q = \frac{\rho^2}{1-\rho}$$

Where, ρ = Utilisation factor

5. Line balancing, Loading and Sequencing

351. A set of five jobs is to processed on a single machine. The processing time in days is given. The holding cost for each job is Rs. K per day.

Job	Processing time
P	5
Q	2
R	3
S	2
T	1

A schedule that minimizes the total inventory cost is

(a) T-S-Q-R-P (b) P-R-S-Q-T
(c) T-R-S-Q-P (d) P-Q-R-S-T

GPSC DEE, Class-2 (GWSSB) 04.07.2021
APGCL AM, 2021, TSPSC AEE 2017
GATE 2008

Ans. (a)

Job	Processing time
P	5
Q	2
R	3
S	2
T	1

For SPT – T S Q R P
In this rules jobs are sequenced in increasing order of there processing time i.e. min time first and max time in last.

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352. A manufacturing industry produces 300 spark plugs per shift of 8 hours. If the standard time per piece is 1.5 minutes, the productivity would be

(a) 3/4 (b) 5/8
(c) 7/16 (d) 15/16

Assam PSC AE (PHED), 18.10.2020
UPPSC AE 13.12.2020, Paper-I
UKPSC AE-2013, Paper-I

Ans. (d) : Actual time required for manufacturing one pieces in minutes $= \frac{8 \times 60}{300}$

$= \frac{8}{5}$

$\text{Productivity} = \frac{\text{Standard time}}{\text{Actual time}}$

$= \frac{1.5}{\frac{8}{5}} = \frac{15}{16}$

353. In an assembly line for assembling toys, five workers are assigned tasks which takes time of 10, 8, 6, 9 and 10 minutes respectively. The balance delay for line is :

(a) 43.3% (b) 14.8%
(c) 14.0% (d) 16.3%

AAI Jr. Executive 26.03.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
OPSC AEE 2019 Paper-I

Ans. : (c) :
Number of work stations (n) = 5
Cycle time (T) = 10 (largest station time)
Sum of all station time
$\sum t_i = 10 + 8 + 6 + 9 + 10 = 43$ minutes

$\text{Line efficiency} = \frac{\sum t_i}{nt} \times 100$

N = no. of work stations = 5

$\text{Line efficiency} = \frac{43}{5 \times 10} \times 100$

$= 86\%$

Balance delay = 100 – (Line efficiency)
Balance delay = 100 – 86
= 14%

354. Time taken for the job from its arrival to the system until its departure is :

(a) completion time (b) flow time
(c) due date (d) processing time

Vadodara Muncipal Corp. DEE, 2018
Kerala PSC IOF 19.04.2016

Ans. (b) : Flow time– Time taken for the job from its arrival to the system until its departure.

355. Johnson's algorithm is used for scheduling of :

(a) n jobs 2 machine case
(b) 2 jobs n machine case
(c) n jobs 3 machine case
(d) 3 jobs n machine case

OPSC AEE 2019 Paper-I
HPPSC Asstt. Prof. 18.11.2016

Ans. (a) : Sequencing rule of n jobs on 2 machines. This model is mainly applicable for flow shop environment where all jobs are processed in same order.

356. Following are said to be the benefits of assembly line balancing:

1. **It minimizes the in-process inventory**
2. **It reduces the work content**
3. **It smoothens the production flow**
4. **It maintains the required rate of output.**

Select the correct answer(s) using the code given below:

(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) 1, 2 and 4

UKPSC AE 2012, 2007 Paper -I, ESE 1998

Ans. (c)

357. The JIT production is essentially

(a) a pull production system
(b) a push production system
(c) a pull and push production system
(d) intermittent production system

TNPSC AE 2019, TRB Asstt. Prof., 2012

Ans. (a) : Pull production system- A method of production control in which downstream activities. Pull production strives to eliminate over production and is one of the three major components of a complete just-in-time production system.
In pull production a downstream operation, whether within the same facility or in a separate facility, provides information to the upstream operation, often via Kanban card, about what part of material is needed, the quantity needed, and when and where it is needed. Nothing a produced by the upstream supplier process until the downstream customer process signals a need. This is the opposite of push production.

358. We have 6 jobs, each of which must go through the machine. A and B in the order AB. Processing times are given in the table

Job	1	2	3	4	5	6
M1	3	12	5	2	9	11
M2	8	10	9	6	3	1

What will be the Minimum making span?

(a) 41 min (b) 42 min
(c) 43 min (d) 44 min

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c)

Job	1	2	3	4	5	6
M1	3	12	5	2	9	11
M2	8	10	9	6	3	1

Sequence order– 4, 1, 3, 2, 5, 6

Job	M1	M2	M1		M2	
			Input	Output	Input	Output
4	2	6	0	2	2	8
1	3	8	2+3	5	8+8	16
3	5	9	5+5	10	16+9	25
2	12	10	10+12	22	25+10	35
5	9	3	9+22	31	35+3	38
6	11	1	31+11	42	42+1	43

Minimum making span = 43

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359. In an assembly line for assembling engine, six workers are assigned tasks take 22, 24, 26, 27, 25 and 26 minutes respectively. The balanced delay for the line is :

(a) 7.4 % (b) 14.8%

(c) 7.0% (d) 10.3%

APPSC Poly Lect. 13.03.2020

Ans. (a) : T_c = 27 minutes

TWC = 22 + 24 + 26 + 27 + 25 + 26

= 150 minutes

n = 6 work station

$$\text{Balance delay} = \left(\frac{nT_c - TWC}{nT_c}\right)\times 100$$

$$= \left(\frac{6\times 27 - 150}{6\times 27}\right)\times 100$$

$$= \left(\frac{162 - 150}{162}\right)\times 100$$

$$= \left(\frac{12}{162}\right)\times 100 = 7.4\%$$

360. Consider a production line with three workstations. Workstation 1 has 5 identical machines in parallel, each with process time of 15 minutes/job. Workstation 2 has 6 identical machines in parallel, each with a process time of 12 minutes/job, and workstation 3 has a single machine with a process time of 3 minutes/job. What is the minimum work in process level at which this production line yields maximum throughout?

(a) 10 jobs (b) 12 jobs

(c) 8 jobs (d) 6 jobs

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) :

Work station	Number of Identical parallel machine	duration by one made
1	05	15 min
2	06	12 min
3	01	3 min

During any time rain number of work is providing = 05+ 06+01 = 12

361. Four jobs are to be processed on a machine as per data listed in the table:

Job	Processing time (in days)	Due date
1	4	6
2	7	9
3	2	19
4	8	17

If the Earliest Due Date (EDD) rule is used to sequence the jobs, the number of jobs delayed is

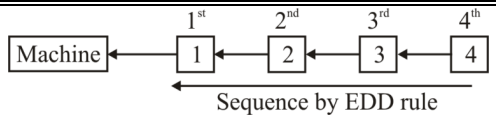
(a) 1 (b) 2

(c) 3 (d) 4

Gujarat PSC AE 2019

Ans. : (c) : Using earliest due date (EDD) rule,

Job	PT	Earliest due date (EDD)	Job flow time	Tardiness
1	4	6	0+4 = 4	–
2	7	9	4+7 = 11	2
3	8	17	11+8 = 19	2
4	2	19	19+2 = 21	2



Number of jobs delayed = 3 (job 2, job 3, job 4)

362. In an assembly line, five workers are assigned a work which take 80, 60, 70, 70, 70 seconds respectively. The balance delay for line is:

(a) 14% (b) 12.5%

(c) 16% (d) 10%

SJVN ET 2019

Ans. (b) : T_P = Processing time of all activity

= 80 + 60 + 70 + 70 + 70 = 350 second.

CL = Cycle time of assembly line

= Max time of individual work station

= Max (80, 60, 70, 70, 70)

= 80 Second

N = No. of station = 5

$$\text{Balance delay (DL)} = \frac{(n\times CL - T_P)}{n\times CL}\times 100$$

$$= \frac{5\times 80 - 350}{5\times 80}\times 100$$

$$= \frac{50}{400} = 12.5\%.$$

363. In an assembly line balancing, 4 tasks time 15, 18, 13, 21 min are to be assigned on the different workstations for the cycle time 25 min. The theoretical and actual number workstations required are

(a) 2 and 4 respectively

(b) 4 and 5 respectively

(c) 3 and 5 respectively

(d) 2 and 3 respectively

HPPSC AE 2018

Ans. (a) :

364. The cycle time, selected in balancing a line must be:

(a) Greater than the smallest time element given in the problem

(b) Less than the highest time element given in the problem

(c) Slightly greater than the highest time element given in the problem

(d) Left to the choice of the problem solver.

GPSC ARTO Pre 30.12.2018

Ans. (c) : The cycle time selected in balancing a line must be slightly greater than the highest time element given in the problem.

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365. Availability which is measure of performance of maintained equipments is expressed as

(a) availability = $\frac{\text{up-time+down-time}}{\text{up-time}}$

(b) availability = $\frac{\text{up-time}}{\text{up-time+down-time}}$

(c) availability = $\frac{\text{up-time}}{\text{up-time-down-time}}$

(d) availability = $\frac{\text{down-time}}{\text{up-time+down-time}}$

MPSC Lect. (Prod.) 2018

Ans. (b)

$$\text{Availability} = \frac{\text{up-time}}{\text{up-time+down-time}}$$

366. Which one of the following scheduling measures is not related to due date or delivery time?

(a) Flow time (b) Tardiness

(c) Lateness (d) Number of tardy jobs

TRB Polytechnic Lecturer 2017

Ans. (a) : Flow time is not related to due date or delivery time.

367. The time required for two operations cutting and binding for 5 jobs are as follows:

Job No.	1	2	3	4	5
Cutting (min)	8	6	2	5	7
Binding (min)	8	7	7	6	4

What is the optimal make span sequence?

(a) 2 - 4 - 1 - 3 - 5 (b) 3 - 4 - 2 - 1 - 5

(c) 1 - 2 - 3 - 4 - 5 (d) 3 - 5 - 2 - 4 – 1

TRB Polytechnic Lecturer 2017

Ans. (b) : Using Johnson's Rule

Job No.	Operation		Sequence
	Cutting (A)	Binding (B)	
1	8	8	4
2	6	7	3
3	2	7	1
4	5	6	2
5	7	4	5

Job – 3 → A < B → (First)
Job – 5 → A > B → Last
Job – 4 → A > B → Second
Job – 2 → A < B → Third
Job – 1 → A = B → Fourth
Then sequence will be
3 – 4 – 2 – 1 – 5

368. Standardization deals with the characteristics of product that include–

(a) dimensions of machine elements

(b) method of testing the product

(c) composition and properties of engineering materials

(d) all the these

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Standardization deal with the characteristics of product that include, dimensions of machine element method of testing the product, composition and properties of engineering materials.

369. Statement (I): Bar chart plots in the time dimension the planned performance of various activities of a project.

Statement (II): One advantage of a bar chart is that the inter sequence and linkage of all activities is indicated there in.

ESE 2017

Ans. (c) : The bar chart is used to deal with complex activities. The bar chart consists of two co-ordinate, the horizontal represents the time elapsed and the vertical represents the job or activities performed.
In any project, there are large numbers of activities which can be started with a certain degree of concurrency; the bar chart can not indicate clearly the independencies among the various activities. This is a major limitation.

370. The assumption made in sequencing problems i.e. No passing rule means.

(a) A job once loaded on a machine should not be removed until it is completed

(b) A job cannot be processed on second machine unless it is processed on first machine

(c) A machine should not be started unless the other is ready to start

(d) No job should be processed unless all other machines are kept ready to start

HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : The assumption made in sequencing problems is No passing rule means a job cannot be processed on second machine unless it is processed on first machine.
No passing rule means the passing is not allowed if each of the job is to be processed through two, machines say M_1 and M_2 in order M_1M_2 then the rule means that each job will go to machine M_1 first and then to machine M_2.

371. The shortest processing time prioritization rule is used for

(a) Reducing a queue size in front of a single server

(b) Reducing work-in-process in a single server system

(c) Reducing average flow time of jobs waiting in front of a server

(d) All of the above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : The shortest processing time prioritization rule is used for –

- Reducing a queue size in front of a single server.
- Reducing work-in-process in a single server system.
- Reducing average flow time of jobs waiting in front of a server.

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372. Weekly production requirements of a product are 1600 items. The cycle time of producing one product on a machine is 12 minutes. The factory works on two shift basis in which total available time is 20 hours. Out of the available time about 20% is expected to be wasted on break down material unavailability and quantity related problems. The factory works for 5 days in a week How many machines are required to fulfill the production requirements?

(a) 4 (b) 6
(c) 3 (d) 2

UPRVUNL AE 21.08.2016

Ans. : (a) Hours utilized per day = 20 (1–0.25)
= 15 hours
Hours utilized in a week = 15×5
= 75 hours
Number of items produced on a machine
$$= \frac{75 \times 60}{12} = 375$$
Therefore number of machines required
$$= \frac{1600}{375}$$
Machine required $= 4.2 \simeq 4$

373. A production line is to be designed to make 2400 items/week for at least the next 3 months. The line operates 40 hours/week. The standard time required to assemble each item is 244 second. What is the smallest number of work station required?

(a) 5 (b) 6
(c) 7 (d) 8

UPPSC AE 12.04.2016 Paper-I

Ans : (a) Minimum number of work station needs
$$= \frac{\text{Assembly time content}}{\text{Time for one product}} = \frac{244}{60} = 4.06$$
Minimum number of work station needed= 5

OR

Let smallest no. of work stations required =x
$40 \times 60 \times 60 \times x = 2400 \times 244$
$$x = \frac{2400 \times 244}{40 \times 60 \times 60} = 4.06$$
So min. no. of work station required = 5

374. Productivity of a production system is analogous to the

(a) output of a machine
(b) input to a machine
(c) efficiency of a machine
(d) efficiency of labour

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Productivity of a production system is analogous to the efficiency of a machine.

375. For solving 'n' jobs and two machines scheduling problem.....algorithm can be used.

(a) Parkinson's (b) Baker's
(c) Taylor's (d) Johnson's

HPPSC AE (PWD) 10.09.2016

Ans. (d) : Johnson's algorithm is used for sequencing of n jobs through two work centres. The purpose is to minimise idle time on machine.

376. Match List-I with List-II and select the correct answer using the codes given below the list :

	List-I	List-II
A.	Work-In-Process likely to be low compared to Output	1. Continuous Production
B.	Flexible Manufacturing Cell	2. Intermittent Production
C.	Just-In-Time inventory and Procurement Techniques	3. Job Production
D.	Scheduling Typically Most Complex Processes	4. Repetitive Production

(a) A-1, B-2, C-4, D-3 (b) A-1, B-2, C-3, D-4
(c) A-3, B-2, C-4, D-1 (d) A-3, B-2, C-1, D-4

UPRVUNL AE 21.08.2016
CSE Pre-2000

Ans. : (a)

List-I	List-II
1. Work-in-process likely to be low compared to output	1.Continous Production
2. Flexible manufacturing cell	2. Intermittent production
3. Just-in-Time inventory and procurement Techniques	3.Repetitive production
4. Scheduling typically most complex processes	4. Job production

377. There are six jobs that to be machined on machine M1 first and then on machine M2. The estimated times for machining are given below:

Job	1	2	3	4	5	6
Machine M1	4	10	2	5	6	1
Machine M2	3	2	5	4	2	6

What is the minimum making span?

(a) 28 (b) 30
(c) 32 (d) 34

NPCIL ET, 2015

Ans. (b) :

Job	M1	M2	
1	4	3	V
2	10	2	II
3	2	5	IV
4	5	4	
5	6	2	III
6	1	5	I
	28	22	

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Sequence-

6 3 4 1 5 2

Job	M_1	M_2	M_1		M_2	
			In	Out	In	Out
6	1	6	0	1	1	7
3	2	5	1+2	3	7+5	12
4	5	4	3+5	8	12+4	16
1	4	3	8+4	12	3+16	19
5	6	2	6+12	18	2+19	21
2	10	2	18+10	28	2+28	30

Minimum make span time = 30 min

378. A job shop incurs a cost of Rs. 600 per day for each day a job is in the shop. At the beginning of a month there are five jobs in the shop with following data-

Job	**1**	**2**	**3**	**4**	**5**
Processing time (days)	**5**	**3**	**8**	**2**	**6**
Due date	**10**	**12**	**20**	**9**	**8**

Which jobs will fail to meet their due dates?

(a) 1 (b) 2

(c) 3 (d) 4

NPCIL ET, 2015

Ans. (b) :

Job	Processing Time	Due Date	Job flow time	Tardiness
4	2	9	0 + 2 = 2	0
2	3	12	2 + 3 = 5	0
1	5	10	5 + 5 = 10	0
5	6	8	10 + 6 = 16	8
3	8	20	16 + 8 = 24	4
			57	12

No. of Tardiness jobs = 2

379. Five jobs are to be processed through two work centres. The time for processing each job is given below. Select the sequence in order to minimize the total processing time

Job	**Work centre I**	**Work centre II**
A	**5**	**2**
B	**3**	**6**
C	**8**	**1**
D	**10**	**7**
E	**7**	**12**

(a) A D B C E (b) B E D C A

(c) B A D C E (d) B D E C A

TNPSC AE 2014

Ans. (*) : Using Johnson's Rule

Job	**Work centre I (A')**	**Work centre II (B')**
A	5	2
B	3	6
C	8	1
D	10	7
E	7	12

If $A'_{min} < B' \rightarrow$ placed first in sequence

If $B'_{min} < A' \rightarrow$ placed last in sequence

Then,

For Job C $\rightarrow B'_{min} < A' \rightarrow$ Job C $\rightarrow 5^{th}$

For Job A $\rightarrow B'_{min} < A' \rightarrow$ Job A $\rightarrow 4^{th}$

For Job B $\rightarrow A'_{min} < B' \rightarrow$ Job B $\rightarrow 1^{st}$

For Job D $\rightarrow B'_{min} < A' \rightarrow$ Job D $\rightarrow 3^{rd}$

For Job E $\rightarrow A'_{min} < B' \rightarrow$ Job E $\rightarrow 2^{nd}$

Then the sequence in order to minimize the total processing time will be $\rightarrow$ B E D A C

380. Five jobs are waiting in a machining centre which are to be assigned to process. The processing times and due dates are given in the following table. Determine the sequence of processing according to Earliest Due Date (EDD)

Job	**Job Processing one day**	**Job due date**
A	**6**	**8**
B	**2**	**6**
C	**8**	**18**
D	**3**	**15**
E	**9**	**23**

(a) A B C D E (b) B D A C E

(c) B A D C E (d) E C A D B

TNPSC AE 2014

Ans. (c) : Using Johnson's Rule

Job	Job Processing one day	Job due date
A	2	6
B	6	8
C	3	15
D	8	18
E	9	23

So, sequence of processing according to Earliest Due Date (EDD) will be BADCE and

sequence of processing according to processing times will be BDACE.

381. Shop loading

(a) means the assignment of dates to specific jobs or operation steps

(b) is typically managed using an assembly chart

(c) means the assignment of jobs to work or processing centers

(d) is oriented toward the management of work in process inventories

TNPSC AE 2014

Ans. (c) : Shop loading means the assignment of jobs to work or processing centers.

382. An assembly line is an example of which one of the following processes?

(a) Product focused process

(b) Customized process

(c) Repetitive process

(d) Specialized process

UPRVUNL AE 2014

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Ans : (c) An assembly line is a manufacturing process in which parts are added as the semifinished assembly moves from work station to another work station where the parts are added in series until the final assembly is produced. Hence, It is a type of Repetitive Process.

383. If work station times are not same, the overall production rate of an assembly line is determined by the:-

(a) Fastest station time
(b) Slowest station time
(c) Average of all station times
(d) Average of slowest and fastest station times

UKPSC AE-2013, Paper-I

Ans. (b) : If work station times are not same, the overall production rate of an assembly line is determined by the slowest station time.

384. Sequencing is a subset of

(a) routing (b) scheduling
(c) expediting (d) none of these

TRB Poly. Lect., 2012

Ans. (b) Sequencing is the order of processing a set of take over available resources. Scheduling involves sequencing task of allocating as well as the determination of process commencement and completing time i.e. time- tabling

385. Match List I correctly with List II and select your answer using the codes given below

List I	List II
(A) OC curve sampling	1. Acceptance
(B) AOQL	2. Dodge-Roaming table
(C) Binomial distribution	3. P-charts
(D) Normal curve	4. Control charts for variable

	A	B	C	D
(a)	1	2	3	4
(b)	1	3	2	4
(c)	4	2	3	1
(d)	4	3	2	1

TNPSC ACF 2012

Ans. (a) : OC Curve – Acceptance simplify
AOAL – Dodge-Roaming table
Binomial distribution – P-charts
Normal curve – Control charts for variable

386. A company has four work centers A, B, C and D with per day capacities of 450 units, 390 units, 360 units and 400 units respectively. The machines are laid in order A, B, C, D and the product has to be operated on all these machines for getting converted into finished product. The actual output turns out to be 306 units. What is the system efficiency?

(a) 70% (b) 85%
(c) 78% (d) 80%

APPSC AEE 2012

Ans. (b) : Input = Smallest of (450, 390, 360, 400) = 360 Actual output = 300 units.

$$\therefore \text{ system efficiency } = \frac{306}{360} = 0.85 = 85\%$$

387. It is used to group work elements into each work station so that all the workers in each workstation are active about the same amount of time. What technique is it?

(a) Mapping
(b) Flexible manufacturing cell
(c) Scheduling
(d) Line balancing

VIZAG STEEL MT 2011

Ans. (d) : Line balancing

388. In a transfer line.

(a) All the machine tools must be automatic
(b) All the machine tools must be of conventional and general purpose type
(c) The work stations must form a closed loop.
(d) The cycle time is the total time taken by all the machining operations

UPSC JWM Advt. No.-52/2010

Ans. (a) : A transfer line is a manufacturing system which consists of a predetermined sequence of machines connected by an automated material handling system and designed for working on a very small family parts.

389. An automated transfer line has 10 stations and an ideal cycle of 1 minute. The probability of a station failure is 0.01 and the average downtime when the breakdown occurs is 10 minutes :

(a) 1.01 (b) 1.1
(c) 2 (d) 10.1

DRDO Scientists 2009

Ans. (b) : $T_p = T_c + FT_d$
$T_c = 1$ mm, $F = 0.01$, $T_d = 10$
$T_p = 1 + 0.01 \times 10$
$T_p = 1 + 0.1$
$\boxed{T_p = 1.1 \text{min}}$

390. Loading consists of

(a) determination of when is to be done
(b) determination of requirements and control of men and machines
(c) determination requirements and control of materials
(d) determination of requirements and control of tools.

TNPSC AE, 2008

Ans. (b) : Loading consists of determination of requirements and control of men and machines.

6. PERT and CPM

391. In PERT critical activity has :

(a) Maximum float (b) Zero float
(c) Minimum float (d) Minimum cost

GPSC DEE, Class-2 (GWSSB) 04.07.2021

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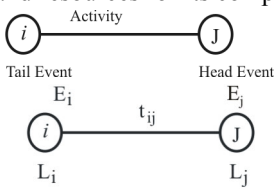
JPSC AE 10.04.2021, Paper-II
Mizoram PSC (Insp. of Factories) 2019 Paper-III
OPSC AEE 2019 Paper-I
MPSC Lect. (Prod.) 2018
Nagaland PSC (CTSE) 2018, Paper-I
TSPSC AEE 2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
Kerala PSC IOF 19.04.2016
Rajasthan Nagar Nigam AE 2016, Shift-I
GPSC Asstt. Prof. 28.08.2016
HPPSC W.S. Poly. 2016
OPSC AEE 2015 Paper I
TSPSC Managers, 2015
GPSC Lect. (Poly.) 07.12.2014
MPSC Poly. Lect. 2014, RPSC AEN Pre-2013
SJVN ET 2013, TRB Poly. Lect., 2012
VIZAG STEEL MT 10.06.2012
BPSC AE 2012 Paper - VI, GATE 2009, 2004

Ans. (b) : Activity - It is a part of project which consume time and resources for its completion.



The activities with zero slack of head event ($L_j - E_j$) and zero slack for the tail event ($L_i - E_i$) are called critical activities the activities with zero float are called critical activities. The sequence of critical activities in a network is called the critical path.

392. Program Evaluation Review Technique (PERT) has following time estimate

(a) Three time estimate (b) Two time estimate
(c) One time estimate (d) Four time estimate

TNPSC AE 2019, MPSC Lect. (Prod.) 2018
ISRO Scientist/Engineer 22.04.2018
RPSC Vice Principal ITI 2018
Sikkim PSC (Under Secretary), 2017
Rajasthan AE (Nagar Nigam) 2016 Shift-3
ISRO Scientist/Engineer (RAC) 29.11.2015
Mizoram PSC AE/SDO 2014 Paper-III
J&K PSC Civil Services Pre, 2013
BPSC AE 2012 Paper - VI
OPSC Civil Services Pre. 2011

Ans. (a) : In project management, the project evaluation review technique (PERT) is used to identify time it takes to finish a particular task or activity there are three estimation times involves in PERT.

1. Optimistic time estimate (t_o)
2. Most likely time estimate (t_m)
3. Pessimistic time estimate (t_p)

Three time estimates are derived for each activity.

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

393. A dummy activity is used in PERT network to describe

(a) Precedence relationship
(b) Necessary time delay
(c) Resource restriction
(d) Resource idleness

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ISRO Scientist/Engineer 12.01.2020
Mizoram PSC (Insp. of Factories) 2019 Paper-III
Gujarat PSC AE 2019
GPSC Asstt. Prof. (Prod.) 21.08.2016
RPSC AE (GWD) 06.05.2016
NPCIL ET, 2015, TNPSC ACF 2012
ESE 2000, 2013

Ans. (a) : Dummy activity- This activity does not consume any time or resources but depicts dependency or relationship with other activity. It is used to maintain the logical sequence. It is indicated by dotted line. So dummy activity is used in PERT network to describe precedence relationship.

394. CPM is the :

(a) time oriented technique
(b) event oriented technique
(c) activity oriented technique
(d) target oriented technique

JPSC AE 10.04.2021, Paper II
TSPSC Manager (Engg.) HMWSSB 12.11.2020
GPSC Asstt. Prof., 28.08.2016
BPSC Poly. Lect. 2016
Rajasthan Nagar Nigam AE 2016, Shift-II
MPSC Dy Engineer 2014
J&K PSC Civil Services Pre, 2013
TRB Asstt. Prof., 2012

Ans. (c) : In CPM technique of project management is manage only certain activities of any project. It is activity oriented technique.

395. In CPM, the crash cost slope is determined by:

(a) $\frac{\text{Crash cost}}{\text{Normal cost}}$

(b) $\frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}}$

(c) $\frac{\text{Normal cost}}{\text{Crash cost}}$

(d) $\frac{\text{Normal cost} - \text{Crash cost}}{\text{Normal time} - \text{Crash time}}$

Nagaland PSC (CTSE) 2018, Paper-I
UPRVUNL AE 21.08.2016, UJVNL AE 2016
UPRVUNL AE 07.10.2016
GPSC Lect. (Poly.) 07.12.2014
ESE 1994

Ans. : (b) In CPM. the crash cost slope is determined by

$$\text{Crash cost slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}}$$

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396. If t_o is the optimistic time, t_p is the pessimistic time and t_m is the most likely time, then the probabilistic time for completion of an activity is given by:

(a) $\frac{t_o + t_p + t_m}{3}$ (b) $\frac{t_o + 2t_p + t_m}{4}$

(c) $\frac{2t_o + t_p + 2t_m}{3}$ (d) $\frac{t_o + t_p + 4t_m}{6}$

(e) $\frac{t_o + 4t_p + t_m}{5}$

SJVN ET 2019
Nagaland PSC (CTSE) 2018, Paper-I
CGPSC ARTO & TSI 07.01.2017
HPPSC Asstt. Prof. 18.11.2016
TRB Asstt. Prof., 2012, GATE 2009

Ans. (d) : Probabilistic Time = $\frac{t_o + t_p + 4t_m}{6}$

When,
t_o = Optistic Time
t_p = Pessimistic Time
t_m = Most Likely time

397. PERT stands for :

(a) Programme Estimation and Review Technique
(b) Programme Evaluation and Review Technique
(c) Projects Estimation and Review Technique
(d) Project Evaluation and Review Technique

Nagaland PSC (CTSE) 2018, Paper-I
GPSC ARTO 01.05.2016, GWSSB AAE, 27.12.2015
TRB Poly. Lect., 2012, RPSC AE GWD, 2011

Ans. (b) : PERT- Programme Evaluation and Review Technique.

398. The critical path of a network represents :

(a) the minimum time required for completion of project
(b) the maximum time required for completion of project
(c) the maximum cost required for completion of project
(d) the minimum cost required for completion of project.

JPSC AE 10.04.2021, Paper-II
Vadodara Municipal Corp. DEE, 2018
Karnataka PSC AE, 10.09.2017
TRB Asstt. Prof., 2012, RPSC ACF-2011

Ans. (a) : The critical path of a network is the longest time of network but it represents the minimum time required for the completion of project.

399. PERT is based on the assumption of :

(a) Normal distribution of time estimates
(b) Exponential distribution of time estimates
(c) Beta distribution of time estimates
(d) Gamma distribution of time estimates

ISRO Scientist/Engineer (RAC), 10.03.2019
Mizoram PSC (Insp. of Factories) 2019 Paper-III
HPPSC Asstt. Prof. 18.11.2016
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
TRB Asstt. Prof., 2012

Ans. (c) : PERT (Programme Evaluation & Review Technique)–It is an event oriented network technique which is mainly associated with probabilities activities. It is beta distribution of time estimates.

400. PERT and CPM are :

(a) techniques to determine project status
(b) decision making techniques
(c) aids to determine cost implications of project
(d) aids to the decision maker

GPSC Lect. 23.10.2016, MPPSC AE 08.11.2015
RPSC ACF-2011, TNPSC AE, 2008

Ans. (d) : PERT is that technique project management which used to manage uncertain (i.e. time is not known) activities of any project and CPM is that technique of project management which used to manage only certain (i.e. time is known) activities of any project.
PERT and CPM are aids to the decision maker.

401. The probability distribution of project completion in PERT follows the distribution

(a) Normal (b) Beta
(c) Gaussian (d) Binomial

CGPSC Poly. Lect. 22.05.2016
Mizoram PSC AE/SDO 2014 Paper-III
APPSC Poly. Lect. 2013, RPSC ACF-2011

Ans. (c) : Probability distribution of project completion in PERT follow Gaussian distribution is a bell-Shaped curve and it is assumed that during any measurement values will follow a normal distribution with an equal number of measurement above and below the mean value.

402. For event 1 the earliest occurrence time is 8 weeks and the latest occurrence time is 26 weeks for event 2 the earliest occurrence time is 32 weeks and the latest occurrence time is 37 weeks. If the activity time is 11 weeks, the total float will be :

(a) 11 (b) 13
(c) 18 (d) 24

APPSC Poly Lect. 13.03.2020, RPSC IOF, 2020
APPSC AEE 2012, ESE 1998

Ans. (c) :

$(LOT)_1 = 26$ $(LOT)_2 = 37$ } LOT = Latest occurance time
$(EOT)_1 = 8$ $(EOT)_2 = 32$ } EOT = Earliest occurance time

$(1) \xrightarrow[\text{Event}]{11\text{water}} (2)$ Event

Total float = LOT - EOT
= 26 – 8
= 18

403. A project has 6 activities (A to F) with respective activity durations 7, 5, 6, 6, 8 and 4 days. The network has three paths A – B, C – D and E – F. All the activities can be crashed with the same crash cost per day. The number of activities needed to be crashed to reduce the project duration by 1 day is

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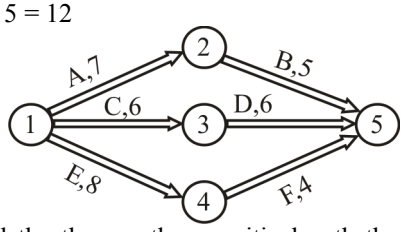
(a) 1 (b) 2
(c) 3 (d) 6

Karnataka PSC Lect., 27.05.2017
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012, GATE 2005

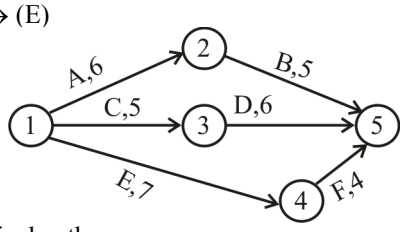
Ans. (c) : All the activities can be crashed with the same crash cost per day

Activity	Duration (days)
A	7
B	5
C	6
D	6
E	8
F	4

Network diagram for the project critical paths:
$1 \rightarrow 2 \rightarrow 5 = 12$
$1 \rightarrow 3 \rightarrow 5 = 12$
$1 \rightarrow 4 \rightarrow 5 = 12$



Since, all the three path are critical path therefore for crashing the project duration by day we need to consider all the 3 paths.
Path 1 → Either A or B need to be crashed
Path 2 → C or D need to be crashed
Path 3 → E or F need to be crashed
∵ The cost slope is same for all activities crashing 1 – 1 activity from 3 paths,
(1 – 2) → (A)
(1 – 3) → (C)
(1 – 4) → (E)



New critical path:
$1 \rightarrow 2 \rightarrow 5 = 11$
$1 \rightarrow 3 \rightarrow 5 = 11$
$1 \rightarrow 4 \rightarrow 5 = 11$

404. Slack is the difference between :
(a) Earliest and latest event time
(b) Earliest and latest activity time
(c) Longest and shortest activity time
(d) None of the above

HPPSC Asstt. Prof. 18.11.2016
TSPSC AEE 2015
JPSC AE 2013, Paper-V
TNPSC AE, 2008

Ans. (a) : Slack–Amount of time by which start of an activity may be delayed without affecting over all duration of project.

405. In PERT chart, the activity time distribution is
(a) Normal (b) Binomial
(c) Poisson (d) Beta

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Gujarat PSC AE 2019, ESE 2012
GATE 2016, 1995

Ans. : (d) : In PERT chart, the activity time distribution is beta.

406. In a PERT network, expected project duration is found to be 36 days from the start of the project. The variance is four days. The probability that the project will be completed in 36 days is :
(a) Zero (b) 34%
(c) 50% (d) 84%

RPSC ACF & FRO, 26.02.2021
CGPSC Asstt. Workshop Supt., 17.07.2016
ESE 1997

Ans. (c) : Given,
Variance = 4 days.
Standard deviation = 2 days.

$$z = \frac{x - \mu}{\sigma}$$

$$= \frac{36 - 36}{2}$$

$$\boxed{z = 0}$$

Probability = 50%

407. A PERT activity has an optimistic time of 3 days, pessimistic time of 15 days and an expected time of 7 days. The most likely time of the activity is
(a) 5 days (b) 6 days
(c) 7 days (d) 9 days

RPSC AE 2018, MPSC Poly. Lect. 2014
TRB Poly. Lect., 2012

Ans. (b) : Optimistic time (t_o)= 3 days, Pessimistic time (t_p) = 15 days expected time (t_e)= 7 days, then, mot likely time (t_m)=?

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

$$t_m = \frac{6t_e - t_o - t_p}{4} = \frac{6x7 - 3 - 15}{4} = \frac{24}{4}$$

$t_m = 6$ days.

408. The time by which the completion of an activity can be delayed beyond the earliest finish time without affecting the earliest start of a subsequent activity is called
(a) Free float (b) Independent float
(c) Total float (d) Interfering float

RPSC AE (GWD) 06.05.2016
GPSC Asst. Prof. (Prod.) 21.08.2016
ESE 2008

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Ans. (a) : The time by which the completion of an activity can be delayed beyond the earliest finish time without affecting the earliest start of a subsequent activity is called free float.
Free float = Total Float – Head event stock

409. The performance of a specific task in CPM is known as:
(a) Dummy (b) Event
(c) Activity (d) Contract

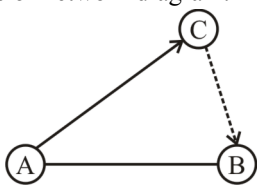
JPSC AE 10.04.2021, Paper-II
Vizag Steel (MT) 2017
GPSC Asstt. Prof. 28.08.2016

Ans. (c) : The performance of a specific task in CPM is known as activity.

410. Dummy activity requires :
(a) Only resource but no time
(b) Only time but no resource
(c) Both resource and time
(d) Neither resource nor time

HPPSC Asstt. Prof. 18.11.2016
RPSC ADE 2016
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. (d) : Dummy Activity–This activity does not consume time or resources only used to maintain a logical sequence of network diagram.



Dummy activity is shown by dotted lines.

411. CPM and PERT techniques are used for :
(a) layout planning
(b) financial management
(c) executing a new project
(d) increasing productivity

Kerala PSC AE 06.08.2015
KPSC AE 2015
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : CPM & PERT technique used for excepting new project. Difference b/w them is given below.

CPM	PERT
• Time is known for any project	• Time is unknown for any project
• It is activity oriented technique	• It is event oriented technique.
• It is a deterministic model	• It is a probability model
• It has repetitive nature	• It has Non-repetition nature of job.
• It uses dummy activities	• It does'nt uses any dummy activities.
• It is suitable for construction projects	• It is suitable for project Which required research & development.

412. Positive slack on a PERT indicates that project is
(a) ahead of schedule (b) Beyond schedule
(c) as per schedule (d) on critical path

TNPSC, AE 2017, GPSC Lect. 23.10.2016
TSPSC AEE 2015

Ans. (a) : A positive slack indicates the amount of time that the task can be delayed without delaying the project finish date. So, it indicates ahead of schedule.

413. Earliest finish time can be regarded as :
(a) EST + duration of activity
(b) EST – duration of activity
(c) LFT + duration of activity
(d) LFT – duration of activity

Mizoram PSC (Insp. of Factories) 2019 Paper-III
TSPSC AEE 2015, ESE 1993

Ans. (a)
- **Earliest Start Time (EST) –**
  It is the earliest event time of tail end event
- **Earliest Finish Time (EFT)**
  It is the sum of earliest start time and activity time.
  EFT = EST + duration of activity

Where EST = Earliest start time
- **Earliest Event Time (EVT) –**
  It is the maximum of the earliest finish times of all activities ending into that event.

414. Fulkerson's rule is associated with :
(a) Transportation problem
(b) Assignment problem
(c) PERT/CPM
(d) Queuing theory

TNPSC AE 2017
TRB Polytechnic Lecturer 2017
Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) : Fulkerson's rule for numbering of network. Activities are generally identified by node numbers, i.e. the numbers of events on either end of the activity arrow. Hence, it is essential to number events or nodes. And network diagram is made for PERT/CPM thus Fulkerson's rule is associated with PERT/CPM for numbering of events or node.
The rule used for numbering events involved in a project scheduling network is called as Fulkerson's rule.

415. Slack represents the difference between the
(a) earliest completion time and latest allowable time
(b) latest allowable time and earliest completion time
(c) earliest completion time and normal expected time
(d) latest allowable time and normal allowable time

Kerala PSC IOF 19.04.2016
TNPSC AE, 2008

Ans. (a) : Slack represents the difference between the earliest completion time and latest allowable time.

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416. CPM has following time estimate

(a) one time estimate
(b) two time estimate
(c) three time estimate
(d) four time estimate

Rajasthan Nagar Nigam AE 2016, Shift-I
RPSC ACF - 2011

Ans. (a) : CPM has one time estimate. CPM stands for critical path method.

417. In the 'PERT' network, the project completion time is considered to follow

(a) Normal Distribution (b) Beta Distribution
(c) Linear Distribution (d) None of the above

Kerala PSC Poly. Lect., 2017
TSPSC Managers, 2015

Ans. (a) : In the 'PERT' network, the project completion time is considered to follow normal distribution.
In the 'PERT' network, the activity completion time is considered to follow beta distribution.

418.

Activity	Predecessors	Duration
a	-	3
b	a	4
c	a	5
d	a	4
e	b	2
f	d	9
g	c, e	6
h	f, g	2

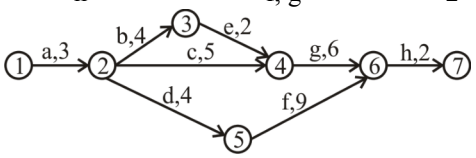
The critical path for the project is

(a) a-b-e-g-h (b) a-c-g-h
(c) a-d-f-h (d) a-b-c-f-h

APPSC Poly. Lect. 2013, GATE 2012

Ans. (c) :

Activity	Predecessors	Duration
a	-	3
b	a	4
c	a	5
d	a	4
e	b	2
f	d	9
g	c, e	6
h	f, g	2



a-b-e-g-h → (3 + 4 + 2 + 6 + 2) = 17
a-c-g-h → (3 + 5 + 6 + 2) = 16
a-d-f-h → (3 + 4 + 9 + 2) = 18
Critical path for project is a-d-f-h.

Critical path- It is a minimum- time consuming and longest path from the first event to last event in the network diagram.

The time taken along the critical path is expected project completion time (T_E). The activity along the Critical path are termed as critical activity & are represented by double line arrow.

419. The variance (V) for critical path

a → b = 4 time units
b → c = 16 time units
c → d = 4 time units
d → e = 1 time units
The standard deviation of the critical path a → e is

(a) 3 (b) 4
(c) 5 (d) 6

TSPSC AEE 2017, ESE 1998

Ans. (c) : We know, Standard deviation

$$(SD) = \sqrt{\text{Sum of variance}}$$

$$= \sqrt{4+16+4+1} = \sqrt{25}$$

SD = 5

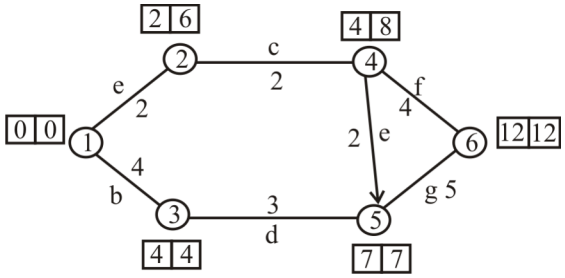
420. The precedence relations and duration (in days) of activities of a project network are given in the following table. The total float (in days) of activities e and f respectively are

Activity	Predecessors	Duration (days)
a	-	2
b	-	4
c	a	2
d	b	3
e	c	2
f	c	4
g	d,e	5

(a) 1 and 4 (b) 0 and 4
(c) 3 and 1 (d) 2 and 3

Assam Engg. College AP/Lect. 18.01.2021
GATE 2014

Ans. (a) : 1 and 4



Total float in e = 7 – (2 + 4) = 1
Total float in f = 12 – (4 + 4) = 4

421. An activity of the project is graphically represented by _____on the Network diagram.

(a) A circle (b) A straight line
(c) An arrow (d) A square

GMB AAE 25.06.2017
HPPSC W.S. Poly. 2016

Ans. (c) : An activity of the project is graphically represented by an arrow on the network diagram (→)

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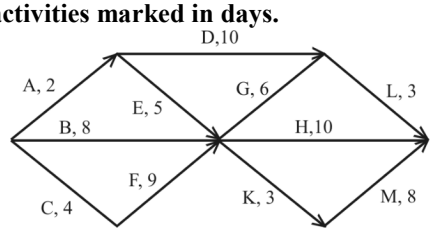
422. A project consists of activities A to M shown in the net in the figure below with the duration of activities marked in days.

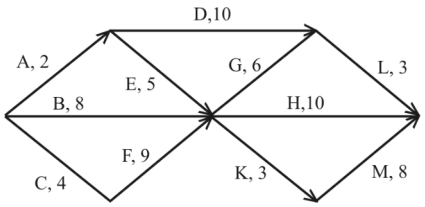


The project can be completed-

(a) Between 18, 19 days
(b) Between 20, 22 days
(c) Between 24, 26 days
(d) Between 60, 70 days

APGENCO AE 2012, GATE 2003

Ans. (c)



Project time –

$\Rightarrow$ Minimum – (A – E – k – M) = 18
$\Rightarrow$ Maximum (C – F – k – M) = 24

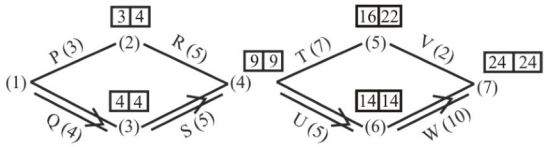
423. The project activities, precedence relationships and durations are described in the table. The critical path of the project is :

Activity	Precedence	Duration (in days)
P	–	3
Q	–	4
R	P	5
S	Q	5
T	R, S	7
U	R, S	5
V	T	2
W	U	10

(a) P-R-T-V (b) Q-S-T-V
(c) P-R-U-W (d) Q-S-U-W

Mizoram PSC (Insp. of Factories) 2019 Paper-III
GATE 2010

Ans. (d)



The critical path of the project is
Q → S → U → W

424. Which one of the following statements is not correct?

(a) PERT is probabilistic and CPM is deterministic
(b) In PERT, events are used and in CPM activities are used
(c) In CPM, crashing is carried out
(d) In CPM, the probability to complete the project in a given duration is calculated

APPSC AEE Screening Test 2016
ESE 2007

Ans. (d)
- PERT is probabilistic and CPM is deterministic.
- In PERT, events are used and CPM activities are used.
- In CPM, crashing is carried out.

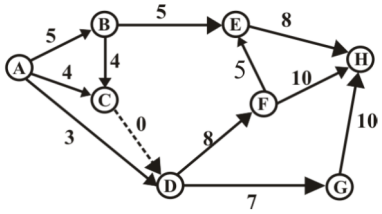
425. Which of the following is a wrong statement?

(a) PERT is suitable for projects having probabilistic time estimates
(b) CPM is suitable for projects having deterministic activities
(c) Both PERT and CPM are event oriented
(d) PERT is event oriented, while CPM is activity oriented

Gujarat PSC AE 2019
BPSC AE Mains 2017 Paper - VI

Ans. : (c) : CPM are activity orientated and PERT are event orientated.

426. For the network shown in the given figure, the earliest expected completion time of the project is



(a) 26 days (b) 27 days
(c) 30 days (d) 28 days
(e) 29 days

CGPSC AE 26.04.2015 Shift-I
ESE 2001

Ans. (c) : Earliest expected completion time of the project is 30 days and critical path will be A–B–C–D–F–E–H.

427. A PERT activity has an optimistic time estimate of 3 days, a pessimistic time estimate of 8 days and a most likely time estimate of 10 days. What is the expected time of this activity?

(a) 5.0 days (b) 7.5 days
(c) 8.0 days (d) 8.5 days

Gujarat PSC AE 2019, ESE 2008

Ans. : (d) :

$$\text{Expected time } (t_e) = \frac{t_p + 4t_m + t_o}{6}$$

$$= \frac{3 + 4 \times 10 + 8}{6}$$

$$= \frac{51}{6} = 8.5 \text{ days.}$$

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428. Time estimates of an activity in a PERT network are :
Optimistic time t_o = 9 days,
pessimistic time (t_p) = 21 days and most likely time t_m = 15 days The approximate probability of completion of this activity in 13 days

(a) 34% (b) 50%
(c) 16% (d) 84%

MPPSC AE 2016, ESE 1999

Ans. : (c) $t_{expected} = \dfrac{t_o + 4t_m + t_p}{6}$

$= \dfrac{9 + (4 \times 15) + 21}{6} = 15$

$S.D. = \dfrac{(t_p - t_o)}{6} = \dfrac{21 - 9}{6} = 2$

$Z = \dfrac{x - \bar{x}}{\sigma} = \dfrac{13 - 15}{2} = -1$

$P(-1) = 0.1586$

$P(-1) = 0.16$ OR 16%

429. The use of CPM lies in

(a) taking corrective measures
(b) scheduling and controlling the project
(c) planning and controlling the most logical sequence of operations
(d) None of the above

BPSC Poly. Lect. 2016, VIZAG STEEL MT 2015

Ans. : (c) The use of CPM lies in planning and Controlling the most logical sequence of operation.

Critical Path Method (CPM)

The critical path method (CPM) was first used by Morgan R. Walker in 1957. It helps in ascertaining time schedules, makes better and detailed planning possible, encourages discipline and provides a standard method for communicating project plans schedules and to time and cost performance.

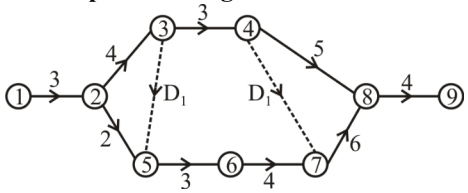
430. PERT is a network analysis mainly based on

(a) event oriented (b) activity oriented
(c) process oriented (d) none of these

GPSC Executive Engineer 23.12.2018
Mizoram PSC AE/SDO 2014 Paper-III

Ans. (a) : PERT (Programme Evaluation and Review Technique)–It is event oriented network technique which is mainly associated with probabilistic activities.

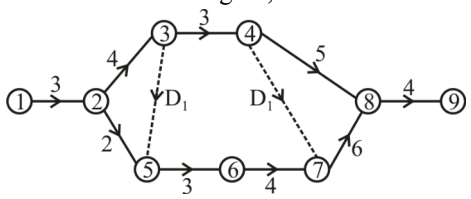
431. In a network shown in the below figure the critical path is along



(a) 1-2-3-4-8-9 (b) 1-2-3-5-6-7-8-9
(c) 1-2-3-4-7-8-9 (d) 1-2-5-6-7-8-9

Nagaland PSC (CTSE) 2018, Paper-I
ESE 1996

Ans. (b) : As, we know the critical path is which has maximum time. So from figure,



So, 1-2-3-5-6-7-8-9 is taking 24 days, & other path 1-2-5-6-7-8-9 takes only 21 day.

432. The variance of the completion time for a projects is the sum of variances of:

(a) All activity times
(b) Non-critical activity times
(c) Critical activity times
(d) Activity times of first and last activities of the project

HPPSC Asstt. Prof. 18.09.2017, ESE 2003

Ans. (c) : Critical activity times

433. Following date refers to the activities of a project, where node 1 refers to the start and node 5 refers to the end of the project

Activity	Duration (days)
1-2	2
2-3	1
4-3	3
1-4	3
2-5	3
3-5	2
4-5	4

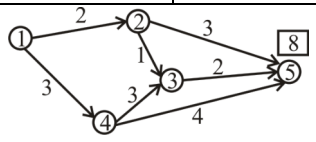
CP in the network is

(a) 1-2-3-5 (b) 1-4-3-5
(c) 1-2-3-4-5 (d) 1-4-5

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GATE 2015

Ans. (b) :

Activity	Duration (days)
1-2	2
2-3	1
4-3	3
1-4	3
2-5	3
3-5	2
4-5	4



Critical path in the network is 1-4-3-5

434. A Product consists of three Parallel Paths with mean and variance of (10,4); (12,4); (12,9) According to the standard PERT the distribution of project duration is

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(a) Beta with mean 10 and standard deviation 2
(b) Beta with mean 12 and standard deviation 3
(c) Normal with mean 10 and standard deviation 3
(d) Normal with mean 12 and standard deviation 3

UJVNL AE 2016, GATE 2002

Ans. (b) : Project distribution is a beta distribution with mean which is equal to duration of critical path.
If two longest path having same completion time, highest variance will decide project completion time.
Critical path is 12.
Standard deviation $= \sqrt{\sigma} = \sqrt{9} = 3$
Hence, Beta with mean 12 and standard deviation 3.

435. In PERT and CPM network the dummy activity
(a) Consumes time
(b) Consume resources
(c) Is used to preserve the logic
(d) Is a real activity

UPPSC AE 13.12.2020, Paper-I, ESE 2010

Ans. (c) : An imaginary activity which does not consume any resource and time is called a dummy activity. Dummy activities are simply used to represent a connection between events in order to maintain a logic in the network.

436. In CPM network, critical path denotes the
(a) Path where maximum resources are used
(b) Path where minimum resources are used
(c) Path where delay of one activity prolongs the duration of completion of project
(d) Path that gets monitored automatically

RPSC Vice Principal ITI 2018
CSE Pre-2002

Ans. (c) : Path where delay of one activity prolongs the duration of completion of project.

437. The _____ path within a network diagram is called critical path
(a) longest (b) shortest
(c) easiest (d) curved

UPRVUNL AE 05.07.2021

Ans. (a) : The longest path within a network diagram is called critical path.

438. A PERT activity has an optimistic time estimate of 4 days, a pessimistic time estimate of 8 days and a most likely time estimate of 12 days. What is the expected time of the activity?
(a) 5 days (b) 8 days
(c) 10 days (d) 12 days

RPSC ACF & FRO, 26.02.2021

Ans. (c) : Given,
$t_o = 4$ days
$t_p = 8$ days
$t_m = 12$ days

Expected time $(t_e) = \dfrac{t_o + 4t_m + t_p}{6}$

$= \dfrac{4 + 4 \times 12 + 8}{6} = \dfrac{60}{6}$

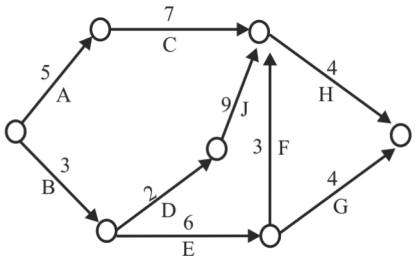
$= 10$ days

439. What is the total float of an activity for which
Earliest Start time ES = 5
Earliest Finish time EF = 15
Latest Start time LS = 35
Latest Finish time LF = 45
(a) 10 (b) 20
(c) 30 (d) 40

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) We know that,
Total float = Latest start date – Early start date
OR
= Latest finish date – Early finish date
Total Float = 35 – 5 = 30 or 45 – 15 = 30

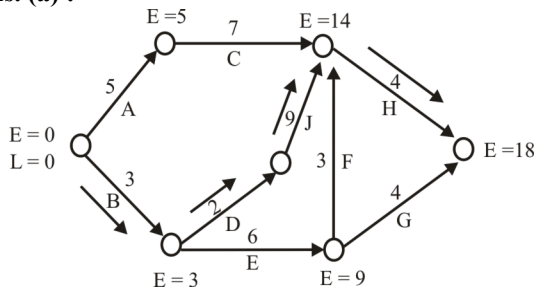
440. For the CPM network shown in the diagram, critical path would be :



(a) B-D-J-H (b) B-E-F-H
(c) B-E-G (d) A-C-H

CGPSC AE 15.01.2021

Ans. (a) :



Critical path would be ⇒ B-D-J-H

441. Slack time of a network equals :
(a) ES + t (b) LS – EF
(c) Zero (d) EF – ES

CGPSC AE 15.01.2021

Ans. (b) : Slack time = Latest start time – Earliest start time.

$\boxed{\text{Slack} = LS - ES = LF - EF}$

LF = Latest Finish time LS = Latest start time
EF = Earliest Finish time ES = Earliest start time

442. PERT
(a) provides an approach for keeping planning up-to-date
(b) provides a way for management to require that planning be done on a uniform and logical basis
(c) permits management of foresee quickly the impact of variations from the plan
(d) All of the above

CGPSC AE 15.01.2021

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Ans. (d) : PERT (Program Evaluation Review Technique)–

(i) PERT provides an approach for keeping planning up to date.

(ii) It provides a way for management to require that planning be done on a uniform and logical basis.

(iii) It permits management to forces quickly the impact of variation from the plan.

443. A dummy activity

(a) is artificially introduced

(b) is represented by a dotted line.

(c) does not consume time

(d) All of the above.

JPSC AE 10.04.2021, Paper-II

Ans. (d) : A Dummy activity is a type of activity in a project network which neither requires any time nor any resources. It is an artificial activity shown in project network to identify the dependence among operation. It is represented by dotted line.

444. PERT analysis is based on :

(a) Optimistic time (b) Pessimistic time

(c) Most likely time (d) All of the above

JPSC AE 10.04.2021, Paper-II

Ans. (d) : PERT analysis is based on three time estimates.

(1) Optimistic time (t_o)

(2) Pessimistic time (t_p)

(3) Most likely time (t_m)

$$\text{Expected time } (T_E) = \frac{t_o + 4t_m + t_p}{6}$$

445. Which one of the following represents an event?

(a) Concrete cured (b) Fixing of door

(c) Plastering of walls (d) Selecting sites

JPSC AE 10.04.2021, Paper-II

Ans. (a) : Event is the point of time which is terminates of certain activity. Concrete cured represents an event. While fixing of door, plastering of walls, selecting sites are activities.

446. Which of the following statements is not correct about PERT?

(a) Network is constructed based on the events.

(b) It does not take uncertainties involved in the estimation of times.

(c) Network deals with uncertainities and hence three time estimations are considered.

(d) As there is no certainty of time, activity duration cannot be reduced.

ESE 18.07.2021

Ans. (b) : In case of PERT, three time estimates, optimistic time (t_o), pessimistic time (t_p) and most likely time (t_m) are considered to take uncertainties involved in the estimation of times.

$$t_E = \frac{t_o + 4t_m + t_p}{6}$$

Hence option (b) is not correct.

447. Critical path method is good for

(a) Small projects only

(b) Large projects only

(c) Both small and large projects equally

(d) Neither small nor large projects

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : CPM also known as critical path analysis, the critical path method widely used technique for analyzing and managing task sequences in large project.

448. Critical path network helps an engineer

(a) to concentrate his attention on critical activities

(b) to divert the resources from non-critical advanced activities to critical activities

(c) to be cautious for avoiding any delay in the critical activities to avoid delay of the whole project

(d) All of the above

Assam PSC AE (PHED), 18.10.2020

Ans. (d) : The critical path is the longest sequence of activities in a project plan which must be completed on time for the project to complete on due date.

449. A PERT network has three activities on critical path with mean time 3, 8, 6 and standard deviation 1, 2 and 2 respectively. The probability that the project will be completed in 20 days is

(a) 0.50 (b) 0.66

(c) 0.84 (d) 0.95

Assam PSC AE (PHED), 18.10.2020
ESE 1993

Ans. (c) : Standard deviation = $\sqrt{\text{variance}}$

$= \sqrt{1\times1 + 2\times2 + 2\times2}$

$= \sqrt{9}$

$= 3$

Days D = 20

S = 3 + 8 + 6 = 17

Probability = (D – S)/standard deviation

$= \frac{20-17}{3}$

$= 1$

Now from z table probability = 0.84

450. For calculating the time of an activity in PERT network maximum weightage is given to

(a) Optimistic time

(b) Pessimistic time

(c) Most likely time

(d) Average time of each activity

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : Most likely time (t_m)- It refers to the estimate of the normal time the activity would take. This assumes normal delays.

The optimistic (t_o)- It is the smallest time taken to complete the activity, if everything goes well.

The pessimistic time (t_p)- It is the longest time that an activity would take, if everything goes wrong.

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Expected time (t_e)– From these three time estimates, we have to calculate the expected time of an activity.

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

Variance of the activity-

$$\sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$$

451. A portion of the total float within which an activity can be delayed for start without affecting the floats of preceding activities is called

(a) Safety float (b) Free float
(c) Independent float (d) Interfering float

ESE 2020

Ans. (c) : Independent Float– It is that portion of the total float within which an activity can be delayed for start without affecting the floats of the preceding activities.

452. Which type of management looks after merchandising, advertising, sales and critical path analysis?

(a) Financial management
(b) Maintenance management
(c) Production management
(d) Distribution management

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Distribution management– Distribution management refers to the process of overseeing the movement of goods from the supplier or manufacturer to point of sale. This type of management looks after merchandising, advertising, sales and critical path analysis.

453. Expected time of an activity when pessimistic time (t_p) = 12 days, Most likely time (t_m) = 9 days, Optimistic time (t_o) = 6 days is:

(a) 7 (b) 9
(c) 6 (d) 8

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

Ans. (b) : Given,
Pessimistic time (t_p) = 12 days.
Most likely time (t_m) = 9 days
Optimistic time (t_o) = 6 days
Then, Expected time (t_e) = ?
We know that

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

$$= \frac{6 + 4 \times 9 + 12}{6} = 9 \text{ days.}$$

454. Which of the following is incorrect with respect to PERT?

(a) PERT is said to be event oriented
(b) PERT is a probabilistic model
(c) PERT is used for non repetitive works
(d) PERT is employed in business and construction problem.

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (d) : PERT (Program Evaluation and Review Technique)–In project management PERT activity time is probabilistic and it is event oriented.

PERT finds application in projects where resources (men, materials and actually money) are always made available and when required.

It especially suitable in defence projects and R & D where activity times cannot be reliably predicated

455. Dijkstra's algorithm is used to determine the shortest path between :

(a) any two nodes of the network
(b) Any node to last node of the network
(c) Starting node to any other node in the network
(d) Any two nodes of unidirectional network

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (c) : Dijkstra's algorithm can be used to determine the shortest path from one node (starting node) in a graph to every other node within the same graph data structure provided that the nodes are reachable from starting node.

456. Critical path is

(a) Shortest path and consumes minimum time
(b) Shortest path and consumes maximum time
(c) Longest path and consumes maximum time
(d) No such relationship exists

HPPSC ADF 05.03.2019

Ans. (c) : Longest path and consumes maximum time

457. A project is expected to take 12 days along the critical path, having the standard deviation of 4 days.

Probability factor (z)	Probability (%)
– 0.5	30.85
1	84.13
– 1	15.87
0	50
3	99.87

The probability of completion of project in 16 days will be: [use the table of probability factor and probability]

(a) 30.85% (b) 84.13%
(c) 15.87% (d) 50%
(e) 99.87%

CGPSC AE 25.02.2018

Ans. (b) : 84.13%

458. In any crash program for a project

(a) both direct and indirect costs increase
(b) indirect costs increase and direct costs decrease
(c) direct costs increase and indirect costs decrease
(d) cost are no criterion

ESE 2018

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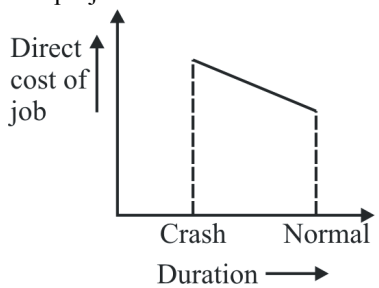
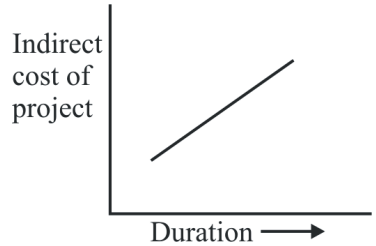
Ans. (c) : Direct Cost–Direct cost increases with the crashing of the project duration.



Indirect Cost–Indirect costs increase with the duration of project.



So during crashing process :

- Direct cost increases.
- Indirect cost decreases.

459. The basic difference between PERT and CPM is that

(a) PERT deals with events and CPM with activities
(b) critical path is determined in PERT only
(c) costs are considered on CPM only and not in PERT
(d) guessed times are used in PERT and evaluated times in CPM

Vadodara Municipal Corp. DEE, 2018

Ans. (a) : CPM (critical path method) is a activity oriented while PERT (Programme evaluation and Review Technique) is a Event oriented method.

460. The conversion from reliability to safety factor depends on the following factors

(a) Interface variable, standard deviation of strength and load, mean strength and mean load
(b) Interface variable and standard deviation of load only
(c) Standard deviation of strength and load only
(d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (a) : Interface variable, standard deviation of strength and load, mean strength and mean load

461. Another term commonly used for activity slack time is

(a) Total float (b) Free float
(c) Independent float (d) All of the above

MPSC Lect. (Prod.) 2018

Ans. (d) : Total float, free float and independent float are the terms commonly used for activity slack time.

462. The purpose of dummy row or column in an assignment problem is to

(a) Obtain balance between total activities and total resources
(b) Prevent a solution from becoming degenerate
(c) Providing a means of representing dummy problem
(d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (a) : The purpose of dummy row or column is an assignment is to obtain balance between total activities and total resources.

463. For an activity, the pessimistic, most likely and optimistic time are 10 days, 6 days and 2 days respectively. The expected duration of activity based on Beta distribution will be:

(a) 6 days (b) 2 days
(c) 3 days (d) 9 days

AAI Jr. Executive 29.11.2018

Ans. (a) : Given that,

$t_p = 10$ days
$t_m = 6$ days
$t_o = 2$ days

$$t_e = \frac{t_o + t_p + 4t_m}{6}$$

$$t_e = \frac{2 + 10 + 4 \times 6}{6} = \frac{36}{6}$$

t_e (Expected duration of activity on Beta distribution) = 6 days

464. For activities on the critical path-

(a) earliest start time (ES) = latest start time (LS)
(b) earliest start time (ES) > latest start time (LS)
(c) earliest start time (ES) < latest start time (LS)
(d) earliest start time (ES)=latest finish time (LF)

RPSC AE 2018

Ans. (a) : Critical path activities—Critical path activities are the projects tasks that must start and finish on time to ensure that the project ends on schedule. A delay in any critical path activity will delay completion of the projects unless the projects plan can be adjusted so that successor tasks finish more quickly than planned.

Earliest start time— The earliest time that an activity can begin.

Earliest finish time— Earliest finish time that an activity can be completed.

Latest start time—The Latest start time that an activity can begin without lengthening the minimum project duration.

465. The following table lists the tasks in a project and the time duration for each task:

Task	Preceding Task	Normal duration (in weeks)
A	--	16
B	--	20
C	A	8
D	A	10
E	B, C	6
F	D, E	12

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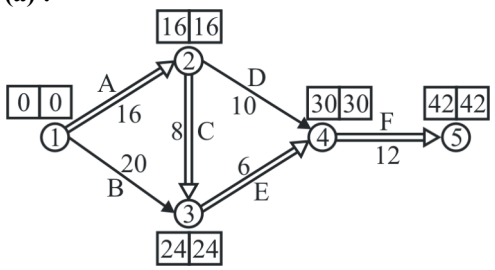
The critical path, the project duration and the free float for activity A are, respectively.

(a) A-C-E-F; 42 weeks and 0 week
(b) B-E-F; 42 weeks and 1 week
(c) B-C-D-F; 50 weeks and 2 weeks
(d) A-C-E-F; 50 weeks and 0 week

ESE 2017

Ans. (a) :



∵ Critical path is longest path of network & its length is equal to project duration.

So, A – C – E – F is critical path with duration 16 + 8 + 6 + 12 = 42 weeks.

∵ Float for critical activity is zero.

So, free float of activity A is zero.

466. The time which results in the least possible direct cost of an activity is known as

(a) Normal time (b) Slow time
(c) Crash time (d) Standard time

TNPSC AE 2017

Ans. (b) : The time which results in the least possible direct cost of an activity is known as slow time.

467. Which of the following is pessimistic time estimate as per PERT?

(a) The most probable time considering all conditions.
(b) The shortest possible time in which an activity can be completed.
(c) The maximum time that would be required to complete an activity.
(d) The minimum time that would be required to complete an activity.

CIL MT 26.03.2017

Ans. (c) : Optimistic time (t_o): If everything in the project goes well. It is the minimum time required to complete an activity.

Most Likely Time (t_m): It is the time for completing an activity which is most likely.

Pessimistic Time (t_p): If everything in the project goes wrong. This is the maximum time required to complete an activity.

468. The additional cost incurred in reducing the activity time in project scheduling is called:

(a) Normal cost (b) Crashing cost
(c) Break-even cost (d) Activity cost

TRB Polytechnic Lecturer 2017

Ans. (b) : The additional cost incurred in reducing the activity time in project scheduling is called crashing cost.

469. Consider a project consisting of nine jobs (A to I). The standard deviation for the jobs are as given below :

Job	A	B	C	D	E	F	G	H	I
Standard deviation	1	4	0	1	0	1	2	1	1

The critical jobs are : A, D, H and I. What is the standard deviation of the project duration?

(a) 2 (b) 1
(c) 4 (d) $\sqrt{11}$

TRB Polytechnic Lecturer 2017

Ans. (a) : Standard deviation of the project duration,

$\sigma^2 = \sigma_A^2 + \sigma_D^2 + \sigma_H^2 + \sigma_A^2$

$= 1^2 + 1^2 + 1^2 + 1^2$

$\sigma^2 = 4$

$\boxed{\sigma = 2}$

470. Which of the following is not true about CPM.

(a) CPM is activity oriented
(b) CPM is used when activity times are deterministic
(c) CPM is a planning device.
(d) CPM uses a probabilistic times

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : CPM not uses probabilistic times, actually it is uses in PERT.

471. Which of the following is not true about PERT?

(a) A control device
(b) Event oriented
(c) Uses a probabilistic times
(d) Directly introduces cost concept analysis

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : PERT not involve directly introduces cost concept analysis, pert basically directly involve in time analysis.

CPM	PERT
(1) It is activity oriented	(1) It is event oriented
(2) It used when the activity times are deterministic	(2) It uses a probabilistic time activity
(3) One time estimate	(3) Three time estimate
(4) It is a planning device	(4) It is control device

472. CPM stands for -

(a) Combined Process Method
(b) Critical Path Method
(c) Common Planning Method
(d) Critical Process Method

GMB AAE 25.06.2017

Ans. (b) : The CPM stands for critical path method or critical path analysis (CPA) is an algorithm for scheduling a set of projects activities.

473. Probabilistic time for completion of any activity can be found out from

(a) optimistic time (b) pessimistic time
(c) most likely time (d) all of these

Sikkim PSC (Under Secretary), 2017

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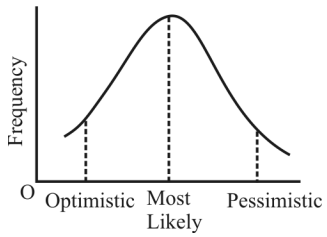
Ans. (d)



The optimistic time (t_o)- It is shortest possible time in which the activity can be finished.

The most likely time (t_m)- Is the estimate of the normal time the activity would take.

The pessimistic time (t_p)- Represent the longest the activity could take every thing wrong.

$$\text{Expected time}\left(t_e\right)=\frac{t_o+4t_m+t_p}{6}$$

474. A minimal spanning tree in network flow models involves

(a) All the nodes with cycle/loop allowed
(b) All the nodes with cycle/loop not allowed
(c) Shortest path between start and end nodes
(d) All the nodes with directed arcs

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : A path forms a loop or cycle if it connects to itself at the end. A spanning tree links all the nodes of network with no cycle/loop allowed.

475. Critical path on PERT/CPM chart is obtained by joining the events having

(a) Maximum slack (b) Minimum slack
(c) Average slack (d) No slack

Vizag Steel (MT) 2017

Ans. (d) : Critical path on PERT/CPM chart is obtained by joining the events having no or zero slack.

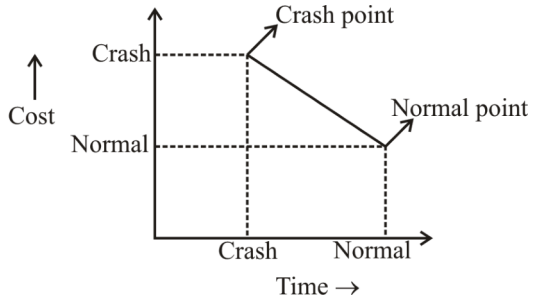
Note-Vizag gives (b) as correct answer.

476. Which of the following is true in time cost trade-off function analysis?

(a) Cost increases linearly at time increases.
(b) Cost decreases exponentially as time increases.
(c) Cost decreases linearly as time increases.
(d) Cost increases exponentially as time increases.
(e) Cost increases linearly as time decreases.

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : So as cost decreases linearly as time increases.



Time-cost trade-off for crashing activities.

477. Negative slack occurs when

(a) Events stick to their schedule
(b) There is deficiency of resources
(c) Dummy activities are large in number
(d) Activities lie on the critical path

RPSC VPITI 14.02.2016

Ans. (b) : When slack of an activity is negative. It represent that a programme falls behind schedule and additional resources are required to complete the project in time.
So there is deficiency of resources

478. In PERT the span of time between the optimistic and pessimistic time estimates of an activity is

(a) 6σ (b) 3σ
(c) 12σ (d) σ

RPSC VPITI 14.02.2016

Ans. (a) : Optimistic time (t_o)- It is the shortest time in which an activity can be completed assuming that everything goes exceptionally well.

Pessimistic time (t_p)- In which an activity will take to complete if every thing turns out to be against expectation.

Most likely time (tm)- This estimate of time lies between optimistic and pessimistic time estimates σ_t

$$=\frac{t_p-t_o}{6}$$

$(t_p - t_o) = 6\sigma_t$

The greater uncertainty in time estimates the greater will be the difference into and t_p and high value of 6t represent high degree of uncertainty.

Standard deviation

$$\sigma=\frac{t_p-t_o}{6}$$

$(t_p - t_o) = 6\sigma$

479. Select the wrongly paired option

(a) CPM- emphasis on minimum project cost
(b) PERT- activity oriented
(c) CPM- deterministic time estimates
(d) PERT- probabilistic time estimates

RPSC AE (GWD) 06.05.2016

Ans. (b) : CPM- emphasis on minimum project cost. Its activity oriented.

PERT- It is a event oriented. Its associated with probabilistic activity.

480. In the context of CPM/PERT, the term crashing of network implies

(a) Project has gone out of control
(b) Diverting resources from non-critical activities to critical activities for timely completion
(c) Short circuiting non- critical events
(d) Abandoning the project

RPSC AE (GWD) 06.05.2016

Ans. (b) : In a CPM/PERT, the term crashing of network implies diverting resources from non-critical activities to critical activities for timely completion.

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481. The PERT technique deals with the project of:

(a) repetitive nature
(b) non-repetitive nature
(c) deterministic nature
(d) either repetitive nature or deterministic nature
(e) None of the other options

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) : PERT is a project management planning tool used to calculate the amount of time it will take to realistically finish a project.

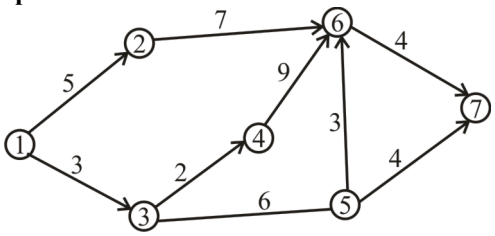
482. CPM was first used in the company :

(a) ONGC
(b) NTPC
(c) NASA
(d) E I Dupont & Company

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : The CPM was developed in the 1950s by James Kelley of Remington Rand and Morgan Walker of Dupont developed a project management technique called the Critical Path Method (CPM).

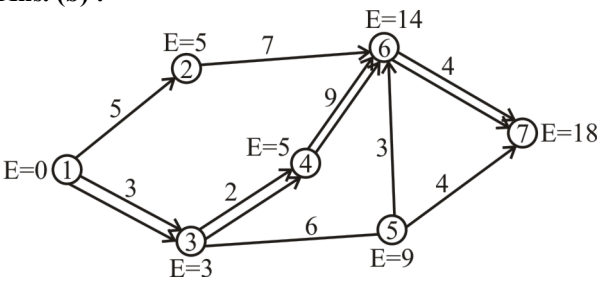
483. For the network shown in the diagram, critical path would be



(a) 1 – 2 – 6 – 7
(b) 1 – 3 – 4 – 6 – 7
(c) 1 – 3 – 5 – 7
(d) 1 – 3 – 5 – 6 – 7
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) :



The critical path is,
$\Rightarrow$ 1 – 3 – 4 – 6 – 7
which is maximum = 3 + 2 + 9 + 4 = 18

484. Variance of an activity may be written as : (where, t_p = pessimistic time, t_o = optimistic time)

(a) $\left(\frac{t_p - t_o}{12}\right)^2$
(b) $\left(\frac{t_p - t_o}{6}\right)^2$
(c) $\left(\frac{t_p - t_o}{12}\right)^{1/2}$
(d) $\left(\frac{t_p - t_o}{6}\right)^{1/2}$

UPRVUNL AE 07.10.2016

Ans. (b) : The average or expected time to complete an activity is given by

$$\mu(\text{or})\, t_e = \frac{t_o + 4t_m + t_p}{6}$$

$$\text{Variance} = (\sigma)^2 = \left(\frac{t_p - t_o}{6}\right)^2$$

485. Find the expected time of an activity with pessimistic time (t_p) = 14 days. Most time (t_m) = 8 days. Optimistic time (t_o) = 2 days.

(a) 6
(b) 7
(c) 8
(d) 9

UPRVUNL AE 07.10.2016

Ans. (c) : Data given,
$t_p = 14$ days
$t_m = 8$ days
$t_0 = 2$ days
then expected time of an activity t_e is given as

$$t_e = \frac{t_p + 4t_m + t_o}{6}, \qquad t_e = \frac{14 + 4 \times 8 + 2}{6}$$

$$t_e = \frac{14 + 32 + 2}{6}$$

$$t_e = \frac{48}{6}$$

$t_e = 8$ days

486. The difference between latest finish time and earliest finish time of activity is called

(a) Total float
(b) Free float
(c) Independent float
(d) None of the above

UJVNL AE 2016

Ans. : (a) The difference between latest finish time and earliest finish time of activity is called total float.

487. PERT is

(a) Program evaluation and review technique
(b) Event oriented.
(c) Able to consider uncertainty in execution timings.
(d) Concerned with all of the above

UJVNL AE 2016

Ans. : (d) following point of PERT
i) Program evaluation and review technique
ii) Event oriented
iii) Able to consider uncertainty in execution timing.

488. The estimated duration of times for an activity in the PERT network under the worst and best environment are as 9 and 3 days. The variance of this activity is

(a) 6 days
(b) 3 days
(c) 2 days
(d) 1 days

GPSC ARTO 01.05.2016

Ans. (d) : Pessimistic duration (t_p) = 9 days
Optimistic duration (t_0) = 3 days

$$\text{Variance (V)} = \left(\frac{t_p - t_o}{6}\right)^2 = \left(\frac{9-3}{6}\right)^2 = 1 \text{ day}$$

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489. The cost corresponding to completion of activity within minimum possible time is known as :

(a) Maximum cost (b) Minimum cost
(c) Normal cost (d) Crash cost

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (d) : Crash Cost–Crash cost is the cost associated with selecting the faster alternative to complete the effort.
Note, i.e. not the extra cost, but the full cost associated with the alternative approach. As such the formula results in a value that represent crash cost per time period.

490. In the network diagram :

(a) an activity and an event are represented by a circle
(b) a activity and an event are represented by an arrow
(c) an activity is represented by a circle and event by an arrow
(d) an activity is represented by an arrow and event by a circle

Kerala PSC IOF 19.04.2016

Ans. (d) an activity is represented by an arrow and event by a circle.

491. Which of the following statements is correct?

(a) When slack of an activity is zero, it falls only on critical path
(b) CPM technique is useful to minimize the direct and indirect expenses
(c) Critical path of a network represents the minimum time required for completion of project
(d) All of these

OPSC AEE 2015 Paper-I

Ans. : (a) Critical Path Method :-In this activity times are known with certainty. For each activity earliest start time (EST) and latest start times (LST) are computed. The length of critical path determines the minimum time in which the entire project can be completed. The activities on the critical path are called 'Critical activities".

492. A network has 3 activities on critical path with mean time as 3, 8 and 6 and variance as 1, 4 and 9. The probability at finishing the project in 17 days is :

(a) 0.5 (b) 0.66
(c) 0.84 (d) 0.95

TSPSC Managers, 2015

Ans. (a) : Standard deviation = $\sqrt{\text{variance}}$

$= \sqrt{(1)^2 + (4)^2 + (9)^2}$

$= \sqrt{1+16+81} = \sqrt{17+81} = \sqrt{98}$

Standard deviation $= \sqrt{98}$
Completed time (D) = 17 days
Total mean time (S) = 3 + 8 + 6 = 17

$$\text{Probability} = \frac{(D-S)}{\text{Standard deviation}} = \frac{17-17}{\sqrt{98}}$$

$= 0$

Now, from table, probability = 0.50.

493. In PERT activity time is considered as Beta distribution because :

(a) It is unimodel
(b) It cuts 2 intercepts on x-axis
(c) Both 1 and 2 are correct.
(d) It is continuous distribution.

TSPSC Managers, 2015

Ans. (c) : In PERT activity time is considered as Beta distribution because it have unimodel and 2 intercepts on X-axis.

494. The primary objective of crashing in CPM is

(a) To decreases cost
(b) To increases cost
(c) To decreases project duration
(d) To optimise the path

TSPSC AEE 2015

Ans. (c) : The primary objective of crashing in CPM is to decreases project duration.

495. In PERT, if the pessimistic time was 14 weeks, the optimistic time was 8 weeks, and the most likely time was 11 weeks.

(a) the expected time would be 6 weeks
(b) the variance would be 11 weeks
(c) the variance would be 1 week
(d) the expected time would be 5½ weeks
(e) the expected time would be 7 weeks

CGPSC AE 26.04.2015 Shift-I

Ans. (c) :

$$\text{Variance} = \left(\frac{\text{Pessimistic time - optimistic time}}{6}\right)^2$$

$$= \left(\frac{14-8}{6}\right)^2 = 1 \text{ week}$$

$$\text{Expected time} = \frac{8+4\times11+14}{6} = 11 \text{ weeks}$$

496. A project consists of three parallel paths with mean durations and variances of (14, 4), (14, 4) and (12, 9) respectively. According to the standard PERT assumptions, the distribution of the project duration is

(a) normal with mean 14 and standard deviation 3
(b) beta with mean 12 and standard deviation 2
(c) beta with mean 14 and standard deviation 2
(d) normal with mean 12 and standard deviation 3
(e) exponential with mean 12 and standard deviation 3

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Project distribution is a beta distribution with mean which is equal to duration of critical path.
If two longest path having same completion time, highest variance will decide project completion time.
Critical path is 14.
Standard deviation $= \sqrt{\sigma} = \sqrt{4} = 2$

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497. PERT was first formally applied to the planning and controlling in

(a) Titan's way
(b) Polaris Weapon System
(c) JP Morgan
(d) Dow Jones

TNPSC AE 2014

Ans. (b) : PERT was first formally applied to the planning and controlling in Polaris Weapon System.

498. A dummy activity is :

(a) Imaginary
(b) True
(c) Alternative to existing pattern
(d) Imaginary and alternative to existing pattern

UPRVUNL AE 2014

Ans. : (a) A dummy activity is a imaginary activity. There is no time consume in dummy activity and it is denoted by dotted line.

499. Consider the following statements about Critical Path Method (CPM) :

(i) It is a method of project evaluation
(ii) It uses four time estimates
(iii) It is a probabilistic method
(iv) It is a tool for managing large projects

Which of the following statements is/are true:

(a) (i) and (ii) are true
(b) (i) alone is true
(c) (i), (ii) and (iv) are true
(d) (i), (ii), (iii) and (iv) are true

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : CPM is a statistical technique of project management that manages well defined activities of a project.

CPM – Critical Path Method.

(1) Activity oriented
(2) Single time based
(3) Deterministic model
(4) Predictable activities
(5) Evolved as construction project

500. Which one of the following is a controlling technique ?

(a) PERT (b) Kaizen
(c) Smed (d) Kan ban

TNPSC AE (Industries) 09.06.2013

Ans. (a) : PERT

501. In a PERT activity, optimistic time is 5 h, pessimistic time is 20 h and the most likely time is 10 h then what is the expected time for completion of the activity (in hours)?

(a) 11.13 (b) 10.83
(c) 9 (d) 10

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : In PERT expected time of an activity is determined by using formula,

$$T_e = \frac{t_o + 4t_m + t_p}{6}$$

Given,

$t_o = 5$ hr, $t_m = 10$ hr, $t_p = 20$ hr

$$T_e = \frac{t_o + 4t_m + t_p}{6}$$

$$T_e = \frac{5 + 4(10) + 20}{6} = \frac{5 + 40 + 20}{6}$$

$$T_e = \frac{65}{6} = 10.83 \text{ hr}$$

502. Which is NOT a controlling technique ?

(a) PERT/CPM (b) Event
(c) Budgeting (d) Gantt chart

TNPSC AE (Industries) 09.06.2013

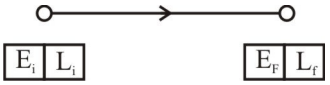
Ans. (b) :

503. The time of completing a project in network analysis is given by following time of the critical activity meeting at the final node

(a) early start (b) early finish
(c) late start (d) late finish

APPSC AEE 2012

Ans. (b) : Critical Path



Where,
E_i = Early start
L_i = late start
E_F = Early finish
L_f = Late finish
The time of completing a project in network analysis is given by following time of the critical activity meeting at the final triode early finish.

504. Crashing is to be applied on

(a) critical activities
(b) non–critical activities
(c) both critical and non-critical activities
(d) critical activities with positive that

TRB Poly. Lect., 2012

Ans. (a) : Crashing analyzes & categorizes activities based on the lowest crash cost per unit time. crashing only works for critical path activities where it is possible to shorten schedules (effective time management)

505. Latest start time of an activity in CPM is

(a) the latest occurrence time of the successor event minus the duration of the activity
(b) earliest occurrence time for the predecessor event plus duration of activity
(c) latest occurrence time of the successor event
(d) earliest occurrence time for the predecessor event

TRB Asstt. Prof., 2012

Ans. (a) : Latest start time of an activity in CPM is the latest occurrence time of the successor event minus the duration of the activity.

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506. The three time estimates of a PERT activity are optimistic time = 8 min, pessimistic time = 14 min and most likely time = 10 min. The expected time of the activity would be

(a) 10.00 min (b) 10.33 min
(c) 10.66 min (d) 11.00 min

TRB Asstt. Prof., 2012, ESE 2002

Ans. (b) : $T = \frac{T_0 + 4T_m + T_p}{6}$

Where T_0 = Optimistic Time
T_p = Pessimistic Time
T_m = Most likely Time
$T_0 = 8$ min, $T_m = 10$ min, $T_p = 14$ min

$$T = \frac{8 + 40 + 14}{6} = \frac{62}{6} = 10.33 \text{ min.}$$

507. Air cargo movements fall under

(a) fixed path system
(b) continuous system
(c) intermittent system
(d) variable path system

TNPSC ACF 2012

Ans. (a) : Air cargo movements fall under fixed path system.

508. The Project completion time can be altered by Changing the activities time which falls on:

(a) The critical path (b) Non-critical path
(c) Dummy activities (d) Any path

RPSC AE GWD, 2011

Ans. (a) : The project completion time can be altered by changing the activities which falls on critical path.

509. With the PERT approach, the variance of the total project completion time is calculated by

(a) adding all the individual activity variances together
(b) adding all the individual activity variances not on the critical path
(c) adding all the individual activity variances on the critical path
(d) choosing the largest variance that occurs from the pool of individual activity variances

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : PERT- Time estimate (t_e) is based on a formula that includes optimistic time (t_o), most likely time (t_M) & pessimistic time estimate (t_p).

$$t_e = \frac{(t_o + 4t_m + t_p)}{6}$$

For variance of total project,
Calculating variance for each path.

$$\sigma^2 = \text{Variance} = \left(\frac{t_p - t_o}{6}\right)^2$$

The variance of the total project completion time is calculated by–
Adding all the individual activity variances on the critical path.

510. Assertion (A) : Cost overrun is preferred to time overrun in project management using PERT technique.

Reason (R) : Early availability of product or service minimizes the penalty costs.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (a) : Cost overrun is preferred to time overrun in project management using PERT technique, but if possible try to eliminate time overrun. Since the project use huge capital investment & the interest on this capital would be much larger than the amount which spent on cost overrun. So if cause early availability which minimizes the penalty costs.

511. The variances of each activity in the critical path of a network are given by 4, 3, 2, 1, 3, 3 and 1. The variance of the entire critical path is :

(a) 17 (b) 7
(c) $\sqrt{(17/7)}$ (d) $\sqrt{(7/17)}$

APPSC IOF, 2009

Ans. (a) : Variance of various activity are added directly

Variance = σ^2 = Variance$_1$ + Variance$_2$ + Variance$_3$ + ...

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2 .....$$

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + ...}$$

Variance = 4 + 3 + 2 + 1 + 3 + 3 + 1

$$\boxed{\text{Variance} = 17}$$

512. The pessimistic, most likely and optimistic times for an activity are 5 days, 3 days and 1 day respectively. Assuming Beta distribution, the expected duration of the activity is

(a) 3 days (b) 3.5 days
(c) 4 days (d) 5 days

DRDO Scientists 2008

Ans. (a) : a = 5, m = 3, b = 1

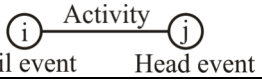
$$T_e = \frac{a + 4m + b}{6} = \frac{5 + 12 + 1}{6} = 3 \text{ days}$$

513. An event is indicated on the network by

(a) an arrow
(b) a straight line with circle at the end
(c) a number enclosed in a circle or a square
(d) a dotted line

TNPSC AE, 2008

Ans. (c) : An event is indicated on the network by a number enclosed in a circle or a square.



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7. Linear Programming, Transportation, Assignments

514. Graphical method, simplex method and transportation method are concerted with

(a) value analysis
(b) linear programming
(c) break-even analysis
(d) queing theory

ISRO Scientist/Engineer 17.12.2017
Vizag Steel (MT) 2017
Nagaland CTSE 2016 Paper-I
OPSC AEE 2015 Paper-I
Mizoram PSC AE/SDO 2014 Paper-III
SJVN ET 2013, APPSC Poly. Lect. 2013
J&K PSC Civil Services Pre, 2013
TRB Asstt. Prof., 2012
UKPSC AE 2012 Paper-I
OPSC Civil Services Pre. 2011
RPSC ACF-2011, TNPSC AE, 2008
OPSC Civil Services Pre 2006

Ans. (b) : Graphical method, simplex method and transportation method are concerted with linear programming.

515. In transportation problem the condition for optimallity test

(a) number of allocations to be (m + n – 1)
(b) allocation are to be independent
(c) both (A) and (B)
(d) none of these

Mizoram PSC (Insp. of Factories) 2019 Paper-III
OPSC AEE 2019 Paper-I
TRB Polytechnic Lecturer 2017
NPCIL ET, 2015
CGPSC AE 26.04.2015 Shift-I
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
UKPSC AE-2013, Paper-I
TRB Poly. Lect., 2012
UKPSC AE 2012 Paper-I

Ans. (c) : In the solution of transportation problem, for application of optimality test, the no, of allocation required is m+n-1 and there should be in independent of positions.

516. Simplex method of solving linear programming problem uses

(a) all the points in the feasible region
(b) only the corner points of the feasible region
(c) intermediate points within the feasible region
(d) only the interior points in the feasible region

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Gujarat PSC AE 2019
GPSC ARTO Pre 30.12.2018
TSPSC AEE 2017
Karnataka PSC Lect., 27.05.2017
VIZAG STEEL MT 18.08.2013
GATE 2010

Ans. (b) : Simplex method is an approach to solving linear programming models by hand using slack variables, tableaus and pivot variables as means to finding the optimal solution of an optimization problem. It uses only corner points of the feasible region.

517. In linear programming the shadow prices are

(a) The values assigned to one unit of capacity
(b) Maximum cost per item
(c) Cost of bought out items
(d) Cost of items manufactured in the plant

TSPSC Manager (Engg.) HMWSSB 12.11.2020
HPPSC ADF 05.03.2019
GPSC Asstt. Director of Transport 05.03.2017
Kerala PSC Poly. Lect., 2017
Kerala PSC AE 06.08.2015
UKPSC AE 2012 Paper-I

Ans. (a) : In linear programming problems the shadow price of a constraint is the difference between the optimised value of the objective function and the value of the objective function, evaluated at the optional basis the shadow prices are values assigned to one unit of capacity.

518. The total number of decision variable in the objective function of an assignment problem size n×n (n job and n machine) is

(a) n^2 (b) 2n
(c) 2n - 1 (d) n

APGCL AM, 2021, Gujarat PSC AE 2019
OPSC AEE 2019 Paper-I
CGPSC Poly. Lect. 22.05.2016
GATE 2014

Ans. : (a) : A, n × n assignment problem, m it is solved as a LPP it will have n^2 variables.

$$[x_{11}, x_{12}, x_{21}, x_{22}, \ldots\ldots\ldots x_{nn}]$$

519. The mathematical technique used for finding the best use of limited resources of a concern in the optimum manner is referred to as

(a) game theory (b) value analysis
(c) queuing theory (d) linear programming

Assam PSC AE (PHED), 18.10.2020
Assam PSC CCE Pre 2015
MPSC Poly. Lect. 2014
ISRO Scientist/Engineer 2006

Ans. (d) : Linear programming is a method to achieve the best outcome in a mathematical model whose requirements are represented by linear relationship linear programming is special case of mathematical programming.
Game theory- It is a powerful frame work that enables managers to analyze systematically the ties among interactions between actors in a market and to develop appropriate competitive strategies.
Value analysis- It is an approach to improving the value of a product or process by understanding its constituent components and their associated costs.
Queuing theory- It is the mathematical study of waiting lines or queues.

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520. A feasible solution to the linear programming problem should-

(a) Satisfy the problem constraints
(b) Optimize the objective function
(c) Satisfy the problem constraints and non-negativity restrictions
(d) Satisfy the non-negativity restrictions

Mizoram PSC (Insp. of Factories) 2019 Paper-III
NPCIL ET, 2015, TNPSC AE 2014, ESE 1994

Ans. (c) : In linear programming problem belongs to the feasible region. The feasible region is the set of all the points satisfy the problem's constraints including inequalities, equalities and integer constraints. In a problem, which maximum or minimum a linear functional subject to a system of linear equations in non-negative variables is called a linear programming problem.

521. Which of the following is NOT the method of finding an initial basic feasible solution of transportation problem?

(a) Stepping stone method
(b) North-west corner method
(c) Vogel's approximation method
(d) Hungarian technique
(e) None of these

CGPSC AE 16.10.2016, RPSC VPITI 14.02.2016
CGPSC AE 26.04.2015 Shift-I, ESE 2006

Ans. (d) : Hungarian model is method of assignment problem not transportation problem.
Hungarian model method is based on the principle that if a constant is added to every element of a row and for a column of cost matrix, the optimum solution of the resulting assignment problem is the same as the original problem and vice versa.
Note-This question has been deleted by CGPSC.

522. Assignment problem is solved by

(a) Hungarian Technique
(b) By simulation
(c) By transportation algorithm
(d) By Flood's technique

TSPSC Manager (Engg.) HMWSSB 12.11.2020
TRB Polytechnic Lecturer 2017
HPPSC AE (PWD) 10.09.2016,
HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : It is a special case of transportation in which the number of columns equal to the number of rows and in the optimal solution there will be one and only one assignment in a given row and column of the assignment value.

- Assignment problem is a special type of LPP where the objective is to minimize the time and cost.
- Hungarian method is used for solving assignment problems.
- Assignment solution is always a degenerate solution.

523. A variable which has no physical meaning, but is used to obtain an initial basic feasible solution to the liner programming problem is referred to as:

(a) Basic variable (b) Non-basic variable
(c) Artificial variable (d) Basis

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
UPRVUNL AE 21.08.2016, ESE 1998

Ans. (c) Artificial variable are the variable that has no physical meaning and is used to obtain an initial basic feasible solution to a linear programming problem.

524. The supply at three sources is 50, 40 and 60 units respectively while the demand at the four destinations is 20, 30, 10 and 50 units, In solving this transportation problem.

(a) Dummy source of capacity 40 units is needed
(b) Dummy destination of capacity 40 units is needed
(c) No solution exists as the problem is infeasible
(d) None solution exists as the problem is degenerate

TSPSC AEE 2017, UJVNL AE 2016
GATE 2002

Ans. (b) : Total supply = 50 + 40 + 60 = 150 units
Total demand = 20 + 30 + 10 + 50 = 110 units

$$\sum_{i=1}^{m} ai \neq \sum_{j=1}^{n} bj \text{, unbalanced}$$

Transportation problems
So, a dummy destination of capacity 40 units is needed.

525. Consider the following linear programming problem :

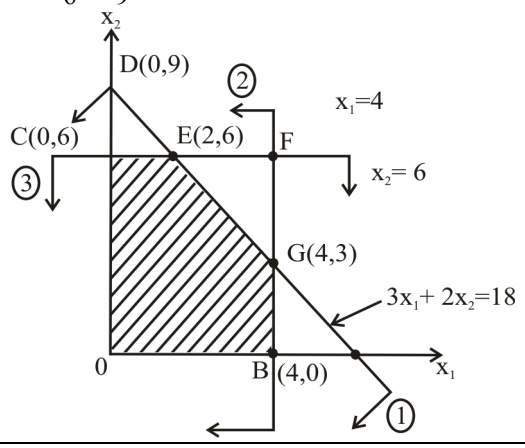
Maximum $Z = 3x_1 + 2x_2$
Subject to $x_1 \le 4$
$x_2 \le 6$
$3x_1 + 2x_2 \le 18$
$x_1 \ge 0, x_2 \ge 0$

(a) The LPP has a unique optimal solution
(b) The LPP is infeasible
(c) The LPP is unbounded
(d) The LPP has multiple optimal solutions

Vizag Steel MT (Re-Exam) 24.11.2013
TRB Asstt. Prof., 2012, GATE 2009

Ans. (d) : Max $z = 3x_1 + 2x_2$

$x_1 \le 4$
$x_2 \le 6$
$3x_1 + 2x_2 \le 18$
$\frac{x_1}{6} + \frac{x_2}{9} \le 1, \quad x_1 \ge 0, x_2 \ge 0$



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Since, at point E (2,6)

$$Z = 3\times2 + 2\times6 = 18$$

At point G (4,3),

$$Z = 3\times4 + 2\times3$$

Hence, It means the LPP has multiple optimal solution because both points have same value of objective junction.

526. A Transportation firm has 4 exit (Supply) points and 5 entry (Demand) points. The total number of entries is greater than exits. So the number of iterations excluding degeneracy would be :

(a) 9 (b) 6
(c) 3 (d) 0

RPSC ADE 2016, UPRVUNL AE 2014
TRB Poly. Lect., 2012

Ans. : (a) For basic feasible solution in transportation problem,
There should be m+n-1 number of allocations.
Since, it is given number of entries is greater than exits.
In order to make it balance transportation one dummy exit with balance entries will be added.
So, the exits and the entries are balanced.
Therefore total number of exits = 4+1=5
Total allocations=m+n-1
=5+5-1
=9

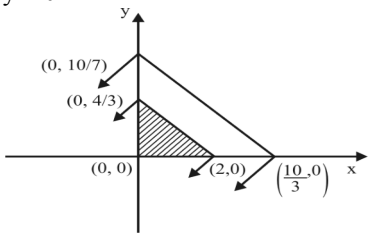
527. A linear programming problem is shown below:

maximum 3x + 7y
Subject to 3x + 7y ≤ 10
4x + 6y ≤ 8
x, y ≥ 0
It has

(a) an unbounded objective function
(b) exactly one optimal solution
(c) exactly two optimal solutions
(d) infinitely many optimal solutions.

MPSC HOD (Govt. Poly. Colleges) 04.10.2014
Gujarat PSC AE 2019, GATE 2013

Ans. (b) : maximum (Z) = 3x + 7y
3x + 7y ≤ 10
4x + 6y ≤ 8
x, y ≥ 0



$$Z_{(0,0)} = 0$$

$$Z_{(2,0)} = 3\times2 + 7\times0 = 6$$

$$Z_{(0,4/3)} = 3\times0 + 7\times\frac{4}{3} = 9.33$$

528. In a transportation problem North-West corner rule would yield

(a) An optimum solution
(b) An initial feasible solution
(c) A Vogel's approximate solution
(d) A minimum cost solution

Mizoram PSC (Insp. of Factories) 2019 Paper-III
(HPPSC LECT. 2016)
Kerala PSC Asstt. Prof. 27.09.2016

Ans. (b) : The North-West corner rule is a method adopted to compute the initial feasible solution of the transportation problem. The name North-West corner is given to this method because the basic variables are selected from the extreme left corner.

529. The graphical method is best suited for solving linear programming problems with _____ variables.

(a) 5 (b) 3
(c) 4 (d) 2

UPRVUNL AE 05.07.2021
Nagaland PSC CTSE 2017, Paper-I
Nagaland CTSE 2016 Paper-I

Ans. (d) : The linear programming with two variable can be solved graphically.

530. The leaving basic variable in simplex method is the basic variable that

(a) has the lowest value
(b) has the smallest coefficient in the key row
(c) has the largest coefficient in the key row
(d) goes to zero first, as the entering basic variable is increased

UPPSC AE 13.12.2020, Paper-I
UKPSC AE 2012 Paper-I
ESE 2010

Ans. (b) : The leaving basic variable in simplex method is the basic variable that has the smallest coefficient in the key row.

531. Which one of the following conditions should be satisfied for the application of optimality test on an initial solution of transportation model?

(a) Number of allocations should be less than m + n – 1
(b) Number of allocations should be equal to m + n – 1
(c) Number of allocation should be equal to m+n
(d) Number of allocations should be more than m+n

GPSC ARTO Pre 30.12.2018
ESE 2004

Ans. (b) : The application of optimality test on an initial solution of transportation model, Number of allocations should be equal to m + n – 1.
The article throws light upon the top two methods used for testing the optimality of transportation solution.
The methods are–
(i) Stepping stone method
(ii) Modified Distribution Method (MODI)

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532. The constraints in a given situation are found to be as follows.

$0 \le x \le 12$

$0 \le y \le 9$

$3x + 6y \le 66$

The objective function which is to be maximized is as follows

$P = 5x + 4y$

The values of x and y are

(a) (11,6) (b) (6,11)

(c) (12,5) (d) (5,12)

RPSC ADE 2016

VIZAG STEEL MT 2015

Ans. (c) : Constraints–

$0 \le x \le 12$ .....(1)

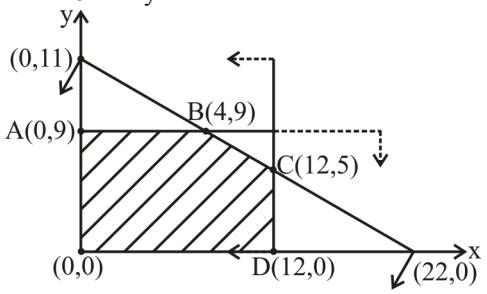
$0 \le y \le 9$ .....(2)

$3x + 6y \le 66$

$\frac{x}{22} + \frac{y}{11} = 1$ .....(3)

Objective function,

$P = 5x + 4y$



$P(A) = P(0, 9) = 5 \times 0 + 4 \times 9 = 36$

$P(B) = P(4, 9) = 5 \times 4 + 4 \times 9 = 56$

$P(C) = P(12, 5) = 5 \times 12 + 4 \times 5 = 80$

$P(D) = P(12, 0) = 5 \times 12 + 4 \times 0 = 60$

So, the maximum value of objective function is 80 at point C(12, 5).

533. In linear programming problem variables are

(a) Probabilistic and continuous

(b) Deterministic and continuous

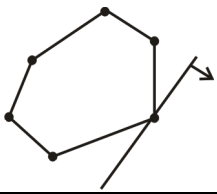
(c) Discrete or continuous and deterministic

(d) All the these

RPSC VPITI 14.02.2016

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : In a linear programming problem, a series of linear constraints produces a convex feasible region of possible value for those variables. In two variable case this region is in the shape of a convex simple polygon linear programming process is deterministic and continuous.



534. In case of solution of a two variables linear programming problem by graphical method one constant line comes parallel to the objective function line. Which one of the following is correct? The problem will have

(a) infeasible solution

(b) unbounded solution

(c) degenerate solution

(d) number of optimal solution

TRB Asstt. Prof., 2012, ESE 2006

Ans. (d) : number of optimal solution

535. Match List I and List II and select the correct answer using codes given below :

List-I	List-II
(a) Vogel's approximation method	1. Assignment model
(b) Flood technique	2. Transportation model
(c) Two phase method	3. PERT and CPM
(d) Crashing	4. Linear programming

	a	b	c	d
(a)	3	4	1	2
(b)	2	1	4	3
(c)	3	1	4	2
(d)	2	4	1	3

TRB Asstt. Prof., 2012, ESE 2004

Ans. (b) :

536. To solve degeneracy in the transportation problem we have to :

(a) Put allocation in one of the empty cells as zero

(b) Put a small element epsilon in any one of the empty cells.

(c) Allocate the smallest element epsilon in such a cell, which will not form a closed loop with other loaded cells.

(d) Allocate the smallest element epsilon in such a cell, which will form a closed loop with other loaded cells.

GPSC ARTO Pre 30.12.2018

HPPSC Asstt. Prof. 20.11.2017

Ans. (c) : To solve degeneracy in transportation problem, we have to allocate smallest element epsilon in such a cell, which will not form a closed loop with other loaded cells.

537. Maximize $z = 11x_1 + 4x_2$

Subject to : $7x_1 + 6x_2 \le 84$

$4x_1 + 2x_2 \le 32$

and $x_1, x_2 \ge 0$

The values of x_1, x_2 which maximizes z are respectively

(a) 2.4 and 11.2 (b) 0 and 14

(c) 8 and 0 (d) 3 and 10

(e) None of these

CGPSC AE 15.01.2021

CGPSC AE 16.10.2016

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Ans. (c) : From solution of subject to,

$x_1 \geq 0$ & $x_2 \geq 0$

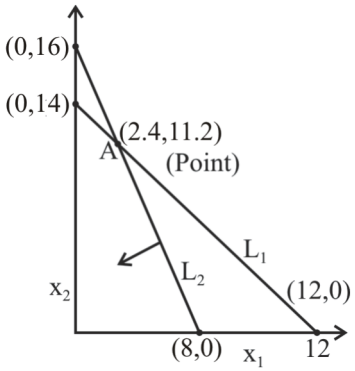
$L_1 \rightarrow 7x_1 + 6x_2 \leq 84$

$$L_1 \rightarrow \frac{x_1}{12} + \frac{x_2}{14} \leq 1$$

We get x_2 & x_1

$L_2 \rightarrow 4x_1 + 2x_2 \leq 32$

$$L_2 \rightarrow \frac{x_1}{8} + \frac{x_2}{16} \leq 1$$



$Z_{(0,14)} = 11(0) + 4(14) = 56$

$Z_{(0,8)} = 11(8) + 0 = 88 = \max$

$Z_{(0,0)} = 0$

$Z_{(2.4,11.2)} = 11(2.4) + 4(11.2) = 71.2$

The values of x_1, x_2 which maximizes z are respectively.

So (8, 0) is the limit.

538. Consider the following statements regarding linear programming:

1. When an artificial variable leaves the basis. its column can be deleted from the subsequent simplex tables.

2. Dual of a dual is the primal.

3. When two minimum ratios of the right-hand side of the coefficient in the key column are equal degeneracy may take place

Select the correct answer from the codes given below

(a) 2 and 3 (b) 1 only

(c) 1 and 3 (d) 1, 2 and 3

UPRVUNL AE 21.08.2016

ESE 2001

Ans. : (d) When an artificial variable leaves the basis, its column can be deleted from the subsequent simplex table.

539. Match List-I with List-II and select the correct answer using the codes given below the list :

	List-I	List-II
A.	**Integer programming**	**1.Bell**
B.	**'C' programming**	**2.Gomory**
C.	**Linear programming**	**3. Dantzing**
D.	**Dynamic programming**	**4. Ritchie**

(a) A-1, B-2, C-3, D-4 (b) A-2, B-1, C-4, D-3

(c) A-2, B-3, C-1, D-4 (d) A-2, B-4, C-3, D-1

UPRVUNL AE 21.08.2016

ESE 1995

Ans. : (d)

List-I	List-II
Integer programming	Gomory
'C' programming	Ritchie
Linear programming	Dantzing
Dynamic programming	Bell

540. Consider the following statements regarding the characteristics of the standard form of a linear programming problem

1. The right hand side of each constraints equation is non-negative

2. All the decision variables are non-negative.

3. All the constraints are expressed in the form of equations

Which of these statements are correct?

(a) 1 and 3 (b) 1 and 2

(c) 1, 2 and 3 (d) 2 and 3

UPRVUNL AE 21.08.2016, ESE 1999

Ans. : (c) Characteristics of the standard form of a linear programming problem

1. The right hand side of each constraints equation is non-negative
2. All the decision variables are non-negative.
3. All the constraints are expressed in the form of equations

541. Linear Programming is

(a) an abstract model, generally using mathematical symbols.

(b) is a technique, which optimizes linear objective function under limited constraints.

(c) is a technique, which ensures only integral values of variables in the problem.

(d) is a technique, which is used to analyze multistage decision process.

Nagaland CTSE 2016, 2017 Paper-I

Ans. (b) : Linear programming is technique, which optimizes linear objectives function under limited constraints.

542. Consider the following statements:

1. A linear programming problem with three variables and two constraints can be solved by graphical method.

2. For solutions of a linear programming problem with mixed constraints. Big-M-method can be employed.

3. In the solution process of a linear programming problem using Big-M-method, when an artificial variable leaves the basis, the column of the artificial variable can be removed from all subsequent tables.

Of above, which are correct?

(a) 1 (b) 1, 2

(c) 1, 2, 3 (d) 1, 3

(e) 2, 3

CGPSC AE 26.04.2015 Shift-I, ESE 2000

Ans. (e) : For solutions of a linear programming problem with mixed constraints. Big-M-method can be employed In the solution process of a linear programming problem using Big-M-method, when an artificial variable leaves the basis, the column of the artificial variable can be removed from all subsequent tables.

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543. In a transportation problem, the materials are transported from 3 plants to 5 warehouses, the basis feasible solution must contain exactly, which one of the following allocated cells?

(a) 3 (b) 5
(c) 7 (d) 8

TNPSC AE 2017, ESE 1998

Ans. (c) : Total number of allocated cells will be (m + n – 1). Then (3 + 5 – 1) = 7

544. In the simplex method, the variables which have not been assigned the value zero, during an iteration, are called

(a) artificial variables (b) basic variables
(c) coded variables (d) actual variables

TNPSC AE 2019, ESE 2012

Ans. (b) : In the simplex method, the variables which have not been assigned the value zero, during an iteration, are called basic variables.

545. In a 6 × 6 transportation problem, degeneracy would arise, if the number of filled slots were :

(a) Equal to thirty six (b) More than twelve
(c) Equal to twelve (d) Less than eleven

RPSC Vice Principal ITI 2018, TSPSC AEE 2015

Ans. (d) :
No. of allocations = m + n –1
= 6 + 6 – 1
= 11

■ In a transportation problem if the number of non-negative in dependent allocations is less than m + n –1, there exists degeneracy.

546. If the primal has an unbounded solution, then the dual has

(a) Optimal solution (b) No solution
(c) Bound solution (d) None of the above

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : If the primal (dual) is unbounded, then the dual (primal) has no feasible solution because such a feasible solution would give a bound on the former problem, which contradictions its unboundedness. It is possible for the primal and dual to be simultaneously infeasible.

547. In the graphical method of linear programming problem the optimum solution would lie in the feasible polygon at:-

(a) Its one corner
(b) Its center
(c) The middle of any side
(d) None of the above

Kerala PSC Asstt. Prof. 27.09.2016
UKPSC AE-2013, Paper-I

Ans. (a) : In the graphical method of linear programming problem the optimum solution would lie in the feasible polygon at its one corner.

548. If 'm' is the number of constraints in a linear programming problem with two variables 'x' and 'y' and non-negativity constraints x & y ≥ 0. The feasible region in the graphical solution will be surrounded by:-

(a) m lines (b) m + 1 lines
(c) m + 2 lines (d) m + 4 lines

UKPSC AE-2013, Paper-I
ESE 2007

Ans. (c) : Number of constraints = m
Number of non-negative constraints x & $y \geq 0$ is 2, then total number of lines of the feasible region in the graphical solution will be m + 2.

549. A solution is not a basic feasible solution in a transportation problem if after allocations

(a) there is no closed loop
(b) there is a closed loop
(c) total number of allocations in one less then the sum of numbers of sources
(d) there is degeneracy

TNPSC ACF 2012, ESE 1996

Ans. (b) : A solution is not a basic feasible solution in transportation problem if there is a closed loop. Further, iteration are necessary to arrive as a basic feasible solution.

550. Simplex method is the method used for

(a) value analysis
(b) network analysis
(c) linear programming
(d) queuing theory

Nagaland PSC (CTSE) 2018, Paper-I
Kerala PSC IOF 19.04.2016

Ans. (c) : Simplex method is an approach to solving linear programming models by hand using slack variables, table are, and pivot variables as a means to finding the optimal solution of an optimization problem.

551. Which of the following subroutines does a computer implementation of linear programming by the simplex method use?

(a) Finding a root of polynominal
(b) Finding a determinant of a matrix
(c) Solving a system of linear equations
(d) Finding the eigen value of matrix

APPSC AEE 2012, ESE 2007

Ans. (c) : Simplex method–
The Simplex method uses an approach that is very efficient. It does not compute the value of the objective function at every paint; instead, it being with a corner point of the feasibility region where all the main variables are zero and then systematically moves from corner point to corner point, while improving the value of the objective function at each stage. The process continues until the optional solution is found. Computer implementation of linear programming by the simplex method use solving a system of linear equation.

552. A tie for leaving in simplex procedure implies

(a) optimatility (b) cycling
(c) no solution (d) degeneracy

GPSC DEE, Class-2 (GWSSB) 04.07.2021
ESE 2005

Ans. (d) : A tie for leaving in simplex procedure implies degeneracy.

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553. Which of the following methods can be used to find basic feasible solution of a transportation problem?

(a) Big-M method
(b) Stepping Stone method
(c) Two Phase method
(d) Vogel's Approximation method

PPSC Asstt. Municipal Engineer 15.06.2021
TRB Poly. Lect., 2012

Ans. (d) : Vogel's approximation method is used to find the best initial basic feasible solution of a transportation problem.

554. Which one of the following is true in case of simplex method of linear programming?

(a) The constants of constraints equation may be positive or negative
(b) Inequalities are not converted into equations
(c) It cannot be used for two-variable problems
(d) The simplex algorithm is an iterative procedure

Mizoram PSC (Insp. of Factories) 2019 Paper-III
ESE 2009

Ans. (d) : The simplex algorithm is an iterative procedure .

555. The supply at three sources is 80, 30 and 40 units respectively while the demand at the four destinations is 30, 20, 25 and 55 units. In solving this transportation problem.

(a) a dummy source of capacity 20 units is needed with unit transportation cost one.
(b) a dummy destination of capacity 20 units is needed with unit transportation cost zero.
(c) a dummy source of capacity 20 units is needed with unit transportation cost zero.
(d) a dummy destination of capacity 20 units is needed with unit transportation cost one.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) :

	D_1	D_2	D_3	D_4
O_1	30 / 80	20	25	55
O_2	30			
O_3	40			

$\Rightarrow$ $D = 30 + 20 + 25 + 55$
$= 130$
$O = O_1 + O_2 + O_3$
$= 80 + 30 + 40$
$= 150$
Dummy of 20
On destination, because O – D = 20

556. If there are m sources and n destinations in a transportation matrix, the total number of basic variables in a basic feasible solution is

(a) m + n (b) m + n–1
(c) m + n +1 (d) m

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : The total number of basic variables in a basic feasible solution is m + n – 1.

557. In Simplex method if in pivot column at the entries are negative or zero when choosing leaving variable then

(a) Solution is Degenerate
(b) Solution is infeasible
(c) Alternative optima
(d) Unbounded

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) A variable which is replaced is known as leaving variable and other one which replaces is known as entering variable. Whenever leaving variable are negative or zero, then it is unbounded.

558. In Northwest corner method the allocations are made

(a) Starting from the left hand side top corner
(b) Starting from the right hand side top corner
(c) Starting from the lowest cost cell
(d) Starting from the lowest requirement and satisfying first

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) In northwest corner method, the largest possible allocation is made to the cell in the upper - left hand corner of the table followed by allocations to adjacent feasible cells.

559. What is the game value of the following payoff matrix?

	X	Y	Z
P	-5	10	20
Q	5	-10	-10
R	5	-20	-20

(a) 5 (b) 2
(c) 8 (d) 0

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a)

	X	Y	Z	Max. value
P	–5	10	20	20 ↓
Q	[5]	–10	–10	[5]
R	[5]	–20	–20	[5]
Max.value	[5]	10	20	

The value of the game is 5 [is common] to Row & column.

560. For these two constraints, which point is in the feasible region of this minimization problem?
$14x + 6y \leq 42$ and $x - y \leq 3$

(a) x = -1, y = 1 (b) x = 0, y = 4
(c) x = 2, y = 1 (d) x = 5, y = 1

CGPSC AE 15.01.2021

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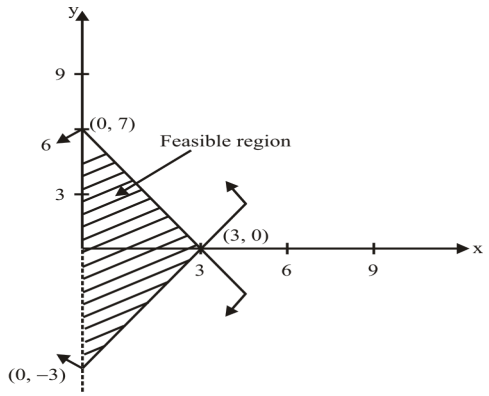
Ans. (c) : $14x + 6y \le 42 \Rightarrow \frac{x}{3}+\frac{y}{7} \le 1$



$x - y \le 3$

$\frac{x}{3}-\frac{y}{7} \le 1$

(2, 1) point is in the feasible region of the minimization problem.

Note-Official answer is (d).

561. Assignment problem helps to find a maximum weight identical in nature in a weighted

(a) Tripartite graph (b) Bipartite graph
(c) Partite graph (d) Both (a) and (b)

APGCL AM, 2021

Ans. (b) : Bipartite graph

562. If $Z = x_1 + x_2$ is subjected to constraints $x_1 + x_2 \le 1, -3x_1 + x_2 \le 3$ and $x_1 \le 0$, $x_2 \ge 0$, then the maximum value of Z occurs at:

(a) $\left(\frac{1}{2},\frac{1}{2}\right)$ (b) (0, 5)
(c) Infeasible solution (d) None of these

JPSC AE 10.04.2021, Paper-II

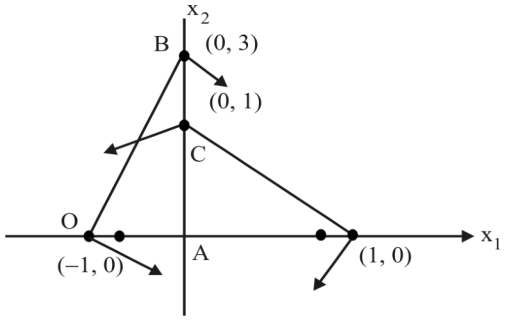
Ans. (c) : $Z = x_1 + x_2$

$x_1 + x_2 \le 1$

$-3x_1 + x_2 \le 3$

$x_1 \le 0$

$x_2 \ge 0$



There is no any closed region so this leads to infeasible solution.

563. When total supply is equal to the total demand for a transportation problem it is termed as a/an____ transportation problem

(a) equal (b) balanced
(c) similar (d) neutral

UPRVUNL AE 05.07.2021

Ans. (b) :

564. As per one of the assumption of linear programming problem in operations research, objective function and constraints are assumed to be _____.

(a) sinusoidal (b) exponential
(c) linear (d) circular

UPRVUNL AE 05.07.2021

Ans. (c) : The assumption of linear programming problem in operations research, objective function and constraints are assumed to be linear.

565. which of the following is NOT a method of solving the transportation problem?

(a) Vogel's approximation method
(b) Minimum cost method
(c) Northwest rule
(d) Minimum space method

UPRVUNL AE 05.07.2021

Ans. (d) :

566. Which one of the following is not a method to obtain the initial basic feasible solution to a transportation problem?

(a) Least Cost Method
(b) Big M Method
(c) Vogel's Approximation Method
(d) North West Corner Rule

VIZAG MT, 14.12.2020

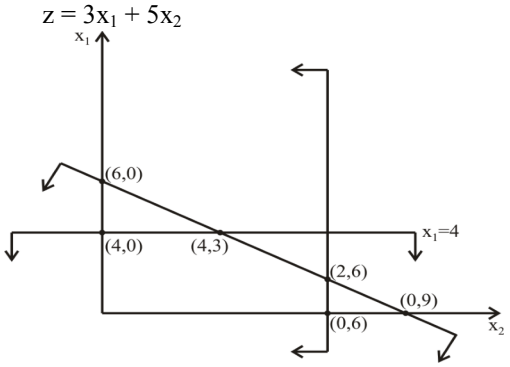
Ans. (b) : In transportation initial basic feasible solution is obtained from–
(1) Least Cost Method
(2) Vogel's Approximation Method
(3) North-West Corner rule

567. Max $z = 3x_1 + 5x_2$
S.t. $x_1 \le 4$; $2x_2 \le 12$
$3x_1 + 2x_2 \le 18$
$x_1, x_2 \ge 0$
What is the optimum value of z?

(a) $x_1 = 2, x_2 = 6, z = 36$
(b) $x_1 = 3, x_2 = 4, z = 29$
(c) $x_1 = 4, x_2 = 3, z = 27$
(d) $x_1 = 5, x_2 = 2, z = 25$

RPSC IOF, 2020

Ans. (a) : Optimum value of z is
$x_1 = 2, x_2 = 6, z = 36$
$z = 3x_1 + 5x_2$



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Max $z = 3x_1 + 5x_2$

The optimum value occurs at the corner point,

At (4, 0) $z = 3(4) + 5(0) = 12$

At (4, 3) $z = 3(4) + 5(3) = 12 + 15 = 27$

At (2, 6) $z = 3(2) + 5(6) = 6 + 30 = 36$

At (0, 6) $z = 3(0) + 5(6) = 30$

The optimum value of z is 36 at point (2, 6).

568. In simplex method:

(a) the selected negative variable is the entering variable and the positive basic variable is the leaving variable

(b) the selected zero variable is the entering variable and the removed basic variable is the leaving variable

(c) the selected zero variable is the leaving variable and the removed basic variable is the entering variable

(d) the selected positive variable is the entering variable and the negative basic variable is the leaving variable

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Simplex method, standard technique in linear programming for solving an optimization problem, typically one involving a function and several constraints expressed as inequalities. In this method, the selected zero variable is the leaving variable and the removed basic variable is the entering variable.

569. In linear programming, dual price represents :

(a) minimum and mean price

(b) minimum and maximum price

(c) mean and maximum price

(d) unit worth of a resource

NLC GET 17.11.2020, Shift-II

Ans. (d) : In linear programming dual price represents unit worth of a resource the dual price gives the improvement in the objective function if the constraint is relaxed by one unit. In the case of a less than or equal constraint, such as a resource constraint, the dual price gives the value of having one more unit of the resource represented by the constraint.

570. In Hungarian method the optimum solution is defined by:

(a) Negative value of total cost

(b) Positive value of total cost

(c) Zero total cost

(d) None of these

HPPSC ADF 05.03.2019

Ans. (c) : An efficient method for solving an assignment problem, known as Hungarian method. In an assignment problem, a solution having zero total cost is an optimum solution.

571. The value of Z_{max} when Z = 3x + 2y subjected to the constraints:

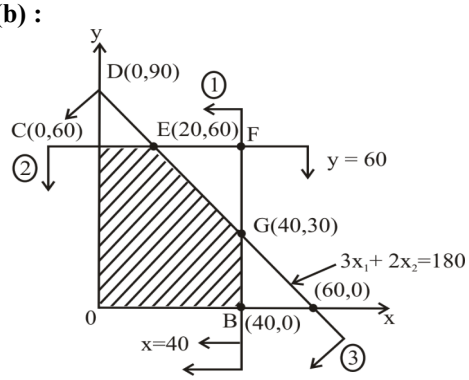
$x \le 40$; $y \le 60$; $3x + 2y \le 180$; $x, y \ge 0$

(a) 120 (b) 180

(c) infinite solution (d) unbounded

HPPSC ADF 05.03.2019

Ans. (b) :



$Zm = 3x + 2y$

Constraints = $x \le 40$ … (i)

$y \le 60$ … (ii)

$3x + 2y \le 180$ … (iii)

$\frac{x}{60} + \frac{y}{90} \le 1$

From constraints, $3x + 2y = 180$ … (iii)

$y = 60$ … (ii)

$3x + 120 = 180$

$3x = 60$ $\boxed{x = 20}$

From constraints, $3x + 2y = 180$ … (ii)

$\boxed{x = 40}$ … (i)

$3 \times 40 + 2y = 180$

$2y = 60$

$\boxed{y = 30}$

Maximum $(Z)_{max} = 3x + 2y$

At, (40, 0) $3\times40 + 2\times30 = 120 + 60 = 180$(max.

At, (40,30) $3\times20 + 60\times2 = 60 + 120 = 180$

At, (20,60) $3\times0 + 60\times2 = 120$

572. The first algorithm for linear Programming was given by :

(a) Bellman (b) Hantzig

(c) Kulm (d) Van Neumann

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) The first algorithm for linear Programming was given by Dantzig.

Dynamic programming $\Rightarrow$ Bellman

573. A toy company products type A and type B dolls. Type A doll requires labour time twice that of type B doll. The company can produce 600 dolls a day, if all the dolls are of type B only. The sale limit in the market is 200 for type A and 250 for type B dolls daily. Assuming that the profits per doll are Rs. 10 for type A and Rs. 6 for type B. Which of the following statements best describes the given LPP problem with constraints?

(a) $2x_1 + x_2 = 600$

$x_1 \ge 200,$

$x_2 \ge 250$

$x_1, x_2 \ge 0$

Find x_1 and x_2 such that profit

$P = 10x_1 + 6x_2$ subjected to the contraint

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(b) $2x_1 + x_2 \geq 600$

$x_1 \geq 600$

$x_2 \geq 250$

$x_1, x_2 \geq 0$

Find x_1 and x_2 such that profit

$P = 10x_1 + 6x_2$ subjected to the contraint

(c) $2x_1 + x_2 \leq 600$

$x_1 \geq 200$

$x_2 \geq 250$

$x_1, x_2 \geq 0$

Find x_1 and x_2 such that profit

$P = 10x_1 + 6x_2$ subjected to the contraint

(d) $2x_1 + x_2 \leq 600$

$x_1 \leq 200$

$x_2 \leq 250$

$x_1, x_2 \geq 0$

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (d) : Given,

Dolls type A– Rs. 10

Dolls type B– Rs. 6

Labour time of A is twice of type B [A = 2B]

$\because$ Company can produce dolls of type B-600 only

$\therefore$ B type dolls $\rightarrow$ 600

A type dolls $\rightarrow$ 1200

Let dolls for type A be x_1 & dolls for type B be x_2 –

1 day type B $\rightarrow$ 600 dolls

1 dolls of type B $\rightarrow \frac{1}{600}$ day

x_2 dolls $\rightarrow \frac{x_2}{600}$ day

1 dolls of type A $\rightarrow \frac{2}{600}$ day

x_1 dolls $\rightarrow \frac{x_1}{300}$ day

$\frac{x_1}{300} + \frac{x_2}{600} \leq 1$

$2x_1 + x_2 \leq 600$

$x_1 \leq 200$

$x_2 \leq 250$

$x_1, x_2 \geq 0$

Maximize profit $[z = 10x_1 + 6x_2]$

574. An objective function is $p(x, y) = 3x + 9y$ and constraints are :

$x + y \leq 8$,

$x + 2y \leq 4$,

$x \geq 0$,

$y \geq 0$

The maximum value of objective function is :

(a) 18 (b) 24

(c) 30 (d) 12

SJVN ET 2019

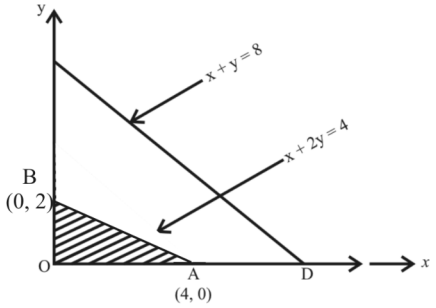
Ans. (a) : Max (z) = 3x + 9y

Subject to,

$x+y \leq 8$ .....(i)

$x+2y \leq 4$ .....(ii)

Solution by graphical method.



Feasible region → (0-A-B-0)

The points lying on the boundary of the feasible region (i.e. point A and B) are considered only because at one of these points the value of objective function (z = 3x + 9y) is maximum.

Since,

At point A (4, 0)

$z = 3 \times 4 + 9 \times 0 = 12$

At point B (0, 2)

Z = 18

Hence, maximum value of objective function,

Max (z) = 18

575. In a transportation problem with 4 supply points and 5 demand points, how many number of constraints are required in its formulation?

(a) 20 (b) 1

(c) 0 (d) 9

BHEL ET 2019

Ans. (d) : Total supply from warehouse = 4

Demand point outcome = 5

Number of functional constraints = 4 + 5

Over constrained structure = 9

576. Two models A and B, of a product earn profits of Rs. 100 and Rs. 80 per piece, respectively. Production times of A and B are 5 hours and 3 hours respectively, while total production time available is 180 hours. For a total batch size of 50, to maximize profit, the number of units A to be produced is

(a) 10 (b) 15

(c) 20 (d) 25

Gujarat PSC AE 2019

Ans. : (b) :

Maximize $\rightarrow$ 100 A + 80 B = Z objective function

Constraints time $\rightarrow$ $5A + 3B \leq 180$

Constraints item $\rightarrow$ $A + B \leq 50$

Maximum value of objective function 'Z' will be at all vertices (corner points) at it will lie max at 'S' when

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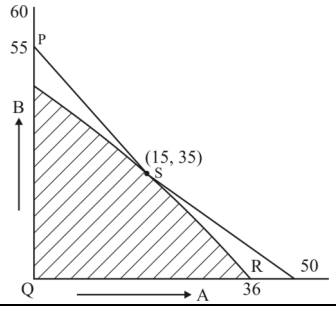
item A = 15
item B = 35



577. Solve the following linear programming problem.

$x_1 - 3x_2 - 6 \le 0,\ 2x_1 + 4x_2 - 8 \ge 0,$

$x_1 - 3x_2 + 6 \ge 0,\ x_1 \ge 0,\ x_2 \ge 0$ **to minimize the objective function $z = 2x_1 + x_2$, z_{min} =**

(a) 2 (b) 6
(c) 7 (d) 8

Vadodara Municipal Corp. DEE, 2018

Ans. (a) Objective function $z = 2x_1 + x_2$

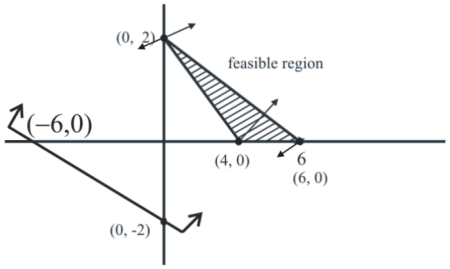
Constraints.

$x_1 - 3x_2 - 6 \le 0$

$2x_1 + 4x_2 - 8 \ge 0$

$x_1 - 3x_2 + 6 \ge 0$

$x_1 \ge 0,\ x_2 \ge 0$



$z = 2x_1 + x_2$

$z(0, 2) = 2 \times 0 + 2 = 2$

$z(4, 0) = 2 \times 4 + 0 = 8$

$z(6, 0) = 2 \times 6 + 0 = 12$

$\boxed{Z_{mini} = 2}$

578. $2x_1 + x_2 + x_3 = 0$,
$x_2 - x_3 = 0$,
$x_1 + x_2 = 0$

(a) A unique solution
(b) No solution
(c) Infinite number of solutions
(d) Five solutions

Vadodara Municipal Corp. DEE, 2018

Ans. (c) : Infinite number of solutions

579. When the elements of net evaluation row of simplex table are equal, the situation is known as :

(a) Tie (b) Degeneracy
(c) Break (d) Shadow price

GPSC ARTO Pre 30.12.2018

Ans. (b) : When the elements of net evaluation row of simplex table are equal, them the situation is known as Degeneracy.

580. If there are more than one optimum solutions for the decision variable, the solutions is

(a) Infeasible (b) Unbounded
(c) Alternative (d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (c) : If there are more than one optimum solutions for the decision variable, the solutions is 'alternative'.
An optimal solution is a feasible solution where the objective function reaches is maximum (or minimum) value for example, the most profit or the least cost.

581. According to transportation problem, number of basic cells will be exactly

(a) m + n – 0 (b) n + m – 2
(c) m + n – 1 (d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (c) : According to transportation,
Number of basic cells = m + n – 1
Total no. of variable = m × n

582. Dynamic programming is a mathematical technique dealing with the optimization of ------ stage decision process.

(a) Multi (b) Single
(c) Both (a) and (b) (d) None of them

MPSC Lect. (Prod.) 2018

Ans. (a) : Dynamic programming is a mathematical technique developed by Rechard Bellman. It is a mathematical technique well suited for the optimization of multi stage decision process.

583. Which of the following statements is INCORRECT?

(a) Assignment model is a special case of linear programming problem
(b) In queuing models, Poisson arrivals and exponential services are assumed
(c) In transportation problem, the non-square matrix is made square by adding a dummy row or dummy column
(d) In linear programming, dual of a dual is primal

APGENCO AE, 2017

Ans. (c) : In transportation problems dummy row or dummy column is added to equate supply and demand.
Hence, given statement in option (c) is incorrect statement.

584. The following statement applies to both transportation model and assignment model :

(a) The inequalities of both problems are related to one type of resource.
(b) Both use VAM for getting basic feasible solution
(c) Both are tested by MODI method for optimality
(d) Both have objective function, structural constraint and non-negativity constraints.

HPPSC Asstt. Prof. 20.11.2017

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Ans. (d) : Both transportation model and assignment model have objective function, structural constraint and non-negativity constraints.

585. In an linear programming problem, the restrictions or limitations under which the objective function is to be optimised are called:

(a) Constraints (b) Objective function
(c) Decision variables (d) None of the above

TRB Polytechnic Lecturer 2017

Ans. (a) : In an linear programming problem, the restrictions or limitations under which the objective function is to be optimised are called

586. The variable that is included in the '$\leq$' type inequality constraint for the purpose of converting general from of LPP to standard from LPP is called :

(a) surplus variable (b) slack variable
(c) artificial variable (d) basic variable

TRB Polytechnic Lecturer 2017

Ans. (b) : The variable that is included in the '$\leq$' type inequality constraint for the purpose of converting general from of LPP to standard from LPP is called slack variable.

587. When there exists a not basic variable whose relative profit $\left(\text{i.e., } \bar{C}j\right)$ is zero in the optimal table, then the nature of solution is?

(a) Unbounded (b) Infeasible
(c) Unique optimal (d) Alternate optimal

TRB Polytechnic Lecturer 2017

Ans. (d) : When there exists a not basic variable whose relative profit $\left(\text{i.e., } \bar{C}j\right)$ is zero in the optimal table, then the nature of solution is alternate optimal.

588. If the i^{th} constraint of a primal (maximisation) is an equation, then the dual (minimisation) variable 'y_i' is :

(a) ≥ 0
(b) ≤ 0
(c) Unrestricted in sign
(d) None of these

TRB Polytechnic Lecturer 2017

Ans. (c) : If the i^{th} constraint of a primal (maximisation) is an equation, then the duel (minimisation) variable 'y_i' is unrestricted in sign.

589. In order that linear programming techniques provide valid results

(a) Relations between factors must be linear (Positive)
(b) Relations between factors must be linear (Negative)
(c) (a) or (b)
(d) None of the above

Vizag Steel (MT) 2017

Ans. (c) : In order that linear programming techniques provide valid results-
i. Relations between factors must be linear (Positive)
ii. Relations between factors must be linear (Negative)

590. While solving a linear programming model, if a redundant constraint is added, then what will be its effect on existing solution?

(a) There will be no effect
(b) The solution space will get further constrained
(c) The solution space becomes concave
(d) The problem no longer remains solvable

BPSC AE Mains 2017 Paper - VI

Ans. : (a) : Constraint which does not become part of the boundary making a feasible region is termed as redundant constraint. Inclusion or exclusion of such constraint does not have any effect on the optimum solution of the problem.

591. In Vogel's approximation method; the opportunity cost associated with a row is determined by:

(a) the difference between the smallest cost and the next smallest cost in the row
(b) the difference between the smallest unused cost and the next smallest unused cost n the row
(c) the difference between the smallest cost and next smallest unused cost in the row
(d) none of the above

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : In vogel's approximation method, the opportunity cost associated with a row is determined by the difference between the smallest unused cost and the next smallest unused cost in the row. It is also known as penalty.

592. In transportation problem a single good is to be shipped from several origins to several destinations at

(a) Minimum overall cost
(b) Optimum overall cost
(c) Minimum price
(d) Optimum price

Nagaland CTSE 2016 Paper-I

Ans. (a) : Transportation problem is a classic operations research problem, where the objective is to determine the schedule for transporting good from source to destination in a way that minimizes the overall shipping cost, while satisfying supply and demand constraints.

593. Modified distribution method is used for

(a) Queuing problem
(b) Assignment problem
(c) Both Queuing problem and Assignment problem
(d) Transportation problem

RPSC LECTURER 16.01.2016

Ans. (d) : Modified distribution method is used for transportation problem.

594. The maximum value of Z = 3x + 4y subjected to the constraints $2x+y\leq 4, x+24\geq 12, x\geq 0, y\geq 0$

(a) 10 (b) 20
(c) 30 (d) No feasible solution

RPSC LECTURER 16.01.2016

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Ans. (d) : z = 3x + 4y (maximum value)

Constraints

$$2x + y \le 4$$

$$x + 2y \ge 12$$

$$x \ge 0$$

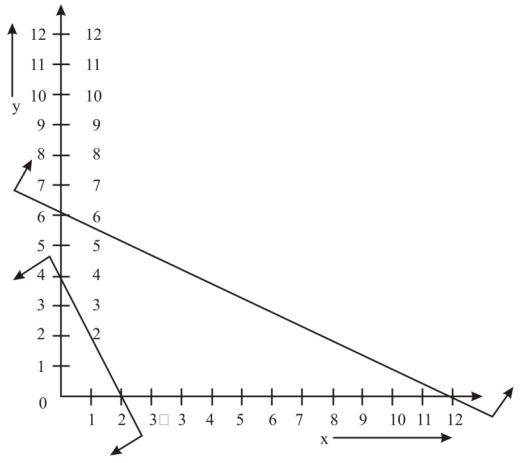
$$y \ge 0$$

$$\frac{x}{2} + \frac{y}{4} \le 1$$

$$\frac{x}{12} + \frac{y}{6} \ge 1$$

$$x \ge 0$$

$$y \ge 0$$



It is a case of no feasible solution.

595. A basic feasible solution in a linear programming problem with m constraints and n variables will have

(a) At the most m variable with non zero values.
(b) At least m variables with non zero
(c) At the most n variables with non zero values.
(d) At least n variables with non zero values.

UPPSC AE 12.04.2016 Paper-I

Ans. : (a) At the most m variable with non zero values.

596. For which of the following situations linear programming can be applied?

(a) Material selection problems
(b) Product design problems
(c) Scheduling of production to meet sales forecast
(d) Quenching problems

BPSC Poly. Lect. 2016

Ans. : (c) Linear programming applied in scheduling of production to meet sales forecast.

597. An initial feasible solution of an optimization problem is

(a) All basic variables are equal to zero
(b) At least one basic variable should have optimum value.
(c) All basic variables should have real values
(d) None of the above

UJVNL AE 2016

Ans. : (a) An initial feasible solution of an optimization problem is all basic variables are equal to zero.

598. A constraint in equation $5x_1 - 3x_2 \le -5$ is converted as $(3x_2 + \alpha_2) - (5x_1 + S) = 5$ then 'S' is called as

(a) basic variable (b) surplus variable
(c) non-basic variable (d) artificial variable

GPSC ARTO 01.05.2016

Ans. (b) :

- A surplus variable is subtracted from a ≥ constraint to convert it to an equation (=)
- A surplus variable represents an excess above a constraint requirement level.
- A surplus variable contributing nothing to the calculated value of the objective function.
- Subtracting surplus variables in the problem.

E.g., $2x_1 + 4x_2 > 16 \Rightarrow 2x_1 + 4x_2 - S = 16$

$4x_1 + 3x_2 > 24 \Rightarrow 4x_1\ 3x_2 - S = 24$

Where, S = Surplus variable

599. While solving linear programming problems, changes in the right hand side constants of constraints is made as one of the ways of making.....

(a) Constraint analysis
(b) Function analysis
(c) Sensitivity analysis
(d) Objective function analysis

HPPSC AE (PWD) 10.09.2016

Ans. (c) : A sensitivity analysis determines how different values of an independent variable affect a particular dependent variable under a given set of assumptions. In other words sensitivity analysis study how various sources of uncertainty in a mathematical model contribute to the model's overall uncertainty.

600. Assignment problems can be solved by using:

(a) Branch-and-bound algorithm
(b) Control based algorithm
(c) Non-linear simplex algorithm
(d) Stochastic algorithm

HPPSC AE (PWD) 10.09.2016

Ans. (a) : Branch and bound is a systematic method for solving optimization problem. Branch and bound is a rather general optimization technique that applies where the greedy method and dynamic programming fail. An important advantages of branch and bound algorithms is that we can control the quality of the solution to be expected even if it is not yet found. The cost of an optimal solution's only upto smaller than the cost of the best computed one.

601. The purpose of a dummy source or a dummy destination in a transportation problem is to

(a) Provide a means of representing a dummy problem
(b) Obtain a balance between total supply and total demand
(c) Prevent the solution from becoming degenerate

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(d) Make certain that the total cost does not exceed some specified figure
(e) None of the above

CGPSC AE 16.10.2016

Ans. (b) : The purpose of a dummy source or dummy destination in a transportation problem is to make certain that the total cost does not exceed some specific figure & to obtain a balance between total supply and total demand.

602. A transportation problem is said not to degenerate when the number of occupied cells are (m = the number of rows, n = number of columns)

(a) m + n (b) m + n + 1
(c) Greater than m + n (d) m + n – 1

GPSC Asstt. Prof. (Prod.) 21.08.2016

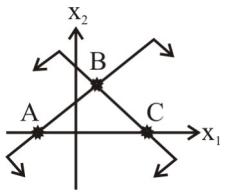
Ans. (d) : If the no. of allocations are m + n – 1 is a transportation problem of m × n then it gives the basic feasible solution.

603. When linear programming problem is solved graphically, every corner value in the feasible region shows

(a) A basic feasible solution
(b) An optimum solution
(c) Solution not depending on constraints
(d) Alternate solution

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (a) :



A, B & C are corner value of graphically solved linear programming this values may give the feasible solution so comes under feasible region.

604. An optimal solution of an assignment problem can be obtained only if:

(a) each row and column has only one zero element
(b) each row and column has at least one zero element
(c) the data is in square matrix
(d) the data is in square matrix and each row has at least one zero element
(e) None of the other options

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) : An optimal solution of an assignment problem can be obtained only if the number of assigned cells is equal to No. of columns/row, then its is an optimal solution.

605. To formulate a problem for solution by simplex method, we must add artificial variable to:

(a) only equality constraints
(b) only greater than constraints
(c) only less than constraints
(d) both equality constraints and greater than constraints
(e) both equality constraints and less than constraints

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : In a linear programming in finding the optimal solution, we normally have the objective function & the constraints, when using simplex method, we will add artificial variables to the constraints & (<) are the constraints.

606. If dual has an unbounded solution, then what can be said about primal?

(a) It has no feasible solution
(b) It has unbounded solution
(c) It has feasible solution
(d) It has degenerate solution
(e) Nothing can be said

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : If the primal (dual) is unbounded, then the dual (primal) has no feasible solution, because such as feasible solution would give a bound on the former problem, which contradicts its unboundness it is possible for the primal and dual to be simultaneously infeasible.

607. In linear programming problem if the new constraint is added, what will change?

(a) Feasibility
(b) Optimality
(c) Both feasibility and optimality
(d) No change

TSPSC Managers, 2015

Ans. (a) : In linear programming problem if the new constraint is added feasibility will change.

608. Simplex problem is considered as unbounded when :

(a) All the variables in entering column are negative
(b) Variables in basis are negative
(c) All the values in (Ej-Cj) row are negative
(d) Artificial variable is in the basis

TSPSC Managers, 2015

Ans. (a) : Simplex problem is considered as unbounded when all the variable in entering column are negative.

609. Constraints in linear programming model represent :

(a) requirements
(b) limitations
(c) balancing requirements and limitations
(d) All of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (d) : Constraints in linear programming model represent
(i) Requirements
(ii) Limitations
(iii) Balancing requirements and limitations.

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610. The optimality of a transportation problem is determined by application of

(a) North west corner method
(b) Modi method
(c) Vogel's application method
(d) Least cost method

MPPSC AE 08.11.2015

Ans. (b) : Modi method
Note- Option (c) is given by the commission.

611. In a linear programming model there are four decision variables and three constraints. During an iteration, by simplex method, the coefficient of the base variable would form

(a) An identity matrix
(b) Slack variables
(c) Basic solution
(d) None of these is correct

MPPSC AE 08.11.2015

Ans. (a) : The coefficient of base variable would form An identity matrix.

612. Consider the following conditions for applying linear programming

1. **There must be a well defined objective function**
2. **The value assigned to each parameter of a linear programming model is assumed to be a known constant**
3. **Decision variables in a linear programming model are allowed to have any values, excluding non-integar values, that satisfy the functional and non negativity constraints.**

Of the above, which is correct

(a) 1 and 2 (b) 1 and 3
(c) 2 only (d) 2 and 3
(e) 1, 2 and 3

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : Let us look at the steps of defining a Linear Programming problem generically
(i) Identify the decision variables.
(ii) Write the objective function.
(iii) Mention the constraints
(iv) Explicitly state the non-negativity restriction.
For a problem to be a linear programming problem, the decision variables, objective function and constraints all have to be linear function.

613. In Graphical solution of maximisation problem, the line which we move from origin to the extreme point of the polygon is:

(a) Any one side of the polygon
(b) Iso profit line
(c) Iso cost line
(d) An imaginary line
(e) Break Even line

CGPSC AE 26.04.2015 Shift-I

Ans. (b) : If the LPP is a maximisation problem, the objective line (iso-profit line) is then moved parallel to the initial objective line, towards the origin until only one point of the feasible region lies on the objective line. This would be the corner point of the feasible region where the objective function has the minimum value.

614. Simplex problem is considered as infeasible when

(a) All the variables in entering column are negative
(b) Variables in the basis are negative
(c) Artificial variable is present in basis
(d) Pivotal value is negative

TSPSC AEE 2015

Ans. (c) : Simplex problem is considered as infeasible when artificial variable is present in basis.

615. Based on final table of simplex LPP said to have alternative solution if in (Ej - Cj) row

(a) One or more basic variable has zero value
(b) Entering variable has negative coefficient
(c) below a non basic variable there is zero
(d) optimum function value is zero

TSPSC AEE 2015

Ans. (c) : below a non basic variable there is zero

616. In linear programming problem when does feasibility change?

(a) change in objective functions coefficient
(b) change in right hand side of feasible region
(c) addiction of new variable
(d) Feasibility does not change

TSPSC AEE 2015

Ans. (b) : change in right hand side of feasible region

617. Any network must have flow between Source and Sink equal to capacity of :

(a) Cut
(b) Minimal cut
(c) Both cut and minimal cut
(d) Either cut or minimal cut

UPRVUNL AE 2014

Ans. : (b) Any network must have flow between source and sink equal to capacity of minimal cut.

618. The simplex method for solving Linear Programming Problem, was developed by

(a) Scarf H (b) Hantzig G B
(c) Jackson J R (d) Miller D W

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Hantzig G B

619. The purpose of the stepping stone is to

(a) develop the initial solution to a transport problem
(b) identify the relevant costs in a transportation problem
(c) determine whether a given solution is a feasible or not
(d) assist one in moving from an initial feasible solution to the optimal solution

TNPSC AE 2014

Ans. (d) : The purpose of the stepping stone is to assist one in moving from an initial feasible solution to the optimal solution.

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620. In order for a transportation matrix which has six rows and four columns not to degenerate, what is the number occupied cells in the matrix?

(a) 6 (b) 7
(c) 8 (d) 9

TNPSC AE 2013

Ans. (d) : Number of cells for non degenerate solution,
$6 + 4 - 1 = 9$

621. Consider the system of equations given below:
$x + y = 2$ and $2x + 2y = 5$
This system has

(a) one solution (b) no solution
(c) infinite solution (d) four solutions

VIZAG STEEL MT 18.08.2013

Ans. (b) $x + y = 2$ ---------(1)
$2x + 2y = 5$ ---------(2)
from equation (2) $2(x+y) = 5$
$x + y = 2.5$ ---------(3)
from (1) & (3) $x + y = 2 \neq 2.5$ so equation has no solutions.

622. Consider the following LPP:
Maximize $z = 60X_1 + 50X_2$
Subject to $X_1 + 2X_2 \leq 40$,
$3X_1 + 2X_2 \leq 60$ where, X_1 and $X_2 \geq 0$

(a) The LPP has a unique optimal solution
(b) The LPP is infeasible
(c) The LPP is unbound
(d) The LPP has multiple optimal solutions

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Maximize, $z = 60x_1 + 50x_2$

$$x_1 + 2x_2 \leq 40$$

$$\frac{x_1}{40} + \frac{x_2}{20} \leq 1 \quad .....(i)$$

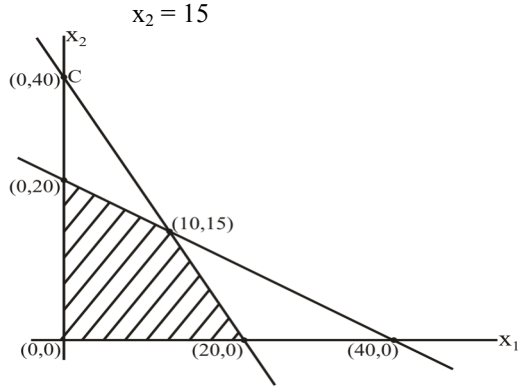
$$3x_1 + 2x_2 \leq 60$$

$$\frac{x_1}{20} + \frac{x_2}{40} \leq 1 \quad .....(ii)$$

$$x_1, x_2 \geq 0$$

From equation (i) & (ii)

$$x_1 + 2x_2 = 40$$
$$3x_1 + 2x_2 = 60$$
$$2x_1 = 20$$
$$x_1 = 10$$
$$x_2 = 15$$



The feasible point are (0,0), (20,0), (0,20) and (10,15)
$z_{(0, 0)} = 0$
$z_{(20, 0)} = 60 \times 20 = 1200$
$z_{(0, 20)} = 50 \times 20 = 1000$
$z_{(10, 15)} = (60 \times 10) + (50 \times 15) = 1350 = \max z$
The given LPP has unique optimal solution.

623. In graphical method, "The Feasible Region" is:

(a) The area which is not giving the answer
(b) The area which will give wrong answer
(c) The area which satisfies all the constraints
(d) The area which does not satisfy the constraints

TNPSC AE (Industries) 09.06.2013

Ans. (c) : The area which satisfies all the constraints is known as feasible region.

624. In simplex method of linear programming the objective row of the matrix consists of:-

(a) Names of the variables
(b) Coefficient of the objective function
(c) Slack variables
(d) None of the above

UKPSC AE-2013, Paper-I

Ans. (c) : In simplex method of linear programming the objective row of the matrix consists of slack variables.

625. Following is not a method of solving a transportation problem:-

(a) Northwest corner method
(b) Least cost method
(c) Vogel's approximation method
(d) Dynamic method

UKPSC AE-2013, Paper-I

Ans. (d) : Northwest corner method, least cost method and Vogel's approximation method are used to solving a transportation problem.

626. If in graphical solution of linear programming problem, the objective function line is parallel to the line representing constraint equation, then the solution of problem is:-

(a) Infeasible solution
(b) Unbound solution
(c) Multiple optimum solution
(d) None of the above

UKPSC AE-2013, Paper-I

Ans. (c) : In linear programming problem, the objective function line is parallel to the line representing constraint equation, then the solution of problem will be multiple optimum solution.

627. Which of the following is true about the initial basic feasible solution in simplex method ?

(a) It is an optimal solution.
(b) All basic variables are zero.
(c) Solution is not possible.
(d) Any one basic variable in zero

UKPSC AE 2012 Paper-I

Ans. (b) : All basic variables are zero.

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628. **Assertion (A) :** Vogel's approximation method yields the best initial basic feasible solution of a transportation problem.
Reason (R) : Vogel's method give allocations to the lowest cost elements of the whole matrix.
Code :
(a) Both (A) and (R) are correct. (R) is the correct explanation of (A).
(b) Both (A) and (R) are correct. (R) is not the correct explanation of (A).
(c) (A) is correct, but (R) is false.
(d) (A) is false, but (R) is correct.

UKPSC AE 2012 Paper-I

Ans. (c) : (A) is correct, but (R) is false.

629. **Which one of the followings statements is not correct regarding simplex method of linear programming ?**
(a) It is an iterative procedure.
(b) It has a trial basic feasible solution to constraints.
(c) The collection of feasible solution does not constitute a convex set.
(d) It improves the first trial solution by a set of rules.

UKPSC AE 2012 Paper-I

Ans. (c) : The collection of feasible solution does not constitute a convex set.

630. **The following is not true for linear programming problems :**
(a) Objective function is expressed as a linear function of variables.
(b) Resources are not limited.
(c) Some alternative course of actions are also available.
(d) Decision variables are inter related.

UKPSC AE 2012 Paper-I

Ans. (b) : Resources are not limited.

631. **Match List I correctly with List II and select your answer using the codes given below**

	List I	**List II**
(A)	Linear programming problem	1.Travelling salesman
(B)	Queuing problem	2. Saddle point
(C)	Dynamic problem	3. Product mix
(D)	Game theory problem	4. Normal distribution

Codes

	A	**B**	**C**	**D**
(a)	3	4	1	2
(b)	4	3	1	2
(c)	3	4	2	1
(d)	4	3	2	1

TNPSC ACF 2012, ESE 1994

Ans. (a) : Linear programming – Produce min
Queuing problem – Normal distribution
Dynamic problem – Traveling salesman
Game theory – Saddle point.

632. **In a transportation problem, North-West method would yield**
I. an optimum solution
II. an initial feasible solution
III. Vogel's approximate solution
Of these
(a) I alone is correct (b) II alone is correct
(c) III alone is correct (d) II and III are correct

TNPSC ACF 2012

Ans. (b) :

Initial Solution	Optimum solution
North west corner rule	Steppings Stone method
Column minima method	or
Least cost method	U–V method.

633. **Assignment of work to manpower and machinery is known as**
(a) scheduling (b) loading
(c) reporting (d) none of these

TRB Poly. Lect., 2012

Ans. (b) : Loading means, assignments of work to manpower, machinery etc. without specifying when the work is to be done.

634. **In the problem $Z = 3x_1 + 2x_2$**
Subject to $2x_1 + x_2 \leq 20$ ..... (i)
$x_1 \leq 10$ ..... (ii)
$x_1.x_2 \geq 0$ ..... (iii)
constraint (iii) is known as
(a) availability constant
(b) non-negativity constraint
(c) objective constraint
(d) none of these

TRB Poly. Lect., 2012

Ans. (b) : Constraint (iii) is known as non-negativity constraint.

635. **Multiple solutions in LPPs indicates that**
(a) more than one solution is available for different objective function values
(b) two solutions are available satisfying all constraints
(c) more than one solution is available for the same objective function value
(d) no solution is available satisfying all the constraints

TRB Poly. Lect., 2012

Ans. (c) : Multiple solutions in LPPs indicate that more than one solution is available for the same objective function value.

636. **Dual of the dual is the**
(a) primal (b) dual
(c) either primal or dual (d) none of these

TRB Poly. Lect., 2012

Ans. (a) : Dual of the dual is known as the primal that optimization problems may be viewed from either of two perspectives, the primal problem or the dual problem, the sole gives a lower bound to two solution of the primal (minimization) problem.

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637. Artificial variable is introduced in the simplex method to

(a) determine the initial basic feasible solution when surplus variable is present

(b) convert the in equation with the sign greater than or equal to, in the form of an equation

(c) apply Big M method for solution to linear programming problems

(d) indicate the sensitivity of the surplus variable

TRB Asstt. Prof., 2012

Ans. (a) : Artificial variable is introduced in the simplex method to determine the initial basic feasible solution when surplus variable is present.

638. At the optimum level (i.e. final iteration) values of slack variables in the primals are

(a) positive (b) negative

(c) zero (d) none of these

TRB Asstt. Prof., 2012

Ans. (a) : At the optimum level (i.e. final iteration) values of slack variables in the primals are positive.

639. In the simplex method, if there are 6 equality constraints and 8 variables, the total number of basic solutions is :

(a) 2 (b) 14

(c) 28 (d) 48

APPSC IOF, 2009

Ans. (d) : Basic solution = Constraints × Variables

$\Rightarrow \quad 6 \times 8 = 48$

640. $\textbf{Max } Z = 6x_1 + 10x_2 + 2x_3$

$2x_1 + 4x_2 + 3x_3 \le 40$

$x_1 + x_2 \le 10$

$2x_1 + x_3 \le 12$

and $x_1, x_2, x_3 \ge 0$

The solution of the L.P is :

(a) $x_1 = 4, x_2 = 6, x_3 = 0$

(b) $x_1 = 6, x_2 = 6, x_3 = 3$

(c) $x_1 = 5, x_2 = 5, x_3 = 12$

(d) None of the above

APPSC IOF, 2009

Ans. (c)

641. In a linear programming problem, if a constraint is inactive, then the unit worth of resource corresponding to that constraint is

(a) zero (b) a high value

(c) negative (d) undefined

DRDO Scientists 2008

Ans. (a) :

642. A transportation problem will have feasible solutions if and only if

(a) $\sum_{i=1}^{m} S_i = \sum_{j=1}^{n} d_j$ (b) $\sum_{i=1}^{m} S_i < \sum_{j=1}^{n} d_j$

(c) $\sum_{i=1}^{m} S_i > \sum_{j=1}^{n} d_j$ (d) none of the above

where S_i denotes number of units being supplied by source and d_j number of units being received by destination

UKPSC AE 2007 Paper -I

Ans. (a) : $\sum_{i=1}^{m} S_i = \sum_{j=1}^{n} d_j$

643. In simplex method, If basic variables satisfy the non-negativity constraints, the basic solution

(a) is not optimal

(b) is basic feasible solution

(c) is basic infeasible solution

(d) does not exist

UKPSC AE 2007 Paper -I

Ans. (b) : Is basic feasible solution

644. The following is not true for the assignment model:

(a) $x_{ij} = 1$

(b) $\sum_{j=1}^{n} x_{ij} = 1$

(c) $\sum_{i=1}^{n} x_{ij} = 1$

(d) it can be stated in the form of n × n cost matrix $[C_{ij}]$ of real numbers

UKPSC AE 2007 Paper -I

Ans. (a) : $x_{ij} = 1$

645. Hungarian Method of solving the Assignment Problems involves

(a) Covering maximum number of zeroes with maximum number of lines

(b) Covering minimum number of zeroes with maximum number of lines

(c) Covering all zeroes with minimum number of lines

(d) Covering maximum number of zeroes with minimum number of lines

OPSC Civil Services Pre 2006

Ans. (c) : Hungarian method of solving the assignment problems in values minimum number of lines covering all zeroes.

8. Quality Control and Analysis

646. A statistical process control chart which shows the number of defects found in a subgroup of fixed size is

(a) C-chart (b) R-chart

(c) Control chart (d) Gantt chart

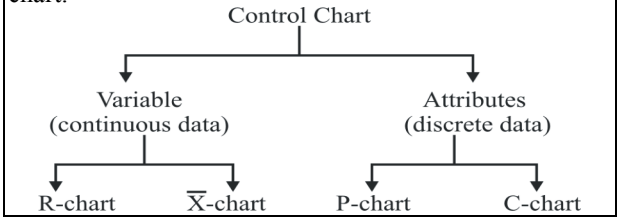
HPPSC Workshop Suptd. 08.07.02021

GPSC ARTO Pre 30.12.2018

ISRO Scientist/Engineer 17.12.2017

UKPSC AE-2013, Paper-I

Ans. (a) : Control Chart–A statistical chart used for controlling manufacturing processes is known as control chart.



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(1) $\bar{X}$ **-chart**–This chart is constructed for controlling the variations in the average quality standards of the products in a production process.

(2) R-chart–The range chart is constructed for controlling the variation in the dispersion or variability of the quality standard of the products in a production process.

(3) P-chart– This chart is constructed for controlling the quality standard in the average fraction defective of the products in a process. Here sample size is varying.

(4) C-chart– This chart is use for the control of number of defects per unit say a price of paper/glass/cloth which may contain more than one defect. Here sample size is fixed.

647. Material handling and plant location is analyzed by:

(a) Gantt chart (b) Bin chart
(c) Activity chart (d) Travel chart

UPPSC AE 13.12.2020, Paper-I
OPSC AEE 2015 PAPER-I
OPSC Civil Services Pre. 2011
RPSC ACF-2011

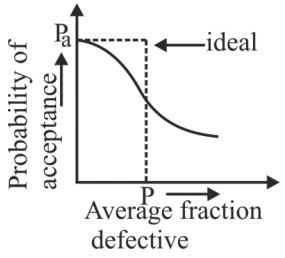
Ans. (d) : A travel chart is a simple table that is useful where there are multiple (and possible irregular) movement between places. It is a variation on the check sheet, indicating movements from and to any combination of a given set of locations.

648. An Operating Characteristics curve (OC-curve) is a plot between

(a) Consumer's risk and producer's risk.
(b) Probability of acceptance and probability of rejection.
(c) Percentage of defective and probability of acceptance.
(d) Average outgoing quality and probability of acceptance.

UPPSC AE 12.04.2016 Paper-I
MPPSC AE 2016
TNPSC AE 2014, ESE 2009

Ans. : (c)



∴ OC curve is a plot between percentage at defective and probability of acceptance

649. Statistical quality control techniques are based on the theory of :

(a) quality (b) statistics
(c) probability (d) control

Mizoram PSC AE/SDO 2014 Paper-III
RPSC ACF-2011
OPSC Civil Services Pre 2006

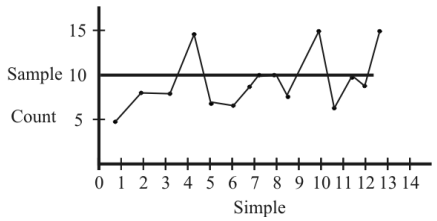
Ans. (c) : Statistical quality control techniques are based on the theory of probability statistical quality control refers to the use of statistical methods in the monitoring and maintaining of the quality of products and services.

650. In a weaving operation, the parameter to be controlled in the number of defects per 10 square yards of material. Control chart appropriate for this task is :

(a) P- chart (b) C- chart
(c) R-chart (d) $\bar{X}$ -chart

RPSC AE 2018, MPSC Poly. Lect. 2014
APGENCO AE 2012

Ans. (b) : In weaving operation the parameter to be controlled in the number of defects per 10 square yards of material control chart appropriate for this task is C – chart.



651. W. A. Shewart introduced the concept of

(a) Control chart (b) Travel chart
(c) Sampling plan (d) Quality circle

OPSC AEE 2019 Paper-I
HPPSC AE 2018
TNPSC AE 2017

Ans. (a) : W.A. Shewart introduced the concept of Control chart.

Control Charts–Control charts, also known as Shewart Charts or Process behavior charts, are a statistical process control tool used to determine if a manufacturing or business process is in a state of control. It is more appropriate to say that the control charts are graphical device for statistical process monitoring (SPM).

652. C-charts are the best example of :

(a) Binomial distribution
(b) Poisson distribution
(c) Normal distribution
(d) None of these

VIZAG STEEL MT 24.01.2021 Shift-II
Kerala PSC AE 06.08.2015
KPSC AE. 2015

Ans. : (b) C-charts are the best example of Poisson distribution.

$\bar{X}$ **- chart** :- X-chart shows the centering of the process e.g. it shows the variation in the average of samples

R-chart:- R- chart shows the uniformity or consistency of the process i.e. shows the variations in the ranges of samples. It is a chart for measured of spread.

C-chart:- It is control chart for defects. When the subgroup size varies from sample to sample, it is necessary to use V- chart.

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653. Which control chart is used to measure "variability of variability" within the samples?

(a) X-bar chart (b) R chart
(c) C chart (d) U chart

VIZAG MT, 14.12.2020
UJJVNL AE 2016
UKPSC AE-2013, 2012 Paper-I

Ans. : (b) R chart is used to measure "variability of variability."

$\bar{X}$ - chart :- X-chart shows the centering of the process e.g. it shows the variation in the average of samples

R-chart:- R-chart shows the uniformity or consistency of the process i.e. shows the variations in the ranges of samples. It is a chart for measured of spread.

C-chart:- It is control chart for defects. When the subgroup size varies from sample to sample, it is necessary to use V- chart.

654. The maximum value of the average outgoing quality for all possible values of proportion defective is called

(a) Average Outgoing Quality (AOQ)
(b) Acceptable Quality Level (AQL)
(c) Average Outgoing Quality Limit (AOQL)
(d) Lot Tolerance Proportion Defective (LTPD)

OPSC AEE 2019 Paper-I
Karnataka PSC AE, 10.09.2017
UPPSC AE 12.04.2016 Paper-I

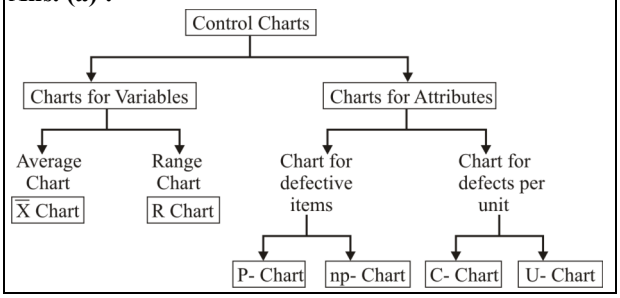
Ans. : (c) The maximum value of the average outgoing quality for all possible value of proportion defective is called Average outgoing quality limit (AOQL).

655. Which pair of notations represent a variable and an attribute, respectively?

(a) $\bar{X}$ and p (b) R and σ
(c) C and σ (d) p and $\bar{X}$

CGPSC AE 15.01.2021, CGPSC AE 16.10.2016

Ans. (a) :



656. Statistical quality control concept was initially developed by :

(a) Francis (b) Otto
(c) Newton (d) Walter Shewart

HPPSC Lect. (Auto) 23.04.2016
GATE 1995

Ans. (d) : Statistical quality control concept was developed by Walter Shewart. It is the use of satisfied of products and services. One method, referred to as acceptance sampling can be used when a decision or items based on the quality found in a sample.

657. The fraction defective chart to record the proportion defective items in a sample is _____ chart.

(a) X-bar (b) R
(c) C (d) P

Kerala PSC Poly. Lect., 2017
Haryana PSC Civil Services Pre, 2014

Ans. (d) The fraction defective chart to record the proportion defective items in a sample is P chart.

658. Six sigma company accepts the following number of defective products in an one million number of products:

(a) 3 (b) 6
(c) 30 (d) 60

GPSC DEE, Class-2 (GWSSB) 04.07.2021
HPPSC Asstt. Prof. 2014

Ans. (a) : In six sigma, 3.4 defects per one million is accepted.

659. Quality control chart for averages was maintained for a dimension of the product. After the control was established it was found that the standard deviation (σ) of the process was 1.00 mm. The dimension of the part is 70 ± 2.5mm Parts above 72.5mm can be reworked but parts below 67.5 mm have to be scrapped what should be the setting of the process to ensure production of no scrap and to minimize the rework?

(a) 70 mm (b) 70.5 mm
(c) 69 mm (d) 68.5 mm

UPRVUNL AE 21.08.2016
ESE 2004

Ans. : (b) Since, parts below 67.5 mm (i.e. LCL) shall have to be scrapped which is more costly, therefore, centering of the process has to be shifted towards (UCL) (i.e. 72.5).

Assuming $\pm 3\sigma$ control limit new centre of the chart is,

$= 67.5 + 3\sigma$
$= 67.5 + 3 \times 1$
$= 70.5$ mm

660. P–charts are based on the

(a) Negative exponential distribution
(b) Normal distribution
(c) Binomial distribution
(d) Poisson distribution

UPRVUNL AE 21.08.2016
UKPSC AE 2012 Paper-I

Ans. : (c) P–charts are based on the Binomial distribution.

C-chart are based on the Poisson distribution.

661. In sampling, AQL stands for:-

(a) Average quality level
(b) Acceptable quality level
(c) Asymmetric quality level
(d) Available quality level

UKPSC AE-2013, Paper-I, SJVN ET 2013

Ans. (b) : In sampling, AQL stands for acceptable quality level.

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662. Fundamental purpose of quality control is to maintain the quality standard of the manufactured items/products at–

(a) almost no cost (b) optimum cost
(c) minimum cost (d) maximum cost

Nagaland PSC CTSE 2017, Paper-I
Nagaland CTSE 2016 Paper-I

Ans. (b) : Quality Control is to maintain the quality standard of the manufactured optimum cast at minimum cost.

663. Choose the false statement :

(a) Control charts indicate whether the process is in control or not
(b) p-chart is a control chart for percent defectives
(c) $\overline{X}$ and R-charts are used to evaluate dispersion of measurements
(d) C-charts are prepared for large and complex components

OPSC AEE 2019 Paper-I
UKPSC AE 2012 Paper-I

Ans. : (c) : The X- bar and R-chart are quality control charts used to monitor the mean and variation of a process based on samples taken in given time R-chart is used to monitor the variability or dispersion of the process.

664. Acceptance sampling is used in:

(a) Job Production (b) Batch Production
(c) Mass Production (d) All of these

OPSC AEE 2015 Paper-I
ISRO Scientist/Engineer 2007

Ans. : (c) Acceptance sampling is used in Mass Production.
Acceptance Sampling :-It is the process of evaluating the quality of a large number of items in a batch or a lot based upon the quality of small sample of items.

665. Flow process chart containg:

(a) Inspection and operation
(b) Inspection, operation and transportation
(c) Inspection, operation, transportation and delay
(d) Inspection, operation, transportation, delay and storage

UJVNL AE 2016
ESE 2001

Ans. : (d) Flow process chart contains Inspection, operation, transportation. delay and storage.

666. The producer's risk is the probability with which a consumer will

(a) Reject a bad lot (b) Reject a good lot
(c) Accept a good lot (d) Accept a bad lot

UPPSC AE 12.04.2016 Paper-I
UJVNL AE 2016

Ans. : (b) The producer's risk is the probability with which a consumer will reject a good lot.

667. Six Sigma means

(a) 3.1 defects per million
(b) 3.4 defects per million
(c) 4.5 defects per million
(d) 5.3 defects per million
(e) 6.5 defects per million

GPSC ARTO Pre 30.12.2018
CGPSC AE 26.04.2015 Shift-I

Ans. (b) : To be clear, 6 sigma as a discipline has a focus around variation. More specifically, ultimately reduction and control over variation in a process.
The term, 6 sigma is used as it describes a target of 3.4 defects per million opportunities which is considered to be world class.

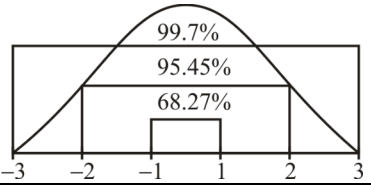
668. The value of standard deviations *v*, are used to decide the upper and lower control limits, 1σ, limits occupy.

(a) 99.73% of the area of the normal curve
(b) 95.45% of the area of the normal curve
(c) 100% of the area of the normal curve
(d) 68.27% of the area of the normal curve

JPSC AE 10.04.2021, Paper-II

Ans. (d) : 1 σ limits – 68.27% of the area of the normal curve.
2σ limits – 95.45% of the area of the normal curve.
3σ limits – 99.73% of the area of the normal curve.
Calculating standard deviation

$$S_x = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$



669. Match List-I (defect/trend) with List II (Chart) and select the correct answer using the codes given below the lists.

List I	List II
A. Trend	1. R-chart
B. Dispersion	2. C-chart
C. Number of defects	3. X-chart
D. Number of defectives	4. np-chart

(a) A-3, B-1,C-4,D-2 (b) A-1, B-2,C-3,D-4
(c) A-1, B-4,C-3,D-2 (d) A-3, B-1,C-2,D-4

RPSC ACF & FRO, 26.02.2021

Ans. (d) : X-chart → Trend
R – chart → Dispersion
Number of defect → C-chart
Number of defectives → np-chart.

670. Which of the following is not a material handling technique for productivity technique?

(a) Work measurement
(b) Inventory control
(c) Quality control
(d) Materials management.

RPSC ACF & FRO, 26.02.2021

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Ans. (c) : Quality control is not a material handling technique for productivity technique. Quality control (QC) is a process through which a business seeks to ensure that product quality is maintained or improved. Quality control involves testing units and determining if they are within the specification for final product.

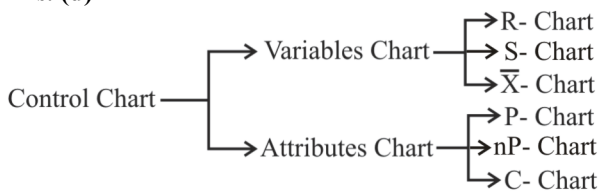
671. Which of the following is not a control chart for variables?

(a) $\bar{x}$ chart (b) R chart

(c) S - chart (d) C - chart

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d)



So, C-Chart is not a control chart.

672. In typing work, the quality parameter is the number of errors committed per page. Which of the following control chart is appropriate for this case?

(a) $\bar{x}$ chart (b) P chart

(c) S chart (d) C chart

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) :

673. Consider a process by which coils are manufactured. Samples of size 5 are randomly selected from the process, and the resistance values (in ohms) of the coils are measured. The data values are given in the form of sum of sample mean x bar as 521 for 25 samples. For sample of size 5 factors $A_2 = 0.577$. Then trial control limits for the x bar charts are

(a) UCL = 20.24 and LCL = 15.12

(b) UCL = 18.34 and LCL = 14.67

(c) UCL = 22.848 and LCL = 18.832

(d) UCL = 20.24 and LCL = 18.832

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c)

674. If all observations in the sample fall within 3 s control limits, we would expect with _____ confidence that the process is in control.

(a) 99.7% (b) 96.5%

(c) 99% (d) 95%

VIZAG STEEL MT 24.01.2021 Shift-II

Ans. (a) : If all observation in the sample fall within 3s control limit, we would expect with 99.7%.

675. In which year Heinrich presented basic concept of Industrial Safety?

(a) 1936 (b) 1933

(c) 1938 (d) 1931

VIZAG STEEL MT 24.01.2021 Shift-II

Ans. (d) : Heinrich presented basic concept of industrial safety in year 1931.

676. The product design plays the lead role in defining the physical form of the product to best meet the _____.

(a) mass production

(b) customer needs

(c) reduction of product cost

(d) sales target

VIZAG STEEL MT 24.01.2021 Shift-II

Ans. (b) : The design function plays the lead role in defining the physical form of the product to best meet customer needs. The design function also generates information relating to many other aspects of the product prior to manufacture, post manufacture and after the product life has expired.

677. A control chart indicates whether the process is in control or not. Following is a process variability chart :

(a) X – chart (b) P- chart

(c) np chart (d) c- chart

JPSC AE 10.04.2021, Paper-II

Ans. (a) : $\bar{X}$ Chart - $\bar{X}$ Charts are used to monitor the mean of a process based. On samples taken from the process at given times. It tells whether the process is in control or not.

P Chart- A P-Chart is an attributes control chart. It shows a proportion on non conforming items rather than the actual account.

np Chart- It shows how the process, measured by the number of number of non conforming items it produces over time.

C-Chart- C- Chart shows how the process, measured by number of non conformities per item or group of items, changes over time.

678. Which one of the following statements is not the correct statement regarding operating characteristics (OC) curve for sampling plan?

(a) It shows ability to distinguish between good and bad lots.

(b) No sampling plans can give complete protection against acceptance of defectives.

(c) Larger the sample size, steeper is the slope of the curve

(d) Acceptance number is zero for ideal sampling plan.

ESE 18.07.2021

Ans. (d)

- Operating curve shows ability to distinguish between good and bad lots.
- No sampling plans can give complete protection against acceptance of defectives.
- Larger the sample size, sleeper is the slope of the curve.
- Acceptance number need not be zero for ideal sampling plan.

679. A fundamental attribute of TQM is

(a) Drawing control charts

(b) Having team meetings

(c) Top management's direct involvement

(d) Meeting ISO 9000 audit

GPSC DEE, Class-2 (GWSSB) 04.07.2021

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Ans. (c) : Fundamental attribute of Total Quality Management (TQM) is direct involvement of top level management.

680. Accuracy can be improved by

(a) Use of X-bar charts
(b) Team meetings
(c) TQM principles
(d) Management talking to workers

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) Accuracy can be improved by use of X-charts.

681. The Baldrige Award is

(a) A ISO 9000 requirement
(b) An Indication of SPC being used
(c) Indication of no competition
(d) Indication that TQM programs are effective

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : The Baldrige Award is indication that TQM programs are effective.

682. Fishbone diagrams are drawn

(a) To find customer needs
(b) To find the cost of quality
(c) To brainstorm causes of an effect
(d) To screen workers suggestions

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Fishbone diagrams are drawn to brainstorm causes of an effect.

683. Which of the following graphical method tells about the root cause of the risk?

(a) Pareto chart
(b) Scatter Diagram
(c) Ishikawa/Fishbone diagram
(d) Monte Carlo analysis

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Ishikawa/Fishbone diagram

684. In quality function deployment, part characteristics and design requirements are part of _____.

(a) production planning (b) process planning
(c) product planning (d) parts deployment

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (d) : During the product development phase, the critical parts and assemblies are identified. The critical product characteristics are cascaded down and translated to critical or key part and assembly or characteristics specifications.

685. A machine has two components, both of which have times to failure that are independent and exponentially distributed with means 20 hours and 25 hours. What is the mean time to failure in hours of the machine?

(a) 5.55 (b) 11.11
(c) 45 (d) 20

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : The machine will fail if any component of machine fails

$$\frac{1}{\text{mean time of failure}} = \frac{1}{20} + \frac{1}{25}$$

mean time to failure = 11.11 hours

686. A process is to be controlled with standard values $\mu = 15$ and $\sigma = 3.6$. The sample size is 9. The control limits for the $\bar{X}$ chart are

(a) 15 ± 10.8 (b) 15 ± 3.6
(c) 0.4 ± 10.8 (d) 0.4 ± 3.6

RPSC IOF, 2020

Ans. (b) : Given,

$\mu = \bar{x} = 15$

Standard deviation, $\sigma = 3.6$

Sample size, $n = 9$

Upper control limit, $UCL = \bar{x} + \frac{3\sigma}{\sqrt{n}}$

$= 15 + \frac{3 \times 3.6}{\sqrt{9}}$

$= 15 + \frac{3 \times 3.6}{3}$

$= 15 + 3.6$

Lower control limit, $LCL = \bar{x} - \frac{3\sigma}{\sqrt{n}}$

$= 15 - \frac{3 \times 3.6}{\sqrt{9}}$

$= 15 - \frac{3 \times 3.6}{3}$

$= 15 - 3.6$

The control limit of $\bar{x}$ chart are 15 ± 3.6

687. The quality function deployment process begins with ____.

(a) customer wants and needs
(b) characteristics of target components
(c) requirements of quality control
(d) parameters of technical design

VIZAG MT, 14.12.2020

Ans. (a) : The Quality Function Deployment (QFD) process begins with customer wants and needs.

QFD is a focused methodology for carefully listening to the voice of customer and then effectively responding to these needs and expectations.

688. The greatest number of readings in a set of data is concentrated as

(a) mean (b) mode
(c) median (d) standard deviation

Assam PSC AE (PHED), 18.10.2020

Ans. (b) : Mode- The greatest number of reading in a set of data is concentrated at mode.

For example, in the following list of numbers 16 is the mode since it appears more times in the set than any other numbers.

3, 3, 6, 9, 16, 16, 16, 27, 27, 37, 48

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Mean- It is the sum of a collection of number divided by the count of numbers in the collection given data set $A= (a_1 + a_2 + ...a_n)$

$$\overline{A} = \frac{a_1 + a_2 + ...a_n}{n}$$

Median- The median is the middle number in a sorted, ascending or descending lists of number and can be more descriptive of that data set than the average.

Example- 1, 3, 3, 6, 7, 8, 9

Median = 6

1, 2, 3, 4, 5, 6, 8, 9

$$\text{Median} = \frac{4+5}{2}$$

$$= 4.5$$

Standard deviation- The standard deviation is a measure of amount of variation or dispersion of a set of values. A low standard deviation indicates that the values tend to be close to the mean of the set, while a high standard deviation indicates that the values are spread out over a wider range.

689. Consider the following data regarding the acceptance sampling process: N = 10,000, n = 89, c = 2, P = 0.01 and P_a = 0.9397 The Average Total Inspection (ATI) will be

(a) 795 (b) 687
(c) 595 (d) 487

ESE 2020

Ans. (b) : ATI $= n + (1 - P_a)(N - n)$
$= 89 + (1 - 0.9397) \times (10000 - 89)$
$= 686.633$
≈ 687

690. Choose the statement which is not true

(a) Break-even analysis helps in taking decision on make or buy
(b) Routing and scheduling are integral part of quality control
(c) Ergonomic deals with the study of man-machine environmental system
(d) Games theory is applicable to the competition between two firms

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : Objectives of quality control-

(i) To set up standards of quality acceptable to the customer and economical to achieve and maintain.
(ii) To locate and identify the process faults in order to control the defectives, scraps and waste.
(iii) To take necessary corrective measure to maintain the quality of the product.
(iv) To ensure that sub-standard do not reach the customers.

691. 10 sets of items selected from an assembly line were found to have the following number of defects, calculates the upper control limit for a 1-Sigma p-chart (to 3 decimal places)

Sample	1	2	3	4	5	6	7	8	9	10
Proportion Defective	0.06	0.06	0.08	0.05	0.04	0.08	0.06	0.04	0.05	0.04

(a) 0.201 (b) 0.129
(c) 0.133 (d) 0.121

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Upper control limit

$$UCL = \overline{p} + 2\sqrt{\frac{\overline{p}(1-\overline{p})}{\overline{n}}}$$

Where, $\overline{p}$ = Average proportion defective

$$\overline{p} = \frac{0.56}{10} = 0.056$$

Sample size of each sample is 10 avg sample size = 10

$$UCL = \overline{p} + 2\sqrt{\frac{\overline{p}(1-\overline{p})}{\overline{n}}}$$

$$= 0.056 + 2\sqrt{\frac{0.056 \times 0.944}{10}}$$

$$= 0.201$$

692. An engine is to be designed to have a minimum reliability of 0.8 and minimum availability of 0.98 over a period of 2×10^3 hr. The MTTR is nearly

(a) 168 hr (b) 174 hr
(c) 183 hr (d) 188 hr

ESE 2019

Ans. (c) : Given,
Reliability (R) = 0.8
Availability (A) = 0.98
Time period (t) = 2×10^3 hr
Reliability $(R) = e^{-\lambda t}$

$$0.8 = e^{-\lambda \times 2000}$$

Taking log on both sides

$$\ell n\,(0.8) = -2000\,\lambda \ell n(e)$$

$$\text{Mean time between failure (MTBF)} = \frac{1}{\lambda} = 8962.84 \text{ hr}$$

$$A = \frac{MTBF}{MTBF + MTTR}$$

$$0.98 = \frac{8962.84}{8962.84 + MTTR}$$

Mean time to repair (MTTR) = 182.91
$\simeq$ 183 hr

693. Kanban is a Japanese word meaning:

(a) Efficient working (b) Reducing wastage
(c) Signal (d) Prompt action

HPPSC ADF 05.03.2019

Ans. (c) : The Japanese word "Kanban" meaning visual board or a sign has been used in the sense of a process definition since 1960. The systems Toyota used to limit their work in progress have been called 'Kanban' systems. Kanban (literally signboard or billboard in Japanese) is a scheduling system for lean and just in time (JIT) production.

694. Consider the following data for quality acceptance process:

N = 10000
n = 89
c = 2
p = 0.01 (incoming lots of quality)
Pa = 0.9397

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The AOQ will be

(a) 0.93% (b) 0.84%
(c) 0.75% (d) 0.66%

ESE 2019

Ans. (a) : Given,

$N = 10,000, \quad n = 89, \quad C = 2$

$P = 0.01$

$P_a = 0.9397$

Average outgoing quality (AOQ)

$$= \left[\frac{(P \times P_a)(N-n)}{N}\right] \times 100$$

$$= \left[\frac{(0.01 \times 0.9397)(10,000-89)}{10,000}\right] \times 100$$

$$= 0.9313\%$$

695. Twenty sample of parts were taken from a production line for 100% inspection, each sample containing 200 parts. The total number of defectives were 300. What will be the fraction defective ($\bar{p}$) during the period?

(a) 0.010 (b) 0.125
(c) 0.085 (d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (d) : Sample size (n) = 200

No of sample (N) = 20

Total No. of defectives (D) = 300

$$\bar{P} = \frac{P_1 + P_2 + ...P_n}{n}$$

$$P = \frac{D}{n \times N}$$

$$P = \frac{300}{200 \times 20} = \frac{3}{40}$$

$$= 0.075$$

696. The age specific failure rate is ----------
where F(t) = Cumulative distribution function
P(t) = Survival probability or reliability
f(t) = Probability density function

(a) F(t)/P(t) (b) P(t)/F(t)
(c) F(t)/F(t) (d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (d) : The age specific failure rate (λ)

$$= \frac{\text{Probability density function (f(t))}}{\text{Reliability function (R(t))}}$$

$$\boxed{\lambda = \frac{f(t)}{R(t)}}$$

697. The internal diameter of certain metallic rings is being controlled. A sample of size 25 is taken for the process control. If the sample mean shows a standard deviation of 0.1 mm, we can infer that the standard deviation for the operation is

(a) 5.0 mm (b) 2.5 mm
(c) 0.5 mm (d) 0.25 mm

MPSC Lect. (Prod.) 2018

Ans. (c) $n = 25, \ \sigma_{\bar{x}} = 0.1$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$$\sigma = \sqrt{n} \times \sigma_{\bar{x}}$$

$$= \sqrt{25} \times 0.1$$

$$= 5 \times 0.1$$

$$\boxed{\sigma = 0.5}$$

698. In a control chart, the upper control limit is set equal to which one of the following?

(a) process mean
(b) process mean plus three standard deviations
(c) upper design tolerance limit
(d) upper value of the maximum range R.

GPSC ARTO Pre 30.12.2018

Ans. (b) : In a control chart,

Upper control limit process mean plus three standard deviation.

Lower control limit process mean minus three standard deviations.

$$UCL = \bar{X} + 3\sigma$$

$$LCL = \bar{X} - 3\sigma$$

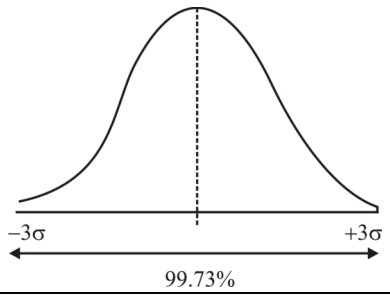
Where σ is standard deviation.

699. The percent of the sample will have values that are within ±3. Standard deviations of the distribution mean is :

(a) 95.5 (b) 96.7
(c) 97.6 (d) 99.7

GPSC ARTO Pre 30.12.2018

Ans. (d) :



700. Which of the following is FALSE about Normal distribution?

(a) Normal Distribution is applied for discrete Random Distribution
(b) The shape of the Normal Curve is Bell Shaped
(c) The area under a standard normal curve is 1
(d) The standard normal curve is symmetric about the value 0

ISRO Scientist/Engineer 17.12.2017

Ans. (a) : Characteristics of normal distribution.

(1) It is 'Bell shaped' and has a single peak at the centre of the distribution.
(2) The arithmetic mean, median and mode are equal.
(3) Normal distribution is applied for continuous random variable.
(4) It is symmetrical about the value 0 i.e. mean.

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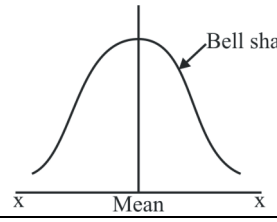
(5) The area under a standard normal curve is 1.



701. In a line organisation
(a) responsibility of each individual is fixed
(b) discipline is strong
(c) quick decisions are taken
(d) all of these

Sikkim PSC (Under Secretary), 2017

Ans. (d) : In a line organisation business or industry structure with self contained departments. Authority upwards from the bottom along the chain of command and each department manager has control over his or her departments affair and employees.

702. The individual human variability in time studies to determine the production standards is taken care of by
(a) personal allowances
(b) work allowances
(c) errors allowances
(d) machine allowances

TNPSC AE 2017, ATE 1996

Ans. (a) : The individual human variability in time studies to determine the production standards is taken care of by personal allowances.

703. The interchangeability can be achieved by:
(a) standardisation
(b) better process planning
(c) bonus plan
(d) better product planning

HPPSC Asstt. Prof. 18.09.2017

Ans. (a) : The interchangeability can be achieved by standardisation.

704. The process capability index C_p is given by
(a) (USL – LSL)/6σ (b) (USL+ LSL)/6σ
(c) 6σ /(USL – LSL) (d) 6σ /(USL + LSL)

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (a) : Process capability- The ability of the process to meet the design specification for a service or product.

$$C_P = \left(\frac{USL - LSL}{6\sigma}\right)$$

705. OC curves are used for the selection of lots by :
(a) Attributes
(b) Variables
(c) Both Attributes and Variables
(d) Non of these

RPSC ADE 2016

Ans. (a) : OC curves are used for selection of lots by a attributes.
The operating characteristics (OC) curve depicts the discriminatory power of an acceptance sampling plan.

706. Which of the following differs in the quality of the management function?
(a) Organizing (b) Leading
(c) Producing (d) Planning

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : Functions of management is systematic way of doing things. Management is a process to emphasize that all manager, irrespective of their aptitude or skills, engage in some inter-related functions to achieve their desired goals. Four functions of management are planning, organizing, leading and controlling that managers perform to accomplish business goals efficiently.
Producing differs in the quality of the management function.

707. Which quality award is international quality award ?
(a) Rajiv Gandhi National Quality Award
(b) Deming Prize
(c) Malcolm Baldrige National Quality Award
(d) Pakistan National Quality Award

HPPSC Asstt. Prof. 18.11.2016

Ans. (b) : Deming prize is quality award is international quality award.

708. Which one of the following is not a type of quality cost ?
(a) Appraisal cost (b) Prevention cost
(c) Internal failure cost (d) Production cost

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : The cost of quality can be divided into four categories. They include prevention appraisal internal failure and external failure. So production cost is not a type of quality cost.

709. Quality means :
(a) more investment in technology
(b) increasing number of inspectors
(c) increasing production
(d) doing first time right

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : Quality is the totality of features and characteristics of product or service that hear on its ability to satisfy green needs it means doing first time right.

710. If 1 million passengers pass through the IGI airport with checked baggage each year a successful Six Sigma program for baggage handling would result in how many passengers with misplaced luggage?
(a) 3.4
(b) 6.0
(c) 34
(d) 2700
(e) 6 times the monthly standard deviation of passengers

CGPSC AE 16.10.2016

Ans. (a) : As we know, for six sigma
Z = 2, for 95.45% (4σ)
Z = 3, for 98.74% (6σ) ⇒ 3.4 defects per million observations

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711. Ettlie Engineering has a new catalyst injection system for your countertop production line. Your process engineering department has conducted experiments and determined that the mean is 8.01 grams with a standard deviation of 0.03. Specifications are $\mu = 8.0$ and $\sigma = 0.04$. What is the C_{pk} performance of the injection system?

(a) 1.44 (b) 1.22
(c) 1.33 (d) 1.11
(e) None of the above

CGPSC AE 16.10.2016

Ans. (b) : As we know,

$$C_{pk} = \text{Minimum of}\left[\frac{\text{Upper specification} - \bar{x}}{36}, \frac{\bar{x} - \text{Lower}}{36}\right]$$

$\bar{x}$ = Process mean, σ = Standard deviation

$$C_{pk} = \text{Minimum of}\left[\frac{8.12 - 8.01}{3 \times 0.03}, \frac{8.01 - 7.88}{3 \times 0.06}\right]$$

$$C_{pk} = \left[\frac{0.11}{0.99} = 1.22, \frac{.13}{.09} = 1.44\right]$$

Note-This question has been deleted by CGPSC.

712. A control chart is established with limits of ± 2 standard errors for use in monitoring samples of size n= 20. Assume the process to be in control. What is the likelihhood of a sample mean falling outside the control limits?

(a) 2.3 % (b) 97.7%
(c) 4.5% (d) 95.5%

UPRVUNL AE 21.08.2016

Ans. : (c) P(2) – P(–2) = 0.9545
Sample falling outside control limit
= 1 – 0.9545
= 0.0455
= 4.55%

713. Quality improvement planning is related to which category of cost?

(a) Cost of failure (b) Cost of appraisal
(c) Cost of prevention (d) Cost of loss

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Quality improvement planning is related to cost of prevention. It is the category of cost.

714. The aim of statistical quality control in industrial applications is to have

(a) Quality improvement
(b) Recording of data related to quality
(c) Cost reduction
(d) All of the above

UJVNL AE 2016

Ans. : (d) The aim of statistical quality control in industrial application is to have
(i) Quality Improvement
(ii) Recording of data related to quality
(iii) Cost reduction.

715. Consistency of a process is checked with the help of preferably by

(a) Sigma chart (b) X bar chart
(c) Cause effect chart (d) None of the above

UJVNL AE 2016

Ans. : (a) Consistency of a process is checked with the help of preferably by sigma chart.

716. Which type of control chart should be used to directly monitor the number of defectives in a process for making iron castings?

(a) $\bar{X}$ – chart (b) $\bar{P}$ – chart
(c) $\bar{C}$ – chart (d) $\bar{R}$ – chart

UPPSC AE 12.04.2016 Paper-I

Ans. : (b) $\bar{P}$ – chart

717. An $\bar{X}$ chart uses the following data

(a) Count Data
(b) Attribute measurement data
(c) Variable measurement data
(d) None of the above

UPPSC AE 12.04.2016 Paper-I

Ans. : (c) An $\bar{X}$ chart uses the variable Measurement data.
$\bar{X}$ chart:- It shows the centering of the process and shows the variation in the averages of individual samples.

718. Which of these would not be a reason for using acceptance sampling?

(a) A very high inspection cost
(b) Boredom and fatigue
(c) A process needing statistical control
(d) Destructive testing

UPPSC AE 12.04.2016 Paper-I

Ans. : (c) A process needing statistical control not be a reason for using acceptance sampling
(i) Acceptance sampling is not process control
(ii) Statistical process control (SPC) is the preferred method to prevent nonconformance.
(iv) Think of SPC as the control method, and acceptance sampling as insurance
(v) You practice good driving techniques, but you don't cancel your insurance policy

719. The error estimate (e) in work sampling varies with sample size (n) as

(a) $e \alpha \frac{1}{n}$ (b) $e \alpha \frac{1}{\sqrt{n}}$
(c) $e \alpha \sqrt{n}$ (d) $e \alpha \frac{1}{n^2}$

UPPSC AE 12.04.2016 Paper-I

Ans. : (b) Error estimate (e) :-

$$e \propto \frac{1}{\sqrt{n}}$$

e = Error estimate, n = Sample size.

720. Tolerances are provided because :

(a) Machines are not accurate
(b) Machine environment follows a normal distribution
(c) Quality of fit will come only when there is range
(d) Ease of manufacturing

TSPSC Managers, 2015

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Ans. (b) : Tolerances are provided to the parts because variations in the material properties introduce errors and production machine themselves have inherent in accuracies. Tolerances are provided because machine environment follows a normal distribution.

721. Who among the following is associated with contributions to quality control in operations management?

(a) Charles Babbage (b) Henry Ford
(c) Frank Gilbreth (d) W. Edwards Deming
(e) Henri Fayol

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : W. Edwards Deming is associated with contributions to quality control in operations management.

722. Consider the following statements

1. The operating characteristic curve of an acceptance sampling plan shows the ability of the plan to distinguish between good and bad lots.
2. No sampling plan can give complete protection against the acceptance of defective products.
3. Double sampling gives protection against the acceptance of defective products
4. Double sampling results in more inspection than single sampling if the incoming quality is very bad
5. C-chart has straight line limits and U chart has zig-zag limits. Which is wrong statement:

(a) 1 and 2 (b) 1, 2 and 3
(c) 3 and 4 (d) 4
(e) 5

CGPSC AE 26.04.2015 Shift-I

Ans. (d) :
(i) The operating characteristic (OC) curve of an acceptance sampling plan reflects the ability of the plan to distinguish between good and bad lots. The OC curve for a sampling plan is a graph of the probability of accepting the lot versus the proportion defective in the lot.
(ii) No sampling plan can give complete protection against the acceptance of defective product.
(iii) In double sampling plan, generally, the decision of acceptance or rejection of a lot is taken on the basis of two sample.
(iii) If the first sample is bad, the lot may be rejected on the first sample and a second sample need not be drawn. If the first sample is good, the lot may be accepted on the first sample and a second sample is not needed.

723. The father of scientific management was

(a) F.W. Taylor (b) Henry Fayol
(c) Gilberth (d) Gantt

TSPSC AEE 2015

Ans. : (a) The father of Scientific management was F.W. Taylor.

724. Assignable variables in statistical quality control results in:

(a) Proper sequencing of operations
(b) Improper operation of the machine
(c) Proper instruments used pro inspection
(d) None of these

OPSC AEE 2015 Paper-I

Ans. : (b) Assignable variables in statistical quality control results in improper operation of the machine.

725. Taguchi loss function is

(a) is a good method of production planning
(b) is commonly used in inventory control
(c) is a good method of production scheduling
(d) a concept used in tolerance design

TSGENCO AE 14.11.2015

Ans. (d) : The Taguchi loss function proposed by the Japanese quality expert Genichi Taguchi states that any deviation from the desired tangent value or specification results in monetary loss to society. In this context, the Taguchi method is based on the hypothesis that the smaller the variation with respect to said target value, the better the product quality.
Note that the desired target value or specification is equipment from the lower specification limit and the upper specification limit.

726. A bad observation which must be ignored can be identified by :

(a) observing the data
(b) using observation to calculate and see if result deviates too much
(c) finding arithmetic mean and seeing which observation deviates most
(d) plotting the result and seeing which observation deviates most from the line

(HPPSC AE 2014)

Ans. : (d) A bad observation which must be ignored can be identified by plotting the result and seeing which observation deviates most from line.

727. Which of the following is not the definition of Quality?

(a) conformance to specification
(b) fitness for use
(c) spare part maintenance
(d) customer delight

Haryana PSC Civil Services Pre, 2014

Ans. (c) : Quality – Quality can be defined as–
1. Conformance to specification
2. Fitness for use
3. Customer satisfaction.

Spare part maintenance do not comes in quality definition.

728. Section 21, 22 and 26 of the factories act and the allied provisions in the factories rules deal with the-

(a) Chemical safety
(b) Precaution against fire
(c) Guarding of machines
(d) Safety in materials handling and storage

MPSC Dy. Engineer 2014

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Ans. (c) : Sections 21, 22 and 26 of the factories act the allied provisions in the factories rules deals with the guarding of machines.

729. Producer's risk in acceptance sampling is

(a) chance of producing defective components
(b) chance of accepting the defective components by the customer
(c) chance of incurring high cost
(d) chance of rejecting the parts by the customer of the specified quality

TNPSC AE 2014

Ans. (d) : Producer's risk in acceptance sampling is chance of rejecting the parts by the customer of the specified quality.

730. Find the traditional non budgetory control device

(a) Gantt charts
(b) Operational audit
(c) Milestone budgeting
(d) Time-event network analysis

TNPSC AE 2014

Ans. (b) : An operational audit is an examination of the manner in which an organization conducts business, with the objective of pointing out improvement that will increase its efficiency and effectiveness.

731. Decision making is defined as :

(a) Selecting missions and objectives and the actions to achieve them
(b) Selection of a course of action from among several alternatives
(c) Identify the basis functions of an organization
(d) Sequence of action, rather than to thinking

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Decision making is defined as selection of a course of action from among several alternatives.

732. Intrapreneur Means :

(a) Employer, who can take risk
(b) Employee, who can't take risk, but do all other activities of employer
(c) Business owner
(d) Employer's relative

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Intrapreneur Means employee, who can't take risk, but do all other activities of employer.

733. What determines the span of management and hence the level of organization ?

(a) Number of managers in the organization
(b) Number of persons a manager can supervise effectively
(c) Qualification of manager
(d) It is determined by employer

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Number of persons a manager can supervise effectively

734. What is the first step in basis control process ?

(a) Correcting variations
(b) Establishing standards
(c) Performance measures
(d) Decision making

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Establishing standards

735. Limitations of 'MBO' [Management by objectives"] is :

(a) MBO leads to better understandins between employer and imployer
(b) If allows greater consistency in decision making
(c) MBO keeps company objectives constantly in view
(d) Management working by objectives may follow too rigid a pattern in thinking and action

TNPSC AE (Industries) 09.06.2013

Ans. (d) : Limitations of 'MBO' [Management by objectives"] is management working by objectives may follow too rigid a pattern in thinking and action.

736. Which one of the following is not a chance cause of variation ?

(a) Variation in raw material
(b) Faulty setup
(c) Small vibration of a machine
(d) Lack of perfection in reading instruments

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Faulty setup

737. Which one of the following is not a Data presentation method ?

(a) Bar diagram (b) Histogram
(c) Pie chart (d) Sampling scheme

TNPSC AE (Industries) 09.06.2013

Ans. (d) : Sampling scheme

738. How the lack of control is indicated on control charts ?

(a) No points are outside control limits
(b) The number of points above and below the centre line is about the same
(c) A run of more than 8 points on the same side of centre line
(d) Most point, but not all are on the center line

TNPSC AE (Industries) 09.06.2013

Ans. (c) :

739. TQM is related to:-

(a) Quality control (b) Control chart
(c) Sampling (d) Work study

UKPSC AE-2013, Paper-I

Ans. (a) : Total quality management (TQM) is related to quality control.

740. Which one of the following shows the percentage of the area in normal distribution curve for ± 2 σ limits?

(a) 99.73% (b) 95.45%
(c) 68.26% (d) None of the above

UKPSC AE-2013, Paper-I

Ans. (b) : 95.45%

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741. Manufacturer's risk is the probability of:-

(a) Rejecting a good lot which otherwise would have been accepted
(b) Defective batch being accepted which otherwise would have been rejected
(c) Bad components in a lot
(d) None of the above

UKPSC AE-2013, Paper-I

Ans. (a) : Manufacturer's risk is the probability of rejecting a good lot which otherwise would have been accepted.

742. In operating characteristics curve, abscissa (x-axis) represents:-

(a) Number of defectives
(b) Percentage defectives
(c) Sample number
(d) Probability of acceptance

UKPSC AE-2013, Paper-I

Ans. (b) : Operation characteristics (OC) curve- The operating characteristic (OC) curve depicts the discriminatory power of an acceptance sampling plan. The OC curve plots the probabilities of accepting a lot [on - y - axis] versus the fraction defective [on - x - axis]

743. Percent idle time for men and machine is found by

(a) Time study (b) ABC analysis
(c) Analytical estimation (d) Activity sampling

TNPSC AE 2013

Ans. (d) : Percent idle time for men and machine is found by activity sampling.

744. Which one of the following techniques is used for determining allowances in time study?

(a) Acceptance sampling
(b) Linear regression
(c) Performance rating
(d) Work sampling

JPSC AE 2013, Paper-V

Ans. : (d) : Work sampling

745. Which one of the following steps would lead to interchangeability?

(a) Quality control (b) Process planning
(c) Operator training (d) Product design

TNPSC ACF 2012

Ans. (a) : Quality control

746. Control chart for variables provides the

I. basic variability of the quality characteristic
II. consistency of performance
III. number of products falling outside the tolerance limits
Which of these statements are correct?

(a) I, II and III (b) I and II
(c) II and III (d) I and III

TNPSC ACF 2012

Ans. (b) : Control chart are used where we can compute only no of defects and its variable provides basic variability of quality characteristic and consistory of performance.

747. Certain pilot study showed that % of occurrence of an activity as 50% with 95% confidence level and an accuracy of ±2%, the no. of observations are:

(a) 2500 (b) 2300
(c) 2200 (d) 2000

APGENCO AE 2012

Ans. (a) : No. of observation

$$n = \frac{Z^2}{L^2}\frac{(1-P)}{P}$$

For 95% confidence level

$Z = 2$

Level of accuracy

$L = 0.04$

Occurrence of an event

$P = 0.5$

$$n = \frac{(2)^2}{(0.04)^2} \times \left(\frac{1-0.5}{0.5}\right)$$

$n = 2500$

748. Choose the Correct relationship between the given statements of Assertion (A) and Reason (R).

Assertion (A) : In case of control charts for variables, if some points fall outside the control limits, it is concluded that process is not under control.

Reason (R) : It was experimentally proved by Shewart that averages of four or more consecutive readings from a universe (population) or from a process, when plotted, will form a normal distribution curve.

Code :

(a) Both (A) and (R) are correct. (R) is the correct explanation of (A).
(b) Both (A) and (R) are correct. (R) is not the correct explanation of (A).
(c) (A) is correct, but (R) is in correct.
(d) (A) is incorrect, but (R) is correct.

UKPSC AE 2012 Paper-I

Ans. (b) : Both (A) and (R) are correct. (R) is not the correct explanation of (A).

749. Which of the following are the quality control limits for p-charts ?

(a) $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})}$ (b) $\bar{p} \pm \sqrt{\bar{p}(1-\bar{p})}$

(c) $\bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$ (d) $\bar{p} \pm 3\sqrt{n\bar{p}(1-\bar{p})}$

UKPSC AE 2012 Paper-I

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Ans. (c) : $\bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$

Chart Type	Upper Control Limit	Lower Control Limit
C	$\bar{C}+3\sqrt{\bar{C}}$	$\bar{C}-3\sqrt{\bar{C}}$
P	$\bar{P}+3\sqrt{\frac{\bar{P}(1-\bar{P})}{\bar{n}}}$	$\bar{P}-3\sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$
R	$D_4\bar{R}$	$D_3\bar{R}$

750. The upper and lower control limits in case of R-chart are given by

(a) $A_2\bar{R}$ & $A_3\bar{R}$ (b) $D_3\bar{R}$ & $D_4\bar{R}$

(c) $\bar{R} \pm D_3\bar{R}$ (d) $\bar{R} \pm A_2\bar{R}$

UKPSC AE 2012 Paper-I

Ans. (b) : $D_3\bar{R}$ & $D_4\bar{R}$

751. Which one of the following would normally be considered as one of the 'costs of quality' ?

(a) Appraisal Costs (b) Marketing Costs

(c) Overhead Costs (d) Transaction Costs

OPSC Civil Services Pre. 2011

Ans. (a) : Appraisal costs are a specific category of quality control costs. Companies pay appraisal costs as part of the quality control process to ensure that their products and services meet customer expectations and regulatory requirements. These costs could include expenses for field tests and inspections.

752. Concepts in "Total Quality Management (TQM)" are attributed to

(a) George Dantzig (b) Taiichi Ohno

(c) W. Edwards Deming (d) Henry Ford

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c): W. Edwards Deming is known as the father of the 'TQM', he attributed the concepts of TQM.

753. Whose book, entitled Quality is Free, emphasized the cost of quality?

(a) Walter Shewhart (b) Phillip Crosby

(c) W. Edwards Deming (d) Joseph Juran

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Philip Crosby book is entitled 'Quality is free', which emphasized the cost of quality. Published in 1979.

754. Which of the following is not true concerning the ISO 9000?

(a) The ISO 9000 committee establishes generic quality standards for manufacturing firms worldwide

(b) The ISO 9000 committee is a government organization that is a subset of the United Nations

(c) The ISO 9000 is a guide for using the other four standards in the series 9001 through 9004

(d) The ISO 9000 does not tell management how to meet requirements but does indicate what is required

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Series 9001 to 9004 is not a government organization. ISO-9000 is defined as a set of international standard on quality management and quality assurance developed to help companies, establishes generic quality standards for manufacturing firms worldwide & having four standards.

755. A p-chart has been prepared. Computations show that the average proportion defective is .032, while the standard deviation is .0176. From this data, what are the 3-sigma control limit for this chart?

(a) LCL = .032 UCL = .085

(b) LCL = .053 UCL = .032

(c) LCL = 0 UCL = .085

(d) not enough information to determine the control limits

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : For 3- sigma,

Given UCL = 3 × (Standard deviation) + Proportion defective

Standard deviation = 0.0176 & Proportion defective = 0.032.

$$= 3 \times (0.0176) + 0.032$$

UCL ≈ 0.085

Similarly

LCL = UCL – 3 × standard deviation – proportion defective

$$= 0.085 - 3 \times (0.0176) - 0.032$$

LCL = 0

756. The term which reflects an acceptable proportion of defects in a lot to the consumer is?

(a) AOQ (b) LTPD

(c) OC (d) AQL

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : AQL (Acceptance Quality Limit), represents the maximum number of defective units, beyond which a batch is rejected. Importers usually set different AQL for critical, major and minor defects.

757. Which one of the following is not true in case of Scientific Management?

(a) It develops harmony between workers and management

(b) It develops co-operation rather than individualism

(c) It manages on the basis of guess work and not on assessment of facts.

(d) It maximizes output with minimum input

RPSC AE GWD, 2011

Ans. (c) : On the basis of guess work and not assessment of facts is not true in case of scientific management.

Scientific management-

(1) It develops harmony between workers and management.

(2) It develops cooperation rather than individualism.

(3) It maximizes output with minimum input.

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758. In scientific recruitment process, Application Blank refers to :

(a) Receiving application from prospective candidate for a given post.

(b) Conducting a test to know level of ability, knowledge, interest and aptitude of candidate.

(c) After preliminary successful interview, taking information in applicant's hand writing about more factual information.

(d) After final selection, taking willingness of joining in applicant's own handwriting and informing job's requirements and responsibility, and salary structure.

RPSC AE GWD, 2011

Ans. (c) : In Scientific recruitment process, application blank refers to after preliminary successful interview, taking information in applicant's hand writing about more factual information.

759. Over Capitalization of firm refers to :

(a) When earnings of the concern on total capital are exceptionally high in relation to returns enjoyed by similar situated concerns in the same field.

(b) When earnings are not adequate enough to yield return on the amount of stock and shares and bonds issued.

(c) When loans and borrowings are more as compared to required investment.

(d) None of the above

RPSC AE GWD, 2011

Ans. (c) : Over capitalization- It is the capitalization under which the actual profits of the company are not sufficient to pay interest on debentures borrowings and a fair rate of dividend to share holders over a period of time.

In other words a company is said to be over capitalized when it is not able to pay interest on debentures, loans and ensure a fair return to share holders.

760. Which one of the following is not the characteristic feature of a good incentive plan?

(a) The incentive plan should be simple and easily understandable

(b) The incentive plan should be profitable to the management only

(c) It should reward workers in proportion to their output and thus high up their morale.

(d) The employees should be assured of a certain minimum wage irrespective of output.

RPSC AE GWD, 2011

Ans. (b) : Characteristic features of a good incentive plan-

(1) The incentive plan should be simple and easily understandable

(2) It should reward workers in proportion to their output and thus high up their morale.

(3) The employees should be assured of a certain minimum wages irrespective of output.

761. In Paired Comparison Method or merit rating (performance appraisal) number of comparisons to be done is given by (If N – Number of employees):

(a) N (N – 1) / 2 (b) N – 1

(c) N (d) (2N – 1)/N

RPSC AE GWD, 2011

Ans. (a) : The number of comparison to be worked out with the formula

$$\text{Number of comparisons} = \frac{N(N-1)}{2}$$

where N = total number of employees to be evaluated.

762. Which one of the following is not a characteristic feature of group behaviour?

(a) Group behaviour is influenced by the type of group i.e. formal or informal

(b) Size of the group i.e. large or small also influence group behaviour

(c) Group influences and changes the attitude and behaviour of an individual towards work and organization.

(d) Group behaviour is always detrimental for the organization and management.

RPSC AE GWD, 2011

Ans. (d) : Group behaviour is always detrimental for the organization and management is not a characteristic feature of group behaviour.

763. Match List-I with List-II and select the correct answer using the code given below the lists :

	List I		List II
A.	X -chart	1.	Precision
B.	R-chart	2.	Number of defects
C.	p-chart	3.	Fractional defect
D.	c-chart	4.	Accuracy

Code:

	A	B	C	D		A	B	C	D
(a)	2	3	1	4	(b)	4	3	1	2
(c)	2	1	3	4	(d)	4	1	3	2

UPSC JWM Advt. No.-50/2010

Ans. (d) : As we know, different chart having different outcome,

Such as,

- X – chart gives us the accuracy of the outcomes.
- R-chart is gives the precision
- P-chart is given the fractional defect in the sample
- C-chart give the No. of defects in a sample

764. Assertion (A) : Quality assurance is superior to quality control as a technique.

Reason (R) : Quality assurance is a down-stream activity and quality control is an upstream activity.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true and R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

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Ans. (b) : Quality assurance can be defined as the part of quality management focused on providing confidence that quality requirements will be fulfilled while quality control is the part of quality management focused on fulfilling quality requirements.

765. The following chart shows the consistency of the process:

(a) $\bar{X}$ (b) R
(c) C (d) P

UKPSC AE 2007 Paper -I

Ans. (b) : R charts are used to monitor the variation of a process based on samples taken from the process at given times (hours, shifts, days, weeks, months, etc.). The measurements of the samples at a given time constitute a subgroup.

766. The upper control limit of the 'C' chart is

(a) $\bar{C}+3\sqrt{\bar{C}}$ (b) $\bar{C}+2\sqrt{\bar{C}}$
(c) $\bar{C}+\sqrt{2\bar{C}}$ (d) $\bar{C}+\sqrt{3\bar{C}}$

UKPSC AE 2007 Paper -I

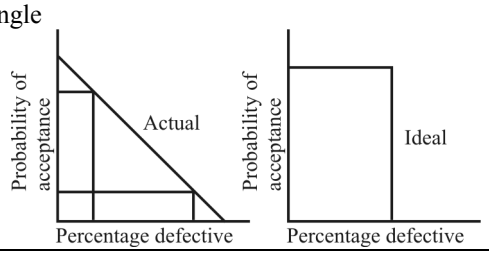
Ans. (a) : $\bar{C}+3\sqrt{\bar{C}}$

767. Ideal 'Operating Characteristic Curve' looks like

(a) Triangle (b) Circle
(c) Rectangle (d) Square

OPSC Civil Services Pre 2006

Ans. (c) : Ideal operating characteristic curve look like rectangle



768. Taguchi's quality philosophy

(a) is connected with reduction in performance variation
(b) is connected with increase in performance variation
(c) independent of performance variation
(d) Helps in cost control of product

OPSC Civil Services Pre 2006

Ans. (b) : The Taguchi method of quality control is an approach to engineering that emphasizes the role of research and development product design and development in reducing the occurrence of defects and failures in manufacturing goods.

769. In the Deming's quality cycle- 'P' stands for

(a) product (b) productivity
(c) process (d) plan

OPSC Civil Services Pre 2006

Ans. (d) : Deming cycle also known as PDCA or PDSA cycle is a continuous quality improvement model consisting out of a logical sequence of four repetitive steps for continuous improvement and learning.
PDCA–Plan, Do, Check, Act.

770. Quality circle leader should be trained in

(a) democratic skills (b) authoritarian skills
(c) interpersonal skills (d) none of the above

OPSC Civil Services Pre 2006

Ans. (c) : The quality circle leader should be trained interpersonal skills.

9. Work Study, Work Measurement and Value Engineering

771. Process charts are generally represented by symbols following symbol :

(a) (b)
(c) (d)

JPSC AE 10.04.2021, Paper-II
APPSC Poly Lect. 13.03.2020
ISRO Scientist/Engineer 17.12.2017
GMB AAE 25.06.2017
GPSC ARTO 01.05.2016
Haryana PSC Civil Services Pre, 2014
GATE 2005

Ans. (b) :

(a) Inspection
(b) Operation
(c) Storage
(d) Delay

772. In value engineering the term 'value' refers to :

(a) Market values
(b) Relation between cost and efficiency
(c) Relation between function and cost
(d) Relation between productibility and cost

VIZAG STEEL MT 24.01.2021 Shift-II
APPSC Poly Lect. 13.03.2020
Nagaland PSC CTSE 2017, Paper-I
Nagaland CTSE 2016 Paper-I
(KPSC AE 2015), PSPCL AE, 2012
APPSC IOF, 2009, ESE 1992

Ans. (c) : Relation between function and cost – Value engineering is a systematic organized approach to providing necessary functions in a project at the lowest cost. Value engineering promotes the substation of material and method with less expensive alternative without sacrificing functionality.

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773. In time study, the rating factor is applied to determine :

(a) standard time of a job
(b) merit rating of the worker
(c) fixation of incentive rate
(d) normal time of a worker

Nagaland PSC (CTSE) 2018, Paper-I
RPSC Vice Principal ITI 2018
APPSC AEE MAINS 2016, PAPER-III
APPSC AEE Screening Test 2016
J&K PSC Civil Services Pre, 2013
APPSC AEE 2012
ESE 1995

Ans. (d) : Rating factor is used to determine normal time of worker

Normal time = $R_f \times$ observation time

Where, R_f is rating factor.

774. The term value in value engineering refers to:

(a) Total cost of the product
(b) Manufacturing cost of the product
(c) Selling price of the product
(d) Utility of the product

UPPSC AE 13.12.2020, Paper-I
OPSC AEE 2019 Paper-I
GPSC Asstt. Prof. 28.08.2016
MPSC Poly. Lect. 2014
J&K PSC Civil Services Pre, 2013
UKPSC AE-2013, Paper-I

Ans. (d) : Value engineering is a systematic, organized approached for providing necessary functions in a project at the lowest cost. Value is defined as the ratio of function to cost. Value refers to utility of the product.

775. The symbol ⇒ in the method study chart refers to :

(a) Transport (b) Storage
(c) Operation (d) Inspection

GPSC DEE, Class-2 (GWSSB) 04.07.2021
APPSC Poly Lect. 13.03.2020
SJVN ET 2019
OPSC AEE 2019 Paper-I
TRB Polytechnic Lecturer 2017
GATE 2005

Ans. (a) : Method study symbol

Transport– ⇨
Storage– ▽
Operation– ○
Inspection– □
Delay– D

776. In process charts, the symbol □ means

(a) Inspection (b) Transportation
(c) Operation (d) Temporary storage

RPSC AE (GWD) 06.05.2016
Mizoram PSC AE/SDO 2014 Paper-III
ISRO Scientist/Engineer 17.12.2017
GMB AAE 25.06.2017
Haryana PSC Civil Services Pre-2014

Ans. (a) :

□ -Inspection
⇦ -Transportation
○ -Operation
D -Temporary storage or delay

777. SIMO charts are used in

(a) method study (b) micro-motion study
(c) process analysis (d) layout analysis

Kerala PSC Poly. Lect., 2017
BPSC Poly. Lect. 2016, GWSSB DEE 07.07.2016
VIZAG STEEL MT 2015, CSE Pre-1996

Ans. : (b) SIMO Charts are used in method study.

SIMO Chart (Simultaneous Motion Chart)

It is used to record therbligs (18 nos.) against the time scale which are performed by the operator or operators at work. SIMO chart is the micro motion form of operator process chart. Because SIMO charts are used primarily for operation of short duration often performed with extreme quickness, it becomes necessary to combine then with films. In this chart movements are recorded against time measured in wink and 1 wink = $\frac{1}{2000}$ the minute.

Note-As per BPSC final answer is (a&b) is correct.

778. Standard time is equal to

(a) normal time plus allowances
(b) normal time minus allowances
(c) normal time plus idle time
(d) normal time minus idle time

BPSC Asstt. Prof. 29.11.2015
TSPSC AEE 2015, TNPSC AE, 2008
ESE 2001

Ans. (a) : Standard time = Normal time + Allowance

779. In micro motion study, 'therblig' is described by

(a) A symbol
(b) An activity
(c) An event
(d) A standard symbol and colour

Karnataka PSC AE, 10.09.2017
Rajasthan Nagar Nigam AE 2016, Shift-I
J&K PSC Civil Services Pre, 2010
OPSC Civil Services Pre 2006

Ans. (d) : Therblig are used in micro motion study having standard symbol and colour.

780. Speed and direction of workers movements can be better recorded by which one of the following

(a) Flow chart (b) Chronocycle graph
(c) Chronometer (d) String Diagram

RPSC ACF & FRO, 26.02.2021
VIZAG MT, 14.12.2020
APPSC AEE MAINS 2016, PAPER-III
Haryana PSC Civil Services Pre, 2014

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Ans. (b) : Chronocycle Graph– A chart for examining fine details of movement over time, such as the hand movements of an shows the movement of hands with notations to indicate how long each movement and pause takes.

781. In work study, 'operation' is represented by

(a)

(b)

(c)

(d)

MPSC Lect. (Prod.) 2018
RPSC LECTURER 16.01.2016
TSPSC AEE 2015, SJVN ET 2013

Ans. (c) :

→ Operation

→ Transport

→ Inspect

→ Delay

→ Storage

→ Handle

→ Decide

782. Time study is carried out to determine the time required to complete job by :

(a) A slow worker (b) A fast worker
(c) An average worker (d) An apprentice

Vadodara Municipal Corp. DEE, 2018
TSPSC AEE 2015
OPSC Civil Services Pre. 2011
RPSC AE GWD, 2011

Ans. (c) : Time study is the appraisal, in terms of time of the value of work involving human effort.

783. Which one of the following charts gives simultaneously, information about the progress of work and machine loading?

(a) Man-machine chart
(b) Process chart
(c) Machine load chart
(d) Gantt chart

OPSC AEE 2019 Paper-I
Nagaland PSC (CTSE) 2018, Paper-I
JPSC AE 2013, Paper-V
CSE Pre-1995

Ans. (a) : Man-Machine chart gives simultaneously information about the progress of work and machine and loading.

784. The aim of value engineering is to:

(a) Find the depreciation value of a machine
(b) Determine the selling price of a product
(c) Minimize the cost without change in quality of the product
(d) All of these

RPSC VPITI 14.02.2016, KPSC ADF 2015
OPSC AEE 2015 PAPER-I
J&K PSC Civil Services Pre, 2013

Ans. : (c) The aim of value engineering is to minimize the cost without change in quality of the product.

Value engineering : The value engineering deals with the concepts that are applied at the design or pre–manufacturing stage so that the unnecessary cost is cut down without impairing the function or utility of the product.

Value of product differs both from price and cost as it is cost proportional to function or utility. That is

$$\text{Value} = \frac{\text{Utility(function)}}{\text{cost}}$$

785. The standard time of an operation while conducting a time study is

(a) Mean observed time + Allowances
(b) Normal time + Allowances
(c) Mean observed time × Rating factor + Allowances
(d) Normal time × Rating factor + allowances

GPSC Executive Engineer 23.12.2018
GPSC ARTO 01.05.2016
TNPSC AE (Industries) 09.06.2013
GATE 2002

Ans. (c) : Standard time = Normal time (1 + Allowance)

Normal time = Observed time × Rating factor

Standard time = Observed time × Rating factor (1 + Allowance)

786. Basic tool in work study is :

(a) graph paper (b) process chart
(c) planning chart (d) stop watch

VIZAG STEEL MT 2015
GWSSB AAE, 27.12.2015
RPSC ACF-2011

Ans. (d) : Work study is mainly aimed at the determining the most efficient method of performing a job. Basic tool in work study is stop watch.

787. Job evaluation is the method of determining the :

(a) relative values of a job
(b) worker's performance on a job
(c) worth of the machine
(d) value of overall production

Kerala PSC Asstt. Prof. 27.09.2016
OPSC AEE 2015 PAPER-I
J&K PSC Civil Services Pre, 2013

Ans. (a) : A Job evaluation is a systematic way of determining the relative value/worth of a job in relation to other Jobs in an organisation.

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788. Time study of an operator with a performance rating of 120% yields a time of 2 minutes. If the allowances of 10% of total available time are to be given, then what is the standard time of the operation.

(a) 2.00 minutes (b) 2.40 minutes
(c) 2.64 minutes (d) 2.67 minutes

CGPSC AE 15.01.2021
RPSC VPITI 14.02.2016
CSE 2009

Ans. (c) : Operation time = 2 minutes
% yield of time performance rating = 120%

then time will be $= 2 \times \frac{120}{100}$

2.4 minutes

Time allowance = 10% of time with yield
Now standard time of operation

$= 2.4 + 2.4 \times \frac{10}{100}$

$= 2.4 + 0.24$

$= 2.64$ minutes

789. Which one of the following is not a technique under Pre-determined Motion Time System (PMTS)?

(a) Work factor
(b) Synthetic data
(c) Stop Watch Time Study
(d) Method Time Measurement

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
MPPSC AE 2016
ESE 2006, 2003

Ans. (c) Techniques under Pre-determined Motion Time System (PMTS)–
1. MTM (Method Time Measurement)
2. Work factor system
3. Synthetic data
4. Motion time analysis

790. According to the Emerson plan, a worker gets a bonus whenever the efficiency of a worker is more than

(a) 50% (b) 55%
(c) 60% (d) 66.67%

APPSC AEE Screening Test 2016
TNPSC AE, 2008
CSE Pre-1996

Ans. (d) : According to Emerson plan– If a worker does a job in 10 hours and if the standard time is also 10 hours, his efficiency is 100%. If he does the job in 20 hours, his efficiency is 50% and if he does the job 8 hours, his efficiency is 125%.
Bonus become payable only when efficiency reaches $66\frac{2}{3}\%$. For efficiency below $66\frac{2}{3}\%$, he gets only the time wage.

791. Work study is concerned with

(a) Improving present method and finding standard time
(b) Motivation of workers
(c) Improving production capability
(d) All of the above

Vizag Steel (MT) 2017
Nagaland CTSE 2016, 2017 Ist Paper
RPSC ACF-2011

Ans. (a) : Improving present method and finding standard time.
Work study is systematic examination of methods of carrying out activities such as improving the use of resources available and to establish standards of performance for activities to be carried out, work study has two components.

792. Basic tool in work study is:

(a) Graph paper (b) Stop watch
(c) Process chart (d) Planning chart

TRB Polytechnic Lecturer 2017
Nagaland PSC CTSE 2017, Paper-I
TNPSC AE 2017

Ans. (b) : Stop watch is a basic tool in work study.

793. What does symbol 'D' imply in work study –

(a) inspection (b) transport
(c) delay or temporary storage
(d) permanent storage

Nagaland PSC CTSE 2017, Paper-I
Rajasthan AE (Nagar Nigam) 2016 Shift-3
GWSSB AAE, 27.12.2015

Ans. (c) : 'D' symbol represent in work study a delay temporary storage.

D → Delay or temporary storage
○ → Operation
△ → Storage
⇨ → Transport
□ → Inspection
◎ → Combined symbol

794. The standard time for an operation has been calculated as 10 minutes. The worker was rated at 80%. If the relaxation and other allowances were 25%. then the normal time would be:

(a) 12. 5 min (b) 10 min
(c) 80 min (d) 08 min

GPSC ARTO 01.05.2016
MPPSC AE 2016
ESE 1994

Ans. : (b) Standard time = 10 min
Worker rated = 80 %
Allowances = 25%

Normal time = Observed time × Rating factor

$$Normal\ time = 10 \times \frac{80}{100} = 8 \text{ min}$$

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$$\text{Standard time} = \text{Normal time}\left(1+\frac{\%\text{Allowances}}{100}\right)$$
$$=8\left(1+\frac{25}{100}\right)$$
$$=8\times1.25$$
$$=10 \text{ min}$$

795. Value engineering technique in which experts of the same rank assemble for product development is called
(a) Delhi
(b) Brain storming
(c) Morphological analysis
(d) Direct expert consultation

UKPSC AE-2013, Paper-I
APPSC AEE 2012, ESE 1993

Ans. (b) : Brain storming – Brain storming is a group creativity technique by which efforts are made to find a conclusion for a specific problem by gathering a list of Ideas spontaneously contributed by its members.

796. In process charts, triangle symbol represents:
(a) delay (b) operation
(c) inspection (d) storage

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
NLC GET 17.11.2020, Shift-II
CGPSC ARTO & TSI 07.01.2017

Ans. (d) :

	Operation name	Symbol	Symbol description
1.	Flow line	→	Process Flow direction
2.	Operation	○	operation process
3.	Process	▭	Process step
4.	Inspection	□	Inspection
5.	Storage	▽	Storage of raw material
6.	Storage	△	storage of finished goods
7.	Start and end	⬭	Process delay or wait
8.	Delay	D	Process delay or wait
9.	Transportation	⇨	For movement of part
10.	Decision	◇	For taking decision during process

797. Mathematically value of a product is expressed as
(a) cost/expenses (b) cost/utility
(c) utility/cost (d) none of the above

NLCIL GET 17.11.2020, Shift-II
UKPSC AE 2007 Paper -I
J & K PSC Screening, 2006

Ans. (c) : Utility/cost

798. The workmen in an engineering company are expected to work for 400 minutes in a shift of 8 hours. The remaining time is meant for rest and personal needs. The standard time (in minutes) per piece of a job, with normal times of 2 minutes is given by :
(a) 2.4 (b) 2.5
(c) 2.6 (d) 2.8

APPSC IOF, 2009
CSE Pre-2006

Ans. (a) : No of piece produced $=\frac{400}{2}=200$ units
Allowance = 8 × 60 – 400
= 80 minutes
Standard time = Normal time + allowance per piece
$$=2+\frac{800}{200}$$
= 2.4 min

799. The cost reduction technique in comparison to the worth of a product is known as :
(a) Reverse Engineering
(b) Value Engineering
(c) Material Engineering
(d) Quality Engineering

JPSC AE 10.04.2021, Paper-II
JPSC AE 2013, Paper-V

Ans. (b) : Value Engineering is a cost reduction technique. It is a systematic method to improve the "Value" of goods or products and services by using an examination of function.

$$\text{Value}=\frac{\text{Function}}{\text{Cost}}$$

800. Which of the following plans guarantees minimum wage to a works and bonus based on fixed percentage of time saved:
(a) Gantt plan (b) Halsey plan
(c) Rowan plan (d) Bedaux plan

PSPCL AE, 2012
ESE 1992

Ans. (c) : Rowan planes guarantees minimum wage to a works and bonus based on fixed percentage of time saved.

801. Work study deals with
(a) Improving production capabilities and the technique
(b) Improving present method and finding standard time
(c) Motivation of workers
(d) Improving production planning and control.

Rajasthan Nagar Nigam AE 2016, Shift-II
OPSC Civil Services Pre 2006

Ans. (b) : Work study deals with –
- Improving production capabilities and technique.
- More effective use of human effort.
- Evaluation of human work.

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Work study is a combination of two groups technique method study and work measurement which are used to examine people's work. The work study is also used in determining the standard time that a qualified worker should take to perform the operation when working in normal place.

Standard Time = Normal Time (1 + Allowance)

Normal Time = Observed Time × Rating Factor.

802. In a time study exercise, the time observed an activity was 54 s. The operator had a performance rating of 120. A personal time allowance of 10% is given. The standard time for the activity, in second, is :

(a) 54 (b) 60.8
(c) 72 (d) 58.32

GPSC ARTO Pre 30.12.2018
GATE 2000

Ans. (c) : Given,

Activity time = 54 seconds

Performance Rating = 120%

$$\text{Normal time} = 54 \times \frac{120}{100} = 64.8 \text{ seconds}$$

Standard time at allowance 10%

ST = NT × Allowance + NT

= 64.8 × 0.1 + 64.8

ST = 71.28 ≈ 72 seconds.

803. The total observed time for an operation is 2 minutes. The rating is 120% and allowances are 10% for the job. The normal time and standard time for the job will be respectively

(a) 2 minutes and 2.20 minutes
(b) 2.4 minutes and 2.64 minutes
(c) 2.2 minutes and 2.42 minutes
(d) 2.6 minutes and 2.86 minutes
(e) None of these

CGPSC AE 16.10.2016
CSE Pre-2005

Ans. (b) : Normal time = Observed time × Rating factor

= 2 × 1.2 = 2.4 min

$$\text{Standard time} = 2.4 \times \frac{110}{100}$$

= 2.64 min

804. Percent idle time for men or machines is found by

(a) time study (b) work sampling
(c) work study (d) method study

MPSC Lect. (Prod.) 2018
RPSC ADE 2016

Ans. (b) : Percent idle time for men or machine is found by work sampling.

805. Value engineering is necessary to be used when the following symptoms are indicated

1. New product designs are to be introduced.
2. The firm is unable to meet delivery date.
3. Rate of return on investment goes down.

Which of the above statement/s is/are correct ?

(a) 1, 2 & 3 (b) 2 only
(c) 1 & 3 only (d) 2 & 3 only

UKPSC AE 2012 Paper-I
ESE 2010

Ans. (a) : 1, 2 & 3

806. Assertion (A) : Value analysis is superior to other conventional cost reduction techniques.

Reason (R) : In conventional cost reduction techniques, value is increased by widening tolerance bands.

Code :

(a) Both (A) and (R) are true. (R) is the correct explanation of (A).
(b) Both (A) and (R) are true. (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

UKPSC AE 2012 Paper-I
ESE 2005

Ans. (b) : Both (A) and (R) are true. (R) is not the correct explanation of (A).

807. Job design is not a process of–

(a) Deciding the contents
(b) Deciding methods to carry out the job.
(c) Deciding the relationship which exists in the organization
(d) Communication which exists in the organization

Nagaland PSC CTSE 2017, Paper-I
Nagaland CTSE 2016 Paper-I

Ans. (d) : Job design is not a process of communication which exists in the Organization.

It, involves in contact of job, And how to carry job & relationship exists in between employer & worker, satisfaction.

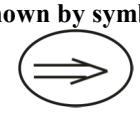
808. The number of fundamental motion involved in micro-motion study is

(a) 12 (b) 8
(c) 20 (d) 18
(e) 16

CGPSC AE 26.04.2015 Shift-I
Mizoram PSC AE/SDO 2014 Paper-III

Ans. (e) : Based on the results of motion study, Gilberth identified 16 basic motions using which any job can be performed.

809. If an object is being painted while transported by chain conveyor can be shown by symbol:

(a)
(b) 
(c)
(d)

UPRVUNL AE 07.10.2016
TNPSC AE (Industries) 09.06.2013
GATE 2003

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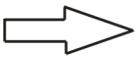
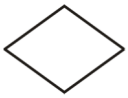
Ans. (b)

Symbol	Name
	Operation
	Transport
	Inspect
	Delay
	Storage
	Handle
	Decide
	Combined Activity (Inspection during operation)
	Operation during transportation

810. In a time study the observed time is 0.75 min, performance rating factor is 110% and allowances are 20% of the normal time. The standard time is.

(a) 0.82 min (b) 0.975 min
(c) 0.99 min (d) 1.03 min

UPPSC AE 12.04.2016 Paper-I
ESE 2009

Ans. : (c) Normal time = observed time ×Rating factor
Normal time = 0.75×1.1
Normal time = 0.825 min
Standard time = Normal time × (1 + % age allowance)
Standard time = 0.825×(1+0.20)=0.99 min.

811. The type of material handling equipment used for handling materials during manufacture of cement is:

(a) Bucket conveyor (b) Belt conveyor
(c) Fork lift (d) Overhead crane

OPSC AEE 2015 Paper-I
TNPSC AE 2013

Ans. : (b) The type of material handling equipment used for handling materials during manufacture of cement is Belt conveyor.

812. A production line is said to be balanced when

(a) There are equal number of machines at each work station
(b) There are equal number of operations at each work station.
(c) The waiting time for service at each station is same
(d) The operation time at each station is same.

APPSC AEE 2012, ESE 1997

Ans. (d) : A production line is said to be balanced when the operation time at each station is same.

813. Which of the following equations represents the wage incentive (W) as per Rowan Plan? (Consider : R – hourly wage rate, T – actual time taken to complete the job and S – standard time.)

(a) $W = R.T + \left(\frac{S-T}{S}\right) \times R.T$

(b) $W = R.T + \left(\frac{S-T}{S}\right)$

(c) $W = R.T + \left(\frac{S-T}{S.R.T}\right)$

(d) $W = R.T \times \left(\frac{S-T}{S}\right)$

APPSC Poly Lect. Automobile Engg., 2020
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : Wage incentive (w) as per Rowan Plan

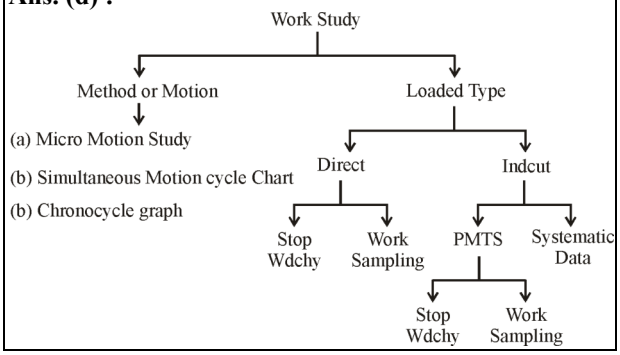
$$W = RT + \left(\frac{S-T}{S}\right) \times R.T$$

814. Which one of following techniques is used for determining allowances in time study?

(a) Acceptance sampling
(b) Linear regression
(c) Performance rating
(d) Work sampling

GPSC Lect. (Poly.) 07.12.2014
TNPSC ACF 2012, CSE Pre-2001

Ans. (d) :



815. Which of the following hand-motion belings to "Therbligs" in motion study?

I. Unavoidable delay
II. Pre-position
III. Select
IV. Reach

(a) I and IV (b) I and II
(c) I, II and III (d) II, III and IV

TNPSC ACF 2012, ESE 2001

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Ans. (c) : Unavoidable delay, pre-position, select are hand motion belongs to therbligs.

816. Fifty observations of a production operation revealed a mean cycle time of 10 min. The worker was evaluated to be performing at 90% efficiency. Assuming the allowances to be 10% of the normal time, the standard time (in second) for the job is

(a) 0.198 (b) 7.3
(c) 9.0 (d) 9.9

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2001

Ans. (d) : Given,
Mean cycle time = Observed time (OT) = 10 min
Performance rating = 90%
Allowance = 10%
Normal time = OT × (Rating factor or performance)
Normal time (N.T.) = 10 × 0.9 = 9 min
Standard time (ST) = NT(1 + %allowance)
= 9(1 + 0.1) = 9.9 min
But official answer is given option (c).

817. Which one of the following is NOT a work measurement technique?

(a) time study (b) work sampling
(c) motion time data (d) micromotion study

GPSC Executive Engineer 23.12.2018
ESE 2009

Ans. (d) : Micromotion study not a work measurement technique.

818. In a stop-watch time study, the observed time as 0.16 minute; the performance rating factor was 125 on the 100 normal (percentage scale). What is the standard time in minutes, if 10% allowances are permitted?

(a) 0.180 (b) 0.200
(c) 0.220 (d) 0.240

HPPSC Asstt. Prof. 18.09.2017
CSE Pre-2004

Ans. (c) : Observed time = 0.16 minute
Allowance = 10

Rating factor = $\frac{125}{100} = 1.25$

Standard time = Observed time × Rating factor (1 + Allowance)

$$= 0.16 \times 1.25\left(1+\frac{10}{100}\right)$$

$= 0.16 \times 1.25\ (1.10)$
$= 0.22$

819. Work study involves

(a) only work measurement
(b) only method study
(c) method study and work measurement
(d) only motion study

APPSC AEE 2012
CSE Pre-1994

Ans. (c) : Work study consists of two techniques namely method study and work measurement. The method study is a technique to simplify the job and develop more economical method of doing it. The work measurement is concerned with the elimination of ineffective time and establishment of time standards for a job.

820. Determination of standard time in complex job system is best done through

(a) Stop watch time study
(b) Group timing techniques
(c) Analysis of micro motions
(d) Analysis of standard data

APPSC AEE 2012
ESE 1996

Ans. (d) : The determination of standard time in a complex job system is best done through analysis of standard data system.

821. In sampling inspection the maximum % defective that can be treated satisfactory as a process average is

(a) Rejectable Quality Level (RQL)
(b) Acceptable Quality Level (AQL)
(c) Average Outgoing Quality Limit (AOQL)
(d) Lot Tolerance Percent Defective (LTPD)

UPPSC AE 13.12.2020, Paper-I
ESE 2020

Ans. (b) : Acceptable Quality Level is the maximum percent defective that is considered satisfactory as a process average by the producer and consumer.

822. A diagram showing the path followed by men and materials while performing a task is known as

(a) String diagram (b) Flow process chart
(c) Travel chart (d) Flow diagram

TNPSC ACF 2012, ESE 1993

Ans. (b) : Flow process chart– In this diagram shows the path followed by men and materials while performing a task.

823. If P = % activity and A = limit of accuracy in work sampling. The number of observations at a confidence level of 95% is given by

(a) $\frac{(1-P)}{A^2P}$ (b) $\frac{2(1-P)}{A^2P}$
(c) $\frac{3(1-P)}{A^2P}$ (d) $\frac{4(1-P)}{A^2P}$

UKPSC AE 2012 Paper-I
UKPSC AE 2007 Paper -I

Ans. (d) : $\frac{4(1-P)}{A^2P}$

824. String diagram is used when

(a) team of workers is working at a place
(b) material handling is to be done
(c) idle time is to be reduced
(d) all of these

J&K PSC Civil Services Pre, 2013
TNPSC AE, 2008

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Ans. (d) : String diagram is used when team of workers is working at a place, material handling is to be done, idle time is to be reduced.

825. Motion study involves analysis of :
(a) actions of operator
(b) layout of work place
(c) tools and equipments
(d) all of the above

Vadodara Municipal Corp. DEE, 2018
RPSC ACF-2011

Ans. (a) : Motion study involves analysis of industrial or work procedures to determine the most efficient methods of operation. Its analysis of actions operators.

826. Work study comprises following main techniques
(a) method study and work measurement
(b) method and time study
(c) time study and work measurement
(d) method study and job evaluation

RPSC ACF-2011, CSE Pre-1994

Ans. (a) : Work study consists of two techniques namely method study and work measurement.
The method study is a technique to simplify the job and develop more economical methods of doing it. Work measurement is concerned with the elimination of ineffective time and establishment of time standards for a job.
Work study is mainly aimed at determining the most efficient method of performing a job. Basic tool in work study is stop watch.

827. A systematic job improvement sequence will consist of :
(a) motion study (b) time study
(c) job enrichment (d) all of these

J&K PSC Civil Services Pre, 2013
CSE Pre-1994

Ans. (d) : Job improvement sequence consist of motion study time study and job enrichment.

828. Which of the following wage incentive plans guarantees minimum wage to a worker and bonus is paid for the fixed percentage of time saved ?
(a) Halsey plan
(b) Gantt plan
(c) Rowan plan
(d) Emerson's efficiency plan

J&K PSC Civil Services Pre, 2013
ESE 1992

Ans. (a) : Halsey plan guarantees a minimum wage to every worker on time basis and incentive is given on the basis of efficiency.

829. In a stop-watch time study, the observed time was 0.16 minute; the performance rating factor was 125 on the 100 normal (percentage scale). What is the standard time in minutes if 10% allowances are permitted?
(a) 0.180 (b) 0.200
(c) 0.220 (d) 0.240

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
CSE Pre-2004

Ans. (c) As we know,
Normal time = Observed time × Performance rating

$$= 0.16 \times \frac{125}{100}$$

$$= 0.2 \text{ min}$$

Standard time = Normal time + Allowance

$$= 0.2 + 0.2 \times \frac{10}{100}$$

$$= 0.220 \text{ min}$$

830. TMU in method time measurement stands for
(a) Time measurement unit
(b) Temperature measurement unit
(c) Time movement unit
(d) Time motion unit

Rajasthan Nagar Nigam AE 2016, Shift-II
UKPSC AE 2012 Paper-I

Ans. (a) : TMU stands for time measuring unit. In methods time measurement (MTM) system, a predetermined time is given to each motion. In motion study the time for each basic element is given in units of TMU.
1 TMU = 0.00001 hr
= 0.0006 minute
= 0.036 second

831. The first four steps of method study are :
(a) select, examine, develop, record
(b) select, record, develop, examine
(c) select, record, examine, develop
(d) select, examine, record, develop

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (c) : Steps of method study are –
(i) Selection of work
(ii) Recording Information
(iii) Examine Information
(iv) Develop new process
(v) Evaluate developing process
(vi) Define after evaluating the process.
(vii) Install new process
(viii) Maintain new method.

832. A soldering operation was work sampled over two days (16 hours) during which an employee soldered 108 joints. Actual working time was 90% of the total time and the performance rating estimated to be 120%. If the contract provides allowance of 20% of the total time available the standard time for the operation would be:
(a) 12 min (b) 8 min
(c) 9 min (d) 10 min

J & K PSC Screening, 2006
GATE 2004

Ans. (a) : Total time = 16 hours = 16 × 60 min

$$\text{Actual time} = \frac{90}{100} \times 16 \times 60 = 864 \text{ min}$$

Normal time = Actual time × Rating factor

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$$\text{Normal time} = \text{Actual time} \times \frac{120}{100}$$
$$= 864 \times 1.2 = 1036.8 \text{ min}$$
$$\text{Standard time} = \text{Normal time} \times \text{Allowances}$$
$$= 1036.8 \times 1.2$$
$$= 1244.16 \text{ min}$$
$$\text{Standard time per joint} = \frac{1244.16}{108} \approx 12 \text{ min}$$

833. Value Engineering is concerned with saving
(a) time
(b) overhead costs
(c) un-necessary costs
(d) administrative difficulties

UKPSC AE 2012, 2007 Paper -I

Ans. (c) : Un-necessary costs

834. _____is defined as the technique in which a statistically competent number of instantaneous observations are taken over a period of time of machines, processes or workers.
(a) Work Sampling (b) Line Balancing
(c) Log study (d) None of the above

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : Work Sampling is defined as the technique in which a statistically competent number of instantaneous observations are taken over a period of time of machines, processes or workers.

835. In performing a task, motion economy refers to the manner in which :
(a) Material movements can be reduced.
(b) Machine movements can be reduced
(c) Human energy can be conserved
(d) Time consumed can be reduced

RPSC ACF & FRO, 26.02.2021

Ans. (c) : Motion economy refers to the manner in which human energy is conserved while performing a task.
The principles of motion economy form a set of rules and suggestions to improve the manual work in manufacturing and reduce fatigue and unnecessary movements by worker.

836. The allowance factor in time study.
(a) adjusts normal time for errors and rework
(b) adjust standard time for lunch breaks
(c) adjusts normal time for personal needs, unvoidable delays and fatigue
(d) allows worker to rest every 20 minutes

CGPSC AE 15.01.2021

Ans. (c) : The allowance factor in time study adjusts normal time for personal needs, unavoidable delays and fatigue.

837. To achieve robust performance, the objective functions must capture performance of _____.
(a) All variable factors
(b) both control and noise factors
(c) controllable factors only
(d) noise factors only

VIZAG STEEL MT 24.01.2021 Shift-II

Ans. (b) : To achieve robust performance, the objective functions must capture performance of both control and noise factors.

838. Select FALSE statement related to Work Study from the given list: Work study …
(a) provides basis for incentive scheme
(b) can determine time required to do a job
(c) is generally well appreciated by workers
(d) helps to assess manpower required

VIZAG STEEL MT 24.01.2021 Shift-II

Ans. (c) Work study is a management tool to achieve higher productivity in any organization weather manufacturing tangible products or offering services to its customers. Work study is related to provides basis for incentive scheme and helps to assesses manpower required.

839. While taking up method study it is desirable to identify short time and long time solutions because :
(a) Management will be happy to delay long time solution
(b) Management gets time for decision making
(c) Changes can be implemented early in short time solution area
(d) Long time solution can wait for implementation

VIZAG STEEL MT 24.01.2021 Shift-II

Ans. (c) : While taking up method study it is desirable to identify short time and long time solution because change can be implemented early in short time solution area.

840. The price paid by the buyer is :
(a) Use value (b) Cost value
(c) Esteem value (d) Exchange value

JPSC AE 10.04.2021, Paper-II

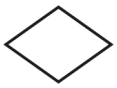
Ans. (a) : Use Value- The price paid by the buyer. It relates the attributes of product which enable it to perform its function.
Esteem Value- Is the additional premium price which a product can attract because of its intrinsic attractiveness to purchaser.
Exchange Value- Is the sum of the attributes which enable the product to be exchanged or sold.
Cost Value- Is the total cost of producing products.

841. Who is named as father of value Analysis?
(a) Lawrence D. Miles (b) Frank Gilberth
(c) F.W. Taylor (d) Philip M. Morse

JPSC AE 10.04.2021, Paper-II

Ans. (a) : Father of value Analysis/ Value engineering- Lawrence D. Miles.

842. Flow chart refers to a pictorial representation of an algorithm. Various symbols are used in a flow chart. Following diamond shaped box denotes:



(a) Initiation (b) Input
(c) Output (d) Decision

JPSC AE 10.04.2021, Paper-II

Ans. (d) : Decision

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843. Value is the cost directly proportionate to :

(a) Price (b) Product material
(c) Function (d) All of the above

JPSC AE 10.04.2021, Paper-II

Ans. (c) : $\text{Value} = \frac{\text{Function}}{\text{Cost}}$

$\text{Value} \propto \text{Function}$

844. Study methods can be improved by

(a) Benchmarking
(b) Improved note taking
(c) Vacations
(d) Sitting in the front row

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Study methods can be improved by benchmarking.

845. A Reaction Plan is

(a) A flow chart
(b) An afterthought
(c) A way to produce good products
(d) A checklist to use when things don't look right

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : A Reaction Plan is a checklist to use when things don't look right.

846. The chart based on film analysis and depicts on a common scale the therbligs performed by both hands of a operator is _____.

(a) Two Handed Process Chart
(b) Simultaneous Motion Cycle Chart
(c) String Diagram
(d) Flow Diagram

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (b) : SIMO-Chart–SIMO chart stands for simultaneous motion chart, which is a recording technique for micro-motion study. It's a chart based on the film analysis used to record simultaneously on a common time scale. The therbligs or group of therbligs performed by different parts of the body of one or more operators.

847. The originator of motion study is _____.

(a) Crossby (b) Gilbreths
(c) Deming (d) F W Taylor

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (b) : Motion study was developed by Frank B. Gilbreth and Lillian M. Gilbreth. This technique aims at removing the unwanted actions/motions so that the work can be completed in a lesser time.

848. The two symbols used in the operation process chart are _____.

(a) operation and storage
(b) operation and transportation
(c) inspection and transportation
(d) operation and inspection

VIZAG MT, 14.12.2020

Ans. (d) : The two symbols used in the operation process chart are operation and inspection.

849. The '5S' are about

(a) JIT
(b) Cost reduction
(c) Smoothening the flow of the operations
(d) Good housekeeping

RPSC IOF, 2020

Ans. (d) : 5S or good housekeeping involves the principle of waste elimination through workplace organisation.

850. In motion study, the Therblig symbol represents :

(a) find (b) search
(c) assembly (d) inspect

APPSC Poly Lect. 13.03.2020

Ans. (a) : – find

851. In which stage of product development process, the value engineering is used as a tool?

(a) design stage (b) production stage
(c) dispatch stage (d) maturity stage

VIZAG MT, 14.12.2020

Ans. (a) : Value engineering tool can be applied during the development or design stage or during the last half of the planning stage. The main aim is to improve the product value, while increase the functionality and lower the cost of proposed components in order to meet the target cost and functionality objectives.

852. Which is not true in context of codification of materials?

(a) done on the basis of source of purchase, nature of item and its use
(b) help in reducing the number of items
(c) increase the clerical work and involves financial loss
(d) helps in maintenance of accurate records

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : Increase the clerical work and involves financial loss.

853. The art of getting work through people, with satisfaction for employer, employees and the public is known as :

(a) aesthetics-ness (b) agronomics
(c) management (d) administration

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Management : The art of getting work through people, with satisfaction for employer and employees and the public is known as management. Management is defined as the process by which a co-operative group directs action towards lommon goals.

Types of management

(1) Financial management,
(2) Maintenance Management,
(3) Production management,
(4) Distribution management

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854. The decisions such as setting basic goals, objectives and establishing policies are the functions of which type of management?

(a) Top management
(b) Lower management
(c) Operating force
(d) Middle management

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : The principal function of the top management can be outlined as follows–
(1) Determination of objective
(2) Formulation of policies
(3) Long range planning and strategy
(4) Organizing for action
(5) Developing of major resources
(6) Selecting key personnel
(7) Co-ordination and controlling

855. Which of the following terms implies taking away of responsibilities of the person during the period of emergency?

(a) Transfer (b) Lay-off
(c) Promotion (d) Dismissal

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : Lay-off : A lay-off is the temporary suspension or permanent termination of employment of an employee or more commonly a group of employees for business reasons, such as personal management or downsizing on organization.

856. According to Emerson's efficiency plan, how is efficiency of a worker defined?

(a) It is the ratio of average time to the actual time taken by the worker to complete the job
(b) It is the ratio of standard time to the actual time taken by the worker to complete the job
(c) It is the ratio of actual time to the average time taken by the worker to complete the job
(d) It is the ratio of actual time to the standard time taken by the worker to complete the job

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : Under Emerson Plan the standard time for the completion of a task is fixed against which the actual performance of the worker is measured. The worker's efficiency can be determined by dividing the time taken by the standard time.

$$\text{Efficiency} = \frac{\text{Standard time}}{\text{Actual time}}$$

857. The ratio of the total number of separations to the average workforce per unit period in terms of percentage in the organisation is known as :

(a) net turnover of employer
(b) net labour turnover
(c) gross domestic people turnover
(d) gross domestic people loss

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The ratio of the total number of separations to the average workforce per unit period in terms of percentage in the organisation is known as net labour turnover.

Turn over rate

$$= \frac{\text{Number of sepration in a period}}{\text{Average number of workers in a period}} \times 100$$

858. The payment made in terms of non-money for the use of labour in industries is known as :

(a) wages (b) incentives
(c) fringe benefits (d) bonus

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Fringe benefits : The payment made in terms of non-money for the use of labour in industries is known as fringe benefit. The most common examples of fringe benefits are health insurance, worker's compensation, retirement plans and family medical leave less common fringe benefits might include paid vacation, meal subsidization commuter, benefits and more.

859. Work measurement is

(a) the systematic recording and critical examination of existing and proposed ways of doing work
(b) the application of techniques to establish time for a qualified worker to carry out a specified job at a defined level of performance
(c) the reduction of work content
(d) the development of new economical and effective processes

TNPSC AE 2019

Ans. (b) : The application of techniques to establish time for a qualified worker to carry out a specified job at a defined level of performance is called work measurement.

860. A box that can represent two different condition in a flow chart.

(a) Circle (b) Square
(c) Diamond (d) Parallelogram

JPSC AE PRE 2019

Ans. (c) : Diamond

861. Both Rowan plan and 50-50 Halsey plan will provide the same earning when the actual time is ..... the standard time

(a) one-fourth (b) one-half
(c) equal to (d) twice

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : As we know,
$S \rightarrow$ Standard time
$T \rightarrow$ Actual time
$R \rightarrow$ Hourly wage
50-50 Halsey plan
$= T * R + 0.5(S - T) * R$
Rowan plan

$$= T * R + \frac{[T * (S - T) * R]}{S}$$

From equation two equation/compare, provide same, earning when actual time is one half the standard time $\left(T = \frac{1}{2}S\right)$.

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862. One time measurement unit (TMU) in method time measurement system equals :

(a) 0.0001 minute (b) 0.0006 minute
(c) 0.006 minute (d) 0.001 minute

Vadodara Municipal Corp. DEE, 2018

Ans. (b) : One time measurement unit (TMU) in method time measurement system equals 0.0006 minute.

863. If the number of items for which material is required is 200 and the standard time per item is 5 minutes, expected efficiency is 50% and the number of hours per machine is 4, the number of machines required is

(a) 18 (b) 10
(c) 6 (d) 9

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Given, No of items = 200

$$\eta = 50\% = \frac{\text{Standard} \times 100}{\text{Actual time}}$$

Standard time = Setup time + (Time per piece × No. of piece)

Total time = 200 × 5 = 1000 min

No. of hour per machine = 4

$$\text{Actual time} = 4 \times \frac{1}{2} = 2\,\text{hr}$$

$$\text{No. of M/c} = \frac{1000\,\text{min}}{120\,\text{min}} \simeq 9$$

864. When the hourly wage rate is `15 and standard time for a particular job is 8 hours, how much can a worker earn daily as per Bedaux plane, if he complete the job in 6 hours?

(a) ` 256 (b) ` 122
(c) ` 82 (d) ` 280

Karnataka PSC AE, 10.09.2017

Ans. (b) Bedaux Plan– It is a wage incentive scheme.

Actual time = 6 hours

Standard time = 8 hours

Hourly wage rate = 15 Rs/hour

Earnings : Time wage = 15 × 6
= 90 Rs

Bonus:

Std. bonus = 8 × 60 = 480

Actual bonus = 6 × 60 = 360

Bonus saved = 480 – 360
= 120

$$\text{Bonus} = \frac{75}{100} \times \frac{120 \times 15}{60} = 22.5$$

Total earnings = 90 + 22.5
= 112.5 Rs

Nearest answer is option (b).

865. More elongated dots with longer gaps between them in a chronocyle graph indicate

(a) Higher working speed
(b) More working time
(c) Improved method
(d) More drop-deliveries

Karnataka PSC AE, 10.09.2017

Ans. (a) : Chronocyle graph is micro motion study graph. More elongated dots with longer gaps between them shows higher working speed.

866. Under the payment of wages act, payment of wages of establishments employing not less than 1000 employees shall be paid within of the wage month

(a) 10^{th} day (b) 15^{th} day
(c) 7^{th} day (d) 2^{nd} day

GPSC EE Pre, 28.01.2017

Ans. (a) : According to provisions for the wage payment for the employees in India.

Wages needs to be paid.

Before the expiry of the 7^{th} day after the last day of the wage period, where there are less than 1000 workers employed and in rest case of on the 10^{th} day.

867. The current erection cost of a structure is Rs. 13,200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in Rs. is

(a) 16,500 (b) 15,180
(c) 11,000 (d) 10,120

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Cost is proportional to wages per day and No. of hours.

Wages are increased by 1/5 and working hours are decreased by 1/24.

$$\text{So, the new cost} = (\text{current cost}) \times \left(1+\frac{1}{5}\right) \times \left(1-\frac{1}{24}\right)$$

$$= 13200 \times \frac{6}{5} \times \frac{23}{24}$$

$$= 15180$$

868. An operator manufactures 75 jobs in 8 hrs. If this time includes the time for setting his machine, calculate the operator's efficiency. Standard time allowed for the job was: setting time 40 min; production time per piece 10 min.

(a) 115.5% (b) 164.6%
(c) 134.7% (d) 124.8%

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Standard time for manufacturing 75 jobs

= Setup time + (Time per piece × No. of piece)

= 40 + (10 × 75) = 790 min

Efficiency of operator

$$= \frac{\text{Standard time} \times 100}{\text{Actual time}} = \frac{790 \times 100}{8 \times 60}$$

= 164.6%

869. The Standard time of an operation is defined as the time taken by a

(a) Faster worker to perform that operation
(b) Slow worker to perform that operation
(c) Trained worker to perform that operation while working at a steady pace
(d) None of the above

GPSC Asstt. Director of Transport 05.03.2017

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Ans. (c) : The standard time of an operation is defined as the time taken by a trained worker to perform that operation while working at a steady pace.

870. Performance rating is equal to

(a) Observed performance + normal performance
(b) Observed performance – normal performance
(c) Observed performance × normal performance
(d) None of the above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Performance rating can be defined as the procedure in which the time study engineer compares the performance of operator (s) under observation to the normal performance and determines a factor called rating factor.

871. Abbreviated work factor data is applied for

(a) Material handling operation
(b) Maintenance operation
(c) Packing and shipping operation
(d) all of the above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Abbreviated work factor data is applied for material handling operation, packing and shipping operation and maintenance operation.

872. The time taken by a normal worker for a specific task or job working under moderate condition and including other allowances such as fatigue, setting of tool and job, repairing of tool and checking of job, etc., is known as

(a) Speed rating (b) Performance rating
(c) Normal time (d) Standard time

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Standard Time– Standard time is the time, which is taken by a normal worker for a specific task or job working under moderate condition and including other allowances such as fatigue, setting of tool and job, repairing of tool and checking of job.

873. If the selected time for an element is 0.30 minute, the passing rating is 110% and if the sum of all secondary adjustment amount to 20%, then the standard time will be

(a) 0.264 min (b) 0.327 min
(c) 0.396 min (d) 0.275 min

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Solution,

Normal time = Observed time × Rating%

$$= 0.30\,\text{min} \times \frac{110}{100} = 0.33\,\text{min}$$

Allowance at 20% = 0.066 min

Standard time = Normal time + Allowance

= 0.33 + 0.066

= 0.396 min

874. If a work content of 10 hrs has to be made at a rate of 400 a week, and the normal working week is 40 hrs, the number of operators required is

(a) 120 (b) 100
(c) 150 (d) 10

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : No. of operators $= \frac{400 \times 10}{40} = 100$

875. Select the following statement.
Bin card will show

(a) The amount received
(b) Amounts issued
(c) The amount balance on hand
(d) The quantity allocated for issue or about quantities on order

TRB Polytechnic Lecturer 2017

Ans. (d) : Bin Card– A BIN CARD is a document that records the status of a good held in stock. A typical retailing business with a large stock warehouse will use a BIN Card to record a running balance of stock on hand, in addition to information about stock received and notes about problems associated with that stock item.

876. Math List - I and List - II and select the correct answer :

List - I	**List - II**
(1) Flatness	**(i)**
(2) Profile of a surface	**(ii)**
(3) Position	**(iii)**
(4) Parallelism	**(iv)**

Codes :

	(1)	**(2)**	**(3)**	**(4)**
(a)	(vi)	(iii)	(i)	(ii)
(b)	(iii)	(iv)	(ii)	(i)
(c)	(ii)	(i)	(iv)	(iii)
(d)	(ii)	(iv)	(i)	(iii)

TRB Polytechnic Lecturer 2017

Ans. (a) : Flatness

Profile of a surface

Position

Parallelism

877. The job of industrial designer is–

(a) to carry out detailed stress analysis of the product
(b) to design industrial products like cranes and hoists
(c) to create aesthetically forms and shapes for the products
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

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Ans. (c) : The job of industrial designer is to create aesthetically forms and shapes for the products. Industrial engineer concerned with design, installation and improves the production system. His objective is to eliminate unproductive operation from production system in order to increase productivity.

878. An 8-hour measurement study in a plant reveals that 320 number of units were produced. If idle time = 15% and performance rating = 120%, with allowance = 12% of normal time, the standard time per unit produced will be

(a) 1.823 minutes (b) 1.714 minutes
(c) 1.645 minutes (d) 1.286 minutes

ESE 2017

Ans. (b) : Total available time = 8 hrs.

Idle time = 15%

Actual available time = 8×0.85

$= 6.8$ hrs.

$$\text{Observed time (OT)} = \frac{\text{Actual available time}}{\text{No. of units to be produced}}$$

$$= \frac{6.8 \times 60}{320} = 1.275 \text{ min}$$

Normal time (NT) = OT × Performance rating factor

$$= 1.275 \times \frac{120}{100} = 1.53 \text{ min}$$

Standard time (ST) = NT × 1.12

$= 1.53 \times 1.12 = 1.728$ min

879. A manufacturing company produces 200 pistons per shift of 8 hours. If the standard time per piece of piston is 2.4 minute the productivity would be

(a) 33.33% (b) 66.66%
(c) 100% (d) 125%

Punjab PSC (Lect.) 06.08.2017

Ans. (c) Total number of piston that can be produced

$$100\% \text{ productivity} = \frac{8 \times 60}{2.4} = 200 \text{ piston}$$

Current production rate = 200 piston

$$\text{Productivity} = \frac{200}{200} \times 100 = 100\%$$

880. Consider the following statements in respect of double sampling plan.

1. Decision to accept or reject the lot is reached sometimes after one sample and sometimes after two samples
2. Decision to accept or reject the lot is taken only after the inspection of both samples
3. Average number of pieces inspected is less than that for single sampling
4. Average number of pieces inspected is double that of single sampling

Which of these statements are correct?

(a) 2 and 4 (b) 1, 2 and 3
(c) 2 and 3 (d) 1 and 4

UPRVUNL AE 21.08.2016

Ans. : (b)

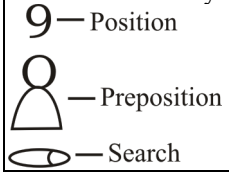
881. Which one among the following is a 'THERBLIG':

(a) Get (b) Step
(c) Put (d) Position

RPSC VPITI 14.02.2016

Ans. (d) : Position and orienting an object in the defined location.

Therblig are 18 kinds of elemental motion used in study of motion economy in the work place.



882. Work sampling is used to find

(a) Percentage occurrence of certain activity
(b) Rating of job
(c) Pre-determined motion time data
(d) Quality control of incoming inventory

RPSC VPITI 14.02.2016

Ans. (a) : Work sampling is the statistical technique used for determining the proportion of time spent by workers in various defined categories of certain activity.

Example- Setting up a machine assembling two parts idle.

883. Which of the following is a guiding principle of scientific management ?

(a) Mathematical models for every aspect
(b) One best way to do a job
(c) Experimentation
(d) All of these

RPSC AE (GWD) 06.05.2016

Ans. (b) : Principles of scientific management- Five Principles of scientific management are not rule of thumb, Harmony not discard, cooperation not individualism and a few others.

Under scientific management each element (or component) of any job and motions required to perform it, are scientifically analyzed to determine the standard time and standard methods of doing it which maximum efficiency.

884. The management approach that relies heavily on analytical mathematical techniques is called

(a) Quantitative approach
(b) System approach
(c) Contingency approach
(d) Operational approach

RPSC AE (GWD) 06.05.2016

Ans. (a) : The management approach that relies heavily on analytical mathematical techniques is called quantitative approach.

885. The function of management that involves identification and grouping of activities to be performed and dividing them among individuals is called

(a) Planning (b) Staffing
(c) Organising (d) Controlling

RPSC AE (GWD) 06.05.2016

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Ans. (c) : The function of management that involves identification and grouping of activities to be performed and dividing them among individuals is called organising.

886. Setting standards and measuring actual performance is an important function of management and is called

(a) Controlling (b) Directing
(c) Standardizing (d) Coordinating

RPSC AE (GWD) 06.05.2016

Ans. (a) : Setting standards and measuring actual performance is an important function of management and is called controlling.

887. Which of the following is a maintenance or hygienic factor according to Herzberg's model of motivation?

(a) Wages (b) Recognition
(c) Advancement (d) Responsibility

RPSC AE (GWD) 06.05.2016

Ans. (a) : According to Herzberg's motivation factors (also called job satisfiers) are primarily intrinsic job elements that lead to satisfaction. Hygiene factors are extrinsic elements of the work environment.

888. Which of the following is not a characteristic of recruitment through internal sources?

(a) Healthy atmosphere
(b) Reduced labour turnover
(c) Higher cost of training
(d) Helpful in maintaining motivation

RPSC AE (GWD) 06.05.2016

Ans. (c) : Higher cost of training

889. The method of merit rating in which each employee is compared with every other employee, is called

(a) Man to man comparison method
(b) Ranking method
(c) Paired comparison method
(d) Grading system

RPSC AE (GWD) 06.05.2016

Ans. (c) : Paired comparison method- In this method every person is compared trade wise, with other persons one at a time, the number of time one person is compared with others is recorded on a piece of paper. There numbers help in yielding rank of employees.

890. Which of the following incentive plans does not guarantee minimum wage?

(a) Gantt Task and Bonus system
(b) Emerson efficiency plan
(c) Merrick's differential piece rate system
(d) All of the these

RPSC AE (GWD) 06.05.2016

Ans. (c) : Halsey plan wage incentive plan guarantees minimum wage to a worker and bonus is paid for the fixed percentage of time saved.

891. Which of the following ratios give an indication of short term financial position and solvency of a firm or company?

(a) Acid test ratio
(b) Debt service ratio
(c) Debt equity ratio
(d) Inventory turn over ratio

RPSC AE (GWD) 06.05.2016

Ans. (a) : The acid test or quick ratio, compares a company's most short-term assets to its most short-term liabilities to see if a company has enough cash to pay its immediate liabilities, such as short term debt. The acid test ratio disregards current assets that are difficult to liquidate quickly such as inventory.

$$\text{Quick ratio} = \frac{\text{Cash + Accounts Recoverable}}{\text{Current Liabilities}}$$

892. Which of the following is NOT a feature of Micro-motion study?

(a) Dividing the activities into small elements called "therbligs"
(b) Recording the motion with a camera
(c) Studying finer details with the microscope
(d) Recording the motions on a SIMO chart

RPSC AE (GWD) 06.05.2016

Ans. (c) : Micro-motion/therbligs- Micro-motion study should be used when it is economical to do so (short cycle highly repetitive operations, large volume production or operation performed by a large number of workers)

Features of micro-motion study is

- Dividing the activities into small elements called therbligs.
- Recording the motion with a camera
- Recording the motion on a SIMO chart.

893. The therblig ⌓ represents the activity of

(a) releasing an object at desired location
(b) holding an object
(c) transportation of an object with crane
(d) none of these

RPSC AE (GWD) 06.05.2016

Ans. (b) : The therblig (micro motion study) "⌓" represents the activity of holding an object.

894. In which of the following replacement analysis methods, the interest rate enters into calculations?

(a) Payback period method
(b) Annual cost method
(c) Total life average method
(d) Rate of return method

RPSC AE (GWD) 06.05.2016

Ans. (b) : The replacement analysis methods, the interest rate enters into calculation by annual cost method.

895. Which of the following replacement methods take into account different types of depreciation and obsolescence?

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(a) MAPI method
(b) Discounted rate of return method
(c) Present worth method
(d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (a) : Procedures for equipment replacement analysis and capital investment analysis from the MAPI (Machinery and Allied Products Institute).

896. Which organisation is used when an organisation has to handle a variety of projects ranging from small to large?

(a) Functional (b) Line & staff
(c) Matrix (d) Project

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : Matrix organisation is used when an organisation has to handle a variety of project ranging from small to large.

Functional Organization– It is based on an organization being divided into smaller groups with specific task.

Line and Staff Organisation– In line organisation top management has complete control, and chain of command is clear and simple.

897. If T is observed time in a time study and R is a performance rating in percentage, then the normal time for a job is given by

(a) $\frac{T \times R}{100}$ (b) $\frac{T \times 100}{R}$

(c) $T\left(1+\frac{R}{100}\right)$ (d) $T\left(1-\frac{R}{100}\right)$

RPSC AE (GWD) 06.05.2016

Ans. (a) : Normal time for a job = $\frac{T \times R}{100}$

Where, T = observed time
R = Performance rating in percentage

898. In Westinghouse system of performance rating, the performance is rated in four factors. Which of the following is NOT a factor employed in this system ?

(a) Skill (b) Effort
(c) Efficiency (d) Conditions

RPSC AE (GWD) 06.05.2016

Ans. (c) : Westinghouse system of Rating- This method consider four factors in evaluating the performance of operator, skill, effort, conditions and consistency.

899. Value is the ratio of :

(a) Productivity/cost (b) Output/cost
(c) Output/input (d) Utility/cost

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : Value = $\frac{\text{Utility}}{\text{Cost}}$

Value is defined as the ratio of function to cost value can therefore be increased by either improving the function or reducing the cost.

900. Time allowed for a job is 18 hours, the worker performs the job in 14 hours. His hourly wage rate is 10 Rs. His incentive, according to Halsey plan and Rowan plan will be respectively

(a) Rs. 20 and Rs. 31.11
(b) Rs. 31.11 and Rs. 20
(c) Rs. 10 and Rs. 21.11
(d) Rs. 21.11 and Rs. 10
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : Halsey Plan–Direct wage, Bonus 50% of (Time served × Rate per hour).

Rowan–

$$\text{Bonus} = \frac{\text{Time saved}}{\text{Time allowed}} \times \text{Time taken} \times \text{Rate per hour}$$

$$= \frac{4}{18} \times 14 \times 10 = 31.11 \text{Rs.}$$

Halsey plan,

Bonus = Time saved × Rate × 50%
= 4 × 10 × 50%
= 20 Rs.

901. Individual motions of a worker are studied in

(a) work analysis (b) operation analysis
(c) motion analysis (d) none of these

GPSC Lect. 23.10.2016

Ans. (c) : Motion analysis is used in analysis of individual motions of worker with the help of image sequences, camera or high speed camera.

902. Only two symbols are used to draw the following chart

(a) Flow process chart
(b) Man-machine chart
(c) Outline process chart
(d) Two-handed chart

APPSC AEE Screening Test 2016

Ans. (c) : An outline process chart is a process chart which gives overall view of a process by recording only the main operations and sequences in proper sequence. So it is obvious that such a chart requires only symbols for "operations and inspection".

903. The subdivision of an operation into Therbligs and their analysis is known as

(a) Work study (b) Time study
(c) Micro motion study (d) None

APPSC AEE Screening Test 2016

Ans. (c)

Micro Motion Study/Therbligs

- Technique of recording and analyzing the timing of basic elements of an operation.
- Developing best possible pattern of movement operator performing minimum effort and fatigue for repeated operations.
- Consists of talking motion pictures of the operation with a clock in the picture (or with a video camera running at a known speed).

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904. Which of the following is not a technique of work measurement?

(a) Time study (b) Synthesis
(c) Work sampling (d) Random sampling

Nagaland CTSE 2016 Paper-I

Ans. (d) : Various techniques of work measurement care:-
(1) Time study (stop watch technique).
(2) Synthesis.
(3) Work sampling.
(4) Predetermined motion and time study.
(5) Analytical estimating.

905. Values for the probabilistic inputs to a simulation are

(a) controlled by decision maker
(b) calculated by fixed mathematical formula
(c) randomly generated based on historical information
(d) selected by the decision maker

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Values for the probabilistic inputs to a simulation are randomly generated based on historical information.

906. The determination of standard time in a complex job system is best done through a

(a) Group timing technique
(b) Analysis of micro motion
(c) Stop watch time and study
(d) Analysis standard date system

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : The determination of standard time in a complex job system is best done through a analysis standard date system.

907. Performance rating of an employee may be expressed as:

(a) (Normal performance) × (Observed performance)

(b) $\sqrt{\left(\frac{\text{Observed performance}}{\text{Normal performance}}\right)}$

(c) $\left(\frac{\text{Observed performance}}{\text{Normal performance}}\right)$

(d) $\left(\frac{\text{Normal performance}}{\text{Observed performance}}\right)^2$

UPRVUNL AE 07.10.2016

Ans. (c) : Performance rating of an employee may be expressed as

$$= \left(\frac{\text{Observed performance}}{\text{Normal performance}}\right)$$

908. A competitive firm will maximize profit at the output, where

(a) the difference between price and marginal cost is highest
(b) the difference between marginal revenue and price is highest
(c) the price is higher than the average total cost by the largest possible amount
(d) the excess of total revenue over the total cost is greatest

BPSC Poly. Lect. 2016

Ans. : (d) A competitive firm will maximize profit at the output, where the excess of total cost by revenue over the total cost is greatest.

909. Analysis is an operation, when carried out in terms of individual motions of worker is known as

(a) work analysis (b) motion analysis
(c) operation analysis (d) None of the above

BPSC Poly. Lect. 2016

Ans. : (b) Analysis is an operation when carried out in terms of individual motion of a worker is known as motion Analysis

910. 'Value engineering' is used in

(a) Understanding customer's requirements.
(b) Designing products according to the customer's requirements.
(c) Producing products according to the customer's requirements.
(d) Providing products to customer with enhanced functionality at no additional cost.

UPPSC AE 12.04.2016 Paper-I

Ans. : (d) Value engineering is used in providing products to customer with enhanced functionality at no additional cost.
Value engineering:- The value engineering deals with the concepts that are applied at the design or pre-manufacturing stage so that the unnecessary cost cut down without Impairing the function or utility of the product.

911. If in a process on the shop floor, the specification are not met, but the charts for variables show control, then which of the following actions should be taken?

(a) change the process
(b) change the method of measurement
(c) change the worker or provide him training
(d) change the specifications or upgrade the process

MPPSC AE 2016

Ans. : (c) If in a process on the shop floor, the specification are not met, but the charts for variables show control then change the worker or provide him training should be taken.

912. Functional organization was introduced by :

(a) F.W Taylor (b) Gilbreth
(c) Henri Fayol (d) Gantt

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (a) : F.W. Taylor, father of scientific management developed the concept of functional organization. He recommended functional organization even at the shop level where workers have to produce goods.
Taylor suggested the division of supervisory functions into two groups, viz:
- Office specialist and;
- Shop specialists.

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913. The main objective of work measurement is to:
(a) plan and schedule or production
(b) estimate the selling prices and delivery dates
(c) formulate a proper incentive scheme
(d) all of the above

Kerala PSC IOF 19.04.2016

Ans. (d) : The main objective of work measurement is to –
- Plan and schedule or production.
- Estimate the selling prices and delivery dates.
- Formulate a proper incentive scheme.

914. An operation process chart represents graphically -
(a) Operation, transport and delay
(b) Operation, inspection and time required for operation
(c) Operation, transportation and inspection
(d) Points at which materials enter the process, operation, inspection and time required for operation

NPCIL ET, 2015

Ans. (d) : An operation process chart represents graphically points at which materials enters the process, operation inspection and time required for operation. This is graphic representation of the sequence of all operations and inspections involved in the process. In this chart, all components or materials entry points are indicated by horizontal lines and the processing of there materials are shown vertically operation process charts provide a compact overall view of the whole system of operations involved in the manufacture of a product.

915. The object of plant layout could be :
(a) to reduce material handling cost
(b) for effective utilization of space
(c) to increase production
(d) any of these

VIZAG STEEL MT 2015

Ans. (d) : The object of plant layout could be
- To reduce material handling cost.
- For effective utilization of space.
- To increase production.

916. Which of these would an Operations Manager not be responsible for ?
(a) Safety and Maintenance
(b) Selecting Suppliers
(c) Recruiting Employees
(d) Sales and Marketing

KPSC ADF 2015

Ans. (d) : Sales and Marketing.

917. An operations strategy is created directly from .....
(a) Corporate strategy (b) Marketing strategy
(c) Business strategy (d) Human strategy

KPSC ADF 2015

Ans. (c) : An operations strategy is created directly from business strategy.

918. An allowance of time given to worker, who is working on more than two machines at a time is called
(a) Contingence allowance
(b) Process allowance
(c) Interference allowance
(d) Fatigue allowance

TSPSC AEE 2015

Ans. : (c) An allowance of time given to worker who is working on more than two machines at a time is called interference allowance.

919. The process of discovering and identifying the pertinent information relative to the nature of a specific job is known as
(a) job analysis (b) job description
(c) job classification (d) job identification

(KPSC AE 2015)

Ans. : (a) The process of discovering and identifying the pertinent information relative to the nature of a specific job is known as job analysis.

920. The most flexible criteria for analysing an economic investment is
(a) rate of return criteria
(b) salvage value
(c) Present value method
(d) future value method

(KPSC AE 2015)

Ans. : (b) The most flexible criteria for analysing an economic investment is salvage value.

921. Indicated results are most accurate when the pointer in the analogue read out system lies
(a) around the mid-third of the scale
(b) in the first quarter of the scale
(c) in the second quarter of the scale
(d) around the full capacity of the scale

(KPSC AE 2015)

Ans. : (a) Around the mid- third of the scale.

922. Work sampling observations are taken on the basis of
(a) detailed calculations
(b) convenience
(c) past experience
(d) table of random numbers

TSPSC AEE 2015

Ans. : (d) Work sampling observations are taken on the basis of table of random numbers.

923. Observed time for an element is 1 (one) minute. The rating factor is 90% and allowance of 20% of normal time. The standard time for this job is
(a) 0.9 minute (b) 1.8 minute
(c) 1.18 minute (d) 1.08 minute

TSPSC AEE 2015

Ans. : (d) Observed time = 1 min
Normal time = observed time × Rating
$= 1 \times 0.9$
$= 0.9$ min

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$$\text{Standard time} = \text{Normal time}\left(1+\frac{\%\text{Allowances}}{100}\right)$$
$$= 0.9\left(1+\frac{20}{100}\right)$$
$$= 0.9(1+0.2)$$
$$\boxed{\text{Standard time} = 1.08 \text{ min}}$$

924. PMTS is a
(a) Predetermined Motion Time Systems
(b) Predetermined Method Time Systems
(c) process of Motion and Time Study
(d) Process of Method and Time Study

TSPSC AEE 2015

Ans. : (a) Predetermined Motion Time Systems

925. One of the following is a drawback in hydraulic drives:
(a) Drive is smooth
(b) Effective of temperature on oil viscosity
(c) Self-lubricating system,
(d) Faster and reverse without shock

OPSC AEE 2015 Paper-I

Ans. : (b) Effective of temperature on oil viscosity is a drawback in hydraulic drives.

926. In a time study exercise, the time observe for an activity was 54 seconds. The operator had a performance rating of 120. A personal allowance of 10% is given. The standard time for the activity in seconds.
(a) 54 (b) 60.8
(c) 58.32 (d) 72

GPSC AE 26.07.2015, GATE 2000

Ans. (d) : Standard time = Observed time × Rating factor (1 + Allowance)
$$= 54\ \ 54\times\frac{120}{100}\times 1.1$$
$$= \frac{6480\times 1.1}{100}$$
$$= 71.28$$

927. Which of the following charts is used as "birds eye view" in plant layout design?
(a) Operation process chart
(b) Man-machine chart
(c) Travel chart
(d) Flow process chart

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Operation process chart is used as "birds eye view" in plant layout design.

928. Which of the following is not a type of flow process chart?
(a) material (b) method
(c) machine (d) man

Haryana PSC Civil Services Pre, 2014

Ans. (b) : Method is not a type of flow process chart.

929. The therblig symbol used for micro motion of 'release' is _____
(a) R (b) RE
(c) RL (d) RS

Haryana PSC Civil Services Pre, 2014

Ans. (c)

Symbol	Name	Abbreviation
	Release Load	RL
	Rest for overcoming fatigue	R

RE & RS are not therblig symbol

930. _____ process chart is a graphic representation of the sequence of all the operations and inspections involved in a process or procedure.
(a) operation (b) outline
(c) travel (d) flow process

Haryana PSC Civil Services Pre, 2014

Ans. (d) :
Note- Official answer given by commission is (a).

931. Formula for calculating standard time (S) when observed time (O), rating factor (r) and PFD allowance are known is :
(a) S = (O/r) + PFD (b) S = (O×r) – PFD
(c) S = (O/r) PFD (d) S = O×r× (1+ PFD)

Haryana PSC Civil Services Pre, 2014

Ans. (d)
$$\text{Standard time} = \underset{(N)}{\text{Normal time}}\left[1+\%\,\text{allowance}\right]$$
Normal time = observation time × Rating factor
$$N = O \times r$$
$$\boxed{\text{Standard time } (S) = O \times r\left[1+PFD\right]}$$
where
PFD → % allowance

932. As per the principle of motion economy related to the sitting standing work place for mates the high clearance should range between :
(a) 10" to 12" (b) 8" to 10"
(c) 6" to 8" (d) 12" to 14"

Haryana PSC Civil Services Pre, 2014

Ans. (b) : 8" to 10"

933. Which one of the following is basic system of time study
(a) Man-machine diagram
(b) Statistical sampling
(c) Threblings
(d) None of these

MPPSC State Forest Service Exam, 2014

Ans. (b) : Statistical sampling is drawing a set of observations randomly from a population distribution often we do not know the nature of the population distribution, so we cannot use standard formulas to generate estimates of one static or another.

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934. In time study, standard time is always

(a) More than normal time
(b) Less than normal time
(c) Equal to allowances
(d) None of these

MPPSC State Forest Service Exam, 2014

Ans. (a) : The standard time is the product of three factors, observed time the time measured to complete the task, performance starting factor, 100% is normal, So a standard time is more than normal time.

935. The strategy that involves production of the amounts demanded at any given time is :

(a) Production level strategy
(b) Chase strategy
(c) Subcontracting strategy
(d) TQM strategy

UPRVUNL AE 2014

Ans. : (b) The strategy that involves production of the amount demanded at any given is chase strategy.
When the company faces variable demand they vary their production as the demand varies.

936. The managerial function controlling is

(a) the process of predicting the demand
(b) the measurement of satisfaction of workers
(c) the measurement and correction of performance
(d) process of making the work environment clean

TNPSC AE 2014

Ans. (c) : The managerial function controlling is the measurement and correction of performance.

937. The time study of a work operation at a restaurant yielded an average observed time of 9 minutes. The analyst rated the observed worker at 90%. The normal time of this operation is

(a) 9 minutes (b) 81 minutes
(c) 8.1 minutes (d) 90 minutes

TNPSC AE 2014

Ans. (c) : The normal time of this operation

$$= 9 \times \frac{90}{100}$$
$$= 8.1 \text{ minutes}$$

938. The factors influencing job design is

(a) individual differences
(b) technology involved
(c) organization structure and internal climate
(d) individual difference, technology involved, organization structure, and internal climate

TNPSC AE 2014

Ans. (d) : Individual difference, technology involved organization structure and internal climate factors are influencing job design.

939. A change within a organization to higher position that has greater responsibility and requiring advanced skills is known as

(a) multiskilling (b) promotion
(c) job enrichment (d) job evaluation

TNPSC AE 2014

Ans. (b) : A change within a organization to higher position that has greater responsibility and requiring advanced skills is known as promotion.

940. Work study is concerned

(a) improving present method and finding standard time
(b) motivation of workers
(c) improving production capability
(d) all of these

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (d) : Work study is concerned with improving present method and finding standard time. The work measurement is concerned with the elimination of ineffective time and establishment of time standards for a job work study is mainly aimed of determining the most efficient method of performing job.

941. Scheduling gives information about

(a) when work should start and how much work should be completed in a certain period
(b) when work should completed
(c) how idle time can be minimized
(d) proper utilization of machines

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (a) : Scheduling gives information about when work should start and how much work should be completed during a certain period.

942. An optimum project schedule implies

(a) optimum utilization of men, machines and materials
(b) lowest possible cost and shorter possible time for project
(c) timely execution of project
(d) to produce best results

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (b) : The aim of project managers is to strike balance between the cost and the time resulting in an optimum project schedule. Hence an optimum project schedule implies the lowest possible cost and the shortest possible optimum time for the project.

943. The allowance factor in a time study is to

(a) adjusts normal time for errors and rework
(b) adjusts standard time for lunch breaks
(c) adjusts normal time for personal needs, unavoidable delays and fatigue
(d) allows workers to rest every 20 minutes

TNPSC AE 2014

Ans. (c) : The allowance factor in a time study is to adjusts normal time for personal needs, unavoidable delays and fatigue.

944. Method study is concerned with :

(a) Finding the best technique to do the job
(b) Finding the best way of doing the job
(c) Increase the time
(d) Finding the time required to complete the job

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Method study is concerned with finding the best way of doing the job.

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945. Work study is a generic term for :
(a) Method study and part study
(b) Work measurement and time study
(c) Method study and work measurement
(d) Time study and activity study

TNPSC AE (Industries) 09.06.2013

Ans. (c) :

946. Why Frederick Taylor has been called "The father of scientific management" ?
(a) He did Hawthorne study
(b) He suggested a comprehensive social system
(c) His primary concern was to increase productivity through application of scientific method
(d) He divides industrial activities into six groups

TNPSC AE (Industries) 09.06.2013

Ans. (c) :

947. When will the string diagram be used in work study ?
(a) To reduced the number of machines
(b) Team of workers are doing multiple activities
(c) When the paths are many and repetitive
(d) When the number of equipments are less

TNPSC AE (Industries) 09.06.2013

Ans. (c) :

948. The main object of scientific layout is :
(a) to produce better quality of product
(b) to utilize maximum floor area
(c) to minimize production delays
(d) all of these

J&K PSC Civil Services Pre, 2013

Ans. (d) : Scientific layout is used to produce better quality of product, utilize maximum floor area and minimize production delay.

949. The procedure of modifying work content to give more meaning and enjoyment to the job by involving employees in planning, organisation and control of their work, is termed as :
(a) job enlargement (b) job enrichment
(c) job rotation (d) job evaluation

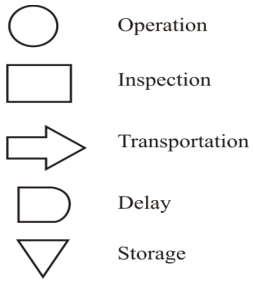
J&K PSC Civil Services Pre, 2013

Ans. (b) : The procedure of modifying work content to give more meaning and enjoyment to the job by involving employees in planning, organisation and control of their work, is termed as job enrichment.

950. What do the symbols and represent in work study?
(a) Delay an storage respectively
(b) Processing and storage respectively
(c) Storage an inspection respectively
(d) Inspection and storage respectively

RPSC AEN Pre-2013

Ans. (d) :



951. For a certain cycle of operations on a job, the observe time is 10 minutes. The performance rating of the worker is 120% and total allowances are 10% of normal time. What will be the standard time for the worker?
(a) 9.17 minutes (b) 13.0 minutes
(c) 13.2 minutes (d) 14.0 minutes

RPSC AEN Pre-2013

Ans. (c) : Given,
Observed time / selected time
= 10 min
Rating = 120%
Allowance = 10%
As we know that,

$$\text{Normal time} = \text{Observed time} \times \frac{\text{Rating}\%}{100}$$

$$= 10 \times \frac{120}{100}$$

$$= 12 \text{ min}$$

$$\text{Allowance } 10\% \ = 12 \times \frac{10}{100} = 1.2 \text{ min}$$

$\therefore$ Standard time = Normal time + Allowances
= 1.2 + 12
= 13.2 min

952. MTM (method-time-measurement) is a work measurement technique by
(a) pat data comparison
(b) pre-determined motion time and system
(c) stop watch study
(d) work sampling study

TNPSC AE 2013

Ans. (b) : MTM (method-time-measurement) is a work measurement technique by pre-determined motion time and system.

953. The Management by Objectives (MBO) involves
(a) setting the goals at workers level
(b) setting the goals at engineers level
(c) setting the goals at all the levels of organization
(d) setting the goals at manager's level

TNPSC AE 2013

Ans. (c) : The Management by Objectives (MBO) involves setting the goals at all the levels of organization.

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954. Advantage of value analysis is

(a) reduction of scrap
(b) better manufacturing
(c) better quality of performance
(d) increase productivity

VIZAG STEEL MT 18.08.2013

Ans. (c) : Advantage of value analysis-

- A systematic approach to solving problems.
- Self satisfaction from contribution their ideas to solution and in participating in an energetic workshop.
- Improvements in their knowledge and ability because workshop team members with different backgrounds.
- Better quality of performance.

955. Benefit of applying value analysis lies in

(a) Decrease of scrap
(b) Increased productivity
(c) Superior quality of performance
(d) Improved manufacturing

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : Advantage of value analysis–
(1) It uses cheaper and better material
(2) It uses an efficient and economical process
(3) It reduces the cost of the product
(4) It improves product design
(5) It increases productivity and thus increase profit

956. Which of the following is not the characteristics of work sampling ?

(a) Any interruption during study will not affect the results.
(b) The study causes less fatigue.
(c) Uneconomical for short cycle jobs.
(d) A stop watch is needed.

UKPSC AE 2012 Paper-I

Ans. (d) : A stop watch is needed.

957. Goodwill of an enterprise is termed as :

(a) Liquid asset (b) Volatile
(c) Fictitious asset (d) Liability

PSPCL AE, 2012

Ans. (c) : Fictitious Asset– Goodwill is usually associated with a business acquisition. The amount paid in excess of the market value is known as good will.

958. Fifty observations of a production operations revealed a mean cycle time of 10 min. The worker was evaluated to be performing at 90% efficiency. Assuming the allowances to be 10% of the normal time, the standard time (in minutes) for the job is:

(a) 0.198 (b) 7.3
(c) 9.0 (d) 9.9

APGENCO AE 2012

Ans. (d) : Observe time = 10 min
Rating factor = 90%
Normal time = OT × RF
$= 10 \times 0.9 = 9$ min
Standard time = NT + 10% NT
$= 9 + \frac{10}{100} \times 9$
$= 9.9$ min

959. Therbligs consist of

(a) 15 basic hand motions
(b) 16 basic hand motions
(c) 18 basic hand motions
(d) 20 basic hand motions

BPSC AE 2012 Paper - VI

Ans. : (c) : Therbligs are 18 kinds of elements motions used in the study of motion economy in the workplace. A workplace task in analyzed by recording each of the therblig units for a process with the results used for optimization of manual labour by eliminating unnecessary movements.

960. Which of the following manufacturing systems has shortest delivery time?

(a) Make to order (b) Assemble to order
(c) Engineer to order (d) Make to stock

BPSC AE 2012 Paper - VI

Ans. : (d) : Make to stock has shortest delivery time.

961. Determination of standard time in complex job system is best done through

(a) stop-watch time study
(b) analysis of micro motions
(c) group timing techniques
(d) analysis of standard data system

TNPSC ACF 2012, ESE 1996

Ans. (a) : Standard time in complex job system is best done through stop watch time study.

962. Method-Time measurement is based on

(a) stop-watch study
(b) work sampling study
(c) pre-determined motion time systems
(d) past data comparison

TNPSC ACF 2012, ESE 1999

Ans. (a) : Method Time measurement (MTM) is a predator mind motion time system that is used primarily in industrial setting to analyze the methods used to perform any manual or task and as a product of that analysis set the standard time in which a wortur should complete task.

963. Which one of the following correctly explains process capability?

(a) Maximum capacity of the machine
(b) Mean value of the measured variable
(c) Lead time of the process
(d) Maximum deviation of the measured variable of the components

TNPSC ACF 2012, ESE 1998

Ans. (d) : Maximum deviation of the measured variable of the components explains the process capability.

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964. Predetermined motion time systems include :

(a) Method Time Measurement
(b) Work Factor Systems
(c) Basic Motion Time Study
(d) All of these

OPSC Civil Services Pre. 2011

Ans. (d) : Predetermined motion time systems include
- Method Time Measurement
- Work Factor Systems
- Basic Motion Time Study

965. Merit rating is the method of determining worth of:

(a) a job
(b) an individual employee
(c) a particular division in workshop
(d) overall quality

RPSC ACF-2011

Ans. (b) : Merit rating is the method of determining worth of an individual employees.

966. Which of the following characteristics is not usually associated with batch production?

(a) products made to customer order
(b) low volume
(c) stable, predictable demand
(d) general purpose equipment

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Characteristics of batch production-
⇒ Work is at repetitive nature & low volume.
⇒ There is a functional layout
⇒ Products made to customer order, not stable, predictable demand.
⇒ One operation is carried out on whole batch & then passes to next operation.
⇒ Same type of machine is arranged at one place.
Note-Official answer is given as (c).

967. A diagram showing the path followed by man and materials while performing a task is known as:

(a) String diagram (b) Flow process chart
(c) Travel chart (d) Flow diagram

RPSC AE GWD, 2011

Ans. (d) : Flow diagram– A diagram showing the path followed by men and materials while performing a task is known as flow diagram.
String diagram – movement of materials
Travel chart – material handling
Flow process chart – activities or operations performed on job.

968. A list of methods for recording the existing and proposed ways of doing work is given below:

(1) Flow diagram
(2) Man and material flow process chart
(3) Two handed process chart
(4) SIMO (Simultaneous Motion) chart

Which of these is used for recording microscopic motion?

(a) (1) only (b) (2) and (3)
(c) (3) and (4) (d) (4) only

RPSC AE GWD, 2011

Ans. (c) : Two handed process chart and SIMO chart is used for recording microscopic motion.

969. Recording techniques in Method Study are:

(a) Flow Process Chart, SIMO chart, Cycle Chart, Travel Chart.
(b) Flow Process Chart, SIMO chart, Activity relationship chart, work sampling chart.
(c) SIMO chart, Travel chart, Chart, MTM chart, Precedence diagram.
(d) SIMO chart, Activity relationship Chart, MTM chart, Precedence diagram.

RPSC AE GWD, 2011

Ans. (a) :

970. Work Sampling involves :

(a) Making of large number of observations at regular constant interval of time over a specified time period.
(b) Making a large number of observations for observed time to calculate standard time.
(c) Making a large number of observations at random time interval over a specified time period.
(d) Making a large number of observations at random time interval for method time measurement.

RPSC AE GWD, 2011

Ans. (b) : Work sampling involves making a large number of observations for observed time to calculate standard time.

$$\text{Standard time} = \text{Normal time}\left(1 + \%\,\text{allowance}\right)$$

Normal time = Rating factor × observation time.

971. Which one of the following symbols will be used for the event 'Reach for bolt' in a two handed process chart for the 'bolt and nut assembly'?

(a) Operation symbol
(b) Inspection symbol
(c) Transportation symbol
(d) Storage symbol

RPSC AE GWD, 2011

Ans. (a) :

972. Match List I with List II and select the correct answer using the code given below the Lists :

List I	List II
A. F.W. Taylor	1. Worked on task and bonus plan
B. Henry L.Gantt	2. Advocated sharing the benefits of improved productivity with the workers

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C. Frank Gilberth — 3. Initiated the concept of time study and applied it to production operations

D. F.A. Halsey — 4. Developed method study as a tool for work analysis

5. Developed the service flow model

Code :	A	B	C	D
(a)	3	1	5	2
(b)	2	5	4	3
(c)	2	3	5	4
(d)	3	1	4	2

UPSC JWM Advt. No.-52/2010

Ans. (d) :

A	B	C	D
↓	↓	↓	↓
3	1	4	2

973. The standard time for a job is 8 hrs, hourly rate is Rs. 10, and the worker completes the job in 6 hrs. The total wages earned by the worker as per Halsey incentive plan is :

(a) Rs. 70 (b) Rs. 10
(c) Rs. 60 (d) Rs. 50

UPSC JWM Advt. No.-52/2010

Ans. (c) : Solution- Given;
Standard time for job =8 hrs
Actual time of worker = 6 hrs.
hourly rate = Rs. 10/hr
According to Halsey incentive plan–
Total wages earned by worker = Actual time × hourly rate
= 6 × 10
= Rs. 60

974. Assertion (A) : Method study, motion analysis and time study are technique for improving productivity.
Reason (R) : All of them are used for establishing experimental time standards.
Codes :
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (a) : Method study, motion analysis & time study are micro analysis methods for increasing productivity & reducing idle time by setting up standard cycles (time).

975. Which one of the following is not a work measurement technique ?

(a) Work supply (b) Simulation
(c) Time study (d) Analytical Estimation

J&K PSC Civil Services Pre, 2010

Ans. (a) : Work supply is not a technique of work measurement.

976. Consider the following statements :
1. To prepare estimates of labour and costs
2. To prepare realistic work schedule
3. To establish better methods of work
4. To compare times required for alternative methods
Which of these are the objectives of work measurement?

(a) 1 only (b) 2 and 3
(c) 1, 2 and 4 (d) 2, 3 and 4

UPSC JWM Adv. No-16/2009

Ans. (c) : Objectives of work measurement-
- To prepare estimates of labour and cost
- To prepare realistic work schedule.
- To compare times required for alternative methods.

977. The stop watch readings of an element recorded by a time study engineer for a typical shop operation are : 10, 9, 11, 10, 11 and 9. How many more cycles should be timed to have the elemental time within ±5% accuracy 95% confidence level ?

(a) 60 (b) 56
(c) 55 (d) 54

APPSC IOF, 2009

Ans. (a) : 60

978. Let AV represent the average observed time for an element and PT be the pre-determined motion time standard for the same element. Then, the performance rating factor is given by:

(a) AV × PT (b) PT/AV
(c) AV/PT (d) AV + PT

APPSC IOF, 2009

Ans. (b) : Normal time or predetermined motion time
= Observed time × Performance rating factor
PT = AV × Performance rating factor

$$\text{Performance rating factor} = \frac{PT}{AV}$$

979. Analysis of an operation, when carried out in terms of individual motions of a worker is known as

(a) work analysis
(b) motion analysis
(c) time and motion analysis
(d) operation analysis

TNPSC AE, 2008

Ans. (b) : Analysis of an operation, when carried out in terms of individual motions of a worker is known as motion analysis.

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980. Time standards are used for

(a) performance evaluation of individual workers
(b) incentive payments
(c) cost estimating
(d) all of these

TNPSC AE, 2008

Ans. (d) : Time standards are used for performance evaluation of individual workers, incentive payments and cost estimating.

981. The incentive wage plan in which savings are expressed as a % of the standard time, is

(a) Halsey plan (b) Bedaux plan
(c) Rowan plan (d) Group plan

TNPSC AE, 2008

Ans. (a) : The incentive wage plan in which savings are expressed as a % of the standard time is Halsey plan.

982. In order to investigate the shortcomings of the existing and to develop an improved procedure, the analysis carried out is

(a) work analysis
(b) motion analysis
(c) time and motion analysis
(d) operation analysis

TNPSC AE, 2008

Ans. (d) : In order to investigate the shortcomings of the existing and to develop an improved procedure, the analysis carried out is operation analysis.

983. Micromotion study is :

(a) enlarged view of motion study
(b) analysis of only one stage of motion study
(c) time study of small components up to microseconds
(d) subdivision of an operation into therbligs and their analysis

J & K PSC Screening, 2006

Ans. (d) : Micromotion study is a techniques of work analysis used for work improvement. It uses motion pictures of different activities or movement with the help of camera.

984. Which one is not a therbligs.

(a) Use (b) Hold
(c) Dispatch (d) Inspection

J & K PSC Screening, 2006

Ans. (c) : Hold, Use and Inspection are therblings but "Dispatch" is not therblings.

985. Taylor's philosophy of scientific management pays attention to :

(a) Conversion of inputs to desired outputs :
(b) Time management
(c) Planning of the appropriate manpower
(d) Motion economy principles

J & K PSC Screening, 2006

Ans. (c) : Taylors philosophy of scientific management pays attention to planning of the appropriate man power.

10. Material Requirement Planning

986. In Industrial Engineering, 'MRP' is _____.

(a) Major Requirement Processing
(b) Maximum Resource Planning
(c) Materials Requirement Planning
(d) Maximum Retail Price

VIZAG STEEL MT 24.01.2021 Shift-II
GWSSB AAE, 27.12.2015
TNPSC AE, 2008

Ans. (c) : In industrial engineering MRP is Material Requirement Planning.

987. In an MRP system, component demand is

(a) forecasted
(b) established by the master production schedule
(c) calculated by the MRP system from the master production schedule
(d) ignored

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2006

Ans. (c) : MRP function is a computational technique with the help of which the master schedule for end products is converted into a detailed schedule for raw materials and components used in end products.

Input to MRP-

(i) Master production schedule
(ii) The bill of material
(iii) Inventory records relating to raw materials.

988. Which of the following term is not associated with material Requirement Planning?

(a) Time phasing
(b) Time bucket
(c) Planned order release
(d) Speed Rating

Nagaland PSC CTSE 2017, Paper-I
Nagaland CTSE 2016 Paper-I

Ans. (d) : Speed rating here is not a part of material required planning, only time phasing, times bucket, & planned order release are the term's of (MRP).

989. Select the full form of ERP in Business Management.

(a) Engineering Resource Planning
(b) Enterprise Resource Planning
(c) Energy Recovery Promotion
(d) Empower Recovery Providers

VIZAG STEEL MT 24.01.2021 Shift-II
UPRVUNL AE 2014

Ans. (b) : Enterprise Resource Planning (ERP) is a system used by companies to manage and integrate the important parts of their businesses. An ERP software system also integrate planning, purchasing inventory, sales, marketing, finance human resources and more.

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990. A 750 hours life test is performed on 10 components. If one component fails after 350 hours of operation and all others survive the test, then the failure per hour is :

(a) 1/7150 (b) 1/7500
(c) 1/750 (d) 1/350

Vadodara Municipal Corp. DEE, 2018
MPSC Poly. Lect. 2014

Ans. (a) : During the test 10 components are tested for 750 hours.

Total time T = 10 × 750 = 7500 hours

Total operating time = 7500 – 350 = 7150 hours.

Number of failure/hour $= \frac{1}{7150}$

991. MRP belongs to the following category :

(a) Dependent demands
(b) Independent demands
(c) Single order
(d) Highly uncertain demands

(HPPSC LECT. 2016), APPSC IOF, 2009

Ans. (a) : MRP belongs to the dependent demands category.

992. Materials requirements planning is driven by

(a) master production schedule
(b) total quality measurement
(c) overall production planning
(d) overall inventory planning

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) A master production schedule (MPS) is an anticipated build schedule for manufacturing end products or product options. It is an important component of production planning systems such as material requirement planning system.

993. Which one of the following is not an input of material requirement planning (MRP)?

(a) Master schedule (b) Order release
(c) Lead time (d) Bill of materials

VIZAG STEEL MT 24.01.2021 Shift-II

Ans. (b) : The input for the MRP (Material Requirement Planning) system includes the master production schedule, inventory records the existing available quantity of an item, and bill of material.

994. In material Requirements Planning, if the inventory holding cost is very high and the setup cost is zero, which one of the following lot sizing approaches should be used ?

(a) EOQ
(b) Lot-for-lot
(c) Base stock level
(d) Fixed order quantity, for 2 periods

APGCL AM, 2021

Ans. (b) : Lot-for-lot production : In this method the ordering quantity is selected to meet the demand for holding cast is very high compare to ordering cost.

995. Which of the following involves least fraction of cost to produce a product?

(a) Product design (b) Manufacturing
(c) Marketing (d) Material

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) :

996. Material requirements Planning (MRP) was first introduced and used by _____.

(a) General Motors (b) Toyota
(c) IBM (d) Microsoft

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (c) : Material requirements planning (MRP) was first introduced and used by IBM.

997. The output of an aggregate plan is

(a) Bill of material (b) Demand forecast
(c) Overtime (d) Total cost

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) Aggregate planning is determine the resource capacity needed to meet demand over an intermediate time horizon, it refers to product lines and it consist of all things.

998. The standardization of products is done to :

(a) reduce material cost
(b) make interchangeable manufacture possible
(c) simplify manufacturing process
(d) eliminate unnecessary varieties in design

APPSC Poly Lect. 13.03.2020

Ans. (b) : The standardization of products is done to make interchangeable manufacture possible.

999. Benefits of material requirement planning include:

(a) increased set-up cost
(b) prolonged production scheduling
(c) increased production lead time.
(d) reduced inventory levels

NLC GET 17.11.2020, Shift-II

Ans. (d) : Benefits of material requirement planning (MRP)

(1) Reduced Inventory levels without reduced customers service.
(2) Increase customer service and statisfaction
(3) Improved utilization of facilities and personnel
(4) Better inventory planning and scheduling.
(5) Faster response to market changes and shifts

1000. The penalty for not having materials when needed is

(a) Loss of materials cost (b) Loss of order cost
(c) Stock out cost (d) General losses

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : The penalty for not having materials when needed is stock out cost.

1001. Which one of the following is not part of material requirement planning?

(a) Bill of material (b) Inventory level
(c) Production schedule (d) Material price

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

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Ans. (d) : Material Requirement Planning (MRP)- It is used for material requirement and production planning. Material requirement planning is computational technique that converts the master. Schedule for the end product into detailed schedule for the raw material and components item. It does not include price of material.

Bill of material (BOM)- Inventory control and master production schedule (MPS) are the part of MRP.

1002. The major objective of Material Requirement Planning (MRP) is to:

(a) Get the right material to the right place at the right time minimizing the inventory cost
(b) Make the best use of resources available in terms of equipment, material and labour
(c) Maintain the finished goods delivery as per the expectation of the customer.
(d) Make the best use of resources available to maximize the inventory cost

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (a) : Objective of material Requirement planning

(i) To ensure that material and components are available for production and final products are ready for dispatch.
(ii) To maintain minimum inventory but also ensure right quantity of material is available at right time to produce right quantity of final products.
(iii) To ensure planning of all manufacturing processes. This scheduling of different job work as to minimize or remove any kind of idle time for machine and workers.
(iv) It helps in maintain minimum inventory levels.
(v) With minimum inventory levels, material planning also reduces associated costs.

1003. Material Requirement planning refers to

A. converting master schedule for end products to detailed schedule for raw materials and components
B. Ensuring that right material in right quantity is available at right time
C. Time phase of loading machines with materials and assigning jobs to operator
D. It is routing and progress control

Which the code for the correct answer from the options given below:

(a) Statement B alone (b) Statement B and C
(c) Statement A and B (d) Statement C and D

Karnataka PSC Lect., 27.05.2017

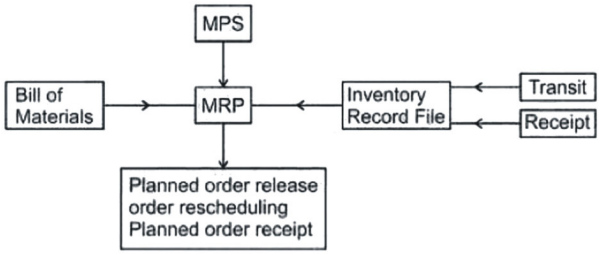
Ans. (c) : Material requirement planning refers to-
(i) Converting master schedule for end products to detailed schedule for raw materials and components.
(ii) Ensuring that right material in right quantity is available at right time.

1004. Which of the following is not an input for material requirement planning?

(a) Bill of materials
(b) Purchase order
(c) Inventory record file
(d) Master production schedule

CIL (MT) 2017 IInd Shift

Ans. (b) : Material Requirement Planning (MRP) is a method of working out the production plan in a multi – stage production system that produces many product and requires their raw material and sub – assemblies. It is used so that all the things needed are available at an appropriate time and production can be carried out without any delay.



Master production Schedule: It is a complete time table of scheduled production in future. It contains information about what product is to be produced, when it is to be produced and in what Quantity.

Inventory Record File: This file gives complete and up to date information about on hand inventory, scheduled receipt and transit inventory.

Bill of material (BOM): It gives information about how each final product is manufactured specifying all sub component item and their sequence of build – up in the final product.

So out of given options, Purchase order is not required for MRP.

1005. A master production schedule specifies

(a) The financial resource required for production
(b) What components are to be made and when
(c) What product is to be made and when
(d) The labour hours required for production

RPSC VPITI 14.02.2016

Ans. (c) : Master production schedule is complete time table of our scheduled production. In future it gives complete information about what product is to be made and in what quantity.

1006. Which of the following is not considered a direct advantage of training ?

(a) Better performance (b) Less wastage
(c) Less accidents (d) Better punctuality

RPSC AE (GWD) 06.05.2016

Ans. (d) : To direct advantage of training

- Better performance
- Less wastage
- Less accidents

1007. Which of the following is NOT associated with on the job training ?

(a) Economic and convenient
(b) Workers get trained in ideal or best way of doing the job
(c) Workers get trained in an actual environment
(d) No special facilities required

RPSC AE (GWD) 06.05.2016

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Ans. (d) : To associated with on the job training
- Economic and convenient
- Workers get trained in ideal or best way of doing the job.
- Workers get trained in on actual environment.

1008. The method of job evaluation in which job is divided into a number of factors which are further subdivided in a number of grades and each grade is assigned certain numerical weightage, is called
(a) Point rating method
(b) Factor comparison method
(c) Grading or ranking method
(d) Classification method

RPSC AE (GWD) 06.05.2016

Ans. (a) : The method of job evaluation in which job is divided into a number of factors which are further subdivided in a number of grades and each grade is assigned certain numerical weightage is called point rating method.

1009. For a manufacturing organisation which of the following is not considered fixed asset?
(a) Buildings
(b) Tools and equipments
(c) Transport vehicles
(d) Finished products

RPSC AE (GWD) 06.05.2016

Ans. (d) : Fixed assets are long term tangible assets which consists of land, building, machinery etc.
In manufacturing organization fixed asset, building tools and equipment, transport vehicles.

1010. Which of the following is called book of original account ?
(a) Ledger (b) Journal
(c) Cash book (d) Main account book

RPSC AE (GWD) 06.05.2016

Ans. (b) : A journal is also named the book of original entry, from when transactions were written in a journal prior to manually posting them to the accounts in the general ledger or subsidiary ledger.

1011. For which of the following industries, the proximity to raw materials will be the most important consideration in selection of location of the plant?
(a) Medicines (b) Televisions
(c) Steel mill (d) Publishing house

RPSC AE (GWD) 06.05.2016

Ans. (c) : Steel mill industries, the proximity to raw materials will be the most important consideration in selection of location of the plant. If raw materials are perishable and to be consumed such as, then the industries always tend to locate nearer to raw material source. Steel and cement industries can be such examples.

1012. Which of the following input data are needed for MRP?
1. Master Production Schedule
2. Inventory Position
3. Bill of material

(a) Only (1) (b) Only (2)
(c) Only (3) (d) Both (1) and (2)
(e) (1), (2) and (3)

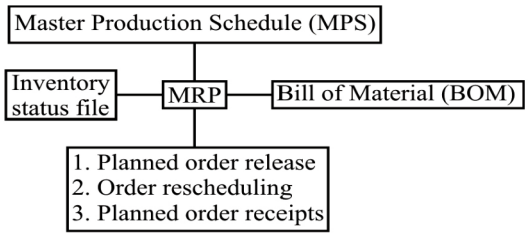
CGPSC Asstt. Workshop Supt., 17.07.2016
ESE 1998

Ans. (e) : The three major inputs of an MRP are as, MPS, the product records & inventory status records.

1013. Which one is not the input to MRP system?
(a) Master production schedule
(b) Bill of materials
(c) Inventory status file
(d) Management information system file

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : Material Requirements Planning (MRP)–It is a simple system of calculating arithmetically the requirements of the input materials at different points of time based on the actual production plan.
It can be seem from the figure that an MRP system has three major input components.



1014. MRP-II includes additional areas which are not part of MRP; are :
(a) Research and Development
(b) Finance and marketing
(c) Purchase and stores
(d) Organization and control

HPPSC Asstt. Prof. 18.11.2016

Ans. (b) : MRPII is an extension of the original materials requirements planning (MRPI) system having additional areas such as finance and marketing. Material requirements planning (MRP) is computer based system that can create detailed production schedules using real time data to coordinate the arrival of component material with machine and labour availability.

1015. Process Planning consists of .....
(a) specifying product details
(b) specifying details of work
(c) specifying quality standards
(d) none of these

GPSC Lect. 23.10.2016

Ans. (b) : Process planning is the relation between design and manufacturing process planning consists in defining the sequence of the steps that should be taken to make the product and specifying details of work.

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1016. A master production schedule contains information about

(a) Quantities and required delivery data of all final products
(b) Quantities and required delivery data of all sub assemblies
(c) Inventory on hand for each such assembly
(d) Inventory on hand for each final project

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : A master production schedule contains information about quantities and required delivery data of all final products.

1017. The key features of material requirement planning system are

(a) Planned order release
(b) Time-phasing of requirement
(c) Provisions for rescheduling
(d) All options are correct

MPPSC AE 08.11.2015

Ans. (d) : The key features of material requirement planning system are-

- Planned order release
- Time-phasing of requirement
- Provisions for rescheduling

1018. Consider the following input data in case of MRP?

1. **Master production schedule**
2. **Inventory position**
3. **Machine capacity**
4. **Details of delivery**
5. **Bill of materials**

Which of the above are needed for determination of MRP?

(a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 2, 5 (d) 1, 3, 4
(e) 1, 3

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : The information input into MRP systems comes from three main sources:
A bill of materials, a master schedule and an inventory records file.

1019. Consider the following statements :

(1) MRP system is useful for dependent demand items
(2) MRP reduces inventory cost.
(3) A Bill of Material (BOM) is a list showing materials and their quantities required to produce the end item.
(4) Master Production Schedule (MPS), bill of materials and inventory status files are three basic input of an MRP system.

Of these statements :

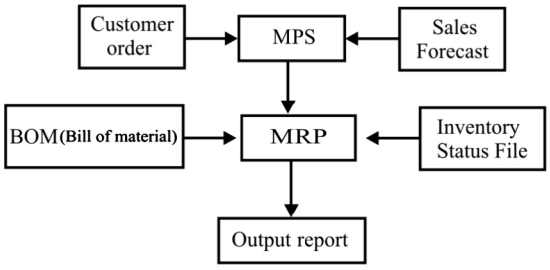
(a) (1), (3) and (4) are true
(b) (1), (2) and (3) are true
(c) (1), (2), (3) and (4) are true
(d) (2), (3) and (4) are true

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : MRP- Material requirement planning (MRP) is a computational technique for determining the quantity and timing for acquisition of dependent items in order to satisfy MPS (Master Production Schedule) requirement. It is used for manufacturing control.

Purpose of MRP –

(1) Inventory reduction.
(2) Reduction in manufacturing time
(3) Reduction in delivery lead time
(4) Reduction delivery commitments.



All statements are true.

1020. The most important input to MRP system is

(a) Forecast
(b) Master production schedule
(c) Safety stock
(d) Plant capacity

GPSC Lect. (Poly.) 07.12.2014

Ans. (b) : The most important input to MRP system is–
(i) Master production schedule
(ii) Bill of material
(iii) Inventory record files
MRP – Material requirement planning

1021. In material handling "Principle of unit load" means :

(a) Instead or moving the small parts one by one, move it by group
(b) Measure the weight of parts
(c) Move/transport the materials carefully
(d) Move one by one with hand

TNPSC AE (Industries) 09.06.2013

Ans. (d) :

1022. MIS does not require

(a) database
(b) hardware and software
(c) procedures for analysis
(d) bills of materials

VIZAG STEEL MT 18.08.2013

Ans. (d) : MIS does not require - Bill of materials.

1023. The MIS structure with one main computer system is called

(a) distributed MIS structure
(b) hierarchical MIS structure
(c) centralized MIS structure
(d) decentralized MIS structure

VIZAG STEEL MT 18.08.2013

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Ans. (c) : Centralization refers to the process in which activities involving planning and decision making within an organization are concentrated to a specific leader or location. the executives and specialists who make critical decision are based in the head office.

1024. Number of management layers in a firm is determined by the

(a) scalar principle
(b) span control
(c) degree of unity on the command
(d) number of managers in the firm

VIZAG STEEL MT 18.08.2013

Ans. (c) : Number of management layers in a firm is determined by the degree of unity of command. Most organization have three management levels. First-level, middle-level and top-level managers. These managers are classified according to hierarchy of authority and perform different tasks.

1025. As opposed to detailed transaction information, the MIS responds to the need for

(a) electronic data processing
(b) managerial information
(c) accounting application
(d) utility billing information

VIZAG STEEL MT 18.08.2013

Ans. (b) : An opposed to detailed transaction information. The MIS responds to the need for managerial information.

1026. Which of the following are needed as input data for material required planning?

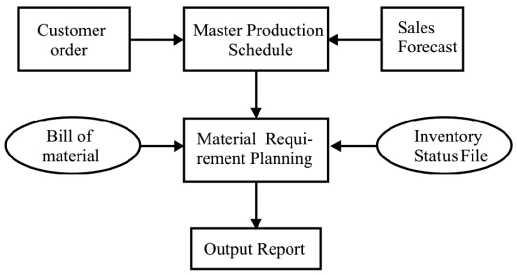
I. Weekly production schedule
II. Bill of material
III. Supplier lead time
IV. Market forecast

Select the correct answer using the codes given below :

(a) I, II and III (b) II, III and IV
(c) I and IV (d) I and III

TRB Asstt. Prof., 2012, ESE 2005

Ans. (a) :



1027. Consider the following data to apply the material requirement planning technique:

1. Bill of material
2. Inventory stock status
3. Lead times for items procured
4. EOQ
5. Master production schedule

Which of these are correct?

(a) 1, 2, 3, 4 and 5 (b) 2, 3, 4 and 5 only
(c) 1, 2, 3 and 4 only (d) 1, 2, 3 and 5 only

UPSC JWM Advt. No.-50/2010

Ans. (d) : As we know, the material requirement planning technique have following data such as–

- Bill of material
- Inventory stock status
- Lead times for items procured
- Master production schedule

1028. Consider the following statements :

1. Volume of purchases
2. Discounts offered
3. Disposal of scrap

Which of these factors will provide economy in materials management?

(a) 1 and 3 (b) 2 and 3
(c) 1 and 2 (d) 3 only

UPSC JWM Adv. No-16/2009

Ans. (a) : (1) Volume of purchesses and
(2) Disposal of Scrap will provide economy in material management.

1029. Consider the following statement :

1. **Reduced in process inventory, with no shortages**
2. **Increased flexibility**
3. **Improved decision making capability.**

Which of the above statements is/are correct for benefits of MRP?

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 3 only

UPSC JWM Adv. No-16/2009

Ans. (c) : <u>Benefits of MRP</u> : (Material Requirement Planning)–

- Increased flexibility
- Improve decision making capability.

1030. Material handling system is affected by the factor

(a) product to be handled
(b) production system
(c) type of building within which material is to be handled
(d) all of these factors

TNPSC AE, 2008

Ans. (d) : Material handling system is affected by the factors –

- Product to be handled
- Production system
- type of building within which material is to be handled
- Economics folder etc

1031. MRP-II means

(a) Materials Requirement Planning
(b) Manufacturing Resource Planning
(c) Non Requirement Planning
(d) Money Requirement Planning

DRDO Scientists 2008

Ans. (b) :

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11. Production Planning and Control

1032. Gantt chart are used for

(a) Forecasting sales
(b) Production schedule
(c) Scheduling and routing
(d) Linear Programming

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
TSPSC Manager (Engg.) HMWSSB 12.11.2020
Nagaland PSC (CTSE) 2018, Paper-I
GPSC ARTO Pre 30.12.2018
Vizag Steel (MT) 2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
Rajasthan Nagar Nigam AE 2016, Shift-I
GPSC Lect. 23.10.2016
Rajasthan AE (Nagar Nigam) 2016 Shift-3
APPSC AEE MAINS 2016, PAPER-III
HPPSC W.S. Poly. 2016
BPSC Poly. Lect. 2016
RPSC AE (GWD) 06.05.2016
Kerala PSC IOF 19.04.2016
OPSC AEE 2015 paper -I
BPSC Asstt. Prof. 29.11.2015
CGPSC AE 26.04.2015 Shift-I
Assam PSC CCE Pre 2015
MPSC Poly. Lect. 2014
J&K PSC Civil Services Pre, 2013
TNPSC AE (Industries) 09.06.2013
APPSC Poly. Lect. 2013
RPSC ACF-2011
RPSC AE GWD, 2011
OPSC Civil Services Pre. 2011

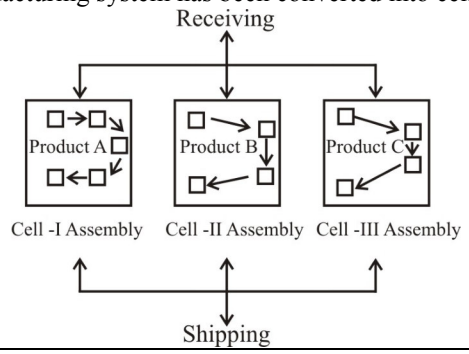
Ans. (b) : A Gantt chart is a popular tool in project management. It basically drills down activities which need to be done by a fixed time period. It is commonly used for tracking project schedule.

1033. For the use of concept, principles and approaches of group technology, the type of layout used is:

(a) product layout
(b) cellular layout
(c) job-shop layout
(d) fixed position layout

AAI Jr. Executive 29.11.2018
GWSSB DEE 07.07.2016
Kerala PSC AE 06.08.2015
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
BPSC AE 2012 Paper - VI
APPSC AEE 2012, ESE 1999

Ans. (b) : it is application of group technology in manufacturing in which all or a portion of a firm's manufacturing system has been converted into cells.



1034. Another name for Military organization is .....

(a) Line organization
(b) Functional organization
(c) Line and staff organization
(d) Hybrid organization

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
RPSC AE 2018
HPPSC Asstt. Prof. 18.09.2017
HPPSC Asstt. Prof. 18.11.2016
Kerala PSC IOF 19.04.2016
RPSC AEN Pre-2013
J&K PSC Civil Services Pre, 2013

Ans. (a) : Another name for Military organization is Line organization.

Line organization—Line organization structure is the oldest and simplest form of organization.

- In these organizations, a supervisor exercise direct supervision over a subordinate. Also, authority flow from the top most person in the organization to the person in the lowest rank. This type of an organization is also called Military organization or a scalar-type organization.

1035. Vehicle manufacturing assembly line is an example of:

(a) product layout (b) process layout
(c) manual layout (d) fixed layout

Assam PSC AE (PHED), 18.10.2020
OPSC AEE 2019 Paper-I
Kerala PSC Poly. Lect., 2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
J&K PSC Civil Services Pre, 2013
RPSC AEN Pre-2013
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2010

Ans. (a) : Product type of layout is generally used in systems where a product has to be manufactured or assembled in large quantities. In product layout the machinery and auxiliary services are located according to the processing sequence of the product without any buffer storage within the line itself.

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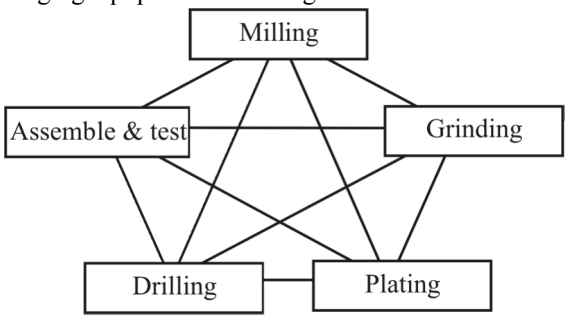
1036. Process layout is employed for

(a) effective utilisation of materials
(b) batch production
(c) effective utilisation of machines
(d) effective utilisation of power

Rajasthan AE (Nagar Nigam) 2016 Shift-3
Kerala PSC Asstt. Prof. 27.09.2016
GWSSB AAE, 27.12.2015
VIZAG STEEL MT 2015
VIZAG STEEL MT 18.08.2013
Vizag Steel MT (Re-Exam) 24.11.2013
J & K PSC Screening, 2006

Ans. (b) : Process layout is a design for the floor shop of a plant which aims to improve efficiency by arranging equipment according to its function.



Process layout is used for batch production.

1037. Routing in production planning and control refers to the :

(a) Balancing of load on machines
(b) Sequence of operations to be performed
(c) Authorization of work to be performed
(d) Progress of work performed

Mizoram PSC (Insp. of Factories) 2019 Paper-III
OPSC AEE 2019 Paper-I
RPSC AE (GWD) 06.05.2016
TNPSC AE 2013, RPSC AE GWD, 2011
ESE 2000

Ans. : (b) : The routing is a vital portion of production control. It lays down the flow of work in the plant. It determines how and where is to be done, what machines are to be used and it prescribes the path and sequence of operations to be followed. This provides a basis for scheduling, dispatching and control functions.

1038. In production, planning and control document which authorize the start operation on the shop floor is :

(a) Dispatch order (b) Route
(c) Loading chart (d) Schedule

Assam PSC CCE Pre 2015
TNPSC AE 2013
TRB Asstt. Prof., 2012
RPSC AE GWD, 2011
TNPSC AE, 2008, ESE 2001

Ans. (a) : In production planning and control dispatching authorize the start operation on the shop floor.

1039. Production scheduling is simpler, and high volume of output and high labour efficiency are achieved in the case of :

(a) Fixed position layout
(b) Process layout
(c) Product layout
(d) A combination of line and process layout

Mizoram PSC (Insp. of Factories) 2019 Paper-III
RPSC Vice Principal ITI 2018
KPSC ADF 2015, TNPSC AE 2014
APPSC AEE 2012, ESE 1993

Ans. (c) :
- Product layout also known as synthetic layout.
- Assembly line such as automobile factory.
- Low variety, high volume production system. It is used for mass production.
- For standardised products, which have quite stable demand in near future. It is used in line balancing.

1040. Fredric W Taylor introduced a system of working known as

(a) line organization
(b) effective organization
(c) functional organization
(d) none of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021
ISRO Scientist/Engineer 17.12.2017
Rajasthan Nagar Nigam AE 2016, Shift-II
HPPSC Asstt. Prof. 18.11.2016
Kerala PSC IOF 19.04.2016
RPSC ACF-2011

Ans. (c) : The concept of functional organization was suggested by F.W. Taylor who recommended the appointment of specialists at important positions. For example, the functional head and marketing head directs the subordinates throughout the organization in his particular area. This means that the subordinates receives orders from several specialists, managing working above them.

1041. Routing prescribes the

(a) Proper utilisation of the machines
(b) Flow of the material in the plant
(c) Proper utilisation of the money
(d) Proper utilisation of the machinery

HPPSC ADF 05.03.2019
Rajasthan Nagar Nigam AE 2016, Shift-II
J&K PSC Civil Services Pre, 2013
RPSC ACF-2011
J & K PSC Screening, 2006

Ans. (b) : Routing prescribes the flow of material in the plant.
It determines what work must be done, where and how routing information is provided by product or process engineering function and it is useful to prepare machine loading charts and schedules.

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1042. The word KANBAN is most appropriately associated with :

(a) EOQ (b) JIT
(c) Capacity planning (d) Product design

OPSC AEE 2019 Paper-I
RPSC-2018
GPSC ARTO Pre 30.12.2018
HPPSC Asstt. Prof. 18.11.2016
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
GATE 2011

Ans. : (b) : Kanban is scheduling system for just-in-time (JIT) production and to control the logistical chain from a production point of view, and is not an inventory, control system (i.e. zero inventory). It is one method through which (JIT) is achieved.

1043. In order to avoid excessive multiplication of facilities, the layout preferred is

(a) Process layout (b) Product layout
(c) Group layout (d) Static layout

Nagaland PSC (CTSE) 2018, Paper-I
GPSC Lect. (Poly.) 07.12.2014
J&K PSC Civil Services Pre, 2013
TNPSC ACF 2012
CSE Pre-1995

Ans. (a) : Process layout is also known as analytical layout. In order to avoid excessive multiplication of facilities, the layout preferred is process layout. In process layout machinery operation are performed in group together and not arranged according to any sequence.

1044. A production line is said to be balanced when

(a) There are equal number of machines at each work station
(b) There are equal number of operators at each work station
(c) The waiting time for service at each station is the same
(d) The operation time at each station is the same

Mizoram PSC (Insp. of Factories) 2019 Paper-III
GPSC Executive Engineer 23.12.2018
HPPSC Asstt. Prof. 18.09.2017
UKPSC AE-2013, Paper-I

Ans. (d) : A production line is said to be balanced when the operation time at each station is the same

1045. Ship building yard represents :

(a) Fixed layout (b) Product layout
(c) Process layout (d) Manual layout

BPSC AE Mains 2017 Paper - VI
SJVN ET 2013, TNPSC AE 2013
CSE Pre-2003

Ans. (a) : Ship building yard represents - Fixed Layout.

1046. Which one of the following type of layout is used for the manufacturing of large aircrafts?

(a) Product layout (b) Process layout
(c) Fixed position layout (d) Combination layout

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
UPPSC AE 13.12.2020, Paper-I
UKPSC AE-2013, Paper-I
ESE 2003

Ans. (c) : Fixed position layout type of layout is used for the manufacturing of large air crafts.

1047. Greater flexibility in the context of work distribution to machines and workers is achieved in

(a) process layout
(b) cellular layout
(c) mixed layout
(d) fixed position layout

TNPSC AE 2013
JPSC AE 2013, Paper-V
UKPSC AE 2012 Paper–I
J & K PSC Screening, 2006

Ans. (a) : Greater flexibility in the context of work distribution to machines and workers is achieved in process layout.

1048. Production Flow Analysis (PFA) is a method of identifying part families that use data from

(a) part programmes
(b) production schedule
(c) route sheets
(d) bill of materials

GPSC DEE, Class-2 (GWSSB) 04.07.2021
APPSC AEE Screening Test 2016
TRB Poly. Lect., 2012, GATE 2001

Ans. (c) Production Flow Analysis (PFA) is a method of identifying part families that use data from route sheets.

1049. The manufacturing area of a plot is divided into four quadrants. Four machines have to be located, one in each quadrant. The total numbers of possible layout are.

(a) 16 (b) 4
(c) 8 (d) 24

TSPSC AEE 2017
GPSC AE 26.07.2015
GATE 1995

Ans. (d) : We know, Total number of possible layouts = $4! = 4\times3\times2$
$= 24$

1050. Aggregate planning is the capacity planning for _____.

(a) Intermediate range
(b) typically one to three months
(c) long range
(d) short range

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I
CGPSC AE 16.10.2016
TNPSC AE 2014

Ans. (a) : Aggregate planning–In production planning, it's the intermediate range capacity planning that typically covers a time horizon 2 to 12 months. The purpose of aggregate planning is planning ahead because it takes time to implement plans.

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1051. In which of the following layouts the lines need to be balanced

(a) process layout
(b) product layout
(c) fixed position layout
(d) plant layout

TSPSC Manager (Engg.) HMWSSB 12.11.2020
RPSC ACF-2011
J & K PSC Screening, 2006

Ans. (b) : Product layout is also known as line layout. In this type of layout the various operations an raw material are performed in a sequence and the machines are placed along the product flow line.

1052. Which of the following is not a decision taken during the aggregate production planning stage?

(a) Scheduling of machines
(b) Amount of labour to be committed
(c) Rate at which production should happen
(d) Inventory to be carried forward

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
APPSC Poly. Lect. 2013, GATE 2012

Ans. (a) : Scheduling of machines is not a decision taken during the aggregate production planning stage.

1053. The layout which is suited for mass production is

(a) Process layout (b) Functional layout
(c) Product layout (d) Plant layout

Rajasthan Nagar Nigam AE 2016, Shift-I
Kerala PSC IOF 19.04.2016
RPSC ACF-2011

Ans. (c) : Product layout (also known as assembly lines arrange activities in a production line a according to a sequence of operations that need to be performed to assemble a particular product.
Product layout are suitable for mass production or repetitive operations in which demand is steady and volume is high.

1054. Product layout is employed for:

(a) Batch production
(b) Continuous production
(c) Effective utilization of machine
(d) All of these

GPSC Asstt. Prof. 28.08.2016
Mizoram PSC AE/SDO 2014 Paper-III

Ans. (b) : A product layout refers to a production system where the work stations and equipment are located along the line of production, as with assembly lines. So, a product layout is employed for continuous production.

1055. The layout with a higher material handling effort is a ..... layout.

(a) Process (b) product
(c) equipment (d) material

(HPPSC LECT. 2016)
GATE 1995

Ans. : (a) The layout with a higher material handling effort is a process layout.
Process layout:- The process layout is also known as functional layout or analytical layout. This type of layout is employed where.
(a) Low volume of production is required
(b) Similar jobs are manufactured on similar machines
(c) Machines are arranged on functional basis

1056. Line balancing is related to

(a) product layout (b) process layout
(c) hybrid layout (d) cellular layout

APPSC AEE MAINS 2016, PAPER-III
BPSC AE 2012 Paper - VI

Ans. : (a) : Suitability of product layout-
(1) Assembly line such as automobile factory.
(2) Low variety and high-volume production system.
(3) For standardized products which have quite stable demand in near future.

1057. Mass production is characterized by

(a) low-volume items with maximum flexibility in their design
(b) high-volume items with maximum flexibility in their design
(c) high-volume items with minimum flexibility in their design
(d) low-volume items with minimum flexibility in their design

Haryana PSC Civil Services Pre 2014
BPSC AE 2012 Paper - VI

Ans. : (c) : Product layout is used for mass production of the product. The product layout is also known as synthetic layout. This type of layout is best suited where one type of product is produced and the product is standardized.

1058. The process layout is best suited where

(a) Automation is employed
(b) Mass production is envisaged
(c) A few number of non-standardized units are to be produced
(d) Machines are arranged according to sequence of operation

ISRO Scientist/Engineer 2009
J & K PSC Screening, 2006

Ans. (c) : The process layout is best suited where a few number of non-standardized units are to be produced.

1059. In a _____ layout all machines or process of the same type are grouped together.

(a) fixed position (b) factory
(c) process (d) product

Haryana PSC Civil Services Pre, 2014
BPSC AE 2012 PAPER VI

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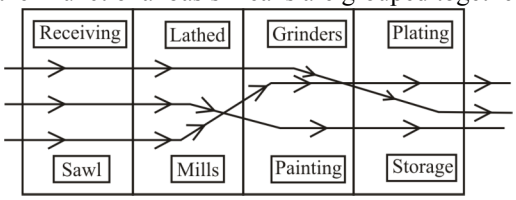
Ans. (c) : In process layout all machines are arranged as per their functional basis means are grouped together.



1060. Materials handling system includes

(a) Automated Guided Vehicles (AGV)
(b) Automated Storage/Retrieval System (AS/RS)
(c) Robots
(d) All of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016
BPSC AE 2012 Paper - VI

Ans. : (d) : Material handling systems includes-
- Automated guided vehicles (AGV)
- Automated storage/Retrieval system (AS/RS)
- Robots.

1061. The correct sequence of operations in production planning and control is:

(a) Routing - Scheduling - Dispatching - Follow up
(b) Scheduling - Follow up - Routing - Dispatching
(c) Dispatching-Scheduling - Routing-Follow up
(d) Routing - Follow up - Dispatching - Scheduling

Vizag MT 14.02.2020
TRB Polytechnic Lecturer 2017

Ans. (a) : Routing - Scheduling - Dispatching - Follow up the correct sequence of operations in production planning and control.

1062. The routing function in a production system design is concerned with

(a) Manpower utilization
(b) Machine utilization
(c) Zero defect manufacturing
(d) Quality assurance

APPSC AEE 2012, ESE 1996

Ans. (b) : The routing function in a production system design is concerned with machine utilization.

1063. Most popular type of organization used for civil engineering constructions is :

(a) line organization
(b) line and staff organization
(c) functional organization
(d) effective organization

Kerala PSC IOF 19.04.2016
RPSC ACF-2011

Ans. (a) : Line organization is the most oldest and simplest method of administrative organization. According to this type of organization, the authority flows from top to bottom in concern the line of command is carried out from top to bottom.

1064. Group Technology brings together and organises

(a) Parts and simulation analysis
(b) Documentation and analysis
(c) Automation and tool production
(d) Common parts, problems and tasks

UPPSC AE 13.12.2020, Paper-I
APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : Group technology brings together and organise common parts, problems and tasks.

1065. Which one of the following software packages is used for plant layout?

(a) SIMSCRIPT (b) DYNAMO
(c) CRAFT (d) MRP

TNPSC ACF 2012
ESE 1995

Ans. (c) : CRAFT software is used for plant layout

1066. Routing and scheduling are integral part of

(a) product planning (b) work study
(c) job analysis (d) quality control

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
J & K PSC Screening, 2006

Ans. (a) : Routing and scheduling are integral part of product planning.

1067. Scheduling

(I) is a general time table or manufacturing
(II) is the time phase of loading
(III) is loading all the work in process on machines according to their capacity

Which of these statements are correct?

(a) I, II and III (b) I and II
(c) II and III (d) I and III

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) A schedule is your projects time table of manufacturing includes time phase of loading, you can also adjust some activities and tasks in case of project running late. It include all, the basic process followed during manufacturing to finished product.

1068. Which one of the following is correct? Production planning and control functions are extremely complex in :

(a) Job-production shop producing small number of pieces only once
(b) Job-production shop producing small number of pieces intermittently
(c) Batch production shop producing a batch only once
(d) Batch production shop producing a batch at irregular intervals

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) Production, planning and control function are extremely complex is batch production shop producing a batch at irregular intervals, due to typical planning, scheduling and many more.

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1069. The function which does *not* come under the preview of production planning is :

(a) Methods planning
(b) Expediting
(c) Scheduling
(d) Materials requirement planning

CGPSC AE 15.01.2021

Ans. (b) : Factors comes under preview of production planning is :
(i) Methods planning
(ii) Scheduling
(iii) Material requirement planning

Follow up or expediting is that branch of production control which regulates the progress of materials and part through the production process.

1070. Every task that is scheduled should be assigned to a specific team member is termed as

(a) Compartmentalization
(b) Defined milestones
(c) Defined responsibilities
(d) Defined outcomes

APGCL AM, 2021

Ans. (c) : Every task that is scheduled should be assigned to a specific team member is termed as defined responsibility.

1071. Aggregate planning deals with a number of constraints. These typically are :

(a) Job assignments, job ordering, dispatching, over-time help
(b) Part-time help, weekly scheduling, SKU production scheduling
(c) Capital investment, expansion or contracting capacity, R & D
(d) Subcontracting, employment levels, inventory level, capacity.

CGPSC AE 15.01.2021

Ans. (d) : Aggregate planning deals with a number of constraints. These typically are :

Subcontracting, Employment levels, inventory levels, capacity.

1072. Which one of the following contains design data on all products, e.g., their constituent component and parts?

(a) Engineering data master file
(b) Process data master file
(c) Inventory master file
(d) Sales master file

ESE 18.07.2021

Ans. (a) : Engineering data master file also known as Bill of Material (BOM) specifies design data and product structure.

1073. OPITZ coding system is used in _____

(a) Welding (b) Casting
(c) Robotics (d) Group Technology

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : OPITZ coding system is used in Group Technology.

1074. Which aspect is not true in related to process layout?

(a) Specialized and strict supervision is required
(b) Product planning and control is difficult and costly
(c) Handling and back-tracking of material is too much
(d) flexibility of equipment and personnel

HPPSC Poly. Lect. 05.07.2021

Ans. (a) :

1075. Which phase of the design includes modelling and simulation?

(a) Product architecture
(b) Configuration design
(c) Parametric design
(d) Detail design

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Configuration design includes modelling and simulation.

1076. Which step in engineering design process involves decision making?

(a) Define problem
(b) Gather information
(c) Concept generation
(d) Evaluation

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : Evaluation

1077. In a matrix organization, which of the following is true?

(a) The project manager is responsible for employee skills improvement
(b) The functional manager is responsible for employee skills improvement
(c) The project manager is responsible for employee annual appraisal
(d) The employee is responsible for his/her own skills improvement

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : In a matrix organization, The functional manager is responsible for employee skills improvement.

1078. In which organization form would the project manager possess the greatest amount of authority?

(a) Classical/traditional (b) Projectized
(c) Strong matrix (d) Weak matrix

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :

1079. What are the major advantages of the functional type of organization?

(a) Single point of contact for the customer
(b) Stable organizational structure
(c) Project orientation
(d) Multifunctional teams are easy to form

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Stable organizational structure

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1080. The best way to resolve a conflict in a project is by

(a) Collaborative (b) Confronting
(c) Compromising (d) Smoothing

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Confronting

1081. At what stage of team formation, the team can work effectively even without a leader?

(a) Forming (b) Storming
(c) Norming (d) Performing

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : Performing

1082. When you have accepted the risk with the ongoing project, also risks have been quantified & their category determine which of the following would work as Plan B, if something goes out of the order?

(a) Fallback plans
(b) Contingency plans
(c) Change request plan
(d) Monte Carlo analysis

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Contingency plans

1083. Railway wagons are manufactured using _____.

(a) fixed position layout (b) process layout
(c) group layout (d) product layout

VIZAG STEEL MT 24.01.2021 Shift-II, Shift-I

Ans. (a) : Railway wagons are manufactured using fixed position layout. In this case the reverse applies, men and equipment are moved to the material lies.
Its used in ship building, aircraft manufacture and big pressure vessel fabrication. It involves higher equipment handling cost.

1084. Which type of organization structure is used in textile industry?

(a) line organization
(b) staff organization
(c) function organization
(d) line and staff organization

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : line organization

1085. When should process modifications be considered during product life cycle?

(a) Introduction (b) Growth
(c) Maturity (d) Decline

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

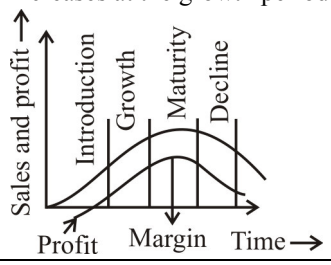
Ans. (a) Introduction

1086. The characteristic life-cycle of a product consists of four periods. The rate of consumption increases rapidly at the beginning of the

(a) Incubation period (b) Growth period
(c) Maturity period (d) Decline period

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 1998

Ans. (b) The life cycle of a product, the rate of consumption increases at the growth period.



1087. All similar machines or operations are grouped together in _____layout.

(a) line (b) process
(c) loop (d) none of the above

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) :

1088. The process which determines the programme for the operations is called:

(a) Loading (b) Routing
(c) Dispatching (d) Scheduling

RPSC ACF & FRO, 26.02.2021

Ans. (d) :
Scheduling–Determines the programme for the operations.
Routing–Prescribes the sequence of operations to be followed
Dispatching–is concerned with starting of processes.
Follow up–regulates the progress of the job through various processes.
Loading–Assignment of work to men & machine.

1089. Kaizen is a ____ process, the purpose of which goes beyond simple productivity improvement

(a) weekly (b) daily
(c) monthly (d) annual

TNCSC AE 2020

Ans. (b) : Kaizen is a Japanese word which refers to continuous improvement. In other words, it means improvement over improvement. In the process of Kaizen, improvement are accomplished gradually in small increments.

1090. Risk priority number is _____, if O is occurrence, S is severity and D is detection.

(a) $\frac{O \times S}{D}$ (b) $O \times (S + D)$
(c) $\frac{O}{(S \times D)}$ (d) $O \times S \times D$

VIZAG MT, 14.12.2020

Ans. (d) : RPN = O × S × D
Where, O = Occurrence
S = Severity
D = Detection
Risk priority number (RPN) is a measure used when assessing risk to help identify critical failure modes associated with your design or process. The RPN values ranges from 1 (absolute best) to 1000 (absolute worst).

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1091. Which is not a capacity option of aggregate planning?

(a) varying production rates through overtime or idle time
(b) inspection and quality control
(c) changing inventory levels
(d) subcontracting

VIZAG MT, 14.12.2020

Ans. (b) : Capacity option of aggregate planning are–
(1) Varying production rates through overtime or idle time.
(2) Changing inventory level.
(3) Subcontracting.

1092. Which of the following is not the objective of plant layout?

(a) To make economical use of floor area
(b) To minimize material handling
(c) To keep the machine breakdown to minimum
(d) To maintain flexibility in the arrangement of operations

Assam PSC AE (PHED), 18.10.2020

Ans. (c) : The primary objective of plant layout is to maximize production at minimum cost. The layout should be designed in such a way that it is flexible to change according to new processes and production technique.
(1) To have most effective and optimum utilization of available floor space.
(2) To introduce system of production control.
(3) To minimize material handling.
(4) To maintain flexibility in the arrangement of operation.

1093. Scheduling deals with

(a) Number of jobs to the done on a machine
(b) Number of machine tools used to do a job
(c) Different materials used in the product
(d) Fixing up starting and finishing times of each operation in doing a job

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Scheduling- Network scheduling is a technique used for planning and scheduling large projects, which aims at completion of project on time and minimization of project cost.
Scheduling also deals with fixing up starting and finishing times of each operation in doing a job.

1094. In aggregate planning one of the methods used to modification of supply is

(a) Advertising and sales promotion
(b) Development of complimentary products
(c) Backlogging
(d) Hiring and lay off of employees depending on the situation

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Hiring and lay off of employees depending on the situation.

1095. The process of planning, organising, directing people for the purpose of contributing to the organisation goals is known as :

(a) process management
(b) procurement management
(c) personnel management
(d) principles management

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Personnel management – The process of planning organising, directing people for the purpose of contributing to the organisation goals is known as personnel management. Personnel management skills are essential for the long-term success of your business personnel management helps. You focus on hiring, training, supervising evaluating terminating and motivating employees.

1096. If an organisation offers a range of products to the market for consumption and service is known :

(a) price mix (b) organisation mix
(c) sale mix (d) product mix

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Product mix– If an organisation offers arrange of product to the market for consumption and service is known as product mix. Product mix, also known as product assortment or product portfolio refers to the complete set of product and or service offered by a firm.

1097. Which of the following actions of personnel management is known as capital punishment of the industry for the employee?

(a) Transfer (b) Demotion
(c) Lay-off (d) Discharge

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Discharge : It is the termination of his employment and may be either the result of some disciplinary action or retrenchment of surplus labour. Thus a discharge involves permanent separation of an employee. From the pay roll for violation of company rules or for inadequate performance. It is initiated by the management.

1098. Which of the following activities is usually done prior to the launch of a new product in the market?

(a) Advertising
(b) Market research
(c) Sales forecasting
(d) Sales and promotion

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : Market research is the process of determining the viability of a new service or product through research conducted directly with potential customers. Market research allows a company to discover the target market and get opinions and other feedback from consumers about their interest in the product or service.

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1099. The first stage in production planning is

(a) Process Planning (b) Factory Planning
(c) Operation Planning (d) Layout Planning

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : The first stage in production planning is factory planning.

1100. In flexible manufacturing system (FMS), automated guided vehicles are used in/by

(a) Machine operators
(b) Material handling system
(c) Strongest system
(d) Control system

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Automated guided vehicle (AGV) system is most widely used with material handling system in FMS.

1101. Which one of the following is true in respect of production control for continuous or assembly line production?

(a) Control is achieved by PERT network
(b) Johnson algorithm is used for sequencing
(c) Control is on one work centre only
(d) Control is on flow of identical components through several operations

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (d) : Production central for continuous or assembly line production, central is an flow of identical components through several operations.

1102. All machines and equipments are grouped together at one location according to their functions in which type of layout?

(a) Product layout
(b) Process layout
(c) Fixed position layout
(d) Hybrid layout

OPSC AEE 2019 Paper-I

Ans. : (b) : Process layout- The process layout is also known as functional layout or analytical layout. This type of layout is employed where,
(i) Low volume of production is required.
(ii) Similar jobs are manufactured on similar machines.
(iii) Machines are arranged on functional basis.

1103. Which one of the following is not an element of material handling?

(a) motion (b) time
(c) quality (d) quantity

TNPSC AE 2019

Ans. (c) : Motion, time and quantity are elements of material handling whereas quality is not an element of material handling.

1104. What is the mode for the following distribution?

Gross profit as percentage of sales	Number of companies
0 – 7	19
7 – 14	25
14 – 21	36
21 – 28	72
28 – 35	51
35 – 42	43
42 – 49	28

(a) 19.55 (b) 21.40
(c) 23.25 (d) 25.10

ESE 2019

Ans. (d) : Here the maximum class frequency is 72 and the class corresponding to this frequency is 21–28.
So the modal class is 21–28.
Now

Modal class = 21–28
Lower limit (ℓ) of modal class = 21
Class size (h) = 7
Frequency (f_1) of the modal class = 72
Frequency (f_0) of class preceding the modal class = 36
Frequency (f_2) of class succeeding the modal class = 51
Now, let us substitute these values in formula

$$\text{Mode} = \ell + \left(\frac{f_1 - f_o}{2f_1 - f_o - f_2}\right) \times h$$

$$= 21 + \left(\frac{72 - 36}{2 \times 72 - 36 - 51}\right) \times 7$$

$$= 25.42$$

1105. The goal of Just in Time (JIT) philosophy is

(a) The elimination of wastes
(b) The elimination of process layout
(c) The elimination of costly machine tools
(d) The elimination of sub assemblies in the inventory

TNPSC AE 2019

Ans. (a) : Just in time [JIT] manufacturing, also known as just - in - production or the Toyota production system (TPS), is a methodology aimed primarily at reducing times with in the production system as well as response time from suppliers and to customers.

1106. Dispatching function of production

(a) Authorizing a production work order to be performed
(b) A dispatch of finished goods on order
(c) Dispatch of bills and invoices to the customer
(d) Movement of in-process material from shop to shop

OPSC AEE 2019 Paper-I
ESE 2001, CSE Pre-1998, 1997

Ans. : (b) : Dispatching function is authorizing a production work order to be launched.

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1107. Kanban in Japanese language indicates

(a) a method of line balancing
(b) information for production and withdrawal of items
(c) priority dispatching
(d) line time employment

TNPSC AE 2019

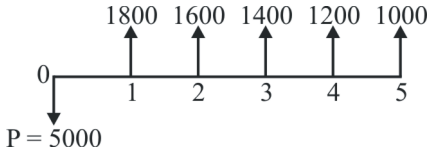
Ans. (b) : Kanban in Japanese language indicates information for production and withdrawal of items.

1108. A project initially costs Rs 5,000 and generates year-end cash inflows of Rs 1,800, Rs 1,600, Rs 1,400, Rs 1,200 and Rs 1,000 respectively in five years of its life. If the rate of return is 10%, the net present value (NPV) will be

(a) Rs 500 (b) Rs 450
(c) Rs 400 (d) Rs 350

ESE 2019

Ans. (b) :



Rate of return (i) = 10% = 0.1

Net present value

$$= -5000 + \frac{1800}{(1+0.1)^1} + \frac{1600}{(1+0.1)^2} + \frac{1400}{(1+0.1)^3} + \frac{1200}{(1+0.1)^4} + \frac{1000}{(1+0.1)^5}$$

$$= -5000 + 1636.36 + 1322.31 + 1051.84 + 819.61 + 620.92$$

$$= 451.04 \simeq ` 450$$

1109. Which layout is most suitable for highest flexibility

(a) Product layout
(b) Process layout
(c) Fixed position layout
(d) Group layout

HPPSC AE 2018

Ans. (d) : Group layout is most suitable for highest flexibility.

Process Layout–In this layout similar items are grouped together. This type of layout is commonly seen in Auto shope departmental stores.

Product Layout–In this type of layout, equipment, tools and machines are located according to how a product is made. Another term for this type of layout is 'assembly line'. Products are passed down the line from station to station as they are being made.

Fixed position layout–In fixed position layout, personal, supplies and equipment are brought to the site where the product will be assembled, rather than the product being moved, through an assembly line or set of assembly stations.

1110. Kanban :

(a) is an information carrying the expression of needs
(b) is a simple way of controlling production.
(c) is a request (card) for production or withdrawal (moment) of parts
(d) all of the above

Vadodara Municipal Corp. DEE, 2018

Ans. (d) : Kanban is an information carrying the expression of needs. It is a simple way of controlling production. It is a request (card) for production or withdrawal of parts.

1111. To meet the weekly fluctuating demand of the customer

(a) Material requirement planning can be used
(b) Economic order quantity can be used
(c) Aggregate production planning can be used
(d) Master production schedule can be used

HPPSC AE 2018

Ans. (c) : To meet the weekly fluctuating demand of the customer aggregate production planning can be used.

Aggregate Production Planning–Aggregate production planning is a marketing activity that does an aggregate plan for the production process, in advance of 6 to 18 months, to give an idea to management as to what quantity of materials and other resources are to be produced and when, so that the total cost of operations of the organization is kept to minimum over that period.

1112. Which of the following layouts is suited to job production?

(a) Process layout (b) Product layout
(c) Fixed position layout (d) Plant layout

MPSC Lect. (Prod.) 2018

Ans. (a) : Process layouts is suited to job production or batch production.

1113. Assembly line

(a) Flow shop type lines with dedicated machines arranged in U-type for best utilization
(b) Machines arranged in T-type
(c) Machines arranged in zig-zag way
(d) None of these

MPSC Lect. (Prod.) 2018

Ans. (a) : Assembly line flow shop type lines with dedicated machines arranged in U-type for best utilization.

1114. JIT is best suited for

(a) Job shop production
(b) Batch production
(c) Repetitive production
(d) Production in very low volume

MPSC Lect. (Prod.) 2018

Ans. (b) : JIT (Just in Time) is best suited for batch production.

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1115. The application of a concurrent design approach results in reducing

(a) Production design and development time
(b) Product design and development time and speeds the product to market
(c) Speeds up the product to market
(d) None of these

MPSC Lect. (Prod.) 2018

Ans. (a) : The application of a concurrent design approach results in reducing production design and development time.

1116. Consider the following statements:

Dispatching authorizes the start of production operation by

1. releasing of material and components from stores to the first process
2. releasing of material from process to process
3. issuing of drawing and instruction sheets

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : Functions of dispatching–

(i) To check the availability of input material and ensure the movement of material from store to first process.
(ii) Releasing of material from process to process.
(iii) To ensure the availability of all production and inspection aids.
(iv) To obtain the requisite drawings, specifications and material list.

1117. As production systems move from projects to batch production to mass production to continuous production-

(a) demand value increases
(b) products become more customized
(c) production systems become less automated
(d) production systems become more flexible

RPSC AE 2018

Ans. (a) : As production systems move from projects to batch production to mass production to continuous production when demand value increases.

1118. Which one of the following is the correct definition of critical ratio in scheduling?

(a) Demand time/supply lead time
(b) Supply lead time/demand time
(c) Demand time/manufacturing lead time
(d) Manufacturing lead time/demand time

HPPSC Asstt. Prof. 18.09.2017

Ans. (a) :

1119. The type of layout used for manufacturing steam turbine, is-

(a) Product Layout
(b) Process Layout
(c) Combination Layout
(d) Fixed Position Layout

GMB AAE 25.06.2017, ESE 1999

Ans. (d) : In a fixed position layout, Personal Supplies and Equipments are brought to the site where the product will be assembled, rather than the product being moved through an assembly line or set of assembly stations. Other Fixed-position layout examples include construction ship building, aircraft, aerospace, farming, drilling for oil, home repair and automated car washes.

1120. Fixed position layout is also known as

(a) analytical layout
(b) synthetic layout
(c) static product layout
(d) none of these

Sikkim PSC (Under Secretary), 2017

Ans. (c) : Fixed position layout involves the movement of man power and machine to the product which remains stationary. The movement of man and machine is advisable as the cost of moving them would be lesser. This type of layout is preferred where the size of the job is bulky and heavy.

Example- Such type layout is locomotives, ship, boilers, generators, wagon, building, aircraft manufacturing.

It involves high equipment cost.

1121. The process of tracing through records and all levels of product structure to identify the effect of change of records of one item is

(a) Pegging (b) Cycle counting
(c) Updating (d) Action bucket

Karnataka PSC AE, 10.09.2017

Ans. (c) : The process of tracing through records and all levels of product structure to identify the effect of change of records of one item is updating.

1122. The CAPP approach that synthesizes a new process plan from scratch for each new part by using Knowledge Base and Inference Engine is

(a) Retrieval CAPP (b) Generative CAPP
(c) Automated CAPP (d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (b) : Generative CAPP.

1123. Among the following which features of the manufacturing system offer flexibilities in FMS?

(a) Expansion (b) Routing
(c) Machine (d) All of the above

Punjab PSC SDE 12.02.2017

Ans. (d) : The following features of the manufacturing system offer flexibilities in FMS –

(i) Expansion
(ii) Routing
(iii) Machine

1124. Expending function consist keeping a watch on

(a) Operators activity
(b) Flow of material and in case of trouble locate source of trouble
(c) Minimizing the delays
(d) Making efficient dispatching

GPSC Asstt. Director of Transport 05.03.2017

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Ans. (b) : Expending function consist keeping a watch on flow of material and in case of trouble locate source of trouble.

1125. Development of surfaces is not useful in industries for constructing _____.

(a) Bins (b) Process vessels
(c) Pipes (d) Hoppers
(e) Chimneys

(CGPSC Polytechnic Lecturer 2017)

Ans. (c) : Pipes, development of surfaces is not useful in industries for constructing.

1126. In concurrent engineering, design and manufacturing are–

(a) sequentially considered
(b) simultaneously considered
(c) separately considered
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : simultaneously considered

1127. The department which take care of both production as well as control is :

(a) Routing (b) Dispatching
(c) Expediting (d) Scheduling

Kerala PSC Poly. Lect., 2017

Ans. (d) : The department which take care of both production as well as control is scheduling.

1128. Which programme is useful when plant is engaged in batch production?

(a) Breakdown of orders
(b) PERT
(c) Cycle schedule
(d) Mass schedule
(e) CPM

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : Batch production is method of manufacturing where the production are made of specified groups or amount with time frame. So for this, cycle schedule is used.

1129. Productivity is defined as

(a) ratio of input to output
(b) product of input and output
(c) difference of input to output
(d) ratio of output to input

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (d) : Productivity is a measure of the efficiency of production.

Productivity is the ratio of what is produced to what is required to produced it.

$$\text{Total productivity} = \frac{\text{Net Output}}{(\text{Labour} + \text{capital})\ \text{Input}}$$

$$= \frac{\text{Total output} - \text{Material and service}}{(\text{Labour} + \text{capital})\ \text{Input}}$$

1130. Which of the following is not an objective of Job evaluation?

(a) To determine relative worth of different jobs.
(b) To develop a rational and consistent wage policy.
(c) To develop a basis for wage differentials.
(d) To determine the relative performance of different workers.

RPSC AE (GWD) 06.05.2016

Ans. (d) : Job evaluation specifies the relative value of each job in an organization. It basically serves the purpose of compensation procedures. Job evaluation is useful tool for making decisions about the compensation to be attached with a particular position.

- To determine relative worth of different jobs.
- To develop a rotational and consistent wage policy.
- To develop a basic for wage differentials.

1131. Which of the following is a characteristic of maturity phase in life cycle of a product?

(a) Volumes are decreasing
(b) Competition is well established
(c) Products are considered for termination
(d) Product variety is large and design is evolving

RPSC AE (GWD) 06.05.2016

Ans. (b) : Product life cycle is the time during which a product is conceived and produced put on to the market and eventually removed from the market. The process has four stages, introduction, growth, maturity and decline.

In the maturity phase, the sales of the product slow down and in some cases stop because market saturation. Competitors are well established offering attractive prices to consumers. All marketing efforts are directed to wipe out competition pressure by lowering their demand.

1132. A company specializes in different types of machines/parts with small volumes and with uncertain demand. Which layout of the plant will be most suitable?

(a) Process layout
(b) Product layout
(c) Fixed position layout
(d) Assembly line type layout

RPSC AE (GWD) 06.05.2016

Ans. (a) : Process layouts are found primarily in job shops or firms that produce customized low volume products that may require different processing requirements and sequence of operations. A manufacturing example would be a machine shop. A machine shop generally has separate departments where general purpose machine are grouped together by function (e.g. milling, grinding, drilling, hydraulic press and lathes)

1133. Which of the following type of plant layout chart/diagram doesn't give information about relative position or closeness of different workstations?

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(a) Flow diagram (b) String diagram
(c) Relationship chart (d) All of these

RPSC AE (GWD) 06.05.2016

Ans. (c) : Relationship chart of a plant layout chart/diagram does not give information about relative position or closeness of different workstation.

1134. What is the most serious disadvantage of product layout?

(a) High material handling
(b) The movement of materials becomes highly inconvenient
(c) It is not robust in the event of disruption
(d) Higher cost per unit

RPSC AE (GWD) 06.05.2016

Ans. (c) : Disadvantage of product layout

- The layout is not flexible. Since the operations are performed in sequential manner, adjustments in the course of production cannot be made.
- It is difficult to expand the capacity of each line of production.
- This layout is costly. Therefore results in duplication of similar type of machine needed for different lines of production.
- Since workers work of specific machines, they have lack of knowledge to work of other machines.

1135. Operation research in management is

(a) contingency approach
(b) quantitative approach
(c) system approach
(d) behavior approach

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (b) : Operation Research– It is a science of optimization; optimization means maximizing cost.

- Operation is the systematic method oriented study of the basic structure characteristic, functions and relationships of an organization to provide the executive with a sound, scientific and quantitative basis for decision making.

1136. What does routing in Production Planning and Control refer to?

1. Balancing of load on machines
2. Progress of work performed
3. Authorization of work
4. Sequence of operations to be performed

(a) Only (1) (b) Only (2)
(c) Only (3) (d) Only (4)
(e) (1), (2), (3) and (4)

CGPSC Asstt. Workshop Supt., 17.07.2016
ESE 2000

Ans. (d) : Routing is the first step in production planning and control. Routing can be defined as the process deciding the path (route) of work and the sequence of operations.

1137. Functional organization is :

(a) Less differentiated and more diffused
(b) More differentiated and longer term
(c) More differentiated and focused
(d) Goal orientation

HPPSC Asstt. Prof. 18.11.2016

Ans. (c) : A functional organization is a common type of organization structure in which the organization is divided into smaller groups based on specialized functional areas.

The features of functional organization are follows–

(1) The entires organizational activities are divided into specified functions, such as operations, finance, marketing, personnel relations.
(2) Each functional area is put under the chase of functional specialist.

1138. Matrix organization includes :

(a) Focusing of undivided human effort on two (or more) essential organizational tasks simultaneously
(b) Formality of structure
(c) Time orientation
(d) Confrontation

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : A matrix organization is a structure in which there is more than one line of reporting managers. Effectively, it means that the employee of the organization have more than one boss. This type of structure is used in organizations which have diverse product lines and devices.

1139. Which one is not a type of plant layout ?

(a) Product layout (b) Process layout
(c) Group layout (d) Jumbled layout

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : Various type of plant layout–

(1) Process layout
(2) Product layout
(3) Fixed layout
(4) Group technology
(5) Hybrid

1140. During the development of a product an entirely new process plan is made based on design logic, examination of geometry and tolerance information. This type of process planning is known as

(a) Retrieval
(b) Generative
(c) Variant
(d) Group technology based
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (b) : The generative approach involves generation of new process plans by means of decision logics and process knowledge. We are discussing only the generative approach of process planning as only in this approach one can develop a process plan at the early design stage where manufacturability evaluation is most effective.

1141. Which of the following activities pertain to a master production schedule?

1. Product design
2. Short term allocation of resources to individual operations

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3. Sizing and timing of specific items production lot
4. Sequencing of individual orders of jobs

(a) 2, 3 and 4 (b) 1, 3 and 4
(c) 1, 2 and 4 (d) 1, 2 and 3

UPRVUNL AE 21.08.2016

Ans. : (d) Some activities pertain to a master production schedule
(i) Product design
(ii) Short term allocation resource to individual operation
(iii) Sizing and timing of specific items production lot.

1142. Which of the following are the characteristices of job order production?
1. A flexible layout is preferred
2. Detailed schedule is needed for each component
3. Division of labor is effective
4. High degree of production control is required.
Which of these statements are correct?

(a) 3 and 4 (b) 1, 3 and 4
(c) 2 and 4 (d) 1 and 3

UPRVUNL AE 21.08.2016

Ans. : (c) Characteristics of job order production-
(i) Detailed schedule is needed for each component.
(ii) High degree of production control is required.

1143. Major elements of planning are

(a) Machines and materials
(b) Men and machines
(c) Men, machines and money
(d) Men, machines, money and material

GPSC Lect. 23.10.2016

Ans. (d) : Planning consists of controlling of five M's → Men, Machines, Methods, Materials and Money.

1144. Material handling is considered as

(a) Economically waste it should be eliminated
(b) Economically waste but cannot be eliminated
(c) Economically profitable so should be increased
(d) It does not cost in any way

UJVNL AE 2016

Ans. : (b) Material handling is considered as Economically waste but cannot be eliminated.

1145. The most important objective behind plant layout is

(a) Overall simplification, safety of integration
(b) Economy in space
(c) Maximum travel time in plant
(d) To provide conveniently located shops

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : The most important objective behind plant layout is overall simplification, safety of integration.

1146. Routing is most difficult in which type of production system?

(a) Mass production
(b) Batch production
(c) Continuous production
(d) Equally difficult in both the job and mass production

RPSC AE (GWD) 06.05.2016

Ans. (b) : The routing procedure is most difficult and complex. In batch order/Intermittent production also generally the equipment is arranged in process type layout. The operations and their sequence differ from batch to batch. The route sheets would require revision whenever the production of the batch changes.
Routing in continuous/mass manufacturing does not present any problem due to product type of layout.

1147. Line and staff organisation is combination of

(a) Staff Organisation & Project organisation.
(b) Scalar organisation and project organisation.
(c) Project organisation & Functional organisation.
(d) Scalar organisation and functional organisation.

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (d) : **Line and Staff Organisation–**The line organisation gradually developed to shape as the line and staff organisation; Taylor's functional organisation hastened its development. A variation of line and staff organisation is scalar organisation and functional organisation.
So line and staff organisation is combination of scalar and functional organisation.
Advantage of Line & Staff Organisation–
1. Expert advice from specialist staff executives.
2. Less wastage of material, man and machine hours.
3. Quality of product improved.
4. There is no confusion as exists in functional organisation.

1148. Benchmarking is used to develop objectives by:

(a) Obtaining reports from the government
(b) Using SMART objectives
(c) Making comparisons with excellent/best practice companies
(d) Listening to suggestions from employees

HPPSC Asstt. Prof. 18.11.2016

Ans. (c) : The objective of benchmarking is to understand and evaluate the current position of a business or organisation in relation to best practice and to identify areas mans of performance improvement.

1149. Production of large turbo generators is an example for

(a) job order production
(b) batch order production
(c) continuous production
(d) selective production

APPSC AEE MAINS 2016, PAPER-III

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Ans. (a) : Production of large turbo generators is an example for job order production.

1150. Product layout is best suited when:
(a) One type of product is produced
(b) Product is standardized
(c) Product is manufactured in large quantities
(d) All of these

OPSC AEE 2015 Paper-I

Ans. : (d) Product layout : In the product layout, the various operation on raw material are performed in a sequence. Accordingly the machines are arranged along the product flow line. This implies that the machines are placed in the sequence in which the raw material is to be operated.

1151. MIS is not used in which of the following applications?
(a) Forecasting (b) Scheduling problem
(c) Inventory (d) Heat transfer

VIZAG STEEL MT 2015

Ans. (d) : following are the applications of MIS (Management Information System).
- Forecasting
- Scheduling problem
- Inventory

1152. Which of the following is the objective of MIS?
(a) To keep the information to up to data
(b) To increase the quantity of the data
(c) To filter the data
(d) To interpret data

VIZAG STEEL MT 2015

Ans. (a) : To keep the information to up to data.

1153. Technology S–curves are helpful in :
(a) product platform planning
(b) concept generation
(c) product architecture
(d) product design

VIZAG STEEL MT 2015

Ans. (a) : Technology S–curves are helpful in product platform planning.

1154. Which of the following is an advantage of using an expert system?
(a) Imposed structure
(b) Knowledge engineering resistance
(c) Rapid prototyping
(d) All options are correct

MPPSC AE 08.11.2015

Ans. (d) : Advantage of using an expert system are-
- Imposed structure
- Knowledge engineering resistance.
- Rapid prototyping etc.

1155. Production planning consists of
(a) Preplanning and routing
(b) Scheduling and dispatching
(c) Expediting
(d) All options are correct

MPPSC AE 08.11.2015

Ans. (a) : Production planning consists of preplanning and routing.

1156. A joint sector undertaking :
(a) is jointly owned by several share holders
(b) is jointly owned by private parties and government
(c) is jointly managed by two directors
(d) is jointly owned by state and central governments

Kerala PSC AE 06.08.2015

Ans. (b) : Joint sector under taking- Joint sector consists of private parties, business under takings wherein the ownership, control and management are shared jointly by the Government. The government, private parties and the investing public jointly own a joint sector enterprise.

1157. Group technology is suitable for production of:
(a) medium range variety and high range quality
(b) high range variety and high range quantity
(c) medium range variety and medium range quantity
(d) low range variety and low range quality

Kerala PSC AE 06.08.2015

Ans. (c) : Group technology is suitable for production of medium range variety and medium range quantity.
Group technology is an approach to organizing manufacture which can be applied in any industry (machining, welding) foundry, press work, forging, plastic molding etc where small-batch variety production is used.

1158. In FMS the tools are identified by means of :
(a) colour code (b) bar code
(c) PLC (d) digital code

Kerala PSC AE 06.08.2015

Ans. (b) : In FMS the tools are identified by bar code Flexible manufacturing system (FMS)-
FMS consists of a group of programmable production machines interconnected and storage system and controlled by an integrated computer system to produce a variety of parts at non-uniform product rates, batch sizes and quantities.

1159. Advantage in a line organization is
(a) strong discipline
(b) quick decision making
(c) simple to start
(d) All of the above

TSPSC AEE 2015

Ans. : (d) Advantage in a line organization
(i) Strong discipline
(ii) Quick decision making
(iii) Simple to start

1160. Consider the following statements about Process Layout
1. Flexible to tolerate breakdown
2. Space utilization is efficient
3. Suitable for high fixed cost and low variable cost
4. Not suitable for automation

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Which of the above are not correct?

(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) 2, 3 and 4
(e) 3 and 4

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : A process layout is one in which all machines or processes of the same type are grouped together.
Features of process layout
(i) More of general layout
(ii) Fixed cost are low, variable costs are high
(iii) Less finished goods stock
(iv) Less automation
(v) Efficient space utilization.

1161. The responsibilities of the operations manager include

(a) planning, organizing, staffing, procuring, and reviewing
(b) forecasting, designing, planning, organizing and controlling
(c) forecasting, designing, operation, procuring and reviewing
(d) planning, organizing, staffing, leading and controlling
(e) designing and operating

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : The responsibilities of the operations manager include planning, organizing, staffing, leading and controlling.

1162. The type of production suitable for making bolts, nuts and washers is

(a) Batch production (b) Job production
(c) Flow production (d) Mass production

TSPSC AEE 2015

Ans. : (d) Mass production suitable for making bolts, nuts and washers.

1163. Jobs going behind the schedule are conveniently shown in

(a) Milestone chart (b) Pi chart
(c) Bar chart (d) Gantt chart

VIZAG STEEL MT 2015

Ans. (d) : Gantt chart

- A Gantt Chart is a type of bar chart that illustrates a project schedule.
- This chart lists the tasks to be performed on the horizontal axis.
- The width of the horizontal bars in the graph shows the duration of each activity.
- Gantt charts illustrate the start and finish dates of the terminal elements and summary elements of project.

1164. Purpose of Scheduling is to:

(a) Prescribe the sequence of operations to be followed
(b) Determine the programme for the operations
(c) Regulate the progress of job through various processes
(d) All of these

OPSC AEE 2015 Paper-I

Ans : (d) Purpose of Scheduling
1. Prescribe the sequence of operations to be followed
2. Determine the programme for the operations
3. Regulate the progress of job through various processes

1165. Decision of plant location will depend on

(a) Transport facility
(b) Availability
(c) Social infrastructure
(d) Types of Machines employed

MPPSC State Forest Service Exam, 2014

Ans. (b) : Availability of raw materials is the most important factor in plant location decisions. Usually manufacturing units where there is the conversion of raw materials into finished goods is the main task. So, this depends on the types of machines employed.

1166. Job shop production comprises of

(a) Special purpose machines
(b) Uninterrupted movement of material
(c) Less job instructions
(d) Large inventory of materials, tools and parts

MPPSC State Forest Service Exam, 2014

Ans. (d) : Job shop production are characterized by manufacturing of one or few quantity of products designed and produced as per the specification of customers within prefixed time and cost. A job shops comprises of general purpose machines arranged into different departments such materials tools.

1167. For a product layout the material handling equipment must

(a) have full flexibility
(b) employ conveyor belts, trucks, tractors etc.
(c) be a general purpose type
(d) be designed as special purpose for a particular application

MPSC Dy. Engineer 2014

Ans. (d) : The product layout can be defined as an arrangement or a mechanism of various machines run to produce the final product in a prescribed order. In this layout the raw material is feet one end and finished products arrive at other end. It is best suitable for mass production as assembly line. So In this material handling equipment must be designed as special purpose for a particular application.

1168. The production planning function, which refers to the determination of the sequence of operations to be performed for the job and allocation of facilities is called

(a) Sequencing (b) Scheduling
(c) Routing (d) Dispatching

MPPSC State Forest Service Exam, 2014

Ans. (c) : Routing is the production planning function, which refers to the determination of the sequence of operations to be performed for the job and allocation of facilities. In short routing determines what work must be done where & flow.

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1169. Which of the following is a subset of scheduling

(a) Routing (b) Sequencing
(c) Dispatching (d) Expediting

MPPSC State Forest Service Exam, 2014

Ans. (b) : Scheduling involves the determination of two types of decisions, allocation decisions which resources will be assigned to perform each of a given sets of jobs. Sequencing decisions in what order & when are each of these jobs performed.

1170. In fixed position layout, which of the following is fixed

(a) Material (b) Machinery
(c) Tools (d) Workmen

MPPSC State Forest Service Exam, 2014

Ans. (a) : A fixed position layout lets the product stay in one place while workers and machinery move to it as needed. e.g ships, airplanes etc. So material is fixed.

1171. Gantt chart does not show

(a) Scheduled start and finish time
(b) The delays that have occurred in each operation
(c) Planned time for each operation
(d) Cost of each operation

MPPSC State Forest Service Exam, 2014

Ans. (d) : A Gantt chart is a graphical depiction of a project schedule. It's a type of bar chart that shows the start & finish date of several elements of a project that include resources, milestones, fasts and dependencies.

1172. Bar charts are suitable for

(a) minor works (b) major works
(c) large projects (d) all of these

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (a) : A bar charts or bar graph is a chart that presents data with rectangular bars. Bar charts suitable for minor work for comparing between categories of data.

1173. The type of organisation preferred for a steel industry is

(a) line organisation
(b) functional organisation
(c) line and staff organisation
(d) line, staff and functional organisation

JPSC AE 2013, Paper-V

Ans. : (d) : Line, staff and functional organisation preferred for a steel industry.

1174. Which one of the following is NOT associated with the process of new product development?

(a) QFD (b) FEMA
(c) KANBAN (d) DFMA

VIZAG STEEL MT 18.08.2013

Ans. (b) : IMI is a family as business made in which buyer as a product provide certain information, FEMA can provide technical assistance for donations, such as layout of warehouse.

1175. Staffing is defind as :

(a) Workers working in an organization
(b) Work done by an employee
(c) Filling, and keeping filled positions in organization structure
(d) The art of getting things done by/with people

TNPSC AE (Industries) 09.06.2013

Ans. (c) : Filling, and keeping filled positions in organization structure

1176. Without objectives and plans, control is not possible, because :

(a) Performance has to be measured against some established criteria
(b) Performance has to be measured against time
(c) Performance has to be measured against some established criteria
(d) Performance has to be measured against person

TNPSC AE (Industries) 09.06.2013

Ans. (c) : Performance has to be measured against some established criteria

1177. Fixed position layout is also known as :

(a) analytical layout
(b) synthetic layout
(c) static product layout
(d) none of these

J&K PSC Civil Services Pre, 2013

Ans. (c) : In fixed position layout the machines are fixed at certain place in particular manner or sequence be also called static product layout.

1178. Which of the following material handling systems will be most suitable for transporting a large number of discrete items on continuous basis in a factory?

(a) Overhead cranes (b) Trucks
(c) Hoists (d) Conveyors

RPSC AEN Pre-2013

Ans. (d) : Conveyors material handling systems will most suitable for transporting a large number of discrete items on continuous basis in a factory.

1179. Which of the following is not a technique for plant layout?

(a) Process charts (b) Travel charts
(c) Man-machine charts (d) Relationship charts

RPSC AEN Pre-2013

Ans. (c) : A man-machine chart graphically represents the relationship between the manual work performed by one or more operations and one or more machines involved in a manufacturing process.

1180. The sequencing of jobs in the ascending order of their processing time is known as

(a) shortest processing time
(b) earliest due date
(c) dispatching
(d) none of these

TRB Poly. Lect., 2012

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Ans. (a) : The Sequencing of jobs in the ascending order of their processing time is known as shortest processing time.

1181. The ideal batch size in JIT production consists of

(a) one part (b) two parts
(c) ten parts (d) five parts

BPSC AE 2012 Paper - VI

Ans : (a) : The ideal batch size in JIT production consists of one part.

1182. Lean production is related to

(a) mass production
(b) batch production
(c) customized production
(d) stock-driven production

BPSC AE 2012 Paper - VI

Ans : (c) : Lean production- Lean production is an approach to management that focuses on cutting out waste whilst ensuring quality. This approach can be applied to all aspects of a business - form design, through production to distribution. Lean production aims to cut costs by making the business more efficient and responsive to market needs.

1183. In manufacturing management the term "Dispatching" is used to describe

(a) dispatch of sales order
(b) dispatch of factory mail
(c) dispatch of finished product to the user
(d) dispatch of work orders through shop floor

TNPSC ACF 2012

Ans. (d) : Dispatching- It refers to the controls of processes of operation planning greving operation order and controlling operation in manufacture site.

1184. The two bin system is concerned with

(a) ordering procedure
(b) forecasting sales
(c) production planning
(d) none of these

TRB Asstt. Prof., 2012

Ans. (a) : The two bin system is concerned with ordering procedure.

1185. What are decision support system used for?

(a) Management decision making
(b) Providing tactical information to management
(c) Providing strategic information to management
(d) Better operation of an organization

VIZAG STEEL MT 2011

Ans. (c) : Decision support system used for Providing strategic information to management.

1186. What is a management information system?

(a) Required by all managers of an organization
(b) Processes data to yield information of value in tactical management
(c) Provides operational information
(d) Allows better management of organizations

VIZAG STEEL MT 2011

Ans. (b) : Management information system is processes data to yield information of value in tactical management.

1187. Which of the following statements concerning a Gantt is true?

(a) Gantt charts are particularly helpful for scheduling and planning large projects
(b) Gantt charts are particularly helpful for scheduling and planning projects with complex precedence relationships
(c) The Gantt chart has been a popular project scheduling tool, but is not widely used now
(d) The Gantt chart indicates where extra time is available and activities can be delayed

RPSC Lect. (Tech. Edu. Dept.) 2011
ESE 2003

Ans. (c) : A Gantt chart is a bar chart that provides is visual view of tasks scheduled over time. A Gantt chart is used for planning projects of all sizes and its a useful way of showing what work is scheduled to be done on a specific day. It can also help to view the start and dates of a project in one simple project. The Gantt chart has been a popular project scheduling tool, but is not widely used now.

1188. Which of the following heuristics in a one-machine shop will always minimize flow time and average number of jobs in the system?

(a) SPT (b) DDATE
(c) CR (d) FCFS

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Shortest processing time, is a priority sequencing route that specifies that the job requiring the shortest processing time, is the next job to be processed. So, it minimize flow time & average no. of jobs in the system.

1189. Which of the following is not basic element of the JIT production system?

(a) cellular layouts
(b) push production system
(c) small-lot production
(d) flexible resources

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : JIT- JIT is an inventory management method whereby labour, material and good (focused in manufacturing) are re-filled or scheduled to arrive exactly when needed. It used in different techniques like cellular layouts, small- lot production, flexible resources etc.

1190. What is the method by which the master schedule is adjusted to balance the due dates of jobs or orders against the capacity of the plant and its individual ..... and facilities called as?

(a) Resource planning (b) Scheduling
(c) Capacity planning (d) Integration

VIZAG STEEL MT 2011

Ans. (c) : Capacity planning

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1191. The act of expanding the process production is referred as :

(a) routing (b) scheduling
(c) follow up (d) dispatching

J&K PSC Civil Services Pre, 2010

Ans. (c)

1192. Consider the following characteristics :

1. Limited time frame
2. Narrow focus
3. Less bureaucratic
4. Used only for providing services

For which of these characteristics, the project management is different from general operations management?

(a) 1, 2, 3 and 4 (b) 1, 3 and 4 only
(c) 2, 3 and 4 only (d) 1, 2 and 3 only

UPSC JWM Advt. No.-50/2010

Ans. (a) : The project management is different from general operations management, such as–
If have limited time frame, narrow focus, less bureaucratic & used only for providing services.

1193. Jobs going ahead of schedule are conveniently shown in :

(a) Pie chart (b) Bar chart
(c) Gantt chart (d) SIMO chart

J&K PSC Civil Services Pre, 2010

Ans. (c) : Jobs going ahead of scheduled are conveniently Gantt chart.

1194. Pre-planning

(a) is the end of all planning
(b) is the beginning of control
(c) culminates in routing
(d) is the substance of control

TNPSC AE, 2008

Ans. (b) : Pre-planning is the beginning of control.

1195. An organisation containing manufacturing, marketing and finance is called

(a) matrix organisation
(b) functional organisation
(c) flow network organisation
(d) modular organisation

TNPSC AE, 2008

Ans. (b) : A function organisation is a common type of organizational structure in which the organization is divided into smaller groups based on specialized functional areas.

1196. The following data pertain to a worker: base rate = Rs. 20 per hour
Time taken for completing the job = 2 hours
Standard time = 3 hours
Under Halsey plan the total earning of the worker is

(a) Rs. 36.67 (b) Rs. 40.67
(c) Rs. 46.67 (d) Rs. 56.57

ISRO Scientist/Engineer 2008
ESE 1993

Ans. (c) : Base rate = 20 Rs.
time taken for completing the job = 2 hours.
Standard time = 3 hours
Under Halsey plan
Assuming 33½% Incentive

T_s = Standard time
T_a = time taken to complete a job

$$E = RT_a + \frac{R}{3}(T_s - T_a)$$

R = 20 Rs. per hour
T_s = 3 hour
T_a = 2 hour

$$\therefore \quad E = 20 \times 2 + \frac{20}{3} \times (3.2)$$

= 46.67 Rs.

1197. Poka-Yoke is a concept of

(a) mistake-proofing systems
(b) rework of jobs
(c) inventory control system
(d) tool for production planning control

OPSC Civil Services Pre 2006

Ans. (a) : Poka yoke is a lean manufacturing tool that refers to "mistake-proofing" or "error-proofing" a process. It was originally coined by Shigeo Shingo in the 1960s and implemented at Toyota as part of the Toyota Production System (TPS).

1198. The main advantage of line organization is its-

(a) Effective command and control
(b) Defined responsibilities at all levels
(c) Rigid discipline in the organization
(d) All of the above

ISRO Scientist/Engineer 2006

Ans. (d) : Merits of line organization–
- Simplicity
- Clear-cut division of authority and responsibility
- Strong discipline
- Unified control
- Prompt decisions
- Flexibility

1199. A combination of individual machine tools arranged sequentially and properly integrated and interlocked is known as

(a) production line
(b) numerically controlled machine
(c) machining centre
(d) transfer machine

WBPSC AE 2003

Ans. (d) : A combination of individual machine tools arranged sequentially and properly integrated and interlocked is known as transfer machine.

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09.

Miscellaneous Questions

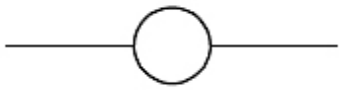
1. Data mining is a process used for _____

(a) designing of database tables
(b) database administration
(c) programming databases
(d) detecting patterns in large volume of operational data

VIZAG Steel MT 24.01.2021 Shift-II
VIZAG STEEL MT 14.12.2020

Ans. (d) : Data mining is a process used by companies to turn raw data into useful information. Data mining is a process used for detecting patterns in large volume of operational data.

2. The following symbol in the ladder logic represents :



(a) Normally open contacts
(b) Normally closed contacts
(c) Outputs loads
(d) Special instruction

ESE 18.07.2021

Ans. (c) : Outputs loads.

3. Statement (I): Wear is an issue whenever two components operate with relative motion between them or when liquids or solids impinge on a surface at high velocity.
Statement (II): Wear is often cumulative and can eventually render the components incapable of delivering their expected performance.

ESE 18.07.2021

Ans. (b) :
Wear is the undesirable removal of solids from a sliding or rolling component. Wear is generally proportional to the applied load and the amount of sliding erosive wear is the cutting of furrows on a surface by hard particles contained in a fluid travelling at high velocity.
It is often cumulative and leads to poor performance of machining parts.

4. Statement (I): Increased productivity, reduced cost of labour and improved quality can be achieved by automation.
Statement (II): Due to automation in process inventory, dependence on operator skills may be increased.

ESE 18.07.2021

Ans. (c) :
- Increased productivity, reduced cost of labour, improved qualities can be achieved by automation.
- Due to automation is process inventory, dependence on operator skills increased.

Hence, statement (II) is incorrect.

5. In 2nd angle projection top view lies _____ and front view lies _____line of intersection of HP and VP.

(a) above, above (b) below, above
(c) below, below (d) above, below

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) :

6. What is a fully equipped computer that is worn as a piece of clothing or attacked to a piece of clothing similar to the way that cell phone is carried on belt?

(a) Wearable computer (b) GPS
(c) Implant chip (d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) :

7. ERP stand for

(a) Enterprise resolution planning
(b) Enterprise reverse planning
(c) Enterprise resource planning
(d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : ERP stand for enterprise resource planning.

8. Only rocket engines can be propelled to SPACE because

(a) They can generate very high thrust
(b) They have high propulsion efficiency
(c) These engines can work on several fuels
(d) They are not air-breathing

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) :

9. The potential which exists in a PN juntion to cause drift of charge carriers is called

(a) Contact potential (b) diffusion potential
(c) ionisation potential (d) threshold potential

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) :

10. The basic rapid prototyping process is

(a) EDM
(b) ECM
(c) Stereo lithography
(d) AJM

PPSC Asstt. Municipal Engineer 15.06.2021

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Ans. (c) : The basic rapid prototyping process is stereo lithography Stereolethography is a form of 3D printing technology used for creating models, prototypes patterns and production parts in a layer by larger fashion using photochemical processes by which light causes chemical misnomers and oligomers to cross link together to form polymers.

11. In 3-D wireframe modeling, the curved surfaces are represented by

(a) suitable under spaced generators.
(b) suitable spaced generators.
(c) randomly spaced generators.
(d) suitably over spaced generators.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : In 3-D wireframe modeling, the curved surfaces are represented by suitably speed generators.

12. Which one of the following is not a psychosocial hazard?

(a) Conflict (b) Work overload
(c) Role uncertainty (d) Allergic reaction

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (d) : Common psychosocial hazards–
- Work-related stress
- Allergic reaction
- Lone or remote working
- Violence in the workplace

When psychosocial hazards are not effectively managed, they can negatively impact on organizational measures including productivity absenteeism and the turnover.

13. ISO 45001 : 2018 refers to _____.

(a) Occupational Health and Safety Management System
(b) Environmental Management System
(c) Qualitative Management System
(d) Energy Management Systems

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (a) : ISO 45001 : 2018 refers to occupational health and safety management systems.

14. The entity number representing plane in IGES (Initial Graphics Exchange Specification) format is _____.

(a) 106 (b) 108
(c) 104 (d) 102

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (b) : The entity number representing plane in IGES (initial graphics Exchanges specification) format is 108. Initial graphics exchange specification (IGES) is neutral file format designed to transfer 2D and 3D drawing data between dissimilar CAD systems.

15. OSHA refers to _____.

(a) Operational Safety and Health Administration
(b) Operational Safety and Health Association
(c) Occupational Safety and Health Association
(d) Occupational Safety and Health Administration

VIZAG Steel MT 24.01.2021 Shift-II, Shift-I

Ans. (d) : OSHA refers to occupational safety and health administration.

16. What material handling equipment is used if high volume of material is to be moved from one fixed location to another?

(a) Trucks (b) Fork lift
(c) Conveyors (d) Cranes

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-I

Ans. (c) Conveyors are used, when material is to be moved frequently between specific points, to move materials over a fixed path, where there is a sufficient flow volume to justify the fixed conveyor investment.

17. A conveyor is preferred to fork lift and trucks when

(a) Material have to be moved on an infrequent basis
(b) Loads are mixed in size and weight
(c) There is horizontal material flow pattern
(d) Route does not vary and continuous movements are required

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-I

Ans. (d) As we know conveyors are used whenever material is to be moved frequently between specific point to move material over a fixed path where there is a sufficient flow volume to justify the fixed conveyor investment.

18. If the speed of a wind stream remains unchanged while passing through the rotor, then :

(a) a large power will be generated
(b) zero power will be generated.
(c) the flow is known as stalled flow.
(d) the speed of the rotor will be extremely high.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (b) : If velocity remains unchanged then the pressure difference on upper and lower surface of blade is not comparable and hence lift force is negligible as a result no power is generated.

19. If x is the mean of data 3, x, 2 and 4, then the mode is_____

(a) 2
(b) Average of 2, 3 and 4
(c) 4
(d) 3

Assam Engg. College AP/Lect. 18.01.2021

Ans. (d) : Average $= \frac{3+x+2+4}{4}$

$$x = \frac{3+x+2+4}{4}$$

$$3x = 9$$

$$\boxed{x = 3}$$

Then, mode of data is 3.

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20. The reasons for the necessity of testing of sewage are

(a) to recommend the type of treatment to be adopted for the particular of sewage
(b) to know the specific gravity of the sewage
(c) to know the sulphur cycle
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : The reasons for the necessity of testing of sewage are to recommend the type of treatment to be adopted for the particular of sewage.

Purpose of sewage treatment–

(1) To kill the pathogenic bacteria present in the sewage which may result in water born diseases like cholera etc.
(2) To avoid unhygienic condition in the area because of highly fouled sewage.

21. The physical impurity like turbidity in water can be measured by

(a) turbidity rod
(b) Jackson turbidimeter
(c) Baylis turbidimeter
(d) All of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : The physical impurity like turbidity in water can be measured by –

(1) Turbidity rod
(2) Jackson turbidimeter
(3) Baylis turbidimeter

The instrument used for measuring physical impurity like turbidity is called turbidimeter.

Which measures the intensity of light scattered at 90 degree as a beam of light passes through a water sample the unit used in the ancient times was Jackson turbidity measured with the Jackson candle turbidimeter.

22. Salient features of water supply schemes are

(a) population forecast and record of industry
(b) assessment of water demand and record of public place
(c) quality and sources of water
(d) All of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (d)

(1) Resource planning and recycling for providing maximum availability.
(2) To give importance to the impact of projects on human settlements and environment.
(3) Population forecast and record of industry.
(4) Assessment of water demand and record of public place.
(5) Quality and sources of water.

These are the features of the water supply scheme.

23. If in a beam, there is one redundant reaction i.e. there is one more reaction than could be determined from the equilibrium equations alone, the beam is said to be :

(a) indeterminate of second order
(b) indeterminate of first order
(c) indeterminate of zeroth order
(d) indeterminate of nth order

CGPSC AE 15.01.2021

Ans. (b) : If in a beam, there is one reduntant reaction, i.e., there is one more reaction than could be determined from the equilibrium equations, alone, the beam is said to be indeterminate of first order.

24. Risk is very clearly defined as the _____.

(a) Probability of causing harm
(b) Attitude in meeting unsafe conditions
(c) Take chance in facing harm
(d) Situation where accidents are bound to occur

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (a) : Risk is the possibility of something bad happening. Risk involves uncertainty about the effect/implications of an activity with respect to something that humans value, often focusing on negative, undesirable consequences.

25. What is the ISO standard with year of approval for OHS management system?

(a) ISO 45001 : 2018 (b) ISO 45002 : 2017
(c) ISO 45003 : 2016 (d) ISO 45005 : 2014

VIZAG Steel MT 24.01.2021 Shift-II

Ans. (a) : The ISO standard with gear of approval for OHS management system.
ISO 45001 : 2018

26. Maximum number of membership in a private limited company can be

(a) 02 (b) 50
(c) 25 (d) 100

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-I

Ans. (b) : According to the provisions of companies act 2013. A private limited company can be started with minimum 2 members and maximum 50 members.

27. What is the minimum number of members required to form a public limited company?

(a) 10 (b) 2
(c) 5 (d) 7

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-I

Ans. (d) As prescribed by the companies act the minimum members to for a public company is 7 and who has a minimum paid up share capital of Rs. 5 Lac.

28. One of the assumptions in the game theory is:

(a) All players act rationally and intelligently
(b) Winner alone acts rationally
(c) Loser acts intelligently
(d) Both the players believe in luck

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-I

Ans. (a) :

29. Star topology refers to a dedicated :

(a) point to point link to a central controller.
(b) tree like link to a central controller.
(c) ring like link to a central controller
(d) bus bar like link to a central controller.

NLC GET 17.11.2020, Shift-II

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Ans. (a) : In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.
The device are not directly linked to one another. Unlike a mesh topology a star topology does not allow direct traffic between devices.

30. Plasticizers used in polymers generally have :
(a) low vapour pressure and low molecular weights
(b) low vapour pressure and high molecular weights
(c) high vapour pressure and low molecular weights
(d) high vapour pressure and high molecular weight

NLC GET 17.11.2020, Shift-II

Ans. (a) : Plasticizers used in polymer generally have low vapour pressure and low molecular weight that increasing the spacing between chains of crystalline polymer to make them more flexible and thereby, toughness plasticizers increase the Flow and thermo-plasticity of a polymer.

31. In brakes, heat dissipation factor or coefficient of heat transfer in $W/m^2/°C$ varies from:
(a) 5 to 20 (b) 20 to 30
(c) 45 to 75 (d) 30 to 45

NLC GET 17.11.2020, Shift-II

Ans. (d) : The value of heat dissipation factor or co-efficient of heat transfer in $W/m^2/°C$ varies from 30 to 45. The expressions for the heat dissipated are quite approximate and should serve only as an indication of the capacity of the brake to dissipate heat.

32. Refractive Index of a material is defined as :
(a) velocity of light in vacuum to that within that material
(b) velocity of sound in vacuum to that within that material
(c) amount of energy reflected by the material to the total amount of energy supplied.
(d) amount of light reflected by the material to the total amount of light impinged.

NLC GET 17.11.2020, Shift-II

Ans. (a) : Refractive Index– The ratio of the speed of light in a vacuum to its speed in a specific medium.
It is a number that indicates the number of times slower than a light wave would be in the material than it is in a vacuum.
The refractive index is represented by symbol n and is given by–

$$n = \frac{c}{v}$$

Where, c is the velocity of light in vacuum.
v is the velocity of light in a substance.

33. Dynamometer is used to find :
(a) torque at constant speed
(b) indicated power at constant speed
(c) volumetric efficiency at constant speed
(d) friction power at constant speed

NLC GET 17.11.2020, Shift-II

Ans. (a) : A dynamometer used to determine the torque at constant speed and power required to operate a driven machine such as a pump. In that case a motoring or driving dynamometer is used. A dynamometer that is designed to be driven is called an absorption or passive dynamometer. A dynamometer that can either drive or absorb is called a universal or active dynamometer

34. STL (Standard Tessellation Language) file format represents the boundary of the material as _____ .
(a) rectangles (b) tetrahedrons
(c) quadrilaterals (d) triangles

VIZAG STEEL MT 14.12.2020

Ans. (d) : STL (Standard Tessellation Language) file format represents the boundary of the material as triangles.
STL is a file format native to the stereo lithography CAD software created by 3D systems. STL has several acronyms such as standard triangle language, and "standard tessellation language".

35. IGES is the abbreviation of _____.
(a) Important Graphics Exchange Specification
(b) Internal G Code Exchange Specification
(c) Initial Graphics Exchange Specification
(d) Intricate G Code Exchange Specification

VIZAG STEEL MT 14.12.2020

Ans. (c) : IGES–Initial Graphics Exchange Specifications.
IGES is a vendor neutral life format that allows the digital exchange of information among computer aided design systems.

36. In a fault tree failure analysis, a primary event is _____.
(a) dependent on events that are lower in the tree
(b) considered as independent, and not to be broken down further
(c) dependent on events that are higher in the tree
(d) the top event of the tree

VIZAG STEEL MT 14.12.2020

Ans. (b) : In a fault tree failure analysis, a primary event is considered as independent, and not to be broken down further.
Fault Tree Analysis (FTA) is a graphical tool to explore the causes of system level failures. It uses Boolean logic to combine a series of lower level and it is basically a top down approach to identify the component level failures that cause the system level failure to occurs. It is an easier method than FMEA (Failure Mode and Effects Analysis).

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37. In HAZOP study, guide word "part of" stands for _____.

(a) quantitative decrease
(b) qualitative increase
(c) quantitative increase
(d) qualitative decrease

VIZAG STEEL MT 14.12.2020

Ans. (d) : In HAZOP study, guide word "part of" stands for qualitative decrease.

HAZOP (Hazard and operability study) is a structured and systematic examination of a complex planned or existing process or operation in order to indentify and evaluate problems that may represent risks to personal or equipment.

38. The sound pressure level in living room should be at or below

(a) 70 decibel (b) 60 decibel
(c) 50 decibel (d) 40 decibel

Assam PSC AE (PHED), 18.10.2020

Ans. (d) : The sound pressure level in living room at or below 40 decibel.

Library, bird call, lowest limit of urban ambient sound is 40 decibel.

Quit rural areas - 30 decibel.

Whisper rusting level - 20 decibel.

Breathing - 10 decibel.

Normal conversation is about 60 decibel.

Passenger car, radio or TV is about 70 decibel.

39. Eddy current dynamometer is _____ type of dynamometer

(a) Absorption (b) Generator
(c) Transmission (d) Reactive

Haryana PSC AE (PHED) 04.09.2020, Paper-I

Ans. (a) : Type of absorption dynamometer –
(i) Eddy current (ii) Magnetic power brake (iii) Hysteresis brake (iv) Electric motor (v) fan brake (vi) hydraulic brake.

40. As per Factories Act, "Child" means a person who has not completed his _____year of age. [Sec 2(c)]

(a) Fourteenth (b) Fifteenth
(c) Sixteenth (d) Eighteenth

RPSC IOF, 2020

Ans. (b) : No child below the age of 14 shall be required to, or allowed to, work in any factory. The act defines a child as a person who is below 15 year of age.

41. The mirror image of a 2D reflection is generated relative to an axis of reflection by rotating the object by:

(a) 270 degree (b) 180 degree
(c) 360 degree (d) 90 degree

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Reflection is a kind of rotation where the angle of rotation is 180 degree.

- The reflected object is always formed on the other side of mirror.
- The size of reflected object is same as the size of original object.

42. Which among the following is NOT a type of crank shaft design?

(a) Overhung crank shaft
(b) Centre crank shaft
(c) Single or multi-throw crank shaft
(d) Rocker arm multi-throw crank shaft

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Type of crank shaft design are
(I) Overhung crank shaft
(II) Centre crank shaft
(III) Single or multi-throw crank shaft

43. A fabric filter is to be constructed using bags that are 0.1 m in diameter and 5.0 m long. The bag house is to receive 5 m^3/s of air. Filtering velocity is 2.0 m/min. Determine the minimum number of bags required for a continuous removal of particulate matter.

(a) 96 (b) 106
(c) 78 (d) 88

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : d = 0.1 m, ℓ = 5 m, Q = 5 m^3/sec, v = 2 m/min $= \frac{1}{30}$ m/sec

$A_{bag} = \pi dh = \pi \times 0.1 \times 5 = 1.5708\ m^2$

$A_c = \frac{Q}{V} = \frac{5}{\left(\frac{1}{30}\right)} = 150\,m^2$

Number of bag required $= \frac{A_c}{A_b} = \frac{150}{1.5708} = 95.4927 \approx 96$

44. Before starting an isometric drawing in AutoCAD, the drafter needs to _____.

(a) Set the grid to isometric
(b) Set the current layer to Defpoints
(c) Turn Object Snap off
(d) Turn Ortho off

MECON MT 2019

Ans. (a) :

45. The basis of reverse osmosis is that

(a) osmotic pressure is greater than hydrostatic pressure
(b) osmotic pressure is equal to hydrostatic pressure
(c) hydrostatic pressure is greater than osmotic pressure
(d) osmotic pressure does not exist

Assam PSC AE (PHED), 18.10.2020

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Ans. (c) : Reverse osmosis occurs when the water is moved across the membrane against the concentration gradient from lower concentration to higher concentration.
If hydrostatic pressure greater than the osmotic pressure is applied to the high concentration the direction of water flow through the membrane can be reversed.

46. A thin and closed diaphragm element is used as a sensor while measuring :
(a) velocity (b) temperature
(c) pressure (d) mass

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : A thin and closed diaphragm element is used as a sensor while measuring pressure.

47. Which of the following is for environment management?
(a) ISO- 9000 (b) ISO-14000
(c) ISO- 26000 (d) ISO- 31000

RPSC IOF, 2020

Ans. (b) : ISO 14000 is the international standard that specifies requirement for an effective Environmental Management System (EMS).

48. Occupational health not concerned merely with the occupational syndromes, rather it also includes
(a) All the factor affecting health of workers
(b) The factors affecting only physical health of workers
(c) The factors affecting only mental health of workers
(d) None of these

RPSC IOF, 2020

Ans. (a) : Four databases are essential to the effectiveness of an occupational health information system. These are job history, work site, environmental agents and heath and safety information all factors affecting health of workers.

49. The main objective of ISO-9000 family of quality management is
(a) Employee satisfaction
(b) Skill enhancement
(c) Customer satisfaction
(d) Environmental issues

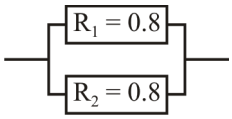
RPSC IOF, 2020

Ans. (c) : ISO- 9000 is quality management standard that intended to increase business efficiency and customers satisfaction.
The goal of ISO-9000 is to embed a quality increasing productivity, reducing unnecessary costs and ensuring quality of processes and products.

50. A system contains 2 components. Reliability of the components is measured for 800 hours and time to failure components are found to be exponential. If each component has reliability of 0.8, then the system reliability at the end of 800 hours will be_____
[system are connected in parallel structure]
(a) 0.22 (b) 0.16
(c) 0.96 (d) 0.32

RPSC IOF, 2020

Ans. (c) :



Reliability of components,
$R_1 = 0.8$
$R_2 = 0.8$
Reliability of system,
$R = 1 - (1 - R_1)(1 - R_2)$
$= 1 - (1 - 0.8)(1 - 0.8)$
$= 1 - (0.2 \times 0.2)$
$= 0.96$

51. A parallel plate capacitor having an area of 6×10^{-4} m^2 and a plate separation of 3×10^{-3} m, across which a potential of 10 V is applied. If a material having a dielectric constant of 5 is positioned within the region between the plate, what will be the capacitance? [Permittivity of vacuum = 8.85×10^{-12} F/m]
(a) 8.85×10^{-12} F (b) 17.7×10^{-12} F
(c) 26.55×10^{-12} F (d) 35.4×10^{-12} F

APPSC Poly Lect. 13.03.2020

Ans. (a) : Capacitance $(C) = \dfrac{K \in_0 A}{d}$
Given;
$K = 5$, $\in_0 = 8.85 \times 10^{-12}$ F/m
$A = 6 \times 10^{-4}$ m^2, $d = 3 \times 10^{-3}$ m
$C = \dfrac{5 \times 8.85 \times 10^{-12} \times 6 \times 10^{-4}}{3 \times 10^{-3}}$
$= 10 \times 8.85 \times 10^{-13}$
$C = 8.85 \times 10^{-12}$ F

52. Which of the following is NOT a part of carriage in the centre lathe?
(a) Tool post (b) Compound rest
(c) Gear box controls (d) Apron

MECON MT 2019

Ans. (c) :

53. A point 'P' is above Horizontal Plane (HP) and in front of Vertical Plane (VP). The point is in the-
(a) First quadrant (b) Second quadrant
(c) Third quadrant (d) Fourth quadrant

MECON MT 2019

Ans. (a) :

54. Which of the following positions is NOT possible for a right solid?
(a) Axis perpendicular to HP and parallel to VP
(b) Axis parallel to VP and perpendicular to HP
(c) Axis parallel to both VP and HP
(d) Axis perpendicular to both VP and HP

MECON MT 2019

Ans. (d) :

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55. The edge formed by the intersection of flute surface and the body clearance is known as-

(a) Heel (b) Face
(c) Flank (d) Web

MECON MT 2019

Ans. (a) :

56. Which of the following is formed where three or more surfaces intersect?

(a) Oblique (b) Line
(c) Edge (d) Vertex

MECON MT 2019

Ans. (d) :

57. _____is the plane upon which the top view is project.

(a) Horizontal (b) Vertical
(c) Profile (d) Base

MECON MT 2019

Ans. (a) :

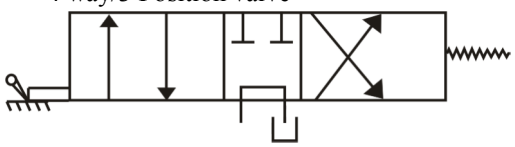
58. The flow direction to and from double acting cylinder is typically controlled by a

(a) 1 way DCV valve (b) 2 way DCV valve
(c) 3 way DCV valve (d) 4 way DCV valve

APPSC AEE SCREENING 17.02.2019

Ans. (d) : Flow direction to and from a double acting cylinder is typically controlled by

4 way/3 Position valve



59. Which is not the correct method for calculating traffic for electrical energy?

(a) Hopkinson demand rate
(b) Block meter rate
(c) Three part tariff rate
(d) Incremental demand rate

OPSC AEE 2019 PAPER - II

Ans. : (d) : Incremental demand date

60. The product is assembled from parts A and B. The probability of defective parts A and B are 0.2 and 0.1 respectively. Then the probability of the assembly of A and B to be non-defective is :

(a) 0.72 (b) 0.7
(c) 0.02 (d) 0.3

OPSC AEE 2019 Paper-I

Ans. : (b) : A and B non defective

$= (1 - 0.2) \times (1 - 0.1) = 0.8 \times 0.9 = 0.72$

61. Consider a system having three subsystems with reliability 0.6, 0.9 and 0.8. If the subsystems are put in series then the reliability of the system would be :

(a) 0.187 (b) 0.655
(c) 0.821 (d) 0.432

OPSC AEE 2019 Paper-I

Ans. : (d) : The reliability of a system consisting of h components connected in series is given by

$R_{sys} = R_A.R_B.....R_n$

$R_{sys} = 0.6 \times 0.9 \times 0.8$

$R_{sys} = 0.432$

62. The 14 principles of continuous improvement are enumerated by

(a) Juren (b) Deming
(c) Shewart (d) Ishikewa

TNPSC AE 2019

Ans. (b) : The 14 principles of continuous improvement are enumerated by deming.

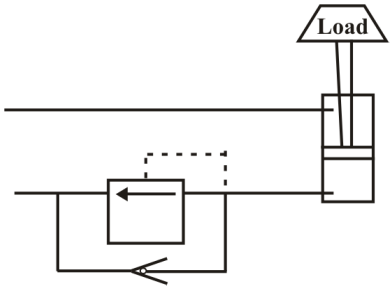
63. The valve used to control as vertical cylinder to prevent it from descending due to external load is

(a) Pressure relief valve
(b) Counterbalance valve
(c) Pressure compensated valve
(d) Directional control valve

APPSC AEE SCREENING 17.02.2019

Ans. (b) : Counter Balance PCV: It is called holding valve, is used to prevent load (weight) from falling unwantedly.

Ex: In crane load lifting operation, vertical presses lift trucks where position (or) hold suspended loads. It is placed in between actuating cylinder and DCV. It consists feedback check valve to sense load sudden fall and works against it, not allow hydraulic oil flow back.



Counter Balance Valve

It creates holding pressure in the cylinder to prevent over slide of the external load while falling.

64. Which one of the following types of tracker uses liquid contained in canisters that can turn easily into vapour?

(a) Active tracker
(b) Passive tracker
(c) Single-axis tracker
(d) Altitude-azimuth tracker

ESE 2019

Ans. (b) : (i) Active trackers require external power to drive tracking mechanism whereas passive trackers do not require external power.

(ii) Passive trackers use low boiling point liquid which vapourises after absorbing solar heat. The tracker can tilt to one or other side due to imbalance caused by vapourisation of liquid.

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65. The duct is said to be low velocity duct if the mean velocity of air in the duct?
(a) 20 m/s < V < 25 m/s
(b) V < 10 m/s
(c) 25 m/s < V < 30 m/s
(d) 15 m/s < V < 20 m/s

SJVN ET 2019

Ans. (b) : V < 10 m/s

66. Global warming is caused by :
(a) Ozone (b) Carbon dioxide
(c) Nitrogen (d) Carbon monoxide

GPSC ARTO Pre 30.12.2018

Ans. (b) : Global warming is caused by carbon dioxide.

67. The frequency of ultrasonic waves is
(a) less than 20 Hz
(b) between 20 to 10000 Hz
(c) between 20 to 20000 Hz
(d) above 20000 Hz

ISRO Scientist/Engineer 22.04.2018

Ans. (d) : Range of frequency of waves
Infrasonic $\leq$ 20 Hz
20 Hz < sonic $\leq$ 20 kHz
Ultrasonic > 20 kHz

68. Loudness of the sound does not change with the change in
(a) frequency
(b) amplitude
(c) distance from the source of sound
(d) none of these

ISRO Scientist/Engineer 22.04.2018

Ans. (a) : Loudness refers to how loud or soft a sound seems to a listener. The loudness of sound is determined in turn, by the intensity of the sound waves.

69. A single-phase full wave rectifier is a
(a) single pulse rectifier
(b) multiple pulse rectifier
(c) two pulse rectifier
(d) three pulse rectifier

GPSC Executive Engineer 23.12.2018

Ans. (c) : Two pulse rectifier.

70. Which of these is the best description of a Zener diode?
(a) it works in the forward region
(b) it is a rectifier diode
(c) it is a constant electric current device
(d) it is a constant voltage device

GPSC Executive Engineer 23.12.2018

Ans. (d) : It is a constant voltage device.

71. A discontinuous real function can be expressed as
(a) Taylor's series and Fourier series
(b) Taylor's series and not by Fourier's series
(c) neither Taylor's series nor Fourier's series
(d) not by Taylor's series, but by Fourier's series

GPSC Executive Engineer 23.12.2018

Ans. (d) :

72. If the doping level of a crystal diode is increased, the breakdown voltage:
(a) remains the same (b) increased
(c) decreased (d) none of the above

GPSC Executive Engineer 23.12.2018

Ans. (c) : decreased

73. Library function pow ( ) belongs to which header file?
(a) mathio.h (b) math.h
(c) square.h (d) stdio.h

GPSC ARTO Pre 30.12.2018

Ans. (b) : Library function pow ( ) belongs to math.h header file.

74. With respect to computer applications CISC stands for :
(a) Compound Instruction Set Computers
(b) Common Instruction Set Computers
(c) Complex Instruction Set Computers
(d) Complex Instruction Set Compilers

GPSC ARTO Pre 30.12.2018

Ans. (c) : CISC stands for Complex Instruction Set Computers. CISC is a computer in which single instructions can execute several low level operation or are capable of multistep operation or addressing modes within single instructions.

75. The maximum efficiency of jet propulsion of a ship with inlet orifices at right angles to the direction of motion of ship, is
(a) 40% (b) 50%
(c) 60% (d) 80%

Nagaland PSC (CTSE) 2018, Paper-II

Ans. (b) : Efficiency of jet propulsion of a ship with inlet orifices at right angles to the direction of motion of ship.

$$\eta = \frac{2vu}{(v+u)^2}$$

Where V – jet velocity
u – ship velocity
For maximum efficiency.

$$\frac{d\eta}{dv} = 0 \Rightarrow 2u\left[(v+u)^{-2}(1) + v(-2)(v+u)^{-2-1} = 0\right]$$

$u = v$

$$\eta_{max} = \left[\frac{2vu}{(v+u)^2}\right]_{v=u}$$

$\eta_{max} = 50\%$.

76. Piezo electric type of load cells can be used for measurement of
(a) dynamic forces only
(b) dynamic forces and static forces provided the load cells have a small time constant
(c) dynamic forces and static forces provided that the load cells have a large time constant
(d) static force only

TNPSC AE 2018

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Ans. (c) : Piezo electric type of load cells can be used for measurement of dynamic forces and static forces provided that the load cells have a large time constant.

77. 'Planning involves the selection of objectives, policies, procedures and programmes from among alternatives' was stated by

(a) Koontz and O' Donnell
(b) Hodge
(c) Alford and Betty
(d) Hurley

TNPSC AE 2017

Ans. (d) : 'Planning involves the selection of objectives, policies, procedures and programmes from among alternatives' was stated by Hurley.

78. Which one of the following element was not involved in directing?

(a) Motivation (b) Leadership
(c) Communication (d) Delegation

TNPSC AE 2017

Ans. (d) : Delegation- Delegation is the assignment of responsibility to another person for the purpose of carrying out specific job-related activities. Delegation is a shift of decision making authority from one organizational level to other.

79. Most shops discourage customers from roaming around the shop work areas because the customers

(a) often want to help
(b) may steal the data and shared it to the competitor
(c) could be in danger without reality it
(d) may find out they are paying for warranty work

TNPSC AE 2017

Ans. (c) : Most shops discourage customers from roaming around the shop work areas because the customers could be in danger without reality it.

80. A material store comprises of three 2-sided aisles, each having corresponding to 50 and 15 storage compartments. The total storage capacity will be

(a) 9000 compartments (b) 4500 compartments
(c) 3500 compartments (d) 2500 compartments

BPSC AE Mains 2017 Paper - VI

Ans. : (b) : Capacity = $2 \times 3 \times 50 \times 15$
= 4500 compartments

81. Localizing an object in an image and selectively analyzing the object in a series of redundant layers is known as

(a) Maxwell pyramid (b) Faraday pyramid
(c) Gaussian pyramid (d) Turning test

TNPSC AE 2017

Ans. (c) : Localizing an object in an image and selectively analyzing the object in a series of redundant layers is known as Gaussian pyramid.

82. Match the List I with List II and choose the correct answer:

List I	List II
(A) Seiko	1. Orderliness
(B) Seiketsu	2. Clean up
(C) Seiso	3. Personal cleanliness
(D) Seiton	4. Proper arrangement

	A	B	C	D
(a)	4	3	2	1
(b)	4	3	1	2
(c)	3	4	2	1
(d)	1	3	2	4

TNPSC AE 2017

Ans. (a) :

Japanese	English
Seiri	Sort
Seiton	Set in order
Seiso	Shine
Seiketsu	Standard size
Shitsuke	Sustain

83. The component of the torque convertor that allows multiplication of torque is the

(a) Turbine (b) Impeller
(c) Pump (d) None of the above

TNPSC AE 2017

Ans. (d) : Stator, the component of the torque convertor that allows multiplication of torque.

84. The stator windings in an alternator are being tested with an ohmmeter. The resistance measured between each of the three windings is nearly 0 ohms. What does it indicate?

(a) The stator windings do not have an open circuit
(b) The stator windings are shorted to the stator frame
(c) The stator windings are open
(d) The stator windings are magnetized

TNPSC AE 2017

Ans. (a) : The stator windings in an alternator are being tested with an ohmmeter. The resistance measured between each of the three windings is nearly 0 ohms it indicate the stator windings do not have an open circuit.

85. A waveform repeats itself 60 times per second. What is the frequency of the waveform?

(a) 120 hertz (b) 1 hertz
(c) 60 hertz (d) 3600 hertz

TNPSC AE 2017

Ans. (c) : A wave form repeats itself 60 times per second it means the frequency of the waveform is 60Hz.

86. Which of the following is NOT a feature of third-angle projection?

(a) The object liens in the third-angle.
(b) The plane of projection lies between the object and the observer.

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(c) The top view is drawn above the front view.
(d) The top view is drawn below the front view.
(e) The right hand side view is drawn to the right side of the front view.

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : Feature of third Angle Projection
1. The object is magnified to be in third Quadrant
2. The plane of projection lies between the observer and object.
3. The plane of projection is assumed to be trans parent.
4. When view are drawn in their relative position. Top view comes above front view Right side view drawn to the right side of elevation.

87. The profile plane is a plane which is
(a) 45º to both H.P. and V.P.
(b) 60º to both H.P. and V.P.
(c) 60º to H.P. inclined to V.P.
(d) 90º to V.P. and inclined to H.P
(e) 90º to both H.P. and V.P.

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Ans. (d) : Plane of Projection:-A third plane perpendicular to both Co-ordinate planes and hence to the ground line is called a profile plane.
This plane is vertical in position, and may be used as a plane of projection.

88. Which of the following indicates the correct relative position of the object, plane and the observer in the first angle projection method?

(a) Observer ⇨ Plane ⇨ Object

(b) Observer ⇨ Object ⇨ Plane

(c) Plane ⇨ Observer ⇨ Object

(d) Object ⇨ Plane ⇨ Observer

(e) Object ⇨ Observer ⇨ Plane

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Ans. (b) :

Observer ⇨ Object ⇨ Plane

89. As per the first-angle projection system, where is the left-hand side view of an object drawn?
(a) To the right-side of the front view
(b) To the left-side of the front view
(c) Below the front view
(d) Above the front view
(e) Top of the front view

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : As per the first angle projection system, where is the lift-hand side view of an object drawn to the right-side of the front view.

90. The isometric projection of an object is one of the axonometric projections of the object in which its principal coordinate axes are:
(a) equally inclined at 30º to the plane of projection
(b) equally inclined at 45º to the plane of projection
(c) equally inclined at 60º to the plane of projection
(d) equally inclined at 45º to the H.P.
(e) equally inclined at 75º to the plane of projection

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Ans. (a) : The isometric projection of an object is one of the axonometric projection of the object in which its principal co-ordinate axes are equally inclined at 30º to the plane of projection.

91. The isometric projection of a circle is:
(a) the same circle itself
(b) an involute
(c) an obloid
(d) a circle of diameter reduced to 70% of the actual diameter
(e) an ellipse

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Ans. (e) : The isometric view or isomeric projection of a circle is an ellipse. It is obtained by using four centre method.

92. In an isometric projection, the dimensions of the object along the isometric axes or isometric lines are reduced to:
(a) 8.26% (b) 14%
(c) 50% (d) 72.6%
(e) 81.6%

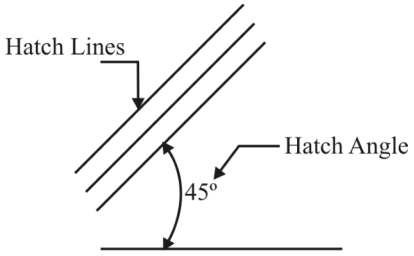
CGPSC ARTO & TSI 07.01.2017

Ans. (d) : In an isometric projection the dimensions of the object along the isometric axes or isometric lines are reduced to 81.6%

93. What is the angle at which the hatching lines are usually inclined with the reference line?
(a) 45º (b) 60º
(c) 75º (d) 90º
(e) 30º

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Hatching lines do not cross each other and they appear parallel to each other mostly at an angle of 45º.



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94. What is the use of dip-stick?

(a) To measure weight
(b) To measure diameter
(c) To measure speed
(d) To check oil level
(e) To clean holes

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : Dip -stick :- A stick or rod used to measure the depth of a liquid often used to check the level at which a liquid in an opaque or inaccessible tank or reservoir stands.

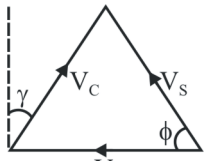
95. Which one of the following is FALSE?

(a) Number of Taps generally used for hand taping is 3
(b) Piispanen's model in machining is used to estimate shear strain
(c) Chip hammering during machining is due to improper chip control
(d) Cemented carbide tools are generally poor in compression

ISRO Scientist/Engineer 17.12.2017

Ans. (d)

(1) Three types of taps are generally used in sequence while doing hand taping
(i) Taper tap
(ii) Plug tap
(iii) Bottoming tap
(2)



Velocity diagram

Figure shows the shear strain **Piispanen** model where the workpiece material shears a deck of cards inclined to the free surface at an angle corresponding to the shear angle.

(3) Chip hammering occurs when cutting materials form a tough or abrasive chip. It occurs when the chip curls back and strikes the tool face away from the cutting edge.
Chip hammering results from improper chip control and can often be eliminated by changing the lead angle, depth of cut, feed rate or tool nose radius to alter the chip flow direction.
(4) Cemented carbide tools are generally poor in shear and cutting tools made out of it can machine much faster than HSS tools.

96. In CAD, the geometric transformation is expressed as P* = [T]P where T is the transformation matrix. If the matrix T is diagonal, then the transformation is called

(a) translation (b) reflection
(c) rotation (d) scaling

ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Translation–Moving an object is called a translation.
Scaling–Changing the size of an object is called a scale.

97. Datum features on the part are

(a) permanent
(b) temporary
(c) either permanent or temporary
(d) none of these

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Datum features are either permanent on the part or temporary (to be removed or altered during the manufacturing process). Temporary datums are often used as in-process datums to establish more permanent datums to be used in the final inspection process.

98. A cubic B-spline curve requires minimum .....control points.

(a) 3 (b) 4
(c) 5 (d) 6

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : B-spline curves are powerful generalization of Bezier curves. They provide local control of the curve shape as opposed to global control by using a special set of blending functions that provide local influence. They also provide the ability to add control points without increasing the degree of curve.
A cubic B-spline curve requires minimum 4 control points while quadratic spline curve requires minimum 3 control points.

99. Isometric projection of a sphere is

(a) Circle (b) Ellipse
(c) Hyperbola (d) Parabola

ISRO Scientist/Engineer 07.05.2017

Ans. (a) : Isometric projection ⇒ 3D of a solid shown in one view projection.

100. The method of development of a right regular prism is

(a) Parallel line method
(b) Radial line method
(c) Triangulation method
(d) Approximate method

ISRO Scientist/Engineer 07.05.2017

Ans. (a) : Parallel Line Method–Its used for developing prism and single curved surface like cylinders, in which all the edge/generation of lateral surface are parallel to each other.
Radial Line Method–Its employed for pyramids and single curved surface like cones in which apex taken centre and slant edge (or) generator as radius its developing.
Triangulation Method–Its used for developing transition pieces.
Approximation Method–Its double employed curved surface like sphere, as they are theoretically develop. The sphere of the developed by approximation method.

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101. Meter is defined as distance between the centre portion of two lines engraved on polished surface of a bar of :

(a) 90% platinum and 10% irridium at 15°C
(b) 90% platinum and 10 % irridium at 10°C
(c) 75% platinum and 25% irridium at 15°C
(d) 75% platinum and 25% irridium at 0°C

TRB Polytechnic Lecturer 2017

Ans. (b) : Meter is defined as distance between the centre portion of two lines engraved on polished surface of a bar of 90% platinum and 10 % irridium at 10°C.

102. In the case of machine drawing, the exploded assembly drawing is used to:

(a) help the mechanics for dismantling the machine for repairing purpose
(b) help the technicians on the shop floor for manufacturing the parts
(c) explain the working principle of each part
(d) prepare the foundation for installing the machine
(e) obtain patent drawing

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : In the case of machine drawing, the exploded assembly drawing is used to help the mechanics for dismantling the machine for repairing purpose.

103. The threshold concentration of sulphur dioxide and benzene in any industrial activity should not be permitted beyond

(a) 2 ppm & 15 ppm respectively
(b) 3 ppm & 20 ppm respectively
(c) 4 ppm & 30 ppm respectively
(d) 5 ppm & 25 ppm respectively

Nagaland PSC CTSE 2017 Paper-2

Ans. (d) : Threshold concentration limit for sulphur dioxide and benzene in any industrial activity should not be permitted beyond 5 ppm & 25 ppm.

104. Ergonomics deals with

(a) design of controls
(b) design of displays
(c) energy expenditure in hand and foot operations
(d) all the these

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : All the these

105. The basic series of preferred numbers are,

(a) R5, R10, R20, R40 and R80
(b) R10, R20, R30, R40 and R50
(c) R5, R10, R15, R20 and R25
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Basic series of preferred numbers are, R5, R10, R20, R40 and R80.

106. Series factor for R20 series is,

(a) $\sqrt[10]{20}$ (b) $\sqrt{20}$
(c) $\sqrt[20]{10}$ (d) $\sqrt[3]{20}$

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Series factor for R20 series is $\sqrt[20]{10}$

107. The external appearance is important in–

(a) consumer durables like refrigerators and audiovisual equipment
(b) industrial products like cranes and hoists
(c) machine elements like gearbox, coupline or pressure vessel
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : The external appearance is important is consumer durables like: refrigerator and audio visual equipment.

108. In cast components,

(a) fiber lines are arranged in a pre-determined way
(b) fiber lines of rolled stock are broken
(c) there are no fiber lines and grains are scattered
(d) grains are arranged in a predetermined way

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : In cast components, there are no fiber lines and grains are scattered.

109. One of the groups is not part of quality circle:

(a) Leaders (b) Management
(c) Facilitators (d) Bankers

Ans. : (d) part of quality circle
(1) Leaders, (2) Management, (3) Facilitators

110. The percentage overshoot of a second order system to a step input depends only on

(a) the value of the step input
(b) the value of the damping ratio
(c) the value of the gain
(d) natural frequency

TNPSC AE 2017

Ans. (b) : The percentage overshoot of a second order system to a step input depends only on the value of the damping ratio.

111. A rectifier diode bridge in an alternator s used to

(a) Convert DC into AC
(b) Regulate voltage output
(c) Bridge the gap between the stator and the rotor
(d) Convert or rectify the negative half of a sine wave into the positive half of a sine wave

TNPSC AE 2017

Ans. (d) : A rectifier diode bridge in an alternator s used to convert or rectify the negative half of a sine wave into the positive half of a sine wave.

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112. The main purpose of the field coils in a DC motor is to

(a) create a stationary magnetic field in the stator
(b) create a magnetic field in the armature
(c) create a CEMF
(d) reverse the polarity in the armature winding just as commutation occurs.

TNPSC AE 2017

Ans. (b) : The main purpose of the field coils in a DC motor is to create a magnetic field in the armature.

113. A power windows motor operates in one direction but not the other direction. Which is the most likely cause of this complaint?

(a) worn brushes
(b) defective permanent magnets
(c) loss of residual magnetism in t he armature
(d) defective power window switch

TNPSC AE 2017

Ans. (d) : Due to defective power window switch. A power windows motor operates in one direction but not the other direction.

114. Technician A says, the conventional body design will have more floor height, hence stability will be increases. Technician B says the weight of the frame is more, hence less vehicles speed. Out of these.

(a) A is correct
(b) B is correct
(c) Both A and B are correct
(d) Neither A nor B are correct

TNPSC AE 2017

Ans. (b) : B is correct.

115. Acute angles between backrest and seat squab results in

(a) Compressed thorax
(b) Numness in arms
(c) Thigs press on the stomach
(d) Numness is feet

TNPSC AE 2017

Ans. (c) : Acute angles between backrest and seat squab results in thigs press on the stomach.

116. What are the gain and natural frequency of the following system transfer function?

$$G(S) = \frac{36}{S^2 + 3S + 36}$$

(a) 36, 6 (b) 6, 6
(c) 1, 6 (d) 6, 1

TNPSC AE 2017

Ans. (c) :

117. To implement the derivative term, we usually use a low-pass filter. The time constant of a low-pass filter should be

(a) much smaller than the derivative time constant
(b) much smaller than the integral time constant
(c) much smaller than the system time constant
(d) much larger than the derivative time constant

TNPSC AE 2017

Ans. (a) : The time constant of a low-pass filter should be much smaller than the derivative time constant.

118. A PID controller has a proportional band of 50%, the proportional gain is

(a) $K_P = 50$ (b) $K_P = PB/50$
(c) $K_P = 50\,PB$ (d) $K_P = 100/PB$

TNPSC AE 2017

Ans. (d) : $K_P = 100/PB$

119. Which of these descriptions is true of the step response of an over damped system?

(a) it rises to a steady state value with no overshoot
(b) it rises to a steady state value with little overshoot
(c) it rises to a steady state value with large overshoot
(d) it does not settle to a steady state value

TNPSC AE 2017

Ans. (a) : It rises to a steady state value with no overshoot is true of the step response of an over damped system.

120. The short hand formula for calculating the closed loop transfer function for simple system is

(a) forward/ (1 + open loop)
(b) forward * feed back/(1 + open loop)
(c) forward/(1 + forward)
(d) loop/(1 + open loop)

TNPSC AE 2017

Ans. (a) : The short hand formula for calculating the closed loop transfer function for simple system is forward/ (1 + open loop).

121. As per factory act, qualified safety officer should be appointed to carry out the prescribed duties if organization is engaging ____or more employees.

(a) 100 (b) 400
(c) 500 (d) 1000

GPSC EE Pre, 28.01.2017

Ans. (d) : As per factory act, qualified safety officer should be appointed to carry out the prescribed duties if organization is engaging 1000 or more employees.

122. Paver finisher (road making machinery) is used for

(a) Better longitudinal level control
(b) Better transverse level control
(c) Uniform pre compaction
(d) All of the mentioned above

GPSC EE Pre, 28.01.2017

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Ans. (d) : A paver (paver finisher, asphalt finisher, paving machine) is a piece of construction equipment : used to lay asphalt on road, bridges, parking lots and others such places. It lays the asphalt flat and provides minor compaction before it is compacted by a roller.

123. The tyre size numbering on the side wall of a car tyre reads 165/80 R14 85 S, then section height of the tyre is

(a) 165 mm (b) 132 mm
(c) 80 mm (d) 85 mm

GPSC EE Pre, 28.01.2017

Ans. (a) : But given option (b)
Tyre size numbering-
165/80 R14 85S
165 – Section which is 165 milli meters. This is the tyre width in mm.
80 – (Aspect Ratio) this represent the height of the tyre side wall as a percentage of section width.
R - (Construction R-Radial)- Radial construction cord piles are arranged at 90 degree of the direction of travel.
14 - (Rim diameter)- 16 inches this is the diameter of tyre's inner rim in inches.
85 - (Load index)- load capacity of tyre maximum weight limit per tyre.
S - (Speed symbol)- speed symbol indicates the maximum speed for the tyre at the full load (in this case up to 240 km/hr).

124. Which of the following earth moving machinery is most suitable for desilting of running large & deep irrigation canal

(a) Regular hydraulic excavator having bucket capacity 0.8 cu.m.
(b) Regular hydraulic excavator having bucket capacity 0.3 cu.m.
(c) Drag line
(d) Dozer

GPSC EE Pre, 28.01.2017

Ans. (c) : Drag line earth moving machinery is most suitable for desilting of running large and deep irrigation canal.

125. If the voltage is reduced to half, the torque developed by an induction motor will be reduced to

(a) 1/16 or original torque
(b) 1/8 of original torque
(c) 1/4 of original torque
(d) 1/2 of original torque

GPSC EE Pre, 28.01.2017

Ans. (d) : If the voltage is reduced of half, the torque developed by an induction motor will be reduced to 1/2 of original torque.

126. "Liquidated damages for delay" is charged for delay in completion of work in which case

(a) For delay in completion of work due to fault of contractor
(b) For delay in completion of work due to fault of owner
(c) For delay in completion of work due to heavy flood
(d) None of the mentioned above

GPSC EE Pre, 28.01.2017

Ans. (a) : The agreement provided that the contractor was to pay $ 500 per day as liquidated damages for every working day elapsing between the agreed and the actual completion dates. Twenty day after the agreed date the plaintiff a banded the work, new bids were advertised and a new contractor started work after a delay of 149 days and completed the work in 264 days.
In holding the liquidated damage provision is application even where there has been a total abandonment of work by the contractor, the principal case rejects the view of the earlier cases and Williston's original view.

127. Two water droplets combine to form a large drop. In this process energy is

(a) Liberated
(b) Absorbed
(c) Neither liberated nor absorbed
(d) Sometimes liberated and sometimes absorbed

GPSC EE Pre, 28.01.2017

Ans. (a) : In the case of a droplet, it is spherical and its surface is exposed to air. The molecules on the surface experience an inward force and hence the surface tries to contract. Since it cannot contract beyond a point the surface remains under tension. This tension is in the form of surface energy. When the two droplets merge, their surface area decreases and hence total a surface energy decreases. The remaining energy is liberated.

128. Thermoplastic materials are those materials which

(a) are formed into shape under heat and pressure and results in a permanently hard product
(b) do not become hard with the application of heat and pressure and no chemical changes occurs
(c) are flexible and can withstand considerable wear under suitable conditions
(d) are used as a friction lining for clutches and brakes

GPSC EE Pre, 28.01.2017

Ans. (b) : Thermoplastic materials are those materials which do not become hard with the application of heat and pressure and no chemical changes occurs. Thermoplastic or thermo softening plastic is a plastic polymer material that becomes moldable at a certain elevated temperature and solidifies upon cooling. Most thermoplastics have a high molecular weight.

129. The time required for the fetching and execution of one simple machine instruction is

(a) Delay time (b) CPU cycle
(c) Real time (d) Seek time

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Ans. (b) : The time required for fetching and execution of one simple machine instruction is CPU cycle.

130. The power factor of a squirrel cage induction motor is
(a) Low at light load
(b) Low at heavy load
(c) Low at light and heavy load
(d) Low at rated load only

GPSC EE Pre, 28.01.2017

Ans. (c) : If the desired power factor for the power distribution system is 0.95, anything below that could be considered to be "low". By that definition, the power factor of a squirrel cage induction motor is "low" at light and heavy loads.

131. In KAIZEN5-S movement meaning of seiton is ____.
(a) Straighten up
(b) Put things in order
(c) Clean up
(d) Displine.

Punjab PSC SDE 12.02.2017

Ans. (b) : In KAIZEN 5-S movement meaning of seiton is put things in order.

132. Which of the following motors are preferred in overhead travelling cranes?
(a) High speed motors
(b) Slow speed motors
(c) Short time rated motors
(d) Continuous duty motors

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) : Intermittent periodic motor is preferred for overhead traveling cranes require high starting torque for a short interval of time intermittent periodic motors have low synchronous speed.

133. Which protocol provides e-mail facility among different hosts?
(a) FTP (b) SMTP
(c) TELNET (d) SNMP

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Simple mail transfer protocol is a TCP/IP protocol used in sending and receiving e-mail, many mail serves now support. ESMTP, which allows multimedia files to be delivered as e-mail.

134. The basic architecture of computer was developed by
(a) John Von Neumann (b) Charles Babbage
(c) Blaise Pascal (d) Garden Moore

GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : The basic architecture of computer was developed by John Von Neumann.

135. Heat calorific value of coal is approximately
(a) 1000 – 2000 kcal/kg
(b) 2000 – 4000 kcal /kg
(c) 5000 – 6500 kcal /kg
(d) 9000 – 10500 kcal /kg

Karnataka PSC Lect., 27.05.2017

Ans. (c) : Heat calorific value of coal is approximately 5000-6500 kcal/kg.

136. A hot wire anemometer is a variable
(a) Inductance transducer
(b) Resistance transducer
(c) Capacitance transducer
(d) Current measure

Karnataka PSC Lect., 27.05.2017

Ans. (b) : A hot wire anemometer is a variable resistance transducer.

137. Which one of the following is a type of particle reinforced composite?
(a) Dispersion- strengthened composite
(b) Continuous composite
(c) Discontinuous composite
(d) Laminar composite

Karnataka PSC Lect., 27.05.2017

Ans. (a) : Dispersion- strengthened composite.

138. In a walking beam transfer mechanism, the total number of cylinders is
(a) one (b) two
(c) three (d) four

Karnataka PSC Lect., 27.05.2017

Ans. (b) : In a walking beam transfer mechanism, the total number of cylinders is two.

The Walking Beam Transfer Bar System–
- The work parts are lifted up from their work station locations by a transfer bar and moved one position ahead to the next station.
- The transfer bar then lowers the pans into nets which position them move accurately for processing.

139. Which one of the following is a rapid prototyping method?
(a) Stereolithography
(b) Primitive instancing
(c) Sweeping
(d) Cell decomposition

Karnataka PSC Lect., 27.05.2017

Ans. (a) : Stereolithography started the rapid prototyping revolution. The technique builds three dimensional models from liquid photosensitive polymers that solidify when exposed to ultravoilet light.

140. What is meant by application factor?
(a) Ratio of maximum torque and average torque
(b) Ratio of maximum load and average load
(c) Both (1) and (2)
(d) None of the above

Karnataka PSC Lect., 27.05.2017

Ans. (c) : Application Factor- Ratio of maximum torque and average torque.
- Ratio of maximum load and average load.

141. What are the assumptions made during stress analysis of curved beams?
(a) The material is non-isotropic
(b) The material is non-homogeneous
(c) The material obeys Hooke's law
(d) All of the above

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Ans. (c) : Assumptions in the stress analysis of curved beam-
(i) Plane sections perpendicular to the axis of the beam remain plane after bending.
(ii) The modulus of elasticity in tension and compression are equal.
(iii) The material is homogeneous and obeys Hooke's law.

142. Which of the following statements is/are true?
a. Gross errors are known as compensating errors.
b. Systematic errors are caused due to system interrelation errors.
c. Probability laws are followed by systematic errors.
Select the code for the correct answer from the options given below:
(a) Only a (b) Only b
(c) Only c (d) All of the above

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Ans. (b) : Systematic errors are caused due to system interrelation errors.

143. The material which after being plastically deformed at room temperature into various shapes, returns to its original shape upon heating is known as
(a) Amorphous alloy
(b) Shape memory alloy
(c) Super alloy
(d) Piezo alloy

Karnataka PSC Lect., 27.05.2017

Ans. (b) : Shape memory alloy

144. When materials are reduced to nano scale
A. Copper becomes transparent
B. Aluminium becomes soluble
C. Gold becomes soluble
D. Silver becomes transparent
Which of the above statements are correct?
Select the code for the correct answer for the options given below:
(a) All statements are correct
(b) Statements A and C only are correct
(c) Statements C and D only are correct
(d) Statements A, B and C are correct

Karnataka PSC Lect., 27.05.2017

Ans. (b) : When materials are reduced to nano scale-
(i) Copper becomes transparent
(ii) Gold becomes soluble

145. The Elastic modulus of a composite material can be given by
(a) $E_c = xE_f + (1-x)\,E_m$
(b) $E_c = \frac{E_f}{x} + (1-x)E_m$
(c) $E_c = \frac{x}{E_f} + \frac{(1-x)}{E_m}$
(d) $E_c = \frac{E_f}{x} + \frac{E_m}{(1-x)}$

Where
E_c = Elastic modulus of composite material
E_f = Elastic modulus of fibre
E_m = Elastic modulus of matrix
x = Fraction of the fibre in composite material

Karnataka PSC Lect., 27.05.2017

Ans. (a) : $E_c = xE_f + (1-x)\,E_m$

146. An involute is defined as which of the following descriptions?
(a) The locus of a point on the circumference of a circle thats rolls without slipping on the circumference of another circle
(b) Locus of a point on a straight line which rolls without slipping on the circumference of a circle.
(c) The circle on which the straight line rolls
(d) The locus of a point on the circumference of a circle that rolls without slipping on the inside circumference of another circle.

Karnataka PSC Lect., 27.05.2017

Ans. (b) : Involute is a curve generated by point on a tangent which rolls on a circle without slipping. The involute profile on a gear will be generated through a generating circle and this generating circle will be known as base circle. It is a fundamental property of a gear its radius will not change in any condition for a gear.

147. In isometric projection, the size of a sphere is equal to which of the following?
(a) 0.816 of the actual size
(b) Actual size
(c) $\frac{\text{Actual size}}{0.816}$
(d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (b) : In Isometric projection 3-D objects are represented in 2D. sphere is represented as circle having same diameter.

148. A square plane lamina of 50 mm side is held parallel to the horizontal plane and perpendicular to the vertical plane and the profile plane. The front view of the square lamina is which of the following?
(a) An inclined line of less than 50 mm length
(b) A regular square of 50 mm side
(c) A straight line of less than 50 mm length
(d) A straight line of more than 50 mm length

Karnataka PSC AE, 10.09.2017

Ans. (*) : None of the option are correct because it will give straight line of 50 mm.
• **Note-** Official answer (d)

149. It is impossible to draw orthographic projection of a line if the line is inclined at
(a) 45° with HP and 30° with VP
(b) 60° with HP and 45° with VP
(c) 30° with HP and 45° with VP
(d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (b) :

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150. The projection in which the rectangular face of an object is projected as a parallelogram and the square face is projected as a rhombus, is

(a) Orthographic projection
(b) Oblique projection
(c) Isometric projection
(d) Perspective projection

Karnataka PSC AE, 10.09.2017

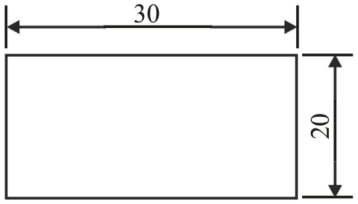
Ans. (c) : Isometric projection

151. The method of dimensioning in which the dimensional figures are always written above the dimensional line at the centre and perpendicular to the dimension line is

(a) Unidirectional system
(b) Aligned system
(c) Vertical dimension system
(d) Both (a) and (b)

Karnataka PSC AE, 10.09.2017

Ans. (b) : Aligned system.



152. In a drawing 1 : 2 is a _____

(a) Full scale (b) Enlarged Scale
(c) Reduced scale (d) None of the above

GMB AAE 25.06.2017

Ans. (c) : There are three types of scales depending upon the properties it indicates as.

(i) Reducing Scale : When the dimensions on the drawing are smaller than the actual dimensions of the object it is represented by the scale or RF.

Scale 1 cm = 100 cm or 1 : 100 and by

$$RF = \frac{1}{100} \quad \text{(less than one)}$$

(ii) Full Scale– Some times the actual dimensions of the object will be adopted on the drawing their in that case off is represented by the scale and RF.

Scale – 1 cm = 1 cm or 1 : 1 and RF = 1/1 (equal to 1)

(iii) Enlarging Scale- The dimensions adopted on the drawing will be bigger than the actual dimensions of the objects then in that case it is represented by enlarging scale and RF.

Scale – 10 cm = 1 cm or 10 : 1 and by RF = 10/1 (greater than 1)

153. _____ is size of A_2 Drawing Sheet as per I.S.

(a) 594 × 841 mm (b) 297 × 420 mm
(c) 210 × 297 mm (d) 420 × 594 mm

GMB AAE 25.06.2017

Ans. (d) : The size of A_2 Drawing sheet as per I.S.

Iso series	mm	Inches.
A_4	594× 841	23.39×33.1
A_2	420× 594	16.54×23.29
A_3	297× 420	11.69×16.54

154. Front view and _____ view should be drawn in vertical line.

(a) Bottom (b) Right side
(c) Left side (d) Rear

GMB AAE 25.06.2017

Ans. (a) : In orthographic Projection, based on parallel line of sight and which the front view of a multiview drawing is projected. The top front and bottom views are all aligned vertically.

155. Which of the following is not included under the definition of wages given under the payment of wages Act, 1936?

(a) Basic wage
(b) Dearness Allowance
(c) Gratuity
(d) Incentive

Kerala PSC Poly. Lect., 2017

Ans. (c) : Gratuity is not included under the definition of wages given under the payment of wages Act, 1936.

156. Which of the following is not a method of depreciation?

(a) Replacement method
(b) Revalution
(c) Reducing balance
(d) Straight line

Kerala PSC Poly. Lect., 2017

Ans. (a) : Method of depreciation are–
(1) Revalution
(2) Reducing balance
(3) Straight line

157. Which of the following method gives uniform depreciation?

(a) Straight line method
(b) Declining balance method
(c) Sum of years digit method
(d) None of the above

UJVNL AE 2016

Ans. : (a) Straight line method gives uniform depreciation.

158. A decomposition method in which the cells which are used to model an object need not have the same shape and size is

(a) Octree model
(b) Part programming
(c) Voxel based model
(d) Cell decomposition

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : In cell decomposition method.

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159. Free trade zone means

(a) two countries without tariff
(b) two manufacturers without tax liability
(c) manufacturer and consumer
(d) manufacturer and wholesaler

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : Free trade zone means two countries without tariff.

160. Combining together of two business parties is termed as

(a) joint stock company
(b) amalgamation
(c) public limited company
(d) private limited company

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : Combining together of two business parties is termed as amalgamation.

161. The acronym DMAIC stands for

(a) Develop, make, analyze, improve, control
(b) Define, measure, analyze, implement, control
(c) Define, make, analyze, import, control
(d) Derive, measure, analyze, implement, create

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : The acronym DMAIC stands for define, measure, analyze, implement, control.

162. The control which will limit pump outlet pressure to a predetermined level and adjust pump outlet flow to the level needed to maintain the set pressure, is

(a) pressure compensator control
(b) Load sense control
(c) Meter-In circuit
(d) Meter-Out circuit

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : The pressure compensator control will limit pump outlet pressure to a predetermined level & adjust pump outlet how to the level needed to maintain the set pressure.

163. One quick way to view the entire drawing area is to use the Zoom command by typing

(a) Type SHOWALL enter
(b) Type Z enter A enter
(c) Type Z enter E enter
(d) Type ALL enter

UPRVUNL AE 21.08.2016

Ans. : (b) Type Z enter A enter to view the entire drawing area ia to use the Zoom.

164. In C-programming, which of the following function sets first n characters of a string to a given character

(a) strinit ()
(b) Strnset ()
(c) strset ()
(d) strcset

GPSC ARTO 01.05.2016

Ans. (b) : Strnset () → It sets the portion of characters in a string to given character.

Strset ()→ Sets all character in a string to given character.

165. In C-programming the keyword used to transfer control from function back to the calling function is

(a) switch
(b) go to
(c) go back
(d) return

GPSC ARTO 01.05.2016

Ans. (d) : Return→ Transfer control function back to the calling function.

166. Which of the following is not a type of inheritance in C++ programming

(a) distributive
(b) hierarchical
(c) multi level
(d) multiple

GPSC ARTO 01.05.2016

Ans. (a) : Type of inheritance in C++ Programming–
(i) Single inheritance
(ii) Multiple inheritance
(iii) Multilevel inheritance
(iv) Hierarchical inheritance
(v) Hybrid inheritance

167. Which classes allow primitive types to be accessed as object in C++ programming?

(a) storage
(b) virtual
(c) friend
(d) wrapper

GPSC ARTO 01.05.2016

Ans. (d) : Wrapper is allow primitive type to be accessed as objected in C++ programming.

168. Which of the following is not a feature of C++

(a) operator overloading
(b) namespaces
(c) inheritance
(d) reflection

GPSC ARTO 01.05.2016

Ans. (d) : C++ is quite similar to C programming In fact, C++ supports all the features altered with C with the addition to various other important feature like-
- Object oriented programming
- Inheritance
- Operator overloading
- Exception and error handling
- Name Space features etc.

169. Which is one of the following not the output device?

(a) Printer
(b) Stylus
(c) Display device
(d) Plotter

MPPSC AE 2016

Ans. : (b) Printer, display device and plotter is the output device

170. For transportation of loose granular material within factory involving both horizontal and vertical displacement, the most suitable material handling device will be

(a) bucket conveyor
(b) roller conveyor
(c) screw conveyor
(d) lift

RPSC AE (GWD) 06.05.2016

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Ans. (a) : For transportation of loose granular material within factory involving both horizontal and vertical displacement, the most suitable material handling device will be bucket conveyer. Bucket conveyors more products (typically aggregate or bulk) vertically via reciprocating buckets. The conveyor is made of an interconnected loop of through/buckets that stay level when moving from horizontal to vertical, preventing their contents from spilling out.

171. In some systems, the working fluids sustain the process on a self sustaining basis, once a system has been started by some external power source. What is the name given to such a systems?

(a) Perpetual system (b) Bootstrap system
(c) Continuous system (d) None of these

KPCL AE 2016

Ans. (b) : Bootstrap air cooling system has two heat exchanger rather than one and a cooling turbine drives a secondary compressor rather than a cooling fan. The air seeped from the fundamental compressor a first cooled by the slam air in the main heat exchangers. The cooled air enters the secondary compressor.

172. Generally automotive microcomputers use….. bits in arithmetic.

(a) 2 (b) 6
(c) 8 (d) 4

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : Generally automotive microcomputers use 8 bits in arithmetic.

173. A fuel cell transforms chemical energy into electrical energy through a series of.....reactions.

(a) Catalyst-aided oxygen-reduction
(b) Catalyst-absent oxidation
(c) Catalyst-aided hydrogen-reduction
(d) None of the above

HPPSC AE (PWD) 10.09.2016

Ans. (a) : Fuel cell transform chemical energy into electrical energy through a series of catalyst aided oxygen reduction reactions.

174. If a triangular plane is parallel to vertical plane and perpendicular to horizontal plane, then in the top view (in first angle projection) it looks like.....

(a) Triangle (b) Horizontal line
(c) Vertical line (d) Inclined line

HPPSC AE (PWD) 10.09.2016

Ans. (b) : If a triangular plane is parallel to vertical plane and perpendicular to horizontal plane then the top view (in first projection) it looks like horizontal line.

175. The ratio of dynamic response of a structure to the static response (for the same load) is known as.....

(a) Impact factor (b) Response factor
(c) Load factor (d) Sensitive factor

HPPSC AE (PWD) 10.09.2016

Ans. (a) : Impact factor- The ratio of dynamic response of a structure to the static response (for the same load) is known as impact factor.

176. Front view of an object lies in.....

(a) Horizontal Plane (HP)
(b) Vertical Plane (VP)
(c) Profile Plane (PP)
(d) Inclined Plane (IP)

HPPSC AE (PWD) 10.09.2016

Ans. (b) : As the object is in first quadrant and the front view projects on vertical plane and top view projects on horizontal plane.

177. The idea about the actual shape of the workpiece can be obtained by plotting the.....graph.

(a) Sequential
(b) Logarithmic incremental
(c) Polar
(d) Cartesian

HPPSC AE (PWD) 10.09.2016

Ans. (c) : The idea about the actual shape of the work piece can be obtained by plotting the polar graph.

178. Wastivity of any system is defined as the ratio of :

(a) Waste/output (b) Waste/input
(c) Output/waste (d) Input/waste

HPPSC Asstt. Prof. 18.11.2016

Ans. (b) : Wastivity of the system is defined as the ratio of the waste to the input.

$$\text{Wastivity} = \frac{\text{Waste}}{\text{Input}}$$

$$\text{Gross wastivity} = \frac{\text{Total waste generated}}{\text{Total input}}$$

179. Induction and orientation are part of the following human resource management functions :

(a) Recruitment (b) Training
(c) Re-training (d) Skill development

HPPSC Asstt. Prof. 18.11.2016

Ans. (b) : Induction and orientation are the part of training.

180. Anthropology is one of the disciplines of human resource management, which deals with:

(a) Measurement and analysis of physical factors in achieving efficiency
(b) Development of a socio-technical model for employees health and safety improvement
(c) Cultural variations and discoverable patterns of behavior from history and environment
(d) Allocation of scarce resources with orientation to future

HPPSC Asstt. Prof. 18.11.2016

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Ans. (c) : Anthropology is one of the disciplines of human resources management which deal with cultural variation and discoverable patterns of behavior from history and environment.

181. ISO stands for :

(a) International Standards Organization
(b) Indian Standards Organization
(c) International Organization for Standards
(d) International Specification for Organizations

HPPSC Asstt. Prof. 18.11.2016

Ans. (c) : ISO Stands for– International Organization for Standards.

182. Number of factors in ISO 9001 are :

(a) 20 (b) 18
(c) 15 (d) 10

HPPSC Asstt. Prof. 18.11.2016

Ans. (a) : Number of factors in ISO 9001 are 20.

183. Warehouse is used to store :

(a) Raw material (b) Work-in-progress
(c) Finished goods (d) Tools

HPPSC Asstt. Prof. 18.11.2016

Ans. (c) : Warehouse is used to store finished goods.

184. The major problem addressed by the warehouse layout strategy is

(a) Minimizing difficulties caused by material flow varying with each product
(b) Requiring frequent contact close to one another
(c) Addressing trade-offs between space and material handling
(d) Balancing product flow from one workstation to the next
(e) None of the above

CGPSC AE 16.10.2016

Ans. (c) : Addressing trade offs between material handling & space in the warehouse is a major problem been tackled in warehouse layout strategy, because it is found that your warehouse or distribution centre capacity is at bursting point & costs are escalating.

185. Orsat apparatus is used to determine the

(a) combustion efficiency
(b) leakage of air into boiler
(c) conductivity of water
(d) analyzing the flue gas on the spot

GPSC Lect. 23.10.2016

Ans. (d) : An Orsat gas analyser is a piece of laboratory equipment used to analyse a gas sample for its oxygen, carbon monoxide and carbon dioxide content.

186. Flexible manufacturing systems are generally applied in

(a) High variety and low volume production
(b) High variety and high volume production
(c) Low variety and low volume production
(d) Medium volume and medium variety production

APPSC AEE Screening Test 2016

Ans. (d) : Flexible manufacturing systems (FMS) are used for medium volume and medium variety production (FMS) is a production system consisting of a set of identical and / or complementary numerically controlled machine which are connected through an automated transportation system. Each process in (FMS) is controlled by a dedicated computer.

187. Which two disciplines are usually tied by a common data base

(a) Documentation and geometric modeling
(b) Drafting and documentation
(c) CAD and CAM
(d) Testing and analysis

APPSC AEE Screening Test 2016

Ans. (c) : CAD/CAM are usually tied by common database because of their joint benefits computer aided design and computer aided manufacturing are often combined into CAD/CAM systems. This combination allows the transfer of information from the design stage the stage of planning for manufacture, without the need to re-enter the data on part geometry manually.

188. Group Technology brings together and organizes

(a) Common parts, problems and tasks
(b) Automation and tool production
(c) Documentation and analysis
(d) Parts and simulation analysis

APPSC AEE Screening Test 2016

Ans. (a) : Group Technology brings together and organizes common parts, problems and tasks. The concept of Group Technology (GT) is drawing increasing interest from manufacturing industry for its potential to enhance productivity and efficiency. Group technology may have a suitable, impact in parts design by standardizing the design and eliminating redundancies. Computer aided process planning together with GT can vastly improve the productivity of process designers and also simplify the design.

189. Which of the following feedback devices can sense both speed and position ?

(a) Resolver (b) tachometer
(c) Encoder (d) All of these

HPPSC W.S. Poly. 2016

Ans. : (c) Encoder feedback devices can sense both speed and position.

190. Which of the following non-contact inspection method used a high frequency sound wave ?

(a) Radiation (b) Reluctance
(c) Ultrasonic (d) Capacitance

HPPSC W.S. Poly. 2016

Ans. : (c) Contact inspection method of high frequency sound waves
(i) Radiation
(ii) Reluctance
(iii) Capacitance.

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191. Which of the following drives is having the maximum power?

(a) DC motor (b) Stepper motor
(c) Hydraulic drive (d) Pneumatic drive

HPPSC W.S. Poly. 2016

Ans. : (c) Hydraulic drives is having the maximum power.

192. The depreciation in any particular year is charged as fixed percentage of the remaining value of asset. This approach is adopted in

(a) Sinking fund
(b) Diminishing balance
(c) Annuity charge
(d) Sum of year digits

RPSC VPITI 14.02.2016

Ans. (b) : According to diminishing balance method, depreciation is charged at a fixed percentage on the book value of the asset. Since the book value reduce every year hence the amount at depreciation also reduces every year.

Under this method, the value of the asset never reduces to zero.

193. For the pay off matrix given below : Player Y 3 – 142 player X – 1 –3 –7 0 4 –6 2 –9

(a) The value of the game = –1
(b) The value of the game = –3
(c) The value of the game = –7
(d) The value of the game = 2

RPSC VPITI 14.02.2016

Ans. (a) :

194. Which principle depends on the personal capacity of the boss and efficient communication system

(a) Principle of unity of command
(b) Principle of Span of Control
(c) Principle of Specification
(d) Principle of Co-ordinate

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (b) : Principle of span of control principles depends on the personal capacity of the boss and efficient communication system.

195. A 20 number crucible:

(a) Can contain 20 m^3 of any metal
(b) Can contain 20 kgf of copper
(c) Can have a height of 20 cm
(d) Has internal diameter of 20 cm

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : A 20 number crucible can contain 20kgf of copper.

196. The fulcrum pin of lever is designed on the basis of:

(a) Torsional moment (b) Axial tensile force
(c) Bending moment (d) Bearing pressure

GPSC Asstt. Prof. 28.08.2016

Ans. (d) :

197. A connecting rod has one end small and the other end big, though the same force transmitted from smaller end to big end because the:

(a) Big end has a higher bending moment
(b) Big end is subjected to centrifugal force
(c) Big end is to be fitted to crank pin of high diameter
(d) Will have same wear as small end side

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : At the ends (piston end), the movement is linear but oscillatory with sudden change in velocity as piston moves back and forth the entire piston assembly must be as light as possible. Same force transmitted from smaller end to big end because they will have same wear as small end side.

198. What is one radians equal to?

(a) $90^\circ/\pi$ (b) $180^\circ/\pi$
(c) $360^\circ/\pi$ (d) None of these

KPCL AE 2016

Ans. (b) : 1 radian = $\frac{180^\circ}{\pi}$

Note-Commission is right answer option (d), but actual correct answer is option (b).

199. One Joule is equal to which of the following?

(a) (Newton) . (Meter) (b) (Coulomb) . (Volt)
(c) (Pascal) . $(Meter)^3$ (d) All the above

KPCL AE 2016

Ans. (d) : 1 Joule = (N – m)

In electrostatic W = qV

1 Joule = Coulomb × Volt

1 Joule = Pascal $(meter)^3$

$$= \frac{N}{m^2} \times m^3 = N - m$$

200. What is Half-Value thickness or Halving thickness

(a) It is the thickness of the material at which the radiating material is reduced to its half-life.
(b) It is the one half of the thickness of the material required to reduce radiation entering it
(c) It is the thickness of the material at which the intensity of radiation entering it is reduced by one half
(d) None of these

KPCL AE 2016

Ans. (c) : A material's half-value layer (HVL), or half value-thickness, is the thickness of the material at which the intensity of radiation entering it is reduced by one half.

201. The electronic device that measures the proportion of oxygen (O_2) in the gas or liquid is called-

(a) Oxygen meter (b) Oxygen sensor
(c) Oxylizer (d) None of these

KPCL AE 2016

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Ans. (b) : An oxygen sensor (or lambda sensor where lambda refers to air-fuel equivalence ratio usually denoted ϕ or λ) is an electronic device that measures the proportion of oxygen (O_2) in the gas or liquid being analysed.

202. Which of the following types of brakes is commonly used in railroad trains and trailers?

(a) Pneumatic brakes
(b) Hydraulic brakes
(c) Both A and B
(d) None of these

KPCL AE 2016

Ans. (a) : A brake which uses air as working fluid is a pneumatic brake. The system actuated to apply this phenomenon is known as pneumatic brake system. An pneumatic brake or a compressed air brake system is a type of friction brake for vehicles in which compressed air pressing on a piston is used to apply the pressure to the brake pad needed to stop the vehicle.
The pneumatic brake is mainly used in large vehicles such as trucks, buses, trailers and semi-trailers in addition to their use in railway train.

203. What is the "aspect ratio" of a tire?

(a) It is sidewall height as a percentage of the inner diameter of the tire
(b) It is sidewall height as a percentage of the radius of the tire
(c) It is sidewall height as a percentage of the total width of the tire
(d) None of these

KPCL AE 2016

Ans. (c) : Aspect ratio of a tire is the side wall height as a percentage of the total width of the tire. Aspect ratio of the tire is 70 percent, so the section height of tire will 129.5 mm. The higher the aspect ratio, the falter the tire sidewall.

204. The quantum of force that produces an acceleration of 1 m/s^2 on a mass of one kg, in the direction of which it acts is equal to

(a) One Watt (b) One Newton
(c) One Joule (d) None of these

KPCL AE 2016

Ans. (b) : Given- acceleration (a) = 1 m/s^2
mass (m) = 1 kg
Quantum force (F) = ma
= 1 × 1 kgm/sec^2
= 1 N

205. If a cube lies behind V.P and above H.P, then cube is in :

(a) I-Quadrant (b) II-Quadrant
(c) III-Quadrant (d) IV-Quadrant

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : If a cube lies behind V.P. and above H.P., then cube is n II-quadrant.

206. The moment of inertia of an area about an axis, which is in a plane perpendicular to the area is called :

(a) Second moment of area
(b) Radius of gyration
(c) Polar moment of inertia
(d) None of these

KPSC ADF 2015

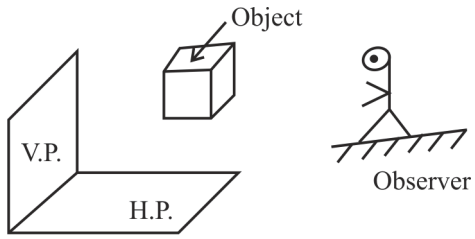
Ans. (c) : Polar moment of inertia

207. In first angle projection method :

(a) The object is placed in the first quadrant
(b) The object lies between the observer and the plane of projection
(c) The plane of projection is assumed to be non-transparent
(d) All the above

KPSC ADF 2015

Ans. (d) :



(Ist quadrant)
(First angle projection method)

208. The top view of a straight line is equal to its actual length when the line is :

(a) Parallel to horizontal plane and inclined to vertical plane
(b) Perpendicular to horizontal plane and inclined to vertical plane
(c) Parallel to vertical plane and perpendicular to horizontal plane
(d) Inclined to horizontal plane and parallel to vertical plane

KPSC ADF 2015

Ans. (a) : The top view of a straight line is equal to its actual length when the line is parallel to horizontal and inclined to vertical plane.

209. Solar energy can be converted into other forms of energy by the process :

(a) Heliochemical process
(b) Helioelectrical process
(c) Heliothermal process
(d) All of the above

KPSC ADF 2015

Ans. (d) : Solar energy cab be converted into other forms of energy by the process :
(a) Heliochemical process
(b) Helioelectrical process
(c) Heliothermal process

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210. Match the correct answers using the correct codes given in List–I with List–II :

List–I (Material)		List–II (Classification)
A.	$NiFe_2O_4$	1. Piezo electric
B.	$PbZrO_3$	2. Ferri magnetic
C.	Nickel	3. Ferro magnetic
D.	Silver	4. Dia magnetic

	A	B	C	D
(a)	2	3	4	1
(b)	2	1	3	4
(c)	3	2	1	4
(d)	4	3	2	1

KPSC ADF 2015

Ans. (b) : A-2, B-1, C-3, D-4

(A)	$NiFe_2O_4$	→	Ferri magnetic properties
(B)	$PbZrO_3$	→	Piezo electric material
(C)	Nickel	→	Ferromagnetic material
(D)	Silver	→	Diamagnetic material

211. If the time required for the phase transformation to proceed half way to completion is 100 s, in a solid state, then rate of transformation is :

(a) $0.001\ s^{-1}$ (b) $0.01\ s^{-1}$
(c) $0.005\ s^{-1}$ (d) $0.02\ s^{-1}$

KPSC ADF 2015

Ans. (b) : To quantitatively describe the rate of a phase transformation, it can be defined as reciprocal of time for transformation to proceed half way to completion :

$$\boxed{\text{rate} = \frac{1}{t_{0.5}}}$$

Given; $t_{0.5} = 100$ second

$$\therefore \quad \text{Rate} = \frac{1}{100} = 0.01 s^{-1}$$

212. The number of degree of freedom for a diatomic molecule is :

(a) 2 (b) 3
(c) 4 (d) 5

KPSC ADF 2015

Ans. (d) : Total number of degree of freedom of a rigid diatomic molecules is 5.

i.e. 3+2 =5

Degree of freedom rigid diatomic molecules have three translation degree of freedom and 2 rotational degree of freedom.

213. The melting point of substances that contract on melting :

(a) decreases with increase in pressure
(b) increases with increase in pressure
(c) decreases at first and then increases with increase in pressure
(d) is independent of pressure

KPSC ADF 2015

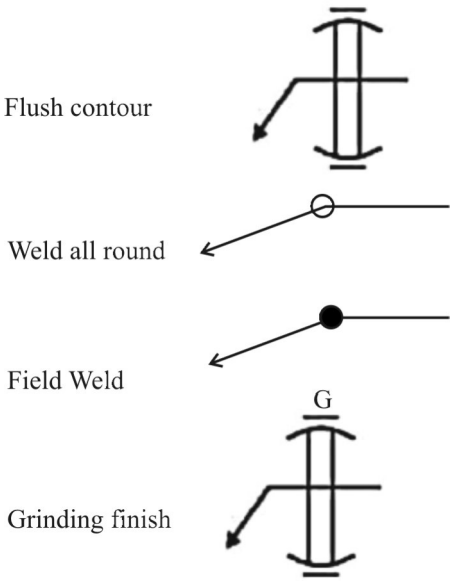
Ans. (a) : The melting point of substances that contract on melting (like ice) decreases with increase in pressure.

214. What the drawing represents in welding?

(a) Weld all round (b) Field weld
(c) Grinding finish (d) Flush contour

KPSC ADF 2015

Ans. (d) :



215. Which type of surface will be produced when the cutting tool is moved perpendicular to the lathe axis?

(a) Cylindrical surface (b) Flat surface
(c) Tapered surface (d) None of the above

KPSC ADF 2015

Ans. (b) : Flat surface will be produced when the cutting tool is moved perpendicular to lathe axis.

216. Clapper box is equipped in all the reciprocating machine tools so as :

(a) To impose relatively less pressure on the work
(b) To produce accurate and distortion free surfaces
(c) To provide problem free reciprocating motion
(d) To rise the tool during return stroke

KPSC ADF 2015

Ans. (d) : Clapper box is equipped in all the reciprocating machine tools so as to rise the tool during return stroke or idle stroke.

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217. Automation is preferred because it :

(a) Offers lesser dependence on workers
(b) Results in reduction in variable cost
(c) Offers easy handling of repetitive works
(d) All of these

KPSC ADF 2015

Ans. (d) : Automation is preferred because it–
- Offers lesser dependence on workers
- Results in reduction in variable cost
- Offers easy handling of repetitive works

218. What type of process would a paper mill be most likely to use?

(a) Continuous flow (b) Job shop
(c) Flow shop (d) Project

KPSC ADF 2015

Ans. (a) : A paper mill would be most likely to use the continuous flow process. Continuous flow is the act of moving a product through the production process from start to finish without stopping. In pure continuous flow, the cycle time equals the load time, as the product never sits in a queue waiting to be worked on.

219. Consider the following statements:

1. Gas cooled thermal reactors use CO, or helium as coolant and require no separate moderator.
2. Fast reactors use heavy water as moderator and coolant.
3. Liquid metal fast breeder reactors use molten sodium as coolant.
4. In Candu type reactors heavy water is used as moderator.

Of these statements, which are correct

(a) 1 and 3 (b) 2 and 4
(c) 3 and 4 (d) 4 and 3

Nagaland PSC CTSE 2015 Paper-II

Ans. (c) : In gas cooled reactor, both natural and enriched uranium fuel with CO_2 as a coolant and graphite as moderator are used.
The best coolants for fast reactors and liquid metals such as sodium, liquid metal have the additional advantage that their boiling point at atmospheric pressure are very high, So, most of them can be used unpressurized in a reactor sodium is most common coolant for fast reactors.

220. In an isometric projection the radius of a sphere it is equal to :

(a) 0.82 R (b) R
(c) 1.22 R (d) 2 R

Kerala PSC AE 06.08.2015

Ans. (b) : The isometric projection of the sphere is simply a circle whose diameter is equal to the true diameter of the sphere.
So, radius of sphere is an isometric projection is equal to its true radius of circle 'R' .

221. In nuclear power plants there are three main sources of radioactive contamination of air. Out of three one of the source is:

(a) step-up transformer
(b) liquid metal
(c) fission of nuclei of nuclear fuels
(d) penstock

MPPSC AE 08.11.2015

Ans. (c) : Fission of nuclei of nuclear fuels is one of the source in nuclear power plants.

222. Which of the following is not considered a method of input control in CAD system?

(a) programmable function box
(b) joystick
(c) plotter
(d) touch material

MPPSC AE 08.11.2015

Ans. (c) : Plotter is not considered a method of input control in a CAD system.

223. The heat production from a normal healthy man when asleep is about

(a) 20 Watts (b) 40 Watts
(c) 60 Watts (d) 80 Watts

MPPSC AE 08.11.2015

Ans. (c) : 60 Watts

224. Hydrogen can play an important role as an alternative fuel to conventional fuel as

(a) An energy carrier
(b) An energy device
(c) An energy system
(d) None of these is correct

MPPSC AE 08.11.2015

Ans. (a) : Hydrogen can play an important role as an alternative fuel to conventional fuel as an energy carrier.

225. The escape velocity on the surface of the earth is-

(a) 11.2 km/sec (b) 1 km/sec
(c) 3.6 km/sec (d) 8.8 km/sec

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (a) Escape velocity-Escape velocity is defined to be the minimum velocity of an object must have in order to escape the gravitational field of the earth, that is , escape the earth without ever falling back. It is 11.2 km/sec.

226. Laser is produced by-

(a) graphite (b) ruby
(c) diamond (d) emerald

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans. : (b) LASER stand for Light Amplification by Stimulated Emission of Radiation.
LASER is device used to produce a very intense, highly directional, coherent, and monochromatic beam of light. A ruby laser is a solid-state laser that uses the synthetic ruby crystal as its laser medium.

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227. A joint sector undertaking :
(a) is jointly owned by several share holders
(b) is jointly owned by private parties and government
(c) is jointly damaged by two directors
(d) is jointly owned by state and central government

(KPSC AE. 2015)

Ans. : (b) A joint sector undertaking is jointly owned by private parties and government.

228. In the Halsey's system of incenctive plan, a worker is
1. paid as per efficiency
2. not paid till his efficiency reaches 60%
3. ensured of a minimum wage
4. forced to do more work
5. improving the straight piece-rate system
Of the above, which are correct
(a) 1 and 3 (b) 2 and 4
(c) 2 and 5 (d) 3 and 5
(e) 1 and 5

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Hasley Plan—Under Hasley plan minimum wages are guaranteed to every worker. A standard time is fixed for the workers. If the workers finish the work before standard time they are given bonus. But no penalty if they fails to do that.

229. In which of the following leadership style, the leader acts as a dictator?
(a) Free-Rein leadership
(b) Democratic leadership
(c) autocratic leadership
(d) all the above

TSPSC AEE 2015

Ans. : (c) Autocratic leadership style, the leader acts as a dictator.

230. Number of bricks of size 20 cm × 10 cm × 10 cm required for constructing a partition wall 10 m long, 1 m high and 10 cm thick is
(a) 200 (b) 300
(c) 500 (d) 600

GPSC AE 26.07.2015

Ans. (c) : Volume of wall = $10 \times 1 \times 0.1 = 1\ cm^3$
Volume of brick = $0.2 \times 0.1 \times 0.1$

$$\text{Number of bricks} = \frac{\text{Volume of wall}}{\text{Volume of brick}}$$

$$= \frac{1}{0.2 \times 0.1 \times 0.1} = 500$$

231. The EMD amount to be deposited with the tender is about _____ of the estimated cost of the work.
(a) 2 to 3% (b) 1 to 2%
(c) 5% (d) 10%

GPSC AE 26.07.2015

Ans. (b) :

232. The weight and volume of 1 bag of cement is
(a) 75 km and 0.055 m^3
(b) 100 km and 0.07m^3
(c) 25 kg and 0.018m^3
(d) 50kmg and 0.035m^3

GPSC AE 26.07.2015

Ans. (d) :

233. Which type of brake is used in EOT crane drive?
(a) Fail safe - Electromagnetic external shoe brake
(b) Fail safe- bank brake
(c) Internal expanding shoe brake
(d) Disc brake

GPSC AE 26.07.2015

Ans. (a) :

234. The amount deducted from the salary of the workers towards Employee's Provident Fund is
(a) Credited into reserves of the company
(b) Deposited in nationalized bank
(c) Deposited in post office
(d) Deposited in the account of worker with Provident Fund commissioner

GPSC AE 26.07.2015

Ans. (d) :

235. According to Factory Act-1948, the permissible noise level in industry is
(a) 60 dB (b) 50 dB
(c) 90 dB (d) 100 dB

GPSC AE 26.07.2015

Ans. (c) :

236. which of the following locking devices has the widest application in automobile industry?
(a) Ring nut (b) Jam nut
(c) swan Nut (d) Castle nut

GPSC AE 26.07.2015

Ans. (d) :

237. If the vacancies move through the crystal lattice, the behaviour is called
(a) Nabarro-Hearing creep
(b) Coble creep
(c) Power law creep
(d) Diffusion creep

TANGEDCO AE 2015

Ans. (d) : Diffusion of vacancies through crystal lattice to cause plastic deformation is called diffusion creep.

238. When analogy is drawn between heat flow and electricity flow in circuits, the heat flow of thermal circuits is equated in the electrical circuit against
(a) voltage (b) current
(c) resistance (d) charge (coulombs)

VIZAG STEEL MT 2015

Ans. (b) : current

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239. Tachometers are used to measure

(a) acceleration
(b) angular velocity
(c) vibration amplitude
(d) displacement

GWSSB AAE, 27.12.2015

Ans. (b) : Tachometers measures angular velocity or rpm of machine.

240. The application of third type levers is found in:

(a) lever loaded safety valve
(b) hand wheel of a punching press
(c) handle of a hand pump
(d) a pair of tongs.

GWSSB AAE, 27.12.2015

Ans. (d) : First Class Lever–This is a type of lever which has the fulcrum in between the weight and the force applied. Its order is represented as force-fulcrum-weight.

Example–

- Our hand pushing an object or seesaws, crowbars.
- Using scissors represents the use of two first-class levers.
- A wheel and axle is also an example.

Second Class Lever–In this, the fulcrum is at one end and the force applied is on the other end. The weight is situated in the middle of these two. The order of this would be fulcrum-weight-force.

Example–

- Wheel barrow
- Staplers
- Doors or gates
- Bottle openers
- Nutcracker
- Nail clippers

Third Class Lever–These are the levers in which the fulcrum is at one end and the force is applied in the middle and the weight is on the other end. The order is represented as a weight-force-fulcrum.

Example–

- Fishing rod
- A broom
- A baseball bat
- A bow and arrow
- Human jaw
- A pair of tongs

241. Basic solution is one which has pH value

(a) equal to 7
(b) less than 7
(c) pH value has nothing to do with basic solution
(d) greater than 7

GWSSB AAE, 27.12.2015

Ans. (d) : greater than 7–

$P_H = 7$ (Neutral)
$P_H > 7$ (Basic Solution)
$P_H < 7$ (Acidic Solution)

242. What should be the shape of an ideal thermometer bulb?

(a) elliptical
(b) spherical
(c) circular
(d) cylindrical

GWSSB AAE, 27.12.2015

Ans. (b) : Spherical- The sphere is the shape with the lowest possible surface area to volume ratio, so all else being equal spheres will heat or cool more slowly than any other shape. That means thermometer with a spherical bulb will respond more slowly than one with different shaped bulb.

243. Simple viscoplastic constitutive equations are used to model:

(a) stress relaxation
(b) creep
(c) low cycle fatigue
(d) ductile fracture

HPPSC Asstt. Prof. 2014

Ans. (b) : Simple viscoplastic constitutive equations are used to model creep.

244. Which of the following is the correct statement?

(a) A stationary point on the graph is any point at which $\frac{dy}{dx} = 0$
(b) A stationary point on the graph is any point at which $\frac{dy}{dx} = \infty$
(c) A stationary point on the graph is any point at which there are gaps or breaks
(d) None of these

MPSC Poly. Lect. 2014

Ans. (a) : A stationary point on a graph is any point at which $\frac{dy}{dx} = 0$

245. If T is the life length of the system, then reliability of a system at time t say R(t) is defined as:

(a) $R(t) = P(T > t)$
(b) $R(t) = P(T < t)$
(c) $R(t) = P(T = t)$
(d) $R(t) = 1 - P(T > t)$

HPPSC Asstt. Prof. 2014

Ans. (a) : Reliability of a system
$R(t) = P(T > t)$
T = life length
t = time
Reliability of a system is defined as the probability. that a product, system or service will perform its intended function adequately for a specified period of time without failure.

246. Select the most suitable statement which is true:

(a) Newton's method allows slower convergence to the minimum point compared to steepest descent, method
(b) Steepest descent direction at any point always passes through the minimum point
(c) In Genetic algorithm mutation probability is more than the crossover probability
(d) GRG method is a variable elimination method

HPPSC Asstt. Prof. 2014

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Ans. (d) : Statement 'd', GRG method is a variable elimination method is true.

247. For a rotating disk to be of uniform strength, the disk should have:

(a) uniform thickness with a hole at the centre
(b) uniform thickness without a hole at the centre
(c) larger tangential stress on outer radius of disk
(d) larger radial stress on outer radius of disk

HPPSC Asstt. Prof. 2014

Ans. (b) : Rotating disk to be of uniform strength, the disk should have uniform thickness without a hole at the centre. Because hole generates stress concentration so for uniform strength hole must not be present.
Disk of uniform strength is defined as one in which the radial and hoop stresses resulting from centrifugal loading are constant and equal to each other.

248. Choose the wrong statement
The most commonly used tests in staffing can be

(a) intelligence tests
(b) proficiency and aptitude tests
(c) personality tests
(d) toughness test

TNPSC AE 2014

Ans. (d) : Intelligence tests, proficiency and aptitude tests and personality tests are most commonly used tests in staffing.

249. Tendency to disperse decision making authority in an organized structure is called

(a) distribution (b) sharing
(c) decentralisation (d) leadership

TNPSC AE 2014

Ans. (c) : Tendency to disperse decision making authority in an organized structure is called decentralisation.

250. Management conflict can be addressed

(a) making the situation that causes conflict
(b) creating motive between persons
(c) compromise
(d) attempts can be made to change the behavior of manager

TNPSC AE 2014

Ans. (c) : With the help of compromise management conflict can be addressed.

251. The 9.9 style managers

(a) who display in their actions the highest possible dedication both to people and to production
(b) who are concerned only with developing an efficient operation and have little concern for people
(c) who have medium concern for production and for people
(d) who have little concern for production but are concerned only for people

TNPSC AE 2014

Ans. (a) : The 9.9 style managers who display in their actions the highest possible dedication both to people and to production.

252. Participative leadership is suitable in

(a) production organization
(b) educational institution
(c) research and development
(d) defence organizations

TNPSC AE 2014

Ans. (c) : Participative leadership is suitable in research and development.

253. The flexible manufacturing systems (FMS) is most suitable for a production system that falls under

(a) mass type (b) batch type
(c) job type (d) all of these

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (b) :

254. The total cost of manufacturing a product is reduced due to

(a) standardization (b) diversification
(c) simplification (d) sophistication

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (a) : Standardization helps companies reduce manufacturing cost and improve …….. their process, produces and procedures.

255. In India, the equipment for thermal power plant is being manufactured by

(a) TATA (b) HEC
(c) NTPC (d) BHEL

GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : In India, the equipment for thermal power plant is being manufactured by BHEL.

256. Which of the following is true if environmentally sound products are made through efficient processes?

(a) It is unprofitable,as long as recyclable materials prices are soft
(b) It is known as lean manufacturing
(c) It can still be profitable
(d) It is easier for repetitive processes than for product-focused processes

UPRVUNL AE 2014

Ans. : (c) If environmentally sound products are made through efficient processes then it can still be profitable because it can be less harmfull to the environment and it can be recycled easily.

257. What is object in C^{++}?

(a) Object is part of syntax of a class
(b) Object is data type of a class
(c) Object is an instance of a class
(d) Object is function of a class

MPPSC State Forest Service Exam, 2014

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Ans. (c) : An object is an instance of a class, when a class is defined, no memory is allocated but when it is instantiated memory is allocated.

258. Which of the following material handling equipment uses gravity

(a) Pulley blocks (b) Pallets
(c) Chutes (d) Conveyors

MPPSC State Forest Service Exam, 2014

Ans. (c) : Transfer chutes are used in many industries to facilitate bulk material transfer from one conveyor belt to another or for redirecting flow from a delivery point into a process or equipment.

259. In hydraulic hoist, gate lifting and lowering operations are carried out by supplying high pressure oil to-

(a) Valve arrangement
(b) Cylinder and piston arrangement
(c) Crank case
(d) Cam and Crank shaft

MPSC Dy. Engineer 2014

Ans. (b) : In hydraulic hoist, gate lifting and lowering operations are carried out by supplying high pressure oil to cylinder and piston arrangement.

260. Match list-I with list-II and select the correct using the codes given below-

List-I	List-II
A. Hydraulic tilt cylinder	**1.** 4-way 4-position control valve
B. Pressure control valve	**2.** 4-way 3-position control valve
C. Blade control valve	**3.** Relief valve
D. Ripper control valve	**4.** Single stage double acting type

(a) A-4 B-3 C-1 D-2
(b) A-3 B-4 C-2 D-1
(c) A-4 B-3 C-1 D-2
(d) A-2 B-1 C-3 D-4

MPSC Dy. Engineer 2014

Ans. (a) : Hydraulic tilt cylinder – single stage double acting type.
Pressure control valve – Relief valve
Blade control valve – 4- way 4- position, control valve.
Ripper control valve – 4 way, 3-position control valve.

261. Gear motors are used where the following conditions exists.

(a) High speed and low torque
(b) Medium speed and low torque
(c) Medium speed and high torque
(d) High speed and high torque.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Medium speed and low torque.

262. The piston motor is used to convert :

(a) Mechanical energy into hydraulic energy
(b) Hydraulic energy into mechanical energy
(c) Mechanical energy into work energy.
(d) Work energy into hydraulic energy.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : The piston motor is used to convert hydraulic energy into mechanical energy. Piston type motors are the most commonly used in hydraulic systems. They are basically the same as hydraulic pumps, except they are used to convert hydraulic energy into mechanical (rotary) energy.

263. The radial piston motor operates

(a) Same as the radial piston pump
(b) In reverse of the radial piston pump
(c) Both the above 1 & 2
(d) None of the above 1 & 2

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : The radial piston motor operates in reverse of the radial piston pump. Radial piston pumps are used for machine tools, such as the displacement of cutting emulsion and as a supply for hydraulic equipment like cylinders. They are also used in High-Pressure Units (HPU) for overload protection and for test rigs.

264. The advantage of tandeming the cylinder is that

(a) Forces are added
(b) Forces are subtracted
(c) Forces are nullified
(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : The advantage of tandeming the cylinder is that forces are added. A tandem cylinder consists of two or more cylinders mounted inline with pistons connected with a common piston rod. A tandem cylinder can provide amplified output force as compared to conventional cylinder of the same bore.

265. The distributor is a stationary assembly that supplies fluid to both the parts of…

(a) Tandem cylinder (b) Telescopic cylinder
(c) Rotating cylinder (d) Stationary cylinder

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : The distributor is a stationary assembly that supplies fluid to both the parts of rotating cylinder.

266. What are the advantages of hydraulic motors?

(a) Compactness
(b) Wide range of speed
(c) Explosion proof in nature
(d) All of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Advantages of hydraulic motors are –
(1) Compactness
(2) Wide range of speed and high force application
(3) Explosion proof in nature.
Disadvantage of hydraulic motors– Hydraulic system run the risk of leaking fluid.

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267. Hydraulic Motor are rated according to their..

(a) Displacement (b) Torque capacity
(c) Maximum pressure (d) All of the above

MPSC Dy. Engg. (GWD) 19.05.2013

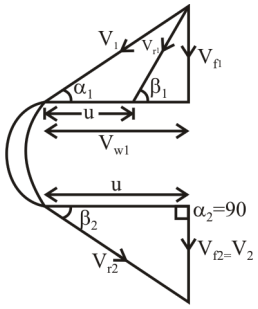
Ans. (d) : Hydraulic motors are rated according to their.
→ Displacement
→ Torque capacity
→ Maximum Pressure limitation.

268. In an impulse steam turbine, the steam enters at a rate of 2 kg/s with a velocity of 500m/s with an axial component of 300 m/s. The peripheral speed of blades is 200 m/s. The steam leaves the blades in axial direction. What is the power developed by the turbine?

(a) 160 kW (b) 200 kW
(c) 100 kW (d) Insufficient data

RPSC AEN Pre-2013

Ans. (a) :



$\dot{m} = 2\,kg/s$

$V_1 = 500$ m/s

$V_{f1} = 300$ m/s

$u = 200$ m/s

$V_{w1}^2 = V_1^2 - V_{f1}^2 = (500)^2 - (300)^2$

$\boxed{V_{w1} = 400\,m/s}$

$\boxed{V_{w2} = 0}$

$P = \dot{m}uV_{w1} = 2 \times 200 \times 400$

$P = 160$ kW

269. Which of the following require training and authorization before you can perform them?

(a) Forklift operation
(b) Lockout/tagout
(c) Repairing electrical equipment
(d) All of these

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (d) : Fork lift operation, lockout/tagout, repairing electrical equipment are require training and authorization before you can perform them.

270. The process of new product development do not incorporate

(a) FEMA (b) KANBAN
(c) DFMA (d) QFD

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : The process of new product development incorporates with KANBAN, DFMA, QFD.

271. FMS production methodology is applicable to situation where the lot size of a product ranges between

(a) 100 to 500 items (b) 100 to 1000 items
(c) 100 to 1500 items (d) 100 to 2000 items

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : FMS production methodology is applicable to situation where the lot size of a product ranges between 100 to 1500 items.

272. Management by exception is based on a

(a) Positive attitude towards problem recognition
(b) Negative attitude towards problem recognition
(c) Indifferent attitude towards problem recognition
(d) Ignoring the problem

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Management by Exception–

- Management by exception (MBE) is an employee empowerment and management style, policy or philosophy wherein managers intervene only when their employees fail to meet their performance standards or when thing go wrong.
- The idea behind it is that management's attention will be focused only on those areas in need of action.
- If the personnel is performing as expected, the manager will take no action.
- MBE normally involves substantial delegation by the manager to his team.

273. Hoists and lifts used for carrying persons shall be fitted with

(a) Electric light
(b) Exhaust fan
(c) At least two ropes or chains separately connected
(d) Gear mechanism

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : Hoists and lifts used for carrying persons shall be fitted with at least two ropes or chains separately connected.

274. An interface between the functional applications and the data is

(a) Management information system
(b) Data base management
(c) Decision support system
(d) Data processing system

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : An interface between the functional applications and the data is data base management.
A Database Management System (DBMS) is a collection of programs that manages the database structure and controls access to the data stored in the database.

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275. For precaution against fire all emergency exists should.

(a) Be closed when not in use
(b) Have a door
(c) Have a fire extinguisher
(d) Be marked properly for easy understanding for all

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (d) : For precaution against five all emergency exists should be marked properly for easy understanding for all.

276. The proportion of "Theoretical lowest amount of parts" to "Tangible number of parts" is called.

(a) Part count ratio
(b) Independent axiom
(c) Zero defect
(d) Assembly efficiency

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : The proportion of 'Theoretical lowest amount of parts' to 'Tangible number of parts' is called assembly efficiency'.
But official answer is given option (a).

277. Accident investigation should be aimed at

(a) Fact finding (b) Fault finding
(c) Interrogation (d) Cause of the injury

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Accident investigation should be aimed at fact-finding.

278. Functional subsystem are connected by

(a) Resource flows (b) Decision flows
(c) Information flows (d) Data flows

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (*) : Functional subsystem are connected by resource flows–
But official answer is option (d).

279. Faulty work habits and careless working are known as

(a) Personal habits (b) Unsafe acts
(c) Awareness (d) Attitude

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : Faulty work habits and careless working are known as unsafe acts.

280. Error reports are an example of

(a) Scheduled report (b) Demand report
(c) Exception report (d) Predictive report

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : Exception Report–

- Exception reports are those reports when a situation is unusual or requires management action.
- An exception report states those instances in which actual performance deviated significantly from expectations, usually in a negative direction which produces an error in the management system.

281. There are four samples P, Q, R and S with natural frequencies 64, 96, 128 and 250 Hz respectively. They are mounted on test setups for conducting vibration experiments. If a loud pure note of frequency 144 Hz is produced by some instrument, which of the samples will show the most perceptible induced vibration?

(a) P (b) Q
(c) R (d) S

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : Execution frequency, f = 144 Hz
Natural frequency, f_n = 64, 96, 128 ad 250

$$\therefore \frac{\text{Execution frequency}}{\text{Natural frequency}} = \frac{f}{f_n} = \frac{144}{64}, \frac{144}{96}, \frac{144}{128}, \frac{144}{250}$$

= 2.25, 1.5, 1.125, 0.576
Since 1.125 is closed to 1, so the sample 'R' will show the most perceptible induced vibration.
But official answer given is option (b).

282. The time taken which elapses after the input to a system or element is abruptly increased from zero to a constant value up to the point at which the system or element gives an output corresponding to some specified % is known as:

(a) Rise time (b) Setting time
(c) Response time (d) Mean time

TNPSC AE (Industries) 09.06.2013

Ans. (c) :

283. A room contains 60 kg of air at 100 kPa and 15°C. The room has a 250-W refrigerator (the refrigerator consumes 250W of electricity when running), a 120-W TV, 1-kW electric resistance heater and a 50-W fan. During a cold winter day, it is observed that the refrigerator, the TV, the fan and the electric resistance heater are running continuously but the air temperature in the room remains constant. The rate of heat loss from the room that day is

(a) 3312 kJ/h (b) 4752 kJ/h
(c) 5112 kJ/h (d) 2052 kJ/h

ISRO Scientist/Engineer 12.05.2013

Ans. (c) : Total heat loss = 250 + 120 + 1000 + 50
= 1420W = 1420 J/s

$$= \frac{1420 \times 3600}{1000} J/s$$

= 1420 × 3.6 kJ/h = 5112 kJ/h.

284. The flow in a constant area duct with friction in the absence of work and heat transfer is :

(a) Fanno flow (b) Iso thermal flow
(c) Rayleigh flow (d) Adiabatic flow

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Ans. (a) :

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285. Flow in a constant area duct with heat transfer is

(a) Rayleigh flow (b) Isentropic flow
(c) Fanno flow (d) Reversible flow

TNPSC AE 2013

Ans. (a) : Flow in a constant area duct with heat transfer is known as Rayleigh flow.

286. What type of valve is a foot valve?

(a) Gate valve
(b) Ball valve
(c) Non return or one-way type of valve
(d) Butterfly valve

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : Foot valve is a Non return or one way type of valve. Foot valve is a lift check valve which is usually installed together with suction strainers in the suction line. It prevents the suction line from running empty after the pump is stopped.

287. Which valve has variable positioning capability?

(a) Solenoid valve
(b) Pressure reducing valve
(c) Needle valve
(d) Servo valve

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Servo valve has a variable positioning capability.

288. In case of dual piston type limited rotary actuator, the linear motion of piston is converted into rotary motion by the.

(a) Use of single cylinder mechanism
(b) Use of rack and pinion mechanism
(c) Use of vanes mechanism
(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : In case of dual piston type limited rotary actuator, the linear motion of piston is converted into rotary motion by the use of rack and pinion mechanism.

289. The telescopic cylinder has a series of…

(a) Gears (b) Pistons
(c) Rams (d) Vanes

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : Telescopic Cylinder–Telescopic cylinders are designed with a series of steel or aluminum tubes of progressively smaller diameter nested within each other. The largest diameter sleeve is called the main or barrel. The smaller inner sleeves are called the stages. The smallest stage is often called the plunger or ram or piston rod.

A typical application of a telescopic cylinder is dump trucks, where a larger force is required to move initially the load and as the load moves up force requirement goes on reducing.

290. In dynamic mounting cylinder has.

(a) Rotary freedom (b) Linear freedom
(c) Angular freedom (d) All freedom

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (c) : In dynamic mounting cylinder has angular freedom.

291. An actuator that gives linear motion as the output, is called as-

(a) Stripper (b) Cylinder or Ram.
(c) Regenerative motor (d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : An actuator that gives linear motion as the output, is called as cylinder or ram. It is a mover. An actuator requires a control signal and a source of energy.

292. 100 tonne press means

(a) the total weight of the press is 100 tonnes
(b) turn-over of pressed jobs per day is 100 tonnes
(c) the press can exert pressure upto 100 tonnes
(d) the foundation of the press can withstand 100 tonnes.

APPSC Poly. Lect. 2013

Ans. (c) : 100 tonne press means the press can exert pressure upto 100 tonnes.

293. Which of the following errors are regularly repetitive in nature?

(a) Systematic errors (b) Random errors
(c) Illegitimate errors (d) Controllable errors

APPSC Poly. Lect. 2013

Ans. (a) : Systematic error are regular and repetitive in nature.

294. Shock strength may be expressed as

(a) $\frac{p_2}{p_1}-1$ (b) $\frac{p_1}{p_2}-1$

(c) $1-\frac{p_2}{p_1}$ (d) $1-\frac{p_1}{p_2}$

APPSC Poly. Lect. 2013

Ans. (a) : Shock strength $(\xi) = \frac{p_2}{p_1}-1$

295. Policies in the management is used for

(a) Decision making
(b) Allocation of resources
(c) Man power management
(d) Budgeting

TNPSC AE 2013

Ans. (a) : Policies in the management is used for decision making.

296. The person who focuses on innovation and creativity and who transforms an idea into a profitable venture by operating within the organizational environment is called

(a) Intrapreneur (b) Entrepreneur
(c) Business man (d) Venture capitalist

TNPSC AE 2013

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Ans. (a) : The person who focuses on innovation and creativity and who transforms an idea into a profitable venture by operating within the organizational environment is called intrapreneur.

297. The 'Scalar Principles' in organization is

(a) The flow of authority from top to bottom
(b) The flow of suggestion from top to bottom
(c) Functional organization
(d) Flow of instructions, informations to horizontal level

TNPSC AE 2013

Ans. (a) : The 'Scalar Principles' in organization is the flow of authority from top to bottom.

298. Which of the following is called Siamese Twins of management?

(a) Planning and Organizing
(b) Planning and Controlling
(c) Planning and Staffing
(d) Planning and Leading

TNPSC AE 2013

Ans. (b) : Planning and Controlling is called Siamese Twins of management.

299. Who was referred as "The Father of the Social System Approach" in management?

(a) Vilfredo Pareto (b) Elton Mayo
(c) Frederick W. Taylor (d) Henry Fayol

TNPSC AE 2013

Ans. (a) : Vilfredo Pareto referred as "The Father of the Social System Approach" in management.

300. Democratic Leader

(a) give rewards and punishment
(b) consults with subordinates on proposed actions and decisions
(c) gives subordinates a high degree of independence in their operations
(d) depend largely on subordinates

TNPSC AE 2013

Ans. (b) : Democratic Leader consults with subordinates on proposed actions and decisions.

301. The budget in which the goals and programs are started from the scratch is

(a) Zero base budget (b) Variable budget
(c) Flexible budget (d) Program budget

TNPSC AE 2013

Ans. (a) : The budget in which the goals and programs are started from the scratch is zero base budget.

302. Equal Pay Act (1963) states that

(a) Equal pay of equal work regardless of sex
(b) Equal pay of equal work based on sex
(c) Equal pay based on races, colours, religions, sex
(d) pay based on origin

TNPSC AE 2013

Ans. (a) : Equal Pay Act (1963) states that equal pay of equal work regardless of sex.

303. The only thing that can keep you safe consistently is

(a) personal protective equipment
(b) your supervisor
(c) obeying all warning signs
(d) your attitude

VIZAG STEEL MT 18.08.2013

Ans. (d) : Only thing that can keep you safe constantly is – your attitude.

304. Foam is used as fire extinguisher in case of

(a) class A fire (b) class B fire
(c) class C fire (d) ordinary fire

VIZAG STEEL MT 18.08.2013

Ans. (b) : Class B fire– From fire extinguishers can be used on class A and B fire. The are most suited to extinguishing the liquid fires such has petrol or diesel and are more versatile than water jet extinguishers because they can also be used on solids such as wood and paper.

305. How many times the logarithm of the ratio of sound pressure in air to the reference sound pressure is the sound pressure level in decibels?

(a) 10 (b) 20
(c) 30 (d) 40

VIZAG STEEL MT 18.08.2013

Ans. (b) : Sound Pressure level– Sound pressure (p) is the average variation in atmospheric pressure caused by the sound pressure level (SPL) is the pressure level as a sound, measured in decibels (dB).

$$SPL = 20.\log_{10}\frac{P}{P_0}$$

It. is 20 times the logarithmic of the ratio of given sound pressure to the reference sound pressure in decibels (dB).

306. Circuit breakers cut off the main supply immediately during

(a) short circuit (b) low voltage supply
(c) spikes (d) load shedding

VIZAG STEEL MT 18.08.2013

Ans. (a) : A short circuit is caused when a "hot" wire comes into contact with a "'neutral wire" in one of your electrical outlets. When ever this happens, a large amount of current will flow through the circuit, creating more heat than what the circuit can handle. When this happen the breaker will trip, shutting of the circuit to prevent dangerous event such as fire.

307. In an air respirator mask, the air supplied to the face should be at least

(a) 90 litre of air/min (b) 200 litre of air/min
(c) 110 litre of air/min (d) 350 litre of air/min

VIZAG STEEL MT 18.08.2013

Ans. (c) : 110 litre of air/min

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308. One kgf/cm^2 when converted into SI units is :

(a) 0.0981 MPa (b) 0.981 Pa
(c) 10^4 Pa (d) 1 Pa

APPSC AE 04.12.2012

Ans. (a) : In SI units -
1 kgf/cm^2 = 0.0981 MPa

309. Consider the following statements

A. C++ is object oriented programming like Java

B. C++ uses the same set of programming concepts as those of C

Which of the statements given above is/are correct?

(a) A only (b) B only
(c) Both A and B (d) Neither A nor B

APPSC AEE 2012

Ans. (b) : C++ uses the same set of programming concepts as those of C.

310. A device used for lifting or lowering objects suspended from a hook at the end of retractable chains or cable is called

(a) Hoist (b) Jib crane
(c) Portable elevator (d) Chain conveyor

TNPSC ACF 2012

Ans. (a) : Hoist is used for lifting or lowering oboists suspended from a hook at the end of retractable chain or cable is called Hoist.

311. Thin cylinders are frequently required to operate under pressures upto

(a) 5 MN/m^2 (b) 15 MN/m^2
(c) 30 MN/m^2 (d) 250 MN/m^2

APPSC AEE 2012

Ans. (c) : Thin cylinders are frequently required to operate under pressure upto 30 MN/m^2 or more, for high pressure such as 250 MN/m^2 or more, the thick cylinders are used.

312. Scarce items are

(a) mostly available in local market
(b) of short supply
(c) semi finished items
(d) none of these

TRB Poly. Lect., 2012

Ans. (b) : Scarce item are limited availability. Scarce goods are those for which the demand would be greater than the supply if their price were zero.

313. One angstrom unit is equal to

(a) 10^{-6} cm (b) 10^{-8} cm
(c) 10^{-4} cm (d) 10^{-10} cm

TRB Poly. Lect., 2012

Ans. (b) : One Angstrom (Å) is equal to (It is unit of Length)

$\rightarrow \frac{1\text{Å}}{\text{meters}} : 10^{-10}$

$\rightarrow \text{cm} : 10^{-8}\,\text{cm}$

314. Simulation is basically a technique which is used for

(a) optimization (b) testing alternatives
(c) cost minimization (d) all of the above

TRB Asstt. Prof., 2012

Ans. (d) :

315. RAM stands for

(a) Network of computers
(b) Software that makes the computer work
(c) Memory used for processing
(d) Memory in which user cannot write anything

APPSC AEE 2012

Ans. (d) : RAM– Random access memory. in memory which user cannot write anything this is computer access memory and it is volatile after shutting the computer.

316. Two rigid plates are clamped by means of bolt and nut with an initial force N. After tightening, a separating force P (P<N) is applied to the lower plate which in turn acts on nut. The tension in the bolt after this will be

(a) (N+P) (b) (N–P)
(c) P (d) N

APPSC AEE 2012

Ans. (a) : Since plate is rigid so tension will be greater than N. So answer should be (N + p).

317. Schlieren flow visualisation technique operates by employing

(a) density variation in flow
(b) density gradients in flow
(c) second derivatives of density
(d) higher derivatives of density

APPSC AEE 2012

Ans. (a) : Schlieren photography similar to the shadowgraph technique and relies on the fact that light rays are best whenever they encounter changes in density of a fluid.

318. The transfer function is the operational relationship of the output and the :

(a) Command (b) Response
(c) Input (d) Error

OPSC Civil Services Pre. 2011

Ans. (a) : The transfer function is the operational relationship of the output and the command.

319. A block diagram is a symbolic outline of a system in which various components or operations are represented by ..... in an ordered sequence.

(a) Circles (b) Rectangular
(c) Triangles (d) Parallelograms

OPSC Civil Services Pre. 2011

Ans. (b) : A block diagram is a symbolic outline of a system in which various components or operations are represented by Rectangular in an ordered sequence.

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320. Which of the following would most commentators regard as one of the 'quality gurus' ?

(a) Joseph Juran (b) Charles Handy
(c) Henri Gantt (d) Joe Pine

OPSC Civil Services Pre. 2011

Ans. (a) : Joseph Juran is the most commentators regard as one of the 'quality gurus'.

321. Depreciation charges are high in case of :

(a) thermal plant (b) diesel plant
(c) hydroelectric plant (d) gas turbine plant

RPSC ACF-2011

Ans. (a) : Depreciation is treated as a revenue loss while is recorded when utility fixed assets such as plant and machinery, building and equipment etc.

322. A 30 ton press means

(a) gross weight of the press is 30 tons
(b) weight of die is 30 tons
(c) pressure exerted by slide is 30 tons
(d) flywheel of the press weighs 30 tons

RPSC ACF-2011

Ans. (c) : A 30 ton press means pressure exerted by slide is 30 tons.

323. Current assets include

(a) manufacturing plant
(b) manufacturing plant and equipment
(c) inventories
(d) common stock held by the company

RPSC ACF-2011

Ans. (d) : Current- assets are on asset that a company holds and can be easily sold or consumed and the further lead to the conversion of liquid cash.

Current assets is a–

- Cash and cash equivalent
- Inventory ongoing projects
- Pre-paid expense
- Account receivable
- Marketable securities.

324. According to McGregor's theory of management :

(a) all managers are complex in behavior and it is difficult to get work done
(b) X managers in a organization will not work and Y managers will work. The ratio X/Y depends on the organizational set up
(c) the ratio X/Y is always 0.5 for any organization
(d) X theory managers presume that the average human being has an inherent dislike of work and will avoid if it he can

RPSC ACF-2011

Ans. (d) : According to McGregor's, theory X management assumes the following

- Work is inherently distasteful to most people and they will attempt to avoid work whenever possible.
- Most people are not ambitions, have little desire for responsibility and prefer to be directed.
- Most people are self centered. A result, they must be closely controlled and often coerced to achieve organized objectives.

325. A public sector undertaking

(a) is fully owned by public through shareholders
(b) is jointly owned by private parties
(c) is jointly owned by private parties and government
(d) fully owned by government

RPSC ACF-2011

Ans. (c) : A public sector undertaking is jointly owned by private parties and government.

326. Salvaging means :

(a) writing off the assets
(b) throwing away the assets
(c) selling the assets
(d) disposing off property which is no longer useful in present situation

RPSC ACF-2011

Ans. (d) : Salvage value is the estimated book value of an asset after depreciation is complete, based on what a company expects to receive in exchange for the asset at the end of its useful life. As such, an assets estimated salvage value is an important component in the calculation of a deprecation schedule.

327. Capital gains are :

(a) unanticipated increase in income
(b) income through interest earned on investment
(c) income on dividends received from companies
(d) unanticipated changes in value of property relative to goods

RPSC ACF-2011

Ans. (b) : A capital gain is an increase in the value of an asset or investment resulting from the price appreciation of the asset or investment. In other words, the gain occurs when the current or sale price of an asset or investment exceeds its purchase price. Capital gains are attributable to all type of capital assets, including, but not limited to stock, bond goodwill and real estate.

328. The financial position of a company is considered to be sound if :

(a) sufficient funds are available under reserve and surplus
(b) company has declared good dividends in the past
(c) sales are increasing
(d) gross profit is reasonable

RPSC ACF-2011

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Ans. (d) : The financial position of a company is measured by the performance it takes in company financial statement a positive and growing cash flow statement, growing profit in the profit and loss statement and a balance of assets, liabilities and owner's equality in the balance sheet.

329. The law of demands states that

(a) when prices rise demand rises
(b) when prices fall demand rises
(c) when income rise demand rises
(d) when income and prices rise demand rises

RPSC ACF-2011

Ans. (b) : The law of demand state that other factors being constant price and quantity demand of any good and service are inversely related to each other. When the price of the product increases, the demand for the same product will fall.

330. Indicate the group of persons who gains from inflation :

(a) stock holders
(b) bond holders
(c) debenture holders
(d) fixed deposit holders

RPSC ACF-2011

Ans. (d) : Inflation is a measure of the rate of rising prices of goods and services in an economy. If inflation is occurring, leading to higher prices for basic necessities such as food, it can have a negative impact on society. Inflation is a measure of the rate of rising prices of goods and services in an economy. Inflation can occur when prices rise due to increases in production costs, such as row material and wages. Fixed income instruments include bonds and certificates of deposit (CP).

331. Economic development of a country depends more on :

(a) natural resources (b) capital formation
(c) availability of market (d) entrepreneurs

RPSC ACF-2011

Ans. (a) : Availability of natural resources and its proper utilization are considered as an important determinant of economic development. If the countries are rich in natural resources and adopted modern technology for its utilization, then they can attains higher level of development at a quicker pace.

332. Performance of a company can be judged by its:

(a) share capital (b) total production
(c) number of employees (d) profits

RPSC ACF-2011

Ans. (d) : Although ratio of analysis are used to analyze the company's past financial performance, but it also can assets management team judging the company's efficiency in terms of its operations and management. They help judge how well the company has been able to utilize its assets and earn profits.

333. During inflation :

(a) prices rise
(b) prices drop
(c) prices remain unchanged
(d) prices fluctuates heavily

RPSC ACF-2011

Ans. (a) : During inflation the price of the product increases, the demand for the same product will fall.

334. Budget is the major instrument of :

(a) fiscal policy (b) monetary policy
(c) economic policy (d) export policy

RPSC ACF-2011

Ans. (a) : In economics and political science, fiscal policy is the use of government budget or revenue collection (taxation) and expenditure (spending) to influence economic. The two main instruments of fiscal policy are government taxation and expenditure.

335. Devaluation affects imports by making it :

(a) costlier (b) cheaper
(c) competitive (d) prohibitive

RPSC ACF-2011

Ans. (b) : The main effects are experts are cheaper to foreign customers. Imports more expensive. In the short-term a devaluation tends to cause of inflation, higher growth and increased demand for experts.

336. Law of demands correlates

(a) quality and quantity
(b) quality and price
(c) quantity and price
(d) price and profit

RPSC ACF-2011

Ans. (c) : The law of demand is an economic principle that explains the negative co-relation between the price of a good or service and its demand. If all other factors remain the same, when the price of a good or service increases the quantity of demand decreases and vice-versa.

337. The term "balance of trade" is associated with:

(a) imports and exports
(b) taxation
(c) gross profits of a company
(d) budgeting

RPSC ACF-2011

Ans. (a) : The balance of trade (BOT) also known as the trade balance refers to the difference between the monetary value of a country's imports and exports over a given time period. A positive trade balance indicates a trade surplus while a negative trade balance indicates a trade deficit.

338. Mixed economy is identified by:

(a) lack of competition
(b) coexistence of public and private sectors
(c) vigorous competition
(d) rising trend in prices

RPSC ACF-2011

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Ans. (b) : A mixed economy is variously defined as economic system blending elements of a market economy with elements of a market economy with elements of a planned economy free markets with state interventationism or private enterprises with enterprises.

339. Which of the following is not a feature of developed economy?

(a) predominance of industries
(b) high per capita income
(c) predominance of indirect taxes
(d) high rate of capital formation.

RPSC ACF-2011

Ans. (d) : The capital formations really signify a very significant part of economic development. This earnings manufacturing and growth of more capital goods, such as machines tools, factories, buildings raw materials, fuels etc. Which are to be additionally used in producing more goods.

Capital formation implies the making of real assets. Low per capital real profits is one of the most defining qualities of developing economies.

340. To compete on speed, a company must

(a) produce a single product
(b) centralize all operations
(c) use only one supplier
(d) have quick feedback mechanisms

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (*) : Note- No answer is given by commission.

341. A product has MTBF of 200 hours and a MTTR of 10 hours. What is its system availability?

(a) .048 (b) .050
(c) .952 (d) 1.050

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : Given,

MTBF = 200 hours

MTTR = 10 hours

$$\text{Availability} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

$$= \frac{200}{200+10} = \frac{200}{210}$$

Availability = 0.952

Note- No answer is given by commission.

342. In decision theory, "expected value of perfect information" is the

(a) average expected payoff
(b) value of a decision given perfect information
(c) value of a decision without perfect information
(d) maximum value a decision maker is willing to pay to purchase perfect information

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : In decision theory "expected value of perfect information" is the maximum value a decision maker is willing to pay to purchase perfect information.

343. Which of the following criteria would a company that produces electronic equipment and components likely consider most important for location analysis?

(a) land and construction costs
(b) availability of skilled workers
(c) transportation costs
(d) proximity to raw materials

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Availability of skilled workers is the most important criteria which is likely to be considered by company producing electronic equipment and components for location analysis.

344. Which one of the following statements is not true for normal human behavior?

(a) Individual variables play a major role and influence the performance of an employee at work.
(b) The behaviour of individual and behaviour of individual as a member of a group is markedly different.
(c) Group cannot influence or change the attitude and behaviour of an individual.
(d) The feeling of uniqueness of isolated entity generates group cohesiveness

RPSC AE GWD, 2011

Ans. (d) :

345. Which of the following statements about the apprenticeship is incorrect?

(a) Apprentices are required to continue to participate in further education for the duration of their vocational studies.
(b) Apprentices combine class room teaching and on-the-job experience.
(c) Full apprenticeship last for two years.
(d) The system is one of the best routes to achieving vocational qualifications.

RPSC AE GWD, 2011

Ans. (c) : Full apprenticeship last for four year not two years.

346. Which one of the following is not an advantage of employee training?

(a) Increased productivity
(b) Increased supervision of employees
(c) Reduced scrap rate
(d) Less fatigue to workers.

RPSC AE GWD, 2011

Ans. (b) : Increased supervision of employees.

Advantage of employee training-

(1) Increased Productivity.
(2) Reduce scrap rate.
(3) Less fatigue to workers.

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347. Placement refers to :
(a) Getting a job after graduation
(b) Assigning a workplace site just after final selection.
(c) Assigning of specific job based on the performance and aptitude in the training.
(d) Shifting of person from one job to another.

RPSC AE GWD, 2011

Ans. (c) : Placement refers to assigning of specific job based on the performance and aptitude in the training.

348. Which one of the following is not a screening technique i.e. to eliminate the unqualified or unsuitable candidate?
(a) Screening of applications
(b) Preliminary interviews
(c) Physical examination
(d) Employment tests.

RPSC AE GWD, 2011

Ans. (d) :

349. Acid Test Ratio is used to supplement the information given by:
(a) Gross Profit Ratio (b) Net Profit Ratio
(c) Current Ratio (d) Debt Equity Ratio

RPSC AE GWD, 2011

Ans. (c) :

350. What is RPO ?
(a) Recruitment Process Output
(b) Recruitment Procedure Outsourcing
(c) Recruitment Process Outsourcing
(d) retirement Process Outsourcing

RPSC AE GWD, 2011

Ans. (c) : Recruitment process outsourcing (RPO) is a form of business process outsourcing (BPO) where an employer transfers all or part of its requirement process to an external service provider.

351. What is meant by a 'Golden hello' ?
(a) Cash lump sum promised to employees who stay on in the organization.
(b) Cash lump sum promised to people who refers a friend
(c) Cash lump sum offered to people leaving the organization
(d) Cash lump sum offered to new recruits

RPSC AE GWD, 2011

Ans. (d) : A golden hello is a signing bonus offered to executive level employees as an inducement to join from a rival company. The payment typically comes in the form of lamp-sum cash payment upon an employee entering the firm.

352. Which one of the following is not a characteristic feature of motivation?
(a) Motivation energizes, directs and sustain human behaviour.
(b) Motivation can be either positive or negative
(c) Individuals differ in their motivation.
(d) Motivation of each individual remains unaffected from time to time, once a head is satisfied.

RPSC AE GWD, 2011

Ans. (d) :

353. The nodal agency for coordinating the energy conservation activities under EC act in India is
(a) Bureau of Indian Standards
(b) Bureau of Energy Efficiency
(c) Bureau of Energy Education
(d) Bureau of Energy and Environment

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : BEE is agency of the government of India, under minister of power, the main function is to developed programme which will increase the conservation & efficient use of energy in India.

354. What type of energy is derived from heated ground water?
(a) Solar energy
(b) Geo-thermal energy
(c) Nuclear energy
(d) OTEC

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Geo-thermal is heat derived within the sub surface of the earth, which heated the ground water, these sources are known as Geo-thermal source of energy.

355. Important waste management techniques
(a) Disposal-biodegradable
(b) Recycling
(c) Landfills
(d) All of the above

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : Important waste management techniques
→ Land fills → Recycling
→ Incineration → Reuse
→ Bioremediation etc. → Biological Reprocessing

356. At constant efficiency horse power of a fan is :
(a) Proportional to r.p.m
(b) Proportional to $(r.p.m)^2$
(c) Proportional to $(r.p.m)^3$
(d) A polynomial function of r.p.m.

J&K PSC Civil Services Pre, 2010

Ans. (c) :

357. For lifting heavy jobs in a shop use is made of :
(a) Conveyors (b) Overhead Crane
(c) Fork Lift (d) Hoist

J&K PSC Civil Services Pre, 2010

Ans. (b) : A crane is a type of machine, generally equipped with a hoist, wire, ropes or chairs and sheaves, that can be used to lift and lower heavy materials and to move them horizontally. Different types that can be found in construction.

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358. Average height of a man in motion study is taken as :

(a) 1.2 m (b) 1.4 m
(c) 1.6 m (d) 1.8 m

J&K PSC Civil Services Pre, 2010

Ans. (b) : Average height of a man in motion study is 1.4 m.

359. For traction applications, the type of motor best suited is

(a) Induction motor
(b) Synchronous motor
(c) DC shunt motor
(d) DC series motor

(KPSC AE. 2015)
ISRO Scientist/Engineer 2010

Ans. (d) : DC series motor have excellent starting torque and best suited for traction application.

360. Across the rotor of an axial-flow reaction gas turbine :

(a) absolute velocity decreases
(b) relative velocity decreases
(c) absolute velocity increases
(d) absolute velocity remains constant

DRDO Scientists 2009

Ans. (a) : In axial flow gas turbine,
Absolute velocity decreases across the rotor.

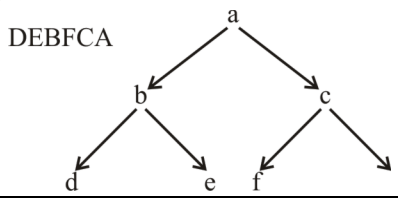
361. Find the post order traversal of binary tree from given preorder and in-order.
Preorder : ABDECF
Inorder : DBEACF

(a) DEBFCA (b) ACFBED
(c) FCADEB (d) EBFCDA

APPSC IOF, 2009

Ans. (a) :

DEBFCA



362.
```
#define SQR(x) x * x
#define CUBE(x) SQR(x) * x
main( )
{
printf("%d\n", CUBE(3+5));
}
```
the above program results :

(a) 512 (b) 64
(c) 38 (d) 48

APPSC IOF, 2009

Ans. (c) : Step 1, int a, b = 3;
a = SQR (b + 2);
a = 3 + 5 * 3 + 5 * 3 + 5;
a = 3 + 15 + 15 + 5 $\Rightarrow$ a = 38

363.
```
extern int a = 25;
    printf("%d\n", a);
```
(a) 0 (b) 25
(c) runtime error (d) compile time error

APPSC IOF, 2009

Ans. (b) : Extern int a = 25;
printf("%d\n".a);
$\Rightarrow$ 25

364. The main features of a programming language is:

(a) readability (b) writability
(c) portability (d) all of the above

APPSC IOF, 2009

Ans. (d) : The main features of a programming language is readability, writability and portability.

365. In FORTRAN an integer variable must starts with :

(a) PQRS (b) IJKLMN
(c) Both (a) and (b) (d) None of the above

APPSC IOF, 2009

Ans. (b) : Variables with names starting I, J, K, L, M and N are integer variables.

366. dBase is a ..... language.

(a) DBMS (b) RDBMS
(c) Both (a) and (b) (d) None of the above

APPSC IOF, 2009

Ans. (c) : dBase is a **R**elational DBMS which itself is a most popular type of DBMS in the market.

367. Cache memory performance will be measured with :

(a) No of Hits
(b) No of Miss
(c) (a) and (b)
(d) $\frac{\text{Total No. of Hits}}{\text{Total No of Hits} + \text{Total No. of Miss}}$

APPSC IOF, 2009

Ans. (d) : The performance of cache memory is frequently measured in terms of a quality called hit ratio. We can improve cache performance using higher cache block size, higher associativity, reduce miss rate, reduce miss.

368. FORTRAN executable statement must start from which column ?

(a) 4 (b) 7
(c) 8 (d) 12

APPSC IOF, 2009

Ans. (b) : Valid FORTRAN statements must begin in column 7 or beyond, cannot extend past column 7.

369. Purpose of linker is :

(a) to check errors
(b) to generate object code
(c) to generate executable object code
(d) none of the above

APPSC IOF, 2009

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Ans. (c) : Linker are called link editors, linking is process of collecting and maintaining piece code and data into a signal life. It takes object modules from assembler as input and forms an executable file as O/P for loader.

370. Which one of the following is NOT an advantage of transfer machine?

(a) Higher production rate
(b) Higher accuracy
(c) Lesser floor space
(d) Low initial investment

UPSC JWM Adv. No-16/2009

Ans. (d) : Low initial investment is Not an advantage of transfer machine.

371. Thermal expansion of solid is employed in

(a) thermocouple
(b) resistance thermometer
(c) bimetal element
(d) zener diode

TNPSC AE, 2008

Ans. (c) : Thermal expansion of solid is employed in bimetal element. The bimetallic strips may be in form of simple straight, helical or spiral. A bimetallic strip consists of two metal strips welded together, each strip mode from a metal having different co-efficient of thermal expansion.

372. Match List I correctly with List II and select your answer using the codes given below :

	List I		List II
(a)	Density	1.	Mass-spring seismic sensor
(b)	Power	2.	Anemometer
(c)	Air flow	3.	Dynamometer
(d)	Acceleration	4.	Resonant elements

	a	b	c	d
(a)	2	1	3	4
(b)	2	3	1	4
(c)	4	2	3	1
(d)	4	3	2	1

TNPSC AE, 2008

Ans. (d) : Density – Resonant elements
Power – Dynamometer
Air flow – Anemometer
Acceleration – Mass-spring seismic sensor

373. In the area of instrumentation and control, PID control stands for

(a) Proportional plus integral plus derivative control
(b) Pneumatic instruments differential control
(c) Pneumatically induced direct control
(d) Poly-inductance division control

TNPSC AE, 2008

Ans. (a) : PID stands for proportional, integral, derivative PID control provides a continuous variation of output within a control loop feedback mechanism to accurately control the process, removing oscillation and increasing process efficiency.

374. When a continuous linear relationship exists between the position of the final control element and the value of the controlled variable, the controller action is

(a) floating (b) differential
(c) integral (d) proportional

TNPSC AE, 2008

Ans. (d) : In the proportional (throttling) mode there is a continuous linear relation between value of the controlled variable and position of the final control element. In this control mode, the output of the controller output p is determined by a constant called proportional gain constant denoted as k_p. The output of the controller is a linear function of e(t).

$$p(t) = k_p\, e(t) + p(0)$$

Where, k_p = Proportional gain constant
$p(0)$ = Controller output with zero error or bias

375. Reference plane is a plane which is

(a) passing through the plane of rotation of the rotating weight
(b) passing through the plane of rotation of the balancing weight
(c) at an angle of 45^0 to the rotating weight
(d) at an angle of 45^0 to the balancing weight

TNPSC AE, 2008

Ans. (b) : Reference plane is a plane which is passing through the plane of rotation of the balancing weight. The Reference planes are the three orthogonal reference plane at the origin of the new part or assembly. They define the Top (XY) and Right (YZ) and Front (XZ) principal planes.

376. The first free trade zone in India was established at

(a) Cochin (b) Madras
(c) Bombay (d) Delhi

TNPSC AE, 2008

Ans. (c) : The first free trade zone in India was established at Bombay.

377. Gantry girders are invariably designed to resist-

(a) Transverse loads only
(b) Lateral loads only
(c) Transverse and lateral loads
(d) Transverse, lateral and axial load

ISRO Scientist/Engineer 2006

Ans. (c) : Gantry girders are used to provide support for overhead travelling cranes in environments such as wear houses and construction workshops. Gantry girders are used to provide support for cranes because of the various directions loads are exerted on the cranes

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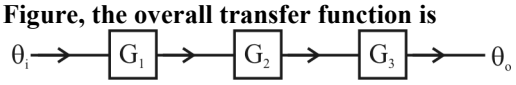
during the moving process for loads. Movement of heavy loads via girder cranes exerts vertical and horizontal load on the gantry girder, with the vertical load applied to the girder by the crane itself. Horizontal movements is exerted on the gantry girder by the surge, or acceleration and braking of the crane operation and the swinging of loads throughout the movement of loads.

378. For elements in series or cascade as shown in Figure, the overall transfer function is



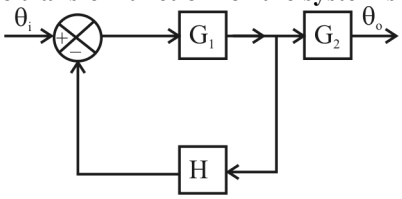
(a) $G_1 + G_2 + G_3$ (b) $G_1 \times G_2 \times G_3$

(c) G_1 (d) G_3

OPSC Civil Services Pre 2006

Ans. (b) : For series system, transfer function is product of transfer function of the individual series elements. For parallel system, over all transfer function is the algebraic sum of all the transfer function.

379. The transfer function for the system shown is



(a) $\dfrac{G_1G_2}{(1+G_1H)}$ (b) $\dfrac{G_1G_2}{(1-G_1H)}$

(c) $\dfrac{G_1}{(1+G_2H)}$ (d) $\dfrac{G_2}{(1+G_1H)}$

OPSC Civil Services Pre 2006

Ans. (a) : Over all transfer function $= \dfrac{G_1G_2}{(1+G_1H)}$

380. Two satellites, of masses m and 2m, are on the same circular orbit around earth. If the velocity of the lighter satellite is v_0, what is the velocity of the heavier satellite?

(a) $\dfrac{1}{2}v_0$ (b) v_0

(c) $2v_0$ (d) $\dfrac{1}{4}v_0$

ISRO Scientist/Engineer 2006

Ans. (b) : Centrifugal force and Gravitational force both are acting on satellites while orbiting around the earth.

For equilibrium condition,

Centrifugal force (F_c) = Gravitational force

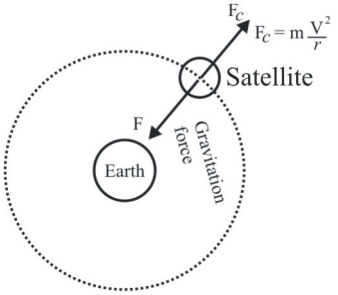
$$m\frac{V_0^2}{r} = \frac{G.Mm}{r^2}$$

$$V_0 = \sqrt{\frac{Gm}{r}}$$

$$V_0 \propto \sqrt{\frac{1}{r}} \qquad \text{....(i)}$$

where m is mass of satellite
M is mass of earth
r is radius of orbit
G is universal gravitational constant

So, Form eqn. (i) we can say that radius of orbit is same for both satellites and there is no effect of change of mass of satellite on speed. So speed will be same for both satellites.



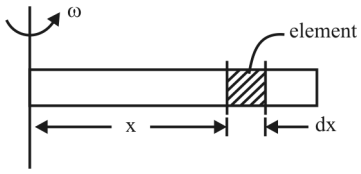
381. A tube of length L is filled completely with an incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocityω. The force exerted by the liquid at the other end is-

(a) $M\omega^2L$ (b) $\dfrac{M\omega^2L}{4}$

(c) $\dfrac{M\omega^2L}{2}$ (d) $\dfrac{M\omega^2L}{16}$

WBPSC AE 2003

Ans. (c) :



take a small element of mass 'm'

$$\therefore \quad m = \frac{Mdx}{L}$$

$$\text{Force} \Rightarrow \quad dF = \left(\frac{Mdx}{L}\right) \times x\omega^2$$

$$\int_0^L dF = \int_0^L \frac{Mdx}{L} \times x\omega^2$$

$$F = \frac{M\omega^2}{L} \times \int_0^L x\,dx$$

$$F = \frac{M\omega^2}{L}\left[\frac{x^2}{2}\right]_0^L$$

$$F = \frac{M\omega^2}{L} \times \frac{L^2}{2}$$

$$F = \frac{M\omega^2}{2} \times L$$

$$\boxed{F = \frac{M\omega^2L}{2}}$$

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